



Vulkan[®] 1.0.301 - A Specification (with all ratified extensions)

The Khronos[®] Vulkan Working Group

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Table of Contents

1. Preamble	1
2. Introduction	3
2.1. Document Conventions	3
3. Fundamentals	6
3.1. Host and Device Environment	6
3.2. Execution Model	6
3.3. Object Model	8
3.4. Application Binary Interface	12
3.5. Command Syntax and Duration	13
3.6. Threading Behavior	16
3.7. Valid Usage	31
3.8. VkResult Return Codes	38
3.9. Numeric Representation and Computation	42
3.10. Fixed-Point Data Conversions	44
3.11. String Representation	45
3.12. Common Object Types	46
3.13. API Name Aliases	76
4. Initialization	77
4.1. Command Function Pointers	77
4.2. Instances	80
5. Devices and Queues	89
5.1. Physical Devices	89
5.2. Devices	119
5.3. Queues	136
6. Command Buffers	143
6.1. Command Buffer Lifecycle	143
6.2. Command Pools	145
6.3. Command Buffer Allocation and Management	152
6.4. Command Buffer Recording	156
6.5. Command Buffer Submission	165
6.6. Queue Forward Progress	187
6.7. Secondary Command Buffer Execution	188
6.8. Nested Command Buffers	194
6.9. Command Buffer Device Mask	195
7. Synchronization and Cache Control	199
7.1. Execution and Memory Dependencies	199
7.2. Implicit Synchronization Guarantees	232
7.3. Fences	234

7.4. Semaphores	259
7.5. Events	288
7.6. Pipeline Barriers	313
7.7. Memory Barriers	323
7.8. Wait Idle Operations	374
7.9. Host Write Ordering Guarantees	376
7.10. Synchronization and Multiple Physical Devices	376
7.11. Calibrated Timestamps	377
8. Render Pass	381
8.1. Render Pass Objects	400
8.2. Render Pass Creation	401
8.3. Render Pass Compatibility	457
8.4. Framebuffers	457
8.5. Render Pass Load Operations	469
8.6. Render Pass Store Operations	470
8.7. Render Pass Multisample Resolve Operations	471
8.8. Render Pass Commands	473
8.9. Common Render Pass Data Races (Informative)	499
9. Shaders	502
9.1. Shader Objects	502
9.2. Shader Modules	522
9.3. Binding Shaders	526
9.4. Shader Execution	527
9.5. Shader Memory Access Ordering	528
9.6. Shader Inputs and Outputs	529
9.7. Vertex Shaders	529
9.8. Tessellation Control Shaders	529
9.9. Tessellation Evaluation Shaders	531
9.10. Geometry Shaders	531
9.11. Fragment Shaders	532
9.12. Compute Shaders	532
9.13. Ray Generation Shaders	532
9.14. Intersection Shaders	532
9.15. Any-Hit Shaders	533
9.16. Closest Hit Shaders	533
9.17. Miss Shaders	534
9.18. Callable Shaders	534
9.19. Interpolation Decorations	534
9.20. Static Use	535
9.21. Scope	535
9.22. Derivative Operations	539

9.23. Helper Invocations	541
9.24. Cooperative Matrices	541
10. Pipelines	546
10.1. Multiple Pipeline Creation	547
10.2. Compute Pipelines	548
10.3. Graphics Pipelines	557
10.4. Ray Tracing Pipelines	600
10.5. Pipeline Destruction	618
10.6. Pipeline Derivatives	619
10.7. Pipeline Cache	619
10.8. Pipeline Binaries	627
10.9. Specialization Constants	640
10.10. Pipeline Libraries	644
10.11. Pipeline Binding	645
10.12. Dynamic State	648
10.13. Pipeline Properties and Shader Information	648
11. Memory Allocation	659
11.1. Host Memory	659
11.2. Device Memory	666
12. Resource Creation	723
12.1. Buffers	723
12.2. Buffer Views	737
12.3. Images	741
12.4. Image Layouts	778
12.5. Image Views	784
12.6. Acceleration Structures	803
12.7. Micromaps	817
12.8. Resource Memory Association	826
12.9. Resource Sharing Mode	858
12.10. Memory Aliasing	860
13. Samplers	863
13.1. Sampler Y'C _B C _R Conversion	873
14. Resource Descriptors	884
14.1. Descriptor Types	884
14.2. Descriptor Sets	888
14.3. Physical Storage Buffer Access	964
15. Shader Interfaces	968
15.1. Shader Input and Output Interfaces	968
15.2. Vertex Input Interface	972
15.3. Fragment Output Interface	972
15.4. Fragment Tile Image Interface	976

15.5. Fragment Input Attachment Interface	977
15.6. Ray Tracing Pipeline Interface	981
15.7. Shader Resource Interface	982
15.8. Built-In Variables	990
16. Image Operations	1019
16.1. Image Operations Overview	1019
16.2. Conversion Formulas	1022
16.3. Texel Input Operations	1024
16.4. Texel Output Operations	1041
16.5. Normalized Texel Coordinate Operations	1043
16.6. Unnormalized Texel Coordinate Operations	1049
16.7. Integer Texel Coordinate Operations	1051
16.8. Image Sample Operations	1052
16.9. Image Operation Steps	1056
16.10. Image Query Instructions	1056
17. Queries	1058
17.1. Query Pools	1058
17.2. Query Operation	1065
17.3. Occlusion Queries	1092
17.4. Pipeline Statistics Queries	1093
17.5. Timestamp Queries	1095
17.6. Performance Queries	1101
17.7. Transform Feedback Queries	1105
17.8. Primitives Generated Queries	1105
17.9. Result Status Queries	1106
17.10. Video Encode Feedback Queries	1106
18. Clear Commands	1109
18.1. Clearing Images Outside a Render Pass Instance	1109
18.2. Clearing Images Inside a Render Pass Instance	1114
18.3. Clear Values	1118
18.4. Filling Buffers	1120
18.5. Updating Buffers	1122
19. Copy Commands	1125
19.1. Copying Data Between Buffers	1125
19.2. Copying Data Between Images	1131
19.3. Copying Data Between Buffers and Images	1149
19.4. Image Copies With Scaling	1191
19.5. Resolving Multisample Images	1206
20. Drawing Commands	1218
20.1. Primitive Topologies	1221
20.2. Primitive Order	1232

20.3. Programmable Primitive Shading	1233
21. Fixed-Function Vertex Processing	1506
21.1. Vertex Attributes	1506
21.2. Vertex Input Description	1510
21.3. Vertex Attribute Divisor in Instanced Rendering	1523
21.4. Vertex Input Address Calculation	1525
22. Tessellation	1527
22.1. Tessellator	1527
22.2. Tessellator Patch Discard	1530
22.3. Tessellator Spacing	1531
22.4. Tessellation Primitive Ordering	1531
22.5. Tessellator Vertex Winding Order	1532
22.6. Triangle Tessellation	1532
22.7. Quad Tessellation	1534
22.8. Isoline Tessellation	1536
22.9. Tessellation Point Mode	1536
22.10. Tessellation Pipeline State	1536
23. Geometry Shading	1541
23.1. Geometry Shader Input Primitives	1541
23.2. Geometry Shader Output Primitives	1542
23.3. Multiple Invocations of Geometry Shaders	1542
23.4. Geometry Shader Primitive Ordering	1542
24. Fixed-Function Vertex Post-Processing	1543
24.1. Transform Feedback	1543
24.2. Flat Shading	1552
24.3. Primitive Clipping	1555
24.4. Clipping Shader Outputs	1562
24.5. Coordinate Transformations	1562
24.6. Controlling the Viewport	1563
25. Rasterization	1573
25.1. Discarding Primitives Before Rasterization	1578
25.2. Controlling the Vertex Stream Used for Rasterization	1579
25.3. Rasterization Order	1582
25.4. Multisampling	1582
25.5. Custom Sample Locations	1588
25.6. Fragment Shading Rates	1593
25.7. Sample Shading	1601
25.8. Barycentric Interpolation	1601
25.9. Points	1604
25.10. Line Segments	1604
25.11. Polygons	1617

26. Fragment Operations	1638
26.1. Discard Rectangles Test	1639
26.2. Scissor Test	1645
26.3. Sample Mask Test	1647
26.4. Fragment Shading	1649
26.5. Multisample Coverage	1653
26.6. Depth and Stencil Operations	1656
26.7. Depth Bounds Test	1658
26.8. Stencil Test	1661
26.9. Depth Test	1670
26.10. Sample Counting	1678
26.11. Coverage Reduction	1678
27. The Framebuffer	1679
27.1. Blending	1679
27.2. Logical Operations	1707
27.3. Color Write Mask	1712
27.4. Color Write Enable	1713
28. Dispatching Commands	1716
29. Device-Generated Commands	1736
29.1. Indirect Commands Layout	1736
29.2. Indirect Commands Generation and Execution	1756
30. Sparse Resources	1810
30.1. Sparse Resource Features	1810
30.2. Sparse Buffers and Fully-Resident Images	1811
30.3. Sparse Partially-Resident Buffers	1812
30.4. Sparse Partially-Resident Images	1812
30.5. Sparse Memory Aliasing	1820
30.6. Sparse Resource Implementation Guidelines (Informative)	1821
30.7. Sparse Resource API	1822
31. Window System Integration (WSI)	1850
31.1. WSI Platform	1850
31.2. WSI Surface	1850
31.3. Presenting Directly to Display Devices	1864
31.4. Querying for WSI Support	1890
31.5. Surface Queries	1894
31.6. Full Screen Exclusive Control	1920
31.7. Device Group Queries	1923
31.8. Present Wait	1929
31.9. WSI Swapchain	1929
31.10. HDR Metadata	1970
32. Deferred Host Operations	1973

32.1. Requesting Deferral	1973
32.2. Deferred Host Operations API	1974
33. Acceleration Structures	1980
33.1. Acceleration Structures	1980
33.2. Host Acceleration Structure Operations	2030
34. Micromap	2046
34.1. Micromaps	2046
34.2. Host Micromap Operations	2065
35. Ray Traversal	2075
35.1. Ray Intersection Candidate Determination	2075
35.2. Ray Intersection Culling	2078
35.3. Ray Intersection Confirmation	2082
35.4. Ray Closest Hit Determination	2084
35.5. Ray Result Determination	2084
36. Ray Tracing	2085
36.1. Shader Call Instructions	2085
36.2. Ray Tracing Commands	2086
36.3. Shader Binding Table	2116
36.4. Ray Tracing Pipeline Stack	2118
36.5. Ray Tracing Capture Replay	2119
37. Video Coding	2120
37.1. Video Picture Resources	2120
37.2. Decoded Picture Buffer	2122
37.3. Video Profiles	2124
37.4. Video Capabilities	2134
37.5. Video Sessions	2143
37.6. Video Profile Compatibility	2155
37.7. Video Session Parameters	2157
37.8. Video Coding Scope	2177
37.9. Video Coding Control	2187
37.10. Inline Queries	2191
37.11. Video Decode Operations	2192
37.12. H.264 Decode Operations	2209
37.13. H.265 Decode Operations	2222
37.14. AV1 Decode Operations	2236
37.15. Video Encode Operations	2247
37.16. Video Encode Rate Control	2276
37.17. H.264 Encode Operations	2285
37.18. H.265 Encode Operations	2318
38. Extending Vulkan	2359
38.1. Instance and Device Functionality	2359

38.2. Core Versions	2359
38.3. Layers	2362
38.4. Extensions	2366
38.5. Extension Dependencies	2370
38.6. Compatibility Guarantees (Informative)	2371
39. Features	2376
39.1. Feature Requirements	2464
40. Limits	2469
40.1. Limit Requirements	2520
40.2. Additional Multisampling Capabilities	2535
41. Formats	2537
41.1. Format Definition	2537
41.2. Format Properties	2582
41.3. Required Format Support	2599
42. Additional Capabilities	2621
42.1. Additional Image Capabilities	2621
42.2. Additional Buffer Capabilities	2642
42.3. Optional Semaphore Capabilities	2645
42.4. Optional Fence Capabilities	2650
42.5. Timestamp Calibration Capabilities	2655
43. Debugging	2657
43.1. Debug Utilities	2659
43.2. Frame Boundary	2681
Appendix A: Vulkan Environment for SPIR-V	2684
Versions and Formats	2684
Capabilities	2684
Validation Rules Within a Module	2693
Precision and Operation of SPIR-V Instructions	2716
Signedness of SPIR-V Image Accesses	2722
Image Format and Type Matching	2723
Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types	2725
Compatibility Between SPIR-V Image Formats and Vulkan Formats	2726
Ray Query Precision and Operation	2727
Appendix B: Memory Model	2728
Agent	2728
Memory Location	2728
Allocation	2728
Memory Operation	2729
Reference	2729
Program-Order	2729
Shader Call Related	2730

Shader Call Order	2730
Scope.....	2730
Atomic Operation	2731
Scoped Modification Order	2731
Memory Semantics	2732
Release Sequence.....	2733
Synchronizes-With	2733
System-Synchronizes-With	2735
Private vs. Non-Private.....	2736
Inter-Thread-Happens-Before	2736
Happens-Before	2737
Availability and Visibility	2737
Availability, Visibility, and Domain Operations	2739
Availability and Visibility Semantics	2740
Per-Instruction Availability and Visibility Semantics.....	2740
Location-Ordered	2741
Data Race.....	2742
Visible-To	2742
Acyclicity	2742
Shader I/O	2743
Deallocation	2743
Descriptions (Informative)	2744
Tessellation Output Ordering	2744
Cooperative Matrix Memory Access	2745
Appendix C: Compressed Image Formats.....	2746
Block-Compressed Image Formats	2747
ETC Compressed Image Formats	2748
ASTC Compressed Image Formats	2749
Appendix D: Core Revisions (Informative)	2751
Vulkan Version 1.0.....	2751
Appendix E: Layers & Extensions (Informative)	2765
Extension Dependencies	2765
Extension Interactions	2765
List of Current Extensions.....	2766
List of Provisional Extensions	3272
List of Deprecated Extensions	3275
Appendix F: API Boilerplate	3278
Vulkan Header Files	3278
Window System-Specific Header Control (Informative)	3282
Provisional Extension Header Control (Informative).....	3284
Video Std Headers	3284

Appendix G: Invariance	3286
Repeatability	3286
Multi-Pass Algorithms	3286
Invariance Rules	3286
Tessellation Invariance	3288
Appendix H: Lexicon.....	3290
Glossary	3290
Common Abbreviations.....	3316
Video-Specific Abbreviations	3317
Prefixes	3319
Appendix I: Credits (Informative)	3321
Working Group Contributors to Vulkan.....	3321
Other Credits.....	3329

Chapter 1. Preamble

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This Specification contains substantially unmodified functionality from, and is a successor to, Khronos specifications including OpenGL, OpenGL ES and OpenCL.

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Some parts of this Specification are purely informative and so are EXCLUDED the Scope of this Specification. The [Document Conventions](#) section of the [Introduction](#) defines how these parts of the Specification are identified.

Where this Specification uses [technical terminology](#), defined in the [Glossary](#) or otherwise, that refer to enabling technologies that are not expressly set forth in this Specification, those enabling technologies are EXCLUDED from the Scope of this Specification. For clarity, enabling technologies not disclosed with particularity in this Specification (e.g. semiconductor manufacturing technology, hardware architecture, processor architecture or microarchitecture, memory architecture, compiler technology, object oriented technology, basic operating system technology, compression technology, algorithms, and so on) are NOT to be considered expressly set forth; only those application program interfaces and data structures disclosed with particularity are included in the Scope of this Specification.

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Chapter 2. Introduction

This document, referred to as the “Vulkan Specification” or just the “Specification” hereafter, describes the Vulkan Application Programming Interface (API). Vulkan is a [C99](#) API designed for explicit control of low-level graphics and compute functionality.

The canonical version of the Specification is available in the official [Vulkan Registry](https://registry.khronos.org/vulkan/) (<https://registry.khronos.org/vulkan/>). The source files used to generate the Vulkan specification are stored in the [Vulkan Documentation Repository](https://github.com/KhronosGroup/Vulkan-Docs) (<https://github.com/KhronosGroup/Vulkan-Docs>).

The source repository additionally has a public issue tracker and allows the submission of pull requests that improve the specification.

2.1. Document Conventions

The Vulkan specification is intended for use by both implementors of the API and application developers seeking to make use of the API, forming a contract between these parties. Specification text may address either party; typically the intended audience can be inferred from context, though some sections are defined to address only one of these parties. (For example, [Valid Usage](#) sections only address application developers). Any requirements, prohibitions, recommendations or options defined by [normative terminology](#) are imposed only on the audience of that text.

NOTE

Structure and enumerated types defined in extensions that were promoted to core in a later version of Vulkan are now defined in terms of the equivalent Vulkan core interfaces. This affects the Vulkan Specification, the Vulkan header files, and the corresponding XML Registry.

2.1.1. Ratification

Ratification of a Vulkan core version or extension is a status conferred by vote of the Khronos Board of Promoters, bringing that core version or extension under the umbrella of the Khronos IP Policy.

All Vulkan core versions and [KHR](#) extensions (including provisional specifications) are ratified, as are some multi-vendor [EXT](#) extensions. Ratification status of extensions is described in the [Layers & Extensions \(Informative\)](#) appendix.

NOTE

Ratification status is primarily of interest to IHVs developing GPU hardware and Vulkan implementations

For developers, ratification does not necessarily mean that an extension is “better”; has a more stable API; or is more widely supported than alternative ways of achieving that functionality.

Interactions between ratified and non-ratified extensions are not themselves ratified.

2.1.2. Informative Language

Some language in the specification is purely informative, intended to give background or suggestions to implementors or developers.

If an entire chapter or section contains only informative language, its title will be suffixed with “(Informative)”.

All NOTES are implicitly informative.

2.1.3. Normative Terminology

Within this specification, the key words **must**, **required**, **should**, **recommended**, **may**, and **optional** are to be interpreted as described in [RFC 2119 - Key words for use in RFCs to Indicate Requirement Levels](https://www.ietf.org/rfc/rfc2119.txt) (<https://www.ietf.org/rfc/rfc2119.txt>). The additional key word **optionally** is an alternate form of **optional**, for use where grammatically appropriate.

These key words are highlighted in the specification for clarity. In text addressing application developers, their use expresses requirements that apply to application behavior. In text addressing implementors, their use expresses requirements that apply to implementations.

In text addressing application developers, the additional key words **can** and **cannot** are to be interpreted as describing the capabilities of an application, as follows:

can

This word means that the application is able to perform the action described.

cannot

This word means that the API and/or the execution environment provide no mechanism through which the application can express or accomplish the action described.

These key words are never used in text addressing implementors.

NOTE

There is an important distinction between **cannot** and **must not**, as used in this Specification. **Cannot** means something the application literally is unable to express or accomplish through the API, while **must not** means something that the application is capable of expressing through the API, but that the consequences of doing so are undefined and potentially unrecoverable for the implementation (see [Valid Usage](#)).

Unless otherwise noted in the section heading, all sections and appendices in this document are normative.

2.1.4. Technical Terminology

The Vulkan Specification makes use of common engineering and graphics terms such as **Pipeline**, **Shader**, and **Host** to identify and describe Vulkan API constructs and their attributes, states, and behaviors. The [Glossary](#) defines the basic meanings of these terms in the context of the Specification. The Specification text provides fuller definitions of the terms and may elaborate,

extend, or clarify the [Glossary](#) definitions. When a term defined in the [Glossary](#) is used in normative language within the Specification, the definitions within the Specification govern and supersede any meanings the terms may have in other technical contexts (i.e. outside the Specification).

2.1.5. Normative References

References to external documents are considered normative references if the Specification uses any of the normative terms defined in [Normative Terminology](#) to refer to them or their requirements, either as a whole or in part.

The following documents are referenced by normative sections of the specification:

IEEE. August, 2008. *IEEE Standard for Floating-Point Arithmetic*. IEEE Std 754-2008. <https://dx.doi.org/10.1109/IEEESTD.2008.4610935> .

Andrew Garrard. *Khronos Data Format Specification, version 1.3*. <https://registry.khronos.org/DataFormat/specs/1.3/dataformat.1.3.html> .

John Kessenich. *SPIR-V Extended Instructions for GLSL, Version 1.00* (February 10, 2016). <https://registry.khronos.org/spir-v/> .

John Kessenich, Boaz Ouriel, and Raun Krisch. *SPIR-V Specification, Version 1.5, Revision 3, Unified* (April 24, 2020). <https://registry.khronos.org/spir-v/> .

ITU-T. *H.264 Advanced Video Coding for Generic Audiovisual Services* (August, 2021). <https://www.itu.int/rec/T-REC-H.264-202108-I/> .

ITU-T. *H.265 High Efficiency Video Coding* (August, 2021). <https://www.itu.int/rec/T-REC-H.265-202108-S/> .

Alliance for Open Media. *AV1 Bitstream & Decoding Process Specification* (January 8, 2019). <https://aomediacodec.github.io/av1-spec/av1-spec.pdf> .

Jon Leech. *The Khronos Vulkan API Registry* (February 26, 2023). <https://registry.khronos.org/vulkan/specs/1.3/registry.html> .

Jon Leech and Tobias Hector. *Vulkan Documentation and Extensions: Procedures and Conventions* (February 26, 2023). <https://registry.khronos.org/vulkan/specs/1.3/styleguide.html> .

Architecture of the Vulkan Loader Interfaces (October, 2021). <https://github.com/KhronosGroup/Vulkan-Loader/blob/main/docs/LoaderInterfaceArchitecture.md> .

Chapter 3. Fundamentals

This chapter introduces fundamental concepts including the Vulkan architecture and execution model, API syntax, queues, pipeline configurations, numeric representation, state and state queries, and the different types of objects and shaders. It provides a framework for interpreting more specific descriptions of commands and behavior in the remainder of the Specification.

3.1. Host and Device Environment

The Vulkan Specification assumes and requires: the following properties of the host environment with respect to Vulkan implementations:

- The host **must** have runtime support for 8, 16, 32 and 64-bit signed and unsigned twos-complement integers, all addressable at the granularity of their size in bytes.
- The host **must** have runtime support for 32- and 64-bit floating-point types satisfying the range and precision constraints in the [Floating-Point Computation](#) section.
- The representation and endianness of these types on the host **must** match the representation and endianness of the same types on every physical device supported.

NOTE

Since a variety of data types and structures in Vulkan **may** be accessible by both host and physical device operations, the implementation **should** be able to access such data efficiently in both paths in order to facilitate writing portable and performant applications.

3.2. Execution Model

This section outlines the execution model of a Vulkan system.

Vulkan exposes one or more *devices*, each of which exposes one or more *queues* which **may** process work asynchronously to one another. The set of queues supported by a device is partitioned into *families*. Each family supports one or more types of functionality and **may** contain multiple queues with similar characteristics. Queues within a single family are considered *compatible* with one another, and work produced for a family of queues **can** be executed on any queue within that family. This specification defines the following types of functionality that queues **may** support: graphics, compute, video decode, video encode, sparse memory management, and transfer.

NOTE

A single device **may** report multiple similar queue families rather than, or as well as, reporting multiple members of one or more of those families. This indicates that while members of those families have similar capabilities, they are *not* directly compatible with one another.

Device memory is explicitly managed by the application. Each device **may** advertise one or more heaps, representing different areas of memory. Memory heaps are either device-local or host-local, but are always visible to the device. Further detail about memory heaps is exposed via memory types available on that heap. Examples of memory areas that **may** be available on an implementation include:

- *device-local* is memory that is physically connected to the device.
- *device-local, host visible* is device-local memory that is visible to the host.
- *host-local, host visible* is memory that is local to the host and visible to the device and host.

On other architectures, there **may** only be a single heap that **can** be used for any purpose.

3.2.1. Queue Operation

Vulkan queues provide an interface to the execution engines of a device. Commands for these execution engines are recorded into command buffers ahead of execution time, and then submitted to a queue for execution. Once submitted to a queue, command buffers will begin and complete execution without further application intervention, though the order of this execution is dependent on a number of [implicit and explicit ordering constraints](#).

Work is submitted to queues using *queue submission commands* that typically take the form `vkQueue*` (e.g. `vkQueueSubmit`, `vkQueueBindSparse`), and **can** take a list of semaphores upon which to wait before work begins and a list of semaphores to signal once work has completed. The work itself, as well as signaling and waiting on the semaphores are all *queue operations*. Queue submission commands return control to the application once queue operations have been submitted - they do not wait for completion.

There are no implicit ordering constraints between queue operations on different queues, or between queues and the host, so these **may** operate in any order with respect to each other. Explicit ordering constraints between different queues or with the host **can** be expressed with [semaphores](#) and [fences](#).

Command buffer submissions to a single queue respect [submission order](#) and other [implicit ordering guarantees](#), but otherwise **may** overlap or execute out of order. Other types of batches and queue submissions against a single queue (e.g. [sparse memory binding](#)) have no implicit ordering constraints with any other queue submission or batch. Additional explicit ordering constraints between queue submissions and individual batches can be expressed with [semaphores](#) and [fences](#).

Before a fence or semaphore is signaled, it is guaranteed that any previously submitted queue operations have completed execution, and that memory writes from those queue operations are [available](#) to future queue operations. Waiting on a signaled semaphore or fence guarantees that previous writes that are available are also [visible](#) to subsequent commands.

Command buffer boundaries, both between primary command buffers of the same or different batches or submissions as well as between primary and secondary command buffers, do not introduce any additional ordering constraints. In other words, submitting the set of command buffers (which **can** include executing secondary command buffers) between any semaphore or fence operations execute the recorded commands as if they had all been recorded into a single primary command buffer, except that the current state is [reset](#) on each boundary. Explicit ordering constraints **can** be expressed with [explicit synchronization primitives](#).

There are a few [implicit ordering guarantees](#) between commands within a command buffer, but only covering a subset of execution. Additional explicit ordering constraints can be expressed with the various [explicit synchronization primitives](#).

NOTE

Implementations have significant freedom to overlap execution of work submitted to a queue, and this is common due to deep pipelining and parallelism in Vulkan devices.

Commands recorded in command buffers can perform actions, set state that persists across commands, synchronize other commands, or indirectly launch other commands, with some commands fulfilling several of these roles. The “Command Properties” section for each such command lists which of these roles the command takes:

Action

Action commands perform operations that can update values in memory. E.g. [draw commands](#), [dispatch commands](#).

State

State setting commands update the current state of a command buffer, affecting the operation of future action commands.

Synchronization

Synchronization commands impose ordering constraints on action commands, by introducing explicit [execution and memory dependencies](#).

Indirection

Indirection commands execute other commands which were not directly recorded in the same command buffer.

NOTE

In the absence of explicit synchronization or [implicit ordering guarantees](#), action commands may overlap execution or execute out of order, potentially leading to data races. However, such reordering does not affect the current state observed by any action command. Each action command uses the state in effect at the point where the command occurs in the command buffer, regardless of when it is executed.

3.3. Object Model

The devices, queues, and other entities in Vulkan are represented by Vulkan objects. At the API level, all objects are referred to by handles. There are two classes of handles, dispatchable and non-dispatchable. *Dispatchable* handle types are a pointer to an opaque type. This pointer **may** be used by layers as part of intercepting API commands, and thus each API command takes a dispatchable type as its first parameter. Each object of a dispatchable type **must** have a unique handle value during its lifetime.

Non-dispatchable handle types are a 64-bit integer type whose meaning is implementation-dependent. Non-dispatchable handles **may** encode object information directly in the handle rather than acting as a reference to an underlying object, and thus **may** not have unique handle values. If handle values are not unique, then destroying one such handle **must** not cause identical handles of other types to become invalid, and **must** not cause identical handles of the same type to become invalid if that handle value has been created more times than it has been destroyed.

All objects created or allocated from a `VkDevice` (i.e. with a `VkDevice` as the first parameter) are private to that device, and **must** not be used on other devices.

3.3.1. Object Lifetime

Objects are created or allocated by `vkCreate*` and `vkAllocate*` commands, respectively. Once an object is created or allocated, its “structure” is considered to be immutable, though the contents of certain object types is still free to change. Objects are destroyed or freed by `vkDestroy*` and `vkFree*` commands, respectively.

Objects that are allocated (rather than created) take resources from an existing pool object or memory heap, and when freed return resources to that pool or heap. While object creation and destruction are generally expected to be low-frequency occurrences during runtime, allocating and freeing objects **can** occur at high frequency. Pool objects help accommodate improved performance of the allocations and frees.

It is an application’s responsibility to track the lifetime of Vulkan objects, and not to destroy them while they are still in use.

The ownership of application-owned memory is immediately acquired by any Vulkan command it is passed into. Ownership of such memory **must** be released back to the application at the end of the duration of the command, unless that command was deferred, so that the application **can** alter or free this memory as soon as all the commands that acquired it have returned. If the command was `deferred`, ownership of such memory is released back to the application when the deferred operation is complete.

The following object types are consumed when they are passed into a Vulkan command and not further accessed by the objects they are used to create. They **must** not be destroyed in the duration of any API command they are passed into:

- `VkShaderModule`
- `VkPipelineCache`

A `VkRenderPass` or `VkPipelineLayout` object passed as a parameter to create another object is not further accessed by that object after the duration of the command it is passed into. A `VkRenderPass` used in a command buffer follows the rules described below.

`VkDescriptorSetLayout` objects **may** be accessed by commands that operate on descriptor sets allocated using that layout, and those descriptor sets **must** not be updated with `vkUpdateDescriptorSets` after the descriptor set layout has been destroyed. Otherwise, a `VkDescriptorSetLayout` object is no longer referenced by an API command it is passed into once host execution of that command completes.

The application **must** not destroy any other type of Vulkan object until all uses of that object by the device (such as via command buffer execution) have completed.

The following Vulkan objects **must** not be destroyed while any command buffers using the object are in the `pending state`:

- `VkEvent`

- `VkQueryPool`
- `VkBuffer`
- `VkBufferView`
- `VkImage`
- `VkImageView`
- `VkPipeline`
- `VkSampler`
- `VkSamplerYcbcrConversion`
- `VkDescriptorPool`
- `VkFramebuffer`
- `VkRenderPass`
- `VkCommandBuffer`
- `VkCommandPool`
- `VkDeviceMemory`
- `VkDescriptorSet`
- `VkIndirectCommandsLayoutEXT`
- `VkIndirectExecutionSetEXT`
- `VkAccelerationStructureKHR`
- `VkVideoSessionKHR`
- `VkVideoSessionParametersKHR`

Destroying these objects will move any command buffers that are in the [recording or executable state](#), and are using those objects, to the [invalid state](#).

The following Vulkan objects **must** not be destroyed while any queue is executing commands that use the object:

- `VkFence`
- `VkSemaphore`
- `VkCommandBuffer`
- `VkCommandPool`

In general, objects **can** be destroyed or freed in any order, even if the object being freed is involved in the use of another object (e.g. use of a resource in a view, use of a view in a descriptor set, use of a [pipeline library](#) in another pipeline, use of a [referenced object](#) for additional shader-related states in a set, use of a [bottom level acceleration structure](#) in an instance referenced by a [top level acceleration structure](#), use of an object in a command buffer, binding of a memory allocation to a resource), as long as any object that uses the freed object is not further used in any way except to be destroyed or to be reset in such a way that it no longer uses the other object (such as resetting a command buffer). If the object has been reset, then it **can** be used as if it never used the freed

object. An exception to this is when there is a parent/child relationship between objects. In this case, the application **must** not destroy a parent object before its children, except when the parent is explicitly defined to free its children when it is destroyed (e.g. for pool objects, as defined below).

`VkCommandPool` objects are parents of `VkCommandBuffer` objects. `VkDescriptorPool` objects are parents of `VkDescriptorSet` objects. `VkDevice` objects are parents of many object types (all that take a `VkDevice` as a parameter to their creation).

The following Vulkan objects have specific restrictions for when they **can** be destroyed:

- `VkQueue` objects **cannot** be explicitly destroyed. Instead, they are implicitly destroyed when the `VkDevice` object they are retrieved from is destroyed.
- Destroying a pool object implicitly frees all objects allocated from that pool. Specifically, destroying `VkCommandPool` frees all `VkCommandBuffer` objects that were allocated from it, and destroying `VkDescriptorPool` frees all `VkDescriptorSet` objects that were allocated from it.
- `VkDevice` objects **can** be destroyed when all `VkQueue` objects retrieved from them are idle, and all objects created from them have been destroyed.
 - This includes the following objects:
 - `VkFence`
 - `VkSemaphore`
 - `VkEvent`
 - `VkQueryPool`
 - `VkBuffer`
 - `VkBufferView`
 - `VkImage`
 - `VkImageView`
 - `VkShaderModule`
 - `VkPipelineCache`
 - `VkPipeline`
 - `VkPipelineLayout`
 - `VkSampler`
 - `VkSamplerYcbcrConversion`
 - `VkDescriptorSetLayout`
 - `VkDescriptorPool`
 - `VkFramebuffer`
 - `VkRenderPass`
 - `VkCommandPool`
 - `VkCommandBuffer`
 - `VkDeviceMemory`

- `VkAccelerationStructureKHR`
- `VkVideoSessionKHR`
- `VkVideoSessionParametersKHR`
- `VkPhysicalDevice` objects **cannot** be explicitly destroyed. Instead, they are implicitly destroyed when the `VkInstance` object they are retrieved from is destroyed.
- `VkInstance` objects **can** be destroyed once all `VkDevice` objects created from any of its `VkPhysicalDevice` objects have been destroyed.

3.3.2. External Object Handles

As defined above, the scope of object handles created or allocated from a `VkDevice` is limited to that logical device. Objects which are not in scope are said to be external. To bring an external object into scope, an external handle **must** be exported from the object in the source scope and imported into the destination scope.

NOTE

The scope of external handles and their associated resources **may** vary according to their type, but they **can** generally be shared across process and API boundaries.

3.4. Application Binary Interface

The mechanism by which Vulkan is made available to applications is platform- or implementation-defined. On many platforms the C interface described in this Specification is provided by a shared library. Since shared libraries can be changed independently of the applications that use them, they present particular compatibility challenges, and this Specification places some requirements on them.

Shared library implementations **must** use the default Application Binary Interface (ABI) of the standard C compiler for the platform, or provide customized API headers that cause application code to use the implementation's non-default ABI. An ABI in this context means the size, alignment, and layout of C data types; the procedure calling convention; and the naming convention for shared library symbols corresponding to C functions. Customizing the calling convention for a platform is usually accomplished by defining [calling convention macros](#) appropriately in `vk_platform.h`.

On platforms where Vulkan is provided as a shared library, library symbols beginning with “vk” and followed by a digit or uppercase letter are reserved for use by the implementation. Applications which use Vulkan **must** not provide definitions of these symbols. This allows the Vulkan shared library to be updated with additional symbols for new API versions or extensions without causing symbol conflicts with existing applications.

Shared library implementations **should** provide library symbols for commands in the highest version of this Specification they support, and for [Window System Integration](#) extensions relevant to the platform. They **may** also provide library symbols for commands defined by additional extensions.

NOTE

These requirements and recommendations are intended to allow implementors to take advantage of platform-specific conventions for SDKs, ABIs, library versioning

mechanisms, etc. while still minimizing the code changes necessary to port applications or libraries between platforms. Platform vendors, or providers of the *de facto* standard Vulkan shared library for a platform, are encouraged to document what symbols the shared library provides and how it will be versioned when new symbols are added.

Applications **should** only rely on shared library symbols for commands in the minimum core version required by the application. [vkGetInstanceProcAddr](#) and [vkGetDeviceProcAddr](#) **should** be used to obtain function pointers for commands in core versions beyond the application's minimum required version.

3.5. Command Syntax and Duration

The Specification describes Vulkan commands as functions or procedures using C99 syntax. Language bindings for other languages such as C++ and JavaScript **may** allow for stricter parameter passing, or object-oriented interfaces.

Vulkan uses the standard C types for the base type of scalar parameters (e.g. types from `<stdint.h>`), with exceptions described below, or elsewhere in the text when appropriate:

[VkBool32](#) represents boolean **True** and **False** values, since C does not have a sufficiently portable built-in boolean type:

```
// Provided by VK_VERSION_1_0
typedef uint32_t VkBool32;
```

VK_TRUE represents a boolean **True** (unsigned integer 1) value, and **VK_FALSE** a boolean **False** (unsigned integer 0) value.

All values returned from a Vulkan implementation in a [VkBool32](#) will be either **VK_TRUE** or **VK_FALSE**.

Applications **must** not pass any other values than **VK_TRUE** or **VK_FALSE** into a Vulkan implementation where a [VkBool32](#) is expected.

VK_TRUE is a constant representing a [VkBool32](#) **True** value.

```
#define VK_TRUE 1U
```

VK_FALSE is a constant representing a [VkBool32](#) **False** value.

```
#define VK_FALSE 0U
```

[VkDeviceSize](#) represents device memory size and offset values:

```
// Provided by VK_VERSION_1_0
typedef uint64_t VkDeviceSize;
```

[VkDeviceAddress](#) represents device buffer address values:

```
// Provided by VK_VERSION_1_0
typedef uint64_t VkDeviceAddress;
```

Commands that create Vulkan objects are of the form `vkCreate*` and take `Vk*CreateInfo` structures with the parameters needed to create the object. These Vulkan objects are destroyed with commands of the form `vkDestroy*`.

The last in-parameter to each command that creates or destroys a Vulkan object is `pAllocator`. The `pAllocator` parameter **can** be a non-NULL value, in which case allocations for the given object are delegated to an application provided callback. Refer to the [Memory Allocation](#) chapter for further details.

Commands that allocate Vulkan objects owned by pool objects are of the form `vkAllocate*`, and take `Vk*AllocateInfo` structures. These Vulkan objects are freed with commands of the form `vkFree*`. These objects do not take allocators; if host memory is needed, they will use the allocator that was specified when their parent pool was created.

Commands are recorded into a command buffer by calling API commands of the form `vkCmd*`. Each such command **may** have different restrictions on where it **can** be used: in a primary and/or secondary command buffer, inside and/or outside a render pass, and in one or more of the supported queue types. These restrictions are documented together with the definition of each such command.

The *duration* of a Vulkan command refers to the interval between calling the command and its return to the caller.

3.5.1. Lifetime of Retrieved Results

Information is retrieved from the implementation with commands of the form `vkGet*` and `vkEnumerate*`.

Unless otherwise specified for an individual command, the results are *invariant*; that is, they will remain unchanged when retrieved again by calling the same command with the same parameters, so long as those parameters themselves all remain valid.

3.5.2. Array Results

Some query commands of the form `vkGet*` and `vkEnumerate*` enable retrieving multiple results in the form of a return array. Such commands typically have two pointer arguments as follows:

- An element count pointer pointing to an integer variable, conventionally named as `p*Count` where `*` is the capitalized singular form of the name of the retrieved values.

- A pointer to an array where the result array is retrieved, conventionally named as **p*** where ***** is the capitalized plural form of the name of the retrieved values.

If such commands are called with the array pointer set to **NULL**, then the number of retrievable elements is returned in the variable pointed to by the element count pointer. Otherwise, the element count pointer **must** point to a variable set by the application to the number of elements in the return array, and on return the variable is overwritten with the number of elements actually written to the return array. If the input element count is less than the number of retrievable array elements, the query will write only as many elements to the return array as specified by the element count variable set by the application, and the command will return **VK_INCOMPLETE** instead of **VK_SUCCESS**, to indicate that not all retrievable array elements were returned.

NOTE

In practice, this means that applications will typically call such query commands twice:

- First, with the array pointer set to **NULL**, to retrieve the number of retrievable elements.
- Second, with the array pointer pointing to an application allocated storage for at least as many elements as indicated by the variable pointed to by the element count pointer, to retrieve at most as many of the retrievable elements.

Query commands that return one or more structures, regardless of whether they return a single or an array of structures with or without a **pNext** chain, **may** also contain arrays within those structures. Such return arrays are typically defined in the form of two members as follows:

- An integer value specifying the element count, conventionally named as ***Count** where ***** is the singular form of the name of the retrieved values.
- A pointer to an array where the result array is retrieved, conventionally named as **p*** where ***** is the capitalized plural form of the name of the retrieved values.

Analogously to query commands that return multiple results, if the command is called with the array pointer member of the output structure in question set to **NULL**, then the number of retrievable elements is returned in the element count member of that output structure. Otherwise, the element count **must** specify the number of elements in the return array, and on return the element count member is overwritten with the number of elements actually written to the return array. If the input element count is less than the number of retrievable array elements, the query will write only as many elements to the return array as specified by the input element count, and the command will return **VK_INCOMPLETE** instead of **VK_SUCCESS**, if the query command has a **VkResult** return type, to indicate that not all retrievable array elements were returned.

NOTE

Applications need to separately track the value they provided as the input element count member for such arrays and compare those with the returned element counts in order to determine whether the actually returned element count is smaller than the size of the return array. Another side effect of this is that it is impossible for the application to determine if the number of retrievable elements has increased beyond the provided input element count so using return arrays in output structures **should** be limited to *invariant* array results. In practice, this means that applications will typically call such query commands multiple times:

- First, with the array pointer member(s) set to **NULL**, to retrieve the number(s) of retrievable elements.
- Second, with the array pointer(s) pointing to an application allocated storage for at least as many elements as indicated by the element count member(s), to retrieve at most as many of the retrievable elements.
- Then the process **may** need to be repeated for all other newly introduced return arrays in any nested output structures indirectly specified through the previously retrieved result arrays.

Regardless of the type of query command, any array pointer member of an output structure **must** either be **NULL**, or point to an application-allocated array. Query commands **must** not return a pointer to implementation allocated storage in any output structure.

3.6. Threading Behavior

Vulkan is intended to provide scalable performance when used on multiple host threads. All commands support being called concurrently from multiple threads, but certain parameters, or components of parameters are defined to be *externally synchronized*. This means that the caller **must** guarantee that no more than one thread is using such a parameter at a given time.

More precisely, Vulkan commands use simple stores to update the state of Vulkan objects. A parameter declared as externally synchronized **may** have its contents updated at any time during the host execution of the command. If two commands operate on the same object and at least one of the commands declares the object to be externally synchronized, then the caller **must** guarantee not only that the commands do not execute simultaneously, but also that the two commands are separated by an appropriate memory barrier (if needed).

NOTE

Memory barriers are particularly relevant for hosts based on the ARM CPU architecture, which is more weakly ordered than many developers are accustomed to from x86/x64 programming. Fortunately, most higher-level synchronization primitives (like the pthread library) perform memory barriers as a part of mutual exclusion, so mutexing Vulkan objects via these primitives will have the desired effect.

Similarly the application **must** avoid any potential data hazard of application-owned memory that has its *ownership temporarily acquired* by a Vulkan command. While the ownership of application-owned memory remains acquired by a command the implementation **may** read the memory at any point, and it **may** write non-**const** qualified memory at any point. Parameters referring to non-**const** qualified application-owned memory are not marked explicitly as *externally synchronized* in the Specification.

If an application is using *deferred host operations* in a command, and that operation is successfully deferred, object parameters and application-owned memory passed to that command **may** be accessed at any time until the deferred operation is complete.

Many object types are *immutable*, meaning the objects **cannot** change once they have been created. These types of objects never need external synchronization, except that they **must** not be destroyed

while they are in use on another thread. In certain special cases mutable object parameters are internally synchronized, making external synchronization unnecessary. Any command parameters that are not labeled as externally synchronized are either not mutated by the command or are internally synchronized. Additionally, certain objects related to a command's parameters (e.g. command pools and descriptor pools) **may** be affected by a command, and **must** also be externally synchronized. These implicit parameters are documented as described below.

Parameters of commands that are externally synchronized are listed below.

Externally Synchronized Parameters

- The **instance** parameter in [vkDestroyInstance](#)
- The **device** parameter in [vkDestroyDevice](#)
- The **queue** parameter in [vkQueueSubmit](#)
- The **fence** parameter in [vkQueueSubmit](#)
- The **queue** parameter in [vkQueueWaitIdle](#)
- The **memory** parameter in [vkFreeMemory](#)
- The **memory** parameter in [vkMapMemory](#)
- The **memory** parameter in [vkUnmapMemory](#)
- The **buffer** parameter in [vkBindBufferMemory](#)
- The **image** parameter in [vkBindImageMemory](#)
- The **queue** parameter in [vkQueueBindSparse](#)
- The **fence** parameter in [vkQueueBindSparse](#)
- The **fence** parameter in [vkDestroyFence](#)
- The **semaphore** parameter in [vkDestroySemaphore](#)
- The **event** parameter in [vkDestroyEvent](#)
- The **event** parameter in [vkSetEvent](#)
- The **event** parameter in [vkResetEvent](#)
- The **queryPool** parameter in [vkDestroyQueryPool](#)
- The **buffer** parameter in [vkDestroyBuffer](#)
- The **bufferView** parameter in [vkDestroyBufferView](#)
- The **image** parameter in [vkDestroyImage](#)
- The **imageView** parameter in [vkDestroyImageView](#)
- The **shaderModule** parameter in [vkDestroyShaderModule](#)
- The **pipelineCache** parameter in [vkDestroyPipelineCache](#)
- The **dstCache** parameter in [vkMergePipelineCaches](#)
- The **pipeline** parameter in [vkDestroyPipeline](#)
- The **pipelineLayout** parameter in [vkDestroyPipelineLayout](#)

- The `sampler` parameter in `vkDestroySampler`
- The `descriptorSetLayout` parameter in `vkDestroyDescriptorSetLayout`
- The `descriptorPool` parameter in `vkDestroyDescriptorPool`
- The `descriptorPool` parameter in `vkResetDescriptorPool`
- The `descriptorPool` member of the `pAllocateInfo` parameter in `vkAllocateDescriptorSets`
- The `descriptorPool` parameter in `vkFreeDescriptorSets`
- The `framebuffer` parameter in `vkDestroyFramebuffer`
- The `renderPass` parameter in `vkDestroyRenderPass`
- The `commandPool` parameter in `vkDestroyCommandPool`
- The `commandPool` parameter in `vkResetCommandPool`
- The `commandPool` member of the `pAllocateInfo` parameter in `vkAllocateCommandBuffers`
- The `commandPool` parameter in `vkFreeCommandBuffers`
- The `commandBuffer` parameter in `vkBeginCommandBuffer`
- The `commandBuffer` parameter in `vkEndCommandBuffer`
- The `commandBuffer` parameter in `vkResetCommandBuffer`
- The `commandBuffer` parameter in `vkCmdBindPipeline`
- The `commandBuffer` parameter in `vkCmdSetViewport`
- The `commandBuffer` parameter in `vkCmdSetScissor`
- The `commandBuffer` parameter in `vkCmdSetLineWidth`
- The `commandBuffer` parameter in `vkCmdSetDepthBias`
- The `commandBuffer` parameter in `vkCmdSetBlendConstants`
- The `commandBuffer` parameter in `vkCmdSetDepthBounds`
- The `commandBuffer` parameter in `vkCmdSetStencilCompareMask`
- The `commandBuffer` parameter in `vkCmdSetStencilWriteMask`
- The `commandBuffer` parameter in `vkCmdSetStencilReference`
- The `commandBuffer` parameter in `vkCmdBindDescriptorSets`
- The `commandBuffer` parameter in `vkCmdBindIndexBuffer`
- The `commandBuffer` parameter in `vkCmdBindVertexBuffers`
- The `commandBuffer` parameter in `vkCmdDraw`
- The `commandBuffer` parameter in `vkCmdDrawIndexed`
- The `commandBuffer` parameter in `vkCmdDrawIndirect`
- The `commandBuffer` parameter in `vkCmdDrawIndexedIndirect`
- The `commandBuffer` parameter in `vkCmdDispatch`
- The `commandBuffer` parameter in `vkCmdDispatchIndirect`
- The `commandBuffer` parameter in `vkCmdCopyBuffer`

- The `commandBuffer` parameter in `vkCmdCopyImage`
- The `commandBuffer` parameter in `vkCmdBlitImage`
- The `commandBuffer` parameter in `vkCmdCopyBufferToImage`
- The `commandBuffer` parameter in `vkCmdCopyImageToBuffer`
- The `commandBuffer` parameter in `vkCmdUpdateBuffer`
- The `commandBuffer` parameter in `vkCmdFillBuffer`
- The `commandBuffer` parameter in `vkCmdClearColorImage`
- The `commandBuffer` parameter in `vkCmdClearDepthStencilImage`
- The `commandBuffer` parameter in `vkCmdClearAttachments`
- The `commandBuffer` parameter in `vkCmdResolveImage`
- The `commandBuffer` parameter in `vkCmdSetEvent`
- The `commandBuffer` parameter in `vkCmdResetEvent`
- The `commandBuffer` parameter in `vkCmdWaitEvents`
- The `commandBuffer` parameter in `vkCmdPipelineBarrier`
- The `commandBuffer` parameter in `vkCmdBeginQuery`
- The `commandBuffer` parameter in `vkCmdEndQuery`
- The `commandBuffer` parameter in `vkCmdResetQueryPool`
- The `commandBuffer` parameter in `vkCmdWriteTimestamp`
- The `commandBuffer` parameter in `vkCmdCopyQueryPoolResults`
- The `commandBuffer` parameter in `vkCmdPushConstants`
- The `commandBuffer` parameter in `vkCmdBeginRenderPass`
- The `commandBuffer` parameter in `vkCmdNextSubpass`
- The `commandBuffer` parameter in `vkCmdEndRenderPass`
- The `commandBuffer` parameter in `vkCmdExecuteCommands`
- The `surface` parameter in `vkDestroySurfaceKHR`
- The `surface` member of the `pCreateInfo` parameter in `vkCreateSwapchainKHR`
- The `oldSwapchain` member of the `pCreateInfo` parameter in `vkCreateSwapchainKHR`
- The `swapchain` parameter in `vkDestroySwapchainKHR`
- The `swapchain` parameter in `vkAcquireNextImageKHR`
- The `semaphore` parameter in `vkAcquireNextImageKHR`
- The `fence` parameter in `vkAcquireNextImageKHR`
- The `queue` parameter in `vkQueuePresentKHR`
- The `surface` parameter in `vkGetDeviceGroupSurfacePresentModesKHR`
- The `surface` parameter in `vkGetPhysicalDevicePresentRectanglesKHR`
- The `display` parameter in `vkCreateDisplayModeKHR`

- The `mode` parameter in `vkGetDisplayPlaneCapabilitiesKHR`
- The `videoSession` parameter in `vkDestroyVideoSessionKHR`
- The `videoSession` parameter in `vkBindVideoSessionMemoryKHR`
- The `videoSessionParameters` parameter in `vkDestroyVideoSessionParametersKHR`
- The `commandBuffer` parameter in `vkCmdBeginVideoCodingKHR`
- The `commandBuffer` parameter in `vkCmdEndVideoCodingKHR`
- The `commandBuffer` parameter in `vkCmdControlVideoCodingKHR`
- The `commandBuffer` parameter in `vkCmdDecodeVideoKHR`
- The `commandBuffer` parameter in `vkCmdBeginRenderingKHR`
- The `commandBuffer` parameter in `vkCmdEndRenderingKHR`
- The `commandBuffer` parameter in `vkCmdSetDeviceMaskKHR`
- The `commandBuffer` parameter in `vkCmdDispatchBaseKHR`
- The `commandPool` parameter in `vkTrimCommandPoolKHR`
- The `commandBuffer` parameter in `vkCmdPushDescriptorSetKHR`
- The `commandBuffer` parameter in `vkCmdPushDescriptorSetWithTemplateKHR`
- The `descriptorUpdateTemplate` parameter in `vkDestroyDescriptorUpdateTemplateKHR`
- The `commandBuffer` parameter in `vkCmdBeginRenderPass2KHR`
- The `commandBuffer` parameter in `vkCmdNextSubpass2KHR`
- The `commandBuffer` parameter in `vkCmdEndRenderPass2KHR`
- The `swapchain` parameter in `vkGetSwapchainStatusKHR`
- The `ycbcrConversion` parameter in `vkDestroySamplerYcbcrConversionKHR`
- The `commandBuffer` parameter in `vkCmdDrawIndirectCountKHR`
- The `commandBuffer` parameter in `vkCmdDrawIndexedIndirectCountKHR`
- The `commandBuffer` parameter in `vkCmdSetFragmentShadingRateKHR`
- The `commandBuffer` parameter in `vkCmdSetRenderingAttachmentLocationsKHR`
- The `commandBuffer` parameter in `vkCmdSetRenderingInputAttachmentIndicesKHR`
- The `swapchain` parameter in `vkWaitForPresentKHR`
- The `operation` parameter in `vkDestroyDeferredOperationKHR`
- The `commandBuffer` parameter in `vkCmdEncodeVideoKHR`
- The `commandBuffer` parameter in `vkCmdSetEvent2KHR`
- The `commandBuffer` parameter in `vkCmdResetEvent2KHR`
- The `commandBuffer` parameter in `vkCmdWaitEvents2KHR`
- The `commandBuffer` parameter in `vkCmdPipelineBarrier2KHR`
- The `commandBuffer` parameter in `vkCmdWriteTimestamp2KHR`
- The `queue` parameter in `vkQueueSubmit2KHR`

- The `fence` parameter in `vkQueueSubmit2KHR`
- The `commandBuffer` parameter in `vkCmdCopyBuffer2KHR`
- The `commandBuffer` parameter in `vkCmdCopyImage2KHR`
- The `commandBuffer` parameter in `vkCmdCopyBufferToImage2KHR`
- The `commandBuffer` parameter in `vkCmdCopyImageToBuffer2KHR`
- The `commandBuffer` parameter in `vkCmdBlitImage2KHR`
- The `commandBuffer` parameter in `vkCmdResolveImage2KHR`
- The `commandBuffer` parameter in `vkCmdTraceRaysIndirect2KHR`
- The `commandBuffer` parameter in `vkCmdBindIndexBuffer2KHR`
- The `pipelineBinary` parameter in `vkDestroyPipelineBinaryKHR`
- The `pipeline` member of the `pInfo` parameter in `vkReleaseCapturedPipelineDataKHR`
- The `commandBuffer` parameter in `vkCmdSetLineStippleKHR`
- The `commandBuffer` parameter in `vkCmdBindDescriptorSets2KHR`
- The `commandBuffer` parameter in `vkCmdPushConstants2KHR`
- The `commandBuffer` parameter in `vkCmdPushDescriptorSet2KHR`
- The `commandBuffer` parameter in `vkCmdPushDescriptorSetWithTemplate2KHR`
- The `commandBuffer` parameter in `vkCmdSetDescriptorBufferOffsets2EXT`
- The `commandBuffer` parameter in `vkCmdBindDescriptorBufferEmbeddedSamplers2EXT`
- The `commandBuffer` parameter in `vkCmdBindTransformFeedbackBuffersEXT`
- The `commandBuffer` parameter in `vkCmdBeginTransformFeedbackEXT`
- The `commandBuffer` parameter in `vkCmdEndTransformFeedbackEXT`
- The `commandBuffer` parameter in `vkCmdBeginQueryIndexedEXT`
- The `commandBuffer` parameter in `vkCmdEndQueryIndexedEXT`
- The `commandBuffer` parameter in `vkCmdDrawIndirectByteCountEXT`
- The `commandBuffer` parameter in `vkCmdSetDiscardRectangleEXT`
- The `commandBuffer` parameter in `vkCmdSetDiscardRectangleEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetDiscardRectangleModeEXT`
- The `objectHandle` member of the `pNameInfo` parameter in `vkSetDebugUtilsObjectNameEXT`
- The `objectHandle` member of the `pTagInfo` parameter in `vkSetDebugUtilsObjectTagEXT`
- The `commandBuffer` parameter in `vkCmdBeginDebugUtilsLabelEXT`
- The `commandBuffer` parameter in `vkCmdEndDebugUtilsLabelEXT`
- The `commandBuffer` parameter in `vkCmdInsertDebugUtilsLabelEXT`
- The `messenger` parameter in `vkDestroyDebugUtilsMessengerEXT`
- The `commandBuffer` parameter in `vkCmdSetSampleLocationsEXT`
- The `commandBuffer` parameter in `vkCmdSetDepthBias2EXT`

- The `commandBuffer` parameter in `vkCmdSetVertexInputEXT`
- The `commandBuffer` parameter in `vkCmdSetColorWriteEnableEXT`
- The `commandBuffer` parameter in `vkCmdDrawMultiEXT`
- The `commandBuffer` parameter in `vkCmdDrawMultiIndexedEXT`
- The `micromap` parameter in `vkDestroyMicromapEXT`
- The `commandBuffer` parameter in `vkCmdBuildMicromapsEXT`
- The `commandBuffer` parameter in `vkCmdCopyMicromapEXT`
- The `commandBuffer` parameter in `vkCmdCopyMicromapToMemoryEXT`
- The `commandBuffer` parameter in `vkCmdCopyMemoryToMicromapEXT`
- The `commandBuffer` parameter in `vkCmdWriteMicromapsPropertiesEXT`
- The `commandBuffer` parameter in `vkCmdSetDepthClampEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetPolygonModeEXT`
- The `commandBuffer` parameter in `vkCmdSetRasterizationSamplesEXT`
- The `commandBuffer` parameter in `vkCmdSetSampleMaskEXT`
- The `commandBuffer` parameter in `vkCmdSetAlphaToCoverageEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetAlphaToOneEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetLogicOpEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetColorBlendEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetColorBlendEquationEXT`
- The `commandBuffer` parameter in `vkCmdSetColorWriteMaskEXT`
- The `commandBuffer` parameter in `vkCmdSetTessellationDomainOriginEXT`
- The `commandBuffer` parameter in `vkCmdSetRasterizationStreamEXT`
- The `commandBuffer` parameter in `vkCmdSetConservativeRasterizationModeEXT`
- The `commandBuffer` parameter in `vkCmdSetExtraPrimitiveOverestimationSizeEXT`
- The `commandBuffer` parameter in `vkCmdSetDepthClipEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetSampleLocationsEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetColorBlendAdvancedEXT`
- The `commandBuffer` parameter in `vkCmdSetProvokingVertexModeEXT`
- The `commandBuffer` parameter in `vkCmdSetLineRasterizationModeEXT`
- The `commandBuffer` parameter in `vkCmdSetLineStippleEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetDepthClipNegativeOneToOneEXT`
- The `commandBuffer` parameter in `vkCmdSetViewportWScalingEnableNV`
- The `commandBuffer` parameter in `vkCmdSetViewportSwizzleNV`
- The `commandBuffer` parameter in `vkCmdSetCoverageToColorEnableNV`
- The `commandBuffer` parameter in `vkCmdSetCoverageToColorLocationNV`

- The `commandBuffer` parameter in `vkCmdSetCoverageModulationModeNV`
- The `commandBuffer` parameter in `vkCmdSetCoverageModulationTableEnableNV`
- The `commandBuffer` parameter in `vkCmdSetCoverageModulationTableNV`
- The `commandBuffer` parameter in `vkCmdSetShadingRateImageEnableNV`
- The `commandBuffer` parameter in `vkCmdSetRepresentativeFragmentTestEnableNV`
- The `commandBuffer` parameter in `vkCmdSetCoverageReductionModeNV`
- The `shader` parameter in `vkDestroyShaderEXT`
- The `commandBuffer` parameter in `vkCmdBindShadersEXT`
- The `commandBuffer` parameter in `vkCmdSetCullModeEXT`
- The `commandBuffer` parameter in `vkCmdSetFrontFaceEXT`
- The `commandBuffer` parameter in `vkCmdSetPrimitiveTopologyEXT`
- The `commandBuffer` parameter in `vkCmdSetViewportWithCountEXT`
- The `commandBuffer` parameter in `vkCmdSetScissorWithCountEXT`
- The `commandBuffer` parameter in `vkCmdBindVertexBuffers2EXT`
- The `commandBuffer` parameter in `vkCmdSetDepthTestEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetDepthWriteEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetDepthCompareOpEXT`
- The `commandBuffer` parameter in `vkCmdSetDepthBoundsTestEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetStencilTestEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetStencilOpEXT`
- The `commandBuffer` parameter in `vkCmdSetPatchControlPointsEXT`
- The `commandBuffer` parameter in `vkCmdSetRasterizerDiscardEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetDepthBiasEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetLogicOpEXT`
- The `commandBuffer` parameter in `vkCmdSetPrimitiveRestartEnableEXT`
- The `commandBuffer` parameter in `vkCmdSetDepthClampRangeEXT`
- The `commandBuffer` parameter in `vkCmdSetAttachmentFeedbackLoopEnableEXT`
- The `commandBuffer` parameter in `vkCmdPreprocessGeneratedCommandsEXT`
- The `stateCommandBuffer` parameter in `vkCmdPreprocessGeneratedCommandsEXT`
- The `commandBuffer` parameter in `vkCmdExecuteGeneratedCommandsEXT`
- The `indirectCommandsLayout` parameter in `vkDestroyIndirectCommandsLayoutEXT`
- The `indirectExecutionSet` parameter in `vkDestroyIndirectExecutionSetEXT`
- The `indirectExecutionSet` parameter in `vkUpdateIndirectExecutionSetPipelineEXT`
- The `indirectExecutionSet` parameter in `vkUpdateIndirectExecutionSetShaderEXT`
- The `accelerationStructure` parameter in `vkDestroyAccelerationStructureKHR`

- The `commandBuffer` parameter in `vkCmdBuildAccelerationStructuresKHR`
- The `commandBuffer` parameter in `vkCmdBuildAccelerationStructuresIndirectKHR`
- The `commandBuffer` parameter in `vkCmdCopyAccelerationStructureKHR`
- The `commandBuffer` parameter in `vkCmdCopyAccelerationStructureToMemoryKHR`
- The `commandBuffer` parameter in `vkCmdCopyMemoryToAccelerationStructureKHR`
- The `commandBuffer` parameter in `vkCmdWriteAccelerationStructuresPropertiesKHR`
- The `commandBuffer` parameter in `vkCmdTraceRaysKHR`
- The `commandBuffer` parameter in `vkCmdTraceRaysIndirectKHR`
- The `commandBuffer` parameter in `vkCmdSetRayTracingPipelineStackSizeKHR`

There are also a few instances where a command **can** take in an application-allocated list whose contents are externally synchronized parameters. In these cases, the caller **must** guarantee that at most one thread is using a given element within the list at a given time. These parameters are listed below.

Externally Synchronized Parameter Lists

- Each element of the `pFences` parameter in `vkResetFences`
- Each element of the `pDescriptorSets` parameter in `vkFreeDescriptorSets`
- Each element of the `pCommandBuffers` parameter in `vkFreeCommandBuffers`
- Each element of the `pWaitSemaphores` member of the `pPresentInfo` parameter in `vkQueuePresentKHR`
- Each element of the `pSwapchains` member of the `pPresentInfo` parameter in `vkQueuePresentKHR`
- The `surface` member of each element of the `pCreateInfos` parameter in `vkCreateSharedSwapchainsKHR`
- The `oldSwapchain` member of each element of the `pCreateInfos` parameter in `vkCreateSharedSwapchainsKHR`

In addition, there are some implicit parameters that need to be externally synchronized. For example, when a `commandBuffer` parameter needs to be externally synchronized, it implies that the `commandPool` from which that command buffer was allocated also needs to be externally synchronized. The implicit parameters and their associated object are listed below.

Implicit Externally Synchronized Parameters

- All `VkPhysicalDevice` objects enumerated from `instance` in `vkDestroyInstance`
- All `VkQueue` objects created from `device` in `vkDestroyDevice`
- All `VkQueue` objects created from `device` in `vkDeviceWaitIdle`

- Any `VkDescriptorSet` objects allocated from `descriptorPool` in `vkResetDescriptorPool`
- The `VkCommandPool` that `commandBuffer` was allocated from in `vkBeginCommandBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from in `vkEndCommandBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from in `vkResetCommandBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindPipeline`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetViewport`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetScissor`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetLineWidth`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthBias`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetBlendConstants`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthBounds`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetStencilCompareMask`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetStencilWriteMask`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetStencilReference`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindDescriptorSets`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindIndexBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindVertexBuffers`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDraw`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDrawIndexed`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDrawIndirect`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDrawIndexedIndirect`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDispatch`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDispatchIndirect`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBlitImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyBufferToImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyImageToBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdUpdateBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdFillBuffer`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdClearColorImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdClearDepthStencilImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdClearAttachments`

- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdResolveImage`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetEvent`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdResetEvent`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdWaitEvents`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdPipelineBarrier`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBeginQuery`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdEndQuery`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdResetQueryPool`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdWriteTimestamp`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyQueryPoolResults`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdPushConstants`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBeginRenderPass`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdNextSubpass`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdEndRenderPass`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdExecuteCommands`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBeginVideoCodingKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdEndVideoCodingKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdControlVideoCodingKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDecodeVideoKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBeginRenderingKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdEndRenderingKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDeviceMaskKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDispatchBaseKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdPushDescriptorSetKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdPushDescriptorSetWithTemplateKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBeginRenderPass2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdNextSubpass2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdEndRenderPass2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDrawIndirectCountKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDrawIndexedIndirectCountKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetFragmentShadingRateKHR`

- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetRenderingAttachmentLocationsKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetRenderingInputAttachmentIndicesKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdEncodeVideoKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetEvent2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdResetEvent2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdWaitEvents2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdPipelineBarrier2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdWriteTimestamp2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyBuffer2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyImage2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyBufferToImage2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyImageToBuffer2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBlitImage2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdResolveImage2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdTraceRaysIndirect2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindIndexBuffer2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetLineStippleKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindDescriptorSets2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdPushConstants2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdPushDescriptorSet2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdPushDescriptorSetWithTemplate2KHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDescriptorBufferOffsets2EXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindDescriptorBufferEmbeddedSamplers2EXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindTransformFeedbackBuffersEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBeginTransformFeedbackEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in

vkCmdEndTransformFeedbackEXT

- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBeginQueryIndexedEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdEndQueryIndexedEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDrawIndirectByteCountEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDiscardRectangleEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDiscardRectangleEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDiscardRectangleModeEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBeginDebugUtilsLabelEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdEndDebugUtilsLabelEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdInsertDebugUtilsLabelEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetSampleLocationsEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthBias2EXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetVertexInputEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetColorWriteEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDrawMultiEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdDrawMultiIndexedEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBuildMicromapsEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyMicromapEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyMicromapToMemoryEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyMemoryToMicromapEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdWriteMicromapsPropertiesEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthClampEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetPolygonModeEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetRasterizationSamplesEXT`

- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetSampleMaskEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetAlphaToCoverageEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetAlphaToOneEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetLogicOpEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetColorBlendEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetColorBlendEquationEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetColorWriteMaskEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetTessellationDomainOriginEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetRasterizationStreamEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetConservativeRasterizationModeEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetExtraPrimitiveOverestimationSizeEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthClipEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetSampleLocationsEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetColorBlendAdvancedEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetProvokingVertexModeEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetLineRasterizationModeEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetLineStippleEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthClipNegativeOneToOneEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetViewportWScalingEnableNV`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetViewportSwizzleNV`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetCoverageToColorEnableNV`
- The `VkCommandPool` that `commandBuffer` was allocated from, in

vkCmdSetCoverageToColorLocationNV

- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetCoverageModulationModeNV`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetCoverageModulationTableEnableNV`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetCoverageModulationTableNV`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetShadingRateImageEnableNV`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetRepresentativeFragmentTestEnableNV`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetCoverageReductionModeNV`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindShadersEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetCullModeEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetFrontFaceEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetPrimitiveTopologyEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetViewportWithCountEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetScissorWithCountEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBindVertexBuffers2EXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthTestEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthWriteEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthCompareOpEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthBoundsTestEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetStencilTestEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetStencilOpEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetPatchControlPointsEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetRasterizerDiscardEnableEXT`

- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthBiasEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetLogicOpEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetPrimitiveRestartEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetDepthClampRangeEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetAttachmentFeedbackLoopEnableEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdPreprocessGeneratedCommandsEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdExecuteGeneratedCommandsEXT`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBuildAccelerationStructuresKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdBuildAccelerationStructuresIndirectKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyAccelerationStructureKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyAccelerationStructureToMemoryKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdCopyMemoryToAccelerationStructureKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdWriteAccelerationStructuresPropertiesKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdTraceRaysKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdTraceRaysIndirectKHR`
- The `VkCommandPool` that `commandBuffer` was allocated from, in `vkCmdSetRayTracingPipelineStackSizeKHR`

3.7. Valid Usage

Valid usage defines a set of conditions which **must** be met in order to achieve well-defined runtime behavior in an application. These conditions depend only on Vulkan state, and the parameters or objects whose usage is constrained by the condition.

The core layer assumes applications are using the API correctly. Except as documented elsewhere in the Specification, the behavior of the core layer to an application using the API incorrectly is undefined, and **may** include program termination. However, implementations **must** ensure that incorrect usage by an application does not affect the integrity of the operating system, the Vulkan implementation, or other applications in the system using Vulkan. In particular, any guarantees

made by an operating system about whether memory from one process **can** be visible to another process or not **must** not be violated by a Vulkan implementation for **any memory allocation**. Vulkan implementations are not **required** to make additional security or integrity guarantees beyond those provided by the OS unless explicitly directed by the application's use of a particular feature or extension.

NOTE

For instance, if an operating system guarantees that data in all its memory allocations are set to zero when newly allocated, the Vulkan implementation **must** make the same guarantees for any allocations it controls (e.g. [VkDeviceMemory](#)).

Similarly, if an operating system guarantees that use-after-free of host allocations will not result in values written by another process becoming visible, the same guarantees **must** be made by the Vulkan implementation for device memory.

Some valid usage conditions have dependencies on runtime limits or feature availability. It is possible to validate these conditions against Vulkan's minimum supported values for these limits and features, or some subset of other known values.

Valid usage conditions do not cover conditions where well-defined behavior (including returning an error code) exists.

Valid usage conditions **should** apply to the command or structure where complete information about the condition would be known during execution of an application. This is such that a validation layer or linter **can** be written directly against these statements at the point they are specified.

NOTE

This does lead to some non-obvious places for valid usage statements. For instance, the valid values for a structure might depend on a separate value in the calling command. In this case, the structure itself will not reference this valid usage as it is impossible to determine validity from the structure that it is invalid - instead this valid usage would be attached to the calling command.

Another example is draw state - the state setters are independent, and can cause a legitimately invalid state configuration between draw calls; so the valid usage statements are attached to the place where all state needs to be valid - at the drawing command.

Valid usage conditions are described in a block labeled "Valid Usage" following each command or structure they apply to.

3.7.1. Usage Validation

Vulkan is a layered API. The lowest layer is the core Vulkan layer, as defined by this Specification. The application **can** use additional layers above the core for debugging, validation, and other purposes.

One of the core principles of Vulkan is that building and submitting command buffers **should** be highly efficient. Thus error checking and validation of state in the core layer is minimal, although more rigorous validation **can** be enabled through the use of layers.

Validation of correct API usage is left to validation layers. Applications **should** be developed with validation layers enabled, to help catch and eliminate errors. Once validated, released applications **should** not enable validation layers by default.

3.7.2. Implicit Valid Usage

Some valid usage conditions apply to all commands and structures in the API, unless explicitly denoted otherwise for a specific command or structure. These conditions are considered *implicit*, and are described in a block labeled “Valid Usage (Implicit)” following each command or structure they apply to. Implicit valid usage conditions are described in detail below.

Valid Usage for Object Handles

Any input parameter to a command that is an object handle **must** be a valid object handle, unless otherwise specified. An object handle is valid if:

- It has been created or allocated by a previous, successful call to the API. Such calls are noted in the Specification.
- It has not been deleted or freed by a previous call to the API. Such calls are noted in the Specification.
- Any objects used by that object, either as part of creation or execution, **must** also be valid.

The reserved values `VK_NULL_HANDLE` and `NULL` **can** be used in place of valid non-dispatchable handles and dispatchable handles, respectively, when *explicitly called out in the Specification*. Any command that creates an object successfully **must** not return these values. It is valid to pass these values to `vkDestroy*` or `vkFree*` commands, which will silently ignore these values.

Valid Usage for Pointers

Any parameter that is a pointer **must** be a *valid pointer* only if it is explicitly called out by a Valid Usage statement.

A pointer is “valid” if it points at memory containing values of the number and type(s) expected by the command, and all fundamental types accessed through the pointer (e.g. as elements of an array or as members of a structure) satisfy the alignment requirements of the host processor.

Valid Usage for Strings

Any parameter that is a pointer to `char` **must** be a finite sequence of values terminated by a null character, or if *explicitly called out in the Specification*, **can** be `NULL`.

Strings specified as UTF-8 encoded **must** not contain invalid UTF-8 sequences. See [String Representation](#) for additional information about strings.

Valid Usage for Enumerated Types

Any parameter of an enumerated type **must** be a valid enumerant for that type. Use of an enumerant is valid if the following conditions are true:

- The enumerant is defined as part of the enumerated type.

- The enumerant is not a value suffixed with `_MAX_ENUM`.
 - This value exists only to ensure that C `enum` types are 32 bits in size and **must** not be used by applications.
- If the enumerant is used in a function that has a `VkInstance` as its first parameter and either:
 - it was added by a core version that is supported and the value of `VkApplicationInfo::apiVersion` is greater than or equal to the version that added it; or
 - it was added by an `instance extension` that was enabled for the instance.
- If the enumerant is used in a function that has a `VkPhysicalDevice` object as its first parameter and either:
 - it was added by a core version that is supported by that device (as reported by `VkPhysicalDeviceProperties::apiVersion`);
 - it was added by an `instance extension` that was enabled for the instance; or
 - it was added by a `device extension` that is supported by that device.
- If the enumerant is used in a function that has any other dispatchable object as its first parameter and either:
 - it was added by a core version that is supported for the device (as reported by `VkPhysicalDeviceProperties::apiVersion`); or
 - it was added by a `device extension` that was enabled for the device.

Additionally, if `maintenance5` is supported, any integer value representable in the range valid for the defined type is valid when used in a function that has a `VkPhysicalDevice` object as its first parameter. Physical device queries will either return results indicating lack of support, or ignore unsupported values when used as a bit flag in a `Vk*Flags*` parameter.

Any enumerated type returned from a query command or otherwise output from Vulkan to the application **must** not have a reserved value. Reserved values are values not defined by any extension for that enumerated type.

NOTE

In some special cases, an enumerant is only meaningful if a feature defined by an extension is also enabled, as well as the extension itself. The global “valid enumerant” rule described here does not address such cases.

NOTE

This language is intended to accommodate cases such as “hidden” extensions known only to driver internals, or layers enabling extensions without knowledge of the application, without allowing return of values not defined by any extension.

NOTE

Application developers are encouraged to be careful when using `switch` statements with Vulkan API enums. This is because new extensions can add new values to existing enums. Using a `default:` statement within a `switch` may avoid future compilation issues.

Valid Usage for Flags

A collection of flags is represented by a bitmask using the type [VkFlags](#):

```
// Provided by VK_VERSION_1_0
typedef uint32_t VkFlags;
```

Bitmasks are passed to many commands and structures to compactly represent options, but [VkFlags](#) is not used directly in the API. Instead, a [Vk*Flags](#) type which is an alias of [VkFlags](#), and whose name matches the corresponding [Vk*FlagBits](#) that are valid for that type, is used.

Any [Vk*Flags](#) member or parameter used in the API as an input **must** be a valid combination of bit flags. A valid combination is either zero or the bitwise OR of valid bit flags.

An individual bit flag is valid for a [Vk*Flags](#) type if it would be a [valid enumerant](#) when used with the equivalent [Vk*FlagBits](#) type, where the bits type is obtained by taking the flag type and replacing the trailing [Flags](#) with [FlagBits](#). For example, a flag value of type [VkColorComponentFlags](#) **must** contain only bit flags defined by [VkColorComponentFlagBits](#).

Any [Vk*Flags](#) member or parameter returned from a query command or otherwise output from Vulkan to the application **may** contain bit flags undefined in its corresponding [Vk*FlagBits](#) type. An application **cannot** rely on the state of these unspecified bits.

Only the low-order 31 bits (bit positions zero through 30) are available for use as flag bits.

NOTE

This restriction is due to poorly defined behavior by C compilers given a C enumerant value of `0x80000000`. In some cases adding this enumerant value may increase the size of the underlying [Vk*FlagBits](#) type, breaking the ABI.

A collection of 64-bit flags is represented by a bitmask using the type [VkFlags64](#):

```
// Provided by VK_KHR_synchronization2
typedef uint64_t VkFlags64;
```

When the 31 bits available in [VkFlags](#) are insufficient, the [VkFlags64](#) type can be passed to commands and structures to represent up to 64 options. [VkFlags64](#) is not used directly in the API. Instead, a [Vk*Flags2](#) type which is an alias of [VkFlags64](#), and whose name matches the corresponding [Vk*FlagBits2](#) that are valid for that type, is used.

Any [Vk*Flags2](#) member or parameter used in the API as an input **must** be a valid combination of bit flags. A valid combination is either zero or the bitwise OR of valid bit flags.

An individual bit flag is valid for a [Vk*Flags2](#) type if it would be a [valid enumerant](#) when used with the equivalent [Vk*FlagBits2](#) type, where the bits type is obtained by taking the flag type and replacing the trailing [Flags2](#) with [FlagBits2](#). For example, a flag value of type [VkAccessFlags2KHR](#) **must** contain only bit flags defined by [VkAccessFlagBits2KHR](#).

Any [Vk*Flags2](#) member or parameter returned from a query command or otherwise output from

Vulkan to the application **may** contain bit flags undefined in its corresponding `Vk*FlagBits2` type. An application **cannot** rely on the state of these unspecified bits.

NOTE

Both the `Vk*FlagBits2` type, and the individual bits defined for that type, are defined as `uint64_t` integers in the C API. This is in contrast to the 32-bit types, where the `Vk*FlagBits` type is defined as a C `enum` and the individual bits as enumerants belonging to that `enum`. As a result, there is less compile time type checking possible for the 64-bit types. This is unavoidable since there is no sufficiently portable way to define a 64-bit `enum` type in C99.

Valid Usage for Structure Types

Any parameter that is a structure containing a `sType` member **must** have a value of `sType` which is a valid `VkStructureType` value matching the type of the structure.

Valid Usage for Structure Pointer Chains

Any parameter that is a structure containing a `void* pNext` member **must** have a value of `pNext` that is either `NULL`, or is a pointer to a valid *extending structure*, containing `sType` and `pNext` members as described in the [Vulkan Documentation and Extensions](#) document in the section “Extending Structures”. The set of structures connected by `pNext` pointers is referred to as a *pNext chain*.

Each structure included in the `pNext` chain **must** be defined at runtime by either:

- a core version which is supported
- an extension which is enabled
- a supported device extension in the case of physical-device-level functionality added by the device extension

Each type of extending structure **must** not appear more than once in a `pNext` chain, including any [aliases](#). This general rule may be explicitly overridden for specific structures.

Any component of the implementation (the loader, any enabled layers, and drivers) **must** skip over, without processing (other than reading the `sType` and `pNext` members) any extending structures in the chain not defined by core versions or extensions supported by that component.

As a convenience to implementations and layers needing to iterate through a structure pointer chain, the Vulkan API provides two *base structures*. These structures allow for some type safety, and can be used by Vulkan API functions that operate on generic inputs and outputs.

The `VkBaseInStructure` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBaseInStructure {
    VkStructureType      sType;
    const struct VkBaseInStructure* pNext;
} VkBaseInStructure;
```

- `sType` is the structure type of the structure being iterated through.
- `pNext` is `NULL` or a pointer to the next structure in a structure chain.

`VkBaseInStructure` can be used to facilitate iterating through a read-only structure pointer chain.

The `VkBaseOutStructure` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBaseOutStructure {
    VkStructureType      sType;
    struct VkBaseOutStructure* pNext;
} VkBaseOutStructure;
```

- `sType` is the structure type of the structure being iterated through.
- `pNext` is `NULL` or a pointer to the next structure in a structure chain.

`VkBaseOutStructure` can be used to facilitate iterating through a structure pointer chain that returns data back to the application.

Valid Usage for Nested Structures

The above conditions also apply recursively to members of structures provided as input to a command, either as a direct argument to the command, or themselves a member of another structure.

Specifics on valid usage of each command are covered in their individual sections.

Valid Usage for Extensions

Instance-level functionality or behavior added by an [instance extension](#) to the API **must** not be used unless that extension is supported by the instance as determined by [vkEnumerateInstanceExtensionProperties](#), and that extension is enabled in [VkInstanceCreateInfo](#).

Physical-device-level functionality or behavior added by an [instance extension](#) to the API **must** not be used unless that extension is supported by the instance as determined by [vkEnumerateInstanceExtensionProperties](#), and that extension is enabled in [VkInstanceCreateInfo](#).

Physical-device-level functionality or behavior added by a [device extension](#) to the API **must** not be used unless the conditions described in [Extending Physical Device From Device Extensions](#) are met.

Device-level functionality added by a [device extension](#) that is dispatched from a [VkDevice](#), or from a child object of a [VkDevice](#) **must** not be used unless that extension is supported by the device as determined by [vkEnumerateDeviceExtensionProperties](#), and that extension is enabled in [VkDeviceCreateInfo](#).

Valid Usage for Newer Core Versions

Physical-device-level functionality or behavior added by a [new core version](#) of the API **must** not be used unless it is supported by the physical device as determined by [VkPhysicalDeviceProperties](#)

`::apiVersion` and the specified version of `VkApplicationInfo::apiVersion`.

Device-level functionality or behavior added by a [new core version](#) of the API **must** not be used unless it is supported by the device as determined by `VkPhysicalDeviceProperties::apiVersion` and the specified version of `VkApplicationInfo::apiVersion`.

3.8. `VkResult` Return Codes

While the core Vulkan API is not designed to capture incorrect usage, some circumstances still require return codes. Commands in Vulkan return their status via return codes that are in one of two categories:

- Successful completion codes are returned when a command needs to communicate success or status information. All successful completion codes are non-negative values.
- Runtime error codes are returned when a command needs to communicate a failure that could only be detected at runtime. All runtime error codes are negative values.

All return codes in Vulkan are reported via `VkResult` return values. The possible codes are:

```
// Provided by VK_VERSION_1_0
typedef enum VkResult {
    VK_SUCCESS = 0,
    VK_NOT_READY = 1,
    VK_TIMEOUT = 2,
    VK_EVENT_SET = 3,
    VK_EVENT_RESET = 4,
    VK_INCOMPLETE = 5,
    VK_ERROR_OUT_OF_HOST_MEMORY = -1,
    VK_ERROR_OUT_OF_DEVICE_MEMORY = -2,
    VK_ERROR_INITIALIZATION_FAILED = -3,
    VK_ERROR_DEVICE_LOST = -4,
    VK_ERROR_MEMORY_MAP_FAILED = -5,
    VK_ERROR_LAYER_NOT_PRESENT = -6,
    VK_ERROR_EXTENSION_NOT_PRESENT = -7,
    VK_ERROR_FEATURE_NOT_PRESENT = -8,
    VK_ERROR_INCOMPATIBLE_DRIVER = -9,
    VK_ERROR_TOO_MANY_OBJECTS = -10,
    VK_ERROR_FORMAT_NOT_SUPPORTED = -11,
    VK_ERROR_FRAGMENTED_POOL = -12,
    VK_ERROR_UNKNOWN = -13,
    VK_ERROR_OUT_OF_POOL_MEMORY = -1000069000,
    VK_ERROR_INVALID_EXTERNAL_HANDLE = -1000072003,
    VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS = -1000257000,
// Provided by VK_KHR_surface
    VK_ERROR_SURFACE_LOST_KHR = -1000000000,
// Provided by VK_KHR_surface
    VK_ERROR_NATIVE_WINDOW_IN_USE_KHR = -1000000001,
// Provided by VK_KHR_swapchain
    VK_SUBOPTIMAL_KHR = 1000001003,
```

```

// Provided by VK_KHR_swapchain
VK_ERROR_OUT_OF_DATE_KHR = -1000001004,
// Provided by VK_KHR_display_swapchain
VK_ERROR_INCOMPATIBLE_DISPLAY_KHR = -1000003001,
// Provided by VK_KHR_video_queue
VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR = -1000023000,
// Provided by VK_KHR_video_queue
VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR = -1000023001,
// Provided by VK_KHR_video_queue
VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR = -1000023002,
// Provided by VK_KHR_video_queue
VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR = -1000023003,
// Provided by VK_KHR_video_queue
VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR = -1000023004,
// Provided by VK_KHR_video_queue
VK_ERROR_VIDEO_STD_VERSION_NOT_SUPPORTED_KHR = -1000023005,
// Provided by VK_EXT_image_drm_format_modifier
VK_ERROR_INVALID_DRM_FORMAT_MODIFIER_PLANE_LAYOUT_EXT = -1000158000,
// Provided by VK_KHR_global_priority
VK_ERROR_NOT_PERMITTED_KHR = -1000174001,
// Provided by VK_EXT_full_screen_exclusive
VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT = -1000255000,
// Provided by VK_KHR_deferred_host_operations
VK_THREAD_IDLE_KHR = 1000268000,
// Provided by VK_KHR_deferred_host_operations
VK_THREAD_DONE_KHR = 1000268001,
// Provided by VK_KHR_deferred_host_operations
VK_OPERATION_DEFERRED_KHR = 1000268002,
// Provided by VK_KHR_deferred_host_operations
VK_OPERATION_NOT_DEFERRED_KHR = 1000268003,
// Provided by VK_KHR_video_encode_queue
VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR = -1000299000,
// Provided by VK_EXT_shader_object
VK_INCOMPATIBLE_SHADER_BINARY_EXT = 1000482000,
// Provided by VK_KHR_pipeline_binary
VK_PIPELINE_BINARY_MISSING_KHR = 1000483000,
// Provided by VK_KHR_pipeline_binary
VK_ERROR_NOT_ENOUGH_SPACE_KHR = -1000483000,
// Provided by VK_KHR_maintenance1
VK_ERROR_OUT_OF_POOL_MEMORY_KHR = VK_ERROR_OUT_OF_POOL_MEMORY,
// Provided by VK_KHR_external_memory
VK_ERROR_INVALID_EXTERNAL_HANDLE_KHR = VK_ERROR_INVALID_EXTERNAL_HANDLE,
// Provided by VK_KHR_buffer_device_address
VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR =
VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS,
// Provided by VK_EXT_shader_object
// VK_ERROR_INCOMPATIBLE_SHADER_BINARY_EXT is a deprecated alias
VK_ERROR_INCOMPATIBLE_SHADER_BINARY_EXT = VK_INCOMPATIBLE_SHADER_BINARY_EXT,
} VkResult;

```

Success Codes

- **VK_SUCCESS** Command successfully completed
- **VK_NOT_READY** A fence or query has not yet completed
- **VK_TIMEOUT** A wait operation has not completed in the specified time
- **VK_EVENT_SET** An event is signaled
- **VK_EVENT_RESET** An event is unsignaled
- **VK_INCOMPLETE** A return array was too small for the result
- **VK_SUBOPTIMAL_KHR** A swapchain no longer matches the surface properties exactly, but **can** still be used to present to the surface successfully.
- **VK_THREAD_IDLE_KHR** A deferred operation is not complete but there is currently no work for this thread to do at the time of this call.
- **VK_THREAD_DONE_KHR** A deferred operation is not complete but there is no work remaining to assign to additional threads.
- **VK_OPERATION_DEFERRED_KHR** A deferred operation was requested and at least some of the work was deferred.
- **VK_OPERATION_NOT_DEFERRED_KHR** A deferred operation was requested and no operations were deferred.
- **VK_PIPELINE_BINARY_MISSING_KHR** The application attempted to create a pipeline binary by querying an internal cache, but the internal cache entry did not exist.
- **VK_INCOMPATIBLE_SHADER_BINARY_EXT** The provided binary shader code is not compatible with this device.

NOTE

In the initial version of the [VK_EXT_shader_object](#) extension, this return code was named **VK_ERROR_INCOMPATIBLE_SHADER_BINARY_EXT** and improperly described as an error code. The name has been changed, but the old name is retained as an alias for compatibility with old code.

Error Codes

- **VK_ERROR_OUT_OF_HOST_MEMORY** A host memory allocation has failed.
- **VK_ERROR_OUT_OF_DEVICE_MEMORY** A device memory allocation has failed.
- **VK_ERROR_INITIALIZATION_FAILED** Initialization of an object could not be completed for implementation-specific reasons.
- **VK_ERROR_DEVICE_LOST** The logical or physical device has been lost. See [Lost Device](#)
- **VK_ERROR_MEMORY_MAP_FAILED** Mapping of a memory object has failed.
- **VK_ERROR_LAYER_NOT_PRESENT** A requested layer is not present or could not be loaded.
- **VK_ERROR_EXTENSION_NOT_PRESENT** A requested extension is not supported.
- **VK_ERROR_FEATURE_NOT_PRESENT** A requested feature is not supported.
- **VK_ERROR_INCOMPATIBLE_DRIVER** The requested version of Vulkan is not supported by the driver or is otherwise incompatible for implementation-specific reasons.

- **VK_ERROR_TOO_MANY_OBJECTS** Too many objects of the type have already been created.
- **VK_ERROR_FORMAT_NOT_SUPPORTED** A requested format is not supported on this device.
- **VK_ERROR_FRAGMENTED_POOL** A pool allocation has failed due to fragmentation of the pool's memory. This **must** only be returned if no attempt to allocate host or device memory was made to accommodate the new allocation. This **should** be returned in preference to **VK_ERROR_OUT_OF_POOL_MEMORY**, but only if the implementation is certain that the pool allocation failure was due to fragmentation.
- **VK_ERROR_SURFACE_LOST_KHR** A surface is no longer available.
- **VK_ERROR_NATIVE_WINDOW_IN_USE_KHR** The requested window is already in use by Vulkan or another API in a manner which prevents it from being used again.
- **VK_ERROR_OUT_OF_DATE_KHR** A surface has changed in such a way that it is no longer compatible with the swapchain, and further presentation requests using the swapchain will fail. Applications **must** query the new surface properties and recreate their swapchain if they wish to continue presenting to the surface.
- **VK_ERROR_INCOMPATIBLE_DISPLAY_KHR** The display used by a swapchain does not use the same presentable image layout, or is incompatible in a way that prevents sharing an image.
- **VK_ERROR_OUT_OF_POOL_MEMORY** A pool memory allocation has failed. This **must** only be returned if no attempt to allocate host or device memory was made to accommodate the new allocation. If the failure was definitely due to fragmentation of the pool, **VK_ERROR_FRAGMENTED_POOL** **should** be returned instead.
- **VK_ERROR_INVALID_EXTERNAL_HANDLE** An external handle is not a valid handle of the specified type.
- **VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS** A buffer creation or memory allocation failed because the requested address is not available. A shader group handle assignment failed because the requested shader group handle information is no longer valid.
- **VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT** An operation on a swapchain created with **VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT** failed as it did not have exclusive full-screen access. This **may** occur due to implementation-dependent reasons, outside of the application's control.
- **VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR** The requested **VkImageUsageFlags** are not supported.
- **VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR** The requested video picture layout is not supported.
- **VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR** A video profile operation specified via **VkVideoProfileInfoKHR::videoCodecOperation** is not supported.
- **VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR** Format parameters in a requested **VkVideoProfileInfoKHR** chain are not supported.
- **VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR** Codec-specific parameters in a requested **VkVideoProfileInfoKHR** chain are not supported.
- **VK_ERROR_VIDEO_STD_VERSION_NOT_SUPPORTED_KHR** The specified video Std header version is not supported.
- **VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR** The specified Video Std parameters do not adhere to the syntactic or semantic requirements of the used video compression standard, or values

derived from parameters according to the rules defined by the used video compression standard do not adhere to the capabilities of the video compression standard or the implementation.

- **VK_ERROR_NOT_PERMITTED_KHR** The driver implementation has denied a request to acquire a priority above the default priority (**VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_EXT**) because the application does not have sufficient privileges.
- **VK_ERROR_NOT_ENOUGH_SPACE_KHR** The application did not provide enough space to return all the required data.
- **VK_ERROR_UNKNOWN** An unknown error has occurred; either the application has provided invalid input, or an implementation failure has occurred.

If a command returns a runtime error, unless otherwise specified any output parameters will have undefined contents, except that if the output parameter is a structure with **sType** and **pNext** fields, those fields will be unmodified. Any structures chained from **pNext** will also have undefined contents, except that **sType** and **pNext** will be unmodified.

VK_ERROR_OUT_OF_*_MEMORY errors do not modify any currently existing Vulkan objects. Objects that have already been successfully created **can** still be used by the application.

NOTE

As a general rule, **Free**, **Release**, and **Reset** commands do not return **VK_ERROR_OUT_OF_HOST_MEMORY**, while any other command with a return code **may** return it. Any exceptions from this rule are described for those commands.

VK_ERROR_UNKNOWN will be returned by an implementation when an unexpected error occurs that cannot be attributed to valid behavior of the application and implementation. Under these conditions, it **may** be returned from any command returning a **VkResult**.

NOTE

VK_ERROR_UNKNOWN is not expected to ever be returned if the application behavior is valid, and if the implementation is bug-free. If **VK_ERROR_UNKNOWN** is received, the application should be checked against the latest validation layers to verify correct behavior as much as possible. If no issues are identified it could be an implementation issue, and the implementor should be contacted for support.

Performance-critical commands generally do not have return codes. If a runtime error occurs in such commands, the implementation will defer reporting the error until a specified point. For commands that record into command buffers (**vkCmd***) runtime errors are reported by **vkEndCommandBuffer**.

3.9. Numeric Representation and Computation

Implementations normally perform computations in floating-point, and **must** meet the range and precision requirements defined under “Floating-Point Computation” below.

These requirements only apply to computations performed in Vulkan operations outside of shader execution, such as texture image specification and sampling, and per-fragment operations. Range and precision requirements during shader execution differ and are specified by the [Precision and Operation of SPIR-V Instructions](#) section.

In some cases, the representation and/or precision of operations is implicitly limited by the specified format of vertex or texel data consumed by Vulkan. Specific floating-point formats are described later in this section.

3.9.1. Floating-Point Computation

Most floating-point computation is performed in SPIR-V shader modules. The properties of computation within shaders are constrained as defined by the [Precision and Operation of SPIR-V Instructions](#) section.

Some floating-point computation is performed outside of shaders, such as viewport and depth range calculations. For these computations, we do not specify how floating-point numbers are to be represented, or the details of how operations on them are performed, but only place minimal requirements on representation and precision as described in the remainder of this section.

We require simply that numbers' floating-point parts contain enough bits and that their exponent fields are large enough so that individual results of floating-point operations are accurate to about 1 part in 10^5 . The maximum representable magnitude for all floating-point values **must** be at least 2^{32} .

$$x \times 0 = 0 \times x = 0 \text{ for any non-infinite and non-NaN } x.$$

$$1 \times x = x \times 1 = x.$$

$$x + 0 = 0 + x = x.$$

$$0^0 = 1.$$

Occasionally, further requirements will be specified. Most single-precision floating-point formats meet these requirements.

The special values Inf and -Inf encode values with magnitudes too large to be represented; the special value NaN encodes “Not A Number” values resulting from undefined arithmetic operations such as $0 / 0$. Implementations **may** support Inf and NaN in their floating-point computations. Any computation which does not support either Inf or NaN, for which that value is an input or output will yield an undefined value.

3.9.2. Floating-Point Format Conversions

When a value is converted to a defined floating-point representation, finite values falling between two representable finite values are rounded to one or the other. The rounding mode is not defined. Finite values whose magnitude is larger than that of any representable finite value may be rounded either to the closest representable finite value or to the appropriately signed infinity. For unsigned destination formats any negative values are converted to zero. Positive infinity is converted to positive infinity; negative infinity is converted to negative infinity in signed formats and to zero in unsigned formats; and any NaN is converted to a NaN.

3.9.3. 16-Bit Floating-Point Numbers

16-bit floating-point numbers are defined in the “16-bit floating-point numbers” section of the [Khronos Data Format Specification](#).

3.9.4. Unsigned 11-Bit Floating-Point Numbers

Unsigned 11-bit floating-point numbers are defined in the “Unsigned 11-bit floating-point numbers” section of the [Khronos Data Format Specification](#).

3.9.5. Unsigned 10-Bit Floating-Point Numbers

Unsigned 10-bit floating-point numbers are defined in the “Unsigned 10-bit floating-point numbers” section of the [Khronos Data Format Specification](#).

3.9.6. General Requirements

Any representable floating-point value in the appropriate format is legal as input to a Vulkan command that requires floating-point data. The result of providing a value that is not a floating-point number to such a command is unspecified, but **must** not lead to Vulkan interruption or termination. For example, providing a negative zero (where applicable) or a denormalized number to a Vulkan command **must** yield deterministic results, while providing a NaN or Inf yields unspecified results.

Some calculations require division. In such cases (including implied divisions performed by vector normalization), division by zero produces an unspecified result but **must** not lead to Vulkan interruption or termination.

3.10. Fixed-Point Data Conversions

When generic vertex attributes and pixel color or depth *components* are represented as integers, they are often (but not always) considered to be *normalized*. Normalized integer values are treated specially when being converted to and from floating-point values, and are usually referred to as *normalized fixed-point*.

In the remainder of this section, b denotes the bit width of the fixed-point integer representation. When the integer is one of the types defined by the API, b is the bit width of that type. When the integer comes from an [image](#) containing color or depth component texels, b is the number of bits allocated to that component in its [specified image format](#).

The signed and unsigned fixed-point representations are assumed to be b -bit binary two's-complement integers and binary unsigned integers, respectively.

3.10.1. Conversion From Normalized Fixed-Point to Floating-Point

Unsigned normalized fixed-point integers represent numbers in the range $[0,1]$. The conversion from an unsigned normalized fixed-point value c to the corresponding floating-point value f is defined as

$$f = \frac{c}{2^b - 1}$$

Signed normalized fixed-point integers represent numbers in the range [-1,1]. The conversion from a signed normalized fixed-point value c to the corresponding floating-point value f is performed using

$$f = \max\left(\frac{c}{2^{b-1} - 1}, -1.0\right)$$

Only the range $[-2^{b-1} + 1, 2^{b-1} - 1]$ is used to represent signed fixed-point values in the range [-1,1]. For example, if $b = 8$, then the integer value -127 corresponds to -1.0 and the value 127 corresponds to 1.0. This equation is used everywhere that signed normalized fixed-point values are converted to floating-point.

Note that while zero is exactly expressible in this representation, one value (-128 in the example) is outside the representable range, and implementations **must** clamp it to -1.0. Where the value is subject to further processing by the implementation, e.g. during texture filtering, values less than -1.0 **may** be used but the result **must** be clamped before the value is returned to shaders.

3.10.2. Conversion From Floating-Point to Normalized Fixed-Point

The conversion from a floating-point value f to the corresponding unsigned normalized fixed-point value c is defined by first clamping f to the range [0,1], then computing

$$c = \text{convertFloatToUint}(f \times (2^b - 1), b)$$

where $\text{convertFloatToUint}(r,b)$ returns one of the two unsigned binary integer values with exactly b bits which are closest to the floating-point value r . Implementations **should** round to nearest. If r is equal to an integer, then that integer value **must** be returned. In particular, if f is equal to 0.0 or 1.0, then c **must** be assigned 0 or $2^b - 1$, respectively.

The conversion from a floating-point value f to the corresponding signed normalized fixed-point value c is performed by clamping f to the range [-1,1], then computing

$$c = \text{convertFloatToInt}(f \times (2^{b-1} - 1), b)$$

where $\text{convertFloatToInt}(r,b)$ returns one of the two signed two's-complement binary integer values with exactly b bits which are closest to the floating-point value r . Implementations **should** round to nearest. If r is equal to an integer, then that integer value **must** be returned. In particular, if f is equal to -1.0, 0.0, or 1.0, then c **must** be assigned $-(2^{b-1} - 1)$, 0, or $2^{b-1} - 1$, respectively.

This equation is used everywhere that floating-point values are converted to signed normalized fixed-point.

3.11. String Representation

Strings passed into and returned from Vulkan API commands are usually defined to be null-

terminated and UTF-8 encoded.

NOTE

Exceptions to this rule exist only when strings are defined or used by operating system APIs where that OS has a different convention. For example, `VkExportMemoryWin32HandleInfoKHR::name` is a null-terminated UTF-16 encoded string used in conjunction with Windows handles.

When a UTF-8 string is **returned from** a Vulkan API query, it is returned in a fixed-length buffer of C `char`. For example, a string returned in `VkPhysicalDeviceProperties::deviceName` has maximum length `VK_MAX_PHYSICAL_DEVICE_NAME_SIZE`, and a string returned in `VkExtensionProperties::extensionName` has maximum length `VK_MAX_EXTENSION_NAME_SIZE`. The string, **including** its null terminator, will always fit completely within this buffer. If the string is shorter than the buffer size, the contents of `char` in the buffer following the null terminator are undefined.

When a UTF-8 string is **passed into** a Vulkan API, such as `VkDeviceCreateInfo::ppEnabledExtensionNames`, there is no explicit limit on the length of that string. However, the string **must** contain a valid UTF-8 encoded string and **must** be null-terminated.

3.12. Common Object Types

Some types of Vulkan objects are used in many different structures and command parameters, and are described here. These types include *offsets*, *extents*, and *rectangles*.

3.12.1. Offsets

Offsets are used to describe a pixel location within an image or framebuffer, as an (x,y) location for two-dimensional images, or an (x,y,z) location for three-dimensional images.

A two-dimensional offset is defined by the structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkOffset2D {
    int32_t    x;
    int32_t    y;
} VkOffset2D;
```

- `x` is the x offset.
- `y` is the y offset.

A three-dimensional offset is defined by the structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkOffset3D {
    int32_t    x;
    int32_t    y;
    int32_t    z;
} VkOffset3D;
```

- **x** is the x offset.
- **y** is the y offset.
- **z** is the z offset.

3.12.2. Extents

Extents are used to describe the size of a rectangular region of pixels within an image or framebuffer, as (width,height) for two-dimensional images, or as (width,height,depth) for three-dimensional images.

A two-dimensional extent is defined by the structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkExtent2D {
    uint32_t    width;
    uint32_t    height;
} VkExtent2D;
```

- **width** is the width of the extent.
- **height** is the height of the extent.

A three-dimensional extent is defined by the structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkExtent3D {
    uint32_t    width;
    uint32_t    height;
    uint32_t    depth;
} VkExtent3D;
```

- **width** is the width of the extent.
- **height** is the height of the extent.
- **depth** is the depth of the extent.

3.12.3. Rectangles

Rectangles are used to describe a specified rectangular region of pixels within an image or framebuffer. Rectangles include both an offset and an extent of the same dimensionality, as

described above. Two-dimensional rectangles are defined by the structure

```
// Provided by VK_VERSION_1_0
typedef struct VkRect2D {
    VkOffset2D    offset;
    VkExtent2D    extent;
} VkRect2D;
```

- **offset** is a [VkOffset2D](#) specifying the rectangle offset.
- **extent** is a [VkExtent2D](#) specifying the rectangle extent.

3.12.4. Structure Types

Each value corresponds to a particular structure with a **sType** member with a matching name. As a general rule, the name of each [VkStructureType](#) value is obtained by taking the name of the structure, stripping the leading **Vk**, prefixing each capital letter with **_**, converting the entire resulting string to upper case, and prefixing it with **VK_STRUCTURE_TYPE_**. For example, structures of type [VkImageCreateInfo](#) correspond to a [VkStructureType](#) value of **VK_STRUCTURE_TYPE_IMAGE_CREATE_INFO**, and thus a structure of this type **must** have its **sType** member set to this value before it is passed to the API.

The values **VK_STRUCTURE_TYPE_LOADER_INSTANCE_CREATE_INFO** and **VK_STRUCTURE_TYPE_LOADER_DEVICE_CREATE_INFO** are reserved for internal use by the loader, and do not have corresponding Vulkan structures in this Specification.

Structure types supported by the Vulkan API include:

```
// Provided by VK_VERSION_1_0
typedef enum VkStructureType {
    VK_STRUCTURE_TYPE_APPLICATION_INFO = 0,
    VK_STRUCTURE_TYPE_INSTANCE_CREATE_INFO = 1,
    VK_STRUCTURE_TYPE_DEVICE_QUEUE_CREATE_INFO = 2,
    VK_STRUCTURE_TYPE_DEVICE_CREATE_INFO = 3,
    VK_STRUCTURE_TYPE_SUBMIT_INFO = 4,
    VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_INFO = 5,
    VK_STRUCTURE_TYPE_MAPPED_MEMORY_RANGE = 6,
    VK_STRUCTURE_TYPE_BIND_SPARSE_INFO = 7,
    VK_STRUCTURE_TYPE_FENCE_CREATE_INFO = 8,
    VK_STRUCTURE_TYPE_SEMAPHORE_CREATE_INFO = 9,
    VK_STRUCTURE_TYPE_EVENT_CREATE_INFO = 10,
    VK_STRUCTURE_TYPE_QUERY_POOL_CREATE_INFO = 11,
    VK_STRUCTURE_TYPE_BUFFER_CREATE_INFO = 12,
    VK_STRUCTURE_TYPE_BUFFER_VIEW_CREATE_INFO = 13,
    VK_STRUCTURE_TYPE_IMAGE_CREATE_INFO = 14,
    VK_STRUCTURE_TYPE_IMAGE_VIEW_CREATE_INFO = 15,
    VK_STRUCTURE_TYPE_SHADER_MODULE_CREATE_INFO = 16,
    VK_STRUCTURE_TYPE_PIPELINE_CACHE_CREATE_INFO = 17,
    VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_CREATE_INFO = 18,
```

```
VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_STATE_CREATE_INFO = 19,  
VK_STRUCTURE_TYPE_PIPELINE_INPUT_ASSEMBLY_STATE_CREATE_INFO = 20,  
VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_STATE_CREATE_INFO = 21,  
VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_STATE_CREATE_INFO = 22,  
VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_CREATE_INFO = 23,  
VK_STRUCTURE_TYPE_PIPELINE_MULTISAMPLE_STATE_CREATE_INFO = 24,  
VK_STRUCTURE_TYPE_PIPELINE_DEPTH_STENCIL_STATE_CREATE_INFO = 25,  
VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_STATE_CREATE_INFO = 26,  
VK_STRUCTURE_TYPE_PIPELINE_DYNAMIC_STATE_CREATE_INFO = 27,  
VK_STRUCTURE_TYPE_GRAPHICS_PIPELINE_CREATE_INFO = 28,  
VK_STRUCTURE_TYPE_COMPUTE_PIPELINE_CREATE_INFO = 29,  
VK_STRUCTURE_TYPE_PIPELINE_LAYOUT_CREATE_INFO = 30,  
VK_STRUCTURE_TYPE_SAMPLER_CREATE_INFO = 31,  
VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO = 32,  
VK_STRUCTURE_TYPE_DESCRIPTOR_POOL_CREATE_INFO = 33,  
VK_STRUCTURE_TYPE_DESCRIPTOR_SET_ALLOCATE_INFO = 34,  
VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET = 35,  
VK_STRUCTURE_TYPE_COPY_DESCRIPTOR_SET = 36,  
VK_STRUCTURE_TYPE_FRAMEBUFFER_CREATE_INFO = 37,  
VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO = 38,  
VK_STRUCTURE_TYPE_COMMAND_POOL_CREATE_INFO = 39,  
VK_STRUCTURE_TYPE_COMMAND_BUFFER_ALLOCATE_INFO = 40,  
VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_INFO = 41,  
VK_STRUCTURE_TYPE_COMMAND_BUFFER_BEGIN_INFO = 42,  
VK_STRUCTURE_TYPE_RENDER_PASS_BEGIN_INFO = 43,  
VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER = 44,  
VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER = 45,  
VK_STRUCTURE_TYPE_MEMORY_BARRIER = 46,  
VK_STRUCTURE_TYPE_LOADER_INSTANCE_CREATE_INFO = 47,  
VK_STRUCTURE_TYPE_LOADER_DEVICE_CREATE_INFO = 48,  
VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_INFO = 1000157000,  
VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_INFO = 1000157001,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_16BIT_STORAGE_FEATURES = 1000083000,  
VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS = 1000127000,  
VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO = 1000127001,  
VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_FLAGS_INFO = 1000060000,  
VK_STRUCTURE_TYPE_DEVICE_GROUP_RENDER_PASS_BEGIN_INFO = 1000060003,  
VK_STRUCTURE_TYPE_DEVICE_GROUP_COMMAND_BUFFER_BEGIN_INFO = 1000060004,  
VK_STRUCTURE_TYPE_DEVICE_GROUP_SUBMIT_INFO = 1000060005,  
VK_STRUCTURE_TYPE_DEVICE_GROUP_BIND_SPARSE_INFO = 1000060006,  
VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_DEVICE_GROUP_INFO = 1000060013,  
VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_DEVICE_GROUP_INFO = 1000060014,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES = 1000070000,  
VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO = 1000070001,  
VK_STRUCTURE_TYPE_BUFFER_MEMORY_REQUIREMENTS_INFO_2 = 1000146000,  
VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2 = 1000146001,  
VK_STRUCTURE_TYPE_IMAGE_SPARSE_MEMORY_REQUIREMENTS_INFO_2 = 1000146002,  
VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2 = 1000146003,  
VK_STRUCTURE_TYPE_SPARSE_IMAGE_MEMORY_REQUIREMENTS_2 = 1000146004,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2 = 1000059000,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROPERTIES_2 = 1000059001,
```



```
VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_2 = 1000059002,  
VK_STRUCTURE_TYPE_IMAGE_FORMAT_PROPERTIES_2 = 1000059003,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_FORMAT_INFO_2 = 1000059004,  
VK_STRUCTURE_TYPE_QUEUE_FAMILY_PROPERTIES_2 = 1000059005,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PROPERTIES_2 = 1000059006,  
VK_STRUCTURE_TYPE_SPARSE_IMAGE_FORMAT_PROPERTIES_2 = 1000059007,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SPARSE_IMAGE_FORMAT_INFO_2 = 1000059008,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_POINT_CLIPPING_PROPERTIES = 1000117000,  
VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO = 1000117001,  
VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_CREATE_INFO = 1000117002,  
VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_DOMAIN_ORIGIN_STATE_CREATE_INFO =  
1000117003,  
VK_STRUCTURE_TYPE_RENDER_PASS_MULTIVIEW_CREATE_INFO = 1000053000,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_FEATURES = 1000053001,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_PROPERTIES = 1000053002,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES = 1000120000,  
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_CREATE_INFO = 1000156000,  
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_INFO = 1000156001,  
VK_STRUCTURE_TYPE_BIND_IMAGE_PLANE_MEMORY_INFO = 1000156002,  
VK_STRUCTURE_TYPE_IMAGE_PLANE_MEMORY_REQUIREMENTS_INFO = 1000156003,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_YCBCR_CONVERSION_FEATURES = 1000156004,  
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_IMAGE_FORMAT_PROPERTIES = 1000156005,  
VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO = 1000085000,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_IMAGE_FORMAT_INFO = 1000071000,  
VK_STRUCTURE_TYPE_EXTERNAL_IMAGE_FORMAT_PROPERTIES = 1000071001,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_BUFFER_INFO = 1000071002,  
VK_STRUCTURE_TYPE_EXTERNAL_BUFFER_PROPERTIES = 1000071003,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES = 1000071004,  
VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_BUFFER_CREATE_INFO = 1000072000,  
VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_IMAGE_CREATE_INFO = 1000072001,  
VK_STRUCTURE_TYPE_EXPORT_MEMORY_ALLOCATE_INFO = 1000072002,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_FENCE_INFO = 1000112000,  
VK_STRUCTURE_TYPE_EXTERNAL_FENCE_PROPERTIES = 1000112001,  
VK_STRUCTURE_TYPE_EXPORT_FENCE_CREATE_INFO = 1000113000,  
VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_CREATE_INFO = 1000077000,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_SEMAPHORE_INFO = 1000076000,  
VK_STRUCTURE_TYPE_EXTERNAL_SEMAPHORE_PROPERTIES = 1000076001,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_3_PROPERTIES = 1000168000,  
VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_SUPPORT = 1000168001,  
VK_STRUCTURE_TYPE_IMAGE_FORMAT_LIST_CREATE_INFO = 1000147000,  
VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_2 = 1000109000,  
VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_2 = 1000109001,  
VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_2 = 1000109002,  
VK_STRUCTURE_TYPE_SUBPASS_DEPENDENCY_2 = 1000109003,  
VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO_2 = 1000109004,  
VK_STRUCTURE_TYPE_SUBPASS_BEGIN_INFO = 1000109005,  
VK_STRUCTURE_TYPE_SUBPASS_END_INFO = 1000109006,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_8BIT_STORAGE_FEATURES = 1000177000,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRIVER_PROPERTIES = 1000196000,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_INT64_FEATURES = 1000180000,  
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES = 1000082000,
```

```
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT_CONTROLS_PROPERTIES = 1000197000,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_STENCIL_RESOLVE_PROPERTIES = 1000199000,
VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_DEPTH_STENCIL_RESOLVE = 1000199001,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_MEMORY_MODEL_FEATURES = 1000211000,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGELESS_FRAMEBUFFER_FEATURES = 1000108000,
VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENTS_CREATE_INFO = 1000108001,
VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENT_IMAGE_INFO = 1000108002,
VK_STRUCTURE_TYPE_RENDER_PASS_ATTACHMENT_BEGIN_INFO = 1000108003,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFORM_BUFFER_STANDARD_LAYOUT_FEATURES =
1000253000,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_EXTENDED_TYPES_FEATURES =
1000175000,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SEPARATE_DEPTH_STENCIL_LAYOUTS_FEATURES =
1000241000,
VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_STENCIL_LAYOUT = 1000241001,
VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_STENCIL_LAYOUT = 1000241002,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_FEATURES = 1000207000,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_PROPERTIES = 1000207001,
VK_STRUCTURE_TYPE_SEMAPHORE_TYPE_CREATE_INFO = 1000207002,
VK_STRUCTURE_TYPE_TIMELINE_SEMAPHORE_SUBMIT_INFO = 1000207003,
VK_STRUCTURE_TYPE_SEMAPHORE_WAIT_INFO = 1000207004,
VK_STRUCTURE_TYPE_SEMAPHORE_SIGNAL_INFO = 1000207005,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BUFFER_DEVICE_ADDRESS_FEATURES = 1000257000,
VK_STRUCTURE_TYPE_BUFFER_DEVICE_ADDRESS_INFO = 1000244001,
VK_STRUCTURE_TYPE_BUFFER_OPAQUE_CAPTURE_ADDRESS_CREATE_INFO = 1000257002,
VK_STRUCTURE_TYPE_MEMORY_OPAQUE_CAPTURE_ADDRESS_ALLOCATE_INFO = 1000257003,
VK_STRUCTURE_TYPE_DEVICE_MEMORY_OPAQUE_CAPTURE_ADDRESS_INFO = 1000257004,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TERMINATE_INVOCATION_FEATURES =
1000215000,
VK_STRUCTURE_TYPE_MEMORY_BARRIER_2 = 1000314000,
VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER_2 = 1000314001,
VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER_2 = 1000314002,
VK_STRUCTURE_TYPE_DEPENDENCY_INFO = 1000314003,
VK_STRUCTURE_TYPE_SUBMIT_INFO_2 = 1000314004,
VK_STRUCTURE_TYPE_SEMAPHORE_SUBMIT_INFO = 1000314005,
VK_STRUCTURE_TYPE_COMMAND_BUFFER_SUBMIT_INFO = 1000314006,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SYNCHRONIZATION_2_FEATURES = 1000314007,
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_WORKGROUP_MEMORY_FEATURES =
1000325000,
VK_STRUCTURE_TYPE_COPY_BUFFER_INFO_2 = 1000337000,
VK_STRUCTURE_TYPE_COPY_IMAGE_INFO_2 = 1000337001,
VK_STRUCTURE_TYPE_COPY_BUFFER_TO_IMAGE_INFO_2 = 1000337002,
VK_STRUCTURE_TYPE_COPY_IMAGE_TO_BUFFER_INFO_2 = 1000337003,
VK_STRUCTURE_TYPE_BLIT_IMAGE_INFO_2 = 1000337004,
VK_STRUCTURE_TYPE_RESOLVE_IMAGE_INFO_2 = 1000337005,
VK_STRUCTURE_TYPE_BUFFER_COPY_2 = 1000337006,
VK_STRUCTURE_TYPE_IMAGE_COPY_2 = 1000337007,
VK_STRUCTURE_TYPE_IMAGE_BLIT_2 = 1000337008,
VK_STRUCTURE_TYPE_BUFFER_IMAGE_COPY_2 = 1000337009,
VK_STRUCTURE_TYPE_IMAGE_RESOLVE_2 = 1000337010,
VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_REQUIRED_SUBGROUP_SIZE_CREATE_INFO =
```

```

1000225001,
    VK_STRUCTURE_TYPE_RENDERING_INFO = 1000044000,
    VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_INFO = 1000044001,
    VK_STRUCTURE_TYPE_PIPELINE_RENDERING_CREATE_INFO = 1000044002,
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_FEATURES = 1000044003,
    VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_RENDERING_INFO = 1000044004,
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_FEATURES =
1000280000,
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_PROPERTIES =
1000280001,
    VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_3 = 1000360000,
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_FEATURES = 1000413000,
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_PROPERTIES = 1000413001,
    VK_STRUCTURE_TYPE_DEVICE_BUFFER_MEMORY_REQUIREMENTS = 1000413002,
    VK_STRUCTURE_TYPE_DEVICE_IMAGE_MEMORY_REQUIREMENTS = 1000413003,
// Provided by VK_KHR_swapchain
    VK_STRUCTURE_TYPE_SWAPCHAIN_CREATE_INFO_KHR = 1000001000,
// Provided by VK_KHR_swapchain
    VK_STRUCTURE_TYPE_PRESENT_INFO_KHR = 1000001001,
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
    VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_CAPABILITIES_KHR = 1000060007,
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
    VK_STRUCTURE_TYPE_IMAGE_SWAPCHAIN_CREATE_INFO_KHR = 1000060008,
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
    VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_SWAPCHAIN_INFO_KHR = 1000060009,
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
    VK_STRUCTURE_TYPE_ACQUIRE_NEXT_IMAGE_INFO_KHR = 1000060010,
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
    VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_INFO_KHR = 1000060011,
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
    VK_STRUCTURE_TYPE_DEVICE_GROUP_SWAPCHAIN_CREATE_INFO_KHR = 1000060012,
// Provided by VK_KHR_display
    VK_STRUCTURE_TYPE_DISPLAY_MODE_CREATE_INFO_KHR = 1000002000,
// Provided by VK_KHR_display
    VK_STRUCTURE_TYPE_DISPLAY_SURFACE_CREATE_INFO_KHR = 1000002001,
// Provided by VK_KHR_display_swapchain
    VK_STRUCTURE_TYPE_DISPLAY_PRESENT_INFO_KHR = 1000003000,
// Provided by VK_KHR_xlib_surface
    VK_STRUCTURE_TYPE_XLIB_SURFACE_CREATE_INFO_KHR = 1000004000,
// Provided by VK_KHR_xcb_surface
    VK_STRUCTURE_TYPE_XCB_SURFACE_CREATE_INFO_KHR = 1000005000,
// Provided by VK_KHR_wayland_surface
    VK_STRUCTURE_TYPE_WAYLAND_SURFACE_CREATE_INFO_KHR = 1000006000,
// Provided by VK_KHR_android_surface
    VK_STRUCTURE_TYPE_ANDROID_SURFACE_CREATE_INFO_KHR = 1000008000,

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// Provided by VK_KHR_win32_surface
VK_STRUCTURE_TYPE_WIN32_SURFACE_CREATE_INFO_KHR = 1000009000,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_PROFILE_INFO_KHR = 1000023000,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_CAPABILITIES_KHR = 1000023001,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_PICTURE_RESOURCE_INFO_KHR = 1000023002,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_SESSION_MEMORY_REQUIREMENTS_KHR = 1000023003,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_BIND_VIDEO_SESSION_MEMORY_INFO_KHR = 1000023004,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_SESSION_CREATE_INFO_KHR = 1000023005,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_CREATE_INFO_KHR = 1000023006,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_UPDATE_INFO_KHR = 1000023007,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_BEGIN_CODING_INFO_KHR = 1000023008,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_END_CODING_INFO_KHR = 1000023009,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_CODING_CONTROL_INFO_KHR = 1000023010,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_REFERENCE_SLOT_INFO_KHR = 1000023011,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_QUEUE_FAMILY_VIDEO_PROPERTIES_KHR = 1000023012,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_PROFILE_LIST_INFO_KHR = 1000023013,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_FORMAT_INFO_KHR = 1000023014,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_VIDEO_FORMAT_PROPERTIES_KHR = 1000023015,
// Provided by VK_KHR_video_queue
VK_STRUCTURE_TYPE_QUEUE_FAMILY_QUERY_RESULT_STATUS_PROPERTIES_KHR = 1000023016,
// Provided by VK_KHR_video_decode_queue
VK_STRUCTURE_TYPE_VIDEO_DECODE_INFO_KHR = 1000024000,
// Provided by VK_KHR_video_decode_queue
VK_STRUCTURE_TYPE_VIDEO_DECODE_CAPABILITIES_KHR = 1000024001,
// Provided by VK_KHR_video_decode_queue
VK_STRUCTURE_TYPE_VIDEO_DECODE_USAGE_INFO_KHR = 1000024002,
// Provided by VK_EXT_transform_feedback
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_FEATURES_EXT = 1000028000,
// Provided by VK_EXT_transform_feedback
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_PROPERTIES_EXT = 1000028001,
// Provided by VK_EXT_transform_feedback
VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_STREAM_CREATE_INFO_EXT =
1000028002,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_CAPABILITIES_KHR = 1000038000,

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// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR =
1000038001,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR = 1000038002,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PICTURE_INFO_KHR = 1000038003,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_DPB_SLOT_INFO_KHR = 1000038004,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_NALU_SLICE_INFO_KHR = 1000038005,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_GOP_REMAINING_FRAME_INFO_KHR = 1000038006,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PROFILE_INFO_KHR = 1000038007,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_INFO_KHR = 1000038008,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_LAYER_INFO_KHR = 1000038009,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_CREATE_INFO_KHR = 1000038010,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_QUALITY_LEVEL_PROPERTIES_KHR = 1000038011,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_GET_INFO_KHR = 1000038012,
// Provided by VK_KHR_video_encode_h264
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_FEEDBACK_INFO_KHR =
1000038013,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_CAPABILITIES_KHR = 1000039000,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR =
1000039001,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR = 1000039002,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PICTURE_INFO_KHR = 1000039003,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_DPB_SLOT_INFO_KHR = 1000039004,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_NALU_SLICE_SEGMENT_INFO_KHR = 1000039005,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_GOP_REMAINING_FRAME_INFO_KHR = 1000039006,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PROFILE_INFO_KHR = 1000039007,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_INFO_KHR = 1000039009,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_LAYER_INFO_KHR = 1000039010,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_CREATE_INFO_KHR = 1000039011,

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// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_QUALITY_LEVEL_PROPERTIES_KHR = 1000039012,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_GET_INFO_KHR = 1000039013,
// Provided by VK_KHR_video_encode_h265
VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_FEEDBACK_INFO_KHR =
1000039014,
// Provided by VK_KHR_video_decode_h264
VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_CAPABILITIES_KHR = 1000040000,
// Provided by VK_KHR_video_decode_h264
VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PICTURE_INFO_KHR = 1000040001,
// Provided by VK_KHR_video_decode_h264
VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PROFILE_INFO_KHR = 1000040003,
// Provided by VK_KHR_video_decode_h264
VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR =
1000040004,
// Provided by VK_KHR_video_decode_h264
VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR = 1000040005,
// Provided by VK_KHR_video_decode_h264
VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_DPB_SLOT_INFO_KHR = 1000040006,
// Provided by VK_EXT_astc_decode_mode
VK_STRUCTURE_TYPE_IMAGE_VIEW_ASTC_DECODE_MODE_EXT = 1000067000,
// Provided by VK_EXT_astc_decode_mode
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ASTC_DECODE_FEATURES_EXT = 1000067001,
// Provided by VK_KHR_external_memory_win32
VK_STRUCTURE_TYPE_IMPORT_MEMORY_WIN32_HANDLE_INFO_KHR = 1000073000,
// Provided by VK_KHR_external_memory_win32
VK_STRUCTURE_TYPE_EXPORT_MEMORY_WIN32_HANDLE_INFO_KHR = 1000073001,
// Provided by VK_KHR_external_memory_win32
VK_STRUCTURE_TYPE_MEMORY_WIN32_HANDLE_PROPERTIES_KHR = 1000073002,
// Provided by VK_KHR_external_memory_win32
VK_STRUCTURE_TYPE_MEMORY_GET_WIN32_HANDLE_INFO_KHR = 1000073003,
// Provided by VK_KHR_external_memory_fd
VK_STRUCTURE_TYPE_IMPORT_MEMORY_FD_INFO_KHR = 1000074000,
// Provided by VK_KHR_external_memory_fd
VK_STRUCTURE_TYPE_MEMORY_FD_PROPERTIES_KHR = 1000074001,
// Provided by VK_KHR_external_memory_fd
VK_STRUCTURE_TYPE_MEMORY_GET_FD_INFO_KHR = 1000074002,
// Provided by VK_KHR_win32_keyed_mutex
VK_STRUCTURE_TYPE_WIN32_KEYED_MUTEX_ACQUIRE_RELEASE_INFO_KHR = 1000075000,
// Provided by VK_KHR_external_semaphore_win32
VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR = 1000078000,
// Provided by VK_KHR_external_semaphore_win32
VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR = 1000078001,
// Provided by VK_KHR_external_semaphore_win32
VK_STRUCTURE_TYPE_D3D12_FENCE_SUBMIT_INFO_KHR = 1000078002,
// Provided by VK_KHR_external_semaphore_win32
VK_STRUCTURE_TYPE_SEMAPHORE_GET_WIN32_HANDLE_INFO_KHR = 1000078003,
// Provided by VK_KHR_external_semaphore_fd
VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_FD_INFO_KHR = 1000079000,
// Provided by VK_KHR_external_semaphore_fd

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VK_STRUCTURE_TYPE_SEMAPHORE_GET_FD_INFO_KHR = 1000079001,
// Provided by VK_KHR_push_descriptor
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PUSH_DESCRIPTOR_PROPERTIES_KHR = 1000080000,
// Provided by VK_KHR_incremental_present
VK_STRUCTURE_TYPE_PRESENT_REGIONS_KHR = 1000084000,
// Provided by VK_EXT_display_surface_counter
VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_EXT = 1000090000,
// Provided by VK_EXT_display_control
VK_STRUCTURE_TYPE_DISPLAY_POWER_INFO_EXT = 1000091000,
// Provided by VK_EXT_display_control
VK_STRUCTURE_TYPE_DEVICE_EVENT_INFO_EXT = 1000091001,
// Provided by VK_EXT_display_control
VK_STRUCTURE_TYPE_DISPLAY_EVENT_INFO_EXT = 1000091002,
// Provided by VK_EXT_display_control
VK_STRUCTURE_TYPE_SWAPCHAIN_COUNTER_CREATE_INFO_EXT = 1000091003,
// Provided by VK_EXT_discard_rectangles
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DISCARD_RECTANGLE_PROPERTIES_EXT = 1000099000,
// Provided by VK_EXT_discard_rectangles
VK_STRUCTURE_TYPE_PIPELINE_DISCARD_RECTANGLE_STATE_CREATE_INFO_EXT = 1000099001,
// Provided by VK_EXT_conservative_rasterization
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CONSERVATIVE_RASTERIZATION_PROPERTIES_EXT =
1000101000,
// Provided by VK_EXT_conservative_rasterization
VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_CONSERVATIVE_STATE_CREATE_INFO_EXT =
1000101001,
// Provided by VK_EXT_depth_clip_enable
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_ENABLE_FEATURES_EXT = 1000102000,
// Provided by VK_EXT_depth_clip_enable
VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_DEPTH_CLIP_STATE_CREATE_INFO_EXT =
1000102001,
// Provided by VK_EXT_hdr_metadata
VK_STRUCTURE_TYPE_HDR_METADATA_EXT = 1000105000,
// Provided by VK_KHR_shared_presentable_image
VK_STRUCTURE_TYPE_SHARED_PRESENT_SURFACE_CAPABILITIES_KHR = 1000111000,
// Provided by VK_KHR_external_fence_win32
VK_STRUCTURE_TYPE_IMPORT_FENCE_WIN32_HANDLE_INFO_KHR = 1000114000,
// Provided by VK_KHR_external_fence_win32
VK_STRUCTURE_TYPE_EXPORT_FENCE_WIN32_HANDLE_INFO_KHR = 1000114001,
// Provided by VK_KHR_external_fence_win32
VK_STRUCTURE_TYPE_FENCE_GET_WIN32_HANDLE_INFO_KHR = 1000114002,
// Provided by VK_KHR_external_fence_fd
VK_STRUCTURE_TYPE_IMPORT_FENCE_FD_INFO_KHR = 1000115000,
// Provided by VK_KHR_external_fence_fd
VK_STRUCTURE_TYPE_FENCE_GET_FD_INFO_KHR = 1000115001,
// Provided by VK_KHR_performance_query
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_FEATURES_KHR = 1000116000,
// Provided by VK_KHR_performance_query
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_PROPERTIES_KHR = 1000116001,
// Provided by VK_KHR_performance_query
VK_STRUCTURE_TYPE_QUERY_POOL_PERFORMANCE_CREATE_INFO_KHR = 1000116002,
// Provided by VK_KHR_performance_query

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VK_STRUCTURE_TYPE_PERFORMANCE_QUERY_SUBMIT_INFO_KHR = 1000116003,
// Provided by VK_KHR_performance_query
VK_STRUCTURE_TYPE_ACQUIRE_PROFILING_LOCK_INFO_KHR = 1000116004,
// Provided by VK_KHR_performance_query
VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_KHR = 1000116005,
// Provided by VK_KHR_performance_query
VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_DESCRIPTION_KHR = 1000116006,
// Provided by VK_KHR_get_surface_capabilities2
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SURFACE_INFO_2_KHR = 1000119000,
// Provided by VK_KHR_get_surface_capabilities2
VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_KHR = 1000119001,
// Provided by VK_KHR_get_surface_capabilities2
VK_STRUCTURE_TYPE_SURFACE_FORMAT_2_KHR = 1000119002,
// Provided by VK_KHR_get_display_properties2
VK_STRUCTURE_TYPE_DISPLAY_PROPERTIES_2_KHR = 1000121000,
// Provided by VK_KHR_get_display_properties2
VK_STRUCTURE_TYPE_DISPLAY_PLANE_PROPERTIES_2_KHR = 1000121001,
// Provided by VK_KHR_get_display_properties2
VK_STRUCTURE_TYPE_DISPLAY_MODE_PROPERTIES_2_KHR = 1000121002,
// Provided by VK_KHR_get_display_properties2
VK_STRUCTURE_TYPE_DISPLAY_PLANE_INFO_2_KHR = 1000121003,
// Provided by VK_KHR_get_display_properties2
VK_STRUCTURE_TYPE_DISPLAY_PLANE_CAPABILITIES_2_KHR = 1000121004,
// Provided by VK_EXT_debug_utils
VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_NAME_INFO_EXT = 1000128000,
// Provided by VK_EXT_debug_utils
VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_TAG_INFO_EXT = 1000128001,
// Provided by VK_EXT_debug_utils
VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT = 1000128002,
// Provided by VK_EXT_debug_utils
VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CALLBACK_DATA_EXT = 1000128003,
// Provided by VK_EXT_debug_utils
VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CREATE_INFO_EXT = 1000128004,
// Provided by VK_EXT_sample_locations
VK_STRUCTURE_TYPE_SAMPLE_LOCATIONS_INFO_EXT = 1000143000,
// Provided by VK_EXT_sample_locations
VK_STRUCTURE_TYPE_RENDER_PASS_SAMPLE_LOCATIONS_BEGIN_INFO_EXT = 1000143001,
// Provided by VK_EXT_sample_locations
VK_STRUCTURE_TYPE_PIPELINE_SAMPLE_LOCATIONS_STATE_CREATE_INFO_EXT = 1000143002,
// Provided by VK_EXT_sample_locations
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLE_LOCATIONS_PROPERTIES_EXT = 1000143003,
// Provided by VK_EXT_sample_locations
VK_STRUCTURE_TYPE_MULTISAMPLE_PROPERTIES_EXT = 1000143004,
// Provided by VK_EXT_blend_operation_advanced
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_FEATURES_EXT =
1000148000,
// Provided by VK_EXT_blend_operation_advanced
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_PROPERTIES_EXT =
1000148001,
// Provided by VK_EXT_blend_operation_advanced
VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_ADVANCED_STATE_CREATE_INFO_EXT =

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1000148002,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET_ACCELERATION_STRUCTURE_KHR = 1000150007,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_GEOMETRY_INFO_KHR = 1000150000,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_DEVICE_ADDRESS_INFO_KHR = 1000150002,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_AABBS_DATA_KHR = 1000150003,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_INSTANCES_DATA_KHR = 1000150004,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_TRIANGLES_DATA_KHR = 1000150005,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_KHR = 1000150006,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_VERSION_INFO_KHR = 1000150009,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_INFO_KHR = 1000150010,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_TO_MEMORY_INFO_KHR = 1000150011,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_COPY_MEMORY_TO_ACCELERATION_STRUCTURE_INFO_KHR = 1000150012,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_FEATURES_KHR =
1000150013,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_PROPERTIES_KHR =
1000150014,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_INFO_KHR = 1000150017,
// Provided by VK_KHR_acceleration_structure
VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_SIZES_INFO_KHR = 1000150020,
// Provided by VK_KHR_ray_tracing_pipeline
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_FEATURES_KHR = 1000347000,
// Provided by VK_KHR_ray_tracing_pipeline
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_PROPERTIES_KHR =
1000347001,
// Provided by VK_KHR_ray_tracing_pipeline
VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_CREATE_INFO_KHR = 1000150015,
// Provided by VK_KHR_ray_tracing_pipeline
VK_STRUCTURE_TYPE_RAY_TRACING_SHADER_GROUP_CREATE_INFO_KHR = 1000150016,
// Provided by VK_KHR_ray_tracing_pipeline
VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_INTERFACE_CREATE_INFO_KHR = 1000150018,
// Provided by VK_KHR_ray_query
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_QUERY_FEATURES_KHR = 1000348013,
// Provided by VK_EXT_image_drm_format_modifier
VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_EXT = 1000158000,
// Provided by VK_EXT_image_drm_format_modifier
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_DRM_FORMAT_MODIFIER_INFO_EXT = 1000158002,
// Provided by VK_EXT_image_drm_format_modifier

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VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_LIST_CREATE_INFO_EXT = 1000158003,
// Provided by VK_EXT_image_drm_format_modifier
VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_EXPLICIT_CREATE_INFO_EXT = 1000158004,
// Provided by VK_EXT_image_drm_format_modifier
VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_PROPERTIES_EXT = 1000158005,
// Provided by VK_EXT_image_drm_format_modifier with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_2_EXT = 1000158006,
#ifdef VK_ENABLE_BETA_EXTENSIONS
// Provided by VK_KHR_portability_subset
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_FEATURES_KHR = 1000163000,
#endif
#ifdef VK_ENABLE_BETA_EXTENSIONS
// Provided by VK_KHR_portability_subset
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_PROPERTIES_KHR = 1000163001,
#endif
// Provided by VK_EXT_filter_cubic
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_IMAGE_FORMAT_INFO_EXT = 1000170000,
// Provided by VK_EXT_filter_cubic
VK_STRUCTURE_TYPE_FILTER_CUBIC_IMAGE_VIEW_IMAGE_FORMAT_PROPERTIES_EXT =
1000170001,
// Provided by VK_EXT_external_memory_host
VK_STRUCTURE_TYPE_IMPORT_MEMORY_HOST_POINTER_INFO_EXT = 1000178000,
// Provided by VK_EXT_external_memory_host
VK_STRUCTURE_TYPE_MEMORY_HOST_POINTER_PROPERTIES_EXT = 1000178001,
// Provided by VK_EXT_external_memory_host
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_MEMORY_HOST_PROPERTIES_EXT =
1000178002,
// Provided by VK_KHR_shader_clock
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_CLOCK_FEATURES_KHR = 1000181000,
// Provided by VK_KHR_video_decode_h265
VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_CAPABILITIES_KHR = 1000187000,
// Provided by VK_KHR_video_decode_h265
VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR =
1000187001,
// Provided by VK_KHR_video_decode_h265
VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR = 1000187002,
// Provided by VK_KHR_video_decode_h265
VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PROFILE_INFO_KHR = 1000187003,
// Provided by VK_KHR_video_decode_h265
VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PICTURE_INFO_KHR = 1000187004,
// Provided by VK_KHR_video_decode_h265
VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_DPB_SLOT_INFO_KHR = 1000187005,
// Provided by VK_KHR_global_priority
VK_STRUCTURE_TYPE_DEVICE_QUEUE_GLOBAL_PRIORITY_CREATE_INFO_KHR = 1000174000,
// Provided by VK_KHR_global_priority
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GLOBAL_PRIORITY_QUERY_FEATURES_KHR = 1000388000,
// Provided by VK_KHR_global_priority
VK_STRUCTURE_TYPE_QUEUE_FAMILY_GLOBAL_PRIORITY_PROPERTIES_KHR = 1000388001,
// Provided by VK_EXT_pci_bus_info
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PCI_BUS_INFO_PROPERTIES_EXT = 1000212000,

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// Provided by VK_KHR_fragment_shading_rate
VK_STRUCTURE_TYPE_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_KHR = 1000226000,
// Provided by VK_KHR_fragment_shading_rate
VK_STRUCTURE_TYPE_PIPELINE_FRAGMENT_SHADING_RATE_STATE_CREATE_INFO_KHR =
1000226001,
// Provided by VK_KHR_fragment_shading_rate
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_PROPERTIES_KHR =
1000226002,
// Provided by VK_KHR_fragment_shading_rate
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_FEATURES_KHR = 1000226003,
// Provided by VK_KHR_fragment_shading_rate
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_KHR = 1000226004,
// Provided by VK_KHR_fragment_shading_rate with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
VK_STRUCTURE_TYPE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_KHR =
1000044006,
// Provided by VK_KHR_dynamic_rendering_local_read
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_LOCAL_READ_FEATURES_KHR =
1000232000,
// Provided by VK_KHR_dynamic_rendering_local_read
VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_LOCATION_INFO_KHR = 1000232001,
// Provided by VK_KHR_dynamic_rendering_local_read
VK_STRUCTURE_TYPE_RENDERING_INPUT_ATTACHMENT_INDEX_INFO_KHR = 1000232002,
// Provided by VK_EXT_shader_image_atomic_int64
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_IMAGE_ATOMIC_INT64_FEATURES_EXT =
1000234000,
// Provided by VK_KHR_shader_quad_control
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_QUAD_CONTROL_FEATURES_KHR = 1000235000,
// Provided by VK_EXT_memory_budget
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_BUDGET_PROPERTIES_EXT = 1000237000,
// Provided by VK_EXT_memory_priority
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PRIORITY_FEATURES_EXT = 1000238000,
// Provided by VK_EXT_memory_priority
VK_STRUCTURE_TYPE_MEMORY_PRIORITY_ALLOCATE_INFO_EXT = 1000238001,
// Provided by VK_KHR_surface_protected_capabilities
VK_STRUCTURE_TYPE_SURFACE_PROTECTED_CAPABILITIES_KHR = 1000239000,
// Provided by VK_KHR_present_wait
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_WAIT_FEATURES_KHR = 1000248000,
// Provided by VK_EXT_fragment_shader_interlock
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_INTERLOCK_FEATURES_EXT =
1000251000,
// Provided by VK_EXT_ycbcr_image_arrays
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_YCBCR_IMAGE_ARRAYS_FEATURES_EXT = 1000252000,
// Provided by VK_EXT_provoking_vertex
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_FEATURES_EXT = 1000254000,
// Provided by VK_EXT_provoking_vertex
VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_PROVOKING_VERTEX_STATE_CREATE_INFO_EXT =
1000254001,
// Provided by VK_EXT_provoking_vertex
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_PROPERTIES_EXT = 1000254002,
// Provided by VK_EXT_full_screen_exclusive

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VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_INFO_EXT = 1000255000,
// Provided by VK_EXT_full_screen_exclusive
VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_FULL_SCREEN_EXCLUSIVE_EXT = 1000255002,
// Provided by VK_KHR_win32_surface with VK_EXT_full_screen_exclusive
VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_WIN32_INFO_EXT = 1000255001,
// Provided by VK_EXT_headless_surface
VK_STRUCTURE_TYPE_HEADLESS_SURFACE_CREATE_INFO_EXT = 1000256000,
// Provided by VK_EXT_shader_atomic_float
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_FEATURES_EXT = 1000260000,
// Provided by VK_KHR_pipeline_executable_properties
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_EXECUTABLE_PROPERTIES_FEATURES_KHR =
1000269000,
// Provided by VK_KHR_pipeline_executable_properties
VK_STRUCTURE_TYPE_PIPELINE_INFO_KHR = 1000269001,
// Provided by VK_KHR_pipeline_executable_properties
VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_PROPERTIES_KHR = 1000269002,
// Provided by VK_KHR_pipeline_executable_properties
VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INFO_KHR = 1000269003,
// Provided by VK_KHR_pipeline_executable_properties
VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_STATISTIC_KHR = 1000269004,
// Provided by VK_KHR_pipeline_executable_properties
VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INTERNAL_REPRESENTATION_KHR = 1000269005,
// Provided by VK_EXT_host_image_copy
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_FEATURES_EXT = 1000270000,
// Provided by VK_EXT_host_image_copy
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_PROPERTIES_EXT = 1000270001,
// Provided by VK_EXT_host_image_copy
VK_STRUCTURE_TYPE_MEMORY_TO_IMAGE_COPY_EXT = 1000270002,
// Provided by VK_EXT_host_image_copy
VK_STRUCTURE_TYPE_IMAGE_TO_MEMORY_COPY_EXT = 1000270003,
// Provided by VK_EXT_host_image_copy
VK_STRUCTURE_TYPE_COPY_IMAGE_TO_MEMORY_INFO_EXT = 1000270004,
// Provided by VK_EXT_host_image_copy
VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INFO_EXT = 1000270005,
// Provided by VK_EXT_host_image_copy
VK_STRUCTURE_TYPE_HOST_IMAGE_LAYOUT_TRANSITION_INFO_EXT = 1000270006,
// Provided by VK_EXT_host_image_copy
VK_STRUCTURE_TYPE_COPY_IMAGE_TO_IMAGE_INFO_EXT = 1000270007,
// Provided by VK_EXT_host_image_copy
VK_STRUCTURE_TYPE_SUBRESOURCE_HOST_MEMCPY_SIZE_EXT = 1000270008,
// Provided by VK_EXT_host_image_copy
VK_STRUCTURE_TYPE_HOST_IMAGE_COPY_DEVICE_PERFORMANCE_QUERY_EXT = 1000270009,
// Provided by VK_KHR_map_memory2
VK_STRUCTURE_TYPE_MEMORY_MAP_INFO_KHR = 1000271000,
// Provided by VK_KHR_map_memory2
VK_STRUCTURE_TYPE_MEMORY_UNMAP_INFO_KHR = 1000271001,
// Provided by VK_EXT_map_memory_placed
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_FEATURES_EXT = 1000272000,
// Provided by VK_EXT_map_memory_placed
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_PROPERTIES_EXT = 1000272001,
// Provided by VK_EXT_map_memory_placed

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VK_STRUCTURE_TYPE_MEMORY_MAP_PLACED_INFO_EXT = 1000272002,
// Provided by VK_EXT_shader_atomic_float2
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_2_FEATURES_EXT = 1000273000,
// Provided by VK_EXT_depth_bias_control
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_BIAS_CONTROL_FEATURES_EXT = 1000283000,
// Provided by VK_EXT_depth_bias_control
VK_STRUCTURE_TYPE_DEPTH_BIAS_INFO_EXT = 1000283001,
// Provided by VK_EXT_depth_bias_control
VK_STRUCTURE_TYPE_DEPTH_BIAS_REPRESENTATION_INFO_EXT = 1000283002,
// Provided by VK_EXT_device_memory_report
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_MEMORY_REPORT_FEATURES_EXT = 1000284000,
// Provided by VK_EXT_device_memory_report
VK_STRUCTURE_TYPE_DEVICE_DEVICE_MEMORY_REPORT_CREATE_INFO_EXT = 1000284001,
// Provided by VK_EXT_device_memory_report
VK_STRUCTURE_TYPE_DEVICE_MEMORY_REPORT_CALLBACK_DATA_EXT = 1000284002,
// Provided by VK_EXT_robustness2
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_FEATURES_EXT = 1000286000,
// Provided by VK_EXT_robustness2
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_PROPERTIES_EXT = 1000286001,
// Provided by VK_EXT_custom_border_color
VK_STRUCTURE_TYPE_SAMPLER_CUSTOM_BORDER_COLOR_CREATE_INFO_EXT = 1000287000,
// Provided by VK_EXT_custom_border_color
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_PROPERTIES_EXT = 1000287001,
// Provided by VK_EXT_custom_border_color
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_FEATURES_EXT = 1000287002,
// Provided by VK_KHR_pipeline_library
VK_STRUCTURE_TYPE_PIPELINE_LIBRARY_CREATE_INFO_KHR = 1000290000,
// Provided by VK_KHR_present_id
VK_STRUCTURE_TYPE_PRESENT_ID_KHR = 1000294000,
// Provided by VK_KHR_present_id
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_ID_FEATURES_KHR = 1000294001,
// Provided by VK_KHR_video_encode_queue
VK_STRUCTURE_TYPE_VIDEO_ENCODE_INFO_KHR = 1000299000,
// Provided by VK_KHR_video_encode_queue
VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_INFO_KHR = 1000299001,
// Provided by VK_KHR_video_encode_queue
VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_LAYER_INFO_KHR = 1000299002,
// Provided by VK_KHR_video_encode_queue
VK_STRUCTURE_TYPE_VIDEO_ENCODE_CAPABILITIES_KHR = 1000299003,
// Provided by VK_KHR_video_encode_queue
VK_STRUCTURE_TYPE_VIDEO_ENCODE_USAGE_INFO_KHR = 1000299004,
// Provided by VK_KHR_video_encode_queue
VK_STRUCTURE_TYPE_QUERY_POOL_VIDEO_ENCODE_FEEDBACK_CREATE_INFO_KHR = 1000299005,
// Provided by VK_KHR_video_encode_queue
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR =
1000299006,
// Provided by VK_KHR_video_encode_queue
VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_PROPERTIES_KHR = 1000299007,
// Provided by VK_KHR_video_encode_queue
VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR = 1000299008,
// Provided by VK_KHR_video_encode_queue

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VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_GET_INFO_KHR = 1000299009,
// Provided by VK_KHR_video_encode_queue
VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_FEEDBACK_INFO_KHR = 1000299010,
// Provided by VK_KHR_fragment_shader_barycentric
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_FEATURES_KHR =
1000203000,
// Provided by VK_KHR_fragment_shader_barycentric
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_PROPERTIES_KHR =
1000322000,
// Provided by VK_KHR_shader_subgroup_uniform_control_flow

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_UNIFORM_CONTROL_FLOW_FEATURES_KHR =
1000323000,
// Provided by VK_KHR_workgroup_memory_explicit_layout
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_WORKGROUP_MEMORY_EXPLICIT_LAYOUT_FEATURES_KHR =
1000336000,
// Provided by VK_EXT_attachment_feedback_loop_layout
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_LAYOUT_FEATURES_EXT =
1000339000,
// Provided by VK_EXT_vertex_input_dynamic_state
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_INPUT_DYNAMIC_STATE_FEATURES_EXT =
1000352000,
// Provided by VK_EXT_shader_object, VK_EXT_vertex_input_dynamic_state
VK_STRUCTURE_TYPE_VERTEX_INPUT_BINDING_DESCRIPTION_2_EXT = 1000352001,
// Provided by VK_EXT_shader_object, VK_EXT_vertex_input_dynamic_state
VK_STRUCTURE_TYPE_VERTEX_INPUT_ATTRIBUTE_DESCRIPTION_2_EXT = 1000352002,
// Provided by VK_EXT_physical_device_drm
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRM_PROPERTIES_EXT = 1000353000,
// Provided by VK_EXT_depth_clip_control
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_CONTROL_FEATURES_EXT = 1000355000,
// Provided by VK_EXT_depth_clip_control
VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLIP_CONTROL_CREATE_INFO_EXT =
1000355001,
// Provided by VK_EXT_primitive_topology_list_restart
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVE_TOPOLOGY_LIST_RESTART_FEATURES_EXT =
1000356000,
// Provided by VK_EXT_present_mode_fifo_latest_ready
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_MODE_FIFO_LATEST_READY_FEATURES_EXT =
1000361000,
// Provided by VK_EXT_frame_boundary
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAME_BOUNDARY_FEATURES_EXT = 1000375000,
// Provided by VK_EXT_frame_boundary
VK_STRUCTURE_TYPE_FRAME_BOUNDARY_EXT = 1000375001,
// Provided by VK_EXT_color_write_enable
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COLOR_WRITE_ENABLE_FEATURES_EXT = 1000381000,
// Provided by VK_EXT_color_write_enable
VK_STRUCTURE_TYPE_PIPELINE_COLOR_WRITE_CREATE_INFO_EXT = 1000381001,
// Provided by VK_EXT_primitives_generated_query
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVES_GENERATED_QUERY_FEATURES_EXT =
1000382000,
// Provided by VK_KHR_ray_tracing_maintenance1

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VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_MAINTENANCE_1_FEATURES_KHR =
1000386000,
// Provided by VK_EXT_image_view_min_lod
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_MIN_LOD_FEATURES_EXT = 1000391000,
// Provided by VK_EXT_image_view_min_lod
VK_STRUCTURE_TYPE_IMAGE_VIEW_MIN_LOD_CREATE_INFO_EXT = 1000391001,
// Provided by VK_EXT_multi_draw
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_FEATURES_EXT = 1000392000,
// Provided by VK_EXT_multi_draw
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_PROPERTIES_EXT = 1000392001,
// Provided by VK_EXT_shader_tile_image
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_FEATURES_EXT = 1000395000,
// Provided by VK_EXT_shader_tile_image
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_PROPERTIES_EXT = 1000395001,
// Provided by VK_EXT_opacity_micromap
VK_STRUCTURE_TYPE_MICROMAP_BUILD_INFO_EXT = 1000396000,
// Provided by VK_EXT_opacity_micromap
VK_STRUCTURE_TYPE_MICROMAP_VERSION_INFO_EXT = 1000396001,
// Provided by VK_EXT_opacity_micromap
VK_STRUCTURE_TYPE_COPY_MICROMAP_INFO_EXT = 1000396002,
// Provided by VK_EXT_opacity_micromap
VK_STRUCTURE_TYPE_COPY_MICROMAP_TO_MEMORY_INFO_EXT = 1000396003,
// Provided by VK_EXT_opacity_micromap
VK_STRUCTURE_TYPE_COPY_MEMORY_TO_MICROMAP_INFO_EXT = 1000396004,
// Provided by VK_EXT_opacity_micromap
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_FEATURES_EXT = 1000396005,
// Provided by VK_EXT_opacity_micromap
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_PROPERTIES_EXT = 1000396006,
// Provided by VK_EXT_opacity_micromap
VK_STRUCTURE_TYPE_MICROMAP_CREATE_INFO_EXT = 1000396007,
// Provided by VK_EXT_opacity_micromap
VK_STRUCTURE_TYPE_MICROMAP_BUILD_SIZES_INFO_EXT = 1000396008,
// Provided by VK_EXT_opacity_micromap
VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_TRIANGLES_OPACITY_MICROMAP_EXT =
1000396009,
// Provided by VK_EXT_border_color_swizzle
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BORDER_COLOR_SWIZZLE_FEATURES_EXT = 1000411000,
// Provided by VK_EXT_border_color_swizzle
VK_STRUCTURE_TYPE_SAMPLER_BORDER_COLOR_COMPONENT_MAPPING_CREATE_INFO_EXT =
1000411001,
// Provided by VK_KHR_shader_subgroup_rotate
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_ROTATE_FEATURES_KHR =
1000416000,
// Provided by VK_EXT_non_seamless_cube_map
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NON_SEAMLESS_CUBE_MAP_FEATURES_EXT = 1000422000,
// Provided by VK_KHR_shader_maximal_reconvergence
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MAXIMAL_RECONVERGENCE_FEATURES_KHR =
1000434000,
// Provided by VK_EXT_nested_command_buffer
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_FEATURES_EXT = 1000451000,
// Provided by VK_EXT_nested_command_buffer

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VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_PROPERTIES_EXT =
1000451001,
// Provided by VK_EXT_external_memory_acquire_unmodified
VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_ACQUIRE_UNMODIFIED_EXT = 1000453000,
// Provided by VK_EXT_extended_dynamic_state3
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_FEATURES_EXT =
1000455000,
// Provided by VK_EXT_extended_dynamic_state3
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_PROPERTIES_EXT =
1000455001,
// Provided by VK_KHR_maintenance5
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_FEATURES_KHR = 1000470000,
// Provided by VK_KHR_maintenance5
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_PROPERTIES_KHR = 1000470001,
// Provided by VK_KHR_maintenance5
VK_STRUCTURE_TYPE_RENDERING_AREA_INFO_KHR = 1000470003,
// Provided by VK_KHR_maintenance5
VK_STRUCTURE_TYPE_DEVICE_IMAGE_SUBRESOURCE_INFO_KHR = 1000470004,
// Provided by VK_KHR_maintenance5
VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2_KHR = 1000338002,
// Provided by VK_KHR_maintenance5
VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2_KHR = 1000338003,
// Provided by VK_KHR_maintenance5
VK_STRUCTURE_TYPE_PIPELINE_CREATE_FLAGS_2_CREATE_INFO_KHR = 1000470005,
// Provided by VK_KHR_maintenance5
VK_STRUCTURE_TYPE_BUFFER_USAGE_FLAGS_2_CREATE_INFO_KHR = 1000470006,
// Provided by VK_KHR_ray_tracing_position_fetch
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_POSITION_FETCH_FEATURES_KHR =
1000481000,
// Provided by VK_EXT_shader_object
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_FEATURES_EXT = 1000482000,
// Provided by VK_EXT_shader_object
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_PROPERTIES_EXT = 1000482001,
// Provided by VK_EXT_shader_object
VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT = 1000482002,
// Provided by VK_KHR_pipeline_binary
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_FEATURES_KHR = 1000483000,
// Provided by VK_KHR_pipeline_binary
VK_STRUCTURE_TYPE_PIPELINE_BINARY_CREATE_INFO_KHR = 1000483001,
// Provided by VK_KHR_pipeline_binary
VK_STRUCTURE_TYPE_PIPELINE_BINARY_INFO_KHR = 1000483002,
// Provided by VK_KHR_pipeline_binary
VK_STRUCTURE_TYPE_PIPELINE_BINARY_KEY_KHR = 1000483003,
// Provided by VK_KHR_pipeline_binary
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_PROPERTIES_KHR = 1000483004,
// Provided by VK_KHR_pipeline_binary
VK_STRUCTURE_TYPE_RELEASE_CAPTURED_PIPELINE_DATA_INFO_KHR = 1000483005,
// Provided by VK_KHR_pipeline_binary
VK_STRUCTURE_TYPE_PIPELINE_BINARY_DATA_INFO_KHR = 1000483006,
// Provided by VK_KHR_pipeline_binary
VK_STRUCTURE_TYPE_PIPELINE_CREATE_INFO_KHR = 1000483007,

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// Provided by VK_KHR_pipeline_binary
VK_STRUCTURE_TYPE_DEVICE_PIPELINE_BINARY_INTERNAL_CACHE_CONTROL_KHR = 1000483008,
// Provided by VK_KHR_pipeline_binary
VK_STRUCTURE_TYPE_PIPELINE_BINARY_HANDLES_INFO_KHR = 1000483009,
// Provided by VK_EXT_legacy_vertex_attributes
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_FEATURES_EXT =
1000495000,
// Provided by VK_EXT_legacy_vertex_attributes
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_PROPERTIES_EXT =
1000495001,
// Provided by VK_EXT_layer_settings
VK_STRUCTURE_TYPE_LAYER_SETTINGS_CREATE_INFO_EXT = 1000496000,
// Provided by VK_EXT_dynamic_rendering_unused_attachments

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_UNUSED_ATTACHMENTS_FEATURES_EXT =
1000499000,
// Provided by VK_KHR_cooperative_matrix
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_FEATURES_KHR = 1000506000,
// Provided by VK_KHR_cooperative_matrix
VK_STRUCTURE_TYPE_COOPERATIVE_MATRIX_PROPERTIES_KHR = 1000506001,
// Provided by VK_KHR_cooperative_matrix
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_PROPERTIES_KHR = 1000506002,
// Provided by VK_KHR_compute_shader_derivatives
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_FEATURES_KHR =
1000201000,
// Provided by VK_KHR_compute_shader_derivatives
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_PROPERTIES_KHR =
1000511000,
// Provided by VK_KHR_video_decode_av1
VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_CAPABILITIES_KHR = 1000512000,
// Provided by VK_KHR_video_decode_av1
VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PICTURE_INFO_KHR = 1000512001,
// Provided by VK_KHR_video_decode_av1
VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PROFILE_INFO_KHR = 1000512003,
// Provided by VK_KHR_video_decode_av1
VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_SESSION_PARAMETERS_CREATE_INFO_KHR =
1000512004,
// Provided by VK_KHR_video_decode_av1
VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_DPB_SLOT_INFO_KHR = 1000512005,
// Provided by VK_KHR_video_maintenance1
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_MAINTENANCE_1_FEATURES_KHR = 1000515000,
// Provided by VK_KHR_video_maintenance1
VK_STRUCTURE_TYPE_VIDEO_INLINE_QUERY_INFO_KHR = 1000515001,
// Provided by VK_EXT_attachment_feedback_loop_dynamic_state

VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_DYNAMIC_STATE_FEATURES_EXT
= 1000524000,
// Provided by VK_KHR_vertex_attribute_divisor
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_PROPERTIES_KHR =
1000525000,
// Provided by VK_KHR_vertex_attribute_divisor

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VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_DIVISOR_STATE_CREATE_INFO_KHR =
1000190001,
// Provided by VK_KHR_vertex_attribute_divisor
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_FEATURES_KHR =
1000190002,
// Provided by VK_KHR_shader_float_controls2
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT_CONTROLS_2_FEATURES_KHR =
1000528000,
// Provided by VK_KHR_index_type_uint8
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INDEX_TYPE_UINT8_FEATURES_KHR = 1000265000,
// Provided by VK_KHR_line_rasterization
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_FEATURES_KHR = 1000259000,
// Provided by VK_KHR_line_rasterization
VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_LINE_STATE_CREATE_INFO_KHR = 1000259001,
// Provided by VK_KHR_line_rasterization
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_PROPERTIES_KHR = 1000259002,
// Provided by VK_KHR_calibrated_timestamps
VK_STRUCTURE_TYPE_CALIBRATED_TIMESTAMP_INFO_KHR = 1000184000,
// Provided by VK_KHR_shader_expect_assume
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_EXPECT_ASSUME_FEATURES_KHR = 1000544000,
// Provided by VK_KHR_maintenance6
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_FEATURES_KHR = 1000545000,
// Provided by VK_KHR_maintenance6
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_PROPERTIES_KHR = 1000545001,
// Provided by VK_KHR_maintenance6
VK_STRUCTURE_TYPE_BIND_MEMORY_STATUS_KHR = 1000545002,
// Provided by VK_KHR_maintenance6
VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_SETS_INFO_KHR = 1000545003,
// Provided by VK_KHR_maintenance6
VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO_KHR = 1000545004,
// Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_INFO_KHR = 1000545005,
// Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_WITH_TEMPLATE_INFO_KHR = 1000545006,
// Provided by VK_KHR_maintenance6 with VK_EXT_descriptor_buffer
VK_STRUCTURE_TYPE_SET_DESCRIPTOR_BUFFER_OFFSETS_INFO_EXT = 1000545007,
// Provided by VK_KHR_maintenance6 with VK_EXT_descriptor_buffer
VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_BUFFER_EMBEDDED_SAMPLERS_INFO_EXT = 1000545008,
// Provided by VK_KHR_shader_relaxed_extended_instruction
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_RELAXED_EXTENDED_INSTRUCTION_FEATURES_KHR
= 1000558000,
// Provided by VK_KHR_maintenance7
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_FEATURES_KHR = 1000562000,
// Provided by VK_KHR_maintenance7
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_PROPERTIES_KHR = 1000562001,
// Provided by VK_KHR_maintenance7
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_LIST_KHR = 1000562002,
// Provided by VK_KHR_maintenance7
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_KHR = 1000562003,
// Provided by VK_KHR_maintenance7
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_VULKAN_PROPERTIES_KHR = 1000562004,

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// Provided by VK_EXT_shader_replicated_composites
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_REPLICATED_COMPOSITES_FEATURES_EXT =
1000564000,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_FEATURES_EXT =
1000572000,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_PROPERTIES_EXT =
1000572001,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_GENERATED_COMMANDS_MEMORY_REQUIREMENTS_INFO_EXT = 1000572002,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_CREATE_INFO_EXT = 1000572003,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_GENERATED_COMMANDS_INFO_EXT = 1000572004,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_CREATE_INFO_EXT = 1000572006,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_TOKEN_EXT = 1000572007,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_PIPELINE_EXT = 1000572008,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_SHADER_EXT = 1000572009,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_PIPELINE_INFO_EXT = 1000572010,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_INFO_EXT = 1000572011,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_LAYOUT_INFO_EXT = 1000572012,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_GENERATED_COMMANDS_PIPELINE_INFO_EXT = 1000572013,
// Provided by VK_EXT_device_generated_commands
VK_STRUCTURE_TYPE_GENERATED_COMMANDS_SHADER_INFO_EXT = 1000572014,
// Provided by VK_EXT_depth_clamp_control
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_CONTROL_FEATURES_EXT = 1000582000,
// Provided by VK_EXT_depth_clamp_control
VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLAMP_CONTROL_CREATE_INFO_EXT =
1000582001,
// Provided by VK_KHR_dynamic_rendering
VK_STRUCTURE_TYPE_RENDERING_INFO_KHR = VK_STRUCTURE_TYPE_RENDERING_INFO,
// Provided by VK_KHR_dynamic_rendering
VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_INFO_KHR =
VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_INFO,
// Provided by VK_KHR_dynamic_rendering
VK_STRUCTURE_TYPE_PIPELINE_RENDERING_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_PIPELINE_RENDERING_CREATE_INFO,
// Provided by VK_KHR_dynamic_rendering
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_FEATURES,
// Provided by VK_KHR_dynamic_rendering
VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_RENDERING_INFO_KHR =

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VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_RENDERING_INFO,
    // Provided by VK_KHR_multiview
    VK_STRUCTURE_TYPE_RENDER_PASS_MULTIVIEW_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_RENDER_PASS_MULTIVIEW_CREATE_INFO,
    // Provided by VK_KHR_multiview
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_FEATURES,
    // Provided by VK_KHR_multiview
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_PROPERTIES,
    // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2,
    // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROPERTIES_2_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROPERTIES_2,
    // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_2_KHR = VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_2,
    // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_IMAGE_FORMAT_PROPERTIES_2_KHR =
VK_STRUCTURE_TYPE_IMAGE_FORMAT_PROPERTIES_2,
    // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_FORMAT_INFO_2_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_FORMAT_INFO_2,
    // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_QUEUE_FAMILY_PROPERTIES_2_KHR =
VK_STRUCTURE_TYPE_QUEUE_FAMILY_PROPERTIES_2,
    // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PROPERTIES_2_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PROPERTIES_2,
    // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_SPARSE_IMAGE_FORMAT_PROPERTIES_2_KHR =
VK_STRUCTURE_TYPE_SPARSE_IMAGE_FORMAT_PROPERTIES_2,
    // Provided by VK_KHR_get_physical_device_properties2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SPARSE_IMAGE_FORMAT_INFO_2_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SPARSE_IMAGE_FORMAT_INFO_2,
    // Provided by VK_KHR_device_group
    VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_FLAGS_INFO_KHR =
VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_FLAGS_INFO,
    // Provided by VK_KHR_device_group
    VK_STRUCTURE_TYPE_DEVICE_GROUP_RENDER_PASS_BEGIN_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_GROUP_RENDER_PASS_BEGIN_INFO,
    // Provided by VK_KHR_device_group
    VK_STRUCTURE_TYPE_DEVICE_GROUP_COMMAND_BUFFER_BEGIN_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_GROUP_COMMAND_BUFFER_BEGIN_INFO,
    // Provided by VK_KHR_device_group
    VK_STRUCTURE_TYPE_DEVICE_GROUP_SUBMIT_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_GROUP_SUBMIT_INFO,
    // Provided by VK_KHR_device_group
    VK_STRUCTURE_TYPE_DEVICE_GROUP_BIND_SPARSE_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_GROUP_BIND_SPARSE_INFO,

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// Provided by VK_KHR_bind_memory2 with VK_KHR_device_group
VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_DEVICE_GROUP_INFO_KHR =
VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_DEVICE_GROUP_INFO,
// Provided by VK_KHR_bind_memory2 with VK_KHR_device_group
VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_DEVICE_GROUP_INFO_KHR =
VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_DEVICE_GROUP_INFO,
// Provided by VK_KHR_device_group_creation
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES,
// Provided by VK_KHR_device_group_creation
VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO,
// Provided by VK_KHR_external_memory_capabilities
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_IMAGE_FORMAT_INFO_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_IMAGE_FORMAT_INFO,
// Provided by VK_KHR_external_memory_capabilities
VK_STRUCTURE_TYPE_EXTERNAL_IMAGE_FORMAT_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_IMAGE_FORMAT_PROPERTIES,
// Provided by VK_KHR_external_memory_capabilities
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_BUFFER_INFO_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_BUFFER_INFO,
// Provided by VK_KHR_external_memory_capabilities
VK_STRUCTURE_TYPE_EXTERNAL_BUFFER_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_BUFFER_PROPERTIES,
// Provided by VK_KHR_external_fence_capabilities,
VK_KHR_external_memory_capabilities, VK_KHR_external_semaphore_capabilities
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES,
// Provided by VK_KHR_external_memory
VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_BUFFER_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_BUFFER_CREATE_INFO,
// Provided by VK_KHR_external_memory
VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_IMAGE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_IMAGE_CREATE_INFO,
// Provided by VK_KHR_external_memory
VK_STRUCTURE_TYPE_EXPORT_MEMORY_ALLOCATE_INFO_KHR =
VK_STRUCTURE_TYPE_EXPORT_MEMORY_ALLOCATE_INFO,
// Provided by VK_KHR_external_semaphore_capabilities
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_SEMAPHORE_INFO_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_SEMAPHORE_INFO,
// Provided by VK_KHR_external_semaphore_capabilities
VK_STRUCTURE_TYPE_EXTERNAL_SEMAPHORE_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_SEMAPHORE_PROPERTIES,
// Provided by VK_KHR_external_semaphore
VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_CREATE_INFO,
// Provided by VK_KHR_shader_float16_int8
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES,
// Provided by VK_KHR_shader_float16_int8
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES_KHR =

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VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES,
    // Provided by VK_KHR_16bit_storage
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_16BIT_STORAGE_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_16BIT_STORAGE_FEATURES,
    // Provided by VK_KHR_descriptor_update_template
    VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO,
    // Provided by VK_EXT_display_surface_counter
    // VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES2_EXT is a deprecated alias
    VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES2_EXT =
VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_EXT,
    // Provided by VK_KHR_imageless_framebuffer
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGELESS_FRAMEBUFFER_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGELESS_FRAMEBUFFER_FEATURES,
    // Provided by VK_KHR_imageless_framebuffer
    VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENTS_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENTS_CREATE_INFO,
    // Provided by VK_KHR_imageless_framebuffer
    VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENT_IMAGE_INFO_KHR =
VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENT_IMAGE_INFO,
    // Provided by VK_KHR_imageless_framebuffer
    VK_STRUCTURE_TYPE_RENDER_PASS_ATTACHMENT_BEGIN_INFO_KHR =
VK_STRUCTURE_TYPE_RENDER_PASS_ATTACHMENT_BEGIN_INFO,
    // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_2_KHR =
VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_2,
    // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_2_KHR =
VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_2,
    // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_2_KHR =
VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_2,
    // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_SUBPASS_DEPENDENCY_2_KHR =
VK_STRUCTURE_TYPE_SUBPASS_DEPENDENCY_2,
    // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO_2_KHR =
VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO_2,
    // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_SUBPASS_BEGIN_INFO_KHR = VK_STRUCTURE_TYPE_SUBPASS_BEGIN_INFO,
    // Provided by VK_KHR_create_renderpass2
    VK_STRUCTURE_TYPE_SUBPASS_END_INFO_KHR = VK_STRUCTURE_TYPE_SUBPASS_END_INFO,
    // Provided by VK_KHR_external_fence_capabilities
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_FENCE_INFO_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_FENCE_INFO,
    // Provided by VK_KHR_external_fence_capabilities
    VK_STRUCTURE_TYPE_EXTERNAL_FENCE_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_EXTERNAL_FENCE_PROPERTIES,
    // Provided by VK_KHR_external_fence
    VK_STRUCTURE_TYPE_EXPORT_FENCE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_EXPORT_FENCE_CREATE_INFO,

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// Provided by VK_KHR_maintenance2
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_POINT_CLIPPING_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_POINT_CLIPPING_PROPERTIES,
// Provided by VK_KHR_maintenance2
VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO,
// Provided by VK_KHR_maintenance2
VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_CREATE_INFO,
// Provided by VK_KHR_maintenance2
VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_DOMAIN_ORIGIN_STATE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_DOMAIN_ORIGIN_STATE_CREATE_INFO,
// Provided by VK_KHR_variable_pointers
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES,
// Provided by VK_KHR_variable_pointers
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTER_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES_KHR,
// Provided by VK_KHR_dedicated_allocation
VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS_KHR =
VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS,
// Provided by VK_KHR_dedicated_allocation
VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO_KHR =
VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO,
// Provided by VK_KHR_get_memory_requirements2
VK_STRUCTURE_TYPE_BUFFER_MEMORY_REQUIREMENTS_INFO_2_KHR =
VK_STRUCTURE_TYPE_BUFFER_MEMORY_REQUIREMENTS_INFO_2,
// Provided by VK_KHR_get_memory_requirements2
VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2_KHR =
VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2,
// Provided by VK_KHR_get_memory_requirements2
VK_STRUCTURE_TYPE_IMAGE_SPARSE_MEMORY_REQUIREMENTS_INFO_2_KHR =
VK_STRUCTURE_TYPE_IMAGE_SPARSE_MEMORY_REQUIREMENTS_INFO_2,
// Provided by VK_KHR_get_memory_requirements2
VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2_KHR =
VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2,
// Provided by VK_KHR_get_memory_requirements2
VK_STRUCTURE_TYPE_SPARSE_IMAGE_MEMORY_REQUIREMENTS_2_KHR =
VK_STRUCTURE_TYPE_SPARSE_IMAGE_MEMORY_REQUIREMENTS_2,
// Provided by VK_KHR_image_format_list
VK_STRUCTURE_TYPE_IMAGE_FORMAT_LIST_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_IMAGE_FORMAT_LIST_CREATE_INFO,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_CREATE_INFO,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_INFO_KHR =
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_INFO,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_STRUCTURE_TYPE_BIND_IMAGE_PLANE_MEMORY_INFO_KHR =
VK_STRUCTURE_TYPE_BIND_IMAGE_PLANE_MEMORY_INFO,

```

```

// Provided by VK_KHR_sampler_ycbcr_conversion
VK_STRUCTURE_TYPE_IMAGE_PLANE_MEMORY_REQUIREMENTS_INFO_KHR =
VK_STRUCTURE_TYPE_IMAGE_PLANE_MEMORY_REQUIREMENTS_INFO,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_YCBCR_CONVERSION_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_YCBCR_CONVERSION_FEATURES,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_IMAGE_FORMAT_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_IMAGE_FORMAT_PROPERTIES,
// Provided by VK_KHR_bind_memory2
VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_INFO_KHR =
VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_INFO,
// Provided by VK_KHR_bind_memory2
VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_INFO_KHR =
VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_INFO,
// Provided by VK_KHR_maintenance3
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_3_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_3_PROPERTIES,
// Provided by VK_KHR_maintenance3
VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_SUPPORT_KHR =
VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_SUPPORT,
// Provided by VK_KHR_shader_subgroup_extended_types
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_EXTENDED_TYPES_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_EXTENDED_TYPES_FEATURES,
// Provided by VK_KHR_8bit_storage
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_8BIT_STORAGE_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_8BIT_STORAGE_FEATURES,
// Provided by VK_KHR_shader_atomic_int64
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_INT64_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_INT64_FEATURES,
// Provided by VK_KHR_driver_properties
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRIVER_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRIVER_PROPERTIES,
// Provided by VK_KHR_shader_float_controls
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT_CONTROLS_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT_CONTROLS_PROPERTIES,
// Provided by VK_KHR_depth_stencil_resolve
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_STENCIL_RESOLVE_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_STENCIL_RESOLVE_PROPERTIES,
// Provided by VK_KHR_depth_stencil_resolve
VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_DEPTH_STENCIL_RESOLVE_KHR =
VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_DEPTH_STENCIL_RESOLVE,
// Provided by VK_KHR_timeline_semaphore
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_FEATURES,
// Provided by VK_KHR_timeline_semaphore
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_PROPERTIES,
// Provided by VK_KHR_timeline_semaphore
VK_STRUCTURE_TYPE_SEMAPHORE_TYPE_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_SEMAPHORE_TYPE_CREATE_INFO,

```

```

// Provided by VK_KHR_timeline_semaphore
VK_STRUCTURE_TYPE_TIMELINE_SEMAPHORE_SUBMIT_INFO_KHR =
VK_STRUCTURE_TYPE_TIMELINE_SEMAPHORE_SUBMIT_INFO,
// Provided by VK_KHR_timeline_semaphore
VK_STRUCTURE_TYPE_SEMAPHORE_WAIT_INFO_KHR = VK_STRUCTURE_TYPE_SEMAPHORE_WAIT_INFO,
// Provided by VK_KHR_timeline_semaphore
VK_STRUCTURE_TYPE_SEMAPHORE_SIGNAL_INFO_KHR =
VK_STRUCTURE_TYPE_SEMAPHORE_SIGNAL_INFO,
// Provided by VK_KHR_vulkan_memory_model
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_MEMORY_MODEL_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_MEMORY_MODEL_FEATURES,
// Provided by VK_KHR_shader_terminate_invocation
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TERMINATE_INVOCATION_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TERMINATE_INVOCATION_FEATURES,
// Provided by VK_KHR_separate_depth_stencil_layouts
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SEPARATE_DEPTH_STENCIL_LAYOUTS_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SEPARATE_DEPTH_STENCIL_LAYOUTS_FEATURES,
// Provided by VK_KHR_separate_depth_stencil_layouts
VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_STENCIL_LAYOUT_KHR =
VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_STENCIL_LAYOUT,
// Provided by VK_KHR_separate_depth_stencil_layouts
VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_STENCIL_LAYOUT_KHR =
VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_STENCIL_LAYOUT,
// Provided by VK_KHR_uniform_buffer_standard_layout
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFORM_BUFFER_STANDARD_LAYOUT_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFORM_BUFFER_STANDARD_LAYOUT_FEATURES,
// Provided by VK_KHR_buffer_device_address
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BUFFER_DEVICE_ADDRESS_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BUFFER_DEVICE_ADDRESS_FEATURES,
// Provided by VK_KHR_buffer_device_address
VK_STRUCTURE_TYPE_BUFFER_DEVICE_ADDRESS_INFO_KHR =
VK_STRUCTURE_TYPE_BUFFER_DEVICE_ADDRESS_INFO,
// Provided by VK_KHR_buffer_device_address
VK_STRUCTURE_TYPE_BUFFER_OPAQUE_CAPTURE_ADDRESS_CREATE_INFO_KHR =
VK_STRUCTURE_TYPE_BUFFER_OPAQUE_CAPTURE_ADDRESS_CREATE_INFO,
// Provided by VK_KHR_buffer_device_address
VK_STRUCTURE_TYPE_MEMORY_OPAQUE_CAPTURE_ADDRESS_ALLOCATE_INFO_KHR =
VK_STRUCTURE_TYPE_MEMORY_OPAQUE_CAPTURE_ADDRESS_ALLOCATE_INFO,
// Provided by VK_KHR_buffer_device_address
VK_STRUCTURE_TYPE_DEVICE_MEMORY_OPAQUE_CAPTURE_ADDRESS_INFO_KHR =
VK_STRUCTURE_TYPE_DEVICE_MEMORY_OPAQUE_CAPTURE_ADDRESS_INFO,
// Provided by VK_KHR_shader_integer_dot_product
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_FEATURES,
// Provided by VK_KHR_shader_integer_dot_product
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_PROPERTIES,
// Provided by VK_KHR_synchronization2
VK_STRUCTURE_TYPE_MEMORY_BARRIER_2_KHR = VK_STRUCTURE_TYPE_MEMORY_BARRIER_2,
// Provided by VK_KHR_synchronization2
VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER_2_KHR =

```

```

VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER_2,
    // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER_2_KHR =
VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER_2,
    // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_DEPENDENCY_INFO_KHR = VK_STRUCTURE_TYPE_DEPENDENCY_INFO,
    // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_SUBMIT_INFO_2_KHR = VK_STRUCTURE_TYPE_SUBMIT_INFO_2,
    // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_SEMAPHORE_SUBMIT_INFO_KHR =
VK_STRUCTURE_TYPE_SEMAPHORE_SUBMIT_INFO,
    // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_COMMAND_BUFFER_SUBMIT_INFO_KHR =
VK_STRUCTURE_TYPE_COMMAND_BUFFER_SUBMIT_INFO,
    // Provided by VK_KHR_synchronization2
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SYNCHRONIZATION_2_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SYNCHRONIZATION_2_FEATURES,
    // Provided by VK_KHR_zero_initialize_workgroup_memory
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_WORKGROUP_MEMORY_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_WORKGROUP_MEMORY_FEATURES,
    // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_COPY_BUFFER_INFO_2_KHR = VK_STRUCTURE_TYPE_COPY_BUFFER_INFO_2,
    // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_COPY_IMAGE_INFO_2_KHR = VK_STRUCTURE_TYPE_COPY_IMAGE_INFO_2,
    // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_COPY_BUFFER_TO_IMAGE_INFO_2_KHR =
VK_STRUCTURE_TYPE_COPY_BUFFER_TO_IMAGE_INFO_2,
    // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_COPY_IMAGE_TO_BUFFER_INFO_2_KHR =
VK_STRUCTURE_TYPE_COPY_IMAGE_TO_BUFFER_INFO_2,
    // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_BLIT_IMAGE_INFO_2_KHR = VK_STRUCTURE_TYPE_BLIT_IMAGE_INFO_2,
    // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_RESOLVE_IMAGE_INFO_2_KHR =
VK_STRUCTURE_TYPE_RESOLVE_IMAGE_INFO_2,
    // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_BUFFER_COPY_2_KHR = VK_STRUCTURE_TYPE_BUFFER_COPY_2,
    // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_IMAGE_COPY_2_KHR = VK_STRUCTURE_TYPE_IMAGE_COPY_2,
    // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_IMAGE_BLIT_2_KHR = VK_STRUCTURE_TYPE_IMAGE_BLIT_2,
    // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_BUFFER_IMAGE_COPY_2_KHR = VK_STRUCTURE_TYPE_BUFFER_IMAGE_COPY_2,
    // Provided by VK_KHR_copy_commands2
    VK_STRUCTURE_TYPE_IMAGE_RESOLVE_2_KHR = VK_STRUCTURE_TYPE_IMAGE_RESOLVE_2,
    // Provided by VK_KHR_format_feature_flags2
    VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_3_KHR = VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_3,
    // Provided by VK_KHR_maintenance4
    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_FEATURES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_FEATURES,
    // Provided by VK_KHR_maintenance4

```

```

    VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_PROPERTIES_KHR =
VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_PROPERTIES,
    // Provided by VK_KHR_maintenance4
    VK_STRUCTURE_TYPE_DEVICE_BUFFER_MEMORY_REQUIREMENTS_KHR =
VK_STRUCTURE_TYPE_DEVICE_BUFFER_MEMORY_REQUIREMENTS,
    // Provided by VK_KHR_maintenance4
    VK_STRUCTURE_TYPE_DEVICE_IMAGE_MEMORY_REQUIREMENTS_KHR =
VK_STRUCTURE_TYPE_DEVICE_IMAGE_MEMORY_REQUIREMENTS,
    // Provided by VK_EXT_shader_object
    VK_STRUCTURE_TYPE_SHADER_REQUIRED_SUBGROUP_SIZE_CREATE_INFO_EXT =
VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_REQUIRED_SUBGROUP_SIZE_CREATE_INFO,
} VkStructureType;

```

3.13. API Name Aliases

A small number of APIs did not follow the [naming conventions](#) when initially defined. For consistency, when we discover an API name that violates the naming conventions, we rename it in the Specification, XML, and header files. For backwards compatibility, the original (incorrect) name is retained as a “typo alias”. The alias is deprecated and should not be used, but will be retained indefinitely.

NOTE

`VK_STENCIL_FRONT_AND_BACK` is an example of a *typo alias*. It was initially defined as part of `VkStencilFaceFlagBits`. Once the naming inconsistency was noticed, it was renamed to `VK_STENCIL_FACE_FRONT_AND_BACK`, and the old name was aliased to the correct name.

Chapter 4. Initialization

Before using Vulkan, an application **must** initialize it by loading the Vulkan commands, and creating a `VkInstance` object.

4.1. Command Function Pointers

Vulkan commands are not necessarily exposed by static linking on a platform. Commands to query function pointers for Vulkan commands are described below.

NOTE

When extensions are [promoted](#) or otherwise incorporated into another extension or Vulkan core version, command [aliases](#) may be included. Whilst the behavior of each command alias is identical, the behavior of retrieving each alias's function pointer is not. A function pointer for a given alias can only be retrieved if the extension or version that introduced that alias is supported and enabled, irrespective of whether any other alias is available.

Function pointers for all Vulkan commands **can** be obtained by calling:

```
// Provided by VK_VERSION_1_0
PFN_vkVoidFunction vkGetInstanceProcAddr(
    VkInstance          instance,
    const char*         pName);
```

- `instance` is the instance that the function pointer will be compatible with, or `NULL` for commands not dependent on any instance.
- `pName` is the name of the command to obtain.

`vkGetInstanceProcAddr` itself is obtained in a platform- and loader- specific manner. Typically, the loader library will export this command as a function symbol, so applications **can** link against the loader library, or load it dynamically and look up the symbol using platform-specific APIs.

The table below defines the various use cases for `vkGetInstanceProcAddr` and expected return value (“fp” is “function pointer”) for each case. A valid returned function pointer (“fp”) **must** not be `NULL`.

The returned function pointer is of type `PFN_vkVoidFunction`, and **must** be cast to the type of the command being queried before use.

Table 1. `vkGetInstanceProcAddr` behavior

<code>instance</code>	<code>pName</code>	return value
* ¹	<code>NULL</code>	undefined
invalid non- <code>NULL</code> instance	* ¹	undefined
<code>NULL</code>	<i>global command</i> ²	fp
instance	<code>vkGetInstanceProcAddr</code>	fp

<i>instance</i>	<i>pName</i>	return value
instance	core <i>dispatchable command</i>	fp ³
instance	enabled instance extension dispatchable command for <i>instance</i>	fp ³
instance	available device extension ⁴ dispatchable command for <i>instance</i>	fp ³
any other case, not covered above		NULL

1

"*" means any representable value for the parameter (including valid values, invalid values, and NULL).

2

The global commands are: [vkEnumerateInstanceExtensionProperties](#), [vkEnumerateInstanceLayerProperties](#), and [vkCreateInstance](#). Dispatchable commands are all other commands which are not global.

3

The returned function pointer **must** only be called with a dispatchable object (the first parameter) that is *instance* or a child of *instance*, e.g. [VkInstance](#), [VkPhysicalDevice](#), [VkDevice](#), [VkQueue](#), or [VkCommandBuffer](#).

4

An “available device extension” is a device extension supported by any physical device enumerated by *instance*.

Valid Usage (Implicit)

- VUID-vkGetInstanceProcAddr-instance-parameter
If *instance* is not NULL, *instance* **must** be a valid [VkInstance](#) handle
- VUID-vkGetInstanceProcAddr-pName-parameter
pName **must** be a null-terminated UTF-8 string

In order to support systems with multiple Vulkan implementations, the function pointers returned by [vkGetInstanceProcAddr](#) **may** point to dispatch code that calls a different real implementation for different [VkDevice](#) objects or their child objects. The overhead of the internal dispatch for [VkDevice](#) objects can be avoided by obtaining device-specific function pointers for any commands that use a device or device-child object as their dispatchable object. Such function pointers **can** be obtained by calling:


```
// Provided by VK_VERSION_1_0
PFN_vkVoidFunction vkGetDeviceProcAddr(
    VkDevice          device,
    const char*       pName);
```

The table below defines the various use cases for `vkGetDeviceProcAddr` and expected return value (“fp” is “function pointer”) for each case. A valid returned function pointer (“fp”) **must** not be `NULL`.

The returned function pointer is of type `PFN_vkVoidFunction`, and **must** be cast to the type of the command being queried before use. The function pointer **must** only be called with a dispatchable object (the first parameter) that is `device` or a child of `device`.

Table 2. `vkGetDeviceProcAddr` behavior

device	pName	return value
<code>NULL</code>	* ¹	undefined
invalid device	* ¹	undefined
device	<code>NULL</code>	undefined
device	requested core version ² device-level dispatchable command ³	fp ⁴
device	enabled extension device-level dispatchable command ³	fp ⁴
any other case, not covered above		<code>NULL</code>

1

“*” means any representable value for the parameter (including valid values, invalid values, and `NULL`).

2

Device-level commands which are part of the core version specified by `VkApplicationInfo::apiVersion` when creating the instance will always return a valid function pointer. If the `maintenance5` feature is enabled, core commands beyond that version which are supported by the implementation will return `NULL`, otherwise the implementation **may** either return `NULL` or a function pointer. If a function pointer is returned, it **must** not be called.

3

In this function, device-level excludes all physical-device-level commands.

4

The returned function pointer **must** only be called with a dispatchable object (the first parameter) that is `device` or a child of `device` e.g. `VkDevice`, `VkQueue`, or `VkCommandBuffer`.

Valid Usage (Implicit)

- VUID-vkGetDeviceProcAddr-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkGetDeviceProcAddr-pName-parameter
pName **must** be a null-terminated UTF-8 string

The definition of [PFN_vkVoidFunction](#) is:

```
// Provided by VK_VERSION_1_0
typedef void (VKAPI_PTR *PFN_vkVoidFunction)(void);
```

This type is returned from command function pointer queries, and **must** be cast to an actual command function pointer before use.

4.1.1. Extending Physical Device From Device Extensions

When the [VK_KHR_get_physical_device_properties2](#) extension is enabled, physical-device-level functionality of a device extension **can** be used with a physical device if the corresponding extension is enumerated by [vkEnumerateDeviceExtensionProperties](#) for that physical device, even before a logical device has been created.

To obtain a function pointer for a physical-device-level command from a device extension, an application **can** use [vkGetInstanceProcAddr](#). This function pointer **may** point to dispatch code, which calls a different real implementation for different [VkPhysicalDevice](#) objects. Applications **must** not use a [VkPhysicalDevice](#) in any command added by an extension or core version that is not supported by that physical device.

Device extensions **may** define structures that **can** be added to the **pNext** chain of physical-device-level commands.

4.2. Instances

There is no global state in Vulkan and all per-application state is stored in a [VkInstance](#) object. Creating a [VkInstance](#) object initializes the Vulkan library and allows the application to pass information about itself to the implementation.

Instances are represented by [VkInstance](#) handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkInstance)
```

To create an instance object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateInstance(
    const VkInstanceCreateInfo*      pCreateInfo,
    const VkAllocationCallbacks*    pAllocator,
    VkInstance*                      pInstance);
```

- **pCreateInfo** is a pointer to a [VkInstanceCreateInfo](#) structure controlling creation of the instance.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pInstance** points a [VkInstance](#) handle in which the resulting instance is returned.

vkCreateInstance verifies that the requested layers exist. If not, **vkCreateInstance** will return **VK_ERROR_LAYER_NOT_PRESENT**. Next **vkCreateInstance** verifies that the requested extensions are supported (e.g. in the implementation or in any enabled instance layer) and if any requested extension is not supported, **vkCreateInstance** **must** return **VK_ERROR_EXTENSION_NOT_PRESENT**. After verifying and enabling the instance layers and extensions the **VkInstance** object is created and returned to the application. If a requested extension is only supported by a layer, both the layer and the extension need to be specified at **vkCreateInstance** time for the creation to succeed.

Valid Usage

- VUID-vkCreateInstance-ppEnabledExtensionNames-01388
All [required extensions](#) for each extension in the [VkInstanceCreateInfo::ppEnabledExtensionNames](#) list **must** also be present in that list

Valid Usage (Implicit)

- VUID-vkCreateInstance-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkInstanceCreateInfo](#) structure
- VUID-vkCreateInstance-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateInstance-pInstance-parameter
pInstance **must** be a valid pointer to a [VkInstance](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

- `VK_ERROR_INITIALIZATION_FAILED`
- `VK_ERROR_LAYER_NOT_PRESENT`
- `VK_ERROR_EXTENSION_NOT_PRESENT`
- `VK_ERROR_INCOMPATIBLE_DRIVER`

The `VkInstanceCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkInstanceCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkInstanceCreateFlags flags;
    const VkApplicationInfo* pApplicationInfo;
    uint32_t              enabledLayerCount;
    const char* const*    ppEnabledLayerNames;
    uint32_t              enabledExtensionCount;
    const char* const*    ppEnabledExtensionNames;
} VkInstanceCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkInstanceCreateFlagBits` indicating the behavior of the instance.
- `pApplicationInfo` is `NULL` or a pointer to a `VkApplicationInfo` structure. If not `NULL`, this information helps implementations recognize behavior inherent to classes of applications. `VkApplicationInfo` is defined in detail below.
- `enabledLayerCount` is the number of global layers to enable.
- `ppEnabledLayerNames` is a pointer to an array of `enabledLayerCount` null-terminated UTF-8 strings containing the names of layers to enable for the created instance. The layers are loaded in the order they are listed in this array, with the first array element being the closest to the application, and the last array element being the closest to the driver. See the [Layers](#) section for further details.
- `enabledExtensionCount` is the number of global extensions to enable.
- `ppEnabledExtensionNames` is a pointer to an array of `enabledExtensionCount` null-terminated UTF-8 strings containing the names of extensions to enable.

To capture events that occur while creating or destroying an instance, an application **can** link a `VkDebugUtilsMessengerCreateInfoEXT` structure to the `pNext` element of the `VkInstanceCreateInfo` structure given to `vkCreateInstance`. This callback is only valid for the duration of the `vkCreateInstance` and the `vkDestroyInstance` call. Use `vkCreateDebugUtilsMessengerEXT` to create persistent callback objects.

Valid Usage

- VUID-VkInstanceCreateInfo-pNext-04926
If the `pNext` chain of `VkInstanceCreateInfo` includes a `VkDebugUtilsMessengerCreateInfoEXT` structure, the list of enabled extensions in `ppEnabledExtensionNames` **must** contain `VK_EXT_debug_utils`
- VUID-VkInstanceCreateInfo-flags-06559
If `flags` has the `VK_INSTANCE_CREATE_ENUMERATE_PORTABILITY_BIT_KHR` bit set, the list of enabled extensions in `ppEnabledExtensionNames` **must** contain `VK_KHR_portability_enumeration`
- VUID-VkInstanceCreateInfo-pNext-10242
If the `pNext` chain of `VkInstanceCreateInfo` includes a `VkLayerSettingsCreateInfoEXT` structure, the list of enabled extensions in `ppEnabledExtensionNames` **must** contain `VK_EXT_layer_settings`

Valid Usage (Implicit)

- VUID-VkInstanceCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_INSTANCE_CREATE_INFO`
- VUID-VkInstanceCreateInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkDebugUtilsMessengerCreateInfoEXT` or `VkLayerSettingsCreateInfoEXT`
- VUID-VkInstanceCreateInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique, with the exception of structures of type `VkDebugUtilsMessengerCreateInfoEXT` or `VkLayerSettingsCreateInfoEXT`
- VUID-VkInstanceCreateInfo-flags-parameter
`flags` **must** be a valid combination of `VkInstanceCreateFlagBits` values
- VUID-VkInstanceCreateInfo-pApplicationInfo-parameter
If `pApplicationInfo` is not `NULL`, `pApplicationInfo` **must** be a valid pointer to a valid `VkApplicationInfo` structure
- VUID-VkInstanceCreateInfo-ppEnabledLayerNames-parameter
If `enabledLayerCount` is not `0`, `ppEnabledLayerNames` **must** be a valid pointer to an array of `enabledLayerCount` null-terminated UTF-8 strings
- VUID-VkInstanceCreateInfo-ppEnabledExtensionNames-parameter
If `enabledExtensionCount` is not `0`, `ppEnabledExtensionNames` **must** be a valid pointer to an array of `enabledExtensionCount` null-terminated UTF-8 strings

```
// Provided by VK_VERSION_1_0
typedef enum VkInstanceCreateFlagBits {
    // Provided by VK_KHR_portability_enumeration
    VK_INSTANCE_CREATE_ENUMERATE_PORTABILITY_BIT_KHR = 0x00000001,
} VkInstanceCreateFlagBits;
```

- `VK_INSTANCE_CREATE_ENUMERATE_PORTABILITY_BIT_KHR` specifies that the instance will enumerate available Vulkan Portability-compliant physical devices and groups in addition to the Vulkan physical devices and groups that are enumerated by default.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkInstanceCreateFlags;
```

`VkInstanceCreateFlags` is a bitmask type for setting a mask of zero or more `VkInstanceCreateFlagBits`.

To create a Vulkan instance with a specific configuration of layer settings, add `VkLayerSettingsCreateInfoEXT` structures to the `pNext` chain of the `VkInstanceCreateInfo` structure, specifying the settings to be configured.

```
// Provided by VK_EXT_layer_settings
typedef struct VkLayerSettingsCreateInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    uint32_t             settingCount;
    const VkLayerSettingEXT* pSettings;
} VkLayerSettingsCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `settingCount` is the number of settings to configure.
- `pSettings` is a pointer to an array of `settingCount` `VkLayerSettingEXT` values specifying the settings to be configured.

Valid Usage (Implicit)

- VUID-VkLayerSettingsCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_LAYER_SETTINGS_CREATE_INFO_EXT`
- VUID-VkLayerSettingsCreateInfoEXT-pSettings-parameter
If `settingCount` is not 0, `pSettings` **must** be a valid pointer to an array of `settingCount` valid `VkLayerSettingEXT` structures

The values of elements of the `VkLayerSettingsCreateInfoEXT::pSettings` array, specifying layer

settings to be configured, are:

```
// Provided by VK_EXT_layer_settings
typedef struct VkLayerSettingEXT {
    const char*      pLayerName;
    const char*      pSettingName;
    VkLayerSettingTypeEXT type;
    uint32_t         valueCount;
    const void*      pValues;
} VkLayerSettingEXT;
```

- **pLayerName** is a pointer to a null-terminated UTF-8 string naming the layer to configure the setting from.
- **pSettingName** is a pointer to a null-terminated UTF-8 string naming the setting to configure. Values of **pSettingName** that are unknown to the layer are ignored.
- **type** is a **VkLayerSettingTypeEXT** value specifying the type of the **pValues** values.
- **valueCount** is the number of values used to configure the layer setting.
- **pValues** is a pointer to an array of **valueCount** values of the type indicated by **type** to configure the layer setting.

When multiple **VkLayerSettingsCreateInfoEXT** structures are chained and the same **pSettingName** is referenced for the same **pLayerName**, the value of the first reference of the layer setting is used.

Valid Usage

- VUID-VkLayerSettingEXT-valueCount-10070
If **valueCount** is not 0, **pValues** **must** be a valid pointer to an array of **valueCount** values of the type indicated by **type**

Valid Usage (Implicit)

- VUID-VkLayerSettingEXT-pLayerName-parameter
pLayerName **must** be a null-terminated UTF-8 string
- VUID-VkLayerSettingEXT-pSettingName-parameter
pSettingName **must** be a null-terminated UTF-8 string
- VUID-VkLayerSettingEXT-type-parameter
type **must** be a valid **VkLayerSettingTypeEXT** value

Possible values of **VkLayerSettingEXT::type**, specifying the type of the data returned in **VkLayerSettingEXT::pValues**, are:


```
// Provided by VK_EXT_layer_settings
typedef enum VkLayerSettingTypeEXT {
    VK_LAYER_SETTING_TYPE_BOOL32_EXT = 0,
    VK_LAYER_SETTING_TYPE_INT32_EXT = 1,
    VK_LAYER_SETTING_TYPE_INT64_EXT = 2,
    VK_LAYER_SETTING_TYPE_UINT32_EXT = 3,
    VK_LAYER_SETTING_TYPE_UINT64_EXT = 4,
    VK_LAYER_SETTING_TYPE_FLOAT32_EXT = 5,
    VK_LAYER_SETTING_TYPE_FLOAT64_EXT = 6,
    VK_LAYER_SETTING_TYPE_STRING_EXT = 7,
} VkLayerSettingTypeEXT;
```

- `VK_LAYER_SETTING_TYPE_BOOL32_EXT` specifies that the layer setting's type is [VkBool32](#).
- `VK_LAYER_SETTING_TYPE_INT32_EXT` specifies that the layer setting's type is signed 32-bit integer.
- `VK_LAYER_SETTING_TYPE_INT64_EXT` specifies that the layer setting's type is signed 64-bit integer.
- `VK_LAYER_SETTING_TYPE_UINT32_EXT` specifies that the layer setting's type is unsigned 32-bit integer.
- `VK_LAYER_SETTING_TYPE_UINT64_EXT` specifies that the layer setting's type is unsigned 64-bit integer.
- `VK_LAYER_SETTING_TYPE_FLOAT32_EXT` specifies that the layer setting's type is 32-bit floating-point.
- `VK_LAYER_SETTING_TYPE_FLOAT64_EXT` specifies that the layer setting's type is 64-bit floating-point.
- `VK_LAYER_SETTING_TYPE_STRING_EXT` specifies that the layer setting's type is a pointer to a null-terminated UTF-8 string.

The `VkApplicationInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkApplicationInfo {
    VkStructureType    sType;
    const void*        pNext;
    const char*        pApplicationName;
    uint32_t           applicationVersion;
    const char*        pEngineName;
    uint32_t           engineVersion;
    uint32_t           apiVersion;
} VkApplicationInfo;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pApplicationName` is `NULL` or is a pointer to a null-terminated UTF-8 string containing the name of the application.
- `applicationVersion` is an unsigned integer variable containing the developer-supplied version number of the application.

- `pEngineName` is `NULL` or is a pointer to a null-terminated UTF-8 string containing the name of the engine (if any) used to create the application.
- `engineVersion` is an unsigned integer variable containing the developer-supplied version number of the engine used to create the application.
- `apiVersion` is the version of the Vulkan API against which the application expects to run, encoded as described in [Version Numbers](#). If `apiVersion` is 0 the implementation **must** ignore it, otherwise if the implementation does not support the requested `apiVersion`, or an effective substitute for `apiVersion`, it **must** return `VK_ERROR_INCOMPATIBLE_DRIVER`. The patch version number specified in `apiVersion` is ignored when creating an instance object. The variant version of the instance **must** match that requested in `apiVersion`.

Valid Usage

- VUID-VkApplicationInfo-apiVersion-04010
If `apiVersion` is not 0, then it **must** be greater than or equal to `VK_API_VERSION_1_0`

Valid Usage (Implicit)

- VUID-VkApplicationInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_APPLICATION_INFO`
- VUID-VkApplicationInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkApplicationInfo-pApplicationName-parameter
If `pApplicationName` is not `NULL`, `pApplicationName` **must** be a null-terminated UTF-8 string
- VUID-VkApplicationInfo-pEngineName-parameter
If `pEngineName` is not `NULL`, `pEngineName` **must** be a null-terminated UTF-8 string

To destroy an instance, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyInstance(
    VkInstance          instance,
    const VkAllocationCallbacks* pAllocator);
```

- `instance` is the handle of the instance to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyInstance-instance-00629
All child objects created using `instance` **must** have been destroyed prior to destroying `instance`

- VUID-vkDestroyInstance-instance-00630

If `VkAllocationCallbacks` were provided when `instance` was created, a compatible set of callbacks **must** be provided here

- VUID-vkDestroyInstance-instance-00631

If no `VkAllocationCallbacks` were provided when `instance` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyInstance-instance-parameter

If `instance` is not `NULL`, `instance` **must** be a valid `VkInstance` handle

- VUID-vkDestroyInstance-pAllocator-parameter

If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

Host Synchronization

- Host access to `instance` **must** be externally synchronized
- Host access to all `VkPhysicalDevice` objects enumerated from `instance` **must** be externally synchronized

Chapter 5. Devices and Queues

Once Vulkan is initialized, devices and queues are the primary objects used to interact with a Vulkan implementation.

Vulkan separates the concept of *physical* and *logical* devices. A physical device usually represents a single complete implementation of Vulkan (excluding instance-level functionality) available to the host, of which there are a finite number. A logical device represents an instance of that implementation with its own state and resources independent of other logical devices.

Physical devices are represented by `VkPhysicalDevice` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkPhysicalDevice)
```

5.1. Physical Devices

To retrieve a list of physical device objects representing the physical devices installed in the system, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumeratePhysicalDevices(
    VkInstance          instance,
    uint32_t*           pPhysicalDeviceCount,
    VkPhysicalDevice*   pPhysicalDevices);
```

- `instance` is a handle to a Vulkan instance previously created with `vkCreateInstance`.
- `pPhysicalDeviceCount` is a pointer to an integer related to the number of physical devices available or queried, as described below.
- `pPhysicalDevices` is either `NULL` or a pointer to an array of `VkPhysicalDevice` handles.

If `pPhysicalDevices` is `NULL`, then the number of physical devices available is returned in `pPhysicalDeviceCount`. Otherwise, `pPhysicalDeviceCount` **must** point to a variable set by the application to the number of elements in the `pPhysicalDevices` array, and on return the variable is overwritten with the number of handles actually written to `pPhysicalDevices`. If `pPhysicalDeviceCount` is less than the number of physical devices available, at most `pPhysicalDeviceCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available physical devices were returned.

Valid Usage (Implicit)

- VUID-vkEnumeratePhysicalDevices-instance-parameter `instance` **must** be a valid `VkInstance` handle
- VUID-vkEnumeratePhysicalDevices-pPhysicalDeviceCount-parameter

`pPhysicalDeviceCount` **must** be a valid pointer to a `uint32_t` value

- VUID-vkEnumeratePhysicalDevices-pPhysicalDevices-parameter

If the value referenced by `pPhysicalDeviceCount` is not 0, and `pPhysicalDevices` is not NULL, `pPhysicalDevices` **must** be a valid pointer to an array of `pPhysicalDeviceCount` `VkPhysicalDevice` handles

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`

To query general properties of physical devices once enumerated, call:

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceProperties(
    VkPhysicalDevice          physicalDevice,
    VkPhysicalDeviceProperties* pProperties);
```

- `physicalDevice` is the handle to the physical device whose properties will be queried.
- `pProperties` is a pointer to a `VkPhysicalDeviceProperties` structure in which properties are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceProperties-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceProperties-pProperties-parameter
`pProperties` **must** be a valid pointer to a `VkPhysicalDeviceProperties` structure

The `VkPhysicalDeviceProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceProperties {
    uint32_t          apiVersion;
    uint32_t          driverVersion;
    uint32_t          vendorID;
    uint32_t          deviceID;
    VkPhysicalDeviceType deviceType;
    char              deviceName[VK_MAX_PHYSICAL_DEVICE_NAME_SIZE];
    uint8_t           pipelineCacheUUID[VK_UUID_SIZE];
    VkPhysicalDeviceLimits limits;
    VkPhysicalDeviceSparseProperties sparseProperties;
} VkPhysicalDeviceProperties;
```

- **apiVersion** is the version of Vulkan supported by the device, encoded as described in [Version Numbers](#).
- **driverVersion** is the vendor-specified version of the driver.
- **vendorID** is a unique identifier for the *vendor* (see below) of the physical device.
- **deviceID** is a unique identifier for the physical device among devices available from the vendor.
- **deviceType** is a [VkPhysicalDeviceType](#) specifying the type of device.
- **deviceName** is an array of `VK_MAX_PHYSICAL_DEVICE_NAME_SIZE` `char` containing a null-terminated UTF-8 string which is the name of the device.
- **pipelineCacheUUID** is an array of `VK_UUID_SIZE` `uint8_t` values representing a universally unique identifier for the device.
- **limits** is the [VkPhysicalDeviceLimits](#) structure specifying device-specific limits of the physical device. See [Limits](#) for details.
- **sparseProperties** is the [VkPhysicalDeviceSparseProperties](#) structure specifying various sparse related properties of the physical device. See [Sparse Properties](#) for details.

NOTE The encoding of **driverVersion** is implementation-defined. It **may** not use the same encoding as **apiVersion**. Applications should follow information from the *vendor* on how to extract the version information from **driverVersion**.

The **vendorID** and **deviceID** fields are provided to allow applications to adapt to device characteristics that are not adequately exposed by other Vulkan queries.

NOTE These **may** include performance profiles, hardware errata, or other characteristics.

The *vendor* identified by **vendorID** is the entity responsible for the most salient characteristics of the underlying implementation of the [VkPhysicalDevice](#) being queried.

NOTE For example, in the case of a discrete GPU implementation, this **should** be the GPU chipset vendor. In the case of a hardware accelerator integrated into a system-on-chip (SoC), this **should** be the supplier of the silicon IP used to create the accelerator.

If the vendor has a [PCI vendor ID](#), the low 16 bits of **vendorID** **must** contain that PCI vendor ID, and the remaining bits **must** be zero. Otherwise, the value returned **must** be a valid Khronos vendor ID, obtained as described in the [Vulkan Documentation and Extensions: Procedures and Conventions](#) document in the section “Registering a Vendor ID with Khronos”. Khronos vendor IDs are allocated starting at 0x10000, to distinguish them from the PCI vendor ID namespace. Khronos vendor IDs are symbolically defined in the [VkVendorId](#) type.

The vendor is also responsible for the value returned in **deviceID**. If the implementation is driven primarily by a [PCI device](#) with a [PCI device ID](#), the low 16 bits of **deviceID** **must** contain that PCI device ID, and the remaining bits **must** be zero. Otherwise, the choice of what values to return **may** be dictated by operating system or platform policies - but **should** uniquely identify both the device version and any major configuration options (for example, core count in the case of multicore devices).

NOTE

The same device ID **should** be used for all physical implementations of that device version and configuration. For example, all uses of a specific silicon IP GPU version and configuration **should** use the same device ID, even if those uses occur in different SoCs.

Khronos vendor IDs which **may** be returned in [VkPhysicalDeviceProperties::vendorID](#) are:

```
// Provided by VK_VERSION_1_0
typedef enum VkVendorId {
    VK_VENDOR_ID_KHRONOS = 0x10000,
    VK_VENDOR_ID_VIV = 0x10001,
    VK_VENDOR_ID_VSI = 0x10002,
    VK_VENDOR_ID_KAZAN = 0x10003,
    VK_VENDOR_ID_CODEPLAY = 0x10004,
    VK_VENDOR_ID_MESA = 0x10005,
    VK_VENDOR_ID_POCL = 0x10006,
    VK_VENDOR_ID_MOBILEYE = 0x10007,
} VkVendorId;
```

NOTE

Khronos vendor IDs may be allocated by vendors at any time. Only the latest canonical versions of this Specification, of the corresponding [vk.xml](#) API Registry, and of the corresponding [vulkan_core.h](#) header file **must** contain all reserved Khronos vendor IDs.

Only Khronos vendor IDs are given symbolic names at present. PCI vendor IDs returned by the implementation can be looked up in the PCI-SIG database.

VK_MAX_PHYSICAL_DEVICE_NAME_SIZE is the length in **char** values of an array containing a physical device name string, as returned in [VkPhysicalDeviceProperties::deviceName](#).

```
#define VK_MAX_PHYSICAL_DEVICE_NAME_SIZE 256U
```

The physical device types which **may** be returned in [VkPhysicalDeviceProperties::deviceType](#) are:


```
// Provided by VK_VERSION_1_0
typedef enum VkPhysicalDeviceType {
    VK_PHYSICAL_DEVICE_TYPE_OTHER = 0,
    VK_PHYSICAL_DEVICE_TYPE_INTEGRATED_GPU = 1,
    VK_PHYSICAL_DEVICE_TYPE_DISCRETE_GPU = 2,
    VK_PHYSICAL_DEVICE_TYPE_VIRTUAL_GPU = 3,
    VK_PHYSICAL_DEVICE_TYPE_CPU = 4,
} VkPhysicalDeviceType;
```

- **VK_PHYSICAL_DEVICE_TYPE_OTHER** - the device does not match any other available types.
- **VK_PHYSICAL_DEVICE_TYPE_INTEGRATED_GPU** - the device is typically one embedded in or tightly coupled with the host.
- **VK_PHYSICAL_DEVICE_TYPE_DISCRETE_GPU** - the device is typically a separate processor connected to the host via an interlink.
- **VK_PHYSICAL_DEVICE_TYPE_VIRTUAL_GPU** - the device is typically a virtual node in a virtualization environment.
- **VK_PHYSICAL_DEVICE_TYPE_CPU** - the device is typically running on the same processors as the host.

The physical device type is advertised for informational purposes only, and does not directly affect the operation of the system. However, the device type **may** correlate with other advertised properties or capabilities of the system, such as how many memory heaps there are.

To query general properties of physical devices once enumerated, call:

```
// Provided by VK_KHR_get_physical_device_properties2
void vkGetPhysicalDeviceProperties2KHR(
    VkPhysicalDevice          physicalDevice,
    VkPhysicalDeviceProperties2* pProperties);
```

- **physicalDevice** is the handle to the physical device whose properties will be queried.
- **pProperties** is a pointer to a **VkPhysicalDeviceProperties2** structure in which properties are returned.

Each structure in **pProperties** and its **pNext** chain contains members corresponding to implementation-dependent properties, behaviors, or limits. **vkGetPhysicalDeviceProperties2KHR** fills in each member to specify the corresponding value for the implementation.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceProperties2-physicalDevice-parameter **physicalDevice** **must** be a valid **VkPhysicalDevice** handle
- VUID-vkGetPhysicalDeviceProperties2-pProperties-parameter **pProperties** **must** be a valid pointer to a **VkPhysicalDeviceProperties2** structure

The `VkPhysicalDeviceProperties2` structure is defined as:

```
typedef struct VkPhysicalDeviceProperties2 {
    VkStructureType      sType;
    void*                pNext;
    VkPhysicalDeviceProperties properties;
} VkPhysicalDeviceProperties2;
```

or the equivalent

```
// Provided by VK_KHR_get_physical_device_properties2
typedef VkPhysicalDeviceProperties2 VkPhysicalDeviceProperties2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `properties` is a `VkPhysicalDeviceProperties` structure describing properties of the physical device. This structure is written with the same values as if it were written by `vkGetPhysicalDeviceProperties`.

The `pNext` chain of this structure is used to extend the structure with properties defined by extensions.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceProperties2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROPERTIES_2`
- VUID-VkPhysicalDeviceProperties2-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkPhysicalDeviceAccelerationStructurePropertiesKHR`, `VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT`, `VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR`, `VkPhysicalDeviceConservativeRasterizationPropertiesEXT`, `VkPhysicalDeviceCooperativeMatrixPropertiesKHR`, `VkPhysicalDeviceCustomBorderColorPropertiesEXT`, `VkPhysicalDeviceDepthStencilResolveProperties`, `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT`, `VkPhysicalDeviceDiscardRectanglePropertiesEXT`, `VkPhysicalDeviceDriverProperties`, `VkPhysicalDeviceDrmPropertiesEXT`, `VkPhysicalDeviceExtendedDynamicState3PropertiesEXT`, `VkPhysicalDeviceExternalMemoryHostPropertiesEXT`, `VkPhysicalDeviceFloatControlsProperties`, `VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR`, `VkPhysicalDeviceFragmentShadingRatePropertiesKHR`, `VkPhysicalDeviceHostImageCopyPropertiesEXT`, `VkPhysicalDeviceIDProperties`,

VkPhysicalDeviceLayeredApiPropertiesListKHR,
 VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT,
 VkPhysicalDeviceLineRasterizationPropertiesKHR,
 VkPhysicalDeviceMaintenance3Properties, VkPhysicalDeviceMaintenance4Properties,
 VkPhysicalDeviceMaintenance5PropertiesKHR,
 VkPhysicalDeviceMaintenance6PropertiesKHR,
 VkPhysicalDeviceMaintenance7PropertiesKHR,
 VkPhysicalDeviceMapMemoryPlacedPropertiesEXT,
 VkPhysicalDeviceMultiDrawPropertiesEXT, VkPhysicalDeviceMultiviewProperties,
 VkPhysicalDeviceNestedCommandBufferPropertiesEXT,
 VkPhysicalDeviceOpacityMicromapPropertiesEXT,
 VkPhysicalDevicePCIBusInfoPropertiesEXT,
 VkPhysicalDevicePerformanceQueryPropertiesKHR,
 VkPhysicalDevicePipelineBinaryPropertiesKHR,
 VkPhysicalDevicePointClippingProperties,
 VkPhysicalDevicePortabilitySubsetPropertiesKHR,
 VkPhysicalDeviceProvokingVertexPropertiesEXT,
 VkPhysicalDevicePushDescriptorPropertiesKHR,
 VkPhysicalDeviceRayTracingPipelinePropertiesKHR,
 VkPhysicalDeviceRobustness2PropertiesEXT,
 VkPhysicalDeviceSampleLocationsPropertiesEXT,
 VkPhysicalDeviceShaderIntegerDotProductProperties,
 VkPhysicalDeviceShaderObjectPropertiesEXT,
 VkPhysicalDeviceShaderTileImagePropertiesEXT,
 VkPhysicalDeviceTimelineSemaphoreProperties,
 VkPhysicalDeviceTransformFeedbackPropertiesEXT,
 VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR

or

- VUID-VkPhysicalDeviceProperties2-sType-unique
 The **sType** value of each struct in the **pNext** chain **must** be unique

The **VkPhysicalDeviceIDProperties** structure is defined as:

```

typedef struct VkPhysicalDeviceIDProperties {
    VkStructureType    sType;
    void*              pNext;
    uint8_t            deviceUUID[VK_UUID_SIZE];
    uint8_t            driverUUID[VK_UUID_SIZE];
    uint8_t            deviceLUID[VK_LUID_SIZE];
    uint32_t           deviceNodeMask;
    VkBool32           deviceLUIDValid;
} VkPhysicalDeviceIDProperties;
  
```

or the equivalent

```
// Provided by VK_KHR_external_fence_capabilities,  
VK_KHR_external_memory_capabilities, VK_KHR_external_semaphore_capabilities  
typedef VkPhysicalDeviceIDProperties VkPhysicalDeviceIDPropertiesKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **deviceUUID** is an array of `VK_UUID_SIZE uint8_t` values representing a universally unique identifier for the device.
- **driverUUID** is an array of `VK_UUID_SIZE uint8_t` values representing a universally unique identifier for the driver build in use by the device.
- **deviceLUID** is an array of `VK_LUID_SIZE uint8_t` values representing a locally unique identifier for the device.
- **deviceNodeMask** is a `uint32_t` bitfield identifying the node within a linked device adapter corresponding to the device.
- **deviceLUIDValid** is a boolean value that will be `VK_TRUE` if **deviceLUID** contains a valid LUID and **deviceNodeMask** contains a valid node mask, and `VK_FALSE` if they do not.

If the `VkPhysicalDeviceIDProperties` structure is included in the **pNext** chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

deviceUUID **must** be immutable for a given device across instances, processes, driver APIs, driver versions, and system reboots.

Applications **can** compare the **driverUUID** value across instance and process boundaries, and **can** make similar queries in external APIs to determine whether they are capable of sharing memory objects and resources using them with the device.

deviceUUID and/or **driverUUID** **must** be used to determine whether a particular external object can be shared between driver components, where such a restriction exists as defined in the compatibility table for the particular object type:

- [External memory handle types compatibility](#)
- [External semaphore handle types compatibility](#)
- [External fence handle types compatibility](#)

If **deviceLUIDValid** is `VK_FALSE`, the values of **deviceLUID** and **deviceNodeMask** are undefined. If **deviceLUIDValid** is `VK_TRUE` and Vulkan is running on the Windows operating system, the contents of **deviceLUID** **can** be cast to an `LUID` object and **must** be equal to the locally unique identifier of a `IDXGIAdapter1` object that corresponds to **physicalDevice**. If **deviceLUIDValid** is `VK_TRUE`, **deviceNodeMask** **must** contain exactly one bit. If Vulkan is running on an operating system that supports the Direct3D 12 API and **physicalDevice** corresponds to an individual device in a linked device adapter, **deviceNodeMask** identifies the Direct3D 12 node corresponding to **physicalDevice**. Otherwise, **deviceNodeMask** **must** be 1.

Although they have identical descriptions, `VkPhysicalDeviceIDProperties::deviceUUID` may differ from `VkPhysicalDeviceProperties2::pipelineCacheUUID`. The former is intended to identify and correlate devices across API and driver boundaries, while the latter is used to identify a compatible device and driver combination to use when serializing and de-serializing pipeline state.

Implementations **should** return `deviceUUID` values which are likely to be unique even in the presence of multiple Vulkan implementations (such as a GPU driver and a software renderer; two drivers for different GPUs; or the same Vulkan driver running on two logically different devices).

NOTE

Khronos' conformance testing is unable to guarantee that `deviceUUID` values are actually unique, so implementors **should** make their own best efforts to ensure this. In particular, hard-coded `deviceUUID` values, especially all-0 bits, **should** never be used.

A combination of values unique to the vendor, the driver, and the hardware environment can be used to provide a `deviceUUID` which is unique to a high degree of certainty. Some possible inputs to such a computation are:

- Information reported by `vkGetPhysicalDeviceProperties`
- PCI device ID (if defined)
- PCI bus ID, or similar system configuration information.
- Driver binary checksums.

NOTE

While `VkPhysicalDeviceIDProperties::deviceUUID` is specified to remain consistent across driver versions and system reboots, it is not intended to be usable as a serializable persistent identifier for a device. It may change when a device is physically added to, removed from, or moved to a different connector in a system while that system is powered down. Further, there is no reasonable way to verify with conformance testing that a given device retains the same UUID in a given system across all driver versions supported in that system. While implementations should make every effort to report consistent device UUIDs across driver versions, applications should avoid relying on the persistence of this value for uses other than identifying compatible devices for external object sharing purposes.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceIDProperties-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES`

`VK_UUID_SIZE` is the length in `uint8_t` values of an array containing a universally unique device or driver build identifier, as returned in `VkPhysicalDeviceIDProperties::deviceUUID` and `VkPhysicalDeviceIDProperties::driverUUID`.

```
#define VK_UUID_SIZE 16U
```

`VK_UUID_SIZE` is the length in `uint8_t` values of an array containing a locally unique device identifier, as returned in `VkPhysicalDeviceIDProperties::deviceLUID`.

```
#define VK_LUID_SIZE 8U
```

or the equivalent

```
#define VK_LUID_SIZE_KHR VK_LUID_SIZE
```

The `VkPhysicalDeviceDriverProperties` structure is defined as:

```
typedef struct VkPhysicalDeviceDriverProperties {  
    VkStructureType    sType;  
    void*              pNext;  
    VkDriverId         driverID;  
    char               driverName[VK_MAX_DRIVER_NAME_SIZE];  
    char               driverInfo[VK_MAX_DRIVER_INFO_SIZE];  
    VkConformanceVersion conformanceVersion;  
} VkPhysicalDeviceDriverProperties;
```

or the equivalent

```
// Provided by VK_KHR_driver_properties  
typedef VkPhysicalDeviceDriverProperties VkPhysicalDeviceDriverPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `driverID` is a unique identifier for the driver of the physical device.
- `driverName` is an array of `VK_MAX_DRIVER_NAME_SIZE` `char` containing a null-terminated UTF-8 string which is the name of the driver.
- `driverInfo` is an array of `VK_MAX_DRIVER_INFO_SIZE` `char` containing a null-terminated UTF-8 string with additional information about the driver.
- `conformanceVersion` is the latest version of the Vulkan conformance test that the implementor has successfully tested this driver against prior to release (see `VkConformanceVersion`).

If the `VkPhysicalDeviceDriverProperties` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

These are properties of the driver corresponding to a physical device.

driverID **must** be immutable for a given driver across instances, processes, driver versions, and system reboots.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDriverProperties-sType-sType
sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRIVER_PROPERTIES`

Khronos driver IDs which **may** be returned in `VkPhysicalDeviceDriverProperties::driverID` are:

```
typedef enum VkDriverId {
    VK_DRIVER_ID_AMD_PROPRIETARY = 1,
    VK_DRIVER_ID_AMD_OPEN_SOURCE = 2,
    VK_DRIVER_ID_MESA_RADV = 3,
    VK_DRIVER_ID_NVIDIA_PROPRIETARY = 4,
    VK_DRIVER_ID_INTEL_PROPRIETARY_WINDOWS = 5,
    VK_DRIVER_ID_INTEL_OPEN_SOURCE_MESA = 6,
    VK_DRIVER_ID_IMAGINATION_PROPRIETARY = 7,
    VK_DRIVER_ID_QUALCOMM_PROPRIETARY = 8,
    VK_DRIVER_ID_ARM_PROPRIETARY = 9,
    VK_DRIVER_ID_GOOGLE_SWIFTSHADER = 10,
    VK_DRIVER_ID_GGP_PROPRIETARY = 11,
    VK_DRIVER_ID_BROADCOM_PROPRIETARY = 12,
    VK_DRIVER_ID_MESA_LLVMPIPE = 13,
    VK_DRIVER_ID_MOLTENVK = 14,
    VK_DRIVER_ID_COREAVI_PROPRIETARY = 15,
    VK_DRIVER_ID_JUICE_PROPRIETARY = 16,
    VK_DRIVER_ID_VERISILICON_PROPRIETARY = 17,
    VK_DRIVER_ID_MESA_TURNIP = 18,
    VK_DRIVER_ID_MESA_V3DV = 19,
    VK_DRIVER_ID_MESA_PANVK = 20,
    VK_DRIVER_ID_SAMSUNG_PROPRIETARY = 21,
    VK_DRIVER_ID_MESA_VENUS = 22,
    VK_DRIVER_ID_MESA_DOZEN = 23,
    VK_DRIVER_ID_MESA_NVK = 24,
    VK_DRIVER_ID_IMAGINATION_OPEN_SOURCE_MESA = 25,
    VK_DRIVER_ID_MESA_HONEYKRISP = 26,
    VK_DRIVER_ID_RESERVED_27 = 27,
    // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_AMD_PROPRIETARY_KHR = VK_DRIVER_ID_AMD_PROPRIETARY,
    // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_AMD_OPEN_SOURCE_KHR = VK_DRIVER_ID_AMD_OPEN_SOURCE,
    // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_MESA_RADV_KHR = VK_DRIVER_ID_MESA_RADV,
    // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_NVIDIA_PROPRIETARY_KHR = VK_DRIVER_ID_NVIDIA_PROPRIETARY,
    // Provided by VK_KHR_driver_properties
    VK_DRIVER_ID_INTEL_PROPRIETARY_WINDOWS_KHR =
    VK_DRIVER_ID_INTEL_PROPRIETARY_WINDOWS,
```



```
// Provided by VK_KHR_driver_properties
VK_DRIVER_ID_INTEL_OPEN_SOURCE_MESA_KHR = VK_DRIVER_ID_INTEL_OPEN_SOURCE_MESA,
// Provided by VK_KHR_driver_properties
VK_DRIVER_ID_IMAGINATION_PROPRIETARY_KHR = VK_DRIVER_ID_IMAGINATION_PROPRIETARY,
// Provided by VK_KHR_driver_properties
VK_DRIVER_ID_QUALCOMM_PROPRIETARY_KHR = VK_DRIVER_ID_QUALCOMM_PROPRIETARY,
// Provided by VK_KHR_driver_properties
VK_DRIVER_ID_ARM_PROPRIETARY_KHR = VK_DRIVER_ID_ARM_PROPRIETARY,
// Provided by VK_KHR_driver_properties
VK_DRIVER_ID_GOOGLE_SWIFTSHADER_KHR = VK_DRIVER_ID_GOOGLE_SWIFTSHADER,
// Provided by VK_KHR_driver_properties
VK_DRIVER_ID_GGP_PROPRIETARY_KHR = VK_DRIVER_ID_GGP_PROPRIETARY,
// Provided by VK_KHR_driver_properties
VK_DRIVER_ID_BROADCOM_PROPRIETARY_KHR = VK_DRIVER_ID_BROADCOM_PROPRIETARY,
} VkDriverId;
```

or the equivalent

```
// Provided by VK_KHR_driver_properties
typedef VkDriverId VkDriverIdKHR;
```

NOTE

Khronos driver IDs may be allocated by vendors at any time. There may be multiple driver IDs for the same vendor, representing different drivers (for e.g. different platforms, proprietary or open source, etc.). Only the latest canonical versions of this Specification, of the corresponding [vk.xml](#) API Registry, and of the corresponding [vulkan_core.h](#) header file **must** contain all reserved Khronos driver IDs.

Only driver IDs registered with Khronos are given symbolic names. There **may** be unregistered driver IDs returned.

[VK_MAX_DRIVER_NAME_SIZE](#) is the length in [char](#) values of an array containing a driver name string, as returned in [VkPhysicalDeviceDriverProperties::driverName](#).

```
#define VK_MAX_DRIVER_NAME_SIZE 256U
```

or the equivalent

```
#define VK_MAX_DRIVER_NAME_SIZE_KHR VK_MAX_DRIVER_NAME_SIZE
```

[VK_MAX_DRIVER_INFO_SIZE](#) is the length in [char](#) values of an array containing a driver information string, as returned in [VkPhysicalDeviceDriverProperties::driverInfo](#).

```
#define VK_MAX_DRIVER_INFO_SIZE 256U
```

or the equivalent

```
#define VK_MAX_DRIVER_INFO_SIZE_KHR    VK_MAX_DRIVER_INFO_SIZE
```

The conformance test suite version an implementation is compliant with is described with the `VkConformanceVersion` structure:

```
typedef struct VkConformanceVersion {
    uint8_t    major;
    uint8_t    minor;
    uint8_t    subminor;
    uint8_t    patch;
} VkConformanceVersion;
```

or the equivalent

```
// Provided by VK_KHR_driver_properties
typedef VkConformanceVersion VkConformanceVersionKHR;
```

- `major` is the major version number of the conformance test suite.
- `minor` is the minor version number of the conformance test suite.
- `subminor` is the subminor version number of the conformance test suite.
- `patch` is the patch version number of the conformance test suite.

The `VkPhysicalDevicePCIBusInfoPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_pci_bus_info
typedef struct VkPhysicalDevicePCIBusInfoPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t            pciDomain;
    uint32_t            pciBus;
    uint32_t            pciDevice;
    uint32_t            pciFunction;
} VkPhysicalDevicePCIBusInfoPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pciDomain` is the PCI bus domain.
- `pciBus` is the PCI bus identifier.
- `pciDevice` is the PCI device identifier.
- `pciFunction` is the PCI device function identifier.

If the `VkPhysicalDevicePCIBusInfoPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

These are properties of the PCI bus information of a physical device.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePCIBusInfoPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PCI_BUS_INFO_PROPERTIES_EXT`

The `VkPhysicalDeviceDrmPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_physical_device_drm
typedef struct VkPhysicalDeviceDrmPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           hasPrimary;
    VkBool32           hasRender;
    int64_t             primaryMajor;
    int64_t             primaryMinor;
    int64_t             renderMajor;
    int64_t             renderMinor;
} VkPhysicalDeviceDrmPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `hasPrimary` is a boolean indicating whether the physical device has a DRM primary node.
- `hasRender` is a boolean indicating whether the physical device has a DRM render node.
- `primaryMajor` is the DRM primary node major number, if any.
- `primaryMinor` is the DRM primary node minor number, if any.
- `renderMajor` is the DRM render node major number, if any.
- `renderMinor` is the DRM render node minor number, if any.

If the `VkPhysicalDeviceDrmPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

These are properties of the DRM information of a physical device.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDrmPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRM_PROPERTIES_EXT`

The `VkPhysicalDeviceShaderIntegerDotProductProperties` structure is defined as:

```
typedef struct VkPhysicalDeviceShaderIntegerDotProductProperties {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           integerDotProduct8BitUnsignedAccelerated;
    VkBool32           integerDotProduct8BitSignedAccelerated;
    VkBool32           integerDotProduct8BitMixedSignednessAccelerated;
    VkBool32           integerDotProduct4x8BitPackedUnsignedAccelerated;
    VkBool32           integerDotProduct4x8BitPackedSignedAccelerated;
    VkBool32           integerDotProduct4x8BitPackedMixedSignednessAccelerated;
    VkBool32           integerDotProduct16BitUnsignedAccelerated;
    VkBool32           integerDotProduct16BitSignedAccelerated;
    VkBool32           integerDotProduct16BitMixedSignednessAccelerated;
    VkBool32           integerDotProduct32BitUnsignedAccelerated;
    VkBool32           integerDotProduct32BitSignedAccelerated;
    VkBool32           integerDotProduct32BitMixedSignednessAccelerated;
    VkBool32           integerDotProduct64BitUnsignedAccelerated;
    VkBool32           integerDotProduct64BitSignedAccelerated;
    VkBool32           integerDotProduct64BitMixedSignednessAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating8BitUnsignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating8BitSignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating8BitMixedSignednessAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating4x8BitPackedUnsignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating4x8BitPackedSignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating4x8BitPackedMixedSignednessAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating16BitUnsignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating16BitSignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating16BitMixedSignednessAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating32BitUnsignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating32BitSignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating32BitMixedSignednessAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating64BitUnsignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating64BitSignedAccelerated;
    VkBool32           integerDotProductAccumulatingSaturating64BitMixedSignednessAccelerated;
} VkPhysicalDeviceShaderIntegerDotProductProperties;
```

or the equivalent

```
// Provided by VK_KHR_shader_integer_dot_product
typedef VkPhysicalDeviceShaderIntegerDotProductProperties
VkPhysicalDeviceShaderIntegerDotProductPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `integerDotProduct8BitUnsignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit unsigned dot product operations using the `OpUDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct8BitSignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit signed dot product operations using the `OpSDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct8BitMixedSignednessAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit mixed signedness dot product operations using the `OpSUDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct4x8BitPackedUnsignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit unsigned dot product operations from operands packed into 32-bit integers using the `OpUDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct4x8BitPackedSignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit signed dot product operations from operands packed into 32-bit integers using the `OpSDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct4x8BitPackedMixedSignednessAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit mixed signedness dot product operations from operands packed into 32-bit integers using the `OpSUDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct16BitUnsignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 16-bit unsigned dot product operations using the `OpUDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct16BitSignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 16-bit signed dot product operations using the `OpSDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct16BitMixedSignednessAccelerated` is a boolean that will be `VK_TRUE` if the support for 16-bit mixed signedness dot product operations using the `OpSUDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct32BitUnsignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 32-bit unsigned dot product operations using the `OpUDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct32BitSignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 32-bit signed dot product operations using the `OpSDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct32BitMixedSignednessAccelerated` is a boolean that will be `VK_TRUE` if the support for 32-bit mixed signedness dot product operations using the `OpSUDotKHR` SPIR-V

instruction is accelerated [as defined below](#).

- `integerDotProduct64BitUnsignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 64-bit unsigned dot product operations using the `OpUDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct64BitSignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 64-bit signed dot product operations using the `OpSDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProduct64BitMixedSignednessAccelerated` is a boolean that will be `VK_TRUE` if the support for 64-bit mixed signedness dot product operations using the `OpSUDotKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating8BitUnsignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit unsigned accumulating saturating dot product operations using the `OpUDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating8BitSignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit signed accumulating saturating dot product operations using the `OpSDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating8BitMixedSignednessAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit mixed signedness accumulating saturating dot product operations using the `OpSUDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating4x8BitPackedUnsignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit unsigned accumulating saturating dot product operations from operands packed into 32-bit integers using the `OpUDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating4x8BitPackedSignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit signed accumulating saturating dot product operations from operands packed into 32-bit integers using the `OpSDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating4x8BitPackedMixedSignednessAccelerated` is a boolean that will be `VK_TRUE` if the support for 8-bit mixed signedness accumulating saturating dot product operations from operands packed into 32-bit integers using the `OpSUDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating16BitUnsignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 16-bit unsigned accumulating saturating dot product operations using the `OpUDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating16BitSignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 16-bit signed accumulating saturating dot product operations using the `OpSDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating16BitMixedSignednessAccelerated` is a boolean that will be `VK_TRUE` if the support for 16-bit mixed signedness accumulating saturating dot product operations using the `OpSUDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating32BitUnsignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 32-bit unsigned accumulating saturating dot product operations using the `OpUDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).

- `integerDotProductAccumulatingSaturating32BitSignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 32-bit signed accumulating saturating dot product operations using the `OpSDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating32BitMixedSignednessAccelerated` is a boolean that will be `VK_TRUE` if the support for 32-bit mixed signedness accumulating saturating dot product operations using the `OpSUDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating64BitUnsignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 64-bit unsigned accumulating saturating dot product operations using the `OpUDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating64BitSignedAccelerated` is a boolean that will be `VK_TRUE` if the support for 64-bit signed accumulating saturating dot product operations using the `OpSDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).
- `integerDotProductAccumulatingSaturating64BitMixedSignednessAccelerated` is a boolean that will be `VK_TRUE` if the support for 64-bit mixed signedness accumulating saturating dot product operations using the `OpSUDotAccSatKHR` SPIR-V instruction is accelerated [as defined below](#).

If the `VkPhysicalDeviceShaderIntegerDotProductProperties` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

These are properties of the integer dot product acceleration information of a physical device.

NOTE

A dot product operation is deemed accelerated if its implementation provides a performance advantage over application-provided code composed from elementary instructions and/or other dot product instructions, either because the implementation uses optimized machine code sequences whose generation from application-provided code cannot be guaranteed or because it uses hardware features that cannot otherwise be targeted from application-provided code.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderIntegerDotProductProperties-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_PROPERTIES`

The `VkPhysicalDeviceShaderTileImagePropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_shader_tile_image
typedef struct VkPhysicalDeviceShaderTileImagePropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderTileImageCoherentReadAccelerated;
    VkBool32           shaderTileImageReadSampleFromPixelRateInvocation;
    VkBool32           shaderTileImageReadFromHelperInvocation;
} VkPhysicalDeviceShaderTileImagePropertiesEXT;
```


- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderTileImageCoherentReadAccelerated` is a boolean that will be `VK_TRUE` if coherent reads of tile image data is accelerated.
- `shaderTileImageReadSampleFromPixelRateInvocation` is a boolean that will be `VK_TRUE` if reading from samples from a pixel rate fragment invocation is supported when `VkPipelineMultisampleStateCreateInfo::rasterizationSamples > 1`.
- `shaderTileImageReadFromHelperInvocation` is a boolean that will be `VK_TRUE` if reads of tile image data from helper fragment invocations result in valid values.

If the `VkPhysicalDeviceShaderTileImagePropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

These are properties of the tile image information of a physical device.

Valid Usage (Implicit)

- `VUID-VkPhysicalDeviceShaderTileImagePropertiesEXT-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_PROPERTIES_EXT`

To query properties of queues available on a physical device, call:

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceQueueFamilyProperties(
    VkPhysicalDevice          physicalDevice,
    uint32_t*                 pQueueFamilyPropertyCount,
    VkQueueFamilyProperties*   pQueueFamilyProperties);
```

- `physicalDevice` is the handle to the physical device whose properties will be queried.
- `pQueueFamilyPropertyCount` is a pointer to an integer related to the number of queue families available or queried, as described below.
- `pQueueFamilyProperties` is either `NULL` or a pointer to an array of `VkQueueFamilyProperties` structures.

If `pQueueFamilyProperties` is `NULL`, then the number of queue families available is returned in `pQueueFamilyPropertyCount`. Implementations **must** support at least one queue family. Otherwise, `pQueueFamilyPropertyCount` **must** point to a variable set by the application to the number of elements in the `pQueueFamilyProperties` array, and on return the variable is overwritten with the number of structures actually written to `pQueueFamilyProperties`. If `pQueueFamilyPropertyCount` is less than the number of queue families available, at most `pQueueFamilyPropertyCount` structures will be written.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceQueueFamilyProperties-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceQueueFamilyProperties-pQueueFamilyPropertyCount-parameter `pQueueFamilyPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceQueueFamilyProperties-pQueueFamilyProperties-parameter
If the value referenced by `pQueueFamilyPropertyCount` is not 0, and `pQueueFamilyProperties` is not `NULL`, `pQueueFamilyProperties` **must** be a valid pointer to an array of `pQueueFamilyPropertyCount` `VkQueueFamilyProperties` structures

The `VkQueueFamilyProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkQueueFamilyProperties {
    VkQueueFlags    queueFlags;
    uint32_t        queueCount;
    uint32_t        timestampValidBits;
    VkExtent3D      minImageTransferGranularity;
} VkQueueFamilyProperties;
```

- `queueFlags` is a bitmask of `VkQueueFlagBits` indicating capabilities of the queues in this queue family.
- `queueCount` is the unsigned integer count of queues in this queue family. Each queue family **must** support at least one queue.
- `timestampValidBits` is the unsigned integer count of meaningful bits in the timestamps written via `vkCmdWriteTimestamp2KHR` or `vkCmdWriteTimestamp`. The valid range for the count is 36 to 64 bits, or a value of 0, indicating no support for timestamps. Bits outside the valid range are guaranteed to be zeros.
- `minImageTransferGranularity` is the minimum granularity supported for image transfer operations on the queues in this queue family.

The value returned in `minImageTransferGranularity` has a unit of compressed texel blocks for images having a block-compressed format, and a unit of texels otherwise.

Possible values of `minImageTransferGranularity` are:

- (0,0,0) specifies that only whole mip levels **must** be transferred using the image transfer operations on the corresponding queues. In this case, the following restrictions apply to all offset and extent parameters of image transfer operations:
 - The `x`, `y`, and `z` members of a `VkOffset3D` parameter **must** always be zero.
 - The `width`, `height`, and `depth` members of a `VkExtent3D` parameter **must** always match the width, height, and depth of the image subresource corresponding to the parameter,

respectively.

- (A_x, A_y, A_z) where A_x , A_y , and A_z are all integer powers of two. In this case the following restrictions apply to all image transfer operations:
 - x , y , and z of a `VkOffset3D` parameter **must** be integer multiples of A_x , A_y , and A_z , respectively.
 - `width` of a `VkExtent3D` parameter **must** be an integer multiple of A_x , or else $x + \text{width}$ **must** equal the width of the image subresource corresponding to the parameter.
 - `height` of a `VkExtent3D` parameter **must** be an integer multiple of A_y , or else $y + \text{height}$ **must** equal the height of the image subresource corresponding to the parameter.
 - `depth` of a `VkExtent3D` parameter **must** be an integer multiple of A_z , or else $z + \text{depth}$ **must** equal the depth of the image subresource corresponding to the parameter.
 - If the format of the image corresponding to the parameters is one of the block-compressed formats then for the purposes of the above calculations the granularity **must** be scaled up by the compressed texel block dimensions.

Queues supporting graphics and/or compute operations **must** report (1,1,1) in `minImageTransferGranularity`, meaning that there are no additional restrictions on the granularity of image transfer operations for these queues. Other queues supporting image transfer operations are only **required** to support whole mip level transfers, thus `minImageTransferGranularity` for queues belonging to such queue families **may** be (0,0,0).

The [Device Memory](#) section describes memory properties queried from the physical device.

For physical device feature queries see the [Features](#) chapter.

Bits which **may** be set in `VkQueueFamilyProperties::queueFlags`, indicating capabilities of queues in a queue family are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueueFlagBits {
    VK_QUEUE_GRAPHICS_BIT = 0x00000001,
    VK_QUEUE_COMPUTE_BIT = 0x00000002,
    VK_QUEUE_TRANSFER_BIT = 0x00000004,
    VK_QUEUE_SPARSE_BINDING_BIT = 0x00000008,
    // Provided by VK_KHR_video_decode_queue
    VK_QUEUE_VIDEO_DECODE_BIT_KHR = 0x00000020,
    // Provided by VK_KHR_video_encode_queue
    VK_QUEUE_VIDEO_ENCODE_BIT_KHR = 0x00000040,
} VkQueueFlagBits;
```

- `VK_QUEUE_GRAPHICS_BIT` specifies that queues in this queue family support graphics operations.
- `VK_QUEUE_COMPUTE_BIT` specifies that queues in this queue family support compute operations.
- `VK_QUEUE_TRANSFER_BIT` specifies that queues in this queue family support transfer operations.
- `VK_QUEUE_SPARSE_BINDING_BIT` specifies that queues in this queue family support sparse memory management operations (see [Sparse Resources](#)). If any of the sparse resource features are

enabled, then at least one queue family **must** support this bit.

- `VK_QUEUE_VIDEO_DECODE_BIT_KHR` specifies that queues in this queue family support [video decode operations](#).
- `VK_QUEUE_VIDEO_ENCODE_BIT_KHR` specifies that queues in this queue family support [video encode operations](#).

If an implementation exposes any queue family that supports graphics operations, at least one queue family of at least one physical device exposed by the implementation **must** support both graphics and compute operations.

NOTE

All commands that are allowed on a queue that supports transfer operations are also allowed on a queue that supports either graphics or compute operations. Thus, if the capabilities of a queue family include `VK_QUEUE_GRAPHICS_BIT` or `VK_QUEUE_COMPUTE_BIT`, then reporting the `VK_QUEUE_TRANSFER_BIT` capability separately for that queue family is **optional**.

For further details see [Queues](#).

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueueFlags;
```

`VkQueueFlags` is a bitmask type for setting a mask of zero or more [VkQueueFlagBits](#).

To query properties of queues available on a physical device, call:

```
// Provided by VK_KHR_get_physical_device_properties2
void vkGetPhysicalDeviceQueueFamilyProperties2KHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t*                 pQueueFamilyPropertyCount,
    VkQueueFamilyProperties2*  pQueueFamilyProperties);
```

- `physicalDevice` is the handle to the physical device whose properties will be queried.
- `pQueueFamilyPropertyCount` is a pointer to an integer related to the number of queue families available or queried, as described in [vkGetPhysicalDeviceQueueFamilyProperties](#).
- `pQueueFamilyProperties` is either `NULL` or a pointer to an array of [VkQueueFamilyProperties2](#) structures.

`vkGetPhysicalDeviceQueueFamilyProperties2KHR` behaves similarly to [vkGetPhysicalDeviceQueueFamilyProperties](#), with the ability to return extended information in a `pNext` chain of output structures.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceQueueFamilyProperties2-physicalDevice-parameter-`physicalDevice` **must** be a valid [VkPhysicalDevice](#) handle

- VUID-vkGetPhysicalDeviceQueueFamilyProperties2-pQueueFamilyPropertyCount-parameter
pQueueFamilyPropertyCount **must** be a valid pointer to a uint32_t value
- VUID-vkGetPhysicalDeviceQueueFamilyProperties2-pQueueFamilyProperties-parameter
If the value referenced by pQueueFamilyPropertyCount is not 0, and pQueueFamilyProperties is not NULL, pQueueFamilyProperties **must** be a valid pointer to an array of pQueueFamilyPropertyCount VkQueueFamilyProperties2 structures

The VkQueueFamilyProperties2 structure is defined as:

```
typedef struct VkQueueFamilyProperties2 {
    VkStructureType      sType;
    void*                pNext;
    VkQueueFamilyProperties queueFamilyProperties;
} VkQueueFamilyProperties2;
```

or the equivalent

```
// Provided by VK_KHR_get_physical_device_properties2
typedef VkQueueFamilyProperties2 VkQueueFamilyProperties2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- queueFamilyProperties is a VkQueueFamilyProperties structure which is populated with the same values as in vkGetPhysicalDeviceQueueFamilyProperties.

Valid Usage (Implicit)

- VUID-VkQueueFamilyProperties2-sType-sType
sType **must** be VK_STRUCTURE_TYPE_QUEUE_FAMILY_PROPERTIES_2
- VUID-VkQueueFamilyProperties2-pNext-pNext
Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkQueueFamilyGlobalPriorityPropertiesKHR, VkQueueFamilyQueryResultStatusPropertiesKHR, or VkQueueFamilyVideoPropertiesKHR
- VUID-VkQueueFamilyProperties2-sType-unique
The sType value of each struct in the pNext chain **must** be unique

The VkQueueFamilyGlobalPriorityPropertiesKHR structure is defined as:

```
// Provided by VK_KHR_global_priority
typedef struct VkQueueFamilyGlobalPriorityPropertiesKHR {
    VkStructureType      sType;
    void*                 pNext;
    uint32_t              priorityCount;
    VkQueueGlobalPriorityKHR priorities[VK_MAX_GLOBAL_PRIORITY_SIZE_KHR];
} VkQueueFamilyGlobalPriorityPropertiesKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **priorityCount** is the number of supported global queue priorities in this queue family, and it **must** be greater than 0.
- **priorities** is an array of [VK_MAX_GLOBAL_PRIORITY_SIZE_KHR](#) [VkQueueGlobalPriorityKHR](#) enums representing all supported global queue priorities in this queue family. The first **priorityCount** elements of the array will be valid.

If the [VkQueueFamilyGlobalPriorityPropertiesKHR](#) structure is included in the **pNext** chain of the [VkQueueFamilyProperties2](#) structure passed to [vkGetPhysicalDeviceQueueFamilyProperties2KHR](#), it is filled in with the list of supported global queue priorities for the indicated family.

The valid elements of **priorities** **must** not contain any duplicate values.

The valid elements of **priorities** **must** be a continuous sequence of [VkQueueGlobalPriorityKHR](#) enums in the ascending order.

NOTE

For example, returning **priorityCount** as 3 with supported **priorities** as [VK_QUEUE_GLOBAL_PRIORITY_LOW_KHR](#), [VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_KHR](#) and [VK_QUEUE_GLOBAL_PRIORITY_REALTIME_KHR](#) is not allowed.

Valid Usage (Implicit)

- VUID-VkQueueFamilyGlobalPriorityPropertiesKHR-sType-sType
sType **must** be [VK_STRUCTURE_TYPE_QUEUE_FAMILY_GLOBAL_PRIORITY_PROPERTIES_KHR](#)

[VK_MAX_GLOBAL_PRIORITY_SIZE_KHR](#) is the length of an array of [VkQueueGlobalPriorityKHR](#) enumerants representing supported queue priorities, as returned in [VkQueueFamilyGlobalPriorityPropertiesKHR::priorities](#).

```
#define VK_MAX_GLOBAL_PRIORITY_SIZE_KHR 16U
```

The [VkQueueFamilyVideoPropertiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkQueueFamilyVideoPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkVideoCodecOperationFlagsKHR videoCodecOperations;
} VkQueueFamilyVideoPropertiesKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **videoCodecOperations** is a bitmask of [VkVideoCodecOperationFlagBitsKHR](#) that indicates the set of video codec operations supported by the queue family.

If this structure is included in the **pNext** chain of the [VkQueueFamilyProperties2](#) structure passed to [vkGetPhysicalDeviceQueueFamilyProperties2KHR](#), then it is filled with the set of video codec operations supported by the specified queue family.

Valid Usage (Implicit)

- VUID-VkQueueFamilyVideoPropertiesKHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_QUEUE_FAMILY_VIDEO_PROPERTIES_KHR**

The [VkQueueFamilyQueryResultStatusPropertiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkQueueFamilyQueryResultStatusPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           queryResultStatusSupport;
} VkQueueFamilyQueryResultStatusPropertiesKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **queryResultStatusSupport** reports **VK_TRUE** if query type **VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR** and use of **VK_QUERY_RESULT_WITH_STATUS_BIT_KHR** are supported.

If this structure is included in the **pNext** chain of the [VkQueueFamilyProperties2](#) structure passed to [vkGetPhysicalDeviceQueueFamilyProperties2KHR](#), then it is filled with information about whether [result status queries](#) are supported by the specified queue family.

Valid Usage (Implicit)

- VUID-VkQueueFamilyQueryResultStatusPropertiesKHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_QUEUE_FAMILY_QUERY_RESULT_STATUS_PROPERTIES_KHR**

To enumerate the performance query counters available on a queue family of a physical device, call:

```
// Provided by VK_KHR_performance_query
VkResult vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t                  queueFamilyIndex,
    uint32_t*                  pCounterCount,
    VkPerformanceCounterKHR*   pCounters,
    VkPerformanceCounterDescriptionKHR* pCounterDescriptions);
```

- **physicalDevice** is the handle to the physical device whose queue family performance query counter properties will be queried.
- **queueFamilyIndex** is the index into the queue family of the physical device we want to get properties for.
- **pCounterCount** is a pointer to an integer related to the number of counters available or queried, as described below.
- **pCounters** is either **NULL** or a pointer to an array of **VkPerformanceCounterKHR** structures.
- **pCounterDescriptions** is either **NULL** or a pointer to an array of **VkPerformanceCounterDescriptionKHR** structures.

If **pCounters** is **NULL** and **pCounterDescriptions** is **NULL**, then the number of counters available is returned in **pCounterCount**. Otherwise, **pCounterCount** **must** point to a variable set by the application to the number of elements in the **pCounters**, **pCounterDescriptions**, or both arrays and on return the variable is overwritten with the number of structures actually written out. If **pCounterCount** is less than the number of counters available, at most **pCounterCount** structures will be written, and **VK_INCOMPLETE** will be returned instead of **VK_SUCCESS**, to indicate that not all the available counters were returned.

Valid Usage (Implicit)

- VUID-vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR-physicalDevice-parameter
physicalDevice **must** be a valid **VkPhysicalDevice** handle
- VUID-vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR-pCounterCount-parameter
pCounterCount **must** be a valid pointer to a **uint32_t** value
- VUID-vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR-pCounters-parameter
If the value referenced by **pCounterCount** is not 0, and **pCounters** is not **NULL**, **pCounters** **must** be a valid pointer to an array of **pCounterCount** **VkPerformanceCounterKHR** structures
- VUID-vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR-pCounterDescriptions-parameter
If the value referenced by **pCounterCount** is not 0, and **pCounterDescriptions** is not **NULL**,

`pCounterDescriptions` **must** be a valid pointer to an array of `pCounterCount` `VkPerformanceCounterDescriptionKHR` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`

The `VkPerformanceCounterKHR` structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkPerformanceCounterKHR {
    VkStructureType      sType;
    void*                pNext;
    VkPerformanceCounterUnitKHR    unit;
    VkPerformanceCounterScopeKHR   scope;
    VkPerformanceCounterStorageKHR storage;
    uint8_t              uuid[VK_UUID_SIZE];
} VkPerformanceCounterKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `unit` is a `VkPerformanceCounterUnitKHR` specifying the unit that the counter data will record.
- `scope` is a `VkPerformanceCounterScopeKHR` specifying the scope that the counter belongs to.
- `storage` is a `VkPerformanceCounterStorageKHR` specifying the storage type that the counter's data uses.
- `uuid` is an array of size `VK_UUID_SIZE`, containing 8-bit values that represent a universally unique identifier for the counter of the physical device.

Valid Usage (Implicit)

- VUID-VkPerformanceCounterKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_KHR`
- VUID-VkPerformanceCounterKHR-pNext-pNext
`pNext` **must** be `NULL`

Performance counters have an associated unit. This unit describes how to interpret the performance counter result.

The performance counter unit types which **may** be returned in `VkPerformanceCounterKHR::unit` are:

```
// Provided by VK_KHR_performance_query
typedef enum VkPerformanceCounterUnitKHR {
    VK_PERFORMANCE_COUNTER_UNIT_GENERIC_KHR = 0,
    VK_PERFORMANCE_COUNTER_UNIT_PERCENTAGE_KHR = 1,
    VK_PERFORMANCE_COUNTER_UNIT_NANOSECONDS_KHR = 2,
    VK_PERFORMANCE_COUNTER_UNIT_BYTES_KHR = 3,
    VK_PERFORMANCE_COUNTER_UNIT_BYTES_PER_SECOND_KHR = 4,
    VK_PERFORMANCE_COUNTER_UNIT_KELVIN_KHR = 5,
    VK_PERFORMANCE_COUNTER_UNIT_WATTS_KHR = 6,
    VK_PERFORMANCE_COUNTER_UNIT_VOLTS_KHR = 7,
    VK_PERFORMANCE_COUNTER_UNIT_AMPS_KHR = 8,
    VK_PERFORMANCE_COUNTER_UNIT_HERTZ_KHR = 9,
    VK_PERFORMANCE_COUNTER_UNIT_CYCLES_KHR = 10,
} VkPerformanceCounterUnitKHR;
```

- `VK_PERFORMANCE_COUNTER_UNIT_GENERIC_KHR` - the performance counter unit is a generic data point.
- `VK_PERFORMANCE_COUNTER_UNIT_PERCENTAGE_KHR` - the performance counter unit is a percentage (%).
- `VK_PERFORMANCE_COUNTER_UNIT_NANOSECONDS_KHR` - the performance counter unit is a value of nanoseconds (ns).
- `VK_PERFORMANCE_COUNTER_UNIT_BYTES_KHR` - the performance counter unit is a value of bytes.
- `VK_PERFORMANCE_COUNTER_UNIT_BYTES_PER_SECOND_KHR` - the performance counter unit is a value of bytes/s.
- `VK_PERFORMANCE_COUNTER_UNIT_KELVIN_KHR` - the performance counter unit is a temperature reported in Kelvin.
- `VK_PERFORMANCE_COUNTER_UNIT_WATTS_KHR` - the performance counter unit is a value of watts (W).
- `VK_PERFORMANCE_COUNTER_UNIT_VOLTS_KHR` - the performance counter unit is a value of volts (V).
- `VK_PERFORMANCE_COUNTER_UNIT_AMPS_KHR` - the performance counter unit is a value of amps (A).
- `VK_PERFORMANCE_COUNTER_UNIT_HERTZ_KHR` - the performance counter unit is a value of hertz (Hz).
- `VK_PERFORMANCE_COUNTER_UNIT_CYCLES_KHR` - the performance counter unit is a value of cycles.

Performance counters have an associated scope. This scope describes the granularity of a performance counter.

The performance counter scope types which **may** be returned in `VkPerformanceCounterKHR::scope` are:

```
// Provided by VK_KHR_performance_query
typedef enum VkPerformanceCounterScopeKHR {
    VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR = 0,
    VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR = 1,
    VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_KHR = 2,
    // VK_QUERY_SCOPE_COMMAND_BUFFER_KHR is a deprecated alias
    VK_QUERY_SCOPE_COMMAND_BUFFER_KHR =
VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR,
    // VK_QUERY_SCOPE_RENDER_PASS_KHR is a deprecated alias
    VK_QUERY_SCOPE_RENDER_PASS_KHR = VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR,
    // VK_QUERY_SCOPE_COMMAND_KHR is a deprecated alias
    VK_QUERY_SCOPE_COMMAND_KHR = VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_KHR,
} VkPerformanceCounterScopeKHR;
```

- **VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR** - the performance counter scope is a single complete command buffer.
- **VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR** - the performance counter scope is zero or more complete render passes. The performance query containing the performance counter **must** begin and end outside a render pass instance.
- **VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_KHR** - the performance counter scope is zero or more commands.

Performance counters have an associated storage. This storage describes the payload of a counter result.

The performance counter storage types which **may** be returned in [VkPerformanceCounterKHR::storage](#) are:

```
// Provided by VK_KHR_performance_query
typedef enum VkPerformanceCounterStorageKHR {
    VK_PERFORMANCE_COUNTER_STORAGE_INT32_KHR = 0,
    VK_PERFORMANCE_COUNTER_STORAGE_INT64_KHR = 1,
    VK_PERFORMANCE_COUNTER_STORAGE_UINT32_KHR = 2,
    VK_PERFORMANCE_COUNTER_STORAGE_UINT64_KHR = 3,
    VK_PERFORMANCE_COUNTER_STORAGE_FLOAT32_KHR = 4,
    VK_PERFORMANCE_COUNTER_STORAGE_FLOAT64_KHR = 5,
} VkPerformanceCounterStorageKHR;
```

- **VK_PERFORMANCE_COUNTER_STORAGE_INT32_KHR** - the performance counter storage is a 32-bit signed integer.
- **VK_PERFORMANCE_COUNTER_STORAGE_INT64_KHR** - the performance counter storage is a 64-bit signed integer.
- **VK_PERFORMANCE_COUNTER_STORAGE_UINT32_KHR** - the performance counter storage is a 32-bit unsigned integer.
- **VK_PERFORMANCE_COUNTER_STORAGE_UINT64_KHR** - the performance counter storage is a 64-bit unsigned integer.

- `VK_PERFORMANCE_COUNTER_STORAGE_FLOAT32_KHR` - the performance counter storage is a 32-bit floating-point.
- `VK_PERFORMANCE_COUNTER_STORAGE_FLOAT64_KHR` - the performance counter storage is a 64-bit floating-point.

The `VkPerformanceCounterDescriptionKHR` structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkPerformanceCounterDescriptionKHR {
    VkStructureType          sType;
    void*                    pNext;
    VkPerformanceCounterDescriptionFlagsKHR flags;
    char                     name[VK_MAX_DESCRIPTION_SIZE];
    char                     category[VK_MAX_DESCRIPTION_SIZE];
    char                     description[VK_MAX_DESCRIPTION_SIZE];
} VkPerformanceCounterDescriptionKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkPerformanceCounterDescriptionFlagBitsKHR` indicating the usage behavior for the counter.
- `name` is an array of size `VK_MAX_DESCRIPTION_SIZE`, containing a null-terminated UTF-8 string specifying the name of the counter.
- `category` is an array of size `VK_MAX_DESCRIPTION_SIZE`, containing a null-terminated UTF-8 string specifying the category of the counter.
- `description` is an array of size `VK_MAX_DESCRIPTION_SIZE`, containing a null-terminated UTF-8 string specifying the description of the counter.

Valid Usage (Implicit)

- VUID-VkPerformanceCounterDescriptionKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_DESCRIPTION_KHR`
- VUID-VkPerformanceCounterDescriptionKHR-pNext-pNext
`pNext` **must** be `NULL`

Bits which **can** be set in `VkPerformanceCounterDescriptionKHR::flags`, specifying usage behavior of a performance counter, are:

```
// Provided by VK_KHR_performance_query
typedef enum VkPerformanceCounterDescriptionFlagBitsKHR {
    VK_PERFORMANCE_COUNTER_DESCRIPTION_PERFORMANCE_IMPACTING_BIT_KHR = 0x00000001,
    VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_BIT_KHR = 0x00000002,
    // VK_PERFORMANCE_COUNTER_DESCRIPTION_PERFORMANCE_IMPACTING_KHR is a deprecated
    alias
    VK_PERFORMANCE_COUNTER_DESCRIPTION_PERFORMANCE_IMPACTING_KHR =
    VK_PERFORMANCE_COUNTER_DESCRIPTION_PERFORMANCE_IMPACTING_BIT_KHR,
    // VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_KHR is a deprecated
    alias
    VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_KHR =
    VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_BIT_KHR,
} VkPerformanceCounterDescriptionFlagBitsKHR;
```

- `VK_PERFORMANCE_COUNTER_DESCRIPTION_PERFORMANCE_IMPACTING_BIT_KHR` specifies that recording the counter **may** have a noticeable performance impact.
- `VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_BIT_KHR` specifies that concurrently recording the counter while other submitted command buffers are running **may** impact the accuracy of the recording.

```
// Provided by VK_KHR_performance_query
typedef VkFlags VkPerformanceCounterDescriptionFlagsKHR;
```

`VkPerformanceCounterDescriptionFlagsKHR` is a bitmask type for setting a mask of zero or more `VkPerformanceCounterDescriptionFlagBitsKHR`.

5.2. Devices

Device objects represent logical connections to physical devices. Each device exposes a number of *queue families* each having one or more *queues*. All queues in a queue family support the same operations.

As described in [Physical Devices](#), a Vulkan application will first query for all physical devices in a system. Each physical device **can** then be queried for its capabilities, including its queue and queue family properties. Once an acceptable physical device is identified, an application will create a corresponding logical device. The created logical device is then the primary interface to the physical device.

How to enumerate the physical devices in a system and query those physical devices for their queue family properties is described in the [Physical Device Enumeration](#) section above.

A single logical device **can** be created from multiple physical devices, if those physical devices belong to the same device group. A *device group* is a set of physical devices that support accessing each other's memory and recording a single command buffer that **can** be executed on all the physical devices. Device groups are enumerated by calling `vkEnumeratePhysicalDeviceGroupsKHR`, and a logical device is created from a subset of the physical devices in a device group by passing the physical devices through `VkDeviceGroupDeviceCreateInfo`. For two physical devices to be in the

same device group, they **must** support identical extensions, features, and properties.

NOTE

Physical devices in the same device group **must** be so similar because there are no rules for how different features/properties would interact. They **must** return the same values for nearly every invariant `vkGetPhysicalDevice*` feature, property, capability, etc., but could potentially differ for certain queries based on things like having a different display connected, or a different compositor. The specification does not attempt to enumerate which state is in each category, because such a list would quickly become out of date.

To retrieve a list of the device groups present in the system, call:

```
// Provided by VK_KHR_device_group_creation
VkResult vkEnumeratePhysicalDeviceGroupsKHR(
    VkInstance          instance,
    uint32_t*           pPhysicalDeviceGroupCount,
    VkPhysicalDeviceGroupProperties* pPhysicalDeviceGroupProperties);
```

- `instance` is a handle to a Vulkan instance previously created with `vkCreateInstance`.
- `pPhysicalDeviceGroupCount` is a pointer to an integer related to the number of device groups available or queried, as described below.
- `pPhysicalDeviceGroupProperties` is either `NULL` or a pointer to an array of `VkPhysicalDeviceGroupProperties` structures.

If `pPhysicalDeviceGroupProperties` is `NULL`, then the number of device groups available is returned in `pPhysicalDeviceGroupCount`. Otherwise, `pPhysicalDeviceGroupCount` **must** point to a variable set by the application to the number of elements in the `pPhysicalDeviceGroupProperties` array, and on return the variable is overwritten with the number of structures actually written to `pPhysicalDeviceGroupProperties`. If `pPhysicalDeviceGroupCount` is less than the number of device groups available, at most `pPhysicalDeviceGroupCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available device groups were returned.

Every physical device **must** be in exactly one device group.

Valid Usage (Implicit)

- VUID-vkEnumeratePhysicalDeviceGroups-instance-parameter `instance` **must** be a valid `VkInstance` handle
- VUID-vkEnumeratePhysicalDeviceGroups-pPhysicalDeviceGroupCount-parameter `pPhysicalDeviceGroupCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkEnumeratePhysicalDeviceGroups-pPhysicalDeviceGroupProperties-parameter
If the value referenced by `pPhysicalDeviceGroupCount` is not `0`, and `pPhysicalDeviceGroupProperties` is not `NULL`, `pPhysicalDeviceGroupProperties` **must** be a valid pointer to an array of `pPhysicalDeviceGroupCount` `VkPhysicalDeviceGroupProperties` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`

The `VkPhysicalDeviceGroupProperties` structure is defined as:

```
typedef struct VkPhysicalDeviceGroupProperties {  
    VkStructureType    sType;  
    void*              pNext;  
    uint32_t           physicalDeviceCount;  
    VkPhysicalDevice    physicalDevices[VK_MAX_DEVICE_GROUP_SIZE];  
    VkBool32           subsetAllocation;  
} VkPhysicalDeviceGroupProperties;
```

or the equivalent

```
// Provided by VK_KHR_device_group_creation  
typedef VkPhysicalDeviceGroupProperties VkPhysicalDeviceGroupPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `physicalDeviceCount` is the number of physical devices in the group.
- `physicalDevices` is an array of `VK_MAX_DEVICE_GROUP_SIZE` `VkPhysicalDevice` handles representing all physical devices in the group. The first `physicalDeviceCount` elements of the array will be valid.
- `subsetAllocation` specifies whether logical devices created from the group support allocating device memory on a subset of devices, via the `deviceMask` member of the `VkMemoryAllocateFlagsInfo`. If this is `VK_FALSE`, then all device memory allocations are made across all physical devices in the group. If `physicalDeviceCount` is 1, then `subsetAllocation` **must** be `VK_FALSE`.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceGroupProperties-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES`

- VUID-VkPhysicalDeviceGroupProperties-pNext-pNext
pNext must be NULL

VK_MAX_DEVICE_GROUP_SIZE is the length of an array containing [VkPhysicalDevice](#) handle values representing all physical devices in a group, as returned in [VkPhysicalDeviceGroupProperties::physicalDevices](#).

```
#define VK_MAX_DEVICE_GROUP_SIZE 32U
```

or the equivalent

```
#define VK_MAX_DEVICE_GROUP_SIZE_KHR VK_MAX_DEVICE_GROUP_SIZE
```

5.2.1. Device Creation

Logical devices are represented by [VkDevice](#) handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkDevice)
```

A logical device is created as a *connection* to a physical device. To create a logical device, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateDevice(
    VkPhysicalDevice          physicalDevice,
    const VkDeviceCreateInfo* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkDevice*                 pDevice);
```

- **physicalDevice must** be one of the device handles returned from a call to [vkEnumeratePhysicalDevices](#) (see [Physical Device Enumeration](#)).
- **pCreateInfo** is a pointer to a [VkDeviceCreateInfo](#) structure containing information about how to create the device.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pDevice** is a pointer to a handle in which the created [VkDevice](#) is returned.

vkCreateDevice verifies that extensions and features requested in the **ppEnabledExtensionNames** and **pEnabledFeatures** members of **pCreateInfo**, respectively, are supported by the implementation. If any requested extension is not supported, **vkCreateDevice must** return **VK_ERROR_EXTENSION_NOT_PRESENT**. If any requested feature is not supported, **vkCreateDevice must** return **VK_ERROR_FEATURE_NOT_PRESENT**. Support for extensions **can** be checked before creating a device by querying [vkEnumerateDeviceExtensionProperties](#). Support for features **can** similarly be checked by querying [vkGetPhysicalDeviceFeatures](#).

After verifying and enabling the extensions the `VkDevice` object is created and returned to the application.

Multiple logical devices **can** be created from the same physical device. Logical device creation **may** fail due to lack of device-specific resources (in addition to other errors). If that occurs, `vkCreateDevice` will return `VK_ERROR_TOO_MANY_OBJECTS`.

Valid Usage

- VUID-vkCreateDevice-ppEnabledExtensionNames-01387

All [required device extensions](#) for each extension in the `VkDeviceCreateInfo::ppEnabledExtensionNames` list **must** also be present in that list

Valid Usage (Implicit)

- VUID-vkCreateDevice-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkCreateDevice-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkDeviceCreateInfo` structure
- VUID-vkCreateDevice-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateDevice-pDevice-parameter
`pDevice` **must** be a valid pointer to a `VkDevice` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`
- `VK_ERROR_EXTENSION_NOT_PRESENT`
- `VK_ERROR_FEATURE_NOT_PRESENT`
- `VK_ERROR_TOO_MANY_OBJECTS`
- `VK_ERROR_DEVICE_LOST`

The `VkDeviceCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDeviceCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkDeviceCreateFlags       flags;
    uint32_t                  queueCreateInfoCount;
    const VkDeviceQueueCreateInfo* pQueueCreateInfos;
    // enabledLayerCount is deprecated and should not be used
    uint32_t                  enabledLayerCount;
    // ppEnabledLayerNames is deprecated and should not be used
    const char* const*        ppEnabledLayerNames;
    uint32_t                  enabledExtensionCount;
    const char* const*        ppEnabledExtensionNames;
    const VkPhysicalDeviceFeatures* pEnabledFeatures;
} VkDeviceCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `queueCreateInfoCount` is the unsigned integer size of the `pQueueCreateInfos` array. Refer to the [Queue Creation](#) section below for further details.
- `pQueueCreateInfos` is a pointer to an array of `VkDeviceQueueCreateInfo` structures describing the queues that are requested to be created along with the logical device. Refer to the [Queue Creation](#) section below for further details.
- `enabledLayerCount` is deprecated and ignored.
- `ppEnabledLayerNames` is deprecated and ignored. See [Device Layer Deprecation](#).
- `enabledExtensionCount` is the number of device extensions to enable.
- `ppEnabledExtensionNames` is a pointer to an array of `enabledExtensionCount` null-terminated UTF-8 strings containing the names of extensions to enable for the created device. See the [Extensions](#) section for further details.
- `pEnabledFeatures` is `NULL` or a pointer to a `VkPhysicalDeviceFeatures` structure containing boolean indicators of all the features to be enabled. Refer to the [Features](#) section for further details.

Valid Usage

- VUID-VkDeviceCreateInfo-queueFamilyIndex-02802
The `queueFamilyIndex` member of each element of `pQueueCreateInfos` **must** be unique within `pQueueCreateInfos`
- VUID-VkDeviceCreateInfo-pNext-00373
If the `pNext` chain includes a `VkPhysicalDeviceFeatures2` structure, then `pEnabledFeatures` **must** be `NULL`
- VUID-VkDeviceCreateInfo-pProperties-04451

If the `VK_KHR_portability_subset` extension is included in `pProperties` of `vkEnumerateDeviceExtensionProperties`, `ppEnabledExtensionNames` **must** include `"VK_KHR_portability_subset"`

- VUID-VkDeviceCreateInfo-None-04896
If `sparseImageInt64Atomics` is enabled, `shaderImageInt64Atomics` **must** be enabled
- VUID-VkDeviceCreateInfo-None-04897
If `sparseImageFloat32Atomics` is enabled, `shaderImageFloat32Atomics` **must** be enabled
- VUID-VkDeviceCreateInfo-None-04898
If `sparseImageFloat32AtomicAdd` is enabled, `shaderImageFloat32AtomicAdd` **must** be enabled
- VUID-VkDeviceCreateInfo-sparseImageFloat32AtomicMinMax-04975
If `sparseImageFloat32AtomicMinMax` is enabled, `shaderImageFloat32AtomicMinMax` **must** be enabled

Valid Usage (Implicit)

- VUID-VkDeviceCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_CREATE_INFO`
- VUID-VkDeviceCreateInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkDeviceDeviceMemoryReportCreateInfoEXT`, `VkDeviceGroupDeviceCreateInfo`, `VkDevicePipelineBinaryInternalCacheControlKHR`, `VkPhysicalDevice16BitStorageFeatures`, `VkPhysicalDevice8BitStorageFeatures`, `VkPhysicalDeviceASTCDecodeFeaturesEXT`, `VkPhysicalDeviceAccelerationStructureFeaturesKHR`, `VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT`, `VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT`, `VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT`, `VkPhysicalDeviceBorderColorSwizzleFeaturesEXT`, `VkPhysicalDeviceBufferDeviceAddressFeatures`, `VkPhysicalDeviceColorWriteEnableFeaturesEXT`, `VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR`, `VkPhysicalDeviceCooperativeMatrixFeaturesKHR`, `VkPhysicalDeviceCustomBorderColorFeaturesEXT`, `VkPhysicalDeviceDepthBiasControlFeaturesEXT`, `VkPhysicalDeviceDepthClampControlFeaturesEXT`, `VkPhysicalDeviceDepthClipControlFeaturesEXT`, `VkPhysicalDeviceDepthClipEnableFeaturesEXT`, `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT`, `VkPhysicalDeviceDeviceMemoryReportFeaturesEXT`, `VkPhysicalDeviceDynamicRenderingFeatures`, `VkPhysicalDeviceDynamicRenderingLocalReadFeaturesKHR`, `VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT`, `VkPhysicalDeviceExtendedDynamicState3FeaturesEXT`, `VkPhysicalDeviceFeatures2`, `VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR`, `VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT`,

VkPhysicalDeviceFragmentShadingRateFeaturesKHR,
VkPhysicalDeviceFrameBoundaryFeaturesEXT,
VkPhysicalDeviceGlobalPriorityQueryFeaturesKHR,
VkPhysicalDeviceHostImageCopyFeaturesEXT,
VkPhysicalDeviceImageViewMinLodFeaturesEXT,
VkPhysicalDeviceImagelessFramebufferFeatures,
VkPhysicalDeviceIndexTypeUint8FeaturesKHR,
VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT,
VkPhysicalDeviceLineRasterizationFeaturesKHR,
VkPhysicalDeviceMaintenance4Features, VkPhysicalDeviceMaintenance5FeaturesKHR,
VkPhysicalDeviceMaintenance6FeaturesKHR,
VkPhysicalDeviceMaintenance7FeaturesKHR,
VkPhysicalDeviceMapMemoryPlacedFeaturesEXT,
VkPhysicalDeviceMemoryPriorityFeaturesEXT, VkPhysicalDeviceMultiDrawFeaturesEXT,
VkPhysicalDeviceMultiviewFeatures,
VkPhysicalDeviceNestedCommandBufferFeaturesEXT,
VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT,
VkPhysicalDeviceOpacityMicromapFeaturesEXT,
VkPhysicalDevicePerformanceQueryFeaturesKHR,
VkPhysicalDevicePipelineBinaryFeaturesKHR,
VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR,
VkPhysicalDevicePortabilitySubsetFeaturesKHR, VkPhysicalDevicePresentIdFeaturesKHR,
VkPhysicalDevicePresentModeFifoLatestReadyFeaturesEXT,
VkPhysicalDevicePresentWaitFeaturesKHR,
VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT,
VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT,
VkPhysicalDeviceProvokingVertexFeaturesEXT, VkPhysicalDeviceRayQueryFeaturesKHR,
VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR,
VkPhysicalDeviceRayTracingPipelineFeaturesKHR,
VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR,
VkPhysicalDeviceRobustness2FeaturesEXT,
VkPhysicalDeviceSamplerYcbcrConversionFeatures,
VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures,
VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT,
VkPhysicalDeviceShaderAtomicFloatFeaturesEXT,
VkPhysicalDeviceShaderAtomicInt64Features,
VkPhysicalDeviceShaderClockFeaturesKHR,
VkPhysicalDeviceShaderExpectAssumeFeaturesKHR,
VkPhysicalDeviceShaderFloat16Int8Features,
VkPhysicalDeviceShaderFloatControls2FeaturesKHR,
VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT,
VkPhysicalDeviceShaderIntegerDotProductFeatures,
VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR,
VkPhysicalDeviceShaderObjectFeaturesEXT,
VkPhysicalDeviceShaderQuadControlFeaturesKHR,
VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR,
VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT,
VkPhysicalDeviceShaderSubgroupExtendedTypesFeatures,

VkPhysicalDeviceShaderSubgroupRotateFeaturesKHR,
 VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR,
 VkPhysicalDeviceShaderTerminateInvocationFeatures,
 VkPhysicalDeviceShaderTileImageFeaturesEXT,
 VkPhysicalDeviceSynchronization2Features,
 VkPhysicalDeviceTimelineSemaphoreFeatures,
 VkPhysicalDeviceTransformFeedbackFeaturesEXT,
 VkPhysicalDeviceUniformBufferStandardLayoutFeatures,
 VkPhysicalDeviceVariablePointersFeatures,
 VkPhysicalDeviceVertexAttributeDivisorFeaturesKHR,
 VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT,
 VkPhysicalDeviceVideoMaintenance1FeaturesKHR,
 VkPhysicalDeviceVulkanMemoryModelFeatures,
 VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR,
 VkPhysicalDeviceYcbcrImageArraysFeaturesEXT,
 VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures

or

- VUID-VkDeviceCreateInfo-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique, with the exception of structures of type **VkDeviceDeviceMemoryReportCreateInfoEXT**
- VUID-VkDeviceCreateInfo-flags-zeroBitmask
flags **must** be 0
- VUID-VkDeviceCreateInfo-pQueueCreateInfos-parameter
pQueueCreateInfos **must** be a valid pointer to an array of **queueCreateInfoCount** valid **VkDeviceQueueCreateInfo** structures
- VUID-VkDeviceCreateInfo-ppEnabledLayerNames-parameter
If **enabledLayerCount** is not 0, **ppEnabledLayerNames** **must** be a valid pointer to an array of **enabledLayerCount** null-terminated UTF-8 strings
- VUID-VkDeviceCreateInfo-ppEnabledExtensionNames-parameter
If **enabledExtensionCount** is not 0, **ppEnabledExtensionNames** **must** be a valid pointer to an array of **enabledExtensionCount** null-terminated UTF-8 strings
- VUID-VkDeviceCreateInfo-pEnabledFeatures-parameter
If **pEnabledFeatures** is not **NULL**, **pEnabledFeatures** **must** be a valid pointer to a valid **VkPhysicalDeviceFeatures** structure
- VUID-VkDeviceCreateInfo-queueCreateInfoCount-arraylength
queueCreateInfoCount **must** be greater than 0

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDeviceCreateFlags;
```

VkDeviceCreateFlags is a bitmask type for setting a mask, but is currently reserved for future use.

A logical device **can** be created that connects to one or more physical devices by adding a **VkDeviceGroupDeviceCreateInfo** structure to the **pNext** chain of **VkDeviceCreateInfo**. The **VkDeviceGroupDeviceCreateInfo** structure is defined as:


```
typedef struct VkDeviceGroupDeviceCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           physicalDeviceCount;
    const VkPhysicalDevice* pPhysicalDevices;
} VkDeviceGroupDeviceCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_device_group_creation
typedef VkDeviceGroupDeviceCreateInfo VkDeviceGroupDeviceCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `physicalDeviceCount` is the number of elements in the `pPhysicalDevices` array.
- `pPhysicalDevices` is a pointer to an array of physical device handles belonging to the same device group.

The elements of the `pPhysicalDevices` array are an ordered list of the physical devices that the logical device represents. These **must** be a subset of a single device group, and need not be in the same order as they were enumerated. The order of the physical devices in the `pPhysicalDevices` array determines the *device index* of each physical device, with element *i* being assigned a device index of *i*. Certain commands and structures refer to one or more physical devices by using device indices or *device masks* formed using device indices.

A logical device created without using `VkDeviceGroupDeviceCreateInfo`, or with `physicalDeviceCount` equal to zero, is equivalent to a `physicalDeviceCount` of one and `pPhysicalDevices` pointing to the `physicalDevice` parameter to `vkCreateDevice`. In particular, the device index of that physical device is zero.

Valid Usage

- VUID-VkDeviceGroupDeviceCreateInfo-pPhysicalDevices-00375
Each element of `pPhysicalDevices` **must** be unique
- VUID-VkDeviceGroupDeviceCreateInfo-pPhysicalDevices-00376
All elements of `pPhysicalDevices` **must** be in the same device group as enumerated by `vkEnumeratePhysicalDeviceGroupsKHR`
- VUID-VkDeviceGroupDeviceCreateInfo-physicalDeviceCount-00377
If `physicalDeviceCount` is not 0, the `physicalDevice` parameter of `vkCreateDevice` **must** be an element of `pPhysicalDevices`

Valid Usage (Implicit)

- VUID-VkDeviceGroupDeviceCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO`
- VUID-VkDeviceGroupDeviceCreateInfo-pPhysicalDevices-parameter
If `physicalDeviceCount` is not `0`, `pPhysicalDevices` **must** be a valid pointer to an array of `physicalDeviceCount` valid `VkPhysicalDevice` handles

To register callbacks for underlying device memory events of type `VkDeviceMemoryReportEventTypeEXT`, add one or multiple `VkDeviceDeviceMemoryReportCreateInfoEXT` structures to the `pNext` chain of the `VkDeviceCreateInfo` structure.

```
// Provided by VK_EXT_device_memory_report
typedef struct VkDeviceDeviceMemoryReportCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceMemoryReportFlagsEXT flags;
    PFN_vkDeviceMemoryReportCallbackEXT pfnUserCallback;
    void*              pUserData;
} VkDeviceDeviceMemoryReportCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is `0` and reserved for future use.
- `pfnUserCallback` is the application callback function to call.
- `pUserData` is user data to be passed to the callback.

The callback **may** be called from multiple threads simultaneously.

The callback **must** be called only once by the implementation when a `VkDeviceMemoryReportEventTypeEXT` event occurs.

NOTE

The callback could be called from a background thread other than the thread calling the Vulkan commands.

Valid Usage (Implicit)

- VUID-VkDeviceDeviceMemoryReportCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_DEVICE_MEMORY_REPORT_CREATE_INFO_EXT`
- VUID-VkDeviceDeviceMemoryReportCreateInfoEXT-flags-zero bitmask
`flags` **must** be `0`
- VUID-VkDeviceDeviceMemoryReportCreateInfoEXT-pfnUserCallback-parameter

`pfnUserCallback` **must** be a valid `PFN_vkDeviceMemoryReportCallbackEXT` value

- `VOID-VkDeviceDeviceMemoryReportCreateInfoEXT-pUserData-parameter`
`pUserData` **must** be a pointer value

The prototype for the `VkDeviceDeviceMemoryReportCreateInfoEXT::pfnUserCallback` function implemented by the application is:

```
// Provided by VK_EXT_device_memory_report
typedef void (VKAPI_PTR *PFN_vkDeviceMemoryReportCallbackEXT)(
    const VkDeviceMemoryReportCallbackDataEXT* pCallbackData,
    void* pUserData);
```

- `pCallbackData` contains all the callback related data in the `VkDeviceMemoryReportCallbackDataEXT` structure.
- `pUserData` is the user data provided when the `VkDeviceDeviceMemoryReportCreateInfoEXT` was created.

The callback **must** not make calls to any Vulkan commands.

The definition of `VkDeviceMemoryReportCallbackDataEXT` is:

```
// Provided by VK_EXT_device_memory_report
typedef struct VkDeviceMemoryReportCallbackDataEXT {
    VkStructureType          sType;
    void*                    pNext;
    VkDeviceMemoryReportFlagsEXT flags;
    VkDeviceMemoryReportEventTypeEXT type;
    uint64_t                 memoryObjectId;
    VkDeviceSize             size;
    VkObjectType             objectType;
    uint64_t                 objectHandle;
    uint32_t                 heapIndex;
} VkDeviceMemoryReportCallbackDataEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is 0 and reserved for future use.
- `type` is a `VkDeviceMemoryReportEventTypeEXT` type specifying the type of event reported in this `VkDeviceMemoryReportCallbackDataEXT` structure.
- `memoryObjectId` is the unique id for the underlying memory object as described below.
- `size` is the size of the memory object in bytes. If `type` is `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT`, `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT` or `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATION_FAILED_EXT`, `size` is a valid `VkDeviceSize` value.

Otherwise, `size` is undefined.

- `objectType` is a `VkObjectType` value specifying the type of the object associated with this device memory report event. If `type` is `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT`, `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_FREE_EXT`, `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT`, `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_UNIMPORT_EXT` or `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATION_FAILED_EXT`, `objectType` is a valid `VkObjectType` enum. Otherwise, `objectType` is undefined.
- `objectHandle` is the object this device memory report event is attributed to. If `type` is `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT`, `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_FREE_EXT`, `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT` or `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_UNIMPORT_EXT`, `objectHandle` is a valid Vulkan handle of the type associated with `objectType` as defined in the `VkObjectType` and `Vulkan Handle Relationship` table. Otherwise, `objectHandle` is undefined.
- `heapIndex` describes which memory heap this device memory allocation is made from. If `type` is `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT` or `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATION_FAILED_EXT`, `heapIndex` corresponds to one of the valid heaps from the `VkPhysicalDeviceMemoryProperties` structure. Otherwise, `heapIndex` is undefined.

`memoryObjectId` is used to avoid double-counting on the same memory object.

If an internally-allocated device memory object or a `VkDeviceMemory` **cannot** be exported, `memoryObjectId` **must** be unique in the `VkDevice`.

If an internally-allocated device memory object or a `VkDeviceMemory` supports being exported, `memoryObjectId` **must** be unique system wide.

If an internal device memory object or a `VkDeviceMemory` is backed by an imported external memory object, `memoryObjectId` **must** be unique system wide.

Implementor's Note

If the heap backing an internally-allocated device memory **cannot** be used to back `VkDeviceMemory`, implementations **can** advertise that heap with no types.

NOTE

This structure should only be considered valid during the lifetime of the triggered callback.

For `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT` and `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT` events, `objectHandle` usually will not yet exist when the application or tool receives the callback. `objectHandle` will only exist when the create or allocate call that triggered the event returns, and if the allocation or import ends up failing `objectHandle` will not ever exist.

Valid Usage (Implicit)

- VUID-VkDeviceMemoryReportCallbackDataEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_MEMORY_REPORT_CALLBACK_DATA_EXT`
- VUID-VkDeviceMemoryReportCallbackDataEXT-pNext-pNext
`pNext` **must** be `NULL`

```
// Provided by VK_EXT_device_memory_report
typedef VkFlags VkDeviceMemoryReportFlagsEXT;
```

`VkDeviceMemoryReportFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

Possible values of `VkDeviceMemoryReportCallbackDataEXT::type`, specifying event types which cause the device driver to call the callback, are:

```
// Provided by VK_EXT_device_memory_report
typedef enum VkDeviceMemoryReportEventTypeEXT {
    VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT = 0,
    VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_FREE_EXT = 1,
    VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT = 2,
    VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_UNIMPORT_EXT = 3,
    VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATION_FAILED_EXT = 4,
} VkDeviceMemoryReportEventTypeEXT;
```

- `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATE_EXT` specifies this event corresponds to the allocation of an internal device memory object or a [VkDeviceMemory](#).
- `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_FREE_EXT` specifies this event corresponds to the deallocation of an internally-allocated device memory object or a [VkDeviceMemory](#).
- `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_IMPORT_EXT` specifies this event corresponds to the import of an external memory object.
- `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_UNIMPORT_EXT` specifies this event is the release of an imported external memory object.
- `VK_DEVICE_MEMORY_REPORT_EVENT_TYPE_ALLOCATION_FAILED_EXT` specifies this event corresponds to the failed allocation of an internal device memory object or a [VkDeviceMemory](#).

To disable the implementation's internal pipeline cache, add a [VkDevicePipelineBinaryInternalCacheControlKHR](#) structure to the `pNext` chain of the [VkDeviceCreateInfo](#) structure.

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkDevicePipelineBinaryInternalCacheControlKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           disableInternalCache;
} VkDevicePipelineBinaryInternalCacheControlKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `disableInternalCache` specifies whether or not to disable the implementation's internal pipeline cache.

If the `VkDeviceCreateInfo::pNext` chain does not include this structure, then `disableInternalCache` defaults to `VK_FALSE`.

Valid Usage

- VUID-VkDevicePipelineBinaryInternalCacheControlKHR-disableInternalCache-09602
If `VkPhysicalDevicePipelineBinaryPropertiesKHR::pipelineBinaryInternalCacheControl` is `VK_FALSE`, `disableInternalCache` **must** be `VK_FALSE`

Valid Usage (Implicit)

- VUID-VkDevicePipelineBinaryInternalCacheControlKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_PIPELINE_BINARY_INTERNAL_CACHE_CONTROL_KHR`

5.2.2. Device Use

The following is a high-level list of `VkDevice` uses along with references on where to find more information:

- Creation of queues. See the [Queues](#) section below for further details.
- Creation and tracking of various synchronization constructs. See [Synchronization and Cache Control](#) for further details.
- Allocating, freeing, and managing memory. See [Memory Allocation](#) and [Resource Creation](#) for further details.
- Creation and destruction of command buffers and command buffer pools. See [Command Buffers](#) for further details.
- Creation, destruction, and management of graphics state. See [Pipelines](#) and [Resource Descriptors](#), among others, for further details.

5.2.3. Lost Device

A logical device **may** become *lost* for a number of implementation-specific reasons, indicating that pending and future command execution **may** fail and cause resources and backing memory to become undefined.

NOTE

Typical reasons for device loss will include things like execution timing out (to prevent denial of service), power management events, platform resource management, implementation errors.

Applications not adhering to [valid usage](#) may also result in device loss being reported, however this is not guaranteed. Even if device loss is reported, the system may be in an unrecoverable state, and further usage of the API is still considered invalid.

When this happens, certain commands will return `VK_ERROR_DEVICE_LOST`. After any such event, the logical device is considered *lost*. It is not possible to reset the logical device to a non-lost state, however the lost state is specific to a logical device (`VkDevice`), and the corresponding physical device (`VkPhysicalDevice`) **may** be otherwise unaffected.

In some cases, the physical device **may** also be lost, and attempting to create a new logical device will fail, returning `VK_ERROR_DEVICE_LOST`. This is usually indicative of a problem with the underlying implementation, or its connection to the host. If the physical device has not been lost, and a new logical device is successfully created from that physical device, it **must** be in the non-lost state.

NOTE

Whilst logical device loss **may** be recoverable, in the case of physical device loss, it is unlikely that an application will be able to recover unless additional, unaffected physical devices exist on the system. The error is largely informational and intended only to inform the application that a platform issue has occurred, and **should** be investigated further. For example, underlying hardware **may** have developed a fault or become physically disconnected from the rest of the system. In many cases, physical device loss **may** cause other more serious issues such as the operating system crashing; in which case it **may** not be reported via the Vulkan API.

When a device is lost, its child objects are not implicitly destroyed and their handles are still valid. Those objects **must** still be destroyed before their parents or the device **can** be destroyed (see the [Object Lifetime](#) section). The host address space corresponding to device memory mapped using [vkMapMemory](#) is still valid, and host memory accesses to these mapped regions are still valid, but the contents are undefined. It is still legal to call any API command on the device and child objects.

Once a device is lost, command execution **may** fail, and certain commands that return a [VkResult](#) **may** return `VK_ERROR_DEVICE_LOST`. These commands can be identified by the inclusion of `VK_ERROR_DEVICE_LOST` in the Return Codes section for each command. Commands that do not allow runtime errors **must** still operate correctly for valid usage and, if applicable, return valid data.

Commands that wait indefinitely for device execution (namely [vkDeviceWaitIdle](#), [vkQueueWaitIdle](#), [vkWaitForFences](#) or [vkAcquireNextImageKHR](#) with a maximum [timeout](#), and [vkGetQueryPoolResults](#) with the `VK_QUERY_RESULT_WAIT_BIT` bit set in [flags](#)) **must** return in finite time even in the case of a lost device, and return either `VK_SUCCESS` or `VK_ERROR_DEVICE_LOST`. For any

command that **may** return `VK_ERROR_DEVICE_LOST`, for the purpose of determining whether a command buffer is in the [pending state](#), or whether resources are considered in-use by the device, a return value of `VK_ERROR_DEVICE_LOST` is equivalent to `VK_SUCCESS`.

If a device was created with the `maintenance5` feature enabled, and any device command returns `VK_ERROR_DEVICE_LOST`, then all device commands for which `VK_ERROR_DEVICE_LOST` is a valid return value and which happen-after it on the same host thread **must** return `VK_ERROR_DEVICE_LOST`.

Device commands executing on other threads **must** begin returning `VK_ERROR_DEVICE_LOST` within finite time.

The content of any external memory objects that have been exported from or imported to a lost device become undefined. Objects on other logical devices or in other APIs which are associated with the same underlying memory resource as the external memory objects on the lost device are unaffected other than their content becoming undefined. The layout of subresources of images on other logical devices that are bound to `VkDeviceMemory` objects associated with the same underlying memory resources as external memory objects on the lost device becomes `VK_IMAGE_LAYOUT_UNDEFINED`.

The state of `VkSemaphore` objects on other logical devices created by [importing a semaphore payload](#) with temporary permanence which was exported from the lost device is undefined. The state of `VkSemaphore` objects on other logical devices that permanently share a semaphore payload with a `VkSemaphore` object on the lost device is undefined, and remains undefined following any subsequent signal operations. Implementations **must** ensure pending and subsequently submitted wait operations on such semaphores behave as defined in [Semaphore State Requirements For Wait Operations](#) for external semaphores not in a valid state for a wait operation.

5.2.4. Device Destruction

To destroy a device, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyDevice(
    VkDevice device,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

To ensure that no work is active on the device, `vkDeviceWaitIdle` **can** be used to gate the destruction of the device. Prior to destroying a device, an application is responsible for destroying/freeing any Vulkan objects that were created using that device as the first parameter of the corresponding `vkCreate*` or `vkAllocate*` command.

NOTE

The lifetime of each of these objects is bound by the lifetime of the `VkDevice` object. Therefore, to avoid resource leaks, it is critical that an application explicitly free all of these resources prior to calling `vkDestroyDevice`.

Valid Usage

- VUID-vkDestroyDevice-device-05137
All child objects created on **device** **must** have been destroyed prior to destroying **device**
- VUID-vkDestroyDevice-device-00379
If **VkAllocationCallbacks** were provided when **device** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDevice-device-00380
If no **VkAllocationCallbacks** were provided when **device** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyDevice-device-parameter
If **device** is not **NULL**, **device** **must** be a valid **VkDevice** handle
- VUID-vkDestroyDevice-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure

Host Synchronization

- Host access to **device** **must** be externally synchronized
- Host access to all **VkQueue** objects created from **device** **must** be externally synchronized

5.3. Queues

5.3.1. Queue Family Properties

As discussed in the [Physical Device Enumeration](#) section above, the [vkGetPhysicalDeviceQueueFamilyProperties](#) command is used to retrieve details about the queue families and queues supported by a device.

Each index in the [pQueueFamilyProperties](#) array returned by [vkGetPhysicalDeviceQueueFamilyProperties](#) describes a unique queue family on that physical device. These indices are used when creating queues, and they correspond directly with the [queueFamilyIndex](#) that is passed to the [vkCreateDevice](#) command via the [VkDeviceQueueCreateInfo](#) structure as described in the [Queue Creation](#) section below.

Grouping of queue families within a physical device is implementation-dependent.

NOTE

The general expectation is that a physical device groups all queues of matching capabilities into a single family. However, while implementations **should** do this, it is possible that a physical device **may** return two separate queue families with the

same capabilities.

Once an application has identified a physical device with the queue(s) that it desires to use, it will create those queues in conjunction with a logical device. This is described in the following section.

5.3.2. Queue Creation

Creating a logical device also creates the queues associated with that device. The queues to create are described by a set of `VkDeviceQueueCreateInfo` structures that are passed to `vkCreateDevice` in `pQueueCreateInfos`.

Queues are represented by `VkQueue` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkQueue)
```

The `VkDeviceQueueCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDeviceQueueCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceQueueCreateFlags flags;
    uint32_t           queueFamilyIndex;
    uint32_t           queueCount;
    const float*        pQueuePriorities;
} VkDeviceQueueCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `queueFamilyIndex` is an unsigned integer indicating the index of the queue family in which to create the queues on this device. This index corresponds to the index of an element of the `pQueueFamilyProperties` array that was returned by `vkGetPhysicalDeviceQueueFamilyProperties`.
- `queueCount` is an unsigned integer specifying the number of queues to create in the queue family indicated by `queueFamilyIndex`, and with the behavior specified by `flags`.
- `pQueuePriorities` is a pointer to an array of `queueCount` normalized floating-point values, specifying priorities of work that will be submitted to each created queue. See [Queue Priority](#) for more information.

Valid Usage

- VUID-VkDeviceQueueCreateInfo-queueFamilyIndex-00381
`queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties`

- VUID-VkDeviceQueueCreateInfo-queueCount-00382
`queueCount` **must** be less than or equal to the `queueCount` member of the `VkQueueFamilyProperties` structure, as returned by `vkGetPhysicalDeviceQueueFamilyProperties` in the `pQueueFamilyProperties[queueFamilyIndex]`
- VUID-VkDeviceQueueCreateInfo-pQueuePriorities-00383
Each element of `pQueuePriorities` **must** be between `0.0` and `1.0` inclusive

Valid Usage (Implicit)

- VUID-VkDeviceQueueCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_QUEUE_CREATE_INFO`
- VUID-VkDeviceQueueCreateInfo-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkDeviceQueueGlobalPriorityCreateInfoKHR`
- VUID-VkDeviceQueueCreateInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkDeviceQueueCreateInfo-flags-zero-bitmask
`flags` **must** be `0`
- VUID-VkDeviceQueueCreateInfo-pQueuePriorities-parameter
`pQueuePriorities` **must** be a valid pointer to an array of `queueCount` float values
- VUID-VkDeviceQueueCreateInfo-queueCount-arraylength
`queueCount` **must** be greater than `0`

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDeviceQueueCreateFlags;
```

`VkDeviceQueueCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

Queues **can** be created with a system-wide priority by adding a `VkDeviceQueueGlobalPriorityCreateInfoKHR` structure to the `pNext` chain of `VkDeviceQueueCreateInfo`.

The `VkDeviceQueueGlobalPriorityCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_global_priority
typedef struct VkDeviceQueueGlobalPriorityCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkQueueGlobalPriorityKHR    globalPriority;
} VkDeviceQueueGlobalPriorityCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `globalPriority` is the system-wide priority associated to these queues as specified by `VkQueueGlobalPriorityKHR`

Queues created without specifying `VkDeviceQueueGlobalPriorityCreateInfoKHR` will default to `VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_KHR`.

Valid Usage (Implicit)

- VUID-VkDeviceQueueGlobalPriorityCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_QUEUE_GLOBAL_PRIORITY_CREATE_INFO_KHR`
- VUID-VkDeviceQueueGlobalPriorityCreateInfoKHR-globalPriority-parameter
`globalPriority` **must** be a valid `VkQueueGlobalPriorityKHR` value

Possible values of `VkDeviceQueueGlobalPriorityCreateInfoKHR::globalPriority`, specifying a system-wide priority level are:

```
// Provided by VK_KHR_global_priority
typedef enum VkQueueGlobalPriorityKHR {
    VK_QUEUE_GLOBAL_PRIORITY_LOW_KHR = 128,
    VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_KHR = 256,
    VK_QUEUE_GLOBAL_PRIORITY_HIGH_KHR = 512,
    VK_QUEUE_GLOBAL_PRIORITY_REALTIME_KHR = 1024,
    VK_QUEUE_GLOBAL_PRIORITY_LOW_EXT = VK_QUEUE_GLOBAL_PRIORITY_LOW_KHR,
    VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_EXT = VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_KHR,
    VK_QUEUE_GLOBAL_PRIORITY_HIGH_EXT = VK_QUEUE_GLOBAL_PRIORITY_HIGH_KHR,
    VK_QUEUE_GLOBAL_PRIORITY_REALTIME_EXT = VK_QUEUE_GLOBAL_PRIORITY_REALTIME_KHR,
} VkQueueGlobalPriorityKHR;
```

Priority values are sorted in ascending order. A comparison operation on the enum values can be used to determine the priority order.

- `VK_QUEUE_GLOBAL_PRIORITY_LOW_KHR` is below the system default. Useful for non-interactive tasks.
- `VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_KHR` is the system default priority.
- `VK_QUEUE_GLOBAL_PRIORITY_HIGH_KHR` is above the system default.
- `VK_QUEUE_GLOBAL_PRIORITY_REALTIME_KHR` is the highest priority. Useful for critical tasks.

Queues with higher system priority **may** be allotted more processing time than queues with lower priority. An implementation **may** allow a higher-priority queue to starve a lower-priority queue until the higher-priority queue has no further commands to execute.

Priorities imply no ordering or scheduling constraints.

No specific guarantees are made about higher priority queues receiving more processing time or

better quality of service than lower priority queues.

The global priority level of a queue takes precedence over the per-process queue priority (`VkDeviceQueueCreateInfo::pQueuePriorities`).

Abuse of this feature **may** result in starving the rest of the system of implementation resources. Therefore, the driver implementation **may** deny requests to acquire a priority above the default priority (`VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_KHR`) if the caller does not have sufficient privileges. In this scenario `VK_ERROR_NOT_PERMITTED_KHR` is returned.

The driver implementation **may** fail the queue allocation request if resources required to complete the operation have been exhausted (either by the same process or a different process). In this scenario `VK_ERROR_INITIALIZATION_FAILED` is returned.

If the `globalPriorityQuery` feature is enabled and the requested global priority is not reported via `VkQueueFamilyGlobalPriorityPropertiesKHR`, the driver implementation **must** fail the queue creation. In this scenario, `VK_ERROR_INITIALIZATION_FAILED` is returned.

To retrieve a handle to a `VkQueue` object, call:

```
// Provided by VK_VERSION_1_0
void vkGetDeviceQueue(
    VkDevice          device,
    uint32_t          queueFamilyIndex,
    uint32_t          queueIndex,
    VkQueue*          pQueue);
```

- `device` is the logical device that owns the queue.
- `queueFamilyIndex` is the index of the queue family to which the queue belongs.
- `queueIndex` is the index within this queue family of the queue to retrieve.
- `pQueue` is a pointer to a `VkQueue` object that will be filled with the handle for the requested queue.

Valid Usage

- VUID-vkGetDeviceQueue-queueFamilyIndex-00384
`queueFamilyIndex` **must** be one of the queue family indices specified when `device` was created, via the `VkDeviceQueueCreateInfo` structure
- VUID-vkGetDeviceQueue-queueIndex-00385
`queueIndex` **must** be less than the value of `VkDeviceQueueCreateInfo::queueCount` for the queue family indicated by `queueFamilyIndex` when `device` was created
- VUID-vkGetDeviceQueue-flags-01841
`VkDeviceQueueCreateInfo::flags` **must** have been zero when `device` was created

Valid Usage (Implicit)

- VUID-vkGetDeviceQueue-device-parameter
device must be a valid [VkDevice](#) handle
- VUID-vkGetDeviceQueue-pQueue-parameter
pQueue must be a valid pointer to a [VkQueue](#) handle

5.3.3. Queue Family Index

The queue family index is used in multiple places in Vulkan in order to tie operations to a specific family of queues.

When retrieving a handle to the queue via [vkGetDeviceQueue](#), the queue family index is used to select which queue family to retrieve the [VkQueue](#) handle from as described in the previous section.

When creating a [VkCommandPool](#) object (see [Command Pools](#)), a queue family index is specified in the [VkCommandPoolCreateInfo](#) structure. Command buffers from this pool **can** only be submitted on queues corresponding to this queue family.

When creating [VkImage](#) (see [Images](#)) and [VkBuffer](#) (see [Buffers](#)) resources, a set of queue families is included in the [VkImageCreateInfo](#) and [VkBufferCreateInfo](#) structures to specify the queue families that **can** access the resource.

When inserting a [VkBufferMemoryBarrier](#) or [VkImageMemoryBarrier](#) (see [Pipeline Barriers](#)), a source and destination queue family index is specified to allow the ownership of a buffer or image to be transferred from one queue family to another. See the [Resource Sharing](#) section for details.

5.3.4. Queue Priority

Each queue is assigned a priority, as set in the [VkDeviceQueueCreateInfo](#) structures when creating the device. The priority of each queue is a normalized floating-point value between 0.0 and 1.0, which is then translated to a discrete priority level by the implementation. Higher values indicate a higher priority, with 0.0 being the lowest priority and 1.0 being the highest.

Within the same device, queues with higher priority **may** be allotted more processing time than queues with lower priority. The implementation makes no guarantees with regards to ordering or scheduling among queues with the same priority, other than the constraints defined by any [explicit synchronization primitives](#). The implementation makes no guarantees with regards to queues across different devices.

An implementation **may** allow a higher-priority queue to starve a lower-priority queue on the same [VkDevice](#) until the higher-priority queue has no further commands to execute. The relationship of queue priorities **must** not cause queues on one [VkDevice](#) to starve queues on another [VkDevice](#).

No specific guarantees are made about higher priority queues receiving more processing time or better quality of service than lower priority queues.

5.3.5. Queue Submission

Work is submitted to a queue via *queue submission* commands such as [vkQueueSubmit2KHR](#) or [vkQueueSubmit](#). Queue submission commands define a set of *queue operations* to be executed by the underlying physical device, including synchronization with semaphores and fences.

Submission commands take as parameters a target queue, zero or more *batches* of work, and an **optional** fence to signal upon completion. Each batch consists of three distinct parts:

1. Zero or more semaphores to wait on before execution of the rest of the batch.
 - If present, these describe a [semaphore wait operation](#).
2. Zero or more work items to execute.
 - If present, these describe a *queue operation* matching the work described.
3. Zero or more semaphores to signal upon completion of the work items.
 - If present, these describe a [semaphore signal operation](#).

If a fence is present in a queue submission, it describes a [fence signal operation](#).

All work described by a queue submission command **must** be submitted to the queue before the command returns.

Sparse Memory Binding

In Vulkan it is possible to sparsely bind memory to buffers and images as described in the [Sparse Resource](#) chapter. Sparse memory binding is a queue operation. A queue whose flags include the [VK_QUEUE_SPARSE_BINDING_BIT](#) **must** be able to support the mapping of a virtual address to a physical address on the device. This causes an update to the page table mappings on the device. This update **must** be synchronized on a queue to avoid corrupting page table mappings during execution of graphics commands. By binding the sparse memory resources on queues, all commands that are dependent on the updated bindings are synchronized to only execute after the binding is updated. See the [Synchronization and Cache Control](#) chapter for how this synchronization is accomplished.

5.3.6. Queue Destruction

Queues are created along with a logical device during [vkCreateDevice](#). All queues associated with a logical device are destroyed when [vkDestroyDevice](#) is called on that device.

Chapter 6. Command Buffers

Command buffers are objects used to record commands which **can** be subsequently submitted to a device queue for execution. There are two levels of command buffers - *primary command buffers*, which **can** execute secondary command buffers, and which are submitted to queues, and *secondary command buffers*, which **can** be executed by primary command buffers, and which are not directly submitted to queues.

Command buffers are represented by `VkCommandBuffer` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_HANDLE(VkCommandBuffer)
```

Recorded commands include commands to bind pipelines and descriptor sets to the command buffer, commands to modify dynamic state, commands to draw (for graphics rendering), commands to dispatch (for compute), commands to execute secondary command buffers (for primary command buffers only), commands to copy buffers and images, and other commands.

Each command buffer manages state independently of other command buffers. There is no inheritance of state across primary and secondary command buffers, or between secondary command buffers. When a command buffer begins recording, all state in that command buffer is undefined. When secondary command buffer(s) are recorded to execute on a primary command buffer, the secondary command buffer inherits no state from the primary command buffer, and all state of the primary command buffer is undefined after an execute secondary command buffer command is recorded. There is one exception to this rule - if the primary command buffer is inside a render pass instance, then the render pass and subpass state is not disturbed by executing secondary command buffers. For state dependent commands (such as draws and dispatches), any state consumed by those commands **must** not be undefined.

Unless otherwise specified, and without explicit synchronization, the various commands submitted to a queue via command buffers **may** execute in arbitrary order relative to each other, and/or concurrently. Also, the memory side effects of those commands **may** not be directly visible to other commands without explicit memory dependencies. This is true within a command buffer, and across command buffers submitted to a given queue. See [the synchronization chapter](#) for information on [implicit](#) and explicit synchronization between commands.

6.1. Command Buffer Lifecycle

Each command buffer is always in one of the following states:

Initial

When a command buffer is [allocated](#), it is in the *initial state*. Some commands are able to *reset* a command buffer (or a set of command buffers) back to this state from any of the executable, recording or invalid state. Command buffers in the initial state **can** only be moved to the recording state, or freed.

Recording

`vkBeginCommandBuffer` changes the state of a command buffer from the initial state to the *recording state*. Once a command buffer is in the recording state, `vkCmd*` commands **can** be used to record to the command buffer.

Executable

`vkEndCommandBuffer` ends the recording of a command buffer, and moves it from the recording state to the *executable state*. Executable command buffers **can** be [submitted](#), reset, or [recorded to another command buffer](#).

Pending

[Queue submission](#) of a command buffer changes the state of a command buffer from the executable state to the *pending state*. Whilst in the pending state, applications **must** not attempt to modify the command buffer in any way - as the device **may** be processing the commands recorded to it. Once execution of a command buffer completes, the command buffer either reverts back to the *executable state*, or if it was recorded with `VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT`, it moves to the *invalid state*. A [synchronization command](#) **should** be used to detect when this occurs.

Invalid

Some operations, such as [modifying or deleting a resource](#) that was used in a command recorded to a command buffer, will transition the state of that command buffer into the *invalid state*. Command buffers in the invalid state **can** only be reset or freed.

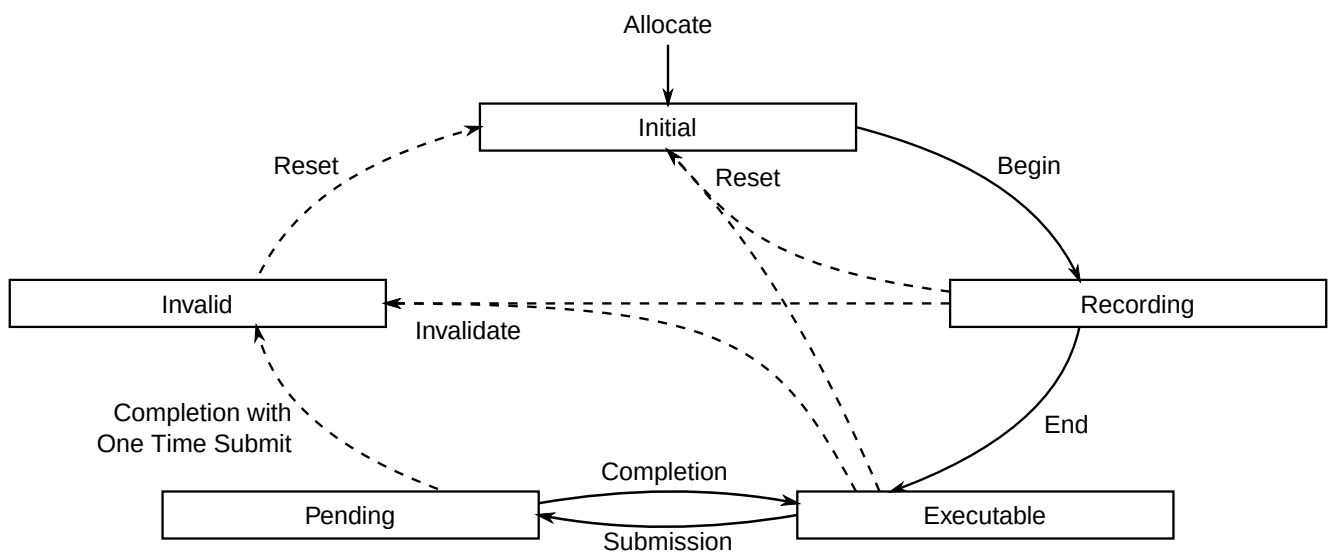


Figure 1. Lifecycle of a command buffer

Any given command that operates on a command buffer has its own requirements on what state a command buffer **must** be in, which are detailed in the valid usage constraints for that command.

Resetting a command buffer is an operation that discards any previously recorded commands and puts a command buffer in the *initial state*. Resetting occurs as a result of `vkResetCommandBuffer` or `vkResetCommandPool`, or as part of `vkBeginCommandBuffer` (which additionally puts the command buffer in the *recording state*).

[Secondary command buffers](#) **can** be recorded to a primary command buffer via

[vkCmdExecuteCommands](#). This partially ties the lifecycle of the two command buffers together - if the primary is submitted to a queue, both the primary and any secondaries recorded to it move to the *pending state*. Once execution of the primary completes, so it does for any secondary recorded within it. After all executions of each command buffer complete, they each move to their appropriate completion state (either to the *executable state* or the *invalid state*, as specified above).

If a secondary moves to the *invalid state* or the *initial state*, then all primary buffers it is recorded in move to the *invalid state*. A primary moving to any other state does not affect the state of a secondary recorded in it.

NOTE

Resetting or freeing a primary command buffer removes the lifecycle linkage to all secondary command buffers that were recorded into it.

6.2. Command Pools

Command pools are opaque objects that command buffer memory is allocated from, and which allow the implementation to amortize the cost of resource creation across multiple command buffers. Command pools are externally synchronized, meaning that a command pool **must** not be used concurrently in multiple threads. That includes use via recording commands on any command buffers allocated from the pool, as well as operations that allocate, free, and reset command buffers or the pool itself.

Command pools are represented by [VkCommandPool](#) handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkCommandPool)
```

To create a command pool, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateCommandPool(
    VkDevice                                device,
    const VkCommandPoolCreateInfo*          pCreateInfo,
    const VkAllocationCallbacks*            pAllocator,
    VkCommandPool*                          pCommandPool);
```

- [device](#) is the logical device that creates the command pool.
- [pCreateInfo](#) is a pointer to a [VkCommandPoolCreateInfo](#) structure specifying the state of the command pool object.
- [pAllocator](#) controls host memory allocation as described in the [Memory Allocation](#) chapter.
- [pCommandPool](#) is a pointer to a [VkCommandPool](#) handle in which the created pool is returned.

Valid Usage

- VUID-vkCreateCommandPool-queueFamilyIndex-01937

`pCreateInfo->queueFamilyIndex` **must** be the index of a queue family available in the logical device `device`

Valid Usage (Implicit)

- VUID-vkCreateCommandPool-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateCommandPool-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkCommandPoolCreateInfo` structure
- VUID-vkCreateCommandPool-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateCommandPool-pCommandPool-parameter
`pCommandPool` **must** be a valid pointer to a `VkCommandPool` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkCommandPoolCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandPoolCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkCommandPoolCreateFlags flags;
    uint32_t           queueFamilyIndex;
} VkCommandPoolCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkCommandPoolCreateFlagBits` indicating usage behavior for the pool and command buffers allocated from it.
- `queueFamilyIndex` designates a queue family as described in section [Queue Family Properties](#). All command buffers allocated from this command pool **must** be submitted on queues from the same queue family.

Valid Usage

Valid Usage (Implicit)

- VUID-VkCommandPoolCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COMMAND_POOL_CREATE_INFO`
- VUID-VkCommandPoolCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCommandPoolCreateInfo-flags-parameter
`flags` **must** be a valid combination of `VkCommandPoolCreateFlagBits` values

Bits which **can** be set in `VkCommandPoolCreateInfo::flags`, specifying usage behavior for a command pool, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandPoolCreateFlagBits {
    VK_COMMAND_POOL_CREATE_TRANSIENT_BIT = 0x00000001,
    VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT = 0x00000002,
} VkCommandPoolCreateFlagBits;
```

- `VK_COMMAND_POOL_CREATE_TRANSIENT_BIT` specifies that command buffers allocated from the pool will be short-lived, meaning that they will be reset or freed in a relatively short timeframe. This flag **may** be used by the implementation to control memory allocation behavior within the pool.
- `VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT` allows any command buffer allocated from a pool to be individually reset to the [initial state](#); either by calling `vkResetCommandBuffer`, or via the implicit reset when calling `vkBeginCommandBuffer`. If this flag is not set on a pool, then `vkResetCommandBuffer` **must** not be called for any command buffer allocated from that pool.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandPoolCreateFlags;
```

`VkCommandPoolCreateFlags` is a bitmask type for setting a mask of zero or more `VkCommandPoolCreateFlagBits`.

To trim a command pool, call:

```
// Provided by VK_KHR_maintenance1
void vkTrimCommandPoolKHR(
    VkDevice          device,
    VkCommandPool     commandPool,
    VkCommandPoolTrimFlags flags);
```

- **device** is the logical device that owns the command pool.
- **commandPool** is the command pool to trim.
- **flags** is reserved for future use.

Trimming a command pool recycles unused memory from the command pool back to the system. Command buffers allocated from the pool are not affected by the command.

This command provides applications with some control over the internal memory allocations used by command pools.

Unused memory normally arises from command buffers that have been recorded and later reset, such that they are no longer using the memory. On reset, a command buffer can return memory to its command pool, but the only way to release memory from a command pool to the system requires calling [vkResetCommandPool](#), which cannot be executed while any command buffers from that pool are still in use. Subsequent recording operations into command buffers will reuse this memory but since total memory requirements fluctuate over time, unused memory can accumulate.

In this situation, trimming a command pool **may** be useful to return unused memory back to the system, returning the total outstanding memory allocated by the pool back to a more “average” value.

NOTE

Implementations utilize many internal allocation strategies that make it impossible to guarantee that all unused memory is released back to the system. For instance, an implementation of a command pool **may** involve allocating memory in bulk from the system and sub-allocating from that memory. In such an implementation any live command buffer that holds a reference to a bulk allocation would prevent that allocation from being freed, even if only a small proportion of the bulk allocation is in use.

In most cases trimming will result in a reduction in allocated but unused memory, but it does not guarantee the “ideal” behavior.

Trimming **may** be an expensive operation, and **should** not be called frequently. Trimming **should** be treated as a way to relieve memory pressure after application-known points when there exists enough unused memory that the cost of trimming is “worth” it.

Valid Usage (Implicit)

- VUID-vkTrimCommandPool-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkTrimCommandPool-commandPool-parameter
commandPool **must** be a valid [VkCommandPool](#) handle
- VUID-vkTrimCommandPool-flags-zeroBitmask
flags **must** be 0

- VUID-vkTrimCommandPool-commandPool-parent
`commandPool` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `commandPool` **must** be externally synchronized

```
typedef VkFlags VkCommandPoolTrimFlags;
```

or the equivalent

```
// Provided by VK_KHR_maintenance1  
typedef VkCommandPoolTrimFlags VkCommandPoolTrimFlagsKHR;
```

`VkCommandPoolTrimFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

To reset a command pool, call:

```
// Provided by VK_VERSION_1_0  
VkResult vkResetCommandPool(  
    VkDevice                device,  
    VkCommandPool           commandPool,  
    VkCommandPoolResetFlags flags);
```

- `device` is the logical device that owns the command pool.
- `commandPool` is the command pool to reset.
- `flags` is a bitmask of `VkCommandPoolResetFlagBits` controlling the reset operation.

Resetting a command pool recycles all of the resources from all of the command buffers allocated from the command pool back to the command pool. All command buffers that have been allocated from the command pool are put in the `initial state`.

Any primary command buffer allocated from another `VkCommandPool` that is in the `recording or executable state` and has a secondary command buffer allocated from `commandPool` recorded into it, becomes `invalid`.

Valid Usage

- VUID-vkResetCommandPool-commandPool-00040
All `VkCommandBuffer` objects allocated from `commandPool` **must** not be in the `pending state`

Valid Usage (Implicit)

- VUID-vkResetCommandPool-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkResetCommandPool-commandPool-parameter
commandPool **must** be a valid [VkCommandPool](#) handle
- VUID-vkResetCommandPool-flags-parameter
flags **must** be a valid combination of [VkCommandPoolResetFlagBits](#) values
- VUID-vkResetCommandPool-commandPool-parent
commandPool **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **commandPool** **must** be externally synchronized

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

Bits which **can** be set in [vkResetCommandPool::flags](#), controlling the reset operation, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandPoolResetFlagBits {
    VK_COMMAND_POOL_RESET_RELEASE_RESOURCES_BIT = 0x00000001,
} VkCommandPoolResetFlagBits;
```

- **VK_COMMAND_POOL_RESET_RELEASE_RESOURCES_BIT** specifies that resetting a command pool recycles all of the resources from the command pool back to the system.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandPoolResetFlags;
```

VkCommandPoolResetFlags is a bitmask type for setting a mask of zero or more [VkCommandPoolResetFlagBits](#).

To destroy a command pool, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyCommandPool(
    VkDevice          device,
    VkCommandPool     commandPool,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the command pool.
- **commandPool** is the handle of the command pool to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

When a pool is destroyed, all command buffers allocated from the pool are [freed](#).

Any primary command buffer allocated from another [VkCommandPool](#) that is in the [recording or executable state](#) and has a secondary command buffer allocated from **commandPool** recorded into it, becomes [invalid](#).

Valid Usage

- VUID-vkDestroyCommandPool-commandPool-00041
All **VkCommandBuffer** objects allocated from **commandPool** **must** not be in the [pending state](#)
- VUID-vkDestroyCommandPool-commandPool-00042
If **VkAllocationCallbacks** were provided when **commandPool** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyCommandPool-commandPool-00043
If no **VkAllocationCallbacks** were provided when **commandPool** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyCommandPool-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkDestroyCommandPool-commandPool-parameter
If **commandPool** is not [VK_NULL_HANDLE](#), **commandPool** **must** be a valid [VkCommandPool](#) handle
- VUID-vkDestroyCommandPool-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkDestroyCommandPool-commandPool-parent
If **commandPool** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **commandPool** **must** be externally synchronized

6.3. Command Buffer Allocation and Management

To allocate command buffers, call:

```
// Provided by VK_VERSION_1_0
VkResult vkAllocateCommandBuffers(
    VkDevice device,
    const VkCommandBufferAllocateInfo* pAllocateInfo,
    VkCommandBuffer* pCommandBuffers);
```

- **device** is the logical device that owns the command pool.
- **pAllocateInfo** is a pointer to a **VkCommandBufferAllocateInfo** structure describing parameters of the allocation.
- **pCommandBuffers** is a pointer to an array of **VkCommandBuffer** handles in which the resulting command buffer objects are returned. The array **must** be at least the length specified by the **commandBufferCount** member of **pAllocateInfo**. Each allocated command buffer begins in the initial state.

vkAllocateCommandBuffers **can** be used to allocate multiple command buffers. If the allocation of any of those command buffers fails, the implementation **must** free all successfully allocated command buffer objects from this command, set all entries of the **pCommandBuffers** array to **NULL** and return the error.

NOTE

Filling **pCommandBuffers** with **NULL** values on failure is an exception to the default error behavior that output parameters will have undefined contents.

When command buffers are first allocated, they are in the **initial state**.

Valid Usage (Implicit)

- VUID-vkAllocateCommandBuffers-device-parameter **device** **must** be a valid **VkDevice** handle
- VUID-vkAllocateCommandBuffers-pAllocateInfo-parameter **pAllocateInfo** **must** be a valid pointer to a valid **VkCommandBufferAllocateInfo** structure
- VUID-vkAllocateCommandBuffers-pCommandBuffers-parameter **pCommandBuffers** **must** be a valid pointer to an array of **pAllocateInfo->commandBufferCount** **VkCommandBuffer** handles
- VUID-vkAllocateCommandBuffers-pAllocateInfo::commandBufferCount-arraylength **pAllocateInfo->commandBufferCount** **must** be greater than 0

Host Synchronization

- Host access to `pAllocateInfo->commandPool` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkCommandBufferAllocateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandBufferAllocateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkCommandPool        commandPool;
    VkCommandBufferLevel level;
    uint32_t             commandBufferCount;
} VkCommandBufferAllocateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `commandPool` is the command pool from which the command buffers are allocated.
- `level` is a `VkCommandBufferLevel` value specifying the command buffer level.
- `commandBufferCount` is the number of command buffers to allocate from the pool.

Valid Usage (Implicit)

- VUID-VkCommandBufferAllocateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COMMAND_BUFFER_ALLOCATE_INFO`
- VUID-VkCommandBufferAllocateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCommandBufferAllocateInfo-commandPool-parameter
`commandPool` **must** be a valid `VkCommandPool` handle
- VUID-VkCommandBufferAllocateInfo-level-parameter
`level` **must** be a valid `VkCommandBufferLevel` value

Possible values of `VkCommandBufferAllocateInfo::level`, specifying the command buffer level, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandBufferLevel {
    VK_COMMAND_BUFFER_LEVEL_PRIMARY = 0,
    VK_COMMAND_BUFFER_LEVEL_SECONDARY = 1,
} VkCommandBufferLevel;
```

- `VK_COMMAND_BUFFER_LEVEL_PRIMARY` specifies a primary command buffer.
- `VK_COMMAND_BUFFER_LEVEL_SECONDARY` specifies a secondary command buffer.

To reset a command buffer, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetCommandBuffer(
    VkCommandBuffer          commandBuffer,
    VkCommandBufferResetFlags flags);
```

- `commandBuffer` is the command buffer to reset. The command buffer **can** be in any state other than `pending`, and is moved into the `initial state`.
- `flags` is a bitmask of `VkCommandBufferResetFlagBits` controlling the reset operation.

Any primary command buffer that is in the `recording or executable state` and has `commandBuffer` recorded into it, becomes `invalid`.

Valid Usage

- VUID-vkResetCommandBuffer-commandBuffer-00045
`commandBuffer` **must** not be in the `pending state`
- VUID-vkResetCommandBuffer-commandBuffer-00046
`commandBuffer` **must** have been allocated from a pool that was created with the `VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT`

Valid Usage (Implicit)

- VUID-vkResetCommandBuffer-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkResetCommandBuffer-flags-parameter
`flags` **must** be a valid combination of `VkCommandBufferResetFlagBits` values

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

Bits which **can** be set in `vkResetCommandBuffer::flags`, controlling the reset operation, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandBufferResetFlagBits {
    VK_COMMAND_BUFFER_RESET_RELEASE_RESOURCES_BIT = 0x00000001,
} VkCommandBufferResetFlagBits;
```

- `VK_COMMAND_BUFFER_RESET_RELEASE_RESOURCES_BIT` specifies that most or all memory resources currently owned by the command buffer **should** be returned to the parent command pool. If this flag is not set, then the command buffer **may** hold onto memory resources and reuse them when recording commands. `commandBuffer` is moved to the [initial state](#).

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandBufferResetFlags;
```

`VkCommandBufferResetFlags` is a bitmask type for setting a mask of zero or more `VkCommandBufferResetFlagBits`.

To free command buffers, call:

```
// Provided by VK_VERSION_1_0
void vkFreeCommandBuffers(
    VkDevice          device,
    VkCommandPool     commandPool,
    uint32_t          commandBufferCount,
    const VkCommandBuffer* pCommandBuffers);
```

- `device` is the logical device that owns the command pool.
- `commandPool` is the command pool from which the command buffers were allocated.
- `commandBufferCount` is the length of the `pCommandBuffers` array.
- `pCommandBuffers` is a pointer to an array of handles of command buffers to free.

Any primary command buffer that is in the [recording or executable state](#) and has any element of

`pCommandBuffers` recorded into it, becomes `invalid`.

Valid Usage

- VUID-vkFreeCommandBuffers-pCommandBuffers-00047
All elements of `pCommandBuffers` **must** not be in the `pending state`
- VUID-vkFreeCommandBuffers-pCommandBuffers-00048
`pCommandBuffers` **must** be a valid pointer to an array of `commandBufferCount` `VkCommandBuffer` handles, each element of which **must** either be a valid handle or `NULL`

Valid Usage (Implicit)

- VUID-vkFreeCommandBuffers-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkFreeCommandBuffers-commandPool-parameter
`commandPool` **must** be a valid `VkCommandPool` handle
- VUID-vkFreeCommandBuffers-commandBufferCount-arraylength
`commandBufferCount` **must** be greater than 0
- VUID-vkFreeCommandBuffers-commandPool-parent
`commandPool` **must** have been created, allocated, or retrieved from `device`
- VUID-vkFreeCommandBuffers-pCommandBuffers-parent
Each element of `pCommandBuffers` that is a valid handle **must** have been created, allocated, or retrieved from `commandPool`

Host Synchronization

- Host access to `commandPool` **must** be externally synchronized
- Host access to each member of `pCommandBuffers` **must** be externally synchronized

6.4. Command Buffer Recording

To begin recording a command buffer, call:

```
// Provided by VK_VERSION_1_0
VkResult vkBeginCommandBuffer(
    VkCommandBuffer          commandBuffer,
    const VkCommandBufferBeginInfo* pBeginInfo);
```

- `commandBuffer` is the handle of the command buffer which is to be put in the recording state.
- `pBeginInfo` is a pointer to a `VkCommandBufferBeginInfo` structure defining additional information about how the command buffer begins recording.

Valid Usage

- VUID-vkBeginCommandBuffer-commandBuffer-00049
`commandBuffer` **must** not be in the `recording or pending` state
- VUID-vkBeginCommandBuffer-commandBuffer-00050
If `commandBuffer` was allocated from a `VkCommandPool` which did not have the `VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT` flag set, `commandBuffer` **must** be in the `initial` state
- VUID-vkBeginCommandBuffer-commandBuffer-00051
If `commandBuffer` is a secondary command buffer, the `pInheritanceInfo` member of `pBeginInfo` **must** be a valid `VkCommandBufferInheritanceInfo` structure
- VUID-vkBeginCommandBuffer-commandBuffer-00052
If `commandBuffer` is a secondary command buffer and either the `occlusionQueryEnable` member of the `pInheritanceInfo` member of `pBeginInfo` is `VK_FALSE`, or the `occlusionQueryPrecise` feature is not enabled, then `pBeginInfo->pInheritanceInfo->queryFlags` **must** not contain `VK_QUERY_CONTROL_PRECISE_BIT`
- VUID-vkBeginCommandBuffer-commandBuffer-02840
If `commandBuffer` is a primary command buffer, then `pBeginInfo->flags` **must** not set both the `VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT` and the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT` flags

Valid Usage (Implicit)

- VUID-vkBeginCommandBuffer-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkBeginCommandBuffer-pBeginInfo-parameter
`pBeginInfo` **must** be a valid pointer to a valid `VkCommandBufferBeginInfo` structure

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`

- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkCommandBufferBeginInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandBufferBeginInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkCommandBufferUsageFlags flags;
    const VkCommandBufferInheritanceInfo* pInheritanceInfo;
} VkCommandBufferBeginInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkCommandBufferUsageFlagBits` specifying usage behavior for the command buffer.
- `pInheritanceInfo` is a pointer to a `VkCommandBufferInheritanceInfo` structure, used if `commandBuffer` is a secondary command buffer. If this is a primary command buffer, then this value is ignored.

Valid Usage

- VUID-VkCommandBufferBeginInfo-flags-09123
If `flags` contains `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT`, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-VkCommandBufferBeginInfo-flags-00055
If `flags` contains `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT`, the `framebuffer` member of `pInheritanceInfo` **must** be either `VK_NULL_HANDLE`, or a valid `VkFramebuffer` that is compatible with the `renderPass` member of `pInheritanceInfo`
- VUID-VkCommandBufferBeginInfo-flags-09240
If `flags` contains `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT` and the `dynamicRendering` feature is not enabled, the `renderPass` member of `pInheritanceInfo` **must** not be `VK_NULL_HANDLE`
- VUID-VkCommandBufferBeginInfo-flags-06002
If `flags` contains `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT` and the `renderPass` member of `pInheritanceInfo` is `VK_NULL_HANDLE`, the `pNext` chain of `pInheritanceInfo` **must** include a `VkCommandBufferInheritanceRenderingInfo` structure
- VUID-VkCommandBufferBeginInfo-flags-06000
If `flags` contains `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT` and the `renderPass` member of `pInheritanceInfo` is not `VK_NULL_HANDLE`, the `renderPass` member of `pInheritanceInfo` **must** be a valid `VkRenderPass`
- VUID-VkCommandBufferBeginInfo-flags-06001
If `flags` contains `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT` and the `renderPass`

member of `pInheritanceInfo` is not `VK_NULL_HANDLE`, the `subpass` member of `pInheritanceInfo` **must** be a valid subpass index within the `renderPass` member of `pInheritanceInfo`

Valid Usage (Implicit)

- VUID-VkCommandBufferBeginInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COMMAND_BUFFER_BEGIN_INFO`
- VUID-VkCommandBufferBeginInfo-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkDeviceGroupCommandBufferBeginInfo`
- VUID-VkCommandBufferBeginInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkCommandBufferBeginInfo-flags-parameter
`flags` **must** be a valid combination of `VkCommandBufferUsageFlagBits` values

Bits which **can** be set in `VkCommandBufferBeginInfo::flags`, specifying usage behavior for a command buffer, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCommandBufferUsageFlagBits {
    VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT = 0x00000001,
    VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT = 0x00000002,
    VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT = 0x00000004,
} VkCommandBufferUsageFlagBits;
```

- `VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT` specifies that each recording of the command buffer will only be submitted once, and the command buffer will be reset and recorded again between each submission.
- `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT` specifies that a secondary command buffer is considered to be entirely inside a render pass. If this is a primary command buffer, then this bit is ignored.
- `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT` specifies that a command buffer **can** be resubmitted to any queue of the same queue family while it is in the *pending state*, and recorded into multiple primary command buffers.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCommandBufferUsageFlags;
```

`VkCommandBufferUsageFlags` is a bitmask type for setting a mask of zero or more `VkCommandBufferUsageFlagBits`.

If the command buffer is a secondary command buffer, then the `VkCommandBufferInheritanceInfo`

structure defines any state that will be inherited from the primary command buffer:

```
// Provided by VK_VERSION_1_0
typedef struct VkCommandBufferInheritanceInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkRenderPass          renderPass;
    uint32_t              subpass;
    VkFramebuffer         framebuffer;
    VkBool32              occlusionQueryEnable;
    VkQueryControlFlags   queryFlags;
    VkQueryPipelineStatisticFlags pipelineStatistics;
} VkCommandBufferInheritanceInfo;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **renderPass** is a **VkRenderPass** object defining which render passes the **VkCommandBuffer** will be **compatible** with and **can** be executed within.
- **subpass** is the index of the subpass within the render pass instance that the **VkCommandBuffer** will be executed within.
- **framebuffer** **can** refer to the **VkFramebuffer** object that the **VkCommandBuffer** will be rendering to if it is executed within a render pass instance. It **can** be **VK_NULL_HANDLE** if the framebuffer is not known.

NOTE

Specifying the exact framebuffer that the secondary command buffer will be executed with **may** result in better performance at command buffer execution time.

- **occlusionQueryEnable** specifies whether the command buffer **can** be executed while an occlusion query is active in the primary command buffer. If this is **VK_TRUE**, then this command buffer **can** be executed whether the primary command buffer has an occlusion query active or not. If this is **VK_FALSE**, then the primary command buffer **must** not have an occlusion query active.
- **queryFlags** specifies the query flags that **can** be used by an active occlusion query in the primary command buffer when this secondary command buffer is executed. If this value includes the **VK_QUERY_CONTROL_PRECISE_BIT** bit, then the active query **can** return boolean results or actual sample counts. If this bit is not set, then the active query **must** not use the **VK_QUERY_CONTROL_PRECISE_BIT** bit.
- **pipelineStatistics** is a bitmask of **VkQueryPipelineStatisticFlagBits** specifying the set of pipeline statistics that **can** be counted by an active query in the primary command buffer when this secondary command buffer is executed. If this value includes a given bit, then this command buffer **can** be executed whether the primary command buffer has a pipeline statistics query active that includes this bit or not. If this value excludes a given bit, then the active pipeline statistics query **must** not be from a query pool that counts that statistic.

If the `VkCommandBuffer` will not be executed within a render pass instance, or if the render pass instance was begun with `vkCmdBeginRenderingKHR`, `renderPass`, `subpass`, and `framebuffer` are ignored.

Valid Usage

- VUID-VkCommandBufferInheritanceInfo-occlusionQueryEnable-00056
If the `inheritedQueries` feature is not enabled, `occlusionQueryEnable` **must** be `VK_FALSE`
- VUID-VkCommandBufferInheritanceInfo-queryFlags-00057
If the `inheritedQueries` feature is enabled, `queryFlags` **must** be a valid combination of `VkQueryControlFlagBits` values
- VUID-VkCommandBufferInheritanceInfo-queryFlags-02788
If the `inheritedQueries` feature is not enabled, `queryFlags` **must** be `0`
- VUID-VkCommandBufferInheritanceInfo-pipelineStatistics-02789
If the `pipelineStatisticsQuery` feature is enabled, `pipelineStatistics` **must** be a valid combination of `VkQueryPipelineStatisticFlagBits` values
- VUID-VkCommandBufferInheritanceInfo-pipelineStatistics-00058
If the `pipelineStatisticsQuery` feature is not enabled, `pipelineStatistics` **must** be `0`

Valid Usage (Implicit)

- VUID-VkCommandBufferInheritanceInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_INFO`
- VUID-VkCommandBufferInheritanceInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkCommandBufferInheritanceRenderingInfo`, `VkRenderingAttachmentLocationInfoKHR`, or `VkRenderingInputAttachmentIndexInfoKHR`
- VUID-VkCommandBufferInheritanceInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkCommandBufferInheritanceInfo-commonparent
Both of `framebuffer`, and `renderPass` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

NOTE	On some implementations, not using the <code>VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT</code> bit enables command buffers to be patched in-place if needed, rather than creating a copy of the command buffer.
-------------	---

If a command buffer is in the `invalid`, or `executable` state, and the command buffer was allocated from a command pool with the `VK_COMMAND_POOL_CREATE_RESET_COMMAND_BUFFER_BIT` flag set, then `vkBeginCommandBuffer` implicitly resets the command buffer, behaving as if `vkResetCommandBuffer` had been called with `VK_COMMAND_BUFFER_RESET_RELEASE_RESOURCES_BIT` not set. After the implicit reset, `commandBuffer` is moved to the `recording` state.

The `VkCommandBufferInheritanceRenderingInfo` structure is defined as:

```
typedef struct VkCommandBufferInheritanceRenderingInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkRenderingFlags    flags;
    uint32_t           viewMask;
    uint32_t           colorAttachmentCount;
    const VkFormat*     pColorAttachmentFormats;
    VkFormat            depthAttachmentFormat;
    VkFormat            stencilAttachmentFormat;
    VkSampleCountFlagBits rasterizationSamples;
} VkCommandBufferInheritanceRenderingInfo;
```

or the equivalent

```
// Provided by VK_KHR_dynamic_rendering
typedef VkCommandBufferInheritanceRenderingInfo
VkCommandBufferInheritanceRenderingInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure
- `flags` is a bitmask of `VkRenderingFlagBits` used by the render pass instance.
- `viewMask` is the view mask used for rendering.
- `colorAttachmentCount` is the number of color attachments specified in the render pass instance.
- `pColorAttachmentFormats` is a pointer to an array of `VkFormat` values defining the format of color attachments.
- `depthAttachmentFormat` is a `VkFormat` value defining the format of the depth attachment.
- `stencilAttachmentFormat` is a `VkFormat` value defining the format of the stencil attachment.
- `rasterizationSamples` is a `VkSampleCountFlagBits` specifying the number of samples used in rasterization.

If the `pNext` chain of `VkCommandBufferInheritanceInfo` includes a `VkCommandBufferInheritanceRenderingInfo` structure, then that structure controls parameters of dynamic render pass instances that the `VkCommandBuffer` can be executed within. If `VkCommandBufferInheritanceInfo::renderPass` is not `VK_NULL_HANDLE`, or `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT` is not specified in `VkCommandBufferBeginInfo::flags`, parameters of this structure are ignored.

If `colorAttachmentCount` is `0` and the `variableMultisampleRate` feature is enabled, `rasterizationSamples` is ignored.

If `depthAttachmentFormat`, `stencilAttachmentFormat`, or any element of `pColorAttachmentFormats` is `VK_FORMAT_UNDEFINED`, it indicates that the corresponding attachment is unused within the render

pass and writes to those attachments are discarded.

Valid Usage

- VUID-VkCommandBufferInheritanceRenderingInfo-colorAttachmentCount-06004
If `colorAttachmentCount` is not 0, `rasterizationSamples` **must** be a valid `VkSampleCountFlagBits` value
- VUID-VkCommandBufferInheritanceRenderingInfo-variableMultisampleRate-06005
If the `variableMultisampleRate` feature is not enabled, `rasterizationSamples` **must** be a valid `VkSampleCountFlagBits` value
- VUID-VkCommandBufferInheritanceRenderingInfo-depthAttachmentFormat-06540
If `depthAttachmentFormat` is not `VK_FORMAT_UNDEFINED`, it **must** be a format that includes a depth component
- VUID-VkCommandBufferInheritanceRenderingInfo-depthAttachmentFormat-06007
If `depthAttachmentFormat` is not `VK_FORMAT_UNDEFINED`, it **must** be a format with `potential format features` that include `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkCommandBufferInheritanceRenderingInfo-pColorAttachmentFormats-06492
If any element of `pColorAttachmentFormats` is not `VK_FORMAT_UNDEFINED`, it **must** be a format with `potential format features` that include `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT`
- VUID-VkCommandBufferInheritanceRenderingInfo-stencilAttachmentFormat-06541
If `stencilAttachmentFormat` is not `VK_FORMAT_UNDEFINED`, it **must** be a format that includes a stencil aspect
- VUID-VkCommandBufferInheritanceRenderingInfo-stencilAttachmentFormat-06199
If `stencilAttachmentFormat` is not `VK_FORMAT_UNDEFINED`, it **must** be a format with `potential format features` that include `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkCommandBufferInheritanceRenderingInfo-depthAttachmentFormat-06200
If `depthAttachmentFormat` is not `VK_FORMAT_UNDEFINED` and `stencilAttachmentFormat` is not `VK_FORMAT_UNDEFINED`, `depthAttachmentFormat` **must** equal `stencilAttachmentFormat`
- VUID-VkCommandBufferInheritanceRenderingInfo-multiview-06008
If the `multiview` feature is not enabled, `viewMask` **must** be 0
- VUID-VkCommandBufferInheritanceRenderingInfo-viewMask-06009
The index of the most significant bit in `viewMask` **must** be less than `maxMultiviewViewCount`

Valid Usage (Implicit)

- VUID-VkCommandBufferInheritanceRenderingInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_RENDERING_INFO`
- VUID-VkCommandBufferInheritanceRenderingInfo-flags-parameter
`flags` **must** be a valid combination of `VkRenderingFlagBits` values
- VUID-VkCommandBufferInheritanceRenderingInfo-pColorAttachmentFormats-parameter
If `colorAttachmentCount` is not 0, `pColorAttachmentFormats` **must** be a valid pointer to an array of `colorAttachmentCount` valid `VkFormat` values

- VUID-VkCommandBufferInheritanceRenderingInfo-depthAttachmentFormat-parameter **depthAttachmentFormat** **must** be a valid [VkFormat](#) value
- VUID-VkCommandBufferInheritanceRenderingInfo-stencilAttachmentFormat-parameter **stencilAttachmentFormat** **must** be a valid [VkFormat](#) value
- VUID-VkCommandBufferInheritanceRenderingInfo-rasterizationSamples-parameter
If **rasterizationSamples** is not 0, **rasterizationSamples** **must** be a valid [VkSampleCountFlagBits](#) value

Once recording starts, an application records a sequence of commands (**vkCmd***) to set state in the command buffer, draw, dispatch, and other commands.

Several commands can also be recorded indirectly from **VkBuffer** content, see [Device-Generated Commands](#).

To complete recording of a command buffer, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEndCommandBuffer(
    VkCommandBuffer                commandBuffer);
```

- **commandBuffer** is the command buffer to complete recording.

The command buffer **must** have been in the [recording state](#), and, if successful, is moved to the [executable state](#).

If there was an error during recording, the application will be notified by an unsuccessful return code returned by **vkEndCommandBuffer**, and the command buffer will be moved to the [invalid state](#).

In case the application recorded one or more [video encode operations](#) into the command buffer, implementations **may** return the **VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR** error if any of the specified Video Std parameters do not adhere to the syntactic or semantic requirements of the used video compression standard, or if values derived from parameters according to the rules defined by the used video compression standard do not adhere to the capabilities of the video compression standard or the implementation.

NOTE

Applications **should** not rely on the **VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR** error being returned by any command as a means to verify Video Std parameters, as implementations are not required to report the error in any specific set of cases.

Valid Usage

- VUID-vkEndCommandBuffer-commandBuffer-00059
commandBuffer **must** be in the [recording state](#)
- VUID-vkEndCommandBuffer-commandBuffer-00060
If **commandBuffer** is a primary command buffer, there **must** not be an active render pass instance

- VUID-vkEndCommandBuffer-commandBuffer-00061
All queries made **active** during the recording of **commandBuffer** **must** have been made inactive
- VUID-vkEndCommandBuffer-None-06991
There **must** be no video session object bound
- VUID-vkEndCommandBuffer-commandBuffer-01815
If **commandBuffer** is a secondary command buffer, there **must** not be an outstanding **vkCmdBeginDebugUtilsLabelEXT** command recorded to **commandBuffer** that has not previously been ended by a call to **vkCmdEndDebugUtilsLabelEXT**

Valid Usage (Implicit)

- VUID-vkEndCommandBuffer-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**
- **VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR**

When a command buffer is in the executable state, it **can** be submitted to a queue for execution.

6.5. Command Buffer Submission

NOTE

Submission can be a high overhead operation, and applications **should** attempt to batch work together into as few calls to **vkQueueSubmit** or **vkQueueSubmit2KHR** as possible.

To submit command buffers to a queue, call:

```
// Provided by VK_KHR_synchronization2
VkResult vkQueueSubmit2KHR(
    VkQueue          queue,
    uint32_t         submitCount,
    const VkSubmitInfo2* pSubmits,
    VkFence          fence);
```

- `queue` is the queue that the command buffers will be submitted to.
- `submitCount` is the number of elements in the `pSubmits` array.
- `pSubmits` is a pointer to an array of `VkSubmitInfo2` structures, each specifying a command buffer submission batch.
- `fence` is an **optional** handle to a fence to be signaled once all submitted command buffers have completed execution. If `fence` is not `VK_NULL_HANDLE`, it defines a [fence signal operation](#).

`vkQueueSubmit2KHR` is a [queue submission command](#), with each batch defined by an element of `pSubmits`.

Semaphore operations submitted with `vkQueueSubmit2KHR` have additional ordering constraints compared to other submission commands, with dependencies involving previous and subsequent queue operations. Information about these additional constraints can be found in the [semaphore](#) section of [the synchronization chapter](#).

If any command buffer submitted to this queue is in the [executable state](#), it is moved to the [pending state](#). Once execution of all submissions of a command buffer complete, it moves from the [pending state](#), back to the [executable state](#). If a command buffer was recorded with the `VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT` flag, it instead moves back to the [invalid state](#).

If `vkQueueSubmit2KHR` fails, it **may** return `VK_ERROR_OUT_OF_HOST_MEMORY` or `VK_ERROR_OUT_OF_DEVICE_MEMORY`. If it does, the implementation **must** ensure that the state and contents of any resources or synchronization primitives referenced by the submitted command buffers and any semaphores referenced by `pSubmits` is unaffected by the call or its failure. If `vkQueueSubmit2KHR` fails in such a way that the implementation is unable to make that guarantee, the implementation **must** return `VK_ERROR_DEVICE_LOST`. See [Lost Device](#).

Valid Usage

- VUID-vkQueueSubmit2-fence-04894
If `fence` is not `VK_NULL_HANDLE`, `fence` **must** be unsignaled
- VUID-vkQueueSubmit2-fence-04895
If `fence` is not `VK_NULL_HANDLE`, `fence` **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-vkQueueSubmit2-synchronization2-03866
The `synchronization2` feature **must** be enabled
- VUID-vkQueueSubmit2-commandBuffer-03867
If a command recorded into the `commandBuffer` member of any element of the

`pCommandBufferInfos` member of any element of `pSubmits` referenced a `VkEvent`, that event **must** not be referenced by a command that has been submitted to another queue and is still in the *pending state*

- VUID-vkQueueSubmit2-semaphore-03868

The `semaphore` member of any binary semaphore element of the `pSignalSemaphoreInfos` member of any element of `pSubmits` **must** be unsignaled when the semaphore signal operation it defines is executed on the device

- VUID-vkQueueSubmit2-stageMask-03869

The `stageMask` member of any element of the `pSignalSemaphoreInfos` member of any element of `pSubmits` **must** only include pipeline stages that are supported by the queue family which `queue` belongs to

- VUID-vkQueueSubmit2-stageMask-03870

The `stageMask` member of any element of the `pWaitSemaphoreInfos` member of any element of `pSubmits` **must** only include pipeline stages that are supported by the queue family which `queue` belongs to

- VUID-vkQueueSubmit2-semaphore-03871

When a semaphore wait operation for a binary semaphore is executed, as defined by the `semaphore` member of any element of the `pWaitSemaphoreInfos` member of any element of `pSubmits`, there **must** be no other queues waiting on the same semaphore

- VUID-vkQueueSubmit2-semaphore-03873

The `semaphore` member of any element of the `pWaitSemaphoreInfos` member of any element of `pSubmits` that was created with a `VkSemaphoreTypeKHR` of `VK_SEMAPHORE_TYPE_BINARY_KHR` **must** reference a semaphore signal operation that has been submitted for execution and any `semaphore signal operations` on which it depends **must** have also been submitted for execution

- VUID-vkQueueSubmit2-commandBuffer-03874

The `commandBuffer` member of any element of the `pCommandBufferInfos` member of any element of `pSubmits` **must** be in the *pending or executable state*

- VUID-vkQueueSubmit2-commandBuffer-03875

If a command recorded into the `commandBuffer` member of any element of the `pCommandBufferInfos` member of any element of `pSubmits` was not recorded with the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT`, it **must** not be in the *pending state*

- VUID-vkQueueSubmit2-commandBuffer-03876

Any *secondary command buffers recorded* into the `commandBuffer` member of any element of the `pCommandBufferInfos` member of any element of `pSubmits` **must** be in the *pending or executable state*

- VUID-vkQueueSubmit2-commandBuffer-03877

If any *secondary command buffers recorded* into the `commandBuffer` member of any element of the `pCommandBufferInfos` member of any element of `pSubmits` was not recorded with the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT`, it **must** not be in the *pending state*

- VUID-vkQueueSubmit2-commandBuffer-03878

The `commandBuffer` member of any element of the `pCommandBufferInfos` member of any element of `pSubmits` **must** have been allocated from a `VkCommandPool` that was created for

the same queue family **queue** belongs to

- VUID-vkQueueSubmit2-commandBuffer-03879

If a command recorded into the **commandBuffer** member of any element of the **pCommandBufferInfos** member of any element of **pSubmits** includes a **Queue Family Ownership Transfer Acquire Operation**, there **must** exist a previously submitted **Queue Family Ownership Transfer Release Operation** on a queue in the queue family identified by the acquire operation, with parameters matching the acquire operation as defined in the definition of such **acquire operations**, and which happens before the acquire operation

- VUID-vkQueueSubmit2-commandBuffer-03880

If a command recorded into the **commandBuffer** member of any element of the **pCommandBufferInfos** member of any element of **pSubmits** was a **vkCmdBeginQuery** whose **queryPool** was created with a **queryType** of **VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR**, the **profiling lock** **must** have been held continuously on the **VkDevice** that **queue** was retrieved from, throughout recording of those command buffers

Valid Usage (Implicit)

- VUID-vkQueueSubmit2-queue-parameter

queue **must** be a valid **VkQueue** handle

- VUID-vkQueueSubmit2-pSubmits-parameter

If **submitCount** is not 0, **pSubmits** **must** be a valid pointer to an array of **submitCount** valid **VkSubmitInfo2** structures

- VUID-vkQueueSubmit2-fence-parameter

If **fence** is not **VK_NULL_HANDLE**, **fence** **must** be a valid **VkFence** handle

- VUID-vkQueueSubmit2-commonparent

Both of **fence**, and **queue** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **queue** **must** be externally synchronized
- Host access to **fence** **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

The `VkSubmitInfo2` structure is defined as:

```
typedef struct VkSubmitInfo2 {  
    VkStructureType      sType;  
    const void*          pNext;  
    VkSubmitFlags        flags;  
    uint32_t             waitSemaphoreInfoCount;  
    const VkSemaphoreSubmitInfo* pWaitSemaphoreInfos;  
    uint32_t             commandBufferInfoCount;  
    const VkCommandBufferSubmitInfo* pCommandBufferInfos;  
    uint32_t             signalSemaphoreInfoCount;  
    const VkSemaphoreSubmitInfo* pSignalSemaphoreInfos;  
} VkSubmitInfo2;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2  
typedef VkSubmitInfo2 VkSubmitInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkSubmitFlagBits`.
- `waitSemaphoreInfoCount` is the number of elements in `pWaitSemaphoreInfos`.
- `pWaitSemaphoreInfos` is a pointer to an array of `VkSemaphoreSubmitInfo` structures defining semaphore wait operations.
- `commandBufferInfoCount` is the number of elements in `pCommandBufferInfos` and the number of command buffers to execute in the batch.
- `pCommandBufferInfos` is a pointer to an array of `VkCommandBufferSubmitInfo` structures describing command buffers to execute in the batch.
- `signalSemaphoreInfoCount` is the number of elements in `pSignalSemaphoreInfos`.
- `pSignalSemaphoreInfos` is a pointer to an array of `VkSemaphoreSubmitInfo` describing semaphore signal operations.

Valid Usage

- VUID-VkSubmitInfo2-semaphore-03881

If the same semaphore is used as the `semaphore` member of both an element of `pSignalSemaphoreInfos` and `pWaitSemaphoreInfos`, and that semaphore is a timeline semaphore, the `value` member of the `pSignalSemaphoreInfos` element **must** be greater than the `value` member of the `pWaitSemaphoreInfos` element

- VUID-VkSubmitInfo2-semaphore-03882

If the `semaphore` member of any element of `pSignalSemaphoreInfos` is a timeline semaphore, the `value` member of that element **must** have a value greater than the current value of the semaphore when the `semaphore signal operation` is executed

- VUID-VkSubmitInfo2-semaphore-03883

If the `semaphore` member of any element of `pSignalSemaphoreInfos` is a timeline semaphore, the `value` member of that element **must** have a value which does not differ from the current value of the semaphore or the value of any outstanding semaphore wait or signal operation on that semaphore by more than `maxTimelineSemaphoreValueDifference`

- VUID-VkSubmitInfo2-semaphore-03884

If the `semaphore` member of any element of `pWaitSemaphoreInfos` is a timeline semaphore, the `value` member of that element **must** have a value which does not differ from the current value of the semaphore or the value of any outstanding semaphore wait or signal operation on that semaphore by more than `maxTimelineSemaphoreValueDifference`

- VUID-VkSubmitInfo2KHR-commandBuffer-06192

If any `commandBuffer` member of an element of `pCommandBufferInfos` contains any `resumed render pass instances`, they **must** be suspended by a render pass instance earlier in submission order within `pCommandBufferInfos`

- VUID-VkSubmitInfo2KHR-commandBuffer-06010

If any `commandBuffer` member of an element of `pCommandBufferInfos` contains any `suspended render pass instances`, they **must** be resumed by a render pass instance later in submission order within `pCommandBufferInfos`

- VUID-VkSubmitInfo2KHR-commandBuffer-06011

If any `commandBuffer` member of an element of `pCommandBufferInfos` contains any `suspended render pass instances`, there **must** be no action or synchronization commands between that render pass instance and the render pass instance that resumes it

- VUID-VkSubmitInfo2KHR-commandBuffer-06012

If any `commandBuffer` member of an element of `pCommandBufferInfos` contains any `suspended render pass instances`, there **must** be no render pass instances between that render pass instance and the render pass instance that resumes it

- VUID-VkSubmitInfo2KHR-variableSampleLocations-06013

If the `variableSampleLocations` limit is not supported, and any `commandBuffer` member of an element of `pCommandBufferInfos` contains any `suspended render pass instances`, where a graphics pipeline has been bound, any pipelines bound in the render pass instance that resumes it, or any subsequent render pass instances that resume from that one and so on, **must** use the same sample locations

Valid Usage (Implicit)

- VUID-VkSubmitInfo2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SUBMIT_INFO_2`
- VUID-VkSubmitInfo2-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkFrameBoundaryEXT`, `VkPerformanceQuerySubmitInfoKHR`, or `VkWin32KeyedMutexAcquireReleaseInfoKHR`
- VUID-VkSubmitInfo2-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkSubmitInfo2-flags-parameter
`flags` **must** be a valid combination of `VkSubmitFlagBits` values
- VUID-VkSubmitInfo2-pWaitSemaphoreInfos-parameter
If `waitSemaphoreInfoCount` is not 0, `pWaitSemaphoreInfos` **must** be a valid pointer to an array of `waitSemaphoreInfoCount` valid `VkSemaphoreSubmitInfo` structures
- VUID-VkSubmitInfo2-pCommandBufferInfos-parameter
If `commandBufferInfoCount` is not 0, `pCommandBufferInfos` **must** be a valid pointer to an array of `commandBufferInfoCount` valid `VkCommandBufferSubmitInfo` structures
- VUID-VkSubmitInfo2-pSignalSemaphoreInfos-parameter
If `signalSemaphoreInfoCount` is not 0, `pSignalSemaphoreInfos` **must** be a valid pointer to an array of `signalSemaphoreInfoCount` valid `VkSemaphoreSubmitInfo` structures

Bits which **can** be set in `VkSubmitInfo2::flags`, specifying submission behavior, are:

```
typedef enum VkSubmitFlagBits {  
    VK_SUBMIT_PROTECTED_BIT = 0x00000001,  
    VK_SUBMIT_PROTECTED_BIT_KHR = VK_SUBMIT_PROTECTED_BIT,  
} VkSubmitFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2  
typedef VkSubmitFlagBits VkSubmitFlagBitsKHR;
```

- `VK_SUBMIT_PROTECTED_BIT` specifies that this batch is a protected submission.

```
typedef VkFlags VkSubmitFlags;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2
typedef VkSubmitFlags VkSubmitFlagsKHR;
```

`VkSubmitFlags` is a bitmask type for setting a mask of zero or more [VkSubmitFlagBits](#).

The `VkSemaphoreSubmitInfo` structure is defined as:

```
typedef struct VkSemaphoreSubmitInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkSemaphore          semaphore;
    uint64_t             value;
    VkPipelineStageFlags2 stageMask;
    uint32_t             deviceIndex;
} VkSemaphoreSubmitInfo;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2
typedef VkSemaphoreSubmitInfo VkSemaphoreSubmitInfoKHR;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `semaphore` is a [VkSemaphore](#) affected by this operation.
- `value` is either the value used to signal `semaphore` or the value waited on by `semaphore`, if `semaphore` is a timeline semaphore. Otherwise it is ignored.
- `stageMask` is a [VkPipelineStageFlags2](#) mask of pipeline stages which limit the first synchronization scope of a semaphore signal operation, or second synchronization scope of a semaphore wait operation as described in the [semaphore wait operation](#) and [semaphore signal operation](#) sections of the [synchronization chapter](#).
- `deviceIndex` is the index of the device within a device group that executes the semaphore wait or signal operation.

Whether this structure defines a semaphore wait or signal operation is defined by how it is used.

Valid Usage

- VUID-VkSemaphoreSubmitInfo-stageMask-03929
If the `geometryShader` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT`
- VUID-VkSemaphoreSubmitInfo-stageMask-03930
If the `tessellationShader` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT` or

VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT

- VUID-VkSemaphoreSubmitInfo-stageMask-03933
If the `transformFeedback` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-VkSemaphoreSubmitInfo-stageMask-07317
If the `attachmentFragmentShadingRate` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-VkSemaphoreSubmitInfo-stageMask-07947
If the `rayTracingPipeline` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkSemaphoreSubmitInfo-device-03888
If the `device` that `semaphore` was created on is not a device group, `deviceIndex` **must** be 0
- VUID-VkSemaphoreSubmitInfo-device-03889
If the `device` that `semaphore` was created on is a device group, `deviceIndex` **must** be a valid device index

Valid Usage (Implicit)

- VUID-VkSemaphoreSubmitInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SEMAPHORE_SUBMIT_INFO`
- VUID-VkSemaphoreSubmitInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkSemaphoreSubmitInfo-semaphore-parameter
`semaphore` **must** be a valid `VkSemaphore` handle
- VUID-VkSemaphoreSubmitInfo-stageMask-parameter
`stageMask` **must** be a valid combination of `VkPipelineStageFlagBits2` values

The `VkCommandBufferSubmitInfo` structure is defined as:

```
typedef struct VkCommandBufferSubmitInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkCommandBuffer     commandBuffer;
    uint32_t           deviceMask;
} VkCommandBufferSubmitInfo;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2
typedef VkCommandBufferSubmitInfo VkCommandBufferSubmitInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `commandBuffer` is a `VkCommandBuffer` to be submitted for execution.
- `deviceMask` is a bitmask indicating which devices in a device group execute the command buffer. A `deviceMask` of `0` is equivalent to setting all bits corresponding to valid devices in the group to `1`.

Valid Usage

- VUID-VkCommandBufferSubmitInfo-commandBuffer-03890
`commandBuffer` **must** not have been allocated with `VK_COMMAND_BUFFER_LEVEL_SECONDARY`
- VUID-VkCommandBufferSubmitInfo-deviceMask-03891
If `deviceMask` is not `0`, it **must** be a valid device mask

Valid Usage (Implicit)

- VUID-VkCommandBufferSubmitInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COMMAND_BUFFER_SUBMIT_INFO`
- VUID-VkCommandBufferSubmitInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCommandBufferSubmitInfo-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle

To submit command buffers to a queue, call:

```
// Provided by VK_VERSION_1_0
VkResult vkQueueSubmit(
    VkQueue          queue,
    uint32_t         submitCount,
    const VkSubmitInfo* pSubmits,
    VkFence          fence);
```

- `queue` is the queue that the command buffers will be submitted to.
- `submitCount` is the number of elements in the `pSubmits` array.
- `pSubmits` is a pointer to an array of `VkSubmitInfo` structures, each specifying a command buffer submission batch.
- `fence` is an **optional** handle to a fence to be signaled once all submitted command buffers have completed execution. If `fence` is not `VK_NULL_HANDLE`, it defines a `fence signal operation`.

`vkQueueSubmit` is a `queue submission command`, with each batch defined by an element of `pSubmits`. Batches begin execution in the order they appear in `pSubmits`, but **may** complete out of order.

Fence and semaphore operations submitted with `vkQueueSubmit` have additional ordering constraints compared to other submission commands, with dependencies involving previous and subsequent queue operations. Information about these additional constraints can be found in the

[semaphore](#) and [fence](#) sections of [the synchronization chapter](#).

Details on the interaction of [pWaitDstStageMask](#) with synchronization are described in the [semaphore wait operation](#) section of [the synchronization chapter](#).

The order that batches appear in [pSubmits](#) is used to determine [submission order](#), and thus all the [implicit ordering guarantees](#) that respect it. Other than these implicit ordering guarantees and any [explicit synchronization primitives](#), these batches **may** overlap or otherwise execute out of order.

If any command buffer submitted to this queue is in the [executable state](#), it is moved to the [pending state](#). Once execution of all submissions of a command buffer complete, it moves from the [pending state](#), back to the [executable state](#). If a command buffer was recorded with the [VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT](#) flag, it instead moves to the [invalid state](#).

If [vkQueueSubmit](#) fails, it **may** return [VK_ERROR_OUT_OF_HOST_MEMORY](#) or [VK_ERROR_OUT_OF_DEVICE_MEMORY](#). If it does, the implementation **must** ensure that the state and contents of any resources or synchronization primitives referenced by the submitted command buffers and any semaphores referenced by [pSubmits](#) is unaffected by the call or its failure. If [vkQueueSubmit](#) fails in such a way that the implementation is unable to make that guarantee, the implementation **must** return [VK_ERROR_DEVICE_LOST](#). See [Lost Device](#).

Valid Usage

- VUID-vkQueueSubmit-fence-00063
If [fence](#) is not [VK_NULL_HANDLE](#), [fence](#) **must** be unsignaled
- VUID-vkQueueSubmit-fence-00064
If [fence](#) is not [VK_NULL_HANDLE](#), [fence](#) **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-vkQueueSubmit-pCommandBuffers-00065
Any calls to [vkCmdSetEvent](#), [vkCmdResetEvent](#) or [vkCmdWaitEvents](#) that have been recorded into any of the command buffer elements of the [pCommandBuffers](#) member of any element of [pSubmits](#), **must** not reference any [VkEvent](#) that is referenced by any of those commands in a command buffer that has been submitted to another queue and is still in the *pending state*
- VUID-vkQueueSubmit-pWaitDstStageMask-00066
Any stage flag included in any element of the [pWaitDstStageMask](#) member of any element of [pSubmits](#) **must** be a pipeline stage supported by one of the capabilities of [queue](#), as specified in the [table of supported pipeline stages](#)
- VUID-vkQueueSubmit-pSignalSemaphores-00067
Each binary semaphore element of the [pSignalSemaphores](#) member of any element of [pSubmits](#) **must** be unsignaled when the semaphore signal operation it defines is executed on the device
- VUID-vkQueueSubmit-pWaitSemaphores-00068
When a semaphore wait operation referring to a binary semaphore defined by any element of the [pWaitSemaphores](#) member of any element of [pSubmits](#) executes on [queue](#), there **must** be no other queues waiting on the same semaphore

- VUID-vkQueueSubmit-pWaitSemaphores-03238
All elements of the `pWaitSemaphores` member of all elements of `pSubmits` created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_BINARY` **must** reference a semaphore signal operation that has been submitted for execution and any semaphore signal operations on which it depends **must** have also been submitted for execution
- VUID-vkQueueSubmit-pCommandBuffers-00070
Each element of the `pCommandBuffers` member of each element of `pSubmits` **must** be in the `pending or executable state`
- VUID-vkQueueSubmit-pCommandBuffers-00071
If any element of the `pCommandBuffers` member of any element of `pSubmits` was not recorded with the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT`, it **must** not be in the `pending state`
- VUID-vkQueueSubmit-pCommandBuffers-00072
Any `secondary command buffers recorded` into any element of the `pCommandBuffers` member of any element of `pSubmits` **must** be in the `pending or executable state`
- VUID-vkQueueSubmit-pCommandBuffers-00073
If any `secondary command buffers recorded` into any element of the `pCommandBuffers` member of any element of `pSubmits` was not recorded with the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT`, it **must** not be in the `pending state`
- VUID-vkQueueSubmit-pCommandBuffers-00074
Each element of the `pCommandBuffers` member of each element of `pSubmits` **must** have been allocated from a `VkCommandPool` that was created for the same queue family `queue` belongs to
- VUID-vkQueueSubmit-pSubmits-02207
If any element of `pSubmits->pCommandBuffers` includes a `Queue Family Ownership Transfer Acquire Operation`, there **must** exist a previously submitted `Queue Family Ownership Transfer Release Operation` on a queue in the queue family identified by the acquire operation, with parameters matching the acquire operation as defined in the definition of such `acquire operations`, and which happens-before the acquire operation
- VUID-vkQueueSubmit-pCommandBuffers-03220
If a command recorded into any element of `pCommandBuffers` was a `vkCmdBeginQuery` whose `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, the `profiling lock` **must** have been held continuously on the `VkDevice` that `queue` was retrieved from, throughout recording of those command buffers
- VUID-vkQueueSubmit-pSubmits-02808
Any resource created with `VK_SHARING_MODE_EXCLUSIVE` that is read by an operation specified by `pSubmits` **must** not be owned by any queue family other than the one which `queue` belongs to, at the time it is executed
- VUID-vkQueueSubmit-pSubmits-04626
Any resource created with `VK_SHARING_MODE_CONCURRENT` that is accessed by an operation specified by `pSubmits` **must** have included the queue family of `queue` at resource creation time

Valid Usage (Implicit)

- VUID-vkQueueSubmit-queue-parameter
`queue` **must** be a valid `VkQueue` handle
- VUID-vkQueueSubmit-pSubmits-parameter
If `submitCount` is not 0, `pSubmits` **must** be a valid pointer to an array of `submitCount` valid `VkSubmitInfo` structures
- VUID-vkQueueSubmit-fence-parameter
If `fence` is not `VK_NULL_HANDLE`, `fence` **must** be a valid `VkFence` handle
- VUID-vkQueueSubmit-commonparent
Both of `fence`, and `queue` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `queue` **must** be externally synchronized
- Host access to `fence` **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

The `VkSubmitInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSubmitInfo {
    VkStructureType           sType;
    const void*               pNext;
    uint32_t                  waitSemaphoreCount;
    const VkSemaphore*        pWaitSemaphores;
    const VkPipelineStageFlags* pWaitDstStageMask;
    uint32_t                  commandBufferCount;
    const VkCommandBuffer*    pCommandBuffers;
    uint32_t                  signalSemaphoreCount;
    const VkSemaphore*        pSignalSemaphores;
} VkSubmitInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `waitSemaphoreCount` is the number of semaphores upon which to wait before executing the command buffers for the batch.
- `pWaitSemaphores` is a pointer to an array of `VkSemaphore` handles upon which to wait before the command buffers for this batch begin execution. If semaphores to wait on are provided, they define a [semaphore wait operation](#).
- `pWaitDstStageMask` is a pointer to an array of pipeline stages at which each corresponding semaphore wait will occur.
- `commandBufferCount` is the number of command buffers to execute in the batch.
- `pCommandBuffers` is a pointer to an array of `VkCommandBuffer` handles to execute in the batch.
- `signalSemaphoreCount` is the number of semaphores to be signaled once the commands specified in `pCommandBuffers` have completed execution.
- `pSignalSemaphores` is a pointer to an array of `VkSemaphore` handles which will be signaled when the command buffers for this batch have completed execution. If semaphores to be signaled are provided, they define a [semaphore signal operation](#).

The order that command buffers appear in `pCommandBuffers` is used to determine [submission order](#), and thus all the [implicit ordering guarantees](#) that respect it. Other than these implicit ordering guarantees and any [explicit synchronization primitives](#), these command buffers **may** overlap or otherwise execute out of order.

Valid Usage

- VUID-VkSubmitInfo-pWaitDstStageMask-04090
If the `geometryShader` feature is not enabled, `pWaitDstStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-VkSubmitInfo-pWaitDstStageMask-04091
If the `tessellationShader` feature is not enabled, `pWaitDstStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`

- VUID-VkSubmitInfo-pWaitDstStageMask-04094
If the `transformFeedback` feature is not enabled, `pWaitDstStageMask` **must** not contain `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-VkSubmitInfo-pWaitDstStageMask-07319
If the `attachmentFragmentShadingRate` feature is not enabled, `pWaitDstStageMask` **must** not contain `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-VkSubmitInfo-pWaitDstStageMask-03937
If the `synchronization2` feature is not enabled, `pWaitDstStageMask` **must** not be 0
- VUID-VkSubmitInfo-pWaitDstStageMask-07950
If the `rayTracingPipeline` feature is not enabled, `pWaitDstStageMask` **must** not contain `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkSubmitInfo-pCommandBuffers-00075
Each element of `pCommandBuffers` **must** not have been allocated with `VK_COMMAND_BUFFER_LEVEL_SECONDARY`
- VUID-VkSubmitInfo-pWaitDstStageMask-00078
Each element of `pWaitDstStageMask` **must** not include `VK_PIPELINE_STAGE_HOST_BIT`
- VUID-VkSubmitInfo-pWaitSemaphores-03239
If any element of `pWaitSemaphores` or `pSignalSemaphores` was created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE`, then the `pNext` chain **must** include a `VkTimelineSemaphoreSubmitInfo` structure
- VUID-VkSubmitInfo-pNext-03240
If the `pNext` chain of this structure includes a `VkTimelineSemaphoreSubmitInfo` structure and any element of `pWaitSemaphores` was created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE`, then its `waitSemaphoreValueCount` member **must** equal `waitSemaphoreCount`
- VUID-VkSubmitInfo-pNext-03241
If the `pNext` chain of this structure includes a `VkTimelineSemaphoreSubmitInfo` structure and any element of `pSignalSemaphores` was created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE`, then its `signalSemaphoreValueCount` member **must** equal `signalSemaphoreCount`
- VUID-VkSubmitInfo-pSignalSemaphores-03242
For each element of `pSignalSemaphores` created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE` the corresponding element of `VkTimelineSemaphoreSubmitInfo::pSignalSemaphoreValues` **must** have a value greater than the current value of the semaphore when the `semaphore signal operation` is executed
- VUID-VkSubmitInfo-pWaitSemaphores-03243
For each element of `pWaitSemaphores` created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE` the corresponding element of `VkTimelineSemaphoreSubmitInfo::pWaitSemaphoreValues` **must** have a value which does not differ from the current value of the semaphore or the value of any outstanding semaphore wait or signal operation on that semaphore by more than `maxTimelineSemaphoreValueDifference`

- VUID-VkSubmitInfo-pSignalSemaphores-03244
For each element of `pSignalSemaphores` created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE` the corresponding element of `VkTimelineSemaphoreSubmitInfo::pSignalSemaphoreValues` **must** have a value which does not differ from the current value of the semaphore or the value of any outstanding semaphore wait or signal operation on that semaphore by more than `maxTimelineSemaphoreValueDifference`
- VUID-VkSubmitInfo-pCommandBuffers-06193
If `pCommandBuffers` contains any `resumed render pass instances`, they **must** be suspended by a render pass instance earlier in submission order within `pCommandBuffers`
- VUID-VkSubmitInfo-pCommandBuffers-06014
If `pCommandBuffers` contains any `suspended render pass instances`, they **must** be resumed by a render pass instance later in submission order within `pCommandBuffers`
- VUID-VkSubmitInfo-pCommandBuffers-06015
If `pCommandBuffers` contains any `suspended render pass instances`, there **must** be no action or synchronization commands executed in a primary or `secondary` command buffer between that render pass instance and the render pass instance that resumes it
- VUID-VkSubmitInfo-pCommandBuffers-06016
If `pCommandBuffers` contains any `suspended render pass instances`, there **must** be no render pass instances between that render pass instance and the render pass instance that resumes it
- VUID-VkSubmitInfo-variableSampleLocations-06017
If the `variableSampleLocations` limit is not supported, and any element of `pCommandBuffers` contains any `suspended render pass instances`, where a graphics pipeline has been bound, any pipelines bound in the render pass instance that resumes it, or any subsequent render pass instances that resume from that one and so on, **must** use the same sample locations

Valid Usage (Implicit)

- VUID-VkSubmitInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SUBMIT_INFO`
- VUID-VkSubmitInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkD3D12FenceSubmitInfoKHR`, `VkDeviceGroupSubmitInfo`, `VkFrameBoundaryEXT`, `VkPerformanceQuerySubmitInfoKHR`, `VkTimelineSemaphoreSubmitInfo`, or `VkWin32KeyedMutexAcquireReleaseInfoKHR`
- VUID-VkSubmitInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkSubmitInfo-pWaitSemaphores-parameter
If `waitSemaphoreCount` is not 0, `pWaitSemaphores` **must** be a valid pointer to an array of `waitSemaphoreCount` valid `VkSemaphore` handles

- VUID-VkSubmitInfo-pWaitDstStageMask-parameter
If `waitSemaphoreCount` is not 0, `pWaitDstStageMask` **must** be a valid pointer to an array of `waitSemaphoreCount` valid combinations of `VkPipelineStageFlagBits` values
- VUID-VkSubmitInfo-pCommandBuffers-parameter
If `commandBufferCount` is not 0, `pCommandBuffers` **must** be a valid pointer to an array of `commandBufferCount` valid `VkCommandBuffer` handles
- VUID-VkSubmitInfo-pSignalSemaphores-parameter
If `signalSemaphoreCount` is not 0, `pSignalSemaphores` **must** be a valid pointer to an array of `signalSemaphoreCount` valid `VkSemaphore` handles
- VUID-VkSubmitInfo-commonparent
Each of the elements of `pCommandBuffers`, the elements of `pSignalSemaphores`, and the elements of `pWaitSemaphores` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

To specify the values to use when waiting for and signaling semaphores created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE`, add a `VkTimelineSemaphoreSubmitInfo` structure to the `pNext` chain of the `VkSubmitInfo` structure when using `vkQueueSubmit` or the `VkBindSparseInfo` structure when using `vkQueueBindSparse`. The `VkTimelineSemaphoreSubmitInfo` structure is defined as:

```
typedef struct VkTimelineSemaphoreSubmitInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           waitSemaphoreValueCount;
    const uint64_t*     pWaitSemaphoreValues;
    uint32_t           signalSemaphoreValueCount;
    const uint64_t*     pSignalSemaphoreValues;
} VkTimelineSemaphoreSubmitInfo;
```

or the equivalent

```
// Provided by VK_KHR_timeline_semaphore
typedef VkTimelineSemaphoreSubmitInfo VkTimelineSemaphoreSubmitInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `waitSemaphoreValueCount` is the number of semaphore wait values specified in `pWaitSemaphoreValues`.
- `pWaitSemaphoreValues` is a pointer to an array of `waitSemaphoreValueCount` values for the corresponding semaphores in `VkSubmitInfo::pWaitSemaphores` to wait for.
- `signalSemaphoreValueCount` is the number of semaphore signal values specified in `pSignalSemaphoreValues`.
- `pSignalSemaphoreValues` is a pointer to an array `signalSemaphoreValueCount` values for the

corresponding semaphores in `VkSubmitInfo::pSignalSemaphores` to set when signaled.

If the semaphore in `VkSubmitInfo::pWaitSemaphores` or `VkSubmitInfo::pSignalSemaphores` corresponding to an entry in `pWaitSemaphoreValues` or `pSignalSemaphoreValues` respectively was not created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE`, the implementation **must** ignore the value in the `pWaitSemaphoreValues` or `pSignalSemaphoreValues` entry.

Valid Usage (Implicit)

- VUID-VkTimelineSemaphoreSubmitInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_TIMELINE_SEMAPHORE_SUBMIT_INFO`
- VUID-VkTimelineSemaphoreSubmitInfo-pWaitSemaphoreValues-parameter
If `waitSemaphoreValueCount` is not 0, and `pWaitSemaphoreValues` is not NULL, `pWaitSemaphoreValues` **must** be a valid pointer to an array of `waitSemaphoreValueCount` `uint64_t` values
- VUID-VkTimelineSemaphoreSubmitInfo-pSignalSemaphoreValues-parameter
If `signalSemaphoreValueCount` is not 0, and `pSignalSemaphoreValues` is not NULL, `pSignalSemaphoreValues` **must** be a valid pointer to an array of `signalSemaphoreValueCount` `uint64_t` values

To specify the values to use when waiting for and signaling semaphores whose **current payload** refers to a Direct3D 12 fence, add a `VkD3D12FenceSubmitInfoKHR` structure to the `pNext` chain of the `VkSubmitInfo` structure. The `VkD3D12FenceSubmitInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_semaphore_win32
typedef struct VkD3D12FenceSubmitInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           waitSemaphoreValuesCount;
    const uint64_t*    pWaitSemaphoreValues;
    uint32_t           signalSemaphoreValuesCount;
    const uint64_t*    pSignalSemaphoreValues;
} VkD3D12FenceSubmitInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is NULL or a pointer to a structure extending this structure.
- `waitSemaphoreValuesCount` is the number of semaphore wait values specified in `pWaitSemaphoreValues`.
- `pWaitSemaphoreValues` is a pointer to an array of `waitSemaphoreValuesCount` values for the corresponding semaphores in `VkSubmitInfo::pWaitSemaphores` to wait for.
- `signalSemaphoreValuesCount` is the number of semaphore signal values specified in `pSignalSemaphoreValues`.
- `pSignalSemaphoreValues` is a pointer to an array of `signalSemaphoreValuesCount` values for the corresponding semaphores in `VkSubmitInfo::pSignalSemaphores` to set when signaled.

If the semaphore in `VkSubmitInfo::pWaitSemaphores` or `VkSubmitInfo::pSignalSemaphores` corresponding to an entry in `pWaitSemaphoreValues` or `pSignalSemaphoreValues` respectively does not currently have a `payload` referring to a Direct3D 12 fence, the implementation **must** ignore the value in the `pWaitSemaphoreValues` or `pSignalSemaphoreValues` entry.

NOTE

As the introduction of the external semaphore handle type `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT` predates that of timeline semaphores, support for importing semaphore payloads from external handles of that type into semaphores created (implicitly or explicitly) with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_BINARY` is preserved for backwards compatibility. However, applications **should** prefer importing such handle types into semaphores created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE`, and use the `VkTimelineSemaphoreSubmitInfo` structure instead of the `VkD3D12FenceSubmitInfoKHR` structure to specify the values to use when waiting for and signaling such semaphores.

Valid Usage

- VUID-VkD3D12FenceSubmitInfoKHR-waitSemaphoreValuesCount-00079
`waitSemaphoreValuesCount` **must** be the same value as `VkSubmitInfo::waitSemaphoreCount`, where this structure is in the `pNext` chain of a `VkSubmitInfo` structure
- VUID-VkD3D12FenceSubmitInfoKHR-signalSemaphoreValuesCount-00080
`signalSemaphoreValuesCount` **must** be the same value as `VkSubmitInfo::signalSemaphoreCount`, where this structure is in the `pNext` chain of a `VkSubmitInfo` structure

Valid Usage (Implicit)

- VUID-VkD3D12FenceSubmitInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_D3D12_FENCE_SUBMIT_INFO_KHR`
- VUID-VkD3D12FenceSubmitInfoKHR-pWaitSemaphoreValues-parameter
If `waitSemaphoreValuesCount` is not 0, and `pWaitSemaphoreValues` is not NULL, `pWaitSemaphoreValues` **must** be a valid pointer to an array of `waitSemaphoreValuesCount` `uint64_t` values
- VUID-VkD3D12FenceSubmitInfoKHR-pSignalSemaphoreValues-parameter
If `signalSemaphoreValuesCount` is not 0, and `pSignalSemaphoreValues` is not NULL, `pSignalSemaphoreValues` **must** be a valid pointer to an array of `signalSemaphoreValuesCount` `uint64_t` values

When submitting work that operates on memory imported from a Direct3D 11 resource to a queue, the keyed mutex mechanism **may** be used in addition to Vulkan semaphores to synchronize the work. Keyed mutexes are a property of a properly created shareable Direct3D 11 resource. They **can** only be used if the imported resource was created with the `D3D11_RESOURCE_MISC_SHARED_KEYEDMUTEX` flag.

To acquire keyed mutexes before submitted work and/or release them after, add a [VkWin32KeyedMutexAcquireReleaseInfoKHR](#) structure to the `pNext` chain of the [VkSubmitInfo](#) structure.

The [VkWin32KeyedMutexAcquireReleaseInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_win32_keyed_mutex
typedef struct VkWin32KeyedMutexAcquireReleaseInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           acquireCount;
    const VkDeviceMemory* pAcquireSyncs;
    const uint64_t*     pAcquireKeys;
    const uint32_t*     pAcquireTimeouts;
    uint32_t           releaseCount;
    const VkDeviceMemory* pReleaseSyncs;
    const uint64_t*     pReleaseKeys;
} VkWin32KeyedMutexAcquireReleaseInfoKHR;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `acquireCount` is the number of entries in the `pAcquireSyncs`, `pAcquireKeys`, and `pAcquireTimeouts` arrays.
- `pAcquireSyncs` is a pointer to an array of [VkDeviceMemory](#) objects which were imported from Direct3D 11 resources.
- `pAcquireKeys` is a pointer to an array of mutex key values to wait for prior to beginning the submitted work. Entries refer to the keyed mutex associated with the corresponding entries in `pAcquireSyncs`.
- `pAcquireTimeouts` is a pointer to an array of timeout values, in millisecond units, for each acquire specified in `pAcquireKeys`.
- `releaseCount` is the number of entries in the `pReleaseSyncs` and `pReleaseKeys` arrays.
- `pReleaseSyncs` is a pointer to an array of [VkDeviceMemory](#) objects which were imported from Direct3D 11 resources.
- `pReleaseKeys` is a pointer to an array of mutex key values to set when the submitted work has completed. Entries refer to the keyed mutex associated with the corresponding entries in `pReleaseSyncs`.

Valid Usage

- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pAcquireSyncs-00081

Each member of `pAcquireSyncs` and `pReleaseSyncs` **must** be a device memory object imported by setting [VkImportMemoryWin32HandleInfoKHR::handleType](#) to `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT` or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT`

Valid Usage (Implicit)

- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_WIN32_KEYED_MUTEX_ACQUIRE_RELEASE_INFO_KHR`
- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pAcquireSyncs-parameter
If `acquireCount` is not 0, `pAcquireSyncs` **must** be a valid pointer to an array of `acquireCount` valid `VkDeviceMemory` handles
- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pAcquireKeys-parameter
If `acquireCount` is not 0, `pAcquireKeys` **must** be a valid pointer to an array of `acquireCount` `uint64_t` values
- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pAcquireTimeouts-parameter
If `acquireCount` is not 0, `pAcquireTimeouts` **must** be a valid pointer to an array of `acquireCount` `uint32_t` values
- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pReleaseSyncs-parameter
If `releaseCount` is not 0, `pReleaseSyncs` **must** be a valid pointer to an array of `releaseCount` valid `VkDeviceMemory` handles
- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-pReleaseKeys-parameter
If `releaseCount` is not 0, `pReleaseKeys` **must** be a valid pointer to an array of `releaseCount` `uint64_t` values
- VUID-VkWin32KeyedMutexAcquireReleaseInfoKHR-commonparent
Both of the elements of `pAcquireSyncs`, and the elements of `pReleaseSyncs` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

If the `pNext` chain of `VkSubmitInfo` includes a `VkDeviceGroupSubmitInfo` structure, then that structure includes device indices and masks specifying which physical devices execute semaphore operations and command buffers.

The `VkDeviceGroupSubmitInfo` structure is defined as:

```
typedef struct VkDeviceGroupSubmitInfo {  
    VkStructureType    sType;  
    const void*        pNext;  
    uint32_t           waitSemaphoreCount;  
    const uint32_t*     pWaitSemaphoreDeviceIndices;  
    uint32_t           commandBufferCount;  
    const uint32_t*     pCommandBufferDeviceMasks;  
    uint32_t           signalSemaphoreCount;  
    const uint32_t*     pSignalSemaphoreDeviceIndices;  
} VkDeviceGroupSubmitInfo;
```

or the equivalent

```
// Provided by VK_KHR_device_group
typedef VkDeviceGroupSubmitInfo VkDeviceGroupSubmitInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `waitSemaphoreCount` is the number of elements in the `pWaitSemaphoreDeviceIndices` array.
- `pWaitSemaphoreDeviceIndices` is a pointer to an array of `waitSemaphoreCount` device indices indicating which physical device executes the semaphore wait operation in the corresponding element of `VkSubmitInfo::pWaitSemaphores`.
- `commandBufferCount` is the number of elements in the `pCommandBufferDeviceMasks` array.
- `pCommandBufferDeviceMasks` is a pointer to an array of `commandBufferCount` device masks indicating which physical devices execute the command buffer in the corresponding element of `VkSubmitInfo::pCommandBuffers`. A physical device executes the command buffer if the corresponding bit is set in the mask.
- `signalSemaphoreCount` is the number of elements in the `pSignalSemaphoreDeviceIndices` array.
- `pSignalSemaphoreDeviceIndices` is a pointer to an array of `signalSemaphoreCount` device indices indicating which physical device executes the semaphore signal operation in the corresponding element of `VkSubmitInfo::pSignalSemaphores`.

If this structure is not present, semaphore operations and command buffers execute on device index zero.

Valid Usage

- VUID-VkDeviceGroupSubmitInfo-waitSemaphoreCount-00082
`waitSemaphoreCount` **must** equal `VkSubmitInfo::waitSemaphoreCount`
- VUID-VkDeviceGroupSubmitInfo-commandBufferCount-00083
`commandBufferCount` **must** equal `VkSubmitInfo::commandBufferCount`
- VUID-VkDeviceGroupSubmitInfo-signalSemaphoreCount-00084
`signalSemaphoreCount` **must** equal `VkSubmitInfo::signalSemaphoreCount`
- VUID-VkDeviceGroupSubmitInfo-pWaitSemaphoreDeviceIndices-00085
All elements of `pWaitSemaphoreDeviceIndices` and `pSignalSemaphoreDeviceIndices` **must** be valid device indices
- VUID-VkDeviceGroupSubmitInfo-pCommandBufferDeviceMasks-00086
All elements of `pCommandBufferDeviceMasks` **must** be valid device masks

Valid Usage (Implicit)

- VUID-VkDeviceGroupSubmitInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_GROUP_SUBMIT_INFO`
- VUID-VkDeviceGroupSubmitInfo-pWaitSemaphoreDeviceIndices-parameter

If `waitSemaphoreCount` is not 0, `pWaitSemaphoreDeviceIndices` **must** be a valid pointer to an array of `waitSemaphoreCount` `uint32_t` values

- VUID-VkDeviceGroupSubmitInfo-pCommandBufferDeviceMasks-parameter
If `commandBufferCount` is not 0, `pCommandBufferDeviceMasks` **must** be a valid pointer to an array of `commandBufferCount` `uint32_t` values
- VUID-VkDeviceGroupSubmitInfo-pSignalSemaphoreDeviceIndices-parameter
If `signalSemaphoreCount` is not 0, `pSignalSemaphoreDeviceIndices` **must** be a valid pointer to an array of `signalSemaphoreCount` `uint32_t` values

If the `pNext` chain of `VkSubmitInfo` includes a `VkPerformanceQuerySubmitInfoKHR` structure, then the structure indicates which counter pass is active for the batch in that submit.

The `VkPerformanceQuerySubmitInfoKHR` structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkPerformanceQuerySubmitInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           counterPassIndex;
} VkPerformanceQuerySubmitInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `counterPassIndex` specifies which counter pass index is active.

If the `VkSubmitInfo::pNext` chain does not include this structure, the batch defaults to use counter pass index 0.

Valid Usage

- VUID-VkPerformanceQuerySubmitInfoKHR-counterPassIndex-03221
`counterPassIndex` **must** be less than the number of counter passes required by any queries within the batch. The required number of counter passes for a performance query is obtained by calling `vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR`

Valid Usage (Implicit)

- VUID-VkPerformanceQuerySubmitInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PERFORMANCE_QUERY_SUBMIT_INFO_KHR`

6.6. Queue Forward Progress

When using binary semaphores, the application **must** ensure that command buffer submissions

will be able to complete without any subsequent operations by the application on any queue. After any call to `vkQueueSubmit` (or other queue operation), for every queued wait on a semaphore created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_BINARY` there **must** be a prior signal of that semaphore that will not be consumed by a different wait on the semaphore.

When using timeline semaphores, wait-before-signal behavior is well-defined and applications **can** submit work via `vkQueueSubmit` defining a [timeline semaphore wait operation](#) before submitting a corresponding [semaphore signal operation](#). For each [timeline semaphore wait operation](#) defined by a call to `vkQueueSubmit`, the application **must** ensure that a corresponding [semaphore signal operation](#) is executed before forward progress can be made.

If a command buffer submission waits for any events to be signaled, the application **must** ensure that command buffer submissions will be able to complete without any subsequent operations by the application. Events signaled by the host **must** be signaled before the command buffer waits on those events.

NOTE

The ability for commands to wait on the host to set an events was originally added to allow low-latency updates to resources between host and device. However, to ensure quality of service, implementations would necessarily detect extended stalls in execution and timeout after a short period. As this period is not defined in the Vulkan specification, it is impossible to correctly validate any application with any wait period. Since the original users of this functionality were highly limited and platform-specific, this functionality is now considered defunct and should not be used.

6.7. Secondary Command Buffer Execution

Secondary command buffers **must** not be directly submitted to a queue. To record a secondary command buffer to execute as part of a primary command buffer, call:

```
// Provided by VK_VERSION_1_0
void vkCmdExecuteCommands(
    VkCommandBuffer          commandBuffer,
    uint32_t                 commandBufferCount,
    const VkCommandBuffer*   pCommandBuffers);
```

- `commandBuffer` is a handle to a primary command buffer that the secondary command buffers are executed in.
- `commandBufferCount` is the length of the `pCommandBuffers` array.
- `pCommandBuffers` is a pointer to an array of `commandBufferCount` secondary command buffer handles, which are recorded to execute in the primary command buffer in the order they are listed in the array.

If any element of `pCommandBuffers` was not recorded with the `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT` flag, and it was recorded into any other primary command buffer which is currently in the [executable or recording state](#), that primary command

buffer becomes [invalid](#).

If the [nestedCommandBuffer](#) feature is enabled it is valid usage for [vkCmdExecuteCommands](#) to also be recorded to a [secondary command buffer](#).

Valid Usage

- VUID-vkCmdExecuteCommands-pCommandBuffers-00088
Each element of [pCommandBuffers](#) **must** have been allocated with a [level](#) of [VK_COMMAND_BUFFER_LEVEL_SECONDARY](#)
- VUID-vkCmdExecuteCommands-pCommandBuffers-00089
Each element of [pCommandBuffers](#) **must** be in the [pending or executable state](#)
- VUID-vkCmdExecuteCommands-pCommandBuffers-00091
If any element of [pCommandBuffers](#) was not recorded with the [VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT](#) flag, it **must** not be in the [pending state](#)
- VUID-vkCmdExecuteCommands-pCommandBuffers-00092
If any element of [pCommandBuffers](#) was not recorded with the [VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT](#) flag, it **must** not have already been recorded to [commandBuffer](#)
- VUID-vkCmdExecuteCommands-pCommandBuffers-00093
If any element of [pCommandBuffers](#) was not recorded with the [VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT](#) flag, it **must** not appear more than once in [pCommandBuffers](#)
- VUID-vkCmdExecuteCommands-pCommandBuffers-00094
Each element of [pCommandBuffers](#) **must** have been allocated from a [VkCommandPool](#) that was created for the same queue family as the [VkCommandPool](#) from which [commandBuffer](#) was allocated
- VUID-vkCmdExecuteCommands-pCommandBuffers-00096
If [vkCmdExecuteCommands](#) is being called within a render pass instance, each element of [pCommandBuffers](#) **must** have been recorded with the [VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT](#)
- VUID-vkCmdExecuteCommands-pCommandBuffers-00099
If [vkCmdExecuteCommands](#) is being called within a render pass instance, and any element of [pCommandBuffers](#) was recorded with [VkCommandBufferInheritanceInfo::framebuffer](#) not equal to [VK_NULL_HANDLE](#), that [VkFramebuffer](#) **must** match the [VkFramebuffer](#) used in the current render pass instance
- VUID-vkCmdExecuteCommands-contents-09680
If [vkCmdExecuteCommands](#) is being called within a render pass instance begun with [vkCmdBeginRenderPass](#), and [vkCmdNextSubpass](#) has not been called in the current render pass instance, the [contents](#) parameter of [vkCmdBeginRenderPass](#) **must** have been [VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS](#), or [VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_EXT](#)
- VUID-vkCmdExecuteCommands-None-09681
If [vkCmdExecuteCommands](#) is being called within a render pass instance begun with

`vkCmdBeginRenderPass`, and `vkCmdNextSubpass` has been called in the current render pass instance, the `contents` parameter of the last call to `vkCmdNextSubpass` **must** have been `VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS`, or `VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR`

- VUID-vkCmdExecuteCommands-pCommandBuffers-06019
If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderPass`, each element of `pCommandBuffers` **must** have been recorded with `VkCommandBufferInheritanceInfo::subpass` set to the index of the subpass which the given command buffer will be executed in
- VUID-vkCmdExecuteCommands-pBeginInfo-06020
If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderPass`, the render passes specified in the `pBeginInfo->pInheritanceInfo->renderPass` members of the `VkBeginCommandBuffer` commands used to begin recording each element of `pCommandBuffers` **must** be `compatible` with the current render pass
- VUID-vkCmdExecuteCommands-pCommandBuffers-00100
If `vkCmdExecuteCommands` is not being called within a render pass instance, each element of `pCommandBuffers` **must** not have been recorded with the `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT`
- VUID-vkCmdExecuteCommands-commandBuffer-00101
If the `inheritedQueries` feature is not enabled, `commandBuffer` **must** not have any queries `active`
- VUID-vkCmdExecuteCommands-commandBuffer-00102
If `commandBuffer` has a `VK_QUERY_TYPE_OCCLUSION` query `active`, then each element of `pCommandBuffers` **must** have been recorded with `VkCommandBufferInheritanceInfo::occlusionQueryEnable` set to `VK_TRUE`
- VUID-vkCmdExecuteCommands-commandBuffer-00103
If `commandBuffer` has a `VK_QUERY_TYPE_OCCLUSION` query `active`, then each element of `pCommandBuffers` **must** have been recorded with `VkCommandBufferInheritanceInfo::queryFlags` having all bits set that are set for the query
- VUID-vkCmdExecuteCommands-commandBuffer-00104
If `commandBuffer` has a `VK_QUERY_TYPE_PIPELINE_STATISTICS` query `active`, then each element of `pCommandBuffers` **must** have been recorded with `VkCommandBufferInheritanceInfo::pipelineStatistics` having all bits set that are set in the `VkQueryPool` the query uses
- VUID-vkCmdExecuteCommands-pCommandBuffers-00105
Each element of `pCommandBuffers` **must** not begin any query types that are `active` in `commandBuffer`
- VUID-vkCmdExecuteCommands-commandBuffer-07594
`commandBuffer` **must** not have any queries other than `VK_QUERY_TYPE_OCCLUSION` and `VK_QUERY_TYPE_PIPELINE_STATISTICS` `active`
- VUID-vkCmdExecuteCommands-None-02286
This command **must** not be recorded when transform feedback is active
- VUID-vkCmdExecuteCommands-commandBuffer-06533
If `vkCmdExecuteCommands` is being called within a render pass instance and any recorded

command in `commandBuffer` in the current subpass will write to an image subresource as an attachment, commands recorded in elements of `pCommandBuffers` **must** not read from the memory backing that image subresource in any other way

- VUID-vkCmdExecuteCommands-commandBuffer-06534

If `vkCmdExecuteCommands` is being called within a render pass instance and any recorded command in `commandBuffer` in the current subpass will read from an image subresource used as an attachment in any way other than as an attachment, commands recorded in elements of `pCommandBuffers` **must** not write to that image subresource as an attachment

- VUID-vkCmdExecuteCommands-pCommandBuffers-06535

If `vkCmdExecuteCommands` is being called within a render pass instance and any recorded command in a given element of `pCommandBuffers` will write to an image subresource as an attachment, commands recorded in elements of `pCommandBuffers` at a higher index **must** not read from the memory backing that image subresource in any other way

- VUID-vkCmdExecuteCommands-pCommandBuffers-06536

If `vkCmdExecuteCommands` is being called within a render pass instance and any recorded command in a given element of `pCommandBuffers` will read from an image subresource used as an attachment in any way other than as an attachment, commands recorded in elements of `pCommandBuffers` at a higher index **must** not write to that image subresource as an attachment

- VUID-vkCmdExecuteCommands-pCommandBuffers-06021

If `pCommandBuffers` contains any `suspended render pass instances`, there **must** be no action or synchronization commands between that render pass instance and any render pass instance that resumes it

- VUID-vkCmdExecuteCommands-pCommandBuffers-06022

If `pCommandBuffers` contains any `suspended render pass instances`, there **must** be no render pass instances between that render pass instance and any render pass instance that resumes it

- VUID-vkCmdExecuteCommands-variableSampleLocations-06023

If the `variableSampleLocations` limit is not supported, and any element of `pCommandBuffers` contains any `suspended render pass instances`, where a graphics pipeline has been bound, any pipelines bound in the render pass instance that resumes it, or any subsequent render pass instances that resume from that one and so on, **must** use the same sample locations

- VUID-vkCmdExecuteCommands-flags-06024

If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR`, its `VkRenderingInfo::flags` parameter **must** have included `VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT`

- VUID-vkCmdExecuteCommands-pBeginInfo-06025

If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR`, the render passes specified in the `pBeginInfo->pInheritanceInfo->renderPass` members of the `vkBeginCommandBuffer` commands used to begin recording each element of `pCommandBuffers` **must** be `VK_NULL_HANDLE`

- VUID-vkCmdExecuteCommands-flags-06026

If `vkCmdExecuteCommands` is being called within a render pass instance begun with

`vkCmdBeginRenderingKHR`, the `flags` member of the `VkCommandBufferInheritanceRenderingInfo` structure included in the `pNext` chain of `VkCommandBufferBeginInfo::pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be equal to the `VkRenderingInfo::flags` parameter to `vkCmdBeginRenderingKHR`, excluding `VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT`

- VUID-vkCmdExecuteCommands-colorAttachmentCount-06027

If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR`, the `colorAttachmentCount` member of the `VkCommandBufferInheritanceRenderingInfo` structure included in the `pNext` chain of `VkCommandBufferBeginInfo::pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be equal to the `VkRenderingInfo::colorAttachmentCount` parameter to `vkCmdBeginRenderingKHR`

- VUID-vkCmdExecuteCommands-imageView-06028

If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR`, if the `imageView` member of an element of the `VkRenderingInfo::pColorAttachments` parameter to `vkCmdBeginRenderingKHR` is not `VK_NULL_HANDLE`, the corresponding element of the `pColorAttachmentFormats` member of the `VkCommandBufferInheritanceRenderingInfo` structure included in the `pNext` chain of `VkCommandBufferBeginInfo::pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be equal to the format used to create that image view

- VUID-vkCmdExecuteCommands-imageView-07606

If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR`, if the `imageView` member of an element of the `VkRenderingInfo::pColorAttachments` parameter to `vkCmdBeginRenderingKHR` is `VK_NULL_HANDLE`, the corresponding element of the `pColorAttachmentFormats` member of the `VkCommandBufferInheritanceRenderingInfo` structure included in the `pNext` chain of `VkCommandBufferBeginInfo::pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdExecuteCommands-pDepthAttachment-06029

If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR`, if the `VkRenderingInfo::pDepthAttachment->imageView` parameter to `vkCmdBeginRenderingKHR` is not `VK_NULL_HANDLE`, the value of the `depthAttachmentFormat` member of the `VkCommandBufferInheritanceRenderingInfo` structure included in the `pNext` chain of `VkCommandBufferBeginInfo::pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be equal to the format used to create that image view

- VUID-vkCmdExecuteCommands-pStencilAttachment-06030

If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR`, if the `VkRenderingInfo::pStencilAttachment->imageView` parameter to `vkCmdBeginRenderingKHR` is not `VK_NULL_HANDLE`, the value of the `stencilAttachmentFormat` member of the `VkCommandBufferInheritanceRenderingInfo` structure included in the `pNext` chain of `VkCommandBufferBeginInfo::pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be equal to the format used to create that image view

- VUID-vkCmdExecuteCommands-pDepthAttachment-06774
If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR` and the `VkRenderingInfo::pDepthAttachment->imageView` parameter to `vkCmdBeginRenderingKHR` was `VK_NULL_HANDLE`, the value of the `depthAttachmentFormat` member of the `VkCommandBufferInheritanceRenderingInfo` structure included in the `pNext` chain of `VkCommandBufferBeginInfo::pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdExecuteCommands-pStencilAttachment-06775
If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR` and the `VkRenderingInfo::pStencilAttachment->imageView` parameter to `vkCmdBeginRenderingKHR` was `VK_NULL_HANDLE`, the value of the `stencilAttachmentFormat` member of the `VkCommandBufferInheritanceRenderingInfo` structure included in the `pNext` chain of `VkCommandBufferBeginInfo::pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdExecuteCommands-viewMask-06031
If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR`, the `viewMask` member of the `VkCommandBufferInheritanceRenderingInfo` structure included in the `pNext` chain of `VkCommandBufferBeginInfo::pInheritanceInfo` used to begin recording each element of `pCommandBuffers` **must** be equal to the `VkRenderingInfo::viewMask` parameter to `vkCmdBeginRenderingKHR`
- VUID-vkCmdExecuteCommands-commandBuffer-09375
`commandBuffer` **must** not be a `secondary command buffer` unless the `nestedCommandBuffer` feature is enabled
- VUID-vkCmdExecuteCommands-nestedCommandBuffer-09376
If the `nestedCommandBuffer` feature is enabled, the `command buffer nesting level` of each element of `pCommandBuffers` **must** be less than `maxCommandBufferNestingLevel`
- VUID-vkCmdExecuteCommands-nestedCommandBufferRendering-09377
If the `nestedCommandBufferRendering` feature is not enabled, and `commandBuffer` is a `secondary command buffer`, `commandBuffer` **must** not have been recorded with `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT`
- VUID-vkCmdExecuteCommands-nestedCommandBufferSimultaneousUse-09378
If the `nestedCommandBufferSimultaneousUse` feature is not enabled, and `commandBuffer` is a `secondary command buffer`, each element of `pCommandBuffers` **must** not have been recorded with `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT`
- VUID-vkCmdExecuteCommands-pCommandBuffers-09504
If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR`, the color attachment mapping state specified by `VkRenderingAttachmentLocationInfoKHR` in the inheritance info of each element of `pCommandBuffers` and in the current state of `commandBuffer` **must** match
- VUID-vkCmdExecuteCommands-pCommandBuffers-09505
If `vkCmdExecuteCommands` is being called within a render pass instance begun with `vkCmdBeginRenderingKHR`, the input attachment mapping state specified by `VkRenderingInputAttachmentIndexInfoKHR` in the inheritance info of each element of `pCommandBuffers` and in the current state of `commandBuffer` **must** match

Valid Usage (Implicit)

- VUID-vkCmdExecuteCommands-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdExecuteCommands-pCommandBuffers-parameter
`pCommandBuffers` **must** be a valid pointer to an array of `commandBufferCount` valid `VkCommandBuffer` handles
- VUID-vkCmdExecuteCommands-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdExecuteCommands-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdExecuteCommands-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdExecuteCommands-commandBufferCount-arraylength
`commandBufferCount` **must** be greater than 0
- VUID-vkCmdExecuteCommands-commonparent
Both of `commandBuffer`, and the elements of `pCommandBuffers` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Transfer Graphics Compute	Indirection

6.8. Nested Command Buffers

In addition to secondary command buffer execution from primary command buffers, an implementation **may** support `nested command buffers`, which enable secondary command buffers to be executed from other secondary command buffers. If the `nestedCommandBuffer` feature is enabled, the implementation supports `nested command buffers`.

Nested command buffer execution works the same as primary-to-secondary execution, except that it is subject to some additional implementation-defined limits.

Each secondary command buffer has a [command buffer nesting level](#), which is determined at [vkEndCommandBuffer](#) time and evaluated at [vkCmdExecuteCommands](#) time. A secondary command buffer that executes no other secondary command buffers has a [command buffer nesting level](#) of zero. Otherwise, the [command buffer nesting level](#) of a secondary command buffer is equal to the maximum nesting level of all secondary command buffers executed by that command buffer plus one. Some implementations **may** have a limit on the maximum nesting level of secondary command buffers that **can** be recorded. This limit is advertised in [maxCommandBufferNestingLevel](#).

If the [nestedCommandBufferRendering](#) feature is enabled, the implementation supports calling [vkCmdExecuteCommands](#) inside secondary command buffers recorded with [VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT](#). If the [nestedCommandBufferSimultaneousUse](#) feature is enabled, the implementation supports calling [vkCmdExecuteCommands](#) with secondary command buffers recorded with [VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT](#).

Whenever [vkCmdExecuteCommands](#) is recorded inside a secondary command buffer recorded with [VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT](#), each member of [pCommandBuffers](#) **must** have been recorded with a [VkCommandBufferBeginInfo](#) with [VkCommandBufferInheritanceInfo](#) compatible with the [VkCommandBufferInheritanceInfo](#) of the command buffer into which the [vkCmdExecuteCommands](#) call is being recorded. The [VkCommandBufferInheritanceRenderingInfo](#) structures are compatible when the [VkCommandBufferInheritanceRenderingInfo::renderpass](#) are [compatible](#), or if they are [VK_NULL_HANDLE](#) then the [VkCommandBufferInheritanceRenderingInfo](#) members match, and all other members of [VkCommandBufferInheritanceRenderingInfo](#) match. This requirement applies recursively, down to the most nested command buffer and up to the command buffer where the render pass was originally begun.

6.9. Command Buffer Device Mask

Each command buffer has a piece of state storing the current device mask of the command buffer. This mask controls which physical devices within the logical device all subsequent commands will execute on, including state-setting commands, action commands, and synchronization commands.

Scissor and viewport state **can** be different values on each physical device (only when set as dynamic state), and each physical device will render using its local copy of the state. Other state is shared between physical devices, such that all physical devices use the most recently set values for the state. However, when recording an action command that uses a piece of state, the most recent command that set that state **must** have included all physical devices that execute the action command in its current device mask.

The command buffer's device mask is orthogonal to the [pCommandBufferDeviceMasks](#) member of [VkDeviceGroupSubmitInfo](#). Commands only execute on a physical device if the device index is set in both device masks.

If the [pNext](#) chain of [VkCommandBufferBeginInfo](#) includes a [VkDeviceGroupCommandBufferBeginInfo](#) structure, then that structure includes an initial device mask for the command buffer.

The `VkDeviceGroupCommandBufferBeginInfo` structure is defined as:

```
typedef struct VkDeviceGroupCommandBufferBeginInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           deviceMask;
} VkDeviceGroupCommandBufferBeginInfo;
```

or the equivalent

```
// Provided by VK_KHR_device_group
typedef VkDeviceGroupCommandBufferBeginInfo VkDeviceGroupCommandBufferBeginInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceMask` is the initial value of the command buffer's device mask.

The initial device mask also acts as an upper bound on the set of devices that **can** ever be in the device mask in the command buffer.

If this structure is not present, the initial value of a command buffer's device mask includes all physical devices in the logical device when the command buffer begins recording.

Valid Usage

- VUID-VkDeviceGroupCommandBufferBeginInfo-deviceMask-00106
`deviceMask` **must** be a valid device mask value
- VUID-VkDeviceGroupCommandBufferBeginInfo-deviceMask-00107
`deviceMask` **must** not be zero

Valid Usage (Implicit)

- VUID-VkDeviceGroupCommandBufferBeginInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_GROUP_COMMAND_BUFFER_BEGIN_INFO`

To update the current device mask of a command buffer, call:

```
// Provided by VK_KHR_device_group
void vkCmdSetDeviceMaskKHR(
    VkCommandBuffer          commandBuffer,
    uint32_t                 deviceMask);
```

- `commandBuffer` is command buffer whose current device mask is modified.

- **deviceMask** is the new value of the current device mask.

deviceMask is used to filter out subsequent commands from executing on all physical devices whose bit indices are not set in the mask, except commands beginning a render pass instance, commands transitioning to the next subpass in the render pass instance, and commands ending a render pass instance, which always execute on the set of physical devices whose bit indices are included in the **deviceMask** member of the [VkDeviceGroupRenderPassBeginInfo](#) structure passed to the command beginning the corresponding render pass instance.

Valid Usage

- VUID-vkCmdSetDeviceMask-deviceMask-00108
deviceMask **must** be a valid device mask value
- VUID-vkCmdSetDeviceMask-deviceMask-00109
deviceMask **must** not be zero
- VUID-vkCmdSetDeviceMask-deviceMask-00110
deviceMask **must** not include any set bits that were not in the [VkDeviceGroupCommandBufferBeginInfo::deviceMask](#) value when the command buffer began recording
- VUID-vkCmdSetDeviceMask-deviceMask-00111
If [vkCmdSetDeviceMaskKHR](#) is called inside a render pass instance, **deviceMask** **must** not include any set bits that were not in the [VkDeviceGroupRenderPassBeginInfo::deviceMask](#) value when the render pass instance began recording

Valid Usage (Implicit)

- VUID-vkCmdSetDeviceMask-commandBuffer-parameter
commandBuffer **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetDeviceMask-commandBuffer-recording
commandBuffer **must** be in the [recording state](#)
- VUID-vkCmdSetDeviceMask-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics, compute, or transfer operations

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	Graphics Compute Transfer	State

Chapter 7. Synchronization and Cache Control

Synchronization of access to resources is primarily the responsibility of the application in Vulkan. The order of execution of commands with respect to the host and other commands on the device has few implicit guarantees, and needs to be explicitly specified. Memory caches and other optimizations are also explicitly managed, requiring that the flow of data through the system is largely under application control.

Whilst some implicit guarantees exist between commands, five explicit synchronization mechanisms are exposed by Vulkan:

Fences

Fences **can** be used to communicate to the host that execution of some task on the device has completed, controlling resource access between host and device.

Semaphores

Semaphores **can** be used to control resource access across multiple queues.

Events

Events provide a fine-grained synchronization primitive which **can** be signaled either within a command buffer or by the host, and **can** be waited upon within a command buffer or queried on the host. Events **can** be used to control resource access within a single queue.

Pipeline Barriers

Pipeline barriers also provide synchronization control within a command buffer, but at a single point, rather than with separate signal and wait operations. Pipeline barriers **can** be used to control resource access within a single queue.

Render Pass Objects

Render pass objects provide a synchronization framework for rendering tasks, built upon the concepts in this chapter. Many cases that would otherwise need an application to use other synchronization primitives **can** be expressed more efficiently as part of a render pass. Render pass objects **can** be used to control resource access within a single queue.

7.1. Execution and Memory Dependencies

An *operation* is an arbitrary amount of work to be executed on the host, a device, or an external entity such as a presentation engine. Synchronization commands introduce explicit *execution dependencies*, and *memory dependencies* between two sets of operations defined by the command's two *synchronization scopes*.

The synchronization scopes define which other operations a synchronization command is able to create execution dependencies with. Any type of operation that is not in a synchronization command's synchronization scopes will not be included in the resulting dependency. For example, for many synchronization commands, the synchronization scopes **can** be limited to just operations executing in specific [pipeline stages](#), which allows other pipeline stages to be excluded from a

dependency. Other scoping options are possible, depending on the particular command.

An *execution dependency* is a guarantee that for two sets of operations, the first set **must** happen-before the second set. If an operation happens-before another operation, then the first operation **must** complete before the second operation is initiated. More precisely:

- Let **Ops₁** and **Ops₂** be separate sets of operations.
- Let **Sync** be a synchronization command.
- Let **Scope_{1st}** and **Scope_{2nd}** be the synchronization scopes of **Sync**.
- Let **ScopedOps₁** be the intersection of sets **Ops₁** and **Scope_{1st}**.
- Let **ScopedOps₂** be the intersection of sets **Ops₂** and **Scope_{2nd}**.
- Submitting **Ops₁**, **Sync** and **Ops₂** for execution, in that order, will result in execution dependency **ExeDep** between **ScopedOps₁** and **ScopedOps₂**.
- Execution dependency **ExeDep** guarantees that **ScopedOps₁** happen-before **ScopedOps₂**.

An *execution dependency chain* is a sequence of execution dependencies that form a happens-before relation between the first dependency's **ScopedOps₁** and the final dependency's **ScopedOps₂**. For each consecutive pair of execution dependencies, a chain exists if the intersection of **Scope_{2nd}** in the first dependency and **Scope_{1st}** in the second dependency is not an empty set. The formation of a single execution dependency from an execution dependency chain can be described by substituting the following in the description of execution dependencies:

- Let **Sync** be a set of synchronization commands that generate an execution dependency chain.
- Let **Scope_{1st}** be the first synchronization scope of the first command in **Sync**.
- Let **Scope_{2nd}** be the second synchronization scope of the last command in **Sync**.

Execution dependencies alone are not sufficient to guarantee that values resulting from writes in one set of operations **can** be read from another set of operations.

Three additional types of operations are used to control memory access. *Availability operations* cause the values generated by specified memory write accesses to become *available* to a memory domain for future access. Any available value remains available until a subsequent write to the same memory location occurs (whether it is made available or not) or the memory is freed. *Memory domain operations* cause writes that are available to a source memory domain to become available to a destination memory domain (an example of this is making writes available to the host domain available to the device domain). *Visibility operations* cause values available to a memory domain to become *visible* to specified memory accesses.

Availability, visibility, memory domains, and memory domain operations are formally defined in the [Availability and Visibility](#) section of the [Memory Model](#) chapter. Which API operations perform each of these operations is defined in [Availability, Visibility, and Domain Operations](#).

A *memory dependency* is an execution dependency which includes availability and visibility operations such that:

- The first set of operations happens-before the availability operation.

- The availability operation happens-before the visibility operation.
- The visibility operation happens-before the second set of operations.

Once written values are made visible to a particular type of memory access, they **can** be read or written by that type of memory access. Most synchronization commands in Vulkan define a memory dependency.

The specific memory accesses that are made available and visible are defined by the *access scopes* of a memory dependency. Any type of access that is in a memory dependency's first access scope and occurs in **ScopedOps₁** is made available. Any type of access that is in a memory dependency's second access scope and occurs in **ScopedOps₂** has any available writes made visible to it. Any type of operation that is not in a synchronization command's access scopes will not be included in the resulting dependency.

A memory dependency enforces availability and visibility of memory accesses and execution order between two sets of operations. Adding to the description of [execution dependency chains](#):

- Let **MemOps₁** be the set of memory accesses performed by **ScopedOps₁**.
- Let **MemOps₂** be the set of memory accesses performed by **ScopedOps₂**.
- Let **AccessScope_{1st}** be the first access scope of the first command in the **Sync** chain.
- Let **AccessScope_{2nd}** be the second access scope of the last command in the **Sync** chain.
- Let **ScopedMemOps₁** be the intersection of sets **MemOps₁** and **AccessScope_{1st}**.
- Let **ScopedMemOps₂** be the intersection of sets **MemOps₂** and **AccessScope_{2nd}**.
- Submitting **Ops₁**, **Sync**, and **Ops₂** for execution, in that order, will result in a memory dependency **MemDep** between **ScopedOps₁** and **ScopedOps₂**.
- Memory dependency **MemDep** guarantees that:
 - Memory writes in **ScopedMemOps₁** are made available.
 - Available memory writes, including those from **ScopedMemOps₁**, are made visible to **ScopedMemOps₂**.

NOTE

Execution and memory dependencies are used to solve data hazards, i.e. to ensure that read and write operations occur in a well-defined order. Write-after-read hazards can be solved with just an execution dependency, but read-after-write and write-after-write hazards need appropriate memory dependencies to be included between them. If an application does not include dependencies to solve these hazards, the results and execution orders of memory accesses are undefined.

7.1.1. Image Layout Transitions

Image subresources **can** be transitioned from one [layout](#) to another as part of a [memory dependency](#) (e.g. by using an [image memory barrier](#)). When a layout transition is specified in a memory dependency, it happens-after the availability operations in the memory dependency, and happens-before the visibility operations. Image layout transitions **may** perform read and write accesses on all memory bound to the image subresource range, so applications **must** ensure that all memory writes have been made [available](#) before a layout transition is executed. Available memory

is automatically made visible to a layout transition, and writes performed by a layout transition are automatically made available.

Layout transitions always apply to a particular image subresource range, and specify both an old layout and new layout. The old layout **must** either be `VK_IMAGE_LAYOUT_UNDEFINED`, or match the current layout of the image subresource range. If the old layout matches the current layout of the image subresource range, the transition preserves the contents of that range. If the old layout is `VK_IMAGE_LAYOUT_UNDEFINED`, the contents of that range **may** be discarded.

NOTE

Image layout transitions with `VK_IMAGE_LAYOUT_UNDEFINED` allow the implementation to discard the image subresource range, which can provide performance or power benefits. Tile-based architectures may be able to avoid flushing tile data to memory, and immediate style renderers may be able to achieve fast metadata clears to reinitialize frame buffer compression state, or similar.

If the contents of an attachment are not needed after a render pass completes, then applications **should** use `VK_ATTACHMENT_STORE_OP_DONT_CARE`.

As image layout transitions **may** perform read and write accesses on the memory bound to the image, if the image subresource affected by the layout transition is bound to peer memory for any device in the current device mask then the memory heap the bound memory comes from **must** support the `VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT` and `VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT` capabilities as returned by `vkGetDeviceGroupPeerMemoryFeaturesKHR`.

NOTE

Applications **must** ensure that layout transitions happen-after all operations accessing the image with the old layout, and happen-before any operations that will access the image with the new layout. Layout transitions are potentially read/write operations, so not defining appropriate memory dependencies to guarantee this will result in a data race.

Image layout transitions interact with [memory aliasing](#).

Layout transitions that are performed via image memory barriers execute in their entirety in [submission order](#), relative to other image layout transitions submitted to the same queue, including those performed by [render passes](#). In effect there is an implicit execution dependency from each such layout transition to all layout transitions previously submitted to the same queue.

The image layout of each image subresource of a depth/stencil image created with `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` is dependent on the last sample locations used to render to the image subresource as a depth/stencil attachment, thus when the `image` member of an [image memory barrier](#) is an image created with this flag the application **can** chain a `VkSampleLocationsInfoEXT` structure to the `pNext` chain of `VkImageMemoryBarrier2` or `VkImageMemoryBarrier` to specify the sample locations to use during any image layout transition.

If the `VkSampleLocationsInfoEXT` structure does not match the sample location state last used to render to the image subresource range specified by `subresourceRange`, or if no `VkSampleLocationsInfoEXT` structure is present, then the contents of the given image subresource range becomes undefined as if `oldLayout` would equal `VK_IMAGE_LAYOUT_UNDEFINED`.

7.1.2. Pipeline Stages

The work performed by an [action command](#) consists of multiple operations, which are performed as a sequence of logically independent steps known as *pipeline stages*. The exact pipeline stages executed depend on the particular command that is used, and current command buffer state when the command was recorded.

NOTE Operations performed by synchronization commands (e.g. [availability and visibility operations](#)) are not executed by a defined pipeline stage. However other commands can still synchronize with them by using the [synchronization scopes](#) to create a [dependency chain](#).

Execution of operations across pipeline stages **must** adhere to [implicit ordering guarantees](#), particularly including [pipeline stage order](#). Otherwise, execution across pipeline stages **may** overlap or execute out of order with regards to other stages, unless otherwise enforced by an execution dependency.

Several of the synchronization commands include pipeline stage parameters, restricting the [synchronization scopes](#) for that command to just those stages. This allows fine grained control over the exact execution dependencies and accesses performed by action commands. Implementations **should** use these pipeline stages to avoid unnecessary stalls or cache flushing.

Bits which **can** be set in a [VkPipelineStageFlags2](#) mask, specifying stages of execution, are:

```
// Flag bits for VkPipelineStageFlagBits2
typedef VkFlags64 VkPipelineStageFlagBits2;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_NONE = 0ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_NONE_KHR = 0ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_TOP_OF_PIPE_BIT =
0x00000001ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_TOP_OF_PIPE_BIT_KHR =
0x00000001ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT =
0x00000002ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT_KHR =
0x00000002ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT =
0x00000004ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT_KHR =
0x00000004ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT =
0x00000008ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT_KHR =
0x00000008ULL;
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT = 0x00000010ULL;
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT_KHR = 0x00000010ULL;
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT = 0x00000020ULL;
```

```

static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT_KHR = 0x00000020U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT =
0x00000040U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT_KHR =
0x00000040U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT =
0x00000080U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT_KHR =
0x00000080U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT =
0x00000100U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT_KHR =
0x00000100U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT =
0x00000200U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT_KHR =
0x00000200U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT =
0x00000400U;
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT_KHR = 0x00000400U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT =
0x00000800U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT_KHR =
0x00000800U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT =
0x00001000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT_KHR =
0x00001000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_TRANSFER_BIT =
0x00001000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_TRANSFER_BIT_KHR =
0x00001000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_BOTTOM_OF_PIPE_BIT =
0x00002000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_BOTTOM_OF_PIPE_BIT_KHR =
0x00002000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_HOST_BIT = 0x00004000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_HOST_BIT_KHR =
0x00004000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT =
0x00008000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT_KHR =
0x00008000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT =
0x00010000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT_KHR =
0x00010000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COPY_BIT = 0x100000000U;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COPY_BIT_KHR =

```



```

0x100000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_RESOLVE_BIT =
0x200000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_RESOLVE_BIT_KHR =
0x200000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_BLIT_BIT = 0x400000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_BLIT_BIT_KHR =
0x400000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_CLEAR_BIT = 0x800000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_CLEAR_BIT_KHR =
0x800000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT =
0x100000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT_KHR =
0x100000000ULL;
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT =
0x200000000ULL;
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT_KHR = 0x200000000ULL;
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_PRE_RASTERIZATION_SHADERS_BIT = 0x400000000ULL;
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_PRE_RASTERIZATION_SHADERS_BIT_KHR = 0x400000000ULL;
// Provided by VK_KHR_video_decode_queue
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR =
0x04000000ULL;
// Provided by VK_KHR_video_encode_queue
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR =
0x08000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_transform_feedback
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT =
0x01000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_conditional_rendering
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT = 0x00040000ULL;
// Provided by VK_KHR_synchronization2 with VK_NV_device_generated_commands
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_NV =
0x00020000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_device_generated_commands
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT =
0x00020000ULL;
// Provided by VK_KHR_fragment_shading_rate with VK_KHR_synchronization2
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR = 0x00400000ULL;
// Provided by VK_KHR_synchronization2 with VK_NV_shading_rate_image
static const VkPipelineStageFlagBits2 VK_PIPELINE_STAGE_2_SHADING_RATE_IMAGE_BIT_NV =
0x00400000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_synchronization2
static const VkPipelineStageFlagBits2
VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR = 0x02000000ULL;
// Provided by VK_KHR_ray_tracing_pipeline with VK_KHR_synchronization2

```

```

static const VkPipelineStageFlagsBits2 VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR =
0x00200000ULL;
// Provided by VK_KHR_synchronization2 with VK_NV_ray_tracing
static const VkPipelineStageFlagsBits2 VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_NV =
0x00200000ULL;
// Provided by VK_KHR_synchronization2 with VK_NV_ray_tracing
static const VkPipelineStageFlagsBits2
VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_NV = 0x02000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_fragment_density_map
static const VkPipelineStageFlagsBits2
VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT = 0x00800000ULL;
// Provided by VK_KHR_synchronization2 with VK_NV_mesh_shader
static const VkPipelineStageFlagsBits2 VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_NV =
0x00080000ULL;
// Provided by VK_KHR_synchronization2 with VK_NV_mesh_shader
static const VkPipelineStageFlagsBits2 VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_NV =
0x00100000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_mesh_shader
static const VkPipelineStageFlagsBits2 VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT =
0x00080000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_mesh_shader
static const VkPipelineStageFlagsBits2 VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT =
0x00100000ULL;
// Provided by VK_KHR_ray_tracing_maintenance1 with VK_KHR_synchronization2 or
VK_VERSION_1_3
static const VkPipelineStageFlagsBits2
VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR = 0x10000000ULL;
// Provided by VK_EXT_opacity_micromap
static const VkPipelineStageFlagsBits2 VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT =
0x40000000ULL;

```

or the equivalent

```

// Provided by VK_KHR_synchronization2
typedef VkPipelineStageFlagsBits2 VkPipelineStageFlagsBits2KHR;

```

- **VK_PIPELINE_STAGE_2_NONE** specifies no stages of execution.
- **VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT** specifies the stage of the pipeline where indirect command parameters are consumed. This stage also includes reading commands written by [vkCmdPreprocessGeneratedCommandsEXT](#).
- **VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT** specifies the stage of the pipeline where index buffers are consumed.
- **VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT** specifies the stage of the pipeline where vertex buffers are consumed.
- **VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT** is equivalent to the logical OR of:
 - **VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT**

- `VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT`
- `VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT` specifies the vertex shader stage.
- `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT` specifies the tessellation control shader stage.
- `VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT` specifies the tessellation evaluation shader stage.
- `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT` specifies the geometry shader stage.
- `VK_PIPELINE_STAGE_2_PRE_RASTERIZATION_SHADERS_BIT` is equivalent to specifying all supported [pre-rasterization shader stages](#):
 - `VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT`
 - `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT`
 - `VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT`
 - `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT`
- `VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT` specifies the fragment shader stage.
- `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT` specifies the stage of the pipeline where early fragment tests (depth and stencil tests before fragment shading) are performed. This stage also includes [render pass load operations](#) for framebuffer attachments with a depth/stencil format.
- `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT` specifies the stage of the pipeline where late fragment tests (depth and stencil tests after fragment shading) are performed. This stage also includes [render pass store operations](#) for framebuffer attachments with a depth/stencil format.
- `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT` specifies the stage of the pipeline where final color values are output from the pipeline. This stage includes [blending](#), [logic operations](#), render pass [load](#) and [store](#) operations for color attachments, [render pass multisample resolve operations](#), and [vkCmdClearAttachments](#).
- `VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT` specifies the compute shader stage.
- `VK_PIPELINE_STAGE_2_HOST_BIT` specifies a pseudo-stage indicating execution on the host of reads/writes of device memory. This stage is not invoked by any commands recorded in a command buffer.
- `VK_PIPELINE_STAGE_2_COPY_BIT` specifies the execution of all [copy commands](#), including [vkCmdCopyQueryPoolResults](#).
- `VK_PIPELINE_STAGE_2_BLIT_BIT` specifies the execution of [vkCmdBlitImage](#).
- `VK_PIPELINE_STAGE_2_RESOLVE_BIT` specifies the execution of [vkCmdResolveImage](#).
- `VK_PIPELINE_STAGE_2_CLEAR_BIT` specifies the execution of [clear commands](#), with the exception of [vkCmdClearAttachments](#).
- `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT` is equivalent to specifying all of:
 - `VK_PIPELINE_STAGE_2_COPY_BIT`
 - `VK_PIPELINE_STAGE_2_BLIT_BIT`
 - `VK_PIPELINE_STAGE_2_RESOLVE_BIT`

- `VK_PIPELINE_STAGE_2_CLEAR_BIT`
- `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`
- `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR` specifies the execution of the ray tracing shader stages.
- `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` specifies the execution of [acceleration structure commands](#) or [acceleration structure copy commands](#).
- `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR` specifies the execution of [acceleration structure copy commands](#).
- `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT` specifies the execution of all graphics pipeline stages, and is equivalent to the logical OR of:
 - `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`
 - `VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT`
 - `VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT`
 - `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT`
 - `VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT`
 - `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT`
 - `VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT`
 - `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`
 - `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`
 - `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`
 - `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`
 - `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT` specifies all operations performed by all commands supported on the queue it is used with.
- `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT` specifies the stage of the pipeline where vertex attribute output values are written to the transform feedback buffers.
- `VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT` specifies the stage of the pipeline where device-side generation of commands via [vkCmdPreprocessGeneratedCommandsEXT](#) is handled.
- `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR` specifies the stage of the pipeline where the [fragment shading rate attachment](#) is read to determine the fragment shading rate for portions of a rasterized primitive.
- `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR` specifies the execution of [video decode operations](#).
- `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR` specifies the execution of [video encode operations](#).
- `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` specifies the execution of [micromap commands](#).
- `VK_PIPELINE_STAGE_2_TOP_OF_PIPE_BIT` is equivalent to `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT` with [VkAccessFlags2](#) set to `0` when specified in the second synchronization scope, but equivalent to `VK_PIPELINE_STAGE_2_NONE` in the first scope.
- `VK_PIPELINE_STAGE_2_BOTTOM_OF_PIPE_BIT` is equivalent to `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

with [VkAccessFlags2](#) set to `0` when specified in the first synchronization scope, but equivalent to `VK_PIPELINE_STAGE_2_NONE` in the second scope.

NOTE

The `TOP` and `BOTTOM` pipeline stages are deprecated, and applications should prefer `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT` and `VK_PIPELINE_STAGE_2_NONE`.

NOTE

The `VkPipelineStageFlags2` bitmask goes beyond the 31 individual bit flags allowable within a C99 enum, which is how [VkPipelineStageFlagBits](#) is defined. The first 31 values are common to both, and are interchangeable.

`VkPipelineStageFlags2` is a bitmask type for setting a mask of zero or more [VkPipelineStageFlagBits2](#) flags:

```
typedef VkFlags64 VkPipelineStageFlags2;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2
typedef VkPipelineStageFlags2 VkPipelineStageFlags2KHR;
```

Bits which **can** be set in a [VkPipelineStageFlags](#) mask, specifying stages of execution, are:

```

// Provided by VK_VERSION_1_0
typedef enum VkPipelineStageFlagBits {
    VK_PIPELINE_STAGE_TOP_OF_PIPE_BIT = 0x00000001,
    VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT = 0x00000002,
    VK_PIPELINE_STAGE_VERTEX_INPUT_BIT = 0x00000004,
    VK_PIPELINE_STAGE_VERTEX_SHADER_BIT = 0x00000008,
    VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT = 0x00000010,
    VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT = 0x00000020,
    VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT = 0x00000040,
    VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT = 0x00000080,
    VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT = 0x00000100,
    VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT = 0x00000200,
    VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT = 0x00000400,
    VK_PIPELINE_STAGE_COMPUTE_SHADER_BIT = 0x00000800,
    VK_PIPELINE_STAGE_TRANSFER_BIT = 0x00001000,
    VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT = 0x00002000,
    VK_PIPELINE_STAGE_HOST_BIT = 0x00004000,
    VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT = 0x00008000,
    VK_PIPELINE_STAGE_ALL_COMMANDS_BIT = 0x00010000,
    VK_PIPELINE_STAGE_NONE = 0,
// Provided by VK_EXT_transform_feedback
    VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT = 0x01000000,
// Provided by VK_KHR_acceleration_structure
    VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR = 0x02000000,
// Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR = 0x00200000,
// Provided by VK_KHR_fragment_shading_rate
    VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR = 0x00400000,
    VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_NV = 0x00020000,
// Provided by VK_KHR_synchronization2
    VK_PIPELINE_STAGE_NONE_KHR = VK_PIPELINE_STAGE_NONE,
// Provided by VK_EXT_device_generated_commands
    VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_EXT =
VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_NV,
} VkPipelineStageFlagBits;

```

These values all have the same meaning as the equivalently named values for [VkPipelineStageFlags2](#).

- **VK_PIPELINE_STAGE_NONE** specifies no stages of execution.
- **VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT** specifies the stage of the pipeline where **VkDrawIndirect*** / **VkDispatchIndirect*** / **VkTraceRaysIndirect*** data structures are consumed. This stage also includes reading commands written by **vkCmdExecuteGeneratedCommandsEXT**.
- **VK_PIPELINE_STAGE_VERTEX_INPUT_BIT** specifies the stage of the pipeline where vertex and index buffers are consumed.
- **VK_PIPELINE_STAGE_VERTEX_SHADER_BIT** specifies the vertex shader stage.
- **VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT** specifies the tessellation control shader

stage.

- `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT` specifies the tessellation evaluation shader stage.
- `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT` specifies the geometry shader stage.
- `VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT` specifies the fragment shader stage.
- `VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT` specifies the stage of the pipeline where early fragment tests (depth and stencil tests before fragment shading) are performed. This stage also includes [render pass load operations](#) for framebuffer attachments with a depth/stencil format.
- `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT` specifies the stage of the pipeline where late fragment tests (depth and stencil tests after fragment shading) are performed. This stage also includes [render pass store operations](#) for framebuffer attachments with a depth/stencil format.
- `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` specifies the stage of the pipeline after blending where the final color values are output from the pipeline. This stage includes [blending](#), [logic operations](#), render pass [load](#) and [store](#) operations for color attachments, [render pass multisample resolve operations](#), and [vkCmdClearAttachments](#).
- `VK_PIPELINE_STAGE_COMPUTE_SHADER_BIT` specifies the execution of a compute shader.
- `VK_PIPELINE_STAGE_TRANSFER_BIT` specifies the following commands:
 - All [copy commands](#), including [vkCmdCopyQueryPoolResults](#)
 - [vkCmdBlitImage2KHR](#) and [vkCmdBlitImage](#)
 - [vkCmdResolveImage2KHR](#) and [vkCmdResolveImage](#)
 - All [clear commands](#), with the exception of [vkCmdClearAttachments](#)
- `VK_PIPELINE_STAGE_HOST_BIT` specifies a pseudo-stage indicating execution on the host of reads/writes of device memory. This stage is not invoked by any commands recorded in a command buffer.
- `VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` specifies the execution of [vkCmdBuildAccelerationStructuresKHR](#), [vkCmdBuildAccelerationStructuresIndirectKHR](#), [vkCmdCopyAccelerationStructureKHR](#), [vkCmdCopyAccelerationStructureToMemoryKHR](#), [vkCmdCopyMemoryToAccelerationStructureKHR](#), and [vkCmdWriteAccelerationStructuresPropertiesKHR](#).
- `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR` specifies the execution of the ray tracing shader stages, via [vkCmdTraceRaysKHR](#), or [vkCmdTraceRaysIndirectKHR](#)
- `VK_PIPELINE_STAGE_ALL_GRAPHICS_BIT` specifies the execution of all graphics pipeline stages, and is equivalent to the logical OR of:
 - `VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT`
 - `VK_PIPELINE_STAGE_VERTEX_INPUT_BIT`
 - `VK_PIPELINE_STAGE_VERTEX_SHADER_BIT`
 - `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT`
 - `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
 - `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`

- `VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT`
- `VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT`
- `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT`
- `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT`
- `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT` specifies all operations performed by all commands supported on the queue it is used with.
- `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT` specifies the stage of the pipeline where vertex attribute output values are written to the transform feedback buffers.
- `VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_EXT` specifies the stage of the pipeline where device-side preprocessing for generated commands via `vkCmdPreprocessGeneratedCommandsEXT` is handled.
- `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR` specifies the stage of the pipeline where the `fragment shading rate attachment` is read to determine the fragment shading rate for portions of a rasterized primitive.
- `VK_PIPELINE_STAGE_TOP_OF_PIPE_BIT` is equivalent to `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT` with `VkAccessFlags` set to `0` when specified in the second synchronization scope, but specifies no stage of execution when specified in the first scope.
- `VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT` is equivalent to `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT` with `VkAccessFlags` set to `0` when specified in the first synchronization scope, but specifies no stage of execution when specified in the second scope.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineStageFlags;
```

`VkPipelineStageFlags` is a bitmask type for setting a mask of zero or more `VkPipelineStageFlagBits`.

If a synchronization command includes a source stage mask, its first `synchronization scope` only includes execution of the pipeline stages specified in that mask and any `logically earlier` stages. Its first `access scope` only includes memory accesses performed by pipeline stages explicitly specified in the source stage mask.

If a synchronization command includes a destination stage mask, its second `synchronization scope` only includes execution of the pipeline stages specified in that mask and any `logically later` stages. Its second `access scope` only includes memory accesses performed by pipeline stages explicitly specified in the destination stage mask.

NOTE

Note that `access scopes` do not interact with the logically earlier or later stages for either scope - only the stages the application specifies are considered part of each access scope.

Certain pipeline stages are only available on queues that support a particular set of operations. The

following table lists, for each pipeline stage flag, which queue capability flag **must** be supported by the queue. When multiple flags are enumerated in the second column of the table, it means that the pipeline stage is supported on the queue if it supports any of the listed capability flags. For further details on queue capabilities see [Physical Device Enumeration](#) and [Queues](#).

Table 3. Supported Pipeline Stage Flags

Pipeline stage flag	Required queue capability flag
VK_PIPELINE_STAGE_2_NONE	None required
VK_PIPELINE_STAGE_2_TOP_OF_PIPE_BIT	None required
VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT	VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_BOTTOM_OF_PIPE_BIT	None required
VK_PIPELINE_STAGE_2_HOST_BIT	None required
VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT	None required
VK_PIPELINE_STAGE_2_COPY_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_RESOLVE_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_BLIT_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_CLEAR_BIT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT	VK_QUEUE_GRAPHICS_BIT

Pipeline stage flag	Required queue capability flag
VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_PRE_RASTERIZATION_SHADERS_BIT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR	VK_QUEUE_VIDEO_DECODE_BIT_KHR
VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR	VK_QUEUE_VIDEO_ENCODE_BIT_KHR
VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_NV	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR	VK_QUEUE_GRAPHICS_BIT
VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR	VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR	VK_QUEUE_COMPUTE_BIT
VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR	VK_QUEUE_GRAPHICS_BIT or VK_QUEUE_COMPUTE_BIT or VK_QUEUE_TRANSFER_BIT
VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT	VK_QUEUE_COMPUTE_BIT

Pipeline stages that execute as a result of a command logically complete execution in a specific order, such that completion of a logically later pipeline stage **must** not happen-before completion of a logically earlier stage. This means that including any stage in the source stage mask for a particular synchronization command also implies that any logically earlier stages are included in **Scope_{1st}** for that command.

Similarly, initiation of a logically earlier pipeline stage **must** not happen-after initiation of a logically later pipeline stage. Including any given stage in the destination stage mask for a particular synchronization command also implies that any logically later stages are included in **Scope_{2nd}** for that command.

NOTE

Implementations **may** not support synchronization at every pipeline stage for every synchronization operation. If a pipeline stage that an implementation does not support synchronization for appears in a source stage mask, it **may** substitute any logically later stage in its place for the first synchronization scope. If a pipeline stage that an implementation does not support synchronization for appears in a destination stage mask, it **may** substitute any logically earlier stage in its place for the second synchronization scope.

For example, if an implementation is unable to signal an event immediately after vertex shader execution is complete, it **may** instead signal the event after color attachment output has completed.

If an implementation makes such a substitution, it **must** not affect the semantics of execution or memory dependencies or image and buffer memory barriers.

Graphics pipelines are executable on queues supporting `VK_QUEUE_GRAPHICS_BIT`. Stages executed by graphics pipelines **can** only be specified in commands recorded for queues supporting `VK_QUEUE_GRAPHICS_BIT`.

The graphics pipeline executes the following stages, with the logical ordering of the stages matching the order specified here:

- `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`
- `VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT`
- `VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT`
- `VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT`
- `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT`
- `VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT`
- `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT`
- `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`
- `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`
- `VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT`
- `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`
- `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`

For the compute pipeline, the following stages occur in this order:

- `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`
- `VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT`

For the transfer pipeline, the following stages occur in this order:

- `VK_PIPELINE_STAGE_2_TRANSFER_BIT`

For host operations, only one pipeline stage occurs, so no order is guaranteed:

- `VK_PIPELINE_STAGE_2_HOST_BIT`

For the command preprocessing pipeline, the following stages occur in this order:

- `VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT`

For acceleration structure build operations, only one pipeline stage occurs, so no order is guaranteed:

- `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`

For acceleration structure copy operations, only one pipeline stage occurs, so no order is guaranteed:

- `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`

For opacity micromap build operations, only one pipeline stage occurs, so no order is guaranteed:

- `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT`

For the ray tracing pipeline, the following stages occur in this order:

- `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`
- `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

For the video decode pipeline, the following stages occur in this order:

- `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`

For the video encode pipeline, the following stages occur in this order:

- `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`

7.1.3. Access Types

Memory in Vulkan **can** be accessed from within shader invocations and via some fixed-function stages of the pipeline. The *access type* is a function of the [descriptor type](#) used, or how a fixed-function stage accesses memory.

Some synchronization commands take sets of access types as parameters to define the [access scopes](#) of a memory dependency. If a synchronization command includes a *source access mask*, its first [access scope](#) only includes accesses via the access types specified in that mask. Similarly, if a synchronization command includes a *destination access mask*, its second [access scope](#) only includes accesses via the access types specified in that mask.

Bits which **can** be set in the `srcAccessMask` and `dstAccessMask` members of `VkMemoryBarrier2KHR`, `VkImageMemoryBarrier2KHR`, and `VkBufferMemoryBarrier2KHR`, specifying access behavior, are:

```
// Flag bits for VkAccessFlagBits2
typedef VkFlags64 VkAccessFlagBits2;
static const VkAccessFlagBits2 VK_ACCESS_2_NONE = 0ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_NONE_KHR = 0ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT = 0x00000001ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT_KHR =
0x00000001ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_INDEX_READ_BIT = 0x00000002ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_INDEX_READ_BIT_KHR = 0x00000002ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT = 0x00000004ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT_KHR =
0x00000004ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_UNIFORM_READ_BIT = 0x00000008ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_UNIFORM_READ_BIT_KHR = 0x00000008ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT = 0x00000010ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT_KHR =
```

```

0x00000010ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_READ_BIT = 0x00000020ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_READ_BIT_KHR = 0x00000020ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_WRITE_BIT = 0x00000040ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_WRITE_BIT_KHR = 0x00000040ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT = 0x00000080ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT_KHR =
0x00000080ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT = 0x00000100ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT_KHR =
0x00000100ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT =
0x00000200ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT_KHR =
0x00000200ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT =
0x00000400ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT_KHR =
0x00000400ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFER_READ_BIT = 0x00000800ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFER_READ_BIT_KHR = 0x00000800ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFER_WRITE_BIT = 0x00001000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFER_WRITE_BIT_KHR = 0x00001000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_HOST_READ_BIT = 0x00002000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_HOST_READ_BIT_KHR = 0x00002000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_HOST_WRITE_BIT = 0x00004000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_HOST_WRITE_BIT_KHR = 0x00004000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_MEMORY_READ_BIT = 0x00008000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_MEMORY_READ_BIT_KHR = 0x00008000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_MEMORY_WRITE_BIT = 0x00010000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_MEMORY_WRITE_BIT_KHR = 0x00010000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_SAMPLED_READ_BIT = 0x100000000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_SAMPLED_READ_BIT_KHR =
0x100000000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_STORAGE_READ_BIT = 0x200000000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_STORAGE_READ_BIT_KHR =
0x200000000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT = 0x400000000ULL;
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT_KHR =
0x400000000ULL;
// Provided by VK_KHR_video_decode_queue
static const VkAccessFlagBits2 VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR = 0x800000000ULL;
// Provided by VK_KHR_video_decode_queue
static const VkAccessFlagBits2 VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR =
0x1000000000ULL;
// Provided by VK_KHR_video_encode_queue
static const VkAccessFlagBits2 VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR =
0x2000000000ULL;
// Provided by VK_KHR_video_encode_queue
static const VkAccessFlagBits2 VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR =
0x4000000000ULL;

```

```

// Provided by VK_KHR_synchronization2 with VK_EXT_transform_feedback
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT =
0x02000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_transform_feedback
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT =
0x04000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_transform_feedback
static const VkAccessFlagBits2 VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT =
0x08000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_conditional_rendering
static const VkAccessFlagBits2 VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT =
0x00100000ULL;
// Provided by VK_KHR_synchronization2 with VK_NV_device_generated_commands
static const VkAccessFlagBits2 VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_NV =
0x00020000ULL;
// Provided by VK_KHR_synchronization2 with VK_NV_device_generated_commands
static const VkAccessFlagBits2 VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_NV =
0x00040000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_device_generated_commands
static const VkAccessFlagBits2 VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_EXT =
0x00020000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_device_generated_commands
static const VkAccessFlagBits2 VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_EXT =
0x00040000ULL;
// Provided by VK_KHR_fragment_shading_rate with VK_KHR_synchronization2
static const VkAccessFlagBits2
VK_ACCESS_2_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR = 0x00800000ULL;
// Provided by VK_KHR_synchronization2 with VK_NV_shading_rate_image
static const VkAccessFlagBits2 VK_ACCESS_2_SHADING_RATE_IMAGE_READ_BIT_NV =
0x00800000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR =
0x00200000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_synchronization2
static const VkAccessFlagBits2 VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR =
0x00400000ULL;
// Provided by VK_KHR_synchronization2 with VK_NV_ray_tracing
static const VkAccessFlagBits2 VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_NV =
0x00200000ULL;
// Provided by VK_KHR_synchronization2 with VK_NV_ray_tracing
static const VkAccessFlagBits2 VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_NV =
0x00400000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_fragment_density_map
static const VkAccessFlagBits2 VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT =
0x01000000ULL;
// Provided by VK_KHR_synchronization2 with VK_EXT_blend_operation_advanced
static const VkAccessFlagBits2 VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT =
0x00080000ULL;
// Provided by VK_KHR_ray_tracing_maintenance1 with (VK_KHR_synchronization2 or
VK_VERSION_1_3) and VK_KHR_ray_tracing_pipeline
static const VkAccessFlagBits2 VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR =

```



```
0x10000000000ULL;
// Provided by VK_EXT_opacity_micromap
static const VkAccessFlagBits2 VK_ACCESS_2_MICROMAP_READ_BIT_EXT = 0x10000000000ULL;
// Provided by VK_EXT_opacity_micromap
static const VkAccessFlagBits2 VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT = 0x20000000000ULL;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2
typedef VkAccessFlagBits2 VkAccessFlagBits2KHR;
```

- **VK_ACCESS_2_NONE** specifies no accesses.
- **VK_ACCESS_2_MEMORY_READ_BIT** specifies all read accesses. It is always valid in any access mask, and is treated as equivalent to setting all **READ** access flags that are valid where it is used.
- **VK_ACCESS_2_MEMORY_WRITE_BIT** specifies all write accesses. It is always valid in any access mask, and is treated as equivalent to setting all **WRITE** access flags that are valid where it is used.
- **VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT** specifies read access to command data read from indirect buffers as part of an indirect build, trace, drawing or dispatch command. Such access occurs in the **VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT** pipeline stage.
- **VK_ACCESS_2_INDEX_READ_BIT** specifies read access to an index buffer as part of an indexed drawing command, bound by **vkCmdBindIndexBuffer2KHR** and **vkCmdBindIndexBuffer**. Such access occurs in the **VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT** pipeline stage.
- **VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT** specifies read access to a vertex buffer as part of a drawing command, bound by **vkCmdBindVertexBuffers**. Such access occurs in the **VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT** pipeline stage.
- **VK_ACCESS_2_UNIFORM_READ_BIT** specifies read access to a **uniform buffer** in any shader pipeline stage.
- **VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT** specifies read access to an **input attachment** within a render pass during fragment shading. Such access occurs in the **VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT** pipeline stage.
- **VK_ACCESS_2_SHADER_SAMPLED_READ_BIT** specifies read access to a **uniform texel buffer** or **sampled image** in any shader pipeline stage.
- **VK_ACCESS_2_SHADER_STORAGE_READ_BIT** specifies read access to a **storage buffer**, **storage texel buffer**, or **storage image** in any shader pipeline stage.
- **VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR** specifies read access to a **shader binding table** in any shader pipeline stage.
- **VK_ACCESS_2_SHADER_READ_BIT** is equivalent to the logical OR of:
 - **VK_ACCESS_2_SHADER_SAMPLED_READ_BIT**
 - **VK_ACCESS_2_SHADER_STORAGE_READ_BIT**
- **VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT** specifies write access to a **storage buffer**, **storage texel buffer**, or **storage image** in any shader pipeline stage.

- `VK_ACCESS_2_SHADER_WRITE_BIT` is equivalent to `VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT`.
- `VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT` specifies read access to a [color attachment](#), such as via [blending](#) (other than [advanced blend operations](#)), [logic operations](#) or certain [render pass load operations](#) in the `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage or via [fragment shader tile image reads](#) in the `VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT` pipeline stage.
- `VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT` specifies write access to a [color attachment](#) during a [render pass](#) or via certain render pass [load](#), [store](#), and [multisample resolve](#) operations. Such access occurs in the `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage.
- `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT` specifies read access to a [depth/stencil attachment](#), via [depth or stencil operations](#) or certain [render pass load operations](#) in the `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT` or `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT` pipeline stages or via [fragment shader tile image reads](#) in the `VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT` pipeline stage.
- `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT` specifies write access to a [depth/stencil attachment](#), via [depth or stencil operations](#) or certain render pass [load](#) and [store](#) operations. Such access occurs in the `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT` or `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT` pipeline stages.
- `VK_ACCESS_2_TRANSFER_READ_BIT` specifies read access to an image or buffer in a [copy](#) operation. Such access occurs in the `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, or `VK_PIPELINE_STAGE_2_RESOLVE_BIT` pipeline stages.
- `VK_ACCESS_2_TRANSFER_WRITE_BIT` specifies write access to an image or buffer in a [clear](#) or [copy](#) operation. Such access occurs in the `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, `VK_PIPELINE_STAGE_2_CLEAR_BIT`, or `VK_PIPELINE_STAGE_2_RESOLVE_BIT` pipeline stages.
- `VK_ACCESS_2_HOST_READ_BIT` specifies read access by a host operation. Accesses of this type are not performed through a resource, but directly on memory. Such access occurs in the `VK_PIPELINE_STAGE_2_HOST_BIT` pipeline stage.
- `VK_ACCESS_2_HOST_WRITE_BIT` specifies write access by a host operation. Accesses of this type are not performed through a resource, but directly on memory. Such access occurs in the `VK_PIPELINE_STAGE_2_HOST_BIT` pipeline stage.
- `VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT` specifies write access to a transform feedback buffer made when transform feedback is active. Such access occurs in the `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT` pipeline stage.
- `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT` specifies read access to a transform feedback counter buffer which is read when [vkCmdBeginTransformFeedbackEXT](#) executes. Such access occurs in the `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT` pipeline stage.
- `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT` specifies write access to a transform feedback counter buffer which is written when [vkCmdEndTransformFeedbackEXT](#) executes. Such access occurs in the `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT` pipeline stage.
- `VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_EXT` specifies reads from buffer inputs to [vkCmdPreprocessGeneratedCommandsEXT](#). Such access occurs in the `VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT` pipeline stage.
- `VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_EXT` specifies writes to the target command buffer

preprocess outputs. Such access occurs in the `VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT` pipeline stage.

- `VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT` specifies read access to `color attachments`, including `advanced blend operations`. Such access occurs in the `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage.
- `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR` specifies read access to an acceleration structure as part of a trace, build, or copy command, or to an `acceleration structure scratch buffer` as part of a build command. Such access occurs in the `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR` pipeline stage or `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` pipeline stage.
- `VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR` specifies write access to an acceleration structure or `acceleration structure scratch buffer` as part of a build or copy command. Such access occurs in the `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` pipeline stage.
- `VK_ACCESS_2_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR` specifies read access to a fragment shading rate attachment during rasterization. Such access occurs in the `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR` pipeline stage.
- `VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR` specifies read access to an image or buffer resource in a `video decode operation`. Such access occurs in the `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR` pipeline stage.
- `VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR` specifies write access to an image or buffer resource in a `video decode operation`. Such access occurs in the `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR` pipeline stage.
- `VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR` specifies read access to an image or buffer resource in a `video encode operation`. Such access occurs in the `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR` pipeline stage.
- `VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR` specifies write access to an image or buffer resource in a `video encode operation`. Such access occurs in the `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR` pipeline stage.
- `VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT` specifies write access to a micromap object. Such access occurs in the `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` pipeline stage.
- `VK_ACCESS_2_MICROMAP_READ_BIT_EXT` specifies read access to a micromap object. Such access occurs in the `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` and `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` pipeline stages.

NOTE

In situations where an application wishes to select all access types for a given set of pipeline stages, `VK_ACCESS_2_MEMORY_READ_BIT` or `VK_ACCESS_2_MEMORY_WRITE_BIT` can be used. This is particularly useful when specifying stages that only have a single access type.

NOTE

The `VkAccessFlags2` bitmask goes beyond the 31 individual bit flags allowable within a C99 enum, which is how `VkAccessFlagBits` is defined. The first 31 values are common to both, and are interchangeable.

VkAccessFlags2 is a bitmask type for setting a mask of zero or more **VkAccessFlagBits2**:

```
typedef VkFlags64 VkAccessFlags2;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2  
typedef VkAccessFlags2 VkAccessFlags2KHR;
```

Bits which **can** be set in the **srcAccessMask** and **dstAccessMask** members of **VkSubpassDependency**, **VkSubpassDependency2**, **VkMemoryBarrier**, **VkBufferMemoryBarrier**, and **VkImageMemoryBarrier**, specifying access behavior, are:

```

// Provided by VK_VERSION_1_0
typedef enum VkAccessFlagBits {
    VK_ACCESS_INDIRECT_COMMAND_READ_BIT = 0x00000001,
    VK_ACCESS_INDEX_READ_BIT = 0x00000002,
    VK_ACCESS_VERTEX_ATTRIBUTE_READ_BIT = 0x00000004,
    VK_ACCESS_UNIFORM_READ_BIT = 0x00000008,
    VK_ACCESS_INPUT_ATTACHMENT_READ_BIT = 0x00000010,
    VK_ACCESS_SHADER_READ_BIT = 0x00000020,
    VK_ACCESS_SHADER_WRITE_BIT = 0x00000040,
    VK_ACCESS_COLOR_ATTACHMENT_READ_BIT = 0x00000080,
    VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT = 0x00000100,
    VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT = 0x00000200,
    VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT = 0x00000400,
    VK_ACCESS_TRANSFER_READ_BIT = 0x00000800,
    VK_ACCESS_TRANSFER_WRITE_BIT = 0x00001000,
    VK_ACCESS_HOST_READ_BIT = 0x00002000,
    VK_ACCESS_HOST_WRITE_BIT = 0x00004000,
    VK_ACCESS_MEMORY_READ_BIT = 0x00008000,
    VK_ACCESS_MEMORY_WRITE_BIT = 0x00010000,
    VK_ACCESS_NONE = 0,
    // Provided by VK_EXT_transform_feedback
    VK_ACCESS_TRANSFORM_FEEDBACK_WRITE_BIT_EXT = 0x02000000,
    // Provided by VK_EXT_transform_feedback
    VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT = 0x04000000,
    // Provided by VK_EXT_transform_feedback
    VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT = 0x08000000,
    // Provided by VK_EXT_blend_operation_advanced
    VK_ACCESS_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT = 0x00080000,
    // Provided by VK_KHR_acceleration_structure
    VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR = 0x00200000,
    // Provided by VK_KHR_acceleration_structure
    VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR = 0x00400000,
    // Provided by VK_KHR_fragment_shading_rate
    VK_ACCESS_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR = 0x00800000,
    VK_ACCESS_COMMAND_PREPROCESS_READ_BIT_NV = 0x00020000,
    VK_ACCESS_COMMAND_PREPROCESS_WRITE_BIT_NV = 0x00040000,
    // Provided by VK_KHR_synchronization2
    VK_ACCESS_NONE_KHR = VK_ACCESS_NONE,
    // Provided by VK_EXT_device_generated_commands
    VK_ACCESS_COMMAND_PREPROCESS_READ_BIT_EXT =
VK_ACCESS_COMMAND_PREPROCESS_READ_BIT_NV,
    // Provided by VK_EXT_device_generated_commands
    VK_ACCESS_COMMAND_PREPROCESS_WRITE_BIT_EXT =
VK_ACCESS_COMMAND_PREPROCESS_WRITE_BIT_NV,
} VkAccessFlagBits;

```

These values all have the same meaning as the equivalently named values for [VkAccessFlags2](#).

- **VK_ACCESS_NONE** specifies no accesses.

- `VK_ACCESS_MEMORY_READ_BIT` specifies all read accesses. It is always valid in any access mask, and is treated as equivalent to setting all `READ` access flags that are valid where it is used.
- `VK_ACCESS_MEMORY_WRITE_BIT` specifies all write accesses. It is always valid in any access mask, and is treated as equivalent to setting all `WRITE` access flags that are valid where it is used.
- `VK_ACCESS_INDIRECT_COMMAND_READ_BIT` specifies read access to indirect command data read as part of an indirect build, trace, drawing or dispatching command. Such access occurs in the `VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT` pipeline stage.
- `VK_ACCESS_INDEX_READ_BIT` specifies read access to an index buffer as part of an indexed drawing command, bound by `vkCmdBindIndexBuffer2KHR` and `vkCmdBindIndexBuffer`. Such access occurs in the `VK_PIPELINE_STAGE_VERTEX_INPUT_BIT` pipeline stage.
- `VK_ACCESS_VERTEX_ATTRIBUTE_READ_BIT` specifies read access to a vertex buffer as part of a drawing command, bound by `vkCmdBindVertexBuffers`. Such access occurs in the `VK_PIPELINE_STAGE_VERTEX_INPUT_BIT` pipeline stage.
- `VK_ACCESS_UNIFORM_READ_BIT` specifies read access to a `uniform buffer` in any shader pipeline stage.
- `VK_ACCESS_INPUT_ATTACHMENT_READ_BIT` specifies read access to an `input attachment` within a render pass during fragment shading. Such access occurs in the `VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT` pipeline stage.
- `VK_ACCESS_SHADER_READ_BIT` specifies read access to a `uniform texel buffer`, `sampled image`, `storage buffer`, `physical storage buffer`, `shader binding table`, `storage texel buffer`, or `storage image` in any shader pipeline stage.
- `VK_ACCESS_SHADER_WRITE_BIT` specifies write access to a `storage buffer`, `physical storage buffer`, `storage texel buffer`, or `storage image` in any shader pipeline stage.
- `VK_ACCESS_COLOR_ATTACHMENT_READ_BIT` specifies read access to a `color attachment`, such as via `blending` (other than `advanced blend operations`), `logic operations` or certain `render pass load operations` in the `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage or via `fragment shader tile image reads` in the `VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT` pipeline stage.
- `VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT` specifies write access to a `color`, `resolve`, or `depth/stencil resolve attachment` during a `render pass` or via certain render pass `load` and `store` operations. Such access occurs in the `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage.
- `VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT` specifies read access to a `depth/stencil attachment`, via `depth or stencil operations` or certain `render pass load operations` in the `VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT` or `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT` pipeline stages or via `fragment shader tile image reads` in the `VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT` pipeline stage.
- `VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT` specifies write access to a `depth/stencil attachment`, via `depth or stencil operations` or certain render pass `load` and `store` operations. Such access occurs in the `VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT` or `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT` pipeline stages.
- `VK_ACCESS_TRANSFER_READ_BIT` specifies read access to an image or buffer in a `copy` operation. Such access occurs in the `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT` pipeline stage.
- `VK_ACCESS_TRANSFER_WRITE_BIT` specifies write access to an image or buffer in a `clear` or `copy`

operation. Such access occurs in the `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT` pipeline stage.

- `VK_ACCESS_HOST_READ_BIT` specifies read access by a host operation. Accesses of this type are not performed through a resource, but directly on memory. Such access occurs in the `VK_PIPELINE_STAGE_HOST_BIT` pipeline stage.
- `VK_ACCESS_HOST_WRITE_BIT` specifies write access by a host operation. Accesses of this type are not performed through a resource, but directly on memory. Such access occurs in the `VK_PIPELINE_STAGE_HOST_BIT` pipeline stage.
- `VK_ACCESS_TRANSFORM_FEEDBACK_WRITE_BIT_EXT` specifies write access to a transform feedback buffer made when transform feedback is active. Such access occurs in the `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT` pipeline stage.
- `VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT` specifies read access to a transform feedback counter buffer which is read when `vkCmdBeginTransformFeedbackEXT` executes. Such access occurs in the `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT` pipeline stage.
- `VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT` specifies write access to a transform feedback counter buffer which is written when `vkCmdEndTransformFeedbackEXT` executes. Such access occurs in the `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT` pipeline stage.
- `VK_ACCESS_COMMAND_PREPROCESS_READ_BIT_EXT` specifies reads from buffer inputs to `vkCmdPreprocessGeneratedCommandsEXT`. Such access occurs in the `VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_EXT` pipeline stage.
- `VK_ACCESS_COMMAND_PREPROCESS_WRITE_BIT_EXT` specifies writes to the target command buffer preprocess outputs in `vkCmdPreprocessGeneratedCommandsEXT`. Such access occurs in the `VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_EXT` pipeline stage.
- `VK_ACCESS_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT` specifies read access to [color attachments](#), including [advanced blend operations](#). Such access occurs in the `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage.
- `VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR` specifies read access to an acceleration structure as part of a trace, build, or copy command, or to an [acceleration structure scratch buffer](#) as part of a build command. Such access occurs in the `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR` pipeline stage or `VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` pipeline stage.
- `VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR` specifies write access to an acceleration structure or [acceleration structure scratch buffer](#) as part of a build or copy command. Such access occurs in the `VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` pipeline stage.
- `VK_ACCESS_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR` specifies read access to a fragment shading rate attachment during rasterization. Such access occurs in the `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR` pipeline stage.

Certain access types are only performed by a subset of pipeline stages. Any synchronization command that takes both stage masks and access masks uses both to define the [access scopes](#) - only the specified access types performed by the specified stages are included in the access scope. An application **must** not specify an access flag in a synchronization command if it does not include a pipeline stage in the corresponding stage mask that is able to perform accesses of that type. The following table lists, for each access flag, which pipeline stages **can** perform that type of access.

Table 4. Supported Access Types

Access flag	Supported pipeline stages
VK_ACCESS_2_NONE	Any
VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT	VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
VK_ACCESS_2_INDEX_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT
VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT
VK_ACCESS_2_UNIFORM_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR,
VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT	VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT,
VK_ACCESS_2_SHADER_READ_BIT	VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR,
VK_ACCESS_2_SHADER_WRITE_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR,
VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT	VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT
VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT	VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT

Access flag	Supported pipeline stages
VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT	VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT
VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT	VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT
VK_ACCESS_2_TRANSFER_READ_BIT	VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
VK_ACCESS_2_TRANSFER_WRITE_BIT	VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT
VK_ACCESS_2_HOST_READ_BIT	VK_PIPELINE_STAGE_2_HOST_BIT
VK_ACCESS_2_HOST_WRITE_BIT	VK_PIPELINE_STAGE_2_HOST_BIT
VK_ACCESS_2_MEMORY_READ_BIT	Any
VK_ACCESS_2_MEMORY_WRITE_BIT	Any
VK_ACCESS_2_SHADER_SAMPLED_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR,

Access flag	Supported pipeline stages
VK_ACCESS_2_SHADER_STORAGE_READ_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR,
VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR,
VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR	VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR	VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR
VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR	VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR	VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR
VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT	VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT	VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT	VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT
VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_NV	VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_NV
VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_NV	VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_NV
VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_EXT	VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT
VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_EXT	VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT
VK_ACCESS_2_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR	VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR

Access flag	Supported pipeline stages
VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR	VK_PIPELINE_STAGE_2_VERTEX_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT, VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT, VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT, VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT, VK_PIPELINE_STAGE_2_COMPUTE_SHADER_BIT, VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR,
VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR	VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR
VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT	VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT
VK_ACCESS_2_MICROMAP_READ_BIT_EXT	VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR
VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT	VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkAccessFlags;
```

VkAccessFlags is a bitmask type for setting a mask of zero or more **VkAccessFlagBits**.

If a memory object does not have the **VK_MEMORY_PROPERTY_HOST_COHERENT_BIT** property, then **vkFlushMappedMemoryRanges** **must** be called in order to guarantee that writes to the memory object from the host are made available to the host domain, where they **can** be further made available to the device domain via a domain operation. Similarly, **vkInvalidateMappedMemoryRanges** **must** be called to guarantee that writes which are available to the host domain are made visible to host operations.

If the memory object does have the **VK_MEMORY_PROPERTY_HOST_COHERENT_BIT** property flag, writes to the memory object from the host are automatically made available to the host domain. Similarly, writes made available to the host domain are automatically made visible to the host.

NOTE

Queue submission commands automatically perform a **domain operation from host to device** for all writes performed before the command executes, so in most cases an explicit memory barrier is not needed for this case. In the few circumstances where a submit does not occur between the host write and the device read access, writes **can** be made available by using an explicit memory barrier.

7.1.4. Framebuffer Region Dependencies

Pipeline stages that operate on, or with respect to, the framebuffer are collectively the *framebuffer-space* pipeline stages. These stages are:

- `VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT`
- `VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT`
- `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT`
- `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT`

For these pipeline stages, an execution or memory dependency from the first set of operations to the second set **can** either be a single *framebuffer-global* dependency, or split into multiple *framebuffer-local* dependencies. A dependency with non-framebuffer-space pipeline stages is neither framebuffer-global nor framebuffer-local.

A *framebuffer region* is a set of sample (x, y, layer, sample) coordinates that is a subset of the entire framebuffer.

Both *synchronization scopes* of a framebuffer-local dependency include only the operations performed within corresponding framebuffer regions (as defined below). No ordering guarantees are made between different framebuffer regions for a framebuffer-local dependency.

Both *synchronization scopes* of a framebuffer-global dependency include operations on all framebuffer-regions.

If the first synchronization scope includes operations on pixels/fragments with N samples and the second synchronization scope includes operations on pixels/fragments with M samples, where N does not equal M, then a framebuffer region containing all samples at a given (x, y, layer) coordinate in the first synchronization scope corresponds to a region containing all samples at the same coordinate in the second synchronization scope. In other words, it is a pixel granularity dependency. If N equals M, then a framebuffer region containing a single (x, y, layer, sample) coordinate in the first synchronization scope corresponds to a region containing the same sample at the same coordinate in the second synchronization scope. In other words, it is a sample granularity dependency.

NOTE

Since fragment shader invocations are not specified to run in any particular groupings, the size of a framebuffer region is implementation-dependent, not known to the application, and **must** be assumed to be no larger than specified above.

NOTE

Practically, the pixel vs. sample granularity dependency means that if an input attachment has a different number of samples than the pipeline's `rasterizationSamples`, then a fragment **can** access any sample in the input attachment's pixel even if it only uses framebuffer-local dependencies. If the input attachment has the same number of samples, then the fragment **can** only access the covered samples in its input `SampleMask` (i.e. the fragment operations happen-after a framebuffer-local dependency for each sample the fragment covers). To access samples that are not covered, a framebuffer-global dependency is required.

If a synchronization command includes a `dependencyFlags` parameter, and specifies the `VK_DEPENDENCY_BY_REGION_BIT` flag, then it defines framebuffer-local dependencies for the framebuffer-space pipeline stages in that synchronization command, for all framebuffer regions. If no `dependencyFlags` parameter is included, or the `VK_DEPENDENCY_BY_REGION_BIT` flag is not specified, then a framebuffer-global dependency is specified for those stages. The `VK_DEPENDENCY_BY_REGION_BIT` flag does not affect the dependencies between non-framebuffer-space pipeline stages, nor does it affect the dependencies between framebuffer-space and non-framebuffer-space pipeline stages.

NOTE

Framebuffer-local dependencies are more efficient for most architectures; particularly tile-based architectures - which can keep framebuffer-regions entirely in on-chip registers and thus avoid external bandwidth across such a dependency. Including a framebuffer-global dependency in your rendering will usually force all implementations to flush data to memory, or to a higher level cache, breaking any potential locality optimizations.

7.1.5. View-Local Dependencies

In a render pass instance that has `multiview` enabled, dependencies **can** be either view-local or view-global.

A view-local dependency only includes operations from a single `source view` from the source subpass in the first synchronization scope, and only includes operations from a single `destination view` from the destination subpass in the second synchronization scope. A view-global dependency includes all views in the view mask of the source and destination subpasses in the corresponding synchronization scopes.

If a synchronization command includes a `dependencyFlags` parameter and specifies the `VK_DEPENDENCY_VIEW_LOCAL_BIT` flag, then it defines view-local dependencies for that synchronization command, for all views. If no `dependencyFlags` parameter is included or the `VK_DEPENDENCY_VIEW_LOCAL_BIT` flag is not specified, then a view-global dependency is specified.

7.1.6. Device-Local Dependencies

Dependencies **can** be either device-local or non-device-local. A device-local dependency acts as multiple separate dependencies, one for each physical device that executes the synchronization command, where each dependency only includes operations from that physical device in both synchronization scopes. A non-device-local dependency is a single dependency where both synchronization scopes include operations from all physical devices that participate in the synchronization command. For subpass dependencies, all physical devices in the `VkDeviceGroupRenderPassBeginInfo::deviceMask` participate in the dependency, and for pipeline barriers all physical devices that are set in the command buffer's current device mask participate in the dependency.

If a synchronization command includes a `dependencyFlags` parameter and specifies the `VK_DEPENDENCY_DEVICE_GROUP_BIT` flag, then it defines a non-device-local dependency for that synchronization command. If no `dependencyFlags` parameter is included or the `VK_DEPENDENCY_DEVICE_GROUP_BIT` flag is not specified, then it defines device-local dependencies for

that synchronization command, for all participating physical devices.

Semaphore and event dependencies are device-local and only execute on the one physical device that performs the dependency.

7.2. Implicit Synchronization Guarantees

A small number of implicit ordering guarantees are provided by Vulkan, ensuring that the order in which commands are submitted is meaningful, and avoiding unnecessary complexity in common operations.

Submission order is a fundamental ordering in Vulkan, giving meaning to the order in which [action and synchronization commands](#) are recorded and submitted to a single queue. Explicit and implicit ordering guarantees between commands in Vulkan all work on the premise that this ordering is meaningful. This order does not itself define any execution or memory dependencies; synchronization commands and other orderings within the API use this ordering to define their scopes.

Submission order for any given set of commands is based on the order in which they were recorded to command buffers and then submitted. This order is determined as follows:

1. The initial order is determined by the order in which [vkQueueSubmit](#) and [vkQueueSubmit2KHR](#) commands are executed on the host, for a single queue, from first to last.
2. The order in which [VkSubmitInfo](#) structures are specified in the [pSubmits](#) parameter of [vkQueueSubmit](#), or in which [VkSubmitInfo2](#) structures are specified in the [pSubmits](#) parameter of [vkQueueSubmit2KHR](#), from lowest index to highest.
3. The order in which command buffers are specified in the [pCommandBuffers](#) member of [VkSubmitInfo](#) or [VkSubmitInfo2](#) from lowest index to highest.
4. The order in which commands outside of a render pass were recorded to a command buffer on the host, from first to last.
5. The order in which commands inside a single subpass were recorded to a command buffer on the host, from first to last.

NOTE

When using a [render pass object](#) with multiple subpasses, commands in different subpasses have no defined submission order relative to each other, regardless of the order in which the subpasses were recorded. Commands within a subpass are still ordered relative to other commands in the same subpass, and those outside of the render pass.

[State commands](#) do not execute any operations on the device, instead they set the state of the command buffer when they execute on the host, in the order that they are recorded. [Action commands](#) consume the current state of the command buffer when they are recorded, and will execute state changes on the device as required to match the recorded state.

[The order of primitives passing through the graphics pipeline](#) and [image layout transitions as part of an image memory barrier](#) provide additional guarantees based on submission order.

Execution of [pipeline stages](#) within a given command also has a loose ordering, dependent only on a single command.

Signal operation order is a fundamental ordering in Vulkan, giving meaning to the order in which semaphore and fence signal operations occur when submitted to a single queue. The signal operation order for queue operations is determined as follows:

1. The initial order is determined by the order in which [vkQueueSubmit](#) and [vkQueueSubmit2KHR](#) commands are executed on the host, for a single queue, from first to last.
2. The order in which [VkSubmitInfo](#) structures are specified in the [pSubmits](#) parameter of [vkQueueSubmit](#), or in which [VkSubmitInfo2](#) structures are specified in the [pSubmits](#) parameter of [vkQueueSubmit2KHR](#), from lowest index to highest.
3. The fence signal operation defined by the [fence](#) parameter of a [vkQueueSubmit](#) or [vkQueueSubmit2KHR](#) or [vkQueueBindSparse](#) command is ordered after all semaphore signal operations defined by that command.

Semaphore signal operations defined by a single [VkSubmitInfo](#) or [VkSubmitInfo2](#) or [VkBindSparseInfo](#) structure are unordered with respect to other semaphore signal operations defined within the same structure.

The [vkSignalSemaphoreKHR](#) command does not execute on a queue but instead performs the signal operation from the host. The semaphore signal operation defined by executing a [vkSignalSemaphoreKHR](#) command happens-after the [vkSignalSemaphoreKHR](#) command is invoked and happens-before the command returns.

When signaling timeline semaphores, it is the responsibility of the application to ensure that they are ordered such that the semaphore value is strictly increasing. Because the first synchronization scope for a semaphore signal operation contains all semaphore signal operations which occur earlier in submission order, all semaphore signal operations contained in any given batch are guaranteed to happen-after all semaphore signal operations contained in any previous batches. However, no ordering guarantee is provided between the semaphore signal operations defined within a single batch. This, combined with the requirement that timeline semaphore values strictly increase, means that it is invalid to signal the same timeline semaphore twice within a single batch.

NOTE

If an application wishes to ensure that some semaphore signal operation happens-after some other semaphore signal operation, it can submit a separate batch containing only semaphore signal operations, which will happen-after the semaphore signal operations in any earlier batches.

When signaling a semaphore from the host, the only ordering guarantee is that the signal operation happens-after when [vkSignalSemaphoreKHR](#) is called and happens-before it returns. Therefore, it is invalid to call [vkSignalSemaphoreKHR](#) while there are any outstanding signal operations on that semaphore from any queue submissions unless those queue submissions have some dependency which ensures that they happen-after the host signal operation. One example of this would be if the pending signal operation is, itself, waiting on the same semaphore at a lower

value and the call to `vkSignalSemaphoreKHR` signals that lower value. Furthermore, if there are two or more processes or threads signaling the same timeline semaphore from the host, the application must ensure that the `vkSignalSemaphoreKHR` with the lower semaphore value returns before `vkSignalSemaphoreKHR` is called with the higher value.

7.3. Fences

Fences are a synchronization primitive that **can** be used to insert a dependency from a queue to the host. Fences have two states - signaled and unsignaled. A fence **can** be signaled as part of the execution of a `queue submission` command. Fences **can** be unsignaled on the host with `vkResetFences`. Fences **can** be waited on by the host with the `vkWaitForFences` command, and the current state **can** be queried with `vkGetFenceStatus`.

The internal data of a fence **may** include a reference to any resources and pending work associated with signal or unsignal operations performed on that fence object, collectively referred to as the fence's *payload*. Mechanisms to import and export that internal data to and from fences are provided [below](#). These mechanisms indirectly enable applications to share fence state between two or more fences and other synchronization primitives across process and API boundaries.

Fences are represented by `VkFence` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkFence)
```

To create a fence, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateFence(
    VkDevice                                device,
    const VkFenceCreateInfo*               pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkFence*                                pFence);
```

- `device` is the logical device that creates the fence.
- `pCreateInfo` is a pointer to a `VkFenceCreateInfo` structure containing information about how the fence is to be created.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pFence` is a pointer to a handle in which the resulting fence object is returned.

Valid Usage (Implicit)

- VUID-vkCreateFence-device-parameter `device` **must** be a valid `VkDevice` handle

- VUID-vkCreateFence-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkFenceCreateInfo](#) structure
- VUID-vkCreateFence-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateFence-pFence-parameter
pFence **must** be a valid pointer to a [VkFence](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

The [VkFenceCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkFenceCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkFenceCreateFlags flags;
} VkFenceCreateInfo;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkFenceCreateFlagBits](#) specifying the initial state and behavior of the fence.

Valid Usage (Implicit)

- VUID-VkFenceCreateInfo-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_FENCE_CREATE_INFO**
- VUID-VkFenceCreateInfo-pNext-pNext
Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either **NULL** or a pointer to a valid instance of [VkExportFenceCreateInfo](#) or [VkExportFenceWin32HandleInfoKHR](#)
- VUID-VkFenceCreateInfo-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkFenceCreateInfo-flags-parameter

flags must be a valid combination of [VkFenceCreateFlagBits](#) values

```
// Provided by VK_VERSION_1_0
typedef enum VkFenceCreateFlagBits {
    VK_FENCE_CREATE_SIGNALED_BIT = 0x00000001,
} VkFenceCreateFlagBits;
```

- **VK_FENCE_CREATE_SIGNALED_BIT** specifies that the fence object is created in the signaled state. Otherwise, it is created in the unsignaled state.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkFenceCreateFlags;
```

VkFenceCreateFlags is a bitmask type for setting a mask of zero or more [VkFenceCreateFlagBits](#).

To create a fence whose payload **can** be exported to external handles, add a [VkExportFenceCreateInfo](#) structure to the **pNext** chain of the [VkFenceCreateInfo](#) structure. The [VkExportFenceCreateInfo](#) structure is defined as:

```
typedef struct VkExportFenceCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkExternalFenceHandleTypeFlags handleTypes;
} VkExportFenceCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_external_fence
typedef VkExportFenceCreateInfo VkExportFenceCreateInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **handleTypes** is a bitmask of [VkExternalFenceHandleTypeFlagBits](#) specifying one or more fence handle types the application **can** export from the resulting fence. The application **can** request multiple handle types for the same fence.

Valid Usage

- VUID-VkExportFenceCreateInfo-handleTypes-01446

The bits in **handleTypes** **must** be supported and compatible, as reported by [VkExternalFenceProperties](#)

Valid Usage (Implicit)

- VUID-VkExportFenceCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EXPORT_FENCE_CREATE_INFO`
- VUID-VkExportFenceCreateInfo-handleTypes-parameter
`handleTypes` **must** be a valid combination of `VkExternalFenceHandleTypeFlagBits` values

To specify additional attributes of NT handles exported from a fence, add a `VkExportFenceWin32HandleInfoKHR` structure to the `pNext` chain of the `VkFenceCreateInfo` structure. The `VkExportFenceWin32HandleInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_fence_win32
typedef struct VkExportFenceWin32HandleInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    const SECURITY_ATTRIBUTES* pAttributes;
    DWORD              dwAccess;
    LPCWSTR            name;
} VkExportFenceWin32HandleInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pAttributes` is a pointer to a Windows `SECURITY_ATTRIBUTES` structure specifying security attributes of the handle.
- `dwAccess` is a `DWORD` specifying access rights of the handle.
- `name` is a null-terminated UTF-16 string to associate with the underlying synchronization primitive referenced by NT handles exported from the created fence.

If `VkExportFenceCreateInfo` is not included in the same `pNext` chain, this structure is ignored.

If `VkExportFenceCreateInfo` is included in the `pNext` chain of `VkFenceCreateInfo` with a Windows `handleType`, but either `VkExportFenceWin32HandleInfoKHR` is not included in the `pNext` chain, or it is included but `pAttributes` is `NULL`, default security descriptor values will be used, and child processes created by the application will not inherit the handle, as described in the MSDN documentation for “Synchronization Object Security and Access Rights”¹. Further, if the structure is not present, the access rights will be

`DXGI_SHARED_RESOURCE_READ | DXGI_SHARED_RESOURCE_WRITE`

for handles of the following types:

`VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT`

Valid Usage

- VUID-VkExportFenceWin32HandleInfoKHR-handleTypes-01447
If `VkExportFenceCreateInfo::handleTypes` does not include `VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT`, a `VkExportFenceWin32HandleInfoKHR` structure **must** not be included in the `pNext` chain of `VkFenceCreateInfo`

Valid Usage (Implicit)

- VUID-VkExportFenceWin32HandleInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EXPORT_FENCE_WIN32_HANDLE_INFO_KHR`
- VUID-VkExportFenceWin32HandleInfoKHR-pAttributes-parameter
If `pAttributes` is not `NULL`, `pAttributes` **must** be a valid pointer to a valid `SECURITY_ATTRIBUTES` value

To export a Windows handle representing the state of a fence, call:

```
// Provided by VK_KHR_external_fence_win32
VkResult vkGetFenceWin32HandleKHR(
    VkDevice device,
    const VkFenceGetWin32HandleInfoKHR* pGetWin32HandleInfo,
    HANDLE* pHandle);
```

- `device` is the logical device that created the fence being exported.
- `pGetWin32HandleInfo` is a pointer to a `VkFenceGetWin32HandleInfoKHR` structure containing parameters of the export operation.
- `pHandle` will return the Windows handle representing the fence state.

For handle types defined as NT handles, the handles returned by `vkGetFenceWin32HandleKHR` are owned by the application. To avoid leaking resources, the application **must** release ownership of them using the `CloseHandle` system call when they are no longer needed.

Exporting a Windows handle from a fence **may** have side effects depending on the transference of the specified handle type, as described in [Importing Fence Payloads](#).

Valid Usage (Implicit)

- VUID-vkGetFenceWin32HandleKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetFenceWin32HandleKHR-pGetWin32HandleInfo-parameter
`pGetWin32HandleInfo` **must** be a valid pointer to a valid `VkFenceGetWin32HandleInfoKHR` structure
- VUID-vkGetFenceWin32HandleKHR-pHandle-parameter

pHandle **must** be a valid pointer to a **HANDLE** value

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_TOO_MANY_OBJECTS**
- **VK_ERROR_OUT_OF_HOST_MEMORY**

The **VkFenceGetWin32HandleInfoKHR** structure is defined as:

```
// Provided by VK_KHR_external_fence_win32
typedef struct VkFenceGetWin32HandleInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkFence                  fence;
    VkExternalFenceHandleTypeFlagBits handleType;
} VkFenceGetWin32HandleInfoKHR;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **fence** is the fence from which state will be exported.
- **handleType** is a **VkExternalFenceHandleTypeFlagBits** value specifying the type of handle requested.

The properties of the handle returned depend on the value of **handleType**. See **VkExternalFenceHandleTypeFlagBits** for a description of the properties of the defined external fence handle types.

Valid Usage

- VUID-VkFenceGetWin32HandleInfoKHR-handleType-01448
handleType **must** have been included in **VkExportFenceCreateInfo::handleTypes** when the **fence**'s current payload was created
- VUID-VkFenceGetWin32HandleInfoKHR-handleType-01449
If **handleType** is defined as an NT handle, **vkGetFenceWin32HandleKHR** **must** be called no more than once for each valid unique combination of **fence** and **handleType**
- VUID-VkFenceGetWin32HandleInfoKHR-fence-01450
fence **must** not currently have its payload replaced by an imported payload as described below in **Importing Fence Payloads** unless that imported payload's handle type was included in **VkExternalFenceProperties::exportFromImportedHandleTypes** for **handleType**

- VUID-VkFenceGetWin32HandleInfoKHR-handleType-01451
If **handleType** refers to a handle type with copy payload transference semantics, **fence** **must** be signaled, or have an associated **fence signal operation** pending execution
- VUID-VkFenceGetWin32HandleInfoKHR-handleType-01452
handleType **must** be defined as an NT handle or a global share handle

Valid Usage (Implicit)

- VUID-VkFenceGetWin32HandleInfoKHR-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_FENCE_GET_WIN32_HANDLE_INFO_KHR**
- VUID-VkFenceGetWin32HandleInfoKHR-pNext-pNext
pNext **must** be **NULL**
- VUID-VkFenceGetWin32HandleInfoKHR-fence-parameter
fence **must** be a valid **VkFence** handle
- VUID-VkFenceGetWin32HandleInfoKHR-handleType-parameter
handleType **must** be a valid **VkExternalFenceHandleTypeFlagBits** value

To export a POSIX file descriptor representing the payload of a fence, call:

```
// Provided by VK_KHR_external_fence_fd
VkResult vkGetFenceFdKHR(
    VkDevice          device,
    const VkFenceGetFdInfoKHR* pGetFdInfo,
    int*              pFd);
```

- **device** is the logical device that created the fence being exported.
- **pGetFdInfo** is a pointer to a **VkFenceGetFdInfoKHR** structure containing parameters of the export operation.
- **pFd** will return the file descriptor representing the fence payload.

Each call to **vkGetFenceFdKHR** **must** create a new file descriptor and transfer ownership of it to the application. To avoid leaking resources, the application **must** release ownership of the file descriptor when it is no longer needed.

NOTE

Ownership can be released in many ways. For example, the application can call **close()** on the file descriptor, or transfer ownership back to Vulkan by using the file descriptor to import a fence payload.

If **pGetFdInfo->handleType** is **VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT** and the fence is signaled at the time **vkGetFenceFdKHR** is called, **pFd** **may** return the value **-1** instead of a valid file descriptor.

Where supported by the operating system, the implementation **must** set the file descriptor to be closed automatically when an **execve** system call is made.

Exporting a file descriptor from a fence **may** have side effects depending on the transference of the specified handle type, as described in [Importing Fence State](#).

Valid Usage (Implicit)

- VUID-vkGetFenceFdKHR-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkGetFenceFdKHR-pGetFdInfo-parameter
pGetFdInfo **must** be a valid pointer to a valid [VkFenceGetFdInfoKHR](#) structure
- VUID-vkGetFenceFdKHR-pFd-parameter
pFd **must** be a valid pointer to an **int** value

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_TOO_MANY_OBJECTS**
- **VK_ERROR_OUT_OF_HOST_MEMORY**

The **VkFenceGetFdInfoKHR** structure is defined as:

```
// Provided by VK_KHR_external_fence_fd
typedef struct VkFenceGetFdInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkFence                  fence;
    VkExternalFenceHandleTypeFlagBits handleType;
} VkFenceGetFdInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **fence** is the fence from which state will be exported.
- **handleType** is a [VkExternalFenceHandleTypeFlagBits](#) value specifying the type of handle requested.

The properties of the file descriptor returned depend on the value of **handleType**. See [VkExternalFenceHandleTypeFlagBits](#) for a description of the properties of the defined external fence handle types.

Valid Usage

- VUID-VkFenceGetFdInfoKHR-handleType-01453
`handleType` **must** have been included in `VkExportFenceCreateInfo::handleTypes` when `fence`'s current payload was created
- VUID-VkFenceGetFdInfoKHR-handleType-01454
If `handleType` refers to a handle type with copy payload transference semantics, `fence` **must** be signaled, or have an associated [fence signal operation](#) pending execution
- VUID-VkFenceGetFdInfoKHR-fence-01455
`fence` **must** not currently have its payload replaced by an imported payload as described below in [Importing Fence Payloads](#) unless that imported payload's handle type was included in `VkExternalFenceProperties::exportFromImportedHandleTypes` for `handleType`
- VUID-VkFenceGetFdInfoKHR-handleType-01456
`handleType` **must** be defined as a POSIX file descriptor handle

Valid Usage (Implicit)

- VUID-VkFenceGetFdInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_FENCE_GET_FD_INFO_KHR`
- VUID-VkFenceGetFdInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkFenceGetFdInfoKHR-fence-parameter
`fence` **must** be a valid [VkFence](#) handle
- VUID-VkFenceGetFdInfoKHR-handleType-parameter
`handleType` **must** be a valid [VkExternalFenceHandleTypeFlagBits](#) value

To destroy a fence, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyFence(
    VkDevice          device,
    VkFence           fence,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the fence.
- `fence` is the handle of the fence to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyFence-fence-01120

All **queue submission** commands that refer to **fence** **must** have completed execution

- VUID-vkDestroyFence-fence-01121

If **VkAllocationCallbacks** were provided when **fence** was created, a compatible set of callbacks **must** be provided here

- VUID-vkDestroyFence-fence-01122

If no **VkAllocationCallbacks** were provided when **fence** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyFence-device-parameter

device **must** be a valid **VkDevice** handle

- VUID-vkDestroyFence-fence-parameter

If **fence** is not **VK_NULL_HANDLE**, **fence** **must** be a valid **VkFence** handle

- VUID-vkDestroyFence-pAllocator-parameter

If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure

- VUID-vkDestroyFence-fence-parent

If **fence** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **fence** **must** be externally synchronized

To query the status of a fence from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkGetFenceStatus(
    VkDevice          device,
    VkFence           fence);
```

- **device** is the logical device that owns the fence.
- **fence** is the handle of the fence to query.

Upon success, **vkGetFenceStatus** returns the status of the fence object, with the following return codes:

Table 5. Fence Object Status Codes

Status	Meaning
VK_SUCCESS	The fence specified by fence is signaled.

Status	Meaning
VK_NOT_READY	The fence specified by <code>fence</code> is unsignaled.
VK_ERROR_DEVICE_LOST	The device has been lost. See Lost Device .

If a [queue submission](#) command is pending execution, then the value returned by this command **may** immediately be out of date.

If the device has been lost (see [Lost Device](#)), `vkGetFenceStatus` **may** return any of the above status codes. If the device has been lost and `vkGetFenceStatus` is called repeatedly, it will eventually return either `VK_SUCCESS` or `VK_ERROR_DEVICE_LOST`.

Valid Usage (Implicit)

- VUID-vkGetFenceStatus-device-parameter
`device` **must** be a valid [VkDevice](#) handle
- VUID-vkGetFenceStatus-fence-parameter
`fence` **must** be a valid [VkFence](#) handle
- VUID-vkGetFenceStatus-fence-parent
`fence` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_NOT_READY`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

To set the state of fences to unsignaled from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetFences(
    VkDevice          device,
    uint32_t          fenceCount,
    const VkFence*    pFences);
```

- `device` is the logical device that owns the fences.

- **fenceCount** is the number of fences to reset.
- **pFences** is a pointer to an array of fence handles to reset.

If any member of **pFences** currently has its **payload imported** with temporary permanence, that fence's prior permanent payload is first restored. The remaining operations described therefore operate on the restored payload.

When **vkResetFences** is executed on the host, it defines a *fence unsignal operation* for each fence, which resets the fence to the unsignaled state.

If any member of **pFences** is already in the unsignaled state when **vkResetFences** is executed, then **vkResetFences** has no effect on that fence.

Valid Usage

- VUID-vkResetFences-pFences-01123
Each element of **pFences** **must** not be currently associated with any queue command that has not yet completed execution on that queue

Valid Usage (Implicit)

- VUID-vkResetFences-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkResetFences-pFences-parameter
pFences **must** be a valid pointer to an array of **fenceCount** valid **VkFence** handles
- VUID-vkResetFences-fenceCount-arraylength
fenceCount **must** be greater than 0
- VUID-vkResetFences-pFences-parent
Each element of **pFences** **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to each member of **pFences** **must** be externally synchronized

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

When a fence is submitted to a queue as part of a **queue submission** command, it defines a memory

dependency on the batches that were submitted as part of that command, and defines a *fence signal operation* which sets the fence to the signaled state.

The first [synchronization scope](#) includes every batch submitted in the same [queue submission](#) command. Fence signal operations that are defined by [vkQueueSubmit](#) or [vkQueueSubmit2KHR](#) additionally include in the first synchronization scope all commands that occur earlier in [submission order](#). Fence signal operations that are defined by [vkQueueSubmit](#) or [vkQueueSubmit2KHR](#) or [vkQueueBindSparse](#) additionally include in the first synchronization scope any semaphore and fence signal operations that occur earlier in [signal operation order](#).

The second [synchronization scope](#) only includes the fence signal operation.

The first [access scope](#) includes all memory access performed by the device.

The second [access scope](#) is empty.

To wait for one or more fences to enter the signaled state on the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkWaitForFences(
    VkDevice          device,
    uint32_t          fenceCount,
    const VkFence*    pFences,
    VkBool32          waitAll,
    uint64_t           timeout);
```

- `device` is the logical device that owns the fences.
- `fenceCount` is the number of fences to wait on.
- `pFences` is a pointer to an array of `fenceCount` fence handles.
- `waitAll` is the condition that **must** be satisfied to successfully unblock the wait. If `waitAll` is `VK_TRUE`, then the condition is that all fences in `pFences` are signaled. Otherwise, the condition is that at least one fence in `pFences` is signaled.
- `timeout` is the timeout period in units of nanoseconds. `timeout` is adjusted to the closest value allowed by the implementation-dependent timeout accuracy, which **may** be substantially longer than one nanosecond, and **may** be longer than the requested period.

If the condition is satisfied when `vkWaitForFences` is called, then `vkWaitForFences` returns immediately. If the condition is not satisfied at the time `vkWaitForFences` is called, then `vkWaitForFences` will block and wait until the condition is satisfied or the `timeout` has expired, whichever is sooner.

If `timeout` is zero, then `vkWaitForFences` does not wait, but simply returns the current state of the fences. `VK_TIMEOUT` will be returned in this case if the condition is not satisfied, even though no actual wait was performed.

If the condition is satisfied before the `timeout` has expired, `vkWaitForFences` returns `VK_SUCCESS`. Otherwise, `vkWaitForFences` returns `VK_TIMEOUT` after the `timeout` has expired.

If device loss occurs (see [Lost Device](#)) before the timeout has expired, `vkWaitForFences` **must** return in finite time with either `VK_SUCCESS` or `VK_ERROR_DEVICE_LOST`.

NOTE

While we guarantee that `vkWaitForFences` **must** return in finite time, no guarantees are made that it returns immediately upon device loss. However, the application can reasonably expect that the delay will be on the order of seconds and that calling `vkWaitForFences` will not result in a permanently (or seemingly permanently) dead process.

Valid Usage (Implicit)

- VUID-vkWaitForFences-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkWaitForFences-pFences-parameter
`pFences` **must** be a valid pointer to an array of `fenceCount` valid `VkFence` handles
- VUID-vkWaitForFences-fenceCount-arraylength
`fenceCount` **must** be greater than 0
- VUID-vkWaitForFences-pFences-parent
Each element of `pFences` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_TIMEOUT`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

An execution dependency is defined by waiting for a fence to become signaled, either via `vkWaitForFences` or by polling on `vkGetFenceStatus`.

The first [synchronization scope](#) includes only the fence signal operation.

The second [synchronization scope](#) includes the host operations of `vkWaitForFences` or `vkGetFenceStatus` indicating that the fence has become signaled.

NOTE

Signaling a fence and waiting on the host does not guarantee that the results of memory accesses will be visible to the host, as the access scope of a memory dependency defined by a fence only includes device access. A [memory barrier](#) or other memory dependency **must** be used to guarantee this. See the description of [host access types](#) for more information.

7.3.1. Alternate Methods to Signal Fences

Besides submitting a fence to a queue as part of a [queue submission](#) command, a fence **may** also be signaled when a particular event occurs on a device or display.

To create a fence that will be signaled when an event occurs on a device, call:

```
// Provided by VK_EXT_display_control
VkResult vkRegisterDeviceEventEXT(
    VkDevice                                device,
    const VkDeviceEventInfoEXT*             pDeviceEventInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkFence*                                pFence);
```

- **device** is a logical device on which the event **may** occur.
- **pDeviceEventInfo** is a pointer to a [VkDeviceEventInfoEXT](#) structure describing the event of interest to the application.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pFence** is a pointer to a handle in which the resulting fence object is returned.

Valid Usage (Implicit)

- VUID-vkRegisterDeviceEventEXT-device-parameter
device must be a valid [VkDevice](#) handle
- VUID-vkRegisterDeviceEventEXT-pDeviceEventInfo-parameter
pDeviceEventInfo must be a valid pointer to a valid [VkDeviceEventInfoEXT](#) structure
- VUID-vkRegisterDeviceEventEXT-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkRegisterDeviceEventEXT-pFence-parameter
pFence must be a valid pointer to a [VkFence](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**

The [VkDeviceEventInfoEXT](#) structure is defined as:

```
// Provided by VK_EXT_display_control
typedef struct VkDeviceEventInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkDeviceEventTypeEXT deviceEvent;
} VkDeviceEventInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `device` is a `VkDeviceEventTypeEXT` value specifying when the fence will be signaled.

Valid Usage (Implicit)

- VUID-VkDeviceEventInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_EVENT_INFO_EXT`
- VUID-VkDeviceEventInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDeviceEventInfoEXT-deviceEvent-parameter
`deviceEvent` **must** be a valid `VkDeviceEventTypeEXT` value

Possible values of `VkDeviceEventInfoEXT::device`, specifying when a fence will be signaled, are:

```
// Provided by VK_EXT_display_control
typedef enum VkDeviceEventTypeEXT {
    VK_DEVICE_EVENT_TYPE_DISPLAY_HOTPLUG_EXT = 0,
} VkDeviceEventTypeEXT;
```

- `VK_DEVICE_EVENT_TYPE_DISPLAY_HOTPLUG_EXT` specifies that the fence is signaled when a display is plugged into or unplugged from the specified device. Applications **can** use this notification to determine when they need to re-enumerate the available displays on a device.

To create a fence that will be signaled when an event occurs on a `VkDisplayKHR` object, call:

```
// Provided by VK_EXT_display_control
VkResult vkRegisterDisplayEventEXT(
    VkDevice          device,
    VkDisplayKHR      display,
    const VkDisplayEventInfoEXT* pDisplayEventInfo,
    const VkAllocationCallbacks* pAllocator,
    VkFence*          pFence);
```

- `device` is a logical device associated with `display`
- `display` is the display on which the event **may** occur.

- `pDisplayEventInfo` is a pointer to a `VkDisplayEventInfoEXT` structure describing the event of interest to the application.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pFence` is a pointer to a handle in which the resulting fence object is returned.

Valid Usage (Implicit)

- VUID-vkRegisterDisplayEventEXT-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkRegisterDisplayEventEXT-display-parameter `display` **must** be a valid `VkDisplayKHR` handle
- VUID-vkRegisterDisplayEventEXT-pDisplayEventInfo-parameter `pDisplayEventInfo` **must** be a valid pointer to a valid `VkDisplayEventInfoEXT` structure
- VUID-vkRegisterDisplayEventEXT-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkRegisterDisplayEventEXT-pFence-parameter `pFence` **must** be a valid pointer to a `VkFence` handle
- VUID-vkRegisterDisplayEventEXT-commonparent
Both of `device`, and `display` **must** have been created, allocated, or retrieved from the same `VkPhysicalDevice`

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`

The `VkDisplayEventInfoEXT` structure is defined as:

```
// Provided by VK_EXT_display_control
typedef struct VkDisplayEventInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkDisplayEventTypeEXT displayEvent;
} VkDisplayEventInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `displayEvent` is a `VkDisplayEventTypeEXT` specifying when the fence will be signaled.

Valid Usage (Implicit)

- VUID-VkDisplayEventInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DISPLAY_EVENT_INFO_EXT`
- VUID-VkDisplayEventInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDisplayEventInfoEXT-displayEvent-parameter
`displayEvent` **must** be a valid `VkDisplayEventTypeEXT` value

Possible values of `VkDisplayEventInfoEXT::displayEvent`, specifying when a fence will be signaled, are:

```
// Provided by VK_EXT_display_control
typedef enum VkDisplayEventTypeEXT {
    VK_DISPLAY_EVENT_TYPE_FIRST_PIXEL_OUT_EXT = 0,
} VkDisplayEventTypeEXT;
```

- `VK_DISPLAY_EVENT_TYPE_FIRST_PIXEL_OUT_EXT` specifies that the fence is signaled when the first pixel of the next display refresh cycle leaves the display engine for the display.

7.3.2. Importing Fence Payloads

Applications **can** import a fence payload into an existing fence using an external fence handle. The effects of the import operation will be either temporary or permanent, as specified by the application. If the import is temporary, the fence will be *restored* to its permanent state the next time that fence is passed to `vkResetFences`.

NOTE

Restoring a fence to its prior permanent payload is a distinct operation from resetting a fence payload. See `vkResetFences` for more detail.

Performing a subsequent temporary import on a fence before resetting it has no effect on this requirement; the next unsignal of the fence **must** still restore its last permanent state. A permanent payload import behaves as if the target fence was destroyed, and a new fence was created with the same handle but the imported payload. Because importing a fence payload temporarily or permanently detaches the existing payload from a fence, similar usage restrictions to those applied to `vkDestroyFence` are applied to any command that imports a fence payload. Which of these import types is used is referred to as the import operation's *permanence*. Each handle type supports either one or both types of permanence.

The implementation **must** perform the import operation by either referencing or copying the payload referred to by the specified external fence handle, depending on the handle's type. The import method used is referred to as the handle type's *transference*. When using handle types with reference transference, importing a payload to a fence adds the fence to the set of all fences sharing that payload. This set includes the fence from which the payload was exported. Fence signaling, waiting, and resetting operations performed on any fence in the set **must** behave as if the set were

a single fence. Importing a payload using handle types with copy transference creates a duplicate copy of the payload at the time of import, but makes no further reference to it. Fence signaling, waiting, and resetting operations performed on the target of copy imports **must** not affect any other fence or payload.

Export operations have the same transference as the specified handle type's import operations. Additionally, exporting a fence payload to a handle with copy transference has the same side effects on the source fence's payload as executing a fence reset operation. If the fence was using a temporarily imported payload, the fence's prior permanent payload will be restored.

NOTE

The tables [Handle Types Supported by VkImportFenceWin32HandleInfoKHR](#) and [Handle Types Supported by VkImportFenceFdInfoKHR](#) define the permanence and transference of each handle type.

[External synchronization](#) allows implementations to modify an object's internal state, i.e. payload, without internal synchronization. However, for fences sharing a payload across processes, satisfying the external synchronization requirements of [VkFence](#) parameters as if all fences in the set were the same object is sometimes infeasible. Satisfying valid usage constraints on the state of a fence would similarly require impractical coordination or levels of trust between processes. Therefore, these constraints only apply to a specific fence handle, not to its payload. For distinct fence objects which share a payload:

- If multiple commands which queue a signal operation, or which unsignal a fence, are called concurrently, behavior will be as if the commands were called in an arbitrary sequential order.
- If a queue submission command is called with a fence that is sharing a payload, and the payload is already associated with another queue command that has not yet completed execution, either one or both of the commands will cause the fence to become signaled when they complete execution.
- If a fence payload is reset while it is associated with a queue command that has not yet completed execution, the payload will become unsignaled, but **may** become signaled again when the command completes execution.
- In the preceding cases, any of the devices associated with the fences sharing the payload **may** be lost, or any of the queue submission or fence reset commands **may** return [VK_ERROR_INITIALIZATION_FAILED](#).

Other than these non-deterministic results, behavior is well defined. In particular:

- The implementation **must** not crash or enter an internally inconsistent state where future valid Vulkan commands might cause undefined results,
- Timeouts on future wait commands on fences sharing the payload **must** be effective.

NOTE

These rules allow processes to synchronize access to shared memory without trusting each other. However, such processes must still be cautious not to use the shared fence for more than synchronizing access to the shared memory. For example, a process should not use a fence with shared payload to tell when commands it submitted to a queue have completed and objects used by those commands may be destroyed, since the other process can accidentally or

maliciously cause the fence to signal before the commands actually complete.

When a fence is using an imported payload, its `VkExportFenceCreateInfo::handleTypes` value is specified when creating the fence from which the payload was exported, rather than specified when creating the fence. Additionally, `VkExternalFenceProperties::exportFromImportedHandleTypes` restricts which handle types **can** be exported from such a fence based on the specific handle type used to import the current payload. Passing a fence to `vkAcquireNextImageKHR` is equivalent to temporarily importing a fence payload to that fence.

NOTE

Because the exportable handle types of an imported fence correspond to its current imported payload, and `vkAcquireNextImageKHR` behaves the same as a temporary import operation for which the source fence is opaque to the application, applications have no way of determining whether any external handle types **can** be exported from a fence in this state. Therefore, applications **must** not attempt to export handles from fences using a temporarily imported payload from `vkAcquireNextImageKHR`.

When importing a fence payload, it is the responsibility of the application to ensure the external handles meet all valid usage requirements. However, implementations **must** perform sufficient validation of external handles to ensure that the operation results in a valid fence which will not cause program termination, device loss, queue stalls, host thread stalls, or corruption of other resources when used as allowed according to its import parameters. If the external handle provided does not meet these requirements, the implementation **must** fail the fence payload import operation with the error code `VK_ERROR_INVALID_EXTERNAL_HANDLE`.

To import a fence payload from a Windows handle, call:

```
// Provided by VK_KHR_external_fence_win32
VkResult vkImportFenceWin32HandleKHR(
    VkDevice device,
    const VkImportFenceWin32HandleInfoKHR* pImportFenceWin32HandleInfo);
```

- `device` is the logical device that created the fence.
- `pImportFenceWin32HandleInfo` is a pointer to a `VkImportFenceWin32HandleInfoKHR` structure specifying the fence and import parameters.

Importing a fence payload from Windows handles does not transfer ownership of the handle to the Vulkan implementation. For handle types defined as NT handles, the application **must** release ownership using the `CloseHandle` system call when the handle is no longer needed.

Applications **can** import the same fence payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance.

Valid Usage

- VUID-vkImportFenceWin32HandleKHR-fence-04448
`fence` **must** not be associated with any queue command that has not yet completed

execution on that queue

Valid Usage (Implicit)

- VUID-vkImportFenceWin32HandleKHR-device-parameter **device** must be a valid [VkDevice](#) handle
- VUID-vkImportFenceWin32HandleKHR-pImportFenceWin32HandleInfo-parameter **pImportFenceWin32HandleInfo** must be a valid pointer to a valid [VkImportFenceWin32HandleInfoKHR](#) structure

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_INVALID_EXTERNAL_HANDLE](#)

The [VkImportFenceWin32HandleInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_external_fence_win32
typedef struct VkImportFenceWin32HandleInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkFence             fence;
    VkFenceImportFlags  flags;
    VkExternalFenceHandleTypeFlagBits handleType;
    HANDLE              handle;
    LPCWSTR             name;
} VkImportFenceWin32HandleInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is [NULL](#) or a pointer to a structure extending this structure.
- **fence** is the fence into which the state will be imported.
- **flags** is a bitmask of [VkFenceImportFlagBits](#) specifying additional parameters for the fence payload import operation.
- **handleType** is a [VkExternalFenceHandleTypeFlagBits](#) value specifying the type of **handle**.
- **handle** is [NULL](#) or the external handle to import.
- **name** is [NULL](#) or a null-terminated UTF-16 string naming the underlying synchronization primitive to import.

The handle types supported by `handleType` are:

Table 6. Handle Types Supported by `VkImportFenceWin32HandleInfoKHR`

Handle Type	Transference	Permanence Supported
<code>VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT</code>	Reference	Temporary,Permanent
<code>VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT</code>	Reference	Temporary,Permanent

Valid Usage

- VUID-VkImportFenceWin32HandleInfoKHR-handleType-01457
`handleType` **must** be a value included in the [Handle Types Supported by VkImportFenceWin32HandleInfoKHR](#) table
- VUID-VkImportFenceWin32HandleInfoKHR-handleType-01459
If `handleType` is not `VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT`, `name` **must** be `NULL`
- VUID-VkImportFenceWin32HandleInfoKHR-handleType-01460
If `handle` is `NULL`, `name` **must** name a valid synchronization primitive of the type specified by `handleType`
- VUID-VkImportFenceWin32HandleInfoKHR-handleType-01461
If `name` is `NULL`, `handle` **must** be a valid handle of the type specified by `handleType`
- VUID-VkImportFenceWin32HandleInfoKHR-handle-01462
If `handle` is not `NULL`, `name` **must** be `NULL`
- VUID-VkImportFenceWin32HandleInfoKHR-handle-01539
If `handle` is not `NULL`, it **must** obey any requirements listed for `handleType` in [external fence handle types compatibility](#)
- VUID-VkImportFenceWin32HandleInfoKHR-name-01540
If `name` is not `NULL`, it **must** obey any requirements listed for `handleType` in [external fence handle types compatibility](#)

Valid Usage (Implicit)

- VUID-VkImportFenceWin32HandleInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMPORT_FENCE_WIN32_HANDLE_INFO_KHR`
- VUID-VkImportFenceWin32HandleInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkImportFenceWin32HandleInfoKHR-fence-parameter
`fence` **must** be a valid [VkFence](#) handle
- VUID-VkImportFenceWin32HandleInfoKHR-flags-parameter
`flags` **must** be a valid combination of [VkFenceImportFlagBits](#) values

Host Synchronization

- Host access to **fence** **must** be externally synchronized

To import a fence payload from a POSIX file descriptor, call:

```
// Provided by VK_KHR_external_fence_fd
VkResult vkImportFenceFdKHR(
    VkDevice device,
    const VkImportFenceFdInfoKHR* pImportFenceFdInfo);
```

- **device** is the logical device that created the fence.
- **pImportFenceFdInfo** is a pointer to a **VkImportFenceFdInfoKHR** structure specifying the fence and import parameters.

Importing a fence payload from a file descriptor transfers ownership of the file descriptor from the application to the Vulkan implementation. The application **must** not perform any operations on the file descriptor after a successful import.

Applications **can** import the same fence payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance.

Valid Usage

- VUID-vkImportFenceFdKHR-fence-01463
fence **must** not be associated with any queue command that has not yet completed execution on that queue

Valid Usage (Implicit)

- VUID-vkImportFenceFdKHR-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkImportFenceFdKHR-pImportFenceFdInfo-parameter
pImportFenceFdInfo **must** be a valid pointer to a valid **VkImportFenceFdInfoKHR** structure

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_INVALID_EXTERNAL_HANDLE**

The `VkImportFenceFdInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_fence_fd
typedef struct VkImportFenceFdInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    VkFence              fence;
    VkFenceImportFlags    flags;
    VkExternalFenceHandleTypeFlagBits handleType;
    int                  fd;
} VkImportFenceFdInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fence` is the fence into which the payload will be imported.
- `flags` is a bitmask of `VkFenceImportFlagBits` specifying additional parameters for the fence payload import operation.
- `handleType` is a `VkExternalFenceHandleTypeFlagBits` value specifying the type of `fd`.
- `fd` is the external handle to import.

The handle types supported by `handleType` are:

Table 7. Handle Types Supported by `VkImportFenceFdInfoKHR`

Handle Type	Transference	Permanence Supported
<code>VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT</code>	Reference	Temporary,Permanent
<code>VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT</code>	Copy	Temporary

Valid Usage

- VUID-VkImportFenceFdInfoKHR-handleType-01464
`handleType` **must** be a value included in the [Handle Types Supported by VkImportFenceFdInfoKHR](#) table
- VUID-VkImportFenceFdInfoKHR-fd-01541
`fd` **must** obey any requirements listed for `handleType` in [external fence handle types compatibility](#)
- VUID-VkImportFenceFdInfoKHR-handleType-07306
If `handleType` refers to a handle type with copy payload transference semantics, `flags` **must** contain `VK_FENCE_IMPORT_TEMPORARY_BIT`

If `handleType` is `VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT`, the special value `-1` for `fd` is treated like a valid sync file descriptor referring to an object that has already signaled. The import operation will succeed and the `VkFence` will have a temporarily imported payload as if a valid file descriptor

had been provided.

NOTE

This special behavior for importing an invalid sync file descriptor allows easier interoperability with other system APIs which use the convention that an invalid sync file descriptor represents work that has already completed and does not need to be waited for. It is consistent with the option for implementations to return a `-1` file descriptor when exporting a `VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT` from a `VkFence` which is signaled.

Valid Usage (Implicit)

- `VUID-VkImportFenceFdInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_IMPORT_FENCE_FD_INFO_KHR`
- `VUID-VkImportFenceFdInfoKHR-pNext-pNext`
`pNext` **must** be `NULL`
- `VUID-VkImportFenceFdInfoKHR-fence-parameter`
`fence` **must** be a valid `VkFence` handle
- `VUID-VkImportFenceFdInfoKHR-flags-parameter`
`flags` **must** be a valid combination of `VkFenceImportFlagBits` values
- `VUID-VkImportFenceFdInfoKHR-handleType-parameter`
`handleType` **must** be a valid `VkExternalFenceHandleTypeFlagBits` value

Host Synchronization

- Host access to `fence` **must** be externally synchronized

Bits which **can** be set in

- `VkImportFenceWin32HandleInfoKHR::flags`
- `VkImportFenceFdInfoKHR::flags`

specifying additional parameters of a fence import operation are:

```
typedef enum VkFenceImportFlagBits {  
    VK_FENCE_IMPORT_TEMPORARY_BIT = 0x00000001,  
    // Provided by VK_KHR_external_fence  
    VK_FENCE_IMPORT_TEMPORARY_BIT_KHR = VK_FENCE_IMPORT_TEMPORARY_BIT,  
} VkFenceImportFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_external_fence
typedef VkFenceImportFlagBits VkFenceImportFlagBitsKHR;
```

- `VK_FENCE_IMPORT_TEMPORARY_BIT` specifies that the fence payload will be imported only temporarily, as described in [Importing Fence Payloads](#), regardless of the permanence of `handleType`.

```
typedef VkFlags VkFenceImportFlags;
```

or the equivalent

```
// Provided by VK_KHR_external_fence
typedef VkFenceImportFlags VkFenceImportFlagsKHR;
```

`VkFenceImportFlags` is a bitmask type for setting a mask of zero or more `VkFenceImportFlagBits`.

7.4. Semaphores

Semaphores are a synchronization primitive that **can** be used to insert a dependency between queue operations or between a queue operation and the host. [Binary semaphores](#) have two states - signaled and unsignaled. [Timeline semaphores](#) have a strictly increasing 64-bit unsigned integer payload and are signaled with respect to a particular reference value. A semaphore **can** be signaled after execution of a queue operation is completed, and a queue operation **can** wait for a semaphore to become signaled before it begins execution. A timeline semaphore **can** additionally be signaled from the host with the [vkSignalSemaphoreKHR](#) command and waited on from the host with the [vkWaitSemaphoresKHR](#) command.

The internal data of a semaphore **may** include a reference to any resources and pending work associated with signal or unsignal operations performed on that semaphore object, collectively referred to as the semaphore's *payload*. Mechanisms to import and export that internal data to and from semaphores are provided [below](#). These mechanisms indirectly enable applications to share semaphore state between two or more semaphores and other synchronization primitives across process and API boundaries.

Semaphores are represented by `VkSemaphore` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSemaphore)
```

To create a semaphore, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateSemaphore(
    VkDevice                                device,
    const VkSemaphoreCreateInfo*            pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkSemaphore*                            pSemaphore);
```

- **device** is the logical device that creates the semaphore.
- **pCreateInfo** is a pointer to a [VkSemaphoreCreateInfo](#) structure containing information about how the semaphore is to be created.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pSemaphore** is a pointer to a handle in which the resulting semaphore object is returned.

Valid Usage (Implicit)

- VUID-vkCreateSemaphore-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCreateSemaphore-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkSemaphoreCreateInfo](#) structure
- VUID-vkCreateSemaphore-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateSemaphore-pSemaphore-parameter
pSemaphore **must** be a valid pointer to a [VkSemaphore](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

The [VkSemaphoreCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSemaphoreCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkSemaphoreCreateFlags flags;
} VkSemaphoreCreateInfo;
```


- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

Valid Usage (Implicit)

- VUID-VkSemaphoreCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SEMAPHORE_CREATE_INFO`
- VUID-VkSemaphoreCreateInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkExportSemaphoreCreateInfo`, `VkExportSemaphoreWin32HandleInfoKHR`, or `VkSemaphoreTypeCreateInfo`
- VUID-VkSemaphoreCreateInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkSemaphoreCreateInfo-flags-zerobitmask
`flags` **must** be `0`

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSemaphoreCreateFlags;
```

`VkSemaphoreCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkSemaphoreTypeCreateInfo` structure is defined as:

```
typedef struct VkSemaphoreTypeCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkSemaphoreType    semaphoreType;
    uint64_t           initialValue;
} VkSemaphoreTypeCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_timeline_semaphore
typedef VkSemaphoreTypeCreateInfo VkSemaphoreTypeCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `semaphoreType` is a `VkSemaphoreType` value specifying the type of the semaphore.
- `initialValue` is the initial payload value if `semaphoreType` is `VK_SEMAPHORE_TYPE_TIMELINE`.

To create a semaphore of a specific type, add a `VkSemaphoreTypeCreateInfo` structure to the

`VkSemaphoreCreateInfo::pNext` chain.

If no `VkSemaphoreCreateInfo` structure is included in the `pNext` chain of `VkSemaphoreCreateInfo`, then the created semaphore will have a default `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_BINARY`.

Valid Usage

- VUID-VkSemaphoreCreateInfo-timelineSemaphore-03252
If the `timelineSemaphore` feature is not enabled, `semaphoreType` **must** not equal `VK_SEMAPHORE_TYPE_TIMELINE`
- VUID-VkSemaphoreCreateInfo-semaphoreType-03279
If `semaphoreType` is `VK_SEMAPHORE_TYPE_BINARY`, `initialValue` **must** be zero

Valid Usage (Implicit)

- VUID-VkSemaphoreCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SEMAPHORE_TYPE_CREATE_INFO`
- VUID-VkSemaphoreCreateInfo-semaphoreType-parameter
`semaphoreType` **must** be a valid `VkSemaphoreType` value

Possible values of `VkSemaphoreCreateInfo::semaphoreType`, specifying the type of a semaphore, are:

```
typedef enum VkSemaphoreType {
    VK_SEMAPHORE_TYPE_BINARY = 0,
    VK_SEMAPHORE_TYPE_TIMELINE = 1,
    // Provided by VK_KHR_timeline_semaphore
    VK_SEMAPHORE_TYPE_BINARY_KHR = VK_SEMAPHORE_TYPE_BINARY,
    // Provided by VK_KHR_timeline_semaphore
    VK_SEMAPHORE_TYPE_TIMELINE_KHR = VK_SEMAPHORE_TYPE_TIMELINE,
} VkSemaphoreType;
```

or the equivalent

```
// Provided by VK_KHR_timeline_semaphore
typedef VkSemaphoreType VkSemaphoreTypeKHR;
```

- `VK_SEMAPHORE_TYPE_BINARY` specifies a *binary semaphore* type that has a boolean payload indicating whether the semaphore is currently signaled or unsignaled. When created, the semaphore is in the unsignaled state.
- `VK_SEMAPHORE_TYPE_TIMELINE` specifies a *timeline semaphore* type that has a strictly increasing 64-bit unsigned integer payload indicating whether the semaphore is signaled with respect to a particular reference value. When created, the semaphore payload has the value given by the `initialValue` field of `VkSemaphoreCreateInfo`.

To create a semaphore whose payload **can** be exported to external handles, add a [VkExportSemaphoreCreateInfo](#) structure to the **pNext** chain of the [VkSemaphoreCreateInfo](#) structure. The [VkExportSemaphoreCreateInfo](#) structure is defined as:

```
typedef struct VkExportSemaphoreCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkExternalSemaphoreHandleTypeFlags handleTypes;
} VkExportSemaphoreCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_external_semaphore
typedef VkExportSemaphoreCreateInfo VkExportSemaphoreCreateInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **handleTypes** is a bitmask of [VkExternalSemaphoreHandleTypeFlagBits](#) specifying one or more semaphore handle types the application **can** export from the resulting semaphore. The application **can** request multiple handle types for the same semaphore.

Valid Usage

- VUID-VkExportSemaphoreCreateInfo-handleTypes-01124
The bits in **handleTypes** **must** be supported and compatible, as reported by [VkExternalSemaphoreProperties](#)

Valid Usage (Implicit)

- VUID-VkExportSemaphoreCreateInfo-sType-sType
sType **must** be `VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_CREATE_INFO`
- VUID-VkExportSemaphoreCreateInfo-handleTypes-parameter
handleTypes **must** be a valid combination of [VkExternalSemaphoreHandleTypeFlagBits](#) values

To specify additional attributes of NT handles exported from a semaphore, add a [VkExportSemaphoreWin32HandleInfoKHR](#) structure to the **pNext** chain of the [VkSemaphoreCreateInfo](#) structure. The [VkExportSemaphoreWin32HandleInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_external_semaphore_win32
typedef struct VkExportSemaphoreWin32HandleInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    const SECURITY_ATTRIBUTES* pAttributes;
    DWORD                dwAccess;
    LPCWSTR               name;
} VkExportSemaphoreWin32HandleInfoKHR;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **pAttributes** is a pointer to a Windows **SECURITY_ATTRIBUTES** structure specifying security attributes of the handle.
- **dwAccess** is a **DWORD** specifying access rights of the handle.
- **name** is a null-terminated UTF-16 string to associate with the underlying synchronization primitive referenced by NT handles exported from the created semaphore.

If **VkExportSemaphoreCreateInfo** is not included in the same **pNext** chain, this structure is ignored.

If **VkExportSemaphoreCreateInfo** is included in the **pNext** chain of **VkSemaphoreCreateInfo** with a Windows **handleType**, but either **VkExportSemaphoreWin32HandleInfoKHR** is not included in the **pNext** chain, or it is included but **pAttributes** is **NULL**, default security descriptor values will be used, and child processes created by the application will not inherit the handle, as described in the MSDN documentation for “Synchronization Object Security and Access Rights”¹. Further, if the structure is not present, the access rights used depend on the handle type.

For handles of the following types:

VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT

The implementation **must** ensure the access rights allow both signal and wait operations on the semaphore.

For handles of the following types:

VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT

The access rights **must** be:

GENERIC_ALL

1

<https://docs.microsoft.com/en-us/windows/win32/sync/synchronization-object-security-and-access-rights>

Valid Usage

- VUID-VkExportSemaphoreWin32HandleInfoKHR-handleTypes-01125

If `VkExportSemaphoreCreateInfo::handleTypes` does not include `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT` or `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT`, `VkExportSemaphoreWin32HandleInfoKHR` **must** not be included in the `pNext` chain of `VkSemaphoreCreateInfo`

Valid Usage (Implicit)

- VUID-VkExportSemaphoreWin32HandleInfoKHR-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR`
- VUID-VkExportSemaphoreWin32HandleInfoKHR-pAttributes-parameter
If `pAttributes` is not `NULL`, `pAttributes` **must** be a valid pointer to a valid `SECURITY_ATTRIBUTES` value

To export a Windows handle representing the payload of a semaphore, call:

```
// Provided by VK_KHR_external_semaphore_win32
VkResult vkGetSemaphoreWin32HandleKHR(
    VkDevice device,
    const VkSemaphoreGetWin32HandleInfoKHR* pGetWin32HandleInfo,
    HANDLE* pHandle);
```

- `device` is the logical device that created the semaphore being exported.
- `pGetWin32HandleInfo` is a pointer to a `VkSemaphoreGetWin32HandleInfoKHR` structure containing parameters of the export operation.
- `pHandle` will return the Windows handle representing the semaphore state.

For handle types defined as NT handles, the handles returned by `vkGetSemaphoreWin32HandleKHR` are owned by the application. To avoid leaking resources, the application **must** release ownership of them using the `CloseHandle` system call when they are no longer needed.

Exporting a Windows handle from a semaphore **may** have side effects depending on the transference of the specified handle type, as described in [Importing Semaphore Payloads](#).

Valid Usage (Implicit)

- VUID-vkGetSemaphoreWin32HandleKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetSemaphoreWin32HandleKHR-pGetWin32HandleInfo-parameter
`pGetWin32HandleInfo` **must** be a valid pointer to a valid `VkSemaphoreGetWin32HandleInfoKHR` structure
- VUID-vkGetSemaphoreWin32HandleKHR-pHandle-parameter
`pHandle` **must** be a valid pointer to a `HANDLE` value

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_TOO_MANY_OBJECTS`
- `VK_ERROR_OUT_OF_HOST_MEMORY`

The `VkSemaphoreGetWin32HandleInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_semaphore_win32
typedef struct VkSemaphoreGetWin32HandleInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkSemaphore              semaphore;
    VkExternalSemaphoreHandleTypeFlagBits handleType;
} VkSemaphoreGetWin32HandleInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `semaphore` is the semaphore from which state will be exported.
- `handleType` is a `VkExternalSemaphoreHandleTypeFlagBits` value specifying the type of handle requested.

The properties of the handle returned depend on the value of `handleType`. See `VkExternalSemaphoreHandleTypeFlagBits` for a description of the properties of the defined external semaphore handle types.

Valid Usage

- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-01126
`handleType` **must** have been included in `VkExportSemaphoreCreateInfo::handleTypes` when the `semaphore`'s current payload was created
- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-01127
If `handleType` is defined as an NT handle, `vkGetSemaphoreWin32HandleKHR` **must** be called no more than once for each valid unique combination of `semaphore` and `handleType`
- VUID-VkSemaphoreGetWin32HandleInfoKHR-semaphore-01128
`semaphore` **must** not currently have its payload replaced by an imported payload as described below in [Importing Semaphore Payloads](#) unless that imported payload's handle type was included in `VkExternalSemaphoreProperties::exportFromImportedHandleTypes` for `handleType`
- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-01129
If `handleType` refers to a handle type with copy payload transference semantics, as defined

below in [Importing Semaphore Payloads](#), there **must** be no queue waiting on **semaphore**

- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-01130

If **handleType** refers to a handle type with copy payload transference semantics, **semaphore** **must** be signaled, or have an associated [semaphore signal operation](#) pending execution

- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-01131
handleType **must** be defined as an NT handle or a global share handle

Valid Usage (Implicit)

- VUID-VkSemaphoreGetWin32HandleInfoKHR-sType-sType
sType **must** be `VK_STRUCTURE_TYPE_SEMAPHORE_GET_WIN32_HANDLE_INFO_KHR`
- VUID-VkSemaphoreGetWin32HandleInfoKHR-pNext-pNext
pNext **must** be `NULL`
- VUID-VkSemaphoreGetWin32HandleInfoKHR-semaphore-parameter
semaphore **must** be a valid [VkSemaphore](#) handle
- VUID-VkSemaphoreGetWin32HandleInfoKHR-handleType-parameter
handleType **must** be a valid [VkExternalSemaphoreHandleTypeFlagBits](#) value

To export a POSIX file descriptor representing the payload of a semaphore, call:

```
// Provided by VK_KHR_external_semaphore_fd
VkResult vkGetSemaphoreFdKHR(
    VkDevice device,
    const VkSemaphoreGetFdInfoKHR* pGetFdInfo,
    int* pFd);
```

- **device** is the logical device that created the semaphore being exported.
- **pGetFdInfo** is a pointer to a [VkSemaphoreGetFdInfoKHR](#) structure containing parameters of the export operation.
- **pFd** will return the file descriptor representing the semaphore payload.

Each call to **vkGetSemaphoreFdKHR** **must** create a new file descriptor and transfer ownership of it to the application. To avoid leaking resources, the application **must** release ownership of the file descriptor when it is no longer needed.

NOTE

Ownership can be released in many ways. For example, the application can call **close()** on the file descriptor, or transfer ownership back to Vulkan by using the file descriptor to import a semaphore payload.

Where supported by the operating system, the implementation **must** set the file descriptor to be closed automatically when an **execve** system call is made.

Exporting a file descriptor from a semaphore **may** have side effects depending on the transference

of the specified handle type, as described in [Importing Semaphore State](#).

Valid Usage (Implicit)

- VUID-vkGetSemaphoreFdKHR-device-parameter **device** must be a valid [VkDevice](#) handle
- VUID-vkGetSemaphoreFdKHR-pGetFdInfo-parameter **pGetFdInfo** must be a valid pointer to a valid [VkSemaphoreGetFdInfoKHR](#) structure
- VUID-vkGetSemaphoreFdKHR-pFd-parameter **pFd** must be a valid pointer to an **int** value

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_TOO_MANY_OBJECTS**
- **VK_ERROR_OUT_OF_HOST_MEMORY**

The [VkSemaphoreGetFdInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_external_semaphore_fd
typedef struct VkSemaphoreGetFdInfoKHR {
    VkStructureType           sType;
    const void*               pNext;
    VkSemaphore               semaphore;
    VkExternalSemaphoreHandleTypeFlagBits handleType;
} VkSemaphoreGetFdInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **semaphore** is the semaphore from which state will be exported.
- **handleType** is a [VkExternalSemaphoreHandleTypeFlagBits](#) value specifying the type of handle requested.

The properties of the file descriptor returned depend on the value of **handleType**. See [VkExternalSemaphoreHandleTypeFlagBits](#) for a description of the properties of the defined external semaphore handle types.

Valid Usage

- VUID-VkSemaphoreGetFdInfoKHR-handleType-01132

`handleType` **must** have been included in `VkExportSemaphoreCreateInfo::handleTypes` when `semaphore`'s current payload was created

- VUID-VkSemaphoreGetFdInfoKHR-semaphore-01133
`semaphore` **must** not currently have its payload replaced by an imported payload as described below in [Importing Semaphore Payloads](#) unless that imported payload's handle type was included in `VkExternalSemaphoreProperties::exportFromImportedHandleTypes` for `handleType`
- VUID-VkSemaphoreGetFdInfoKHR-handleType-01134
If `handleType` refers to a handle type with copy payload transference semantics, as defined below in [Importing Semaphore Payloads](#), there **must** be no queue waiting on `semaphore`
- VUID-VkSemaphoreGetFdInfoKHR-handleType-01135
If `handleType` refers to a handle type with copy payload transference semantics, `semaphore` **must** be signaled, or have an associated [semaphore signal operation](#) pending execution
- VUID-VkSemaphoreGetFdInfoKHR-handleType-01136
`handleType` **must** be defined as a POSIX file descriptor handle
- VUID-VkSemaphoreGetFdInfoKHR-handleType-03253
If `handleType` refers to a handle type with copy payload transference semantics, `semaphore` **must** have been created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_BINARY`
- VUID-VkSemaphoreGetFdInfoKHR-handleType-03254
If `handleType` refers to a handle type with copy payload transference semantics, `semaphore` **must** have an associated semaphore signal operation that has been submitted for execution and any semaphore signal operations on which it depends **must** have also been submitted for execution

Valid Usage (Implicit)

- VUID-VkSemaphoreGetFdInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SEMAPHORE_GET_FD_INFO_KHR`
- VUID-VkSemaphoreGetFdInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkSemaphoreGetFdInfoKHR-semaphore-parameter
`semaphore` **must** be a valid `VkSemaphore` handle
- VUID-VkSemaphoreGetFdInfoKHR-handleType-parameter
`handleType` **must** be a valid `VkExternalSemaphoreHandleTypeFlagBits` value

To destroy a semaphore, call:

```
// Provided by VK_VERSION_1_0
void vkDestroySemaphore(
    VkDevice          device,
    VkSemaphore       semaphore,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the semaphore.
- **semaphore** is the handle of the semaphore to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroySemaphore-semaphore-05149
All submitted batches that refer to **semaphore** **must** have completed execution
- VUID-vkDestroySemaphore-semaphore-01138
If **VkAllocationCallbacks** were provided when **semaphore** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroySemaphore-semaphore-01139
If no **VkAllocationCallbacks** were provided when **semaphore** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroySemaphore-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroySemaphore-semaphore-parameter
If **semaphore** is not **VK_NULL_HANDLE**, **semaphore** **must** be a valid **VkSemaphore** handle
- VUID-vkDestroySemaphore-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroySemaphore-semaphore-parent
If **semaphore** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **semaphore** **must** be externally synchronized

7.4.1. Semaphore Signaling

When a batch is submitted to a queue via a [queue submission](#), and it includes semaphores to be signaled, it defines a memory dependency on the batch, and defines *semaphore signal operations* which set the semaphores to the signaled state.

In case of semaphores created with a **VkSemaphoreType** of **VK_SEMAPHORE_TYPE_TIMELINE** the semaphore is considered signaled with respect to the counter value set to be signaled as specified in [VkTimelineSemaphoreSubmitInfo](#) or [VkSemaphoreSignalInfo](#).

The first [synchronization scope](#) includes every command submitted in the same batch. In the case of [vkQueueSubmit2KHR](#), the first synchronization scope is limited to the pipeline stage specified by [VkSemaphoreSubmitInfo::stageMask](#). Semaphore signal operations that are defined by [vkQueueSubmit](#) or [vkQueueSubmit2KHR](#) additionally include all commands that occur earlier in [submission order](#). Semaphore signal operations that are defined by [vkQueueSubmit](#) or [vkQueueSubmit2KHR](#) or [vkQueueBindSparse](#) additionally include in the first synchronization scope any semaphore and fence signal operations that occur earlier in [signal operation order](#).

The second [synchronization scope](#) includes only the semaphore signal operation.

The first [access scope](#) includes all memory access performed by the device.

The second [access scope](#) is empty.

7.4.2. Semaphore Waiting

When a batch is submitted to a queue via a [queue submission](#), and it includes semaphores to be waited on, it defines a memory dependency between prior semaphore signal operations and the batch, and defines *semaphore wait operations*.

Such semaphore wait operations set the semaphores created with a [VkSemaphoreType](#) of [VK_SEMAPHORE_TYPE_BINARY](#) to the unsignaled state. In case of semaphores created with a [VkSemaphoreType](#) of [VK_SEMAPHORE_TYPE_TIMELINE](#) a prior semaphore signal operation defines a memory dependency with a semaphore wait operation if the value the semaphore is signaled with is greater than or equal to the value the semaphore is waited with, thus the semaphore will continue to be considered signaled with respect to the counter value waited on as specified in [VkTimelineSemaphoreSubmitInfo](#).

The first [synchronization scope](#) includes one [semaphore signal operation](#) for each semaphore waited on by this batch. The specific signal operation waited on for each semaphore **must** meet the following criteria:

- for binary semaphores, the signal operation is either earlier in [submission order](#) on the same queue, or is submitted by a command whose host operation happens-before this batch is submitted on the host
- for binary semaphores, no wait operation exists that happens-after the signal operation and happens-before this wait operation
- the signal operation is not guaranteed to happen-after the semaphore wait operation in this batch
- for timeline semaphores, the signal value is greater than or equal to the wait value

If multiple semaphore signal operations meet these criteria, any of those operations **may** be included in the first [synchronization scope](#). When waiting on a binary semaphore, applications **must** ensure that exactly one semaphore signal operation meets these criteria.

The second [synchronization scope](#) includes every command submitted in the same batch. In the case of [vkQueueSubmit](#), the second synchronization scope is limited to operations on the pipeline stages determined by the [destination stage mask](#) specified by the corresponding element of

`pWaitDstStageMask`. In the case of `vkQueueSubmit2KHR`, the second synchronization scope is limited to the pipeline stage specified by `VkSemaphoreSubmitInfo::stageMask`. Also, in the case of either `vkQueueSubmit2KHR` or `vkQueueSubmit`, the second synchronization scope additionally includes all commands that occur later in `submission order`.

The first `access scope` is empty.

The second `access scope` includes all memory access performed by the device.

The semaphore wait operation happens-after the first set of operations in the execution dependency, and happens-before the second set of operations in the execution dependency.

NOTE

Unlike timeline semaphores, fences or events, waiting for a binary semaphore also unsignals that semaphore when the wait completes. Applications **must** ensure that between two such wait operations, the semaphore is signaled again, with execution dependencies used to ensure these occur in order. Binary semaphore waits and signals should thus occur in discrete 1:1 pairs.

A common scenario for using `pWaitDstStageMask` with values other than `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT` is when synchronizing a window system presentation operation against subsequent command buffers which render the next frame. In this case, a presentation image **must** not be overwritten until the presentation operation completes, but other pipeline stages **can** execute without waiting. A mask of `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` prevents subsequent color attachment writes from executing until the semaphore signals. Some implementations **may** be able to execute transfer operations and/or pre-rasterization work before the semaphore is signaled.

If an image layout transition needs to be performed on a presentable image before it is used in a framebuffer, that **can** be performed as the first operation submitted to the queue after acquiring the image, and **should** not prevent other work from overlapping with the presentation operation. For example, a `VkImageMemoryBarrier` could use:

NOTE

- `srcStageMask = VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT`
- `srcAccessMask = 0`
- `dstStageMask = VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT`
- `dstAccessMask = VK_ACCESS_COLOR_ATTACHMENT_READ_BIT | VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT.`
- `oldLayout = VK_IMAGE_LAYOUT_PRESENT_SRC_KHR`
- `newLayout = VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL`

Alternatively, `oldLayout` **can** be `VK_IMAGE_LAYOUT_UNDEFINED`, if the image's contents need not be preserved.

This barrier accomplishes a dependency chain between previous presentation operations and subsequent color attachment output operations, with the layout

transition performed in between, and does not introduce a dependency between previous work and any [pre-rasterization shader stages](#). More precisely, the semaphore signals after the presentation operation completes, the semaphore wait stalls the [VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT](#) stage, and there is a dependency from that same stage to itself with the layout transition performed in between.

7.4.3. Semaphore State Requirements for Wait Operations

Before waiting on a semaphore, the application **must** ensure the semaphore is in a valid state for a wait operation. Specifically, when a [semaphore wait operation](#) is submitted to a queue:

- A binary semaphore **must** be signaled, or have an associated [semaphore signal operation](#) that is pending execution.
- Any [semaphore signal operations](#) on which the pending binary semaphore signal operation depends **must** also be completed or pending execution.
- There **must** be no other queue waiting on the same binary semaphore when the operation executes.

7.4.4. Host Operations on Semaphores

In addition to [semaphore signal operations](#) and [semaphore wait operations](#) submitted to device queues, timeline semaphores support the following host operations:

- Query the current counter value of the semaphore using the [vkGetSemaphoreCounterValueKHR](#) command.
- Wait for a set of semaphores to reach particular counter values using the [vkWaitSemaphoresKHR](#) command.
- Signal the semaphore with a particular counter value from the host using the [vkSignalSemaphoreKHR](#) command.

To query the current counter value of a semaphore created with a [VkSemaphoreType](#) of [VK_SEMAPHORE_TYPE_TIMELINE](#) from the host, call:

```
// Provided by VK_KHR_timeline_semaphore
VkResult vkGetSemaphoreCounterValueKHR(
    VkDevice          device,
    VkSemaphore        semaphore,
    uint64_t*         pValue);
```

- [device](#) is the logical device that owns the semaphore.
- [semaphore](#) is the handle of the semaphore to query.
- [pValue](#) is a pointer to a 64-bit integer value in which the current counter value of the semaphore is returned.

NOTE

If a [queue submission](#) command is pending execution, then the value returned by this command **may** immediately be out of date.

Valid Usage

- VUID-vkGetSemaphoreCounterValue-semaphore-03255
semaphore **must** have been created with a [VkSemaphoreType](#) of `VK_SEMAPHORE_TYPE_TIMELINE`

Valid Usage (Implicit)

- VUID-vkGetSemaphoreCounterValue-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkGetSemaphoreCounterValue-semaphore-parameter
semaphore **must** be a valid [VkSemaphore](#) handle
- VUID-vkGetSemaphoreCounterValue-pValue-parameter
pValue **must** be a valid pointer to a `uint64_t` value
- VUID-vkGetSemaphoreCounterValue-semaphore-parent
semaphore **must** have been created, allocated, or retrieved from **device**

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

To wait for a set of semaphores created with a [VkSemaphoreType](#) of `VK_SEMAPHORE_TYPE_TIMELINE` to reach particular counter values on the host, call:

```
// Provided by VK_KHR_timeline_semaphore
VkResult vkWaitSemaphoresKHR(
    VkDevice                                device,
    const VkSemaphoreWaitInfo*             pWaitInfo,
    uint64_t                                timeout);
```

- **device** is the logical device that owns the semaphores.
- **pWaitInfo** is a pointer to a [VkSemaphoreWaitInfo](#) structure containing information about the

wait condition.

- **timeout** is the timeout period in units of nanoseconds. **timeout** is adjusted to the closest value allowed by the implementation-dependent timeout accuracy, which **may** be substantially longer than one nanosecond, and **may** be longer than the requested period.

If the condition is satisfied when **vkWaitSemaphoresKHR** is called, then **vkWaitSemaphoresKHR** returns immediately. If the condition is not satisfied at the time **vkWaitSemaphoresKHR** is called, then **vkWaitSemaphoresKHR** will block and wait until the condition is satisfied or the **timeout** has expired, whichever is sooner.

If **timeout** is zero, then **vkWaitSemaphoresKHR** does not wait, but simply returns information about the current state of the semaphores. **VK_TIMEOUT** will be returned in this case if the condition is not satisfied, even though no actual wait was performed.

If the condition is satisfied before the **timeout** has expired, **vkWaitSemaphoresKHR** returns **VK_SUCCESS**. Otherwise, **vkWaitSemaphoresKHR** returns **VK_TIMEOUT** after the **timeout** has expired.

If device loss occurs (see [Lost Device](#)) before the timeout has expired, **vkWaitSemaphoresKHR** **must** return in finite time with either **VK_SUCCESS** or **VK_ERROR_DEVICE_LOST**.

Valid Usage (Implicit)

- VUID-vkWaitSemaphores-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkWaitSemaphores-pWaitInfo-parameter **pWaitInfo** **must** be a valid pointer to a valid [VkSemaphoreWaitInfo](#) structure

Return Codes

Success

- **VK_SUCCESS**
- **VK_TIMEOUT**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**
- **VK_ERROR_DEVICE_LOST**

The **VkSemaphoreWaitInfo** structure is defined as:

```
typedef struct VkSemaphoreWaitInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkSemaphoreWaitFlags flags;
    uint32_t             semaphoreCount;
    const VkSemaphore*   pSemaphores;
    const uint64_t*      pValues;
} VkSemaphoreWaitInfo;
```

or the equivalent

```
// Provided by VK_KHR_timeline_semaphore
typedef VkSemaphoreWaitInfo VkSemaphoreWaitInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkSemaphoreWaitFlagBits` specifying additional parameters for the semaphore wait operation.
- `semaphoreCount` is the number of semaphores to wait on.
- `pSemaphores` is a pointer to an array of `semaphoreCount` semaphore handles to wait on.
- `pValues` is a pointer to an array of `semaphoreCount` timeline semaphore values.

Valid Usage

- VUID-VkSemaphoreWaitInfo-pSemaphores-03256
All of the elements of `pSemaphores` **must** reference a semaphore that was created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE`

Valid Usage (Implicit)

- VUID-VkSemaphoreWaitInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SEMAPHORE_WAIT_INFO`
- VUID-VkSemaphoreWaitInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkSemaphoreWaitInfo-flags-parameter
`flags` **must** be a valid combination of `VkSemaphoreWaitFlagBits` values
- VUID-VkSemaphoreWaitInfo-pSemaphores-parameter
`pSemaphores` **must** be a valid pointer to an array of `semaphoreCount` valid `VkSemaphore` handles
- VUID-VkSemaphoreWaitInfo-pValues-parameter
`pValues` **must** be a valid pointer to an array of `semaphoreCount` `uint64_t` values

- VUID-VkSemaphoreWaitInfo-semaphoreCount-arraylength
`semaphoreCount` **must** be greater than 0

Bits which **can** be set in `VkSemaphoreWaitInfo::flags`, specifying additional parameters of a semaphore wait operation, are:

```
typedef enum VkSemaphoreWaitFlagBits {
    VK_SEMAPHORE_WAIT_ANY_BIT = 0x00000001,
    // Provided by VK_KHR_timeline_semaphore
    VK_SEMAPHORE_WAIT_ANY_BIT_KHR = VK_SEMAPHORE_WAIT_ANY_BIT,
} VkSemaphoreWaitFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_timeline_semaphore
typedef VkSemaphoreWaitFlagBits VkSemaphoreWaitFlagBitsKHR;
```

- `VK_SEMAPHORE_WAIT_ANY_BIT` specifies that the semaphore wait condition is that at least one of the semaphores in `VkSemaphoreWaitInfo::pSemaphores` has reached the value specified by the corresponding element of `VkSemaphoreWaitInfo::pValues`. If `VK_SEMAPHORE_WAIT_ANY_BIT` is not set, the semaphore wait condition is that all of the semaphores in `VkSemaphoreWaitInfo::pSemaphores` have reached the value specified by the corresponding element of `VkSemaphoreWaitInfo::pValues`.

```
typedef VkFlags VkSemaphoreWaitFlags;
```

or the equivalent

```
// Provided by VK_KHR_timeline_semaphore
typedef VkSemaphoreWaitFlags VkSemaphoreWaitFlagsKHR;
```

`VkSemaphoreWaitFlags` is a bitmask type for setting a mask of zero or more `VkSemaphoreWaitFlagBits`.

To signal a semaphore created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE` with a particular counter value, on the host, call:

```
// Provided by VK_KHR_timeline_semaphore
VkResult vkSignalSemaphoreKHR(
    VkDevice device,
    const VkSemaphoreSignalInfo* pSignalInfo);
```

- `device` is the logical device that owns the semaphore.
- `pSignalInfo` is a pointer to a `VkSemaphoreSignalInfo` structure containing information about the

signal operation.

When `vkSignalSemaphoreKHR` is executed on the host, it defines and immediately executes a *semaphore signal operation* which sets the timeline semaphore to the given value.

The first synchronization scope is defined by the host execution model, but includes execution of `vkSignalSemaphoreKHR` on the host and anything that happened-before it.

The second synchronization scope is empty.

Valid Usage (Implicit)

- VUID-vkSignalSemaphore-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkSignalSemaphore-pSignalInfo-parameter
`pSignalInfo` **must** be a valid pointer to a valid `VkSemaphoreSignalInfo` structure

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkSemaphoreSignalInfo` structure is defined as:

```
typedef struct VkSemaphoreSignalInfo {  
    VkStructureType    sType;  
    const void*        pNext;  
    VkSemaphore        semaphore;  
    uint64_t           value;  
} VkSemaphoreSignalInfo;
```

or the equivalent

```
// Provided by VK_KHR_timeline_semaphore  
typedef VkSemaphoreSignalInfo VkSemaphoreSignalInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `semaphore` is the handle of the semaphore to signal.

- **value** is the value to signal.

Valid Usage

- VUID-VkSemaphoreSignalInfo-semaphore-03257
semaphore **must** have been created with a **VkSemaphoreType** of **VK_SEMAPHORE_TYPE_TIMELINE**
- VUID-VkSemaphoreSignalInfo-value-03258
value **must** have a value greater than the current value of the semaphore
- VUID-VkSemaphoreSignalInfo-value-03259
value **must** be less than the value of any pending semaphore signal operations
- VUID-VkSemaphoreSignalInfo-value-03260
value **must** have a value which does not differ from the current value of the semaphore or the value of any outstanding semaphore wait or signal operation on **semaphore** by more than **maxTimelineSemaphoreValueDifference**

Valid Usage (Implicit)

- VUID-VkSemaphoreSignalInfo-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_SEMAPHORE_SIGNAL_INFO**
- VUID-VkSemaphoreSignalInfo-pNext-pNext
pNext **must** be **NULL**
- VUID-VkSemaphoreSignalInfo-semaphore-parameter
semaphore **must** be a valid **VkSemaphore** handle

7.4.5. Importing Semaphore Payloads

Applications **can** import a semaphore payload into an existing semaphore using an external semaphore handle. The effects of the import operation will be either temporary or permanent, as specified by the application. If the import is temporary, the implementation **must** restore the semaphore to its prior permanent state after submitting the next semaphore wait operation. Performing a subsequent temporary import on a semaphore before performing a semaphore wait has no effect on this requirement; the next wait submitted on the semaphore **must** still restore its last permanent state. A permanent payload import behaves as if the target semaphore was destroyed, and a new semaphore was created with the same handle but the imported payload. Because importing a semaphore payload temporarily or permanently detaches the existing payload from a semaphore, similar usage restrictions to those applied to **vkDestroySemaphore** are applied to any command that imports a semaphore payload. Which of these import types is used is referred to as the import operation's *permanence*. Each handle type supports either one or both types of permanence.

The implementation **must** perform the import operation by either referencing or copying the payload referred to by the specified external semaphore handle, depending on the handle's type. The import method used is referred to as the handle type's *transference*. When using handle types

with reference transference, importing a payload to a semaphore adds the semaphore to the set of all semaphores sharing that payload. This set includes the semaphore from which the payload was exported. Semaphore signaling and waiting operations performed on any semaphore in the set **must** behave as if the set were a single semaphore. Importing a payload using handle types with copy transference creates a duplicate copy of the payload at the time of import, but makes no further reference to it. Semaphore signaling and waiting operations performed on the target of copy imports **must** not affect any other semaphore or payload.

Export operations have the same transference as the specified handle type's import operations. Additionally, exporting a semaphore payload to a handle with copy transference has the same side effects on the source semaphore's payload as executing a semaphore wait operation. If the semaphore was using a temporarily imported payload, the semaphore's prior permanent payload will be restored.

NOTE

The permanence and transference of handle types can be found in:

- [Handle Types Supported by `VkImportSemaphoreWin32HandleInfoKHR`](#)
- [Handle Types Supported by `VkImportSemaphoreFdInfoKHR`](#)

[External synchronization](#) allows implementations to modify an object's internal state, i.e. payload, without internal synchronization. However, for semaphores sharing a payload across processes, satisfying the external synchronization requirements of `VkSemaphore` parameters as if all semaphores in the set were the same object is sometimes infeasible. Satisfying the [wait operation state requirements](#) would similarly require impractical coordination or levels of trust between processes. Therefore, these constraints only apply to a specific semaphore handle, not to its payload. For distinct semaphore objects which share a payload, if the semaphores are passed to separate queue submission commands concurrently, behavior will be as if the commands were called in an arbitrary sequential order. If the [wait operation state requirements](#) are violated for the shared payload by a queue submission command, or if a signal operation is queued for a shared payload that is already signaled or has a pending signal operation, effects **must** be limited to one or more of the following:

- Returning `VK_ERROR_INITIALIZATION_FAILED` from the command which resulted in the violation.
- Losing the logical device on which the violation occurred immediately or at a future time, resulting in a `VK_ERROR_DEVICE_LOST` error from subsequent commands, including the one causing the violation.
- Continuing execution of the violating command or operation as if the semaphore wait completed successfully after an implementation-dependent timeout. In this case, the state of the payload becomes undefined, and future operations on semaphores sharing the payload will be subject to these same rules. The semaphore **must** be destroyed or have its payload replaced by an import operation to again have a well-defined state.

NOTE

These rules allow processes to synchronize access to shared memory without trusting each other. However, such processes must still be cautious not to use the shared semaphore for more than synchronizing access to the shared memory. For example, a process should not use a shared semaphore as part of an execution dependency chain that, when complete, leads to objects being destroyed, if it does

not trust other processes sharing the semaphore payload.

When a semaphore is using an imported payload, its `VkExportSemaphoreCreateInfo::handleTypes` value is specified when creating the semaphore from which the payload was exported, rather than specified when creating the semaphore. Additionally, `VkExternalSemaphoreProperties::exportFromImportedHandleTypes` restricts which handle types **can** be exported from such a semaphore based on the specific handle type used to import the current payload. Passing a semaphore to `vkAcquireNextImageKHR` is equivalent to temporarily importing a semaphore payload to that semaphore.

NOTE

Because the exportable handle types of an imported semaphore correspond to its current imported payload, and `vkAcquireNextImageKHR` behaves the same as a temporary import operation for which the source semaphore is opaque to the application, applications have no way of determining whether any external handle types **can** be exported from a semaphore in this state. Therefore, applications **must** not attempt to export external handles from semaphores using a temporarily imported payload from `vkAcquireNextImageKHR`.

When importing a semaphore payload, it is the responsibility of the application to ensure the external handles meet all valid usage requirements. However, implementations **must** perform sufficient validation of external handles to ensure that the operation results in a valid semaphore which will not cause program termination, device loss, queue stalls, or corruption of other resources when used as allowed according to its import parameters, and excepting those side effects allowed for violations of the `valid semaphore state for wait operations` rules. If the external handle provided does not meet these requirements, the implementation **must** fail the semaphore payload import operation with the error code `VK_ERROR_INVALID_EXTERNAL_HANDLE`.

In addition, when importing a semaphore payload that is not compatible with the payload type corresponding to the `VkSemaphoreType` the semaphore was created with, the implementation **may** fail the semaphore payload import operation with the error code `VK_ERROR_INVALID_EXTERNAL_HANDLE`.

NOTE

As the introduction of the external semaphore handle type `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT` predates that of timeline semaphores, support for importing semaphore payloads from external handles of that type into semaphores created (implicitly or explicitly) with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_BINARY` is preserved for backwards compatibility. However, applications **should** prefer importing such handle types into semaphores created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE`.

To import a semaphore payload from a Windows handle, call:

```
// Provided by VK_KHR_external_semaphore_win32
VkResult vkImportSemaphoreWin32HandleKHR(
    VkDevice device,
    const VkImportSemaphoreWin32HandleInfoKHR* pImportSemaphoreWin32HandleInfo);
```

- `device` is the logical device that created the semaphore.

- `pImportSemaphoreWin32HandleInfo` is a pointer to a `VkImportSemaphoreWin32HandleInfoKHR` structure specifying the semaphore and import parameters.

Importing a semaphore payload from Windows handles does not transfer ownership of the handle to the Vulkan implementation. For handle types defined as NT handles, the application **must** release ownership using the `CloseHandle` system call when the handle is no longer needed.

Applications **can** import the same semaphore payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance.

Valid Usage (Implicit)

- VUID-vkImportSemaphoreWin32HandleKHR-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkImportSemaphoreWin32HandleKHR-pImportSemaphoreWin32HandleInfo-parameter `pImportSemaphoreWin32HandleInfo` **must** be a valid pointer to a valid `VkImportSemaphoreWin32HandleInfoKHR` structure

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_INVALID_EXTERNAL_HANDLE`

The `VkImportSemaphoreWin32HandleInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_semaphore_win32
typedef struct VkImportSemaphoreWin32HandleInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkSemaphore         semaphore;
    VkSemaphoreImportFlags flags;
    VkExternalSemaphoreHandleTypeFlagBits handleType;
    HANDLE              handle;
    LPCWSTR             name;
} VkImportSemaphoreWin32HandleInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `semaphore` is the semaphore into which the payload will be imported.

- **flags** is a bitmask of [VkSemaphoreImportFlagBits](#) specifying additional parameters for the semaphore payload import operation.
- **handleType** is a [VkExternalSemaphoreHandleTypeFlagBits](#) value specifying the type of **handle**.
- **handle** is **NULL** or the external handle to import.
- **name** is **NULL** or a null-terminated UTF-16 string naming the underlying synchronization primitive to import.

The handle types supported by **handleType** are:

Table 8. Handle Types Supported by [VkImportSemaphoreWin32HandleInfoKHR](#)

Handle Type	Transference	Permanence Supported
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT	Reference	Temporary,Permanent
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT	Reference	Temporary,Permanent
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT	Reference	Temporary,Permanent

Valid Usage

- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-01140
handleType **must** be a value included in the [Handle Types Supported by VkImportSemaphoreWin32HandleInfoKHR](#) table
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-01466
If **handleType** is not [VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT](#) or [VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT](#), **name** **must** be **NULL**
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-01467
If **handle** is **NULL**, **name** **must** name a valid synchronization primitive of the type specified by **handleType**
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-01468
If **name** is **NULL**, **handle** **must** be a valid handle of the type specified by **handleType**
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handle-01469
If **handle** is not **NULL**, **name** **must** be **NULL**
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handle-01542
If **handle** is not **NULL**, it **must** obey any requirements listed for **handleType** in [external semaphore handle types compatibility](#)
- VUID-VkImportSemaphoreWin32HandleInfoKHR-name-01543
If **name** is not **NULL**, it **must** obey any requirements listed for **handleType** in [external semaphore handle types compatibility](#)
- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-03261
If **handleType** is [VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT](#) or

`VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT`, the `VkSemaphoreCreateInfo::flags` field **must** match that of the semaphore from which `handle` or `name` was exported

- VUID-VkImportSemaphoreWin32HandleInfoKHR-handleType-03262
If `handleType` is `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT` or `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT`, the `VkSemaphoreTypeCreateInfo::semaphoreType` field **must** match that of the semaphore from which `handle` or `name` was exported
- VUID-VkImportSemaphoreWin32HandleInfoKHR-flags-03322
If `flags` contains `VK_SEMAPHORE_IMPORT_TEMPORARY_BIT`, the `VkSemaphoreTypeCreateInfo::semaphoreType` field of the semaphore from which `handle` or `name` was exported **must** not be `VK_SEMAPHORE_TYPE_TIMELINE`

Valid Usage (Implicit)

- VUID-VkImportSemaphoreWin32HandleInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR`
- VUID-VkImportSemaphoreWin32HandleInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkImportSemaphoreWin32HandleInfoKHR-semaphore-parameter
`semaphore` **must** be a valid `VkSemaphore` handle
- VUID-VkImportSemaphoreWin32HandleInfoKHR-flags-parameter
`flags` **must** be a valid combination of `VkSemaphoreImportFlagBits` values

Host Synchronization

- Host access to `semaphore` **must** be externally synchronized

To import a semaphore payload from a POSIX file descriptor, call:

```
// Provided by VK_KHR_external_semaphore_fd
VkResult vkImportSemaphoreFdKHR(
    VkDevice device,
    const VkImportSemaphoreFdInfoKHR* pImportSemaphoreFdInfo);
```

- `device` is the logical device that created the semaphore.
- `pImportSemaphoreFdInfo` is a pointer to a `VkImportSemaphoreFdInfoKHR` structure specifying the semaphore and import parameters.

Importing a semaphore payload from a file descriptor transfers ownership of the file descriptor from the application to the Vulkan implementation. The application **must** not perform any operations on the file descriptor after a successful import.

Applications **can** import the same semaphore payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance.

Valid Usage

- VUID-vkImportSemaphoreFdKHR-semaphore-01142
semaphore **must** not be associated with any queue command that has not yet completed execution on that queue

Valid Usage (Implicit)

- VUID-vkImportSemaphoreFdKHR-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkImportSemaphoreFdKHR-pImportSemaphoreFdInfo-parameter
pImportSemaphoreFdInfo **must** be a valid pointer to a valid [VkImportSemaphoreFdInfoKHR](#) structure

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_INVALID_EXTERNAL_HANDLE](#)

The [VkImportSemaphoreFdInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_external_semaphore_fd
typedef struct VkImportSemaphoreFdInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkSemaphore        semaphore;
    VkSemaphoreImportFlags flags;
    VkExternalSemaphoreHandleTypeFlagBits handleType;
    int                fd;
} VkImportSemaphoreFdInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is [NULL](#) or a pointer to a structure extending this structure.
- **semaphore** is the semaphore into which the payload will be imported.
- **flags** is a bitmask of [VkSemaphoreImportFlagBits](#) specifying additional parameters for the semaphore payload import operation.

- `handleType` is a `VkExternalSemaphoreHandleTypeFlagBits` value specifying the type of `fd`.
- `fd` is the external handle to import.

The handle types supported by `handleType` are:

Table 9. Handle Types Supported by `VkImportSemaphoreFdInfoKHR`

Handle Type	Transference	Permanence Supported
<code>VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT</code>	Reference	Temporary,Permanent
<code>VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT</code>	Copy	Temporary

Valid Usage

- VUID-VkImportSemaphoreFdInfoKHR-handleType-01143
`handleType` **must** be a value included in the [Handle Types Supported by VkImportSemaphoreFdInfoKHR](#) table
- VUID-VkImportSemaphoreFdInfoKHR-fd-01544
`fd` **must** obey any requirements listed for `handleType` in [external semaphore handle types compatibility](#)
- VUID-VkImportSemaphoreFdInfoKHR-handleType-03263
If `handleType` is `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT`, the `VkSemaphoreCreateInfo::flags` field **must** match that of the semaphore from which `fd` was exported
- VUID-VkImportSemaphoreFdInfoKHR-handleType-07307
If `handleType` refers to a handle type with copy payload transference semantics, `flags` **must** contain `VK_SEMAPHORE_IMPORT_TEMPORARY_BIT`
- VUID-VkImportSemaphoreFdInfoKHR-handleType-03264
If `handleType` is `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT`, the `VkSemaphoreTypeCreateInfo::semaphoreType` field **must** match that of the semaphore from which `fd` was exported
- VUID-VkImportSemaphoreFdInfoKHR-flags-03323
If `flags` contains `VK_SEMAPHORE_IMPORT_TEMPORARY_BIT`, the `VkSemaphoreTypeCreateInfo::semaphoreType` field of the semaphore from which `fd` was exported **must** not be `VK_SEMAPHORE_TYPE_TIMELINE`

If `handleType` is `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT`, the special value `-1` for `fd` is treated like a valid sync file descriptor referring to an object that has already signaled. The import operation will succeed and the `VkSemaphore` will have a temporarily imported payload as if a valid file descriptor had been provided.

NOTE

This special behavior for importing an invalid sync file descriptor allows easier interoperability with other system APIs which use the convention that an invalid sync file descriptor represents work that has already completed and does not need

to be waited for. It is consistent with the option for implementations to return a -1 file descriptor when exporting a `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT` from a `VkSemaphore` which is signaled.

Valid Usage (Implicit)

- `VUID-VkImportSemaphoreFdInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_FD_INFO_KHR`
- `VUID-VkImportSemaphoreFdInfoKHR-pNext-pNext`
`pNext` **must** be `NULL`
- `VUID-VkImportSemaphoreFdInfoKHR-semaphore-parameter`
`semaphore` **must** be a valid `VkSemaphore` handle
- `VUID-VkImportSemaphoreFdInfoKHR-flags-parameter`
`flags` **must** be a valid combination of `VkSemaphoreImportFlagBits` values
- `VUID-VkImportSemaphoreFdInfoKHR-handleType-parameter`
`handleType` **must** be a valid `VkExternalSemaphoreHandleTypeFlagBits` value

Host Synchronization

- Host access to `semaphore` **must** be externally synchronized

Bits which **can** be set in

- `VkImportSemaphoreWin32HandleInfoKHR::flags`
- `VkImportSemaphoreFdInfoKHR::flags`

specifying additional parameters of a semaphore import operation are:

```
typedef enum VkSemaphoreImportFlagBits {  
    VK_SEMAPHORE_IMPORT_TEMPORARY_BIT = 0x00000001,  
    // Provided by VK_KHR_external_semaphore  
    VK_SEMAPHORE_IMPORT_TEMPORARY_BIT_KHR = VK_SEMAPHORE_IMPORT_TEMPORARY_BIT,  
} VkSemaphoreImportFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_external_semaphore  
typedef VkSemaphoreImportFlagBits VkSemaphoreImportFlagBitsKHR;
```

These bits have the following meanings:

- `VK_SEMAPHORE_IMPORT_TEMPORARY_BIT` specifies that the semaphore payload will be imported only temporarily, as described in [Importing Semaphore Payloads](#), regardless of the permanence of

`handleType`.

```
typedef VkFlags VkSemaphoreImportFlags;
```

or the equivalent

```
// Provided by VK_KHR_external_semaphore
typedef VkSemaphoreImportFlags VkSemaphoreImportFlagsKHR;
```

`VkSemaphoreImportFlags` is a bitmask type for setting a mask of zero or more `VkSemaphoreImportFlagBits`.

7.5. Events

Events are a synchronization primitive that **can** be used to insert a fine-grained dependency between commands submitted to the same queue, or between the host and a queue. Events **must** not be used to insert a dependency between commands submitted to different queues. Events have two states - signaled and unsignaled. An application **can** signal or unsignal an event either on the host or on the device. A device **can** be made to wait for an event to become signaled before executing further operations. No command exists to wait for an event to become signaled on the host, but the current state of an event **can** be queried.

Events are represented by `VkEvent` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkEvent)
```

To create an event, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateEvent(
    VkDevice                                device,
    const VkEventCreateInfo*                pCreateInfo,
    const VkAllocationCallbacks*            pAllocator,
    VkEvent*                                 pEvent);
```

- `device` is the logical device that creates the event.
- `pCreateInfo` is a pointer to a `VkEventCreateInfo` structure containing information about how the event is to be created.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pEvent` is a pointer to a handle in which the resulting event object is returned.

When created, the event object is in the unsignaled state.

Valid Usage

- VUID-vkCreateEvent-device-09672

device **must** support at least one queue family with one of the `VK_QUEUE_VIDEO_ENCODE_BIT_KHR`, `VK_QUEUE_VIDEO_DECODE_BIT_KHR`, `VK_QUEUE_COMPUTE_BIT`, or `VK_QUEUE_GRAPHICS_BIT` capabilities

- VUID-vkCreateEvent-events-04468

If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::events` is `VK_FALSE`, then the implementation does not support `events`, and `vkCreateEvent` **must** not be used

Valid Usage (Implicit)

- VUID-vkCreateEvent-device-parameter

device **must** be a valid `VkDevice` handle

- VUID-vkCreateEvent-pCreateInfo-parameter

pCreateInfo **must** be a valid pointer to a valid `VkEventCreateInfo` structure

- VUID-vkCreateEvent-pAllocator-parameter

If **pAllocator** is not `NULL`, **pAllocator** **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

- VUID-vkCreateEvent-pEvent-parameter

pEvent **must** be a valid pointer to a `VkEvent` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkEventCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkEventCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkEventCreateFlags  flags;
} VkEventCreateInfo;
```

- **sType** is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkEventCreateFlagBits` defining additional creation parameters.

Valid Usage (Implicit)

- VUID-VkEventCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EVENT_CREATE_INFO`
- VUID-VkEventCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkEventCreateInfo-flags-parameter
`flags` **must** be a valid combination of `VkEventCreateFlagBits` values

```
// Provided by VK_VERSION_1_0
typedef enum VkEventCreateFlagBits {
    VK_EVENT_CREATE_DEVICE_ONLY_BIT = 0x00000001,
    // Provided by VK_KHR_synchronization2
    VK_EVENT_CREATE_DEVICE_ONLY_BIT_KHR = VK_EVENT_CREATE_DEVICE_ONLY_BIT,
} VkEventCreateFlagBits;
```

- `VK_EVENT_CREATE_DEVICE_ONLY_BIT` specifies that host event commands will not be used with this event.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkEventCreateFlags;
```

`VkEventCreateFlags` is a bitmask type for setting a mask of `VkEventCreateFlagBits`.

To destroy an event, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyEvent(
    VkDevice          device,
    VkEvent           event,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the event.
- `event` is the handle of the event to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyEvent-event-01145
All submitted commands that refer to `event` **must** have completed execution

- VUID-vkDestroyEvent-event-01146
If `VkAllocationCallbacks` were provided when `event` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyEvent-event-01147
If no `VkAllocationCallbacks` were provided when `event` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyEvent-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyEvent-event-parameter
If `event` is not `VK_NULL_HANDLE`, `event` **must** be a valid `VkEvent` handle
- VUID-vkDestroyEvent-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyEvent-event-parent
If `event` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `event` **must** be externally synchronized

To query the state of an event from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkGetEventStatus(
    VkDevice      device,
    VkEvent       event);
```

- `device` is the logical device that owns the event.
- `event` is the handle of the event to query.

Upon success, `vkGetEventStatus` returns the state of the event object with the following return codes:

Table 10. Event Object Status Codes

Status	Meaning
<code>VK_EVENT_SET</code>	The event specified by <code>event</code> is signaled.
<code>VK_EVENT_RESET</code>	The event specified by <code>event</code> is unsignaled.

If a `vkCmdSetEvent` or `vkCmdResetEvent` command is in a command buffer that is in the `pending` state, then the value returned by this command **may** immediately be out of date.

The state of an event **can** be updated by the host. The state of the event is immediately changed, and subsequent calls to `vkGetEventStatus` will return the new state. If an event is already in the requested state, then updating it to the same state has no effect.

Valid Usage

- VUID-vkGetEventStatus-event-03940
`event` **must** not have been created with `VK_EVENT_CREATE_DEVICE_ONLY_BIT`

Valid Usage (Implicit)

- VUID-vkGetEventStatus-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetEventStatus-event-parameter
`event` **must** be a valid `VkEvent` handle
- VUID-vkGetEventStatus-event-parent
`event` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_EVENT_SET`
- `VK_EVENT_RESET`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

To set the state of an event to signaled from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkSetEvent(
    VkDevice          device,
    VkEvent           event);
```

- `device` is the logical device that owns the event.
- `event` is the event to set.

When `vkSetEvent` is executed on the host, it defines an *event signal operation* which sets the event to the signaled state.

If `event` is already in the signaled state when `vkSetEvent` is executed, then `vkSetEvent` has no effect, and no event signal operation occurs.

NOTE

If a command buffer is waiting for an event to be signaled from the host, the application must signal the event before submitting the command buffer, as described in the [queue forward progress](#) section.

Valid Usage

- VUID-vkSetEvent-event-03941
`event` **must** not have been created with `VK_EVENT_CREATE_DEVICE_ONLY_BIT`
- VUID-vkSetEvent-event-09543
`event` **must** not be waited on by a command buffer in the [pending state](#)

Valid Usage (Implicit)

- VUID-vkSetEvent-device-parameter
`device` **must** be a valid [VkDevice](#) handle
- VUID-vkSetEvent-event-parameter
`event` **must** be a valid [VkEvent](#) handle
- VUID-vkSetEvent-event-parent
`event` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `event` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

To set the state of an event to unsignaled from the host, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetEvent(
    VkDevice          device,
    VkEvent            event);
```

- **device** is the logical device that owns the event.
- **event** is the event to reset.

When **vkResetEvent** is executed on the host, it defines an *event unsignal operation* which resets the event to the unsignaled state.

If **event** is already in the unsignaled state when **vkResetEvent** is executed, then **vkResetEvent** has no effect, and no event unsignal operation occurs.

Valid Usage

- VUID-vkResetEvent-event-03821
There **must** be an execution dependency between **vkResetEvent** and the execution of any **vkCmdWaitEvents** that includes **event** in its **pEvents** parameter
- VUID-vkResetEvent-event-03822
There **must** be an execution dependency between **vkResetEvent** and the execution of any **vkCmdWaitEvents2KHR** that includes **event** in its **pEvents** parameter
- VUID-vkResetEvent-event-03823
event **must** not have been created with **VK_EVENT_CREATE_DEVICE_ONLY_BIT**

Valid Usage (Implicit)

- VUID-vkResetEvent-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkResetEvent-event-parameter
event **must** be a valid **VkEvent** handle
- VUID-vkResetEvent-event-parent
event **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **event** **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The state of an event **can** also be updated on the device by commands inserted in command buffers.

To signal an event from a device, call:

```
// Provided by VK_KHR_synchronization2
void vkCmdSetEvent2KHR(
    VkCommandBuffer          commandBuffer,
    VkEvent                  event,
    const VkDependencyInfo*  pDependencyInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `event` is the event that will be signaled.
- `pDependencyInfo` is a pointer to a `VkDependencyInfo` structure defining the first scopes of this operation.

When `vkCmdSetEvent2KHR` is submitted to a queue, it defines the first half of memory dependencies defined by `pDependencyInfo`, as well as an event signal operation which sets the event to the signaled state. A memory dependency is defined between the event signal operation and commands that occur earlier in submission order.

The first `synchronization scope` and `access scope` are defined by the union of all the memory dependencies defined by `pDependencyInfo`, and are applied to all operations that occur earlier in `submission order`. `Queue family ownership transfers` and `image layout transitions` defined by `pDependencyInfo` are also included in the first scopes.

The second `synchronization scope` includes only the event signal operation, and any `queue family ownership transfers` and `image layout transitions` defined by `pDependencyInfo`.

The second `access scope` includes only `queue family ownership transfers` and `image layout transitions`.

Future `vkCmdWaitEvents2KHR` commands rely on all values of each element in `pDependencyInfo` matching exactly with those used to signal the corresponding event. `vkCmdWaitEvents` **must** not be used to wait on the result of a signal operation defined by `vkCmdSetEvent2KHR`.

NOTE

The extra information provided by `vkCmdSetEvent2KHR` compared to `vkCmdSetEvent` allows implementations to more efficiently schedule the operations required to satisfy the requested dependencies. With `vkCmdSetEvent`, the full

dependency information is not known until [vkCmdWaitEvents](#) is recorded, forcing implementations to insert the required operations at that point and not before.

If [event](#) is already in the signaled state when [vkCmdSetEvent2KHR](#) is executed on the device, then [vkCmdSetEvent2KHR](#) has no effect, no event signal operation occurs, and no dependency is generated.

Valid Usage

- VUID-vkCmdSetEvent2-synchronization2-03824
The [synchronization2](#) feature **must** be enabled
- VUID-vkCmdSetEvent2-dependencyFlags-03825
The [dependencyFlags](#) member of [pDependencyInfo](#) **must** be 0
- VUID-vkCmdSetEvent2-srcStageMask-09391
The [srcStageMask](#) member of any element of the [pMemoryBarriers](#), [pBufferMemoryBarriers](#), or [pImageMemoryBarriers](#) members of [pDependencyInfo](#) **must** not include [VK_PIPELINE_STAGE_2_HOST_BIT](#)
- VUID-vkCmdSetEvent2-dstStageMask-09392
The [dstStageMask](#) member of any element of the [pMemoryBarriers](#), [pBufferMemoryBarriers](#), or [pImageMemoryBarriers](#) members of [pDependencyInfo](#) **must** not include [VK_PIPELINE_STAGE_2_HOST_BIT](#)
- VUID-vkCmdSetEvent2-commandBuffer-03826
The current device mask of [commandBuffer](#) **must** include exactly one physical device
- VUID-vkCmdSetEvent2-srcStageMask-03827
The [srcStageMask](#) member of any element of the [pMemoryBarriers](#), [pBufferMemoryBarriers](#), or [pImageMemoryBarriers](#) members of [pDependencyInfo](#) **must** only include pipeline stages valid for the queue family that was used to create the command pool that [commandBuffer](#) was allocated from
- VUID-vkCmdSetEvent2-dstStageMask-03828
The [dstStageMask](#) member of any element of the [pMemoryBarriers](#), [pBufferMemoryBarriers](#), or [pImageMemoryBarriers](#) members of [pDependencyInfo](#) **must** only include pipeline stages valid for the queue family that was used to create the command pool that [commandBuffer](#) was allocated from

Valid Usage (Implicit)

- VUID-vkCmdSetEvent2-commandBuffer-parameter
[commandBuffer](#) **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetEvent2-event-parameter
[event](#) **must** be a valid [VkEvent](#) handle
- VUID-vkCmdSetEvent2-pDependencyInfo-parameter
[pDependencyInfo](#) **must** be a valid pointer to a valid [VkDependencyInfo](#) structure
- VUID-vkCmdSetEvent2-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdSetEvent2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, compute, decode, or encode operations
- VUID-vkCmdSetEvent2-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdSetEvent2-commonparent
Both of `commandBuffer`, and `event` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Both	Graphics Compute Decode Encode	Synchronization

The `VkDependencyInfo` structure is defined as:

```
typedef struct VkDependencyInfo {  
    VkStructureType      sType;  
    const void*          pNext;  
    VkDependencyFlags    dependencyFlags;  
    uint32_t             memoryBarrierCount;  
    const VkMemoryBarrier2* pMemoryBarriers;  
    uint32_t             bufferMemoryBarrierCount;  
    const VkBufferMemoryBarrier2* pBufferMemoryBarriers;  
    uint32_t             imageMemoryBarrierCount;  
    const VkImageMemoryBarrier2* pImageMemoryBarriers;  
} VkDependencyInfo;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2
typedef VkDependencyInfo VkDependencyInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `dependencyFlags` is a bitmask of `VkDependencyFlagBits` specifying how execution and memory dependencies are formed.
- `memoryBarrierCount` is the length of the `pMemoryBarriers` array.
- `pMemoryBarriers` is a pointer to an array of `VkMemoryBarrier2` structures defining memory dependencies between any memory accesses.
- `bufferMemoryBarrierCount` is the length of the `pBufferMemoryBarriers` array.
- `pBufferMemoryBarriers` is a pointer to an array of `VkBufferMemoryBarrier2` structures defining memory dependencies between buffer ranges.
- `imageMemoryBarrierCount` is the length of the `pImageMemoryBarriers` array.
- `pImageMemoryBarriers` is a pointer to an array of `VkImageMemoryBarrier2` structures defining memory dependencies between image subresources.

This structure defines a set of [memory dependencies](#), as well as [queue family ownership transfer operations](#) and [image layout transitions](#).

Each member of `pMemoryBarriers`, `pBufferMemoryBarriers`, and `pImageMemoryBarriers` defines a separate [memory dependency](#).

Valid Usage (Implicit)

- VUID-VkDependencyInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEPENDENCY_INFO`
- VUID-VkDependencyInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDependencyInfo-dependencyFlags-parameter
`dependencyFlags` **must** be a valid combination of `VkDependencyFlagBits` values
- VUID-VkDependencyInfo-pMemoryBarriers-parameter
If `memoryBarrierCount` is not 0, `pMemoryBarriers` **must** be a valid pointer to an array of `memoryBarrierCount` valid `VkMemoryBarrier2` structures
- VUID-VkDependencyInfo-pBufferMemoryBarriers-parameter
If `bufferMemoryBarrierCount` is not 0, `pBufferMemoryBarriers` **must** be a valid pointer to an array of `bufferMemoryBarrierCount` valid `VkBufferMemoryBarrier2` structures
- VUID-VkDependencyInfo-pImageMemoryBarriers-parameter
If `imageMemoryBarrierCount` is not 0, `pImageMemoryBarriers` **must** be a valid pointer to an array of `imageMemoryBarrierCount` valid `VkImageMemoryBarrier2` structures

To set the state of an event to signaled from a device, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetEvent(
    VkCommandBuffer          commandBuffer,
    VkEvent                  event,
    VkPipelineStageFlags     stageMask);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `event` is the event that will be signaled.
- `stageMask` specifies the [source stage mask](#) used to determine the first [synchronization scope](#).

`vkCmdSetEvent` behaves identically to [vkCmdSetEvent2KHR](#), except that it does not define an access scope, and **must** only be used with [vkCmdWaitEvents](#), not [vkCmdWaitEvents2KHR](#).

Valid Usage

- VUID-vkCmdSetEvent-stageMask-04090
If the `geometryShader` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-vkCmdSetEvent-stageMask-04091
If the `tessellationShader` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-vkCmdSetEvent-stageMask-04094
If the `transformFeedback` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-vkCmdSetEvent-stageMask-07319
If the `attachmentFragmentShadingRate` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-vkCmdSetEvent-stageMask-03937
If the `synchronization2` feature is not enabled, `stageMask` **must** not be `0`
- VUID-vkCmdSetEvent-stageMask-07950
If the `rayTracingPipeline` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-vkCmdSetEvent-stageMask-06457
Any pipeline stage included in `stageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the [VkCommandPoolCreateInfo](#) structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the [table of supported pipeline stages](#)
- VUID-vkCmdSetEvent-stageMask-01149
`stageMask` **must** not include `VK_PIPELINE_STAGE_HOST_BIT`
- VUID-vkCmdSetEvent-commandBuffer-01152

The current device mask of **commandBuffer** **must** include exactly one physical device

Valid Usage (Implicit)

- VUID-vkCmdSetEvent-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdSetEvent-event-parameter
event **must** be a valid **VkEvent** handle
- VUID-vkCmdSetEvent-stageMask-parameter
stageMask **must** be a valid combination of **VkPipelineStageFlagBits** values
- VUID-vkCmdSetEvent-commandBuffer-recording
commandBuffer **must** be in the **recording state**
- VUID-vkCmdSetEvent-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics, compute, decode, or encode operations
- VUID-vkCmdSetEvent-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdSetEvent-commonparent
Both of **commandBuffer**, and **event** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Both	Graphics Compute Decode Encode	Synchronization

To unsignal the event from a device, call:

```
// Provided by VK_KHR_synchronization2
void vkCmdResetEvent2KHR(
    VkCommandBuffer          commandBuffer,
    VkEvent                  event,
    VkPipelineStageFlags2    stageMask);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `event` is the event that will be unsignaled.
- `stageMask` is a `VkPipelineStageFlags2` mask of pipeline stages used to determine the first `synchronization scope`.

When `vkCmdResetEvent2KHR` is submitted to a queue, it defines an execution dependency on commands that were submitted before it, and defines an event unsignal operation which resets the event to the unsignaled state.

The first `synchronization scope` includes all commands that occur earlier in `submission order`. The synchronization scope is limited to operations by `stageMask` or stages that are `logically earlier` than `stageMask`.

The second `synchronization scope` includes only the event unsignal operation.

If `event` is already in the unsignaled state when `vkCmdResetEvent2KHR` is executed on the device, then this command has no effect, no event unsignal operation occurs, and no execution dependency is generated.

Valid Usage

- VUID-vkCmdResetEvent2-stageMask-03929
If the `geometryShader` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT`
- VUID-vkCmdResetEvent2-stageMask-03930
If the `tessellationShader` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-vkCmdResetEvent2-stageMask-03933
If the `transformFeedback` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-vkCmdResetEvent2-stageMask-07317
If the `attachmentFragmentShadingRate` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-vkCmdResetEvent2-stageMask-07947
If the `rayTracingPipeline` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`
- VUID-vkCmdResetEvent2-synchronization2-03829
The `synchronization2` feature **must** be enabled

- VUID-vkCmdResetEvent2-stageMask-03830
`stageMask` **must** not include `VK_PIPELINE_STAGE_2_HOST_BIT`
- VUID-vkCmdResetEvent2-event-03831
There **must** be an execution dependency between `vkCmdResetEvent2KHR` and the execution of any `vkCmdWaitEvents` that includes `event` in its `pEvents` parameter
- VUID-vkCmdResetEvent2-event-03832
There **must** be an execution dependency between `vkCmdResetEvent2KHR` and the execution of any `vkCmdWaitEvents2KHR` that includes `event` in its `pEvents` parameter
- VUID-vkCmdResetEvent2-commandBuffer-03833
`commandBuffer`'s current device mask **must** include exactly one physical device

Valid Usage (Implicit)

- VUID-vkCmdResetEvent2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdResetEvent2-event-parameter
`event` **must** be a valid `VkEvent` handle
- VUID-vkCmdResetEvent2-stageMask-parameter
`stageMask` **must** be a valid combination of `VkPipelineStageFlagBits2` values
- VUID-vkCmdResetEvent2-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdResetEvent2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, compute, decode, or encode operations
- VUID-vkCmdResetEvent2-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdResetEvent2-commonparent
Both of `commandBuffer`, and `event` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Both	Graphics Compute Decode Encode	Synchronization

To set the state of an event to unsignaled from a device, call:

```
// Provided by VK_VERSION_1_0
void vkCmdResetEvent(
    VkCommandBuffer          commandBuffer,
    VkEvent                  event,
    VkPipelineStageFlags     stageMask);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `event` is the event that will be unsignaled.
- `stageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the `source stage mask` used to determine when the `event` is unsignaled.

`vkCmdResetEvent` behaves identically to `vkCmdResetEvent2KHR`.

Valid Usage

- VUID-vkCmdResetEvent-stageMask-04090
If the `geometryShader` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-vkCmdResetEvent-stageMask-04091
If the `tessellationShader` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-vkCmdResetEvent-stageMask-04094
If the `transformFeedback` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-vkCmdResetEvent-stageMask-07319
If the `attachmentFragmentShadingRate` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-vkCmdResetEvent-stageMask-03937
If the `synchronization2` feature is not enabled, `stageMask` **must** not be `0`
- VUID-vkCmdResetEvent-stageMask-07950

If the `rayTracingPipeline` feature is not enabled, `stageMask` **must** not contain `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`

- VUID-vkCmdResetEvent-stageMask-06458

Any pipeline stage included in `stageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the [table of supported pipeline stages](#)

- VUID-vkCmdResetEvent-stageMask-01153

`stageMask` **must** not include `VK_PIPELINE_STAGE_HOST_BIT`

- VUID-vkCmdResetEvent-event-03834

There **must** be an execution dependency between `vkCmdResetEvent` and the execution of any `vkCmdWaitEvents` that includes `event` in its `pEvents` parameter

- VUID-vkCmdResetEvent-event-03835

There **must** be an execution dependency between `vkCmdResetEvent` and the execution of any `vkCmdWaitEvents2KHR` that includes `event` in its `pEvents` parameter

- VUID-vkCmdResetEvent-commandBuffer-01157

`commandBuffer`'s current device mask **must** include exactly one physical device

Valid Usage (Implicit)

- VUID-vkCmdResetEvent-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdResetEvent-event-parameter

`event` **must** be a valid `VkEvent` handle

- VUID-vkCmdResetEvent-stageMask-parameter

`stageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values

- VUID-vkCmdResetEvent-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdResetEvent-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, compute, decode, or encode operations

- VUID-vkCmdResetEvent-renderpass

This command **must** only be called outside of a render pass instance

- VUID-vkCmdResetEvent-commonparent

Both of `commandBuffer`, and `event` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Both	Graphics Compute Decode Encode	Synchronization

To wait for one or more events to enter the signaled state on a device, call:

```
// Provided by VK_KHR_synchronization2
void vkCmdWaitEvents2KHR(
    VkCommandBuffer          commandBuffer,
    uint32_t                 eventCount,
    const VkEvent*           pEvents,
    const VkDependencyInfo*  pDependencyInfos);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `eventCount` is the length of the `pEvents` array.
- `pEvents` is a pointer to an array of `eventCount` events to wait on.
- `pDependencyInfos` is a pointer to an array of `eventCount` `VkDependencyInfo` structures, defining the second `synchronization scope`.

When `vkCmdWaitEvents2KHR` is submitted to a queue, it inserts memory dependencies according to the elements of `pDependencyInfos` and each corresponding element of `pEvents`. `vkCmdWaitEvents2KHR` **must** not be used to wait on event signal operations occurring on other queues, or signal operations executed by `vkCmdSetEvent`.

The first `synchronization scope` and `access scope` of each memory dependency defined by any element `i` of `pDependencyInfos` are applied to operations that occurred earlier in `submission order` than the last event signal operation on element `i` of `pEvents`.

Signal operations for an event at index `i` are only included if:

- The event was signaled by a `vkCmdSetEvent2KHR` command that occurred earlier in `submission order` with a `dependencyInfo` parameter exactly equal to the element of `pDependencyInfos` at index `i`; or
- The event was created without `VK_EVENT_CREATE_DEVICE_ONLY_BIT`, and the first `synchronization scope` defined by the element of `pDependencyInfos` at index `i` only includes host operations (`VK_PIPELINE_STAGE_2_HOST_BIT`).

The second [synchronization scope](#) and [access scope](#) of each memory dependency defined by any element *i* of [pDependencyInfos](#) are applied to operations that occurred later in [submission order](#) than [vkCmdWaitEvents2KHR](#).

NOTE

[vkCmdWaitEvents2KHR](#) is used with [vkCmdSetEvent2KHR](#) to define a memory dependency between two sets of action commands, roughly in the same way as pipeline barriers, but split into two commands such that work between the two **may** execute unhindered.

NOTE

Applications should be careful to avoid race conditions when using events. There is no direct ordering guarantee between [vkCmdSetEvent2KHR](#) and [vkCmdResetEvent2KHR](#), [vkCmdResetEvent](#), or [vkCmdSetEvent](#). Another execution dependency (e.g. a pipeline barrier or semaphore with [VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT](#)) is needed to prevent such a race condition.

Valid Usage

- VUID-vkCmdWaitEvents2-synchronization2-03836
The [synchronization2](#) feature **must** be enabled
- VUID-vkCmdWaitEvents2-pEvents-03837
Members of [pEvents](#) **must** not have been signaled by [vkCmdSetEvent](#)
- VUID-vkCmdWaitEvents2-pEvents-03838
For any element *i* of [pEvents](#), if that event is signaled by [vkCmdSetEvent2KHR](#), that command's [dependencyInfo](#) parameter **must** be exactly equal to the *i*th element of [pDependencyInfos](#)
- VUID-vkCmdWaitEvents2-pEvents-03839
For any element *i* of [pEvents](#), if that event is signaled by [vkSetEvent](#), barriers in the *i*th element of [pDependencyInfos](#) **must** include only host operations in their first [synchronization scope](#)
- VUID-vkCmdWaitEvents2-pEvents-03840
For any element *i* of [pEvents](#), if barriers in the *i*th element of [pDependencyInfos](#) include only host operations, the *i*th element of [pEvents](#) **must** be signaled before [vkCmdWaitEvents2KHR](#) is executed
- VUID-vkCmdWaitEvents2-pEvents-03841
For any element *i* of [pEvents](#), if barriers in the *i*th element of [pDependencyInfos](#) do not include host operations, the *i*th element of [pEvents](#) **must** be signaled by a corresponding [vkCmdSetEvent2KHR](#) that occurred earlier in [submission order](#)
- VUID-vkCmdWaitEvents2-srcStageMask-03842
The [srcStageMask](#) member of any element of the [pMemoryBarriers](#), [pBufferMemoryBarriers](#), or [pImageMemoryBarriers](#) members of [pDependencyInfos](#) **must** only include pipeline stages valid for the queue family that was used to create the command pool that [commandBuffer](#) was allocated from
- VUID-vkCmdWaitEvents2-dstStageMask-03843
The [dstStageMask](#) member of any element of the [pMemoryBarriers](#), [pBufferMemoryBarriers](#),

or `pImageMemoryBarriers` members of `pDependencyInfos` **must** only include pipeline stages valid for the queue family that was used to create the command pool that `commandBuffer` was allocated from

- VUID-vkCmdWaitEvents2-dependencyFlags-03844

If `vkCmdWaitEvents2KHR` is being called inside a render pass instance, the `srcStageMask` member of any element of the `pMemoryBarriers`, `pBufferMemoryBarriers`, or `pImageMemoryBarriers` members of `pDependencyInfos` **must** not include `VK_PIPELINE_STAGE_2_HOST_BIT`

- VUID-vkCmdWaitEvents2-commandBuffer-03846

`commandBuffer`'s current device mask **must** include exactly one physical device

Valid Usage (Implicit)

- VUID-vkCmdWaitEvents2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdWaitEvents2-pEvents-parameter
`pEvents` **must** be a valid pointer to an array of `eventCount` valid `VkEvent` handles
- VUID-vkCmdWaitEvents2-pDependencyInfos-parameter
`pDependencyInfos` **must** be a valid pointer to an array of `eventCount` valid `VkDependencyInfo` structures
- VUID-vkCmdWaitEvents2-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdWaitEvents2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, compute, decode, or encode operations
- VUID-vkCmdWaitEvents2-eventCount-arraylength
`eventCount` **must** be greater than 0
- VUID-vkCmdWaitEvents2-commonparent
Both of `commandBuffer`, and the elements of `pEvents` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	Graphics Compute Decode Encode	Synchronization

To wait for one or more events to enter the signaled state on a device, call:

```
// Provided by VK_VERSION_1_0
void vkCmdWaitEvents(
    VkCommandBuffer                commandBuffer,
    uint32_t                       eventCount,
    const VkEvent*                 pEvents,
    VkPipelineStageFlags           srcStageMask,
    VkPipelineStageFlags           dstStageMask,
    uint32_t                       memoryBarrierCount,
    const VkMemoryBarrier*         pMemoryBarriers,
    uint32_t                       bufferMemoryBarrierCount,
    const VkBufferMemoryBarrier*   pBufferMemoryBarriers,
    uint32_t                       imageMemoryBarrierCount,
    const VkImageMemoryBarrier*    pImageMemoryBarriers);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `eventCount` is the length of the `pEvents` array.
- `pEvents` is a pointer to an array of event object handles to wait on.
- `srcStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [source stage mask](#).
- `dstStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [destination stage mask](#).
- `memoryBarrierCount` is the length of the `pMemoryBarriers` array.
- `pMemoryBarriers` is a pointer to an array of `VkMemoryBarrier` structures.
- `bufferMemoryBarrierCount` is the length of the `pBufferMemoryBarriers` array.
- `pBufferMemoryBarriers` is a pointer to an array of `VkBufferMemoryBarrier` structures.
- `imageMemoryBarrierCount` is the length of the `pImageMemoryBarriers` array.
- `pImageMemoryBarriers` is a pointer to an array of `VkImageMemoryBarrier` structures.

`vkCmdWaitEvents` is largely similar to `vkCmdWaitEvents2KHR`, but **can** only wait on signal operations defined by `vkCmdSetEvent`. As `vkCmdSetEvent` does not define any access scopes, `vkCmdWaitEvents` defines the first access scope for each event signal operation in addition to its own access scopes.

NOTE

Since `vkCmdSetEvent` does not have any dependency information beyond a stage

mask, implementations do not have the same opportunity to perform [availability and visibility operations](#) or [image layout transitions](#) in advance as they do with [vkCmdSetEvent2KHR](#) and [vkCmdWaitEvents2KHR](#).

When [vkCmdWaitEvents](#) is submitted to a queue, it defines a memory dependency between prior event signal operations on the same queue or the host, and subsequent commands. [vkCmdWaitEvents](#) **must** not be used to wait on event signal operations occurring on other queues.

The first synchronization scope only includes event signal operations that operate on members of [pEvents](#), and the operations that happened-before the event signal operations. Event signal operations performed by [vkCmdSetEvent](#) that occur earlier in [submission order](#) are included in the first synchronization scope, if the [logically latest](#) pipeline stage in their [stageMask](#) parameter is [logically earlier](#) than or equal to the [logically latest](#) pipeline stage in [srcStageMask](#). Event signal operations performed by [vkSetEvent](#) are only included in the first synchronization scope if [VK_PIPELINE_STAGE_HOST_BIT](#) is included in [srcStageMask](#).

The second [synchronization scope](#) includes all commands that occur later in [submission order](#). The second synchronization scope is limited to operations on the pipeline stages determined by the [destination stage mask](#) specified by [dstStageMask](#).

The first [access scope](#) is limited to accesses in the pipeline stages determined by the [source stage mask](#) specified by [srcStageMask](#). Within that, the first access scope only includes the first access scopes defined by elements of the [pMemoryBarriers](#), [pBufferMemoryBarriers](#) and [pImageMemoryBarriers](#) arrays, which each define a set of [memory barriers](#). If no memory barriers are specified, then the first access scope includes no accesses.

The second [access scope](#) is limited to accesses in the pipeline stages determined by the [destination stage mask](#) specified by [dstStageMask](#). Within that, the second access scope only includes the second access scopes defined by elements of the [pMemoryBarriers](#), [pBufferMemoryBarriers](#) and [pImageMemoryBarriers](#) arrays, which each define a set of [memory barriers](#). If no memory barriers are specified, then the second access scope includes no accesses.

Valid Usage

- VUID-vkCmdWaitEvents-srcStageMask-04090
If the [geometryShader](#) feature is not enabled, [srcStageMask](#) **must** not contain [VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT](#)
- VUID-vkCmdWaitEvents-srcStageMask-04091
If the [tessellationShader](#) feature is not enabled, [srcStageMask](#) **must** not contain [VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT](#) or [VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT](#)
- VUID-vkCmdWaitEvents-srcStageMask-04094
If the [transformFeedback](#) feature is not enabled, [srcStageMask](#) **must** not contain [VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT](#)
- VUID-vkCmdWaitEvents-srcStageMask-07319
If the [attachmentFragmentShadingRate](#) feature is not enabled, [srcStageMask](#) **must** not contain [VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR](#)

- VUID-vkCmdWaitEvents-srcStageMask-03937
If the `synchronization2` feature is not enabled, `srcStageMask` **must** not be 0
- VUID-vkCmdWaitEvents-srcStageMask-07950
If the `rayTracingPipeline` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-vkCmdWaitEvents-srcAccessMask-06257
If the `rayQuery` feature is not enabled and a memory barrier `srcAccessMask` includes `VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `srcStageMask` **must** not include any of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages except `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-vkCmdWaitEvents-dstStageMask-04090
If the `geometryShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-vkCmdWaitEvents-dstStageMask-04091
If the `tessellationShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-vkCmdWaitEvents-dstStageMask-04094
If the `transformFeedback` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-vkCmdWaitEvents-dstStageMask-07319
If the `attachmentFragmentShadingRate` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-vkCmdWaitEvents-dstStageMask-03937
If the `synchronization2` feature is not enabled, `dstStageMask` **must** not be 0
- VUID-vkCmdWaitEvents-dstStageMask-07950
If the `rayTracingPipeline` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-vkCmdWaitEvents-dstAccessMask-06257
If the `rayQuery` feature is not enabled and a memory barrier `dstAccessMask` includes `VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `dstStageMask` **must** not include any of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages except `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-vkCmdWaitEvents-srcAccessMask-02815
The `srcAccessMask` member of each element of `pMemoryBarriers` **must** only include access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)
- VUID-vkCmdWaitEvents-dstAccessMask-02816
The `dstAccessMask` member of each element of `pMemoryBarriers` **must** only include access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdWaitEvents-pBufferMemoryBarriers-02817
For any element of `pBufferMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `srcQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `srcAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)
- VUID-vkCmdWaitEvents-pBufferMemoryBarriers-02818
For any element of `pBufferMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `dstQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `dstAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)
- VUID-vkCmdWaitEvents-pImageMemoryBarriers-02819
For any element of `pImageMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `srcQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `srcAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)
- VUID-vkCmdWaitEvents-pImageMemoryBarriers-02820
For any element of `pImageMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `dstQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `dstAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)
- VUID-vkCmdWaitEvents-srcStageMask-06459
Any pipeline stage included in `srcStageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the [table of supported pipeline stages](#)
- VUID-vkCmdWaitEvents-dstStageMask-06460
Any pipeline stage included in `dstStageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the [table of supported pipeline stages](#)
- VUID-vkCmdWaitEvents-srcStageMask-01158
`srcStageMask` **must** be the bitwise OR of the `stageMask` parameter used in previous calls to `vkCmdSetEvent` with any of the elements of `pEvents` and `VK_PIPELINE_STAGE_HOST_BIT` if any of the elements of `pEvents` was set using `vkSetEvent`
- VUID-vkCmdWaitEvents-srcStageMask-07308
If `vkCmdWaitEvents` is being called inside a render pass instance, `srcStageMask` **must** not include `VK_PIPELINE_STAGE_HOST_BIT`

- VUID-vkCmdWaitEvents-srcQueueFamilyIndex-02803
The `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members of any element of `pBufferMemoryBarriers` or `pImageMemoryBarriers` **must** be equal
- VUID-vkCmdWaitEvents-commandBuffer-01167
`commandBuffer`'s current device mask **must** include exactly one physical device
- VUID-vkCmdWaitEvents-pEvents-03847
Elements of `pEvents` **must** not have been signaled by `vkCmdSetEvent2KHR`

Valid Usage (Implicit)

- VUID-vkCmdWaitEvents-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdWaitEvents-pEvents-parameter
`pEvents` **must** be a valid pointer to an array of `eventCount` valid `VkEvent` handles
- VUID-vkCmdWaitEvents-srcStageMask-parameter
`srcStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values
- VUID-vkCmdWaitEvents-dstStageMask-parameter
`dstStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values
- VUID-vkCmdWaitEvents-pMemoryBarriers-parameter
If `memoryBarrierCount` is not 0, `pMemoryBarriers` **must** be a valid pointer to an array of `memoryBarrierCount` valid `VkMemoryBarrier` structures
- VUID-vkCmdWaitEvents-pBufferMemoryBarriers-parameter
If `bufferMemoryBarrierCount` is not 0, `pBufferMemoryBarriers` **must** be a valid pointer to an array of `bufferMemoryBarrierCount` valid `VkBufferMemoryBarrier` structures
- VUID-vkCmdWaitEvents-pImageMemoryBarriers-parameter
If `imageMemoryBarrierCount` is not 0, `pImageMemoryBarriers` **must** be a valid pointer to an array of `imageMemoryBarrierCount` valid `VkImageMemoryBarrier` structures
- VUID-vkCmdWaitEvents-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdWaitEvents-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, compute, decode, or encode operations
- VUID-vkCmdWaitEvents-eventCount-arraylength
`eventCount` **must** be greater than 0
- VUID-vkCmdWaitEvents-commonparent
Both of `commandBuffer`, and the elements of `pEvents` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	Graphics Compute Decode Encode	Synchronization

7.6. Pipeline Barriers

To record a pipeline barrier, call:

```
// Provided by VK_KHR_synchronization2
void vkCmdPipelineBarrier2KHR(
    VkCommandBuffer                commandBuffer,
    const VkDependencyInfo*        pDependencyInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pDependencyInfo` is a pointer to a `VkDependencyInfo` structure defining the scopes of this operation.

When `vkCmdPipelineBarrier2KHR` is submitted to a queue, it defines memory dependencies between commands that were submitted to the same queue before it, and those submitted to the same queue after it.

The first `synchronization scope` and `access scope` of each memory dependency defined by `pDependencyInfo` are applied to operations that occurred earlier in `submission order`.

The second `synchronization scope` and `access scope` of each memory dependency defined by `pDependencyInfo` are applied to operations that occurred later in `submission order`.

If `vkCmdPipelineBarrier2KHR` is recorded within a render pass instance, the synchronization scopes are limited to a subset of operations within the same subpass or render pass instance.

Valid Usage

- VUID-vkCmdPipelineBarrier2-None-07889
If `vkCmdPipelineBarrier2KHR` is called within a render pass instance using a `VkRenderPass` object, the render pass **must** have been created with at least one subpass dependency that expresses a dependency from the current subpass to itself, does not include

`VK_DEPENDENCY_BY_REGION_BIT` if this command does not, does not include `VK_DEPENDENCY_VIEW_LOCAL_BIT` if this command does not, and has [synchronization scopes](#) and [access scopes](#) that are all supersets of the scopes defined in this command

- VUID-vkCmdPipelineBarrier2-bufferMemoryBarrierCount-01178
If `vkCmdPipelineBarrier2KHR` is called within a render pass instance using a [VkRenderPass](#) object, it **must** not include any buffer memory barriers
- VUID-vkCmdPipelineBarrier2-image-04073
If `vkCmdPipelineBarrier2KHR` is called within a render pass instance using a [VkRenderPass](#) object, the `image` member of any image memory barrier included in this command **must** be an attachment used in the current subpass both as an input attachment, and as either a color, or depth/stencil attachment
- VUID-vkCmdPipelineBarrier2-oldLayout-01181
If `vkCmdPipelineBarrier2KHR` is called within a render pass instance, the `oldLayout` and `newLayout` members of any image memory barrier included in this command **must** be equal
- VUID-vkCmdPipelineBarrier2-srcQueueFamilyIndex-01182
If `vkCmdPipelineBarrier2KHR` is called within a render pass instance, the `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members of any memory barrier included in this command **must** be equal
- VUID-vkCmdPipelineBarrier2-None-07890
If `vkCmdPipelineBarrier2KHR` is called within a render pass instance, and the source stage masks of any memory barriers include [framebuffer-space stages](#), destination stage masks of all memory barriers **must** only include [framebuffer-space stages](#)
- VUID-vkCmdPipelineBarrier2-dependencyFlags-07891
If `vkCmdPipelineBarrier2KHR` is called within a render pass instance, and the source stage masks of any memory barriers include [framebuffer-space stages](#), then `dependencyFlags` **must** include `VK_DEPENDENCY_BY_REGION_BIT`
- VUID-vkCmdPipelineBarrier2-None-07892
If `vkCmdPipelineBarrier2KHR` is called within a render pass instance, the source and destination stage masks of any memory barriers **must** only include graphics pipeline stages
- VUID-vkCmdPipelineBarrier2-dependencyFlags-01186
If `vkCmdPipelineBarrier2KHR` is called outside of a render pass instance, the dependency flags **must** not include `VK_DEPENDENCY_VIEW_LOCAL_BIT`
- VUID-vkCmdPipelineBarrier2-None-07893
If `vkCmdPipelineBarrier2KHR` is called inside a render pass instance, and there is more than one view in the current subpass, dependency flags **must** include `VK_DEPENDENCY_VIEW_LOCAL_BIT`
- VUID-vkCmdPipelineBarrier2-None-09553
If none of the `shaderTileImageColorReadAccess`, `shaderTileImageStencilReadAccess`, or `shaderTileImageDepthReadAccess` features are enabled, and the `dynamicRenderingLocalRead` feature is not enabled, `vkCmdPipelineBarrier2KHR` **must** not be called within a render pass instance started with [vkCmdBeginRenderingKHR](#)

- VUID-vkCmdPipelineBarrier2-None-09554
If the `dynamicRenderingLocalRead` feature is not enabled, and `vkCmdPipelineBarrier2KHR` is called within a render pass instance started with `vkCmdBeginRenderingKHR`, there **must** be no buffer or image memory barriers specified by this command
- VUID-vkCmdPipelineBarrier2-None-09586
If the `dynamicRenderingLocalRead` feature is not enabled, and `vkCmdPipelineBarrier2KHR` is called within a render pass instance started with `vkCmdBeginRenderingKHR`, memory barriers specified by this command **must** only include `VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT`, `VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT`, `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT`, or `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT` in their access masks
- VUID-vkCmdPipelineBarrier2-image-09555
If `vkCmdPipelineBarrier2KHR` is called within a render pass instance started with `vkCmdBeginRenderingKHR`, and the `image` member of any image memory barrier is used as an attachment in the current render pass instance, it **must** be in the `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR` or `VK_IMAGE_LAYOUT_GENERAL` layout
- VUID-vkCmdPipelineBarrier2-srcStageMask-09556
If `vkCmdPipelineBarrier2KHR` is called within a render pass instance started with `vkCmdBeginRenderingKHR`, this command **must** only specify `framebuffer-space` stages in `srcStageMask` and `dstStageMask`
- VUID-vkCmdPipelineBarrier2-synchronization2-03848
The `synchronization2` feature **must** be enabled
- VUID-vkCmdPipelineBarrier2-srcStageMask-09673
The `srcStageMask` member of any element of the `pMemoryBarriers` member of `pDependencyInfo` **must** only include pipeline stages valid for the queue family that was used to create the command pool that `commandBuffer` was allocated from
- VUID-vkCmdPipelineBarrier2-dstStageMask-09674
The `dstStageMask` member of any element of the `pMemoryBarriers` member of `pDependencyInfo` **must** only include pipeline stages valid for the queue family that was used to create the command pool that `commandBuffer` was allocated from
- VUID-vkCmdPipelineBarrier2-srcStageMask-09675
If a buffer or image memory barrier does not specify an `acquire operation`, the respective `srcStageMask` member of the element of the `pBufferMemoryBarriers` or `pImageMemoryBarriers` members of `pDependencyInfo` **must** only include pipeline stages valid for the queue family that was used to create the command pool that `commandBuffer` was allocated from
- VUID-vkCmdPipelineBarrier2-dstStageMask-09676
If a buffer or image memory barrier does not specify an `release operation`, the respective `dstStageMask` member of the element of the `pBufferMemoryBarriers` or `pImageMemoryBarriers` members of `pDependencyInfo` **must** only include pipeline stages valid for the queue family that was used to create the command pool that `commandBuffer` was allocated from

Valid Usage (Implicit)

- VUID-vkCmdPipelineBarrier2-commandBuffer-parameter **commandBuffer** **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdPipelineBarrier2-pDependencyInfo-parameter **pDependencyInfo** **must** be a valid pointer to a valid [VkDependencyInfo](#) structure
- VUID-vkCmdPipelineBarrier2-commandBuffer-recording **commandBuffer** **must** be in the [recording state](#)
- VUID-vkCmdPipelineBarrier2-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support transfer, graphics, compute, decode, or encode operations

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	Transfer Graphics Compute Decode Encode	Synchronization

To record a pipeline barrier, call:

```
// Provided by VK_VERSION_1_0
void vkCmdPipelineBarrier(
    VkCommandBuffer          commandBuffer,
    VkPipelineStageFlags     srcStageMask,
    VkPipelineStageFlags     dstStageMask,
    VkDependencyFlags        dependencyFlags,
    uint32_t                 memoryBarrierCount,
    const VkMemoryBarrier*   pMemoryBarriers,
    uint32_t                 bufferMemoryBarrierCount,
    const VkBufferMemoryBarrier* pBufferMemoryBarriers,
    uint32_t                 imageMemoryBarrierCount,
    const VkImageMemoryBarrier* pImageMemoryBarriers);
```


- `commandBuffer` is the command buffer into which the command is recorded.
- `srcStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [source stages](#).
- `dstStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [destination stages](#).
- `dependencyFlags` is a bitmask of `VkDependencyFlagBits` specifying how execution and memory dependencies are formed.
- `memoryBarrierCount` is the length of the `pMemoryBarriers` array.
- `pMemoryBarriers` is a pointer to an array of `VkMemoryBarrier` structures.
- `bufferMemoryBarrierCount` is the length of the `pBufferMemoryBarriers` array.
- `pBufferMemoryBarriers` is a pointer to an array of `VkBufferMemoryBarrier` structures.
- `imageMemoryBarrierCount` is the length of the `pImageMemoryBarriers` array.
- `pImageMemoryBarriers` is a pointer to an array of `VkImageMemoryBarrier` structures.

`vkCmdPipelineBarrier` operates almost identically to `vkCmdPipelineBarrier2KHR`, except that the scopes and barriers are defined as direct parameters rather than being defined by a `VkDependencyInfo`.

When `vkCmdPipelineBarrier` is submitted to a queue, it defines a memory dependency between commands that were submitted to the same queue before it, and those submitted to the same queue after it.

If `vkCmdPipelineBarrier` was recorded outside a render pass instance, the first [synchronization scope](#) includes all commands that occur earlier in [submission order](#). If `vkCmdPipelineBarrier` was recorded inside a render pass instance, the first synchronization scope includes only commands that occur earlier in [submission order](#) within the same subpass. In either case, the first synchronization scope is limited to operations on the pipeline stages determined by the [source stage mask](#) specified by `srcStageMask`.

If `vkCmdPipelineBarrier` was recorded outside a render pass instance, the second [synchronization scope](#) includes all commands that occur later in [submission order](#). If `vkCmdPipelineBarrier` was recorded inside a render pass instance, the second synchronization scope includes only commands that occur later in [submission order](#) within the same subpass. In either case, the second synchronization scope is limited to operations on the pipeline stages determined by the [destination stage mask](#) specified by `dstStageMask`.

The first [access scope](#) is limited to accesses in the pipeline stages determined by the [source stage mask](#) specified by `srcStageMask`. Within that, the first access scope only includes the first access scopes defined by elements of the `pMemoryBarriers`, `pBufferMemoryBarriers` and `pImageMemoryBarriers` arrays, which each define a set of [memory barriers](#). If no memory barriers are specified, then the first access scope includes no accesses.

The second [access scope](#) is limited to accesses in the pipeline stages determined by the [destination stage mask](#) specified by `dstStageMask`. Within that, the second access scope only includes the second access scopes defined by elements of the `pMemoryBarriers`, `pBufferMemoryBarriers` and `pImageMemoryBarriers` arrays, which each define a set of [memory barriers](#). If no memory barriers are specified, then the second access scope includes no accesses.

If `dependencyFlags` includes `VK_DEPENDENCY_BY_REGION_BIT`, then any dependency between `framebuffer-space` pipeline stages is `framebuffer-local` - otherwise it is `framebuffer-global`.

Valid Usage

- VUID-vkCmdPipelineBarrier-srcStageMask-04090
If the `geometryShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-vkCmdPipelineBarrier-srcStageMask-04091
If the `tessellationShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-vkCmdPipelineBarrier-srcStageMask-04094
If the `transformFeedback` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-vkCmdPipelineBarrier-srcStageMask-07319
If the `attachmentFragmentShadingRate` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-vkCmdPipelineBarrier-srcStageMask-03937
If the `synchronization2` feature is not enabled, `srcStageMask` **must** not be `0`
- VUID-vkCmdPipelineBarrier-srcStageMask-07950
If the `rayTracingPipeline` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-vkCmdPipelineBarrier-srcAccessMask-06257
If the `rayQuery` feature is not enabled and a memory barrier `srcAccessMask` includes `VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `srcStageMask` **must** not include any of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages except `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-vkCmdPipelineBarrier-dstStageMask-04090
If the `geometryShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-vkCmdPipelineBarrier-dstStageMask-04091
If the `tessellationShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-vkCmdPipelineBarrier-dstStageMask-04094
If the `transformFeedback` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-vkCmdPipelineBarrier-dstStageMask-07319
If the `attachmentFragmentShadingRate` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-vkCmdPipelineBarrier-dstStageMask-03937

If the `synchronization2` feature is not enabled, `dstStageMask` **must** not be 0

- VUID-vkCmdPipelineBarrier-dstStageMask-07950

If the `rayTracingPipeline` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`

- VUID-vkCmdPipelineBarrier-dstAccessMask-06257

If the `rayQuery` feature is not enabled and a memory barrier `dstAccessMask` includes `VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `dstStageMask` **must** not include any of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages except `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`

- VUID-vkCmdPipelineBarrier-srcAccessMask-02815

The `srcAccessMask` member of each element of `pMemoryBarriers` **must** only include access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdPipelineBarrier-dstAccessMask-02816

The `dstAccessMask` member of each element of `pMemoryBarriers` **must** only include access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdPipelineBarrier-pBufferMemoryBarriers-02817

For any element of `pBufferMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `srcQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `srcAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdPipelineBarrier-pBufferMemoryBarriers-02818

For any element of `pBufferMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `dstQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `dstAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdPipelineBarrier-pImageMemoryBarriers-02819

For any element of `pImageMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `srcQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `srcAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)

- VUID-vkCmdPipelineBarrier-pImageMemoryBarriers-02820

For any element of `pImageMemoryBarriers`, if its `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members are equal, or if its `dstQueueFamilyIndex` is the queue family index that was used to create the command pool that `commandBuffer` was allocated from, then its `dstAccessMask` member **must** only contain access flags that are supported by one or more of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)

types

- VUID-vkCmdPipelineBarrier-None-07889
If `vkCmdPipelineBarrier` is called within a render pass instance using a `VkRenderPass` object, the render pass **must** have been created with at least one subpass dependency that expresses a dependency from the current subpass to itself, does not include `VK_DEPENDENCY_BY_REGION_BIT` if this command does not, does not include `VK_DEPENDENCY_VIEW_LOCAL_BIT` if this command does not, and has `synchronization scopes` and `access scopes` that are all supersets of the scopes defined in this command
- VUID-vkCmdPipelineBarrier-bufferMemoryBarrierCount-01178
If `vkCmdPipelineBarrier` is called within a render pass instance using a `VkRenderPass` object, it **must** not include any buffer memory barriers
- VUID-vkCmdPipelineBarrier-image-04073
If `vkCmdPipelineBarrier` is called within a render pass instance using a `VkRenderPass` object, the `image` member of any image memory barrier included in this command **must** be an attachment used in the current subpass both as an input attachment, and as either a color, or depth/stencil attachment
- VUID-vkCmdPipelineBarrier-oldLayout-01181
If `vkCmdPipelineBarrier` is called within a render pass instance, the `oldLayout` and `newLayout` members of any image memory barrier included in this command **must** be equal
- VUID-vkCmdPipelineBarrier-srcQueueFamilyIndex-01182
If `vkCmdPipelineBarrier` is called within a render pass instance, the `srcQueueFamilyIndex` and `dstQueueFamilyIndex` members of any memory barrier included in this command **must** be equal
- VUID-vkCmdPipelineBarrier-None-07890
If `vkCmdPipelineBarrier` is called within a render pass instance, and the source stage masks of any memory barriers include `framebuffer-space stages`, destination stage masks of all memory barriers **must** only include `framebuffer-space stages`
- VUID-vkCmdPipelineBarrier-dependencyFlags-07891
If `vkCmdPipelineBarrier` is called within a render pass instance, and the source stage masks of any memory barriers include `framebuffer-space stages`, then `dependencyFlags` **must** include `VK_DEPENDENCY_BY_REGION_BIT`
- VUID-vkCmdPipelineBarrier-None-07892
If `vkCmdPipelineBarrier` is called within a render pass instance, the source and destination stage masks of any memory barriers **must** only include graphics pipeline stages
- VUID-vkCmdPipelineBarrier-dependencyFlags-01186
If `vkCmdPipelineBarrier` is called outside of a render pass instance, the dependency flags **must** not include `VK_DEPENDENCY_VIEW_LOCAL_BIT`
- VUID-vkCmdPipelineBarrier-None-07893
If `vkCmdPipelineBarrier` is called inside a render pass instance, and there is more than one view in the current subpass, dependency flags **must** include `VK_DEPENDENCY_VIEW_LOCAL_BIT`
- VUID-vkCmdPipelineBarrier-None-09553
If none of the `shaderTileImageColorReadAccess`, `shaderTileImageStencilReadAccess`, or

`shaderTileImageDepthReadAccess` features are enabled, and the `dynamicRenderingLocalRead` feature is not enabled, `vkCmdPipelineBarrier` **must** not be called within a render pass instance started with `vkCmdBeginRenderingKHR`

- VUID-vkCmdPipelineBarrier-None-09554

If the `dynamicRenderingLocalRead` feature is not enabled, and `vkCmdPipelineBarrier` is called within a render pass instance started with `vkCmdBeginRenderingKHR`, there **must** be no buffer or image memory barriers specified by this command

- VUID-vkCmdPipelineBarrier-None-09586

If the `dynamicRenderingLocalRead` feature is not enabled, and `vkCmdPipelineBarrier` is called within a render pass instance started with `vkCmdBeginRenderingKHR`, memory barriers specified by this command **must** only include `VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT`, `VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT`, `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT`, or `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT` in their access masks

- VUID-vkCmdPipelineBarrier-image-09555

If `vkCmdPipelineBarrier` is called within a render pass instance started with `vkCmdBeginRenderingKHR`, and the `image` member of any image memory barrier is used as an attachment in the current render pass instance, it **must** be in the `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR` or `VK_IMAGE_LAYOUT_GENERAL` layout

- VUID-vkCmdPipelineBarrier-srcStageMask-09556

If `vkCmdPipelineBarrier` is called within a render pass instance started with `vkCmdBeginRenderingKHR`, this command **must** only specify `framebuffer-space` stages in `srcStageMask` and `dstStageMask`

- VUID-vkCmdPipelineBarrier-srcStageMask-06461

Any pipeline stage included in `srcStageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the [table of supported pipeline stages](#)

- VUID-vkCmdPipelineBarrier-dstStageMask-06462

Any pipeline stage included in `dstStageMask` **must** be supported by the capabilities of the queue family specified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` structure that was used to create the `VkCommandPool` that `commandBuffer` was allocated from, as specified in the [table of supported pipeline stages](#)

- VUID-vkCmdPipelineBarrier-srcStageMask-09633

If either `srcStageMask` or `dstStageMask` includes `VK_PIPELINE_STAGE_HOST_BIT`, for any element of `pImageMemoryBarriers`, `srcQueueFamilyIndex` and `dstQueueFamilyIndex` **must** be equal

- VUID-vkCmdPipelineBarrier-srcStageMask-09634

If either `srcStageMask` or `dstStageMask` includes `VK_PIPELINE_STAGE_HOST_BIT`, for any element of `pBufferMemoryBarriers`, `srcQueueFamilyIndex` and `dstQueueFamilyIndex` **must** be equal

Valid Usage (Implicit)

- VUID-vkCmdPipelineBarrier-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdPipelineBarrier-srcStageMask-parameter
`srcStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values
- VUID-vkCmdPipelineBarrier-dstStageMask-parameter
`dstStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values
- VUID-vkCmdPipelineBarrier-dependencyFlags-parameter
`dependencyFlags` **must** be a valid combination of `VkDependencyFlagBits` values
- VUID-vkCmdPipelineBarrier-pMemoryBarriers-parameter
If `memoryBarrierCount` is not 0, `pMemoryBarriers` **must** be a valid pointer to an array of `memoryBarrierCount` valid `VkMemoryBarrier` structures
- VUID-vkCmdPipelineBarrier-pBufferMemoryBarriers-parameter
If `bufferMemoryBarrierCount` is not 0, `pBufferMemoryBarriers` **must** be a valid pointer to an array of `bufferMemoryBarrierCount` valid `VkBufferMemoryBarrier` structures
- VUID-vkCmdPipelineBarrier-pImageMemoryBarriers-parameter
If `imageMemoryBarrierCount` is not 0, `pImageMemoryBarriers` **must** be a valid pointer to an array of `imageMemoryBarrierCount` valid `VkImageMemoryBarrier` structures
- VUID-vkCmdPipelineBarrier-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdPipelineBarrier-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, compute, decode, or encode operations

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	Transfer Graphics Compute Decode Encode	Synchronization

Bits which **can** be set in `vkCmdPipelineBarrier::dependencyFlags`, specifying how execution and memory dependencies are formed, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkDependencyFlagBits {
    VK_DEPENDENCY_BY_REGION_BIT = 0x00000001,
    VK_DEPENDENCY_DEVICE_GROUP_BIT = 0x00000004,
    VK_DEPENDENCY_VIEW_LOCAL_BIT = 0x00000002,
    // Provided by VK_EXT_attachment_feedback_loop_layout
    VK_DEPENDENCY_FEEDBACK_LOOP_BIT_EXT = 0x00000008,
    // Provided by VK_KHR_multiview
    VK_DEPENDENCY_VIEW_LOCAL_BIT_KHR = VK_DEPENDENCY_VIEW_LOCAL_BIT,
    // Provided by VK_KHR_device_group
    VK_DEPENDENCY_DEVICE_GROUP_BIT_KHR = VK_DEPENDENCY_DEVICE_GROUP_BIT,
} VkDependencyFlagBits;
```

- `VK_DEPENDENCY_BY_REGION_BIT` specifies that dependencies will be [framebuffer-local](#).
- `VK_DEPENDENCY_VIEW_LOCAL_BIT` specifies that dependencies will be [view-local](#).
- `VK_DEPENDENCY_DEVICE_GROUP_BIT` specifies that dependencies are [non-device-local](#).
- `VK_DEPENDENCY_FEEDBACK_LOOP_BIT_EXT` specifies that the render pass will write to and read from the same image using the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` layout.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDependencyFlags;
```

`VkDependencyFlags` is a bitmask type for setting a mask of zero or more `VkDependencyFlagBits`.

7.7. Memory Barriers

Memory barriers are used to explicitly control access to buffer and image subresource ranges. Memory barriers are used to [transfer ownership between queue families](#), [change image layouts](#), and define [availability and visibility operations](#). They explicitly define the [access types](#) and buffer and image subresource ranges that are included in the [access scopes](#) of a memory dependency that is created by a synchronization command that includes them.

7.7.1. Global Memory Barriers

Global memory barriers apply to memory accesses involving all memory objects that exist at the time of its execution.

The `VkMemoryBarrier2` structure is defined as:


```
typedef struct VkMemoryBarrier2 {
    VkStructureType      sType;
    const void*          pNext;
    VkPipelineStageFlags2 srcStageMask;
    VkAccessFlags2       srcAccessMask;
    VkPipelineStageFlags2 dstStageMask;
    VkAccessFlags2       dstAccessMask;
} VkMemoryBarrier2;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2
typedef VkMemoryBarrier2 VkMemoryBarrier2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcStageMask` is a `VkPipelineStageFlags2` mask of pipeline stages to be included in the [first synchronization scope](#).
- `srcAccessMask` is a `VkAccessFlags2` mask of access flags to be included in the [first access scope](#).
- `dstStageMask` is a `VkPipelineStageFlags2` mask of pipeline stages to be included in the [second synchronization scope](#).
- `dstAccessMask` is a `VkAccessFlags2` mask of access flags to be included in the [second access scope](#).

This structure defines a [memory dependency](#) affecting all device memory.

The first [synchronization scope](#) and [access scope](#) described by this structure include only operations and memory accesses specified by `srcStageMask` and `srcAccessMask`.

The second [synchronization scope](#) and [access scope](#) described by this structure include only operations and memory accesses specified by `dstStageMask` and `dstAccessMask`.

Valid Usage

- VUID-VkMemoryBarrier2-srcStageMask-03929
If the `geometryShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT`
- VUID-VkMemoryBarrier2-srcStageMask-03930
If the `tessellationShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-VkMemoryBarrier2-srcStageMask-03933
If the `transformFeedback` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-VkMemoryBarrier2-srcStageMask-07317

If the `attachmentFragmentShadingRate` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-VkMemoryBarrier2-srcStageMask-07947

If the `rayTracingPipeline` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

- VUID-VkMemoryBarrier2-srcAccessMask-03900

If `srcAccessMask` includes `VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03901

If `srcAccessMask` includes `VK_ACCESS_2_INDEX_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03902

If `srcAccessMask` includes `VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT`, `VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03903

If `srcAccessMask` includes `VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT`, `VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03904

If `srcAccessMask` includes `VK_ACCESS_2_UNIFORM_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-srcAccessMask-03905

If `srcAccessMask` includes `VK_ACCESS_2_SHADER_SAMPLED_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-srcAccessMask-03906

If `srcAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-srcAccessMask-03907

If `srcAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-srcAccessMask-07454

If `srcAccessMask` includes `VK_ACCESS_2_SHADER_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`,

VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR,
VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, or one of the VK_PIPELINE_STAGE_*_SHADER_BIT stages

- VUID-VkMemoryBarrier2-srcAccessMask-03909

If `srcAccessMask` includes `VK_ACCESS_2_SHADER_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-srcAccessMask-03910

If `srcAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03911

If `srcAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03912

If `srcAccessMask` includes `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03913

If `srcAccessMask` includes `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03914

If `srcAccessMask` includes `VK_ACCESS_2_TRANSFER_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, `VK_PIPELINE_STAGE_2_RESOLVE_BIT`, `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03915

If `srcAccessMask` includes `VK_ACCESS_2_TRANSFER_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, `VK_PIPELINE_STAGE_2_RESOLVE_BIT`, `VK_PIPELINE_STAGE_2_CLEAR_BIT`, `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, or `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03916

If `srcAccessMask` includes `VK_ACCESS_2_HOST_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_HOST_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03917

If `srcAccessMask` includes `VK_ACCESS_2_HOST_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_HOST_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03920

If `srcAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-04747

If `srcAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`, `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03922

If `srcAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03926

If `srcAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-03927

If `srcAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-srcAccessMask-03928

If `srcAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-srcAccessMask-06256

If the `rayQuery` feature is not enabled and `srcAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `srcStageMask` **must** not include any of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages except `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

- VUID-VkMemoryBarrier2-srcAccessMask-07272

If `srcAccessMask` includes `VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT` or `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

- VUID-VkMemoryBarrier2-srcAccessMask-04858

If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`

- VUID-VkMemoryBarrier2-srcAccessMask-04859

If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR`, `srcStageMask` **must**

include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`

- VUID-VkMemoryBarrier2-srcAccessMask-04860

If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`

- VUID-VkMemoryBarrier2-srcAccessMask-04861

If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`

- VUID-VkMemoryBarrier2-srcAccessMask-07457

If `srcAccessMask` includes `VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT`

- VUID-VkMemoryBarrier2-srcAccessMask-07458

If `srcAccessMask` includes `VK_ACCESS_2_MICROMAP_READ_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` or `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`

- VUID-VkMemoryBarrier2-dstStageMask-03929

If the `geometryShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT`

- VUID-VkMemoryBarrier2-dstStageMask-03930

If the `tessellationShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT`

- VUID-VkMemoryBarrier2-dstStageMask-03933

If the `transformFeedback` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`

- VUID-VkMemoryBarrier2-dstStageMask-07317

If the `attachmentFragmentShadingRate` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-VkMemoryBarrier2-dstStageMask-07947

If the `rayTracingPipeline` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

- VUID-VkMemoryBarrier2-dstAccessMask-03900

If `dstAccessMask` includes `VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03901

If `dstAccessMask` includes `VK_ACCESS_2_INDEX_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03902

If `dstAccessMask` includes `VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT`,

VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or
VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryBarrier2-dstAccessMask-03903

If `dstAccessMask` includes `VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT`, `VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03904

If `dstAccessMask` includes `VK_ACCESS_2_UNIFORM_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-dstAccessMask-03905

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_SAMPLED_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-dstAccessMask-03906

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-dstAccessMask-03907

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-dstAccessMask-07454

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-dstAccessMask-03909

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-dstAccessMask-03910

If `dstAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03911

If `dstAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03912

If `dstAccessMask` includes `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`,

VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryBarrier2-dstAccessMask-03913

If `dstAccessMask` includes `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03914

If `dstAccessMask` includes `VK_ACCESS_2_TRANSFER_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, `VK_PIPELINE_STAGE_2_RESOLVE_BIT`, `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03915

If `dstAccessMask` includes `VK_ACCESS_2_TRANSFER_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, `VK_PIPELINE_STAGE_2_RESOLVE_BIT`, `VK_PIPELINE_STAGE_2_CLEAR_BIT`, `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, or `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03916

If `dstAccessMask` includes `VK_ACCESS_2_HOST_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_HOST_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03917

If `dstAccessMask` includes `VK_ACCESS_2_HOST_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_HOST_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03920

If `dstAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-04747

If `dstAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`, `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03922

If `dstAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-03926

If `dstAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`

VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkMemoryBarrier2-dstAccessMask-03927

If `dstAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkMemoryBarrier2-dstAccessMask-03928

If `dstAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkMemoryBarrier2-dstAccessMask-06256

If the `rayQuery` feature is not enabled and `dstAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `dstStageMask` **must** not include any of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages except `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

- VUID-VkMemoryBarrier2-dstAccessMask-07272

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT` or `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

- VUID-VkMemoryBarrier2-dstAccessMask-04858

If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`

- VUID-VkMemoryBarrier2-dstAccessMask-04859

If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`

- VUID-VkMemoryBarrier2-dstAccessMask-04860

If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`

- VUID-VkMemoryBarrier2-dstAccessMask-04861

If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`

- VUID-VkMemoryBarrier2-dstAccessMask-07457

If `dstAccessMask` includes `VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT`

- VUID-VkMemoryBarrier2-dstAccessMask-07458

If `dstAccessMask` includes `VK_ACCESS_2_MICROMAP_READ_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` or `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`

Valid Usage (Implicit)

- VUID-VkMemoryBarrier2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_BARRIER_2`
- VUID-VkMemoryBarrier2-srcStageMask-parameter
`srcStageMask` **must** be a valid combination of `VkPipelineStageFlagBits2` values
- VUID-VkMemoryBarrier2-srcAccessMask-parameter
`srcAccessMask` **must** be a valid combination of `VkAccessFlagBits2` values
- VUID-VkMemoryBarrier2-dstStageMask-parameter
`dstStageMask` **must** be a valid combination of `VkPipelineStageFlagBits2` values
- VUID-VkMemoryBarrier2-dstAccessMask-parameter
`dstAccessMask` **must** be a valid combination of `VkAccessFlagBits2` values

The `VkMemoryBarrier` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryBarrier {
    VkStructureType    sType;
    const void*        pNext;
    VkAccessFlags       srcAccessMask;
    VkAccessFlags       dstAccessMask;
} VkMemoryBarrier;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcAccessMask` is a bitmask of `VkAccessFlagBits` specifying a `source access mask`.
- `dstAccessMask` is a bitmask of `VkAccessFlagBits` specifying a `destination access mask`.

The first `access scope` is limited to access types in the `source access mask` specified by `srcAccessMask`.

The second `access scope` is limited to access types in the `destination access mask` specified by `dstAccessMask`.

Valid Usage (Implicit)

- VUID-VkMemoryBarrier-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_BARRIER`
- VUID-VkMemoryBarrier-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkMemoryBarrier-srcAccessMask-parameter
`srcAccessMask` **must** be a valid combination of `VkAccessFlagBits` values
- VUID-VkMemoryBarrier-dstAccessMask-parameter

dstAccessMask must be a valid combination of [VkAccessFlagBits](#) values

7.7.2. Buffer Memory Barriers

Buffer memory barriers only apply to memory accesses involving a specific buffer range. That is, a memory dependency formed from a buffer memory barrier is [scoped](#) to access via the specified buffer range. Buffer memory barriers **can** also be used to define a [queue family ownership transfer](#) for the specified buffer range.

The `VkBufferMemoryBarrier2` structure is defined as:

```
typedef struct VkBufferMemoryBarrier2 {
    VkStructureType    sType;
    const void*        pNext;
    VkPipelineStageFlags2 srcStageMask;
    VkAccessFlags2      srcAccessMask;
    VkPipelineStageFlags2 dstStageMask;
    VkAccessFlags2      dstAccessMask;
    uint32_t            srcQueueFamilyIndex;
    uint32_t            dstQueueFamilyIndex;
    VkBuffer            buffer;
    VkDeviceSize        offset;
    VkDeviceSize        size;
} VkBufferMemoryBarrier2;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2
typedef VkBufferMemoryBarrier2 VkBufferMemoryBarrier2KHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **srcStageMask** is a [VkPipelineStageFlags2](#) mask of pipeline stages to be included in the [first synchronization scope](#).
- **srcAccessMask** is a [VkAccessFlags2](#) mask of access flags to be included in the [first access scope](#).
- **dstStageMask** is a [VkPipelineStageFlags2](#) mask of pipeline stages to be included in the [second synchronization scope](#).
- **dstAccessMask** is a [VkAccessFlags2](#) mask of access flags to be included in the [second access scope](#).
- **srcQueueFamilyIndex** is the source queue family for a [queue family ownership transfer](#).
- **dstQueueFamilyIndex** is the destination queue family for a [queue family ownership transfer](#).
- **buffer** is a handle to the buffer whose backing memory is affected by the barrier.
- **offset** is an offset in bytes into the backing memory for **buffer**; this is relative to the base offset as bound to the buffer (see [vkBindBufferMemory](#)).

- **size** is a size in bytes of the affected area of backing memory for **buffer**, or **VK_WHOLE_SIZE** to use the range from **offset** to the end of the buffer.

This structure defines a **memory dependency** limited to a range of a buffer, and **can** define a **queue family ownership transfer operation** for that range.

The first **synchronization scope** and **access scope** described by this structure include only operations and memory accesses specified by **srcStageMask** and **srcAccessMask**.

The second **synchronization scope** and **access scope** described by this structure include only operations and memory accesses specified by **dstStageMask** and **dstAccessMask**.

Both **access scopes** are limited to only memory accesses to **buffer** in the range defined by **offset** and **size**.

If **buffer** was created with **VK_SHARING_MODE_EXCLUSIVE**, and **srcQueueFamilyIndex** is not equal to **dstQueueFamilyIndex**, this memory barrier defines a **queue family ownership transfer operation**. When executed on a queue in the family identified by **srcQueueFamilyIndex**, this barrier defines a **queue family release operation** for the specified buffer range, and the second synchronization scope does not apply to this operation. When executed on a queue in the family identified by **dstQueueFamilyIndex**, this barrier defines a **queue family acquire operation** for the specified buffer range, and the first synchronization scope does not apply to this operation.

A **queue family ownership transfer operation** is also defined if the values are not equal, and either is one of the special queue family values reserved for external memory ownership transfers, as described in **Queue Family Ownership Transfer**. A **queue family release operation** is defined when **dstQueueFamilyIndex** is one of those values, and a **queue family acquire operation** is defined when **srcQueueFamilyIndex** is one of those values.

Valid Usage

- VUID-VkBufferMemoryBarrier2-srcStageMask-03929
If the **geometryShader** feature is not enabled, **srcStageMask** **must** not contain **VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT**
- VUID-VkBufferMemoryBarrier2-srcStageMask-03930
If the **tessellationShader** feature is not enabled, **srcStageMask** **must** not contain **VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT** or **VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT**
- VUID-VkBufferMemoryBarrier2-srcStageMask-03933
If the **transformFeedback** feature is not enabled, **srcStageMask** **must** not contain **VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT**
- VUID-VkBufferMemoryBarrier2-srcStageMask-07317
If the **attachmentFragmentShadingRate** feature is not enabled, **srcStageMask** **must** not contain **VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR**
- VUID-VkBufferMemoryBarrier2-srcStageMask-07947
If the **rayTracingPipeline** feature is not enabled, **srcStageMask** **must** not contain **VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR**

- VUID-VkBufferMemoryBarrier2-srcAccessMask-03900
If `srcAccessMask` includes `VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03901
If `srcAccessMask` includes `VK_ACCESS_2_INDEX_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03902
If `srcAccessMask` includes `VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT`, `VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03903
If `srcAccessMask` includes `VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT`, `VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03904
If `srcAccessMask` includes `VK_ACCESS_2_UNIFORM_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03905
If `srcAccessMask` includes `VK_ACCESS_2_SHADER_SAMPLED_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03906
If `srcAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03907
If `srcAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-07454
If `srcAccessMask` includes `VK_ACCESS_2_SHADER_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03909
If `srcAccessMask` includes `VK_ACCESS_2_SHADER_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of

the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkBufferMemoryBarrier2-srcAccessMask-03910
If `srcAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03911
If `srcAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03912
If `srcAccessMask` includes `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03913
If `srcAccessMask` includes `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03914
If `srcAccessMask` includes `VK_ACCESS_2_TRANSFER_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, `VK_PIPELINE_STAGE_2_RESOLVE_BIT`, `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03915
If `srcAccessMask` includes `VK_ACCESS_2_TRANSFER_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, `VK_PIPELINE_STAGE_2_RESOLVE_BIT`, `VK_PIPELINE_STAGE_2_CLEAR_BIT`, `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03916
If `srcAccessMask` includes `VK_ACCESS_2_HOST_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_HOST_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03917
If `srcAccessMask` includes `VK_ACCESS_2_HOST_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_HOST_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03920
If `srcAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkBufferMemoryBarrier2-srcAccessMask-04747
If `srcAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`, `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03922
If `srcAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03926
If `srcAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03927
If `srcAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-srcAccessMask-03928
If `srcAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-06256
If the `rayQuery` feature is not enabled and `srcAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `srcStageMask` **must** not include any of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages except `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-07272
If `srcAccessMask` includes `VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT` or `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-04858
If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-04859
If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-04860
If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-srcAccessMask-04861
If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR`, `srcStageMask` **must**

include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`

- VUID-VkBufferMemoryBarrier2-srcAccessMask-07457

If `srcAccessMask` includes `VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT`

- VUID-VkBufferMemoryBarrier2-srcAccessMask-07458

If `srcAccessMask` includes `VK_ACCESS_2_MICROMAP_READ_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` or `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`

- VUID-VkBufferMemoryBarrier2-dstStageMask-03929

If the `geometryShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT`

- VUID-VkBufferMemoryBarrier2-dstStageMask-03930

If the `tessellationShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT`

- VUID-VkBufferMemoryBarrier2-dstStageMask-03933

If the `transformFeedback` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`

- VUID-VkBufferMemoryBarrier2-dstStageMask-07317

If the `attachmentFragmentShadingRate` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-VkBufferMemoryBarrier2-dstStageMask-07947

If the `rayTracingPipeline` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

- VUID-VkBufferMemoryBarrier2-dstAccessMask-03900

If `dstAccessMask` includes `VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkBufferMemoryBarrier2-dstAccessMask-03901

If `dstAccessMask` includes `VK_ACCESS_2_INDEX_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkBufferMemoryBarrier2-dstAccessMask-03902

If `dstAccessMask` includes `VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT`, `VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkBufferMemoryBarrier2-dstAccessMask-03903

If `dstAccessMask` includes `VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT`, `VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkBufferMemoryBarrier2-dstAccessMask-03904
If `dstAccessMask` includes `VK_ACCESS_2_UNIFORM_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03905
If `dstAccessMask` includes `VK_ACCESS_2_SHADER_SAMPLED_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03906
If `dstAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03907
If `dstAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-07454
If `dstAccessMask` includes `VK_ACCESS_2_SHADER_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03909
If `dstAccessMask` includes `VK_ACCESS_2_SHADER_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03910
If `dstAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03911
If `dstAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03912
If `dstAccessMask` includes `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03913
If `dstAccessMask` includes `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkBufferMemoryBarrier2-dstAccessMask-03914
If `dstAccessMask` includes `VK_ACCESS_2_TRANSFER_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, `VK_PIPELINE_STAGE_2_RESOLVE_BIT`, `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03915
If `dstAccessMask` includes `VK_ACCESS_2_TRANSFER_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, `VK_PIPELINE_STAGE_2_RESOLVE_BIT`, `VK_PIPELINE_STAGE_2_CLEAR_BIT`, `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, or `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03916
If `dstAccessMask` includes `VK_ACCESS_2_HOST_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_HOST_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03917
If `dstAccessMask` includes `VK_ACCESS_2_HOST_WRITE_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_HOST_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03920
If `dstAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-04747
If `dstAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`, `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03922
If `dstAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03926
If `dstAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-03927
If `dstAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkBufferMemoryBarrier2-dstAccessMask-03928
If `dstAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-06256
If the `rayQuery` feature is not enabled and `dstAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `dstStageMask` **must** not include any of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages except `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-07272
If `dstAccessMask` includes `VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT` or `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-04858
If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-04859
If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-04860
If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-04861
If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-07457
If `dstAccessMask` includes `VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT`
- VUID-VkBufferMemoryBarrier2-dstAccessMask-07458
If `dstAccessMask` includes `VK_ACCESS_2_MICROMAP_READ_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` or `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`
- VUID-VkBufferMemoryBarrier2-offset-01187
`offset` **must** be less than the size of `buffer`
- VUID-VkBufferMemoryBarrier2-size-01188
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be greater than 0
- VUID-VkBufferMemoryBarrier2-size-01189
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be less than or equal to the size of `buffer` minus `offset`
- VUID-VkBufferMemoryBarrier2-buffer-01931
If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-VkBufferMemoryBarrier2-buffer-09095
If `buffer` was created with a sharing mode of `VK_SHARING_MODE_EXCLUSIVE`, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `srcQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_EXTERNAL`, `VK_QUEUE_FAMILY_FOREIGN_EXT`, or a valid queue family
- VUID-VkBufferMemoryBarrier2-buffer-09096
If `buffer` was created with a sharing mode of `VK_SHARING_MODE_EXCLUSIVE`, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `dstQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_EXTERNAL`, `VK_QUEUE_FAMILY_FOREIGN_EXT`, or a valid queue family
- VUID-VkBufferMemoryBarrier2-srcQueueFamilyIndex-04087
If `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, at least one of `srcQueueFamilyIndex` or `dstQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_EXTERNAL` or `VK_QUEUE_FAMILY_FOREIGN_EXT`
- VUID-VkBufferMemoryBarrier2-None-09097
If the `VK_KHR_external_memory` extension is not enabled, `srcQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_EXTERNAL`
- VUID-VkBufferMemoryBarrier2-None-09098
If the `VK_KHR_external_memory` extension is not enabled, `dstQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_EXTERNAL`
- VUID-VkBufferMemoryBarrier2-srcQueueFamilyIndex-09099
If the `VK_EXT_queue_family_foreign` extension is not enabled `srcQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_FOREIGN_EXT`
- VUID-VkBufferMemoryBarrier2-dstQueueFamilyIndex-09100
If the `VK_EXT_queue_family_foreign` extension is not enabled `dstQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_FOREIGN_EXT`
- VUID-VkBufferMemoryBarrier2-srcStageMask-03851
If either `srcStageMask` or `dstStageMask` includes `VK_PIPELINE_STAGE_2_HOST_BIT`, `srcQueueFamilyIndex` and `dstQueueFamilyIndex` **must** be equal

Valid Usage (Implicit)

- VUID-VkBufferMemoryBarrier2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER_2`
- VUID-VkBufferMemoryBarrier2-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkExternalMemoryAcquireUnmodifiedEXT`
- VUID-VkBufferMemoryBarrier2-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkBufferMemoryBarrier2-srcStageMask-parameter
`srcStageMask` **must** be a valid combination of `VkPipelineStageFlagBits2` values
- VUID-VkBufferMemoryBarrier2-srcAccessMask-parameter
`srcAccessMask` **must** be a valid combination of `VkAccessFlagBits2` values
- VUID-VkBufferMemoryBarrier2-dstStageMask-parameter

dstStageMask **must** be a valid combination of [VkPipelineStageFlagBits2](#) values

- `VOID-VkBufferMemoryBarrier2-dstAccessMask-parameter`

dstAccessMask **must** be a valid combination of [VkAccessFlagBits2](#) values

- `VOID-VkBufferMemoryBarrier2-buffer-parameter`

buffer **must** be a valid [VkBuffer](#) handle

The [VkBufferMemoryBarrier](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferMemoryBarrier {
    VkStructureType    sType;
    const void*        pNext;
    VkAccessFlags       srcAccessMask;
    VkAccessFlags       dstAccessMask;
    uint32_t            srcQueueFamilyIndex;
    uint32_t            dstQueueFamilyIndex;
    VkBuffer            buffer;
    VkDeviceSize        offset;
    VkDeviceSize        size;
} VkBufferMemoryBarrier;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **srcAccessMask** is a bitmask of [VkAccessFlagBits](#) specifying a [source access mask](#).
- **dstAccessMask** is a bitmask of [VkAccessFlagBits](#) specifying a [destination access mask](#).
- **srcQueueFamilyIndex** is the source queue family for a [queue family ownership transfer](#).
- **dstQueueFamilyIndex** is the destination queue family for a [queue family ownership transfer](#).
- **buffer** is a handle to the buffer whose backing memory is affected by the barrier.
- **offset** is an offset in bytes into the backing memory for **buffer**; this is relative to the base offset as bound to the buffer (see [vkBindBufferMemory](#)).
- **size** is a size in bytes of the affected area of backing memory for **buffer**, or `VK_WHOLE_SIZE` to use the range from **offset** to the end of the buffer.

The first [access scope](#) is limited to access to memory through the specified buffer range, via access types in the [source access mask](#) specified by **srcAccessMask**. If **srcAccessMask** includes `VK_ACCESS_HOST_WRITE_BIT`, a [memory domain operation](#) is performed where available memory in the host domain is also made available to the device domain.

The second [access scope](#) is limited to access to memory through the specified buffer range, via access types in the [destination access mask](#) specified by **dstAccessMask**. If **dstAccessMask** includes `VK_ACCESS_HOST_WRITE_BIT` or `VK_ACCESS_HOST_READ_BIT`, a [memory domain operation](#) is performed where available memory in the device domain is also made available to the host domain.

NOTE

When `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT` is used, available memory in host

domain is automatically made visible to host domain, and any host write is automatically made available to host domain.

If `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, and `srcQueueFamilyIndex` is equal to the current queue family, then the memory barrier defines a [queue family release operation](#) for the specified buffer range, and the second synchronization scope of the calling command does not apply to this operation.

If `dstQueueFamilyIndex` is not equal to `srcQueueFamilyIndex`, and `dstQueueFamilyIndex` is equal to the current queue family, then the memory barrier defines a [queue family acquire operation](#) for the specified buffer range, and the first synchronization scope of the calling command does not apply to this operation.

Valid Usage

- VUID-VkBufferMemoryBarrier-offset-01187
`offset` **must** be less than the size of `buffer`
- VUID-VkBufferMemoryBarrier-size-01188
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be greater than 0
- VUID-VkBufferMemoryBarrier-size-01189
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be less than or equal to than the size of `buffer` minus `offset`
- VUID-VkBufferMemoryBarrier-buffer-01931
If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkBufferMemoryBarrier-buffer-09095
If `buffer` was created with a sharing mode of `VK_SHARING_MODE_EXCLUSIVE`, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `srcQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_EXTERNAL`, `VK_QUEUE_FAMILY_FOREIGN_EXT`, or a valid queue family
- VUID-VkBufferMemoryBarrier-buffer-09096
If `buffer` was created with a sharing mode of `VK_SHARING_MODE_EXCLUSIVE`, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `dstQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_EXTERNAL`, `VK_QUEUE_FAMILY_FOREIGN_EXT`, or a valid queue family
- VUID-VkBufferMemoryBarrier-srcQueueFamilyIndex-04087
If `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, at least one of `srcQueueFamilyIndex` or `dstQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_EXTERNAL` or `VK_QUEUE_FAMILY_FOREIGN_EXT`
- VUID-VkBufferMemoryBarrier-None-09097
If the [VK_KHR_external_memory](#) extension is not enabled, `srcQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_EXTERNAL`
- VUID-VkBufferMemoryBarrier-None-09098
If the [VK_KHR_external_memory](#) extension is not enabled, `dstQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_EXTERNAL`
- VUID-VkBufferMemoryBarrier-srcQueueFamilyIndex-09099

If the `VK_EXT_queue_family_foreign` extension is not enabled `srcQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_FOREIGN_EXT`

- VUID-VkBufferMemoryBarrier-dstQueueFamilyIndex-09100

If the `VK_EXT_queue_family_foreign` extension is not enabled `dstQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_FOREIGN_EXT`

- VUID-VkBufferMemoryBarrier-None-09049

If the `synchronization2` feature is not enabled, and `buffer` was created with a sharing mode of `VK_SHARING_MODE_CONCURRENT`, at least one of `srcQueueFamilyIndex` and `dstQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_IGNORED`

- VUID-VkBufferMemoryBarrier-None-09050

If the `synchronization2` feature is not enabled, and `buffer` was created with a sharing mode of `VK_SHARING_MODE_CONCURRENT`, `srcQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_IGNORED` or `VK_QUEUE_FAMILY_EXTERNAL`

- VUID-VkBufferMemoryBarrier-None-09051

If the `synchronization2` feature is not enabled, and `buffer` was created with a sharing mode of `VK_SHARING_MODE_CONCURRENT`, `dstQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_IGNORED` or `VK_QUEUE_FAMILY_EXTERNAL`

Valid Usage (Implicit)

- VUID-VkBufferMemoryBarrier-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER`

- VUID-VkBufferMemoryBarrier-pNext-pNext

`pNext` **must** be `NULL` or a pointer to a valid instance of `VkExternalMemoryAcquireUnmodifiedEXT`

- VUID-VkBufferMemoryBarrier-sType-unique

The `sType` value of each struct in the `pNext` chain **must** be unique

- VUID-VkBufferMemoryBarrier-buffer-parameter

`buffer` **must** be a valid `VkBuffer` handle

`VK_WHOLE_SIZE` is a special value indicating that the entire remaining length of a buffer following a given `offset` should be used. It **can** be specified for `VkBufferMemoryBarrier::size` and other structures.

```
#define VK_WHOLE_SIZE (~0ULL)
```

7.7.3. Image Memory Barriers

Image memory barriers only apply to memory accesses involving a specific image subresource range. That is, a memory dependency formed from an image memory barrier is `scoped` to access via the specified image subresource range. Image memory barriers **can** also be used to define `image layout transitions` or a `queue family ownership transfer` for the specified image subresource

range.

The `VkImageMemoryBarrier2` structure is defined as:

```
typedef struct VkImageMemoryBarrier2 {
    VkStructureType    sType;
    const void*        pNext;
    VkPipelineStageFlags2 srcStageMask;
    VkAccessFlags2      srcAccessMask;
    VkPipelineStageFlags2 dstStageMask;
    VkAccessFlags2      dstAccessMask;
    VkImageLayout        oldLayout;
    VkImageLayout        newLayout;
    uint32_t             srcQueueFamilyIndex;
    uint32_t             dstQueueFamilyIndex;
    VkImage              image;
    VkImageSubresourceRange subresourceRange;
} VkImageMemoryBarrier2;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2
typedef VkImageMemoryBarrier2 VkImageMemoryBarrier2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcStageMask` is a `VkPipelineStageFlags2` mask of pipeline stages to be included in the [first synchronization scope](#).
- `srcAccessMask` is a `VkAccessFlags2` mask of access flags to be included in the [first access scope](#).
- `dstStageMask` is a `VkPipelineStageFlags2` mask of pipeline stages to be included in the [second synchronization scope](#).
- `dstAccessMask` is a `VkAccessFlags2` mask of access flags to be included in the [second access scope](#).
- `oldLayout` is the old layout in an [image layout transition](#).
- `newLayout` is the new layout in an [image layout transition](#).
- `srcQueueFamilyIndex` is the source queue family for a [queue family ownership transfer](#).
- `dstQueueFamilyIndex` is the destination queue family for a [queue family ownership transfer](#).
- `image` is a handle to the image affected by this barrier.
- `subresourceRange` describes the [image subresource range](#) within `image` that is affected by this barrier.

This structure defines a [memory dependency](#) limited to an image subresource range, and **can** define a [queue family ownership transfer operation](#) and [image layout transition](#) for that subresource range.

The first [synchronization scope](#) and [access scope](#) described by this structure include only operations and memory accesses specified by [srcStageMask](#) and [srcAccessMask](#).

The second [synchronization scope](#) and [access scope](#) described by this structure include only operations and memory accesses specified by [dstStageMask](#) and [dstAccessMask](#).

Both [access scopes](#) are limited to only memory accesses to [image](#) in the subresource range defined by [subresourceRange](#).

If [image](#) was created with [VK_SHARING_MODE_EXCLUSIVE](#), and [srcQueueFamilyIndex](#) is not equal to [dstQueueFamilyIndex](#), this memory barrier defines a [queue family ownership transfer operation](#). When executed on a queue in the family identified by [srcQueueFamilyIndex](#), this barrier defines a [queue family release operation](#) for the specified image subresource range, and the second synchronization scope does not apply to this operation. When executed on a queue in the family identified by [dstQueueFamilyIndex](#), this barrier defines a [queue family acquire operation](#) for the specified image subresource range, and the first synchronization, the first synchronization scope does not apply to this operation.

A [queue family ownership transfer operation](#) is also defined if the values are not equal, and either is one of the special queue family values reserved for external memory ownership transfers, as described in [Queue Family Ownership Transfer](#). A [queue family release operation](#) is defined when [dstQueueFamilyIndex](#) is one of those values, and a [queue family acquire operation](#) is defined when [srcQueueFamilyIndex](#) is one of those values.

If [oldLayout](#) is not equal to [newLayout](#), then the memory barrier defines an [image layout transition](#) for the specified image subresource range. If this memory barrier defines a [queue family ownership transfer operation](#), the layout transition is only executed once between the queues.

NOTE

When the old and new layout are equal, the layout values are ignored - data is preserved no matter what values are specified, or what layout the image is currently in.

If [image](#) has a multi-planar format and the image is *disjoint*, then including [VK_IMAGE_ASPECT_COLOR_BIT](#) in the [aspectMask](#) member of [subresourceRange](#) is equivalent to including [VK_IMAGE_ASPECT_PLANE_0_BIT](#), [VK_IMAGE_ASPECT_PLANE_1_BIT](#), and (for three-plane formats only) [VK_IMAGE_ASPECT_PLANE_2_BIT](#).

Valid Usage

- VUID-VkImageMemoryBarrier2-srcStageMask-03929
If the [geometryShader](#) feature is not enabled, [srcStageMask](#) **must** not contain [VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT](#)
- VUID-VkImageMemoryBarrier2-srcStageMask-03930
If the [tessellationShader](#) feature is not enabled, [srcStageMask](#) **must** not contain [VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT](#) or [VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT](#)
- VUID-VkImageMemoryBarrier2-srcStageMask-03933
If the [transformFeedback](#) feature is not enabled, [srcStageMask](#) **must** not contain

VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT

- VUID-VkImageMemoryBarrier2-srcStageMask-07317

If the `attachmentFragmentShadingRate` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-VkImageMemoryBarrier2-srcStageMask-07947

If the `rayTracingPipeline` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03900

If `srcAccessMask` includes `VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03901

If `srcAccessMask` includes `VK_ACCESS_2_INDEX_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03902

If `srcAccessMask` includes `VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT`, `VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03903

If `srcAccessMask` includes `VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT`, `VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03904

If `srcAccessMask` includes `VK_ACCESS_2_UNIFORM_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-srcAccessMask-03905

If `srcAccessMask` includes `VK_ACCESS_2_SHADER_SAMPLED_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-srcAccessMask-03906

If `srcAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-srcAccessMask-03907

If `srcAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-srcAccessMask-07454

If `srcAccessMask` includes `VK_ACCESS_2_SHADER_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-srcAccessMask-03909

If `srcAccessMask` includes `VK_ACCESS_2_SHADER_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-srcAccessMask-03910

If `srcAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03911

If `srcAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03912

If `srcAccessMask` includes `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03913

If `srcAccessMask` includes `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03914

If `srcAccessMask` includes `VK_ACCESS_2_TRANSFER_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, `VK_PIPELINE_STAGE_2_RESOLVE_BIT`, `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03915

If `srcAccessMask` includes `VK_ACCESS_2_TRANSFER_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COPY_BIT`, `VK_PIPELINE_STAGE_2_BLIT_BIT`, `VK_PIPELINE_STAGE_2_RESOLVE_BIT`, `VK_PIPELINE_STAGE_2_CLEAR_BIT`, `VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, or `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03916

If `srcAccessMask` includes `VK_ACCESS_2_HOST_READ_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_HOST_BIT`

- VUID-VkImageMemoryBarrier2-srcAccessMask-03917
If `srcAccessMask` includes `VK_ACCESS_2_HOST_WRITE_BIT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_HOST_BIT`
- VUID-VkImageMemoryBarrier2-srcAccessMask-03920
If `srcAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkImageMemoryBarrier2-srcAccessMask-04747
If `srcAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`, `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkImageMemoryBarrier2-srcAccessMask-03922
If `srcAccessMask` includes `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkImageMemoryBarrier2-srcAccessMask-03926
If `srcAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkImageMemoryBarrier2-srcAccessMask-03927
If `srcAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages
- VUID-VkImageMemoryBarrier2-srcAccessMask-03928
If `srcAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkImageMemoryBarrier2-srcAccessMask-06256
If the `rayQuery` feature is not enabled and `srcAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `srcStageMask` **must** not include any of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages except `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcAccessMask-07272
If `srcAccessMask` includes `VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT` or `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcAccessMask-04858
If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcAccessMask-04859

- If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcAccessMask-04860
If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcAccessMask-04861
If `srcAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcAccessMask-07457
If `srcAccessMask` includes `VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT`
- VUID-VkImageMemoryBarrier2-srcAccessMask-07458
If `srcAccessMask` includes `VK_ACCESS_2_MICROMAP_READ_BIT_EXT`, `srcStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` or `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`
- VUID-VkImageMemoryBarrier2-dstStageMask-03929
If the `geometryShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT`
- VUID-VkImageMemoryBarrier2-dstStageMask-03930
If the `tessellationShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-VkImageMemoryBarrier2-dstStageMask-03933
If the `transformFeedback` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-VkImageMemoryBarrier2-dstStageMask-07317
If the `attachmentFragmentShadingRate` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-VkImageMemoryBarrier2-dstStageMask-07947
If the `rayTracingPipeline` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkImageMemoryBarrier2-dstAccessMask-03900
If `dstAccessMask` includes `VK_ACCESS_2_INDIRECT_COMMAND_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkImageMemoryBarrier2-dstAccessMask-03901
If `dstAccessMask` includes `VK_ACCESS_2_INDEX_READ_BIT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_INDEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`
- VUID-VkImageMemoryBarrier2-dstAccessMask-03902
If `dstAccessMask` includes `VK_ACCESS_2_VERTEX_ATTRIBUTE_READ_BIT`, `dstStageMask` **must**

include VK_PIPELINE_STAGE_2_VERTEX_ATTRIBUTE_INPUT_BIT,
VK_PIPELINE_STAGE_2_VERTEX_INPUT_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or
VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03903

If `dstAccessMask` includes `VK_ACCESS_2_INPUT_ATTACHMENT_READ_BIT`, `dstStageMask` **must** include
VK_PIPELINE_STAGE_2_FRAGMENT_SHADER_BIT,
VK_PIPELINE_STAGE_2_SUBPASS_SHADER_BIT_HUAWEI, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or
VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03904

If `dstAccessMask` includes `VK_ACCESS_2_UNIFORM_READ_BIT`, `dstStageMask` **must** include
VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of
the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-dstAccessMask-03905

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_SAMPLED_READ_BIT`, `dstStageMask` **must** include
VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of
the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-dstAccessMask-03906

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_READ_BIT`, `dstStageMask` **must** include
VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of
the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-dstAccessMask-03907

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_STORAGE_WRITE_BIT`, `dstStageMask` **must**
include VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or
one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-dstAccessMask-07454

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_READ_BIT`, `dstStageMask` **must** include
VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT,
VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR,
VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT`
stages

- VUID-VkImageMemoryBarrier2-dstAccessMask-03909

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_WRITE_BIT`, `dstStageMask` **must** include
VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT, or one of
the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-dstAccessMask-03910

If `dstAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_READ_BIT`, `dstStageMask` **must**
include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT
VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03911

If `dstAccessMask` includes `VK_ACCESS_2_COLOR_ATTACHMENT_WRITE_BIT`, `dstStageMask` **must**
include VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT
VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03912

If `dstAccessMask` includes `VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_READ_BIT`, `dstStageMask`

must include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03913

If `dstAccessMask` includes VK_ACCESS_2_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_EARLY_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_LATE_FRAGMENT_TESTS_BIT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03914

If `dstAccessMask` includes VK_ACCESS_2_TRANSFER_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03915

If `dstAccessMask` includes VK_ACCESS_2_TRANSFER_WRITE_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_COPY_BIT, VK_PIPELINE_STAGE_2_BLIT_BIT, VK_PIPELINE_STAGE_2_RESOLVE_BIT, VK_PIPELINE_STAGE_2_CLEAR_BIT, VK_PIPELINE_STAGE_2_ALL_TRANSFER_BIT, VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR, or VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR, VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03916

If `dstAccessMask` includes VK_ACCESS_2_HOST_READ_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_HOST_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03917

If `dstAccessMask` includes VK_ACCESS_2_HOST_WRITE_BIT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_HOST_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03920

If `dstAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-04747

If `dstAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_DRAW_INDIRECT_BIT, VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03922

If `dstAccessMask` includes VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT, `dstStageMask` **must** include VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT, VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT, or VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT

- VUID-VkImageMemoryBarrier2-dstAccessMask-03926

If `dstAccessMask` includes VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT,

`dstStageMask` **must** include `VK_PIPELINE_STAGE_2_COLOR_ATTACHMENT_OUTPUT_BIT`, `VK_PIPELINE_STAGE_2_ALL_GRAPHICS_BIT`, or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-dstAccessMask-03927

If `dstAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`, or one of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages

- VUID-VkImageMemoryBarrier2-dstAccessMask-03928

If `dstAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`, `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` or `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT`

- VUID-VkImageMemoryBarrier2-dstAccessMask-06256

If the `rayQuery` feature is not enabled and `dstAccessMask` includes `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`, `dstStageMask` **must** not include any of the `VK_PIPELINE_STAGE_*_SHADER_BIT` stages except `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

- VUID-VkImageMemoryBarrier2-dstAccessMask-07272

If `dstAccessMask` includes `VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_ALL_COMMANDS_BIT` or `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

- VUID-VkImageMemoryBarrier2-dstAccessMask-04858

If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`

- VUID-VkImageMemoryBarrier2-dstAccessMask-04859

If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`

- VUID-VkImageMemoryBarrier2-dstAccessMask-04860

If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`

- VUID-VkImageMemoryBarrier2-dstAccessMask-04861

If `dstAccessMask` includes `VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR`

- VUID-VkImageMemoryBarrier2-dstAccessMask-07457

If `dstAccessMask` includes `VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT`

- VUID-VkImageMemoryBarrier2-dstAccessMask-07458

If `dstAccessMask` includes `VK_ACCESS_2_MICROMAP_READ_BIT_EXT`, `dstStageMask` **must** include `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` or `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`

- VUID-VkImageMemoryBarrier2-oldLayout-01208

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#)

or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier2-oldLayout-01209

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier2-oldLayout-01210

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier2-oldLayout-01211

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_SAMPLED_BIT` or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier2-oldLayout-01212

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT`

- VUID-VkImageMemoryBarrier2-oldLayout-01213

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT`

- VUID-VkImageMemoryBarrier2-oldLayout-01197

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), `oldLayout` **must** be `VK_IMAGE_LAYOUT_UNDEFINED` or the current layout of the image subresources affected by the barrier

- VUID-VkImageMemoryBarrier2-newLayout-01198

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), `newLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`

- VUID-VkImageMemoryBarrier2-oldLayout-01658

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier2-oldLayout-01659

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#)

or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier2-synchronization2-07793

If the `synchronization2` feature is not enabled, `oldLayout` **must** not be `VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR` or `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`

- VUID-VkImageMemoryBarrier2-synchronization2-07794

If the `synchronization2` feature is not enabled, `newLayout` **must** not be `VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR` or `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`

- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-03938

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL`, `image` **must** have been created with `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-03939

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL`, `image` **must** have been created with at least one of `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_SAMPLED_BIT`, or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier2-oldLayout-02088

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR` set

- VUID-VkImageMemoryBarrier2-image-09117

If `image` was created with a sharing mode of `VK_SHARING_MODE_EXCLUSIVE`, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `srcQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_EXTERNAL`, `VK_QUEUE_FAMILY_FOREIGN_EXT`, or a valid queue family

- VUID-VkImageMemoryBarrier2-image-09118

If `image` was created with a sharing mode of `VK_SHARING_MODE_EXCLUSIVE`, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are not equal, `dstQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_EXTERNAL`, `VK_QUEUE_FAMILY_FOREIGN_EXT`, or a valid queue family

- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-04070

If `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, at least one of `srcQueueFamilyIndex` or `dstQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_EXTERNAL` or `VK_QUEUE_FAMILY_FOREIGN_EXT`

- VUID-VkImageMemoryBarrier2-None-09119

If the [VK_KHR_external_memory](#) extension is not enabled, `srcQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_EXTERNAL`

- VUID-VkImageMemoryBarrier2-None-09120

If the [VK_KHR_external_memory](#) extension is not enabled, `dstQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_EXTERNAL`

- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-09121
If the `VK_EXT_queue_family_foreign` extension is not enabled `srcQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_FOREIGN_EXT`
- VUID-VkImageMemoryBarrier2-dstQueueFamilyIndex-09122
If the `VK_EXT_queue_family_foreign` extension is not enabled `dstQueueFamilyIndex` **must** not be `VK_QUEUE_FAMILY_FOREIGN_EXT`
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07120
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a `queue family ownership transfer` or `oldLayout` and `newLayout` define an `image layout transition`, and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07121
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a `queue family ownership transfer` or `oldLayout` and `newLayout` define an `image layout transition`, and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07122
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a `queue family ownership transfer` or `oldLayout` and `newLayout` define an `image layout transition`, and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07123
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a `queue family ownership transfer` or `oldLayout` and `newLayout` define an `image layout transition`, and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07124
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a `queue family ownership transfer` or `oldLayout` and `newLayout` define an `image layout transition`, and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07125
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a `queue family ownership transfer` or `oldLayout` and `newLayout` define an `image layout transition`, and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR`
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-07006
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a `queue family ownership transfer` or `oldLayout` and `newLayout` define an `image layout transition`, and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` then `image` **must** have been created with either the `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT` usage bits, and the `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_SAMPLED_BIT` usage bits, and the `VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` usage bit

- VUID-VkImageMemoryBarrier2-attachmentFeedbackLoopLayout-07313
If the `attachmentFeedbackLoopLayout` feature is not enabled, `newLayout` **must** not be `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT`
- VUID-VkImageMemoryBarrier2-srcQueueFamilyIndex-09550
If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a `queue family ownership transfer` or `oldLayout` and `newLayout` define an `image layout transition`, and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR` then `image` **must** have been created with either `VK_IMAGE_USAGE_STORAGE_BIT`, or with both `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` and either of `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkImageMemoryBarrier2-dynamicRenderingLocalRead-09551
If the `dynamicRenderingLocalRead` feature is not enabled, `oldLayout` **must** not be `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR`
- VUID-VkImageMemoryBarrier2-dynamicRenderingLocalRead-09552
If the `dynamicRenderingLocalRead` feature is not enabled, `newLayout` **must** not be `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR`
- VUID-VkImageMemoryBarrier2-subresourceRange-01486
`subresourceRange.baseMipLevel` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkImageMemoryBarrier2-subresourceRange-01724
If `subresourceRange.levelCount` is not `VK_REMAINING_MIP_LEVELS`, `subresourceRange.baseMipLevel + subresourceRange.levelCount` **must** be less than or equal to the `mipLevels` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkImageMemoryBarrier2-subresourceRange-01488
`subresourceRange.baseArrayLayer` **must** be less than the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkImageMemoryBarrier2-subresourceRange-01725
If `subresourceRange.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `subresourceRange.baseArrayLayer + subresourceRange.layerCount` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkImageMemoryBarrier2-image-01932
If `image` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkImageMemoryBarrier2-image-09241
If `image` has a color format that is single-plane, then the `aspectMask` member of `subresourceRange` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkImageMemoryBarrier2-image-09242
If `image` has a color format and is not *disjoint*, then the `aspectMask` member of `subresourceRange` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkImageMemoryBarrier2-image-01672
If `image` has a multi-planar format and the image is *disjoint*, then the `aspectMask` member of `subresourceRange` **must** include at least one `multi-planar aspect mask` bit or `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkImageMemoryBarrier2-image-03320

If `image` has a depth/stencil format with both depth and stencil and the `separateDepthStencilLayouts` feature is not enabled, then the `aspectMask` member of `subresourceRange` **must** include both `VK_IMAGE_ASPECT_DEPTH_BIT` and `VK_IMAGE_ASPECT_STENCIL_BIT`

- VUID-VkImageMemoryBarrier2-image-03319

If `image` has a depth/stencil format with both depth and stencil and the `separateDepthStencilLayouts` feature is enabled, then the `aspectMask` member of `subresourceRange` **must** include either or both `VK_IMAGE_ASPECT_DEPTH_BIT` and `VK_IMAGE_ASPECT_STENCIL_BIT`

- VUID-VkImageMemoryBarrier2-aspectMask-08702

If the `aspectMask` member of `subresourceRange` includes `VK_IMAGE_ASPECT_DEPTH_BIT`, `oldLayout` and `newLayout` **must** not be one of `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`

- VUID-VkImageMemoryBarrier2-aspectMask-08703

If the `aspectMask` member of `subresourceRange` includes `VK_IMAGE_ASPECT_STENCIL_BIT`, `oldLayout` and `newLayout` **must** not be one of `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`

- VUID-VkImageMemoryBarrier2-subresourceRange-09601

`subresourceRange.aspectMask` **must** be valid for the `format` the `image` was created with

- VUID-VkImageMemoryBarrier2-srcStageMask-03854

If either `srcStageMask` or `dstStageMask` includes `VK_PIPELINE_STAGE_2_HOST_BIT`, `srcQueueFamilyIndex` and `dstQueueFamilyIndex` **must** be equal

- VUID-VkImageMemoryBarrier2-srcStageMask-03855

If `srcStageMask` includes `VK_PIPELINE_STAGE_2_HOST_BIT`, and `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), `oldLayout` **must** be one of `VK_IMAGE_LAYOUT_PREINITIALIZED`, `VK_IMAGE_LAYOUT_UNDEFINED`, or `VK_IMAGE_LAYOUT_GENERAL`

Valid Usage (Implicit)

- VUID-VkImageMemoryBarrier2-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER_2`

- VUID-VkImageMemoryBarrier2-pNext-pNext

Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of [VkExternalMemoryAcquireUnmodifiedEXT](#) or [VkSampleLocationsInfoEXT](#)

- VUID-VkImageMemoryBarrier2-sType-unique

The `sType` value of each struct in the `pNext` chain **must** be unique

- VUID-VkImageMemoryBarrier2-srcStageMask-parameter

`srcStageMask` **must** be a valid combination of [VkPipelineStageFlagBits2](#) values

- VUID-VkImageMemoryBarrier2-srcAccessMask-parameter

`srcAccessMask` **must** be a valid combination of [VkAccessFlagBits2](#) values

- VUID-VkImageMemoryBarrier2-dstStageMask-parameter
dstStageMask **must** be a valid combination of [VkPipelineStageFlagBits2](#) values
- VUID-VkImageMemoryBarrier2-dstAccessMask-parameter
dstAccessMask **must** be a valid combination of [VkAccessFlagBits2](#) values
- VUID-VkImageMemoryBarrier2-oldLayout-parameter
oldLayout **must** be a valid [VkImageLayout](#) value
- VUID-VkImageMemoryBarrier2-newLayout-parameter
newLayout **must** be a valid [VkImageLayout](#) value
- VUID-VkImageMemoryBarrier2-image-parameter
image **must** be a valid [VkImage](#) handle
- VUID-VkImageMemoryBarrier2-subresourceRange-parameter
subresourceRange **must** be a valid [VkImageSubresourceRange](#) structure

The [VkImageMemoryBarrier](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageMemoryBarrier {
    VkStructureType      sType;
    const void*          pNext;
    VkAccessFlags         srcAccessMask;
    VkAccessFlags         dstAccessMask;
    VkImageLayout         oldLayout;
    VkImageLayout         newLayout;
    uint32_t              srcQueueFamilyIndex;
    uint32_t              dstQueueFamilyIndex;
    VkImage               image;
    VkImageSubresourceRange subresourceRange;
} VkImageMemoryBarrier;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **srcAccessMask** is a bitmask of [VkAccessFlagBits](#) specifying a [source access mask](#).
- **dstAccessMask** is a bitmask of [VkAccessFlagBits](#) specifying a [destination access mask](#).
- **oldLayout** is the old layout in an [image layout transition](#).
- **newLayout** is the new layout in an [image layout transition](#).
- **srcQueueFamilyIndex** is the source queue family for a [queue family ownership transfer](#).
- **dstQueueFamilyIndex** is the destination queue family for a [queue family ownership transfer](#).
- **image** is a handle to the image affected by this barrier.
- **subresourceRange** describes the [image subresource range](#) within **image** that is affected by this barrier.

The first [access scope](#) is limited to access to memory through the specified image subresource

range, via access types in the [source access mask](#) specified by `srcAccessMask`. If `srcAccessMask` includes `VK_ACCESS_HOST_WRITE_BIT`, memory writes performed by that access type are also made visible, as that access type is not performed through a resource.

The second [access scope](#) is limited to access to memory through the specified image subresource range, via access types in the [destination access mask](#) specified by `dstAccessMask`. If `dstAccessMask` includes `VK_ACCESS_HOST_WRITE_BIT` or `VK_ACCESS_HOST_READ_BIT`, available memory writes are also made visible to accesses of those types, as those access types are not performed through a resource.

If `srcQueueFamilyIndex` is not equal to `dstQueueFamilyIndex`, and `srcQueueFamilyIndex` is equal to the current queue family, then the memory barrier defines a [queue family release operation](#) for the specified image subresource range, and the second synchronization scope of the calling command does not apply to this operation.

If `dstQueueFamilyIndex` is not equal to `srcQueueFamilyIndex`, and `dstQueueFamilyIndex` is equal to the current queue family, then the memory barrier defines a [queue family acquire operation](#) for the specified image subresource range, and the first synchronization scope of the calling command does not apply to this operation.

If the `synchronization2` feature is not enabled or `oldLayout` is not equal to `newLayout`, `oldLayout` and `newLayout` define an [image layout transition](#) for the specified image subresource range.

NOTE

If the `synchronization2` feature is enabled, when the old and new layout are equal, the layout values are ignored - data is preserved no matter what values are specified, or what layout the image is currently in.

If `image` has a multi-planar format and the image is *disjoint*, then including `VK_IMAGE_ASPECT_COLOR_BIT` in the `aspectMask` member of `subresourceRange` is equivalent to including `VK_IMAGE_ASPECT_PLANE_0_BIT`, `VK_IMAGE_ASPECT_PLANE_1_BIT`, and (for three-plane formats only) `VK_IMAGE_ASPECT_PLANE_2_BIT`.

Valid Usage

- VUID-VkImageMemoryBarrier-oldLayout-01208

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier-oldLayout-01209

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier-oldLayout-01210

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL` then `image` **must** have been created with

VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT

- VUID-VkImageMemoryBarrier-oldLayout-01211

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_SAMPLED_BIT` or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier-oldLayout-01212

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT`

- VUID-VkImageMemoryBarrier-oldLayout-01213

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT`

- VUID-VkImageMemoryBarrier-oldLayout-01197

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), `oldLayout` **must** be `VK_IMAGE_LAYOUT_UNDEFINED` or the current layout of the image subresources affected by the barrier

- VUID-VkImageMemoryBarrier-newLayout-01198

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), `newLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`

- VUID-VkImageMemoryBarrier-oldLayout-01658

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier-oldLayout-01659

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL` then `image` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier-synchronization2-07793

If the `synchronization2` feature is not enabled, `oldLayout` **must** not be `VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR` or `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`

- VUID-VkImageMemoryBarrier-synchronization2-07794

If the `synchronization2` feature is not enabled, `newLayout` **must** not be `VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR` or `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-03938

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is

VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL, **image must** have been created with VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT or VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-03939

If **srcQueueFamilyIndex** and **dstQueueFamilyIndex** define a [queue family ownership transfer](#) or **oldLayout** and **newLayout** define an [image layout transition](#), and **oldLayout** or **newLayout** is VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL, **image must** have been created with at least one of VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT, VK_IMAGE_USAGE_SAMPLED_BIT, or VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT

- VUID-VkImageMemoryBarrier-oldLayout-02088

If **srcQueueFamilyIndex** and **dstQueueFamilyIndex** define a [queue family ownership transfer](#) or **oldLayout** and **newLayout** define an [image layout transition](#), and **oldLayout** or **newLayout** is VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR then **image must** have been created with VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR set

- VUID-VkImageMemoryBarrier-image-09117

If **image** was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and **srcQueueFamilyIndex** and **dstQueueFamilyIndex** are not equal, **srcQueueFamilyIndex must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue family

- VUID-VkImageMemoryBarrier-image-09118

If **image** was created with a sharing mode of VK_SHARING_MODE_EXCLUSIVE, and **srcQueueFamilyIndex** and **dstQueueFamilyIndex** are not equal, **dstQueueFamilyIndex must** be VK_QUEUE_FAMILY_EXTERNAL, VK_QUEUE_FAMILY_FOREIGN_EXT, or a valid queue family

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-04070

If **srcQueueFamilyIndex** is not equal to **dstQueueFamilyIndex**, at least one of **srcQueueFamilyIndex** or **dstQueueFamilyIndex must** not be VK_QUEUE_FAMILY_EXTERNAL or VK_QUEUE_FAMILY_FOREIGN_EXT

- VUID-VkImageMemoryBarrier-None-09119

If the [VK_KHR_external_memory](#) extension is not enabled, **srcQueueFamilyIndex must** not be VK_QUEUE_FAMILY_EXTERNAL

- VUID-VkImageMemoryBarrier-None-09120

If the [VK_KHR_external_memory](#) extension is not enabled, **dstQueueFamilyIndex must** not be VK_QUEUE_FAMILY_EXTERNAL

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-09121

If the [VK_EXT_queue_family_foreign](#) extension is not enabled **srcQueueFamilyIndex must** not be VK_QUEUE_FAMILY_FOREIGN_EXT

- VUID-VkImageMemoryBarrier-dstQueueFamilyIndex-09122

If the [VK_EXT_queue_family_foreign](#) extension is not enabled **dstQueueFamilyIndex must** not be VK_QUEUE_FAMILY_FOREIGN_EXT

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07120

If **srcQueueFamilyIndex** and **dstQueueFamilyIndex** define a [queue family ownership transfer](#) or **oldLayout** and **newLayout** define an [image layout transition](#), and **oldLayout** or **newLayout** is VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR then **image must** have been created with VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07121

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07122

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07123

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR`

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07124

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR`

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07125

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR` then `image` **must** have been created with `VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR`

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-07006

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` then `image` **must** have been created with either the `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT` usage bits, and the `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_SAMPLED_BIT` usage bits, and the `VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` usage bit

- VUID-VkImageMemoryBarrier-attachmentFeedbackLoopLayout-07313

If the `attachmentFeedbackLoopLayout` feature is not enabled, `newLayout` **must** not be `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT`

- VUID-VkImageMemoryBarrier-srcQueueFamilyIndex-09550

If `srcQueueFamilyIndex` and `dstQueueFamilyIndex` define a [queue family ownership transfer](#) or `oldLayout` and `newLayout` define an [image layout transition](#), and `oldLayout` or `newLayout` is `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR` then `image` **must** have been created with either `VK_IMAGE_USAGE_STORAGE_BIT`, or with both `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` and either of `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkImageMemoryBarrier-dynamicRenderingLocalRead-09551

If the `dynamicRenderingLocalRead` feature is not enabled, `oldLayout` **must** not be `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR`

- VUID-VkImageMemoryBarrier-dynamicRenderingLocalRead-09552
If the `dynamicRenderingLocalRead` feature is not enabled, `newLayout` **must** not be `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR`
- VUID-VkImageMemoryBarrier-subresourceRange-01486
`subresourceRange.baseMipLevel` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkImageMemoryBarrier-subresourceRange-01724
If `subresourceRange.levelCount` is not `VK_REMAINING_MIP_LEVELS`, `subresourceRange.baseMipLevel + subresourceRange.levelCount` **must** be less than or equal to the `mipLevels` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkImageMemoryBarrier-subresourceRange-01488
`subresourceRange.baseArrayLayer` **must** be less than the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkImageMemoryBarrier-subresourceRange-01725
If `subresourceRange.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `subresourceRange.baseArrayLayer + subresourceRange.layerCount` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkImageMemoryBarrier-image-01932
If `image` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkImageMemoryBarrier-image-09241
If `image` has a color format that is single-plane, then the `aspectMask` member of `subresourceRange` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkImageMemoryBarrier-image-09242
If `image` has a color format and is not *disjoint*, then the `aspectMask` member of `subresourceRange` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkImageMemoryBarrier-image-01672
If `image` has a multi-planar format and the image is *disjoint*, then the `aspectMask` member of `subresourceRange` **must** include at least one `multi-planar aspect mask` bit or `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkImageMemoryBarrier-image-03320
If `image` has a depth/stencil format with both depth and stencil and the `separateDepthStencilLayouts` feature is not enabled, then the `aspectMask` member of `subresourceRange` **must** include both `VK_IMAGE_ASPECT_DEPTH_BIT` and `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-VkImageMemoryBarrier-image-03319
If `image` has a depth/stencil format with both depth and stencil and the `separateDepthStencilLayouts` feature is enabled, then the `aspectMask` member of `subresourceRange` **must** include either or both `VK_IMAGE_ASPECT_DEPTH_BIT` and `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-VkImageMemoryBarrier-aspectMask-08702
If the `aspectMask` member of `subresourceRange` includes `VK_IMAGE_ASPECT_DEPTH_BIT`, `oldLayout` and `newLayout` **must** not be one of `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`

or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`

- VUID-VkImageMemoryBarrier-aspectMask-08703

If the `aspectMask` member of `subresourceRange` includes `VK_IMAGE_ASPECT_STENCIL_BIT`, `oldLayout` and `newLayout` **must** not be one of `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`

- VUID-VkImageMemoryBarrier-subresourceRange-09601

`subresourceRange.aspectMask` **must** be valid for the `format` the `image` was created with

- VUID-VkImageMemoryBarrier-None-09052

If the `synchronization2` feature is not enabled, and `image` was created with a sharing mode of `VK_SHARING_MODE_CONCURRENT`, at least one of `srcQueueFamilyIndex` and `dstQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_IGNORED`

- VUID-VkImageMemoryBarrier-None-09053

If the `synchronization2` feature is not enabled, and `image` was created with a sharing mode of `VK_SHARING_MODE_CONCURRENT`, `srcQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_IGNORED` or `VK_QUEUE_FAMILY_EXTERNAL`

- VUID-VkImageMemoryBarrier-None-09054

If the `synchronization2` feature is not enabled, and `image` was created with a sharing mode of `VK_SHARING_MODE_CONCURRENT`, `dstQueueFamilyIndex` **must** be `VK_QUEUE_FAMILY_IGNORED` or `VK_QUEUE_FAMILY_EXTERNAL`

Valid Usage (Implicit)

- VUID-VkImageMemoryBarrier-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER`

- VUID-VkImageMemoryBarrier-pNext-pNext

Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkExternalMemoryAcquireUnmodifiedEXT` or `VkSampleLocationsInfoEXT`

- VUID-VkImageMemoryBarrier-sType-unique

The `sType` value of each struct in the `pNext` chain **must** be unique

- VUID-VkImageMemoryBarrier-oldLayout-parameter

`oldLayout` **must** be a valid `VkImageLayout` value

- VUID-VkImageMemoryBarrier-newLayout-parameter

`newLayout` **must** be a valid `VkImageLayout` value

- VUID-VkImageMemoryBarrier-image-parameter

`image` **must** be a valid `VkImage` handle

- VUID-VkImageMemoryBarrier-subresourceRange-parameter

`subresourceRange` **must** be a valid `VkImageSubresourceRange` structure

To facilitate usage of images whose memory is initialized on the host, Vulkan allows image layout transitions to be performed by the host as well, albeit supporting limited layouts.

To perform an image layout transition on the host, call:

```
// Provided by VK_EXT_host_image_copy
VkResult vkTransitionImageLayoutEXT(
    VkDevice          device,
    uint32_t          transitionCount,
    const VkHostImageLayoutTransitionInfoEXT* pTransitions);
```

- `device` is the device which owns `pTransitions[i].image`.
- `transitionCount` is the number of image layout transitions to perform.
- `pTransitions` is a pointer to an array of `VkHostImageLayoutTransitionInfoEXT` structures specifying the image and `subresource ranges` within them to transition.

Valid Usage (Implicit)

- VUID-vkTransitionImageLayoutEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkTransitionImageLayoutEXT-pTransitions-parameter
`pTransitions` **must** be a valid pointer to an array of `transitionCount` valid `VkHostImageLayoutTransitionInfoEXT` structures
- VUID-vkTransitionImageLayoutEXT-transitionCount-arraylength
`transitionCount` **must** be greater than `0`

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_INITIALIZATION_FAILED`
- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_MEMORY_MAP_FAILED`

The `VkHostImageLayoutTransitionInfoEXT` structure is defined as:

```
// Provided by VK_EXT_host_image_copy
typedef struct VkHostImageLayoutTransitionInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkImage               image;
    VkImageLayout         oldLayout;
    VkImageLayout         newLayout;
    VkImageSubresourceRange subresourceRange;
} VkHostImageLayoutTransitionInfoEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **image** is a handle to the image affected by this layout transition.
- **oldLayout** is the old layout in an [image layout transition](#).
- **newLayout** is the new layout in an [image layout transition](#).
- **subresourceRange** describes the [image subresource range](#) within **image** that is affected by this layout transition.

vkTransitionImageLayoutEXT does not check whether the device memory associated with an image is currently in use before performing the layout transition. The application **must** guarantee that any previously submitted command that reads from or writes to this subresource has completed before the host performs the layout transition. The memory of **image** is accessed by the host as if [coherent](#).

NOTE

Image layout transitions performed on the host do not require queue family ownership transfers as the physical layout of the image will not vary between queue families for the layouts supported by this function.

NOTE

If the device has written to the image memory, it is not automatically made available to the host. Before this command can be called, a memory barrier for this image **must** have been issued on the device with the second [synchronization scope](#) including **VK_PIPELINE_STAGE_HOST_BIT** and **VK_ACCESS_HOST_READ_BIT**.

Because queue submissions [automatically make host memory visible to the device](#), there would not be a need for a memory barrier before using the results of this layout transition on the device.

Valid Usage

- VUID-VkHostImageLayoutTransitionInfoEXT-image-09055
image **must** have been created with **VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT**
- VUID-VkHostImageLayoutTransitionInfoEXT-subresourceRange-01486
subresourceRange.baseMipLevel **must** be less than the **mipLevels** specified in [VkImageCreateInfo](#) when **image** was created
- VUID-VkHostImageLayoutTransitionInfoEXT-subresourceRange-01724

If `subresourceRange.levelCount` is not `VK_REMAINING_MIP_LEVELS`, `subresourceRange.baseMipLevel + subresourceRange.levelCount` **must** be less than or equal to the `mipLevels` specified in `VkImageCreateInfo` when `image` was created

- VUID-VkHostImageLayoutTransitionInfoEXT-subresourceRange-01488
`subresourceRange.baseArrayLayer` **must** be less than the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkHostImageLayoutTransitionInfoEXT-subresourceRange-01725
If `subresourceRange.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `subresourceRange.baseArrayLayer + subresourceRange.layerCount` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkHostImageLayoutTransitionInfoEXT-image-01932
If `image` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkHostImageLayoutTransitionInfoEXT-image-09241
If `image` has a color format that is single-plane, then the `aspectMask` member of `subresourceRange` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkHostImageLayoutTransitionInfoEXT-image-09242
If `image` has a color format and is not *disjoint*, then the `aspectMask` member of `subresourceRange` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkHostImageLayoutTransitionInfoEXT-image-01672
If `image` has a multi-planar format and the image is *disjoint*, then the `aspectMask` member of `subresourceRange` **must** include at least one `multi-planar aspect mask` bit or `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkHostImageLayoutTransitionInfoEXT-image-03320
If `image` has a depth/stencil format with both depth and stencil and the `separateDepthStencilLayouts` feature is not enabled, then the `aspectMask` member of `subresourceRange` **must** include both `VK_IMAGE_ASPECT_DEPTH_BIT` and `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-VkHostImageLayoutTransitionInfoEXT-image-03319
If `image` has a depth/stencil format with both depth and stencil and the `separateDepthStencilLayouts` feature is enabled, then the `aspectMask` member of `subresourceRange` **must** include either or both `VK_IMAGE_ASPECT_DEPTH_BIT` and `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-VkHostImageLayoutTransitionInfoEXT-aspectMask-08702
If the `aspectMask` member of `subresourceRange` includes `VK_IMAGE_ASPECT_DEPTH_BIT`, `oldLayout` and `newLayout` **must** not be one of `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkHostImageLayoutTransitionInfoEXT-aspectMask-08703
If the `aspectMask` member of `subresourceRange` includes `VK_IMAGE_ASPECT_STENCIL_BIT`, `oldLayout` and `newLayout` **must** not be one of `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`
- VUID-VkHostImageLayoutTransitionInfoEXT-subresourceRange-09601
`subresourceRange.aspectMask` **must** be valid for the `format` the `image` was created with

- VUID-VkHostImageLayoutTransitionInfoEXT-oldLayout-09229
`oldLayout` **must** be either `VK_IMAGE_LAYOUT_UNDEFINED` or the current layout of the image subresources as specified in `subresourceRange`
- VUID-VkHostImageLayoutTransitionInfoEXT-oldLayout-09230
If `oldLayout` is not `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`, it **must** be one of the layouts in `VkPhysicalDeviceHostImageCopyPropertiesEXT::pCopySrcLayouts`
- VUID-VkHostImageLayoutTransitionInfoEXT-newLayout-09057
`newLayout` **must** be one of the layouts in `VkPhysicalDeviceHostImageCopyPropertiesEXT::pCopyDstLayouts`

Valid Usage (Implicit)

- VUID-VkHostImageLayoutTransitionInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_HOST_IMAGE_LAYOUT_TRANSITION_INFO_EXT`
- VUID-VkHostImageLayoutTransitionInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkHostImageLayoutTransitionInfoEXT-image-parameter
`image` **must** be a valid `VkImage` handle
- VUID-VkHostImageLayoutTransitionInfoEXT-oldLayout-parameter
`oldLayout` **must** be a valid `VkImageLayout` value
- VUID-VkHostImageLayoutTransitionInfoEXT-newLayout-parameter
`newLayout` **must** be a valid `VkImageLayout` value
- VUID-VkHostImageLayoutTransitionInfoEXT-subresourceRange-parameter
`subresourceRange` **must** be a valid `VkImageSubresourceRange` structure

7.7.4. Queue Family Ownership Transfer

Resources created with a `VkSharingMode` of `VK_SHARING_MODE_EXCLUSIVE` **must** have their ownership explicitly transferred from one queue family to another in order to access their content in a well-defined manner on a queue in a different queue family.

The special queue family index `VK_QUEUE_FAMILY_IGNORED` indicates that a queue family parameter or member is ignored.

```
#define VK_QUEUE_FAMILY_IGNORED          (~0U)
```

Resources shared with external APIs or instances using external memory **must** also explicitly manage ownership transfers between local and external queues (or equivalent constructs in external APIs) regardless of the `VkSharingMode` specified when creating them.

The special queue family index `VK_QUEUE_FAMILY_EXTERNAL` represents any queue external to the resource's current Vulkan instance, as long as the queue uses the same underlying device group or physical device, and the same driver version as the resource's `VkDevice`, as indicated by

VkPhysicalDeviceIDProperties::deviceUUID and VkPhysicalDeviceIDProperties::driverUUID.

```
#define VK_QUEUE_FAMILY_EXTERNAL          (~1U)
```

or the equivalent

```
#define VK_QUEUE_FAMILY_EXTERNAL_KHR      VK_QUEUE_FAMILY_EXTERNAL
```

The special queue family index `VK_QUEUE_FAMILY_FOREIGN_EXT` represents any queue external to the resource's current Vulkan instance, regardless of the queue's underlying physical device or driver version. This includes, for example, queues for fixed-function image processing devices, media codec devices, and display devices, as well as all queues that use the same underlying device group or physical device, and the same driver version as the resource's `VkDevice`.

```
#define VK_QUEUE_FAMILY_FOREIGN_EXT      (~2U)
```

If memory dependencies are correctly expressed between uses of such a resource between two queues in different families, but no ownership transfer is defined, the contents of that resource are undefined for any read accesses performed by the second queue family.

NOTE	If an application does not need the contents of a resource to remain valid when transferring from one queue family to another, then the ownership transfer should be skipped.
NOTE	Applications should expect transfers to/from <code>VK_QUEUE_FAMILY_FOREIGN_EXT</code> to be more expensive than transfers to/from <code>VK_QUEUE_FAMILY_EXTERNAL_KHR</code> .

A queue family ownership transfer consists of two distinct parts:

1. Release exclusive ownership from the source queue family
2. Acquire exclusive ownership for the destination queue family

An application **must** ensure that these operations occur in the correct order by defining an execution dependency between them, e.g. using a semaphore.

A *release operation* is used to release exclusive ownership of a range of a buffer or image subresource range. A release operation is defined by executing a `buffer memory barrier` (for a buffer range) or an `image memory barrier` (for an image subresource range) using a pipeline barrier command, on a queue from the source queue family. The `srcQueueFamilyIndex` parameter of the barrier **must** be the source queue family index, and the `dstQueueFamilyIndex` parameter to the destination queue family index. `dstAccessMask` is ignored for such a barrier, such that no visibility operation is executed - the value of this mask does not affect the validity of the barrier. The release operation happens-after the availability operation. `dstStageMask` is also ignored for such a barrier as defined by `buffer memory ownership transfer` and `image memory ownership transfer`.

An *acquire operation* is used to acquire exclusive ownership of a range of a buffer or image subresource range. An acquire operation is defined by executing a [buffer memory barrier](#) (for a buffer range) or an [image memory barrier](#) (for an image subresource range) using a pipeline barrier command, on a queue from the destination queue family. The buffer range or image subresource range specified in an acquire operation **must** match exactly that of a previous release operation. The `srcQueueFamilyIndex` parameter of the barrier **must** be the source queue family index, and the `dstQueueFamilyIndex` parameter to the destination queue family index. `srcAccessMask` is ignored for such a barrier, such that no availability operation is executed - the value of this mask does not affect the validity of the barrier. The acquire operation happens-before the visibility operation. `srcStageMask` is also ignored for such a barrier as defined by [buffer memory ownership transfer](#) and [image memory ownership transfer](#). As the first synchronization scope for an acquire operation is empty there is no happens-before dependency. Such a dependency **can** be introduced by using `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT`.

NOTE

Whilst it is not invalid to provide destination or source access masks for memory barriers used for release or acquire operations, respectively, they have no practical effect. Access after a release operation has undefined results, and so visibility for those accesses has no practical effect. Similarly, write access before an acquire operation will produce undefined results for future access, so availability of those writes has no practical use. In an earlier version of the specification, these were required to match on both sides - but this was subsequently relaxed. These masks **should** be set to 0.

NOTE

Since a release and acquire operation does not synchronize with second and first scopes respectively, the `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT` stage must be used to wait for a release operation to complete. Typically, a release and acquire pair is performed by a [VkSemaphore](#) signal and wait in their respective queues. Signaling a semaphore with `vkQueueSubmit` waits for `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT`. With `vkQueueSubmit2KHR`, `stageMask` for the signal semaphore must be `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT`. Similarly, for the acquire operation, waiting for a semaphore must use `VK_PIPELINE_STAGE_ALL_COMMANDS_BIT` to make sure the acquire operation is synchronized.

If the transfer is via an image memory barrier, and an [image layout transition](#) is desired, then the values of `oldLayout` and `newLayout` in the *release operation*'s memory barrier **must** be equal to values of `oldLayout` and `newLayout` in the *acquire operation*'s memory barrier. Although the image layout transition is submitted twice, it will only be executed once. A layout transition specified in this way happens-after the *release operation* and happens-before the *acquire operation*.

If the values of `srcQueueFamilyIndex` and `dstQueueFamilyIndex` are equal, no ownership transfer is performed, and the barrier operates as if they were both set to `VK_QUEUE_FAMILY_IGNORED`.

Queue family ownership transfers **may** perform read and write accesses on all memory bound to the image subresource or buffer range, so applications **must** ensure that all memory writes have been made [available](#) before a queue family ownership transfer is executed. Available memory is automatically made visible to queue family release and acquire operations, and writes performed by those operations are automatically made available.

Once a queue family has acquired ownership of a buffer range or image subresource range of a `VK_SHARING_MODE_EXCLUSIVE` resource, its contents are undefined to other queue families unless ownership is transferred. The contents of any portion of another resource which aliases memory that is bound to the transferred buffer or image subresource range are undefined after a release or acquire operation.

NOTE

Because `events` **cannot** be used directly for inter-queue synchronization, and because `vkCmdSetEvent` does not have the queue family index or memory barrier parameters needed by a *release operation*, the release and acquire operations of a queue family ownership transfer **can** only be performed using `vkCmdPipelineBarrier`.

An *acquire operation* **may** have a performance penalty when acquiring ownership of a subresource range from one of the special queue families reserved for external memory ownership transfers described above. The application **can** reduce the performance penalty in some cases by adding a `VkExternalMemoryAcquireUnmodifiedEXT` structure to the `pNext` chain of the *acquire operation's* memory barrier structure.

The `VkExternalMemoryAcquireUnmodifiedEXT` structure is defined as:

```
// Provided by VK_EXT_external_memory_acquire_unmodified
typedef struct VkExternalMemoryAcquireUnmodifiedEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           acquireUnmodifiedMemory;
} VkExternalMemoryAcquireUnmodifiedEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `acquireUnmodifiedMemory` specifies, if `VK_TRUE`, that no range of `VkDeviceMemory` bound to the resource of the memory barrier's subresource range was modified at any time since the resource's most recent release of ownership to the queue family specified by the memory barrier's `srcQueueFamilyIndex`. If `VK_FALSE`, it specifies nothing.

If the application releases ownership of the subresource range to one of the special queue families reserved for external memory ownership transfers with a memory barrier structure, and later re-acquires ownership from the same queue family with a memory barrier structure, and if no range of `VkDeviceMemory` bound to the resource was modified at any time between the *release operation* and the *acquire operation*, then the application **should** add a `VkExternalMemoryAcquireUnmodifiedEXT` structure to the `pNext` chain of the *acquire operation's* memory barrier structure because this **may** reduce the performance penalty.

This struct is ignored if `acquireUnmodifiedMemory` is `VK_FALSE`. In particular, `VK_FALSE` does *not* specify that memory was modified.

This struct is ignored if the memory barrier's `srcQueueFamilyIndex` is not a special queue family reserved for external memory ownership transfers.

NOTE

The method by which the application determines whether memory was modified between the *release operation* and *acquire operation* is outside the scope of Vulkan.

For any Vulkan operation that accesses a resource, the application **must** not assume the implementation accesses the resource's memory as read-only, even for *apparently* read-only operations such as transfer commands and shader reads.

The validity of `VkExternalMemoryAcquireUnmodifiedEXT::acquireUnmodifiedMemory` is independent of memory ranges outside the ranges of `VkDeviceMemory` bound to the resource. In particular, it is independent of any implementation-private memory associated with the resource.

Valid Usage

- VUID-VkExternalMemoryAcquireUnmodifiedEXT-acquireUnmodifiedMemory-08922

If `acquireUnmodifiedMemory` is `VK_TRUE`, and the memory barrier's `srcQueueFamilyIndex` is a special queue family reserved for external memory ownership transfers (as described in [Queue Family Ownership Transfer](#)), then each range of `VkDeviceMemory` bound to the resource **must** have remained unmodified during all time since the resource's most recent release of ownership to the queue family

Valid Usage (Implicit)

- VUID-VkExternalMemoryAcquireUnmodifiedEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_ACQUIRE_UNMODIFIED_EXT`

7.8. Wait Idle Operations

To wait on the host for the completion of outstanding queue operations for a given queue, call:

```
// Provided by VK_VERSION_1_0
VkResult vkQueueWaitIdle(
    VkQueue queue);
```

- `queue` is the queue on which to wait.

`vkQueueWaitIdle` is equivalent to having submitted a valid fence to every previously executed [queue submission command](#) that accepts a fence, then waiting for all of those fences to signal using [vkWaitForFences](#) with an infinite timeout and `waitAll` set to `VK_TRUE`.

Valid Usage (Implicit)

- VUID-vkQueueWaitIdle-queue-parameter
`queue` **must** be a valid `VkQueue` handle

Host Synchronization

- Host access to **queue** **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**
- **VK_ERROR_DEVICE_LOST**

To wait on the host for the completion of outstanding queue operations for all queues on a given logical device, call:

```
// Provided by VK_VERSION_1_0
VkResult vkDeviceWaitIdle(
    VkDevice device);
```

- **device** is the logical device to idle.

vkDeviceWaitIdle is equivalent to calling **vkQueueWaitIdle** for all queues owned by **device**.

Valid Usage (Implicit)

- VUID-vkDeviceWaitIdle-device-parameter **device** **must** be a valid **VkDevice** handle

Host Synchronization

- Host access to all **VkQueue** objects created from **device** **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

7.9. Host Write Ordering Guarantees

When batches of command buffers are submitted to a queue via a [queue submission command](#), it defines a memory dependency with prior host operations, and execution of command buffers submitted to the queue.

The first [synchronization scope](#) includes execution of `vkQueueSubmit` on the host and anything that happened-before it, as defined by the host memory model.

NOTE

Some systems allow writes that do not directly integrate with the host memory model; these have to be synchronized by the application manually. One example of this is non-temporal store instructions on x86; to ensure these happen-before submission, applications should call `_mm_sfence()`.

The second [synchronization scope](#) includes all commands submitted in the same [queue submission](#), and all commands that occur later in [submission order](#).

The first [access scope](#) includes all host writes to mappable device memory that are available to the host memory domain.

The second [access scope](#) includes all memory access performed by the device.

7.10. Synchronization and Multiple Physical Devices

If a logical device includes more than one physical device, then fences, semaphores, and events all still have a single instance of the signaled state.

A fence becomes signaled when all physical devices complete the necessary queue operations.

Semaphore wait and signal operations all include a device index that is the sole physical device that performs the operation. These indices are provided in the [VkDeviceGroupSubmitInfo](#) and [VkDeviceGroupBindSparseInfo](#) structures. Semaphores are not exclusively owned by any physical device. For example, a semaphore can be signaled by one physical device and then waited on by a different physical device.

An event **can** only be waited on by the same physical device that signaled it (or the host).

7.11. Calibrated Timestamps

In order to be able to correlate the time a particular operation took place at on timelines of different time domains (e.g. a device operation vs. a host operation), Vulkan allows querying calibrated timestamps from multiple time domains.

To query calibrated timestamps from a set of time domains, call:

```
// Provided by VK_KHR_calibrated_timestamps
VkResult vkGetCalibratedTimestampsKHR(
    VkDevice          device,
    uint32_t          timestampCount,
    const VkCalibratedTimestampInfoKHR* pTimestampInfos,
    uint64_t*          pTimestamps,
    uint64_t*          pMaxDeviation);
```

- **device** is the logical device used to perform the query.
- **timestampCount** is the number of timestamps to query.
- **pTimestampInfos** is a pointer to an array of **timestampCount** [VkCalibratedTimestampInfoKHR](#) structures, describing the time domains the calibrated timestamps should be captured from.
- **pTimestamps** is a pointer to an array of **timestampCount** 64-bit unsigned integer values in which the requested calibrated timestamp values are returned.
- **pMaxDeviation** is a pointer to a 64-bit unsigned integer value in which the strictly positive maximum deviation, in nanoseconds, of the calibrated timestamp values is returned.

NOTE

The maximum deviation **may** vary between calls to [vkGetCalibratedTimestampsKHR](#) even for the same set of time domains due to implementation and platform specific reasons. It is the application's responsibility to assess whether the returned maximum deviation makes the timestamp values suitable for any particular purpose and **can** choose to re-issue the timestamp calibration call pursuing a lower deviation value.

Calibrated timestamp values **can** be extrapolated to estimate future coinciding timestamp values, however, depending on the nature of the time domains and other properties of the platform extrapolating values over a sufficiently long period of time **may** no longer be accurate enough to fit any particular purpose, so applications are expected to re-calibrate the timestamps on a regular basis.

Valid Usage

- VUID-vkGetCalibratedTimestampsEXT-timeDomain-09246
The **timeDomain** value of each [VkCalibratedTimestampInfoKHR](#) in **pTimestampInfos** **must** be unique

Valid Usage (Implicit)

- VUID-vkGetCalibratedTimestampsKHR-device-parameter **device** must be a valid [VkDevice](#) handle
- VUID-vkGetCalibratedTimestampsKHR-pTimestampInfos-parameter **pTimestampInfos** must be a valid pointer to an array of **timestampCount** valid [VkCalibratedTimestampInfoKHR](#) structures
- VUID-vkGetCalibratedTimestampsKHR-pTimestamps-parameter **pTimestamps** must be a valid pointer to an array of **timestampCount** `uint64_t` values
- VUID-vkGetCalibratedTimestampsKHR-pMaxDeviation-parameter **pMaxDeviation** must be a valid pointer to a `uint64_t` value
- VUID-vkGetCalibratedTimestampsKHR-timestampCount-arraylength **timestampCount** must be greater than 0

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The [VkCalibratedTimestampInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_calibrated_timestamps
typedef struct VkCalibratedTimestampInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkTimeDomainKHR    timeDomain;
} VkCalibratedTimestampInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **timeDomain** is a [VkTimeDomainKHR](#) value specifying the time domain from which the calibrated timestamp value should be returned.

Valid Usage

- VUID-VkCalibratedTimestampInfoEXT-timeDomain-02354 **timeDomain** must be one of the [VkTimeDomainKHR](#) values returned by [vkGetPhysicalDeviceCalibratableTimeDomainsKHR](#)

Valid Usage (Implicit)

- VUID-VkCalibratedTimestampInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_CALIBRATED_TIMESTAMP_INFO_KHR`
- VUID-VkCalibratedTimestampInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCalibratedTimestampInfoKHR-timeDomain-parameter
`timeDomain` **must** be a valid `VkTimeDomainKHR` value

The set of supported time domains consists of:

```
// Provided by VK_KHR_calibrated_timestamps
typedef enum VkTimeDomainKHR {
    VK_TIME_DOMAIN_DEVICE_KHR = 0,
    VK_TIME_DOMAIN_CLOCK_MONOTONIC_KHR = 1,
    VK_TIME_DOMAIN_CLOCK_MONOTONIC_RAW_KHR = 2,
    VK_TIME_DOMAIN_QUERY_PERFORMANCE_COUNTER_KHR = 3,
} VkTimeDomainKHR;
```

- `VK_TIME_DOMAIN_DEVICE_KHR` specifies the device time domain. Timestamp values in this time domain use the same units and are comparable with device timestamp values captured using `vkCmdWriteTimestamp` or `vkCmdWriteTimestamp2KHR` and are defined to be incrementing according to the `timestampPeriod` of the device.
- `VK_TIME_DOMAIN_CLOCK_MONOTONIC_KHR` specifies the `CLOCK_MONOTONIC` time domain available on POSIX platforms. Timestamp values in this time domain are in units of nanoseconds and are comparable with platform timestamp values captured using the POSIX `clock_gettime` API as computed by this example:

NOTE

An implementation supporting `VK_KHR_calibrated_timestamps` will use the same time domain for all its `VkQueue` so that timestamp values reported for `VK_TIME_DOMAIN_DEVICE_KHR` can be matched to any timestamp captured through `vkCmdWriteTimestamp` or `vkCmdWriteTimestamp2KHR`.

```
struct timespec tv;
clock_gettime(CLOCK_MONOTONIC, &tv);
return tv.tv_nsec + tv.tv_sec*1000000000ull;
```

- `VK_TIME_DOMAIN_CLOCK_MONOTONIC_RAW_KHR` specifies the `CLOCK_MONOTONIC_RAW` time domain available on POSIX platforms. Timestamp values in this time domain are in units of nanoseconds and are comparable with platform timestamp values captured using the POSIX `clock_gettime` API as computed by this example:

```
struct timespec tv;
clock_gettime(CLOCK_MONOTONIC_RAW, &tv);
```



```
return tv.tv_nsec + tv.tv_sec*1000000000ull;
```

- `VK_TIME_DOMAIN_QUERY_PERFORMANCE_COUNTER_KHR` specifies the performance counter (QPC) time domain available on Windows. Timestamp values in this time domain are in the same units as those provided by the Windows QueryPerformanceCounter API and are comparable with platform timestamp values captured using that API as computed by this example:

```
LARGE_INTEGER counter;  
QueryPerformanceCounter(&counter);  
return counter.QuadPart;
```

Chapter 8. Render Pass

Draw commands **must** be recorded within a *render pass instance*. Each render pass instance defines a set of image resources, referred to as *attachments*, used during rendering.

To begin a render pass instance, call:

```
// Provided by VK_KHR_dynamic_rendering
void vkCmdBeginRenderingKHR(
    VkCommandBuffer                commandBuffer,
    const VkRenderingInfo*         pRenderingInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pRenderingInfo` is a pointer to a `VkRenderingInfo` structure specifying details of the render pass instance to begin.

After beginning a render pass instance, the command buffer is ready to record **draw commands**.

If `pRenderingInfo->flags` includes `VK_RENDERING_RESUMING_BIT` then this render pass is resumed from a render pass instance that has been suspended earlier in **submission order**.

Valid Usage

- VUID-vkCmdBeginRendering-dynamicRendering-06446
The `dynamicRendering` feature **must** be enabled
- VUID-vkCmdBeginRendering-commandBuffer-06068
If `commandBuffer` is a secondary command buffer, and the `nestedCommandBuffer` feature is not enabled, `pRenderingInfo->flags` **must** not include `VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT`
- VUID-vkCmdBeginRendering-pRenderingInfo-09588
If `pRenderingInfo->pDepthAttachment` is not `NULL` and `pRenderingInfo->pDepthAttachment->imageView` is not `VK_NULL_HANDLE`, `pRenderingInfo->pDepthAttachment->imageView` **must** be in the layout specified by `pRenderingInfo->pDepthAttachment->imageLayout`
- VUID-vkCmdBeginRendering-pRenderingInfo-09589
If `pRenderingInfo->pDepthAttachment` is not `NULL`, `pRenderingInfo->pDepthAttachment->imageView` is not `VK_NULL_HANDLE`, `pRenderingInfo->pDepthAttachment->imageResolveMode` is not `VK_RESOLVE_MODE_NONE`, and `pRenderingInfo->pDepthAttachment->resolveImageView` is not `VK_NULL_HANDLE`, `pRenderingInfo->pDepthAttachment->resolveImageView` **must** be in the layout specified by `pRenderingInfo->pDepthAttachment->resolveImageLayout`
- VUID-vkCmdBeginRendering-pRenderingInfo-09590
If `pRenderingInfo->pStencilAttachment` is not `NULL` and `pRenderingInfo->pStencilAttachment->imageView` is not `VK_NULL_HANDLE`, `pRenderingInfo->pStencilAttachment->imageView` **must** be in the layout specified by `pRenderingInfo->pStencilAttachment->imageLayout`

- VUID-vkCmdBeginRendering-pRenderingInfo-09591
If `pRenderingInfo->pStencilAttachment` is not `NULL`, `pRenderingInfo->pStencilAttachment->imageView` is not `VK_NULL_HANDLE`, `pRenderingInfo->pStencilAttachment->imageResolveMode` is not `VK_RESOLVE_MODE_NONE`, and `pRenderingInfo->pStencilAttachment->resolveImageView` is not `VK_NULL_HANDLE`, `pRenderingInfo->pStencilAttachment->resolveImageLayout` **must** be in the layout specified by `pRenderingInfo->pStencilAttachment->resolveImageLayout`
- VUID-vkCmdBeginRendering-pRenderingInfo-09592
For any element of `pRenderingInfo->pColorAttachments`, if `imageView` is not `VK_NULL_HANDLE`, that image view **must** be in the layout specified by `imageLayout`
- VUID-vkCmdBeginRendering-pRenderingInfo-09593
For any element of `pRenderingInfo->pColorAttachments`, if `imageView` is not `VK_NULL_HANDLE` and `resolveMode` is not `VK_RESOLVE_MODE_NONE`, and `resolveImageView` is not `VK_NULL_HANDLE`, `resolveImageLayout` **must** be in the layout specified by `resolveImageLayout`

Valid Usage (Implicit)

- VUID-vkCmdBeginRendering-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBeginRendering-pRenderingInfo-parameter
`pRenderingInfo` **must** be a valid pointer to a valid `VkRenderingInfo` structure
- VUID-vkCmdBeginRendering-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdBeginRendering-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBeginRendering-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdBeginRendering-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Graphics	Action State

The `VkRenderingInfo` structure is defined as:

```
typedef struct VkRenderingInfo {  
    VkStructureType      sType;  
    const void*          pNext;  
    VkRenderingFlags      flags;  
    VkRect2D              renderArea;  
    uint32_t              layerCount;  
    uint32_t              viewMask;  
    uint32_t              colorAttachmentCount;  
    const VkRenderingAttachmentInfo* pColorAttachments;  
    const VkRenderingAttachmentInfo* pDepthAttachment;  
    const VkRenderingAttachmentInfo* pStencilAttachment;  
} VkRenderingInfo;
```

or the equivalent

```
// Provided by VK_KHR_dynamic_rendering  
typedef VkRenderingInfo VkRenderingInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkRenderingFlagBits`.
- `renderArea` is the render area that is affected by the render pass instance.
- `layerCount` is the number of layers rendered to in each attachment when `viewMask` is `0`.
- `viewMask` is the view mask indicating the indices of attachment layers that will be rendered when it is not `0`.
- `colorAttachmentCount` is the number of elements in `pColorAttachments`.
- `pColorAttachments` is a pointer to an array of `colorAttachmentCount` `VkRenderingAttachmentInfo` structures describing any color attachments used.
- `pDepthAttachment` is a pointer to a `VkRenderingAttachmentInfo` structure describing a depth attachment.
- `pStencilAttachment` is a pointer to a `VkRenderingAttachmentInfo` structure describing a stencil attachment.

If `viewMask` is not 0, multiview is enabled.

If there is an instance of `VkDeviceGroupRenderPassBeginInfo` included in the `pNext` chain and its `deviceRenderAreaCount` member is not 0, then `renderArea` is ignored, and the render area is defined per-device by that structure.

Each element of the `pColorAttachments` array corresponds to an output location in the shader, i.e. if the shader declares an output variable decorated with a `Location` value of `X`, then it uses the attachment provided in `pColorAttachments[X]`. If the `imageView` member of any element of `pColorAttachments` is `VK_NULL_HANDLE`, writes to the corresponding location by a fragment are discarded.

Valid Usage

- VUID-VkRenderingInfo-viewMask-06069
If `viewMask` is 0, `layerCount` **must** not be 0
- VUID-VkRenderingInfo-multisampledRenderToSingleSampled-06857
`imageView` members of `pDepthAttachment`, `pStencilAttachment`, and elements of `pColorAttachments` that are not `VK_NULL_HANDLE` **must** have been created with the same `sampleCount`
- VUID-VkRenderingInfo-imageView-09429
`imageView` members of elements of `pColorAttachments` that are not `VK_NULL_HANDLE` **must** have been created with the same `sampleCount`
- VUID-VkRenderingInfo-None-08994
If `VkDeviceGroupRenderPassBeginInfo::deviceRenderAreaCount` is 0, `renderArea.extent.width` **must** be greater than 0
- VUID-VkRenderingInfo-None-08995
If `VkDeviceGroupRenderPassBeginInfo::deviceRenderAreaCount` is 0, `renderArea.extent.height` **must** be greater than 0
- VUID-VkRenderingInfo-pNext-06077
If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, `renderArea.offset.x` **must** be greater than or equal to 0
- VUID-VkRenderingInfo-pNext-06078
If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, `renderArea.offset.y` **must** be greater than or equal to 0
- VUID-VkRenderingInfo-pNext-07815
If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, the sum of `renderArea.extent.width` and `renderArea.offset.x` **must** be less than or equal to `maxFramebufferWidth`
- VUID-VkRenderingInfo-pNext-07816
If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, the sum of `renderArea.extent.height` and `renderArea.offset.y` **must** be less than or equal to `maxFramebufferHeight`

- VUID-VkRenderingInfo-pNext-06079
If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, the width of the `imageView` member of any element of `pColorAttachments`, `pDepthAttachment`, or `pStencilAttachment` that is not `VK_NULL_HANDLE` **must** be greater than or equal to `renderArea.offset.x + renderArea.extent.width`
- VUID-VkRenderingInfo-pNext-06080
If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, the height of the `imageView` member of any element of `pColorAttachments`, `pDepthAttachment`, or `pStencilAttachment` that is not `VK_NULL_HANDLE` **must** be greater than or equal to `renderArea.offset.y + renderArea.extent.height`
- VUID-VkRenderingInfo-pNext-06083
If the `pNext` chain contains `VkDeviceGroupRenderPassBeginInfo`, the width of the `imageView` member of any element of `pColorAttachments`, `pDepthAttachment`, or `pStencilAttachment` that is not `VK_NULL_HANDLE` **must** be greater than or equal to the sum of the `offset.x` and `extent.width` members of each element of `pDeviceRenderAreas`
- VUID-VkRenderingInfo-pNext-06084
If the `pNext` chain contains `VkDeviceGroupRenderPassBeginInfo`, the height of the `imageView` member of any element of `pColorAttachments`, `pDepthAttachment`, or `pStencilAttachment` that is not `VK_NULL_HANDLE` **must** be greater than or equal to the sum of the `offset.y` and `extent.height` members of each element of `pDeviceRenderAreas`
- VUID-VkRenderingInfo-pDepthAttachment-06085
If neither `pDepthAttachment` or `pStencilAttachment` are `NULL` and the `imageView` member of either structure is not `VK_NULL_HANDLE`, the `imageView` member of each structure **must** be the same
- VUID-VkRenderingInfo-pDepthAttachment-06086
If neither `pDepthAttachment` or `pStencilAttachment` are `NULL`, and the `resolveMode` member of each is not `VK_RESOLVE_MODE_NONE`, the `resolveImageView` member of each structure **must** be the same
- VUID-VkRenderingInfo-colorAttachmentCount-06087
If `colorAttachmentCount` is not 0 and the `imageView` member of an element of `pColorAttachments` is not `VK_NULL_HANDLE`, that `imageView` **must** have been created with `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`
- VUID-VkRenderingInfo-colorAttachmentCount-09476
If `colorAttachmentCount` is not 0 and there is an element of `pColorAttachments` with its `imageView` member not `VK_NULL_HANDLE`, and its `resolveMode` member not set to `VK_RESOLVE_MODE_NONE`, the `resolveImageView` member of that element of `pColorAttachments` **must** have been created with `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`
- VUID-VkRenderingInfo-pDepthAttachment-06547
If `pDepthAttachment` is not `NULL` and `pDepthAttachment->imageView` is not `VK_NULL_HANDLE`, `pDepthAttachment->imageView` **must** have been created with a format that includes a depth component
- VUID-VkRenderingInfo-pDepthAttachment-06088

If `pDepthAttachment` is not `NULL` and `pDepthAttachment->imageView` is not `VK_NULL_HANDLE`, `pDepthAttachment->imageView` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkRenderingInfo-pDepthAttachment-09477

If `pDepthAttachment` is not `NULL` and `pDepthAttachment->resolveMode` is not `VK_RESOLVE_MODE_NONE`, `pDepthAttachment->resolveImageView` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkRenderingInfo-pStencilAttachment-06548

If `pStencilAttachment` is not `NULL` and `pStencilAttachment->imageView` is not `VK_NULL_HANDLE`, `pStencilAttachment->imageView` **must** have been created with a format that includes a stencil aspect

- VUID-VkRenderingInfo-pStencilAttachment-06089

If `pStencilAttachment` is not `NULL` and `pStencilAttachment->imageView` is not `VK_NULL_HANDLE`, `pStencilAttachment->imageView` **must** have been created with a stencil usage including `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkRenderingInfo-pStencilAttachment-09478

If `pStencilAttachment` is not `NULL` and `pStencilAttachment->resolveMode` is not `VK_RESOLVE_MODE_NONE`, `pStencilAttachment->resolveImageView` **must** have been created with `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkRenderingInfo-colorAttachmentCount-06090

If `colorAttachmentCount` is not `0` and the `imageView` member of an element of `pColorAttachments` is not `VK_NULL_HANDLE`, the `layout` member of that element of `pColorAttachments` **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`

- VUID-VkRenderingInfo-colorAttachmentCount-06091

If `colorAttachmentCount` is not `0` and the `imageView` member of an element of `pColorAttachments` is not `VK_NULL_HANDLE`, if the `resolveMode` member of that element of `pColorAttachments` is not `VK_RESOLVE_MODE_NONE`, its `resolveImageLayout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`

- VUID-VkRenderingInfo-pDepthAttachment-06092

If `pDepthAttachment` is not `NULL` and `pDepthAttachment->imageView` is not `VK_NULL_HANDLE`, `pDepthAttachment->layout` **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL`

- VUID-VkRenderingInfo-pDepthAttachment-06093

If `pDepthAttachment` is not `NULL`, `pDepthAttachment->imageView` is not `VK_NULL_HANDLE`, and `pDepthAttachment->resolveMode` is not `VK_RESOLVE_MODE_NONE`, `pDepthAttachment->resolveImageLayout` **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL`

- VUID-VkRenderingInfo-pStencilAttachment-06094

If `pStencilAttachment` is not `NULL` and `pStencilAttachment->imageView` is not `VK_NULL_HANDLE`, `pStencilAttachment->layout` **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL`

- VUID-VkRenderingInfo-pStencilAttachment-06095

If `pStencilAttachment` is not `NULL`, `pStencilAttachment->imageView` is not `VK_NULL_HANDLE`, and `pStencilAttachment->resolveMode` is not `VK_RESOLVE_MODE_NONE`, `pStencilAttachment->`

>resolveImageLayout **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL

- VUID-VkRenderingInfo-colorAttachmentCount-06096

If colorAttachmentCount is not 0 and the imageView member of an element of pColorAttachments is not VK_NULL_HANDLE, the layout member of that element of pColorAttachments **must** not be VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkRenderingInfo-colorAttachmentCount-06097

If colorAttachmentCount is not 0 and the imageView member of an element of pColorAttachments is not VK_NULL_HANDLE, if the resolveMode member of that element of pColorAttachments is not VK_RESOLVE_MODE_NONE, its resolveImageLayout member **must** not be VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkRenderingInfo-pDepthAttachment-06098

If pDepthAttachment is not NULL, pDepthAttachment->imageView is not VK_NULL_HANDLE, and pDepthAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, pDepthAttachment->resolveImageLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL

- VUID-VkRenderingInfo-pStencilAttachment-06099

If pStencilAttachment is not NULL, pStencilAttachment->imageView is not VK_NULL_HANDLE, and pStencilAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, pStencilAttachment->resolveImageLayout **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkRenderingInfo-colorAttachmentCount-06100

If colorAttachmentCount is not 0 and the imageView member of an element of pColorAttachments is not VK_NULL_HANDLE, the layout member of that element of pColorAttachments **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkRenderingInfo-colorAttachmentCount-06101

If colorAttachmentCount is not 0 and the imageView member of an element of pColorAttachments is not VK_NULL_HANDLE, if the resolveMode member of that element of pColorAttachments is not VK_RESOLVE_MODE_NONE, its resolveImageLayout member **must** not be VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL, VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkRenderingInfo-pDepthAttachment-07732

If pDepthAttachment is not NULL and pDepthAttachment->imageView is not VK_NULL_HANDLE, pDepthAttachment->layout **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkRenderingInfo-pDepthAttachment-07733

If pDepthAttachment is not NULL, pDepthAttachment->imageView is not VK_NULL_HANDLE, and pDepthAttachment->resolveMode is not VK_RESOLVE_MODE_NONE, pDepthAttachment->resolveImageLayout **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkRenderingInfo-pStencilAttachment-07734
If `pStencilAttachment` is not `NULL` and `pStencilAttachment->imageView` is not `VK_NULL_HANDLE`, `pStencilAttachment->layout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`
- VUID-VkRenderingInfo-pStencilAttachment-07735
If `pStencilAttachment` is not `NULL`, `pStencilAttachment->imageView` is not `VK_NULL_HANDLE`, and `pStencilAttachment->resolveMode` is not `VK_RESOLVE_MODE_NONE`, `pStencilAttachment->resolveImageLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`
- VUID-VkRenderingInfo-pDepthAttachment-06102
If `pDepthAttachment` is not `NULL` and `pDepthAttachment->imageView` is not `VK_NULL_HANDLE`, `pDepthAttachment->resolveMode` **must** be one of the bits set in `VkPhysicalDeviceDepthStencilResolveProperties::supportedDepthResolveModes`
- VUID-VkRenderingInfo-pStencilAttachment-06103
If `pStencilAttachment` is not `NULL` and `pStencilAttachment->imageView` is not `VK_NULL_HANDLE`, `pStencilAttachment->resolveMode` **must** be one of the bits set in `VkPhysicalDeviceDepthStencilResolveProperties::supportedStencilResolveModes`
- VUID-VkRenderingInfo-pDepthAttachment-06104
If `pDepthAttachment` or `pStencilAttachment` are both not `NULL`, `pDepthAttachment->imageView` and `pStencilAttachment->imageView` are both not `VK_NULL_HANDLE`, and `VkPhysicalDeviceDepthStencilResolveProperties::independentResolveNone` is `VK_FALSE`, the `resolveMode` of both structures **must** be the same value
- VUID-VkRenderingInfo-pDepthAttachment-06105
If `pDepthAttachment` or `pStencilAttachment` are both not `NULL`, `pDepthAttachment->imageView` and `pStencilAttachment->imageView` are both not `VK_NULL_HANDLE`, `VkPhysicalDeviceDepthStencilResolveProperties::independentResolve` is `VK_FALSE`, and the `resolveMode` of neither structure is `VK_RESOLVE_MODE_NONE`, the `resolveMode` of both structures **must** be the same value
- VUID-VkRenderingInfo-colorAttachmentCount-06106
`colorAttachmentCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxColorAttachments`
- VUID-VkRenderingInfo-pNext-06119
If `maintenance7` is not enabled or the `robustFragmentShadingRateAttachmentAccess` limit is `VK_FALSE` or the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure was created with `VkImageSubresourceRange::baseMipLevel` greater than 0, the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, and the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure included in the `pNext` chain is not `VK_NULL_HANDLE`, `imageView` **must** have a width greater than or equal to $\lceil \frac{renderArea_x + renderArea_{width}}{shadingRateAttachmentTexelSize_{width}} \rceil$
- VUID-VkRenderingInfo-pNext-06121
If `maintenance7` is not enabled or the `robustFragmentShadingRateAttachmentAccess` limit is `VK_FALSE` or the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure was created with `VkImageSubresourceRange::baseMipLevel` greater than 0, the `pNext` chain does not contain

`VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0 and the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure included in the `pNext` chain is not `VK_NULL_HANDLE`, `imageView` **must** have a height greater than or equal to $\lceil \frac{renderArea_y + renderArea_{height}}{shadingRateAttachmentTexelSize_{height}} \rceil$

- VUID-VkRenderingInfo-pNext-06120

If `maintenance7` is not enabled or the `robustFragmentShadingRateAttachmentAccess` limit is `VK_FALSE` or the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure was created with `VkImageSubresourceRange::baseMipLevel` greater than 0, the `pNext` chain contains a `VkDeviceGroupRenderPassBeginInfo` structure, its `deviceRenderAreaCount` member is not 0, and the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure included in the `pNext` chain is not `VK_NULL_HANDLE`, `imageView` **must** have a width greater than or equal to $\lceil \frac{pDeviceRenderAreas_x + pDeviceRenderAreas_{width}}{shadingRateAttachmentTexelSize_{width}} \rceil$ for each element of `pDeviceRenderAreas`

- VUID-VkRenderingInfo-pNext-06122

If `maintenance7` is not enabled or the `robustFragmentShadingRateAttachmentAccess` limit is `VK_FALSE` or the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure was created with `VkImageSubresourceRange::baseMipLevel` greater than 0, the `pNext` chain contains a `VkDeviceGroupRenderPassBeginInfo` structure, its `deviceRenderAreaCount` member is not 0, and the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure included in the `pNext` chain is not `VK_NULL_HANDLE`, `imageView` **must** have a height greater than or equal to $\lceil \frac{pDeviceRenderAreas_y + pDeviceRenderAreas_{height}}{shadingRateAttachmentTexelSize_{height}} \rceil$ for each element of `pDeviceRenderAreas`

- VUID-VkRenderingInfo-layerCount-07817

`layerCount` **must** be less than or equal to `maxFramebufferLayers`

- VUID-VkRenderingInfo-imageView-06123

If the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure included in the `pNext` chain is not `VK_NULL_HANDLE`, and `viewMask` is 0, `imageView` **must** have a `layerCount` that is either equal to 1 or greater than or equal to `layerCount`

- VUID-VkRenderingInfo-imageView-06124

If the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure included in the `pNext` chain is not `VK_NULL_HANDLE`, and `viewMask` is not 0, `imageView` **must** have a `layerCount` that either equal to 1 or greater than or equal to the index of the most significant bit in `viewMask`

- VUID-VkRenderingInfo-imageView-06125

If the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure included in the `pNext` chain is not `VK_NULL_HANDLE`, it **must** not be equal to the `imageView` or `resolveImageView` member of `pDepthAttachment`, `pStencilAttachment`, or any element of `pColorAttachments`

- VUID-VkRenderingInfo-multiview-06127

If the `multiview` feature is not enabled, `viewMask` **must** be 0

- VUID-VkRenderingInfo-viewMask-06128

The index of the most significant bit in `viewMask` **must** be less than `maxMultiviewViewCount`

- VUID-VkRenderingInfo-None-09044
Valid attachments specified by this structure **must** not be bound to memory locations that are bound to any other valid attachments specified by this structure
- VUID-VkRenderingInfo-flags-10012
If `flags` includes `VK_RENDERING_CONTENTS_INLINE_BIT_KHR` then at least one of the following features **must** be enabled
 - `maintenance7`
 - `nestedCommandBuffer`
- VUID-VkRenderingInfo-colorAttachmentCount-09479
If `colorAttachmentCount` is not `0` and the `imageView` member of an element of `pColorAttachments` is not `VK_NULL_HANDLE`, that `imageView` **must** have been created with the `identity swizzle`
- VUID-VkRenderingInfo-colorAttachmentCount-09480
If `colorAttachmentCount` is not `0`, and there is an element of `pColorAttachments` with its `imageView` member not set to `VK_NULL_HANDLE` and its `resolveMode` member not set to `VK_RESOLVE_MODE_NONE`, the `resolveImageView` member of that element of `pColorAttachments` **must** have been created with the `identity swizzle`
- VUID-VkRenderingInfo-pDepthAttachment-09481
If `pDepthAttachment` is not `NULL` and `pDepthAttachment->imageView` is not `VK_NULL_HANDLE`, `pDepthAttachment->imageView` **must** have been created with the `identity swizzle`
- VUID-VkRenderingInfo-pDepthAttachment-09482
If `pDepthAttachment` is not `NULL`, `pDepthAttachment->imageView` is not `VK_NULL_HANDLE`, and `pDepthAttachment->resolveMode` is not `VK_RESOLVE_MODE_NONE`, `pDepthAttachment->resolveImageView` **must** have been created with the `identity swizzle`
- VUID-VkRenderingInfo-pStencilAttachment-09483
If `pStencilAttachment` is not `NULL` and `pStencilAttachment->imageView` is not `VK_NULL_HANDLE`, `pStencilAttachment->imageView` **must** have been created with the `identity swizzle`
- VUID-VkRenderingInfo-pStencilAttachment-09484
If `pStencilAttachment` is not `NULL`, `pStencilAttachment->imageView` is not `VK_NULL_HANDLE`, and `pStencilAttachment->resolveMode` is not `VK_RESOLVE_MODE_NONE`, `pStencilAttachment->resolveImageView` **must** have been created with the `identity swizzle`
- VUID-VkRenderingInfo-imageView-09485
If the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure included in the `pNext` chain is not `VK_NULL_HANDLE`, it **must** have been created with the `identity swizzle`

Valid Usage (Implicit)

- VUID-VkRenderingInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RENDERING_INFO`
- VUID-VkRenderingInfo-pNext-pNext

Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either **NULL** or a pointer to a valid instance of **VkDeviceGroupRenderPassBeginInfo** or **VkRenderingFragmentShadingRateAttachmentInfoKHR**

- VUID-VkRenderingInfo-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkRenderingInfo-flags-parameter
flags **must** be a valid combination of **VkRenderingFlagBits** values
- VUID-VkRenderingInfo-pColorAttachments-parameter
If **colorAttachmentCount** is not 0, **pColorAttachments** **must** be a valid pointer to an array of **colorAttachmentCount** valid **VkRenderingAttachmentInfo** structures
- VUID-VkRenderingInfo-pDepthAttachment-parameter
If **pDepthAttachment** is not **NULL**, **pDepthAttachment** **must** be a valid pointer to a valid **VkRenderingAttachmentInfo** structure
- VUID-VkRenderingInfo-pStencilAttachment-parameter
If **pStencilAttachment** is not **NULL**, **pStencilAttachment** **must** be a valid pointer to a valid **VkRenderingAttachmentInfo** structure

Bits which **can** be set in **VkRenderingInfo::flags** describing additional properties of the render pass are:

```
typedef enum VkRenderingFlagBits {
    VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT = 0x00000001,
    VK_RENDERING_SUSPENDING_BIT = 0x00000002,
    VK_RENDERING_RESUMING_BIT = 0x00000004,
    // Provided by VK_KHR_maintenance7
    VK_RENDERING_CONTENTS_INLINE_BIT_KHR = 0x00000010,
    VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT_KHR =
VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT,
    VK_RENDERING_SUSPENDING_BIT_KHR = VK_RENDERING_SUSPENDING_BIT,
    VK_RENDERING_RESUMING_BIT_KHR = VK_RENDERING_RESUMING_BIT,
    // Provided by VK_EXT_nested_command_buffer
    VK_RENDERING_CONTENTS_INLINE_BIT_EXT = VK_RENDERING_CONTENTS_INLINE_BIT_KHR,
} VkRenderingFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_dynamic_rendering
typedef VkRenderingFlagBits VkRenderingFlagBitsKHR;
```

- **VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT** specifies that draw calls for the render pass instance will be recorded in secondary command buffers. If the **nestedCommandBuffer** feature is enabled, the draw calls **can** come from both inline and **vkCmdExecuteCommands**.
- **VK_RENDERING_RESUMING_BIT** specifies that the render pass instance is resuming an earlier suspended render pass instance.

- `VK_RENDERING_SUSPENDING_BIT` specifies that the render pass instance will be suspended.
- `VK_RENDERING_CONTENTS_INLINE_BIT_KHR` specifies that draw calls for the render pass instance **can** be recorded inline within the current command buffer. This **can** be combined with the `VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT` bit to allow draw calls to be recorded both inline and in secondary command buffers.

The contents of `pRenderingInfo` **must** match between suspended render pass instances and the render pass instances that resume them, other than the presence or absence of the `VK_RENDERING_RESUMING_BIT`, `VK_RENDERING_SUSPENDING_BIT`, and `VK_RENDERING_CONTENTS_SECONDARY_COMMAND_BUFFERS_BIT` flags. No action or synchronization commands, or other render pass instances, are allowed between suspending and resuming render pass instances.

```
typedef VkFlags VkRenderingFlags;
```

or the equivalent

```
// Provided by VK_KHR_dynamic_rendering
typedef VkRenderingFlags VkRenderingFlagsKHR;
```

`VkRenderingFlags` is a bitmask type for setting a mask of zero or more `VkRenderingFlagBits`.

The `VkRenderingAttachmentInfo` structure is defined as:

```
typedef struct VkRenderingAttachmentInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkImageView         imageView;
    VkImageLayout       imageLayout;
    VkResolveModeFlagBits resolveMode;
    VkImageView         resolveImageView;
    VkImageLayout       resolveImageLayout;
    VkAttachmentLoadOp   loadOp;
    VkAttachmentStoreOp  storeOp;
    VkClearValue         clearValue;
} VkRenderingAttachmentInfo;
```

or the equivalent

```
// Provided by VK_KHR_dynamic_rendering
typedef VkRenderingAttachmentInfo VkRenderingAttachmentInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `imageView` is the image view that will be used for rendering.
- `imageLayout` is the layout that `imageView` will be in during rendering.
- `resolveMode` is a `VkResolveModeFlagBits` value defining how data written to `imageView` will be resolved into `resolveImageView`.
- `resolveImageView` is an image view used to write resolved data at the end of rendering.
- `resolveImageLayout` is the layout that `resolveImageView` will be in during rendering.
- `loadOp` is a `VkAttachmentLoadOp` value defining the `load operation` for the attachment.
- `storeOp` is a `VkAttachmentStoreOp` value defining the `store operation` for the attachment.
- `clearValue` is a `VkClearValue` structure defining values used to clear `imageView` when `loadOp` is `VK_ATTACHMENT_LOAD_OP_CLEAR`.

Values in `imageView` are loaded and stored according to the values of `loadOp` and `storeOp`, within the render area for each device specified in `VkRenderingInfo`. If `imageView` is `VK_NULL_HANDLE`, other members of this structure are ignored; writes to this attachment will be discarded, and no `load`, `store`, or `multisample resolve` operations will be performed.

If `resolveMode` is `VK_RESOLVE_MODE_NONE`, then `resolveImageView` is ignored. If `resolveMode` is not `VK_RESOLVE_MODE_NONE`, and `resolveImageView` is not `VK_NULL_HANDLE`, a `render pass multisample resolve operation` is defined for the attachment subresource.

NOTE

The resolve mode and store operation are independent; it is valid to write both resolved and unresolved values, and equally valid to discard the unresolved values while writing the resolved ones.

Store and resolve operations are only performed at the end of a render pass instance that does not specify the `VK_RENDERING_SUSPENDING_BIT_KHR` flag.

Load operations are only performed at the beginning of a render pass instance that does not specify the `VK_RENDERING_RESUMING_BIT_KHR` flag.

Image contents at the end of a suspended render pass instance remain defined for access by a resuming render pass instance.

Valid Usage

- VUID-VkRenderingAttachmentInfo-imageView-06129
If `imageView` is not `VK_NULL_HANDLE` and has a non-integer color format, `resolveMode` **must** be `VK_RESOLVE_MODE_NONE` or `VK_RESOLVE_MODE_AVERAGE_BIT`
- VUID-VkRenderingAttachmentInfo-imageView-06130
If `imageView` is not `VK_NULL_HANDLE` and has an integer color format, `resolveMode` **must** be `VK_RESOLVE_MODE_NONE` or `VK_RESOLVE_MODE_SAMPLE_ZERO_BIT`
- VUID-VkRenderingAttachmentInfo-imageView-06861
`imageView` **must** not have a sample count of `VK_SAMPLE_COUNT_1_BIT` if all of the following hold:

- `imageView` is not `VK_NULL_HANDLE`
- `resolveMode` is not `VK_RESOLVE_MODE_NONE`
- VUID-VkRenderingAttachmentInfo-imageView-06862
`resolveImageView` **must** not be `VK_NULL_HANDLE` if all of the following hold:
 - `imageView` is not `VK_NULL_HANDLE`
 - `resolveMode` is not `VK_RESOLVE_MODE_NONE`
- VUID-VkRenderingAttachmentInfo-imageView-06864
 If `imageView` is not `VK_NULL_HANDLE`, `resolveImageView` is not `VK_NULL_HANDLE`, and `resolveMode` is not `VK_RESOLVE_MODE_NONE`, `resolveImageView` **must** have a sample count of `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkRenderingAttachmentInfo-imageView-06865
 If `imageView` is not `VK_NULL_HANDLE`, `resolveImageView` is not `VK_NULL_HANDLE`, and `resolveMode` is not `VK_RESOLVE_MODE_NONE`, `imageView` and `resolveImageView` **must** have the same `VkFormat`
- VUID-VkRenderingAttachmentInfo-imageView-06135
 If `imageView` is not `VK_NULL_HANDLE`, `imageLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`, `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL`, `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`, or `VK_IMAGE_LAYOUT_PREINITIALIZED`
- VUID-VkRenderingAttachmentInfo-imageView-06136
 If `imageView` is not `VK_NULL_HANDLE` and `resolveMode` is not `VK_RESOLVE_MODE_NONE`, `resolveImageLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`, `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL`, `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`, or `VK_IMAGE_LAYOUT_PREINITIALIZED`
- VUID-VkRenderingAttachmentInfo-imageView-06137
 If `imageView` is not `VK_NULL_HANDLE` and `resolveMode` is not `VK_RESOLVE_MODE_NONE`, `resolveImageLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL` or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkRenderingAttachmentInfo-imageView-06142
 If `imageView` is not `VK_NULL_HANDLE` and `resolveMode` is not `VK_RESOLVE_MODE_NONE`, `resolveImageLayout` **must** not be `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`
- VUID-VkRenderingAttachmentInfo-imageView-06143
 If `imageView` is not `VK_NULL_HANDLE`, `imageLayout` **must** not be `VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR`
- VUID-VkRenderingAttachmentInfo-imageView-06144
 If `imageView` is not `VK_NULL_HANDLE` and `resolveMode` is not `VK_RESOLVE_MODE_NONE`, `resolveImageLayout` **must** not be `VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR`
- VUID-VkRenderingAttachmentInfo-imageView-06145
 If `imageView` is not `VK_NULL_HANDLE`, `imageLayout` **must** not be `VK_IMAGE_LAYOUT_PRESENT_SRC_KHR`
- VUID-VkRenderingAttachmentInfo-imageView-06146

If `imageView` is not `VK_NULL_HANDLE` and `resolveMode` is not `VK_RESOLVE_MODE_NONE`, `resolveImageLayout` **must** not be `VK_IMAGE_LAYOUT_PRESENT_SRC_KHR`

Valid Usage (Implicit)

- `VUID-VkRenderingAttachmentInfo-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_INFO`
- `VUID-VkRenderingAttachmentInfo-pNext-pNext`
`pNext` **must** be `NULL`
- `VUID-VkRenderingAttachmentInfo-imageView-parameter`
If `imageView` is not `VK_NULL_HANDLE`, `imageView` **must** be a valid `VkImageView` handle
- `VUID-VkRenderingAttachmentInfo-imageLayout-parameter`
`imageLayout` **must** be a valid `VkImageLayout` value
- `VUID-VkRenderingAttachmentInfo-resolveMode-parameter`
If `resolveMode` is not `0`, `resolveMode` **must** be a valid `VkResolveModeFlagBits` value
- `VUID-VkRenderingAttachmentInfo-resolveImageView-parameter`
If `resolveImageView` is not `VK_NULL_HANDLE`, `resolveImageView` **must** be a valid `VkImageView` handle
- `VUID-VkRenderingAttachmentInfo-resolveImageLayout-parameter`
`resolveImageLayout` **must** be a valid `VkImageLayout` value
- `VUID-VkRenderingAttachmentInfo-loadOp-parameter`
`loadOp` **must** be a valid `VkAttachmentLoadOp` value
- `VUID-VkRenderingAttachmentInfo-storeOp-parameter`
`storeOp` **must** be a valid `VkAttachmentStoreOp` value
- `VUID-VkRenderingAttachmentInfo-clearValue-parameter`
`clearValue` **must** be a valid `VkClearValue` union
- `VUID-VkRenderingAttachmentInfo-commonparent`
Both of `imageView`, and `resolveImageView` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

The `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure is defined as:

```
// Provided by VK_KHR_fragment_shading_rate with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
typedef struct VkRenderingFragmentShadingRateAttachmentInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkImageView         imageView;
    VkImageLayout       imageLayout;
    VkExtent2D          shadingRateAttachmentTexelSize;
} VkRenderingFragmentShadingRateAttachmentInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imageView` is the image view that will be used as a fragment shading rate attachment.
- `imageLayout` is the layout that `imageView` will be in during rendering.
- `shadingRateAttachmentTexelSize` specifies the number of pixels corresponding to each texel in `imageView`.

This structure can be included in the `pNext` chain of `VkRenderingInfo` to define a `fragment shading rate attachment`. If `imageView` is `VK_NULL_HANDLE`, or if this structure is not specified, the implementation behaves as if a valid shading rate attachment was specified with all texels specifying a single pixel per fragment.

Valid Usage

- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06147
If `imageView` is not `VK_NULL_HANDLE`, `layout` **must** be `VK_IMAGE_LAYOUT_GENERAL` or `VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06148
If `imageView` is not `VK_NULL_HANDLE`, it **must** have been created with `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06149
If `imageView` is not `VK_NULL_HANDLE`, `shadingRateAttachmentTexelSize.width` **must** be a power of two value
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06150
If `imageView` is not `VK_NULL_HANDLE`, `shadingRateAttachmentTexelSize.width` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSize.width`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06151
If `imageView` is not `VK_NULL_HANDLE`, `shadingRateAttachmentTexelSize.width` **must** be greater than or equal to `minFragmentShadingRateAttachmentTexelSize.width`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06152
If `imageView` is not `VK_NULL_HANDLE`, `shadingRateAttachmentTexelSize.height` **must** be a power of two value
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06153
If `imageView` is not `VK_NULL_HANDLE`, `shadingRateAttachmentTexelSize.height` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSize.height`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06154
If `imageView` is not `VK_NULL_HANDLE`, `shadingRateAttachmentTexelSize.height` **must** be greater than or equal to `minFragmentShadingRateAttachmentTexelSize.height`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06155
If `imageView` is not `VK_NULL_HANDLE`, the quotient of `shadingRateAttachmentTexelSize.width` and `shadingRateAttachmentTexelSize.height` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSizeAspectRatio`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-06156

If `imageView` is not `VK_NULL_HANDLE`, the quotient of `shadingRateAttachmentTexelSize.height` and `shadingRateAttachmentTexelSize.width` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSizeAspectRatio`

Valid Usage (Implicit)

- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_KHR`
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageView-parameter If `imageView` is not `VK_NULL_HANDLE`, `imageView` **must** be a valid `VkImageView` handle
- VUID-VkRenderingFragmentShadingRateAttachmentInfoKHR-imageLayout-parameter `imageLayout` **must** be a valid `VkImageLayout` value

To query the render area granularity for a render pass instance, call:

```
// Provided by VK_KHR_maintenance5
void vkGetRenderingAreaGranularityKHR(
    VkDevice device,
    const VkRenderingAreaInfoKHR* pRenderingAreaInfo,
    VkExtent2D* pGranularity);
```

- `device` is the logical device that owns the render pass instance.
- `pRenderingAreaInfo` is a pointer to a `VkRenderingAreaInfoKHR` structure specifying details of the render pass instance to query the render area granularity for.
- `pGranularity` is a pointer to a `VkExtent2D` structure in which the granularity is returned.

The conditions leading to an optimal `renderArea` are:

- the `offset.x` member in `renderArea` is a multiple of the `width` member of the returned `VkExtent2D` (the horizontal granularity).
- the `offset.y` member in `renderArea` is a multiple of the `height` member of the returned `VkExtent2D` (the vertical granularity).
- either the `extent.width` member in `renderArea` is a multiple of the horizontal granularity or `offset.x+extent.width` is equal to the `width` of each attachment used in the render pass instance.
- either the `extent.height` member in `renderArea` is a multiple of the vertical granularity or `offset.y+extent.height` is equal to the `height` of each attachment used in the render pass instance.

Valid Usage (Implicit)

- VUID-vkGetRenderingAreaGranularityKHR-device-parameter `device` **must** be a valid `VkDevice` handle

- VUID-vkGetRenderingAreaGranularityKHR-pRenderingAreaInfo-parameter
`pRenderingAreaInfo` **must** be a valid pointer to a valid `VkRenderingAreaInfoKHR` structure
- VUID-vkGetRenderingAreaGranularityKHR-pGranularity-parameter
`pGranularity` **must** be a valid pointer to a `VkExtent2D` structure

The `VkRenderingAreaInfoKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance5
typedef struct VkRenderingAreaInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           viewMask;
    uint32_t           colorAttachmentCount;
    const VkFormat*     pColorAttachmentFormats;
    VkFormat            depthAttachmentFormat;
    VkFormat            stencilAttachmentFormat;
} VkRenderingAreaInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `viewMask` is the viewMask used for rendering.
- `colorAttachmentCount` is the number of entries in `pColorAttachmentFormats`
- `pColorAttachmentFormats` is a pointer to an array of `VkFormat` values defining the format of color attachments used in the render pass instance.
- `depthAttachmentFormat` is a `VkFormat` value defining the format of the depth attachment used in the render pass instance.
- `stencilAttachmentFormat` is a `VkFormat` value defining the format of the stencil attachment used in the render pass instance.

Valid Usage (Implicit)

- VUID-VkRenderingAreaInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RENDERING_AREA_INFO_KHR`
- VUID-VkRenderingAreaInfoKHR-pNext-pNext
`pNext` **must** be `NULL`

To end a render pass instance, call:

```
// Provided by VK_KHR_dynamic_rendering
void vkCmdEndRenderingKHR(
    VkCommandBuffer                                commandBuffer);
```

- `commandBuffer` is the command buffer in which to record the command.

If the value of `pRenderingInfo->flags` used to begin this render pass instance included `VK_RENDERING_SUSPENDING_BIT`, then this render pass is suspended and will be resumed later in [submission order](#).

Valid Usage

- VUID-vkCmdEndRendering-None-06161
The current render pass instance **must** have been begun with `vkCmdBeginRenderingKHR`
- VUID-vkCmdEndRendering-commandBuffer-06162
The current render pass instance **must** have been begun in `commandBuffer`
- VUID-vkCmdEndRendering-None-06781
This command **must** not be recorded when transform feedback is active
- VUID-vkCmdEndRendering-None-06999
If `vkCmdBeginQuery*` was called within the render pass, the corresponding `vkCmdEndQuery*` **must** have been called subsequently within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndRendering-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdEndRendering-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdEndRendering-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdEndRendering-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdEndRendering-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	Action State

NOTE

For more complex rendering graphs, it is possible to pre-define a static *render pass* object, which as well as allowing draw commands, allows the definition of framebuffer-local dependencies between multiple subpasses. These objects have a lot of setup cost compared to [vkCmdBeginRenderingKHR](#), but use of subpass dependencies can confer important performance benefits on some devices.

8.1. Render Pass Objects

A render pass object represents a collection of attachments, subpasses, and dependencies between the subpasses, and describes how the attachments are used over the course of the subpasses.

Render passes are represented by [VkRenderPass](#) handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkRenderPass)
```

An *attachment description* describes the properties of an attachment including its format, sample count, and how its contents are treated at the beginning and end of each render pass instance.

A *subpass* represents a phase of rendering that reads and writes a subset of the attachments in a render pass. Rendering commands are recorded into a particular subpass of a render pass instance.

A *subpass description* describes the subset of attachments that is involved in the execution of a subpass. Each subpass **can** read from some attachments as *input attachments*, write to some as *color attachments* or *depth/stencil attachments*, and perform *multisample resolve operations* to *resolve attachments*. A subpass description **can** also include a set of *preserve attachments*, which are attachments that are not read or written by the subpass but whose contents **must** be preserved throughout the subpass.

A subpass *uses an attachment* if the attachment is a color, depth/stencil, resolve, depth/stencil resolve, fragment shading rate, or input attachment for that subpass (as determined by the [pColorAttachments](#), [pDepthStencilAttachment](#), [pResolveAttachments](#), [VkSubpassDescriptionDepthStencilResolve::pDepthStencilResolveAttachment](#), [VkFragmentShadingRateAttachmentInfoKHR::pFragmentShadingRateAttachment->attachment](#), and [pInputAttachments](#) members of [VkSubpassDescription](#), respectively). A subpass does not use an attachment if that attachment is preserved by the subpass. The *first use of an attachment* is in the lowest numbered subpass that uses that attachment. Similarly, the *last use of an attachment* is in the highest numbered subpass that uses that attachment.

The subpasses in a render pass all render to the same dimensions, and fragments for pixel (x,y,layer) in one subpass **can** only read attachment contents written by previous subpasses at that same (x,y,layer) location. For multi-pixel fragments, the pixel read from an input attachment is selected from the pixels covered by that fragment in an implementation-dependent manner. However, this selection **must** be made consistently for any fragment with the same shading rate for the lifetime of the [VkDevice](#).

NOTE

By describing a complete set of subpasses in advance, render passes provide the implementation an opportunity to optimize the storage and transfer of attachment data between subpasses.

In practice, this means that subpasses with a simple framebuffer-space dependency **may** be merged into a single tiled rendering pass, keeping the attachment data on-chip for the duration of a render pass instance. However, it is also quite common for a render pass to only contain a single subpass.

Subpass dependencies describe [execution and memory dependencies](#) between subpasses.

A *subpass dependency chain* is a sequence of subpass dependencies in a render pass, where the source subpass of each subpass dependency (after the first) equals the destination subpass of the previous dependency.

Execution of subpasses **may** overlap or execute out of order with regards to other subpasses, unless otherwise enforced by an execution dependency. Each subpass only respects [submission order](#) for commands recorded in the same subpass, and the [vkCmdBeginRenderPass](#) and [vkCmdEndRenderPass](#) commands that delimit the render pass - commands within other subpasses are not included. This affects most other [implicit ordering guarantees](#).

A render pass describes the structure of subpasses and attachments independent of any specific image views for the attachments. The specific image views that will be used for the attachments, and their dimensions, are specified in [VkFramebuffer](#) objects. Framebuffers are created with respect to a specific render pass that the framebuffer is compatible with (see [Render Pass Compatibility](#)). Collectively, a render pass and a framebuffer define the complete render target state for one or more subpasses as well as the algorithmic dependencies between the subpasses.

The various pipeline stages of the drawing commands for a given subpass **may** execute concurrently and/or out of order, both within and across drawing commands, whilst still respecting [pipeline order](#). However for a given (x,y,layer,sample) sample location, certain per-sample operations are performed in [rasterization order](#).

[VK_ATTACHMENT_UNUSED](#) is a constant indicating that a render pass attachment is not used.

```
#define VK_ATTACHMENT_UNUSED (~0U)
```

8.2. Render Pass Creation

To create a render pass, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateRenderPass(
    VkDevice                                device,
    const VkRenderPassCreateInfo*          pCreateInfo,
    const VkAllocationCallbacks*          pAllocator,
    VkRenderPass*                          pRenderPass);
```

- **device** is the logical device that creates the render pass.
- **pCreateInfo** is a pointer to a [VkRenderPassCreateInfo](#) structure describing the parameters of the render pass.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pRenderPass** is a pointer to a [VkRenderPass](#) handle in which the resulting render pass object is returned.

Valid Usage

- VUID-vkCreateRenderPass-device-10000
device must support at least one queue family with the [VK_QUEUE_GRAPHICS_BIT](#) capability

Valid Usage (Implicit)

- VUID-vkCreateRenderPass-device-parameter
device must be a valid [VkDevice](#) handle
- VUID-vkCreateRenderPass-pCreateInfo-parameter
pCreateInfo must be a valid pointer to a valid [VkRenderPassCreateInfo](#) structure
- VUID-vkCreateRenderPass-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateRenderPass-pRenderPass-parameter
pRenderPass must be a valid pointer to a [VkRenderPass](#) handle

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

The [VkRenderPassCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkRenderPassCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkRenderPassCreateFlags  flags;
    uint32_t                 attachmentCount;
    const VkAttachmentDescription* pAttachments;
    uint32_t                 subpassCount;
    const VkSubpassDescription* pSubpasses;
    uint32_t                 dependencyCount;
    const VkSubpassDependency* pDependencies;
} VkRenderPassCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `attachmentCount` is the number of attachments used by this render pass.
- `pAttachments` is a pointer to an array of `attachmentCount` `VkAttachmentDescription` structures describing the attachments used by the render pass.
- `subpassCount` is the number of subpasses to create.
- `pSubpasses` is a pointer to an array of `subpassCount` `VkSubpassDescription` structures describing each subpass.
- `dependencyCount` is the number of memory dependencies between pairs of subpasses.
- `pDependencies` is a pointer to an array of `dependencyCount` `VkSubpassDependency` structures describing dependencies between pairs of subpasses.

NOTE

Care should be taken to avoid a data race here; if any subpasses access attachments with overlapping memory locations, and one of those accesses is a write, a subpass dependency needs to be included between them.

Valid Usage

- VUID-VkRenderPassCreateInfo-attachment-00834
If the `attachment` member of any element of `pInputAttachments`, `pColorAttachments`, `pResolveAttachments` or `pDepthStencilAttachment`, or any element of `pPreserveAttachments` in any element of `pSubpasses` is not `VK_ATTACHMENT_UNUSED`, then it **must** be less than `attachmentCount`
- VUID-VkRenderPassCreateInfo-pAttachments-00836
For any member of `pAttachments` with a `loadOp` equal to `VK_ATTACHMENT_LOAD_OP_CLEAR`, the first use of that attachment **must** not specify a `layout` equal to `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkRenderPassCreateInfo-pAttachments-02511

For any member of `pAttachments` with a `stencilLoadOp` equal to `VK_ATTACHMENT_LOAD_OP_CLEAR`, the first use of that attachment **must** not specify a `layout` equal to `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`

- VUID-VkRenderPassCreateInfo-pAttachments-01566

For any member of `pAttachments` with a `loadOp` equal to `VK_ATTACHMENT_LOAD_OP_CLEAR`, the first use of that attachment **must** not specify a `layout` equal to `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`

- VUID-VkRenderPassCreateInfo-pAttachments-01567

For any member of `pAttachments` with a `stencilLoadOp` equal to `VK_ATTACHMENT_LOAD_OP_CLEAR`, the first use of that attachment **must** not specify a `layout` equal to `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`

- VUID-VkRenderPassCreateInfo-pNext-01926

If the `pNext` chain includes a `VkRenderPassInputAttachmentAspectCreateInfo` structure, the `subpass` member of each element of its `pAspectReferences` member **must** be less than `subpassCount`

- VUID-VkRenderPassCreateInfo-pNext-01927

If the `pNext` chain includes a `VkRenderPassInputAttachmentAspectCreateInfo` structure, the `inputAttachmentIndex` member of each element of its `pAspectReferences` member **must** be less than the value of `inputAttachmentCount` in the element of `pSubpasses` identified by its `subpass` member

- VUID-VkRenderPassCreateInfo-pNext-01963

If the `pNext` chain includes a `VkRenderPassInputAttachmentAspectCreateInfo` structure, for any element of the `pInputAttachments` member of any element of `pSubpasses` where the `attachment` member is not `VK_ATTACHMENT_UNUSED`, the `aspectMask` member of the corresponding element of `VkRenderPassInputAttachmentAspectCreateInfo::pAspectReferences` **must** only include aspects that are present in images of the format specified by the element of `pAttachments` at `attachment`

- VUID-VkRenderPassCreateInfo-pNext-01928

If the `pNext` chain includes a `VkRenderPassMultiviewCreateInfo` structure, and its `subpassCount` member is not zero, that member **must** be equal to the value of `subpassCount`

- VUID-VkRenderPassCreateInfo-pNext-01929

If the `pNext` chain includes a `VkRenderPassMultiviewCreateInfo` structure, if its `dependencyCount` member is not zero, it **must** be equal to `dependencyCount`

- VUID-VkRenderPassCreateInfo-pNext-01930

If the `pNext` chain includes a `VkRenderPassMultiviewCreateInfo` structure, for each non-zero element of `pViewOffsets`, the `srcSubpass` and `dstSubpass` members of `pDependencies` at the same index **must** not be equal

- VUID-VkRenderPassCreateInfo-pNext-02512

If the `pNext` chain includes a `VkRenderPassMultiviewCreateInfo` structure, for any element of `pDependencies` with a `dependencyFlags` member that does not include `VK_DEPENDENCY_VIEW_LOCAL_BIT`, the corresponding element of the `pViewOffsets` member of that `VkRenderPassMultiviewCreateInfo` instance **must** be 0

- VUID-VkRenderPassCreateInfo-pNext-02513

If the `pNext` chain includes a `VkRenderPassMultiviewCreateInfo` structure, elements of its `pViewMasks` member **must** either all be 0, or all not be 0

- VUID-VkRenderPassCreateInfo-pNext-02514

If the `pNext` chain includes a `VkRenderPassMultiviewCreateInfo` structure, and each element of its `pViewMasks` member is 0, the `dependencyFlags` member of each element of `pDependencies` **must** not include `VK_DEPENDENCY_VIEW_LOCAL_BIT`

- VUID-VkRenderPassCreateInfo-pNext-02515

If the `pNext` chain includes a `VkRenderPassMultiviewCreateInfo` structure, and each element of its `pViewMasks` member is 0, its `correlationMaskCount` member **must** be 0

- VUID-VkRenderPassCreateInfo-pDependencies-00837

For any element of `pDependencies`, if the `srcSubpass` is not `VK_SUBPASS_EXTERNAL`, all stage flags included in the `srcStageMask` member of that dependency **must** be a pipeline stage supported by the `pipeline` identified by the `pipelineBindPoint` member of the source subpass

- VUID-VkRenderPassCreateInfo-pDependencies-00838

For any element of `pDependencies`, if the `dstSubpass` is not `VK_SUBPASS_EXTERNAL`, all stage flags included in the `dstStageMask` member of that dependency **must** be a pipeline stage supported by the `pipeline` identified by the `pipelineBindPoint` member of the destination subpass

- VUID-VkRenderPassCreateInfo-pDependencies-06866

For any element of `pDependencies`, if its `srcSubpass` is not `VK_SUBPASS_EXTERNAL`, it **must** be less than `subpassCount`

- VUID-VkRenderPassCreateInfo-pDependencies-06867

For any element of `pDependencies`, if its `dstSubpass` is not `VK_SUBPASS_EXTERNAL`, it **must** be less than `subpassCount`

Valid Usage (Implicit)

- VUID-VkRenderPassCreateInfo-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO`

- VUID-VkRenderPassCreateInfo-pNext-pNext

Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkRenderPassInputAttachmentAspectCreateInfo` or `VkRenderPassMultiviewCreateInfo`

- VUID-VkRenderPassCreateInfo-sType-unique

The `sType` value of each struct in the `pNext` chain **must** be unique

- VUID-VkRenderPassCreateInfo-flags-zerobitmask

`flags` **must** be 0

- VUID-VkRenderPassCreateInfo-pAttachments-parameter

If `attachmentCount` is not 0, `pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid `VkAttachmentDescription` structures

- VUID-VkRenderPassCreateInfo-pSubpasses-parameter

`pSubpasses` **must** be a valid pointer to an array of `subpassCount` valid `VkSubpassDescription` structures

- VUID-VkRenderPassCreateInfo-pDependencies-parameter
If `dependencyCount` is not 0, `pDependencies` **must** be a valid pointer to an array of `dependencyCount` valid `VkSubpassDependency` structures
- VUID-VkRenderPassCreateInfo-subpassCount-arraylength
`subpassCount` **must** be greater than 0

Bits which **can** be set in `VkRenderPassCreateInfo::flags`, describing additional properties of the render pass, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkRenderPassCreateFlagBits {
} VkRenderPassCreateFlagBits;
```

NOTE

All bits for this type are defined by extensions, and none of those extensions are enabled in this build of the specification.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkRenderPassCreateFlags;
```

`VkRenderPassCreateFlags` is a bitmask type for setting a mask of zero or more `VkRenderPassCreateFlagBits`.

If the `VkRenderPassCreateInfo::pNext` chain includes a `VkRenderPassMultiviewCreateInfo` structure, then that structure includes an array of view masks, view offsets, and correlation masks for the render pass.

The `VkRenderPassMultiviewCreateInfo` structure is defined as:

```
typedef struct VkRenderPassMultiviewCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t            subpassCount;
    const uint32_t*     pViewMasks;
    uint32_t            dependencyCount;
    const int32_t*      pViewOffsets;
    uint32_t            correlationMaskCount;
    const uint32_t*     pCorrelationMasks;
} VkRenderPassMultiviewCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_multiview
typedef VkRenderPassMultiviewCreateInfo VkRenderPassMultiviewCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `subpassCount` is zero or the number of subpasses in the render pass.
- `pViewMasks` is a pointer to an array of `subpassCount` view masks, where each mask is a bitfield of view indices describing which views rendering is broadcast to in each subpass, when multiview is enabled. If `subpassCount` is zero, each view mask is treated as zero.
- `dependencyCount` is zero or the number of dependencies in the render pass.
- `pViewOffsets` is a pointer to an array of `dependencyCount` view offsets, one for each dependency. If `dependencyCount` is zero, each dependency's view offset is treated as zero. Each view offset controls which views in the source subpass the views in the destination subpass depend on.
- `correlationMaskCount` is zero or the number of correlation masks.
- `pCorrelationMasks` is a pointer to an array of `correlationMaskCount` view masks indicating sets of views that **may** be more efficient to render concurrently.

When a subpass uses a non-zero view mask, *multiview* functionality is considered to be enabled. Multiview is all-or-nothing for a render pass - that is, either all subpasses **must** have a non-zero view mask (though some subpasses **may** have only one view) or all **must** be zero. Multiview causes all drawing and clear commands in the subpass to behave as if they were broadcast to each view, where a view is represented by one layer of the framebuffer attachments. All draws and clears are broadcast to each *view index* whose bit is set in the view mask. The view index is provided in the `ViewIndex` shader input variable, and color, depth/stencil, and input attachments all read/write the layer of the framebuffer corresponding to the view index.

If the view mask is zero for all subpasses, multiview is considered to be disabled and all drawing commands execute normally, without this additional broadcasting.

Some implementations **may** not support multiview in conjunction with [geometry shaders](#) or [tessellation shaders](#).

When multiview is enabled, the `VK_DEPENDENCY_VIEW_LOCAL_BIT` bit in a dependency **can** be used to express a view-local dependency, meaning that each view in the destination subpass depends on a single view in the source subpass. Unlike pipeline barriers, a subpass dependency **can** potentially have a different view mask in the source subpass and the destination subpass. If the dependency is view-local, then each view (`dstView`) in the destination subpass depends on the view `dstView + pViewOffsets[dependency]` in the source subpass. If there is not such a view in the source subpass, then this dependency does not affect that view in the destination subpass. If the dependency is not view-local, then all views in the destination subpass depend on all views in the source subpass, and the view offset is ignored. A non-zero view offset is not allowed in a self-dependency.

The elements of `pCorrelationMasks` are a set of masks of views indicating that views in the same mask **may** exhibit spatial coherency between the views, making it more efficient to render them concurrently. Correlation masks **must** not have a functional effect on the results of the multiview

rendering.

When multiview is enabled, at the beginning of each subpass all non-render pass state is undefined. In particular, each time [vkCmdBeginRenderPass](#) or [vkCmdNextSubpass](#) is called the graphics pipeline **must** be bound, any relevant descriptor sets or vertex/index buffers **must** be bound, and any relevant dynamic state or push constants **must** be set before they are used.

Valid Usage

- VUID-VkRenderPassMultiviewCreateInfo-pCorrelationMasks-00841
Each view index **must** not be set in more than one element of [pCorrelationMasks](#)
- VUID-VkRenderPassMultiviewCreateInfo-multiview-06555
If the [multiview](#) feature is not enabled, each element of [pViewMasks](#) **must** be 0
- VUID-VkRenderPassMultiviewCreateInfo-pViewMasks-06697
The index of the most significant bit in each element of [pViewMasks](#) **must** be less than [maxMultiviewViewCount](#)

Valid Usage (Implicit)

- VUID-VkRenderPassMultiviewCreateInfo-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_RENDER_PASS_MULTIVIEW_CREATE_INFO](#)
- VUID-VkRenderPassMultiviewCreateInfo-pViewMasks-parameter
If [subpassCount](#) is not 0, [pViewMasks](#) **must** be a valid pointer to an array of [subpassCount](#) [uint32_t](#) values
- VUID-VkRenderPassMultiviewCreateInfo-pViewOffsets-parameter
If [dependencyCount](#) is not 0, [pViewOffsets](#) **must** be a valid pointer to an array of [dependencyCount](#) [int32_t](#) values
- VUID-VkRenderPassMultiviewCreateInfo-pCorrelationMasks-parameter
If [correlationMaskCount](#) is not 0, [pCorrelationMasks](#) **must** be a valid pointer to an array of [correlationMaskCount](#) [uint32_t](#) values

The [VkAttachmentDescription](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkAttachmentDescription {
    VkAttachmentDescriptionFlags    flags;
    VkFormat                       format;
    VkSampleCountFlagBits          samples;
    VkAttachmentLoadOp              loadOp;
    VkAttachmentStoreOp             storeOp;
    VkAttachmentLoadOp              stencilLoadOp;
    VkAttachmentStoreOp             stencilStoreOp;
    VkImageLayout                   initialLayout;
    VkImageLayout                   finalLayout;
} VkAttachmentDescription;
```

- **flags** is a bitmask of **VkAttachmentDescriptionFlagBits** specifying additional properties of the attachment.
- **format** is a **VkFormat** value specifying the format of the image view that will be used for the attachment.
- **samples** is a **VkSampleCountFlagBits** value specifying the number of samples of the image.
- **loadOp** is a **VkAttachmentLoadOp** value specifying how the contents of color and depth components of the attachment are treated at the beginning of the subpass where it is first used.
- **storeOp** is a **VkAttachmentStoreOp** value specifying how the contents of color and depth components of the attachment are treated at the end of the subpass where it is last used.
- **stencilLoadOp** is a **VkAttachmentLoadOp** value specifying how the contents of stencil components of the attachment are treated at the beginning of the subpass where it is first used.
- **stencilStoreOp** is a **VkAttachmentStoreOp** value specifying how the contents of stencil components of the attachment are treated at the end of the last subpass where it is used.
- **initialLayout** is the layout the attachment image subresource will be in when a render pass instance begins.
- **finalLayout** is the layout the attachment image subresource will be transitioned to when a render pass instance ends.

If the attachment uses a color format, then **loadOp** and **storeOp** are used, and **stencilLoadOp** and **stencilStoreOp** are ignored. If the format has depth and/or stencil components, **loadOp** and **storeOp** apply only to the depth data, while **stencilLoadOp** and **stencilStoreOp** define how the stencil data is handled. **loadOp** and **stencilLoadOp** define the **load operations** for the attachment. **storeOp** and **stencilStoreOp** define the **store operations** for the attachment. If an attachment is not used by any subpass, **loadOp**, **storeOp**, **stencilStoreOp**, and **stencilLoadOp** will be ignored for that attachment, and no load or store ops will be performed. However, any transition specified by **initialLayout** and **finalLayout** will still be executed.

If **flags** includes **VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT**, then the attachment is treated as if it shares physical memory with another attachment in the same render pass. This information limits the ability of the implementation to reorder certain operations (like layout transitions and the **loadOp**) such that it is not improperly reordered against other uses of the same physical memory via a different attachment. This is described in more detail below.

If a render pass uses multiple attachments that alias the same device memory, those attachments **must** each include the `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT` bit in their attachment description flags. Attachments aliasing the same memory occurs in multiple ways:

- Multiple attachments being assigned the same image view as part of framebuffer creation.
- Attachments using distinct image views that correspond to the same image subresource of an image.
- Attachments using views of distinct image subresources which are bound to overlapping memory ranges.

NOTE

Render passes **must** include subpass dependencies (either directly or via a subpass dependency chain) between any two subpasses that operate on the same attachment or aliasing attachments and those subpass dependencies **must** include execution and memory dependencies separating uses of the aliases, if at least one of those subpasses writes to one of the aliases. These dependencies **must** not include the `VK_DEPENDENCY_BY_REGION_BIT` if the aliases are views of distinct image subresources which overlap in memory.

Multiple attachments that alias the same memory **must** not be used in a single subpass. A given attachment index **must** not be used multiple times in a single subpass, with one exception: two subpass attachments **can** use the same attachment index if at least one use is as an input attachment and neither use is as a resolve or preserve attachment. In other words, the same view **can** be used simultaneously as an input and color or depth/stencil attachment, but **must** not be used as multiple color or depth/stencil attachments nor as resolve or preserve attachments.

If a set of attachments alias each other, then all except the first to be used in the render pass **must** use an `initialLayout` of `VK_IMAGE_LAYOUT_UNDEFINED`, since the earlier uses of the other aliases make their contents undefined. Once an alias has been used and a different alias has been used after it, the first alias **must** not be used in any later subpasses. However, an application **can** assign the same image view to multiple aliasing attachment indices, which allows that image view to be used multiple times even if other aliases are used in between.

NOTE

Once an attachment needs the `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT` bit, there **should** be no additional cost of introducing additional aliases, and using these additional aliases **may** allow more efficient clearing of the attachments on multiple uses via `VK_ATTACHMENT_LOAD_OP_CLEAR`.

Valid Usage

- VUID-VkAttachmentDescription-format-06699
If `format` includes a color or depth component and `loadOp` is `VK_ATTACHMENT_LOAD_OP_LOAD`, then `initialLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`
- VUID-VkAttachmentDescription-finalLayout-00843
`finalLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`
- VUID-VkAttachmentDescription-format-03280
If `format` is a color format, `initialLayout` **must** not be

VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or
VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription-format-03281

If `format` is a depth/stencil format, `initialLayout` **must** not be
VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL

- VUID-VkAttachmentDescription-format-03282

If `format` is a color format, `finalLayout` **must** not be
VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL or
VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription-format-03283

If `format` is a depth/stencil format, `finalLayout` **must** not be
VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL

- VUID-VkAttachmentDescription-format-06487

If `format` is a color format, `initialLayout` **must** not be
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL or
VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL

- VUID-VkAttachmentDescription-format-06488

If `format` is a color format, `finalLayout` **must** not be
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL or
VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL

- VUID-VkAttachmentDescription-separateDepthStencilLayouts-03284

If the `separateDepthStencilLayouts` feature is not enabled, `initialLayout` **must** not be
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL,
VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL,

- VUID-VkAttachmentDescription-separateDepthStencilLayouts-03285

If the `separateDepthStencilLayouts` feature is not enabled, `finalLayout` **must** not be
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL,
VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL,

- VUID-VkAttachmentDescription-format-03286

If `format` is a color format, `initialLayout` **must** not be
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL,
VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription-format-03287

If `format` is a color format, `finalLayout` **must** not be
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL,
VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription-format-06906

If `format` is a depth/stencil format which includes both depth and stencil components,
`initialLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription-format-06907
If **format** is a depth/stencil format which includes both depth and stencil components, **finalLayout** **must** not be **VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL** or **VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL**
- VUID-VkAttachmentDescription-format-03290
If **format** is a depth/stencil format which includes only the depth component, **initialLayout** **must** not be **VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL** or **VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL**
- VUID-VkAttachmentDescription-format-03291
If **format** is a depth/stencil format which includes only the depth component, **finalLayout** **must** not be **VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL** or **VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL**
- VUID-VkAttachmentDescription-synchronization2-06908
If the **synchronization2** feature is not enabled, **initialLayout** **must** not be **VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR** or **VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR**
- VUID-VkAttachmentDescription-synchronization2-06909
If the **synchronization2** feature is not enabled, **finalLayout** **must** not be **VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR** or **VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR**
- VUID-VkAttachmentDescription-attachmentFeedbackLoopLayout-07309
If the **attachmentFeedbackLoopLayout** feature is not enabled, **initialLayout** **must** not be **VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT**
- VUID-VkAttachmentDescription-attachmentFeedbackLoopLayout-07310
If the **attachmentFeedbackLoopLayout** feature is not enabled, **finalLayout** **must** not be **VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT**
- VUID-VkAttachmentDescription-samples-08745
samples **must** be a valid **VkSampleCountFlagBits** value that is set in **imageCreateSampleCounts** (as defined in **Image Creation Limits**) for the given **format**
- VUID-VkAttachmentDescription-dynamicRenderingLocalRead-09544
If the **dynamicRenderingLocalRead** feature is not enabled, **initialLayout** **must** not be **VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR**
- VUID-VkAttachmentDescription-dynamicRenderingLocalRead-09545
If the **dynamicRenderingLocalRead** feature is not enabled, **finalLayout** **must** not be **VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR**
- VUID-VkAttachmentDescription-format-06698
format **must** not be **VK_FORMAT_UNDEFINED**
- VUID-VkAttachmentDescription-format-06700
If **format** includes a stencil component and **stencilLoadOp** is **VK_ATTACHMENT_LOAD_OP_LOAD**, then **initialLayout** **must** not be **VK_IMAGE_LAYOUT_UNDEFINED**
- VUID-VkAttachmentDescription-format-03292
If **format** is a depth/stencil format which includes only the stencil component, **initialLayout** **must** not be **VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL** or **VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL**
- VUID-VkAttachmentDescription-format-03293

If **format** is a depth/stencil format which includes only the stencil component, **finalLayout** **must** not be **VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL** or **VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL**

- VUID-VkAttachmentDescription-format-06242

If **format** is a depth/stencil format which includes both depth and stencil components, **initialLayout** **must** not be **VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL** or **VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL**

- VUID-VkAttachmentDescription-format-06243

If **format** is a depth/stencil format which includes both depth and stencil components, **finalLayout** **must** not be **VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL** or **VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL**

Valid Usage (Implicit)

- VUID-VkAttachmentDescription-flags-parameter
flags **must** be a valid combination of **VkAttachmentDescriptionFlagBits** values
- VUID-VkAttachmentDescription-format-parameter
format **must** be a valid **VkFormat** value
- VUID-VkAttachmentDescription-samples-parameter
samples **must** be a valid **VkSampleCountFlagBits** value
- VUID-VkAttachmentDescription-loadOp-parameter
loadOp **must** be a valid **VkAttachmentLoadOp** value
- VUID-VkAttachmentDescription-storeOp-parameter
storeOp **must** be a valid **VkAttachmentStoreOp** value
- VUID-VkAttachmentDescription-stencilLoadOp-parameter
stencilLoadOp **must** be a valid **VkAttachmentLoadOp** value
- VUID-VkAttachmentDescription-stencilStoreOp-parameter
stencilStoreOp **must** be a valid **VkAttachmentStoreOp** value
- VUID-VkAttachmentDescription-initialLayout-parameter
initialLayout **must** be a valid **VkImageLayout** value
- VUID-VkAttachmentDescription-finalLayout-parameter
finalLayout **must** be a valid **VkImageLayout** value

Bits which **can** be set in **VkAttachmentDescription::flags**, describing additional properties of the attachment, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkAttachmentDescriptionFlagBits {
    VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT = 0x00000001,
} VkAttachmentDescriptionFlagBits;
```

- **VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT** specifies that the attachment aliases the same device

memory as other attachments.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkAttachmentDescriptionFlags;
```

`VkAttachmentDescriptionFlags` is a bitmask type for setting a mask of zero or more `VkAttachmentDescriptionFlagBits`.

The `VkRenderPassInputAttachmentAspectCreateInfo` structure is defined as:

```
typedef struct VkRenderPassInputAttachmentAspectCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    uint32_t                  aspectReferenceCount;
    const VkInputAttachmentAspectReference* pAspectReferences;
} VkRenderPassInputAttachmentAspectCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_maintenance2
typedef VkRenderPassInputAttachmentAspectCreateInfo
VkRenderPassInputAttachmentAspectCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `aspectReferenceCount` is the number of elements in the `pAspectReferences` array.
- `pAspectReferences` is a pointer to an array of `aspectReferenceCount` `VkInputAttachmentAspectReference` structures containing a mask describing which aspect(s) **can** be accessed for a given input attachment within a given subpass.

To specify which aspects of an input attachment **can** be read, add a `VkRenderPassInputAttachmentAspectCreateInfo` structure to the `pNext` chain of the `VkRenderPassCreateInfo` structure:

An application **can** access any aspect of an input attachment that does not have a specified aspect mask in the `pAspectReferences` array. Otherwise, an application **must** not access aspect(s) of an input attachment other than those in its specified aspect mask.

Valid Usage (Implicit)

- VUID-VkRenderPassInputAttachmentAspectCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO`
- VUID-VkRenderPassInputAttachmentAspectCreateInfo-pAspectReferences-parameter
`pAspectReferences` **must** be a valid pointer to an array of `aspectReferenceCount` valid

VkInputAttachmentAspectReference structures

- VUID-VkRenderPassInputAttachmentAspectCreateInfo-aspectReferenceCount-arraylength **aspectReferenceCount** **must** be greater than 0

The **VkInputAttachmentAspectReference** structure is defined as:

```
typedef struct VkInputAttachmentAspectReference {  
    uint32_t      subpass;  
    uint32_t      inputAttachmentIndex;  
    VkImageAspectFlags aspectMask;  
} VkInputAttachmentAspectReference;
```

or the equivalent

```
// Provided by VK_KHR_maintenance2  
typedef VkInputAttachmentAspectReference VkInputAttachmentAspectReferenceKHR;
```

- **subpass** is an index into the **pSubpasses** array of the parent **VkRenderPassCreateInfo** structure.
- **inputAttachmentIndex** is an index into the **pInputAttachments** of the specified subpass.
- **aspectMask** is a mask of which aspect(s) **can** be accessed within the specified subpass.

This structure specifies an aspect mask for a specific input attachment of a specific subpass in the render pass.

subpass and **inputAttachmentIndex** index into the render pass as:

```
pCreateInfo->pSubpasses[subpass].pInputAttachments[inputAttachmentIndex]
```

Valid Usage

- VUID-VkInputAttachmentAspectReference-aspectMask-01964 **aspectMask** **must** not include **VK_IMAGE_ASPECT_METADATA_BIT**
- VUID-VkInputAttachmentAspectReference-aspectMask-02250 **aspectMask** **must** not include **VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT** for any index *i*

Valid Usage (Implicit)

- VUID-VkInputAttachmentAspectReference-aspectMask-parameter **aspectMask** **must** be a valid combination of **VkImageAspectFlagBits** values
- VUID-VkInputAttachmentAspectReference-aspectMask-requiredbitmask **aspectMask** **must** not be 0

The `VkSubpassDescription` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSubpassDescription {
    VkSubpassDescriptionFlags    flags;
    VkPipelineBindPoint          pipelineBindPoint;
    uint32_t                     inputAttachmentCount;
    const VkAttachmentReference* pInputAttachments;
    uint32_t                     colorAttachmentCount;
    const VkAttachmentReference* pColorAttachments;
    const VkAttachmentReference* pResolveAttachments;
    const VkAttachmentReference* pDepthStencilAttachment;
    uint32_t                     preserveAttachmentCount;
    const uint32_t*              pPreserveAttachments;
} VkSubpassDescription;
```

- `flags` is a bitmask of `VkSubpassDescriptionFlagBits` specifying usage of the subpass.
- `pipelineBindPoint` is a `VkPipelineBindPoint` value specifying the pipeline type supported for this subpass.
- `inputAttachmentCount` is the number of input attachments.
- `pInputAttachments` is a pointer to an array of `VkAttachmentReference` structures defining the input attachments for this subpass and their layouts.
- `colorAttachmentCount` is the number of color attachments.
- `pColorAttachments` is a pointer to an array of `colorAttachmentCount` `VkAttachmentReference` structures defining the color attachments for this subpass and their layouts.
- `pResolveAttachments` is `NULL` or a pointer to an array of `colorAttachmentCount` `VkAttachmentReference` structures defining the resolve attachments for this subpass and their layouts.
- `pDepthStencilAttachment` is a pointer to a `VkAttachmentReference` structure specifying the depth/stencil attachment for this subpass and its layout.
- `preserveAttachmentCount` is the number of preserved attachments.
- `pPreserveAttachments` is a pointer to an array of `preserveAttachmentCount` render pass attachment indices identifying attachments that are not used by this subpass, but whose contents **must** be preserved throughout the subpass.

Each element of the `pInputAttachments` array corresponds to an input attachment index in a fragment shader, i.e. if a shader declares an image variable decorated with a `InputAttachmentIndex` value of `X`, then it uses the attachment provided in `pInputAttachments[X]`. Input attachments **must** also be bound to the pipeline in a descriptor set. If the `attachment` member of any element of `pInputAttachments` is `VK_ATTACHMENT_UNUSED`, the application **must** not read from the corresponding input attachment index. Fragment shaders **can** use subpass input variables to access the contents of an input attachment at the fragment's (x, y, layer) framebuffer coordinates.

Each element of the `pColorAttachments` array corresponds to an output location in the shader, i.e. if the shader declares an output variable decorated with a `Location` value of `X`, then it uses the

attachment provided in `pColorAttachments[X]`. If the `attachment` member of any element of `pColorAttachments` is `VK_ATTACHMENT_UNUSED`, or if `Color Write Enable` has been disabled for the corresponding attachment index, then writes to the corresponding location by a fragment shader are discarded.

If `pResolveAttachments` is not `NULL`, each of its elements corresponds to a color attachment (the element in `pColorAttachments` at the same index), and a `multisample resolve operation` is defined for each attachment unless the resolve attachment index is `VK_ATTACHMENT_UNUSED`.

Similarly, if `VkSubpassDescriptionDepthStencilResolve::pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED`, it corresponds to the depth/stencil attachment in `pDepthStencilAttachment`, and `multisample resolve operation` for depth and stencil are defined by `VkSubpassDescriptionDepthStencilResolve::depthResolveMode` and `VkSubpassDescriptionDepthStencilResolve::stencilResolveMode`, respectively. If `VkSubpassDescriptionDepthStencilResolve::depthResolveMode` is `VK_RESOLVE_MODE_NONE` or the `pDepthStencilResolveAttachment` does not have a depth aspect, no resolve operation is performed for the depth attachment. If `VkSubpassDescriptionDepthStencilResolve::stencilResolveMode` is `VK_RESOLVE_MODE_NONE` or the `pDepthStencilResolveAttachment` does not have a stencil aspect, no resolve operation is performed for the stencil attachment.

If the image subresource range referenced by the depth/stencil attachment is created with `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT`, then the `multisample resolve operation` uses the sample locations state specified in the `sampleLocationsInfo` member of the element of the `VkRenderPassSampleLocationsBeginInfoEXT::pPostSubpassSampleLocations` for the subpass.

If `pDepthStencilAttachment` is `NULL`, or if its attachment index is `VK_ATTACHMENT_UNUSED`, it indicates that no depth/stencil attachment will be used in the subpass.

The contents of an attachment within the render area become undefined at the start of a subpass **S** if all of the following conditions are true:

- The attachment is used as a color, depth/stencil, or resolve attachment in any subpass in the render pass.
- There is a subpass **S**₁ that uses or preserves the attachment, and a subpass dependency from **S**₁ to **S**.
- The attachment is not used or preserved in subpass **S**.

Once the contents of an attachment become undefined in subpass **S**, they remain undefined for subpasses in subpass dependency chains starting with subpass **S** until they are written again. However, they remain valid for subpasses in other subpass dependency chains starting with subpass **S**₁ if those subpasses use or preserve the attachment.

Valid Usage

- VUID-VkSubpassDescription-attachment-06912

If the `attachment` member of an element of `pInputAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` or

VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL

- VUID-VkSubpassDescription-attachment-06913

If the `attachment` member of an element of `pColorAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`

- VUID-VkSubpassDescription-attachment-06914

If the `attachment` member of an element of `pResolveAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`

- VUID-VkSubpassDescription-attachment-06915

If the `attachment` member of `pDepthStencilAttachment` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`

- VUID-VkSubpassDescription-attachment-06916

If the `attachment` member of an element of `pColorAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`

- VUID-VkSubpassDescription-attachment-06917

If the `attachment` member of an element of `pResolveAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`

- VUID-VkSubpassDescription-attachment-06918

If the `attachment` member of an element of `pInputAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`

- VUID-VkSubpassDescription-attachment-06919

If the `attachment` member of an element of `pColorAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`

- VUID-VkSubpassDescription-attachment-06920

If the `attachment` member of an element of `pResolveAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`

- VUID-VkSubpassDescription-attachment-06921

If the `attachment` member of an element of `pInputAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR`

- VUID-VkSubpassDescription-attachment-06922

If the `attachment` member of an element of `pColorAttachments` is not `VK_ATTACHMENT_UNUSED`,

its `layout` member **must** not be `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`

- VUID-VkSubpassDescription-attachment-06923

If the `attachment` member of an element of `pResolveAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`

- VUID-VkSubpassDescription-pipelineBindPoint-04952

`pipelineBindPoint` **must** be `VK_PIPELINE_BIND_POINT_GRAPHICS`

- VUID-VkSubpassDescription-colorAttachmentCount-00845

`colorAttachmentCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxColorAttachments`

- VUID-VkSubpassDescription-loadOp-00846

If the first use of an attachment in this render pass is as an input attachment, and the attachment is not also used as a color or depth/stencil attachment in the same subpass, then `loadOp` **must** not be `VK_ATTACHMENT_LOAD_OP_CLEAR`

- VUID-VkSubpassDescription-pResolveAttachments-00847

If `pResolveAttachments` is not `NULL`, for each resolve attachment that is not `VK_ATTACHMENT_UNUSED`, the corresponding color attachment **must** not be `VK_ATTACHMENT_UNUSED`

- VUID-VkSubpassDescription-pResolveAttachments-00848

If `pResolveAttachments` is not `NULL`, for each resolve attachment that is not `VK_ATTACHMENT_UNUSED`, the corresponding color attachment **must** not have a sample count of `VK_SAMPLE_COUNT_1_BIT`

- VUID-VkSubpassDescription-pResolveAttachments-00849

If `pResolveAttachments` is not `NULL`, each resolve attachment that is not `VK_ATTACHMENT_UNUSED` **must** have a sample count of `VK_SAMPLE_COUNT_1_BIT`

- VUID-VkSubpassDescription-pResolveAttachments-00850

If `pResolveAttachments` is not `NULL`, each resolve attachment that is not `VK_ATTACHMENT_UNUSED` **must** have the same `VkFormat` as its corresponding color attachment

- VUID-VkSubpassDescription-pColorAttachments-09430

All attachments in `pColorAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have the same sample count

- VUID-VkSubpassDescription-pInputAttachments-02647

All attachments in `pInputAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have image formats whose `potential format features` contain at least `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT` or `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkSubpassDescription-pColorAttachments-02648

All attachments in `pColorAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have image formats whose `potential format features` contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT`

- VUID-VkSubpassDescription-pResolveAttachments-02649

All attachments in `pResolveAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have image formats whose `potential format features` contain

VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT

- VUID-VkSubpassDescription-pDepthStencilAttachment-02650
If `pDepthStencilAttachment` is not `NULL` and the attachment is not `VK_ATTACHMENT_UNUSED` then it **must** have an image format whose `potential format features` contain `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkSubpassDescription-pDepthStencilAttachment-01418
If `pDepthStencilAttachment` is not `VK_ATTACHMENT_UNUSED` and any attachments in `pColorAttachments` are not `VK_ATTACHMENT_UNUSED`, they **must** have the same sample count
- VUID-VkSubpassDescription-attachment-00853
Each element of `pPreserveAttachments` **must** not be `VK_ATTACHMENT_UNUSED`
- VUID-VkSubpassDescription-pPreserveAttachments-00854
Each element of `pPreserveAttachments` **must** not also be an element of any other member of the subpass description
- VUID-VkSubpassDescription-layout-02519
If any attachment is used by more than one `VkAttachmentReference` member, then each use **must** use the same `layout`
- VUID-VkSubpassDescription-pDepthStencilAttachment-04438
`pDepthStencilAttachment` and `pColorAttachments` **must** not contain references to the same attachment

Valid Usage (Implicit)

- VUID-VkSubpassDescription-flags-zerobitmask
`flags` **must** be `0`
- VUID-VkSubpassDescription-pipelineBindPoint-parameter
`pipelineBindPoint` **must** be a valid `VkPipelineBindPoint` value
- VUID-VkSubpassDescription-pInputAttachments-parameter
If `inputAttachmentCount` is not `0`, `pInputAttachments` **must** be a valid pointer to an array of `inputAttachmentCount` valid `VkAttachmentReference` structures
- VUID-VkSubpassDescription-pColorAttachments-parameter
If `colorAttachmentCount` is not `0`, `pColorAttachments` **must** be a valid pointer to an array of `colorAttachmentCount` valid `VkAttachmentReference` structures
- VUID-VkSubpassDescription-pResolveAttachments-parameter
If `colorAttachmentCount` is not `0`, and `pResolveAttachments` is not `NULL`, `pResolveAttachments` **must** be a valid pointer to an array of `colorAttachmentCount` valid `VkAttachmentReference` structures
- VUID-VkSubpassDescription-pDepthStencilAttachment-parameter
If `pDepthStencilAttachment` is not `NULL`, `pDepthStencilAttachment` **must** be a valid pointer to a valid `VkAttachmentReference` structure
- VUID-VkSubpassDescription-pPreserveAttachments-parameter
If `preserveAttachmentCount` is not `0`, `pPreserveAttachments` **must** be a valid pointer to an array of `preserveAttachmentCount` `uint32_t` values

Bits which **can** be set in `VkSubpassDescription::flags`, specifying usage of the subpass, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSubpassDescriptionFlagBits {
} VkSubpassDescriptionFlagBits;
```

NOTE

All bits for this type are defined by extensions, and none of those extensions are enabled in this build of the specification.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSubpassDescriptionFlags;
```

`VkSubpassDescriptionFlags` is a bitmask type for setting a mask of zero or more `VkSubpassDescriptionFlagBits`.

The `VkAttachmentReference` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkAttachmentReference {
    uint32_t      attachment;
    VkImageLayout layout;
} VkAttachmentReference;
```

- `attachment` is either an integer value identifying an attachment at the corresponding index in `VkRenderPassCreateInfo::pAttachments`, or `VK_ATTACHMENT_UNUSED` to signify that this attachment is not used.
- `layout` is a `VkImageLayout` value specifying the layout the attachment uses during the subpass.

Valid Usage

- VUID-VkAttachmentReference-layout-03077
If `attachment` is not `VK_ATTACHMENT_UNUSED`, `layout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`, `VK_IMAGE_LAYOUT_PREINITIALIZED`, or `VK_IMAGE_LAYOUT_PRESENT_SRC_KHR`
- VUID-VkAttachmentReference-separateDepthStencilLayouts-03313
If the `separateDepthStencilLayouts` feature is not enabled, and `attachment` is not `VK_ATTACHMENT_UNUSED`, `layout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`,
- VUID-VkAttachmentReference-synchronization2-06910
If the `synchronization2` feature is not enabled, `layout` **must** not be `VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR` or `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`
- VUID-VkAttachmentReference-attachmentFeedbackLoopLayout-07311
If the `attachmentFeedbackLoopLayout` feature is not enabled, `layout` **must** not be

VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT

- VUID-VkAttachmentReference-dynamicRenderingLocalRead-09546

If the `dynamicRenderingLocalRead` feature is not enabled, `layout` **must** not be `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR`

Valid Usage (Implicit)

- VUID-VkAttachmentReference-layout-parameter
`layout` **must** be a valid `VkImageLayout` value

`VK_SUBPASS_EXTERNAL` is a special subpass index value expanding synchronization scope outside a subpass. It is described in more detail by [VkSubpassDependency](#).

```
#define VK_SUBPASS_EXTERNAL          (~0U)
```

The `VkSubpassDependency` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSubpassDependency {
    uint32_t          srcSubpass;
    uint32_t          dstSubpass;
    VkPipelineStageFlags srcStageMask;
    VkPipelineStageFlags dstStageMask;
    VkAccessFlags      srcAccessMask;
    VkAccessFlags      dstAccessMask;
    VkDependencyFlags  dependencyFlags;
} VkSubpassDependency;
```

- `srcSubpass` is the subpass index of the first subpass in the dependency, or `VK_SUBPASS_EXTERNAL`.
- `dstSubpass` is the subpass index of the second subpass in the dependency, or `VK_SUBPASS_EXTERNAL`.
- `srcStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [source stage mask](#).
- `dstStageMask` is a bitmask of `VkPipelineStageFlagBits` specifying the [destination stage mask](#).
- `srcAccessMask` is a bitmask of `VkAccessFlagBits` specifying a [source access mask](#).
- `dstAccessMask` is a bitmask of `VkAccessFlagBits` specifying a [destination access mask](#).
- `dependencyFlags` is a bitmask of `VkDependencyFlagBits`.

If `srcSubpass` is equal to `dstSubpass` then the `VkSubpassDependency` does not directly define a [dependency](#). Instead, it enables pipeline barriers to be used in a render pass instance within the identified subpass, where the scopes of one pipeline barrier **must** be a subset of those described by one subpass dependency. Subpass dependencies specified in this way that include [framebuffer-space stages](#) in the `srcStageMask` **must** only include [framebuffer-space stages](#) in `dstStageMask`, and

must include `VK_DEPENDENCY_BY_REGION_BIT`. When a subpass dependency is specified in this way for a subpass that has more than one view in its view mask, its `dependencyFlags` **must** include `VK_DEPENDENCY_VIEW_LOCAL_BIT`.

If `srcSubpass` and `dstSubpass` are not equal, when a render pass instance which includes a subpass dependency is submitted to a queue, it defines a `dependency` between the subpasses identified by `srcSubpass` and `dstSubpass`.

If `srcSubpass` is equal to `VK_SUBPASS_EXTERNAL`, the first `synchronization scope` includes commands that occur earlier in `submission order` than the `vkCmdBeginRenderPass` used to begin the render pass instance. Otherwise, the first set of commands includes all commands submitted as part of the subpass instance identified by `srcSubpass` and any `load`, `store`, or `multisample resolve` operations on attachments used in `srcSubpass`. In either case, the first synchronization scope is limited to operations on the pipeline stages determined by the `source stage mask` specified by `srcStageMask`.

If `dstSubpass` is equal to `VK_SUBPASS_EXTERNAL`, the second `synchronization scope` includes commands that occur later in `submission order` than the `vkCmdEndRenderPass` used to end the render pass instance. Otherwise, the second set of commands includes all commands submitted as part of the subpass instance identified by `dstSubpass` and any `load`, `store`, and `multisample resolve` operations on attachments used in `dstSubpass`. In either case, the second synchronization scope is limited to operations on the pipeline stages determined by the `destination stage mask` specified by `dstStageMask`.

The first `access scope` is limited to accesses in the pipeline stages determined by the `source stage mask` specified by `srcStageMask`. It is also limited to access types in the `source access mask` specified by `srcAccessMask`.

The second `access scope` is limited to accesses in the pipeline stages determined by the `destination stage mask` specified by `dstStageMask`. It is also limited to access types in the `destination access mask` specified by `dstAccessMask`.

The `availability and visibility operations` defined by a subpass dependency affect the execution of `image layout transitions` within the render pass.

For non-attachment resources, the memory dependency expressed by subpass dependency is nearly identical to that of a `VkMemoryBarrier` (with matching `srcAccessMask` and `dstAccessMask` parameters) submitted as a part of a `vkCmdPipelineBarrier` (with matching `srcStageMask` and `dstStageMask` parameters). The only difference being that its scopes are limited to the identified subpasses rather than potentially affecting everything before and after.

NOTE

For attachments however, subpass dependencies work more like a `VkImageMemoryBarrier` defined similarly to the `VkMemoryBarrier` above, the queue family indices set to `VK_QUEUE_FAMILY_IGNORED`, and layouts as follows:

- The equivalent to `oldLayout` is the attachment's layout according to the subpass description for `srcSubpass`.
- The equivalent to `newLayout` is the attachment's layout according to the subpass description for `dstSubpass`.

Valid Usage

- VUID-VkSubpassDependency-srcStageMask-04090
If the `geometryShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-VkSubpassDependency-srcStageMask-04091
If the `tessellationShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-VkSubpassDependency-srcStageMask-04094
If the `transformFeedback` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-VkSubpassDependency-srcStageMask-07319
If the `attachmentFragmentShadingRate` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-VkSubpassDependency-srcStageMask-03937
If the `synchronization2` feature is not enabled, `srcStageMask` **must** not be `0`
- VUID-VkSubpassDependency-srcStageMask-07950
If the `rayTracingPipeline` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkSubpassDependency-dstStageMask-04090
If the `geometryShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-VkSubpassDependency-dstStageMask-04091
If the `tessellationShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-VkSubpassDependency-dstStageMask-04094
If the `transformFeedback` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-VkSubpassDependency-dstStageMask-07319
If the `attachmentFragmentShadingRate` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-VkSubpassDependency-dstStageMask-03937
If the `synchronization2` feature is not enabled, `dstStageMask` **must** not be `0`
- VUID-VkSubpassDependency-dstStageMask-07950
If the `rayTracingPipeline` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkSubpassDependency-srcSubpass-00864
`srcSubpass` **must** be less than or equal to `dstSubpass`, unless one of them is `VK_SUBPASS_EXTERNAL`, to avoid cyclic dependencies and ensure a valid execution order
- VUID-VkSubpassDependency-srcSubpass-00865

`srcSubpass` and `dstSubpass` **must** not both be equal to `VK_SUBPASS_EXTERNAL`

- VUID-VkSubpassDependency-srcSubpass-06809

If `srcSubpass` is equal to `dstSubpass` and `srcStageMask` includes a `framebuffer-space` stage, `dstStageMask` **must** only contain `framebuffer-space` stages

- VUID-VkSubpassDependency-srcAccessMask-00868

Any access flag included in `srcAccessMask` **must** be supported by one of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)

- VUID-VkSubpassDependency-dstAccessMask-00869

Any access flag included in `dstAccessMask` **must** be supported by one of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)

- VUID-VkSubpassDependency-srcSubpass-02243

If `srcSubpass` equals `dstSubpass`, and `srcStageMask` and `dstStageMask` both include a `framebuffer-space` stage, then `dependencyFlags` **must** include `VK_DEPENDENCY_BY_REGION_BIT`

- VUID-VkSubpassDependency-dependencyFlags-02520

If `dependencyFlags` includes `VK_DEPENDENCY_VIEW_LOCAL_BIT`, `srcSubpass` **must** not be equal to `VK_SUBPASS_EXTERNAL`

- VUID-VkSubpassDependency-dependencyFlags-02521

If `dependencyFlags` includes `VK_DEPENDENCY_VIEW_LOCAL_BIT`, `dstSubpass` **must** not be equal to `VK_SUBPASS_EXTERNAL`

- VUID-VkSubpassDependency-srcSubpass-00872

If `srcSubpass` equals `dstSubpass` and that subpass has more than one bit set in the view mask, then `dependencyFlags` **must** include `VK_DEPENDENCY_VIEW_LOCAL_BIT`

Valid Usage (Implicit)

- VUID-VkSubpassDependency-srcStageMask-parameter

`srcStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values

- VUID-VkSubpassDependency-dstStageMask-parameter

`dstStageMask` **must** be a valid combination of `VkPipelineStageFlagBits` values

- VUID-VkSubpassDependency-srcAccessMask-parameter

`srcAccessMask` **must** be a valid combination of `VkAccessFlagBits` values

- VUID-VkSubpassDependency-dstAccessMask-parameter

`dstAccessMask` **must** be a valid combination of `VkAccessFlagBits` values

- VUID-VkSubpassDependency-dependencyFlags-parameter

`dependencyFlags` **must** be a valid combination of `VkDependencyFlagBits` values

When multiview is enabled, the execution of the multiple views of one subpass **may** not occur simultaneously or even back-to-back, and rather **may** be interleaved with the execution of other subpasses. The load and store operations apply to attachments on a per-view basis. For example, an attachment using `VK_ATTACHMENT_LOAD_OP_CLEAR` will have each view cleared on first use, but the first use of one view may be temporally distant from the first use of another view.

NOTE

A good mental model for multiview is to think of a multiview subpass as if it were a collection of individual (per-view) subpasses that are logically grouped together and described as a single multiview subpass in the API. Similarly, a multiview attachment can be thought of like several individual attachments that happen to be layers in a single image. A view-local dependency between two multiview subpasses acts like a set of one-to-one dependencies between corresponding pairs of per-view subpasses. A view-global dependency between two multiview subpasses acts like a set of $N \times M$ dependencies between all pairs of per-view subpasses in the source and destination. Thus, it is a more compact representation which also makes clear the commonality and reuse that is present between views in a subpass. This interpretation motivates the answers to questions like “when does the load op apply” - it is on the first use of each view of an attachment, as if each view was a separate attachment.

The content of each view follows the description in [attachment content behavior](#). In particular, if an attachment is preserved, all views within the attachment are preserved.

If any two subpasses of a render pass activate transform feedback to the same bound transform feedback buffers, a subpass dependency **must** be included (either directly or via some intermediate subpasses) between them.

If there is no subpass dependency from `VK_SUBPASS_EXTERNAL` to the first subpass that uses an attachment, then an implicit subpass dependency exists from `VK_SUBPASS_EXTERNAL` to the first subpass it is used in. The implicit subpass dependency only exists if there exists an automatic layout transition away from `initialLayout`. The subpass dependency operates as if defined with the following parameters:

```
VkSubpassDependency implicitDependency = {  
    .srcSubpass = VK_SUBPASS_EXTERNAL,  
    .dstSubpass = firstSubpass, // First subpass attachment is used in  
    .srcStageMask = VK_PIPELINE_STAGE_NONE,  
    .dstStageMask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,  
    .srcAccessMask = 0,  
    .dstAccessMask = VK_ACCESS_INPUT_ATTACHMENT_READ_BIT |  
                     VK_ACCESS_COLOR_ATTACHMENT_READ_BIT |  
                     VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT |  
                     VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT |  
                     VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT,  
    .dependencyFlags = 0  
};
```

Similarly, if there is no subpass dependency from the last subpass that uses an attachment to `VK_SUBPASS_EXTERNAL`, then an implicit subpass dependency exists from the last subpass it is used in to `VK_SUBPASS_EXTERNAL`. The implicit subpass dependency only exists if there exists an automatic layout transition into `finalLayout`. The subpass dependency operates as if defined with the following parameters:

```

VkSubpassDependency implicitDependency = {
    .srcSubpass = lastSubpass, // Last subpass attachment is used in
    .dstSubpass = VK_SUBPASS_EXTERNAL,
    .srcStageMask = VK_PIPELINE_STAGE_ALL_COMMANDS_BIT,
    .dstStageMask = VK_PIPELINE_STAGE_NONE,
    .srcAccessMask = VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT |
                    VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT,
    .dstAccessMask = 0,
    .dependencyFlags = 0
};

```

As subpasses **may** overlap or execute out of order with regards to other subpasses unless a subpass dependency chain describes otherwise, the layout transitions required between subpasses **cannot** be known to an application. Instead, an application provides the layout that each attachment **must** be in at the start and end of a render pass, and the layout it **must** be in during each subpass it is used in. The implementation then **must** execute layout transitions between subpasses in order to guarantee that the images are in the layouts required by each subpass, and in the final layout at the end of the render pass.

Automatic layout transitions apply to the entire image subresource attached to the framebuffer. If multiview is not enabled and the attachment is a view of a 1D or 2D image, the automatic layout transitions apply to the number of layers specified by [VkFramebufferCreateInfo::layers](#). If multiview is enabled and the attachment is a view of a 1D or 2D image, the automatic layout transitions apply to the layers corresponding to views which are used by some subpass in the render pass, even if that subpass does not reference the given attachment. If the attachment view is a 2D or 2D array view of a 3D image, even if the attachment view only refers to a subset of the slices of the selected mip level of the 3D image, automatic layout transitions apply to the entire subresource referenced which is the entire mip level in this case.

Automatic layout transitions away from the layout used in a subpass happen-after the availability operations for all dependencies with that subpass as the [srcSubpass](#).

Automatic layout transitions into the layout used in a subpass happen-before the visibility operations for all dependencies with that subpass as the [dstSubpass](#).

Automatic layout transitions away from [initialLayout](#) happen-after the availability operations for all dependencies with a [srcSubpass](#) equal to [VK_SUBPASS_EXTERNAL](#), where [dstSubpass](#) uses the attachment that will be transitioned. For attachments created with [VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT](#), automatic layout transitions away from [initialLayout](#) happen-after the availability operations for all dependencies with a [srcSubpass](#) equal to [VK_SUBPASS_EXTERNAL](#), where [dstSubpass](#) uses any aliased attachment.

Automatic layout transitions into [finalLayout](#) happen-before the visibility operations for all dependencies with a [dstSubpass](#) equal to [VK_SUBPASS_EXTERNAL](#), where [srcSubpass](#) uses the attachment that will be transitioned. For attachments created with [VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT](#), automatic layout transitions into [finalLayout](#) happen-before the visibility operations for all dependencies with a [dstSubpass](#) equal to [VK_SUBPASS_EXTERNAL](#), where [srcSubpass](#) uses any aliased attachment.

The image layout of the depth aspect of a depth/stencil attachment referring to an image created with `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` is dependent on the last sample locations used to render to the attachment, thus automatic layout transitions use the sample locations state specified in `VkRenderPassSampleLocationsBeginInfoEXT`.

Automatic layout transitions of an attachment referring to a depth/stencil image created with `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` use the sample locations the image subresource range referenced by the attachment was last rendered with. If the current render pass does not use the attachment as a depth/stencil attachment in any subpass that happens-before, the automatic layout transition uses the sample locations state specified in the `sampleLocationsInfo` member of the element of the `VkRenderPassSampleLocationsBeginInfoEXT::pAttachmentInitialSampleLocations` array for which the `attachmentIndex` member equals the attachment index of the attachment, if one is specified. Otherwise, the automatic layout transition uses the sample locations state specified in the `sampleLocationsInfo` member of the element of the `VkRenderPassSampleLocationsBeginInfoEXT::pPostSubpassSampleLocations` array for which the `subpassIndex` member equals the index of the subpass that last used the attachment as a depth/stencil attachment, if one is specified.

If no sample locations state has been specified for an automatic layout transition performed on an attachment referring to a depth/stencil image created with `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` the contents of the depth aspect of the depth/stencil attachment become undefined as if the layout of the attachment was transitioned from the `VK_IMAGE_LAYOUT_UNDEFINED` layout.

If two subpasses use the same attachment, and both subpasses use the attachment in a read-only layout, no subpass dependency needs to be specified between those subpasses. If an implementation treats those layouts separately, it **must** insert an implicit subpass dependency between those subpasses to separate the uses in each layout. The subpass dependency operates as if defined with the following parameters:

```
// Used for input attachments
VkPipelineStageFlags inputAttachmentStages = VK_PIPELINE_STAGE_FRAGMENT_SHADER_BIT;
VkAccessFlags inputAttachmentDstAccess = VK_ACCESS_INPUT_ATTACHMENT_READ_BIT;

// Used for depth/stencil attachments
VkPipelineStageFlags depthStencilAttachmentStages =
VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT |
VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT;
VkAccessFlags depthStencilAttachmentDstAccess =
VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT;

VkSubpassDependency implicitDependency = {
    .srcSubpass = firstSubpass;
    .dstSubpass = secondSubpass;
    .srcStageMask = inputAttachmentStages | depthStencilAttachmentStages;
    .dstStageMask = inputAttachmentStages | depthStencilAttachmentStages;
    .srcAccessMask = 0;
    .dstAccessMask = inputAttachmentDstAccess | depthStencilAttachmentDstAccess;
    .dependencyFlags = 0;
```



```
};
```

When drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, the application **must** specify which types of attachments that are written to during a render pass will also be accessed as non-attachments in the render pass.

To [dynamically set](#) whether a pipeline **can** access a resource as a non-attachment while it is also used as an attachment that is written to, call:

```
// Provided by VK_EXT_attachment_feedback_loop_dynamic_state
void vkCmdSetAttachmentFeedbackLoopEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkImageAspectFlags       aspectMask);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `aspectMask` specifies the types of attachments for which feedback loops will be enabled. Attachment types whose aspects are not included in `aspectMask` will have feedback loops disabled.

For attachments that are written to in a render pass, only attachments with the aspects specified in `aspectMask` **can** be accessed as non-attachments by subsequent [drawing commands](#).

Valid Usage

- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-attachmentFeedbackLoopDynamicState-08862
The `attachmentFeedbackLoopDynamicState` feature **must** be enabled
- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-aspectMask-08863
`aspectMask` **must** only include `VK_IMAGE_ASPECT_NONE`, `VK_IMAGE_ASPECT_COLOR_BIT`, `VK_IMAGE_ASPECT_DEPTH_BIT`, and `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-attachmentFeedbackLoopLayout-08864
If the `attachmentFeedbackLoopLayout` feature is not enabled, `aspectMask` **must** be `VK_IMAGE_ASPECT_NONE`

Valid Usage (Implicit)

- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-aspectMask-parameter
`aspectMask` **must** be a valid combination of `VkImageAspectFlagBits` values
- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)

- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetAttachmentFeedbackLoopEnableEXT-vidcoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

A more extensible version of render pass creation is also defined below.

To create a render pass, call:

```
// Provided by VK_KHR_create_renderpass2
VkResult vkCreateRenderPass2KHR(
    VkDevice device,
    const VkRenderPassCreateInfo2* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkRenderPass* pRenderPass);
```

- `device` is the logical device that creates the render pass.
- `pCreateInfo` is a pointer to a `VkRenderPassCreateInfo2` structure describing the parameters of the render pass.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pRenderPass` is a pointer to a `VkRenderPass` handle in which the resulting render pass object is returned.

This command is functionally identical to `vkCreateRenderPass`, but includes extensible sub-structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage

- VUID-vkCreateRenderPass2-device-10001
device must support at least one queue family with the `VK_QUEUE_GRAPHICS_BIT` capability

Valid Usage (Implicit)

- VUID-vkCreateRenderPass2-device-parameter
device must be a valid `VkDevice` handle
- VUID-vkCreateRenderPass2-pCreateInfo-parameter
pCreateInfo must be a valid pointer to a valid `VkRenderPassCreateInfo2` structure
- VUID-vkCreateRenderPass2-pAllocator-parameter
If **pAllocator** is not `NULL`, **pAllocator must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateRenderPass2-pRenderPass-parameter
pRenderPass must be a valid pointer to a `VkRenderPass` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkRenderPassCreateInfo2` structure is defined as:

```
typedef struct VkRenderPassCreateInfo2 {
    VkStructureType    sType;
    const void*        pNext;
    VkRenderPassCreateFlags flags;
    uint32_t            attachmentCount;
    const VkAttachmentDescription2* pAttachments;
    uint32_t            subpassCount;
    const VkSubpassDescription2* pSubpasses;
    uint32_t            dependencyCount;
    const VkSubpassDependency2* pDependencies;
    uint32_t            correlatedViewMaskCount;
    const uint32_t*      pCorrelatedViewMasks;
} VkRenderPassCreateInfo2;
```

or the equivalent

```
// Provided by VK_KHR_create_renderpass2
typedef VkRenderPassCreateInfo2 VkRenderPassCreateInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `attachmentCount` is the number of attachments used by this render pass.
- `pAttachments` is a pointer to an array of `attachmentCount` `VkAttachmentDescription2` structures describing the attachments used by the render pass.
- `subpassCount` is the number of subpasses to create.
- `pSubpasses` is a pointer to an array of `subpassCount` `VkSubpassDescription2` structures describing each subpass.
- `dependencyCount` is the number of dependencies between pairs of subpasses.
- `pDependencies` is a pointer to an array of `dependencyCount` `VkSubpassDependency2` structures describing dependencies between pairs of subpasses.
- `correlatedViewMaskCount` is the number of correlation masks.
- `pCorrelatedViewMasks` is a pointer to an array of view masks indicating sets of views that **may** be more efficient to render concurrently.

Parameters defined by this structure with the same name as those in `VkRenderPassCreateInfo` have the identical effect to those parameters; the child structures are variants of those used in `VkRenderPassCreateInfo` which add `sType` and `pNext` parameters, allowing them to be extended.

If the `VkSubpassDescription2::viewMask` member of any element of `pSubpasses` is not zero, *multiview* functionality is considered to be enabled for this render pass.

`correlatedViewMaskCount` and `pCorrelatedViewMasks` have the same effect as `VkRenderPassMultiviewCreateInfo::correlationMaskCount` and `VkRenderPassMultiviewCreateInfo::pCorrelationMasks`, respectively.

Valid Usage

- VUID-VkRenderPassCreateInfo2-None-03049
If any two subpasses operate on attachments with overlapping ranges of the same `VkDeviceMemory` object, and at least one subpass writes to that area of `VkDeviceMemory`, a subpass dependency **must** be included (either directly or via some intermediate subpasses) between them
- VUID-VkRenderPassCreateInfo2-attachment-03050
If the `attachment` member of any element of `pInputAttachments`, `pColorAttachments`, `pResolveAttachments` or `pDepthStencilAttachment`, or the attachment indexed by any element of `pPreserveAttachments` in any element of `pSubpasses` is bound to a range of a `VkDeviceMemory` object that overlaps with any other attachment in any subpass (including the same subpass), the `VkAttachmentDescription2` structures describing them **must** include

VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT in flags

- VUID-VkRenderPassCreateInfo2-attachment-03051

If the `attachment` member of any element of `pInputAttachments`, `pColorAttachments`, `pResolveAttachments` or `pDepthStencilAttachment`, or any element of `pPreserveAttachments` in any element of `pSubpasses` is not `VK_ATTACHMENT_UNUSED`, then it **must** be less than `attachmentCount`

- VUID-VkRenderPassCreateInfo2-pSubpasses-06473

If the `pSubpasses` `pNext` chain includes a `VkSubpassDescriptionDepthStencilResolve` structure and the `pDepthStencilResolveAttachment` member is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED`, then `attachment` **must** be less than `attachmentCount`

- VUID-VkRenderPassCreateInfo2-pAttachments-02522

For any member of `pAttachments` with a `loadOp` equal to `VK_ATTACHMENT_LOAD_OP_CLEAR`, the first use of that attachment **must** not specify a `layout` equal to `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`,
`VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, or
`VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`

- VUID-VkRenderPassCreateInfo2-pAttachments-02523

For any member of `pAttachments` with a `stencilLoadOp` equal to `VK_ATTACHMENT_LOAD_OP_CLEAR`, the first use of that attachment **must** not specify a `layout` equal to `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`,
`VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, or
`VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`

- VUID-VkRenderPassCreateInfo2-pDependencies-03054

For any element of `pDependencies`, if the `srcSubpass` is not `VK_SUBPASS_EXTERNAL`, all stage flags included in the `srcStageMask` member of that dependency **must** be a pipeline stage supported by the `pipeline` identified by the `pipelineBindPoint` member of the source subpass

- VUID-VkRenderPassCreateInfo2-pDependencies-03055

For any element of `pDependencies`, if the `dstSubpass` is not `VK_SUBPASS_EXTERNAL`, all stage flags included in the `dstStageMask` member of that dependency **must** be a pipeline stage supported by the `pipeline` identified by the `pipelineBindPoint` member of the destination subpass

- VUID-VkRenderPassCreateInfo2-pCorrelatedViewMasks-03056

The set of bits included in any element of `pCorrelatedViewMasks` **must** not overlap with the set of bits included in any other element of `pCorrelatedViewMasks`

- VUID-VkRenderPassCreateInfo2-viewMask-03057

If the `VkSubpassDescription2::viewMask` member of all elements of `pSubpasses` is `0`, `correlatedViewMaskCount` **must** be `0`

- VUID-VkRenderPassCreateInfo2-viewMask-03058

The `VkSubpassDescription2::viewMask` member of all elements of `pSubpasses` **must** either all be `0`, or all not be `0`

- VUID-VkRenderPassCreateInfo2-viewMask-03059

If the `VkSubpassDescription2::viewMask` member of all elements of `pSubpasses` is `0`, the `dependencyFlags` member of any element of `pDependencies` **must** not include

VK_DEPENDENCY_VIEW_LOCAL_BIT

- VUID-VkRenderPassCreateInfo2-pDependencies-03060

For any element of `pDependencies` where its `srcSubpass` member equals its `dstSubpass` member, if the `viewMask` member of the corresponding element of `pSubpasses` includes more than one bit, its `dependencyFlags` member **must** include `VK_DEPENDENCY_VIEW_LOCAL_BIT`

- VUID-VkRenderPassCreateInfo2-attachment-02525

If the `attachment` member of any element of the `pInputAttachments` member of any element of `pSubpasses` is not `VK_ATTACHMENT_UNUSED`, the `aspectMask` member of that element of `pInputAttachments` **must** only include aspects that are present in images of the format specified by the element of `pAttachments` specified by `attachment`

- VUID-VkRenderPassCreateInfo2-srcSubpass-02526

The `srcSubpass` member of each element of `pDependencies` **must** be less than `subpassCount`

- VUID-VkRenderPassCreateInfo2-dstSubpass-02527

The `dstSubpass` member of each element of `pDependencies` **must** be less than `subpassCount`

- VUID-VkRenderPassCreateInfo2-pAttachments-04585

If any element of `pAttachments` is used as a fragment shading rate attachment in any subpass, it **must** not be used as any other attachment in the render pass

- VUID-VkRenderPassCreateInfo2-pAttachments-09387

If any element of `pAttachments` is used as a fragment shading rate attachment, the `loadOp` for that attachment **must** not be `VK_ATTACHMENT_LOAD_OP_CLEAR`

- VUID-VkRenderPassCreateInfo2-pAttachments-04586

If any element of `pAttachments` is used as a fragment shading rate attachment in any subpass, it **must** have an image format whose `potential format features` contain `VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-VkRenderPassCreateInfo2-attachment-06244

If the `attachment` member of the `pDepthStencilAttachment` member of an element of `pSubpasses` is not `VK_ATTACHMENT_UNUSED`, the `layout` member of that same structure is either `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, and the `pNext` chain of that structure does not include a `VkAttachmentReferenceStencilLayout` structure, then the element of `pAttachments` with an index equal to `attachment` **must** not have a `format` that includes both depth and stencil components

- VUID-VkRenderPassCreateInfo2-attachment-06245

If the `attachment` member of the `pDepthStencilAttachment` member of an element of `pSubpasses` is not `VK_ATTACHMENT_UNUSED` and the `layout` member of that same structure is either `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`, then the element of `pAttachments` with an index equal to `attachment` **must** have a `format` that includes only a stencil component

- VUID-VkRenderPassCreateInfo2-attachment-06246

If the `attachment` member of the `pDepthStencilAttachment` member of an element of `pSubpasses` is not `VK_ATTACHMENT_UNUSED` and the `layout` member of that same structure is either `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, then the element of `pAttachments` with an index

equal to `attachment` **must** not have a `format` that includes only a stencil component

Valid Usage (Implicit)

- `VUID-VkRenderPassCreateInfo2-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO_2`
- `VUID-VkRenderPassCreateInfo2-pNext-pNext`
`pNext` **must** be `NULL`
- `VUID-VkRenderPassCreateInfo2-flags-zeroBitmask`
`flags` **must** be `0`
- `VUID-VkRenderPassCreateInfo2-pAttachments-parameter`
If `attachmentCount` is not `0`, `pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid `VkAttachmentDescription2` structures
- `VUID-VkRenderPassCreateInfo2-pSubpasses-parameter`
`pSubpasses` **must** be a valid pointer to an array of `subpassCount` valid `VkSubpassDescription2` structures
- `VUID-VkRenderPassCreateInfo2-pDependencies-parameter`
If `dependencyCount` is not `0`, `pDependencies` **must** be a valid pointer to an array of `dependencyCount` valid `VkSubpassDependency2` structures
- `VUID-VkRenderPassCreateInfo2-pCorrelatedViewMasks-parameter`
If `correlatedViewMaskCount` is not `0`, `pCorrelatedViewMasks` **must** be a valid pointer to an array of `correlatedViewMaskCount` `uint32_t` values
- `VUID-VkRenderPassCreateInfo2-subpassCount-arraylength`
`subpassCount` **must** be greater than `0`

The `VkAttachmentDescription2` structure is defined as:

```
typedef struct VkAttachmentDescription2 {
    VkStructureType      sType;
    const void*          pNext;
    VkAttachmentDescriptionFlags flags;
    VkFormat              format;
    VkSampleCountFlagBits samples;
    VkAttachmentLoadOp    loadOp;
    VkAttachmentStoreOp   storeOp;
    VkAttachmentLoadOp    stencilLoadOp;
    VkAttachmentStoreOp   stencilStoreOp;
    VkImageLayout         initialLayout;
    VkImageLayout         finalLayout;
} VkAttachmentDescription2;
```

or the equivalent


```
// Provided by VK_KHR_create_renderpass2
typedef VkAttachmentDescription2 VkAttachmentDescription2KHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkAttachmentDescriptionFlagBits](#) specifying additional properties of the attachment.
- **format** is a [VkFormat](#) value specifying the format of the image that will be used for the attachment.
- **samples** is a [VkSampleCountFlagBits](#) value specifying the number of samples of the image.
- **loadOp** is a [VkAttachmentLoadOp](#) value specifying how the contents of color and depth components of the attachment are treated at the beginning of the subpass where it is first used.
- **storeOp** is a [VkAttachmentStoreOp](#) value specifying how the contents of color and depth components of the attachment are treated at the end of the subpass where it is last used.
- **stencilLoadOp** is a [VkAttachmentLoadOp](#) value specifying how the contents of stencil components of the attachment are treated at the beginning of the subpass where it is first used.
- **stencilStoreOp** is a [VkAttachmentStoreOp](#) value specifying how the contents of stencil components of the attachment are treated at the end of the last subpass where it is used.
- **initialLayout** is the layout the attachment image subresource will be in when a render pass instance begins.
- **finalLayout** is the layout the attachment image subresource will be transitioned to when a render pass instance ends.

Parameters defined by this structure with the same name as those in [VkAttachmentDescription](#) have the identical effect to those parameters.

If the [separateDepthStencilLayouts](#) feature is enabled, and **format** is a depth/stencil format, **initialLayout** and **finalLayout** **can** be set to a layout that only specifies the layout of the depth aspect.

If the **pNext** chain includes a [VkAttachmentDescriptionStencilLayout](#) structure, then the **stencilInitialLayout** and **stencilFinalLayout** members specify the initial and final layouts of the stencil aspect of a depth/stencil format, and **initialLayout** and **finalLayout** only apply to the depth aspect. For depth-only formats, the [VkAttachmentDescriptionStencilLayout](#) structure is ignored. For stencil-only formats, the initial and final layouts of the stencil aspect are taken from the [VkAttachmentDescriptionStencilLayout](#) structure if present, or **initialLayout** and **finalLayout** if not present.

If **format** is a depth/stencil format, and either **initialLayout** or **finalLayout** does not specify a layout for the stencil aspect, then the application **must** specify the initial and final layouts of the stencil aspect by including a [VkAttachmentDescriptionStencilLayout](#) structure in the **pNext** chain.

loadOp and **storeOp** are ignored for fragment shading rate attachments. No access to the shading rate attachment is performed in **loadOp** and **storeOp**. Instead, access to

VK_ACCESS_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR is performed as fragments are rasterized.

Valid Usage

- VUID-VkAttachmentDescription2-format-06699
If `format` includes a color or depth component and `loadOp` is `VK_ATTACHMENT_LOAD_OP_LOAD`, then `initialLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`
- VUID-VkAttachmentDescription2-finalLayout-00843
`finalLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`
- VUID-VkAttachmentDescription2-format-03280
If `format` is a color format, `initialLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkAttachmentDescription2-format-03281
If `format` is a depth/stencil format, `initialLayout` **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL`
- VUID-VkAttachmentDescription2-format-03282
If `format` is a color format, `finalLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkAttachmentDescription2-format-03283
If `format` is a depth/stencil format, `finalLayout` **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL`
- VUID-VkAttachmentDescription2-format-06487
If `format` is a color format, `initialLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`
- VUID-VkAttachmentDescription2-format-06488
If `format` is a color format, `finalLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`
- VUID-VkAttachmentDescription2-separateDepthStencilLayouts-03284
If the `separateDepthStencilLayouts` feature is not enabled, `initialLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`,
- VUID-VkAttachmentDescription2-separateDepthStencilLayouts-03285
If the `separateDepthStencilLayouts` feature is not enabled, `finalLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`,
- VUID-VkAttachmentDescription2-format-03286
If `format` is a color format, `initialLayout` **must** not be

VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL,
VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription2-format-03287

If `format` is a color format, `finalLayout` **must** not be
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL,
VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL, or
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription2-format-06906

If `format` is a depth/stencil format which includes both depth and stencil components,
`initialLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription2-format-06907

If `format` is a depth/stencil format which includes both depth and stencil components,
`finalLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription2-format-03290

If `format` is a depth/stencil format which includes only the depth component,
`initialLayout` **must** not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription2-format-03291

If `format` is a depth/stencil format which includes only the depth component, `finalLayout`
must not be VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL or
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL

- VUID-VkAttachmentDescription2-synchronization2-06908

If the `synchronization2` feature is not enabled, `initialLayout` **must** not be
VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR

- VUID-VkAttachmentDescription2-synchronization2-06909

If the `synchronization2` feature is not enabled, `finalLayout` **must** not be
VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR or VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR

- VUID-VkAttachmentDescription2-attachmentFeedbackLoopLayout-07309

If the `attachmentFeedbackLoopLayout` feature is not enabled, `initialLayout` **must** not be
VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT

- VUID-VkAttachmentDescription2-attachmentFeedbackLoopLayout-07310

If the `attachmentFeedbackLoopLayout` feature is not enabled, `finalLayout` **must** not be
VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT

- VUID-VkAttachmentDescription2-samples-08745

`samples` **must** be a valid [VkSampleCountFlagBits](#) value that is set in
`imageCreateSampleCounts` (as defined in [Image Creation Limits](#)) for the given `format`

- VUID-VkAttachmentDescription2-dynamicRenderingLocalRead-09544

If the `dynamicRenderingLocalRead` feature is not enabled, `initialLayout` **must** not be
VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR

- VUID-VkAttachmentDescription2-dynamicRenderingLocalRead-09545

If the `dynamicRenderingLocalRead` feature is not enabled, `finalLayout` **must** not be `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR`

- VUID-VkAttachmentDescription2-pNext-06704

If the `pNext` chain does not include a `VkAttachmentDescriptionStencilLayout` structure, `format` includes a stencil component, and `stencilLoadOp` is `VK_ATTACHMENT_LOAD_OP_LOAD`, then `initialLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`

- VUID-VkAttachmentDescription2-pNext-06705

If the `pNext` chain includes a `VkAttachmentDescriptionStencilLayout` structure, `format` includes a stencil component, and `stencilLoadOp` is `VK_ATTACHMENT_LOAD_OP_LOAD`, then `VkAttachmentDescriptionStencilLayout::stencilInitialLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`

- VUID-VkAttachmentDescription2-format-06249

If `format` is a depth/stencil format which includes both depth and stencil components, and `initialLayout` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, the `pNext` chain **must** include a `VkAttachmentDescriptionStencilLayout` structure

- VUID-VkAttachmentDescription2-format-06250

If `format` is a depth/stencil format which includes both depth and stencil components, and `finalLayout` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, the `pNext` chain **must** include a `VkAttachmentDescriptionStencilLayout` structure

- VUID-VkAttachmentDescription2-format-06247

If the `pNext` chain does not include a `VkAttachmentDescriptionStencilLayout` structure and `format` only includes a stencil component, `initialLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`

- VUID-VkAttachmentDescription2-format-06248

If the `pNext` chain does not include a `VkAttachmentDescriptionStencilLayout` structure and `format` only includes a stencil component, `finalLayout` **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`

- VUID-VkAttachmentDescription2-format-09332

`format` **must** not be `VK_FORMAT_UNDEFINED`

Valid Usage (Implicit)

- VUID-VkAttachmentDescription2-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_2`

- VUID-VkAttachmentDescription2-pNext-pNext

`pNext` **must** be `NULL` or a pointer to a valid instance of `VkAttachmentDescriptionStencilLayout`

- VUID-VkAttachmentDescription2-sType-unique

The `sType` value of each struct in the `pNext` chain **must** be unique

- VUID-VkAttachmentDescription2-flags-parameter

`flags` **must** be a valid combination of `VkAttachmentDescriptionFlagBits` values

- VUID-VkAttachmentDescription2-format-parameter **format** **must** be a valid [VkFormat](#) value
- VUID-VkAttachmentDescription2-samples-parameter **samples** **must** be a valid [VkSampleCountFlagBits](#) value
- VUID-VkAttachmentDescription2-loadOp-parameter **loadOp** **must** be a valid [VkAttachmentLoadOp](#) value
- VUID-VkAttachmentDescription2-storeOp-parameter **storeOp** **must** be a valid [VkAttachmentStoreOp](#) value
- VUID-VkAttachmentDescription2-stencilLoadOp-parameter **stencilLoadOp** **must** be a valid [VkAttachmentLoadOp](#) value
- VUID-VkAttachmentDescription2-stencilStoreOp-parameter **stencilStoreOp** **must** be a valid [VkAttachmentStoreOp](#) value
- VUID-VkAttachmentDescription2-initialLayout-parameter **initialLayout** **must** be a valid [VkImageLayout](#) value
- VUID-VkAttachmentDescription2-finalLayout-parameter **finalLayout** **must** be a valid [VkImageLayout](#) value

The [VkAttachmentDescriptionStencilLayout](#) structure is defined as:

```
typedef struct VkAttachmentDescriptionStencilLayout {
    VkStructureType    sType;
    void*              pNext;
    VkImageLayout       stencilInitialLayout;
    VkImageLayout       stencilFinalLayout;
} VkAttachmentDescriptionStencilLayout;
```

or the equivalent

```
// Provided by VK_KHR_separate_depth_stencil_layouts
typedef VkAttachmentDescriptionStencilLayout VkAttachmentDescriptionStencilLayoutKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **stencilInitialLayout** is the layout the stencil aspect of the attachment image subresource will be in when a render pass instance begins.
- **stencilFinalLayout** is the layout the stencil aspect of the attachment image subresource will be transitioned to when a render pass instance ends.

Valid Usage

- VUID-VkAttachmentDescriptionStencilLayout-stencilInitialLayout-03308
stencilInitialLayout **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL`,

VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL,
 VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL,
 VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL,
 VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL, or
 VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL

- VUID-VkAttachmentDescriptionStencilLayout-stencilFinalLayout-03309

stencilFinalLayout **must** not be VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL,
 VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL, VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL,
 VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL,
 VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL,
 VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL, or
 VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL

- VUID-VkAttachmentDescriptionStencilLayout-stencilFinalLayout-03310

stencilFinalLayout **must** not be VK_IMAGE_LAYOUT_UNDEFINED or
 VK_IMAGE_LAYOUT_PREINITIALIZED

Valid Usage (Implicit)

- VUID-VkAttachmentDescriptionStencilLayout-sType-sType
 sType **must** be VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_STENCIL_LAYOUT
- VUID-VkAttachmentDescriptionStencilLayout-stencilInitialLayout-parameter
 stencilInitialLayout **must** be a valid [VkImageLayout](#) value
- VUID-VkAttachmentDescriptionStencilLayout-stencilFinalLayout-parameter
 stencilFinalLayout **must** be a valid [VkImageLayout](#) value

The [VkSubpassDescription2](#) structure is defined as:

```
typedef struct VkSubpassDescription2 {
    VkStructureType           sType;
    const void*               pNext;
    VkSubpassDescriptionFlags flags;
    VkPipelineBindPoint       pipelineBindPoint;
    uint32_t                  viewMask;
    uint32_t                  inputAttachmentCount;
    const VkAttachmentReference2* pInputAttachments;
    uint32_t                  colorAttachmentCount;
    const VkAttachmentReference2* pColorAttachments;
    const VkAttachmentReference2* pResolveAttachments;
    const VkAttachmentReference2* pDepthStencilAttachment;
    uint32_t                  preserveAttachmentCount;
    const uint32_t*           pPreserveAttachments;
} VkSubpassDescription2;
```

or the equivalent

```
// Provided by VK_KHR_create_renderpass2
typedef VkSubpassDescription2 VkSubpassDescription2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkSubpassDescriptionFlagBits` specifying usage of the subpass.
- `pipelineBindPoint` is a `VkPipelineBindPoint` value specifying the pipeline type supported for this subpass.
- `viewMask` is a bitfield of view indices describing which views rendering is broadcast to in this subpass, when multiview is enabled.
- `inputAttachmentCount` is the number of input attachments.
- `pInputAttachments` is a pointer to an array of `VkAttachmentReference2` structures defining the input attachments for this subpass and their layouts.
- `colorAttachmentCount` is the number of color attachments.
- `pColorAttachments` is a pointer to an array of `colorAttachmentCount` `VkAttachmentReference2` structures defining the color attachments for this subpass and their layouts.
- `pResolveAttachments` is `NULL` or a pointer to an array of `colorAttachmentCount` `VkAttachmentReference2` structures defining the resolve attachments for this subpass and their layouts.
- `pDepthStencilAttachment` is a pointer to a `VkAttachmentReference2` structure specifying the depth/stencil attachment for this subpass and its layout.
- `preserveAttachmentCount` is the number of preserved attachments.
- `pPreserveAttachments` is a pointer to an array of `preserveAttachmentCount` render pass attachment indices identifying attachments that are not used by this subpass, but whose contents **must** be preserved throughout the subpass.

Parameters defined by this structure with the same name as those in `VkSubpassDescription` have the identical effect to those parameters.

`viewMask` has the same effect for the described subpass as `VkRenderPassMultiviewCreateInfo::pViewMasks` has on each corresponding subpass.

If a `VkFragmentShadingRateAttachmentInfoKHR` structure is included in the `pNext` chain, `pFragmentShadingRateAttachment` is not `NULL`, and its `attachment` member is not `VK_ATTACHMENT_UNUSED`, the identified attachment defines a fragment shading rate attachment for that subpass.

Valid Usage

- VUID-VkSubpassDescription2-attachment-06912

If the `attachment` member of an element of `pInputAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL`

- VUID-VkSubpassDescription2-attachment-06913
If the `attachment` member of an element of `pColorAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- VUID-VkSubpassDescription2-attachment-06914
If the `attachment` member of an element of `pResolveAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- VUID-VkSubpassDescription2-attachment-06915
If the `attachment` member of `pDepthStencilAttachment` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- VUID-VkSubpassDescription2-attachment-06916
If the `attachment` member of an element of `pColorAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`
- VUID-VkSubpassDescription2-attachment-06917
If the `attachment` member of an element of `pResolveAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL` or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`
- VUID-VkSubpassDescription2-attachment-06918
If the `attachment` member of an element of `pInputAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`
- VUID-VkSubpassDescription2-attachment-06919
If the `attachment` member of an element of `pColorAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkSubpassDescription2-attachment-06920
If the `attachment` member of an element of `pResolveAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkSubpassDescription2-attachment-06921
If the `attachment` member of an element of `pInputAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR`
- VUID-VkSubpassDescription2-attachment-06922
If the `attachment` member of an element of `pColorAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`

- VUID-VkSubpassDescription2-attachment-06923
If the `attachment` member of an element of `pResolveAttachments` is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** not be `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`
- VUID-VkSubpassDescription2-attachment-06251
If the `attachment` member of `pDepthStencilAttachment` is not `VK_ATTACHMENT_UNUSED` and its `pNext` chain includes a `VkAttachmentReferenceStencilLayout` structure, the `layout` member of `pDepthStencilAttachment` **must** not be `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL` or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`
- VUID-VkSubpassDescription2-pipelineBindPoint-04953
`pipelineBindPoint` **must** be `VK_PIPELINE_BIND_POINT_GRAPHICS`
- VUID-VkSubpassDescription2-colorAttachmentCount-03063
`colorAttachmentCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxColorAttachments`
- VUID-VkSubpassDescription2-loadOp-03064
If the first use of an attachment in this render pass is as an input attachment, and the attachment is not also used as a color or depth/stencil attachment in the same subpass, then `loadOp` **must** not be `VK_ATTACHMENT_LOAD_OP_CLEAR`
- VUID-VkSubpassDescription2-pResolveAttachments-03065
If `pResolveAttachments` is not `NULL`, for each resolve attachment that does not have the value `VK_ATTACHMENT_UNUSED`, the corresponding color attachment **must** not have the value `VK_ATTACHMENT_UNUSED`
- VUID-VkSubpassDescription2-pResolveAttachments-03066
If `pResolveAttachments` is not `NULL`, for each resolve attachment that is not `VK_ATTACHMENT_UNUSED`, the corresponding color attachment **must** not have a sample count of `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkSubpassDescription2-pResolveAttachments-03068
Each element of `pResolveAttachments` **must** have the same `VkFormat` as its corresponding color attachment
- VUID-VkSubpassDescription2-pResolveAttachments-03067
If `pResolveAttachments` is not `NULL`, each resolve attachment that is not `VK_ATTACHMENT_UNUSED` **must** have a sample count of `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkSubpassDescription2-pInputAttachments-02897
All attachments in `pInputAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have image formats whose `potential format features` contain at least `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT` or `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkSubpassDescription2-pColorAttachments-02898
All attachments in `pColorAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have image formats whose `potential format features` contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT`
- VUID-VkSubpassDescription2-pResolveAttachments-02899
All attachments in `pResolveAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have

image formats whose potential format features contain
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT

- VUID-VkSubpassDescription2-pDepthStencilAttachment-02900
If `pDepthStencilAttachment` is not `NULL` and the attachment is not `VK_ATTACHMENT_UNUSED` then it **must** have an image format whose potential format features contain `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkSubpassDescription2-multisampledRenderToSingleSampled-06872
All attachments in `pDepthStencilAttachment` or `pColorAttachments` that are not `VK_ATTACHMENT_UNUSED` **must** have the same sample count
- VUID-VkSubpassDescription2-attachment-03073
Each element of `pPreserveAttachments` **must** not be `VK_ATTACHMENT_UNUSED`
- VUID-VkSubpassDescription2-pPreserveAttachments-03074
Each element of `pPreserveAttachments` **must** not also be an element of any other member of the subpass description
- VUID-VkSubpassDescription2-layout-02528
If any attachment is used by more than one `VkAttachmentReference2` member, then each use **must** use the same layout
- VUID-VkSubpassDescription2-attachment-02799
If the `attachment` member of any element of `pInputAttachments` is not `VK_ATTACHMENT_UNUSED`, then the `aspectMask` member **must** be a valid combination of `VkImageAspectFlagBits`
- VUID-VkSubpassDescription2-attachment-02800
If the `attachment` member of any element of `pInputAttachments` is not `VK_ATTACHMENT_UNUSED`, then the `aspectMask` member **must** not be `0`
- VUID-VkSubpassDescription2-attachment-02801
If the `attachment` member of any element of `pInputAttachments` is not `VK_ATTACHMENT_UNUSED`, then the `aspectMask` member **must** not include `VK_IMAGE_ASPECT_METADATA_BIT`
- VUID-VkSubpassDescription2-attachment-04563
If the `attachment` member of any element of `pInputAttachments` is not `VK_ATTACHMENT_UNUSED`, then the `aspectMask` member **must** not include `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT` for any index `i`
- VUID-VkSubpassDescription2-pDepthStencilAttachment-04440
An attachment **must** not be used in both `pDepthStencilAttachment` and `pColorAttachments`
- VUID-VkSubpassDescription2-multiview-06558
If the `multiview` feature is not enabled, `viewMask` **must** be `0`
- VUID-VkSubpassDescription2-viewMask-06706
The index of the most significant bit in `viewMask` **must** be less than `maxMultiviewViewCount`

Valid Usage (Implicit)

- VUID-VkSubpassDescription2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_2`
- VUID-VkSubpassDescription2-pNext-pNext

Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either **NULL** or a pointer to a valid instance of **VkFragmentShadingRateAttachmentInfoKHR** or **VkSubpassDescriptionDepthStencilResolve**

- **VOID-VkSubpassDescription2-sType-unique**
The **sType** value of each struct in the **pNext** chain **must** be unique
- **VOID-VkSubpassDescription2-flags-zero bitmask**
flags **must** be 0
- **VOID-VkSubpassDescription2-pipelineBindPoint-parameter**
pipelineBindPoint **must** be a valid **VkPipelineBindPoint** value
- **VOID-VkSubpassDescription2-pInputAttachments-parameter**
If **inputAttachmentCount** is not 0, **pInputAttachments** **must** be a valid pointer to an array of **inputAttachmentCount** valid **VkAttachmentReference2** structures
- **VOID-VkSubpassDescription2-pColorAttachments-parameter**
If **colorAttachmentCount** is not 0, **pColorAttachments** **must** be a valid pointer to an array of **colorAttachmentCount** valid **VkAttachmentReference2** structures
- **VOID-VkSubpassDescription2-pResolveAttachments-parameter**
If **colorAttachmentCount** is not 0, and **pResolveAttachments** is not **NULL**, **pResolveAttachments** **must** be a valid pointer to an array of **colorAttachmentCount** valid **VkAttachmentReference2** structures
- **VOID-VkSubpassDescription2-pDepthStencilAttachment-parameter**
If **pDepthStencilAttachment** is not **NULL**, **pDepthStencilAttachment** **must** be a valid pointer to a valid **VkAttachmentReference2** structure
- **VOID-VkSubpassDescription2-pPreserveAttachments-parameter**
If **preserveAttachmentCount** is not 0, **pPreserveAttachments** **must** be a valid pointer to an array of **preserveAttachmentCount** **uint32_t** values

The **VkSubpassDescriptionDepthStencilResolve** structure is defined as:

```
typedef struct VkSubpassDescriptionDepthStencilResolve {  
    VkStructureType          sType;  
    const void*              pNext;  
    VkResolveModeFlagBits    depthResolveMode;  
    VkResolveModeFlagBits    stencilResolveMode;  
    const VkAttachmentReference2* pDepthStencilResolveAttachment;  
} VkSubpassDescriptionDepthStencilResolve;
```

or the equivalent

```
// Provided by VK_KHR_depth_stencil_resolve  
typedef VkSubpassDescriptionDepthStencilResolve  
VkSubpassDescriptionDepthStencilResolveKHR;
```

- **sType** is a **VkStructureType** value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthResolveMode` is a `VkResolveModeFlagBits` value describing the depth resolve mode.
- `stencilResolveMode` is a `VkResolveModeFlagBits` value describing the stencil resolve mode.
- `pDepthStencilResolveAttachment` is `NULL` or a pointer to a `VkAttachmentReference2` structure defining the depth/stencil resolve attachment for this subpass and its layout.

If the `pNext` chain of `VkSubpassDescription2` includes a `VkSubpassDescriptionDepthStencilResolve` structure, then that structure describes `multisample resolve operations` for the depth/stencil attachment in a subpass. If this structure is not included in the `pNext` chain of `VkSubpassDescription2`, or if it is and either `pDepthStencilResolveAttachment` is `NULL` or its attachment index is `VK_ATTACHMENT_UNUSED`, it indicates that no depth/stencil resolve attachment will be used in the subpass.

Valid Usage

- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03177
If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED`, `pDepthStencilAttachment` **must** not be `NULL` or have the value `VK_ATTACHMENT_UNUSED`
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03179
If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED`, `pDepthStencilAttachment` **must** not have a sample count of `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03180
If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED`, `pDepthStencilResolveAttachment` **must** have a sample count of `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-02651
If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED` then it **must** have an image format whose `potential format features` contain `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03181
If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED` and `VkFormat` of `pDepthStencilResolveAttachment` has a depth component, then the `VkFormat` of `pDepthStencilAttachment` **must** have a depth component with the same number of bits and `numeric format`
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03182
If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED`, and `VkFormat` of `pDepthStencilResolveAttachment` has a stencil component, then the `VkFormat` of `pDepthStencilAttachment` **must** have a stencil component with the same number of bits and `numeric format`
- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03178
If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED`, `depthResolveMode` and `stencilResolveMode` **must** not both be

VK_RESOLVE_MODE_NONE

- VUID-VkSubpassDescriptionDepthStencilResolve-depthResolveMode-03183

If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED` and the `VkFormat` of `pDepthStencilResolveAttachment` has a depth component, then the value of `depthResolveMode` **must** be one of the bits set in `VkPhysicalDeviceDepthStencilResolveProperties::supportedDepthResolveModes` or `VK_RESOLVE_MODE_NONE`

- VUID-VkSubpassDescriptionDepthStencilResolve-stencilResolveMode-03184

If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED` and the `VkFormat` of `pDepthStencilResolveAttachment` has a stencil component, then the value of `stencilResolveMode` **must** be one of the bits set in `VkPhysicalDeviceDepthStencilResolveProperties::supportedStencilResolveModes` or `VK_RESOLVE_MODE_NONE`

- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03185

If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED`, the `VkFormat` of `pDepthStencilResolveAttachment` has both depth and stencil components, `VkPhysicalDeviceDepthStencilResolveProperties::independentResolve` is `VK_FALSE`, and `VkPhysicalDeviceDepthStencilResolveProperties::independentResolveNone` is `VK_FALSE`, then the values of `depthResolveMode` and `stencilResolveMode` **must** be identical

- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-03186

If `pDepthStencilResolveAttachment` is not `NULL` and does not have the value `VK_ATTACHMENT_UNUSED`, the `VkFormat` of `pDepthStencilResolveAttachment` has both depth and stencil components, `VkPhysicalDeviceDepthStencilResolveProperties::independentResolve` is `VK_FALSE` and `VkPhysicalDeviceDepthStencilResolveProperties::independentResolveNone` is `VK_TRUE`, then the values of `depthResolveMode` and `stencilResolveMode` **must** be identical or one of them **must** be `VK_RESOLVE_MODE_NONE`

Valid Usage (Implicit)

- VUID-VkSubpassDescriptionDepthStencilResolve-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_DEPTH_STENCIL_RESOLVE`

- VUID-VkSubpassDescriptionDepthStencilResolve-pDepthStencilResolveAttachment-parameter

If `pDepthStencilResolveAttachment` is not `NULL`, `pDepthStencilResolveAttachment` **must** be a valid pointer to a valid `VkAttachmentReference2` structure

The `VkFragmentShadingRateAttachmentInfoKHR` structure is defined as:


```
// Provided by VK_KHR_fragment_shading_rate
typedef struct VkFragmentShadingRateAttachmentInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    const VkAttachmentReference2* pFragmentShadingRateAttachment;
    VkExtent2D           shadingRateAttachmentTexelSize;
} VkFragmentShadingRateAttachmentInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pFragmentShadingRateAttachment` is `NULL` or a pointer to a `VkAttachmentReference2` structure defining the fragment shading rate attachment for this subpass.
- `shadingRateAttachmentTexelSize` specifies the size of the portion of the framebuffer corresponding to each texel in `pFragmentShadingRateAttachment`.

If no shading rate attachment is specified, or if this structure is not specified, the implementation behaves as if a valid shading rate attachment was specified with all texels specifying a single pixel per fragment.

Valid Usage

- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04524
If `pFragmentShadingRateAttachment` is not `NULL` and its `attachment` member is not `VK_ATTACHMENT_UNUSED`, its `layout` member **must** be equal to `VK_IMAGE_LAYOUT_GENERAL` or `VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR`
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04525
If `pFragmentShadingRateAttachment` is not `NULL` and its `attachment` member is not `VK_ATTACHMENT_UNUSED`, `shadingRateAttachmentTexelSize.width` **must** be a power of two value
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04526
If `pFragmentShadingRateAttachment` is not `NULL` and its `attachment` member is not `VK_ATTACHMENT_UNUSED`, `shadingRateAttachmentTexelSize.width` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSize.width`
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04527
If `pFragmentShadingRateAttachment` is not `NULL` and its `attachment` member is not `VK_ATTACHMENT_UNUSED`, `shadingRateAttachmentTexelSize.width` **must** be greater than or equal to `minFragmentShadingRateAttachmentTexelSize.width`
- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04528
If `pFragmentShadingRateAttachment` is not `NULL` and its `attachment` member is not

`VK_ATTACHMENT_UNUSED`, `shadingRateAttachmentTexelSize.height` **must** be a power of two value

- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04529

If `pFragmentShadingRateAttachment` is not `NULL` and its `attachment` member is not `VK_ATTACHMENT_UNUSED`, `shadingRateAttachmentTexelSize.height` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSize.height`

- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04530

If `pFragmentShadingRateAttachment` is not `NULL` and its `attachment` member is not `VK_ATTACHMENT_UNUSED`, `shadingRateAttachmentTexelSize.height` **must** be greater than or equal to `minFragmentShadingRateAttachmentTexelSize.height`

- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04531

If `pFragmentShadingRateAttachment` is not `NULL` and its `attachment` member is not `VK_ATTACHMENT_UNUSED`, the quotient of `shadingRateAttachmentTexelSize.width` and `shadingRateAttachmentTexelSize.height` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSizeAspectRatio`

- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-04532

If `pFragmentShadingRateAttachment` is not `NULL` and its `attachment` member is not `VK_ATTACHMENT_UNUSED`, the quotient of `shadingRateAttachmentTexelSize.height` and `shadingRateAttachmentTexelSize.width` **must** be less than or equal to `maxFragmentShadingRateAttachmentTexelSizeAspectRatio`

Valid Usage (Implicit)

- VUID-VkFragmentShadingRateAttachmentInfoKHR-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_KHR`

- VUID-VkFragmentShadingRateAttachmentInfoKHR-pFragmentShadingRateAttachment-parameter

If `pFragmentShadingRateAttachment` is not `NULL`, `pFragmentShadingRateAttachment` **must** be a valid pointer to a valid [VkAttachmentReference2](#) structure

The [VkAttachmentReference2](#) structure is defined as:

```
typedef struct VkAttachmentReference2 {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           attachment;
    VkImageLayout       layout;
    VkImageAspectFlags  aspectMask;
} VkAttachmentReference2;
```

or the equivalent

```
// Provided by VK_KHR_create_renderpass2
typedef VkAttachmentReference2 VkAttachmentReference2KHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **attachment** is either an integer value identifying an attachment at the corresponding index in [VkRenderPassCreateInfo2::pAttachments](#), or `VK_ATTACHMENT_UNUSED` to signify that this attachment is not used.
- **layout** is a [VkImageLayout](#) value specifying the layout the attachment uses during the subpass.
- **aspectMask** is a mask of which aspect(s) **can** be accessed within the specified subpass as an input attachment.

Parameters defined by this structure with the same name as those in [VkAttachmentReference](#) have the identical effect to those parameters.

aspectMask is ignored when this structure is used to describe anything other than an input attachment reference.

If the [separateDepthStencilLayouts](#) feature is enabled, and **attachment** has a depth/stencil format, **layout** **can** be set to a layout that only specifies the layout of the depth aspect.

If **layout** only specifies the layout of the depth aspect of the attachment, the layout of the stencil aspect is specified by the **stencilLayout** member of a [VkAttachmentReferenceStencilLayout](#) structure included in the **pNext** chain. Otherwise, **layout** describes the layout for all relevant image aspects.

Valid Usage

- VUID-VkAttachmentReference2-layout-03077
If **attachment** is not `VK_ATTACHMENT_UNUSED`, **layout** **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`, `VK_IMAGE_LAYOUT_PREINITIALIZED`, or `VK_IMAGE_LAYOUT_PRESENT_SRC_KHR`
- VUID-VkAttachmentReference2-separateDepthStencilLayouts-03313
If the [separateDepthStencilLayouts](#) feature is not enabled, and **attachment** is not `VK_ATTACHMENT_UNUSED`, **layout** **must** not be `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`,
- VUID-VkAttachmentReference2-synchronization2-06910
If the [synchronization2](#) feature is not enabled, **layout** **must** not be `VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR` or `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`
- VUID-VkAttachmentReference2-attachmentFeedbackLoopLayout-07311
If the [attachmentFeedbackLoopLayout](#) feature is not enabled, **layout** **must** not be `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT`

- VUID-VkAttachmentReference2-dynamicRenderingLocalRead-09546

If the `dynamicRenderingLocalRead` feature is not enabled, `layout` **must** not be `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR`

Valid Usage (Implicit)

- VUID-VkAttachmentReference2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_2`
- VUID-VkAttachmentReference2-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkAttachmentReferenceStencilLayout`
- VUID-VkAttachmentReference2-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkAttachmentReference2-layout-parameter
`layout` **must** be a valid `VkImageLayout` value

The `VkAttachmentReferenceStencilLayout` structure is defined as:

```
typedef struct VkAttachmentReferenceStencilLayout {
    VkStructureType    sType;
    void*              pNext;
    VkImageLayout      stencilLayout;
} VkAttachmentReferenceStencilLayout;
```

or the equivalent

```
// Provided by VK_KHR_separate_depth_stencil_layouts
typedef VkAttachmentReferenceStencilLayout VkAttachmentReferenceStencilLayoutKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stencilLayout` is a `VkImageLayout` value specifying the layout the stencil aspect of the attachment uses during the subpass.

Valid Usage

- VUID-VkAttachmentReferenceStencilLayout-stencilLayout-03318
`stencilLayout` **must** not be `VK_IMAGE_LAYOUT_UNDEFINED`, `VK_IMAGE_LAYOUT_PREINITIALIZED`, `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`,

VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL,
VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL,
VK_IMAGE_LAYOUT_PRESENT_SRC_KHR

or

Valid Usage (Implicit)

- VUID-VkAttachmentReferenceStencilLayout-sType-sType
sType **must** be VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_STENCIL_LAYOUT
- VUID-VkAttachmentReferenceStencilLayout-stencilLayout-parameter
stencilLayout **must** be a valid VkImageLayout value

The VkSubpassDependency2 structure is defined as:

```
typedef struct VkSubpassDependency2 {  
    VkStructureType    sType;  
    const void*        pNext;  
    uint32_t           srcSubpass;  
    uint32_t           dstSubpass;  
    VkPipelineStageFlags srcStageMask;  
    VkPipelineStageFlags dstStageMask;  
    VkAccessFlags       srcAccessMask;  
    VkAccessFlags       dstAccessMask;  
    VkDependencyFlags   dependencyFlags;  
    int32_t             viewOffset;  
} VkSubpassDependency2;
```

or the equivalent

```
// Provided by VK_KHR_create_renderpass2  
typedef VkSubpassDependency2 VkSubpassDependency2KHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- srcSubpass is the subpass index of the first subpass in the dependency, or VK_SUBPASS_EXTERNAL.
- dstSubpass is the subpass index of the second subpass in the dependency, or VK_SUBPASS_EXTERNAL.
- srcStageMask is a bitmask of VkPipelineStageFlagBits specifying the source stage mask.
- dstStageMask is a bitmask of VkPipelineStageFlagBits specifying the destination stage mask.
- srcAccessMask is a bitmask of VkAccessFlagBits specifying a source access mask.
- dstAccessMask is a bitmask of VkAccessFlagBits specifying a destination access mask.
- dependencyFlags is a bitmask of VkDependencyFlagBits.

- `viewOffset` controls which views in the source subpass the views in the destination subpass depend on.

Parameters defined by this structure with the same name as those in `VkSubpassDependency` have the identical effect to those parameters.

`viewOffset` has the same effect for the described subpass dependency as `VkRenderPassMultiviewCreateInfo::pViewOffsets` has on each corresponding subpass dependency.

If a `VkMemoryBarrier2` is included in the `pNext` chain, `srcStageMask`, `dstStageMask`, `srcAccessMask`, and `dstAccessMask` parameters are ignored. The synchronization and access scopes instead are defined by the parameters of `VkMemoryBarrier2`.

Valid Usage

- VUID-VkSubpassDependency2-srcStageMask-04090
If the `geometryShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-VkSubpassDependency2-srcStageMask-04091
If the `tessellationShader` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-VkSubpassDependency2-srcStageMask-04094
If the `transformFeedback` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-VkSubpassDependency2-srcStageMask-07319
If the `attachmentFragmentShadingRate` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-VkSubpassDependency2-srcStageMask-03937
If the `synchronization2` feature is not enabled, `srcStageMask` **must** not be `0`
- VUID-VkSubpassDependency2-srcStageMask-07950
If the `rayTracingPipeline` feature is not enabled, `srcStageMask` **must** not contain `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkSubpassDependency2-dstStageMask-04090
If the `geometryShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT`
- VUID-VkSubpassDependency2-dstStageMask-04091
If the `tessellationShader` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-VkSubpassDependency2-dstStageMask-04094
If the `transformFeedback` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-VkSubpassDependency2-dstStageMask-07319

If the `attachmentFragmentShadingRate` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-VkSubpassDependency2-dstStageMask-03937
If the `synchronization2` feature is not enabled, `dstStageMask` **must** not be `0`
- VUID-VkSubpassDependency2-dstStageMask-07950
If the `rayTracingPipeline` feature is not enabled, `dstStageMask` **must** not contain `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR`
- VUID-VkSubpassDependency2-srcSubpass-03084
`srcSubpass` **must** be less than or equal to `dstSubpass`, unless one of them is `VK_SUBPASS_EXTERNAL`, to avoid cyclic dependencies and ensure a valid execution order
- VUID-VkSubpassDependency2-srcSubpass-03085
`srcSubpass` and `dstSubpass` **must** not both be equal to `VK_SUBPASS_EXTERNAL`
- VUID-VkSubpassDependency2-srcSubpass-06810
If `srcSubpass` is equal to `dstSubpass` and `srcStageMask` includes a `framebuffer-space` stage, `dstStageMask` **must** only contain `framebuffer-space` stages
- VUID-VkSubpassDependency2-srcAccessMask-03088
Any access flag included in `srcAccessMask` **must** be supported by one of the pipeline stages in `srcStageMask`, as specified in the [table of supported access types](#)
- VUID-VkSubpassDependency2-dstAccessMask-03089
Any access flag included in `dstAccessMask` **must** be supported by one of the pipeline stages in `dstStageMask`, as specified in the [table of supported access types](#)
- VUID-VkSubpassDependency2-dependencyFlags-03090
If `dependencyFlags` includes `VK_DEPENDENCY_VIEW_LOCAL_BIT`, `srcSubpass` **must** not be equal to `VK_SUBPASS_EXTERNAL`
- VUID-VkSubpassDependency2-dependencyFlags-03091
If `dependencyFlags` includes `VK_DEPENDENCY_VIEW_LOCAL_BIT`, `dstSubpass` **must** not be equal to `VK_SUBPASS_EXTERNAL`
- VUID-VkSubpassDependency2-srcSubpass-02245
If `srcSubpass` equals `dstSubpass`, and `srcStageMask` and `dstStageMask` both include a `framebuffer-space` stage, then `dependencyFlags` **must** include `VK_DEPENDENCY_BY_REGION_BIT`
- VUID-VkSubpassDependency2-viewOffset-02530
If `viewOffset` is not equal to `0`, `srcSubpass` **must** not be equal to `dstSubpass`
- VUID-VkSubpassDependency2-dependencyFlags-03092
If `dependencyFlags` does not include `VK_DEPENDENCY_VIEW_LOCAL_BIT`, `viewOffset` **must** be `0`

Valid Usage (Implicit)

- VUID-VkSubpassDependency2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SUBPASS_DEPENDENCY_2`
- VUID-VkSubpassDependency2-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkMemoryBarrier2`

- VUID-VkSubpassDependency2-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkSubpassDependency2-srcStageMask-parameter
srcStageMask **must** be a valid combination of [VkPipelineStageFlagBits](#) values
- VUID-VkSubpassDependency2-dstStageMask-parameter
dstStageMask **must** be a valid combination of [VkPipelineStageFlagBits](#) values
- VUID-VkSubpassDependency2-srcAccessMask-parameter
srcAccessMask **must** be a valid combination of [VkAccessFlagBits](#) values
- VUID-VkSubpassDependency2-dstAccessMask-parameter
dstAccessMask **must** be a valid combination of [VkAccessFlagBits](#) values
- VUID-VkSubpassDependency2-dependencyFlags-parameter
dependencyFlags **must** be a valid combination of [VkDependencyFlagBits](#) values

To destroy a render pass, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyRenderPass(
    VkDevice                device,
    VkRenderPass            renderPass,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the render pass.
- **renderPass** is the handle of the render pass to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyRenderPass-renderPass-00873
All submitted commands that refer to **renderPass** **must** have completed execution
- VUID-vkDestroyRenderPass-renderPass-00874
If **VkAllocationCallbacks** were provided when **renderPass** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyRenderPass-renderPass-00875
If no **VkAllocationCallbacks** were provided when **renderPass** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyRenderPass-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkDestroyRenderPass-renderPass-parameter

If `renderPass` is not `VK_NULL_HANDLE`, `renderPass` **must** be a valid `VkRenderPass` handle

- `VUID-vkDestroyRenderPass-pAllocator-parameter`

If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

- `VUID-vkDestroyRenderPass-renderPass-parent`

If `renderPass` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `renderPass` **must** be externally synchronized

8.3. Render Pass Compatibility

Framebuffers and graphics pipelines are created based on a specific render pass object. They **must** only be used with that render pass object, or one compatible with it.

Two attachment references are compatible if they have matching format and sample count, or are both `VK_ATTACHMENT_UNUSED` or the pointer that would contain the reference is `NULL`.

Two arrays of attachment references are compatible if all corresponding pairs of attachments are compatible. If the arrays are of different lengths, attachment references not present in the smaller array are treated as `VK_ATTACHMENT_UNUSED`.

Two render passes are compatible if their corresponding color, input, resolve, and depth/stencil attachment references are compatible and if they are otherwise identical except for:

- Initial and final image layout in attachment descriptions
- Load and store operations in attachment descriptions
- Image layout in attachment references

As an additional special case, if two render passes have a single subpass, the resolve attachment reference and depth/stencil resolve mode compatibility requirements are ignored.

A framebuffer is compatible with a render pass if it was created using the same render pass or a compatible render pass.

8.4. Framebuffers

Render passes operate in conjunction with *framebuffers*. Framebuffers represent a collection of specific memory attachments that a render pass instance uses.

Framebuffers are represented by `VkFramebuffer` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkFramebuffer)
```

To create a framebuffer, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateFramebuffer(
    VkDevice                                device,
    const VkFramebufferCreateInfo*          pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkFramebuffer*                          pFramebuffer);
```

- **device** is the logical device that creates the framebuffer.
- **pCreateInfo** is a pointer to a [VkFramebufferCreateInfo](#) structure describing additional information about framebuffer creation.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pFramebuffer** is a pointer to a [VkFramebuffer](#) handle in which the resulting framebuffer object is returned.

Valid Usage

- VUID-vkCreateFramebuffer-device-10002
device **must** support at least one queue family with the **VK_QUEUE_GRAPHICS_BIT** capability
- VUID-vkCreateFramebuffer-pCreateInfo-02777
If **pCreateInfo->flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, and **attachmentCount** is not 0, each element of **pCreateInfo->pAttachments** **must** have been created on **device**

Valid Usage (Implicit)

- VUID-vkCreateFramebuffer-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCreateFramebuffer-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkFramebufferCreateInfo](#) structure
- VUID-vkCreateFramebuffer-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateFramebuffer-pFramebuffer-parameter
pFramebuffer **must** be a valid pointer to a [VkFramebuffer](#) handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkFramebufferCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkFramebufferCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkFramebufferCreateFlags flags;
    VkRenderPass          renderPass;
    uint32_t              attachmentCount;
    const VkImageView*    pAttachments;
    uint32_t              width;
    uint32_t              height;
    uint32_t              layers;
} VkFramebufferCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkFramebufferCreateFlagBits`
- `renderPass` is a render pass defining what render passes the framebuffer will be compatible with. See [Render Pass Compatibility](#) for details.
- `attachmentCount` is the number of attachments.
- `pAttachments` is a pointer to an array of `VkImageView` handles, each of which will be used as the corresponding attachment in a render pass instance. If `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, this parameter is ignored.
- `width`, `height` and `layers` define the dimensions of the framebuffer. If the render pass uses multiview, then `layers` **must** be one and each attachment requires a number of layers that is greater than the maximum bit index set in the view mask in the subpasses in which it is used.

It is legal for a subpass to use no color or depth/stencil attachments, either because it has no attachment references or because all of them are `VK_ATTACHMENT_UNUSED`. This kind of subpass **can** use shader side effects such as image stores and atomics to produce an output. In this case, the subpass continues to use the `width`, `height`, and `layers` of the framebuffer to define the dimensions of the rendering area, and the `rasterizationSamples` from each pipeline's `VkPipelineMultisampleStateCreateInfo` to define the number of samples used in rasterization; however, if `VkPhysicalDeviceFeatures::variableMultisampleRate` is `VK_FALSE`, then all pipelines to be

bound with the subpass **must** have the same value for `VkPipelineMultisampleStateCreateInfo::rasterizationSamples`. In all such cases, `rasterizationSamples` **must** be a valid `VkSampleCountFlagBits` value that is set in `VkPhysicalDeviceLimits::framebufferNoAttachmentsSampleCounts`.

Valid Usage

- VUID-VkFramebufferCreateInfo-attachmentCount-00876
`attachmentCount` **must** be equal to the attachment count specified in `renderPass`
- VUID-VkFramebufferCreateInfo-flags-02778
If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT` and `attachmentCount` is not 0, `pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid `VkImageView` handles
- VUID-VkFramebufferCreateInfo-pAttachments-00877
If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of `pAttachments` that is used as a color attachment or resolve attachment by `renderPass` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`
- VUID-VkFramebufferCreateInfo-pAttachments-02633
If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of `pAttachments` that is used as a depth/stencil attachment by `renderPass` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkFramebufferCreateInfo-pAttachments-02634
If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of `pAttachments` that is used as a depth/stencil resolve attachment by `renderPass` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-VkFramebufferCreateInfo-pAttachments-00879
If `renderpass` is not `VK_NULL_HANDLE`, `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of `pAttachments` that is used as an input attachment by `renderPass` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`
- VUID-VkFramebufferCreateInfo-pAttachments-00880
If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of `pAttachments` **must** have been created with a `VkFormat` value that matches the `VkFormat` specified by the corresponding `VkAttachmentDescription` in `renderPass`
- VUID-VkFramebufferCreateInfo-pAttachments-00881
If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of `pAttachments` **must** have been created with a `samples` value that matches the `samples` value specified by the corresponding `VkAttachmentDescription` in `renderPass`
- VUID-VkFramebufferCreateInfo-flags-04533
If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of `pAttachments` that is used as an input, color, resolve, or depth/stencil attachment by `renderPass` **must** have been created with a `VkImageCreateInfo::extent.width` greater than or equal to `width`
- VUID-VkFramebufferCreateInfo-flags-04534

If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of `pAttachments` that is used as an input, color, resolve, or depth/stencil attachment by `renderPass` **must** have been created with a `VkImageCreateInfo::extent.height` greater than or equal to `height`

- VUID-VkFramebufferCreateInfo-flags-04535

If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of `pAttachments` that is used as an input, color, resolve, or depth/stencil attachment by `renderPass` **must** have been created with a `VkImageViewCreateInfo::subresourceRange.layerCount` greater than or equal to `layers`

- VUID-VkFramebufferCreateInfo-renderPass-04536

If `renderPass` was specified with non-zero view masks, each element of `pAttachments` that is used as an input, color, resolve, or depth/stencil attachment by `renderPass` **must** have a `layerCount` greater than the index of the most significant bit set in any of those view masks

- VUID-VkFramebufferCreateInfo-flags-04537

If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, and `renderPass` was specified with non-zero view masks, each element of `pAttachments` that is used as a `fragment shading rate attachment` by `renderPass` **must** have a `layerCount` that is either 1, or greater than the index of the most significant bit set in any of those view masks

- VUID-VkFramebufferCreateInfo-flags-04538

If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, and `renderPass` was not specified with non-zero view masks, each element of `pAttachments` that is used as a `fragment shading rate attachment` by `renderPass` **must** have a `layerCount` that is either 1, or greater than `layers`

- VUID-VkFramebufferCreateInfo-flags-04539

If `maintenance7` is not enabled or the `robustFragmentShadingRateAttachmentAccess` limit is `VK_FALSE` or the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure was created with `VkImageSubresourceRange::baseMipLevel` greater than 0, `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, an element of `pAttachments` that is used as a `fragment shading rate attachment` **must** have a width at least as large as $\lceil \text{width} / \text{texelWidth} \rceil$, where `texelWidth` is the largest value of `shadingRateAttachmentTexelSize.width` in a `VkFragmentShadingRateAttachmentInfoKHR` which references that attachment

- VUID-VkFramebufferCreateInfo-flags-04540

If `maintenance7` is not enabled or the `robustFragmentShadingRateAttachmentAccess` limit is `VK_FALSE` or the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure was created with `VkImageSubresourceRange::baseMipLevel` greater than 0, `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, an element of `pAttachments` that is used as a `fragment shading rate attachment` **must** have a height at least as large as $\lceil \text{height} / \text{texelHeight} \rceil$, where `texelHeight` is the largest value of `shadingRateAttachmentTexelSize.height` in a `VkFragmentShadingRateAttachmentInfoKHR` which references that attachment

- VUID-VkFramebufferCreateInfo-pAttachments-00883

If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of

pAttachments **must** only specify a single mip level

- VUID-VkFramebufferCreateInfo-pAttachments-00884

If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** **must** have been created with the identity swizzle

- VUID-VkFramebufferCreateInfo-width-00885

width **must** be greater than 0

- VUID-VkFramebufferCreateInfo-width-00886

width **must** be less than or equal to **maxFramebufferWidth**

- VUID-VkFramebufferCreateInfo-height-00887

height **must** be greater than 0

- VUID-VkFramebufferCreateInfo-height-00888

height **must** be less than or equal to **maxFramebufferHeight**

- VUID-VkFramebufferCreateInfo-layers-00889

layers **must** be greater than 0

- VUID-VkFramebufferCreateInfo-layers-00890

layers **must** be less than or equal to **maxFramebufferLayers**

- VUID-VkFramebufferCreateInfo-renderPass-02531

If **renderPass** was specified with non-zero view masks, **layers** **must** be 1

- VUID-VkFramebufferCreateInfo-pAttachments-00891

If **flags** does not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, each element of **pAttachments** that is a 2D or 2D array image view taken from a 3D image **must** not be a depth/stencil format

- VUID-VkFramebufferCreateInfo-flags-03189

If the **imagelessFramebuffer** feature is not enabled, **flags** **must** not include **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**

- VUID-VkFramebufferCreateInfo-flags-03190

If **flags** includes **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, the **pNext** chain **must** include a **VkFramebufferAttachmentsCreateInfo** structure

- VUID-VkFramebufferCreateInfo-flags-03191

If **flags** includes **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, the **attachmentImageInfoCount** member of a **VkFramebufferAttachmentsCreateInfo** structure in the **pNext** chain **must** be equal to either zero or **attachmentCount**

- VUID-VkFramebufferCreateInfo-flags-04541

If **flags** includes **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, the **width** member of any element of the **pAttachmentImageInfos** member of a **VkFramebufferAttachmentsCreateInfo** structure in the **pNext** chain that is used as an input, color, resolve or depth/stencil attachment in **renderPass** **must** be greater than or equal to **width**

- VUID-VkFramebufferCreateInfo-flags-04542

If **flags** includes **VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT**, the **height** member of any element of the **pAttachmentImageInfos** member of a **VkFramebufferAttachmentsCreateInfo** structure in the **pNext** chain that is used as an input, color, resolve or depth/stencil attachment in **renderPass** **must** be greater than or equal to **height**

- VUID-VkFramebufferCreateInfo-flags-04543

If `maintenance7` is not enabled or the `robustFragmentShadingRateAttachmentAccess` limit is `VK_FALSE` or the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure was created with `VkImageSubresourceRange::baseMipLevel` greater than 0, and `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, the `width` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure in the `pNext` chain that is used as a `fragment shading rate attachment` **must** be greater than or equal to $\lceil \text{width} / \text{texelWidth} \rceil$, where `texelWidth` is the largest value of `shadingRateAttachmentTexelSize.width` in a `VkFragmentShadingRateAttachmentInfoKHR` which references that attachment

- VUID-VkFramebufferCreateInfo-flags-04544

If `maintenance7` is not enabled or the `robustFragmentShadingRateAttachmentAccess` limit is `VK_FALSE` or the `imageView` member of a `VkRenderingFragmentShadingRateAttachmentInfoKHR` structure was created with `VkImageSubresourceRange::baseMipLevel` greater than 0, and `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, the `height` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure in the `pNext` chain that is used as a `fragment shading rate attachment` **must** be greater than or equal to $\lceil \text{height} / \text{texelHeight} \rceil$, where `texelHeight` is the largest value of `shadingRateAttachmentTexelSize.height` in a `VkFragmentShadingRateAttachmentInfoKHR` which references that attachment

- VUID-VkFramebufferCreateInfo-flags-04545

If `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, the `layerCount` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure in the `pNext` chain that is used as a `fragment shading rate attachment` **must** be either 1, or greater than or equal to `layers`

- VUID-VkFramebufferCreateInfo-flags-04587

If `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT` and `renderPass` was specified with non-zero view masks, the `layerCount` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure in the `pNext` chain that is used as a `fragment shading rate attachment` **must** be either 1, or greater than the index of the most significant bit set in any of those view masks

- VUID-VkFramebufferCreateInfo-renderPass-03198

If `multiview` is enabled for `renderPass` and `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, the `layerCount` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure included in the `pNext` chain used as an input, color, resolve, or depth/stencil attachment in `renderPass` **must** be greater than the maximum bit index set in the view mask in the subpasses in which it is used in `renderPass`

- VUID-VkFramebufferCreateInfo-renderPass-04546

If `multiview` is not enabled for `renderPass` and `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, the `layerCount` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure included in the `pNext` chain used as an input, color, resolve, or depth/stencil attachment in `renderPass` **must** be greater than or equal to `layers`

- VUID-VkFramebufferCreateInfo-flags-03201

If `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, the `usage` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure included in the `pNext` chain that refers to an attachment used as a color attachment or resolve attachment by `renderPass` **must** include `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`

- VUID-VkFramebufferCreateInfo-flags-03202

If `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, the `usage` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure included in the `pNext` chain that refers to an attachment used as a depth/stencil attachment by `renderPass` **must** include `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkFramebufferCreateInfo-flags-03203

If `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, the `usage` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure included in the `pNext` chain that refers to an attachment used as a depth/stencil resolve attachment by `renderPass` **must** include `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-VkFramebufferCreateInfo-flags-03204

If `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, the `usage` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure included in the `pNext` chain that refers to an attachment used as an input attachment by `renderPass` **must** include `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`

- VUID-VkFramebufferCreateInfo-flags-03205

If `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, at least one element of the `pViewFormats` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure included in the `pNext` chain **must** be equal to the corresponding value of `VkAttachmentDescription::format` used to create `renderPass`

- VUID-VkFramebufferCreateInfo-flags-04113

If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of `pAttachments` **must** have been created with `VkImageViewCreateInfo::viewType` not equal to `VK_IMAGE_VIEW_TYPE_3D`

- VUID-VkFramebufferCreateInfo-flags-04548

If `flags` does not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of `pAttachments` that is used as a fragment shading rate attachment by `renderPass` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-VkFramebufferCreateInfo-flags-04549

If `flags` includes `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, the `usage` member of any element of the `pAttachmentImageInfos` member of a `VkFramebufferAttachmentsCreateInfo` structure included in the `pNext` chain that refers to an attachment used as a fragment shading rate attachment by `renderPass` **must** include `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

Valid Usage (Implicit)

- VUID-VkFramebufferCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_FRAMEBUFFER_CREATE_INFO`
- VUID-VkFramebufferCreateInfo-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkFramebufferAttachmentsCreateInfo`
- VUID-VkFramebufferCreateInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkFramebufferCreateInfo-flags-parameter
`flags` **must** be a valid combination of `VkFramebufferCreateFlagBits` values
- VUID-VkFramebufferCreateInfo-renderPass-parameter
`renderPass` **must** be a valid `VkRenderPass` handle
- VUID-VkFramebufferCreateInfo-commonparent
Both of `renderPass`, and the elements of `pAttachments` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

The `VkFramebufferAttachmentsCreateInfo` structure is defined as:

```
typedef struct VkFramebufferAttachmentsCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           attachmentImageInfoCount;
    const VkFramebufferAttachmentImageInfo* pAttachmentImageInfos;
} VkFramebufferAttachmentsCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_imageless_framebuffer
typedef VkFramebufferAttachmentsCreateInfo VkFramebufferAttachmentsCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `attachmentImageInfoCount` is the number of attachments being described.
- `pAttachmentImageInfos` is a pointer to an array of `VkFramebufferAttachmentImageInfo` structures, each structure describing a number of parameters of the corresponding attachment in a render pass instance.

Valid Usage (Implicit)

- VUID-VkFramebufferAttachmentsCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENTS_CREATE_INFO`

- VUID-VkFramebufferAttachmentsCreateInfo-pAttachmentImageInfos-parameter
If `attachmentImageInfoCount` is not 0, `pAttachmentImageInfos` **must** be a valid pointer to an array of `attachmentImageInfoCount` valid `VkFramebufferAttachmentImageInfo` structures

The `VkFramebufferAttachmentImageInfo` structure is defined as:

```
typedef struct VkFramebufferAttachmentImageInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkImageCreateFlags  flags;
    VkImageUsageFlags   usage;
    uint32_t           width;
    uint32_t           height;
    uint32_t           layerCount;
    uint32_t           viewFormatCount;
    const VkFormat*     pViewFormats;
} VkFramebufferAttachmentImageInfo;
```

or the equivalent

```
// Provided by VK_KHR_imageless_framebuffer
typedef VkFramebufferAttachmentImageInfo VkFramebufferAttachmentImageInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkImageCreateFlagBits`, matching the value of `VkImageCreateInfo::flags` used to create an image that will be used with this framebuffer.
- `usage` is a bitmask of `VkImageUsageFlagBits`, matching the value of `VkImageCreateInfo::usage` used to create an image used with this framebuffer.
- `width` is the width of the image view used for rendering.
- `height` is the height of the image view used for rendering.
- `layerCount` is the number of array layers of the image view used for rendering.
- `viewFormatCount` is the number of entries in the `pViewFormats` array, matching the value of `VkImageFormatListCreateInfo::viewFormatCount` used to create an image used with this framebuffer.
- `pViewFormats` is a pointer to an array of `VkFormat` values specifying all of the formats which **can** be used when creating views of the image, matching the value of `VkImageFormatListCreateInfo::pViewFormats` used to create an image used with this framebuffer.

Images that **can** be used with the framebuffer when beginning a render pass, as specified by `VkRenderPassAttachmentBeginInfo`, **must** be created with parameters that are identical to those specified here.

Valid Usage

- VUID-VkFramebufferAttachmentImageInfo-viewFormatCount-09536
If `viewFormatCount` is not 0, each element of `pViewFormats` **must** not be `VK_FORMAT_UNDEFINED`

Valid Usage (Implicit)

- VUID-VkFramebufferAttachmentImageInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENT_IMAGE_INFO`
- VUID-VkFramebufferAttachmentImageInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkFramebufferAttachmentImageInfo-flags-parameter
`flags` **must** be a valid combination of `VkImageCreateFlagBits` values
- VUID-VkFramebufferAttachmentImageInfo-usage-parameter
`usage` **must** be a valid combination of `VkImageUsageFlagBits` values
- VUID-VkFramebufferAttachmentImageInfo-usage-requiredbitmask
`usage` **must** not be 0
- VUID-VkFramebufferAttachmentImageInfo-pViewFormats-parameter
If `viewFormatCount` is not 0, `pViewFormats` **must** be a valid pointer to an array of `viewFormatCount` valid `VkFormat` values

Bits which **can** be set in `VkFramebufferCreateInfo::flags`, specifying options for framebuffers, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkFramebufferCreateFlagBits {
    VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT = 0x00000001,
    // Provided by VK_KHR_imageless_framebuffer
    VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT_KHR = VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT,
} VkFramebufferCreateFlagBits;
```

- `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT` specifies that image views are not specified, and only attachment compatibility information will be provided via a `VkFramebufferAttachmentImageInfo` structure.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkFramebufferCreateFlags;
```

`VkFramebufferCreateFlags` is a bitmask type for setting a mask of zero or more `VkFramebufferCreateFlagBits`.

To destroy a framebuffer, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyFramebuffer(
    VkDevice          device,
    VkFramebuffer     framebuffer,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the framebuffer.
- **framebuffer** is the handle of the framebuffer to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyFramebuffer-framebuffer-00892
All submitted commands that refer to **framebuffer** **must** have completed execution
- VUID-vkDestroyFramebuffer-framebuffer-00893
If **VkAllocationCallbacks** were provided when **framebuffer** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyFramebuffer-framebuffer-00894
If no **VkAllocationCallbacks** were provided when **framebuffer** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyFramebuffer-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroyFramebuffer-framebuffer-parameter
If **framebuffer** is not **VK_NULL_HANDLE**, **framebuffer** **must** be a valid **VkFramebuffer** handle
- VUID-vkDestroyFramebuffer-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroyFramebuffer-framebuffer-parent
If **framebuffer** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **framebuffer** **must** be externally synchronized

8.5. Render Pass Load Operations

Render pass load operations define the initial values of an attachment during a render pass instance.

Load operations for attachments with a depth/stencil format execute in the `VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT` pipeline stage. Load operations for attachments with a color format execute in the `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage. The load operation for each sample in an attachment happens-before any recorded command which accesses the sample in that render pass instance via that attachment or an alias.

NOTE

Because load operations always happen first, external synchronization with attachment access only needs to synchronize the load operations with previous commands; not the operations within the render pass instance. This does not apply when using `VK_ATTACHMENT_LOAD_OP_NONE_KHR`.

Load operations only update values within the defined render area for the render pass instance. However, any writes performed by a load operation (as defined by its access masks) to a given attachment **may** read and write back any memory locations within the image subresource bound for that attachment. For depth/stencil images, if [maintenance7](#) is not enabled on the device or `separateDepthStencilAttachmentAccess` is `VK_FALSE`, writes to one aspect **may** also result in read-modify-write operations for the other aspect. If the subresource is in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` layout, implementations **must** not access pixels outside of the render area.

NOTE

As entire subresources could be accessed by load operations, applications cannot safely access values outside of the render area during a render pass instance when a load operation that modifies values is used.

Load operations that **can** be used for a render pass are:

```
// Provided by VK_VERSION_1_0
typedef enum VkAttachmentLoadOp {
    VK_ATTACHMENT_LOAD_OP_LOAD = 0,
    VK_ATTACHMENT_LOAD_OP_CLEAR = 1,
    VK_ATTACHMENT_LOAD_OP_DONT_CARE = 2,
    // Provided by VK_KHR_load_store_op_none
    VK_ATTACHMENT_LOAD_OP_NONE_KHR = 1000400000,
    // Provided by VK_EXT_load_store_op_none
    VK_ATTACHMENT_LOAD_OP_NONE_EXT = VK_ATTACHMENT_LOAD_OP_NONE_KHR,
} VkAttachmentLoadOp;
```

- `VK_ATTACHMENT_LOAD_OP_LOAD` specifies that the previous contents of the image within the render area will be preserved as the initial values. For attachments with a depth/stencil format, this uses the access type `VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_READ_BIT`. For attachments with a color format, this uses the access type `VK_ACCESS_COLOR_ATTACHMENT_READ_BIT`.
- `VK_ATTACHMENT_LOAD_OP_CLEAR` specifies that the contents within the render area will be cleared to

a uniform value, which is specified when a render pass instance is begun. For attachments with a depth/stencil format, this uses the access type `VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT`. For attachments with a color format, this uses the access type `VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT`.

- `VK_ATTACHMENT_LOAD_OP_DONT_CARE` specifies that the previous contents within the area need not be preserved; the contents of the attachment will be undefined inside the render area. For attachments with a depth/stencil format, this uses the access type `VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT`. For attachments with a color format, this uses the access type `VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT`.
- `VK_ATTACHMENT_LOAD_OP_NONE_KHR` specifies that the previous contents of the image will be undefined inside the render pass. No access type is used as the image is not accessed.

During a render pass instance, input and color attachments with color formats that have a component size of 8, 16, or 32 bits **must** be represented in the attachment's format throughout the instance. Attachments with other floating- or fixed-point color formats, or with depth components **may** be represented in a format with a precision higher than the attachment format, but **must** be represented with the same range. When such a component is loaded via the `loadOp`, it will be converted into an implementation-dependent format used by the render pass. Such components **must** be converted from the render pass format, to the format of the attachment, before they are resolved or stored at the end of a render pass instance via `storeOp`. Conversions occur as described in [Numeric Representation and Computation](#) and [Fixed-Point Data Conversions](#).

8.6. Render Pass Store Operations

Render pass store operations define how values written to an attachment during a render pass instance are stored to memory.

Store operations for attachments with a depth/stencil format execute in the `VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT` pipeline stage. Store operations for attachments with a color format execute in the `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage. The store operation for each sample in an attachment happens-after any recorded command which accesses the sample via that attachment or an alias.

NOTE

Because store operations always happen after other accesses in a render pass instance, external synchronization with attachment access in an earlier render pass only needs to synchronize with the store operations; not the operations within the render pass instance. This does not apply when using `VK_ATTACHMENT_STORE_OP_NONE`.

Store operations only update values within the defined render area for the render pass instance. However, any writes performed by a store operation (as defined by its access masks) to a given attachment **may** read and write back any memory locations within the image subresource bound for that attachment. For depth/stencil images, if `separateDepthStencilAttachmentAccess` is `VK_FALSE`, writes to one aspect **may** also result in read-modify-write operations for the other aspect. If the subresource is in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` layout, implementations **must** not access pixels outside of the render area.

NOTE

As entire subresources could be accessed by store operations, applications cannot

safely access values outside of the render area via aliased resources during a render pass instance when a store operation that modifies values is used.

Possible values of `VkAttachmentDescription::storeOp` and `stencilStoreOp`, specifying how the contents of the attachment are treated, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkAttachmentStoreOp {
    VK_ATTACHMENT_STORE_OP_STORE = 0,
    VK_ATTACHMENT_STORE_OP_DONT_CARE = 1,
    VK_ATTACHMENT_STORE_OP_NONE = 1000301000,
    // Provided by VK_KHR_dynamic_rendering, VK_KHR_load_store_op_none
    VK_ATTACHMENT_STORE_OP_NONE_KHR = VK_ATTACHMENT_STORE_OP_NONE,
    // Provided by VK_EXT_load_store_op_none
    VK_ATTACHMENT_STORE_OP_NONE_EXT = VK_ATTACHMENT_STORE_OP_NONE,
} VkAttachmentStoreOp;
```

- `VK_ATTACHMENT_STORE_OP_STORE` specifies the contents generated during the render pass and within the render area are written to memory. For attachments with a depth/stencil format, this uses the access type `VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT`. For attachments with a color format, this uses the access type `VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT`.
- `VK_ATTACHMENT_STORE_OP_DONT_CARE` specifies the contents within the render area are not needed after rendering, and **may** be discarded; the contents of the attachment will be undefined inside the render area. For attachments with a depth/stencil format, this uses the access type `VK_ACCESS_DEPTH_STENCIL_ATTACHMENT_WRITE_BIT`. For attachments with a color format, this uses the access type `VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT`.
- `VK_ATTACHMENT_STORE_OP_NONE` specifies the contents within the render area are not accessed by the store operation as long as no values are written to the attachment during the render pass. If values are written during the render pass, this behaves identically to `VK_ATTACHMENT_STORE_OP_DONT_CARE` and with matching access semantics.

NOTE

`VK_ATTACHMENT_STORE_OP_DONT_CARE` **can** cause contents generated during previous render passes to be discarded before reaching memory, even if no write to the attachment occurs during the current render pass.

8.7. Render Pass Multisample Resolve Operations

Render pass multisample resolve operations combine sample values from a single pixel in a multisample attachment and store the result to the corresponding pixel in a single sample attachment.

Multisample resolve operations for attachments execute in the `VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT` pipeline stage. A final resolve operation for all pixels in the render area happens-after any recorded command which writes a pixel via the multisample attachment to be resolved or an explicit alias of it in the subpass that it is specified. Any single sample attachment specified for use in a multisample resolve operation **may** have its

contents modified at any point once rendering begins for the render pass instance. Reads from the multisample attachment can be synchronized with `VK_ACCESS_COLOR_ATTACHMENT_READ_BIT`. Access to the single sample attachment can be synchronized with `VK_ACCESS_COLOR_ATTACHMENT_READ_BIT` and `VK_ACCESS_COLOR_ATTACHMENT_WRITE_BIT`. These pipeline stage and access types are used whether the attachments are color or depth/stencil attachments.

When using render pass objects, a subpass dependency specified with the above pipeline stages and access flags will ensure synchronization with multisample resolve operations for any attachments that were last accessed by that subpass. This allows later subpasses to read resolved values as input attachments.

Resolve operations only update values within the defined render area for the render pass instance. However, any writes performed by a resolve operation (as defined by its access masks) to a given attachment **may** read and write back any memory locations within the image subresource bound for that attachment. For depth/stencil images, if `separateDepthStencilAttachmentAccess` is `VK_FALSE`, writes to one aspect **may** also result in read-modify-write operations for the other aspect. If the subresource is in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` layout, implementations **must** not access pixels outside of the render area.

NOTE

As entire subresources could be accessed by multisample resolve operations, applications cannot safely access values outside of the render area via aliased resources during a render pass instance when a multisample resolve operation is performed.

Multisample values in a multisample attachment are combined according to the resolve mode used:

```
typedef enum VkResolveModeFlagBits {
    VK_RESOLVE_MODE_NONE = 0,
    VK_RESOLVE_MODE_SAMPLE_ZERO_BIT = 0x00000001,
    VK_RESOLVE_MODE_AVERAGE_BIT = 0x00000002,
    VK_RESOLVE_MODE_MIN_BIT = 0x00000004,
    VK_RESOLVE_MODE_MAX_BIT = 0x00000008,
    // Provided by VK_KHR_depth_stencil_resolve
    VK_RESOLVE_MODE_NONE_KHR = VK_RESOLVE_MODE_NONE,
    // Provided by VK_KHR_depth_stencil_resolve
    VK_RESOLVE_MODE_SAMPLE_ZERO_BIT_KHR = VK_RESOLVE_MODE_SAMPLE_ZERO_BIT,
    // Provided by VK_KHR_depth_stencil_resolve
    VK_RESOLVE_MODE_AVERAGE_BIT_KHR = VK_RESOLVE_MODE_AVERAGE_BIT,
    // Provided by VK_KHR_depth_stencil_resolve
    VK_RESOLVE_MODE_MIN_BIT_KHR = VK_RESOLVE_MODE_MIN_BIT,
    // Provided by VK_KHR_depth_stencil_resolve
    VK_RESOLVE_MODE_MAX_BIT_KHR = VK_RESOLVE_MODE_MAX_BIT,
} VkResolveModeFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_depth_stencil_resolve
typedef VkResolveModeFlagBits VkResolveModeFlagBitsKHR;
```

- `VK_RESOLVE_MODE_NONE` indicates that no resolve operation is done.
- `VK_RESOLVE_MODE_SAMPLE_ZERO_BIT` indicates that result of the resolve operation is equal to the value of sample 0.
- `VK_RESOLVE_MODE_AVERAGE_BIT` indicates that result of the resolve operation is the average of the sample values.
- `VK_RESOLVE_MODE_MIN_BIT` indicates that result of the resolve operation is the minimum of the sample values.
- `VK_RESOLVE_MODE_MAX_BIT` indicates that result of the resolve operation is the maximum of the sample values.

If no resolve mode is otherwise specified, `VK_RESOLVE_MODE_AVERAGE_BIT` is used.

```
typedef VkFlags VkResolveModeFlags;
```

or the equivalent

```
// Provided by VK_KHR_depth_stencil_resolve
typedef VkResolveModeFlags VkResolveModeFlagsKHR;
```

`VkResolveModeFlags` is a bitmask type for setting a mask of zero or more `VkResolveModeFlagBits`.

8.8. Render Pass Commands

An application records the commands for a render pass instance one subpass at a time, by beginning a render pass instance, iterating over the subpasses to record commands for that subpass, and then ending the render pass instance.

To begin a render pass instance, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBeginRenderPass(
    VkCommandBuffer                commandBuffer,
    const VkRenderPassBeginInfo*   pRenderPassBegin,
    VkSubpassContents              contents);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pRenderPassBegin` is a pointer to a `VkRenderPassBeginInfo` structure specifying the render pass to begin an instance of, and the framebuffer the instance uses.
- `contents` is a `VkSubpassContents` value specifying how the commands in the first subpass will be

provided.

After beginning a render pass instance, the command buffer is ready to record the commands for the first subpass of that render pass.

Valid Usage

- VUID-vkCmdBeginRenderPass-initialLayout-00895

If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`

- VUID-vkCmdBeginRenderPass-initialLayout-01758

If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-vkCmdBeginRenderPass-initialLayout-02842

If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-vkCmdBeginRenderPass-stencilInitialLayout-02843

If any of the `stencilInitialLayout` or `stencilFinalLayout` member of the `VkAttachmentDescriptionStencilLayout` structures or the `stencilLayout` member of the `VkAttachmentReferenceStencilLayout` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-vkCmdBeginRenderPass-initialLayout-00897

If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_SAMPLED_BIT` or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`

- VUID-vkCmdBeginRenderPass-initialLayout-00898

If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_TRANSFER_SRC_BIT`

- VUID-vkCmdBeginRenderPass-initialLayout-00899

If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_TRANSFER_DST_BIT`

- VUID-vkCmdBeginRenderPass-initialLayout-00900

If the `initialLayout` member of any of the `VkAttachmentDescription` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is not `VK_IMAGE_LAYOUT_UNDEFINED`, then each such `initialLayout` **must** be equal to the current layout of the corresponding attachment image subresource of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin`

- VUID-vkCmdBeginRenderPass-srcStageMask-06451

The `srcStageMask` members of any element of the `pDependencies` member of `VkRenderPassCreateInfo` used to create `renderPass` **must** be supported by the capabilities of the queue family identified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` used to create the command pool which `commandBuffer` was allocated from

- VUID-vkCmdBeginRenderPass-dstStageMask-06452

The `dstStageMask` members of any element of the `pDependencies` member of `VkRenderPassCreateInfo` used to create `renderPass` **must** be supported by the capabilities of the queue family identified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` used to create the command pool which `commandBuffer` was allocated from

- VUID-vkCmdBeginRenderPass-framebuffer-02532

For any attachment in `framebuffer` that is used by `renderPass` and is bound to memory locations that are also bound to another attachment used by `renderPass`, and if at least one of those uses causes either attachment to be written to, both attachments **must** have had the `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT` set

- VUID-vkCmdBeginRenderPass-framebuffer-09045

If any attachments specified in `framebuffer` are used by `renderPass` and are bound to overlapping memory locations, there **must** be only one that is used as a color attachment, depth/stencil, or resolve attachment in any subpass

- VUID-vkCmdBeginRenderPass-initialLayout-07000

If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including either the `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT` and either the `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_SAMPLED_BIT` usage bits

- VUID-vkCmdBeginRenderPass-initialLayout-07001

If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value the `VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` usage bit

- VUID-vkCmdBeginRenderPass-initialLayout-09537

If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including either `VK_IMAGE_USAGE_STORAGE_BIT`, or both `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` and either of `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-vkCmdBeginRenderPass-contents-09640

If `contents` is `VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR`, then at least one of the following features **must** be enabled:

- `maintenance7`
- `nestedCommandBuffer`

Valid Usage (Implicit)

- VUID-vkCmdBeginRenderPass-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBeginRenderPass-pRenderPassBegin-parameter
`pRenderPassBegin` **must** be a valid pointer to a valid `VkRenderPassBeginInfo` structure
- VUID-vkCmdBeginRenderPass-contents-parameter
`contents` **must** be a valid `VkSubpassContents` value

- VUID-vkCmdBeginRenderPass-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdBeginRenderPass-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBeginRenderPass-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdBeginRenderPass-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBeginRenderPass-bufferlevel
`commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Outside	Graphics	Action State Synchronization

Alternatively to begin a render pass, call:

```
// Provided by VK_KHR_create_renderpass2
void vkCmdBeginRenderPass2KHR(
    VkCommandBuffer                commandBuffer,
    const VkRenderPassBeginInfo*   pRenderPassBegin,
    const VkSubpassBeginInfo*      pSubpassBeginInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pRenderPassBegin` is a pointer to a `VkRenderPassBeginInfo` structure specifying the render pass to begin an instance of, and the framebuffer the instance uses.
- `pSubpassBeginInfo` is a pointer to a `VkSubpassBeginInfo` structure containing information about the subpass which is about to begin rendering.

After beginning a render pass instance, the command buffer is ready to record the commands for

the first subpass of that render pass.

Valid Usage

- VUID-vkCmdBeginRenderPass2-framebuffer-02779
Both the `framebuffer` and `renderPass` members of `pRenderPassBegin` **must** have been created on the same `VkDevice` that `commandBuffer` was allocated on
- VUID-vkCmdBeginRenderPass2-initialLayout-03094
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`
- VUID-vkCmdBeginRenderPass2-initialLayout-03096
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`,
`VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`,
`VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL`, or
`VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-vkCmdBeginRenderPass2-initialLayout-02844
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL`, or `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`,
`VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or
`VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`
- VUID-vkCmdBeginRenderPass2-stencilInitialLayout-02845
If any of the `stencilInitialLayout` or `stencilFinalLayout` member of the `VkAttachmentDescriptionStencilLayout` structures or the `stencilLayout` member of the `VkAttachmentReferenceStencilLayout` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL`, or
`VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

- VUID-vkCmdBeginRenderPass2-initialLayout-03097
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_SAMPLED_BIT` or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`
- VUID-vkCmdBeginRenderPass2-initialLayout-03098
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_TRANSFER_SRC_BIT`
- VUID-vkCmdBeginRenderPass2-initialLayout-03099
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including `VK_IMAGE_USAGE_TRANSFER_DST_BIT`
- VUID-vkCmdBeginRenderPass2-initialLayout-03100
If the `initialLayout` member of any of the `VkAttachmentDescription` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is not `VK_IMAGE_LAYOUT_UNDEFINED`, then each such `initialLayout` **must** be equal to the current layout of the corresponding attachment image subresource of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin`
- VUID-vkCmdBeginRenderPass2-srcStageMask-06453
The `srcStageMask` members of any element of the `pDependencies` member of `VkRenderPassCreateInfo` used to create `renderPass` **must** be supported by the capabilities of the queue family identified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` used to create the command pool which `commandBuffer` was allocated from
- VUID-vkCmdBeginRenderPass2-dstStageMask-06454
The `dstStageMask` members of any element of the `pDependencies` member of `VkRenderPassCreateInfo` used to create `renderPass` **must** be supported by the capabilities of the queue family identified by the `queueFamilyIndex` member of the `VkCommandPoolCreateInfo` used to create the command pool which `commandBuffer` was allocated from
- VUID-vkCmdBeginRenderPass2-framebuffer-02533
For any attachment in `framebuffer` that is used by `renderPass` and is bound to memory locations that are also bound to another attachment used by `renderPass`, and if at least one of those uses causes either attachment to be written to, both attachments **must** have had the `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT` set

- VUID-vkCmdBeginRenderPass2-framebuffer-09046
If any attachments specified in `framebuffer` are used by `renderPass` and are bound to overlapping memory locations, there **must** be only one that is used as a color attachment, depth/stencil, or resolve attachment in any subpass
- VUID-vkCmdBeginRenderPass2-initialLayout-07002
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including either the `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT` and either the `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_SAMPLED_BIT` usage bits
- VUID-vkCmdBeginRenderPass2-initialLayout-07003
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value the `VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` usage bit
- VUID-vkCmdBeginRenderPass2-initialLayout-09538
If any of the `initialLayout` or `finalLayout` member of the `VkAttachmentDescription` structures or the `layout` member of the `VkAttachmentReference` structures specified when creating the render pass specified in the `renderPass` member of `pRenderPassBegin` is `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR` then the corresponding attachment image view of the framebuffer specified in the `framebuffer` member of `pRenderPassBegin` **must** have been created with a `usage` value including either `VK_IMAGE_USAGE_STORAGE_BIT`, or both `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` and either of `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`

Valid Usage (Implicit)

- VUID-vkCmdBeginRenderPass2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBeginRenderPass2-pRenderPassBegin-parameter
`pRenderPassBegin` **must** be a valid pointer to a valid `VkRenderPassBeginInfo` structure
- VUID-vkCmdBeginRenderPass2-pSubpassBeginInfo-parameter
`pSubpassBeginInfo` **must** be a valid pointer to a valid `VkSubpassBeginInfo` structure
- VUID-vkCmdBeginRenderPass2-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdBeginRenderPass2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdBeginRenderPass2-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdBeginRenderPass2-vidcoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBeginRenderPass2-bufferlevel
commandBuffer **must** be a primary **VkCommandBuffer**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Outside	Graphics	Action State Synchronization

The **VkRenderPassBeginInfo** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkRenderPassBeginInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkRenderPass        renderPass;
    VkFramebuffer        framebuffer;
    VkRect2D            renderArea;
    uint32_t            clearValueCount;
    const VkClearValue* pClearValues;
} VkRenderPassBeginInfo;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **renderPass** is the render pass to begin an instance of.
- **framebuffer** is the framebuffer containing the attachments that are used with the render pass.
- **renderArea** is the render area that is affected by the render pass instance, and is described in more detail below.
- **clearValueCount** is the number of elements in **pClearValues**.

- `pClearValues` is a pointer to an array of `clearValueCount` `VkClearValue` structures containing clear values for each attachment, if the attachment uses a `loadOp` value of `VK_ATTACHMENT_LOAD_OP_CLEAR` or if the attachment has a depth/stencil format and uses a `stencilLoadOp` value of `VK_ATTACHMENT_LOAD_OP_CLEAR`. The array is indexed by attachment number. Only elements corresponding to cleared attachments are used. Other elements of `pClearValues` are ignored.

`renderArea` is the render area that is affected by the render pass instance. The effects of attachment load, store and multisample resolve operations are restricted to the pixels whose x and y coordinates fall within the render area on all attachments. The render area extends to all layers of `framebuffer`. The application **must** ensure (using scissor if necessary) that all rendering is contained within the render area. The render area **must** be contained within the framebuffer dimensions.

NOTE

There **may** be a performance cost for using a render area smaller than the framebuffer, unless it matches the render area granularity for the render pass.

Valid Usage

- VUID-VkRenderPassBeginInfo-clearValueCount-00902
`clearValueCount` **must** be greater than the largest attachment index in `renderPass` specifying a `loadOp` (or `stencilLoadOp`, if the attachment has a depth/stencil format) of `VK_ATTACHMENT_LOAD_OP_CLEAR`
- VUID-VkRenderPassBeginInfo-clearValueCount-04962
If `clearValueCount` is not 0, `pClearValues` **must** be a valid pointer to an array of `clearValueCount` `VkClearValue` unions
- VUID-VkRenderPassBeginInfo-renderPass-00904
`renderPass` **must** be `compatible` with the `renderPass` member of the `VkFramebufferCreateInfo` structure specified when creating `framebuffer`
- VUID-VkRenderPassBeginInfo-None-08996
If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, `renderArea.extent.width` **must** be greater than 0
- VUID-VkRenderPassBeginInfo-None-08997
If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, `renderArea.extent.height` **must** be greater than 0
- VUID-VkRenderPassBeginInfo-pNext-02850
If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, `renderArea.offset.x` **must** be greater than or equal to 0
- VUID-VkRenderPassBeginInfo-pNext-02851
If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, `renderArea.offset.y` **must** be greater than or equal to 0
- VUID-VkRenderPassBeginInfo-pNext-02852

If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, `renderArea.offset.x + renderArea.extent.width` **must** be less than or equal to `VkFramebufferCreateInfo::width` the `framebuffer` was created with

- VUID-VkRenderPassBeginInfo-pNext-02853

If the `pNext` chain does not contain `VkDeviceGroupRenderPassBeginInfo` or its `deviceRenderAreaCount` member is equal to 0, `renderArea.offset.y + renderArea.extent.height` **must** be less than or equal to `VkFramebufferCreateInfo::height` the `framebuffer` was created with

- VUID-VkRenderPassBeginInfo-pNext-02856

If the `pNext` chain contains `VkDeviceGroupRenderPassBeginInfo`, `offset.x + extent.width` of each element of `pDeviceRenderAreas` **must** be less than or equal to `VkFramebufferCreateInfo::width` the `framebuffer` was created with

- VUID-VkRenderPassBeginInfo-pNext-02857

If the `pNext` chain contains `VkDeviceGroupRenderPassBeginInfo`, `offset.y + extent.height` of each element of `pDeviceRenderAreas` **must** be less than or equal to `VkFramebufferCreateInfo::height` the `framebuffer` was created with

- VUID-VkRenderPassBeginInfo-framebuffer-03207

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that did not include `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, and the `pNext` chain includes a `VkRenderPassAttachmentBeginInfo` structure, its `attachmentCount` **must** be zero

- VUID-VkRenderPassBeginInfo-framebuffer-03208

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that included `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, the `attachmentCount` of a `VkRenderPassAttachmentBeginInfo` structure included in the `pNext` chain **must** be equal to the value of `VkFramebufferAttachmentsCreateInfo::attachmentImageInfoCount` used to create `framebuffer`

- VUID-VkRenderPassBeginInfo-framebuffer-02780

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that included `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of the `pAttachments` member of a `VkRenderPassAttachmentBeginInfo` structure included in the `pNext` chain **must** have been created on the same `VkDevice` as `framebuffer` and `renderPass`

- VUID-VkRenderPassBeginInfo-framebuffer-03209

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that included `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of the `pAttachments` member of a `VkRenderPassAttachmentBeginInfo` structure included in the `pNext` chain **must** be a `VkImageView` of an image created with a value of `VkImageCreateInfo::flags` equal to the `flags` member of the corresponding element of `VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos` used to create `framebuffer`

- VUID-VkRenderPassBeginInfo-framebuffer-04627

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that included `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of the `pAttachments` member of a `VkRenderPassAttachmentBeginInfo` structure included in the `pNext` chain **must** be a `VkImageView` with an `inherited usage` equal to the `usage` member of the corresponding element of `VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos` used to create

framebuffer

- VUID-VkRenderPassBeginInfo-framebuffer-03211

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that included `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of the `pAttachments` member of a `VkRenderPassAttachmentBeginInfo` structure included in the `pNext` chain **must** be a `VkImageView` with a width equal to the `width` member of the corresponding element of `VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos` used to create `framebuffer`

- VUID-VkRenderPassBeginInfo-framebuffer-03212

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that included `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of the `pAttachments` member of a `VkRenderPassAttachmentBeginInfo` structure included in the `pNext` chain **must** be a `VkImageView` with a height equal to the `height` member of the corresponding element of `VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos` used to create `framebuffer`

- VUID-VkRenderPassBeginInfo-framebuffer-03213

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that included `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of the `pAttachments` member of a `VkRenderPassAttachmentBeginInfo` structure included in the `pNext` chain **must** be a `VkImageView` of an image created with a value of `VkImageViewCreateInfo::subresourceRange.layerCount` equal to the `layerCount` member of the corresponding element of `VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos` used to create `framebuffer`

- VUID-VkRenderPassBeginInfo-framebuffer-03214

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that included `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of the `pAttachments` member of a `VkRenderPassAttachmentBeginInfo` structure included in the `pNext` chain **must** be a `VkImageView` of an image created with a value of `VkImageFormatListCreateInfo::viewFormatCount` equal to the `viewFormatCount` member of the corresponding element of `VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos` used to create `framebuffer`

- VUID-VkRenderPassBeginInfo-framebuffer-03215

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that included `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of the `pAttachments` member of a `VkRenderPassAttachmentBeginInfo` structure included in the `pNext` chain **must** be a `VkImageView` of an image created with a set of elements in `VkImageFormatListCreateInfo::pViewFormats` equal to the set of elements in the `pViewFormats` member of the corresponding element of `VkFramebufferAttachmentsCreateInfo::pAttachmentImageInfos` used to create `framebuffer`

- VUID-VkRenderPassBeginInfo-framebuffer-03216

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that included `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of the `pAttachments` member of a `VkRenderPassAttachmentBeginInfo` structure included in the `pNext` chain **must** be a `VkImageView` of an image created with a value of `VkImageViewCreateInfo::format` equal to the corresponding value of `VkAttachmentDescription::format` in `renderPass`

- VUID-VkRenderPassBeginInfo-framebuffer-09047

If `framebuffer` was created with a `VkFramebufferCreateInfo::flags` value that included `VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT`, each element of the `pAttachments` member of a

`VkRenderPassAttachmentBeginInfo` structure included in the `pNext` chain **must** be a `VkImageView` of an image created with a value of `VkImageCreateInfo::samples` equal to the corresponding value of `VkAttachmentDescription::samples` in `renderPass`

Valid Usage (Implicit)

- VUID-VkRenderPassBeginInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RENDER_PASS_BEGIN_INFO`
- VUID-VkRenderPassBeginInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkDeviceGroupRenderPassBeginInfo`, `VkRenderPassAttachmentBeginInfo`, or `VkRenderPassSampleLocationsBeginInfoEXT`
- VUID-VkRenderPassBeginInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkRenderPassBeginInfo-renderPass-parameter
`renderPass` **must** be a valid `VkRenderPass` handle
- VUID-VkRenderPassBeginInfo-framebuffer-parameter
`framebuffer` **must** be a valid `VkFramebuffer` handle
- VUID-VkRenderPassBeginInfo-commonparent
Both of `framebuffer`, and `renderPass` **must** have been created, allocated, or retrieved from the same `VkDevice`

The image layout of the depth aspect of a depth/stencil attachment referring to an image created with `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` is dependent on the last sample locations used to render to the image subresource, thus preserving the contents of such depth/stencil attachments across subpass boundaries requires the application to specify these sample locations whenever a layout transition of the attachment **may** occur. This information **can** be provided by adding a `VkRenderPassSampleLocationsBeginInfoEXT` structure to the `pNext` chain of `VkRenderPassBeginInfo`.

The `VkRenderPassSampleLocationsBeginInfoEXT` structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkRenderPassSampleLocationsBeginInfoEXT {
    VkStructureType             sType;
    const void*                 pNext;
    uint32_t                    attachmentInitialSampleLocationsCount;
    const VkAttachmentSampleLocationsEXT* pAttachmentInitialSampleLocations;
    uint32_t                    postSubpassSampleLocationsCount;
    const VkSubpassSampleLocationsEXT* pPostSubpassSampleLocations;
} VkRenderPassSampleLocationsBeginInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `attachmentInitialSampleLocationsCount` is the number of elements in the `pAttachmentInitialSampleLocations` array.
- `pAttachmentInitialSampleLocations` is a pointer to an array of `attachmentInitialSampleLocationsCount` `VkAttachmentSampleLocationsEXT` structures specifying the attachment indices and their corresponding sample location state. Each element of `pAttachmentInitialSampleLocations` **can** specify the sample location state to use in the automatic layout transition performed to transition a depth/stencil attachment from the initial layout of the attachment to the image layout specified for the attachment in the first subpass using it.
- `postSubpassSampleLocationsCount` is the number of elements in the `pPostSubpassSampleLocations` array.
- `pPostSubpassSampleLocations` is a pointer to an array of `postSubpassSampleLocationsCount` `VkSubpassSampleLocationsEXT` structures specifying the subpass indices and their corresponding sample location state. Each element of `pPostSubpassSampleLocations` **can** specify the sample location state to use in the automatic layout transition performed to transition the depth/stencil attachment used by the specified subpass to the image layout specified in a dependent subpass or to the final layout of the attachment in case the specified subpass is the last subpass using that attachment. In addition, if `VkPhysicalDeviceSampleLocationsPropertiesEXT::variableSampleLocations` is `VK_FALSE`, each element of `pPostSubpassSampleLocations` **must** specify the sample location state that matches the sample locations used by all pipelines that will be bound to a command buffer during the specified subpass. If `variableSampleLocations` is `VK_TRUE`, the sample locations used for rasterization do not depend on `pPostSubpassSampleLocations`.

Valid Usage (Implicit)

- VUID-VkRenderPassSampleLocationsBeginInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RENDER_PASS_SAMPLE_LOCATIONS_BEGIN_INFO_EXT`
- VUID-VkRenderPassSampleLocationsBeginInfoEXT-pAttachmentInitialSampleLocations-parameter
If `attachmentInitialSampleLocationsCount` is not 0, `pAttachmentInitialSampleLocations` **must** be a valid pointer to an array of `attachmentInitialSampleLocationsCount` valid `VkAttachmentSampleLocationsEXT` structures
- VUID-VkRenderPassSampleLocationsBeginInfoEXT-pPostSubpassSampleLocations-parameter
If `postSubpassSampleLocationsCount` is not 0, `pPostSubpassSampleLocations` **must** be a valid pointer to an array of `postSubpassSampleLocationsCount` valid `VkSubpassSampleLocationsEXT` structures

The `VkAttachmentSampleLocationsEXT` structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkAttachmentSampleLocationsEXT {
    uint32_t          attachmentIndex;
    VkSampleLocationsInfoEXT sampleLocationsInfo;
} VkAttachmentSampleLocationsEXT;
```

- **attachmentIndex** is the index of the attachment for which the sample locations state is provided.
- **sampleLocationsInfo** is the sample locations state to use for the layout transition of the given attachment from the initial layout of the attachment to the image layout specified for the attachment in the first subpass using it.

If the image referenced by the framebuffer attachment at index **attachmentIndex** was not created with **VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT** then the values specified in **sampleLocationsInfo** are ignored.

Valid Usage

- VUID-VkAttachmentSampleLocationsEXT-attachmentIndex-01531
attachmentIndex **must** be less than the **attachmentCount** specified in **VkRenderPassCreateInfo** the render pass specified by **VkRenderPassBeginInfo::renderPass** was created with

Valid Usage (Implicit)

- VUID-VkAttachmentSampleLocationsEXT-sampleLocationsInfo-parameter
sampleLocationsInfo **must** be a valid **VkSampleLocationsInfoEXT** structure

The **VkSubpassSampleLocationsEXT** structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkSubpassSampleLocationsEXT {
    uint32_t          subpassIndex;
    VkSampleLocationsInfoEXT sampleLocationsInfo;
} VkSubpassSampleLocationsEXT;
```

- **subpassIndex** is the index of the subpass for which the sample locations state is provided.
- **sampleLocationsInfo** is the sample locations state to use for the layout transition of the depth/stencil attachment away from the image layout the attachment is used with in the subpass specified in **subpassIndex**.

If the image referenced by the depth/stencil attachment used in the subpass identified by **subpassIndex** was not created with **VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT** or if the subpass does not use a depth/stencil attachment, and **VkPhysicalDeviceSampleLocationsPropertiesEXT::variableSampleLocations** is **VK_TRUE** then the

values specified in `sampleLocationsInfo` are ignored.

Valid Usage

- VUID-VkSubpassSampleLocationsEXT-subpassIndex-01532
`subpassIndex` **must** be less than the `subpassCount` specified in `VkRenderPassCreateInfo` the render pass specified by `VkRenderPassBeginInfo::renderPass` was created with

Valid Usage (Implicit)

- VUID-VkSubpassSampleLocationsEXT-sampleLocationsInfo-parameter
`sampleLocationsInfo` **must** be a valid `VkSampleLocationsInfoEXT` structure

The `VkSubpassBeginInfo` structure is defined as:

```
typedef struct VkSubpassBeginInfo {  
    VkStructureType    sType;  
    const void*        pNext;  
    VkSubpassContents  contents;  
} VkSubpassBeginInfo;
```

or the equivalent

```
// Provided by VK_KHR_create_renderpass2  
typedef VkSubpassBeginInfo VkSubpassBeginInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `contents` is a `VkSubpassContents` value specifying how the commands in the next subpass will be provided.

Valid Usage

- VUID-VkSubpassBeginInfo-contents-09382
If `contents` is `VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR`, then at least one of the following features **must** be enabled:
 - `maintenance7`
 - `nestedCommandBuffer`

Valid Usage (Implicit)

- VUID-VkSubpassBeginInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SUBPASS_BEGIN_INFO`
- VUID-VkSubpassBeginInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkSubpassBeginInfo-contents-parameter
`contents` **must** be a valid `VkSubpassContents` value

Possible values of `vkCmdBeginRenderPass::contents`, specifying how the commands in the first subpass will be provided, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSubpassContents {
    VK_SUBPASS_CONTENTS_INLINE = 0,
    VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS = 1,
    // Provided by VK_KHR_maintenance7
    VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR = 1000451000,
    // Provided by VK_EXT_nested_command_buffer
    VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_EXT =
VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR,
} VkSubpassContents;
```

- `VK_SUBPASS_CONTENTS_INLINE` specifies that the contents of the subpass will be recorded inline in the primary command buffer, and secondary command buffers **must** not be executed within the subpass.
- `VK_SUBPASS_CONTENTS_SECONDARY_COMMAND_BUFFERS` specifies that the contents are recorded in secondary command buffers that will be called from the primary command buffer, and `vkCmdExecuteCommands` is the only valid command in the command buffer until `vkCmdNextSubpass` or `vkCmdEndRenderPass`.
- `VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR` specifies that the contents of the subpass **can** be recorded both inline and in secondary command buffers executed from this command buffer with `vkCmdExecuteCommands`.

If the `pNext` chain of `VkRenderPassBeginInfo` or `VkRenderingInfo` includes a `VkDeviceGroupRenderPassBeginInfo` structure, then that structure includes a device mask and set of render areas for the render pass instance.

The `VkDeviceGroupRenderPassBeginInfo` structure is defined as:

```
typedef struct VkDeviceGroupRenderPassBeginInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           deviceMask;
    uint32_t           deviceRenderAreaCount;
    const VkRect2D*    pDeviceRenderAreas;
} VkDeviceGroupRenderPassBeginInfo;
```

or the equivalent

```
// Provided by VK_KHR_device_group
typedef VkDeviceGroupRenderPassBeginInfo VkDeviceGroupRenderPassBeginInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceMask` is the device mask for the render pass instance.
- `deviceRenderAreaCount` is the number of elements in the `pDeviceRenderAreas` array.
- `pDeviceRenderAreas` is a pointer to an array of `VkRect2D` structures defining the render area for each physical device.

The `deviceMask` serves several purposes. It is an upper bound on the set of physical devices that **can** be used during the render pass instance, and the initial device mask when the render pass instance begins. In addition, commands transitioning to the next subpass in a render pass instance and commands ending the render pass instance, and, accordingly render pass `load`, `store`, and `multisample resolve` operations and subpass dependencies corresponding to the render pass instance, are executed on the physical devices included in the device mask provided here.

If `deviceRenderAreaCount` is not zero, then the elements of `pDeviceRenderAreas` override the value of `VkRenderPassBeginInfo::renderArea`, and provide a render area specific to each physical device. These render areas serve the same purpose as `VkRenderPassBeginInfo::renderArea`, including controlling the region of attachments that are cleared by `VK_ATTACHMENT_LOAD_OP_CLEAR` and that are resolved into resolve attachments.

If this structure is not present, the render pass instance's device mask is the value of `VkDeviceGroupCommandBufferBeginInfo::deviceMask`. If this structure is not present or if `deviceRenderAreaCount` is zero, `VkRenderPassBeginInfo::renderArea` is used for all physical devices.

Valid Usage

- VUID-VkDeviceGroupRenderPassBeginInfo-deviceMask-00905
`deviceMask` **must** be a valid device mask value
- VUID-VkDeviceGroupRenderPassBeginInfo-deviceMask-00906
`deviceMask` **must** not be zero
- VUID-VkDeviceGroupRenderPassBeginInfo-deviceMask-00907

deviceMask **must** be a subset of the command buffer's initial device mask

- VUID-VkDeviceGroupRenderPassBeginInfo-deviceRenderAreaCount-00908
deviceRenderAreaCount **must** either be zero or equal to the number of physical devices in the logical device
- VUID-VkDeviceGroupRenderPassBeginInfo-offset-06166
The **offset.x** member of any element of **pDeviceRenderAreas** **must** be greater than or equal to 0
- VUID-VkDeviceGroupRenderPassBeginInfo-offset-06167
The **offset.y** member of any element of **pDeviceRenderAreas** **must** be greater than or equal to 0
- VUID-VkDeviceGroupRenderPassBeginInfo-offset-06168
The sum of the **offset.x** and **extent.width** members of any element of **pDeviceRenderAreas** **must** be less than or equal to **maxFramebufferWidth**
- VUID-VkDeviceGroupRenderPassBeginInfo-offset-06169
The sum of the **offset.y** and **extent.height** members of any element of **pDeviceRenderAreas** **must** be less than or equal to **maxFramebufferHeight**
- VUID-VkDeviceGroupRenderPassBeginInfo-extent-08998
The **extent.width** member of any element of **pDeviceRenderAreas** **must** be greater than 0
- VUID-VkDeviceGroupRenderPassBeginInfo-extent-08999
The **extent.height** member of any element of **pDeviceRenderAreas** **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkDeviceGroupRenderPassBeginInfo-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_DEVICE_GROUP_RENDER_PASS_BEGIN_INFO**
- VUID-VkDeviceGroupRenderPassBeginInfo-pDeviceRenderAreas-parameter
If **deviceRenderAreaCount** is not 0, **pDeviceRenderAreas** **must** be a valid pointer to an array of **deviceRenderAreaCount** **VkRect2D** structures

The **VkRenderPassAttachmentBeginInfo** structure is defined as:

```
typedef struct VkRenderPassAttachmentBeginInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           attachmentCount;
    const VkImageView* pAttachments;
} VkRenderPassAttachmentBeginInfo;
```

or the equivalent


```
// Provided by VK_KHR_imageless_framebuffer
typedef VkRenderPassAttachmentBeginInfo VkRenderPassAttachmentBeginInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `attachmentCount` is the number of attachments.
- `pAttachments` is a pointer to an array of `VkImageView` handles, each of which will be used as the corresponding attachment in the render pass instance.

Valid Usage

- VUID-VkRenderPassAttachmentBeginInfo-pAttachments-03218
Each element of `pAttachments` **must** only specify a single mip level
- VUID-VkRenderPassAttachmentBeginInfo-pAttachments-03219
Each element of `pAttachments` **must** have been created with the `identity swizzle`
- VUID-VkRenderPassAttachmentBeginInfo-pAttachments-04114
Each element of `pAttachments` **must** have been created with `VkImageViewCreateInfo::viewType` not equal to `VK_IMAGE_VIEW_TYPE_3D`

Valid Usage (Implicit)

- VUID-VkRenderPassAttachmentBeginInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RENDER_PASS_ATTACHMENT_BEGIN_INFO`
- VUID-VkRenderPassAttachmentBeginInfo-pAttachments-parameter
If `attachmentCount` is not 0, `pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid `VkImageView` handles

To query the render area granularity, call:

```
// Provided by VK_VERSION_1_0
void vkGetRenderAreaGranularity(
    VkDevice          device,
    VkRenderPass      renderPass,
    VkExtent2D*       pGranularity);
```

- `device` is the logical device that owns the render pass.
- `renderPass` is a handle to a render pass.
- `pGranularity` is a pointer to a `VkExtent2D` structure in which the granularity is returned.

The conditions leading to an optimal `renderArea` are:

- the `offset.x` member in `renderArea` is a multiple of the `width` member of the returned `VkExtent2D` (the horizontal granularity).
- the `offset.y` member in `renderArea` is a multiple of the `height` member of the returned `VkExtent2D` (the vertical granularity).
- either the `extent.width` member in `renderArea` is a multiple of the horizontal granularity or `offset.x+extent.width` is equal to the `width` of the `framebuffer` in the `VkRenderPassBeginInfo`.
- either the `extent.height` member in `renderArea` is a multiple of the vertical granularity or `offset.y+extent.height` is equal to the `height` of the `framebuffer` in the `VkRenderPassBeginInfo`.

Subpass dependencies are not affected by the render area, and apply to the entire image subresources attached to the framebuffer as specified in the description of [automatic layout transitions](#). Similarly, pipeline barriers are valid even if their effect extends outside the render area.

Valid Usage (Implicit)

- VUID-vkGetRenderAreaGranularity-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkGetRenderAreaGranularity-renderPass-parameter `renderPass` **must** be a valid `VkRenderPass` handle
- VUID-vkGetRenderAreaGranularity-pGranularity-parameter `pGranularity` **must** be a valid pointer to a `VkExtent2D` structure
- VUID-vkGetRenderAreaGranularity-renderPass-parent `renderPass` **must** have been created, allocated, or retrieved from `device`

To transition to the next subpass in the render pass instance after recording the commands for a subpass, call:

```
// Provided by VK_VERSION_1_0
void vkCmdNextSubpass(
    VkCommandBuffer          commandBuffer,
    VkSubpassContents        contents);
```

- `commandBuffer` is the command buffer in which to record the command.
- `contents` specifies how the commands in the next subpass will be provided, in the same fashion as the corresponding parameter of `vkCmdBeginRenderPass`.

The subpass index for a render pass begins at zero when `vkCmdBeginRenderPass` is recorded, and increments each time `vkCmdNextSubpass` is recorded.

After transitioning to the next subpass, the application **can** record the commands for that subpass.

Valid Usage

- VUID-vkCmdNextSubpass-None-00909
The current subpass index **must** be less than the number of subpasses in the render pass minus one
- VUID-vkCmdNextSubpass-None-02349
This command **must** not be recorded when transform feedback is active

Valid Usage (Implicit)

- VUID-vkCmdNextSubpass-commandBuffer-parameter
commandBuffer **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdNextSubpass-contents-parameter
contents **must** be a valid [VkSubpassContents](#) value
- VUID-vkCmdNextSubpass-commandBuffer-recording
commandBuffer **must** be in the [recording state](#)
- VUID-vkCmdNextSubpass-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdNextSubpass-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdNextSubpass-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdNextSubpass-bufferlevel
commandBuffer **must** be a primary [VkCommandBuffer](#)

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Inside	Outside	Graphics	Action State Synchronization

To transition to the next subpass in the render pass instance after recording the commands for a subpass, call:

```
// Provided by VK_KHR_create_renderpass2
void vkCmdNextSubpass2KHR(
    VkCommandBuffer                      commandBuffer,
    const VkSubpassBeginInfo*           pSubpassBeginInfo,
    const VkSubpassEndInfo*             pSubpassEndInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pSubpassBeginInfo` is a pointer to a `VkSubpassBeginInfo` structure containing information about the subpass which is about to begin rendering.
- `pSubpassEndInfo` is a pointer to a `VkSubpassEndInfo` structure containing information about how the previous subpass will be ended.

`vkCmdNextSubpass2KHR` is semantically identical to `vkCmdNextSubpass`, except that it is extensible, and that `contents` is provided as part of an extensible structure instead of as a flat parameter.

Valid Usage

- VUID-vkCmdNextSubpass2-None-03102
The current subpass index **must** be less than the number of subpasses in the render pass minus one
- VUID-vkCmdNextSubpass2-None-02350
This command **must** not be recorded when transform feedback is active

Valid Usage (Implicit)

- VUID-vkCmdNextSubpass2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdNextSubpass2-pSubpassBeginInfo-parameter
`pSubpassBeginInfo` **must** be a valid pointer to a valid `VkSubpassBeginInfo` structure
- VUID-vkCmdNextSubpass2-pSubpassEndInfo-parameter
`pSubpassEndInfo` **must** be a valid pointer to a valid `VkSubpassEndInfo` structure
- VUID-vkCmdNextSubpass2-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdNextSubpass2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdNextSubpass2-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdNextSubpass2-encoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdNextSubpass2-bufferlevel
`commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Inside	Outside	Graphics	Action State Synchronization

To record a command to end a render pass instance after recording the commands for the last subpass, call:

```
// Provided by VK_VERSION_1_0
void vkCmdEndRenderPass(
    VkCommandBuffer                                commandBuffer);
```

- `commandBuffer` is the command buffer in which to end the current render pass instance.

Ending a render pass instance performs any multisample resolve operations on the final subpass.

Valid Usage

- VUID-vkCmdEndRenderPass-None-00910
The current subpass index **must** be equal to the number of subpasses in the render pass minus one
- VUID-vkCmdEndRenderPass-None-02351
This command **must** not be recorded when transform feedback is active
- VUID-vkCmdEndRenderPass-None-06170
The current render pass instance **must** not have been begun with `vkCmdBeginRenderingKHR`
- VUID-vkCmdEndRenderPass-None-07004
If `vkCmdBeginQuery*` was called within a subpass of the render pass, the corresponding

`vkCmdEndQuery*` **must** have been called subsequently within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndRenderPass-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdEndRenderPass-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdEndRenderPass-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdEndRenderPass-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdEndRenderPass-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdEndRenderPass-bufferlevel
`commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Inside	Outside	Graphics	Action State Synchronization

To record a command to end a render pass instance after recording the commands for the last subpass, call:

```
// Provided by VK_KHR_create_renderpass2
void vkCmdEndRenderPass2KHR(
    VkCommandBuffer                      commandBuffer,
    const VkSubpassEndInfo*              pSubpassEndInfo);
```

- `commandBuffer` is the command buffer in which to end the current render pass instance.
- `pSubpassEndInfo` is a pointer to a `VkSubpassEndInfo` structure containing information about how the last subpass will be ended.

`vkCmdEndRenderPass2KHR` is semantically identical to `vkCmdEndRenderPass`, except that it is extensible.

Valid Usage

- VUID-vkCmdEndRenderPass2-None-03103
The current subpass index **must** be equal to the number of subpasses in the render pass minus one
- VUID-vkCmdEndRenderPass2-None-02352
This command **must** not be recorded when transform feedback is active
- VUID-vkCmdEndRenderPass2-None-06171
The current render pass instance **must** not have been begun with `vkCmdBeginRenderingKHR`
- VUID-vkCmdEndRenderPass2-None-07005
If `vkCmdBeginQuery*` was called within a subpass of the render pass, the corresponding `vkCmdEndQuery*` **must** have been called subsequently within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndRenderPass2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdEndRenderPass2-pSubpassEndInfo-parameter
`pSubpassEndInfo` **must** be a valid pointer to a valid `VkSubpassEndInfo` structure
- VUID-vkCmdEndRenderPass2-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdEndRenderPass2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdEndRenderPass2-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdEndRenderPass2-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdEndRenderPass2-bufferlevel
`commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Inside	Outside	Graphics	Action State Synchronization

The `VkSubpassEndInfo` structure is defined as:

```
typedef struct VkSubpassEndInfo {
    VkStructureType    sType;
    const void*        pNext;
} VkSubpassEndInfo;
```

or the equivalent

```
// Provided by VK_KHR_create_renderpass2
typedef VkSubpassEndInfo VkSubpassEndInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

Valid Usage (Implicit)

- VUID-VkSubpassEndInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SUBPASS_END_INFO`
- VUID-VkSubpassEndInfo-pNext-pNext
`pNext` **must** be `NULL`

8.9. Common Render Pass Data Races (Informative)

Due to the complexity of how rendering is performed, there are several ways an application can accidentally introduce a data race, usually by doing something that may seem benign but actually cannot be supported. This section indicates a number of the more common cases as guidelines to help avoid them.

8.9.1. Sampling From a Read-only Attachment

Vulkan includes read-only layouts for depth/stencil images, that allow the images to be both read during a render pass for the purposes of depth/stencil tests, and read as a non-attachment.

However, because `VK_ATTACHMENT_STORE_OP_STORE` and `VK_ATTACHMENT_STORE_OP_DONT_CARE` may perform write operations, even if no recorded command writes to an attachment, reading from an image while also using it as an attachment with these store operations can result in a data race. If the reads from the non-attachment are performed in a fragment shader where the accessed samples match those covered by the fragment shader, no data race will occur as store operations are guaranteed to operate after fragment shader execution for the set of samples the fragment covers. Notably, input attachments can also be used for this case. Reading other samples or in any other shader stage can result in unexpected behavior due to the potential for a data race, and validation errors should be generated for doing so. In practice, many applications have shipped reading samples outside of the covered fragment without any observable issue, but there is no guarantee that this will always work, and it is not advisable to rely on this in new or re-worked code bases. As `VK_ATTACHMENT_STORE_OP_NONE` is guaranteed to perform no writes, applications wishing to read an image as both an attachment and a non-attachment should make use of this store operation, coupled with a load operation that also performs no writes.

8.9.2. Non-overlapping Access Between Resources

When relying on non-overlapping accesses between attachments and other resources, it is important to note that `load` and `store` operations have fairly wide alignment requirements - potentially affecting entire subresources and adjacent depth/stencil aspects. This makes it invalid to access a non-attachment subresource that is simultaneously being used as an attachment where either access performs a write operation.

The only exception to this is if a subresource is in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, in which case the overlap is defined to occur at a per-pixel granularity, and applications can read data from pixels outside the render area without introducing a data race.

8.9.3. Depth/Stencil and Input Attachments

When rendering to only the depth OR stencil aspect of an image, an input attachment accessing the other aspect will not cause a data race only under very specific conditions. To avoid a data race, the aspect not being written **must** be in a read-only layout, and writes to it **must** be disabled in the draw state. For example, to read from stencil while writing depth, the attachment **must** be in `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL` (or equivalent), and the stencil write mask **must** be 0. Similarly to read from depth while writing stencil, the attachment **must** be in `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL` (or equivalent), and depth write enable **must** be `VK_FALSE`.

8.9.4. Synchronization Options

There are several synchronization options available to synchronize between accesses to resources within a render pass. Some of the options are outlined below:

- A [VkSubpassDependency](#) in a render pass object can synchronize attachment writes and [multisample resolve operations](#) from a prior subpass for subsequent input attachment reads.
- A [vkCmdPipelineBarrier](#) inside a subpass can synchronize prior attachment writes in the subpass with subsequent input attachment reads.
- A [vkCmdPipelineBarrier](#) inside a subpass can synchronize prior attachment writes in the subpass with subsequent non-attachment reads if the attachment is in the [VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT](#) image layout.
- If a subresource is used as two separate non-attachment resources, writes to a pixel or individual sample in a fragment shader can be synchronized with access to the same pixel or sample in another fragment shader by using one of the [fragment interlock](#) execution modes.

Chapter 9. Shaders

A shader specifies programmable operations that execute for each vertex, control point, tessellated vertex, primitive, fragment, or workgroup in the corresponding stage(s) of the graphics and compute pipelines.

Graphics pipelines include vertex shader execution as a result of [primitive assembly](#), followed, if enabled, by tessellation control and evaluation shaders operating on [patches](#), geometry shaders, if enabled, operating on primitives, and fragment shaders, if present, operating on fragments generated by [Rasterization](#). In this specification, vertex, tessellation control, tessellation evaluation and geometry shaders are collectively referred to as [pre-rasterization shader stages](#) and occur in the logical pipeline before rasterization. The fragment shader occurs logically after rasterization.

Only the compute shader stage is included in a compute pipeline. Compute shaders operate on compute invocations in a workgroup.

Shaders **can** read from input variables, and read from and write to output variables. Input and output variables **can** be used to transfer data between shader stages, or to allow the shader to interact with values that exist in the execution environment. Similarly, the execution environment provides constants describing capabilities.

Shader variables are associated with execution environment-provided inputs and outputs using *built-in* decorations in the shader. The available decorations for each stage are documented in the following subsections.

9.1. Shader Objects

Shaders **may** be compiled and linked into pipeline objects as described in [Pipelines](#) chapter, or if the `shaderObject` feature is enabled they **may** be compiled into individual per-stage *shader objects* which **can** be bound on a command buffer independently from one another. Unlike pipelines, shader objects are not intrinsically tied to any specific set of state. Instead, state is specified dynamically in the command buffer.

Each shader object represents a single compiled shader stage, which **may optionally** be linked with one or more other stages.

Shader objects are represented by `VkShaderEXT` handles:

```
// Provided by VK_EXT_shader_object
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkShaderEXT)
```

9.1.1. Shader Object Creation

Shader objects **may** be created from shader code provided as SPIR-V, or in an opaque, implementation-defined binary format specific to the physical device.

To create one or more shader objects, call:

```
// Provided by VK_EXT_shader_object
VkResult vkCreateShadersEXT(
    VkDevice                                device,
    uint32_t                                createInfoCount,
    const VkShaderCreateInfoEXT*            pCreateInfos,
    const VkAllocationCallbacks*           pAllocator,
    VkShaderEXT*                            pShaders);
```

- **device** is the logical device that creates the shader objects.
- **createInfoCount** is the length of the **pCreateInfos** and **pShaders** arrays.
- **pCreateInfos** is a pointer to an array of **VkShaderCreateInfoEXT** structures.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pShaders** is a pointer to an array of **VkShaderEXT** handles in which the resulting shader objects are returned.

When this function returns, whether or not it succeeds, it is guaranteed that every element of **pShaders** will have been overwritten by either **VK_NULL_HANDLE** or a valid **VkShaderEXT** handle.

This means that whenever shader creation fails, the application **can** determine which shader the returned error pertains to by locating the first **VK_NULL_HANDLE** element in **pShaders**. It also means that an application **can** reliably clean up from a failed call by iterating over the **pShaders** array and destroying every element that is not **VK_NULL_HANDLE**.

Valid Usage

- VUID-vkCreateShadersEXT-device-09669
device **must** support at least one queue family with one of the **VK_QUEUE_COMPUTE_BIT** or **VK_QUEUE_GRAPHICS_BIT** capabilities
- VUID-vkCreateShadersEXT-stage-09670
If the **stage** member of any element of **pCreateInfos** is **VK_SHADER_STAGE_COMPUTE_BIT**, **device** **must** support at least one queue family with the **VK_QUEUE_COMPUTE_BIT** capability
- VUID-vkCreateShadersEXT-stage-09671
If the **stage** member of any element of **pCreateInfos** is **VK_SHADER_STAGE_VERTEX_BIT**, **VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT**, **VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT**, **VK_SHADER_STAGE_GEOMETRY_BIT**, or **VK_SHADER_STAGE_FRAGMENT_BIT**, **device** **must** support at least one queue family with the **VK_QUEUE_GRAPHICS_BIT** capability
- VUID-vkCreateShadersEXT-None-08400
The **shaderObject** feature **must** be enabled
- VUID-vkCreateShadersEXT-pCreateInfos-08402
If the **flags** member of any element of **pCreateInfos** includes **VK_SHADER_CREATE_LINK_STAGE_BIT_EXT**, the **flags** member of all other elements of **pCreateInfos** whose **stage** is **VK_SHADER_STAGE_VERTEX_BIT**, **VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT**, **VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT**, **VK_SHADER_STAGE_GEOMETRY_BIT**, or **VK_SHADER_STAGE_FRAGMENT_BIT** **must** also include

VK_SHADER_CREATE_LINK_STAGE_BIT_EXT

- VUID-vkCreateShadersEXT-pCreateInfos-08409

For each element of `pCreateInfos` whose `flags` member includes `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT`, if there is any other element of `pCreateInfos` whose `stage` is logically later than the `stage` of the former and whose `flags` member also includes `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT`, the `nextStage` of the former **must** be equal to the `stage` of the element with the logically earliest `stage` following the `stage` of the former whose `flags` member also includes `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT`

- VUID-vkCreateShadersEXT-pCreateInfos-08410

The `stage` member of each element of `pCreateInfos` whose `flags` member includes `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` **must** be unique

- VUID-vkCreateShadersEXT-pCreateInfos-08411

The `codeType` member of all elements of `pCreateInfos` whose `flags` member includes `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` **must** be the same

- VUID-vkCreateShadersEXT-pCreateInfos-08867

If `pCreateInfos` contains elements with both `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` and `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, both elements' `flags` include `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT`, both elements' `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and the `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` stage's `pCode` contains an `OpExecutionMode` instruction specifying the type of subdivision, it **must** match the subdivision type specified in the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage

- VUID-vkCreateShadersEXT-pCreateInfos-08868

If `pCreateInfos` contains elements with both `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` and `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, both elements' `flags` include `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT`, both elements' `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and the `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` stage's `pCode` contains an `OpExecutionMode` instruction specifying the orientation of triangles, it **must** match the triangle orientation specified in the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage

- VUID-vkCreateShadersEXT-pCreateInfos-08869

If `pCreateInfos` contains elements with both `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` and `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, both elements' `flags` include `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT`, both elements' `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and the `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` stage's `pCode` contains an `OpExecutionMode` instruction specifying `PointMode`, the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage **must** also contain an `OpExecutionMode` instruction specifying `PointMode`

- VUID-vkCreateShadersEXT-pCreateInfos-08870

If `pCreateInfos` contains elements with both `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` and `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, both elements' `flags` include `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT`, both elements' `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and the `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` stage's `pCode` contains an `OpExecutionMode` instruction specifying the spacing of segments on the edges of tessellated primitives, it **must** match the segment spacing specified in the

VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT stage

- VUID-vkCreateShadersEXT-pCreateInfo-08871

If `pCreateInfo` contains elements with both `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` and `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, both elements' `flags` include `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT`, both elements' `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and the `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` stage's `pCode` contains an `OpExecutionMode` instruction specifying the output patch size, it **must** match the output patch size specified in the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage

Valid Usage (Implicit)

- VUID-vkCreateShadersEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateShadersEXT-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to an array of `createInfoCount` valid `VkShaderCreateInfoEXT` structures
- VUID-vkCreateShadersEXT-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateShadersEXT-pShaders-parameter
`pShaders` **must** be a valid pointer to an array of `createInfoCount` `VkShaderEXT` handles
- VUID-vkCreateShadersEXT-createInfoCount-arraylength
`createInfoCount` **must** be greater than 0

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPATIBLE_SHADER_BINARY_EXT`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`

The `VkShaderCreateInfoEXT` structure is defined as:


```
// Provided by VK_EXT_shader_object
typedef struct VkShaderCreateInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkShaderCreateFlagsEXT   flags;
    VkShaderStageFlagBits    stage;
    VkShaderStageFlags       nextStage;
    VkShaderCodeTypeEXT      codeType;
    size_t                   codeSize;
    const void*              pCode;
    const char*              pName;
    uint32_t                 setLayoutCount;
    const VkDescriptorSetLayout* pSetLayouts;
    uint32_t                 pushConstantRangeCount;
    const VkPushConstantRange* pPushConstantRanges;
    const VkSpecializationInfo* pSpecializationInfo;
} VkShaderCreateInfoEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkShaderCreateFlagBitsEXT](#) describing additional parameters of the shader.
- **stage** is a [VkShaderStageFlagBits](#) value specifying a single shader stage.
- **nextStage** is a bitmask of [VkShaderStageFlagBits](#) specifying zero or stages which **may** be used as a logically next bound stage when drawing with the shader bound.
- **codeType** is a [VkShaderCodeTypeEXT](#) value specifying the type of the shader code pointed to by **pCode**.
- **codeSize** is the size in bytes of the shader code pointed to by **pCode**.
- **pCode** is a pointer to the shader code to use to create the shader.
- **pName** is a pointer to a null-terminated UTF-8 string specifying the entry point name of the shader for this stage.
- **setLayoutCount** is the number of descriptor set layouts pointed to by **pSetLayouts**.
- **pSetLayouts** is a pointer to an array of [VkDescriptorSetLayout](#) objects used by the shader stage.
- **pushConstantRangeCount** is the number of push constant ranges pointed to by **pPushConstantRanges**.
- **pPushConstantRanges** is a pointer to an array of [VkPushConstantRange](#) structures used by the shader stage.
- **pSpecializationInfo** is a pointer to a [VkSpecializationInfo](#) structure, as described in [Specialization Constants](#), or `NULL`.

Valid Usage

- VUID-VkShaderCreateInfoEXT-codeSize-08735
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, `codeSize` **must** be a multiple of 4
- VUID-VkShaderCreateInfoEXT-pCode-08736
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, `pCode` **must** point to valid SPIR-V code, formatted and packed as described by the [Khronos SPIR-V Specification](#)
- VUID-VkShaderCreateInfoEXT-pCode-08737
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, `pCode` **must** adhere to the validation rules described by the [Validation Rules within a Module](#) section of the [SPIR-V Environment](#) appendix
- VUID-VkShaderCreateInfoEXT-pCode-08738
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, `pCode` **must** declare the `Shader` capability for SPIR-V code
- VUID-VkShaderCreateInfoEXT-pCode-08739
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, `pCode` **must** not declare any capability that is not supported by the API, as described by the [Capabilities](#) section of the [SPIR-V Environment](#) appendix
- VUID-VkShaderCreateInfoEXT-pCode-08740
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `pCode` declares any of the capabilities listed in the [SPIR-V Environment](#) appendix, one of the corresponding requirements **must** be satisfied
- VUID-VkShaderCreateInfoEXT-pCode-08741
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, `pCode` **must** not declare any SPIR-V extension that is not supported by the API, as described by the [Extension](#) section of the [SPIR-V Environment](#) appendix
- VUID-VkShaderCreateInfoEXT-pCode-08742
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `pCode` declares any of the SPIR-V extensions listed in the [SPIR-V Environment](#) appendix, one of the corresponding requirements **must** be satisfied
- VUID-VkShaderCreateInfoEXT-flags-08412
If `stage` is not `VK_SHADER_STAGE_TASK_BIT_EXT`, `VK_SHADER_STAGE_MESH_BIT_EXT`, `VK_SHADER_STAGE_VERTEX_BIT`, `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`, `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, `VK_SHADER_STAGE_GEOMETRY_BIT`, or `VK_SHADER_STAGE_FRAGMENT_BIT`, `flags` **must** not include `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT`
- VUID-VkShaderCreateInfoEXT-flags-08486
If `stage` is not `VK_SHADER_STAGE_FRAGMENT_BIT`, `flags` **must** not include `VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT`
- VUID-VkShaderCreateInfoEXT-flags-08487
If the `attachmentFragmentShadingRate` feature is not enabled, `flags` **must** not include `VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT`
- VUID-VkShaderCreateInfoEXT-flags-08485

If `stage` is not `VK_SHADER_STAGE_COMPUTE_BIT`, `flags` **must** not include `VK_SHADER_CREATE_DISPATCH_BASE_BIT_EXT`

- VUID-VkShaderCreateInfoEXT-stage-08418

`stage` **must** not be `VK_SHADER_STAGE_ALL_GRAPHICS` or `VK_SHADER_STAGE_ALL`

- VUID-VkShaderCreateInfoEXT-stage-08419

If the `tessellationShader` feature is not enabled, `stage` **must** not be `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` or `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`

- VUID-VkShaderCreateInfoEXT-stage-08420

If the `geometryShader` feature is not enabled, `stage` **must** not be `VK_SHADER_STAGE_GEOMETRY_BIT`

- VUID-VkShaderCreateInfoEXT-nextStage-08427

If `stage` is `VK_SHADER_STAGE_VERTEX_BIT`, `nextStage` **must** not include any bits other than `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`, `VK_SHADER_STAGE_GEOMETRY_BIT`, and `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-VkShaderCreateInfoEXT-nextStage-08428

If the `tessellationShader` feature is not enabled, `nextStage` **must** not include `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` or `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`

- VUID-VkShaderCreateInfoEXT-nextStage-08429

If the `geometryShader` feature is not enabled, `nextStage` **must** not include `VK_SHADER_STAGE_GEOMETRY_BIT`

- VUID-VkShaderCreateInfoEXT-nextStage-08430

If `stage` is `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`, `nextStage` **must** not include any bits other than `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`

- VUID-VkShaderCreateInfoEXT-nextStage-08431

If `stage` is `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, `nextStage` **must** not include any bits other than `VK_SHADER_STAGE_GEOMETRY_BIT` and `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-VkShaderCreateInfoEXT-nextStage-08433

If `stage` is `VK_SHADER_STAGE_GEOMETRY_BIT`, `nextStage` **must** not include any bits other than `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-VkShaderCreateInfoEXT-nextStage-08434

If `stage` is `VK_SHADER_STAGE_FRAGMENT_BIT` or `VK_SHADER_STAGE_COMPUTE_BIT`, `nextStage` **must** be 0

- VUID-VkShaderCreateInfoEXT-pName-08440

If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, `pName` **must** be the name of an `OpEntryPoint` in `pCode` with an execution model that matches `stage`

- VUID-VkShaderCreateInfoEXT-pCode-08492

If `codeType` is `VK_SHADER_CODE_TYPE_BINARY_EXT`, `pCode` **must** be aligned to 16 bytes

- VUID-VkShaderCreateInfoEXT-pCode-08493

If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, `pCode` **must** be aligned to 4 bytes

- VUID-VkShaderCreateInfoEXT-pCode-08448

If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and the identified entry point includes any variable in its interface that is declared with the `ClipDistance BuiltIn` decoration, that variable **must** not have an array size greater than `VkPhysicalDeviceLimits::maxClipDistances`

- VUID-VkShaderCreateInfoEXT-pCode-08449

If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and the identified entry point includes any variable in its interface that is declared with the `CullDistance BuiltIn` decoration, that variable **must** not have an array size greater than `VkPhysicalDeviceLimits::maxCullDistances`

- VUID-VkShaderCreateInfoEXT-pCode-08450

If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and the identified entry point includes variables in its interface that are declared with the `ClipDistance BuiltIn` decoration and variables in its interface that are declared with the `CullDistance BuiltIn` decoration, those variables **must** not have array sizes which sum to more than `VkPhysicalDeviceLimits::maxCombinedClipAndCullDistances`

- VUID-VkShaderCreateInfoEXT-pCode-08451

If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and the identified entry point includes any variable in its interface that is declared with the `SampleMask BuiltIn` decoration, that variable **must** not have an array size greater than `VkPhysicalDeviceLimits::maxSampleMaskWords`

- VUID-VkShaderCreateInfoEXT-pCode-08452

If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is `VK_SHADER_STAGE_VERTEX_BIT`, the identified entry point **must** not include any input variable in its interface that is decorated with `CullDistance`

- VUID-VkShaderCreateInfoEXT-pCode-08453

If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` or `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, and the identified entry point has an `OpExecutionMode` instruction specifying a patch size with `OutputVertices`, the patch size **must** be greater than `0` and less than or equal to `VkPhysicalDeviceLimits::maxTessellationPatchSize`

- VUID-VkShaderCreateInfoEXT-pCode-08454

If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is `VK_SHADER_STAGE_GEOMETRY_BIT`, the identified entry point **must** have an `OpExecutionMode` instruction specifying a maximum output vertex count that is greater than `0` and less than or equal to `VkPhysicalDeviceLimits::maxGeometryOutputVertices`

- VUID-VkShaderCreateInfoEXT-pCode-08455

If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is `VK_SHADER_STAGE_GEOMETRY_BIT`, the identified entry point **must** have an `OpExecutionMode` instruction specifying an invocation count that is greater than `0` and less than or equal to `VkPhysicalDeviceLimits::maxGeometryShaderInvocations`

- VUID-VkShaderCreateInfoEXT-pCode-08456

If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is a `pre-rasterization shader stage`, and the identified entry point writes to `Layer` for any primitive, it **must** write the same value to `Layer` for all vertices of a given primitive

- VUID-VkShaderCreateInfoEXT-pCode-08457
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is a [pre-rasterization shader stage](#), and the identified entry point writes to `ViewportIndex` for any primitive, it **must** write the same value to `ViewportIndex` for all vertices of a given primitive
- VUID-VkShaderCreateInfoEXT-pCode-08458
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is `VK_SHADER_STAGE_FRAGMENT_BIT`, the identified entry point **must** not include any output variables in its interface decorated with `CullDistance`
- VUID-VkShaderCreateInfoEXT-pCode-08459
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is `VK_SHADER_STAGE_FRAGMENT_BIT`, and the identified entry point writes to `FragDepth` in any execution path, all execution paths that are not exclusive to helper invocations **must** either discard the fragment, or write or initialize the value of `FragDepth`
- VUID-VkShaderCreateInfoEXT-pCode-08460
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, the shader code in `pCode` **must** be valid as described by the [Khronos SPIR-V Specification](#) after applying the specializations provided in `pSpecializationInfo`, if any, and then converting all specialization constants into fixed constants
- VUID-VkShaderCreateInfoEXT-codeType-08872
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, `pCode` **must** contain an `OpExecutionMode` instruction specifying the type of subdivision
- VUID-VkShaderCreateInfoEXT-codeType-08873
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, `pCode` **must** contain an `OpExecutionMode` instruction specifying the orientation of triangles generated by the tessellator
- VUID-VkShaderCreateInfoEXT-codeType-08874
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, `pCode` **must** contain an `OpExecutionMode` instruction specifying the spacing of segments on the edges of tessellated primitives
- VUID-VkShaderCreateInfoEXT-codeType-08875
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and `stage` is `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, `pCode` **must** contain an `OpExecutionMode` instruction specifying the output patch size
- VUID-VkShaderCreateInfoEXT-pPushConstantRanges-10063
Any two elements of `pPushConstantRanges` **must** not include the same stage in `stageFlags`
- VUID-VkShaderCreateInfoEXT-codeType-10064
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and if a push constant block is declared in a shader, then an element of `pPushConstantRanges::stageFlags` **must** match `pname::stage`
- VUID-VkShaderCreateInfoEXT-codeType-10065
If `codeType` is `VK_SHADER_CODE_TYPE_SPIRV_EXT`, and if a push constant block is declared in a shader, the block **must** be contained inside the element of `pPushConstantRanges` that matches the stage

Valid Usage (Implicit)

- VUID-VkShaderCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT`
- VUID-VkShaderCreateInfoEXT-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkPipelineShaderStageRequiredSubgroupSizeCreateInfo`
- VUID-VkShaderCreateInfoEXT-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkShaderCreateInfoEXT-flags-parameter
`flags` **must** be a valid combination of `VkShaderCreateFlagBitsEXT` values
- VUID-VkShaderCreateInfoEXT-stage-parameter
`stage` **must** be a valid `VkShaderStageFlagBits` value
- VUID-VkShaderCreateInfoEXT-nextStage-parameter
`nextStage` **must** be a valid combination of `VkShaderStageFlagBits` values
- VUID-VkShaderCreateInfoEXT-codeType-parameter
`codeType` **must** be a valid `VkShaderCodeTypeEXT` value
- VUID-VkShaderCreateInfoEXT-pCode-parameter
`pCode` **must** be a valid pointer to an array of `codeSize` bytes
- VUID-VkShaderCreateInfoEXT-pName-parameter
If `pName` is not `NULL`, `pName` **must** be a null-terminated UTF-8 string
- VUID-VkShaderCreateInfoEXT-pSetLayouts-parameter
If `setLayoutCount` is not `0`, and `pSetLayouts` is not `NULL`, `pSetLayouts` **must** be a valid pointer to an array of `setLayoutCount` valid `VkDescriptorSetLayout` handles
- VUID-VkShaderCreateInfoEXT-pPushConstantRanges-parameter
If `pushConstantRangeCount` is not `0`, and `pPushConstantRanges` is not `NULL`, `pPushConstantRanges` **must** be a valid pointer to an array of `pushConstantRangeCount` valid `VkPushConstantRange` structures
- VUID-VkShaderCreateInfoEXT-pSpecializationInfo-parameter
If `pSpecializationInfo` is not `NULL`, `pSpecializationInfo` **must** be a valid pointer to a valid `VkSpecializationInfo` structure
- VUID-VkShaderCreateInfoEXT-codeSize-arraylength
`codeSize` **must** be greater than `0`

```
// Provided by VK_EXT_shader_object
typedef VkFlags VkShaderCreateFlagsEXT;
```

`VkShaderCreateFlagsEXT` is a bitmask type for setting a mask of zero or more `VkShaderCreateFlagBitsEXT`.

Possible values of the `flags` member of `VkShaderCreateInfoEXT` specifying how a shader object is

created, are:

```
// Provided by VK_EXT_shader_object
typedef enum VkShaderCreateFlagBitsEXT {
    VK_SHADER_CREATE_LINK_STAGE_BIT_EXT = 0x00000001,
    // Provided by VK_EXT_shader_object with VK_EXT_subgroup_size_control or
    VK_VERSION_1_3
    VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT = 0x00000002,
    // Provided by VK_EXT_shader_object with VK_EXT_subgroup_size_control or
    VK_VERSION_1_3
    VK_SHADER_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT = 0x00000004,
    // Provided by VK_EXT_shader_object with VK_EXT_mesh_shader or VK_NV_mesh_shader
    VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT = 0x00000008,
    // Provided by VK_EXT_shader_object with VK_KHR_device_group or VK_VERSION_1_1
    VK_SHADER_CREATE_DISPATCH_BASE_BIT_EXT = 0x00000010,
    // Provided by VK_KHR_fragment_shading_rate with VK_EXT_shader_object
    VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT = 0x00000020,
    // Provided by VK_EXT_fragment_density_map with VK_EXT_shader_object
    VK_SHADER_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT = 0x00000040,
    // Provided by VK_EXT_device_generated_commands
    VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT = 0x00000080,
} VkShaderCreateFlagBitsEXT;
```

- **VK_SHADER_CREATE_LINK_STAGE_BIT_EXT** specifies that a shader is linked to all other shaders created in the same [vkCreateShadersEXT](#) call whose [VkShaderCreateInfoEXT](#) structures' **flags** include **VK_SHADER_CREATE_LINK_STAGE_BIT_EXT**.
- **VK_SHADER_CREATE_DISPATCH_BASE_BIT_EXT** specifies that a compute shader **can** be used with [vkCmdDispatchBaseKHR](#) with a non-zero base workgroup.
- **VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT** specifies that a fragment shader **can** be used with a fragment shading rate attachment.
- **VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT** specifies that the shader **can** be used in combination with [Device-Generated Commands](#).

NOTE

The behavior of **VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT** differs subtly from the behavior of **VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR** in that the shader bit allows, but does not require the shader to be used with that type of attachment. This means that the application need not create multiple shaders when it does not know in advance whether the shader will be used with or without the attachment type, or when it needs the same shader to be compatible with usage both with and without. This **may** come at some performance cost on some implementations, so applications **should** still only set bits that are actually necessary.

Shader objects **can** be created using different types of shader code. Possible values of [VkShaderCreateInfoEXT::codeType](#), are:


```
// Provided by VK_EXT_shader_object
typedef enum VkShaderCodeTypeEXT {
    VK_SHADER_CODE_TYPE_BINARY_EXT = 0,
    VK_SHADER_CODE_TYPE_SPIRV_EXT = 1,
} VkShaderCodeTypeEXT;
```

- `VK_SHADER_CODE_TYPE_BINARY_EXT` specifies shader code in an opaque, implementation-defined binary format specific to the physical device.
- `VK_SHADER_CODE_TYPE_SPIRV_EXT` specifies shader code in SPIR-V format.

9.1.2. Binary Shader Code

Binary shader code **can** be retrieved from a shader object using the command:

```
// Provided by VK_EXT_shader_object
VkResult vkGetShaderBinaryDataEXT(
    VkDevice          device,
    VkShaderEXT       shader,
    size_t*           pDataSize,
    void*             pData);
```

- `device` is the logical device that shader object was created from.
- `shader` is the shader object to retrieve binary shader code from.
- `pDataSize` is a pointer to a `size_t` value related to the size of the binary shader code, as described below.
- `pData` is either `NULL` or a pointer to a buffer.

If `pData` is `NULL`, then the size of the binary shader code of the shader object, in bytes, is returned in `pDataSize`. Otherwise, `pDataSize` **must** point to a variable set by the application to the size of the buffer, in bytes, pointed to by `pData`, and on return the variable is overwritten with the amount of data actually written to `pData`. If `pDataSize` is less than the size of the binary shader code, nothing is written to `pData`, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`.

NOTE

The behavior of this command when `pDataSize` is too small differs from how some other getter-type commands work in Vulkan. Because shader binary data is only usable in its entirety, it would never be useful for the implementation to return partial data. Because of this, nothing is written to `pData` unless `pDataSize` is large enough to fit the data in its entirety.

Binary shader code retrieved using `vkGetShaderBinaryDataEXT` **can** be passed to a subsequent call to `vkCreateShadersEXT` on a compatible physical device by specifying `VK_SHADER_CODE_TYPE_BINARY_EXT` in the `codeType` member of `VkShaderCreateInfoEXT`.

The shader code returned by repeated calls to this function with the same `VkShaderEXT` is guaranteed to be invariant for the lifetime of the `VkShaderEXT` object.

Valid Usage

- VUID-vkGetShaderBinaryDataEXT-None-08461
The `shaderObject` feature **must** be enabled
- VUID-vkGetShaderBinaryDataEXT-None-08499
If `pData` is not `NULL`, it **must** be aligned to 16 bytes

Valid Usage (Implicit)

- VUID-vkGetShaderBinaryDataEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetShaderBinaryDataEXT-shader-parameter
`shader` **must** be a valid `VkShaderEXT` handle
- VUID-vkGetShaderBinaryDataEXT-pDataSize-parameter
`pDataSize` **must** be a valid pointer to a `size_t` value
- VUID-vkGetShaderBinaryDataEXT-pData-parameter
If the value referenced by `pDataSize` is not 0, and `pData` is not `NULL`, `pData` **must** be a valid pointer to an array of `pDataSize` bytes
- VUID-vkGetShaderBinaryDataEXT-shader-parent
`shader` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

9.1.3. Binary Shader Compatibility

Binary shader compatibility means that binary shader code returned from a call to `vkGetShaderBinaryDataEXT` **can** be passed to a later call to `vkCreateShadersEXT`, potentially on a different logical and/or physical device, and that this will result in the successful creation of a shader object functionally equivalent to the shader object that the code was originally queried from.

Binary shader code queried from `vkGetShaderBinaryDataEXT` is not guaranteed to be compatible across all devices, but implementations are required to provide some compatibility guarantees. Applications **may** determine binary shader compatibility using either (or both) of two mechanisms.

Guaranteed compatibility of shader binaries is expressed through a combination of the `shaderBinaryUUID` and `shaderBinaryVersion` members of the `VkPhysicalDeviceShaderObjectPropertiesEXT` structure queried from a physical device. Binary shaders retrieved from a physical device with a certain `shaderBinaryUUID` are guaranteed to be compatible with all other physical devices reporting the same `shaderBinaryUUID` and the same or higher `shaderBinaryVersion`.

Whenever a new version of an implementation incorporates any changes that affect the output of `vkGetShaderBinaryDataEXT`, the implementation **should** either increment `shaderBinaryVersion` if binary shader code retrieved from older versions remains compatible with the new implementation, or else replace `shaderBinaryUUID` with a new value if backward compatibility has been broken. Binary shader code queried from a device with a matching `shaderBinaryUUID` and lower `shaderBinaryVersion` relative to the device on which `vkCreateShadersEXT` is being called **may** be suboptimal for the new device in ways that do not change shader functionality, but it is still guaranteed to be usable to successfully create the shader object(s).

NOTE

Implementations are encouraged to share `shaderBinaryUUID` between devices and driver versions to the maximum extent their hardware naturally allows, and are **strongly** discouraged from ever changing the `shaderBinaryUUID` for the same hardware except unless absolutely necessary.

In addition to the shader compatibility guarantees described above, it is valid for an application to call `vkCreateShadersEXT` with binary shader code created on a device with a different or unknown `shaderBinaryUUID` and/or higher `shaderBinaryVersion`. In this case, the implementation **may** use any unspecified means of its choosing to determine whether the provided binary shader code is usable. If it is, `vkCreateShadersEXT` **must** return `VK_SUCCESS`, and the created shader object is guaranteed to be valid. Otherwise, in the absence of some error, `vkCreateShadersEXT` **must** return `VK_INCOMPATIBLE_SHADER_BINARY_EXT` to indicate that the provided binary shader code is not compatible with the device.

9.1.4. Binding Shader Objects

Once shader objects have been created, they **can** be bound to the command buffer using the command:

```
// Provided by VK_EXT_shader_object
void vkCmdBindShadersEXT(
    VkCommandBuffer          commandBuffer,
    uint32_t                 stageCount,
    const VkShaderStageFlags* pStages,
    const VkShaderEXT*        pShaders);
```

- `commandBuffer` is the command buffer that the shader object will be bound to.
- `stageCount` is the length of the `pStages` and `pShaders` arrays.
- `pStages` is a pointer to an array of `VkShaderStageFlags` values specifying one stage per array index that is affected by the corresponding value in the `pShaders` array.

- `pShaders` is a pointer to an array of `VkShaderEXT` handles and/or `VK_NULL_HANDLE` values describing the shader binding operations to be performed on each stage in `pStages`.

When binding linked shaders, an application **may** bind them in any combination of one or more calls to `vkCmdBindShadersEXT` (i.e., shaders that were created linked together do not need to be bound in the same `vkCmdBindShadersEXT` call).

Any shader object bound to a particular stage **may** be unbound by setting its value in `pShaders` to `VK_NULL_HANDLE`. If `pShaders` is `NULL`, `vkCmdBindShadersEXT` behaves as if `pShaders` was an array of `stageCount` `VK_NULL_HANDLE` values (i.e., any shaders bound to the stages specified in `pStages` are unbound).

Valid Usage

- VUID-vkCmdBindShadersEXT-None-08462
The `shaderObject` feature **must** be enabled
- VUID-vkCmdBindShadersEXT-pStages-08463
Every element of `pStages` **must** be unique
- VUID-vkCmdBindShadersEXT-pStages-08464
`pStages` **must** not contain `VK_SHADER_STAGE_ALL_GRAPHICS` or `VK_SHADER_STAGE_ALL`
- VUID-vkCmdBindShadersEXT-pStages-08465
`pStages` **must** not contain `VK_SHADER_STAGE_RAYGEN_BIT_KHR`,
`VK_SHADER_STAGE_ANY_HIT_BIT_KHR`, `VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR`,
`VK_SHADER_STAGE_MISS_BIT_KHR`, `VK_SHADER_STAGE_INTERSECTION_BIT_KHR`, or
`VK_SHADER_STAGE_CALLABLE_BIT_KHR`
- VUID-vkCmdBindShadersEXT-pShaders-08469
For each element of `pStages`, if `pShaders` is not `NULL`, and the element of the `pShaders` array with the same index is not `VK_NULL_HANDLE`, it **must** have been created with a `stage` equal to the corresponding element of `pStages`
- VUID-vkCmdBindShadersEXT-pShaders-08474
If the `tessellationShader` feature is not enabled, and `pStages` contains `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` or
`VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, and `pShaders` is not `NULL`, the same index or indices in `pShaders` **must** be `VK_NULL_HANDLE`
- VUID-vkCmdBindShadersEXT-pShaders-08475
If the `geometryShader` feature is not enabled, and `pStages` contains `VK_SHADER_STAGE_GEOMETRY_BIT`, and `pShaders` is not `NULL`, the same index in `pShaders` **must** be `VK_NULL_HANDLE`
- VUID-vkCmdBindShadersEXT-pShaders-08476
If `pStages` contains `VK_SHADER_STAGE_COMPUTE_BIT`, the `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdBindShadersEXT-pShaders-08477
If `pStages` contains `VK_SHADER_STAGE_VERTEX_BIT`, `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`,
`VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, `VK_SHADER_STAGE_GEOMETRY_BIT`, or
`VK_SHADER_STAGE_FRAGMENT_BIT`, the `VkCommandPool` that `commandBuffer` was allocated from

must support graphics operations

Valid Usage (Implicit)

- VUID-vkCmdBindShadersEXT-commandBuffer-parameter
commandBuffer **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdBindShadersEXT-pStages-parameter
pStages **must** be a valid pointer to an array of **stageCount** valid [VkShaderStageFlagBits](#) values
- VUID-vkCmdBindShadersEXT-pShaders-parameter
If **pShaders** is not **NULL**, **pShaders** **must** be a valid pointer to an array of **stageCount** valid or [VK_NULL_HANDLE](#) [VkShaderEXT](#) handles
- VUID-vkCmdBindShadersEXT-commandBuffer-recording
commandBuffer **must** be in the [recording state](#)
- VUID-vkCmdBindShadersEXT-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics, or compute operations
- VUID-vkCmdBindShadersEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindShadersEXT-stageCount-arraylength
stageCount **must** be greater than 0
- VUID-vkCmdBindShadersEXT-commonparent
Both of **commandBuffer**, and the elements of **pShaders** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	State

9.1.5. Setting State

Whenever shader objects are used to issue drawing commands, the appropriate [dynamic state](#) setting commands **must** have been called to set the relevant state in the command buffer prior to drawing:

- [vkCmdSetViewportWithCountEXT](#)
- [vkCmdSetScissorWithCountEXT](#)
- [vkCmdSetRasterizerDiscardEnableEXT](#)
- [vkCmdSetVertexInputEXT](#)
- [vkCmdSetPrimitiveTopologyEXT](#)
- [vkCmdSetPrimitiveRestartEnableEXT](#)

If a shader is bound to the [VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT](#) stage, the following command **must** have been called in the command buffer prior to drawing:

- [vkCmdSetPatchControlPointsEXT](#), if [primitiveTopology](#) is [VK_PRIMITIVE_TOPOLOGY_PATCH_LIST](#)

If a shader is bound to the [VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT](#) stage, the following command **must** have been called in the command buffer prior to drawing:

- [vkCmdSetTessellationDomainOriginEXT](#)

If [rasterizerDiscardEnable](#) is [VK_FALSE](#), the following commands **must** have been called in the command buffer prior to drawing:

- [vkCmdSetRasterizationSamplesEXT](#)
- [vkCmdSetSampleMaskEXT](#)
- [vkCmdSetAlphaToCoverageEnableEXT](#)
- [vkCmdSetAlphaToOneEnableEXT](#), if the [alphaToOne](#) feature is enabled on the device
- [vkCmdSetPolygonModeEXT](#)
- [vkCmdSetLineWidth](#), if [polygonMode](#) is [VK_POLYGON_MODE_LINE](#), or if [primitiveTopology](#) is a line topology, or if a shader which outputs line primitives is bound to the [VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT](#) or [VK_SHADER_STAGE_GEOMETRY_BIT](#) stage
- [vkCmdSetCullModeEXT](#)
- [vkCmdSetFrontFaceEXT](#)
- [vkCmdSetDepthTestEnableEXT](#)
- [vkCmdSetDepthWriteEnableEXT](#)
- [vkCmdSetDepthCompareOpEXT](#), if [depthTestEnable](#) is [VK_TRUE](#)
- [vkCmdSetDepthBoundsTestEnableEXT](#), if the [depthBounds](#) feature is enabled on the device
- [vkCmdSetDepthBounds](#), if [depthBoundsTestEnable](#) is [VK_TRUE](#)
- [vkCmdSetDepthBiasEnableEXT](#)

- `vkCmdSetDepthBias` or `vkCmdSetDepthBias2EXT`, if `depthBiasEnable` is `VK_TRUE`
- `vkCmdSetDepthClampEnableEXT`, if the `depthClamp` feature is enabled on the device
- `vkCmdSetStencilTestEnableEXT`
- `vkCmdSetStencilOpEXT`, if `stencilTestEnable` is `VK_TRUE`
- `vkCmdSetStencilCompareMask`, if `stencilTestEnable` is `VK_TRUE`
- `vkCmdSetStencilWriteMask`, if `stencilTestEnable` is `VK_TRUE`
- `vkCmdSetStencilReference`, if `stencilTestEnable` is `VK_TRUE`

If a shader is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and `rasterizerDiscardEnable` is `VK_FALSE`, the following commands **must** have been called in the command buffer prior to drawing:

- `vkCmdSetLogicOpEnableEXT`, if the `logicOp` feature is enabled on the device
- `vkCmdSetLogicOpEXT`, if `logicOpEnable` is `VK_TRUE`
- `vkCmdSetColorBlendEnableEXT` and `vkCmdSetColorWriteMaskEXT`, if color attachments are bound, with values set for every color attachment in the render pass instance active at draw time
- `vkCmdSetColorBlendEquationEXT` or `vkCmdSetColorBlendAdvancedEXT`, if color attachments are bound, for every attachment whose index in `pColorBlendEnables` is a pointer to a value of `VK_TRUE`
- `vkCmdSetBlendConstants`, if any index in `pColorBlendEnables` is `VK_TRUE`, and the same index in `pColorBlendEquations` is a `VkColorBlendEquationEXT` structure with any `VkBlendFactor` member with a value of `VK_BLEND_FACTOR_CONSTANT_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR`, `VK_BLEND_FACTOR_CONSTANT_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`

If the `pipelineFragmentShadingRate` feature is enabled on the device, and a shader is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and `rasterizerDiscardEnable` is `VK_FALSE`, the following command **must** have been called in the command buffer prior to drawing:

- `vkCmdSetFragmentShadingRateKHR`

If the `geometryStreams` feature is enabled on the device, and a shader is bound to the `VK_SHADER_STAGE_GEOMETRY_BIT` stage, the following command **must** have been called in the command buffer prior to drawing:

- `vkCmdSetRasterizationStreamEXT`

If the `VK_EXT_discard_rectangles` extension is enabled on the device, and `rasterizerDiscardEnable` is `VK_FALSE`, the following commands **must** have been called in the command buffer prior to drawing:

- `vkCmdSetDiscardRectangleEnableEXT`
- `vkCmdSetDiscardRectangleModeEXT`, if `discardRectangleEnable` is `VK_TRUE`
- `vkCmdSetDiscardRectangleEXT`, if `discardRectangleEnable` is `VK_TRUE`

If `VK_EXT_conservative_rasterization` extension is enabled on the device, and `rasterizerDiscardEnable` is `VK_FALSE`, the following commands **must** have been called in the

command buffer prior to drawing:

- `vkCmdSetConservativeRasterizationModeEXT`
- `vkCmdSetExtraPrimitiveOverestimationSizeEXT`, if `conservativeRasterizationMode` is `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`

If the `depthClipEnable` feature is enabled on the device, the following command **must** have been called in the command buffer prior to drawing:

- `vkCmdSetDepthClipEnableEXT`

If the `VK_EXT_sample_locations` extension is enabled on the device, and `rasterizerDiscardEnable` is `VK_FALSE`, the following commands **must** have been called in the command buffer prior to drawing:

- `vkCmdSetSampleLocationsEnableEXT`
- `vkCmdSetSampleLocationsEXT`, if `sampleLocationsEnable` is `VK_TRUE`

If the `VK_EXT_provoking_vertex` extension is enabled on the device, and `rasterizerDiscardEnable` is `VK_FALSE`, and a shader is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, the following command **must** have been called in the command buffer prior to drawing:

- `vkCmdSetProvokingVertexModeEXT`

If any of the `<features-stippledRectangularLines, stippledRectangularLines>>`, `<features-stippledBresenhamLines, stippledBresenhamLines>>`, or `<features-stippledSmoothLines, stippledSmoothLines>>` features are enabled on the device, and `rasterizerDiscardEnable` is `VK_FALSE`, and if `polygonMode` is `VK_POLYGON_MODE_LINE` or a shader is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage and `primitiveTopology` is a line topology or a shader which outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, the following commands **must** have been called in the command buffer prior to drawing:

- `vkCmdSetLineRasterizationModeEXT`
- `vkCmdSetLineStippleEnableEXT`
- `vkCmdSetLineStippleKHR`, if `stippledLineEnable` is `VK_TRUE`

If the `depthClipControl` feature is enabled on the device, the following command **must** have been called in the command buffer prior to drawing:

- `vkCmdSetDepthClipNegativeOneToOneEXT`

If the `colorWriteEnable` feature is enabled on the device, and a shader is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and `rasterizerDiscardEnable` is `VK_FALSE`, the following command **must** have been called in the command buffer prior to drawing:

- `vkCmdSetColorWriteEnableEXT`, with values set for every color attachment in the render pass instance active at draw time

If the `attachmentFeedbackLoopDynamicState` feature is enabled on the device, and a shader is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and `rasterizerDiscardEnable` is `VK_FALSE`, the

following command **must** have been called in the command buffer prior to drawing:

- [vkCmdSetAttachmentFeedbackLoopEnableEXT](#)

If the [depthClamp](#) and [depthClampControl](#) feature are enabled on the device, and [depthClampEnable](#) is [VK_TRUE](#), the following command **must** have been called in the command buffer prior to drawing:

- [vkCmdSetDepthClampRangeEXT](#)

State **can** be set either at any time before or after shader objects are bound, but all required state **must** be set prior to issuing drawing commands.

9.1.6. Interaction With Pipelines

Calling [vkCmdBindShadersEXT](#) causes the pipeline bind points [corresponding to each stage](#) in [pStages](#) to be disturbed, meaning that any [pipelines](#) that had previously been bound to those pipeline bind points are no longer bound.

If [VK_PIPELINE_BIND_POINT_GRAPHICS](#) is disturbed (i.e., if [pStages](#) contains any graphics stage), any graphics pipeline state that the previously bound pipeline did not specify as [dynamic](#) becomes undefined, and **must** be set in the command buffer before issuing drawing commands using shader objects.

Calls to [vkCmdBindPipeline](#) likewise disturb the shader stage(s) corresponding to [pipelineBindPoint](#), meaning that any shaders that had previously been bound to any of those stages are no longer bound, even if the pipeline was created without shaders for some of those stages.

9.1.7. Shader Object Destruction

To destroy a shader object, call:

```
// Provided by VK_EXT_shader_object
void vkDestroyShaderEXT(
    VkDevice          device,
    VkShaderEXT       shader,
    const VkAllocationCallbacks* pAllocator);
```

- [device](#) is the logical device that destroys the shader object.
- [shader](#) is the handle of the shader object to destroy.
- [pAllocator](#) controls host memory allocation as described in the [Memory Allocation](#) chapter.

Destroying a shader object used by one or more command buffers in the [recording or executable state](#) causes those command buffers to move into the *invalid state*.

Valid Usage

- VUID-vkDestroyShaderEXT-None-08481
The [shaderObject](#) feature **must** be enabled

- VUID-vkDestroyShaderEXT-shader-08482
All submitted commands that refer to **shader** **must** have completed execution
- VUID-vkDestroyShaderEXT-pAllocator-08483
If **VkAllocationCallbacks** were provided when **shader** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyShaderEXT-pAllocator-08484
If no **VkAllocationCallbacks** were provided when **shader** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyShaderEXT-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroyShaderEXT-shader-parameter
If **shader** is not **VK_NULL_HANDLE**, **shader** **must** be a valid **VkShaderEXT** handle
- VUID-vkDestroyShaderEXT-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroyShaderEXT-shader-parent
If **shader** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **shader** **must** be externally synchronized

9.2. Shader Modules

Shader modules contain *shader code* and one or more entry points. Shaders are selected from a shader module by specifying an entry point as part of **pipeline** creation. The stages of a pipeline **can** use shaders that come from different modules. The shader code defining a shader module **must** be in the SPIR-V format, as described by the **Vulkan Environment for SPIR-V** appendix.

Shader modules are represented by **VkShaderModule** handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkShaderModule)
```

To create a shader module, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateShaderModule(
    VkDevice                                device,
    const VkShaderModuleCreateInfo*         pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkShaderModule*                         pShaderModule);
```

- **device** is the logical device that creates the shader module.
- **pCreateInfo** is a pointer to a [VkShaderModuleCreateInfo](#) structure.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pShaderModule** is a pointer to a [VkShaderModule](#) handle in which the resulting shader module object is returned.

Once a shader module has been created, any entry points it contains **can** be used in pipeline shader stages as described in [Compute Pipelines](#) and [Graphics Pipelines](#).

NOTE

If the **maintenance5** feature is enabled, shader module creation can be omitted entirely. Instead, applications should provide the [VkShaderModuleCreateInfo](#) structure directly in to pipeline creation by chaining it to [VkPipelineShaderStageCreateInfo](#). This avoids the overhead of creating and managing an additional object.

Valid Usage

- VUID-vkCreateShaderModule-pCreateInfo-06904
If **pCreateInfo** is not **NULL**, **pCreateInfo->pNext** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkCreateShaderModule-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCreateShaderModule-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkShaderModuleCreateInfo](#) structure
- VUID-vkCreateShaderModule-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateShaderModule-pShaderModule-parameter
pShaderModule **must** be a valid pointer to a [VkShaderModule](#) handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkShaderModuleCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkShaderModuleCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkShaderModuleCreateFlags flags;
    size_t               codeSize;
    const uint32_t*       pCode;
} VkShaderModuleCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `codeSize` is the size, in bytes, of the code pointed to by `pCode`.
- `pCode` is a pointer to code that is used to create the shader module. The type and format of the code is determined from the content of the memory addressed by `pCode`.

Valid Usage

- VUID-VkShaderModuleCreateInfo-codeSize-08735
`codeSize` **must** be a multiple of 4
- VUID-VkShaderModuleCreateInfo-pCode-08736
`pCode` **must** point to valid SPIR-V code, formatted and packed as described by the [Khronos SPIR-V Specification](#)
- VUID-VkShaderModuleCreateInfo-pCode-08737
`pCode` **must** adhere to the validation rules described by the [Validation Rules within a Module](#) section of the [SPIR-V Environment](#) appendix
- VUID-VkShaderModuleCreateInfo-pCode-08738
`pCode` **must** declare the `Shader` capability for SPIR-V code
- VUID-VkShaderModuleCreateInfo-pCode-08739
`pCode` **must** not declare any capability that is not supported by the API, as described by the [Capabilities](#) section of the [SPIR-V Environment](#) appendix

- VUID-VkShaderModuleCreateInfo-pCode-08740
and **pCode** declares any of the capabilities listed in the [SPIR-V Environment](#) appendix, one of the corresponding requirements **must** be satisfied
- VUID-VkShaderModuleCreateInfo-pCode-08741
pCode **must** not declare any SPIR-V extension that is not supported by the API, as described by the [Extension](#) section of the [SPIR-V Environment](#) appendix
- VUID-VkShaderModuleCreateInfo-pCode-08742
and **pCode** declares any of the SPIR-V extensions listed in the [SPIR-V Environment](#) appendix, one of the corresponding requirements **must** be satisfied
- VUID-VkShaderModuleCreateInfo-codeSize-01085
codeSize **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkShaderModuleCreateInfo-sType-sType
sType **must** be `VK_STRUCTURE_TYPE_SHADER_MODULE_CREATE_INFO`
- VUID-VkShaderModuleCreateInfo-flags-zeroBitmask
flags **must** be 0
- VUID-VkShaderModuleCreateInfo-pCode-parameter
pCode **must** be a valid pointer to an array of $\frac{\text{codeSize}}{4}$ `uint32_t` values

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkShaderModuleCreateFlags;
```

`VkShaderModuleCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

To destroy a shader module, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyShaderModule(
    VkDevice          device,
    VkShaderModule    shaderModule,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the shader module.
- **shaderModule** is the handle of the shader module to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

A shader module **can** be destroyed while pipelines created using its shaders are still in use.

Valid Usage

- VUID-vkDestroyShaderModule-shaderModule-01092
If `VkAllocationCallbacks` were provided when `shaderModule` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyShaderModule-shaderModule-01093
If no `VkAllocationCallbacks` were provided when `shaderModule` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyShaderModule-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyShaderModule-shaderModule-parameter
If `shaderModule` is not `VK_NULL_HANDLE`, `shaderModule` **must** be a valid `VkShaderModule` handle
- VUID-vkDestroyShaderModule-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyShaderModule-shaderModule-parent
If `shaderModule` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `shaderModule` **must** be externally synchronized

9.3. Binding Shaders

Before a shader can be used it **must** be first bound to the command buffer.

Calling `vkCmdBindPipeline` binds all stages corresponding to the `VkPipelineBindPoint`. Calling `vkCmdBindShadersEXT` binds all stages in `pStages`

The following table describes the relationship between shader stages and pipeline bind points:

Shader stage	Pipeline bind point	behavior controlled
• VK_SHADER_STAGE_VERTEX_BIT • VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT • VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT • VK_SHADER_STAGE_GEOMETRY_BIT • VK_SHADER_STAGE_FRAGMENT_BIT	VK_PIPELINE_BIND_POINT_GRAPHICS	all drawing commands
• VK_SHADER_STAGE_COMPUTE_BIT	VK_PIPELINE_BIND_POINT_COMPUTE	all dispatch commands
• VK_SHADER_STAGE_ANY_HIT_BIT_KHR • VK_SHADER_STAGE_CALLABLE_BIT_KHR • VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR • VK_SHADER_STAGE_INTERSECTION_BIT_KHR • VK_SHADER_STAGE_MISS_BIT_KHR • VK_SHADER_STAGE_RAYGEN_BIT_KHR	VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR	vkCmdTraceRaysKHR and vkCmdTraceRaysIndirectKHR

9.4. Shader Execution

At each stage of the pipeline, multiple invocations of a shader **may** execute simultaneously. Further, invocations of a single shader produced as the result of different commands **may** execute simultaneously. The relative execution order of invocations of the same shader type is undefined. Shader invocations **may** complete in a different order than that in which the primitives they originated from were drawn or dispatched by the application. However, fragment shader outputs are written to attachments in [rasterization order](#).

The relative execution order of invocations of different shader types is largely undefined. However, when invoking a shader whose inputs are generated from a previous pipeline stage, the shader invocations from the previous stage are guaranteed to have executed far enough to generate input values for all required inputs.

9.4.1. Shader Termination

A shader invocation that is *terminated* has finished executing instructions.

Executing `OpReturn` in the entry point, or executing `OpTerminateInvocation` in any function will

terminate an invocation. Implementations **may** also terminate a shader invocation when `OpKill` is executed in any function; otherwise it becomes a [helper invocation](#).

In addition to the above conditions, [helper invocations](#) **may** be terminated when all non-helper invocations in the same [derivative group](#) either terminate or become [helper invocations](#).

A shader stage for a given command completes execution when all invocations for that stage have terminated.

NOTE

`OpKill` will behave the same as either `OpTerminateInvocation` or `OpDemoteToHelperInvocation` depending on the implementation. It is recommended that shader authors use `OpTerminateInvocation` or `OpDemoteToHelperInvocation` instead of `OpKill` whenever possible to produce more predictable behavior.

9.5. Shader Memory Access Ordering

The order in which image or buffer memory is read or written by shaders is largely undefined. For some shader types (vertex, tessellation evaluation, and in some cases, fragment), even the number of shader invocations that **may** perform loads and stores is undefined.

In particular, the following rules apply:

- [Vertex](#) and [tessellation evaluation](#) shaders will be invoked at least once for each unique vertex, as defined in those sections.
- [Fragment](#) shaders will be invoked zero or more times, as defined in that section.
- The relative execution order of invocations of the same shader type is undefined. A store issued by a shader when working on primitive B might complete prior to a store for primitive A, even if primitive A is specified prior to primitive B. This applies even to fragment shaders; while fragment shader outputs are always written to the framebuffer in [rasterization order](#), stores executed by fragment shader invocations are not.
- The relative execution order of invocations of different shader types is largely undefined.

NOTE

The above limitations on shader invocation order make some forms of synchronization between shader invocations within a single set of primitives unimplementable. For example, having one invocation poll memory written by another invocation assumes that the other invocation has been launched and will complete its writes in finite time.

The [Memory Model](#) appendix defines the terminology and rules for how to correctly communicate between shader invocations, such as when a write is [Visible-To](#) a read, and what constitutes a [Data Race](#).

Applications **must** not cause a data race.

The SPIR-V **SubgroupMemory**, **CrossWorkgroupMemory**, and **AtomicCounterMemory** memory semantics are ignored. Sequentially consistent atomics and barriers are not supported and **SequentiallyConsistent** is treated as **AcquireRelease**. **SequentiallyConsistent** **should** not be

used.

9.6. Shader Inputs and Outputs

Data is passed into and out of shaders using variables with input or output storage class, respectively. User-defined inputs and outputs are connected between stages by matching their **Location** decorations. Additionally, data **can** be provided by or communicated to special functions provided by the execution environment using **BuiltIn** decorations.

In many cases, the same **BuiltIn** decoration **can** be used in multiple shader stages with similar meaning. The specific behavior of variables decorated as **BuiltIn** is documented in the following sections.

9.7. Vertex Shaders

Each vertex shader invocation operates on one vertex and its associated **vertex attribute** data, and outputs one vertex and associated data. Graphics pipelines **must** include a vertex shader, and the vertex shader stage is always the first shader stage in the graphics pipeline.

9.7.1. Vertex Shader Execution

A vertex shader **must** be executed at least once for each vertex specified by a drawing command. If the subpass includes multiple views in its view mask, the shader **may** be invoked separately for each view. During execution, the shader is presented with the index of the vertex and instance for which it has been invoked. Input variables declared in the vertex shader are filled by the implementation with the values of vertex attributes associated with the invocation being executed.

If the same vertex is specified multiple times in a drawing command (e.g. by including the same index value multiple times in an index buffer) the implementation **may** reuse the results of vertex shading if it can statically determine that the vertex shader invocations will produce identical results.

NOTE

It is implementation-dependent when and if results of vertex shading are reused, and thus how many times the vertex shader will be executed. This is true also if the vertex shader contains stores or atomic operations (see **vertexPipelineStoresAndAtomics**).

9.8. Tessellation Control Shaders

The tessellation control shader is used to read an input patch provided by the application and to produce an output patch. Each tessellation control shader invocation operates on an input patch (after all control points in the patch are processed by a vertex shader) and its associated data, and outputs a single control point of the output patch and its associated data, and **can** also output additional per-patch data. The input patch is sized according to the **patchControlPoints** member of **VkPipelineTessellationStateCreateInfo**, as part of input assembly.

The input patch can also be dynamically sized with **patchControlPoints** parameter of

[vkCmdSetPatchControlPointsEXT](#).

To [dynamically set](#) the number of control points per patch, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetPatchControlPointsEXT(
    VkCommandBuffer          commandBuffer,
    uint32_t                 patchControlPoints);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `patchControlPoints` specifies the number of control points per patch.

This command sets the number of control points per patch for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the [VkPipelineTessellationStateCreateInfo::patchControlPoints](#) value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetPatchControlPointsEXT-None-09422
At least one of the following **must** be true:
 - The `shaderObject` feature is enabled
- VUID-vkCmdSetPatchControlPointsEXT-patchControlPoints-04874
`patchControlPoints` **must** be greater than zero and less than or equal to `VkPhysicalDeviceLimits::maxTessellationPatchSize`

Valid Usage (Implicit)

- VUID-vkCmdSetPatchControlPointsEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetPatchControlPointsEXT-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdSetPatchControlPointsEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetPatchControlPointsEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

The size of the output patch is controlled by the `OpExecutionMode OutputVertices` specified in the tessellation control or tessellation evaluation shaders, which **must** be specified in at least one of the shaders. The size of the input and output patches **must** each be greater than zero and less than or equal to `VkPhysicalDeviceLimits::maxTessellationPatchSize`.

9.8.1. Tessellation Control Shader Execution

A tessellation control shader is invoked at least once for each *output* vertex in a patch. If the subpass includes multiple views in its view mask, the shader **may** be invoked separately for each view.

Inputs to the tessellation control shader are generated by the vertex shader. Each invocation of the tessellation control shader **can** read the attributes of any incoming vertices and their associated data. The invocations corresponding to a given patch execute logically in parallel, with undefined relative execution order. However, the `OpControlBarrier` instruction **can** be used to provide limited control of the execution order by synchronizing invocations within a patch, effectively dividing tessellation control shader execution into a set of phases. Tessellation control shaders will read undefined values if one invocation reads a per-vertex or per-patch output written by another invocation at any point during the same phase, or if two invocations attempt to write different values to the same per-patch output in a single phase.

9.9. Tessellation Evaluation Shaders

The Tessellation Evaluation Shader operates on an input patch of control points and their associated data, and a single input barycentric coordinate indicating the invocation's relative position within the subdivided patch, and outputs a single vertex and its associated data.

9.9.1. Tessellation Evaluation Shader Execution

A tessellation evaluation shader is invoked at least once for each unique vertex generated by the tessellator. If the subpass includes multiple views in its view mask, the shader **may** be invoked separately for each view.

9.10. Geometry Shaders

The geometry shader operates on a group of vertices and their associated data assembled from a

single input primitive, and emits zero or more output primitives and the group of vertices and their associated data required for each output primitive.

9.10.1. Geometry Shader Execution

A geometry shader is invoked at least once for each primitive produced by the tessellation stages, or at least once for each primitive generated by [primitive assembly](#) when tessellation is not in use. A shader can request that the geometry shader runs multiple [instances](#). A geometry shader is invoked at least once for each instance. If the subpass includes multiple views in its view mask, the shader **may** be invoked separately for each view.

9.11. Fragment Shaders

Fragment shaders are invoked as a [fragment operation](#) in a graphics pipeline. Each fragment shader invocation operates on a single fragment and its associated data. With few exceptions, fragment shaders do not have access to any data associated with other fragments and are considered to execute in isolation of fragment shader invocations associated with other fragments.

9.12. Compute Shaders

Compute shaders are invoked via [vkCmdDispatch](#) and [vkCmdDispatchIndirect](#) commands. In general, they have access to similar resources as shader stages executing as part of a graphics pipeline.

Compute workloads are formed from groups of work items called workgroups and processed by the compute shader in the current compute pipeline. A workgroup is a collection of shader invocations that execute the same shader, potentially in parallel. Compute shaders execute in *global workgroups* which are divided into a number of *local workgroups* with a size that **can** be set by assigning a value to the [LocalSize](#) or [LocalSizeId](#) execution mode or via an object decorated by the [WorkgroupSize](#) decoration. An invocation within a local workgroup **can** share data with other members of the local workgroup through shared variables and issue memory and control flow barriers to synchronize with other members of the local workgroup.

9.13. Ray Generation Shaders

A ray generation shader is similar to a compute shader. Its main purpose is to execute ray tracing queries using [pipeline trace ray](#) instructions (such as [OpTraceRayKHR](#)) and process the results.

9.13.1. Ray Generation Shader Execution

One ray generation shader is executed per ray tracing dispatch. Its location in the shader binding table (see [Shader Binding Table](#) for details) is passed directly into [vkCmdTraceRaysKHR](#) using the [pRaygenShaderBindingTable](#) parameter or .

9.14. Intersection Shaders

Intersection shaders enable the implementation of arbitrary, application defined geometric

primitives. An intersection shader for a primitive is executed whenever its axis-aligned bounding box is hit by a ray.

Like other ray tracing shader domains, an intersection shader operates on a single ray at a time. It also operates on a single primitive at a time. It is therefore the purpose of an intersection shader to compute the ray-primitive intersections and report them. To report an intersection, the shader calls the `OpReportIntersectionKHR` instruction.

An intersection shader communicates with any-hit and closest shaders by generating attribute values that they **can** read. Intersection shaders **cannot** read or modify the ray payload.

9.14.1. Intersection Shader Execution

The order in which intersections are found along a ray, and therefore the order in which intersection shaders are executed, is unspecified.

The intersection shader of the closest AABB which intersects the ray is guaranteed to be executed at some point during traversal, unless the ray is forcibly terminated.

9.15. Any-Hit Shaders

The any-hit shader is executed after the intersection shader reports an intersection that lies within the current $[t_{\min}, t_{\max}]$ of the ray. The main use of any-hit shaders is to programmatically decide whether or not an intersection will be accepted. The intersection will be accepted unless the shader calls the `OpIgnoreIntersectionKHR` instruction. Any-hit shaders have read-only access to the attributes generated by the corresponding intersection shader, and **can** read or modify the ray payload.

9.15.1. Any-Hit Shader Execution

The order in which intersections are found along a ray, and therefore the order in which any-hit shaders are executed, is unspecified.

The any-hit shader of the closest hit is guaranteed to be executed at some point during traversal, unless the ray is forcibly terminated.

9.16. Closest Hit Shaders

Closest hit shaders have read-only access to the attributes generated by the corresponding intersection shader, and **can** read or modify the ray payload. They also have access to a number of system-generated values. Closest hit shaders **can** call `pipeline trace ray` instructions to recursively trace rays.

9.16.1. Closest Hit Shader Execution

Exactly one closest hit shader is executed when traversal is finished and an intersection has been found and accepted.

9.17. Miss Shaders

Miss shaders **can** access the ray payload and **can** trace new rays through the [pipeline trace ray](#) instructions, but **cannot** access attributes since they are not associated with an intersection.

9.17.1. Miss Shader Execution

A miss shader is executed instead of a closest hit shader if no intersection was found during traversal.

9.18. Callable Shaders

Callable shaders **can** access a callable payload that works similarly to ray payloads to do subroutine work.

9.18.1. Callable Shader Execution

A callable shader is executed by calling `OpExecuteCallableKHR` from an allowed shader stage.

9.19. Interpolation Decorations

Variables in the `Input` storage class in a fragment shader's interface are interpolated from the values specified by the primitive being rasterized.

NOTE

Interpolation decorations can be present on input and output variables in pre-rasterization shaders but have no effect on the interpolation performed.

An undecorated input variable will be interpolated with perspective-correct interpolation according to the primitive type being rasterized. `Lines` and `polygons` are interpolated in the same way as the primitive's clip coordinates. If the `NoPerspective` decoration is present, linear interpolation is instead used for `lines` and `polygons`. For points, as there is only a single vertex, input values are never interpolated and instead take the value written for the single vertex.

If the `Flat` decoration is present on an input variable, the value is not interpolated, and instead takes its value directly from the `provoking vertex`. Fragment shader inputs that are signed or unsigned integers, integer vectors, or any double-precision floating-point type **must** be decorated with `Flat`.

Interpolation of input variables is performed at an implementation-defined position within the fragment area being shaded. The position is further constrained as follows:

- If the `Centroid` decoration is used, the interpolation position used for the variable **must** also fall within the bounds of the primitive being rasterized.
- If the `Sample` decoration is used, the interpolation position used for the variable **must** be at the position of the sample being shaded by the current fragment shader invocation.
- If a sample count of 1 is used, the interpolation position **must** be at the center of the fragment area.

NOTE

As **Centroid** restricts the possible interpolation position to the covered area of the primitive, the position can be forced to vary between neighboring fragments when it otherwise would not. Derivatives calculated based on these differing locations can produce inconsistent results compared to undecorated inputs. It is recommended that input variables used in derivative calculations are not decorated with **Centroid**.

If the **PerVertexKHR** decoration is present on an input variable, the value is not interpolated, and instead values from all input vertices are available in an array. Each index of the array corresponds to one of the vertices of the primitive that produced the fragment.

9.20. Static Use

A SPIR-V module declares a global object in memory using the **OpVariable** instruction, which results in a pointer **x** to that object. A specific entry point in a SPIR-V module is said to *statically use* that object if that entry point's call tree contains a function containing a instruction with **x** as an **id** operand. A shader entry point also *statically uses* any variables explicitly declared in its interface.

9.21. Scope

A *scope* describes a set of shader invocations, where each such set is a *scope instance*. Each invocation belongs to one or more scope instances, but belongs to no more than one scope instance for each scope.

The operations available between invocations in a given scope instance vary, with smaller scopes generally able to perform more operations, and with greater efficiency.

9.21.1. Cross Device

All invocations executed in a Vulkan instance fall into a single *cross device scope instance*.

Whilst the **CrossDevice** scope is defined in SPIR-V, it is disallowed in Vulkan. API [synchronization](#) commands **can** be used to communicate between devices.

9.21.2. Device

All invocations executed on a single device form a *device scope instance*.

If the **vulkanMemoryModel** and **vulkanMemoryModelDeviceScope** features are enabled, this scope is represented in SPIR-V by the **Device Scope**, which **can** be used as a **Memory Scope** for barrier and atomic operations.

If both the **shaderDeviceClock** and **vulkanMemoryModelDeviceScope** features are enabled, using the **Device Scope** with the **OpReadClockKHR** instruction will read from a clock that is consistent across invocations in the same device scope instance.

There is no method to synchronize the execution of these invocations within SPIR-V, and this **can** only be done with API synchronization primitives.

Invocations executing on different devices in a device group operate in separate device scope instances.

9.21.3. Queue Family

Invocations executed by queues in a given queue family form a *queue family scope instance*.

This scope is identified in SPIR-V as the **QueueFamily Scope** if the **vulkanMemoryModel** feature is enabled, or if not, the **Device Scope**, which **can** be used as a **Memory Scope** for barrier and atomic operations.

If the **shaderDeviceClock** feature is enabled, but the **vulkanMemoryModelDeviceScope** feature is not enabled, using the **Device Scope** with the **OpReadClockKHR** instruction will read from a clock that is consistent across invocations in the same queue family scope instance.

There is no method to synchronize the execution of these invocations within SPIR-V, and this **can** only be done with API synchronization primitives.

Each invocation in a queue family scope instance **must** be in the same **device scope instance**.

9.21.4. Command

Any shader invocations executed as the result of a single command such as **vkCmdDispatch** or **vkCmdDraw** form a *command scope instance*. For indirect drawing commands with **drawCount** greater than one, invocations from separate draws are in separate command scope instances. For ray tracing shaders, an invocation group is an implementation-dependent subset of the set of shader invocations of a given shader stage which are produced by a single trace rays command.

There is no specific **Scope** for communication across invocations in a command scope instance. As this has a clear boundary at the API level, coordination here **can** be performed in the API, rather than in SPIR-V.

Each invocation in a command scope instance **must** be in the same **queue-family scope instance**.

For shaders without defined **workgroups**, this set of invocations forms an *invocation group* as defined in the **SPIR-V specification**.

9.21.5. Primitive

Any fragment shader invocations executed as the result of rasterization of a single primitive form a *primitive scope instance*.

There is no specific **Scope** for communication across invocations in a primitive scope instance.

Any generated **helper invocations** are included in this scope instance.

Each invocation in a primitive scope instance **must** be in the same **command scope instance**.

Any input variables decorated with **Flat** are uniform within a primitive scope instance.

9.21.6. Shader Call

Any [shader-call-related](#) invocations that are executed in one or more ray tracing execution models form a *shader call scope instance*.

The [ShaderCallKHR Scope](#) can be used as [Memory Scope](#) for barrier and atomic operations.

Each invocation in a shader call scope instance **must** be in the same [queue family scope instance](#).

9.21.7. Workgroup

A *local workgroup* is a set of invocations that can synchronize and share data with each other using memory in the [Workgroup](#) storage class.

The [Workgroup Scope](#) can be used as both an [Execution Scope](#) and [Memory Scope](#) for barrier and atomic operations.

Each invocation in a local workgroup **must** be in the same [command scope instance](#).

Only compute shaders have defined workgroups - other shader types **cannot** use workgroup functionality. For shaders that have defined workgroups, this set of invocations forms an *invocation group* as defined in the [SPIR-V specification](#).

When variables declared with the [Workgroup](#) storage class are explicitly laid out (hence they are also decorated with [Block](#)), the amount of storage consumed is the size of the largest Block variable, not counting any padding at the end. The amount of storage consumed by the non-Block variables declared with the [Workgroup](#) storage class is implementation-dependent. However, the amount of storage consumed may not exceed the largest block size that would be obtained if all active non-Block variables declared with [Workgroup](#) storage class were assigned offsets in an arbitrary order by successively taking the smallest valid offset according to the [Standard Storage Buffer Layout](#) rules, and with [Boolean](#) values considered as 32-bit integer values for the purpose of this calculation. (This is equivalent to using the GLSL std430 layout rules.)

9.21.8. Quad

A *quad scope instance* is formed of four shader invocations.

In a fragment shader, each invocation in a quad scope instance is formed of invocations in neighboring framebuffer locations (x_i, y_i) , where:

- i is the index of the invocation within the scope instance.
- w and h are the number of pixels the fragment covers in the x and y axes.
- w and h are identical for all participating invocations.
- $(x_0) = (x_1 - w) = (x_2) = (x_3 - w)$
- $(y_0) = (y_1) = (y_2 - h) = (y_3 - h)$
- Each invocation has the same layer and sample indices.

In a mesh, task, or compute shader, if the [DerivativeGroupQuadsKHR](#) execution mode is specified, each

invocation in a quad scope instance is formed of invocations with adjacent local invocation IDs (x_i , y_i), where:

- i is the index of the invocation within the quad scope instance.
- $(x_0) = (x_1 - 1) = (x_2) = (x_3 - 1)$
- $(y_0) = (y_1) = (y_2 - 1) = (y_3 - 1)$
- x_0 and y_0 are integer multiples of 2.
- Each invocation has the same z coordinate.

In a mesh, task, or compute shader, if the `DerivativeGroupLinearKHR` execution mode is specified, each invocation in a quad scope instance is formed of invocations with adjacent local invocation indices (l_i), where:

- i is the index of the invocation within the quad scope instance.
- $(l_0) = (l_1 - 1) = (l_2 - 2) = (l_3 - 3)$
- l_0 is an integer multiple of 4.

The specific set of invocations that make up a quad scope instance in other shader stages is undefined.

In a fragment shader, each invocation in a quad scope instance **must** be in the same [primitive scope instance](#).

For [shaders that have defined workgroups](#), each invocation in a quad scope instance **must** be in the same [local workgroup](#).

In other shader stages, each invocation in a quad scope instance **must** be in the same [device scope instance](#).

Fragment, mesh, task, and compute shaders have defined quad scope instances.

9.21.9. Fragment Interlock

A *fragment interlock scope instance* is formed of fragment shader invocations based on their framebuffer locations ($x, y, layer, sample$), executed by commands inside a single [subpass](#).

The specific set of invocations included varies based on the execution mode as follows:

- If the `SampleInterlockOrderedEXT` or `SampleInterlockUnorderedEXT` execution modes are used, only invocations with identical framebuffer locations ($x, y, layer, sample$) are included.
- If the `PixelInterlockOrderedEXT` or `PixelInterlockUnorderedEXT` execution modes are used, fragments with different sample ids are also included.
- If the `ShadingRateInterlockOrderedEXT` or `ShadingRateInterlockUnorderedEXT` execution modes are used, fragments from neighboring framebuffer locations are also included. The [fragment shading rate](#) determines these fragments.

Only fragment shaders with one of the above execution modes have defined fragment interlock

scope instances.

There is no specific **Scope** value for communication across invocations in a fragment interlock scope instance. However, this is implicitly used as a memory scope by **OpBeginInvocationInterlockEXT** and **OpEndInvocationInterlockEXT**.

Each invocation in a fragment interlock scope instance **must** be in the same [queue family scope instance](#).

9.21.10. Invocation

The smallest *scope* is a single invocation; this is represented by the **Invocation Scope** in SPIR-V.

Fragment shader invocations **must** be in a [primitive scope instance](#).

Invocations in [fragment shaders that have a defined fragment interlock scope](#) **must** be in a [fragment interlock scope instance](#).

Invocations in [shaders that have defined workgroups](#) **must** be in a [local workgroup](#).

Invocations in [shaders that have a defined quad scope](#) **must** be in a [quad scope instance](#).

All invocations in all stages **must** be in a [command scope instance](#).

9.22. Derivative Operations

Derivative operations calculate the partial derivative for an expression P as a function of an invocation's x and y coordinates.

Derivative operations operate on a set of invocations known as a *derivative group* as defined in the [SPIR-V specification](#).

A derivative group in a fragment shader is equivalent to the [quad scope instance](#) if the **QuadDerivativesKHR** execution mode is specified, otherwise it is equivalent to the [primitive scope instance](#). A derivative group in a mesh, task, or compute shader is equivalent to the [quad scope instance](#).

Derivatives are calculated assuming that P is piecewise linear and continuous within the derivative group.

The following control-flow restrictions apply to derivative operations:

- If the **QuadDerivativesKHR** execution mode is specified, dynamic instances of any derivative operations **must** be executed in control flow that is uniform within the current [quad scope instance](#).
- If the **QuadDerivativesKHR** execution mode is not specified:
 - dynamic instances of explicit derivative instructions (**OpDPdx***, **OpDPdy***, and **OpFwidth***) **must** be executed in control flow that is uniform within a derivative group.
 - dynamic instances of implicit derivative operations **can** be executed in control flow that is

not uniform within the derivative group, but results are undefined.

Fragment shaders that statically execute derivative operations **must** launch sufficient invocations to ensure their correct operation; additional [helper invocations](#) are launched for framebuffer locations not covered by rasterized fragments if necessary.

NOTE

In a mesh, task, or compute shader, it is the application's responsibility to ensure that sufficient invocations are launched.

Derivative operations calculate their results as the difference between the result of P across invocations in the quad. For fine derivative operations ([OpDPdxFine](#) and [OpDPdyFine](#)), the values of DPdx(P_i) are calculated as

$$\text{DPdx}(P_0) = \text{DPdx}(P_1) = P_1 - P_0$$

$$\text{DPdx}(P_2) = \text{DPdx}(P_3) = P_3 - P_2$$

and the values of DPdy(P_i) are calculated as

$$\text{DPdy}(P_0) = \text{DPdy}(P_2) = P_2 - P_0$$

$$\text{DPdy}(P_1) = \text{DPdy}(P_3) = P_3 - P_1$$

where i is the index of each invocation as described in [Quad](#).

Coarse derivative operations ([OpDPdxCoarse](#) and [OpDPdyCoarse](#)), calculate their results in roughly the same manner, but **may** only calculate two values instead of four (one for each of DPdx and DPdy), reusing the same result no matter the originating invocation. If an implementation does this, it **should** use the fine derivative calculations described for P₀.

NOTE

Derivative values are calculated between fragments rather than pixels. If the fragment shader invocations involved in the calculation cover multiple pixels, these operations cover a wider area, resulting in larger derivative values. This in turn will result in a coarser LOD being selected for image sampling operations using derivatives.

Applications may want to account for this when using multi-pixel fragments; if pixel derivatives are desired, applications should use explicit derivative operations and divide the results by the size of the fragment in each dimension as follows:

$$\text{DPdx}(P_n)' = \text{DPdx}(P_n) / w$$

$$\text{DPdy}(P_n)' = \text{DPdy}(P_n) / h$$

where w and h are the size of the fragments in the quad, and $DPdx(P_n)$ and $DPdy(P_n)$ are the pixel derivatives.

The results for `OpDPdx` and `OpDPdy` **may** be calculated as either fine or coarse derivatives, with implementations favoring the most efficient approach. Implementations **must** choose coarse or fine consistently between the two.

Executing `OpFwidthFine`, `OpFwidthCoarse`, or `OpFwidth` is equivalent to executing the corresponding `OpDPdx*` and `OpDPdy*` instructions, taking the absolute value of the results, and summing them.

Executing an `OpImage*Sample*ImplicitLod` instruction is equivalent to executing `OpDPdx(Coordinate)` and `OpDPdy(Coordinate)`, and passing the results as the `Grad` operands `dx` and `dy`.

NOTE

It is expected that using the `ImplicitLod` variants of sampling functions will be substantially more efficient than using the `ExplicitLod` variants with explicitly generated derivatives.

9.23. Helper Invocations

When performing `derivative` operations in a fragment shader, additional invocations **may** be spawned in order to ensure correct results. These additional invocations are known as *helper invocations* and **can** be identified by a non-zero value in the `HelperInvocation` built-in. Stores and atomics performed by helper invocations **must** not have any effect on memory except for the `Function`, `Private` and `Output` storage classes, and values returned by atomic instructions in helper invocations are undefined.

NOTE

While storage to `Output` storage class has an effect even in helper invocations, it does not mean that helper invocations have an effect on the framebuffer. `Output` variables in fragment shaders can be read from as well, and they behave more like `Private` variables for the duration of the shader invocation.

9.24. Cooperative Matrices

A *cooperative matrix* type is a SPIR-V type where the storage for and computations performed on the matrix are spread across the invocations in a scope instance. These types give the implementation freedom in how to optimize matrix multiplies.

SPIR-V defines the types and instructions, but does not specify rules about what sizes/combinations are valid, and it is expected that different implementations **may** support different sizes.

To enumerate the supported cooperative matrix types and operations, call:

```
// Provided by VK_KHR_cooperative_matrix
VkResult vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t*                 pPropertyCount,
    VkCooperativeMatrixPropertiesKHR* pProperties);
```

- `physicalDevice` is the physical device.
- `pPropertyCount` is a pointer to an integer related to the number of cooperative matrix properties available or queried.
- `pProperties` is either `NULL` or a pointer to an array of `VkCooperativeMatrixPropertiesKHR` structures.

If `pProperties` is `NULL`, then the number of cooperative matrix properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of cooperative matrix properties available, at most `pPropertyCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available cooperative matrix properties were returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR-pPropertyCount-parameter `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR-pProperties-parameter
If the value referenced by `pPropertyCount` is not 0, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkCooperativeMatrixPropertiesKHR` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

Each `VkCooperativeMatrixPropertiesKHR` structure describes a single supported combination of types for a matrix multiply/add operation (`OpCooperativeMatrixMulAddKHR`). The multiply **can** be

described in terms of the following variables and types (in SPIR-V pseudocode):

```
%A is of type OpTypeCooperativeMatrixKHR %AType %scope %MSize %KSize %MatrixAKHR
%B is of type OpTypeCooperativeMatrixKHR %BType %scope %KSize %NSize %MatrixBKHR
%C is of type OpTypeCooperativeMatrixKHR %CType %scope %MSize %NSize
%MatrixAccumulatorKHR
%Result is of type OpTypeCooperativeMatrixKHR %ResultType %scope %MSize %NSize
%MatrixAccumulatorKHR

%Result = %A * %B + %C // using OpCooperativeMatrixMulAddKHR
```

A matrix multiply with these dimensions is known as an $M \times N \times K$ matrix multiply.

The `VkCooperativeMatrixPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_cooperative_matrix
typedef struct VkCooperativeMatrixPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           MSize;
    uint32_t           NSize;
    uint32_t           KSize;
    VkComponentTypeKHR AType;
    VkComponentTypeKHR BType;
    VkComponentTypeKHR CType;
    VkComponentTypeKHR ResultType;
    VkBool32           saturatingAccumulation;
    VkScopeKHR         scope;
} VkCooperativeMatrixPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `MSize` is the number of rows in matrices `A`, `C`, and `Result`.
- `KSize` is the number of columns in matrix `A` and rows in matrix `B`.
- `NSize` is the number of columns in matrices `B`, `C`, `Result`.
- `AType` is the component type of matrix `A`, of type `VkComponentTypeKHR`.
- `BType` is the component type of matrix `B`, of type `VkComponentTypeKHR`.
- `CType` is the component type of matrix `C`, of type `VkComponentTypeKHR`.
- `ResultType` is the component type of matrix `Result`, of type `VkComponentTypeKHR`.
- `saturatingAccumulation` indicates whether the `SaturatingAccumulation` operand to `OpCooperativeMatrixMulAddKHR` **must** be present or not. If it is `VK_TRUE`, the `SaturatingAccumulation` operand **must** be present. If it is `VK_FALSE`, the `SaturatingAccumulation` operand **must** not be present.
- `scope` is the scope of all the matrix types, of type `VkScopeKHR`.

If some types are preferred over other types (e.g. for performance), they **should** appear earlier in the list enumerated by `vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR`.

At least one entry in the list **must** have power of two values for all of `MSize`, `KSize`, and `NSize`.

`scope` **must** be `VK_SCOPE_SUBGROUP_KHR`.

Valid Usage (Implicit)

- `VUID-VkCooperativeMatrixPropertiesKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_COOPERATIVE_MATRIX_PROPERTIES_KHR`
- `VUID-VkCooperativeMatrixPropertiesKHR-pNext-pNext`
`pNext` **must** be `NULL`

Possible values for `VkScopeKHR` include:

```
// Provided by VK_KHR_cooperative_matrix
typedef enum VkScopeKHR {
    VK_SCOPE_DEVICE_KHR = 1,
    VK_SCOPE_WORKGROUP_KHR = 2,
    VK_SCOPE_SUBGROUP_KHR = 3,
    VK_SCOPE_QUEUE_FAMILY_KHR = 5,
} VkScopeKHR;
```

- `VK_SCOPE_DEVICE_KHR` corresponds to SPIR-V `Device` scope.
- `VK_SCOPE_WORKGROUP_KHR` corresponds to SPIR-V `Workgroup` scope.
- `VK_SCOPE_SUBGROUP_KHR` corresponds to SPIR-V `Subgroup` scope.
- `VK_SCOPE_QUEUE_FAMILY_KHR` corresponds to SPIR-V `QueueFamily` scope.

All enum values match the corresponding SPIR-V value.

Possible values for `VkComponentTypeKHR` include:

```
// Provided by VK_KHR_cooperative_matrix
typedef enum VkComponentTypeKHR {
    VK_COMPONENT_TYPE_FLOAT16_KHR = 0,
    VK_COMPONENT_TYPE_FLOAT32_KHR = 1,
    VK_COMPONENT_TYPE_FLOAT64_KHR = 2,
    VK_COMPONENT_TYPE_SINT8_KHR = 3,
    VK_COMPONENT_TYPE_SINT16_KHR = 4,
    VK_COMPONENT_TYPE_SINT32_KHR = 5,
    VK_COMPONENT_TYPE_SINT64_KHR = 6,
    VK_COMPONENT_TYPE_UINT8_KHR = 7,
    VK_COMPONENT_TYPE_UINT16_KHR = 8,
    VK_COMPONENT_TYPE_UINT32_KHR = 9,
    VK_COMPONENT_TYPE_UINT64_KHR = 10,
} VkComponentTypeKHR;
```

- `VK_COMPONENT_TYPE_FLOAT16_KHR` corresponds to SPIR-V `OpTypeFloat 16`.
- `VK_COMPONENT_TYPE_FLOAT32_KHR` corresponds to SPIR-V `OpTypeFloat 32`.
- `VK_COMPONENT_TYPE_FLOAT64_KHR` corresponds to SPIR-V `OpTypeFloat 64`.
- `VK_COMPONENT_TYPE_SINT8_KHR` corresponds to SPIR-V `OpTypeInt 8 0/1`.
- `VK_COMPONENT_TYPE_SINT16_KHR` corresponds to SPIR-V `OpTypeInt 16 0/1`.
- `VK_COMPONENT_TYPE_SINT32_KHR` corresponds to SPIR-V `OpTypeInt 32 0/1`.
- `VK_COMPONENT_TYPE_SINT64_KHR` corresponds to SPIR-V `OpTypeInt 64 0/1`.
- `VK_COMPONENT_TYPE_UINT8_KHR` corresponds to SPIR-V `OpTypeInt 8 0/1`.
- `VK_COMPONENT_TYPE_UINT16_KHR` corresponds to SPIR-V `OpTypeInt 16 0/1`.
- `VK_COMPONENT_TYPE_UINT32_KHR` corresponds to SPIR-V `OpTypeInt 32 0/1`.
- `VK_COMPONENT_TYPE_UINT64_KHR` corresponds to SPIR-V `OpTypeInt 64 0/1`.

Chapter 10. Pipelines

The following [figure](#) shows a block diagram of the Vulkan pipelines. Some Vulkan commands specify geometric objects to be drawn or computational work to be performed, while others specify state controlling how objects are handled by the various pipeline stages, or control data transfer between memory organized as images and buffers. Commands are effectively sent through a processing pipeline, either a *graphics pipeline*, a *ray tracing pipeline*, or a *compute pipeline*.

The first stage of the [graphics pipeline](#) ([Input Assembler](#)) assembles vertices to form geometric primitives such as points, lines, and triangles, based on a requested primitive topology. In the next stage ([Vertex Shader](#)) vertices **can** be transformed, computing positions and attributes for each vertex. If [tessellation](#) and/or [geometry](#) shaders are supported, they **can** then generate multiple primitives from a single input primitive, possibly changing the primitive topology or generating additional attribute data in the process.

The final resulting primitives are [clipped](#) to a clip volume in preparation for the next stage, [Rasterization](#). The rasterizer produces a series of *fragments* associated with a region of the framebuffer, from a two-dimensional description of a point, line segment, or triangle. These fragments are processed by [fragment operations](#) to determine whether generated values will be written to the framebuffer. [Fragment shading](#) determines the values to be written to the framebuffer attachments. Framebuffer operations then read and write the color and depth/stencil attachments of the framebuffer for a given subpass of a [render pass instance](#). The attachments **can** be used as input attachments in the fragment shader in a later subpass of the same render pass.

The [compute pipeline](#) is a separate pipeline from the graphics pipeline, which operates on one-, two-, or three-dimensional workgroups which **can** read from and write to buffer and image memory.

This ordering is meant only as a tool for describing Vulkan, not as a strict rule of how Vulkan is implemented, and we present it only as a means to organize the various operations of the pipelines. Actual ordering guarantees between pipeline stages are explained in detail in the [synchronization chapter](#).

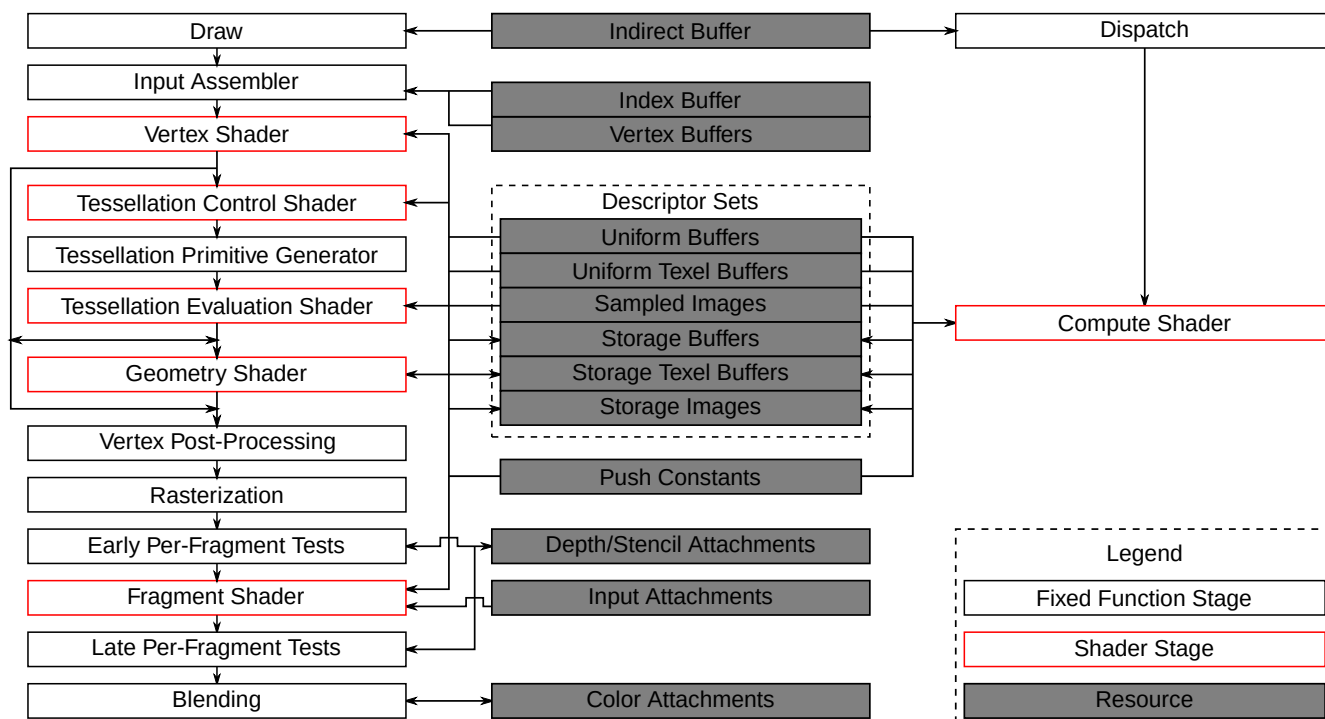


Figure 2. Block diagram of the Vulkan pipeline

Each pipeline is controlled by a monolithic object created from a description of all of the shader stages and any relevant fixed-function stages. [Linking](#) the whole pipeline together allows the optimization of shaders based on their input/outputs and eliminates expensive draw time state validation.

A pipeline object is bound to the current state using [vkCmdBindPipeline](#). Any pipeline object state that is specified as [dynamic](#) is not applied to the current state when the pipeline object is bound, but is instead set by dynamic state setting commands.

No state, including dynamic state, is inherited from one command buffer to another.

Compute, ray tracing, and graphics pipelines are each represented by [VkPipeline](#) handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPipeline)
```

10.1. Multiple Pipeline Creation

Multiple pipelines **can** be created in a single call by commands such as [vkCreateRayTracingPipelinesKHR](#), [vkCreateComputePipelines](#), and [vkCreateGraphicsPipelines](#).

The creation commands are passed an array [pCreateInfos](#) of [Vk*PipelineCreateInfo](#) structures specifying parameters of each pipeline to be created, and return a corresponding array of handles in [pPipelines](#). Each element index i of [pPipelines](#) is created based on the corresponding element i of [pCreateInfos](#).

Applications **can** group together similar pipelines to be created in a single call, and implementations are encouraged to look for reuse opportunities when creating a group.

When attempting to create many pipelines in a single command, it is possible that creation **may** fail for a subset of them. In this case, the corresponding elements of `pPipelines` will be `VK_NULL_HANDLE`. If creation fails for a pipeline despite valid arguments (for example, due to out of memory errors), the `VkResult` code returned by the pipeline creation command will indicate why. The implementation will attempt to create all pipelines, and only return `VK_NULL_HANDLE` values for those that actually failed.

If creation fails for multiple pipelines, the returned `VkResult` **must** be the return value of any one of the pipelines which did not succeed. An application **can** reliably clean up from a failed call by iterating over the `pPipelines` array and destroying every element that is not `VK_NULL_HANDLE`.

If the entire command fails and no pipelines are created, all elements of `pPipelines` will be `VK_NULL_HANDLE`.

10.2. Compute Pipelines

Compute pipelines consist of a single static compute shader stage and the pipeline layout.

The compute pipeline represents a compute shader and is created by calling `vkCreateComputePipelines` with `module` and `pName` selecting an entry point from a shader module, where that entry point defines a valid compute shader, in the `VkPipelineShaderStageCreateInfo` structure contained within the `VkComputePipelineCreateInfo` structure.

To create compute pipelines, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateComputePipelines(
    VkDevice                device,
    VkPipelineCache         pipelineCache,
    uint32_t               createInfoCount,
    const VkComputePipelineCreateInfo* pCreateInfos,
    const VkAllocationCallbacks* pAllocator,
    VkPipeline*             pPipelines);
```

- `device` is the logical device that creates the compute pipelines.
- `pipelineCache` is either `VK_NULL_HANDLE`, indicating that pipeline caching is disabled; or the handle of a valid `pipeline cache` object, in which case use of that cache is enabled for the duration of the command.
- `createInfoCount` is the length of the `pCreateInfos` and `pPipelines` arrays.
- `pCreateInfos` is a pointer to an array of `VkComputePipelineCreateInfo` structures.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pPipelines` is a pointer to an array of `VkPipeline` handles in which the resulting compute pipeline objects are returned.

Pipelines are created and returned as described for [Multiple Pipeline Creation](#).

Valid Usage

- VUID-vkCreateComputePipelines-device-09661
`device` **must** support at least one queue family with the `VK_QUEUE_COMPUTE_BIT` capability
- VUID-vkCreateComputePipelines-flags-00695
If the `flags` member of any element of `pCreateInfo`s contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, and the `basePipelineIndex` member of that same element is not `-1`, `basePipelineIndex` **must** be less than the index into `pCreateInfo`s that corresponds to that element
- VUID-vkCreateComputePipelines-flags-00696
If the `flags` member of any element of `pCreateInfo`s contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, the base pipeline **must** have been created with the `VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT` flag set
- VUID-vkCreateComputePipelines-pNext-09616
If `VkPipelineBinaryInfoKHR::binaryCount` is not `0` for any element of `pCreateInfo`s, `pipelineCache` **must** be `VK_NULL_HANDLE`
- VUID-vkCreateComputePipelines-pNext-09617
If a `VkPipelineCreateFlags2CreateInfoKHR` structure with the `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR` flag set is included in the `pNext` chain of any element of `pCreateInfo`s, `pipelineCache` **must** be `VK_NULL_HANDLE`
- VUID-vkCreateComputePipelines-binaryCount-09620
If `VkPipelineBinaryInfoKHR::binaryCount` is not `0` for any element of `pCreateInfo`s, `VK_PIPELINE_CREATION_FEEDBACK_APPLICATION_PIPELINE_CACHE_HIT_BIT` **must** not be set in the `flags` of that element
- VUID-vkCreateComputePipelines-binaryCount-09621
If `VkPipelineBinaryInfoKHR::binaryCount` is not `0` for any element of `pCreateInfo`s, `VK_PIPELINE_CREATION_FEEDBACK_BASE_PIPELINE_ACCELERATION_BIT` **must** not be set in the `flags` of that element
- VUID-vkCreateComputePipelines-binaryCount-09622
If `VkPipelineBinaryInfoKHR::binaryCount` is not `0` for any element of `pCreateInfo`s, `VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT_EXT` **must** not be set in the `flags` of that element

Valid Usage (Implicit)

- VUID-vkCreateComputePipelines-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateComputePipelines-pipelineCache-parameter
If `pipelineCache` is not `VK_NULL_HANDLE`, `pipelineCache` **must** be a valid `VkPipelineCache` handle
- VUID-vkCreateComputePipelines-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to an array of `createInfoCount` valid

VkComputePipelineCreateInfo structures

- VUID-vkCreateComputePipelines-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateComputePipelines-pPipelines-parameter
`pPipelines` **must** be a valid pointer to an array of `createInfoCount` `VkPipeline` handles
- VUID-vkCreateComputePipelines-createInfoCount-arraylength
`createInfoCount` **must** be greater than `0`
- VUID-vkCreateComputePipelines-pipelineCache-parent
If `pipelineCache` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkComputePipelineCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkComputePipelineCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkPipelineCreateFlags flags;
    VkPipelineShaderStageCreateInfo stage;
    VkPipelineLayout    layout;
    VkPipeline          basePipelineHandle;
    int32_t             basePipelineIndex;
} VkComputePipelineCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkPipelineCreateFlagBits` specifying how the pipeline will be generated.
- `stage` is a `VkPipelineShaderStageCreateInfo` structure describing the compute shader.
- `layout` is the description of binding locations used by both the pipeline and descriptor sets used with the pipeline.
- `basePipelineHandle` is a pipeline to derive from.
- `basePipelineIndex` is an index into the `pCreateInfos` parameter to use as a pipeline to derive

from.

The parameters `basePipelineHandle` and `basePipelineIndex` are described in more detail in [Pipeline Derivatives](#).

If the `pNext` chain includes a `VkPipelineCreateFlags2CreateInfoKHR` structure, `VkPipelineCreateFlags2CreateInfoKHR::flags` from that structure is used instead of `flags` from this structure.

Valid Usage

- VUID-VkComputePipelineCreateInfo-None-09497
If the `pNext` chain does not include a `VkPipelineCreateFlags2CreateInfoKHR` structure, `flags` **must** be a valid combination of `VkPipelineCreateFlagBits` values
- VUID-VkComputePipelineCreateInfo-flags-07984
If `flags` contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, and `basePipelineIndex` is -1, `basePipelineHandle` **must** be a valid compute `VkPipeline` handle
- VUID-VkComputePipelineCreateInfo-flags-07985
If `flags` contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, and `basePipelineHandle` is `VK_NULL_HANDLE`, `basePipelineIndex` **must** be a valid index into the calling command's `pCreateInfos` parameter
- VUID-VkComputePipelineCreateInfo-flags-07986
If `flags` contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, `basePipelineIndex` **must** be -1 or `basePipelineHandle` **must** be `VK_NULL_HANDLE`
- VUID-VkComputePipelineCreateInfo-layout-07987
If a push constant block is declared in a shader, a push constant range in `layout` **must** match the shader stage
- VUID-VkComputePipelineCreateInfo-layout-10069
If a push constant block is declared in a shader, the block must be contained inside the push constant range in `layout` that matches the stage
- VUID-VkComputePipelineCreateInfo-layout-07988
If a `resource variables` is declared in a shader, a descriptor slot in `layout` **must** match the shader stage
- VUID-VkComputePipelineCreateInfo-layout-07990
If a `resource variables` is declared in a shader, a descriptor slot in `layout` **must** match the descriptor type
- VUID-VkComputePipelineCreateInfo-layout-07991
If a `resource variables` is declared in a shader as an array, a descriptor slot in `layout` **must** match the descriptor count
- VUID-VkComputePipelineCreateInfo-flags-03365
`flags` **must** not include `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR`
- VUID-VkComputePipelineCreateInfo-flags-03366

<code>flags</code>	must	not	include
<code>VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR</code>			

- VUID-VkComputePipelineCreateInfo-flags-03367
flags must not include `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR`
- VUID-VkComputePipelineCreateInfo-flags-03368
flags must not include `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR`
- VUID-VkComputePipelineCreateInfo-flags-03369
flags must not include `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR`
- VUID-VkComputePipelineCreateInfo-flags-03370
flags must not include `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR`
- VUID-VkComputePipelineCreateInfo-flags-03576
flags must not include `VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR`
- VUID-VkComputePipelineCreateInfo-flags-11007
 If **flags** includes `VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT`, then the `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands` feature **must** be enabled
- VUID-VkComputePipelineCreateInfo-stage-00701
 The **stage** member of **stage must** be `VK_SHADER_STAGE_COMPUTE_BIT`
- VUID-VkComputePipelineCreateInfo-stage-00702
 The shader code for the entry point identified by **stage** and the rest of the state identified by this structure **must** adhere to the pipeline linking rules described in the [Shader Interfaces](#) chapter
- VUID-VkComputePipelineCreateInfo-layout-01687
 The number of resources in **layout** accessible to the compute shader stage **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageResources`
- VUID-VkComputePipelineCreateInfo-shaderEnqueue-09177
flags must not include `VK_PIPELINE_CREATE_LIBRARY_BIT_KHR`
- VUID-VkComputePipelineCreateInfo-flags-07367
flags must not include `VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT`

Valid Usage (Implicit)

- VUID-VkComputePipelineCreateInfo-sType-sType
sType must be `VK_STRUCTURE_TYPE_COMPUTE_PIPELINE_CREATE_INFO`
- VUID-VkComputePipelineCreateInfo-pNext-pNext
 Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either `NULL` or a pointer to a valid instance of `VkPipelineBinaryInfoKHR` or `VkPipelineCreateFlags2CreateInfoKHR`
- VUID-VkComputePipelineCreateInfo-sType-unique
 The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkComputePipelineCreateInfo-stage-parameter
stage must be a valid `VkPipelineShaderStageCreateInfo` structure

- VUID-VkComputePipelineCreateInfo-layout-parameter
layout **must** be a valid [VkPipelineLayout](#) handle
- VUID-VkComputePipelineCreateInfo-commonparent
Both of **basePipelineHandle**, and **layout** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same [VkDevice](#)

The [VkPipelineShaderStageCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineShaderStageCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineShaderStageCreateFlags flags;
    VkShaderStageFlagBits    stage;
    VkShaderModule            module;
    const char*              pName;
    const VkSpecializationInfo* pSpecializationInfo;
} VkPipelineShaderStageCreateInfo;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkPipelineShaderStageCreateFlagBits](#) specifying how the pipeline shader stage will be generated.
- **stage** is a [VkShaderStageFlagBits](#) value specifying a single pipeline stage.
- **module** is optionally a [VkShaderModule](#) object containing the shader code for this stage.
- **pName** is a pointer to a null-terminated UTF-8 string specifying the entry point name of the shader for this stage.
- **pSpecializationInfo** is a pointer to a [VkSpecializationInfo](#) structure, as described in [Specialization Constants](#), or **NULL**.

If **module** is not [VK_NULL_HANDLE](#), the shader code used by the pipeline is defined by **module**. If **module** is [VK_NULL_HANDLE](#), the shader code is defined by the chained [VkShaderModuleCreateInfo](#) if present.

Valid Usage

- VUID-VkPipelineShaderStageCreateInfo-stage-00704
If the **geometryShader** feature is not enabled, **stage** **must** not be **VK_SHADER_STAGE_GEOMETRY_BIT**
- VUID-VkPipelineShaderStageCreateInfo-stage-00705
If the **tessellationShader** feature is not enabled, **stage** **must** not be **VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT** or **VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT**

- VUID-VkPipelineShaderStageCreateInfo-stage-00706
stage **must** not be `VK_SHADER_STAGE_ALL_GRAPHICS`, or `VK_SHADER_STAGE_ALL`
- VUID-VkPipelineShaderStageCreateInfo-pName-00707
pName **must** be the name of an `OpEntryPoint` in module with an execution model that matches stage
- VUID-VkPipelineShaderStageCreateInfo-maxClipDistances-00708
If the identified entry point includes any variable in its interface that is declared with the `ClipDistance BuiltIn` decoration, that variable **must** not have an array size greater than `VkPhysicalDeviceLimits::maxClipDistances`
- VUID-VkPipelineShaderStageCreateInfo-maxCullDistances-00709
If the identified entry point includes any variable in its interface that is declared with the `CullDistance BuiltIn` decoration, that variable **must** not have an array size greater than `VkPhysicalDeviceLimits::maxCullDistances`
- VUID-VkPipelineShaderStageCreateInfo-maxCombinedClipAndCullDistances-00710
If the identified entry point includes variables in its interface that are declared with the `ClipDistance BuiltIn` decoration and variables in its interface that are declared with the `CullDistance BuiltIn` decoration, those variables **must** not have array sizes which sum to more than `VkPhysicalDeviceLimits::maxCombinedClipAndCullDistances`
- VUID-VkPipelineShaderStageCreateInfo-maxSampleMaskWords-00711
If the identified entry point includes any variable in its interface that is declared with the `SampleMask BuiltIn` decoration, that variable **must** not have an array size greater than `VkPhysicalDeviceLimits::maxSampleMaskWords`
- VUID-VkPipelineShaderStageCreateInfo-stage-00713
If stage is `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` or `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, and the identified entry point has an `OpExecutionMode` instruction specifying a patch size with `OutputVertices`, the patch size **must** be greater than 0 and less than or equal to `VkPhysicalDeviceLimits::maxTessellationPatchSize`
- VUID-VkPipelineShaderStageCreateInfo-stage-00714
If stage is `VK_SHADER_STAGE_GEOMETRY_BIT`, the identified entry point **must** have an `OpExecutionMode` instruction specifying a maximum output vertex count that is greater than 0 and less than or equal to `VkPhysicalDeviceLimits::maxGeometryOutputVertices`
- VUID-VkPipelineShaderStageCreateInfo-stage-00715
If stage is `VK_SHADER_STAGE_GEOMETRY_BIT`, the identified entry point **must** have an `OpExecutionMode` instruction specifying an invocation count that is greater than 0 and less than or equal to `VkPhysicalDeviceLimits::maxGeometryShaderInvocations`
- VUID-VkPipelineShaderStageCreateInfo-stage-02596
If stage is either `VK_SHADER_STAGE_VERTEX_BIT`, `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`, `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, or `VK_SHADER_STAGE_GEOMETRY_BIT`, and the identified entry point writes to Layer for any primitive, it **must** write the same value to Layer for all vertices of a given primitive
- VUID-VkPipelineShaderStageCreateInfo-stage-02597
If stage is either `VK_SHADER_STAGE_VERTEX_BIT`, `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`, `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`, or `VK_SHADER_STAGE_GEOMETRY_BIT`, and the

identified entry point writes to `ViewportIndex` for any primitive, it **must** write the same value to `ViewportIndex` for all vertices of a given primitive

- VUID-VkPipelineShaderStageCreateInfo-stage-06685
If `stage` is `VK_SHADER_STAGE_FRAGMENT_BIT`, and the identified entry point writes to `FragDepth` in any execution path, all execution paths that are not exclusive to helper invocations **must** either discard the fragment, or write or initialize the value of `FragDepth`
- VUID-VkPipelineShaderStageCreateInfo-stage-06686
If `stage` is `VK_SHADER_STAGE_FRAGMENT_BIT`, and the identified entry point writes to `FragStencilRefEXT` in any execution path, all execution paths that are not exclusive to helper invocations **must** either discard the fragment, or write or initialize the value of `FragStencilRefEXT`
- VUID-VkPipelineShaderStageCreateInfo-stage-08771
`module` **must** be a valid `VkShaderModule`, or the `pNext` chain of the parent `Vk*CreateInfo` structure **must** set `VkPipelineBinaryInfoKHR::binaryCount` to a value greater than 0, if none of the following features are enabled:
 - `maintenance5`
- VUID-VkPipelineShaderStageCreateInfo-stage-06845
If `module` is `VK_NULL_HANDLE` there **must** be a valid `VkShaderModuleCreateInfo` structure in the `pNext` chain, or the `pNext` chain of the parent `Vk*CreateInfo` structure **must** set `VkPipelineBinaryInfoKHR::binaryCount` to a value greater than 0,
- VUID-VkPipelineShaderStageCreateInfo-pSpecializationInfo-06849
The shader code used by the pipeline **must** be valid as described by the [Khronos SPIR-V Specification](#) after applying the specializations provided in `pSpecializationInfo`, if any, and then converting all specialization constants into fixed constants

Valid Usage (Implicit)

- VUID-VkPipelineShaderStageCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_SHADER_STAGE_CREATE_INFO`
- VUID-VkPipelineShaderStageCreateInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkDebugUtilsObjectNameInfoEXT`, `VkPipelineShaderStageRequiredSubgroupSizeCreateInfo`, or `VkShaderModuleCreateInfo`
- VUID-VkPipelineShaderStageCreateInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkPipelineShaderStageCreateInfo-flags-zeroBitmask
`flags` **must** be 0
- VUID-VkPipelineShaderStageCreateInfo-stage-parameter
`stage` **must** be a valid `VkShaderStageFlagBits` value
- VUID-VkPipelineShaderStageCreateInfo-module-parameter
If `module` is not `VK_NULL_HANDLE`, `module` **must** be a valid `VkShaderModule` handle
- VUID-VkPipelineShaderStageCreateInfo-pName-parameter

pName **must** be a null-terminated UTF-8 string

- VUID-VkPipelineShaderStageCreateInfo-pSpecializationInfo-parameter

If **pSpecializationInfo** is not **NULL**, **pSpecializationInfo** **must** be a valid pointer to a valid **VkSpecializationInfo** structure

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineShaderStageCreateFlags;
```

VkPipelineShaderStageCreateFlags is a bitmask type for setting a mask of zero or more **VkPipelineShaderStageCreateFlagBits**.

Possible values of the **flags** member of **VkPipelineShaderStageCreateInfo** specifying how a pipeline shader stage is created, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineShaderStageCreateFlagBits {
} VkPipelineShaderStageCreateFlagBits;
```

Bits which **can** be set by commands and structures, specifying one or more shader stages, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkShaderStageFlagBits {
    VK_SHADER_STAGE_VERTEX_BIT = 0x00000001,
    VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT = 0x00000002,
    VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT = 0x00000004,
    VK_SHADER_STAGE_GEOMETRY_BIT = 0x00000008,
    VK_SHADER_STAGE_FRAGMENT_BIT = 0x00000010,
    VK_SHADER_STAGE_COMPUTE_BIT = 0x00000020,
    VK_SHADER_STAGE_ALL_GRAPHICS = 0x0000001F,
    VK_SHADER_STAGE_ALL = 0x7FFFFFFF,
// Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_RAYGEN_BIT_KHR = 0x00000100,
// Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_ANY_HIT_BIT_KHR = 0x00000200,
// Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR = 0x00000400,
// Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_MISS_BIT_KHR = 0x00000800,
// Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_INTERSECTION_BIT_KHR = 0x00001000,
// Provided by VK_KHR_ray_tracing_pipeline
    VK_SHADER_STAGE_CALLABLE_BIT_KHR = 0x00002000,
} VkShaderStageFlagBits;
```

- **VK_SHADER_STAGE_VERTEX_BIT** specifies the vertex stage.

- `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT` specifies the tessellation control stage.
- `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` specifies the tessellation evaluation stage.
- `VK_SHADER_STAGE_GEOMETRY_BIT` specifies the geometry stage.
- `VK_SHADER_STAGE_FRAGMENT_BIT` specifies the fragment stage.
- `VK_SHADER_STAGE_COMPUTE_BIT` specifies the compute stage.
- `VK_SHADER_STAGE_ALL_GRAPHICS` is a combination of bits used as shorthand to specify all graphics stages defined above (excluding the compute stage).
- `VK_SHADER_STAGE_ALL` is a combination of bits used as shorthand to specify all shader stages supported by the device, including all additional stages which are introduced by extensions.
- `VK_SHADER_STAGE_RAYGEN_BIT_KHR` specifies the ray generation stage.
- `VK_SHADER_STAGE_ANY_HIT_BIT_KHR` specifies the any-hit stage.
- `VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR` specifies the closest hit stage.
- `VK_SHADER_STAGE_MISS_BIT_KHR` specifies the miss stage.
- `VK_SHADER_STAGE_INTERSECTION_BIT_KHR` specifies the intersection stage.
- `VK_SHADER_STAGE_CALLABLE_BIT_KHR` specifies the callable stage.

NOTE

`VK_SHADER_STAGE_ALL_GRAPHICS` only includes the original five graphics stages included in Vulkan 1.0, and not any stages added by extensions. Thus, it may not have the desired effect in all cases.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkShaderStageFlags;
```

`VkShaderStageFlags` is a bitmask type for setting a mask of zero or more `VkShaderStageFlagBits`.

10.3. Graphics Pipelines

Graphics pipelines consist of multiple shader stages, multiple fixed-function pipeline stages, and a pipeline layout.

To create graphics pipelines, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateGraphicsPipelines(
    VkDevice          device,
    VkPipelineCache   pipelineCache,
    uint32_t          createInfoCount,
    const VkGraphicsPipelineCreateInfo* pCreateInfos,
    const VkAllocationCallbacks*       pAllocator,
    VkPipeline*        pPipelines);
```

- **device** is the logical device that creates the graphics pipelines.
- **pipelineCache** is either `VK_NULL_HANDLE`, indicating that pipeline caching is disabled; or the handle of a valid `pipeline cache` object, in which case use of that cache is enabled for the duration of the command.
- **createInfoCount** is the length of the **pCreateInfo**s and **pPipelines** arrays.
- **pCreateInfo**s is a pointer to an array of `VkGraphicsPipelineCreateInfo` structures.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pPipelines** is a pointer to an array of `VkPipeline` handles in which the resulting graphics pipeline objects are returned.

The `VkGraphicsPipelineCreateInfo` structure includes an array of `VkPipelineShaderStageCreateInfo` structures for each of the desired active shader stages, as well as creation information for all relevant fixed-function stages, and a pipeline layout.

Pipelines are created and returned as described for [Multiple Pipeline Creation](#).

Valid Usage

- VUID-vkCreateGraphicsPipelines-device-09662
device **must** support at least one queue family with the `VK_QUEUE_GRAPHICS_BIT` capability
- VUID-vkCreateGraphicsPipelines-flags-00720
If the **flags** member of any element of **pCreateInfo**s contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, and the **basePipelineIndex** member of that same element is not `-1`, **basePipelineIndex** **must** be less than the index into **pCreateInfo**s that corresponds to that element
- VUID-vkCreateGraphicsPipelines-flags-00721
If the **flags** member of any element of **pCreateInfo**s contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, the base pipeline **must** have been created with the `VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT` flag set
- VUID-vkCreateGraphicsPipelines-pNext-09616
If `VkPipelineBinaryInfoKHR::binaryCount` is not `0` for any element of **pCreateInfo**s, **pipelineCache** **must** be `VK_NULL_HANDLE`
- VUID-vkCreateGraphicsPipelines-pNext-09617
If a `VkPipelineCreateFlags2CreateInfoKHR` structure with the `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR` flag set is included in the **pNext** chain of any element of **pCreateInfo**s, **pipelineCache** **must** be `VK_NULL_HANDLE`
- VUID-vkCreateGraphicsPipelines-binaryCount-09620
If `VkPipelineBinaryInfoKHR::binaryCount` is not `0` for any element of **pCreateInfo**s, `VK_PIPELINE_CREATION_FEEDBACK_APPLICATION_PIPELINE_CACHE_HIT_BIT` **must** not be set in the **flags** of that element
- VUID-vkCreateGraphicsPipelines-binaryCount-09621
If `VkPipelineBinaryInfoKHR::binaryCount` is not `0` for any element of **pCreateInfo**s, `VK_PIPELINE_CREATION_FEEDBACK_BASE_PIPELINE_ACCELERATION_BIT` **must** not be set in the

flags of that element

- VUID-vkCreateGraphicsPipelines-binaryCount-09622

If `VkPipelineBinaryInfoKHR::binaryCount` is not 0 for any element of `pCreateInfos`, `VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT_EXT` **must** not be set in the flags of that element

Valid Usage (Implicit)

- VUID-vkCreateGraphicsPipelines-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateGraphicsPipelines-pipelineCache-parameter
If `pipelineCache` is not `VK_NULL_HANDLE`, `pipelineCache` **must** be a valid `VkPipelineCache` handle
- VUID-vkCreateGraphicsPipelines-pCreateInfos-parameter
`pCreateInfos` **must** be a valid pointer to an array of `createInfoCount` valid `VkGraphicsPipelineCreateInfo` structures
- VUID-vkCreateGraphicsPipelines-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateGraphicsPipelines-pPipelines-parameter
`pPipelines` **must** be a valid pointer to an array of `createInfoCount` `VkPipeline` handles
- VUID-vkCreateGraphicsPipelines-createInfoCount-arraylength
`createInfoCount` **must** be greater than 0
- VUID-vkCreateGraphicsPipelines-pipelineCache-parent
If `pipelineCache` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkGraphicsPipelineCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkGraphicsPipelineCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineCreateFlags     flags;
    uint32_t                  stageCount;
    const VkPipelineShaderStageCreateInfo* pStages;
    const VkPipelineVertexInputStateCreateInfo* pVertexInputState;
    const VkPipelineInputAssemblyStateCreateInfo* pInputAssemblyState;
    const VkPipelineTessellationStateCreateInfo* pTessellationState;
    const VkPipelineViewportStateCreateInfo* pViewportState;
    const VkPipelineRasterizationStateCreateInfo* pRasterizationState;
    const VkPipelineMultisampleStateCreateInfo* pMultisampleState;
    const VkPipelineDepthStencilStateCreateInfo* pDepthStencilState;
    const VkPipelineColorBlendStateCreateInfo* pColorBlendState;
    const VkPipelineDynamicStateCreateInfo* pDynamicState;
    VkPipelineLayout          layout;
    VkRenderPass              renderPass;
    uint32_t                  subpass;
    VkPipeline                basePipelineHandle;
    int32_t                   basePipelineIndex;
} VkGraphicsPipelineCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkPipelineCreateFlagBits` specifying how the pipeline will be generated.
- `stageCount` is the number of entries in the `pStages` array.
- `pStages` is a pointer to an array of `stageCount` `VkPipelineShaderStageCreateInfo` structures describing the set of the shader stages to be included in the graphics pipeline.
- `pVertexInputState` is a pointer to a `VkPipelineVertexInputStateCreateInfo` structure. It **can** be `NULL` if the pipeline is created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state set.
- `pInputAssemblyState` is a pointer to a `VkPipelineInputAssemblyStateCreateInfo` structure which determines input assembly behavior for vertex shading, as described in [Drawing Commands](#). If the `VK_EXT_extended_dynamic_state3` extension is enabled, it **can** be `NULL` if the pipeline is created with both `VK_DYNAMIC_STATE_PRIMITIVE_RESTART_ENABLE`, and `VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY` dynamic states set and `dynamicPrimitiveTopologyUnrestricted` is `VK_TRUE`.
- `pTessellationState` is a pointer to a `VkPipelineTessellationStateCreateInfo` structure defining tessellation state used by tessellation shaders.
- `pViewportState` is a pointer to a `VkPipelineViewportStateCreateInfo` structure defining viewport state used when rasterization is enabled. If the `VK_EXT_extended_dynamic_state3` extension is enabled, it **can** be `NULL` if the pipeline is created with both `VK_DYNAMIC_STATE_VIEWPORT_WITH_COUNT`, and `VK_DYNAMIC_STATE_SCISSOR_WITH_COUNT` dynamic states set.
- `pRasterizationState` is a pointer to a `VkPipelineRasterizationStateCreateInfo` structure defining

rasterization state. If the [VK_EXT_extended_dynamic_state3](#) extension is enabled, it **can** be `NULL` if the pipeline is created with all of `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT`, `VK_DYNAMIC_STATE_RASTERIZER_DISCARD_ENABLE`, `VK_DYNAMIC_STATE_POLYGON_MODE_EXT`, `VK_DYNAMIC_STATE_CULL_MODE`, `VK_DYNAMIC_STATE_FRONT_FACE`, `VK_DYNAMIC_STATE_DEPTH_BIAS_ENABLE`, `VK_DYNAMIC_STATE_DEPTH_BIAS`, and `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic states set.

- `pMultisampleState` is a pointer to a [VkPipelineMultisampleStateCreateInfo](#) structure defining multisample state used when rasterization is enabled. If the [VK_EXT_extended_dynamic_state3](#) extension is enabled, it **can** be `NULL` if the pipeline is created with all of `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT`, `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT`, and `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic states set, and either [alphaToOne](#) is disabled on the device or `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT` is set, in which case [VkPipelineMultisampleStateCreateInfo::sampleShadingEnable](#) is assumed to be `VK_FALSE`.
- `pDepthStencilState` is a pointer to a [VkPipelineDepthStencilStateCreateInfo](#) structure defining depth/stencil state used when rasterization is enabled for depth or stencil attachments accessed during rendering. If the [VK_EXT_extended_dynamic_state3](#) extension is enabled, it **can** be `NULL` if the pipeline is created with all of `VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE`, `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE`, `VK_DYNAMIC_STATE_DEPTH_COMPARE_OP`, `VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE`, `VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE`, `VK_DYNAMIC_STATE_STENCIL_OP`, and `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic states set.
- `pColorBlendState` is a pointer to a [VkPipelineColorBlendStateCreateInfo](#) structure defining color blend state used when rasterization is enabled for any color attachments accessed during rendering. If the [VK_EXT_extended_dynamic_state3](#) extension is enabled, it **can** be `NULL` if the pipeline is created with all of `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT`, `VK_DYNAMIC_STATE_LOGIC_OP_EXT`, `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT`, `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT`, `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and `VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic states set.
- `pDynamicState` is a pointer to a [VkPipelineDynamicStateCreateInfo](#) structure defining which properties of the pipeline state object are dynamic and **can** be changed independently of the pipeline state. This **can** be `NULL`, which means no state in the pipeline is considered dynamic.
- `layout` is the description of binding locations used by both the pipeline and descriptor sets used with the pipeline.
- `renderPass` is a handle to a render pass object describing the environment in which the pipeline will be used. The pipeline **must** only be used with a render pass instance compatible with the one provided. See [Render Pass Compatibility](#) for more information.
- `subpass` is the index of the subpass in the render pass where this pipeline will be used.
- `basePipelineHandle` is a pipeline to derive from.
- `basePipelineIndex` is an index into the `pCreateInfos` parameter to use as a pipeline to derive from.

The parameters `basePipelineHandle` and `basePipelineIndex` are described in more detail in [Pipeline Derivatives](#).

NOTE

With [VK_EXT_extended_dynamic_state3](#), it is possible that many of the `VkGraphicsPipelineCreateInfo` members above **can** be `NULL` because all their state is

dynamic and therefore ignored. This is optional so the application **can** still use a valid pointer if it needs to set the **pNext** or **flags** fields to specify state for other extensions.

The state required for a graphics pipeline is divided into **vertex input state**, **pre-rasterization shader state**, **fragment shader state**, and **fragment output state**.

Vertex Input State

Vertex input state is defined by:

- [VkPipelineVertexInputStateCreateInfo](#)
- [VkPipelineInputAssemblyStateCreateInfo](#)

This state **must** be specified to create a **complete graphics pipeline**.

Pre-Rasterization Shader State

Pre-rasterization shader state is defined by:

- [VkPipelineShaderStageCreateInfo](#) entries for:
 - Vertex shaders
 - Tessellation control shaders
 - Tessellation evaluation shaders
 - Geometry shaders
- Within the [VkPipelineLayout](#), the full pipeline layout **must** be specified.
- [VkPipelineViewportStateCreateInfo](#)
- [VkPipelineRasterizationStateCreateInfo](#)
- [VkPipelineTessellationStateCreateInfo](#)
- [VkRenderPass](#) and **subpass** parameter
- The **viewMask** parameter of [VkPipelineRenderingCreateInfo](#) (formats are ignored)
- [VkPipelineDiscardRectangleStateCreateInfoEXT](#)
- [VkPipelineFragmentShadingRateStateCreateInfoKHR](#)

This state **must** be specified to create a **complete graphics pipeline**.

Fragment Shader State

Fragment shader state is defined by:

- A [VkPipelineShaderStageCreateInfo](#) entry for the fragment shader
- Within the [VkPipelineLayout](#), the full pipeline layout **must** be specified.
- [VkPipelineMultisampleStateCreateInfo](#) if **sample shading** is enabled or **renderpass** is not **VK_NULL_HANDLE**
- [VkPipelineDepthStencilStateCreateInfo](#)

- [VkRenderPass](#) and [subpass](#) parameter
- The [viewMask](#) parameter of [VkPipelineRenderingCreateInfo](#) (formats are ignored)
- [VkPipelineFragmentShadingRateStateCreateInfoKHR](#)
- Inclusion/omission of the [VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR](#) flag
- [VkRenderingInputAttachmentIndexInfoKHR](#)

If [rasterizerDiscardEnable](#) is [VK_FALSE](#) this state **must** be specified to create a [complete graphics pipeline](#).

Fragment Output State

Fragment output state is defined by:

- [VkPipelineColorBlendStateCreateInfo](#)
- [VkRenderPass](#) and [subpass](#) parameter
- [VkPipelineMultisampleStateCreateInfo](#)
- [VkPipelineRenderingCreateInfo](#)
- Inclusion/omission of the [VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT](#) and [VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT](#) flags
- [VkRenderingAttachmentLocationInfoKHR](#)

If [rasterizerDiscardEnable](#) is [VK_FALSE](#) this state **must** be specified to create a [complete graphics pipeline](#).

Dynamic State

Dynamic state values set via [pDynamicState](#) **must** be ignored if the state they correspond to is not otherwise statically set by one of the state subsets used to create the pipeline. For example, if a pipeline only included [pre-rasterization shader state](#), then any dynamic state value corresponding to depth or stencil testing has no effect.

Complete Graphics Pipelines

A complete graphics pipeline always includes [pre-rasterization shader state](#), with other subsets included depending on that state as specified in the above sections.

If the [pNext](#) chain includes a [VkPipelineCreateFlags2CreateInfoKHR](#) structure, [VkPipelineCreateFlags2CreateInfoKHR::flags](#) from that structure is used instead of [flags](#) from this structure.

Valid Usage

- VUID-VkGraphicsPipelineCreateInfo-None-09497
If the [pNext](#) chain does not include a [VkPipelineCreateFlags2CreateInfoKHR](#) structure, [flags](#) **must** be a valid combination of [VkPipelineCreateFlagBits](#) values
- VUID-VkGraphicsPipelineCreateInfo-flags-07984

If **flags** contains the **VK_PIPELINE_CREATE_DERIVATIVE_BIT** flag, and **basePipelineIndex** is -1, **basePipelineHandle** **must** be a valid graphics **VkPipeline** handle

- VUID-VkGraphicsPipelineCreateInfo-flags-07985

If **flags** contains the **VK_PIPELINE_CREATE_DERIVATIVE_BIT** flag, and **basePipelineHandle** is **VK_NULL_HANDLE**, **basePipelineIndex** **must** be a valid index into the calling command's **pCreateInfos** parameter

- VUID-VkGraphicsPipelineCreateInfo-flags-07986

If **flags** contains the **VK_PIPELINE_CREATE_DERIVATIVE_BIT** flag, **basePipelineIndex** **must** be -1 or **basePipelineHandle** **must** be **VK_NULL_HANDLE**

- VUID-VkGraphicsPipelineCreateInfo-layout-07987

If a push constant block is declared in a shader, a push constant range in **layout** **must** match the shader stage

- VUID-VkGraphicsPipelineCreateInfo-layout-10069

If a push constant block is declared in a shader, the block must be contained inside the push constant range in **layout** that matches the stage

- VUID-VkGraphicsPipelineCreateInfo-layout-07988

If a **resource variables** is declared in a shader, a descriptor slot in **layout** **must** match the shader stage

- VUID-VkGraphicsPipelineCreateInfo-layout-07990

If a **resource variables** is declared in a shader, a descriptor slot in **layout** **must** match the descriptor type

- VUID-VkGraphicsPipelineCreateInfo-layout-07991

If a **resource variables** is declared in a shader as an array, a descriptor slot in **layout** **must** match the descriptor count

- VUID-VkGraphicsPipelineCreateInfo-stage-02096

If the pipeline requires **pre-rasterization shader state** the **stage** member of one element of **pStages** **must** be **VK_SHADER_STAGE_VERTEX_BIT**

- VUID-VkGraphicsPipelineCreateInfo-pStages-00729

If the pipeline requires **pre-rasterization shader state** and **pStages** includes a tessellation control shader stage, it **must** include a tessellation evaluation shader stage

- VUID-VkGraphicsPipelineCreateInfo-pStages-00730

If the pipeline requires **pre-rasterization shader state** and **pStages** includes a tessellation evaluation shader stage, it **must** include a tessellation control shader stage

- VUID-VkGraphicsPipelineCreateInfo-pStages-09022

If the pipeline requires **pre-rasterization shader state** and **pStages** includes a tessellation control shader stage, and the **VK_EXT_extended_dynamic_state3** extension is not enabled or the **VK_DYNAMIC_STATE_PATCH_CONTROL_POINTS_EXT** dynamic state is not set, **pTessellationState** **must** be a valid pointer to a valid **VkPipelineTessellationStateCreateInfo** structure

- VUID-VkGraphicsPipelineCreateInfo-pTessellationState-09023

If **pTessellationState** is not **NULL** it **must** be a pointer to a valid **VkPipelineTessellationStateCreateInfo** structure

- VUID-VkGraphicsPipelineCreateInfo-pStages-00732

If the pipeline requires [pre-rasterization shader state](#) and `pStages` includes tessellation shader stages, the shader code of at least one stage **must** contain an `OpExecutionMode` instruction specifying the type of subdivision in the pipeline

- VUID-VkGraphicsPipelineCreateInfo-pStages-00733

If the pipeline requires [pre-rasterization shader state](#) and `pStages` includes tessellation shader stages, and the shader code of both stages contain an `OpExecutionMode` instruction specifying the type of subdivision in the pipeline, they **must** both specify the same subdivision mode

- VUID-VkGraphicsPipelineCreateInfo-pStages-00734

If the pipeline requires [pre-rasterization shader state](#) and `pStages` includes tessellation shader stages, the shader code of at least one stage **must** contain an `OpExecutionMode` instruction specifying the output patch size in the pipeline

- VUID-VkGraphicsPipelineCreateInfo-pStages-00735

If the pipeline requires [pre-rasterization shader state](#) and `pStages` includes tessellation shader stages, and the shader code of both contain an `OpExecutionMode` instruction specifying the out patch size in the pipeline, they **must** both specify the same patch size

- VUID-VkGraphicsPipelineCreateInfo-pStages-08888

If the pipeline is being created with [pre-rasterization shader state](#) and [vertex input state](#) and `pStages` includes tessellation shader stages, and either `VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY` dynamic state is not enabled or `dynamicPrimitiveTopologyUnrestricted` is `VK_FALSE`, the `topology` member of `pInputAssembly` **must** be `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST`

- VUID-VkGraphicsPipelineCreateInfo-topology-08889

If the pipeline is being created with [pre-rasterization shader state](#) and [vertex input state](#) and the `topology` member of `pInputAssembly` is `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST`, and either `VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY` dynamic state is not enabled or `dynamicPrimitiveTopologyUnrestricted` is `VK_FALSE`, then `pStages` **must** include tessellation shader stages

- VUID-VkGraphicsPipelineCreateInfo-TessellationEvaluation-07723

If the pipeline is being created with a `TessellationEvaluation Execution Model`, no `Geometry Execution Model`, uses the `PointMode Execution Mode`, and `shaderTessellationAndGeometryPointSize` is enabled, a `PointSize` decorated variable **must** be written to if `maintenance5` is not enabled

- VUID-VkGraphicsPipelineCreateInfo-topology-08773

If the pipeline is being created with a `Vertex Execution Model` and no `TessellationEvaluation` or `Geometry Execution Model`, and the `topology` member of `pInputAssembly` is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, and either `VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY` dynamic state is not enabled or `dynamicPrimitiveTopologyUnrestricted` is `VK_FALSE`, a `PointSize` decorated variable **must** be written to if `maintenance5` is not enabled

- VUID-VkGraphicsPipelineCreateInfo-maintenance5-08775

If `maintenance5` is enabled and a `PointSize` decorated variable is written to, all execution paths **must** write to a `PointSize` decorated variable

- VUID-VkGraphicsPipelineCreateInfo-TessellationEvaluation-07724

If the pipeline is being created with a **TessellationEvaluation Execution Model**, no **Geometry Execution Model**, uses the **PointMode Execution Mode**, and **shaderTessellationAndGeometryPointSize** is not enabled, a **PointSize** decorated variable **must** not be written to

- VUID-VkGraphicsPipelineCreateInfo-shaderTessellationAndGeometryPointSize-08776

If the pipeline is being created with a **Geometry Execution Model**, uses the **OutputPoints Execution Mode**, and **shaderTessellationAndGeometryPointSize** is enabled, a **PointSize** decorated variable **must** be written to for every vertex emitted if **maintenance5** is not enabled

- VUID-VkGraphicsPipelineCreateInfo-Geometry-07726

If the pipeline is being created with a **Geometry Execution Model**, uses the **OutputPoints Execution Mode**, and **shaderTessellationAndGeometryPointSize** is not enabled, a **PointSize** decorated variable **must** not be written to

- VUID-VkGraphicsPipelineCreateInfo-pStages-00738

If the pipeline requires **pre-rasterization shader state** and **pStages** includes a geometry shader stage, and does not include any tessellation shader stages, its shader code **must** contain an **OpExecutionMode** instruction specifying an input primitive type that is **compatible** with the primitive topology specified in **pInputAssembly**

- VUID-VkGraphicsPipelineCreateInfo-pStages-00739

If the pipeline requires **pre-rasterization shader state** and **pStages** includes a geometry shader stage, and also includes tessellation shader stages, its shader code **must** contain an **OpExecutionMode** instruction specifying an input primitive type that is **compatible** with the primitive topology that is output by the tessellation stages

- VUID-VkGraphicsPipelineCreateInfo-pStages-00740

If the pipeline requires **pre-rasterization shader state** and **fragment shader state**, it includes both a fragment shader and a geometry shader, and the fragment shader code reads from an input variable that is decorated with **PrimitiveId**, then the geometry shader code **must** write to a matching output variable, decorated with **PrimitiveId**, in all execution paths

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06038

If **renderPass** is not **VK_NULL_HANDLE** and the pipeline is being created with **fragment shader state** the fragment shader **must** not read from any input attachment that is defined as **VK_ATTACHMENT_UNUSED** in **subpass**

- VUID-VkGraphicsPipelineCreateInfo-pStages-00742

If the pipeline requires **pre-rasterization shader state** and multiple pre-rasterization shader stages are included in **pStages**, the shader code for the entry points identified by those **pStages** and the rest of the state identified by this structure **must** adhere to the pipeline linking rules described in the **Shader Interfaces** chapter

- VUID-VkGraphicsPipelineCreateInfo-None-04889

If the pipeline requires **pre-rasterization shader state** and **fragment shader state**, the fragment shader and last **pre-rasterization shader stage** and any relevant state **must** adhere to the pipeline linking rules described in the **Shader Interfaces** chapter

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06041

If **renderPass** is not **VK_NULL_HANDLE**, and the pipeline is being created with **fragment**

output interface state, then for each color attachment in the subpass, if the potential format features of the format of the corresponding attachment description do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the blendEnable member of the corresponding element of the pAttachments member of pColorBlendState **must** be VK_FALSE

- VUID-VkGraphicsPipelineCreateInfo-renderPass-07609

If renderPass is not VK_NULL_HANDLE, the pipeline is being created with fragment output interface state, the pColorBlendState pointer is not NULL, the attachmentCount member of pColorBlendState is not ignored, and the subpass uses color attachments, the attachmentCount member of pColorBlendState **must** be equal to the colorAttachmentCount used to create subpass

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04130

If the pipeline requires pre-rasterization shader state, and pViewportState->pViewports is not dynamic, then pViewportState->pViewports **must** be a valid pointer to an array of pViewportState->viewportCount valid VkViewport structures

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04131

If the pipeline requires pre-rasterization shader state, and pViewportState->pScissors is not dynamic, then pViewportState->pScissors **must** be a valid pointer to an array of pViewportState->scissorCount VkRect2D structures

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-00749

If the pipeline requires pre-rasterization shader state, and the wideLines feature is not enabled, and no element of the pDynamicStates member of pDynamicState is VK_DYNAMIC_STATE_LINE_WIDTH, the lineWidth member of pRasterizationState **must** be 1.0

- VUID-VkGraphicsPipelineCreateInfo-rasterizerDiscardEnable-09024

If the pipeline requires pre-rasterization shader state, and the rasterizerDiscardEnable member of pRasterizationState is VK_FALSE, and related dynamic state is not set, pViewportState **must** be a valid pointer to a valid VkPipelineViewportStateCreateInfo structure

- VUID-VkGraphicsPipelineCreateInfo-pViewportState-09025

If pViewportState is not NULL it **must** be a valid pointer to a valid VkPipelineViewportStateCreateInfo structure

- VUID-VkGraphicsPipelineCreateInfo-pMultisampleState-09026

If the pipeline requires fragment output interface state, and the VK_EXT_extended_dynamic_state3 extension is not enabled or any of the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT, VK_DYNAMIC_STATE_SAMPLE_MASK_EXT, or VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic states is not set, or alphaToOne is enabled on the device and VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT is not set, pMultisampleState **must** be a valid pointer to a valid VkPipelineMultisampleStateCreateInfo structure

- VUID-VkGraphicsPipelineCreateInfo-pMultisampleState-09027

If pMultisampleState is not NULL it **must** be a valid pointer to a valid VkPipelineMultisampleStateCreateInfo structure

- VUID-VkGraphicsPipelineCreateInfo-alphaToCoverageEnable-08891

If the pipeline is being created with fragment shader state, the VkPipelineMultisampleStateCreateInfo::alphaToCoverageEnable is not ignored and is

VK_TRUE, then the [Fragment Output Interface](#) **must** contain a variable for the alpha Component word in Location 0 at Index 0

- VUID-VkGraphicsPipelineCreateInfo-renderPass-09028
If `renderPass` is not `VK_NULL_HANDLE`, the pipeline is being created with [fragment shader state](#), and `subpass` uses a depth/stencil attachment, and [related dynamic state is not set](#), `pDepthStencilState` **must** be a valid pointer to a valid [VkPipelineDepthStencilStateCreateInfo](#) structure
- VUID-VkGraphicsPipelineCreateInfo-pDepthStencilState-09029
If `pDepthStencilState` is not `NULL` it **must** be a valid pointer to a valid [VkPipelineDepthStencilStateCreateInfo](#) structure
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09030
If `renderPass` is not `VK_NULL_HANDLE`, the pipeline is being created with [fragment output interface state](#), and `subpass` uses color attachments, and [related dynamic state is not set](#), `pColorBlendState` **must** be a valid pointer to a valid [VkPipelineColorBlendStateCreateInfo](#) structure
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-00754
If the pipeline requires [pre-rasterization shader state](#), the `depthBiasClamp` feature is not enabled, no element of the `pDynamicStates` member of `pDynamicState` is `VK_DYNAMIC_STATE_DEPTH_BIAS`, and the `depthBiasEnable` member of `pRasterizationState` is `VK_TRUE`, the `depthBiasClamp` member of `pRasterizationState` **must** be `0.0`
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-02510
If the pipeline requires [fragment shader state](#), the `VK_EXT_depth_range_unrestricted` extension is not enabled and no element of the `pDynamicStates` member of `pDynamicState` is `VK_DYNAMIC_STATE_DEPTH_BOUNDS`, and the `depthBoundsTestEnable` member of `pDepthStencilState` is `VK_TRUE`, the `minDepthBounds` and `maxDepthBounds` members of `pDepthStencilState` **must** be between `0.0` and `1.0`, inclusive
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07610
If the pipeline requires [fragment shader state](#) or [fragment output interface state](#), and `rasterizationSamples` and `sampleLocationsInfo` are not dynamic, and [VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable](#) included in the `pNext` chain of `pMultisampleState` is `VK_TRUE`, `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide [VkMultisamplePropertiesEXT::sampleLocationGridSize.width](#) as returned by [vkGetPhysicalDeviceMultisamplePropertiesEXT](#) with a `samples` parameter equaling `rasterizationSamples`
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07611
If the pipeline requires [fragment shader state](#) or [fragment output interface state](#), and `rasterizationSamples` and `sampleLocationsInfo` are not dynamic, and [VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable](#) the included in the `pNext` chain of `pMultisampleState` is `VK_TRUE` or `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` is used, `sampleLocationsInfo.sampleLocationGridSize.height` **must** evenly divide [VkMultisamplePropertiesEXT::sampleLocationGridSize.height](#) as returned by [vkGetPhysicalDeviceMultisamplePropertiesEXT](#) with a `samples` parameter equaling `rasterizationSamples`

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07612
If the pipeline requires `fragment shader state` or `fragment output interface state`, and `rasterizationSamples` and `sampleLocationsInfo` are not dynamic, and `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable` included in the `pNext` chain of `pMultisampleState` is `VK_TRUE` or `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` is used, `sampleLocationsInfo.sampleLocationsPerPixel` **must** equal `rasterizationSamples`
- VUID-VkGraphicsPipelineCreateInfo-sampleLocationsEnable-01524
If the pipeline requires `fragment shader state`, and the `sampleLocationsEnable` member of a `VkPipelineSampleLocationsStateCreateInfoEXT` structure included in the `pNext` chain of `pMultisampleState` is `VK_TRUE`, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`
- VUID-VkGraphicsPipelineCreateInfo-subpass-00758
If the pipeline requires `fragment output interface state`, `rasterizationSamples` is not dynamic, and `subpass` does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` member of `pMultisampleState` **must** follow the rules for a `zero-attachment subpass`
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06046
If `renderPass` is not `VK_NULL_HANDLE`, `subpass` **must** be a valid subpass within `renderPass`
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06047
If `renderPass` is not `VK_NULL_HANDLE`, the pipeline is being created with `pre-rasterization shader state`, `subpass` viewMask is not `0`, and `multiviewTessellationShader` is not enabled, then `pStages` **must** not include tessellation shaders
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06048
If `renderPass` is not `VK_NULL_HANDLE`, the pipeline is being created with `pre-rasterization shader state`, `subpass` viewMask is not `0`, and `multiviewGeometryShader` is not enabled, then `pStages` **must** not include a geometry shader
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06050
If `renderPass` is not `VK_NULL_HANDLE` and the pipeline is being created with `pre-rasterization shader state`, and `subpass` viewMask is not `0`, then all of the shaders in the pipeline **must** not include variables decorated with the `Layer` built-in decoration in their interfaces
- VUID-VkGraphicsPipelineCreateInfo-flags-00764
`flags` **must** not contain the `VK_PIPELINE_CREATE_DISPATCH_BASE` flag
- VUID-VkGraphicsPipelineCreateInfo-pStages-01565
If the pipeline requires `fragment shader state` and an input attachment was referenced by an `aspectMask` at `renderPass` creation time, the fragment shader **must** only read from the aspects that were specified for that input attachment
- VUID-VkGraphicsPipelineCreateInfo-layout-01688
The number of resources in `layout` accessible to each shader stage that is used by the pipeline **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageResources`
- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04058
If the pipeline requires `pre-rasterization shader state`, and no element of the `pDynamicStates` member of `pDynamicState` is `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT`, and if

`pNext` chain includes a `VkPipelineDiscardRectangleStateCreateInfoEXT` structure, and if its `discardRectangleCount` member is not 0, then its `pDiscardRectangles` member **must** be a valid pointer to an array of `discardRectangleCount` `VkRect2D` structures

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07855

If `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` is included in the `pDynamicStates` array then the implementation **must** support at least `specVersion 2` of the `VK_EXT_discard_rectangles` extension

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-07856

If `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` is included in the `pDynamicStates` array then the implementation **must** support at least `specVersion 2` of the `VK_EXT_discard_rectangles` extension

- VUID-VkGraphicsPipelineCreateInfo-pStages-02097

If the pipeline requires `vertex input state`, and `pVertexInputState` is not dynamic, then `pVertexInputState` **must** be a valid pointer to a valid `VkPipelineVertexInputStateCreateInfo` structure

- VUID-VkGraphicsPipelineCreateInfo-Input-07904

If the pipeline is being created with `vertex input state` and `pVertexInputState` is not dynamic, then all variables with the `Input` storage class decorated with `Location` in the `Vertex Execution Model OpEntryPoint` **must** contain a location in `VkVertexInputAttributeDescription::location`

- VUID-VkGraphicsPipelineCreateInfo-Input-08733

If the pipeline requires `vertex input state` and `pVertexInputState` is not dynamic, then the numeric type associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be the same as `VkVertexInputAttributeDescription::format`

- VUID-VkGraphicsPipelineCreateInfo-pVertexInputState-08929

If the pipeline is being created with `vertex input state` and `pVertexInputState` is not dynamic, and `VkVertexInputAttributeDescription::format` has a 64-bit component, then the scalar width associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be 64-bit

- VUID-VkGraphicsPipelineCreateInfo-pVertexInputState-08930

If the pipeline is being created with `vertex input state` and `pVertexInputState` is not dynamic, and the scalar width associated with a `Location` decorated `Input` variable in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the corresponding `VkVertexInputAttributeDescription::format` **must** have a 64-bit component

- VUID-VkGraphicsPipelineCreateInfo-pVertexInputState-09198

If the pipeline is being created with `vertex input state` and `pVertexInputState` is not dynamic, and `VkVertexInputAttributeDescription::format` has a 64-bit component, then all `Input` variables at the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** not use components that are not present in the format

- VUID-VkGraphicsPipelineCreateInfo-dynamicPrimitiveTopologyUnrestricted-09031

If the pipeline requires `vertex input state`, and `related dynamic state is not set`, `pInputAssemblyState` **must** be a valid pointer to a valid `VkPipelineInputAssemblyStateCreateInfo` structure

- VUID-VkGraphicsPipelineCreateInfo-pInputAssemblyState-09032
If `pInputAssemblyState` is not `NULL` it **must** be a valid pointer to a valid `VkPipelineInputAssemblyStateCreateInfo` structure
- VUID-VkGraphicsPipelineCreateInfo-pStages-02317
If the pipeline requires `pre-rasterization shader state`, the `Xfb` execution mode **can** be specified by no more than one shader stage in `pStages`
- VUID-VkGraphicsPipelineCreateInfo-pStages-02318
If the pipeline requires `pre-rasterization shader state`, and any shader stage in `pStages` specifies `Xfb` execution mode it **must** be the last `pre-rasterization shader stage`
- VUID-VkGraphicsPipelineCreateInfo-rasterizationStream-02319
If the pipeline requires `pre-rasterization shader state`, and a `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream` value other than zero is specified, all variables in the output interface of the entry point being compiled decorated with `Position`, `PointSize`, `ClipDistance`, or `CullDistance` **must** be decorated with identical `Stream` values that match the `rasterizationStream`
- VUID-VkGraphicsPipelineCreateInfo-rasterizationStream-02320
If the pipeline requires `pre-rasterization shader state`, and `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream` is zero, or not specified, all variables in the output interface of the entry point being compiled decorated with `Position`, `PointSize`, `ClipDistance`, or `CullDistance` **must** be decorated with a `Stream` value of zero, or **must** not specify the `Stream` decoration
- VUID-VkGraphicsPipelineCreateInfo-geometryStreams-02321
If the pipeline requires `pre-rasterization shader state`, and the last `pre-rasterization shader stage` is a geometry shader, and that geometry shader uses the `GeometryStreams` capability, then `VkPhysicalDeviceTransformFeedbackFeaturesEXT::geometryStreams` feature **must** be enabled
- VUID-VkGraphicsPipelineCreateInfo-lineRasterizationMode-02766
If the pipeline requires `pre-rasterization shader state` and at least one of `fragment output interface state` or `fragment shader state`, and `pMultisampleState` is not `NULL`, the `lineRasterizationMode` member of a `VkPipelineRasterizationLineStateCreateInfoKHR` structure included in the `pNext` chain of `pRasterizationState` is `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR` or `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, then the `alphaToCoverageEnable`, `alphaToOneEnable`, and `sampleShadingEnable` members of `pMultisampleState` **must** all be `VK_FALSE`
- VUID-VkGraphicsPipelineCreateInfo-stippledLineEnable-02767
If the pipeline requires `pre-rasterization shader state`, the `stippledLineEnable` member of `VkPipelineRasterizationLineStateCreateInfoKHR` is `VK_TRUE`, and no element of the `pDynamicStates` member of `pDynamicState` is `VK_DYNAMIC_STATE_LINE_STIPPLE_EXT`, then the `lineStippleFactor` member of `VkPipelineRasterizationLineStateCreateInfoKHR` **must** be in the range [1,256]
- VUID-VkGraphicsPipelineCreateInfo-flags-03372
`flags` **must** not include `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR`
- VUID-VkGraphicsPipelineCreateInfo-flags-03373

flags **must** **not** **include**
VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR

- VUID-VkGraphicsPipelineCreateInfo-flags-03374

flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR

- VUID-VkGraphicsPipelineCreateInfo-flags-03375

flags **must** **not** **include**
VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR

- VUID-VkGraphicsPipelineCreateInfo-flags-03376

flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR

- VUID-VkGraphicsPipelineCreateInfo-flags-03377

flags **must** not include VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR

- VUID-VkGraphicsPipelineCreateInfo-flags-03577

flags **must** **not** **include**
VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR

- VUID-VkGraphicsPipelineCreateInfo-flags-11000

If **flags** includes VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT, then the VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands feature **must** be enabled

- VUID-VkGraphicsPipelineCreateInfo-flags-11001

If the pipeline requires **pre-rasterization shader state** and **flags** includes VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT, then all stages **must** not specify Xfb execution mode

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04494

If the pipeline requires **pre-rasterization shader state** or **fragment shader state** and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.width **must** be greater than or equal to 1

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04495

If the pipeline requires **pre-rasterization shader state** or **fragment shader state** and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.height **must** be greater than or equal to 1

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04496

If the pipeline requires **pre-rasterization shader state** or **fragment shader state** and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.width **must** be a power-of-two value

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04497

If the pipeline requires **pre-rasterization shader state** or **fragment shader state** and VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR is not included in pDynamicState->pDynamicStates, VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.height **must** be a power-of-two value

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04498

If the pipeline requires [pre-rasterization shader state](#) or [fragment shader state](#) and `VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR` is not included in `pDynamicState->pDynamicStates`, `VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.width` **must** be less than or equal to 4

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04499

If the pipeline requires [pre-rasterization shader state](#) or [fragment shader state](#) and `VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR` is not included in `pDynamicState->pDynamicStates`, `VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.height` **must** be less than or equal to 4

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04500

If the pipeline requires [pre-rasterization shader state](#) or [fragment shader state](#) and `VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR` is not included in `pDynamicState->pDynamicStates`, and the `pipelineFragmentShadingRate` feature is not enabled, `VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.width` and `VkPipelineFragmentShadingRateStateCreateInfoKHR::fragmentSize.height` **must** both be equal to 1

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-06567

If the pipeline requires [pre-rasterization shader state](#) or [fragment shader state](#) and `VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR` is not included in `pDynamicState->pDynamicStates`, `VkPipelineFragmentShadingRateStateCreateInfoKHR::combinerOps[0]` **must** be a valid `VkFragmentShadingRateCombinerOpKHR` value

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-06568

If the pipeline requires [pre-rasterization shader state](#) or [fragment shader state](#) and `VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR` is not included in `pDynamicState->pDynamicStates`, `VkPipelineFragmentShadingRateStateCreateInfoKHR::combinerOps[1]` **must** be a valid `VkFragmentShadingRateCombinerOpKHR` value

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04501

If the pipeline requires [pre-rasterization shader state](#) or [fragment shader state](#) and `VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR` is not included in `pDynamicState->pDynamicStates`, and the `primitiveFragmentShadingRate` feature is not enabled, `VkPipelineFragmentShadingRateStateCreateInfoKHR::combinerOps[0]` **must** be `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR`

- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-04502

If the pipeline requires [pre-rasterization shader state](#) or [fragment shader state](#) and `VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR` is not included in `pDynamicState->pDynamicStates`, and the `attachmentFragmentShadingRate` feature is not enabled, `VkPipelineFragmentShadingRateStateCreateInfoKHR::combinerOps[1]` **must** be `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR`

- VUID-VkGraphicsPipelineCreateInfo-primitiveFragmentShadingRateWithMultipleViewports-04504

If the pipeline requires [pre-rasterization shader state](#) and the `primitiveFragmentShadingRateWithMultipleViewports` limit is not supported, and entry points specified in `pStages` write to the `ViewportIndex` built-in, they **must** not also write to the `PrimitiveShadingRateKHR` built-in

- VUID-VkGraphicsPipelineCreateInfo-fragmentShadingRateNonTrivialCombinerOps-04506

If the pipeline requires [pre-rasterization shader state](#) or [fragment shader state](#), the [fragmentShadingRateNonTrivialCombinerOps](#) limit is not supported, and [VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR](#) is not included in [pDynamicState->pDynamicStates](#), elements of [VkPipelineFragmentShadingRateStateCreateInfoKHR::combinerOps](#) **must** be [VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR](#) or [VK_FRAGMENT_SHADING_RATE_COMBINER_OP_REPLACE_KHR](#)

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-03578

All elements of the [pDynamicStates](#) member of [pDynamicState](#) **must** not be [VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR](#)

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04807

If the pipeline requires [pre-rasterization shader state](#) and the [vertexInputDynamicState](#) feature is not enabled, there **must** be no element of the [pDynamicStates](#) member of [pDynamicState](#) set to [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#)

- VUID-VkGraphicsPipelineCreateInfo-pDynamicStates-04800

If the [colorWriteEnable](#) feature is not enabled, there **must** be no element of the [pDynamicStates](#) member of [pDynamicState](#) set to [VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT](#)

- VUID-VkGraphicsPipelineCreateInfo-dynamicRendering-06576

If the [dynamicRendering](#) feature is not enabled and the pipeline requires [pre-rasterization shader state](#), [fragment shader state](#), or [fragment output interface state](#), [renderPass](#) **must** not be [VK_NULL_HANDLE](#)

- VUID-VkGraphicsPipelineCreateInfo-multiview-06577

If the [multiview](#) feature is not enabled, the pipeline requires [pre-rasterization shader state](#), [fragment shader state](#), or [fragment output interface state](#), and [renderPass](#) is [VK_NULL_HANDLE](#), [VkPipelineRenderingCreateInfo::viewMask](#) **must** be 0

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06578

If the pipeline requires [pre-rasterization shader state](#), [fragment shader state](#), or [fragment output interface state](#), and [renderPass](#) is [VK_NULL_HANDLE](#), the index of the most significant bit in [VkPipelineRenderingCreateInfo::viewMask](#) **must** be less than [maxMultiviewViewCount](#)

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06579

If the pipeline requires [fragment output interface state](#), and [renderPass](#) is [VK_NULL_HANDLE](#), and [VkPipelineRenderingCreateInfo::colorAttachmentCount](#) is not 0, [VkPipelineRenderingCreateInfo::pColorAttachmentFormats](#) **must** be a valid pointer to an array of [colorAttachmentCount](#) valid [VkFormat](#) values

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06580

If the pipeline requires [fragment output interface state](#), and [renderPass](#) is [VK_NULL_HANDLE](#), each element of [VkPipelineRenderingCreateInfo::pColorAttachmentFormats](#) **must** be a valid [VkFormat](#) value

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06582

If the pipeline requires [fragment output interface state](#), [renderPass](#) is [VK_NULL_HANDLE](#), and any element of [VkPipelineRenderingCreateInfo::pColorAttachmentFormats](#) is not [VK_FORMAT_UNDEFINED](#), that format **must** be a format with [potential format features](#) that include [VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT](#)

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06583

If the pipeline requires **fragment output interface state**, and **renderPass** is **VK_NULL_HANDLE**, **VkPipelineRenderingCreateInfo::depthAttachmentFormat** **must** be a valid **VkFormat** value

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06584

If the pipeline requires **fragment output interface state**, and **renderPass** is **VK_NULL_HANDLE**, **VkPipelineRenderingCreateInfo::stencilAttachmentFormat** **must** be a valid **VkFormat** value

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06585

If the pipeline requires **fragment output interface state**, **renderPass** is **VK_NULL_HANDLE**, and **VkPipelineRenderingCreateInfo::depthAttachmentFormat** is not **VK_FORMAT_UNDEFINED**, it **must** be a format with **potential format features** that include **VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT**

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06586

If the pipeline requires **fragment output interface state**, **renderPass** is **VK_NULL_HANDLE**, and **VkPipelineRenderingCreateInfo::stencilAttachmentFormat** is not **VK_FORMAT_UNDEFINED**, it **must** be a format with **potential format features** that include **VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT**

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06587

If the pipeline requires **fragment output interface state**, **renderPass** is **VK_NULL_HANDLE**, and **VkPipelineRenderingCreateInfo::depthAttachmentFormat** is not **VK_FORMAT_UNDEFINED**, it **must** be a format that includes a depth component

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06588

If the pipeline requires **fragment output interface state**, **renderPass** is **VK_NULL_HANDLE**, and **VkPipelineRenderingCreateInfo::stencilAttachmentFormat** is not **VK_FORMAT_UNDEFINED**, it **must** be a format that includes a stencil component

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06589

If the pipeline requires **fragment output interface state**, **renderPass** is **VK_NULL_HANDLE**, **VkPipelineRenderingCreateInfo::depthAttachmentFormat** is not **VK_FORMAT_UNDEFINED**, and **VkPipelineRenderingCreateInfo::stencilAttachmentFormat** is not **VK_FORMAT_UNDEFINED**, **depthAttachmentFormat** **must** equal **stencilAttachmentFormat**

- VUID-VkGraphicsPipelineCreateInfo-renderPass-09033

If **renderPass** is **VK_NULL_HANDLE**, the pipeline is being created with **fragment shader state** and **fragment output interface state**, and either of **VkPipelineRenderingCreateInfo::depthAttachmentFormat** or **VkPipelineRenderingCreateInfo::stencilAttachmentFormat** are not **VK_FORMAT_UNDEFINED**, and the **VK_EXT_extended_dynamic_state3** extension is not enabled or any of the **VK_DYNAMIC_STATE_DEPTH_TEST_ENABLE**, **VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE**, **VK_DYNAMIC_STATE_DEPTH_COMPARE_OP**, **VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE**, **VK_DYNAMIC_STATE_STENCIL_TEST_ENABLE**, **VK_DYNAMIC_STATE_STENCIL_OP**, or **VK_DYNAMIC_STATE_DEPTH_BOUNDS** dynamic states are not set, **pDepthStencilState** **must** be a valid pointer to a valid **VkPipelineDepthStencilStateCreateInfo** structure

- VUID-VkGraphicsPipelineCreateInfo-pDepthStencilState-09034

If **pDepthStencilState** is not **NULL** it **must** be a valid pointer to a valid **VkPipelineDepthStencilStateCreateInfo** structure

- VUID-VkGraphicsPipelineCreateInfo-renderPass-09037

If `renderPass` is `VK_NULL_HANDLE`, the pipeline is being created with `fragment output interface state`, and any element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` is not `VK_FORMAT_UNDEFINED`, and the `VK_EXT_extended_dynamic_state3` extension is not enabled, or any of the `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT`, `VK_DYNAMIC_STATE_LOGIC_OP_EXT`, `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT`, `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT`, `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, or `VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic states are not set, `pColorBlendState` **must** be a valid pointer to a valid `VkPipelineColorBlendStateCreateInfo` structure

- VUID-VkGraphicsPipelineCreateInfo-pColorBlendState-09038

If `pColorBlendState` is not `NULL` it **must** be a valid pointer to a valid `VkPipelineColorBlendStateCreateInfo` structure

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06055

If `renderPass` is `VK_NULL_HANDLE`, `pColorBlendState` is not dynamic, and the pipeline is being created with `fragment output interface state`, `pColorBlendState->attachmentCount` **must** be equal to `VkPipelineRenderingCreateInfo::colorAttachmentCount`

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06057

If `renderPass` is `VK_NULL_HANDLE`, the pipeline is being created with `pre-rasterization shader state`, `VkPipelineRenderingCreateInfo::viewMask` is not `0`, and the `multiviewTessellationShader` feature is not enabled, then `pStages` **must** not include tessellation shaders

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06058

If `renderPass` is `VK_NULL_HANDLE`, the pipeline is being created with `pre-rasterization shader state`, `VkPipelineRenderingCreateInfo::viewMask` is not `0`, and the `multiviewGeometryShader` feature is not enabled, then `pStages` **must** not include a geometry shader

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06059

If `renderPass` is `VK_NULL_HANDLE`, the pipeline is being created with `pre-rasterization shader state`, and `VkPipelineRenderingCreateInfo::viewMask` is not `0`, all of the shaders in the pipeline **must** not include variables decorated with the `Layer` built-in decoration in their interfaces

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06061

If the `dynamicRenderingLocalRead` feature is not enabled, the pipeline requires `fragment shader state`, and `renderPass` is `VK_NULL_HANDLE`, fragment shaders in `pStages` **must** not include the `InputAttachment` capability

- VUID-VkGraphicsPipelineCreateInfo-renderPass-08710

If the pipeline requires `fragment shader state` and `renderPass` is not `VK_NULL_HANDLE`, fragment shaders in `pStages` **must** not include any of the `TileImageColorReadAccessEXT`, `TileImageDepthReadAccessEXT`, or `TileImageStencilReadAccessEXT` capabilities

- VUID-VkGraphicsPipelineCreateInfo-renderPass-06062

If the pipeline requires `fragment output interface state` and `renderPass` is `VK_NULL_HANDLE`, for each color attachment format defined by the `pColorAttachmentFormats` member of `VkPipelineRenderingCreateInfo`, if its `potential format features` do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the `blendEnable` member of the corresponding element of the `pAttachments` member of

`pColorBlendState` **must** be `VK_FALSE`

- VUID-VkGraphicsPipelineCreateInfo-pStages-06600
If the pipeline requires `pre-rasterization shader state` or `fragment shader state`, `pStages` **must** be a valid pointer to an array of `stageCount` valid `VkPipelineShaderStageCreateInfo` structures
- VUID-VkGraphicsPipelineCreateInfo-stageCount-09587
If the pipeline does not require `pre-rasterization shader state` or `fragment shader state`, `stageCount` **must** be zero
- VUID-VkGraphicsPipelineCreateInfo-pRasterizationState-06601
If the pipeline requires `pre-rasterization shader state`, and `related dynamic state is not set`, `pRasterizationState` **must** be a valid pointer to a valid `VkPipelineRasterizationStateCreateInfo` structure
- VUID-VkGraphicsPipelineCreateInfo-pRasterizationState-09039
If `related dynamic state is not set`, then `pMultisampleState` **must** be a valid pointer to a valid `VkPipelineMultisampleStateCreateInfo` structure
- VUID-VkGraphicsPipelineCreateInfo-pRasterizationState-09040
If `pRasterizationState` is not `NULL` it **must** be a valid pointer to a valid `VkPipelineRasterizationStateCreateInfo` structure
- VUID-VkGraphicsPipelineCreateInfo-layout-06602
If the pipeline requires `fragment shader state` or `pre-rasterization shader state`, `layout` **must** be a valid `VkPipelineLayout` handle
- VUID-VkGraphicsPipelineCreateInfo-renderPass-06603
If the pipeline requires `pre-rasterization shader state`, `fragment shader state`, or `fragment output state`, and `renderPass` is not `VK_NULL_HANDLE`, `renderPass` **must** be a valid `VkRenderPass` handle
- VUID-VkGraphicsPipelineCreateInfo-stageCount-09530
If the pipeline requires `pre-rasterization shader state`, `stageCount` **must** be greater than 0
- VUID-VkGraphicsPipelineCreateInfo-graphicsPipelineLibrary-06606
`flags` **must** not include `VK_PIPELINE_CREATE_LIBRARY_BIT_KHR`
- VUID-VkGraphicsPipelineCreateInfo-conservativePointAndLineRasterization-08892
If `conservativePointAndLineRasterization` is not supported; the pipeline is being created with `vertex input state` and `pre-rasterization shader state`; the pipeline does not include a geometry shader; and the value of `VkPipelineInputAssemblyStateCreateInfo::topology` is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, or `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP`, and either `VK_DYNAMIC_STATE_PRIMITIVE_TOPOLOGY` dynamic state is not enabled or `dynamicPrimitiveTopologyUnrestricted` is `VK_FALSE`, then `VkPipelineRasterizationConservativeStateCreateInfoEXT::conservativeRasterizationMode` **must** be `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`
- VUID-VkGraphicsPipelineCreateInfo-conservativePointAndLineRasterization-06760
If `conservativePointAndLineRasterization` is not supported, the pipeline requires `pre-rasterization shader state`, and the pipeline includes a geometry shader with either the `OutputPoints` or `OutputLineStrip` execution modes, `VkPipelineRasterizationConservativeStateCreateInfoEXT::conservativeRasterizationMode`

must be `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`

- VUID-VkGraphicsPipelineCreateInfo-pStages-06894

If the pipeline requires `pre-rasterization shader state` but not `fragment shader state`, elements of `pStages` **must** not have `stage` set to `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-VkGraphicsPipelineCreateInfo-pStages-06895

If the pipeline requires `fragment shader state` but not `pre-rasterization shader state`, elements of `pStages` **must** not have `stage` set to a shader stage which participates in pre-rasterization

- VUID-VkGraphicsPipelineCreateInfo-pStages-06896

If the pipeline requires `pre-rasterization shader state`, all elements of `pStages` **must** have a `stage` set to a shader stage which participates in `fragment shader state` or `pre-rasterization shader state`

- VUID-VkGraphicsPipelineCreateInfo-stage-06897

If the pipeline requires `fragment shader state` and/or `pre-rasterization shader state`, any value of `stage` **must** not be set in more than one element of `pStages`

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3TessellationDomainOrigin-07370

If the `extendedDynamicState3TessellationDomainOrigin` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT`

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3DepthClampEnable-07371

If the `extendedDynamicState3DepthClampEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT`

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3PolygonMode-07372

If the `extendedDynamicState3PolygonMode` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_POLYGON_MODE_EXT`

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3RasterizationSamples-07373

If the `extendedDynamicState3RasterizationSamples` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT`

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3SampleMask-07374

If the `extendedDynamicState3SampleMask` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT`

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3AlphaToCoverageEnable-07375

If the `extendedDynamicState3AlphaToCoverageEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT`

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3AlphaToOneEnable-07376

If the `extendedDynamicState3AlphaToOneEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT`

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3LogicOpEnable-07377
If the `extendedDynamicState3LogicOpEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ColorBlendEnable-07378
If the `extendedDynamicState3ColorBlendEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ColorBlendEquation-07379
If the `extendedDynamicState3ColorBlendEquation` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ColorWriteMask-07380
If the `extendedDynamicState3ColorWriteMask` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3RasterizationStream-07381
If the `extendedDynamicState3RasterizationStream` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ConservativeRasterizationMode-07382
If the `extendedDynamicState3ConservativeRasterizationMode` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ExtraPrimitiveOverestimationSize-07383
If the `extendedDynamicState3ExtraPrimitiveOverestimationSize` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT`
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-09639
If the pipeline requires `pre-rasterization shader state`, `pDynamicState` includes `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT`, and `pDynamicState` does not include `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT`, `pRasterizationState` **must** include a `VkPipelineRasterizationConservativeStateCreateInfoEXT` in its `pNext` chain
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3DepthClipEnable-07384
If the `extendedDynamicState3DepthClipEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3SampleLocationsEnable-07385
If the `extendedDynamicState3SampleLocationsEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT`

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ColorBlendAdvanced-07386
If the `extendedDynamicState3ColorBlendAdvanced` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ProvokingVertexMode-07387
If the `extendedDynamicState3ProvokingVertexMode` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3LineRasterizationMode-07388
If the `extendedDynamicState3LineRasterizationMode` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3LineStippleEnable-07389
If the `extendedDynamicState3LineStippleEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3DepthClipNegativeOneToOne-07390
If the `extendedDynamicState3DepthClipNegativeOneToOne` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ViewportWScalingEnable-07391
If the `extendedDynamicState3ViewportWScalingEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_VIEWPORT_W_SCALING_ENABLE_NV`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ViewportSwizzle-07392
If the `extendedDynamicState3ViewportSwizzle` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_VIEWPORT_SWIZZLE_NV`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageToColorEnable-07393
If the `extendedDynamicState3CoverageToColorEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_ENABLE_NV`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageToColorLocation-07394
If the `extendedDynamicState3CoverageToColorLocation` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_LOCATION_NV`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageModulationMode-07395
If the `extendedDynamicState3CoverageModulationMode` feature is not enabled, there **must** be

no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_COVERAGE_MODULATION_MODE_NV`

- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageModulationTableEnable-07396
If the `extendedDynamicState3CoverageModulationTableEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_ENABLE_NV`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageModulationTable-07397
If the `extendedDynamicState3CoverageModulationTable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_NV`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3CoverageReductionMode-07398
If the `extendedDynamicState3CoverageReductionMode` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_COVERAGE_REDUCTION_MODE_NV`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3RepresentativeFragmentTestEnable-07399
If the `extendedDynamicState3RepresentativeFragmentTestEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_REPRESENTATIVE_FRAGMENT_TEST_ENABLE_NV`
- VUID-VkGraphicsPipelineCreateInfo-extendedDynamicState3ShadingRateImageEnable-07400
If the `extendedDynamicState3ShadingRateImageEnable` feature is not enabled, there **must** be no element of the `pDynamicStates` member of `pDynamicState` set to `VK_DYNAMIC_STATE_SHADING_RATE_IMAGE_ENABLE_NV`
- VUID-VkGraphicsPipelineCreateInfo-flags-07401
`flags` **must** not include `VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT`
- VUID-VkGraphicsPipelineCreateInfo-pStages-08711
If `pStages` includes a fragment shader stage, `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE` is not set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` member of `VkPipelineDepthStencilStateCreateInfo` **must** be `VK_FALSE`
- VUID-VkGraphicsPipelineCreateInfo-pStages-08712
If `pStages` includes a fragment shader stage, `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` is not set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the value of `VkStencilOpState::writeMask` for both `front` and `back` in `VkPipelineDepthStencilStateCreateInfo` **must** be `0`
- VUID-VkGraphicsPipelineCreateInfo-renderPass-08744
If `renderPass` is `VK_NULL_HANDLE`, the pipeline requires `fragment output state` or `fragment shader state`, the pipeline enables `sample shading`, `rasterizationSamples` is not dynamic, and the `pNext` chain includes a `VkPipelineRenderingCreateInfo` structure,

`rasterizationSamples` **must** be a valid `VkSampleCountFlagBits` value that is set in `imageCreateSampleCounts` (as defined in [Image Creation Limits](#)) for every element of `depthAttachmentFormat`, `stencilAttachmentFormat` and the `pColorAttachmentFormats` array which is not `VK_FORMAT_UNDEFINED`

- VUID-VkGraphicsPipelineCreateInfo-None-08893
The pipeline **must** be created with [pre-rasterization shader state](#)
- VUID-VkGraphicsPipelineCreateInfo-pStages-08894
If `pStages` includes a vertex shader stage, the pipeline **must** be created with [vertex input state](#)
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-08896
If `pRasterizationState->rasterizerDiscardEnable` is `VK_FALSE`, the pipeline **must** be created with [fragment shader state](#) and [fragment output interface state](#)
- VUID-VkGraphicsPipelineCreateInfo-None-09043
If `pDynamicState->pDynamicStates` does not include `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and the format of any color attachment is `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`, the `colorWriteMask` member of the corresponding element of `pColorBlendState->pAttachments` **must** either include all of `VK_COLOR_COMPONENT_R_BIT`, `VK_COLOR_COMPONENT_G_BIT`, and `VK_COLOR_COMPONENT_B_BIT`, or none of them
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09531
If the pipeline is being created with [fragment shader state](#) and [fragment output state](#), the value of `renderPass` is `VK_NULL_HANDLE`, and [VkRenderingInputAttachmentIndexInfoKHR](#) is included, [VkRenderingInputAttachmentIndexInfoKHR::colorAttachmentCount](#) **must** be equal to [VkPipelineRenderingCreateInfo::colorAttachmentCount](#)
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09652
If the pipeline is being created with [fragment shader state](#) and [fragment output state](#), the value of `renderPass` is `VK_NULL_HANDLE`, and [VkRenderingInputAttachmentIndexInfoKHR](#) is not included, the fragment shader **must** not contain any input attachments with a `InputAttachmentIndex` greater than or equal to [VkPipelineRenderingCreateInfo::colorAttachmentCount](#)
- VUID-VkGraphicsPipelineCreateInfo-renderPass-09532
If the pipeline is being created with [fragment output state](#), and the value of `renderPass` is `VK_NULL_HANDLE`, [VkRenderingAttachmentLocationInfoKHR::colorAttachmentCount](#) **must** be equal to [VkPipelineRenderingCreateInfo::colorAttachmentCount](#)

Valid Usage (Implicit)

- VUID-VkGraphicsPipelineCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_GRAPHICS_PIPELINE_CREATE_INFO`
- VUID-VkGraphicsPipelineCreateInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of [VkPipelineBinaryInfoKHR](#), [VkPipelineCreateFlags2CreateInfoKHR](#), [VkPipelineDiscardRectangleStateCreateInfoEXT](#),

VkPipelineFragmentShadingRateStateCreateInfoKHR, VkPipelineLibraryCreateInfoKHR, VkPipelineRenderingCreateInfo, VkRenderingAttachmentLocationInfoKHR, or VkRenderingInputAttachmentIndexInfoKHR

- VUID-VkGraphicsPipelineCreateInfo-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkGraphicsPipelineCreateInfo-pDynamicState-parameter
If **pDynamicState** is not **NULL**, **pDynamicState** **must** be a valid pointer to a valid **VkPipelineDynamicStateCreateInfo** structure
- VUID-VkGraphicsPipelineCreateInfo-commonparent
Each of **basePipelineHandle**, **layout**, and **renderPass** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same **VkDevice**

The **VkPipelineRenderingCreateInfo** structure is defined as:

```
typedef struct VkPipelineRenderingCreateInfo {  
    VkStructureType    sType;  
    const void*        pNext;  
    uint32_t           viewMask;  
    uint32_t           colorAttachmentCount;  
    const VkFormat*    pColorAttachmentFormats;  
    VkFormat           depthAttachmentFormat;  
    VkFormat           stencilAttachmentFormat;  
} VkPipelineRenderingCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_dynamic_rendering  
typedef VkPipelineRenderingCreateInfo VkPipelineRenderingCreateInfoKHR;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **viewMask** is the viewMask used for rendering.
- **colorAttachmentCount** is the number of entries in **pColorAttachmentFormats**
- **pColorAttachmentFormats** is a pointer to an array of **VkFormat** values defining the format of color attachments used in this pipeline.
- **depthAttachmentFormat** is a **VkFormat** value defining the format of the depth attachment used in this pipeline.
- **stencilAttachmentFormat** is a **VkFormat** value defining the format of the stencil attachment used in this pipeline.

When a pipeline is created without a **VkRenderPass**, if the **pNext** chain of **VkGraphicsPipelineCreateInfo** includes this structure, it specifies the view mask and format of attachments used for rendering. If this structure is not specified, and the pipeline does not include

a `VkRenderPass`, `viewMask` and `colorAttachmentCount` are 0, and `depthAttachmentFormat` and `stencilAttachmentFormat` are `VK_FORMAT_UNDEFINED`. If a graphics pipeline is created with a valid `VkRenderPass`, parameters of this structure are ignored.

If `depthAttachmentFormat`, `stencilAttachmentFormat`, or any element of `pColorAttachmentFormats` is `VK_FORMAT_UNDEFINED`, it indicates that the corresponding attachment is unused within the render pass. Valid formats indicate that an attachment **can** be used - but it is still valid to set the attachment to `NULL` when beginning rendering.

Valid Usage

- VUID-VkPipelineRenderingCreateInfo-colorAttachmentCount-09533
`colorAttachmentCount` **must** be less than or equal to `maxColorAttachments`

Valid Usage (Implicit)

- VUID-VkPipelineRenderingCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_RENDERING_CREATE_INFO`

The `VkPipelineCreateFlags2CreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance5
typedef struct VkPipelineCreateFlags2CreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkPipelineCreateFlags2KHR    flags;
} VkPipelineCreateFlags2CreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkPipelineCreateFlagBits2KHR` specifying how a pipeline will be generated.

If this structure is included in the `pNext` chain of a pipeline creation structure, `flags` is used instead of the corresponding `flags` value passed in that creation structure, allowing additional creation flags to be specified.

Valid Usage (Implicit)

- VUID-VkPipelineCreateFlags2CreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_CREATE_FLAGS_2_CREATE_INFO_KHR`
- VUID-VkPipelineCreateFlags2CreateInfoKHR-flags-parameter
`flags` **must** be a valid combination of `VkPipelineCreateFlagBits2KHR` values
- VUID-VkPipelineCreateFlags2CreateInfoKHR-flags-requiredbitmask
`flags` **must** not be 0

Bits which **can** be set in `VkPipelineCreateFlags2CreateInfoKHR::flags`, specifying how a pipeline is created, are:

```
// Provided by VK_KHR_maintenance5
// Flag bits for VkPipelineCreateFlagBits2KHR
typedef VkFlags64 VkPipelineCreateFlagBits2KHR;
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_DISABLE_OPTIMIZATION_BIT_KHR = 0x00000001ULL;
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_ALLOW_DERIVATIVES_BIT_KHR = 0x00000002ULL;
static const VkPipelineCreateFlagBits2KHR VK_PIPELINE_CREATE_2_DERIVATIVE_BIT_KHR =
0x00000004ULL;
// Provided by VK_KHR_maintenance5 with VK_VERSION_1_1 or VK_KHR_device_group
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_VIEW_INDEX_FROM_DEVICE_INDEX_BIT_KHR = 0x00000008ULL;
// Provided by VK_KHR_maintenance5 with VK_VERSION_1_1 or VK_KHR_device_group
static const VkPipelineCreateFlagBits2KHR VK_PIPELINE_CREATE_2_DISPATCH_BASE_BIT_KHR =
0x00000010ULL;
// Provided by VK_KHR_maintenance5 with VK_NV_ray_tracing
static const VkPipelineCreateFlagBits2KHR VK_PIPELINE_CREATE_2_DEFER_COMPILE_BIT_NV =
0x00000020ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_pipeline_executable_properties
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_CAPTURE_STATISTICS_BIT_KHR = 0x00000040ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_pipeline_executable_properties
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR = 0x00000080ULL;
// Provided by VK_KHR_maintenance5 with VK_VERSION_1_3 or
VK_EXT_pipeline_creation_cache_control
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT_KHR = 0x00000100ULL;
// Provided by VK_KHR_maintenance5 with VK_VERSION_1_3 or
VK_EXT_pipeline_creation_cache_control
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_EARLY_RETURN_ON_FAILURE_BIT_KHR = 0x00000200ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_graphics_pipeline_library
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_LINK_TIME_OPTIMIZATION_BIT_EXT = 0x00000400ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_graphics_pipeline_library
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT = 0x00800000ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_pipeline_library
static const VkPipelineCreateFlagBits2KHR VK_PIPELINE_CREATE_2_LIBRARY_BIT_KHR =
0x00000800ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR = 0x00001000ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_AABBS_BIT_KHR = 0x00002000ULL;
```

```

// Provided by VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR = 0x00004000ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR = 0x00008000ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR = 0x00010000ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR = 0x00020000ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR =
0x00080000ULL;
// Provided by VK_KHR_maintenance5 with VK_NV_device_generated_commands
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_NV = 0x00040000ULL;
// Provided by VK_KHR_maintenance5 with VK_NV_ray_tracing_motion_blur
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_RAY_TRACING_ALLOW_MOTION_BIT_NV = 0x00100000ULL;
// Provided by VK_KHR_maintenance5 with (VK_KHR_dynamic_rendering or VK_VERSION_1_3)
and VK_KHR_fragment_shading_rate
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR =
0x00200000ULL;
// Provided by VK_KHR_maintenance5 with (VK_KHR_dynamic_rendering or VK_VERSION_1_3)
and VK_EXT_fragment_density_map
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT =
0x00400000ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_opacity_micromap
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT = 0x01000000ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_attachment_feedback_loop_layout
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT = 0x02000000ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_attachment_feedback_loop_layout
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT = 0x04000000ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_pipeline_protected_access
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_NO_PROTECTED_ACCESS_BIT_EXT = 0x08000000ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_pipeline_protected_access
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_PROTECTED_ACCESS_ONLY_BIT_EXT = 0x40000000ULL;
// Provided by VK_KHR_maintenance5 with VK_NV_displacement_micromap
static const VkPipelineCreateFlagsBits2KHR
VK_PIPELINE_CREATE_2_RAY_TRACING_DISPLACEMENT_MICROMAP_BIT_NV = 0x10000000ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_descriptor_buffer

```

```

static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_DESCRIPTOR_BUFFER_BIT_EXT = 0x20000000ULL;
// Provided by VK_KHR_pipeline_binary
static const VkPipelineCreateFlagBits2KHR VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR =
0x80000000ULL;
// Provided by VK_EXT_device_generated_commands
static const VkPipelineCreateFlagBits2KHR
VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT = 0x400000000ULL;

```

- **VK_PIPELINE_CREATE_2_DISABLE_OPTIMIZATION_BIT_KHR** specifies that the created pipeline will not be optimized. Using this flag **may** reduce the time taken to create the pipeline.
- **VK_PIPELINE_CREATE_2_ALLOW_DERIVATIVES_BIT_KHR** specifies that the pipeline to be created is allowed to be the parent of a pipeline that will be created in a subsequent pipeline creation call.
- **VK_PIPELINE_CREATE_2_DERIVATIVE_BIT_KHR** specifies that the pipeline to be created will be a child of a previously created parent pipeline.
- **VK_PIPELINE_CREATE_2_VIEW_INDEX_FROM_DEVICE_INDEX_BIT_KHR** specifies that any shader input variables decorated as **ViewIndex** will be assigned values as if they were decorated as **DeviceIndex**.
- **VK_PIPELINE_CREATE_2_DISPATCH_BASE_BIT_KHR** specifies that a compute pipeline **can** be used with **vkCmdDispatchBaseKHR** with a non-zero base workgroup.
- **VK_PIPELINE_CREATE_2_CAPTURE_STATISTICS_BIT_KHR** specifies that the shader compiler should capture statistics for the pipeline executables produced by the compile process which **can** later be retrieved by calling **vkGetPipelineExecutableStatisticsKHR**. Enabling this flag **must** not affect the final compiled pipeline but **may** disable pipeline caching or otherwise affect pipeline creation time.
- **VK_PIPELINE_CREATE_2_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR** specifies that the shader compiler should capture the internal representations of pipeline executables produced by the compile process which **can** later be retrieved by calling **vkGetPipelineExecutableInternalRepresentationsKHR**. Enabling this flag **must** not affect the final compiled pipeline but **may** disable pipeline caching or otherwise affect pipeline creation time. When capturing IR from pipelines created with pipeline libraries, there is no guarantee that IR from libraries **can** be retrieved from the linked pipeline. Applications **should** retrieve IR from each library, and any linked pipelines, separately.
- **VK_PIPELINE_CREATE_2_LIBRARY_BIT_KHR** specifies that the pipeline **cannot** be used directly, and instead defines a *pipeline library* that **can** be combined with other pipelines using the **VkPipelineLibraryCreateInfoKHR** structure. This is available in ray tracing pipelines.
- **VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR** specifies that an any-hit shader will always be present when an any-hit shader would be executed. A NULL any-hit shader is an any-hit shader which is effectively **VK_SHADER_UNUSED_KHR**, such as from a shader group consisting entirely of zeros.
- **VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR** specifies that a closest hit shader will always be present when a closest hit shader would be executed. A NULL closest hit shader is a closest hit shader which is effectively **VK_SHADER_UNUSED_KHR**, such as from a shader group consisting entirely of zeros.

- `VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR` specifies that a miss shader will always be present when a miss shader would be executed. A NULL miss shader is a miss shader which is effectively `VK_SHADER_UNUSED_KHR`, such as from a shader group consisting entirely of zeros.
- `VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR` specifies that an intersection shader will always be present when an intersection shader would be executed. A NULL intersection shader is an intersection shader which is effectively `VK_SHADER_UNUSED_KHR`, such as from a shader group consisting entirely of zeros.
- `VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR` specifies that triangle primitives will be skipped during traversal using [pipeline trace ray](#) instructions.
- `VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_AABBS_BIT_KHR` specifies that AABB primitives will be skipped during traversal using [pipeline trace ray](#) instructions.
- `VK_PIPELINE_CREATE_2_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR` specifies that the shader group handles **can** be saved and reused on a subsequent run (e.g. for trace capture and replay).
- `VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT` specifies that the pipeline **can** be used in a `VkIndirectExecutionSetEXT`.
- `VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR` specifies that the pipeline will be used with a fragment shading rate attachment.
- `VK_PIPELINE_CREATE_2_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` specifies that the pipeline **may** be used with an attachment feedback loop including color attachments.
- `VK_PIPELINE_CREATE_2_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` specifies that the pipeline **may** be used with an attachment feedback loop including depth-stencil attachments.
- `VK_PIPELINE_CREATE_2_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT` specifies that the ray tracing pipeline **can** be used with acceleration structures which reference an opacity micromap array.
- `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR` specifies that `VkPipelineBinaryKHR` objects **can** be created from the pipeline. If `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR` is used, implementations **should** not store pipeline data to an internal cache, if such a cache exists as stated by `pipelineBinaryInternalCache`. If `pipelineBinaryPrefersInternalCache` is `VK_TRUE`, applications **should** not use `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR`.

It is valid to set both `VK_PIPELINE_CREATE_2_ALLOW_DERIVATIVES_BIT_KHR` and `VK_PIPELINE_CREATE_2_DERIVATIVE_BIT_KHR`. This allows a pipeline to be both a parent and possibly a child in a pipeline hierarchy. See [Pipeline Derivatives](#) for more information.

```
// Provided by VK_KHR_maintenance5
typedef VkFlags64 VkPipelineCreateFlags2KHR;
```

`VkPipelineCreateFlags2KHR` is a bitmask type for setting a mask of zero or more [VkPipelineCreateFlagBits2KHR](#).

Bits which **can** be set in

- [VkGraphicsPipelineCreateInfo::flags](#)

- `VkComputePipelineCreateInfo::flags`
- `VkRayTracingPipelineCreateInfoKHR::flags`

specify how a pipeline is created, and are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineCreateFlagBits {
    VK_PIPELINE_CREATE_DISABLE_OPTIMIZATION_BIT = 0x00000001,
    VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT = 0x00000002,
    VK_PIPELINE_CREATE_DERIVATIVE_BIT = 0x00000004,
    VK_PIPELINE_CREATE_VIEW_INDEX_FROM_DEVICE_INDEX_BIT = 0x00000008,
    // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR = 0x00004000,
    // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR = 0x00008000,
    // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR = 0x00010000,
    // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR = 0x00020000,
    // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR = 0x00001000,
    // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR = 0x00002000,
    // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR =
0x00080000,
    // Provided by VK_KHR_fragment_shading_rate with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
    VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR =
0x00200000,
    // Provided by VK_KHR_pipeline_executable_properties
    VK_PIPELINE_CREATE_CAPTURE_STATISTICS_BIT_KHR = 0x00000040,
    // Provided by VK_KHR_pipeline_executable_properties
    VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR = 0x00000080,
    // Provided by VK_KHR_pipeline_library
    VK_PIPELINE_CREATE_LIBRARY_BIT_KHR = 0x00000800,
    // Provided by VK_EXT_attachment_feedback_loop_layout
    VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT = 0x02000000,
    // Provided by VK_EXT_attachment_feedback_loop_layout
    VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT = 0x04000000,
    // Provided by VK_EXT_opacity_micromap
    VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT = 0x01000000,
    VK_PIPELINE_CREATE_DISPATCH_BASE = VK_PIPELINE_CREATE_DISPATCH_BASE_BIT,
    // Provided by VK_KHR_device_group
    VK_PIPELINE_CREATE_VIEW_INDEX_FROM_DEVICE_INDEX_BIT_KHR =
VK_PIPELINE_CREATE_VIEW_INDEX_FROM_DEVICE_INDEX_BIT,
    // Provided by VK_KHR_device_group
    VK_PIPELINE_CREATE_DISPATCH_BASE_KHR = VK_PIPELINE_CREATE_DISPATCH_BASE,
    // Provided by VK_KHR_fragment_shading_rate with VK_VERSION_1_3 or
VK_KHR_dynamic_rendering
```



```
// VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
is a deprecated alias
VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR =
VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR,
} VkPipelineCreateFlagBits;
```

- **VK_PIPELINE_CREATE_DISABLE_OPTIMIZATION_BIT** specifies that the created pipeline will not be optimized. Using this flag **may** reduce the time taken to create the pipeline.
- **VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT** specifies that the pipeline to be created is allowed to be the parent of a pipeline that will be created in a subsequent pipeline creation call.
- **VK_PIPELINE_CREATE_DERIVATIVE_BIT** specifies that the pipeline to be created will be a child of a previously created parent pipeline.
- **VK_PIPELINE_CREATE_VIEW_INDEX_FROM_DEVICE_INDEX_BIT** specifies that any shader input variables decorated as **ViewIndex** will be assigned values as if they were decorated as **DeviceIndex**.
- **VK_PIPELINE_CREATE_DISPATCH_BASE** specifies that a compute pipeline **can** be used with **vkCmdDispatchBaseKHR** with a non-zero base workgroup.
- **VK_PIPELINE_CREATE_CAPTURE_STATISTICS_BIT_KHR** specifies that the shader compiler should capture statistics for the pipeline executables produced by the compile process which **can** later be retrieved by calling **vkGetPipelineExecutableStatisticsKHR**. Enabling this flag **must** not affect the final compiled pipeline but **may** disable pipeline caching or otherwise affect pipeline creation time.
- **VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR** specifies that the shader compiler should capture the internal representations of pipeline executables produced by the compile process which **can** later be retrieved by calling **vkGetPipelineExecutableInternalRepresentationsKHR**. Enabling this flag **must** not affect the final compiled pipeline but **may** disable pipeline caching or otherwise affect pipeline creation time. When capturing IR from pipelines created with pipeline libraries, there is no guarantee that IR from libraries **can** be retrieved from the linked pipeline. Applications **should** retrieve IR from each library, and any linked pipelines, separately.
- **VK_PIPELINE_CREATE_LIBRARY_BIT_KHR** specifies that the pipeline **cannot** be used directly, and instead defines a *pipeline library* that **can** be combined with other pipelines using the **VkPipelineLibraryCreateInfoKHR** structure. This is available in ray tracing pipelines.
- **VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR** specifies that an any-hit shader will always be present when an any-hit shader would be executed. A NULL any-hit shader is an any-hit shader which is effectively **VK_SHADER_UNUSED_KHR**, such as from a shader group consisting entirely of zeros.
- **VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR** specifies that a closest hit shader will always be present when a closest hit shader would be executed. A NULL closest hit shader is a closest hit shader which is effectively **VK_SHADER_UNUSED_KHR**, such as from a shader group consisting entirely of zeros.
- **VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR** specifies that a miss shader will always be present when a miss shader would be executed. A NULL miss shader is a miss shader which is effectively **VK_SHADER_UNUSED_KHR**, such as from a shader group consisting entirely of zeros.

- `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR` specifies that an intersection shader will always be present when an intersection shader would be executed. A NULL intersection shader is an intersection shader which is effectively `VK_SHADER_UNUSED_KHR`, such as from a shader group consisting entirely of zeros.
- `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR` specifies that triangle primitives will be skipped during traversal using [pipeline trace ray](#) instructions.
- `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR` specifies that AABB primitives will be skipped during traversal using [pipeline trace ray](#) instructions.
- `VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR` specifies that the shader group handles **can** be saved and reused on a subsequent run (e.g. for trace capture and replay).
- `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR` specifies that the pipeline will be used with a fragment shading rate attachment and dynamic rendering.
- `VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` specifies that the pipeline **may** be used with an attachment feedback loop including color attachments. It is ignored if `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` is set in `pDynamicStates`.
- `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` specifies that the pipeline **may** be used with an attachment feedback loop including depth-stencil attachments. It is ignored if `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` is set in `pDynamicStates`.
- `VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT` specifies that the ray tracing pipeline **can** be used with acceleration structures which reference an opacity micromap array.

It is valid to set both `VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT` and `VK_PIPELINE_CREATE_DERIVATIVE_BIT`. This allows a pipeline to be both a parent and possibly a child in a pipeline hierarchy. See [Pipeline Derivatives](#) for more information.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineCreateFlags;
```

`VkPipelineCreateFlags` is a bitmask type for setting a mask of zero or more [VkPipelineCreateFlagBits](#).

The `VkPipelineBinaryInfoKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    uint32_t             binaryCount;
    const VkPipelineBinaryKHR* pPipelineBinaries;
} VkPipelineBinaryInfoKHR;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `binaryCount` is the number of elements in the `pPipelineBinaries` array.
- `pPipelineBinaries` is a pointer to an array of `VkPipelineBinaryKHR` handles.

If a `VkPipelineBinaryInfoKHR` structure with a `binaryCount` greater than 0 is included in the `pNext` chain of any `Vk*PipelineCreateInfo` structure when creating a pipeline, implementations **must** use the data in `pPipelineBinaries` instead of recalculating it. Any shader module identifiers or shader modules declared in `VkPipelineShaderStageCreateInfo` instances are ignored.

If this structure is not included in the `pNext` chain, it is equivalent to specifying this structure with a `binaryCount` of 0.

Valid Usage

- VUID-VkPipelineBinaryInfoKHR-binaryCount-09603
`binaryCount` and the order of the elements in `pPipelineBinaries` **must** exactly match that returned by `vkCreatePipelineBinariesKHR` for the matching `Vk*PipelineCreateInfo` structure and its `pNext` chain, ignoring the presence of the `VkPipelineBinaryInfoKHR` structure, the presence of the `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR` flag, and absence of any shader module identifiers or shader modules, for the same `global pipeline key`, from either:
 - `VkPipelineBinaryCreateInfoKHR::pPipelineCreateInfo`, or
 - `VkPipelineBinaryCreateInfoKHR::pipeline`

Valid Usage (Implicit)

- VUID-VkPipelineBinaryInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_BINARY_INFO_KHR`
- VUID-VkPipelineBinaryInfoKHR-pPipelineBinaries-parameter
 If `binaryCount` is not 0, `pPipelineBinaries` **must** be a valid pointer to an array of `binaryCount` valid `VkPipelineBinaryKHR` handles

The `VkPipelineDynamicStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineDynamicStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineDynamicStateCreateFlags flags;
    uint32_t                  dynamicStateCount;
    const VkDynamicState*      pDynamicStates;
} VkPipelineDynamicStateCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `flags` is reserved for future use.
- `dynamicStateCount` is the number of elements in the `pDynamicStates` array.
- `pDynamicStates` is a pointer to an array of `VkDynamicState` values specifying which pieces of pipeline state will use the values from dynamic state commands rather than from pipeline state creation information.

Valid Usage

- VUID-VkPipelineDynamicStateCreateInfo-pDynamicStates-01442
Each element of `pDynamicStates` **must** be unique

Valid Usage (Implicit)

- VUID-VkPipelineDynamicStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_DYNAMIC_STATE_CREATE_INFO`
- VUID-VkPipelineDynamicStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineDynamicStateCreateInfo-flags-zero bitmask
`flags` **must** be `0`
- VUID-VkPipelineDynamicStateCreateInfo-pDynamicStates-parameter
If `dynamicStateCount` is not `0`, `pDynamicStates` **must** be a valid pointer to an array of `dynamicStateCount` valid `VkDynamicState` values

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineDynamicStateCreateFlags;
```

`VkPipelineDynamicStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The source of different pieces of dynamic state is specified by the `VkPipelineDynamicStateCreateInfo::pDynamicStates` property of the currently active pipeline, each of whose elements **must** be one of the values:

```
// Provided by VK_VERSION_1_0
typedef enum VkDynamicState {
    VK_DYNAMIC_STATE_VIEWPORT = 0,
    VK_DYNAMIC_STATE_SCISSOR = 1,
    VK_DYNAMIC_STATE_LINE_WIDTH = 2,
    VK_DYNAMIC_STATE_DEPTH_BIAS = 3,
    VK_DYNAMIC_STATE_BLEND_CONSTANTS = 4,
    VK_DYNAMIC_STATE_DEPTH_BOUNDS = 5,
    VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK = 6,
    VK_DYNAMIC_STATE_STENCIL_WRITE_MASK = 7,
```

```

VK_DYNAMIC_STATE_STENCIL_REFERENCE = 8,
// Provided by VK_EXT_discard_rectangles
VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT = 1000099000,
// Provided by VK_EXT_discard_rectangles
VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT = 1000099001,
// Provided by VK_EXT_discard_rectangles
VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT = 1000099002,
// Provided by VK_EXT_sample_locations
VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT = 1000143000,
// Provided by VK_KHR_ray_tracing_pipeline
VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR = 1000347000,
// Provided by VK_KHR_fragment_shading_rate
VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR = 1000226000,
// Provided by VK_EXT_vertex_input_dynamic_state
VK_DYNAMIC_STATE_VERTEX_INPUT_EXT = 1000352000,
// Provided by VK_EXT_color_write_enable
VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT = 1000381000,
// Provided by VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT = 1000455003,
// Provided by VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_POLYGON_MODE_EXT = 1000455004,
// Provided by VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT = 1000455005,
// Provided by VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_SAMPLE_MASK_EXT = 1000455006,
// Provided by VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT = 1000455007,
// Provided by VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT = 1000455008,
// Provided by VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT = 1000455009,
// Provided by VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT = 1000455010,
// Provided by VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT = 1000455011,
// Provided by VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT = 1000455012,
// Provided by VK_EXT_extended_dynamic_state3 with VK_KHR_maintenance2 or
VK_VERSION_1_1
VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT = 1000455002,
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_transform_feedback
VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT = 1000455013,
// Provided by VK_EXT_conservative_rasterization with VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT = 1000455014,
// Provided by VK_EXT_conservative_rasterization with VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT = 1000455015,
// Provided by VK_EXT_depth_clip_enable with VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT = 1000455016,
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_sample_locations
VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT = 1000455017,
// Provided by VK_EXT_blend_operation_advanced with VK_EXT_extended_dynamic_state3

```

```

VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT = 1000455018,
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_provoking_vertex
VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT = 1000455019,
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_line_rasterization
VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT = 1000455020,
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_line_rasterization
VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT = 1000455021,
// Provided by VK_EXT_depth_clip_control with VK_EXT_extended_dynamic_state3
VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT = 1000455022,
// Provided by VK_EXT_extended_dynamic_state3 with VK_NV_clip_space_w_scaling
VK_DYNAMIC_STATE_VIEWPORT_W_SCALING_ENABLE_NV = 1000455023,
// Provided by VK_EXT_extended_dynamic_state3 with VK_NV_viewport_swizzle
VK_DYNAMIC_STATE_VIEWPORT_SWIZZLE_NV = 1000455024,
// Provided by VK_EXT_extended_dynamic_state3 with VK_NV_fragment_coverage_to_color
VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_ENABLE_NV = 1000455025,
// Provided by VK_EXT_extended_dynamic_state3 with VK_NV_fragment_coverage_to_color
VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_LOCATION_NV = 1000455026,
// Provided by VK_EXT_extended_dynamic_state3 with VK_NV_framebuffer_mixed_samples
VK_DYNAMIC_STATE_COVERAGE_MODULATION_MODE_NV = 1000455027,
// Provided by VK_EXT_extended_dynamic_state3 with VK_NV_framebuffer_mixed_samples
VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_ENABLE_NV = 1000455028,
// Provided by VK_EXT_extended_dynamic_state3 with VK_NV_framebuffer_mixed_samples
VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_NV = 1000455029,
// Provided by VK_EXT_extended_dynamic_state3 with VK_NV_shading_rate_image
VK_DYNAMIC_STATE_SHADING_RATE_IMAGE_ENABLE_NV = 1000455030,
// Provided by VK_EXT_extended_dynamic_state3 with
VK_NV_representative_fragment_test
VK_DYNAMIC_STATE_REPRESENTATIVE_FRAGMENT_TEST_ENABLE_NV = 1000455031,
// Provided by VK_EXT_extended_dynamic_state3 with VK_NV_coverage_reduction_mode
VK_DYNAMIC_STATE_COVERAGE_REDUCTION_MODE_NV = 1000455032,
// Provided by VK_EXT_attachment_feedback_loop_dynamic_state
VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT = 1000524000,
// Provided by VK_KHR_line_rasterization
VK_DYNAMIC_STATE_LINE_STIPPLE_KHR = 1000259000,
// Provided by VK_EXT_depth_clamp_control
VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT = 1000582000,
} VkDynamicState;

```

- **VK_DYNAMIC_STATE_VIEWPORT** specifies that the **pViewports** state in **VkPipelineViewportStateCreateInfo** will be ignored and **must** be set dynamically with **vkCmdSetViewport** before any drawing commands. The number of viewports used by a pipeline is still specified by the **viewportCount** member of **VkPipelineViewportStateCreateInfo**.
- **VK_DYNAMIC_STATE_SCISSOR** specifies that the **pScissors** state in **VkPipelineViewportStateCreateInfo** will be ignored and **must** be set dynamically with **vkCmdSetScissor** before any drawing commands. The number of scissor rectangles used by a pipeline is still specified by the **scissorCount** member of **VkPipelineViewportStateCreateInfo**.
- **VK_DYNAMIC_STATE_LINE_WIDTH** specifies that the **lineWidth** state in **VkPipelineRasterizationStateCreateInfo** will be ignored and **must** be set dynamically with **vkCmdSetLineWidth** before any drawing commands that generate line primitives for the

rasterizer.

- `VK_DYNAMIC_STATE_DEPTH_BIAS` specifies that any instance of `VkDepthBiasRepresentationInfoEXT` included in the `pNext` chain of `VkPipelineRasterizationStateCreateInfo` as well as the `depthBiasConstantFactor`, `depthBiasClamp` and `depthBiasSlopeFactor` states in `VkPipelineRasterizationStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetDepthBias` or `vkCmdSetDepthBias2EXT` before any draws are performed with `depth bias` enabled.
- `VK_DYNAMIC_STATE_BLEND_CONSTANTS` specifies that the `blendConstants` state in `VkPipelineColorBlendStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetBlendConstants` before any draws are performed with a pipeline state with `VkPipelineColorBlendAttachmentState` member `blendEnable` set to `VK_TRUE` and any of the blend functions using a constant blend color.
- `VK_DYNAMIC_STATE_DEPTH_BOUNDS` specifies that the `minDepthBounds` and `maxDepthBounds` states of `VkPipelineDepthStencilStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetDepthBounds` before any draws are performed with a pipeline state with `VkPipelineDepthStencilStateCreateInfo` member `depthBoundsTestEnable` set to `VK_TRUE`.
- `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` specifies that the `compareMask` state in `VkPipelineDepthStencilStateCreateInfo` for both `front` and `back` will be ignored and **must** be set dynamically with `vkCmdSetStencilCompareMask` before any draws are performed with a pipeline state with `VkPipelineDepthStencilStateCreateInfo` member `stencilTestEnable` set to `VK_TRUE`.
- `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` specifies that the `writeMask` state in `VkPipelineDepthStencilStateCreateInfo` for both `front` and `back` will be ignored and **must** be set dynamically with `vkCmdSetStencilWriteMask` before any draws are performed with a pipeline state with `VkPipelineDepthStencilStateCreateInfo` member `stencilTestEnable` set to `VK_TRUE`.
- `VK_DYNAMIC_STATE_STENCIL_REFERENCE` specifies that the `reference` state in `VkPipelineDepthStencilStateCreateInfo` for both `front` and `back` will be ignored and **must** be set dynamically with `vkCmdSetStencilReference` before any draws are performed with a pipeline state with `VkPipelineDepthStencilStateCreateInfo` member `stencilTestEnable` set to `VK_TRUE`.
- `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` specifies that the `pDiscardRectangles` state in `VkPipelineDiscardRectangleStateCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetDiscardRectangleEXT` before any draw or clear commands.
- `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` specifies that the presence of the `VkPipelineDiscardRectangleStateCreateInfoEXT` structure in the `VkGraphicsPipelineCreateInfo` chain with a `discardRectangleCount` greater than zero does not implicitly enable discard rectangles and they **must** be enabled dynamically with `vkCmdSetDiscardRectangleEnableEXT` before any draw commands. This is available on implementations that support at least `specVersion 2` of the `VK_EXT_discard_rectangles` extension.
- `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` specifies that the `discardRectangleMode` state in `VkPipelineDiscardRectangleStateCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetDiscardRectangleModeEXT` before any draw commands. This is available on implementations that support at least `specVersion 2` of the `VK_EXT_discard_rectangles` extension.
- `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` specifies that the `sampleLocationsInfo` state in `VkPipelineSampleLocationsStateCreateInfoEXT` will be ignored and **must** be set dynamically

with `vkCmdSetSampleLocationsEXT` before any draw or clear commands. Enabling custom sample locations is still indicated by the `sampleLocationsEnable` member of `VkPipelineSampleLocationsStateCreateInfoEXT`.

- `VK_DYNAMIC_STATE_LINE_STIPPLE_EXT` specifies that the `lineStippleFactor` and `lineStipplePattern` state in `VkPipelineRasterizationLineStateCreateInfoKHR` will be ignored and **must** be set dynamically with `vkCmdSetLineStippleKHR` before any draws are performed with a pipeline state with `VkPipelineRasterizationLineStateCreateInfoKHR` member `stippledLineEnable` set to `VK_TRUE`.
- `VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR` specifies that state in `VkPipelineFragmentShadingRateStateCreateInfoKHR` will be ignored and **must** be set dynamically with `vkCmdSetFragmentShadingRateKHR` before any drawing commands.
- `VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR` specifies that the default stack size computation for the pipeline will be ignored and **must** be set dynamically with `vkCmdSetRayTracingPipelineStackSizeKHR` before any ray tracing calls are performed.
- `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` specifies that the `pVertexInputState` state will be ignored and **must** be set dynamically with `vkCmdSetVertexInputEXT` before any drawing commands
- `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` specifies that the `pColorWriteEnables` state in `VkPipelineColorWriteCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetColorWriteEnableEXT` before any draw call.
- `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` specifies that the `domainOrigin` state in `VkPipelineTessellationDomainOriginStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetTessellationDomainOriginEXT` before any draw call.
- `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT` specifies that the `depthClampEnable` state in `VkPipelineRasterizationStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetDepthClampEnableEXT` before any draw call.
- `VK_DYNAMIC_STATE_POLYGON_MODE_EXT` specifies that the `polygonMode` state in `VkPipelineRasterizationStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetPolygonModeEXT` before any draw call.
- `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` specifies that the `rasterizationSamples` state in `VkPipelineMultisampleStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetRasterizationSamplesEXT` before any draw call.
- `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` specifies that the `pSampleMask` state in `VkPipelineMultisampleStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetSampleMaskEXT` before any draw call.
- `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` specifies that the `alphaToCoverageEnable` state in `VkPipelineMultisampleStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetAlphaToCoverageEnableEXT` before any draw call.
- `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT` specifies that the `alphaToOneEnable` state in `VkPipelineMultisampleStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetAlphaToOneEnableEXT` before any draw call.
- `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` specifies that the `logicOpEnable` state in `VkPipelineColorBlendStateCreateInfo` will be ignored and **must** be set dynamically with `vkCmdSetLogicOpEnableEXT` before any draw call.

- `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` specifies that the `blendEnable` state in `VkPipelineColorBlendAttachmentState` will be ignored and **must** be set dynamically with `vkCmdSetColorBlendEnableEXT` before any draw call.
- `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` specifies that the `srcColorBlendFactor`, `dstColorBlendFactor`, `colorBlendOp`, `srcAlphaBlendFactor`, `dstAlphaBlendFactor`, and `alphaBlendOp` states in `VkPipelineColorBlendAttachmentState` will be ignored and **must** be set dynamically with `vkCmdSetColorBlendEquationEXT` before any draw call.
- `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` specifies that the `colorWriteMask` state in `VkPipelineColorBlendAttachmentState` will be ignored and **must** be set dynamically with `vkCmdSetColorWriteMaskEXT` before any draw call.
- `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` specifies that the `rasterizationStream` state in `VkPipelineRasterizationStateStreamCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetRasterizationStreamEXT` before any draw call.
- `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` specifies that the `conservativeRasterizationMode` state in `VkPipelineRasterizationConservativeStateCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetConservativeRasterizationModeEXT` before any draw call.
- `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT` specifies that the `extraPrimitiveOverestimationSize` state in `VkPipelineRasterizationConservativeStateCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetExtraPrimitiveOverestimationSizeEXT` before any draw call.
- `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT` specifies that the `depthClipEnable` state in `VkPipelineRasterizationDepthClipStateCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetDepthClipEnableEXT` before any draw call.
- `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` specifies that the `sampleLocationsEnable` state in `VkPipelineSampleLocationsStateCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetSampleLocationsEnableEXT` before any draw call.
- `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` specifies that the `colorBlendOp` state in `VkPipelineColorBlendAttachmentState`, and `srcPremultiplied`, `dstPremultiplied`, and `blendOverlap` states in `VkPipelineColorBlendAdvancedStateCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetColorBlendAdvancedEXT` before any draw call.
- `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT` specifies that the `provokingVertexMode` state in `VkPipelineRasterizationProvokingVertexStateCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetProvokingVertexModeEXT` before any draw call.
- `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` specifies that the `lineRasterizationMode` state in `VkPipelineRasterizationLineStateCreateInfoKHR` will be ignored and **must** be set dynamically with `vkCmdSetLineRasterizationModeEXT` before any draw call.
- `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` specifies that the `stippledLineEnable` state in `VkPipelineRasterizationLineStateCreateInfoKHR` will be ignored and **must** be set dynamically with `vkCmdSetLineStippleEnableEXT` before any draw call.
- `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` specifies that the `negativeOneToOne` state in `VkPipelineViewportDepthClipControlCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetDepthClipNegativeOneToOneEXT` before any draw call.

- `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` specifies that the `depthClampMode` and `pDepthClampRange` state in `VkPipelineViewportDepthClampControlCreateInfoEXT` will be ignored and **must** be set dynamically with `vkCmdSetDepthClampRangeEXT` before any draw call.
- `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` specifies that the `VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` and `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` flags will be ignored and **must** be set dynamically with `vkCmdSetAttachmentFeedbackLoopEnableEXT` before any draw call.

10.3.1. Valid Combinations of Stages for Graphics Pipelines

If tessellation shader stages are omitted, the tessellation shading and fixed-function stages of the pipeline are skipped.

If a geometry shader is omitted, the geometry shading stage is skipped.

If a fragment shader is omitted, fragment color outputs have undefined values, and the fragment depth value is determined by [Fragment Operations](#) state. This **can** be useful for depth-only rendering.

Presence of a shader stage in a pipeline is indicated by including a valid `VkPipelineShaderStageCreateInfo` with `module` and `pName` selecting an entry point from a shader module, where that entry point is valid for the stage specified by `stage`.

Presence of some of the fixed-function stages in the pipeline is implicitly derived from enabled shaders and provided state. For example, the fixed-function tessellator is always present when the pipeline has valid Tessellation Control and Tessellation Evaluation shaders.

For example:

- Depth/stencil-only rendering in a subpass with no color attachments
 - Active Pipeline Shader Stages
 - Vertex Shader
 - Required: Fixed-Function Pipeline Stages
 - `VkPipelineVertexInputStateCreateInfo`
 - `VkPipelineInputAssemblyStateCreateInfo`
 - `VkPipelineViewportStateCreateInfo`
 - `VkPipelineRasterizationStateCreateInfo`
 - `VkPipelineMultisampleStateCreateInfo`
 - `VkPipelineDepthStencilStateCreateInfo`
- Color-only rendering in a subpass with no depth/stencil attachment
 - Active Pipeline Shader Stages
 - Vertex Shader
 - Fragment Shader

- Required: Fixed-Function Pipeline Stages
 - [VkPipelineVertexInputStateCreateInfo](#)
 - [VkPipelineInputAssemblyStateCreateInfo](#)
 - [VkPipelineViewportStateCreateInfo](#)
 - [VkPipelineRasterizationStateCreateInfo](#)
 - [VkPipelineMultisampleStateCreateInfo](#)
 - [VkPipelineColorBlendStateCreateInfo](#)
- Rendering pipeline with tessellation and geometry shaders
 - Active Pipeline Shader Stages
 - Vertex Shader
 - Tessellation Control Shader
 - Tessellation Evaluation Shader
 - Geometry Shader
 - Fragment Shader
 - Required: Fixed-Function Pipeline Stages
 - [VkPipelineVertexInputStateCreateInfo](#)
 - [VkPipelineInputAssemblyStateCreateInfo](#)
 - [VkPipelineTessellationStateCreateInfo](#)
 - [VkPipelineViewportStateCreateInfo](#)
 - [VkPipelineRasterizationStateCreateInfo](#)
 - [VkPipelineMultisampleStateCreateInfo](#)
 - [VkPipelineDepthStencilStateCreateInfo](#)
 - [VkPipelineColorBlendStateCreateInfo](#)

10.4. Ray Tracing Pipelines

Ray tracing pipelines consist of multiple shader stages, fixed-function traversal stages, and a pipeline layout.

`VK_SHADER_UNUSED_KHR` is a special shader index used to indicate that a ray generation, miss, or callable shader member is not used.

```
#define VK_SHADER_UNUSED_KHR          (~0U)
```

To create ray tracing pipelines, call:

```
// Provided by VK_KHR_ray_tracing_pipeline
VkResult vkCreateRayTracingPipelinesKHR(
    VkDevice                                device,
    VkDeferredOperationKHR                  deferredOperation,
    VkPipelineCache                          pipelineCache,
    uint32_t                                createInfoCount,
    const VkRayTracingPipelineCreateInfoKHR* pCreateInfos,
    const VkAllocationCallbacks*            pAllocator,
    VkPipeline*                              pPipelines);
```

- **device** is the logical device that creates the ray tracing pipelines.
- **deferredOperation** is `VK_NULL_HANDLE` or the handle of a valid `VkDeferredOperationKHR` request deferral object for this command.
- **pipelineCache** is either `VK_NULL_HANDLE`, indicating that pipeline caching is disabled, or the handle of a valid `pipeline cache` object, in which case use of that cache is enabled for the duration of the command.
- **createInfoCount** is the length of the **pCreateInfos** and **pPipelines** arrays.
- **pCreateInfos** is a pointer to an array of `VkRayTracingPipelineCreateInfoKHR` structures.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pPipelines** is a pointer to an array in which the resulting ray tracing pipeline objects are returned.

The `VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS` error is returned if the implementation is unable to reuse the shader group handles provided in `VkRayTracingShaderGroupCreateInfoKHR::pShaderGroupCaptureReplayHandle` when `VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipelineShaderGroupHandleCaptureReplay` is enabled.

Pipelines are created and returned as described for [Multiple Pipeline Creation](#).

Valid Usage

- VUID-vkCreateRayTracingPipelinesKHR-device-09677
device **must** support at least one queue family with the `VK_QUEUE_COMPUTE_BIT` capability
- VUID-vkCreateRayTracingPipelinesKHR-flags-03415
If the **flags** member of any element of **pCreateInfos** contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, and the **basePipelineIndex** member of that same element is not `-1`, **basePipelineIndex** **must** be less than the index into **pCreateInfos** that corresponds to that element
- VUID-vkCreateRayTracingPipelinesKHR-flags-03416
If the **flags** member of any element of **pCreateInfos** contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, the base pipeline **must** have been created with the `VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT` flag set
- VUID-vkCreateRayTracingPipelinesKHR-flags-03816
flags **must** not contain the `VK_PIPELINE_CREATE_DISPATCH_BASE` flag

- VUID-vkCreateRayTracingPipelinesKHR-deferredOperation-03678
Any previous deferred operation that was associated with **deferredOperation** **must** be complete
- VUID-vkCreateRayTracingPipelinesKHR-pNext-09616
If **VkPipelineBinaryInfoKHR::binaryCount** is not 0 for any element of **pCreateInfos**, **pipelineCache** **must** be **VK_NULL_HANDLE**
- VUID-vkCreateRayTracingPipelinesKHR-pNext-09617
If a **VkPipelineCreateFlags2CreateInfoKHR** structure with the **VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR** flag set is included in the **pNext** chain of any element of **pCreateInfos**, **pipelineCache** **must** be **VK_NULL_HANDLE**
- VUID-vkCreateRayTracingPipelinesKHR-binaryCount-09620
If **VkPipelineBinaryInfoKHR::binaryCount** is not 0 for any element of **pCreateInfos**, **VK_PIPELINE_CREATION_FEEDBACK_APPLICATION_PIPELINE_CACHE_HIT_BIT** **must** not be set in the **flags** of that element
- VUID-vkCreateRayTracingPipelinesKHR-binaryCount-09621
If **VkPipelineBinaryInfoKHR::binaryCount** is not 0 for any element of **pCreateInfos**, **VK_PIPELINE_CREATION_FEEDBACK_BASE_PIPELINE_ACCELERATION_BIT** **must** not be set in the **flags** of that element
- VUID-vkCreateRayTracingPipelinesKHR-binaryCount-09622
If **VkPipelineBinaryInfoKHR::binaryCount** is not 0 for any element of **pCreateInfos**, **VK_PIPELINE_CREATE_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT_EXT** **must** not be set in the **flags** of that element
- VUID-vkCreateRayTracingPipelinesKHR-rayTracingPipeline-03586
The **rayTracingPipeline** feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateRayTracingPipelinesKHR-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkCreateRayTracingPipelinesKHR-deferredOperation-parameter
If **deferredOperation** is not **VK_NULL_HANDLE**, **deferredOperation** **must** be a valid **VkDeferredOperationKHR** handle
- VUID-vkCreateRayTracingPipelinesKHR-pipelineCache-parameter
If **pipelineCache** is not **VK_NULL_HANDLE**, **pipelineCache** **must** be a valid **VkPipelineCache** handle
- VUID-vkCreateRayTracingPipelinesKHR-pCreateInfos-parameter
pCreateInfos **must** be a valid pointer to an array of **createInfoCount** valid **VkRayTracingPipelineCreateInfoKHR** structures
- VUID-vkCreateRayTracingPipelinesKHR-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkCreateRayTracingPipelinesKHR-pPipelines-parameter

`pPipelines` **must** be a valid pointer to an array of `createInfoCount` `VkPipeline` handles

- VUID-vkCreateRayTracingPipelinesKHR-createInfoCount-arraylength
`createInfoCount` **must** be greater than 0
- VUID-vkCreateRayTracingPipelinesKHR-deferredOperation-parent
If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`
- VUID-vkCreateRayTracingPipelinesKHR-pipelineCache-parent
If `pipelineCache` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_OPERATION_DEFERRED_KHR`
- `VK_OPERATION_NOT_DEFERRED_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS`

The `VkRayTracingPipelineCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkRayTracingPipelineCreateInfoKHR {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineCreateFlags     flags;
    uint32_t                  stageCount;
    const VkPipelineShaderStageCreateInfo* pStages;
    uint32_t                  groupCount;
    const VkRayTracingShaderGroupCreateInfoKHR* pGroups;
    uint32_t                  maxPipelineRayRecursionDepth;
    const VkPipelineLibraryCreateInfoKHR* pLibraryInfo;
    const VkRayTracingPipelineInterfaceCreateInfoKHR* pLibraryInterface;
    const VkPipelineDynamicStateCreateInfo* pDynamicState;
    VkPipelineLayout          layout;
    VkPipeline                basePipelineHandle;
    int32_t                   basePipelineIndex;
} VkRayTracingPipelineCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkPipelineCreateFlagBits` specifying how the pipeline will be generated.
- `stageCount` is the number of entries in the `pStages` array.
- `pStages` is a pointer to an array of `stageCount` `VkPipelineShaderStageCreateInfo` structures describing the set of the shader stages to be included in the ray tracing pipeline.
- `groupCount` is the number of entries in the `pGroups` array.
- `pGroups` is a pointer to an array of `groupCount` `VkRayTracingShaderGroupCreateInfoKHR` structures describing the set of the shader stages to be included in each shader group in the ray tracing pipeline.
- `maxPipelineRayRecursionDepth` is the `maximum recursion depth` of shaders executed by this pipeline.
- `pLibraryInfo` is a pointer to a `VkPipelineLibraryCreateInfoKHR` structure defining pipeline libraries to include.
- `pLibraryInterface` is a pointer to a `VkRayTracingPipelineInterfaceCreateInfoKHR` structure defining additional information when using pipeline libraries.
- `pDynamicState` is a pointer to a `VkPipelineDynamicStateCreateInfo` structure, and is used to indicate which properties of the pipeline state object are dynamic and **can** be changed independently of the pipeline state. This **can** be `NULL`, which means no state in the pipeline is considered dynamic.
- `layout` is the description of binding locations used by both the pipeline and descriptor sets used with the pipeline.
- `basePipelineHandle` is a pipeline to derive from.
- `basePipelineIndex` is an index into the `pCreateInfos` parameter to use as a pipeline to derive from.

The parameters `basePipelineHandle` and `basePipelineIndex` are described in more detail in [Pipeline Derivatives](#).

When `VK_PIPELINE_CREATE_LIBRARY_BIT_KHR` is specified, this pipeline defines a *pipeline library* which **cannot** be bound as a ray tracing pipeline directly. Instead, pipeline libraries define common shaders and shader groups which **can** be included in future pipeline creation.

If pipeline libraries are included in `pLibraryInfo`, shaders defined in those libraries are treated as if they were defined as additional entries in `pStages`, appended in the order they appear in the `pLibraries` array and in the `pStages` array when those libraries were defined.

When referencing shader groups in order to obtain a shader group handle, groups defined in those libraries are treated as if they were defined as additional entries in `pGroups`, appended in the order they appear in the `pLibraries` array and in the `pGroups` array when those libraries were defined. The shaders these groups reference are set when the pipeline library is created, referencing those specified in the pipeline library, not in the pipeline that includes it.

The default stack size for a pipeline if `VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR` is not provided is computed as described in [Ray Tracing Pipeline Stack](#).

If the `pNext` chain includes a `VkPipelineCreateFlags2CreateInfoKHR` structure, `VkPipelineCreateFlags2CreateInfoKHR::flags` from that structure is used instead of `flags` from this structure.

Valid Usage

- VUID-VkRayTracingPipelineCreateInfoKHR-None-09497
If the `pNext` chain does not include a `VkPipelineCreateFlags2CreateInfoKHR` structure, `flags` **must** be a valid combination of `VkPipelineCreateFlagBits` values
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-07984
If `flags` contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, and `basePipelineIndex` is -1, `basePipelineHandle` **must** be a valid ray tracing `VkPipeline` handle
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-07985
If `flags` contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, and `basePipelineHandle` is `VK_NULL_HANDLE`, `basePipelineIndex` **must** be a valid index into the calling command's `pCreateInfos` parameter
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-07986
If `flags` contains the `VK_PIPELINE_CREATE_DERIVATIVE_BIT` flag, `basePipelineIndex` **must** be -1 or `basePipelineHandle` **must** be `VK_NULL_HANDLE`
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-07987
If a push constant block is declared in a shader, a push constant range in `layout` **must** match the shader stage
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-10069
If a push constant block is declared in a shader, the block must be contained inside the push constant range in `layout` that matches the stage
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-07988
If a `resource variables` is declared in a shader, a descriptor slot in `layout` **must** match the shader stage
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-07990
If a `resource variables` is declared in a shader, a descriptor slot in `layout` **must** match the descriptor type
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-07991
If a `resource variables` is declared in a shader as an array, a descriptor slot in `layout` **must** match the descriptor count
- VUID-VkRayTracingPipelineCreateInfoKHR-pStages-03426
The shader code for the entry points identified by `pStages`, and the rest of the state identified by this structure **must** adhere to the pipeline linking rules described in the [Shader Interfaces](#) chapter
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-03428
The number of resources in `layout` accessible to each shader stage that is used by the pipeline **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageResources`
- VUID-VkRayTracingPipelineCreateInfoKHR-stage-03425
If `flags` does not include `VK_PIPELINE_CREATE_LIBRARY_BIT_KHR`, the `stage` member of at least

one element of `pStages`, including those implicitly added by `pLibraryInfo`, **must** be `VK_SHADER_STAGE_RAYGEN_BIT_KHR`

- VUID-VkRayTracingPipelineCreateInfoKHR-maxPipelineRayRecursionDepth-03589
`maxPipelineRayRecursionDepth` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxRayRecursionDepth`
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-03465
If `flags` includes `VK_PIPELINE_CREATE_LIBRARY_BIT_KHR`, `pLibraryInterface` **must** not be `NULL`
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-03590
If `pLibraryInfo` is not `NULL` and its `libraryCount` member is greater than 0, `pLibraryInterface` **must** not be `NULL`
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraries-03591
Each element of `pLibraryInfo->pLibraries` **must** have been created with the value of `maxPipelineRayRecursionDepth` equal to that in this pipeline
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-03592
If `pLibraryInfo` is not `NULL`, each element of its `pLibraries` member **must** have been created with a `layout` that is compatible with the `layout` in this pipeline
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-03593
If `pLibraryInfo` is not `NULL`, each element of its `pLibraries` member **must** have been created with values of the `maxPipelineRayPayloadSize` and `maxPipelineRayHitAttributeSize` members of `pLibraryInterface` equal to those in this pipeline
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-03594
If `flags` includes `VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR`, each element of `pLibraryInfo->pLibraries` **must** have been created with the `VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR` bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04718
If `flags` includes `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR`, each element of `pLibraryInfo->pLibraries` **must** have been created with the `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR` bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04719
If `flags` includes `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR`, each element of `pLibraryInfo->pLibraries` **must** have been created with the `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR` bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04720
If `flags` includes `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR`, each element of `pLibraryInfo->pLibraries` **must** have been created with the `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR` bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04721
If `flags` includes `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR`, each element of `pLibraryInfo->pLibraries` **must** have been created with the `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR` bit set
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04722
If `flags` includes `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR`,

each element of `pLibraryInfo->pLibraries` **must** have been created with the `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR` bit set

- VUID-VkRayTracingPipelineCreateInfoKHR-flags-04723

If `flags` includes `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR`, each element of `pLibraryInfo->pLibraries` **must** have been created with the `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR` bit set

- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-03595

If the `VK_KHR_pipeline_library` extension is not enabled, `pLibraryInfo` and `pLibraryInterface` **must** be NULL

- VUID-VkRayTracingPipelineCreateInfoKHR-flags-03470

If `flags` includes `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR`, for any element of `pGroups` with a type of `VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR` or `VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR`, the `anyHitShader` of that element **must** not be `VK_SHADER_UNUSED_KHR`

- VUID-VkRayTracingPipelineCreateInfoKHR-flags-03471

If `flags` includes `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR`, for any element of `pGroups` with a type of `VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR` or `VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR`, the `closestHitShader` of that element **must** not be `VK_SHADER_UNUSED_KHR`

- VUID-VkRayTracingPipelineCreateInfoKHR-rayTraversalPrimitiveCulling-03596

If the `rayTraversalPrimitiveCulling` feature is not enabled, `flags` **must** not include `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR`

- VUID-VkRayTracingPipelineCreateInfoKHR-rayTraversalPrimitiveCulling-03597

If the `rayTraversalPrimitiveCulling` feature is not enabled, `flags` **must** not include `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR`

- VUID-VkRayTracingPipelineCreateInfoKHR-flags-06546

`flags` **must** not include both `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR` and `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR`

- VUID-VkRayTracingPipelineCreateInfoKHR-flags-03598

If `flags` includes `VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR`, `rayTracingPipelineShaderGroupHandleCaptureReplay` **must** be enabled

- VUID-VkRayTracingPipelineCreateInfoKHR-rayTracingPipelineShaderGroupHandleCaptureReplay-03599

If `VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipelineShaderGroupHandleCaptureReplay` is `VK_TRUE` and the `pShaderGroupCaptureReplayHandle` member of any element of `pGroups` is not NULL, `flags` **must** include `VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR`

- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-07999

If `pLibraryInfo` is NULL or its `libraryCount` is 0, `stageCount` **must** not be 0

- VUID-VkRayTracingPipelineCreateInfoKHR-flags-08700
If **flags** does not include **VK_PIPELINE_CREATE_LIBRARY_BIT_KHR** and either **pLibraryInfo** is **NULL** or its **libraryCount** is **0**, **groupCount** **must** not be **0**
- VUID-VkRayTracingPipelineCreateInfoKHR-pDynamicStates-03602
Any element of the **pDynamicStates** member of **pDynamicState** **must** be **VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR**
- VUID-VkRayTracingPipelineCreateInfoKHR-stage-06899
The **stage** value in all **pStages** elements **must** be one of **VK_SHADER_STAGE_RAYGEN_BIT_KHR**, **VK_SHADER_STAGE_ANY_HIT_BIT_KHR**, **VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR**, **VK_SHADER_STAGE_MISS_BIT_KHR**, **VK_SHADER_STAGE_INTERSECTION_BIT_KHR**, or **VK_SHADER_STAGE_CALLABLE_BIT_KHR**
- VUID-VkRayTracingPipelineCreateInfoKHR-flags-07403
If **flags** includes **VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT**, each element of **pLibraryInfo->pLibraries** **must** have been created with the **VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT** bit set

Valid Usage (Implicit)

- VUID-VkRayTracingPipelineCreateInfoKHR-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_CREATE_INFO_KHR**
- VUID-VkRayTracingPipelineCreateInfoKHR-pNext-pNext
Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either **NULL** or a pointer to a valid instance of **VkPipelineBinaryInfoKHR** or **VkPipelineCreateFlags2CreateInfoKHR**
- VUID-VkRayTracingPipelineCreateInfoKHR-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkRayTracingPipelineCreateInfoKHR-pStages-parameter
If **stageCount** is not **0**, **pStages** **must** be a valid pointer to an array of **stageCount** valid **VkPipelineShaderStageCreateInfo** structures
- VUID-VkRayTracingPipelineCreateInfoKHR-pGroups-parameter
If **groupCount** is not **0**, **pGroups** **must** be a valid pointer to an array of **groupCount** valid **VkRayTracingShaderGroupCreateInfoKHR** structures
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInfo-parameter
If **pLibraryInfo** is not **NULL**, **pLibraryInfo** **must** be a valid pointer to a valid **VkPipelineLibraryCreateInfoKHR** structure
- VUID-VkRayTracingPipelineCreateInfoKHR-pLibraryInterface-parameter
If **pLibraryInterface** is not **NULL**, **pLibraryInterface** **must** be a valid pointer to a valid **VkRayTracingPipelineInterfaceCreateInfoKHR** structure
- VUID-VkRayTracingPipelineCreateInfoKHR-pDynamicState-parameter
If **pDynamicState** is not **NULL**, **pDynamicState** **must** be a valid pointer to a valid **VkPipelineDynamicStateCreateInfo** structure
- VUID-VkRayTracingPipelineCreateInfoKHR-layout-parameter

layout must be a valid [VkPipelineLayout](#) handle

- VUID-VkRayTracingPipelineCreateInfoKHR-commonparent

Both of **basePipelineHandle**, and **layout** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same [VkDevice](#)

The [VkRayTracingShaderGroupCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkRayTracingShaderGroupCreateInfoKHR {
    VkStructureType           sType;
    const void*               pNext;
    VkRayTracingShaderGroupTypeKHR type;
    uint32_t                  generalShader;
    uint32_t                  closestHitShader;
    uint32_t                  anyHitShader;
    uint32_t                  intersectionShader;
    const void*               pShaderGroupCaptureReplayHandle;
} VkRayTracingShaderGroupCreateInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **type** is the type of hit group specified in this structure.
- **generalShader** is the index of the ray generation, miss, or callable shader from [VkRayTracingPipelineCreateInfoKHR::pStages](#) in the group if the shader group has **type** of **VK_RAY_TRACING_SHADER_GROUP_TYPE_GENERAL_KHR**, and **VK_SHADER_UNUSED_KHR** otherwise.
- **closestHitShader** is the optional index of the closest hit shader from [VkRayTracingPipelineCreateInfoKHR::pStages](#) in the group if the shader group has **type** of **VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR** or **VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR**, and **VK_SHADER_UNUSED_KHR** otherwise.
- **anyHitShader** is the optional index of the any-hit shader from [VkRayTracingPipelineCreateInfoKHR::pStages](#) in the group if the shader group has **type** of **VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR** or **VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR**, and **VK_SHADER_UNUSED_KHR** otherwise.
- **intersectionShader** is the index of the intersection shader from [VkRayTracingPipelineCreateInfoKHR::pStages](#) in the group if the shader group has **type** of **VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR**, and **VK_SHADER_UNUSED_KHR** otherwise.
- **pShaderGroupCaptureReplayHandle** is **NULL** or a pointer to replay information for this shader group queried from [vkGetRayTracingCaptureReplayShaderGroupHandlesKHR](#), as described in [Ray Tracing Capture Replay](#). Ignored if [VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipelineShaderGroupHandleCaptureReplay](#) is **VK_FALSE**.

Valid Usage

- VUID-VkRayTracingShaderGroupCreateInfoKHR-type-03474
If `type` is `VK_RAY_TRACING_SHADER_GROUP_TYPE_GENERAL_KHR` then `generalShader` **must** be a valid index into `VkRayTracingPipelineCreateInfoKHR::pStages` referring to a shader of `VK_SHADER_STAGE_RAYGEN_BIT_KHR`, `VK_SHADER_STAGE_MISS_BIT_KHR`, or `VK_SHADER_STAGE_CALLABLE_BIT_KHR`
- VUID-VkRayTracingShaderGroupCreateInfoKHR-type-03475
If `type` is `VK_RAY_TRACING_SHADER_GROUP_TYPE_GENERAL_KHR` then `closestHitShader`, `anyHitShader`, and `intersectionShader` **must** be `VK_SHADER_UNUSED_KHR`
- VUID-VkRayTracingShaderGroupCreateInfoKHR-type-03476
If `type` is `VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR` then `intersectionShader` **must** be a valid index into `VkRayTracingPipelineCreateInfoKHR::pStages` referring to a shader of `VK_SHADER_STAGE_INTERSECTION_BIT_KHR`
- VUID-VkRayTracingShaderGroupCreateInfoKHR-type-03477
If `type` is `VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR` then `intersectionShader` **must** be `VK_SHADER_UNUSED_KHR`
- VUID-VkRayTracingShaderGroupCreateInfoKHR-closestHitShader-03478
`closestHitShader` **must** be either `VK_SHADER_UNUSED_KHR` or a valid index into `VkRayTracingPipelineCreateInfoKHR::pStages` referring to a shader of `VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR`
- VUID-VkRayTracingShaderGroupCreateInfoKHR-anyHitShader-03479
`anyHitShader` **must** be either `VK_SHADER_UNUSED_KHR` or a valid index into `VkRayTracingPipelineCreateInfoKHR::pStages` referring to a shader of `VK_SHADER_STAGE_ANY_HIT_BIT_KHR`
- VUID-VkRayTracingShaderGroupCreateInfoKHR-rayTracingPipelineShaderGroupHandleCaptureReplayMixed-03603
If `VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipelineShaderGroupHandleCaptureReplayMixed` is `VK_FALSE` then `pShaderGroupCaptureReplayHandle` **must** not be provided if it has not been provided on a previous call to ray tracing pipeline creation
- VUID-VkRayTracingShaderGroupCreateInfoKHR-rayTracingPipelineShaderGroupHandleCaptureReplayMixed-03604
If `VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipelineShaderGroupHandleCaptureReplayMixed` is `VK_FALSE` then the caller **must** guarantee that no ray tracing pipeline creation commands with `pShaderGroupCaptureReplayHandle` provided execute simultaneously with ray tracing pipeline creation commands without `pShaderGroupCaptureReplayHandle` provided

Valid Usage (Implicit)

- VUID-VkRayTracingShaderGroupCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RAY_TRACING_SHADER_GROUP_CREATE_INFO_KHR`

- VUID-VkRayTracingShaderGroupCreateInfoKHR-pNext-pNext **pNext must** be **NULL**
- VUID-VkRayTracingShaderGroupCreateInfoKHR-type-parameter **type must** be a valid **VkRayTracingShaderGroupTypeKHR** value

The **VkRayTracingShaderGroupTypeKHR** enumeration is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef enum VkRayTracingShaderGroupTypeKHR {
    VK_RAY_TRACING_SHADER_GROUP_TYPE_GENERAL_KHR = 0,
    VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR = 1,
    VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR = 2,
} VkRayTracingShaderGroupTypeKHR;
```

- **VK_RAY_TRACING_SHADER_GROUP_TYPE_GENERAL_KHR** indicates a shader group with a single **VK_SHADER_STAGE_RAYGEN_BIT_KHR**, **VK_SHADER_STAGE_MISS_BIT_KHR**, or **VK_SHADER_STAGE_CALLABLE_BIT_KHR** shader in it.
- **VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR** specifies a shader group that only hits triangles and **must** not contain an intersection shader, only closest hit and any-hit shaders.
- **VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR** specifies a shader group that only intersects with custom geometry and **must** contain an intersection shader and **may** contain closest hit and any-hit shaders.

NOTE

For current group types, the hit group type could be inferred from the presence or absence of the intersection shader, but we provide the type explicitly for future hit groups that do not have that property.

The **VkRayTracingPipelineInterfaceCreateInfoKHR** structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkRayTracingPipelineInterfaceCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           maxPipelineRayPayloadSize;
    uint32_t           maxPipelineRayHitAttributeSize;
} VkRayTracingPipelineInterfaceCreateInfoKHR;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **maxPipelineRayPayloadSize** is the maximum payload size in bytes used by any shader in the pipeline.
- **maxPipelineRayHitAttributeSize** is the maximum attribute structure size in bytes used by any shader in the pipeline.

`maxPipelineRayPayloadSize` is calculated as the maximum number of bytes used by any block declared in the `RayPayloadKHR` or `IncomingRayPayloadKHR` storage classes. `maxPipelineRayHitAttributeSize` is calculated as the maximum number of bytes used by any block declared in the `HitAttributeKHR` storage class. As variables in these storage classes do not have explicit offsets, the size should be calculated as if each variable has a [scalar alignment](#) equal to the largest scalar alignment of any of the block's members.

NOTE

There is no explicit upper limit for `maxPipelineRayPayloadSize`, but in practice it should be kept as small as possible. Similar to invocation local memory, it must be allocated for each shader invocation and for devices which support many simultaneous invocations, this storage can rapidly be exhausted, resulting in failure.

Valid Usage

- VUID-VkRayTracingPipelineInterfaceCreateInfoKHR-maxPipelineRayHitAttributeSize-03605
`maxPipelineRayHitAttributeSize` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxRayHitAttributeSize`

Valid Usage (Implicit)

- VUID-VkRayTracingPipelineInterfaceCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_INTERFACE_CREATE_INFO_KHR`
- VUID-VkRayTracingPipelineInterfaceCreateInfoKHR-pNext-pNext
`pNext` **must** be `NULL`

To query the opaque handles of shaders in the ray tracing pipeline, call:

```
// Provided by VK_KHR_ray_tracing_pipeline
VkResult vkGetRayTracingShaderGroupHandlesKHR(
    VkDevice          device,
    VkPipeline        pipeline,
    uint32_t          firstGroup,
    uint32_t          groupCount,
    size_t            dataSize,
    void*             pData);
```

- `device` is the logical device containing the ray tracing pipeline.
- `pipeline` is the ray tracing pipeline object containing the shaders.
- `firstGroup` is the index of the first group to retrieve a handle for from the `VkRayTracingPipelineCreateInfoKHR::pGroups` array.
- `groupCount` is the number of shader handles to retrieve.

- `dataSize` is the size in bytes of the buffer pointed to by `pData`.
- `pData` is a pointer to an application-allocated buffer where the results will be written.

On success, an array of `groupCount` shader handles will be written to `pData`, with each element being of size `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleSize`.

Valid Usage

- VUID-vkGetRayTracingShaderGroupHandlesKHR-pipeline-04619
`pipeline` **must** be a ray tracing pipeline
- VUID-vkGetRayTracingShaderGroupHandlesKHR-firstGroup-04050
`firstGroup` **must** be less than the number of shader groups in `pipeline`
- VUID-vkGetRayTracingShaderGroupHandlesKHR-firstGroup-02419
The sum of `firstGroup` and `groupCount` **must** be less than or equal to the number of shader groups in `pipeline`
- VUID-vkGetRayTracingShaderGroupHandlesKHR-dataSize-02420
`dataSize` **must** be at least `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleSize × groupCount`
- VUID-vkGetRayTracingShaderGroupHandlesKHR-pipeline-07828
`pipeline` **must** not have been created with `VK_PIPELINE_CREATE_LIBRARY_BIT_KHR`

Valid Usage (Implicit)

- VUID-vkGetRayTracingShaderGroupHandlesKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetRayTracingShaderGroupHandlesKHR-pipeline-parameter
`pipeline` **must** be a valid `VkPipeline` handle
- VUID-vkGetRayTracingShaderGroupHandlesKHR-pData-parameter
`pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkGetRayTracingShaderGroupHandlesKHR-dataSize-arraylength
`dataSize` **must** be greater than 0
- VUID-vkGetRayTracingShaderGroupHandlesKHR-pipeline-parent
`pipeline` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

To query the opaque capture data of shader groups in a ray tracing pipeline, call:

```
// Provided by VK_KHR_ray_tracing_pipeline
VkResult vkGetRayTracingCaptureReplayShaderGroupHandlesKHR(
    VkDevice          device,
    VkPipeline        pipeline,
    uint32_t          firstGroup,
    uint32_t          groupCount,
    size_t            dataSize,
    void*             pData);
```

- **device** is the logical device containing the ray tracing pipeline.
- **pipeline** is the ray tracing pipeline object containing the shaders.
- **firstGroup** is the index of the first group to retrieve a handle for from the `VkRayTracingPipelineCreateInfoKHR::pGroups` array.
- **groupCount** is the number of shader handles to retrieve.
- **dataSize** is the size in bytes of the buffer pointed to by **pData**.
- **pData** is a pointer to an application-allocated buffer where the results will be written.

On success, an array of **groupCount** shader handles will be written to **pData**, with each element being of size `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleCaptureReplaySize`.

Once queried, this opaque data **can** be provided at pipeline creation time (in a subsequent execution), using `VkRayTracingShaderGroupCreateInfoKHR::pShaderGroupCaptureReplayHandle`, as described in [Ray Tracing Capture Replay](#).

Valid Usage

- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pipeline-04620 **pipeline must** be a ray tracing pipeline
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-firstGroup-04051 **firstGroup must** be less than the number of shader groups in **pipeline**
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-firstGroup-03483 The sum of **firstGroup** and **groupCount** **must** be less than or equal to the number of shader groups in **pipeline**
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-dataSize-03484 **dataSize must** be at least `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleCaptureReplaySize × groupCount`
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-rayTracingPipelineShaderGroupHandleCaptureReplay-03606 `VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipelineShaderGroupHandleCaptureReplay` **must** be enabled to call this function
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pipeline-03607 **pipeline must** have been created with a **flags** that included

VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR

- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pipeline-07829
pipeline must not have been created with VK_PIPELINE_CREATE_LIBRARY_BIT_KHR

Valid Usage (Implicit)

- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-device-parameter
device must be a valid [VkDevice](#) handle
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pipeline-parameter
pipeline must be a valid [VkPipeline](#) handle
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pData-parameter
pData must be a valid pointer to an array of **dataSize** bytes
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-dataSize-arraylength
dataSize must be greater than 0
- VUID-vkGetRayTracingCaptureReplayShaderGroupHandlesKHR-pipeline-parent
pipeline must have been created, allocated, or retrieved from **device**

Return Codes

Success

- VK_SUCCESS

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_OUT_OF_DEVICE_MEMORY

To query the pipeline stack size of shaders in a shader group in the ray tracing pipeline, call:

```
// Provided by VK_KHR_ray_tracing_pipeline
VkDeviceSize vkGetRayTracingShaderGroupStackSizeKHR(
    VkDevice          device,
    VkPipeline        pipeline,
    uint32_t          group,
    VkShaderGroupShaderKHR groupShader);
```

- **device** is the logical device containing the ray tracing pipeline.
- **pipeline** is the ray tracing pipeline object containing the shaders groups.
- **group** is the index of the shader group to query.
- **groupShader** is the type of shader from the group to query.

The return value is the ray tracing pipeline stack size in bytes for the specified shader as called

from the specified shader group.

Valid Usage

- VUID-vkGetRayTracingShaderGroupStackSizeKHR-pipeline-04622
`pipeline` **must** be a ray tracing pipeline
- VUID-vkGetRayTracingShaderGroupStackSizeKHR-group-03608
The value of `group` **must** be less than the number of shader groups in `pipeline`
- VUID-vkGetRayTracingShaderGroupStackSizeKHR-groupShader-03609
The shader identified by `groupShader` in `group` **must** not be `VK_SHADER_UNUSED_KHR`

Valid Usage (Implicit)

- VUID-vkGetRayTracingShaderGroupStackSizeKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetRayTracingShaderGroupStackSizeKHR-pipeline-parameter
`pipeline` **must** be a valid `VkPipeline` handle
- VUID-vkGetRayTracingShaderGroupStackSizeKHR-groupShader-parameter
`groupShader` **must** be a valid `VkShaderGroupShaderKHR` value
- VUID-vkGetRayTracingShaderGroupStackSizeKHR-pipeline-parent
`pipeline` **must** have been created, allocated, or retrieved from `device`

Possible values of `groupShader` in `vkGetRayTracingShaderGroupStackSizeKHR` are:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef enum VkShaderGroupShaderKHR {
    VK_SHADER_GROUP_SHADER_GENERAL_KHR = 0,
    VK_SHADER_GROUP_SHADER_CLOSEST_HIT_KHR = 1,
    VK_SHADER_GROUP_SHADER_ANY_HIT_KHR = 2,
    VK_SHADER_GROUP_SHADER_INTERSECTION_KHR = 3,
} VkShaderGroupShaderKHR;
```

- `VK_SHADER_GROUP_SHADER_GENERAL_KHR` uses the shader specified in the group with `VkRayTracingShaderGroupCreateInfoKHR::generalShader`
- `VK_SHADER_GROUP_SHADER_CLOSEST_HIT_KHR` uses the shader specified in the group with `VkRayTracingShaderGroupCreateInfoKHR::closestHitShader`
- `VK_SHADER_GROUP_SHADER_ANY_HIT_KHR` uses the shader specified in the group with `VkRayTracingShaderGroupCreateInfoKHR::anyHitShader`
- `VK_SHADER_GROUP_SHADER_INTERSECTION_KHR` uses the shader specified in the group with `VkRayTracingShaderGroupCreateInfoKHR::intersectionShader`

To [dynamically set](#) the stack size for a ray tracing pipeline, call:

```
// Provided by VK_KHR_ray_tracing_pipeline
void vkCmdSetRayTracingPipelineStackSizeKHR(
    VkCommandBuffer                      commandBuffer,
    uint32_t                             pipelineStackSize);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **pipelineStackSize** is the stack size to use for subsequent ray tracing trace commands.

This command sets the stack size for subsequent ray tracing commands when the ray tracing pipeline is created with **VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR** set in **VkPipelineDynamicStateCreateInfo::pDynamicStates**. Otherwise, the stack size is computed as described in [Ray Tracing Pipeline Stack](#).

Valid Usage

- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-pipelineStackSize-03610
pipelineStackSize **must** be large enough for any dynamic execution through the shaders in the ray tracing pipeline used by a subsequent trace call

Valid Usage (Implicit)

- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-commandBuffer-recording
commandBuffer **must** be in the **recording state**
- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support compute operations
- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdSetRayTracingPipelineStackSizeKHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	State

10.5. Pipeline Destruction

To destroy a pipeline, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyPipeline(
    VkDevice          device,
    VkPipeline        pipeline,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the pipeline.
- **pipeline** is the handle of the pipeline to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyPipeline-pipeline-00765
All submitted commands that refer to **pipeline** **must** have completed execution
- VUID-vkDestroyPipeline-pipeline-00766
If **VkAllocationCallbacks** were provided when **pipeline** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyPipeline-pipeline-00767
If no **VkAllocationCallbacks** were provided when **pipeline** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyPipeline-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkDestroyPipeline-pipeline-parameter
If **pipeline** is not [VK_NULL_HANDLE](#), **pipeline** **must** be a valid [VkPipeline](#) handle
- VUID-vkDestroyPipeline-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure

- VUID-vkDestroyPipeline-pipeline-parent

If **pipeline** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **pipeline** **must** be externally synchronized

10.6. Pipeline Derivatives

A pipeline derivative is a child pipeline created from a parent pipeline, where the child and parent are expected to have much commonality.

The goal of derivative pipelines is that they be cheaper to create using the parent as a starting point, and that it be more efficient (on either host or device) to switch/bind between children of the same parent.

A derivative pipeline is created by setting the **VK_PIPELINE_CREATE_DERIVATIVE_BIT** flag in the **Vk*PipelineCreateInfo** structure. If this is set, then exactly one of **basePipelineHandle** or **basePipelineIndex** members of the structure **must** have a valid handle/index, and specifies the parent pipeline. If **basePipelineHandle** is used, the parent pipeline **must** have already been created. If **basePipelineIndex** is used, then the parent is being created in the same command. **VK_NULL_HANDLE** acts as the invalid handle for **basePipelineHandle**, and -1 is the invalid index for **basePipelineIndex**. If **basePipelineIndex** is used, the base pipeline **must** appear earlier in the array. The base pipeline **must** have been created with the **VK_PIPELINE_CREATE_ALLOW_DERIVATIVES_BIT** flag set.

10.7. Pipeline Cache

Pipeline cache objects allow the result of pipeline construction to be reused between pipelines and between runs of an application. Reuse between pipelines is achieved by passing the same pipeline cache object when creating multiple related pipelines. Reuse across runs of an application is achieved by retrieving pipeline cache contents in one run of an application, saving the contents, and using them to preinitialize a pipeline cache on a subsequent run. The contents of the pipeline cache objects are managed by the implementation. Applications **can** manage the host memory consumed by a pipeline cache object and control the amount of data retrieved from a pipeline cache object.

Pipeline cache objects are represented by **VkPipelineCache** handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPipelineCache)
```

10.7.1. Creating a Pipeline Cache

To create pipeline cache objects, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreatePipelineCache(
    VkDevice device,
    const VkPipelineCacheCreateInfo* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkPipelineCache* pPipelineCache);
```

- **device** is the logical device that creates the pipeline cache object.
- **pCreateInfo** is a pointer to a [VkPipelineCacheCreateInfo](#) structure containing initial parameters for the pipeline cache object.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pPipelineCache** is a pointer to a [VkPipelineCache](#) handle in which the resulting pipeline cache object is returned.

NOTE

Applications **can** track and manage the total host memory size of a pipeline cache object using the **pAllocator**. Applications **can** limit the amount of data retrieved from a pipeline cache object in [vkGetPipelineCacheData](#). Implementations **should** not internally limit the total number of entries added to a pipeline cache object or the total host memory consumed.

Once created, a pipeline cache **can** be passed to the [vkCreateGraphicsPipelines](#), [vkCreateRayTracingPipelinesKHR](#), and [vkCreateComputePipelines](#) commands. If the pipeline cache passed into these commands is not [VK_NULL_HANDLE](#), the implementation will query it for possible reuse opportunities and update it with new content. The use of the pipeline cache object in these commands is internally synchronized, and the same pipeline cache object **can** be used in multiple threads simultaneously.

NOTE

Implementations **should** make every effort to limit any critical sections to the actual accesses to the cache, which is expected to be significantly shorter than the duration of the [vkCreate*Pipelines](#) commands.

Valid Usage (Implicit)

- VUID-vkCreatePipelineCache-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCreatePipelineCache-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkPipelineCacheCreateInfo](#) structure
- VUID-vkCreatePipelineCache-pAllocator-parameter
If **pAllocator** is not [NULL](#), **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreatePipelineCache-pPipelineCache-parameter
pPipelineCache **must** be a valid pointer to a [VkPipelineCache](#) handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkPipelineCacheCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineCacheCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkPipelineCacheCreateFlags flags;
    size_t               initialDataSize;
    const void*          pInitialData;
} VkPipelineCacheCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `initialDataSize` is the number of bytes in `pInitialData`. If `initialDataSize` is zero, the pipeline cache will initially be empty.
- `pInitialData` is a pointer to previously retrieved pipeline cache data. If the pipeline cache data is incompatible (as defined below) with the device, the pipeline cache will be initially empty. If `initialDataSize` is zero, `pInitialData` is ignored.

Valid Usage

- VUID-VkPipelineCacheCreateInfo-initialDataSize-00768
If `initialDataSize` is not 0, it **must** be equal to the size of `pInitialData`, as returned by `vkGetPipelineCacheData` when `pInitialData` was originally retrieved
- VUID-VkPipelineCacheCreateInfo-initialDataSize-00769
If `initialDataSize` is not 0, `pInitialData` **must** have been retrieved from a previous call to `vkGetPipelineCacheData`

Valid Usage (Implicit)

- VUID-VkPipelineCacheCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_CACHE_CREATE_INFO`

- VUID-VkPipelineCacheCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineCacheCreateInfo-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkPipelineCacheCreateInfo-pInitialData-parameter
If `initialDataSize` is not `0`, `pInitialData` **must** be a valid pointer to an array of `initialDataSize` bytes

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineCacheCreateFlags;
```

`VkPipelineCacheCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

10.7.2. Merging Pipeline Caches

Pipeline cache objects **can** be merged using the command:

```
// Provided by VK_VERSION_1_0
VkResult vkMergePipelineCaches(
    VkDevice          device,
    VkPipelineCache    dstCache,
    uint32_t          srcCacheCount,
    const VkPipelineCache* pSrcCaches);
```

- `device` is the logical device that owns the pipeline cache objects.
- `dstCache` is the handle of the pipeline cache to merge results into.
- `srcCacheCount` is the length of the `pSrcCaches` array.
- `pSrcCaches` is a pointer to an array of pipeline cache handles, which will be merged into `dstCache`. The previous contents of `dstCache` are included after the merge.

NOTE The details of the merge operation are implementation-dependent, but implementations **should** merge the contents of the specified pipelines and prune duplicate entries.

Valid Usage

- VUID-vkMergePipelineCaches-dstCache-00770
`dstCache` **must** not appear in the list of source caches

Valid Usage (Implicit)

- VUID-vkMergePipelineCaches-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkMergePipelineCaches-dstCache-parameter
dstCache **must** be a valid [VkPipelineCache](#) handle
- VUID-vkMergePipelineCaches-pSrcCaches-parameter
pSrcCaches **must** be a valid pointer to an array of **srcCacheCount** valid [VkPipelineCache](#) handles
- VUID-vkMergePipelineCaches-srcCacheCount-arraylength
srcCacheCount **must** be greater than 0
- VUID-vkMergePipelineCaches-dstCache-parent
dstCache **must** have been created, allocated, or retrieved from **device**
- VUID-vkMergePipelineCaches-pSrcCaches-parent
Each element of **pSrcCaches** **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **dstCache** **must** be externally synchronized

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

10.7.3. Retrieving Pipeline Cache Data

Data **can** be retrieved from a pipeline cache object using the command:

```
// Provided by VK_VERSION_1_0
VkResult vkGetPipelineCacheData(
    VkDevice          device,
    VkPipelineCache    pipelineCache,
    size_t*           pDataSize,
    void*             pData);
```

- **device** is the logical device that owns the pipeline cache.

- `pipelineCache` is the pipeline cache to retrieve data from.
- `pDataSize` is a pointer to a `size_t` value related to the amount of data in the pipeline cache, as described below.
- `pData` is either `NULL` or a pointer to a buffer.

If `pData` is `NULL`, then the maximum size of the data that **can** be retrieved from the pipeline cache, in bytes, is returned in `pDataSize`. Otherwise, `pDataSize` **must** point to a variable set by the application to the size of the buffer, in bytes, pointed to by `pData`, and on return the variable is overwritten with the amount of data actually written to `pData`. If `pDataSize` is less than the maximum size that **can** be retrieved by the pipeline cache, at most `pDataSize` bytes will be written to `pData`, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all of the pipeline cache was returned.

Any data written to `pData` is valid and **can** be provided as the `pInitialData` member of the `VkPipelineCacheCreateInfo` structure passed to `vkCreatePipelineCache`.

Two calls to `vkGetPipelineCacheData` with the same parameters **must** retrieve the same data unless a command that modifies the contents of the cache is called between them.

The initial bytes written to `pData` **must** be a header as described in the [Pipeline Cache Header](#) section.

If `pDataSize` is less than what is necessary to store this header, nothing will be written to `pData` and zero will be written to `pDataSize`.

Valid Usage (Implicit)

- VUID-vkGetPipelineCacheData-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkGetPipelineCacheData-pipelineCache-parameter `pipelineCache` **must** be a valid `VkPipelineCache` handle
- VUID-vkGetPipelineCacheData-pDataSize-parameter `pDataSize` **must** be a valid pointer to a `size_t` value
- VUID-vkGetPipelineCacheData-pData-parameter
If the value referenced by `pDataSize` is not 0, and `pData` is not `NULL`, `pData` **must** be a valid pointer to an array of `pDataSize` bytes
- VUID-vkGetPipelineCacheData-pipelineCache-parent `pipelineCache` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

10.7.4. Pipeline Cache Header

Applications **can** store the data retrieved from the pipeline cache, and use these data, possibly in a future run of the application, to populate new pipeline cache objects. The results of pipeline compiles, however, **may** depend on the vendor ID, device ID, driver version, and other details of the device. To enable applications to detect when previously retrieved data is incompatible with the device, the pipeline cache data **must** begin with a valid pipeline cache header.

NOTE

Structures described in this section are not part of the Vulkan API and are only used to describe the representation of data elements in pipeline cache data. Accordingly, the valid usage clauses defined for structures defined in this section do not define valid usage conditions for APIs accepting pipeline cache data as input, as providing invalid pipeline cache data as input to any Vulkan API commands will result in the provided pipeline cache data being ignored.

Version one of the pipeline cache header is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineCacheHeaderVersionOne {
    uint32_t          headerSize;
    VkPipelineCacheHeaderVersion headerVersion;
    uint32_t          vendorID;
    uint32_t          deviceID;
    uint8_t           pipelineCacheUUID[VK_UUID_SIZE];
} VkPipelineCacheHeaderVersionOne;
```

- `headerSize` is the length in bytes of the pipeline cache header.
- `headerVersion` is a `VkPipelineCacheHeaderVersion` value specifying the version of the header. A consumer of the pipeline cache **should** use the cache version to interpret the remainder of the cache header.
- `vendorID` is the `VkPhysicalDeviceProperties::vendorID` of the implementation.
- `deviceID` is the `VkPhysicalDeviceProperties::deviceID` of the implementation.
- `pipelineCacheUUID` is the `VkPhysicalDeviceProperties::pipelineCacheUUID` of the implementation.

Unlike most structures declared by the Vulkan API, all fields of this structure are written with the least significant byte first, regardless of host byte-order.

The C language specification does not define the packing of structure members. This layout assumes tight structure member packing, with members laid out in the order listed in the structure, and the intended size of the structure is 32 bytes. If a compiler produces code that diverges from that pattern, applications **must** employ another method to set values at the correct offsets.

Valid Usage

- VUID-VkPipelineCacheHeaderVersionOne-headerSize-04967
`headerSize` **must** be 32
- VUID-VkPipelineCacheHeaderVersionOne-headerVersion-04968
`headerVersion` **must** be `VK_PIPELINE_CACHE_HEADER_VERSION_ONE`
- VUID-VkPipelineCacheHeaderVersionOne-headerSize-08990
`headerSize` **must** not exceed the size of the pipeline cache

Valid Usage (Implicit)

- VUID-VkPipelineCacheHeaderVersionOne-headerVersion-parameter
`headerVersion` **must** be a valid `VkPipelineCacheHeaderVersion` value

Possible values of the `headerVersion` value of the pipeline cache header are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineCacheHeaderVersion {
    VK_PIPELINE_CACHE_HEADER_VERSION_ONE = 1,
} VkPipelineCacheHeaderVersion;
```

- `VK_PIPELINE_CACHE_HEADER_VERSION_ONE` specifies version one of the pipeline cache, described by `VkPipelineCacheHeaderVersionOne`.

10.7.5. Destroying a Pipeline Cache

To destroy a pipeline cache, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyPipelineCache(
    VkDevice                device,
    VkPipelineCache         pipelineCache,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the pipeline cache object.
- `pipelineCache` is the handle of the pipeline cache to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyPipelineCache-pipelineCache-00771
If `VkAllocationCallbacks` were provided when `pipelineCache` was created, a compatible set

of callbacks **must** be provided here

- VUID-vkDestroyPipelineCache-pipelineCache-00772

If no `VkAllocationCallbacks` were provided when `pipelineCache` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyPipelineCache-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyPipelineCache-pipelineCache-parameter
If `pipelineCache` is not `VK_NULL_HANDLE`, `pipelineCache` **must** be a valid `VkPipelineCache` handle
- VUID-vkDestroyPipelineCache-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyPipelineCache-pipelineCache-parent
If `pipelineCache` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `pipelineCache` **must** be externally synchronized

10.8. Pipeline Binaries

Pipeline binary objects allow the result of pipeline construction to be reused between pipelines and between runs of an application. Reuse is achieved by extracting pipeline binaries from a `VkPipeline` object, associating them with a corresponding `VkPipelineBinaryKeyKHR` and then adding a `VkPipelineBinaryInfoKHR` to the `pNext` chain of any `Vk*PipelineCreateInfo` when creating a pipeline. Pipeline binaries can be reused between runs by extracting `VkPipelineBinaryDataKHR` from `VkPipelineBinaryKHR` objects, saving the contents, and then using them to create a `VkPipelineBinaryKHR` object on subsequent runs.

When creating a pipeline that includes `VkPipelineBinaryInfoKHR` in the `pNext` chain, or has the `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR` flag set, the use of `VkPipelineCache` objects is not allowed.

Pipeline binary objects are represented by `VkPipelineBinaryKHR` handles:

```
// Provided by VK_KHR_pipeline_binary
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPipelineBinaryKHR)
```

10.8.1. Generating the Pipeline Key

To generate the key for a particular pipeline creation info, call:

```
// Provided by VK_KHR_pipeline_binary
VkResult vkGetPipelineKeyKHR(
    VkDevice device,
    const VkPipelineCreateInfoKHR* pPipelineCreateInfo,
    VkPipelineBinaryKeyKHR* pPipelineKey);
```

- **device** is the logical device that creates the pipeline object.
- **pPipelineCreateInfo** is **NULL** or a pointer to a **VkPipelineCreateInfoKHR** structure.
- **pPipelineKey** is a pointer to a **VkPipelineBinaryKeyKHR** structure in which the resulting key is returned.

If **pPipelineCreateInfo** is **NULL**, then the implementation **must** return the global key that applies to all pipelines. If the key obtained in this way changes between saving and restoring data obtained from **vkGetPipelineBinaryDataKHR** in a different **VkDevice**, then the application **must** assume that the restored data is invalid and cannot be passed to **vkCreatePipelineBinariesKHR**. Otherwise the application **can** assume the data is still valid.

If **pPipelineCreateInfo** is not **NULL**, the key obtained functions as a method to compare two pipeline creation info structures. Implementations **may** not compare parts of a pipeline creation info which would not contribute to the final binary output. If a shader module identifier is used instead of a shader module, the **pPipelineKey** generated **must** be equal to the key generated when using the shader module from which the identifier was queried. If the content of two **pPipelineKey** are equal, pipelines created with the two **pPipelineCreateInfo->pname:pNext** create infos **must** produce the same **VkPipelineBinaryKHR** contents.

The pipeline key is distinct from pipeline binary key. Pipeline binary keys **can** only be obtained after compilation. The pipeline key is intended to optionally allow associating pipeline create info with multiple pipeline binary keys.

Valid Usage

- VUID-vkGetPipelineKeyKHR-pNext-09605
The **pNext** chain of **pPipelineCreateInfo** **must** not set **VkPipelineBinaryInfoKHR::binaryCount** to a value greater than 0

Valid Usage (Implicit)

- VUID-vkGetPipelineKeyKHR-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkGetPipelineKeyKHR-pPipelineCreateInfo-parameter
If **pPipelineCreateInfo** is not **NULL**, **pPipelineCreateInfo** **must** be a valid pointer to a valid

[VkPipelineCreateInfoKHR](#) structure

- VUID-vkGetPipelineKeyKHR-pPipelineKey-parameter
[pPipelineKey](#) **must** be a valid pointer to a [VkPipelineBinaryKeyKHR](#) structure

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

The [VkPipelineCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineCreateInfoKHR {
    VkStructureType    sType;
    void*              pNext;
} VkPipelineCreateInfoKHR;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is a pointer to a structure extending this structure.

Valid Usage

- VUID-VkPipelineCreateInfoKHR-pNext-09604
[pNext](#) **must** be pointer to a valid instance of [VkGraphicsPipelineCreateInfo](#), [VkRayTracingPipelineCreateInfoKHR](#), or [VkComputePipelineCreateInfo](#)

Valid Usage (Implicit)

- VUID-VkPipelineCreateInfoKHR-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_PIPELINE_CREATE_INFO_KHR](#)

The [VkPipelineBinaryKeyKHR](#) structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryKeyKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t            keySize;
    uint8_t             key[VK_MAX_PIPELINE_BINARY_KEY_SIZE_KHR];
} VkPipelineBinaryKeyKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **keySize** is the size, in bytes, of valid data returned in **key**.
- **key** is a buffer of opaque data specifying a pipeline binary key.

Any returned values beyond the first **keySize** bytes are undefined. Implementations **must** return a **keySize** greater than 0, and less-or-equal to **VK_MAX_PIPELINE_BINARY_KEY_SIZE_KHR**.

Two keys are considered equal if **keySize** is equal and the first **keySize** bytes of **key** compare equal.

Implementations **may** return a different **keySize** for different binaries.

Implementations **should** ensure that **keySize** is large enough to uniquely identify a pipeline binary.

Valid Usage (Implicit)

- VUID-VkPipelineBinaryKeyKHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_PIPELINE_BINARY_KEY_KHR**
- VUID-VkPipelineBinaryKeyKHR-pNext-pNext
pNext must be **NULL**

VK_MAX_PIPELINE_BINARY_KEY_SIZE_KHR is the length in bytes of a binary key, as returned in [VkPipelineBinaryKeyKHR::keySize](#).

```
#define VK_MAX_PIPELINE_BINARY_KEY_SIZE_KHR 32U
```

10.8.2. Creating Pipeline Binaries

To create pipeline binary objects, call:

```
// Provided by VK_KHR_pipeline_binary
VkResult vkCreatePipelineBinariesKHR(
    VkDevice                                device,
    const VkPipelineBinaryCreateInfoKHR*    pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkPipelineBinaryHandlesInfoKHR*         pBinaries);
```

- **device** is the logical device that creates the pipeline binary objects.
- **pCreateInfo** is a pointer to a [VkPipelineBinaryCreateInfoKHR](#) structure that contains the data to create the pipeline binaries from.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pBinaries** is a pointer to a [VkPipelineBinaryHandlesInfoKHR](#) structure in which the resulting pipeline binaries are returned.

The implementation will attempt to create all pipeline binaries. If creation fails for any pipeline binary, then:

- The corresponding entry in the **pPipelineBinaries** output array will be filled with [VK_NULL_HANDLE](#).
- The **VkResult** returned by [vkCreatePipelineBinariesKHR](#) will contain the error value for the first entry in the output array in **pBinaries** containing [VK_NULL_HANDLE](#).

Valid Usage (Implicit)

- VUID-vkCreatePipelineBinariesKHR-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCreatePipelineBinariesKHR-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkPipelineBinaryCreateInfoKHR](#) structure
- VUID-vkCreatePipelineBinariesKHR-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreatePipelineBinariesKHR-pBinaries-parameter
pBinaries **must** be a valid pointer to a [VkPipelineBinaryHandlesInfoKHR](#) structure

Return Codes

Success

- **VK_SUCCESS**
- **VK_INCOMPLETE**
- **VK_PIPELINE_BINARY_MISSING_KHR**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**
- **VK_ERROR_INITIALIZATION_FAILED**

The [VkPipelineBinaryHandlesInfoKHR](#) structure is defined as:


```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryHandlesInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    uint32_t             pipelineBinaryCount;
    VkPipelineBinaryKHR* pPipelineBinaries;
} VkPipelineBinaryHandlesInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pipelineBinaryCount` is the number of binaries associated with this pipeline or the number of entries in the `pPipelineBinaries` array.
- `pPipelineBinaries` is `NULL` or a pointer to an array of `VkPipelineBinaryKHR` handles in which the resulting pipeline binaries are returned.

If `pPipelineBinaries` is `NULL`, the number of binaries that would be created is returned in `pipelineBinaryCount`. Otherwise, `pipelineBinaryCount` **must** be the number of entries in the `pPipelineBinaries` array, and on return from `vkCreatePipelineBinariesKHR` `pipelineBinaryCount` is overwritten with the number of handles actually written to `pPipelineBinaries`. If the value of `pipelineBinaryCount` is less than the number of binaries that would have been created, at most `pipelineBinaryCount` handles will be written to `pPipelineBinaries` and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that `pPipelineBinaries` was not large enough to create all the binaries.

Valid Usage (Implicit)

- VUID-VkPipelineBinaryHandlesInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_BINARY_HANDLES_INFO_KHR`
- VUID-VkPipelineBinaryHandlesInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineBinaryHandlesInfoKHR-pPipelineBinaries-parameter
If `pipelineBinaryCount` is not 0, and `pPipelineBinaries` is not `NULL`, `pPipelineBinaries` **must** be a valid pointer to an array of `pipelineBinaryCount` `VkPipelineBinaryKHR` handles

The `VkPipelineBinaryCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryCreateInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    const VkPipelineBinaryKeysAndDataKHR* pKeysAndDataInfo;
    VkPipeline           pipeline;
    const VkPipelineCreateInfoKHR* pPipelineCreateInfo;
} VkPipelineBinaryCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pKeysAndDataInfo` is `NULL` or a pointer to a `VkPipelineBinaryKeysAndDataKHR` structure that contains keys and data to create the pipeline binaries from.
- `pipeline` is `VK_NULL_HANDLE` or a `VkPipeline` that contains data to create the pipeline binaries from.
- `pPipelineCreateInfo` is `NULL` or a pointer to a `VkPipelineCreateInfoKHR` structure with the pipeline creation info. This is used to probe the implementation's internal cache for pipeline binaries.

When `pPipelineCreateInfo` is not `NULL`, an implementation will attempt to retrieve pipeline binary data from an internal cache external to the application if `pipelineBinaryInternalCache` is `VK_TRUE`. Applications **can** use this to determine if a pipeline **can** be created without compilation. If the implementation fails to create a pipeline binary due to missing an internal cache entry, `VK_PIPELINE_BINARY_MISSING_KHR` is returned. If creation succeeds, the resulting binary **can** be used to create a pipeline. `VK_PIPELINE_BINARY_MISSING_KHR` **may** be returned for any reason in this situation, even if creating a pipeline binary with the same parameters that succeeded earlier.

If `pipelineBinaryPrecompiledInternalCache` is `VK_TRUE`, the implementation **may** be able to create pipeline binaries even when `pPipelineCreateInfo` has not been used to create binaries before by the application.

NOTE

On some platforms, internal pipeline caches may be pre-populated before running the application.

Valid Usage

- VUID-VkPipelineBinaryCreateInfoKHR-pipeline-09607
If `pipeline` is not `VK_NULL_HANDLE`, `pipeline` **must** have been created with `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR`
- VUID-VkPipelineBinaryCreateInfoKHR-pipeline-09608
If `pipeline` is not `VK_NULL_HANDLE`, `vkReleaseCapturedPipelineDataKHR` **must** not have been called on `pipeline` prior to this command
- VUID-VkPipelineBinaryCreateInfoKHR-pipelineBinaryInternalCache-09609
If `pipelineBinaryInternalCache` is `VK_FALSE` `pPipelineCreateInfo` **must** be `NULL`
- VUID-VkPipelineBinaryCreateInfoKHR-device-09610
If `device` was created with `VkDevicePipelineBinaryInternalCacheControlKHR::disableInternalCache` set to `VK_TRUE`, `pPipelineCreateInfo` **must** be `NULL`
- VUID-VkPipelineBinaryCreateInfoKHR-pKeysAndDataInfo-09619
One and only one of `pKeysAndDataInfo`, `pipeline`, or `pPipelineCreateInfo` **must** be non-`NULL`
- VUID-VkPipelineBinaryCreateInfoKHR-pPipelineCreateInfo-09606
If `pPipelineCreateInfo` is not `NULL`, the `pNext` chain of `pPipelineCreateInfo` **must** not set `VkPipelineBinaryInfoKHR::binaryCount` to a value greater than 0

Valid Usage (Implicit)

- VUID-VkPipelineBinaryCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_BINARY_CREATE_INFO_KHR`
- VUID-VkPipelineBinaryCreateInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineBinaryCreateInfoKHR-pKeysAndDataInfo-parameter
If `pKeysAndDataInfo` is not `NULL`, `pKeysAndDataInfo` **must** be a valid pointer to a valid `VkPipelineBinaryKeysAndDataKHR` structure
- VUID-VkPipelineBinaryCreateInfoKHR-pipeline-pipeline-parameter
If `pipeline` is not `VK_NULL_HANDLE`, `pipeline` **must** be a valid `VkPipeline` handle
- VUID-VkPipelineBinaryCreateInfoKHR-pPipelineCreateInfo-parameter
If `pPipelineCreateInfo` is not `NULL`, `pPipelineCreateInfo` **must** be a valid pointer to a valid `VkPipelineCreateInfoKHR` structure

The `VkPipelineBinaryKeysAndDataKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryKeysAndDataKHR {
    uint32_t                binaryCount;
    const VkPipelineBinaryKeyKHR* pPipelineBinaryKeys;
    const VkPipelineBinaryDataKHR* pPipelineBinaryData;
} VkPipelineBinaryKeysAndDataKHR;
```

- `binaryCount` is the size of the `pPipelineBinaryKeys` and `pPipelineBinaryData` arrays
- `pPipelineBinaryKeys` is a pointer to an array of `VkPipelineBinaryKeyKHR` structures containing the pipeline binary keys
- `pPipelineBinaryData` is a pointer to an array of `VkPipelineBinaryDataKHR` structures containing the pipeline binary data

Valid Usage (Implicit)

- VUID-VkPipelineBinaryKeysAndDataKHR-pPipelineBinaryKeys-parameter
`pPipelineBinaryKeys` **must** be a valid pointer to an array of `binaryCount` valid `VkPipelineBinaryKeyKHR` structures
- VUID-VkPipelineBinaryKeysAndDataKHR-pPipelineBinaryData-parameter
`pPipelineBinaryData` **must** be a valid pointer to an array of `binaryCount` valid `VkPipelineBinaryDataKHR` structures
- VUID-VkPipelineBinaryKeysAndDataKHR-binaryCount-arraylength
`binaryCount` **must** be greater than 0

The `VkPipelineBinaryDataKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryDataKHR {
    size_t    dataSize;
    void*     pData;
} VkPipelineBinaryDataKHR;
```

- `dataSize` is the size of the `pData` buffer in bytes.
- `pData` is a pointer to a buffer of `size` bytes that contains pipeline binary data obtained from `vkGetPipelineBinaryDataKHR`.

Valid Usage (Implicit)

- VUID-VkPipelineBinaryDataKHR-pData-parameter
`pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-VkPipelineBinaryDataKHR-dataSize-arraylength
`dataSize` **must** be greater than 0

10.8.3. Retrieving Pipeline Binary Data

Data **can** be retrieved from a pipeline binary object using the command:

```
// Provided by VK_KHR_pipeline_binary
VkResult vkGetPipelineBinaryDataKHR(
    VkDevice device,
    const VkPipelineBinaryDataInfoKHR* pInfo,
    VkPipelineBinaryKeyKHR* pPipelineBinaryKey,
    size_t* pPipelineBinaryDataSize,
    void* pPipelineBinaryData);
```

- `device` is the logical device that created the pipeline binary.
- `pInfo` is a pointer to a `VkPipelineBinaryDataInfoKHR` structure which describes the pipeline binary to get data from.
- `pPipelineBinaryKey` is a pointer to a `VkPipelineBinaryKeyKHR` structure where the key for this binary will be written.
- `pPipelineBinaryDataSize` is a pointer to a `size_t` value related to the amount of data in the pipeline binary, as described below.
- `pPipelineBinaryData` is either `NULL` or a pointer to a buffer.

If `pPipelineBinaryData` is `NULL`, then the size of the data, in bytes, that is required to store the binary is returned in `pPipelineBinaryDataSize`. Otherwise, `pPipelineBinaryDataSize` **must** contain the size of the buffer, in bytes, pointed to by `pPipelineBinaryData`, and on return `pPipelineBinaryDataSize` is overwritten with the size of the data, in bytes, that is required to store the binary. If `pPipelineBinaryDataSize` is less than the size that is required to store the binary, nothing is written

to `pPipelineBinaryData` and `VK_ERROR_NOT_ENOUGH_SPACE_KHR` will be returned, instead of `VK_SUCCESS`.

If `pipelineBinaryCompressedData` is `VK_FALSE`, implementations **should** not return compressed pipeline binary data to the application.

Valid Usage (Implicit)

- VUID-vkGetPipelineBinaryDataKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetPipelineBinaryDataKHR-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkPipelineBinaryDataInfoKHR` structure
- VUID-vkGetPipelineBinaryDataKHR-pPipelineBinaryKey-parameter
`pPipelineBinaryKey` **must** be a valid pointer to a `VkPipelineBinaryKeyKHR` structure
- VUID-vkGetPipelineBinaryDataKHR-pPipelineBinaryDataSize-parameter
`pPipelineBinaryDataSize` **must** be a valid pointer to a `size_t` value
- VUID-vkGetPipelineBinaryDataKHR-pPipelineBinaryData-parameter
If the value referenced by `pPipelineBinaryDataSize` is not 0, and `pPipelineBinaryData` is not `NULL`, `pPipelineBinaryData` **must** be a valid pointer to an array of `pPipelineBinaryDataSize` bytes

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_NOT_ENOUGH_SPACE_KHR`

The `VkPipelineBinaryDataInfoKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPipelineBinaryDataInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    VkPipelineBinaryKHR pipelineBinary;
} VkPipelineBinaryDataInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pipelineBinary` is the pipeline binary to get data from.

Valid Usage (Implicit)

- VUID-VkPipelineBinaryDataInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_BINARY_DATA_INFO_KHR`
- VUID-VkPipelineBinaryDataInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineBinaryDataInfoKHR-pipelineBinary-parameter
`pipelineBinary` **must** be a valid `VkPipelineBinaryKHR` handle

10.8.4. Releasing Captured Pipeline Binary Data

To release pipeline resources captured with `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR`, call:

```
// Provided by VK_KHR_pipeline_binary
VkResult vkReleaseCapturedPipelineDataKHR(
    VkDevice device,
    const VkReleaseCapturedPipelineDataInfoKHR* pInfo,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that created the pipeline object.
- `pInfo` is a pointer to a `VkReleaseCapturedPipelineDataInfoKHR` structure which describes the pipeline to release the data from.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

The implementation **may** free any resources captured as a result of creating the pipeline with `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR` and put the pipeline into a state as if `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR` had not been provided at pipeline creation time.

NOTE

Any resources captured as a result of creating the pipeline with `VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR` are implicitly freed by `vkDestroyPipeline`.

Valid Usage

- VUID-vkReleaseCapturedPipelineDataKHR-pipeline-09611
If `VkAllocationCallbacks` were provided when `pipeline` was created, a compatible set of callbacks **must** be provided in `pAllocator`
- VUID-vkReleaseCapturedPipelineDataKHR-pipeline-09612
If no `VkAllocationCallbacks` were provided when `pipeline` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkReleaseCapturedPipelineDataKHR-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkReleaseCapturedPipelineDataKHR-pInfo-parameter **pInfo** **must** be a valid pointer to a valid [VkReleaseCapturedPipelineDataInfoKHR](#) structure
- VUID-vkReleaseCapturedPipelineDataKHR-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure

Host Synchronization

- Host access to **pInfo->pipeline** **must** be externally synchronized

Return Codes

Success

- **VK_SUCCESS**

Failure

None

The [VkReleaseCapturedPipelineDataInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkReleaseCapturedPipelineDataInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    VkPipeline         pipeline;
} VkReleaseCapturedPipelineDataInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **pipeline** the handle of the pipeline object to release the data from.

Valid Usage

- VUID-VkReleaseCapturedPipelineDataInfoKHR-pipeline-09613
pipeline **must** have been created with **VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR**
- VUID-VkReleaseCapturedPipelineDataInfoKHR-pipeline-09618
pipeline **must** not have been used in a previous call to **vkReleaseCapturedPipelineDataKHR**

Valid Usage (Implicit)

- VUID-VkReleaseCapturedPipelineDataInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RELEASE_CAPTURED_PIPELINE_DATA_INFO_KHR`
- VUID-VkReleaseCapturedPipelineDataInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkReleaseCapturedPipelineDataInfoKHR-pipeline-parameter
`pipeline` **must** be a valid `VkPipeline` handle

10.8.5. Destroying Pipeline Binaries

To destroy a `VkPipelineBinaryKHR`, call:

```
// Provided by VK_KHR_pipeline_binary
void vkDestroyPipelineBinaryKHR(
    VkDevice          device,
    VkPipelineBinaryKHR pipelineBinary,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that created the pipeline binary object.
- `pipelineBinary` is the handle of the pipeline binary object to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyPipelineBinaryKHR-pipelineBinary-09614
If `VkAllocationCallbacks` were provided when `pipelineBinary` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyPipelineBinaryKHR-pipelineBinary-09615
If no `VkAllocationCallbacks` were provided when `pipelineBinary` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyPipelineBinaryKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyPipelineBinaryKHR-pipelineBinary-parameter
If `pipelineBinary` is not `VK_NULL_HANDLE`, `pipelineBinary` **must** be a valid `VkPipelineBinaryKHR` handle
- VUID-vkDestroyPipelineBinaryKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

- VUID-vkDestroyPipelineBinaryKHR-pipelineBinary-parent
If `pipelineBinary` is a valid handle, it **must** have been created, allocated, or retrieved from device

Host Synchronization

- Host access to `pipelineBinary` **must** be externally synchronized

10.9. Specialization Constants

Specialization constants are a mechanism whereby constants in a SPIR-V module **can** have their constant value specified at the time the `VkPipeline` is created. This allows a SPIR-V module to have constants that **can** be modified while executing an application that uses the Vulkan API.

NOTE

Specialization constants are useful to allow a compute shader to have its local workgroup size changed at runtime by the user, for example.

Each `VkPipelineShaderStageCreateInfo` structure contains a `pSpecializationInfo` member, which **can** be `NULL` to indicate no specialization constants, or point to a `VkSpecializationInfo` structure.

The `VkSpecializationInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSpecializationInfo {
    uint32_t          mapEntryCount;
    const VkSpecializationMapEntry* pMapEntries;
    size_t            dataSize;
    const void*        pData;
} VkSpecializationInfo;
```

- `mapEntryCount` is the number of entries in the `pMapEntries` array.
- `pMapEntries` is a pointer to an array of `VkSpecializationMapEntry` structures, which map constant IDs to offsets in `pData`.
- `dataSize` is the byte size of the `pData` buffer.
- `pData` contains the actual constant values to specialize with.

Valid Usage

- VUID-VkSpecializationInfo-offset-00773
The `offset` member of each element of `pMapEntries` **must** be less than `dataSize`
- VUID-VkSpecializationInfo-pMapEntries-00774
The `size` member of each element of `pMapEntries` **must** be less than or equal to `dataSize` minus `offset`

- VUID-VkSpecializationInfo-constantID-04911

The **constantID** value of each element of **pMapEntries** **must** be unique within **pMapEntries**

Valid Usage (Implicit)

- VUID-VkSpecializationInfo-pMapEntries-parameter
If **mapEntryCount** is not 0, **pMapEntries** **must** be a valid pointer to an array of **mapEntryCount** valid **VkSpecializationMapEntry** structures
- VUID-VkSpecializationInfo-pData-parameter
If **dataSize** is not 0, **pData** **must** be a valid pointer to an array of **dataSize** bytes

The **VkSpecializationMapEntry** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSpecializationMapEntry {
    uint32_t    constantID;
    uint32_t    offset;
    size_t      size;
} VkSpecializationMapEntry;
```

- **constantID** is the ID of the specialization constant in SPIR-V.
- **offset** is the byte offset of the specialization constant value within the supplied data buffer.
- **size** is the byte size of the specialization constant value within the supplied data buffer.

If a **constantID** value is not a specialization constant ID used in the shader, that map entry does not affect the behavior of the pipeline.

Valid Usage

- VUID-VkSpecializationMapEntry-constantID-00776
For a **constantID** specialization constant declared in a shader, **size** **must** match the byte size of the **constantID**. If the specialization constant is of type **boolean**, **size** **must** be the byte size of **VkBool32**

In human readable SPIR-V:

```
OpDecorate %x SpecId 13 ; decorate .x component of WorkgroupSize with ID 13
OpDecorate %y SpecId 42 ; decorate .y component of WorkgroupSize with ID 42
OpDecorate %z SpecId 3 ; decorate .z component of WorkgroupSize with ID 3
OpDecorate %wgsz BuiltIn WorkgroupSize ; decorate WorkgroupSize onto constant
%i32 = OpTypeInt 32 0 ; declare an unsigned 32-bit type
%uvec3 = OpTypeVector %i32 3 ; declare a 3 element vector type of unsigned 32-bit
%x = OpSpecConstant %i32 1 ; declare the .x component of WorkgroupSize
%y = OpSpecConstant %i32 1 ; declare the .y component of WorkgroupSize
```

```
%z = OpSpecConstant %i32 1 ; declare the .z component of WorkgroupSize
%wgsiz = OpSpecConstantComposite %uvec3 %x %y %z ; declare WorkgroupSize
```

From the above we have three specialization constants, one for each of the x, y & z elements of the WorkgroupSize vector.

Now to specialize the above via the specialization constants mechanism:

```
const VkSpecializationMapEntry entries[] =
{
    {
        .constantID = 13,
        .offset = 0 * sizeof(uint32_t),
        .size = sizeof(uint32_t)
    },
    {
        .constantID = 42,
        .offset = 1 * sizeof(uint32_t),
        .size = sizeof(uint32_t)
    },
    {
        .constantID = 3,
        .offset = 2 * sizeof(uint32_t),
        .size = sizeof(uint32_t)
    }
};

const uint32_t data[] = { 16, 8, 4 }; // our workgroup size is 16x8x4

const VkSpecializationInfo info =
{
    .mapEntryCount = 3,
    .pMapEntries = entries,
    .dataSize = 3 * sizeof(uint32_t),
    .pData = data,
};
```

Then when calling `vkCreateComputePipelines`, and passing the `VkSpecializationInfo` we defined as the `pSpecializationInfo` parameter of `VkPipelineShaderStageCreateInfo`, we will create a compute pipeline with the runtime specified local workgroup size.

Another example would be that an application has a SPIR-V module that has some platform-dependent constants they wish to use.

In human readable SPIR-V:

```
OpDecorate %1 SpecId 0 ; decorate our signed 32-bit integer constant
OpDecorate %2 SpecId 12 ; decorate our 32-bit floating-point constant
%i32 = OpTypeInt 32 1 ; declare a signed 32-bit type
```

```
%float = OpTypeFloat 32 ; declare a 32-bit floating-point type
%1 = OpSpecConstant %i32 -1 ; some signed 32-bit integer constant
%2 = OpSpecConstant %float 0.5 ; some 32-bit floating-point constant
```

From the above we have two specialization constants, one is a signed 32-bit integer and the second is a 32-bit floating-point value.

Now to specialize the above via the specialization constants mechanism:

```
struct SpecializationData {
    int32_t data0;
    float data1;
};

const VkSpecializationMapEntry entries[] =
{
    {
        .constantID = 0,
        .offset = offsetof(SpecializationData, data0),
        .size = sizeof(SpecializationData::data0)
    },
    {
        .constantID = 12,
        .offset = offsetof(SpecializationData, data1),
        .size = sizeof(SpecializationData::data1)
    }
};

SpecializationData data;
data.data0 = -42;    // set the data for the 32-bit integer
data.data1 = 42.0f; // set the data for the 32-bit floating-point

const VkSpecializationInfo info =
{
    .mapEntryCount = 2,
    .pMapEntries = entries,
    .dataSize = sizeof(data),
    .pdata = &data,
};
```

It is legal for a SPIR-V module with specializations to be compiled into a pipeline where no specialization information was provided. SPIR-V specialization constants contain default values such that if a specialization is not provided, the default value will be used. In the examples above, it would be valid for an application to only specialize some of the specialization constants within the SPIR-V module, and let the other constants use their default values encoded within the OpSpecConstant declarations.

10.10. Pipeline Libraries

A pipeline library is a special pipeline that was created using the `VK_PIPELINE_CREATE_LIBRARY_BIT_KHR` and cannot be bound, instead it defines a set of pipeline state which can be linked into other pipelines. For ray tracing pipelines this includes shaders and shader groups. The application **must** maintain the lifetime of a pipeline library based on the pipelines that link with it.

This linkage is achieved by using the following structure within the appropriate creation mechanisms:

The `VkPipelineLibraryCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_library
typedef struct VkPipelineLibraryCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           libraryCount;
    const VkPipeline*   pLibraries;
} VkPipelineLibraryCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `libraryCount` is the number of pipeline libraries in `pLibraries`.
- `pLibraries` is a pointer to an array of `VkPipeline` structures specifying pipeline libraries to use when creating a pipeline.

Valid Usage

- VUID-VkPipelineLibraryCreateInfoKHR-pLibraries-03381

Each element of `pLibraries` **must** have been created with `VK_PIPELINE_CREATE_LIBRARY_BIT_KHR`

Valid Usage (Implicit)

- VUID-VkPipelineLibraryCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_LIBRARY_CREATE_INFO_KHR`
- VUID-VkPipelineLibraryCreateInfoKHR-pLibraries-parameter
If `libraryCount` is not 0, `pLibraries` **must** be a valid pointer to an array of `libraryCount` valid `VkPipeline` handles

Pipelines created with `VK_PIPELINE_CREATE_LIBRARY_BIT_KHR` libraries **can** depend on other pipeline libraries in `VkPipelineLibraryCreateInfoKHR`.

A pipeline library is considered in-use, as long as one of the linking pipelines is in-use. This applies recursively if a pipeline library includes other pipeline libraries.

10.11. Pipeline Binding

Once a pipeline has been created, it **can** be bound to the command buffer using the command:

```
// Provided by VK_VERSION_1_0
void vkCmdBindPipeline(
    VkCommandBuffer          commandBuffer,
    VkPipelineBindPoint      pipelineBindPoint,
    VkPipeline                pipeline);
```

- `commandBuffer` is the command buffer that the pipeline will be bound to.
- `pipelineBindPoint` is a `VkPipelineBindPoint` value specifying to which bind point the pipeline is bound. Binding one does not disturb the others.
- `pipeline` is the pipeline to be bound.

Once bound, a pipeline binding affects subsequent commands that interact with the given pipeline type in the command buffer until a different pipeline of the same type is bound to the bind point, or until the pipeline bind point is disturbed by binding a [shader object](#) as described in [Interaction with Pipelines](#). Commands that do not interact with the [given pipeline](#) type **must** not be affected by the pipeline state.

Valid Usage

- VUID-vkCmdBindPipeline-pipelineBindPoint-00777
If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_COMPUTE`, the `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdBindPipeline-pipelineBindPoint-00778
If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_GRAPHICS`, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBindPipeline-pipelineBindPoint-00779
If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_COMPUTE`, `pipeline` **must** be a compute pipeline
- VUID-vkCmdBindPipeline-pipelineBindPoint-00780
If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_GRAPHICS`, `pipeline` **must** be a graphics pipeline
- VUID-vkCmdBindPipeline-pipeline-00781
If the `variableMultisampleRate` feature is not supported, `pipeline` is a graphics pipeline, the current subpass [uses no attachments](#), and this is not the first call to this function with a graphics pipeline after transitioning to the current subpass, then the sample count specified by this pipeline **must** match that set in the previous pipeline
- VUID-vkCmdBindPipeline-variableSampleLocations-01525

If `VkPhysicalDeviceSampleLocationsPropertiesEXT::variableSampleLocations` is `VK_FALSE`, and `pipeline` is a graphics pipeline created with a `VkPipelineSampleLocationsStateCreateInfoEXT` structure having its `sampleLocationsEnable` member set to `VK_TRUE` but without `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` enabled then the current render pass instance **must** have been begun by specifying a `VkRenderPassSampleLocationsBeginInfoEXT` structure whose `pPostSubpassSampleLocations` member contains an element with a `subpassIndex` matching the current subpass index and the `sampleLocationsInfo` member of that element **must** match the `sampleLocationsInfo` specified in `VkPipelineSampleLocationsStateCreateInfoEXT` when the pipeline was created

- VUID-vkCmdBindPipeline-None-02323

This command **must** not be recorded when transform feedback is active

- VUID-vkCmdBindPipeline-pipelineBindPoint-02391

If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR`, the `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations

- VUID-vkCmdBindPipeline-pipelineBindPoint-02392

If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR`, `pipeline` **must** be a ray tracing pipeline

- VUID-vkCmdBindPipeline-pipelineBindPoint-06721

If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR`, `commandBuffer` **must** not be a protected command buffer

- VUID-vkCmdBindPipeline-pipeline-03382

`pipeline` **must** not have been created with `VK_PIPELINE_CREATE_LIBRARY_BIT_KHR` set

- VUID-vkCmdBindPipeline-commandBuffer-04809

If `commandBuffer` is a secondary command buffer with `VkCommandBufferInheritanceViewportScissorInfoNV::viewportScissor2D` enabled and `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_GRAPHICS` and `pipeline` was created with `VkPipelineDiscardRectangleStateCreateInfoEXT` structure and its `discardRectangleCount` member is not `0`, or the pipeline was created with `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` enabled, then the pipeline **must** have been created with `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` enabled

- VUID-vkCmdBindPipeline-pipelineBindPoint-04881

If `pipelineBindPoint` is `VK_PIPELINE_BIND_POINT_GRAPHICS` and the `provokingVertexModePerPipeline` limit is `VK_FALSE`, then pipeline's `VkPipelineRasterizationProvokingVertexStateCreateInfoEXT::provokingVertexMode` **must** be the same as that of any other pipelines previously bound to this bind point within the current render pass instance, including any pipeline already bound when beginning the render pass instance

Valid Usage (Implicit)

- VUID-vkCmdBindPipeline-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdBindPipeline-pipelineBindPoint-parameter
`pipelineBindPoint` **must** be a valid `VkPipelineBindPoint` value
- VUID-vkCmdBindPipeline-pipeline-parameter
`pipeline` **must** be a valid `VkPipeline` handle
- VUID-vkCmdBindPipeline-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdBindPipeline-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdBindPipeline-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindPipeline-commonparent
Both of `commandBuffer`, and `pipeline` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	State

Possible values of `vkCmdBindPipeline::pipelineBindPoint`, specifying the bind point of a pipeline object, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPipelineBindPoint {
    VK_PIPELINE_BIND_POINT_GRAPHICS = 0,
    VK_PIPELINE_BIND_POINT_COMPUTE = 1,
    // Provided by VK_KHR_ray_tracing_pipeline
    VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR = 1000165000,
} VkPipelineBindPoint;
```

- `VK_PIPELINE_BIND_POINT_COMPUTE` specifies binding as a compute pipeline.
- `VK_PIPELINE_BIND_POINT_GRAPHICS` specifies binding as a graphics pipeline.

- `VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR` specifies binding as a ray tracing pipeline.

10.11.1. Interaction With Shader Objects

If the `shaderObject` feature is enabled, applications **can** use both pipelines and `shader objects` at the same time. The interaction between pipelines and shader objects is described in [Interaction with Pipelines](#).

10.12. Dynamic State

When a pipeline object is bound, any pipeline object state that is not specified as dynamic is applied to the command buffer state. Pipeline object state that is specified as dynamic is not applied to the command buffer state at this time.

Instead, dynamic state **can** be modified at any time and persists for the lifetime of the command buffer, or until modified by another dynamic state setting command, or made invalid by binding a pipeline in which that state is statically specified.

When a pipeline object is bound, the following applies to each state parameter:

- If the state is not specified as dynamic in the new pipeline object, then that command buffer state is overwritten by the state in the new pipeline object. Before any draw or dispatch call with this pipeline there **must** not have been any calls to any of the corresponding dynamic state setting commands after this pipeline was bound.
- If the state is specified as dynamic in the new pipeline object, then that command buffer state is not disturbed. Before any draw or dispatch call with this pipeline there **must** have been at least one call to each of the corresponding dynamic state setting commands. The state-setting commands **must** be recorded after command buffer recording was begun, or after the last command binding a pipeline object with that state specified as static, whichever was the latter.
- If the state is not included (corresponding pointer in `VkGraphicsPipelineCreateInfo` was `NULL` or was ignored) in the new pipeline object, then that command buffer state is not disturbed.

Dynamic state that does not affect the result of operations **can** be left undefined.

NOTE

For example, if blending is disabled by the pipeline object state then the dynamic color blend constants do not need to be specified in the command buffer, even if this state is specified as dynamic in the pipeline object.

NOTE

Applications running on Vulkan implementations advertising an `VkPhysicalDeviceDriverProperties::conformanceVersion` less than 1.3.8.0 should be aware that rebinding the currently bound pipeline object may not reapply static state.

10.13. Pipeline Properties and Shader Information

When a pipeline is created, its state and shaders are compiled into zero or more device-specific executables, which are used when executing commands against that pipeline. To query the

properties of these pipeline executables, call:

```
// Provided by VK_KHR_pipeline_executable_properties
VkResult vkGetPipelineExecutablePropertiesKHR(
    VkDevice device,
    const VkPipelineInfoKHR* pPipelineInfo,
    uint32_t* pExecutableCount,
    VkPipelineExecutablePropertiesKHR* pProperties);
```

- **device** is the device that created the pipeline.
- **pPipelineInfo** describes the pipeline being queried.
- **pExecutableCount** is a pointer to an integer related to the number of pipeline executables available or queried, as described below.
- **pProperties** is either **NULL** or a pointer to an array of **VkPipelineExecutablePropertiesKHR** structures.

If **pProperties** is **NULL**, then the number of pipeline executables associated with the pipeline is returned in **pExecutableCount**. Otherwise, **pExecutableCount** **must** point to a variable set by the application to the number of elements in the **pProperties** array, and on return the variable is overwritten with the number of structures actually written to **pProperties**. If **pExecutableCount** is less than the number of pipeline executables associated with the pipeline, at most **pExecutableCount** structures will be written, and **VK_INCOMPLETE** will be returned instead of **VK_SUCCESS**, to indicate that not all the available properties were returned.

Valid Usage

- VUID-vkGetPipelineExecutablePropertiesKHR-pipelineExecutableInfo-03270
The **pipelineExecutableInfo** feature **must** be enabled
- VUID-vkGetPipelineExecutablePropertiesKHR-pipeline-03271
The **pipeline** member of **pPipelineInfo** **must** have been created with **device**

Valid Usage (Implicit)

- VUID-vkGetPipelineExecutablePropertiesKHR-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkGetPipelineExecutablePropertiesKHR-pPipelineInfo-parameter
pPipelineInfo **must** be a valid pointer to a valid **VkPipelineInfoKHR** structure
- VUID-vkGetPipelineExecutablePropertiesKHR-pExecutableCount-parameter
pExecutableCount **must** be a valid pointer to a **uint32_t** value
- VUID-vkGetPipelineExecutablePropertiesKHR-pProperties-parameter
If the value referenced by **pExecutableCount** is not 0, and **pProperties** is not **NULL**, **pProperties** **must** be a valid pointer to an array of **pExecutableCount** **VkPipelineExecutablePropertiesKHR** structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkPipelineExecutablePropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPipelineExecutablePropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkShaderStageFlags stages;
    char               name[VK_MAX_DESCRIPTION_SIZE];
    char               description[VK_MAX_DESCRIPTION_SIZE];
    uint32_t           subgroupSize;
} VkPipelineExecutablePropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stages` is a bitmask of zero or more `VkShaderStageFlagBits` indicating which shader stages (if any) were principally used as inputs to compile this pipeline executable.
- `name` is an array of `VK_MAX_DESCRIPTION_SIZE` `char` containing a null-terminated UTF-8 string which is a short human readable name for this pipeline executable.
- `description` is an array of `VK_MAX_DESCRIPTION_SIZE` `char` containing a null-terminated UTF-8 string which is a human readable description for this pipeline executable.
- `subgroupSize` is the subgroup size with which this pipeline executable is dispatched.

Not all implementations have a 1:1 mapping between shader stages and pipeline executables and some implementations **may** reduce a given shader stage to fixed function hardware programming such that no pipeline executable is available. No guarantees are provided about the mapping between shader stages and pipeline executables and `stages` **should** be considered a best effort hint. Because the application **cannot** rely on the `stages` field to provide an exact description, `name` and `description` provide a human readable name and description which more accurately describes the given pipeline executable.

Valid Usage (Implicit)

- `VUID-VkPipelineExecutablePropertiesKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_PROPERTIES_KHR`

- VUID-VkPipelineExecutablePropertiesKHR-pNext-pNext
pNext must be NULL

The `VkPipelineInfoKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPipelineInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkPipeline         pipeline;
} VkPipelineInfoKHR;
```

- **sType** is a `VkStructureType` value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **pipeline** is a `VkPipeline` handle.

Valid Usage (Implicit)

- VUID-VkPipelineInfoKHR-sType-sType
sType must be VK_STRUCTURE_TYPE_PIPELINE_INFO_KHR
- VUID-VkPipelineInfoKHR-pNext-pNext
pNext must be NULL
- VUID-VkPipelineInfoKHR-pipeline-parameter
pipeline must be a valid `VkPipeline` handle

Each pipeline executable **may** have a set of statistics associated with it that are generated by the pipeline compilation process. These statistics **may** include things such as instruction counts, amount of spilling (if any), maximum number of simultaneous threads, or anything else which **may** aid developers in evaluating the expected performance of a shader. To query the compile time statistics associated with a pipeline executable, call:

```
// Provided by VK_KHR_pipeline_executable_properties
VkResult vkGetPipelineExecutableStatisticsKHR(
    VkDevice                device,
    const VkPipelineExecutableInfoKHR* pExecutableInfo,
    uint32_t*               pStatisticCount,
    VkPipelineExecutableStatisticKHR* pStatistics);
```

- **device** is the device that created the pipeline.
- **pExecutableInfo** describes the pipeline executable being queried.
- **pStatisticCount** is a pointer to an integer related to the number of statistics available or queried, as described below.

- `pStatistics` is either `NULL` or a pointer to an array of `VkPipelineExecutableStatisticKHR` structures.

If `pStatistics` is `NULL`, then the number of statistics associated with the pipeline executable is returned in `pStatisticCount`. Otherwise, `pStatisticCount` **must** point to a variable set by the application to the number of elements in the `pStatistics` array, and on return the variable is overwritten with the number of structures actually written to `pStatistics`. If `pStatisticCount` is less than the number of statistics associated with the pipeline executable, at most `pStatisticCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available statistics were returned.

Valid Usage

- VUID-vkGetPipelineExecutableStatisticsKHR-pipelineExecutableInfo-03272
The `pipelineExecutableInfo` feature **must** be enabled
- VUID-vkGetPipelineExecutableStatisticsKHR-pipeline-03273
The `pipeline` member of `pExecutableInfo` **must** have been created with `device`
- VUID-vkGetPipelineExecutableStatisticsKHR-pipeline-03274
The `pipeline` member of `pExecutableInfo` **must** have been created with `VK_PIPELINE_CREATE_CAPTURE_STATISTICS_BIT_KHR`

Valid Usage (Implicit)

- VUID-vkGetPipelineExecutableStatisticsKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetPipelineExecutableStatisticsKHR-pExecutableInfo-parameter
`pExecutableInfo` **must** be a valid pointer to a valid `VkPipelineExecutableInfoKHR` structure
- VUID-vkGetPipelineExecutableStatisticsKHR-pStatisticCount-parameter
`pStatisticCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPipelineExecutableStatisticsKHR-pStatistics-parameter
If the value referenced by `pStatisticCount` is not `0`, and `pStatistics` is not `NULL`, `pStatistics` **must** be a valid pointer to an array of `pStatisticCount` `VkPipelineExecutableStatisticKHR` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`

- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkPipelineExecutableInfoKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPipelineExecutableInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkPipeline          pipeline;
    uint32_t            executableIndex;
} VkPipelineExecutableInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pipeline` is the pipeline to query.
- `executableIndex` is the index of the pipeline executable to query in the array of executable properties returned by `vkGetPipelineExecutablePropertiesKHR`.

Valid Usage

- VUID-VkPipelineExecutableInfoKHR-executableIndex-03275
`executableIndex` **must** be less than the number of pipeline executables associated with `pipeline` as returned in the `pExecutableCount` parameter of `vkGetPipelineExecutablePropertiesKHR`

Valid Usage (Implicit)

- VUID-VkPipelineExecutableInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INFO_KHR`
- VUID-VkPipelineExecutableInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineExecutableInfoKHR-pipeline-parameter
`pipeline` **must** be a valid `VkPipeline` handle

The `VkPipelineExecutableStatisticKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPipelineExecutableStatisticKHR {
    VkStructureType          sType;
    void*                    pNext;
    char                     name[VK_MAX_DESCRIPTION_SIZE];
    char                     description[VK_MAX_DESCRIPTION_SIZE];
    VkPipelineExecutableStatisticFormatKHR format;
    VkPipelineExecutableStatisticValueKHR value;
} VkPipelineExecutableStatisticKHR;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **name** is an array of **VK_MAX_DESCRIPTION_SIZE char** containing a null-terminated UTF-8 string which is a short human readable name for this statistic.
- **description** is an array of **VK_MAX_DESCRIPTION_SIZE char** containing a null-terminated UTF-8 string which is a human readable description for this statistic.
- **format** is a **VkPipelineExecutableStatisticFormatKHR** value specifying the format of the data found in **value**.
- **value** is the value of this statistic.

Valid Usage (Implicit)

- VUID-VkPipelineExecutableStatisticKHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_STATISTIC_KHR**
- VUID-VkPipelineExecutableStatisticKHR-pNext-pNext
pNext must be **NULL**

The **VkPipelineExecutableStatisticFormatKHR** enum is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef enum VkPipelineExecutableStatisticFormatKHR {
    VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_BOOL32_KHR = 0,
    VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_INT64_KHR = 1,
    VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_UINT64_KHR = 2,
    VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_FLOAT64_KHR = 3,
} VkPipelineExecutableStatisticFormatKHR;
```

- **VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_BOOL32_KHR** specifies that the statistic is returned as a 32-bit boolean value which **must** be either **VK_TRUE** or **VK_FALSE** and **should** be read from the **b32** field of **VkPipelineExecutableStatisticValueKHR**.
- **VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_INT64_KHR** specifies that the statistic is returned as a signed 64-bit integer and **should** be read from the **i64** field of **VkPipelineExecutableStatisticValueKHR**.

- `VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_UINT64_KHR` specifies that the statistic is returned as an unsigned 64-bit integer and **should** be read from the `u64` field of `VkPipelineExecutableStatisticValueKHR`.
- `VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_FLOAT64_KHR` specifies that the statistic is returned as a 64-bit floating-point value and **should** be read from the `f64` field of `VkPipelineExecutableStatisticValueKHR`.

The `VkPipelineExecutableStatisticValueKHR` union is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef union VkPipelineExecutableStatisticValueKHR {
    VkBool32      b32;
    int64_t       i64;
    uint64_t      u64;
    double        f64;
} VkPipelineExecutableStatisticValueKHR;
```

- `b32` is the 32-bit boolean value if the `VkPipelineExecutableStatisticFormatKHR` is `VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_BOOL32_KHR`.
- `i64` is the signed 64-bit integer value if the `VkPipelineExecutableStatisticFormatKHR` is `VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_INT64_KHR`.
- `u64` is the unsigned 64-bit integer value if the `VkPipelineExecutableStatisticFormatKHR` is `VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_UINT64_KHR`.
- `f64` is the 64-bit floating-point value if the `VkPipelineExecutableStatisticFormatKHR` is `VK_PIPELINE_EXECUTABLE_STATISTIC_FORMAT_FLOAT64_KHR`.

Each pipeline executable **may** have one or more text or binary internal representations associated with it which are generated as part of the compile process. These **may** include the final shader assembly, a binary form of the compiled shader, or the shader compiler's internal representation at any number of intermediate compile steps. To query the internal representations associated with a pipeline executable, call:

```
// Provided by VK_KHR_pipeline_executable_properties
VkResult vkGetPipelineExecutableInternalRepresentationsKHR(
    VkDevice device,
    const VkPipelineExecutableInfoKHR* pExecutableInfo,
    uint32_t* pInternalRepresentationCount,
    VkPipelineExecutableInternalRepresentationKHR* pInternalRepresentations);
```

- `device` is the device that created the pipeline.
- `pExecutableInfo` describes the pipeline executable being queried.
- `pInternalRepresentationCount` is a pointer to an integer related to the number of internal representations available or queried, as described below.
- `pInternalRepresentations` is either `NULL` or a pointer to an array of

[VkPipelineExecutableInternalRepresentationKHR](#) structures.

If `pInternalRepresentations` is `NULL`, then the number of internal representations associated with the pipeline executable is returned in `pInternalRepresentationCount`. Otherwise, `pInternalRepresentationCount` **must** point to a variable set by the application to the number of elements in the `pInternalRepresentations` array, and on return the variable is overwritten with the number of structures actually written to `pInternalRepresentations`. If `pInternalRepresentationCount` is less than the number of internal representations associated with the pipeline executable, at most `pInternalRepresentationCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available representations were returned.

While the details of the internal representations remain implementation-dependent, the implementation **should** order the internal representations in the order in which they occur in the compiled pipeline with the final shader assembly (if any) last.

Valid Usage

- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pipelineExecutableInfo-03276
The `pipelineExecutableInfo` feature **must** be enabled
- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pipeline-03277
The `pipeline` member of `pExecutableInfo` **must** have been created with `device`
- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pipeline-03278
The `pipeline` member of `pExecutableInfo` **must** have been created with `VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR`

Valid Usage (Implicit)

- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-device-parameter
`device` **must** be a valid [VkDevice](#) handle
- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pExecutableInfo-parameter
`pExecutableInfo` **must** be a valid pointer to a valid [VkPipelineExecutableInfoKHR](#) structure
- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pInternalRepresentationCount-parameter
`pInternalRepresentationCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPipelineExecutableInternalRepresentationsKHR-pInternalRepresentations-parameter
If the value referenced by `pInternalRepresentationCount` is not `0`, and `pInternalRepresentations` is not `NULL`, `pInternalRepresentations` **must** be a valid pointer to an array of `pInternalRepresentationCount` [VkPipelineExecutableInternalRepresentationKHR](#) structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkPipelineExecutableInternalRepresentationKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPipelineExecutableInternalRepresentationKHR {
    VkStructureType    sType;
    void*              pNext;
    char               name[VK_MAX_DESCRIPTION_SIZE];
    char               description[VK_MAX_DESCRIPTION_SIZE];
    VkBool32           isText;
    size_t             dataSize;
    void*              pData;
} VkPipelineExecutableInternalRepresentationKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `name` is an array of `VK_MAX_DESCRIPTION_SIZE` `char` containing a null-terminated UTF-8 string which is a short human readable name for this internal representation.
- `description` is an array of `VK_MAX_DESCRIPTION_SIZE` `char` containing a null-terminated UTF-8 string which is a human readable description for this internal representation.
- `isText` specifies whether the returned data is text or opaque data. If `isText` is `VK_TRUE` then the data returned in `pData` is text and is guaranteed to be a null-terminated UTF-8 string.
- `dataSize` is an integer related to the size, in bytes, of the internal representation's data, as described below.
- `pData` is either `NULL` or a pointer to a block of data into which the implementation will write the internal representation.

If `pData` is `NULL`, then the size, in bytes, of the internal representation data is returned in `dataSize`. Otherwise, `dataSize` **must** be the size of the buffer, in bytes, pointed to by `pData` and on return `dataSize` is overwritten with the number of bytes of data actually written to `pData` including any trailing null character. If `dataSize` is less than the size, in bytes, of the internal representation's data, at most `dataSize` bytes of data will be written to `pData`, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available representation was returned.

If `isText` is `VK_TRUE` and `pData` is not `NULL` and `dataSize` is not zero, the last byte written to `pData` will

be a null character.

Valid Usage (Implicit)

- VUID-VkPipelineExecutableInternalRepresentationKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INTERNAL_REPRESENTATION_KHR`
- VUID-VkPipelineExecutableInternalRepresentationKHR-pNext-pNext
`pNext` **must** be `NULL`

Chapter 11. Memory Allocation

Vulkan memory is broken up into two categories, *host memory* and *device memory*.

11.1. Host Memory

Host memory is memory needed by the Vulkan implementation for non-device-visible storage.

NOTE This memory **may** be used to store the implementation's representation and state of Vulkan objects.

Vulkan provides applications the opportunity to perform host memory allocations on behalf of the Vulkan implementation. If this feature is not used, the implementation will perform its own memory allocations. Since most memory allocations are off the critical path, this is not meant as a performance feature. Rather, this **can** be useful for certain embedded systems, for debugging purposes (e.g. putting a guard page after all host allocations), or for memory allocation logging.

Allocators are provided by the application as a pointer to a `VkAllocationCallbacks` structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkAllocationCallbacks {
    void*                                pUserData;
    PFN_vkAllocationFunction              pfnAllocation;
    PFN_vkReallocationFunction            pfnReallocation;
    PFN_vkFreeFunction                   pfnFree;
    PFN_vkInternalAllocationNotification pfnInternalAllocation;
    PFN_vkInternalFreeNotification       pfnInternalFree;
} VkAllocationCallbacks;
```

- `pUserData` is a value to be interpreted by the implementation of the callbacks. When any of the callbacks in `VkAllocationCallbacks` are called, the Vulkan implementation will pass this value as the first parameter to the callback. This value **can** vary each time an allocator is passed into a command, even when the same object takes an allocator in multiple commands.
- `pfnAllocation` is a `PFN_vkAllocationFunction` pointer to an application-defined memory allocation function.
- `pfnReallocation` is a `PFN_vkReallocationFunction` pointer to an application-defined memory reallocation function.
- `pfnFree` is a `PFN_vkFreeFunction` pointer to an application-defined memory free function.
- `pfnInternalAllocation` is a `PFN_vkInternalAllocationNotification` pointer to an application-defined function that is called by the implementation when the implementation makes internal allocations.
- `pfnInternalFree` is a `PFN_vkInternalFreeNotification` pointer to an application-defined function that is called by the implementation when the implementation frees internal allocations.

Valid Usage

- VUID-VkAllocationCallbacks-pfnAllocation-00632
`pfnAllocation` **must** be a valid pointer to a valid application-defined [PFN_vkAllocationFunction](#)
- VUID-VkAllocationCallbacks-pfnReallocation-00633
`pfnReallocation` **must** be a valid pointer to a valid application-defined [PFN_vkReallocationFunction](#)
- VUID-VkAllocationCallbacks-pfnFree-00634
`pfnFree` **must** be a valid pointer to a valid application-defined [PFN_vkFreeFunction](#)
- VUID-VkAllocationCallbacks-pfnInternalAllocation-00635
If either of `pfnInternalAllocation` or `pfnInternalFree` is not `NULL`, both **must** be valid callbacks

The type of `pfnAllocation` is:

```
// Provided by VK_VERSION_1_0
typedef void* (VKAPI_PTR *PFN_vkAllocationFunction)(
    void*                pUserData,
    size_t               size,
    size_t               alignment,
    VkSystemAllocationScope allocationScope);
```

- `pUserData` is the value specified for [VkAllocationCallbacks::pUserData](#) in the allocator specified by the application.
- `size` is the size in bytes of the requested allocation.
- `alignment` is the requested alignment of the allocation in bytes and **must** be a power of two.
- `allocationScope` is a [VkSystemAllocationScope](#) value specifying the allocation scope of the lifetime of the allocation, as described [here](#).

If `pfnAllocation` is unable to allocate the requested memory, it **must** return `NULL`. If the allocation was successful, it **must** return a valid pointer to memory allocation containing at least `size` bytes, and with the pointer value being a multiple of `alignment`.

NOTE

Correct Vulkan operation **cannot** be assumed if the application does not follow these rules.

For example, `pfnAllocation` (or `pfnReallocation`) could cause termination of running Vulkan instance(s) on a failed allocation for debugging purposes, either directly or indirectly. In these circumstances, it **cannot** be assumed that any part of any affected [VkInstance](#) objects are going to operate correctly (even [vkDestroyInstance](#)), and the application **must** ensure it cleans up properly via other means (e.g. process termination).

If `pfnAllocation` returns `NULL`, and if the implementation is unable to continue correct processing of the current command without the requested allocation, it **must** treat this as a runtime error, and generate `VK_ERROR_OUT_OF_HOST_MEMORY` at the appropriate time for the command in which the condition was detected, as described in [Return Codes](#).

If the implementation is able to continue correct processing of the current command without the requested allocation, then it **may** do so, and **must** not generate `VK_ERROR_OUT_OF_HOST_MEMORY` as a result of this failed allocation.

The type of `pfnReallocation` is:

```
// Provided by VK_VERSION_1_0
typedef void* (VKAPI_PTR *PFN_vkReallocationFunction)(
    void*                pUserData,
    void*                pOriginal,
    size_t               size,
    size_t               alignment,
    VkSystemAllocationScope allocationScope);
```

- `pUserData` is the value specified for `VkAllocationCallbacks::pUserData` in the allocator specified by the application.
- `pOriginal` **must** be either `NULL` or a pointer previously returned by `pfnReallocation` or `pfnAllocation` of a compatible allocator.
- `size` is the size in bytes of the requested allocation.
- `alignment` is the requested alignment of the allocation in bytes and **must** be a power of two.
- `allocationScope` is a `VkSystemAllocationScope` value specifying the allocation scope of the lifetime of the allocation, as described [here](#).

If the reallocation was successful, `pfnReallocation` **must** return an allocation with enough space for `size` bytes, and the contents of the original allocation from bytes zero to `min(original size, new size) - 1` **must** be preserved in the returned allocation. If `size` is larger than the old size, the contents of the additional space are undefined. If satisfying these requirements involves creating a new allocation, then the old allocation **should** be freed.

If `pOriginal` is `NULL`, then `pfnReallocation` **must** behave equivalently to a call to `PFN_vkAllocationFunction` with the same parameter values (without `pOriginal`).

If `size` is zero, then `pfnReallocation` **must** behave equivalently to a call to `PFN_vkFreeFunction` with the same `pUserData` parameter value, and `pMemory` equal to `pOriginal`.

If `pOriginal` is non-`NULL`, the implementation **must** ensure that `alignment` is equal to the `alignment` used to originally allocate `pOriginal`.

If this function fails and `pOriginal` is non-`NULL` the application **must** not free the old allocation.

`pfnReallocation` **must** follow the same [rules for return values](#) as `PFN_vkAllocationFunction`.

The type of `pfnFree` is:

```
// Provided by VK_VERSION_1_0
typedef void (VKAPI_PTR *PFN_vkFreeFunction)(
    void*                                pUserData,
    void*                                pMemory);
```

- `pUserData` is the value specified for `VkAllocationCallbacks::pUserData` in the allocator specified by the application.
- `pMemory` is the allocation to be freed.

`pMemory` **may** be `NULL`, which the callback **must** handle safely. If `pMemory` is non-`NULL`, it **must** be a pointer previously allocated by `pfnAllocation` or `pfnReallocation`. The application **should** free this memory.

The type of `pfnInternalAllocation` is:

```
// Provided by VK_VERSION_1_0
typedef void (VKAPI_PTR *PFN_vkInternalAllocationNotification)(
    void*                                pUserData,
    size_t                                size,
    VkInternalAllocationType             allocationType,
    VkSystemAllocationScope              allocationScope);
```

- `pUserData` is the value specified for `VkAllocationCallbacks::pUserData` in the allocator specified by the application.
- `size` is the requested size of an allocation.
- `allocationType` is a `VkInternalAllocationType` value specifying the requested type of an allocation.
- `allocationScope` is a `VkSystemAllocationScope` value specifying the allocation scope of the lifetime of the allocation, as described [here](#).

This is a purely informational callback.

The type of `pfnInternalFree` is:

```
// Provided by VK_VERSION_1_0
typedef void (VKAPI_PTR *PFN_vkInternalFreeNotification)(
    void*                                pUserData,
    size_t                                size,
    VkInternalAllocationType             allocationType,
    VkSystemAllocationScope              allocationScope);
```

- `pUserData` is the value specified for `VkAllocationCallbacks::pUserData` in the allocator specified by the application.
- `size` is the requested size of an allocation.

- `allocationType` is a `VkInternalAllocationType` value specifying the requested type of an allocation.
- `allocationScope` is a `VkSystemAllocationScope` value specifying the allocation scope of the lifetime of the allocation, as described [here](#).

Each allocation has an *allocation scope* defining its lifetime and which object it is associated with. Possible values passed to the `allocationScope` parameter of the callback functions specified by `VkAllocationCallbacks`, indicating the allocation scope, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSystemAllocationScope {
    VK_SYSTEM_ALLOCATION_SCOPE_COMMAND = 0,
    VK_SYSTEM_ALLOCATION_SCOPE_OBJECT = 1,
    VK_SYSTEM_ALLOCATION_SCOPE_CACHE = 2,
    VK_SYSTEM_ALLOCATION_SCOPE_DEVICE = 3,
    VK_SYSTEM_ALLOCATION_SCOPE_INSTANCE = 4,
} VkSystemAllocationScope;
```

- `VK_SYSTEM_ALLOCATION_SCOPE_COMMAND` specifies that the allocation is scoped to the duration of the Vulkan command.
- `VK_SYSTEM_ALLOCATION_SCOPE_OBJECT` specifies that the allocation is scoped to the lifetime of the Vulkan object that is being created or used.
- `VK_SYSTEM_ALLOCATION_SCOPE_CACHE` specifies that the allocation is scoped to the lifetime of a `VkPipelineCache` object.
- `VK_SYSTEM_ALLOCATION_SCOPE_DEVICE` specifies that the allocation is scoped to the lifetime of the Vulkan device.
- `VK_SYSTEM_ALLOCATION_SCOPE_INSTANCE` specifies that the allocation is scoped to the lifetime of the Vulkan instance.

Most Vulkan commands operate on a single object, or there is a sole object that is being created or manipulated. When an allocation uses an allocation scope of `VK_SYSTEM_ALLOCATION_SCOPE_OBJECT` or `VK_SYSTEM_ALLOCATION_SCOPE_CACHE`, the allocation is scoped to the object being created or manipulated.

When an implementation requires host memory, it will make callbacks to the application using the most specific allocator and allocation scope available:

- If an allocation is scoped to the duration of a command, the allocator will use the `VK_SYSTEM_ALLOCATION_SCOPE_COMMAND` allocation scope. The most specific allocator available is used: if the object being created or manipulated has an allocator, that object's allocator will be used, else if the parent `VkDevice` has an allocator it will be used, else if the parent `VkInstance` has an allocator it will be used. Else,
- If an allocation is associated with a `VkPipelineCache` object, the allocator will use the `VK_SYSTEM_ALLOCATION_SCOPE_CACHE` allocation scope. The most specific allocator available is used (cache, else device, else instance). Else,

- If an allocation is scoped to the lifetime of an object, that object is being created or manipulated by the command, and that object's type is not `VkDevice` or `VkInstance`, the allocator will use an allocation scope of `VK_SYSTEM_ALLOCATION_SCOPE_OBJECT`. The most specific allocator available is used (object, else device, else instance). Else,
- If an allocation is scoped to the lifetime of a device, the allocator will use an allocation scope of `VK_SYSTEM_ALLOCATION_SCOPE_DEVICE`. The most specific allocator available is used (device, else instance). Else,
- If the allocation is scoped to the lifetime of an instance and the instance has an allocator, its allocator will be used with an allocation scope of `VK_SYSTEM_ALLOCATION_SCOPE_INSTANCE`.
- Otherwise an implementation will allocate memory through an alternative mechanism that is unspecified.

Objects that are allocated from pools do not specify their own allocator. When an implementation requires host memory for such an object, that memory is sourced from the object's parent pool's allocator.

The application is not expected to handle allocating memory that is intended for execution by the host due to the complexities of differing security implementations across multiple platforms. The implementation will allocate such memory internally and invoke an application provided informational callback when these *internal allocations* are allocated and freed. Upon allocation of executable memory, `pfnInternalAllocation` will be called. Upon freeing executable memory, `pfnInternalFree` will be called. An implementation will only call an informational callback for executable memory allocations and frees.

The `allocationType` parameter to the `pfnInternalAllocation` and `pfnInternalFree` functions **may** be one of the following values:

```
// Provided by VK_VERSION_1_0
typedef enum VkInternalAllocationType {
    VK_INTERNAL_ALLOCATION_TYPE_EXECUTABLE = 0,
} VkInternalAllocationType;
```

- `VK_INTERNAL_ALLOCATION_TYPE_EXECUTABLE` specifies that the allocation is intended for execution by the host.

An implementation **must** only make calls into an application-provided allocator during the execution of an API command. An implementation **must** only make calls into an application-provided allocator from the same thread that called the provoking API command. The implementation **should** not synchronize calls to any of the callbacks. If synchronization is needed, the callbacks **must** provide it themselves. The informational callbacks are subject to the same restrictions as the allocation callbacks.

If an implementation intends to make calls through a `VkAllocationCallbacks` structure between the time a `vkCreate*` command returns and the time a corresponding `vkDestroy*` command begins, that implementation **must** save a copy of the allocator before the `vkCreate*` command returns. The callback functions and any data structures they rely upon **must** remain valid for the lifetime of the object they are associated with.

If an allocator is provided to a `vkCreate*` command, a *compatible* allocator **must** be provided to the corresponding `vkDestroy*` command. Two `VkAllocationCallbacks` structures are compatible if memory allocated with `pfnAllocation` or `pfnReallocation` in each **can** be freed with `pfnReallocation` or `pfnFree` in the other. An allocator **must** not be provided to a `vkDestroy*` command if an allocator was not provided to the corresponding `vkCreate*` command.

If a non-NULL allocator is used, the `pfnAllocation`, `pfnReallocation` and `pfnFree` members **must** be non-NULL and point to valid implementations of the callbacks. An application **can** choose to not provide informational callbacks by setting both `pfnInternalAllocation` and `pfnInternalFree` to NULL. `pfnInternalAllocation` and `pfnInternalFree` **must** either both be NULL or both be non-NULL.

If `pfnAllocation` or `pfnReallocation` fail, the implementation **may** fail object creation and/or generate a `VK_ERROR_OUT_OF_HOST_MEMORY` error, as appropriate.

Allocation callbacks **must** not call any Vulkan commands.

The following sets of rules define when an implementation is permitted to call the allocator callbacks.

`pfnAllocation` or `pfnReallocation` **may** be called in the following situations:

- Allocations scoped to a `VkDevice` or `VkInstance` **may** be allocated from any API command.
- Allocations scoped to a command **may** be allocated from any API command.
- Allocations scoped to a `VkPipelineCache` **may** only be allocated from:
 - `vkCreatePipelineCache`
 - `vkMergePipelineCaches` for `dstCache`
 - `vkCreateGraphicsPipelines` for `pipelineCache`
 - `vkCreateComputePipelines` for `pipelineCache`
- Allocations scoped to a `VkDescriptorPool` **may** only be allocated from:
 - any command that takes the pool as a direct argument
 - `vkAllocateDescriptorSets` for the `descriptorPool` member of its `pAllocateInfo` parameter
 - `vkCreateDescriptorPool`
- Allocations scoped to a `VkCommandPool` **may** only be allocated from:
 - any command that takes the pool as a direct argument
 - `vkCreateCommandPool`
 - `vkAllocateCommandBuffers` for the `commandPool` member of its `pAllocateInfo` parameter
 - any `vkCmd*` command whose `commandBuffer` was allocated from that `VkCommandPool`
- Allocations scoped to any other object **may** only be allocated in that object's `vkCreate*` command.

`pfnFree`, or `pfnReallocation` with zero `size`, **may** be called in the following situations:

- Allocations scoped to a `VkDevice` or `VkInstance` **may** be freed from any API command.

- Allocations scoped to a command **must** be freed by any API command which allocates such memory.
- Allocations scoped to a `VkPipelineCache` **may** be freed from `vkDestroyPipelineCache`.
- Allocations scoped to a `VkDescriptorPool` **may** be freed from
 - any command that takes the pool as a direct argument
- Allocations scoped to a `VkCommandPool` **may** be freed from:
 - any command that takes the pool as a direct argument
 - `vkResetCommandBuffer` whose `commandBuffer` was allocated from that `VkCommandPool`
- Allocations scoped to any other object **may** be freed in that object's `vkDestroy*` command.
- Any command that allocates host memory **may** also free host memory of the same scope.

11.2. Device Memory

Device memory is memory that is visible to the device—for example the contents of the image or buffer objects, which **can** be natively used by the device.

11.2.1. Device Memory Properties

Memory properties of a physical device describe the memory heaps and memory types available.

To query memory properties, call:

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceMemoryProperties(
    VkPhysicalDevice          physicalDevice,
    VkPhysicalDeviceMemoryProperties* pMemoryProperties);
```

- `physicalDevice` is the handle to the device to query.
- `pMemoryProperties` is a pointer to a `VkPhysicalDeviceMemoryProperties` structure in which the properties are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceMemoryProperties-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceMemoryProperties-pMemoryProperties-parameter `pMemoryProperties` **must** be a valid pointer to a `VkPhysicalDeviceMemoryProperties` structure

The `VkPhysicalDeviceMemoryProperties` structure is defined as:


```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceMemoryProperties {
    uint32_t      memoryTypeCount;
    VkMemoryType  memoryTypes[VK_MAX_MEMORY_TYPES];
    uint32_t      memoryHeapCount;
    VkMemoryHeap  memoryHeaps[VK_MAX_MEMORY_HEAPS];
} VkPhysicalDeviceMemoryProperties;
```

- `memoryTypeCount` is the number of valid elements in the `memoryTypes` array.
- `memoryTypes` is an array of `VK_MAX_MEMORY_TYPES` `VkMemoryType` structures describing the *memory types* that **can** be used to access memory allocated from the heaps specified by `memoryHeaps`.
- `memoryHeapCount` is the number of valid elements in the `memoryHeaps` array.
- `memoryHeaps` is an array of `VK_MAX_MEMORY_HEAPS` `VkMemoryHeap` structures describing the *memory heaps* from which memory **can** be allocated.

The `VkPhysicalDeviceMemoryProperties` structure describes a number of *memory heaps* as well as a number of *memory types* that **can** be used to access memory allocated in those heaps. Each heap describes a memory resource of a particular size, and each memory type describes a set of memory properties (e.g. host cached vs. uncached) that **can** be used with a given memory heap. Allocations using a particular memory type will consume resources from the heap indicated by that memory type's heap index. More than one memory type **may** share each heap, and the heaps and memory types provide a mechanism to advertise an accurate size of the physical memory resources while allowing the memory to be used with a variety of different properties.

The number of memory heaps is given by `memoryHeapCount` and is less than or equal to `VK_MAX_MEMORY_HEAPS`. Each heap is described by an element of the `memoryHeaps` array as a `VkMemoryHeap` structure. The number of memory types available across all memory heaps is given by `memoryTypeCount` and is less than or equal to `VK_MAX_MEMORY_TYPES`. Each memory type is described by an element of the `memoryTypes` array as a `VkMemoryType` structure.

At least one heap **must** include `VK_MEMORY_HEAP_DEVICE_LOCAL_BIT` in `VkMemoryHeap::flags`. If there are multiple heaps that all have similar performance characteristics, they **may** all include `VK_MEMORY_HEAP_DEVICE_LOCAL_BIT`. In a unified memory architecture (UMA) system there is often only a single memory heap which is considered to be equally “local” to the host and to the device, and such an implementation **must** advertise the heap as device-local.

Each memory type returned by `vkGetPhysicalDeviceMemoryProperties` **must** have its `propertyFlags` set to one of the following values:

- 0
- `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` | `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT`
- `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` | `VK_MEMORY_PROPERTY_HOST_CACHED_BIT`
- `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` |

- VK_MEMORY_PROPERTY_HOST_CACHED_BIT |
 VK_MEMORY_PROPERTY_HOST_COHERENT_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT |
 VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT |
 VK_MEMORY_PROPERTY_HOST_COHERENT_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT |
 VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT |
 VK_MEMORY_PROPERTY_HOST_CACHED_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT |
 VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT |
 VK_MEMORY_PROPERTY_HOST_CACHED_BIT |
 VK_MEMORY_PROPERTY_HOST_COHERENT_BIT
- VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT |
 VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT

There **must** be at least one memory type with both the `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` and `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT` bits set in its `propertyFlags`. There **must** be at least one memory type with the `VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT` bit set in its `propertyFlags`.

For each pair of elements **X** and **Y** returned in `memoryTypes`, **X** **must** be placed at a lower index position than **Y** if:

- the set of bit flags returned in the `propertyFlags` member of **X** is a strict subset of the set of bit flags returned in the `propertyFlags` member of **Y**; or
- the `propertyFlags` members of **X** and **Y** are equal, and **X** belongs to a memory heap with greater performance (as determined in an implementation-specific manner)

NOTE

There is no ordering requirement between **X** and **Y** elements for the case their `propertyFlags` members are not in a subset relation. That potentially allows more than one possible way to order the same set of memory types. Notice that the [list of all allowed memory property flag combinations](#) is written in a valid order. But if instead `VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT` was before `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT | VK_MEMORY_PROPERTY_HOST_COHERENT_BIT`, the list would still be in a valid order.

This ordering requirement enables applications to use a simple search loop to select the desired memory type along the lines of:

```
// Find a memory in 'memoryTypeBitsRequirement' that includes all of
// 'requiredProperties'
int32_t findProperties(const VkPhysicalDeviceMemoryProperties* pMemoryProperties,
                     uint32_t memoryTypeBitsRequirement,
                     VkMemoryPropertyFlags requiredProperties) {
    const uint32_t memoryCount = pMemoryProperties->memoryTypeCount;
    for (uint32_t memoryIndex = 0; memoryIndex < memoryCount; ++memoryIndex) {
```

```

const uint32_t memoryTypeBits = (1 << memoryIndex);
const bool isRequiredMemoryType = memoryTypeBitsRequirement & memoryTypeBits;

const VkMemoryPropertyFlags properties =
    pMemoryProperties->memoryTypes[memoryIndex].propertyFlags;
const bool hasRequiredProperties =
    (properties & requiredProperties) == requiredProperties;

if (isRequiredMemoryType && hasRequiredProperties)
    return static_cast<int32_t>(memoryIndex);
}

// failed to find memory type
return -1;
}

// Try to find an optimal memory type, or if it does not exist try fallback memory
type
// `device` is the VkDevice
// `image` is the VkImage that requires memory to be bound
// `memoryProperties` properties as returned by vkGetPhysicalDeviceMemoryProperties
// `requiredProperties` are the property flags that must be present
// `optimalProperties` are the property flags that are preferred by the application
VkMemoryRequirements memoryRequirements;
vkGetImageMemoryRequirements(device, image, &memoryRequirements);
int32_t memoryType =
    findProperties(&memoryProperties, memoryRequirements.memoryTypeBits,
optimalProperties);
if (memoryType == -1) // not found; try fallback properties
    memoryType =
        findProperties(&memoryProperties, memoryRequirements.memoryTypeBits,
requiredProperties);

```

`VK_MAX_MEMORY_TYPES` is the length of an array of [VkMemoryType](#) structures describing memory types, as returned in [VkPhysicalDeviceMemoryProperties::memoryTypes](#).

```
#define VK_MAX_MEMORY_TYPES 32U
```

`VK_MAX_MEMORY_HEAPS` is the length of an array of [VkMemoryHeap](#) structures describing memory heaps, as returned in [VkPhysicalDeviceMemoryProperties::memoryHeaps](#).

```
#define VK_MAX_MEMORY_HEAPS 16U
```

To query memory properties, call:

```
// Provided by VK_KHR_get_physical_device_properties2
void vkGetPhysicalDeviceMemoryProperties2KHR(
    VkPhysicalDevice          physicalDevice,
    VkPhysicalDeviceMemoryProperties2* pMemoryProperties);
```

- `physicalDevice` is the handle to the device to query.
- `pMemoryProperties` is a pointer to a `VkPhysicalDeviceMemoryProperties2` structure in which the properties are returned.

`vkGetPhysicalDeviceMemoryProperties2KHR` behaves similarly to `vkGetPhysicalDeviceMemoryProperties`, with the ability to return extended information in a `pNext` chain of output structures.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceMemoryProperties2-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceMemoryProperties2-pMemoryProperties-parameter `pMemoryProperties` **must** be a valid pointer to a `VkPhysicalDeviceMemoryProperties2` structure

The `VkPhysicalDeviceMemoryProperties2` structure is defined as:

```
typedef struct VkPhysicalDeviceMemoryProperties2 {
    VkStructureType    sType;
    void*              pNext;
    VkPhysicalDeviceMemoryProperties    memoryProperties;
} VkPhysicalDeviceMemoryProperties2;
```

or the equivalent

```
// Provided by VK_KHR_get_physical_device_properties2
typedef VkPhysicalDeviceMemoryProperties2 VkPhysicalDeviceMemoryProperties2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryProperties` is a `VkPhysicalDeviceMemoryProperties` structure which is populated with the same values as in `vkGetPhysicalDeviceMemoryProperties`.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMemoryProperties2-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PROPERTIES_2`

- VUID-VkPhysicalDeviceMemoryProperties2-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkPhysicalDeviceMemoryBudgetPropertiesEXT`
- VUID-VkPhysicalDeviceMemoryProperties2-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique

The `VkMemoryHeap` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryHeap {
    VkDeviceSize      size;
    VkMemoryHeapFlags flags;
} VkMemoryHeap;
```

- `size` is the total memory size in bytes in the heap.
- `flags` is a bitmask of `VkMemoryHeapFlagBits` specifying attribute flags for the heap.

Bits which **may** be set in `VkMemoryHeap::flags`, indicating attribute flags for the heap, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkMemoryHeapFlagBits {
    VK_MEMORY_HEAP_DEVICE_LOCAL_BIT = 0x00000001,
    VK_MEMORY_HEAP_MULTI_INSTANCE_BIT = 0x00000002,
    // Provided by VK_KHR_device_group_creation
    VK_MEMORY_HEAP_MULTI_INSTANCE_BIT_KHR = VK_MEMORY_HEAP_MULTI_INSTANCE_BIT,
} VkMemoryHeapFlagBits;
```

- `VK_MEMORY_HEAP_DEVICE_LOCAL_BIT` specifies that the heap corresponds to device-local memory. Device-local memory **may** have different performance characteristics than host-local memory, and **may** support different memory property flags.
- `VK_MEMORY_HEAP_MULTI_INSTANCE_BIT` specifies that in a logical device representing more than one physical device, there is a per-physical device instance of the heap memory. By default, an allocation from such a heap will be replicated to each physical device's instance of the heap.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkMemoryHeapFlags;
```

`VkMemoryHeapFlags` is a bitmask type for setting a mask of zero or more `VkMemoryHeapFlagBits`.

The `VkMemoryType` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryType {
    VkMemoryPropertyFlags    propertyFlags;
    uint32_t                 heapIndex;
} VkMemoryType;
```

- `heapIndex` describes which memory heap this memory type corresponds to, and **must** be less than `memoryHeapCount` from the `VkPhysicalDeviceMemoryProperties` structure.
- `propertyFlags` is a bitmask of `VkMemoryPropertyFlagBits` of properties for this memory type.

Bits which **may** be set in `VkMemoryType::propertyFlags`, indicating properties of a memory type, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkMemoryPropertyFlagBits {
    VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT = 0x00000001,
    VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT = 0x00000002,
    VK_MEMORY_PROPERTY_HOST_COHERENT_BIT = 0x00000004,
    VK_MEMORY_PROPERTY_HOST_CACHED_BIT = 0x00000008,
    VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT = 0x00000010,
} VkMemoryPropertyFlagBits;
```

- `VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT` bit specifies that memory allocated with this type is the most efficient for device access. This property will be set if and only if the memory type belongs to a heap with the `VK_MEMORY_HEAP_DEVICE_LOCAL_BIT` set.
- `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` bit specifies that memory allocated with this type **can** be mapped for host access using `vkMapMemory`.
- `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT` bit specifies that the host cache management commands `vkFlushMappedMemoryRanges` and `vkInvalidateMappedMemoryRanges` are not needed to manage **availability and visibility** on the host.
- `VK_MEMORY_PROPERTY_HOST_CACHED_BIT` bit specifies that memory allocated with this type is cached on the host. Host memory accesses to uncached memory are slower than to cached memory, however uncached memory is always host coherent.
- `VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT` bit specifies that the memory type only allows device access to the memory. Memory types **must** not have both `VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT` and `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` set. Additionally, the object's backing memory **may** be provided by the implementation lazily as specified in [Lazily Allocated Memory](#).

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkMemoryPropertyFlags;
```

`VkMemoryPropertyFlags` is a bitmask type for setting a mask of zero or more `VkMemoryPropertyFlagBits`.

If the `VkPhysicalDeviceMemoryBudgetPropertiesEXT` structure is included in the `pNext` chain of `VkPhysicalDeviceMemoryProperties2`, it is filled with the current memory budgets and usages.

The `VkPhysicalDeviceMemoryBudgetPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_memory_budget
typedef struct VkPhysicalDeviceMemoryBudgetPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize        heapBudget[VK_MAX_MEMORY_HEAPS];
    VkDeviceSize        heapUsage[VK_MAX_MEMORY_HEAPS];
} VkPhysicalDeviceMemoryBudgetPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `heapBudget` is an array of `VK_MAX_MEMORY_HEAPS` `VkDeviceSize` values in which memory budgets are returned, with one element for each memory heap. A heap's budget is a rough estimate of how much memory the process **can** allocate from that heap before allocations **may** fail or cause performance degradation. The budget includes any currently allocated device memory.
- `heapUsage` is an array of `VK_MAX_MEMORY_HEAPS` `VkDeviceSize` values in which memory usages are returned, with one element for each memory heap. A heap's usage is an estimate of how much memory the process is currently using in that heap.

The values returned in this structure are not invariant. The `heapBudget` and `heapUsage` values **must** be zero for array elements greater than or equal to `VkPhysicalDeviceMemoryProperties::memoryHeapCount`. The `heapBudget` value **must** be non-zero for array elements less than `VkPhysicalDeviceMemoryProperties::memoryHeapCount`. The `heapBudget` value **must** be less than or equal to `VkMemoryHeap::size` for each heap.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMemoryBudgetPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_BUDGET_PROPERTIES_EXT`

11.2.2. Device Memory Objects

A Vulkan device operates on data in device memory via memory objects that are represented in the API by a `VkDeviceMemory` handle:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDeviceMemory)
```


11.2.3. Device Memory Allocation

To allocate memory objects, call:

```
// Provided by VK_VERSION_1_0
VkResult vkAllocateMemory(
    VkDevice                                device,
    const VkMemoryAllocateInfo*             pAllocateInfo,
    const VkAllocationCallbacks*            pAllocator,
    VkDeviceMemory*                         pMemory);
```

- **device** is the logical device that owns the memory.
- **pAllocateInfo** is a pointer to a [VkMemoryAllocateInfo](#) structure describing parameters of the allocation. A successfully returned allocation **must** use the requested parameters—no substitution is permitted by the implementation.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pMemory** is a pointer to a [VkDeviceMemory](#) handle in which information about the allocated memory is returned.

Allocations returned by **vkAllocateMemory** are guaranteed to meet any alignment requirement of the implementation. For example, if an implementation requires 128 byte alignment for images and 64 byte alignment for buffers, the device memory returned through this mechanism would be 128-byte aligned. This ensures that applications **can** correctly suballocate objects of different types (with potentially different alignment requirements) in the same memory object.

When memory is allocated, its contents are undefined.

The maximum number of valid memory allocations that **can** exist simultaneously within a [VkDevice](#) **may** be restricted by implementation- or platform-dependent limits. The **maxMemoryAllocationCount** feature describes the number of allocations that **can** exist simultaneously before encountering these internal limits.

NOTE

For historical reasons, if **maxMemoryAllocationCount** is exceeded, some implementations may return **VK_ERROR_TOO_MANY_OBJECTS**. Exceeding this limit will result in undefined behavior, and an application should not rely on the use of the returned error code in order to identify when the limit is reached.

NOTE

Many protected memory implementations involve complex hardware and system software support, and often have additional and much lower limits on the number of simultaneous protected memory allocations (from memory types with the **VK_MEMORY_PROPERTY_PROTECTED_BIT** property) than for non-protected memory allocations. These limits can be system-wide, and depend on a variety of factors outside of the Vulkan implementation, so they cannot be queried in Vulkan. Applications **should** use as few allocations as possible from such memory types by suballocating aggressively, and be prepared for allocation failure even when there is apparently plenty of capacity remaining in the memory heap. As a guideline, the Vulkan conformance test suite requires that at least 80 minimum-size allocations

can exist concurrently when no other uses of protected memory are active in the system.

Some platforms **may** have a limit on the maximum size of a single allocation. For example, certain systems **may** fail to create allocations with a size greater than or equal to 4GB. Such a limit is implementation-dependent, and if such a failure occurs then the error `VK_ERROR_OUT_OF_DEVICE_MEMORY` **must** be returned. This limit is advertised in `VkPhysicalDeviceMaintenance3Properties::maxMemoryAllocationSize`.

Valid Usage

- VUID-vkAllocateMemory-pAllocateInfo-01713
`pAllocateInfo->allocationSize` **must** be less than or equal to `VkPhysicalDeviceMemoryProperties::memoryHeaps[memindex].size` where `memindex` = `VkPhysicalDeviceMemoryProperties::memoryTypes[pAllocateInfo->memoryTypeIndex].heapIndex` as returned by `vkGetPhysicalDeviceMemoryProperties` for the `VkPhysicalDevice` that `device` was created from
- VUID-vkAllocateMemory-pAllocateInfo-01714
`pAllocateInfo->memoryTypeIndex` **must** be less than `VkPhysicalDeviceMemoryProperties::memoryTypeCount` as returned by `vkGetPhysicalDeviceMemoryProperties` for the `VkPhysicalDevice` that `device` was created from
- VUID-vkAllocateMemory-maxMemoryAllocationCount-04101
There **must** be less than `VkPhysicalDeviceLimits::maxMemoryAllocationCount` device memory allocations currently allocated on the device

Valid Usage (Implicit)

- VUID-vkAllocateMemory-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkAllocateMemory-pAllocateInfo-parameter
`pAllocateInfo` **must** be a valid pointer to a valid `VkMemoryAllocateInfo` structure
- VUID-vkAllocateMemory-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkAllocateMemory-pMemory-parameter
`pMemory` **must** be a valid pointer to a `VkDeviceMemory` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INVALID_EXTERNAL_HANDLE`
- `VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR`

The `VkMemoryAllocateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryAllocateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceSize        allocationSize;
    uint32_t           memoryTypeIndex;
} VkMemoryAllocateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `allocationSize` is the size of the allocation in bytes.
- `memoryTypeIndex` is an index identifying a memory type from the `memoryTypes` array of the `VkPhysicalDeviceMemoryProperties` structure.

The internal data of an allocated device memory object **must** include a reference to implementation-specific resources, referred to as the memory object's *payload*. Applications **can** also import and export that internal data to and from device memory objects to share data between Vulkan instances and other compatible APIs. A `VkMemoryAllocateInfo` structure defines a memory import operation if its `pNext` chain includes one of the following structures:

- `VkImportMemoryWin32HandleInfoKHR` with a non-zero `handleType` value
- `VkImportMemoryFdInfoKHR` with a non-zero `handleType` value
- `VkImportMemoryHostPointerInfoEXT` with a non-zero `handleType` value

If the parameters define an import operation and the external handle type is `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT`, or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT`, `allocationSize` is ignored. The implementation **must** query the size of these allocations from the OS.

Whether device memory objects constructed via a memory import operation hold a reference to their payload depends on the properties of the handle type used to perform the import, as defined below for each valid handle type. Importing memory **must** not modify the content of the memory. Implementations **must** ensure that importing memory does not enable the importing Vulkan instance to access any memory or resources in other Vulkan instances other than that corresponding to the memory object imported. Implementations **must** also ensure accessing

imported memory which has not been initialized does not allow the importing Vulkan instance to obtain data from the exporting Vulkan instance or vice-versa.

NOTE

How exported and imported memory is isolated is left to the implementation, but applications should be aware that such isolation **may** prevent implementations from placing multiple exportable memory objects in the same physical or virtual page. Hence, applications **should** avoid creating many small external memory objects whenever possible.

Importing memory **must** not increase overall heap usage within a system. However, it **must** affect the following per-process values:

- `VkPhysicalDeviceLimits::maxMemoryAllocationCount`
- `VkPhysicalDeviceMemoryBudgetPropertiesEXT::heapUsage`

When performing a memory import operation, it is the responsibility of the application to ensure the external handles and their associated payloads meet all valid usage requirements. However, implementations **must** perform sufficient validation of external handles and payloads to ensure that the operation results in a valid memory object which will not cause program termination, device loss, queue stalls, or corruption of other resources when used as allowed according to its allocation parameters. If the external handle provided does not meet these requirements, the implementation **must** fail the memory import operation with the error code `VK_ERROR_INVALID_EXTERNAL_HANDLE`.

Valid Usage

- VUID-VkMemoryAllocateInfo-allocationSize-07897
If the parameters do not define an `import` or `export` operation, `allocationSize` **must** be greater than 0
- VUID-VkMemoryAllocateInfo-None-06657
The parameters **must** not define more than one `import` operation
- VUID-VkMemoryAllocateInfo-allocationSize-07899
If the parameters define an `export` operation, `allocationSize` **must** be greater than 0
- VUID-VkMemoryAllocateInfo-pNext-00639
If the `pNext` chain includes a `VkExportMemoryAllocateInfo` structure, and any of the handle types specified in `VkExportMemoryAllocateInfo::handleTypes` require a dedicated allocation, as reported by `vkGetPhysicalDeviceImageFormatProperties2KHR` in `VkExternalImageFormatProperties::externalMemoryProperties.externalMemoryFeatures`, or by `vkGetPhysicalDeviceExternalBufferPropertiesKHR` in `VkExternalBufferProperties::externalMemoryProperties.externalMemoryFeatures`, the `pNext` chain **must** include a `VkMemoryDedicatedAllocateInfo` or `VkDedicatedAllocationMemoryAllocateInfoNV` structure with either its `image` or `buffer` member set to a value other than `VK_NULL_HANDLE`
- VUID-VkMemoryAllocateInfo-allocationSize-01742
If the parameters define an `import` operation, the external handle specified was created by the Vulkan API, and the external handle type is `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT`, then the values of `allocationSize` and

`memoryTypeIndex` **must** match those specified when the payload being imported was created

- VUID-VkMemoryAllocateInfo-None-00643

If the parameters define an import operation and the external handle specified was created by the Vulkan API, the device mask specified by `VkMemoryAllocateFlagsInfo` **must** match the mask specified when the payload being imported was allocated

- VUID-VkMemoryAllocateInfo-None-00644

If the parameters define an import operation and the external handle specified was created by the Vulkan API, the list of physical devices that comprise the logical device passed to `vkAllocateMemory` **must** match the list of physical devices that comprise the logical device on which the payload was originally allocated

- VUID-VkMemoryAllocateInfo-memoryTypeIndex-00645

If the parameters define an import operation and the external handle is an NT handle or a global share handle created outside of the Vulkan API, the value of `memoryTypeIndex` **must** be one of those returned by `vkGetMemoryWin32HandlePropertiesKHR`

- VUID-VkMemoryAllocateInfo-allocationSize-01743

If the parameters define an import operation, the external handle was created by the Vulkan API, and the external handle type is `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT` or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT`, then the values of `allocationSize` and `memoryTypeIndex` **must** match those specified when the payload being imported was created

- VUID-VkMemoryAllocateInfo-allocationSize-00647

If the parameters define an import operation and the external handle type is `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT`, `allocationSize` **must** match the size specified when creating the Direct3D 12 heap from which the payload was extracted

- VUID-VkMemoryAllocateInfo-memoryTypeIndex-00648

If the parameters define an import operation and the external handle is a POSIX file descriptor created outside of the Vulkan API, the value of `memoryTypeIndex` **must** be one of those returned by `vkGetMemoryFdPropertiesKHR`

- VUID-VkMemoryAllocateInfo-memoryTypeIndex-01744

If the parameters define an import operation and the external handle is a host pointer, the value of `memoryTypeIndex` **must** be one of those returned by `vkGetMemoryHostPointerPropertiesEXT`

- VUID-VkMemoryAllocateInfo-allocationSize-01745

If the parameters define an import operation and the external handle is a host pointer, `allocationSize` **must** be an integer multiple of `VkPhysicalDeviceExternalMemoryHostPropertiesEXT::minImportedHostPointerAlignment`

- VUID-VkMemoryAllocateInfo-pNext-02806

If the parameters define an import operation and the external handle is a host pointer, the `pNext` chain **must** not include a `VkMemoryDedicatedAllocateInfo` structure with either its `image` or `buffer` field set to a value other than `VK_NULL_HANDLE`

- VUID-VkMemoryAllocateInfo-opaqueCaptureAddress-03329

If `VkMemoryOpaqueCaptureAddressAllocateInfo::opaqueCaptureAddress` is not zero,

`VkMemoryAllocateFlagsInfo::flags` **must** include
`VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT`

- VUID-VkMemoryAllocateInfo-flags-03330
If `VkMemoryAllocateFlagsInfo::flags` includes `VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT`, the `bufferDeviceAddressCaptureReplay` feature **must** be enabled
- VUID-VkMemoryAllocateInfo-flags-03331
If `VkMemoryAllocateFlagsInfo::flags` includes `VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT`, the `bufferDeviceAddress` feature **must** be enabled
- VUID-VkMemoryAllocateInfo-pNext-03332
If the `pNext` chain includes a `VkImportMemoryHostPointerInfoEXT` structure, `VkMemoryOpaqueCaptureAddressAllocateInfo::opaqueCaptureAddress` **must** be zero
- VUID-VkMemoryAllocateInfo-opaqueCaptureAddress-03333
If the parameters define an import operation, `VkMemoryOpaqueCaptureAddressAllocateInfo::opaqueCaptureAddress` **must** be zero

Valid Usage (Implicit)

- VUID-VkMemoryAllocateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_INFO`
- VUID-VkMemoryAllocateInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkExportMemoryAllocateInfo`, `VkExportMemoryWin32HandleInfoKHR`, `VkImportMemoryFdInfoKHR`, `VkImportMemoryHostPointerInfoEXT`, `VkImportMemoryWin32HandleInfoKHR`, `VkMemoryAllocateFlagsInfo`, `VkMemoryDedicatedAllocateInfo`, `VkMemoryOpaqueCaptureAddressAllocateInfo`, or `VkMemoryPriorityAllocateInfoEXT`
- VUID-VkMemoryAllocateInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique

If the `pNext` chain includes a `VkMemoryDedicatedAllocateInfo` structure, then that structure includes a handle of the sole buffer or image resource that the memory **can** be bound to.

The `VkMemoryDedicatedAllocateInfo` structure is defined as:

```
typedef struct VkMemoryDedicatedAllocateInfo {  
    VkStructureType    sType;  
    const void*        pNext;  
    VkImage             image;  
    VkBuffer            buffer;  
} VkMemoryDedicatedAllocateInfo;
```

or the equivalent


```
// Provided by VK_KHR_dedicated_allocation
typedef VkMemoryDedicatedAllocateInfo VkMemoryDedicatedAllocateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `image` is `VK_NULL_HANDLE` or a handle of an image which this memory will be bound to.
- `buffer` is `VK_NULL_HANDLE` or a handle of a buffer which this memory will be bound to.

Valid Usage

- VUID-VkMemoryDedicatedAllocateInfo-image-01432
At least one of `image` and `buffer` **must** be `VK_NULL_HANDLE`
- VUID-VkMemoryDedicatedAllocateInfo-image-02964
If `image` is not `VK_NULL_HANDLE`, `VkMemoryAllocateInfo::allocationSize` **must** equal the `VkMemoryRequirements::size` of the image
- VUID-VkMemoryDedicatedAllocateInfo-image-01434
If `image` is not `VK_NULL_HANDLE`, `image` **must** have been created without `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` set in `VkImageCreateInfo::flags`
- VUID-VkMemoryDedicatedAllocateInfo-buffer-02965
If `buffer` is not `VK_NULL_HANDLE`, `VkMemoryAllocateInfo::allocationSize` **must** equal the `VkMemoryRequirements::size` of the buffer
- VUID-VkMemoryDedicatedAllocateInfo-buffer-01436
If `buffer` is not `VK_NULL_HANDLE`, `buffer` **must** have been created without `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` set in `VkBufferCreateInfo::flags`
- VUID-VkMemoryDedicatedAllocateInfo-image-01876
If `image` is not `VK_NULL_HANDLE` and `VkMemoryAllocateInfo` defines a memory import operation with handle type `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT`, or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT`, and the external handle was created by the Vulkan API, then the memory being imported **must** also be a dedicated image allocation and `image` **must** be identical to the image associated with the imported memory
- VUID-VkMemoryDedicatedAllocateInfo-buffer-01877
If `buffer` is not `VK_NULL_HANDLE` and `VkMemoryAllocateInfo` defines a memory import operation with handle type `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT`, or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT`, and the external handle was created by the Vulkan API, then the memory being imported **must** also be a dedicated buffer

allocation and **buffer** **must** be identical to the buffer associated with the imported memory

- VUID-VkMemoryDedicatedAllocateInfo-image-01878

If **image** is not `VK_NULL_HANDLE` and `VkMemoryAllocateInfo` defines a memory import operation with handle type `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT`, the memory being imported **must** also be a dedicated image allocation and **image** **must** be identical to the image associated with the imported memory

- VUID-VkMemoryDedicatedAllocateInfo-buffer-01879

If **buffer** is not `VK_NULL_HANDLE` and `VkMemoryAllocateInfo` defines a memory import operation with handle type `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT`, the memory being imported **must** also be a dedicated buffer allocation and **buffer** **must** be identical to the buffer associated with the imported memory

- VUID-VkMemoryDedicatedAllocateInfo-image-01797

If **image** is not `VK_NULL_HANDLE`, **image** **must** not have been created with `VK_IMAGE_CREATE_DISJOINT_BIT` set in `VkImageCreateInfo::flags`

Valid Usage (Implicit)

- VUID-VkMemoryDedicatedAllocateInfo-sType-sType

sType **must** be `VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO`

- VUID-VkMemoryDedicatedAllocateInfo-image-parameter

If **image** is not `VK_NULL_HANDLE`, **image** **must** be a valid `VkImage` handle

- VUID-VkMemoryDedicatedAllocateInfo-buffer-parameter

If **buffer** is not `VK_NULL_HANDLE`, **buffer** **must** be a valid `VkBuffer` handle

- VUID-VkMemoryDedicatedAllocateInfo-commonparent

Both of **buffer**, and **image** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

If the **pNext** chain includes a `VkMemoryPriorityAllocateInfoEXT` structure, then that structure includes a priority for the memory.

The `VkMemoryPriorityAllocateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_memory_priority
typedef struct VkMemoryPriorityAllocateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    float               priority;
} VkMemoryPriorityAllocateInfoEXT;
```

- **sType** is a `VkStructureType` value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **priority** is a floating-point value between 0 and 1, indicating the priority of the allocation

relative to other memory allocations. Larger values are higher priority. The granularity of the priorities is implementation-dependent.

Memory allocations with higher priority **may** be more likely to stay in device-local memory when the system is under memory pressure.

If this structure is not included, it is as if the `priority` value were `0.5`.

Valid Usage

- VUID-VkMemoryPriorityAllocateInfoEXT-priority-02602
`priority` **must** be between `0` and `1`, inclusive

Valid Usage (Implicit)

- VUID-VkMemoryPriorityAllocateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_PRIORITY_ALLOCATE_INFO_EXT`

When allocating memory whose payload **may** be exported to another process or Vulkan instance, add a `VkExportMemoryAllocateInfo` structure to the `pNext` chain of the `VkMemoryAllocateInfo` structure, specifying the handle types that **may** be exported.

The `VkExportMemoryAllocateInfo` structure is defined as:

```
typedef struct VkExportMemoryAllocateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkExternalMemoryHandleTypeFlags handleTypes;
} VkExportMemoryAllocateInfo;
```

or the equivalent

```
// Provided by VK_KHR_external_memory
typedef VkExportMemoryAllocateInfo VkExportMemoryAllocateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleTypes` is zero or a bitmask of `VkExternalMemoryHandleTypeFlagBits` specifying one or more memory handle types the application **can** export from the resulting allocation. The application **can** request multiple handle types for the same allocation.

Valid Usage

- VUID-VkExportMemoryAllocateInfo-handleTypes-00656

The bits in `handleTypes` **must** be supported and compatible, as reported by `VkExternalImageFormatProperties` or `VkExternalBufferProperties`

Valid Usage (Implicit)

- VUID-VkExportMemoryAllocateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EXPORT_MEMORY_ALLOCATE_INFO`
- VUID-VkExportMemoryAllocateInfo-handleTypes-parameter
`handleTypes` **must** be a valid combination of `VkExternalMemoryHandleTypeFlagBits` values

11.2.4. Win32 External Memory

To specify additional attributes of NT handles exported from a memory object, add a `VkExportMemoryWin32HandleInfoKHR` structure to the `pNext` chain of the `VkMemoryAllocateInfo` structure. The `VkExportMemoryWin32HandleInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_memory_win32
typedef struct VkExportMemoryWin32HandleInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    const SECURITY_ATTRIBUTES* pAttributes;
    DWORD              dwAccess;
    LPCWSTR             name;
} VkExportMemoryWin32HandleInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pAttributes` is a pointer to a Windows `SECURITY_ATTRIBUTES` structure specifying security attributes of the handle.
- `dwAccess` is a `DWORD` specifying access rights of the handle.
- `name` is a null-terminated UTF-16 string to associate with the payload referenced by NT handles exported from the created memory.

If `VkExportMemoryAllocateInfo` is not included in the same `pNext` chain, this structure is ignored.

If `VkExportMemoryAllocateInfo` is included in the `pNext` chain of `VkMemoryAllocateInfo` with a Windows `handleType`, but either `VkExportMemoryWin32HandleInfoKHR` is not included in the `pNext` chain, or it is included but `pAttributes` is set to `NULL`, default security descriptor values will be used, and child processes created by the application will not inherit the handle, as described in the MSDN documentation for “Synchronization Object Security and Access Rights”¹. Further, if the structure is not present, the access rights used depend on the handle type.

For handles of the following types:

- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT`

The implementation **must** ensure the access rights allow read and write access to the memory.

1

<https://docs.microsoft.com/en-us/windows/win32/sync/synchronization-object-security-and-access-rights>

Valid Usage

- VUID-VkExportMemoryWin32HandleInfoKHR-handleTypes-00657
If `VkExportMemoryAllocateInfo::handleTypes` does not include `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT`, a `VkExportMemoryWin32HandleInfoKHR` structure **must** not be included in the `pNext` chain of `VkMemoryAllocateInfo`

Valid Usage (Implicit)

- VUID-VkExportMemoryWin32HandleInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EXPORT_MEMORY_WIN32_HANDLE_INFO_KHR`
- VUID-VkExportMemoryWin32HandleInfoKHR-pAttributes-parameter
If `pAttributes` is not `NULL`, `pAttributes` **must** be a valid pointer to a valid `SECURITY_ATTRIBUTES` value

To import memory from a Windows handle, add a `VkImportMemoryWin32HandleInfoKHR` structure to the `pNext` chain of the `VkMemoryAllocateInfo` structure.

The `VkImportMemoryWin32HandleInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_memory_win32
typedef struct VkImportMemoryWin32HandleInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkExternalMemoryHandleTypeFlagBits handleType;
    HANDLE                   handle;
    LPCWSTR                  name;
} VkImportMemoryWin32HandleInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleType` is a `VkExternalMemoryHandleTypeFlagBits` value specifying the type of `handle` or `name`.
- `handle` is `NULL` or the external handle to import.
- `name` is `NULL` or a null-terminated UTF-16 string naming the payload to import.

Importing memory object payloads from Windows handles does not transfer ownership of the handle to the Vulkan implementation. For handle types defined as NT handles, the application **must** release handle ownership using the `CloseHandle` system call when the handle is no longer needed. For handle types defined as NT handles, the imported memory object holds a reference to its payload.

NOTE

Non-NT handle import operations do not add a reference to their associated payload. If the original object owning the payload is destroyed, all resources and handles sharing that payload will become invalid.

Applications **can** import the same payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance. In all cases, each import operation **must** create a distinct `VkDeviceMemory` object.

Valid Usage

- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-00658
If `handleType` is not `0`, it **must** be supported for import, as reported by `VkExternalImageFormatProperties` or `VkExternalBufferProperties`
- VUID-VkImportMemoryWin32HandleInfoKHR-handle-00659
The memory from which `handle` was exported, or the memory named by `name` **must** have been created on the same underlying physical device as `device`
- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-00660
If `handleType` is not `0`, it **must** be defined as an NT handle or a global share handle
- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-01439
If `handleType` is not `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT`, or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT`, `name` **must** be `NULL`
- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-01440
If `handleType` is not `0` and `handle` is `NULL`, `name` **must** name a valid memory resource of the type specified by `handleType`
- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-00661
If `handleType` is not `0` and `name` is `NULL`, `handle` **must** be a valid handle of the type specified by `handleType`
- VUID-VkImportMemoryWin32HandleInfoKHR-handle-01441
If `handle` is not `NULL`, `name` **must** be `NULL`
- VUID-VkImportMemoryWin32HandleInfoKHR-handle-01518
If `handle` is not `NULL`, it **must** obey any requirements listed for `handleType` in [external memory handle types compatibility](#)
- VUID-VkImportMemoryWin32HandleInfoKHR-name-01519
If `name` is not `NULL`, it **must** obey any requirements listed for `handleType` in [external memory handle types compatibility](#)

Valid Usage (Implicit)

- VUID-VkImportMemoryWin32HandleInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMPORT_MEMORY_WIN32_HANDLE_INFO_KHR`
- VUID-VkImportMemoryWin32HandleInfoKHR-handleType-parameter
If `handleType` is not `0`, `handleType` **must** be a valid `VkExternalMemoryHandleTypeFlagBits` value

To export a Windows handle representing the payload of a Vulkan device memory object, call:

```
// Provided by VK_KHR_external_memory_win32
VkResult vkGetMemoryWin32HandleKHR(
    VkDevice device,
    const VkMemoryGetWin32HandleInfoKHR* pGetWin32HandleInfo,
    HANDLE* pHandle);
```

- `device` is the logical device that created the device memory being exported.
- `pGetWin32HandleInfo` is a pointer to a `VkMemoryGetWin32HandleInfoKHR` structure containing parameters of the export operation.
- `pHandle` will return the Windows handle representing the payload of the device memory object.

For handle types defined as NT handles, the handles returned by `vkGetMemoryWin32HandleKHR` are owned by the application and hold a reference to their payload. To avoid leaking resources, the application **must** release ownership of them using the `CloseHandle` system call when they are no longer needed.

NOTE

Non-NT handle types do not add a reference to their associated payload. If the original object owning the payload is destroyed, all resources and handles sharing that payload will become invalid.

Valid Usage (Implicit)

- VUID-vkGetMemoryWin32HandleKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetMemoryWin32HandleKHR-pGetWin32HandleInfo-parameter
`pGetWin32HandleInfo` **must** be a valid pointer to a valid `VkMemoryGetWin32HandleInfoKHR` structure
- VUID-vkGetMemoryWin32HandleKHR-pHandle-parameter
`pHandle` **must** be a valid pointer to a `HANDLE` value

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_TOO_MANY_OBJECTS`
- `VK_ERROR_OUT_OF_HOST_MEMORY`

The `VkMemoryGetWin32HandleInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_memory_win32
typedef struct VkMemoryGetWin32HandleInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkDeviceMemory           memory;
    VkExternalMemoryHandleTypeFlagBits handleType;
} VkMemoryGetWin32HandleInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memory` is the memory object from which the handle will be exported.
- `handleType` is a `VkExternalMemoryHandleTypeFlagBits` value specifying the type of handle requested.

The properties of the handle returned depend on the value of `handleType`. See `VkExternalMemoryHandleTypeFlagBits` for a description of the properties of the defined external memory handle types.

Valid Usage

- VUID-VkMemoryGetWin32HandleInfoKHR-handleType-00662
`handleType` **must** have been included in `VkExportMemoryAllocateInfo::handleTypes` when `memory` was created
- VUID-VkMemoryGetWin32HandleInfoKHR-handleType-00663
If `handleType` is defined as an NT handle, `vkGetMemoryWin32HandleKHR` **must** be called no more than once for each valid unique combination of `memory` and `handleType`
- VUID-VkMemoryGetWin32HandleInfoKHR-handleType-00664
`handleType` **must** be defined as an NT handle or a global share handle

Valid Usage (Implicit)

- VUID-VkMemoryGetWin32HandleInfoKHR-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_GET_WIN32_HANDLE_INFO_KHR`

- VUID-VkMemoryGetWin32HandleInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkMemoryGetWin32HandleInfoKHR-memory-parameter
`memory` **must** be a valid `VkDeviceMemory` handle
- VUID-VkMemoryGetWin32HandleInfoKHR-handleType-parameter
`handleType` **must** be a valid `VkExternalMemoryHandleTypeFlagBits` value

Windows memory handles compatible with Vulkan **may** also be created by non-Vulkan APIs using methods beyond the scope of this specification. To determine the correct parameters to use when importing such handles, call:

```
// Provided by VK_KHR_external_memory_win32
VkResult vkGetMemoryWin32HandlePropertiesKHR(
    VkDevice                device,
    VkExternalMemoryHandleTypeFlagBits handleType,
    HANDLE                  handle,
    VkMemoryWin32HandlePropertiesKHR* pMemoryWin32HandleProperties);
```

- `device` is the logical device that will be importing `handle`.
- `handleType` is a `VkExternalMemoryHandleTypeFlagBits` value specifying the type of the handle `handle`.
- `handle` is the handle which will be imported.
- `pMemoryWin32HandleProperties` is a pointer to a `VkMemoryWin32HandlePropertiesKHR` structure in which properties of `handle` are returned.

Valid Usage

- VUID-vkGetMemoryWin32HandlePropertiesKHR-handle-00665
`handle` **must** point to a valid Windows memory handle
- VUID-vkGetMemoryWin32HandlePropertiesKHR-handleType-00666
`handleType` **must** not be one of the handle types defined as opaque

Valid Usage (Implicit)

- VUID-vkGetMemoryWin32HandlePropertiesKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetMemoryWin32HandlePropertiesKHR-handleType-parameter
`handleType` **must** be a valid `VkExternalMemoryHandleTypeFlagBits` value
- VUID-vkGetMemoryWin32HandlePropertiesKHR-pMemoryWin32HandleProperties-parameter
`pMemoryWin32HandleProperties` **must** be a valid pointer to a

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_INVALID_EXTERNAL_HANDLE`

The `VkMemoryWin32HandlePropertiesKHR` structure returned is defined as:

```
// Provided by VK_KHR_external_memory_win32
typedef struct VkMemoryWin32HandlePropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           memoryTypeBits;
} VkMemoryWin32HandlePropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryTypeBits` is a bitmask containing one bit set for every memory type which the specified windows handle **can** be imported as.

Valid Usage (Implicit)

- VUID-VkMemoryWin32HandlePropertiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_WIN32_HANDLE_PROPERTIES_KHR`
- VUID-VkMemoryWin32HandlePropertiesKHR-pNext-pNext
`pNext` **must** be `NULL`

11.2.5. File Descriptor External Memory

To import memory from a POSIX file descriptor handle, add a `VkImportMemoryFdInfoKHR` structure to the `pNext` chain of the `VkMemoryAllocateInfo` structure. The `VkImportMemoryFdInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_memory_fd
typedef struct VkImportMemoryFdInfoKHR {
    VkStructureType           sType;
    const void*               pNext;
    VkExternalMemoryHandleTypeFlagBits handleType;
    int                        fd;
} VkImportMemoryFdInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **handleType** is a [VkExternalMemoryHandleTypeFlagBits](#) value specifying the handle type of **fd**.
- **fd** is the external handle to import.

Importing memory from a file descriptor transfers ownership of the file descriptor from the application to the Vulkan implementation. The application **must** not perform any operations on the file descriptor after a successful import. The imported memory object holds a reference to its payload.

Applications **can** import the same payload into multiple instances of Vulkan, into the same instance from which it was exported, and multiple times into a given Vulkan instance. In all cases, each import operation **must** create a distinct [VkDeviceMemory](#) object.

Valid Usage

- VUID-VkImportMemoryFdInfoKHR-handleType-00667
If **handleType** is not **0**, it **must** be supported for import, as reported by [VkExternalImageFormatProperties](#) or [VkExternalBufferProperties](#)
- VUID-VkImportMemoryFdInfoKHR-fd-00668
The memory from which **fd** was exported **must** have been created on the same underlying physical device as **device**
- VUID-VkImportMemoryFdInfoKHR-handleType-00669
If **handleType** is not **0**, it **must** be **VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT** or **VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT**
- VUID-VkImportMemoryFdInfoKHR-handleType-00670
If **handleType** is not **0**, **fd** **must** be a valid handle of the type specified by **handleType**
- VUID-VkImportMemoryFdInfoKHR-fd-01746
The memory represented by **fd** **must** have been created from a physical device and driver that is compatible with **device** and **handleType**, as described in [External Memory Handle Types Compatibility](#)
- VUID-VkImportMemoryFdInfoKHR-fd-01520
fd **must** obey any requirements listed for **handleType** in [external memory handle types compatibility](#)

Valid Usage (Implicit)

- VUID-VkImportMemoryFdInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMPORT_MEMORY_FD_INFO_KHR`
- VUID-VkImportMemoryFdInfoKHR-handleType-parameter
If `handleType` is not `0`, `handleType` **must** be a valid `VkExternalMemoryHandleTypeFlagBits` value

To export a POSIX file descriptor referencing the payload of a Vulkan device memory object, call:

```
// Provided by VK_KHR_external_memory_fd
VkResult vkGetMemoryFdKHR(
    VkDevice device,
    const VkMemoryGetFdInfoKHR* pGetFdInfo,
    int* pFd);
```

- `device` is the logical device that created the device memory being exported.
- `pGetFdInfo` is a pointer to a `VkMemoryGetFdInfoKHR` structure containing parameters of the export operation.
- `pFd` will return a file descriptor referencing the payload of the device memory object.

Each call to `vkGetMemoryFdKHR` **must** create a new file descriptor holding a reference to the memory object's payload and transfer ownership of the file descriptor to the application. To avoid leaking resources, the application **must** release ownership of the file descriptor using the `close` system call when it is no longer needed, or by importing a Vulkan memory object from it. Where supported by the operating system, the implementation **must** set the file descriptor to be closed automatically when an `execve` system call is made.

Valid Usage (Implicit)

- VUID-vkGetMemoryFdKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetMemoryFdKHR-pGetFdInfo-parameter
`pGetFdInfo` **must** be a valid pointer to a valid `VkMemoryGetFdInfoKHR` structure
- VUID-vkGetMemoryFdKHR-pFd-parameter
`pFd` **must** be a valid pointer to an `int` value

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_TOO_MANY_OBJECTS`
- `VK_ERROR_OUT_OF_HOST_MEMORY`

The `VkMemoryGetFdInfoKHR` structure is defined as:

```
// Provided by VK_KHR_external_memory_fd
typedef struct VkMemoryGetFdInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceMemory     memory;
    VkExternalMemoryHandleTypeFlagBits handleType;
} VkMemoryGetFdInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memory` is the memory object from which the handle will be exported.
- `handleType` is a `VkExternalMemoryHandleTypeFlagBits` value specifying the type of handle requested.

The properties of the file descriptor exported depend on the value of `handleType`. See `VkExternalMemoryHandleTypeFlagBits` for a description of the properties of the defined external memory handle types.

NOTE

The size of the exported file **may** be larger than the size requested by `VkMemoryAllocateInfo::allocationSize`. If `handleType` is `VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT`, then the application **can** query the file's actual size with `lseek`.

Valid Usage

- VUID-VkMemoryGetFdInfoKHR-handleType-00671
`handleType` **must** have been included in `VkExportMemoryAllocateInfo::handleTypes` when `memory` was created
- VUID-VkMemoryGetFdInfoKHR-handleType-00672
`handleType` **must** be `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT` or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT`

Valid Usage (Implicit)

- VUID-VkMemoryGetFdInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_GET_FD_INFO_KHR`
- VUID-VkMemoryGetFdInfoKHR-pNext-pNext

pNext must be NULL

- VUID-VkMemoryGetFdInfoKHR-memory-parameter
memory must be a valid [VkDeviceMemory](#) handle
- VUID-VkMemoryGetFdInfoKHR-handleType-parameter
handleType must be a valid [VkExternalMemoryHandleTypeFlagBits](#) value

POSIX file descriptor memory handles compatible with Vulkan **may** also be created by non-Vulkan APIs using methods beyond the scope of this specification. To determine the correct parameters to use when importing such handles, call:

```
// Provided by VK_KHR_external_memory_fd
VkResult vkGetMemoryFdPropertiesKHR(
    VkDevice                device,
    VkExternalMemoryHandleTypeFlagBits handleType,
    int                     fd,
    VkMemoryFdPropertiesKHR* pMemoryFdProperties);
```

- **device** is the logical device that will be importing **fd**.
- **handleType** is a [VkExternalMemoryHandleTypeFlagBits](#) value specifying the type of the handle **fd**.
- **fd** is the handle which will be imported.
- **pMemoryFdProperties** is a pointer to a [VkMemoryFdPropertiesKHR](#) structure in which the properties of the handle **fd** are returned.

Valid Usage

- VUID-vkGetMemoryFdPropertiesKHR-fd-00673
fd must point to a valid POSIX file descriptor memory handle
- VUID-vkGetMemoryFdPropertiesKHR-handleType-00674
handleType must not be [VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT](#)

Valid Usage (Implicit)

- VUID-vkGetMemoryFdPropertiesKHR-device-parameter
device must be a valid [VkDevice](#) handle
- VUID-vkGetMemoryFdPropertiesKHR-handleType-parameter
handleType must be a valid [VkExternalMemoryHandleTypeFlagBits](#) value
- VUID-vkGetMemoryFdPropertiesKHR-pMemoryFdProperties-parameter
pMemoryFdProperties must be a valid pointer to a [VkMemoryFdPropertiesKHR](#) structure

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_INVALID_EXTERNAL_HANDLE`

The `VkMemoryFdPropertiesKHR` structure returned is defined as:

```
// Provided by VK_KHR_external_memory_fd
typedef struct VkMemoryFdPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           memoryTypeBits;
} VkMemoryFdPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryTypeBits` is a bitmask containing one bit set for every memory type which the specified file descriptor **can** be imported as.

Valid Usage (Implicit)

- VUID-VkMemoryFdPropertiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_FD_PROPERTIES_KHR`
- VUID-VkMemoryFdPropertiesKHR-pNext-pNext
`pNext` **must** be `NULL`

11.2.6. Host External Memory

To import memory from a host pointer, add a `VkImportMemoryHostPointerInfoEXT` structure to the `pNext` chain of the `VkMemoryAllocateInfo` structure. The `VkImportMemoryHostPointerInfoEXT` structure is defined as:

```
// Provided by VK_EXT_external_memory_host
typedef struct VkImportMemoryHostPointerInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkExternalMemoryHandleTypeFlagBits handleType;
    void*              pHostPointer;
} VkImportMemoryHostPointerInfoEXT;
```


- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleType` is a `VkExternalMemoryHandleTypeFlagBits` value specifying the handle type.
- `pHostPointer` is the host pointer to import from.

Importing memory from a host pointer shares ownership of the memory between the host and the Vulkan implementation. The application **can** continue to access the memory through the host pointer but it is the application's responsibility to synchronize device and non-device access to the payload as defined in [Host Access to Device Memory Objects](#).

Applications **can** import the same payload into multiple instances of Vulkan and multiple times into a given Vulkan instance. However, implementations **may** fail to import the same payload multiple times into a given physical device due to platform constraints.

Importing memory from a particular host pointer **may** not be possible due to additional platform-specific restrictions beyond the scope of this specification in which case the implementation **must** fail the memory import operation with the error code `VK_ERROR_INVALID_EXTERNAL_HANDLE_KHR`.

Whether device memory objects imported from a host pointer hold a reference to their payload is undefined. As such, the application **must** ensure that the imported memory range remains valid and accessible for the lifetime of the imported memory object.

Implementations **may** support importing host pointers for memory types which are not host-visible. In this case, after a successful call to `vkAllocateMemory`, the memory range imported from `pHostPointer` **must** not be accessed by the application until the `VkDeviceMemory` has been destroyed. Memory contents for the host memory becomes undefined on import, and is left undefined after the `VkDeviceMemory` has been destroyed. Applications **must** also not access host memory which is mapped to the same physical memory as `pHostPointer`, but mapped to a different host pointer while the `VkDeviceMemory` handle is valid. Implementations running on general-purpose operating systems **should** not support importing host pointers for memory types which are not host-visible.

NOTE

Using host pointers to back non-host visible allocations is a platform-specific use case, and applications should not attempt to do this unless instructed by the platform.

Valid Usage

- VUID-VkImportMemoryHostPointerInfoEXT-handleType-01747
If `handleType` is not `0`, it **must** be supported for import, as reported in [VkExternalMemoryProperties](#)
- VUID-VkImportMemoryHostPointerInfoEXT-handleType-01748
If `handleType` is not `0`, it **must** be `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT` or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT`
- VUID-VkImportMemoryHostPointerInfoEXT-pHostPointer-01749
`pHostPointer` **must** be a pointer aligned to an integer multiple of

VkPhysicalDeviceExternalMemoryHostPropertiesEXT::minImportedHostPointerAlignment

- VUID-VkImportMemoryHostPointerInfoEXT-handleType-01750

If `handleType` is `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT`, `pHostPointer` **must** be a pointer to `allocationSize` number of bytes of host memory, where `allocationSize` is the member of the `VkMemoryAllocateInfo` structure this structure is chained to

- VUID-VkImportMemoryHostPointerInfoEXT-handleType-01751

If `handleType` is `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT`, `pHostPointer` **must** be a pointer to `allocationSize` number of bytes of host mapped foreign memory, where `allocationSize` is the member of the `VkMemoryAllocateInfo` structure this structure is chained to

Valid Usage (Implicit)

- VUID-VkImportMemoryHostPointerInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMPORT_MEMORY_HOST_POINTER_INFO_EXT`
- VUID-VkImportMemoryHostPointerInfoEXT-handleType-parameter
`handleType` **must** be a valid `VkExternalMemoryHandleTypeFlagBits` value
- VUID-VkImportMemoryHostPointerInfoEXT-pHostPointer-parameter
`pHostPointer` **must** be a pointer value

To determine the correct parameters to use when importing host pointers, call:

```
// Provided by VK_EXT_external_memory_host
VkResult vkGetMemoryHostPointerPropertiesEXT(
    VkDevice device,
    VkExternalMemoryHandleTypeFlagBits handleType,
    const void* pHostPointer,
    VkMemoryHostPointerPropertiesEXT* pMemoryHostPointerProperties);
```

- `device` is the logical device that will be importing `pHostPointer`.
- `handleType` is a `VkExternalMemoryHandleTypeFlagBits` value specifying the type of the handle `pHostPointer`.
- `pHostPointer` is the host pointer to import from.
- `pMemoryHostPointerProperties` is a pointer to a `VkMemoryHostPointerPropertiesEXT` structure in which the host pointer properties are returned.

Valid Usage

- VUID-vkGetMemoryHostPointerPropertiesEXT-handleType-01752

`handleType` **must** be `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT` or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT`

- VUID-vkGetMemoryHostPointerPropertiesEXT-pHostPointer-01753
`pHostPointer` **must** be a pointer aligned to an integer multiple of `VkPhysicalDeviceExternalMemoryHostPropertiesEXT::minImportedHostPointerAlignment`
- VUID-vkGetMemoryHostPointerPropertiesEXT-handleType-01754
If `handleType` is `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT`, `pHostPointer` **must** be a pointer to host memory
- VUID-vkGetMemoryHostPointerPropertiesEXT-handleType-01755
If `handleType` is `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT`, `pHostPointer` **must** be a pointer to host mapped foreign memory

Valid Usage (Implicit)

- VUID-vkGetMemoryHostPointerPropertiesEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetMemoryHostPointerPropertiesEXT-handleType-parameter
`handleType` **must** be a valid `VkExternalMemoryHandleTypeFlagBits` value
- VUID-vkGetMemoryHostPointerPropertiesEXT-pHostPointer-parameter
`pHostPointer` **must** be a pointer value
- VUID-vkGetMemoryHostPointerPropertiesEXT-pMemoryHostPointerProperties-parameter
`pMemoryHostPointerProperties` **must** be a valid pointer to a `VkMemoryHostPointerPropertiesEXT` structure

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_INVALID_EXTERNAL_HANDLE`

The `VkMemoryHostPointerPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_external_memory_host
typedef struct VkMemoryHostPointerPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           memoryTypeBits;
} VkMemoryHostPointerPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryTypeBits` is a bitmask containing one bit set for every memory type which the specified host pointer **can** be imported as.

The value returned by `memoryTypeBits` **should** only include bits that identify memory types which are host visible. Implementations **may** include bits that identify memory types which are not host visible. Behavior for imported pointers of such types is defined by [VkImportMemoryHostPointerInfoEXT](#).

Valid Usage (Implicit)

- `VUID-VkMemoryHostPointerPropertiesEXT-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_HOST_POINTER_PROPERTIES_EXT`
- `VUID-VkMemoryHostPointerPropertiesEXT-pNext-pNext`
`pNext` **must** be `NULL`

11.2.7. Device Group Memory Allocations

If the `pNext` chain of [VkMemoryAllocateInfo](#) includes a `VkMemoryAllocateFlagsInfo` structure, then that structure includes flags and a device mask controlling how many instances of the memory will be allocated.

The `VkMemoryAllocateFlagsInfo` structure is defined as:

```
typedef struct VkMemoryAllocateFlagsInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkMemoryAllocateFlags flags;
    uint32_t           deviceMask;
} VkMemoryAllocateFlagsInfo;
```

or the equivalent

```
// Provided by VK_KHR_device_group
typedef VkMemoryAllocateFlagsInfo VkMemoryAllocateFlagsInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of [VkMemoryAllocateFlagBits](#) controlling the allocation.
- `deviceMask` is a mask of physical devices in the logical device, indicating that memory **must** be allocated on each device in the mask, if `VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT` is set in `flags`.

If `VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT` is not set, the number of instances allocated depends on whether `VK_MEMORY_HEAP_MULTI_INSTANCE_BIT` is set in the memory heap. If

`VK_MEMORY_HEAP_MULTI_INSTANCE_BIT` is set, then memory is allocated for every physical device in the logical device (as if `deviceMask` has bits set for all device indices). If `VK_MEMORY_HEAP_MULTI_INSTANCE_BIT` is not set, then a single instance of memory is allocated (as if `deviceMask` is set to one).

On some implementations, allocations from a multi-instance heap **may** consume memory on all physical devices even if the `deviceMask` excludes some devices. If `VkPhysicalDeviceGroupProperties::subsetAllocation` is `VK_TRUE`, then memory is only consumed for the devices in the device mask.

NOTE

In practice, most allocations on a multi-instance heap will be allocated across all physical devices. Unicast allocation support is an optional optimization for a minority of allocations.

Valid Usage

- VUID-VkMemoryAllocateFlagsInfo-deviceMask-00675
If `VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT` is set, `deviceMask` **must** be a valid device mask
- VUID-VkMemoryAllocateFlagsInfo-deviceMask-00676
If `VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT` is set, `deviceMask` **must** not be zero

Valid Usage (Implicit)

- VUID-VkMemoryAllocateFlagsInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_FLAGS_INFO`
- VUID-VkMemoryAllocateFlagsInfo-flags-parameter
`flags` **must** be a valid combination of `VkMemoryAllocateFlagBits` values

Bits which **can** be set in `VkMemoryAllocateFlagsInfo::flags`, controlling device memory allocation, are:

```
typedef enum VkMemoryAllocateFlagBits {
    VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT = 0x00000001,
    VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT = 0x00000002,
    VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT = 0x00000004,
    // Provided by VK_KHR_device_group
    VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT_KHR = VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT,
    // Provided by VK_KHR_buffer_device_address
    VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT_KHR = VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT,
    // Provided by VK_KHR_buffer_device_address
    VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR =
    VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT,
} VkMemoryAllocateFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_device_group
typedef VkMemoryAllocateFlagBits VkMemoryAllocateFlagBitsKHR;
```

- `VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT` specifies that memory will be allocated for the devices in `VkMemoryAllocateFlagsInfo::deviceMask`.
- `VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT` specifies that the memory **can** be attached to a buffer object created with the `VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT` bit set in `usage`, and that the memory handle **can** be used to retrieve an opaque address via `vkGetDeviceMemoryOpaqueCaptureAddressKHR`.
- `VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT` specifies that the memory's address **can** be saved and reused on a subsequent run (e.g. for trace capture and replay), see `VkBufferOpaqueCaptureAddressCreateInfo` for more detail.

```
typedef VkFlags VkMemoryAllocateFlags;
```

or the equivalent

```
// Provided by VK_KHR_device_group
typedef VkMemoryAllocateFlags VkMemoryAllocateFlagsKHR;
```

`VkMemoryAllocateFlags` is a bitmask type for setting a mask of zero or more `VkMemoryAllocateFlagBits`.

11.2.8. Opaque Capture Address Allocation

To request a specific device address for a memory allocation, add a `VkMemoryOpaqueCaptureAddressAllocateInfo` structure to the `pNext` chain of the `VkMemoryAllocateInfo` structure. The `VkMemoryOpaqueCaptureAddressAllocateInfo` structure is defined as:

```
typedef struct VkMemoryOpaqueCaptureAddressAllocateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint64_t           opaqueCaptureAddress;
} VkMemoryOpaqueCaptureAddressAllocateInfo;
```

or the equivalent

```
// Provided by VK_KHR_buffer_device_address
typedef VkMemoryOpaqueCaptureAddressAllocateInfo
VkMemoryOpaqueCaptureAddressAllocateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `opaqueCaptureAddress` is the opaque capture address requested for the memory allocation.

If `opaqueCaptureAddress` is zero, no specific address is requested.

If `opaqueCaptureAddress` is not zero, it **should** be an address retrieved from `vkGetDeviceMemoryOpaqueCaptureAddressKHR` on an identically created memory allocation on the same implementation.

NOTE

In most cases, it is expected that a non-zero `opaqueAddress` is an address retrieved from `vkGetDeviceMemoryOpaqueCaptureAddressKHR` on an identically created memory allocation. If this is not the case, it is likely that `VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS` errors will occur.

This is, however, not a strict requirement because trace capture/replay tools may need to adjust memory allocation parameters for imported memory.

If this structure is not present, it is as if `opaqueCaptureAddress` is zero.

Valid Usage (Implicit)

- `VOID-VkMemoryOpaqueCaptureAddressAllocateInfo-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_OPAQUE_CAPTURE_ADDRESS_ALLOCATE_INFO`

11.2.9. Freeing Device Memory

To free a memory object, call:

```
// Provided by VK_VERSION_1_0
void vkFreeMemory(
    VkDevice          device,
    VkDeviceMemory    memory,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that owns the memory.
- `memory` is the `VkDeviceMemory` object to be freed.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Before freeing a memory object, an application **must** ensure the memory object is no longer in use by the device—for example by command buffers in the *pending state*. Memory **can** be freed whilst still bound to resources, but those resources **must** not be used afterwards. Freeing a memory object releases the reference it held, if any, to its payload. If there are still any bound images or buffers, the memory object's payload **may** not be immediately released by the implementation, but **must** be released by the time all bound images and buffers have been destroyed. Once all references to a payload are released, it is returned to the heap from which it was allocated.

How memory objects are bound to Images and Buffers is described in detail in the [Resource Memory Association](#) section.

If a memory object is mapped at the time it is freed, it is implicitly unmapped.

NOTE

As described [below](#), host writes are not implicitly flushed when the memory object is unmapped, but the implementation **must** guarantee that writes that have not been flushed do not affect any other memory.

Valid Usage

- VUID-vkFreeMemory-memory-00677
All submitted commands that refer to **memory** (via images or buffers) **must** have completed execution

Valid Usage (Implicit)

- VUID-vkFreeMemory-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkFreeMemory-memory-parameter
If **memory** is not [VK_NULL_HANDLE](#), **memory** **must** be a valid [VkDeviceMemory](#) handle
- VUID-vkFreeMemory-pAllocator-parameter
If **pAllocator** is not [NULL](#), **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkFreeMemory-memory-parent
If **memory** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **memory** **must** be externally synchronized

11.2.10. Host Access to Device Memory Objects

Memory objects created with [vkAllocateMemory](#) are not directly host accessible.

Memory objects created with the memory property [VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT](#) are considered *mappable*. Memory objects **must** be mappable in order to be successfully mapped on the host.

To retrieve a host virtual address pointer to a region of a mappable memory object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkMapMemory(
    VkDevice                device,
    VkDeviceMemory          memory,
    VkDeviceSize            offset,
    VkDeviceSize            size,
    VkMemoryMapFlags        flags,
    void**                  ppData);
```

- **device** is the logical device that owns the memory.
- **memory** is the `VkDeviceMemory` object to be mapped.
- **offset** is a zero-based byte offset from the beginning of the memory object.
- **size** is the size of the memory range to map, or `VK_WHOLE_SIZE` to map from **offset** to the end of the allocation.
- **flags** is a bitmask of `VkMemoryMapFlagBits` specifying additional parameters of the memory map operation.
- **ppData** is a pointer to a `void*` variable in which a host-accessible pointer to the beginning of the mapped range is returned. The value of the returned pointer minus **offset** **must** be aligned to `VkPhysicalDeviceLimits::minMemoryMapAlignment`.

After a successful call to `vkMapMemory` the memory object **memory** is considered to be currently *host mapped*.

NOTE

It is an application error to call `vkMapMemory` on a memory object that is already *host mapped*.

NOTE

`vkMapMemory` will fail if the implementation is unable to allocate an appropriately sized contiguous virtual address range, e.g. due to virtual address space fragmentation or platform limits. In such cases, `vkMapMemory` **must** return `VK_ERROR_MEMORY_MAP_FAILED`. The application **can** improve the likelihood of success by reducing the size of the mapped range and/or removing unneeded mappings using `vkUnmapMemory`.

`vkMapMemory` does not check whether the device memory is currently in use before returning the host-accessible pointer. The application **must** guarantee that any previously submitted command that writes to this range has completed before the host reads from or writes to that range, and that any previously submitted command that reads from that range has completed before the host writes to that region (see [here](#) for details on fulfilling such a guarantee). If the device memory was allocated without the `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT` set, these guarantees **must** be made for an extended range: the application **must** round down the start of the range to the nearest multiple of `VkPhysicalDeviceLimits::nonCoherentAtomSize`, and round the end of the range up to the nearest multiple of `VkPhysicalDeviceLimits::nonCoherentAtomSize`.

While a range of device memory is host mapped, the application is responsible for synchronizing both device and host access to that memory range.

NOTE

It is important for the application developer to become meticulously familiar with all of the mechanisms described in the chapter on [Synchronization and Cache Control](#) as they are crucial to maintaining memory access ordering.

Calling `vkMapMemory` is equivalent to calling `vkMapMemory2KHR` with an empty `pNext` chain.

Valid Usage

- VUID-vkMapMemory-memory-00678
`memory` **must** not be currently host mapped
- VUID-vkMapMemory-offset-00679
`offset` **must** be less than the size of `memory`
- VUID-vkMapMemory-size-00680
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be greater than `0`
- VUID-vkMapMemory-size-00681
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be less than or equal to the size of the `memory` minus `offset`
- VUID-vkMapMemory-memory-00682
`memory` **must** have been created with a memory type that reports `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT`
- VUID-vkMapMemory-memory-00683
`memory` **must** not have been allocated with multiple instances
- VUID-vkMapMemory-flags-09568
`VK_MEMORY_MAP_PLACED_BIT_EXT` **must** not be set in `flags`

Valid Usage (Implicit)

- VUID-vkMapMemory-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkMapMemory-memory-parameter
`memory` **must** be a valid `VkDeviceMemory` handle
- VUID-vkMapMemory-flags-parameter
`flags` **must** be a valid combination of `VkMemoryMapFlagBits` values
- VUID-vkMapMemory-ppData-parameter
`ppData` **must** be a valid pointer to a pointer value
- VUID-vkMapMemory-memory-parent
`memory` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `memory` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_MEMORY_MAP_FAILED`

Bits which **can** be set in `vkMapMemory::flags` and `VkMemoryMapInfoKHR::flags`, specifying additional properties of a memory map, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkMemoryMapFlagBits {
    // Provided by VK_EXT_map_memory_placed
    VK_MEMORY_MAP_PLACED_BIT_EXT = 0x00000001,
} VkMemoryMapFlagBits;
```

- `VK_MEMORY_MAP_PLACED_BIT_EXT` requests that the implementation place the memory map at the virtual address specified by the application via `VkMemoryMapPlacedInfoEXT::pPlacedAddress`, replacing any existing mapping at that address. This flag **must** not be used with `vkMapMemory` as there is no way to specify the placement address.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkMemoryMapFlags;
```

`VkMemoryMapFlags` is a bitmask type for setting a mask of zero or more `VkMemoryMapFlagBits`.

Alternatively, to retrieve a host virtual address pointer to a region of a mappable memory object, call:

```
// Provided by VK_KHR_map_memory2
VkResult vkMapMemory2KHR(
    VkDevice device,
    const VkMemoryMapInfoKHR* pMemoryMapInfo,
    void** ppData);
```

- `device` is the logical device that owns the memory.
- `pMemoryMapInfo` is a pointer to a `VkMemoryMapInfoKHR` structure describing parameters of the map.
- `ppData` is a pointer to a `void *` variable in which is returned a host-accessible pointer to the beginning of the mapped range. This pointer minus `VkMemoryMapInfoKHR::offset` **must** be

aligned to at least `VkPhysicalDeviceLimits::minMemoryMapAlignment`.

This function behaves identically to `vkMapMemory` except that it gets its parameters via an extensible structure pointer rather than directly as function arguments.

Valid Usage (Implicit)

- VUID-vkMapMemory2KHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkMapMemory2KHR-pMemoryMapInfo-parameter
`pMemoryMapInfo` **must** be a valid pointer to a valid `VkMemoryMapInfoKHR` structure
- VUID-vkMapMemory2KHR-ppData-parameter
`ppData` **must** be a valid pointer to a pointer value

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_MEMORY_MAP_FAILED`

The `VkMemoryMapInfoKHR` structure is defined as:

```
// Provided by VK_KHR_map_memory2
typedef struct VkMemoryMapInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkMemoryMapFlags    flags;
    VkDeviceMemory      memory;
    VkDeviceSize        offset;
    VkDeviceSize        size;
} VkMemoryMapInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkMemoryMapFlagBits` specifying additional parameters of the memory map operation.
- `memory` is the `VkDeviceMemory` object to be mapped.
- `offset` is a zero-based byte offset from the beginning of the memory object.

- `size` is the size of the memory range to map, or `VK_WHOLE_SIZE` to map from `offset` to the end of the allocation.

Valid Usage

- VUID-VkMemoryMapInfoKHR-memory-07958
`memory` **must** not be currently host mapped
- VUID-VkMemoryMapInfoKHR-offset-07959
`offset` **must** be less than the size of `memory`
- VUID-VkMemoryMapInfoKHR-size-07960
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be greater than 0
- VUID-VkMemoryMapInfoKHR-size-07961
If `size` is not equal to `VK_WHOLE_SIZE`, `size` **must** be less than or equal to the size of the `memory` minus `offset`
- VUID-VkMemoryMapInfoKHR-memory-07962
`memory` **must** have been created with a memory type that reports `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT`
- VUID-VkMemoryMapInfoKHR-memory-07963
`memory` **must** not have been allocated with multiple instances
- VUID-VkMemoryMapInfoKHR-flags-09569
If `VK_MEMORY_MAP_PLACED_BIT_EXT` is set in `flags`, the `memoryMapPlaced` feature **must** be enabled
- VUID-VkMemoryMapInfoKHR-flags-09570
If `VK_MEMORY_MAP_PLACED_BIT_EXT` is set in `flags`, the `pNext` chain **must** include a `VkMemoryMapPlacedInfoEXT` structure and `VkMemoryMapPlacedInfoEXT::pPlacedAddress` **must** not be `NULL`
- VUID-VkMemoryMapInfoKHR-flags-09571
If `VK_MEMORY_MAP_PLACED_BIT_EXT` is set in `flags` and the `memoryMapRangePlaced` feature is not enabled, `offset` **must** be zero
- VUID-VkMemoryMapInfoKHR-flags-09572
If `VK_MEMORY_MAP_PLACED_BIT_EXT` is set in `flags` and the `memoryMapRangePlaced` feature is not enabled, `size` **must** be `VK_WHOLE_SIZE` or `VkMemoryAllocateInfo::allocationSize`
- VUID-VkMemoryMapInfoKHR-flags-09573
If `VK_MEMORY_MAP_PLACED_BIT_EXT` is set in `flags` and the `memoryMapRangePlaced` feature is enabled, `offset` **must** be aligned to an integer multiple of `VkPhysicalDeviceMapMemoryPlacedPropertiesEXT::minPlacedMemoryMapAlignment`
- VUID-VkMemoryMapInfoKHR-flags-09574
If `VK_MEMORY_MAP_PLACED_BIT_EXT` is set in `flags` and `size` is not `VK_WHOLE_SIZE`, `size` **must** be aligned to an integer multiple of `VkPhysicalDeviceMapMemoryPlacedPropertiesEXT::minPlacedMemoryMapAlignment`
- VUID-VkMemoryMapInfoKHR-flags-09651
If `VK_MEMORY_MAP_PLACED_BIT_EXT` is set in `flags` and `size` is `VK_WHOLE_SIZE`, `VkMemoryAllocateInfo::allocationSize` **must** be aligned to an integer multiple of

VkPhysicalDeviceMapMemoryPlacedPropertiesEXT::minPlacedMemoryMapAlignment

- VUID-VkMemoryMapInfoKHR-flags-09575

If `VK_MEMORY_MAP_PLACED_BIT_EXT` is set in `flags`, the memory object **must** not have been imported from a handle type of `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT` or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT`

Valid Usage (Implicit)

- VUID-VkMemoryMapInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_MAP_INFO_KHR`
- VUID-VkMemoryMapInfoKHR-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkMemoryMapPlacedInfoEXT`
- VUID-VkMemoryMapInfoKHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkMemoryMapInfoKHR-flags-parameter
`flags` **must** be a valid combination of `VkMemoryMapFlagBits` values
- VUID-VkMemoryMapInfoKHR-memory-parameter
`memory` **must** be a valid `VkDeviceMemory` handle

Host Synchronization

- Host access to `memory` **must** be externally synchronized

If `VK_MEMORY_MAP_PLACED_BIT_EXT` is set in `VkMemoryMapInfoKHR::flags` and the `pNext` chain of `VkMemoryMapInfoKHR` includes a `VkMemoryMapPlacedInfoEXT` structure, then that structure specifies the placement address of the memory map. The implementation will place the memory map at the specified address, replacing any existing maps in the specified memory range. Replacing memory maps in this way does not implicitly unmap Vulkan memory objects. Instead, the application **must** ensure no other Vulkan memory objects are mapped anywhere in the specified virtual address range. If successful, `ppData` will be set to the same value as `VkMemoryMapPlacedInfoEXT::pPlacedAddress` and `vkMapMemory2KHR` will return `VK_SUCCESS`. If it cannot place the map at the requested address for any reason, the memory object is left unmapped and `vkMapMemory2KHR` will return `VK_ERROR_MEMORY_MAP_FAILED`.

The `VkMemoryMapPlacedInfoEXT` structure is defined as:

```
// Provided by VK_EXT_map_memory_placed
typedef struct VkMemoryMapPlacedInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    void*              pPlacedAddress;
} VkMemoryMapPlacedInfoEXT;
```


- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pPlacedAddress` is the virtual address at which to place the address. If `VkMemoryMapInfoKHR::flags` does not contain `VK_MEMORY_MAP_PLACED_BIT_EXT`, this value is ignored.

Valid Usage

- VUID-VkMemoryMapPlacedInfoEXT-flags-09576
If `VkMemoryMapInfoKHR::flags` contains `VK_MEMORY_MAP_PLACED_BIT_EXT`, `pPlacedAddress` **must** not be `NULL`
- VUID-VkMemoryMapPlacedInfoEXT-pPlacedAddress-09577
`pPlacedAddress` **must** be aligned to an integer multiple of `VkPhysicalDeviceMapMemoryPlacedPropertiesEXT::minPlacedMemoryMapAlignment`
- VUID-VkMemoryMapPlacedInfoEXT-pPlacedAddress-09578
The address range specified by `pPlacedAddress` and `VkMemoryMapInfoKHR::size` **must** not overlap any existing Vulkan memory object mapping

Valid Usage (Implicit)

- VUID-VkMemoryMapPlacedInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_MAP_PLACED_INFO_EXT`

Two commands are provided to enable applications to work with non-coherent memory allocations: `vkFlushMappedMemoryRanges` and `vkInvalidateMappedMemoryRanges`.

NOTE

If the memory object was created with the `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT` set, `vkFlushMappedMemoryRanges` and `vkInvalidateMappedMemoryRanges` are unnecessary and **may** have a performance cost. However, [availability and visibility operations](#) still need to be managed on the device. See the description of [host access types](#) for more information.

NOTE

While memory objects imported from a handle type of `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT` or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT` are inherently mapped to host address space, they are not considered to be host mapped device memory unless they are explicitly host mapped using `vkMapMemory`. That means flushing or invalidating host caches with respect to host accesses performed on such memory through the original host pointer specified at import time is the responsibility of the application and **must** be performed with appropriate synchronization primitives provided by the platform which are outside the scope of Vulkan. `vkFlushMappedMemoryRanges` and `vkInvalidateMappedMemoryRanges`, however, **can** still be used on such memory objects to synchronize host accesses performed through the host pointer of the host mapped device memory range returned by `vkMapMemory`.

After a successful call to `vkMapMemory` or `vkMapMemory2KHR` the memory object `memory` is considered to be currently *host mapped*.

To flush ranges of non-coherent memory from the host caches, call:

```
// Provided by VK_VERSION_1_0
VkResult vkFlushMappedMemoryRanges(
    VkDevice          device,
    uint32_t          memoryRangeCount,
    const VkMappedMemoryRange* pMemoryRanges);
```

- `device` is the logical device that owns the memory ranges.
- `memoryRangeCount` is the length of the `pMemoryRanges` array.
- `pMemoryRanges` is a pointer to an array of `VkMappedMemoryRange` structures describing the memory ranges to flush.

`vkFlushMappedMemoryRanges` guarantees that host writes to the memory ranges described by `pMemoryRanges` are made available to the host memory domain, such that they **can** be made available to the device memory domain via [memory domain operations](#) using the `VK_ACCESS_HOST_WRITE_BIT` access type.

The first [synchronization scope](#) includes all host operations that happened-before it, as defined by the host memory model.

NOTE	<i>Note</i> Some systems allow writes that do not directly integrate with the host memory model; these have to be synchronized by the application manually. One example of this is non-temporal store instructions on x86; to ensure these happen-before submission, applications should call <code>_mm_sfence()</code>.
-------------	--

The second [synchronization scope](#) is empty.

The first [access scope](#) includes host writes to the specified memory ranges.

NOTE	<i>Note</i> When a host write to a memory location is made available in this way, each whole aligned set of <code>nonCoherentAtomSize</code> bytes that the memory location exists in will also be made available as if they were written by the host. For example, with a <code>nonCoherentAtomSize</code> of 128, if an application writes to the first byte of a memory object via a host mapping, the first 128 bytes of the memory object will be made available by this command. While the value of the following 127 bytes will be unchanged, this does count as an access for the purpose of synchronization, so care must be taken to avoid data races.
-------------	--

The second [access scope](#) is empty.

Unmapping non-coherent memory does not implicitly flush the host mapped memory, and host

writes that have not been flushed **may** not ever be visible to the device. However, implementations **must** ensure that writes that have not been flushed do not become visible to any other memory.

NOTE

The above guarantee avoids a potential memory corruption in scenarios where host writes to a mapped memory object have not been flushed before the memory is unmapped (or freed), and the virtual address range is subsequently reused for a different mapping (or memory allocation).

Valid Usage (Implicit)

- VUID-vkFlushMappedMemoryRanges-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkFlushMappedMemoryRanges-pMemoryRanges-parameter
pMemoryRanges **must** be a valid pointer to an array of [memoryRangeCount](#) valid [VkMappedMemoryRange](#) structures
- VUID-vkFlushMappedMemoryRanges-memoryRangeCount-arraylength
memoryRangeCount **must** be greater than 0

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

To invalidate ranges of non-coherent memory from the host caches, call:

```
// Provided by VK_VERSION_1_0
VkResult vkInvalidateMappedMemoryRanges(
    VkDevice          device,
    uint32_t          memoryRangeCount,
    const VkMappedMemoryRange* pMemoryRanges);
```

- **device** is the logical device that owns the memory ranges.
- **memoryRangeCount** is the length of the **pMemoryRanges** array.
- **pMemoryRanges** is a pointer to an array of [VkMappedMemoryRange](#) structures describing the memory ranges to invalidate.

[vkInvalidateMappedMemoryRanges](#) guarantees that device writes to the memory ranges described by **pMemoryRanges**, which have been made available to the host memory domain using the [VK_ACCESS_HOST_WRITE_BIT](#) and [VK_ACCESS_HOST_READ_BIT](#) [access types](#), are made visible to the host. If

a range of non-coherent memory is written by the host and then invalidated without first being flushed, its contents are undefined.

The first [synchronization scope](#) includes all host operations that happened-before it, as defined by the host memory model.

Note

This function does not synchronize with device operations directly - other host [synchronization operations](#) that depend on device operations such as [vkWaitForFences](#) must be executed beforehand. So for any non-coherent device write to be made visible to the host, there has to be a dependency chain along the following lines:

NOTE

1. Device write
2. Device memory barrier including host reads in its second scope
3. Signal on the device (e.g. a [fence signal operation](#))
4. Wait on the host (e.g. [vkWaitForFences](#))
5. [vkInvalidateMappedMemoryRanges](#)

The second [synchronization scope](#) includes all host operations that happen-after it, as defined by the host memory model.

The first [access scope](#) is empty.

The second [access scope](#) includes host reads to the specified memory ranges.

Note

When a device write to a memory location is made visible to the host in this way, each whole aligned set of [nonCoherentAtomSize](#) bytes that the memory location exists in will also be made visible as if they were written by the device. For example, with a [nonCoherentAtomSize](#) of 128, if an application writes to the first byte of a memory object on the device, the first 128 bytes of the memory object will be made visible by this command. While the value of the following 127 bytes will be unchanged, this does count as an access for the purpose of synchronization, so care must be taken to avoid data races.

NOTE

NOTE

Mapping non-coherent memory does not implicitly invalidate that memory.

Valid Usage (Implicit)

- VUID-vkInvalidateMappedMemoryRanges-device-parameter [device](#) **must** be a valid [VkDevice](#) handle
- VUID-vkInvalidateMappedMemoryRanges-pMemoryRanges-parameter [pMemoryRanges](#) **must** be a valid pointer to an array of [memoryRangeCount](#) valid [VkMappedMemoryRange](#) structures

- VUID-vkInvalidateMappedMemoryRanges-memoryRangeCount-arraylength
`memoryRangeCount` **must** be greater than 0

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkMappedMemoryRange` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMappedMemoryRange {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceMemory      memory;
    VkDeviceSize        offset;
    VkDeviceSize        size;
} VkMappedMemoryRange;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memory` is the memory object to which this range belongs.
- `offset` is the zero-based byte offset from the beginning of the memory object.
- `size` is either the size of range, or `VK_WHOLE_SIZE` to affect the range from `offset` to the end of the current mapping of the allocation.

Valid Usage

- VUID-VkMappedMemoryRange-memory-00684
`memory` **must** be currently host mapped
- VUID-VkMappedMemoryRange-size-00685
If `size` is not equal to `VK_WHOLE_SIZE`, `offset` and `size` **must** specify a range contained within the currently mapped range of `memory`
- VUID-VkMappedMemoryRange-size-00686
If `size` is equal to `VK_WHOLE_SIZE`, `offset` **must** be within the currently mapped range of `memory`
- VUID-VkMappedMemoryRange-offset-00687
`offset` **must** be a multiple of `VkPhysicalDeviceLimits::nonCoherentAtomSize`

- VUID-VkMappedMemoryRange-size-01389

If **size** is equal to **VK_WHOLE_SIZE**, the end of the current mapping of **memory** **must** either be a multiple of **VkPhysicalDeviceLimits::nonCoherentAtomSize** bytes from the beginning of the memory object, or be equal to the end of the memory object

- VUID-VkMappedMemoryRange-size-01390

If **size** is not equal to **VK_WHOLE_SIZE**, **size** **must** either be a multiple of **VkPhysicalDeviceLimits::nonCoherentAtomSize**, or **offset** plus **size** **must** equal the size of **memory**

Valid Usage (Implicit)

- VUID-VkMappedMemoryRange-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_MAPPED_MEMORY_RANGE**
- VUID-VkMappedMemoryRange-pNext-pNext
pNext **must** be **NULL**
- VUID-VkMappedMemoryRange-memory-parameter
memory **must** be a valid **VkDeviceMemory** handle

To unmap a memory object once host access to it is no longer needed by the application, call:

```
// Provided by VK_VERSION_1_0
void vkUnmapMemory(
    VkDevice          device,
    VkDeviceMemory    memory);
```

- **device** is the logical device that owns the memory.
- **memory** is the memory object to be unmapped.

Calling **vkUnmapMemory** is equivalent to calling **vkUnmapMemory2KHR** with an empty **pNext** chain and **flags** set to zero.

Valid Usage

- VUID-vkUnmapMemory-memory-00689
memory **must** be currently host mapped

Valid Usage (Implicit)

- VUID-vkUnmapMemory-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkUnmapMemory-memory-parameter
memory **must** be a valid **VkDeviceMemory** handle

- VUID-vkUnmapMemory-memory-parent
memory must have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **memory must** be externally synchronized

Alternatively, to unmap a memory object once host access to it is no longer needed by the application, call:

```
// Provided by VK_KHR_map_memory2
VkResult vkUnmapMemory2KHR(
    VkDevice                                device,
    const VkMemoryUnmapInfoKHR*             pMemoryUnmapInfo);
```

- **device** is the logical device that owns the memory.
- **pMemoryUnmapInfo** is a pointer to a [VkMemoryUnmapInfoKHR](#) structure describing parameters of the unmap.

This function behaves identically to [vkUnmapMemory](#) except that it gets its parameters via an extensible structure pointer rather than directly as function arguments.

Valid Usage (Implicit)

- VUID-vkUnmapMemory2KHR-device-parameter
device must be a valid [VkDevice](#) handle
- VUID-vkUnmapMemory2KHR-pMemoryUnmapInfo-parameter
pMemoryUnmapInfo must be a valid pointer to a valid [VkMemoryUnmapInfoKHR](#) structure

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_MEMORY_MAP_FAILED**

The [VkMemoryUnmapInfoKHR](#) structure is defined as:


```
// Provided by VK_KHR_map_memory2
typedef struct VkMemoryUnmapInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkMemoryUnmapFlagsKHR flags;
    VkDeviceMemory     memory;
} VkMemoryUnmapInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkMemoryUnmapFlagBitsKHR](#) specifying additional parameters of the memory map operation.
- **memory** is the [VkDeviceMemory](#) object to be unmapped.

Valid Usage

- VUID-VkMemoryUnmapInfoKHR-memory-07964
memory must be currently host mapped
- VUID-VkMemoryUnmapInfoKHR-flags-09579
If **VK_MEMORY_UNMAP_RESERVE_BIT_EXT** is set in **flags**, the **memoryUnmapReserve must** be enabled
- VUID-VkMemoryUnmapInfoKHR-flags-09580
If **VK_MEMORY_UNMAP_RESERVE_BIT_EXT** is set in **flags**, the memory object **must** not have been imported from a handle type of **VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT** or **VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT**

Valid Usage (Implicit)

- VUID-VkMemoryUnmapInfoKHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_MEMORY_UNMAP_INFO_KHR**
- VUID-VkMemoryUnmapInfoKHR-pNext-pNext
pNext must be **NULL**
- VUID-VkMemoryUnmapInfoKHR-flags-parameter
flags must be a valid combination of [VkMemoryUnmapFlagBitsKHR](#) values
- VUID-VkMemoryUnmapInfoKHR-memory-parameter
memory must be a valid [VkDeviceMemory](#) handle

Host Synchronization

- Host access to **memory must** be externally synchronized

Bits which **can** be set in [VkMemoryUnmapInfoKHR::flags](#), specifying additional properties of a

memory unmap, are:

```
// Provided by VK_KHR_map_memory2
typedef enum VkMemoryUnmapFlagBitsKHR {
    // Provided by VK_EXT_map_memory_placed
    VK_MEMORY_UNMAP_RESERVE_BIT_EXT = 0x00000001,
} VkMemoryUnmapFlagBitsKHR;
```

- **VK_MEMORY_UNMAP_RESERVE_BIT_EXT** requests that virtual address range currently occupied by the memory map remain reserved after the **vkUnmapMemory2KHR** call completes. Future system memory map operations or calls to **vkMapMemory** or **vkMapMemory2KHR** will not return addresses in that range unless the range has since been unreserved by the application or the mapping is explicitly placed in that range by calling **vkMapMemory2KHR** with **VK_MEMORY_MAP_PLACED_BIT_EXT**, or doing the system memory map equivalent. When **VK_MEMORY_UNMAP_RESERVE_BIT_EXT** is set, the memory unmap operation **may** fail, in which case the memory object will remain host mapped and **vkUnmapMemory2KHR** will return **VK_ERROR_MEMORY_MAP_FAILED**.

```
// Provided by VK_KHR_map_memory2
typedef VkFlags VkMemoryUnmapFlagsKHR;
```

VkMemoryUnmapFlagsKHR is a bitmask type for setting a mask of zero or more **VkMemoryUnmapFlagBitsKHR**.

11.2.11. Lazily Allocated Memory

If the memory object is allocated from a heap with the **VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT** bit set, that object's backing memory **may** be provided by the implementation lazily. The actual committed size of the memory **may** initially be as small as zero (or as large as the requested size), and monotonically increases as additional memory is needed.

A memory type with this flag set is only allowed to be bound to a **VkImage** whose usage flags include **VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT**.

NOTE

Using lazily allocated memory objects for framebuffer attachments that are not needed once a render pass instance has completed **may** allow some implementations to never allocate memory for such attachments.

To determine the amount of lazily-allocated memory that is currently committed for a memory object, call:

```
// Provided by VK_VERSION_1_0
void vkGetDeviceMemoryCommitment(
    VkDevice                device,
    VkDeviceMemory          memory,
    VkDeviceSize*           pCommittedMemoryInBytes);
```

- **device** is the logical device that owns the memory.
- **memory** is the memory object being queried.
- **pCommittedMemoryInBytes** is a pointer to a **VkDeviceSize** value in which the number of bytes currently committed is returned, on success.

The implementation **may** update the commitment at any time, and the value returned by this query **may** be out of date.

The implementation guarantees to allocate any committed memory from the **heapIndex** indicated by the memory type that the memory object was created with.

Valid Usage

- VUID-vkGetDeviceMemoryCommitment-memory-00690
memory must have been created with a memory type that reports **VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT**

Valid Usage (Implicit)

- VUID-vkGetDeviceMemoryCommitment-device-parameter
device must be a valid **VkDevice** handle
- VUID-vkGetDeviceMemoryCommitment-memory-parameter
memory must be a valid **VkDeviceMemory** handle
- VUID-vkGetDeviceMemoryCommitment-pCommittedMemoryInBytes-parameter
pCommittedMemoryInBytes must be a valid pointer to a **VkDeviceSize** value
- VUID-vkGetDeviceMemoryCommitment-memory-parent
memory must have been created, allocated, or retrieved from **device**

11.2.12. Peer Memory Features

Peer memory is memory that is allocated for a given physical device and then bound to a resource and accessed by a different physical device, in a logical device that represents multiple physical devices. Some ways of reading and writing peer memory **may** not be supported by a device.

To determine how peer memory **can** be accessed, call:

```
// Provided by VK_KHR_device_group
void vkGetDeviceGroupPeerMemoryFeaturesKHR(
    VkDevice          device,
    uint32_t          heapIndex,
    uint32_t          localDeviceIndex,
    uint32_t          remoteDeviceIndex,
    VkPeerMemoryFeatureFlags* pPeerMemoryFeatures);
```

- **device** is the logical device that owns the memory.
- **heapIndex** is the index of the memory heap from which the memory is allocated.
- **localDeviceIndex** is the device index of the physical device that performs the memory access.
- **remoteDeviceIndex** is the device index of the physical device that the memory is allocated for.
- **pPeerMemoryFeatures** is a pointer to a **VkPeerMemoryFeatureFlags** bitmask indicating which types of memory accesses are supported for the combination of heap, local, and remote devices.

Valid Usage

- VUID-vkGetDeviceGroupPeerMemoryFeatures-heapIndex-00691
heapIndex must be less than **memoryHeapCount**
- VUID-vkGetDeviceGroupPeerMemoryFeatures-localDeviceIndex-00692
localDeviceIndex must be a valid device index
- VUID-vkGetDeviceGroupPeerMemoryFeatures-remoteDeviceIndex-00693
remoteDeviceIndex must be a valid device index
- VUID-vkGetDeviceGroupPeerMemoryFeatures-localDeviceIndex-00694
localDeviceIndex must not equal **remoteDeviceIndex**

Valid Usage (Implicit)

- VUID-vkGetDeviceGroupPeerMemoryFeatures-device-parameter
device must be a valid **VkDevice** handle
- VUID-vkGetDeviceGroupPeerMemoryFeatures-pPeerMemoryFeatures-parameter
pPeerMemoryFeatures must be a valid pointer to a **VkPeerMemoryFeatureFlags** value

Bits which **may** be set in **vkGetDeviceGroupPeerMemoryFeaturesKHR::pPeerMemoryFeatures**, indicating supported peer memory features, are:

```
typedef enum VkPeerMemoryFeatureFlagBits {
    VK_PEER_MEMORY_FEATURE_COPY_SRC_BIT = 0x00000001,
    VK_PEER_MEMORY_FEATURE_COPY_DST_BIT = 0x00000002,
    VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT = 0x00000004,
    VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT = 0x00000008,
    // Provided by VK_KHR_device_group
    VK_PEER_MEMORY_FEATURE_COPY_SRC_BIT_KHR = VK_PEER_MEMORY_FEATURE_COPY_SRC_BIT,
    // Provided by VK_KHR_device_group
    VK_PEER_MEMORY_FEATURE_COPY_DST_BIT_KHR = VK_PEER_MEMORY_FEATURE_COPY_DST_BIT,
    // Provided by VK_KHR_device_group
    VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT_KHR =
VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT,
    // Provided by VK_KHR_device_group
    VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT_KHR =
VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT,
} VkPeerMemoryFeatureFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_device_group
typedef VkPeerMemoryFeatureFlagBits VkPeerMemoryFeatureFlagBitsKHR;
```

- **VK_PEER_MEMORY_FEATURE_COPY_SRC_BIT** specifies that the memory **can** be accessed as the source of any **vkCmdCopy*** command.
- **VK_PEER_MEMORY_FEATURE_COPY_DST_BIT** specifies that the memory **can** be accessed as the destination of any **vkCmdCopy*** command.
- **VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT** specifies that the memory **can** be read as any memory access type.
- **VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT** specifies that the memory **can** be written as any memory access type. Shader atomics are considered to be writes.

NOTE The peer memory features of a memory heap also apply to any accesses that **may** be performed during [image layout transitions](#).

VK_PEER_MEMORY_FEATURE_COPY_DST_BIT **must** be supported for all host local heaps and for at least one device-local memory heap.

If a device does not support a peer memory feature, it is still valid to use a resource that includes both local and peer memory bindings with the corresponding access type as long as only the local bindings are actually accessed. For example, an application doing split-frame rendering would use framebuffer attachments that include both local and peer memory bindings, but would scissor the rendering to only update local memory.

```
typedef VkFlags VkPeerMemoryFeatureFlags;
```

or the equivalent

```
// Provided by VK_KHR_device_group
typedef VkPeerMemoryFeatureFlags VkPeerMemoryFeatureFlagsKHR;
```

`VkPeerMemoryFeatureFlags` is a bitmask type for setting a mask of zero or more `VkPeerMemoryFeatureFlagBits`.

11.2.13. Opaque Capture Address Query

To query a 64-bit opaque capture address value from a memory object, call:

```
// Provided by VK_KHR_buffer_device_address
uint64_t vkGetDeviceMemoryOpaqueCaptureAddressKHR(
    VkDevice device,
    const VkDeviceMemoryOpaqueCaptureAddressInfo* pInfo);
```

- `device` is the logical device that the memory object was allocated on.
- `pInfo` is a pointer to a `VkDeviceMemoryOpaqueCaptureAddressInfo` structure specifying the memory object to retrieve an address for.

The 64-bit return value is an opaque address representing the start of `pInfo->memory`.

If the memory object was allocated with a non-zero value of `VkMemoryOpaqueCaptureAddressAllocateInfo::opaqueCaptureAddress`, the return value **must** be the same address.

NOTE	The expected usage for these opaque addresses is only for trace capture/replay tools to store these addresses in a trace and subsequently specify them during replay.
-------------	---

Valid Usage

- VUID-vkGetDeviceMemoryOpaqueCaptureAddress-None-03334
The `bufferDeviceAddress` feature **must** be enabled
- VUID-vkGetDeviceMemoryOpaqueCaptureAddress-device-03335
If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetDeviceMemoryOpaqueCaptureAddress-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetDeviceMemoryOpaqueCaptureAddress-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkDeviceMemoryOpaqueCaptureAddressInfo`

structure

The `VkDeviceMemoryOpaqueCaptureAddressInfo` structure is defined as:

```
typedef struct VkDeviceMemoryOpaqueCaptureAddressInfo {  
    VkStructureType    sType;  
    const void*        pNext;  
    VkDeviceMemory     memory;  
} VkDeviceMemoryOpaqueCaptureAddressInfo;
```

or the equivalent

```
// Provided by VK_KHR_buffer_device_address  
typedef VkDeviceMemoryOpaqueCaptureAddressInfo  
VkDeviceMemoryOpaqueCaptureAddressInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memory` specifies the memory whose address is being queried.

Valid Usage

- VUID-VkDeviceMemoryOpaqueCaptureAddressInfo-memory-03336
`memory` **must** have been allocated with `VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT`

Valid Usage (Implicit)

- VUID-VkDeviceMemoryOpaqueCaptureAddressInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_MEMORY_OPAQUE_CAPTURE_ADDRESS_INFO`
- VUID-VkDeviceMemoryOpaqueCaptureAddressInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDeviceMemoryOpaqueCaptureAddressInfo-memory-parameter
`memory` **must** be a valid `VkDeviceMemory` handle

Chapter 12. Resource Creation

Vulkan supports two primary resource types: *buffers* and *images*. Resources are views of memory with associated formatting and dimensionality. Buffers provide access to raw arrays of bytes, whereas images **can** be multidimensional and **may** have associated metadata.

Other resource types, such as [acceleration structures](#) and [micromaps](#) use buffers as the backing store for opaque data structures.

12.1. Buffers

Buffers represent linear arrays of data which are used for various purposes by binding them to a graphics or compute pipeline via descriptor sets or certain commands, or by directly specifying them as parameters to certain commands.

Buffers are represented by `VkBuffer` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkBuffer)
```

To create buffers, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateBuffer(
    VkDevice                                device,
    const VkBufferCreateInfo*               pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkBuffer*                               pBuffer);
```

- `device` is the logical device that creates the buffer object.
- `pCreateInfo` is a pointer to a [VkBufferCreateInfo](#) structure containing parameters affecting creation of the buffer.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pBuffer` is a pointer to a [VkBuffer](#) handle in which the resulting buffer object is returned.

Valid Usage

- VUID-vkCreateBuffer-device-09664
`device` **must** support at least one queue family with one of the `VK_QUEUE_VIDEO_ENCODE_BIT_KHR`, `VK_QUEUE_VIDEO_DECODE_BIT_KHR`, `VK_QUEUE_SPARSE_BINDING_BIT`, `VK_QUEUE_TRANSFER_BIT`, `VK_QUEUE_COMPUTE_BIT`, or `VK_QUEUE_GRAPHICS_BIT` capabilities
- VUID-vkCreateBuffer-flags-00911
If the `flags` member of `pCreateInfo` includes `VK_BUFFER_CREATE_SPARSE_BINDING_BIT`,

creating this **VkBuffer** **must** not cause the total required sparse memory for all currently valid sparse resources on the device to exceed **VkPhysicalDeviceLimits::sparseAddressSpaceSize**

Valid Usage (Implicit)

- VUID-vkCreateBuffer-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkCreateBuffer-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid **VkBufferCreateInfo** structure
- VUID-vkCreateBuffer-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkCreateBuffer-pBuffer-parameter
pBuffer **must** be a valid pointer to a **VkBuffer** handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**
- **VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR**

The **VkBufferCreateInfo** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkBufferCreateFlags flags;
    VkDeviceSize        size;
    VkBufferUsageFlags  usage;
    VkSharingMode        sharingMode;
    uint32_t            queueFamilyIndexCount;
    const uint32_t*      pQueueFamilyIndices;
} VkBufferCreateInfo;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.

- **flags** is a bitmask of [VkBufferCreateFlagBits](#) specifying additional parameters of the buffer.
- **size** is the size in bytes of the buffer to be created.
- **usage** is a bitmask of [VkBufferUsageFlagBits](#) specifying allowed usages of the buffer.
- **sharingMode** is a [VkSharingMode](#) value specifying the sharing mode of the buffer when it will be accessed by multiple queue families.
- **queueFamilyIndexCount** is the number of entries in the **pQueueFamilyIndices** array.
- **pQueueFamilyIndices** is a pointer to an array of queue families that will access this buffer. It is ignored if **sharingMode** is not `VK_SHARING_MODE_CONCURRENT`.

If the **pNext** chain includes a [VkBufferUsageFlags2CreateInfoKHR](#) structure, [VkBufferUsageFlags2CreateInfoKHR::usage](#) from that structure is used instead of **usage** from this structure.

Valid Usage

- VUID-VkBufferCreateInfo-None-09499
If the **pNext** chain does not include a [VkBufferUsageFlags2CreateInfoKHR](#) structure, **usage** **must** be a valid combination of [VkBufferUsageFlagBits](#) values
- VUID-VkBufferCreateInfo-None-09500
If the **pNext** chain does not include a [VkBufferUsageFlags2CreateInfoKHR](#) structure, **usage** **must** not be 0
- VUID-VkBufferCreateInfo-size-00912
size **must** be greater than 0
- VUID-VkBufferCreateInfo-sharingMode-00913
If **sharingMode** is `VK_SHARING_MODE_CONCURRENT`, **pQueueFamilyIndices** **must** be a valid pointer to an array of **queueFamilyIndexCount** `uint32_t` values
- VUID-VkBufferCreateInfo-sharingMode-00914
If **sharingMode** is `VK_SHARING_MODE_CONCURRENT`, **queueFamilyIndexCount** **must** be greater than 1
- VUID-VkBufferCreateInfo-sharingMode-01419
If **sharingMode** is `VK_SHARING_MODE_CONCURRENT`, each element of **pQueueFamilyIndices** **must** be unique and **must** be less than **pQueueFamilyPropertyCount** returned by either [vkGetPhysicalDeviceQueueFamilyProperties2KHR](#) or [vkGetPhysicalDeviceQueueFamilyProperties](#) for the **physicalDevice** that was used to create **device**
- VUID-VkBufferCreateInfo-flags-00915
If the **sparseBinding** feature is not enabled, **flags** **must** not contain `VK_BUFFER_CREATE_SPARSE_BINDING_BIT`
- VUID-VkBufferCreateInfo-flags-00916
If the **sparseResidencyBuffer** feature is not enabled, **flags** **must** not contain `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT`
- VUID-VkBufferCreateInfo-flags-00917
If the **sparseResidencyAliased** feature is not enabled, **flags** **must** not contain

VK_BUFFER_CREATE_SPARSE_ALIASED_BIT

- VUID-VkBufferCreateInfo-flags-00918

If `flags` contains `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` or `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT`, it **must** also contain `VK_BUFFER_CREATE_SPARSE_BINDING_BIT`

- VUID-VkBufferCreateInfo-pNext-00920

If the `pNext` chain includes a `VkExternalMemoryBufferCreateInfo` structure, its `handleTypes` member **must** only contain bits that are also in `VkExternalBufferProperties::externalMemoryProperties.compatibleHandleTypes`, as returned by `vkGetPhysicalDeviceExternalBufferPropertiesKHR` with `pExternalBufferInfo->handleType` equal to any one of the handle types specified in `VkExternalMemoryBufferCreateInfo::handleTypes`

- VUID-VkBufferCreateInfo-opaqueCaptureAddress-03337

If `VkBufferOpaqueCaptureAddressCreateInfo::opaqueCaptureAddress` is not zero, `flags` **must** include `VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT`

- VUID-VkBufferCreateInfo-flags-03338

If `flags` includes `VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT`, the `bufferDeviceAddressCaptureReplay` feature **must** be enabled

- VUID-VkBufferCreateInfo-usage-04813

If `usage` includes `VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR` or `VK_BUFFER_USAGE_VIDEO_DECODE_DST_BIT_KHR`, and `flags` does not include `VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoProfileListInfoKHR` structure with `profileCount` greater than 0 and `pProfiles` including at least one `VkVideoProfileInfoKHR` structure with a `videoCodecOperation` member specifying a decode operation

- VUID-VkBufferCreateInfo-usage-04814

If `usage` includes `VK_BUFFER_USAGE_VIDEO_ENCODE_SRC_BIT_KHR` or `VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR`, and `flags` does not include `VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoProfileListInfoKHR` structure with `profileCount` greater than 0 and `pProfiles` including at least one `VkVideoProfileInfoKHR` structure with a `videoCodecOperation` member specifying an encode operation

- VUID-VkBufferCreateInfo-flags-08325

If `flags` includes `VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`, then `videoMaintenance1` **must** be enabled

- VUID-VkBufferCreateInfo-size-06409

`size` **must** be less than or equal to `VkPhysicalDeviceMaintenance4Properties::maxBufferSize`

- VUID-VkBufferCreateInfo-flags-09641

If `flags` includes `VK_BUFFER_CREATE_PROTECTED_BIT`, then `usage` **must** not contain any of the following bits

- `VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT`
- `VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT`

- `VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR`
- `VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR`
- `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR`
- `VK_BUFFER_USAGE_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT`
- `VK_BUFFER_USAGE_MICROMAP_STORAGE_BIT_EXT`

Valid Usage (Implicit)

- `VUID-VkBufferCreateInfo-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_BUFFER_CREATE_INFO`
- `VUID-VkBufferCreateInfo-pNext-pNext`
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkBufferOpaqueCaptureAddressCreateInfo`, `VkBufferUsageFlags2CreateInfoKHR`, `VkExternalMemoryBufferCreateInfo`, or `VkVideoProfileListInfoKHR`
- `VUID-VkBufferCreateInfo-sType-unique`
The `sType` value of each struct in the `pNext` chain **must** be unique
- `VUID-VkBufferCreateInfo-flags-parameter`
`flags` **must** be a valid combination of `VkBufferCreateFlagBits` values
- `VUID-VkBufferCreateInfo-sharingMode-parameter`
`sharingMode` **must** be a valid `VkSharingMode` value

The `VkBufferUsageFlags2CreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance5
typedef struct VkBufferUsageFlags2CreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBufferUsageFlags2KHR  usage;
} VkBufferUsageFlags2CreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `usage` is a bitmask of `VkBufferUsageFlagBits2KHR` specifying allowed usages of the buffer.

If this structure is included in the `pNext` chain of a buffer creation structure, `usage` is used instead of the corresponding `usage` value passed in that creation structure, allowing additional usage flags to be specified. If this structure is included in the `pNext` chain of a buffer query structure, the usage flags of the buffer are returned in `usage` of this structure, and the usage flags representable in `usage` of the buffer query structure are also returned in that field.

Valid Usage (Implicit)

- VUID-VkBufferUsageFlags2CreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BUFFER_USAGE_FLAGS_2_CREATE_INFO_KHR`
- VUID-VkBufferUsageFlags2CreateInfoKHR-usage-parameter
`usage` **must** be a valid combination of `VkBufferUsageFlagBits2KHR` values
- VUID-VkBufferUsageFlags2CreateInfoKHR-usage-requiredbitmask
`usage` **must** not be `0`

Bits which **can** be set in `VkBufferUsageFlags2CreateInfoKHR::usage`, specifying usage behavior of a buffer, are:

```
// Provided by VK_KHR_maintenance5
// Flag bits for VkBufferUsageFlagBits2KHR
typedef VkFlags64 VkBufferUsageFlagBits2KHR;
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_TRANSFER_SRC_BIT_KHR =
0x00000001ULL;
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_TRANSFER_DST_BIT_KHR =
0x00000002ULL;
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_UNIFORM_TEXEL_BUFFER_BIT_KHR
= 0x00000004ULL;
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_STORAGE_TEXEL_BUFFER_BIT_KHR
= 0x00000008ULL;
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_UNIFORM_BUFFER_BIT_KHR =
0x00000010ULL;
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_STORAGE_BUFFER_BIT_KHR =
0x00000020ULL;
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_INDEX_BUFFER_BIT_KHR =
0x00000040ULL;
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_VERTEX_BUFFER_BIT_KHR =
0x00000080ULL;
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_INDIRECT_BUFFER_BIT_KHR =
0x00000100ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_conditional_rendering
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_CONDITIONAL_RENDERING_BIT_EXT
= 0x00000200ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_ray_tracing_pipeline
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_SHADER_BINDING_TABLE_BIT_KHR
= 0x00000400ULL;
// Provided by VK_KHR_maintenance5 with VK_NV_ray_tracing
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_RAY_TRACING_BIT_NV =
0x00000400ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_transform_feedback
static const VkBufferUsageFlagBits2KHR
VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT = 0x00000800ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_transform_feedback
static const VkBufferUsageFlagBits2KHR
VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT = 0x00001000ULL;
```

```

// Provided by VK_KHR_maintenance5 with VK_KHR_video_decode_queue
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_VIDEO_DECODE_SRC_BIT_KHR =
0x00002000ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_video_decode_queue
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_VIDEO_DECODE_DST_BIT_KHR =
0x00004000ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_video_encode_queue
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_VIDEO_ENCODE_DST_BIT_KHR =
0x00008000ULL;
// Provided by VK_KHR_maintenance5 with VK_KHR_video_encode_queue
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_VIDEO_ENCODE_SRC_BIT_KHR =
0x00010000ULL;
// Provided by VK_KHR_maintenance5 with (VK_VERSION_1_2 or
VK_KHR_buffer_device_address) or VK_EXT_buffer_device_address
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_SHADER_DEVICE_ADDRESS_BIT_KHR
= 0x00020000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_maintenance5
static const VkBufferUsageFlagBits2KHR
VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR =
0x00080000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_maintenance5
static const VkBufferUsageFlagBits2KHR
VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR = 0x00100000ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_descriptor_buffer
static const VkBufferUsageFlagBits2KHR
VK_BUFFER_USAGE_2_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT = 0x00200000ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_descriptor_buffer
static const VkBufferUsageFlagBits2KHR
VK_BUFFER_USAGE_2_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT = 0x00400000ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_descriptor_buffer
static const VkBufferUsageFlagBits2KHR
VK_BUFFER_USAGE_2_PUSH_DESCRIPTOR_DESCRIPTOR_BUFFER_BIT_EXT = 0x00400000ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_opacity_micromap
static const VkBufferUsageFlagBits2KHR
VK_BUFFER_USAGE_2_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT = 0x00800000ULL;
// Provided by VK_KHR_maintenance5 with VK_EXT_opacity_micromap
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_MICROMAP_STORAGE_BIT_EXT =
0x01000000ULL;
// Provided by VK_EXT_device_generated_commands
static const VkBufferUsageFlagBits2KHR VK_BUFFER_USAGE_2_PREPROCESS_BUFFER_BIT_EXT =
0x80000000ULL;

```

- `VK_BUFFER_USAGE_2_TRANSFER_SRC_BIT_KHR` specifies that the buffer **can** be used as the source of a *transfer command* (see the definition of `VK_PIPELINE_STAGE_TRANSFER_BIT`).
- `VK_BUFFER_USAGE_2_TRANSFER_DST_BIT_KHR` specifies that the buffer **can** be used as the destination of a transfer command.
- `VK_BUFFER_USAGE_2_UNIFORM_TEXEL_BUFFER_BIT_KHR` specifies that the buffer **can** be used to create a `VkBufferView` suitable for occupying a `VkDescriptorSet` slot of type `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER`.

- `VK_BUFFER_USAGE_2_STORAGE_TEXEL_BUFFER_BIT_KHR` specifies that the buffer **can** be used to create a `VkBufferView` suitable for occupying a `VkDescriptorSet` slot of type `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`.
- `VK_BUFFER_USAGE_2_UNIFORM_BUFFER_BIT_KHR` specifies that the buffer **can** be used in a `VkDescriptorBufferInfo` suitable for occupying a `VkDescriptorSet` slot either of type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`.
- `VK_BUFFER_USAGE_2_STORAGE_BUFFER_BIT_KHR` specifies that the buffer **can** be used in a `VkDescriptorBufferInfo` suitable for occupying a `VkDescriptorSet` slot either of type `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`.
- `VK_BUFFER_USAGE_2_INDEX_BUFFER_BIT_KHR` specifies that the buffer is suitable for passing as the `buffer` parameter to `vkCmdBindIndexBuffer2KHR` and `vkCmdBindIndexBuffer`.
- `VK_BUFFER_USAGE_2_VERTEX_BUFFER_BIT_KHR` specifies that the buffer is suitable for passing as an element of the `pBuffers` array to `vkCmdBindVertexBuffers`.
- `VK_BUFFER_USAGE_2_INDIRECT_BUFFER_BIT_KHR` specifies that the buffer is suitable for passing as the `buffer` parameter to `vkCmdDrawIndirect`, `vkCmdDrawIndexedIndirect`, or `vkCmdDispatchIndirect`. It is also suitable for passing as the underlying buffer of either the `preprocessAddress` or `sequenceCountAddress` members of `VkGeneratedCommandsInfoEXT`.
- `VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT` specifies that the buffer is suitable for using for binding as a transform feedback buffer with `vkCmdBindTransformFeedbackBuffersEXT`.
- `VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT` specifies that the buffer is suitable for using as a counter buffer with `vkCmdBeginTransformFeedbackEXT` and `vkCmdEndTransformFeedbackEXT`.
- `VK_BUFFER_USAGE_2_SHADER_BINDING_TABLE_BIT_KHR` specifies that the buffer is suitable for use as a [Shader Binding Table](#).
- `VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR` specifies that the buffer is suitable for use as a read-only input to an [acceleration structure build](#).
- `VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR` specifies that the buffer is suitable for storage space for a [VkAccelerationStructureKHR](#).
- `VK_BUFFER_USAGE_2_SHADER_DEVICE_ADDRESS_BIT_KHR` specifies that the buffer **can** be used to retrieve a buffer device address via `vkGetBufferDeviceAddressKHR` and use that address to access the buffer's memory from a shader.
- `VK_BUFFER_USAGE_2_VIDEO_DECODE_SRC_BIT_KHR` specifies that the buffer **can** be used as the source video bitstream buffer in a [video decode operation](#).
- `VK_BUFFER_USAGE_2_VIDEO_DECODE_DST_BIT_KHR` is reserved for future use.
- `VK_BUFFER_USAGE_2_VIDEO_ENCODE_DST_BIT_KHR` specifies that the buffer **can** be used as the destination video bitstream buffer in a [video encode operation](#).
- `VK_BUFFER_USAGE_2_VIDEO_ENCODE_SRC_BIT_KHR` is reserved for future use.
- `VK_BUFFER_USAGE_2_PREPROCESS_BUFFER_BIT_EXT` specifies that the buffer **can** be used as a preprocess buffer for [Device-Generated Commands](#).

```
// Provided by VK_KHR_maintenance5
typedef VkFlags64 VkBufferUsageFlags2KHR;
```

`VkBufferUsageFlags2KHR` is a bitmask type for setting a mask of zero or more `VkBufferUsageFlagBits2KHR`.

Bits which **can** be set in `VkBufferCreateInfo::usage`, specifying usage behavior of a buffer, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkBufferUsageFlagBits {
    VK_BUFFER_USAGE_TRANSFER_SRC_BIT = 0x00000001,
    VK_BUFFER_USAGE_TRANSFER_DST_BIT = 0x00000002,
    VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT = 0x00000004,
    VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT = 0x00000008,
    VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT = 0x00000010,
    VK_BUFFER_USAGE_STORAGE_BUFFER_BIT = 0x00000020,
    VK_BUFFER_USAGE_INDEX_BUFFER_BIT = 0x00000040,
    VK_BUFFER_USAGE_VERTEX_BUFFER_BIT = 0x00000080,
    VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT = 0x00000100,
    VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT = 0x00020000,
    // Provided by VK_KHR_video_decode_queue
    VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR = 0x00002000,
    // Provided by VK_KHR_video_decode_queue
    VK_BUFFER_USAGE_VIDEO_DECODE_DST_BIT_KHR = 0x00004000,
    // Provided by VK_EXT_transform_feedback
    VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT = 0x00000800,
    // Provided by VK_EXT_transform_feedback
    VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT = 0x00001000,
    // Provided by VK_KHR_acceleration_structure
    VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR = 0x00080000,
    // Provided by VK_KHR_acceleration_structure
    VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR = 0x00100000,
    // Provided by VK_KHR_ray_tracing_pipeline
    VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR = 0x00000400,
    // Provided by VK_KHR_video_encode_queue
    VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR = 0x00008000,
    // Provided by VK_KHR_video_encode_queue
    VK_BUFFER_USAGE_VIDEO_ENCODE_SRC_BIT_KHR = 0x00010000,
    // Provided by VK_EXT_opacity_micromap
    VK_BUFFER_USAGE_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT = 0x00800000,
    // Provided by VK_EXT_opacity_micromap
    VK_BUFFER_USAGE_MICROMAP_STORAGE_BIT_EXT = 0x01000000,
    // Provided by VK_KHR_buffer_device_address
    VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT_KHR =
VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT,
} VkBufferUsageFlagBits;
```

- `VK_BUFFER_USAGE_TRANSFER_SRC_BIT` specifies that the buffer **can** be used as the source of a

transfer command (see the definition of `VK_PIPELINE_STAGE_TRANSFER_BIT`).

- `VK_BUFFER_USAGE_TRANSFER_DST_BIT` specifies that the buffer **can** be used as the destination of a transfer command.
- `VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT` specifies that the buffer **can** be used to create a `VkBufferView` suitable for occupying a `VkDescriptorSet` slot of type `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER`.
- `VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT` specifies that the buffer **can** be used to create a `VkBufferView` suitable for occupying a `VkDescriptorSet` slot of type `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`.
- `VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT` specifies that the buffer **can** be used in a `VkDescriptorBufferInfo` suitable for occupying a `VkDescriptorSet` slot either of type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`.
- `VK_BUFFER_USAGE_STORAGE_BUFFER_BIT` specifies that the buffer **can** be used in a `VkDescriptorBufferInfo` suitable for occupying a `VkDescriptorSet` slot either of type `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`.
- `VK_BUFFER_USAGE_INDEX_BUFFER_BIT` specifies that the buffer is suitable for passing as the `buffer` parameter to `vkCmdBindIndexBuffer2KHR` and `vkCmdBindIndexBuffer`.
- `VK_BUFFER_USAGE_VERTEX_BUFFER_BIT` specifies that the buffer is suitable for passing as an element of the `pBuffers` array to `vkCmdBindVertexBuffers`.
- `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` specifies that the buffer is suitable for passing as the `buffer` parameter to `vkCmdDrawIndirect`, `vkCmdDrawIndexedIndirect`, or `vkCmdDispatchIndirect`. It is also suitable for passing as the underlying buffer of either the `preprocessAddress` or `sequenceCountAddress` members of `VkGeneratedCommandsInfoEXT`.
- `VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT` specifies that the buffer is suitable for using for binding as a transform feedback buffer with `vkCmdBindTransformFeedbackBuffersEXT`.
- `VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT` specifies that the buffer is suitable for using as a counter buffer with `vkCmdBeginTransformFeedbackEXT` and `vkCmdEndTransformFeedbackEXT`.
- `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` specifies that the buffer is suitable for use as a `Shader Binding Table`.
- `VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR` specifies that the buffer is suitable for use as a read-only input to an `acceleration structure build`.
- `VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR` specifies that the buffer is suitable for storage space for a `VkAccelerationStructureKHR`.
- `VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT` specifies that the buffer **can** be used to retrieve a buffer device address via `vkGetBufferDeviceAddressKHR` and use that address to access the buffer's memory from a shader.
- `VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR` specifies that the buffer **can** be used as the source video bitstream buffer in a `video decode operation`.
- `VK_BUFFER_USAGE_VIDEO_DECODE_DST_BIT_KHR` is reserved for future use.
- `VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR` specifies that the buffer **can** be used as the

destination video bitstream buffer in a [video encode operation](#).

- `VK_BUFFER_USAGE_VIDEO_ENCODE_SRC_BIT_KHR` is reserved for future use.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkBufferUsageFlags;
```

`VkBufferUsageFlags` is a bitmask type for setting a mask of zero or more `VkBufferUsageFlagBits`.

Bits which **can** be set in `VkBufferCreateInfo::flags`, specifying additional parameters of a buffer, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkBufferCreateFlagBits {
    VK_BUFFER_CREATE_SPARSE_BINDING_BIT = 0x00000001,
    VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT = 0x00000002,
    VK_BUFFER_CREATE_SPARSE_ALIASED_BIT = 0x00000004,
    VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT = 0x00000010,
    // Provided by VK_KHR_video_maintenance1
    VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR = 0x00000040,
    // Provided by VK_KHR_buffer_device_address
    VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR =
VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT,
} VkBufferCreateFlagBits;
```

- `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` specifies that the buffer will be backed using sparse memory binding.
- `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` specifies that the buffer **can** be partially backed using sparse memory binding. Buffers created with this flag **must** also be created with the `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` flag.
- `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT` specifies that the buffer will be backed using sparse memory binding with memory ranges that might also simultaneously be backing another buffer (or another portion of the same buffer). Buffers created with this flag **must** also be created with the `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` flag.
- `VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT` specifies that the buffer's address **can** be saved and reused on a subsequent run (e.g. for trace capture and replay), see [VkBufferOpaqueCaptureAddressCreateInfo](#) for more detail.
- `VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR` specifies that the buffer **can** be used in [video coding operations](#) without having to specify at buffer creation time the set of video profiles the buffer will be used with.

See [Sparse Resource Features](#) and [Physical Device Features](#) for details of the sparse memory features supported on a device.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkBufferCreateFlags;
```

`VkBufferCreateFlags` is a bitmask type for setting a mask of zero or more `VkBufferCreateFlagBits`.

To define a set of external memory handle types that **may** be used as backing store for a buffer, add a `VkExternalMemoryBufferCreateInfo` structure to the `pNext` chain of the `VkBufferCreateInfo` structure. The `VkExternalMemoryBufferCreateInfo` structure is defined as:

```
typedef struct VkExternalMemoryBufferCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkExternalMemoryHandleTypeFlags handleTypes;
} VkExternalMemoryBufferCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_external_memory
typedef VkExternalMemoryBufferCreateInfo VkExternalMemoryBufferCreateInfoKHR;
```

NOTE A `VkExternalMemoryBufferCreateInfo` structure with a non-zero `handleTypes` field must be included in the creation parameters for a buffer that will be bound to memory that is either exported or imported.

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleTypes` is zero or a bitmask of `VkExternalMemoryHandleTypeFlagBits` specifying one or more external memory handle types.

Valid Usage (Implicit)

- VUID-VkExternalMemoryBufferCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_BUFFER_CREATE_INFO`
- VUID-VkExternalMemoryBufferCreateInfo-handleTypes-parameter
`handleTypes` **must** be a valid combination of `VkExternalMemoryHandleTypeFlagBits` values

To request a specific device address for a buffer, add a `VkBufferOpaqueCaptureAddressCreateInfo` structure to the `pNext` chain of the `VkBufferCreateInfo` structure. The `VkBufferOpaqueCaptureAddressCreateInfo` structure is defined as:

```
typedef struct VkBufferOpaqueCaptureAddressCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint64_t           opaqueCaptureAddress;
} VkBufferOpaqueCaptureAddressCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_buffer_device_address
typedef VkBufferOpaqueCaptureAddressCreateInfo
VkBufferOpaqueCaptureAddressCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `opaqueCaptureAddress` is the opaque capture address requested for the buffer.

If `opaqueCaptureAddress` is zero, no specific address is requested.

If `opaqueCaptureAddress` is not zero, then it **should** be an address retrieved from `vkGetBufferOpaqueCaptureAddressKHR` for an identically created buffer on the same implementation.

If this structure is not present, it is as if `opaqueCaptureAddress` is zero.

Apps **should** avoid creating buffers with app-provided addresses and implementation-provided addresses in the same process, to reduce the likelihood of `VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS` errors.

NOTE

The expected usage for this is that a trace capture/replay tool will add the `VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT` flag to all buffers that use `VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT`, and during capture will save the queried opaque device addresses in the trace. During replay, the buffers will be created specifying the original address so any address values stored in the trace data will remain valid.

Implementations are expected to separate such buffers in the GPU address space so normal allocations will avoid using these addresses. Apps/tools should avoid mixing app-provided and implementation-provided addresses for buffers created with `VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT`, to avoid address space allocation conflicts.

Valid Usage (Implicit)

- VUID-VkBufferOpaqueCaptureAddressCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BUFFER_OPAQUE_CAPTURE_ADDRESS_CREATE_INFO`

To destroy a buffer, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyBuffer(
    VkDevice          device,
    VkBuffer           buffer,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the buffer.
- **buffer** is the buffer to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyBuffer-buffer-00922
All submitted commands that refer to **buffer**, either directly or via a **VkBufferView**, **must** have completed execution
- VUID-vkDestroyBuffer-buffer-00923
If **VkAllocationCallbacks** were provided when **buffer** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyBuffer-buffer-00924
If no **VkAllocationCallbacks** were provided when **buffer** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyBuffer-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroyBuffer-buffer-parameter
If **buffer** is not **VK_NULL_HANDLE**, **buffer** **must** be a valid **VkBuffer** handle
- VUID-vkDestroyBuffer-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroyBuffer-buffer-parent
If **buffer** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **buffer** **must** be externally synchronized

12.2. Buffer Views

A *buffer view* represents a contiguous range of a buffer and a specific format to be used to interpret the data. Buffer views are used to enable shaders to access buffer contents using [image operations](#). In order to create a valid buffer view, the buffer **must** have been created with at least one of the following usage flags:

- `VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT`
- `VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT`

Buffer views are represented by `VkBufferView` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkBufferView)
```

To create a buffer view, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateBufferView(
    VkDevice                                device,
    const VkBufferViewCreateInfo*           pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkBufferView*                           pView);
```

- `device` is the logical device that creates the buffer view.
- `pCreateInfo` is a pointer to a `VkBufferViewCreateInfo` structure containing parameters to be used to create the buffer view.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pView` is a pointer to a `VkBufferView` handle in which the resulting buffer view object is returned.

Valid Usage

- VUID-vkCreateBufferView-device-09665
`device` **must** support at least one queue family with one of the `VK_QUEUE_COMPUTE_BIT` or `VK_QUEUE_GRAPHICS_BIT` capabilities

Valid Usage (Implicit)

- VUID-vkCreateBufferView-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateBufferView-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkBufferViewCreateInfo` structure

- VUID-vkCreateBufferView-pAllocator-parameter

If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

- VUID-vkCreateBufferView-pView-parameter

`pView` **must** be a valid pointer to a `VkBufferView` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkBufferViewCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferViewCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkBufferViewCreateFlags flags;
    VkBuffer            buffer;
    VkFormat            format;
    VkDeviceSize        offset;
    VkDeviceSize        range;
} VkBufferViewCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `buffer` is a `VkBuffer` on which the view will be created.
- `format` is a `VkFormat` describing the format of the data elements in the buffer.
- `offset` is an offset in bytes from the base address of the buffer. Accesses to the buffer view from shaders use addressing that is relative to this starting offset.
- `range` is a size in bytes of the buffer view. If `range` is equal to `VK_WHOLE_SIZE`, the range from `offset` to the end of the buffer is used. If `VK_WHOLE_SIZE` is used and the remaining size of the buffer is not a multiple of the `texel block size` of `format`, the nearest smaller multiple is used.

The buffer view has a *buffer view usage* identifying which descriptor types can be created from it. This usage **can** be defined by including the `VkBufferUsageFlags2CreateInfoKHR` structure in the `pNext` chain, and specifying the `usage` value there. If this structure is not included, it is equal to the `VkBufferCreateInfo::usage` value used to create `buffer`.

Valid Usage

- VUID-VkBufferViewCreateInfo-offset-00925
offset **must** be less than the size of **buffer**
- VUID-VkBufferViewCreateInfo-range-00928
If **range** is not equal to **VK_WHOLE_SIZE**, **range** **must** be greater than 0
- VUID-VkBufferViewCreateInfo-range-00929
If **range** is not equal to **VK_WHOLE_SIZE**, **range** **must** be an integer multiple of the texel block size of **format**
- VUID-VkBufferViewCreateInfo-range-00930
If **range** is not equal to **VK_WHOLE_SIZE**, the number of texel buffer elements given by $(\lceil \text{range} / (\text{texel block size}) \rceil \times (\text{texels per block}))$ where texel block size and texels per block are as defined in the [Compatible Formats](#) table for **format**, **must** be less than or equal to **VkPhysicalDeviceLimits::maxTexelBufferElements**
- VUID-VkBufferViewCreateInfo-offset-00931
If **range** is not equal to **VK_WHOLE_SIZE**, the sum of **offset** and **range** **must** be less than or equal to the size of **buffer**
- VUID-VkBufferViewCreateInfo-range-04059
If **range** is equal to **VK_WHOLE_SIZE**, the number of texel buffer elements given by $(\lceil (\text{size} - \text{offset}) / (\text{texel block size}) \rceil \times (\text{texels per block}))$ where size is the size of **buffer**, and texel block size and texels per block are as defined in the [Compatible Formats](#) table for **format**, **must** be less than or equal to **VkPhysicalDeviceLimits::maxTexelBufferElements**
- VUID-VkBufferViewCreateInfo-buffer-00932
buffer **must** have been created with a **usage** value containing at least one of **VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT** or **VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT**
- VUID-VkBufferViewCreateInfo-format-08778
If the **buffer view usage** contains **VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT**, then **format features** of **format** **must** contain **VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT**
- VUID-VkBufferViewCreateInfo-format-08779
If the **buffer view usage** contains **VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT**, then **format features** of **format** **must** contain **VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT**
- VUID-VkBufferViewCreateInfo-buffer-00935
If **buffer** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-VkBufferViewCreateInfo-offset-02749
offset **must** be a multiple of **VkPhysicalDeviceLimits::minTexelBufferOffsetAlignment**
- VUID-VkBufferViewCreateInfo-pNext-08780
If the **pNext** chain includes a **VkBufferUsageFlags2CreateInfoKHR**, its **usage** **must** not contain any other bit than **VK_BUFFER_USAGE_2_UNIFORM_TEXEL_BUFFER_BIT_KHR** or **VK_BUFFER_USAGE_2_STORAGE_TEXEL_BUFFER_BIT_KHR**
- VUID-VkBufferViewCreateInfo-pNext-08781
If the **pNext** chain includes a **VkBufferUsageFlags2CreateInfoKHR**, its **usage** **must** be a

subset of the [VkBufferCreateInfo::usage](#) specified or [VkBufferUsageFlags2CreateInfoKHR::usage](#) from [VkBufferCreateInfo::pNext](#) when creating [buffer](#)

Valid Usage (Implicit)

- VUID-VkBufferViewCreateInfo-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_BUFFER_VIEW_CREATE_INFO](#)
- VUID-VkBufferViewCreateInfo-pNext-pNext
[pNext](#) **must** be [NULL](#) or a pointer to a valid instance of [VkBufferUsageFlags2CreateInfoKHR](#)
- VUID-VkBufferViewCreateInfo-sType-unique
The [sType](#) value of each struct in the [pNext](#) chain **must** be unique
- VUID-VkBufferViewCreateInfo-flags-zerobitmask
[flags](#) **must** be 0
- VUID-VkBufferViewCreateInfo-buffer-parameter
[buffer](#) **must** be a valid [VkBuffer](#) handle
- VUID-VkBufferViewCreateInfo-format-parameter
[format](#) **must** be a valid [VkFormat](#) value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkBufferViewCreateFlags;
```

[VkBufferViewCreateFlags](#) is a bitmask type for setting a mask, but is currently reserved for future use.

To destroy a buffer view, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyBufferView(
    VkDevice          device,
    VkBufferView      bufferView,
    const VkAllocationCallbacks* pAllocator);
```

- [device](#) is the logical device that destroys the buffer view.
- [bufferView](#) is the buffer view to destroy.
- [pAllocator](#) controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyBufferView-bufferView-00936
All submitted commands that refer to [bufferView](#) **must** have completed execution
- VUID-vkDestroyBufferView-bufferView-00937

If `VkAllocationCallbacks` were provided when `bufferView` was created, a compatible set of callbacks **must** be provided here

- VUID-vkDestroyBufferView-bufferView-00938

If no `VkAllocationCallbacks` were provided when `bufferView` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyBufferView-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyBufferView-bufferView-parameter
If `bufferView` is not `VK_NULL_HANDLE`, `bufferView` **must** be a valid `VkBufferView` handle
- VUID-vkDestroyBufferView-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyBufferView-bufferView-parent
If `bufferView` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `bufferView` **must** be externally synchronized

12.2.1. Buffer View Format Features

Valid uses of a `VkBufferView` **may** depend on the buffer view's *format features*, defined below. Such constraints are documented in the affected valid usage statement.

- If Vulkan 1.3 is supported or the `VK_KHR_format_feature_flags2` extension is supported, then the buffer view's set of *format features* is the value of `VkFormatProperties3::bufferFeatures` found by calling `vkGetPhysicalDeviceFormatProperties2KHR` on the same `format` as `VkBufferViewCreateInfo::format`.

12.3. Images

Images represent multidimensional - up to 3 - arrays of data which **can** be used for various purposes (e.g. attachments, textures), by binding them to a graphics or compute pipeline via descriptor sets, or by directly specifying them as parameters to certain commands.

Images are represented by `VkImage` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkImage)
```

To create images, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateImage(
    VkDevice                                device,
    const VkImageCreateInfo*                pCreateInfo,
    const VkAllocationCallbacks*            pAllocator,
    VkImage*                                 pImage);
```

- **device** is the logical device that creates the image.
- **pCreateInfo** is a pointer to a [VkImageCreateInfo](#) structure containing parameters to be used to create the image.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pImage** is a pointer to a [VkImage](#) handle in which the resulting image object is returned.

Valid Usage

- VUID-vkCreateImage-device-09666
device **must** support at least one queue family with one of the [VK_QUEUE_VIDEO_ENCODE_BIT_KHR](#), [VK_QUEUE_VIDEO_DECODE_BIT_KHR](#), [VK_QUEUE_SPARSE_BINDING_BIT](#), [VK_QUEUE_TRANSFER_BIT](#), [VK_QUEUE_COMPUTE_BIT](#), or [VK_QUEUE_GRAPHICS_BIT](#) capabilities
- VUID-vkCreateImage-flags-00939
If the **flags** member of **pCreateInfo** includes [VK_IMAGE_CREATE_SPARSE_BINDING_BIT](#), creating this **VkImage** **must** not cause the total required sparse memory for all currently valid sparse resources on the device to exceed [VkPhysicalDeviceLimits::sparseAddressSpaceSize](#)

Valid Usage (Implicit)

- VUID-vkCreateImage-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCreateImage-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkImageCreateInfo](#) structure
- VUID-vkCreateImage-pAllocator-parameter
If **pAllocator** is not [NULL](#), **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateImage-pImage-parameter
pImage **must** be a valid pointer to a [VkImage](#) handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR`

The `VkImageCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkImageCreateFlags    flags;
    VkImageType           imageType;
    VkFormat              format;
    VkExtent3D            extent;
    uint32_t              mipLevels;
    uint32_t              arrayLayers;
    VkSampleCountFlagBits samples;
    VkImageTiling          tiling;
    VkImageUsageFlags      usage;
    VkSharingMode          sharingMode;
    uint32_t              queueFamilyIndexCount;
    const uint32_t*        pQueueFamilyIndices;
    VkImageLayout          initialLayout;
} VkImageCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkImageCreateFlagBits` describing additional parameters of the image.
- `imageType` is a `VkImageType` value specifying the basic dimensionality of the image. Layers in array textures do not count as a dimension for the purposes of the image type.
- `format` is a `VkFormat` describing the format and type of the texel blocks that will be contained in the image.
- `extent` is a `VkExtent3D` describing the number of data elements in each dimension of the base level.
- `mipLevels` describes the number of levels of detail available for minified sampling of the image.
- `arrayLayers` is the number of layers in the image.
- `samples` is a `VkSampleCountFlagBits` value specifying the number of `samples per texel`.

- `tiling` is a `VkImageTiling` value specifying the tiling arrangement of the texel blocks in memory.
- `usage` is a bitmask of `VkImageUsageFlagBits` describing the intended usage of the image.
- `sharingMode` is a `VkSharingMode` value specifying the sharing mode of the image when it will be accessed by multiple queue families.
- `queueFamilyIndexCount` is the number of entries in the `pQueueFamilyIndices` array.
- `pQueueFamilyIndices` is a pointer to an array of queue families that will access this image. It is ignored if `sharingMode` is not `VK_SHARING_MODE_CONCURRENT`.
- `initialLayout` is a `VkImageLayout` value specifying the initial `VkImageLayout` of all image subresources of the image. See [Image Layouts](#).

Images created with `tiling` equal to `VK_IMAGE_TILING_LINEAR` have further restrictions on their limits and capabilities compared to images created with `tiling` equal to `VK_IMAGE_TILING_OPTIMAL`. Creation of images with tiling `VK_IMAGE_TILING_LINEAR` **may** not be supported unless other parameters meet all of the constraints:

- `imageType` is `VK_IMAGE_TYPE_2D`
- `format` is not a depth/stencil format
- `mipLevels` is 1
- `arrayLayers` is 1
- `samples` is `VK_SAMPLE_COUNT_1_BIT`
- `usage` only includes `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` and/or `VK_IMAGE_USAGE_TRANSFER_DST_BIT`

Images created with one of the [formats that require a sampler Y'C_BC_R conversion](#), have further restrictions on their limits and capabilities compared to images created with other formats. Creation of images with a format requiring [Y'C_BC_R conversion](#) **may** not be supported unless other parameters meet all of the constraints:

- `imageType` is `VK_IMAGE_TYPE_2D`
- `mipLevels` is 1
- `arrayLayers` is 1, unless the `ycbcrImageArrays` feature is enabled, or otherwise indicated by `VkImageFormatProperties::maxArrayLayers`, as returned by `vkGetPhysicalDeviceImageFormatProperties`
- `samples` is `VK_SAMPLE_COUNT_1_BIT`

Implementations **may** support additional limits and capabilities beyond those listed above.

To determine the set of valid `usage` bits for a given format, call `vkGetPhysicalDeviceFormatProperties`.

If the size of the resultant image would exceed `maxResourceSize`, then `vkCreateImage` **must** fail and return `VK_ERROR_OUT_OF_DEVICE_MEMORY`. This failure **may** occur even when all image creation parameters satisfy their valid usage requirements.

If the implementation reports `VK_TRUE` in `VkPhysicalDeviceHostImageCopyPropertiesEXT::identicalMemoryTypeRequirements`, usage of `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` **must** not affect

the memory type requirements of the image as described in [Sparse Resource Memory Requirements](#) and [Resource Memory Association](#).

NOTE

For images created without `VK_IMAGE_CREATE_EXTENDED_USAGE_BIT` a `usage` bit is valid if it is supported for the format the image is created with.

For images created with `VK_IMAGE_CREATE_EXTENDED_USAGE_BIT` a `usage` bit is valid if it is supported for at least one of the formats a `VkImageView` created from the image **can** have (see [Image Views](#) for more detail).

Image Creation Limits

Valid values for some image creation parameters are limited by a numerical upper bound or by inclusion in a bitset. For example, `VkImageCreateInfo::arrayLayers` is limited by `imageCreateMaxArrayLayers`, defined below; and `VkImageCreateInfo::samples` is limited by `imageCreateSampleCounts`, also defined below.

Several limiting values are defined below, as well as assisting values from which the limiting values are derived. The limiting values are referenced by the relevant valid usage statements of `VkImageCreateInfo`.

- Let `uint64_t imageCreateDrmFormatModifiers[]` be the set of [Linux DRM format modifiers](#) that the resultant image **may** have.
 - If `tiling` is not `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then `imageCreateDrmFormatModifiers` is empty.
 - If `VkImageCreateInfo::pNext` contains `VkImageDrmFormatModifierExplicitCreateInfoEXT`, then `imageCreateDrmFormatModifiers` contains exactly one modifier, `VkImageDrmFormatModifierExplicitCreateInfoEXT::drmFormatModifier`.
 - If `VkImageCreateInfo::pNext` contains `VkImageDrmFormatModifierListCreateInfoEXT`, then `imageCreateDrmFormatModifiers` contains the entire array `VkImageDrmFormatModifierListCreateInfoEXT::pDrmFormatModifiers`.
- Let `VkBool32 imageCreateMaybeLinear` indicate if the resultant image may be [linear](#).
 - If `tiling` is `VK_IMAGE_TILING_LINEAR`, then `imageCreateMaybeLinear` is `VK_TRUE`.
 - If `tiling` is `VK_IMAGE_TILING_OPTIMAL`, then `imageCreateMaybeLinear` is `VK_FALSE`.
 - If `tiling` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then `imageCreateMaybeLinear` is `VK_TRUE` if and only if `imageCreateDrmFormatModifiers` contains `DRM_FORMAT_MOD_LINEAR`.
- Let `VkFormatFeatureFlags imageCreateFormatFeatures` be the set of valid *format features* available during image creation.
 - If `tiling` is `VK_IMAGE_TILING_LINEAR`, then `imageCreateFormatFeatures` is the value of `VkFormatProperties::linearTilingFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` with parameter `format` equal to `VkImageCreateInfo::format`.
 - If `tiling` is `VK_IMAGE_TILING_OPTIMAL`, then `imageCreateFormatFeatures` is the value of

`VkFormatProperties::optimalTilingFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` with parameter `format` equal to `VkImageCreateInfo::format`.

- If `tiling` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then the value of `imageCreateFormatFeatures` is found by calling `vkGetPhysicalDeviceFormatProperties2KHR` with `VkImageFormatProperties::format` equal to `VkImageCreateInfo::format` and with `VkDrmFormatModifierPropertiesListEXT` chained into `VkFormatProperties2`; by collecting all members of the returned array `VkDrmFormatModifierPropertiesListEXT::pDrmFormatModifierProperties` whose `drmFormatModifier` belongs to `imageCreateDrmFormatModifiers`; and by taking the bitwise intersection, over the collected array members, of `drmFormatModifierTilingFeatures`. (The resultant `imageCreateFormatFeatures` may be empty).
- Let `VkImageFormatProperties2 imageCreateImageFormatPropertiesList[]` be the list of structures obtained by calling `vkGetPhysicalDeviceImageFormatProperties2KHR`, possibly multiple times, as follows:
 - The parameters `VkPhysicalDeviceImageFormatInfo2::format`, `imageType`, `tiling`, `usage`, and `flags` **must** be equal to those in `VkImageCreateInfo`.
 - If `VkImageCreateInfo::pNext` contains a `VkExternalMemoryImageCreateInfo` structure whose `handleTypes` is not `0`, then `VkPhysicalDeviceImageFormatInfo2::pNext` **must** contain a `VkPhysicalDeviceExternalImageFormatInfo` structure whose `handleType` is not `0`; and `vkGetPhysicalDeviceImageFormatProperties2KHR` **must** be called for each handle type in `VkExternalMemoryImageCreateInfo::handleTypes`, successively setting `VkPhysicalDeviceExternalImageFormatInfo::handleType` on each call.
 - If `VkImageCreateInfo::pNext` contains no `VkExternalMemoryImageCreateInfo` structure, or contains a structure whose `handleTypes` is `0`, then `VkPhysicalDeviceImageFormatInfo2::pNext` **must** either contain no `VkPhysicalDeviceExternalImageFormatInfo` structure, or contain a structure whose `handleType` is `0`.
 - If `VkImageCreateInfo::pNext` contains a `VkVideoProfileListInfoKHR` structure then `VkPhysicalDeviceImageFormatInfo2::pNext` **must** also contain the same `VkVideoProfileListInfoKHR` structure on each call.
 - If `tiling` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then:
 - `VkPhysicalDeviceImageFormatInfo2::pNext` **must** contain a `VkPhysicalDeviceImageDrmFormatModifierInfoEXT` structure where `sharingMode` is equal to `VkImageCreateInfo::sharingMode`;
 - if `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, then `queueFamilyIndexCount` and `pQueueFamilyIndices` **must** be equal to those in `VkImageCreateInfo`;
 - if `flags` contains `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT`, then the `VkImageFormatListCreateInfo` structure included in the `pNext` chain of `VkPhysicalDeviceImageFormatInfo2` **must** be equivalent to the one included in the `pNext` chain of `VkImageCreateInfo`;
 - `vkGetPhysicalDeviceImageFormatProperties2KHR` **must** be called for each

modifier in `imageCreateDrmFormatModifiers`, successively setting `VkPhysicalDeviceImageDrmFormatModifierInfoEXT::drmFormatModifier` on each call.

- If `tiling` is not `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then `VkPhysicalDeviceImageFormatInfo2::pNext` must contain no `VkPhysicalDeviceImageDrmFormatModifierInfoEXT` structure.
- If any call to `vkGetPhysicalDeviceImageFormatProperties2KHR` returns an error, then `imageCreateImageFormatPropertiesList` is defined to be the empty list.
- Let `uint32_t imageCreateMaxMipLevels` be the minimum value of `VkImageFormatProperties::maxMipLevels` in `imageCreateImageFormatPropertiesList`. The value is undefined if `imageCreateImageFormatPropertiesList` is empty.
- Let `uint32_t imageCreateMaxArrayLayers` be the minimum value of `VkImageFormatProperties::maxArrayLayers` in `imageCreateImageFormatPropertiesList`. The value is undefined if `imageCreateImageFormatPropertiesList` is empty.
- Let `VkExtent3D imageCreateMaxExtent` be the component-wise minimum over all `VkImageFormatProperties::maxExtent` values in `imageCreateImageFormatPropertiesList`. The value is undefined if `imageCreateImageFormatPropertiesList` is empty.
- Let `VkSampleCountFlags imageCreateSampleCounts` be the intersection of each `VkImageFormatProperties::sampleCounts` in `imageCreateImageFormatPropertiesList`. The value is undefined if `imageCreateImageFormatPropertiesList` is empty.
- Let `VkVideoFormatPropertiesKHR videoFormatProperties[]` be defined as follows.
 - If `VkImageCreateInfo::pNext` contains a `VkVideoProfileListInfoKHR` structure, then `videoFormatProperties` is the list of structures obtained by calling `vkGetPhysicalDeviceVideoFormatPropertiesKHR` with `VkPhysicalDeviceVideoFormatInfoKHR::imageUsage` equal to the `usage` member of `VkImageCreateInfo` and `VkPhysicalDeviceVideoFormatInfoKHR::pNext` containing the same `VkVideoProfileListInfoKHR` structure chained to `VkImageCreateInfo`.
 - If `VkImageCreateInfo::pNext` contains no `VkVideoProfileListInfoKHR` structure, then `videoFormatProperties` is an empty list.
- Let `VkBool32 supportedVideoFormat` indicate if the image parameters are supported by the specified video profiles.
 - `supportedVideoFormat` is `VK_TRUE` if there exists an element in the `videoFormatProperties` list for which all of the following conditions are true:
 - `VkImageCreateInfo::format` equals `VkVideoFormatPropertiesKHR::format`.
 - `VkImageCreateInfo::flags` only contains `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR` and/or bits also set in `VkVideoFormatPropertiesKHR::imageCreateFlags`.
 - `VkImageCreateInfo::imageType` equals `VkVideoFormatPropertiesKHR::imageType`.
 - `VkImageCreateInfo::tiling` equals `VkVideoFormatPropertiesKHR::imageTiling`.
 - `VkImageCreateInfo::usage` only contains bits also set in `VkVideoFormatPropertiesKHR::imageUsageFlags`, or `VkImageCreateInfo::flags`

includes `VK_IMAGE_CREATE_EXTENDED_USAGE_BIT`.

- Otherwise `supportedVideoFormat` is `VK_FALSE`.

Valid Usage

- VUID-VkImageCreateInfo-imageCreateMaxMipLevels-02251

Each of the following values (as described in [Image Creation Limits](#)) **must** not be undefined : `imageCreateMaxMipLevels`, `imageCreateMaxArrayLayers`, `imageCreateMaxExtent`, and `imageCreateSampleCounts`

- VUID-VkImageCreateInfo-sharingMode-00941

If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, `pQueueFamilyIndices` **must** be a valid pointer to an array of `queueFamilyIndexCount` `uint32_t` values

- VUID-VkImageCreateInfo-sharingMode-00942

If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, `queueFamilyIndexCount` **must** be greater than 1

- VUID-VkImageCreateInfo-sharingMode-01420

If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, each element of `pQueueFamilyIndices` **must** be unique and **must** be less than `pQueueFamilyPropertyCount` returned by either [vkGetPhysicalDeviceQueueFamilyProperties](#) or [vkGetPhysicalDeviceQueueFamilyProperties2KHR](#) for the `physicalDevice` that was used to create `device`

- VUID-VkImageCreateInfo-format-00943

`format` **must** not be `VK_FORMAT_UNDEFINED`

- VUID-VkImageCreateInfo-extent-00944

`extent.width` **must** be greater than 0

- VUID-VkImageCreateInfo-extent-00945

`extent.height` **must** be greater than 0

- VUID-VkImageCreateInfo-extent-00946

`extent.depth` **must** be greater than 0

- VUID-VkImageCreateInfo-mipLevels-00947

`mipLevels` **must** be greater than 0

- VUID-VkImageCreateInfo-arrayLayers-00948

`arrayLayers` **must** be greater than 0

- VUID-VkImageCreateInfo-flags-00949

If `flags` contains `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`, `imageType` **must** be `VK_IMAGE_TYPE_2D`

- VUID-VkImageCreateInfo-flags-08865

If `flags` contains `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`, `extent.width` and `extent.height` **must** be equal

- VUID-VkImageCreateInfo-flags-08866

If `flags` contains `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`, `arrayLayers` **must** be greater than or equal to 6

- VUID-VkImageCreateInfo-flags-00950
If `flags` contains `VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT`, `imageType` **must** be `VK_IMAGE_TYPE_3D`
- VUID-VkImageCreateInfo-flags-09403
If `flags` contains `VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT`, `flags` **must** not include `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT`, `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`, or `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`
- VUID-VkImageCreateInfo-extent-02252
`extent.width` **must** be less than or equal to `imageCreateMaxExtent.width` (as defined in [Image Creation Limits](#))
- VUID-VkImageCreateInfo-extent-02253
`extent.height` **must** be less than or equal to `imageCreateMaxExtent.height` (as defined in [Image Creation Limits](#))
- VUID-VkImageCreateInfo-extent-02254
`extent.depth` **must** be less than or equal to `imageCreateMaxExtent.depth` (as defined in [Image Creation Limits](#))
- VUID-VkImageCreateInfo-imageType-00956
If `imageType` is `VK_IMAGE_TYPE_1D`, both `extent.height` and `extent.depth` **must** be 1
- VUID-VkImageCreateInfo-imageType-00957
If `imageType` is `VK_IMAGE_TYPE_2D`, `extent.depth` **must** be 1
- VUID-VkImageCreateInfo-mipLevels-00958
`mipLevels` **must** be less than or equal to the number of levels in the complete mipmap chain based on `extent.width`, `extent.height`, and `extent.depth`
- VUID-VkImageCreateInfo-mipLevels-02255
`mipLevels` **must** be less than or equal to `imageCreateMaxMipLevels` (as defined in [Image Creation Limits](#))
- VUID-VkImageCreateInfo-arrayLayers-02256
`arrayLayers` **must** be less than or equal to `imageCreateMaxArrayLayers` (as defined in [Image Creation Limits](#))
- VUID-VkImageCreateInfo-imageType-00961
If `imageType` is `VK_IMAGE_TYPE_3D`, `arrayLayers` **must** be 1
- VUID-VkImageCreateInfo-samples-02257
If `samples` is not `VK_SAMPLE_COUNT_1_BIT`, then `imageType` **must** be `VK_IMAGE_TYPE_2D`, `flags` **must** not contain `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`, `mipLevels` **must** be equal to 1, and `imageCreateMaybeLinear` (as defined in [Image Creation Limits](#)) **must** be `VK_FALSE`,
- VUID-VkImageCreateInfo-usage-00963
If `usage` includes `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT`, then bits other than `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, and `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` **must** not be set
- VUID-VkImageCreateInfo-usage-00964
If `usage` includes `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT`, or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`, `extent.width` **must** be less than or equal to

VkPhysicalDeviceLimits::maxFramebufferWidth

- VUID-VkImageCreateInfo-usage-00965

If `usage` includes `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT`, or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`, `extent.height` **must** be less than or equal to `VkPhysicalDeviceLimits::maxFramebufferHeight`

- VUID-VkImageCreateInfo-usage-00966

If `usage` includes `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT`, `usage` **must** also contain at least one of `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`, `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`

- VUID-VkImageCreateInfo-samples-02258

`samples` **must** be a valid `VkSampleCountFlagBits` value that is set in `imageCreateSampleCounts` (as defined in [Image Creation Limits](#))

- VUID-VkImageCreateInfo-usage-00968

If the `shaderStorageImageMultisample` feature is not enabled, and `usage` contains `VK_IMAGE_USAGE_STORAGE_BIT`, `samples` **must** be `VK_SAMPLE_COUNT_1_BIT`

- VUID-VkImageCreateInfo-flags-00969

If the `sparseBinding` feature is not enabled, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`

- VUID-VkImageCreateInfo-flags-01924

If the `sparseResidencyAliased` feature is not enabled, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT`

- VUID-VkImageCreateInfo-tiling-04121

If `tiling` is `VK_IMAGE_TILING_LINEAR`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`

- VUID-VkImageCreateInfo-imageType-00970

If `imageType` is `VK_IMAGE_TYPE_1D`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`

- VUID-VkImageCreateInfo-imageType-00971

If the `sparseResidencyImage2D` feature is not enabled, and `imageType` is `VK_IMAGE_TYPE_2D`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`

- VUID-VkImageCreateInfo-imageType-00972

If the `sparseResidencyImage3D` feature is not enabled, and `imageType` is `VK_IMAGE_TYPE_3D`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`

- VUID-VkImageCreateInfo-imageType-00973

If the `sparseResidency2Samples` feature is not enabled, `imageType` is `VK_IMAGE_TYPE_2D`, and `samples` is `VK_SAMPLE_COUNT_2_BIT`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`

- VUID-VkImageCreateInfo-imageType-00974

If the `sparseResidency4Samples` feature is not enabled, `imageType` is `VK_IMAGE_TYPE_2D`, and `samples` is `VK_SAMPLE_COUNT_4_BIT`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`

- VUID-VkImageCreateInfo-imageType-00975

If the `sparseResidency8Samples` feature is not enabled, `imageType` is `VK_IMAGE_TYPE_2D`, and `samples` is `VK_SAMPLE_COUNT_8_BIT`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`

- VUID-VkImageCreateInfo-imageType-00976

If the `sparseResidency16Samples` feature is not enabled, `imageType` is `VK_IMAGE_TYPE_2D`, and `samples` is `VK_SAMPLE_COUNT_16_BIT`, `flags` **must** not contain `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`

- VUID-VkImageCreateInfo-flags-00987

If `flags` contains `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` or `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT`, it **must** also contain `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`

- VUID-VkImageCreateInfo-None-01925

If any of the bits `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`, `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`, or `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` are set, `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT` **must** not also be set

- VUID-VkImageCreateInfo-pNext-00990

If the `pNext` chain includes a `VkExternalMemoryImageCreateInfo` structure, its `handleTypes` member **must** only contain bits that are also in `VkExternalImageFormatProperties::externalMemoryProperties.compatibleHandleTypes`, as returned by `vkGetPhysicalDeviceImageFormatProperties2KHR` with `format`, `imageType`, `tiling`, `usage`, and `flags` equal to those in this structure, and with a `VkPhysicalDeviceExternalImageFormatInfo` structure included in the `pNext` chain, with a `handleType` equal to any one of the handle types specified in `VkExternalMemoryImageCreateInfo::handleTypes`

- VUID-VkImageCreateInfo-physicalDeviceCount-01421

If the logical device was created with `VkDeviceGroupDeviceCreateInfo::physicalDeviceCount` equal to 1, `flags` **must** not contain `VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT`

- VUID-VkImageCreateInfo-flags-02259

If `flags` contains `VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT`, then `mipLevels` **must** be one, `arrayLayers` **must** be one, `imageType` **must** be `VK_IMAGE_TYPE_2D`. and `imageCreateMayBeLinear` (as defined in [Image Creation Limits](#)) **must** be `VK_FALSE`

- VUID-VkImageCreateInfo-flags-01572

If `flags` contains `VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT`, then `format` **must** be a [compressed image format](#)

- VUID-VkImageCreateInfo-flags-01573

If `flags` contains `VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT`, then `flags` **must** also contain `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT`

- VUID-VkImageCreateInfo-initialLayout-00993

`initialLayout` **must** be `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`

- VUID-VkImageCreateInfo-pNext-01443

If the `pNext` chain includes a `VkExternalMemoryImageCreateInfo` or `VkExternalMemoryImageCreateInfoNV` structure whose `handleTypes` member is not 0, `initialLayout` **must** be `VK_IMAGE_LAYOUT_UNDEFINED`

- VUID-VkImageCreateInfo-format-06410
If the image **format** is one of the **formats that require a sampler Y'C_BC_R conversion**, **mipLevels must** be 1
- VUID-VkImageCreateInfo-format-06411
If the image **format** is one of the **formats that require a sampler Y'C_BC_R conversion**, **samples must** be **VK_SAMPLE_COUNT_1_BIT**
- VUID-VkImageCreateInfo-format-06412
If the image **format** is one of the **formats that require a sampler Y'C_BC_R conversion**, **imageType must** be **VK_IMAGE_TYPE_2D**
- VUID-VkImageCreateInfo-imageCreateFormatFeatures-02260
If **format** is a *multi-planar* format, and if **imageCreateFormatFeatures** (as defined in **Image Creation Limits**) does not contain **VK_FORMAT_FEATURE_DISJOINT_BIT**, then **flags must** not contain **VK_IMAGE_CREATE_DISJOINT_BIT**
- VUID-VkImageCreateInfo-format-01577
If **format** is not a *multi-planar* format, and **flags** does not include **VK_IMAGE_CREATE_ALIAS_BIT**, **flags must** not contain **VK_IMAGE_CREATE_DISJOINT_BIT**
- VUID-VkImageCreateInfo-format-04712
If **format** has a **_422** or **_420** suffix, **extent.width must** be a multiple of 2
- VUID-VkImageCreateInfo-format-04713
If **format** has a **_420** suffix, **extent.height must** be a multiple of 2
- VUID-VkImageCreateInfo-tiling-02261
If **tiling** is **VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT**, then the **pNext** chain **must** include exactly one of **VkImageDrmFormatModifierListCreateInfoEXT** or **VkImageDrmFormatModifierExplicitCreateInfoEXT** structures
- VUID-VkImageCreateInfo-pNext-02262
If the **pNext** chain includes a **VkImageDrmFormatModifierListCreateInfoEXT** or **VkImageDrmFormatModifierExplicitCreateInfoEXT** structure, then **tiling must** be **VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT**
- VUID-VkImageCreateInfo-tiling-02353
If **tiling** is **VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT** and **flags** contains **VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT**, then the **pNext** chain **must** include a **VkImageFormatListCreateInfo** structure with non-zero **viewFormatCount**
- VUID-VkImageCreateInfo-flags-01533
If **flags** contains **VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT** **format must** be a depth or depth/stencil format
- VUID-VkImageCreateInfo-imageType-02082
If **usage** includes **VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR**, **imageType must** be **VK_IMAGE_TYPE_2D**
- VUID-VkImageCreateInfo-samples-02083
If **usage** includes **VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR**, **samples must** be **VK_SAMPLE_COUNT_1_BIT**
- VUID-VkImageCreateInfo-imageView2DOn3DImage-04459
If the **VK_KHR_portability_subset** extension is enabled, and

`VkPhysicalDevicePortabilitySubsetFeaturesKHR::imageView2DOn3DImage` is `VK_FALSE`, `flags` **must** not contain `VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT`

- VUID-VkImageCreateInfo-multisampleArrayImage-04460

If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::multisampleArrayImage` is `VK_FALSE`, and `samples` is not `VK_SAMPLE_COUNT_1_BIT`, then `arrayLayers` **must** be 1

- VUID-VkImageCreateInfo-pNext-06722

If a `VkImageFormatListCreateInfo` structure was included in the `pNext` chain and `format` is not a `multi-planar` format and `VkImageFormatListCreateInfo::viewFormatCount` is not zero, then each format in `VkImageFormatListCreateInfo::pViewFormats` **must** either be compatible with the `format` as described in the `compatibility table` or, if `flags` contains `VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT`, be an uncompressed format that is size-compatible with `format`

- VUID-VkImageCreateInfo-pNext-10062

If a `VkImageFormatListCreateInfo` structure was included in the `pNext` chain and `format` is a `multi-planar` format and `flags` contains `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT` and `VkImageFormatListCreateInfo::viewFormatCount` is not zero, then each format in `VkImageFormatListCreateInfo::pViewFormats` **must** be compatible with the `VkFormat` for the plane of the image format

- VUID-VkImageCreateInfo-flags-04738

If `flags` does not contain `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT` and the `pNext` chain includes a `VkImageFormatListCreateInfo` structure, then `VkImageFormatListCreateInfo::viewFormatCount` **must** be 0 or 1

- VUID-VkImageCreateInfo-usage-04815

If `usage` includes `VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR`, `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`, or `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`, and `flags` does not include `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoProfileListInfoKHR` structure with `profileCount` greater than 0 and `pProfiles` including at least one `VkVideoProfileInfoKHR` structure with a `videoCodecOperation` member specifying a decode operation

- VUID-VkImageCreateInfo-usage-04816

If `usage` includes `VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR`, `VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR`, or `VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR`, and `flags` does not include `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoProfileListInfoKHR` structure with `profileCount` greater than 0 and `pProfiles` including at least one `VkVideoProfileInfoKHR` structure with a `videoCodecOperation` member specifying an encode operation

- VUID-VkImageCreateInfo-flags-08328

If `flags` includes `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`, then `videoMaintenance1` **must** be enabled

- VUID-VkImageCreateInfo-flags-08329

If `flags` includes `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR` and `usage` does not include `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`, then `usage` **must** not include `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`

- VUID-VkImageCreateInfo-flags-08331
If **flags** includes `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`, then **usage** **must** not include `VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR`
- VUID-VkImageCreateInfo-pNext-06811
If the **pNext** chain includes a `VkVideoProfileListInfoKHR` structure with **profileCount** greater than 0, then **supportedVideoFormat** **must** be `VK_TRUE`
- VUID-VkImageCreateInfo-imageCreateFormatFeatures-09048
If **imageCreateFormatFeatures** (as defined in [Image Creation Limits](#)) does not contain `VK_FORMAT_FEATURE_2_HOST_IMAGE_TRANSFER_BIT_EXT`, then **usage** **must** not contain `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT`
- VUID-VkImageCreateInfo-usage-10245
If **usage** includes `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT`, then the **hostImageCopy** feature **must** be enabled

Valid Usage (Implicit)

- VUID-VkImageCreateInfo-sType-sType
sType **must** be `VK_STRUCTURE_TYPE_IMAGE_CREATE_INFO`
- VUID-VkImageCreateInfo-pNext-pNext
Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either `NULL` or a pointer to a valid instance of `VkExternalMemoryImageCreateInfo`, `VkImageDrmFormatModifierExplicitCreateInfoEXT`, `VkImageDrmFormatModifierListCreateInfoEXT`, `VkImageFormatListCreateInfo`, `VkImageSwapchainCreateInfoKHR`, or `VkVideoProfileListInfoKHR`
- VUID-VkImageCreateInfo-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkImageCreateInfo-flags-parameter
flags **must** be a valid combination of `VkImageCreateFlagBits` values
- VUID-VkImageCreateInfo-imageType-parameter
imageType **must** be a valid `VkImageType` value
- VUID-VkImageCreateInfo-format-parameter
format **must** be a valid `VkFormat` value
- VUID-VkImageCreateInfo-samples-parameter
samples **must** be a valid `VkSampleCountFlagBits` value
- VUID-VkImageCreateInfo-tiling-parameter
tiling **must** be a valid `VkImageTiling` value
- VUID-VkImageCreateInfo-usage-parameter
usage **must** be a valid combination of `VkImageUsageFlagBits` values
- VUID-VkImageCreateInfo-usage-requiredbitmask
usage **must** not be 0
- VUID-VkImageCreateInfo-sharingMode-parameter
sharingMode **must** be a valid `VkSharingMode` value

- VUID-VkImageCreateInfo-initialLayout-parameter
initialLayout **must** be a valid [VkImageLayout](#) value

To define a set of external memory handle types that **may** be used as backing store for an image, add a [VkExternalMemoryImageCreateInfo](#) structure to the **pNext** chain of the [VkImageCreateInfo](#) structure. The [VkExternalMemoryImageCreateInfo](#) structure is defined as:

```
typedef struct VkExternalMemoryImageCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkExternalMemoryHandleTypeFlags handleTypes;
} VkExternalMemoryImageCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_external_memory
typedef VkExternalMemoryImageCreateInfo VkExternalMemoryImageCreateInfoKHR;
```

NOTE

A [VkExternalMemoryImageCreateInfo](#) structure with a non-zero **handleTypes** field must be included in the creation parameters for an image that will be bound to memory that is either exported or imported.

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **handleTypes** is zero or a bitmask of [VkExternalMemoryHandleTypeFlagBits](#) specifying one or more external memory handle types.

Valid Usage (Implicit)

- VUID-VkExternalMemoryImageCreateInfo-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_IMAGE_CREATE_INFO**
- VUID-VkExternalMemoryImageCreateInfo-handleTypes-parameter
handleTypes **must** be a valid combination of [VkExternalMemoryHandleTypeFlagBits](#) values

If the **pNext** chain of [VkImageCreateInfo](#) includes a [VkImageSwapchainCreateInfoKHR](#) structure, then that structure includes a swapchain handle indicating that the image will be bound to memory from that swapchain.

The [VkImageSwapchainCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
typedef struct VkImageSwapchainCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkSwapchainKHR     swapchain;
} VkImageSwapchainCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchain` is `VK_NULL_HANDLE` or a handle of a swapchain that the image will be bound to.

Valid Usage

- VUID-VkImageSwapchainCreateInfoKHR-swapchain-00995
If `swapchain` is not `VK_NULL_HANDLE`, the fields of `VkImageCreateInfo` **must** match the [implied image creation parameters](#) of the swapchain

Valid Usage (Implicit)

- VUID-VkImageSwapchainCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_SWAPCHAIN_CREATE_INFO_KHR`
- VUID-VkImageSwapchainCreateInfoKHR-swapchain-parameter
If `swapchain` is not `VK_NULL_HANDLE`, `swapchain` **must** be a valid `VkSwapchainKHR` handle

If the `pNext` chain of `VkImageCreateInfo` includes a `VkImageFormatListCreateInfo` structure, then that structure contains a list of all formats that **can** be used when creating views of this image.

The `VkImageFormatListCreateInfo` structure is defined as:

```
typedef struct VkImageFormatListCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           viewFormatCount;
    const VkFormat*    pViewFormats;
} VkImageFormatListCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_image_format_list
typedef VkImageFormatListCreateInfo VkImageFormatListCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `viewFormatCount` is the number of entries in the `pViewFormats` array.
- `pViewFormats` is a pointer to an array of `VkFormat` values specifying all formats which **can** be used when creating views of this image.

If `viewFormatCount` is zero, `pViewFormats` is ignored and the image is created as if the `VkImageFormatListCreateInfo` structure were not included in the `pNext` chain of `VkImageCreateInfo`.

Valid Usage

- VUID-VkImageFormatListCreateInfo-viewFormatCount-09540
If `viewFormatCount` is not 0, each element of `pViewFormats` **must** not be `VK_FORMAT_UNDEFINED`

Valid Usage (Implicit)

- VUID-VkImageFormatListCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_FORMAT_LIST_CREATE_INFO`
- VUID-VkImageFormatListCreateInfo-pViewFormats-parameter
If `viewFormatCount` is not 0, `pViewFormats` **must** be a valid pointer to an array of `viewFormatCount` valid `VkFormat` values

If the `pNext` chain of `VkImageCreateInfo` includes a `VkImageDrmFormatModifierListCreateInfoEXT` structure, then the image will be created with one of the [Linux DRM format modifiers](#) listed in the structure. The choice of modifier is implementation-dependent.

The `VkImageDrmFormatModifierListCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkImageDrmFormatModifierListCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           drmFormatModifierCount;
    const uint64_t*     pDrmFormatModifiers;
} VkImageDrmFormatModifierListCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `drmFormatModifierCount` is the length of the `pDrmFormatModifiers` array.
- `pDrmFormatModifiers` is a pointer to an array of *Linux DRM format modifiers*.

Valid Usage

- VUID-VkImageDrmFormatModifierListCreateInfoEXT-pDrmFormatModifiers-02263

Each *modifier* in `pDrmFormatModifiers` **must** be compatible with the parameters in `VkImageCreateInfo` and its `pNext` chain, as determined by querying `VkPhysicalDeviceImageFormatInfo2` extended with `VkPhysicalDeviceImageDrmFormatModifierInfoEXT`

Valid Usage (Implicit)

- VUID-VkImageDrmFormatModifierListCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_LIST_CREATE_INFO_EXT`
- VUID-VkImageDrmFormatModifierListCreateInfoEXT-pDrmFormatModifiers-parameter
`pDrmFormatModifiers` **must** be a valid pointer to an array of `drmFormatModifierCount` `uint64_t` values
- VUID-VkImageDrmFormatModifierListCreateInfoEXT-drmFormatModifierCount-arraylength
`drmFormatModifierCount` **must** be greater than 0

If the `pNext` chain of `VkImageCreateInfo` includes a `VkImageDrmFormatModifierExplicitCreateInfoEXT` structure, then the image will be created with the `Linux DRM format modifier` and memory layout defined by the structure.

The `VkImageDrmFormatModifierExplicitCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkImageDrmFormatModifierExplicitCreateInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    uint64_t             drmFormatModifier;
    uint32_t             drmFormatModifierPlaneCount;
    const VkSubresourceLayout* pPlaneLayouts;
} VkImageDrmFormatModifierExplicitCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `drmFormatModifier` is the *Linux DRM format modifier* with which the image will be created.
- `drmFormatModifierPlaneCount` is the number of *memory planes* in the image (as reported by `VkDrmFormatModifierPropertiesEXT`) as well as the length of the `pPlaneLayouts` array.
- `pPlaneLayouts` is a pointer to an array of `VkSubresourceLayout` structures describing the image's *memory planes*.

The i^{th} member of `pPlaneLayouts` describes the layout of the image's i^{th} *memory plane* (that is,

VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT). In each element of `pPlaneLayouts`, the implementation **must** ignore `size`. The implementation calculates the size of each plane, which the application **can** query with `vkGetImageSubresourceLayout`.

When creating an image with `VkImageDrmFormatModifierExplicitCreateInfoEXT`, it is the application's responsibility to satisfy all valid usage requirements. However, the implementation **must** validate that the provided `pPlaneLayouts`, when combined with the provided `drmFormatModifier` and other creation parameters in `VkImageCreateInfo` and its `pNext` chain, produce a valid image. (This validation is necessarily implementation-dependent and outside the scope of Vulkan, and therefore not described by valid usage requirements). If this validation fails, then `vkCreateImage` returns `VK_ERROR_INVALID_DRM_FORMAT_MODIFIER_PLANE_LAYOUT_EXT`.

Valid Usage

- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-drmFormatModifier-02264
`drmFormatModifier` **must** be compatible with the parameters in `VkImageCreateInfo` and its `pNext` chain, as determined by querying `VkPhysicalDeviceImageFormatInfo2` extended with `VkPhysicalDeviceImageDrmFormatModifierInfoEXT`
- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-drmFormatModifierPlaneCount-02265
`drmFormatModifierPlaneCount` **must** be equal to the `VkDrmFormatModifierPropertiesEXT::drmFormatModifierPlaneCount` associated with `VkImageCreateInfo::format` and `drmFormatModifier`, as found by querying `VkDrmFormatModifierPropertiesListEXT`
- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-size-02267
For each element of `pPlaneLayouts`, `size` **must** be 0
- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-arrayPitch-02268
For each element of `pPlaneLayouts`, `arrayPitch` **must** be 0 if `VkImageCreateInfo::arrayLayers` is 1
- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-depthPitch-02269
For each element of `pPlaneLayouts`, `depthPitch` **must** be 0 if `VkImageCreateInfo::extent.depth` is 1

Valid Usage (Implicit)

- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_EXPLICIT_CREATE_INFO_EXT`
- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-pPlaneLayouts-parameter
`pPlaneLayouts` **must** be a valid pointer to an array of `drmFormatModifierPlaneCount` `VkSubresourceLayout` structures
- VUID-VkImageDrmFormatModifierExplicitCreateInfoEXT-drmFormatModifierPlaneCount-arraylength
`drmFormatModifierPlaneCount` **must** be greater than 0

Bits which **can** be set in

- `VkImageViewUsageCreateInfo::usage`
- `VkImageCreateInfo::usage`

specify intended usage of an image, and are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageUsageFlagBits {
    VK_IMAGE_USAGE_TRANSFER_SRC_BIT = 0x00000001,
    VK_IMAGE_USAGE_TRANSFER_DST_BIT = 0x00000002,
    VK_IMAGE_USAGE_SAMPLED_BIT = 0x00000004,
    VK_IMAGE_USAGE_STORAGE_BIT = 0x00000008,
    VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT = 0x00000010,
    VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT = 0x00000020,
    VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT = 0x00000040,
    VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT = 0x00000080,
    // Provided by VK_KHR_video_decode_queue
    VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR = 0x00000400,
    // Provided by VK_KHR_video_decode_queue
    VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR = 0x00000800,
    // Provided by VK_KHR_video_decode_queue
    VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR = 0x00001000,
    // Provided by VK_KHR_fragment_shading_rate
    VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR = 0x00000100,
    // Provided by VK_EXT_host_image_copy
    VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT = 0x00400000,
    // Provided by VK_KHR_video_encode_queue
    VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR = 0x00002000,
    // Provided by VK_KHR_video_encode_queue
    VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR = 0x00004000,
    // Provided by VK_KHR_video_encode_queue
    VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR = 0x00008000,
    // Provided by VK_EXT_attachment_feedback_loop_layout
    VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT = 0x00080000,
} VkImageUsageFlagBits;
```

- `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` specifies that the image **can** be used as the source of a transfer command.
- `VK_IMAGE_USAGE_TRANSFER_DST_BIT` specifies that the image **can** be used as the destination of a transfer command.
- `VK_IMAGE_USAGE_SAMPLED_BIT` specifies that the image **can** be used to create a `VkImageView` suitable for occupying a `VkDescriptorSet` slot either of type `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and be sampled by a shader.
- `VK_IMAGE_USAGE_STORAGE_BIT` specifies that the image **can** be used to create a `VkImageView` suitable for occupying a `VkDescriptorSet` slot of type `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`.
- `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` specifies that the image **can** be used to create a `VkImageView` suitable for use as a color or resolve attachment in a `VkFramebuffer`.
- `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT` specifies that the image **can** be used to create a

`VkImageView` suitable for use as a depth/stencil or depth/stencil resolve attachment in a `VkFramebuffer`.

- `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT` specifies that implementations **may** support using [memory allocations](#) with the `VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT` to back an image with this usage. This bit **can** be set for any image that **can** be used to create a `VkImageView` suitable for use as a color, resolve, depth/stencil, or input attachment.
- `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` specifies that the image **can** be used to create a `VkImageView` suitable for occupying `VkDescriptorSet` slot of type `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`; be read from a shader as an input attachment; and be used as an input attachment in a framebuffer.
- `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR` specifies that the image **can** be used to create a `VkImageView` suitable for use as a [fragment shading rate attachment](#)
- `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR` specifies that the image **can** be used as a [decode output picture](#) in a [video decode operation](#).
- `VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR` is reserved for future use.
- `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR` specifies that the image **can** be used as an output [reconstructed picture](#) or an input [reference picture](#) in a [video decode operation](#).
- `VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR` is reserved for future use.
- `VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR` specifies that the image **can** be used as an [encode input picture](#) in a [video encode operation](#).
- `VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR` specifies that the image **can** be used as an output [reconstructed picture](#) or an input [reference picture](#) in a [video encode operation](#).
- `VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` specifies that the image **can** be transitioned to the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` layout to be used as a color or depth/stencil attachment in a `VkFramebuffer` and/or as a read-only input resource in a shader (sampled image, combined image sampler or input attachment) in the same render pass.
- `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` specifies that the image **can** be used with host copy commands and host layout transitions.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkImageUsageFlags;
```

`VkImageUsageFlags` is a bitmask type for setting a mask of zero or more [VkImageUsageFlagBits](#).

When creating a `VkImageView` one of the following [VkImageUsageFlagBits](#) **must** be set:

- `VK_IMAGE_USAGE_SAMPLED_BIT`
- `VK_IMAGE_USAGE_STORAGE_BIT`
- `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT`
- `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`
- `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT`

- `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT`
- `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`
- `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`
- `VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR`
- `VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR`

Bits which **can** be set in `VkImageCreateInfo::flags`, specifying additional parameters of an image, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageCreateFlagBits {
    VK_IMAGE_CREATE_SPARSE_BINDING_BIT = 0x00000001,
    VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT = 0x00000002,
    VK_IMAGE_CREATE_SPARSE_ALIASED_BIT = 0x00000004,
    VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT = 0x00000008,
    VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT = 0x00000010,
    VK_IMAGE_CREATE_ALIAS_BIT = 0x00000400,
    VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT = 0x00000040,
    VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT = 0x00000020,
    VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT = 0x00000080,
    VK_IMAGE_CREATE_EXTENDED_USAGE_BIT = 0x00000100,
    VK_IMAGE_CREATE_DISJOINT_BIT = 0x00000200,
    // Provided by VK_EXT_sample_locations
    VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT = 0x00001000,
    // Provided by VK_KHR_video_maintenance1
    VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR = 0x00100000,
    // Provided by VK_KHR_bind_memory2 with VK_KHR_device_group
    VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR =
VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT,
    // Provided by VK_KHR_maintenance1
    VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT_KHR =
VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT,
    // Provided by VK_KHR_maintenance2
    VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT_KHR =
VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT,
    // Provided by VK_KHR_maintenance2
    VK_IMAGE_CREATE_EXTENDED_USAGE_BIT_KHR = VK_IMAGE_CREATE_EXTENDED_USAGE_BIT,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_IMAGE_CREATE_DISJOINT_BIT_KHR = VK_IMAGE_CREATE_DISJOINT_BIT,
    // Provided by VK_KHR_bind_memory2
    VK_IMAGE_CREATE_ALIAS_BIT_KHR = VK_IMAGE_CREATE_ALIAS_BIT,
} VkImageCreateFlagBits;
```

- `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` specifies that the image will be backed using sparse memory binding.
- `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` specifies that the image **can** be partially backed using

sparse memory binding. Images created with this flag **must** also be created with the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` flag.

- `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` specifies that the image will be backed using sparse memory binding with memory ranges that might also simultaneously be backing another image (or another portion of the same image). Images created with this flag **must** also be created with the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` flag.
- `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT` specifies that the image **can** be used to create a `VkImageView` with a different format from the image. For [multi-planar](#) formats, `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT` specifies that a `VkImageView` can be created of a *plane* of the image.
- `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT` specifies that the image **can** be used to create a `VkImageView` of type `VK_IMAGE_VIEW_TYPE_CUBE` or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`.
- `VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT` specifies that the image **can** be used to create a `VkImageView` of type `VK_IMAGE_VIEW_TYPE_2D` or `VK_IMAGE_VIEW_TYPE_2D_ARRAY`.
- `VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT` specifies that the image **can** be used with a non-zero value of the `splitInstanceBindRegionCount` member of a `VkBindImageMemoryDeviceGroupInfo` structure passed into `vkBindImageMemory2KHR`. This flag also has the effect of making the image use the standard sparse image block dimensions.
- `VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT` specifies that the image having a compressed format **can** be used to create a `VkImageView` with an uncompressed format where each texel in the image view corresponds to a compressed texel block of the image.
- `VK_IMAGE_CREATE_EXTENDED_USAGE_BIT` specifies that the image **can** be created with usage flags that are not supported for the format the image is created with but are supported for at least one format a `VkImageView` created from the image **can** have.
- `VK_IMAGE_CREATE_DISJOINT_BIT` specifies that an image with a [multi-planar format](#) **must** have each plane separately bound to memory, rather than having a single memory binding for the whole image; the presence of this bit distinguishes a *disjoint image* from an image without this bit set.
- `VK_IMAGE_CREATE_ALIAS_BIT` specifies that two images created with the same creation parameters and aliased to the same memory **can** interpret the contents of the memory consistently with each other, subject to the rules described in the [Memory Aliasing](#) section. This flag further specifies that each plane of a *disjoint* image **can** share an in-memory non-linear representation with single-plane images, and that a single-plane image **can** share an in-memory non-linear representation with a plane of a multi-planar disjoint image, according to the rules in [Compatible Formats of Planes of Multi-Planar Formats](#). If the `pNext` chain includes a `VkExternalMemoryImageCreateInfo` structure whose `handleTypes` member is not `0`, it is as if `VK_IMAGE_CREATE_ALIAS_BIT` is set.
- `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` specifies that an image with a depth or depth/stencil format **can** be used with custom sample locations when used as a depth/stencil attachment.
- `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR` specifies that the image **can** be used in [video coding operations](#) without having to specify at image creation time the set of video profiles the image will be used with, except for images used only as [DPB](#) pictures, as long as the

image is otherwise [compatible](#) with the video profile in question.

NOTE

This enables exchanging video picture data without additional copies or conversions when used as:

- [Decode output pictures](#), indifferent of the video profile used to produce them.
- [Encode input pictures](#), indifferent of the video profile used to consume them.

This includes images created with both `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR` and `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`, which is necessary to use the same video picture as the [reconstructed picture](#) and [decode output picture](#) in a video decode operation on implementations supporting `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR`.

However, images with only DPB usage remain tied to the video profiles the image was created with, as the data layout of such DPB-only images **may** be implementation- and codec-dependent.

If an application would like to share or reuse the device memory backing such images (e.g. for the purposes of temporal aliasing), then it **should** create separate image objects for each video profile and bind them to the same underlying device memory range, similar to how memory resources **can** be shared across separate video sessions or any other memory-backed resource.

See [Sparse Resource Features](#) and [Sparse Physical Device Features](#) for more details.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkImageCreateFlags;
```

`VkImageCreateFlags` is a bitmask type for setting a mask of zero or more [VkImageCreateFlagBits](#).

Possible values of `VkImageCreateInfo::imageType`, specifying the basic dimensionality of an image, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageType {
    VK_IMAGE_TYPE_1D = 0,
    VK_IMAGE_TYPE_2D = 1,
    VK_IMAGE_TYPE_3D = 2,
} VkImageType;
```

- `VK_IMAGE_TYPE_1D` specifies a one-dimensional image.
- `VK_IMAGE_TYPE_2D` specifies a two-dimensional image.
- `VK_IMAGE_TYPE_3D` specifies a three-dimensional image.

Possible values of `VkImageCreateInfo::tiling`, specifying the tiling arrangement of texel blocks in

an image, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageTiling {
    VK_IMAGE_TILING_OPTIMAL = 0,
    VK_IMAGE_TILING_LINEAR = 1,
    // Provided by VK_EXT_image_drm_format_modifier
    VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT = 1000158000,
} VkImageTiling;
```

- `VK_IMAGE_TILING_OPTIMAL` specifies optimal tiling (texels are laid out in an implementation-dependent arrangement, for more efficient memory access).
- `VK_IMAGE_TILING_LINEAR` specifies linear tiling (texels are laid out in memory in row-major order, possibly with some padding on each row).
- `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT` indicates that the image's tiling is defined by a [Linux DRM format modifier](#). The modifier is specified at image creation with `VkImageDrmFormatModifierListCreateInfoEXT` or `VkImageDrmFormatModifierExplicitCreateInfoEXT`, and **can** be queried with `vkGetImageDrmFormatModifierPropertiesEXT`.

To query the memory layout of an image subresource, call:

```
// Provided by VK_VERSION_1_0
void vkGetImageSubresourceLayout(
    VkDevice          device,
    VkImage           image,
    const VkImageSubresource* pSubresource,
    VkSubresourceLayout* pLayout);
```

- `device` is the logical device that owns the image.
- `image` is the image whose layout is being queried.
- `pSubresource` is a pointer to a `VkImageSubresource` structure selecting a specific image subresource from the image.
- `pLayout` is a pointer to a `VkSubresourceLayout` structure in which the layout is returned.

If the image is [linear](#), then the returned layout is valid for [host access](#).

If the image's tiling is `VK_IMAGE_TILING_LINEAR` and its format is a [multi-planar format](#), then `vkGetImageSubresourceLayout` describes one *format plane* of the image. If the image's tiling is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then `vkGetImageSubresourceLayout` describes one *memory plane* of the image. If the image's tiling is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT` and the image is [non-linear](#), then the returned layout has an implementation-dependent meaning; the vendor of the image's [DRM format modifier](#) **may** provide documentation that explains how to interpret the returned layout.

`vkGetImageSubresourceLayout` is invariant for the lifetime of a single image.

Valid Usage

- VUID-vkGetImageSubresourceLayout-image-07790
The `image` **must** have been created with `tiling` equal to `VK_IMAGE_TILING_LINEAR` or `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`
- VUID-vkGetImageSubresourceLayout-aspectMask-00997
The `aspectMask` member of `pSubresource` **must** only have a single bit set
- VUID-vkGetImageSubresourceLayout-mipLevel-01716
The `mipLevel` member of `pSubresource` **must** be less than the `mipLevels` specified in `image`
- VUID-vkGetImageSubresourceLayout-arrayLayer-01717
The `arrayLayer` member of `pSubresource` **must** be less than the `arrayLayers` specified in `image`
- VUID-vkGetImageSubresourceLayout-format-08886
If `format` of the `image` is a color format that is not a `multi-planar image format`, and `tiling` of the `image` is `VK_IMAGE_TILING_LINEAR` or `VK_IMAGE_TILING_OPTIMAL`, the `aspectMask` member of `pSubresource` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-vkGetImageSubresourceLayout-format-04462
If `format` of the `image` has a depth component, the `aspectMask` member of `pSubresource` **must** contain `VK_IMAGE_ASPECT_DEPTH_BIT`
- VUID-vkGetImageSubresourceLayout-format-04463
If `format` of the `image` has a stencil component, the `aspectMask` member of `pSubresource` **must** contain `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-vkGetImageSubresourceLayout-format-04464
If `format` of the `image` does not contain a stencil or depth component, the `aspectMask` member of `pSubresource` **must** not contain `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-vkGetImageSubresourceLayout-tiling-08717
If the `tiling` of the `image` is `VK_IMAGE_TILING_LINEAR` and has a `multi-planar image format`, then the `aspectMask` member of `pSubresource` **must** be a single valid `multi-planar aspect mask` bit
- VUID-vkGetImageSubresourceLayout-tiling-09433
If the `tiling` of the `image` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then the `aspectMask` member of `pSubresource` **must** be `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT` and the index `i` **must** be less than the `VkDrmFormatModifierPropertiesEXT::drmFormatModifierPlaneCount` associated with the image's `format` and `VkImageDrmFormatModifierPropertiesEXT::drmFormatModifier`

Valid Usage (Implicit)

- VUID-vkGetImageSubresourceLayout-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetImageSubresourceLayout-image-parameter

image **must** be a valid [VkImage](#) handle

- VUID-vkGetImageSubresourceLayout-pSubresource-parameter
pSubresource **must** be a valid pointer to a valid [VkImageSubresource](#) structure
- VUID-vkGetImageSubresourceLayout-pLayout-parameter
pLayout **must** be a valid pointer to a [VkSubresourceLayout](#) structure
- VUID-vkGetImageSubresourceLayout-image-parent
image **must** have been created, allocated, or retrieved from **device**

The [VkImageSubresource](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageSubresource {
    VkImageAspectFlags    aspectMask;
    uint32_t              mipLevel;
    uint32_t              arrayLayer;
} VkImageSubresource;
```

- **aspectMask** is a [VkImageAspectFlags](#) value selecting the image *aspect*.
- **mipLevel** selects the mipmap level.
- **arrayLayer** selects the array layer.

Valid Usage (Implicit)

- VUID-VkImageSubresource-aspectMask-parameter
aspectMask **must** be a valid combination of [VkImageAspectFlagBits](#) values
- VUID-VkImageSubresource-aspectMask-requiredbitmask
aspectMask **must** not be 0

Information about the layout of the image subresource is returned in a [VkSubresourceLayout](#) structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSubresourceLayout {
    VkDeviceSize    offset;
    VkDeviceSize    size;
    VkDeviceSize    rowPitch;
    VkDeviceSize    arrayPitch;
    VkDeviceSize    depthPitch;
} VkSubresourceLayout;
```

- **offset** is the byte offset from the start of the image or the plane where the image subresource begins.
- **size** is the size in bytes of the image subresource. **size** includes any extra memory that is

required based on `rowPitch`.

- `rowPitch` describes the number of bytes between each row of texels in an image.
- `arrayPitch` describes the number of bytes between each array layer of an image.
- `depthPitch` describes the number of bytes between each slice of 3D image.

If the image is `linear`, then `rowPitch`, `arrayPitch` and `depthPitch` describe the layout of the image subresource in linear memory. For uncompressed formats, `rowPitch` is the number of bytes between texels with the same x coordinate in adjacent rows (y coordinates differ by one). `arrayPitch` is the number of bytes between texels with the same x and y coordinate in adjacent array layers of the image (array layer values differ by one). `depthPitch` is the number of bytes between texels with the same x and y coordinate in adjacent slices of a 3D image (z coordinates differ by one). Expressed as an addressing formula, the starting byte of a texel in the image subresource has address:

```
// (x,y,z,layer) are in texel coordinates
address(x,y,z,layer) = layer*arrayPitch + z*depthPitch + y*rowPitch + x*elementSize +
offset
```

For compressed formats, the `rowPitch` is the number of bytes between compressed texel blocks in adjacent rows. `arrayPitch` is the number of bytes between compressed texel blocks in adjacent array layers. `depthPitch` is the number of bytes between compressed texel blocks in adjacent slices of a 3D image.

```
// (x,y,z,layer) are in compressed texel block coordinates
address(x,y,z,layer) = layer*arrayPitch + z*depthPitch + y*rowPitch + x
*compressedTexelBlockByteSize + offset;
```

The value of `arrayPitch` is undefined for images that were not created as arrays. `depthPitch` is defined only for 3D images.

If the image has a *single-plane* color format and its tiling is `VK_IMAGE_TILING_LINEAR`, then the `aspectMask` member of `VkImageSubresource` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`.

If the image has a depth/stencil format and its tiling is `VK_IMAGE_TILING_LINEAR`, then `aspectMask` **must** be either `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`. On implementations that store depth and stencil aspects separately, querying each of these image subresource layouts will return a different `offset` and `size` representing the region of memory used for that aspect. On implementations that store depth and stencil aspects interleaved, the same `offset` and `size` are returned and represent the interleaved memory allocation.

If the image has a `multi-planar format` and its tiling is `VK_IMAGE_TILING_LINEAR`, then the `aspectMask` member of `VkImageSubresource` **must** be `VK_IMAGE_ASPECT_PLANE_0_BIT`, `VK_IMAGE_ASPECT_PLANE_1_BIT`, or (for 3-plane formats only) `VK_IMAGE_ASPECT_PLANE_2_BIT`. Querying each of these image subresource layouts will return a different `offset` and `size` representing the region of memory used for that plane. If the image is *disjoint*, then the `offset` is relative to the base address of the plane. If the image is *non-disjoint*, then the `offset` is relative to the base address of the image.

If the image's tiling is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then the `aspectMask` member of `VkImageSubresource` **must** be one of `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT`, where the maximum allowed plane index *i* is defined by the `VkDrmFormatModifierPropertiesEXT::drmFormatModifierPlaneCount` associated with the image's `VkImageCreateInfo::format` and `modifier`. The memory range used by the subresource is described by `offset` and `size`. If the image is *disjoint*, then the `offset` is relative to the base address of the *memory plane*. If the image is *non-disjoint*, then the `offset` is relative to the base address of the image. If the image is *non-linear*, then `rowPitch`, `arrayPitch`, and `depthPitch` have an implementation-dependent meaning.

To query the memory layout of an image subresource, call:

```
// Provided by VK_KHR_maintenance5
void vkGetImageSubresourceLayout2KHR(
    VkDevice          device,
    VkImage           image,
    const VkImageSubresource2KHR* pSubresource,
    VkSubresourceLayout2KHR*      pLayout);
```

or the equivalent command

```
// Provided by VK_EXT_host_image_copy
void vkGetImageSubresourceLayout2EXT(
    VkDevice          device,
    VkImage           image,
    const VkImageSubresource2KHR* pSubresource,
    VkSubresourceLayout2KHR*      pLayout);
```

- `device` is the logical device that owns the image.
- `image` is the image whose layout is being queried.
- `pSubresource` is a pointer to a `VkImageSubresource2KHR` structure selecting a specific image for the image subresource.
- `pLayout` is a pointer to a `VkSubresourceLayout2KHR` structure in which the layout is returned.

`vkGetImageSubresourceLayout2KHR` behaves similarly to `vkGetImageSubresourceLayout`, with the ability to specify extended inputs via chained input structures, and to return extended information via chained output structures.

It is legal to call `vkGetImageSubresourceLayout2KHR` with an `image` created with `tiling` equal to `VK_IMAGE_TILING_OPTIMAL`, but the members of `VkSubresourceLayout2KHR::subresourceLayout` will have undefined values in this case.

NOTE

Structures chained from `VkImageSubresource2KHR::pNext` will also be updated when `tiling` is equal to `VK_IMAGE_TILING_OPTIMAL`.

Valid Usage

- VUID-vkGetImageSubresourceLayout2KHR-aspectMask-00997
The **aspectMask** member of **pSubresource** **must** only have a single bit set
- VUID-vkGetImageSubresourceLayout2KHR-mipLevel-01716
The **mipLevel** member of **pSubresource** **must** be less than the **mipLevels** specified in **image**
- VUID-vkGetImageSubresourceLayout2KHR-arrayLayer-01717
The **arrayLayer** member of **pSubresource** **must** be less than the **arrayLayers** specified in **image**
- VUID-vkGetImageSubresourceLayout2KHR-format-08886
If **format** of the **image** is a color format that is not a **multi-planar image format**, and **tiling** of the **image** is **VK_IMAGE_TILING_LINEAR** or **VK_IMAGE_TILING_OPTIMAL**, the **aspectMask** member of **pSubresource** **must** be **VK_IMAGE_ASPECT_COLOR_BIT**
- VUID-vkGetImageSubresourceLayout2KHR-format-04462
If **format** of the **image** has a depth component, the **aspectMask** member of **pSubresource** **must** contain **VK_IMAGE_ASPECT_DEPTH_BIT**
- VUID-vkGetImageSubresourceLayout2KHR-format-04463
If **format** of the **image** has a stencil component, the **aspectMask** member of **pSubresource** **must** contain **VK_IMAGE_ASPECT_STENCIL_BIT**
- VUID-vkGetImageSubresourceLayout2KHR-format-04464
If **format** of the **image** does not contain a stencil or depth component, the **aspectMask** member of **pSubresource** **must** not contain **VK_IMAGE_ASPECT_DEPTH_BIT** or **VK_IMAGE_ASPECT_STENCIL_BIT**
- VUID-vkGetImageSubresourceLayout2KHR-tiling-08717
If the **tiling** of the **image** is **VK_IMAGE_TILING_LINEAR** and has a **multi-planar image format**, then the **aspectMask** member of **pSubresource** **must** be a single valid **multi-planar aspect mask** bit
- VUID-vkGetImageSubresourceLayout2KHR-tiling-09435
If the **tiling** of the **image** is **VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT**, then the **aspectMask** member of **pSubresource** **must** be **VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT** and the index **i** **must** be less than the **VkDrmFormatModifierPropertiesEXT::drmFormatModifierPlaneCount** associated with the image's **format** and **VkImageDrmFormatModifierPropertiesEXT::drmFormatModifier**

Valid Usage (Implicit)

- VUID-vkGetImageSubresourceLayout2KHR-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkGetImageSubresourceLayout2KHR-image-parameter
image **must** be a valid **VkImage** handle
- VUID-vkGetImageSubresourceLayout2KHR-pSubresource-parameter
pSubresource **must** be a valid pointer to a valid **VkImageSubresource2KHR** structure

- VUID-vkGetImageSubresourceLayout2KHR-pLayout-parameter
pLayout must be a valid pointer to a [VkSubresourceLayout2KHR](#) structure
- VUID-vkGetImageSubresourceLayout2KHR-image-parent
image must have been created, allocated, or retrieved from **device**

The [VkImageSubresource2KHR](#) structure is defined as:

```
// Provided by VK_KHR_maintenance5
typedef struct VkImageSubresource2KHR {
    VkStructureType    sType;
    void*              pNext;
    VkImageSubresource imageSubresource;
} VkImageSubresource2KHR;
```

or the equivalent

```
// Provided by VK_EXT_host_image_copy
typedef VkImageSubresource2KHR VkImageSubresource2EXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **imageSubresource** is a [VkImageSubresource](#) structure.

Valid Usage (Implicit)

- VUID-VkImageSubresource2KHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2_KHR**
- VUID-VkImageSubresource2KHR-pNext-pNext
pNext must be **NULL**
- VUID-VkImageSubresource2KHR-imageSubresource-parameter
imageSubresource must be a valid [VkImageSubresource](#) structure

Information about the layout of the image subresource is returned in a [VkSubresourceLayout2KHR](#) structure:

```
// Provided by VK_KHR_maintenance5
typedef struct VkSubresourceLayout2KHR {
    VkStructureType    sType;
    void*              pNext;
    VkSubresourceLayout subresourceLayout;
} VkSubresourceLayout2KHR;
```


or the equivalent

```
// Provided by VK_EXT_host_image_copy
typedef VkSubresourceLayout2KHR VkSubresourceLayout2EXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **subresourceLayout** is a [VkSubresourceLayout](#) structure.

Valid Usage (Implicit)

- VUID-VkSubresourceLayout2KHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2_KHR**
- VUID-VkSubresourceLayout2KHR-pNext-pNext
pNext must be **NULL** or a pointer to a valid instance of [VkSubresourceHostMemcpySizeEXT](#)
- VUID-VkSubresourceLayout2KHR-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique

To query the memory size needed to copy to or from an image using [vkCopyMemoryToImageEXT](#) or [vkCopyImageToMemoryEXT](#) when the **VK_HOST_IMAGE_COPY_MEMCPY_EXT** flag is specified, add a [VkSubresourceHostMemcpySizeEXT](#) structure to the **pNext** chain of the [VkSubresourceLayout2EXT](#) structure in a call to [vkGetImageSubresourceLayout2EXT](#).

The [VkSubresourceHostMemcpySizeEXT](#) structure is defined as:

```
// Provided by VK_EXT_host_image_copy
typedef struct VkSubresourceHostMemcpySizeEXT {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize        size;
} VkSubresourceHostMemcpySizeEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **size** is the size in bytes of the image subresource.

Valid Usage (Implicit)

- VUID-VkSubresourceHostMemcpySizeEXT-sType-sType
sType must be **VK_STRUCTURE_TYPE_SUBRESOURCE_HOST_MEMCPY_SIZE_EXT**

To query the memory layout of an image subresource, without an image object, call:

```
// Provided by VK_KHR_maintenance5
void vkGetDeviceImageSubresourceLayoutKHR(
    VkDevice device,
    const VkDeviceImageSubresourceInfoKHR* pInfo,
    VkSubresourceLayout2KHR* pLayout);
```

- **device** is the logical device that owns the image.
- **pInfo** is a pointer to a [VkDeviceImageSubresourceInfoKHR](#) structure containing parameters required for the subresource layout query.
- **pLayout** is a pointer to a [VkSubresourceLayout2KHR](#) structure in which the layout is returned.

[vkGetDeviceImageSubresourceLayoutKHR](#) behaves similarly to [vkGetImageSubresourceLayout2KHR](#), but uses a [VkImageCreateInfo](#) structure to specify the image rather than a [VkImage](#) object.

Valid Usage (Implicit)

- VUID-vkGetDeviceImageSubresourceLayoutKHR-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkGetDeviceImageSubresourceLayoutKHR-pInfo-parameter **pInfo** **must** be a valid pointer to a valid [VkDeviceImageSubresourceInfoKHR](#) structure
- VUID-vkGetDeviceImageSubresourceLayoutKHR-pLayout-parameter **pLayout** **must** be a valid pointer to a [VkSubresourceLayout2KHR](#) structure

The [VkDeviceImageSubresourceInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_maintenance5
typedef struct VkDeviceImageSubresourceInfoKHR {
    VkStructureType sType;
    const void* pNext;
    const VkImageCreateInfo* pCreateInfo;
    const VkImageSubresource2KHR* pSubresource;
} VkDeviceImageSubresourceInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **pCreateInfo** is a pointer to a [VkImageCreateInfo](#) structure containing parameters affecting creation of the image to query.
- **pSubresource** **pSubresource** is a pointer to a [VkImageSubresource2KHR](#) structure selecting a specific image subresource for the query.

Valid Usage

- VUID-VkDeviceImageSubresourceInfoKHR-aspectMask-00997

The `aspectMask` member of `pSubresource` **must** only have a single bit set

- VUID-VkDeviceImageSubresourceInfoKHR-mipLevel-01716

The `mipLevel` member of `pSubresource` **must** be less than the `mipLevels` specified in `pCreateInfo`

- VUID-VkDeviceImageSubresourceInfoKHR-arrayLayer-01717

The `arrayLayer` member of `pSubresource` **must** be less than the `arrayLayers` specified in `pCreateInfo`

- VUID-VkDeviceImageSubresourceInfoKHR-format-08886

If `format` of the `image` is a color format that is not a `multi-planar image format`, and `tiling` of the `pCreateInfo` is `VK_IMAGE_TILING_LINEAR` or `VK_IMAGE_TILING_OPTIMAL`, the `aspectMask` member of `pSubresource` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`

- VUID-VkDeviceImageSubresourceInfoKHR-format-04462

If `format` of the `pCreateInfo` has a depth component, the `aspectMask` member of `pSubresource` **must** contain `VK_IMAGE_ASPECT_DEPTH_BIT`

- VUID-VkDeviceImageSubresourceInfoKHR-format-04463

If `format` of the `pCreateInfo` has a stencil component, the `aspectMask` member of `pSubresource` **must** contain `VK_IMAGE_ASPECT_STENCIL_BIT`

- VUID-VkDeviceImageSubresourceInfoKHR-format-04464

If `format` of the `pCreateInfo` does not contain a stencil or depth component, the `aspectMask` member of `pSubresource` **must** not contain `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`

- VUID-VkDeviceImageSubresourceInfoKHR-tiling-08717

If the `tiling` of the `pCreateInfo` is `VK_IMAGE_TILING_LINEAR` and has a `multi-planar image format`, then the `aspectMask` member of `pSubresource` **must** be a single valid `multi-planar aspect mask` bit

Valid Usage (Implicit)

- VUID-VkDeviceImageSubresourceInfoKHR-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_IMAGE_SUBRESOURCE_INFO_KHR`

- VUID-VkDeviceImageSubresourceInfoKHR-pNext-pNext

`pNext` **must** be `NULL`

- VUID-VkDeviceImageSubresourceInfoKHR-pCreateInfo-parameter

`pCreateInfo` **must** be a valid pointer to a valid `VkImageCreateInfo` structure

- VUID-VkDeviceImageSubresourceInfoKHR-pSubresource-parameter

`pSubresource` **must** be a valid pointer to a valid `VkImageSubresource2KHR` structure

If an image was created with `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then the image has a `Linux DRM format modifier`. To query the *modifier*, call:

```
// Provided by VK_EXT_image_drm_format_modifier
VkResult vkGetImageDrmFormatModifierPropertiesEXT(
    VkDevice          device,
    VkImage           image,
    VkImageDrmFormatModifierPropertiesEXT* pProperties);
```

- **device** is the logical device that owns the image.
- **image** is the queried image.
- **pProperties** is a pointer to a [VkImageDrmFormatModifierPropertiesEXT](#) structure in which properties of the image's *DRM format modifier* are returned.

Valid Usage

- VUID-vkGetImageDrmFormatModifierPropertiesEXT-image-02272
image must have been created with **tiling** equal to **VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT**

Valid Usage (Implicit)

- VUID-vkGetImageDrmFormatModifierPropertiesEXT-device-parameter
device must be a valid [VkDevice](#) handle
- VUID-vkGetImageDrmFormatModifierPropertiesEXT-image-parameter
image must be a valid [VkImage](#) handle
- VUID-vkGetImageDrmFormatModifierPropertiesEXT-pProperties-parameter
pProperties must be a valid pointer to a [VkImageDrmFormatModifierPropertiesEXT](#) structure
- VUID-vkGetImageDrmFormatModifierPropertiesEXT-image-parent
image must have been created, allocated, or retrieved from **device**

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**

The [VkImageDrmFormatModifierPropertiesEXT](#) structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkImageDrmFormatModifierPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint64_t           drmFormatModifier;
} VkImageDrmFormatModifierPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `drmFormatModifier` returns the image's [Linux DRM format modifier](#).

If the `image` was created with `VkImageDrmFormatModifierListCreateInfoEXT`, then the returned `drmFormatModifier` **must** belong to the list of modifiers provided at time of image creation in `VkImageDrmFormatModifierListCreateInfoEXT::pDrmFormatModifiers`. If the `image` was created with `VkImageDrmFormatModifierExplicitCreateInfoEXT`, then the returned `drmFormatModifier` **must** be the modifier provided at time of image creation in `VkImageDrmFormatModifierExplicitCreateInfoEXT::drmFormatModifier`.

Valid Usage (Implicit)

- VUID-VkImageDrmFormatModifierPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_PROPERTIES_EXT`
- VUID-VkImageDrmFormatModifierPropertiesEXT-pNext-pNext
`pNext` **must** be `NULL`

To destroy an image, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyImage(
    VkDevice          device,
    VkImage           image,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the image.
- `image` is the image to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyImage-image-01000
All submitted commands that refer to `image`, either directly or via a `VkImageView`, **must** have completed execution
- VUID-vkDestroyImage-image-01001

If `VkAllocationCallbacks` were provided when `image` was created, a compatible set of callbacks **must** be provided here

- VUID-vkDestroyImage-image-01002

If no `VkAllocationCallbacks` were provided when `image` was created, `pAllocator` **must** be `NULL`

- VUID-vkDestroyImage-image-04882

`image` **must** not have been acquired from `vkGetSwapchainImagesKHR`

Valid Usage (Implicit)

- VUID-vkDestroyImage-device-parameter

`device` **must** be a valid `VkDevice` handle

- VUID-vkDestroyImage-image-parameter

If `image` is not `VK_NULL_HANDLE`, `image` **must** be a valid `VkImage` handle

- VUID-vkDestroyImage-pAllocator-parameter

If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

- VUID-vkDestroyImage-image-parent

If `image` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `image` **must** be externally synchronized

12.3.1. Image Format Features

Valid uses of a `VkImage` **may** depend on the image's *format features*, defined below. Such constraints are documented in the affected valid usage statement.

- If the image was created with `VK_IMAGE_TILING_LINEAR`, then its set of *format features* is the value of `VkFormatProperties::linearTilingFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` on the same `format` as `VkImageCreateInfo::format`.
- If the image was created with `VK_IMAGE_TILING_OPTIMAL`, then its set of *format features* is the value of `VkFormatProperties::optimalTilingFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` on the same `format` as `VkImageCreateInfo::format`.
- If the image was created with `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then:
 - The image's DRM format modifier is the value of `VkImageDrmFormatModifierPropertiesEXT::drmFormatModifier` found by calling `vkGetImageDrmFormatModifierPropertiesEXT`.
 - Let `VkDrmFormatModifierPropertiesListEXT::pDrmFormatModifierProperties` be the array found by calling `vkGetPhysicalDeviceFormatProperties2KHR` on the same `format` as `VkImageCreateInfo::format`.

- Let `VkDrmFormatModifierPropertiesEXT prop` be the array element whose `drmFormatModifier` member is the value of the image's DRM format modifier.
- Then the image's set of *format features* is the value of `prop::drmFormatModifierTilingFeatures`.

12.3.2. Image Mip Level Sizing

A *complete mipmap chain* is the full set of mip levels, from the largest mip level provided, down to the *minimum mip level size*.

Conventional Images

For conventional images, the dimensions of each successive mip level, $n+1$, are:

$$\text{width}_{n+1} = \max(\lfloor \text{width}_n / 2 \rfloor, 1)$$

$$\text{height}_{n+1} = \max(\lfloor \text{height}_n / 2 \rfloor, 1)$$

$$\text{depth}_{n+1} = \max(\lfloor \text{depth}_n / 2 \rfloor, 1)$$

where width_n , height_n , and depth_n are the dimensions of the next larger mip level, n .

The minimum mip level size is:

- 1 for one-dimensional images,
- 1x1 for two-dimensional images, and
- 1x1x1 for three-dimensional images.

The number of levels in a complete mipmap chain is:

$$\lceil \log_2(\max(\text{width}_0, \text{height}_0, \text{depth}_0)) \rceil + 1$$

where width_0 , height_0 , and depth_0 are the dimensions of the largest (most detailed) mip level, 0.

12.4. Image Layouts

Images are stored in implementation-dependent opaque layouts in memory. Each layout has limitations on what kinds of operations are supported for image subresources using the layout. At any given time, the data representing an image subresource in memory exists in a particular layout which is determined by the most recent layout transition that was performed on that image subresource. Applications have control over which layout each image subresource uses, and **can** transition an image subresource from one layout to another. Transitions **can** happen with an image memory barrier, included as part of a `vkCmdPipelineBarrier` or a `vkCmdWaitEvents` command buffer command (see [Image Memory Barriers](#)), or as part of a subpass dependency within a render pass (see [VkSubpassDependency](#)).

Image layout is per-image subresource. Separate image subresources of the same image **can** be in different layouts at the same time, with the exception that depth and stencil aspects of a given image subresource **can** only be in different layouts if the `separateDepthStencilLayouts` feature is enabled.

NOTE

Each layout **may** offer optimal performance for a specific usage of image memory. For example, an image with a layout of `VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL` **may** provide optimal performance for use as a color attachment, but be unsupported for use in transfer commands. Applications **can** transition an image subresource from one layout to another in order to achieve optimal performance when the image subresource is used for multiple kinds of operations. After initialization, applications need not use any layout other than the general layout, though this **may** produce suboptimal performance on some implementations.

Upon creation, all image subresources of an image are initially in the same layout, where that layout is selected by the `VkImageCreateInfo::initialLayout` member. The `initialLayout` **must** be either `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`. If it is `VK_IMAGE_LAYOUT_PREINITIALIZED`, then the image data **can** be preinitialized by the host while using this layout, and the transition away from this layout will preserve that data. If it is `VK_IMAGE_LAYOUT_UNDEFINED`, then the contents of the data are considered to be undefined, and the transition away from this layout is not guaranteed to preserve that data. For either of these initial layouts, any image subresources **must** be transitioned to another layout before they are accessed by the device.

Host access to image memory is only well-defined for `linear` images and for image subresources of those images which are currently in either the `VK_IMAGE_LAYOUT_PREINITIALIZED` or `VK_IMAGE_LAYOUT_GENERAL` layout. Calling `vkGetImageSubresourceLayout` for a linear image returns a subresource layout mapping that is valid for either of those image layouts.

The set of image layouts consists of:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageLayout {
    VK_IMAGE_LAYOUT_UNDEFINED = 0,
    VK_IMAGE_LAYOUT_GENERAL = 1,
    VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL = 2,
    VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL = 3,
    VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL = 4,
    VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL = 5,
    VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL = 6,
    VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL = 7,
    VK_IMAGE_LAYOUT_PREINITIALIZED = 8,
    VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL = 1000117000,
    VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL = 1000117001,
    VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL = 1000241000,
    VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL = 1000241001,
    VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL = 1000241002,
    VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL = 1000241003,
    VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL = 1000314000,
```

```

    VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL = 1000314001,
// Provided by VK_KHR_swapchain
    VK_IMAGE_LAYOUT_PRESENT_SRC_KHR = 1000001002,
// Provided by VK_KHR_video_decode_queue
    VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR = 1000024000,
// Provided by VK_KHR_video_decode_queue
    VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR = 1000024001,
// Provided by VK_KHR_video_decode_queue
    VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR = 1000024002,
// Provided by VK_KHR_shared_presentable_image
    VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR = 1000111000,
// Provided by VK_KHR_fragment_shading_rate
    VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR = 1000164003,
// Provided by VK_KHR_dynamic_rendering_local_read
    VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR = 1000232000,
// Provided by VK_KHR_video_encode_queue
    VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR = 1000299000,
// Provided by VK_KHR_video_encode_queue
    VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR = 1000299001,
// Provided by VK_KHR_video_encode_queue
    VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR = 1000299002,
// Provided by VK_EXT_attachment_feedback_loop_layout
    VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT = 1000339000,
// Provided by VK_KHR_maintenance2
    VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL,
// Provided by VK_KHR_maintenance2
    VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL,
// Provided by VK_KHR_separate_depth_stencil_layouts
    VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL,
// Provided by VK_KHR_separate_depth_stencil_layouts
    VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL,
// Provided by VK_KHR_separate_depth_stencil_layouts
    VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL,
// Provided by VK_KHR_separate_depth_stencil_layouts
    VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL_KHR =
VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL,
// Provided by VK_KHR_synchronization2
    VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR = VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL,
// Provided by VK_KHR_synchronization2
    VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR = VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL,
} VkImageLayout;

```

The type(s) of device access supported by each layout are:

- **VK_IMAGE_LAYOUT_UNDEFINED** specifies that the layout is unknown. Image memory **cannot** be

transitioned into this layout. This layout **can** be used as the **initialLayout** member of **VkImageCreateInfo**. This layout **can** be used in place of the current image layout in a layout transition, but doing so will cause the contents of the image's memory to be undefined.

- **VK_IMAGE_LAYOUT_PREINITIALIZED** specifies that an image's memory is in a defined layout and **can** be populated by data, but that it has not yet been initialized by the driver. Image memory **cannot** be transitioned into this layout. This layout **can** be used as the **initialLayout** member of **VkImageCreateInfo**. This layout is intended to be used as the initial layout for an image whose contents are written by the host, and hence the data **can** be written to memory immediately, without first executing a layout transition. Currently, **VK_IMAGE_LAYOUT_PREINITIALIZED** is only useful with **linear** images because there is not a standard layout defined for **VK_IMAGE_TILING_OPTIMAL** images.
- **VK_IMAGE_LAYOUT_GENERAL** supports all types of device access.
- **VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL** specifies a layout that **must** only be used with attachment accesses in the graphics pipeline.
- **VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL** specifies a layout allowing read only access as an attachment, or in shaders as a sampled image, combined image/sampler, or input attachment.
- **VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL** **must** only be used as a color or resolve attachment in a **VkFramebuffer**. This layout is valid only for image subresources of images created with the **VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT** usage bit enabled.
- **VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL** specifies a layout for both the depth and stencil aspects of a depth/stencil format image allowing read and write access as a depth/stencil attachment. It is equivalent to **VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL** and **VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL**.
- **VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL** specifies a layout for both the depth and stencil aspects of a depth/stencil format image allowing read only access as a depth/stencil attachment or in shaders as a sampled image, combined image/sampler, or input attachment. It is equivalent to **VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL** and **VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL**.
- **VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL** specifies a layout for depth/stencil format images allowing read and write access to the stencil aspect as a stencil attachment, and read only access to the depth aspect as a depth attachment or in shaders as a sampled image, combined image/sampler, or input attachment. It is equivalent to **VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL** and **VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL**.
- **VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL** specifies a layout for depth/stencil format images allowing read and write access to the depth aspect as a depth attachment, and read only access to the stencil aspect as a stencil attachment or in shaders as a sampled image, combined image/sampler, or input attachment. It is equivalent to **VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL** and **VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL**.
- **VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL** specifies a layout for the depth aspect of a depth/stencil format image allowing read and write access as a depth attachment.
- **VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL** specifies a layout for the depth aspect of a depth/stencil format image allowing read-only access as a depth attachment or in shaders as a sampled image, combined image/sampler, or input attachment.

- `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL` specifies a layout for the stencil aspect of a depth/stencil format image allowing read and write access as a stencil attachment.
- `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL` specifies a layout for the stencil aspect of a depth/stencil format image allowing read-only access as a stencil attachment or in shaders as a sampled image, combined image/sampler, or input attachment.
- `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL` specifies a layout allowing read-only access in a shader as a sampled image, combined image/sampler, or input attachment. This layout is valid only for image subresources of images created with the `VK_IMAGE_USAGE_SAMPLED_BIT` or `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` usage bits enabled.
- `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` **must** only be used as a source image of a transfer command (see the definition of `VK_PIPELINE_STAGE_TRANSFER_BIT`). This layout is valid only for image subresources of images created with the `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` usage bit enabled.
- `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` **must** only be used as a destination image of a transfer command. This layout is valid only for image subresources of images created with the `VK_IMAGE_USAGE_TRANSFER_DST_BIT` usage bit enabled.
- `VK_IMAGE_LAYOUT_PRESENT_SRC_KHR` **must** only be used for presenting a presentable image for display.
- `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR` is valid only for shared presentable images, and **must** be used for any usage the image supports.
- `VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR` **must** only be used as a [fragment shading rate attachment](#) or This layout is valid only for image subresources of images created with the `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR` usage bit enabled.
- `VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR` **must** only be used as a [decode output picture](#) in a [video decode operation](#). This layout is valid only for image subresources of images created with the `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR` usage bit enabled.
- `VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR` is reserved for future use.
- `VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR` **must** only be used as an output [reconstructed picture](#) or an input [reference picture](#) in a [video decode operation](#). This layout is valid only for image subresources of images created with the `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR` usage bit enabled.
- `VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR` is reserved for future use.
- `VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR` **must** only be used as an [encode input picture](#) in a [video encode operation](#). This layout is valid only for image subresources of images created with the `VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR` usage bit enabled.
- `VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR` **must** only be used as an output [reconstructed picture](#) or an input [reference picture](#) in a [video encode operation](#). This layout is valid only for image subresources of images created with the `VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR` usage bit enabled.
- `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` **must** only be used as either a color attachment or depth/stencil attachment in a `VkFramebuffer` and/or read-only access in a shader as a sampled image, combined image/sampler, or input attachment. This layout is valid only for

image subresources of images created with the `VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` usage bit enabled and either the `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT` and either the `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_SAMPLED_BIT` usage bits enabled.

- `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR` **must** only be used as either a storage image, or a color or depth/stencil attachment and an input attachment. This layout is valid only for image subresources of images created with either `VK_IMAGE_USAGE_STORAGE_BIT`, or both `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` and either of `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` or `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`.

The layout of each image subresource is not a state of the image subresource itself, but is rather a property of how the data in memory is organized, and thus for each mechanism of accessing an image in the API the application **must** specify a parameter or structure member that indicates which image layout the image subresource(s) are considered to be in when the image will be accessed. For transfer commands, this is a parameter to the command (see [Clear Commands](#) and [Copy Commands](#)). For use as a framebuffer attachment, this is a member in the substructures of the `VkRenderPassCreateInfo` (see [Render Pass](#)). For use in a descriptor set, this is a member in the `VkDescriptorImageInfo` structure (see [Descriptor Set Updates](#)).

12.4.1. Image Layout Matching Rules

At the time that any command buffer command accessing an image executes on any queue, the layouts of the image subresources that are accessed **must** all match exactly the layout specified via the API controlling those accesses, except in case of accesses to an image with a depth/stencil format performed through descriptors referring to only a single aspect of the image, where the following relaxed matching rules apply:

- Descriptors referring just to the depth aspect of a depth/stencil image only need to match in the image layout of the depth aspect, thus `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL` and `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL` are considered to match.
- Descriptors referring just to the stencil aspect of a depth/stencil image only need to match in the image layout of the stencil aspect, thus `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL` and `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL` are considered to match.

When performing a layout transition on an image subresource, the old layout value **must** either equal the current layout of the image subresource (at the time the transition executes), or else be `VK_IMAGE_LAYOUT_UNDEFINED` (implying that the contents of the image subresource need not be preserved). The new layout used in a transition **must** not be `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED`.

The image layout of each image subresource of a depth/stencil image created with `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` is dependent on the last sample locations used to render to the image subresource as a depth/stencil attachment, thus applications **must** provide the same sample locations that were last used to render to the given image subresource whenever a layout transition of the image subresource happens, otherwise the contents of the depth aspect of the image subresource become undefined.

In addition, depth reads from a depth/stencil attachment referring to an image subresource range

of a depth/stencil image created with `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` using different sample locations than what have been last used to perform depth writes to the image subresources of the same image subresource range return undefined values.

Similarly, depth writes to a depth/stencil attachment referring to an image subresource range of a depth/stencil image created with `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` using different sample locations than what have been last used to perform depth writes to the image subresources of the same image subresource range make the contents of the depth aspect of those image subresources undefined.

12.5. Image Views

Image objects are not directly accessed by pipeline shaders for reading or writing image data. Instead, *image views* representing contiguous ranges of the image subresources and containing additional metadata are used for that purpose. Views **must** be created on images of compatible types, and **must** represent a valid subset of image subresources.

Image views are represented by `VkImageView` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkImageView)
```

`VK_REMAINING_ARRAY_LAYERS` is a special constant value used for image views to indicate that all remaining array layers in an image after the base layer should be included in the view.

```
#define VK_REMAINING_ARRAY_LAYERS        (~0U)
```

`VK_REMAINING_MIP_LEVELS` is a special constant value used for image views to indicate that all remaining mipmap levels in an image after the base level should be included in the view.

```
#define VK_REMAINING_MIP_LEVELS          (~0U)
```

The types of image views that **can** be created are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageViewType {
    VK_IMAGE_VIEW_TYPE_1D = 0,
    VK_IMAGE_VIEW_TYPE_2D = 1,
    VK_IMAGE_VIEW_TYPE_3D = 2,
    VK_IMAGE_VIEW_TYPE_CUBE = 3,
    VK_IMAGE_VIEW_TYPE_1D_ARRAY = 4,
    VK_IMAGE_VIEW_TYPE_2D_ARRAY = 5,
    VK_IMAGE_VIEW_TYPE_CUBE_ARRAY = 6,
} VkImageViewType;
```


To create an image view, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateImageView(
    VkDevice                                device,
    const VkImageViewCreateInfo*            pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkImageView*                            pView);
```

- **device** is the logical device that creates the image view.
- **pCreateInfo** is a pointer to a **VkImageViewCreateInfo** structure containing parameters to be used to create the image view.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pView** is a pointer to a **VkImageView** handle in which the resulting image view object is returned.

Valid Usage

- VUID-vkCreateImageView-device-09667
device must support at least one queue family with one of the **VK_QUEUE_VIDEO_ENCODE_BIT_KHR**, **VK_QUEUE_VIDEO_DECODE_BIT_KHR**, **VK_QUEUE_COMPUTE_BIT**, or **VK_QUEUE_GRAPHICS_BIT** capabilities
- VUID-vkCreateImageView-image-09179
VkImageViewCreateInfo::image must have been created from **device**

Valid Usage (Implicit)

- VUID-vkCreateImageView-device-parameter
device must be a valid **VkDevice** handle
- VUID-vkCreateImageView-pCreateInfo-parameter
pCreateInfo must be a valid pointer to a valid **VkImageViewCreateInfo** structure
- VUID-vkCreateImageView-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkCreateImageView-pView-parameter
pView must be a valid pointer to a **VkImageView** handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR`

The `VkImageViewCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageViewCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkImageViewCreateFlags flags;
    VkImage               image;
    VkImageViewType       viewType;
    VkFormat              format;
    VkComponentMapping     components;
    VkImageSubresourceRange subresourceRange;
} VkImageViewCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkImageViewCreateFlagBits` specifying additional parameters of the image view.
- `image` is a `VkImage` on which the view will be created.
- `viewType` is a `VkImageViewType` value specifying the type of the image view.
- `format` is a `VkFormat` specifying the format and type used to interpret texel blocks of the image.
- `components` is a `VkComponentMapping` structure specifying a remapping of color components (or of depth or stencil components after they have been converted into color components).
- `subresourceRange` is a `VkImageSubresourceRange` structure selecting the set of mipmap levels and array layers to be accessible to the view.

Some of the `image` creation parameters are inherited by the view. In particular, image view creation inherits the implicit parameter `usage` specifying the allowed usages of the image view that, by default, takes the value of the corresponding `usage` parameter specified in `VkImageCreateInfo` at image creation time. The implicit `usage` **can** be overridden by adding a `VkImageViewUsageCreateInfo` structure to the `pNext` chain, but the view usage **must** be a subset of the image usage.

If `image` was created with the `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT` flag, and if the `format` of the image is not `multi-planar`, `format` **can** be different from the image's format, but if `image` was created without the `VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT` flag and they are not equal they **must** be *compatible*. Image format compatibility is defined in the [Format Compatibility Classes](#) section. Views of compatible formats will have the same mapping between texel coordinates and memory locations irrespective of the `format`, with only the interpretation of the bit pattern changing.

If **image** was created with a **multi-planar** format, and the image view's **aspectMask** is one of **VK_IMAGE_ASPECT_PLANE_0_BIT**, **VK_IMAGE_ASPECT_PLANE_1_BIT** or **VK_IMAGE_ASPECT_PLANE_2_BIT**, the view's aspect mask is considered to be equivalent to **VK_IMAGE_ASPECT_COLOR_BIT** when used as a framebuffer attachment.

NOTE

Values intended to be used with one view format **may** not be exactly preserved when written or read through a different format. For example, an integer value that happens to have the bit pattern of a floating-point denorm or NaN **may** be flushed or canonicalized when written or read through a view with a floating-point format. Similarly, a value written through a signed normalized format that has a bit pattern exactly equal to -2^b **may** be changed to $-2^b + 1$ as described in [Conversion from Normalized Fixed-Point to Floating-Point](#).

If **image** was created with the **VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT** flag, **format** **must** be *compatible* with the image's format as described above; or **must** be an uncompressed format, in which case it **must** be *size-compatible* with the image's format. In this case, the resulting image view's texel dimensions equal the dimensions of the selected mip level divided by the compressed texel block size and rounded up.

The **VkComponentMapping** **components** member describes a remapping from components of the image to components of the vector returned by shader image instructions. This remapping **must** be the identity swizzle for storage image descriptors, input attachment descriptors, framebuffer attachments, and any **VkImageView** used with a combined image sampler that enables **sampler Y'C_BC_R conversion**.

If the image view is to be used with a sampler which supports **sampler Y'C_BC_R conversion**, an *identically defined object* of type **VkSamplerYcbcrConversion** to that used to create the sampler **must** be passed to **vkCreateImageView** in a **VkSamplerYcbcrConversionInfo** included in the **pNext** chain of **VkImageViewCreateInfo**. Conversely, if a **VkSamplerYcbcrConversion** object is passed to **vkCreateImageView**, an identically defined **VkSamplerYcbcrConversion** object **must** be used when sampling the image.

If the image has a **multi-planar** format, **subresourceRange.aspectMask** is **VK_IMAGE_ASPECT_COLOR_BIT**, and **usage** includes **VK_IMAGE_USAGE_SAMPLED_BIT**, then the **format** **must** be identical to the image **format** and the sampler to be used with the image view **must** enable **sampler Y'C_BC_R conversion**.

When such an image is used in a **video coding** operation, the **sampler Y'C_BC_R conversion** has no effect.

If **image** was created with the **VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT** and the image has a **multi-planar** format, and if **subresourceRange.aspectMask** is **VK_IMAGE_ASPECT_PLANE_0_BIT**, **VK_IMAGE_ASPECT_PLANE_1_BIT**, or **VK_IMAGE_ASPECT_PLANE_2_BIT**, **format** **must** be *compatible* with the corresponding plane of the image, and the sampler to be used with the image view **must** not enable **sampler Y'C_BC_R conversion**. The **width** and **height** of the single-plane image view **must** be derived from the multi-planar image's dimensions in the manner listed for **plane compatibility** for the plane.

Any view of an image plane will have the same mapping between texel coordinates and memory locations as used by the components of the color aspect, subject to the formulae relating texel

coordinates to lower-resolution planes as described in [Chroma Reconstruction](#). That is, if an R or B plane has a reduced resolution relative to the G plane of the multi-planar image, the image view operates using the (u_{plane}, v_{plane}) unnormalized coordinates of the reduced-resolution plane, and these coordinates access the same memory locations as the (u_{color}, v_{color}) unnormalized coordinates of the color aspect for which chroma reconstruction operations operate on the same (u_{plane}, v_{plane}) or (i_{plane}, j_{plane}) coordinates.

Table 11. Image Type and Image View Type Compatibility Requirements

Image View Type	Compatible Image Types
VK_IMAGE_VIEW_TYPE_1D	VK_IMAGE_TYPE_1D
VK_IMAGE_VIEW_TYPE_1D_ARRAY	VK_IMAGE_TYPE_1D
VK_IMAGE_VIEW_TYPE_2D	VK_IMAGE_TYPE_2D , VK_IMAGE_TYPE_3D
VK_IMAGE_VIEW_TYPE_2D_ARRAY	VK_IMAGE_TYPE_2D , VK_IMAGE_TYPE_3D
VK_IMAGE_VIEW_TYPE_CUBE	VK_IMAGE_TYPE_2D
VK_IMAGE_VIEW_TYPE_CUBE_ARRAY	VK_IMAGE_TYPE_2D
VK_IMAGE_VIEW_TYPE_3D	VK_IMAGE_TYPE_3D

Valid Usage

- VUID-VkImageViewCreateInfo-image-01003
If **image** was not created with **VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT** then **viewType** **must** not be **VK_IMAGE_VIEW_TYPE_CUBE** or **VK_IMAGE_VIEW_TYPE_CUBE_ARRAY**
- VUID-VkImageViewCreateInfo-viewType-01004
If the **imageCubeArray** feature is not enabled, **viewType** **must** not be **VK_IMAGE_VIEW_TYPE_CUBE_ARRAY**
- VUID-VkImageViewCreateInfo-image-06723
If **image** was created with **VK_IMAGE_TYPE_3D** but without **VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT** set then **viewType** **must** not be **VK_IMAGE_VIEW_TYPE_2D_ARRAY**
- VUID-VkImageViewCreateInfo-image-06727
If **image** was created with **VK_IMAGE_TYPE_3D** but without **VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT** set then **viewType** **must** not be **VK_IMAGE_VIEW_TYPE_2D**
- VUID-VkImageViewCreateInfo-image-04970
If **image** was created with **VK_IMAGE_TYPE_3D** and **viewType** is **VK_IMAGE_VIEW_TYPE_2D** or **VK_IMAGE_VIEW_TYPE_2D_ARRAY** then **subresourceRange.levelCount** **must** be 1
- VUID-VkImageViewCreateInfo-image-04972
If **image** was created with a **samples** value not equal to **VK_SAMPLE_COUNT_1_BIT** then **viewType** **must** be either **VK_IMAGE_VIEW_TYPE_2D** or **VK_IMAGE_VIEW_TYPE_2D_ARRAY**
- VUID-VkImageViewCreateInfo-image-04441
image **must** have been created with a **usage** value containing at least one of the usages defined in the [valid image usage](#) list for image views

- VUID-VkImageViewCreateInfo-None-02273
The **format features** of the resultant image view **must** contain at least one bit
- VUID-VkImageViewCreateInfo-usage-02274
If **usage** contains **VK_IMAGE_USAGE_SAMPLED_BIT**, then the **format features** of the resultant image view **must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT**
- VUID-VkImageViewCreateInfo-usage-02275
If **usage** contains **VK_IMAGE_USAGE_STORAGE_BIT**, then the image view's **format features must** contain **VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT**
- VUID-VkImageViewCreateInfo-usage-02276
If **usage** contains **VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT**, then the image view's **format features must** contain **VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT**
- VUID-VkImageViewCreateInfo-usage-02277
If **usage** contains **VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT**, then the image view's **format features must** contain **VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT**
- VUID-VkImageViewCreateInfo-image-08333
If **image** was created with **VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR** and **usage** contains **VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR**, then the image view's **format features must** contain **VK_FORMAT_FEATURE_VIDEO_DECODE_OUTPUT_BIT_KHR**
- VUID-VkImageViewCreateInfo-image-08334
If **image** was created with **VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR** and **usage** contains **VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR**, then the image view's **format features must** contain **VK_FORMAT_FEATURE_VIDEO_DECODE_DPB_BIT_KHR**
- VUID-VkImageViewCreateInfo-image-08335
If **image** was created with **VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR**, then **usage must** not include **VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR**
- VUID-VkImageViewCreateInfo-image-08336
If **image** was created with **VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR** and **usage** contains **VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR**, then the image view's **format features must** contain **VK_FORMAT_FEATURE_VIDEO_ENCODE_INPUT_BIT_KHR**
- VUID-VkImageViewCreateInfo-image-08337
If **image** was created with **VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR** and **usage** contains **VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR**, then the image view's **format features must** contain **VK_FORMAT_FEATURE_VIDEO_ENCODE_DPB_BIT_KHR**
- VUID-VkImageViewCreateInfo-image-08338
If **image** was created with **VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR**, then **usage must** not include **VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR**
- VUID-VkImageViewCreateInfo-usage-08932
If **usage** contains **VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT**,

then the image view's **format features must** contain at least one of
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT or
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
- VUID-VkImageViewCreateInfo-subresourceRange-01478

`subresourceRange.baseMipLevel` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `image` was created

- VUID-VkImageViewCreateInfo-subresourceRange-01718

If `subresourceRange.levelCount` is not `VK_REMAINING_MIP_LEVELS`, `subresourceRange.baseMipLevel + subresourceRange.levelCount` **must** be less than or equal to the `mipLevels` specified in `VkImageCreateInfo` when `image` was created

- VUID-VkImageViewCreateInfo-image-01482

If `image` is not a 3D image created with `VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT` set, or `viewType` is not `VK_IMAGE_VIEW_TYPE_2D` or `VK_IMAGE_VIEW_TYPE_2D_ARRAY`, `subresourceRange.baseArrayLayer` **must** be less than the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created

- VUID-VkImageViewCreateInfo-subresourceRange-01483

If `subresourceRange.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `image` is not a 3D image created with `VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT` set, or `viewType` is not `VK_IMAGE_VIEW_TYPE_2D` or `VK_IMAGE_VIEW_TYPE_2D_ARRAY`, `subresourceRange.layerCount` **must** be non-zero and `subresourceRange.baseArrayLayer + subresourceRange.layerCount` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created

- VUID-VkImageViewCreateInfo-image-02724

If `image` is a 3D image created with `VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT` set, and `viewType` is `VK_IMAGE_VIEW_TYPE_2D` or `VK_IMAGE_VIEW_TYPE_2D_ARRAY`, `subresourceRange.baseArrayLayer` **must** be less than the depth computed from `baseMipLevel` and `extent.depth` specified in `VkImageCreateInfo` when `image` was created, according to the formula defined in [Image Mip Level Sizing](#)

- VUID-VkImageViewCreateInfo-subresourceRange-02725

If `subresourceRange.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `image` is a 3D image created with `VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT` set, and `viewType` is `VK_IMAGE_VIEW_TYPE_2D` or `VK_IMAGE_VIEW_TYPE_2D_ARRAY`, `subresourceRange.layerCount` **must** be non-zero and `subresourceRange.baseArrayLayer + subresourceRange.layerCount` **must** be less than or equal to the depth computed from `baseMipLevel` and `extent.depth` specified in `VkImageCreateInfo` when `image` was created, according to the formula defined in [Image Mip Level Sizing](#)

- VUID-VkImageViewCreateInfo-image-01761

If `image` was created with the `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT` flag, but without the `VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT` flag, and if the `format` of the `image` is not a [multi-planar](#) format, `format` **must** be compatible with the `format` used to create `image`, as defined in [Format Compatibility Classes](#)

- VUID-VkImageViewCreateInfo-image-01583

If `image` was created with the `VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT` flag, `format` **must** be compatible with, or **must** be an uncompressed format that is size-compatible with, the `format` used to create `image`

- VUID-VkImageViewCreateInfo-image-07072

If `image` was created with the `VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT` flag and `format` is a non-compressed format, the `levelCount` member of `subresourceRange` **must** be 1

- VUID-VkImageViewCreateInfo-image-09487
If **image** was created with the **VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT** flag, the **VkPhysicalDeviceMaintenance6PropertiesKHR::blockTexelViewCompatibleMultipleLayers** property is not **VK_TRUE**, and **format** is a non-compressed format, then the **layerCount** member of **subresourceRange** **must** be 1
- VUID-VkImageViewCreateInfo-pNext-01585
If a **VkImageFormatListCreateInfo** structure was included in the **pNext** chain of the **VkImageCreateInfo** structure used when creating **image** and **VkImageFormatListCreateInfo::viewFormatCount** is not zero then **format** **must** be one of the formats in **VkImageFormatListCreateInfo::pViewFormats**
- VUID-VkImageViewCreateInfo-image-01586
If **image** was created with the **VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT** flag, if the **format** of the **image** is a **multi-planar** format, and if **subresourceRange.aspectMask** is one of the **multi-planar aspect mask** bits, then **format** **must** be compatible with the **VkFormat** for the plane of the **image format** indicated by **subresourceRange.aspectMask**, as defined in **Compatible Formats of Planes of Multi-Planar Formats**
- VUID-VkImageViewCreateInfo-subresourceRange-07818
subresourceRange.aspectMask **must** only have at most 1 valid **multi-planar aspect mask** bit
- VUID-VkImageViewCreateInfo-image-01762
If **image** was not created with the **VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT** flag, or if the **format** of the **image** is a **multi-planar** format and if **subresourceRange.aspectMask** is **VK_IMAGE_ASPECT_COLOR_BIT**, **format** **must** be identical to the **format** used to create **image**
- VUID-VkImageViewCreateInfo-format-06415
If the image view **requires a sampler Y'C_BC_R conversion** and **usage** contains **VK_IMAGE_USAGE_SAMPLED_BIT**, then the **pNext** chain **must** include a **VkSamplerYcbcrConversionInfo** structure with a conversion value other than **VK_NULL_HANDLE**
- VUID-VkImageViewCreateInfo-format-04714
If **format** has a **_422** or **_420** suffix then **image** **must** have been created with a width that is a multiple of 2
- VUID-VkImageViewCreateInfo-format-04715
If **format** has a **_420** suffix then **image** **must** have been created with a height that is a multiple of 2
- VUID-VkImageViewCreateInfo-pNext-01970
If the **pNext** chain includes a **VkSamplerYcbcrConversionInfo** structure with a **conversion** value other than **VK_NULL_HANDLE**, all members of **components** **must** have the **identity swizzle**
- VUID-VkImageViewCreateInfo-pNext-06658
If the **pNext** chain includes a **VkSamplerYcbcrConversionInfo** structure with a **conversion** value other than **VK_NULL_HANDLE**, **format** **must** be the same used in **VkSamplerYcbcrConversionCreateInfo::format**
- VUID-VkImageViewCreateInfo-image-01020
If **image** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object

- VUID-VkImageViewCreateInfo-subResourceRange-01021
`viewType` **must** be compatible with the type of `image` as shown in the [view type compatibility table](#)
- VUID-VkImageViewCreateInfo-image-02086
 If `image` was created with `usage` containing `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`, `viewType` **must** be `VK_IMAGE_VIEW_TYPE_2D` or `VK_IMAGE_VIEW_TYPE_2D_ARRAY`
- VUID-VkImageViewCreateInfo-usage-04550
 If the `attachmentFragmentShadingRate` feature is enabled, and the `usage` for the image view includes `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`, then the image view's `format` `features` **must** contain `VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-VkImageViewCreateInfo-usage-04551
 If the `attachmentFragmentShadingRate` feature is enabled, the `usage` for the image view includes `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`, and `layeredShadingRateAttachments` is `VK_FALSE`, `subresourceRange.layerCount` **must** be 1
- VUID-VkImageViewCreateInfo-pNext-02661
 If the `pNext` chain includes a `VkImageViewUsageCreateInfo` structure, its `usage` member **must** not include any bits that were not set in the `usage` member of the `VkImageCreateInfo` structure used to create `image`
- VUID-VkImageViewCreateInfo-imageViewType-04973
 If `viewType` is `VK_IMAGE_VIEW_TYPE_1D`, `VK_IMAGE_VIEW_TYPE_2D`, or `VK_IMAGE_VIEW_TYPE_3D`; and `subresourceRange.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, then `subresourceRange.layerCount` **must** be 1
- VUID-VkImageViewCreateInfo-imageViewType-04974
 If `viewType` is `VK_IMAGE_VIEW_TYPE_1D`, `VK_IMAGE_VIEW_TYPE_2D`, or `VK_IMAGE_VIEW_TYPE_3D`; and `subresourceRange.layerCount` is `VK_REMAINING_ARRAY_LAYERS`, then the remaining number of layers **must** be 1
- VUID-VkImageViewCreateInfo-viewType-02960
 If `viewType` is `VK_IMAGE_VIEW_TYPE_CUBE` and `subresourceRange.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `subresourceRange.layerCount` **must** be 6
- VUID-VkImageViewCreateInfo-viewType-02961
 If `viewType` is `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY` and `subresourceRange.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `subresourceRange.layerCount` **must** be a multiple of 6
- VUID-VkImageViewCreateInfo-viewType-02962
 If `viewType` is `VK_IMAGE_VIEW_TYPE_CUBE` and `subresourceRange.layerCount` is `VK_REMAINING_ARRAY_LAYERS`, the remaining number of layers **must** be 6
- VUID-VkImageViewCreateInfo-viewType-02963
 If `viewType` is `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY` and `subresourceRange.layerCount` is `VK_REMAINING_ARRAY_LAYERS`, the remaining number of layers **must** be a multiple of 6
- VUID-VkImageViewCreateInfo-imageViewFormatSwizzle-04465
 If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::imageViewFormatSwizzle` is `VK_FALSE`, all

elements of **components** **must** have the **identity swizzle**

- VUID-VkImageViewCreateInfo-imageViewFormatReinterpretation-04466

If the **VK_KHR_portability_subset** extension is enabled, and **VkPhysicalDevicePortabilitySubsetFeaturesKHR::imageViewFormatReinterpretation** is **VK_FALSE**, the **VkFormat** in **format** **must** not contain a different number of components, or a different number of bits in each component, than the format of the **VkImage** in **image**

- VUID-VkImageViewCreateInfo-image-04817

If **image** was created with **usage** containing **VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR**, **VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR**, or **VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR**, then the **viewType** **must** be **VK_IMAGE_VIEW_TYPE_2D** or **VK_IMAGE_VIEW_TYPE_2D_ARRAY**

- VUID-VkImageViewCreateInfo-image-04818

If **image** was created with **usage** containing **VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR**, **VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR**, or **VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR**, then the **viewType** **must** be **VK_IMAGE_VIEW_TYPE_2D** or **VK_IMAGE_VIEW_TYPE_2D_ARRAY**

- VUID-VkImageViewCreateInfo-subresourceRange-09594

subresourceRange.aspectMask **must** be valid for the **format** the **image** was created with

Valid Usage (Implicit)

- VUID-VkImageViewCreateInfo-sType-sType

sType **must** be **VK_STRUCTURE_TYPE_IMAGE_VIEW_CREATE_INFO**

- VUID-VkImageViewCreateInfo-pNext-pNext

Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either **NULL** or a pointer to a valid instance of **VkImageViewASTCDecodeModeEXT**, **VkImageViewMinLodCreateInfoEXT**, **VkImageViewUsageCreateInfo**, or **VkSamplerYcbcrConversionInfo**

- VUID-VkImageViewCreateInfo-sType-unique

The **sType** value of each struct in the **pNext** chain **must** be unique

- VUID-VkImageViewCreateInfo-flags-zeroBitmask

flags **must** be 0

- VUID-VkImageViewCreateInfo-image-parameter

image **must** be a valid **VkImage** handle

- VUID-VkImageViewCreateInfo-viewType-parameter

viewType **must** be a valid **VkImageViewType** value

- VUID-VkImageViewCreateInfo-format-parameter

format **must** be a valid **VkFormat** value

- VUID-VkImageViewCreateInfo-components-parameter

components **must** be a valid **VkComponentMapping** structure

- VUID-VkImageViewCreateInfo-subresourceRange-parameter

subresourceRange **must** be a valid **VkImageSubresourceRange** structure

Bits which **can** be set in `VkImageViewCreateInfo::flags`, specifying additional parameters of an image view, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkImageViewCreateFlagBits {
} VkImageViewCreateFlagBits;
```

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkImageViewCreateFlags;
```

`VkImageViewCreateFlags` is a bitmask type for setting a mask of zero or more `VkImageViewCreateFlagBits`.

The set of usages for the created image view **can** be restricted compared to the parent image's `usage` flags by adding a `VkImageViewUsageCreateInfo` structure to the `pNext` chain of `VkImageViewCreateInfo`.

The `VkImageViewUsageCreateInfo` structure is defined as:

```
typedef struct VkImageViewUsageCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkImageUsageFlags  usage;
} VkImageViewUsageCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_maintenance2
typedef VkImageViewUsageCreateInfo VkImageViewUsageCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `usage` is a bitmask of `VkImageUsageFlagBits` specifying allowed usages of the image view.

When this structure is chained to `VkImageViewCreateInfo` the `usage` field overrides the implicit `usage` parameter inherited from image creation time and its value is used instead for the purposes of determining the valid usage conditions of `VkImageViewCreateInfo`.

Valid Usage (Implicit)

- VUID-VkImageViewUsageCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_CREATE_INFO`
- VUID-VkImageViewUsageCreateInfo-usage-parameter
`usage` **must** be a valid combination of `VkImageUsageFlagBits` values

- VUID-VkImageViewUsageCreateInfo-usage-requiredbitmask
usage must not be 0

The `VkImageSubresourceRange` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageSubresourceRange {
    VkImageAspectFlags    aspectMask;
    uint32_t              baseMipLevel;
    uint32_t              levelCount;
    uint32_t              baseArrayLayer;
    uint32_t              layerCount;
} VkImageSubresourceRange;
```

- **aspectMask** is a bitmask of `VkImageAspectFlagBits` specifying which aspect(s) of the image are included in the view.
- **baseMipLevel** is the first mipmap level accessible to the view.
- **levelCount** is the number of mipmap levels (starting from **baseMipLevel**) accessible to the view.
- **baseArrayLayer** is the first array layer accessible to the view.
- **layerCount** is the number of array layers (starting from **baseArrayLayer**) accessible to the view.

The number of mipmap levels and array layers **must** be a subset of the image subresources in the image. If an application wants to use all mip levels or layers in an image after the **baseMipLevel** or **baseArrayLayer**, it **can** set **levelCount** and **layerCount** to the special values `VK_REMAINING_MIP_LEVELS` and `VK_REMAINING_ARRAY_LAYERS` without knowing the exact number of mip levels or layers.

For cube and cube array image views, the layers of the image view starting at **baseArrayLayer** correspond to faces in the order +X, -X, +Y, -Y, +Z, -Z. For cube arrays, each set of six sequential layers is a single cube, so the number of cube maps in a cube map array view is $\text{layerCount} / 6$, and image array layer (**baseArrayLayer** + i) is face index (i mod 6) of cube $i / 6$. If the number of layers in the view, whether set explicitly in **layerCount** or implied by `VK_REMAINING_ARRAY_LAYERS`, is not a multiple of 6, the last cube map in the array **must** not be accessed.

aspectMask **must** be only `VK_IMAGE_ASPECT_COLOR_BIT`, `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT` if **format** is a color, depth-only or stencil-only format, respectively, except if **format** is a **multi-planar format**. If using a depth/stencil format with both depth and stencil components, **aspectMask** **must** include at least one of `VK_IMAGE_ASPECT_DEPTH_BIT` and `VK_IMAGE_ASPECT_STENCIL_BIT`, and **can** include both.

When the `VkImageSubresourceRange` structure is used to select a subset of the slices of a 3D image's mip level in order to create a 2D or 2D array image view of a 3D image created with `VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT`, **baseArrayLayer** and **layerCount** specify the first slice index and the number of slices to include in the created image view. Such an image view **can** be used as a framebuffer attachment that refers only to the specified range of slices of the selected mip level. However, any layout transitions performed on such an attachment view during a render pass instance still apply to the entire subresource referenced which includes all the slices of the selected

mip level.

When using an image view of a depth/stencil image to populate a descriptor set (e.g. for sampling in the shader, or for use as an input attachment), the **aspectMask** **must** only include one bit, which selects whether the image view is used for depth reads (i.e. using a floating-point sampler or input attachment in the shader) or stencil reads (i.e. using an unsigned integer sampler or input attachment in the shader). When an image view of a depth/stencil image is used as a depth/stencil framebuffer attachment, the **aspectMask** is ignored and both depth and stencil image subresources are used.

When creating a **VkImageView**, if **sampler Y'C_BC_R conversion** is enabled in the sampler, the **aspectMask** of a **subresourceRange** used by the **VkImageView** **must** be **VK_IMAGE_ASPECT_COLOR_BIT**.

When creating a **VkImageView**, if **sampler Y'C_BC_R conversion** is not enabled in the sampler and the image **format** is **multi-planar**, the image **must** have been created with **VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT**, and the **aspectMask** of the **VkImageView**'s **subresourceRange** **must** be **VK_IMAGE_ASPECT_PLANE_0_BIT**, **VK_IMAGE_ASPECT_PLANE_1_BIT** or **VK_IMAGE_ASPECT_PLANE_2_BIT**.

Valid Usage

- VUID-VkImageSubresourceRange-levelCount-01720
If **levelCount** is not **VK_REMAINING_MIP_LEVELS**, it **must** be greater than 0
- VUID-VkImageSubresourceRange-layerCount-01721
If **layerCount** is not **VK_REMAINING_ARRAY_LAYERS**, it **must** be greater than 0
- VUID-VkImageSubresourceRange-aspectMask-01670
If **aspectMask** includes **VK_IMAGE_ASPECT_COLOR_BIT**, then it **must** not include any of **VK_IMAGE_ASPECT_PLANE_0_BIT**, **VK_IMAGE_ASPECT_PLANE_1_BIT**, or **VK_IMAGE_ASPECT_PLANE_2_BIT**
- VUID-VkImageSubresourceRange-aspectMask-02278
aspectMask **must** not include **VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT** for any index *i*

Valid Usage (Implicit)

- VUID-VkImageSubresourceRange-aspectMask-parameter
aspectMask **must** be a valid combination of **VkImageAspectFlagBits** values
- VUID-VkImageSubresourceRange-aspectMask-requiredbitmask
aspectMask **must** not be 0

Bits which **can** be set in an aspect mask to specify aspects of an image for purposes such as identifying a subresource, are:

```

// Provided by VK_VERSION_1_0
typedef enum VkImageAspectFlagBits {
    VK_IMAGE_ASPECT_COLOR_BIT = 0x00000001,
    VK_IMAGE_ASPECT_DEPTH_BIT = 0x00000002,
    VK_IMAGE_ASPECT_STENCIL_BIT = 0x00000004,
    VK_IMAGE_ASPECT_METADATA_BIT = 0x00000008,
    VK_IMAGE_ASPECT_PLANE_0_BIT = 0x00000010,
    VK_IMAGE_ASPECT_PLANE_1_BIT = 0x00000020,
    VK_IMAGE_ASPECT_PLANE_2_BIT = 0x00000040,
    VK_IMAGE_ASPECT_NONE = 0,
    // Provided by VK_EXT_image_drm_format_modifier
    VK_IMAGE_ASPECT_MEMORY_PLANE_0_BIT_EXT = 0x00000080,
    // Provided by VK_EXT_image_drm_format_modifier
    VK_IMAGE_ASPECT_MEMORY_PLANE_1_BIT_EXT = 0x00000100,
    // Provided by VK_EXT_image_drm_format_modifier
    VK_IMAGE_ASPECT_MEMORY_PLANE_2_BIT_EXT = 0x00000200,
    // Provided by VK_EXT_image_drm_format_modifier
    VK_IMAGE_ASPECT_MEMORY_PLANE_3_BIT_EXT = 0x00000400,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_IMAGE_ASPECT_PLANE_0_BIT_KHR = VK_IMAGE_ASPECT_PLANE_0_BIT,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_IMAGE_ASPECT_PLANE_1_BIT_KHR = VK_IMAGE_ASPECT_PLANE_1_BIT,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_IMAGE_ASPECT_PLANE_2_BIT_KHR = VK_IMAGE_ASPECT_PLANE_2_BIT,
    // Provided by VK_KHR_maintenance4
    VK_IMAGE_ASPECT_NONE_KHR = VK_IMAGE_ASPECT_NONE,
} VkImageAspectFlagBits;

```

- **VK_IMAGE_ASPECT_NONE** specifies no image aspect, or the image aspect is not applicable.
- **VK_IMAGE_ASPECT_COLOR_BIT** specifies the color aspect.
- **VK_IMAGE_ASPECT_DEPTH_BIT** specifies the depth aspect.
- **VK_IMAGE_ASPECT_STENCIL_BIT** specifies the stencil aspect.
- **VK_IMAGE_ASPECT_METADATA_BIT** specifies the metadata aspect used for [sparse resource](#) operations.
- **VK_IMAGE_ASPECT_PLANE_0_BIT** specifies plane 0 of a *multi-planar* image format.
- **VK_IMAGE_ASPECT_PLANE_1_BIT** specifies plane 1 of a *multi-planar* image format.
- **VK_IMAGE_ASPECT_PLANE_2_BIT** specifies plane 2 of a *multi-planar* image format.
- **VK_IMAGE_ASPECT_MEMORY_PLANE_0_BIT_EXT** specifies *memory plane 0*.
- **VK_IMAGE_ASPECT_MEMORY_PLANE_1_BIT_EXT** specifies *memory plane 1*.
- **VK_IMAGE_ASPECT_MEMORY_PLANE_2_BIT_EXT** specifies *memory plane 2*.
- **VK_IMAGE_ASPECT_MEMORY_PLANE_3_BIT_EXT** specifies *memory plane 3*.

```

// Provided by VK_VERSION_1_0
typedef VkFlags VkImageAspectFlags;

```

VkImageAspectFlags is a bitmask type for setting a mask of zero or more **VkImageAspectFlagBits**.

The **VkComponentMapping** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkComponentMapping {
    VkComponentSwizzle    r;
    VkComponentSwizzle    g;
    VkComponentSwizzle    b;
    VkComponentSwizzle    a;
} VkComponentMapping;
```

- **r** is a **VkComponentSwizzle** specifying the component value placed in the R component of the output vector.
- **g** is a **VkComponentSwizzle** specifying the component value placed in the G component of the output vector.
- **b** is a **VkComponentSwizzle** specifying the component value placed in the B component of the output vector.
- **a** is a **VkComponentSwizzle** specifying the component value placed in the A component of the output vector.

Valid Usage (Implicit)

- VUID-VkComponentMapping-r-parameter
r must be a valid **VkComponentSwizzle** value
- VUID-VkComponentMapping-g-parameter
g must be a valid **VkComponentSwizzle** value
- VUID-VkComponentMapping-b-parameter
b must be a valid **VkComponentSwizzle** value
- VUID-VkComponentMapping-a-parameter
a must be a valid **VkComponentSwizzle** value

Possible values of the members of **VkComponentMapping**, specifying the component values placed in each component of the output vector, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkComponentSwizzle {
    VK_COMPONENT_SWIZZLE_IDENTITY = 0,
    VK_COMPONENT_SWIZZLE_ZERO = 1,
    VK_COMPONENT_SWIZZLE_ONE = 2,
    VK_COMPONENT_SWIZZLE_R = 3,
    VK_COMPONENT_SWIZZLE_G = 4,
    VK_COMPONENT_SWIZZLE_B = 5,
    VK_COMPONENT_SWIZZLE_A = 6,
} VkComponentSwizzle;
```

- **VK_COMPONENT_SWIZZLE_IDENTITY** specifies that the component is set to the identity swizzle.
- **VK_COMPONENT_SWIZZLE_ZERO** specifies that the component is set to zero.
- **VK_COMPONENT_SWIZZLE_ONE** specifies that the component is set to either 1 or 1.0, depending on whether the type of the image view format is integer or floating-point respectively, as determined by the [Format Definition](#) section for each [VkFormat](#).
- **VK_COMPONENT_SWIZZLE_R** specifies that the component is set to the value of the R component of the image.
- **VK_COMPONENT_SWIZZLE_G** specifies that the component is set to the value of the G component of the image.
- **VK_COMPONENT_SWIZZLE_B** specifies that the component is set to the value of the B component of the image.
- **VK_COMPONENT_SWIZZLE_A** specifies that the component is set to the value of the A component of the image.

Setting the identity swizzle on a component is equivalent to setting the identity mapping on that component. That is:

*Table 12. Component Mappings Equivalent To **VK_COMPONENT_SWIZZLE_IDENTITY***

Component	Identity Mapping
components.r	VK_COMPONENT_SWIZZLE_R
components.g	VK_COMPONENT_SWIZZLE_G
components.b	VK_COMPONENT_SWIZZLE_B
components.a	VK_COMPONENT_SWIZZLE_A

If the **pNext** chain includes a **VkImageViewASTCDecodeModeEXT** structure, then that structure includes a parameter specifying the decode mode for image views using ASTC compressed formats.

The **VkImageViewASTCDecodeModeEXT** structure is defined as:


```
// Provided by VK_EXT_astc_decode_mode
typedef struct VkImageViewASTCDecodeModeEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkFormat            decodeMode;
} VkImageViewASTCDecodeModeEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `decodeMode` is the intermediate format used to decode ASTC compressed formats.

Valid Usage

- VUID-VkImageViewASTCDecodeModeEXT-decodeMode-02230
`decodeMode` **must** be one of `VK_FORMAT_R16G16B16A16_SFLOAT`, `VK_FORMAT_R8G8B8A8_UNORM`, or `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`
- VUID-VkImageViewASTCDecodeModeEXT-decodeMode-02231
If the `decodeModeSharedExponent` feature is not enabled, `decodeMode` **must** not be `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`
- VUID-VkImageViewASTCDecodeModeEXT-decodeMode-02232
If `decodeMode` is `VK_FORMAT_R8G8B8A8_UNORM` the image view **must** not include blocks using any of the ASTC HDR modes
- VUID-VkImageViewASTCDecodeModeEXT-format-04084
`format` of the image view **must** be one of the [ASTC Compressed Image Formats](#)

If `format` uses sRGB encoding then the `decodeMode` has no effect.

Valid Usage (Implicit)

- VUID-VkImageViewASTCDecodeModeEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_VIEW_ASTC_DECODE_MODE_EXT`
- VUID-VkImageViewASTCDecodeModeEXT-decodeMode-parameter
`decodeMode` **must** be a valid `VkFormat` value

To destroy an image view, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyImageView(
    VkDevice            device,
    VkImageView         imageView,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the image view.
- **imageView** is the image view to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyImageView-imageView-01026
All submitted commands that refer to **imageView** **must** have completed execution
- VUID-vkDestroyImageView-imageView-01027
If **VkAllocationCallbacks** were provided when **imageView** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyImageView-imageView-01028
If no **VkAllocationCallbacks** were provided when **imageView** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyImageView-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroyImageView-imageView-parameter
If **imageView** is not **VK_NULL_HANDLE**, **imageView** **must** be a valid **VkImageView** handle
- VUID-vkDestroyImageView-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkDestroyImageView-imageView-parent
If **imageView** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **imageView** **must** be externally synchronized

12.5.1. Image View Format Features

Valid uses of a **VkImageView** **may** depend on the image view's *format features*, defined below. Such constraints are documented in the affected valid usage statement.

- If Vulkan 1.3 is supported or the **VK_KHR_format_feature_flags2** extension is supported, and **VkImageViewCreateInfo::image** was created with **VK_IMAGE_TILING_LINEAR**, then the image view's set of *format features* is the value of **VkFormatProperties3::linearTilingFeatures** found by calling **vkGetPhysicalDeviceFormatProperties2KHR** on the same **format** as **VkImageViewCreateInfo::format**.

- If Vulkan 1.3 is not supported and the `VK_KHR_format_feature_flags2` extension is not supported, and `VkImageViewCreateInfo::image` was created with `VK_IMAGE_TILING_LINEAR`, then the image view's set of *format features* is the union of the value of `VkFormatProperties::linearTilingFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` on the same `format` as `VkImageViewCreateInfo::format`, with:
 - `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT` if the format is a depth/stencil format and the image view features also contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT`.
 - `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT` if the format is one of the `extended storage formats` and `shaderStorageImageReadWithoutFormat` is enabled on the device.
 - `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT` if the format is one of the `extended storage formats` and `shaderStorageImageWriteWithoutFormat` is enabled on the device.
- If Vulkan 1.3 is supported or the `VK_KHR_format_feature_flags2` extension is supported, and `VkImageViewCreateInfo::image` was created with `VK_IMAGE_TILING_OPTIMAL`, then the image view's set of *format features* is the value of `VkFormatProperties::optimalTilingFeatures` or `VkFormatProperties3::optimalTilingFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` or `vkGetPhysicalDeviceImageFormatProperties2KHR` on the same `format` as `VkImageViewCreateInfo::format`.
- If Vulkan 1.3 is not supported and the `VK_KHR_format_feature_flags2` extension is not supported, and `VkImageViewCreateInfo::image` was created with `VK_IMAGE_TILING_OPTIMAL`, then the image view's set of *format features* is the union of the value of `VkFormatProperties::optimalTilingFeatures` found by calling `vkGetPhysicalDeviceFormatProperties` on the same `format` as `VkImageViewCreateInfo::format`, with:
 - `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT` if the format is a depth/stencil format and the image view features also contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT`.
 - `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT` if the format is one of the `extended storage formats` and `shaderStorageImageReadWithoutFormat` is enabled on the device.
 - `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT` if the format is one of the `extended storage formats` and `shaderStorageImageWriteWithoutFormat` is enabled on the device.
- If `VkImageViewCreateInfo::image` was created with `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then:
 - The image's DRM format modifier is the value of `VkImageDrmFormatModifierPropertiesEXT::drmFormatModifier` found by calling `vkGetImageDrmFormatModifierPropertiesEXT`.
 - Let `VkDrmFormatModifierPropertiesListEXT::pDrmFormatModifierProperties` be the array found by calling `vkGetPhysicalDeviceFormatProperties2KHR` on the same `format` as `VkImageViewCreateInfo::format`.
 - Let `VkDrmFormatModifierPropertiesEXT prop` be the array element whose `drmFormatModifier` member is the value of the image's DRM format modifier.
 - Then the image view's set of *format features* is `prop::drmFormatModifierTilingFeatures`.

The `VkImageViewMinLodCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_image_view_min_lod
typedef struct VkImageViewMinLodCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    float              minLod;
} VkImageViewMinLodCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `minLod` is the value to clamp the minimum LOD accessible by this `VkImageView`.

If the `pNext` chain includes a `VkImageViewMinLodCreateInfoEXT` structure, then that structure includes a parameter specifying a value to clamp the minimum LOD value during [Image Level\(s\) Selection](#), [Texel Gathering](#) and [Integer Texel Coordinate Operations](#).

If the image view contains `VkImageViewMinLodCreateInfoEXT` and it is used as part of a sampling operation:

`minLodFloatimageView = minLod`

otherwise:

`minLodFloatimageView = 0.0`

An integer variant of this parameter is also defined for sampling operations which access integer mipmap levels:

`minLodIntegerimageView = ⌊minLodFloatimageView⌋`

Valid Usage

- VUID-VkImageViewMinLodCreateInfoEXT-minLod-06455
If the `minLod` feature is not enabled, `minLod` **must** be `0.0`
- VUID-VkImageViewMinLodCreateInfoEXT-minLod-06456
`minLod` **must** be less or equal to the index of the last mipmap level accessible to the view

Valid Usage (Implicit)

- VUID-VkImageViewMinLodCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_VIEW_MIN_LOD_CREATE_INFO_EXT`

12.6. Acceleration Structures

Acceleration structures are opaque data structures that are built by the implementation to more efficiently perform spatial queries on the provided geometric data. For these extensions, an

acceleration structure is either a top-level acceleration structure containing a set of bottom-level acceleration structures or a bottom-level acceleration structure containing either a set of axis-aligned bounding boxes for custom geometry or a set of triangles.

Each instance in the top-level acceleration structure contains a reference to a bottom-level acceleration structure as well as an instance transform plus information required to index into the shader bindings. The top-level acceleration structure is what is bound to the acceleration descriptor, for example to trace inside the shader in the ray tracing pipeline.

Acceleration structures for the [VK_KHR_acceleration_structure extension](#) are represented by [VkAccelerationStructureKHR](#) handles:

```
// Provided by VK_KHR_acceleration_structure
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkAccelerationStructureKHR)
```

To create an acceleration structure, call:

```
// Provided by VK_KHR_acceleration_structure
VkResult vkCreateAccelerationStructureKHR(
    VkDevice device,
    const VkAccelerationStructureCreateInfoKHR* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkAccelerationStructureKHR* pAccelerationStructure);
```

- [device](#) is the logical device that creates the acceleration structure object.
- [pCreateInfo](#) is a pointer to a [VkAccelerationStructureCreateInfoKHR](#) structure containing parameters affecting creation of the acceleration structure.
- [pAllocator](#) controls host memory allocation as described in the [Memory Allocation](#) chapter.
- [pAccelerationStructure](#) is a pointer to a [VkAccelerationStructureKHR](#) handle in which the resulting acceleration structure object is returned.

Similar to other objects in Vulkan, the acceleration structure creation merely creates an object with a specific “shape”. The type and quantity of geometry that can be built into an acceleration structure is determined by the parameters of [VkAccelerationStructureCreateInfoKHR](#).

The acceleration structure data is stored in the object referred to by [VkAccelerationStructureCreateInfoKHR::buffer](#). Once memory has been bound to that buffer, it **must** be populated by acceleration structure build or acceleration structure copy commands such as [vkCmdBuildAccelerationStructuresKHR](#), [vkBuildAccelerationStructuresKHR](#), [vkCmdCopyAccelerationStructureKHR](#), and [vkCopyAccelerationStructureKHR](#).

NOTE

The expected usage for a trace capture/replay tool is that it will serialize and later deserialize the acceleration structure data using acceleration structure copy commands. During capture the tool will use [vkCopyAccelerationStructureToMemoryKHR](#) or [vkCmdCopyAccelerationStructureToMemoryKHR](#) with a [mode](#) of

`VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR`, `vkCopyMemoryToAccelerationStructureKHR` and
`vkCmdCopyMemoryToAccelerationStructureKHR` with a `mode` of
`VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR` during replay.

NOTE

Memory does not need to be bound to the underlying buffer when `vkCreateAccelerationStructureKHR` is called.

The input buffers passed to acceleration structure build commands will be referenced by the implementation for the duration of the command. After the command completes, the acceleration structure **may** hold a reference to any acceleration structure specified by an active instance contained therein. Apart from this referencing, acceleration structures **must** be fully self-contained. The application **can** reuse or free any memory which was used by the command as an input or as scratch without affecting the results of ray traversal.

Valid Usage

- VUID-vkCreateAccelerationStructureKHR-accelerationStructure-03611
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkCreateAccelerationStructureKHR-deviceAddress-03488
If `VkAccelerationStructureCreateInfoKHR::deviceAddress` is not zero, the `accelerationStructureCaptureReplay` feature **must** be enabled
- VUID-vkCreateAccelerationStructureKHR-device-03489
If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateAccelerationStructureKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateAccelerationStructureKHR-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkAccelerationStructureCreateInfoKHR` structure
- VUID-vkCreateAccelerationStructureKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateAccelerationStructureKHR-pAccelerationStructure-parameter
`pAccelerationStructure` **must** be a valid pointer to a `VkAccelerationStructureKHR` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR`

The `VkAccelerationStructureCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureCreateInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkAccelerationStructureCreateFlagsKHR createFlags;
    VkBuffer                  buffer;
    VkDeviceSize              offset;
    VkDeviceSize              size;
    VkAccelerationStructureTypeKHR type;
    VkDeviceAddress           deviceAddress;
} VkAccelerationStructureCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `createFlags` is a bitmask of `VkAccelerationStructureCreateFlagBitsKHR` specifying additional creation parameters of the acceleration structure.
- `buffer` is the buffer on which the acceleration structure will be stored.
- `offset` is an offset in bytes from the base address of the buffer at which the acceleration structure will be stored, and **must** be a multiple of `256`.
- `size` is the size required for the acceleration structure.
- `type` is a `VkAccelerationStructureTypeKHR` value specifying the type of acceleration structure that will be created.
- `deviceAddress` is the device address requested for the acceleration structure if the `accelerationStructureCaptureReplay` feature is being used. If `deviceAddress` is zero, no specific address is requested.

Applications **should** avoid creating acceleration structures with application-provided addresses and implementation-provided addresses in the same process, to reduce the likelihood of `VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR` errors.

NOTE

The expected usage for this is that a trace capture/replay tool will add the `VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT` flag to all buffers that use `VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT`, and will add

`VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT` to all buffers used as storage for an acceleration structure where `deviceAddress` is not zero. This also means that the tool will need to add `VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT` to memory allocations to allow the flag to be set where the application may not have otherwise required it. During capture the tool will save the queried opaque device addresses in the trace. During replay, the buffers will be created specifying the original address so any address values stored in the trace data will remain valid.

Implementations are expected to separate such buffers in the GPU address space so normal allocations will avoid using these addresses. Apps/tools should avoid mixing app-provided and implementation-provided addresses for buffers created with `VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT`, to avoid address space allocation conflicts.

Applications **should** create an acceleration structure with a specific `VkAccelerationStructureTypeKHR` other than `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR`.

NOTE

`VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR` is intended to be used by API translation layers. This can be used at acceleration structure creation time in cases where the actual acceleration structure type (top or bottom) is not yet known. The actual acceleration structure type must be specified as `VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR` or `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR` when the build is performed.

If the acceleration structure will be the target of a build operation, the required size for an acceleration structure **can** be queried with `vkGetAccelerationStructureBuildSizesKHR`. If the acceleration structure is going to be the target of a compacting copy, `vkCmdWriteAccelerationStructuresPropertiesKHR` or `vkWriteAccelerationStructuresPropertiesKHR` **can** be used to obtain the compacted size required.

Valid Usage

- VUID-VkAccelerationStructureCreateInfoKHR-deviceAddress-03612
If `deviceAddress` is not zero, `createFlags` **must** include `VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR`
- VUID-VkAccelerationStructureCreateInfoKHR-deviceAddress-09488
If `deviceAddress` is not zero, it **must** have been retrieved from an identically created acceleration structure, except for `buffer` and `deviceAddress`
- VUID-VkAccelerationStructureCreateInfoKHR-deviceAddress-09489
If `deviceAddress` is not zero, `buffer` **must** have been created identically to the `buffer` used to create the acceleration structure from which `deviceAddress` was retrieved, except for `VkBufferOpaqueCaptureAddressCreateInfo::opaqueCaptureAddress`
- VUID-VkAccelerationStructureCreateInfoKHR-deviceAddress-09490
If `deviceAddress` is not zero, `buffer` **must** have been created with a `VkBufferOpaqueCaptureAddressCreateInfo::opaqueCaptureAddress` that was retrieved from `vkGetBufferOpaqueCaptureAddressKHR` for the `buffer` that was used to create the

acceleration structure from which `deviceAddress` was retrieved

- VUID-VkAccelerationStructureCreateInfoKHR-createFlags-03613
If `createFlags` includes `VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR`, `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureCaptureReplay` **must** be `VK_TRUE`
- VUID-VkAccelerationStructureCreateInfoKHR-buffer-03614
`buffer` **must** have been created with a `usage` value containing `VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR`
- VUID-VkAccelerationStructureCreateInfoKHR-buffer-03615
`buffer` **must** not have been created with `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT`
- VUID-VkAccelerationStructureCreateInfoKHR-offset-03616
The sum of `offset` and `size` **must** be less than the size of `buffer`
- VUID-VkAccelerationStructureCreateInfoKHR-offset-03734
`offset` **must** be a multiple of 256 bytes

Valid Usage (Implicit)

- VUID-VkAccelerationStructureCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_INFO_KHR`
- VUID-VkAccelerationStructureCreateInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkAccelerationStructureCreateInfoKHR-createFlags-parameter
`createFlags` **must** be a valid combination of `VkAccelerationStructureCreateFlagBitsKHR` values
- VUID-VkAccelerationStructureCreateInfoKHR-buffer-parameter
`buffer` **must** be a valid `VkBuffer` handle
- VUID-VkAccelerationStructureCreateInfoKHR-type-parameter
`type` **must** be a valid `VkAccelerationStructureTypeKHR` value

To get the build sizes for an acceleration structure, call:

```
// Provided by VK_KHR_acceleration_structure
void vkGetAccelerationStructureBuildSizesKHR(
    VkDevice device,
    VkAccelerationStructureBuildTypeKHR buildType,
    const VkAccelerationStructureBuildGeometryInfoKHR* pBuildInfo,
    const uint32_t* pMaxPrimitiveCounts,
    VkAccelerationStructureBuildSizesInfoKHR* pSizeInfo);
```

- `device` is the logical device that will be used for creating the acceleration structure.
- `buildType` defines whether host or device operations (or both) are being queried for.

- `pBuildInfo` is a pointer to a `VkAccelerationStructureBuildGeometryInfoKHR` structure describing parameters of a build operation.
- `pMaxPrimitiveCounts` is a pointer to an array of `pBuildInfo->geometryCount` `uint32_t` values defining the number of primitives built into each geometry.
- `pSizeInfo` is a pointer to a `VkAccelerationStructureBuildSizesInfoKHR` structure which returns the size required for an acceleration structure and the sizes required for the scratch buffers, given the build parameters.

The `srcAccelerationStructure`, `dstAccelerationStructure`, and `mode` members of `pBuildInfo` are ignored. Any `VkDeviceOrHostAddressKHR` or `VkDeviceOrHostAddressConstKHR` members of `pBuildInfo` are ignored by this command, except that the `hostAddress` member of `VkAccelerationStructureGeometryTrianglesDataKHR::transformData` will be examined to check if it is `NULL`.

An acceleration structure created with the `accelerationStructureSize` returned by this command supports any build or update with a `VkAccelerationStructureBuildGeometryInfoKHR` structure and array of `VkAccelerationStructureBuildRangeInfoKHR` structures subject to the following properties:

- The build command is a host build command, and `buildType` is `VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_KHR` or `VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR`
- The build command is a device build command, and `buildType` is `VK_ACCELERATION_STRUCTURE_BUILD_TYPE_DEVICE_KHR` or `VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR`
- For `VkAccelerationStructureBuildGeometryInfoKHR`:
 - Its `type`, and `flags` members are equal to `pBuildInfo->type` and `pBuildInfo->flags`, respectively.
 - `geometryCount` is less than or equal to `pBuildInfo->geometryCount`.
 - For each element of either `pGeometries` or `ppGeometries` at a given index, its `geometryType` member is equal to `pBuildInfo->geometryType`.
 - For each element of either `pGeometries` or `ppGeometries` at a given index, its `flags` member is equal to the corresponding member of the same element in `pBuildInfo`.
 - For each element of either `pGeometries` or `ppGeometries` at a given index, with a `geometryType` member equal to `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, the `vertexFormat` and `indexType` members of `geometry.triangles` are equal to the corresponding members of the same element in `pBuildInfo`.
 - For each element of either `pGeometries` or `ppGeometries` at a given index, with a `geometryType` member equal to `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, the `maxVertex` member of `geometry.triangles` is less than or equal to the corresponding member of the same element in `pBuildInfo`.
 - For each element of either `pGeometries` or `ppGeometries` at a given index, with a `geometryType` member equal to `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if the applicable address in the `transformData` member of `geometry.triangles` is not `NULL`, the corresponding `transformData.hostAddress` parameter in `pBuildInfo` is not `NULL`.

- For each `VkAccelerationStructureBuildRangeInfoKHR` corresponding to the `VkAccelerationStructureBuildGeometryInfoKHR`:
 - Its `primitiveCount` member is less than or equal to the corresponding element of `pMaxPrimitiveCounts`.
 - For each element of either `pGeometries` or `ppGeometries` at a given index, with a `geometryType` member equal to `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if the `pNext` chain contains `VkAccelerationStructureTrianglesOpacityMicromapEXT` the corresponding member of `pBuildInfo` also contains `VkAccelerationStructureTrianglesOpacityMicromapEXT` and with an equivalent `micromap`.

Similarly, the `updateScratchSize` value will support any build command specifying the `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR` mode under the above conditions, and the `buildScratchSize` value will support any build command specifying the `VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR` mode under the above conditions.

Valid Usage

- VUID-vkGetAccelerationStructureBuildSizesKHR-accelerationStructure-08933
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkGetAccelerationStructureBuildSizesKHR-device-03618
If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled
- VUID-vkGetAccelerationStructureBuildSizesKHR-pBuildInfo-03619
If `pBuildInfo->geometryCount` is not 0, `pMaxPrimitiveCounts` **must** be a valid pointer to an array of `pBuildInfo->geometryCount` `uint32_t` values
- VUID-vkGetAccelerationStructureBuildSizesKHR-pBuildInfo-03785
If `pBuildInfo->pGeometries` or `pBuildInfo->ppGeometries` has a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, each `pMaxPrimitiveCounts[i]` **must** be less than or equal to `VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxInstanceCount`

Valid Usage (Implicit)

- VUID-vkGetAccelerationStructureBuildSizesKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetAccelerationStructureBuildSizesKHR-buildType-parameter
`buildType` **must** be a valid `VkAccelerationStructureBuildTypeKHR` value
- VUID-vkGetAccelerationStructureBuildSizesKHR-pBuildInfo-parameter
`pBuildInfo` **must** be a valid pointer to a valid `VkAccelerationStructureBuildGeometryInfoKHR` structure
- VUID-vkGetAccelerationStructureBuildSizesKHR-pMaxPrimitiveCounts-parameter
If `pMaxPrimitiveCounts` is not NULL, `pMaxPrimitiveCounts` **must** be a valid pointer to an array of `pBuildInfo->geometryCount` `uint32_t` values

- VUID-vkGetAccelerationStructureBuildSizesKHR-pSizeInfo-parameter
pSizeInfo **must** be a valid pointer to a [VkAccelerationStructureBuildSizesInfoKHR](#) structure

The [VkAccelerationStructureBuildSizesInfoKHR](#) structure describes the required build sizes for an acceleration structure and scratch buffers and is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureBuildSizesInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceSize        accelerationStructureSize;
    VkDeviceSize        updateScratchSize;
    VkDeviceSize        buildScratchSize;
} VkAccelerationStructureBuildSizesInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **accelerationStructureSize** is the size in bytes required in a [VkAccelerationStructureKHR](#) for a build or update operation.
- **updateScratchSize** is the size in bytes required in a scratch buffer for an update operation.
- **buildScratchSize** is the size in bytes required in a scratch buffer for a build operation.

Valid Usage (Implicit)

- VUID-VkAccelerationStructureBuildSizesInfoKHR-sType-sType
sType **must** be [VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_SIZES_INFO_KHR](#)
- VUID-VkAccelerationStructureBuildSizesInfoKHR-pNext-pNext
pNext **must** be **NULL**

Values which **can** be set in [VkAccelerationStructureCreateInfoKHR::type](#) specifying the type of acceleration structure, are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkAccelerationStructureTypeKHR {
    VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR = 0,
    VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR = 1,
    VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR = 2,
} VkAccelerationStructureTypeKHR;
```

- [VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR](#) is a top-level acceleration structure containing instance data referring to bottom-level acceleration structures.
- [VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR](#) is a bottom-level acceleration structure

containing the AABBs or geometry to be intersected.

- `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR` is an acceleration structure whose type is determined at build time used for special circumstances. In these cases, the acceleration structure type is not known at creation time, but **must** be specified at build time as either top or bottom.

Bits which **can** be set in `VkAccelerationStructureCreateInfoKHR::createFlags`, specifying additional creation parameters for acceleration structures, are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkAccelerationStructureCreateFlagBitsKHR {
    VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR =
0x00000001,
} VkAccelerationStructureCreateFlagBitsKHR;
```

- `VK_ACCELERATION_STRUCTURE_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR` specifies that the acceleration structure's address **can** be saved and reused on a subsequent run.

```
// Provided by VK_KHR_acceleration_structure
typedef VkFlags VkAccelerationStructureCreateFlagsKHR;
```

`VkAccelerationStructureCreateFlagsKHR` is a bitmask type for setting a mask of zero or more `VkAccelerationStructureCreateFlagBitsKHR`.

Bits which **can** be set in `VkAccelerationStructureBuildGeometryInfoKHR::flags` specifying additional parameters for acceleration structure builds, are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkBuildAccelerationStructureFlagBitsKHR {
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR = 0x00000001,
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR = 0x00000002,
    VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_TRACE_BIT_KHR = 0x00000004,
    VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_BUILD_BIT_KHR = 0x00000008,
    VK_BUILD_ACCELERATION_STRUCTURE_LOW_MEMORY_BIT_KHR = 0x00000010,
    // Provided by VK_EXT_opacity_micromap
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_EXT = 0x00000040,
    // Provided by VK_EXT_opacity_micromap
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_EXT = 0x00000080,
    // Provided by VK_EXT_opacity_micromap
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_EXT =
0x00000100,
    // Provided by VK_KHR_ray_tracing_position_fetch
    VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_KHR = 0x00000800,
} VkBuildAccelerationStructureFlagBitsKHR;
```

- `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR` indicates that the specified acceleration

structure **can** be updated with a `mode` of `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR` in `VkAccelerationStructureBuildGeometryInfoKHR`.

- `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR` indicates that the specified acceleration structure **can** act as the source for a copy acceleration structure command with `mode` of `VK_COPY_ACCELERATION_STRUCTURE_MODE_COMPACT_KHR` to produce a compacted acceleration structure.
- `VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_TRACE_BIT_KHR` indicates that the given acceleration structure build **should** prioritize trace performance over build time.
- `VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_BUILD_BIT_KHR` indicates that the given acceleration structure build **should** prioritize build time over trace performance.
- `VK_BUILD_ACCELERATION_STRUCTURE_LOW_MEMORY_BIT_KHR` indicates that this acceleration structure **should** minimize the size of the scratch memory and the final result acceleration structure, potentially at the expense of build time or trace performance.
- `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_EXT` indicates that the opacity micromaps associated with the specified acceleration structure **may** change with an acceleration structure update.
- `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_EXT` indicates that the data of the opacity micromaps associated with the specified acceleration structure **may** change with an acceleration structure update.
- `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_EXT` indicates that the specified acceleration structure **may** be referenced in an instance with `VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_EXT` set.
- `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_KHR` indicates that the specified acceleration structure **can** be used when fetching the vertex positions of a hit triangle.

NOTE

`VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR` and `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR` **may** take more time and memory than a normal build, and so **should** only be used when those features are needed.

NOTE

`VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR` and `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR` are allowed to be used together. In that case, the result of the compaction copy is used as the source of a build with `mode` of `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR` to perform the compacted update.

```
// Provided by VK_KHR_acceleration_structure
typedef VkFlags VkBuildAccelerationStructureFlagsKHR;
```

`VkBuildAccelerationStructureFlagsKHR` is a bitmask type for setting a mask of zero or more `VkBuildAccelerationStructureFlagBitsKHR`.

Geometry types are specified by `VkGeometryTypeKHR`, which takes values:


```
// Provided by VK_KHR_acceleration_structure
typedef enum VkGeometryTypeKHR {
    VK_GEOMETRY_TYPE_TRIANGLES_KHR = 0,
    VK_GEOMETRY_TYPE_AABBS_KHR = 1,
    VK_GEOMETRY_TYPE_INSTANCES_KHR = 2,
} VkGeometryTypeKHR;
```

- `VK_GEOMETRY_TYPE_TRIANGLES_KHR` specifies a geometry type consisting of triangles.
- `VK_GEOMETRY_TYPE_AABBS_KHR` specifies a geometry type consisting of axis-aligned bounding boxes.
- `VK_GEOMETRY_TYPE_INSTANCES_KHR` specifies a geometry type consisting of acceleration structure instances.

Bits specifying additional parameters for geometries in acceleration structure builds, are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkGeometryFlagBitsKHR {
    VK_GEOMETRY_OPAQUE_BIT_KHR = 0x00000001,
    VK_GEOMETRY_NO_DUPLICATE_ANY_HIT_INVOCATION_BIT_KHR = 0x00000002,
} VkGeometryFlagBitsKHR;
```

- `VK_GEOMETRY_OPAQUE_BIT_KHR` indicates that this geometry does not invoke the any-hit shaders even if present in a hit group.
- `VK_GEOMETRY_NO_DUPLICATE_ANY_HIT_INVOCATION_BIT_KHR` indicates that the implementation **must** only call the any-hit shader a single time for each primitive in this geometry. If this bit is absent an implementation **may** invoke the any-hit shader more than once for this geometry.

```
// Provided by VK_KHR_acceleration_structure
typedef VkFlags VkGeometryFlagsKHR;
```

`VkGeometryFlagsKHR` is a bitmask type for setting a mask of zero or more `VkGeometryFlagBitsKHR`.

To destroy an acceleration structure, call:

```
// Provided by VK_KHR_acceleration_structure
void vkDestroyAccelerationStructureKHR(
    VkDevice device,
    VkAccelerationStructureKHR accelerationStructure,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the acceleration structure.
- `accelerationStructure` is the acceleration structure to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-08934
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-02442
All submitted commands that refer to `accelerationStructure` **must** have completed execution
- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-02443
If `VkAllocationCallbacks` were provided when `accelerationStructure` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-02444
If no `VkAllocationCallbacks` were provided when `accelerationStructure` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyAccelerationStructureKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-parameter
If `accelerationStructure` is not `VK_NULL_HANDLE`, `accelerationStructure` **must** be a valid `VkAccelerationStructureKHR` handle
- VUID-vkDestroyAccelerationStructureKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyAccelerationStructureKHR-accelerationStructure-parent
If `accelerationStructure` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `accelerationStructure` **must** be externally synchronized

Possible values of `buildType` in `vkGetAccelerationStructureBuildSizesKHR` are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkAccelerationStructureBuildTypeKHR {
    VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_KHR = 0,
    VK_ACCELERATION_STRUCTURE_BUILD_TYPE_DEVICE_KHR = 1,
    VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR = 2,
} VkAccelerationStructureBuildTypeKHR;
```

- `VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_KHR` requests the memory requirement for operations performed by the host.
- `VK_ACCELERATION_STRUCTURE_BUILD_TYPE_DEVICE_KHR` requests the memory requirement for operations performed by the device.
- `VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR` requests the memory requirement for operations performed by either the host, or the device.

To query the 64-bit device address for an acceleration structure, call:

```
// Provided by VK_KHR_acceleration_structure
VkDeviceAddress vkGetAccelerationStructureDeviceAddressKHR(
    VkDevice device,
    const VkAccelerationStructureDeviceAddressInfoKHR* pInfo);
```

- `device` is the logical device that the acceleration structure was created on.
- `pInfo` is a pointer to a `VkAccelerationStructureDeviceAddressInfoKHR` structure specifying the acceleration structure to retrieve an address for.

The 64-bit return value is an address of the acceleration structure, which can be used for device and shader operations that involve acceleration structures, such as ray traversal and acceleration structure building.

If the acceleration structure was created with a non-zero value of `VkAccelerationStructureCreateInfoKHR::deviceAddress`, the return value will be the same address.

If the acceleration structure was created with a `type` of `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR`, the returned address **must** be consistent with the relative offset to other acceleration structures with `type VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR` allocated with the same `VkBuffer`. That is, the difference in returned addresses between the two **must** be the same as the difference in offsets provided at acceleration structure creation.

The returned address **must** be aligned to 256 bytes.

NOTE

The acceleration structure device address **may** be different from the buffer device address corresponding to the acceleration structure's start offset in its storage buffer for acceleration structure types other than `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR`.

Valid Usage

- VUID-vkGetAccelerationStructureDeviceAddressKHR-accelerationStructure-08935
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkGetAccelerationStructureDeviceAddressKHR-device-03504
If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled

- VUID-vkGetAccelerationStructureDeviceAddressKHR-pInfo-09541
If the buffer on which `pInfo->accelerationStructure` was placed is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkGetAccelerationStructureDeviceAddressKHR-pInfo-09542
The buffer on which `pInfo->accelerationStructure` was placed **must** have been created with the `VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT` usage flag

Valid Usage (Implicit)

- VUID-vkGetAccelerationStructureDeviceAddressKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetAccelerationStructureDeviceAddressKHR-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkAccelerationStructureDeviceAddressInfoKHR` structure

The `VkAccelerationStructureDeviceAddressInfoKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureDeviceAddressInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    VkAccelerationStructureKHR accelerationStructure;
} VkAccelerationStructureDeviceAddressInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `accelerationStructure` specifies the acceleration structure whose address is being queried.

Valid Usage (Implicit)

- VUID-VkAccelerationStructureDeviceAddressInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_DEVICE_ADDRESS_INFO_KHR`
- VUID-VkAccelerationStructureDeviceAddressInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkAccelerationStructureDeviceAddressInfoKHR-accelerationStructure-parameter
`accelerationStructure` **must** be a valid `VkAccelerationStructureKHR` handle

12.7. Micromaps

Micromaps are opaque data structures that are built by the implementation to encode sub-triangle data to be included in an acceleration structure.

Micromaps are represented by `VkMicromapEXT` handles:

```
// Provided by VK_EXT_opacity_micromap
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkMicromapEXT)
```

To create a micromap, call:

```
// Provided by VK_EXT_opacity_micromap
VkResult vkCreateMicromapEXT(
    VkDevice                                device,
    const VkMicromapCreateInfoEXT*         pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkMicromapEXT*                          pMicromap);
```

- `device` is the logical device that creates the acceleration structure object.
- `pCreateInfo` is a pointer to a `VkMicromapCreateInfoEXT` structure containing parameters affecting creation of the micromap.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pMicromap` is a pointer to a `VkMicromapEXT` handle in which the resulting micromap object is returned.

Similar to other objects in Vulkan, the micromap creation merely creates an object with a specific “shape”. The type and quantity of geometry that can be built into a micromap is determined by the parameters of `VkMicromapCreateInfoEXT`.

The micromap data is stored in the object referred to by `VkMicromapCreateInfoEXT::buffer`. Once memory has been bound to that buffer, it **must** be populated by micromap build or micromap copy commands such as `vkCmdBuildMicromapsEXT`, `vkBuildMicromapsEXT`, `vkCmdCopyMicromapEXT`, and `vkCopyMicromapEXT`.

NOTE	The expected usage for a trace capture/replay tool is that it will serialize and later deserialize the micromap data using micromap copy commands. During capture the tool will use <code>vkCopyMicromapToMemoryEXT</code> or <code>vkCmdCopyMicromapToMemoryEXT</code> with a <code>mode</code> of <code>VK_COPY_MICROMAP_MODE_SERIALIZE_EXT</code> , and <code>vkCopyMemoryToMicromapEXT</code> or <code>vkCmdCopyMemoryToMicromapEXT</code> with a <code>mode</code> of <code>VK_COPY_MICROMAP_MODE_DESERIALIZE_EXT</code> during replay.
-------------	--

The input buffers passed to micromap build commands will be referenced by the implementation for the duration of the command. Micromaps **must** be fully self-contained. The application **can** reuse or free any memory which was used by the command as an input or as scratch without affecting the results of a subsequent acceleration structure build using the micromap or traversal of that acceleration structure.

Valid Usage

- VUID-vkCreateMicromapEXT-micromap-07430
The **micromap** feature **must** be enabled
- VUID-vkCreateMicromapEXT-deviceAddress-07431
If **VkMicromapCreateInfoEXT::deviceAddress** is not zero, the **micromapCaptureReplay** feature **must** be enabled
- VUID-vkCreateMicromapEXT-device-07432
If **device** was created with multiple physical devices, then the **bufferDeviceAddressMultiDevice** feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateMicromapEXT-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkCreateMicromapEXT-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid **VkMicromapCreateInfoEXT** structure
- VUID-vkCreateMicromapEXT-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkCreateMicromapEXT-pMicromap-parameter
pMicromap **must** be a valid pointer to a **VkMicromapEXT** handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR**

The **VkMicromapCreateInfoEXT** structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkMicromapCreateInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkMicromapCreateFlagsEXT createFlags;
    VkBuffer              buffer;
    VkDeviceSize          offset;
    VkDeviceSize          size;
    VkMicromapTypeEXT     type;
    VkDeviceAddress        deviceAddress;
} VkMicromapCreateInfoEXT;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **createFlags** is a bitmask of **VkMicromapCreateFlagBitsEXT** specifying additional creation parameters of the micromap.
- **buffer** is the buffer on which the micromap will be stored.
- **offset** is an offset in bytes from the base address of the buffer at which the micromap will be stored, and **must** be a multiple of **256**.
- **size** is the size required for the micromap.
- **type** is a **VkMicromapTypeEXT** value specifying the type of micromap that will be created.
- **deviceAddress** is the device address requested for the micromap if the **micromapCaptureReplay** feature is being used.

If **deviceAddress** is zero, no specific address is requested.

If **deviceAddress** is not zero, **deviceAddress** **must** be an address retrieved from an identically created micromap on the same implementation. The micromap **must** also be placed on an identically created **buffer** and at the same **offset**.

Applications **should** avoid creating micromaps with application-provided addresses and implementation-provided addresses in the same process, to reduce the likelihood of **VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR** errors.

NOTE

The expected usage for this is that a trace capture/replay tool will add the **VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT** flag to all buffers that use **VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT**, and will add **VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT** to all buffers used as storage for a micromap where **deviceAddress** is not zero. This also means that the tool will need to add **VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT** to memory allocations to allow the flag to be set where the application may not have otherwise required it. During capture the tool will save the queried opaque device addresses in the trace. During replay, the buffers will be created specifying the original address so any address values stored in the trace data will remain valid.

Implementations are expected to separate such buffers in the GPU address space so normal allocations will avoid using these addresses. Apps/tools should avoid mixing app-provided and implementation-provided addresses for buffers created with `VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT`, to avoid address space allocation conflicts.

If the micromap will be the target of a build operation, the required size for a micromap **can** be queried with `vkGetMicromapBuildSizesEXT`.

Valid Usage

- VUID-VkMicromapCreateInfoEXT-deviceAddress-07433
If `deviceAddress` is not zero, `createFlags` **must** include `VK_MICROMAP_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_EXT`
- VUID-VkMicromapCreateInfoEXT-createFlags-07434
If `createFlags` includes `VK_MICROMAP_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_EXT`, `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapCaptureReplay` **must** be `VK_TRUE`
- VUID-VkMicromapCreateInfoEXT-buffer-07435
`buffer` **must** have been created with a `usage` value containing `VK_BUFFER_USAGE_MICROMAP_STORAGE_BIT_EXT`
- VUID-VkMicromapCreateInfoEXT-buffer-07436
`buffer` **must** not have been created with `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT`
- VUID-VkMicromapCreateInfoEXT-offset-07437
The sum of `offset` and `size` **must** be less than the size of `buffer`
- VUID-VkMicromapCreateInfoEXT-offset-07438
`offset` **must** be a multiple of 256 bytes

Valid Usage (Implicit)

- VUID-VkMicromapCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MICROMAP_CREATE_INFO_EXT`
- VUID-VkMicromapCreateInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkMicromapCreateInfoEXT-createFlags-parameter
`createFlags` **must** be a valid combination of `VkMicromapCreateFlagBitsEXT` values
- VUID-VkMicromapCreateInfoEXT-buffer-parameter
`buffer` **must** be a valid `VkBuffer` handle
- VUID-VkMicromapCreateInfoEXT-type-parameter
`type` **must** be a valid `VkMicromapTypeEXT` value

To get the build sizes for a micromap, call:

```
// Provided by VK_EXT_opacity_micromap
void vkGetMicromapBuildSizesEXT(
    VkDevice                                device,
    VkAccelerationStructureBuildTypeKHR     buildType,
    const VkMicromapBuildInfoEXT*          pBuildInfo,
    VkMicromapBuildSizesInfoEXT*           pSizeInfo);
```

- `device` is the logical device that will be used for creating the micromap.
- `buildType` defines whether host or device operations (or both) are being queried for.
- `pBuildInfo` is a pointer to a `VkMicromapBuildInfoEXT` structure describing parameters of a build operation.
- `pSizeInfo` is a pointer to a `VkMicromapBuildSizesInfoEXT` structure which returns the size required for a micromap and the sizes required for the scratch buffers, given the build parameters.

The `dstMicromap` and `mode` members of `pBuildInfo` are ignored. Any `VkDeviceOrHostAddressKHR` members of `pBuildInfo` are ignored by this command.

A micromap created with the `micromapSize` returned by this command supports any build with a `VkMicromapBuildInfoEXT` structure subject to the following properties:

- The build command is a host build command, and `buildType` is `VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_KHR` or `VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR`
- The build command is a device build command, and `buildType` is `VK_ACCELERATION_STRUCTURE_BUILD_TYPE_DEVICE_KHR` or `VK_ACCELERATION_STRUCTURE_BUILD_TYPE_HOST_OR_DEVICE_KHR`
- For `VkMicromapBuildInfoEXT`:
 - Its `type`, and `flags` members are equal to `pBuildInfo->type` and `pBuildInfo->flags`, respectively.
 - The sum of usage information in either `pUsageCounts` or `ppUsageCounts` is equal to the sum of usage information in either `pBuildInfo->pUsageCounts` or `pBuildInfo->ppUsageCounts`.

Similarly, the `buildScratchSize` value will support any build command specifying the `VK_BUILD_MICROMAP_MODE_BUILD_EXT mode` under the above conditions.

Valid Usage

- VUID-vkGetMicromapBuildSizesEXT-dstMicromap-09180
`VkMicromapBuildInfoEXT::dstMicromap` **must** have been created from `device`
- VUID-vkGetMicromapBuildSizesEXT-micromap-07439
The `micromap` feature **must** be enabled
- VUID-vkGetMicromapBuildSizesEXT-device-07440
If `device` was created with multiple physical devices, then the

`bufferDeviceAddressMultiDevice` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetMicromapBuildSizesEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetMicromapBuildSizesEXT-buildType-parameter
`buildType` **must** be a valid `VkAccelerationStructureBuildTypeKHR` value
- VUID-vkGetMicromapBuildSizesEXT-pBuildInfo-parameter
`pBuildInfo` **must** be a valid pointer to a valid `VkMicromapBuildInfoEXT` structure
- VUID-vkGetMicromapBuildSizesEXT-pSizeInfo-parameter
`pSizeInfo` **must** be a valid pointer to a `VkMicromapBuildSizesInfoEXT` structure

The `VkMicromapBuildSizesInfoEXT` structure describes the required build sizes for a micromap and scratch buffers and is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkMicromapBuildSizesInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceSize        micromapSize;
    VkDeviceSize        buildScratchSize;
    VkBool32            discardable;
} VkMicromapBuildSizesInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `micromapSize` is the size in bytes required in a `VkMicromapEXT` for a build or update operation.
- `buildScratchSize` is the size in bytes required in a scratch buffer for a build operation.
- `discardable` indicates whether or not the micromap object may be destroyed after an acceleration structure build or update. A false value means that acceleration structures built with this micromap **may** contain references to the data contained therein, and the application **must** not destroy the micromap until ray traversal has concluded. A true value means that the information in the micromap will be copied by value into the acceleration structure, and the micromap **may** be destroyed after the acceleration structure build concludes.

Valid Usage (Implicit)

- VUID-VkMicromapBuildSizesInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MICROMAP_BUILD_SIZES_INFO_EXT`
- VUID-VkMicromapBuildSizesInfoEXT-pNext-pNext
`pNext` **must** be `NULL`

Values which **can** be set in `VkMicromapCreateInfoEXT::type` specifying the type of micromap, are:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkMicromapTypeEXT {
    VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT = 0,
} VkMicromapTypeEXT;
```

- `VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT` is a micromap containing data to control the opacity of a triangle.

Bits which **can** be set in `VkMicromapCreateInfoEXT::createFlags`, specifying additional creation parameters for micromaps, are:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkMicromapCreateFlagBitsEXT {
    VK_MICROMAP_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_EXT = 0x00000001,
} VkMicromapCreateFlagBitsEXT;
```

- `VK_MICROMAP_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_EXT` specifies that the micromap's address **can** be saved and reused on a subsequent run.

```
// Provided by VK_EXT_opacity_micromap
typedef VkFlags VkMicromapCreateFlagsEXT;
```

`VkMicromapCreateFlagsEXT` is a bitmask type for setting a mask of zero or more `VkMicromapCreateFlagBitsEXT`.

Bits which **can** be set in `VkMicromapBuildInfoEXT::flags` specifying additional parameters for micromap builds, are:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkBuildMicromapFlagBitsEXT {
    VK_BUILD_MICROMAP_PREFER_FAST_TRACE_BIT_EXT = 0x00000001,
    VK_BUILD_MICROMAP_PREFER_FAST_BUILD_BIT_EXT = 0x00000002,
    VK_BUILD_MICROMAP_ALLOW_COMPACTION_BIT_EXT = 0x00000004,
} VkBuildMicromapFlagBitsEXT;
```

- `VK_BUILD_MICROMAP_PREFER_FAST_TRACE_BIT_EXT` indicates that the given micromap build **should** prioritize trace performance over build time.
- `VK_BUILD_MICROMAP_PREFER_FAST_BUILD_BIT_EXT` indicates that the given micromap build **should** prioritize build time over trace performance.

```
// Provided by VK_EXT_opacity_micromap
typedef VkFlags VkBuildMicromapFlagsEXT;
```

`VkBuildMicromapFlagsEXT` is a bitmask type for setting a mask of zero or more `VkBuildMicromapFlagBitsEXT`.

To destroy a micromap, call:

```
// Provided by VK_EXT_opacity_micromap
void vkDestroyMicromapEXT(
    VkDevice          device,
    VkMicromapEXT     micromap,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the micromap.
- `micromap` is the micromap to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyMicromapEXT-micromap-07441
All submitted commands that refer to `micromap` **must** have completed execution
- VUID-vkDestroyMicromapEXT-micromap-07442
If `VkAllocationCallbacks` were provided when `micromap` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyMicromapEXT-micromap-07443
If no `VkAllocationCallbacks` were provided when `micromap` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyMicromapEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyMicromapEXT-micromap-parameter
If `micromap` is not `VK_NULL_HANDLE`, `micromap` **must** be a valid `VkMicromapEXT` handle
- VUID-vkDestroyMicromapEXT-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyMicromapEXT-micromap-parent
If `micromap` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `micromap` **must** be externally synchronized

12.8. Resource Memory Association

Resources are initially created as *virtual allocations* with no backing memory. Device memory is allocated separately (see [Device Memory](#)) and then associated with the resource. This association is done differently for sparse and non-sparse resources.

Resources created with any of the sparse creation flags are considered sparse resources. Resources created without these flags are non-sparse. The details on resource memory association for sparse resources is described in [Sparse Resources](#).

Non-sparse resources **must** be bound completely and contiguously to a single `VkDeviceMemory` object before the resource is passed as a parameter to any of the following operations:

- creating image or buffer views
- updating descriptor sets
- recording commands in a command buffer

Once bound, the memory binding is immutable for the lifetime of the resource.

In a logical device representing more than one physical device, buffer and image resources exist on all physical devices but **can** be bound to memory differently on each. Each such replicated resource is an *instance* of the resource. For sparse resources, each instance **can** be bound to memory arbitrarily differently. For non-sparse resources, each instance **can** either be bound to the local or a peer instance of the memory, or for images **can** be bound to rectangular regions from the local and/or peer instances. When a resource is used in a descriptor set, each physical device interprets the descriptor according to its own instance's binding to memory.

NOTE

There are no new copy commands to transfer data between physical devices. Instead, an application **can** create a resource with a peer mapping and use it as the source or destination of a transfer command executed by a single physical device to copy the data from one physical device to another.

To determine the memory requirements for a buffer resource, call:

```
// Provided by VK_VERSION_1_0
void vkGetBufferMemoryRequirements(
    VkDevice          device,
    VkBuffer          buffer,
    VkMemoryRequirements* pMemoryRequirements);
```

- `device` is the logical device that owns the buffer.
- `buffer` is the buffer to query.
- `pMemoryRequirements` is a pointer to a `VkMemoryRequirements` structure in which the memory requirements of the buffer object are returned.

Valid Usage (Implicit)

- VUID-vkGetBufferMemoryRequirements-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkGetBufferMemoryRequirements-buffer-parameter **buffer** **must** be a valid [VkBuffer](#) handle
- VUID-vkGetBufferMemoryRequirements-pMemoryRequirements-parameter **pMemoryRequirements** **must** be a valid pointer to a [VkMemoryRequirements](#) structure
- VUID-vkGetBufferMemoryRequirements-buffer-parent **buffer** **must** have been created, allocated, or retrieved from **device**

To determine the memory requirements for an image resource which is not created with the [VK_IMAGE_CREATE_DISJOINT_BIT](#) flag set, call:

```
// Provided by VK_VERSION_1_0
void vkGetImageMemoryRequirements(
    VkDevice          device,
    VkImage           image,
    VkMemoryRequirements* pMemoryRequirements);
```

- **device** is the logical device that owns the image.
- **image** is the image to query.
- **pMemoryRequirements** is a pointer to a [VkMemoryRequirements](#) structure in which the memory requirements of the image object are returned.

Valid Usage

- VUID-vkGetImageMemoryRequirements-image-01588 **image** **must** not have been created with the [VK_IMAGE_CREATE_DISJOINT_BIT](#) flag set

Valid Usage (Implicit)

- VUID-vkGetImageMemoryRequirements-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkGetImageMemoryRequirements-image-parameter **image** **must** be a valid [VkImage](#) handle
- VUID-vkGetImageMemoryRequirements-pMemoryRequirements-parameter **pMemoryRequirements** **must** be a valid pointer to a [VkMemoryRequirements](#) structure
- VUID-vkGetImageMemoryRequirements-image-parent **image** **must** have been created, allocated, or retrieved from **device**

The `VkMemoryRequirements` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkMemoryRequirements {
    VkDeviceSize    size;
    VkDeviceSize    alignment;
    uint32_t        memoryTypeBits;
} VkMemoryRequirements;
```

- `size` is the size, in bytes, of the memory allocation **required** for the resource.
- `alignment` is the alignment, in bytes, of the offset within the allocation **required** for the resource.
- `memoryTypeBits` is a bitmask and contains one bit set for every supported memory type for the resource. Bit `i` is set if and only if the memory type `i` in the `VkPhysicalDeviceMemoryProperties` structure for the physical device is supported for the resource.

If the resource being queried was created with the `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT`, `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT`, or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT` external memory handle type, the value of `size` has no meaning and **should** be ignored.

The implementation guarantees certain properties about the memory requirements returned by `vkGetBufferMemoryRequirements2KHR`, `vkGetImageMemoryRequirements2KHR`, `vkGetDeviceBufferMemoryRequirementsKHR`, `vkGetDeviceImageMemoryRequirementsKHR`, `vkGetBufferMemoryRequirements` and `vkGetImageMemoryRequirements`:

- The `memoryTypeBits` member always contains at least one bit set.
- If `buffer` is a `VkBuffer` not created with the `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` or `VK_BUFFER_CREATE_PROTECTED_BIT` bits set, or if `image` is a `linear` image that was not created with the `VK_IMAGE_CREATE_PROTECTED_BIT` bit set, then the `memoryTypeBits` member always contains at least one bit set corresponding to a `VkMemoryType` with a `propertyFlags` that has both the `VK_MEMORY_PROPERTY_HOST_VISIBLE_BIT` bit and the `VK_MEMORY_PROPERTY_HOST_COHERENT_BIT` bit set. In other words, mappable coherent memory **can** always be attached to these objects.
- If `buffer` was created with `VkExternalMemoryBufferCreateInfo::handleTypes` set to `0` or `image` was created with `VkExternalMemoryImageCreateInfo::handleTypes` set to `0`, the `memoryTypeBits` member always contains at least one bit set corresponding to a `VkMemoryType` with a `propertyFlags` that has the `VK_MEMORY_PROPERTY_DEVICE_LOCAL_BIT` bit set.
- The `memoryTypeBits` member is identical for all `VkBuffer` objects created with the same value for the `flags` and `usage` members in the `VkBufferCreateInfo` structure and the `handleTypes` member of the `VkExternalMemoryBufferCreateInfo` structure passed to `vkCreateBuffer`. Further, if `usage1` and `usage2` of type `VkBufferUsageFlags` are such that the bits set in `usage2` are a subset of the bits set in `usage1`, and they have the same `flags` and `VkExternalMemoryBufferCreateInfo::handleTypes`, then the bits set in `memoryTypeBits` returned for `usage1` **must** be a subset of the bits set in `memoryTypeBits` returned for `usage2`, for all values of `flags`.

- The `alignment` member is a power of two.
- The `alignment` member is identical for all `VkBuffer` objects created with the same combination of values for the `usage` and `flags` members in the `VkBufferCreateInfo` structure passed to `vkCreateBuffer`.
- If the `maintenance4` feature is enabled, then the `alignment` member is identical for all `VkImage` objects created with the same combination of values for the `flags`, `imageType`, `format`, `extent`, `mipLevels`, `arrayLayers`, `samples`, `tiling` and `usage` members in the `VkImageCreateInfo` structure passed to `vkCreateImage`.
- The `alignment` member satisfies the buffer descriptor offset alignment requirements associated with the `VkBuffer`'s `usage`:
 - If `usage` included `VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT` or `VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT`, `alignment` **must** be an integer multiple of `VkPhysicalDeviceLimits::minTexelBufferOffsetAlignment`.
 - If `usage` included `VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT`, `alignment` **must** be an integer multiple of `VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment`.
 - If `usage` included `VK_BUFFER_USAGE_STORAGE_BUFFER_BIT`, `alignment` **must** be an integer multiple of `VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment`.
- For images created with a color format, the `memoryTypeBits` member is identical for all `VkImage` objects created with the same combination of values for the `tiling` member, the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` bit of the `flags` member, the `VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT` bit of the `flags` member, the `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` bit of the `usage` member if the `VkPhysicalDeviceHostImageCopyPropertiesEXT::identicalMemoryTypeRequirements` property is `VK_FALSE`, `handleTypes` member of `VkExternalMemoryImageCreateInfo`, and the `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT` of the `usage` member in the `VkImageCreateInfo` structure passed to `vkCreateImage`.
- For images created with a depth/stencil format, the `memoryTypeBits` member is identical for all `VkImage` objects created with the same combination of values for the `format` member, the `tiling` member, the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` bit of the `flags` member, the `VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT` bit of the `flags` member, the `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` bit of the `usage` member if the `VkPhysicalDeviceHostImageCopyPropertiesEXT::identicalMemoryTypeRequirements` property is `VK_FALSE`, `handleTypes` member of `VkExternalMemoryImageCreateInfo`, and the `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT` of the `usage` member in the `VkImageCreateInfo` structure passed to `vkCreateImage`.
- If the memory requirements are for a `VkImage`, the `memoryTypeBits` member **must** not refer to a `VkMemoryType` with a `propertyFlags` that has the `VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT` bit set if the `image` did not have `VK_IMAGE_USAGE_TRANSIENT_ATTACHMENT_BIT` bit set in the `usage` member of the `VkImageCreateInfo` structure passed to `vkCreateImage`.
- If the memory requirements are for a `VkBuffer`, the `memoryTypeBits` member **must** not refer to a `VkMemoryType` with a `propertyFlags` that has the `VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT` bit set.

NOTE

The implication of this requirement is that lazily allocated memory is disallowed

for buffers in all cases.

- The **size** member is identical for all **VkBuffer** objects created with the same combination of creation parameters specified in **VkBufferCreateInfo** and its **pNext** chain.
- The **size** member is identical for all **VkImage** objects created with the same combination of creation parameters specified in **VkImageCreateInfo** and its **pNext** chain.

NOTE

This, however, does not imply that they interpret the contents of the bound memory identically with each other. That additional guarantee, however, **can** be explicitly requested using **VK_IMAGE_CREATE_ALIAS_BIT**.

- If the **maintenance4** feature is enabled, these additional guarantees apply:
 - For a **VkBuffer**, the **size** memory requirement is never greater than that of another **VkBuffer** created with a greater or equal **size** specified in **VkBufferCreateInfo**, all other creation parameters being identical.
 - For a **VkBuffer**, the **size** memory requirement is never greater than the result of aligning **VkBufferCreateInfo::size** with the **alignment** memory requirement.
 - For a **VkImage**, the **size** memory requirement is never greater than that of another **VkImage** created with a greater or equal value in each of **extent.width**, **extent.height**, and **extent.depth**; all other creation parameters being identical.
 - The memory requirements returned by **vkGetDeviceBufferMemoryRequirementsKHR** are identical to those that would be returned by **vkGetBufferMemoryRequirements2KHR** if it were called with a **VkBuffer** created with the same **VkBufferCreateInfo** values.
 - The memory requirements returned by **vkGetDeviceImageMemoryRequirementsKHR** are identical to those that would be returned by **vkGetImageMemoryRequirements2KHR** if it were called with a **VkImage** created with the same **VkImageCreateInfo** values.

To determine the memory requirements for a buffer resource, call:

```
// Provided by VK_KHR_get_memory_requirements2
void vkGetBufferMemoryRequirements2KHR(
    VkDevice          device,
    const VkBufferMemoryRequirementsInfo2* pInfo,
    VkMemoryRequirements2* pMemoryRequirements);
```

- **device** is the logical device that owns the buffer.
- **pInfo** is a pointer to a **VkBufferMemoryRequirementsInfo2** structure containing parameters required for the memory requirements query.
- **pMemoryRequirements** is a pointer to a **VkMemoryRequirements2** structure in which the memory requirements of the buffer object are returned.

Valid Usage (Implicit)

- VUID-vkGetBufferMemoryRequirements2-device-parameter

device must be a valid [VkDevice](#) handle

- VUID-vkGetBufferMemoryRequirements2-pInfo-parameter
pInfo must be a valid pointer to a valid [VkBufferMemoryRequirementsInfo2](#) structure
- VUID-vkGetBufferMemoryRequirements2-pMemoryRequirements-parameter
pMemoryRequirements must be a valid pointer to a [VkMemoryRequirements2](#) structure

To determine the memory requirements for a buffer resource without creating an object, call:

```
// Provided by VK_KHR_maintenance4
void vkGetDeviceBufferMemoryRequirementsKHR(
    VkDevice                device,
    const VkDeviceBufferMemoryRequirements* pInfo,
    VkMemoryRequirements2* pMemoryRequirements);
```

- **device** is the logical device intended to own the buffer.
- **pInfo** is a pointer to a [VkDeviceBufferMemoryRequirements](#) structure containing parameters required for the memory requirements query.
- **pMemoryRequirements** is a pointer to a [VkMemoryRequirements2](#) structure in which the memory requirements of the buffer object are returned.

Valid Usage (Implicit)

- VUID-vkGetDeviceBufferMemoryRequirements-device-parameter
device must be a valid [VkDevice](#) handle
- VUID-vkGetDeviceBufferMemoryRequirements-pInfo-parameter
pInfo must be a valid pointer to a valid [VkDeviceBufferMemoryRequirements](#) structure
- VUID-vkGetDeviceBufferMemoryRequirements-pMemoryRequirements-parameter
pMemoryRequirements must be a valid pointer to a [VkMemoryRequirements2](#) structure

The [VkBufferMemoryRequirementsInfo2](#) structure is defined as:

```
typedef struct VkBufferMemoryRequirementsInfo2 {
    VkStructureType    sType;
    const void*        pNext;
    VkBuffer            buffer;
} VkBufferMemoryRequirementsInfo2;
```

or the equivalent

```
// Provided by VK_KHR_get_memory_requirements2
typedef VkBufferMemoryRequirementsInfo2 VkBufferMemoryRequirementsInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `buffer` is the buffer to query.

Valid Usage (Implicit)

- VUID-VkBufferMemoryRequirementsInfo2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BUFFER_MEMORY_REQUIREMENTS_INFO_2`
- VUID-VkBufferMemoryRequirementsInfo2-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkBufferMemoryRequirementsInfo2-buffer-parameter
`buffer` **must** be a valid `VkBuffer` handle

The `VkDeviceBufferMemoryRequirements` structure is defined as:

```
typedef struct VkDeviceBufferMemoryRequirements {
    VkStructureType    sType;
    const void*        pNext;
    const VkBufferCreateInfo* pCreateInfo;
} VkDeviceBufferMemoryRequirements;
```

or the equivalent

```
// Provided by VK_KHR_maintenance4
typedef VkDeviceBufferMemoryRequirements VkDeviceBufferMemoryRequirementsKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pCreateInfo` is a pointer to a `VkBufferCreateInfo` structure containing parameters affecting creation of the buffer to query.

Valid Usage (Implicit)

- VUID-VkDeviceBufferMemoryRequirements-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_BUFFER_MEMORY_REQUIREMENTS`
- VUID-VkDeviceBufferMemoryRequirements-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDeviceBufferMemoryRequirements-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkBufferCreateInfo` structure

To determine the memory requirements for an image resource, call:

```
// Provided by VK_KHR_get_memory_requirements2
void vkGetImageMemoryRequirements2KHR(
    VkDevice device,
    const VkImageMemoryRequirementsInfo2* pInfo,
    VkMemoryRequirements2* pMemoryRequirements);
```

- **device** is the logical device that owns the image.
- **pInfo** is a pointer to a [VkImageMemoryRequirementsInfo2](#) structure containing parameters required for the memory requirements query.
- **pMemoryRequirements** is a pointer to a [VkMemoryRequirements2](#) structure in which the memory requirements of the image object are returned.

Valid Usage (Implicit)

- VUID-vkGetImageMemoryRequirements2-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkGetImageMemoryRequirements2-pInfo-parameter **pInfo** **must** be a valid pointer to a valid [VkImageMemoryRequirementsInfo2](#) structure
- VUID-vkGetImageMemoryRequirements2-pMemoryRequirements-parameter **pMemoryRequirements** **must** be a valid pointer to a [VkMemoryRequirements2](#) structure

To determine the memory requirements for an image resource without creating an object, call:

```
// Provided by VK_KHR_maintenance4
void vkGetDeviceImageMemoryRequirementsKHR(
    VkDevice device,
    const VkDeviceImageMemoryRequirements* pInfo,
    VkMemoryRequirements2* pMemoryRequirements);
```

- **device** is the logical device intended to own the image.
- **pInfo** is a pointer to a [VkDeviceImageMemoryRequirements](#) structure containing parameters required for the memory requirements query.
- **pMemoryRequirements** is a pointer to a [VkMemoryRequirements2](#) structure in which the memory requirements of the image object are returned.

Valid Usage (Implicit)

- VUID-vkGetDeviceImageMemoryRequirements-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkGetDeviceImageMemoryRequirements-pInfo-parameter **pInfo** **must** be a valid pointer to a valid [VkDeviceImageMemoryRequirements](#) structure
- VUID-vkGetDeviceImageMemoryRequirements-pMemoryRequirements-parameter

`pMemoryRequirements` **must** be a valid pointer to a `VkMemoryRequirements2` structure

The `VkImageMemoryRequirementsInfo2` structure is defined as:

```
typedef struct VkImageMemoryRequirementsInfo2 {
    VkStructureType    sType;
    const void*        pNext;
    VkImage             image;
} VkImageMemoryRequirementsInfo2;
```

or the equivalent

```
// Provided by VK_KHR_get_memory_requirements2
typedef VkImageMemoryRequirementsInfo2 VkImageMemoryRequirementsInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `image` is the image to query.

Valid Usage

- VUID-VkImageMemoryRequirementsInfo2-image-01589
If `image` was created with a *multi-planar* format and the `VK_IMAGE_CREATE_DISJOINT_BIT` flag, there **must** be a `VkImagePlaneMemoryRequirementsInfo` included in the `pNext` chain of the `VkImageMemoryRequirementsInfo2` structure
- VUID-VkImageMemoryRequirementsInfo2-image-02279
If `image` was created with `VK_IMAGE_CREATE_DISJOINT_BIT` and with `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then there **must** be a `VkImagePlaneMemoryRequirementsInfo` included in the `pNext` chain of the `VkImageMemoryRequirementsInfo2` structure
- VUID-VkImageMemoryRequirementsInfo2-image-01590
If `image` was not created with the `VK_IMAGE_CREATE_DISJOINT_BIT` flag, there **must** not be a `VkImagePlaneMemoryRequirementsInfo` included in the `pNext` chain of the `VkImageMemoryRequirementsInfo2` structure
- VUID-VkImageMemoryRequirementsInfo2-image-02280
If `image` was created with a single-plane format and with any *tiling* other than `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then there **must** not be a `VkImagePlaneMemoryRequirementsInfo` included in the `pNext` chain of the `VkImageMemoryRequirementsInfo2` structure

Valid Usage (Implicit)

- VUID-VkImageMemoryRequirementsInfo2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2`
- VUID-VkImageMemoryRequirementsInfo2-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkImagePlaneMemoryRequirementsInfo`
- VUID-VkImageMemoryRequirementsInfo2-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkImageMemoryRequirementsInfo2-image-parameter
`image` **must** be a valid `VkImage` handle

The `VkDeviceImageMemoryRequirements` structure is defined as:

```
typedef struct VkDeviceImageMemoryRequirements {  
    VkStructureType      sType;  
    const void*          pNext;  
    const VkImageCreateInfo* pCreateInfo;  
    VkImageAspectFlagBits planeAspect;  
} VkDeviceImageMemoryRequirements;
```

or the equivalent

```
// Provided by VK_KHR_maintenance4  
typedef VkDeviceImageMemoryRequirements VkDeviceImageMemoryRequirementsKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pCreateInfo` is a pointer to a `VkImageCreateInfo` structure containing parameters affecting creation of the image to query.
- `planeAspect` is a `VkImageAspectFlagBits` value specifying the aspect corresponding to the image plane to query. This parameter is ignored unless `pCreateInfo->tiling` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, or `pCreateInfo->flags` has `VK_IMAGE_CREATE_DISJOINT_BIT` set.

Valid Usage

- VUID-VkDeviceImageMemoryRequirements-pCreateInfo-06416
The `pCreateInfo->pNext` chain **must** not contain a `VkImageSwapchainCreateInfoKHR` structure
- VUID-VkDeviceImageMemoryRequirements-pCreateInfo-06776
The `pCreateInfo->pNext` chain **must** not contain a

[VkImageDrmFormatModifierExplicitCreateInfoEXT](#) structure

- VUID-VkDeviceImageMemoryRequirements-pCreateInfo-06417
If `pCreateInfo->format` specifies a *multi-planar* format and `pCreateInfo->flags` has `VK_IMAGE_CREATE_DISJOINT_BIT` set then `planeAspect` **must** not be `VK_IMAGE_ASPECT_NONE_KHR`
- VUID-VkDeviceImageMemoryRequirements-pCreateInfo-06419
If `pCreateInfo->flags` has `VK_IMAGE_CREATE_DISJOINT_BIT` set and if the `pCreateInfo->tiling` is `VK_IMAGE_TILING_LINEAR` or `VK_IMAGE_TILING_OPTIMAL`, then `planeAspect` **must** be a single valid [multi-planar aspect mask](#) bit
- VUID-VkDeviceImageMemoryRequirements-pCreateInfo-06420
If `pCreateInfo->tiling` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then `planeAspect` **must** be a single valid *memory plane* for the image (that is, `aspectMask` **must** specify a plane index that is less than the [VkDrmFormatModifierPropertiesEXT::drmFormatModifierPlaneCount](#) associated with the image's `format` and [VkImageDrmFormatModifierPropertiesEXT::drmFormatModifier](#))

Valid Usage (Implicit)

- VUID-VkDeviceImageMemoryRequirements-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_IMAGE_MEMORY_REQUIREMENTS`
- VUID-VkDeviceImageMemoryRequirements-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDeviceImageMemoryRequirements-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid [VkImageCreateInfo](#) structure
- VUID-VkDeviceImageMemoryRequirements-planeAspect-parameter
If `planeAspect` is not 0, `planeAspect` **must** be a valid [VkImageAspectFlagBits](#) value

To determine the memory requirements for a plane of a disjoint image, add a [VkImagePlaneMemoryRequirementsInfo](#) structure to the `pNext` chain of the [VkImageMemoryRequirementsInfo2](#) structure.

The [VkImagePlaneMemoryRequirementsInfo](#) structure is defined as:

```
typedef struct VkImagePlaneMemoryRequirementsInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkImageAspectFlagBits planeAspect;
} VkImagePlaneMemoryRequirementsInfo;
```

or the equivalent

```
// Provided by VK_KHR_sampler_ycbcr_conversion
typedef VkImagePlaneMemoryRequirementsInfo VkImagePlaneMemoryRequirementsInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `planeAspect` is a `VkImageAspectFlagBits` value specifying the aspect corresponding to the image plane to query.

Valid Usage

- VUID-VkImagePlaneMemoryRequirementsInfo-planeAspect-02281
If the image's `tiling` is `VK_IMAGE_TILING_LINEAR` or `VK_IMAGE_TILING_OPTIMAL`, then `planeAspect` **must** be a single valid `multi-planar aspect mask` bit
- VUID-VkImagePlaneMemoryRequirementsInfo-planeAspect-02282
If the image's `tiling` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then `planeAspect` **must** be a single valid *memory plane* for the image (that is, `aspectMask` **must** specify a plane index that is less than the `VkDrmFormatModifierPropertiesEXT::drmFormatModifierPlaneCount` associated with the image's `format` and `VkImageDrmFormatModifierPropertiesEXT::drmFormatModifier`)

Valid Usage (Implicit)

- VUID-VkImagePlaneMemoryRequirementsInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_PLANE_MEMORY_REQUIREMENTS_INFO`
- VUID-VkImagePlaneMemoryRequirementsInfo-planeAspect-parameter
`planeAspect` **must** be a valid `VkImageAspectFlagBits` value

The `VkMemoryRequirements2` structure is defined as:

```
typedef struct VkMemoryRequirements2 {
    VkStructureType    sType;
    void*              pNext;
    VkMemoryRequirements memoryRequirements;
} VkMemoryRequirements2;
```

or the equivalent

```
// Provided by VK_KHR_get_memory_requirements2
typedef VkMemoryRequirements2 VkMemoryRequirements2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryRequirements` is a `VkMemoryRequirements` structure describing the memory requirements of the resource.

Valid Usage (Implicit)

- VUID-VkMemoryRequirements2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2`
- VUID-VkMemoryRequirements2-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkMemoryDedicatedRequirements`
- VUID-VkMemoryRequirements2-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique

The `VkMemoryDedicatedRequirements` structure is defined as:

```
typedef struct VkMemoryDedicatedRequirements {  
    VkStructureType    sType;  
    void*              pNext;  
    VkBool32           prefersDedicatedAllocation;  
    VkBool32           requiresDedicatedAllocation;  
} VkMemoryDedicatedRequirements;
```

or the equivalent

```
// Provided by VK_KHR_dedicated_allocation  
typedef VkMemoryDedicatedRequirements VkMemoryDedicatedRequirementsKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `prefersDedicatedAllocation` specifies that the implementation would prefer a dedicated allocation for this resource. The application is still free to suballocate the resource but it **may** get better performance if a dedicated allocation is used.
- `requiresDedicatedAllocation` specifies that a dedicated allocation is required for this resource.

To determine the dedicated allocation requirements of a buffer or image resource, add a `VkMemoryDedicatedRequirements` structure to the `pNext` chain of the `VkMemoryRequirements2` structure passed as the `pMemoryRequirements` parameter of `vkGetBufferMemoryRequirements2KHR` or `vkGetImageMemoryRequirements2KHR`, respectively.

Constraints on the values returned for buffer resources are:

- `requiresDedicatedAllocation` **may** be `VK_TRUE` if the `pNext` chain of `VkBufferCreateInfo` for the call to `vkCreateBuffer` used to create the buffer being queried included a `VkExternalMemoryBufferCreateInfo` structure, and any of the handle types specified in `VkExternalMemoryBufferCreateInfo::handleTypes` requires dedicated allocation, as reported by `vkGetPhysicalDeviceExternalBufferPropertiesKHR` in `VkExternalBufferProperties::externalMemoryProperties.externalMemoryFeatures`. Otherwise, `requiresDedicatedAllocation` will be `VK_FALSE`.

- When the implementation sets `requiresDedicatedAllocation` to `VK_TRUE`, it **must** also set `prefersDedicatedAllocation` to `VK_TRUE`.
- If `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` was set in `VkBufferCreateInfo::flags` when `buffer` was created, then both `prefersDedicatedAllocation` and `requiresDedicatedAllocation` will be `VK_FALSE`.

Constraints on the values returned for image resources are:

- `requiresDedicatedAllocation` **may** be `VK_TRUE` if the `pNext` chain of `VkImageCreateInfo` for the call to `vkCreateImage` used to create the image being queried included a `VkExternalMemoryImageCreateInfo` structure, and any of the handle types specified in `VkExternalMemoryImageCreateInfo::handleTypes` requires dedicated allocation, as reported by `vkGetPhysicalDeviceImageFormatProperties2KHR` in `VkExternalImageFormatProperties::externalMemoryProperties.externalMemoryFeatures`.
- `requiresDedicatedAllocation` **may** be `VK_TRUE` if the image's tiling is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`.
- `requiresDedicatedAllocation` will otherwise be `VK_FALSE`.
- If `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` was set in `VkImageCreateInfo::flags` when `image` was created, then both `prefersDedicatedAllocation` and `requiresDedicatedAllocation` will be `VK_FALSE`.

Valid Usage (Implicit)

- VUID-VkMemoryDedicatedRequirements-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS`

To attach memory to a buffer object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkBindBufferMemory(
    VkDevice          device,
    VkBuffer          buffer,
    VkDeviceMemory    memory,
    VkDeviceSize      memoryOffset);
```

- `device` is the logical device that owns the buffer and memory.
- `buffer` is the buffer to be attached to memory.
- `memory` is a `VkDeviceMemory` object describing the device memory to attach.
- `memoryOffset` is the start offset of the region of `memory` which is to be bound to the buffer. The number of bytes returned in the `VkMemoryRequirements::size` member in `memory`, starting from `memoryOffset` bytes, will be bound to the specified buffer.

`vkBindBufferMemory` is equivalent to passing the same parameters through `VkBindBufferMemoryInfo` to `vkBindBufferMemory2KHR`.

Valid Usage

- VUID-vkBindBufferMemory-buffer-07459
`buffer` **must** not have been bound to a memory object
- VUID-vkBindBufferMemory-buffer-01030
`buffer` **must** not have been created with any sparse memory binding flags
- VUID-vkBindBufferMemory-memoryOffset-01031
`memoryOffset` **must** be less than the size of `memory`
- VUID-vkBindBufferMemory-memory-01035
`memory` **must** have been allocated using one of the memory types allowed in the `memoryTypeBits` member of the `VkMemoryRequirements` structure returned from a call to `vkGetBufferMemoryRequirements` with `buffer`
- VUID-vkBindBufferMemory-memoryOffset-01036
`memoryOffset` **must** be an integer multiple of the `alignment` member of the `VkMemoryRequirements` structure returned from a call to `vkGetBufferMemoryRequirements` with `buffer`
- VUID-vkBindBufferMemory-size-01037
The `size` member of the `VkMemoryRequirements` structure returned from a call to `vkGetBufferMemoryRequirements` with `buffer` **must** be less than or equal to the size of `memory` minus `memoryOffset`
- VUID-vkBindBufferMemory-buffer-01444
If `buffer` requires a dedicated allocation (as reported by `vkGetBufferMemoryRequirements2KHR` in `VkMemoryDedicatedRequirements::requiresDedicatedAllocation` for `buffer`), `memory` **must** have been allocated with `VkMemoryDedicatedAllocateInfo::buffer` equal to `buffer`
- VUID-vkBindBufferMemory-memory-01508
If the `VkMemoryAllocateInfo` provided when `memory` was allocated included a `VkMemoryDedicatedAllocateInfo` structure in its `pNext` chain, and `VkMemoryDedicatedAllocateInfo::buffer` was not `VK_NULL_HANDLE`, then `buffer` **must** equal `VkMemoryDedicatedAllocateInfo::buffer`, and `memoryOffset` **must** be zero
- VUID-vkBindBufferMemory-memory-02726
If the value of `VkExportMemoryAllocateInfo::handleTypes` used to allocate `memory` is not `0`, it **must** include at least one of the handles set in `VkExternalMemoryBufferCreateInfo::handleTypes` when `buffer` was created
- VUID-vkBindBufferMemory-memory-02985
If `memory` was allocated by a memory import operation, the external handle type of the imported memory **must** also have been set in `VkExternalMemoryBufferCreateInfo::handleTypes` when `buffer` was created
- VUID-vkBindBufferMemory-bufferDeviceAddress-03339
If the `VkPhysicalDeviceBufferDeviceAddressFeatures::bufferDeviceAddress` feature is enabled and `buffer` was created with the `VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT` bit set, `memory` **must** have been allocated with the `VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT` bit set

- VUID-vkBindBufferMemory-bufferDeviceAddressCaptureReplay-09200

If the [VkPhysicalDeviceBufferDeviceAddressFeatures::bufferDeviceAddressCaptureReplay](#) feature is enabled and **buffer** was created with the [VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT](#) bit set, **memory** **must** have been allocated with the [VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT](#) bit set

Valid Usage (Implicit)

- VUID-vkBindBufferMemory-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkBindBufferMemory-buffer-parameter
buffer **must** be a valid [VkBuffer](#) handle
- VUID-vkBindBufferMemory-memory-parameter
memory **must** be a valid [VkDeviceMemory](#) handle
- VUID-vkBindBufferMemory-buffer-parent
buffer **must** have been created, allocated, or retrieved from **device**
- VUID-vkBindBufferMemory-memory-parent
memory **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **buffer** **must** be externally synchronized

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)
- [VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR](#)

To attach memory to buffer objects for one or more buffers at a time, call:

```
// Provided by VK_KHR_bind_memory2
VkResult vkBindBufferMemory2KHR(
    VkDevice          device,
    uint32_t          bindInfoCount,
    const VkBindBufferMemoryInfo* pBindInfos);
```


- **device** is the logical device that owns the buffers and memory.
- **bindInfoCount** is the number of elements in **pBindInfos**.
- **pBindInfos** is a pointer to an array of **bindInfoCount** **VkBindBufferMemoryInfo** structures describing buffers and memory to bind.

On some implementations, it **may** be more efficient to batch memory bindings into a single command.

If the **maintenance6** feature is enabled, this command **must** attempt to perform all of the memory binding operations described by **pBindInfos**, and **must** not early exit on the first failure.

If any of the memory binding operations described by **pBindInfos** fail, the **VkResult** returned by this command **must** be the return value of any one of the memory binding operations which did not return **VK_SUCCESS**.

NOTE

If the **vkBindBufferMemory2KHR** command failed, **VkBindMemoryStatusKHR** structures were not included in the **pNext** chains of each element of **pBindInfos**, and **bindInfoCount** was greater than one, then the buffers referenced by **pBindInfos** will be in an indeterminate state, and must not be used.

Applications should destroy these buffers.

Valid Usage (Implicit)

- VUID-vkBindBufferMemory2-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkBindBufferMemory2-pBindInfos-parameter
pBindInfos **must** be a valid pointer to an array of **bindInfoCount** valid **VkBindBufferMemoryInfo** structures
- VUID-vkBindBufferMemory2-bindInfoCount-arraylength
bindInfoCount **must** be greater than 0

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**
- **VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR**

VkBindBufferMemoryInfo contains members corresponding to the parameters of **vkBindBufferMemory**.

The `VkBindBufferMemoryInfo` structure is defined as:

```
typedef struct VkBindBufferMemoryInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkBuffer            buffer;
    VkDeviceMemory      memory;
    VkDeviceSize        memoryOffset;
} VkBindBufferMemoryInfo;
```

or the equivalent

```
// Provided by VK_KHR_bind_memory2
typedef VkBindBufferMemoryInfo VkBindBufferMemoryInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `buffer` is the buffer to be attached to memory.
- `memory` is a `VkDeviceMemory` object describing the device memory to attach.
- `memoryOffset` is the start offset of the region of `memory` which is to be bound to the buffer. The number of bytes returned in the `VkMemoryRequirements::size` member in `memory`, starting from `memoryOffset` bytes, will be bound to the specified buffer.

Valid Usage

- VUID-VkBindBufferMemoryInfo-buffer-07459
`buffer` **must** not have been bound to a memory object
- VUID-VkBindBufferMemoryInfo-buffer-01030
`buffer` **must** not have been created with any sparse memory binding flags
- VUID-VkBindBufferMemoryInfo-memoryOffset-01031
`memoryOffset` **must** be less than the size of `memory`
- VUID-VkBindBufferMemoryInfo-memory-01035
`memory` **must** have been allocated using one of the memory types allowed in the `memoryTypeBits` member of the `VkMemoryRequirements` structure returned from a call to `vkGetBufferMemoryRequirements` with `buffer`
- VUID-VkBindBufferMemoryInfo-memoryOffset-01036
`memoryOffset` **must** be an integer multiple of the `alignment` member of the `VkMemoryRequirements` structure returned from a call to `vkGetBufferMemoryRequirements` with `buffer`
- VUID-VkBindBufferMemoryInfo-size-01037
The `size` member of the `VkMemoryRequirements` structure returned from a call to `vkGetBufferMemoryRequirements` with `buffer` **must** be less than or equal to the size of `memory` minus `memoryOffset`

- VUID-VkBindBufferMemoryInfo-buffer-01444
If **buffer** requires a dedicated allocation (as reported by `vkGetBufferMemoryRequirements2KHR` in `VkMemoryDedicatedRequirements::requiresDedicatedAllocation` for **buffer**), **memory** **must** have been allocated with `VkMemoryDedicatedAllocateInfo::buffer` equal to **buffer**
- VUID-VkBindBufferMemoryInfo-memory-01508
If the `VkMemoryAllocateInfo` provided when **memory** was allocated included a `VkMemoryDedicatedAllocateInfo` structure in its **pNext** chain, and `VkMemoryDedicatedAllocateInfo::buffer` was not `VK_NULL_HANDLE`, then **buffer** **must** equal `VkMemoryDedicatedAllocateInfo::buffer`, and **memoryOffset** **must** be zero
- VUID-VkBindBufferMemoryInfo-memory-02726
If the value of `VkExportMemoryAllocateInfo::handleTypes` used to allocate **memory** is not 0, it **must** include at least one of the handles set in `VkExternalMemoryBufferCreateInfo::handleTypes` when **buffer** was created
- VUID-VkBindBufferMemoryInfo-memory-02985
If **memory** was allocated by a memory import operation, the external handle type of the imported memory **must** also have been set in `VkExternalMemoryBufferCreateInfo::handleTypes` when **buffer** was created
- VUID-VkBindBufferMemoryInfo-bufferDeviceAddress-03339
If the `VkPhysicalDeviceBufferDeviceAddressFeatures::bufferDeviceAddress` feature is enabled and **buffer** was created with the `VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT` bit set, **memory** **must** have been allocated with the `VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT` bit set
- VUID-VkBindBufferMemoryInfo-bufferDeviceAddressCaptureReplay-09200
If the `VkPhysicalDeviceBufferDeviceAddressFeatures::bufferDeviceAddressCaptureReplay` feature is enabled and **buffer** was created with the `VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT` bit set, **memory** **must** have been allocated with the `VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT` bit set
- VUID-VkBindBufferMemoryInfo-pNext-01605
If the **pNext** chain includes a `VkBindBufferMemoryDeviceGroupInfo` structure, all instances of **memory** specified by `VkBindBufferMemoryDeviceGroupInfo::pDeviceIndices` **must** have been allocated

Valid Usage (Implicit)

- VUID-VkBindBufferMemoryInfo-sType-sType
sType **must** be `VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_INFO`
- VUID-VkBindBufferMemoryInfo-pNext-pNext
Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either `NULL` or a pointer to a valid instance of `VkBindBufferMemoryDeviceGroupInfo` or `VkBindMemoryStatusKHR`
- VUID-VkBindBufferMemoryInfo-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique

- VUID-VkBindBufferMemoryInfo-buffer-parameter
buffer **must** be a valid [VkBuffer](#) handle
- VUID-VkBindBufferMemoryInfo-memory-parameter
memory **must** be a valid [VkDeviceMemory](#) handle
- VUID-VkBindBufferMemoryInfo-commonparent
Both of **buffer**, and **memory** **must** have been created, allocated, or retrieved from the same [VkDevice](#)

The [VkBindBufferMemoryDeviceGroupInfo](#) structure is defined as:

```
typedef struct VkBindBufferMemoryDeviceGroupInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           deviceIndexCount;
    const uint32_t*     pDeviceIndices;
} VkBindBufferMemoryDeviceGroupInfo;
```

or the equivalent

```
// Provided by VK_KHR_bind_memory2 with VK_KHR_device_group
typedef VkBindBufferMemoryDeviceGroupInfo VkBindBufferMemoryDeviceGroupInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **deviceIndexCount** is the number of elements in **pDeviceIndices**.
- **pDeviceIndices** is a pointer to an array of device indices.

If the **pNext** chain of [VkBindBufferMemoryInfo](#) includes a [VkBindBufferMemoryDeviceGroupInfo](#) structure, then that structure determines how memory is bound to buffers across multiple devices in a device group.

If **deviceIndexCount** is greater than zero, then on device index *i* the buffer is attached to the instance of **memory** on the physical device with device index **pDeviceIndices[i]**.

If **deviceIndexCount** is zero and **memory** comes from a memory heap with the `VK_MEMORY_HEAP_MULTI_INSTANCE_BIT` bit set, then it is as if **pDeviceIndices** contains consecutive indices from zero to the number of physical devices in the logical device, minus one. In other words, by default each physical device attaches to its own instance of **memory**.

If **deviceIndexCount** is zero and **memory** comes from a memory heap without the `VK_MEMORY_HEAP_MULTI_INSTANCE_BIT` bit set, then it is as if **pDeviceIndices** contains an array of zeros. In other words, by default each physical device attaches to instance zero.

Valid Usage

- VUID-VkBindBufferMemoryDeviceGroupInfo-deviceIndexCount-01606
`deviceIndexCount` **must** either be zero or equal to the number of physical devices in the logical device
- VUID-VkBindBufferMemoryDeviceGroupInfo-pDeviceIndices-01607
All elements of `pDeviceIndices` **must** be valid device indices

Valid Usage (Implicit)

- VUID-VkBindBufferMemoryDeviceGroupInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_DEVICE_GROUP_INFO`
- VUID-VkBindBufferMemoryDeviceGroupInfo-pDeviceIndices-parameter
If `deviceIndexCount` is not 0, `pDeviceIndices` **must** be a valid pointer to an array of `deviceIndexCount` `uint32_t` values

The `VkBindMemoryStatusKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance6
typedef struct VkBindMemoryStatusKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkResult*          pResult;
} VkBindMemoryStatusKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pResult` is a pointer to a `VkResult` value.

If the `pNext` chain of `VkBindBufferMemoryInfo` or `VkBindImageMemoryInfo` includes a `VkBindMemoryStatusKHR` structure, then the `VkBindMemoryStatusKHR::pResult` will be populated with a value describing the result of the corresponding memory binding operation.

Valid Usage (Implicit)

- VUID-VkBindMemoryStatusKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BIND_MEMORY_STATUS_KHR`
- VUID-VkBindMemoryStatusKHR-pResult-parameter
`pResult` **must** be a valid pointer to a `VkResult` value

To attach memory to a `VkImage` object created without the `VK_IMAGE_CREATE_DISJOINT_BIT` set, call:

```
// Provided by VK_VERSION_1_0
VkResult vkBindImageMemory(
    VkDevice          device,
    VkImage           image,
    VkDeviceMemory    memory,
    VkDeviceSize      memoryOffset);
```

- **device** is the logical device that owns the image and memory.
- **image** is the image.
- **memory** is the [VkDeviceMemory](#) object describing the device memory to attach.
- **memoryOffset** is the start offset of the region of **memory** which is to be bound to the image. The number of bytes returned in the [VkMemoryRequirements::size](#) member in **memory**, starting from **memoryOffset** bytes, will be bound to the specified image.

[vkBindImageMemory](#) is equivalent to passing the same parameters through [VkBindImageMemoryInfo](#) to [vkBindImageMemory2KHR](#).

Valid Usage

- VUID-vkBindImageMemory-image-07460
image must not have been bound to a memory object
- VUID-vkBindImageMemory-image-01045
image must not have been created with any sparse memory binding flags
- VUID-vkBindImageMemory-memoryOffset-01046
memoryOffset must be less than the size of **memory**
- VUID-vkBindImageMemory-image-01445
If **image** requires a dedicated allocation (as reported by [vkGetImageMemoryRequirements2KHR](#) in [VkMemoryDedicatedRequirements::requiresDedicatedAllocation](#) for **image**), **memory must** have been created with [VkMemoryDedicatedAllocateInfo::image](#) equal to **image**
- VUID-vkBindImageMemory-memory-02628
If the [VkMemoryAllocateInfo](#) provided when **memory** was allocated included a [VkMemoryDedicatedAllocateInfo](#) structure in its **pNext** chain, and [VkMemoryDedicatedAllocateInfo::image](#) was not [VK_NULL_HANDLE](#), then **image must** equal [VkMemoryDedicatedAllocateInfo::image](#) and **memoryOffset must** be zero
- VUID-vkBindImageMemory-memory-02728
If the value of [VkExportMemoryAllocateInfo::handleTypes](#) used to allocate **memory** is not 0, it **must** include at least one of the handles set in [VkExternalMemoryImageCreateInfo::handleTypes](#) when **image** was created
- VUID-vkBindImageMemory-memory-02989
If **memory** was created by a memory import operation, the external handle type of the imported memory **must** also have been set in [VkExternalMemoryImageCreateInfo::handleTypes](#) when **image** was created

- VUID-vkBindImageMemory-image-01608
`image` **must** not have been created with the `VK_IMAGE_CREATE_DISJOINT_BIT` set
- VUID-vkBindImageMemory-memory-01047
`memory` **must** have been allocated using one of the memory types allowed in the `memoryTypeBits` member of the `VkMemoryRequirements` structure returned from a call to `vkGetImageMemoryRequirements` with `image`
- VUID-vkBindImageMemory-memoryOffset-01048
`memoryOffset` **must** be an integer multiple of the `alignment` member of the `VkMemoryRequirements` structure returned from a call to `vkGetImageMemoryRequirements` with `image`
- VUID-vkBindImageMemory-size-01049
The difference of the size of `memory` and `memoryOffset` **must** be greater than or equal to the `size` member of the `VkMemoryRequirements` structure returned from a call to `vkGetImageMemoryRequirements` with the same `image`

Valid Usage (Implicit)

- VUID-vkBindImageMemory-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkBindImageMemory-image-parameter
`image` **must** be a valid `VkImage` handle
- VUID-vkBindImageMemory-memory-parameter
`memory` **must** be a valid `VkDeviceMemory` handle
- VUID-vkBindImageMemory-image-parent
`image` **must** have been created, allocated, or retrieved from `device`
- VUID-vkBindImageMemory-memory-parent
`memory` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `image` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

To attach memory to image objects for one or more images at a time, call:

```
// Provided by VK_KHR_bind_memory2
VkResult vkBindImageMemory2KHR(
    VkDevice                                device,
    uint32_t                                bindInfoCount,
    const VkBindImageMemoryInfo*            pBindInfos);
```

- **device** is the logical device that owns the images and memory.
- **bindInfoCount** is the number of elements in **pBindInfos**.
- **pBindInfos** is a pointer to an array of **VkBindImageMemoryInfo** structures, describing images and memory to bind.

On some implementations, it **may** be more efficient to batch memory bindings into a single command.

If the **maintenance6** feature is enabled, this command **must** attempt to perform all of the memory binding operations described by **pBindInfos**, and **must** not early exit on the first failure.

If any of the memory binding operations described by **pBindInfos** fail, the **VkResult** returned by this command **must** be the return value of any one of the memory binding operations which did not return **VK_SUCCESS**.

NOTE

If the **vkBindImageMemory2KHR** command failed, **VkBindMemoryStatusKHR** structures were not included in the **pNext** chains of each element of **pBindInfos**, and **bindInfoCount** was greater than one, then the images referenced by **pBindInfos** will be in an indeterminate state, and must not be used.

Applications should destroy these images.

Valid Usage

- VUID-vkBindImageMemory2-pBindInfos-02858
If any **VkBindImageMemoryInfo::image** was created with **VK_IMAGE_CREATE_DISJOINT_BIT** then all planes of **VkBindImageMemoryInfo::image** **must** be bound individually in separate **pBindInfos**
- VUID-vkBindImageMemory2-pBindInfos-04006
pBindInfos **must** not refer to the same image subresource more than once

Valid Usage (Implicit)

- VUID-vkBindImageMemory2-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkBindImageMemory2-pBindInfos-parameter
pBindInfos **must** be a valid pointer to an array of **bindInfoCount** valid

VkBindImageMemoryInfo structures

- VUID-vkBindImageMemory2-bindInfoCount-arraylength
`bindInfoCount` **must** be greater than 0

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

`VkBindImageMemoryInfo` contains members corresponding to the parameters of `vkBindImageMemory`.

The `VkBindImageMemoryInfo` structure is defined as:

```
typedef struct VkBindImageMemoryInfo {  
    VkStructureType    sType;  
    const void*        pNext;  
    VkImage             image;  
    VkDeviceMemory      memory;  
    VkDeviceSize        memoryOffset;  
} VkBindImageMemoryInfo;
```

or the equivalent

```
// Provided by VK_KHR_bind_memory2  
typedef VkBindImageMemoryInfo VkBindImageMemoryInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `image` is the image to be attached to memory.
- `memory` is a `VkDeviceMemory` object describing the device memory to attach.
- `memoryOffset` is the start offset of the region of `memory` which is to be bound to the image. The number of bytes returned in the `VkMemoryRequirements::size` member in `memory`, starting from `memoryOffset` bytes, will be bound to the specified image.

Valid Usage

- VUID-VkBindImageMemoryInfo-image-07460
`image` **must** not have been bound to a memory object

- VUID-VkBindImageMemoryInfo-image-01045
image **must** not have been created with any sparse memory binding flags
- VUID-VkBindImageMemoryInfo-memoryOffset-01046
memoryOffset **must** be less than the size of **memory**
- VUID-VkBindImageMemoryInfo-image-01445
If **image** requires a dedicated allocation (as reported by `vkGetImageMemoryRequirements2KHR` in `VkMemoryDedicatedRequirements::requiresDedicatedAllocation` for **image**), **memory** **must** have been created with `VkMemoryDedicatedAllocateInfo::image` equal to **image**
- VUID-VkBindImageMemoryInfo-memory-02628
If the `VkMemoryAllocateInfo` provided when **memory** was allocated included a `VkMemoryDedicatedAllocateInfo` structure in its **pNext** chain, and `VkMemoryDedicatedAllocateInfo::image` was not `VK_NULL_HANDLE`, then **image** **must** equal `VkMemoryDedicatedAllocateInfo::image` and **memoryOffset** **must** be zero
- VUID-VkBindImageMemoryInfo-memory-02728
If the value of `VkExportMemoryAllocateInfo::handleTypes` used to allocate **memory** is not `0`, it **must** include at least one of the handles set in `VkExternalMemoryImageCreateInfo::handleTypes` when **image** was created
- VUID-VkBindImageMemoryInfo-memory-02989
If **memory** was created by a memory import operation, the external handle type of the imported memory **must** also have been set in `VkExternalMemoryImageCreateInfo::handleTypes` when **image** was created
- VUID-VkBindImageMemoryInfo-pNext-01615
If the **pNext** chain does not include a `VkBindImagePlaneMemoryInfo` structure, **memory** **must** have been allocated using one of the memory types allowed in the **memoryTypeBits** member of the `VkMemoryRequirements` structure returned from a call to `vkGetImageMemoryRequirements2KHR` with **image**
- VUID-VkBindImageMemoryInfo-pNext-01616
If the **pNext** chain does not include a `VkBindImagePlaneMemoryInfo` structure, **memoryOffset** **must** be an integer multiple of the **alignment** member of the `VkMemoryRequirements` structure returned from a call to `vkGetImageMemoryRequirements2KHR` with **image**
- VUID-VkBindImageMemoryInfo-pNext-01617
If the **pNext** chain does not include a `VkBindImagePlaneMemoryInfo` structure, the difference of the size of **memory** and **memoryOffset** **must** be greater than or equal to the **size** member of the `VkMemoryRequirements` structure returned from a call to `vkGetImageMemoryRequirements2KHR` with the same **image**
- VUID-VkBindImageMemoryInfo-pNext-01618
If the **pNext** chain includes a `VkBindImagePlaneMemoryInfo` structure, **image** **must** have been created with the `VK_IMAGE_CREATE_DISJOINT_BIT` bit set
- VUID-VkBindImageMemoryInfo-image-07736
If **image** was created with the `VK_IMAGE_CREATE_DISJOINT_BIT` bit set, then the **pNext** chain **must** include a `VkBindImagePlaneMemoryInfo` structure

- VUID-VkBindImageMemoryInfo-pNext-01619
If the `pNext` chain includes a `VkBindImagePlaneMemoryInfo` structure, `memory` **must** have been allocated using one of the memory types allowed in the `memoryTypeBits` member of the `VkMemoryRequirements` structure returned from a call to `vkGetImageMemoryRequirements2KHR` with `image` and where `VkBindImagePlaneMemoryInfo::planeAspect` corresponds to the `VkImagePlaneMemoryRequirementsInfo::planeAspect` in the `VkImageMemoryRequirementsInfo2` structure's `pNext` chain
- VUID-VkBindImageMemoryInfo-pNext-01620
If the `pNext` chain includes a `VkBindImagePlaneMemoryInfo` structure, `memoryOffset` **must** be an integer multiple of the `alignment` member of the `VkMemoryRequirements` structure returned from a call to `vkGetImageMemoryRequirements2KHR` with `image` and where `VkBindImagePlaneMemoryInfo::planeAspect` corresponds to the `VkImagePlaneMemoryRequirementsInfo::planeAspect` in the `VkImageMemoryRequirementsInfo2` structure's `pNext` chain
- VUID-VkBindImageMemoryInfo-pNext-01621
If the `pNext` chain includes a `VkBindImagePlaneMemoryInfo` structure, the difference of the size of `memory` and `memoryOffset` **must** be greater than or equal to the `size` member of the `VkMemoryRequirements` structure returned from a call to `vkGetImageMemoryRequirements2KHR` with the same `image` and where `VkBindImagePlaneMemoryInfo::planeAspect` corresponds to the `VkImagePlaneMemoryRequirementsInfo::planeAspect` in the `VkImageMemoryRequirementsInfo2` structure's `pNext` chain
- VUID-VkBindImageMemoryInfo-pNext-01626
If the `pNext` chain includes a `VkBindImageMemoryDeviceGroupInfo` structure, all instances of `memory` specified by `VkBindImageMemoryDeviceGroupInfo::pDeviceIndices` **must** have been allocated
- VUID-VkBindImageMemoryInfo-pNext-01627
If the `pNext` chain includes a `VkBindImageMemoryDeviceGroupInfo` structure, and `VkBindImageMemoryDeviceGroupInfo::splitInstanceBindRegionCount` is not zero, then `image` **must** have been created with the `VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT` bit set
- VUID-VkBindImageMemoryInfo-pNext-01628
If the `pNext` chain includes a `VkBindImageMemoryDeviceGroupInfo` structure, all elements of `VkBindImageMemoryDeviceGroupInfo::pSplitInstanceBindRegions` **must** be valid rectangles contained within the dimensions of `image`
- VUID-VkBindImageMemoryInfo-pNext-01629
If the `pNext` chain includes a `VkBindImageMemoryDeviceGroupInfo` structure, the union of the areas of all elements of `VkBindImageMemoryDeviceGroupInfo::pSplitInstanceBindRegions` that correspond to the same instance of `image` **must** cover the entire image
- VUID-VkBindImageMemoryInfo-image-01630
If `image` was created with a valid swapchain handle in `VkImageSwapchainCreateInfoKHR::swapchain`, then the `pNext` chain **must** include a `VkBindImageMemorySwapchainInfoKHR` structure containing the same swapchain

handle

- VUID-VkBindImageMemoryInfo-pNext-01631

If the **pNext** chain includes a **VkBindImageMemorySwapchainInfoKHR** structure, **memory** **must** be **VK_NULL_HANDLE**

- VUID-VkBindImageMemoryInfo-pNext-01632

If the **pNext** chain does not include a **VkBindImageMemorySwapchainInfoKHR** structure, **memory** **must** be a valid **VkDeviceMemory** handle

Valid Usage (Implicit)

- VUID-VkBindImageMemoryInfo-sType-sType

sType **must** be **VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_INFO**

- VUID-VkBindImageMemoryInfo-pNext-pNext

Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either **NULL** or a pointer to a valid instance of **VkBindImageMemoryDeviceGroupInfo**, **VkBindImageMemorySwapchainInfoKHR**, **VkBindImagePlaneMemoryInfo**, or **VkBindMemoryStatusKHR**

- VUID-VkBindImageMemoryInfo-sType-unique

The **sType** value of each struct in the **pNext** chain **must** be unique

- VUID-VkBindImageMemoryInfo-image-parameter

image **must** be a valid **VkImage** handle

- VUID-VkBindImageMemoryInfo-commonparent

Both of **image**, and **memory** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same **VkDevice**

The **VkBindImageMemoryDeviceGroupInfo** structure is defined as:

```
typedef struct VkBindImageMemoryDeviceGroupInfo {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           deviceIndexCount;
    const uint32_t*     pDeviceIndices;
    uint32_t           splitInstanceBindRegionCount;
    const VkRect2D*     pSplitInstanceBindRegions;
} VkBindImageMemoryDeviceGroupInfo;
```

or the equivalent

```
// Provided by VK_KHR_bind_memory2 with VK_KHR_device_group
typedef VkBindImageMemoryDeviceGroupInfo VkBindImageMemoryDeviceGroupInfoKHR;
```

- **sType** is a **VkStructureType** value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceIndexCount` is the number of elements in `pDeviceIndices`.
- `pDeviceIndices` is a pointer to an array of device indices.
- `splitInstanceBindRegionCount` is the number of elements in `pSplitInstanceBindRegions`.
- `pSplitInstanceBindRegions` is a pointer to an array of `VkRect2D` structures describing which regions of the image are attached to each instance of memory.

If the `pNext` chain of `VkBindImageMemoryInfo` includes a `VkBindImageMemoryDeviceGroupInfo` structure, then that structure determines how memory is bound to images across multiple devices in a device group.

If `deviceIndexCount` is greater than zero, then on device index `i` `image` is attached to the instance of the memory on the physical device with device index `pDeviceIndices[i]`.

Let `N` be the number of physical devices in the logical device. If `splitInstanceBindRegionCount` is greater than zero, then `pSplitInstanceBindRegions` is a pointer to an array of N^2 rectangles, where the image region specified by the rectangle at element `i*N+j` in resource instance `i` is bound to the memory instance `j`. The blocks of the memory that are bound to each sparse image block region use an offset in memory, relative to `memoryOffset`, computed as if the whole image was being bound to a contiguous range of memory. In other words, horizontally adjacent image blocks use consecutive blocks of memory, vertically adjacent image blocks are separated by the number of bytes per block multiplied by the width in blocks of `image`, and the block at (0,0) corresponds to memory starting at `memoryOffset`.

If `splitInstanceBindRegionCount` and `deviceIndexCount` are zero and the memory comes from a memory heap with the `VK_MEMORY_HEAP_MULTI_INSTANCE_BIT` bit set, then it is as if `pDeviceIndices` contains consecutive indices from zero to the number of physical devices in the logical device, minus one. In other words, by default each physical device attaches to its own instance of the memory.

If `splitInstanceBindRegionCount` and `deviceIndexCount` are zero and the memory comes from a memory heap without the `VK_MEMORY_HEAP_MULTI_INSTANCE_BIT` bit set, then it is as if `pDeviceIndices` contains an array of zeros. In other words, by default each physical device attaches to instance zero.

Valid Usage

- VUID-VkBindImageMemoryDeviceGroupInfo-deviceIndexCount-01633
At least one of `deviceIndexCount` and `splitInstanceBindRegionCount` **must** be zero
- VUID-VkBindImageMemoryDeviceGroupInfo-deviceIndexCount-01634
`deviceIndexCount` **must** either be zero or equal to the number of physical devices in the logical device
- VUID-VkBindImageMemoryDeviceGroupInfo-pDeviceIndices-01635
All elements of `pDeviceIndices` **must** be valid device indices
- VUID-VkBindImageMemoryDeviceGroupInfo-splitInstanceBindRegionCount-01636
`splitInstanceBindRegionCount` **must** either be zero or equal to the number of physical devices in the logical device squared

- VUID-VkBindImageMemoryDeviceGroupInfo-pSplitInstanceBindRegions-01637
Elements of `pSplitInstanceBindRegions` that correspond to the same instance of an image **must** not overlap
- VUID-VkBindImageMemoryDeviceGroupInfo-offset-01638
The `offset.x` member of any element of `pSplitInstanceBindRegions` **must** be a multiple of the sparse image block width (`VkSparseImageFormatProperties::imageGranularity.width`) of all non-metadata aspects of the image
- VUID-VkBindImageMemoryDeviceGroupInfo-offset-01639
The `offset.y` member of any element of `pSplitInstanceBindRegions` **must** be a multiple of the sparse image block height (`VkSparseImageFormatProperties::imageGranularity.height`) of all non-metadata aspects of the image
- VUID-VkBindImageMemoryDeviceGroupInfo-extent-01640
The `extent.width` member of any element of `pSplitInstanceBindRegions` **must** either be a multiple of the sparse image block width of all non-metadata aspects of the image, or else `extent.width + offset.x` **must** equal the width of the image subresource
- VUID-VkBindImageMemoryDeviceGroupInfo-extent-01641
The `extent.height` member of any element of `pSplitInstanceBindRegions` **must** either be a multiple of the sparse image block height of all non-metadata aspects of the image, or else `extent.height + offset.y` **must** equal the height of the image subresource

Valid Usage (Implicit)

- VUID-VkBindImageMemoryDeviceGroupInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_DEVICE_GROUP_INFO`
- VUID-VkBindImageMemoryDeviceGroupInfo-pDeviceIndices-parameter
If `deviceIndexCount` is not 0, `pDeviceIndices` **must** be a valid pointer to an array of `deviceIndexCount` `uint32_t` values
- VUID-VkBindImageMemoryDeviceGroupInfo-pSplitInstanceBindRegions-parameter
If `splitInstanceBindRegionCount` is not 0, `pSplitInstanceBindRegions` **must** be a valid pointer to an array of `splitInstanceBindRegionCount` `VkRect2D` structures

If the `pNext` chain of `VkBindImageMemoryInfo` includes a `VkBindImageMemorySwapchainInfoKHR` structure, then that structure includes a swapchain handle and image index indicating that the image will be bound to memory from that swapchain.

The `VkBindImageMemorySwapchainInfoKHR` structure is defined as:


```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
typedef struct VkBindImageMemorySwapchainInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkSwapchainKHR     swapchain;
    uint32_t           imageIndex;
} VkBindImageMemorySwapchainInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchain` is `VK_NULL_HANDLE` or a swapchain handle.
- `imageIndex` is an image index within `swapchain`.

If `swapchain` is not `NULL`, the `swapchain` and `imageIndex` are used to determine the memory that the image is bound to, instead of `memory` and `memoryOffset`.

Memory **can** be bound to a swapchain and use the `pDeviceIndices` or `pSplitInstanceBindRegions` members of `VkBindImageMemoryDeviceGroupInfo`.

Valid Usage

- VUID-VkBindImageMemorySwapchainInfoKHR-imageIndex-01644
`imageIndex` **must** be less than the number of images in `swapchain`

Valid Usage (Implicit)

- VUID-VkBindImageMemorySwapchainInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_SWAPCHAIN_INFO_KHR`
- VUID-VkBindImageMemorySwapchainInfoKHR-swapchain-parameter
`swapchain` **must** be a valid `VkSwapchainKHR` handle

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

In order to bind *planes* of a *disjoint image*, add a `VkBindImagePlaneMemoryInfo` structure to the `pNext` chain of `VkBindImageMemoryInfo`.

The `VkBindImagePlaneMemoryInfo` structure is defined as:

```
typedef struct VkBindImagePlaneMemoryInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkImageAspectFlagBits planeAspect;
} VkBindImagePlaneMemoryInfo;
```

or the equivalent

```
// Provided by VK_KHR_sampler_ycbcr_conversion
typedef VkBindImagePlaneMemoryInfo VkBindImagePlaneMemoryInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `planeAspect` is a `VkImageAspectFlagBits` value specifying the aspect of the disjoint image plane to bind.

Valid Usage

- VUID-VkBindImagePlaneMemoryInfo-planeAspect-02283
If the image's `tiling` is `VK_IMAGE_TILING_LINEAR` or `VK_IMAGE_TILING_OPTIMAL`, then `planeAspect` **must** be a single valid `multi-planar aspect mask` bit
- VUID-VkBindImagePlaneMemoryInfo-planeAspect-02284
If the image's `tiling` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then `planeAspect` **must** be a single valid *memory plane* for the image (that is, `aspectMask` **must** specify a plane index that is less than the `VkDrmFormatModifierPropertiesEXT::drmFormatModifierPlaneCount` associated with the image's `format` and `VkImageDrmFormatModifierPropertiesEXT::drmFormatModifier`)

Valid Usage (Implicit)

- VUID-VkBindImagePlaneMemoryInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BIND_IMAGE_PLANE_MEMORY_INFO`
- VUID-VkBindImagePlaneMemoryInfo-planeAspect-parameter
`planeAspect` **must** be a valid `VkImageAspectFlagBits` value

Buffer-Image Granularity

The implementation-dependent limit `bufferImageGranularity` specifies a page-like granularity at which linear and non-linear resources **must** be placed in adjacent memory locations to avoid aliasing. Two resources which do not satisfy this granularity requirement are said to *alias*. `bufferImageGranularity` is specified in bytes, and **must** be a power of two. Implementations which do not impose a granularity restriction **may** report a `bufferImageGranularity` value of one.

NOTE

Despite its name, `bufferImageGranularity` is really a granularity between “linear” and “non-linear” resources.

Given resourceA at the lower memory offset and resourceB at the higher memory offset in the same `VkDeviceMemory` object, where one resource is linear and the other is non-linear (as defined in the [Glossary](#)), and the following:

```
resourceA.end      = resourceA.memoryOffset + resourceA.size - 1
resourceA.endPage  = resourceA.end & ~(bufferImageGranularity-1)
resourceB.start    = resourceB.memoryOffset
resourceB.startPage = resourceB.start & ~(bufferImageGranularity-1)
```

The following property **must** hold:

```
resourceA.endPage < resourceB.startPage
```

That is, the end of the first resource (A) and the beginning of the second resource (B) **must** be on separate “pages” of size `bufferImageGranularity`. `bufferImageGranularity` **may** be different than the physical page size of the memory heap. This restriction is only needed when a linear resource and a non-linear resource are adjacent in memory and will be used simultaneously. The memory ranges of adjacent resources **can** be closer than `bufferImageGranularity`, provided they meet the **alignment** requirement for the objects in question.

Sparse block size in bytes and sparse image and buffer memory alignments **must** all be multiples of the `bufferImageGranularity`. Therefore, memory bound to sparse resources naturally satisfies the `bufferImageGranularity`.

12.9. Resource Sharing Mode

Buffer and image objects are created with a *sharing mode* controlling how they **can** be accessed from queues. The supported sharing modes are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSharingMode {
    VK_SHARING_MODE_EXCLUSIVE = 0,
    VK_SHARING_MODE_CONCURRENT = 1,
} VkSharingMode;
```

- `VK_SHARING_MODE_EXCLUSIVE` specifies that access to any range or image subresource of the object will be exclusive to a single queue family at a time.
- `VK_SHARING_MODE_CONCURRENT` specifies that concurrent access to any range or image subresource of the object from multiple queue families is supported.

NOTE

`VK_SHARING_MODE_CONCURRENT` **may** result in lower performance access to the buffer or

image than `VK_SHARING_MODE_EXCLUSIVE`.

Ranges of buffers and image subresources of image objects created using `VK_SHARING_MODE_EXCLUSIVE` **must** only be accessed by queues in the queue family that has *ownership* of the resource. Upon creation, such resources are not owned by any queue family; ownership is implicitly acquired upon first use within a queue. Once a resource using `VK_SHARING_MODE_EXCLUSIVE` is owned by some queue family, the application **must** perform a [queue family ownership transfer](#) to make the memory contents of a range or image subresource accessible to a different queue family.

NOTE

Images still require a [layout transition](#) from `VK_IMAGE_LAYOUT_UNDEFINED` or `VK_IMAGE_LAYOUT_PREINITIALIZED` before being used on the first queue.

A queue family **can** take ownership of an image subresource or buffer range of a resource created with `VK_SHARING_MODE_EXCLUSIVE`, without an ownership transfer, in the same way as for a resource that was just created; however, taking ownership in this way has the effect that the contents of the image subresource or buffer range are undefined.

Ranges of buffers and image subresources of image objects created using `VK_SHARING_MODE_CONCURRENT` **must** only be accessed by queues from the queue families specified through the `queueFamilyIndexCount` and `pQueueFamilyIndices` members of the corresponding create info structures.

12.9.1. External Resource Sharing

Resources **should** only be accessed in the Vulkan instance that has exclusive ownership of their underlying memory. Only one Vulkan instance has exclusive ownership of a resource's underlying memory at a given time, regardless of whether the resource was created using `VK_SHARING_MODE_EXCLUSIVE` or `VK_SHARING_MODE_CONCURRENT`. Applications can transfer ownership of a resource's underlying memory only if the memory has been imported from or exported to another instance or external API using external memory handles. The semantics for transferring ownership outside of the instance are similar to those used for transferring ownership of `VK_SHARING_MODE_EXCLUSIVE` resources between queues, and is also accomplished using [VkBufferMemoryBarrier](#) or [VkImageMemoryBarrier](#) operations. To make the contents of the underlying memory accessible in the destination instance or API, applications **must**

1. Release exclusive ownership from the source instance or API.
2. Ensure the release operation has completed using semaphores or fences.
3. Acquire exclusive ownership in the destination instance or API

Unlike queue family ownership transfers, the destination instance or API is not specified explicitly when releasing ownership, nor is the source instance or API specified when acquiring ownership. Instead, the image or memory barrier's `dstQueueFamilyIndex` or `srcQueueFamilyIndex` parameters are set to the reserved queue family index `VK_QUEUE_FAMILY_EXTERNAL` or `VK_QUEUE_FAMILY_FOREIGN_EXT` to represent the external destination or source respectively.

Binding a resource to a memory object shared between multiple Vulkan instances or other APIs does not change the ownership of the underlying memory. The first entity to access the resource implicitly acquires ownership. An entity **can** also implicitly take ownership from another entity in

the same way without an explicit ownership transfer. However, taking ownership in this way has the effect that the contents of the underlying memory are undefined.

Accessing a resource backed by memory that is owned by a particular instance or API has the same semantics as accessing a `VK_SHARING_MODE_EXCLUSIVE` resource, with one exception: Implementations **must** ensure layout transitions performed on one member of a set of identical subresources of identical images that alias the same range of an underlying memory object affect the layout of all the subresources in the set.

As a corollary, writes to any image subresources in such a set **must** not make the contents of memory used by other subresources in the set undefined. An application **can** define the content of a subresource of one image by performing device writes to an identical subresource of another image provided both images are bound to the same region of external memory. Applications **may** also add resources to such a set after the content of the existing set members has been defined without making the content undefined by creating a new image with the initial layout `VK_IMAGE_LAYOUT_UNDEFINED` and binding it to the same region of external memory as the existing images.

NOTE

Because layout transitions apply to all identical images aliasing the same region of external memory, the actual layout of the memory backing a new image as well as an existing image with defined content will not be undefined. Such an image is not usable until it acquires ownership of its memory from the existing owner. Therefore, the layout specified as part of this transition will be the true initial layout of the image. The undefined layout specified when creating it is a placeholder to simplify valid usage requirements.

12.10. Memory Aliasing

A range of a `VkDeviceMemory` allocation is *aliased* if it is bound to multiple resources simultaneously, as described below, via `vkBindImageMemory`, `vkBindBufferMemory`, via [sparse memory bindings](#), or by binding the memory to resources in multiple Vulkan instances or external APIs using external memory handle export and import mechanisms.

Consider two resources, `resourceA` and `resourceB`, bound respectively to memory `rangeA` and `rangeB`. Let `paddedRangeA` and `paddedRangeB` be, respectively, `rangeA` and `rangeB` aligned to `bufferImageGranularity`. If the resources are both linear or both non-linear (as defined in the [Glossary](#)), then the resources *alias* the memory in the intersection of `rangeA` and `rangeB`. If one resource is linear and the other is non-linear, then the resources *alias* the memory in the intersection of `paddedRangeA` and `paddedRangeB`.

Applications **can** alias memory, but use of multiple aliases is subject to several constraints.

NOTE

Memory aliasing **can** be useful to reduce the total device memory footprint of an application, if some large resources are used for disjoint periods of time.

When a [non-linear](#), non-`VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` image is bound to an aliased range, all image subresources of the image *overlap* the range. When a linear image is bound to an aliased range, the image subresources that (according to the image's advertised layout) include bytes from

the aliased range overlap the range. When a `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` image has sparse image blocks bound to an aliased range, only image subresources including those sparse image blocks overlap the range, and when the memory bound to the image's mip tail overlaps an aliased range all image subresources in the mip tail overlap the range.

Buffers, and linear image subresources in either the `VK_IMAGE_LAYOUT_PREINITIALIZED` or `VK_IMAGE_LAYOUT_GENERAL` layouts, are *host-accessible subresources*. That is, the host has a well-defined addressing scheme to interpret the contents, and thus the layout of the data in memory **can** be consistently interpreted across aliases if each of those aliases is a host-accessible subresource. Non-linear images, and linear image subresources in other layouts, are not host-accessible.

If two aliases are both host-accessible, then they interpret the contents of the memory in consistent ways, and data written to one alias **can** be read by the other alias.

If two aliases are both images that were created with identical creation parameters, both were created with the `VK_IMAGE_CREATE_ALIAS_BIT` flag set, and both are bound identically to memory except for `VkBindImageMemoryDeviceGroupInfo::pDeviceIndices` and `VkBindImageMemoryDeviceGroupInfo::pSplitInstanceBindRegions`, then they interpret the contents of the memory in consistent ways, and data written to one alias **can** be read by the other alias.

Additionally, if an individual plane of a multi-planar image and a single-plane image alias the same memory, then they also interpret the contents of the memory in consistent ways under the same conditions, but with the following modifications:

- Both **must** have been created with the `VK_IMAGE_CREATE_DISJOINT_BIT` flag.
- The single-plane image **must** have a `VkFormat` that is *equivalent* to that of the multi-planar image's individual plane.
- The single-plane image and the individual plane of the multi-planar image **must** be bound identically to memory except for `VkBindImageMemoryDeviceGroupInfo::pDeviceIndices` and `VkBindImageMemoryDeviceGroupInfo::pSplitInstanceBindRegions`.
- The `width` and `height` of the single-plane image are derived from the multi-planar image's dimensions in the manner listed for *plane compatibility* for the aliased plane.
- If either image's `tiling` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then both images **must** be *linear*.
- All other creation parameters **must** be identical

Aliases created by binding the same memory to resources in multiple Vulkan instances or external APIs using external memory handle export and import mechanisms interpret the contents of the memory in consistent ways, and data written to one alias **can** be read by the other alias.

Otherwise, the aliases interpret the contents of the memory differently, and writes via one alias make the contents of memory partially or completely undefined to the other alias. If the first alias is a host-accessible subresource, then the bytes affected are those written by the memory operations according to its addressing scheme. If the first alias is not host-accessible, then the bytes affected are those overlapped by the image subresources that were written. If the second alias is a host-accessible subresource, the affected bytes become undefined. If the second alias is not host-accessible, all sparse image blocks (for sparse partially-resident images) or all image subresources

(for non-sparse image and fully resident sparse images) that overlap the affected bytes become undefined.

If any image subresources are made undefined due to writes to an alias, then each of those image subresources **must** have its layout transitioned from `VK_IMAGE_LAYOUT_UNDEFINED` to a valid layout before it is used, or from `VK_IMAGE_LAYOUT_PREINITIALIZED` if the memory has been written by the host. If any sparse blocks of a sparse image have been made undefined, then only the image subresources containing them **must** be transitioned.

Use of an overlapping range by two aliases **must** be separated by a memory dependency using the appropriate [access types](#) if at least one of those uses performs writes, whether the aliases interpret memory consistently or not. If buffer or image memory barriers are used, the scope of the barrier **must** contain the entire range and/or set of image subresources that overlap.

If two aliasing image views are used in the same framebuffer, then the render pass **must** declare the attachments using the `VK_ATTACHMENT_DESCRIPTION_MAY_ALIAS_BIT`, and follow the other rules listed in that section.

NOTE

Memory recycled via an application suballocator (i.e. without freeing and reallocating the memory objects) is not substantially different from memory aliasing. However, a suballocator usually waits on a fence before recycling a region of memory, and signaling a fence involves sufficient implicit dependencies to satisfy all the above requirements.

12.10.1. Resource Memory Overlap

Applications **can** safely access resources concurrently via separate device and host operations as long as the accessed memory locations are guaranteed to not overlap, as defined in [Memory Location](#), and the operation, resource, and access are otherwise independently valid.

Some operations have alignment requirements or access ambiguous [memory locations](#), so the semantics of a particular operation **should** be considered when determining the overlap. Such requirements will be described alongside the operation. Operations between host and device when using non-coherent memory are aligned to `nonCoherentAtomSize`, as defined by [vkFlushMappedMemoryRanges](#) and [vkInvalidateMappedMemoryRanges](#).

NOTE

The intent is that buffers (or linear images) can be accessed concurrently, even when they share cache lines, but otherwise do not access the same memory range. The concept of a device cache line size is not exposed in the memory model.

Chapter 13. Samplers

VkSampler objects represent the state of an image sampler which is used by the implementation to read image data and apply filtering and other transformations for the shader.

Samplers are represented by **VkSampler** handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSampler)
```

To create a sampler object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateSampler(
    VkDevice                                device,
    const VkSamplerCreateInfo*              pCreateInfo,
    const VkAllocationCallbacks*            pAllocator,
    VkSampler*                              pSampler);
```

- **device** is the logical device that creates the sampler.
- **pCreateInfo** is a pointer to a **VkSamplerCreateInfo** structure specifying the state of the sampler object.
- **pAllocator** controls host memory allocation as described in the **Memory Allocation** chapter.
- **pSampler** is a pointer to a **VkSampler** handle in which the resulting sampler object is returned.

Valid Usage

- VUID-vkCreateSampler-device-09668
device **must** support at least one queue family with one of the **VK_QUEUE_COMPUTE_BIT** or **VK_QUEUE_GRAPHICS_BIT** capabilities
- VUID-vkCreateSampler-maxSamplerAllocationCount-04110
There **must** be less than **VkPhysicalDeviceLimits::maxSamplerAllocationCount** **VkSampler** objects currently created on the device

Valid Usage (Implicit)

- VUID-vkCreateSampler-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkCreateSampler-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid **VkSamplerCreateInfo** structure
- VUID-vkCreateSampler-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid

[VkAllocationCallbacks](#) structure

- VUID-vkCreateSampler-pSampler-parameter
[pSampler](#) **must** be a valid pointer to a [VkSampler](#) handle

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)
- [VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR](#)

The [VkSamplerCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSamplerCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkSamplerCreateFlags  flags;
    VkFilter              magFilter;
    VkFilter              minFilter;
    VkSamplerMipmapMode   mipmapMode;
    VkSamplerAddressMode  addressModeU;
    VkSamplerAddressMode  addressModeV;
    VkSamplerAddressMode  addressModeW;
    float                 mipLodBias;
    VkBool32              anisotropyEnable;
    float                 maxAnisotropy;
    VkBool32              compareEnable;
    VkCompareOp           compareOp;
    float                 minLod;
    float                 maxLod;
    VkBorderColor         borderColor;
    VkBool32              unnormalizedCoordinates;
} VkSamplerCreateInfo;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [flags](#) is a bitmask of [VkSamplerCreateFlagBits](#) describing additional parameters of the sampler.
- [magFilter](#) is a [VkFilter](#) value specifying the magnification filter to apply to lookups.
- [minFilter](#) is a [VkFilter](#) value specifying the minification filter to apply to lookups.
- [mipmapMode](#) is a [VkSamplerMipmapMode](#) value specifying the mipmap filter to apply to lookups.

- `addressModeU` is a `VkSamplerAddressMode` value specifying the addressing mode for U coordinates outside [0,1).
- `addressModeV` is a `VkSamplerAddressMode` value specifying the addressing mode for V coordinates outside [0,1).
- `addressModeW` is a `VkSamplerAddressMode` value specifying the addressing mode for W coordinates outside [0,1).
- `mipLodBias` is the bias to be added to mipmap LOD calculation and bias provided by image sampling functions in SPIR-V, as described in the [LOD Operation](#) section.
- `anisotropyEnable` is `VK_TRUE` to enable anisotropic filtering, as described in the [Texel Anisotropic Filtering](#) section, or `VK_FALSE` otherwise.
- `maxAnisotropy` is the anisotropy value clamp used by the sampler when `anisotropyEnable` is `VK_TRUE`. If `anisotropyEnable` is `VK_FALSE`, `maxAnisotropy` is ignored.
- `compareEnable` is `VK_TRUE` to enable comparison against a reference value during lookups, or `VK_FALSE` otherwise.
 - Note: Some implementations will default to shader state if this member does not match.
- `compareOp` is a `VkCompareOp` value specifying the comparison operator to apply to fetched data before filtering as described in the [Depth Compare Operation](#) section.
- `minLod` is used to clamp the [minimum of the computed LOD value](#).
- `maxLod` is used to clamp the [maximum of the computed LOD value](#). To avoid clamping the maximum value, set `maxLod` to the constant `VK_LOD_CLAMP_NONE`.
- `borderColor` is a `VkBorderColor` value specifying the predefined border color to use.
- `unnormalizedCoordinates` controls whether to use unnormalized or normalized texel coordinates to address texels of the image. When `unnormalizedCoordinates` is `VK_TRUE`, the range of the image coordinates used to lookup the texel is in the range of zero to the image size in each dimension. When `unnormalizedCoordinates` is `VK_FALSE`, the range of image coordinates is zero to one.

When `unnormalizedCoordinates` is `VK_TRUE`, images the sampler is used with in the shader have the following requirements:

- The `viewType` **must** be either `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`.
- The image view **must** have a single layer and a single mip level.

When `unnormalizedCoordinates` is `VK_TRUE`, image built-in functions in the shader that use the sampler have the following requirements:

- The functions **must** not use projection.
- The functions **must** not use offsets.

NOTE

Mapping of OpenGL to Vulkan Filter Modes

`magFilter` values of `VK_FILTER_NEAREST` and `VK_FILTER_LINEAR` directly correspond to `GL_NEAREST` and `GL_LINEAR` magnification filters. `minFilter` and `mipmapMode` combine to correspond to the similarly named OpenGL minification filter of `GL_minFilter_MIPMAP_mipmapMode` (e.g. `minFilter` of `VK_FILTER_LINEAR` and `mipmapMode` of

`VK_SAMPLER_MIPMAP_MODE_NEAREST` correspond to `GL_LINEAR_MIPMAP_NEAREST`).

There are no Vulkan filter modes that directly correspond to OpenGL minification filters of `GL_LINEAR` or `GL_NEAREST`, but they **can** be emulated using `VK_SAMPLER_MIPMAP_MODE_NEAREST`, `minLod = 0`, and `maxLod = 0.25`, and using `minFilter = VK_FILTER_LINEAR` or `minFilter = VK_FILTER_NEAREST`, respectively.

Note that using a `maxLod` of zero would cause **magnification** to always be performed, and the `magFilter` to always be used. This is valid, just not an exact match for OpenGL behavior. Clamping the maximum LOD to 0.25 allows the λ value to be non-zero and minification to be performed, while still always rounding down to the base level. If the `minFilter` and `magFilter` are equal, then using a `maxLod` of zero also works.

The maximum number of sampler objects which **can** be simultaneously created on a device is implementation-dependent and specified by the `maxSamplerAllocationCount` member of the `VkPhysicalDeviceLimits` structure.

NOTE

For historical reasons, if `maxSamplerAllocationCount` is exceeded, some implementations may return `VK_ERROR_TOO_MANY_OBJECTS`. Exceeding this limit will result in undefined behavior, and an application should not rely on the use of the returned error code in order to identify when the limit is reached.

Since `VkSampler` is a non-dispatchable handle type, implementations **may** return the same handle for sampler state vectors that are identical. In such cases, all such objects would only count once against the `maxSamplerAllocationCount` limit.

Valid Usage

- VUID-VkSamplerCreateInfo-mipLodBias-01069
The absolute value of `mipLodBias` **must** be less than or equal to `VkPhysicalDeviceLimits::maxSamplerLodBias`
- VUID-VkSamplerCreateInfo-samplerMipLodBias-04467
If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::samplerMipLodBias` is `VK_FALSE`, `mipLodBias` **must** be zero
- VUID-VkSamplerCreateInfo-maxLod-01973
`maxLod` **must** be greater than or equal to `minLod`
- VUID-VkSamplerCreateInfo-anisotropyEnable-01070
If the `samplerAnisotropy` feature is not enabled, `anisotropyEnable` **must** be `VK_FALSE`
- VUID-VkSamplerCreateInfo-anisotropyEnable-01071
If `anisotropyEnable` is `VK_TRUE`, `maxAnisotropy` **must** be between `1.0` and `VkPhysicalDeviceLimits::maxSamplerAnisotropy`, inclusive
- VUID-VkSamplerCreateInfo-minFilter-01645
If `sampler YCBCR conversion` is enabled and the `potential format features` of the sampler `YCBCR` conversion do not support

VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT, minFilter and magFilter **must** be equal to the sampler Y'C_BC_R conversion's chromaFilter

- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01072
If unnormalizedCoordinates is VK_TRUE, minFilter and magFilter **must** be equal
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01073
If unnormalizedCoordinates is VK_TRUE, mipmapMode **must** be VK_SAMPLER_MIPMAP_MODE_NEAREST
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01074
If unnormalizedCoordinates is VK_TRUE, minLod and maxLod **must** be zero
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01075
If unnormalizedCoordinates is VK_TRUE, addressModeU and addressModeV **must** each be either VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE or VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01076
If unnormalizedCoordinates is VK_TRUE, anisotropyEnable **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-unnormalizedCoordinates-01077
If unnormalizedCoordinates is VK_TRUE, compareEnable **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-addressModeU-01078
If any of addressModeU, addressModeV or addressModeW are VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER, borderColor **must** be a valid VkBorderColor value
- VUID-VkSamplerCreateInfo-addressModeU-01646
If sampler Y'C_BC_R conversion is enabled, addressModeU, addressModeV, and addressModeW **must** be VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE, anisotropyEnable **must** be VK_FALSE, and unnormalizedCoordinates **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-addressModeU-01079
If samplerMirrorClampToEdge is not enabled, and if the VK_KHR_sampler_mirror_clamp_to_edge extension is not enabled, addressModeU, addressModeV and addressModeW **must** not be VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE
- VUID-VkSamplerCreateInfo-compareEnable-01080
If compareEnable is VK_TRUE, compareOp **must** be a valid VkCompareOp value
- VUID-VkSamplerCreateInfo-magFilter-01081
If either magFilter or minFilter is VK_FILTER_CUBIC_EXT, anisotropyEnable **must** be VK_FALSE
- VUID-VkSamplerCreateInfo-nonSeamlessCubeMap-06788
If the nonSeamlessCubeMap feature is not enabled, flags **must** not include VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT
- VUID-VkSamplerCreateInfo-borderColor-04011
If borderColor is one of VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT, then a VkSamplerCustomBorderColorCreateInfoEXT **must** be included in the pNext chain
- VUID-VkSamplerCreateInfo-customBorderColors-04085
If the customBorderColors feature is not enabled, borderColor **must** not be VK_BORDER_COLOR_FLOAT_CUSTOM_EXT or VK_BORDER_COLOR_INT_CUSTOM_EXT
- VUID-VkSamplerCreateInfo-borderColor-04442

If `borderColor` is one of `VK_BORDER_COLOR_FLOAT_CUSTOM_EXT` or `VK_BORDER_COLOR_INT_CUSTOM_EXT`, and `VkSamplerCustomBorderColorCreateInfoEXT::format` is not `VK_FORMAT_UNDEFINED`, `VkSamplerCustomBorderColorCreateInfoEXT::customBorderColor` **must** be within the range of values representable in `format`

- VUID-VkSamplerCreateInfo-None-04012

The maximum number of samplers with custom border colors which **can** be simultaneously created on a device is implementation-dependent and specified by the `maxCustomBorderColorSamplers` member of the `VkPhysicalDeviceCustomBorderColorPropertiesEXT` structure

Valid Usage (Implicit)

- VUID-VkSamplerCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SAMPLER_CREATE_INFO`
- VUID-VkSamplerCreateInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkSamplerBorderColorComponentMappingCreateInfoEXT`, `VkSamplerCustomBorderColorCreateInfoEXT`, or `VkSamplerYcbcrConversionInfo`
- VUID-VkSamplerCreateInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkSamplerCreateInfo-flags-parameter
`flags` **must** be a valid combination of `VkSamplerCreateFlagBits` values
- VUID-VkSamplerCreateInfo-magFilter-parameter
`magFilter` **must** be a valid `VkFilter` value
- VUID-VkSamplerCreateInfo-minFilter-parameter
`minFilter` **must** be a valid `VkFilter` value
- VUID-VkSamplerCreateInfo-mipmapMode-parameter
`mipmapMode` **must** be a valid `VkSamplerMipmapMode` value
- VUID-VkSamplerCreateInfo-addressModeU-parameter
`addressModeU` **must** be a valid `VkSamplerAddressMode` value
- VUID-VkSamplerCreateInfo-addressModeV-parameter
`addressModeV` **must** be a valid `VkSamplerAddressMode` value
- VUID-VkSamplerCreateInfo-addressModeW-parameter
`addressModeW` **must** be a valid `VkSamplerAddressMode` value

`VK_LOD_CLAMP_NONE` is a special constant value used for `VkSamplerCreateInfo::maxLod` to indicate that maximum LOD clamping should not be performed.

```
#define VK_LOD_CLAMP_NONE 1000.0F
```

Bits which **can** be set in `VkSamplerCreateInfo::flags`, specifying additional parameters of a sampler, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSamplerCreateFlagBits {
    // Provided by VK_EXT_non_seamless_cube_map
    VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT = 0x00000004,
} VkSamplerCreateFlagBits;
```

- `VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT` specifies that [cube map edge handling](#) is not performed.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSamplerCreateFlags;
```

`VkSamplerCreateFlags` is a bitmask type for setting a mask of zero or more `VkSamplerCreateFlagBits`.

Possible values of the `VkSamplerCreateInfo::magFilter` and `minFilter` parameters, specifying filters used for texture lookups, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkFilter {
    VK_FILTER_NEAREST = 0,
    VK_FILTER_LINEAR = 1,
    // Provided by VK_EXT_filter_cubic
    VK_FILTER_CUBIC_EXT = 1000015000,
} VkFilter;
```

- `VK_FILTER_NEAREST` specifies nearest filtering.
- `VK_FILTER_LINEAR` specifies linear filtering.
- `VK_FILTER_CUBIC_EXT` specifies cubic filtering.

These filters are described in detail in [Texel Filtering](#).

Possible values of the `VkSamplerCreateInfo::mipmapMode`, specifying the mipmap mode used for texture lookups, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSamplerMipmapMode {
    VK_SAMPLER_MIPMAP_MODE_NEAREST = 0,
    VK_SAMPLER_MIPMAP_MODE_LINEAR = 1,
} VkSamplerMipmapMode;
```

- `VK_SAMPLER_MIPMAP_MODE_NEAREST` specifies nearest filtering.
- `VK_SAMPLER_MIPMAP_MODE_LINEAR` specifies linear filtering.

These modes are described in detail in [Texel Filtering](#).

Possible values of the `VkSamplerCreateInfo::addressMode*` parameters, specifying the behavior of sampling with coordinates outside the range [0,1] for the respective u, v, or w coordinate as defined in the [Wrapping Operation](#) section, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSamplerAddressMode {
    VK_SAMPLER_ADDRESS_MODE_REPEAT = 0,
    VK_SAMPLER_ADDRESS_MODE_MIRRORED_REPEAT = 1,
    VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE = 2,
    VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER = 3,
    // Provided by VK_KHR_sampler_mirror_clamp_to_edge
    VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE = 4,
    // Provided by VK_KHR_sampler_mirror_clamp_to_edge
    // VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE_KHR is a deprecated alias
    VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE_KHR =
VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE,
} VkSamplerAddressMode;
```

- `VK_SAMPLER_ADDRESS_MODE_REPEAT` specifies that the repeat wrap mode will be used.
- `VK_SAMPLER_ADDRESS_MODE_MIRRORED_REPEAT` specifies that the mirrored repeat wrap mode will be used.
- `VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE` specifies that the clamp to edge wrap mode will be used.
- `VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER` specifies that the clamp to border wrap mode will be used.
- `VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE` specifies that the mirror clamp to edge wrap mode will be used. This is only valid if the `VK_KHR_sampler_mirror_clamp_to_edge` extension is enabled.

Comparison operators compare a *reference* and a *test* value, and return a true (“passed”) or false (“failed”) value depending on the comparison operator chosen. The supported operators are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCompareOp {
    VK_COMPARE_OP_NEVER = 0,
    VK_COMPARE_OP_LESS = 1,
    VK_COMPARE_OP_EQUAL = 2,
    VK_COMPARE_OP_LESS_OR_EQUAL = 3,
    VK_COMPARE_OP_GREATER = 4,
    VK_COMPARE_OP_NOT_EQUAL = 5,
    VK_COMPARE_OP_GREATER_OR_EQUAL = 6,
    VK_COMPARE_OP_ALWAYS = 7,
} VkCompareOp;
```

- `VK_COMPARE_OP_NEVER` specifies that the comparison always evaluates false.

- `VK_COMPARE_OP_LESS` specifies that the comparison evaluates *reference* < *test*.
- `VK_COMPARE_OP_EQUAL` specifies that the comparison evaluates *reference* = *test*.
- `VK_COMPARE_OP_LESS_OR_EQUAL` specifies that the comparison evaluates *reference* ≤ *test*.
- `VK_COMPARE_OP_GREATER` specifies that the comparison evaluates *reference* > *test*.
- `VK_COMPARE_OP_NOT_EQUAL` specifies that the comparison evaluates *reference* ≠ *test*.
- `VK_COMPARE_OP_GREATER_OR_EQUAL` specifies that the comparison evaluates *reference* ≥ *test*.
- `VK_COMPARE_OP_ALWAYS` specifies that the comparison always evaluates true.

Comparison operators are used for:

- The `Depth Compare Operation` operator for a sampler, specified by `VkSamplerCreateInfo::compareOp`.
- The stencil comparison operator for the `stencil test`, specified by `VkStencilOpState::compareOp`.
- The `Depth Comparison` operator for the `depth test`, specified by `VkPipelineDepthStencilStateCreateInfo::depthCompareOp`.

Each such use describes how the *reference* and *test* values for that comparison are determined.

Possible values of `VkSamplerCreateInfo::borderColor`, specifying the border color used for texture lookups, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkBorderColor {
    VK_BORDER_COLOR_FLOAT_TRANSPARENT_BLACK = 0,
    VK_BORDER_COLOR_INT_TRANSPARENT_BLACK = 1,
    VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK = 2,
    VK_BORDER_COLOR_INT_OPAQUE_BLACK = 3,
    VK_BORDER_COLOR_FLOAT_OPAQUE_WHITE = 4,
    VK_BORDER_COLOR_INT_OPAQUE_WHITE = 5,
    // Provided by VK_EXT_custom_border_color
    VK_BORDER_COLOR_FLOAT_CUSTOM_EXT = 1000287003,
    // Provided by VK_EXT_custom_border_color
    VK_BORDER_COLOR_INT_CUSTOM_EXT = 1000287004,
} VkBorderColor;
```

- `VK_BORDER_COLOR_FLOAT_TRANSPARENT_BLACK` specifies a transparent, floating-point format, black color.
- `VK_BORDER_COLOR_INT_TRANSPARENT_BLACK` specifies a transparent, integer format, black color.
- `VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK` specifies an opaque, floating-point format, black color.
- `VK_BORDER_COLOR_INT_OPAQUE_BLACK` specifies an opaque, integer format, black color.
- `VK_BORDER_COLOR_FLOAT_OPAQUE_WHITE` specifies an opaque, floating-point format, white color.
- `VK_BORDER_COLOR_INT_OPAQUE_WHITE` specifies an opaque, integer format, white color.
- `VK_BORDER_COLOR_FLOAT_CUSTOM_EXT` indicates that a `VkSamplerCustomBorderColorCreateInfoEXT`

structure is included in the `VkSamplerCreateInfo::pNext` chain containing the color data in floating-point format.

- `VK_BORDER_COLOR_INT_CUSTOM_EXT` indicates that a `VkSamplerCustomBorderColorCreateInfoEXT` structure is included in the `VkSamplerCreateInfo::pNext` chain containing the color data in integer format.

These colors are described in detail in [Texel Replacement](#).

To destroy a sampler, call:

```
// Provided by VK_VERSION_1_0
void vkDestroySampler(
    VkDevice          device,
    VkSampler          sampler,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the sampler.
- `sampler` is the sampler to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroySampler-sampler-01082
All submitted commands that refer to `sampler` **must** have completed execution
- VUID-vkDestroySampler-sampler-01083
If `VkAllocationCallbacks` were provided when `sampler` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroySampler-sampler-01084
If no `VkAllocationCallbacks` were provided when `sampler` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroySampler-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroySampler-sampler-parameter
If `sampler` is not `VK_NULL_HANDLE`, `sampler` **must** be a valid `VkSampler` handle
- VUID-vkDestroySampler-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroySampler-sampler-parent
If `sampler` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to **sampler** **must** be externally synchronized

13.1. Sampler Y'C_BC_R Conversion

To create a sampler with Y'C_BC_R conversion enabled, add a [VkSamplerYcbcrConversionInfo](#) structure to the **pNext** chain of the [VkSamplerCreateInfo](#) structure. To create a sampler Y'C_BC_R conversion, the **samplerYcbcrConversion** feature **must** be enabled. Conversion **must** be fixed at pipeline creation time, through use of a combined image sampler with an immutable sampler in [VkDescriptorSetLayoutBinding](#).

A [VkSamplerYcbcrConversionInfo](#) **must** be provided for samplers to be used with image views that access [VK_IMAGE_ASPECT_COLOR_BIT](#) if the format is one of the [formats that require a sampler Y'C_BC_R conversion](#).

The [VkSamplerYcbcrConversionInfo](#) structure is defined as:

```
typedef struct VkSamplerYcbcrConversionInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkSamplerYcbcrConversion conversion;
} VkSamplerYcbcrConversionInfo;
```

or the equivalent

```
// Provided by VK_KHR_sampler_ycbcr_conversion
typedef VkSamplerYcbcrConversionInfo VkSamplerYcbcrConversionInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **conversion** is a [VkSamplerYcbcrConversion](#) handle created with [vkCreateSamplerYcbcrConversionKHR](#).

Valid Usage (Implicit)

- VUID-VkSamplerYcbcrConversionInfo-sType-sType
sType **must** be [VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_INFO](#)
- VUID-VkSamplerYcbcrConversionInfo-conversion-parameter
conversion **must** be a valid [VkSamplerYcbcrConversion](#) handle

A sampler Y'C_BC_R conversion is an opaque representation of a device-specific sampler Y'C_BC_R conversion description, represented as a [VkSamplerYcbcrConversion](#) handle:

```
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSamplerYcbcrConversion)
```

or the equivalent

```
// Provided by VK_KHR_sampler_ycbcr_conversion  
typedef VkSamplerYcbcrConversion VkSamplerYcbcrConversionKHR;
```

To create a [VkSamplerYcbcrConversion](#), call:

```
// Provided by VK_KHR_sampler_ycbcr_conversion  
VkResult vkCreateSamplerYcbcrConversionKHR(  
    VkDevice device,  
    const VkSamplerYcbcrConversionCreateInfo* pCreateInfo,  
    const VkAllocationCallbacks* pAllocator,  
    VkSamplerYcbcrConversion* pYcbcrConversion);
```

- **device** is the logical device that creates the sampler Y'C_BC_R conversion.
- **pCreateInfo** is a pointer to a [VkSamplerYcbcrConversionCreateInfo](#) structure specifying the requested sampler Y'C_BC_R conversion.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pYcbcrConversion** is a pointer to a [VkSamplerYcbcrConversion](#) handle in which the resulting sampler Y'C_BC_R conversion is returned.

The interpretation of the configured sampler Y'C_BC_R conversion is described in more detail in [the description of sampler Y'C_BC_R conversion](#) in the [Image Operations](#) chapter.

Valid Usage

- VUID-vkCreateSamplerYcbcrConversion-None-01648
The **samplerYcbcrConversion** feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateSamplerYcbcrConversion-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCreateSamplerYcbcrConversion-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkSamplerYcbcrConversionCreateInfo](#) structure
- VUID-vkCreateSamplerYcbcrConversion-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateSamplerYcbcrConversion-pYcbcrConversion-parameter

`pYcbcrConversion` **must** be a valid pointer to a `VkSamplerYcbcrConversion` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkSamplerYcbcrConversionCreateInfo` structure is defined as:

```
typedef struct VkSamplerYcbcrConversionCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkFormat            format;
    VkSamplerYcbcrModelConversion ycbcrModel;
    VkSamplerYcbcrRange ycbcrRange;
    VkComponentMapping  components;
    VkChromaLocation    xChromaOffset;
    VkChromaLocation    yChromaOffset;
    VkFilter            chromaFilter;
    VkBool32            forceExplicitReconstruction;
} VkSamplerYcbcrConversionCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_sampler_ycbcr_conversion
typedef VkSamplerYcbcrConversionCreateInfo VkSamplerYcbcrConversionCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `format` is the format of the image from which color information will be retrieved.
- `ycbcrModel` describes the color matrix for conversion between color models.
- `ycbcrRange` describes whether the encoded values have headroom and foot room, or whether the encoding uses the full numerical range.
- `components` applies a *swizzle* based on `VkComponentSwizzle` enums prior to range expansion and color model conversion.
- `xChromaOffset` describes the `sample location` associated with downsampled chroma components in the x dimension. `xChromaOffset` has no effect for formats in which chroma components are not downsampled horizontally.

- `yChromaOffset` describes the `sample location` associated with downsampled chroma components in the y dimension. `yChromaOffset` has no effect for formats in which the chroma components are not downsampled vertically.
- `chromaFilter` is the filter for chroma reconstruction.
- `forceExplicitReconstruction` **can** be used to ensure that reconstruction is done explicitly, if supported.

NOTE

Setting `forceExplicitReconstruction` to `VK_TRUE` **may** have a performance penalty on implementations where explicit reconstruction is not the default mode of operation.

If `format` supports `VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT` the `forceExplicitReconstruction` value behaves as if it were `VK_TRUE`.

Sampler Y'C_BC_R conversion objects do not support *external format conversion* without additional extensions defining *external formats*.

Valid Usage

- VUID-VkSamplerYcbcrConversionCreateInfo-format-04061
`format` **must** represent unsigned normalized values (i.e. the format **must** be a `UNORM` format)
- VUID-VkSamplerYcbcrConversionCreateInfo-format-01650
The `potential format features` of the sampler Y'C_BC_R conversion **must** support `VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT` or `VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT`
- VUID-VkSamplerYcbcrConversionCreateInfo-xChromaOffset-01651
If the `potential format features` of the sampler Y'C_BC_R conversion do not support `VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT`, `xChromaOffset` and `yChromaOffset` **must** not be `VK_CHROMA_LOCATION_COSITED_EVEN` if the corresponding components are `downsampled`
- VUID-VkSamplerYcbcrConversionCreateInfo-xChromaOffset-01652
If the `potential format features` of the sampler Y'C_BC_R conversion do not support `VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT`, `xChromaOffset` and `yChromaOffset` **must** not be `VK_CHROMA_LOCATION_MIDPOINT` if the corresponding components are `downsampled`
- VUID-VkSamplerYcbcrConversionCreateInfo-components-02581
If the format has a `_422` or `_420` suffix, then `components.g` **must** be the `identity swizzle`
- VUID-VkSamplerYcbcrConversionCreateInfo-components-02582
If the format has a `_422` or `_420` suffix, then `components.a` **must** be the `identity swizzle`, `VK_COMPONENT_SWIZZLE_ONE`, or `VK_COMPONENT_SWIZZLE_ZERO`
- VUID-VkSamplerYcbcrConversionCreateInfo-components-02583
If the format has a `_422` or `_420` suffix, then `components.r` **must** be the `identity swizzle` or `VK_COMPONENT_SWIZZLE_B`
- VUID-VkSamplerYcbcrConversionCreateInfo-components-02584
If the format has a `_422` or `_420` suffix, then `components.b` **must** be the `identity swizzle` or

VK_COMPONENT_SWIZZLE_R

- VUID-VkSamplerYcbcrConversionCreateInfo-components-02585
If the format has a `_422` or `_420` suffix, and if either `components.r` or `components.b` is the `identity swizzle`, both values **must** be the identity swizzle
- VUID-VkSamplerYcbcrConversionCreateInfo-ycbcrModel-01655
If `ycbcrModel` is not `VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY`, then `components.r`, `components.g`, and `components.b` **must** correspond to components of the `format`; that is, `components.r`, `components.g`, and `components.b` **must** not be `VK_COMPONENT_SWIZZLE_ZERO` or `VK_COMPONENT_SWIZZLE_ONE`, and **must** not correspond to a component containing zero or one as a consequence of `conversion to RGBA`
- VUID-VkSamplerYcbcrConversionCreateInfo-ycbcrRange-02748
If `ycbcrRange` is `VK_SAMPLER_YCBCR_RANGE_ITU_NARROW` then the R, G and B components obtained by applying the `component` swizzle to `format` **must** each have a bit-depth greater than or equal to 8
- VUID-VkSamplerYcbcrConversionCreateInfo-forceExplicitReconstruction-01656
If the `potential format features` of the sampler $Y'C_B C_R$ conversion do not support `VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT` `forceExplicitReconstruction` **must** be `VK_FALSE`
- VUID-VkSamplerYcbcrConversionCreateInfo-chromaFilter-01657
If the `potential format features` of the sampler $Y'C_B C_R$ conversion do not support `VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT`, `chromaFilter` **must** not be `VK_FILTER_LINEAR`

Valid Usage (Implicit)

- VUID-VkSamplerYcbcrConversionCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_CREATE_INFO`
- VUID-VkSamplerYcbcrConversionCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkSamplerYcbcrConversionCreateInfo-format-parameter
`format` **must** be a valid `VkFormat` value
- VUID-VkSamplerYcbcrConversionCreateInfo-ycbcrModel-parameter
`ycbcrModel` **must** be a valid `VkSamplerYcbcrModelConversion` value
- VUID-VkSamplerYcbcrConversionCreateInfo-ycbcrRange-parameter
`ycbcrRange` **must** be a valid `VkSamplerYcbcrRange` value
- VUID-VkSamplerYcbcrConversionCreateInfo-components-parameter
`components` **must** be a valid `VkComponentMapping` structure
- VUID-VkSamplerYcbcrConversionCreateInfo-xChromaOffset-parameter
`xChromaOffset` **must** be a valid `VkChromaLocation` value
- VUID-VkSamplerYcbcrConversionCreateInfo-yChromaOffset-parameter
`yChromaOffset` **must** be a valid `VkChromaLocation` value
- VUID-VkSamplerYcbcrConversionCreateInfo-chromaFilter-parameter

chromaFilter must be a valid **VkFilter** value

If **chromaFilter** is **VK_FILTER_NEAREST**, chroma samples are reconstructed to luma component resolution using nearest-neighbour sampling. Otherwise, chroma samples are reconstructed using interpolation. More details can be found in [the description of sampler Y'C_BC_R conversion](#) in the [Image Operations](#) chapter.

VkSamplerYcbcrModelConversion defines the conversion from the source color model to the shader color model. Possible values are:

```
typedef enum VkSamplerYcbcrModelConversion {
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY = 0,
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY = 1,
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709 = 2,
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601 = 3,
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020 = 4,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY_KHR =
VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY_KHR =
VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709_KHR =
VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601_KHR =
VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020_KHR =
VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020,
} VkSamplerYcbcrModelConversion;
```

or the equivalent

```
// Provided by VK_KHR_sampler_ycbcr_conversion
typedef VkSamplerYcbcrModelConversion VkSamplerYcbcrModelConversionKHR;
```

- **VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY** specifies that the input values to the conversion are unmodified.
- **VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY** specifies no model conversion but the inputs are range expanded as for Y'C_BC_R.
- **VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709** specifies the color model conversion from Y'C_BC_R to R'G'B' defined in BT.709 and described in the “BT.709 Y'C_BC_R conversion” section of the [Khronos Data Format Specification](#).
- **VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601** specifies the color model conversion from Y'C_BC_R

to R'G'B' defined in BT.601 and described in the “BT.601 Y'C_BC_R conversion” section of the [Khronos Data Format Specification](#).

- `VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020` specifies the color model conversion from Y'C_BC_R to R'G'B' defined in BT.2020 and described in the “BT.2020 Y'C_BC_R conversion” section of the [Khronos Data Format Specification](#).

In the `VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_*` color models, for the input to the sampler Y'C_BC_R range expansion and model conversion:

- the Y (Y' luma) component corresponds to the G component of an RGB image.
- the CB (C_B or “U” blue color difference) component corresponds to the B component of an RGB image.
- the CR (C_R or “V” red color difference) component corresponds to the R component of an RGB image.
- the alpha component, if present, is not modified by color model conversion.

These rules reflect the mapping of components after the component swizzle operation (controlled by `VkSamplerYcbcrConversionCreateInfo::components`).

NOTE

For example, an “YUVA” 32-bit format comprising four 8-bit components can be implemented as `VK_FORMAT_R8G8B8A8_UNORM` with a component mapping:

- `components.a = VK_COMPONENT_SWIZZLE_IDENTITY`
- `components.r = VK_COMPONENT_SWIZZLE_B`
- `components.g = VK_COMPONENT_SWIZZLE_R`
- `components.b = VK_COMPONENT_SWIZZLE_G`

The `VkSamplerYcbcrRange` enum describes whether color components are encoded using the full range of numerical values or whether values are reserved for headroom and foot room. `VkSamplerYcbcrRange` is defined as:

```
typedef enum VkSamplerYcbcrRange {
    VK_SAMPLER_YCBCR_RANGE_ITU_FULL = 0,
    VK_SAMPLER_YCBCR_RANGE_ITU_NARROW = 1,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_RANGE_ITU_FULL_KHR = VK_SAMPLER_YCBCR_RANGE_ITU_FULL,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_SAMPLER_YCBCR_RANGE_ITU_NARROW_KHR = VK_SAMPLER_YCBCR_RANGE_ITU_NARROW,
} VkSamplerYcbcrRange;
```

or the equivalent

```
// Provided by VK_KHR_sampler_ycbcr_conversion
typedef VkSamplerYcbcrRange VkSamplerYcbcrRangeKHR;
```

- `VK_SAMPLER_YCBCR_RANGE_ITU_FULL` specifies that the full range of the encoded values are valid and interpreted according to the ITU “full range” quantization rules.
- `VK_SAMPLER_YCBCR_RANGE_ITU_NARROW` specifies that headroom and foot room are reserved in the numerical range of encoded values, and the remaining values are expanded according to the ITU “narrow range” quantization rules.

The formulae for these conversions is described in the [Sampler Y'C_BC_R Range Expansion](#) section of the [Image Operations](#) chapter.

No range modification takes place if `ycbcrModel` is `VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY`; the `ycbcrRange` field of `VkSamplerYcbcrConversionCreateInfo` is ignored in this case.

The `VkChromaLocation` enum defines the location of downsampled chroma component samples relative to the luma samples, and is defined as:

```
typedef enum VkChromaLocation {
    VK_CHROMA_LOCATION_COSITED_EVEN = 0,
    VK_CHROMA_LOCATION_MIDPOINT = 1,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_CHROMA_LOCATION_COSITED_EVEN_KHR = VK_CHROMA_LOCATION_COSITED_EVEN,
    // Provided by VK_KHR_sampler_ycbcr_conversion
    VK_CHROMA_LOCATION_MIDPOINT_KHR = VK_CHROMA_LOCATION_MIDPOINT,
} VkChromaLocation;
```

or the equivalent

```
// Provided by VK_KHR_sampler_ycbcr_conversion
typedef VkChromaLocation VkChromaLocationKHR;
```

- `VK_CHROMA_LOCATION_COSITED_EVEN` specifies that downsampled chroma samples are aligned with luma samples with even coordinates.
- `VK_CHROMA_LOCATION_MIDPOINT` specifies that downsampled chroma samples are located half way between each even luma sample and the nearest higher odd luma sample.

To destroy a sampler Y'C_BC_R conversion, call:

```
// Provided by VK_KHR_sampler_ycbcr_conversion
void vkDestroySamplerYcbcrConversionKHR(
    VkDevice          device,
    VkSamplerYcbcrConversion ycbcrConversion,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the Y'C_BC_R conversion.
- `ycbcrConversion` is the conversion to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage (Implicit)

- VUID-vkDestroySamplerYcbcrConversion-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkDestroySamplerYcbcrConversion-ycbcrConversion-parameter
If **ycbcrConversion** is not [VK_NULL_HANDLE](#), **ycbcrConversion** **must** be a valid [VkSamplerYcbcrConversion](#) handle
- VUID-vkDestroySamplerYcbcrConversion-pAllocator-parameter
If **pAllocator** is not [NULL](#), **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkDestroySamplerYcbcrConversion-ycbcrConversion-parent
If **ycbcrConversion** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **ycbcrConversion** **must** be externally synchronized

In addition to the predefined border color values, applications **can** provide a custom border color value by including the [VkSamplerCustomBorderColorCreateInfoEXT](#) structure in the [VkSamplerCreateInfo::pNext](#) chain.

The [VkSamplerCustomBorderColorCreateInfoEXT](#) structure is defined as:

```
// Provided by VK_EXT_custom_border_color
typedef struct VkSamplerCustomBorderColorCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkClearColorValue  customBorderColor;
    VkFormat            format;
} VkSamplerCustomBorderColorCreateInfoEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is [NULL](#) or a pointer to a structure extending this structure.
- **customBorderColor** is a [VkClearColorValue](#) representing the desired custom sampler border color.
- **format** is a [VkFormat](#) representing the format of the sampled image view(s). This field may be [VK_FORMAT_UNDEFINED](#) if the [customBorderColorWithoutFormat](#) feature is enabled.

NOTE

If **format** is a depth/stencil format, the aspect is determined by the value of [VkSamplerCreateInfo::borderColor](#). If [VkSamplerCreateInfo::borderColor](#) is [VK_BORDER_COLOR_FLOAT_CUSTOM_EXT](#), the depth aspect is considered. If [VkSamplerCreateInfo::borderColor](#) is [VK_BORDER_COLOR_INT_CUSTOM_EXT](#), the stencil

aspect is considered.

If `format` is `VK_FORMAT_UNDEFINED`, the `VkSamplerCreateInfo::borderColor` is `VK_BORDER_COLOR_INT_CUSTOM_EXT`, and the sampler is used with an image with a stencil format, then the implementation **must** source the custom border color from either the first or second components of `VkSamplerCreateInfo::customBorderColor` and **should** source it from the first component.

Valid Usage

- VUID-VkSamplerCustomBorderColorCreateInfoEXT-format-07605
If `format` is not `VK_FORMAT_UNDEFINED` and `format` is not a depth/stencil format then the `VkSamplerCreateInfo::borderColor` type **must** match the sampled type of the provided `format`, as shown in the *SPIR-V Type* column of the [Interpretation of Numeric Format](#) table
- VUID-VkSamplerCustomBorderColorCreateInfoEXT-format-04014
If the `customBorderColorWithoutFormat` feature is not enabled then `format` **must** not be `VK_FORMAT_UNDEFINED`
- VUID-VkSamplerCustomBorderColorCreateInfoEXT-format-04015
If the sampler is used to sample an image view of `VK_FORMAT_B4G4R4A4_UNORM_PACK16`, `VK_FORMAT_B5G6R5_UNORM_PACK16`, `VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR`, or `VK_FORMAT_B5G5R5A1_UNORM_PACK16` format then `format` **must** not be `VK_FORMAT_UNDEFINED`

Valid Usage (Implicit)

- VUID-VkSamplerCustomBorderColorCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SAMPLER_CUSTOM_BORDER_COLOR_CREATE_INFO_EXT`
- VUID-VkSamplerCustomBorderColorCreateInfoEXT-format-parameter
`format` **must** be a valid `VkFormat` value

If the sampler is created with `VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK`, `VK_BORDER_COLOR_INT_OPAQUE_BLACK`, `VK_BORDER_COLOR_FLOAT_CUSTOM_EXT`, or `VK_BORDER_COLOR_INT_CUSTOM_EXT` `borderColor`, and that sampler will be combined with an image view that does not have an [identity swizzle](#), and `VkPhysicalDeviceBorderColorSwizzleFeaturesEXT::borderColorSwizzleFromImage` is not enabled, then it is necessary to specify the component mapping of the border color, by including the `VkSamplerBorderColorComponentMappingCreateInfoEXT` structure in the `VkSamplerCreateInfo::pNext` chain, to get defined results.

The `VkSamplerBorderColorComponentMappingCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_border_color_swizzle
typedef struct VkSamplerBorderColorComponentMappingCreateInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkComponentMapping    components;
    VkBool32              srgb;
} VkSamplerBorderColorComponentMappingCreateInfoEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **components** is a [VkComponentMapping](#) structure specifying a remapping of the border color components.
- **srgb** indicates that the sampler will be combined with an image view that has an image format which is sRGB encoded.

The [VkComponentMapping](#) **components** member describes a remapping from components of the border color to components of the vector returned by shader image instructions when the border color is used.

Valid Usage

- VUID-VkSamplerBorderColorComponentMappingCreateInfoEXT-borderColorSwizzle-06437
The **borderColorSwizzle** feature **must** be enabled

Valid Usage (Implicit)

- VUID-VkSamplerBorderColorComponentMappingCreateInfoEXT-sType-sType **must** be [VK_STRUCTURE_TYPE_SAMPLER_BORDER_COLOR_COMPONENT_MAPPING_CREATE_INFO_EXT](#)
- VUID-VkSamplerBorderColorComponentMappingCreateInfoEXT-components-parameter **components** **must** be a valid [VkComponentMapping](#) structure

Chapter 14. Resource Descriptors

A *descriptor* is an opaque data structure representing a shader resource such as a buffer, buffer view, image view, sampler, or combined image sampler. Descriptors are organized into *descriptor sets*, which are bound during command recording for use in subsequent drawing commands. The arrangement of content in each descriptor set is determined by a *descriptor set layout*, which determines what descriptors can be stored within it. The sequence of descriptor set layouts that **can** be used by a pipeline is specified in a *pipeline layout*. Each pipeline object **can** use up to `maxBoundDescriptorSets` (see [Limits](#)) descriptor sets.

Shaders access resources via variables decorated with a descriptor set and binding number that link them to a descriptor in a descriptor set. The shader interface mapping to bound descriptor sets is described in the [Shader Resource Interface](#) section.

Shaders **can** also access buffers without going through descriptors by using [Physical Storage Buffer Access](#) to access them through 64-bit addresses.

14.1. Descriptor Types

There are a number of different types of descriptor supported by Vulkan, corresponding to different resources or usage. The following sections describe the API definitions of each descriptor type. The mapping of each type to SPIR-V is listed in the [Shader Resource and Descriptor Type Correspondence](#) and [Shader Resource and Storage Class Correspondence](#) tables in the [Shader Interfaces](#) chapter.

14.1.1. Storage Image

A *storage image* (`VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`) is a descriptor type associated with an [image resource](#) via an [image view](#) that load, store, and atomic operations **can** be performed on.

Storage image loads are supported in all shader stages for image views whose [format features](#) contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT`.

Stores to storage images are supported in compute shaders for image views whose [format features](#) contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT`.

Atomic operations on storage images are supported in compute shaders for image views whose [format features](#) contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`.

When the `fragmentStoresAndAtomics` feature is enabled, stores and atomic operations are also supported for storage images in fragment shaders with the same set of image formats as supported in compute shaders. When the `vertexPipelineStoresAndAtomics` feature is enabled, stores and atomic operations are also supported in vertex, tessellation, and geometry shaders with the same set of image formats as supported in compute shaders.

The image subresources for a storage image **must** be in the `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR` or `VK_IMAGE_LAYOUT_GENERAL` layout in order to access its data in a shader.

14.1.2. Sampler

A *sampler descriptor* (`VK_DESCRIPTOR_TYPE_SAMPLER`) is a descriptor type associated with a [sampler](#) object, used to control the behavior of [sampling operations](#) performed on a [sampled image](#).

14.1.3. Sampled Image

A *sampled image* (`VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`) is a descriptor type associated with an [image resource](#) via an [image view](#) that [sampling operations](#) **can** be performed on.

Shaders combine a sampled image variable and a sampler variable to perform sampling operations.

Sampled images are supported in all shader stages for image views whose [format features](#) contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`.

An image subresources for a sampled image **must** be in one of the following layouts:

- `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_GENERAL`
- `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`
- `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`
- `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`
- `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT`

14.1.4. Combined Image Sampler

A *combined image sampler* (`VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`) is a single descriptor type associated with both a [sampler](#) and an [image resource](#), combining both a [sampler](#) and [sampled image](#) descriptor into a single descriptor.

If the descriptor refers to a sampler that performs [Y'C_BC_R conversion](#), the sampler **must** only be used to sample the image in the same descriptor. Otherwise, the sampler and image in this type of descriptor **can** be used freely with any other samplers and images.

An image subresources for a combined image sampler **must** be in one of the following layouts:

- `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_GENERAL`
- `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`

- `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`
- `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`
- `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT`

NOTE

On some implementations, it **may** be more efficient to sample from an image using a combination of sampler and sampled image that are stored together in the descriptor set in a combined descriptor.

14.1.5. Uniform Texel Buffer

A *uniform texel buffer* (`VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER`) is a descriptor type associated with a [buffer resource](#) via a [buffer view](#) that [image sampling operations](#) **can** be performed on.

Uniform texel buffers define a tightly-packed 1-dimensional linear array of texels, with texels going through format conversion when read in a shader in the same way as they are for an image.

Load operations from uniform texel buffers are supported in all shader stages for buffer view formats which report [format features](#) support for `VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT`

14.1.6. Storage Texel Buffer

A *storage texel buffer* (`VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`) is a descriptor type associated with a [buffer resource](#) via a [buffer view](#) that [image load, store, and atomic operations](#) **can** be performed on.

Storage texel buffers define a tightly-packed 1-dimensional linear array of texels, with texels going through format conversion when read in a shader in the same way as they are for an image. Unlike [uniform texel buffers](#), these buffers can also be written to in the same way as for [storage images](#).

Storage texel buffer loads are supported in all shader stages for texel buffer view formats which report [format features](#) support for `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT`

Stores to storage texel buffers are supported in compute shaders for texel buffer formats which report [format features](#) support for `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT`

Atomic operations on storage texel buffers are supported in compute shaders for texel buffer formats which report [format features](#) support for `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`

When the `fragmentStoresAndAtomics` feature is enabled, stores and atomic operations are also supported for storage texel buffers in fragment shaders with the same set of texel buffer formats as supported in compute shaders. When the `vertexPipelineStoresAndAtomics` feature is enabled, stores and atomic operations are also supported in vertex, tessellation, and geometry shaders with the same set of texel buffer formats as supported in compute shaders.

14.1.7. Storage Buffer

A *storage buffer* (`VK_DESCRIPTOR_TYPE_STORAGE_BUFFER`) is a descriptor type associated with a [buffer resource](#) directly, described in a shader as a structure with various members that load, store, and atomic operations **can** be performed on.

NOTE

Atomic operations **can** only be performed on members of certain types as defined in the [SPIR-V environment appendix](#).

14.1.8. Uniform Buffer

A *uniform buffer* (`VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER`) is a descriptor type associated with a [buffer resource](#) directly, described in a shader as a structure with various members that load operations **can** be performed on.

14.1.9. Dynamic Uniform Buffer

A *dynamic uniform buffer* (`VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`) is almost identical to a [uniform buffer](#), and differs only in how the offset into the buffer is specified. The base offset calculated by the `VkDescriptorBufferInfo` when initially [updating the descriptor set](#) is added to a [dynamic offset](#) when binding the descriptor set.

14.1.10. Dynamic Storage Buffer

A *dynamic storage buffer* (`VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`) is almost identical to a [storage buffer](#), and differs only in how the offset into the buffer is specified. The base offset calculated by the `VkDescriptorBufferInfo` when initially [updating the descriptor set](#) is added to a [dynamic offset](#) when binding the descriptor set.

14.1.11. Input Attachment

An *input attachment* (`VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`) is a descriptor type associated with an [image resource](#) via an [image view](#) that **can** be used for [framebuffer local](#) load operations in fragment shaders.

All image formats that are supported for color attachments (`VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT`) or depth/stencil attachments (`VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT`) for a given image tiling mode are also supported for input attachments.

An image view used as an input attachment **must** be in one of the following layouts:

- `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_GENERAL`
- `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`
- `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`

- `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`
- `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR`
- `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT`
- `VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR`

14.1.12. Acceleration Structure

An *acceleration structure* (`VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR`) is a descriptor type that is used to retrieve scene geometry from within shaders that are used for ray traversal. Shaders have read-only access to the memory.

14.2. Descriptor Sets

Descriptors are grouped together into descriptor set objects. A descriptor set object is an opaque object containing storage for a set of descriptors, where the types and number of descriptors is defined by a descriptor set layout. The layout object **may** be used to define the association of each descriptor binding with memory or other implementation resources. The layout is used both for determining the resources that need to be associated with the descriptor set, and determining the interface between shader stages and shader resources.

14.2.1. Descriptor Set Layout

A descriptor set layout object is defined by an array of zero or more descriptor bindings. Each individual descriptor binding is specified by a descriptor type, a count (array size) of the number of descriptors in the binding, a set of shader stages that **can** access the binding, and (if using immutable samplers) an array of sampler descriptors.

Descriptor set layout objects are represented by `VkDescriptorSetLayout` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDescriptorSetLayout)
```

To create descriptor set layout objects, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateDescriptorSetLayout(
    VkDevice                                device,
    const VkDescriptorSetLayoutCreateInfo*  pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkDescriptorSetLayout*                  pSetLayout);
```

- `device` is the logical device that creates the descriptor set layout.
- `pCreateInfo` is a pointer to a `VkDescriptorSetLayoutCreateInfo` structure specifying the state of the descriptor set layout object.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

- `pSetLayout` is a pointer to a `VkDescriptorSetLayout` handle in which the resulting descriptor set layout object is returned.

Valid Usage

- VUID-vkCreateDescriptorSetLayout-support-09582
If the descriptor layout exceeds the limits reported through the [physical device limits](#), then `vkGetDescriptorSetLayoutSupportKHR` **must** have returned `VkDescriptorSetLayoutSupport` with `support` equal to `VK_TRUE` for `pCreateInfo`

Valid Usage (Implicit)

- VUID-vkCreateDescriptorSetLayout-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateDescriptorSetLayout-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkDescriptorSetLayoutCreateInfo` structure
- VUID-vkCreateDescriptorSetLayout-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateDescriptorSetLayout-pSetLayout-parameter
`pSetLayout` **must** be a valid pointer to a `VkDescriptorSetLayout` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

Information about the descriptor set layout is passed in a `VkDescriptorSetLayoutCreateInfo` structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorSetLayoutCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkDescriptorSetLayoutCreateFlags flags;
    uint32_t                  bindingCount;
    const VkDescriptorSetLayoutBinding* pBindings;
} VkDescriptorSetLayoutCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkDescriptorSetLayoutCreateFlagBits` specifying options for descriptor set layout creation.
- `bindingCount` is the number of elements in `pBindings`.
- `pBindings` is a pointer to an array of `VkDescriptorSetLayoutBinding` structures.

Valid Usage

- VUID-VkDescriptorSetLayoutCreateInfo-binding-00279
The `VkDescriptorSetLayoutBinding::binding` members of the elements of the `pBindings` array **must** each have different values
- VUID-VkDescriptorSetLayoutCreateInfo-flags-00280
If `flags` contains `VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR`, then all elements of `pBindings` **must** not have a `descriptorType` of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`
- VUID-VkDescriptorSetLayoutCreateInfo-flags-00281
If `flags` contains `VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR`, then the total number of elements of all bindings **must** be less than or equal to `VkPhysicalDevicePushDescriptorPropertiesKHR::maxPushDescriptors`

Valid Usage (Implicit)

- VUID-VkDescriptorSetLayoutCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO`
- VUID-VkDescriptorSetLayoutCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDescriptorSetLayoutCreateInfo-flags-parameter
`flags` **must** be a valid combination of `VkDescriptorSetLayoutCreateFlagBits` values
- VUID-VkDescriptorSetLayoutCreateInfo-pBindings-parameter
If `bindingCount` is not 0, `pBindings` **must** be a valid pointer to an array of `bindingCount` valid `VkDescriptorSetLayoutBinding` structures

Bits which **can** be set in `VkDescriptorSetLayoutCreateInfo::flags`, specifying options for descriptor set layout, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkDescriptorSetLayoutCreateFlagBits {
    // Provided by VK_KHR_push_descriptor
    VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR = 0x00000001,
} VkDescriptorSetLayoutCreateFlagBits;
```


- `VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR` specifies that descriptor sets **must** not be allocated using this layout, and descriptors are instead pushed by `VkCmdPushDescriptorSetKHR`.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDescriptorSetLayoutCreateFlags;
```

`VkDescriptorSetLayoutCreateFlags` is a bitmask type for setting a mask of zero or more `VkDescriptorSetLayoutCreateFlagBits`.

The `VkDescriptorSetLayoutBinding` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorSetLayoutBinding {
    uint32_t          binding;
    VkDescriptorType   descriptorType;
    uint32_t          descriptorCount;
    VkShaderStageFlags stageFlags;
    const VkSampler*   pImmutableSamplers;
} VkDescriptorSetLayoutBinding;
```

- `binding` is the binding number of this entry and corresponds to a resource of the same binding number in the shader stages.
- `descriptorType` is a `VkDescriptorType` specifying which type of resource descriptors are used for this binding.
- `descriptorCount` is the number of descriptors contained in the binding, accessed in a shader as an array. If `descriptorCount` is zero this binding entry is reserved and the resource **must** not be accessed from any stage via this binding within any pipeline using the set layout.
- `stageFlags` member is a bitmask of `VkShaderStageFlagBits` specifying which pipeline shader stages **can** access a resource for this binding. `VK_SHADER_STAGE_ALL` is a shorthand specifying that all defined shader stages, including any additional stages defined by extensions, **can** access the resource.

If a shader stage is not included in `stageFlags`, then a resource **must** not be accessed from that stage via this binding within any pipeline using the set layout. Other than input attachments which are limited to the fragment shader, there are no limitations on what combinations of stages **can** use a descriptor binding, and in particular a binding **can** be used by both graphics stages and the compute stage.

- `pImmutableSamplers` affects initialization of samplers. If `descriptorType` specifies a `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` type descriptor, then `pImmutableSamplers` **can** be used to initialize a set of *immutable samplers*. Immutable samplers are permanently bound into the set layout and **must** not be changed; updating a `VK_DESCRIPTOR_TYPE_SAMPLER` descriptor with immutable samplers is not allowed and updates to a `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` descriptor with immutable samplers does not modify the samplers (the image views are updated, but the sampler updates are ignored). If

`pImmutableSamplers` is not `NULL`, then it is a pointer to an array of sampler handles that will be copied into the set layout and used for the corresponding binding. Only the sampler handles are copied; the sampler objects **must** not be destroyed before the final use of the set layout and any descriptor pools and sets created using it. If `pImmutableSamplers` is `NULL`, then the sampler slots are dynamic and sampler handles **must** be bound into descriptor sets using this layout. If `descriptorType` is not one of these descriptor types, then `pImmutableSamplers` is ignored.

The above layout definition allows the descriptor bindings to be specified sparsely such that not all binding numbers between 0 and the maximum binding number need to be specified in the `pBindings` array. Bindings that are not specified have a `descriptorCount` and `stageFlags` of zero, and the value of `descriptorType` is undefined. However, all binding numbers between 0 and the maximum binding number in the `VkDescriptorSetLayoutCreateInfo::pBindings` array **may** consume memory in the descriptor set layout even if not all descriptor bindings are used, though it **should** not consume additional memory from the descriptor pool.

NOTE

The maximum binding number specified **should** be as compact as possible to avoid wasted memory.

Valid Usage

- VUID-VkDescriptorSetLayoutBinding-descriptorType-00282
If `descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and `descriptorCount` is not 0 and `pImmutableSamplers` is not `NULL`, `pImmutableSamplers` **must** be a valid pointer to an array of `descriptorCount` valid `VkSampler` handles
- VUID-VkDescriptorSetLayoutBinding-descriptorCount-09465
If `descriptorCount` is not 0, `stageFlags` **must** be `VK_SHADER_STAGE_ALL` or a valid combination of other `VkShaderStageFlags` values
- VUID-VkDescriptorSetLayoutBinding-descriptorType-01510
If `descriptorType` is `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` and `descriptorCount` is not 0, then `stageFlags` **must** be 0 or `VK_SHADER_STAGE_FRAGMENT_BIT`
- VUID-VkDescriptorSetLayoutBinding-pImmutableSamplers-04009
The sampler objects indicated by `pImmutableSamplers` **must** not have a `borderColor` with one of the values `VK_BORDER_COLOR_FLOAT_CUSTOM_EXT` or `VK_BORDER_COLOR_INT_CUSTOM_EXT`

Valid Usage (Implicit)

- VUID-VkDescriptorSetLayoutBinding-descriptorType-parameter
`descriptorType` **must** be a valid `VkDescriptorType` value

To query information about whether a descriptor set layout **can** be created, call:

```
// Provided by VK_KHR_maintenance3
void vkGetDescriptorSetLayoutSupportKHR(
    VkDevice device,
    const VkDescriptorSetLayoutCreateInfo* pCreateInfo,
    VkDescriptorSetLayoutSupport* pSupport);
```

- **device** is the logical device that would create the descriptor set layout.
- **pCreateInfo** is a pointer to a [VkDescriptorSetLayoutCreateInfo](#) structure specifying the state of the descriptor set layout object.
- **pSupport** is a pointer to a [VkDescriptorSetLayoutSupport](#) structure, in which information about support for the descriptor set layout object is returned.

Some implementations have limitations on what fits in a descriptor set which are not easily expressible in terms of existing limits like **maxDescriptorSet***, for example if all descriptor types share a limited space in memory but each descriptor is a different size or alignment. This command returns information about whether a descriptor set satisfies this limit. If the descriptor set layout satisfies the [VkPhysicalDeviceMaintenance3Properties::maxPerSetDescriptors](#) limit, this command is guaranteed to return **VK_TRUE** in [VkDescriptorSetLayoutSupport::supported](#). If the descriptor set layout exceeds the [VkPhysicalDeviceMaintenance3Properties::maxPerSetDescriptors](#) limit, whether the descriptor set layout is supported is implementation-dependent and **may** depend on whether the descriptor sizes and alignments cause the layout to exceed an internal limit.

This command does not consider other limits such as **maxPerStageDescriptor***, and so a descriptor set layout that is supported according to this command **must** still satisfy the pipeline layout limits such as **maxPerStageDescriptor*** in order to be used in a pipeline layout.

NOTE	This is a VkDevice query rather than VkPhysicalDevice because the answer may depend on enabled features.
-------------	---

Valid Usage (Implicit)

- VUID-vkGetDescriptorSetLayoutSupport-device-parameter **device must** be a valid [VkDevice](#) handle
- VUID-vkGetDescriptorSetLayoutSupport-pCreateInfo-parameter **pCreateInfo must** be a valid pointer to a valid [VkDescriptorSetLayoutCreateInfo](#) structure
- VUID-vkGetDescriptorSetLayoutSupport-pSupport-parameter **pSupport must** be a valid pointer to a [VkDescriptorSetLayoutSupport](#) structure

Information about support for the descriptor set layout is returned in a [VkDescriptorSetLayoutSupport](#) structure:

```
typedef struct VkDescriptorSetLayoutSupport {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           supported;
} VkDescriptorSetLayoutSupport;
```

or the equivalent

```
// Provided by VK_KHR_maintenance3
typedef VkDescriptorSetLayoutSupport VkDescriptorSetLayoutSupportKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `supported` specifies whether the descriptor set layout **can** be created.

`supported` will be `VK_TRUE` if the descriptor set **can** be created, or else `VK_FALSE`.

Valid Usage (Implicit)

- VUID-VkDescriptorSetLayoutSupport-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_SUPPORT`
- VUID-VkDescriptorSetLayoutSupport-pNext-pNext
`pNext` **must** be `NULL`

The following examples show a shader snippet using two descriptor sets, and application code that creates corresponding descriptor set layouts.

GLSL Example

```
//
// binding to a single sampled image descriptor in set 0
//
layout (set=0, binding=0) uniform texture2D mySampledImage;

//
// binding to an array of sampled image descriptors in set 0
//
layout (set=0, binding=1) uniform texture2D myArrayOfSampledImages[12];

//
// binding to a single uniform buffer descriptor in set 1
//
layout (set=1, binding=0) uniform myUniformBuffer
{
    vec4 myElement[32];
}
```

```
};
```

SPIR-V Example

```
...
%1 = OpExtInstImport "GLSL.std.450"
...
OpName %9 "mySampledImage"
OpName %14 "myArrayOfSampledImages"
OpName %18 "myUniformBuffer"
OpMemberName %18 0 "myElement"
OpName %20 ""
OpDecorate %9 DescriptorSet 0
OpDecorate %9 Binding 0
OpDecorate %14 DescriptorSet 0
OpDecorate %14 Binding 1
OpDecorate %17 ArrayStride 16
OpMemberDecorate %18 0 Offset 0
OpDecorate %18 Block
OpDecorate %20 DescriptorSet 1
OpDecorate %20 Binding 0
%2 = OpTypeVoid
%3 = OpTypeFunction %2
%6 = OpTypeFloat 32
%7 = OpTypeImage %6 2D 0 0 0 1 Unknown
%8 = OpTypePointer UniformConstant %7
%9 = OpVariable %8 UniformConstant
%10 = OpTypeInt 32 0
%11 = OpConstant %10 12
%12 = OpTypeArray %7 %11
%13 = OpTypePointer UniformConstant %12
%14 = OpVariable %13 UniformConstant
%15 = OpTypeVector %6 4
%16 = OpConstant %10 32
%17 = OpTypeArray %15 %16
%18 = OpTypeStruct %17
%19 = OpTypePointer Uniform %18
%20 = OpVariable %19 Uniform
...
```

API Example

```
VkResult myResult;

const VkDescriptorSetLayoutBinding myDescriptorSetLayoutBinding[] =
{
    // binding to a single image descriptor
    {
        .binding = 0,
        .descriptorType = VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE,
```

```

        .descriptorCount = 1,
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .pImmutableSamplers = NULL
    },

    // binding to an array of image descriptors
    {
        .binding = 1,
        .descriptorType = VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE,
        .descriptorCount = 12,
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .pImmutableSamplers = NULL
    },

    // binding to a single uniform buffer descriptor
    {
        .binding = 0,
        .descriptorType = VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER,
        .descriptorCount = 1,
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .pImmutableSamplers = NULL
    }
};

const VkDescriptorSetLayoutCreateInfo myDescriptorSetLayoutCreateInfo[] =
{
    // Information for first descriptor set with two descriptor bindings
    {
        .sType = VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO,
        .pNext = NULL,
        .flags = 0,
        .bindingCount = 2,
        .pBindings = &myDescriptorSetLayoutBinding[0]
    },

    // Information for second descriptor set with one descriptor binding
    {
        .sType = VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_CREATE_INFO,
        .pNext = NULL,
        .flags = 0,
        .bindingCount = 1,
        .pBindings = &myDescriptorSetLayoutBinding[2]
    }
};

VkDescriptorSetLayout myDescriptorSetLayout[2];

//
// Create first descriptor set layout
//
myResult = vkCreateDescriptorSetLayout(

```

```

myDevice,
&myDescriptorSetLayoutCreateInfo[0],
NULL,
&myDescriptorSetLayout[0]);

//
// Create second descriptor set layout
//
myResult = vkCreateDescriptorSetLayout(
    myDevice,
    &myDescriptorSetLayoutCreateInfo[1],
    NULL,
    &myDescriptorSetLayout[1]);

```

To destroy a descriptor set layout, call:

```

// Provided by VK_VERSION_1_0
void vkDestroyDescriptorSetLayout(
    VkDevice                device,
    VkDescriptorSetLayout    descriptorSetLayout,
    const VkAllocationCallbacks* pAllocator);

```

- **device** is the logical device that destroys the descriptor set layout.
- **descriptorSetLayout** is the descriptor set layout to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-00284
If **VkAllocationCallbacks** were provided when **descriptorSetLayout** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-00285
If no **VkAllocationCallbacks** were provided when **descriptorSetLayout** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyDescriptorSetLayout-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-parameter
If **descriptorSetLayout** is not **VK_NULL_HANDLE**, **descriptorSetLayout** **must** be a valid **VkDescriptorSetLayout** handle
- VUID-vkDestroyDescriptorSetLayout-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid

[VkAllocationCallbacks](#) structure

- VUID-vkDestroyDescriptorSetLayout-descriptorSetLayout-parent

If [descriptorSetLayout](#) is a valid handle, it **must** have been created, allocated, or retrieved from [device](#)

Host Synchronization

- Host access to [descriptorSetLayout](#) **must** be externally synchronized

14.2.2. Pipeline Layouts

Access to descriptor sets from a pipeline is accomplished through a *pipeline layout*. Zero or more descriptor set layouts and zero or more push constant ranges are combined to form a pipeline layout object describing the complete set of resources that **can** be accessed by a pipeline. The pipeline layout represents a sequence of descriptor sets with each having a specific layout. This sequence of layouts is used to determine the interface between shader stages and shader resources. Each pipeline is created using a pipeline layout.

Pipeline layout objects are represented by [VkPipelineLayout](#) handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkPipelineLayout)
```

To create a pipeline layout, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreatePipelineLayout(
    VkDevice                                device,
    const VkPipelineLayoutCreateInfo*       pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkPipelineLayout*                       pPipelineLayout);
```

- [device](#) is the logical device that creates the pipeline layout.
- [pCreateInfo](#) is a pointer to a [VkPipelineLayoutCreateInfo](#) structure specifying the state of the pipeline layout object.
- [pAllocator](#) controls host memory allocation as described in the [Memory Allocation](#) chapter.
- [pPipelineLayout](#) is a pointer to a [VkPipelineLayout](#) handle in which the resulting pipeline layout object is returned.

Valid Usage (Implicit)

- VUID-vkCreatePipelineLayout-device-parameter [device](#) **must** be a valid [VkDevice](#) handle

- VUID-vkCreatePipelineLayout-pCreateInfo-parameter
pCreateInfo must be a valid pointer to a valid [VkPipelineLayoutCreateInfo](#) structure
- VUID-vkCreatePipelineLayout-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** must be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreatePipelineLayout-pPipelineLayout-parameter
pPipelineLayout must be a valid pointer to a [VkPipelineLayout](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

The [VkPipelineLayoutCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineLayoutCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkPipelineLayoutCreateFlags flags;
    uint32_t              setLayoutCount;
    const VkDescriptorSetLayout* pSetLayouts;
    uint32_t              pushConstantRangeCount;
    const VkPushConstantRange* pPushConstantRanges;
} VkPipelineLayoutCreateInfo;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkPipelineLayoutCreateFlagBits](#) specifying options for pipeline layout creation.
- **setLayoutCount** is the number of descriptor sets included in the pipeline layout.
- **pSetLayouts** is a pointer to an array of [VkDescriptorSetLayout](#) objects.
- **pushConstantRangeCount** is the number of push constant ranges included in the pipeline layout.
- **pPushConstantRanges** is a pointer to an array of [VkPushConstantRange](#) structures defining a set of push constant ranges for use in a single pipeline layout. In addition to descriptor set layouts, a pipeline layout also describes how many push constants **can** be accessed by each stage of the pipeline.

NOTE

Push constants represent a high speed path to modify constant data in pipelines

that is expected to outperform memory-backed resource updates.

Valid Usage

- VUID-VkPipelineLayoutCreateInfo-setLayoutCount-00286
`setLayoutCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxBoundDescriptorSets`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03016
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_SAMPLER` and `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorSamplers`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03017
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` and `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorUniformBuffers`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03018
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` and `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorStorageBuffers`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-06939
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, and `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER`, accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorSampledImages`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03020
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, and `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorStorageImages`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03021
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxPerStageDescriptorInputAttachments`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03028
The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_SAMPLER` and `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetSamplers`
- VUID-VkPipelineLayoutCreateInfo-descriptorType-03029

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetUniformBuffers`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03030

If the `maintenance7` feature is not enabled, the total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetUniformBuffersDynamic`

- VUID-VkPipelineLayoutCreateInfo-maintenance7-10003

If the `maintenance7` feature is enabled, the total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceMaintenance7PropertiesKHR::maxDescriptorSetTotalUniformBuffersDynamic`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03031

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetStorageBuffers`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03032

If the `maintenance7` feature is not enabled, the total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetStorageBuffersDynamic`

- VUID-VkPipelineLayoutCreateInfo-maintenance7-10004

If the `maintenance7` feature is enabled, the total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceMaintenance7PropertiesKHR::maxDescriptorSetTotalStorageBuffersDynamic`

- VUID-VkPipelineLayoutCreateInfo-None-10005

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceMaintenance7PropertiesKHR::maxDescriptorSetTotalBuffersDynamic`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03033

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, and `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetSampledImages`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03034

The total number of descriptors in descriptor set layouts with a `descriptorType` of

VK_DESCRIPTOR_TYPE_STORAGE_IMAGE, and VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetStorageImages`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03035

The total number of descriptors in descriptor set layouts with a `descriptorType` of `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDescriptorSetInputAttachments`

- VUID-VkPipelineLayoutCreateInfo-pPushConstantRanges-00292

Any two elements of `pPushConstantRanges` **must** not include the same stage in `stageFlags`

- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-00293

`pSetLayouts` **must** not contain more than one descriptor set layout that was created with `VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR` set

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03571

The total number of bindings in descriptor set layouts created without the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set with a `descriptorType` of `VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxPerStageDescriptorAccelerationStructures`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03572

The total number of bindings with a `descriptorType` of `VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR` accessible to any given shader stage across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxPerStageDescriptorUpdateAfterBindAccelerationStructures`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03573

The total number of bindings in descriptor set layouts created without the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set with a `descriptorType` of `VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxDescriptorSetAccelerationStructures`

- VUID-VkPipelineLayoutCreateInfo-descriptorType-03574

The total number of bindings with a `descriptorType` of `VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR` accessible across all shader stages and across all elements of `pSetLayouts` **must** be less than or equal to `VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxDescriptorSetUpdateAfterBindAccelerationStructures`

- VUID-VkPipelineLayoutCreateInfo-graphicsPipelineLibrary-06753

Elements of `pSetLayouts` **must** be valid `VkDescriptorSetLayout` objects

Valid Usage (Implicit)

- VUID-VkPipelineLayoutCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_LAYOUT_CREATE_INFO`
- VUID-VkPipelineLayoutCreateInfo-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkPipelineLayoutCreateInfo-pSetLayouts-parameter
If `setLayoutCount` is not `0`, `pSetLayouts` **must** be a valid pointer to an array of `setLayoutCount` valid or `VK_NULL_HANDLE` `VkDescriptorSetLayout` handles
- VUID-VkPipelineLayoutCreateInfo-pPushConstantRanges-parameter
If `pushConstantRangeCount` is not `0`, `pPushConstantRanges` **must** be a valid pointer to an array of `pushConstantRangeCount` valid `VkPushConstantRange` structures

```
typedef enum VkPipelineLayoutCreateFlagBits {  
} VkPipelineLayoutCreateFlagBits;
```

All values for this enum are defined by extensions.

```
// Provided by VK_VERSION_1_0  
typedef VkFlags VkPipelineLayoutCreateFlags;
```

`VkPipelineLayoutCreateFlags` is a bitmask type for setting a mask of `VkPipelineLayoutCreateFlagBits`.

The `VkPushConstantRange` structure is defined as:

```
// Provided by VK_VERSION_1_0  
typedef struct VkPushConstantRange {  
    VkShaderStageFlags    stageFlags;  
    uint32_t              offset;  
    uint32_t              size;  
} VkPushConstantRange;
```

- `stageFlags` is a set of stage flags describing the shader stages that will access a range of push constants. If a particular stage is not included in the range, then accessing members of that range of push constants from the corresponding shader stage will return undefined values.
- `offset` and `size` are the start offset and size, respectively, consumed by the range. Both `offset` and `size` are in units of bytes and **must** be a multiple of 4. The layout of the push constant variables is specified in the shader.

Valid Usage

- VUID-VkPushConstantRange-offset-00294
`offset` **must** be less than `VkPhysicalDeviceLimits::maxPushConstantsSize`

- VUID-VkPushConstantRange-offset-00295
offset must be a multiple of 4
- VUID-VkPushConstantRange-size-00296
size must be greater than 0
- VUID-VkPushConstantRange-size-00297
size must be a multiple of 4
- VUID-VkPushConstantRange-size-00298
size must be less than or equal to `VkPhysicalDeviceLimits::maxPushConstantsSize` minus **offset**

Valid Usage (Implicit)

- VUID-VkPushConstantRange-stageFlags-parameter
stageFlags must be a valid combination of `VkShaderStageFlagBits` values
- VUID-VkPushConstantRange-stageFlags-requiredbitmask
stageFlags must not be 0

Once created, pipeline layouts are used as part of pipeline creation (see [Pipelines](#)), as part of binding descriptor sets (see [Descriptor Set Binding](#)), and as part of setting push constants (see [Push Constant Updates](#)). Pipeline creation accepts a pipeline layout as input, and the layout **may** be used to map (set, binding, arrayElement) tuples to implementation resources or memory locations within a descriptor set. The assignment of implementation resources depends only on the bindings defined in the descriptor sets that comprise the pipeline layout, and not on any shader source.

All resource variables [statically used](#) in all shaders in a pipeline **must** be declared with a (set, binding, arrayElement) that exists in the corresponding descriptor set layout and is of an appropriate descriptor type and includes the set of shader stages it is used by in **stageFlags**. The pipeline layout **can** include entries that are not used by a particular pipeline. The pipeline layout allows the application to provide a consistent set of bindings across multiple pipeline compiles, which enables those pipelines to be compiled in a way that the implementation **may** cheaply switch pipelines without reprogramming the bindings.

Similarly, the push constant block declared in each shader (if present) **must** only place variables at offsets that are each included in a push constant range with **stageFlags** including the bit corresponding to the shader stage that uses it. The pipeline layout **can** include ranges or portions of ranges that are not used by a particular pipeline.

There is a limit on the total number of resources of each type that **can** be included in bindings in all descriptor set layouts in a pipeline layout as shown in [Pipeline Layout Resource Limits](#). The “Total Resources Available” column gives the limit on the number of each type of resource that **can** be included in bindings in all descriptor sets in the pipeline layout. Some resource types count against multiple limits. Additionally, there are limits on the total number of each type of resource that **can** be used in any pipeline stage as described in [Shader Resource Limits](#).

Table 13. Pipeline Layout Resource Limits

Total Resources Available	Resource Types
maxDescriptorSetSamplers	sampler
	combined image sampler
maxDescriptorSetSampledImages	sampled image
	combined image sampler
	uniform texel buffer
maxDescriptorSetStorageImages	storage image
	storage texel buffer
maxDescriptorSetUniformBuffers	uniform buffer
	uniform buffer dynamic
maxDescriptorSetUniformBuffersDynamic or maxDescriptorSetUpdateAfterBindTotalUniformBuf fersDynamic	uniform buffer dynamic
maxDescriptorSetStorageBuffers	storage buffer
	storage buffer dynamic
maxDescriptorSetStorageBuffersDynamic or maxDescriptorSetUpdateAfterBindTotalStorageBuf fersDynamic	storage buffer dynamic
maxDescriptorSetInputAttachments	input attachment
maxDescriptorSetAccelerationStructures or maxDescriptorSetUpdateAfterBindAccelerationStr uctures	acceleration structure

To destroy a pipeline layout, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyPipelineLayout(
    VkDevice          device,
    VkPipelineLayout  pipelineLayout,
    const VkAllocationCallbacks* pAllocator);
```

- **device** is the logical device that destroys the pipeline layout.
- **pipelineLayout** is the pipeline layout to destroy.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyPipelineLayout-pipelineLayout-00299
If **VkAllocationCallbacks** were provided when **pipelineLayout** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyPipelineLayout-pipelineLayout-00300

If no `VkAllocationCallbacks` were provided when `pipelineLayout` was created, `pAllocator` **must** be `NULL`

- VUID-vkDestroyPipelineLayout-pipelineLayout-02004
`pipelineLayout` **must** not have been passed to any `vkCmd*` command for any command buffers that are still in the `recording state` when `vkDestroyPipelineLayout` is called

Valid Usage (Implicit)

- VUID-vkDestroyPipelineLayout-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyPipelineLayout-pipelineLayout-parameter
If `pipelineLayout` is not `VK_NULL_HANDLE`, `pipelineLayout` **must** be a valid `VkPipelineLayout` handle
- VUID-vkDestroyPipelineLayout-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyPipelineLayout-pipelineLayout-parent
If `pipelineLayout` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `pipelineLayout` **must** be externally synchronized

Pipeline Layout Compatibility

Two pipeline layouts are defined to be “compatible for `push constants`” if they were created with identical push constant ranges. Two pipeline layouts are defined to be “compatible for set N” if they were created with *identically defined* descriptor set layouts for sets zero through N, and if they were created with identical push constant ranges.

When binding a descriptor set (see [Descriptor Set Binding](#)) to set number N, a previously bound descriptor set bound with lower index M than N is disturbed if the pipeline layouts for set M and N are not compatible for set M. Otherwise, the bound descriptor set in M is not disturbed.

If, additionally, the previously bound descriptor set for set N was bound using a pipeline layout not compatible for set N, then all bindings in sets numbered greater than N are disturbed.

When binding a pipeline, the pipeline **can** correctly access any previously bound descriptor set N if it was bound with compatible pipeline layout for set N, and it was not disturbed.

Layout compatibility means that descriptor sets **can** be bound to a command buffer for use by any pipeline created with a compatible pipeline layout, and without having bound a particular pipeline first. It also means that descriptor sets **can** remain valid across a pipeline change, and the same

resources will be accessible to the newly bound pipeline.

When a descriptor set is disturbed by binding descriptor sets, the disturbed set is considered to contain undefined descriptors bound with the same pipeline layout as the disturbing descriptor set.

Implementor's Note

A consequence of layout compatibility is that when the implementation compiles a pipeline layout and maps pipeline resources to implementation resources, the mechanism for set N **should** only be a function of sets [0..N].

NOTE

Place the least frequently changing descriptor sets near the start of the pipeline layout, and place the descriptor sets representing the most frequently changing resources near the end. When pipelines are switched, only the descriptor set bindings that have been invalidated will need to be updated and the remainder of the descriptor set bindings will remain in place.

The maximum number of descriptor sets that **can** be bound to a pipeline layout is queried from physical device properties (see `maxBoundDescriptorSets` in [Limits](#)).

API Example

```
const VkDescriptorSetLayout layouts[] = { layout1, layout2 };

const VkPushConstantRange ranges[] =
{
    {
        .stageFlags = VK_SHADER_STAGE_VERTEX_BIT,
        .offset = 0,
        .size = 4
    },
    {
        .stageFlags = VK_SHADER_STAGE_FRAGMENT_BIT,
        .offset = 4,
        .size = 4
    },
};

const VkPipelineLayoutCreateInfo createInfo =
{
    .sType = VK_STRUCTURE_TYPE_PIPELINE_LAYOUT_CREATE_INFO,
    .pNext = NULL,
    .flags = 0,
    .setLayoutCount = 2,
    .pSetLayouts = layouts,
    .pushConstantRangeCount = 2,
    .pPushConstantRanges = ranges
};
```

```
VkPipelineLayout myPipelineLayout;
myResult = vkCreatePipelineLayout(
    myDevice,
    &createInfo,
    NULL,
    &myPipelineLayout);
```

14.2.3. Allocation of Descriptor Sets

A *descriptor pool* maintains a pool of descriptors, from which descriptor sets are allocated. Descriptor pools are externally synchronized, meaning that the application **must** not allocate and/or free descriptor sets from the same pool in multiple threads simultaneously.

Descriptor pools are represented by `VkDescriptorPool` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDescriptorPool)
```

To create a descriptor pool object, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateDescriptorPool(
    VkDevice device,
    const VkDescriptorPoolCreateInfo* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkDescriptorPool* pDescriptorPool);
```

- `device` is the logical device that creates the descriptor pool.
- `pCreateInfo` is a pointer to a `VkDescriptorPoolCreateInfo` structure specifying the state of the descriptor pool object.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pDescriptorPool` is a pointer to a `VkDescriptorPool` handle in which the resulting descriptor pool object is returned.

The created descriptor pool is returned in `pDescriptorPool`.

Valid Usage (Implicit)

- VUID-vkCreateDescriptorPool-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateDescriptorPool-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkDescriptorPoolCreateInfo` structure
- VUID-vkCreateDescriptorPool-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid

[VkAllocationCallbacks](#) structure

- VUID-vkCreateDescriptorPool-pDescriptorPool-parameter
[pDescriptorPool](#) **must** be a valid pointer to a [VkDescriptorPool](#) handle

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

Additional information about the pool is passed in a [VkDescriptorPoolCreateInfo](#) structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorPoolCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkDescriptorPoolCreateFlags flags;
    uint32_t           maxSets;
    uint32_t           poolSizeCount;
    const VkDescriptorPoolSize* pPoolSizes;
} VkDescriptorPoolCreateInfo;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [flags](#) is a bitmask of [VkDescriptorPoolCreateFlagBits](#) specifying certain supported operations on the pool.
- [maxSets](#) is the maximum number of descriptor sets that **can** be allocated from the pool.
- [poolSizeCount](#) is the number of elements in [pPoolSizes](#).
- [pPoolSizes](#) is a pointer to an array of [VkDescriptorPoolSize](#) structures, each containing a descriptor type and number of descriptors of that type to be allocated in the pool.

If multiple [VkDescriptorPoolSize](#) structures containing the same descriptor type appear in the [pPoolSizes](#) array then the pool will be created with enough storage for the total number of descriptors of each type.

Fragmentation of a descriptor pool is possible and **may** lead to descriptor set allocation failures. A failure due to fragmentation is defined as failing a descriptor set allocation despite the sum of all outstanding descriptor set allocations from the pool plus the requested allocation requiring no more than the total number of descriptors requested at pool creation. Implementations provide certain guarantees of when fragmentation **must** not cause allocation failure, as described below.

If a descriptor pool has not had any descriptor sets freed since it was created or most recently reset then fragmentation **must** not cause an allocation failure (note that this is always the case for a pool created without the `VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT` bit set). Additionally, if all sets allocated from the pool since it was created or most recently reset use the same number of descriptors (of each type) and the requested allocation also uses that same number of descriptors (of each type), then fragmentation **must** not cause an allocation failure.

If an allocation failure occurs due to fragmentation, an application **can** create an additional descriptor pool to perform further descriptor set allocations.

Valid Usage

- VUID-VkDescriptorPoolCreateInfo-descriptorPoolOverallocation-09227
`maxSets` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkDescriptorPoolCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DESCRIPTOR_POOL_CREATE_INFO`
- VUID-VkDescriptorPoolCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDescriptorPoolCreateInfo-flags-parameter
`flags` **must** be a valid combination of `VkDescriptorPoolCreateFlagBits` values
- VUID-VkDescriptorPoolCreateInfo-pPoolSizes-parameter
If `poolSizeCount` is not 0, `pPoolSizes` **must** be a valid pointer to an array of `poolSizeCount` valid `VkDescriptorPoolSize` structures

Bits which **can** be set in `VkDescriptorPoolCreateInfo::flags`, enabling operations on a descriptor pool, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkDescriptorPoolCreateFlagBits {
    VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT = 0x00000001,
} VkDescriptorPoolCreateFlagBits;
```

- `VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT` specifies that descriptor sets **can** return their individual allocations to the pool, i.e. all of `vkAllocateDescriptorSets`, `vkFreeDescriptorSets`, and `vkResetDescriptorPool` are allowed. Otherwise, descriptor sets allocated from the pool **must** not be individually freed back to the pool, i.e. only `vkAllocateDescriptorSets` and `vkResetDescriptorPool` are allowed.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDescriptorPoolCreateFlags;
```

`VkDescriptorPoolCreateFlags` is a bitmask type for setting a mask of zero or more `VkDescriptorPoolCreateFlagBits`.

The `VkDescriptorPoolSize` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorPoolSize {
    VkDescriptorType    type;
    uint32_t            descriptorCount;
} VkDescriptorPoolSize;
```

- `type` is the type of descriptor.
- `descriptorCount` is the number of descriptors of that type to allocate.

NOTE

When creating a descriptor pool that will contain descriptors for combined image samplers of multi-planar formats, an application needs to account for non-trivial descriptor consumption when choosing the `descriptorCount` value, as indicated by `VkSamplerYcbcrConversionImageFormatProperties::combinedImageSamplerDescriptorCount`.

For simplicity the application **can** use the `VkPhysicalDeviceMaintenance6PropertiesKHR::maxCombinedImageSamplerDescriptorCount` property, which is sized to accommodate any and all **formats that require a sampler Y'C_BC_R conversion** supported by the implementation.

Valid Usage

- VUID-VkDescriptorPoolSize-descriptorCount-00302
`descriptorCount` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkDescriptorPoolSize-type-parameter
`type` **must** be a valid `VkDescriptorType` value

To destroy a descriptor pool, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyDescriptorPool(
    VkDevice                device,
    VkDescriptorPool         descriptorPool,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the descriptor pool.

- `descriptorPool` is the descriptor pool to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

When a pool is destroyed, all descriptor sets allocated from the pool are implicitly freed and become invalid. Descriptor sets allocated from a given pool do not need to be freed before destroying that descriptor pool.

Valid Usage

- VUID-vkDestroyDescriptorPool-descriptorPool-00303
All submitted commands that refer to `descriptorPool` (via any allocated descriptor sets) **must** have completed execution
- VUID-vkDestroyDescriptorPool-descriptorPool-00304
If `VkAllocationCallbacks` were provided when `descriptorPool` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDescriptorPool-descriptorPool-00305
If no `VkAllocationCallbacks` were provided when `descriptorPool` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyDescriptorPool-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyDescriptorPool-descriptorPool-parameter
If `descriptorPool` is not `VK_NULL_HANDLE`, `descriptorPool` **must** be a valid `VkDescriptorPool` handle
- VUID-vkDestroyDescriptorPool-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyDescriptorPool-descriptorPool-parent
If `descriptorPool` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `descriptorPool` **must** be externally synchronized

Descriptor sets are allocated from descriptor pool objects, and are represented by `VkDescriptorSet` handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDescriptorSet)
```

To allocate descriptor sets from a descriptor pool, call:

```
// Provided by VK_VERSION_1_0
VkResult vkAllocateDescriptorSets(
    VkDevice device,
    const VkDescriptorSetAllocateInfo* pAllocateInfo,
    VkDescriptorSet* pDescriptorSets);
```

- **device** is the logical device that owns the descriptor pool.
- **pAllocateInfo** is a pointer to a [VkDescriptorSetAllocateInfo](#) structure describing parameters of the allocation.
- **pDescriptorSets** is a pointer to an array of [VkDescriptorSet](#) handles in which the resulting descriptor set objects are returned.

The allocated descriptor sets are returned in **pDescriptorSets**.

When a descriptor set is allocated, the initial state is largely uninitialized and all descriptors are undefined, with the exception that samplers with a non-null **pImmutableSamplers** are initialized on allocation. Descriptors also become undefined if the underlying resource or view object is destroyed. Descriptor sets containing undefined descriptors **can** still be bound and used, subject to the following conditions:

- Descriptors that are **statically used** **must** have been populated before the descriptor set is **consumed**.
- Entries that are not used by a pipeline **can** have undefined descriptors.

If a call to **vkAllocateDescriptorSets** would cause the total number of descriptor sets allocated from the pool to exceed the value of [VkDescriptorPoolCreateInfo::maxSets](#) used to create **pAllocateInfo->descriptorPool**, then the allocation **may** fail due to lack of space in the descriptor pool. Similarly, the allocation **may** fail due to lack of space if the call to **vkAllocateDescriptorSets** would cause the number of any given descriptor type to exceed the sum of all the **descriptorCount** members of each element of [VkDescriptorPoolCreateInfo::pPoolSizes](#) with a **type** equal to that type.

If the allocation fails due to no more space in the descriptor pool, and not because of system or device memory exhaustion, then **VK_ERROR_OUT_OF_POOL_MEMORY** **must** be returned.

vkAllocateDescriptorSets **can** be used to create multiple descriptor sets. If the creation of any of those descriptor sets fails, then the implementation **must** destroy all successfully created descriptor set objects from this command, set all entries of the **pDescriptorSets** array to [VK_NULL_HANDLE](#) and return the error.

Valid Usage (Implicit)

- VUID-vkAllocateDescriptorSets-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkAllocateDescriptorSets-pAllocateInfo-parameter `pAllocateInfo` **must** be a valid pointer to a valid `VkDescriptorSetAllocateInfo` structure
- VUID-vkAllocateDescriptorSets-pDescriptorSets-parameter `pDescriptorSets` **must** be a valid pointer to an array of `pAllocateInfo->descriptorSetCount` `VkDescriptorSet` handles
- VUID-vkAllocateDescriptorSets-pAllocateInfo::descriptorSetCount-arraylength `pAllocateInfo->descriptorSetCount` **must** be greater than 0

Host Synchronization

- Host access to `pAllocateInfo->descriptorPool` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_FRAGMENTED_POOL`
- `VK_ERROR_OUT_OF_POOL_MEMORY`

The `VkDescriptorSetAllocateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorSetAllocateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkDescriptorPool    descriptorPool;
    uint32_t           descriptorSetCount;
    const VkDescriptorSetLayout* pSetLayouts;
} VkDescriptorSetAllocateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `descriptorPool` is the pool which the sets will be allocated from.

- `descriptorSetCount` determines the number of descriptor sets to be allocated from the pool.
- `pSetLayouts` is a pointer to an array of descriptor set layouts, with each member specifying how the corresponding descriptor set is allocated.

Valid Usage

- VUID-VkDescriptorSetAllocateInfo-apiVersion-07895
If the `VK_KHR_maintenance1` extension is not enabled and `VkPhysicalDeviceProperties::apiVersion` is less than Vulkan 1.1, `descriptorSetCount` **must** not be greater than the number of sets that are currently available for allocation in `descriptorPool`
- VUID-VkDescriptorSetAllocateInfo-apiVersion-07896
If the `VK_KHR_maintenance1` extension is not enabled and `VkPhysicalDeviceProperties::apiVersion` is less than Vulkan 1.1, `descriptorPool` **must** have enough free descriptor capacity remaining to allocate the descriptor sets of the specified layouts
- VUID-VkDescriptorSetAllocateInfo-pSetLayouts-00308
Each element of `pSetLayouts` **must** not have been created with `VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR` set

Valid Usage (Implicit)

- VUID-VkDescriptorSetAllocateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DESCRIPTOR_SET_ALLOCATE_INFO`
- VUID-VkDescriptorSetAllocateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDescriptorSetAllocateInfo-descriptorPool-parameter
`descriptorPool` **must** be a valid `VkDescriptorPool` handle
- VUID-VkDescriptorSetAllocateInfo-pSetLayouts-parameter
`pSetLayouts` **must** be a valid pointer to an array of `descriptorSetCount` valid `VkDescriptorSetLayout` handles
- VUID-VkDescriptorSetAllocateInfo-descriptorSetCount-arraylength
`descriptorSetCount` **must** be greater than 0
- VUID-VkDescriptorSetAllocateInfo-commonparent
Both of `descriptorPool`, and the elements of `pSetLayouts` **must** have been created, allocated, or retrieved from the same `VkDevice`

To free allocated descriptor sets, call:

```
// Provided by VK_VERSION_1_0
VkResult vkFreeDescriptorSets(
    VkDevice          device,
    VkDescriptorPool   descriptorPool,
    uint32_t          descriptorSetCount,
    const VkDescriptorSet* pDescriptorSets);
```

- **device** is the logical device that owns the descriptor pool.
- **descriptorPool** is the descriptor pool from which the descriptor sets were allocated.
- **descriptorSetCount** is the number of elements in the **pDescriptorSets** array.
- **pDescriptorSets** is a pointer to an array of handles to **VkDescriptorSet** objects.

After calling **vkFreeDescriptorSets**, all descriptor sets in **pDescriptorSets** are invalid.

Valid Usage

- VUID-vkFreeDescriptorSets-pDescriptorSets-00309
All submitted commands that refer to any element of **pDescriptorSets** **must** have completed execution
- VUID-vkFreeDescriptorSets-pDescriptorSets-00310
pDescriptorSets **must** be a valid pointer to an array of **descriptorSetCount** **VkDescriptorSet** handles, each element of which **must** either be a valid handle or **VK_NULL_HANDLE**
- VUID-vkFreeDescriptorSets-descriptorPool-00312
descriptorPool **must** have been created with the **VK_DESCRIPTOR_POOL_CREATE_FREE_DESCRIPTOR_SET_BIT** flag

Valid Usage (Implicit)

- VUID-vkFreeDescriptorSets-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkFreeDescriptorSets-descriptorPool-parameter
descriptorPool **must** be a valid **VkDescriptorPool** handle
- VUID-vkFreeDescriptorSets-descriptorSetCount-arraylength
descriptorSetCount **must** be greater than 0
- VUID-vkFreeDescriptorSets-descriptorPool-parent
descriptorPool **must** have been created, allocated, or retrieved from **device**
- VUID-vkFreeDescriptorSets-pDescriptorSets-parent
Each element of **pDescriptorSets** that is a valid handle **must** have been created, allocated, or retrieved from **descriptorPool**

Host Synchronization

- Host access to **descriptorPool** **must** be externally synchronized
- Host access to each member of **pDescriptorSets** **must** be externally synchronized

Return Codes

Success

- **VK_SUCCESS**

Failure

None

To return all descriptor sets allocated from a given pool to the pool, rather than freeing individual descriptor sets, call:

```
// Provided by VK_VERSION_1_0
VkResult vkResetDescriptorPool(
    VkDevice          device,
    VkDescriptorPool   descriptorPool,
    VkDescriptorPoolResetFlags flags);
```

- **device** is the logical device that owns the descriptor pool.
- **descriptorPool** is the descriptor pool to be reset.
- **flags** is reserved for future use.

Resetting a descriptor pool recycles all of the resources from all of the descriptor sets allocated from the descriptor pool back to the descriptor pool, and the descriptor sets are implicitly freed.

Valid Usage

- VUID-vkResetDescriptorPool-descriptorPool-00313
All uses of **descriptorPool** (via any allocated descriptor sets) **must** have completed execution

Valid Usage (Implicit)

- VUID-vkResetDescriptorPool-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkResetDescriptorPool-descriptorPool-parameter
descriptorPool **must** be a valid **VkDescriptorPool** handle
- VUID-vkResetDescriptorPool-flags-zero bitmask

flags must be 0

- VUID-vkResetDescriptorPool-descriptorPool-parent
descriptorPool must have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **descriptorPool must** be externally synchronized
- Host access to any **VkDescriptorSet** objects allocated from **descriptorPool must** be externally synchronized

Return Codes

Success

- **VK_SUCCESS**

Failure

None

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkDescriptorPoolResetFlags;
```

VkDescriptorPoolResetFlags is a bitmask type for setting a mask, but is currently reserved for future use.

14.2.4. Descriptor Set Updates

Once allocated, descriptor sets **can** be updated with a combination of write and copy operations. To update descriptor sets, call:

```
// Provided by VK_VERSION_1_0
void vkUpdateDescriptorSets(
    VkDevice          device,
    uint32_t          descriptorWriteCount,
    const VkWriteDescriptorSet* pDescriptorWrites,
    uint32_t          descriptorCopyCount,
    const VkCopyDescriptorSet* pDescriptorCopies);
```

- **device** is the logical device that updates the descriptor sets.
- **descriptorWriteCount** is the number of elements in the **pDescriptorWrites** array.
- **pDescriptorWrites** is a pointer to an array of **VkWriteDescriptorSet** structures describing the descriptor sets to write to.
- **descriptorCopyCount** is the number of elements in the **pDescriptorCopies** array.

- `pDescriptorCopies` is a pointer to an array of `VkCopyDescriptorSet` structures describing the descriptor sets to copy between.

The operations described by `pDescriptorWrites` are performed first, followed by the operations described by `pDescriptorCopies`. Within each array, the operations are performed in the order they appear in the array.

Each element in the `pDescriptorWrites` array describes an operation updating the descriptor set using descriptors for resources specified in the structure.

Each element in the `pDescriptorCopies` array is a `VkCopyDescriptorSet` structure describing an operation copying descriptors between sets.

If the `dstSet` member of any element of `pDescriptorWrites` or `pDescriptorCopies` is bound, accessed, or modified by any command that was recorded to a command buffer which is currently in the `recording or executable state`, that command buffer becomes `invalid`.

Valid Usage

- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06236
For each element `i` where `pDescriptorWrites[i].descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`, elements of the `pTexelBufferView` member of `pDescriptorWrites[i]` **must** have been created on `device`
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06237
For each element `i` where `pDescriptorWrites[i].descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER`, `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER`, `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, the `buffer` member of any element of the `pBufferInfo` member of `pDescriptorWrites[i]` **must** have been created on `device`
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06238
For each element `i` where `pDescriptorWrites[i].descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and `dstSet` was not allocated with a layout that included immutable samplers for `dstBinding` with `descriptorType`, the `sampler` member of any element of the `pImageInfo` member of `pDescriptorWrites[i]` **must** have been created on `device`
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06239
For each element `i` where `pDescriptorWrites[i].descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` the `imageView` member of any element of `pDescriptorWrites[i]` **must** have been created on `device`
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06240
For each element `i` where `pDescriptorWrites[i].descriptorType` is `VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR`, elements of the `pAccelerationStructures` member of a `VkWriteDescriptorSetAccelerationStructureKHR` structure in the `pNext` chain of `pDescriptorWrites[i]` **must** have been created on `device`

- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06493
For each element `i` where `pDescriptorWrites[i].descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLER`, `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, `pDescriptorWrites[i].pImageInfo` **must** be a valid pointer to an array of `pDescriptorWrites[i].descriptorCount` valid `VkDescriptorImageInfo` structures
- VUID-vkUpdateDescriptorSets-None-03047
The `dstSet` member of each element of `pDescriptorWrites` or `pDescriptorCopies` **must** not be used by any command that was recorded to a command buffer which is in the `pending` state
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-06993
Host access to `pDescriptorWrites[i].dstSet` and `pDescriptorCopies[i].dstSet` **must** be `externally synchronized`

Valid Usage (Implicit)

- VUID-vkUpdateDescriptorSets-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkUpdateDescriptorSets-pDescriptorWrites-parameter
If `descriptorWriteCount` is not `0`, `pDescriptorWrites` **must** be a valid pointer to an array of `descriptorWriteCount` valid `VkWriteDescriptorSet` structures
- VUID-vkUpdateDescriptorSets-pDescriptorCopies-parameter
If `descriptorCopyCount` is not `0`, `pDescriptorCopies` **must** be a valid pointer to an array of `descriptorCopyCount` valid `VkCopyDescriptorSet` structures

The `VkWriteDescriptorSet` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkWriteDescriptorSet {
    VkStructureType    sType;
    const void*        pNext;
    VkDescriptorSet     dstSet;
    uint32_t           dstBinding;
    uint32_t           dstArrayElement;
    uint32_t           descriptorCount;
    VkDescriptorType    descriptorType;
    const VkDescriptorImageInfo* pImageInfo;
    const VkDescriptorBufferInfo* pBufferInfo;
    const VkBufferView* pTexelBufferView;
} VkWriteDescriptorSet;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `dstSet` is the destination descriptor set to update.
- `dstBinding` is the descriptor binding within that set.
- `dstArrayElement` is the starting element in that array.
- `descriptorCount` is the number of descriptors to update. `descriptorCount` is one of
 - the number of elements in `pImageInfo`
 - the number of elements in `pBufferInfo`
 - the number of elements in `pTexelBufferView`
 - a value matching the `accelerationStructureCount` of a `VkWriteDescriptorSetAccelerationStructureKHR` structure in the `pNext` chain
- `descriptorType` is a `VkDescriptorType` specifying the type of each descriptor in `pImageInfo`, `pBufferInfo`, or `pTexelBufferView`, as described below. It **must** be the same type as the `descriptorType` specified in `VkDescriptorSetLayoutBinding` for `dstSet` at `dstBinding`. The type of the descriptor also controls which array the descriptors are taken from.
- `pImageInfo` is a pointer to an array of `VkDescriptorImageInfo` structures or is ignored, as described below.
- `pBufferInfo` is a pointer to an array of `VkDescriptorBufferInfo` structures or is ignored, as described below.
- `pTexelBufferView` is a pointer to an array of `VkBufferView` handles as described in the [Buffer Views](#) section or is ignored, as described below.

Only one of `pImageInfo`, `pBufferInfo`, or `pTexelBufferView` members is used according to the descriptor type specified in the `descriptorType` member of the containing `VkWriteDescriptorSet` structure, or if `descriptorType` is `VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR`, in which case the source data for the descriptor writes is taken from the `VkWriteDescriptorSetAccelerationStructureKHR` structure in the `pNext` chain of `VkWriteDescriptorSet`, as specified below.

If the `nullDescriptor` feature is enabled, the buffer, acceleration structure, imageView, or bufferView **can** be `VK_NULL_HANDLE`. Loads from a null descriptor return zero values and stores and atomics to a null descriptor are discarded. A null acceleration structure descriptor results in the miss shader being invoked.

Consecutive Binding Updates

If the `dstBinding` has fewer than `descriptorCount` array elements remaining starting from `dstArrayElement`, then the remainder will be used to update the subsequent binding - `dstBinding+1` starting at array element zero. If a binding has a `descriptorCount` of zero, it is skipped. This behavior applies recursively, with the update affecting consecutive bindings as needed to update all `descriptorCount` descriptors. Consecutive bindings **must** have identical `VkDescriptorType`, `VkShaderStageFlags`, and immutable samplers references.

Valid Usage

- VUID-VkWriteDescriptorSet-dstBinding-00315
`dstBinding` **must** be less than or equal to the maximum value of `binding` of all

`VkDescriptorSetLayoutBinding` structures specified when `dstSet`'s descriptor set layout was created

- VUID-VkWriteDescriptorSet-dstBinding-00316
`dstBinding` **must** be a binding with a non-zero `descriptorCount`
- VUID-VkWriteDescriptorSet-dstBinding-10009
`dstBinding` **must** be a binding with a non-zero `VkDescriptorSetLayoutCreateInfo::bindingCount`
- VUID-VkWriteDescriptorSet-descriptorCount-00317
All consecutive bindings updated via a single `VkWriteDescriptorSet` structure, except those with a `descriptorCount` of zero, **must** have identical `descriptorType` and `stageFlags`
- VUID-VkWriteDescriptorSet-descriptorCount-00318
All consecutive bindings updated via a single `VkWriteDescriptorSet` structure, except those with a `descriptorCount` of zero, **must** all either use immutable samplers or **must** all not use immutable samplers
- VUID-VkWriteDescriptorSet-descriptorType-00319
`descriptorType` **must** match the type of `dstBinding` within `dstSet`
- VUID-VkWriteDescriptorSet-dstSet-00320
`dstSet` **must** be a valid `VkDescriptorSet` handle
- VUID-VkWriteDescriptorSet-dstArrayElement-00321
The sum of `dstArrayElement` and `descriptorCount` **must** be less than or equal to the number of array elements in the descriptor set binding specified by `dstBinding`, and all applicable consecutive bindings
- VUID-VkWriteDescriptorSet-descriptorType-02994
If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`, each element of `pTexelBufferView` **must** be either a valid `VkBufferView` handle or `VK_NULL_HANDLE`
- VUID-VkWriteDescriptorSet-descriptorType-02995
If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` and the `nullDescriptor` feature is not enabled, each element of `pTexelBufferView` **must** not be `VK_NULL_HANDLE`
- VUID-VkWriteDescriptorSet-descriptorType-00324
If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER`, `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER`, `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, `pBufferInfo` **must** be a valid pointer to an array of `descriptorCount` valid `VkDescriptorBufferInfo` structures
- VUID-VkWriteDescriptorSet-descriptorType-00325
If `descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and `dstSet` was not allocated with a layout that included immutable samplers for `dstBinding` with `descriptorType`, the `sampler` member of each element of `pImageInfo` **must** be a valid `VkSampler` object
- VUID-VkWriteDescriptorSet-descriptorType-02996
If `descriptorType` is `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, or `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, the `imageView`

member of each element of `pImageInfo` **must** be either a valid `VkImageView` handle or `VK_NULL_HANDLE`

- VUID-VkWriteDescriptorSet-descriptorType-02997

If `descriptorType` is `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, or `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, and the `nullDescriptor` feature is not enabled, the `imageView` member of each element of `pImageInfo` **must** not be `VK_NULL_HANDLE`

- VUID-VkWriteDescriptorSet-descriptorType-07683

If `descriptorType` is `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, the `imageView` member of each element of `pImageInfo` **must** not be `VK_NULL_HANDLE`

- VUID-VkWriteDescriptorSet-descriptorType-02382

If `descriptorType` is `VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR`, the `pNext` chain **must** include a `VkWriteDescriptorSetAccelerationStructureKHR` structure whose `accelerationStructureCount` member equals `descriptorCount`

- VUID-VkWriteDescriptorSet-descriptorType-01946

If `descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, then the `imageView` member of each `pImageInfo` element **must** have been created without a `VkSamplerYcbcrConversionInfo` structure in its `pNext` chain

- VUID-VkWriteDescriptorSet-descriptorType-02738

If `descriptorType` is `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and if any element of `pImageInfo` has an `imageView` member that was created with a `VkSamplerYcbcrConversionInfo` structure in its `pNext` chain, then `dstSet` **must** have been allocated with a layout that included immutable samplers for `dstBinding`, and the corresponding immutable sampler **must** have been created with an *identically defined* `VkSamplerYcbcrConversionInfo` object

- VUID-VkWriteDescriptorSet-descriptorType-01948

If `descriptorType` is `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, and `dstSet` was allocated with a layout that included immutable samplers for `dstBinding`, then the `imageView` member of each element of `pImageInfo` which corresponds to an immutable sampler that enables `sampler Y'CBCR conversion` **must** have been created with a `VkSamplerYcbcrConversionInfo` structure in its `pNext` chain with an *identically defined* `VkSamplerYcbcrConversionInfo` to the corresponding immutable sampler

- VUID-VkWriteDescriptorSet-descriptorType-09506

If `descriptorType` is `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `dstSet` was allocated with a layout that included immutable samplers for `dstBinding`, and those samplers enable `sampler Y'CBCR conversion`, then `imageView` **must** not be `VK_NULL_HANDLE`

- VUID-VkWriteDescriptorSet-descriptorType-00327

If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, the `offset` member of each element of `pBufferInfo` **must** be a multiple of `VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment`

- VUID-VkWriteDescriptorSet-descriptorType-00328

If `descriptorType` is `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, the `offset` member of each element of

`pBufferInfo` **must** be a multiple of `VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment`

- VUID-VkWriteDescriptorSet-descriptorType-00329
If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER`, `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER`, or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, and the `buffer` member of any element of `pBufferInfo` is the handle of a non-sparse buffer, then that buffer **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkWriteDescriptorSet-descriptorType-00330
If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, the `buffer` member of each element of `pBufferInfo` **must** have been created with `VK_BUFFER_USAGE_UNIFORM_BUFFER_BIT` set
- VUID-VkWriteDescriptorSet-descriptorType-00331
If `descriptorType` is `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, the `buffer` member of each element of `pBufferInfo` **must** have been created with `VK_BUFFER_USAGE_STORAGE_BUFFER_BIT` set
- VUID-VkWriteDescriptorSet-descriptorType-00332
If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, the `range` member of each element of `pBufferInfo`, or the `effective range` if `range` is `VK_WHOLE_SIZE`, **must** be less than or equal to `VkPhysicalDeviceLimits::maxUniformBufferRange`
- VUID-VkWriteDescriptorSet-descriptorType-00333
If `descriptorType` is `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, the `range` member of each element of `pBufferInfo`, or the `effective range` if `range` is `VK_WHOLE_SIZE`, **must** be less than or equal to `VkPhysicalDeviceLimits::maxStorageBufferRange`
- VUID-VkWriteDescriptorSet-descriptorType-08765
If `descriptorType` is `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER`, the `pTexelBufferView` `buffer view usage` **must** include `VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT`
- VUID-VkWriteDescriptorSet-descriptorType-08766
If `descriptorType` is `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`, the `pTexelBufferView` `buffer view usage` **must** include `VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT`
- VUID-VkWriteDescriptorSet-descriptorType-00336
If `descriptorType` is `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE` or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, the `imageView` member of each element of `pImageInfo` **must** have been created with the `identity swizzle`
- VUID-VkWriteDescriptorSet-descriptorType-00337
If `descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, the `imageView` member of each element of `pImageInfo` **must** have been created with `VK_IMAGE_USAGE_SAMPLED_BIT` set
- VUID-VkWriteDescriptorSet-descriptorType-04149
If `descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE` the `imageLayout` member of each element of `pImageInfo` **must** be a member of the list given in `Sampled Image`

- VUID-VkWriteDescriptorSet-descriptorType-04150
If `descriptorType` is `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` the `imageLayout` member of each element of `pImageInfo` **must** be a member of the list given in [Combined Image Sampler](#)
- VUID-VkWriteDescriptorSet-descriptorType-04151
If `descriptorType` is `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` the `imageLayout` member of each element of `pImageInfo` **must** be a member of the list given in [Input Attachment](#)
- VUID-VkWriteDescriptorSet-descriptorType-04152
If `descriptorType` is `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE` the `imageLayout` member of each element of `pImageInfo` **must** be a member of the list given in [Storage Image](#)
- VUID-VkWriteDescriptorSet-descriptorType-00338
If `descriptorType` is `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, the `imageView` member of each element of `pImageInfo` **must** have been created with `VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT` set
- VUID-VkWriteDescriptorSet-descriptorType-00339
If `descriptorType` is `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, the `imageView` member of each element of `pImageInfo` **must** have been created with `VK_IMAGE_USAGE_STORAGE_BIT` set
- VUID-VkWriteDescriptorSet-descriptorType-02752
If `descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLER`, then `dstSet` **must** not have been allocated with a layout that included immutable samplers for `dstBinding`
- VUID-VkWriteDescriptorSet-descriptorType-06450
If `descriptorType` is `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, the `imageView` member of each element of `pImageInfo` **must** have either been created without a [VkImageViewMinLodCreateInfoEXT](#) included in the `pNext` chain or with a [VkImageViewMinLodCreateInfoEXT::minLod](#) of `0.0`

Valid Usage (Implicit)

- VUID-VkWriteDescriptorSet-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET`
- VUID-VkWriteDescriptorSet-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of [VkWriteDescriptorSetAccelerationStructureKHR](#)
- VUID-VkWriteDescriptorSet-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkWriteDescriptorSet-descriptorType-parameter
`descriptorType` **must** be a valid [VkDescriptorType](#) value
- VUID-VkWriteDescriptorSet-descriptorCount-arraylength
`descriptorCount` **must** be greater than `0`
- VUID-VkWriteDescriptorSet-commonparent
Both of `dstSet`, and the elements of `pTexelBufferView` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same [VkDevice](#)

The type of descriptors in a descriptor set is specified by `VkWriteDescriptorSet::descriptorType`, which **must** be one of the values:

```
// Provided by VK_VERSION_1_0
typedef enum VkDescriptorType {
    VK_DESCRIPTOR_TYPE_SAMPLER = 0,
    VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER = 1,
    VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE = 2,
    VK_DESCRIPTOR_TYPE_STORAGE_IMAGE = 3,
    VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER = 4,
    VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER = 5,
    VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER = 6,
    VK_DESCRIPTOR_TYPE_STORAGE_BUFFER = 7,
    VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC = 8,
    VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC = 9,
    VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT = 10,
    // Provided by VK_KHR_acceleration_structure
    VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR = 1000150000,
} VkDescriptorType;
```

- `VK_DESCRIPTOR_TYPE_SAMPLER` specifies a [sampler descriptor](#).
- `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` specifies a [combined image sampler descriptor](#).
- `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE` specifies a [sampled image descriptor](#).
- `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE` specifies a [storage image descriptor](#).
- `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` specifies a [uniform texel buffer descriptor](#).
- `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` specifies a [storage texel buffer descriptor](#).
- `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` specifies a [uniform buffer descriptor](#).
- `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` specifies a [storage buffer descriptor](#).
- `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` specifies a [dynamic uniform buffer descriptor](#).
- `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` specifies a [dynamic storage buffer descriptor](#).
- `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` specifies an [input attachment descriptor](#).

When a descriptor set is updated via elements of `VkWriteDescriptorSet`, members of `pImageInfo`, `pBufferInfo` and `pTexelBufferView` are only accessed by the implementation when they correspond to descriptor type being defined - otherwise they are ignored. The members accessed are as follows for each descriptor type:

- For `VK_DESCRIPTOR_TYPE_SAMPLER`, only the `sampler` member of each element of `VkWriteDescriptorSet::pImageInfo` is accessed.
- For `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, only the `imageView` and `imageLayout` members of each element of `VkWriteDescriptorSet::pImageInfo` are accessed.
- For `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, all members of each element of `VkWriteDescriptorSet::pImageInfo` are accessed.

- For `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER`, `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER`, `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`, all members of each element of `VkWriteDescriptorSet::pBufferInfo` are accessed.
- For `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`, each element of `VkWriteDescriptorSet::pTexelBufferView` is accessed.

When updating descriptors with a `descriptorType` of `VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR`, none of the `pImageInfo`, `pBufferInfo`, or `pTexelBufferView` members are accessed, instead the source data of the descriptor update operation is taken from the `VkWriteDescriptorSetAccelerationStructureKHR` structure in the `pNext` chain of `VkWriteDescriptorSet`.

The `VkDescriptorBufferInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorBufferInfo {
    VkBuffer      buffer;
    VkDeviceSize  offset;
    VkDeviceSize  range;
} VkDescriptorBufferInfo;
```

- `buffer` is `VK_NULL_HANDLE` or the buffer resource.
- `offset` is the offset in bytes from the start of `buffer`. Access to buffer memory via this descriptor uses addressing that is relative to this starting offset.
- `range` is the size in bytes that is used for this descriptor update, or `VK_WHOLE_SIZE` to use the range from `offset` to the end of the buffer.

NOTE

When setting `range` to `VK_WHOLE_SIZE`, the **effective range** **must** not be larger than the maximum range for the descriptor type (`maxUniformBufferRange` or `maxStorageBufferRange`). This means that `VK_WHOLE_SIZE` is not typically useful in the common case where uniform buffer descriptors are suballocated from a buffer that is much larger than `maxUniformBufferRange`.

For `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` and `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` descriptor types, `offset` is the base offset from which the dynamic offset is applied and `range` is the static size used for all dynamic offsets.

When `range` is `VK_WHOLE_SIZE` the effective range is calculated at `vkUpdateDescriptorSets` is by taking the size of `buffer` minus the `offset`.

Valid Usage

- VUID-VkDescriptorBufferInfo-offset-00340
`offset` **must** be less than the size of `buffer`
- VUID-VkDescriptorBufferInfo-range-00341
If `range` is not equal to `VK_WHOLE_SIZE`, `range` **must** be greater than 0

- VUID-VkDescriptorBufferInfo-range-00342
If **range** is not equal to **VK_WHOLE_SIZE**, **range** **must** be less than or equal to the size of **buffer** minus **offset**
- VUID-VkDescriptorBufferInfo-buffer-02998
If the **nullDescriptor** feature is not enabled, **buffer** **must** not be **VK_NULL_HANDLE**
- VUID-VkDescriptorBufferInfo-buffer-02999
If **buffer** is **VK_NULL_HANDLE**, **offset** **must** be zero and **range** **must** be **VK_WHOLE_SIZE**

Valid Usage (Implicit)

- VUID-VkDescriptorBufferInfo-buffer-parameter
If **buffer** is not **VK_NULL_HANDLE**, **buffer** **must** be a valid **VkBuffer** handle

The **VkDescriptorImageInfo** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDescriptorImageInfo {
    VkSampler      sampler;
    VkImageView     imageView;
    VkImageLayout  imageLayout;
} VkDescriptorImageInfo;
```

- **sampler** is a sampler handle, and is used in descriptor updates for types **VK_DESCRIPTOR_TYPE_SAMPLER** and **VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER** if the binding being updated does not use immutable samplers.
- **imageView** is **VK_NULL_HANDLE** or an image view handle, and is used in descriptor updates for types **VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE**, **VK_DESCRIPTOR_TYPE_STORAGE_IMAGE**, **VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER**, and **VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT**.
- **imageLayout** is the layout that the image subresources accessible from **imageView** will be in at the time this descriptor is accessed. **imageLayout** is used in descriptor updates for types **VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE**, **VK_DESCRIPTOR_TYPE_STORAGE_IMAGE**, **VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER**, and **VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT**.

Members of **VkDescriptorImageInfo** that are not used in an update (as described above) are ignored.

Valid Usage

- VUID-VkDescriptorImageInfo-imageView-06712
imageView **must** not be a 2D array image view created from a 3D image
- VUID-VkDescriptorImageInfo-descriptorType-06713
imageView **must** not be a 2D view created from a 3D image
- VUID-VkDescriptorImageInfo-descriptorType-06714
imageView **must** not be a 2D view created from a 3D image

- VUID-VkDescriptorImageInfo-imageView-01976
If **imageView** is created from a depth/stencil image, the **aspectMask** used to create the **imageView** **must** include either **VK_IMAGE_ASPECT_DEPTH_BIT** or **VK_IMAGE_ASPECT_STENCIL_BIT** but not both
- VUID-VkDescriptorImageInfo-imageLayout-09425
If **imageLayout** is **VK_IMAGE_LAYOUT_COLOR_ATTACHMENT_OPTIMAL**, then the **aspectMask** used to create **imageView** **must** not include either **VK_IMAGE_ASPECT_DEPTH_BIT** or **VK_IMAGE_ASPECT_STENCIL_BIT**
- VUID-VkDescriptorImageInfo-imageLayout-09426
If **imageLayout** is **VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL**, **VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL**, **VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL**, **VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL**, **VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL**, **VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL**, **VK_IMAGE_LAYOUT_DEPTH_STENCIL_ATTACHMENT_OPTIMAL** or **VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL**, then the **aspectMask** used to create **imageView** **must** not include **VK_IMAGE_ASPECT_COLOR_BIT**
- VUID-VkDescriptorImageInfo-imageLayout-00344
imageLayout **must** match the actual **VkImageLayout** of each subresource accessible from **imageView** at the time this descriptor is accessed as defined by the **image layout matching rules**
- VUID-VkDescriptorImageInfo-sampler-01564
If **sampler** is used and the **VkFormat** of the image is a **multi-planar format**, the image **must** have been created with **VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT**, and the **aspectMask** of the **imageView** **must** be a valid **multi-planar aspect mask** bit
- VUID-VkDescriptorImageInfo-mutableComparisonSamplers-04450
If the **VK_KHR_portability_subset** extension is enabled, and **VkPhysicalDevicePortabilitySubsetFeaturesKHR::mutableComparisonSamplers** is **VK_FALSE**, then **sampler** **must** have been created with **VkSamplerCreateInfo::compareEnable** set to **VK_FALSE**

Valid Usage (Implicit)

- VUID-VkDescriptorImageInfo-commonparent
Both of **imageView**, and **sampler** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same **VkDevice**

The **VkWriteDescriptorSetAccelerationStructureKHR** structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkWriteDescriptorSetAccelerationStructureKHR {
    VkStructureType          sType;
    const void*              pNext;
    uint32_t                 accelerationStructureCount;
    const VkAccelerationStructureKHR* pAccelerationStructures;
} VkWriteDescriptorSetAccelerationStructureKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `accelerationStructureCount` is the number of elements in `pAccelerationStructures`.
- `pAccelerationStructures` is a pointer to an array of `VkAccelerationStructureKHR` structures specifying the acceleration structures to update.

Valid Usage

- VUID-VkWriteDescriptorSetAccelerationStructureKHR-accelerationStructureCount-02236
`accelerationStructureCount` **must** be equal to `descriptorCount` in the extended structure
- VUID-VkWriteDescriptorSetAccelerationStructureKHR-pAccelerationStructures-03579
Each acceleration structure in `pAccelerationStructures` **must** have been created with a type of `VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR` or `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR`
- VUID-VkWriteDescriptorSetAccelerationStructureKHR-pAccelerationStructures-03580
If the `nullDescriptor` feature is not enabled, each element of `pAccelerationStructures` **must** not be `VK_NULL_HANDLE`

Valid Usage (Implicit)

- VUID-VkWriteDescriptorSetAccelerationStructureKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET_ACCELERATION_STRUCTURE_KHR`
- VUID-VkWriteDescriptorSetAccelerationStructureKHR-pAccelerationStructures-parameter
`pAccelerationStructures` **must** be a valid pointer to an array of `accelerationStructureCount` valid or `VK_NULL_HANDLE` `VkAccelerationStructureKHR` handles
- VUID-VkWriteDescriptorSetAccelerationStructureKHR-accelerationStructureCount-arraylength
`accelerationStructureCount` **must** be greater than 0

The `VkCopyDescriptorSet` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkCopyDescriptorSet {
    VkStructureType    sType;
    const void*        pNext;
    VkDescriptorSet     srcSet;
    uint32_t            srcBinding;
    uint32_t            srcArrayElement;
    VkDescriptorSet     dstSet;
    uint32_t            dstBinding;
    uint32_t            dstArrayElement;
    uint32_t            descriptorCount;
} VkCopyDescriptorSet;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcSet`, `srcBinding`, and `srcArrayElement` are the source set, binding, and array element, respectively.
- `dstSet`, `dstBinding`, and `dstArrayElement` are the destination set, binding, and array element, respectively.
- `descriptorCount` is the number of descriptors to copy from the source to destination. If `descriptorCount` is greater than the number of remaining array elements in the source or destination binding, those affect consecutive bindings in a manner similar to `VkWriteDescriptorSet` above.

Valid Usage

- VUID-VkCopyDescriptorSet-srcBinding-00345
`srcBinding` **must** be a valid binding within `srcSet`
- VUID-VkCopyDescriptorSet-srcArrayElement-00346
The sum of `srcArrayElement` and `descriptorCount` **must** be less than or equal to the number of array elements in the descriptor set binding specified by `srcBinding`, and all applicable `consecutive bindings`
- VUID-VkCopyDescriptorSet-dstBinding-00347
`dstBinding` **must** be a valid binding within `dstSet`
- VUID-VkCopyDescriptorSet-dstArrayElement-00348
The sum of `dstArrayElement` and `descriptorCount` **must** be less than or equal to the number of array elements in the descriptor set binding specified by `dstBinding`, and all applicable `consecutive bindings`
- VUID-VkCopyDescriptorSet-dstBinding-02632
The type of `dstBinding` within `dstSet` **must** be equal to the type of `srcBinding` within `srcSet`
- VUID-VkCopyDescriptorSet-srcSet-00349
If `srcSet` is equal to `dstSet`, then the source and destination ranges of descriptors **must** not overlap, where the ranges **may** include array elements from `consecutive bindings`

- VUID-VkCopyDescriptorSet-dstBinding-02753

If the descriptor type of the descriptor set binding specified by `dstBinding` is `VK_DESCRIPTOR_TYPE_SAMPLER`, then `dstSet` **must** not have been allocated with a layout that included immutable samplers for `dstBinding`

Valid Usage (Implicit)

- VUID-VkCopyDescriptorSet-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COPY_DESCRIPTOR_SET`
- VUID-VkCopyDescriptorSet-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCopyDescriptorSet-srcSet-parameter
`srcSet` **must** be a valid `VkDescriptorSet` handle
- VUID-VkCopyDescriptorSet-dstSet-parameter
`dstSet` **must** be a valid `VkDescriptorSet` handle
- VUID-VkCopyDescriptorSet-commonparent
Both of `dstSet`, and `srcSet` **must** have been created, allocated, or retrieved from the same `VkDevice`

14.2.5. Descriptor Update Templates

A descriptor update template specifies a mapping from descriptor update information in host memory to descriptors in a descriptor set. It is designed to avoid passing redundant information to the driver when frequently updating the same set of descriptors in descriptor sets.

Descriptor update template objects are represented by `VkDescriptorUpdateTemplate` handles:

```
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDescriptorUpdateTemplate)
```

or the equivalent

```
// Provided by VK_KHR_descriptor_update_template
typedef VkDescriptorUpdateTemplate VkDescriptorUpdateTemplateKHR;
```

14.2.6. Descriptor Set Updates With Templates

Updating a large `VkDescriptorSet` array **can** be an expensive operation since an application **must** specify one `VkWriteDescriptorSet` structure for each descriptor or descriptor array to update, each of which re-specifies the same state when updating the same descriptor in multiple descriptor sets. For cases when an application wishes to update the same set of descriptors in multiple descriptor sets allocated using the same `VkDescriptorSetLayout`, `vkUpdateDescriptorSetWithTemplateKHR` **can** be used as a replacement for `vkUpdateDescriptorSets`.

`VkDescriptorUpdateTemplate` allows implementations to convert a set of descriptor update operations on a single descriptor set to an internal format that, in conjunction with `vkUpdateDescriptorSetWithTemplateKHR` or `vkCmdPushDescriptorSetWithTemplateKHR`, **can** be more efficient compared to calling `vkUpdateDescriptorSets` or `vkCmdPushDescriptorSetKHR`. The descriptors themselves are not specified in the `VkDescriptorUpdateTemplate`, rather, offsets into an application provided pointer to host memory are specified, which are combined with a pointer passed to `vkUpdateDescriptorSetWithTemplateKHR` or `vkCmdPushDescriptorSetWithTemplateKHR`. This allows large batches of updates to be executed without having to convert application data structures into a strictly-defined Vulkan data structure.

To create a descriptor update template, call:

```
// Provided by VK_KHR_descriptor_update_template
VkResult vkCreateDescriptorUpdateTemplateKHR(
    VkDevice device,
    const VkDescriptorUpdateTemplateCreateInfo* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkDescriptorUpdateTemplate* pDescriptorUpdateTemplate);
```

- `device` is the logical device that creates the descriptor update template.
- `pCreateInfo` is a pointer to a `VkDescriptorUpdateTemplateCreateInfo` structure specifying the set of descriptors to update with a single call to `vkCmdPushDescriptorSetWithTemplateKHR` or `vkUpdateDescriptorSetWithTemplateKHR`.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pDescriptorUpdateTemplate` is a pointer to a `VkDescriptorUpdateTemplate` handle in which the resulting descriptor update template object is returned.

Valid Usage (Implicit)

- VUID-vkCreateDescriptorUpdateTemplate-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkCreateDescriptorUpdateTemplate-pCreateInfo-parameter `pCreateInfo` **must** be a valid pointer to a valid `VkDescriptorUpdateTemplateCreateInfo` structure
- VUID-vkCreateDescriptorUpdateTemplate-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateDescriptorUpdateTemplate-pDescriptorUpdateTemplate-parameter `pDescriptorUpdateTemplate` **must** be a valid pointer to a `VkDescriptorUpdateTemplate` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkDescriptorUpdateTemplateCreateInfo` structure is defined as:

```
typedef struct VkDescriptorUpdateTemplateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkDescriptorUpdateTemplateCreateFlags flags;
    uint32_t                  descriptorUpdateEntryCount;
    const VkDescriptorUpdateTemplateEntry* pDescriptorUpdateEntries;
    VkDescriptorUpdateTemplateType templateType;
    VkDescriptorSetLayout      descriptorSetLayout;
    VkPipelineBindPoint        pipelineBindPoint;
    VkPipelineLayout           pipelineLayout;
    uint32_t                   set;
} VkDescriptorUpdateTemplateCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_descriptor_update_template
typedef VkDescriptorUpdateTemplateCreateInfo VkDescriptorUpdateTemplateCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `descriptorUpdateEntryCount` is the number of elements in the `pDescriptorUpdateEntries` array.
- `pDescriptorUpdateEntries` is a pointer to an array of `VkDescriptorUpdateTemplateEntry` structures describing the descriptors to be updated by the descriptor update template.
- `templateType` Specifies the type of the descriptor update template. If set to `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET` it **can** only be used to update descriptor sets with a fixed `descriptorSetLayout`. If set to `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_SET_KHR` it **can** only be used to push descriptor sets using the provided `pipelineBindPoint`, `pipelineLayout`, and `set` number.
- `descriptorSetLayout` is the descriptor set layout used to build the descriptor update template. All descriptor sets which are going to be updated through the newly created descriptor update template **must** be created with a layout that matches (is the same as, or defined identically to)

this layout. This parameter is ignored if `templateType` is not `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET`.

- `pipelineBindPoint` is a `VkPipelineBindPoint` indicating the type of the pipeline that will use the descriptors. This parameter is ignored if `templateType` is not `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_KHR`
- `pipelineLayout` is a `VkPipelineLayout` object used to program the bindings. This parameter is ignored if `templateType` is not `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_KHR`
- `set` is the set number of the descriptor set in the pipeline layout that will be updated. This parameter is ignored if `templateType` is not `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_KHR`

Valid Usage

- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-00350
If `templateType` is `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET`, `descriptorSetLayout` **must** be a valid `VkDescriptorSetLayout` handle
- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-00351
If `templateType` is `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_KHR`, `pipelineBindPoint` **must** be a valid `VkPipelineBindPoint` value
- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-00352
If `templateType` is `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_KHR`, `pipelineLayout` **must** be a valid `VkPipelineLayout` handle
- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-00353
If `templateType` is `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_KHR`, `set` **must** be the unique set number in the pipeline layout that uses a descriptor set layout that was created with `VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR`

Valid Usage (Implicit)

- VUID-VkDescriptorUpdateTemplateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO`
- VUID-VkDescriptorUpdateTemplateCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDescriptorUpdateTemplateCreateInfo-flags-zerobitmask
`flags` **must** be `0`
- VUID-VkDescriptorUpdateTemplateCreateInfo-pDescriptorUpdateEntries-parameter
`pDescriptorUpdateEntries` **must** be a valid pointer to an array of `descriptorUpdateEntryCount` valid `VkDescriptorUpdateTemplateEntry` structures
- VUID-VkDescriptorUpdateTemplateCreateInfo-templateType-parameter
`templateType` **must** be a valid `VkDescriptorUpdateTemplateType` value
- VUID-VkDescriptorUpdateTemplateCreateInfo-descriptorUpdateEntryCount-arraylength
`descriptorUpdateEntryCount` **must** be greater than `0`

- VUID-VkDescriptorUpdateTemplateCreateInfo-commonparent
Both of `descriptorSetLayout`, and `pipelineLayout` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

```
typedef VkFlags VkDescriptorUpdateTemplateCreateFlags;
```

or the equivalent

```
// Provided by VK_KHR_descriptor_update_template
typedef VkDescriptorUpdateTemplateCreateFlags
VkDescriptorUpdateTemplateCreateFlagsKHR;
```

`VkDescriptorUpdateTemplateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The descriptor update template type is determined by the `VkDescriptorUpdateTemplateCreateInfo::templateType` property, which takes the following values:

```
typedef enum VkDescriptorUpdateTemplateType {
    VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET = 0,
    // Provided by VK_KHR_descriptor_update_template with VK_KHR_push_descriptor,
    // VK_KHR_push_descriptor with VK_VERSION_1_1 or VK_KHR_descriptor_update_template
    VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_SET_KHR = 1,
    // Provided by VK_KHR_descriptor_update_template
    VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET_KHR =
    VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET,
} VkDescriptorUpdateTemplateType;
```

or the equivalent

```
// Provided by VK_KHR_descriptor_update_template
typedef VkDescriptorUpdateTemplateType VkDescriptorUpdateTemplateTypeKHR;
```

- `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET` specifies that the descriptor update template will be used for descriptor set updates only.
- `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_SET_KHR` specifies that the descriptor update template will be used for push descriptor updates only.

The `VkDescriptorUpdateTemplateEntry` structure is defined as:

```
typedef struct VkDescriptorUpdateTemplateEntry {
    uint32_t      dstBinding;
    uint32_t      dstArrayElement;
    uint32_t      descriptorCount;
    VkDescriptorType descriptorType;
    size_t        offset;
    size_t        stride;
} VkDescriptorUpdateTemplateEntry;
```

or the equivalent

```
// Provided by VK_KHR_descriptor_update_template
typedef VkDescriptorUpdateTemplateEntry VkDescriptorUpdateTemplateEntryKHR;
```

- **dstBinding** is the descriptor binding to update when using this descriptor update template.
- **dstArrayElement** is the starting element in the array belonging to **dstBinding**.
- **descriptorCount** is the number of descriptors to update. If **descriptorCount** is greater than the number of remaining array elements in the destination binding, those affect consecutive bindings in a manner similar to [VkWriteDescriptorSet](#) above.
- **descriptorType** is a [VkDescriptorType](#) specifying the type of the descriptor.
- **offset** is the offset in bytes of the first binding in the raw data structure.
- **stride** is the stride in bytes between two consecutive array elements of the descriptor update information in the raw data structure. The actual pointer ptr for each array element j of update entry i is computed using the following formula:

```
const char *ptr = (const char *)pData + pDescriptorUpdateEntries[i].offset + j
* pDescriptorUpdateEntries[i].stride
```

The stride is useful in case the bindings are stored in structs along with other data.

Valid Usage

- VUID-VkDescriptorUpdateTemplateEntry-dstBinding-00354
dstBinding **must** be a valid binding in the descriptor set layout implicitly specified when using a descriptor update template to update descriptors
- VUID-VkDescriptorUpdateTemplateEntry-dstArrayElement-00355
dstArrayElement and **descriptorCount** **must** be less than or equal to the number of array elements in the descriptor set binding implicitly specified when using a descriptor update template to update descriptors, and all applicable [consecutive bindings](#)

Valid Usage (Implicit)

- VUID-VkDescriptorUpdateTemplateEntry-descriptorType-parameter
`descriptorType` **must** be a valid `VkDescriptorType` value

To destroy a descriptor update template, call:

```
// Provided by VK_KHR_descriptor_update_template
void vkDestroyDescriptorUpdateTemplateKHR(
    VkDevice                device,
    VkDescriptorUpdateTemplate descriptorUpdateTemplate,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that has been used to create the descriptor update template
- `descriptorUpdateTemplate` is the descriptor update template to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyDescriptorUpdateTemplate-descriptorSetLayout-00356
If `VkAllocationCallbacks` were provided when `descriptorUpdateTemplate` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDescriptorUpdateTemplate-descriptorSetLayout-00357
If no `VkAllocationCallbacks` were provided when `descriptorUpdateTemplate` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyDescriptorUpdateTemplate-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyDescriptorUpdateTemplate-descriptorUpdateTemplate-parameter
If `descriptorUpdateTemplate` is not `VK_NULL_HANDLE`, `descriptorUpdateTemplate` **must** be a valid `VkDescriptorUpdateTemplate` handle
- VUID-vkDestroyDescriptorUpdateTemplate-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyDescriptorUpdateTemplate-descriptorUpdateTemplate-parent
If `descriptorUpdateTemplate` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `descriptorUpdateTemplate` **must** be externally synchronized

Once a `VkDescriptorUpdateTemplate` has been created, descriptor sets **can** be updated by calling:

```
// Provided by VK_KHR_descriptor_update_template
void vkUpdateDescriptorSetWithTemplateKHR(
    VkDevice                device,
    VkDescriptorSet          descriptorSet,
    VkDescriptorUpdateTemplate descriptorUpdateTemplate,
    const void*              pData);
```

- `device` is the logical device that updates the descriptor set.
- `descriptorSet` is the descriptor set to update
- `descriptorUpdateTemplate` is a `VkDescriptorUpdateTemplate` object specifying the update mapping between `pData` and the descriptor set to update.
- `pData` is a pointer to memory containing one or more `VkDescriptorImageInfo`, `VkDescriptorBufferInfo`, or `VkBufferView` structures or `VkAccelerationStructureKHR` handles used to write the descriptors.

Valid Usage

- VUID-vkUpdateDescriptorSetWithTemplate-pData-01685
`pData` **must** be a valid pointer to a memory containing one or more valid instances of `VkDescriptorImageInfo`, `VkDescriptorBufferInfo`, or `VkBufferView` in a layout defined by `descriptorUpdateTemplate` when it was created with `vkCreateDescriptorUpdateTemplateKHR`
- VUID-vkUpdateDescriptorSetWithTemplate-descriptorSet-06995
Host access to `descriptorSet` **must** be externally synchronized

Valid Usage (Implicit)

- VUID-vkUpdateDescriptorSetWithTemplate-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkUpdateDescriptorSetWithTemplate-descriptorSet-parameter
`descriptorSet` **must** be a valid `VkDescriptorSet` handle
- VUID-vkUpdateDescriptorSetWithTemplate-descriptorUpdateTemplate-parameter
`descriptorUpdateTemplate` **must** be a valid `VkDescriptorUpdateTemplate` handle
- VUID-vkUpdateDescriptorSetWithTemplate-descriptorSet-parent
`descriptorSet` **must** have been created, allocated, or retrieved from `device`
- VUID-vkUpdateDescriptorSetWithTemplate-descriptorUpdateTemplate-parent

`descriptorUpdateTemplate` **must** have been created, allocated, or retrieved from `device`

API Example

```
struct AppBufferView {
    VkBufferView bufferView;
    uint32_t      applicationRelatedInformation;
};

struct AppDataStructure
{
    VkDescriptorImageInfo  imageInfo;           // a single image info
    VkDescriptorBufferInfo bufferInfoArray[3]; // 3 buffer infos in an array
    AppBufferView          bufferView[2];      // An application-defined structure
    containing a bufferView
    // ... some more application-related data
};

const VkDescriptorUpdateTemplateEntry descriptorUpdateTemplateEntries[] =
{
    // binding to a single image descriptor
    {
        .binding = 0,
        .dstArrayElement = 0,
        .descriptorCount = 1,
        .descriptorType = VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER,
        .offset = offsetof(AppDataStructure, imageInfo),
        .stride = 0           // stride not required if descriptorCount is 1
    },

    // binding to an array of buffer descriptors
    {
        .binding = 1,
        .dstArrayElement = 0,
        .descriptorCount = 3,
        .descriptorType = VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER,
        .offset = offsetof(AppDataStructure, bufferInfoArray),
        .stride = sizeof(VkDescriptorBufferInfo) // descriptor buffer infos are
compact
    },

    // binding to an array of buffer views
    {
        .binding = 2,
        .dstArrayElement = 0,
        .descriptorCount = 2,
        .descriptorType = VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER,
        .offset = offsetof(AppDataStructure, bufferView) +
            offsetof(AppBufferView, bufferView),
        .stride = sizeof(AppBufferView) // bufferViews do not have to be
```

```

compact
    },
};

// create a descriptor update template for descriptor set updates
const VkDescriptorUpdateTemplateCreateInfo createInfo =
{
    .sType = VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO,
    .pNext = NULL,
    .flags = 0,
    .descriptorUpdateEntryCount = 3,
    .pDescriptorUpdateEntries = descriptorUpdateTemplateEntries,
    .templateType = VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET,
    .descriptorSetLayout = myLayout,
    .pipelineBindPoint = 0,      // ignored by given templateType
    .pipelineLayout = 0,        // ignored by given templateType
    .set = 0,                   // ignored by given templateType
};

VkDescriptorUpdateTemplate myDescriptorUpdateTemplate;
myResult = vkCreateDescriptorUpdateTemplate(
    myDevice,
    &createInfo,
    NULL,
    &myDescriptorUpdateTemplate);

AppDataStructure appData;

// fill appData here or cache it in your engine
vkUpdateDescriptorSetWithTemplate(myDevice, myDescriptorSet,
myDescriptorUpdateTemplate, &appData);

```

14.2.7. Descriptor Set Binding

To bind one or more descriptor sets to a command buffer, call:

```

// Provided by VK_VERSION_1_0
void vkCmdBindDescriptorSets(
    VkCommandBuffer          commandBuffer,
    VkPipelineBindPoint      pipelineBindPoint,
    VkPipelineLayout         layout,
    uint32_t                 firstSet,
    uint32_t                 descriptorSetCount,
    const VkDescriptorSet*   pDescriptorSets,
    uint32_t                 dynamicOffsetCount,
    const uint32_t*          pDynamicOffsets);

```

- `commandBuffer` is the command buffer that the descriptor sets will be bound to.

- `pipelineBindPoint` is a `VkPipelineBindPoint` indicating the type of the pipeline that will use the descriptors. There is a separate set of bind points for each pipeline type, so binding one does not disturb the others.
- `layout` is a `VkPipelineLayout` object used to program the bindings.
- `firstSet` is the set number of the first descriptor set to be bound.
- `descriptorSetCount` is the number of elements in the `pDescriptorSets` array.
- `pDescriptorSets` is a pointer to an array of handles to `VkDescriptorSet` objects describing the descriptor sets to bind to.
- `dynamicOffsetCount` is the number of dynamic offsets in the `pDynamicOffsets` array.
- `pDynamicOffsets` is a pointer to an array of `uint32_t` values specifying dynamic offsets.

`vkCmdBindDescriptorSets` binds descriptor sets `pDescriptorSets[0..descriptorSetCount-1]` to set numbers `[firstSet..firstSet+descriptorSetCount-1]` for subsequent `bound pipeline commands` set by `pipelineBindPoint`. Any bindings that were previously applied via these sets are no longer valid.

Once bound, a descriptor set affects rendering of subsequent commands that interact with the given pipeline type in the command buffer until either a different set is bound to the same set number, or the set is disturbed as described in [Pipeline Layout Compatibility](#).

A compatible descriptor set **must** be bound for all set numbers that any shaders in a pipeline access, at the time that a drawing or dispatching command is recorded to execute using that pipeline. However, if none of the shaders in a pipeline statically use any bindings with a particular set number, then no descriptor set need be bound for that set number, even if the pipeline layout includes a non-trivial descriptor set layout for that set number.

When consuming a descriptor, a descriptor is considered valid if the descriptor is not undefined as described by [descriptor set allocation](#). If the `nullDescriptor` feature is enabled, a null descriptor is also considered valid. A descriptor that was disturbed by [Pipeline Layout Compatibility](#), or was never bound by `vkCmdBindDescriptorSets` is not considered valid. If a pipeline accesses a descriptor either statically or dynamically depending on the `VkDescriptorBindingFlagBits`, the consuming descriptor type in the pipeline **must** match the `VkDescriptorType` in `VkDescriptorSetLayoutCreateInfo` for the descriptor to be considered valid.

NOTE

Further validation may be carried out beyond validation for descriptor types, e.g. [Texel Input Validation](#).

If any of the sets being bound include dynamic uniform or storage buffers, then `pDynamicOffsets` includes one element for each array element in each dynamic descriptor type binding in each set. Values are taken from `pDynamicOffsets` in an order such that all entries for set N come before set N+1; within a set, entries are ordered by the binding numbers in the descriptor set layouts; and within a binding array, elements are in order. `dynamicOffsetCount` **must** equal the total number of dynamic descriptors in the sets being bound.

The effective offset used for dynamic uniform and storage buffer bindings is the sum of the relative offset taken from `pDynamicOffsets`, and the base address of the buffer plus base offset in the descriptor set. The range of the dynamic uniform and storage buffer bindings is the buffer range as

specified in the descriptor set.

Each of the `pDescriptorSets` **must** be compatible with the pipeline layout specified by `layout`. The layout used to program the bindings **must** also be compatible with the pipeline used in subsequent `bound pipeline commands` with that pipeline type, as defined in the [Pipeline Layout Compatibility](#) section.

The descriptor set contents bound by a call to `vkCmdBindDescriptorSets` **may** be consumed at the following times:

- during host execution of the command, or during shader execution of the resulting draws and dispatches, or any time in between.

Thus, the contents of a descriptor set binding **must** not be altered (overwritten by an update command, or freed) between the first point in time that it **may** be consumed, and when the command completes executing on the queue.

The contents of `pDynamicOffsets` are consumed immediately during execution of `vkCmdBindDescriptorSets`. Once all pending uses have completed, it is legal to update and reuse a descriptor set.

Valid Usage

- VUID-vkCmdBindDescriptorSets-pDescriptorSets-00358
Each element of `pDescriptorSets` **must** have been allocated with a `VkDescriptorSetLayout` that matches (is the same as, or identically defined as) the `VkDescriptorSetLayout` at set n in `layout`, where n is the sum of `firstSet` and the index into `pDescriptorSets`
- VUID-vkCmdBindDescriptorSets-dynamicOffsetCount-00359
`dynamicOffsetCount` **must** be equal to the total number of dynamic descriptors in `pDescriptorSets`
- VUID-vkCmdBindDescriptorSets-firstSet-00360
The sum of `firstSet` and `descriptorSetCount` **must** be less than or equal to `VkPipelineLayoutCreateInfo::setLayoutCount` provided when `layout` was created
- VUID-vkCmdBindDescriptorSets-pDynamicOffsets-01971
Each element of `pDynamicOffsets` which corresponds to a descriptor binding with type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` **must** be a multiple of `VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment`
- VUID-vkCmdBindDescriptorSets-pDynamicOffsets-01972
Each element of `pDynamicOffsets` which corresponds to a descriptor binding with type `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` **must** be a multiple of `VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment`
- VUID-vkCmdBindDescriptorSets-pDescriptorSets-01979
For each dynamic uniform or storage buffer binding in `pDescriptorSets`, the sum of the `effective offset` and the range of the binding **must** be less than or equal to the size of the buffer
- VUID-vkCmdBindDescriptorSets-pDescriptorSets-06715

For each dynamic uniform or storage buffer binding in `pDescriptorSets`, if the range was set with `VK_WHOLE_SIZE` then `pDynamicOffsets` which corresponds to the descriptor binding **must** be 0

- VUID-vkCmdBindDescriptorSets-pDescriptorSets-06563
Each element of `pDescriptorSets` **must** be a valid `VkDescriptorSet`
- VUID-vkCmdBindDescriptorSets-pipelineBindPoint-00361
`pipelineBindPoint` **must** be supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family

Valid Usage (Implicit)

- VUID-vkCmdBindDescriptorSets-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBindDescriptorSets-pipelineBindPoint-parameter
`pipelineBindPoint` **must** be a valid `VkPipelineBindPoint` value
- VUID-vkCmdBindDescriptorSets-layout-parameter
`layout` **must** be a valid `VkPipelineLayout` handle
- VUID-vkCmdBindDescriptorSets-pDescriptorSets-parameter
`pDescriptorSets` **must** be a valid pointer to an array of `descriptorSetCount` valid or `VK_NULL_HANDLE` `VkDescriptorSet` handles
- VUID-vkCmdBindDescriptorSets-pDynamicOffsets-parameter
If `dynamicOffsetCount` is not 0, `pDynamicOffsets` **must** be a valid pointer to an array of `dynamicOffsetCount` `uint32_t` values
- VUID-vkCmdBindDescriptorSets-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdBindDescriptorSets-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdBindDescriptorSets-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindDescriptorSets-descriptorSetCount-arraylength
`descriptorSetCount` **must** be greater than 0
- VUID-vkCmdBindDescriptorSets-commonparent
Each of `commandBuffer`, `layout`, and the elements of `pDescriptorSets` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	State

Alternatively, to bind one or more descriptor sets to a command buffer, call:

```
// Provided by VK_KHR_maintenance6
void vkCmdBindDescriptorSets2KHR(
    VkCommandBuffer          commandBuffer,
    const VkBindDescriptorSetsInfoKHR* pBindDescriptorSetsInfo);
```

- `commandBuffer` is the command buffer that the descriptor sets will be bound to.
- `pBindDescriptorSetsInfo` is a pointer to a `VkBindDescriptorSetsInfoKHR` structure.

Valid Usage

- VUID-vkCmdBindDescriptorSets2KHR-pBindDescriptorSetsInfo-09467
Each bit in `pBindDescriptorSetsInfo->stageFlags` **must** be a stage supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family

Valid Usage (Implicit)

- VUID-vkCmdBindDescriptorSets2KHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBindDescriptorSets2KHR-pBindDescriptorSetsInfo-parameter
`pBindDescriptorSetsInfo` **must** be a valid pointer to a valid `VkBindDescriptorSetsInfoKHR` structure
- VUID-vkCmdBindDescriptorSets2KHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdBindDescriptorSets2KHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdBindDescriptorSets2KHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	State

The `VkBindDescriptorSetsInfoKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance6
typedef struct VkBindDescriptorSetsInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    VkShaderStageFlags   stageFlags;
    VkPipelineLayout      layout;
    uint32_t             firstSet;
    uint32_t             descriptorSetCount;
    const VkDescriptorSet* pDescriptorSets;
    uint32_t             dynamicOffsetCount;
    const uint32_t*       pDynamicOffsets;
} VkBindDescriptorSetsInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stageFlags` is a bitmask of `VkShaderStageFlagBits` specifying the shader stages the descriptor sets will be bound to.
- `layout` is a `VkPipelineLayout` object used to program the bindings.
- `firstSet` is the set number of the first descriptor set to be bound.
- `descriptorSetCount` is the number of elements in the `pDescriptorSets` array.
- `pDescriptorSets` is a pointer to an array of handles to `VkDescriptorSet` objects describing the descriptor sets to bind to.
- `dynamicOffsetCount` is the number of dynamic offsets in the `pDynamicOffsets` array.
- `pDynamicOffsets` is a pointer to an array of `uint32_t` values specifying dynamic offsets.

If `stageFlags` specifies a subset of all stages corresponding to one or more pipeline bind points, the

binding operation still affects all stages corresponding to the given pipeline bind point(s) as if the equivalent original version of this command had been called with the same parameters. For example, specifying a `stageFlags` value of `VK_SHADER_STAGE_VERTEX_BIT | VK_SHADER_STAGE_FRAGMENT_BIT | VK_SHADER_STAGE_COMPUTE_BIT` is equivalent to calling the original version of this command once with `VK_PIPELINE_BIND_POINT_GRAPHICS` and once with `VK_PIPELINE_BIND_POINT_COMPUTE`.

Valid Usage

- VUID-VkBindDescriptorSetsInfoKHR-pDescriptorSets-00358
Each element of `pDescriptorSets` **must** have been allocated with a `VkDescriptorSetLayout` that matches (is the same as, or identically defined as) the `VkDescriptorSetLayout` at set n in `layout`, where n is the sum of `firstSet` and the index into `pDescriptorSets`
- VUID-VkBindDescriptorSetsInfoKHR-dynamicOffsetCount-00359
`dynamicOffsetCount` **must** be equal to the total number of dynamic descriptors in `pDescriptorSets`
- VUID-VkBindDescriptorSetsInfoKHR-firstSet-00360
The sum of `firstSet` and `descriptorSetCount` **must** be less than or equal to `VkPipelineLayoutCreateInfo::setLayoutCount` provided when `layout` was created
- VUID-VkBindDescriptorSetsInfoKHR-pDynamicOffsets-01971
Each element of `pDynamicOffsets` which corresponds to a descriptor binding with type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` **must** be a multiple of `VkPhysicalDeviceLimits::minUniformBufferOffsetAlignment`
- VUID-VkBindDescriptorSetsInfoKHR-pDynamicOffsets-01972
Each element of `pDynamicOffsets` which corresponds to a descriptor binding with type `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` **must** be a multiple of `VkPhysicalDeviceLimits::minStorageBufferOffsetAlignment`
- VUID-VkBindDescriptorSetsInfoKHR-pDescriptorSets-01979
For each dynamic uniform or storage buffer binding in `pDescriptorSets`, the sum of the `effective offset` and the range of the binding **must** be less than or equal to the size of the buffer
- VUID-VkBindDescriptorSetsInfoKHR-pDescriptorSets-06715
For each dynamic uniform or storage buffer binding in `pDescriptorSets`, if the range was set with `VK_WHOLE_SIZE` then `pDynamicOffsets` which corresponds to the descriptor binding **must** be 0
- VUID-VkBindDescriptorSetsInfoKHR-pDescriptorSets-06563
Each element of `pDescriptorSets` **must** be a valid `VkDescriptorSet`
- VUID-VkBindDescriptorSetsInfoKHR-None-09495
`layout` **must** be a valid `VkPipelineLayout` handle

Valid Usage (Implicit)

- VUID-VkBindDescriptorSetsInfoKHR-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_SETS_INFO_KHR`

- `VUID-VkBindDescriptorSetsInfoKHR-pNext-pNext`
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkPipelineLayoutCreateInfo`
- `VUID-VkBindDescriptorSetsInfoKHR-sType-unique`
The `sType` value of each struct in the `pNext` chain **must** be unique
- `VUID-VkBindDescriptorSetsInfoKHR-stageFlags-parameter`
`stageFlags` **must** be a valid combination of `VkShaderStageFlagBits` values
- `VUID-VkBindDescriptorSetsInfoKHR-stageFlags-requiredbitmask`
`stageFlags` **must** not be `0`
- `VUID-VkBindDescriptorSetsInfoKHR-layout-parameter`
If `layout` is not `VK_NULL_HANDLE`, `layout` **must** be a valid `VkPipelineLayout` handle
- `VUID-VkBindDescriptorSetsInfoKHR-pDescriptorSets-parameter`
`pDescriptorSets` **must** be a valid pointer to an array of `descriptorSetCount` valid `VkDescriptorSet` handles
- `VUID-VkBindDescriptorSetsInfoKHR-pDynamicOffsets-parameter`
If `dynamicOffsetCount` is not `0`, and `pDynamicOffsets` is not `NULL`, `pDynamicOffsets` **must** be a valid pointer to an array of `dynamicOffsetCount` or `VK_NULL_HANDLE` `uint32_t` values
- `VUID-VkBindDescriptorSetsInfoKHR-descriptorSetCount-arraylength`
`descriptorSetCount` **must** be greater than `0`
- `VUID-VkBindDescriptorSetsInfoKHR-commonparent`
Both of `layout`, and the elements of `pDescriptorSets` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

14.2.8. Push Descriptor Updates

In addition to allocating descriptor sets and binding them to a command buffer, an application **can** record descriptor updates into the command buffer.

To push descriptor updates into a command buffer, call:

```
// Provided by VK_KHR_push_descriptor
void vkCmdPushDescriptorSetKHR(
    VkCommandBuffer          commandBuffer,
    VkPipelineBindPoint      pipelineBindPoint,
    VkPipelineLayout         layout,
    uint32_t                 set,
    uint32_t                 descriptorWriteCount,
    const VkWriteDescriptorSet* pDescriptorWrites);
```

- `commandBuffer` is the command buffer that the descriptors will be recorded in.
- `pipelineBindPoint` is a `VkPipelineBindPoint` indicating the type of the pipeline that will use the descriptors. There is a separate set of push descriptor bindings for each pipeline type, so binding one does not disturb the others.

- `layout` is a `VkPipelineLayout` object used to program the bindings.
- `set` is the set number of the descriptor set in the pipeline layout that will be updated.
- `descriptorWriteCount` is the number of elements in the `pDescriptorWrites` array.
- `pDescriptorWrites` is a pointer to an array of `VkWriteDescriptorSet` structures describing the descriptors to be updated.

Push descriptors are a small bank of descriptors whose storage is internally managed by the command buffer rather than being written into a descriptor set and later bound to a command buffer. Push descriptors allow for incremental updates of descriptors without managing the lifetime of descriptor sets.

When a command buffer begins recording, all push descriptors are undefined. Push descriptors **can** be updated incrementally and cause shaders to use the updated descriptors for subsequent `bound pipeline commands` with the pipeline type set by `pipelineBindPoint` until the descriptor is overwritten, or else until the set is disturbed as described in [Pipeline Layout Compatibility](#). When the set is disturbed or push descriptors with a different descriptor set layout are set, all push descriptors are undefined.

Push descriptors that are `statically used` by a pipeline **must** not be undefined at the time that a drawing or dispatching command is recorded to execute using that pipeline. This includes immutable sampler descriptors, which **must** be pushed before they are accessed by a pipeline (the immutable samplers are pushed, rather than the samplers in `pDescriptorWrites`). Push descriptors that are not statically used **can** remain undefined.

Push descriptors do not use dynamic offsets. Instead, the corresponding non-dynamic descriptor types **can** be used and the `offset` member of `VkDescriptorBufferInfo` **can** be changed each time the descriptor is written.

Each element of `pDescriptorWrites` is interpreted as in `VkWriteDescriptorSet`, except the `dstSet` member is ignored.

To push an immutable sampler, use a `VkWriteDescriptorSet` with `dstBinding` and `dstArrayElement` selecting the immutable sampler's binding. If the descriptor type is `VK_DESCRIPTOR_TYPE_SAMPLER`, the `pImageInfo` parameter is ignored and the immutable sampler is taken from the push descriptor set layout in the pipeline layout. If the descriptor type is `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, the `sampler` member of the `pImageInfo` parameter is ignored and the immutable sampler is taken from the push descriptor set layout in the pipeline layout.

Valid Usage

- VUID-vkCmdPushDescriptorSetKHR-set-00364
`set` **must** be less than `VkPipelineLayoutCreateInfo::setLayoutCount` provided when `layout` was created
- VUID-vkCmdPushDescriptorSetKHR-set-00365
`set` **must** be the unique set number in the pipeline layout that uses a descriptor set layout that was created with `VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR`
- VUID-vkCmdPushDescriptorSetKHR-pDescriptorWrites-06494

For each element `i` where `pDescriptorWrites[i].descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLER`, `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, `pDescriptorWrites[i].pImageInfo` **must** be a valid pointer to an array of `pDescriptorWrites[i].descriptorCount` valid `VkDescriptorImageInfo` structures

- VUID-vkCmdPushDescriptorSetKHR-pipelineBindPoint-00363
`pipelineBindPoint` **must** be supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family

Valid Usage (Implicit)

- VUID-vkCmdPushDescriptorSetKHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdPushDescriptorSetKHR-pipelineBindPoint-parameter
`pipelineBindPoint` **must** be a valid `VkPipelineBindPoint` value
- VUID-vkCmdPushDescriptorSetKHR-layout-parameter
`layout` **must** be a valid `VkPipelineLayout` handle
- VUID-vkCmdPushDescriptorSetKHR-pDescriptorWrites-parameter
`pDescriptorWrites` **must** be a valid pointer to an array of `descriptorWriteCount` valid `VkWriteDescriptorSet` structures
- VUID-vkCmdPushDescriptorSetKHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdPushDescriptorSetKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdPushDescriptorSetKHR-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdPushDescriptorSetKHR-descriptorWriteCount-arraylength
`descriptorWriteCount` **must** be greater than 0
- VUID-vkCmdPushDescriptorSetKHR-commonparent
Both of `commandBuffer`, and `layout` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	State

Alternatively, to push descriptor updates into a command buffer, call:

```
// Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
void vkCmdPushDescriptorSet2KHR(
    VkCommandBuffer                      commandBuffer,
    const VkPushDescriptorSetInfoKHR*    pPushDescriptorSetInfo);
```

- `commandBuffer` is the command buffer that the descriptors will be recorded in.
- `pPushDescriptorSetInfo` is a pointer to a `VkPushDescriptorSetInfoKHR` structure.

Valid Usage

- VUID-vkCmdPushDescriptorSet2KHR-pPushDescriptorSetInfo-09468
Each bit in `pPushDescriptorSetInfo->stageFlags` **must** be a stage supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family

Valid Usage (Implicit)

- VUID-vkCmdPushDescriptorSet2KHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdPushDescriptorSet2KHR-pPushDescriptorSetInfo-parameter
`pPushDescriptorSetInfo` **must** be a valid pointer to a valid `VkPushDescriptorSetInfoKHR` structure
- VUID-vkCmdPushDescriptorSet2KHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdPushDescriptorSet2KHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdPushDescriptorSet2KHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	State

The `VkPushDescriptorSetInfoKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
typedef struct VkPushDescriptorSetInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkShaderStageFlags       stageFlags;
    VkPipelineLayout          layout;
    uint32_t                  set;
    uint32_t                  descriptorWriteCount;
    const VkWriteDescriptorSet* pDescriptorWrites;
} VkPushDescriptorSetInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stageFlags` is a bitmask of `VkShaderStageFlagBits` specifying the shader stages that will use the descriptors.
- `layout` is a `VkPipelineLayout` object used to program the bindings.
- `set` is the set number of the descriptor set in the pipeline layout that will be updated.
- `descriptorWriteCount` is the number of elements in the `pDescriptorWrites` array.
- `pDescriptorWrites` is a pointer to an array of `VkWriteDescriptorSet` structures describing the descriptors to be updated.

If `stageFlags` specifies a subset of all stages corresponding to one or more pipeline bind points, the binding operation still affects all stages corresponding to the given pipeline bind point(s) as if the equivalent original version of this command had been called with the same parameters. For example, specifying a `stageFlags` value of `VK_SHADER_STAGE_VERTEX_BIT | VK_SHADER_STAGE_FRAGMENT_BIT | VK_SHADER_STAGE_COMPUTE_BIT` is equivalent to calling the original version of this command once with `VK_PIPELINE_BIND_POINT_GRAPHICS` and once with `VK_PIPELINE_BIND_POINT_COMPUTE`.

Valid Usage

- VUID-VkPushDescriptorSetInfoKHR-set-00364
`set` **must** be less than `VkPipelineLayoutCreateInfo::setLayoutCount` provided when `layout` was created
- VUID-VkPushDescriptorSetInfoKHR-set-00365
`set` **must** be the unique set number in the pipeline layout that uses a descriptor set layout that was created with `VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR`
- VUID-VkPushDescriptorSetInfoKHR-pDescriptorWrites-06494
For each element `i` where `pDescriptorWrites[i].descriptorType` is `VK_DESCRIPTOR_TYPE_SAMPLER`, `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT`, `pDescriptorWrites[i].pImageInfo` **must** be a valid pointer to an array of `pDescriptorWrites[i].descriptorCount` valid `VkDescriptorImageInfo` structures
- VUID-VkPushDescriptorSetInfoKHR-None-09495
`layout` **must** be a valid `VkPipelineLayout` handle

Valid Usage (Implicit)

- VUID-VkPushDescriptorSetInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_INFO_KHR`
- VUID-VkPushDescriptorSetInfoKHR-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkPipelineLayoutCreateInfo`
- VUID-VkPushDescriptorSetInfoKHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkPushDescriptorSetInfoKHR-stageFlags-parameter
`stageFlags` **must** be a valid combination of `VkShaderStageFlagBits` values
- VUID-VkPushDescriptorSetInfoKHR-stageFlags-requiredbitmask
`stageFlags` **must** not be `0`
- VUID-VkPushDescriptorSetInfoKHR-layout-parameter
If `layout` is not `VK_NULL_HANDLE`, `layout` **must** be a valid `VkPipelineLayout` handle
- VUID-VkPushDescriptorSetInfoKHR-pDescriptorWrites-parameter
`pDescriptorWrites` **must** be a valid pointer to an array of `descriptorWriteCount` valid `VkWriteDescriptorSet` structures
- VUID-VkPushDescriptorSetInfoKHR-descriptorWriteCount-arraylength
`descriptorWriteCount` **must** be greater than `0`

14.2.9. Push Descriptor Updates With Descriptor Update Templates

It is also possible to use a descriptor update template to specify the push descriptors to update. To

do so, call:

```
// Provided by VK_KHR_descriptor_update_template with VK_KHR_push_descriptor,
VK_KHR_push_descriptor with VK_VERSION_1_1 or VK_KHR_descriptor_update_template
void vkCmdPushDescriptorSetWithTemplateKHR(
    VkCommandBuffer          commandBuffer,
    VkDescriptorUpdateTemplate descriptorUpdateTemplate,
    VkPipelineLayout         layout,
    uint32_t                 set,
    const void*              pData);
```

- `commandBuffer` is the command buffer that the descriptors will be recorded in.
- `descriptorUpdateTemplate` is a descriptor update template defining how to interpret the descriptor information in `pData`.
- `layout` is a `VkPipelineLayout` object used to program the bindings. It **must** be compatible with the layout used to create the `descriptorUpdateTemplate` handle.
- `set` is the set number of the descriptor set in the pipeline layout that will be updated. This **must** be the same number used to create the `descriptorUpdateTemplate` handle.
- `pData` is a pointer to memory containing descriptors for the templated update.

Valid Usage

- VUID-vkCmdPushDescriptorSetWithTemplateKHR-commandBuffer-00366
The `pipelineBindPoint` specified during the creation of the descriptor update template **must** be supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-pData-01686
`pData` **must** be a valid pointer to a memory containing one or more valid instances of `VkDescriptorImageInfo`, `VkDescriptorBufferInfo`, or `VkBufferView` in a layout defined by `descriptorUpdateTemplate` when it was created with `vkCreateDescriptorUpdateTemplateKHR`
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-layout-07993
`layout` **must** be compatible with the layout used to create `descriptorUpdateTemplate`
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-descriptorUpdateTemplate-07994
`descriptorUpdateTemplate` **must** have been created with a `templateType` of `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_SET_KHR`
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-set-07995
`set` **must** be the same value used to create `descriptorUpdateTemplate`
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-set-07304
`set` **must** be less than `VkPipelineLayoutCreateInfo::setLayoutCount` provided when `layout` was created
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-set-07305
`set` **must** be the unique set number in the pipeline layout that uses a descriptor set layout that was created with `VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR`

Valid Usage (Implicit)

- VUID-vkCmdPushDescriptorSetWithTemplateKHR-commandBuffer-parameter **commandBuffer** **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-descriptorUpdateTemplate-parameter **descriptorUpdateTemplate** **must** be a valid [VkDescriptorUpdateTemplate](#) handle
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-layout-parameter **layout** **must** be a valid [VkPipelineLayout](#) handle
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-commandBuffer-recording **commandBuffer** **must** be in the [recording state](#)
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics, or compute operations
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdPushDescriptorSetWithTemplateKHR-commonparent
Each of **commandBuffer**, **descriptorUpdateTemplate**, and **layout** **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	State

API Example

```
struct AppDataStructure
{
    VkDescriptorImageInfo  imageInfo;          // a single image info
    // ... some more application-related data
};

const VkDescriptorUpdateTemplateEntry descriptorUpdateTemplateEntries[] =
{
```

```

// binding to a single image descriptor
{
    .binding = 0,
    .dstArrayElement = 0,
    .descriptorCount = 1,
    .descriptorType = VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER,
    .offset = offsetof(AppDataStructure, imageInfo),
    .stride = 0    // not required if descriptorCount is 1
}
};

// create a descriptor update template for push descriptor set updates
const VkDescriptorUpdateTemplateCreateInfo createInfo =
{
    .sType = VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO,
    .pNext = NULL,
    .flags = 0,
    .descriptorUpdateEntryCount = 1,
    .pDescriptorUpdateEntries = descriptorUpdateTemplateEntries,
    .templateType = VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_KHR,
    .descriptorSetLayout = 0,    // ignored by given templateType
    .pipelineBindPoint = VK_PIPELINE_BIND_POINT_GRAPHICS,
    .pipelineLayout = myPipelineLayout,
    .set = 0,
};

VkDescriptorUpdateTemplate myDescriptorUpdateTemplate;
myResult = vkCreateDescriptorUpdateTemplate(
    myDevice,
    &createInfo,
    NULL,
    &myDescriptorUpdateTemplate);

AppDataStructure appData;
// fill appData here or cache it in your engine
vkCmdPushDescriptorSetWithTemplateKHR(myCmdBuffer, myDescriptorUpdateTemplate,
myPipelineLayout, 0, &appData);

```

Alternatively, to use a descriptor update template to specify the push descriptors to update, call:

```

// Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
void vkCmdPushDescriptorSetWithTemplate2KHR(
    VkCommandBuffer                commandBuffer,
    const VkPushDescriptorSetWithTemplateInfoKHR* pPushDescriptorSetWithTemplateInfo);

```

- `commandBuffer` is the command buffer that the descriptors will be recorded in.
- `pPushDescriptorSetWithTemplateInfo` is a pointer to a `VkPushDescriptorSetWithTemplateInfoKHR` structure.

Valid Usage (Implicit)

- VUID-vkCmdPushDescriptorSetWithTemplate2KHR-commandBuffer-parameter **commandBuffer** **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdPushDescriptorSetWithTemplate2KHR-pPushDescriptorSetWithTemplateInfo-parameter **pPushDescriptorSetWithTemplateInfo** **must** be a valid pointer to a valid [VkPushDescriptorSetWithTemplateInfoKHR](#) structure
- VUID-vkCmdPushDescriptorSetWithTemplate2KHR-commandBuffer-recording **commandBuffer** **must** be in the [recording state](#)
- VUID-vkCmdPushDescriptorSetWithTemplate2KHR-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics, or compute operations
- VUID-vkCmdPushDescriptorSetWithTemplate2KHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	State

The [VkPushDescriptorSetWithTemplateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_maintenance6 with VK_KHR_push_descriptor
typedef struct VkPushDescriptorSetWithTemplateInfoKHR {
    VkStructureType             sType;
    const void*                 pNext;
    VkDescriptorUpdateTemplate   descriptorUpdateTemplate;
    VkPipelineLayout             layout;
    uint32_t                     set;
    const void*                 pData;
} VkPushDescriptorSetWithTemplateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `descriptorUpdateTemplate` is a descriptor update template defining how to interpret the descriptor information in `pData`.
- `layout` is a `VkPipelineLayout` object used to program the bindings. It **must** be compatible with the layout used to create the `descriptorUpdateTemplate` handle.
- `set` is the set number of the descriptor set in the pipeline layout that will be updated. This **must** be the same number used to create the `descriptorUpdateTemplate` handle.
- `pData` is a pointer to memory containing descriptors for the templated update.

Valid Usage

- VUID-VkPushDescriptorSetWithTemplateInfoKHR-commandBuffer-00366
The `pipelineBindPoint` specified during the creation of the descriptor update template **must** be supported by the `commandBuffer`'s parent `VkCommandPool`'s queue family
- VUID-VkPushDescriptorSetWithTemplateInfoKHR-pData-01686
`pData` **must** be a valid pointer to a memory containing one or more valid instances of `VkDescriptorImageInfo`, `VkDescriptorBufferInfo`, or `VkBufferView` in a layout defined by `descriptorUpdateTemplate` when it was created with `vkCreateDescriptorUpdateTemplateKHR`
- VUID-VkPushDescriptorSetWithTemplateInfoKHR-layout-07993
`layout` **must** be compatible with the layout used to create `descriptorUpdateTemplate`
- VUID-VkPushDescriptorSetWithTemplateInfoKHR-descriptorUpdateTemplate-07994
`descriptorUpdateTemplate` **must** have been created with a `templateType` of `VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_SET_KHR`
- VUID-VkPushDescriptorSetWithTemplateInfoKHR-set-07995
`set` **must** be the same value used to create `descriptorUpdateTemplate`
- VUID-VkPushDescriptorSetWithTemplateInfoKHR-set-07304
`set` **must** be less than `VkPipelineLayoutCreateInfo::setLayoutCount` provided when `layout` was created
- VUID-VkPushDescriptorSetWithTemplateInfoKHR-set-07305
`set` **must** be the unique set number in the pipeline layout that uses a descriptor set layout that was created with `VK_DESCRIPTOR_SET_LAYOUT_CREATE_DESCRIPTOR_BIT_KHR`
- VUID-VkPushDescriptorSetWithTemplateInfoKHR-None-09495
`layout` **must** be a valid `VkPipelineLayout` handle

Valid Usage (Implicit)

- VUID-VkPushDescriptorSetWithTemplateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DESCRIPTOR_SET_WITH_DESCRIPTOR_UPDATE_TEMPLATE_KHR`
- VUID-VkPushDescriptorSetWithTemplateInfoKHR-pNext-pNext

pNext **must** be **NULL** or a pointer to a valid instance of [VkPipelineLayoutCreateInfo](#)

- **VOID-VkPushDescriptorSetWithTemplateInfoKHR-sType-unique**
The **sType** value of each struct in the **pNext** chain **must** be unique
- **VOID-VkPushDescriptorSetWithTemplateInfoKHR-descriptorUpdateTemplate-parameter**
descriptorUpdateTemplate **must** be a valid [VkDescriptorUpdateTemplate](#) handle
- **VOID-VkPushDescriptorSetWithTemplateInfoKHR-layout-parameter**
If **layout** is not [VK_NULL_HANDLE](#), **layout** **must** be a valid [VkPipelineLayout](#) handle
- **VOID-VkPushDescriptorSetWithTemplateInfoKHR-pData-parameter**
pData **must** be a pointer value
- **VOID-VkPushDescriptorSetWithTemplateInfoKHR-commonparent**
Both of **descriptorUpdateTemplate**, and **layout** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same [VkDevice](#)

14.2.10. Push Constant Updates

As described above in section [Pipeline Layouts](#), the pipeline layout defines shader push constants which are updated via Vulkan commands rather than via writes to memory or copy commands.

NOTE

Push constants represent a high speed path to modify constant data in pipelines that is expected to outperform memory-backed resource updates.

To update push constants, call:

```
// Provided by VK_VERSION_1_0
void vkCmdPushConstants(
    VkCommandBuffer          commandBuffer,
    VkPipelineLayout         layout,
    VkShaderStageFlags       stageFlags,
    uint32_t                 offset,
    uint32_t                 size,
    const void*              pValues);
```

- **commandBuffer** is the command buffer in which the push constant update will be recorded.
- **layout** is the pipeline layout used to program the push constant updates.
- **stageFlags** is a bitmask of [VkShaderStageFlagBits](#) specifying the shader stages that will use the push constants in the updated range.
- **offset** is the start offset of the push constant range to update, in units of bytes.
- **size** is the size of the push constant range to update, in units of bytes.
- **pValues** is a pointer to an array of **size** bytes containing the new push constant values.

When a command buffer begins recording, all push constant values are undefined. Reads of undefined push constant values by the executing shader return undefined values.

Push constant values **can** be updated incrementally, causing shader stages in **stageFlags** to read the new data from **pValues** for push constants modified by this command, while still reading the previous data for push constants not modified by this command. When a **bound pipeline command** is issued, the bound pipeline's layout **must** be compatible with the layouts used to set the values of all push constants in the pipeline layout's push constant ranges, as described in **Pipeline Layout Compatibility**. Binding a pipeline with a layout that is not compatible with the push constant layout does not disturb the push constant values.

NOTE

As **stageFlags** needs to include all flags the relevant push constant ranges were created with, any flags that are not supported by the queue family that the **VkCommandPool** used to allocate **commandBuffer** was created on are ignored.

Valid Usage

- VUID-vkCmdPushConstants-offset-01795
For each byte in the range specified by **offset** and **size** and for each shader stage in **stageFlags**, there **must** be a push constant range in **layout** that includes that byte and that stage
- VUID-vkCmdPushConstants-offset-01796
For each byte in the range specified by **offset** and **size** and for each push constant range that overlaps that byte, **stageFlags** **must** include all stages in that push constant range's **VkPushConstantRange::stageFlags**
- VUID-vkCmdPushConstants-offset-00368
offset **must** be a multiple of 4
- VUID-vkCmdPushConstants-size-00369
size **must** be a multiple of 4
- VUID-vkCmdPushConstants-offset-00370
offset **must** be less than **VkPhysicalDeviceLimits::maxPushConstantsSize**
- VUID-vkCmdPushConstants-size-00371
size **must** be less than or equal to **VkPhysicalDeviceLimits::maxPushConstantsSize** minus **offset**

Valid Usage (Implicit)

- VUID-vkCmdPushConstants-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdPushConstants-layout-parameter
layout **must** be a valid **VkPipelineLayout** handle
- VUID-vkCmdPushConstants-stageFlags-parameter
stageFlags **must** be a valid combination of **VkShaderStageFlagBits** values
- VUID-vkCmdPushConstants-stageFlags-requiredbitmask
stageFlags **must** not be 0

- VUID-vkCmdPushConstants-pValues-parameter
pValues must be a valid pointer to an array of **size** bytes
- VUID-vkCmdPushConstants-commandBuffer-recording
commandBuffer must be in the **recording state**
- VUID-vkCmdPushConstants-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics, or compute operations
- VUID-vkCmdPushConstants-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdPushConstants-size-arraylength
size must be greater than 0
- VUID-vkCmdPushConstants-commonparent
Both of **commandBuffer**, and **layout must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	State

Alternatively, to update push constants, call:

```
// Provided by VK_KHR_maintenance6
void vkCmdPushConstants2KHR(
    VkCommandBuffer          commandBuffer,
    const VkPushConstantsInfoKHR* pPushConstantsInfo);
```

- **commandBuffer** is the command buffer in which the push constant update will be recorded.
- **pPushConstantsInfo** is a pointer to a **VkPushConstantsInfoKHR** structure.

Valid Usage (Implicit)

- VUID-vkCmdPushConstants2KHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdPushConstants2KHR-pPushConstantsInfo-parameter
`pPushConstantsInfo` **must** be a valid pointer to a valid `VkPushConstantsInfoKHR` structure
- VUID-vkCmdPushConstants2KHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdPushConstants2KHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdPushConstants2KHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	State

The `VkPushConstantsInfoKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance6
typedef struct VkPushConstantsInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkPipelineLayout    layout;
    VkShaderStageFlags stageFlags;
    uint32_t            offset;
    uint32_t            size;
    const void*        pValues;
} VkPushConstantsInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `layout` is the pipeline layout used to program the push constant updates.

- **stageFlags** is a bitmask of **VkShaderStageFlagBits** specifying the shader stages that will use the push constants in the updated range.
- **offset** is the start offset of the push constant range to update, in units of bytes.
- **size** is the size of the push constant range to update, in units of bytes.
- **pValues** is a pointer to an array of **size** bytes containing the new push constant values.

Valid Usage

- VUID-VkPushConstantsInfoKHR-offset-01795
For each byte in the range specified by **offset** and **size** and for each shader stage in **stageFlags**, there **must** be a push constant range in **layout** that includes that byte and that stage
- VUID-VkPushConstantsInfoKHR-offset-01796
For each byte in the range specified by **offset** and **size** and for each push constant range that overlaps that byte, **stageFlags** **must** include all stages in that push constant range's **VkPushConstantRange::stageFlags**
- VUID-VkPushConstantsInfoKHR-offset-00368
offset **must** be a multiple of 4
- VUID-VkPushConstantsInfoKHR-size-00369
size **must** be a multiple of 4
- VUID-VkPushConstantsInfoKHR-offset-00370
offset **must** be less than **VkPhysicalDeviceLimits::maxPushConstantsSize**
- VUID-VkPushConstantsInfoKHR-size-00371
size **must** be less than or equal to **VkPhysicalDeviceLimits::maxPushConstantsSize** minus **offset**
- VUID-VkPushConstantsInfoKHR-None-09495
layout **must** be a valid **VkPipelineLayout** handle

Valid Usage (Implicit)

- VUID-VkPushConstantsInfoKHR-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO_KHR**
- VUID-VkPushConstantsInfoKHR-pNext-pNext
pNext **must** be **NULL** or a pointer to a valid instance of **VkPipelineLayoutCreateInfo**
- VUID-VkPushConstantsInfoKHR-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkPushConstantsInfoKHR-layout-parameter
If **layout** is not **VK_NULL_HANDLE**, **layout** **must** be a valid **VkPipelineLayout** handle
- VUID-VkPushConstantsInfoKHR-stageFlags-parameter
stageFlags **must** be a valid combination of **VkShaderStageFlagBits** values

- VUID-VkPushConstantsInfoKHR-stageFlags-requiredbitmask
`stageFlags` **must** not be 0
- VUID-VkPushConstantsInfoKHR-pValues-parameter
`pValues` **must** be a valid pointer to an array of `size` bytes
- VUID-VkPushConstantsInfoKHR-size-arraylength
`size` **must** be greater than 0

14.3. Physical Storage Buffer Access

To query a 64-bit buffer device address value through which buffer memory **can** be accessed in a shader, call:

```
// Provided by VK_KHR_buffer_device_address
VkDeviceAddress vkGetBufferDeviceAddressKHR(
    VkDevice device,
    const VkBufferDeviceAddressInfo* pInfo);
```

- `device` is the logical device that the buffer was created on.
- `pInfo` is a pointer to a `VkBufferDeviceAddressInfo` structure specifying the buffer to retrieve an address for.

The 64-bit return value is an address of the start of `pInfo->buffer`. The address range starting at this value and whose size is the size of the buffer **can** be used in a shader to access the memory bound to that buffer, using the `SPV_KHR_physical_storage_buffer` extension and the `PhysicalStorageBuffer` storage class. For example, this value **can** be stored in a uniform buffer, and the shader **can** read the value from the uniform buffer and use it to do a dependent read/write to this buffer. A value of zero is reserved as a “null” pointer and **must** not be returned as a valid buffer device address. All loads, stores, and atomics in a shader through `PhysicalStorageBuffer` pointers **must** access addresses in the address range of some buffer.

If the buffer was created with a non-zero value of `VkBufferOpaqueCaptureAddressCreateInfo::opaqueCaptureAddress`, the return value will be the same address that was returned at capture time.

The returned address **must** satisfy the alignment requirement specified by `VkMemoryRequirements::alignment` for the buffer in `VkBufferDeviceAddressInfo::buffer`.

If multiple `VkBuffer` objects are bound to overlapping ranges of `VkDeviceMemory`, implementations **may** return address ranges which overlap. In this case, it is ambiguous which `VkBuffer` is associated with any given device address. For purposes of valid usage, if multiple `VkBuffer` objects **can** be attributed to a device address, a `VkBuffer` is selected such that valid usage passes, if it exists.

Valid Usage

- VUID-vkGetBufferDeviceAddress-bufferDeviceAddress-03324

The `bufferDeviceAddress` feature **must** be enabled

- VUID-vkGetBufferDeviceAddress-device-03325

If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetBufferDeviceAddress-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetBufferDeviceAddress-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkBufferDeviceAddressInfo` structure

The `VkBufferDeviceAddressInfo` structure is defined as:

```
typedef struct VkBufferDeviceAddressInfo {  
    VkStructureType    sType;  
    const void*        pNext;  
    VkBuffer            buffer;  
} VkBufferDeviceAddressInfo;
```

or the equivalent

```
// Provided by VK_KHR_buffer_device_address  
typedef VkBufferDeviceAddressInfo VkBufferDeviceAddressInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `buffer` specifies the buffer whose address is being queried.

Valid Usage

- VUID-VkBufferDeviceAddressInfo-buffer-02600
If `buffer` is non-sparse and was not created with the `VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT` flag, then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkBufferDeviceAddressInfo-buffer-02601
`buffer` **must** have been created with `VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT`

Valid Usage (Implicit)

- VUID-VkBufferDeviceAddressInfo-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_BUFFER_DEVICE_ADDRESS_INFO`

- `VUID-VkBufferDeviceAddressInfo-pNext-pNext`
`pNext` **must** be `NULL`
- `VUID-VkBufferDeviceAddressInfo-buffer-parameter`
`buffer` **must** be a valid `VkBuffer` handle

To query a 64-bit buffer opaque capture address, call:

```
// Provided by VK_KHR_buffer_device_address
uint64_t vkGetBufferOpaqueCaptureAddressKHR(
    VkDevice device,
    const VkBufferDeviceAddressInfo* pInfo);
```

- `device` is the logical device that the buffer was created on.
- `pInfo` is a pointer to a `VkBufferDeviceAddressInfo` structure specifying the buffer to retrieve an address for.

The 64-bit return value is an opaque capture address of the start of `pInfo->buffer`.

If the buffer was created with a non-zero value of `VkBufferOpaqueCaptureAddressCreateInfo::opaqueCaptureAddress` the return value **must** be the same address.

Valid Usage

- `VUID-vkGetBufferOpaqueCaptureAddress-None-03326`
The `bufferDeviceAddress` feature **must** be enabled
- `VUID-vkGetBufferOpaqueCaptureAddress-device-03327`
If `device` was created with multiple physical devices, then the `bufferDeviceAddressMultiDevice` feature **must** be enabled

Valid Usage (Implicit)

- `VUID-vkGetBufferOpaqueCaptureAddress-device-parameter`
`device` **must** be a valid `VkDevice` handle
- `VUID-vkGetBufferOpaqueCaptureAddress-pInfo-parameter`
`pInfo` **must** be a valid pointer to a valid `VkBufferDeviceAddressInfo` structure

The `VkStridedDeviceAddressRegionKHR` structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkStridedDeviceAddressRegionKHR {
    VkDeviceAddress    deviceAddress;
    VkDeviceSize       stride;
    VkDeviceSize       size;
} VkStridedDeviceAddressRegionKHR;
```

- **deviceAddress** is the device address (as returned by the [vkGetBufferDeviceAddressKHR](#) command) at which the region starts, or zero if the region is unused.
- **stride** is the byte stride between consecutive elements.
- **size** is the size in bytes of the region starting at **deviceAddress**.

Valid Usage

- VUID-VkStridedDeviceAddressRegionKHR-size-04631
If **size** is not zero, all addresses between **deviceAddress** and **deviceAddress + size - 1** **must** be in the buffer device address range of the same buffer
- VUID-VkStridedDeviceAddressRegionKHR-size-04632
If **size** is not zero, **stride** **must** be less than or equal to the size of the buffer from which **deviceAddress** was queried

Chapter 15. Shader Interfaces

When a pipeline is created, the set of shaders specified in the corresponding `VkPipelineCreateInfo` structure are implicitly linked at a number of different interfaces.

- [Shader Input and Output Interface](#)
- [Vertex Input Interface](#)
- [Fragment Output Interface](#)
- [Fragment Tile Image Interface](#)
- [Fragment Input Attachment Interface](#)
- [Ray Tracing Pipeline Interface](#)
- [Shader Resource Interface](#)

This chapter describes valid uses for a set of SPIR-V decorations. Any other use of one of these decorations is invalid, with the exception that, when using SPIR-V versions 1.4 and earlier: `Block`, `BufferBlock`, `Offset`, `ArrayStride`, and `MatrixStride` can also decorate types and type members used by variables in the `Private` and `Function` storage classes.

NOTE

In this chapter, there are references to SPIR-V terms such as the `MeshNV` execution model. These terms will appear even in a build of the specification which does not support any extensions. This is as intended, since these terms appear in the unified SPIR-V specification without such qualifiers.

15.1. Shader Input and Output Interfaces

When multiple stages are present in a pipeline, the outputs of one stage form an interface with the inputs of the next stage. When such an interface involves a shader, shader outputs are matched against the inputs of the next stage, and shader inputs are matched against the outputs of the previous stage.

All the variables forming the shader input and output *interfaces* are listed as operands to the `OpEntryPoint` instruction and are declared with the `Input` or `Output` storage classes, respectively, in the SPIR-V module. These generally form the interfaces between consecutive shader stages, regardless of any non-shader stages between the consecutive shader stages.

There are two classes of variables that **can** be matched between shader stages, built-in variables and user-defined variables. Each class has a different set of matching criteria.

`Output` variables of a shader stage have undefined values until the shader writes to them or uses the `Initializer` operand when declaring the variable.

15.1.1. Built-in Interface Block

Shader `built-in` variables meeting the following requirements define the *built-in interface block*. They **must**

- be explicitly declared (there are no implicit built-ins),
- be identified with a **BuiltIn** decoration,
- form object types as described in the **Built-in Variables** section, and
- be declared in a block whose top-level members are the built-ins.

There **must** be no more than one built-in interface block per shader per interface .

Built-ins **must** not have any **Location** or **Component** decorations.

15.1.2. User-defined Variable Interface

The non-built-in variables listed by **OpEntryPoint** with the **Input** or **Output** storage class form the *user-defined variable interface*. These **must** have **numeric type** or, recursively, composite types of such types. If an implementation supports **storageInputOutput16**, components **can** have a width of 16 bits. These variables **must** be identified with a **Location** decoration and **can** also be identified with a **Component** decoration.

15.1.3. Interface Matching

An output variable, block, or structure member in a given shader stage has an interface match with an input variable, block, or structure member in a subsequent shader stage if they both adhere to the following conditions:

- They have equivalent decorations, other than:
 - **XfbBuffer**, **XfbStride**, **Offset**, and **Stream**
 - one is not decorated with **Component** and the other is declared with a **Component** of 0
 - **Interpolation decorations**
 - **RelaxedPrecision** if one is an input variable and the other an output variable
- Their types match as follows:
 - if the input is declared in a tessellation control or geometry shader as an **OpTypeArray** with an **Element Type** equivalent to the **OpType*** declaration of the output, and neither is a structure member; or
 - if the **maintenance4** feature is enabled, they are declared as **OpTypeVector** variables, and the output has a **Component Count** value higher than that of the input but the same **Component Type**; or
 - if the input is decorated with **PerVertexKHR**, and is declared in a fragment shader as an **OpTypeArray** with an **Element Type** equivalent to the **OpType*** declaration of the output, and neither the input nor the output is a structure member; or
 - if in any other case they are declared with an equivalent **OpType*** declaration.
- If both are structures and every member has an interface match.

NOTE

The word “structure” above refers to both variables that have an **OpTypeStruct** type and interface blocks (which are also declared as **OpTypeStruct**).

All input variables and blocks **must** have an interface match in the preceding shader stage, except for built-in variables in fragment shaders. Shaders **can** declare and write to output variables that are not declared or read by the subsequent stage.

The value of an input variable is undefined if the preceding stage does not write to a matching output variable, as described above.

15.1.4. Location Assignment

This section describes **Location** assignments for user-defined variables and how many **Location** slots are consumed by a given user-variable type. As mentioned above, some inputs and outputs have an additional level of arrayness relative to other shader inputs and outputs. This outer array level is removed from the type before considering how many **Location** slots the type consumes.

The **Location** value specifies an interface slot comprised of a 32-bit four-component vector conveyed between stages. The **Component** specifies **word components** within these vector **Location** slots. Only types with widths of 16, 32 or 64 are supported in shader interfaces.

Inputs and outputs of the following types consume a single interface **Location**:

- 16-bit scalar and vector types, and
- 32-bit scalar and vector types, and
- 64-bit scalar and 2-component vector types.

64-bit three- and four-component vectors consume two consecutive **Location** slots.

If a declared input or output is an array of size n and each element takes m **Location** slots, it will be assigned $m \times n$ consecutive **Location** slots starting with the specified **Location**.

If the declared input or output is an $n \times m$ 16-, 32- or 64-bit matrix, it will be assigned multiple **Location** slots starting with the specified **Location**. The number of **Location** slots assigned for each matrix will be the same as for an n -element array of m -component vectors.

An **OpVariable** with a structure type that is not a block **must** be decorated with a **Location**.

When an **OpVariable** with a structure type (either block or non-block) is decorated with a **Location**, the members in the structure type **must** not be decorated with a **Location**. The **OpVariable**'s members are assigned consecutive **Location** slots in declaration order, starting from the first member, which is assigned the **Location** decoration from the **OpVariable**.

When a block-type **OpVariable** is declared without a **Location** decoration, each member in its structure type **must** be decorated with a **Location**. Types nested deeper than the top-level members **must** not have **Location** decorations.

The **Location** slots consumed by block and structure members are determined by applying the rules above in a depth-first traversal of the instantiated members as though the structure or block member were declared as an input or output variable of the same type.

Any two inputs listed as operands on the same **OpEntryPoint** **must** not be assigned the same **Location** slot and **Component** word, either explicitly or implicitly. Any two outputs listed as operands

on the same **OpEntryPoint** **must** not be assigned the same **Location** slot and **Component** word, either explicitly or implicitly.

The number of input and output **Location** slots available for a shader input or output interface is limited, and dependent on the shader stage as described in [Shader Input and Output Locations](#). All variables in both the [built-in interface block](#) and the [user-defined variable interface](#) count against these limits. Each effective **Location** **must** have a value less than the number of **Location** slots available for the given interface, as specified in the “Locations Available” column in [Shader Input and Output Locations](#).

Table 14. Shader Input and Output Locations

Shader Interface	Locations Available
vertex input	<code>maxVertexInputAttributes</code>
vertex output	<code>maxVertexOutputComponents</code> / 4
tessellation control input	<code>maxTessellationControlPerVertexInputComponents</code> / 4
tessellation control output	<code>maxTessellationControlPerVertexOutputComponents</code> / 4
tessellation evaluation input	<code>maxTessellationEvaluationInputComponents</code> / 4
tessellation evaluation output	<code>maxTessellationEvaluationOutputComponents</code> / 4
geometry input	<code>maxGeometryInputComponents</code> / 4
geometry output	<code>maxGeometryOutputComponents</code> / 4
fragment input	<code>maxFragmentInputComponents</code> / 4
fragment output	<code>maxFragmentOutputAttachments</code>

15.1.5. Component Assignment

The **Component** decoration allows the **Location** to be more finely specified for scalars and vectors, down to the individual **Component** word within a **Location** slot that are consumed. The **Component** word within a **Location** are 0, 1, 2, and 3. A variable or block member starting at **Component** N will consume **Component** words N, N+1, N+2, ... up through its size. For 16-, and 32-bit types, it is invalid if this sequence of **Component** words gets larger than 3. A scalar 64-bit type will consume two of these **Component** words in sequence, and a two-component 64-bit vector type will consume all four **Component** words available within a **Location**. A three- or four-component 64-bit vector type **must** not specify a non-zero **Component** decoration. A three-component 64-bit vector type will consume all four **Component** words of the first **Location** and **Component** 0 and 1 of the second **Location**. This leaves **Component** 2 and 3 available for other component-qualified declarations.

A scalar or two-component 64-bit data type **must** not specify a **Component** decoration of 1 or 3. A **Component** decoration **must** not be specified for any type that is not a scalar or vector.

A four-component 64-bit data type will consume all four **Component** words of the first **Location** and all four **Component** words of the second **Location**.

15.2. Vertex Input Interface

When the vertex stage is present in a pipeline, the vertex shader input variables form an interface with the vertex input attributes. The vertex shader input variables are matched by the **Location** and **Component** decorations to the vertex input attributes specified in the **pVertexInputState** member of the **VkGraphicsPipelineCreateInfo** structure.

The vertex shader input variables listed by **OpEntryPoint** with the **Input** storage class form the *vertex input interface*. These variables **must** be identified with a **Location** decoration and **can** also be identified with a **Component** decoration.

For the purposes of interface matching: variables declared without a **Component** decoration are considered to have a **Component** decoration of zero. The number of available vertex input **Location** slots is given by the **maxVertexInputAttributes** member of the **VkPhysicalDeviceLimits** structure.

See [Attribute Location and Component Assignment](#) for details.

All vertex shader inputs declared as above **must** have a corresponding attribute and binding in the pipeline.

15.3. Fragment Output Interface

When the fragment stage is present in a pipeline, the fragment shader outputs form an interface with the output attachments defined by a **render pass instance**. The fragment shader output variables are matched by the **Location** and **Component** decorations to specified color attachments.

The fragment shader output variables listed by **OpEntryPoint** with the **Output** storage class form the *fragment output interface*. These variables **must** be identified with a **Location** decoration. They **can** also be identified with a **Component** decoration and/or an **Index** decoration. For the purposes of interface matching: variables declared without a **Component** decoration are considered to have a **Component** decoration of zero, and variables declared without an **Index** decoration are considered to have an **Index** decoration of zero.

A fragment shader output variable identified with a **Location** decoration of *i* is associated with the color attachment indicated by **VkRenderingInfo::pColorAttachments[i]**. When using render pass objects, it is associated with the color attachment indicated by **VkSubpassDescription::pColorAttachments[i]**. Values are written to those attachments after passing through the blending unit as described in [Blending](#), if enabled. Locations are consumed as described in [Location Assignment](#). The number of available fragment output **Location** slots is given by the **maxFragmentOutputAttachments** member of the **VkPhysicalDeviceLimits** structure.

If the **dynamicRenderingLocalRead** feature is supported, fragment output locations **can** be remapped when using dynamic rendering.

To set the fragment output location mappings during rendering, call:

```
// Provided by VK_KHR_dynamic_rendering_local_read
void vkCmdSetRenderingAttachmentLocationsKHR(
    VkCommandBuffer                commandBuffer,
    const VkRenderingAttachmentLocationInfoKHR* pLocationInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pLocationInfo` is a `VkRenderingAttachmentLocationInfoKHR` structure indicating the new mappings.

This command sets the attachment location mappings for subsequent drawing commands, and **must** match the mappings provided to the currently bound pipeline, if one is bound, which **can** be set by chaining `VkRenderingAttachmentLocationInfoKHR` to `VkGraphicsPipelineCreateInfo`.

Until this command is called, mappings in the command buffer state are treated as each color attachment specified in `vkCmdBeginRenderingKHR` having a location equal to its index in `VkRenderingInfo::pColorAttachments`. This state is reset whenever `vkCmdBeginRenderingKHR` is called.

Valid Usage

- VUID-vkCmdSetRenderingAttachmentLocationsKHR-dynamicRenderingLocalRead-09509 `dynamicRenderingLocalRead` **must** be enabled
- VUID-vkCmdSetRenderingAttachmentLocationsKHR-pLocationInfo-09510 `pLocationInfo->colorAttachmentCount` **must** be equal to the value of `VkRenderingInfo::colorAttachmentCount` used to begin the current render pass instance
- VUID-vkCmdSetRenderingAttachmentLocationsKHR-commandBuffer-09511 The current render pass instance **must** have been started or resumed by `vkCmdBeginRenderingKHR` in this `commandBuffer`

Valid Usage (Implicit)

- VUID-vkCmdSetRenderingAttachmentLocationsKHR-commandBuffer-parameter `commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetRenderingAttachmentLocationsKHR-pLocationInfo-parameter `pLocationInfo` **must** be a valid pointer to a valid `VkRenderingAttachmentLocationInfoKHR` structure
- VUID-vkCmdSetRenderingAttachmentLocationsKHR-commandBuffer-recording `commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetRenderingAttachmentLocationsKHR-commandBuffer-cmdpool The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetRenderingAttachmentLocationsKHR-renderpass This command **must** only be called inside of a render pass instance

- VUID-vkCmdSetRenderingAttachmentLocationsKHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	State

The `VkRenderingAttachmentLocationInfoKHR` structure is defined as:

```
// Provided by VK_KHR_dynamic_rendering_local_read
typedef struct VkRenderingAttachmentLocationInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           colorAttachmentCount;
    const uint32_t*    pColorAttachmentLocations;
} VkRenderingAttachmentLocationInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `colorAttachmentCount` is the number of elements in `pColorAttachmentLocations`.
- `pColorAttachmentLocations` is a pointer to an array of `colorAttachmentCount` `uint32_t` values defining remapped locations for color attachments.

This structure allows applications to remap the locations of color attachments to different fragment shader output locations.

Each element of `pColorAttachmentLocations` set to `VK_ATTACHMENT_UNUSED` will be inaccessible to this pipeline as a color attachment; no location will map to it. Each element of `pColorAttachmentLocations` set to any other value will map the specified location value to the color attachment specified in the render pass at the corresponding index in the `pColorAttachmentLocations` array. Any writes to a fragment output location that is not mapped to an attachment **must** be discarded.

If `pColorAttachmentLocations` is `NULL`, it is equivalent to setting each element to its index within the

array.

This structure **can** be included in the **pNext** chain of a [VkGraphicsPipelineCreateInfo](#) structure to set this state for a pipeline. If this structure is not included in the **pNext** chain of [VkGraphicsPipelineCreateInfo](#), it is equivalent to specifying this structure with the following properties:

- **colorAttachmentCount** set to [VkPipelineRenderingCreateInfo::colorAttachmentCount](#).
- **pColorAttachmentLocations** set to **NULL**.

This structure **can** be included in the **pNext** chain of a [VkCommandBufferInheritanceInfo](#) structure to specify inherited state from the primary command buffer. If [VkCommandBufferInheritanceInfo::renderPass](#) is not [VK_NULL_HANDLE](#), or [VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT](#) is not specified in [VkCommandBufferBeginInfo::flags](#), members of this structure are ignored. If this structure is not included in the **pNext** chain of [VkCommandBufferInheritanceInfo](#), it is equivalent to specifying this structure with the following properties:

- **colorAttachmentCount** set to [VkCommandBufferInheritanceRenderingInfo::colorAttachmentCount](#).
- **pColorAttachmentLocations** set to **NULL**.

Valid Usage

- VUID-VkRenderingAttachmentLocationInfoKHR-dynamicRenderingLocalRead-09512
If the **dynamicRenderingLocalRead** feature is not enabled, and **pColorAttachmentLocations** is not **NULL**, each element **must** be the value of its index within the array
- VUID-VkRenderingAttachmentLocationInfoKHR-pColorAttachmentLocations-09513
Elements of **pColorAttachmentLocations** that are not **VK_ATTACHMENT_UNUSED** **must** each be unique
- VUID-VkRenderingAttachmentLocationInfoKHR-colorAttachmentCount-09514
colorAttachmentCount **must** be less than or equal to **maxColorAttachments**
- VUID-VkRenderingAttachmentLocationInfoKHR-pColorAttachmentLocations-09515
Each element of **pColorAttachmentLocations** **must** be less than **maxColorAttachments**

Valid Usage (Implicit)

- VUID-VkRenderingAttachmentLocationInfoKHR-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_LOCATION_INFO_KHR**

When an active fragment shader invocation finishes, the values of all fragment shader outputs are copied out and used as blend inputs or color attachments writes. If the invocation does not set a value for them, the input values to those blending or color attachment writes are undefined.

Components of the output variables are assigned as described in [Component Assignment](#). Output **Component** words identified as 0, 1, 2, and 3 will be directed to the R, G, B, and A inputs to the

blending unit, respectively, or to the output attachment if blending is disabled. If two variables are placed within the same **Location**, they **must** have the same underlying type (floating-point or integer). **Component** words which do not correspond to any fragment shader output will also result in undefined values for blending or color attachment writes.

Fragment outputs identified with an **Index** of zero are directed to the first input of the blending unit associated with the corresponding **Location**. Outputs identified with an **Index** of one are directed to the second input of the corresponding blending unit.

There **must** be no output variable which has the same **Location**, **Component**, and **Index** as any other, either explicitly declared or implied.

Output values written by a fragment shader **must** be declared with either **OpTypeFloat** or **OpTypeInt**, and a **Width** of 32. If **storageInputOutput16** is supported, output values written by a fragment shader **can** be also declared with either **OpTypeFloat** or **OpTypeInt** and a **Width** of 16. Composites of these types are also permitted. If the color attachment has a signed or unsigned normalized fixed-point format, color values are assumed to be floating-point and are converted to fixed-point as described in [Conversion From Floating-Point to Normalized Fixed-Point](#); If the color attachment has an integer format, color values are assumed to be integers and converted to the bit-depth of the target. Any value that cannot be represented in the attachment's format is undefined. For any other attachment format no conversion is performed. If the type of the values written by the fragment shader do not match the format of the corresponding color attachment, the resulting values are undefined for those components.

15.4. Fragment Tile Image Interface

When a fragment stage is present in a pipeline, the fragment shader tile image variables decorated with **Location** form an interface with the color attachments defined by the render pass instance. The fragment shader tile image variables are matched by **Location** decorations to the color attachments specified in the **pColorAttachments** array of the **VkRenderingInfoKHR** structure describing the render pass instance the fragment shader is executed in.

The fragment shader variables listed by **OpEntryPoint** with the **TileImageEXT** storage class and a decoration of **Location** form the *fragment tile image interface*. These variables **must** be declared with a type of **OpTypeImage**, and a **Dim** operand of **TileImageDataEXT**. The **Component** decoration is not supported for these variables.

Reading from a tile image variable with a **Location** decoration of i reads from the color attachment identified by the element of **VkRenderingInfoKHR::pColorAttachments** with a **location** equal to i . If the tile image variable is declared as an array of size N , it consumes N consecutive tile image locations, starting with the index specified. There **must** not be more than one tile image variable with the same **Location** whether explicitly declared or implied by an array declaration. The number of available tile image locations is the same as the number of available fragment output locations as given by the **maxFragmentOutputAttachments** member of the **VkPhysicalDeviceLimits** structure.

The basic data type (floating-point, integer, unsigned integer) of the tile image variable **must** match the basic format of the corresponding color attachment, or the values read from the tile image variables are undefined.

15.5. Fragment Input Attachment Interface

When a fragment stage is present in a pipeline, the fragment shader subpass inputs form an interface with the input attachments of the current subpass. The fragment shader subpass input variables are matched by `InputAttachmentIndex` decorations to the input attachments specified in the `pInputAttachments` array of the `VkSubpassDescription` structure describing the subpass that the fragment shader is executed in.

The fragment shader subpass input variables with the `UniformConstant` storage class and a decoration of `InputAttachmentIndex` that are statically used by `OpEntryPoint` form the *fragment input attachment interface*. These variables **must** be declared with a type of `OpTypeImage`, a `Dim` operand of `SubpassData`, an `Arrayed` operand of 0, and a `Sampled` operand of 2. The `MS` operand of the `OpTypeImage` **must** be 0 if the `samples` field of the corresponding `VkAttachmentDescription` is `VK_SAMPLE_COUNT_1_BIT` and 1 otherwise.

A subpass input variable identified with an `InputAttachmentIndex` decoration of *i* reads from the input attachment indicated by `pInputAttachments[i]` member of `VkSubpassDescription`. If the subpass input variable is declared as an array of size *N*, it consumes *N* consecutive input attachments, starting with the index specified. There **must** not be more than one input variable with the same `InputAttachmentIndex` whether explicitly declared or implied by an array declaration per image aspect. A multi-aspect image (e.g. a depth/stencil format) **can** use the same input variable. The number of available input attachment indices is given by the `maxPerStageDescriptorInputAttachments` member of the `VkPhysicalDeviceLimits` structure.

When using dynamic rendering with the `dynamicRenderingLocalRead` feature enabled, a subpass input variable with a `InputAttachmentIndex` decoration of *i* can be mapped to a color, depth, or stencil attachment.

To set the input attachment index mappings during dynamic rendering, call:

```
// Provided by VK_KHR_dynamic_rendering_local_read
void vkCmdSetRenderingInputAttachmentIndicesKHR(
    VkCommandBuffer                                commandBuffer,
    const VkRenderingInputAttachmentIndexInfoKHR* pInputAttachmentIndexInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pInputAttachmentIndexInfo` is a `VkRenderingInputAttachmentIndexInfoKHR` structure indicating the new mappings.

This command sets the input attachment index mappings for subsequent drawing commands, and **must** match the mappings provided to the currently bound pipeline, if one is bound, which **can** be set by chaining `VkRenderingInputAttachmentIndexInfoKHR` to `VkGraphicsPipelineCreateInfo`.

Until this command is called, mappings in the command buffer state are treated as each color attachment specified in `vkCmdBeginRenderingKHR` mapping to subpass inputs with a `InputAttachmentIndex` equal to its index in `VkRenderingInfo::pColorAttachments`, and depth/stencil attachments mapping to input attachments without these decorations. This state is reset whenever `vkCmdBeginRenderingKHR` is called.

Valid Usage

- VUID-vkCmdSetRenderingInputAttachmentIndicesKHR-dynamicRenderingLocalRead-09516
`dynamicRenderingLocalRead` **must** be enabled
- VUID-vkCmdSetRenderingInputAttachmentIndicesKHR-pInputAttachmentIndexInfo-09517
`pInputAttachmentIndexInfo->colorAttachmentCount` **must** be equal to the value of `VkRenderingInfo::colorAttachmentCount` used to begin the current render pass instance
- VUID-vkCmdSetRenderingInputAttachmentIndicesKHR-commandBuffer-09518
The current render pass instance **must** have been started or resumed by `vkCmdBeginRenderingKHR` in this `commandBuffer`

Valid Usage (Implicit)

- VUID-vkCmdSetRenderingInputAttachmentIndicesKHR-commandBuffer-parameter-`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetRenderingInputAttachmentIndicesKHR-pInputAttachmentIndexInfo-parameter
`pInputAttachmentIndexInfo` **must** be a valid pointer to a valid `VkRenderingInputAttachmentIndexInfoKHR` structure
- VUID-vkCmdSetRenderingInputAttachmentIndicesKHR-commandBuffer-recording-`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdSetRenderingInputAttachmentIndicesKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetRenderingInputAttachmentIndicesKHR-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdSetRenderingInputAttachmentIndicesKHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	State

The `VkRenderingInputAttachmentIndexInfoKHR` structure is defined as:

```
// Provided by VK_KHR_dynamic_rendering_local_read
typedef struct VkRenderingInputAttachmentIndexInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           colorAttachmentCount;
    const uint32_t*     pColorAttachmentInputIndices;
    const uint32_t*     pDepthInputAttachmentIndex;
    const uint32_t*     pStencilInputAttachmentIndex;
} VkRenderingInputAttachmentIndexInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `colorAttachmentCount` is the number of elements in `pColorAttachmentInputIndices`.
- `pColorAttachmentInputIndices` is a pointer to an array of `colorAttachmentCount` `uint32_t` values defining indices for color attachments to be used as input attachments.
- `pDepthInputAttachmentIndex` is either `NULL`, or a pointer to a `uint32_t` value defining the index for the depth attachment to be used as an input attachment.
- `pStencilInputAttachmentIndex` is either `NULL`, or a pointer to a `uint32_t` value defining the index for the stencil attachment to be used as an input attachment.

This structure allows applications to remap attachments to different input attachment indices.

Each element of `pColorAttachmentInputIndices` set to a value of `VK_ATTACHMENT_UNUSED` indicates that the corresponding attachment will not be used as an input attachment in this pipeline. Any other value in each of those elements will map the corresponding attachment to a `InputAttachmentIndex` value defined in shader code.

If `pColorAttachmentInputIndices` is `NULL`, it is equivalent to setting each element to its index within the array.

If `pDepthInputAttachmentIndex` or `pStencilInputAttachmentIndex` are set to `NULL`, they map to input attachments without a `InputAttachmentIndex` decoration. If they point to a value of `VK_ATTACHMENT_UNUSED`, it indicates that the corresponding attachment will not be used as an input attachment in this pipeline. If they point to any other value it maps the corresponding attachment to a `InputAttachmentIndex` value defined in shader code.

This structure **can** be included in the **pNext** chain of a **VkGraphicsPipelineCreateInfo** structure to set this state for a pipeline. If this structure is not included in the **pNext** chain of **VkGraphicsPipelineCreateInfo**, it is equivalent to specifying this structure with the following properties:

- **colorAttachmentCount** set to **VkPipelineRenderingCreateInfo::colorAttachmentCount**.
- **pColorAttachmentInputIndices** set to **NULL**.
- **pDepthInputAttachmentIndex** set to **NULL**.
- **pStencilInputAttachmentIndex** set to **NULL**.

This structure **can** be included in the **pNext** chain of a **VkCommandBufferInheritanceInfo** structure to specify inherited state from the primary command buffer. If this structure is not included in the **pNext** chain of **VkCommandBufferInheritanceInfo**, it is equivalent to specifying this structure with the following properties:

- **colorAttachmentCount** set to **VkCommandBufferInheritanceRenderingInfo::colorAttachmentCount**.
- **pColorAttachmentInputIndices** set to **NULL**.
- **pDepthInputAttachmentIndex** set to **NULL**.
- **pStencilInputAttachmentIndex** set to **NULL**.

Valid Usage

- VUID-VkRenderingInputAttachmentIndexInfoKHR-dynamicRenderingLocalRead-09519
If the **dynamicRenderingLocalRead** feature is not enabled, and **pColorAttachmentInputIndices** is not **NULL**, each element **must** be **VK_ATTACHMENT_UNUSED**
- VUID-VkRenderingInputAttachmentIndexInfoKHR-dynamicRenderingLocalRead-09520
If the **dynamicRenderingLocalRead** feature is not enabled, **pDepthInputAttachmentIndex** **must** be a valid pointer to a value of **VK_ATTACHMENT_UNUSED**
- VUID-VkRenderingInputAttachmentIndexInfoKHR-dynamicRenderingLocalRead-09521
If the **dynamicRenderingLocalRead** feature is not enabled, **pStencilInputAttachmentIndex** **must** be a valid pointer to a value of **VK_ATTACHMENT_UNUSED**
- VUID-VkRenderingInputAttachmentIndexInfoKHR-pColorAttachmentInputIndices-09522
Elements of **pColorAttachmentInputIndices** that are not **VK_ATTACHMENT_UNUSED** **must** each be unique
- VUID-VkRenderingInputAttachmentIndexInfoKHR-pColorAttachmentInputIndices-09523
Elements of **pColorAttachmentInputIndices** that are not **VK_ATTACHMENT_UNUSED** **must** not take the same value as the content of **pDepthInputAttachmentIndex**
- VUID-VkRenderingInputAttachmentIndexInfoKHR-pColorAttachmentInputIndices-09524
Elements of **pColorAttachmentInputIndices** that are not **VK_ATTACHMENT_UNUSED** **must** not take the same value as the content of **pStencilInputAttachmentIndex**
- VUID-VkRenderingInputAttachmentIndexInfoKHR-colorAttachmentCount-09525
colorAttachmentCount **must** be less than or equal to **maxColorAttachments**

Valid Usage (Implicit)

- VUID-VkRenderingInputAttachmentIndexInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_RENDERING_INPUT_ATTACHMENT_INDEX_INFO_KHR`
- VUID-VkRenderingInputAttachmentIndexInfoKHR-pColorAttachmentInputIndices-parameter
If `colorAttachmentCount` is not `0`, and `pColorAttachmentInputIndices` is not `NULL`, `pColorAttachmentInputIndices` **must** be a valid pointer to an array of `colorAttachmentCount` `uint32_t` values
- VUID-VkRenderingInputAttachmentIndexInfoKHR-pDepthInputAttachmentIndex-parameter
If `pDepthInputAttachmentIndex` is not `NULL`, `pDepthInputAttachmentIndex` **must** be a valid pointer to a valid `uint32_t` value
- VUID-VkRenderingInputAttachmentIndexInfoKHR-pStencilInputAttachmentIndex-parameter
If `pStencilInputAttachmentIndex` is not `NULL`, `pStencilInputAttachmentIndex` **must** be a valid pointer to a valid `uint32_t` value

Variables identified with the `InputAttachmentIndex` **must** only be used by a fragment stage. The [numeric format](#) of the subpass input **must** match the format of the corresponding input attachment, or the values of subpass loads from these variables are undefined. If the framebuffer attachment contains both depth and stencil aspects, the numeric format of the subpass input determines if depth or stencil aspect is accessed by the shader.

See [Input Attachment](#) for more details.

15.5.1. Fragment Input Attachment Compatibility

An input attachment that is statically accessed by a fragment shader **must** be backed by a descriptor that is equivalent to the `VkImageView` in the `VkFramebuffer`, except for `subresourceRange.aspectMask`. The `aspectMask` **must** be equal to the aspect accessed by the shader.

15.6. Ray Tracing Pipeline Interface

Ray tracing pipelines **may** have more stages than other pipelines with multiple instances of each stage and more dynamic interactions between the stages, but still have interface structures that obey the same general rules as interfaces between shader stages in other pipelines. The three types of inter-stage interface variables for ray tracing pipelines are:

- Ray payloads containing data tracked for the entire lifetime of the ray.
- Hit attributes containing data about a specific hit for the duration of its processing.
- Callable data for passing data into and out of a callable shader.

Ray payloads and callable data are used in explicit shader call instructions, so they have an incoming variant to distinguish the parameter passed to the invocation from any other payloads or

data being used by subsequent shader call instructions.

An interface structure used between stages **must** match between the stages using it. Specifically:

- The hit attribute structure read in an any-hit or closest hit shader **must** be the same structure as the hit attribute structure written in the corresponding intersection shader in the same hit group.
- The incoming callable data for a callable shader **must** be the same structure as the callable data referenced by the execute callable instruction in the calling shader.
- The ray payload for a shader invoked by a ray tracing command **must** be the same structure for all shader stages using the payload for that ray.

Any shader with an incoming ray payload, incoming callable data, or hit attribute **must** only declare one variable of that type.

Table 15. Ray Pipeline Shader Interface

Shader Stage	Ray Payload	Incoming Ray Payload	Hit Attribute	Callable Data	Incoming Callable Data
Ray Generation	r/w			r/w	
Intersection			r/w		
Any-Hit		r/w	r		
Closest Hit	r/w	r/w	r	r/w	
Miss	r/w	r/w		r/w	
Callable				r/w	r/w

15.7. Shader Resource Interface

When a shader stage accesses buffer or image resources, as described in the [Resource Descriptors](#) section, the shader resource variables **must** be matched with the [pipeline layout](#) that is provided at pipeline creation time.

The set of shader variables that form the *shader resource interface* for a stage are the variables statically used by that stage's `OpEntryPoint` with a storage class of `Uniform`, `UniformConstant`, `StorageBuffer`, or `PushConstant`. For the fragment shader, this includes the [fragment input attachment interface](#).

The shader resource interface consists of two sub-interfaces: the push constant interface and the descriptor set interface.

15.7.1. Push Constant Interface

The shader variables defined with a storage class of `PushConstant` that are statically used by the shader entry points for the pipeline define the *push constant interface*. They **must** be:

- typed as `OpTypeStruct`,
- identified with a `Block` decoration, and
- laid out explicitly using the `Offset`, `ArrayStride`, and `MatrixStride` decorations as specified in [Offset and Stride Assignment](#).

There **must** be no more than one push constant block statically used per shader entry point.

Each statically used member of a push constant block **must** be placed at an `Offset` such that the entire member is entirely contained within the `VkPushConstantRange` for each `OpEntryPoint` that uses it, and the `stageFlags` for that range **must** specify the appropriate `VkShaderStageFlagBits` for that stage. The `Offset` decoration for any member of a push constant block **must** not cause the space required for that member to extend outside the range `[0, maxPushConstantsSize)`.

Any member of a push constant block that is declared as an array **must** only be accessed with *dynamically uniform* indices.

15.7.2. Descriptor Set Interface

The *descriptor set interface* is comprised of the shader variables with the storage class of `StorageBuffer`, `Uniform` or `UniformConstant` (including the variables in the [fragment input attachment interface](#)) that are statically used by the shader entry points for the pipeline.

These variables **must** have `DescriptorSet` and `Binding` decorations specified, which are assigned and matched with the `VkDescriptorSetLayout` objects in the pipeline layout as described in [DescriptorSet and Binding Assignment](#).

The `Image Format` of an `OpTypeImage` declaration **must** not be `Unknown`, for variables which are used for `OpImageRead`, `OpImageSparseRead`, or `OpImageWrite` operations, except under the following conditions:

- For `OpImageWrite`, if the image format is listed in the [storage without format](#) list and if the `shaderStorageImageWriteWithoutFormat` feature is enabled and the shader module declares the `StorageImageWriteWithoutFormat` capability.
- For `OpImageWrite`, if the image format supports `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT` and the shader module declares the `StorageImageWriteWithoutFormat` capability.
- For `OpImageRead` or `OpImageSparseRead`, if the image format is listed in the [storage without format](#) list and if the `shaderStorageImageReadWithoutFormat` feature is enabled and the shader module declares the `StorageImageReadWithoutFormat` capability.
- For `OpImageRead` or `OpImageSparseRead`, if the image format supports `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT` and the shader module declares the `StorageImageReadWithoutFormat` capability.
- For `OpImageRead`, if `Dim` is `SubpassData` (indicating a read from an input attachment).

The `Image Format` of an `OpTypeImage` declaration **must** not be `Unknown`, for variables which are used for `OpAtomic*` operations.

Variables identified with the **Uniform** storage class are used to access transparent buffer backed resources. Such variables **must** be:

- typed as **OpTypeStruct**, or an array of this type,
- identified with a **Block** or **BufferBlock** decoration, and
- laid out explicitly using the **Offset**, **ArrayStride**, and **MatrixStride** decorations as specified in [Offset and Stride Assignment](#).

Variables identified with the **StorageBuffer** storage class are used to access transparent buffer backed resources. Such variables **must** be:

- typed as **OpTypeStruct**, or an array of this type,
- identified with a **Block** decoration, and
- laid out explicitly using the **Offset**, **ArrayStride**, and **MatrixStride** decorations as specified in [Offset and Stride Assignment](#).

The **Offset** decoration for any member of a **Block**-decorated variable in the **Uniform** storage class **must** not cause the space required for that variable to extend outside the range [0, **maxUniformBufferRange**). The **Offset** decoration for any member of a **Block**-decorated variable in the **StorageBuffer** storage class **must** not cause the space required for that variable to extend outside the range [0, **maxStorageBufferRange**).

Variables identified with a storage class of **UniformConstant** and a decoration of **InputAttachmentIndex** **must** be declared as described in [Fragment Input Attachment Interface](#).

SPIR-V variables decorated with a descriptor set and binding that identify a **combined image sampler descriptor** **can** have a type of **OpTypeImage**, **OpTypeSampler** (**Sampled=1**), or **OpTypeSampledImage**.

When accessing a resource through such a variable, the resource **must** be selected via compile time constant expressions unless features are enabled to allow dynamically uniform or non-uniform expressions, as described below:

- Storage images (except storage texel buffers and input attachments):
 - Dynamically uniform: **shaderStorageImageArrayDynamicIndexing** and **StorageImageArrayDynamicIndexing**
- Sampled images (except uniform texel buffers), samplers and combined image samplers:
 - Dynamically uniform: **shaderSampledImageArrayDynamicIndexing** and **SampledImageArrayDynamicIndexing**
- Uniform buffers:
 - Dynamically uniform: **shaderUniformBufferArrayDynamicIndexing** and **UniformBufferArrayDynamicIndexing**
- Storage buffers:
 - Dynamically uniform: **shaderStorageBufferArrayDynamicIndexing** and **StorageBufferArrayDynamicIndexing**
- Acceleration structures:

- Dynamically uniform: Always supported.
- Non-uniform: Always supported.

A combined image sampler in an array that enables sampler Y'C_BC_R conversion **must** only be indexed by constant integral expressions.

Table 16. Shader Resource and Descriptor Type Correspondence

Resource type	Descriptor Type
sampler	VK_DESCRIPTOR_TYPE_SAMPLER or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER
sampled image	VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE or VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER
storage image	VK_DESCRIPTOR_TYPE_STORAGE_IMAGE
combined image sampler	VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER
uniform texel buffer	VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER
storage texel buffer	VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER
uniform buffer	VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER or VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC
storage buffer	VK_DESCRIPTOR_TYPE_STORAGE_BUFFER or VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC
input attachment	VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT
acceleration structure	VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR

Table 17. Shader Resource and Storage Class Correspondence

Resource type	Storage Class	Type ¹	Decoration(s) ²
sampler	UniformConstant	OpTypeSampler	
sampled image	UniformConstant	OpTypeImage (Sampled=1)	
storage image	UniformConstant	OpTypeImage (Sampled=2)	
combined image sampler	UniformConstant	OpTypeSampledImage OpTypeImage (Sampled=1) OpTypeSampler	
uniform texel buffer	UniformConstant	OpTypeImage (Dim=Buffer, Sampled=1)	
storage texel buffer	UniformConstant	OpTypeImage (Dim=Buffer, Sampled=2)	
uniform buffer	Uniform	OpTypeStruct	Block, Offset, (ArrayStride), (MatrixStride)

Resource type	Storage Class	Type ¹	Decoration(s) ²
storage buffer	Uniform	OpTypeStruct	BufferBlock, Offset, (ArrayStride), (MatrixStride)
	StorageBuffer		Block, Offset, (ArrayStride), (MatrixStride)
input attachment	UniformConstant	OpTypeImage (Dim =SubpassData, Sampled=2)	InputAttachmentIndex
acceleration structure	UniformConstant	OpTypeAccelerationStructureKHR	

1

Where **OpTypeImage** is referenced, the **Dim** values **Buffer** and **Subpassdata** are only accepted where they are specifically referenced. They do not correspond to resource types where a generic **OpTypeImage** is specified.

2

In addition to **DescriptorSet** and **Binding**.

15.7.3. DescriptorSet and Binding Assignment

A variable decorated with a **DescriptorSet** decoration of *s* and a **Binding** decoration of *b* indicates that this variable is associated with the **VkDescriptorSetLayoutBinding** that has a **binding** equal to *b* in **pSetLayouts[s]** that was specified in **VkPipelineLayoutCreateInfo**.

DescriptorSet decoration values **must** be between zero and **maxBoundDescriptorSets** minus one, inclusive. **Binding** decoration values **can** be any 32-bit unsigned integer value, as described in **Descriptor Set Layout**. Each descriptor set has its own binding name space.

If the **Binding** decoration is used with an array, the entire array is assigned that binding value. The array **must** be a single-dimensional array and size of the array **must** be no larger than the number of descriptors in the binding. The array **must** not be runtime-sized. The index of each element of the array is referred to as the *arrayElement*. For the purposes of interface matching and descriptor set **operations**, if a resource variable is not an array, it is treated as if it has an *arrayElement* of zero.

There is a limit on the number of resources of each type that **can** be accessed by a pipeline stage as shown in **Shader Resource Limits**. The “Resources Per Stage” column gives the limit on the number each type of resource that **can** be statically used for an entry point in any given stage in a pipeline. The “Resource Types” column lists which resource types are counted against the limit. Some resource types count against multiple limits.

The pipeline layout **may** include descriptor sets and bindings which are not referenced by any variables statically used by the entry points for the shader stages in the binding’s **stageFlags**.

However, if a variable assigned to a given **DescriptorSet** and **Binding** is statically used by the entry point for a shader stage, the pipeline layout **must** contain a descriptor set layout binding in that descriptor set layout and for that binding number, and that binding’s **stageFlags** **must** include the appropriate **VkShaderStageFlagBits** for that stage. The variable **must** be of a valid resource type

determined by its SPIR-V type and storage class, as defined in [Shader Resource and Storage Class Correspondence](#). The descriptor set layout binding **must** be of a corresponding descriptor type, as defined in [Shader Resource and Descriptor Type Correspondence](#).

NOTE

There are no limits on the number of shader variables that can have overlapping set and binding values in a shader; but which resources are [statically used](#) has an impact. If any shader variable identifying a resource is [statically used](#) in a shader, then the underlying descriptor bound at the declared set and binding **must support the declared type in the shader** when the shader executes.

If multiple shader variables are declared with the same set and binding values, and with the same underlying descriptor type, they can all be statically used within the same shader. However, accesses are not automatically synchronized, and [Aliased](#) decorations should be used to avoid data hazards (see [section 2.18.2 Aliasing in the SPIR-V specification](#)).

If multiple shader variables with the same set and binding values are declared in a single shader, but with different declared types, where any of those are not supported by the relevant bound descriptor, that shader can only be executed if the variables with the unsupported type are not statically used.

A noteworthy example of using multiple statically-used shader variables sharing the same descriptor set and binding values is a descriptor of type [VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER](#) that has multiple corresponding shader variables in the [UniformConstant](#) storage class, where some could be [OpTypeImage](#) ([Sampled=1](#)), some could be [OpTypeSampler](#), and some could be [OpTypeSampledImage](#).

Table 18. Shader Resource Limits

Resources per Stage	Resource Types
maxPerStageDescriptorSamplers	sampler
	combined image sampler
maxPerStageDescriptorSampledImages	sampled image
	combined image sampler
	uniform texel buffer
maxPerStageDescriptorStorageImages	storage image
	storage texel buffer
maxPerStageDescriptorUniformBuffers	uniform buffer
	uniform buffer dynamic
maxPerStageDescriptorStorageBuffers	storage buffer
	storage buffer dynamic
maxPerStageDescriptorInputAttachments	input attachment ¹

Resources per Stage	Resource Types
<code>maxPerStageDescriptorAccelerationStructures</code> or <code>maxPerStageDescriptorUpdateAfterBindAccelerationStructures</code>	acceleration structure

1

Input attachments **can** only be used in the fragment shader stage

15.7.4. Offset and Stride Assignment

Certain objects **must** be explicitly laid out using the `Offset`, `ArrayStride`, and `MatrixStride`, as described in [SPIR-V explicit layout validation rules](#). All such layouts also **must** conform to the following requirements.

NOTE

The numeric order of `Offset` decorations does not need to follow member declaration order.

Alignment Requirements

There are different alignment requirements depending on the specific resources and on the features enabled on the device.

Matrix types are defined in terms of arrays as follows:

- A column-major matrix with C columns and R rows is equivalent to a C element array of vectors with R components.
- A row-major matrix with C columns and R rows is equivalent to an R element array of vectors with C components.

The *scalar alignment* of the type of an `OpTypeStruct` member is defined recursively as follows:

- A scalar of size N has a scalar alignment of N.
- A vector type has a scalar alignment equal to that of its component type.
- An array type has a scalar alignment equal to that of its element type.
- A structure has a scalar alignment equal to the largest scalar alignment of any of its members.
- A matrix type inherits *scalar alignment* from the equivalent array declaration.

The *base alignment* of the type of an `OpTypeStruct` member is defined recursively as follows:

- A scalar has a base alignment equal to its scalar alignment.
- A two-component vector has a base alignment equal to twice its scalar alignment.
- A three- or four-component vector has a base alignment equal to four times its scalar alignment.
- An array has a base alignment equal to the base alignment of its element type.
- A structure has a base alignment equal to the largest base alignment of any of its members. An empty structure has a base alignment equal to the size of the smallest scalar type permitted by

the capabilities declared in the SPIR-V module. (e.g., for a 1 byte aligned empty struct in the `StorageBuffer` storage class, `StorageBuffer8BitAccess` or `UniformAndStorageBuffer8BitAccess` **must** be declared in the SPIR-V module.)

- A matrix type inherits *base alignment* from the equivalent array declaration.

The *extended alignment* of the type of an `OpTypeStruct` member is similarly defined as follows:

- A scalar or vector type has an extended alignment equal to its base alignment.
- An array or structure type has an extended alignment equal to the largest extended alignment of any of its members, rounded up to a multiple of 16.
- A matrix type inherits extended alignment from the equivalent array declaration.

A member is defined to *improperly straddle* if either of the following are true:

- It is a vector with total size less than or equal to 16 bytes, and has `Offset` decorations placing its first byte at F and its last byte at L, where $\text{floor}(F / 16) \neq \text{floor}(L / 16)$.
- It is a vector with total size greater than 16 bytes and has its `Offset` decorations placing its first byte at a non-integer multiple of 16.

Standard Buffer Layout

Every member of an `OpTypeStruct` that is required to be explicitly laid out **must** be aligned according to the first matching rule as follows. If the struct is contained in pointer types of multiple storage classes, it **must** satisfy the requirements for every storage class used to reference it.

1. If the `workgroupMemoryExplicitLayoutScalarBlockLayout` feature is enabled on the device and the storage class is `Workgroup` then every member **must** be aligned according to its scalar alignment.
2. All vectors **must** be aligned according to their scalar alignment.
3. If the `uniformBufferStandardLayout` feature is not enabled on the device, then any member of an `OpTypeStruct` with a storage class of `Uniform` and a decoration of `Block` **must** be aligned according to its extended alignment.
4. Every other member **must** be aligned according to its base alignment.

The memory layout **must** obey the following rules:

- The `Offset` decoration of any member **must** be a multiple of its alignment.
- Any `ArrayStride` or `MatrixStride` decoration **must** be a multiple of the alignment of the array or matrix as defined above.

If one of the conditions below applies

- The storage class is `Workgroup`, and either the struct member is not part of a `Block` or the `workgroupMemoryExplicitLayoutScalarBlockLayout` feature is not enabled on the device.
- The storage class is any other storage class.

the memory layout **must** also obey the following rules:

- Vectors **must** not improperly straddle, as defined above.
- The **Offset** decoration of a member **must** not place it between the end of a structure, an array or a matrix and the next multiple of the alignment of that structure, array or matrix.

NOTE

The **std430 layout** in GLSL satisfies these rules for types using the base alignment.
The **std140 layout** satisfies the rules for types using the extended alignment.

15.8. Built-In Variables

Built-in variables are accessed in shaders by declaring a variable decorated with a **BuiltIn** SPIR-V decoration. The meaning of each **BuiltIn** decoration is as follows. In the remainder of this section, the name of a built-in is used interchangeably with a term equivalent to a variable decorated with that particular built-in. Built-ins that represent integer values **can** be declared as either signed or unsigned 32-bit integers.

As mentioned above, some inputs and outputs have an additional level of arrayness relative to other shader inputs and outputs. This level of arrayness is not included in the type descriptions below, but **must** be included when declaring the built-in.

Any two **Input** storage class **OpVariable** declarations listed as operands on the same **OpEntryPoint** **must** not have the same **BuiltIn** decoration. Any two **Output** storage class **OpVariable** declarations listed as operands on the same **OpEntryPoint** **must** not have the same **BuiltIn** decoration.

BaryCoordKHR

The **BaryCoordKHR** decoration **can** be used to decorate a fragment shader input variable. This variable will contain a three-component floating-point vector with barycentric weights that indicate the location of the fragment relative to the screen-space locations of vertices of its primitive, obtained using perspective interpolation.

Valid Usage

- VUID-BaryCoordKHR-BaryCoordKHR-04154
The **BaryCoordKHR** decoration **must** be used only within the **Fragment Execution Model**
- VUID-BaryCoordKHR-BaryCoordKHR-04155
The variable decorated with **BaryCoordKHR** **must** be declared using the **Input Storage Class**
- VUID-BaryCoordKHR-BaryCoordKHR-04156
The variable decorated with **BaryCoordKHR** **must** be declared as a three-component vector of 32-bit floating-point values

BaryCoordNoPerspKHR

The **BaryCoordNoPerspKHR** decoration **can** be used to decorate a fragment shader input variable. This variable will contain a three-component floating-point vector with barycentric weights that indicate the location of the fragment relative to the screen-space locations of vertices of its primitive, obtained using linear interpolation.

Valid Usage

- VUID-BaryCoordNoPerspKHR-BaryCoordNoPerspKHR-04160
The **BaryCoordNoPerspKHR** decoration **must** be used only within the **Fragment Execution Model**
- VUID-BaryCoordNoPerspKHR-BaryCoordNoPerspKHR-04161
The variable decorated with **BaryCoordNoPerspKHR** **must** be declared using the **Input Storage Class**
- VUID-BaryCoordNoPerspKHR-BaryCoordNoPerspKHR-04162
The variable decorated with **BaryCoordNoPerspKHR** **must** be declared as a three-component vector of 32-bit floating-point values

BaseInstance

Decorating a variable with the **BaseInstance** built-in will make that variable contain the integer value corresponding to the first instance that was passed to the command that invoked the current vertex shader invocation. **BaseInstance** is the **firstInstance** parameter to a *direct drawing command* or the **firstInstance** member of a structure consumed by an *indirect drawing command*.

Valid Usage

- VUID-BaseInstance-BaseInstance-04181
The **BaseInstance** decoration **must** be used only within the **Vertex Execution Model**
- VUID-BaseInstance-BaseInstance-04182
The variable decorated with **BaseInstance** **must** be declared using the **Input Storage Class**
- VUID-BaseInstance-BaseInstance-04183
The variable decorated with **BaseInstance** **must** be declared as a scalar 32-bit integer value

BaseVertex

Decorating a variable with the **BaseVertex** built-in will make that variable contain the integer value corresponding to the first vertex or vertex offset that was passed to the command that invoked the current vertex shader invocation. For *non-indexed drawing commands*, this variable is the **firstVertex** parameter to a *direct drawing command* or the **firstVertex** member of the structure consumed by an *indirect drawing command*. For *indexed drawing commands*, this variable is the **vertexOffset** parameter to a *direct drawing command* or the **vertexOffset** member of the structure consumed by an *indirect drawing command*.

Valid Usage

- VUID-BaseVertex-BaseVertex-04184
The **BaseVertex** decoration **must** be used only within the **Vertex Execution Model**
- VUID-BaseVertex-BaseVertex-04185
The variable decorated with **BaseVertex** **must** be declared using the **Input Storage Class**

- VUID-BaseVertex-BaseVertex-04186

The variable decorated with **BaseVertex** **must** be declared as a scalar 32-bit integer value

ClipDistance

Decorating a variable with the **ClipDistance** built-in decoration will make that variable contain the mechanism for controlling user clipping. **ClipDistance** is an array such that the i^{th} element of the array specifies the clip distance for plane i . A clip distance of 0 means the vertex is on the plane, a positive distance means the vertex is inside the clip half-space, and a negative distance means the vertex is outside the clip half-space.

NOTE The array variable decorated with **ClipDistance** is explicitly sized by the shader.

NOTE In the last **pre-rasterization shader stage**, these values will be linearly interpolated across the primitive and the portion of the primitive with interpolated distances less than 0 will be considered outside the clip volume. If **ClipDistance** is then used by a fragment shader, **ClipDistance** contains these linearly interpolated values.

Valid Usage

- VUID-ClipDistance-ClipDistance-04187
The **ClipDistance** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **Vertex**, **Fragment**, **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model**
- VUID-ClipDistance-ClipDistance-04188
The variable decorated with **ClipDistance** within the **MeshEXT**, **MeshNV**, or **Vertex Execution Model** **must** be declared using the **Output Storage Class**
- VUID-ClipDistance-ClipDistance-04189
The variable decorated with **ClipDistance** within the **Fragment Execution Model** **must** be declared using the **Input Storage Class**
- VUID-ClipDistance-ClipDistance-04190
The variable decorated with **ClipDistance** within the **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model** **must** not be declared in a **Storage Class** other than **Input** or **Output**
- VUID-ClipDistance-ClipDistance-04191
The variable decorated with **ClipDistance** **must** be declared as an array of 32-bit floating-point values

CullDistance

Decorating a variable with the **CullDistance** built-in decoration will make that variable contain the mechanism for controlling user culling. If any member of this array is assigned a negative value for all vertices belonging to a primitive, then the primitive is discarded before rasterization.

NOTE In fragment shaders, the values of the **CullDistance** array are linearly interpolated

across each primitive.

NOTE

If **CullDistance** decorates an input variable, that variable will contain the corresponding value from the **CullDistance** decorated output variable from the previous shader stage.

Valid Usage

- VUID-CullDistance-CullDistance-04196
The **CullDistance** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **Vertex**, **Fragment**, **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model**
- VUID-CullDistance-CullDistance-04197
The variable decorated with **CullDistance** within the **MeshEXT**, **MeshNV** or **Vertex Execution Model** **must** be declared using the **Output Storage Class**
- VUID-CullDistance-CullDistance-04198
The variable decorated with **CullDistance** within the **Fragment Execution Model** **must** be declared using the **Input Storage Class**
- VUID-CullDistance-CullDistance-04199
The variable decorated with **CullDistance** within the **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model** **must** not be declared using a **Storage Class** other than **Input** or **Output**
- VUID-CullDistance-CullDistance-04200
The variable decorated with **CullDistance** **must** be declared as an array of 32-bit floating-point values

CullMaskKHR

A variable decorated with the **CullMaskKHR** decoration will specify the cull mask of the ray being processed. The value is given by the **Cull Mask** parameter passed into one of the **OpTrace*** instructions.

Valid Usage

- VUID-CullMaskKHR-CullMaskKHR-06735
The **CullMaskKHR** decoration **must** be used only within the **IntersectionKHR**, **AnyHitKHR**, **ClosestHitKHR**, or **MissKHR Execution Model**
- VUID-CullMaskKHR-CullMaskKHR-06736
The variable decorated with **CullMaskKHR** **must** be declared using the **Input Storage Class**
- VUID-CullMaskKHR-CullMaskKHR-06737
The variable decorated with **CullMaskKHR** **must** be declared as a scalar 32-bit integer value

DeviceIndex

The **DeviceIndex** decoration **can** be applied to a shader input which will be filled with the device index of the physical device that is executing the current shader invocation. This value will be in

the range $[0, \max(1, \text{physicalDeviceCount})]$, where `physicalDeviceCount` is the `physicalDeviceCount` member of `VkDeviceGroupDeviceCreateInfo`.

Valid Usage

- VUID-DeviceIndex-DeviceIndex-04205
The variable decorated with `DeviceIndex` **must** be declared using the `Input Storage Class`
- VUID-DeviceIndex-DeviceIndex-04206
The variable decorated with `DeviceIndex` **must** be declared as a scalar 32-bit integer value

DrawIndex

Decorating a variable with the `DrawIndex` built-in will make that variable contain the integer value corresponding to the zero-based index of the draw that invoked the current vertex shader invocation. For *indirect drawing commands*, `DrawIndex` begins at zero and increments by one for each draw executed. The number of draws is given by the `drawCount` parameter. For *direct drawing commands*, if `vkCmdDrawMultiEXT` or `vkCmdDrawMultiIndexedEXT` is used, this variable contains the integer value corresponding to the zero-based index of the draw. Otherwise `DrawIndex` is always zero. `DrawIndex` is dynamically uniform.

Valid Usage

- VUID-DrawIndex-DrawIndex-04207
The `DrawIndex` decoration **must** be used only within the `Vertex`, `MeshEXT`, `TaskEXT`, `MeshNV`, or `TaskNV Execution Model`
- VUID-DrawIndex-DrawIndex-04208
The variable decorated with `DrawIndex` **must** be declared using the `Input Storage Class`
- VUID-DrawIndex-DrawIndex-04209
The variable decorated with `DrawIndex` **must** be declared as a scalar 32-bit integer value

FragCoord

Decorating a variable with the `FragCoord` built-in decoration will make that variable contain the framebuffer coordinate $(x, y, z, \frac{1}{w})$ of the fragment being processed. The (x,y) coordinate $(0,0)$ is the upper left corner of the upper left pixel in the framebuffer.

When `Sample Shading` is enabled, the x and y components of `FragCoord` reflect the location of one of the samples corresponding to the shader invocation.

Otherwise, the x and y components of `FragCoord` reflect the location of the center of the fragment.

The z component of `FragCoord` is the interpolated depth value of the primitive.

The w component is the interpolated $\frac{1}{w}$.

The `Centroid` interpolation decoration is ignored, but allowed, on `FragCoord`.

Valid Usage

- VUID-FragCoord-FragCoord-04210
The **FragCoord** decoration **must** be used only within the **Fragment Execution Model**
- VUID-FragCoord-FragCoord-04211
The variable decorated with **FragCoord** **must** be declared using the **Input Storage Class**
- VUID-FragCoord-FragCoord-04212
The variable decorated with **FragCoord** **must** be declared as a four-component vector of 32-bit floating-point values

FragDepth

To have a shader supply a fragment-depth value, the shader **must** declare the **DepthReplacing** execution mode. Such a shader's fragment-depth value will come from the variable decorated with the **FragDepth** built-in decoration.

This value will be used for any subsequent depth testing performed by the implementation or writes to the depth attachment. See [fragment shader depth replacement](#) for details.

Valid Usage

- VUID-FragDepth-FragDepth-04213
The **FragDepth** decoration **must** be used only within the **Fragment Execution Model**
- VUID-FragDepth-FragDepth-04214
The variable decorated with **FragDepth** **must** be declared using the **Output Storage Class**
- VUID-FragDepth-FragDepth-04215
The variable decorated with **FragDepth** **must** be declared as a scalar 32-bit floating-point value
- VUID-FragDepth-FragDepth-04216
If the shader dynamically writes to the variable decorated with **FragDepth**, the **DepthReplacing Execution Mode** **must** be declared

FragStencilRefEXT

Decorating a variable with the **FragStencilRefEXT** built-in decoration will make that variable contain the new stencil reference value for all samples covered by the fragment. This value will be used as the stencil reference value used in stencil testing.

To write to **FragStencilRefEXT**, a shader **must** declare the **StencilRefReplacingEXT** execution mode. If a shader declares the **StencilRefReplacingEXT** execution mode and there is an execution path through the shader that does not set **FragStencilRefEXT**, then the fragment's stencil reference value is undefined for executions of the shader that take that path.

Only the least significant **s** bits of the integer value of the variable decorated with **FragStencilRefEXT** are considered for stencil testing, where **s** is the number of bits in the stencil framebuffer attachment, and higher order bits are discarded.

See [fragment shader stencil reference replacement](#) for more details.

Valid Usage

- VUID-FragStencilRefEXT-FragStencilRefEXT-04223
The **FragStencilRefEXT** decoration **must** be used only within the **Fragment Execution Model**
- VUID-FragStencilRefEXT-FragStencilRefEXT-04224
The variable decorated with **FragStencilRefEXT** **must** be declared using the **Output Storage Class**
- VUID-FragStencilRefEXT-FragStencilRefEXT-04225
The variable decorated with **FragStencilRefEXT** **must** be declared as a scalar integer value

FrontFacing

Decorating a variable with the **FrontFacing** built-in decoration will make that variable contain whether the fragment is front or back facing. This variable is non-zero if the current fragment is considered to be part of a **front-facing** polygon primitive or of a non-polygon primitive and is zero if the fragment is considered to be part of a back-facing polygon primitive.

Valid Usage

- VUID-FrontFacing-FrontFacing-04229
The **FrontFacing** decoration **must** be used only within the **Fragment Execution Model**
- VUID-FrontFacing-FrontFacing-04230
The variable decorated with **FrontFacing** **must** be declared using the **Input Storage Class**
- VUID-FrontFacing-FrontFacing-04231
The variable decorated with **FrontFacing** **must** be declared as a boolean value

FullyCoveredEXT

Decorating a variable with the **FullyCoveredEXT** built-in decoration will make that variable indicate whether the **fragment area** is fully covered by the generating primitive. This variable is non-zero if conservative rasterization is enabled and the current fragment area is fully covered by the generating primitive, and is zero if the fragment is not covered or partially covered, or conservative rasterization is disabled.

Valid Usage

- VUID-FullyCoveredEXT-FullyCoveredEXT-04232
The **FullyCoveredEXT** decoration **must** be used only within the **Fragment Execution Model**
- VUID-FullyCoveredEXT-FullyCoveredEXT-04233
The variable decorated with **FullyCoveredEXT** **must** be declared using the **Input Storage Class**
- VUID-FullyCoveredEXT-FullyCoveredEXT-04234
The variable decorated with **FullyCoveredEXT** **must** be declared as a boolean value

- VUID-FullyCoveredEXT-conservativeRasterizationPostDepthCoverage-04235
If `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::conservativeRasterizationPostDepthCoverage` is not supported the `PostDepthCoverage Execution Mode` **must** not be declared, when a variable with the `FullyCoveredEXT` decoration is declared

GlobalInvocationId

Decorating a variable with the `GlobalInvocationId` built-in decoration will make that variable contain the location of the current invocation within the global workgroup. Each component is equal to the index of the local workgroup multiplied by the size of the local workgroup plus `LocalInvocationId`.

Valid Usage

- VUID-GlobalInvocationId-GlobalInvocationId-04236
The `GlobalInvocationId` decoration **must** be used only within the `GLCompute`, `MeshEXT`, `TaskEXT`, `MeshNV`, or `TaskNV Execution Model`
- VUID-GlobalInvocationId-GlobalInvocationId-04237
The variable decorated with `GlobalInvocationId` **must** be declared using the `Input Storage Class`
- VUID-GlobalInvocationId-GlobalInvocationId-04238
The variable decorated with `GlobalInvocationId` **must** be declared as a three-component vector of 32-bit integer values

HelperInvocation

Decorating a variable with the `HelperInvocation` built-in decoration will make that variable contain whether the current invocation is a helper invocation. This variable is non-zero if the current fragment being shaded is a helper invocation and zero otherwise. A helper invocation is an invocation of the shader that is produced to satisfy internal requirements such as the generation of derivatives.

NOTE It is very likely that a helper invocation will have a value of `SampleMask` fragment shader input value that is zero.

Valid Usage

- VUID-HelperInvocation-HelperInvocation-04239
The `HelperInvocation` decoration **must** be used only within the `Fragment Execution Model`
- VUID-HelperInvocation-HelperInvocation-04240
The variable decorated with `HelperInvocation` **must** be declared using the `Input Storage Class`
- VUID-HelperInvocation-HelperInvocation-04241
The variable decorated with `HelperInvocation` **must** be declared as a boolean value

HitKindKHR

A variable decorated with the `HitKindKHR` decoration will describe the intersection that triggered the execution of the current shader. The values are determined by the intersection shader. For user-defined intersection shaders this is the value that was passed to the “Hit Kind” operand of `OpReportIntersectionKHR`. For triangle intersection candidates, this will be one of `HitKindFrontFacingTriangleKHR` or `HitKindBackFacingTriangleKHR`.

Valid Usage

- VUID-HitKindKHR-HitKindKHR-04242
The `HitKindKHR` decoration **must** be used only within the `AnyHitKHR` or `ClosestHitKHR Execution Model`
- VUID-HitKindKHR-HitKindKHR-04243
The variable decorated with `HitKindKHR` **must** be declared using the `Input Storage Class`
- VUID-HitKindKHR-HitKindKHR-04244
The variable decorated with `HitKindKHR` **must** be declared as a scalar 32-bit integer value

HitTriangleVertexPositionsKHR

A variable decorated with the `HitTriangleVertexPositionsKHR` decoration will specify the object space vertices of the triangle at the current intersection in application-provided order. The positions returned are transformed by the geometry transform, which is performed at standard `floating-point` precision, but without a specifically defined order of floating-point operations to perform the matrix multiplication.

Valid Usage

- VUID-HitTriangleVertexPositionsKHR-HitTriangleVertexPositionsKHR-08747
The `HitTriangleVertexPositionsKHR` decoration **must** be used only within the `AnyHitKHR` or `ClosestHitKHR Execution Model`
- VUID-HitTriangleVertexPositionsKHR-HitTriangleVertexPositionsKHR-08748
The variable decorated with `HitTriangleVertexPositionsKHR` **must** be declared using the `Input Storage Class`
- VUID-HitTriangleVertexPositionsKHR-HitTriangleVertexPositionsKHR-08749
The variable decorated with `HitTriangleVertexPositionsKHR` **must** be declared as an array of three vectors of three 32-bit float values
- VUID-HitTriangleVertexPositionsKHR-HitTriangleVertexPositionsKHR-08750
The variable decorated with `HitTriangleVertexPositionsKHR` **must** be used only if the value of `HitKindKHR` is `HitKindFrontFacingTriangleKHR` or `HitKindBackFacingTriangleKHR`
- VUID-HitTriangleVertexPositionsKHR-None-08751
The acceleration structure corresponding to the current intersection **must** have been built with `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_KHR`

IncomingRayFlagsKHR

A variable with the `IncomingRayFlagsKHR` decoration will contain the ray flags passed in to the trace call that invoked this particular shader. Setting pipeline flags on the ray tracing pipeline **must** not cause any corresponding flags to be set in variables with this decoration.

Valid Usage

- VUID-IncomingRayFlagsKHR-IncomingRayFlagsKHR-04248
The `IncomingRayFlagsKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, `ClosestHitKHR`, or `MissKHR Execution Model`
- VUID-IncomingRayFlagsKHR-IncomingRayFlagsKHR-04249
The variable decorated with `IncomingRayFlagsKHR` **must** be declared using the `Input Storage Class`
- VUID-IncomingRayFlagsKHR-IncomingRayFlagsKHR-04250
The variable decorated with `IncomingRayFlagsKHR` **must** be declared as a scalar 32-bit integer value

InstanceCustomIndexKHR

A variable decorated with the `InstanceCustomIndexKHR` decoration will contain the application-defined value of the instance that intersects the current ray. This variable contains the value that was specified in `VkAccelerationStructureInstanceKHR::instanceCustomIndex` for the current acceleration structure instance in the lower 24 bits and the upper 8 bits will be zero.

Valid Usage

- VUID-InstanceCustomIndexKHR-InstanceCustomIndexKHR-04251
The `InstanceCustomIndexKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, or `ClosestHitKHR Execution Model`
- VUID-InstanceCustomIndexKHR-InstanceCustomIndexKHR-04252
The variable decorated with `InstanceCustomIndexKHR` **must** be declared using the `Input Storage Class`
- VUID-InstanceCustomIndexKHR-InstanceCustomIndexKHR-04253
The variable decorated with `InstanceCustomIndexKHR` **must** be declared as a scalar 32-bit integer value

InstanceId

Decorating a variable in an intersection, any-hit, or closest hit shader with the `InstanceId` decoration will make that variable contain the index of the instance that intersects the current ray.

Valid Usage

- VUID-InstanceId-InstanceId-04254

The **InstanceId** decoration **must** be used only within the **IntersectionKHR**, **AnyHitKHR**, or **ClosestHitKHR Execution Model**

- VUID-InstanceId-InstanceId-04255

The variable decorated with **InstanceId** **must** be declared using the **Input Storage Class**

- VUID-InstanceId-InstanceId-04256

The variable decorated with **InstanceId** **must** be declared as a scalar 32-bit integer value

InvocationId

Decorating a variable with the **InvocationId** built-in decoration will make that variable contain the index of the current shader invocation in a geometry shader, or the index of the output patch vertex in a tessellation control shader.

In a geometry shader, the index of the current shader invocation ranges from zero to the number of **instances** declared in the shader minus one. If the instance count of the geometry shader is one or is not specified, then **InvocationId** will be zero.

Valid Usage

- VUID-InvocationId-InvocationId-04257

The **InvocationId** decoration **must** be used only within the **TessellationControl** or **Geometry Execution Model**

- VUID-InvocationId-InvocationId-04258

The variable decorated with **InvocationId** **must** be declared using the **Input Storage Class**

- VUID-InvocationId-InvocationId-04259

The variable decorated with **InvocationId** **must** be declared as a scalar 32-bit integer value

InstanceIndex

Decorating a variable in a vertex shader with the **InstanceIndex** built-in decoration will make that variable contain the index of the instance that is being processed by the current vertex shader invocation. **InstanceIndex** begins at the **firstInstance** parameter to **vkCmdDraw** or **vkCmdDrawIndexed** or at the **firstInstance** member of a structure consumed by **vkCmdDrawIndirect** or **vkCmdDrawIndexedIndirect**.

Valid Usage

- VUID-InstanceIndex-InstanceIndex-04263

The **InstanceIndex** decoration **must** be used only within the **Vertex Execution Model**

- VUID-InstanceIndex-InstanceIndex-04264

The variable decorated with **InstanceIndex** **must** be declared using the **Input Storage Class**

- VUID-InstanceIndex-InstanceIndex-04265

The variable decorated with **InstanceIndex** **must** be declared as a scalar 32-bit integer value

LaunchIdKHR

A variable decorated with the **LaunchIdKHR** decoration will specify the index of the work item being processed. One work item is generated for each of the **width** × **height** × **depth** items dispatched by a **vkCmdTraceRaysKHR** command. All shader invocations inherit the same value for variables decorated with **LaunchIdKHR**.

Valid Usage

- VUID-LaunchIdKHR-LaunchIdKHR-04266
The **LaunchIdKHR** decoration **must** be used only within the **RayGenerationKHR**, **IntersectionKHR**, **AnyHitKHR**, **ClosestHitKHR**, **MissKHR**, or **CallableKHR Execution Model**
- VUID-LaunchIdKHR-LaunchIdKHR-04267
The variable decorated with **LaunchIdKHR** **must** be declared using the **Input Storage Class**
- VUID-LaunchIdKHR-LaunchIdKHR-04268
The variable decorated with **LaunchIdKHR** **must** be declared as a three-component vector of 32-bit integer values

LaunchSizeKHR

A variable decorated with the **LaunchSizeKHR** decoration will contain the **width**, **height**, and **depth** dimensions passed to the **vkCmdTraceRaysKHR** command that initiated this shader execution. The **width** is in the first component, the **height** is in the second component, and the **depth** is in the third component.

Valid Usage

- VUID-LaunchSizeKHR-LaunchSizeKHR-04269
The **LaunchSizeKHR** decoration **must** be used only within the **RayGenerationKHR**, **IntersectionKHR**, **AnyHitKHR**, **ClosestHitKHR**, **MissKHR**, or **CallableKHR Execution Model**
- VUID-LaunchSizeKHR-LaunchSizeKHR-04270
The variable decorated with **LaunchSizeKHR** **must** be declared using the **Input Storage Class**
- VUID-LaunchSizeKHR-LaunchSizeKHR-04271
The variable decorated with **LaunchSizeKHR** **must** be declared as a three-component vector of 32-bit integer values

Layer

Decorating a variable with the **Layer** built-in decoration will make that variable contain the select layer of a multi-layer framebuffer attachment.

In a geometry shader, any variable decorated with **Layer** can be written with the framebuffer layer index to which the primitive produced by that shader will be directed.

If the last active **pre-rasterization shader stage** shader entry point's interface does not include a variable decorated with **Layer**, then the first layer is used. If a **pre-rasterization shader stage** shader entry point's interface includes a variable decorated with **Layer**, it **must** write the same

value to **Layer** for all output vertices of a given primitive. If the **Layer** value is less than 0 or greater than or equal to the number of layers in the framebuffer, then primitives **may** still be rasterized, fragment shaders **may** be executed, and the framebuffer values for all layers are undefined.

In a fragment shader, a variable decorated with **Layer** contains the layer index of the primitive that the fragment invocation belongs to.

Valid Usage

- VUID-Layer-Layer-04272
The **Layer** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **Vertex**, **TessellationEvaluation**, **Geometry**, or **Fragment Execution Model**
- VUID-Layer-Layer-04274
The variable decorated with **Layer** within the **MeshEXT**, **MeshNV**, **Vertex**, **TessellationEvaluation**, or **Geometry Execution Model** **must** be declared using the **Output Storage Class**
- VUID-Layer-Layer-04275
The variable decorated with **Layer** within the **Fragment Execution Model** **must** be declared using the **Input Storage Class**
- VUID-Layer-Layer-04276
The variable decorated with **Layer** **must** be declared as a scalar 32-bit integer value
- VUID-Layer-Layer-07039
The variable decorated with **Layer** within the **MeshEXT Execution Model** **must** also be decorated with the **PerPrimitiveEXT** decoration

LocalInvocationId

Decorating a variable with the **LocalInvocationId** built-in decoration will make that variable contain the location of the current compute shader invocation within the local workgroup. Each component ranges from zero through to the size of the workgroup in that dimension minus one.

NOTE

If the size of the workgroup in a particular dimension is one, then the **LocalInvocationId** in that dimension will be zero. If the workgroup is effectively two-dimensional, then **LocalInvocationId.z** will be zero. If the workgroup is effectively one-dimensional, then both **LocalInvocationId.y** and **LocalInvocationId.z** will be zero.

Valid Usage

- VUID-LocalInvocationId-LocalInvocationId-04281
The **LocalInvocationId** decoration **must** be used only within the **GLCompute**, **MeshEXT**, **TaskEXT**, **MeshNV**, or **TaskNV Execution Model**
- VUID-LocalInvocationId-LocalInvocationId-04282
The variable decorated with **LocalInvocationId** **must** be declared using the **Input Storage**

Class

- VUID-LocalInvocationId-LocalInvocationId-04283

The variable decorated with **LocalInvocationId** **must** be declared as a three-component vector of 32-bit integer values

LocalInvocationIndex

Decorating a variable with the **LocalInvocationIndex** built-in decoration will make that variable contain a one-dimensional representation of **LocalInvocationId**. This is computed as:

```
LocalInvocationIndex =  
    LocalInvocationId.z * WorkgroupSize.x * WorkgroupSize.y +  
    LocalInvocationId.y * WorkgroupSize.x +  
    LocalInvocationId.x;
```

Valid Usage

- VUID-LocalInvocationIndex-LocalInvocationIndex-04284
The **LocalInvocationIndex** decoration **must** be used only within the **GLCompute**, **MeshEXT**, **TaskEXT**, **MeshNV**, or **TaskNV Execution Model**
- VUID-LocalInvocationIndex-LocalInvocationIndex-04285
The variable decorated with **LocalInvocationIndex** **must** be declared using the **Input Storage Class**
- VUID-LocalInvocationIndex-LocalInvocationIndex-04286
The variable decorated with **LocalInvocationIndex** **must** be declared as a scalar 32-bit integer value

NumWorkgroups

Decorating a variable with the **NumWorkgroups** built-in decoration will make that variable contain the number of local workgroups that are part of the dispatch that the invocation belongs to. Each component is equal to the values of the workgroup count parameters passed into the dispatching commands.

Valid Usage

- VUID-NumWorkgroups-NumWorkgroups-04296
The **NumWorkgroups** decoration **must** be used only within the **GLCompute**, **MeshEXT**, or **TaskEXT Execution Model**
- VUID-NumWorkgroups-NumWorkgroups-04297
The variable decorated with **NumWorkgroups** **must** be declared using the **Input Storage Class**
- VUID-NumWorkgroups-NumWorkgroups-04298
The variable decorated with **NumWorkgroups** **must** be declared as a three-component vector of 32-bit integer values

ObjectRayDirectionKHR

A variable decorated with the **ObjectRayDirectionKHR** decoration will specify the direction of the ray being processed, in object space.

Valid Usage

- VUID-ObjectRayDirectionKHR-ObjectRayDirectionKHR-04299
The **ObjectRayDirectionKHR** decoration **must** be used only within the **IntersectionKHR**, **AnyHitKHR**, or **ClosestHitKHR Execution Model**
- VUID-ObjectRayDirectionKHR-ObjectRayDirectionKHR-04300
The variable decorated with **ObjectRayDirectionKHR** **must** be declared using the **Input Storage Class**
- VUID-ObjectRayDirectionKHR-ObjectRayDirectionKHR-04301
The variable decorated with **ObjectRayDirectionKHR** **must** be declared as a three-component vector of 32-bit floating-point values

ObjectRayOriginKHR

A variable decorated with the **ObjectRayOriginKHR** decoration will specify the origin of the ray being processed, in object space.

Valid Usage

- VUID-ObjectRayOriginKHR-ObjectRayOriginKHR-04302
The **ObjectRayOriginKHR** decoration **must** be used only within the **IntersectionKHR**, **AnyHitKHR**, or **ClosestHitKHR Execution Model**
- VUID-ObjectRayOriginKHR-ObjectRayOriginKHR-04303
The variable decorated with **ObjectRayOriginKHR** **must** be declared using the **Input Storage Class**
- VUID-ObjectRayOriginKHR-ObjectRayOriginKHR-04304
The variable decorated with **ObjectRayOriginKHR** **must** be declared as a three-component vector of 32-bit floating-point values

ObjectToWorldKHR

A variable decorated with the **ObjectToWorldKHR** decoration will contain the current object-to-world transformation matrix, which is determined by the instance of the current intersection.

Valid Usage

- VUID-ObjectToWorldKHR-ObjectToWorldKHR-04305
The **ObjectToWorldKHR** decoration **must** be used only within the **IntersectionKHR**, **AnyHitKHR**, or **ClosestHitKHR Execution Model**
- VUID-ObjectToWorldKHR-ObjectToWorldKHR-04306
The variable decorated with **ObjectToWorldKHR** **must** be declared using the **Input Storage**

Class

- VUID-ObjectToWorldKHR-ObjectToWorldKHR-04307

The variable decorated with **ObjectToWorldKHR** **must** be declared as a matrix with four columns of three-component vectors of 32-bit floating-point values

PatchVertices

Decorating a variable with the **PatchVertices** built-in decoration will make that variable contain the number of vertices in the input patch being processed by the shader. In a Tessellation Control Shader, this is the same as the `name:patchControlPoints` member of [VkPipelineTessellationStateCreateInfo](#). In a Tessellation Evaluation Shader, **PatchVertices** is equal to the tessellation control output patch size. When the same shader is used in different pipelines where the patch sizes are configured differently, the value of the **PatchVertices** variable will also differ.

Valid Usage

- VUID-PatchVertices-PatchVertices-04308

The **PatchVertices** decoration **must** be used only within the **TessellationControl** or **TessellationEvaluation Execution Model**

- VUID-PatchVertices-PatchVertices-04309

The variable decorated with **PatchVertices** **must** be declared using the **Input Storage Class**

- VUID-PatchVertices-PatchVertices-04310

The variable decorated with **PatchVertices** **must** be declared as a scalar 32-bit integer value

PointCoord

Decorating a variable with the **PointCoord** built-in decoration will make that variable contain the coordinate of the current fragment within the point being rasterized, normalized to the size of the point with origin in the upper left corner of the point, as described in [Basic Point Rasterization](#). If the primitive the fragment shader invocation belongs to is not a point, then the variable decorated with **PointCoord** contains an undefined value.

NOTE

Depending on how the point is rasterized, **PointCoord** **may** never reach (0,0) or (1,1).

Valid Usage

- VUID-PointCoord-PointCoord-04311

The **PointCoord** decoration **must** be used only within the **Fragment Execution Model**

- VUID-PointCoord-PointCoord-04312

The variable decorated with **PointCoord** **must** be declared using the **Input Storage Class**

- VUID-PointCoord-PointCoord-04313

The variable decorated with **PointCoord** **must** be declared as a two-component vector of 32-bit floating-point values

PointSize

Decorating a variable with the **PointSize** built-in decoration will make that variable contain the size of point primitives or the final rasterization of polygons if **polygon mode** is **VK_POLYGON_MODE_POINT** when **VkPhysicalDeviceMaintenance5PropertiesKHR::polygonModePointSize** is set to **VK_TRUE**. The value written to the variable decorated with **PointSize** by the last **pre-rasterization shader stage** in the pipeline is used as the framebuffer-space size of points produced by rasterization. If **maintenance5** is enabled and a value is not written to a variable decorated with **PointSize**, a value of 1.0 is used as the size of points.

NOTE

When **PointSize** decorates a variable in the **Input Storage Class**, it contains the data written to the output variable decorated with **PointSize** from the previous shader stage.

Valid Usage

- VUID-PointSize-PointSize-04314
The **PointSize** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **Vertex**, **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model**
- VUID-PointSize-PointSize-04315
The variable decorated with **PointSize** within the **MeshEXT**, **MeshNV**, or **Vertex Execution Model** **must** be declared using the **Output Storage Class**
- VUID-PointSize-PointSize-04316
The variable decorated with **PointSize** within the **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model** **must** not be declared using a **Storage Class** other than **Input** or **Output**
- VUID-PointSize-PointSize-04317
The variable decorated with **PointSize** **must** be declared as a scalar 32-bit floating-point value

Position

Decorating a variable with the **Position** built-in decoration will make that variable contain the position of the current vertex. In the last **pre-rasterization shader stage**, the value of the variable decorated with **Position** is used in subsequent primitive assembly, clipping, and rasterization operations.

NOTE

When **Position** decorates a variable in the **Input Storage Class**, it contains the data written to the output variable decorated with **Position** from the previous shader stage.

Valid Usage

- VUID-Position-Position-04318
The **Position** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **Vertex**, **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model**

- VUID-Position-Position-04319
The variable decorated with **Position** within the **MeshEXT**, **MeshNV**, or **Vertex Execution Model** **must** be declared using the **Output Storage Class**
- VUID-Position-Position-04320
The variable decorated with **Position** within the **TessellationControl**, **TessellationEvaluation**, or **Geometry Execution Model** **must** not be declared using a **Storage Class** other than **Input** or **Output**
- VUID-Position-Position-04321
The variable decorated with **Position** **must** be declared as a four-component vector of 32-bit floating-point values

PrimitiveId

Decorating a variable with the **PrimitiveId** built-in decoration will make that variable contain the index of the current primitive.

The index of the first primitive generated by a drawing command is zero, and the index is incremented after every individual point, line, or triangle primitive is processed.

For triangles drawn as points or line segments (see **Polygon Mode**), the primitive index is incremented only once, even if multiple points or lines are eventually drawn.

Variables decorated with **PrimitiveId** are reset to zero between each instance drawn.

Restarting a primitive topology using primitive restart has no effect on the value of variables decorated with **PrimitiveId**.

In tessellation control and tessellation evaluation shaders, it will contain the index of the patch within the current set of rendering primitives that corresponds to the shader invocation.

In a geometry shader, it will contain the number of primitives presented as input to the shader since the current set of rendering primitives was started.

In a fragment shader, it will contain the primitive index written by the geometry shader if a geometry shader is present, or with the value that would have been presented as input to the geometry shader had it been present.

In an intersection, any-hit, or closest hit shader, it will contain the index within the geometry of the triangle or bounding box being processed.

NOTE

When the **PrimitiveId** decoration is applied to an output variable in the geometry shader, the resulting value is seen through the **PrimitiveId** decorated input variable in the fragment shader.

The fragment shader using **PrimitiveId** will need to declare either the **Geometry** or **Tessellation** capability to satisfy the requirement SPIR-V has to use **PrimitiveId**.

Valid Usage

- VUID-PrimitiveId-PrimitiveId-04330

The **PrimitiveId** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **IntersectionKHR**, **AnyHitKHR**, **ClosestHitKHR**, **TessellationControl**, **TessellationEvaluation**, **Geometry**, or **Fragment Execution Model**

- VUID-PrimitiveId-Fragment-04331

If pipeline contains both the **Fragment** and **Geometry Execution Model** and a variable decorated with **PrimitiveId** is read from **Fragment** shader, then the **Geometry** shader **must** write to the output variables decorated with **PrimitiveId** in all execution paths

- VUID-PrimitiveId-Fragment-04332

If pipeline contains both the **Fragment** and **MeshEXT** or **MeshNV Execution Model** and a variable decorated with **PrimitiveId** is read from **Fragment** shader, then the **MeshEXT** or **MeshNV** shader **must** write to the output variables decorated with **PrimitiveId** in all execution paths

- VUID-PrimitiveId-Fragment-04333

If **Fragment Execution Model** contains a variable decorated with **PrimitiveId**, then either the **MeshShadingEXT**, **MeshShadingNV**, **Geometry** or **Tessellation** capability **must** also be declared

- VUID-PrimitiveId-PrimitiveId-04334

The variable decorated with **PrimitiveId** within the **TessellationControl**, **TessellationEvaluation**, **Fragment**, **IntersectionKHR**, **AnyHitKHR**, or **ClosestHitKHR Execution Model** **must** be declared using the **Input Storage Class**

- VUID-PrimitiveId-PrimitiveId-04335

The variable decorated with **PrimitiveId** within the **Geometry Execution Model** **must** be declared using the **Input** or **Output Storage Class**

- VUID-PrimitiveId-PrimitiveId-04336

The variable decorated with **PrimitiveId** within the **MeshEXT** or **MeshNV Execution Model** **must** be declared using the **Output Storage Class**

- VUID-PrimitiveId-PrimitiveId-04337

The variable decorated with **PrimitiveId** **must** be declared as a scalar 32-bit integer value

- VUID-PrimitiveId-PrimitiveId-07040

The variable decorated with **PrimitiveId** within the **MeshEXT Execution Model** **must** also be decorated with the **PerPrimitiveEXT** decoration

PrimitiveShadingRateKHR

Decorating a variable with the **PrimitiveShadingRateKHR** built-in decoration will make that variable contain the **primitive fragment shading rate**.

The value written to the variable decorated with **PrimitiveShadingRateKHR** by the last **pre-rasterization shader stage** in the pipeline is used as the **primitive fragment shading rate**. Outputs in previous shader stages are ignored.

If the last active **pre-rasterization shader stage** shader entry point's interface does not include a variable decorated with **PrimitiveShadingRateKHR**, then it is as if the shader specified a fragment

shading rate value of 0, indicating a horizontal and vertical rate of 1 pixel.

If a shader has `PrimitiveShadingRateKHR` in the output interface and there is an execution path through the shader that does not write to it, its value is undefined for executions of the shader that take that path.

Valid Usage

- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04484
The `PrimitiveShadingRateKHR` decoration **must** be used only within the `MeshEXT`, `MeshNV`, `Vertex`, or `Geometry Execution Model`
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04485
The variable decorated with `PrimitiveShadingRateKHR` **must** be declared using the `Output Storage Class`
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04486
The variable decorated with `PrimitiveShadingRateKHR` **must** be declared as a scalar 32-bit integer value
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04487
The value written to `PrimitiveShadingRateKHR` **must** include no more than one of `Vertical2Pixels` and `Vertical4Pixels`
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04488
The value written to `PrimitiveShadingRateKHR` **must** include no more than one of `Horizontal2Pixels` and `Horizontal4Pixels`
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-04489
The value written to `PrimitiveShadingRateKHR` **must** not have any bits set other than those defined by `Fragment Shading Rate Flags` enumerants in the SPIR-V specification
- VUID-PrimitiveShadingRateKHR-PrimitiveShadingRateKHR-07059
The variable decorated with `PrimitiveShadingRateKHR` within the `MeshEXT Execution Model` **must** also be decorated with the `PerPrimitiveEXT` decoration

`RayGeometryIndexKHR`

A variable decorated with the `RayGeometryIndexKHR` decoration will contain the `geometry index` for the acceleration structure geometry currently being shaded.

Valid Usage

- VUID-RayGeometryIndexKHR-RayGeometryIndexKHR-04345
The `RayGeometryIndexKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, or `ClosestHitKHR Execution Model`
- VUID-RayGeometryIndexKHR-RayGeometryIndexKHR-04346
The variable decorated with `RayGeometryIndexKHR` **must** be declared using the `Input Storage Class`
- VUID-RayGeometryIndexKHR-RayGeometryIndexKHR-04347

The variable decorated with **RayGeometryIndexKHR** **must** be declared as a scalar 32-bit integer value

RayTmaxKHR

A variable decorated with the **RayTmaxKHR** decoration will contain the parametric $t_{\max}$ value of the ray being processed. The value is independent of the space in which the ray origin and direction exist. The value is initialized to the parameter passed into the **pipeline trace ray** instruction.

The $t_{\max}$ value changes throughout the lifetime of the ray that produced the intersection. In the closest hit shader, the value reflects the closest distance to the intersected primitive. In the any-hit shader, it reflects the distance to the primitive currently being intersected. In the intersection shader, it reflects the distance to the closest primitive intersected so far or the initial value. The value can change in the intersection shader after calling **OpReportIntersectionKHR** if the corresponding any-hit shader does not ignore the intersection. In a miss shader, the value is identical to the parameter passed into the **pipeline trace ray** instruction.

Valid Usage

- VUID-RayTmaxKHR-RayTmaxKHR-04348
The **RayTmaxKHR** decoration **must** be used only within the **IntersectionKHR**, **AnyHitKHR**, **ClosestHitKHR**, or **MissKHR Execution Model**
- VUID-RayTmaxKHR-RayTmaxKHR-04349
The variable decorated with **RayTmaxKHR** **must** be declared using the **Input Storage Class**
- VUID-RayTmaxKHR-RayTmaxKHR-04350
The variable decorated with **RayTmaxKHR** **must** be declared as a scalar 32-bit floating-point value

RayTminKHR

A variable decorated with the **RayTminKHR** decoration will contain the parametric $t_{\min}$ value of the ray being processed. The value is independent of the space in which the ray origin and direction exist. The value is the parameter passed into the **pipeline trace ray** instruction.

The $t_{\min}$ value remains constant for the duration of the ray query.

Valid Usage

- VUID-RayTminKHR-RayTminKHR-04351
The **RayTminKHR** decoration **must** be used only within the **IntersectionKHR**, **AnyHitKHR**, **ClosestHitKHR**, or **MissKHR Execution Model**
- VUID-RayTminKHR-RayTminKHR-04352
The variable decorated with **RayTminKHR** **must** be declared using the **Input Storage Class**
- VUID-RayTminKHR-RayTminKHR-04353
The variable decorated with **RayTminKHR** **must** be declared as a scalar 32-bit floating-point value

SampleId

Decorating a variable with the **SampleId** built-in decoration will make that variable contain the **coverage index** for the current fragment shader invocation. **SampleId** ranges from zero to the number of samples in the framebuffer minus one. If a fragment shader entry point's interface includes an input variable decorated with **SampleId**, **Sample Shading** is considered enabled with a **minSampleShading** value of 1.0.

Valid Usage

- VUID-SampleId-SampleId-04354
The **SampleId** decoration **must** be used only within the **Fragment Execution Model**
- VUID-SampleId-SampleId-04355
The variable decorated with **SampleId** **must** be declared using the **Input Storage Class**
- VUID-SampleId-SampleId-04356
The variable decorated with **SampleId** **must** be declared as a scalar 32-bit integer value

SampleMask

Decorating a variable with the **SampleMask** built-in decoration will make any variable contain the **sample mask** for the current fragment shader invocation.

A variable in the **Input** storage class decorated with **SampleMask** will contain a bitmask of the set of samples covered by the primitive generating the fragment during rasterization. It has a sample bit set if and only if the sample is considered covered for this fragment shader invocation. **SampleMask[]** is an array of integers. Bits are mapped to samples in a manner where bit B of mask M (**SampleMask[M]**) corresponds to sample $32 \times M + B$.

A variable in the **Output** storage class decorated with **SampleMask** is an array of integers forming a bit array in a manner similar to an input variable decorated with **SampleMask**, but where each bit represents coverage as computed by the shader. This computed **SampleMask** is combined with the generated coverage mask in the **multisample coverage** operation.

Variables decorated with **SampleMask** **must** be either an unsized array, or explicitly sized to be no larger than the implementation-dependent maximum sample-mask (as an array of 32-bit elements), determined by the maximum number of samples.

If a fragment shader entry point's interface includes an output variable decorated with **SampleMask**, the sample mask will be undefined for any array elements of any fragment shader invocations that fail to assign a value. If a fragment shader entry point's interface does not include an output variable decorated with **SampleMask**, the sample mask has no effect on the processing of a fragment.

Valid Usage

- VUID-SampleMask-SampleMask-04357
The **SampleMask** decoration **must** be used only within the **Fragment Execution Model**
- VUID-SampleMask-SampleMask-04358

The variable decorated with **SampleMask** **must** be declared using the **Input** or **Output Storage Class**

- VUID-SampleMask-SampleMask-04359

The variable decorated with **SampleMask** **must** be declared as an array of 32-bit integer values

SamplePosition

Decorating a variable with the **SamplePosition** built-in decoration will make that variable contain the sub-pixel position of the sample being shaded. The top left of the pixel is considered to be at coordinate (0,0) and the bottom right of the pixel is considered to be at coordinate (1,1).

If a fragment shader entry point's interface includes an input variable decorated with **SamplePosition**, **Sample Shading** is considered enabled with a **minSampleShading** value of 1.0.

If the current pipeline uses **custom sample locations** the value of any variable decorated with the **SamplePosition** built-in decoration is undefined.

Valid Usage

- VUID-SamplePosition-SamplePosition-04360

The **SamplePosition** decoration **must** be used only within the **Fragment Execution Model**

- VUID-SamplePosition-SamplePosition-04361

The variable decorated with **SamplePosition** **must** be declared using the **Input Storage Class**

- VUID-SamplePosition-SamplePosition-04362

The variable decorated with **SamplePosition** **must** be declared as a two-component vector of 32-bit floating-point values

ShadingRateKHR

Decorating a variable with the **ShadingRateKHR** built-in decoration will make that variable contain the **fragment shading rate** for the current fragment invocation.

Valid Usage

- VUID-ShadingRateKHR-ShadingRateKHR-04490

The **ShadingRateKHR** decoration **must** be used only within the **Fragment Execution Model**

- VUID-ShadingRateKHR-ShadingRateKHR-04491

The variable decorated with **ShadingRateKHR** **must** be declared using the **Input Storage Class**

- VUID-ShadingRateKHR-ShadingRateKHR-04492

The variable decorated with **ShadingRateKHR** **must** be declared as a scalar 32-bit integer value

TessCoord

Decorating a variable with the **TessCoord** built-in decoration will make that variable contain the three-dimensional (u,v,w) barycentric coordinate of the tessellated vertex within the patch. u, v, and w are in the range [0,1] and vary linearly across the primitive being subdivided. For the tessellation modes of **Quads** or **Isolines**, the third component is always zero.

Valid Usage

- VUID-TessCoord-TessCoord-04387
The **TessCoord** decoration **must** be used only within the **TessellationEvaluation Execution Model**
- VUID-TessCoord-TessCoord-04388
The variable decorated with **TessCoord** **must** be declared using the **Input Storage Class**
- VUID-TessCoord-TessCoord-04389
The variable decorated with **TessCoord** **must** be declared as a three-component vector of 32-bit floating-point values

TessLevelOuter

Decorating a variable with the **TessLevelOuter** built-in decoration will make that variable contain the outer tessellation levels for the current patch.

In tessellation control shaders, the variable decorated with **TessLevelOuter** **can** be written to, controlling the tessellation factors for the resulting patch. These values are used by the tessellator to control primitive tessellation and **can** be read by tessellation evaluation shaders.

In tessellation evaluation shaders, the variable decorated with **TessLevelOuter** **can** read the values written by the tessellation control shader.

Valid Usage

- VUID-TessLevelOuter-TessLevelOuter-04390
The **TessLevelOuter** decoration **must** be used only within the **TessellationControl** or **TessellationEvaluation Execution Model**
- VUID-TessLevelOuter-TessLevelOuter-04391
The variable decorated with **TessLevelOuter** within the **TessellationControl Execution Model** **must** be declared using the **Output Storage Class**
- VUID-TessLevelOuter-TessLevelOuter-04392
The variable decorated with **TessLevelOuter** within the **TessellationEvaluation Execution Model** **must** be declared using the **Input Storage Class**
- VUID-TessLevelOuter-TessLevelOuter-04393
The variable decorated with **TessLevelOuter** **must** be declared as an array of size four, containing 32-bit floating-point values

TessLevelInner

Decorating a variable with the **TessLevelInner** built-in decoration will make that variable contain the inner tessellation levels for the current patch.

In tessellation control shaders, the variable decorated with **TessLevelInner** **can** be written to, controlling the tessellation factors for the resulting patch. These values are used by the tessellator to control primitive tessellation and **can** be read by tessellation evaluation shaders.

In tessellation evaluation shaders, the variable decorated with **TessLevelInner** **can** read the values written by the tessellation control shader.

Valid Usage

- VUID-TessLevelInner-TessLevelInner-04394
The **TessLevelInner** decoration **must** be used only within the **TessellationControl** or **TessellationEvaluation Execution Model**
- VUID-TessLevelInner-TessLevelInner-04395
The variable decorated with **TessLevelInner** within the **TessellationControl Execution Model** **must** be declared using the **Output Storage Class**
- VUID-TessLevelInner-TessLevelInner-04396
The variable decorated with **TessLevelInner** within the **TessellationEvaluation Execution Model** **must** be declared using the **Input Storage Class**
- VUID-TessLevelInner-TessLevelInner-04397
The variable decorated with **TessLevelInner** **must** be declared as an array of size two, containing 32-bit floating-point values

VertexIndex

Decorating a variable with the **VertexIndex** built-in decoration will make that variable contain the index of the vertex that is being processed by the current vertex shader invocation. For non-indexed draws, this variable begins at the **firstVertex** parameter to **vkCmdDraw** or the **firstVertex** member of a structure consumed by **vkCmdDrawIndirect** and increments by one for each vertex in the draw. For indexed draws, its value is the content of the index buffer for the vertex plus the **vertexOffset** parameter to **vkCmdDrawIndexed** or the **vertexOffset** member of the structure consumed by **vkCmdDrawIndexedIndirect**.

NOTE

VertexIndex starts at the same starting value for each instance.

Valid Usage

- VUID-VertexIndex-VertexIndex-04398
The **VertexIndex** decoration **must** be used only within the **Vertex Execution Model**
- VUID-VertexIndex-VertexIndex-04399
The variable decorated with **VertexIndex** **must** be declared using the **Input Storage Class**
- VUID-VertexIndex-VertexIndex-04400

The variable decorated with **VertexIndex** **must** be declared as a scalar 32-bit integer value

ViewIndex

The **ViewIndex** decoration **can** be applied to a shader input which will be filled with the index of the view that is being processed by the current shader invocation.

If multiview is enabled in the render pass, this value will be the index of one of the bits set in the view mask of the subpass the pipeline is compiled against. If multiview is not enabled in the render pass, this value will be zero.

Valid Usage

- VUID-ViewIndex-ViewIndex-04401

The **ViewIndex** decoration **must** be used only within the **MeshEXT**, **Vertex**, **Geometry**, **TessellationControl**, **TessellationEvaluation** or **Fragment Execution Model**

- VUID-ViewIndex-ViewIndex-04402

The variable decorated with **ViewIndex** **must** be declared using the **Input Storage Class**

- VUID-ViewIndex-ViewIndex-04403

The variable decorated with **ViewIndex** **must** be declared as a scalar 32-bit integer value

ViewportIndex

Decorating a variable with the **ViewportIndex** built-in decoration will make that variable contain the index of the viewport.

In a geometry shader, the variable decorated with **ViewportIndex** can be written to with the viewport index to which the primitive produced by that shader will be directed.

The selected viewport index is used to select the viewport transform and scissor rectangle.

If the last active **pre-rasterization shader stage** shader entry point's interface does not include a variable decorated with **ViewportIndex** then the first viewport is used. If a **pre-rasterization shader stage** shader entry point's interface includes a variable decorated with **ViewportIndex**, it **must** write the same value to **ViewportIndex** for all output vertices of a given primitive.

In a fragment shader, the variable decorated with **ViewportIndex** contains the viewport index of the primitive that the fragment invocation belongs to.

Valid Usage

- VUID-ViewportIndex-ViewportIndex-04404

The **ViewportIndex** decoration **must** be used only within the **MeshEXT**, **MeshNV**, **Vertex**, **TessellationEvaluation**, **Geometry**, or **Fragment Execution Model**

- VUID-ViewportIndex-ViewportIndex-04406

The variable decorated with **ViewportIndex** within the **MeshEXT**, **MeshNV**, **Vertex**, **TessellationEvaluation**, or **Geometry Execution Model** **must** be declared using the **Output Storage Class**

- VUID-ViewportIndex-ViewportIndex-04407
The variable decorated with **ViewportIndex** within the **Fragment Execution Model** **must** be declared using the **Input Storage Class**
- VUID-ViewportIndex-ViewportIndex-04408
The variable decorated with **ViewportIndex** **must** be declared as a scalar 32-bit integer value
- VUID-ViewportIndex-ViewportIndex-07060
The variable decorated with **ViewportIndex** within the **MeshEXT Execution Model** **must** also be decorated with the **PerPrimitiveEXT** decoration

WorkgroupId

Decorating a variable with the **WorkgroupId** built-in decoration will make that variable contain the global workgroup that the current invocation is a member of. Each component ranges from a base value to a base + count value, based on the parameters passed into the dispatching commands.

Valid Usage

- VUID-WorkgroupId-WorkgroupId-04422
The **WorkgroupId** decoration **must** be used only within the **GLCompute**, **MeshEXT**, **TaskEXT**, **MeshNV**, or **TaskNV Execution Model**
- VUID-WorkgroupId-WorkgroupId-04423
The variable decorated with **WorkgroupId** **must** be declared using the **Input Storage Class**
- VUID-WorkgroupId-WorkgroupId-04424
The variable decorated with **WorkgroupId** **must** be declared as a three-component vector of 32-bit integer values

WorkgroupSize

Decorating an object with the **WorkgroupSize** built-in decoration will make that object contain the dimensions of a local workgroup. If an object is decorated with the **WorkgroupSize** decoration, this takes precedence over any **LocalSize** or **LocalSizeId** execution mode.

Valid Usage

- VUID-WorkgroupSize-WorkgroupSize-04425
The **WorkgroupSize** decoration **must** be used only within the **GLCompute**, **MeshEXT**, **TaskEXT**, **MeshNV**, or **TaskNV Execution Model**
- VUID-WorkgroupSize-WorkgroupSize-04426
The variable decorated with **WorkgroupSize** **must** be a specialization constant or a constant
- VUID-WorkgroupSize-WorkgroupSize-04427
The variable decorated with **WorkgroupSize** **must** be declared as a three-component vector of 32-bit integer values

WorldRayDirectionKHR

A variable decorated with the `WorldRayDirectionKHR` decoration will specify the direction of the ray being processed, in world space. The value is the parameter passed into the `pipeline trace ray` instruction.

Valid Usage

- VUID-WorldRayDirectionKHR-WorldRayDirectionKHR-04428
The `WorldRayDirectionKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, `ClosestHitKHR`, or `MissKHR Execution Model`
- VUID-WorldRayDirectionKHR-WorldRayDirectionKHR-04429
The variable decorated with `WorldRayDirectionKHR` **must** be declared using the `Input Storage Class`
- VUID-WorldRayDirectionKHR-WorldRayDirectionKHR-04430
The variable decorated with `WorldRayDirectionKHR` **must** be declared as a three-component vector of 32-bit floating-point values

WorldRayOriginKHR

A variable decorated with the `WorldRayOriginKHR` decoration will specify the origin of the ray being processed, in world space. The value is the parameter passed into the `pipeline trace ray` instruction.

Valid Usage

- VUID-WorldRayOriginKHR-WorldRayOriginKHR-04431
The `WorldRayOriginKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, `ClosestHitKHR`, or `MissKHR Execution Model`
- VUID-WorldRayOriginKHR-WorldRayOriginKHR-04432
The variable decorated with `WorldRayOriginKHR` **must** be declared using the `Input Storage Class`
- VUID-WorldRayOriginKHR-WorldRayOriginKHR-04433
The variable decorated with `WorldRayOriginKHR` **must** be declared as a three-component vector of 32-bit floating-point values

WorldToObjectKHR

A variable decorated with the `WorldToObjectKHR` decoration will contain the current world-to-object transformation matrix, which is determined by the instance of the current intersection.

Valid Usage

- VUID-WorldToObjectKHR-WorldToObjectKHR-04434
The `WorldToObjectKHR` decoration **must** be used only within the `IntersectionKHR`, `AnyHitKHR`, or `ClosestHitKHR Execution Model`

- VUID-WorldToObjectKHR-WorldToObjectKHR-04435

The variable decorated with **WorldToObjectKHR** **must** be declared using the **Input Storage Class**

- VUID-WorldToObjectKHR-WorldToObjectKHR-04436

The variable decorated with **WorldToObjectKHR** **must** be declared as a matrix with four columns of three-component vectors of 32-bit floating-point values

Chapter 16. Image Operations

16.1. Image Operations Overview

Vulkan Image Operations are operations performed by those SPIR-V Image Instructions which take an `OpTypeImage` (representing a `VkImageView`) or `OpTypeSampledImage` (representing a (`VkImageView`, `VkSampler`) pair). Read, write, and atomic operations also take texel coordinates as operands, and return a value based on a neighborhood of texture elements (*texels*) within the image. Query operations return properties of the bound image or of the lookup itself. The “Depth” operand of `OpTypeImage` is ignored.

NOTE

Texel is a term which is a combination of the words texture and element. Early interactive computer graphics supported texture operations on textures, a small subset of the image operations on images described here. The discrete samples remain essentially equivalent, however, so we retain the historical term texel to refer to them.

Image Operations include the functionality of the following SPIR-V Image Instructions:

- `OpImageSample*` and `OpImageSparseSample*` read one or more neighboring texels of the image, and `filter` the texel values based on the state of the sampler.
 - Instructions with `ImplicitLod` in the name `determine` the LOD used in the sampling operation based on the coordinates used in neighboring fragments.
 - Instructions with `ExplicitLod` in the name `determine` the LOD used in the sampling operation based on additional coordinates.
 - Instructions with `Proj` in the name apply homogeneous `projection` to the coordinates.
- `OpImageFetch` and `OpImageSparseFetch` return a single texel of the image. No sampler is used.
- `OpImage*Gather` and `OpImageSparse*Gather` read neighboring texels and `return a single component` of each.
- `OpImageRead` (and `OpImageSparseRead`) and `OpImageWrite` read and write, respectively, a texel in the image. No sampler is used.
- `OpImage*Dref*` instructions apply `depth comparison` on the texel values.
- `OpImageSparse*` instructions additionally return a `sparse residency` code.
- `OpImageQuerySize`, `OpImageQuerySizeLod`, `OpImageQueryLevels`, and `OpImageQuerySamples` return properties of the image descriptor that would be accessed. The image itself is not accessed.
- `OpImageQueryLod` returns the LOD parameters that would be used in a sample operation. The actual operation is not performed.

16.1.1. Texel Coordinate Systems

Images are addressed by *texel coordinates*. There are three *texel coordinate systems*:

- normalized texel coordinates [0.0, 1.0]

- unnormalized texel coordinates [0.0, width / height / depth)
- integer texel coordinates [0, width / height / depth)

SPIR-V `OpImageFetch`, `OpImageSparseFetch`, `OpImageRead`, `OpImageSparseRead`, and `OpImageWrite` instructions use integer texel coordinates.

Other image instructions **can** use either normalized or unnormalized texel coordinates (selected by the `unnormalizedCoordinates` state of the sampler used in the instruction), but there are [limitations](#) on what operations, image state, and sampler state is supported. Normalized coordinates are logically [converted](#) to unnormalized as part of image operations, and [certain steps](#) are only performed on normalized coordinates. The array layer coordinate is always treated as unnormalized even when other coordinates are normalized.

Normalized texel coordinates are referred to as (s,t,r,q,a), with the coordinates having the following meanings:

- s: Coordinate in the first dimension of an image.
- t: Coordinate in the second dimension of an image.
- r: Coordinate in the third dimension of an image.
 - (s,t,r) are interpreted as a direction vector for Cube images.
- q: Fourth coordinate, for homogeneous (projective) coordinates.
- a: Coordinate for array layer.

The coordinates are extracted from the SPIR-V operand based on the dimensionality of the image variable and type of instruction. For `Proj` instructions, the components are in order (s, [t,] [r,] q), with t and r being conditionally present based on the `Dim` of the image. For non-`Proj` instructions, the coordinates are (s [t] [r] [a]), with t and r being conditionally present based on the `Dim` of the image and a being conditionally present based on the `Arrayed` property of the image. Projective image instructions are not supported on `Arrayed` images.

Unnormalized texel coordinates are referred to as (u,v,w,a), with the coordinates having the following meanings:

- u: Coordinate in the first dimension of an image.
- v: Coordinate in the second dimension of an image.
- w: Coordinate in the third dimension of an image.
- a: Coordinate for array layer.

Only the u and v coordinates are directly extracted from the SPIR-V operand, because only 1D and 2D (non-`Arrayed`) dimensionalities support unnormalized coordinates. The components are in order (u [v]), with v being conditionally present when the dimensionality is 2D. When normalized coordinates are converted to unnormalized coordinates, all four coordinates are used.

Integer texel coordinates are referred to as (i,j,k,l,n), with the coordinates having the following meanings:

- i: Coordinate in the first dimension of an image.

- j: Coordinate in the second dimension of an image.
- k: Coordinate in the third dimension of an image.
- l: Coordinate for array layer.
- n: Index of the sample within the texel.

They are extracted from the SPIR-V operand in order (i [j] [k] [l] [n]), with j and k conditionally present based on the **Dim** of the image, and l conditionally present based on the **Arrayed** property of the image. n is conditionally present and is taken from the **Sample** image operand.

For all coordinate types, unused coordinates are assigned a value of zero.

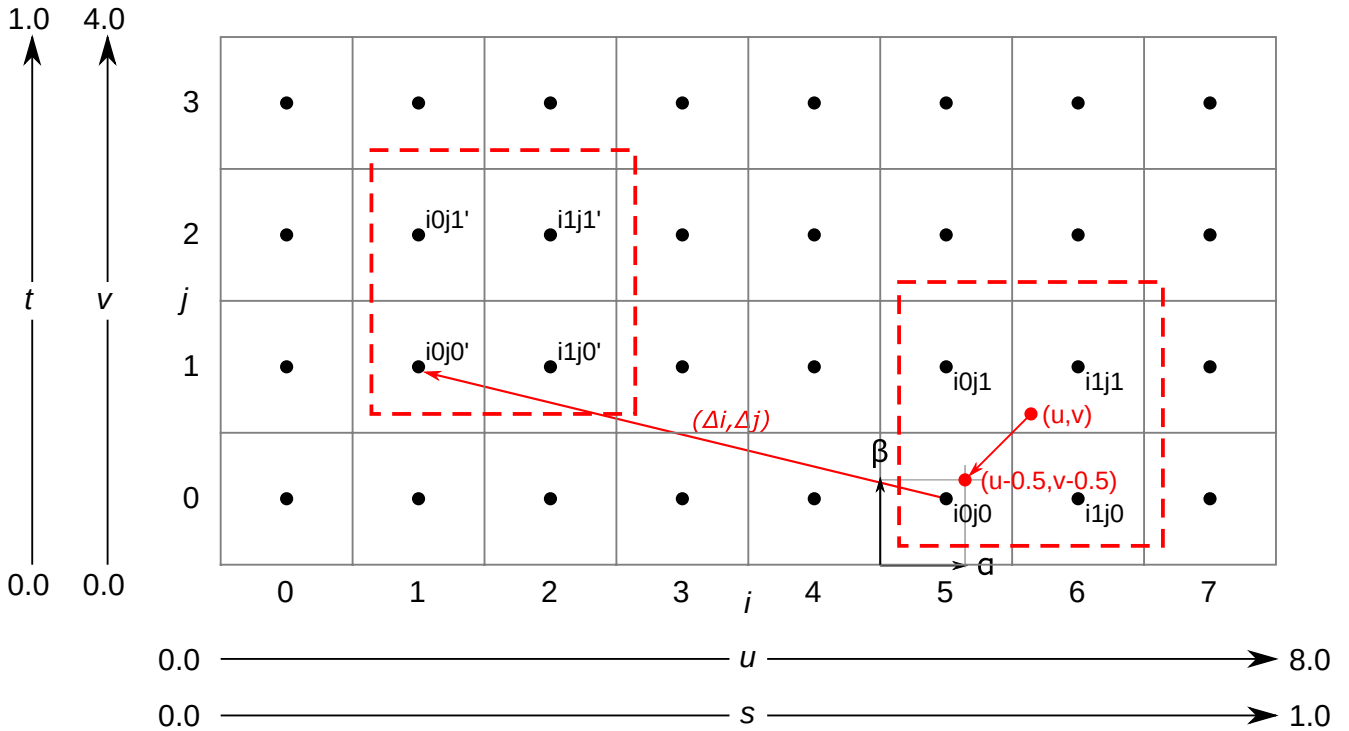


Figure 3. Texel Coordinate Systems, Linear Filtering

The Texel Coordinate Systems - For the example shown of an 8×4 texel two dimensional image.

- Normalized texel coordinates:
 - The s coordinate goes from 0.0 to 1.0.
 - The t coordinate goes from 0.0 to 1.0.
- Unnormalized texel coordinates:
 - The u coordinate within the range 0.0 to 8.0 is within the image, otherwise it is outside the image.
 - The v coordinate within the range 0.0 to 4.0 is within the image, otherwise it is outside the image.
- Integer texel coordinates:
 - The i coordinate within the range 0 to 7 addresses texels within the image, otherwise it is outside the image.
 - The j coordinate within the range 0 to 3 addresses texels within the image, otherwise it is

outside the image.

- Also shown for linear filtering:
 - Given the unnormalized coordinates (u,v) , the four texels selected are i_0j_0 , i_1j_0 , i_0j_1 , and i_1j_1 .
 - The fractions α and β .
 - Given the offset Δ_i and Δ_j , the four texels selected by the offset are $i_0j'_0$, $i_1j'_0$, $i_0j'_1$, and $i_1j'_1$.

NOTE

For formats with reduced-resolution components, Δ_i and Δ_j are relative to the resolution of the highest-resolution component, and therefore may be divided by two relative to the unnormalized coordinate space of the lower-resolution components.

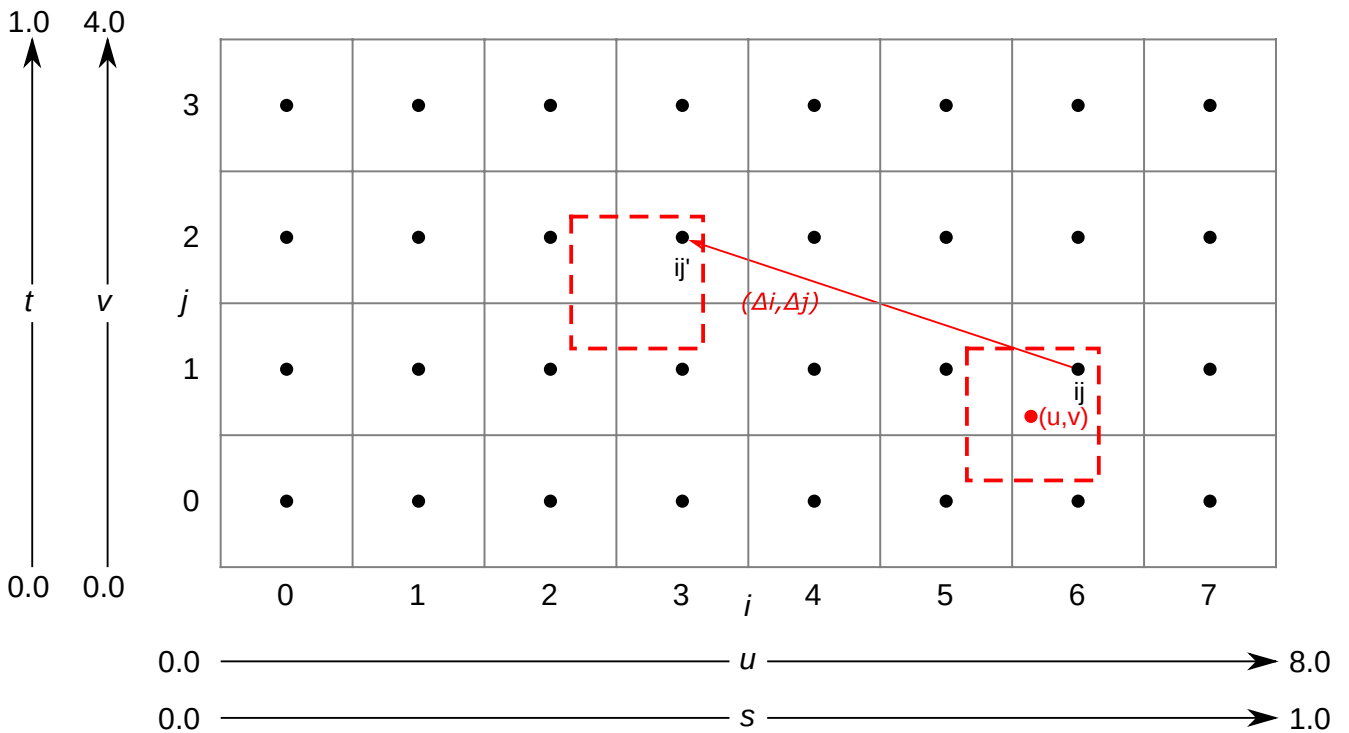


Figure 4. Texel Coordinate Systems, Nearest Filtering

The Texel Coordinate Systems - For the example shown of an 8×4 texel two dimensional image.

- Texel coordinates as above. Also shown for nearest filtering:
 - Given the unnormalized coordinates (u,v) , the texel selected is ij .
 - Given the offset Δ_i and Δ_j , the texel selected by the offset is ij' .

16.2. Conversion Formulas

16.2.1. RGB to Shared Exponent Conversion

An RGB color (red, green, blue) is transformed to a shared exponent color (red_{shared} , $green_{shared}$, $blue_{shared}$, exp_{shared}) as follows:

First, the components (red, green, blue) are clamped to ($red_{clamped}$, $green_{clamped}$, $blue_{clamped}$) as:

$$\text{red}_{\text{clamped}} = \max(0, \min(\text{sharedexp}_{\text{max}}, \text{red}))$$

$$\text{green}_{\text{clamped}} = \max(0, \min(\text{sharedexp}_{\text{max}}, \text{green}))$$

$$\text{blue}_{\text{clamped}} = \max(0, \min(\text{sharedexp}_{\text{max}}, \text{blue}))$$

where:

$$N = 9$$

number of mantissa bits per component

$$B = 15$$

exponent bias

$$E_{\text{max}} = 31$$

maximum possible biased exponent value

$$\text{sharedexp}_{\text{max}} = \frac{(2^N - 1)}{2^N} \times 2^{(E_{\text{max}} - B)}$$

NOTE

NaN, if supported, is handled as in

[IEEE 754-2008](#) `minNum()` and `maxNum()`. This results in any NaN being mapped to zero.

The largest clamped component, $\text{max}_{\text{clamped}}$ is determined:

$$\text{max}_{\text{clamped}} = \max(\text{red}_{\text{clamped}}, \text{green}_{\text{clamped}}, \text{blue}_{\text{clamped}})$$

A preliminary shared exponent exp' is computed:

$$\text{exp}' = \begin{cases} \lfloor \log_2(\text{max}_{\text{clamped}}) \rfloor + (B + 1) & \text{for } \text{max}_{\text{clamped}} > 2^{-(B+1)} \\ 0 & \text{for } \text{max}_{\text{clamped}} \leq 2^{-(B+1)} \end{cases}$$

The shared exponent $\text{exp}_{\text{shared}}$ is computed:

$$\text{max}_{\text{shared}} = \lfloor \frac{\text{max}_{\text{clamped}}}{2^{(\text{exp}' - B - N)}} + \frac{1}{2} \rfloor$$

$$\text{exp}_{\text{shared}} = \begin{cases} \text{exp}' & \text{for } 0 \leq \text{max}_{\text{shared}} < 2^N \\ \text{exp}' + 1 & \text{for } \text{max}_{\text{shared}} = 2^N \end{cases}$$

Finally, three integer values in the range 0 to 2^N are computed:

$$red_{shared} = \lfloor \frac{red_{clamped}}{2^{(exp_{shared} - B - N)}} + \frac{1}{2} \rfloor$$

$$green_{shared} = \lfloor \frac{green_{clamped}}{2^{(exp_{shared} - B - N)}} + \frac{1}{2} \rfloor$$

$$blue_{shared} = \lfloor \frac{blue_{clamped}}{2^{(exp_{shared} - B - N)}} + \frac{1}{2} \rfloor$$

16.2.2. Shared Exponent to RGB

A shared exponent color (red_{shared} , $green_{shared}$, $blue_{shared}$, exp_{shared}) is transformed to an RGB color (red, green, blue) as follows:

$$red = red_{shared} \times 2^{(exp_{shared} - B - N)}$$

$$green = green_{shared} \times 2^{(exp_{shared} - B - N)}$$

$$blue = blue_{shared} \times 2^{(exp_{shared} - B - N)}$$

where:

$N = 9$ (number of mantissa bits per component)

$B = 15$ (exponent bias)

16.3. Texel Input Operations

Texel input instructions are SPIR-V image instructions that read from an image. *Texel input operations* are a set of steps that are performed on state, coordinates, and texel values while processing a texel input instruction, and which are common to some or all texel input instructions. They include the following steps, which are performed in the listed order:

- [Validation operations](#)
 - [Instruction/Sampler/Image validation](#)
 - [Coordinate validation](#)
 - [Sparse validation](#)
 - [Layout validation](#)
- [Format conversion](#)
- [Texel replacement](#)
- [Depth comparison](#)

- Conversion to RGBA
- Component swizzle
- Chroma reconstruction
- $Y'C_B C_R$ conversion

For texel input instructions involving multiple texels (for sampling or gathering), these steps are applied for each texel that is used in the instruction. Depending on the type of image instruction, other steps are conditionally performed between these steps or involving multiple coordinate or texel values.

If [Chroma Reconstruction](#) is implicit, [Texel Filtering](#) instead takes place during chroma reconstruction, before [sampler \$Y'C_B C_R\$ conversion](#) occurs.

16.3.1. Texel Input Validation Operations

Texel input validation operations inspect instruction/image/sampler state or coordinates, and in certain circumstances cause the texel value to be replaced or become undefined. There are a series of validations that the texel undergoes.

Instruction/Sampler/Image View Validation

There are a number of cases where a SPIR-V instruction **can** mismatch with the sampler, the image view, or both, and a number of further cases where the sampler **can** mismatch with the image view. In such cases the value of the texel returned is undefined.

These cases include:

- The sampler `borderColor` is an integer type and the image view `format` is not one of the [VkFormat](#) integer types or a stencil component of a depth/stencil format.
- The sampler `borderColor` is a float type and the image view `format` is not one of the [VkFormat](#) float types or a depth component of a depth/stencil format.
- The sampler `borderColor` is one of the opaque black colors (`VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK` or `VK_BORDER_COLOR_INT_OPAQUE_BLACK`) and the image view [VkComponentSwizzle](#) for any of the [VkComponentMapping](#) components is not the [identity swizzle](#), and [VkPhysicalDeviceBorderColorSwizzleFeaturesEXT::borderColorSwizzleFromImage](#) feature is not enabled, and [VkSamplerBorderColorComponentMappingCreateInfoEXT](#) is not specified.
- [VkSamplerBorderColorComponentMappingCreateInfoEXT::components](#), if specified, has a component swizzle that does not match the component swizzle of the image view, and either component swizzle is not a form of identity swizzle.
- [VkSamplerBorderColorComponentMappingCreateInfoEXT::srgb](#), if specified, does not match the sRGB encoding of the image view.
- The sampler `borderColor` is a custom color (`VK_BORDER_COLOR_FLOAT_CUSTOM_EXT` or `VK_BORDER_COLOR_INT_CUSTOM_EXT`) and the supplied [VkSamplerCustomBorderColorCreateInfoEXT::customBorderColor](#) is outside the bounds of the values representable in the image view's `format`.
- The sampler `borderColor` is a custom color (`VK_BORDER_COLOR_FLOAT_CUSTOM_EXT` or

VK_BORDER_COLOR_INT_CUSTOM_EXT) and the image view `VkComponentSwizzle` for any of the `VkComponentMapping` components is not the `identity swizzle`, and `VkPhysicalDeviceBorderColorSwizzleFeaturesEXT::borderColorSwizzleFromImage` feature is not enabled, and `VkSamplerBorderColorComponentMappingCreateInfoEXT` is not specified.

- The `VkImageLayout` of any subresource in the image view does not match the `VkDescriptorImageInfo::imageLayout` used to write the image descriptor.
- The SPIR-V Image Format is not `compatible` with the image view's `format`.
- The sampler `unnormalizedCoordinates` is `VK_TRUE` and any of the `limitations of unnormalized coordinates` are violated.
- The SPIR-V instruction is one of the `OpImage*Dref*` instructions and the sampler `compareEnable` is `VK_FALSE`
- The SPIR-V instruction is not one of the `OpImage*Dref*` instructions and the sampler `compareEnable` is `VK_TRUE`
- The SPIR-V instruction is one of the `OpImage*Dref*` instructions, the image view `format` is one of the depth/stencil formats, and the image view aspect is not `VK_IMAGE_ASPECT_DEPTH_BIT`.
- The SPIR-V instruction's image variable's properties are not compatible with the image view:
 - If the image view's `viewType` is one of `VK_IMAGE_VIEW_TYPE_1D_ARRAY`, `VK_IMAGE_VIEW_TYPE_2D_ARRAY`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY` then the instruction **must** have `Arrayed` = 1. Otherwise the instruction **must** have `Arrayed` = 0.
 - If the image was created with `VkImageCreateInfo::samples` equal to `VK_SAMPLE_COUNT_1_BIT`, the instruction **must** have `MS` = 0.
 - If the image was created with `VkImageCreateInfo::samples` not equal to `VK_SAMPLE_COUNT_1_BIT`, the instruction **must** have `MS` = 1.
 - If the `Sampled Type` of the `OpTypeImage` does not match the `SPIR-V Type`.
 - If the `signedness of any read or sample operation` does not match the signedness of the image's format.
- The sampler was created with a specified `VkSamplerCustomBorderColorCreateInfoEXT::format` which does not match the `VkFormat` of the image view(s) it is sampling.
- The sampler is sampling an image view of `VK_FORMAT_B4G4R4A4_UNORM_PACK16`, `VK_FORMAT_B5G6R5_UNORM_PACK16`, or `VK_FORMAT_B5G5R5A1_UNORM_PACK16` format without a specified `VkSamplerCustomBorderColorCreateInfoEXT::format`.

Only `OpImageSample*` and `OpImageSparseSample*` **can** be used with a sampler or image view that enables `sampler Y'CBCR conversion`.

`OpImageFetch`, `OpImageSparseFetch`, `OpImage*Gather`, and `OpImageSparse*Gather` **must** not be used with a sampler or image view that enables `sampler Y'CBCR conversion`.

The `ConstOffset` and `Offset` operands **must** not be used with a sampler or image view that enables `sampler Y'CBCR conversion`.

If the underlying `VkImage` format has an X component in its format description, undefined values are read from those bits.

NOTE

If the `VkImage` format and `VkImageView` format are the same, these bits will be unused by format conversion and this will have no effect. However, if the `VkImageView` format is different, then some bits of the result may be undefined. For example, when a `VK_FORMAT_R10X6_UNORM_PACK16` `VkImage` is sampled via a `VK_FORMAT_R16_UNORM` `VkImageView`, the low 6 bits of the value before format conversion are undefined and format conversion may return a range of different values.

NOTE

Some implementations will return undefined values in the case where a sampler uses a `VkSamplerAddressMode` of `VK_SAMPLER_ADDRESS_MODE_MIRRORED_REPEAT`, the sampler is used with operands `Offset`, `ConstOffset`, or `ConstOffsets`, and the value of the offset is larger than or equal to the corresponding width, height, or depth of any accessed image level.

This behavior was not tested prior to Vulkan conformance test suite version 1.3.8.0. Affected implementations will have a conformance test waiver for this issue.

Integer Texel Coordinate Validation

Integer texel coordinates are validated against the size of the image level, and the number of layers and number of samples in the image. For SPIR-V instructions that use integer texel coordinates, this is performed directly on the integer coordinates. For instructions that use normalized or unnormalized texel coordinates, this is performed on the coordinates that result after [conversion](#) to integer texel coordinates.

If the integer texel coordinates do not satisfy all of the conditions

$$0 \leq i < w_s$$

$$0 \leq j < h_s$$

$$0 \leq k < d_s$$

$$0 \leq l < \text{layers}$$

$$0 \leq n < \text{samples}$$

where:

w_s = width of the image level

h_s = height of the image level

d_s = depth of the image level

layers = number of layers in the image

samples = number of samples per texel in the image

then the texel fails integer texel coordinate validation.

There are four cases to consider:

1. Valid Texel Coordinates

- If the texel coordinates pass validation (that is, the coordinates lie within the image),
then the texel value comes from the value in image memory.

2. Border Texel

- If the texel coordinates fail validation, and
- If the read is the result of an image sample instruction or image gather instruction, and
- If the image is not a cube image, or if a sampler created with `VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT` is used,

then the texel is a border texel and [texel replacement](#) is performed.

3. Invalid Texel

- If the texel coordinates fail validation, and
- If the read is the result of an image fetch instruction, image read instruction, or atomic instruction,

then the texel is an invalid texel and [texel replacement](#) is performed.

4. Cube Map Edge or Corner

Otherwise the texel coordinates lie beyond the edges or corners of the selected cube map face, and [Cube map edge handling](#) is performed.

Cube Map Edge Handling

If the texel coordinates lie beyond the edges or corners of the selected cube map face (as described in the prior section), the following steps are performed. Note that this does not occur when using `VK_FILTER_NEAREST` filtering within a mip level, since `VK_FILTER_NEAREST` is treated as using `VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE`.

• Cube Map Edge Texel

- If the texel lies beyond the selected cube map face in either only i or only j , then the coordinates (i,j) and the array layer l are transformed to select the adjacent texel from the

appropriate neighboring face.

- Cube Map Corner Texel
 - If the texel lies beyond the selected cube map face in both i and j , then there is no unique neighboring face from which to read that texel. The texel **should** be replaced by the average of the three values of the adjacent texels in each incident face. However, implementations **may** replace the cube map corner texel by other methods. The methods are subject to the constraint that for linear filtering if the three available texels have the same value, the resulting filtered texel **must** have that value, and for cubic filtering if the twelve available samples have the same value, the resulting filtered texel **must** have that value.

Sparse Validation

If the texel reads from an unbound region of a sparse image, the texel is a *sparse unbound texel*, and processing continues with [texel replacement](#).

Layout Validation

If all planes of a *disjoint multi-planar* image are not in the same [image layout](#), the image **must** not be sampled with [sampler Y'C_BC_R conversion](#) enabled.

16.3.2. Format Conversion

Texels undergo a format conversion from the [VkFormat](#) of the image view to a vector of either floating-point or signed or unsigned integer components, with the number of components based on the number of components present in the format.

- Color formats have one, two, three, or four components, according to the format.
- Depth/stencil formats are one component. The depth or stencil component is selected by the [aspectMask](#) of the image view.

Each component is converted based on its type and size (as defined in the [Format Definition](#) section for each [VkFormat](#)), using the appropriate equations in [16-Bit Floating-Point Numbers](#), [Unsigned 11-Bit Floating-Point Numbers](#), [Unsigned 10-Bit Floating-Point Numbers](#), [Fixed-Point Data Conversion](#), and [Shared Exponent to RGB](#). Signed integer components smaller than 32 bits are sign-extended.

If the image view format is sRGB, the color components are first converted as if they are UNORM, and then sRGB to linear conversion is applied to the R, G, and B components as described in the “sRGB EOTF” section of the [Khronos Data Format Specification](#). The A component, if present, is unchanged.

If the image view format is block-compressed, then the texel value is first decoded, then converted based on the type and number of components defined by the compressed format.

16.3.3. Texel Replacement

A texel is replaced if it is one (and only one) of:

- a border texel,

- an invalid texel, or
- a sparse unbound texel.

Border texels are replaced with a value based on the image format and the `borderColor` of the sampler. The border color is:

Table 19. Border Color B , Custom Border Color `VkSamplerCustomBorderColorCreateInfoEXT`

`::customBorderColor` U

Sampler <code>borderColor</code>	Corresponding Border Color
<code>VK_BORDER_COLOR_FLOAT_TRANSPARENT_BLACK</code>	$[B_r, B_g, B_b, B_a] = [0.0, 0.0, 0.0, 0.0]$
<code>VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK</code>	$[B_r, B_g, B_b, B_a] = [0.0, 0.0, 0.0, 1.0]$
<code>VK_BORDER_COLOR_FLOAT_OPAQUE_WHITE</code>	$[B_r, B_g, B_b, B_a] = [1.0, 1.0, 1.0, 1.0]$
<code>VK_BORDER_COLOR_INT_TRANSPARENT_BLACK</code>	$[B_r, B_g, B_b, B_a] = [0, 0, 0, 0]$
<code>VK_BORDER_COLOR_INT_OPAQUE_BLACK</code>	$[B_r, B_g, B_b, B_a] = [0, 0, 0, 1]$
<code>VK_BORDER_COLOR_INT_OPAQUE_WHITE</code>	$[B_r, B_g, B_b, B_a] = [1, 1, 1, 1]$
<code>VK_BORDER_COLOR_FLOAT_CUSTOM_EXT</code>	$[B_r, B_g, B_b, B_a] = [U_r, U_g, U_b, U_a]$
<code>VK_BORDER_COLOR_INT_CUSTOM_EXT</code>	$[B_r, B_g, B_b, B_a] = [U_r, U_g, U_b, U_a]$

The custom border color (U) **may** be rounded by implementations prior to texel replacement, but the error introduced by such a rounding **must** not exceed one ULP of the image's `format`.

NOTE

The names `VK_BORDER_COLOR_*_TRANSPARENT_BLACK`, `VK_BORDER_COLOR_*_OPAQUE_BLACK`, and `VK_BORDER_COLOR_*_OPAQUE_WHITE` are meant to describe which components are zeros and ones in the vocabulary of compositing, and are not meant to imply that the numerical value of `VK_BORDER_COLOR_INT_OPAQUE_WHITE` is a saturating value for integers.

This is substituted for the texel value by replacing the number of components in the image format

Table 20. Border Texel Components After Replacement

Texel Aspect or Format	Component Assignment
Depth aspect	$D = B_r$
Stencil aspect	$S = B_r^\dagger$
One component color format	$\text{Color}_r = B_r$
Two component color format	$[\text{Color}_r, \text{Color}_g] = [B_r, B_g]$
Three component color format	$[\text{Color}_r, \text{Color}_g, \text{Color}_b] = [B_r, B_g, B_b]$
Four component color format	$[\text{Color}_r, \text{Color}_g, \text{Color}_b, \text{Color}_a] = [B_r, B_g, B_b, B_a]$
Single component alpha format	$[\text{Color}_r, \text{Color}_g, \text{Color}_b, \text{Color}_a] = [0, 0, 0, B_a]$

$\dagger S = B_g$ **may** be substituted as the replacement method by the implementation when `VkSamplerCreateInfo::borderColor` is `VK_BORDER_COLOR_INT_CUSTOM_EXT` and `VkSamplerCustomBorderColorCreateInfoEXT::format` is `VK_FORMAT_UNDEFINED`. Implementations

should use $S = B_r$ as the replacement method.

The value returned by a read of an invalid texel is undefined, unless that read operation is from a buffer resource and the **robustBufferAccess** feature is enabled. In that case, an invalid texel is replaced as described by the **robustBufferAccess** feature. If the access is to an image resource and the x, y, z, or layer coordinate validation fails and If the **robustImageAccess2** feature is enabled, zero values **must** be returned. If only the sample index was invalid, the values returned are undefined.

Loads from a null descriptor return a four component color value of all zeros. However, for storage images and storage texel buffers using an explicit SPIR-V Image Format, loads from a null descriptor **may** return an alpha value of 1 (float or integer, depending on format) if the format does not include alpha.

If the **VkPhysicalDeviceSparseProperties::residencyNonResidentStrict** property is **VK_TRUE**, a sparse unbound texel is replaced with 0 or 0.0 values for integer and floating-point components of the image format, respectively.

If **residencyNonResidentStrict** is **VK_FALSE**, the value of the sparse unbound texel is undefined.

16.3.4. Depth Compare Operation

If the image view has a depth/stencil format, the depth component is selected by the **aspectMask**, and the operation is an **OpImage*Dref*** instruction, a depth comparison is performed. The result is 1.0 if the comparison evaluates to true, and 0.0 otherwise. This value replaces the depth component D.

The compare operation is selected by the **VkCompareOp** value set by **VkSamplerCreateInfo::compareOp**. The reference value from the SPIR-V operand D_{ref} and the texel depth value D_{tex} are used as the *reference* and *test* values, respectively, in that operation.

If the image being sampled has an unsigned normalized fixed-point format, then D_{ref} is clamped to [0,1] before the compare operation.

16.3.5. Conversion to RGBA

The texel is expanded from one, two, or three components to four components based on the image base color:

Table 21. Texel Color After Conversion To RGBA

Texel Aspect or Format	RGBA Color
Depth aspect	$[Color_r, Color_g, Color_b, Color_a] = [D, 0, 0, one]$
Stencil aspect	$[Color_r, Color_g, Color_b, Color_a] = [S, 0, 0, one]$
One component color format	$[Color_r, Color_g, Color_b, Color_a] = [Color_r, 0, 0, one]$
Two component color format	$[Color_r, Color_g, Color_b, Color_a] = [Color_r, Color_g, 0, one]$
Three component color format	$[Color_r, Color_g, Color_b, Color_a] = [Color_r, Color_g, Color_b, one]$
Four component color format	$[Color_r, Color_g, Color_b, Color_a] = [Color_r, Color_g, Color_b, Color_a]$
One alpha component color format	$[Color_r, Color_g, Color_b, Color_a] = [0, 0, 0, Color_a]$

where one = 1.0f for floating-point formats and depth aspects, and one = 1 for integer formats and stencil aspects.

16.3.6. Component Swizzle

All texel input instructions apply a *swizzle* based on:

- the [VkComponentSwizzle](#) enums in the **components** member of the [VkImageViewCreateInfo](#) structure for the image being read if [sampler Y'C_BC_R conversion](#) is not enabled, and
- the [VkComponentSwizzle](#) enums in the **components** member of the [VkSamplerYcbcrConversionCreateInfo](#) structure for the [sampler Y'C_BC_R conversion](#) if sampler Y'C_BC_R conversion is enabled.

The swizzle **can** rearrange the components of the texel, or substitute zero or one for any components. It is defined as follows for each color component:

$$Color_{component} = \begin{cases} Color_r & \text{for RED swizzle} \\ Color_g & \text{for GREEN swizzle} \\ Color_b & \text{for BLUE swizzle} \\ Color_a & \text{for ALPHA swizzle} \\ 0 & \text{for ZERO swizzle} \\ one & \text{for ONE swizzle} \\ identity & \text{for IDENTITY swizzle} \end{cases}$$

where:

$$one = \begin{cases} 1.0f & \text{for floating-point components} \\ 1 & \text{for integer components} \end{cases}$$

$$identity = \begin{cases} Color_r & \text{for } component = r \\ Color_g & \text{for } component = g \\ Color_b & \text{for } component = b \\ Color_a & \text{for } component = a \end{cases}$$

If the border color is one of the [VK_BORDER_COLOR_*_OPAQUE_BLACK](#) enums and the [VkComponentSwizzle](#) is not the [identity swizzle](#) for all components, the value of the texel after swizzle is undefined.

If the image view has a depth/stencil format and the [VkComponentSwizzle](#) is [VK_COMPONENT_SWIZZLE_ONE](#), and [VkPhysicalDeviceMaintenance5PropertiesKHR::depthStencilSwizzleOneSupport](#) is not [VK_TRUE](#), the value of the texel after swizzle is undefined.

16.3.7. Sparse Residency

[OpImageSparse*](#) instructions return a structure which includes a *residency code* indicating whether any texels accessed by the instruction are sparse unbound texels. This code **can** be interpreted by the [OpImageSparseTexelsResident](#) instruction which converts the residency code to a boolean value.

16.3.8. Chroma Reconstruction

In some color models, the color representation is defined in terms of monochromatic light intensity (often called “luma”) and color differences relative to this intensity, often called “chroma”. It is common for color models other than RGB to represent the chroma components at lower spatial resolution than the luma component. This approach is used to take advantage of the eye’s lower spatial sensitivity to color compared with its sensitivity to brightness. Less commonly, the same approach is used with additive color, since the green component dominates the eye’s sensitivity to light intensity and the spatial sensitivity to color introduced by red and blue is lower.

Lower-resolution components are “downsampled” by resizing them to a lower spatial resolution than the component representing luminance. This process is also commonly known as “chroma subsampling”. There is one luminance sample in each texture texel, but each chrominance sample may be shared among several texels in one or both texture dimensions.

- “_444” formats do not spatially downsample chroma values compared with luma: there are unique chroma samples for each texel.
- “_422” formats have downsampling in the x dimension (corresponding to u or s coordinates): they are sampled at half the resolution of luma in that dimension.
- “_420” formats have downsampling in the x dimension (corresponding to u or s coordinates) and the y dimension (corresponding to v or t coordinates): they are sampled at half the resolution of luma in both dimensions.

The process of reconstructing a full color value for texture access involves accessing both chroma and luma values at the same location. To generate the color accurately, the values of the lower-resolution components at the location of the luma samples are reconstructed from the lower-resolution sample locations, an operation known here as “chroma reconstruction” irrespective of the actual color model.

The location of the chroma samples relative to the luma coordinates is determined by the `xChromaOffset` and `yChromaOffset` members of the `VkSamplerYcbcrConversionCreateInfo` structure used to create the sampler $Y'C_B C_R$ conversion.

The following diagrams show the relationship between unnormalized (u,v) coordinates and (i,j) integer texel positions in the luma component (shown in black, with circles showing integer sample positions) and the texel coordinates of reduced-resolution chroma components, shown as crosses in red.

NOTE

If the chroma values are reconstructed at the locations of the luma samples by means of interpolation, chroma samples from outside the image bounds are needed; these are determined according to [Wrapping Operation](#). These diagrams represent this by showing the bounds of the “chroma texel” extending beyond the image bounds, and including additional chroma sample positions where required for interpolation. The limits of a sample for `NEAREST` sampling is shown as a grid.

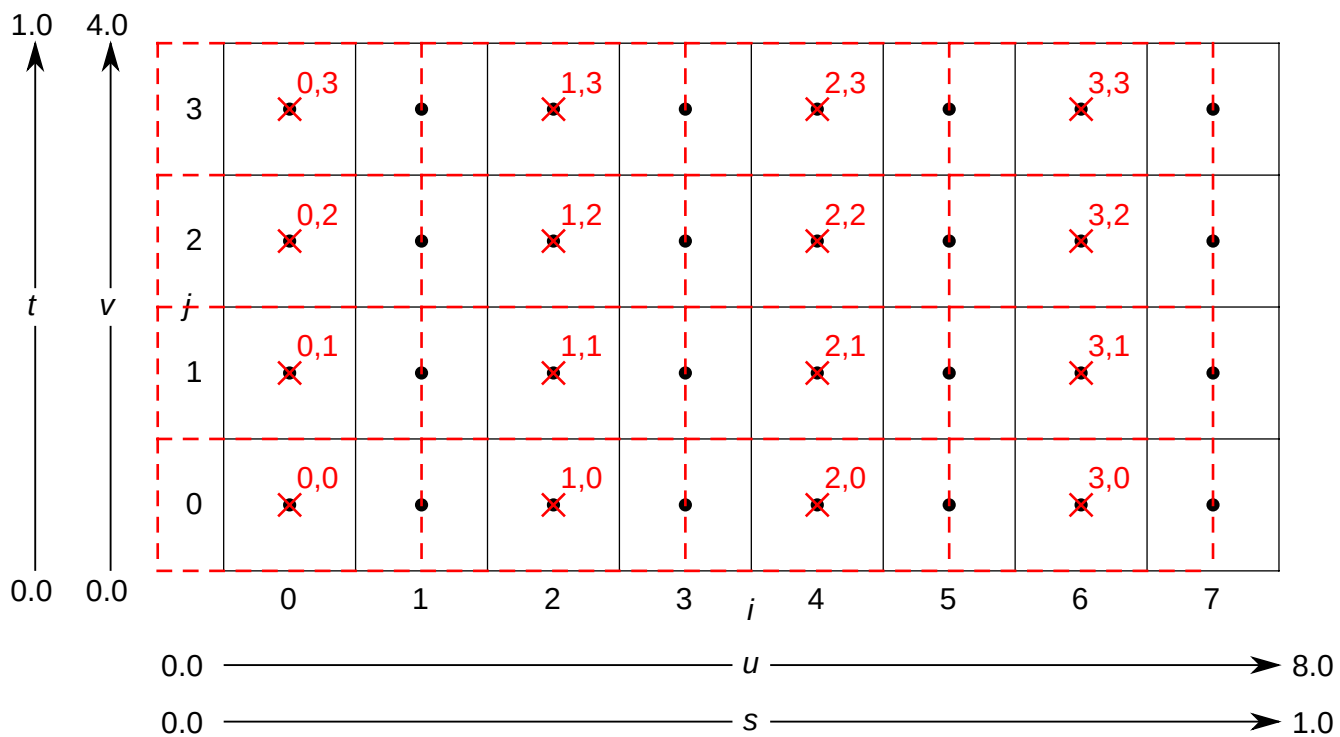


Figure 5. 422 downsampling, $xChromaOffset=COSITED_EVEN$

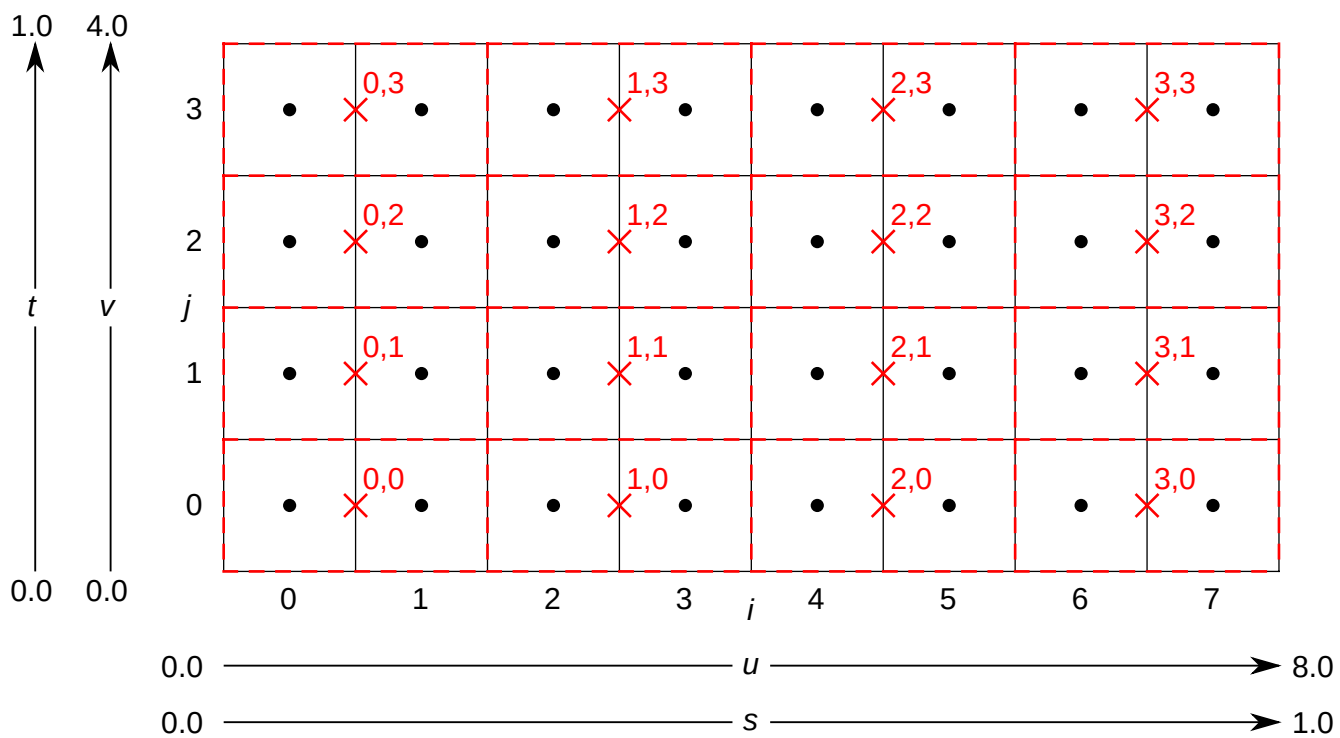


Figure 6. 422 downsampling, $xChromaOffset=MIDPOINT$

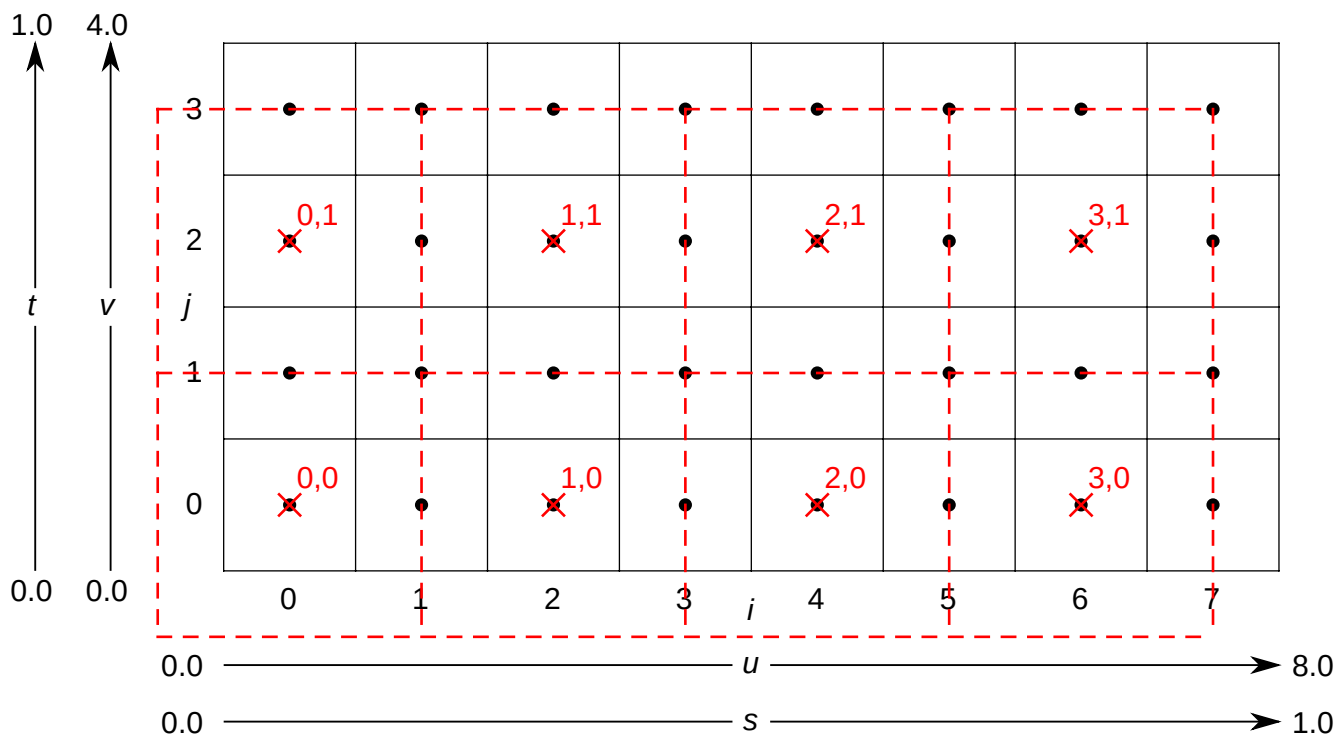


Figure 7. 420 downsampling, $x\text{ChromaOffset}=\text{COSITED_EVEN}$, $y\text{ChromaOffset}=\text{COSITED_EVEN}$

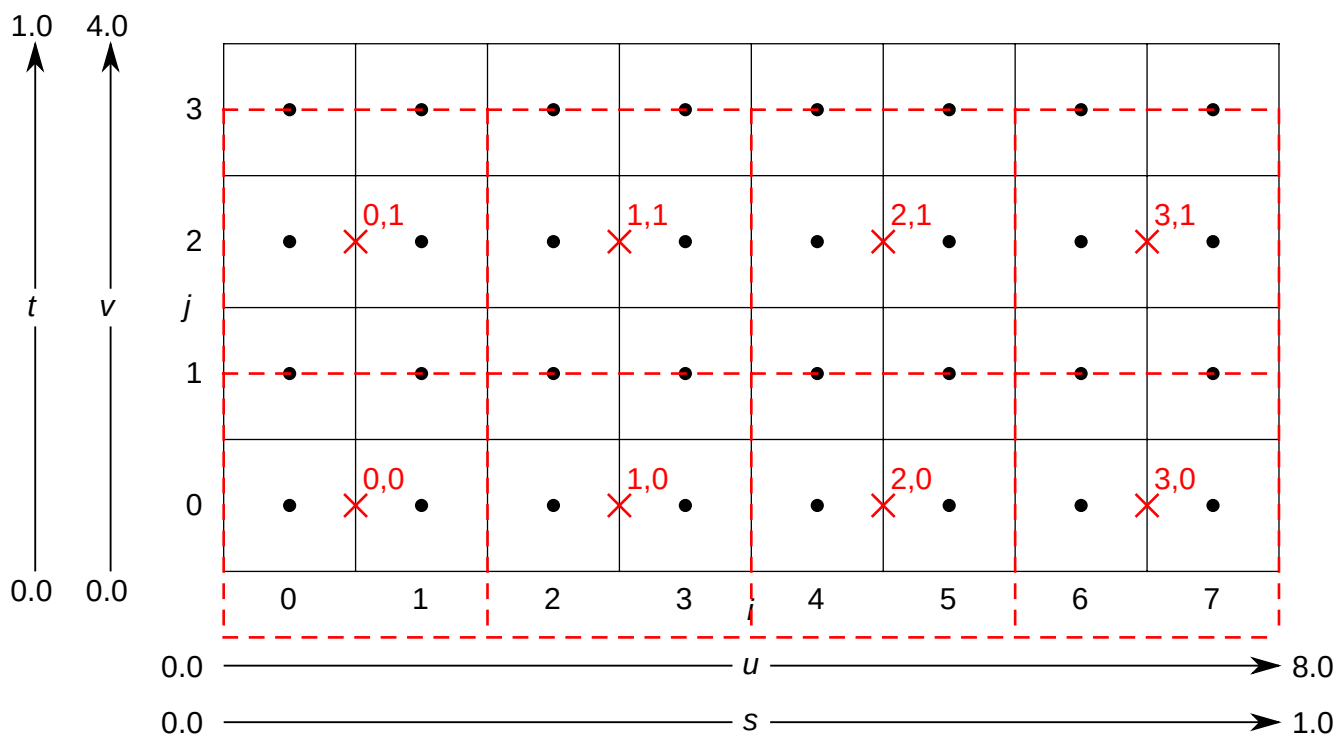


Figure 8. 420 downsampling, $x\text{ChromaOffset}=\text{MIDPOINT}$, $y\text{ChromaOffset}=\text{COSITED_EVEN}$

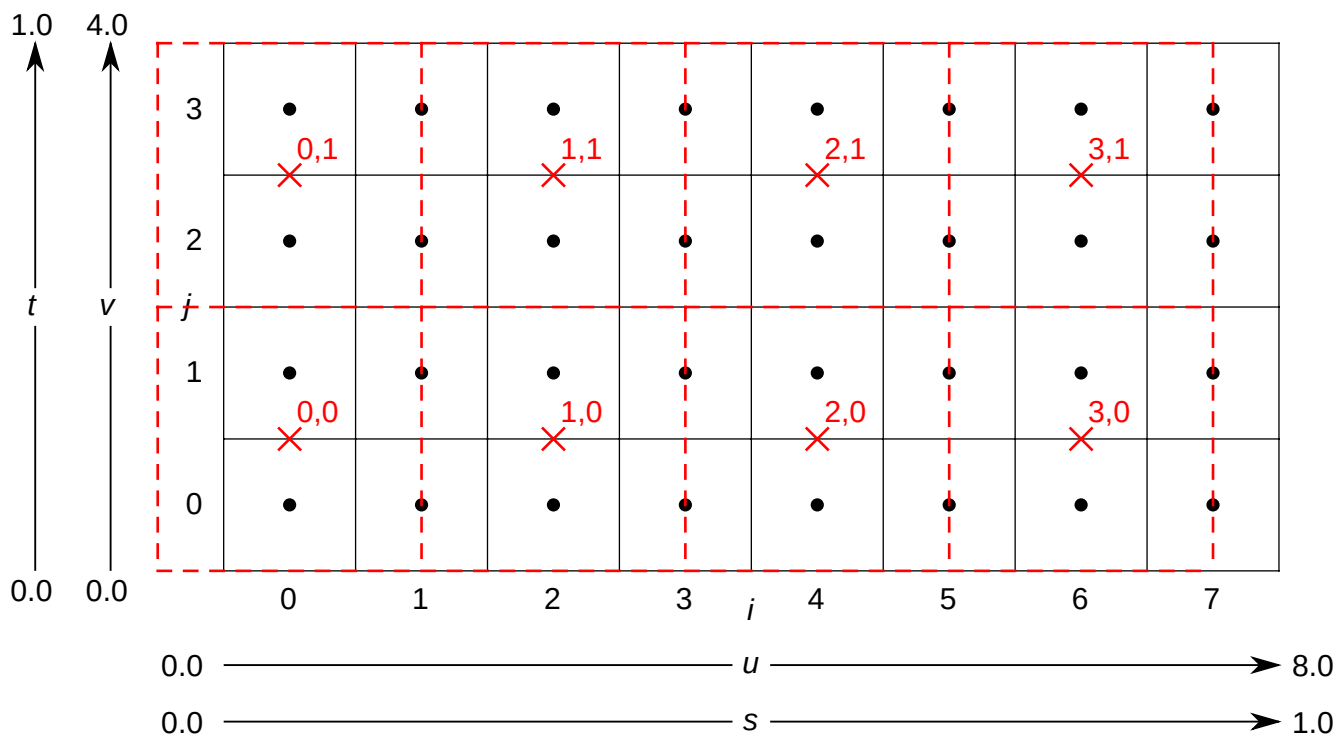


Figure 9. 420 downsampling, $xChromaOffset=COSITED_EVEN$, $yChromaOffset=MIDPOINT$

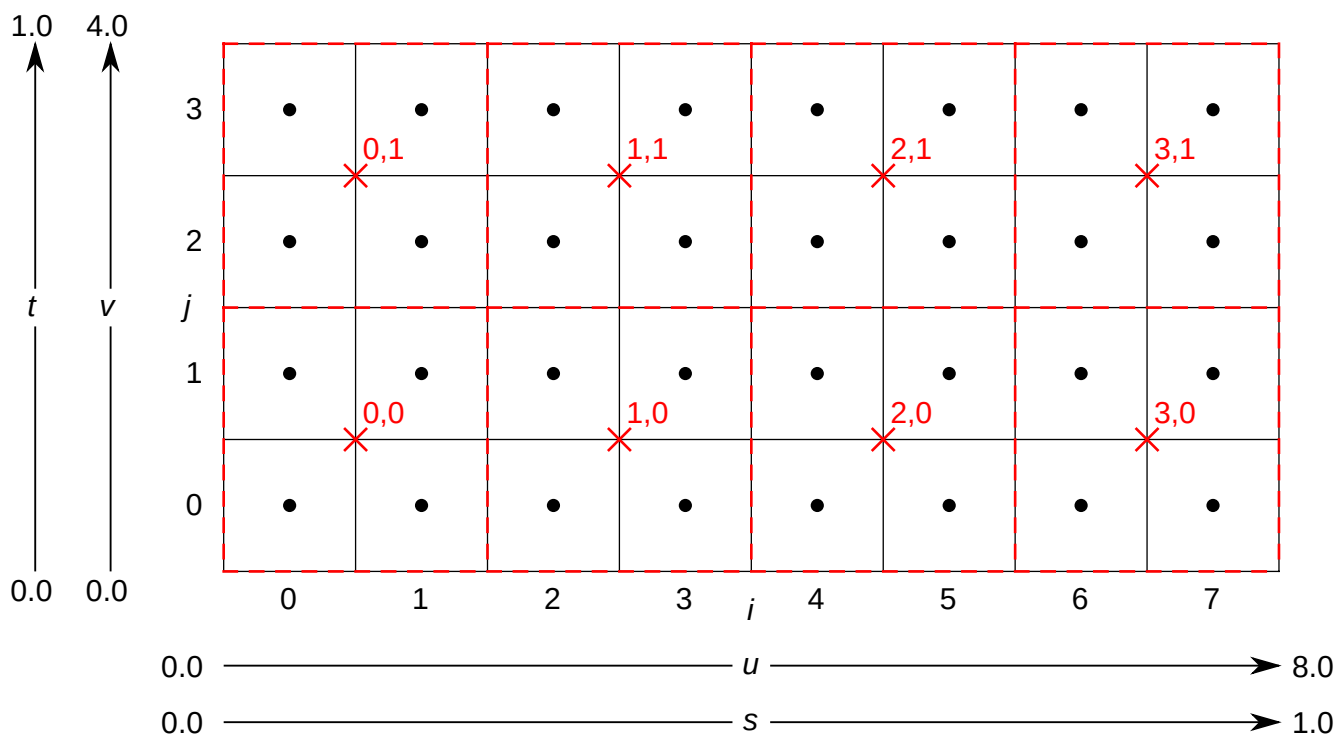


Figure 10. 420 downsampling, $xChromaOffset=MIDPOINT$, $yChromaOffset=MIDPOINT$

Reconstruction is implemented in one of two ways:

If the format of the image that is to be sampled sets `VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT`, or the `VkSamplerYcbcrConversionCreateInfo`'s `forceExplicitReconstruction` is `VK_TRUE`, reconstruction is performed as an explicit step independent of filtering, described in the [Explicit Reconstruction](#) section.

If the format of the image that is to be sampled does not set `VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT` and if the

`VkSamplerYcbcrConversionCreateInfo`'s `forceExplicitReconstruction` is `VK_FALSE`, reconstruction is performed as an implicit part of filtering prior to color model conversion, with no separate post-conversion texel filtering step, as described in the [Implicit Reconstruction](#) section.

Explicit Reconstruction

- If the `chromaFilter` member of the `VkSamplerYcbcrConversionCreateInfo` structure is `VK_FILTER_NEAREST`:
 - If the format's R and B components are reduced in resolution in just width by a factor of two relative to the G component (i.e. this is a “_422” format), the $\tau_{ijk}[level]$ values accessed by [texel filtering](#) are reconstructed as follows:

$$\begin{aligned}\tau_R'(i, j) &= \tau_R(\lfloor i \times 0.5 \rfloor, j)[level] \\ \tau_B'(i, j) &= \tau_B(\lfloor i \times 0.5 \rfloor, j)[level]\end{aligned}$$

- If the format's R and B components are reduced in resolution in width and height by a factor of two relative to the G component (i.e. this is a “_420” format), the $\tau_{ijk}[level]$ values accessed by [texel filtering](#) are reconstructed as follows:

$$\begin{aligned}\tau_R'(i, j) &= \tau_R(\lfloor i \times 0.5 \rfloor, \lfloor j \times 0.5 \rfloor)[level] \\ \tau_B'(i, j) &= \tau_B(\lfloor i \times 0.5 \rfloor, \lfloor j \times 0.5 \rfloor)[level]\end{aligned}$$

NOTE

`xChromaOffset` and `yChromaOffset` have no effect if `chromaFilter` is `VK_FILTER_NEAREST` for explicit reconstruction.

- If the `chromaFilter` member of the `VkSamplerYcbcrConversionCreateInfo` structure is `VK_FILTER_LINEAR`:
 - If the format's R and B components are reduced in resolution in just width by a factor of two relative to the G component (i.e. this is a “_422” format):
 - If `xChromaOffset` is `VK_CHROMA_LOCATION_COSITED_EVEN`:

$$\tau_{RB}'(i, j) = \begin{cases} \tau_{RB}(\lfloor i \times 0.5 \rfloor, j)[level], & 0.5 \times i = \lfloor 0.5 \times i \rfloor \\ 0.5 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor, j)[level] + \\ 0.5 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor + 1, j)[level], & 0.5 \times i \neq \lfloor 0.5 \times i \rfloor \end{cases}$$

- If `xChromaOffset` is `VK_CHROMA_LOCATION_MIDPOINT`:

$$\tau_{RB}'(i, j) = \begin{cases} 0.25 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor - 1, j)[level] + \\ 0.75 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor, j)[level], & 0.5 \times i = \lfloor 0.5 \times i \rfloor \\ 0.75 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor, j)[level] + \\ 0.25 \times \tau_{RB}(\lfloor i \times 0.5 \rfloor + 1, j)[level], & 0.5 \times i \neq \lfloor 0.5 \times i \rfloor \end{cases}$$

- If the format's R and B components are reduced in resolution in width and height by a factor of two relative to the G component (i.e. this is a “_420” format), a similar relationship applies. Due to the number of options, these formulae are expressed more concisely as follows:

$$\begin{aligned}
i_{RB} &= \begin{cases} 0.5 \times (i) & \text{xChromaOffset=COSITED_EVEN} \\ 0.5 \times (i - 0.5) & \text{xChromaOffset=MIDPOINT} \end{cases} \\
j_{RB} &= \begin{cases} 0.5 \times (j) & \text{yChromaOffset=COSITED_EVEN} \\ 0.5 \times (j - 0.5) & \text{yChromaOffset=MIDPOINT} \end{cases} \\
i_{floor} &= \lfloor i_{RB} \rfloor \\
j_{floor} &= \lfloor j_{RB} \rfloor \\
i_{frac} &= i_{RB} - i_{floor} \\
j_{frac} &= j_{RB} - j_{floor}
\end{aligned}$$

$$\begin{aligned}
\tau_{RB}'(i, j) &= \tau_{RB}(i_{floor}, j_{floor})[level] \times (1 - i_{frac}) \times (1 - j_{frac}) + \\
&\quad \tau_{RB}(1 + i_{floor}, j_{floor})[level] \times (i_{frac}) \times (1 - j_{frac}) + \\
&\quad \tau_{RB}(i_{floor}, 1 + j_{floor})[level] \times (1 - i_{frac}) \times (j_{frac}) + \\
&\quad \tau_{RB}(1 + i_{floor}, 1 + j_{floor})[level] \times (i_{frac}) \times (j_{frac})
\end{aligned}$$

NOTE

In the case where the texture itself is bilinearly interpolated as described in [Texel Filtering](#), thus requiring four full-color samples for the filtering operation, and where the reconstruction of these samples uses bilinear interpolation in the chroma components due to `chromaFilter=VK_FILTER_LINEAR`, up to nine chroma samples may be required, depending on the sample location.

Implicit Reconstruction

Implicit reconstruction takes place by the samples being interpolated, as required by the filter settings of the sampler, except that `chromaFilter` takes precedence for the chroma samples.

If `chromaFilter` is `VK_FILTER_NEAREST`, an implementation **may** behave as if `xChromaOffset` and `yChromaOffset` were both `VK_CHROMA_LOCATION_MIDPOINT`, irrespective of the values set.

NOTE

This will not have any visible effect if the locations of the luma samples coincide with the location of the samples used for rasterization.

The sample coordinates are adjusted by the downsample factor of the component (such that, for example, the sample coordinates are divided by two if the component has a downsample factor of two relative to the luma component):

$$\begin{aligned}
u_{RB}'(422/420) &= \begin{cases} 0.5 \times (u + 0.5), & \text{xChromaOffset=COSITED_EVEN} \\ 0.5 \times u, & \text{xChromaOffset=MIDPOINT} \end{cases} \\
v_{RB}'(420) &= \begin{cases} 0.5 \times (v + 0.5), & \text{yChromaOffset=COSITED_EVEN} \\ 0.5 \times v, & \text{yChromaOffset=MIDPOINT} \end{cases}
\end{aligned}$$

16.3.9. Sampler Y'C_BC_R Conversion

Sampler Y'C_BC_R conversion performs the following operations, which an implementation **may** combine into a single mathematical operation:

- [Sampler Y'C_BC_R Range Expansion](#)
- [Sampler Y'C_BC_R Model Conversion](#)

Sampler Y'C_BC_R Range Expansion

Sampler Y'C_BC_R range expansion is applied to color component values after all texel input operations which are not specific to sampler Y'C_BC_R conversion. For example, the input values to this stage have been converted using the normal [format conversion](#) rules.

Sampler Y'C_BC_R range expansion is not applied if `ycbcrModel` is `VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY`. That is, the shader receives the vector C'_{rgba} as output by the Component Swizzle stage without further modification.

For other values of `ycbcrModel`, range expansion is applied to the texel component values output by the [Component Swizzle](#) defined by the `components` member of [VkSamplerYcbcrConversionCreateInfo](#). Range expansion applies independently to each component of the image. For the purposes of range expansion and Y'C_BC_R model conversion, the R and B components contain color difference (chroma) values and the G component contains luma. The A component is not modified by sampler Y'C_BC_R range expansion.

The range expansion to be applied is defined by the `ycbcrRange` member of the [VkSamplerYcbcrConversionCreateInfo](#) structure:

- If `ycbcrRange` is `VK_SAMPLER_YCBCR_RANGE_ITU_FULL`, the following transformations are applied:

$$\begin{aligned} Y' &= C'_{rgba}[G] \\ C_B &= C'_{rgba}[B] - \frac{2^{(n-1)}}{(2^n) - 1} \\ C_R &= C'_{rgba}[R] - \frac{2^{(n-1)}}{(2^n) - 1} \end{aligned}$$

NOTE

These formulae correspond to the “full range” encoding in the “Quantization schemes” chapter of the [Khronos Data Format Specification](#).

Should any future amendments be made to the ITU specifications from which these equations are derived, the formulae used by Vulkan **may** also be updated to maintain parity.

- If `ycbcrRange` is `VK_SAMPLER_YCBCR_RANGE_ITU_NARROW`, the following transformations are applied:

$$\begin{aligned} Y' &= \frac{C'_{rgba}[G] \times (2^n - 1) - 16 \times 2^{n-8}}{219 \times 2^{n-8}} \\ C_B &= \frac{C'_{rgba}[B] \times (2^n - 1) - 128 \times 2^{n-8}}{224 \times 2^{n-8}} \\ C_R &= \frac{C'_{rgba}[R] \times (2^n - 1) - 128 \times 2^{n-8}}{224 \times 2^{n-8}} \end{aligned}$$

NOTE

These formulae correspond to the “narrow range” encoding in the “Quantization schemes” chapter of the [Khronos Data Format Specification](#).

- n is the bit-depth of the components in the format.

The precision of the operations performed during range expansion **must** be at least that of the source format.

An implementation **may** clamp the results of these range expansion operations such that Y' falls in the range $[0,1]$, and/or such that C_B and C_R fall in the range $[-0.5,0.5]$.

Sampler Y'C_BC_R Model Conversion

The range-expanded values are converted between color models, according to the color model conversion specified in the `ycbcrModel` member:

VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY

The color components are not modified by the color model conversion since they are assumed already to represent the desired color model in which the shader is operating; Y'C_BC_R range expansion is also ignored.

VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY

The color components are not modified by the color model conversion and are assumed to be treated as though in Y'C_BC_R form both in memory and in the shader; Y'C_BC_R range expansion is applied to the components as for other Y'C_BC_R models, with the vector (C_R, Y', C_B, A) provided to the shader.

VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709

The color components are transformed from a Y'C_BC_R representation to an R'G'B' representation as described in the “BT.709 Y'C_BC_R conversion” section of the [Khronos Data Format Specification](#).

VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601

The color components are transformed from a Y'C_BC_R representation to an R'G'B' representation as described in the “BT.601 Y'C_BC_R conversion” section of the [Khronos Data Format Specification](#).

VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020

The color components are transformed from a Y'C_BC_R representation to an R'G'B' representation as described in the “BT.2020 Y'C_BC_R conversion” section of the [Khronos Data Format Specification](#).

In this operation, each output component is dependent on each input component.

An implementation **may** clamp the R'G'B' results of these conversions to the range $[0,1]$.

The precision of the operations performed during model conversion **must** be at least that of the source format.

The alpha component is not modified by these model conversions.

Sampling operations in a non-linear color space can introduce color and intensity shifts at sharp transition boundaries. To avoid this issue, the technically precise color correction sequence described in the “Introduction to Color Conversions” chapter of the [Khronos Data Format Specification](#) may be performed as follows:

- Calculate the [unnormalized texel coordinates](#) corresponding to the desired sample position.
- For a [minFilter](#) or [magFilter](#) of [VK_FILTER_NEAREST](#):
 1. Calculate (i,j) for the sample location as described under the “nearest filtering” formulae in [\(u,v,w,a\) to \(i,j,k,l,n\) Transformation and Array Layer Selection](#)
 2. Calculate the normalized texel coordinates corresponding to these integer coordinates.
 3. Sample using [sampler Y'C_BC_R conversion](#) at this location.
- For a [minFilter](#) or [magFilter](#) of [VK_FILTER_LINEAR](#):
 1. Calculate $(i_{[0,1]},j_{[0,1]})$ for the sample location as described under the “linear filtering” formulae in [\(u,v,w,a\) to \(i,j,k,l,n\) Transformation and Array Layer Selection](#)
 2. Calculate the normalized texel coordinates corresponding to these integer coordinates.
 3. Sample using [sampler Y'C_BC_R conversion](#) at each of these locations.
 4. Convert the non-linear A'R'G'B' outputs of the Y'C_BC_R conversions to linear ARGB values as described in the “Transfer Functions” chapter of the [Khronos Data Format Specification](#).
 5. Interpolate the linear ARGB values using the α and β values described in the “linear filtering” section of [\(u,v,w,a\) to \(i,j,k,l,n\) Transformation and Array Layer Selection](#) and the equations in [Texel Filtering](#).

NOTE

The additional calculations and, especially, additional number of sampling operations in the [VK_FILTER_LINEAR](#) case can be expected to have a performance impact compared with using the outputs directly. Since the variations from “correct” results are subtle for most content, the application author should determine whether a more costly implementation is strictly necessary.

If [chromaFilter](#), and [minFilter](#) or [magFilter](#) are both [VK_FILTER_NEAREST](#), these operations are redundant and sampling using [sampler Y'C_BC_R conversion](#) at the desired sample coordinates will produce the “correct” results without further processing.

16.4. Texel Output Operations

Texel output instructions are SPIR-V image instructions that write to an image. *Texel output operations* are a set of steps that are performed on state, coordinates, and texel values while processing a texel output instruction, and which are common to some or all texel output

instructions. They include the following steps, which are performed in the listed order:

- [Validation operations](#)
 - [Format validation](#)
 - [Type validation](#)
 - [Coordinate validation](#)
 - [Sparse validation](#)
- [Texel output format conversion](#)

16.4.1. Texel Output Validation Operations

Texel output validation operations inspect instruction/image state or coordinates, and in certain circumstances cause the write to have no effect. There are a series of validations that the texel undergoes.

Texel Format Validation

If the image format of the `OpTypeImage` is not [compatible](#) with the `VkImageView`'s `format`, the write causes the contents of the image's memory to become undefined.

Texel Type Validation

If the `Sampled Type` of the `OpTypeImage` does not match the [SPIR-V Type](#), the write causes the value of the texel to become undefined. For integer types, if the [signedness of the access](#) does not match the signedness of the accessed resource, the write causes the value of the texel to become undefined.

16.4.2. Integer Texel Coordinate Validation

The integer texel coordinates are validated according to the same rules as for texel input [coordinate validation](#).

If the texel fails integer texel coordinate validation, then the write has no effect.

16.4.3. Sparse Texel Operation

If the texel attempts to write to an unbound region of a sparse image, the texel is a sparse unbound texel. In such a case, if the `VkPhysicalDeviceSparseProperties::residencyNonResidentStrict` property is `VK_TRUE`, the sparse unbound texel write has no effect. If `residencyNonResidentStrict` is `VK_FALSE`, the write **may** have a side effect that becomes visible to other accesses to unbound texels in any resource, but will not be visible to any device memory allocated by the application.

16.4.4. Texel Output Format Conversion

If the image format is sRGB, a linear to sRGB conversion is applied to the R, G, and B components as described in the "sRGB EOTF" section of the [Khronos Data Format Specification](#). The A component, if present, is unchanged.

Texels then undergo a format conversion from the floating-point, signed, or unsigned integer type

of the texel data to the [VkFormat](#) of the image view. If the number of components in the texel data is larger than the number of components in the format, additional components are discarded.

Each component is converted based on its type and size (as defined in the [Format Definition](#) section for each [VkFormat](#)). Floating-point outputs are converted as described in [Floating-Point Format Conversions](#) and [Fixed-Point Data Conversion](#). Integer outputs are converted such that their value is preserved. The converted value of any integer that cannot be represented in the target format is undefined.

If the [VkImageView](#) format has an X component in its format description, undefined values are written to those bits.

If the underlying [VkImage](#) format has an X component in its format description, undefined values are also written to those bits, even if result format conversion produces a valid value for those bits because the [VkImageView](#) format is different.

16.5. Normalized Texel Coordinate Operations

If the image sampler instruction provides normalized texel coordinates, some of the following operations are performed.

16.5.1. Projection Operation

For [Proj](#) image operations, the normalized texel coordinates (s,t,r,q,a) and (if present) the D_{ref} coordinate are transformed as follows:

$$\begin{aligned} s &= \frac{s}{q}, & \text{for 1D, 2D, or 3D image} \\ t &= \frac{t}{q}, & \text{for 2D or 3D image} \\ r &= \frac{r}{q}, & \text{for 3D image} \\ D_{ref} &= \frac{D_{ref}}{q}, & \text{if provided} \end{aligned}$$

16.5.2. Derivative Image Operations

Derivatives are used for LOD selection. These derivatives are either implicit (in an [ImplicitLod](#) image instruction in a mesh, task, compute, or fragment shader) or explicit (provided explicitly by shader to the image instruction in any shader).

For implicit derivatives image instructions, the derivatives of texel coordinates are calculated in the same manner as [derivative operations](#). That is:

$$\begin{aligned} \partial s / \partial x &= dPdx(s), & \partial s / \partial y &= dPdy(s), & \text{for 1D, 2D, Cube, or 3D image} \\ \partial t / \partial x &= dPdx(t), & \partial t / \partial y &= dPdy(t), & \text{for 2D, Cube, or 3D image} \\ \partial r / \partial x &= dPdx(r), & \partial r / \partial y &= dPdy(r), & \text{for Cube or 3D image} \end{aligned}$$

Partial derivatives not defined above for certain image dimensionalities are set to zero.

For explicit LOD image instructions, if the **optional** SPIR-V operand **Grad** is provided, then the operand values are used for the derivatives. The number of components present in each derivative for a given image dimensionality matches the number of partial derivatives computed above.

If the **optional** SPIR-V operand **Lod** is provided, then derivatives are set to zero, the cube map derivative transformation is skipped, and the scale factor operation is skipped. Instead, the floating-point scalar coordinate is directly assigned to λ_{base} as described in [LOD Operation](#).

16.5.3. Cube Map Face Selection and Transformations

For cube map image instructions, the (s,t,r) coordinates are treated as a direction vector (r_x, r_y, r_z). The direction vector is used to select a cube map face. The direction vector is transformed to a per-face texel coordinate system ($s_{\text{face}}, t_{\text{face}}$). The direction vector is also used to transform the derivatives to per-face derivatives.

16.5.4. Cube Map Face Selection

The direction vector selects one of the cube map's faces based on the largest magnitude coordinate direction (the major axis direction). Since two or more coordinates **can** have identical magnitude, the implementation **must** have rules to disambiguate this situation.

The rules **should** have as the first rule that r_z wins over r_y and r_x , and the second rule that r_y wins over r_x . An implementation **may** choose other rules, but the rules **must** be deterministic and depend only on (r_x, r_y, r_z).

The layer number (corresponding to a cube map face), the coordinate selections for s_c , t_c , r_c , and the selection of derivatives, are determined by the major axis direction as specified in the following two tables.

Table 22. Cube Map Face and Coordinate Selection

Major Axis Direction	Layer Number	Cube Map Face	s_c	t_c	r_c
$+r_x$	0	Positive X	$-r_z$	$-r_y$	r_x
$-r_x$	1	Negative X	$+r_z$	$-r_y$	r_x
$+r_y$	2	Positive Y	$+r_x$	$+r_z$	r_y
$-r_y$	3	Negative Y	$+r_x$	$-r_z$	r_y
$+r_z$	4	Positive Z	$+r_x$	$-r_y$	r_z
$-r_z$	5	Negative Z	$-r_x$	$-r_y$	r_z

Table 23. Cube Map Derivative Selection

Major Axis Direction	$\partial s_c / \partial x$	$\partial s_c / \partial y$	$\partial t_c / \partial x$	$\partial t_c / \partial y$	$\partial r_c / \partial x$	$\partial r_c / \partial y$
$+r_x$	$-\partial r_z / \partial x$	$-\partial r_z / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$
$-r_x$	$+\partial r_z / \partial x$	$+\partial r_z / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$-\partial r_x / \partial x$	$-\partial r_x / \partial y$
$+r_y$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$	$+\partial r_z / \partial x$	$+\partial r_z / \partial y$	$+\partial r_y / \partial x$	$+\partial r_y / \partial y$
$-r_y$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$	$-\partial r_z / \partial x$	$-\partial r_z / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$
$+r_z$	$+\partial r_x / \partial x$	$+\partial r_x / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$+\partial r_z / \partial x$	$+\partial r_z / \partial y$
$-r_z$	$-\partial r_x / \partial x$	$-\partial r_x / \partial y$	$-\partial r_y / \partial x$	$-\partial r_y / \partial y$	$-\partial r_z / \partial x$	$-\partial r_z / \partial y$

16.5.5. Cube Map Coordinate Transformation

$$s_{face} = \frac{1}{2} \times \frac{s_c}{|r_c|} + \frac{1}{2}$$

$$t_{face} = \frac{1}{2} \times \frac{t_c}{|r_c|} + \frac{1}{2}$$

16.5.6. Cube Map Derivative Transformation

The partial derivatives of the [Cube Map Coordinate Transformations](#) can be computed as:

$$\begin{aligned} \frac{\partial s_{face}}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{1}{2} \times \frac{s_c}{|r_c|} + \frac{1}{2} \right) \\ &= \frac{1}{2} \times \frac{\partial}{\partial x} \left(\frac{s_c}{|r_c|} \right) \\ &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial s_c / \partial x - s_c \times \partial r_c / \partial x}{(r_c)^2} \right) \end{aligned}$$

The other derivatives are simplified similarly, resulting in

$$\begin{aligned} \frac{\partial s_{face}}{\partial y} &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial s_c / \partial y - s_c \times \partial r_c / \partial y}{(r_c)^2} \right) \\ \frac{\partial t_{face}}{\partial x} &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial t_c / \partial x - t_c \times \partial r_c / \partial x}{(r_c)^2} \right) \\ \frac{\partial t_{face}}{\partial y} &= \frac{1}{2} \times \left(\frac{|r_c| \times \partial t_c / \partial y - t_c \times \partial r_c / \partial y}{(r_c)^2} \right) \end{aligned}$$

16.5.7. Scale Factor Operation, LOD Operation and Image Level(s) Selection

LOD selection **can** be either explicit (provided explicitly by the image instruction) or implicit (determined from a scale factor calculated from the derivatives). The LOD **must** be computed with `mipmapPrecisionBits` of accuracy.

Scale Factor Operation

The magnitude of the derivatives are calculated by:

$$m_{ux} = |\partial s / \partial x| \times w_{base}$$

$$m_{vx} = |\partial t / \partial x| \times h_{base}$$

$$m_{wx} = |\partial r / \partial x| \times d_{base}$$

$$m_{uy} = |\partial s / \partial y| \times w_{base}$$

$$m_{vy} = |\partial t / \partial y| \times h_{base}$$

$$m_{wy} = |\partial r / \partial y| \times d_{base}$$

where:

$$\partial t / \partial x = \partial t / \partial y = 0 \text{ (for 1D images)}$$

$$\partial r / \partial x = \partial r / \partial y = 0 \text{ (for 1D, 2D or Cube images)}$$

and:

$$w_{base} = \text{image.w}$$

$$h_{base} = \text{image.h}$$

$$d_{base} = \text{image.d}$$

(for the **baseMipLevel**, from the image descriptor).

A point sampled in screen space has an elliptical footprint in texture space. The minimum and maximum scale factors (ρ_{min} , ρ_{max}) **should** be the minor and major axes of this ellipse.

The *scale factors* ρ_x and ρ_y , calculated from the magnitude of the derivatives in x and y, are used to compute the minimum and maximum scale factors.

ρ_x and ρ_y **may** be approximated with functions f_x and f_y , subject to the following constraints:

f_x is continuous and monotonically increasing in each of m_{ux} , m_{vx} , and m_{wx}
 f_y is continuous and monotonically increasing in each of m_{uy} , m_{vy} , and m_{wy}

$$\max(|m_{ux}|, |m_{vx}|, |m_{wx}|) \leq f_x \leq \sqrt{2}(|m_{ux}| + |m_{vx}| + |m_{wx}|)$$

$$\max(|m_{uy}|, |m_{vy}|, |m_{wy}|) \leq f_y \leq \sqrt{2}(|m_{uy}| + |m_{vy}| + |m_{wy}|)$$

The minimum and maximum scale factors ($\rho_{\min}, \rho_{\max}$) are determined by:

$$\rho_{\max} = \max(\rho_x, \rho_y)$$

$$\rho_{\min} = \min(\rho_x, \rho_y)$$

The ratio of anisotropy is determined by:

$$\eta = \min(\rho_{\max}/\rho_{\min}, \max_{\text{Aniso}})$$

where:

$$\text{sampler.max}_{\text{Aniso}} = \text{maxAnisotropy} \text{ (from sampler descriptor)}$$

$$\text{limits.max}_{\text{Aniso}} = \text{maxSamplerAnisotropy} \text{ (from physical device limits)}$$

$$\max_{\text{Aniso}} = \min(\text{sampler.max}_{\text{Aniso}}, \text{limits.max}_{\text{Aniso}})$$

If $\rho_{\max} = \rho_{\min} = 0$, then all the partial derivatives are zero, the fragment's footprint in texel space is a point, and η **should** be treated as 1. If $\rho_{\max} \neq 0$ and $\rho_{\min} = 0$ then all partial derivatives along one axis are zero, the fragment's footprint in texel space is a line segment, and η **should** be treated as $\max_{\text{Aniso}}$. However, anytime the footprint is small in texel space the implementation **may** use a smaller value of η , even when $\rho_{\min}$ is zero or close to zero. If either [VkPhysicalDeviceFeatures::samplerAnisotropy](#) or [VkSamplerCreateInfo::anisotropyEnable](#) are [VK_FALSE](#), $\max_{\text{Aniso}}$ is set to 1.

If $\eta = 1$, sampling is isotropic. If $\eta > 1$, sampling is anisotropic.

The sampling rate (N) is derived as:

$$N = \lceil \eta \rceil$$

An implementation **may** round N up to the nearest supported sampling rate. An implementation **may** use the value of N as an approximation of η .

LOD Operation

The LOD parameter λ is computed as follows:

$$\lambda_{base}(x, y) = \begin{cases} shaderOp.Lod & \text{(from optional SPIR-V operand)} \\ \log_2\left(\frac{\rho_{max}}{\eta}\right) & \text{otherwise} \end{cases}$$
$$\lambda'(x, y) = \lambda_{base} + \text{clamp}(sampler.bias + shaderOp.bias, -maxSamplerLodBias, maxSamplerLodBias)$$
$$\lambda = \begin{cases} lod_{max}, & \lambda' > lod_{max} \\ \lambda', & lod_{min} \leq \lambda' \leq lod_{max} \\ lod_{min}, & \lambda' < lod_{min} \\ undefined, & lod_{min} > lod_{max} \end{cases}$$

where:

$$sampler.bias = mipLodBias \quad \text{(from sampler descriptor)}$$
$$shaderOp.bias = \begin{cases} Bias & \text{(from optional SPIR-V operand)} \\ 0 & \text{otherwise} \end{cases}$$
$$sampler.lod_{min} = minLod \quad \text{(from sampler descriptor)}$$
$$shaderOp.lod_{min} = \begin{cases} MinLod & \text{(from optional SPIR-V operand)} \\ 0 & \text{otherwise} \end{cases}$$
$$lod_{min} = \max(sampler.lod_{min}, shaderOp.lod_{min})$$
$$lod_{max} = maxLod \quad \text{(from sampler descriptor)}$$

and `maxSamplerLodBias` is the value of the [VkPhysicalDeviceLimits](#) feature `maxSamplerLodBias`.

Image Level(s) Selection

The image level(s) d , d_{hi} , and d_{lo} which texels are read from are determined by an image-level parameter d_l , which is computed based on the LOD parameter, as follows:

$$d_l = \begin{cases} nearest(d'), & \text{mipmapMode is VK_SAMPLER_MIPMAP_MODE_NEAREST} \\ d', & \text{otherwise} \end{cases}$$

where:

$$d' = \max(level_{base} + \text{clamp}(\lambda, 0, q), minLod_{imageView})$$

$$nearest(d') = \begin{cases} [d' + 0.5] - 1, & \text{preferred} \\ [d' + 0.5], & \text{alternative} \end{cases}$$

and:

$$minLod_{imageView} = \begin{cases} minLodFloat_{imageView}, & \text{preferred} \\ minLodInteger_{imageView}, & \text{alternative} \end{cases}$$
$$level_{base} = baseMipLevel$$
$$q = levelCount - 1$$

`baseMipLevel` and `levelCount` are taken from the `subresourceRange` of the image view.

$\text{minLod}_{\text{imageView}}$ **must** be less or equal to $\text{level}_{\text{base}} + q$.

If the sampler's `mipmapMode` is `VK_SAMPLER_MIPMAP_MODE_NEAREST`, then the level selected is $d = d_l$.

If the sampler's `mipmapMode` is `VK_SAMPLER_MIPMAP_MODE_LINEAR`, two neighboring levels are selected:

$$\begin{aligned}d_{hi} &= \lfloor d_l \rfloor \\d_{lo} &= \min(d_{hi} + 1, \text{level}_{\text{base}} + q) \\ \delta &= d_l - d_{hi}\end{aligned}$$

δ is the fractional value, quantized to the number of [mipmap precision bits](#), used for [linear filtering](#) between levels.

16.5.8. (s,t,r,q,a) to (u,v,w,a) Transformation

The normalized texel coordinates are scaled by the image level dimensions and the array layer is selected.

This transformation is performed once for each level used in [filtering](#) (either d , or d_{hi} and d_{lo}).

$$\begin{aligned}u(x, y) &= s(x, y) \times \text{width}_{\text{scale}} + \Delta_i \\v(x, y) &= \begin{cases} 0 & \text{for 1D images} \\ t(x, y) \times \text{height}_{\text{scale}} + \Delta_j & \text{otherwise} \end{cases} \\w(x, y) &= \begin{cases} 0 & \text{for 2D or Cube images} \\ r(x, y) \times \text{depth}_{\text{scale}} + \Delta_k & \text{otherwise} \end{cases} \\a(x, y) &= \begin{cases} a(x, y) & \text{for array images} \\ 0 & \text{otherwise} \end{cases}\end{aligned}$$

where:

$$\text{width}_{\text{scale}} = \text{width}_{\text{level}}$$

$$\text{height}_{\text{scale}} = \text{height}_{\text{level}}$$

$$\text{depth}_{\text{scale}} = \text{depth}_{\text{level}}$$

and where $(\Delta_i, \Delta_j, \Delta_k)$ are taken from the image instruction if it includes a `ConstOffset` or `Offset` operand, otherwise they are taken to be zero.

Operations then proceed to Unnormalized Texel Coordinate Operations.

16.6. Unnormalized Texel Coordinate Operations

16.6.1. (u,v,w,a) to (i,j,k,l,n) Transformation and Array Layer Selection

The unnormalized texel coordinates are transformed to integer texel coordinates relative to the selected mipmap level.

The layer index l is computed as:

$$l = \text{clamp}(\text{RNE}(a), 0, \text{layerCount} - 1) + \text{baseArrayLayer}$$

where layerCount is the number of layers in the image subresource range of the image view, baseArrayLayer is the first layer from the subresource range, and where:

$$\text{RNE}(a) = \begin{cases} \text{roundTiesToEven}(a) & \text{preferred, from IEEE Std 754-2008 Floating-Point Arithmetic} \\ \lfloor a + 0.5 \rfloor & \text{alternative} \end{cases}$$

The sample index n is assigned the value 0.

Nearest filtering (VK_FILTER_NEAREST) computes the integer texel coordinates that the unnormalized coordinates lie within:

$$\begin{aligned} i &= \lfloor u + \text{shift} \rfloor \\ j &= \lfloor v + \text{shift} \rfloor \\ k &= \lfloor w + \text{shift} \rfloor \end{aligned}$$

where:

$$\text{shift} = 0.0$$

Linear filtering (VK_FILTER_LINEAR) computes a set of neighboring coordinates which bound the unnormalized coordinates. The integer texel coordinates are combinations of i_0 or i_1 , j_0 or j_1 , k_0 or k_1 , as well as weights α , β , and γ .

$$\begin{aligned} i_0 &= \lfloor u - \text{shift} \rfloor \\ i_1 &= i_0 + 1 \\ j_0 &= \lfloor v - \text{shift} \rfloor \\ j_1 &= j_0 + 1 \\ k_0 &= \lfloor w - \text{shift} \rfloor \\ k_1 &= k_0 + 1 \end{aligned}$$

$$\alpha = \text{frac}(u - \text{shift})$$

$$\beta = \text{frac}(v - \text{shift})$$

$$\gamma = \text{frac}(w - \text{shift})$$

where:

shift = 0.5

and where:

$$\text{frac}(x) = x - \lfloor x \rfloor$$

where the number of fraction bits retained is specified by `VkPhysicalDeviceLimits::subTexelPrecisionBits`.

Cubic filtering (`VK_FILTER_CUBIC_EXT`) computes a set of neighboring coordinates which bound the unnormalized coordinates. The integer texel coordinates are combinations of i_0, i_1, i_2 or i_3, j_0, j_1, j_2 or j_3, k_0, k_1, k_2 or k_3 , as well as weights α, β , and γ .

$$\begin{aligned} i_0 &= \lfloor u - \frac{3}{2} \rfloor & i_1 &= i_0 + 1 & i_2 &= i_1 + 1 & i_3 &= i_2 + 1 \\ j_0 &= \lfloor v - \frac{3}{2} \rfloor & j_1 &= j_0 + 1 & j_2 &= j_1 + 1 & j_3 &= j_2 + 1 \\ k_0 &= \lfloor w - \frac{3}{2} \rfloor & k_1 &= k_0 + 1 & k_2 &= k_1 + 1 & k_3 &= k_2 + 1 \end{aligned}$$

$$\alpha = \text{frac}\left(u - \frac{1}{2}\right)$$

$$\beta = \text{frac}\left(v - \frac{1}{2}\right)$$

$$\gamma = \text{frac}\left(w - \frac{1}{2}\right)$$

where:

$$\text{frac}(x) = x - \lfloor x \rfloor$$

where the number of fraction bits retained is specified by `VkPhysicalDeviceLimits::subTexelPrecisionBits`.

16.7. Integer Texel Coordinate Operations

The `OpImageFetch` and `OpImageFetchSparse` SPIR-V instructions **may** supply a LOD from which texels are to be fetched using the optional SPIR-V operand `Lod`. Other integer-coordinate operations **must** not. If the `Lod` is provided then it **must** be an integer.

The image level selected is:

$$d = level_{base} + \begin{cases} Lod & \text{(from optional SPIR-V operand)} \\ 0 & \text{otherwise} \end{cases}$$

If d does not lie in the range $[\text{baseMipLevel}, \text{baseMipLevel} + \text{levelCount})$ or d is less than $\text{minLodInteger}_{\text{imageView}}$, then any values fetched are zero if the `robustImageAccess2` feature is enabled, otherwise are undefined, and any writes (if supported) are discarded.

16.8. Image Sample Operations

16.8.1. Wrapping Operation

If the used sampler was created without `VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT`, Cube images ignore the wrap modes specified in the sampler. Instead, if `VK_FILTER_NEAREST` is used within a mip level then `VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE` is used, and if `VK_FILTER_LINEAR` is used within a mip level then sampling at the edges is performed as described earlier in the [Cube map edge handling](#) section.

The first integer texel coordinate i is transformed based on the `addressModeU` parameter of the sampler.

$$i = \begin{cases} i \bmod \text{size} & \text{for repeat} \\ (\text{size} - 1) - \text{mirror}((i \bmod (2 \times \text{size})) - \text{size}) & \text{for mirrored repeat} \\ \text{clamp}(i, 0, \text{size} - 1) & \text{for clamp to edge} \\ \text{clamp}(i, -1, \text{size}) & \text{for clamp to border} \\ \text{clamp}(\text{mirror}(i), 0, \text{size} - 1) & \text{for mirror clamp to edge} \end{cases}$$

where:

$$\text{mirror}(n) = \begin{cases} n & \text{for } n \geq 0 \\ -(1 + n) & \text{otherwise} \end{cases}$$

j (for 2D and Cube image) and k (for 3D image) are similarly transformed based on the `addressModeV` and `addressModeW` parameters of the sampler, respectively.

16.8.2. Texel Gathering

SPIR-V instructions with `Gather` in the name return a vector derived from 4 texels in the base level of the image view. The rules for the `VK_FILTER_LINEAR` minification filter are applied to identify the four selected texels. Each texel is then converted to an RGBA value according to [conversion to RGBA](#) and then [swizzled](#). A four-component vector is then assembled by taking the component indicated by the `Component` value in the instruction from the swizzled color value of the four texels. If the operation does not use the `ConstOffsets` image operand then the four texels form the 2×2 rectangle used for texture filtering:

$$\begin{aligned} \tau[R] &= \tau_{i0j1}[\text{level}_{\text{base}}][\text{comp}] \\ \tau[G] &= \tau_{i1j1}[\text{level}_{\text{base}}][\text{comp}] \\ \tau[B] &= \tau_{i1j0}[\text{level}_{\text{base}}][\text{comp}] \\ \tau[A] &= \tau_{i0j0}[\text{level}_{\text{base}}][\text{comp}] \end{aligned}$$

If the operation does use the `ConstOffsets` image operand then the offsets allow a custom filter to be

defined:

$$\begin{aligned}\tau[R] &= \tau_{i_0 j_0} + \Delta_0[level_{base}][comp] \\ \tau[G] &= \tau_{i_0 j_0} + \Delta_1[level_{base}][comp] \\ \tau[B] &= \tau_{i_0 j_0} + \Delta_2[level_{base}][comp] \\ \tau[A] &= \tau_{i_0 j_0} + \Delta_3[level_{base}][comp]\end{aligned}$$

where:

$$\tau[level_{base}][comp] = \begin{cases} \tau[level_{base}][R], & \text{for } comp = 0 \\ \tau[level_{base}][G], & \text{for } comp = 1 \\ \tau[level_{base}][B], & \text{for } comp = 2 \\ \tau[level_{base}][A], & \text{for } comp = 3 \end{cases}$$

comp from SPIR-V operand Component

OpImage*Gather must not be used on a sampled image with **sampler Y'C_BC_R conversion** enabled.

If $level_{base} < minLodInteger_{imageView}$, then any values fetched are zero if **robustImageAccess2** is enabled. Otherwise values are undefined.

16.8.3. Texel Filtering

Texel filtering is first performed for each level (either d or d_{hi} and d_{lo}).

If λ is less than or equal to zero, the texture is said to be *magnified*, and the filter mode within a mip level is selected by the **magFilter** in the sampler. If λ is greater than zero, the texture is said to be *minified*, and the filter mode within a mip level is selected by the **minFilter** in the sampler.

Texel Nearest Filtering

Within a mip level, **VK_FILTER_NEAREST** filtering selects a single value using the (i, j, k) texel coordinates, with all texels taken from layer 1.

$$\tau[level] = \begin{cases} \tau_{ijk}[level], & \text{for 3D image} \\ \tau_{ij}[level], & \text{for 2D or Cube image} \\ \tau_i[level], & \text{for 1D image} \end{cases}$$

Texel Linear Filtering

Within a mip level, **VK_FILTER_LINEAR** filtering combines 8 (for 3D), 4 (for 2D or Cube), or 2 (for 1D) texel values, together with their linear weights. The linear weights are derived from the fractions computed earlier:

$$\begin{aligned}
w_{i_0} &= (1 - \alpha) \\
w_{i_1} &= (\alpha) \\
w_{j_0} &= (1 - \beta) \\
w_{j_1} &= (\beta) \\
w_{k_0} &= (1 - \gamma) \\
w_{k_1} &= (\gamma)
\end{aligned}$$

The values of multiple texels, together with their weights, are combined using a weighted average to produce a filtered value:

$$\begin{aligned}
\tau_{3D} &= \sum_{k=k_0}^{k_1} \sum_{j=j_0}^{j_1} \sum_{i=i_0}^{i_1} (w_i)(w_j)(w_k)\tau_{ijk} \\
\tau_{2D} &= \sum_{j=j_0}^{j_1} \sum_{i=i_0}^{i_1} (w_i)(w_j)\tau_{ij} \\
\tau_{1D} &= \sum_{i=i_0}^{i_1} (w_i)\tau_i
\end{aligned}$$

Texel Cubic Filtering

Within a mip level, `VK_FILTER_CUBIC_EXT`, filtering computes a weighted average of 64 (for 3D), 16 (for 2D), or 4 (for 1D) texel values, together with their Catmull-Rom weights.

Catmull-Rom weights are derived from the fractions computed earlier.

$$\begin{aligned}
\begin{bmatrix} w_{i_0} & w_{i_1} & w_{i_2} & w_{i_3} \end{bmatrix} &= \frac{1}{2} \begin{bmatrix} 1 & \alpha & \alpha^2 & \alpha^3 \end{bmatrix} \begin{bmatrix} 0 & 2 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 2 & -5 & 4 & -1 \\ -1 & 3 & -3 & 1 \end{bmatrix} \\
\begin{bmatrix} w_{j_0} & w_{j_1} & w_{j_2} & w_{j_3} \end{bmatrix} &= \frac{1}{2} \begin{bmatrix} 1 & \beta & \beta^2 & \beta^3 \end{bmatrix} \begin{bmatrix} 0 & 2 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 2 & -5 & 4 & -1 \\ -1 & 3 & -3 & 1 \end{bmatrix} \\
\begin{bmatrix} w_{k_0} & w_{k_1} & w_{k_2} & w_{k_3} \end{bmatrix} &= \frac{1}{2} \begin{bmatrix} 1 & \gamma & \gamma^2 & \gamma^3 \end{bmatrix} \begin{bmatrix} 0 & 2 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 2 & -5 & 4 & -1 \\ -1 & 3 & -3 & 1 \end{bmatrix}
\end{aligned}$$

The values of multiple texels, together with their weights, are combined to produce a filtered value.

The `VkSamplerReductionModeCreateInfo::reductionMode` **can** control the process by which multiple texels, together with their weights, are combined to produce a filtered texture value.

When the `reductionMode` is set (explicitly or implicitly) to `VK_SAMPLER_REDUCTION_MODE_WEIGHTED_AVERAGE`, a weighted average is computed:

$$\tau_{3D} = \sum_{k=j_0}^{k_3} \sum_{j=j_0}^{j_3} \sum_{i=i_0}^{i_3} (w_i)(w_j)(w_k)\tau_{ijk}$$

$$\tau_{2D} = \sum_{j=j_0}^{j_3} \sum_{i=i_0}^{i_3} (w_i)(w_j)\tau_{ij}$$

$$\tau_{1D} = \sum_{i=i_0}^{i_3} (w_i)\tau_i$$

Texel Mipmap Filtering

`VK_SAMPLER_MIPMAP_MODE_NEAREST` filtering returns the value of a single mipmap level,

$\tau = \tau[d]$.

`VK_SAMPLER_MIPMAP_MODE_LINEAR` filtering combines the values of multiple mipmap levels ($\tau[hi]$ and $\tau[lo]$), together with their linear weights.

The linear weights are derived from the fraction computed earlier:

$$w_{hi} = (1 - \delta)$$

$$w_{lo} = (\delta)$$

The values of multiple mipmap levels together with their linear weights, are combined using a weighted average to produce a final filtered value:

$$\tau = (w_{hi})\tau[hi] + (w_{lo})\tau[lo]$$

Texel Anisotropic Filtering

Anisotropic filtering is enabled by the `anisotropyEnable` in the sampler. When enabled, the image filtering scheme accounts for a degree of anisotropy.

The particular scheme for anisotropic texture filtering is implementation-dependent. Implementations **should** consider the `magFilter`, `minFilter` and `mipmapMode` of the sampler to control the specifics of the anisotropic filtering scheme used. In addition, implementations **should** consider `minLod` and `maxLod` of the sampler.

NOTE

For historical reasons, vendor implementations of anisotropic filtering interpret these sampler parameters in different ways, particularly in corner cases such as `magFilter`, `minFilter` of `NEAREST` or `maxAnisotropy` equal to 1.0. Applications should not expect consistent behavior in such cases, and should use anisotropic filtering only with parameters which are expected to give a quality improvement relative to `LINEAR` filtering.

The following describes one particular approach to implementing anisotropic filtering for the 2D Image case; implementations **may** choose other methods:

Given a `magFilter`, `minFilter` of `VK_FILTER_LINEAR` and a `mipmapMode` of

VK_SAMPLER_MIPMAP_MODE_NEAREST:

Instead of a single isotropic sample, N isotropic samples are sampled within the image footprint of the image level d to approximate an anisotropic filter. The sum $\tau_{2D_{aniso}}$ is defined using the single isotropic $\tau_{2D}(u,v)$ at level d .

$$\tau_{2D_{aniso}} = \frac{1}{N} \sum_{i=1}^N \tau_{2D} \left(u \left(x - \frac{1}{2} + \frac{i}{N+1}, y \right), v \left(x - \frac{1}{2} + \frac{i}{N+1}, y \right) \right), \quad \text{when } \rho_x > \rho_y$$
$$\tau_{2D_{aniso}} = \frac{1}{N} \sum_{i=1}^N \tau_{2D} \left(u \left(x, y - \frac{1}{2} + \frac{i}{N+1} \right), v \left(x, y - \frac{1}{2} + \frac{i}{N+1} \right) \right), \quad \text{when } \rho_y \geq \rho_x$$

16.9. Image Operation Steps

Each step described in this chapter is performed by a subset of the image instructions:

- Texel Input Validation Operations, Format Conversion, Texel Replacement, Conversion to RGBA, and Component Swizzle: Performed by all instructions except **OpImageWrite**.
- Depth Comparison: Performed by **OpImage*Dref** instructions.
- All Texel output operations: Performed by **OpImageWrite**.
- Projection: Performed by all **OpImage*Proj** instructions.
- Derivative Image Operations, Cube Map Operations, Scale Factor Operation, LOD Operation and Image Level(s) Selection, and Texel Anisotropic Filtering: Performed by all **OpImageSample*** and **OpImageSparseSample*** instructions.
- (s,t,r,q,a) to (u,v,w,a) Transformation, Wrapping, and (u,v,w,a) to (i,j,k,l,n) Transformation And Array Layer Selection: Performed by all **OpImageSample**, **OpImageSparseSample**, and **OpImage*Gather** instructions.
- Texel Gathering: Performed by **OpImage*Gather** instructions.
- Texel Filtering: Performed by all **OpImageSample*** and **OpImageSparseSample*** instructions.
- Sparse Residency: Performed by all **OpImageSparse*** instructions.

16.10. Image Query Instructions

16.10.1. Image Property Queries

OpImageQuerySize, **OpImageQuerySizeLod**, **OpImageQueryLevels**, and **OpImageQuerySamples** query properties of the image descriptor that would be accessed by a shader image operation. They return 0 if the bound descriptor is a null descriptor.

OpImageQuerySizeLod returns the size of the image level identified by the **Level of Detail** operand. If that level does not exist in the image, and the descriptor is not null, then the value returned is undefined.

16.10.2. LOD Query

`OpImageQueryLod` returns the Lod parameters that would be used in an image operation with the given image and coordinates. If the descriptor that would be accessed is a null descriptor then (0,0) is returned. Otherwise, the steps described in this chapter are performed as if for `OpImageSampleImplicitLod`, up to [Scale Factor Operation](#), [LOD Operation](#) and [Image Level\(s\) Selection](#). The return value is the vector $(\lambda', d_l - \text{level}_{\text{base}})$. These values **may** be subject to implementation-specific maxima and minima for very large, out-of-range values.

Chapter 17. Queries

Queries provide a mechanism to return information about the processing of a sequence of Vulkan commands. Query operations are asynchronous, and as such, their results are not returned immediately. Instead, their results, and their availability status are stored in a [Query Pool](#). The state of these queries **can** be read back on the host, or copied to a buffer object on the device.

The supported query types are [Occlusion Queries](#), [Pipeline Statistics Queries](#), [Result Status Queries](#), [Video Encode Feedback Queries](#) and [Timestamp Queries](#). [Performance Queries](#) are supported if the associated extension is available. [Transform Feedback Queries](#) are supported if the associated extension is available.

Several additional queries with specific purposes associated with ray tracing are available if the corresponding extensions are supported, as described for [VkQueryType](#).

17.1. Query Pools

Queries are managed using *query pool* objects. Each query pool is a collection of a specific number of queries of a particular type.

Query pools are represented by [VkQueryPool](#) handles:

```
// Provided by VK_VERSION_1_0
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkQueryPool)
```

To create a query pool, call:

```
// Provided by VK_VERSION_1_0
VkResult vkCreateQueryPool(
    VkDevice                                device,
    const VkQueryPoolCreateInfo*             pCreateInfo,
    const VkAllocationCallbacks*            pAllocator,
    VkQueryPool*                             pQueryPool);
```

- [device](#) is the logical device that creates the query pool.
- [pCreateInfo](#) is a pointer to a [VkQueryPoolCreateInfo](#) structure containing the number and type of queries to be managed by the pool.
- [pAllocator](#) controls host memory allocation as described in the [Memory Allocation](#) chapter.
- [pQueryPool](#) is a pointer to a [VkQueryPool](#) handle in which the resulting query pool object is returned.

Valid Usage

- VUID-vkCreateQueryPool-device-09663

[device](#) must support at least one queue family with one of the

VK_QUEUE_VIDEO_ENCODE_BIT_KHR, VK_QUEUE_VIDEO_DECODE_BIT_KHR, VK_QUEUE_COMPUTE_BIT, or VK_QUEUE_GRAPHICS_BIT capabilities

Valid Usage (Implicit)

- VUID-vkCreateQueryPool-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCreateQueryPool-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkQueryPoolCreateInfo](#) structure
- VUID-vkCreateQueryPool-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateQueryPool-pQueryPool-parameter
pQueryPool **must** be a valid pointer to a [VkQueryPool](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

The [VkQueryPoolCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkQueryPoolCreateInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkQueryPoolCreateFlags flags;
    VkQueryType        queryType;
    uint32_t           queryCount;
    VkQueryPipelineStatisticFlags pipelineStatistics;
} VkQueryPoolCreateInfo;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is reserved for future use.
- **queryType** is a [VkQueryType](#) value specifying the type of queries managed by the pool.
- **queryCount** is the number of queries managed by the pool.

- `pipelineStatistics` is a bitmask of `VkQueryPipelineStatisticFlagBits` specifying which counters will be returned in queries on the new pool, as described below in [Pipeline Statistics Queries](#).

`pipelineStatistics` is ignored if `queryType` is not `VK_QUERY_TYPE_PIPELINE_STATISTICS`.

Valid Usage

- VUID-VkQueryPoolCreateInfo-queryType-00791
If the `pipelineStatisticsQuery` feature is not enabled, `queryType` **must** not be `VK_QUERY_TYPE_PIPELINE_STATISTICS`
- VUID-VkQueryPoolCreateInfo-queryType-00792
If `queryType` is `VK_QUERY_TYPE_PIPELINE_STATISTICS`, `pipelineStatistics` **must** be a valid combination of `VkQueryPipelineStatisticFlagBits` values
- VUID-VkQueryPoolCreateInfo-queryType-09534
If `queryType` is `VK_QUERY_TYPE_PIPELINE_STATISTICS`, `pipelineStatistics` **must** not be zero
- VUID-VkQueryPoolCreateInfo-queryType-03222
If `queryType` is `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, the `pNext` chain **must** include a `VkQueryPoolPerformanceCreateInfoKHR` structure
- VUID-VkQueryPoolCreateInfo-queryCount-02763
`queryCount` **must** be greater than 0
- VUID-VkQueryPoolCreateInfo-queryType-07133
If `queryType` is `VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR`, then the `pNext` chain **must** include a `VkVideoProfileInfoKHR` structure with `videoCodecOperation` specifying an encode operation
- VUID-VkQueryPoolCreateInfo-queryType-07906
If `queryType` is `VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR`, then the `pNext` chain **must** include a `VkQueryPoolVideoEncodeFeedbackCreateInfoKHR` structure
- VUID-VkQueryPoolCreateInfo-queryType-07907
If `queryType` is `VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR`, and the `pNext` chain includes a `VkVideoProfileInfoKHR` structure and a `VkQueryPoolVideoEncodeFeedbackCreateInfoKHR` structure, then `VkQueryPoolVideoEncodeFeedbackCreateInfoKHR::encodeFeedbackFlags` **must** not contain any bits that are not set in `VkVideoEncodeCapabilitiesKHR::supportedEncodeFeedbackFlags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile described by `VkVideoProfileInfoKHR` and its `pNext` chain

Valid Usage (Implicit)

- VUID-VkQueryPoolCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_QUERY_POOL_CREATE_INFO`
- VUID-VkQueryPoolCreateInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkQueryPoolPerformanceCreateInfoKHR`,

[VkQueryPoolVideoEncodeFeedbackCreateInfoKHR](#), [VkVideoDecodeAV1ProfileInfoKHR](#),
[VkVideoDecodeH264ProfileInfoKHR](#), [VkVideoDecodeH265ProfileInfoKHR](#),
[VkVideoDecodeUsageInfoKHR](#), [VkVideoEncodeH264ProfileInfoKHR](#),
[VkVideoEncodeH265ProfileInfoKHR](#), [VkVideoEncodeUsageInfoKHR](#), or
[VkVideoProfileInfoKHR](#)

- VUID-VkQueryPoolCreateInfo-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkQueryPoolCreateInfo-flags-zero-bitmask
flags **must** be 0
- VUID-VkQueryPoolCreateInfo-queryType-parameter
queryType **must** be a valid [VkQueryType](#) value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryPoolCreateFlags;
```

VkQueryPoolCreateFlags is a bitmask type for setting a mask, but is currently reserved for future use.

The **VkQueryPoolPerformanceCreateInfoKHR** structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkQueryPoolPerformanceCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           queueFamilyIndex;
    uint32_t           counterIndexCount;
    const uint32_t*     pCounterIndices;
} VkQueryPoolPerformanceCreateInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **queueFamilyIndex** is the queue family index to create this performance query pool for.
- **counterIndexCount** is the length of the **pCounterIndices** array.
- **pCounterIndices** is a pointer to an array of indices into the [vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR::pCounters](#) to enable in this performance query pool.

Valid Usage

- VUID-VkQueryPoolPerformanceCreateInfoKHR-queueFamilyIndex-03236
queueFamilyIndex **must** be a valid queue family index of the device
- VUID-VkQueryPoolPerformanceCreateInfoKHR-performanceCounterQueryPools-03237
The **performanceCounterQueryPools** feature **must** be enabled

- VUID-VkQueryPoolPerformanceCreateInfoKHR-pCounterIndices-03321

Each element of `pCounterIndices` **must** be in the range of counters reported by `vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR` for the queue family specified in `queueFamilyIndex`

Valid Usage (Implicit)

- VUID-VkQueryPoolPerformanceCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_QUERY_POOL_PERFORMANCE_CREATE_INFO_KHR`
- VUID-VkQueryPoolPerformanceCreateInfoKHR-pCounterIndices-parameter
`pCounterIndices` **must** be a valid pointer to an array of `counterIndexCount` `uint32_t` values
- VUID-VkQueryPoolPerformanceCreateInfoKHR-counterIndexCount-arraylength
`counterIndexCount` **must** be greater than 0

To query the number of passes required to query a performance query pool on a physical device, call:

```
// Provided by VK_KHR_performance_query
void vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR(
    VkPhysicalDevice          physicalDevice,
    const VkQueryPoolPerformanceCreateInfoKHR* pPerformanceQueryCreateInfo,
    uint32_t*                 pNumPasses);
```

- `physicalDevice` is the handle to the physical device whose queue family performance query counter properties will be queried.
- `pPerformanceQueryCreateInfo` is a pointer to a `VkQueryPoolPerformanceCreateInfoKHR` of the performance query that is to be created.
- `pNumPasses` is a pointer to an integer related to the number of passes required to query the performance query pool, as described below.

The `pPerformanceQueryCreateInfo` member `VkQueryPoolPerformanceCreateInfoKHR::queueFamilyIndex` **must** be a queue family of `physicalDevice`. The number of passes required to capture the counters specified in the `pPerformanceQueryCreateInfo` member `VkQueryPoolPerformanceCreateInfoKHR::pCounters` is returned in `pNumPasses`.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR-pPerformanceQueryCreateInfo-parameter
`pPerformanceQueryCreateInfo` **must** be a valid pointer to a valid

[VkQueryPoolPerformanceCreateInfoKHR](#) structure

- VUID-vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR-pNumPasses-parameter
`pNumPasses` **must** be a valid pointer to a `uint32_t` value

To destroy a query pool, call:

```
// Provided by VK_VERSION_1_0
void vkDestroyQueryPool(
    VkDevice          device,
    VkQueryPool       queryPool,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the query pool.
- `queryPool` is the query pool to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyQueryPool-queryPool-00793
All submitted commands that refer to `queryPool` **must** have completed execution
- VUID-vkDestroyQueryPool-queryPool-00794
If `VkAllocationCallbacks` were provided when `queryPool` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyQueryPool-queryPool-00795
If no `VkAllocationCallbacks` were provided when `queryPool` was created, `pAllocator` **must** be `NULL`

NOTE

Applications **can** verify that `queryPool` **can** be destroyed by checking that `vkGetQueryPoolResults()` without the `VK_QUERY_RESULT_PARTIAL_BIT` flag returns `VK_SUCCESS` for all queries that are used in command buffers submitted for execution.

Valid Usage (Implicit)

- VUID-vkDestroyQueryPool-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyQueryPool-queryPool-parameter
If `queryPool` is not `VK_NULL_HANDLE`, `queryPool` **must** be a valid `VkQueryPool` handle
- VUID-vkDestroyQueryPool-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

- VUID-vkDestroyQueryPool-queryPool-parent

If `queryPool` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `queryPool` **must** be externally synchronized

Possible values of `VkQueryPoolCreateInfo::queryType`, specifying the type of queries managed by the pool, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryType {
    VK_QUERY_TYPE_OCCLUSION = 0,
    VK_QUERY_TYPE_PIPELINE_STATISTICS = 1,
    VK_QUERY_TYPE_TIMESTAMP = 2,
    // Provided by VK_KHR_video_queue
    VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR = 1000023000,
    // Provided by VK_EXT_transform_feedback
    VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT = 1000028004,
    // Provided by VK_KHR_performance_query
    VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR = 1000116000,
    // Provided by VK_KHR_acceleration_structure
    VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR = 1000150000,
    // Provided by VK_KHR_acceleration_structure
    VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR = 1000150001,
    // Provided by VK_KHR_video_encode_queue
    VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR = 1000299000,
    // Provided by VK_EXT_primitives_generated_query
    VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT = 1000382000,
    // Provided by VK_KHR_ray_tracing_maintenance1
    VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR =
1000386000,
    // Provided by VK_KHR_ray_tracing_maintenance1
    VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR = 1000386001,
    // Provided by VK_EXT_opacity_micromap
    VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT = 1000396000,
    // Provided by VK_EXT_opacity_micromap
    VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT = 1000396001,
} VkQueryType;
```

- `VK_QUERY_TYPE_OCCLUSION` specifies an `occlusion query`.
- `VK_QUERY_TYPE_PIPELINE_STATISTICS` specifies a `pipeline statistics query`.
- `VK_QUERY_TYPE_TIMESTAMP` specifies a `timestamp query`.
- `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` specifies a `performance query`.

- `VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT` specifies a transform feedback query.
- `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` specifies a primitives generated query.
- `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR` specifies an acceleration structure size query for use with `vkCmdWriteAccelerationStructuresPropertiesKHR` or `vkWriteAccelerationStructuresPropertiesKHR`.
- `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR` specifies a serialization acceleration structure size query.
- `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR` specifies an acceleration structure size query for use with `vkCmdWriteAccelerationStructuresPropertiesKHR` or `vkWriteAccelerationStructuresPropertiesKHR`.
- `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR` specifies a serialization acceleration structure pointer count query.
- `VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR` specifies a result status query.
- `VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR` specifies a video encode feedback query.

17.2. Query Operation

The operation of queries is controlled by the commands `vkCmdBeginQuery`, `vkCmdEndQuery`, `vkCmdBeginQueryIndexedEXT`, `vkCmdEndQueryIndexedEXT`, `vkCmdResetQueryPool`, `vkCmdCopyQueryPoolResults`, `vkCmdWriteTimestamp2KHR`, and `vkCmdWriteTimestamp`.

In order for a `VkCommandBuffer` to record query management commands, the queue family for which its `VkCommandPool` was created **must** support the appropriate type of operations (graphics, compute) suitable for the query type of a given query pool.

Each query in a query pool has a status that is either *unavailable* or *available*, and also has state to store the numerical results of a query operation of the type requested when the query pool was created. Resetting a query via `vkCmdResetQueryPool` sets the status to unavailable and makes the numerical results undefined. A query is made available by the operation of `vkCmdEndQuery`, `vkCmdEndQueryIndexedEXT`, `vkCmdWriteTimestamp2KHR`, or `vkCmdWriteTimestamp`. Both the availability status and numerical results **can** be retrieved by calling either `vkGetQueryPoolResults` or `vkCmdCopyQueryPoolResults`.

After query pool creation, each query is in an uninitialized state and **must** be reset before it is used. Queries **must** also be reset between uses.

If a logical device includes multiple physical devices, then each command that writes a query **must** execute on a single physical device, and any call to `vkCmdBeginQuery` **must** execute the corresponding `vkCmdEndQuery` command on the same physical device.

To reset a range of queries in a query pool on a queue, call:


```
// Provided by VK_VERSION_1_0
void vkCmdResetQueryPool(
    VkCommandBuffer          commandBuffer,
    VkQueryPool              queryPool,
    uint32_t                 firstQuery,
    uint32_t                 queryCount);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the handle of the query pool managing the queries being reset.
- `firstQuery` is the initial query index to reset.
- `queryCount` is the number of queries to reset.

When executed on a queue, this command sets the status of query indices [`firstQuery`, `firstQuery` + `queryCount` - 1] to unavailable.

This command defines an execution dependency between other query commands that reference the same query.

The first [synchronization scope](#) includes all commands which reference the queries in `queryPool` indicated by `firstQuery` and `queryCount` that occur earlier in [submission order](#).

The second [synchronization scope](#) includes all commands which reference the queries in `queryPool` indicated by `firstQuery` and `queryCount` that occur later in [submission order](#).

The operation of this command happens after the first scope and happens before the second scope.

If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, this command sets the status of query indices [`firstQuery`, `firstQuery` + `queryCount` - 1] to unavailable for each pass of `queryPool`, as indicated by a call to [vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR](#).

NOTE

Because `vkCmdResetQueryPool` resets all the passes of the indicated queries, applications must not record a `vkCmdResetQueryPool` command for a `queryPool` created with `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` in a command buffer that needs to be submitted multiple times as indicated by a call to [vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR](#). Otherwise applications will never be able to complete the recorded queries.

Valid Usage

- VUID-vkCmdResetQueryPool-firstQuery-09436
`firstQuery` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdResetQueryPool-firstQuery-09437
The sum of `firstQuery` and `queryCount` **must** be less than or equal to the number of queries in `queryPool`

- VUID-vkCmdResetQueryPool-None-02841
All queries used by the command **must** not be active
- VUID-vkCmdResetQueryPool-firstQuery-02862
If **queryPool** was created with **VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR**, this command **must** not be recorded in a command buffer that, either directly or through secondary command buffers, also contains begin commands for a query from the set of queries [**firstQuery**, **firstQuery** + **queryCount** - 1]

Valid Usage (Implicit)

- VUID-vkCmdResetQueryPool-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdResetQueryPool-queryPool-parameter
queryPool **must** be a valid **VkQueryPool** handle
- VUID-vkCmdResetQueryPool-commandBuffer-recording
commandBuffer **must** be in the **recording state**
- VUID-vkCmdResetQueryPool-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics, compute, decode, or encode operations
- VUID-vkCmdResetQueryPool-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdResetQueryPool-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdResetQueryPool-commonparent
Both of **commandBuffer**, and **queryPool** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Graphics Compute Decode Encode	Action

Once queries are reset and ready for use, query commands **can** be issued to a command buffer. Occlusion queries and pipeline statistics queries count events - drawn samples and pipeline stage invocations, respectively - resulting from commands that are recorded between a [vkCmdBeginQuery](#) command and a [vkCmdEndQuery](#) command within a specified command buffer, effectively scoping a set of drawing and/or dispatching commands. Timestamp queries write timestamps to a query pool. Performance queries record performance counters to a query pool.

A query **must** begin and end in the same command buffer, although if it is a primary command buffer, and the [inheritedQueries](#) feature is enabled, it **can** execute secondary command buffers during the query operation. For a secondary command buffer to be executed while a query is active, it **must** set the [occlusionQueryEnable](#), [queryFlags](#), and/or [pipelineStatistics](#) members of [VkCommandBufferInheritanceInfo](#) to conservative values, as described in the [Command Buffer Recording](#) section. A query **must** either begin and end inside the same subpass of a render pass instance, or **must** both begin and end outside of a render pass instance (i.e. contain entire render pass instances).

If queries are used while executing a render pass instance that has multiview enabled, the query uses N consecutive query indices in the query pool (starting at [query](#)) where N is the number of bits set in the view mask in the subpass the query is used in. How the numerical results of the query are distributed among the queries is implementation-dependent. For example, some implementations **may** write each view's results to a distinct query, while other implementations **may** write the total result to the first query and write zero to the other queries. However, the sum of the results in all the queries **must** accurately reflect the total result of the query summed over all views. Applications **can** sum the results from all the queries to compute the total result.

Queries used with multiview rendering **must** not span subpasses, i.e. they **must** begin and end in the same subpass.

A query **must** either begin and end inside the same video coding scope, or **must** both begin and end outside of a video coding scope and **must** not contain entire video coding scopes.

To begin a query, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBeginQuery(
    VkCommandBuffer          commandBuffer,
    VkQueryPool              queryPool,
    uint32_t                 query,
    VkQueryControlFlags      flags);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the query pool that will manage the results of the query.
- `query` is the query index within the query pool that will contain the results.
- `flags` is a bitmask of `VkQueryControlFlagBits` specifying constraints on the types of queries that **can** be performed.

If the `queryType` of the pool is `VK_QUERY_TYPE_OCCLUSION` and `flags` contains `VK_QUERY_CONTROL_PRECISE_BIT`, an implementation **must** return a result that matches the actual number of samples passed. This is described in more detail in [Occlusion Queries](#).

Calling `vkCmdBeginQuery` is equivalent to calling `vkCmdBeginQueryIndexedEXT` with the `index` parameter set to zero.

After beginning a query, that query is considered *active* within the command buffer it was called in until that same query is ended. Queries active in a primary command buffer when secondary command buffers are executed are considered active for those secondary command buffers.

Furthermore, if the query is started within a video coding scope, the following command buffer states are initialized for the query type:

- The `active_query_index` is set to the value specified by `query`.
- The *last activatable query index* is also set to the value specified by `query`.

Each [video coding operation](#) stores a result to the query corresponding to the current active query index, followed by incrementing the active query index. If the active query index gets incremented past the last activatable query index, issuing any further video coding operations results in undefined behavior.

NOTE

In practice, this means that currently no more than a single video coding operation **must** be issued between a begin and end query pair.

This command defines an execution dependency between other query commands that reference the same query.

The first [synchronization scope](#) includes all commands which reference the queries in `queryPool` indicated by `query` that occur earlier in [submission order](#).

The second [synchronization scope](#) includes all commands which reference the queries in `queryPool` indicated by `query` that occur later in [submission order](#).

The operation of this command happens after the first scope and happens before the second scope.

Valid Usage

- VUID-vkCmdBeginQuery-None-00807
All queries used by the command **must** be *unavailable*
- VUID-vkCmdBeginQuery-queryType-02804
The `queryType` used to create `queryPool` **must** not be `VK_QUERY_TYPE_TIMESTAMP`
- VUID-vkCmdBeginQuery-queryType-04728
The `queryType` used to create `queryPool` **must** not be `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR` or `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR`
- VUID-vkCmdBeginQuery-queryType-06741
The `queryType` used to create `queryPool` **must** not be `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR` or `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR`
- VUID-vkCmdBeginQuery-queryType-00800
If the `occlusionQueryPrecise` feature is not enabled, or the `queryType` used to create `queryPool` was not `VK_QUERY_TYPE_OCCLUSION`, `flags` **must** not contain `VK_QUERY_CONTROL_PRECISE_BIT`
- VUID-vkCmdBeginQuery-query-00802
`query` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdBeginQuery-queryType-00803
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_OCCLUSION`, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBeginQuery-queryType-00804
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PIPELINE_STATISTICS` and any of the `pipelineStatistics` indicate graphics operations, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBeginQuery-queryType-00805
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PIPELINE_STATISTICS` and any of the `pipelineStatistics` indicate compute operations, the `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdBeginQuery-query-00808
If called within a render pass instance, the sum of `query` and the number of bits set in the current subpass's view mask **must** be less than or equal to the number of queries in `queryPool`
- VUID-vkCmdBeginQuery-queryType-07126
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR`, then the `VkCommandPool` that `commandBuffer` was allocated from **must** have been created with a queue family index that supports `result status queries`, as indicated by `VkQueueFamilyQueryResultStatusPropertiesKHR::queryResultStatusSupport`
- VUID-vkCmdBeginQuery-None-07127

If there is a bound video session, then there **must** be no [active](#) queries

- VUID-vkCmdBeginQuery-None-08370

If there is a bound video session, then it **must** not have been created with [VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR](#)

- VUID-vkCmdBeginQuery-queryType-07128

If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR](#) and there is a bound video session, then [queryPool](#) **must** have been created with a [VkVideoProfileInfoKHR](#) structure included in the [pNext](#) chain of [VkQueryPoolCreateInfo](#) identical to the one specified in [VkVideoSessionCreateInfoKHR::pVideoProfile](#) the bound video session was created with

- VUID-vkCmdBeginQuery-queryType-07129

If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR](#), then there **must** be a bound video session

- VUID-vkCmdBeginQuery-queryType-07130

If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR](#) and there is a bound video session, then [queryPool](#) **must** have been created with a [VkVideoProfileInfoKHR](#) structure included in the [pNext](#) chain of [VkQueryPoolCreateInfo](#) identical to the one specified in [VkVideoSessionCreateInfoKHR::pVideoProfile](#) the bound video session was created with

- VUID-vkCmdBeginQuery-queryType-07131

If the [queryType](#) used to create [queryPool](#) was not [VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR](#) or [VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR](#), then there **must** be no bound video session

- VUID-vkCmdBeginQuery-queryPool-01922

[queryPool](#) **must** have been created with a [queryType](#) that differs from that of any queries that are [active](#) within [commandBuffer](#)

- VUID-vkCmdBeginQuery-queryType-02327

If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT](#) the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support graphics operations

- VUID-vkCmdBeginQuery-queryType-02328

If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT](#) then [VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackQueries](#) **must** be supported

- VUID-vkCmdBeginQuery-queryType-06687

If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT](#) the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support graphics operations

- VUID-vkCmdBeginQuery-queryType-06688

If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT](#) then [primitivesGeneratedQuery](#) **must** be enabled

- VUID-vkCmdBeginQuery-queryPool-07289

If [queryPool](#) was created with a [queryType](#) of [VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR](#), then the [VkQueryPoolPerformanceCreateInfoKHR::queueFamilyIndex](#) [queryPool](#) was created with **must** equal the queue family index of the [VkCommandPool](#) that [commandBuffer](#) was allocated

from

- VUID-vkCmdBeginQuery-queryPool-03223

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, the `profiling lock` **must** have been held before `vkBeginCommandBuffer` was called on `commandBuffer`

- VUID-vkCmdBeginQuery-queryPool-03224

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` and one of the counters used to create `queryPool` was `VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR`, the query begin **must** be the first recorded command in `commandBuffer`

- VUID-vkCmdBeginQuery-queryPool-03225

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` and one of the counters used to create `queryPool` was `VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR`, the begin command **must** not be recorded within a render pass instance

- VUID-vkCmdBeginQuery-queryPool-03226

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` and another query pool with a `queryType` `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` has been used within `commandBuffer`, its parent primary command buffer or secondary command buffer recorded within the same parent primary command buffer as `commandBuffer`, the `performanceCounterMultipleQueryPools` feature **must** be enabled

- VUID-vkCmdBeginQuery-None-02863

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, this command **must** not be recorded in a command buffer that, either directly or through secondary command buffers, also contains a `vkCmdResetQueryPool` command affecting the same query

Valid Usage (Implicit)

- VUID-vkCmdBeginQuery-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdBeginQuery-queryPool-parameter

`queryPool` **must** be a valid `VkQueryPool` handle

- VUID-vkCmdBeginQuery-flags-parameter

`flags` **must** be a valid combination of `VkQueryControlFlagBits` values

- VUID-vkCmdBeginQuery-commandBuffer-recording

`commandBuffer` **must** be in the `recording state`

- VUID-vkCmdBeginQuery-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, compute, decode, or encode operations

- VUID-vkCmdBeginQuery-commonparent

Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	Graphics Compute Decode Encode	Action State

To begin an indexed query, call:

```
// Provided by VK_EXT_transform_feedback
void vkCmdBeginQueryIndexedEXT(
    VkCommandBuffer          commandBuffer,
    VkQueryPool              queryPool,
    uint32_t                 query,
    VkQueryControlFlags      flags,
    uint32_t                 index);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the query pool that will manage the results of the query.
- `query` is the query index within the query pool that will contain the results.
- `flags` is a bitmask of `VkQueryControlFlagBits` specifying constraints on the types of queries that **can** be performed.
- `index` is the query type specific index. When the query type is `VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT` or `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT`, the index represents the vertex stream.

The `vkCmdBeginQueryIndexedEXT` command operates the same as the `vkCmdBeginQuery` command, except that it also accepts a query type specific `index` parameter.

This command defines an execution dependency between other query commands that reference the same query index.

The first `synchronization scope` includes all commands which reference the queries in `queryPool` indicated by `query` and `index` that occur earlier in `submission order`.

The second [synchronization scope](#) includes all commands which reference the queries in [queryPool](#) indicated by [query](#) and [index](#) that occur later in [submission order](#).

The operation of this command happens after the first scope and happens before the second scope.

Valid Usage

- VUID-vkCmdBeginQueryIndexedEXT-None-00807
All queries used by the command **must** be *unavailable*
- VUID-vkCmdBeginQueryIndexedEXT-queryType-02804
The [queryType](#) used to create [queryPool](#) **must** not be [VK_QUERY_TYPE_TIMESTAMP](#)
- VUID-vkCmdBeginQueryIndexedEXT-queryType-04728
The [queryType](#) used to create [queryPool](#) **must** not be [VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR](#) or [VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR](#)
- VUID-vkCmdBeginQueryIndexedEXT-queryType-06741
The [queryType](#) used to create [queryPool](#) **must** not be [VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR](#) or [VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR](#)
- VUID-vkCmdBeginQueryIndexedEXT-queryType-00800
If the [occlusionQueryPrecise](#) feature is not enabled, or the [queryType](#) used to create [queryPool](#) was not [VK_QUERY_TYPE_OCCLUSION](#), [flags](#) **must** not contain [VK_QUERY_CONTROL_PRECISE_BIT](#)
- VUID-vkCmdBeginQueryIndexedEXT-query-00802
[query](#) **must** be less than the number of queries in [queryPool](#)
- VUID-vkCmdBeginQueryIndexedEXT-queryType-00803
If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_OCCLUSION](#), the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support graphics operations
- VUID-vkCmdBeginQueryIndexedEXT-queryType-00804
If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_PIPELINE_STATISTICS](#) and any of the [pipelineStatistics](#) indicate graphics operations, the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support graphics operations
- VUID-vkCmdBeginQueryIndexedEXT-queryType-00805
If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_PIPELINE_STATISTICS](#) and any of the [pipelineStatistics](#) indicate compute operations, the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support compute operations
- VUID-vkCmdBeginQueryIndexedEXT-query-00808
If called within a render pass instance, the sum of [query](#) and the number of bits set in the current subpass's view mask **must** be less than or equal to the number of queries in [queryPool](#)
- VUID-vkCmdBeginQueryIndexedEXT-queryType-07126
If the [queryType](#) used to create [queryPool](#) was [VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR](#), then the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** have been created with a queue family index that supports [result status queries](#), as indicated by

VkQueueFamilyQueryResultStatusPropertiesKHR::queryResultStatusSupport

- VUID-vkCmdBeginQueryIndexedEXT-None-07127
If there is a bound video session, then there **must** be no **active** queries
- VUID-vkCmdBeginQueryIndexedEXT-None-08370
If there is a bound video session, then it **must** not have been created with **VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR**
- VUID-vkCmdBeginQueryIndexedEXT-queryType-07128
If the **queryType** used to create **queryPool** was **VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR** and there is a bound video session, then **queryPool** **must** have been created with a **VkVideoProfileInfoKHR** structure included in the **pNext** chain of **VkQueryPoolCreateInfo** identical to the one specified in **VkVideoSessionCreateInfoKHR::pVideoProfile** the bound video session was created with
- VUID-vkCmdBeginQueryIndexedEXT-queryType-07129
If the **queryType** used to create **queryPool** was **VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR**, then there **must** be a bound video session
- VUID-vkCmdBeginQueryIndexedEXT-queryType-07130
If the **queryType** used to create **queryPool** was **VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR** and there is a bound video session, then **queryPool** **must** have been created with a **VkVideoProfileInfoKHR** structure included in the **pNext** chain of **VkQueryPoolCreateInfo** identical to the one specified in **VkVideoSessionCreateInfoKHR::pVideoProfile** the bound video session was created with
- VUID-vkCmdBeginQueryIndexedEXT-queryType-07131
If the **queryType** used to create **queryPool** was not **VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR** or **VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR**, then there **must** be no bound video session
- VUID-vkCmdBeginQueryIndexedEXT-queryPool-04753
If the **queryPool** was created with the same **queryType** as that of another **active** query within **commandBuffer**, then **index** **must** not match the index used for the active query
- VUID-vkCmdBeginQueryIndexedEXT-queryType-02338
If the **queryType** used to create **queryPool** was **VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT** the **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdBeginQueryIndexedEXT-queryType-02339
If the **queryType** used to create **queryPool** was **VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT** the **index** parameter **must** be less than **VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams**
- VUID-vkCmdBeginQueryIndexedEXT-queryType-06692
If the **queryType** used to create **queryPool** was not **VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT** and not **VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT**, the **index** **must** be zero
- VUID-vkCmdBeginQueryIndexedEXT-queryType-06689
If the **queryType** used to create **queryPool** was **VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT** the **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdBeginQueryIndexedEXT-queryType-06690

If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` the `index` parameter **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams`

- VUID-vkCmdBeginQueryIndexedEXT-queryType-06691

If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` and the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled, the `index` parameter **must** be zero

- VUID-vkCmdBeginQueryIndexedEXT-queryType-06693

If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` then `primitivesGeneratedQuery` **must** be enabled

- VUID-vkCmdBeginQueryIndexedEXT-queryType-02341

If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT` then `VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackQueries` **must** be supported

- VUID-vkCmdBeginQueryIndexedEXT-queryPool-07289

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, then the `VkQueryPoolPerformanceCreateInfoKHR::queueFamilyIndex` `queryPool` was created with **must** equal the queue family index of the `VkCommandPool` that `commandBuffer` was allocated from

- VUID-vkCmdBeginQueryIndexedEXT-queryPool-03223

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, the `profiling lock` **must** have been held before `vkBeginCommandBuffer` was called on `commandBuffer`

- VUID-vkCmdBeginQueryIndexedEXT-queryPool-03224

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` and one of the counters used to create `queryPool` was `VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR`, the query begin **must** be the first recorded command in `commandBuffer`

- VUID-vkCmdBeginQueryIndexedEXT-queryPool-03225

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` and one of the counters used to create `queryPool` was `VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR`, the begin command **must** not be recorded within a render pass instance

- VUID-vkCmdBeginQueryIndexedEXT-queryPool-03226

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` and another query pool with a `queryType` `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` has been used within `commandBuffer`, its parent primary command buffer or secondary command buffer recorded within the same parent primary command buffer as `commandBuffer`, the `performanceCounterMultipleQueryPools` feature **must** be enabled

- VUID-vkCmdBeginQueryIndexedEXT-None-02863

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, this command **must** not be recorded in a command buffer that, either directly or through secondary command buffers, also contains a `vkCmdResetQueryPool` command affecting the same query

Valid Usage (Implicit)

- VUID-vkCmdBeginQueryIndexedEXT-commandBuffer-parameter **commandBuffer** **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdBeginQueryIndexedEXT-queryPool-parameter **queryPool** **must** be a valid [VkQueryPool](#) handle
- VUID-vkCmdBeginQueryIndexedEXT-flags-parameter **flags** **must** be a valid combination of [VkQueryControlFlagBits](#) values
- VUID-vkCmdBeginQueryIndexedEXT-commandBuffer-recording **commandBuffer** **must** be in the [recording state](#)
- VUID-vkCmdBeginQueryIndexedEXT-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics, compute, decode, or encode operations
- VUID-vkCmdBeginQueryIndexedEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBeginQueryIndexedEXT-commonparent
Both of **commandBuffer**, and **queryPool** **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute Decode Encode	Action State

Bits which **can** be set in [vkCmdBeginQuery::flags](#), specifying constraints on the types of queries that **can** be performed, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryControlFlagBits {
    VK_QUERY_CONTROL_PRECISE_BIT = 0x00000001,
} VkQueryControlFlagBits;
```

- `VK_QUERY_CONTROL_PRECISE_BIT` specifies the precision of [occlusion queries](#).

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryControlFlags;
```

`VkQueryControlFlags` is a bitmask type for setting a mask of zero or more [VkQueryControlFlagBits](#).

To end a query after the set of desired drawing or dispatching commands is executed, call:

```
// Provided by VK_VERSION_1_0
void vkCmdEndQuery(
    VkCommandBuffer          commandBuffer,
    VkQueryPool              queryPool,
    uint32_t                 query);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the query pool that is managing the results of the query.
- `query` is the query index within the query pool where the result is stored.

The command completes the query in `queryPool` identified by `query`, and marks it as available.

This command defines an execution dependency between other query commands that reference the same query.

The first [synchronization scope](#) includes all commands which reference the queries in `queryPool` indicated by `query` that occur earlier in [submission order](#).

The second [synchronization scope](#) includes only the operation of this command.

Calling `vkCmdEndQuery` is equivalent to calling `vkCmdEndQueryIndexedEXT` with the `index` parameter set to zero.

Valid Usage

- VUID-vkCmdEndQuery-None-01923
All queries used by the command **must** be [active](#)
- VUID-vkCmdEndQuery-query-00810
`query` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdEndQuery-query-00812
If `vkCmdEndQuery` is called within a render pass instance, the sum of `query` and the number

of bits set in the current subpass's view mask **must** be less than or equal to the number of queries in `queryPool`

- VUID-vkCmdEndQuery-queryPool-03227

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` and one or more of the counters used to create `queryPool` was `VK_PERFORMANCE_COUNTER_SCOPE_COMMAND_BUFFER_KHR`, the `vkCmdEndQuery` **must** be the last recorded command in `commandBuffer`

- VUID-vkCmdEndQuery-queryPool-03228

If `queryPool` was created with a `queryType` of `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR` and one or more of the counters used to create `queryPool` was `VK_PERFORMANCE_COUNTER_SCOPE_RENDER_PASS_KHR`, the `vkCmdEndQuery` **must** not be recorded within a render pass instance

- VUID-vkCmdEndQuery-None-07007

If called within a subpass of a render pass instance, the corresponding `vkCmdBeginQuery*` command **must** have been called previously within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndQuery-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdEndQuery-queryPool-parameter

`queryPool` **must** be a valid `VkQueryPool` handle

- VUID-vkCmdEndQuery-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdEndQuery-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, compute, decode, or encode operations

- VUID-vkCmdEndQuery-commonparent

Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	Graphics Compute Decode Encode	Action State

To end an indexed query after the set of desired drawing or dispatching commands is recorded, call:

```
// Provided by VK_EXT_transform_feedback
void vkCmdEndQueryIndexedEXT(
    VkCommandBuffer          commandBuffer,
    VkQueryPool              queryPool,
    uint32_t                 query,
    uint32_t                 index);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the query pool that is managing the results of the query.
- `query` is the query index within the query pool where the result is stored.
- `index` is the query type specific index.

The command completes the query in `queryPool` identified by `query` and `index`, and marks it as available.

The `vkCmdEndQueryIndexedEXT` command operates the same as the `vkCmdEndQuery` command, except that it also accepts a query type specific `index` parameter.

This command defines an execution dependency between other query commands that reference the same query index.

The first [synchronization scope](#) includes all commands which reference the queries in `queryPool` indicated by `query` that occur earlier in [submission order](#).

The second [synchronization scope](#) includes only the operation of this command.

Valid Usage

- VUID-vkCmdEndQueryIndexedEXT-None-02342
All queries used by the command **must** be [active](#)
- VUID-vkCmdEndQueryIndexedEXT-query-02343
`query` **must** be less than the number of queries in `queryPool`

- VUID-vkCmdEndQueryIndexedEXT-query-02345

If `vkCmdEndQueryIndexedEXT` is called within a render pass instance, the sum of `query` and the number of bits set in the current subpass's view mask **must** be less than or equal to the number of queries in `queryPool`

- VUID-vkCmdEndQueryIndexedEXT-queryType-06694

If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT` or `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT`, the `index` parameter **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams`

- VUID-vkCmdEndQueryIndexedEXT-queryType-06695

If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT` and `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT`, the `index` **must** be zero

- VUID-vkCmdEndQueryIndexedEXT-queryType-06696

If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT` or `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT`, `index` **must** equal the `index` used to begin the query

- VUID-vkCmdEndQueryIndexedEXT-None-07007

If called within a subpass of a render pass instance, the corresponding `vkCmdBeginQuery*` command **must** have been called previously within the same subpass

Valid Usage (Implicit)

- VUID-vkCmdEndQueryIndexedEXT-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdEndQueryIndexedEXT-queryPool-parameter

`queryPool` **must** be a valid `VkQueryPool` handle

- VUID-vkCmdEndQueryIndexedEXT-commandBuffer-recording

`commandBuffer` **must** be in the `recording state`

- VUID-vkCmdEndQueryIndexedEXT-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, compute, decode, or encode operations

- VUID-vkCmdEndQueryIndexedEXT-videocoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdEndQueryIndexedEXT-commonparent

Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute Decode Encode	Action State

An application **can** retrieve results either by requesting they be written into application-provided memory, or by requesting they be copied into a `VkBuffer`. In either case, the layout in memory is defined as follows:

- The first query's result is written starting at the first byte requested by the command, and each subsequent query's result begins `stride` bytes later.
- Occlusion queries, pipeline statistics queries, transform feedback queries, primitives generated queries, video encode feedback queries, and timestamp queries store results in a tightly packed array of unsigned integers, either 32- or 64-bits as requested by the command, storing the numerical results and, if requested, the availability status.
- Performance queries store results in a tightly packed array whose type is determined by the `unit` member of the corresponding `VkPerformanceCounterKHR`.
- If `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT` is used, the final element of each query's result is an integer indicating whether the query's result is available, with any non-zero value indicating that it is available.
- If `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR` is used, the final element of each query's result is an integer value indicating that status of the query result. Positive values indicate success, negative values indicate failure, and 0 indicates that the result is not yet available. Specific error codes are encoded in the `VkQueryResultStatusKHR` enumeration.
- Occlusion queries write one integer value - the number of samples passed. Pipeline statistics queries write one integer value for each bit that is enabled in the `pipelineStatistics` when the pool is created, and the statistics values are written in bit order starting from the least significant bit. Timestamp queries write one integer value. Performance queries write one `VkPerformanceCounterResultKHR` value for each `VkPerformanceCounterKHR` in the query. Transform feedback queries write two integers; the first integer is the number of primitives successfully written to the corresponding transform feedback buffer and the second is the number of primitives output to the vertex stream, regardless of whether they were successfully captured or not. In other words, if the transform feedback buffer was sized too small for the number of primitives output by the vertex stream, the first integer represents the number of primitives actually written and the second is the number that would have been written if all the transform feedback buffers associated with that vertex stream were large enough. Primitives generated queries write the number of primitives output to the vertex stream, regardless of whether transform feedback is active or not, or whether they were successfully captured by

transform feedback or not. This is identical to the second integer of the transform feedback queries if transform feedback is active. Video encode feedback queries write one or more integer values for each bit that is enabled in `VkQueryPoolVideoEncodeFeedbackCreateInfoKHR::encodeFeedbackFlags` when the pool is created, and the feedback values are written in bit order starting from the least significant bit, as described [here](#).

- If more than one query is retrieved and `stride` is not at least as large as the size of the array of values corresponding to a single query, the values written to memory are undefined.

To retrieve status and results for a set of queries, call:

```
// Provided by VK_VERSION_1_0
VkResult vkGetQueryPoolResults(
    VkDevice          device,
    VkQueryPool       queryPool,
    uint32_t          firstQuery,
    uint32_t          queryCount,
    size_t            dataSize,
    void*             pData,
    VkDeviceSize       stride,
    VkQueryResultFlags flags);
```

- `device` is the logical device that owns the query pool.
- `queryPool` is the query pool managing the queries containing the desired results.
- `firstQuery` is the initial query index.
- `queryCount` is the number of queries to read.
- `dataSize` is the size in bytes of the buffer pointed to by `pData`.
- `pData` is a pointer to an application-allocated buffer where the results will be written
- `stride` is the stride in bytes between results for individual queries within `pData`.
- `flags` is a bitmask of `VkQueryResultFlagBits` specifying how and when results are returned.

Any results written for a query are written according to [a layout dependent on the query type](#).

If no bits are set in `flags`, and all requested queries are in the available state, results are written as an array of 32-bit unsigned integer values. Behavior when not all queries are available is described [below](#).

If `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT` is set, results for all queries in `queryPool` identified by `firstQuery` and `queryCount` are copied to `pData`, along with an extra availability or status value written directly after the results of each query and interpreted as an unsigned integer. A value of zero indicates that the results are not yet available, otherwise the query is complete and results are available. The size of the availability or status values is 64 bits if `VK_QUERY_RESULT_64_BIT` is set in `flags`. Otherwise, it is 32 bits.

If `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR` is set, results for all queries in `queryPool` identified by

`firstQuery` and `queryCount` are copied to `pData`, along with an extra status value written directly after the results of each query and interpreted as a signed integer. A value of zero indicates that the results are not yet available. Positive values indicate that the operations within the query completed successfully, and the query results are valid. Negative values indicate that the operations within the query completed unsuccessfully.

`VkQueryResultStatusKHR` defines specific meaning for values returned here, though implementations are free to return other values.

If the status value written is negative, indicating that the operations within the query completed unsuccessfully, then all other results written by this command are undefined unless otherwise specified for any of the results of the used query type.

NOTE

If `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT` or `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR` is set, the layout of data in the buffer is a *(result,availability)* or *(result,status)* pair for each query returned, and `stride` is the stride between each pair.

Results for any available query written by this command are final and represent the final result of the query. If `VK_QUERY_RESULT_PARTIAL_BIT` is set, then for any query that is unavailable, an intermediate result between zero and the final result value is written for that query. Otherwise, any result written by this command is undefined.

If `VK_QUERY_RESULT_64_BIT` is set, results and, if returned, availability or status values for all queries are written as an array of 64-bit values. If the `queryPool` was created with `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, results for each query are written as an array of the type indicated by `VkPerformanceCounterKHR::storage` for the counter being queried. Otherwise, results and availability or status values are written as an array of 32-bit values. If an unsigned integer query's value overflows the result type, the value **may** either wrap or saturate. If the `maintenance7` feature is enabled, for an unsigned integer query, the 32-bit result value **must** be equal to the 32 least significant bits of the equivalent 64-bit result value. If a signed integer query's value overflows the result type, the value is undefined. If a floating-point query's value is not representable as the result type, the value is undefined.

If `VK_QUERY_RESULT_WAIT_BIT` is set, this command defines an execution dependency with any earlier commands that writes one of the identified queries. The first `synchronization scope` includes all instances of `vkCmdEndQuery`, `vkCmdEndQueryIndexedEXT`, `vkCmdWriteTimestamp2KHR`, and `vkCmdWriteTimestamp` that reference any query in `queryPool` indicated by `firstQuery` and `queryCount`. The second `synchronization scope` includes the host operations of this command.

If `VK_QUERY_RESULT_WAIT_BIT` is not set, `vkGetQueryPoolResults` **may** return `VK_NOT_READY` if there are queries in the unavailable state.

NOTE

Applications **must** take care to ensure that use of the `VK_QUERY_RESULT_WAIT_BIT` bit has the desired effect.

For example, if a query has been used previously and a command buffer records the commands `vkCmdResetQueryPool`, `vkCmdBeginQuery`, and `vkCmdEndQuery` for that query, then the query will remain in the available state until the `vkCmdResetQueryPool` command executes on a queue. Applications **can** use fences or

events to ensure that a query has already been reset before checking for its results or availability status. Otherwise, a stale value could be returned from a previous use of the query.

The above also applies when `VK_QUERY_RESULT_WAIT_BIT` is used in combination with `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT`. In this case, the returned availability status **may** reflect the result of a previous use of the query unless the `vkCmdResetQueryPool` command has been executed since the last use of the query.

A similar situation can arise with the `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR` flag.

NOTE

Applications **can** double-buffer query pool usage, with a pool per frame, and reset queries at the end of the frame in which they are read.

Valid Usage

- VUID-vkGetQueryPoolResults-firstQuery-09436
`firstQuery` **must** be less than the number of queries in `queryPool`
- VUID-vkGetQueryPoolResults-firstQuery-09437
The sum of `firstQuery` and `queryCount` **must** be less than or equal to the number of queries in `queryPool`
- VUID-vkGetQueryPoolResults-queryCount-09438
If `queryCount` is greater than 1, `stride` **must** not be zero
- VUID-vkGetQueryPoolResults-queryType-09439
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_TIMESTAMP`, `flags` **must** not contain `VK_QUERY_RESULT_PARTIAL_BIT`
- VUID-vkGetQueryPoolResults-queryType-09440
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, `flags` **must** not contain `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT`, `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR`, `VK_QUERY_RESULT_PARTIAL_BIT`, or `VK_QUERY_RESULT_64_BIT`
- VUID-vkGetQueryPoolResults-queryType-09441
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, the `queryPool` **must** have been recorded once for each pass as retrieved via a call to `vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR`
- VUID-vkGetQueryPoolResults-queryType-09442
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR`, then `flags` **must** include `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR`
- VUID-vkGetQueryPoolResults-flags-09443
If `flags` includes `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR`, then it **must** not include `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT`
- VUID-vkGetQueryPoolResults-None-09401
All queries used by the command **must** not be uninitialized

- VUID-vkGetQueryPoolResults-flags-02828
If `VK_QUERY_RESULT_64_BIT` is not set in `flags` and the `queryType` used to create `queryPool` was not `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, then `pData` and `stride` **must** be multiples of 4
- VUID-vkGetQueryPoolResults-flags-00815
If `VK_QUERY_RESULT_64_BIT` is set in `flags` then `pData` and `stride` **must** be multiples of 8
- VUID-vkGetQueryPoolResults-stride-08993
If `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT` is set, `stride` **must** be large enough to contain the unsigned integer representing availability or status in addition to the query result
- VUID-vkGetQueryPoolResults-queryType-03229
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, then `pData` and `stride` **must** be multiples of the size of `VkPerformanceCounterResultKHR`
- VUID-vkGetQueryPoolResults-queryType-04519
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, then `stride` **must** be large enough to contain the `VkQueryPoolPerformanceCreateInfoKHR::counterIndexCount` used to create `queryPool` times the size of `VkPerformanceCounterResultKHR`
- VUID-vkGetQueryPoolResults-dataSize-00817
`dataSize` **must** be large enough to contain the result of each query, as described [here](#)

Valid Usage (Implicit)

- VUID-vkGetQueryPoolResults-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetQueryPoolResults-queryPool-parameter
`queryPool` **must** be a valid `VkQueryPool` handle
- VUID-vkGetQueryPoolResults-pData-parameter
`pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkGetQueryPoolResults-flags-parameter
`flags` **must** be a valid combination of `VkQueryResultFlagBits` values
- VUID-vkGetQueryPoolResults-dataSize-arraylength
`dataSize` **must** be greater than 0
- VUID-vkGetQueryPoolResults-queryPool-parent
`queryPool` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_NOT_READY`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

Bits which **can** be set in `vkGetQueryPoolResults::flags` and `vkCmdCopyQueryPoolResults::flags`, specifying how and when results are returned, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryResultFlagBits {
    VK_QUERY_RESULT_64_BIT = 0x00000001,
    VK_QUERY_RESULT_WAIT_BIT = 0x00000002,
    VK_QUERY_RESULT_WITH_AVAILABILITY_BIT = 0x00000004,
    VK_QUERY_RESULT_PARTIAL_BIT = 0x00000008,
    // Provided by VK_KHR_video_queue
    VK_QUERY_RESULT_WITH_STATUS_BIT_KHR = 0x00000010,
} VkQueryResultFlagBits;
```

- `VK_QUERY_RESULT_64_BIT` specifies the results will be written as an array of 64-bit unsigned integer values. If this bit is not set, the results will be written as an array of 32-bit unsigned integer values.
- `VK_QUERY_RESULT_WAIT_BIT` specifies that Vulkan will wait for each query's status to become available before retrieving its results.
- `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT` specifies that the availability status accompanies the results.
- `VK_QUERY_RESULT_PARTIAL_BIT` specifies that returning partial results is acceptable.
- `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR` specifies that the last value returned in the query is a `VkQueryResultStatusKHR` value. See [result status query](#) for information on how an application can determine whether the use of this flag bit is supported.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryResultFlags;
```

`VkQueryResultFlags` is a bitmask type for setting a mask of zero or more `VkQueryResultFlagBits`.

Specific status codes that **can** be returned from a query are:

```
// Provided by VK_KHR_video_queue
typedef enum VkQueryResultStatusKHR {
    VK_QUERY_RESULT_STATUS_ERROR_KHR = -1,
    VK_QUERY_RESULT_STATUS_NOT_READY_KHR = 0,
    VK_QUERY_RESULT_STATUS_COMPLETE_KHR = 1,
    // Provided by VK_KHR_video_encode_queue
    VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR = -1000299000,
} VkQueryResultStatusKHR;
```

- `VK_QUERY_RESULT_STATUS_NOT_READY_KHR` indicates that the query result is not yet available.
- `VK_QUERY_RESULT_STATUS_ERROR_KHR` indicates that operations did not complete successfully.
- `VK_QUERY_RESULT_STATUS_COMPLETE_KHR` indicates that operations completed successfully and the query result is available.
- `VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR` indicates that a video encode operation did not complete successfully due to the destination video bitstream buffer range not being sufficiently large to fit the encoded bitstream data.

To copy query statuses and numerical results directly to buffer memory, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyQueryPoolResults(
    VkCommandBuffer          commandBuffer,
    VkQueryPool              queryPool,
    uint32_t                 firstQuery,
    uint32_t                 queryCount,
    VkBuffer                 dstBuffer,
    VkDeviceSize             dstOffset,
    VkDeviceSize             stride,
    VkQueryResultFlags       flags);
```

- `commandBuffer` is the command buffer into which this command will be recorded.
- `queryPool` is the query pool managing the queries containing the desired results.
- `firstQuery` is the initial query index.
- `queryCount` is the number of queries. `firstQuery` and `queryCount` together define a range of queries.
- `dstBuffer` is a `VkBuffer` object that will receive the results of the copy command.
- `dstOffset` is an offset into `dstBuffer`.
- `stride` is the stride in bytes between results for individual queries within `dstBuffer`. The required size of the backing memory for `dstBuffer` is determined as described above for `vkGetQueryPoolResults`.
- `flags` is a bitmask of `VkQueryResultFlagBits` specifying how and when results are returned.

Any results written for a query are written according to [a layout dependent on the query type](#).

Results for any query in `queryPool` identified by `firstQuery` and `queryCount` that is available are copied to `dstBuffer`.

If `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT` is set, results for all queries in `queryPool` identified by `firstQuery` and `queryCount` are copied to `dstBuffer`, along with an extra availability value written directly after the results of each query and interpreted as an unsigned integer. A value of zero indicates that the results are not yet available, otherwise the query is complete and results are available.

If `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR` is set, results for all queries in `queryPool` identified by `firstQuery` and `queryCount` are copied to `dstBuffer`, along with an extra status value written directly after the results of each query and interpreted as a signed integer. A value of zero indicates that the results are not yet available. Positive values indicate that the operations within the query completed successfully, and the query results are valid. Negative values indicate that the operations within the query completed unsuccessfully.

`VkQueryResultStatusKHR` defines specific meaning for values returned here, though implementations are free to return other values.

If the status value written is negative, indicating that the operations within the query completed unsuccessfully, then all other results written by this command are undefined unless otherwise specified for any of the results of the used query type.

Results for any available query written by this command are final and represent the final result of the query. If `VK_QUERY_RESULT_PARTIAL_BIT` is set, then for any query that is unavailable, an intermediate result between zero and the final result value is written for that query. Otherwise, any result written by this command is undefined.

If `VK_QUERY_RESULT_64_BIT` is set, results and availability or status values for all queries are written as an array of 64-bit values. If the `queryPool` was created with `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, results for each query are written as an array of the type indicated by `VkPerformanceCounterKHR::storage` for the counter being queried. Otherwise, results and availability or status values are written as an array of 32-bit values. If an unsigned integer query's value overflows the result type, the value **may** either wrap or saturate. If the `maintenance7` feature is enabled, for an unsigned integer query, the 32-bit result value **must** be equal to the 32 least significant bits of the equivalent 64-bit result value. If a signed integer query's value overflows the result type, the value is undefined. If a floating-point query's value is not representable as the result type, the value is undefined.

This command defines an execution dependency between other query commands that reference the same query.

The first `synchronization scope` includes all commands which reference the queries in `queryPool` indicated by `query` that occur earlier in `submission order`. If `flags` does not include `VK_QUERY_RESULT_WAIT_BIT`, `vkCmdEndQueryIndexedEXT`, `vkCmdWriteTimestamp2KHR`, `vkCmdEndQuery`, and `vkCmdWriteTimestamp` are excluded from this scope.

The second `synchronization scope` includes all commands which reference the queries in `queryPool` indicated by `query` that occur later in `submission order`.

The operation of this command happens after the first scope and happens before the second scope.

`vkCmdCopyQueryPoolResults` is considered to be a transfer operation, and its writes to buffer memory **must** be synchronized using `VK_PIPELINE_STAGE_TRANSFER_BIT` and `VK_ACCESS_TRANSFER_WRITE_BIT` before using the results.

Valid Usage

- VUID-vkCmdCopyQueryPoolResults-firstQuery-09436
`firstQuery` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdCopyQueryPoolResults-firstQuery-09437
The sum of `firstQuery` and `queryCount` **must** be less than or equal to the number of queries in `queryPool`
- VUID-vkCmdCopyQueryPoolResults-queryCount-09438
If `queryCount` is greater than 1, `stride` **must** not be zero
- VUID-vkCmdCopyQueryPoolResults-queryType-09439
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_TIMESTAMP`, `flags` **must** not contain `VK_QUERY_RESULT_PARTIAL_BIT`
- VUID-vkCmdCopyQueryPoolResults-queryType-09440
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, `flags` **must** not contain `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT`, `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR`, `VK_QUERY_RESULT_PARTIAL_BIT`, or `VK_QUERY_RESULT_64_BIT`
- VUID-vkCmdCopyQueryPoolResults-queryType-09441
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, the `queryPool` **must** have been recorded once for each pass as retrieved via a call to `vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR`
- VUID-vkCmdCopyQueryPoolResults-queryType-09442
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR`, then `flags` **must** include `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR`
- VUID-vkCmdCopyQueryPoolResults-flags-09443
If `flags` includes `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR`, then it **must** not include `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT`
- VUID-vkCmdCopyQueryPoolResults-None-09402
All queries used by the command **must** not be uninitialized when the command is executed
- VUID-vkCmdCopyQueryPoolResults-dstOffset-00819
`dstOffset` **must** be less than the size of `dstBuffer`
- VUID-vkCmdCopyQueryPoolResults-flags-00822
If `VK_QUERY_RESULT_64_BIT` is not set in `flags` then `dstOffset` and `stride` **must** be multiples of 4
- VUID-vkCmdCopyQueryPoolResults-flags-00823
If `VK_QUERY_RESULT_64_BIT` is set in `flags` then `dstOffset` and `stride` **must** be multiples of 8

- VUID-vkCmdCopyQueryPoolResults-dstBuffer-00824
`dstBuffer` **must** have enough storage, from `dstOffset`, to contain the result of each query, as described [here](#)
- VUID-vkCmdCopyQueryPoolResults-dstBuffer-00825
`dstBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-vkCmdCopyQueryPoolResults-dstBuffer-00826
If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyQueryPoolResults-queryType-03232
If the `queryType` used to create `queryPool` was `VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR`, `VkPhysicalDevicePerformanceQueryPropertiesKHR::allowCommandBufferQueryCopies` **must** be `VK_TRUE`
- VUID-vkCmdCopyQueryPoolResults-None-07429
All queries used by the command **must** not be active
- VUID-vkCmdCopyQueryPoolResults-None-08752
All queries used by the command **must** have been made *available* by prior executed commands

Valid Usage (Implicit)

- VUID-vkCmdCopyQueryPoolResults-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyQueryPoolResults-queryPool-parameter
`queryPool` **must** be a valid `VkQueryPool` handle
- VUID-vkCmdCopyQueryPoolResults-dstBuffer-parameter
`dstBuffer` **must** be a valid `VkBuffer` handle
- VUID-vkCmdCopyQueryPoolResults-flags-parameter
`flags` **must** be a valid combination of `VkQueryResultFlagBits` values
- VUID-vkCmdCopyQueryPoolResults-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdCopyQueryPoolResults-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdCopyQueryPoolResults-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyQueryPoolResults-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdCopyQueryPoolResults-commonparent
Each of `commandBuffer`, `dstBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Graphics Compute	Action

Rendering operations such as clears, MSAA resolves, attachment load/store operations, and blits **may** count towards the results of queries. This behavior is implementation-dependent and **may** vary depending on the path used within an implementation. For example, some implementations have several types of clears, some of which **may** include vertices and some not.

17.3. Occlusion Queries

Occlusion queries track the number of samples that pass the per-fragment tests for a set of drawing commands. As such, occlusion queries are only available on queue families supporting graphics operations. The application **can** then use these results to inform future rendering decisions. An occlusion query is begun and ended by calling `vkCmdBeginQuery` and `vkCmdEndQuery`, respectively. When an occlusion query begins, the count of passing samples always starts at zero. For each drawing command, the count is incremented as described in [Sample Counting](#). If `flags` does not contain `VK_QUERY_CONTROL_PRECISE_BIT` an implementation **may** generate any non-zero result value for the query if the count of passing samples is non-zero.

Not setting `VK_QUERY_CONTROL_PRECISE_BIT` mode **may** be more efficient on some implementations, and **should** be used where it is sufficient to know a boolean result on whether any samples passed the per-fragment tests. In this case, some implementations **may** only return zero or one, indifferent to the actual number of samples passing the per-fragment tests.

NOTE

Setting `VK_QUERY_CONTROL_PRECISE_BIT` does not guarantee that different implementations return the same number of samples in an occlusion query. Some implementations may kill fragments in the [pre-rasterization shader stage](#), and these killed fragments do not contribute to the final result of the query. It is possible that some implementations generate a zero result value for the query, while others generate a non-zero value.

When an occlusion query finishes, the result for that query is marked as available. The application **can** then either copy the result to a buffer (via `vkCmdCopyQueryPoolResults`) or request it be put into

host memory (via `vkGetQueryPoolResults`).

NOTE

If occluding geometry is not drawn first, samples **can** pass the depth test, but still not be visible in a final image.

17.4. Pipeline Statistics Queries

Pipeline statistics queries allow the application to sample a specified set of `VkPipeline` counters. These counters are accumulated by Vulkan for a set of either drawing or dispatching commands while a pipeline statistics query is active. As such, pipeline statistics queries are available on queue families supporting either graphics or compute operations. The availability of pipeline statistics queries is indicated by the `pipelineStatisticsQuery` member of the `VkPhysicalDeviceFeatures` object (see `vkGetPhysicalDeviceFeatures` and `vkCreateDevice` for detecting and requesting this query type on a `VkDevice`).

A pipeline statistics query is begun and ended by calling `vkCmdBeginQuery` and `vkCmdEndQuery`, respectively. When a pipeline statistics query begins, all statistics counters are set to zero. While the query is active, the pipeline type determines which set of statistics are available, but these **must** be configured on the query pool when it is created. If a statistic counter is issued on a command buffer that does not support the corresponding operation, or the counter corresponds to a shading stage which is missing from any of the pipelines used while the query is active, the value of that counter is undefined after the query has been made available. At least one statistic counter relevant to the operations supported on the recording command buffer **must** be enabled.

Bits which **can** be set in `VkQueryPoolCreateInfo::pipelineStatistics` for query pools and in `VkCommandBufferInheritanceInfo::pipelineStatistics` for secondary command buffers, individually enabling pipeline statistics counters, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkQueryPipelineStatisticFlagBits {
    VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_VERTICES_BIT = 0x00000001,
    VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_PRIMITIVES_BIT = 0x00000002,
    VK_QUERY_PIPELINE_STATISTIC_VERTEX_SHADER_INVOCATIONS_BIT = 0x00000004,
    VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_INVOCATIONS_BIT = 0x00000008,
    VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_PRIMITIVES_BIT = 0x00000010,
    VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT = 0x00000020,
    VK_QUERY_PIPELINE_STATISTIC_CLIPPING_PRIMITIVES_BIT = 0x00000040,
    VK_QUERY_PIPELINE_STATISTIC_FRAGMENT_SHADER_INVOCATIONS_BIT = 0x00000080,
    VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_CONTROL_SHADER_PATCHES_BIT = 0x00000100,
    VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_EVALUATION_SHADER_INVOCATIONS_BIT =
0x00000200,
    VK_QUERY_PIPELINE_STATISTIC_COMPUTE_SHADER_INVOCATIONS_BIT = 0x00000400,
} VkQueryPipelineStatisticFlagBits;
```

- `VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_VERTICES_BIT` specifies that queries managed by the pool will count the number of vertices processed by the `input assembly` stage. Vertices corresponding to incomplete primitives **may** contribute to the count.

- `VK_QUERY_PIPELINE_STATISTIC_INPUT_ASSEMBLY_PRIMITIVES_BIT` specifies that queries managed by the pool will count the number of primitives processed by the [input assembly](#) stage. If primitive restart is enabled, restarting the primitive topology has no effect on the count. Incomplete primitives **may** be counted.
- `VK_QUERY_PIPELINE_STATISTIC_VERTEX_SHADER_INVOCATIONS_BIT` specifies that queries managed by the pool will count the number of vertex shader invocations. This counter's value is incremented each time a vertex shader is [invoked](#).
- `VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_INVOCATIONS_BIT` specifies that queries managed by the pool will count the number of geometry shader invocations. This counter's value is incremented each time a geometry shader is [invoked](#). In the case of [instanced geometry shaders](#), the geometry shader invocations count is incremented for each separate instanced invocation.
- `VK_QUERY_PIPELINE_STATISTIC_GEOMETRY_SHADER_PRIMITIVES_BIT` specifies that queries managed by the pool will count the number of primitives generated by geometry shader invocations. The counter's value is incremented each time the geometry shader emits a primitive. Restarting primitive topology using the SPIR-V instructions `OpEndPrimitive` or `OpEndStreamPrimitive` has no effect on the geometry shader output primitives count.
- `VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT` specifies that queries managed by the pool will count the number of primitives processed by the [Primitive Clipping](#) stage of the pipeline. The counter's value is incremented each time a primitive reaches the primitive clipping stage.
- `VK_QUERY_PIPELINE_STATISTIC_CLIPPING_PRIMITIVES_BIT` specifies that queries managed by the pool will count the number of primitives output by the [Primitive Clipping](#) stage of the pipeline. The counter's value is incremented each time a primitive passes the primitive clipping stage. The actual number of primitives output by the primitive clipping stage for a particular input primitive is implementation-dependent but **must** satisfy the following conditions:
 - If at least one vertex of the input primitive lies inside the clipping volume, the counter is incremented by one or more.
 - Otherwise, the counter is incremented by zero or more.
- `VK_QUERY_PIPELINE_STATISTIC_FRAGMENT_SHADER_INVOCATIONS_BIT` specifies that queries managed by the pool will count the number of fragment shader invocations. The counter's value is incremented each time the fragment shader is [invoked](#).
- `VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_CONTROL_SHADER_PATCHES_BIT` specifies that queries managed by the pool will count the number of patches processed by the tessellation control shader. The counter's value is incremented once for each patch for which a tessellation control shader is [invoked](#).
- `VK_QUERY_PIPELINE_STATISTIC_TESSELLATION_EVALUATION_SHADER_INVOCATIONS_BIT` specifies that queries managed by the pool will count the number of invocations of the tessellation evaluation shader. The counter's value is incremented each time the tessellation evaluation shader is [invoked](#).
- `VK_QUERY_PIPELINE_STATISTIC_COMPUTE_SHADER_INVOCATIONS_BIT` specifies that queries managed by the pool will count the number of compute shader invocations. The counter's value is incremented every time the compute shader is invoked. Implementations **may** skip the

execution of certain compute shader invocations or execute additional compute shader invocations for implementation-dependent reasons as long as the results of rendering otherwise remain unchanged.

These values are intended to measure relative statistics on one implementation. Various device architectures will count these values differently. Any or all counters **may** be affected by the issues described in [Query Operation](#).

NOTE

For example, tile-based rendering devices **may** need to replay the scene multiple times, affecting some of the counts.

If a pipeline has `rasterizerDiscardEnable` enabled, implementations **may** discard primitives after the final [pre-rasterization shader stage](#). As a result, if `rasterizerDiscardEnable` is enabled, the clipping input and output primitives counters **may** not be incremented.

When a pipeline statistics query finishes, the result for that query is marked as available. The application **can** copy the result to a buffer (via `vkCmdCopyQueryPoolResults`), or request it be put into host memory (via `vkGetQueryPoolResults`).

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkQueryPipelineStatisticFlags;
```

`VkQueryPipelineStatisticFlags` is a bitmask type for setting a mask of zero or more [VkQueryPipelineStatisticFlags](#).

17.5. Timestamp Queries

Timestamps provide applications with a mechanism for timing the execution of commands. A timestamp is an integer value generated by the `VkPhysicalDevice`. Unlike other queries, timestamps do not operate over a range, and so do not use `vkCmdBeginQuery` or `vkCmdEndQuery`. The mechanism is built around a set of commands that allow the application to tell the `VkPhysicalDevice` to write timestamp values to a [query pool](#) and then either read timestamp values on the host (using `vkGetQueryPoolResults`) or copy timestamp values to a `VkBuffer` (using `vkCmdCopyQueryPoolResults`). The application **can** then compute differences between timestamps to determine execution time.

The number of valid bits in a timestamp value is determined by the `VkQueueFamilyProperties::timestampValidBits` property of the queue on which the timestamp is written. Timestamps are supported on any queue which reports a non-zero value for `timestampValidBits` via `vkGetPhysicalDeviceQueueFamilyProperties`. If the `timestampComputeAndGraphics` limit is `VK_TRUE`, timestamps are supported by every queue family that supports either graphics or compute operations (see [VkQueueFamilyProperties](#)).

The number of nanoseconds it takes for a timestamp value to be incremented by 1 **can** be obtained from `VkPhysicalDeviceLimits::timestampPeriod` after a call to `vkGetPhysicalDeviceProperties`.

To request a timestamp and write the value to memory, call:

```
// Provided by VK_KHR_synchronization2
void vkCmdWriteTimestamp2KHR(
    VkCommandBuffer                      commandBuffer,
    VkPipelineStageFlags2                stage,
    VkQueryPool                          queryPool,
    uint32_t                             query);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `stage` specifies a stage of the pipeline.
- `queryPool` is the query pool that will manage the timestamp.
- `query` is the query within the query pool that will contain the timestamp.

When `vkCmdWriteTimestamp2KHR` is submitted to a queue, it defines an execution dependency on commands that were submitted before it, and writes a timestamp to a query pool.

The first [synchronization scope](#) includes all commands that occur earlier in [submission order](#). The synchronization scope is limited to operations on the pipeline stage specified by `stage`.

The second [synchronization scope](#) includes only the timestamp write operation.

NOTE

Implementations may write the timestamp at any stage that is [logically later](#) than `stage`.

Any timestamp write that [happens-after](#) another timestamp write in the same submission **must** not have a lower value unless its value overflows the maximum supported integer bit width of the query. If `VK_KHR_calibrated_timestamps` is enabled, this extends to timestamp writes across all submissions on the same logical device: any timestamp write that [happens-after](#) another **must** not have a lower value unless its value overflows the maximum supported integer bit width of the query. Timestamps written by this command **must** be in the `VK_TIME_DOMAIN_DEVICE_KHR` [time domain](#). If an overflow occurs, the timestamp value **must** wrap back to zero.

NOTE

Comparisons between timestamps should be done between timestamps where they are guaranteed to not decrease. For example, subtracting an older timestamp from a newer one to determine the execution time of a sequence of commands is only a reliable measurement if the two timestamp writes were performed in the same submission, or if the writes were performed on the same logical device and `VK_KHR_calibrated_timestamps` is enabled.

If `vkCmdWriteTimestamp2KHR` is called while executing a render pass instance that has multiview enabled, the timestamp uses N consecutive query indices in the query pool (starting at `query`) where N is the number of bits set in the view mask of the subpass the command is executed in. The resulting query values are determined by an implementation-dependent choice of one of the following behaviors:

- The first query is a timestamp value and (if more than one bit is set in the view mask) zero is written to the remaining queries. If two timestamps are written in the same subpass, the sum of the execution time of all views between those commands is the difference between the first

query written by each command.

- All N queries are timestamp values. If two timestamps are written in the same subpass, the sum of the execution time of all views between those commands is the sum of the difference between corresponding queries written by each command. The difference between corresponding queries **may** be the execution time of a single view.

In either case, the application **can** sum the differences between all N queries to determine the total execution time.

Valid Usage

- VUID-vkCmdWriteTimestamp2-stage-03929
If the `geometryShader` feature is not enabled, `stage` **must** not contain `VK_PIPELINE_STAGE_2_GEOMETRY_SHADER_BIT`
- VUID-vkCmdWriteTimestamp2-stage-03930
If the `tessellationShader` feature is not enabled, `stage` **must** not contain `VK_PIPELINE_STAGE_2_TESSELLATION_CONTROL_SHADER_BIT` or `VK_PIPELINE_STAGE_2_TESSELLATION_EVALUATION_SHADER_BIT`
- VUID-vkCmdWriteTimestamp2-stage-03933
If the `transformFeedback` feature is not enabled, `stage` **must** not contain `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`
- VUID-vkCmdWriteTimestamp2-stage-07317
If the `attachmentFragmentShadingRate` feature is not enabled, `stage` **must** not contain `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-vkCmdWriteTimestamp2-stage-07947
If the `rayTracingPipeline` feature is not enabled, `stage` **must** not contain `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`
- VUID-vkCmdWriteTimestamp2-synchronization2-03858
The `synchronization2` feature **must** be enabled
- VUID-vkCmdWriteTimestamp2-stage-03859
`stage` **must** only include a single pipeline stage
- VUID-vkCmdWriteTimestamp2-stage-03860
`stage` **must** only include stages valid for the queue family that was used to create the command pool that `commandBuffer` was allocated from
- VUID-vkCmdWriteTimestamp2-queryPool-03861
`queryPool` **must** have been created with a `queryType` of `VK_QUERY_TYPE_TIMESTAMP`
- VUID-vkCmdWriteTimestamp2-timestampValidBits-03863
The command pool's queue family **must** support a non-zero `timestampValidBits`
- VUID-vkCmdWriteTimestamp2-query-04903
`query` **must** be less than the number of queries in `queryPool`
- VUID-vkCmdWriteTimestamp2-None-03864
All queries used by the command **must** be *unavailable*
- VUID-vkCmdWriteTimestamp2-query-03865

If `vkCmdWriteTimestamp2KHR` is called within a render pass instance, the sum of `query` and the number of bits set in the current subpass's view mask **must** be less than or equal to the number of queries in `queryPool`

Valid Usage (Implicit)

- VUID-vkCmdWriteTimestamp2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdWriteTimestamp2-stage-parameter
`stage` **must** be a valid combination of `VkPipelineStageFlagBits2` values
- VUID-vkCmdWriteTimestamp2-queryPool-parameter
`queryPool` **must** be a valid `VkQueryPool` handle
- VUID-vkCmdWriteTimestamp2-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdWriteTimestamp2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, compute, decode, or encode operations
- VUID-vkCmdWriteTimestamp2-commonparent
Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	Transfer Graphics Compute Decode Encode	Action

To request a timestamp and write the value to memory, call:

```
// Provided by VK_VERSION_1_0
void vkCmdWriteTimestamp(
    VkCommandBuffer          commandBuffer,
    VkPipelineStageFlagBits pipelineStage,
    VkQueryPool              queryPool,
    uint32_t                 query);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pipelineStage` is a `VkPipelineStageFlagBits` value, specifying a stage of the pipeline.
- `queryPool` is the query pool that will manage the timestamp.
- `query` is the query within the query pool that will contain the timestamp.

When `vkCmdWriteTimestamp` is submitted to a queue, it defines an execution dependency on commands that were submitted before it, and writes a timestamp to a query pool.

The first [synchronization scope](#) includes all commands that occur earlier in [submission order](#). The synchronization scope is limited to operations on the pipeline stage specified by `pipelineStage`.

The second [synchronization scope](#) includes only the timestamp write operation.

NOTE

Implementations may write the timestamp at any stage that is [logically later](#) than [stage](#).

Any timestamp write that [happens-after](#) another timestamp write in the same submission **must** not have a lower value unless its value overflows the maximum supported integer bit width of the query. If `VK_KHR_calibrated_timestamps` is enabled, this extends to timestamp writes across all submissions on the same logical device: any timestamp write that [happens-after](#) another **must** not have a lower value unless its value overflows the maximum supported integer bit width of the query. Timestamps written by this command **must** be in the `VK_TIME_DOMAIN_DEVICE_KHR` [time domain](#). If an overflow occurs, the timestamp value **must** wrap back to zero.

NOTE

Comparisons between timestamps should be done between timestamps where they are guaranteed to not decrease. For example, subtracting an older timestamp from a newer one to determine the execution time of a sequence of commands is only a reliable measurement if the two timestamp writes were performed in the same submission, or if the writes were performed on the same logical device and `VK_KHR_calibrated_timestamps` is enabled.

If `vkCmdWriteTimestamp` is called while executing a render pass instance that has multiview enabled, the timestamp uses N consecutive query indices in the query pool (starting at `query`) where N is the number of bits set in the view mask of the subpass the command is executed in. The resulting query values are determined by an implementation-dependent choice of one of the following behaviors:

- The first query is a timestamp value and (if more than one bit is set in the view mask) zero is written to the remaining queries. If two timestamps are written in the same subpass, the sum of the execution time of all views between those commands is the difference between the first query written by each command.

- All N queries are timestamp values. If two timestamps are written in the same subpass, the sum of the execution time of all views between those commands is the sum of the difference between corresponding queries written by each command. The difference between corresponding queries **may** be the execution time of a single view.

In either case, the application **can** sum the differences between all N queries to determine the total execution time.

Valid Usage

- VUID-vkCmdWriteTimestamp-pipelineStage-04074
pipelineStage **must** be a valid stage for the queue family that was used to create the command pool that commandBuffer was allocated from
- VUID-vkCmdWriteTimestamp-pipelineStage-04075
If the geometryShader feature is not enabled, pipelineStage **must** not be VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT
- VUID-vkCmdWriteTimestamp-pipelineStage-04076
If the tessellationShader feature is not enabled, pipelineStage **must** not be VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT or VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT
- VUID-vkCmdWriteTimestamp-pipelineStage-04079
If the transformFeedback feature is not enabled, pipelineStage **must** not be VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT
- VUID-vkCmdWriteTimestamp-fragmentShadingRate-07315
If the attachmentFragmentShadingRate feature is not enabled, pipelineStage **must** not be VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
- VUID-vkCmdWriteTimestamp-synchronization2-06489
If the synchronization2 feature is not enabled, pipelineStage **must** not be VK_PIPELINE_STAGE_NONE
- VUID-vkCmdWriteTimestamp-rayTracingPipeline-07944
If the rayTracingPipeline feature is not enabled, pipelineStage **must** not be VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- VUID-vkCmdWriteTimestamp-queryPool-01416
queryPool **must** have been created with a queryType of VK_QUERY_TYPE_TIMESTAMP
- VUID-vkCmdWriteTimestamp-timestampValidBits-00829
The command pool's queue family **must** support a non-zero timestampValidBits
- VUID-vkCmdWriteTimestamp-query-04904
query **must** be less than the number of queries in queryPool
- VUID-vkCmdWriteTimestamp-None-00830
All queries used by the command **must** be unavailable
- VUID-vkCmdWriteTimestamp-query-00831
If vkCmdWriteTimestamp is called within a render pass instance, the sum of query and the number of bits set in the current subpass's view mask **must** be less than or equal to the

number of queries in `queryPool`

Valid Usage (Implicit)

- `VOID-vkCmdWriteTimestamp-commandBuffer-parameter`
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- `VOID-vkCmdWriteTimestamp-pipelineStage-parameter`
`pipelineStage` **must** be a valid `VkPipelineStageFlagBits` value
- `VOID-vkCmdWriteTimestamp-queryPool-parameter`
`queryPool` **must** be a valid `VkQueryPool` handle
- `VOID-vkCmdWriteTimestamp-commandBuffer-recording`
`commandBuffer` **must** be in the `recording state`
- `VOID-vkCmdWriteTimestamp-commandBuffer-cmdpool`
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, compute, decode, or encode operations
- `VOID-vkCmdWriteTimestamp-commonparent`
Both of `commandBuffer`, and `queryPool` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Both	Transfer Graphics Compute Decode Encode	Action

17.6. Performance Queries

Performance queries provide applications with a mechanism for getting performance counter information about the execution of command buffers, render passes, and commands.

Each queue family advertises the performance counters that **can** be queried on a queue of that family via a call to [vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR](#). Implementations **may** limit access to performance counters based on platform requirements or only to specialized drivers for development purposes.

NOTE

This may include no performance counters being enumerated, or a reduced set. Please refer to platform-specific documentation for guidance on any such restrictions.

Performance queries use the existing [vkCmdBeginQuery](#) and [vkCmdEndQuery](#) to control what command buffers, render passes, or commands to get performance information for.

Implementations **may** require multiple passes where the command buffer, render passes, or commands being recorded are the same and are executed on the same queue to record performance counter data. This is achieved by submitting the same batch and providing a [VkPerformanceQuerySubmitInfoKHR](#) structure containing a counter pass index. The number of passes required for a given performance query pool **can** be queried via a call to [vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR](#).

NOTE

Command buffers created with [VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT](#) **must** not be re-submitted. Changing command buffer usage bits **may** affect performance. To avoid this, the application **should** re-record any command buffers with the [VK_COMMAND_BUFFER_USAGE_ONE_TIME_SUBMIT_BIT](#) when multiple counter passes are required.

Performance counter results from a performance query pool **can** be obtained with the command [vkGetQueryPoolResults](#).

The [VkPerformanceCounterResultKHR](#) union is defined as:

```
// Provided by VK_KHR_performance_query
typedef union VkPerformanceCounterResultKHR {
    int32_t      int32;
    int64_t      int64;
    uint32_t     uint32;
    uint64_t     uint64;
    float        float32;
    double       float64;
} VkPerformanceCounterResultKHR;
```

- [int32](#) is a 32-bit signed integer value.
- [int64](#) is a 64-bit signed integer value.
- [uint32](#) is a 32-bit unsigned integer value.
- [uint64](#) is a 64-bit unsigned integer value.
- [float32](#) is a 32-bit floating-point value.
- [float64](#) is a 64-bit floating-point value.

Performance query results are returned in an array of `VkPerformanceCounterResultKHR` unions containing the data associated with each counter in the query, stored in the same order as the counters supplied in `pCounterIndices` when creating the performance query. `VkPerformanceCounterKHR::storage` specifies how to parse the counter data.

17.6.1. Profiling Lock

To record and submit a command buffer containing a performance query pool the profiling lock **must** be held. The profiling lock **must** be acquired prior to any call to `vkBeginCommandBuffer` that will be using a performance query pool. The profiling lock **must** be held while any command buffer containing a performance query pool is in the *recording*, *executable*, or *pending state*. To acquire the profiling lock, call:

```
// Provided by VK_KHR_performance_query
VkResult vkAcquireProfilingLockKHR(
    VkDevice device,
    const VkAcquireProfilingLockInfoKHR* pInfo);
```

- `device` is the logical device to profile.
- `pInfo` is a pointer to a `VkAcquireProfilingLockInfoKHR` structure containing information about how the profiling is to be acquired.

Implementations **may** allow multiple actors to hold the profiling lock concurrently.

Valid Usage (Implicit)

- VUID-vkAcquireProfilingLockKHR-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkAcquireProfilingLockKHR-pInfo-parameter `pInfo` **must** be a valid pointer to a valid `VkAcquireProfilingLockInfoKHR` structure

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_TIMEOUT`

The `VkAcquireProfilingLockInfoKHR` structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkAcquireProfilingLockInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    VkAcquireProfilingLockFlagsKHR flags;
    uint64_t             timeout;
} VkAcquireProfilingLockInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is reserved for future use.
- **timeout** indicates how long the function waits, in nanoseconds, if the profiling lock is not available.

Valid Usage (Implicit)

- VUID-VkAcquireProfilingLockInfoKHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_ACQUIRE_PROFILING_LOCK_INFO_KHR**
- VUID-VkAcquireProfilingLockInfoKHR-pNext-pNext
pNext must be **NULL**
- VUID-VkAcquireProfilingLockInfoKHR-flags-zeroBitmask
flags must be **0**

If **timeout** is 0, **vkAcquireProfilingLockKHR** will not block while attempting to acquire the profiling lock. If **timeout** is **UINT64_MAX**, the function will not return until the profiling lock was acquired.

```
// Provided by VK_KHR_performance_query
typedef enum VkAcquireProfilingLockFlagBitsKHR {
} VkAcquireProfilingLockFlagBitsKHR;
```

```
// Provided by VK_KHR_performance_query
typedef VkFlags VkAcquireProfilingLockFlagsKHR;
```

[VkAcquireProfilingLockFlagsKHR](#) is a bitmask type for setting a mask, but is currently reserved for future use.

To release the profiling lock, call:

```
// Provided by VK_KHR_performance_query
void vkReleaseProfilingLockKHR(
    VkDevice device);
```

- **device** is the logical device to cease profiling on.

Valid Usage

- VUID-vkReleaseProfilingLockKHR-device-03235

The profiling lock of **device** **must** have been held via a previous successful call to [vkAcquireProfilingLockKHR](#)

Valid Usage (Implicit)

- VUID-vkReleaseProfilingLockKHR-device-parameter
device **must** be a valid [VkDevice](#) handle

17.7. Transform Feedback Queries

Transform feedback queries track the number of primitives attempted to be written and actually written, by the vertex stream being captured, to a transform feedback buffer. This query is updated during drawing commands while transform feedback is active. The number of primitives actually written will be less than the number attempted to be written if the bound transform feedback buffer size was too small for the number of primitives actually drawn. Primitives are not written beyond the bound range of the transform feedback buffer. A transform feedback query is begun and ended by calling [vkCmdBeginQuery](#) and [vkCmdEndQuery](#), respectively to query for vertex stream zero. [vkCmdBeginQueryIndexedEXT](#) and [vkCmdEndQueryIndexedEXT](#) **can** be used to begin and end transform feedback queries for any supported vertex stream. When a transform feedback query begins, the count of primitives written and primitives needed starts from zero. For each drawing command, the count is incremented as vertex attribute outputs are captured to the transform feedback buffers while transform feedback is active.

When a transform feedback query finishes, the result for that query is marked as available. The application **can** then either copy the result to a buffer (via [vkCmdCopyQueryPoolResults](#)) or request it be put into host memory (via [vkGetQueryPoolResults](#)).

17.8. Primitives Generated Queries

When a generated primitive query for a vertex stream is active, the primitives-generated count is incremented every time a primitive emitted to that stream reaches the transform feedback stage, whether or not transform feedback is active. A primitives generated query is begun and ended by calling [vkCmdBeginQuery](#) and [vkCmdEndQuery](#), respectively to query for vertex stream zero. [vkCmdBeginQueryIndexedEXT](#) and [vkCmdEndQueryIndexedEXT](#) **can** be used to begin and end primitives generated queries for any supported vertex stream. When a primitives generated query begins, the count of primitives generated starts from zero.

When a primitives generated query finishes, the result for that query is marked as available. The application **can** then either copy the result to a buffer (via [vkCmdCopyQueryPoolResults](#)) or request it be put into host memory (via [vkGetQueryPoolResults](#)).

NOTE

The result of this query is typically identical to `VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT`, but the primitives generated query is deterministic, i.e. it must be identical to the number of primitives processed. `VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT` may vary for implementation-dependent reasons, e.g. the same primitive may be processed multiple times for purposes of clipping.

17.9. Result Status Queries

Result status queries serve a single purpose: allowing the application to determine whether a set of operations have completed successfully or not, as indicated by the `VkQueryResultStatusKHR` value written when retrieving the result of a query using the `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR` flag.

Unlike other query types, result status queries do not track or maintain any other data beyond the completion status, thus no other data is written when retrieving their results.

Support for result status queries is indicated by `VkQueueFamilyQueryResultStatusPropertiesKHR::queryResultStatusSupport`, as returned by `vkGetPhysicalDeviceQueueFamilyProperties2KHR` for the queue family in question.

17.10. Video Encode Feedback Queries

Video encode feedback queries allow the application to capture feedback values generated by video encode operations. As such, video encode feedback queries are available on queue families supporting video encode operations. The availability of individual video encode feedback values is indicated by the bits of `VkVideoEncodeCapabilitiesKHR::supportedEncodeFeedbackFlags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the `video profile` the queries are intended to be used with.

The set of enabled video encode feedback values **must** be configured on the query pool when it is created using the `encodeFeedbackFlags` member of the `VkQueryPoolVideoEncodeFeedbackCreateInfoKHR` included in the `pNext` chain of `VkQueryPoolCreateInfo`.

The `VkQueryPoolVideoEncodeFeedbackCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkQueryPoolVideoEncodeFeedbackCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkVideoEncodeFeedbackFlagsKHR encodeFeedbackFlags;
} VkQueryPoolVideoEncodeFeedbackCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `encodeFeedbackFlags` is a bitmask of `VkVideoEncodeFeedbackFlagBitsKHR` values specifying the

set of enabled video encode feedback values captured by queries of the new pool.

Valid Usage (Implicit)

- VUID-VkQueryPoolVideoEncodeFeedbackCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_QUERY_POOL_VIDEO_ENCODE_FEEDBACK_CREATE_INFO_KHR`
- VUID-VkQueryPoolVideoEncodeFeedbackCreateInfoKHR-encodeFeedbackFlags-parameter
`encodeFeedbackFlags` **must** be a valid combination of `VkVideoEncodeFeedbackFlagBitsKHR` values
- VUID-VkQueryPoolVideoEncodeFeedbackCreateInfoKHR-encodeFeedbackFlags-requiredbitmask
`encodeFeedbackFlags` **must** not be 0

Bits which **can** be set in `VkQueryPoolVideoEncodeFeedbackCreateInfoKHR::encodeFeedbackFlags` for video encode feedback query pools are:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeFeedbackFlagBitsKHR {
    VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR = 0x00000004,
} VkVideoEncodeFeedbackFlagBitsKHR;
```

- `VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR` specifies that queries managed by the pool will capture the byte offset of the bitstream data written by the video encode operation to the bitstream buffer specified in `VkVideoEncodeInfoKHR::dstBuffer` relative to the offset specified in `VkVideoEncodeInfoKHR::dstBufferOffset`. For the first video encode operation issued by any `video encode command`, this value will always be zero, meaning that bitstream data is always written to the buffer specified in `VkVideoEncodeInfoKHR::dstBuffer` starting from the offset specified in `VkVideoEncodeInfoKHR::dstBufferOffset`.
- `VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR` specifies that queries managed by the pool will capture the number of bytes written by the video encode operation to the bitstream buffer specified in `VkVideoEncodeInfoKHR::dstBuffer`.
- `VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR` specifies that queries managed by the pool will capture a boolean value indicating that the data written to the bitstream buffer specified in `VkVideoEncodeInfoKHR::dstBuffer` contains `overridden parameters`.

When retrieving the results of video encode feedback queries, the values corresponding to each enabled video encode feedback are written in the order of the bits defined above, followed by an optional value indicating availability or result status if `VK_QUERY_RESULT_WITH_AVAILABILITY_BIT` or `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR` is specified, respectively.

If the result status of a video encode feedback query is negative, then the results of all enabled video encode feedback values will be undefined.

NOTE

Thus it is recommended that applications always specify `VK_QUERY_RESULT_WITH_STATUS_BIT_KHR` when retrieving the results of video encode feedback queries and ignore such undefined video encode feedback values for any `unsuccessfully` completed video encode operations.

```
// Provided by VK_KHR_video_encode_queue  
typedef VkFlags VkVideoEncodeFeedbackFlagsKHR;
```

`VkVideoEncodeFeedbackFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoEncodeFeedbackFlagBitsKHR`.

Chapter 18. Clear Commands

18.1. Clearing Images Outside a Render Pass Instance

Color and depth/stencil images **can** be cleared outside a render pass instance using [vkCmdClearColorImage](#) or [vkCmdClearDepthStencilImage](#), respectively. These commands are only allowed outside of a render pass instance.

To clear one or more subranges of a color image, call:

```
// Provided by VK_VERSION_1_0
void vkCmdClearColorImage(
    VkCommandBuffer          commandBuffer,
    VkImage                  image,
    VkImageLayout             imageLayout,
    const VkClearColorValue* pColor,
    uint32_t                  rangeCount,
    const VkImageSubresourceRange* pRanges);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `image` is the image to be cleared.
- `imageLayout` specifies the current layout of the image subresource ranges to be cleared, and **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_GENERAL` or `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`.
- `pColor` is a pointer to a [VkClearColorValue](#) structure containing the values that the image subresource ranges will be cleared to (see [Clear Values](#) below).
- `rangeCount` is the number of image subresource range structures in `pRanges`.
- `pRanges` is a pointer to an array of [VkImageSubresourceRange](#) structures describing a range of mipmap levels, array layers, and aspects to be cleared, as described in [Image Views](#).

Each specified range in `pRanges` is cleared to the value specified by `pColor`.

Valid Usage

- VUID-vkCmdClearColorImage-image-01993
The [format features](#) of `image` **must** contain `VK_FORMAT_FEATURE_TRANSFER_DST_BIT`
- VUID-vkCmdClearColorImage-image-00002
`image` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-vkCmdClearColorImage-image-01545
`image` **must** not use any of the [formats that require a sampler Y'C_BC_R conversion](#)
- VUID-vkCmdClearColorImage-image-00003
If `image` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdClearColorImage-imageLayout-00004
`imageLayout` **must** specify the layout of the image subresource ranges of `image` specified in `pRanges` at the time this command is executed on a `VkDevice`
- VUID-vkCmdClearColorImage-imageLayout-01394
`imageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdClearColorImage-aspectMask-02498
The `VkImageSubresourceRange::aspectMask` members of the elements of the `pRanges` array **must** each only include `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-vkCmdClearColorImage-baseMipLevel-01470
The `VkImageSubresourceRange::baseMipLevel` members of the elements of the `pRanges` array **must** each be less than the `mipLevels` specified in `VkImageCreateInfo` when `image` was created
- VUID-vkCmdClearColorImage-pRanges-01692
For each `VkImageSubresourceRange` element of `pRanges`, if the `levelCount` member is not `VK_REMAINING_MIP_LEVELS`, then `baseMipLevel` + `levelCount` **must** be less than or equal to the `mipLevels` specified in `VkImageCreateInfo` when `image` was created
- VUID-vkCmdClearColorImage-baseArrayLayer-01472
The `VkImageSubresourceRange::baseArrayLayer` members of the elements of the `pRanges` array **must** each be less than the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-vkCmdClearColorImage-pRanges-01693
For each `VkImageSubresourceRange` element of `pRanges`, if the `layerCount` member is not `VK_REMAINING_ARRAY_LAYERS`, then `baseArrayLayer` + `layerCount` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-vkCmdClearColorImage-image-00007
`image` **must** not have a compressed or depth/stencil format
- VUID-vkCmdClearColorImage-pColor-04961
`pColor` **must** be a valid pointer to a `VkClearColorValue` union
- VUID-vkCmdClearColorImage-image-09678
If `image`'s format has components other than R and G, it **must** not have a 64-bit component width

Valid Usage (Implicit)

- VUID-vkCmdClearColorImage-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdClearColorImage-image-parameter
`image` **must** be a valid `VkImage` handle
- VUID-vkCmdClearColorImage-imageLayout-parameter
`imageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdClearColorImage-pRanges-parameter

`pRanges` **must** be a valid pointer to an array of `rangeCount` valid `VkImageSubresourceRange` structures

- VUID-vkCmdClearColorImage-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdClearColorImage-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdClearColorImage-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdClearColorImage-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdClearColorImage-rangeCount-arraylength
`rangeCount` **must** be greater than 0
- VUID-vkCmdClearColorImage-commonparent
Both of `commandBuffer`, and `image` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Graphics Compute	Action

To clear one or more subranges of a depth/stencil image, call:

```
// Provided by VK_VERSION_1_0
void vkCmdClearDepthStencilImage(
    VkCommandBuffer          commandBuffer,
    VkImage                  image,
    VkImageLayout             imageLayout,
    const VkClearDepthStencilValue* pDepthStencil,
    uint32_t                  rangeCount,
    const VkImageSubresourceRange* pRanges);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `image` is the image to be cleared.
- `imageLayout` specifies the current layout of the image subresource ranges to be cleared, and **must** be `VK_IMAGE_LAYOUT_GENERAL` or `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`.
- `pDepthStencil` is a pointer to a `VkClearDepthStencilValue` structure containing the values that the depth and stencil image subresource ranges will be cleared to (see [Clear Values](#) below).
- `rangeCount` is the number of image subresource range structures in `pRanges`.
- `pRanges` is a pointer to an array of `VkImageSubresourceRange` structures describing a range of mipmap levels, array layers, and aspects to be cleared, as described in [Image Views](#).

Valid Usage

- VUID-vkCmdClearDepthStencilImage-image-01994
The `format features` of `image` **must** contain `VK_FORMAT_FEATURE_TRANSFER_DST_BIT`
- VUID-vkCmdClearDepthStencilImage-pRanges-02659
If the `aspect` member of any element of `pRanges` includes `VK_IMAGE_ASPECT_STENCIL_BIT`, `VK_IMAGE_USAGE_TRANSFER_DST_BIT` **must** have been included in the `VkImageCreateInfo::usage` used to create `image`
- VUID-vkCmdClearDepthStencilImage-pRanges-02660
If the `aspect` member of any element of `pRanges` includes `VK_IMAGE_ASPECT_DEPTH_BIT`, `VK_IMAGE_USAGE_TRANSFER_DST_BIT` **must** have been included in the `VkImageCreateInfo::usage` used to create `image`
- VUID-vkCmdClearDepthStencilImage-image-00010
If `image` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdClearDepthStencilImage-imageLayout-00011
`imageLayout` **must** specify the layout of the image subresource ranges of `image` specified in `pRanges` at the time this command is executed on a `VkDevice`
- VUID-vkCmdClearDepthStencilImage-imageLayout-00012
`imageLayout` **must** be either of `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdClearDepthStencilImage-aspectMask-02824
The `VkImageSubresourceRange::aspectMask` member of each element of the `pRanges` array **must** not include bits other than `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-vkCmdClearDepthStencilImage-image-02825
If the `image`'s format does not have a stencil component, then the `VkImageSubresourceRange::aspectMask` member of each element of the `pRanges` array **must** not include the `VK_IMAGE_ASPECT_STENCIL_BIT` bit
- VUID-vkCmdClearDepthStencilImage-image-02826
If the `image`'s format does not have a depth component, then the `VkImageSubresourceRange::aspectMask` member of each element of the `pRanges` array

must not include the `VK_IMAGE_ASPECT_DEPTH_BIT` bit

- VUID-vkCmdClearDepthStencilImage-baseMipLevel-01474

The `VkImageSubresourceRange::baseMipLevel` members of the elements of the `pRanges` array **must** each be less than the `mipLevels` specified in `VkImageCreateInfo` when `image` was created

- VUID-vkCmdClearDepthStencilImage-pRanges-01694

For each `VkImageSubresourceRange` element of `pRanges`, if the `levelCount` member is not `VK_REMAINING_MIP_LEVELS`, then `baseMipLevel + levelCount` **must** be less than or equal to the `mipLevels` specified in `VkImageCreateInfo` when `image` was created

- VUID-vkCmdClearDepthStencilImage-baseArrayLayer-01476

The `VkImageSubresourceRange::baseArrayLayer` members of the elements of the `pRanges` array **must** each be less than the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created

- VUID-vkCmdClearDepthStencilImage-pRanges-01695

For each `VkImageSubresourceRange` element of `pRanges`, if the `layerCount` member is not `VK_REMAINING_ARRAY_LAYERS`, then `baseArrayLayer + layerCount` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created

- VUID-vkCmdClearDepthStencilImage-image-00014

`image` **must** have a depth/stencil format

Valid Usage (Implicit)

- VUID-vkCmdClearDepthStencilImage-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdClearDepthStencilImage-image-parameter

`image` **must** be a valid `VkImage` handle

- VUID-vkCmdClearDepthStencilImage-imageLayout-parameter

`imageLayout` **must** be a valid `VkImageLayout` value

- VUID-vkCmdClearDepthStencilImage-pDepthStencil-parameter

`pDepthStencil` **must** be a valid pointer to a valid `VkClearDepthStencilValue` structure

- VUID-vkCmdClearDepthStencilImage-pRanges-parameter

`pRanges` **must** be a valid pointer to an array of `rangeCount` valid `VkImageSubresourceRange` structures

- VUID-vkCmdClearDepthStencilImage-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdClearDepthStencilImage-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdClearDepthStencilImage-renderpass

This command **must** only be called outside of a render pass instance

- VUID-vkCmdClearDepthStencilImage-videocoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdClearDepthStencilImage-rangeCount-arraylength
`rangeCount` **must** be greater than 0
- VUID-vkCmdClearDepthStencilImage-commonparent
Both of `commandBuffer`, and `image` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Graphics	Action

Clears outside render pass instances are treated as transfer operations for the purposes of memory barriers.

18.2. Clearing Images Inside a Render Pass Instance

To clear one or more regions of color and depth/stencil attachments inside a render pass instance, call:

```
// Provided by VK_VERSION_1_0
void vkCmdClearAttachments(
    VkCommandBuffer          commandBuffer,
    uint32_t                 attachmentCount,
    const VkClearAttachment* pAttachments,
    uint32_t                 rectCount,
    const VkClearRect*       pRects);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `attachmentCount` is the number of entries in the `pAttachments` array.
- `pAttachments` is a pointer to an array of `VkClearAttachment` structures defining the attachments to clear and the clear values to use.
- `rectCount` is the number of entries in the `pRects` array.
- `pRects` is a pointer to an array of `VkClearRect` structures defining regions within each selected

attachment to clear.

Unlike other [clear commands](#), [vkCmdClearAttachments](#) is not a transfer command. It performs its operations in [rasterization order](#). For color attachments, the operations are executed as color attachment writes, by the [VK_PIPELINE_STAGE_COLOR_ATTACHMENT_OUTPUT_BIT](#) stage. For depth/stencil attachments, the operations are executed as [depth writes](#) and [stencil writes](#) by the [VK_PIPELINE_STAGE_EARLY_FRAGMENT_TESTS_BIT](#) and [VK_PIPELINE_STAGE_LATE_FRAGMENT_TESTS_BIT](#) stages.

[vkCmdClearAttachments](#) is not affected by the bound pipeline state.

NOTE	It is generally preferable to clear attachments by using the VK_ATTACHMENT_LOAD_OP_CLEAR load operation at the start of rendering, as it is more efficient on some implementations.
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If any attachment's [aspectMask](#) to be cleared is not backed by an image view, the clear has no effect on that aspect.

If an attachment being cleared refers to an image view created with an [aspectMask](#) equal to one of [VK_IMAGE_ASPECT_PLANE_0_BIT](#), [VK_IMAGE_ASPECT_PLANE_1_BIT](#) or [VK_IMAGE_ASPECT_PLANE_2_BIT](#), it is considered to be [VK_IMAGE_ASPECT_COLOR_BIT](#) for purposes of this command, and **must** be cleared with the [VK_IMAGE_ASPECT_COLOR_BIT](#) aspect as specified by [image view creation](#).

Valid Usage

- VUID-vkCmdClearAttachments-aspectMask-07884
If the current render pass instance does not use dynamic rendering, and the [aspectMask](#) member of any element of [pAttachments](#) contains [VK_IMAGE_ASPECT_DEPTH_BIT](#), the current subpass instance's depth-stencil attachment **must** be either [VK_ATTACHMENT_UNUSED](#) or the attachment [format](#) **must** contain a depth component
- VUID-vkCmdClearAttachments-aspectMask-07885
If the current render pass instance does not use dynamic rendering, and the [aspectMask](#) member of any element of [pAttachments](#) contains [VK_IMAGE_ASPECT_STENCIL_BIT](#), the current subpass instance's depth-stencil attachment **must** be either [VK_ATTACHMENT_UNUSED](#) or the attachment [format](#) **must** contain a stencil component
- VUID-vkCmdClearAttachments-aspectMask-07271
If the [aspectMask](#) member of any element of [pAttachments](#) contains [VK_IMAGE_ASPECT_COLOR_BIT](#), the [colorAttachment](#) **must** be a valid color attachment index in the current render pass instance
- VUID-vkCmdClearAttachments-rect-02682
The [rect](#) member of each element of [pRects](#) **must** have an [extent.width](#) greater than 0
- VUID-vkCmdClearAttachments-rect-02683
The [rect](#) member of each element of [pRects](#) **must** have an [extent.height](#) greater than 0
- VUID-vkCmdClearAttachments-pRects-00016
The rectangular region specified by each element of [pRects](#) **must** be contained within the render area of the current render pass instance
- VUID-vkCmdClearAttachments-pRects-06937

The layers specified by each element of **pRects** **must** be contained within every attachment that **pAttachments** refers to, i.e. for each element of **pRects**, **VkClearRect::baseArrayLayer** + **VkClearRect::layerCount** **must** be less than or equal to the number of layers rendered to in the current render pass instance

- VUID-vkCmdClearAttachments-layerCount-01934

The **layerCount** member of each element of **pRects** **must** not be 0

- VUID-vkCmdClearAttachments-baseArrayLayer-00018

If the render pass instance this is recorded in uses multiview, then **baseArrayLayer** **must** be zero and **layerCount** **must** be one

- VUID-vkCmdClearAttachments-colorAttachment-09503

The **colorAttachment** member of each element of **pAttachments** **must** not identify a color attachment that is currently mapped to **VK_ATTACHMENT_UNUSED** in **commandBuffer** via **VkRenderingAttachmentLocationInfoKHR**

- VUID-vkCmdClearAttachments-None-09679

If the attachment format has components other than R and G, it **must** not have a 64-bit component width

Valid Usage (Implicit)

- VUID-vkCmdClearAttachments-commandBuffer-parameter

commandBuffer **must** be a valid **VkCommandBuffer** handle

- VUID-vkCmdClearAttachments-pAttachments-parameter

pAttachments **must** be a valid pointer to an array of **attachmentCount** valid **VkClearAttachment** structures

- VUID-vkCmdClearAttachments-pRects-parameter

pRects **must** be a valid pointer to an array of **rectCount** **VkClearRect** structures

- VUID-vkCmdClearAttachments-commandBuffer-recording

commandBuffer **must** be in the **recording state**

- VUID-vkCmdClearAttachments-commandBuffer-cmdpool

The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics operations

- VUID-vkCmdClearAttachments-renderpass

This command **must** only be called inside of a render pass instance

- VUID-vkCmdClearAttachments-videocoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdClearAttachments-attachmentCount-arraylength

attachmentCount **must** be greater than 0

- VUID-vkCmdClearAttachments-rectCount-arraylength

rectCount **must** be greater than 0

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	Action

The `VkClearRect` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkClearRect {
    VkRect2D    rect;
    uint32_t    baseArrayLayer;
    uint32_t    layerCount;
} VkClearRect;
```

- `rect` is the two-dimensional region to be cleared.
- `baseArrayLayer` is the first layer to be cleared.
- `layerCount` is the number of layers to clear.

The layers [`baseArrayLayer`, `baseArrayLayer` + `layerCount`) counting from the base layer of the attachment image view are cleared.

The `VkClearAttachment` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkClearAttachment {
    VkImageAspectFlags    aspectMask;
    uint32_t              colorAttachment;
    VkClearColorValue     clearColor;
} VkClearAttachment;
```

- `aspectMask` is a mask selecting the color, depth and/or stencil aspects of the attachment to be cleared.
- `colorAttachment` is only meaningful if `VK_IMAGE_ASPECT_COLOR_BIT` is set in `aspectMask`, in which case it is an index into the currently bound color attachments.

- `clearValue` is the color or depth/stencil value to clear the attachment to, as described in [Clear Values](#) below.

Valid Usage

- VUID-VkClearAttachment-aspectMask-00019
If `aspectMask` includes `VK_IMAGE_ASPECT_COLOR_BIT`, it **must** not include `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`
- VUID-VkClearAttachment-aspectMask-00020
`aspectMask` **must** not include `VK_IMAGE_ASPECT_METADATA_BIT`
- VUID-VkClearAttachment-aspectMask-02246
`aspectMask` **must** not include `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT` for any index *i*

Valid Usage (Implicit)

- VUID-VkClearAttachment-aspectMask-parameter
`aspectMask` **must** be a valid combination of [VkImageAspectFlagBits](#) values
- VUID-VkClearAttachment-aspectMask-requiredbitmask
`aspectMask` **must** not be 0

18.3. Clear Values

The `VkClearColorValue` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef union VkClearColorValue {
    float      float32[4];
    int32_t    int32[4];
    uint32_t   uint32[4];
} VkClearColorValue;
```

- `float32` are the color clear values when the format of the image or attachment is one of the [numeric formats](#) with a numeric type that is floating-point. Floating-point values are automatically converted to the format of the image, with the clear value being treated as linear if the image is sRGB.
- `int32` are the color clear values when the format of the image or attachment has a numeric type that is signed integer (`SINT`). Signed integer values are converted to the format of the image by casting to the smaller type (with negative 32-bit values mapping to negative values in the smaller type). If the integer clear value is not representable in the target type (e.g. would overflow in conversion to that type), the clear value is undefined.
- `uint32` are the color clear values when the format of the image or attachment has a numeric type that is unsigned integer (`UINT`). Unsigned integer values are converted to the format of the image by casting to the integer type with fewer bits.

The four array elements of the clear color map to R, G, B, and A components of image formats, in order.

If the image has more than one sample, the same value is written to all samples for any pixels being cleared.

If the image or attachment format has a 64-bit component width, the first 2 array elements of each of the arrays above are reinterpreted as a single 64-bit element for the R component. The next 2 array elements are used in the same way for the G component. In other words, the union behaves as if it had the following additional members:

```
double float64[2];
int64_t int64[2];
uint64_t uint64[2];
```

The `VkClearDepthStencilValue` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkClearDepthStencilValue {
    float        depth;
    uint32_t     stencil;
} VkClearDepthStencilValue;
```

- `depth` is the clear value for the depth aspect of the depth/stencil attachment. It is a floating-point value which is automatically converted to the attachment's format.
- `stencil` is the clear value for the stencil aspect of the depth/stencil attachment. It is a 32-bit integer value which is converted to the attachment's format by taking the appropriate number of LSBs.

Valid Usage

- VUID-VkClearDepthStencilValue-depth-00022
Unless the `VK_EXT_depth_range_unrestricted` extension is enabled `depth` **must** be between `0.0` and `1.0`, inclusive

The `VkClearValue` union is defined as:

```
// Provided by VK_VERSION_1_0
typedef union VkClearValue {
    VkClearColorValue        color;
    VkClearDepthStencilValue depthStencil;
} VkClearValue;
```

- `color` specifies the color image clear values to use when clearing a color image or attachment.
- `depthStencil` specifies the depth and stencil clear values to use when clearing a depth/stencil

image or attachment.

This union is used where part of the API requires either color or depth/stencil clear values, depending on the attachment, and defines the initial clear values in the [VkRenderPassBeginInfo](#) structure.

18.4. Filling Buffers

To clear buffer data, call:

```
// Provided by VK_VERSION_1_0
void vkCmdFillBuffer(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 dstBuffer,
    VkDeviceSize             dstOffset,
    VkDeviceSize             size,
    uint32_t                 data);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **dstBuffer** is the buffer to be filled.
- **dstOffset** is the byte offset into the buffer at which to start filling, and **must** be a multiple of 4.
- **size** is the number of bytes to fill, and **must** be either a multiple of 4, or **VK_WHOLE_SIZE** to fill the range from **offset** to the end of the buffer. If **VK_WHOLE_SIZE** is used and the remaining size of the buffer is not a multiple of 4, then the nearest smaller multiple is used.
- **data** is the 4-byte word written repeatedly to the buffer to fill **size** bytes of data. The data word is written to memory according to the host endianness.

vkCmdFillBuffer is treated as a “transfer” operation for the purposes of synchronization barriers. The **VK_BUFFER_USAGE_TRANSFER_DST_BIT** **must** be specified in **usage** of [VkBufferCreateInfo](#) in order for the buffer to be compatible with **vkCmdFillBuffer**.

Valid Usage

- VUID-vkCmdFillBuffer-dstOffset-00024
dstOffset **must** be less than the size of **dstBuffer**
- VUID-vkCmdFillBuffer-dstOffset-00025
dstOffset **must** be a multiple of 4
- VUID-vkCmdFillBuffer-size-00026
If **size** is not equal to **VK_WHOLE_SIZE**, **size** **must** be greater than 0
- VUID-vkCmdFillBuffer-size-00027
If **size** is not equal to **VK_WHOLE_SIZE**, **size** **must** be less than or equal to the size of **dstBuffer** minus **dstOffset**
- VUID-vkCmdFillBuffer-size-00028
If **size** is not equal to **VK_WHOLE_SIZE**, **size** **must** be a multiple of 4

- VUID-vkCmdFillBuffer-dstBuffer-00029
`dstBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-vkCmdFillBuffer-apiVersion-07894
If the `VK_KHR_maintenance1` extension is not enabled and `VkPhysicalDeviceProperties::apiVersion` is less than Vulkan 1.1, the `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics or compute operations
- VUID-vkCmdFillBuffer-dstBuffer-00031
If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

Valid Usage (Implicit)

- VUID-vkCmdFillBuffer-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdFillBuffer-dstBuffer-parameter
`dstBuffer` **must** be a valid `VkBuffer` handle
- VUID-vkCmdFillBuffer-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdFillBuffer-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics or compute operations
- VUID-vkCmdFillBuffer-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdFillBuffer-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdFillBuffer-commonparent
Both of `commandBuffer`, and `dstBuffer` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Transfer Graphics Compute	Action

18.5. Updating Buffers

To update buffer data inline in a command buffer, call:

```
// Provided by VK_VERSION_1_0
void vkCmdUpdateBuffer(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 dstBuffer,
    VkDeviceSize             dstOffset,
    VkDeviceSize             dataSize,
    const void*              pData);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `dstBuffer` is a handle to the buffer to be updated.
- `dstOffset` is the byte offset into the buffer to start updating, and **must** be a multiple of 4.
- `dataSize` is the number of bytes to update, and **must** be a multiple of 4.
- `pData` is a pointer to the source data for the buffer update, and **must** be at least `dataSize` bytes in size.

`dataSize` **must** be less than or equal to 65536 bytes. For larger updates, applications **can** use buffer to buffer [copies](#).

NOTE

Buffer updates performed with `vkCmdUpdateBuffer` first copy the data into command buffer memory when the command is recorded (which requires additional storage and may incur an additional allocation), and then copy the data from the command buffer into `dstBuffer` when the command is executed on a device.

The additional cost of this functionality compared to [buffer to buffer copies](#) means it is only recommended for very small amounts of data, and is why it is limited to only 65536 bytes.

Applications **can** work around this by issuing multiple `vkCmdUpdateBuffer` commands to different ranges of the same buffer, but it is strongly recommended that they **should** not.

The source data is copied from `pData` to the command buffer when the command is called.

`vkCmdUpdateBuffer` is only allowed outside of a render pass. This command is treated as a “transfer” operation for the purposes of synchronization barriers. The `VK_BUFFER_USAGE_TRANSFER_DST_BIT` **must** be specified in `usage` of `VkBufferCreateInfo` in order for the buffer to be compatible with `vkCmdUpdateBuffer`.

Valid Usage

- VUID-vkCmdUpdateBuffer-dstOffset-00032
`dstOffset` **must** be less than the size of `dstBuffer`
- VUID-vkCmdUpdateBuffer-dataSize-00033
`dataSize` **must** be less than or equal to the size of `dstBuffer` minus `dstOffset`
- VUID-vkCmdUpdateBuffer-dstBuffer-00034
`dstBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-vkCmdUpdateBuffer-dstBuffer-00035
If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdUpdateBuffer-dstOffset-00036
`dstOffset` **must** be a multiple of 4
- VUID-vkCmdUpdateBuffer-dataSize-00037
`dataSize` **must** be less than or equal to 65536
- VUID-vkCmdUpdateBuffer-dataSize-00038
`dataSize` **must** be a multiple of 4

Valid Usage (Implicit)

- VUID-vkCmdUpdateBuffer-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdUpdateBuffer-dstBuffer-parameter
`dstBuffer` **must** be a valid `VkBuffer` handle
- VUID-vkCmdUpdateBuffer-pData-parameter
`pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkCmdUpdateBuffer-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdUpdateBuffer-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdUpdateBuffer-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdUpdateBuffer-vidcoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdUpdateBuffer-dataSize-arraylength

`dataSize` **must** be greater than 0

- VUID-vkCmdUpdateBuffer-commonparent

Both of `commandBuffer`, and `dstBuffer` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Transfer Graphics Compute	Action

NOTE

The `pData` parameter was of type `uint32_t*` instead of `void*` prior to version 1.0.19 of the Specification and `VK_HEADER_VERSION` 19 of the [Vulkan Header Files](#). This was a historical anomaly, as the source data may be of other types.

Chapter 19. Copy Commands

An application **can** copy buffer and image data using several methods described in this chapter, depending on the type of data transfer.

All copy commands are treated as “transfer” operations for the purposes of synchronization barriers.

All copy commands that have a source format with an X component in its format description read undefined values from those bits.

All copy commands that have a destination format with an X component in its format description write undefined values to those bits.

19.1. Copying Data Between Buffers

To copy data between buffer objects, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyBuffer(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 srcBuffer,
    VkBuffer                 dstBuffer,
    uint32_t                 regionCount,
    const VkBufferCopy*      pRegions);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **srcBuffer** is the source buffer.
- **dstBuffer** is the destination buffer.
- **regionCount** is the number of regions to copy.
- **pRegions** is a pointer to an array of **VkBufferCopy** structures specifying the regions to copy.

Each source region specified by **pRegions** is copied from the source buffer to the destination region of the destination buffer. If any of the specified regions in **srcBuffer** overlaps in memory with any of the specified regions in **dstBuffer**, values read from those overlapping regions are undefined.

Valid Usage

- VUID-vkCmdCopyBuffer-srcOffset-00113
The **srcOffset** member of each element of **pRegions** **must** be less than the size of **srcBuffer**
- VUID-vkCmdCopyBuffer-dstOffset-00114
The **dstOffset** member of each element of **pRegions** **must** be less than the size of **dstBuffer**
- VUID-vkCmdCopyBuffer-size-00115
The **size** member of each element of **pRegions** **must** be less than or equal to the size of **srcBuffer** minus **srcOffset**

- VUID-vkCmdCopyBuffer-size-00116
The **size** member of each element of **pRegions** **must** be less than or equal to the size of **dstBuffer** minus **dstOffset**
- VUID-vkCmdCopyBuffer-pRegions-00117
The union of the source regions, and the union of the destination regions, specified by the elements of **pRegions**, **must** not overlap in memory
- VUID-vkCmdCopyBuffer-srcBuffer-00118
srcBuffer **must** have been created with **VK_BUFFER_USAGE_TRANSFER_SRC_BIT** usage flag
- VUID-vkCmdCopyBuffer-srcBuffer-00119
If **srcBuffer** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-vkCmdCopyBuffer-dstBuffer-00120
dstBuffer **must** have been created with **VK_BUFFER_USAGE_TRANSFER_DST_BIT** usage flag
- VUID-vkCmdCopyBuffer-dstBuffer-00121
If **dstBuffer** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object

Valid Usage (Implicit)

- VUID-vkCmdCopyBuffer-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdCopyBuffer-srcBuffer-parameter
srcBuffer **must** be a valid **VkBuffer** handle
- VUID-vkCmdCopyBuffer-dstBuffer-parameter
dstBuffer **must** be a valid **VkBuffer** handle
- VUID-vkCmdCopyBuffer-pRegions-parameter
pRegions **must** be a valid pointer to an array of **regionCount** valid **VkBufferCopy** structures
- VUID-vkCmdCopyBuffer-commandBuffer-recording
commandBuffer **must** be in the **recording state**
- VUID-vkCmdCopyBuffer-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdCopyBuffer-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyBuffer-vidcoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdCopyBuffer-regionCount-arraylength
regionCount **must** be greater than 0
- VUID-vkCmdCopyBuffer-commonparent
Each of **commandBuffer**, **dstBuffer**, and **srcBuffer** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Transfer Graphics Compute	Action

The `VkBufferCopy` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferCopy {
    VkDeviceSize    srcOffset;
    VkDeviceSize    dstOffset;
    VkDeviceSize    size;
} VkBufferCopy;
```

- `srcOffset` is the starting offset in bytes from the start of `srcBuffer`.
- `dstOffset` is the starting offset in bytes from the start of `dstBuffer`.
- `size` is the number of bytes to copy.

Valid Usage

- VUID-VkBufferCopy-size-01988
The `size` **must** be greater than 0

A more extensible version of the copy buffer command is defined below.

To copy data between buffer objects, call:

```
// Provided by VK_KHR_copy_commands2
void vkCmdCopyBuffer2KHR(
    VkCommandBuffer          commandBuffer,
    const VkCopyBufferInfo2* pCopyBufferInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.

- `pCopyBufferInfo` is a pointer to a `VkCopyBufferInfo2` structure describing the copy parameters.

Each source region specified by `pCopyBufferInfo->pRegions` is copied from the source buffer to the destination region of the destination buffer. If any of the specified regions in `pCopyBufferInfo->srcBuffer` overlaps in memory with any of the specified regions in `pCopyBufferInfo->dstBuffer`, values read from those overlapping regions are undefined.

Valid Usage

Valid Usage (Implicit)

- VUID-vkCmdCopyBuffer2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyBuffer2-pCopyBufferInfo-parameter
`pCopyBufferInfo` **must** be a valid pointer to a valid `VkCopyBufferInfo2` structure
- VUID-vkCmdCopyBuffer2-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdCopyBuffer2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdCopyBuffer2-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyBuffer2-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Transfer Graphics Compute	Action

The `VkCopyBufferInfo2` structure is defined as:


```
typedef struct VkCopyBufferInfo2 {
    VkStructureType      sType;
    const void*          pNext;
    VkBuffer              srcBuffer;
    VkBuffer              dstBuffer;
    uint32_t             regionCount;
    const VkBufferCopy2*  pRegions;
} VkCopyBufferInfo2;
```

or the equivalent

```
// Provided by VK_KHR_copy_commands2
typedef VkCopyBufferInfo2 VkCopyBufferInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcBuffer` is the source buffer.
- `dstBuffer` is the destination buffer.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of `VkBufferCopy2` structures specifying the regions to copy.

Valid Usage

- VUID-VkCopyBufferInfo2-srcOffset-00113
The `srcOffset` member of each element of `pRegions` **must** be less than the size of `srcBuffer`
- VUID-VkCopyBufferInfo2-dstOffset-00114
The `dstOffset` member of each element of `pRegions` **must** be less than the size of `dstBuffer`
- VUID-VkCopyBufferInfo2-size-00115
The `size` member of each element of `pRegions` **must** be less than or equal to the size of `srcBuffer` minus `srcOffset`
- VUID-VkCopyBufferInfo2-size-00116
The `size` member of each element of `pRegions` **must** be less than or equal to the size of `dstBuffer` minus `dstOffset`
- VUID-VkCopyBufferInfo2-pRegions-00117
The union of the source regions, and the union of the destination regions, specified by the elements of `pRegions`, **must** not overlap in memory
- VUID-VkCopyBufferInfo2-srcBuffer-00118
`srcBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_SRC_BIT` usage flag
- VUID-VkCopyBufferInfo2-srcBuffer-00119
If `srcBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-VkCopyBufferInfo2-dstBuffer-00120
`dstBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-VkCopyBufferInfo2-dstBuffer-00121
If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

Valid Usage (Implicit)

- VUID-VkCopyBufferInfo2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COPY_BUFFER_INFO_2`
- VUID-VkCopyBufferInfo2-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCopyBufferInfo2-srcBuffer-parameter
`srcBuffer` **must** be a valid `VkBuffer` handle
- VUID-VkCopyBufferInfo2-dstBuffer-parameter
`dstBuffer` **must** be a valid `VkBuffer` handle
- VUID-VkCopyBufferInfo2-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkBufferCopy2` structures
- VUID-VkCopyBufferInfo2-regionCount-arraylength
`regionCount` **must** be greater than 0
- VUID-VkCopyBufferInfo2-commonparent
Both of `dstBuffer`, and `srcBuffer` **must** have been created, allocated, or retrieved from the same `VkDevice`

The `VkBufferCopy2` structure is defined as:

```
typedef struct VkBufferCopy2 {
    VkStructureType    sType;
    const void*        pNext;
    VkDeviceSize        srcOffset;
    VkDeviceSize        dstOffset;
    VkDeviceSize        size;
} VkBufferCopy2;
```

or the equivalent

```
// Provided by VK_KHR_copy_commands2
typedef VkBufferCopy2 VkBufferCopy2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `srcOffset` is the starting offset in bytes from the start of `srcBuffer`.
- `dstOffset` is the starting offset in bytes from the start of `dstBuffer`.
- `size` is the number of bytes to copy.

Valid Usage

- VUID-VkBufferCopy2-size-01988
The `size` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkBufferCopy2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BUFFER_COPY_2`
- VUID-VkBufferCopy2-pNext-pNext
`pNext` **must** be `NULL`

19.2. Copying Data Between Images

To copy data between image objects, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyImage(
    VkCommandBuffer          commandBuffer,
    VkImage                  srcImage,
    VkImageLayout            srcImageLayout,
    VkImage                  dstImage,
    VkImageLayout            dstImageLayout,
    uint32_t                 regionCount,
    const VkImageCopy*       pRegions);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `srcImage` is the source image.
- `srcImageLayout` is the current layout of the source image subresource.
- `dstImage` is the destination image.
- `dstImageLayout` is the current layout of the destination image subresource.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of `VkImageCopy` structures specifying the regions to copy.

Each source region specified by `pRegions` is copied from the source image to the destination region of the destination image. If any of the specified regions in `srcImage` overlaps in memory with any of the specified regions in `dstImage`, values read from those overlapping regions are undefined.

Multi-planar images **can** only be copied on a per-plane basis, and the subresources used in each region when copying to or from such images **must** specify only one plane, though different regions **can** specify different planes. When copying planes of multi-planar images, the format considered is the compatible format for that plane, rather than the format of the multi-planar image.

If the format of the destination image has a different block extent than the source image (e.g. one is a compressed format), the offset and extent for each of the regions specified is scaled according to the block extents of each format to match in size. Copy regions for each image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Image data **can** be copied between images with different image types. If one image is `VK_IMAGE_TYPE_3D` and the other image is `VK_IMAGE_TYPE_2D` with multiple layers, then each slice is copied to or from a different layer; `depth` slices in the 3D image correspond to `layerCount` layers in the 2D image, with an effective `depth` of 1 used for the 2D image. If `maintenance5` is enabled, all other combinations are allowed and function as if 1D images are 2D images with a height of 1. Otherwise, other combinations of image types are disallowed.

Valid Usage

- VUID-vkCmdCopyImage-pRegions-00124
The union of all source regions, and the union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory
- VUID-vkCmdCopyImage-srcImage-01995
The `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_TRANSFER_SRC_BIT`
- VUID-vkCmdCopyImage-srcImageLayout-00128
`srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdCopyImage-srcImageLayout-01917
`srcImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL`, or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdCopyImage-srcImage-09460
If `srcImage` and `dstImage` are the same, and any elements of `pRegions` contains the `srcSubresource` and `dstSubresource` with matching `mipLevel` and overlapping array layers, then the `srcImageLayout` and `dstImageLayout` **must** be `VK_IMAGE_LAYOUT_GENERAL` or `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`
- VUID-vkCmdCopyImage-dstImage-01996
The `format features` of `dstImage` **must** contain `VK_FORMAT_FEATURE_TRANSFER_DST_BIT`
- VUID-vkCmdCopyImage-dstImageLayout-00133
`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdCopyImage-dstImageLayout-01395
`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`, or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdCopyImage-srcImage-01548

If the `VkFormat` of each of `srcImage` and `dstImage` is not a *multi-planar format*, the `VkFormat` of each of `srcImage` and `dstImage` **must** be *size-compatible*

- VUID-vkCmdCopyImage-None-01549

In a copy to or from a plane of a *multi-planar image*, the `VkFormat` of the image and plane **must** be compatible according to *the description of compatible planes* for the plane being copied

- VUID-vkCmdCopyImage-srcImage-09247

If the `VkFormat` of each of `srcImage` and `dstImage` is a *compressed image format*, the formats **must** have the same texel block extent

- VUID-vkCmdCopyImage-srcImage-00136

The sample count of `srcImage` and `dstImage` **must** match

- VUID-vkCmdCopyImage-srcOffset-01783

The `srcOffset` and `extent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in *VkQueueFamilyProperties*

- VUID-vkCmdCopyImage-dstOffset-01784

The `dstOffset` and `extent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in *VkQueueFamilyProperties*

- VUID-vkCmdCopyImage-srcImage-01551

If neither `srcImage` nor `dstImage` has a *multi-planar image format* then for each element of `pRegions`, `srcSubresource.aspectMask` and `dstSubresource.aspectMask` **must** match

- VUID-vkCmdCopyImage-srcImage-08713

If `srcImage` has a *multi-planar image format*, then for each element of `pRegions`, `srcSubresource.aspectMask` **must** be a single valid *multi-planar aspect mask* bit

- VUID-vkCmdCopyImage-dstImage-08714

If `dstImage` has a *multi-planar image format*, then for each element of `pRegions`, `dstSubresource.aspectMask` **must** be a single valid *multi-planar aspect mask* bit

- VUID-vkCmdCopyImage-srcImage-01556

If `srcImage` has a *multi-planar image format* and the `dstImage` does not have a multi-planar image format, then for each element of `pRegions`, `dstSubresource.aspectMask` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`

- VUID-vkCmdCopyImage-dstImage-01557

If `dstImage` has a *multi-planar image format* and the `srcImage` does not have a multi-planar image format, then for each element of `pRegions`, `srcSubresource.aspectMask` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`

- VUID-vkCmdCopyImage-apiVersion-07932

If the `VK_KHR_maintenance1` extension is not enabled, or and either `srcImage` or `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcSubresource.baseArrayLayer` and `dstSubresource.baseArrayLayer` **must** both be 0, and `srcSubresource.layerCount` and `dstSubresource.layerCount` **must** both be 1

- VUID-vkCmdCopyImage-srcImage-04443

If `srcImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`,

`srcSubresource.baseArrayLayer` **must** be 0 and `srcSubresource.layerCount` **must** be 1

- VUID-vkCmdCopyImage-dstImage-04444

If `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `dstSubresource.baseArrayLayer` **must** be 0 and `dstSubresource.layerCount` **must** be 1

- VUID-vkCmdCopyImage-aspectMask-00142

For each element of `pRegions`, `srcSubresource.aspectMask` **must** specify aspects present in `srcImage`

- VUID-vkCmdCopyImage-aspectMask-00143

For each element of `pRegions`, `dstSubresource.aspectMask` **must** specify aspects present in `dstImage`

- VUID-vkCmdCopyImage-srcOffset-00144

For each element of `pRegions`, `srcOffset.x` and `(extent.width + srcOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `srcSubresource` of `srcImage`

- VUID-vkCmdCopyImage-srcOffset-00145

For each element of `pRegions`, `srcOffset.y` and `(extent.height + srcOffset.y)` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `srcSubresource` of `srcImage`

- VUID-vkCmdCopyImage-srcImage-00146

If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffset.y` **must** be 0 and `extent.height` **must** be 1

- VUID-vkCmdCopyImage-srcOffset-00147

If `srcImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcOffset.z` and `(extent.depth + srcOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `srcSubresource` of `srcImage`

- VUID-vkCmdCopyImage-srcImage-01785

If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffset.z` **must** be 0 and `extent.depth` **must** be 1

- VUID-vkCmdCopyImage-dstImage-01786

If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffset.z` **must** be 0 and `extent.depth` **must** be 1

- VUID-vkCmdCopyImage-srcImage-01787

If `srcImage` is of type `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `srcOffset.z` **must** be 0

- VUID-vkCmdCopyImage-dstImage-01788

If `dstImage` is of type `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `dstOffset.z` **must** be 0

- VUID-vkCmdCopyImage-apiVersion-07933

If the [VK_KHR_maintenance1](#) extension is not enabled, and `srcImage` and `dstImage` **must** have the same `VkImageType`

- VUID-vkCmdCopyImage-apiVersion-08969

If the [VK_KHR_maintenance1](#) extension is not enabled, and `srcImage` or `dstImage` is of type `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `extent.depth` **must** be 1

- VUID-vkCmdCopyImage-srcImage-07743
If `srcImage` and `dstImage` have a different `VkImageType`, and `maintenance5` is not enabled, one **must** be `VK_IMAGE_TYPE_3D` and the other **must** be `VK_IMAGE_TYPE_2D`
- VUID-vkCmdCopyImage-srcImage-08793
If `srcImage` and `dstImage` have the same `VkImageType`, for each element of `pRegions`, if neither of the `layerCount` members of `srcSubresource` or `dstSubresource` are `VK_REMAINING_ARRAY_LAYERS`, the `layerCount` members of `srcSubresource` or `dstSubresource` **must** match
- VUID-vkCmdCopyImage-srcImage-08794
If `srcImage` and `dstImage` have the same `VkImageType`, and one of the `layerCount` members of `srcSubresource` or `dstSubresource` is `VK_REMAINING_ARRAY_LAYERS`, the other member **must** be either `VK_REMAINING_ARRAY_LAYERS` or equal to the `arrayLayers` member of the `VkImageCreateInfo` used to create the image minus `baseArrayLayer`
- VUID-vkCmdCopyImage-srcImage-01790
If `srcImage` and `dstImage` are both of type `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `extent.depth` **must** be 1
- VUID-vkCmdCopyImage-srcImage-01791
If `srcImage` is of type `VK_IMAGE_TYPE_2D`, and `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `extent.depth` **must** equal `srcSubresource.layerCount`
- VUID-vkCmdCopyImage-dstImage-01792
If `dstImage` is of type `VK_IMAGE_TYPE_2D`, and `srcImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `extent.depth` **must** equal `dstSubresource.layerCount`
- VUID-vkCmdCopyImage-dstOffset-00150
For each element of `pRegions`, `dstOffset.x` and `(extent.width + dstOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdCopyImage-dstOffset-00151
For each element of `pRegions`, `dstOffset.y` and `(extent.height + dstOffset.y)` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdCopyImage-dstImage-00152
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffset.y` **must** be 0 and `extent.height` **must** be 1
- VUID-vkCmdCopyImage-dstOffset-00153
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `dstOffset.z` and `(extent.depth + dstOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdCopyImage-pRegions-07278
For each element of `pRegions`, `srcOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImage-pRegions-07279
For each element of `pRegions`, `srcOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`

- VUID-vkCmdCopyImage-pRegions-07280
For each element of `pRegions`, `srcOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImage-pRegions-07281
For each element of `pRegions`, `dstOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyImage-pRegions-07282
For each element of `pRegions`, `dstOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyImage-pRegions-07283
For each element of `pRegions`, `dstOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyImage-srcImage-01728
For each element of `pRegions`, if the sum of `srcOffset.x` and `extent.width` does not equal the width of the subresource specified by `srcSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImage-srcImage-01729
For each element of `pRegions`, if the sum of `srcOffset.y` and `extent.height` does not equal the height of the subresource specified by `srcSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImage-srcImage-01730
For each element of `pRegions`, if the sum of `srcOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImage-dstImage-01732
For each element of `pRegions`, if the sum of `dstOffset.x` and `extent.width` does not equal the width of the subresource specified by `dstSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyImage-dstImage-01733
For each element of `pRegions`, if the sum of `dstOffset.y` and `extent.height` does not equal the height of the subresource specified by `dstSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyImage-dstImage-01734
For each element of `pRegions`, if the sum of `dstOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `dstSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-vkCmdCopyImage-aspect-06662
`VK_IMAGE_USAGE_TRANSFER_SRC_BIT` **must** have been included in the `VkImageCreateInfo::usage` used to create `srcImage`
- VUID-vkCmdCopyImage-aspect-06663
`VK_IMAGE_USAGE_TRANSFER_DST_BIT` **must** have been included in the `VkImageCreateInfo::usage` used to create `dstImage`
- VUID-vkCmdCopyImage-srcImage-07966

If `srcImage` is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdCopyImage-srcSubresource-07967

The `srcSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `srcImage` was created

- VUID-vkCmdCopyImage-srcSubresource-07968

If `srcSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `srcSubresource.baseArrayLayer + srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `srcImage` was created

- VUID-vkCmdCopyImage-dstImage-07966

If `dstImage` is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdCopyImage-dstSubresource-07967

The `dstSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created

- VUID-vkCmdCopyImage-dstSubresource-07968

If `dstSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `dstSubresource.baseArrayLayer + dstSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created

Valid Usage (Implicit)

- VUID-vkCmdCopyImage-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdCopyImage-srcImage-parameter
`srcImage` **must** be a valid `VkImage` handle

- VUID-vkCmdCopyImage-srcImageLayout-parameter
`srcImageLayout` **must** be a valid `VkImageLayout` value

- VUID-vkCmdCopyImage-dstImage-parameter
`dstImage` **must** be a valid `VkImage` handle

- VUID-vkCmdCopyImage-dstImageLayout-parameter
`dstImageLayout` **must** be a valid `VkImageLayout` value

- VUID-vkCmdCopyImage-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkImageCopy` structures

- VUID-vkCmdCopyImage-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdCopyImage-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations

- VUID-vkCmdCopyImage-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyImage-vidcoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdCopyImage-regionCount-arraylength
regionCount **must** be greater than 0
- VUID-vkCmdCopyImage-commonparent
Each of **commandBuffer**, **dstImage**, and **srcImage** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Transfer Graphics Compute	Action

The **VkImageCopy** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageCopy {
    VkImageSubresourceLayers srcSubresource;
    VkOffset3D srcOffset;
    VkImageSubresourceLayers dstSubresource;
    VkOffset3D dstOffset;
    VkExtent3D extent;
} VkImageCopy;
```

- **srcSubresource** and **dstSubresource** are **VkImageSubresourceLayers** structures specifying the image subresources of the images used for the source and destination image data, respectively.
- **srcOffset** and **dstOffset** select the initial **x**, **y**, and **z** offsets in texels of the sub-regions of the source and destination image data.
- **extent** is the size in texels of the image to copy in **width**, **height** and **depth**.

Valid Usage

- VUID-VkImageCopy-apiVersion-07940
If the [VK_KHR_sampler_ycbcr_conversion](#) extension is not enabled, and the [aspectMask](#) member of [srcSubresource](#) and [dstSubresource](#) **must** match
- VUID-VkImageCopy-apiVersion-07941
If the [VK_KHR_maintenance1](#) extension is not enabled, and the [layerCount](#) member of [srcSubresource](#) and [dstSubresource](#) **must** match
- VUID-VkImageCopy-extent-06668
[extent.width](#) **must** not be 0
- VUID-VkImageCopy-extent-06669
[extent.height](#) **must** not be 0
- VUID-VkImageCopy-extent-06670
[extent.depth](#) **must** not be 0

Valid Usage (Implicit)

- VUID-VkImageCopy-srcSubresource-parameter
[srcSubresource](#) **must** be a valid [VkImageSubresourceLayers](#) structure
- VUID-VkImageCopy-dstSubresource-parameter
[dstSubresource](#) **must** be a valid [VkImageSubresourceLayers](#) structure

The [VkImageSubresourceLayers](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageSubresourceLayers {
    VkImageAspectFlags    aspectMask;
    uint32_t              mipLevel;
    uint32_t              baseArrayLayer;
    uint32_t              layerCount;
} VkImageSubresourceLayers;
```

- [aspectMask](#) is a combination of [VkImageAspectFlagBits](#), selecting the color, depth and/or stencil aspects to be copied.
- [mipLevel](#) is the mipmap level to copy
- [baseArrayLayer](#) and [layerCount](#) are the starting layer and number of layers to copy.

Valid Usage

- VUID-VkImageSubresourceLayers-aspectMask-00167
If [aspectMask](#) contains [VK_IMAGE_ASPECT_COLOR_BIT](#), it **must** not contain either of [VK_IMAGE_ASPECT_DEPTH_BIT](#) or [VK_IMAGE_ASPECT_STENCIL_BIT](#)

- VUID-VkImageSubresourceLayers-aspectMask-00168
`aspectMask` **must** not contain `VK_IMAGE_ASPECT_METADATA_BIT`
- VUID-VkImageSubresourceLayers-aspectMask-02247
`aspectMask` **must** not include `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT` for any index *i*
- VUID-VkImageSubresourceLayers-layerCount-09243
If the `maintenance5` feature is not enabled, `layerCount` **must** not be `VK_REMAINING_ARRAY_LAYERS`
- VUID-VkImageSubresourceLayers-layerCount-01700
If `layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, it **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkImageSubresourceLayers-aspectMask-parameter
`aspectMask` **must** be a valid combination of `VkImageAspectFlagBits` values
- VUID-VkImageSubresourceLayers-aspectMask-requiredbitmask
`aspectMask` **must** not be 0

A more extensible version of the copy image command is defined below.

To copy data between image objects, call:

```
// Provided by VK_KHR_copy_commands2
void vkCmdCopyImage2KHR(
    VkCommandBuffer                commandBuffer,
    const VkCopyImageInfo2*       pCopyImageInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pCopyImageInfo` is a pointer to a `VkCopyImageInfo2` structure describing the copy parameters.

This command is functionally identical to `vkCmdCopyImage`, but includes extensible sub-structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage

Valid Usage (Implicit)

- VUID-vkCmdCopyImage2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyImage2-pCopyImageInfo-parameter
`pCopyImageInfo` **must** be a valid pointer to a valid `VkCopyImageInfo2` structure
- VUID-vkCmdCopyImage2-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdCopyImage2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdCopyImage2-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyImage2-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Transfer Graphics Compute	Action

The `VkCopyImageInfo2` structure is defined as:

```
typedef struct VkCopyImageInfo2 {  
    VkStructureType    sType;  
    const void*        pNext;  
    VkImage             srcImage;  
    VkImageLayout       srcImageLayout;  
    VkImage             dstImage;  
    VkImageLayout       dstImageLayout;  
    uint32_t            regionCount;  
    const VkImageCopy2* pRegions;  
} VkCopyImageInfo2;
```

or the equivalent

```
// Provided by VK_KHR_copy_commands2  
typedef VkCopyImageInfo2 VkCopyImageInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcImage` is the source image.
- `srcImageLayout` is the current layout of the source image subresource.
- `dstImage` is the destination image.
- `dstImageLayout` is the current layout of the destination image subresource.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of `VkImageCopy2` structures specifying the regions to copy.

Valid Usage

- VUID-VkCopyImageInfo2-pRegions-00124
The union of all source regions, and the union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory
- VUID-VkCopyImageInfo2-srcImage-01995
The `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_TRANSFER_SRC_BIT`
- VUID-VkCopyImageInfo2-srcImageLayout-00128
`srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-VkCopyImageInfo2-srcImageLayout-01917
`srcImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL`, or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-VkCopyImageInfo2-srcImage-09460
If `srcImage` and `dstImage` are the same, and any elements of `pRegions` contains the `srcSubresource` and `dstSubresource` with matching `mipLevel` and overlapping array layers, then the `srcImageLayout` and `dstImageLayout` **must** be `VK_IMAGE_LAYOUT_GENERAL` or `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`
- VUID-VkCopyImageInfo2-dstImage-01996
The `format features` of `dstImage` **must** contain `VK_FORMAT_FEATURE_TRANSFER_DST_BIT`
- VUID-VkCopyImageInfo2-dstImageLayout-00133
`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-VkCopyImageInfo2-dstImageLayout-01395
`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`, or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-VkCopyImageInfo2-srcImage-01548
If the `VkFormat` of each of `srcImage` and `dstImage` is not a *multi-planar format*, the `VkFormat` of each of `srcImage` and `dstImage` **must** be *size-compatible*
- VUID-VkCopyImageInfo2-None-01549
In a copy to or from a plane of a *multi-planar image*, the `VkFormat` of the image and plane **must** be compatible according to *the description of compatible planes* for the plane being copied

- VUID-VkCopyImageInfo2-srcImage-09247
If the `VkFormat` of each of `srcImage` and `dstImage` is a `compressed image format`, the formats **must** have the same texel block extent
- VUID-VkCopyImageInfo2-srcImage-00136
The sample count of `srcImage` and `dstImage` **must** match
- VUID-VkCopyImageInfo2-srcOffset-01783
The `srcOffset` and `extent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in `VkQueueFamilyProperties`
- VUID-VkCopyImageInfo2-dstOffset-01784
The `dstOffset` and `extent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in `VkQueueFamilyProperties`
- VUID-VkCopyImageInfo2-srcImage-01551
If neither `srcImage` nor `dstImage` has a `multi-planar image format` then for each element of `pRegions`, `srcSubresource.aspectMask` and `dstSubresource.aspectMask` **must** match
- VUID-VkCopyImageInfo2-srcImage-08713
If `srcImage` has a `multi-planar image format`, then for each element of `pRegions`, `srcSubresource.aspectMask` **must** be a single valid `multi-planar aspect mask` bit
- VUID-VkCopyImageInfo2-dstImage-08714
If `dstImage` has a `multi-planar image format`, then for each element of `pRegions`, `dstSubresource.aspectMask` **must** be a single valid `multi-planar aspect mask` bit
- VUID-VkCopyImageInfo2-srcImage-01556
If `srcImage` has a `multi-planar image format` and the `dstImage` does not have a multi-planar image format, then for each element of `pRegions`, `dstSubresource.aspectMask` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkCopyImageInfo2-dstImage-01557
If `dstImage` has a `multi-planar image format` and the `srcImage` does not have a multi-planar image format, then for each element of `pRegions`, `srcSubresource.aspectMask` **must** be `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkCopyImageInfo2-apiVersion-07932
If the `VK_KHR_maintenance1` extension is not enabled, or and either `srcImage` or `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcSubresource.baseArrayLayer` and `dstSubresource.baseArrayLayer` **must** both be 0, and `srcSubresource.layerCount` and `dstSubresource.layerCount` **must** both be 1
- VUID-VkCopyImageInfo2-srcImage-04443
If `srcImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcSubresource.baseArrayLayer` **must** be 0 and `srcSubresource.layerCount` **must** be 1
- VUID-VkCopyImageInfo2-dstImage-04444
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `dstSubresource.baseArrayLayer` **must** be 0 and `dstSubresource.layerCount` **must** be 1
- VUID-VkCopyImageInfo2-aspectMask-00142
For each element of `pRegions`, `srcSubresource.aspectMask` **must** specify aspects present in

`srcImage`

- VUID-VkCopyImageInfo2-aspectMask-00143
For each element of `pRegions`, `dstSubresource.aspectMask` **must** specify aspects present in `dstImage`
- VUID-VkCopyImageInfo2-srcOffset-00144
For each element of `pRegions`, `srcOffset.x` and `(extent.width + srcOffset.x)` **must** both be greater than or equal to `0` and less than or equal to the width of the specified `srcSubresource` of `srcImage`
- VUID-VkCopyImageInfo2-srcOffset-00145
For each element of `pRegions`, `srcOffset.y` and `(extent.height + srcOffset.y)` **must** both be greater than or equal to `0` and less than or equal to the height of the specified `srcSubresource` of `srcImage`
- VUID-VkCopyImageInfo2-srcImage-00146
If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffset.y` **must** be `0` and `extent.height` **must** be `1`
- VUID-VkCopyImageInfo2-srcOffset-00147
If `srcImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcOffset.z` and `(extent.depth + srcOffset.z)` **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `srcSubresource` of `srcImage`
- VUID-VkCopyImageInfo2-srcImage-01785
If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffset.z` **must** be `0` and `extent.depth` **must** be `1`
- VUID-VkCopyImageInfo2-dstImage-01786
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffset.z` **must** be `0` and `extent.depth` **must** be `1`
- VUID-VkCopyImageInfo2-srcImage-01787
If `srcImage` is of type `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `srcOffset.z` **must** be `0`
- VUID-VkCopyImageInfo2-dstImage-01788
If `dstImage` is of type `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `dstOffset.z` **must** be `0`
- VUID-VkCopyImageInfo2-apiVersion-07933
If the `VK_KHR_maintenance1` extension is not enabled, and `srcImage` and `dstImage` **must** have the same `VkImageType`
- VUID-VkCopyImageInfo2-apiVersion-08969
If the `VK_KHR_maintenance1` extension is not enabled, and `srcImage` or `dstImage` is of type `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `extent.depth` **must** be `1`
- VUID-VkCopyImageInfo2-srcImage-07743
If `srcImage` and `dstImage` have a different `VkImageType`, and `maintenance5` is not enabled, one **must** be `VK_IMAGE_TYPE_3D` and the other **must** be `VK_IMAGE_TYPE_2D`
- VUID-VkCopyImageInfo2-srcImage-08793
If `srcImage` and `dstImage` have the same `VkImageType`, for each element of `pRegions`, if neither of the `layerCount` members of `srcSubresource` or `dstSubresource` are

VK_REMAINING_ARRAY_LAYERS, the `layerCount` members of `srcSubresource` or `dstSubresource` **must** match

- VUID-VkCopyImageInfo2-srcImage-08794
If `srcImage` and `dstImage` have the same `VkImageType`, and one of the `layerCount` members of `srcSubresource` or `dstSubresource` is `VK_REMAINING_ARRAY_LAYERS`, the other member **must** be either `VK_REMAINING_ARRAY_LAYERS` or equal to the `arrayLayers` member of the `VkImageCreateInfo` used to create the image minus `baseArrayLayer`
- VUID-VkCopyImageInfo2-srcImage-01790
If `srcImage` and `dstImage` are both of type `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `extent.depth` **must** be 1
- VUID-VkCopyImageInfo2-srcImage-01791
If `srcImage` is of type `VK_IMAGE_TYPE_2D`, and `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `extent.depth` **must** equal `srcSubresource.layerCount`
- VUID-VkCopyImageInfo2-dstImage-01792
If `dstImage` is of type `VK_IMAGE_TYPE_2D`, and `srcImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `extent.depth` **must** equal `dstSubresource.layerCount`
- VUID-VkCopyImageInfo2-dstOffset-00150
For each element of `pRegions`, `dstOffset.x` and `(extent.width + dstOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `dstSubresource` of `dstImage`
- VUID-VkCopyImageInfo2-dstOffset-00151
For each element of `pRegions`, `dstOffset.y` and `(extent.height + dstOffset.y)` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `dstSubresource` of `dstImage`
- VUID-VkCopyImageInfo2-dstImage-00152
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffset.y` **must** be 0 and `extent.height` **must** be 1
- VUID-VkCopyImageInfo2-dstOffset-00153
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `dstOffset.z` and `(extent.depth + dstOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `dstSubresource` of `dstImage`
- VUID-VkCopyImageInfo2-pRegions-07278
For each element of `pRegions`, `srcOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageInfo2-pRegions-07279
For each element of `pRegions`, `srcOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageInfo2-pRegions-07280
For each element of `pRegions`, `srcOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageInfo2-pRegions-07281
For each element of `pRegions`, `dstOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`

- VUID-VkCopyImageInfo2-pRegions-07282
For each element of `pRegions`, `dstOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-VkCopyImageInfo2-pRegions-07283
For each element of `pRegions`, `dstOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-VkCopyImageInfo2-srcImage-01728
For each element of `pRegions`, if the sum of `srcOffset.x` and `extent.width` does not equal the width of the subresource specified by `srcSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageInfo2-srcImage-01729
For each element of `pRegions`, if the sum of `srcOffset.y` and `extent.height` does not equal the height of the subresource specified by `srcSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageInfo2-srcImage-01730
For each element of `pRegions`, if the sum of `srcOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageInfo2-dstImage-01732
For each element of `pRegions`, if the sum of `dstOffset.x` and `extent.width` does not equal the width of the subresource specified by `dstSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-VkCopyImageInfo2-dstImage-01733
For each element of `pRegions`, if the sum of `dstOffset.y` and `extent.height` does not equal the height of the subresource specified by `dstSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-VkCopyImageInfo2-dstImage-01734
For each element of `pRegions`, if the sum of `dstOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `dstSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-VkCopyImageInfo2-aspect-06662
`VK_IMAGE_USAGE_TRANSFER_SRC_BIT` **must** have been included in the `VkImageCreateInfo::usage` used to create `srcImage`
- VUID-VkCopyImageInfo2-aspect-06663
`VK_IMAGE_USAGE_TRANSFER_DST_BIT` **must** have been included in the `VkImageCreateInfo::usage` used to create `dstImage`
- VUID-VkCopyImageInfo2-srcImage-07966
If `srcImage` is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyImageInfo2-srcSubresource-07967
The `srcSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-VkCopyImageInfo2-srcSubresource-07968

If `srcSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `srcSubresource.baseArrayLayer + srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `srcImage` was created

- VUID-VkCopyImageInfo2-dstImage-07966

If `dstImage` is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-VkCopyImageInfo2-dstSubresource-07967

The `dstSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created

- VUID-VkCopyImageInfo2-dstSubresource-07968

If `dstSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `dstSubresource.baseArrayLayer + dstSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created

Valid Usage (Implicit)

- VUID-VkCopyImageInfo2-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_COPY_IMAGE_INFO_2`

- VUID-VkCopyImageInfo2-pNext-pNext

`pNext` **must** be `NULL`

- VUID-VkCopyImageInfo2-srcImage-parameter

`srcImage` **must** be a valid `VkImage` handle

- VUID-VkCopyImageInfo2-srcImageLayout-parameter

`srcImageLayout` **must** be a valid `VkImageLayout` value

- VUID-VkCopyImageInfo2-dstImage-parameter

`dstImage` **must** be a valid `VkImage` handle

- VUID-VkCopyImageInfo2-dstImageLayout-parameter

`dstImageLayout` **must** be a valid `VkImageLayout` value

- VUID-VkCopyImageInfo2-pRegions-parameter

`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkImageCopy2` structures

- VUID-VkCopyImageInfo2-regionCount-arraylength

`regionCount` **must** be greater than 0

- VUID-VkCopyImageInfo2-commonparent

Both of `dstImage`, and `srcImage` **must** have been created, allocated, or retrieved from the same `VkDevice`

The `VkImageCopy2` structure is defined as:

```
typedef struct VkImageCopy2 {
    VkStructureType    sType;
    const void*        pNext;
    VkImageSubresourceLayers srcSubresource;
    VkOffset3D         srcOffset;
    VkImageSubresourceLayers dstSubresource;
    VkOffset3D         dstOffset;
    VkExtent3D         extent;
} VkImageCopy2;
```

or the equivalent

```
// Provided by VK_KHR_copy_commands2
typedef VkImageCopy2 VkImageCopy2KHR;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcSubresource` and `dstSubresource` are [VkImageSubresourceLayers](#) structures specifying the image subresources of the images used for the source and destination image data, respectively.
- `srcOffset` and `dstOffset` select the initial `x`, `y`, and `z` offsets in texels of the sub-regions of the source and destination image data.
- `extent` is the size in texels of the image to copy in `width`, `height` and `depth`.

Valid Usage

- VUID-VkImageCopy2-apiVersion-07940
If the [VK_KHR_sampler_ycbcr_conversion](#) extension is not enabled, and the `aspectMask` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageCopy2-apiVersion-07941
If the [VK_KHR_maintenance1](#) extension is not enabled, and the `layerCount` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageCopy2-extent-06668
`extent.width` **must** not be 0
- VUID-VkImageCopy2-extent-06669
`extent.height` **must** not be 0
- VUID-VkImageCopy2-extent-06670
`extent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkImageCopy2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_COPY_2`

- VUID-VkImageCopy2-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkImageCopy2-srcSubresource-parameter
`srcSubresource` **must** be a valid `VkImageSubresourceLayers` structure
- VUID-VkImageCopy2-dstSubresource-parameter
`dstSubresource` **must** be a valid `VkImageSubresourceLayers` structure

19.3. Copying Data Between Buffers and Images

Data **can** be copied between buffers and images, enabling applications to load and store data between images and application-defined offsets in buffer memory.

When copying between a buffer and an image, texels in the image and bytes in the buffer are accessed as follows.

Texels at each coordinate (x,y,z,layer) in the image subresource are accessed, where:

x is in the range [`imageOffset.x`, `imageOffset.x` + `imageExtent.width`),

y is in the range [`imageOffset.y`, `imageOffset.y` + `imageExtent.height`),

z is in the range [`imageOffset.z`, `imageOffset.z` + `imageExtent.depth`),

layer is in the range [`imageSubresource.baseArrayLayer`, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount`)

For each (x,y,z,layer) coordinate in the image, bytes in the buffer are accessed at offsets in the range [`texelOffset`, `texelOffset` + `blockSize`), where:

$$\text{texelOffset} = \text{bufferOffset} + ((x - \text{imageOffset.x}) / \text{blockWidth} \times \text{blockSize}) + ((y - \text{imageOffset.y}) / \text{blockHeight} \times \text{rowExtent}) + ((z - \text{imageOffset.z}) / \text{blockDepth} \times \text{sliceExtent}) + ((\text{layer} - \text{imageSubresource.baseArrayLayer}) \times \text{layerExtent})$$

$$\text{rowExtent} = \lceil \max(\text{bufferRowLength}, \text{imageExtent.width}) / \text{blockWidth} \rceil \times \text{blockSize}$$

$$\text{sliceExtent} = \lceil \max(\text{bufferImageHeight}, \text{imageExtent.height}) / \text{blockHeight} \rceil \times \text{rowExtent}$$

$$\text{layerExtent} = \lceil \text{imageExtent.depth} / \text{blockDepth} \rceil \times \text{sliceExtent}$$

and where blockSize, blockWidth, blockHeight, and blockDepth are the [texel block size and extents](#) of the image's format.

When copying between a buffer and the depth or stencil aspect of an image, data in the buffer is assumed to be laid out as separate planes rather than interleaved. Addressing calculations are thus performed for a different format than the base image, according to the aspect, as described in the following table:

Table 24. Depth/Stencil Aspect Copy Table

Base Format	Depth Aspect Format	Stencil Aspect Format
VK_FORMAT_D16_UNORM	VK_FORMAT_D16_UNORM	-
VK_FORMAT_X8_D24_UNORM_PACK32	VK_FORMAT_X8_D24_UNORM_PACK32	-
VK_FORMAT_D32_SFLOAT	VK_FORMAT_D32_SFLOAT	-
VK_FORMAT_S8_UINT	-	VK_FORMAT_S8_UINT
VK_FORMAT_D16_UNORM_S8_UINT	VK_FORMAT_D16_UNORM	VK_FORMAT_S8_UINT
VK_FORMAT_D24_UNORM_S8_UINT	VK_FORMAT_X8_D24_UNORM_PACK32	VK_FORMAT_S8_UINT
VK_FORMAT_D32_SFLOAT_S8_UINT	VK_FORMAT_D32_SFLOAT	VK_FORMAT_S8_UINT

When copying between a buffer and any plane of a [multi-planar image](#), addressing calculations are performed using the [compatible format for that plane](#), rather than the format of the multi-planar image.

Each texel block is copied from one resource to the other according to the above addressing equations.

To copy data from a buffer object to an image object, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyBufferToImage(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 srcBuffer,
    VkImage                  dstImage,
    VkImageLayout             dstImageLayout,
    uint32_t                 regionCount,
    const VkBufferImageCopy* pRegions);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `srcBuffer` is the source buffer.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the copy.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of [VkBufferImageCopy](#) structures specifying the regions to copy.

Each source region specified by `pRegions` is copied from the source buffer to the destination region

of the destination image according to the [addressing calculations](#) for each resource. If any of the specified regions in `srcBuffer` overlaps in memory with any of the specified regions in `dstImage`, values read from those overlapping regions are undefined. If any region accesses a depth aspect in `dstImage` and the `VK_EXT_depth_range_unrestricted` extension is not enabled, values copied from `srcBuffer` outside of the range [0,1] will be written as undefined values to the destination image.

Copy regions for the image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Valid Usage

- VUID-vkCmdCopyBufferToImage-dstImage-07966
If `dstImage` is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyBufferToImage-imageSubresource-07967
The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-vkCmdCopyBufferToImage-imageSubresource-07968
If `imageSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-vkCmdCopyBufferToImage-imageSubresource-07970
The image region specified by each element of `pRegions` **must** be contained within the specified `imageSubresource` of `dstImage`
- VUID-vkCmdCopyBufferToImage-imageSubresource-07971
For each element of `pRegions`, `imageOffset.x` and `(imageExtent.width + imageOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `imageSubresource` of `dstImage`
- VUID-vkCmdCopyBufferToImage-imageSubresource-07972
For each element of `pRegions`, `imageOffset.y` and `(imageExtent.height + imageOffset.y)` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `imageSubresource` of `dstImage`
- VUID-vkCmdCopyBufferToImage-dstImage-07973
`dstImage` **must** have a sample count equal to `VK_SAMPLE_COUNT_1_BIT`
- VUID-vkCmdCopyBufferToImage-commandBuffer-07737
If the queue family used to create the `VkCommandPool` which `commandBuffer` was allocated from does not support `VK_QUEUE_GRAPHICS_BIT` or `VK_QUEUE_COMPUTE_BIT`, the `bufferOffset` member of any element of `pRegions` **must** be a multiple of 4
- VUID-vkCmdCopyBufferToImage-imageOffset-07738
The `imageOffset` and `imageExtent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family,

as described in [VkQueueFamilyProperties](#)

- VUID-vkCmdCopyBufferToImage-commandBuffer-07739

If the queue family used to create the [VkCommandPool](#) which `commandBuffer` was allocated from does not support `VK_QUEUE_GRAPHICS_BIT`, for each element of `pRegions`, the `aspectMask` member of `imageSubresource` **must** not be `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`

- VUID-vkCmdCopyBufferToImage-pRegions-00171

`srcBuffer` **must** be large enough to contain all buffer locations that are accessed according to [Buffer and Image Addressing](#), for each element of `pRegions`

- VUID-vkCmdCopyBufferToImage-pRegions-00173

The union of all source regions, and the union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory

- VUID-vkCmdCopyBufferToImage-srcBuffer-00174

`srcBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_SRC_BIT` usage flag

- VUID-vkCmdCopyBufferToImage-dstImage-01997

The [format features](#) of `dstImage` **must** contain `VK_FORMAT_FEATURE_TRANSFER_DST_BIT`

- VUID-vkCmdCopyBufferToImage-srcBuffer-00176

If `srcBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdCopyBufferToImage-dstImage-00177

`dstImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT` usage flag

- VUID-vkCmdCopyBufferToImage-dstImageLayout-00180

`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`

- VUID-vkCmdCopyBufferToImage-dstImageLayout-01396

`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`, or `VK_IMAGE_LAYOUT_GENERAL`

- VUID-vkCmdCopyBufferToImage-pRegions-07931

If `VK_EXT_depth_range_unrestricted` is not enabled, for each element of `pRegions` whose `imageSubresource` contains a depth aspect, the data in `srcBuffer` **must** be in the range [0,1]

- VUID-vkCmdCopyBufferToImage-dstImage-07979

If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `imageOffset.y` **must** be 0 and `imageExtent.height` **must** be 1

- VUID-vkCmdCopyBufferToImage-imageOffset-09104

For each element of `pRegions`, `imageOffset.z` and $(\text{imageExtent.depth} + \text{imageOffset.z})$ **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `imageSubresource` of `dstImage`

- VUID-vkCmdCopyBufferToImage-dstImage-07980

If `dstImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `imageOffset.z` **must** be 0 and `imageExtent.depth` **must** be 1

- VUID-vkCmdCopyBufferToImage-dstImage-07274

For each element of `pRegions`, `imageOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyBufferToImage-dstImage-07275

For each element of `pRegions`, `imageOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyBufferToImage-dstImage-07276

For each element of `pRegions`, `imageOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyBufferToImage-dstImage-00207

For each element of `pRegions`, if the sum of `imageOffset.x` and `extent.width` does not equal the width of the subresource specified by `imageSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyBufferToImage-dstImage-00208

For each element of `pRegions`, if the sum of `imageOffset.y` and `extent.height` does not equal the height of the subresource specified by `imageSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyBufferToImage-dstImage-00209

For each element of `pRegions`, if the sum of `imageOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyBufferToImage-imageSubresource-09105

For each element of `pRegions`, `imageSubresource.aspectMask` **must** specify aspects present in `dstImage`

- VUID-vkCmdCopyBufferToImage-dstImage-07981

If `dstImage` has a `multi-planar image format`, then for each element of `pRegions`, `imageSubresource.aspectMask` **must** be a single valid `multi-planar aspect mask` bit

- VUID-vkCmdCopyBufferToImage-dstImage-07983

If `dstImage` is of type `VK_IMAGE_TYPE_3D`, for each element of `pRegions`, `imageSubresource.baseArrayLayer` **must** be 0 and `imageSubresource.layerCount` **must** be 1

- VUID-vkCmdCopyBufferToImage-bufferRowLength-09106

For each element of `pRegions`, `bufferRowLength` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyBufferToImage-bufferImageHeight-09107

For each element of `pRegions`, `bufferImageHeight` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`

- VUID-vkCmdCopyBufferToImage-bufferRowLength-09108

For each element of `pRegions`, `bufferRowLength` divided by the `texel block extent width` and then multiplied by the texel block size of `dstImage` **must** be less than or equal to $2^{31}-1$

- VUID-vkCmdCopyBufferToImage-dstImage-07975

If `dstImage` does not have either a depth/stencil format or a `multi-planar format`, then for each element of `pRegions`, `bufferOffset` **must** be a multiple of the `texel block size`

- VUID-vkCmdCopyBufferToImage-dstImage-07976

If **dstImage** has a **multi-planar format**, then for each element of **pRegions**, **bufferOffset** **must** be a multiple of the element size of the compatible format for the format and the **aspectMask** of the **imageSubresource** as defined in **Compatible Formats of Planes of Multi-Planar Formats**

- VUID-vkCmdCopyBufferToImage-dstImage-07978

If **dstImage** has a depth/stencil format, the **bufferOffset** member of any element of **pRegions** **must** be a multiple of 4

Valid Usage (Implicit)

- VUID-vkCmdCopyBufferToImage-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdCopyBufferToImage-srcBuffer-parameter
srcBuffer **must** be a valid **VkBuffer** handle
- VUID-vkCmdCopyBufferToImage-dstImage-parameter
dstImage **must** be a valid **VkImage** handle
- VUID-vkCmdCopyBufferToImage-dstImageLayout-parameter
dstImageLayout **must** be a valid **VkImageLayout** value
- VUID-vkCmdCopyBufferToImage-pRegions-parameter
pRegions **must** be a valid pointer to an array of **regionCount** valid **VkBufferImageCopy** structures
- VUID-vkCmdCopyBufferToImage-commandBuffer-recording
commandBuffer **must** be in the **recording state**
- VUID-vkCmdCopyBufferToImage-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdCopyBufferToImage-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyBufferToImage-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdCopyBufferToImage-regionCount-arraylength
regionCount **must** be greater than 0
- VUID-vkCmdCopyBufferToImage-commonparent
Each of **commandBuffer**, **dstImage**, and **srcBuffer** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Transfer Graphics Compute	Action

To copy data from an image object to a buffer object, call:

```
// Provided by VK_VERSION_1_0
void vkCmdCopyImageToBuffer(
    VkCommandBuffer          commandBuffer,
    VkImage                  srcImage,
    VkImageLayout            srcImageLayout,
    VkBuffer                 dstBuffer,
    uint32_t                 regionCount,
    const VkBufferImageCopy* pRegions);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the copy.
- `dstBuffer` is the destination buffer.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of `VkBufferImageCopy` structures specifying the regions to copy.

Each source region specified by `pRegions` is copied from the source image to the destination region of the destination buffer according to the [addressing calculations](#) for each resource. If any of the specified regions in `srcImage` overlaps in memory with any of the specified regions in `dstBuffer`, values read from those overlapping regions are undefined.

Copy regions for the image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Valid Usage

- VUID-vkCmdCopyImageToBuffer-srcImage-07966
If `srcImage` is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyImageToBuffer-imageSubresource-07967
The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `srcImage` was created

- VUID-vkCmdCopyImageToBuffer-imageSubresource-07968
If `imageSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `imageSubresource.baseArrayLayer + imageSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-vkCmdCopyImageToBuffer-imageSubresource-07970
The image region specified by each element of `pRegions` **must** be contained within the specified `imageSubresource` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-imageSubresource-07971
For each element of `pRegions`, `imageOffset.x` and `(imageExtent.width + imageOffset.x)` **must** both be greater than or equal to `0` and less than or equal to the width of the specified `imageSubresource` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-imageSubresource-07972
For each element of `pRegions`, `imageOffset.y` and `(imageExtent.height + imageOffset.y)` **must** both be greater than or equal to `0` and less than or equal to the height of the specified `imageSubresource` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-07973
`srcImage` **must** have a sample count equal to `VK_SAMPLE_COUNT_1_BIT`
- VUID-vkCmdCopyImageToBuffer-commandBuffer-07746
If the queue family used to create the `VkCommandPool` which `commandBuffer` was allocated from does not support `VK_QUEUE_GRAPHICS_BIT` or `VK_QUEUE_COMPUTE_BIT`, the `bufferOffset` member of any element of `pRegions` **must** be a multiple of `4`
- VUID-vkCmdCopyImageToBuffer-imageOffset-07747
The `imageOffset` and `imageExtent` members of each element of `pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in `VkQueueFamilyProperties`
- VUID-vkCmdCopyImageToBuffer-pRegions-00183
`dstBuffer` **must** be large enough to contain all buffer locations that are accessed according to `Buffer and Image Addressing`, for each element of `pRegions`
- VUID-vkCmdCopyImageToBuffer-pRegions-00184
The union of all source regions, and the union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory
- VUID-vkCmdCopyImageToBuffer-srcImage-00186
`srcImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` usage flag
- VUID-vkCmdCopyImageToBuffer-srcImage-01998
The `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_TRANSFER_SRC_BIT`
- VUID-vkCmdCopyImageToBuffer-dstBuffer-00191
`dstBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-vkCmdCopyImageToBuffer-dstBuffer-00192
If `dstBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdCopyImageToBuffer-srcImageLayout-00189
`srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdCopyImageToBuffer-srcImageLayout-01397
`srcImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL`, or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdCopyImageToBuffer-srcImage-07979
If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `imageOffset.y` **must** be 0 and `imageExtent.height` **must** be 1
- VUID-vkCmdCopyImageToBuffer-imageOffset-09104
For each element of `pRegions`, `imageOffset.z` and `(imageExtent.depth + imageOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `imageSubresource` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-07980
If `srcImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `imageOffset.z` **must** be 0 and `imageExtent.depth` **must** be 1
- VUID-vkCmdCopyImageToBuffer-srcImage-07274
For each element of `pRegions`, `imageOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-07275
For each element of `pRegions`, `imageOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-07276
For each element of `pRegions`, `imageOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-00207
For each element of `pRegions`, if the sum of `imageOffset.x` and `extent.width` does not equal the width of the subresource specified by `imageSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-00208
For each element of `pRegions`, if the sum of `imageOffset.y` and `extent.height` does not equal the height of the subresource specified by `imageSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-00209
For each element of `pRegions`, if the sum of `imageOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-imageSubresource-09105
For each element of `pRegions`, `imageSubresource.aspectMask` **must** specify aspects present in `srcImage`
- VUID-vkCmdCopyImageToBuffer-srcImage-07981
If `srcImage` has a `multi-planar image format`, then for each element of `pRegions`, `imageSubresource.aspectMask` **must** be a single valid `multi-planar aspect mask` bit

- VUID-vkCmdCopyImageToBuffer-srcImage-07983
If `srcImage` is of type `VK_IMAGE_TYPE_3D`, for each element of `pRegions`, `imageSubresource.baseArrayLayer` **must** be 0 and `imageSubresource.layerCount` **must** be 1
- VUID-vkCmdCopyImageToBuffer-bufferRowLength-09106
For each element of `pRegions`, `bufferRowLength` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-bufferImageHeight-09107
For each element of `pRegions`, `bufferImageHeight` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-vkCmdCopyImageToBuffer-bufferRowLength-09108
For each element of `pRegions`, `bufferRowLength` divided by the `texel block extent width` and then multiplied by the texel block size of `srcImage` **must** be less than or equal to $2^{31}-1$
- VUID-vkCmdCopyImageToBuffer-srcImage-07975
If `srcImage` does not have either a depth/stencil format or a `multi-planar format`, then for each element of `pRegions`, `bufferOffset` **must** be a multiple of the `texel block size`
- VUID-vkCmdCopyImageToBuffer-srcImage-07976
If `srcImage` has a `multi-planar format`, then for each element of `pRegions`, `bufferOffset` **must** be a multiple of the element size of the compatible format for the format and the `aspectMask` of the `imageSubresource` as defined in `Compatible Formats of Planes of Multi-Planar Formats`
- VUID-vkCmdCopyImageToBuffer-srcImage-07978
If `srcImage` has a depth/stencil format, the `bufferOffset` member of any element of `pRegions` **must** be a multiple of 4

Valid Usage (Implicit)

- VUID-vkCmdCopyImageToBuffer-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyImageToBuffer-srcImage-parameter
`srcImage` **must** be a valid `VkImage` handle
- VUID-vkCmdCopyImageToBuffer-srcImageLayout-parameter
`srcImageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdCopyImageToBuffer-dstBuffer-parameter
`dstBuffer` **must** be a valid `VkBuffer` handle
- VUID-vkCmdCopyImageToBuffer-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkBufferImageCopy` structures
- VUID-vkCmdCopyImageToBuffer-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdCopyImageToBuffer-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics,

or compute operations

- VUID-vkCmdCopyImageToBuffer-renderpass

This command **must** only be called outside of a render pass instance

- VUID-vkCmdCopyImageToBuffer-vidcoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdCopyImageToBuffer-regionCount-arraylength

regionCount **must** be greater than 0

- VUID-vkCmdCopyImageToBuffer-commonparent

Each of **commandBuffer**, **dstBuffer**, and **srcImage** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Transfer Graphics Compute	Action

For both **vkCmdCopyBufferToImage** and **vkCmdCopyImageToBuffer**, each element of **pRegions** is a structure defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBufferImageCopy {
    VkDeviceSize      bufferOffset;
    uint32_t          bufferRowLength;
    uint32_t          bufferImageHeight;
    VkImageSubresourceLayers imageSubresource;
    VkOffset3D         imageOffset;
    VkExtent3D         imageExtent;
} VkBufferImageCopy;
```

- **bufferOffset** is the offset in bytes from the start of the buffer object where the image data is copied from or to.
- **bufferRowLength** and **bufferImageHeight** specify in texels a subregion of a larger two- or three-dimensional image in buffer memory, and control the addressing calculations. If either of these

values is zero, that aspect of the buffer memory is considered to be tightly packed according to the `imageExtent`.

- `imageSubresource` is a `VkImageSubresourceLayers` used to specify the specific image subresources of the image used for the source or destination image data.
- `imageOffset` selects the initial `x`, `y`, `z` offsets in texels of the sub-region of the source or destination image data.
- `imageExtent` is the size in texels of the image to copy in `width`, `height` and `depth`.

Valid Usage

- VUID-VkBufferImageCopy-bufferRowLength-09101
`bufferRowLength` **must** be 0, or greater than or equal to the `width` member of `imageExtent`
- VUID-VkBufferImageCopy-bufferImageHeight-09102
`bufferImageHeight` **must** be 0, or greater than or equal to the `height` member of `imageExtent`
- VUID-VkBufferImageCopy-aspectMask-09103
The `aspectMask` member of `imageSubresource` **must** only have a single bit set
- VUID-VkBufferImageCopy-imageExtent-06659
`imageExtent.width` **must** not be 0
- VUID-VkBufferImageCopy-imageExtent-06660
`imageExtent.height` **must** not be 0
- VUID-VkBufferImageCopy-imageExtent-06661
`imageExtent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkBufferImageCopy-imageSubresource-parameter
`imageSubresource` **must** be a valid `VkImageSubresourceLayers` structure

More extensible versions of the commands to copy between buffers and images are defined below.

To copy data from a buffer object to an image object, call:

```
// Provided by VK_KHR_copy_commands2
void vkCmdCopyBufferToImage2KHR(
    VkCommandBuffer                commandBuffer,
    const VkCopyBufferToImageInfo2* pCopyBufferToImageInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pCopyBufferToImageInfo` is a pointer to a `VkCopyBufferToImageInfo2` structure describing the copy parameters.

This command is functionally identical to `vkCmdCopyBufferToImage`, but includes extensible sub-

structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage

- VUID-vkCmdCopyBufferToImage2-commandBuffer-07737
If the queue family used to create the `VkCommandPool` which `commandBuffer` was allocated from does not support `VK_QUEUE_GRAPHICS_BIT` or `VK_QUEUE_COMPUTE_BIT`, the `bufferOffset` member of any element of `pCopyBufferToImageInfo->pRegions` **must** be a multiple of 4
- VUID-vkCmdCopyBufferToImage2-imageOffset-07738
The `imageOffset` and `imageExtent` members of each element of `pCopyBufferToImageInfo->pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in `VkQueueFamilyProperties`
- VUID-vkCmdCopyBufferToImage2-commandBuffer-07739
If the queue family used to create the `VkCommandPool` which `commandBuffer` was allocated from does not support `VK_QUEUE_GRAPHICS_BIT`, for each element of `pCopyBufferToImageInfo->pRegions`, the `aspectMask` member of `imageSubresource` **must** not be `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT`

Valid Usage (Implicit)

- VUID-vkCmdCopyBufferToImage2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyBufferToImage2-pCopyBufferToImageInfo-parameter
`pCopyBufferToImageInfo` **must** be a valid pointer to a valid `VkCopyBufferToImageInfo2` structure
- VUID-vkCmdCopyBufferToImage2-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdCopyBufferToImage2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdCopyBufferToImage2-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyBufferToImage2-encoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Transfer Graphics Compute	Action

The `VkCopyBufferToImageInfo2` structure is defined as:

```
typedef struct VkCopyBufferToImageInfo2 {  
    VkStructureType    sType;  
    const void*        pNext;  
    VkBuffer            srcBuffer;  
    VkImage             dstImage;  
    VkImageLayout       dstImageLayout;  
    uint32_t            regionCount;  
    const VkBufferImageCopy2* pRegions;  
} VkCopyBufferToImageInfo2;
```

or the equivalent

```
// Provided by VK_KHR_copy_commands2  
typedef VkCopyBufferToImageInfo2 VkCopyBufferToImageInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcBuffer` is the source buffer.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the copy.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of `VkBufferImageCopy2` structures specifying the regions to copy.

Valid Usage

- VUID-VkCopyBufferToImageInfo2-pRegions-04565
The image region specified by each element of `pRegions` **must** be contained within the specified `imageSubresource` of `dstImage`
- VUID-VkCopyBufferToImageInfo2-pRegions-00171
`srcBuffer` **must** be large enough to contain all buffer locations that are accessed according

to [Buffer and Image Addressing](#), for each element of `pRegions`

- VUID-VkCopyBufferToImageInfo2-pRegions-00173
The union of all source regions, and the union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory
- VUID-VkCopyBufferToImageInfo2-srcBuffer-00174
`srcBuffer` **must** have been created with `VK_BUFFER_USAGE_TRANSFER_SRC_BIT` usage flag
- VUID-VkCopyBufferToImageInfo2-dstImage-01997
The [format features](#) of `dstImage` **must** contain `VK_FORMAT_FEATURE_TRANSFER_DST_BIT`
- VUID-VkCopyBufferToImageInfo2-srcBuffer-00176
If `srcBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyBufferToImageInfo2-dstImage-00177
`dstImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-VkCopyBufferToImageInfo2-dstImageLayout-00180
`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-VkCopyBufferToImageInfo2-dstImageLayout-01396
`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL`, or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-VkCopyBufferToImageInfo2-pRegions-07931
If `VK_EXT_depth_range_unrestricted` is not enabled, for each element of `pRegions` whose `imageSubresource` contains a depth aspect, the data in `srcBuffer` **must** be in the range [0,1]
- VUID-VkCopyBufferToImageInfo2-dstImage-07966
If `dstImage` is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyBufferToImageInfo2-imageSubresource-07967
The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-VkCopyBufferToImageInfo2-imageSubresource-07968
If `imageSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-VkCopyBufferToImageInfo2-dstImage-07973
`dstImage` **must** have a sample count equal to `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkCopyBufferToImageInfo2-dstImage-07979
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `imageOffset.y` **must** be 0 and `imageExtent.height` **must** be 1
- VUID-VkCopyBufferToImageInfo2-imageOffset-09104
For each element of `pRegions`, `imageOffset.z` and `(imageExtent.depth + imageOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified

`imageSubresource` of `dstImage`

- VUID-VkCopyBufferToImageInfo2-dstImage-07980
If `dstImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `imageOffset.z` **must** be 0 and `imageExtent.depth` **must** be 1
- VUID-VkCopyBufferToImageInfo2-dstImage-07274
For each element of `pRegions`, `imageOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-VkCopyBufferToImageInfo2-dstImage-07275
For each element of `pRegions`, `imageOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-VkCopyBufferToImageInfo2-dstImage-07276
For each element of `pRegions`, `imageOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-VkCopyBufferToImageInfo2-dstImage-00207
For each element of `pRegions`, if the sum of `imageOffset.x` and `extent.width` does not equal the width of the subresource specified by `imageSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-VkCopyBufferToImageInfo2-dstImage-00208
For each element of `pRegions`, if the sum of `imageOffset.y` and `extent.height` does not equal the height of the subresource specified by `imageSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-VkCopyBufferToImageInfo2-dstImage-00209
For each element of `pRegions`, if the sum of `imageOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-VkCopyBufferToImageInfo2-imageSubresource-09105
For each element of `pRegions`, `imageSubresource.aspectMask` **must** specify aspects present in `dstImage`
- VUID-VkCopyBufferToImageInfo2-dstImage-07981
If `dstImage` has a `multi-planar image format`, then for each element of `pRegions`, `imageSubresource.aspectMask` **must** be a single valid `multi-planar aspect mask` bit
- VUID-VkCopyBufferToImageInfo2-dstImage-07983
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, for each element of `pRegions`, `imageSubresource.baseArrayLayer` **must** be 0 and `imageSubresource.layerCount` **must** be 1
- VUID-VkCopyBufferToImageInfo2-bufferRowLength-09106
For each element of `pRegions`, `bufferRowLength` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-VkCopyBufferToImageInfo2-bufferImageHeight-09107
For each element of `pRegions`, `bufferImageHeight` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-VkCopyBufferToImageInfo2-bufferRowLength-09108
For each element of `pRegions`, `bufferRowLength` divided by the `texel block extent width` and

then multiplied by the texel block size of **dstImage** **must** be less than or equal to $2^{31}-1$

- VUID-VkCopyBufferToImageInfo2-dstImage-07975

If **dstImage** does not have either a depth/stencil format or a **multi-planar format**, then for each element of **pRegions**, **bufferOffset** **must** be a multiple of the **texel block size**

- VUID-VkCopyBufferToImageInfo2-dstImage-07976

If **dstImage** has a **multi-planar format**, then for each element of **pRegions**, **bufferOffset** **must** be a multiple of the element size of the compatible format for the format and the **aspectMask** of the **imageSubresource** as defined in **Compatible Formats of Planes of Multi-Planar Formats**

- VUID-VkCopyBufferToImageInfo2-dstImage-07978

If **dstImage** has a depth/stencil format, the **bufferOffset** member of any element of **pRegions** **must** be a multiple of 4

- VUID-VkCopyBufferToImageInfo2-pRegions-06223

For each element of **pRegions** not containing **VkCopyCommandTransformInfoQCOM** in its **pNext** chain, **imageOffset.x** and **(imageExtent.width + imageOffset.x)** **must** both be greater than or equal to 0 and less than or equal to the width of the specified **imageSubresource** of **dstImage**

- VUID-VkCopyBufferToImageInfo2-pRegions-06224

For each element of **pRegions** not containing **VkCopyCommandTransformInfoQCOM** in its **pNext** chain, **imageOffset.y** and **(imageExtent.height + imageOffset.y)** **must** both be greater than or equal to 0 and less than or equal to the height of the specified **imageSubresource** of **dstImage**

Valid Usage (Implicit)

- VUID-VkCopyBufferToImageInfo2-sType-sType

sType **must** be **VK_STRUCTURE_TYPE_COPY_BUFFER_TO_IMAGE_INFO_2**

- VUID-VkCopyBufferToImageInfo2-pNext-pNext

pNext **must** be **NULL**

- VUID-VkCopyBufferToImageInfo2-srcBuffer-parameter

srcBuffer **must** be a valid **VkBuffer** handle

- VUID-VkCopyBufferToImageInfo2-dstImage-parameter

dstImage **must** be a valid **VkImage** handle

- VUID-VkCopyBufferToImageInfo2-dstImageLayout-parameter

dstImageLayout **must** be a valid **VkImageLayout** value

- VUID-VkCopyBufferToImageInfo2-pRegions-parameter

pRegions **must** be a valid pointer to an array of **regionCount** valid **VkBufferImageCopy2** structures

- VUID-VkCopyBufferToImageInfo2-regionCount-arraylength

regionCount **must** be greater than 0

- VUID-VkCopyBufferToImageInfo2-commonparent

Both of **dstImage**, and **srcBuffer** **must** have been created, allocated, or retrieved from the same **VkDevice**

To copy data from an image object to a buffer object, call:

```
// Provided by VK_KHR_copy_commands2
void vkCmdCopyImageToBuffer2KHR(
    VkCommandBuffer                      commandBuffer,
    const VkCopyImageToBufferInfo2*     pCopyImageToBufferInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pCopyImageToBufferInfo` is a pointer to a `VkCopyImageToBufferInfo2` structure describing the copy parameters.

This command is functionally identical to `vkCmdCopyImageToBuffer`, but includes extensible sub-structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage

- VUID-vkCmdCopyImageToBuffer2-commandBuffer-07746
If the queue family used to create the `VkCommandPool` which `commandBuffer` was allocated from does not support `VK_QUEUE_GRAPHICS_BIT` or `VK_QUEUE_COMPUTE_BIT`, the `bufferOffset` member of any element of `pCopyImageToBufferInfo->pRegions` **must** be a multiple of 4
- VUID-vkCmdCopyImageToBuffer2-imageOffset-07747
The `imageOffset` and `imageExtent` members of each element of `pCopyImageToBufferInfo->pRegions` **must** respect the image transfer granularity requirements of `commandBuffer`'s command pool's queue family, as described in `VkQueueFamilyProperties`

Valid Usage (Implicit)

- VUID-vkCmdCopyImageToBuffer2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyImageToBuffer2-pCopyImageToBufferInfo-parameter
`pCopyImageToBufferInfo` **must** be a valid pointer to a valid `VkCopyImageToBufferInfo2` structure
- VUID-vkCmdCopyImageToBuffer2-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdCopyImageToBuffer2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support transfer, graphics, or compute operations
- VUID-vkCmdCopyImageToBuffer2-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyImageToBuffer2-encoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Transfer Graphics Compute	Action

The `VkCopyImageToBufferInfo2` structure is defined as:

```
typedef struct VkCopyImageToBufferInfo2 {  
    VkStructureType      sType;  
    const void*          pNext;  
    VkImage               srcImage;  
    VkImageLayout         srcImageLayout;  
    VkBuffer              dstBuffer;  
    uint32_t              regionCount;  
    const VkBufferImageCopy2* pRegions;  
} VkCopyImageToBufferInfo2;
```

or the equivalent

```
// Provided by VK_KHR_copy_commands2  
typedef VkCopyImageToBufferInfo2 VkCopyImageToBufferInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the copy.
- `dstBuffer` is the destination buffer.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of `VkBufferImageCopy2` structures specifying the regions to copy.

Valid Usage

- VUID-VkCopyImageToBufferInfo2-pRegions-04566
The image region specified by each element of **pRegions** **must** be contained within the specified **imageSubresource** of **srcImage**
- VUID-VkCopyImageToBufferInfo2-pRegions-00183
dstBuffer **must** be large enough to contain all buffer locations that are accessed according to **Buffer and Image Addressing**, for each element of **pRegions**
- VUID-VkCopyImageToBufferInfo2-pRegions-00184
The union of all source regions, and the union of all destination regions, specified by the elements of **pRegions**, **must** not overlap in memory
- VUID-VkCopyImageToBufferInfo2-srcImage-00186
srcImage **must** have been created with **VK_IMAGE_USAGE_TRANSFER_SRC_BIT** usage flag
- VUID-VkCopyImageToBufferInfo2-srcImage-01998
The **format features** of **srcImage** **must** contain **VK_FORMAT_FEATURE_TRANSFER_SRC_BIT**
- VUID-VkCopyImageToBufferInfo2-dstBuffer-00191
dstBuffer **must** have been created with **VK_BUFFER_USAGE_TRANSFER_DST_BIT** usage flag
- VUID-VkCopyImageToBufferInfo2-dstBuffer-00192
If **dstBuffer** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-VkCopyImageToBufferInfo2-srcImageLayout-00189
srcImageLayout **must** specify the layout of the image subresources of **srcImage** specified in **pRegions** at the time this command is executed on a **VkDevice**
- VUID-VkCopyImageToBufferInfo2-srcImageLayout-01397
srcImageLayout **must** be **VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR**, **VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL**, or **VK_IMAGE_LAYOUT_GENERAL**
- VUID-VkCopyImageToBufferInfo2-srcImage-07966
If **srcImage** is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-VkCopyImageToBufferInfo2-imageSubresource-07967
The **imageSubresource.mipLevel** member of each element of **pRegions** **must** be less than the **mipLevels** specified in **VkImageCreateInfo** when **srcImage** was created
- VUID-VkCopyImageToBufferInfo2-imageSubresource-07968
If **imageSubresource.layerCount** is not **VK_REMAINING_ARRAY_LAYERS**, **imageSubresource.baseArrayLayer** + **imageSubresource.layerCount** of each element of **pRegions** **must** be less than or equal to the **arrayLayers** specified in **VkImageCreateInfo** when **srcImage** was created
- VUID-VkCopyImageToBufferInfo2-srcImage-07973
srcImage **must** have a sample count equal to **VK_SAMPLE_COUNT_1_BIT**
- VUID-VkCopyImageToBufferInfo2-srcImage-07979
If **srcImage** is of type **VK_IMAGE_TYPE_1D**, then for each element of **pRegions**, **imageOffset.y**

must be 0 and **imageExtent.height** **must** be 1

- VUID-VkCopyImageToBufferInfo2-imageOffset-09104

For each element of **pRegions**, **imageOffset.z** and (**imageExtent.depth** + **imageOffset.z**) **must** both be greater than or equal to 0 and less than or equal to the depth of the specified **imageSubresource** of **srcImage**

- VUID-VkCopyImageToBufferInfo2-srcImage-07980

If **srcImage** is of type **VK_IMAGE_TYPE_1D** or **VK_IMAGE_TYPE_2D**, then for each element of **pRegions**, **imageOffset.z** **must** be 0 and **imageExtent.depth** **must** be 1

- VUID-VkCopyImageToBufferInfo2-srcImage-07274

For each element of **pRegions**, **imageOffset.x** **must** be a multiple of the **texel block extent width** of the **VkFormat** of **srcImage**

- VUID-VkCopyImageToBufferInfo2-srcImage-07275

For each element of **pRegions**, **imageOffset.y** **must** be a multiple of the **texel block extent height** of the **VkFormat** of **srcImage**

- VUID-VkCopyImageToBufferInfo2-srcImage-07276

For each element of **pRegions**, **imageOffset.z** **must** be a multiple of the **texel block extent depth** of the **VkFormat** of **srcImage**

- VUID-VkCopyImageToBufferInfo2-srcImage-00207

For each element of **pRegions**, if the sum of **imageOffset.x** and **extent.width** does not equal the width of the subresource specified by **imageSubresource**, **extent.width** **must** be a multiple of the **texel block extent width** of the **VkFormat** of **srcImage**

- VUID-VkCopyImageToBufferInfo2-srcImage-00208

For each element of **pRegions**, if the sum of **imageOffset.y** and **extent.height** does not equal the height of the subresource specified by **imageSubresource**, **extent.height** **must** be a multiple of the **texel block extent height** of the **VkFormat** of **srcImage**

- VUID-VkCopyImageToBufferInfo2-srcImage-00209

For each element of **pRegions**, if the sum of **imageOffset.z** and **extent.depth** does not equal the depth of the subresource specified by **srcSubresource**, **extent.depth** **must** be a multiple of the **texel block extent depth** of the **VkFormat** of **srcImage**

- VUID-VkCopyImageToBufferInfo2-imageSubresource-09105

For each element of **pRegions**, **imageSubresource.aspectMask** **must** specify aspects present in **srcImage**

- VUID-VkCopyImageToBufferInfo2-srcImage-07981

If **srcImage** has a **multi-planar image format**, then for each element of **pRegions**, **imageSubresource.aspectMask** **must** be a single valid **multi-planar aspect mask** bit

- VUID-VkCopyImageToBufferInfo2-srcImage-07983

If **srcImage** is of type **VK_IMAGE_TYPE_3D**, for each element of **pRegions**, **imageSubresource.baseArrayLayer** **must** be 0 and **imageSubresource.layerCount** **must** be 1

- VUID-VkCopyImageToBufferInfo2-bufferRowLength-09106

For each element of **pRegions**, **bufferRowLength** **must** be a multiple of the **texel block extent width** of the **VkFormat** of **srcImage**

- VUID-VkCopyImageToBufferInfo2-bufferImageHeight-09107

For each element of **pRegions**, **bufferImageHeight** **must** be a multiple of the **texel block extent height** of the **VkFormat** of **srcImage**

- VUID-VkCopyImageToBufferInfo2-bufferRowLength-09108

For each element of **pRegions**, **bufferRowLength** divided by the **texel block extent width** and then multiplied by the texel block size of **srcImage** **must** be less than or equal to $2^{31}-1$

- VUID-VkCopyImageToBufferInfo2-srcImage-07975

If **srcImage** does not have either a depth/stencil format or a **multi-planar format**, then for each element of **pRegions**, **bufferOffset** **must** be a multiple of the **texel block size**

- VUID-VkCopyImageToBufferInfo2-srcImage-07976

If **srcImage** has a **multi-planar format**, then for each element of **pRegions**, **bufferOffset** **must** be a multiple of the element size of the compatible format for the format and the **aspectMask** of the **imageSubresource** as defined in **Compatible Formats of Planes of Multi-Planar Formats**

- VUID-VkCopyImageToBufferInfo2-srcImage-07978

If **srcImage** has a depth/stencil format, the **bufferOffset** member of any element of **pRegions** **must** be a multiple of 4

- VUID-VkCopyImageToBufferInfo2-imageOffset-00197

For each element of **pRegions** not containing **VkCopyCommandTransformInfoQCOM** in its **pNext** chain, **imageOffset.x** and **(imageExtent.width + imageOffset.x)** **must** both be greater than or equal to 0 and less than or equal to the width of the specified **imageSubresource** of **srcImage**

- VUID-VkCopyImageToBufferInfo2-imageOffset-00198

For each element of **pRegions** not containing **VkCopyCommandTransformInfoQCOM** in its **pNext** chain, **imageOffset.y** and **(imageExtent.height + imageOffset.y)** **must** both be greater than or equal to 0 and less than or equal to the height of the specified **imageSubresource** of **srcImage**

Valid Usage (Implicit)

- VUID-VkCopyImageToBufferInfo2-sType-sType

sType **must** be **VK_STRUCTURE_TYPE_COPY_IMAGE_TO_BUFFER_INFO_2**

- VUID-VkCopyImageToBufferInfo2-pNext-pNext

pNext **must** be **NULL**

- VUID-VkCopyImageToBufferInfo2-srcImage-parameter

srcImage **must** be a valid **VkImage** handle

- VUID-VkCopyImageToBufferInfo2-srcImageLayout-parameter

srcImageLayout **must** be a valid **VkImageLayout** value

- VUID-VkCopyImageToBufferInfo2-dstBuffer-parameter

dstBuffer **must** be a valid **VkBuffer** handle

- VUID-VkCopyImageToBufferInfo2-pRegions-parameter

pRegions **must** be a valid pointer to an array of **regionCount** valid **VkBufferImageCopy2** structures

- VUID-VkCopyImageToBufferInfo2-regionCount-arraylength

regionCount **must** be greater than 0

- VUID-VkCopyImageToBufferInfo2-commonparent

Both of **dstBuffer**, and **srcImage** **must** have been created, allocated, or retrieved from the same **VkDevice**

For both **vkCmdCopyBufferToImage2KHR** and **vkCmdCopyImageToBuffer2KHR**, each element of **pRegions** is a structure defined as:

```
typedef struct VkBufferImageCopy2 {  
    VkStructureType      sType;  
    const void*          pNext;  
    VkDeviceSize         bufferOffset;  
    uint32_t             bufferRowLength;  
    uint32_t             bufferImageHeight;  
    VkImageSubresourceLayers imageSubresource;  
    VkOffset3D           imageOffset;  
    VkExtent3D           imageExtent;  
} VkBufferImageCopy2;
```

or the equivalent

```
// Provided by VK_KHR_copy_commands2  
typedef VkBufferImageCopy2 VkBufferImageCopy2KHR;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **bufferOffset** is the offset in bytes from the start of the buffer object where the image data is copied from or to.
- **bufferRowLength** and **bufferImageHeight** specify in texels a subregion of a larger two- or three-dimensional image in buffer memory, and control the addressing calculations. If either of these values is zero, that aspect of the buffer memory is considered to be tightly packed according to the **imageExtent**.
- **imageSubresource** is a **VkImageSubresourceLayers** used to specify the specific image subresources of the image used for the source or destination image data.
- **imageOffset** selects the initial **x**, **y**, **z** offsets in texels of the sub-region of the source or destination image data.
- **imageExtent** is the size in texels of the image to copy in **width**, **height** and **depth**.

This structure is functionally identical to **VkBufferImageCopy**, but adds **sType** and **pNext** parameters, allowing it to be more easily extended.

Valid Usage

- VUID-VkBufferImageCopy2-bufferRowLength-09101
`bufferRowLength` **must** be 0, or greater than or equal to the `width` member of `imageExtent`
- VUID-VkBufferImageCopy2-bufferImageHeight-09102
`bufferImageHeight` **must** be 0, or greater than or equal to the `height` member of `imageExtent`
- VUID-VkBufferImageCopy2-aspectMask-09103
The `aspectMask` member of `imageSubresource` **must** only have a single bit set
- VUID-VkBufferImageCopy2-imageExtent-06659
`imageExtent.width` **must** not be 0
- VUID-VkBufferImageCopy2-imageExtent-06660
`imageExtent.height` **must** not be 0
- VUID-VkBufferImageCopy2-imageExtent-06661
`imageExtent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkBufferImageCopy2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BUFFER_IMAGE_COPY_2`
- VUID-VkBufferImageCopy2-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkBufferImageCopy2-imageSubresource-parameter
`imageSubresource` **must** be a valid `VkImageSubresourceLayers` structure

The following commands can be used to copy between host memory and images.

To copy data from host memory to an image object, call:

```
// Provided by VK_EXT_host_image_copy
VkResult vkCopyMemoryToImageEXT(
    VkDevice device,
    const VkCopyMemoryToImageInfoEXT* pCopyMemoryToImageInfo);
```

- `device` is the device which owns `pCopyMemoryToImageInfo->dstImage`.
- `pCopyMemoryToImageInfo` is a pointer to a `VkCopyMemoryToImageInfoEXT` structure describing the copy parameters.

This command is functionally similar to `vkCmdCopyBufferToImage2KHR`, except it is executed on the host and reads from host memory instead of a buffer. The memory of `pCopyMemoryToImageInfo->dstImage` is accessed by the host as if `coherent`.

NOTE

Because queue submissions `automatically make host memory visible to the device`,

there would not be a need for a memory barrier before using the results of this copy operation on the device.

Valid Usage (Implicit)

- VUID-vkCopyMemoryToImageEXT-device-parameter **device** must be a valid [VkDevice](#) handle
- VUID-vkCopyMemoryToImageEXT-pCopyMemoryToImageInfo-parameter **pCopyMemoryToImageInfo** must be a valid pointer to a valid [VkCopyMemoryToImageInfoEXT](#) structure

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_INITIALIZATION_FAILED](#)
- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)
- [VK_ERROR_MEMORY_MAP_FAILED](#)

The [VkCopyMemoryToImageInfoEXT](#) structure is defined as:

```
// Provided by VK_EXT_host_image_copy
typedef struct VkCopyMemoryToImageInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkHostImageCopyFlagsEXT flags;
    VkImage              dstImage;
    VkImageLayout        dstImageLayout;
    uint32_t             regionCount;
    const VkMemoryToImageCopyEXT* pRegions;
} VkCopyMemoryToImageInfoEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is [NULL](#) or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkHostImageCopyFlagBitsEXT](#) values describing additional copy parameters.
- **dstImage** is the destination image.
- **dstImageLayout** is the layout of the destination image subresources for the copy.
- **regionCount** is the number of regions to copy.

- `pRegions` is a pointer to an array of `VkMemoryToImageCopyEXT` structures specifying the regions to copy.

`vkCopyMemoryToImageEXT` does not check whether the device memory associated with `dstImage` is currently in use before performing the copy. The application **must** guarantee that any previously submitted command that reads from or writes to the copy regions has completed before the host performs the copy.

Copy regions for the image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Valid Usage

- VUID-VkCopyMemoryToImageInfoEXT-dstImage-09109
If `dstImage` is sparse then all memory ranges accessed by the copy command **must** be bound as described in [Binding Resource Memory](#)
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-09111
`dstImage` **must** have been created with `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` set in `VkImageCreateInfo::usage`
- VUID-VkCopyMemoryToImageInfoEXT-imageOffset-09114
If `flags` contains `VK_HOST_IMAGE_COPY_MEMCPY_EXT`, the `x`, `y`, and `z` members of the `imageOffset` member of each element of `pRegions` **must** be 0
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-09115
If `flags` contains `VK_HOST_IMAGE_COPY_MEMCPY_EXT`, the `imageExtent` member of each element of `pRegions` **must** equal the extents of `dstImage` identified by `imageSubresource`
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-07966
If `dstImage` is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyMemoryToImageInfoEXT-imageSubresource-07967
The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-VkCopyMemoryToImageInfoEXT-imageSubresource-07968
If `imageSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-VkCopyMemoryToImageInfoEXT-imageSubresource-07970
The image region specified by each element of `pRegions` **must** be contained within the specified `imageSubresource` of `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-imageSubresource-07971
For each element of `pRegions`, `imageOffset.x` and `(imageExtent.width + imageOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `imageSubresource` of `dstImage`

- VUID-VkCopyMemoryToImageInfoEXT-imageSubresource-07972
For each element of `pRegions`, `imageOffset.y` and `(imageExtent.height + imageOffset.y)` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `imageSubresource` of `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-07973
`dstImage` **must** have a sample count equal to `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-07979
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `imageOffset.y` **must** be 0 and `imageExtent.height` **must** be 1
- VUID-VkCopyMemoryToImageInfoEXT-imageOffset-09104
For each element of `pRegions`, `imageOffset.z` and `(imageExtent.depth + imageOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `imageSubresource` of `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-07980
If `dstImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `imageOffset.z` **must** be 0 and `imageExtent.depth` **must** be 1
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-07274
For each element of `pRegions`, `imageOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-07275
For each element of `pRegions`, `imageOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-07276
For each element of `pRegions`, `imageOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-00207
For each element of `pRegions`, if the sum of `imageOffset.x` and `extent.width` does not equal the width of the subresource specified by `imageSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-00208
For each element of `pRegions`, if the sum of `imageOffset.y` and `extent.height` does not equal the height of the subresource specified by `imageSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-00209
For each element of `pRegions`, if the sum of `imageOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-imageSubresource-09105
For each element of `pRegions`, `imageSubresource.aspectMask` **must** specify aspects present in `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-07981
If `dstImage` has a `multi-planar image format`, then for each element of `pRegions`,

`imageSubresource.aspectMask` **must** be a single valid [multi-planar aspect mask](#) bit

- VUID-VkCopyMemoryToImageInfoEXT-dstImage-07983
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, for each element of `pRegions`, `imageSubresource.baseArrayLayer` **must** be 0 and `imageSubresource.layerCount` **must** be 1
- VUID-VkCopyMemoryToImageInfoEXT-memoryRowLength-09106
For each element of `pRegions`, `memoryRowLength` **must** be a multiple of the [texel block extent width](#) of the `VkFormat` of `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-memoryImageHeight-09107
For each element of `pRegions`, `memoryImageHeight` **must** be a multiple of the [texel block extent height](#) of the `VkFormat` of `dstImage`
- VUID-VkCopyMemoryToImageInfoEXT-memoryRowLength-09108
For each element of `pRegions`, `memoryRowLength` divided by the [texel block extent width](#) and then multiplied by the texel block size of `dstImage` **must** be less than or equal to $2^{31}-1$
- VUID-VkCopyMemoryToImageInfoEXT-dstImageLayout-09059
`dstImageLayout` **must** specify the current layout of the image subresources of `dstImage` specified in `pRegions`
- VUID-VkCopyMemoryToImageInfoEXT-dstImageLayout-09060
`dstImageLayout` **must** be one of the image layouts returned in `VkPhysicalDeviceHostImageCopyPropertiesEXT::pCopyDstLayouts`
- VUID-VkCopyMemoryToImageInfoEXT-flags-09393
If `flags` includes `VK_HOST_IMAGE_COPY_MEMCPY_EXT`, for each region in `pRegions`, `memoryRowLength` and `memoryImageHeight` **must** both be 0

Valid Usage (Implicit)

- VUID-VkCopyMemoryToImageInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INFO_EXT`
- VUID-VkCopyMemoryToImageInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCopyMemoryToImageInfoEXT-flags-parameter
`flags` **must** be a valid combination of [VkHostImageCopyFlagBitsEXT](#) values
- VUID-VkCopyMemoryToImageInfoEXT-dstImage-parameter
`dstImage` **must** be a valid [VkImage](#) handle
- VUID-VkCopyMemoryToImageInfoEXT-dstImageLayout-parameter
`dstImageLayout` **must** be a valid [VkImageLayout](#) value
- VUID-VkCopyMemoryToImageInfoEXT-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid [VkMemoryToImageCopyEXT](#) structures
- VUID-VkCopyMemoryToImageInfoEXT-regionCount-arraylength
`regionCount` **must** be greater than 0

Each element of [VkCopyMemoryToImageInfoEXT::pRegions](#) is a structure defined as:

```
// Provided by VK_EXT_host_image_copy
typedef struct VkMemoryToImageCopyEXT {
    VkStructureType      sType;
    const void*          pNext;
    const void*          pHostPointer;
    uint32_t              memoryRowLength;
    uint32_t              memoryImageHeight;
    VkImageSubresourceLayers imageSubresource;
    VkOffset3D            imageOffset;
    VkExtent3D            imageExtent;
} VkMemoryToImageCopyEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **pHostPointer** is the host memory address which is the source of the copy.
- **memoryRowLength** and **memoryImageHeight** specify in texels a subregion of a larger two- or three-dimensional image in host memory, and control the addressing calculations. If either of these values is zero, that aspect of the host memory is considered to be tightly packed according to the **imageExtent**.
- **imageSubresource** is a [VkImageSubresourceLayers](#) used to specify the specific image subresources of the image used for the source or destination image data.
- **imageOffset** selects the initial **x**, **y**, **z** offsets in texels of the sub-region of the destination image data.
- **imageExtent** is the size in texels of the image to copy in **width**, **height** and **depth**.

This structure is functionally similar to [VkBufferImageCopy2](#), except it defines host memory as the source of copy instead of a buffer. In particular, the same data packing rules and restrictions as that structure apply here as well.

Valid Usage

- VUID-VkMemoryToImageCopyEXT-pHostPointer-09061
pHostPointer **must** point to memory that is large enough to contain all memory locations that are accessed according to [Buffer and Image Addressing](#), for each element of **pRegions**
- VUID-VkMemoryToImageCopyEXT-pRegions-09062
The union of all source regions, and the union of all destination regions, specified by the elements of **pRegions**, **must** not overlap in memory
- VUID-VkMemoryToImageCopyEXT-memoryRowLength-09101
memoryRowLength **must** be 0, or greater than or equal to the **width** member of **imageExtent**
- VUID-VkMemoryToImageCopyEXT-memoryImageHeight-09102
memoryImageHeight **must** be 0, or greater than or equal to the **height** member of **imageExtent**

- VUID-VkMemoryToImageCopyEXT-aspectMask-09103
The `aspectMask` member of `imageSubresource` **must** only have a single bit set
- VUID-VkMemoryToImageCopyEXT-imageExtent-06659
`imageExtent.width` **must** not be 0
- VUID-VkMemoryToImageCopyEXT-imageExtent-06660
`imageExtent.height` **must** not be 0
- VUID-VkMemoryToImageCopyEXT-imageExtent-06661
`imageExtent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkMemoryToImageCopyEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MEMORY_TO_IMAGE_COPY_EXT`
- VUID-VkMemoryToImageCopyEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkMemoryToImageCopyEXT-pHostPointer-parameter
`pHostPointer` **must** be a pointer value
- VUID-VkMemoryToImageCopyEXT-imageSubresource-parameter
`imageSubresource` **must** be a valid `VkImageSubresourceLayers` structure

To copy data from an image object to host memory, call:

```
// Provided by VK_EXT_host_image_copy
VkResult vkCopyImageToMemoryEXT(
    VkDevice device,
    const VkCopyImageToMemoryInfoEXT* pCopyImageToMemoryInfo);
```

- `device` is the device which owns `pCopyImageToMemoryInfo->srcImage`.
- `pCopyImageToMemoryInfo` is a pointer to a `VkCopyImageToMemoryInfoEXT` structure describing the copy parameters.

This command is functionally similar to `vkCmdCopyImageToBuffer2KHR`, except it is executed on the host and writes to host memory instead of a buffer. The memory of `pCopyImageToMemoryInfo->srcImage` is accessed by the host as if `coherent`.

NOTE

If the device has written to the image memory, it is not automatically made available to the host. Before this copy command can be called, a memory barrier for this image **must** have been issued on the device with the second `synchronization scope` including `VK_PIPELINE_STAGE_HOST_BIT` and `VK_ACCESS_HOST_READ_BIT`.

Valid Usage (Implicit)

- VUID-vkCopyImageToMemoryEXT-device-parameter **device** must be a valid [VkDevice](#) handle
- VUID-vkCopyImageToMemoryEXT-pCopyImageToMemoryInfo-parameter **pCopyImageToMemoryInfo** must be a valid pointer to a valid [VkCopyImageToMemoryInfoEXT](#) structure

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_INITIALIZATION_FAILED](#)
- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)
- [VK_ERROR_MEMORY_MAP_FAILED](#)

The [VkCopyImageToMemoryInfoEXT](#) structure is defined as:

```
// Provided by VK_EXT_host_image_copy
typedef struct VkCopyImageToMemoryInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkHostImageCopyFlagsEXT flags;
    VkImage              srcImage;
    VkImageLayout        srcImageLayout;
    uint32_t             regionCount;
    const VkImageToMemoryCopyEXT* pRegions;
} VkCopyImageToMemoryInfoEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is [NULL](#) or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkHostImageCopyFlagBitsEXT](#) values describing additional copy parameters.
- **srcImage** is the source image.
- **srcImageLayout** is the layout of the source image subresources for the copy.
- **regionCount** is the number of regions to copy.
- **pRegions** is a pointer to an array of [VkImageToMemoryCopyEXT](#) structures specifying the regions to copy.

`vkCopyImageToMemoryEXT` does not check whether the device memory associated with `srcImage` is currently in use before performing the copy. The application **must** guarantee that any previously submitted command that writes to the copy regions has completed before the host performs the copy.

Copy regions for the image **must** be aligned to a multiple of the texel block extent in each dimension, except at the edges of the image, where region extents **must** match the edge of the image.

Valid Usage

- VUID-VkCopyImageToMemoryInfoEXT-srcImage-09109
If `srcImage` is sparse then all memory ranges accessed by the copy command **must** be bound as described in [Binding Resource Memory](#)
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-09111
`srcImage` **must** have been created with `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` set in `VkImageCreateInfo::usage`
- VUID-VkCopyImageToMemoryInfoEXT-imageOffset-09114
If `flags` contains `VK_HOST_IMAGE_COPY_MEMCPY_EXT`, the `x`, `y`, and `z` members of the `imageOffset` member of each element of `pRegions` **must** be 0
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-09115
If `flags` contains `VK_HOST_IMAGE_COPY_MEMCPY_EXT`, the `imageExtent` member of each element of `pRegions` **must** equal the extents of `srcImage` identified by `imageSubresource`
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-07966
If `srcImage` is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyImageToMemoryInfoEXT-imageSubresource-07967
The `imageSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-VkCopyImageToMemoryInfoEXT-imageSubresource-07968
If `imageSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `imageSubresource.baseArrayLayer` + `imageSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-VkCopyImageToMemoryInfoEXT-imageSubresource-07970
The image region specified by each element of `pRegions` **must** be contained within the specified `imageSubresource` of `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-imageSubresource-07971
For each element of `pRegions`, `imageOffset.x` and `(imageExtent.width + imageOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `imageSubresource` of `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-imageSubresource-07972
For each element of `pRegions`, `imageOffset.y` and `(imageExtent.height + imageOffset.y)`

must both be greater than or equal to 0 and less than or equal to the height of the specified `imageSubresource` of `srcImage`

- VUID-VkCopyImageToMemoryInfoEXT-srcImage-07973
`srcImage` **must** have a sample count equal to `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-07979
If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `imageOffset.y` **must** be 0 and `imageExtent.height` **must** be 1
- VUID-VkCopyImageToMemoryInfoEXT-imageOffset-09104
For each element of `pRegions`, `imageOffset.z` and `(imageExtent.depth + imageOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `imageSubresource` of `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-07980
If `srcImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `imageOffset.z` **must** be 0 and `imageExtent.depth` **must** be 1
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-07274
For each element of `pRegions`, `imageOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-07275
For each element of `pRegions`, `imageOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-07276
For each element of `pRegions`, `imageOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-00207
For each element of `pRegions`, if the sum of `imageOffset.x` and `extent.width` does not equal the width of the subresource specified by `imageSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-00208
For each element of `pRegions`, if the sum of `imageOffset.y` and `extent.height` does not equal the height of the subresource specified by `imageSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-00209
For each element of `pRegions`, if the sum of `imageOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-imageSubresource-09105
For each element of `pRegions`, `imageSubresource.aspectMask` **must** specify aspects present in `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-07981
If `srcImage` has a `multi-planar image format`, then for each element of `pRegions`, `imageSubresource.aspectMask` **must** be a single valid `multi-planar aspect mask` bit
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-07983

If `srcImage` is of type `VK_IMAGE_TYPE_3D`, for each element of `pRegions`, `imageSubresource.baseArrayLayer` **must** be 0 and `imageSubresource.layerCount` **must** be 1

- VUID-VkCopyImageToMemoryInfoEXT-memoryRowLength-09106
For each element of `pRegions`, `memoryRowLength` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-memoryImageHeight-09107
For each element of `pRegions`, `memoryImageHeight` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToMemoryInfoEXT-memoryRowLength-09108
For each element of `pRegions`, `memoryRowLength` divided by the `texel block extent width` and then multiplied by the texel block size of `srcImage` **must** be less than or equal to $2^{31}-1$
- VUID-VkCopyImageToMemoryInfoEXT-srcImageLayout-09064
`srcImageLayout` **must** specify the current layout of the image subresources of `srcImage` specified in `pRegions`
- VUID-VkCopyImageToMemoryInfoEXT-srcImageLayout-09065
`srcImageLayout` **must** be one of the image layouts returned in `VkPhysicalDeviceHostImageCopyPropertiesEXT::pCopySrcLayouts`
- VUID-VkCopyImageToMemoryInfoEXT-flags-09394
If `flags` includes `VK_HOST_IMAGE_COPY_MEMCPY_EXT`, for each region in `pRegions`, `memoryRowLength` and `memoryImageHeight` **must** both be 0

Valid Usage (Implicit)

- VUID-VkCopyImageToMemoryInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COPY_IMAGE_TO_MEMORY_INFO_EXT`
- VUID-VkCopyImageToMemoryInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCopyImageToMemoryInfoEXT-flags-parameter
`flags` **must** be a valid combination of `VkHostImageCopyFlagBitsEXT` values
- VUID-VkCopyImageToMemoryInfoEXT-srcImage-parameter
`srcImage` **must** be a valid `VkImage` handle
- VUID-VkCopyImageToMemoryInfoEXT-srcImageLayout-parameter
`srcImageLayout` **must** be a valid `VkImageLayout` value
- VUID-VkCopyImageToMemoryInfoEXT-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkImageToMemoryCopyEXT` structures
- VUID-VkCopyImageToMemoryInfoEXT-regionCount-arraylength
`regionCount` **must** be greater than 0

Each element of `VkCopyImageToMemoryInfoEXT::pRegions` is a structure defined as:

```
// Provided by VK_EXT_host_image_copy
typedef struct VkImageToMemoryCopyEXT {
    VkStructureType      sType;
    const void*          pNext;
    void*                pHostPointer;
    uint32_t             memoryRowLength;
    uint32_t             memoryImageHeight;
    VkImageSubresourceLayers imageSubresource;
    VkOffset3D           imageOffset;
    VkExtent3D           imageExtent;
} VkImageToMemoryCopyEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **pHostPointer** is the host memory address which is the destination of the copy.
- **memoryRowLength** and **memoryImageHeight** specify in texels a subregion of a larger two- or three-dimensional image in host memory, and control the addressing calculations. If either of these values is zero, that aspect of the host memory is considered to be tightly packed according to the **imageExtent**.
- **imageSubresource** is a [VkImageSubresourceLayers](#) used to specify the specific image subresources of the image used for the source or destination image data.
- **imageOffset** selects the initial **x**, **y**, **z** offsets in texels of the sub-region of the source image data.
- **imageExtent** is the size in texels of the image to copy in **width**, **height** and **depth**.

This structure is functionally similar to [VkBufferImageCopy2](#), except it defines host memory as the target of copy instead of a buffer. In particular, the same data packing rules and restrictions as that structure apply here as well.

Valid Usage

- VUID-VkImageToMemoryCopyEXT-pHostPointer-09066
pHostPointer **must** point to memory that is large enough to contain all memory locations that are accessed according to [Buffer and Image Addressing](#), for each element of **pRegions**
- VUID-VkImageToMemoryCopyEXT-pRegions-09067
The union of all source regions, and the union of all destination regions, specified by the elements of **pRegions**, **must** not overlap in memory
- VUID-VkImageToMemoryCopyEXT-memoryRowLength-09101
memoryRowLength **must** be 0, or greater than or equal to the **width** member of **imageExtent**
- VUID-VkImageToMemoryCopyEXT-memoryImageHeight-09102
memoryImageHeight **must** be 0, or greater than or equal to the **height** member of **imageExtent**
- VUID-VkImageToMemoryCopyEXT-aspectMask-09103
The **aspectMask** member of **imageSubresource** **must** only have a single bit set

- VUID-VkImageToMemoryCopyEXT-imageExtent-06659
`imageExtent.width` **must** not be 0
- VUID-VkImageToMemoryCopyEXT-imageExtent-06660
`imageExtent.height` **must** not be 0
- VUID-VkImageToMemoryCopyEXT-imageExtent-06661
`imageExtent.depth` **must** not be 0

Valid Usage (Implicit)

- VUID-VkImageToMemoryCopyEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_TO_MEMORY_COPY_EXT`
- VUID-VkImageToMemoryCopyEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkImageToMemoryCopyEXT-pHostPointer-parameter
`pHostPointer` **must** be a pointer value
- VUID-VkImageToMemoryCopyEXT-imageSubresource-parameter
`imageSubresource` **must** be a valid `VkImageSubresourceLayers` structure

Bits which **can** be set in `VkCopyMemoryToImageInfoEXT::flags`, `VkCopyImageToMemoryInfoEXT::flags`, and `VkCopyImageToImageInfoEXT::flags`, specifying additional copy parameters are:

```
// Provided by VK_EXT_host_image_copy
typedef enum VkHostImageCopyFlagBitsEXT {
    VK_HOST_IMAGE_COPY_MEMCPY_EXT = 0x00000001,
} VkHostImageCopyFlagBitsEXT;
```

- `VK_HOST_IMAGE_COPY_MEMCPY_EXT` specifies that no memory layout swizzling is to be applied during data copy. For copies between memory and images, this flag indicates that image data in host memory is swizzled in exactly the same way as the image data on the device. Using this flag indicates that the implementations **may** use a simple memory copy to transfer the data between the host memory and the device memory. The format of the swizzled data in host memory is platform dependent and is not defined in this specification.

```
// Provided by VK_EXT_host_image_copy
typedef VkFlags VkHostImageCopyFlagsEXT;
```

`VkHostImageCopyFlagsEXT` is a bitmask type for setting a mask of zero or more `VkHostImageCopyFlagBitsEXT`.

To copy data from an image object to another image object using the host, call:


```
// Provided by VK_EXT_host_image_copy
VkResult vkCopyImageToImageEXT(
    VkDevice device,
    const VkCopyImageToImageInfoEXT* pCopyImageToImageInfo);
```

- `device` is the device which owns `pCopyImageToImageInfo->srcImage` and `pCopyImageToImageInfo->dstImage`.
- `pCopyImageToImageInfo` is a pointer to a `VkCopyImageToImageInfoEXT` structure describing the copy parameters.

This command is functionally similar to `vkCmdCopyImage2KHR`, except it is executed on the host. The memory of `pCopyImageToImageInfo->srcImage` and `pCopyImageToImageInfo->dstImage` is accessed by the host as if `coherent`.

NOTE

If the device has written to the memory of `pCopyImageToImageInfo->srcImage`, it is not automatically made available to the host. Before this copy command can be called, a memory barrier for this image **must** have been issued on the device with the second `synchronization scope` including `VK_PIPELINE_STAGE_HOST_BIT` and `VK_ACCESS_HOST_READ_BIT`.

Because queue submissions `automatically make host memory visible to the device`, there would not be a need for a memory barrier before using the results of this copy operation in `pCopyMemoryToImageInfo->dstImage` on the device.

Valid Usage (Implicit)

- VUID-vkCopyImageToImageEXT-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkCopyImageToImageEXT-pCopyImageToImageInfo-parameter `pCopyImageToImageInfo` **must** be a valid pointer to a valid `VkCopyImageToImageInfoEXT` structure

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_INITIALIZATION_FAILED`
- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_MEMORY_MAP_FAILED`

The `VkCopyImageToImageInfoEXT` structure is defined as:

```
// Provided by VK_EXT_host_image_copy
typedef struct VkCopyImageToImageInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkHostImageCopyFlagsEXT flags;
    VkImage               srcImage;
    VkImageLayout         srcImageLayout;
    VkImage               dstImage;
    VkImageLayout         dstImageLayout;
    uint32_t              regionCount;
    const VkImageCopy2*   pRegions;
} VkCopyImageToImageInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkHostImageCopyFlagBitsEXT` values describing additional copy parameters.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the copy.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the copy.
- `regionCount` is the number of regions to copy.
- `pRegions` is a pointer to an array of `VkImageCopy2` structures specifying the regions to copy.

`vkCopyImageToImageEXT` does not check whether the device memory associated with `srcImage` or `dstImage` is currently in use before performing the copy. The application **must** guarantee that any previously submitted command that writes to the copy regions has completed before the host performs the copy.

Valid Usage

- VUID-VkCopyImageToImageInfoEXT-srcImage-09069
`srcImage` and `dstImage` **must** have been created with identical image creation parameters
- VUID-VkCopyImageToImageInfoEXT-srcImage-09109
If `srcImage` is sparse then all memory ranges accessed by the copy command **must** be bound as described in [Binding Resource Memory](#)
- VUID-VkCopyImageToImageInfoEXT-srcImage-09111
`srcImage` **must** have been created with `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` set in `VkImageCreateInfo::usage`
- VUID-VkCopyImageToImageInfoEXT-srcOffset-09114
If `flags` contains `VK_HOST_IMAGE_COPY_MEMCPY_EXT`, the `x`, `y`, and `z` members of the `srcOffset`

member of each element of `pRegions` **must** be 0

- VUID-VkCopyImageToImageInfoEXT-srcImage-09115

If `flags` contains `VK_HOST_IMAGE_COPY_MEMCPY_EXT`, the `extent` member of each element of `pRegions` **must** equal the extents of `srcImage` identified by `srcSubresource`

- VUID-VkCopyImageToImageInfoEXT-srcImage-07966

If `srcImage` is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-VkCopyImageToImageInfoEXT-srcSubresource-07967

The `srcSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `srcImage` was created

- VUID-VkCopyImageToImageInfoEXT-srcSubresource-07968

If `srcSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `srcSubresource.baseArrayLayer + srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `srcImage` was created

- VUID-VkCopyImageToImageInfoEXT-srcSubresource-07970

The image region specified by each element of `pRegions` **must** be contained within the specified `srcSubresource` of `srcImage`

- VUID-VkCopyImageToImageInfoEXT-srcSubresource-07971

For each element of `pRegions`, `srcOffset.x` and `(extent.width + srcOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `srcSubresource` of `srcImage`

- VUID-VkCopyImageToImageInfoEXT-srcSubresource-07972

For each element of `pRegions`, `srcOffset.y` and `(extent.height + srcOffset.y)` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `srcSubresource` of `srcImage`

- VUID-VkCopyImageToImageInfoEXT-srcImage-07979

If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffset.y` **must** be 0 and `extent.height` **must** be 1

- VUID-VkCopyImageToImageInfoEXT-srcOffset-09104

For each element of `pRegions`, `srcOffset.z` and `(extent.depth + srcOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `srcSubresource` of `srcImage`

- VUID-VkCopyImageToImageInfoEXT-srcImage-07980

If `srcImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `srcOffset.z` **must** be 0 and `extent.depth` **must** be 1

- VUID-VkCopyImageToImageInfoEXT-srcImage-07274

For each element of `pRegions`, `srcOffset.x` **must** be a multiple of the *texel block extent width* of the `VkFormat` of `srcImage`

- VUID-VkCopyImageToImageInfoEXT-srcImage-07275

For each element of `pRegions`, `srcOffset.y` **must** be a multiple of the *texel block extent height* of the `VkFormat` of `srcImage`

- VUID-VkCopyImageToImageInfoEXT-srcImage-07276
For each element of `pRegions`, `srcOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToImageInfoEXT-srcImage-00207
For each element of `pRegions`, if the sum of `srcOffset.x` and `extent.width` does not equal the width of the subresource specified by `srcSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToImageInfoEXT-srcImage-00208
For each element of `pRegions`, if the sum of `srcOffset.y` and `extent.height` does not equal the height of the subresource specified by `srcSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToImageInfoEXT-srcImage-00209
For each element of `pRegions`, if the sum of `srcOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `srcImage`
- VUID-VkCopyImageToImageInfoEXT-srcSubresource-09105
For each element of `pRegions`, `srcSubresource.aspectMask` **must** specify aspects present in `srcImage`
- VUID-VkCopyImageToImageInfoEXT-srcImage-07981
If `srcImage` has a `multi-planar image format`, then for each element of `pRegions`, `srcSubresource.aspectMask` **must** be a single valid `multi-planar aspect mask` bit
- VUID-VkCopyImageToImageInfoEXT-srcImage-07983
If `srcImage` is of type `VK_IMAGE_TYPE_3D`, for each element of `pRegions`, `srcSubresource.baseArrayLayer` **must** be 0 and `srcSubresource.layerCount` **must** be 1
- VUID-VkCopyImageToImageInfoEXT-dstImage-09109
If `dstImage` is sparse then all memory ranges accessed by the copy command **must** be bound as described in `Binding Resource Memory`
- VUID-VkCopyImageToImageInfoEXT-dstImage-09111
`dstImage` **must** have been created with `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` set in `VkImageCreateInfo::usage`
- VUID-VkCopyImageToImageInfoEXT-dstOffset-09114
If `flags` contains `VK_HOST_IMAGE_COPY_MEMCPY_EXT`, the `x`, `y`, and `z` members of the `dstOffset` member of each element of `pRegions` **must** be 0
- VUID-VkCopyImageToImageInfoEXT-dstImage-09115
If `flags` contains `VK_HOST_IMAGE_COPY_MEMCPY_EXT`, the `extent` member of each element of `pRegions` **must** equal the extents of `dstImage` identified by `dstSubresource`
- VUID-VkCopyImageToImageInfoEXT-dstImage-07966
If `dstImage` is non-sparse then the image or the specified *disjoint* plane **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkCopyImageToImageInfoEXT-dstSubresource-07967
The `dstSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created

- VUID-VkCopyImageToImageInfoEXT-dstSubresource-07968
If `dstSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `dstSubresource.baseArrayLayer + dstSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-VkCopyImageToImageInfoEXT-dstSubresource-07970
The image region specified by each element of `pRegions` **must** be contained within the specified `dstSubresource` of `dstImage`
- VUID-VkCopyImageToImageInfoEXT-dstSubresource-07971
For each element of `pRegions`, `dstOffset.x` and `(extent.width + dstOffset.x)` **must** both be greater than or equal to `0` and less than or equal to the width of the specified `dstSubresource` of `dstImage`
- VUID-VkCopyImageToImageInfoEXT-dstSubresource-07972
For each element of `pRegions`, `dstOffset.y` and `(extent.height + dstOffset.y)` **must** both be greater than or equal to `0` and less than or equal to the height of the specified `dstSubresource` of `dstImage`
- VUID-VkCopyImageToImageInfoEXT-dstImage-07979
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffset.y` **must** be `0` and `extent.height` **must** be `1`
- VUID-VkCopyImageToImageInfoEXT-dstOffset-09104
For each element of `pRegions`, `dstOffset.z` and `(extent.depth + dstOffset.z)` **must** both be greater than or equal to `0` and less than or equal to the depth of the specified `dstSubresource` of `dstImage`
- VUID-VkCopyImageToImageInfoEXT-dstImage-07980
If `dstImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `dstOffset.z` **must** be `0` and `extent.depth` **must** be `1`
- VUID-VkCopyImageToImageInfoEXT-dstImage-07274
For each element of `pRegions`, `dstOffset.x` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-VkCopyImageToImageInfoEXT-dstImage-07275
For each element of `pRegions`, `dstOffset.y` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`
- VUID-VkCopyImageToImageInfoEXT-dstImage-07276
For each element of `pRegions`, `dstOffset.z` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-VkCopyImageToImageInfoEXT-dstImage-00207
For each element of `pRegions`, if the sum of `dstOffset.x` and `extent.width` does not equal the width of the subresource specified by `dstSubresource`, `extent.width` **must** be a multiple of the `texel block extent width` of the `VkFormat` of `dstImage`
- VUID-VkCopyImageToImageInfoEXT-dstImage-00208
For each element of `pRegions`, if the sum of `dstOffset.y` and `extent.height` does not equal the height of the subresource specified by `dstSubresource`, `extent.height` **must** be a multiple of the `texel block extent height` of the `VkFormat` of `dstImage`

- VUID-VkCopyImageToImageInfoEXT-dstImage-00209
For each element of `pRegions`, if the sum of `dstOffset.z` and `extent.depth` does not equal the depth of the subresource specified by `srcSubresource`, `extent.depth` **must** be a multiple of the `texel block extent depth` of the `VkFormat` of `dstImage`
- VUID-VkCopyImageToImageInfoEXT-dstSubresource-09105
For each element of `pRegions`, `dstSubresource.aspectMask` **must** specify aspects present in `dstImage`
- VUID-VkCopyImageToImageInfoEXT-dstImage-07981
If `dstImage` has a `multi-planar image format`, then for each element of `pRegions`, `dstSubresource.aspectMask` **must** be a single valid `multi-planar aspect mask` bit
- VUID-VkCopyImageToImageInfoEXT-dstImage-07983
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, for each element of `pRegions`, `dstSubresource.baseArrayLayer` **must** be 0 and `dstSubresource.layerCount` **must** be 1
- VUID-VkCopyImageToImageInfoEXT-srcImageLayout-09070
`srcImageLayout` **must** specify the current layout of the image subresources of `srcImage` specified in `pRegions`
- VUID-VkCopyImageToImageInfoEXT-dstImageLayout-09071
`dstImageLayout` **must** specify the current layout of the image subresources of `dstImage` specified in `pRegions`
- VUID-VkCopyImageToImageInfoEXT-srcImageLayout-09072
`srcImageLayout` **must** be one of the image layouts returned in `VkPhysicalDeviceHostImageCopyPropertiesEXT::pCopySrcLayouts`
- VUID-VkCopyImageToImageInfoEXT-dstImageLayout-09073
`dstImageLayout` **must** be one of the image layouts returned in `VkPhysicalDeviceHostImageCopyPropertiesEXT::pCopyDstLayouts`

Valid Usage (Implicit)

- VUID-VkCopyImageToImageInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COPY_IMAGE_TO_IMAGE_INFO_EXT`
- VUID-VkCopyImageToImageInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCopyImageToImageInfoEXT-flags-parameter
`flags` **must** be a valid combination of `VkHostImageCopyFlagBitsEXT` values
- VUID-VkCopyImageToImageInfoEXT-srcImage-parameter
`srcImage` **must** be a valid `VkImage` handle
- VUID-VkCopyImageToImageInfoEXT-srcImageLayout-parameter
`srcImageLayout` **must** be a valid `VkImageLayout` value
- VUID-VkCopyImageToImageInfoEXT-dstImage-parameter
`dstImage` **must** be a valid `VkImage` handle
- VUID-VkCopyImageToImageInfoEXT-dstImageLayout-parameter
`dstImageLayout` **must** be a valid `VkImageLayout` value

- VUID-VkCopyImageToImageInfoEXT-pRegions-parameter
pRegions **must** be a valid pointer to an array of **regionCount** valid [VkImageCopy2](#) structures
- VUID-VkCopyImageToImageInfoEXT-regionCount-arraylength
regionCount **must** be greater than 0
- VUID-VkCopyImageToImageInfoEXT-commonparent
Both of **dstImage**, and **srcImage** **must** have been created, allocated, or retrieved from the same [VkDevice](#)

19.4. Image Copies With Scaling

To copy regions of a source image into a destination image, potentially performing format conversion, arbitrary scaling, and filtering, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBlitImage(
    VkCommandBuffer          commandBuffer,
    VkImage                  srcImage,
    VkImageLayout            srcImageLayout,
    VkImage                  dstImage,
    VkImageLayout            dstImageLayout,
    uint32_t                 regionCount,
    const VkImageBlit*       pRegions,
    VkFilter                 filter);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **srcImage** is the source image.
- **srcImageLayout** is the layout of the source image subresources for the blit.
- **dstImage** is the destination image.
- **dstImageLayout** is the layout of the destination image subresources for the blit.
- **regionCount** is the number of regions to blit.
- **pRegions** is a pointer to an array of [VkImageBlit](#) structures specifying the regions to blit.
- **filter** is a [VkFilter](#) specifying the filter to apply if the blits require scaling.

vkCmdBlitImage **must** not be used for multisampled source or destination images. Use [vkCmdResolveImage](#) for this purpose.

As the sizes of the source and destination extents **can** differ in any dimension, texels in the source extent are scaled and filtered to the destination extent. Scaling occurs via the following operations:

- For each destination texel, the integer coordinate of that texel is converted to an unnormalized texture coordinate, using the effective inverse of the equations described in [unnormalized to integer conversion](#):

$$u_{\text{base}} = i + \frac{1}{2}$$

$$v_{\text{base}} = j + \frac{1}{2}$$

$$w_{\text{base}} = k + \frac{1}{2}$$

- These base coordinates are then offset by the first destination offset:

$$u_{\text{offset}} = u_{\text{base}} - x_{\text{dst0}}$$

$$v_{\text{offset}} = v_{\text{base}} - y_{\text{dst0}}$$

$$w_{\text{offset}} = w_{\text{base}} - z_{\text{dst0}}$$

$$a_{\text{offset}} = a - \text{baseArrayCount}_{\text{dst}}$$

- The scale is determined from the source and destination regions, and applied to the offset coordinates:

$$\text{scale}_u = (x_{\text{src1}} - x_{\text{src0}}) / (x_{\text{dst1}} - x_{\text{dst0}})$$

$$\text{scale}_v = (y_{\text{src1}} - y_{\text{src0}}) / (y_{\text{dst1}} - y_{\text{dst0}})$$

$$\text{scale}_w = (z_{\text{src1}} - z_{\text{src0}}) / (z_{\text{dst1}} - z_{\text{dst0}})$$

$$u_{\text{scaled}} = u_{\text{offset}} \times \text{scale}_u$$

$$v_{\text{scaled}} = v_{\text{offset}} \times \text{scale}_v$$

$$w_{\text{scaled}} = w_{\text{offset}} \times \text{scale}_w$$

- Finally the source offset is added to the scaled coordinates, to determine the final unnormalized coordinates used to sample from `srcImage`:

$$u = u_{\text{scaled}} + x_{\text{src0}}$$

$$V = V_{\text{scaled}} + Y_{\text{src0}}$$

$$W = W_{\text{scaled}} + Z_{\text{src0}}$$

$$q = \text{mipLevel}$$

$$a = a_{\text{offset}} + \text{baseArrayCount}_{\text{src}}$$

These coordinates are used to sample from the source image, as described in [Image Operations chapter](#), with the filter mode equal to that of `filter`, a mipmap mode of `VK_SAMPLER_MIPMAP_MODE_NEAREST` and an address mode of `VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_EDGE`. Implementations **must** clamp at the edge of the source image, and **may** additionally clamp to the edge of the source region.

NOTE

Due to allowable rounding errors in the generation of the source texture coordinates, it is not always possible to guarantee exactly which source texels will be sampled for a given blit. As rounding errors are implementation-dependent, the exact results of a blitting operation are also implementation-dependent.

Blits are done layer by layer starting with the `baseArrayLayer` member of `srcSubresource` for the source and `dstSubresource` for the destination. `layerCount` layers are blitted to the destination image.

When blitting 3D textures, slices in the destination region bounded by `dstOffsets[0].z` and `dstOffsets[1].z` are sampled from slices in the source region bounded by `srcOffsets[0].z` and `srcOffsets[1].z`. If the `filter` parameter is `VK_FILTER_LINEAR` then the value sampled from the source image is taken by doing linear filtering using the interpolated `z` coordinate represented by `w` in the previous equations. If the `filter` parameter is `VK_FILTER_NEAREST` then the value sampled from the source image is taken from the single nearest slice, with an implementation-dependent arithmetic rounding mode.

The following filtering and conversion rules apply:

- Integer formats **can** only be converted to other integer formats with the same signedness.
- No format conversion is supported between depth/stencil images. The formats **must** match.
- Format conversions on unorm, snorm, scaled and packed float formats of the copied aspect of the image are performed by first converting the pixels to float values.
- For sRGB source formats, nonlinear RGB values are converted to linear representation prior to filtering.
- After filtering, the float values are first clamped and then cast to the destination image format. In case of sRGB destination format, linear RGB values are converted to nonlinear representation before writing the pixel to the image.

Signed and unsigned integers are converted by first clamping to the representable range of the

destination format, then casting the value.

Valid Usage

- VUID-vkCmdBlitImage-pRegions-00215
The source region specified by each element of **pRegions** **must** be a region that is contained within **srcImage**
- VUID-vkCmdBlitImage-pRegions-00216
The destination region specified by each element of **pRegions** **must** be a region that is contained within **dstImage**
- VUID-vkCmdBlitImage-pRegions-00217
The union of all destination regions, specified by the elements of **pRegions**, **must** not overlap in memory with any texel that **may** be sampled during the blit operation
- VUID-vkCmdBlitImage-srcImage-01999
The **format features** of **srcImage** **must** contain **VK_FORMAT_FEATURE_BLIT_SRC_BIT**
- VUID-vkCmdBlitImage-srcImage-06421
srcImage **must** not use a **format that requires a sampler Y'C_BC_R conversion**
- VUID-vkCmdBlitImage-srcImage-00219
srcImage **must** have been created with **VK_IMAGE_USAGE_TRANSFER_SRC_BIT** usage flag
- VUID-vkCmdBlitImage-srcImage-00220
If **srcImage** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-vkCmdBlitImage-srcImageLayout-00221
srcImageLayout **must** specify the layout of the image subresources of **srcImage** specified in **pRegions** at the time this command is executed on a **VkDevice**
- VUID-vkCmdBlitImage-srcImageLayout-01398
srcImageLayout **must** be **VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR**, **VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL** or **VK_IMAGE_LAYOUT_GENERAL**
- VUID-vkCmdBlitImage-srcImage-09459
If **srcImage** and **dstImage** are the same, and an elements of **pRegions** contains the **srcSubresource** and **dstSubresource** with matching **mipLevel** and overlapping array layers, then the **srcImageLayout** and **dstImageLayout** **must** be **VK_IMAGE_LAYOUT_GENERAL** or **VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR**
- VUID-vkCmdBlitImage-dstImage-02000
The **format features** of **dstImage** **must** contain **VK_FORMAT_FEATURE_BLIT_DST_BIT**
- VUID-vkCmdBlitImage-dstImage-06422
dstImage **must** not use a **format that requires a sampler Y'C_BC_R conversion**
- VUID-vkCmdBlitImage-dstImage-00224
dstImage **must** have been created with **VK_IMAGE_USAGE_TRANSFER_DST_BIT** usage flag
- VUID-vkCmdBlitImage-dstImage-00225
If **dstImage** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object

- VUID-vkCmdBlitImage-dstImageLayout-00226
`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdBlitImage-dstImageLayout-01399
`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdBlitImage-srcImage-00229
If either of `srcImage` or `dstImage` was created with a signed integer `VkFormat`, the other **must** also have been created with a signed integer `VkFormat`
- VUID-vkCmdBlitImage-srcImage-00230
If either of `srcImage` or `dstImage` was created with an unsigned integer `VkFormat`, the other **must** also have been created with an unsigned integer `VkFormat`
- VUID-vkCmdBlitImage-srcImage-00231
If either of `srcImage` or `dstImage` was created with a depth/stencil format, the other **must** have exactly the same format
- VUID-vkCmdBlitImage-srcImage-00232
If `srcImage` was created with a depth/stencil format, `filter` **must** be `VK_FILTER_NEAREST`
- VUID-vkCmdBlitImage-srcImage-00233
`srcImage` **must** have been created with a `samples` value of `VK_SAMPLE_COUNT_1_BIT`
- VUID-vkCmdBlitImage-dstImage-00234
`dstImage` **must** have been created with a `samples` value of `VK_SAMPLE_COUNT_1_BIT`
- VUID-vkCmdBlitImage-filter-02001
If `filter` is `VK_FILTER_LINEAR`, then the `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdBlitImage-filter-02002
If `filter` is `VK_FILTER_CUBIC_EXT`, then the `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`
- VUID-vkCmdBlitImage-filter-00237
If `filter` is `VK_FILTER_CUBIC_EXT`, `srcImage` **must** be of type `VK_IMAGE_TYPE_2D`
- VUID-vkCmdBlitImage-srcSubresource-01705
The `srcSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-vkCmdBlitImage-dstSubresource-01706
The `dstSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-vkCmdBlitImage-srcSubresource-01707
If `srcSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `srcSubresource.baseArrayLayer + srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-vkCmdBlitImage-dstSubresource-01708
If `srcSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`,

`dstSubresource.baseArrayLayer + dstSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created

- VUID-vkCmdBlitImage-srcImage-00240

If either `srcImage` or `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcSubresource.baseArrayLayer` and `dstSubresource.baseArrayLayer` **must** each be 0, and `srcSubresource.layerCount` and `dstSubresource.layerCount` **must** each be 1

- VUID-vkCmdBlitImage-aspectMask-00241

For each element of `pRegions`, `srcSubresource.aspectMask` **must** specify aspects present in `srcImage`

- VUID-vkCmdBlitImage-aspectMask-00242

For each element of `pRegions`, `dstSubresource.aspectMask` **must** specify aspects present in `dstImage`

- VUID-vkCmdBlitImage-srcOffset-00243

For each element of `pRegions`, `srcOffsets[0].x` and `srcOffsets[1].x` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `srcSubresource` of `srcImage`

- VUID-vkCmdBlitImage-srcOffset-00244

For each element of `pRegions`, `srcOffsets[0].y` and `srcOffsets[1].y` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `srcSubresource` of `srcImage`

- VUID-vkCmdBlitImage-srcImage-00245

If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffsets[0].y` **must** be 0 and `srcOffsets[1].y` **must** be 1

- VUID-vkCmdBlitImage-srcOffset-00246

For each element of `pRegions`, `srcOffsets[0].z` and `srcOffsets[1].z` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `srcSubresource` of `srcImage`

- VUID-vkCmdBlitImage-srcImage-00247

If `srcImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `srcOffsets[0].z` **must** be 0 and `srcOffsets[1].z` **must** be 1

- VUID-vkCmdBlitImage-dstOffset-00248

For each element of `pRegions`, `dstOffsets[0].x` and `dstOffsets[1].x` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `dstSubresource` of `dstImage`

- VUID-vkCmdBlitImage-dstOffset-00249

For each element of `pRegions`, `dstOffsets[0].y` and `dstOffsets[1].y` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `dstSubresource` of `dstImage`

- VUID-vkCmdBlitImage-dstImage-00250

If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffsets[0].y` **must** be 0 and `dstOffsets[1].y` **must** be 1

- VUID-vkCmdBlitImage-dstOffset-00251

For each element of `pRegions`, `dstOffsets[0].z` and `dstOffsets[1].z` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `dstSubresource` of `dstImage`

- VUID-vkCmdBlitImage-dstImage-00252

If `dstImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `dstOffsets[0].z` **must** be 0 and `dstOffsets[1].z` **must** be 1

Valid Usage (Implicit)

- VUID-vkCmdBlitImage-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBlitImage-srcImage-parameter
`srcImage` **must** be a valid `VkImage` handle
- VUID-vkCmdBlitImage-srcImageLayout-parameter
`srcImageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdBlitImage-dstImage-parameter
`dstImage` **must** be a valid `VkImage` handle
- VUID-vkCmdBlitImage-dstImageLayout-parameter
`dstImageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdBlitImage-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkImageBlit` structures
- VUID-vkCmdBlitImage-filter-parameter
`filter` **must** be a valid `VkFilter` value
- VUID-vkCmdBlitImage-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdBlitImage-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBlitImage-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdBlitImage-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBlitImage-regionCount-arraylength
`regionCount` **must** be greater than 0
- VUID-vkCmdBlitImage-commonparent
Each of `commandBuffer`, `dstImage`, and `srcImage` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Graphics	Action

The **VkImageBlit** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageBlit {
    VkImageSubresourceLayers    srcSubresource;
    VkOffset3D                  srcOffsets[2];
    VkImageSubresourceLayers    dstSubresource;
    VkOffset3D                  dstOffsets[2];
} VkImageBlit;
```

- **srcSubresource** is the subresource to blit from.
- **srcOffsets** is a pointer to an array of two **VkOffset3D** structures specifying the bounds of the source region within **srcSubresource**.
- **dstSubresource** is the subresource to blit into.
- **dstOffsets** is a pointer to an array of two **VkOffset3D** structures specifying the bounds of the destination region within **dstSubresource**.

For each element of the **pRegions** array, a blit operation is performed for the specified source and destination regions.

Valid Usage

- VUID-VkImageBlit-aspectMask-00238
The **aspectMask** member of **srcSubresource** and **dstSubresource** **must** match
- VUID-VkImageBlit-layerCount-08800
If neither of the **layerCount** members of **srcSubresource** or **dstSubresource** are **VK_REMAINING_ARRAY_LAYERS**, the **layerCount** members of **srcSubresource** or **dstSubresource** **must** match
- VUID-VkImageBlit-layerCount-08801

If one of the `layerCount` members of `srcSubresource` or `dstSubresource` is `VK_REMAINING_ARRAY_LAYERS`, the other member **must** be either `VK_REMAINING_ARRAY_LAYERS` or equal to the `arrayLayers` member of the `VkImageCreateInfo` used to create the image minus `baseArrayLayer`

Valid Usage (Implicit)

- VUID-VkImageBlit-srcSubresource-parameter
`srcSubresource` **must** be a valid `VkImageSubresourceLayers` structure
- VUID-VkImageBlit-dstSubresource-parameter
`dstSubresource` **must** be a valid `VkImageSubresourceLayers` structure

A more extensible version of the blit image command is defined below.

To copy regions of a source image into a destination image, potentially performing format conversion, arbitrary scaling, and filtering, call:

```
// Provided by VK_KHR_copy_commands2
void vkCmdBlitImage2KHR(
    VkCommandBuffer          commandBuffer,
    const VkBlitImageInfo2* pBlitImageInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pBlitImageInfo` is a pointer to a `VkBlitImageInfo2` structure describing the blit parameters.

This command is functionally identical to `vkCmdBlitImage`, but includes extensible sub-structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage

Valid Usage (Implicit)

- VUID-vkCmdBlitImage2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBlitImage2-pBlitImageInfo-parameter
`pBlitImageInfo` **must** be a valid pointer to a valid `VkBlitImageInfo2` structure
- VUID-vkCmdBlitImage2-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdBlitImage2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdBlitImage2-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdBlitImage2-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Graphics	Action

The `VkBlitImageInfo2` structure is defined as:

```
typedef struct VkBlitImageInfo2 {
    VkStructureType    sType;
    const void*        pNext;
    VkImage             srcImage;
    VkImageLayout       srcImageLayout;
    VkImage             dstImage;
    VkImageLayout       dstImageLayout;
    uint32_t           regionCount;
    const VkImageBlit2* pRegions;
    VkFilter            filter;
} VkBlitImageInfo2;
```

or the equivalent

```
// Provided by VK_KHR_copy_commands2
typedef VkBlitImageInfo2 VkBlitImageInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the blit.

- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the blit.
- `regionCount` is the number of regions to blit.
- `pRegions` is a pointer to an array of `VkImageBlit2` structures specifying the regions to blit.
- `filter` is a `VkFilter` specifying the filter to apply if the blits require scaling.

Valid Usage

- VUID-VkBlitImageInfo2-pRegions-00215
The source region specified by each element of `pRegions` **must** be a region that is contained within `srcImage`
- VUID-VkBlitImageInfo2-pRegions-00216
The destination region specified by each element of `pRegions` **must** be a region that is contained within `dstImage`
- VUID-VkBlitImageInfo2-pRegions-00217
The union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory with any texel that **may** be sampled during the blit operation
- VUID-VkBlitImageInfo2-srcImage-01999
The `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_BLIT_SRC_BIT`
- VUID-VkBlitImageInfo2-srcImage-06421
`srcImage` **must** not use a `format that requires a sampler Y'CBCR conversion`
- VUID-VkBlitImageInfo2-srcImage-00219
`srcImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` usage flag
- VUID-VkBlitImageInfo2-srcImage-00220
If `srcImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkBlitImageInfo2-srcImageLayout-00221
`srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-VkBlitImageInfo2-srcImageLayout-01398
`srcImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-VkBlitImageInfo2-srcImage-09459
If `srcImage` and `dstImage` are the same, and an elements of `pRegions` contains the `srcSubresource` and `dstSubresource` with matching `mipLevel` and overlapping array layers, then the `srcImageLayout` and `dstImageLayout` **must** be `VK_IMAGE_LAYOUT_GENERAL` or `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`
- VUID-VkBlitImageInfo2-dstImage-02000
The `format features` of `dstImage` **must** contain `VK_FORMAT_FEATURE_BLIT_DST_BIT`
- VUID-VkBlitImageInfo2-dstImage-06422
`dstImage` **must** not use a `format that requires a sampler Y'CBCR conversion`

- VUID-VkBlitImageInfo2-dstImage-00224
`dstImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-VkBlitImageInfo2-dstImage-00225
If `dstImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkBlitImageInfo2-dstImageLayout-00226
`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-VkBlitImageInfo2-dstImageLayout-01399
`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-VkBlitImageInfo2-srcImage-00229
If either of `srcImage` or `dstImage` was created with a signed integer `VkFormat`, the other **must** also have been created with a signed integer `VkFormat`
- VUID-VkBlitImageInfo2-srcImage-00230
If either of `srcImage` or `dstImage` was created with an unsigned integer `VkFormat`, the other **must** also have been created with an unsigned integer `VkFormat`
- VUID-VkBlitImageInfo2-srcImage-00231
If either of `srcImage` or `dstImage` was created with a depth/stencil format, the other **must** have exactly the same format
- VUID-VkBlitImageInfo2-srcImage-00232
If `srcImage` was created with a depth/stencil format, `filter` **must** be `VK_FILTER_NEAREST`
- VUID-VkBlitImageInfo2-srcImage-00233
`srcImage` **must** have been created with a `samples` value of `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkBlitImageInfo2-dstImage-00234
`dstImage` **must** have been created with a `samples` value of `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkBlitImageInfo2-filter-02001
If `filter` is `VK_FILTER_LINEAR`, then the `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-VkBlitImageInfo2-filter-02002
If `filter` is `VK_FILTER_CUBIC_EXT`, then the `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`
- VUID-VkBlitImageInfo2-filter-00237
If `filter` is `VK_FILTER_CUBIC_EXT`, `srcImage` **must** be of type `VK_IMAGE_TYPE_2D`
- VUID-VkBlitImageInfo2-srcSubresource-01705
The `srcSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-VkBlitImageInfo2-dstSubresource-01706
The `dstSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-VkBlitImageInfo2-srcSubresource-01707
If `srcSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`,

`srcSubresource.baseArrayLayer + srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `srcImage` was created

- VUID-VkBlitImageInfo2-dstSubresource-01708

If `srcSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `dstSubresource.baseArrayLayer + dstSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created

- VUID-VkBlitImageInfo2-srcImage-00240

If either `srcImage` or `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcSubresource.baseArrayLayer` and `dstSubresource.baseArrayLayer` **must** each be 0, and `srcSubresource.layerCount` and `dstSubresource.layerCount` **must** each be 1

- VUID-VkBlitImageInfo2-aspectMask-00241

For each element of `pRegions`, `srcSubresource.aspectMask` **must** specify aspects present in `srcImage`

- VUID-VkBlitImageInfo2-aspectMask-00242

For each element of `pRegions`, `dstSubresource.aspectMask` **must** specify aspects present in `dstImage`

- VUID-VkBlitImageInfo2-srcOffset-00243

For each element of `pRegions`, `srcOffsets[0].x` and `srcOffsets[1].x` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `srcSubresource` of `srcImage`

- VUID-VkBlitImageInfo2-srcOffset-00244

For each element of `pRegions`, `srcOffsets[0].y` and `srcOffsets[1].y` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `srcSubresource` of `srcImage`

- VUID-VkBlitImageInfo2-srcImage-00245

If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffsets[0].y` **must** be 0 and `srcOffsets[1].y` **must** be 1

- VUID-VkBlitImageInfo2-srcOffset-00246

For each element of `pRegions`, `srcOffsets[0].z` and `srcOffsets[1].z` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `srcSubresource` of `srcImage`

- VUID-VkBlitImageInfo2-srcImage-00247

If `srcImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `srcOffsets[0].z` **must** be 0 and `srcOffsets[1].z` **must** be 1

- VUID-VkBlitImageInfo2-dstOffset-00248

For each element of `pRegions`, `dstOffsets[0].x` and `dstOffsets[1].x` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `dstSubresource` of `dstImage`

- VUID-VkBlitImageInfo2-dstOffset-00249

For each element of `pRegions`, `dstOffsets[0].y` and `dstOffsets[1].y` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `dstSubresource` of `dstImage`

`dstImage`

- VUID-VkBlitImageInfo2-dstImage-00250
If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffsets[0].y` **must** be 0 and `dstOffsets[1].y` **must** be 1
- VUID-VkBlitImageInfo2-dstOffset-00251
For each element of `pRegions`, `dstOffsets[0].z` and `dstOffsets[1].z` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `dstSubresource` of `dstImage`
- VUID-VkBlitImageInfo2-dstImage-00252
If `dstImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `dstOffsets[0].z` **must** be 0 and `dstOffsets[1].z` **must** be 1

Valid Usage (Implicit)

- VUID-VkBlitImageInfo2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_BLIT_IMAGE_INFO_2`
- VUID-VkBlitImageInfo2-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkBlitImageInfo2-srcImage-parameter
`srcImage` **must** be a valid `VkImage` handle
- VUID-VkBlitImageInfo2-srcImageLayout-parameter
`srcImageLayout` **must** be a valid `VkImageLayout` value
- VUID-VkBlitImageInfo2-dstImage-parameter
`dstImage` **must** be a valid `VkImage` handle
- VUID-VkBlitImageInfo2-dstImageLayout-parameter
`dstImageLayout` **must** be a valid `VkImageLayout` value
- VUID-VkBlitImageInfo2-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkImageBlit2` structures
- VUID-VkBlitImageInfo2-filter-parameter
`filter` **must** be a valid `VkFilter` value
- VUID-VkBlitImageInfo2-regionCount-arraylength
`regionCount` **must** be greater than 0
- VUID-VkBlitImageInfo2-commonparent
Both of `dstImage`, and `srcImage` **must** have been created, allocated, or retrieved from the same `VkDevice`

The `VkImageBlit2` structure is defined as:

```
typedef struct VkImageBlit2 {
    VkStructureType    sType;
    const void*        pNext;
    VkImageSubresourceLayers srcSubresource;
    VkOffset3D          srcOffsets[2];
    VkImageSubresourceLayers dstSubresource;
    VkOffset3D          dstOffsets[2];
} VkImageBlit2;
```

or the equivalent

```
// Provided by VK_KHR_copy_commands2
typedef VkImageBlit2 VkImageBlit2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcSubresource` is the subresource to blit from.
- `srcOffsets` is a pointer to an array of two `VkOffset3D` structures specifying the bounds of the source region within `srcSubresource`.
- `dstSubresource` is the subresource to blit into.
- `dstOffsets` is a pointer to an array of two `VkOffset3D` structures specifying the bounds of the destination region within `dstSubresource`.

For each element of the `pRegions` array, a blit operation is performed for the specified source and destination regions.

Valid Usage

- VUID-VkImageBlit2-aspectMask-00238
The `aspectMask` member of `srcSubresource` and `dstSubresource` **must** match
- VUID-VkImageBlit2-layerCount-08800
If neither of the `layerCount` members of `srcSubresource` or `dstSubresource` are `VK_REMAINING_ARRAY_LAYERS`, the `layerCount` members of `srcSubresource` or `dstSubresource` **must** match
- VUID-VkImageBlit2-layerCount-08801
If one of the `layerCount` members of `srcSubresource` or `dstSubresource` is `VK_REMAINING_ARRAY_LAYERS`, the other member **must** be either `VK_REMAINING_ARRAY_LAYERS` or equal to the `arrayLayers` member of the `VkImageCreateInfo` used to create the image minus `baseArrayLayer`

Valid Usage (Implicit)

- VUID-VkImageBlit2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_BLIT_2`
- VUID-VkImageBlit2-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkImageBlit2-srcSubresource-parameter
`srcSubresource` **must** be a valid `VkImageSubresourceLayers` structure
- VUID-VkImageBlit2-dstSubresource-parameter
`dstSubresource` **must** be a valid `VkImageSubresourceLayers` structure

19.5. Resolving Multisample Images

To resolve a multisample color image to a non-multisample color image, call:

```
// Provided by VK_VERSION_1_0
void vkCmdResolveImage(
    VkCommandBuffer          commandBuffer,
    VkImage                  srcImage,
    VkImageLayout            srcImageLayout,
    VkImage                  dstImage,
    VkImageLayout            dstImageLayout,
    uint32_t                 regionCount,
    const VkImageResolve*    pRegions);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the resolve.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the resolve.
- `regionCount` is the number of regions to resolve.
- `pRegions` is a pointer to an array of `VkImageResolve` structures specifying the regions to resolve.

During the resolve the samples corresponding to each pixel location in the source are converted to a single sample before being written to the destination. If the source formats are floating-point or normalized types, the sample values for each pixel are resolved in an implementation-dependent manner. If the source formats are integer types, a single sample's value is selected for each pixel.

`srcOffset` and `dstOffset` select the initial `x`, `y`, and `z` offsets in texels of the sub-regions of the source and destination image data. `extent` is the size in texels of the source image to resolve in `width`, `height` and `depth`. Each element of `pRegions` **must** be a region that is contained within its corresponding image.

Resolves are done layer by layer starting with `baseArrayLayer` member of `srcSubresource` for the source and `dstSubresource` for the destination. `layerCount` layers are resolved to the destination image.

Valid Usage

- VUID-vkCmdResolveImage-pRegions-00255
The union of all source regions, and the union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory
- VUID-vkCmdResolveImage-srcImage-00256
If `srcImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdResolveImage-srcImage-00257
`srcImage` **must** have a sample count equal to any valid sample count value other than `VK_SAMPLE_COUNT_1_BIT`
- VUID-vkCmdResolveImage-dstImage-00258
If `dstImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdResolveImage-dstImage-00259
`dstImage` **must** have a sample count equal to `VK_SAMPLE_COUNT_1_BIT`
- VUID-vkCmdResolveImage-srcImageLayout-00260
`srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdResolveImage-srcImageLayout-01400
`srcImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdResolveImage-dstImageLayout-00262
`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-vkCmdResolveImage-dstImageLayout-01401
`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-vkCmdResolveImage-dstImage-02003
The `format features` of `dstImage` **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT`
- VUID-vkCmdResolveImage-srcImage-01386
`srcImage` and `dstImage` **must** have been created with the same image format
- VUID-vkCmdResolveImage-srcSubresource-01709
The `srcSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-vkCmdResolveImage-dstSubresource-01710
The `dstSubresource.mipLevel` member of each element of `pRegions` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `dstImage` was created

- VUID-vkCmdResolveImage-srcSubresource-01711
If `srcSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `srcSubresource.baseArrayLayer + srcSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `srcImage` was created
- VUID-vkCmdResolveImage-dstSubresource-01712
If `dstSubresource.layerCount` is not `VK_REMAINING_ARRAY_LAYERS`, `dstSubresource.baseArrayLayer + dstSubresource.layerCount` of each element of `pRegions` **must** be less than or equal to the `arrayLayers` specified in `VkImageCreateInfo` when `dstImage` was created
- VUID-vkCmdResolveImage-srcImage-04446
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `srcSubresource.layerCount` **must** be 1
- VUID-vkCmdResolveImage-srcImage-04447
If `dstImage` is of type `VK_IMAGE_TYPE_3D`, then for each element of `pRegions`, `dstSubresource.baseArrayLayer` **must** be 0 and `dstSubresource.layerCount` **must** be 1
- VUID-vkCmdResolveImage-srcOffset-00269
For each element of `pRegions`, `srcOffset.x` and `(extent.width + srcOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `srcSubresource` of `srcImage`
- VUID-vkCmdResolveImage-srcOffset-00270
For each element of `pRegions`, `srcOffset.y` and `(extent.height + srcOffset.y)` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `srcSubresource` of `srcImage`
- VUID-vkCmdResolveImage-srcImage-00271
If `srcImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `srcOffset.y` **must** be 0 and `extent.height` **must** be 1
- VUID-vkCmdResolveImage-srcOffset-00272
For each element of `pRegions`, `srcOffset.z` and `(extent.depth + srcOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `srcSubresource` of `srcImage`
- VUID-vkCmdResolveImage-srcImage-00273
If `srcImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `srcOffset.z` **must** be 0 and `extent.depth` **must** be 1
- VUID-vkCmdResolveImage-dstOffset-00274
For each element of `pRegions`, `dstOffset.x` and `(extent.width + dstOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdResolveImage-dstOffset-00275
For each element of `pRegions`, `dstOffset.y` and `(extent.height + dstOffset.y)` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdResolveImage-dstImage-00276

If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffset.y` **must** be 0 and `extent.height` **must** be 1

- VUID-vkCmdResolveImage-dstOffset-00277
For each element of `pRegions`, `dstOffset.z` and `(extent.depth + dstOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `dstSubresource` of `dstImage`
- VUID-vkCmdResolveImage-dstImage-00278
If `dstImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `dstOffset.z` **must** be 0 and `extent.depth` **must** be 1
- VUID-vkCmdResolveImage-srcImage-06762
`srcImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` usage flag
- VUID-vkCmdResolveImage-srcImage-06763
The `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_TRANSFER_SRC_BIT`
- VUID-vkCmdResolveImage-dstImage-06764
`dstImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT` usage flag
- VUID-vkCmdResolveImage-dstImage-06765
The `format features` of `dstImage` **must** contain `VK_FORMAT_FEATURE_TRANSFER_DST_BIT`

Valid Usage (Implicit)

- VUID-vkCmdResolveImage-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdResolveImage-srcImage-parameter
`srcImage` **must** be a valid `VkImage` handle
- VUID-vkCmdResolveImage-srcImageLayout-parameter
`srcImageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdResolveImage-dstImage-parameter
`dstImage` **must** be a valid `VkImage` handle
- VUID-vkCmdResolveImage-dstImageLayout-parameter
`dstImageLayout` **must** be a valid `VkImageLayout` value
- VUID-vkCmdResolveImage-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkImageResolve` structures
- VUID-vkCmdResolveImage-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdResolveImage-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdResolveImage-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdResolveImage-encoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdResolveImage-regionCount-arraylength
regionCount **must** be greater than 0
- VUID-vkCmdResolveImage-commonparent

Each of **commandBuffer**, **dstImage**, and **srcImage** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Graphics	Action

The **VkImageResolve** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageResolve {
    VkImageSubresourceLayers  srcSubresource;
    VkOffset3D                srcOffset;
    VkImageSubresourceLayers  dstSubresource;
    VkOffset3D                dstOffset;
    VkExtent3D                extent;
} VkImageResolve;
```

- **srcSubresource** and **dstSubresource** are **VkImageSubresourceLayers** structures specifying the image subresources of the images used for the source and destination image data, respectively. Resolve of depth/stencil images is not supported.
- **srcOffset** and **dstOffset** select the initial **x**, **y**, and **z** offsets in texels of the sub-regions of the source and destination image data.
- **extent** is the size in texels of the source image to resolve in **width**, **height** and **depth**.

Valid Usage

- VUID-VkImageResolve-aspectMask-00266

The `aspectMask` member of `srcSubresource` and `dstSubresource` **must** only contain `VK_IMAGE_ASPECT_COLOR_BIT`

- VUID-VkImageResolve-layerCount-08803

If neither of the `layerCount` members of `srcSubresource` or `dstSubresource` are `VK_REMAINING_ARRAY_LAYERS`, the `layerCount` member of `srcSubresource` and `dstSubresource` **must** match

- VUID-VkImageResolve-layerCount-08804

If one of the `layerCount` members of `srcSubresource` or `dstSubresource` is `VK_REMAINING_ARRAY_LAYERS`, the other member **must** be either `VK_REMAINING_ARRAY_LAYERS` or equal to the `arrayLayers` member of the `VkImageCreateInfo` used to create the image minus `baseArrayLayer`

Valid Usage (Implicit)

- VUID-VkImageResolve-srcSubresource-parameter
`srcSubresource` **must** be a valid `VkImageSubresourceLayers` structure
- VUID-VkImageResolve-dstSubresource-parameter
`dstSubresource` **must** be a valid `VkImageSubresourceLayers` structure

A more extensible version of the resolve image command is defined below.

To resolve a multisample image to a non-multisample image, call:

```
// Provided by VK_KHR_copy_commands2
void vkCmdResolveImage2KHR(
    VkCommandBuffer                      commandBuffer,
    const VkResolveImageInfo2*          pResolveImageInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pResolveImageInfo` is a pointer to a `VkResolveImageInfo2` structure describing the resolve parameters.

This command is functionally identical to `vkCmdResolveImage`, but includes extensible sub-structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage

Valid Usage (Implicit)

- VUID-vkCmdResolveImage2-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdResolveImage2-pResolveImageInfo-parameter
`pResolveImageInfo` **must** be a valid pointer to a valid `VkResolveImageInfo2` structure
- VUID-vkCmdResolveImage2-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdResolveImage2-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdResolveImage2-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdResolveImage2-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Graphics	Action

The `VkResolveImageInfo2` structure is defined as:

```
typedef struct VkResolveImageInfo2 {
    VkStructureType    sType;
    const void*        pNext;
    VkImage             srcImage;
    VkImageLayout       srcImageLayout;
    VkImage             dstImage;
    VkImageLayout       dstImageLayout;
    uint32_t            regionCount;
    const VkImageResolve2* pRegions;
} VkResolveImageInfo2;
```

or the equivalent


```
// Provided by VK_KHR_copy_commands2
typedef VkResolveImageInfo2 VkResolveImageInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcImage` is the source image.
- `srcImageLayout` is the layout of the source image subresources for the resolve.
- `dstImage` is the destination image.
- `dstImageLayout` is the layout of the destination image subresources for the resolve.
- `regionCount` is the number of regions to resolve.
- `pRegions` is a pointer to an array of `VkImageResolve2` structures specifying the regions to resolve.

Valid Usage

- VUID-VkResolveImageInfo2-pRegions-00255
The union of all source regions, and the union of all destination regions, specified by the elements of `pRegions`, **must** not overlap in memory
- VUID-VkResolveImageInfo2-srcImage-00256
If `srcImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkResolveImageInfo2-srcImage-00257
`srcImage` **must** have a sample count equal to any valid sample count value other than `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkResolveImageInfo2-dstImage-00258
If `dstImage` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkResolveImageInfo2-dstImage-00259
`dstImage` **must** have a sample count equal to `VK_SAMPLE_COUNT_1_BIT`
- VUID-VkResolveImageInfo2-srcImageLayout-00260
`srcImageLayout` **must** specify the layout of the image subresources of `srcImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-VkResolveImageInfo2-srcImageLayout-01400
`srcImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_SRC_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`
- VUID-VkResolveImageInfo2-dstImageLayout-00262
`dstImageLayout` **must** specify the layout of the image subresources of `dstImage` specified in `pRegions` at the time this command is executed on a `VkDevice`
- VUID-VkResolveImageInfo2-dstImageLayout-01401
`dstImageLayout` **must** be `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`, `VK_IMAGE_LAYOUT_TRANSFER_DST_OPTIMAL` or `VK_IMAGE_LAYOUT_GENERAL`

- VUID-VkResolveImageInfo2-dstImage-02003
The **format features** of **dstImage** **must** contain **VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT**
- VUID-VkResolveImageInfo2-srcImage-01386
srcImage and **dstImage** **must** have been created with the same image format
- VUID-VkResolveImageInfo2-srcSubresource-01709
The **srcSubresource.mipLevel** member of each element of **pRegions** **must** be less than the **mipLevels** specified in **VkImageCreateInfo** when **srcImage** was created
- VUID-VkResolveImageInfo2-dstSubresource-01710
The **dstSubresource.mipLevel** member of each element of **pRegions** **must** be less than the **mipLevels** specified in **VkImageCreateInfo** when **dstImage** was created
- VUID-VkResolveImageInfo2-srcSubresource-01711
If **srcSubresource.layerCount** is not **VK_REMAINING_ARRAY_LAYERS**, **srcSubresource.baseArrayLayer** + **srcSubresource.layerCount** of each element of **pRegions** **must** be less than or equal to the **arrayLayers** specified in **VkImageCreateInfo** when **srcImage** was created
- VUID-VkResolveImageInfo2-dstSubresource-01712
If **dstSubresource.layerCount** is not **VK_REMAINING_ARRAY_LAYERS**, **dstSubresource.baseArrayLayer** + **dstSubresource.layerCount** of each element of **pRegions** **must** be less than or equal to the **arrayLayers** specified in **VkImageCreateInfo** when **dstImage** was created
- VUID-VkResolveImageInfo2-srcImage-04446
If **dstImage** is of type **VK_IMAGE_TYPE_3D**, then for each element of **pRegions**, **srcSubresource.layerCount** **must** be 1
- VUID-VkResolveImageInfo2-srcImage-04447
If **dstImage** is of type **VK_IMAGE_TYPE_3D**, then for each element of **pRegions**, **dstSubresource.baseArrayLayer** **must** be 0 and **dstSubresource.layerCount** **must** be 1
- VUID-VkResolveImageInfo2-srcOffset-00269
For each element of **pRegions**, **srcOffset.x** and **(extent.width + srcOffset.x)** **must** both be greater than or equal to 0 and less than or equal to the width of the specified **srcSubresource** of **srcImage**
- VUID-VkResolveImageInfo2-srcOffset-00270
For each element of **pRegions**, **srcOffset.y** and **(extent.height + srcOffset.y)** **must** both be greater than or equal to 0 and less than or equal to the height of the specified **srcSubresource** of **srcImage**
- VUID-VkResolveImageInfo2-srcImage-00271
If **srcImage** is of type **VK_IMAGE_TYPE_1D**, then for each element of **pRegions**, **srcOffset.y** **must** be 0 and **extent.height** **must** be 1
- VUID-VkResolveImageInfo2-srcOffset-00272
For each element of **pRegions**, **srcOffset.z** and **(extent.depth + srcOffset.z)** **must** both be greater than or equal to 0 and less than or equal to the depth of the specified **srcSubresource** of **srcImage**
- VUID-VkResolveImageInfo2-srcImage-00273
If **srcImage** is of type **VK_IMAGE_TYPE_1D** or **VK_IMAGE_TYPE_2D**, then for each element of

`pRegions`, `srcOffset.z` **must** be 0 and `extent.depth` **must** be 1

- VUID-VkResolveImageInfo2-dstOffset-00274

For each element of `pRegions`, `dstOffset.x` and `(extent.width + dstOffset.x)` **must** both be greater than or equal to 0 and less than or equal to the width of the specified `dstSubresource` of `dstImage`

- VUID-VkResolveImageInfo2-dstOffset-00275

For each element of `pRegions`, `dstOffset.y` and `(extent.height + dstOffset.y)` **must** both be greater than or equal to 0 and less than or equal to the height of the specified `dstSubresource` of `dstImage`

- VUID-VkResolveImageInfo2-dstImage-00276

If `dstImage` is of type `VK_IMAGE_TYPE_1D`, then for each element of `pRegions`, `dstOffset.y` **must** be 0 and `extent.height` **must** be 1

- VUID-VkResolveImageInfo2-dstOffset-00277

For each element of `pRegions`, `dstOffset.z` and `(extent.depth + dstOffset.z)` **must** both be greater than or equal to 0 and less than or equal to the depth of the specified `dstSubresource` of `dstImage`

- VUID-VkResolveImageInfo2-dstImage-00278

If `dstImage` is of type `VK_IMAGE_TYPE_1D` or `VK_IMAGE_TYPE_2D`, then for each element of `pRegions`, `dstOffset.z` **must** be 0 and `extent.depth` **must** be 1

- VUID-VkResolveImageInfo2-srcImage-06762

`srcImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` usage flag

- VUID-VkResolveImageInfo2-srcImage-06763

The `format features` of `srcImage` **must** contain `VK_FORMAT_FEATURE_TRANSFER_SRC_BIT`

- VUID-VkResolveImageInfo2-dstImage-06764

`dstImage` **must** have been created with `VK_IMAGE_USAGE_TRANSFER_DST_BIT` usage flag

- VUID-VkResolveImageInfo2-dstImage-06765

The `format features` of `dstImage` **must** contain `VK_FORMAT_FEATURE_TRANSFER_DST_BIT`

Valid Usage (Implicit)

- VUID-VkResolveImageInfo2-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_RESOLVE_IMAGE_INFO_2`

- VUID-VkResolveImageInfo2-pNext-pNext

`pNext` **must** be `NULL`

- VUID-VkResolveImageInfo2-srcImage-parameter

`srcImage` **must** be a valid `VkImage` handle

- VUID-VkResolveImageInfo2-srcImageLayout-parameter

`srcImageLayout` **must** be a valid `VkImageLayout` value

- VUID-VkResolveImageInfo2-dstImage-parameter

`dstImage` **must** be a valid `VkImage` handle

- VUID-VkResolveImageInfo2-dstImageLayout-parameter

`dstImageLayout` **must** be a valid `VkImageLayout` value

- VUID-VkResolveImageInfo2-pRegions-parameter
`pRegions` **must** be a valid pointer to an array of `regionCount` valid `VkImageResolve2` structures
- VUID-VkResolveImageInfo2-regionCount-arraylength
`regionCount` **must** be greater than 0
- VUID-VkResolveImageInfo2-commonparent
Both of `dstImage`, and `srcImage` **must** have been created, allocated, or retrieved from the same `VkDevice`

The `VkImageResolve2` structure is defined as:

```
typedef struct VkImageResolve2 {  
    VkStructureType      sType;  
    const void*          pNext;  
    VkImageSubresourceLayers srcSubresource;  
    VkOffset3D           srcOffset;  
    VkImageSubresourceLayers dstSubresource;  
    VkOffset3D           dstOffset;  
    VkExtent3D           extent;  
} VkImageResolve2;
```

or the equivalent

```
// Provided by VK_KHR_copy_commands2  
typedef VkImageResolve2 VkImageResolve2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcSubresource` and `dstSubresource` are `VkImageSubresourceLayers` structures specifying the image subresources of the images used for the source and destination image data, respectively. Resolve of depth/stencil images is not supported.
- `srcOffset` and `dstOffset` select the initial `x`, `y`, and `z` offsets in texels of the sub-regions of the source and destination image data.
- `extent` is the size in texels of the source image to resolve in `width`, `height` and `depth`.

Valid Usage

- VUID-VkImageResolve2-aspectMask-00266
The `aspectMask` member of `srcSubresource` and `dstSubresource` **must** only contain `VK_IMAGE_ASPECT_COLOR_BIT`
- VUID-VkImageResolve2-layerCount-08803
If neither of the `layerCount` members of `srcSubresource` or `dstSubresource` are

VK_REMAINING_ARRAY_LAYERS, the `layerCount` member of `srcSubresource` and `dstSubresource` **must** match

- VUID-VkImageResolve2-layerCount-08804

If one of the `layerCount` members of `srcSubresource` or `dstSubresource` is VK_REMAINING_ARRAY_LAYERS, the other member **must** be either VK_REMAINING_ARRAY_LAYERS or equal to the `arrayLayers` member of the `VkImageCreateInfo` used to create the image minus `baseArrayLayer`

Valid Usage (Implicit)

- VUID-VkImageResolve2-sType-sType
`sType` **must** be VK_STRUCTURE_TYPE_IMAGE_RESOLVE_2
- VUID-VkImageResolve2-pNext-pNext
`pNext` **must** be NULL
- VUID-VkImageResolve2-srcSubresource-parameter
`srcSubresource` **must** be a valid `VkImageSubresourceLayers` structure
- VUID-VkImageResolve2-dstSubresource-parameter
`dstSubresource` **must** be a valid `VkImageSubresourceLayers` structure

Chapter 20. Drawing Commands

Drawing commands (commands with **Draw** in the name) provoke work in a graphics pipeline. Drawing commands are recorded into a command buffer and when executed by a queue, will produce work which executes according to the bound graphics pipeline, or if the **shaderObject** feature is enabled, any **shader objects** bound to graphics stages. A graphics pipeline or a combination of one or more graphics shader objects **must** be bound to a command buffer before any drawing commands are recorded in that command buffer.

Each draw is made up of zero or more vertices and zero or more instances, which are processed by the device and result in the assembly of primitives. Primitives are assembled according to the **pInputAssemblyState** member of the **VkGraphicsPipelineCreateInfo** structure, which is of type **VkPipelineInputAssemblyStateCreateInfo**:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineInputAssemblyStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineInputAssemblyStateCreateFlags flags;
    VkPrimitiveTopology        topology;
    VkBool32                  primitiveRestartEnable;
} VkPipelineInputAssemblyStateCreateInfo;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is reserved for future use.
- **topology** is a **VkPrimitiveTopology** defining the primitive topology, as described below.
- **primitiveRestartEnable** controls whether a special vertex index value is treated as restarting the assembly of primitives. This enable only applies to indexed draws (**vkCmdDrawIndexed**, **vkCmdDrawMultiIndexedEXT**, and **vkCmdDrawIndexedIndirect**), and the special index value is either **0xFFFFFFFF** when the **indexType** parameter of **vkCmdBindIndexBuffer2KHR** or **vkCmdBindIndexBuffer** is equal to **VK_INDEX_TYPE_UINT32**, **0xFF** when **indexType** is equal to **VK_INDEX_TYPE_UINT8_KHR**, or **0xFFFF** when **indexType** is equal to **VK_INDEX_TYPE_UINT16**. Primitive restart is not allowed for “list” topologies, unless one of the features **primitiveTopologyPatchListRestart** (for **VK_PRIMITIVE_TOPOLOGY_PATCH_LIST**) or **primitiveTopologyListRestart** (for all other list topologies) is enabled.

Restarting the assembly of primitives discards the most recent index values if those elements formed an incomplete primitive, and restarts the primitive assembly using the subsequent indices, but only assembling the immediately following element through the end of the originally specified elements. The primitive restart index value comparison is performed before adding the **vertexOffset** value to the index value.

Valid Usage

- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-06252

If the `primitiveTopologyListRestart` feature is not enabled, and `topology` is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, `primitiveRestartEnable` **must** be `VK_FALSE`

- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-06253

If the `primitiveTopologyPatchListRestart` feature is not enabled, and `topology` is `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST`, `primitiveRestartEnable` **must** be `VK_FALSE`

- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-00429

If the `geometryShader` feature is not enabled, `topology` **must** not be any of `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY` or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY`

- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-00430

If the `tessellationShader` feature is not enabled, `topology` **must** not be `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST`

- VUID-VkPipelineInputAssemblyStateCreateInfo-triangleFans-04452

If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::triangleFans` is `VK_FALSE`, `topology` **must** not be `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN`

Valid Usage (Implicit)

- VUID-VkPipelineInputAssemblyStateCreateInfo-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_INPUT_ASSEMBLY_STATE_CREATE_INFO`

- VUID-VkPipelineInputAssemblyStateCreateInfo-pNext-pNext

`pNext` **must** be `NULL`

- VUID-VkPipelineInputAssemblyStateCreateInfo-flags-zeroBitmask

`flags` **must** be 0

- VUID-VkPipelineInputAssemblyStateCreateInfo-topology-parameter

`topology` **must** be a valid `VkPrimitiveTopology` value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineInputAssemblyStateCreateFlags;
```

`VkPipelineInputAssemblyStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

To [dynamically control](#) whether a special vertex index value is treated as restarting the assembly of primitives, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetPrimitiveRestartEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                 primitiveRestartEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `primitiveRestartEnable` controls whether a special vertex index value is treated as restarting the assembly of primitives. It behaves in the same way as `VkPipelineInputAssemblyStateCreateInfo::primitiveRestartEnable`

This command sets the primitive restart enable for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the `VkPipelineInputAssemblyStateCreateInfo::primitiveRestartEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetPrimitiveRestartEnable-None-08970
At least one of the following **must** be true:
 - the `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetPrimitiveRestartEnable-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetPrimitiveRestartEnable-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdSetPrimitiveRestartEnable-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetPrimitiveRestartEnable-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

20.1. Primitive Topologies

Primitive topology determines how consecutive vertices are organized into primitives, and determines the type of primitive that is used at the beginning of the graphics pipeline. The effective topology for later stages of the pipeline is altered by tessellation or geometry shading (if either is in use) and depends on the execution modes of those shaders.

The primitive topologies defined by [VkPrimitiveTopology](#) are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPrimitiveTopology {
    VK_PRIMITIVE_TOPOLOGY_POINT_LIST = 0,
    VK_PRIMITIVE_TOPOLOGY_LINE_LIST = 1,
    VK_PRIMITIVE_TOPOLOGY_LINE_STRIP = 2,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST = 3,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP = 4,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN = 5,
    VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY = 6,
    VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY = 7,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY = 8,
    VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY = 9,
    VK_PRIMITIVE_TOPOLOGY_PATCH_LIST = 10,
} VkPrimitiveTopology;
```

- `VK_PRIMITIVE_TOPOLOGY_POINT_LIST` specifies a series of [separate point primitives](#).
- `VK_PRIMITIVE_TOPOLOGY_LINE_LIST` specifies a series of [separate line primitives](#).
- `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP` specifies a series of [connected line primitives](#) with consecutive lines sharing a vertex.
- `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST` specifies a series of [separate triangle primitives](#).
- `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP` specifies a series of [connected triangle primitives](#) with consecutive triangles sharing an edge.
- `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN` specifies a series of [connected triangle primitives](#) with all triangles sharing a common vertex. If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::triangleFans` is `VK_FALSE`, then triangle fans are not supported by the implementation, and `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN` **must** not be used.

- `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY` specifies a series of [separate line primitives with adjacency](#).
- `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY` specifies a series of [connected line primitives with adjacency](#), with consecutive primitives sharing three vertices.
- `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY` specifies a series of [separate triangle primitives with adjacency](#).
- `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY` specifies [connected triangle primitives with adjacency](#), with consecutive triangles sharing an edge.
- `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST` specifies [separate patch primitives](#).

Each primitive topology, and its construction from a list of vertices, is described in detail below with a supporting diagram, according to the following key:

	Vertex	A point in 3-dimensional space. Positions chosen within the diagrams are arbitrary and for illustration only.
5	Vertex Number	Sequence position of a vertex within the provided vertex data.
	Provoking Vertex	Provoking vertex within the main primitive. The tail is angled towards the relevant primitive. Used in flat shading .
	Primitive Edge	An edge connecting the points of a main primitive.
	Adjacency Edge	Points connected by these lines do not contribute to a main primitive, and are only accessible in a geometry shader .
	Winding Order	The relative order in which vertices are defined within a primitive, used in the facing determination . This ordering has no specific start or end point.

The diagrams are supported with mathematical definitions where the vertices (v) and primitives (p) are numbered starting from 0; v_0 is the first vertex in the provided data and p_0 is the first primitive in the set of primitives defined by the vertices and topology.

To [dynamically set](#) primitive topology, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetPrimitiveTopologyEXT(
    VkCommandBuffer          commandBuffer,
    VkPrimitiveTopology       primitiveTopology);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `primitiveTopology` specifies the primitive topology to use for drawing.

This command sets the primitive topology for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the `VkPipelineInputAssemblyStateCreateInfo::topology` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetPrimitiveTopology-None-08971

At least one of the following **must** be true:

- the `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetPrimitiveTopology-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetPrimitiveTopology-primitiveTopology-parameter
`primitiveTopology` **must** be a valid `VkPrimitiveTopology` value
- VUID-vkCmdSetPrimitiveTopology-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdSetPrimitiveTopology-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetPrimitiveTopology-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

20.1.1. Topology Class

The primitive topologies are grouped into the following topology classes:

Table 25. Topology Classes

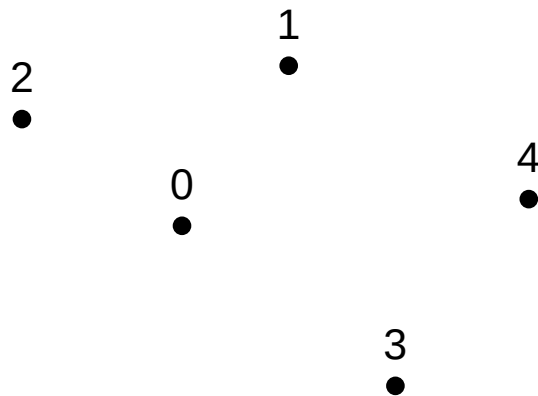
Topology Class	Primitive Topology
Point	VK_PRIMITIVE_TOPOLOGY_POINT_LIST
Line	VK_PRIMITIVE_TOPOLOGY_LINE_LIST, VK_PRIMITIVE_TOPOLOGY_LINE_STRIP, VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY, VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY
Triangle	VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY, VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY
Patch	VK_PRIMITIVE_TOPOLOGY_PATCH_LIST

20.1.2. Point Lists

When the topology is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, each consecutive vertex defines a single point primitive, according to the equation:

$$p_i = \{v_i\}$$

As there is only one vertex, that vertex is the provoking vertex. The number of primitives generated is equal to `vertexCount`.



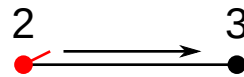
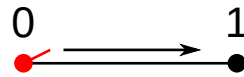
20.1.3. Line Lists

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, each consecutive pair of vertices defines a single line primitive, according to the equation:

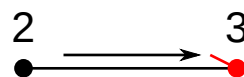
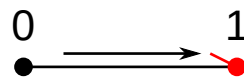
$$p_i = \{v_{2i}, v_{2i+1}\}$$

The number of primitives generated is equal to $\lceil \text{vertexCount} / 2 \rceil$.

When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT`, the provoking vertex for p_i is v_{2i} .



When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, the provoking vertex for p_i is v_{2i+1} .



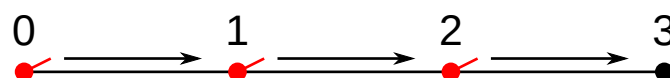
20.1.4. Line Strips

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP`, one line primitive is defined by each vertex and the following vertex, according to the equation:

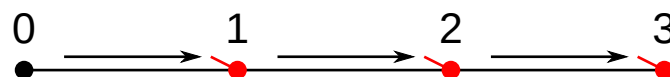
$$p_i = \{v_i, v_{i+1}\}$$

The number of primitives generated is equal to $\max(0, \text{vertexCount}-1)$.

When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT`, the provoking vertex for p_i is v_i .



When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, the provoking vertex for p_i is v_{i+1} .



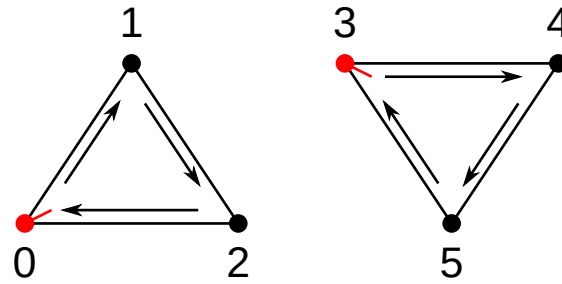
20.1.5. Triangle Lists

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, each consecutive set of three vertices defines a single triangle primitive, according to the equation:

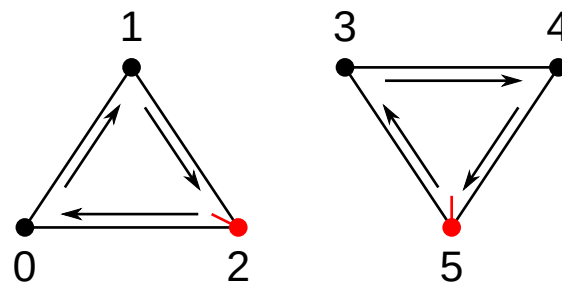
$$p_i = \{v_{3i}, v_{3i+1}, v_{3i+2}\}$$

The number of primitives generated is equal to $\lceil \text{vertexCount}/3 \rceil$.

When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT`, the provoking vertex for p_i is v_{3i} .



When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, the provoking vertex for p_i is v_{3i+2} .



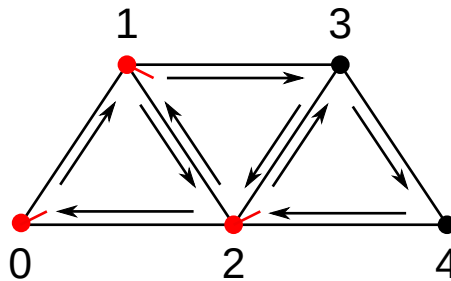
20.1.6. Triangle Strips

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP`, one triangle primitive is defined by each vertex and the two vertices that follow it, according to the equation:

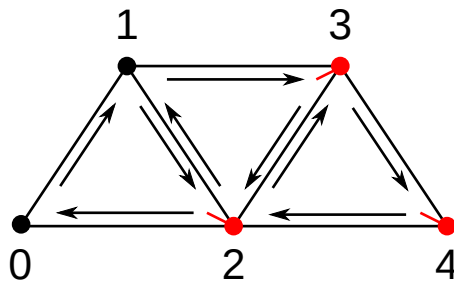
$$p_i = \{v_i, v_{i+(1+i\%2)}, v_{i+(2-i\%2)}\}$$

The number of primitives generated is equal to $\max(0, \text{vertexCount}-2)$.

When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT`, the provoking vertex for p_i is v_i .



When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, the provoking vertex for p_i is v_{i+2} .



NOTE

The ordering of the vertices in each successive triangle is reversed, so that the winding order is consistent throughout the strip.

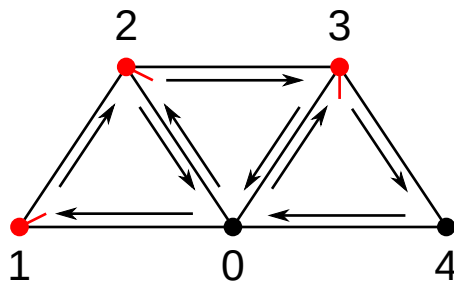
20.1.7. Triangle Fans

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN`, triangle primitives are defined around a shared common vertex, according to the equation:

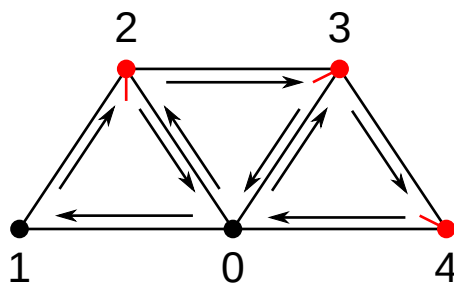
$$p_i = \{v_{i+1}, v_{i+2}, v_0\}$$

The number of primitives generated is equal to $\max(0, \text{vertexCount} - 2)$.

When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT`, the provoking vertex for p_i is v_{i+1} .



When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, the provoking vertex for p_i is v_{i+2} .



NOTE

If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::triangleFans` is `VK_FALSE`, then triangle fans are not supported by the implementation, and `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN` **must** not be used.

20.1.8. Line Lists With Adjacency

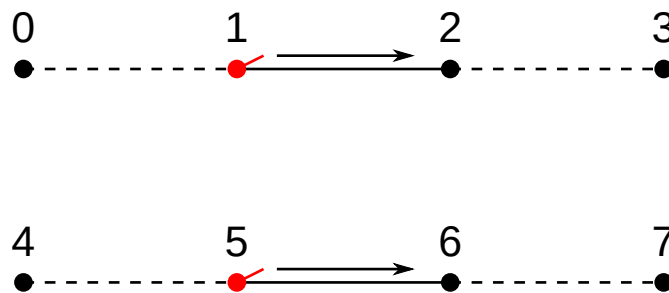
When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, each consecutive set of four vertices defines a single line primitive with adjacency, according to the equation:

$$p_i = \{v_{4i}, v_{4i+1}, v_{4i+2}, v_{4i+3}\}$$

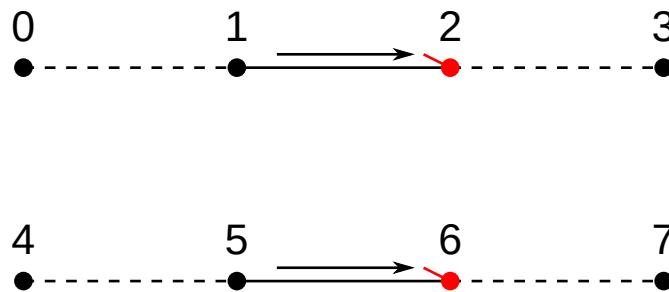
A line primitive is described by the second and third vertices of the total primitive, with the remaining two vertices only accessible in a [geometry shader](#).

The number of primitives generated is equal to $\lfloor \text{vertexCount}/4 \rfloor$.

When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT`, the provoking vertex for p_i is v_{4i+1} .



When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, the provoking vertex for p_i is v_{4i+2} .



20.1.9. Line Strips With Adjacency

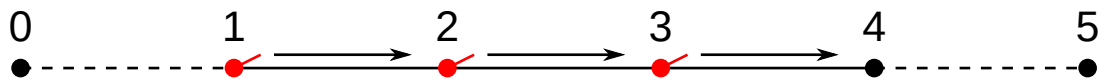
When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY`, one line primitive with adjacency is defined by each vertex and the following vertex, according to the equation:

$$p_i = \{v_i, v_{i+1}, v_{i+2}, v_{i+3}\}$$

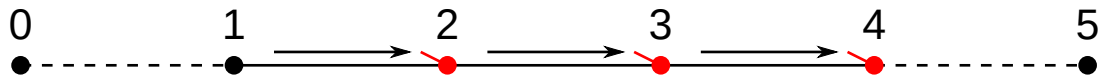
A line primitive is described by the second and third vertices of the total primitive, with the remaining two vertices only accessible in a [geometry shader](#).

The number of primitives generated is equal to $\max(0, \text{vertexCount}-3)$.

When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT`, the provoking vertex for p_i is v_{i+1} .



When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, the provoking vertex for p_i is v_{i+2} .



20.1.10. Triangle Lists With Adjacency

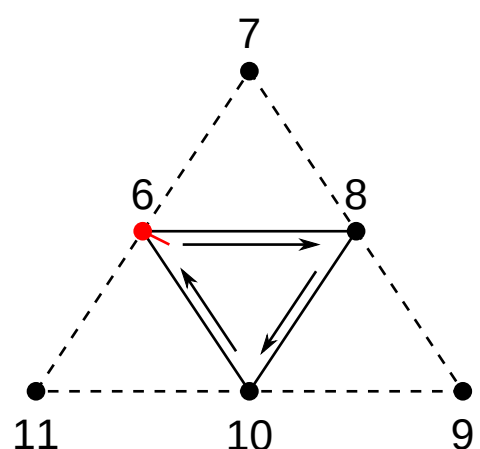
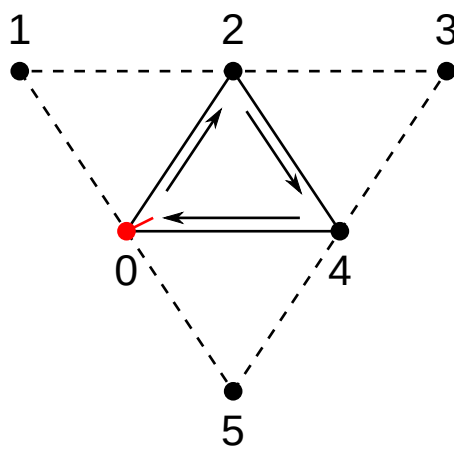
When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, each consecutive set of six vertices defines a single triangle primitive with adjacency, according to the equations:

$$p_i = \{v_{6i}, v_{6i+1}, v_{6i+2}, v_{6i+3}, v_{6i+4}, v_{6i+5}\}$$

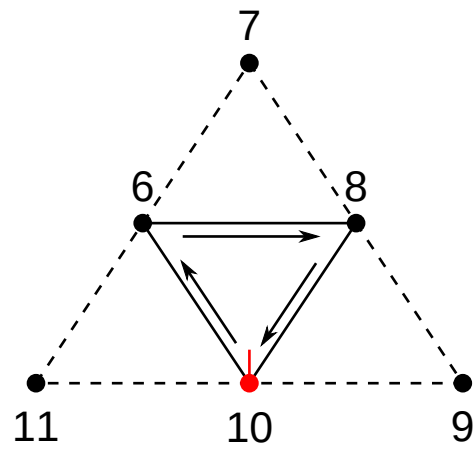
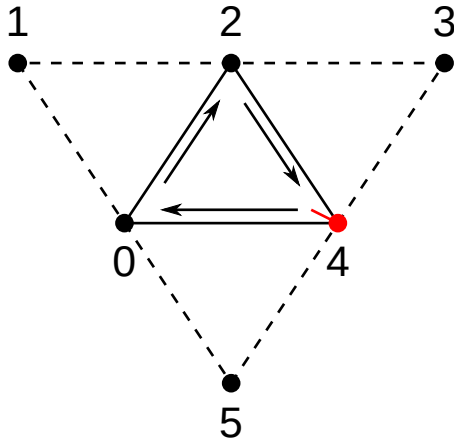
A triangle primitive is described by the first, third, and fifth vertices of the total primitive, with the remaining three vertices only accessible in a [geometry shader](#).

The number of primitives generated is equal to $\lfloor \text{vertexCount} / 6 \rfloor$.

When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT`, the provoking vertex for p_i is v_{6i} .



When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, the provoking vertex for p_i is v_{6i+4} .



20.1.11. Triangle Strips With Adjacency

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY`, one triangle primitive with adjacency is defined by each vertex and the following 5 vertices.

The number of primitives generated, n , is equal to $\lceil \max(0, \text{vertexCount} - 4) / 2 \rceil$.

If $n=1$, the primitive is defined as:

$$p = \{v_0, v_1, v_2, v_5, v_4, v_3\}$$

If $n>1$, the total primitive consists of different vertices according to where it is in the strip:

$$p_i = \{v_{2i}, v_{2i+1}, v_{2i+2}, v_{2i+6}, v_{2i+4}, v_{2i+3}\} \text{ when } i=0$$

$$p_i = \{v_{2i}, v_{2i+3}, v_{2i+4}, v_{2i+6}, v_{2i+2}, v_{2i-2}\} \text{ when } i>0, i<n-1, \text{ and } i\%2=1$$

$$p_i = \{v_{2i}, v_{2i-2}, v_{2i+2}, v_{2i+6}, v_{2i+4}, v_{2i+3}\} \text{ when } i>0, i<n-1, \text{ and } i\%2=0$$

$$p_i = \{v_{2i}, v_{2i+3}, v_{2i+4}, v_{2i+5}, v_{2i+2}, v_{2i-2}\} \text{ when } i=n-1 \text{ and } i\%2=1$$

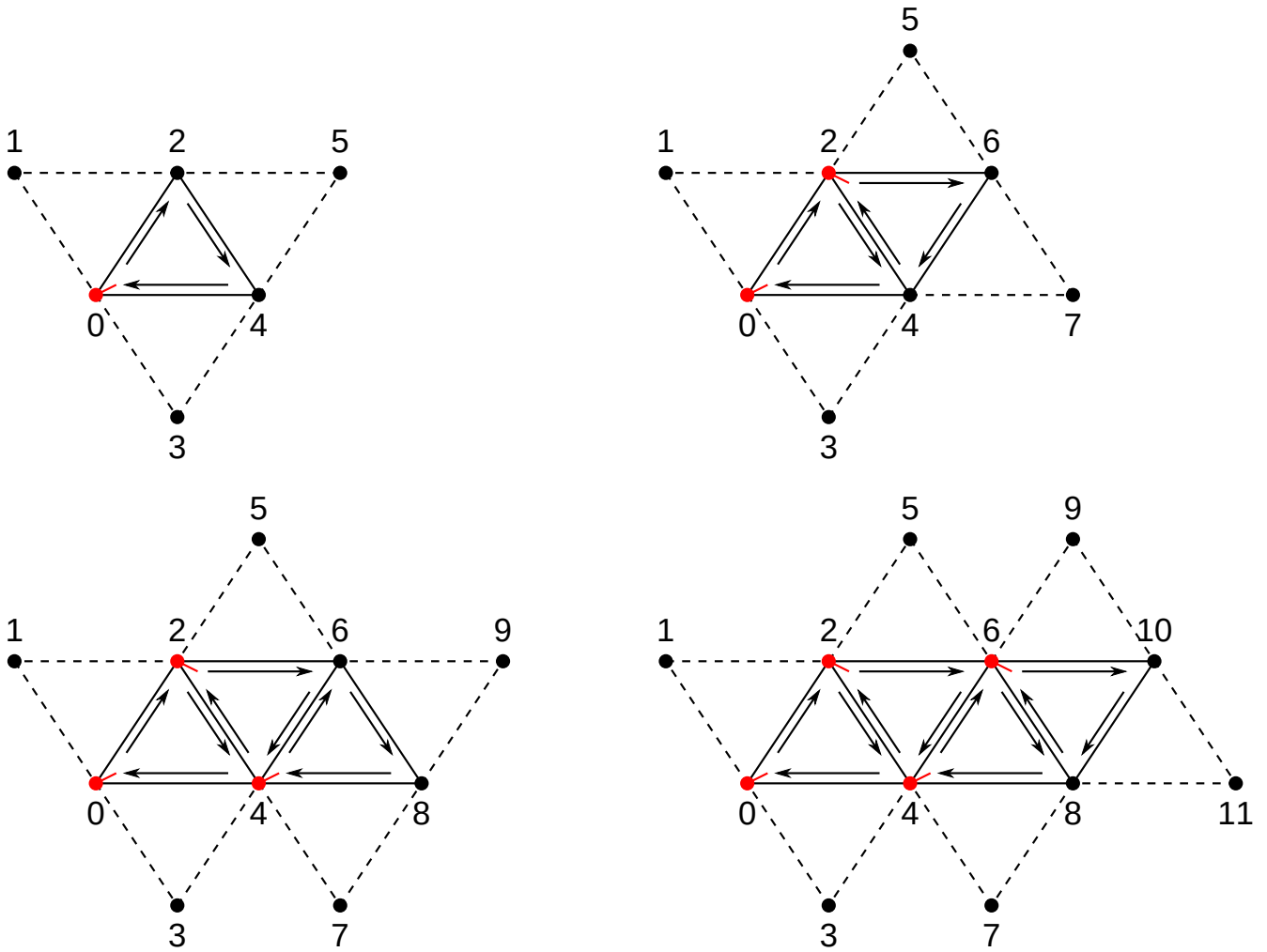
$$p_i = \{v_{2i}, v_{2i-2}, v_{2i+2}, v_{2i+5}, v_{2i+4}, v_{2i+3}\} \text{ when } i=n-1 \text{ and } i\%2=0$$

A triangle primitive is described by the first, third, and fifth vertices of the total primitive in all cases, with the remaining three vertices only accessible in a [geometry shader](#).

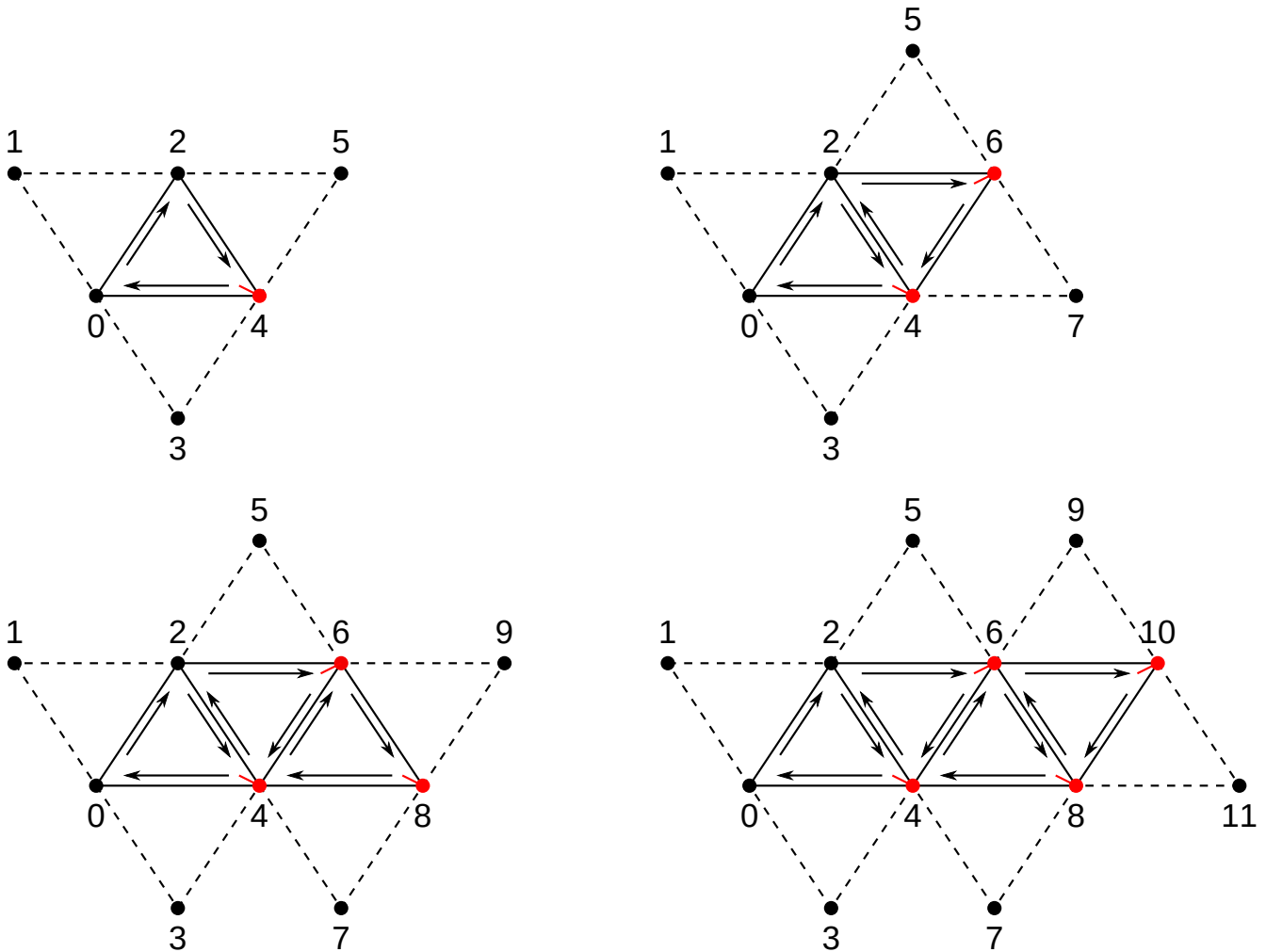
NOTE

The ordering of the vertices in each successive triangle is altered so that the winding order is consistent throughout the strip.

When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT`, the provoking vertex for p_i is always v_{2i} .



When the `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, the provoking vertex for p_i is always v_{2i+4} .



20.1.12. Patch Lists

When the primitive topology is `VK_PRIMITIVE_TOPOLOGY_PATCH_LIST`, each consecutive set of m vertices defines a single patch primitive, according to the equation:

$$p_i = \{v_{mi}, v_{mi+1}, \dots, v_{mi+(m-2)}, v_{mi+(m-1)}\}$$

where m is equal to `VkPipelineTessellationStateCreateInfo::patchControlPoints`.

Patch lists are never passed to [vertex post-processing](#), and as such no provoking vertex is defined for patch primitives. The number of primitives generated is equal to $\lfloor \text{vertexCount}/m \rfloor$.

The vertices comprising a patch have no implied geometry, and are used as inputs to tessellation shaders and the fixed-function tessellator to generate new point, line, or triangle primitives.

20.2. Primitive Order

Primitives generated by [drawing commands](#) progress through the stages of the [graphics pipeline](#) in *primitive order*. Primitive order is initially determined in the following way:

1. Submission order determines the initial ordering
2. For indirect drawing commands, the order in which accessed instances of the

`VkDrawIndirectCommand` are stored in `buffer`, from lower indirect buffer addresses to higher addresses.

3. If a drawing command includes multiple instances, the order in which instances are executed, from lower numbered instances to higher.
4. The order in which primitives are specified by a drawing command:
 - For non-indexed draws, from vertices with a lower numbered `vertexIndex` to a higher numbered `vertexIndex`.
 - For indexed draws, vertices sourced from a lower index buffer addresses to higher addresses.

Within this order implementations further sort primitives:

5. If tessellation shading is active, by an implementation-dependent order of new primitives generated by `tessellation`.
6. If geometry shading is active, by the order new primitives are generated by `geometry shading`.
7. If the `polygon mode` is not `VK_POLYGON_MODE_FILL`, by an implementation-dependent ordering of the new primitives generated within the original primitive.

Primitive order is later used to define `rasterization order`, which determines the order in which fragments output results to a framebuffer.

20.3. Programmable Primitive Shading

Once primitives are assembled, they proceed to the vertex shading stage of the pipeline. If the draw includes multiple instances, then the set of primitives is sent to the vertex shading stage multiple times, once for each instance.

It is implementation-dependent whether vertex shading occurs on vertices that are discarded as part of incomplete primitives, but if it does occur then it operates as if they were vertices in complete primitives and such invocations **can** have side effects.

Vertex shading receives two per-vertex inputs from the primitive assembly stage - the `vertexIndex` and the `instanceIndex`. How these values are generated is defined below, with each command.

Drawing commands fall roughly into two categories:

- Non-indexed drawing commands present a sequential `vertexIndex` to the vertex shader. The sequential index is generated automatically by the device (see `Fixed-Function Vertex Processing` for details on both specifying the vertex attributes indexed by `vertexIndex`, as well as binding vertex buffers containing those attributes to a command buffer). These commands are:
 - `vkCmdDraw`
 - `vkCmdDrawIndirect`
 - `vkCmdDrawIndirectCountKHR`
 - `vkCmdDrawMultiEXT`
- Indexed drawing commands read index values from an *index buffer* and use this to compute the

`vertexIndex` value for the vertex shader. These commands are:

- `vkCmdDrawIndexed`
- `vkCmdDrawIndexedIndirect`
- `vkCmdDrawIndexedIndirectCountKHR`
- `vkCmdDrawMultiIndexedEXT`

To bind an index buffer to a command buffer, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBindIndexBuffer(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 buffer,
    VkDeviceSize             offset,
    VkIndexType              indexType);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer being bound.
- `offset` is the starting offset in bytes within `buffer` used in index buffer address calculations.
- `indexType` is a `VkIndexType` value specifying the size of the indices.

If the `maintenance6` feature is enabled, `buffer` can be `VK_NULL_HANDLE`. If `buffer` is `VK_NULL_HANDLE` and the `nullDescriptor` feature is enabled, every index fetched results in a value of zero.

Valid Usage

- VUID-vkCmdBindIndexBuffer-offset-08782
`offset` **must** be less than the size of `buffer`
- VUID-vkCmdBindIndexBuffer-offset-08783
The sum of `offset` and the base address of the range of `VkDeviceMemory` object that is backing `buffer`, **must** be a multiple of the size of the type indicated by `indexType`
- VUID-vkCmdBindIndexBuffer-buffer-08784
`buffer` **must** have been created with the `VK_BUFFER_USAGE_INDEX_BUFFER_BIT` flag
- VUID-vkCmdBindIndexBuffer-buffer-08785
If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBindIndexBuffer-indexType-08786
`indexType` **must** not be `VK_INDEX_TYPE_NONE_KHR`
- VUID-vkCmdBindIndexBuffer-indexType-08787
If `indexType` is `VK_INDEX_TYPE_UINT8_KHR`, the `indexTypeUint8` feature **must** be enabled
- VUID-vkCmdBindIndexBuffer-None-09493
If `maintenance6` is not enabled, `buffer` **must** not be `VK_NULL_HANDLE`

- VUID-vkCmdBindIndexBuffer-buffer-09494
If **buffer** is [VK_NULL_HANDLE](#), offset **must** be zero

Valid Usage (Implicit)

- VUID-vkCmdBindIndexBuffer-commandBuffer-parameter
commandBuffer **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdBindIndexBuffer-buffer-parameter
If **buffer** is not [VK_NULL_HANDLE](#), **buffer** **must** be a valid [VkBuffer](#) handle
- VUID-vkCmdBindIndexBuffer-indexType-parameter
indexType **must** be a valid [VkIndexType](#) value
- VUID-vkCmdBindIndexBuffer-commandBuffer-recording
commandBuffer **must** be in the [recording state](#)
- VUID-vkCmdBindIndexBuffer-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdBindIndexBuffer-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindIndexBuffer-commonparent
Both of **buffer**, and **commandBuffer** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To bind an index buffer, along with its size, to a command buffer, call:

```
// Provided by VK_KHR_maintenance5
void vkCmdBindIndexBuffer2KHR(
    VkCommandBuffer          commandBuffer,
    VkBuffer                  buffer,
    VkDeviceSize              offset,
    VkDeviceSize              size,
    VkIndexType               indexType);
```

- **commandBuffer** is the command buffer into which the command is recorded.
- **buffer** is the buffer being bound.
- **offset** is the starting offset in bytes within **buffer** used in index buffer address calculations.
- **size** is the size in bytes of index data bound from **buffer**.
- **indexType** is a [VkIndexType](#) value specifying the size of the indices.

size specifies the bound size of the index buffer starting from **offset**. If **size** is [VK_WHOLE_SIZE](#) then the bound size is from **offset** to the end of the **buffer**.

If the [maintenance6](#) feature is enabled, **buffer** can be [VK_NULL_HANDLE](#). If **buffer** is [VK_NULL_HANDLE](#) and the [nullDescriptor](#) feature is enabled, every index fetched results in a value of zero.

Valid Usage

- VUID-vkCmdBindIndexBuffer2KHR-offset-08782
offset **must** be less than the size of **buffer**
- VUID-vkCmdBindIndexBuffer2KHR-offset-08783
The sum of **offset** and the base address of the range of [VkDeviceMemory](#) object that is backing **buffer**, **must** be a multiple of the size of the type indicated by **indexType**
- VUID-vkCmdBindIndexBuffer2KHR-buffer-08784
buffer **must** have been created with the [VK_BUFFER_USAGE_INDEX_BUFFER_BIT](#) flag
- VUID-vkCmdBindIndexBuffer2KHR-buffer-08785
If **buffer** is non-sparse then it **must** be bound completely and contiguously to a single [VkDeviceMemory](#) object
- VUID-vkCmdBindIndexBuffer2KHR-indexType-08786
indexType **must** not be [VK_INDEX_TYPE_NONE_KHR](#)
- VUID-vkCmdBindIndexBuffer2KHR-indexType-08787
If **indexType** is [VK_INDEX_TYPE_UINT8_KHR](#), the [indexTypeUint8](#) feature **must** be enabled
- VUID-vkCmdBindIndexBuffer2KHR-None-09493
If [maintenance6](#) is not enabled, **buffer** **must** not be [VK_NULL_HANDLE](#)
- VUID-vkCmdBindIndexBuffer2KHR-buffer-09494
If **buffer** is [VK_NULL_HANDLE](#), **offset** **must** be zero
- VUID-vkCmdBindIndexBuffer2KHR-size-08767

If **size** is not **VK_WHOLE_SIZE**, **size** **must** be a multiple of the size of the type indicated by **indexType**

- VUID-vkCmdBindIndexBuffer2KHR-size-08768

If **size** is not **VK_WHOLE_SIZE**, the sum of **offset** and **size** **must** be less than or equal to the size of **buffer**

Valid Usage (Implicit)

- VUID-vkCmdBindIndexBuffer2KHR-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdBindIndexBuffer2KHR-buffer-parameter
If **buffer** is not **VK_NULL_HANDLE**, **buffer** **must** be a valid **VkBuffer** handle
- VUID-vkCmdBindIndexBuffer2KHR-indexType-parameter
indexType **must** be a valid **VkIndexType** value
- VUID-vkCmdBindIndexBuffer2KHR-commandBuffer-recording
commandBuffer **must** be in the **recording state**
- VUID-vkCmdBindIndexBuffer2KHR-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdBindIndexBuffer2KHR-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindIndexBuffer2KHR-commonparent
Both of **buffer**, and **commandBuffer** that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Possible values of **vkCmdBindIndexBuffer2KHR::indexType** and **vkCmdBindIndexBuffer::indexType**, specifying the size of indices, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkIndexType {
    VK_INDEX_TYPE_UINT16 = 0,
    VK_INDEX_TYPE_UINT32 = 1,
    // Provided by VK_KHR_acceleration_structure
    VK_INDEX_TYPE_NONE_KHR = 1000165000,
    // Provided by VK_KHR_index_type_uint8
    VK_INDEX_TYPE_UINT8_KHR = 1000265000,
} VkIndexType;
```

- `VK_INDEX_TYPE_UINT16` specifies that indices are 16-bit unsigned integer values.
- `VK_INDEX_TYPE_UINT32` specifies that indices are 32-bit unsigned integer values.
- `VK_INDEX_TYPE_NONE_KHR` specifies that no indices are provided.
- `VK_INDEX_TYPE_UINT8_KHR` specifies that indices are 8-bit unsigned integer values.

The parameters for each drawing command are specified directly in the command or read from buffer memory, depending on the command. Drawing commands that source their parameters from buffer memory are known as *indirect* drawing commands.

All drawing commands interact with the `robustBufferAccess` feature.

To record a non-indexed draw, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDraw(
    VkCommandBuffer          commandBuffer,
    uint32_t                 vertexCount,
    uint32_t                 instanceCount,
    uint32_t                 firstVertex,
    uint32_t                 firstInstance);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `vertexCount` is the number of vertices to draw.
- `instanceCount` is the number of instances to draw.
- `firstVertex` is the index of the first vertex to draw.
- `firstInstance` is the instance ID of the first instance to draw.

When the command is executed, primitives are assembled using the current primitive topology and `vertexCount` consecutive vertex indices with the first `vertexIndex` value equal to `firstVertex`. The primitives are drawn `instanceCount` times with `instanceIndex` starting with `firstInstance` and increasing sequentially for each instance. The assembled primitives execute the bound graphics pipeline.

Valid Usage

- VUID-vkCmdDraw-magFilter-04553

If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`

- VUID-vkCmdDraw-mipmapMode-04770

If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`

- VUID-vkCmdDraw-unnormalizedCoordinates-09635

If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1

- VUID-vkCmdDraw-unnormalizedCoordinates-09636

If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`

- VUID-vkCmdDraw-None-06479

If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`

- VUID-vkCmdDraw-None-02691

If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`

- VUID-vkCmdDraw-None-07888

If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`

- VUID-vkCmdDraw-None-02692

If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`

- VUID-vkCmdDraw-None-02693

If the `VK_EXT_filter_cubic` extension is not enabled and any `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, it **must** not have a `VkImageViewType` of `VK_IMAGE_VIEW_TYPE_3D`, `VK_IMAGE_VIEW_TYPE_CUBE`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`

- VUID-vkCmdDraw-filterCubic-02694

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdDraw-filterCubicMinmax-02695
Any [VkImageView](#) being sampled with [VK_FILTER_CUBIC_EXT](#) with a reduction mode of either [VK_SAMPLER_REDUCTION_MODE_MIN](#) or [VK_SAMPLER_REDUCTION_MODE_MAX](#) as a result of this command **must** have a [VkImageViewType](#) and format that supports cubic filtering together with minmax filtering, as specified by [VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax](#) returned by [vkGetPhysicalDeviceImageFormatProperties2KHR](#)
- VUID-vkCmdDraw-OpTypeImage-07027
For any [VkImageView](#) being written as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDraw-OpTypeImage-07028
For any [VkImageView](#) being read as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDraw-OpTypeImage-07029
For any [VkBufferView](#) being written as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDraw-OpTypeImage-07030
Any [VkBufferView](#) being read as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#) then the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDraw-None-08600
For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDraw-None-08601
For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDraw-None-10068
For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDraw-maintenance4-08602
If the [maintenance4](#) feature is not enabled, then for each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) and [VkPushConstantRange](#) arrays used to create the current [VkShaderEXT](#), as described in

Pipeline Layout Compatibility

- VUID-vkCmdDraw-None-08114
Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)
- VUID-vkCmdDraw-None-08606
If the [shaderObject](#) feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDraw-None-08608
If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDraw-None-08609
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage
- VUID-vkCmdDraw-None-08610
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage
- VUID-vkCmdDraw-None-08611
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage
- VUID-vkCmdDraw-None-08607
If the [shaderObject](#) feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and [VK_NULL_HANDLE](#) shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command
- VUID-vkCmdDraw-uniformBuffers-06935
If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDraw-None-08612
If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform

buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDraw-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDraw-None-08613

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDraw-None-06550

If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions

- VUID-vkCmdDraw-ConstOffset-06551

If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands

- VUID-vkCmdDraw-viewType-07752

If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDraw-format-07753

If a [VkImageView](#) is accessed as a result of this command, then the [numeric type](#) of the image view's [format](#) and the [Sampled Type](#) operand of the [OpTypeImage](#) **must** match

- VUID-vkCmdDraw-OpImageWrite-08795

If a [VkImageView](#) created with a format other than [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDraw-OpImageWrite-08796

If a [VkImageView](#) created with the format [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have four components

- VUID-vkCmdDraw-OpImageWrite-04469

If a [VkBufferView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdDraw-SampledType-04470

If a [VkImageView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64

- VUID-vkCmdDraw-SampledType-04471

If a [VkImageView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32

- VUID-vkCmdDraw-SampledType-04472

If a [VkBufferView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64

- VUID-vkCmdDraw-SampledType-04473

If a [VkBufferView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32

- VUID-vkCmdDraw-sparseImageInt64Atomics-04474

If the [sparseImageInt64Atomics](#) feature is not enabled, [VkImage](#) objects created with the [VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT](#) flag **must** not be accessed by atomic instructions through an [OpTypeImage](#) with a [SampledType](#) with a [Width](#) of 64 by this command

- VUID-vkCmdDraw-sparseImageInt64Atomics-04475

If the [sparseImageInt64Atomics](#) feature is not enabled, [VkBuffer](#) objects created with the [VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT](#) flag **must** not be accessed by atomic instructions through an [OpTypeImage](#) with a [SampledType](#) with a [Width](#) of 64 by this command

- VUID-vkCmdDraw-None-07288

Any shader invocation executed by this command **must** [terminate](#)

- VUID-vkCmdDraw-None-09600

If a descriptor with type equal to any of [VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE](#), [VK_DESCRIPTOR_TYPE_STORAGE_IMAGE](#), or [VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT](#) is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDraw-renderPass-02684

The current render pass **must** be [compatible](#) with the [renderPass](#) member of the [VkGraphicsPipelineCreateInfo](#) structure specified when creating the [VkPipeline](#) bound to [VK_PIPELINE_BIND_POINT_GRAPHICS](#)

- VUID-vkCmdDraw-subpass-02685

The subpass index of the current render pass **must** be equal to the [subpass](#) member of the [VkGraphicsPipelineCreateInfo](#) structure specified when creating the [VkPipeline](#) bound to [VK_PIPELINE_BIND_POINT_GRAPHICS](#)

- VUID-vkCmdDraw-None-07748

If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set

- VUID-vkCmdDraw-OpTypeImage-07468

If any shader executed by this pipeline accesses an [OpTypeImage](#) variable with a [Dim](#) operand of [SubpassData](#), it **must** be decorated with an [InputAttachmentIndex](#) that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDraw-None-07469

Input attachment views accessed in a subpass **must** be created with the same [VkFormat](#)

as the corresponding subpass definition, and be created with a [VkImageView](#) that is compatible with the attachment referenced by the subpass' [pInputAttachments](#) [[InputAttachmentIndex](#)] in the currently bound [VkFramebuffer](#) as specified by [Fragment Input Attachment Compatibility](#)

- VUID-vkCmdDraw-pDepthInputAttachmentIndex-09595

Input attachment views accessed in a dynamic render pass with a [InputAttachmentIndex](#) referenced by [VkRenderingInputAttachmentIndexInfoKHR](#), or no [InputAttachmentIndex](#) if [VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex](#) or [VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex](#) are **NULL**, **must** be created with a [VkImageView](#) that is compatible with the corresponding color, depth, or stencil attachment in [VkRenderingInfo](#)

- VUID-vkCmdDraw-pDepthInputAttachmentIndex-09596

Input attachment views accessed in a dynamic render pass via a shader object **must** have an [InputAttachmentIndex](#) if both [VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex](#) and [VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex](#) are non-NULL

- VUID-vkCmdDraw-InputAttachmentIndex-09597

If an input attachment view accessed in a dynamic render pass via a shader object has an [InputAttachmentIndex](#), the [InputAttachmentIndex](#) **must** match an index in [VkRenderingInputAttachmentIndexInfoKHR](#)

- VUID-vkCmdDraw-None-06537

Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDraw-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the [VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT](#) image layout, and either:

- the [VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT](#) is set on the currently bound pipeline or
- the last call to [vkCmdSetAttachmentFeedbackLoopEnableEXT](#) included [VK_IMAGE_ASPECT_COLOR_BIT](#) and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with [VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT](#)

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDraw-None-09001

If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the [VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT](#) image layout, and either:

- the [VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT](#) is set on the currently bound pipeline or

- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_DEPTH_BIT` and

- there is no currently bound graphics pipeline or
- the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDraw-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_STENCIL_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDraw-None-09003

If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command

- VUID-vkCmdDraw-None-06539

If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment

- VUID-vkCmdDraw-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, `depth writes` **must** be disabled

- VUID-vkCmdDraw-None-06887

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, `all stencil ops` **must** be `VK_STENCIL_OP_KEEP`

- VUID-vkCmdDraw-None-07831

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VIEWPORT` dynamic state enabled then `vkCmdSetViewport` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07832

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SCISSOR` dynamic state enabled then `vkCmdSetScissor` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07833

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic state enabled then `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08617

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08618

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08619

If a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07834

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BIAS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBiasEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` or `vkCmdSetDepthBias2EXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07835

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic state enabled then `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08621

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer set any element of `pColorBlendEnables` to `VK_TRUE`, and the most recent call to `vkCmdSetColorBlendEquationEXT` in the current command buffer set the same element of `pColorBlendEquations` to a `VkColorBlendEquationEXT` structure with any `VkBlendFactor`

member with a value of `VK_BLEND_FACTOR_CONSTANT_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR`, `VK_BLEND_FACTOR_CONSTANT_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`, `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07836

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBoundsTestEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07837

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilCompareMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilWriteMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07839

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_REFERENCE` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilReference` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-maxMultiviewInstanceIndex-02688

If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to `VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex`

- VUID-vkCmdDraw-sampleLocationsEnable-02689

If the bound graphics pipeline was created with `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable` set to `VK_TRUE` and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDraw-None-07634

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the

`VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleLocationsEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-06666

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `vkCmdSetSampleLocationsEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's `format features` do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the `blendEnable` member of the corresponding element of the `pAttachments` member of `pColorBlendState` **must** be `VK_FALSE`

- VUID-vkCmdDraw-None-08643

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then for each color attachment in the render pass, if the corresponding image view's `format features` do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the corresponding member of `pColorBlendEnables` in the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer that affected that attachment index **must** have been `VK_FALSE`

- VUID-vkCmdDraw-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then `rasterizationSamples` for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDraw-None-08644

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`,

then the most recent call to `vkCmdSetRasterizationSamplesEXT` in the current command buffer **must** have set `rasterizationSamples` to be the same as the number of samples for the current render pass color and/or depth/stencil attachments

- VUID-vkCmdDraw-None-08876

If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with `vkCmdBeginRenderingKHR`

- VUID-vkCmdDraw-imageView-06172

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, this command

must not write any values to the depth attachment

- VUID-vkCmdDraw-imageView-06173

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDraw-imageView-06174

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDraw-imageView-06175

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDraw-imageView-06176

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDraw-imageView-06177

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDraw-viewMask-06178

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::viewMask` equal to `VkRenderingInfo::viewMask`

- VUID-vkCmdDraw-colorAttachmentCount-06179

If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::colorAttachmentCount` equal to `VkRenderingInfo::colorAttachmentCount`

- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08910

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline

- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08912
If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` equal to `VK_NULL_HANDLE` **must** have the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound pipeline equal to `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08911
If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline, or the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats`, if it exists, **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDraw-None-07749
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-None-08646
If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDraw-attachmentCount-07750
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then the `attachmentCount` parameter of `vkCmdSetColorWriteEnableEXT` **must** be greater than or equal to the `VkPipelineColorBlendStateCreateInfo::attachmentCount` of the currently bound graphics pipeline
- VUID-vkCmdDraw-None-08647
If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then the `attachmentCount` parameter of most recent call to `vkCmdSetColorWriteEnableEXT` in the current command buffer **must** be greater than or equal to the number of color attachments in the current render pass instance
- VUID-vkCmdDraw-None-07751
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state enabled then

`vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command for each discard rectangle in `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount`

- VUID-vkCmdDraw-None-07880

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDiscardRectangleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07881

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `discardRectangleEnable` is `VK_TRUE`, then `vkCmdSetDiscardRectangleModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-rasterizerDiscardEnable-09236

If the `VK_EXT_discard_rectangles` extension is enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetDiscardRectangleEnableEXT` in the current command buffer set `discardRectangleEnable` to `VK_TRUE`, then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08913

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08914

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08915

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08916
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08917
If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`
- VUID-vkCmdDraw-dynamicRenderingUnusedAttachments-08918
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDraw-imageView-06183
If the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView` was not `VK_NULL_HANDLE`, the currently bound graphics pipeline **must** have been created with `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-vkCmdDraw-multisampledRenderToSingleSampled-07285
If the current render pass instance was begun with `vkCmdBeginRenderingKHR` with a `VkRenderingInfo::colorAttachmentCount` parameter greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with a `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a sample count equal to the value of `rasterizationSamples` for the currently bound graphics pipeline
- VUID-vkCmdDraw-multisampledRenderToSingleSampled-07286
If `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pDepthAttachment->imageView`
- VUID-vkCmdDraw-multisampledRenderToSingleSampled-07287
If `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pStencilAttachment->imageView`
- VUID-vkCmdDraw-renderPass-06198
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound pipeline **must** have been created with a `VkGraphicsPipelineCreateInfo::renderPass` equal to `VK_NULL_HANDLE`
- VUID-vkCmdDraw-pColorAttachments-08963
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a

graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the `VkRenderingInfo::pColorAttachments->imageView` was not `VK_NULL_HANDLE`, then the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDraw-pDepthAttachment-08964

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, depth test is enabled, depth write is enabled, and the `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDraw-pStencilAttachment-08965

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, stencil test is enabled and the `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDraw-primitivesGeneratedQueryWithRasterizerDiscard-06708

If the `primitivesGeneratedQueryWithRasterizerDiscard` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, `rasterization discard` **must** not be enabled

- VUID-vkCmdDraw-primitivesGeneratedQueryWithNonZeroStreams-06709

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, the bound graphics pipeline **must** not have been created with a non-zero value in `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream`

- VUID-vkCmdDraw-None-07619

If a shader object is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` dynamic state enabled, then `vkCmdSetTessellationDomainOriginEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07620

If the `depthClamp` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDepthClampEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07621

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_POLYGON_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetPolygonModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07622

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetRasterizationSamplesEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07623

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-alphaToCoverageEnable-08919

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and `alphaToCoverageEnable` was `VK_TRUE` in the last call to `vkCmdSetAlphaToCoverageEnableEXT`, then the `Fragment Output Interface` **must** contain a variable for the alpha `Component` word in `Location 0` at `Index 0`

- VUID-vkCmdDraw-alphaToCoverageEnable-08920

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetAlphaToCoverageEnableEXT` in the current command buffer set `alphaToCoverageEnable` to `VK_TRUE`, then the `Fragment Output Interface` **must** contain a variable for the alpha `Component` word in `Location 0` at `Index 0`

- VUID-vkCmdDraw-None-07624

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToCoverageEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07625

If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToOneEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07626

If the `logicOp` feature is enabled, a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetLogicOpEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07627

If the bound graphics pipeline state was created with the

`VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08657

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07628

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08658

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` for any attachment set that attachment's value in `pColorBlendEnables` to `VK_TRUE`, then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07629

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08659

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07630

If the `geometryStreams` feature is enabled, and a shader object is bound to the `VK_SHADER_STAGE_GEOMETRY_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` dynamic state enabled, then `vkCmdSetRasterizationStreamEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07631

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetConservativeRasterizationModeEXT` **must** have been called and not

subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07632

If the **VK_EXT_conservative_rasterization** extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **conservativeRasterizationMode** is **VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT**, then **vkCmdSetExtraPrimitiveOverestimationSizeEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07633

If the **depthClipEnable** feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT** dynamic state, then **vkCmdSetDepthClipEnableEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07635

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT** dynamic state enabled then **vkCmdSetColorBlendAdvancedEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-rasterizerDiscardEnable-09416

If the **VK_EXT_blend_operation_advanced** extension is enabled, and a shader object is bound to the **VK_SHADER_STAGE_FRAGMENT_BIT** stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to **VK_FALSE**, then at least one of **vkCmdSetColorBlendEquationEXT** and **vkCmdSetColorBlendAdvancedEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07636

If the **VK_EXT_provoking_vertex** extension is enabled, a shader object is bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage or a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT** dynamic state enabled, and the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, then **vkCmdSetProvokingVertexModeEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07637

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT** dynamic state enabled then **vkCmdSetLineRasterizationModeEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08666

If any of the **stippledRectangularLines**, **stippledBresenhamLines** or **stippledSmoothLines** features are enabled, and a shader object is bound to any graphics stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to **VK_FALSE**, and the most recent call to

`vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08667

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08668

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07638

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` dynamic state enabled then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08669

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08670

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08671

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines`

features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07849

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_KHR` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stippledLineEnable` is `VK_TRUE`, then `vkCmdSetLineStippleKHR` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-07639

If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` dynamic state enabled, then `vkCmdSetDepthClipNegativeOneToOneEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-09650

If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` dynamic state enabled, and the `current value` of `depthClampEnable` is `VK_TRUE`, then `vkCmdSetDepthClampRangeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-pColorBlendEnables-07470

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` state enabled and the last call to `vkCmdSetColorBlendEnableEXT` set `pColorBlendEnables` for any attachment to `VK_TRUE`, then for those attachments in the subpass the corresponding image view's `format features` **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`

- VUID-vkCmdDraw-rasterizationSamples-07471

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** follow the rules for a `zero-attachment subpass`

- VUID-vkCmdDraw-samples-07472

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `VkPipelineMultisampleStateCreateInfo::rasterizationSamples` parameter used to create

the bound graphics pipeline

- VUID-vkCmdDraw-samples-07473

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state and `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` states enabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `rasterizationSamples` parameter in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDraw-rasterizationSamples-07474

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and neither the `VK_AMD_mixed_attachment_samples` nor the `VK_NV_framebuffer_mixed_samples` extensions are enabled, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDraw-firstAttachment-07476

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDraw-rasterizerDiscardEnable-09417

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDraw-firstAttachment-07477

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDraw-rasterizerDiscardEnable-09418

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is

enabled

- VUID-vkCmdDraw-firstAttachment-07478

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDraw-rasterizerDiscardEnable-09419

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDraw-firstAttachment-07479

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendAdvancedEXT` calls **must** specify the advanced blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDraw-advancedBlendMaxColorAttachments-07480

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` and `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic states enabled and the last calls to `vkCmdSetColorBlendEnableEXT` and `vkCmdSetColorBlendAdvancedEXT` have enabled advanced blending, then the number of active color attachments in the current subpass **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdDraw-primitivesGeneratedQueryWithNonZeroStreams-07481

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, and the bound graphics pipeline was created with `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` state enabled, the last call to `vkCmdSetRasterizationStreamEXT` **must** have set the `rasterizationStream` to zero

- VUID-vkCmdDraw-sampleLocationsPerPixel-07482

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` member of the `VkPipelineMultisampleStateCreateInfo` structure the bound graphics pipeline has been created with

- VUID-vkCmdDraw-sampleLocationsPerPixel-07483

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current value of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` parameter of the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDraw-sampleLocationsEnable-07484

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDraw-sampleLocationsEnable-07485

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDraw-sampleLocationsEnable-07486

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDraw-sampleLocationsEnable-07487

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`

- VUID-vkCmdDraw-sampleLocationsEnable-07936

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current value of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.width`

must evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDraw-sampleLocationsEnable-07937

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the **current value** of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.height` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDraw-sampleLocationsEnable-07938

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the **current value** of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationsPerPixel` **must** equal `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDraw-stippledLineEnable-07495

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR`, then the `stippledRectangularLines` feature **must** be enabled

- VUID-vkCmdDraw-stippledLineEnable-07496

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR`, then the `stippledBresenhamLines` feature **must** be enabled

- VUID-vkCmdDraw-stippledLineEnable-07497

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, then the `stippledSmoothLines` feature **must** be enabled

- VUID-vkCmdDraw-stippledLineEnable-07498

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is

VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR, then the `stippledRectangularLines` feature **must** be enabled and `VkPhysicalDeviceLimits::strictLines` **must** be `VK_TRUE`

- VUID-vkCmdDraw-conservativePointAndLineRasterization-07499

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, `conservativePointAndLineRasterization` is not supported, and the effective primitive topology output by the last pre-rasterization shader stage is a line or point, then the `conservativeRasterizationMode` set by the last call to `vkCmdSetConservativeRasterizationModeEXT` **must** be `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`

- VUID-vkCmdDraw-None-08877

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` dynamic state enabled, and the `current` value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAttachmentFeedbackLoopEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-08684

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_VERTEX_BIT`

- VUID-vkCmdDraw-None-08685

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`

- VUID-vkCmdDraw-None-08686

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`

- VUID-vkCmdDraw-None-08687

If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_GEOMETRY_BIT`

- VUID-vkCmdDraw-None-08688

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-vkCmdDraw-None-08698

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, then all shaders created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag in the same `vkCreateShadersEXT` call **must** also be bound

- VUID-vkCmdDraw-None-08699

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, any stages in between stages whose shaders which did not create a shader with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag as part of

the same `vkCreateShadersEXT` call **must** not have any `VkShaderEXT` bound

- VUID-vkCmdDraw-None-08878

All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges

- VUID-vkCmdDraw-None-08879

All bound graphics shader objects **must** have been created with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdDraw-pDynamicStates-08715

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to `vkCmdSetDepthWriteEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDraw-pDynamicStates-08716

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to `vkCmdSetStencilWriteMask` **must** be 0

- VUID-vkCmdDraw-None-09116

If a shader object is bound to any graphics stage or the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and the format of any color attachment is `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`, the corresponding element of the `pColorWriteMasks` parameter of `vkCmdSetColorWriteMaskEXT` **must** either include all of `VK_COLOR_COMPONENT_R_BIT`, `VK_COLOR_COMPONENT_G_BIT`, and `VK_COLOR_COMPONENT_B_BIT`, or none of them

- VUID-vkCmdDraw-maxFragmentDualSrcAttachments-09239

If `blending` is enabled for any attachment where either the source or destination blend factors for that attachment [use the secondary color input](#), the maximum value of `Location` for any output attachment [statically used](#) in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDraw-None-09548

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, the value of each element of `VkRenderingAttachmentLocationInfoKHR::pColorAttachmentLocations` set by `vkCmdSetRenderingAttachmentLocationsKHR` **must** match the value set for the corresponding element in the currently bound pipeline

- VUID-vkCmdDraw-None-09549

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, input attachment index mappings in the currently bound pipeline **must** match those set for the current render pass instance via `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDraw-None-04007

All vertex input bindings accessed via vertex input variables declared in the vertex

shader entry point's interface **must** have either valid or [VK_NULL_HANDLE](#) buffers bound

- VUID-vkCmdDraw-None-04008

If the [nullDescriptor](#) feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be [VK_NULL_HANDLE](#)

- VUID-vkCmdDraw-None-02721

If [robustBufferAccess](#) is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)

- VUID-vkCmdDraw-None-04914

If there is a shader object bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage or the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#) dynamic state enabled then [vkCmdSetVertexInputEXT](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this draw command

- VUID-vkCmdDraw-Input-07939

If there is a shader object bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage or the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#) dynamic state enabled then all variables with the [Input](#) storage class decorated with [Location](#) in the [Vertex Execution Model OpEntryPoint](#) **must** contain a location in [VkVertexInputAttributeDescription2EXT::location](#)

- VUID-vkCmdDraw-Input-08734

If there is a shader object bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage or the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#) dynamic state enabled and either [legacyVertexAttributes](#) is not enabled or the SPIR-V Type associated with a given [Input](#) variable of the corresponding [Location](#) in the [Vertex Execution Model OpEntryPoint](#) is 64-bit, then the numeric type associated with all [Input](#) variables of the corresponding [Location](#) in the [Vertex Execution Model OpEntryPoint](#) **must** be the same as [VkVertexInputAttributeDescription2EXT::format](#)

- VUID-vkCmdDraw-format-08936

If there is a shader object bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage or the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#) dynamic state enabled and [VkVertexInputAttributeDescription2EXT::format](#) has a 64-bit component, then the scalar width associated with all [Input](#) variables of the corresponding [Location](#) in the [Vertex Execution Model OpEntryPoint](#) **must** be 64-bit

- VUID-vkCmdDraw-format-08937

If there is a shader object bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage or the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#) dynamic state enabled and the scalar width associated with a [Location](#) decorated [Input](#) variable in the [Vertex Execution Model OpEntryPoint](#) is 64-bit, then the corresponding [VkVertexInputAttributeDescription2EXT::format](#) **must** have a 64-bit component

- VUID-vkCmdDraw-None-09203

If there is a shader object bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage or the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#) dynamic state enabled and [VkVertexInputAttributeDescription2EXT::format](#) has a 64-bit

component, then all **Input** variables at the corresponding **Location** in the **Vertex Execution Model OpEntryPoint** **must** not use components that are not present in the format

- VUID-vkCmdDraw-None-04879

If there is a shader object bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage then **vkCmdSetPrimitiveRestartEnableEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDraw-None-09637

If the **primitiveTopologyListRestart** feature is not enabled, the topology is **VK_PRIMITIVE_TOPOLOGY_POINT_LIST**, **VK_PRIMITIVE_TOPOLOGY_LINE_LIST**, **VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST**, **VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY**, or **VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY**, there is a shader object bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage then **vkCmdSetPrimitiveRestartEnableEXT** **must** be **VK_FALSE**

- VUID-vkCmdDraw-pNext-09461

If the bound graphics pipeline state was created with **VkPipelineVertexInputDivisorStateCreateInfoKHR** in the **pNext** chain of **VkGraphicsPipelineCreateInfo::pVertexInputState**, any member of **VkPipelineVertexInputDivisorStateCreateInfoKHR::pVertexBindingDivisors** has a value other than **1** in **divisor**, and **VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance** is **VK_FALSE**, then **firstInstance** **must** be **0**

- VUID-vkCmdDraw-None-09462

If **shader objects** are used for drawing or the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_VERTEX_INPUT_EXT** dynamic state enabled, any member of the **pVertexBindingDescriptions** parameter to the **vkCmdSetVertexInputEXT** call that sets this dynamic state has a value other than **1** in **divisor**, and **VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance** is **VK_FALSE**, then **firstInstance** **must** be **0**

Valid Usage (Implicit)

- VUID-vkCmdDraw-commandBuffer-parameter

commandBuffer **must** be a valid **VkCommandBuffer** handle

- VUID-vkCmdDraw-commandBuffer-recording

commandBuffer **must** be in the **recording state**

- VUID-vkCmdDraw-commandBuffer-cmdpool

The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics operations

- VUID-vkCmdDraw-renderpass

This command **must** only be called inside of a render pass instance

- VUID-vkCmdDraw-videocoding

This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	Action

To record an indexed draw, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDrawIndexed(
    VkCommandBuffer          commandBuffer,
    uint32_t                 indexCount,
    uint32_t                 instanceCount,
    uint32_t                 firstIndex,
    int32_t                  vertexOffset,
    uint32_t                 firstInstance);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `indexCount` is the number of vertices to draw.
- `instanceCount` is the number of instances to draw.
- `firstIndex` is the base index within the index buffer.
- `vertexOffset` is the value added to the vertex index before indexing into the vertex buffer.
- `firstInstance` is the instance ID of the first instance to draw.

When the command is executed, primitives are assembled using the current primitive topology and `indexCount` vertices whose indices are retrieved from the index buffer. The index buffer is treated as an array of tightly packed unsigned integers of size defined by the `vkCmdBindIndexBuffer2KHR::indexType` or the `vkCmdBindIndexBuffer::indexType` parameter with which the buffer was bound.

The first vertex index is at an offset of $\text{firstIndex} \times \text{indexSize} + \text{offset}$ within the bound index buffer, where `offset` is the offset specified by `vkCmdBindIndexBuffer` or `vkCmdBindIndexBuffer2KHR`, and `indexSize` is the byte size of the type specified by `indexType`. Subsequent index values are retrieved from consecutive locations in the index buffer. Indices are first compared to the primitive restart value, then zero extended to 32 bits (if the `indexType` is `VK_INDEX_TYPE_UINT8_KHR` or `VK_INDEX_TYPE_UINT16`) and have `vertexOffset` added to them, before being supplied as the

`vertexIndex` value.

The primitives are drawn `instanceCount` times with `instanceIndex` starting with `firstInstance` and increasing sequentially for each instance. The assembled primitives execute the bound graphics pipeline.

Valid Usage

- VUID-vkCmdDrawIndexed-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndexed-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndexed-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDrawIndexed-unnormalizedCoordinates-09636
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`
- VUID-vkCmdDrawIndexed-None-06479
If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`
- VUID-vkCmdDrawIndexed-None-02691
If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`
- VUID-vkCmdDrawIndexed-None-07888
If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`
- VUID-vkCmdDrawIndexed-None-02692
If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`
- VUID-vkCmdDrawIndexed-None-02693
If the `VK_EXT_filter_cubic` extension is not enabled and any `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, it **must** not have a `VkImageViewType` of

VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY

- VUID-vkCmdDrawIndexed-filterCubic-02694

Any [VkImageView](#) being sampled with [VK_FILTER_CUBIC_EXT](#) as a result of this command **must** have a [VkImageViewType](#) and format that supports cubic filtering, as specified by [VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic](#) returned by [vkGetPhysicalDeviceImageFormatProperties2KHR](#)

- VUID-vkCmdDrawIndexed-filterCubicMinmax-02695

Any [VkImageView](#) being sampled with [VK_FILTER_CUBIC_EXT](#) with a reduction mode of either [VK_SAMPLER_REDUCTION_MODE_MIN](#) or [VK_SAMPLER_REDUCTION_MODE_MAX](#) as a result of this command **must** have a [VkImageViewType](#) and format that supports cubic filtering together with minmax filtering, as specified by [VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax](#) returned by [vkGetPhysicalDeviceImageFormatProperties2KHR](#)

- VUID-vkCmdDrawIndexed-OpTypeImage-07027

For any [VkImageView](#) being written as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDrawIndexed-OpTypeImage-07028

For any [VkImageView](#) being read as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDrawIndexed-OpTypeImage-07029

For any [VkBufferView](#) being written as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDrawIndexed-OpTypeImage-07030

Any [VkBufferView](#) being read as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#) then the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDrawIndexed-None-08600

For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndexed-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndexed-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in

the descriptor sets used by this command

- VUID-vkCmdDrawIndexed-maintenance4-08602

If the **maintenance4** feature is not enabled, then for each push constant that is statically used by a **bound shader**, a push constant value **must** have been set for the same pipeline bind point, with a **VkPipelineLayout** that is compatible for push constants, with the **VkPipelineLayout** used to create the current **VkPipeline** or the **VkDescriptorSetLayout** and **VkPushConstantRange** arrays used to create the current **VkShaderEXT**, as described in **Pipeline Layout Compatibility**

- VUID-vkCmdDrawIndexed-None-08114

Descriptors in each bound descriptor set, specified via **vkCmdBindDescriptorSets**, **must** be valid as described by **descriptor validity** if they are statically used by a **bound shader**

- VUID-vkCmdDrawIndexed-None-08606

If the **shaderObject** feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDrawIndexed-None-08608

If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the **VkPipeline** object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDrawIndexed-None-08609

If the **VkPipeline** object bound to the pipeline bind point used by this command or any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a **VkSampler** object that uses unnormalized coordinates, that sampler **must** not be used to sample from any **VkImage** with a **VkImageView** of the type **VK_IMAGE_VIEW_TYPE_3D**, **VK_IMAGE_VIEW_TYPE_CUBE**, **VK_IMAGE_VIEW_TYPE_1D_ARRAY**, **VK_IMAGE_VIEW_TYPE_2D_ARRAY** or **VK_IMAGE_VIEW_TYPE_CUBE_ARRAY**, in any shader stage

- VUID-vkCmdDrawIndexed-None-08610

If the **VkPipeline** object bound to the pipeline bind point used by this command or any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a **VkSampler** object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V **OpImageSample*** or **OpImageSparseSample*** instructions with **ImplicitLod**, **Dref** or **Proj** in their name, in any shader stage

- VUID-vkCmdDrawIndexed-None-08611

If the **VkPipeline** object bound to the pipeline bind point used by this command or any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a **VkSampler** object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V **OpImageSample*** or **OpImageSparseSample*** instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDrawIndexed-None-08607

If the **shaderObject** feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and **VK_NULL_HANDLE** shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command

- VUID-vkCmdDrawIndexed-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexed-None-08612

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexed-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexed-None-08613

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexed-None-06550

If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y_C_R conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions

- VUID-vkCmdDrawIndexed-ConstOffset-06551

If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y_C_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands

- VUID-vkCmdDrawIndexed-viewType-07752

If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDrawIndexed-format-07753

If a [VkImageView](#) is accessed as a result of this command, then the [numeric type](#) of the image view's [format](#) and the [Sampled Type](#) operand of the [OpTypeImage](#) **must** match

- VUID-vkCmdDrawIndexed-OpImageWrite-08795

If a [VkImageView](#) created with a format other than [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDrawIndexed-OpImageWrite-08796

If a [VkImageView](#) created with the format [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have four components

- VUID-vkCmdDrawIndexed-OpImageWrite-04469

If a [VkBufferView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as

the buffer view's format

- VUID-vkCmdDrawIndexed-SampledType-04470

If a `VkImageView` with a `VkFormat` that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64

- VUID-vkCmdDrawIndexed-SampledType-04471

If a `VkImageView` with a `VkFormat` that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32

- VUID-vkCmdDrawIndexed-SampledType-04472

If a `VkBufferView` with a `VkFormat` that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64

- VUID-vkCmdDrawIndexed-SampledType-04473

If a `VkBufferView` with a `VkFormat` that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32

- VUID-vkCmdDrawIndexed-sparseImageInt64Atomics-04474

If the `sparseImageInt64Atomics` feature is not enabled, `VkImage` objects created with the `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command

- VUID-vkCmdDrawIndexed-sparseImageInt64Atomics-04475

If the `sparseImageInt64Atomics` feature is not enabled, `VkBuffer` objects created with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command

- VUID-vkCmdDrawIndexed-None-07288

Any shader invocation executed by this command **must** `terminate`

- VUID-vkCmdDrawIndexed-None-09600

If a descriptor with type equal to any of `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDrawIndexed-renderPass-02684

The current render pass **must** be `compatible` with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`

- VUID-vkCmdDrawIndexed-subpass-02685

The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`

- VUID-vkCmdDrawIndexed-None-07748

If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set

- VUID-vkCmdDrawIndexed-OpTypeImage-07468

If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDrawIndexed-None-07469

Input attachment views accessed in a subpass **must** be created with the same `VkFormat` as the corresponding subpass definition, and be created with a `VkImageView` that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the currently bound `VkFramebuffer` as specified by [Fragment Input Attachment Compatibility](#)

- VUID-vkCmdDrawIndexed-pDepthInputAttachmentIndex-09595

Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by `VkRenderingInputAttachmentIndexInfoKHR`, or no `InputAttachmentIndex` if `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` or `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are `NULL`, **must** be created with a `VkImageView` that is compatible with the corresponding color, depth, or stencil attachment in `VkRenderingInfo`

- VUID-vkCmdDrawIndexed-pDepthInputAttachmentIndex-09596

Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` and `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are non-`NULL`

- VUID-vkCmdDrawIndexed-InputAttachmentIndex-09597

If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDrawIndexed-None-06537

Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexed-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_COLOR_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexed-None-09001

If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_DEPTH_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexed-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_STENCIL_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexed-None-09003

If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command

- VUID-vkCmdDrawIndexed-None-06539

If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment

- VUID-vkCmdDrawIndexed-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, `depth writes` **must** be disabled

- VUID-vkCmdDrawIndexed-None-06887

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, `all stencil ops` **must** be `VK_STENCIL_OP_KEEP`

- VUID-vkCmdDrawIndexed-None-07831

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VIEWPORT`

dynamic state enabled then `vkCmdSetViewport` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07832

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SCISSOR` dynamic state enabled then `vkCmdSetScissor` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07833

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic state enabled then `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08617

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08618

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08619

If a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07834

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BIAS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBiasEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` or `vkCmdSetDepthBias2EXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07835

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic state enabled then `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08621

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent

call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer set any element of `pColorBlendEnables` to `VK_TRUE`, and the most recent call to `vkCmdSetColorBlendEquationEXT` in the current command buffer set the same element of `pColorBlendEquations` to a `VkColorBlendEquationEXT` structure with any `VkBlendFactor` member with a value of `VK_BLEND_FACTOR_CONSTANT_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR`, `VK_BLEND_FACTOR_CONSTANT_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`, `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07836

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBoundsTestEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07837

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilCompareMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilWriteMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07839

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_REFERENCE` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilReference` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-maxMultiviewInstanceIndex-02688

If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to `VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex`

- VUID-vkCmdDrawIndexed-sampleLocationsEnable-02689

If the bound graphics pipeline was created with

VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable set to VK_TRUE and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT bit set

- VUID-vkCmdDrawIndexed-None-07634

If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT dynamic state enabled, and the **current value** of rasterizerDiscardEnable is VK_FALSE, then vkCmdSetSampleLocationsEnableEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-06666

If the VK_EXT_sample_locations extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the **current value** of rasterizerDiscardEnable is VK_FALSE, and the **current value** of sampleLocationsEnable is VK_TRUE, then vkCmdSetSampleLocationsEXT **must** have been called and not subsequently invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's **format features** do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the blendEnable member of the corresponding element of the pAttachments member of pColorBlendState **must** be VK_FALSE

- VUID-vkCmdDrawIndexed-None-08643

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, and the most recent call to vkCmdSetRasterizerDiscardEnableEXT in the current command buffer set rasterizerDiscardEnable to VK_FALSE, then for each color attachment in the render pass, if the corresponding image view's **format features** do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding member of pColorBlendEnables in the most recent call to vkCmdSetColorBlendEnableEXT in the current command buffer that affected that attachment index **must** have been VK_FALSE

- VUID-vkCmdDrawIndexed-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then rasterizationSamples for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexed-None-08644

If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetRasterizerDiscardEnableEXT in the current command buffer set rasterizerDiscardEnable to VK_FALSE,

then the most recent call to vkCmdSetRasterizationSamplesEXT in the current command buffer **must** have set rasterizationSamples to be the same as the number of samples for the current render pass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexed-None-08876

If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with [vkCmdBeginRenderingKHR](#)

- VUID-vkCmdDrawIndexed-imageView-06172

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pDepthAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pDepthAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndexed-imageView-06173

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pStencilAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pStencilAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndexed-imageView-06174

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pDepthAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pDepthAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL](#), this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndexed-imageView-06175

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pStencilAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pStencilAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndexed-imageView-06176

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pDepthAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pDepthAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndexed-imageView-06177

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pStencilAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pStencilAttachment](#) is [VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndexed-viewMask-06178

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the currently bound graphics pipeline **must** have been created with a [VkPipelineRenderingCreateInfo::viewMask](#) equal to [VkRenderingInfo::viewMask](#)

- VUID-vkCmdDrawIndexed-colorAttachmentCount-06179

If the [dynamicRenderingUnusedAttachments](#) feature is not enabled and the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the currently bound graphics pipeline **must** have been created with a [VkPipelineRenderingCreateInfo::colorAttachmentCount](#) equal to [VkRenderingInfo::colorAttachmentCount](#)

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08910

If the [dynamicRenderingUnusedAttachments](#) feature is not enabled, and the current render

pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08912

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` equal to `VK_NULL_HANDLE` **must** have the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound pipeline equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08911

If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline, or the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats`, if it exists, **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexed-None-07749

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08646

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-attachmentCount-07750

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then the `attachmentCount` parameter of `vkCmdSetColorWriteEnableEXT` **must** be greater than or equal to the `VkPipelineColorBlendStateCreateInfo::attachmentCount` of the currently bound graphics pipeline

- VUID-vkCmdDrawIndexed-None-08647

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set

`rasterizerDiscardEnable` to `VK_FALSE`, then the `attachmentCount` parameter of most recent call to `vkCmdSetColorWriteEnableEXT` in the current command buffer **must** be greater than or equal to the number of color attachments in the current render pass instance

- VUID-vkCmdDrawIndexed-None-07751

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state enabled then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command for each discard rectangle in `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount`

- VUID-vkCmdDrawIndexed-None-07880

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDiscardRectangleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07881

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `discardRectangleEnable` is `VK_TRUE`, then `vkCmdSetDiscardRectangleModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-rasterizerDiscardEnable-09236

If the `VK_EXT_discard_rectangles` extension is enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetDiscardRectangleEnableEXT` in the current command buffer set `discardRectangleEnable` to `VK_TRUE`, then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08913

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08914

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08915

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08916

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08917

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdDrawIndexed-dynamicRenderingUnusedAttachments-08918

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexed-imageView-06183

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView` was not `VK_NULL_HANDLE`, the currently bound graphics pipeline **must** have been created with `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-vkCmdDrawIndexed-multisampledRenderToSingleSampled-07285

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` with a `VkRenderingInfo::colorAttachmentCount` parameter greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with a `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a sample count equal to the value of `rasterizationSamples` for the currently bound graphics pipeline

- VUID-vkCmdDrawIndexed-multisampledRenderToSingleSampled-07286

If `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDrawIndexed-multisampledRenderToSingleSampled-07287

If `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdDrawIndexed-renderPass-06198
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound pipeline **must** have been created with a `VkGraphicsPipelineCreateInfo::renderPass` equal to `VK_NULL_HANDLE`
- VUID-vkCmdDrawIndexed-pColorAttachments-08963
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the `VkRenderingInfo::pColorAttachments->imageView` was not `VK_NULL_HANDLE`, then the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndexed-pDepthAttachment-08964
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, depth test is enabled, depth write is enabled, and the `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndexed-pStencilAttachment-08965
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, stencil test is enabled and the `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndexed-primitivesGeneratedQueryWithRasterizerDiscard-06708
If the `primitivesGeneratedQueryWithRasterizerDiscard` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, `rasterization discard` **must** not be enabled
- VUID-vkCmdDrawIndexed-primitivesGeneratedQueryWithNonZeroStreams-06709
If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, the bound graphics pipeline **must** not have been created with a non-zero value in `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream`
- VUID-vkCmdDrawIndexed-None-07619
If a shader object is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` dynamic state enabled, then `vkCmdSetTessellationDomainOriginEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexed-None-07620
If the `depthClamp` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDepthClampEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07621

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_POLYGON_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetPolygonModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07622

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetRasterizationSamplesEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07623

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-alphaToCoverageEnable-08919

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and `alphaToCoverageEnable` was `VK_TRUE` in the last call to `vkCmdSetAlphaToCoverageEnableEXT`, then the `Fragment Output Interface` **must** contain a variable for the alpha `Component` word in `Location 0` at `Index 0`

- VUID-vkCmdDrawIndexed-alphaToCoverageEnable-08920

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetAlphaToCoverageEnableEXT` in the current command buffer set `alphaToCoverageEnable` to `VK_TRUE`, then the `Fragment Output Interface` **must** contain a variable for the alpha `Component` word in `Location 0` at `Index 0`

- VUID-vkCmdDrawIndexed-None-07624

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToCoverageEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07625

If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToOneEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07626

If the `logicOp` feature is enabled, a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created

with the `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetLogicOpEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07627

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08657

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07628

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08658

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` for any attachment set that attachment's value in `pColorBlendEnables` to `VK_TRUE`, then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07629

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08659

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07630

If the `geometryStreams` feature is enabled, and a shader object is bound to the `VK_SHADER_STAGE_GEOMETRY_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` dynamic state enabled, then `vkCmdSetRasterizationStreamEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07631

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetConservativeRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07632

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT` dynamic state enabled, the current value of `rasterizerDiscardEnable` is `VK_FALSE`, and the current value of `conservativeRasterizationMode` is `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`, then `vkCmdSetExtraPrimitiveOverestimationSizeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07633

If the `depthClipEnable` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT` dynamic state, then `vkCmdSetDepthClipEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07635

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-rasterizerDiscardEnable-09416

If the `VK_EXT_blend_operation_advanced` extension is enabled, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then at least one of `vkCmdSetColorBlendEquationEXT` and `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07636

If the `VK_EXT_provoking_vertex` extension is enabled, a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetProvokingVertexModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07637

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic state enabled then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently

[invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08666

If any of the [stippledRectangularLines](#), [stippledBresenhamLines](#) or [stippledSmoothLines](#) features are enabled, and a shader object is bound to any graphics stage, and the most recent call to [vkCmdSetRasterizerDiscardEnableEXT](#) in the current command buffer set [rasterizerDiscardEnable](#) to [VK_FALSE](#), and the most recent call to [vkCmdSetPolygonModeEXT](#) in the current command buffer set [polygonMode](#) to [VK_POLYGON_MODE_LINE](#), then [vkCmdSetLineRasterizationModeEXT](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08667

If any of the [stippledRectangularLines](#), [stippledBresenhamLines](#) or [stippledSmoothLines](#) features are enabled, and a shader object is bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage, and the most recent call to [vkCmdSetRasterizerDiscardEnableEXT](#) in the current command buffer set [rasterizerDiscardEnable](#) to [VK_FALSE](#), and the most recent call to [vkCmdSetPrimitiveTopologyEXT](#) in the current command buffer set [primitiveTopology](#) to any line topology, then [vkCmdSetLineRasterizationModeEXT](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08668

If any of the [stippledRectangularLines](#), [stippledBresenhamLines](#) or [stippledSmoothLines](#) features are enabled, and a shader object that outputs line primitives is bound to the [VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT](#) or [VK_SHADER_STAGE_GEOMETRY_BIT](#) stage, and the most recent call to [vkCmdSetRasterizerDiscardEnableEXT](#) in the current command buffer set [rasterizerDiscardEnable](#) to [VK_FALSE](#), then [vkCmdSetLineRasterizationModeEXT](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07638

If the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT](#) dynamic state enabled then [vkCmdSetLineStippleEnableEXT](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08669

If any of the [stippledRectangularLines](#), [stippledBresenhamLines](#) or [stippledSmoothLines](#) features are enabled, and a shader object is bound to any graphics stage, and the most recent call to [vkCmdSetRasterizerDiscardEnableEXT](#) in the current command buffer set [rasterizerDiscardEnable](#) to [VK_FALSE](#), and the most recent call to [vkCmdSetPolygonModeEXT](#) in the current command buffer set [polygonMode](#) to [VK_POLYGON_MODE_LINE](#), then [vkCmdSetLineStippleEnableEXT](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08670

If any of the [stippledRectangularLines](#), [stippledBresenhamLines](#) or [stippledSmoothLines](#) features are enabled, and a shader object is bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage, and the most recent call to [vkCmdSetRasterizerDiscardEnableEXT](#) in the current

command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08671

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07849

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_KHR` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stippledLineEnable` is `VK_TRUE`, then `vkCmdSetLineStippleKHR` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-07639

If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` dynamic state enabled, then `vkCmdSetDepthClipNegativeOneToOneEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-09650

If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` dynamic state enabled, and the `current value` of `depthClampEnable` is `VK_TRUE`, then `vkCmdSetDepthClampRangeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-pColorBlendEnables-07470

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` state enabled and the last call to `vkCmdSetColorBlendEnableEXT` set `pColorBlendEnables` for any attachment to `VK_TRUE`, then for those attachments in the subpass the corresponding image view's `format features` **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`

- VUID-vkCmdDrawIndexed-rasterizationSamples-07471

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** follow the rules for a `zero-attachment` subpass

- VUID-vkCmdDrawIndexed-samples-07472

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `VkPipelineMultisampleStateCreateInfo::rasterizationSamples` parameter used to create the bound graphics pipeline

- VUID-vkCmdDrawIndexed-samples-07473

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state and `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` states enabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `rasterizationSamples` parameter in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndexed-rasterizationSamples-07474

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and neither the `VK_AMD_mixed_attachment_samples` nor the `VK_NV_framebuffer_mixed_samples` extensions are enabled, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexed-firstAttachment-07476

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexed-rasterizerDiscardEnable-09417

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexed-firstAttachment-07477

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndexed-rasterizerDiscardEnable-09418

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most

recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndexed-firstAttachment-07478

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexed-rasterizerDiscardEnable-09419

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexed-firstAttachment-07479

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendAdvancedEXT` calls **must** specify the advanced blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndexed-advancedBlendMaxColorAttachments-07480

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` and `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic states enabled and the last calls to `vkCmdSetColorBlendEnableEXT` and `vkCmdSetColorBlendAdvancedEXT` have enabled advanced blending, then the number of active color attachments in the current subpass **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdDrawIndexed-primitivesGeneratedQueryWithNonZeroStreams-07481

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, and the bound graphics pipeline was created with `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` state enabled, the last call to `vkCmdSetRasterizationStreamEXT` **must** have set the `rasterizationStream` to zero

- VUID-vkCmdDrawIndexed-sampleLocationsPerPixel-07482

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the

`VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` member of the `VkPipelineMultisampleStateCreateInfo` structure the bound graphics pipeline has been created with

- VUID-vkCmdDrawIndexed-sampleLocationsPerPixel-07483

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` parameter of the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07484

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07485

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07486

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07487

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, the fragment shader

code **must** not statically use the extended instruction `InterpolateAtSample`

- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07936

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current value of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07937

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current value of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.height` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndexed-sampleLocationsEnable-07938

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current value of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationsPerPixel` **must** equal `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndexed-stippledLineEnable-07495

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR`, then the `stippledRectangularLines` feature **must** be enabled

- VUID-vkCmdDrawIndexed-stippledLineEnable-07496

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR`, then the `stippledBresenhamLines` feature **must** be enabled

- VUID-vkCmdDrawIndexed-stippledLineEnable-07497

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, then the `stippledSmoothLines` feature

must be enabled

- VUID-vkCmdDrawIndexed-stippledLineEnable-07498

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR`, then the `stippledRectangularLines` feature **must** be enabled and `VkPhysicalDeviceLimits::strictLines` **must** be `VK_TRUE`

- VUID-vkCmdDrawIndexed-conservativePointAndLineRasterization-07499

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, `conservativePointAndLineRasterization` is not supported, and the effective primitive topology output by the last pre-rasterization shader stage is a line or point, then the `conservativeRasterizationMode` set by the last call to `vkCmdSetConservativeRasterizationModeEXT` **must** be `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`

- VUID-vkCmdDrawIndexed-None-08877

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAttachmentFeedbackLoopEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-08684

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_VERTEX_BIT`

- VUID-vkCmdDrawIndexed-None-08685

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`

- VUID-vkCmdDrawIndexed-None-08686

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`

- VUID-vkCmdDrawIndexed-None-08687

If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_GEOMETRY_BIT`

- VUID-vkCmdDrawIndexed-None-08688

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-vkCmdDrawIndexed-None-08698

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, then all shaders created with the

`VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag in the same `vkCreateShadersEXT` call **must** also be bound

- VUID-vkCmdDrawIndexed-None-08699

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, any stages in between stages whose shaders which did not create a shader with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag as part of the same `vkCreateShadersEXT` call **must** not have any `VkShaderEXT` bound

- VUID-vkCmdDrawIndexed-None-08878

All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges

- VUID-vkCmdDrawIndexed-None-08879

All bound graphics shader objects **must** have been created with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdDrawIndexed-pDynamicStates-08715

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to `vkCmdSetDepthWriteEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndexed-pDynamicStates-08716

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to `vkCmdSetStencilWriteMask` **must** be 0

- VUID-vkCmdDrawIndexed-None-09116

If a shader object is bound to any graphics stage or the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and the format of any color attachment is `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`, the corresponding element of the `pColorWriteMasks` parameter of `vkCmdSetColorWriteMaskEXT` **must** either include all of `VK_COLOR_COMPONENT_R_BIT`, `VK_COLOR_COMPONENT_G_BIT`, and `VK_COLOR_COMPONENT_B_BIT`, or none of them

- VUID-vkCmdDrawIndexed-maxFragmentDualSrcAttachments-09239

If **blending** is enabled for any attachment where either the source or destination blend factors for that attachment **use the secondary color input**, the maximum value of `Location` for any output attachment **statically used** in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDrawIndexed-None-09548

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, the value of each element of `VkRenderingAttachmentLocationInfoKHR::pColorAttachmentLocations` set by `vkCmdSetRenderingAttachmentLocationsKHR` **must** match the value set for the corresponding element in the currently bound pipeline

- VUID-vkCmdDrawIndexed-None-09549

If the current render pass was begun with [vkCmdBeginRenderingKHR](#), and there is no shader object bound to any graphics stage, input attachment index mappings in the currently bound pipeline **must** match those set for the current render pass instance via [VkRenderingInputAttachmentIndexInfoKHR](#)

- VUID-vkCmdDrawIndexed-None-04007

All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or [VK_NULL_HANDLE](#) buffers bound

- VUID-vkCmdDrawIndexed-None-04008

If the [nullDescriptor](#) feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be [VK_NULL_HANDLE](#)

- VUID-vkCmdDrawIndexed-None-02721

If [robustBufferAccess](#) is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)

- VUID-vkCmdDrawIndexed-None-04914

If there is a shader object bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage or the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#) dynamic state enabled then [vkCmdSetVertexInputEXT](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this draw command

- VUID-vkCmdDrawIndexed-Input-07939

If there is a shader object bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage or the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#) dynamic state enabled then all variables with the [Input](#) storage class decorated with [Location](#) in the [Vertex Execution Model OpEntryPoint](#) **must** contain a location in [VkVertexInputAttributeDescription2EXT::location](#)

- VUID-vkCmdDrawIndexed-Input-08734

If there is a shader object bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage or the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#) dynamic state enabled and either [legacyVertexAttributes](#) is not enabled or the SPIR-V Type associated with a given [Input](#) variable of the corresponding [Location](#) in the [Vertex Execution Model OpEntryPoint](#) is 64-bit, then the numeric type associated with all [Input](#) variables of the corresponding [Location](#) in the [Vertex Execution Model OpEntryPoint](#) **must** be the same as [VkVertexInputAttributeDescription2EXT::format](#)

- VUID-vkCmdDrawIndexed-format-08936

If there is a shader object bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage or the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#) dynamic state enabled and [VkVertexInputAttributeDescription2EXT::format](#) has a 64-bit component, then the scalar width associated with all [Input](#) variables of the corresponding [Location](#) in the [Vertex Execution Model OpEntryPoint](#) **must** be 64-bit

- VUID-vkCmdDrawIndexed-format-08937

If there is a shader object bound to the [VK_SHADER_STAGE_VERTEX_BIT](#) stage or the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#) dynamic

state enabled and the scalar width associated with a `Location` decorated `Input` variable in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the corresponding `VkVertexInputAttributeDescription2EXT::format` **must** have a 64-bit component

- VUID-vkCmdDrawIndexed-None-09203

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and `VkVertexInputAttributeDescription2EXT::format` has a 64-bit component, then all `Input` variables at the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** not use components that are not present in the format

- VUID-vkCmdDrawIndexed-None-04879

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexed-None-09637

If the `primitiveTopologyListRestart` feature is not enabled, the topology is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndexed-None-07312

If `maintenance6` is not enabled, a valid index buffer **must** be bound

- VUID-vkCmdDrawIndexed-pNext-09461

If the bound graphics pipeline state was created with `VkPipelineVertexInputDivisorStateCreateInfoKHR` in the `pNext` chain of `VkGraphicsPipelineCreateInfo::pVertexInputState`, any member of `VkPipelineVertexInputDivisorStateCreateInfoKHR::pVertexBindingDivisors` has a value other than 1 in `divisor`, and `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance` is `VK_FALSE`, then `firstInstance` **must** be 0

- VUID-vkCmdDrawIndexed-None-09462

If `shader objects` are used for drawing or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled, any member of the `pVertexBindingDescriptions` parameter to the `vkCmdSetVertexInputEXT` call that sets this dynamic state has a value other than 1 in `divisor`, and `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance` is `VK_FALSE`, then `firstInstance` **must** be 0

- VUID-vkCmdDrawIndexed-robustBufferAccess2-08798

If `robustBufferAccess2` is not enabled, $(\text{indexSize} \times (\text{firstIndex} + \text{indexCount}) + \text{offset})$ **must** be less than or equal to the size of the bound index buffer, with `indexSize` being based on the type specified by `indexType`, where the index buffer, `indexType`, and `offset` are specified via `vkCmdBindIndexBuffer` or `vkCmdBindIndexBuffer2KHR`. If `vkCmdBindIndexBuffer2KHR` is used to bind the index buffer, the size of the bound index buffer is `vkCmdBindIndexBuffer2KHR::size`

Valid Usage (Implicit)

- VUID-vkCmdDrawIndexed-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdDrawIndexed-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdDrawIndexed-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdDrawIndexed-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndexed-vidcoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	Action

To record an ordered sequence of draws which have no state changes between them, call:

```
// Provided by VK_EXT_multi_draw
void vkCmdDrawMultiEXT(
    VkCommandBuffer          commandBuffer,
    uint32_t                 drawCount,
    const VkMultiDrawInfoEXT* pVertexInfo,
    uint32_t                 instanceCount,
    uint32_t                 firstInstance,
    uint32_t                 stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `drawCount` is the number of draws to execute, and **can** be zero.

- `pVertexInfo` is a pointer to an array of `VkMultiDrawInfoEXT` with vertex information to be drawn.
- `instanceCount` is the number of instances per draw.
- `firstInstance` is the instance ID of the first instance in each draw.
- `stride` is the byte stride between consecutive elements of `pVertexInfo`.

The number of draws recorded is `drawCount`, with each draw reading, sequentially, a `firstVertex` and a `vertexCount` from `pVertexInfo`. For each recorded draw, primitives are assembled as for `vkCmdDraw`, and drawn `instanceCount` times with `instanceIndex` starting with `firstInstance` and sequentially for each instance.

Valid Usage

- VUID-vkCmdDrawMultiEXT-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawMultiEXT-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawMultiEXT-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDrawMultiEXT-unnormalizedCoordinates-09636
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`
- VUID-vkCmdDrawMultiEXT-None-06479
If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`
- VUID-vkCmdDrawMultiEXT-None-02691
If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`
- VUID-vkCmdDrawMultiEXT-None-07888
If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`
- VUID-vkCmdDrawMultiEXT-None-02692
If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then

the image view's format features **must** contain
`VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`

- VUID-vkCmdDrawMultiEXT-None-02693

If the `VK_EXT_filter_cubic` extension is not enabled and any `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, it **must** not have a `VkImageViewType` of `VK_IMAGE_VIEW_TYPE_3D`, `VK_IMAGE_VIEW_TYPE_CUBE`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`

- VUID-vkCmdDrawMultiEXT-filterCubic-02694

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdDrawMultiEXT-filterCubicMinmax-02695

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` with a reduction mode of either `VK_SAMPLER_REDUCTION_MODE_MIN` or `VK_SAMPLER_REDUCTION_MODE_MAX` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering together with minmax filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdDrawMultiEXT-OpTypeImage-07027

For any `VkImageView` being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawMultiEXT-OpTypeImage-07028

For any `VkImageView` being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's format features **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawMultiEXT-OpTypeImage-07029

For any `VkBufferView` being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's buffer features **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawMultiEXT-OpTypeImage-07030

Any `VkBufferView` being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown` then the view's buffer features **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawMultiEXT-None-08600

For each set n that is statically used by a bound shader, a descriptor set **must** have been bound to n at the same pipeline bind point, with a `VkPipelineLayout` that is compatible for set n , with the `VkPipelineLayout` used to create the current `VkPipeline` or the `VkDescriptorSetLayout` array used to create the current `VkShaderEXT`, as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawMultiEXT-None-08601

For each push constant that is statically used by a bound shader, a push constant value **must** have been set for the same pipeline bind point, with a `VkPipelineLayout` that is compatible for push constants, with the `VkPipelineLayout` used to create the current

[VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#) , as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawMultiEXT-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDrawMultiEXT-maintenance4-08602

If the [maintenance4](#) feature is not enabled, then for each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) and [VkPushConstantRange](#) arrays used to create the current [VkShaderEXT](#) , as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawMultiEXT-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDrawMultiEXT-None-08606

If the [shaderObject](#) feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDrawMultiEXT-None-08608

If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDrawMultiEXT-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDrawMultiEXT-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage

- VUID-vkCmdDrawMultiEXT-None-08611

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDrawMultiEXT-None-08607

If the `shaderObject` feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and `VK_NULL_HANDLE` shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command

- VUID-vkCmdDrawMultiEXT-uniformBuffers-06935

If any stage of the `VkPipeline` object bound to the pipeline bind point used by this command accesses a uniform buffer, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawMultiEXT-None-08612

If the `robustBufferAccess` feature is not enabled, and any `VkShaderEXT` bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawMultiEXT-storageBuffers-06936

If any stage of the `VkPipeline` object bound to the pipeline bind point used by this command accesses a storage buffer, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawMultiEXT-None-08613

If the `robustBufferAccess` feature is not enabled, and any `VkShaderEXT` bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawMultiEXT-None-06550

If a bound shader accesses a `VkSampler` or `VkImageView` object that enables `sampler Y'CBCR conversion`, that object **must** only be used with `OpImageSample*` or `OpImageSparseSample*` instructions

- VUID-vkCmdDrawMultiEXT-ConstOffset-06551

If a bound shader accesses a `VkSampler` or `VkImageView` object that enables `sampler Y'CBCR conversion`, that object **must** not use the `ConstOffset` and `Offset` operands

- VUID-vkCmdDrawMultiEXT-viewType-07752

If a `VkImageView` is accessed as a result of this command, then the image view's `viewType` **must** match the `Dim` operand of the `OpTypeImage` as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDrawMultiEXT-format-07753

If a `VkImageView` is accessed as a result of this command, then the `numeric type` of the image view's `format` and the `Sampled Type` operand of the `OpTypeImage` **must** match

- VUID-vkCmdDrawMultiEXT-OpImageWrite-08795

If a `VkImageView` created with a format other than `VK_FORMAT_A8_UNORM_KHR` is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDrawMultiEXT-OpImageWrite-08796

If a `VkImageView` created with the format `VK_FORMAT_A8_UNORM_KHR` is accessed using

`OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components

- VUID-vkCmdDrawMultiEXT-OpImageWrite-04469

If a `VkBufferView` is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdDrawMultiEXT-SampledType-04470

If a `VkImageView` with a `VkFormat` that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64

- VUID-vkCmdDrawMultiEXT-SampledType-04471

If a `VkImageView` with a `VkFormat` that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32

- VUID-vkCmdDrawMultiEXT-SampledType-04472

If a `VkBufferView` with a `VkFormat` that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64

- VUID-vkCmdDrawMultiEXT-SampledType-04473

If a `VkBufferView` with a `VkFormat` that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32

- VUID-vkCmdDrawMultiEXT-sparseImageInt64Atomics-04474

If the `sparseImageInt64Atomics` feature is not enabled, `VkImage` objects created with the `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command

- VUID-vkCmdDrawMultiEXT-sparseImageInt64Atomics-04475

If the `sparseImageInt64Atomics` feature is not enabled, `VkBuffer` objects created with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command

- VUID-vkCmdDrawMultiEXT-None-07288

Any shader invocation executed by this command **must** `terminate`

- VUID-vkCmdDrawMultiEXT-None-09600

If a descriptor with type equal to any of `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDrawMultiEXT-renderPass-02684

The current render pass **must** be `compatible` with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`

- VUID-vkCmdDrawMultiEXT-subpass-02685

The subpass index of the current render pass **must** be equal to the `subpass` member of the

`VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`

- VUID-vkCmdDrawMultiEXT-None-07748

If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set

- VUID-vkCmdDrawMultiEXT-OpTypeImage-07468

If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDrawMultiEXT-None-07469

Input attachment views accessed in a subpass **must** be created with the same `VkFormat` as the corresponding subpass definition, and be created with a `VkImageView` that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the currently bound `VkFramebuffer` as specified by [Fragment Input Attachment Compatibility](#)

- VUID-vkCmdDrawMultiEXT-pDepthInputAttachmentIndex-09595

Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by `VkRenderingInputAttachmentIndexInfoKHR`, or no `InputAttachmentIndex` if `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` or `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are `NULL`, **must** be created with a `VkImageView` that is compatible with the corresponding color, depth, or stencil attachment in `VkRenderingInfo`

- VUID-vkCmdDrawMultiEXT-pDepthInputAttachmentIndex-09596

Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` and `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are non-`NULL`

- VUID-vkCmdDrawMultiEXT-InputAttachmentIndex-09597

If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDrawMultiEXT-None-06537

Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDrawMultiEXT-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_COLOR_BIT` and
 - there is no currently bound graphics pipeline or

- the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawMultiEXT-None-09001

If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_DEPTH_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawMultiEXT-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_STENCIL_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawMultiEXT-None-09003

If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command

- VUID-vkCmdDrawMultiEXT-None-06539

If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment

- VUID-vkCmdDrawMultiEXT-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, `depth writes` **must** be disabled

- VUID-vkCmdDrawMultiEXT-None-06887

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, **all stencil ops must** be `VK_STENCIL_OP_KEEP`

- VUID-vkCmdDrawMultiEXT-None-07831

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VIEWPORT` dynamic state enabled then `vkCmdSetViewport` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07832

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SCISSOR` dynamic state enabled then `vkCmdSetScissor` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07833

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic state enabled then `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08617

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08618

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08619

If a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07834

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BIAS` dynamic state enabled, the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, and the **current value** of `depthBiasEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` or `vkCmdSetDepthBias2EXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07835

If the bound graphics pipeline state was created with the

`VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic state enabled then `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08621

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer set any element of `pColorBlendEnables` to `VK_TRUE`, and the most recent call to `vkCmdSetColorBlendEquationEXT` in the current command buffer set the same element of `pColorBlendEquations` to a `VkColorBlendEquationEXT` structure with any `VkBlendFactor` member with a value of `VK_BLEND_FACTOR_CONSTANT_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR`, `VK_BLEND_FACTOR_CONSTANT_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`, `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07836

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBoundsTestEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07837

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilCompareMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilWriteMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07839

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_REFERENCE` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilReference` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-maxMultiviewInstanceIndex-02688

If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to `VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex`

- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-02689

If the bound graphics pipeline was created with `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable` set to `VK_TRUE` and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawMultiEXT-None-07634

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleLocationsEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-06666

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` dynamic state enabled, the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, and the **current value** of `sampleLocationsEnable` is `VK_TRUE`, then `vkCmdSetSampleLocationsEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's **format features** do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the `blendEnable` member of the corresponding element of the `pAttachments` member of `pColorBlendState` **must** be `VK_FALSE`

- VUID-vkCmdDrawMultiEXT-None-08643

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then for each color attachment in the render pass, if the corresponding image view's **format features** do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the corresponding member of `pColorBlendEnables` in the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer that affected that attachment index **must** have been `VK_FALSE`

- VUID-vkCmdDrawMultiEXT-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then `rasterizationSamples` for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawMultiEXT-None-08644

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`,

then the most recent call to [vkCmdSetRasterizationSamplesEXT](#) in the current command buffer **must** have set `rasterizationSamples` to be the same as the number of samples for the current render pass color and/or depth/stencil attachments

- VUID-vkCmdDrawMultiEXT-None-08876

If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with [vkCmdBeginRenderingKHR](#)

- VUID-vkCmdDrawMultiEXT-imageView-06172

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawMultiEXT-imageView-06173

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawMultiEXT-imageView-06174

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawMultiEXT-imageView-06175

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawMultiEXT-imageView-06176

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawMultiEXT-imageView-06177

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawMultiEXT-viewMask-06178

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the currently bound graphics pipeline **must** have been created with a [VkPipelineRenderingCreateInfo::viewMask](#) equal to `VkRenderingInfo::viewMask`

- VUID-vkCmdDrawMultiEXT-colorAttachmentCount-06179

If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the currently bound graphics

pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::colorAttachmentCount` equal to `VkRenderingInfo::colorAttachmentCount`

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08910

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08912

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` equal to `VK_NULL_HANDLE` **must** have the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound pipeline equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08911

If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline, or the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats`, if it exists, **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiEXT-None-07749

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08646

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-attachmentCount-07750

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then the `attachmentCount` parameter of `vkCmdSetColorWriteEnableEXT` **must** be greater than or equal to the `VkPipelineColorBlendStateCreateInfo::attachmentCount` of the currently bound graphics pipeline

- VUID-vkCmdDrawMultiEXT-None-08647
If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then the `attachmentCount` parameter of most recent call to `vkCmdSetColorWriteEnableEXT` in the current command buffer **must** be greater than or equal to the number of color attachments in the current render pass instance
- VUID-vkCmdDrawMultiEXT-None-07751
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state enabled then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command for each discard rectangle in `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount`
- VUID-vkCmdDrawMultiEXT-None-07880
If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDiscardRectangleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07881
If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `discardRectangleEnable` is `VK_TRUE`, then `vkCmdSetDiscardRectangleModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-rasterizerDiscardEnable-09236
If the `VK_EXT_discard_rectangles` extension is enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetDiscardRectangleEnableEXT` in the current command buffer set `discardRectangleEnable` to `VK_TRUE`, then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08913
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08914
If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of

`VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08915

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08916

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08917

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdDrawMultiEXT-dynamicRenderingUnusedAttachments-08918

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiEXT-imageView-06183

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView` was not `VK_NULL_HANDLE`, the currently bound graphics pipeline **must** have been created with `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-vkCmdDrawMultiEXT-multisampledRenderToSingleSampled-07285

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` with a `VkRenderingInfo::colorAttachmentCount` parameter greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with a `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a sample count equal to the value of `rasterizationSamples` for the currently bound graphics pipeline

- VUID-vkCmdDrawMultiEXT-multisampledRenderToSingleSampled-07286

If `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDrawMultiEXT-multisampledRenderToSingleSampled-07287
If `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pStencilAttachment->imageView`
- VUID-vkCmdDrawMultiEXT-renderPass-06198
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound pipeline **must** have been created with a `VkGraphicsPipelineCreateInfo::renderPass` equal to `VK_NULL_HANDLE`
- VUID-vkCmdDrawMultiEXT-pColorAttachments-08963
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the `VkRenderingInfo::pColorAttachments->imageView` was not `VK_NULL_HANDLE`, then the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawMultiEXT-pDepthAttachment-08964
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, depth test is enabled, depth write is enabled, and the `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawMultiEXT-pStencilAttachment-08965
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, stencil test is enabled and the `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawMultiEXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
If the `primitivesGeneratedQueryWithRasterizerDiscard` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, `rasterization discard` **must** not be enabled
- VUID-vkCmdDrawMultiEXT-primitivesGeneratedQueryWithNonZeroStreams-06709
If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, the bound graphics pipeline **must** not have been created with a non-zero value in `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream`
- VUID-vkCmdDrawMultiEXT-None-07619
If a shader object is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` dynamic state enabled, then `vkCmdSetTessellationDomainOriginEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiEXT-None-07620
If the `depthClamp` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the

VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetDepthClampEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07621

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetPolygonModeEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07622

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetRasterizationSamplesEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07623

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetSampleMaskEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-alphaToCoverageEnable-08919

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to `vkCmdSetAlphaToCoverageEnableEXT`, then the **Fragment Output Interface** **must** contain a variable for the alpha **Component** word in **Location** 0 at **Index** 0

- VUID-vkCmdDrawMultiEXT-alphaToCoverageEnable-08920

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetAlphaToCoverageEnableEXT` in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the **Fragment Output Interface** **must** contain a variable for the alpha **Component** word in **Location** 0 at **Index** 0

- VUID-vkCmdDrawMultiEXT-None-07624

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetAlphaToCoverageEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07625

If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetAlphaToOneEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this

drawing command

- VUID-vkCmdDrawMultiEXT-None-07626

If the `logicOp` feature is enabled, a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetLogicOpEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07627

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08657

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07628

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08658

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` for any attachment set that attachment's value in `pColorBlendEnables` to `VK_TRUE`, then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07629

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08659

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07630

If the `geometryStreams` feature is enabled, and a shader object is bound to the

`VK_SHADER_STAGE_GEOMETRY_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` dynamic state enabled, then `vkCmdSetRasterizationStreamEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07631

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetConservativeRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07632

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT` dynamic state enabled, the current value of `rasterizerDiscardEnable` is `VK_FALSE`, and the current value of `conservativeRasterizationMode` is `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`, then `vkCmdSetExtraPrimitiveOverestimationSizeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07633

If the `depthClipEnable` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT` dynamic state, then `vkCmdSetDepthClipEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07635

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-rasterizerDiscardEnable-09416

If the `VK_EXT_blend_operation_advanced` extension is enabled, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then at least one of `vkCmdSetColorBlendEquationEXT` and `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07636

If the `VK_EXT_provoking_vertex` extension is enabled, a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetProvokingVertexModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07637

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic state enabled then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08666

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08667

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08668

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07638

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` dynamic state enabled then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08669

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08670

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08671

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07849

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_KHR` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stippledLineEnable` is `VK_TRUE`, then `vkCmdSetLineStippleKHR` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-07639

If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` dynamic state enabled, then `vkCmdSetDepthClipNegativeOneToOneEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-09650

If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` dynamic state enabled, and the `current value` of `depthClampEnable` is `VK_TRUE`, then `vkCmdSetDepthClampRangeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-pColorBlendEnables-07470

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` state enabled and the last call to `vkCmdSetColorBlendEnableEXT` set `pColorBlendEnables` for any attachment to `VK_TRUE`, then for those attachments in the subpass the corresponding image view's `format features` **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`

- VUID-vkCmdDrawMultiEXT-rasterizationSamples-07471

If the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** follow the rules for a `zero-attachment` subpass

- VUID-vkCmdDrawMultiEXT-samples-07472

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `VkPipelineMultisampleStateCreateInfo::rasterizationSamples` parameter used to create the bound graphics pipeline

- VUID-vkCmdDrawMultiEXT-samples-07473

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state and `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` states enabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `rasterizationSamples` parameter in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawMultiEXT-rasterizationSamples-07474

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and neither the `VK_AMD_mixed_attachment_samples` nor the `VK_NV_framebuffer_mixed_samples` extensions are enabled, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawMultiEXT-firstAttachment-07476

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawMultiEXT-rasterizerDiscardEnable-09417

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawMultiEXT-firstAttachment-07477

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend

equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawMultiEXT-rasterizerDiscardEnable-09418

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawMultiEXT-firstAttachment-07478

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawMultiEXT-rasterizerDiscardEnable-09419

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawMultiEXT-firstAttachment-07479

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendAdvancedEXT` calls **must** specify the advanced blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawMultiEXT-advancedBlendMaxColorAttachments-07480

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` and `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic states enabled and the last calls to `vkCmdSetColorBlendEnableEXT` and `vkCmdSetColorBlendAdvancedEXT` have enabled advanced blending, then the number of active color attachments in the current subpass **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdDrawMultiEXT-primitivesGeneratedQueryWithNonZeroStreams-07481

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, and the bound graphics pipeline was created with `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` state enabled, the last call to

`vkCmdSetRasterizationStreamEXT` **must** have set the `rasterizationStream` to zero

- VUID-vkCmdDrawMultiEXT-sampleLocationsPerPixel-07482

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` member of the `VkPipelineMultisampleStateCreateInfo` structure the bound graphics pipeline has been created with

- VUID-vkCmdDrawMultiEXT-sampleLocationsPerPixel-07483

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` parameter of the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07484

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07485

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07486

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07487

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`

- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07936

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07937

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.height` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawMultiEXT-sampleLocationsEnable-07938

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationsPerPixel` **must** equal `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawMultiEXT-stippledLineEnable-07495

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR`, then the `stippledRectangularLines` feature **must** be enabled

- VUID-vkCmdDrawMultiEXT-stippledLineEnable-07496

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR`, then the `stippledBresenhamLines` feature **must** be enabled

- VUID-vkCmdDrawMultiEXT-stippledLineEnable-07497

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, then the `stippledSmoothLines` feature **must** be enabled

- VUID-vkCmdDrawMultiEXT-stippledLineEnable-07498

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR`, then the `stippledRectangularLines` feature **must** be enabled and `VkPhysicalDeviceLimits::strictLines` **must** be `VK_TRUE`

- VUID-vkCmdDrawMultiEXT-conservativePointAndLineRasterization-07499

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, `conservativePointAndLineRasterization` is not supported, and the effective primitive topology output by the last pre-rasterization shader stage is a line or point, then the `conservativeRasterizationMode` set by the last call to `vkCmdSetConservativeRasterizationModeEXT` **must** be `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`

- VUID-vkCmdDrawMultiEXT-None-08877

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAttachmentFeedbackLoopEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-08684

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_VERTEX_BIT`

- VUID-vkCmdDrawMultiEXT-None-08685

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`

- VUID-vkCmdDrawMultiEXT-None-08686

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`

- VUID-vkCmdDrawMultiEXT-None-08687

If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_GEOMETRY_BIT`

- VUID-vkCmdDrawMultiEXT-None-08688

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the

current command buffer with `pStages` with an element of `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-vkCmdDrawMultiEXT-None-08698

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, then all shaders created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag in the same `vkCreateShadersEXT` call **must** also be bound

- VUID-vkCmdDrawMultiEXT-None-08699

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, any stages in between stages whose shaders which did not create a shader with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag as part of the same `vkCreateShadersEXT` call **must** not have any `VkShaderEXT` bound

- VUID-vkCmdDrawMultiEXT-None-08878

All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges

- VUID-vkCmdDrawMultiEXT-None-08879

All bound graphics shader objects **must** have been created with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdDrawMultiEXT-pDynamicStates-08715

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to `vkCmdSetDepthWriteEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawMultiEXT-pDynamicStates-08716

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to `vkCmdSetStencilWriteMask` **must** be 0

- VUID-vkCmdDrawMultiEXT-None-09116

If a shader object is bound to any graphics stage or the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and the format of any color attachment is `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`, the corresponding element of the `pColorWriteMasks` parameter of `vkCmdSetColorWriteMaskEXT` **must** either include all of `VK_COLOR_COMPONENT_R_BIT`, `VK_COLOR_COMPONENT_G_BIT`, and `VK_COLOR_COMPONENT_B_BIT`, or none of them

- VUID-vkCmdDrawMultiEXT-maxFragmentDualSrcAttachments-09239

If `blending` is enabled for any attachment where either the source or destination blend factors for that attachment `use the secondary color input`, the maximum value of `Location` for any output attachment `statically used` in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDrawMultiEXT-None-09548

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, the value of each element of

`VkRenderingAttachmentLocationInfoKHR::pColorAttachmentLocations` set by `vkCmdSetRenderingAttachmentLocationsKHR` **must** match the value set for the corresponding element in the currently bound pipeline

- VUID-vkCmdDrawMultiEXT-None-09549

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, input attachment index mappings in the currently bound pipeline **must** match those set for the current render pass instance via `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDrawMultiEXT-None-04007

All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or `VK_NULL_HANDLE` buffers bound

- VUID-vkCmdDrawMultiEXT-None-04008

If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be `VK_NULL_HANDLE`

- VUID-vkCmdDrawMultiEXT-None-02721

If `robustBufferAccess` is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)

- VUID-vkCmdDrawMultiEXT-None-04914

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then `vkCmdSetVertexInputEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this draw command

- VUID-vkCmdDrawMultiEXT-Input-07939

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then all variables with the `Input` storage class decorated with `Location` in the `Vertex Execution Model OpEntryPoint` **must** contain a location in `VkVertexInputAttributeDescription2EXT::location`

- VUID-vkCmdDrawMultiEXT-Input-08734

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and either `legacyVertexAttributes` is not enabled or the SPIR-V Type associated with a given `Input` variable of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the numeric type associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be the same as `VkVertexInputAttributeDescription2EXT::format`

- VUID-vkCmdDrawMultiEXT-format-08936

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and `VkVertexInputAttributeDescription2EXT::format` has a 64-bit component, then the scalar width associated with all `Input` variables of the corresponding

Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit

- VUID-vkCmdDrawMultiEXT-format-08937

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding `VkVertexInputAttributeDescription2EXT::format` **must** have a 64-bit component

- VUID-vkCmdDrawMultiEXT-None-09203

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and `VkVertexInputAttributeDescription2EXT::format` has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format

- VUID-vkCmdDrawMultiEXT-None-04879

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiEXT-None-09637

If the `primitiveTopologyListRestart` feature is not enabled, the topology is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawMultiEXT-pNext-09461

If the bound graphics pipeline state was created with `VkPipelineVertexInputDivisorStateCreateInfoKHR` in the `pNext` chain of `VkGraphicsPipelineCreateInfo::pVertexInputState`, any member of `VkPipelineVertexInputDivisorStateCreateInfoKHR::pVertexBindingDivisors` has a value other than 1 in `divisor`, and `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance` is `VK_FALSE`, then `firstInstance` **must** be 0

- VUID-vkCmdDrawMultiEXT-None-09462

If `shader objects` are used for drawing or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled, any member of the `pVertexBindingDescriptions` parameter to the `vkCmdSetVertexInputEXT` call that sets this dynamic state has a value other than 1 in `divisor`, and `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance` is `VK_FALSE`, then `firstInstance` **must** be 0

- VUID-vkCmdDrawMultiEXT-None-04933

The `multiDraw` feature **must** be enabled

- VUID-vkCmdDrawMultiEXT-drawCount-04934

`drawCount` **must** be less than `VkPhysicalDeviceMultiDrawPropertiesEXT::maxMultiDrawCount`

- VUID-vkCmdDrawMultiEXT-drawCount-04935

If `drawCount` is greater than zero, `pVertexInfo` **must** be a valid pointer to memory containing one or more valid instances of `VkMultiDrawInfoEXT` structures

- VUID-vkCmdDrawMultiEXT-drawCount-09628

If `drawCount` is greater than 1, `stride` **must** be a multiple of 4 and **must** be greater than or equal to `sizeof(VkMultiDrawInfoEXT)`

Valid Usage (Implicit)

- VUID-vkCmdDrawMultiEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdDrawMultiEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdDrawMultiEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdDrawMultiEXT-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawMultiEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	Action

To record an ordered sequence of indexed draws which have no state changes between them, call:

```
// Provided by VK_EXT_multi_draw
void vkCmdDrawMultiIndexedEXT(
    VkCommandBuffer                commandBuffer,
    uint32_t                       drawCount,
    const VkMultiDrawIndexedInfoEXT* pIndexInfo,
    uint32_t                       instanceCount,
    uint32_t                       firstInstance,
    uint32_t                       stride,
    const int32_t*                 pVertexOffset);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `drawCount` is the number of draws to execute, and **can** be zero.
- `pIndexInfo` is a pointer to an array of `VkMultiDrawIndexedInfoEXT` with index information to be drawn.
- `instanceCount` is the number of instances per draw.
- `firstInstance` is the instance ID of the first instance in each draw.
- `stride` is the byte stride between consecutive elements of `pIndexInfo`.
- `pVertexOffset` is `NULL` or a pointer to the value added to the vertex index before indexing into the vertex buffer. When specified, `VkMultiDrawIndexedInfoEXT::offset` is ignored.

The number of draws recorded is `drawCount`, with each draw reading, sequentially, a `firstIndex` and an `indexCount` from `pIndexInfo`. For each recorded draw, primitives are assembled as for `vkCmdDrawIndexed`, and drawn `instanceCount` times with `instanceIndex` starting with `firstInstance` and sequentially for each instance. If `pVertexOffset` is `NULL`, a `vertexOffset` is also read from `pIndexInfo`, otherwise the value from dereferencing `pVertexOffset` is used.

Valid Usage

- VUID-vkCmdDrawMultiIndexedEXT-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawMultiIndexedEXT-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawMultiIndexedEXT-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDrawMultiIndexedEXT-unnormalizedCoordinates-09636

If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`

- VUID-vkCmdDrawMultiIndexedEXT-None-06479

If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-None-02691

If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-None-07888

If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-None-02692

If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`

- VUID-vkCmdDrawMultiIndexedEXT-None-02693

If the `VK_EXT_filter_cubic` extension is not enabled and any `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, it **must** not have a `VkImageViewType` of `VK_IMAGE_VIEW_TYPE_3D`, `VK_IMAGE_VIEW_TYPE_CUBE`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`

- VUID-vkCmdDrawMultiIndexedEXT-filterCubic-02694

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdDrawMultiIndexedEXT-filterCubicMinmax-02695

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` with a reduction mode of either `VK_SAMPLER_REDUCTION_MODE_MIN` or `VK_SAMPLER_REDUCTION_MODE_MAX` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering together with minmax filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdDrawMultiIndexedEXT-OpTypeImage-07027

For any `VkImageView` being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-OpTypeImage-07028

For any `VkImageView` being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-OpTypeImage-07029

For any `VkBufferView` being written as a storage texel buffer where the image format

field of the `OpTypeImage` is `Unknown`, the view's `buffer features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-OpTypeImage-07030

Any `VkBufferView` being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown` then the view's `buffer features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-None-08600

For each set n that is statically used by a `bound shader`, a descriptor set **must** have been bound to n at the same pipeline bind point, with a `VkPipelineLayout` that is compatible for set n , with the `VkPipelineLayout` used to create the current `VkPipeline` or the `VkDescriptorSetLayout` array used to create the current `VkShaderEXT`, as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawMultiIndexedEXT-None-08601

For each push constant that is statically used by a `bound shader`, a push constant value **must** have been set for the same pipeline bind point, with a `VkPipelineLayout` that is compatible for push constants, with the `VkPipelineLayout` used to create the current `VkPipeline` or the `VkDescriptorSetLayout` array used to create the current `VkShaderEXT`, as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawMultiIndexedEXT-None-10068

For each array of resources that is used by a `bound shader`, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDrawMultiIndexedEXT-maintenance4-08602

If the `maintenance4` feature is not enabled, then for each push constant that is statically used by a `bound shader`, a push constant value **must** have been set for the same pipeline bind point, with a `VkPipelineLayout` that is compatible for push constants, with the `VkPipelineLayout` used to create the current `VkPipeline` or the `VkDescriptorSetLayout` and `VkPushConstantRange` arrays used to create the current `VkShaderEXT`, as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawMultiIndexedEXT-None-08114

Descriptors in each bound descriptor set, specified via `vkCmdBindDescriptorSets`, **must** be valid as described by [descriptor validity](#) if they are statically used by a `bound shader`

- VUID-vkCmdDrawMultiIndexedEXT-None-08606

If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDrawMultiIndexedEXT-None-08608

If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the `VkPipeline` object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDrawMultiIndexedEXT-None-08609

If the `VkPipeline` object bound to the pipeline bind point used by this command or any `VkShaderEXT` bound to a stage corresponding to the pipeline bind point used by this command accesses a `VkSampler` object that uses unnormalized coordinates, that sampler

must not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDrawMultiIndexedEXT-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage

- VUID-vkCmdDrawMultiIndexedEXT-None-08611

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDrawMultiIndexedEXT-None-08607

If the [shaderObject](#) feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and [VK_NULL_HANDLE](#) shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command

- VUID-vkCmdDrawMultiIndexedEXT-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawMultiIndexedEXT-None-08612

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawMultiIndexedEXT-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawMultiIndexedEXT-None-08613

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawMultiIndexedEXT-None-06550

If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler YCbCr conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions

- VUID-vkCmdDrawMultiIndexedEXT-ConstOffset-06551
If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands
- VUID-vkCmdDrawMultiIndexedEXT-viewType-07752
If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)
- VUID-vkCmdDrawMultiIndexedEXT-format-07753
If a [VkImageView](#) is accessed as a result of this command, then the [numeric type](#) of the image view's [format](#) and the [Sampled Type](#) operand of the [OpTypeImage](#) **must** match
- VUID-vkCmdDrawMultiIndexedEXT-OpImageWrite-08795
If a [VkImageView](#) created with a format other than [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawMultiIndexedEXT-OpImageWrite-08796
If a [VkImageView](#) created with the format [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have four components
- VUID-vkCmdDrawMultiIndexedEXT-OpImageWrite-04469
If a [VkBufferView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawMultiIndexedEXT-SampledType-04470
If a [VkImageView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64
- VUID-vkCmdDrawMultiIndexedEXT-SampledType-04471
If a [VkImageView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32
- VUID-vkCmdDrawMultiIndexedEXT-SampledType-04472
If a [VkBufferView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64
- VUID-vkCmdDrawMultiIndexedEXT-SampledType-04473
If a [VkBufferView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32
- VUID-vkCmdDrawMultiIndexedEXT-sparseImageInt64Atomics-04474
If the [sparseImageInt64Atomics](#) feature is not enabled, [VkImage](#) objects created with the [VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT](#) flag **must** not be accessed by atomic instructions through an [OpTypeImage](#) with a [SampledType](#) with a [Width](#) of 64 by this command
- VUID-vkCmdDrawMultiIndexedEXT-sparseImageInt64Atomics-04475

If the `sparseImageInt64Atomics` feature is not enabled, `VkBuffer` objects created with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command

- VUID-vkCmdDrawMultiIndexedEXT-None-07288

Any shader invocation executed by this command **must** [terminate](#)

- VUID-vkCmdDrawMultiIndexedEXT-None-09600

If a descriptor with type equal to any of `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDrawMultiIndexedEXT-renderPass-02684

The current render pass **must** be [compatible](#) with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`

- VUID-vkCmdDrawMultiIndexedEXT-subpass-02685

The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`

- VUID-vkCmdDrawMultiIndexedEXT-None-07748

If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set

- VUID-vkCmdDrawMultiIndexedEXT-OpTypeImage-07468

If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDrawMultiIndexedEXT-None-07469

Input attachment views accessed in a subpass **must** be created with the same `VkFormat` as the corresponding subpass definition, and be created with a `VkImageView` that is compatible with the attachment referenced by the subpass' `pInputAttachments[InputAttachmentIndex]` in the currently bound `VkFramebuffer` as specified by [Fragment Input Attachment Compatibility](#)

- VUID-vkCmdDrawMultiIndexedEXT-pDepthInputAttachmentIndex-09595

Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by `VkRenderingInputAttachmentIndexInfoKHR`, or no `InputAttachmentIndex` if `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` or `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are `NULL`, **must** be created with a `VkImageView` that is compatible with the corresponding color, depth, or stencil attachment in `VkRenderingInfo`

- VUID-vkCmdDrawMultiIndexedEXT-pDepthInputAttachmentIndex-09596

Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` and `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are non-`NULL`

- VUID-vkCmdDrawMultiIndexedEXT-InputAttachmentIndex-09597

If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDrawMultiIndexedEXT-None-06537

Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDrawMultiIndexedEXT-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_COLOR_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawMultiIndexedEXT-None-09001

If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_DEPTH_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawMultiIndexedEXT-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_STENCIL_BIT` and
 - there is no currently bound graphics pipeline or

- the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawMultiIndexedEXT-None-09003

If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command

- VUID-vkCmdDrawMultiIndexedEXT-None-06539

If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment

- VUID-vkCmdDrawMultiIndexedEXT-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, `depth writes` **must** be disabled

- VUID-vkCmdDrawMultiIndexedEXT-None-06887

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, `all stencil ops` **must** be `VK_STENCIL_OP_KEEP`

- VUID-vkCmdDrawMultiIndexedEXT-None-07831

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VIEWPORT` dynamic state enabled then `vkCmdSetViewport` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07832

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SCISSOR` dynamic state enabled then `vkCmdSetScissor` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07833

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic state enabled then `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08617

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08618

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, `vkCmdSetLineWidth` **must** have been called and not subsequently

invalidated in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08619

If a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, `vkCmdSetLineWidth` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07834

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BIAS` dynamic state enabled, the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, and the **current value** of `depthBiasEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` or `vkCmdSetDepthBias2EXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07835

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic state enabled then `vkCmdSetBlendConstants` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08621

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer set any element of `pColorBlendEnables` to `VK_TRUE`, and the most recent call to `vkCmdSetColorBlendEquationEXT` in the current command buffer set the same element of `pColorBlendEquations` to a `VkColorBlendEquationEXT` structure with any `VkBlendFactor` member with a value of `VK_BLEND_FACTOR_CONSTANT_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR`, `VK_BLEND_FACTOR_CONSTANT_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`, `vkCmdSetBlendConstants` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07836

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic state enabled, the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, and the **current value** of `depthBoundsTestEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07837

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` dynamic state enabled, the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, and the **current value** of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilCompareMask` **must** have been called

and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_STENCIL_WRITE_MASK** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **stencilTestEnable** is **VK_TRUE**, then **vkCmdSetStencilWriteMask** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07839

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_STENCIL_REFERENCE** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, the **current value** of **stencilTestEnable** is **VK_TRUE**, then **vkCmdSetStencilReference** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-maxMultiviewInstanceIndex-02688

If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to **VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex**

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-02689

If the bound graphics pipeline was created with **VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable** set to **VK_TRUE** and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the **VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT** bit set

- VUID-vkCmdDrawMultiIndexedEXT-None-07634

If the **VK_EXT_sample_locations** extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT** dynamic state enabled, and the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, then **vkCmdSetSampleLocationsEnableEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-06666

If the **VK_EXT_sample_locations** extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **sampleLocationsEnable** is **VK_TRUE**, then **vkCmdSetSampleLocationsEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's **format features** do not contain **VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT**, then the **blendEnable** member of the corresponding element of the **pAttachments** member of **pColorBlendState** **must** be **VK_FALSE**

- VUID-vkCmdDrawMultiIndexedEXT-None-08643

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then for each color attachment in the render pass, if the corresponding image view's `format features` do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the corresponding member of `pColorBlendEnables` in the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer that affected that attachment index **must** have been `VK_FALSE`

- VUID-vkCmdDrawMultiIndexedEXT-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then `rasterizationSamples` for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawMultiIndexedEXT-None-08644

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`,

then the most recent call to `vkCmdSetRasterizationSamplesEXT` in the current command buffer **must** have set `rasterizationSamples` to be the same as the number of samples for the current render pass color and/or depth/stencil attachments

- VUID-vkCmdDrawMultiIndexedEXT-None-08876

If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with `vkCmdBeginRenderingKHR`

- VUID-vkCmdDrawMultiIndexedEXT-imageView-06172

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawMultiIndexedEXT-imageView-06173

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawMultiIndexedEXT-imageView-06174

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawMultiIndexedEXT-imageView-06175

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawMultiIndexedEXT-imageView-06176
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, this command **must** not write any values to the depth attachment
- VUID-vkCmdDrawMultiIndexedEXT-imageView-06177
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment
- VUID-vkCmdDrawMultiIndexedEXT-viewMask-06178
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::viewMask` equal to `VkRenderingInfo::viewMask`
- VUID-vkCmdDrawMultiIndexedEXT-colorAttachmentCount-06179
If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::colorAttachmentCount` equal to `VkRenderingInfo::colorAttachmentCount`
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08910
If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08912
If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` equal to `VK_NULL_HANDLE` **must** have the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound pipeline equal to `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08911
If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline, or the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats`, if it exists, **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawMultiIndexedEXT-None-07749

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08646

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-attachmentCount-07750

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then the `attachmentCount` parameter of `vkCmdSetColorWriteEnableEXT` **must** be greater than or equal to the `VkPipelineColorBlendStateCreateInfo::attachmentCount` of the currently bound graphics pipeline

- VUID-vkCmdDrawMultiIndexedEXT-None-08647

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then the `attachmentCount` parameter of most recent call to `vkCmdSetColorWriteEnableEXT` in the current command buffer **must** be greater than or equal to the number of color attachments in the current render pass instance

- VUID-vkCmdDrawMultiIndexedEXT-None-07751

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state enabled then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command for each discard rectangle in `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount`

- VUID-vkCmdDrawMultiIndexedEXT-None-07880

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDiscardRectangleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07881

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `discardRectangleEnable` is `VK_TRUE`, then `vkCmdSetDiscardRectangleModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-rasterizerDiscardEnable-09236

If the `VK_EXT_discard_rectangles` extension is enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetDiscardRectangleEnableEXT` in the current command buffer set `discardRectangleEnable` to `VK_TRUE`, then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08913

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08914

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08915

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08916

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08917

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdDrawMultiIndexedEXT-dynamicRenderingUnusedAttachments-08918

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently

bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiIndexedEXT-imageView-06183

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView` was not `VK_NULL_HANDLE`, the currently bound graphics pipeline **must** have been created with `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-vkCmdDrawMultiIndexedEXT-multisampledRenderToSingleSampled-07285

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` with a `VkRenderingInfo::colorAttachmentCount` parameter greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with a `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a sample count equal to the value of `rasterizationSamples` for the currently bound graphics pipeline

- VUID-vkCmdDrawMultiIndexedEXT-multisampledRenderToSingleSampled-07286

If `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDrawMultiIndexedEXT-multisampledRenderToSingleSampled-07287

If `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdDrawMultiIndexedEXT-renderPass-06198

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound pipeline **must** have been created with a `VkGraphicsPipelineCreateInfo::renderPass` equal to `VK_NULL_HANDLE`

- VUID-vkCmdDrawMultiIndexedEXT-pColorAttachments-08963

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the `VkRenderingInfo::pColorAttachments->imageView` was not `VK_NULL_HANDLE`, then the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiIndexedEXT-pDepthAttachment-08964

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, depth test is enabled, depth write is enabled, and the `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiIndexedEXT-pStencilAttachment-08965

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, stencil test is enabled and the `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawMultiIndexedEXT-primitivesGeneratedQueryWithRasterizerDiscard-06708
If the `primitivesGeneratedQueryWithRasterizerDiscard` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, `rasterization discard` **must** not be enabled
- VUID-vkCmdDrawMultiIndexedEXT-primitivesGeneratedQueryWithNonZeroStreams-06709
If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, the bound graphics pipeline **must** not have been created with a non-zero value in `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream`
- VUID-vkCmdDrawMultiIndexedEXT-None-07619
If a shader object is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` dynamic state enabled, then `vkCmdSetTessellationDomainOriginEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07620
If the `depthClamp` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDepthClampEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07621
If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_POLYGON_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetPolygonModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07622
If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetRasterizationSamplesEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-None-07623
If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawMultiIndexedEXT-alphaToCoverageEnable-08919
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and

`alphaToCoverageEnable` was `VK_TRUE` in the last call to `vkCmdSetAlphaToCoverageEnableEXT`, then the **Fragment Output Interface** **must** contain a variable for the alpha **Component** word in **Location 0** at **Index 0**

- VUID-vkCmdDrawMultiIndexedEXT-alphaToCoverageEnable-08920

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetAlphaToCoverageEnableEXT` in the current command buffer set `alphaToCoverageEnable` to `VK_TRUE`, then the **Fragment Output Interface** **must** contain a variable for the alpha **Component** word in **Location 0** at **Index 0**

- VUID-vkCmdDrawMultiIndexedEXT-None-07624

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToCoverageEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07625

If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToOneEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07626

If the `logicOp` feature is enabled, a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetLogicOpEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07627

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08657

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07628

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08658

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` for any attachment set that attachment's value in `pColorBlendEnables` to `VK_TRUE`, then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07629

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08659

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07630

If the `geometryStreams` feature is enabled, and a shader object is bound to the `VK_SHADER_STAGE_GEOMETRY_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` dynamic state enabled, then `vkCmdSetRasterizationStreamEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07631

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetConservativeRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07632

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT` dynamic state enabled, the current value of `rasterizerDiscardEnable` is `VK_FALSE`, and the current value of `conservativeRasterizationMode` is `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`, then `vkCmdSetExtraPrimitiveOverestimationSizeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07633

If the `depthClipEnable` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT` dynamic state, then `vkCmdSetDepthClipEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07635

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-rasterizerDiscardEnable-09416

If the `VK_EXT_blend_operation_advanced` extension is enabled, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then at least one of `vkCmdSetColorBlendEquationEXT` and `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07636

If the `VK_EXT_provoking_vertex` extension is enabled, a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetProvokingVertexModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07637

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic state enabled then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08666

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08667

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08668

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines`

features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07638

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` dynamic state enabled then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08669

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08670

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08671

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07849

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_KHR` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stippledLineEnable` is `VK_TRUE`, then `vkCmdSetLineStippleKHR` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-07639

If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` dynamic state enabled, then `vkCmdSetDepthClipNegativeOneToOneEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-09650

If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` dynamic state enabled, and the `current value` of `depthClampEnable` is `VK_TRUE`, then `vkCmdSetDepthClampRangeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-pColorBlendEnables-07470

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` state enabled and the last call to `vkCmdSetColorBlendEnableEXT` set `pColorBlendEnables` for any attachment to `VK_TRUE`, then for those attachments in the subpass the corresponding image view's `format features` **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-rasterizationSamples-07471

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** follow the rules for a `zero-attachment` subpass

- VUID-vkCmdDrawMultiIndexedEXT-samples-07472

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `VkPipelineMultisampleStateCreateInfo::rasterizationSamples` parameter used to create the bound graphics pipeline

- VUID-vkCmdDrawMultiIndexedEXT-samples-07473

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state and `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` states enabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `rasterizationSamples` parameter in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawMultiIndexedEXT-rasterizationSamples-07474

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and neither the `VK_AMD_mixed_attachment_samples` nor the `VK_NV_framebuffer_mixed_samples` extensions are enabled, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawMultiIndexedEXT-firstAttachment-07476

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawMultiIndexedEXT-rasterizerDiscardEnable-09417

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawMultiIndexedEXT-firstAttachment-07477

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawMultiIndexedEXT-rasterizerDiscardEnable-09418

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawMultiIndexedEXT-firstAttachment-07478

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawMultiIndexedEXT-rasterizerDiscardEnable-09419

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawMultiIndexedEXT-firstAttachment-07479

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendAdvancedEXT` calls **must** specify the advanced blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawMultiIndexedEXT-advancedBlendMaxColorAttachments-07480

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` and `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic states enabled and the last calls to `vkCmdSetColorBlendEnableEXT` and `vkCmdSetColorBlendAdvancedEXT` have enabled advanced blending, then the number of active color attachments in the current subpass **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdDrawMultiIndexedEXT-primitivesGeneratedQueryWithNonZeroStreams-07481

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, and the bound graphics pipeline was created with `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` state enabled, the last call to `vkCmdSetRasterizationStreamEXT` **must** have set the `rasterizationStream` to zero

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsPerPixel-07482

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` member of the `VkPipelineMultisampleStateCreateInfo` structure the bound graphics pipeline has been created with

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsPerPixel-07483

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` parameter of the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07484

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07485

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound

graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07486

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07487

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07936

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07937

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.height` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawMultiIndexedEXT-sampleLocationsEnable-07938

If the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawMultiIndexedEXT-stippledLineEnable-07495

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR, then the stippledRectangularLines feature **must** be enabled

- VUID-vkCmdDrawMultiIndexedEXT-stippledLineEnable-07496

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR, then the stippledBresenhamLines feature **must** be enabled

- VUID-vkCmdDrawMultiIndexedEXT-stippledLineEnable-07497

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR, then the stippledSmoothLines feature **must** be enabled

- VUID-vkCmdDrawMultiIndexedEXT-stippledLineEnable-07498

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE

- VUID-vkCmdDrawMultiIndexedEXT-conservativePointAndLineRasterization-07499

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, conservativePointAndLineRasterization is not supported, and the effective primitive topology output by the last pre-rasterization shader stage is a line or point, then the conservativeRasterizationMode set by the last call to vkCmdSetConservativeRasterizationModeEXT **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT

- VUID-vkCmdDrawMultiIndexedEXT-None-08877

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then

`vkCmdSetAttachmentFeedbackLoopEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-08684

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_VERTEX_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-None-08685

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-None-08686

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-None-08687

If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_GEOMETRY_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-None-08688

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-vkCmdDrawMultiIndexedEXT-None-08698

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, then all shaders created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag in the same `vkCreateShadersEXT` call **must** also be bound

- VUID-vkCmdDrawMultiIndexedEXT-None-08699

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, any stages in between stages whose shaders which did not create a shader with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag as part of the same `vkCreateShadersEXT` call **must** not have any `VkShaderEXT` bound

- VUID-vkCmdDrawMultiIndexedEXT-None-08878

All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges

- VUID-vkCmdDrawMultiIndexedEXT-None-08879

All bound graphics shader objects **must** have been created with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdDrawMultiIndexedEXT-pDynamicStates-08715

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to `vkCmdSetDepthWriteEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawMultiIndexedEXT-pDynamicStates-08716

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to `vkCmdSetStencilWriteMask` **must** be 0

- VUID-vkCmdDrawMultiIndexedEXT-None-09116

If a shader object is bound to any graphics stage or the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and the format of any color attachment is `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`, the corresponding element of the `pColorWriteMasks` parameter of `vkCmdSetColorWriteMaskEXT` **must** either include all of `VK_COLOR_COMPONENT_R_BIT`, `VK_COLOR_COMPONENT_G_BIT`, and `VK_COLOR_COMPONENT_B_BIT`, or none of them

- VUID-vkCmdDrawMultiIndexedEXT-maxFragmentDualSrcAttachments-09239

If `blending` is enabled for any attachment where either the source or destination blend factors for that attachment [use the secondary color input](#), the maximum value of `Location` for any output attachment [statically used](#) in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDrawMultiIndexedEXT-None-09548

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, the value of each element of `VkRenderingAttachmentLocationInfoKHR::pColorAttachmentLocations` set by `vkCmdSetRenderingAttachmentLocationsKHR` **must** match the value set for the corresponding element in the currently bound pipeline

- VUID-vkCmdDrawMultiIndexedEXT-None-09549

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, input attachment index mappings in the currently bound pipeline **must** match those set for the current render pass instance via `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDrawMultiIndexedEXT-None-04007

All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or `VK_NULL_HANDLE` buffers bound

- VUID-vkCmdDrawMultiIndexedEXT-None-04008

If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be `VK_NULL_HANDLE`

- VUID-vkCmdDrawMultiIndexedEXT-None-02721

If `robustBufferAccess` is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)

- VUID-vkCmdDrawMultiIndexedEXT-None-04914

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then `vkCmdSetVertexInputEXT` **must** have been called and not subsequently

[invalidated](#) in the current command buffer prior to this draw command

- VUID-vkCmdDrawMultiIndexedEXT-Input-07939

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then all variables with the `Input` storage class decorated with `Location` in the `Vertex Execution Model OpEntryPoint` **must** contain a location in `VkVertexInputAttributeDescription2EXT::location`

- VUID-vkCmdDrawMultiIndexedEXT-Input-08734

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and either `legacyVertexAttributes` is not enabled or the SPIR-V Type associated with a given `Input` variable of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the numeric type associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be the same as `VkVertexInputAttributeDescription2EXT::format`

- VUID-vkCmdDrawMultiIndexedEXT-format-08936

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and `VkVertexInputAttributeDescription2EXT::format` has a 64-bit component, then the scalar width associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be 64-bit

- VUID-vkCmdDrawMultiIndexedEXT-format-08937

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and the scalar width associated with a `Location` decorated `Input` variable in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the corresponding `VkVertexInputAttributeDescription2EXT::format` **must** have a 64-bit component

- VUID-vkCmdDrawMultiIndexedEXT-None-09203

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and `VkVertexInputAttributeDescription2EXT::format` has a 64-bit component, then all `Input` variables at the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** not use components that are not present in the format

- VUID-vkCmdDrawMultiIndexedEXT-None-04879

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawMultiIndexedEXT-None-09637

If the `primitiveTopologyListRestart` feature is not enabled, the topology is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawMultiIndexedEXT-None-07312
If **maintenance6** is not enabled, a valid index buffer **must** be bound
- VUID-vkCmdDrawMultiIndexedEXT-pNext-09461
If the bound graphics pipeline state was created with **VkPipelineVertexInputDivisorStateCreateInfoKHR** in the **pNext** chain of **VkGraphicsPipelineCreateInfo::pVertexInputState**, any member of **VkPipelineVertexInputDivisorStateCreateInfoKHR::pVertexBindingDivisors** has a value other than **1** in **divisor**, and **VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance** is **VK_FALSE**, then **firstInstance** **must** be **0**
- VUID-vkCmdDrawMultiIndexedEXT-None-09462
If **shader objects** are used for drawing or the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_VERTEX_INPUT_EXT** dynamic state enabled, any member of the **pVertexBindingDescriptions** parameter to the **vkCmdSetVertexInputEXT** call that sets this dynamic state has a value other than **1** in **divisor**, and **VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance** is **VK_FALSE**, then **firstInstance** **must** be **0**
- VUID-vkCmdDrawMultiIndexedEXT-robustBufferAccess2-08798
If **robustBufferAccess2** is not enabled, $(\text{indexSize} \times (\text{firstIndex} + \text{indexCount}) + \text{offset})$ **must** be less than or equal to the size of the bound index buffer, with **indexSize** being based on the type specified by **indexType**, where the index buffer, **indexType**, and **offset** are specified via **vkCmdBindIndexBuffer** or **vkCmdBindIndexBuffer2KHR**. If **vkCmdBindIndexBuffer2KHR** is used to bind the index buffer, the size of the bound index buffer is **vkCmdBindIndexBuffer2KHR::size**
- VUID-vkCmdDrawMultiIndexedEXT-None-04937
The **multiDraw** feature **must** be enabled
- VUID-vkCmdDrawMultiIndexedEXT-drawCount-04939
drawCount **must** be less than **VkPhysicalDeviceMultiDrawPropertiesEXT::maxMultiDrawCount**
- VUID-vkCmdDrawMultiIndexedEXT-drawCount-04940
If **drawCount** is greater than zero, **pIndexInfo** **must** be a valid pointer to memory containing one or more valid instances of **VkMultiDrawIndexedInfoEXT** structures
- VUID-vkCmdDrawMultiIndexedEXT-drawCount-09629
If **drawCount** is greater than **1**, **stride** **must** be a multiple of **4** and **must** be greater than or equal to **sizeof(VkMultiDrawIndexedInfoEXT)**

Valid Usage (Implicit)

- VUID-vkCmdDrawMultiIndexedEXT-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdDrawMultiIndexedEXT-pVertexOffset-parameter
If **pVertexOffset** is not **NULL**, **pVertexOffset** **must** be a valid pointer to a valid **int32_t** value
- VUID-vkCmdDrawMultiIndexedEXT-commandBuffer-recording
commandBuffer **must** be in the **recording state**

- VUID-vkCmdDrawMultiIndexedEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdDrawMultiIndexedEXT-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawMultiIndexedEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	Action

The `VkMultiDrawInfoEXT` structure is defined as:

```
// Provided by VK_EXT_multi_draw
typedef struct VkMultiDrawInfoEXT {
    uint32_t    firstVertex;
    uint32_t    vertexCount;
} VkMultiDrawInfoEXT;
```

- `firstVertex` is the first vertex to draw.
- `vertexCount` is the number of vertices to draw.

The members of `VkMultiDrawInfoEXT` have the same meaning as the `firstVertex` and `vertexCount` parameters in `vkCmdDraw`.

The `VkMultiDrawIndexedInfoEXT` structure is defined as:

```
// Provided by VK_EXT_multi_draw
typedef struct VkMultiDrawIndexedInfoEXT {
    uint32_t    firstIndex;
    uint32_t    indexCount;
    int32_t     vertexOffset;
} VkMultiDrawIndexedInfoEXT;
```

- `firstIndex` is the first index to draw.
- `indexCount` is the number of vertices to draw.
- `vertexOffset` is the value added to the vertex index before indexing into the vertex buffer for indexed multidraws.

The `firstIndex`, `indexCount`, and `vertexOffset` members of `VkMultiDrawIndexedInfoEXT` have the same meaning as the `firstIndex`, `indexCount`, and `vertexOffset` parameters, respectively, of `vkCmdDrawIndexed`.

To record a non-indexed indirect drawing command, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDrawIndirect(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 buffer,
    VkDeviceSize             offset,
    uint32_t                 drawCount,
    uint32_t                 stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.
- `offset` is the byte offset into `buffer` where parameters begin.
- `drawCount` is the number of draws to execute, and **can** be zero.
- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawIndirect` behaves similarly to `vkCmdDraw` except that the parameters are read by the device from a buffer during execution. `drawCount` draws are executed by the command, with parameters taken from `buffer` starting at `offset` and increasing by `stride` bytes for each successive draw. The parameters of each draw are encoded in an array of `VkDrawIndirectCommand` structures. If `drawCount` is less than or equal to one, `stride` is ignored.

Valid Usage

- VUID-vkCmdDrawIndirect-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain

VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT

- VUID-vkCmdDrawIndirect-mipmapMode-04770

If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`

- VUID-vkCmdDrawIndirect-unnormalizedCoordinates-09635

If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1

- VUID-vkCmdDrawIndirect-unnormalizedCoordinates-09636

If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`

- VUID-vkCmdDrawIndirect-None-06479

If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`

- VUID-vkCmdDrawIndirect-None-02691

If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`

- VUID-vkCmdDrawIndirect-None-07888

If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`

- VUID-vkCmdDrawIndirect-None-02692

If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`

- VUID-vkCmdDrawIndirect-None-02693

If the `VK_EXT_filter_cubic` extension is not enabled and any `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, it **must** not have a `VkImageViewType` of `VK_IMAGE_VIEW_TYPE_3D`, `VK_IMAGE_VIEW_TYPE_CUBE`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`

- VUID-vkCmdDrawIndirect-filterCubic-02694

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdDrawIndirect-filterCubicMinmax-02695

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` with a reduction mode of either `VK_SAMPLER_REDUCTION_MODE_MIN` or `VK_SAMPLER_REDUCTION_MODE_MAX` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering together with minmax filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax` returned by

vkGetPhysicalDeviceImageFormatProperties2KHR

- VUID-vkCmdDrawIndirect-OpTypeImage-07027

For any [VkImageView](#) being written as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDrawIndirect-OpTypeImage-07028

For any [VkImageView](#) being read as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDrawIndirect-OpTypeImage-07029

For any [VkBufferView](#) being written as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDrawIndirect-OpTypeImage-07030

Any [VkBufferView](#) being read as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#) then the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDrawIndirect-None-08600

For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndirect-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndirect-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDrawIndirect-maintenance4-08602

If the [maintenance4](#) feature is not enabled, then for each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) and [VkPushConstantRange](#) arrays used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndirect-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDrawIndirect-None-08606

If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDrawIndirect-None-08608

If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the `VkPipeline` object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDrawIndirect-None-08609

If the `VkPipeline` object bound to the pipeline bind point used by this command or any `VkShaderEXT` bound to a stage corresponding to the pipeline bind point used by this command accesses a `VkSampler` object that uses unnormalized coordinates, that sampler **must** not be used to sample from any `VkImage` with a `VkImageView` of the type `VK_IMAGE_VIEW_TYPE_3D`, `VK_IMAGE_VIEW_TYPE_CUBE`, `VK_IMAGE_VIEW_TYPE_1D_ARRAY`, `VK_IMAGE_VIEW_TYPE_2D_ARRAY` or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`, in any shader stage

- VUID-vkCmdDrawIndirect-None-08610

If the `VkPipeline` object bound to the pipeline bind point used by this command or any `VkShaderEXT` bound to a stage corresponding to the pipeline bind point used by this command accesses a `VkSampler` object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions with `ImplicitLod`, `Dref` or `Proj` in their name, in any shader stage

- VUID-vkCmdDrawIndirect-None-08611

If the `VkPipeline` object bound to the pipeline bind point used by this command or any `VkShaderEXT` bound to a stage corresponding to the pipeline bind point used by this command accesses a `VkSampler` object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V `OpImageSample*` or `OpImageSparseSample*` instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDrawIndirect-None-08607

If the `shaderObject` feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and `VK_NULL_HANDLE` shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command

- VUID-vkCmdDrawIndirect-uniformBuffers-06935

If any stage of the `VkPipeline` object bound to the pipeline bind point used by this command accesses a uniform buffer, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirect-None-08612

If the `robustBufferAccess` feature is not enabled, and any `VkShaderEXT` bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirect-storageBuffers-06936

If any stage of the `VkPipeline` object bound to the pipeline bind point used by this command accesses a storage buffer, and the `robustBufferAccess` feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the

descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirect-None-08613

If the **robustBufferAccess** feature is not enabled, and any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirect-None-06550

If a **bound shader** accesses a **VkSampler** or **VkImageView** object that enables **sampler Y'C_BC_R conversion**, that object **must** only be used with **OpImageSample*** or **OpImageSparseSample*** instructions

- VUID-vkCmdDrawIndirect-ConstOffset-06551

If a **bound shader** accesses a **VkSampler** or **VkImageView** object that enables **sampler Y'C_BC_R conversion**, that object **must** not use the **ConstOffset** and **Offset** operands

- VUID-vkCmdDrawIndirect-viewType-07752

If a **VkImageView** is accessed as a result of this command, then the image view's **viewType** **must** match the **Dim** operand of the **OpTypeImage** as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDrawIndirect-format-07753

If a **VkImageView** is accessed as a result of this command, then the **numeric type** of the image view's **format** and the **Sampled Type** operand of the **OpTypeImage** **must** match

- VUID-vkCmdDrawIndirect-OpImageWrite-08795

If a **VkImageView** created with a format other than **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDrawIndirect-OpImageWrite-08796

If a **VkImageView** created with the format **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have four components

- VUID-vkCmdDrawIndirect-OpImageWrite-04469

If a **VkBufferView** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdDrawIndirect-SampledType-04470

If a **VkImageView** with a **VkFormat** that has a 64-bit component width is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 64

- VUID-vkCmdDrawIndirect-SampledType-04471

If a **VkImageView** with a **VkFormat** that has a component width less than 64-bit is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 32

- VUID-vkCmdDrawIndirect-SampledType-04472

If a **VkBufferView** with a **VkFormat** that has a 64-bit component width is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction

must have a **Width** of 64

- VUID-vkCmdDrawIndirect-SampledType-04473

If a **VkBufferView** with a **VkFormat** that has a component width less than 64-bit is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 32

- VUID-vkCmdDrawIndirect-sparseImageInt64Atomics-04474

If the **sparseImageInt64Atomics** feature is not enabled, **VkImage** objects created with the **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command

- VUID-vkCmdDrawIndirect-sparseImageInt64Atomics-04475

If the **sparseImageInt64Atomics** feature is not enabled, **VkBuffer** objects created with the **VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command

- VUID-vkCmdDrawIndirect-None-07288

Any shader invocation executed by this command **must** **terminate**

- VUID-vkCmdDrawIndirect-None-09600

If a descriptor with type equal to any of **VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE**, **VK_DESCRIPTOR_TYPE_STORAGE_IMAGE**, or **VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT** is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDrawIndirect-renderPass-02684

The current render pass **must** be **compatible** with the **renderPass** member of the **VkGraphicsPipelineCreateInfo** structure specified when creating the **VkPipeline** bound to **VK_PIPELINE_BIND_POINT_GRAPHICS**

- VUID-vkCmdDrawIndirect-subpass-02685

The subpass index of the current render pass **must** be equal to the **subpass** member of the **VkGraphicsPipelineCreateInfo** structure specified when creating the **VkPipeline** bound to **VK_PIPELINE_BIND_POINT_GRAPHICS**

- VUID-vkCmdDrawIndirect-None-07748

If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set

- VUID-vkCmdDrawIndirect-OpTypeImage-07468

If any shader executed by this pipeline accesses an **OpTypeImage** variable with a **Dim** operand of **SubpassData**, it **must** be decorated with an **InputAttachmentIndex** that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDrawIndirect-None-07469

Input attachment views accessed in a subpass **must** be created with the same **VkFormat** as the corresponding subpass definition, and be created with a **VkImageView** that is compatible with the attachment referenced by the subpass' **pInputAttachments** [**InputAttachmentIndex**] in the currently bound **VkFramebuffer** as specified by **Fragment Input Attachment Compatibility**

- VUID-vkCmdDrawIndirect-pDepthInputAttachmentIndex-09595

Input attachment views accessed in a dynamic render pass with a **InputAttachmentIndex**

referenced by `VkRenderingInputAttachmentIndexInfoKHR`, or no `InputAttachmentIndex` if `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` or `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are `NULL`, **must** be created with a `VkImageView` that is compatible with the corresponding color, depth, or stencil attachment in `VkRenderingInfo`

- VUID-vkCmdDrawIndirect-pDepthInputAttachmentIndex-09596

Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` and `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are non-`NULL`

- VUID-vkCmdDrawIndirect-InputAttachmentIndex-09597

If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDrawIndirect-None-06537

Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirect-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_COLOR_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirect-None-09001

If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_DEPTH_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirect-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_STENCIL_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirect-None-09003

If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command

- VUID-vkCmdDrawIndirect-None-06539

If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment

- VUID-vkCmdDrawIndirect-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, `depth writes` **must** be disabled

- VUID-vkCmdDrawIndirect-None-06887

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, `all stencil ops` **must** be `VK_STENCIL_OP_KEEP`

- VUID-vkCmdDrawIndirect-None-07831

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VIEWPORT` dynamic state enabled then `vkCmdSetViewport` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07832

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SCISSOR` dynamic state enabled then `vkCmdSetScissor` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07833

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic state enabled then `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08617

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set

`rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08618

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08619

If a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07834

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BIAS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBiasEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` or `vkCmdSetDepthBias2EXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07835

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic state enabled then `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08621

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer set any element of `pColorBlendEnables` to `VK_TRUE`, and the most recent call to `vkCmdSetColorBlendEquationEXT` in the current command buffer set the same element of `pColorBlendEquations` to a `VkColorBlendEquationEXT` structure with any `VkBlendFactor` member with a value of `VK_BLEND_FACTOR_CONSTANT_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR`, `VK_BLEND_FACTOR_CONSTANT_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`, `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07836

If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBoundsTestEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07837

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilCompareMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilWriteMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07839

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_REFERENCE` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilReference` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-maxMultiviewInstanceIndex-02688

If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to `VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex`

- VUID-vkCmdDrawIndirect-sampleLocationsEnable-02689

If the bound graphics pipeline was created with `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable` set to `VK_TRUE` and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawIndirect-None-07634

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleLocationsEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-06666

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the

VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is VK_FALSE, and the **current value** of **sampleLocationsEnable** is VK_TRUE, then **vkCmdSetSampleLocationsEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's **format features** do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the **blendEnable** member of the corresponding element of the **pAttachments** member of **pColorBlendState** **must** be VK_FALSE

- VUID-vkCmdDrawIndirect-None-08643

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to VK_FALSE, then for each color attachment in the render pass, if the corresponding image view's **format features** do not contain VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT, then the corresponding member of **pColorBlendEnables** in the most recent call to **vkCmdSetColorBlendEnableEXT** in the current command buffer that affected that attachment index **must** have been VK_FALSE

- VUID-vkCmdDrawIndirect-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then **rasterizationSamples** for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndirect-None-08644

If a shader object is bound to any graphics stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to VK_FALSE,

then the most recent call to **vkCmdSetRasterizationSamplesEXT** in the current command buffer **must** have set **rasterizationSamples** to be the same as the number of samples for the current render pass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndirect-None-08876

If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with **vkCmdBeginRenderingKHR**

- VUID-vkCmdDrawIndirect-imageView-06172

If the current render pass instance was begun with **vkCmdBeginRenderingKHR**, the **imageView** member of **pDepthAttachment** is not VK_NULL_HANDLE, and the **layout** member of **pDepthAttachment** is VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndirect-imageView-06173

If the current render pass instance was begun with **vkCmdBeginRenderingKHR**, the **imageView** member of **pStencilAttachment** is not VK_NULL_HANDLE, and the **layout** member of **pStencilAttachment** is VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndirect-imageView-06174
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`, this command **must** not write any values to the depth attachment
- VUID-vkCmdDrawIndirect-imageView-06175
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment
- VUID-vkCmdDrawIndirect-imageView-06176
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, this command **must** not write any values to the depth attachment
- VUID-vkCmdDrawIndirect-imageView-06177
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment
- VUID-vkCmdDrawIndirect-viewMask-06178
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::viewMask` equal to `VkRenderingInfo::viewMask`
- VUID-vkCmdDrawIndirect-colorAttachmentCount-06179
If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::colorAttachmentCount` equal to `VkRenderingInfo::colorAttachmentCount`
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08910
If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline
- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08912
If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` equal to `VK_NULL_HANDLE` **must** have the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound pipeline equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08911
If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline, or the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats`, if it exists, **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndirect-None-07749
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-None-08646
If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirect-attachmentCount-07750
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then the `attachmentCount` parameter of `vkCmdSetColorWriteEnableEXT` **must** be greater than or equal to the `VkPipelineColorBlendStateCreateInfo::attachmentCount` of the currently bound graphics pipeline
- VUID-vkCmdDrawIndirect-None-08647
If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then the `attachmentCount` parameter of most recent call to `vkCmdSetColorWriteEnableEXT` in the current command buffer **must** be greater than or equal to the number of color attachments in the current render pass instance
- VUID-vkCmdDrawIndirect-None-07751
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state enabled then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command for each discard rectangle in `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount`
- VUID-vkCmdDrawIndirect-None-07880
If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` dynamic state enabled, the `current value`

of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDiscardRectangleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07881

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `discardRectangleEnable` is `VK_TRUE`, then `vkCmdSetDiscardRectangleModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-rasterizerDiscardEnable-09236

If the `VK_EXT_discard_rectangles` extension is enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetDiscardRectangleEnableEXT` in the current command buffer set `discardRectangleEnable` to `VK_TRUE`, then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08913

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08914

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08915

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08916

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08917

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdDrawIndirect-dynamicRenderingUnusedAttachments-08918

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirect-imageView-06183

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView` was not `VK_NULL_HANDLE`, the currently bound graphics pipeline **must** have been created with `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-vkCmdDrawIndirect-multisampledRenderToSingleSampled-07285

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` with a `VkRenderingInfo::colorAttachmentCount` parameter greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with a `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a sample count equal to the value of `rasterizationSamples` for the currently bound graphics pipeline

- VUID-vkCmdDrawIndirect-multisampledRenderToSingleSampled-07286

If `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDrawIndirect-multisampledRenderToSingleSampled-07287

If `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdDrawIndirect-renderPass-06198

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound pipeline **must** have been created with a `VkGraphicsPipelineCreateInfo::renderPass` equal to `VK_NULL_HANDLE`

- VUID-vkCmdDrawIndirect-pColorAttachments-08963

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the `VkRenderingInfo::pColorAttachments->imageView` was not `VK_NULL_HANDLE`, then the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirect-pDepthAttachment-08964

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a

graphics pipeline bound, depth test is enabled, depth write is enabled, and the `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirect-pStencilAttachment-08965

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, stencil test is enabled and the `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirect-primitivesGeneratedQueryWithRasterizerDiscard-06708

If the `primitivesGeneratedQueryWithRasterizerDiscard` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, `rasterization discard` **must** not be enabled

- VUID-vkCmdDrawIndirect-primitivesGeneratedQueryWithNonZeroStreams-06709

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, the bound graphics pipeline **must** not have been created with a non-zero value in `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream`

- VUID-vkCmdDrawIndirect-None-07619

If a shader object is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` dynamic state enabled, then `vkCmdSetTessellationDomainOriginEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07620

If the `depthClamp` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDepthClampEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07621

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_POLYGON_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetPolygonModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07622

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetRasterizationSamplesEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07623

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleMaskEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-alphaToCoverageEnable-08919

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and `alphaToCoverageEnable` was `VK_TRUE` in the last call to `vkCmdSetAlphaToCoverageEnableEXT`, then the **Fragment Output Interface** **must** contain a variable for the alpha **Component** word in **Location** 0 at **Index** 0

- VUID-vkCmdDrawIndirect-alphaToCoverageEnable-08920

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetAlphaToCoverageEnableEXT` in the current command buffer set `alphaToCoverageEnable` to `VK_TRUE`, then the **Fragment Output Interface** **must** contain a variable for the alpha **Component** word in **Location** 0 at **Index** 0

- VUID-vkCmdDrawIndirect-None-07624

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToCoverageEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07625

If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToOneEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07626

If the `logicOp` feature is enabled, a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetLogicOpEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07627

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08657

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then

`vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07628

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08658

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` for any attachment set that attachment's value in `pColorBlendEnables` to `VK_TRUE`, then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07629

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08659

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07630

If the `geometryStreams` feature is enabled, and a shader object is bound to the `VK_SHADER_STAGE_GEOMETRY_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` dynamic state enabled, then `vkCmdSetRasterizationStreamEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07631

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetConservativeRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07632

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT` dynamic state enabled, the current value of `rasterizerDiscardEnable` is `VK_FALSE`, and the current value of `conservativeRasterizationMode` is `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`,

then `vkCmdSetExtraPrimitiveOverestimationSizeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07633

If the `depthClipEnable` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT` dynamic state, then `vkCmdSetDepthClipEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07635

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-rasterizerDiscardEnable-09416

If the `VK_EXT_blend_operation_advanced` extension is enabled, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then at least one of `vkCmdSetColorBlendEquationEXT` and `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07636

If the `VK_EXT_provoking_vertex` extension is enabled, a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetProvokingVertexModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07637

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic state enabled then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08666

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08667

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT`

stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08668

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07638

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` dynamic state enabled then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08669

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08670

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08671

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07849

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_KHR` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stippledLineEnable` is `VK_TRUE`, then `vkCmdSetLineStippleKHR` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-07639

If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` dynamic state enabled, then `vkCmdSetDepthClipNegativeOneToOneEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-09650

If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` dynamic state enabled, and the `current value` of `depthClampEnable` is `VK_TRUE`, then `vkCmdSetDepthClampRangeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-pColorBlendEnables-07470

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` state enabled and the last call to `vkCmdSetColorBlendEnableEXT` set `pColorBlendEnables` for any attachment to `VK_TRUE`, then for those attachments in the subpass the corresponding image view's `format features` **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`

- VUID-vkCmdDrawIndirect-rasterizationSamples-07471

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** follow the rules for a `zero-attachment` subpass

- VUID-vkCmdDrawIndirect-samples-07472

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `VkPipelineMultisampleStateCreateInfo::rasterizationSamples` parameter used to create the bound graphics pipeline

- VUID-vkCmdDrawIndirect-samples-07473

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state and `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` states enabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `rasterizationSamples` parameter in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndirect-rasterizationSamples-07474

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and neither the `VK_AMD_mixed_attachment_samples` nor the `VK_NV_framebuffer_mixed_samples` extensions are enabled, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndirect-firstAttachment-07476

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndirect-rasterizerDiscardEnable-09417

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndirect-firstAttachment-07477

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndirect-rasterizerDiscardEnable-09418

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndirect-firstAttachment-07478

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawIndirect-rasterizerDiscardEnable-09419

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawIndirect-firstAttachment-07479

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendAdvancedEXT` calls **must** specify the advanced blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndirect-advancedBlendMaxColorAttachments-07480

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` and `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic states enabled and the last calls to `vkCmdSetColorBlendEnableEXT` and `vkCmdSetColorBlendAdvancedEXT` have enabled advanced blending, then the number of active color attachments in the current subpass **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdDrawIndirect-primitivesGeneratedQueryWithNonZeroStreams-07481

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, and the bound graphics pipeline was created with `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` state enabled, the last call to `vkCmdSetRasterizationStreamEXT` **must** have set the `rasterizationStream` to zero

- VUID-vkCmdDrawIndirect-sampleLocationsPerPixel-07482

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` member of the `VkPipelineMultisampleStateCreateInfo` structure the bound graphics pipeline has been created with

- VUID-vkCmdDrawIndirect-sampleLocationsPerPixel-07483

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` parameter of the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07484

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound

graphics pipeline was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07485

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07486

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07487

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`

- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07936

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07937

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of

`sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.height` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndirect-sampleLocationsEnable-07938

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current value of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationsPerPixel` **must** equal `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndirect-stippledLineEnable-07495

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR`, then the `stippledRectangularLines` feature **must** be enabled

- VUID-vkCmdDrawIndirect-stippledLineEnable-07496

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR`, then the `stippledBresenhamLines` feature **must** be enabled

- VUID-vkCmdDrawIndirect-stippledLineEnable-07497

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, then the `stippledSmoothLines` feature **must** be enabled

- VUID-vkCmdDrawIndirect-stippledLineEnable-07498

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR`, then the `stippledRectangularLines` feature **must** be enabled and `VkPhysicalDeviceLimits::strictLines` **must** be `VK_TRUE`

- VUID-vkCmdDrawIndirect-conservativePointAndLineRasterization-07499

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, `conservativePointAndLineRasterization` is not supported, and the effective primitive topology output by the last pre-rasterization shader stage is a line or point, then the `conservativeRasterizationMode` set by the last call to

`vkCmdSetConservativeRasterizationModeEXT`
`VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`

must

be

- VUID-vkCmdDrawIndirect-None-08877

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAttachmentFeedbackLoopEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-08684

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_VERTEX_BIT`

- VUID-vkCmdDrawIndirect-None-08685

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`

- VUID-vkCmdDrawIndirect-None-08686

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`

- VUID-vkCmdDrawIndirect-None-08687

If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_GEOMETRY_BIT`

- VUID-vkCmdDrawIndirect-None-08688

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-vkCmdDrawIndirect-None-08698

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, then all shaders created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag in the same `vkCreateShadersEXT` call **must** also be bound

- VUID-vkCmdDrawIndirect-None-08699

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, any stages in between stages whose shaders which did not create a shader with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag as part of the same `vkCreateShadersEXT` call **must** not have any `VkShaderEXT` bound

- VUID-vkCmdDrawIndirect-None-08878

All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges

- VUID-vkCmdDrawIndirect-None-08879

All bound graphics shader objects **must** have been created with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdDrawIndirect-pDynamicStates-08715
If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to `vkCmdSetDepthWriteEnableEXT` **must** be `VK_FALSE`
- VUID-vkCmdDrawIndirect-pDynamicStates-08716
If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to `vkCmdSetStencilWriteMask` **must** be `0`
- VUID-vkCmdDrawIndirect-None-09116
If a shader object is bound to any graphics stage or the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and the format of any color attachment is `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`, the corresponding element of the `pColorWriteMasks` parameter of `vkCmdSetColorWriteMaskEXT` **must** either include all of `VK_COLOR_COMPONENT_R_BIT`, `VK_COLOR_COMPONENT_G_BIT`, and `VK_COLOR_COMPONENT_B_BIT`, or none of them
- VUID-vkCmdDrawIndirect-maxFragmentDualSrcAttachments-09239
If `blending` is enabled for any attachment where either the source or destination blend factors for that attachment [use the secondary color input](#), the maximum value of `Location` for any output attachment [statically used](#) in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdDrawIndirect-None-09548
If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, the value of each element of `VkRenderingAttachmentLocationInfoKHR::pColorAttachmentLocations` set by `vkCmdSetRenderingAttachmentLocationsKHR` **must** match the value set for the corresponding element in the currently bound pipeline
- VUID-vkCmdDrawIndirect-None-09549
If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, input attachment index mappings in the currently bound pipeline **must** match those set for the current render pass instance via `VkRenderingInputAttachmentIndexInfoKHR`
- VUID-vkCmdDrawIndirect-None-04007
All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or `VK_NULL_HANDLE` buffers bound
- VUID-vkCmdDrawIndirect-None-04008
If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be `VK_NULL_HANDLE`
- VUID-vkCmdDrawIndirect-None-02721

If `robustBufferAccess` is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)

- VUID-vkCmdDrawIndirect-None-04914

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then `vkCmdSetVertexInputEXT` **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this draw command

- VUID-vkCmdDrawIndirect-Input-07939

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then all variables with the `Input` storage class decorated with `Location` in the `Vertex Execution Model OpEntryPoint` **must** contain a location in [VkVertexInputAttributeDescription2EXT::location](#)

- VUID-vkCmdDrawIndirect-Input-08734

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and either `legacyVertexAttributes` is not enabled or the SPIR-V Type associated with a given `Input` variable of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the numeric type associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be the same as [VkVertexInputAttributeDescription2EXT::format](#)

- VUID-vkCmdDrawIndirect-format-08936

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and [VkVertexInputAttributeDescription2EXT::format](#) has a 64-bit component, then the scalar width associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be 64-bit

- VUID-vkCmdDrawIndirect-format-08937

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and the scalar width associated with a `Location` decorated `Input` variable in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the corresponding [VkVertexInputAttributeDescription2EXT::format](#) **must** have a 64-bit component

- VUID-vkCmdDrawIndirect-None-09203

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and [VkVertexInputAttributeDescription2EXT::format](#) has a 64-bit component, then all `Input` variables at the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** not use components that are not present in the format

- VUID-vkCmdDrawIndirect-None-04879

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirect-None-09637

If the `primitiveTopologyListRestart` feature is not enabled, the topology is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndirect-buffer-02708

If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdDrawIndirect-buffer-02709

`buffer` **must** have been created with the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set

- VUID-vkCmdDrawIndirect-offset-02710

`offset` **must** be a multiple of 4

- VUID-vkCmdDrawIndirect-drawCount-02718

If the `multiDrawIndirect` feature is not enabled, `drawCount` **must** be 0 or 1

- VUID-vkCmdDrawIndirect-drawCount-02719

`drawCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDrawIndirectCount`

- VUID-vkCmdDrawIndirect-drawCount-00476

If `drawCount` is greater than 1, `stride` **must** be a multiple of 4 and **must** be greater than or equal to `sizeof(VkDrawIndirectCommand)`

- VUID-vkCmdDrawIndirect-drawCount-00487

If `drawCount` is equal to 1, `(offset + sizeof(VkDrawIndirectCommand))` **must** be less than or equal to the size of `buffer`

- VUID-vkCmdDrawIndirect-drawCount-00488

If `drawCount` is greater than 1, `(stride × (drawCount - 1) + offset + sizeof(VkDrawIndirectCommand))` **must** be less than or equal to the size of `buffer`

Valid Usage (Implicit)

- VUID-vkCmdDrawIndirect-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdDrawIndirect-buffer-parameter

`buffer` **must** be a valid `VkBuffer` handle

- VUID-vkCmdDrawIndirect-commandBuffer-recording

`commandBuffer` **must** be in the `recording state`

- VUID-vkCmdDrawIndirect-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdDrawIndirect-renderpass

This command **must** only be called inside of a render pass instance

- VUID-vkCmdDrawIndirect-vidcoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdDrawIndirect-commonparent

Both of **buffer**, and **commandBuffer** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	Action

The **VkDrawIndirectCommand** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDrawIndirectCommand {
    uint32_t    vertexCount;
    uint32_t    instanceCount;
    uint32_t    firstVertex;
    uint32_t    firstInstance;
} VkDrawIndirectCommand;
```

- **vertexCount** is the number of vertices to draw.
- **instanceCount** is the number of instances to draw.
- **firstVertex** is the index of the first vertex to draw.
- **firstInstance** is the instance ID of the first instance to draw.

The members of **VkDrawIndirectCommand** have the same meaning as the similarly named parameters of **vkCmdDraw**.

Valid Usage

- VUID-VkDrawIndirectCommand-pNext-09461

If the bound graphics pipeline state was created with **VkPipelineVertexInputDivisorStateCreateInfoKHR** in the **pNext** chain of **VkGraphicsPipelineCreateInfo::pVertexInputState**, any member of

`VkPipelineVertexInputDivisorStateCreateInfoKHR::pVertexBindingDivisors` has a value other than 1 in `divisor`, and `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance` is `VK_FALSE`, then `firstInstance` **must** be 0

- VUID-VkDrawIndirectCommand-None-09462

If `shader objects` are used for drawing or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled, any member of the `pVertexBindingDescriptions` parameter to the `vkCmdSetVertexInputEXT` call that sets this dynamic state has a value other than 1 in `divisor`, and `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance` is `VK_FALSE`, then `firstInstance` **must** be 0

- VUID-VkDrawIndirectCommand-firstInstance-00501

If the `drawIndirectFirstInstance` feature is not enabled, `firstInstance` **must** be 0

To record a non-indexed draw call with a draw call count sourced from a buffer, call:

```
// Provided by VK_KHR_draw_indirect_count
void vkCmdDrawIndirectCountKHR(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 buffer,
    VkDeviceSize             offset,
    VkBuffer                 countBuffer,
    VkDeviceSize             countBufferOffset,
    uint32_t                 maxDrawCount,
    uint32_t                 stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.
- `offset` is the byte offset into `buffer` where parameters begin.
- `countBuffer` is the buffer containing the draw count.
- `countBufferOffset` is the byte offset into `countBuffer` where the draw count begins.
- `maxDrawCount` specifies the maximum number of draws that will be executed. The actual number of executed draw calls is the minimum of the count specified in `countBuffer` and `maxDrawCount`.
- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawIndirectCountKHR` behaves similarly to `vkCmdDrawIndirect` except that the draw count is read by the device from a buffer during execution. The command will read an unsigned 32-bit integer from `countBuffer` located at `countBufferOffset` and use this as the draw count.

Valid Usage

- VUID-vkCmdDrawIndirectCount-magFilter-04553

If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this

command, then the image view's **format features must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT**

- VUID-vkCmdDrawIndirectCount-mipmapMode-04770

If a **VkSampler** created with **mipmapMode** equal to **VK_SAMPLER_MIPMAP_MODE_LINEAR**, and **compareEnable** equal to **VK_FALSE** is used to sample a **VkImageView** as a result of this command, then the image view's **format features must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT**

- VUID-vkCmdDrawIndirectCount-unnormalizedCoordinates-09635

If a **VkSampler** created with **unnormalizedCoordinates** equal to **VK_TRUE** is used to sample a **VkImageView** as a result of this command, then the image view's **levelCount** and **layerCount must** be 1

- VUID-vkCmdDrawIndirectCount-unnormalizedCoordinates-09636

If a **VkSampler** created with **unnormalizedCoordinates** equal to **VK_TRUE** is used to sample a **VkImageView** as a result of this command, then the image view's **viewType must** be **VK_IMAGE_VIEW_TYPE_1D** or **VK_IMAGE_VIEW_TYPE_2D**

- VUID-vkCmdDrawIndirectCount-None-06479

If a **VkImageView** is sampled with **depth comparison**, the image view's **format features must** contain **VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT**

- VUID-vkCmdDrawIndirectCount-None-02691

If a **VkImageView** is accessed using atomic operations as a result of this command, then the image view's **format features must** contain **VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT**

- VUID-vkCmdDrawIndirectCount-None-07888

If a **VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER** descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's **format features must** contain **VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT**

- VUID-vkCmdDrawIndirectCount-None-02692

If a **VkImageView** is sampled with **VK_FILTER_CUBIC_EXT** as a result of this command, then the image view's **format features must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT**

- VUID-vkCmdDrawIndirectCount-None-02693

If the **VK_EXT_filter_cubic** extension is not enabled and any **VkImageView** is sampled with **VK_FILTER_CUBIC_EXT** as a result of this command, it **must** not have a **VkImageViewType** of **VK_IMAGE_VIEW_TYPE_3D**, **VK_IMAGE_VIEW_TYPE_CUBE**, or **VK_IMAGE_VIEW_TYPE_CUBE_ARRAY**

- VUID-vkCmdDrawIndirectCount-filterCubic-02694

Any **VkImageView** being sampled with **VK_FILTER_CUBIC_EXT** as a result of this command **must** have a **VkImageViewType** and format that supports cubic filtering, as specified by **VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic** returned by **vkGetPhysicalDeviceImageFormatProperties2KHR**

- VUID-vkCmdDrawIndirectCount-filterCubicMinmax-02695

Any **VkImageView** being sampled with **VK_FILTER_CUBIC_EXT** with a reduction mode of either **VK_SAMPLER_REDUCTION_MODE_MIN** or **VK_SAMPLER_REDUCTION_MODE_MAX** as a result of this command **must** have a **VkImageViewType** and format that supports cubic filtering together with minmax filtering, as specified by

VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax returned by vkGetPhysicalDeviceImageFormatProperties2KHR

- VUID-vkCmdDrawIndirectCount-OpTypeImage-07027
For any [VkImageView](#) being written as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDrawIndirectCount-OpTypeImage-07028
For any [VkImageView](#) being read as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDrawIndirectCount-OpTypeImage-07029
For any [VkBufferView](#) being written as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDrawIndirectCount-OpTypeImage-07030
Any [VkBufferView](#) being read as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#) then the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDrawIndirectCount-None-08600
For each set *n* that is statically used by a [bound shader](#), a descriptor set **must** have been bound to *n* at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set *n*, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDrawIndirectCount-None-08601
For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDrawIndirectCount-None-10068
For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndirectCount-maintenance4-08602
If the [maintenance4](#) feature is not enabled, then for each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) and [VkPushConstantRange](#) arrays used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDrawIndirectCount-None-08114
Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDrawIndirectCount-None-08606
If the **shaderObject** feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndirectCount-None-08608
If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the **VkPipeline** object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndirectCount-None-08609
If the **VkPipeline** object bound to the pipeline bind point used by this command or any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a **VkSampler** object that uses unnormalized coordinates, that sampler **must** not be used to sample from any **VkImage** with a **VkImageView** of the type **VK_IMAGE_VIEW_TYPE_3D**, **VK_IMAGE_VIEW_TYPE_CUBE**, **VK_IMAGE_VIEW_TYPE_1D_ARRAY**, **VK_IMAGE_VIEW_TYPE_2D_ARRAY** or **VK_IMAGE_VIEW_TYPE_CUBE_ARRAY**, in any shader stage
- VUID-vkCmdDrawIndirectCount-None-08610
If the **VkPipeline** object bound to the pipeline bind point used by this command or any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a **VkSampler** object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V **OpImageSample*** or **OpImageSparseSample*** instructions with **ImplicitLod**, **Dref** or **Proj** in their name, in any shader stage
- VUID-vkCmdDrawIndirectCount-None-08611
If the **VkPipeline** object bound to the pipeline bind point used by this command or any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a **VkSampler** object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V **OpImageSample*** or **OpImageSparseSample*** instructions that includes a LOD bias or any offset values, in any shader stage
- VUID-vkCmdDrawIndirectCount-None-08607
If the **shaderObject** feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and **VK_NULL_HANDLE** shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command
- VUID-vkCmdDrawIndirectCount-uniformBuffers-06935
If any stage of the **VkPipeline** object bound to the pipeline bind point used by this command accesses a uniform buffer, and the **robustBufferAccess** feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectCount-None-08612
If the **robustBufferAccess** feature is not enabled, and any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDrawIndirectCount-storageBuffers-06936
If any stage of the **VkPipeline** object bound to the pipeline bind point used by this command accesses a storage buffer, and the **robustBufferAccess** feature is not enabled,

that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirectCount-None-08613

If the **robustBufferAccess** feature is not enabled, and any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirectCount-None-06550

If a bound shader accesses a **VkSampler** or **VkImageView** object that enables **sampler Y'C_BC_R conversion**, that object **must** only be used with **OpImageSample*** or **OpImageSparseSample*** instructions

- VUID-vkCmdDrawIndirectCount-ConstOffset-06551

If a bound shader accesses a **VkSampler** or **VkImageView** object that enables **sampler Y'C_BC_R conversion**, that object **must** not use the **ConstOffset** and **Offset** operands

- VUID-vkCmdDrawIndirectCount-viewType-07752

If a **VkImageView** is accessed as a result of this command, then the image view's **viewType** **must** match the **Dim** operand of the **OpTypeImage** as described in **Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types**

- VUID-vkCmdDrawIndirectCount-format-07753

If a **VkImageView** is accessed as a result of this command, then the **numeric type** of the image view's **format** and the **Sampled Type** operand of the **OpTypeImage** **must** match

- VUID-vkCmdDrawIndirectCount-OpImageWrite-08795

If a **VkImageView** created with a format other than **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDrawIndirectCount-OpImageWrite-08796

If a **VkImageView** created with the format **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have four components

- VUID-vkCmdDrawIndirectCount-OpImageWrite-04469

If a **VkBufferView** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdDrawIndirectCount-SampledType-04470

If a **VkImageView** with a **VkFormat** that has a 64-bit component width is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 64

- VUID-vkCmdDrawIndirectCount-SampledType-04471

If a **VkImageView** with a **VkFormat** that has a component width less than 64-bit is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 32

- VUID-vkCmdDrawIndirectCount-SampledType-04472

If a **VkBufferView** with a **VkFormat** that has a 64-bit component width is accessed as a

result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 64

- VUID-vkCmdDrawIndirectCount-SampledType-04473

If a **VkBufferView** with a **VkFormat** that has a component width less than 64-bit is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 32

- VUID-vkCmdDrawIndirectCount-sparseImageInt64Atomics-04474

If the **sparseImageInt64Atomics** feature is not enabled, **VkImage** objects created with the **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command

- VUID-vkCmdDrawIndirectCount-sparseImageInt64Atomics-04475

If the **sparseImageInt64Atomics** feature is not enabled, **VkBuffer** objects created with the **VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command

- VUID-vkCmdDrawIndirectCount-None-07288

Any shader invocation executed by this command **must** **terminate**

- VUID-vkCmdDrawIndirectCount-None-09600

If a descriptor with type equal to any of **VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE**, **VK_DESCRIPTOR_TYPE_STORAGE_IMAGE**, or **VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT** is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdDrawIndirectCount-renderPass-02684

The current render pass **must** be **compatible** with the **renderPass** member of the **VkGraphicsPipelineCreateInfo** structure specified when creating the **VkPipeline** bound to **VK_PIPELINE_BIND_POINT_GRAPHICS**

- VUID-vkCmdDrawIndirectCount-subpass-02685

The subpass index of the current render pass **must** be equal to the **subpass** member of the **VkGraphicsPipelineCreateInfo** structure specified when creating the **VkPipeline** bound to **VK_PIPELINE_BIND_POINT_GRAPHICS**

- VUID-vkCmdDrawIndirectCount-None-07748

If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set

- VUID-vkCmdDrawIndirectCount-OpTypeImage-07468

If any shader executed by this pipeline accesses an **OpTypeImage** variable with a **Dim** operand of **SubpassData**, it **must** be decorated with an **InputAttachmentIndex** that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDrawIndirectCount-None-07469

Input attachment views accessed in a subpass **must** be created with the same **VkFormat** as the corresponding subpass definition, and be created with a **VkImageView** that is compatible with the attachment referenced by the subpass' **pInputAttachments** [**InputAttachmentIndex**] in the currently bound **VkFramebuffer** as specified by **Fragment Input Attachment Compatibility**

- VUID-vkCmdDrawIndirectCount-pDepthInputAttachmentIndex-09595

Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by `VkRenderingInputAttachmentIndexInfoKHR`, or no `InputAttachmentIndex` if `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` or `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are `NULL`, **must** be created with a `VkImageView` that is compatible with the corresponding color, depth, or stencil attachment in `VkRenderingInfo`

- VUID-vkCmdDrawIndirectCount-pDepthInputAttachmentIndex-09596

Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` and `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are non-`NULL`

- VUID-vkCmdDrawIndirectCount-InputAttachmentIndex-09597

If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDrawIndirectCount-None-06537

Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirectCount-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_COLOR_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirectCount-None-09001

If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_DEPTH_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirectCount-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_STENCIL_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirectCount-None-09003

If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command

- VUID-vkCmdDrawIndirectCount-None-06539

If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment

- VUID-vkCmdDrawIndirectCount-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, `depth writes` **must** be disabled

- VUID-vkCmdDrawIndirectCount-None-06887

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, `all stencil ops` **must** be `VK_STENCIL_OP_KEEP`

- VUID-vkCmdDrawIndirectCount-None-07831

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VIEWPORT` dynamic state enabled then `vkCmdSetViewport` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07832

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SCISSOR` dynamic state enabled then `vkCmdSetScissor` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07833

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic state enabled then `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08617

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08618

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08619

If a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07834

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BIAS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBiasEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` or `vkCmdSetDepthBias2EXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07835

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic state enabled then `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08621

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer set any element of `pColorBlendEnables` to `VK_TRUE`, and the most recent call to `vkCmdSetColorBlendEquationEXT` in the current command buffer set the same element of `pColorBlendEquations` to a `VkColorBlendEquationEXT` structure with any `VkBlendFactor` member with a value of `VK_BLEND_FACTOR_CONSTANT_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR`, `VK_BLEND_FACTOR_CONSTANT_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`, `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07836

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBoundsTestEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07837

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilCompareMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilWriteMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07839

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_REFERENCE` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilReference` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-maxMultiviewInstanceIndex-02688

If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to `VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex`

- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-02689

If the bound graphics pipeline was created with `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable` set to `VK_TRUE` and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawIndirectCount-None-07634

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleLocationsEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-06666

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `vkCmdSetSampleLocationsEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's `format features` do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the `blendEnable` member of the corresponding element of the `pAttachments` member of `pColorBlendState` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndirectCount-None-08643

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then for each color attachment in the render pass, if the corresponding image view's `format features` do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the corresponding member of `pColorBlendEnables` in the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer that affected that attachment index **must** have been `VK_FALSE`

- VUID-vkCmdDrawIndirectCount-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then `rasterizationSamples` for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndirectCount-None-08644

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`,

then the most recent call to `vkCmdSetRasterizationSamplesEXT` in the current command buffer **must** have set `rasterizationSamples` to be the same as the number of samples for the current render pass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndirectCount-None-08876

If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with `vkCmdBeginRenderingKHR`

- VUID-vkCmdDrawIndirectCount-imageView-06172

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndirectCount-imageView-06173

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, this

command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndirectCount-imageView-06174

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndirectCount-imageView-06175

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndirectCount-imageView-06176

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndirectCount-imageView-06177

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndirectCount-viewMask-06178

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::viewMask` equal to `VkRenderingInfo::viewMask`

- VUID-vkCmdDrawIndirectCount-colorAttachmentCount-06179

If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::colorAttachmentCount` equal to `VkRenderingInfo::colorAttachmentCount`

- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08910

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline

- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08912

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` equal to `VK_NULL_HANDLE` **must** have the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used

to create the currently bound pipeline equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08911

If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline, or the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats`, if it exists, **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectCount-None-07749

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08646

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-attachmentCount-07750

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then the `attachmentCount` parameter of `vkCmdSetColorWriteEnableEXT` **must** be greater than or equal to the `VkPipelineColorBlendStateCreateInfo::attachmentCount` of the currently bound graphics pipeline

- VUID-vkCmdDrawIndirectCount-None-08647

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then the `attachmentCount` parameter of most recent call to `vkCmdSetColorWriteEnableEXT` in the current command buffer **must** be greater than or equal to the number of color attachments in the current render pass instance

- VUID-vkCmdDrawIndirectCount-None-07751

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state enabled then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command for each discard rectangle in `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount`

- VUID-vkCmdDrawIndirectCount-None-07880

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the

VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT dynamic state enabled, the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDiscardRectangleEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07881

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` dynamic state enabled, the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, and the **current value** of `discardRectangleEnable` is `VK_TRUE`, then `vkCmdSetDiscardRectangleModeEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-rasterizerDiscardEnable-09236

If the `VK_EXT_discard_rectangles` extension is enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetDiscardRectangleEnableEXT` in the current command buffer set `discardRectangleEnable` to `VK_TRUE`, then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08913

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08914

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08915

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08916

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08917
If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`
- VUID-vkCmdDrawIndirectCount-dynamicRenderingUnusedAttachments-08918
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndirectCount-imageView-06183
If the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView` was not `VK_NULL_HANDLE`, the currently bound graphics pipeline **must** have been created with `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-vkCmdDrawIndirectCount-multisampledRenderToSingleSampled-07285
If the current render pass instance was begun with `vkCmdBeginRenderingKHR` with a `VkRenderingInfo::colorAttachmentCount` parameter greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with a `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a sample count equal to the value of `rasterizationSamples` for the currently bound graphics pipeline
- VUID-vkCmdDrawIndirectCount-multisampledRenderToSingleSampled-07286
If `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pDepthAttachment->imageView`
- VUID-vkCmdDrawIndirectCount-multisampledRenderToSingleSampled-07287
If `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pStencilAttachment->imageView`
- VUID-vkCmdDrawIndirectCount-renderPass-06198
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound pipeline **must** have been created with a `VkGraphicsPipelineCreateInfo::renderPass` equal to `VK_NULL_HANDLE`
- VUID-vkCmdDrawIndirectCount-pColorAttachments-08963
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the `VkRenderingInfo::pColorAttachments->imageView` was not `VK_NULL_HANDLE`, then the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndirectCount-pDepthAttachment-08964

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, depth test is enabled, depth write is enabled, and the `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectCount-pStencilAttachment-08965

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, stencil test is enabled and the `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectCount-primitivesGeneratedQueryWithRasterizerDiscard-06708

If the `primitivesGeneratedQueryWithRasterizerDiscard` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, `rasterization discard` **must** not be enabled

- VUID-vkCmdDrawIndirectCount-primitivesGeneratedQueryWithNonZeroStreams-06709

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, the bound graphics pipeline **must** not have been created with a non-zero value in `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream`

- VUID-vkCmdDrawIndirectCount-None-07619

If a shader object is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` dynamic state enabled, then `vkCmdSetTessellationDomainOriginEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07620

If the `depthClamp` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDepthClampEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07621

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_POLYGON_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetPolygonModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07622

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetRasterizationSamplesEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07623

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-alphaToCoverageEnable-08919

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and `alphaToCoverageEnable` was `VK_TRUE` in the last call to `vkCmdSetAlphaToCoverageEnableEXT`, then the `Fragment Output Interface` **must** contain a variable for the alpha `Component` word in `Location 0` at `Index 0`

- VUID-vkCmdDrawIndirectCount-alphaToCoverageEnable-08920

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetAlphaToCoverageEnableEXT` in the current command buffer set `alphaToCoverageEnable` to `VK_TRUE`, then the `Fragment Output Interface` **must** contain a variable for the alpha `Component` word in `Location 0` at `Index 0`

- VUID-vkCmdDrawIndirectCount-None-07624

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToCoverageEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07625

If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToOneEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07626

If the `logicOp` feature is enabled, a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetLogicOpEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07627

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08657

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set

`rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07628

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08658

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` for any attachment set that attachment's value in `pColorBlendEnables` to `VK_TRUE`, then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07629

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08659

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07630

If the `geometryStreams` feature is enabled, and a shader object is bound to the `VK_SHADER_STAGE_GEOMETRY_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` dynamic state enabled, then `vkCmdSetRasterizationStreamEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07631

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetConservativeRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07632

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT` dynamic state enabled, the current value of `rasterizerDiscardEnable` is `VK_FALSE`, and the current value of

`conservativeRasterizationMode` is `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`, then `vkCmdSetExtraPrimitiveOverestimationSizeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07633

If the `depthClipEnable` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT` dynamic state, then `vkCmdSetDepthClipEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07635

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-rasterizerDiscardEnable-09416

If the `VK_EXT_blend_operation_advanced` extension is enabled, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then at least one of `vkCmdSetColorBlendEquationEXT` and `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07636

If the `VK_EXT_provoking_vertex` extension is enabled, a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetProvokingVertexModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07637

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic state enabled then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08666

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08667

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines`

features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08668

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07638

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` dynamic state enabled then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08669

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08670

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08671

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07849

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_KHR` dynamic state enabled, the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, and the **current value** of `stippledLineEnable` is `VK_TRUE`, then `vkCmdSetLineStippleKHR` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-07639

If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` dynamic state enabled, then `vkCmdSetDepthClipNegativeOneToOneEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-09650

If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` dynamic state enabled, and the **current value** of `depthClampEnable` is `VK_TRUE`, then `vkCmdSetDepthClampRangeEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-pColorBlendEnables-07470

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` state enabled and the last call to `vkCmdSetColorBlendEnableEXT` set `pColorBlendEnables` for any attachment to `VK_TRUE`, then for those attachments in the subpass the corresponding image view's **format features** **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`

- VUID-vkCmdDrawIndirectCount-rasterizationSamples-07471

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** follow the rules for a **zero-attachment subpass**

- VUID-vkCmdDrawIndirectCount-samples-07472

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `VkPipelineMultisampleStateCreateInfo::rasterizationSamples` parameter used to create the bound graphics pipeline

- VUID-vkCmdDrawIndirectCount-samples-07473

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state and `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` states enabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `rasterizationSamples` parameter in the last call to

vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndirectCount-rasterizationSamples-07474

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and neither the `VK_AMD_mixed_attachment_samples` nor the `VK_NV_framebuffer_mixed_samples` extensions are enabled, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndirectCount-firstAttachment-07476

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndirectCount-rasterizerDiscardEnable-09417

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndirectCount-firstAttachment-07477

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndirectCount-rasterizerDiscardEnable-09418

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndirectCount-firstAttachment-07478

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and

`attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawIndirectCount-rasterizerDiscardEnable-09419

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawIndirectCount-firstAttachment-07479

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendAdvancedEXT` calls **must** specify the advanced blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndirectCount-advancedBlendMaxColorAttachments-07480

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` and `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic states enabled and the last calls to `vkCmdSetColorBlendEnableEXT` and `vkCmdSetColorBlendAdvancedEXT` have enabled advanced blending, then the number of active color attachments in the current subpass **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdDrawIndirectCount-primitivesGeneratedQueryWithNonZeroStreams-07481

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, and the bound graphics pipeline was created with `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` state enabled, the last call to `vkCmdSetRasterizationStreamEXT` **must** have set the `rasterizationStream` to zero

- VUID-vkCmdDrawIndirectCount-sampleLocationsPerPixel-07482

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` member of the `VkPipelineMultisampleStateCreateInfo` structure the bound graphics pipeline has been created with

- VUID-vkCmdDrawIndirectCount-sampleLocationsPerPixel-07483

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` parameter of the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07484

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07485

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07486

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07487

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`

- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07936

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07937

If the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationGridSize.height **must** evenly divide VkMultisamplePropertiesEXT::sampleLocationGridSize.height as returned by vkGetPhysicalDeviceMultisamplePropertiesEXT with a samples parameter equaling the value of rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndirectCount-sampleLocationsEnable-07938

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndirectCount-stippledLineEnable-07495

If the bound graphics pipeline state was created with the or VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR, then the stippledRectangularLines feature **must** be enabled

- VUID-vkCmdDrawIndirectCount-stippledLineEnable-07496

If the bound graphics pipeline state was created with the or VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR, then the stippledBresenhamLines feature **must** be enabled

- VUID-vkCmdDrawIndirectCount-stippledLineEnable-07497

If the bound graphics pipeline state was created with the or VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR, then the stippledSmoothLines feature **must** be enabled

- VUID-vkCmdDrawIndirectCount-stippledLineEnable-07498

If the bound graphics pipeline state was created with the or VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE

- VUID-vkCmdDrawIndirectCount-conservativePointAndLineRasterization-07499

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, conservativePointAndLineRasterization is not supported, and the effective primitive

topology output by the last pre-rasterization shader stage is a line or point, then the `conservativeRasterizationMode` set by the last call to `vkCmdSetConservativeRasterizationModeEXT` **must** be `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`

- VUID-vkCmdDrawIndirectCount-None-08877

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAttachmentFeedbackLoopEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-08684

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_VERTEX_BIT`

- VUID-vkCmdDrawIndirectCount-None-08685

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`

- VUID-vkCmdDrawIndirectCount-None-08686

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`

- VUID-vkCmdDrawIndirectCount-None-08687

If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_GEOMETRY_BIT`

- VUID-vkCmdDrawIndirectCount-None-08688

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-vkCmdDrawIndirectCount-None-08698

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, then all shaders created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag in the same `vkCreateShadersEXT` call **must** also be bound

- VUID-vkCmdDrawIndirectCount-None-08699

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, any stages in between stages whose shaders which did not create a shader with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag as part of the same `vkCreateShadersEXT` call **must** not have any `VkShaderEXT` bound

- VUID-vkCmdDrawIndirectCount-None-08878

All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges

- VUID-vkCmdDrawIndirectCount-None-08879

All bound graphics shader objects **must** have been created with identical or identically

defined arrays of descriptor set layouts

- VUID-vkCmdDrawIndirectCount-pDynamicStates-08715

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to `vkCmdSetDepthWriteEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndirectCount-pDynamicStates-08716

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to `vkCmdSetStencilWriteMask` **must** be `0`

- VUID-vkCmdDrawIndirectCount-None-09116

If a shader object is bound to any graphics stage or the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and the format of any color attachment is `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`, the corresponding element of the `pColorWriteMasks` parameter of `vkCmdSetColorWriteMaskEXT` **must** either include all of `VK_COLOR_COMPONENT_R_BIT`, `VK_COLOR_COMPONENT_G_BIT`, and `VK_COLOR_COMPONENT_B_BIT`, or none of them

- VUID-vkCmdDrawIndirectCount-maxFragmentDualSrcAttachments-09239

If `blending` is enabled for any attachment where either the source or destination blend factors for that attachment [use the secondary color input](#), the maximum value of `Location` for any output attachment [statically used](#) in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDrawIndirectCount-None-09548

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, the value of each element of `VkRenderingAttachmentLocationInfoKHR::pColorAttachmentLocations` set by `vkCmdSetRenderingAttachmentLocationsKHR` **must** match the value set for the corresponding element in the currently bound pipeline

- VUID-vkCmdDrawIndirectCount-None-09549

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, input attachment index mappings in the currently bound pipeline **must** match those set for the current render pass instance via `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDrawIndirectCount-None-04007

All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or `VK_NULL_HANDLE` buffers bound

- VUID-vkCmdDrawIndirectCount-None-04008

If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be `VK_NULL_HANDLE`

- VUID-vkCmdDrawIndirectCount-None-02721

If **robustBufferAccess** is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)

- VUID-vkCmdDrawIndirectCount-None-04914

If there is a shader object bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage or the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_VERTEX_INPUT_EXT** dynamic state enabled then **vkCmdSetVertexInputEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this draw command

- VUID-vkCmdDrawIndirectCount-Input-07939

If there is a shader object bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage or the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_VERTEX_INPUT_EXT** dynamic state enabled then all variables with the **Input** storage class decorated with **Location** in the **Vertex Execution Model OpEntryPoint** **must** contain a location in **VkVertexInputAttributeDescription2EXT::location**

- VUID-vkCmdDrawIndirectCount-Input-08734

If there is a shader object bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage or the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_VERTEX_INPUT_EXT** dynamic state enabled and either **legacyVertexAttributes** is not enabled or the SPIR-V Type associated with a given **Input** variable of the corresponding **Location** in the **Vertex Execution Model OpEntryPoint** is 64-bit, then the numeric type associated with all **Input** variables of the corresponding **Location** in the **Vertex Execution Model OpEntryPoint** **must** be the same as **VkVertexInputAttributeDescription2EXT::format**

- VUID-vkCmdDrawIndirectCount-format-08936

If there is a shader object bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage or the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_VERTEX_INPUT_EXT** dynamic state enabled and **VkVertexInputAttributeDescription2EXT::format** has a 64-bit component, then the scalar width associated with all **Input** variables of the corresponding **Location** in the **Vertex Execution Model OpEntryPoint** **must** be 64-bit

- VUID-vkCmdDrawIndirectCount-format-08937

If there is a shader object bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage or the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_VERTEX_INPUT_EXT** dynamic state enabled and the scalar width associated with a **Location** decorated **Input** variable in the **Vertex Execution Model OpEntryPoint** is 64-bit, then the corresponding **VkVertexInputAttributeDescription2EXT::format** **must** have a 64-bit component

- VUID-vkCmdDrawIndirectCount-None-09203

If there is a shader object bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage or the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_VERTEX_INPUT_EXT** dynamic state enabled and **VkVertexInputAttributeDescription2EXT::format** has a 64-bit component, then all **Input** variables at the corresponding **Location** in the **Vertex Execution Model OpEntryPoint** **must** not use components that are not present in the format

- VUID-vkCmdDrawIndirectCount-None-04879

If there is a shader object bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage then **vkCmdSetPrimitiveRestartEnableEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectCount-None-09637

If the `primitiveTopologyListRestart` feature is not enabled, the topology is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndirectCount-buffer-02708

If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdDrawIndirectCount-buffer-02709

`buffer` **must** have been created with the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set

- VUID-vkCmdDrawIndirectCount-offset-02710

`offset` **must** be a multiple of 4

- VUID-vkCmdDrawIndirectCount-countBuffer-02714

If `countBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdDrawIndirectCount-countBuffer-02715

`countBuffer` **must** have been created with the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set

- VUID-vkCmdDrawIndirectCount-countBufferOffset-02716

`countBufferOffset` **must** be a multiple of 4

- VUID-vkCmdDrawIndirectCount-countBuffer-02717

The count stored in `countBuffer` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDrawIndirectCount`

- VUID-vkCmdDrawIndirectCount-countBufferOffset-04129

$(\text{countBufferOffset} + \text{sizeof}(\text{uint32_t}))$ **must** be less than or equal to the size of `countBuffer`

- VUID-vkCmdDrawIndirectCount-stride-03110

`stride` **must** be a multiple of 4 and **must** be greater than or equal to `sizeof(VkDrawIndirectCommand)`

- VUID-vkCmdDrawIndirectCount-maxDrawCount-03111

If `maxDrawCount` is greater than or equal to 1, $(\text{stride} \times (\text{maxDrawCount} - 1) + \text{offset} + \text{sizeof}(\text{VkDrawIndirectCommand}))$ **must** be less than or equal to the size of `buffer`

Valid Usage (Implicit)

- VUID-vkCmdDrawIndirectCount-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdDrawIndirectCount-buffer-parameter

`buffer` **must** be a valid `VkBuffer` handle

- VUID-vkCmdDrawIndirectCount-countBuffer-parameter

`countBuffer` **must** be a valid `VkBuffer` handle

- VUID-vkCmdDrawIndirectCount-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdDrawIndirectCount-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdDrawIndirectCount-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndirectCount-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdDrawIndirectCount-commonparent
Each of `buffer`, `commandBuffer`, and `countBuffer` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	Action

To record an indexed indirect drawing command, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDrawIndexedIndirect(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 buffer,
    VkDeviceSize             offset,
    uint32_t                 drawCount,
    uint32_t                 stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.
- `offset` is the byte offset into `buffer` where parameters begin.
- `drawCount` is the number of draws to execute, and **can** be zero.

- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawIndexedIndirect` behaves similarly to `vkCmdDrawIndexed` except that the parameters are read by the device from a buffer during execution. `drawCount` draws are executed by the command, with parameters taken from `buffer` starting at `offset` and increasing by `stride` bytes for each successive draw. The parameters of each draw are encoded in an array of `VkDrawIndexedIndirectCommand` structures. If `drawCount` is less than or equal to one, `stride` is ignored.

Valid Usage

- VUID-vkCmdDrawIndexedIndirect-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndexedIndirect-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndexedIndirect-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDrawIndexedIndirect-unnormalizedCoordinates-09636
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`
- VUID-vkCmdDrawIndexedIndirect-None-06479
If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`
- VUID-vkCmdDrawIndexedIndirect-None-02691
If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`
- VUID-vkCmdDrawIndexedIndirect-None-07888
If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`
- VUID-vkCmdDrawIndexedIndirect-None-02692
If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`

- VUID-vkCmdDrawIndexedIndirect-None-02693
If the [VK_EXT_filter_cubic](#) extension is not enabled and any [VkImageView](#) is sampled with [VK_FILTER_CUBIC_EXT](#) as a result of this command, it **must** not have a [VkImageViewType](#) of [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#)
- VUID-vkCmdDrawIndexedIndirect-filterCubic-02694
Any [VkImageView](#) being sampled with [VK_FILTER_CUBIC_EXT](#) as a result of this command **must** have a [VkImageViewType](#) and format that supports cubic filtering, as specified by [VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic](#) returned by [vkGetPhysicalDeviceImageFormatProperties2KHR](#)
- VUID-vkCmdDrawIndexedIndirect-filterCubicMinmax-02695
Any [VkImageView](#) being sampled with [VK_FILTER_CUBIC_EXT](#) with a reduction mode of either [VK_SAMPLER_REDUCTION_MODE_MIN](#) or [VK_SAMPLER_REDUCTION_MODE_MAX](#) as a result of this command **must** have a [VkImageViewType](#) and format that supports cubic filtering together with minmax filtering, as specified by [VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax](#) returned by [vkGetPhysicalDeviceImageFormatProperties2KHR](#)
- VUID-vkCmdDrawIndexedIndirect-OpTypeImage-07027
For any [VkImageView](#) being written as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDrawIndexedIndirect-OpTypeImage-07028
For any [VkImageView](#) being read as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDrawIndexedIndirect-OpTypeImage-07029
For any [VkBufferView](#) being written as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDrawIndexedIndirect-OpTypeImage-07030
Any [VkBufferView](#) being read as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#) then the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDrawIndexedIndirect-None-08600
For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDrawIndexedIndirect-None-08601
For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndexedIndirect-None-10068
For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDrawIndexedIndirect-maintenance4-08602
If the [maintenance4](#) feature is not enabled, then for each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) and [VkPushConstantRange](#) arrays used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDrawIndexedIndirect-None-08114
Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)
- VUID-vkCmdDrawIndexedIndirect-None-08606
If the [shaderObject](#) feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDrawIndexedIndirect-None-08608
If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDrawIndexedIndirect-None-08609
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage
- VUID-vkCmdDrawIndexedIndirect-None-08610
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage
- VUID-vkCmdDrawIndexedIndirect-None-08611
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage
- VUID-vkCmdDrawIndexedIndirect-None-08607
If the [shaderObject](#) feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and [VK_NULL_HANDLE](#) shader objects **must** be bound to every supported shader stage

corresponding to the pipeline bind point used by this command

- VUID-vkCmdDrawIndexedIndirect-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirect-None-08612

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirect-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirect-None-08613

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirect-None-06550

If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions

- VUID-vkCmdDrawIndexedIndirect-ConstOffset-06551

If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands

- VUID-vkCmdDrawIndexedIndirect-viewType-07752

If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDrawIndexedIndirect-format-07753

If a [VkImageView](#) is accessed as a result of this command, then the [numeric type](#) of the image view's [format](#) and the [Sampled Type](#) operand of the [OpTypeImage](#) **must** match

- VUID-vkCmdDrawIndexedIndirect-OpImageWrite-08795

If a [VkImageView](#) created with a format other than [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDrawIndexedIndirect-OpImageWrite-08796

If a [VkImageView](#) created with the format [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have four components

- VUID-vkCmdDrawIndexedIndirect-OpImageWrite-04469
If a [VkBufferView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndexedIndirect-SampledType-04470
If a [VkImageView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64
- VUID-vkCmdDrawIndexedIndirect-SampledType-04471
If a [VkImageView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32
- VUID-vkCmdDrawIndexedIndirect-SampledType-04472
If a [VkBufferView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64
- VUID-vkCmdDrawIndexedIndirect-SampledType-04473
If a [VkBufferView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32
- VUID-vkCmdDrawIndexedIndirect-sparseImageInt64Atomics-04474
If the [sparseImageInt64Atomics](#) feature is not enabled, [VkImage](#) objects created with the [VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT](#) flag **must** not be accessed by atomic instructions through an [OpTypeImage](#) with a [SampledType](#) with a [Width](#) of 64 by this command
- VUID-vkCmdDrawIndexedIndirect-sparseImageInt64Atomics-04475
If the [sparseImageInt64Atomics](#) feature is not enabled, [VkBuffer](#) objects created with the [VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT](#) flag **must** not be accessed by atomic instructions through an [OpTypeImage](#) with a [SampledType](#) with a [Width](#) of 64 by this command
- VUID-vkCmdDrawIndexedIndirect-None-07288
Any shader invocation executed by this command **must** [terminate](#)
- VUID-vkCmdDrawIndexedIndirect-None-09600
If a descriptor with type equal to any of [VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE](#), [VK_DESCRIPTOR_TYPE_STORAGE_IMAGE](#), or [VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT](#) is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndexedIndirect-renderPass-02684
The current render pass **must** be [compatible](#) with the [renderPass](#) member of the [VkGraphicsPipelineCreateInfo](#) structure specified when creating the [VkPipeline](#) bound to [VK_PIPELINE_BIND_POINT_GRAPHICS](#)
- VUID-vkCmdDrawIndexedIndirect-subpass-02685
The subpass index of the current render pass **must** be equal to the [subpass](#) member of the [VkGraphicsPipelineCreateInfo](#) structure specified when creating the [VkPipeline](#) bound to [VK_PIPELINE_BIND_POINT_GRAPHICS](#)

- VUID-vkCmdDrawIndexedIndirect-None-07748

If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set

- VUID-vkCmdDrawIndexedIndirect-OpTypeImage-07468

If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdDrawIndexedIndirect-None-07469

Input attachment views accessed in a subpass **must** be created with the same `VkFormat` as the corresponding subpass definition, and be created with a `VkImageView` that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the currently bound `VkFramebuffer` as specified by [Fragment Input Attachment Compatibility](#)

- VUID-vkCmdDrawIndexedIndirect-pDepthInputAttachmentIndex-09595

Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by `VkRenderingInputAttachmentIndexInfoKHR`, or no `InputAttachmentIndex` if `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` or `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are `NULL`, **must** be created with a `VkImageView` that is compatible with the corresponding color, depth, or stencil attachment in [VkRenderingInfo](#)

- VUID-vkCmdDrawIndexedIndirect-pDepthInputAttachmentIndex-09596

Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` and `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are non-`NULL`

- VUID-vkCmdDrawIndexedIndirect-InputAttachmentIndex-09597

If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in [VkRenderingInputAttachmentIndexInfoKHR](#)

- VUID-vkCmdDrawIndexedIndirect-None-06537

Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexedIndirect-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_COLOR_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexedIndirect-None-09001

If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_DEPTH_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexedIndirect-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_STENCIL_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexedIndirect-None-09003

If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command

- VUID-vkCmdDrawIndexedIndirect-None-06539

If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment

- VUID-vkCmdDrawIndexedIndirect-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, `depth writes` **must** be disabled

- VUID-vkCmdDrawIndexedIndirect-None-06887

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, `all stencil ops` **must** be `VK_STENCIL_OP_KEEP`

- VUID-vkCmdDrawIndexedIndirect-None-07831
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VIEWPORT` dynamic state enabled then `vkCmdSetViewport` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07832
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SCISSOR` dynamic state enabled then `vkCmdSetScissor` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07833
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic state enabled then `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-08617
If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-08618
If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-08619
If a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07834
If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BIAS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBiasEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` or `vkCmdSetDepthBias2EXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-07835
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic state enabled then `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08621

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer set any element of `pColorBlendEnables` to `VK_TRUE`, and the most recent call to `vkCmdSetColorBlendEquationEXT` in the current command buffer set the same element of `pColorBlendEquations` to a `VkColorBlendEquationEXT` structure with any `VkBlendFactor` member with a value of `VK_BLEND_FACTOR_CONSTANT_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR`, `VK_BLEND_FACTOR_CONSTANT_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`, `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07836

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBoundsTestEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07837

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilCompareMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilWriteMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07839

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_REFERENCE` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilReference` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-maxMultiviewInstanceIndex-02688

If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to `VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex`

- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-02689

If the bound graphics pipeline was created with `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable` set to `VK_TRUE` and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawIndexedIndirect-None-07634

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleLocationsEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-06666

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` dynamic state enabled, the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, and the **current value** of `sampleLocationsEnable` is `VK_TRUE`, then `vkCmdSetSampleLocationsEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's **format features** do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the `blendEnable` member of the corresponding element of the `pAttachments` member of `pColorBlendState` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndexedIndirect-None-08643

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then for each color attachment in the render pass, if the corresponding image view's **format features** do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the corresponding member of `pColorBlendEnables` in the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer that affected that attachment index **must** have been `VK_FALSE`

- VUID-vkCmdDrawIndexedIndirect-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then `rasterizationSamples` for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexedIndirect-None-08644

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`,

then the most recent call to `vkCmdSetRasterizationSamplesEXT` in the current command buffer **must** have set `rasterizationSamples` to be the same as the number of samples for

the current render pass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexedIndirect-None-08876

If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with [vkCmdBeginRenderingKHR](#)

- VUID-vkCmdDrawIndexedIndirect-imageView-06172

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pDepthAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pDepthAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndexedIndirect-imageView-06173

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pStencilAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pStencilAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndexedIndirect-imageView-06174

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pDepthAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pDepthAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL](#), this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndexedIndirect-imageView-06175

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pStencilAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pStencilAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndexedIndirect-imageView-06176

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pDepthAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pDepthAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndexedIndirect-imageView-06177

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pStencilAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pStencilAttachment](#) is [VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndexedIndirect-viewMask-06178

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the currently bound graphics pipeline **must** have been created with a [VkPipelineRenderingCreateInfo::viewMask](#) equal to [VkRenderingInfo::viewMask](#)

- VUID-vkCmdDrawIndexedIndirect-colorAttachmentCount-06179

If the [dynamicRenderingUnusedAttachments](#) feature is not enabled and the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the currently bound graphics pipeline **must** have been created with a [VkPipelineRenderingCreateInfo::colorAttachmentCount](#) equal to [VkRenderingInfo::colorAttachmentCount](#)

- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08910
If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08912
If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` equal to `VK_NULL_HANDLE` **must** have the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound pipeline equal to `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08911
If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline, or the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats`, if it exists, **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndexedIndirect-None-07749
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-None-08646
If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirect-attachmentCount-07750
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then the `attachmentCount` parameter of `vkCmdSetColorWriteEnableEXT` **must** be greater than or equal to the `VkPipelineColorBlendStateCreateInfo::attachmentCount` of the currently bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect-None-08647
If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to

the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then the `attachmentCount` parameter of most recent call to `vkCmdSetColorWriteEnableEXT` in the current command buffer **must** be greater than or equal to the number of color attachments in the current render pass instance

- VUID-vkCmdDrawIndexedIndirect-None-07751

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state enabled then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command for each discard rectangle in `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount`

- VUID-vkCmdDrawIndexedIndirect-None-07880

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDiscardRectangleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07881

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `discardRectangleEnable` is `VK_TRUE`, then `vkCmdSetDiscardRectangleModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-rasterizerDiscardEnable-09236

If the `VK_EXT_discard_rectangles` extension is enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetDiscardRectangleEnableEXT` in the current command buffer set `discardRectangleEnable` to `VK_TRUE`, then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08913

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08914

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo`

`::pDepthAttachment->imageView`

- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08915
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08916
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08917
If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`
- VUID-vkCmdDrawIndexedIndirect-dynamicRenderingUnusedAttachments-08918
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndexedIndirect-imageView-06183
If the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView` was not `VK_NULL_HANDLE`, the currently bound graphics pipeline **must** have been created with `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-vkCmdDrawIndexedIndirect-multisampledRenderToSingleSampled-07285
If the current render pass instance was begun with `vkCmdBeginRenderingKHR` with a `VkRenderingInfo::colorAttachmentCount` parameter greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with a `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a sample count equal to the value of `rasterizationSamples` for the currently bound graphics pipeline
- VUID-vkCmdDrawIndexedIndirect-multisampledRenderToSingleSampled-07286
If `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pDepthAttachment->imageView`
- VUID-vkCmdDrawIndexedIndirect-multisampledRenderToSingleSampled-07287
If `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value

of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdDrawIndexedIndirect-renderPass-06198

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound pipeline **must** have been created with a `VkGraphicsPipelineCreateInfo::renderPass` equal to `VK_NULL_HANDLE`

- VUID-vkCmdDrawIndexedIndirect-pColorAttachments-08963

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the `VkRenderingInfo::pColorAttachments->imageView` was not `VK_NULL_HANDLE`, then the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexedIndirect-pDepthAttachment-08964

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, depth test is enabled, depth write is enabled, and the `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexedIndirect-pStencilAttachment-08965

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, stencil test is enabled and the `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexedIndirect-primitivesGeneratedQueryWithRasterizerDiscard-06708

If the `primitivesGeneratedQueryWithRasterizerDiscard` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, `rasterization discard` **must** not be enabled

- VUID-vkCmdDrawIndexedIndirect-primitivesGeneratedQueryWithNonZeroStreams-06709

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, the bound graphics pipeline **must** not have been created with a non-zero value in `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream`

- VUID-vkCmdDrawIndexedIndirect-None-07619

If a shader object is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` dynamic state enabled, then `vkCmdSetTessellationDomainOriginEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07620

If the `depthClamp` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the

VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetDepthClampEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07621

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_POLYGON_MODE_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetPolygonModeEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07622

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetRasterizationSamplesEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07623

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_SAMPLE_MASK_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetSampleMaskEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-alphaToCoverageEnable-08919

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and `alphaToCoverageEnable` was VK_TRUE in the last call to `vkCmdSetAlphaToCoverageEnableEXT`, then the **Fragment Output Interface** **must** contain a variable for the alpha **Component** word in **Location** 0 at **Index** 0

- VUID-vkCmdDrawIndexedIndirect-alphaToCoverageEnable-08920

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetAlphaToCoverageEnableEXT` in the current command buffer set `alphaToCoverageEnable` to VK_TRUE, then the **Fragment Output Interface** **must** contain a variable for the alpha **Component** word in **Location** 0 at **Index** 0

- VUID-vkCmdDrawIndexedIndirect-None-07624

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetAlphaToCoverageEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07625

If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is VK_FALSE, then `vkCmdSetAlphaToOneEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this

drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07626

If the `logicOp` feature is enabled, a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetLogicOpEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07627

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08657

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07628

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08658

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` for any attachment set that attachment's value in `pColorBlendEnables` to `VK_TRUE`, then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07629

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08659

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07630

If the `geometryStreams` feature is enabled, and a shader object is bound to the

`VK_SHADER_STAGE_GEOMETRY_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` dynamic state enabled, then `vkCmdSetRasterizationStreamEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07631

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetConservativeRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07632

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT` dynamic state enabled, the current value of `rasterizerDiscardEnable` is `VK_FALSE`, and the current value of `conservativeRasterizationMode` is `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`, then `vkCmdSetExtraPrimitiveOverestimationSizeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07633

If the `depthClipEnable` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT` dynamic state, then `vkCmdSetDepthClipEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07635

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-rasterizerDiscardEnable-09416

If the `VK_EXT_blend_operation_advanced` extension is enabled, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then at least one of `vkCmdSetColorBlendEquationEXT` and `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07636

If the `VK_EXT_provoking_vertex` extension is enabled, a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetProvokingVertexModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07637

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic state enabled then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08666

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08667

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08668

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07638

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` dynamic state enabled then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08669

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08670

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08671

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07849

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_KHR` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stippledLineEnable` is `VK_TRUE`, then `vkCmdSetLineStippleKHR` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-07639

If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` dynamic state enabled, then `vkCmdSetDepthClipNegativeOneToOneEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-09650

If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` dynamic state enabled, and the `current value` of `depthClampEnable` is `VK_TRUE`, then `vkCmdSetDepthClampRangeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-pColorBlendEnables-07470

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` state enabled and the last call to `vkCmdSetColorBlendEnableEXT` set `pColorBlendEnables` for any attachment to `VK_TRUE`, then for those attachments in the subpass the corresponding image view's `format features` **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`

- VUID-vkCmdDrawIndexedIndirect-rasterizationSamples-07471

If the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** follow the rules for a `zero-attachment` subpass

- VUID-vkCmdDrawIndexedIndirect-samples-07472

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `VkPipelineMultisampleStateCreateInfo::rasterizationSamples` parameter used to create the bound graphics pipeline

- VUID-vkCmdDrawIndexedIndirect-samples-07473

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state and `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` states enabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `rasterizationSamples` parameter in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndexedIndirect-rasterizationSamples-07474

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and neither the `VK_AMD_mixed_attachment_samples` nor the `VK_NV_framebuffer_mixed_samples` extensions are enabled, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexedIndirect-firstAttachment-07476

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexedIndirect-rasterizerDiscardEnable-09417

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexedIndirect-firstAttachment-07477

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend

equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndexedIndirect-rasterizerDiscardEnable-09418

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndexedIndirect-firstAttachment-07478

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexedIndirect-rasterizerDiscardEnable-09419

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexedIndirect-firstAttachment-07479

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendAdvancedEXT` calls **must** specify the advanced blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndexedIndirect-advancedBlendMaxColorAttachments-07480

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` and `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic states enabled and the last calls to `vkCmdSetColorBlendEnableEXT` and `vkCmdSetColorBlendAdvancedEXT` have enabled advanced blending, then the number of active color attachments in the current subpass **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdDrawIndexedIndirect-primitivesGeneratedQueryWithNonZeroStreams-07481

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, and the bound graphics pipeline

was created with `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` state enabled, the last call to `vkCmdSetRasterizationStreamEXT` **must** have set the `rasterizationStream` to zero

- VUID-vkCmdDrawIndexedIndirect-sampleLocationsPerPixel-07482

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` member of the `VkPipelineMultisampleStateCreateInfo` structure the bound graphics pipeline has been created with

- VUID-vkCmdDrawIndexedIndirect-sampleLocationsPerPixel-07483

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` parameter of the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07484

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07485

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07486

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07487
If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`
- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07936
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`
- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07937
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.height` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`
- VUID-vkCmdDrawIndexedIndirect-sampleLocationsEnable-07938
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationsPerPixel` **must** equal `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`
- VUID-vkCmdDrawIndexedIndirect-stippledLineEnable-07495
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR`, then the `stippledRectangularLines` feature **must** be enabled
- VUID-vkCmdDrawIndexedIndirect-stippledLineEnable-07496
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR`, then the `stippledBresenhamLines` feature **must** be enabled
- VUID-vkCmdDrawIndexedIndirect-stippledLineEnable-07497

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, then the `stippledSmoothLines` feature **must** be enabled

- VUID-vkCmdDrawIndexedIndirect-stippledLineEnable-07498

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR`, then the `stippledRectangularLines` feature **must** be enabled and `VkPhysicalDeviceLimits::strictLines` **must** be `VK_TRUE`

- VUID-vkCmdDrawIndexedIndirect-conservativePointAndLineRasterization-07499

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, `conservativePointAndLineRasterization` is not supported, and the effective primitive topology output by the last pre-rasterization shader stage is a line or point, then the `conservativeRasterizationMode` set by the last call to `vkCmdSetConservativeRasterizationModeEXT` **must** be `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`

- VUID-vkCmdDrawIndexedIndirect-None-08877

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAttachmentFeedbackLoopEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-08684

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_VERTEX_BIT`

- VUID-vkCmdDrawIndexedIndirect-None-08685

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`

- VUID-vkCmdDrawIndexedIndirect-None-08686

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`

- VUID-vkCmdDrawIndexedIndirect-None-08687

If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_GEOMETRY_BIT`

- VUID-vkCmdDrawIndexedIndirect-None-08688

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the

current command buffer with `pStages` with an element of `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-vkCmdDrawIndexedIndirect-None-08698

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, then all shaders created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag in the same `vkCreateShadersEXT` call **must** also be bound

- VUID-vkCmdDrawIndexedIndirect-None-08699

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, any stages in between stages whose shaders which did not create a shader with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag as part of the same `vkCreateShadersEXT` call **must** not have any `VkShaderEXT` bound

- VUID-vkCmdDrawIndexedIndirect-None-08878

All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges

- VUID-vkCmdDrawIndexedIndirect-None-08879

All bound graphics shader objects **must** have been created with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdDrawIndexedIndirect-pDynamicStates-08715

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to `vkCmdSetDepthWriteEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndexedIndirect-pDynamicStates-08716

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to `vkCmdSetStencilWriteMask` **must** be 0

- VUID-vkCmdDrawIndexedIndirect-None-09116

If a shader object is bound to any graphics stage or the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and the format of any color attachment is `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`, the corresponding element of the `pColorWriteMasks` parameter of `vkCmdSetColorWriteMaskEXT` **must** either include all of `VK_COLOR_COMPONENT_R_BIT`, `VK_COLOR_COMPONENT_G_BIT`, and `VK_COLOR_COMPONENT_B_BIT`, or none of them

- VUID-vkCmdDrawIndexedIndirect-maxFragmentDualSrcAttachments-09239

If **blending** is enabled for any attachment where either the source or destination blend factors for that attachment **use the secondary color input**, the maximum value of `Location` for any output attachment **statically used** in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDrawIndexedIndirect-None-09548

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, the value of each element of

`VkRenderingAttachmentLocationInfoKHR::pColorAttachmentLocations` set by `vkCmdSetRenderingAttachmentLocationsKHR` **must** match the value set for the corresponding element in the currently bound pipeline

- VUID-vkCmdDrawIndexedIndirect-None-09549

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, input attachment index mappings in the currently bound pipeline **must** match those set for the current render pass instance via `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDrawIndexedIndirect-None-04007

All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or `VK_NULL_HANDLE` buffers bound

- VUID-vkCmdDrawIndexedIndirect-None-04008

If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be `VK_NULL_HANDLE`

- VUID-vkCmdDrawIndexedIndirect-None-02721

If `robustBufferAccess` is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)

- VUID-vkCmdDrawIndexedIndirect-None-04914

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then `vkCmdSetVertexInputEXT` **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this draw command

- VUID-vkCmdDrawIndexedIndirect-Input-07939

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then all variables with the `Input` storage class decorated with `Location` in the `Vertex Execution Model OpEntryPoint` **must** contain a location in `VkVertexInputAttributeDescription2EXT::location`

- VUID-vkCmdDrawIndexedIndirect-Input-08734

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and either `legacyVertexAttributes` is not enabled or the SPIR-V Type associated with a given `Input` variable of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the numeric type associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be the same as `VkVertexInputAttributeDescription2EXT::format`

- VUID-vkCmdDrawIndexedIndirect-format-08936

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and `VkVertexInputAttributeDescription2EXT::format` has a 64-bit component, then the scalar width associated with all `Input` variables of the corresponding

Location in the Vertex Execution Model OpEntryPoint **must** be 64-bit

- VUID-vkCmdDrawIndexedIndirect-format-08937

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and the scalar width associated with a Location decorated Input variable in the Vertex Execution Model OpEntryPoint is 64-bit, then the corresponding `VkVertexInputAttributeDescription2EXT::format` **must** have a 64-bit component

- VUID-vkCmdDrawIndexedIndirect-None-09203

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and `VkVertexInputAttributeDescription2EXT::format` has a 64-bit component, then all Input variables at the corresponding Location in the Vertex Execution Model OpEntryPoint **must** not use components that are not present in the format

- VUID-vkCmdDrawIndexedIndirect-None-04879

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirect-None-09637

If the `primitiveTopologyListRestart` feature is not enabled, the topology is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndexedIndirect-buffer-02708

If `buffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdDrawIndexedIndirect-buffer-02709

`buffer` **must** have been created with the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set

- VUID-vkCmdDrawIndexedIndirect-offset-02710

`offset` **must** be a multiple of 4

- VUID-vkCmdDrawIndexedIndirect-drawCount-02718

If the `multiDrawIndirect` feature is not enabled, `drawCount` **must** be 0 or 1

- VUID-vkCmdDrawIndexedIndirect-drawCount-02719

`drawCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxDrawIndirectCount`

- VUID-vkCmdDrawIndexedIndirect-None-07312

If `maintenance6` is not enabled, a valid index buffer **must** be bound

- VUID-vkCmdDrawIndexedIndirect-drawCount-00528

If `drawCount` is greater than 1, `stride` **must** be a multiple of 4 and **must** be greater than or equal to `sizeof(VkDrawIndexedIndirectCommand)`

- VUID-vkCmdDrawIndexedIndirect-drawCount-00539

If `drawCount` is equal to 1, (`offset + sizeof(VkDrawIndexedIndirectCommand)`) **must** be less

than or equal to the size of `buffer`

- VUID-vkCmdDrawIndexedIndirect-drawCount-00540

If `drawCount` is greater than 1, $(\text{stride} \times (\text{drawCount} - 1) + \text{offset} + \text{sizeof}(\text{VkDrawIndexedIndirectCommand}))$ **must** be less than or equal to the size of `buffer`

Valid Usage (Implicit)

- VUID-vkCmdDrawIndexedIndirect-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdDrawIndexedIndirect-buffer-parameter
`buffer` **must** be a valid `VkBuffer` handle
- VUID-vkCmdDrawIndexedIndirect-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdDrawIndexedIndirect-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdDrawIndexedIndirect-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndexedIndirect-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdDrawIndexedIndirect-commonparent
Both of `buffer`, and `commandBuffer` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	Action

The `VkDrawIndexedIndirectCommand` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDrawIndexedIndirectCommand {
    uint32_t    indexCount;
    uint32_t    instanceCount;
    uint32_t    firstIndex;
    int32_t     vertexOffset;
    uint32_t    firstInstance;
} VkDrawIndexedIndirectCommand;
```

- `indexCount` is the number of vertices to draw.
- `instanceCount` is the number of instances to draw.
- `firstIndex` is the base index within the index buffer.
- `vertexOffset` is the value added to the vertex index before indexing into the vertex buffer.
- `firstInstance` is the instance ID of the first instance to draw.

The members of `VkDrawIndexedIndirectCommand` have the same meaning as the similarly named parameters of `vkCmdDrawIndexed`.

Valid Usage

- VUID-VkDrawIndexedIndirectCommand-pNext-09461

If the bound graphics pipeline state was created with `VkPipelineVertexInputDivisorStateCreateInfoKHR` in the `pNext` chain of `VkGraphicsPipelineCreateInfo::pVertexInputState`, any member of `VkPipelineVertexInputDivisorStateCreateInfoKHR::pVertexBindingDivisors` has a value other than 1 in `divisor`, and `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance` is `VK_FALSE`, then `firstInstance` **must** be 0

- VUID-VkDrawIndexedIndirectCommand-None-09462

If `shader objects` are used for drawing or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled, any member of the `pVertexBindingDescriptions` parameter to the `vkCmdSetVertexInputEXT` call that sets this dynamic state has a value other than 1 in `divisor`, and `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance` is `VK_FALSE`, then `firstInstance` **must** be 0

- VUID-VkDrawIndexedIndirectCommand-robustBufferAccess2-08798

If `robustBufferAccess2` is not enabled, $(\text{indexSize} \times (\text{firstIndex} + \text{indexCount}) + \text{offset})$ **must** be less than or equal to the size of the bound index buffer, with `indexSize` being based on the type specified by `indexType`, where the index buffer, `indexType`, and `offset` are specified via `vkCmdBindIndexBuffer` or `vkCmdBindIndexBuffer2KHR`. If `vkCmdBindIndexBuffer2KHR` is used to bind the index buffer, the size of the bound index buffer is `vkCmdBindIndexBuffer2KHR::size`

- VUID-VkDrawIndexedIndirectCommand-firstInstance-00554

If the `drawIndirectFirstInstance` feature is not enabled, `firstInstance` **must** be 0

To record an indexed draw call with a draw call count sourced from a buffer, call:

```
// Provided by VK_KHR_draw_indirect_count
void vkCmdDrawIndexedIndirectCountKHR(
    VkCommandBuffer          commandBuffer,
    VkBuffer                 buffer,
    VkDeviceSize             offset,
    VkBuffer                 countBuffer,
    VkDeviceSize             countBufferOffset,
    uint32_t                 maxDrawCount,
    uint32_t                 stride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `buffer` is the buffer containing draw parameters.
- `offset` is the byte offset into `buffer` where parameters begin.
- `countBuffer` is the buffer containing the draw count.
- `countBufferOffset` is the byte offset into `countBuffer` where the draw count begins.
- `maxDrawCount` specifies the maximum number of draws that will be executed. The actual number of executed draw calls is the minimum of the count specified in `countBuffer` and `maxDrawCount`.
- `stride` is the byte stride between successive sets of draw parameters.

`vkCmdDrawIndexedIndirectCountKHR` behaves similarly to `vkCmdDrawIndexedIndirect` except that the draw count is read by the device from a buffer during execution. The command will read an unsigned 32-bit integer from `countBuffer` located at `countBufferOffset` and use this as the draw count.

Valid Usage

- VUID-vkCmdDrawIndexedIndirectCount-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndexedIndirectCount-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndexedIndirectCount-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDrawIndexedIndirectCount-unnormalizedCoordinates-09636
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a

`VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`

- VUID-vkCmdDrawIndexedIndirectCount-None-06479

If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`

- VUID-vkCmdDrawIndexedIndirectCount-None-02691

If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`

- VUID-vkCmdDrawIndexedIndirectCount-None-07888

If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`

- VUID-vkCmdDrawIndexedIndirectCount-None-02692

If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`

- VUID-vkCmdDrawIndexedIndirectCount-None-02693

If the `VK_EXT_filter_cubic` extension is not enabled and any `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, it **must** not have a `VkImageViewType` of `VK_IMAGE_VIEW_TYPE_3D`, `VK_IMAGE_VIEW_TYPE_CUBE`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`

- VUID-vkCmdDrawIndexedIndirectCount-filterCubic-02694

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdDrawIndexedIndirectCount-filterCubicMinmax-02695

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` with a reduction mode of either `VK_SAMPLER_REDUCTION_MODE_MIN` or `VK_SAMPLER_REDUCTION_MODE_MAX` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering together with minmax filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdDrawIndexedIndirectCount-OpTypeImage-07027

For any `VkImageView` being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawIndexedIndirectCount-OpTypeImage-07028

For any `VkImageView` being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawIndexedIndirectCount-OpTypeImage-07029

For any `VkBufferView` being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's `buffer features` **must** contain

VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT

- VUID-vkCmdDrawIndexedIndirectCount-OpTypeImage-07030

Any [VkBufferView](#) being read as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#) then the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDrawIndexedIndirectCount-None-08600

For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndexedIndirectCount-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndexedIndirectCount-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDrawIndexedIndirectCount-maintenance4-08602

If the [maintenance4](#) feature is not enabled, then for each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) and [VkPushConstantRange](#) arrays used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndexedIndirectCount-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDrawIndexedIndirectCount-None-08606

If the [shaderObject](#) feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDrawIndexedIndirectCount-None-08608

If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDrawIndexedIndirectCount-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type

VK_IMAGE_VIEW_TYPE_3D, VK_IMAGE_VIEW_TYPE_CUBE, VK_IMAGE_VIEW_TYPE_1D_ARRAY, VK_IMAGE_VIEW_TYPE_2D_ARRAY or VK_IMAGE_VIEW_TYPE_CUBE_ARRAY, in any shader stage

- VUID-vkCmdDrawIndexedIndirectCount-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage

- VUID-vkCmdDrawIndexedIndirectCount-None-08611

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDrawIndexedIndirectCount-None-08607

If the [shaderObject](#) feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and [VK_NULL_HANDLE](#) shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command

- VUID-vkCmdDrawIndexedIndirectCount-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirectCount-None-08612

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirectCount-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirectCount-None-08613

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndexedIndirectCount-None-06550

If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler YCbCr conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions

- VUID-vkCmdDrawIndexedIndirectCount-ConstOffset-06551

If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands

- VUID-vkCmdDrawIndexedIndirectCount-viewType-07752

If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdDrawIndexedIndirectCount-format-07753

If a [VkImageView](#) is accessed as a result of this command, then the [numeric type](#) of the image view's [format](#) and the [Sampled Type](#) operand of the [OpTypeImage](#) **must** match

- VUID-vkCmdDrawIndexedIndirectCount-OpImageWrite-08795

If a [VkImageView](#) created with a format other than [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdDrawIndexedIndirectCount-OpImageWrite-08796

If a [VkImageView](#) created with the format [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have four components

- VUID-vkCmdDrawIndexedIndirectCount-OpImageWrite-04469

If a [VkBufferView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdDrawIndexedIndirectCount-SampledType-04470

If a [VkImageView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64

- VUID-vkCmdDrawIndexedIndirectCount-SampledType-04471

If a [VkImageView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32

- VUID-vkCmdDrawIndexedIndirectCount-SampledType-04472

If a [VkBufferView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64

- VUID-vkCmdDrawIndexedIndirectCount-SampledType-04473

If a [VkBufferView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32

- VUID-vkCmdDrawIndexedIndirectCount-sparseImageInt64Atomics-04474

If the [sparseImageInt64Atomics](#) feature is not enabled, [VkImage](#) objects created with the [VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT](#) flag **must** not be accessed by atomic instructions through an [OpTypeImage](#) with a [SampledType](#) with a [Width](#) of 64 by this command

- VUID-vkCmdDrawIndexedIndirectCount-sparseImageInt64Atomics-04475

If the [sparseImageInt64Atomics](#) feature is not enabled, [VkBuffer](#) objects created with the

`VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command

- VUID-vkCmdDrawIndexedIndirectCount-None-07288
Any shader invocation executed by this command **must** [terminate](#)
- VUID-vkCmdDrawIndexedIndirectCount-None-09600
If a descriptor with type equal to any of `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndexedIndirectCount-renderPass-02684
The current render pass **must** be [compatible](#) with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`
- VUID-vkCmdDrawIndexedIndirectCount-subpass-02685
The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`
- VUID-vkCmdDrawIndexedIndirectCount-None-07748
If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set
- VUID-vkCmdDrawIndexedIndirectCount-OpTypeImage-07468
If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndexedIndirectCount-None-07469
Input attachment views accessed in a subpass **must** be created with the same `VkFormat` as the corresponding subpass definition, and be created with a `VkImageView` that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the currently bound `VkFramebuffer` as specified by [Fragment Input Attachment Compatibility](#)
- VUID-vkCmdDrawIndexedIndirectCount-pDepthInputAttachmentIndex-09595
Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by `VkRenderingInputAttachmentIndexInfoKHR`, or no `InputAttachmentIndex` if `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` or `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are `NULL`, **must** be created with a `VkImageView` that is compatible with the corresponding color, depth, or stencil attachment in `VkRenderingInfo`
- VUID-vkCmdDrawIndexedIndirectCount-pDepthInputAttachmentIndex-09596
Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` and `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are non-`NULL`
- VUID-vkCmdDrawIndexedIndirectCount-InputAttachmentIndex-09597
If an input attachment view accessed in a dynamic render pass via a shader object has an

`InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDrawIndexedIndirectCount-None-06537

Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexedIndirectCount-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_COLOR_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexedIndirectCount-None-09001

If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_DEPTH_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexedIndirectCount-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_STENCIL_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with

VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndexedIndirectCount-None-09003
If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command
- VUID-vkCmdDrawIndexedIndirectCount-None-06539
If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment
- VUID-vkCmdDrawIndexedIndirectCount-None-06886
If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, **depth writes must** be disabled
- VUID-vkCmdDrawIndexedIndirectCount-None-06887
If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back **writeMask** are not zero, and stencil test is enabled, **all stencil ops must** be **VK_STENCIL_OP_KEEP**
- VUID-vkCmdDrawIndexedIndirectCount-None-07831
If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_VIEWPORT** dynamic state enabled then **vkCmdSetViewport must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07832
If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_SCISSOR** dynamic state enabled then **vkCmdSetScissor must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-07833
If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_LINE_WIDTH** dynamic state enabled then **vkCmdSetLineWidth must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-08617
If a shader object is bound to any graphics stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to **VK_FALSE**, and the most recent call to **vkCmdSetPolygonModeEXT** in the current command buffer set **polygonMode** to **VK_POLYGON_MODE_LINE**, **vkCmdSetLineWidth must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndexedIndirectCount-None-08618
If a shader object is bound to any graphics stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to **VK_FALSE**, and the most recent call to **vkCmdSetPrimitiveTopologyEXT** in the current command buffer set **primitiveTopology** to any line topology, **vkCmdSetLineWidth must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08619

If a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, `vkCmdSetLineWidth` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07834

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BIAS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBiasEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` or `vkCmdSetDepthBias2EXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07835

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_BLEND_CONSTANTS` dynamic state enabled then `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08621

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer set any element of `pColorBlendEnables` to `VK_TRUE`, and the most recent call to `vkCmdSetColorBlendEquationEXT` in the current command buffer set the same element of `pColorBlendEquations` to a `VkColorBlendEquationEXT` structure with any `VkBlendFactor` member with a value of `VK_BLEND_FACTOR_CONSTANT_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR`, `VK_BLEND_FACTOR_CONSTANT_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`, `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07836

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBoundsTestEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07837

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilCompareMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilWriteMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07839

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_REFERENCE` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilReference` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-maxMultiviewInstanceIndex-02688

If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to `VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex`

- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-02689

If the bound graphics pipeline was created with `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable` set to `VK_TRUE` and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawIndexedIndirectCount-None-07634

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleLocationsEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-06666

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `vkCmdSetSampleLocationsEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's `format features` do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the `blendEnable` member of the corresponding element of the `pAttachments` member of `pColorBlendState` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndexedIndirectCount-None-08643

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent

call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then for each color attachment in the render pass, if the corresponding image view's `format features` do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the corresponding member of `pColorBlendEnables` in the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer that affected that attachment index **must** have been `VK_FALSE`

- VUID-vkCmdDrawIndexedIndirectCount-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then `rasterizationSamples` for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexedIndirectCount-None-08644

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`,

then the most recent call to `vkCmdSetRasterizationSamplesEXT` in the current command buffer **must** have set `rasterizationSamples` to be the same as the number of samples for the current render pass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexedIndirectCount-None-08876

If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with `vkCmdBeginRenderingKHR`

- VUID-vkCmdDrawIndexedIndirectCount-imageView-06172

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndexedIndirectCount-imageView-06173

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndexedIndirectCount-imageView-06174

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndexedIndirectCount-imageView-06175

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndexedIndirectCount-imageView-06176

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndexedIndirectCount-imageView-06177

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndexedIndirectCount-viewMask-06178

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::viewMask` equal to `VkRenderingInfo::viewMask`

- VUID-vkCmdDrawIndexedIndirectCount-colorAttachmentCount-06179

If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::colorAttachmentCount` equal to `VkRenderingInfo::colorAttachmentCount`

- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08910

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline

- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08912

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` equal to `VK_NULL_HANDLE` **must** have the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound pipeline equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08911

If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline, or the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats`, if it exists, **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexedIndirectCount-None-07749

If the bound graphics pipeline state was created with the

`VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08646

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-attachmentCount-07750

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then the `attachmentCount` parameter of `vkCmdSetColorWriteEnableEXT` **must** be greater than or equal to the `VkPipelineColorBlendStateCreateInfo::attachmentCount` of the currently bound graphics pipeline

- VUID-vkCmdDrawIndexedIndirectCount-None-08647

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then the `attachmentCount` parameter of most recent call to `vkCmdSetColorWriteEnableEXT` in the current command buffer **must** be greater than or equal to the number of color attachments in the current render pass instance

- VUID-vkCmdDrawIndexedIndirectCount-None-07751

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state enabled then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command for each discard rectangle in `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount`

- VUID-vkCmdDrawIndexedIndirectCount-None-07880

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDiscardRectangleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07881

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `discardRectangleEnable` is `VK_TRUE`, then `vkCmdSetDiscardRectangleModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-rasterizerDiscardEnable-09236

If the `VK_EXT_discard_rectangles` extension is enabled, and a shader object is bound to any

graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetDiscardRectangleEnableEXT` in the current command buffer set `discardRectangleEnable` to `VK_TRUE`, then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08913
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08914
If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08915
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08916
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08917
If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`
- VUID-vkCmdDrawIndexedIndirectCount-dynamicRenderingUnusedAttachments-08918
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo`

::pStencilAttachment->imageView, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexedIndirectCount-imageView-06183

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView` was not `VK_NULL_HANDLE`, the currently bound graphics pipeline **must** have been created with `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-vkCmdDrawIndexedIndirectCount-multisampledRenderToSingleSampled-07285

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` with a `VkRenderingInfo::colorAttachmentCount` parameter greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with a `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a sample count equal to the value of `rasterizationSamples` for the currently bound graphics pipeline

- VUID-vkCmdDrawIndexedIndirectCount-multisampledRenderToSingleSampled-07286

If `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDrawIndexedIndirectCount-multisampledRenderToSingleSampled-07287

If `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdDrawIndexedIndirectCount-renderPass-06198

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound pipeline **must** have been created with a `VkGraphicsPipelineCreateInfo::renderPass` equal to `VK_NULL_HANDLE`

- VUID-vkCmdDrawIndexedIndirectCount-pColorAttachments-08963

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the `VkRenderingInfo::pColorAttachments->imageView` was not `VK_NULL_HANDLE`, then the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexedIndirectCount-pDepthAttachment-08964

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, depth test is enabled, depth write is enabled, and the `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexedIndirectCount-pStencilAttachment-08965

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, stencil test is enabled and the `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndexedIndirectCount-

primitivesGeneratedQueryWithRasterizerDiscard-06708

If the `primitivesGeneratedQueryWithRasterizerDiscard` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, `rasterizationDiscard` **must** not be enabled

- VUID-vkCmdDrawIndexedIndirectCount-

primitivesGeneratedQueryWithNonZeroStreams-06709

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, the bound graphics pipeline **must** not have been created with a non-zero value in `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream`

- VUID-vkCmdDrawIndexedIndirectCount-None-07619

If a shader object is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` dynamic state enabled, then `vkCmdSetTessellationDomainOriginEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07620

If the `depthClamp` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDepthClampEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07621

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_POLYGON_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetPolygonModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07622

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetRasterizationSamplesEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07623

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-alphaToCoverageEnable-08919

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and `alphaToCoverageEnable` was `VK_TRUE` in the last call to

`vkCmdSetAlphaToCoverageEnableEXT`, then the **Fragment Output Interface** **must** contain a variable for the alpha **Component** word in **Location** 0 at **Index** 0

- VUID-vkCmdDrawIndexedIndirectCount-alphaToCoverageEnable-08920

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetAlphaToCoverageEnableEXT` in the current command buffer set `alphaToCoverageEnable` to `VK_TRUE`, then the **Fragment Output Interface** **must** contain a variable for the alpha **Component** word in **Location** 0 at **Index** 0

- VUID-vkCmdDrawIndexedIndirectCount-None-07624

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToCoverageEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07625

If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToOneEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07626

If the `logicOp` feature is enabled, a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetLogicOpEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07627

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08657

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07628

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08658

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent

call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` for any attachment set that attachment's value in `pColorBlendEnables` to `VK_TRUE`, then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07629

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08659

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07630

If the `geometryStreams` feature is enabled, and a shader object is bound to the `VK_SHADER_STAGE_GEOMETRY_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` dynamic state enabled, then `vkCmdSetRasterizationStreamEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07631

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetConservativeRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07632

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT` dynamic state enabled, the current value of `rasterizerDiscardEnable` is `VK_FALSE`, and the current value of `conservativeRasterizationMode` is `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`, then `vkCmdSetExtraPrimitiveOverestimationSizeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07633

If the `depthClipEnable` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT` dynamic state, then `vkCmdSetDepthClipEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07635

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-rasterizerDiscardEnable-09416

If the `VK_EXT_blend_operation_advanced` extension is enabled, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then at least one of `vkCmdSetColorBlendEquationEXT` and `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07636

If the `VK_EXT_provoking_vertex` extension is enabled, a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetProvokingVertexModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07637

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic state enabled then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08666

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08667

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08668

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the

VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT or VK_SHADER_STAGE_GEOMETRY_BIT stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07638

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` dynamic state enabled then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08669

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08670

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08671

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07849

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_KHR` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stippledLineEnable` is `VK_TRUE`, then `vkCmdSetLineStippleKHR` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-07639

If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics

stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` dynamic state enabled, then `vkCmdSetDepthClipNegativeOneToOneEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-09650

If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` dynamic state enabled, and the `current value` of `depthClampEnable` is `VK_TRUE`, then `vkCmdSetDepthClampRangeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-pColorBlendEnables-07470

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` state enabled and the last call to `vkCmdSetColorBlendEnableEXT` set `pColorBlendEnables` for any attachment to `VK_TRUE`, then for those attachments in the subpass the corresponding image view's `format features` **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`

- VUID-vkCmdDrawIndexedIndirectCount-rasterizationSamples-07471

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** follow the rules for a `zero-attachment` subpass

- VUID-vkCmdDrawIndexedIndirectCount-samples-07472

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `VkPipelineMultisampleStateCreateInfo::rasterizationSamples` parameter used to create the bound graphics pipeline

- VUID-vkCmdDrawIndexedIndirectCount-samples-07473

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state and `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` states enabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `rasterizationSamples` parameter in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndexedIndirectCount-rasterizationSamples-07474

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and neither the `VK_AMD_mixed_attachment_samples` nor the `VK_NV_framebuffer_mixed_samples` extensions are enabled, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndexedIndirectCount-firstAttachment-07476

If the bound graphics pipeline state was created with the

`VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexedIndirectCount-rasterizerDiscardEnable-09417

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexedIndirectCount-firstAttachment-07477

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndexedIndirectCount-rasterizerDiscardEnable-09418

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndexedIndirectCount-firstAttachment-07478

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexedIndirectCount-rasterizerDiscardEnable-09419

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawIndexedIndirectCount-firstAttachment-07479

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendAdvancedEXT` calls **must** specify the advanced blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndexedIndirectCount-advancedBlendMaxColorAttachments-07480

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` and `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic states enabled and the last calls to `vkCmdSetColorBlendEnableEXT` and `vkCmdSetColorBlendAdvancedEXT` have enabled advanced blending, then the number of active color attachments in the current subpass **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdDrawIndexedIndirectCount-primitivesGeneratedQueryWithNonZeroStreams-07481

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, and the bound graphics pipeline was created with `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` state enabled, the last call to `vkCmdSetRasterizationStreamEXT` **must** have set the `rasterizationStream` to zero

- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsPerPixel-07482

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` member of the `VkPipelineMultisampleStateCreateInfo` structure the bound graphics pipeline has been created with

- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsPerPixel-07483

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` parameter of the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07484

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07485

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound

graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07486

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07487

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`

- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07936

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07937

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.height` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndexedIndirectCount-sampleLocationsEnable-07938

If the bound graphics pipeline state was created with the

VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the current value of sampleLocationsEnable is VK_TRUE, then sampleLocationsInfo.sampleLocationsPerPixel **must** equal rasterizationSamples in the last call to vkCmdSetRasterizationSamplesEXT

- VUID-vkCmdDrawIndexedIndirectCount-stippledLineEnable-07495

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR, then the stippledRectangularLines feature **must** be enabled

- VUID-vkCmdDrawIndexedIndirectCount-stippledLineEnable-07496

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR, then the stippledBresenhamLines feature **must** be enabled

- VUID-vkCmdDrawIndexedIndirectCount-stippledLineEnable-07497

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR, then the stippledSmoothLines feature **must** be enabled

- VUID-vkCmdDrawIndexedIndirectCount-stippledLineEnable-07498

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current stippledLineEnable state is VK_TRUE and the current lineRasterizationMode state is VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR, then the stippledRectangularLines feature **must** be enabled and VkPhysicalDeviceLimits::strictLines **must** be VK_TRUE

- VUID-vkCmdDrawIndexedIndirectCount-conservativePointAndLineRasterization-07499

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, conservativePointAndLineRasterization is not supported, and the effective primitive topology output by the last pre-rasterization shader stage is a line or point, then the conservativeRasterizationMode set by the last call to vkCmdSetConservativeRasterizationModeEXT **must** be VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT

- VUID-vkCmdDrawIndexedIndirectCount-None-08877

If a shader object is bound to the VK_SHADER_STAGE_FRAGMENT_BIT stage or a graphics pipeline is bound which was created with the VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT dynamic state enabled, and the current value of rasterizerDiscardEnable is VK_FALSE, then

`vkCmdSetAttachmentFeedbackLoopEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-08684
If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_VERTEX_BIT`
- VUID-vkCmdDrawIndexedIndirectCount-None-08685
If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`
- VUID-vkCmdDrawIndexedIndirectCount-None-08686
If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`
- VUID-vkCmdDrawIndexedIndirectCount-None-08687
If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_GEOMETRY_BIT`
- VUID-vkCmdDrawIndexedIndirectCount-None-08688
If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_FRAGMENT_BIT`
- VUID-vkCmdDrawIndexedIndirectCount-None-08698
If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, then all shaders created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag in the same `vkCreateShadersEXT` call **must** also be bound
- VUID-vkCmdDrawIndexedIndirectCount-None-08699
If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, any stages in between stages whose shaders which did not create a shader with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag as part of the same `vkCreateShadersEXT` call **must** not have any `VkShaderEXT` bound
- VUID-vkCmdDrawIndexedIndirectCount-None-08878
All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndexedIndirectCount-None-08879
All bound graphics shader objects **must** have been created with identical or identically defined arrays of descriptor set layouts
- VUID-vkCmdDrawIndexedIndirectCount-pDynamicStates-08715
If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to `vkCmdSetDepthWriteEnableEXT` **must** be `VK_FALSE`
- VUID-vkCmdDrawIndexedIndirectCount-pDynamicStates-08716

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to `vkCmdSetStencilWriteMask` **must** be 0

- VUID-vkCmdDrawIndexedIndirectCount-None-09116

If a shader object is bound to any graphics stage or the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and the format of any color attachment is `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`, the corresponding element of the `pColorWriteMasks` parameter of `vkCmdSetColorWriteMaskEXT` **must** either include all of `VK_COLOR_COMPONENT_R_BIT`, `VK_COLOR_COMPONENT_G_BIT`, and `VK_COLOR_COMPONENT_B_BIT`, or none of them

- VUID-vkCmdDrawIndexedIndirectCount-maxFragmentDualSrcAttachments-09239

If `blending` is enabled for any attachment where either the source or destination blend factors for that attachment [use the secondary color input](#), the maximum value of `Location` for any output attachment [statically used](#) in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`

- VUID-vkCmdDrawIndexedIndirectCount-None-09548

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, the value of each element of `VkRenderingAttachmentLocationInfoKHR::pColorAttachmentLocations` set by `vkCmdSetRenderingAttachmentLocationsKHR` **must** match the value set for the corresponding element in the currently bound pipeline

- VUID-vkCmdDrawIndexedIndirectCount-None-09549

If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, input attachment index mappings in the currently bound pipeline **must** match those set for the current render pass instance via `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdDrawIndexedIndirectCount-None-04007

All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or `VK_NULL_HANDLE` buffers bound

- VUID-vkCmdDrawIndexedIndirectCount-None-04008

If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be `VK_NULL_HANDLE`

- VUID-vkCmdDrawIndexedIndirectCount-None-02721

If `robustBufferAccess` is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)

- VUID-vkCmdDrawIndexedIndirectCount-None-04914

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then `vkCmdSetVertexInputEXT` **must** have been called and not subsequently

[invalidated](#) in the current command buffer prior to this draw command

- VUID-vkCmdDrawIndexedIndirectCount-Input-07939

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then all variables with the `Input` storage class decorated with `Location` in the `Vertex Execution Model OpEntryPoint` **must** contain a location in `VkVertexInputAttributeDescription2EXT::location`

- VUID-vkCmdDrawIndexedIndirectCount-Input-08734

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and either `legacyVertexAttributes` is not enabled or the SPIR-V Type associated with a given `Input` variable of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the numeric type associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be the same as `VkVertexInputAttributeDescription2EXT::format`

- VUID-vkCmdDrawIndexedIndirectCount-format-08936

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and `VkVertexInputAttributeDescription2EXT::format` has a 64-bit component, then the scalar width associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be 64-bit

- VUID-vkCmdDrawIndexedIndirectCount-format-08937

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and the scalar width associated with a `Location` decorated `Input` variable in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the corresponding `VkVertexInputAttributeDescription2EXT::format` **must** have a 64-bit component

- VUID-vkCmdDrawIndexedIndirectCount-None-09203

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and `VkVertexInputAttributeDescription2EXT::format` has a 64-bit component, then all `Input` variables at the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** not use components that are not present in the format

- VUID-vkCmdDrawIndexedIndirectCount-None-04879

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndexedIndirectCount-None-09637

If the `primitiveTopologyListRestart` feature is not enabled, the topology is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndexedIndirectCount-buffer-02708
If **buffer** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-vkCmdDrawIndexedIndirectCount-buffer-02709
buffer **must** have been created with the **VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT** bit set
- VUID-vkCmdDrawIndexedIndirectCount-offset-02710
offset **must** be a multiple of 4
- VUID-vkCmdDrawIndexedIndirectCount-countBuffer-02714
If **countBuffer** is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object
- VUID-vkCmdDrawIndexedIndirectCount-countBuffer-02715
countBuffer **must** have been created with the **VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT** bit set
- VUID-vkCmdDrawIndexedIndirectCount-countBufferOffset-02716
countBufferOffset **must** be a multiple of 4
- VUID-vkCmdDrawIndexedIndirectCount-countBuffer-02717
The count stored in **countBuffer** **must** be less than or equal to **VkPhysicalDeviceLimits::maxDrawIndirectCount**
- VUID-vkCmdDrawIndexedIndirectCount-countBufferOffset-04129
(**countBufferOffset** + **sizeof(uint32_t)**) **must** be less than or equal to the size of **countBuffer**
- VUID-vkCmdDrawIndexedIndirectCount-None-07312
If **maintenance6** is not enabled, a valid index buffer **must** be bound
- VUID-vkCmdDrawIndexedIndirectCount-stride-03142
stride **must** be a multiple of 4 and **must** be greater than or equal to **sizeof(VkDrawIndexedIndirectCommand)**
- VUID-vkCmdDrawIndexedIndirectCount-maxDrawCount-03143
If **maxDrawCount** is greater than or equal to 1, (**stride** × (**maxDrawCount** - 1) + **offset** + **sizeof(VkDrawIndexedIndirectCommand)**) **must** be less than or equal to the size of **buffer**

Valid Usage (Implicit)

- VUID-vkCmdDrawIndexedIndirectCount-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdDrawIndexedIndirectCount-buffer-parameter
buffer **must** be a valid **VkBuffer** handle
- VUID-vkCmdDrawIndexedIndirectCount-countBuffer-parameter
countBuffer **must** be a valid **VkBuffer** handle
- VUID-vkCmdDrawIndexedIndirectCount-commandBuffer-recording
commandBuffer **must** be in the **recording state**
- VUID-vkCmdDrawIndexedIndirectCount-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics

operations

- VUID-vkCmdDrawIndexedIndirectCount-renderpass

This command **must** only be called inside of a render pass instance

- VUID-vkCmdDrawIndexedIndirectCount-videocoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdDrawIndexedIndirectCount-commonparent

Each of **buffer**, **commandBuffer**, and **countBuffer** **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	Action

20.3.1. Drawing Transform Feedback

It is possible to draw vertex data that was previously captured during active [transform feedback](#) by binding one or more of the transform feedback buffers as vertex buffers. A pipeline barrier is required between using the buffers as transform feedback buffers and vertex buffers to ensure all writes to the transform feedback buffers are visible when the data is read as vertex attributes. The source access is [VK_ACCESS_TRANSFORM_FEEDBACK_WRITE_BIT_EXT](#) and the destination access is [VK_ACCESS_VERTEX_ATTRIBUTE_READ_BIT](#) for the pipeline stages [VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT](#) and [VK_PIPELINE_STAGE_VERTEX_INPUT_BIT](#) respectively. The value written to the counter buffer by [vkCmdEndTransformFeedbackEXT](#) **can** be used to determine the vertex count for the draw. A pipeline barrier is required between using the counter buffer for [vkCmdEndTransformFeedbackEXT](#) and [vkCmdDrawIndirectByteCountEXT](#) where the source access is [VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT](#) and the destination access is [VK_ACCESS_INDIRECT_COMMAND_READ_BIT](#) for the pipeline stages [VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT](#) and [VK_PIPELINE_STAGE_DRAW_INDIRECT_BIT](#) respectively.

To record a non-indexed draw call, where the vertex count is based on a byte count read from a buffer and the passed in vertex stride parameter, call:

```
// Provided by VK_EXT_transform_feedback
void vkCmdDrawIndirectByteCountEXT(
    VkCommandBuffer                                commandBuffer,
    uint32_t                                        instanceCount,
    uint32_t                                        firstInstance,
    VkBuffer                                         counterBuffer,
    VkDeviceSize                                    counterBufferOffset,
    uint32_t                                        counterOffset,
    uint32_t                                        vertexStride);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `instanceCount` is the number of instances to draw.
- `firstInstance` is the instance ID of the first instance to draw.
- `counterBuffer` is the buffer handle from where the byte count is read.
- `counterBufferOffset` is the offset into the buffer used to read the byte count, which is used to calculate the vertex count for this draw call.
- `counterOffset` is subtracted from the byte count read from the `counterBuffer` at the `counterBufferOffset`
- `vertexStride` is the stride in bytes between each element of the vertex data that is used to calculate the vertex count from the counter value. This value is typically the same value that was used in the graphics pipeline state when the transform feedback was captured as the `XfbStride`.

When the command is executed, primitives are assembled in the same way as done with `vkCmdDraw` except the `vertexCount` is calculated based on the byte count read from `counterBuffer` at offset `counterBufferOffset`. The assembled primitives execute the bound graphics pipeline.

The effective `vertexCount` is calculated as follows:

```
const uint32_t * counterBufferPtr = (const uint8_t *)counterBuffer.address +
counterBufferOffset;
vertexCount = floor(max(0, (*counterBufferPtr - counterOffset)) / vertexStride);
```

The effective `firstVertex` is zero.

Valid Usage

- VUID-vkCmdDrawIndirectByteCountEXT-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDrawIndirectByteCountEXT-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and

`compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`

- VUID-vkCmdDrawIndirectByteCountEXT-unnormalizedCoordinates-09635

If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1

- VUID-vkCmdDrawIndirectByteCountEXT-unnormalizedCoordinates-09636

If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`

- VUID-vkCmdDrawIndirectByteCountEXT-None-06479

If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`

- VUID-vkCmdDrawIndirectByteCountEXT-None-02691

If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`

- VUID-vkCmdDrawIndirectByteCountEXT-None-07888

If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`

- VUID-vkCmdDrawIndirectByteCountEXT-None-02692

If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`

- VUID-vkCmdDrawIndirectByteCountEXT-None-02693

If the `VK_EXT_filter_cubic` extension is not enabled and any `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, it **must** not have a `VkImageViewType` of `VK_IMAGE_VIEW_TYPE_3D`, `VK_IMAGE_VIEW_TYPE_CUBE`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`

- VUID-vkCmdDrawIndirectByteCountEXT-filterCubic-02694

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdDrawIndirectByteCountEXT-filterCubicMinmax-02695

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` with a reduction mode of either `VK_SAMPLER_REDUCTION_MODE_MIN` or `VK_SAMPLER_REDUCTION_MODE_MAX` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering together with minmax filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdDrawIndirectByteCountEXT-OpTypeImage-07027

For any `VkImageView` being written as a storage image where the image format field of

the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawIndirectByteCountEXT-OpTypeImage-07028

For any `VkImageView` being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawIndirectByteCountEXT-OpTypeImage-07029

For any `VkBufferView` being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's `buffer features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawIndirectByteCountEXT-OpTypeImage-07030

Any `VkBufferView` being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown` then the view's `buffer features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`

- VUID-vkCmdDrawIndirectByteCountEXT-None-08600

For each set n that is statically used by a `bound shader`, a descriptor set **must** have been bound to n at the same pipeline bind point, with a `VkPipelineLayout` that is compatible for set n , with the `VkPipelineLayout` used to create the current `VkPipeline` or the `VkDescriptorSetLayout` array used to create the current `VkShaderEXT`, as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndirectByteCountEXT-None-08601

For each push constant that is statically used by a `bound shader`, a push constant value **must** have been set for the same pipeline bind point, with a `VkPipelineLayout` that is compatible for push constants, with the `VkPipelineLayout` used to create the current `VkPipeline` or the `VkDescriptorSetLayout` array used to create the current `VkShaderEXT`, as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndirectByteCountEXT-None-10068

For each array of resources that is used by a `bound shader`, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDrawIndirectByteCountEXT-maintenance4-08602

If the `maintenance4` feature is not enabled, then for each push constant that is statically used by a `bound shader`, a push constant value **must** have been set for the same pipeline bind point, with a `VkPipelineLayout` that is compatible for push constants, with the `VkPipelineLayout` used to create the current `VkPipeline` or the `VkDescriptorSetLayout` and `VkPushConstantRange` arrays used to create the current `VkShaderEXT`, as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDrawIndirectByteCountEXT-None-08114

Descriptors in each bound descriptor set, specified via `vkCmdBindDescriptorSets`, **must** be valid as described by [descriptor validity](#) if they are statically used by a `bound shader`

- VUID-vkCmdDrawIndirectByteCountEXT-None-08606

If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08608

If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDrawIndirectByteCountEXT-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDrawIndirectByteCountEXT-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage

- VUID-vkCmdDrawIndirectByteCountEXT-None-08611

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdDrawIndirectByteCountEXT-None-08607

If the [shaderObject](#) feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and [VK_NULL_HANDLE](#) shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command

- VUID-vkCmdDrawIndirectByteCountEXT-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirectByteCountEXT-None-08612

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirectByteCountEXT-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirectByteCountEXT-None-08613

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage

corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdDrawIndirectByteCountEXT-None-06550
If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions
- VUID-vkCmdDrawIndirectByteCountEXT-ConstOffset-06551
If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands
- VUID-vkCmdDrawIndirectByteCountEXT-viewType-07752
If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)
- VUID-vkCmdDrawIndirectByteCountEXT-format-07753
If a [VkImageView](#) is accessed as a result of this command, then the [numeric type](#) of the image view's [format](#) and the [Sampled Type](#) operand of the [OpTypeImage](#) **must** match
- VUID-vkCmdDrawIndirectByteCountEXT-OpImageWrite-08795
If a [VkImageView](#) created with a format other than [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDrawIndirectByteCountEXT-OpImageWrite-08796
If a [VkImageView](#) created with the format [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have four components
- VUID-vkCmdDrawIndirectByteCountEXT-OpImageWrite-04469
If a [VkBufferView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDrawIndirectByteCountEXT-SampledType-04470
If a [VkImageView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64
- VUID-vkCmdDrawIndirectByteCountEXT-SampledType-04471
If a [VkImageView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32
- VUID-vkCmdDrawIndirectByteCountEXT-SampledType-04472
If a [VkBufferView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64
- VUID-vkCmdDrawIndirectByteCountEXT-SampledType-04473
If a [VkBufferView](#) with a [VkFormat](#) that has a component width less than 64-bit is

accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 32

- VUID-vkCmdDrawIndirectByteCountEXT-sparseImageInt64Atomics-04474
If the **sparseImageInt64Atomics** feature is not enabled, **VkImage** objects created with the **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command
- VUID-vkCmdDrawIndirectByteCountEXT-sparseImageInt64Atomics-04475
If the **sparseImageInt64Atomics** feature is not enabled, **VkBuffer** objects created with the **VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command
- VUID-vkCmdDrawIndirectByteCountEXT-None-07288
Any shader invocation executed by this command **must** **terminate**
- VUID-vkCmdDrawIndirectByteCountEXT-None-09600
If a descriptor with type equal to any of **VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE**, **VK_DESCRIPTOR_TYPE_STORAGE_IMAGE**, or **VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT** is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDrawIndirectByteCountEXT-renderPass-02684
The current render pass **must** be **compatible** with the **renderPass** member of the **VkGraphicsPipelineCreateInfo** structure specified when creating the **VkPipeline** bound to **VK_PIPELINE_BIND_POINT_GRAPHICS**
- VUID-vkCmdDrawIndirectByteCountEXT-subpass-02685
The subpass index of the current render pass **must** be equal to the **subpass** member of the **VkGraphicsPipelineCreateInfo** structure specified when creating the **VkPipeline** bound to **VK_PIPELINE_BIND_POINT_GRAPHICS**
- VUID-vkCmdDrawIndirectByteCountEXT-None-07748
If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set
- VUID-vkCmdDrawIndirectByteCountEXT-OpTypeImage-07468
If any shader executed by this pipeline accesses an **OpTypeImage** variable with a **Dim** operand of **SubpassData**, it **must** be decorated with an **InputAttachmentIndex** that corresponds to a valid input attachment in the current subpass
- VUID-vkCmdDrawIndirectByteCountEXT-None-07469
Input attachment views accessed in a subpass **must** be created with the same **VkFormat** as the corresponding subpass definition, and be created with a **VkImageView** that is compatible with the attachment referenced by the subpass' **pInputAttachments** [**InputAttachmentIndex**] in the currently bound **VkFramebuffer** as specified by **Fragment Input Attachment Compatibility**
- VUID-vkCmdDrawIndirectByteCountEXT-pDepthInputAttachmentIndex-09595
Input attachment views accessed in a dynamic render pass with a **InputAttachmentIndex** referenced by **VkRenderingInputAttachmentIndexInfoKHR**, or no **InputAttachmentIndex** if **VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex** or **VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex** are **NULL**, **must** be created with a **VkImageView** that is compatible with the corresponding color, depth, or

stencil attachment in [VkRenderingInfo](#)

- VUID-vkCmdDrawIndirectByteCountEXT-pDepthInputAttachmentIndex-09596

Input attachment views accessed in a dynamic render pass via a shader object **must** have an [InputAttachmentIndex](#) if both [VkRenderingInputAttachmentIndexInfoKHR](#):[pDepthInputAttachmentIndex](#) and [VkRenderingInputAttachmentIndexInfoKHR](#):[pStencilInputAttachmentIndex](#) are non-NULL

- VUID-vkCmdDrawIndirectByteCountEXT-InputAttachmentIndex-09597

If an input attachment view accessed in a dynamic render pass via a shader object has an [InputAttachmentIndex](#), the [InputAttachmentIndex](#) **must** match an index in [VkRenderingInputAttachmentIndexInfoKHR](#)

- VUID-vkCmdDrawIndirectByteCountEXT-None-06537

Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirectByteCountEXT-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the [VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT](#) image layout, and either:

- the [VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT](#) is set on the currently bound pipeline or
- the last call to [vkCmdSetAttachmentFeedbackLoopEnableEXT](#) included [VK_IMAGE_ASPECT_COLOR_BIT](#) and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with [VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT](#)

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirectByteCountEXT-None-09001

If a depth attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the [VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT](#) image layout, and either:

- the [VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT](#) is set on the currently bound pipeline or
- the last call to [vkCmdSetAttachmentFeedbackLoopEnableEXT](#) included [VK_IMAGE_ASPECT_DEPTH_BIT](#) and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with [VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT](#)

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirectByteCountEXT-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the

VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT image layout, and either:

- the VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT is set on the currently bound pipeline or
- the last call to vkCmdSetAttachmentFeedbackLoopEnableEXT included VK_IMAGE_ASPECT_STENCIL_BIT and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdDrawIndirectByteCountEXT-None-09003

If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command

- VUID-vkCmdDrawIndirectByteCountEXT-None-06539

If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment

- VUID-vkCmdDrawIndirectByteCountEXT-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, **depth writes must** be disabled

- VUID-vkCmdDrawIndirectByteCountEXT-None-06887

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back **writeMask** are not zero, and stencil test is enabled, **all stencil ops must** be VK_STENCIL_OP_KEEP

- VUID-vkCmdDrawIndirectByteCountEXT-None-07831

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_VIEWPORT dynamic state enabled then vkCmdSetViewport **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07832

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SCISSOR dynamic state enabled then vkCmdSetScissor **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07833

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_WIDTH dynamic state enabled then vkCmdSetLineWidth **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08617

If a shader object is bound to any graphics stage, and the most recent call to vkCmdSetRasterizerDiscardEnableEXT in the current command buffer set **rasterizerDiscardEnable** to VK_FALSE, and the most recent call to vkCmdSetPolygonModeEXT in the current command buffer set **polygonMode** to VK_POLYGON_MODE_LINE, vkCmdSetLineWidth **must** have been called and not subsequently

[invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08618

If a shader object is bound to any graphics stage, and the most recent call to [vkCmdSetRasterizerDiscardEnableEXT](#) in the current command buffer set [rasterizerDiscardEnable](#) to [VK_FALSE](#), and the most recent call to [vkCmdSetPrimitiveTopologyEXT](#) in the current command buffer set [primitiveTopology](#) to any line topology, [vkCmdSetLineWidth](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08619

If a shader object that outputs line primitives is bound to the [VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT](#) or [VK_SHADER_STAGE_GEOMETRY_BIT](#) stage, and the most recent call to [vkCmdSetRasterizerDiscardEnableEXT](#) in the current command buffer set [rasterizerDiscardEnable](#) to [VK_FALSE](#), [vkCmdSetLineWidth](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07834

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the [VK_DYNAMIC_STATE_DEPTH_BIAS](#) dynamic state enabled, the [current value](#) of [rasterizerDiscardEnable](#) is [VK_FALSE](#), and the [current value](#) of [depthBiasEnable](#) is [VK_TRUE](#), then [vkCmdSetDepthBounds](#) or [vkCmdSetDepthBias2EXT](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07835

If the bound graphics pipeline state was created with the [VK_DYNAMIC_STATE_BLEND_CONSTANTS](#) dynamic state enabled then [vkCmdSetBlendConstants](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08621

If a shader object is bound to the [VK_SHADER_STAGE_FRAGMENT_BIT](#) stage, and the most recent call to [vkCmdSetRasterizerDiscardEnableEXT](#) in the current command buffer set [rasterizerDiscardEnable](#) to [VK_FALSE](#), and the most recent call to [vkCmdSetColorBlendEnableEXT](#) in the current command buffer set any element of [pColorBlendEnables](#) to [VK_TRUE](#), and the most recent call to [vkCmdSetColorBlendEquationEXT](#) in the current command buffer set the same element of [pColorBlendEquations](#) to a [VkColorBlendEquationEXT](#) structure with any [VkBlendFactor](#) member with a value of [VK_BLEND_FACTOR_CONSTANT_COLOR](#), [VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR](#), [VK_BLEND_FACTOR_CONSTANT_ALPHA](#), or [VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA](#), [vkCmdSetBlendConstants](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07836

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the [VK_DYNAMIC_STATE_DEPTH_BOUNDS](#) dynamic state enabled, the [current value](#) of [rasterizerDiscardEnable](#) is [VK_FALSE](#), and the [current value](#) of [depthBoundsTestEnable](#) is [VK_TRUE](#), then [vkCmdSetDepthBounds](#) **must** have been called and

not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07837

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the [VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK](#) dynamic state enabled, the [current value](#) of [rasterizerDiscardEnable](#) is [VK_FALSE](#), and the [current value](#) of [stencilTestEnable](#) is [VK_TRUE](#), then [vkCmdSetStencilCompareMask](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the [VK_DYNAMIC_STATE_STENCIL_WRITE_MASK](#) dynamic state enabled, the [current value](#) of [rasterizerDiscardEnable](#) is [VK_FALSE](#), and the [current value](#) of [stencilTestEnable](#) is [VK_TRUE](#), then [vkCmdSetStencilWriteMask](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07839

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the [VK_DYNAMIC_STATE_STENCIL_REFERENCE](#) dynamic state enabled, the [current value](#) of [rasterizerDiscardEnable](#) is [VK_FALSE](#), the [current value](#) of [stencilTestEnable](#) is [VK_TRUE](#), then [vkCmdSetStencilReference](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-maxMultiviewInstanceIndex-02688

If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to [VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex](#)

- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-02689

If the bound graphics pipeline was created with [VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable](#) set to [VK_TRUE](#) and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the [VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT](#) bit set

- VUID-vkCmdDrawIndirectByteCountEXT-None-07634

If the [VK_EXT_sample_locations](#) extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the [VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT](#) dynamic state enabled, and the [current value](#) of [rasterizerDiscardEnable](#) is [VK_FALSE](#), then [vkCmdSetSampleLocationsEnableEXT](#) **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-06666

If the [VK_EXT_sample_locations](#) extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the [VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT](#) dynamic state enabled, the [current value](#) of [rasterizerDiscardEnable](#) is [VK_FALSE](#), and the [current value](#) of [sampleLocationsEnable](#) is [VK_TRUE](#), then [vkCmdSetSampleLocationsEXT](#) **must** have been called and not subsequently

[invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's [format features](#) do not contain [VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT](#), then the [blendEnable](#) member of the corresponding element of the [pAttachments](#) member of [pColorBlendState](#) **must** be [VK_FALSE](#)

- VUID-vkCmdDrawIndirectByteCountEXT-None-08643

If a shader object is bound to the [VK_SHADER_STAGE_FRAGMENT_BIT](#) stage, and the most recent call to [vkCmdSetRasterizerDiscardEnableEXT](#) in the current command buffer set [rasterizerDiscardEnable](#) to [VK_FALSE](#), then for each color attachment in the render pass, if the corresponding image view's [format features](#) do not contain [VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT](#), then the corresponding member of [pColorBlendEnables](#) in the most recent call to [vkCmdSetColorBlendEnableEXT](#) in the current command buffer that affected that attachment index **must** have been [VK_FALSE](#)

- VUID-vkCmdDrawIndirectByteCountEXT-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then [rasterizationSamples](#) for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndirectByteCountEXT-None-08644

If a shader object is bound to any graphics stage, and the most recent call to [vkCmdSetRasterizerDiscardEnableEXT](#) in the current command buffer set [rasterizerDiscardEnable](#) to [VK_FALSE](#),

then the most recent call to [vkCmdSetRasterizationSamplesEXT](#) in the current command buffer **must** have set [rasterizationSamples](#) to be the same as the number of samples for the current render pass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndirectByteCountEXT-None-08876

If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with [vkCmdBeginRenderingKHR](#)

- VUID-vkCmdDrawIndirectByteCountEXT-imageView-06172

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pDepthAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pDepthAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndirectByteCountEXT-imageView-06173

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pStencilAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pStencilAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndirectByteCountEXT-imageView-06174

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pDepthAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member

of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndirectByteCountEXT-imageView-06175

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndirectByteCountEXT-imageView-06176

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pDepthAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pDepthAttachment` is `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL`, this command **must** not write any values to the depth attachment

- VUID-vkCmdDrawIndirectByteCountEXT-imageView-06177

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `imageView` member of `pStencilAttachment` is not `VK_NULL_HANDLE`, and the `layout` member of `pStencilAttachment` is `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL`, this command **must** not write any values to the stencil attachment

- VUID-vkCmdDrawIndirectByteCountEXT-viewMask-06178

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::viewMask` equal to `VkRenderingInfo::viewMask`

- VUID-vkCmdDrawIndirectByteCountEXT-colorAttachmentCount-06179

If the `dynamicRenderingUnusedAttachments` feature is not enabled and the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound graphics pipeline **must** have been created with a `VkPipelineRenderingCreateInfo::colorAttachmentCount` equal to `VkRenderingInfo::colorAttachmentCount`

- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08910

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline

- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08912

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` equal to `VK_NULL_HANDLE` **must** have the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound pipeline equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08911

If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo`

`::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline, or the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats`, if it exists, **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectByteCountEXT-None-07749

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08646

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-attachmentCount-07750

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then the `attachmentCount` parameter of `vkCmdSetColorWriteEnableEXT` **must** be greater than or equal to the `VkPipelineColorBlendStateCreateInfo::attachmentCount` of the currently bound graphics pipeline

- VUID-vkCmdDrawIndirectByteCountEXT-None-08647

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then the `attachmentCount` parameter of most recent call to `vkCmdSetColorWriteEnableEXT` in the current command buffer **must** be greater than or equal to the number of color attachments in the current render pass instance

- VUID-vkCmdDrawIndirectByteCountEXT-None-07751

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state enabled then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command for each discard rectangle in `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount`

- VUID-vkCmdDrawIndirectByteCountEXT-None-07880

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDiscardRectangleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07881

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `discardRectangleEnable` is `VK_TRUE`, then `vkCmdSetDiscardRectangleModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-rasterizerDiscardEnable-09236

If the `VK_EXT_discard_rectangles` extension is enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetDiscardRectangleEnableEXT` in the current command buffer set `discardRectangleEnable` to `VK_TRUE`, then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08913

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08914

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08915

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08916

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08917

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of

`VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdDrawIndirectByteCountEXT-dynamicRenderingUnusedAttachments-08918
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndirectByteCountEXT-imageView-06183
If the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView` was not `VK_NULL_HANDLE`, the currently bound graphics pipeline **must** have been created with `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- VUID-vkCmdDrawIndirectByteCountEXT-multisampledRenderToSingleSampled-07285
If the current render pass instance was begun with `vkCmdBeginRenderingKHR` with a `VkRenderingInfo::colorAttachmentCount` parameter greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with a `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a sample count equal to the value of `rasterizationSamples` for the currently bound graphics pipeline
- VUID-vkCmdDrawIndirectByteCountEXT-multisampledRenderToSingleSampled-07286
If `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pDepthAttachment->imageView`
- VUID-vkCmdDrawIndirectByteCountEXT-multisampledRenderToSingleSampled-07287
If `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pStencilAttachment->imageView`
- VUID-vkCmdDrawIndirectByteCountEXT-renderPass-06198
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound pipeline **must** have been created with a `VkGraphicsPipelineCreateInfo::renderPass` equal to `VK_NULL_HANDLE`
- VUID-vkCmdDrawIndirectByteCountEXT-pColorAttachments-08963
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the `VkRenderingInfo::pColorAttachments->imageView` was not `VK_NULL_HANDLE`, then the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`
- VUID-vkCmdDrawIndirectByteCountEXT-pDepthAttachment-08964
If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, depth test is enabled, depth write is enabled, and the `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the pipeline **must**

not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectByteCountEXT-pStencilAttachment-08965

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, stencil test is enabled and the `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdDrawIndirectByteCountEXT-primitivesGeneratedQueryWithRasterizerDiscard-06708

If the `primitivesGeneratedQueryWithRasterizerDiscard` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, `rasterization discard` **must** not be enabled

- VUID-vkCmdDrawIndirectByteCountEXT-primitivesGeneratedQueryWithNonZeroStreams-06709

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, the bound graphics pipeline **must** not have been created with a non-zero value in `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream`

- VUID-vkCmdDrawIndirectByteCountEXT-None-07619

If a shader object is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` dynamic state enabled, then `vkCmdSetTessellationDomainOriginEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07620

If the `depthClamp` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDepthClampEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07621

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_POLYGON_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetPolygonModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07622

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetRasterizationSamplesEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07623

If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-alphaToCoverageEnable-08919

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and `alphaToCoverageEnable` was `VK_TRUE` in the last call to `vkCmdSetAlphaToCoverageEnableEXT`, then the `Fragment Output Interface` **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0

- VUID-vkCmdDrawIndirectByteCountEXT-alphaToCoverageEnable-08920

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetAlphaToCoverageEnableEXT` in the current command buffer set `alphaToCoverageEnable` to `VK_TRUE`, then the `Fragment Output Interface` **must** contain a variable for the alpha `Component` word in `Location` 0 at `Index` 0

- VUID-vkCmdDrawIndirectByteCountEXT-None-07624

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToCoverageEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07625

If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToOneEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07626

If the `logicOp` feature is enabled, a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetLogicOpEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07627

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08657

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated`

in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07628

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08658

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` for any attachment set that attachment's value in `pColorBlendEnables` to `VK_TRUE`, then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07629

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08659

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07630

If the `geometryStreams` feature is enabled, and a shader object is bound to the `VK_SHADER_STAGE_GEOMETRY_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` dynamic state enabled, then `vkCmdSetRasterizationStreamEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07631

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetConservativeRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07632

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT` dynamic state enabled, the current value of `rasterizerDiscardEnable` is `VK_FALSE`, and the current value of `conservativeRasterizationMode` is `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`, then `vkCmdSetExtraPrimitiveOverestimationSizeEXT` **must** have been called and not

subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07633

If the **depthClipEnable** feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT** dynamic state, then **vkCmdSetDepthClipEnableEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07635

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT** dynamic state enabled then **vkCmdSetColorBlendAdvancedEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-rasterizerDiscardEnable-09416

If the **VK_EXT_blend_operation_advanced** extension is enabled, and a shader object is bound to the **VK_SHADER_STAGE_FRAGMENT_BIT** stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to **VK_FALSE**, then at least one of **vkCmdSetColorBlendEquationEXT** and **vkCmdSetColorBlendAdvancedEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07636

If the **VK_EXT_provoking_vertex** extension is enabled, a shader object is bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage or a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT** dynamic state enabled, and the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, then **vkCmdSetProvokingVertexModeEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07637

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT** dynamic state enabled then **vkCmdSetLineRasterizationModeEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08666

If any of the **stippledRectangularLines**, **stippledBresenhamLines** or **stippledSmoothLines** features are enabled, and a shader object is bound to any graphics stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to **VK_FALSE**, and the most recent call to **vkCmdSetPolygonModeEXT** in the current command buffer set **polygonMode** to **VK_POLYGON_MODE_LINE**, then **vkCmdSetLineRasterizationModeEXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08667

If any of the **stippledRectangularLines**, **stippledBresenhamLines** or **stippledSmoothLines** features are enabled, and a shader object is bound to the **VK_SHADER_STAGE_VERTEX_BIT** stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current

command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08668

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07638

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` dynamic state enabled then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08669

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08670

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-08671

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07849

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines`

features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_KHR` dynamic state enabled, the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, and the **current value** of `stippledLineEnable` is `VK_TRUE`, then `vkCmdSetLineStippleKHR` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-07639

If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` dynamic state enabled, then `vkCmdSetDepthClipNegativeOneToOneEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-09650

If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` dynamic state enabled, and the **current value** of `depthClampEnable` is `VK_TRUE`, then `vkCmdSetDepthClampRangeEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-pColorBlendEnables-07470

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` state enabled and the last call to `vkCmdSetColorBlendEnableEXT` set `pColorBlendEnables` for any attachment to `VK_TRUE`, then for those attachments in the subpass the corresponding image view's **format features** **must** contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`

- VUID-vkCmdDrawIndirectByteCountEXT-rasterizationSamples-07471

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** follow the rules for a **zero-attachment subpass**

- VUID-vkCmdDrawIndirectByteCountEXT-samples-07472

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `VkPipelineMultisampleStateCreateInfo::rasterizationSamples` parameter used to create the bound graphics pipeline

- VUID-vkCmdDrawIndirectByteCountEXT-samples-07473

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` state and `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` states enabled, then the `samples` parameter in the last call to `vkCmdSetSampleMaskEXT` **must** be greater or equal to the `rasterizationSamples` parameter in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndirectByteCountEXT-rasterizationSamples-07474

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and neither the `VK_AMD_mixed_attachment_samples` nor the `VK_NV_framebuffer_mixed_samples` extensions are enabled, then the `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT` **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdDrawIndirectByteCountEXT-firstAttachment-07476

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndirectByteCountEXT-rasterizerDiscardEnable-09417

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorBlendEnableEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdDrawIndirectByteCountEXT-firstAttachment-07477

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndirectByteCountEXT-rasterizerDiscardEnable-09418

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdDrawIndirectByteCountEXT-firstAttachment-07478

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdDrawIndirectByteCountEXT-rasterizerDiscardEnable-09419
If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass
- VUID-vkCmdDrawIndirectByteCountEXT-firstAttachment-07479
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendAdvancedEXT` calls **must** specify the advanced blend equations for all active color attachments in the current subpass where blending is enabled
- VUID-vkCmdDrawIndirectByteCountEXT-advancedBlendMaxColorAttachments-07480
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` and `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic states enabled and the last calls to `vkCmdSetColorBlendEnableEXT` and `vkCmdSetColorBlendAdvancedEXT` have enabled advanced blending, then the number of active color attachments in the current subpass **must** not exceed `advancedBlendMaxColorAttachments`
- VUID-vkCmdDrawIndirectByteCountEXT-primitivesGeneratedQueryWithNonZeroStreams-07481
If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, and the bound graphics pipeline was created with `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` state enabled, the last call to `vkCmdSetRasterizationStreamEXT` **must** have set the `rasterizationStream` to zero
- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsPerPixel-07482
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` member of the `VkPipelineMultisampleStateCreateInfo` structure the bound graphics pipeline has been created with
- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsPerPixel-07483
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` parameter of the last call to `vkCmdSetRasterizationSamplesEXT`
- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07484

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07485

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07486

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07487

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`

- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07936

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07937

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the

VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the **current value** of `sampleLocationsEnable` is VK_TRUE, then `sampleLocationsInfo.sampleLocationGridSize.height` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndirectByteCountEXT-sampleLocationsEnable-07938

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT state disabled and the VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT state enabled, and the **current value** of `sampleLocationsEnable` is VK_TRUE, then `sampleLocationsInfo.sampleLocationsPerPixel` **must** equal `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdDrawIndirectByteCountEXT-stippledLineEnable-07495

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current `stippledLineEnable` state is VK_TRUE and the current `lineRasterizationMode` state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR, then the `stippledRectangularLines` feature **must** be enabled

- VUID-vkCmdDrawIndirectByteCountEXT-stippledLineEnable-07496

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current `stippledLineEnable` state is VK_TRUE and the current `lineRasterizationMode` state is VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR, then the `stippledBresenhamLines` feature **must** be enabled

- VUID-vkCmdDrawIndirectByteCountEXT-stippledLineEnable-07497

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current `stippledLineEnable` state is VK_TRUE and the current `lineRasterizationMode` state is VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR, then the `stippledSmoothLines` feature **must** be enabled

- VUID-vkCmdDrawIndirectByteCountEXT-stippledLineEnable-07498

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT or VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT dynamic states enabled, and if the current `stippledLineEnable` state is VK_TRUE and the current `lineRasterizationMode` state is VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR, then the `stippledRectangularLines` feature **must** be enabled and `VkPhysicalDeviceLimits::strictLines` **must** be VK_TRUE

- VUID-vkCmdDrawIndirectByteCountEXT-conservativePointAndLineRasterization-07499

If the bound graphics pipeline state was created with the VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT dynamic state enabled, `conservativePointAndLineRasterization` is not supported, and the effective primitive topology output by the last pre-rasterization shader stage is a line or point, then the

`conservativeRasterizationMode` set by the last call to `vkCmdSetConservativeRasterizationModeEXT` **must** be `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`

- VUID-vkCmdDrawIndirectByteCountEXT-None-08877
If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAttachmentFeedbackLoopEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command
- VUID-vkCmdDrawIndirectByteCountEXT-None-08684
If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_VERTEX_BIT`
- VUID-vkCmdDrawIndirectByteCountEXT-None-08685
If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`
- VUID-vkCmdDrawIndirectByteCountEXT-None-08686
If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`
- VUID-vkCmdDrawIndirectByteCountEXT-None-08687
If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_GEOMETRY_BIT`
- VUID-vkCmdDrawIndirectByteCountEXT-None-08688
If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_FRAGMENT_BIT`
- VUID-vkCmdDrawIndirectByteCountEXT-None-08698
If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, then all shaders created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag in the same `vkCreateShadersEXT` call **must** also be bound
- VUID-vkCmdDrawIndirectByteCountEXT-None-08699
If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, any stages in between stages whose shaders which did not create a shader with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag as part of the same `vkCreateShadersEXT` call **must** not have any `VkShaderEXT` bound
- VUID-vkCmdDrawIndirectByteCountEXT-None-08878
All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges
- VUID-vkCmdDrawIndirectByteCountEXT-None-08879
All bound graphics shader objects **must** have been created with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdDrawIndirectByteCountEXT-pDynamicStates-08715
If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to `vkCmdSetDepthWriteEnableEXT` **must** be `VK_FALSE`
- VUID-vkCmdDrawIndirectByteCountEXT-pDynamicStates-08716
If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to `vkCmdSetStencilWriteMask` **must** be `0`
- VUID-vkCmdDrawIndirectByteCountEXT-None-09116
If a shader object is bound to any graphics stage or the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and the format of any color attachment is `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`, the corresponding element of the `pColorWriteMasks` parameter of `vkCmdSetColorWriteMaskEXT` **must** either include all of `VK_COLOR_COMPONENT_R_BIT`, `VK_COLOR_COMPONENT_G_BIT`, and `VK_COLOR_COMPONENT_B_BIT`, or none of them
- VUID-vkCmdDrawIndirectByteCountEXT-maxFragmentDualSrcAttachments-09239
If `blending` is enabled for any attachment where either the source or destination blend factors for that attachment [use the secondary color input](#), the maximum value of `Location` for any output attachment [statically used](#) in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdDrawIndirectByteCountEXT-None-09548
If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, the value of each element of `VkRenderingAttachmentLocationInfoKHR::pColorAttachmentLocations` set by `vkCmdSetRenderingAttachmentLocationsKHR` **must** match the value set for the corresponding element in the currently bound pipeline
- VUID-vkCmdDrawIndirectByteCountEXT-None-09549
If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, input attachment index mappings in the currently bound pipeline **must** match those set for the current render pass instance via `VkRenderingInputAttachmentIndexInfoKHR`
- VUID-vkCmdDrawIndirectByteCountEXT-None-04007
All vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** have either valid or `VK_NULL_HANDLE` buffers bound
- VUID-vkCmdDrawIndirectByteCountEXT-None-04008
If the `nullDescriptor` feature is not enabled, all vertex input bindings accessed via vertex input variables declared in the vertex shader entry point's interface **must** not be `VK_NULL_HANDLE`
- VUID-vkCmdDrawIndirectByteCountEXT-None-02721

If `robustBufferAccess` is not enabled, then for a given vertex buffer binding, any attribute data fetched **must** be entirely contained within the corresponding vertex buffer binding, as described in [Vertex Input Description](#)

- VUID-vkCmdDrawIndirectByteCountEXT-None-04914

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then `vkCmdSetVertexInputEXT` **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this draw command

- VUID-vkCmdDrawIndirectByteCountEXT-Input-07939

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then all variables with the `Input` storage class decorated with `Location` in the `Vertex Execution Model OpEntryPoint` **must** contain a location in [VkVertexInputAttributeDescription2EXT::location](#)

- VUID-vkCmdDrawIndirectByteCountEXT-Input-08734

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and either `legacyVertexAttributes` is not enabled or the SPIR-V Type associated with a given `Input` variable of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the numeric type associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be the same as [VkVertexInputAttributeDescription2EXT::format](#)

- VUID-vkCmdDrawIndirectByteCountEXT-format-08936

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and [VkVertexInputAttributeDescription2EXT::format](#) has a 64-bit component, then the scalar width associated with all `Input` variables of the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** be 64-bit

- VUID-vkCmdDrawIndirectByteCountEXT-format-08937

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and the scalar width associated with a `Location` decorated `Input` variable in the `Vertex Execution Model OpEntryPoint` is 64-bit, then the corresponding [VkVertexInputAttributeDescription2EXT::format](#) **must** have a 64-bit component

- VUID-vkCmdDrawIndirectByteCountEXT-None-09203

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled and [VkVertexInputAttributeDescription2EXT::format](#) has a 64-bit component, then all `Input` variables at the corresponding `Location` in the `Vertex Execution Model OpEntryPoint` **must** not use components that are not present in the format

- VUID-vkCmdDrawIndirectByteCountEXT-None-04879

If there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** have been called and not subsequently [invalidated](#) in the current command buffer prior to this drawing command

- VUID-vkCmdDrawIndirectByteCountEXT-None-09637

If the `primitiveTopologyListRestart` feature is not enabled, the topology is `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY`, there is a shader object bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage then `vkCmdSetPrimitiveRestartEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdDrawIndirectByteCountEXT-pNext-09461

If the bound graphics pipeline state was created with `VkPipelineVertexInputDivisorStateCreateInfoKHR` in the `pNext` chain of `VkGraphicsPipelineCreateInfo::pVertexInputState`, any member of `VkPipelineVertexInputDivisorStateCreateInfoKHR::pVertexBindingDivisors` has a value other than 1 in `divisor`, and `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance` is `VK_FALSE`, then `firstInstance` **must** be 0

- VUID-vkCmdDrawIndirectByteCountEXT-None-09462

If `shader objects` are used for drawing or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled, any member of the `pVertexBindingDescriptions` parameter to the `vkCmdSetVertexInputEXT` call that sets this dynamic state has a value other than 1 in `divisor`, and `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::supportsNonZeroFirstInstance` is `VK_FALSE`, then `firstInstance` **must** be 0

- VUID-vkCmdDrawIndirectByteCountEXT-transformFeedback-02287

`VkPhysicalDeviceTransformFeedbackFeaturesEXT::transformFeedback` **must** be enabled

- VUID-vkCmdDrawIndirectByteCountEXT-transformFeedbackDraw-02288

The implementation **must** support `VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackDraw`

- VUID-vkCmdDrawIndirectByteCountEXT-vertexStride-02289

`vertexStride` **must** be greater than 0 and less than or equal to `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBufferDataStride`

- VUID-vkCmdDrawIndirectByteCountEXT-counterBuffer-04567

If `counterBuffer` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdDrawIndirectByteCountEXT-counterBuffer-02290

`counterBuffer` **must** have been created with the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set

- VUID-vkCmdDrawIndirectByteCountEXT-counterBufferOffset-04568

`counterBufferOffset` **must** be a multiple of 4

- VUID-vkCmdDrawIndirectByteCountEXT-counterOffset-09474

`counterOffset` **must** be a multiple of 4

- VUID-vkCmdDrawIndirectByteCountEXT-vertexStride-09475

`vertexStride` **must** be a multiple of 4

Valid Usage (Implicit)

- VUID-vkCmdDrawIndirectByteCountEXT-commandBuffer-parameter **commandBuffer** **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdDrawIndirectByteCountEXT-counterBuffer-parameter **counterBuffer** **must** be a valid **VkBuffer** handle
- VUID-vkCmdDrawIndirectByteCountEXT-commandBuffer-recording **commandBuffer** **must** be in the **recording state**
- VUID-vkCmdDrawIndirectByteCountEXT-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdDrawIndirectByteCountEXT-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdDrawIndirectByteCountEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdDrawIndirectByteCountEXT-commonparent
Both of **commandBuffer**, and **counterBuffer** **must** have been created, allocated, or retrieved from the same **VkDevice**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	Action

Chapter 21. Fixed-Function Vertex Processing

Vertex fetching is controlled via configurable state, as a logically distinct graphics pipeline stage.

21.1. Vertex Attributes

Vertex shaders **can** define input variables, which receive *vertex attribute* data transferred from one or more `VkBuffer`(s) by drawing commands. Vertex shader input variables are bound to buffers via an indirect binding where the vertex shader associates a *vertex input attribute* number with each variable, vertex input attributes are associated to *vertex input bindings* on a per-pipeline basis, and vertex input bindings are associated with specific buffers on a per-draw basis via the `vkCmdBindVertexBuffers` command. Vertex input attribute and vertex input binding descriptions also contain format information controlling how data is extracted from buffer memory and converted to the format expected by the vertex shader.

There are `VkPhysicalDeviceLimits::maxVertexInputAttributes` number of vertex input attributes and `VkPhysicalDeviceLimits::maxVertexInputBindings` number of vertex input bindings (each referred to by zero-based indices), where there are at least as many vertex input attributes as there are vertex input bindings. Applications **can** store multiple vertex input attributes interleaved in a single buffer, and use a single vertex input binding to access those attributes.

In GLSL, vertex shaders associate input variables with a vertex input attribute number using the `location` layout qualifier. The `Component` layout qualifier associates components of a vertex shader input variable with components of a vertex input attribute.

GLSL Example

```
// Assign location M to variableName
layout (location=M, component=2) in vec2 variableName;

// Assign locations [N,N+L) to the array elements of variableNameArray
layout (location=N) in vec4 variableNameArray[L];
```

In SPIR-V, vertex shaders associate input variables with a vertex input attribute number using the `Location` decoration. The `Component` decoration associates components of a vertex shader input variable with components of a vertex input attribute. The `Location` and `Component` decorations are specified via the `OpDecorate` instruction.

SPIR-V Example

```
...
%1 = OpExtInstImport "GLSL.std.450"
...
OpName %9 "variableName"
OpName %15 "variableNameArray"
OpDecorate %18 BuiltIn VertexIndex
```



```

OpDecorate %19 BuiltIn InstanceIndex
OpDecorate %9 Location M
OpDecorate %9 Component 2
OpDecorate %15 Location N
...
%2 = OpTypeVoid
%3 = OpTypeFunction %2
%6 = OpTypeFloat 32
%7 = OpTypeVector %6 2
%8 = OpTypePointer Input %7
%9 = OpVariable %8 Input
%10 = OpTypeVector %6 4
%11 = OpTypeInt 32 0
%12 = OpConstant %11 L
%13 = OpTypeArray %10 %12
%14 = OpTypePointer Input %13
%15 = OpVariable %14 Input
...

```

21.1.1. Attribute Location and Component Assignment

The **Location** decoration specifies which vertex input attribute is used to read and interpret the data that a variable will consume.

When a vertex shader input variable declared using a 16- or 32-bit scalar or vector data type is assigned a **Location**, its value(s) are taken from the components of the input attribute specified with the corresponding `VkVertexInputAttributeDescription::location`. The components used depend on the type of variable and the **Component** decoration specified in the variable declaration, as identified in [Input Attribute Components Accessed By 16-Bit and 32-Bit Input Variables](#). Any 16-bit or 32-bit scalar or vector input will consume a single **Location**. For 16-bit and 32-bit data types, missing components are filled in with default values as described [below](#).

If an implementation supports `storageInputOutput16`, vertex shader input variables **can** have a width of 16 bits.

Table 26. Input Attribute Components Accessed By 16-Bit and 32-Bit Input Variables

16-bit or 32-bit data type	Component decoration	Components consumed
scalar	0 or unspecified	(x, o, o, o)
scalar	1	(o, y, o, o)
scalar	2	(o, o, z, o)
scalar	3	(o, o, o, w)
two-component vector	0 or unspecified	(x, y, o, o)
two-component vector	1	(o, y, z, o)
two-component vector	2	(o, o, z, w)

16-bit or 32-bit data type	Component decoration	Components consumed
three-component vector	0 or unspecified	(x, y, z, o)
three-component vector	1	(o, y, z, w)
four-component vector	0 or unspecified	(x, y, z, w)

Components indicated by “o” are available for use by other input variables which are sourced from the same attribute, and if used, are either filled with the corresponding component from the input format (if present), or the default value.

When a vertex shader input variable declared using a 32-bit floating-point matrix type is assigned a **Location** *i*, its values are taken from consecutive input attributes starting with the corresponding `VkVertexInputAttributeDescription::location`. Such matrices are treated as an array of column vectors with values taken from the input attributes identified in [Input Attributes Accessed by 32-Bit Input Matrix Variables](#). The `VkVertexInputAttributeDescription::format` **must** be specified with a `VkFormat` that corresponds to the appropriate type of column vector. The **Component** decoration **must** not be used with matrix types.

Table 27. Input Attributes Accessed by 32-Bit Input Matrix Variables

Data type	Column vector type	Locations consumed	Components consumed
mat2	two-component vector	i, i+1	(x, y, o, o), (x, y, o, o)
mat2x3	three-component vector	i, i+1	(x, y, z, o), (x, y, z, o)
mat2x4	four-component vector	i, i+1	(x, y, z, w), (x, y, z, w)
mat3x2	two-component vector	i, i+1, i+2	(x, y, o, o), (x, y, o, o), (x, y, o, o)
mat3	three-component vector	i, i+1, i+2	(x, y, z, o), (x, y, z, o), (x, y, z, o)
mat3x4	four-component vector	i, i+1, i+2	(x, y, z, w), (x, y, z, w), (x, y, z, w)
mat4x2	two-component vector	i, i+1, i+2, i+3	(x, y, o, o), (x, y, o, o), (x, y, o, o), (x, y, o, o)
mat4x3	three-component vector	i, i+1, i+2, i+3	(x, y, z, o), (x, y, z, o), (x, y, z, o), (x, y, z, o)
mat4	four-component vector	i, i+1, i+2, i+3	(x, y, z, w), (x, y, z, w), (x, y, z, w), (x, y, z, w)

Components indicated by “o” are available for use by other input variables which are sourced from the same attribute, and if used, are either filled with the corresponding component from the input (if present), or the default value.

When a vertex shader input variable declared using a scalar or vector 64-bit data type is assigned a **Location** *i*, its values are taken from consecutive input attributes starting with the corresponding

VkVertexInputAttributeDescription::location. The **Location** slots and **Component** words used depend on the type of variable and the **Component** decoration specified in the variable declaration, as identified in [Input Attribute Locations and Components Accessed by 64-Bit Input Variables](#). For 64-bit data types, no default attribute values are provided. Input variables **must** not use more components than provided by the attribute.

Table 28. Input Attribute Locations and Components Accessed by 64-Bit Input Variables

Input format	Locations consumed	64-bit data type	Location decoration	Component decoration	32-bit component s consumed
R64	i	scalar	i	0 or unspecified	(x, y, -, -)
R64G64	i	scalar	i	0 or unspecified	(x, y, o, o)
		scalar	i	2	(o, o, z, w)
		two-component vector	i	0 or unspecified	(x, y, z, w)
R64G64B64	i, i+1	scalar	i	0 or unspecified	(x, y, o, o), (o, o, -, -)
		scalar	i	2	(o, o, z, w), (o, o, -, -)
		scalar	i+1	0 or unspecified	(o, o, o, o), (x, y, -, -)
		two-component vector	i	0 or unspecified	(x, y, z, w), (o, o, -, -)
		three-component vector	i	unspecified	(x, y, z, w), (x, y, -, -)
R64G64B64A64	i, i+1	scalar	i	0 or unspecified	(x, y, o, o), (o, o, o, o)
		scalar	i	2	(o, o, z, w), (o, o, o, o)
		scalar	i+1	0 or unspecified	(o, o, o, o), (x, y, o, o)
		scalar	i+1	2	(o, o, o, o), (o, o, z, w)
		two-component vector	i	0 or unspecified	(x, y, z, w), (o, o, o, o)
		two-component vector	i+1	0 or unspecified	(o, o, o, o), (x, y, z, w)
		three-component vector	i	unspecified	(x, y, z, w), (x, y, o, o)
		four-component vector	i	unspecified	(x, y, z, w), (x, y, z, w)

Components indicated by “o” are available for use by other input variables which are sourced from the same attribute. Components indicated by “-” are not available for input variables as there are no default values provided for 64-bit data types, and there is no data provided by the input format.

When a vertex shader input variable declared using a 64-bit floating-point matrix type is assigned a **Location** *i*, its values are taken from consecutive input attribute locations. Such matrices are treated as an array of column vectors with values taken from the input attributes as shown in [Input Attribute Locations and Components Accessed by 64-Bit Input Variables](#). Each column vector starts at the **Location** immediately following the last **Location** of the previous column vector. The number of attributes and components assigned to each matrix is determined by the matrix dimensions and ranges from two to eight locations.

When a vertex shader input variable declared using an array type is assigned a location, its values are taken from consecutive input attributes starting with the corresponding **VkVertexInputAttributeDescription::location**. The number of attributes and components assigned to each element are determined according to the data type of the array elements and **Component** decoration (if any) specified in the declaration of the array, as described above. Each element of the array, in order, is assigned to consecutive locations, but all at the same specified component within each location.

Only input variables declared with the data types and component decorations as specified above are supported. Two variables are allowed to share the same **Location** slot only if their **Component** words do not overlap. If multiple variables share the same **Location** slot, they **must** all have the same SPIR-V floating-point component type or all have the same width scalar type components.

21.2. Vertex Input Description

Applications specify vertex input attribute and vertex input binding descriptions as part of graphics pipeline creation by setting the **VkGraphicsPipelineCreateInfo::pVertexInputState** pointer to a **VkPipelineVertexInputStateCreateInfo** structure. Alternatively, if the graphics pipeline is created with the **VK_DYNAMIC_STATE_VERTEX_INPUT_EXT** dynamic state enabled, then the vertex input attribute and vertex input binding descriptions are specified dynamically with **vkCmdSetVertexInputEXT**, and the **VkGraphicsPipelineCreateInfo::pVertexInputState** pointer is ignored.

The **VkPipelineVertexInputStateCreateInfo** structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineVertexInputStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineVertexInputStateCreateFlags flags;
    uint32_t                  vertexBindingDescriptionCount;
    const VkVertexInputBindingDescription* pVertexBindingDescriptions;
    uint32_t                  vertexAttributeDescriptionCount;
    const VkVertexInputAttributeDescription* pVertexAttributeDescriptions;
} VkPipelineVertexInputStateCreateInfo;
```

- **sType** is a **VkStructureType** value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `vertexBindingDescriptionCount` is the number of vertex binding descriptions provided in `pVertexBindingDescriptions`.
- `pVertexBindingDescriptions` is a pointer to an array of `VkVertexInputBindingDescription` structures.
- `vertexAttributeDescriptionCount` is the number of vertex attribute descriptions provided in `pVertexAttributeDescriptions`.
- `pVertexAttributeDescriptions` is a pointer to an array of `VkVertexInputAttributeDescription` structures.

Valid Usage

- VUID-VkPipelineVertexInputStateCreateInfo-vertexBindingDescriptionCount-00613
`vertexBindingDescriptionCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-VkPipelineVertexInputStateCreateInfo-vertexAttributeDescriptionCount-00614
`vertexAttributeDescriptionCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputAttributes`
- VUID-VkPipelineVertexInputStateCreateInfo-binding-00615
For every `binding` specified by each element of `pVertexAttributeDescriptions`, a `VkVertexInputBindingDescription` **must** exist in `pVertexBindingDescriptions` with the same value of `binding`
- VUID-VkPipelineVertexInputStateCreateInfo-pVertexBindingDescriptions-00616
All elements of `pVertexBindingDescriptions` **must** describe distinct binding numbers
- VUID-VkPipelineVertexInputStateCreateInfo-pVertexAttributeDescriptions-00617
All elements of `pVertexAttributeDescriptions` **must** describe distinct attribute locations

Valid Usage (Implicit)

- VUID-VkPipelineVertexInputStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_STATE_CREATE_INFO`
- VUID-VkPipelineVertexInputStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkPipelineVertexInputDivisorStateCreateInfoKHR`
- VUID-VkPipelineVertexInputStateCreateInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkPipelineVertexInputStateCreateInfo-flags-zero bitmask
`flags` **must** be 0
- VUID-VkPipelineVertexInputStateCreateInfo-pVertexBindingDescriptions-parameter
If `vertexBindingDescriptionCount` is not 0, `pVertexBindingDescriptions` **must** be a valid

pointer to an array of `vertexBindingDescriptionCount` valid `VkVertexInputBindingDescription` structures

- VUID-VkPipelineVertexInputStateCreateInfo-pVertexAttributeDescriptions-parameter
If `vertexAttributeDescriptionCount` is not 0, `pVertexAttributeDescriptions` **must** be a valid pointer to an array of `vertexAttributeDescriptionCount` valid `VkVertexInputAttributeDescription` structures

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineVertexInputStateCreateFlags;
```

`VkPipelineVertexInputStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

Each vertex input binding is specified by the `VkVertexInputBindingDescription` structure, defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkVertexInputBindingDescription {
    uint32_t      binding;
    uint32_t      stride;
    VkVertexInputRate inputRate;
} VkVertexInputBindingDescription;
```

- `binding` is the binding number that this structure describes.
- `stride` is the byte stride between consecutive elements within the buffer.
- `inputRate` is a `VkVertexInputRate` value specifying whether vertex attribute addressing is a function of the vertex index or of the instance index.

Valid Usage

- VUID-VkVertexInputBindingDescription-binding-00618
`binding` **must** be less than `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-VkVertexInputBindingDescription-stride-00619
`stride` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputBindingStride`
- VUID-VkVertexInputBindingDescription-stride-04456
If the `VK_KHR_portability_subset` extension is enabled, `stride` **must** be a multiple of, and at least as large as, `VkPhysicalDevicePortabilitySubsetPropertiesKHR::minVertexInputBindingStrideAlignment`

Valid Usage (Implicit)

- VUID-VkVertexInputBindingDescription-inputRate-parameter
`inputRate` **must** be a valid `VkVertexInputRate` value

Possible values of `VkVertexInputBindingDescription::inputRate`, specifying the rate at which vertex attributes are pulled from buffers, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkVertexInputRate {
    VK_VERTEX_INPUT_RATE_VERTEX = 0,
    VK_VERTEX_INPUT_RATE_INSTANCE = 1,
} VkVertexInputRate;
```

- `VK_VERTEX_INPUT_RATE_VERTEX` specifies that vertex attribute addressing is a function of the vertex index.
- `VK_VERTEX_INPUT_RATE_INSTANCE` specifies that vertex attribute addressing is a function of the instance index.

Each vertex input attribute is specified by the `VkVertexInputAttributeDescription` structure, defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkVertexInputAttributeDescription {
    uint32_t    location;
    uint32_t    binding;
    VkFormat     format;
    uint32_t    offset;
} VkVertexInputAttributeDescription;
```

- `location` is the shader input location number for this attribute.
- `binding` is the binding number which this attribute takes its data from.
- `format` is the size and type of the vertex attribute data.
- `offset` is a byte offset of this attribute relative to the start of an element in the vertex input binding.

Valid Usage

- VUID-VkVertexInputAttributeDescription-location-00620
`location` **must** be less than `VkPhysicalDeviceLimits::maxVertexInputAttributes`
- VUID-VkVertexInputAttributeDescription-binding-00621
`binding` **must** be less than `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-VkVertexInputAttributeDescription-offset-00622
`offset` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputAttributeOffset`
- VUID-VkVertexInputAttributeDescription-format-00623
The `format` features of `format` **must** contain `VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT`
- VUID-VkVertexInputAttributeDescription-vertexAttributeAccessBeyondStride-04457

If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::vertexAttributeAccessBeyondStride` is `VK_FALSE`, the sum of `offset` plus the size of the vertex attribute data described by `format` **must** not be greater than `stride` in the `VkVertexInputBindingDescription` referenced in `binding`

Valid Usage (Implicit)

- VUID-VkVertexInputAttributeDescription-format-parameter `format` **must** be a valid `VkFormat` value

To [dynamically set](#) the vertex input attribute and vertex input binding descriptions, call:

```
// Provided by VK_EXT_shader_object, VK_EXT_vertex_input_dynamic_state
void vkCmdSetVertexInputEXT(
    VkCommandBuffer                      commandBuffer,
    uint32_t                             vertexBindingDescriptionCount,
    const VkVertexInputBindingDescription2EXT* pVertexBindingDescriptions,
    uint32_t                             vertexAttributeDescriptionCount,
    const VkVertexInputAttributeDescription2EXT* pVertexAttributeDescriptions);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `vertexBindingDescriptionCount` is the number of vertex binding descriptions provided in `pVertexBindingDescriptions`.
- `pVertexBindingDescriptions` is a pointer to an array of `VkVertexInputBindingDescription2EXT` structures.
- `vertexAttributeDescriptionCount` is the number of vertex attribute descriptions provided in `pVertexAttributeDescriptions`.
- `pVertexAttributeDescriptions` is a pointer to an array of `VkVertexInputAttributeDescription2EXT` structures.

This command sets the vertex input attribute and vertex input binding descriptions state for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkGraphicsPipelineCreateInfo::pVertexInputState` values used to create the currently active pipeline.

If drawing using [shader objects](#), then `vkCmdBindVertexBuffers2EXT` can be used instead of `vkCmdSetVertexInputEXT` to dynamically set the stride.

Valid Usage

- VUID-vkCmdSetVertexInputEXT-None-08546
Either the `vertexInputDynamicState` feature or the `shaderObject` feature or both **must** be

enabled

- VUID-vkCmdSetVertexInputEXT-vertexBindingDescriptionCount-04791
`vertexBindingDescriptionCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-vkCmdSetVertexInputEXT-vertexAttributeDescriptionCount-04792
`vertexAttributeDescriptionCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputAttributes`
- VUID-vkCmdSetVertexInputEXT-binding-04793
For every `binding` specified by each element of `pVertexAttributeDescriptions`, a `VkVertexInputBindingDescription2EXT` **must** exist in `pVertexBindingDescriptions` with the same value of `binding`
- VUID-vkCmdSetVertexInputEXT-pVertexBindingDescriptions-04794
All elements of `pVertexBindingDescriptions` **must** describe distinct binding numbers
- VUID-vkCmdSetVertexInputEXT-pVertexAttributeDescriptions-04795
All elements of `pVertexAttributeDescriptions` **must** describe distinct attribute locations

Valid Usage (Implicit)

- VUID-vkCmdSetVertexInputEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetVertexInputEXT-pVertexBindingDescriptions-parameter
If `vertexBindingDescriptionCount` is not 0, `pVertexBindingDescriptions` **must** be a valid pointer to an array of `vertexBindingDescriptionCount` valid `VkVertexInputBindingDescription2EXT` structures
- VUID-vkCmdSetVertexInputEXT-pVertexAttributeDescriptions-parameter
If `vertexAttributeDescriptionCount` is not 0, `pVertexAttributeDescriptions` **must** be a valid pointer to an array of `vertexAttributeDescriptionCount` valid `VkVertexInputAttributeDescription2EXT` structures
- VUID-vkCmdSetVertexInputEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetVertexInputEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetVertexInputEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

The `VkVertexInputBindingDescription2EXT` structure is defined as:

```
// Provided by VK_EXT_shader_object, VK_EXT_vertex_input_dynamic_state
typedef struct VkVertexInputBindingDescription2EXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           binding;
    uint32_t           stride;
    VkVertexInputRate   inputRate;
    uint32_t           divisor;
} VkVertexInputBindingDescription2EXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `binding` is the binding number that this structure describes.
- `stride` is the byte stride between consecutive elements within the buffer.
- `inputRate` is a `VkVertexInputRate` value specifying whether vertex attribute addressing is a function of the vertex index or of the instance index.
- `divisor` is the number of successive instances that will use the same value of the vertex attribute when instanced rendering is enabled. This member **can** be a value other than `1` if the `vertexAttributeInstanceRateDivisor` feature is enabled. For example, if the divisor is `N`, the same vertex attribute will be applied to `N` successive instances before moving on to the next vertex attribute. The maximum value of `divisor` is implementation-dependent and can be queried using `VkPhysicalDeviceVertexAttributeDivisorPropertiesEXT::maxVertexAttribDivisor`. A value of `0` **can** be used for the divisor if the `vertexAttributeInstanceRateZeroDivisor` feature is enabled. In this case, the same vertex attribute will be applied to all instances.

Valid Usage

- VUID-VkVertexInputBindingDescription2EXT-binding-04796
`binding` **must** be less than `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-VkVertexInputBindingDescription2EXT-stride-04797
`stride` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputBindingStride`
- VUID-VkVertexInputBindingDescription2EXT-divisor-04798
If the `vertexAttributeInstanceRateZeroDivisor` feature is not enabled, `divisor` **must** not be

0

- VUID-VkVertexInputBindingDescription2EXT-divisor-04799
If the `vertexAttributeInstanceRateDivisor` feature is not enabled, `divisor` **must** be 1
- VUID-VkVertexInputBindingDescription2EXT-divisor-06226
`divisor` **must** be a value between 0 and `VkPhysicalDeviceVertexAttributeDivisorPropertiesEXT::maxVertexAttribDivisor`, inclusive
- VUID-VkVertexInputBindingDescription2EXT-divisor-06227
If `divisor` is not 1 then `inputRate` **must** be of type `VK_VERTEX_INPUT_RATE_INSTANCE`

Valid Usage (Implicit)

- VUID-VkVertexInputBindingDescription2EXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VERTEX_INPUT_BINDING_DESCRIPTION_2_EXT`
- VUID-VkVertexInputBindingDescription2EXT-inputRate-parameter
`inputRate` **must** be a valid `VkVertexInputRate` value

The `VkVertexInputAttributeDescription2EXT` structure is defined as:

```
// Provided by VK_EXT_shader_object, VK_EXT_vertex_input_dynamic_state
typedef struct VkVertexInputAttributeDescription2EXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           location;
    uint32_t           binding;
    VkFormat            format;
    uint32_t           offset;
} VkVertexInputAttributeDescription2EXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `location` is the shader input location number for this attribute.
- `binding` is the binding number which this attribute takes its data from.
- `format` is the size and type of the vertex attribute data.
- `offset` is a byte offset of this attribute relative to the start of an element in the vertex input binding.

Valid Usage

- VUID-VkVertexInputAttributeDescription2EXT-location-06228
`location` **must** be less than `VkPhysicalDeviceLimits::maxVertexInputAttributes`
- VUID-VkVertexInputAttributeDescription2EXT-binding-06229
`binding` **must** be less than `VkPhysicalDeviceLimits::maxVertexInputBindings`

- VUID-VkVertexInputAttributeDescription2EXT-offset-06230
`offset` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputAttributeOffset`
- VUID-VkVertexInputAttributeDescription2EXT-format-04805
The `format` features of `format` **must** contain `VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT`
- VUID-VkVertexInputAttributeDescription2EXT-vertexAttributeAccessBeyondStride-04806
If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::vertexAttributeAccessBeyondStride` is `VK_FALSE`, the sum of `offset` plus the size of the vertex attribute data described by `format` **must** not be greater than `stride` in the `VkVertexInputBindingDescription2EXT` referenced in `binding`

Valid Usage (Implicit)

- VUID-VkVertexInputAttributeDescription2EXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VERTEX_INPUT_ATTRIBUTE_DESCRIPTION_2_EXT`
- VUID-VkVertexInputAttributeDescription2EXT-format-parameter
`format` **must** be a valid `VkFormat` value

To bind vertex buffers to a command buffer for use in subsequent drawing commands, call:

```
// Provided by VK_VERSION_1_0
void vkCmdBindVertexBuffers(
    VkCommandBuffer          commandBuffer,
    uint32_t                 firstBinding,
    uint32_t                 bindingCount,
    const VkBuffer*          pBuffer,
    const VkDeviceSize*      pOffsets);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `firstBinding` is the index of the first vertex input binding whose state is updated by the command.
- `bindingCount` is the number of vertex input bindings whose state is updated by the command.
- `pBuffers` is a pointer to an array of buffer handles.
- `pOffsets` is a pointer to an array of buffer offsets.

The values taken from elements `i` of `pBuffers` and `pOffsets` replace the current state for the vertex input binding `firstBinding + i`, for `i` in `[0, bindingCount)`. The vertex input binding is updated to start at the offset indicated by `pOffsets[i]` from the start of the buffer `pBuffers[i]`. All vertex input attributes that use each of these bindings will use these updated addresses in their address calculations for subsequent drawing commands. If the `nullDescriptor` feature is enabled, elements of `pBuffers` **can** be `VK_NULL_HANDLE`, and **can** be used by the vertex shader. If a vertex input attribute is bound to a vertex input binding that is `VK_NULL_HANDLE`, the values taken from

memory are considered to be zero, and missing G, B, or A components are filled with (0,0,1).

Valid Usage

- VUID-vkCmdBindVertexBuffers-firstBinding-00624
`firstBinding` **must** be less than `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-vkCmdBindVertexBuffers-firstBinding-00625
The sum of `firstBinding` and `bindingCount` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-vkCmdBindVertexBuffers-pOffsets-00626
All elements of `pOffsets` **must** be less than the size of the corresponding element in `pBuffers`
- VUID-vkCmdBindVertexBuffers-pBuffers-00627
All elements of `pBuffers` **must** have been created with the `VK_BUFFER_USAGE_VERTEX_BUFFER_BIT` flag
- VUID-vkCmdBindVertexBuffers-pBuffers-00628
Each element of `pBuffers` that is non-sparse **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBindVertexBuffers-pBuffers-04001
If the `nullDescriptor` feature is not enabled, all elements of `pBuffers` **must** not be `VK_NULL_HANDLE`
- VUID-vkCmdBindVertexBuffers-pBuffers-04002
If an element of `pBuffers` is `VK_NULL_HANDLE`, then the corresponding element of `pOffsets` **must** be zero

Valid Usage (Implicit)

- VUID-vkCmdBindVertexBuffers-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBindVertexBuffers-pBuffers-parameter
`pBuffers` **must** be a valid pointer to an array of `bindingCount` valid or `VK_NULL_HANDLE` `VkBuffer` handles
- VUID-vkCmdBindVertexBuffers-pOffsets-parameter
`pOffsets` **must** be a valid pointer to an array of `bindingCount` `VkDeviceSize` values
- VUID-vkCmdBindVertexBuffers-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdBindVertexBuffers-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBindVertexBuffers-vidcoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindVertexBuffers-bindingCount-arraylength

`bindingCount` **must** be greater than 0

- VUID-vkCmdBindVertexBuffers-commonparent

Both of `commandBuffer`, and the elements of `pBuffers` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Alternatively, to bind vertex buffers, along with their sizes and strides, to a command buffer for use in subsequent drawing commands, call:

```
// Provided by VK_EXT_shader_object
void vkCmdBindVertexBuffers2EXT(
    VkCommandBuffer                commandBuffer,
    uint32_t                       firstBinding,
    uint32_t                       bindingCount,
    const VkBuffer*                pBuffers,
    const VkDeviceSize*            pOffsets,
    const VkDeviceSize*            pSizes,
    const VkDeviceSize*            pStrides);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `firstBinding` is the index of the first vertex input binding whose state is updated by the command.
- `bindingCount` is the number of vertex input bindings whose state is updated by the command.
- `pBuffers` is a pointer to an array of buffer handles.
- `pOffsets` is a pointer to an array of buffer offsets.
- `pSizes` is `NULL` or a pointer to an array of the size in bytes of vertex data bound from `pBuffers`.
- `pStrides` is `NULL` or a pointer to an array of buffer strides.

The values taken from elements `i` of `pBuffers` and `pOffsets` replace the current state for the vertex

input binding `firstBinding + i`, for `i` in `[0, bindingCount)`. The vertex input binding is updated to start at the offset indicated by `pOffsets[i]` from the start of the buffer `pBuffers[i]`. If `pSizes` is not `NULL` then `pSizes[i]` specifies the bound size of the vertex buffer starting from the corresponding elements of `pBuffers[i]` plus `pOffsets[i]`. If `pSizes[i]` is `VK_WHOLE_SIZE` then the bound size is from `pBuffers[i]` plus `pOffsets[i]` to the end of the buffer `pBuffers[i]`. All vertex input attributes that use each of these bindings will use these updated addresses in their address calculations for subsequent drawing commands. If the `nullDescriptor` feature is enabled, elements of `pBuffers` can be `VK_NULL_HANDLE`, and can be used by the vertex shader. If a vertex input attribute is bound to a vertex input binding that is `VK_NULL_HANDLE`, the values taken from memory are considered to be zero, and missing G, B, or A components are filled with (0,0,1).

This command also dynamically sets the byte strides between consecutive elements within buffer `pBuffers[i]` to the corresponding `pStrides[i]` value when drawing using shader objects. Otherwise, strides are specified by the `VkVertexInputBindingDescription::stride` values used to create the currently active pipeline.

If drawing using shader objects or if the bound pipeline state object was also created with the `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT` dynamic state enabled then `vkCmdSetVertexInputEXT` can be used instead of `vkCmdBindVertexBuffers2EXT` to set the stride.

NOTE

Unlike the static state to set the same, `pStrides` must be between 0 and the maximum extent of the attributes in the binding. `vkCmdSetVertexInputEXT` does not have this restriction so can be used if other stride values are desired.

Valid Usage

- VUID-vkCmdBindVertexBuffers2-firstBinding-03355
`firstBinding` must be less than `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-vkCmdBindVertexBuffers2-firstBinding-03356
The sum of `firstBinding` and `bindingCount` must be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-vkCmdBindVertexBuffers2-pOffsets-03357
If `pSizes` is not `NULL`, all elements of `pOffsets` must be less than the size of the corresponding element in `pBuffers`
- VUID-vkCmdBindVertexBuffers2-pSizes-03358
If `pSizes` is not `NULL`, all elements of `pOffsets` plus `pSizes`, where `pSizes` is not `VK_WHOLE_SIZE`, must be less than or equal to the size of the corresponding element in `pBuffers`
- VUID-vkCmdBindVertexBuffers2-pBuffers-03359
All elements of `pBuffers` must have been created with the `VK_BUFFER_USAGE_VERTEX_BUFFER_BIT` flag
- VUID-vkCmdBindVertexBuffers2-pBuffers-03360
Each element of `pBuffers` that is non-sparse must be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBindVertexBuffers2-pBuffers-04111

If the `nullDescriptor` feature is not enabled, all elements of `pBuffers` **must** not be `VK_NULL_HANDLE`

- VUID-vkCmdBindVertexBuffers2-pBuffers-04112

If an element of `pBuffers` is `VK_NULL_HANDLE`, then the corresponding element of `pOffsets` **must** be zero

- VUID-vkCmdBindVertexBuffers2-pStrides-03362

If `pStrides` is not `NULL` each element of `pStrides` **must** be less than or equal to `VkPhysicalDeviceLimits::maxVertexInputBindingStride`

- VUID-vkCmdBindVertexBuffers2-pStrides-06209

If `pStrides` is not `NULL` each element of `pStrides` **must** be either 0 or greater than or equal to the maximum extent of all vertex input attributes fetched from the corresponding binding, where the extent is calculated as the `VkVertexInputAttributeDescription::offset` plus `VkVertexInputAttributeDescription::format` size

Valid Usage (Implicit)

- VUID-vkCmdBindVertexBuffers2-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdBindVertexBuffers2-pBuffers-parameter

`pBuffers` **must** be a valid pointer to an array of `bindingCount` valid or `VK_NULL_HANDLE` `VkBuffer` handles

- VUID-vkCmdBindVertexBuffers2-pOffsets-parameter

`pOffsets` **must** be a valid pointer to an array of `bindingCount` `VkDeviceSize` values

- VUID-vkCmdBindVertexBuffers2-pSizes-parameter

If `pSizes` is not `NULL`, `pSizes` **must** be a valid pointer to an array of `bindingCount` `VkDeviceSize` values

- VUID-vkCmdBindVertexBuffers2-pStrides-parameter

If `pStrides` is not `NULL`, `pStrides` **must** be a valid pointer to an array of `bindingCount` `VkDeviceSize` values

- VUID-vkCmdBindVertexBuffers2-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdBindVertexBuffers2-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdBindVertexBuffers2-vidcoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdBindVertexBuffers2-bindingCount-arraylength

If any of `pSizes`, or `pStrides` are not `NULL`, `bindingCount` **must** be greater than 0

- VUID-vkCmdBindVertexBuffers2-commonparent

Both of `commandBuffer`, and the elements of `pBuffers` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

21.3. Vertex Attribute Divisor in Instanced Rendering

If the `vertexAttributeInstanceRateDivisor` feature is enabled and the `pNext` chain of `VkPipelineVertexInputStateCreateInfo` includes a `VkPipelineVertexInputDivisorStateCreateInfoKHR` structure, then that structure controls how vertex attributes are assigned to an instance when instanced rendering is enabled.

The `VkPipelineVertexInputDivisorStateCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_vertex_attribute_divisor
typedef struct VkPipelineVertexInputDivisorStateCreateInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    uint32_t                 vertexBindingDivisorCount;
    const VkVertexInputBindingDivisorDescriptionKHR* pVertexBindingDivisors;
} VkPipelineVertexInputDivisorStateCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `vertexBindingDivisorCount` is the number of elements in the `pVertexBindingDivisors` array.
- `pVertexBindingDivisors` is a pointer to an array of `VkVertexInputBindingDivisorDescriptionKHR` structures specifying the divisor value for each binding.

Valid Usage (Implicit)

- VUID-VkPipelineVertexInputDivisorStateCreateInfoKHR-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_DIVISOR_STATE_CREATE_INFO_KHR`
- VUID-VkPipelineVertexInputDivisorStateCreateInfoKHR-pVertexBindingDivisors-parameter

`pVertexBindingDivisors` **must** be a valid pointer to an array of `vertexBindingDivisorCount` `VkVertexInputBindingDivisorDescriptionKHR` structures

- VUID-VkPipelineVertexInputDivisorStateCreateInfoKHR-vertexBindingDivisorCount-arraylength
`vertexBindingDivisorCount` **must** be greater than 0

The individual divisor values per binding are specified using the `VkVertexInputBindingDivisorDescriptionKHR` structure which is defined as:

```
// Provided by VK_KHR_vertex_attribute_divisor
typedef struct VkVertexInputBindingDivisorDescriptionKHR {
    uint32_t    binding;
    uint32_t    divisor;
} VkVertexInputBindingDivisorDescriptionKHR;
```

- `binding` is the binding number for which the divisor is specified.
- `divisor` is the number of successive instances that will use the same value of the vertex attribute when instanced rendering is enabled. For example, if the divisor is N, the same vertex attribute will be applied to N successive instances before moving on to the next vertex attribute. The maximum value of `divisor` is implementation-dependent and can be queried using `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::maxVertexAttribDivisor`. A value of 0 **can** be used for the divisor if the `vertexAttributeInstanceRateZeroDivisor` feature is enabled. In this case, the same vertex attribute will be applied to all instances.

If this structure is not used to define a divisor value for an attribute, then the divisor has a logical default value of 1.

Valid Usage

- VUID-VkVertexInputBindingDivisorDescriptionKHR-binding-01869
`binding` **must** be less than `VkPhysicalDeviceLimits::maxVertexInputBindings`
- VUID-VkVertexInputBindingDivisorDescriptionKHR-vertexAttributeInstanceRateZeroDivisor-02228
If the `vertexAttributeInstanceRateZeroDivisor` feature is not enabled, `divisor` **must** not be 0
- VUID-VkVertexInputBindingDivisorDescriptionKHR-vertexAttributeInstanceRateDivisor-02229
If the `vertexAttributeInstanceRateDivisor` feature is not enabled, `divisor` **must** be 1
- VUID-VkVertexInputBindingDivisorDescriptionKHR-divisor-01870
`divisor` **must** be a value between 0 and `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR::maxVertexAttribDivisor`, inclusive
- VUID-VkVertexInputBindingDivisorDescriptionKHR-inputRate-01871
`VkVertexInputBindingDescription::inputRate` **must** be of type `VK_VERTEX_INPUT_RATE_INSTANCE` for this binding

21.4. Vertex Input Address Calculation

The address of each attribute for each `vertexIndex` and `instanceIndex` is calculated as follows:

- Let `attribDesc` be the member of `VkPipelineVertexInputStateCreateInfo::pVertexAttributeDescriptions` with `VkVertexInputAttributeDescription::location` equal to the vertex input attribute number.
- Let `bindingDesc` be the member of `VkPipelineVertexInputStateCreateInfo::pVertexBindingDescriptions` with `VkVertexInputAttributeDescription::binding` equal to `attribDesc.binding`.
- Let `vertexIndex` be the index of the vertex within the draw (a value between `firstVertex` and `firstVertex+vertexCount` for `vkCmdDraw`, or a value taken from the index buffer plus `vertexOffset` for `vkCmdDrawIndexed`), and let `instanceIndex` be the instance number of the draw (a value between `firstInstance` and `firstInstance+instanceCount`).
- Let `offset` be an array of offsets into the currently bound vertex buffers specified during `vkCmdBindVertexBuffers` with `pOffsets`.
- Let `divisor` be the member of `VkPipelineVertexInputDivisorStateCreateInfoKHR::pVertexBindingDivisors` with `VkVertexInputBindingDivisorDescriptionKHR::binding` equal to `attribDesc.binding`. If the vertex binding state is dynamically set, instead let `divisor` be the member of the `pVertexBindingDescriptions` parameter to the `vkCmdSetVertexInputEXT` call with `VkVertexInputBindingDescription2EXT::binding` equal to `attribDesc.binding`.
- Let `stride` be the member of `VkPipelineVertexInputStateCreateInfo::pVertexBindingDescriptions->stride` unless there is dynamic state causing the value to be ignored. In this case the value is set from the last value from one of the following
 - `vkCmdSetVertexInputEXT::pVertexBindingDescriptions->stride`
 - `vkCmdBindVertexBuffers2EXT::pStride`, if not `NULL`

```
bufferBindingAddress = buffer[binding].baseAddress + offset[binding];

if (bindingDesc.inputRate == VK_VERTEX_INPUT_RATE_VERTEX)
    effectiveVertexOffset = vertexIndex * stride;
else
    if (divisor == 0)
        effectiveVertexOffset = firstInstance * stride;
    else
        effectiveVertexOffset = (firstInstance + ((instanceIndex - firstInstance) /
divisor)) * stride;

attribAddress = bufferBindingAddress + effectiveVertexOffset + attribDesc.offset;
```

21.4.1. Vertex Input Extraction

For each attribute, raw data is extracted starting at `attribAddress` and is converted from the `VkVertexInputAttributeDescription`'s `format` to either floating-point, unsigned integer, or signed integer based on the `numeric type` of `format`. The numeric type of `format` **must** match the numeric

type of the input variable in the shader. The input variable in the shader **must** be declared as a 64-bit data type if and only if `format` is a 64-bit data type. If either `format` is a 64-bit format or `legacyVertexAttributes` is not enabled, and `format` is a packed format, `attribAddress` **must** be a multiple of the size in bytes of the whole attribute data type as described in [Packed Formats](#). Otherwise, if either `format` is a 64-bit format or `legacyVertexAttributes` is not enabled, `attribAddress` **must** be a multiple of the size in bytes of the component type indicated by `format` (see [Formats](#)). For attributes that are not 64-bit data types, each component is converted to the format of the input variable based on its type and size (as defined in the [Format Definition](#) section for each `VkFormat`), using the appropriate equations in [16-Bit Floating-Point Numbers](#), [Unsigned 11-Bit Floating-Point Numbers](#), [Unsigned 10-Bit Floating-Point Numbers](#), [Fixed-Point Data Conversion](#), and [Shared Exponent to RGB](#). Signed integer components smaller than 32 bits are sign-extended. Attributes that are not 64-bit data types are expanded to four components in the same way as described in [conversion to RGBA](#). The number of components in the vertex shader input variable need not exactly match the number of components in the format. If the vertex shader has fewer components, the extra components are discarded.

Chapter 22. Tessellation

Tessellation involves three pipeline stages. First, a [tessellation control shader](#) transforms control points of a patch and **can** produce per-patch data. Second, a fixed-function tessellator generates multiple primitives corresponding to a tessellation of the patch in (u,v) or (u,v,w) parameter space. Third, a [tessellation evaluation shader](#) transforms the vertices of the tessellated patch, for example to compute their positions and attributes as part of the tessellated surface. The tessellator is enabled when the pipeline contains both a tessellation control shader and a tessellation evaluation shader.

22.1. Tessellator

If a pipeline includes both tessellation shaders (control and evaluation), the tessellator consumes each input patch (after vertex shading) and produces a new set of independent primitives (points, lines, or triangles). These primitives are logically produced by subdividing a geometric primitive (rectangle or triangle) according to the per-patch outer and inner tessellation levels written by the tessellation control shader. These levels are specified using the [built-in variables](#) `TessLevelOuter` and `TessLevelInner`, respectively. This subdivision is performed in an implementation-dependent manner. If no tessellation shaders are present in the pipeline, the tessellator is disabled and incoming primitives are passed through without modification.

The type of subdivision performed by the tessellator is specified by an `OpExecutionMode` instruction using one of the `Triangles`, `Quads`, or `Isolines` execution modes. When using [shader objects](#), this instruction **must** be specified in the tessellation evaluation shader, and **may** also be specified in the tessellation control shader. When using pipelines, this instruction **may** be specified in either the tessellation evaluation or tessellation control shader. When using shader objects, tessellation-related modes that are **required must** be specified in the tessellation evaluation shader, and **may** also be specified in the tessellation control shader. Other tessellation-related modes **may** be specified in the tessellation evaluation shader. When using pipelines, other tessellation-related execution modes **can** also be specified in either the tessellation control or tessellation evaluation shaders.

Any tessellation-related modes specified in both the tessellation control and tessellation evaluation shaders **must** be the same.

Tessellation execution modes include:

- `Triangles`, `Quads`, and `Isolines`. These control the type of subdivision and topology of the output primitives. When using [shader objects](#), one mode **must** be set in at least the tessellation evaluation stage. When using pipelines, one mode **must** be set in at least one of the tessellation shader stages. If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::tessellationIsolines` is `VK_FALSE`, then isoline tessellation is not supported by the implementation, and `Isolines` **must** not be used in either tessellation shader stage.
- `VertexOrderCw` and `VertexOrderCcw`. These control the orientation of triangles generated by the tessellator. When using [shader objects](#), one mode **must** be set in at least the tessellation evaluation stage. When using pipelines, one mode **must** be set in at least one of the tessellation

shader stages.

- **PointMode**. Controls generation of points rather than triangles or lines. This functionality defaults to disabled, and is enabled if either shader stage includes the execution mode. When using [shader objects](#), if **PointMode** is set in the tessellation control stage, it **must** be identically set in the tessellation evaluation stage. If the [VK_KHR_portability_subset](#) extension is enabled, and [VkPhysicalDevicePortabilitySubsetFeaturesKHR::tessellationPointMode](#) is **VK_FALSE**, then point mode tessellation is not supported by the implementation, and **PointMode** **must** not be used in either tessellation shader stage.
- **SpacingEqual**, **SpacingFractionalEven**, and **SpacingFractionalOdd**. Controls the spacing of segments on the edges of tessellated primitives. When using [shader objects](#), one mode **must** be set in at least the tessellation evaluation stage. When using pipelines, one mode **must** be set in at least one of the tessellation shader stages.
- **OutputVertices**. Controls the size of the output patch of the tessellation control shader. When using [shader objects](#), one value **must** be set in at least the tessellation control stage. When using pipelines, one value **must** be set in at least one of the tessellation shader stages.

For triangles, the tessellator subdivides a triangle primitive into smaller triangles. For quads, the tessellator subdivides a rectangle primitive into smaller triangles. For isolines, the tessellator subdivides a rectangle primitive into a collection of line segments arranged in strips stretching across the rectangle in the u dimension (i.e. the coordinates in **TessCoord** are of the form (0,x) through (1,x) for all tessellation evaluation shader invocations that share a line).

Each vertex produced by the tessellator has an associated (u,v,w) or (u,v) position in a normalized parameter space, with parameter values in the range [0,1], as illustrated in figures [Domain parameterization for tessellation primitive modes \(upper-left origin\)](#) and [Domain parameterization for tessellation primitive modes \(lower-left origin\)](#). The domain space **can** have either an upper-left or lower-left origin, selected by the **domainOrigin** member of [VkPipelineTessellationDomainOriginStateCreateInfo](#).

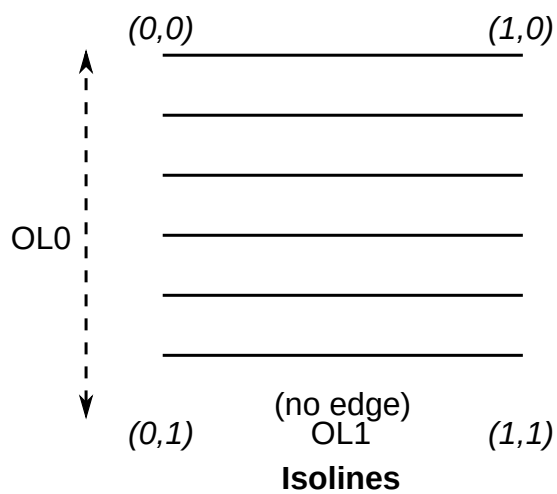
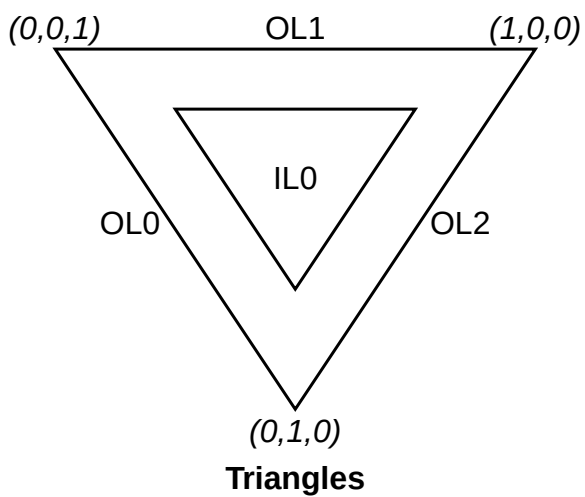
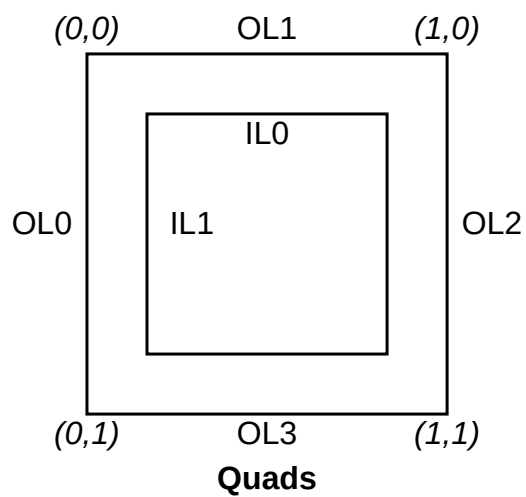


Figure 11. Domain parameterization for tessellation primitive modes (upper-left origin)

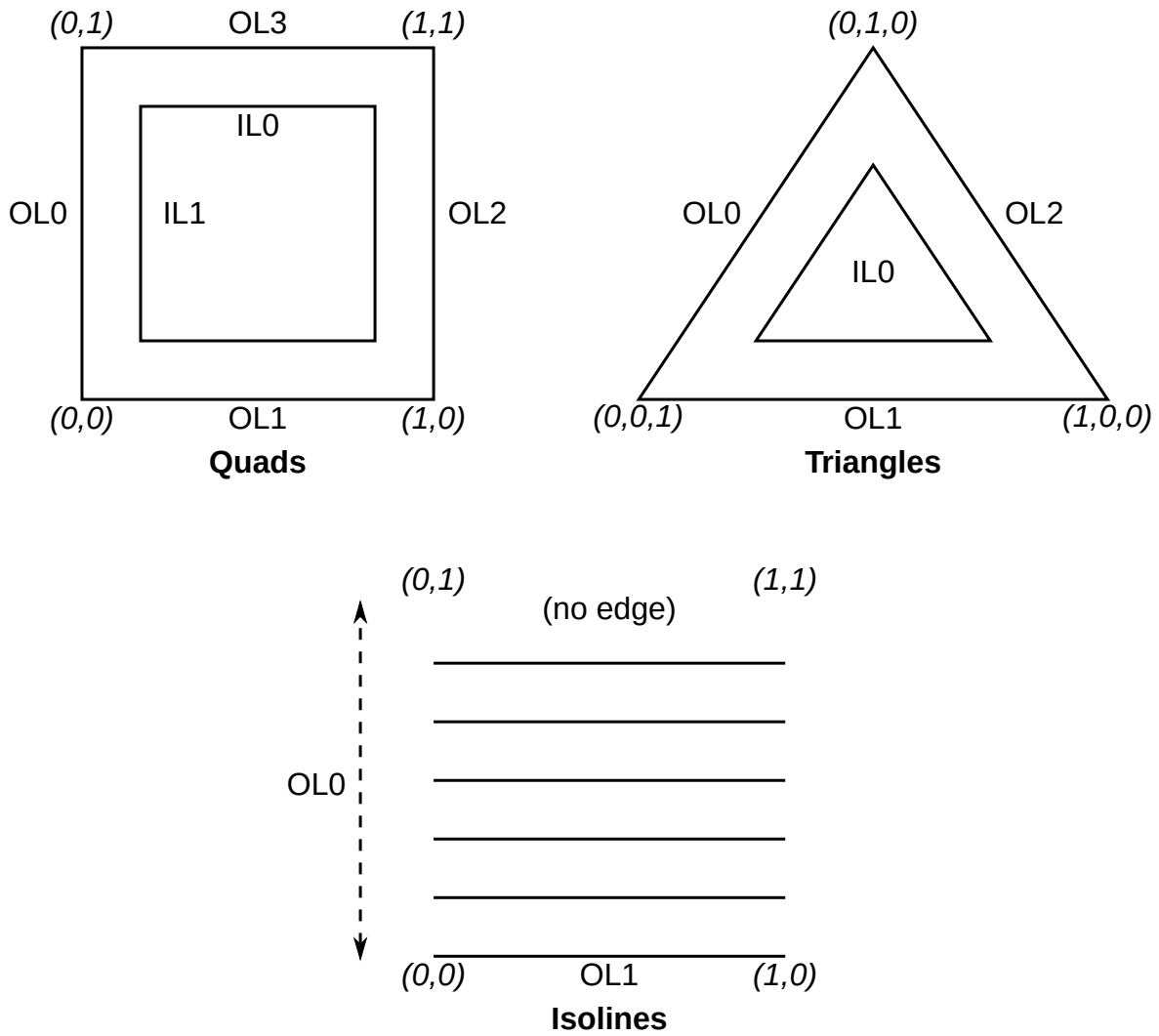


Figure 12. Domain parameterization for tessellation primitive modes (lower-left origin)

Caption

In the domain parameterization diagrams, the coordinates illustrate the value of **TessCoord** at the corners of the domain. The labels on the edges indicate the inner (IL0 and IL1) and outer (OL0 through OL3) tessellation level values used to control the number of subdivisions along each edge of the domain.

For triangles, the vertex's position is a barycentric coordinate (u,v,w) , where $u + v + w = 1.0$, and indicates the relative influence of the three vertices of the triangle on the position of the vertex. For quads and isolines, the position is a (u,v) coordinate indicating the relative horizontal and vertical position of the vertex relative to the subdivided rectangle. The subdivision process is explained in more detail in subsequent sections.

22.2. Tessellator Patch Discard

A patch is discarded by the tessellator if any relevant outer tessellation level is less than or equal to zero.

Patches will also be discarded if any relevant outer tessellation level corresponds to a floating-point

NaN (not a number) in implementations supporting NaN.

No new primitives are generated and the tessellation evaluation shader is not executed for patches that are discarded. For **Quads**, all four outer levels are relevant. For **Triangles** and **Isolines**, only the first three or two outer levels, respectively, are relevant. Negative inner levels will not cause a patch to be discarded; they will be clamped as described below.

22.3. Tessellator Spacing

Each of the tessellation levels is used to determine the number and spacing of segments used to subdivide a corresponding edge. The method used to derive the number and spacing of segments is specified by an **OpExecutionMode** in the tessellation control or tessellation evaluation shader using one of the identifiers **SpacingEqual**, **SpacingFractionalEven**, or **SpacingFractionalOdd**.

If **SpacingEqual** is used, the floating-point tessellation level is first clamped to $[1, \text{maxLevel}]$, where **maxLevel** is the implementation-dependent maximum tessellation level (**VkPhysicalDeviceLimits::maxTessellationGenerationLevel**). The result is rounded up to the nearest integer n , and the corresponding edge is divided into n segments of equal length in (u,v) space.

If **SpacingFractionalEven** is used, the tessellation level is first clamped to $[2, \text{maxLevel}]$ and then rounded up to the nearest even integer n . If **SpacingFractionalOdd** is used, the tessellation level is clamped to $[1, \text{maxLevel} - 1]$ and then rounded up to the nearest odd integer n . If n is one, the edge will not be subdivided. Otherwise, the corresponding edge will be divided into $n - 2$ segments of equal length, and two additional segments of equal length that are typically shorter than the other segments. The length of the two additional segments relative to the others will decrease monotonically with $n - f$, where f is the clamped floating-point tessellation level. When $n - f$ is zero, the additional segments will have equal length to the other segments. As $n - f$ approaches 2.0, the relative length of the additional segments approaches zero. The two additional segments **must** be placed symmetrically on opposite sides of the subdivided edge. The relative location of these two segments is implementation-dependent, but **must** be identical for any pair of subdivided edges with identical values of f .

When tessellating triangles or quads using **point mode** with fractional odd spacing, the tessellator **may** produce *interior vertices* that are positioned on the edge of the patch if an inner tessellation level is less than or equal to one. Such vertices are considered distinct from vertices produced by subdividing the outer edge of the patch, even if there are pairs of vertices with identical coordinates.

22.4. Tessellation Primitive Ordering

Few guarantees are provided for the relative ordering of primitives produced by tessellation, as they pertain to **primitive order**.

- The output primitives generated from each input primitive are passed to subsequent pipeline stages in an implementation-dependent order.
- All output primitives generated from a given input primitive are passed to subsequent pipeline stages before any output primitives generated from subsequent input primitives.

22.5. Tessellator Vertex Winding Order

When the tessellator produces triangles (in the **Triangles** or **Quads** modes), the orientation of all triangles is specified with an **OpExecutionMode** of **VertexOrderCw** or **VertexOrderCcw** in the tessellation control or tessellation evaluation shaders. If the order is **VertexOrderCw**, the vertices of all generated triangles will have clockwise ordering in (u,v) or (u,v,w) space. If the order is **VertexOrderCcw**, the vertices will have counter-clockwise ordering in that space.

If the tessellation domain has an upper-left origin, the vertices of a triangle have counter-clockwise ordering if

$$a = u_0 v_1 - u_1 v_0 + u_1 v_2 - u_2 v_1 + u_2 v_0 - u_0 v_2$$

is negative, and clockwise ordering if a is positive. u_i and v_i are the u and v coordinates in normalized parameter space of the i th vertex of the triangle. If the tessellation domain has a lower-left origin, the vertices of a triangle have counter-clockwise ordering if a is positive, and clockwise ordering if a is negative.

NOTE

The value a is proportional (with a positive factor) to the signed area of the triangle.

In **Triangles** mode, even though the vertex coordinates have a w value, it does not participate directly in the computation of a , being an affine combination of u and v .

22.6. Triangle Tessellation

If the tessellation primitive mode is **Triangles**, an equilateral triangle is subdivided into a collection of triangles covering the area of the original triangle. First, the original triangle is subdivided into a collection of concentric equilateral triangles. The edges of each of these triangles are subdivided, and the area between each triangle pair is filled by triangles produced by joining the vertices on the subdivided edges. The number of concentric triangles and the number of subdivisions along each triangle except the outermost is derived from the first inner tessellation level. The edges of the outermost triangle are subdivided independently, using the first, second, and third outer tessellation levels to control the number of subdivisions of the $u = 0$ (left), $v = 0$ (bottom), and $w = 0$ (right) edges, respectively. The second inner tessellation level and the fourth outer tessellation level have no effect in this mode.

If the first inner tessellation level and all three outer tessellation levels are exactly one after clamping and rounding, only a single triangle with (u,v,w) coordinates of (0,0,1), (1,0,0), and (0,1,0) is generated. If the inner tessellation level is one and any of the outer tessellation levels is greater than one, the inner tessellation level is treated as though it were originally specified as $1 + \epsilon$ and will result in a two- or three-segment subdivision depending on the tessellation spacing. When used with fractional odd spacing, the three-segment subdivision **may** produce *inner vertices* positioned on the edge of the triangle.

If any tessellation level is greater than one, tessellation begins by producing a set of concentric inner triangles and subdividing their edges. First, the three outer edges are temporarily subdivided using the clamped and rounded first inner tessellation level and the specified tessellation spacing,

generating n segments. For the outermost inner triangle, the inner triangle is degenerate — a single point at the center of the triangle — if n is two. Otherwise, for each corner of the outer triangle, an inner triangle corner is produced at the intersection of two lines extended perpendicular to the corner's two adjacent edges running through the vertex of the subdivided outer edge nearest that corner. If n is three, the edges of the inner triangle are not subdivided and it is the final triangle in the set of concentric triangles. Otherwise, each edge of the inner triangle is divided into $n - 2$ segments, with the $n - 1$ vertices of this subdivision produced by intersecting the inner edge with lines perpendicular to the edge running through the $n - 1$ innermost vertices of the subdivision of the outer edge. Once the outermost inner triangle is subdivided, the previous subdivision process repeats itself, using the generated triangle as an outer triangle. This subdivision process is illustrated in [Inner Triangle Tessellation](#).

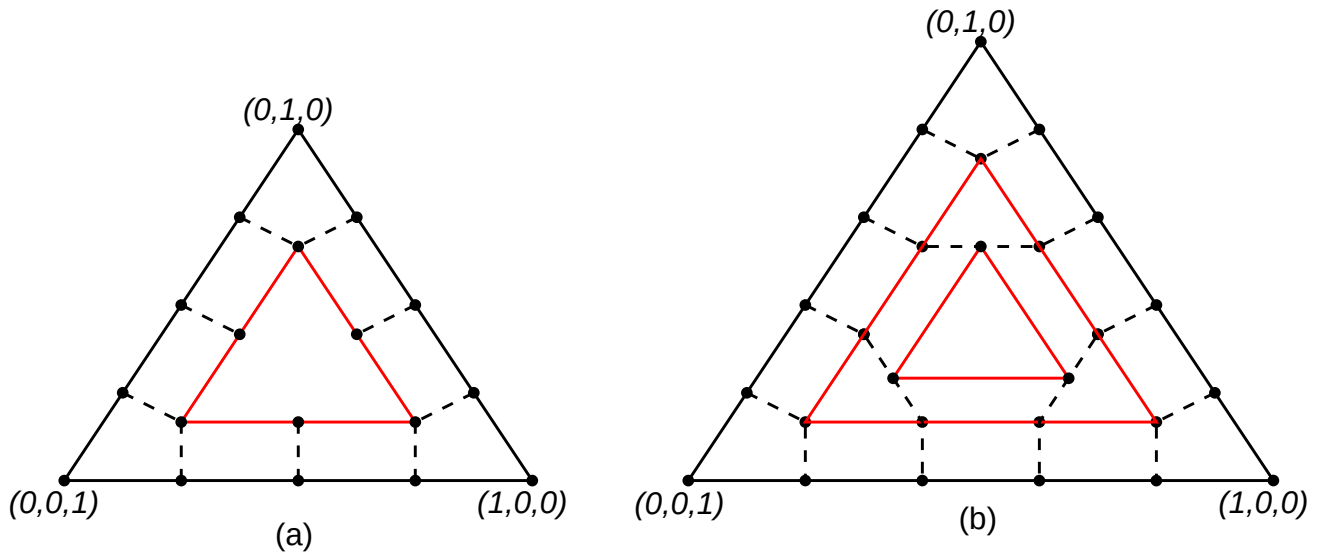


Figure 13. Inner Triangle Tessellation

Caption

In the [Inner Triangle Tessellation](#) diagram, inner tessellation levels of (a) four and (b) five are shown (not to scale). Solid black circles depict vertices along the edges of the concentric triangles. The edges of inner triangles are subdivided by intersecting the edge with segments perpendicular to the edge passing through each inner vertex of the subdivided outer edge. Dotted lines depict edges connecting corresponding vertices on the inner and outer triangle edges.

Once all the concentric triangles are produced and their edges are subdivided, the area between each pair of adjacent inner triangles is filled completely with a set of non-overlapping triangles. In this subdivision, two of the three vertices of each triangle are taken from adjacent vertices on a subdivided edge of one triangle; the third is one of the vertices on the corresponding edge of the other triangle. If the innermost triangle is degenerate (i.e., a point), the triangle containing it is subdivided into six triangles by connecting each of the six vertices on that triangle with the center point. If the innermost triangle is not degenerate, that triangle is added to the set of generated triangles as-is.

After the area corresponding to any inner triangles is filled, the tessellator generates triangles to cover the area between the outermost triangle and the outermost inner triangle. To do this, the

temporary subdivision of the outer triangle edge above is discarded. Instead, the $u = 0$, $v = 0$, and $w = 0$ edges are subdivided according to the first, second, and third outer tessellation levels, respectively, and the tessellation spacing. The original subdivision of the first inner triangle is retained. The area between the outer and first inner triangles is completely filled by non-overlapping triangles as described above. If the first (and only) inner triangle is degenerate, a set of triangles is produced by connecting each vertex on the outer triangle edges with the center point.

After all triangles are generated, each vertex in the subdivided triangle is assigned a barycentric (u,v,w) coordinate based on its location relative to the three vertices of the outer triangle.

The algorithm used to subdivide the triangular domain in (u,v,w) space into individual triangles is implementation-dependent. However, the set of triangles produced will completely cover the domain, and no portion of the domain will be covered by multiple triangles.

Output triangles are generated with a topology similar to [triangle lists](#), except that the order in which each triangle is generated, and the order in which the vertices are generated for each triangle, are implementation-dependent. However, the order of vertices in each triangle is consistent across the domain as described in [Tessellator Vertex Winding Order](#).

22.7. Quad Tessellation

If the tessellation primitive mode is **Quads**, a rectangle is subdivided into a collection of triangles covering the area of the original rectangle. First, the original rectangle is subdivided into a regular mesh of rectangles, where the number of rectangles along the $u = 0$ and $u = 1$ (vertical) and $v = 0$ and $v = 1$ (horizontal) edges are derived from the first and second inner tessellation levels, respectively. All rectangles, except those adjacent to one of the outer rectangle edges, are decomposed into triangle pairs. The outermost rectangle edges are subdivided independently, using the first, second, third, and fourth outer tessellation levels to control the number of subdivisions of the $u = 0$ (left), $v = 0$ (bottom), $u = 1$ (right), and $v = 1$ (top) edges, respectively. The area between the inner rectangles of the mesh and the outer rectangle edges are filled by triangles produced by joining the vertices on the subdivided outer edges to the vertices on the edge of the inner rectangle mesh.

If both clamped inner tessellation levels and all four clamped outer tessellation levels are exactly one, only a single triangle pair covering the outer rectangle is generated. Otherwise, if either clamped inner tessellation level is one, that tessellation level is treated as though it was originally specified as $1 + \epsilon$ and will result in a two- or three-segment subdivision depending on the tessellation spacing. When used with fractional odd spacing, the three-segment subdivision **may** produce *inner vertices* positioned on the edge of the rectangle.

If any tessellation level is greater than one, tessellation begins by subdividing the $u = 0$ and $u = 1$ edges of the outer rectangle into m segments using the clamped and rounded first inner tessellation level and the tessellation spacing. The $v = 0$ and $v = 1$ edges are subdivided into n segments using the second inner tessellation level. Each vertex on the $u = 0$ and $v = 0$ edges are joined with the corresponding vertex on the $u = 1$ and $v = 1$ edges to produce a set of vertical and horizontal lines that divide the rectangle into a grid of smaller rectangles. The primitive generator emits a pair of non-overlapping triangles covering each such rectangle not adjacent to an edge of the outer rectangle. The boundary of the region covered by these triangles forms an inner rectangle, the

edges of which are subdivided by the grid vertices that lie on the edge. If either m or n is two, the inner rectangle is degenerate, and one or both of the rectangle's *edges* consist of a single point. This subdivision is illustrated in Figure [Inner Quad Tessellation](#).

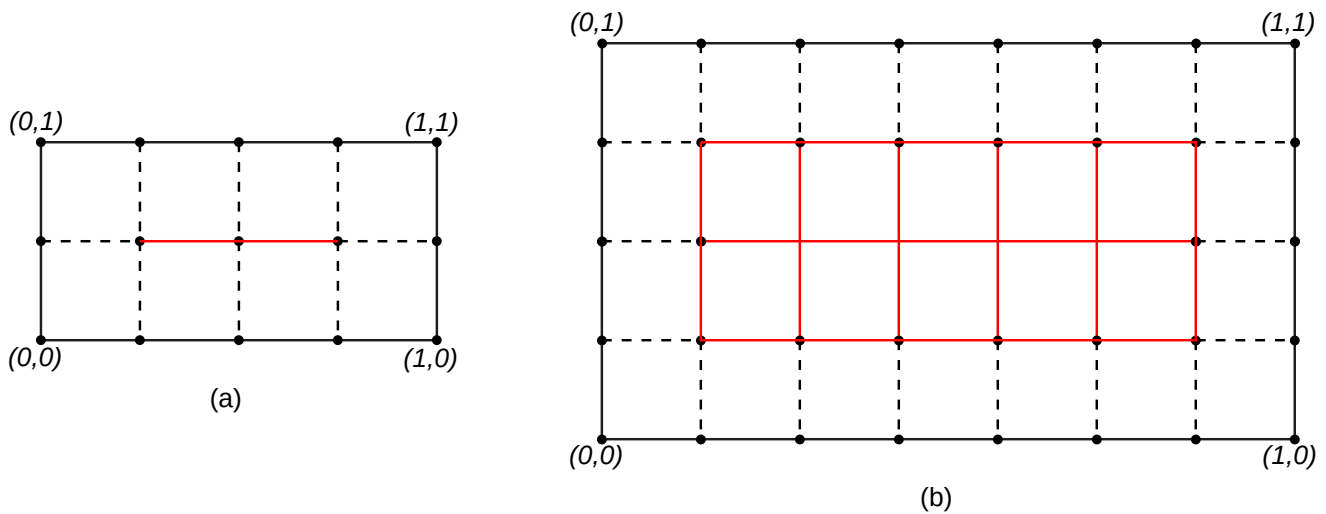


Figure 14. Inner Quad Tessellation

Caption

In the [Inner Quad Tessellation](#) diagram, inner quad tessellation levels of (a) (4,2) and (b) (7,4) are shown. The regions highlighted in red in figure (b) depict the 10 inner rectangles, each of which will be subdivided into two triangles. Solid black circles depict vertices on the boundary of the outer and inner rectangles, where the inner rectangle of figure (a) is degenerate (a single line segment). Dotted lines depict the horizontal and vertical edges connecting corresponding vertices on the inner and outer rectangle edges.

After the area corresponding to the inner rectangle is filled, the tessellator **must** produce triangles to cover the area between the inner and outer rectangles. To do this, the subdivision of the outer rectangle edge above is discarded. Instead, the $u = 0$, $v = 0$, $u = 1$, and $v = 1$ edges are subdivided according to the first, second, third, and fourth outer tessellation levels, respectively, and the tessellation spacing. The original subdivision of the inner rectangle is retained. The area between the outer and inner rectangles is completely filled by non-overlapping triangles. Two of the three vertices of each triangle are adjacent vertices on a subdivided edge of one rectangle; the third is one of the vertices on the corresponding edge of the other rectangle. If either edge of the innermost rectangle is degenerate, the area near the corresponding outer edges is filled by connecting each vertex on the outer edge with the single vertex making up the *inner edge*.

The algorithm used to subdivide the rectangular domain in (u,v) space into individual triangles is implementation-dependent. However, the set of triangles produced will completely cover the domain, and no portion of the domain will be covered by multiple triangles.

Output triangles are generated with a topology similar to [triangle lists](#), except that the order in which each triangle is generated, and the order in which the vertices are generated for each triangle, are implementation-dependent. However, the order of vertices in each triangle is consistent across the domain as described in [Tessellator Vertex Winding Order](#).

22.8. Isoline Tessellation

If the tessellation primitive mode is `Isolines`, a set of independent horizontal line segments is drawn. The segments are arranged into connected strips called *isolines*, where the vertices of each isoline have a constant *v* coordinate and *u* coordinates covering the full range [0,1]. The number of isolines generated is derived from the first outer tessellation level; the number of segments in each isoline is derived from the second outer tessellation level. Both inner tessellation levels and the third and fourth outer tessellation levels have no effect in this mode.

As with quad tessellation above, isoline tessellation begins with a rectangle. The *u* = 0 and *u* = 1 edges of the rectangle are subdivided according to the first outer tessellation level. For the purposes of this subdivision, the tessellation spacing mode is ignored and treated as `equal_spacing`. An isoline is drawn connecting each vertex on the *u* = 0 rectangle edge to the corresponding vertex on the *u* = 1 rectangle edge, except that no line is drawn between (0,1) and (1,1). If the number of isolines on the subdivided *u* = 0 and *u* = 1 edges is *n*, this process will result in *n* equally spaced lines with constant *v* coordinates of $0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}$.

Each of the *n* isolines is then subdivided according to the second outer tessellation level and the tessellation spacing, resulting in *m* line segments. Each segment of each line is emitted by the tessellator. These line segments are generated with a topology similar to [line lists](#), except that the order in which each line is generated, and the order in which the vertices are generated for each line segment, are implementation-dependent.

NOTE	If the <code>VK_KHR_portability_subset</code> extension is enabled, and <code>VkPhysicalDevicePortabilitySubsetFeaturesKHR::tessellationIsolines</code> is <code>VK_FALSE</code> , then isoline tessellation is not supported by the implementation.
-------------	--

22.9. Tessellation Point Mode

For all primitive modes, the tessellator is capable of generating points instead of lines or triangles. If the tessellation control or tessellation evaluation shader specifies the `OpExecutionMode PointMode`, the primitive generator will generate one point for each distinct vertex produced by tessellation, rather than emitting triangles or lines. Otherwise, the tessellator will produce a collection of line segments or triangles according to the primitive mode. These points are generated with a topology similar to [point lists](#), except the order in which the points are generated for each input primitive is undefined.

NOTE	If the <code>VK_KHR_portability_subset</code> extension is enabled, and <code>VkPhysicalDevicePortabilitySubsetFeaturesKHR::tessellationPointMode</code> is <code>VK_FALSE</code> , then tessellation point mode is not supported by the implementation.
-------------	--

22.10. Tessellation Pipeline State

The `pTessellationState` member of `VkGraphicsPipelineCreateInfo` is a pointer to a `VkPipelineTessellationStateCreateInfo` structure.

The `VkPipelineTessellationStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineTessellationStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineTessellationStateCreateFlags flags;
    uint32_t                  patchControlPoints;
} VkPipelineTessellationStateCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `patchControlPoints` is the number of control points per patch.

Valid Usage

- VUID-VkPipelineTessellationStateCreateInfo-patchControlPoints-01214
`patchControlPoints` **must** be greater than zero and less than or equal to `VkPhysicalDeviceLimits::maxTessellationPatchSize`

Valid Usage (Implicit)

- VUID-VkPipelineTessellationStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_STATE_CREATE_INFO`
- VUID-VkPipelineTessellationStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkPipelineTessellationDomainOriginStateCreateInfo`
- VUID-VkPipelineTessellationStateCreateInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkPipelineTessellationStateCreateInfo-flags-zero-bitmask
`flags` **must** be 0

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineTessellationStateCreateFlags;
```

`VkPipelineTessellationStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkPipelineTessellationDomainOriginStateCreateInfo` structure is defined as:

```
typedef struct VkPipelineTessellationDomainOriginStateCreateInfo {
    VkStructureType      sType;
    const void*          pNext;
    VkTessellationDomainOrigin domainOrigin;
} VkPipelineTessellationDomainOriginStateCreateInfo;
```

or the equivalent

```
// Provided by VK_KHR_maintenance2
typedef VkPipelineTessellationDomainOriginStateCreateInfo
VkPipelineTessellationDomainOriginStateCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `domainOrigin` is a `VkTessellationDomainOrigin` value controlling the origin of the tessellation domain space.

If the `VkPipelineTessellationDomainOriginStateCreateInfo` structure is included in the `pNext` chain of `VkPipelineTessellationStateCreateInfo`, it controls the origin of the tessellation domain. If this structure is not present, it is as if `domainOrigin` was `VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT`.

Valid Usage (Implicit)

- VUID-VkPipelineTessellationDomainOriginStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_DOMAIN_ORIGIN_STATE_CREATE_INFO`
- VUID-VkPipelineTessellationDomainOriginStateCreateInfo-domainOrigin-parameter
`domainOrigin` **must** be a valid `VkTessellationDomainOrigin` value

The possible tessellation domain origins are specified by the `VkTessellationDomainOrigin` enumeration:

```
typedef enum VkTessellationDomainOrigin {
    VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT = 0,
    VK_TESSELLATION_DOMAIN_ORIGIN_LOWER_LEFT = 1,
    // Provided by VK_KHR_maintenance2
    VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT_KHR =
VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT,
    // Provided by VK_KHR_maintenance2
    VK_TESSELLATION_DOMAIN_ORIGIN_LOWER_LEFT_KHR =
VK_TESSELLATION_DOMAIN_ORIGIN_LOWER_LEFT,
} VkTessellationDomainOrigin;
```

or the equivalent

```
// Provided by VK_KHR_maintenance2
typedef VkTessellationDomainOrigin VkTessellationDomainOriginKHR;
```

- `VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT` specifies that the origin of the domain space is in the upper left corner, as shown in figure [Domain parameterization for tessellation primitive modes \(upper-left origin\)](#).
- `VK_TESSELLATION_DOMAIN_ORIGIN_LOWER_LEFT` specifies that the origin of the domain space is in the lower left corner, as shown in figure [Domain parameterization for tessellation primitive modes \(lower-left origin\)](#).

This enum affects how the `VertexOrderCw` and `VertexOrderCcw` tessellation execution modes are interpreted, since the winding is defined relative to the orientation of the domain.

To [dynamically set](#) the origin of the tessellation domain space, call:

```
// Provided by VK_EXT_extended_dynamic_state3 with VK_KHR_maintenance2 or
VK_VERSION_1_1, VK_EXT_shader_object
void vkCmdSetTessellationDomainOriginEXT(
    VkCommandBuffer                commandBuffer,
    VkTessellationDomainOrigin     domainOrigin);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `domainOrigin` specifies the origin of the tessellation domain space.

This command sets the origin of the tessellation domain space for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineTessellationDomainOriginStateCreateInfo::domainOrigin` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetTessellationDomainOriginEXT-None-09423

At least one of the following **must** be true:

- The `extendedDynamicState3TessellationDomainOrigin` feature is enabled
- The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetTessellationDomainOriginEXT-commandBuffer-parameter `commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetTessellationDomainOriginEXT-domainOrigin-parameter

`domainOrigin` **must** be a valid `VkTessellationDomainOrigin` value

- VUID-vkCmdSetTessellationDomainOriginEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetTessellationDomainOriginEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetTessellationDomainOriginEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Chapter 23. Geometry Shading

The geometry shader operates on a group of vertices and their associated data assembled from a single input primitive, and emits zero or more output primitives and the group of vertices and their associated data required for each output primitive. Geometry shading is enabled when a geometry shader is included in the pipeline.

23.1. Geometry Shader Input Primitives

Each geometry shader invocation has access to all vertices in the primitive (and their associated data), which are presented to the shader as an array of inputs.

The input primitive type expected by the geometry shader is specified with an `OpExecutionMode` instruction in the geometry shader, and **must** match the incoming primitive type specified by either the pipeline's `primitive topology` if tessellation is inactive, or the `tessellation mode` if tessellation is active, as follows:

- An input primitive type of `InputPoints` **must** only be used with a pipeline topology of `VK_PRIMITIVE_TOPOLOGY_POINT_LIST`, or with a tessellation shader specifying `PointMode`. The input arrays always contain one element, as described by the `point list topology` or `tessellation in point mode`.
- An input primitive type of `InputLines` **must** only be used with a pipeline topology of `VK_PRIMITIVE_TOPOLOGY_LINE_LIST` or `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP`, or with a tessellation shader specifying `IsoLines` that does not specify `PointMode`. The input arrays always contain two elements, as described by the `line list topology` or `line strip topology`, or by `isoline tessellation`.
- An input primitive type of `InputLinesAdjacency` **must** only be used when tessellation is inactive, with a pipeline topology of `VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY` or `VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY`. The input arrays always contain four elements, as described by the `line list with adjacency topology` or `line strip with adjacency topology`.
- An input primitive type of `Triangles` **must** only be used with a pipeline topology of `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST`, `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP`, or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN`; or with a tessellation shader specifying `Quads` or `Triangles` that does not specify `PointMode`. The input arrays always contain three elements, as described by the `triangle list topology`, `triangle strip topology`, or `triangle fan topology`, or by `triangle` or `quad tessellation`. Vertices **may** be in a different absolute order than specified by the topology, but **must** adhere to the specified winding order.
- An input primitive type of `InputTrianglesAdjacency` **must** only be used when tessellation is inactive, with a pipeline topology of `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY` or `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY`. The input arrays always contain six elements, as described by the `triangle list with adjacency topology` or `triangle strip with adjacency topology`. Vertices **may** be in a different absolute order than specified by the topology, but **must** adhere to the specified winding order, and the vertices making up the main primitive **must** still occur at the first, third, and fifth index.

23.2. Geometry Shader Output Primitives

A geometry shader generates primitives in one of three output modes: points, line strips, or triangle strips. The primitive mode is specified in the shader using an `OpExecutionMode` instruction with the `OutputPoints`, `OutputLineStrip` or `OutputTriangleStrip` modes, respectively. Each geometry shader **must** include exactly one output primitive mode.

The vertices output by the geometry shader are assembled into points, lines, or triangles based on the output primitive type and the resulting primitives are then further processed as described in [Rasterization](#). If the number of vertices emitted by the geometry shader is not sufficient to produce a single primitive, vertices corresponding to incomplete primitives are not processed by subsequent pipeline stages. The number of vertices output by the geometry shader is limited to a maximum count specified in the shader.

The maximum output vertex count is specified in the shader using an `OpExecutionMode` instruction with the mode set to `OutputVertices` and the maximum number of vertices that will be produced by the geometry shader specified as a literal. Each geometry shader **must** specify a maximum output vertex count.

23.3. Multiple Invocations of Geometry Shaders

Geometry shaders **can** be invoked more than one time for each input primitive. This is known as *geometry shader instancing* and is requested by including an `OpExecutionMode` instruction with `mode` specified as `Invocations` and the number of invocations specified as an integer literal.

In this mode, the geometry shader will execute at least *n* times for each input primitive, where *n* is the number of invocations specified in the `OpExecutionMode` instruction. The instance number is available to each invocation as a built-in input using `InvocationId`.

23.4. Geometry Shader Primitive Ordering

Limited guarantees are provided for the relative ordering of primitives produced by a geometry shader, as they pertain to [primitive order](#).

- For instanced geometry shaders, the output primitives generated from each input primitive are passed to subsequent pipeline stages using the invocation number to order the primitives, from least to greatest.
- All output primitives generated from a given input primitive are passed to subsequent pipeline stages before any output primitives generated from subsequent input primitives.

Chapter 24. Fixed-Function Vertex Post-Processing

After [pre-rasterization shader stages](#), the following fixed-function operations are applied to vertices of the resulting primitives:

- Transform feedback (see [Transform Feedback](#))
- Flat shading (see [Flat Shading](#)).
- Primitive clipping, including application-defined half-spaces (see [Primitive Clipping](#)).
- Shader output attribute clipping (see [Clipping Shader Outputs](#)).
- Perspective division on clip coordinates (see [Coordinate Transformations](#)).
- Viewport mapping, including depth range scaling (see [Controlling the Viewport](#)).
- Front face determination for polygon primitives (see [Basic Polygon Rasterization](#)).

Next, rasterization is performed on primitives as described in chapter [Rasterization](#).

24.1. Transform Feedback

Before any other fixed-function vertex post-processing, vertex outputs from the last shader in the [pre-rasterization shader stage](#) **can** be written out to one or more transform feedback buffers bound to the command buffer. To capture vertex outputs the last [pre-rasterization shader stage](#) shader **must** be declared with the `Xfb` execution mode. Outputs decorated with `XfbBuffer` will be written out to the corresponding transform feedback buffers bound to the command buffer when transform feedback is active. Transform feedback buffers are bound to the command buffer by using `vkCmdBindTransformFeedbackBuffersEXT`. Transform feedback is made active by calling `vkCmdBeginTransformFeedbackEXT` and made inactive by calling `vkCmdEndTransformFeedbackEXT`. After vertex data is written it is possible to use `vkCmdDrawIndirectByteCountEXT` to start a new draw where the `vertexCount` is derived from the number of bytes written by a previous transform feedback.

When an individual point, line, or triangle primitive reaches the transform feedback stage while transform feedback is active, the values of the specified output variables are assembled into primitives and appended to the bound transform feedback buffers. After activating transform feedback, the values of the first assembled primitive are written at the starting offsets of the bound transform feedback buffers, and subsequent primitives are appended to the buffer. If the optional `pCounterBuffers` and `pCounterBufferOffsets` parameters are specified, the starting points within the transform feedback buffers are adjusted so data is appended to the previously written values indicated by the value stored by the implementation in the counter buffer.

For multi-vertex primitives, all values for a given vertex are written before writing values for any other vertex. When `transformFeedbackPreservesProvokingVertex` is not enabled, implementations **may** write out any vertex within the primitive first, but all subsequent vertices for that primitive **must** be written out in a consistent winding order defined as follows:

- If neither [geometry](#) or [tessellation shading](#) is active, vertices within a primitive are appended

according to the winding order described by the [primitive topology](#) defined by the [VkPipelineInputAssemblyStateCreateInfo:topology](#) used to execute the [drawing command](#).

- If [geometry shading](#) is active, vertices within a primitive are appended according to the winding order described by the [primitive topology](#) defined by the [OutputPoints](#), [OutputLineStrip](#), or [OutputTriangleStrip](#) execution mode.
- If [tessellation shading](#) is active but [geometry shading](#) is not, vertices within a primitive are appended according to the winding order defined by [triangle tessellation](#), [quad tessellation](#), and [isoline tessellation](#).

When [transformFeedbackPreservesProvokingVertex](#) is enabled, then in addition to writing vertices with a consistent winding order, the vertex order **must** preserve the [provoking vertex](#) of each primitive:

- When the [pipeline's provoking vertex mode](#) is [VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT](#), the primitive's provoking vertex **must** be the first vertex written.
- When the [pipeline's provoking vertex mode](#) is [VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT](#), the primitive's provoking vertex **must** be the last vertex written.

If [transformFeedbackPreservesTriangleFanProvokingVertex](#) is [VK_FALSE](#), neither [geometry](#) nor [tessellation](#) shading is active, and the [primitive topology](#) is [VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN](#), then the first vertex written from each primitive is implementation-defined even when [transformFeedbackPreservesProvokingVertex](#) is enabled.

When capturing vertices, the stride associated with each transform feedback buffer, as indicated by the [XfbStride](#) decoration, indicates the number of bytes of storage reserved for each vertex in the transform feedback buffer. For every vertex captured, each output attribute with a [Offset](#) decoration will be written to the storage reserved for the vertex at the associated transform feedback buffer. When writing output variables that are arrays or structures, individual array elements or structure members are written tightly packed in order. For vector types, individual components are written in order. For matrix types, outputs are written as an array of column vectors.

If any component of an output with an assigned transform feedback offset was not written to by its shader, the value recorded for that component is undefined. All components of an output variable **must** be written at an offset aligned to the size of the component. The size of each component of an output variable **must** be at least 32-bits. When capturing a vertex, any portion of the reserved storage not associated with an output variable with an assigned transform feedback offset will be unmodified.

When transform feedback is inactive, no vertices are recorded. If there is a valid counter buffer handle and counter buffer offset in the [pCounterBuffers](#) and [pCounterBufferOffsets](#) arrays, writes to the corresponding transform feedback buffer will start at the byte offset represented by the value stored in the counter buffer location.

Individual lines or triangles of a strip or fan primitive will be extracted and recorded separately. Incomplete primitives are not recorded.

When using a geometry shader that emits vertices to multiple vertex streams, a primitive will be

assembled and output for each stream when there are enough vertices emitted for the output primitive type. All outputs assigned to a given transform feedback buffer are required to come from a single vertex stream.

The sizes of the transform feedback buffers are defined by the `vkCmdBindTransformFeedbackBuffersEXT` `pSizes` parameter for each of the bound buffers, or the size of the bound buffer, whichever is the lesser. If there is less space remaining in any of the transform feedback buffers than the size of all of the vertex data for that primitive based on the `XfbStride` for that `XfbBuffer` then no vertex data of that primitive is recorded in any transform feedback buffer, and the value for the number of primitives written in the corresponding `VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT` query for all transform feedback buffers is no longer incremented.

Any outputs made to a `XfbBuffer` that is not bound to a transform feedback buffer is ignored.

To bind transform feedback buffers to a command buffer for use in subsequent drawing commands, call:

```
// Provided by VK_EXT_transform_feedback
void vkCmdBindTransformFeedbackBuffersEXT(
    VkCommandBuffer          commandBuffer,
    uint32_t                 firstBinding,
    uint32_t                 bindingCount,
    const VkBuffer*          pBuffers,
    const VkDeviceSize*      pOffsets,
    const VkDeviceSize*      pSizes);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `firstBinding` is the index of the first transform feedback binding whose state is updated by the command.
- `bindingCount` is the number of transform feedback bindings whose state is updated by the command.
- `pBuffers` is a pointer to an array of buffer handles.
- `pOffsets` is a pointer to an array of buffer offsets.
- `pSizes` is `NULL` or a pointer to an array of `VkDeviceSize` buffer sizes, specifying the maximum number of bytes to capture to the corresponding transform feedback buffer. If `pSizes` is `NULL`, or the value of the `pSizes` array element is `VK_WHOLE_SIZE`, then the maximum number of bytes captured will be the size of the corresponding buffer minus the buffer offset.

The values taken from elements `i` of `pBuffers`, `pOffsets` and `pSizes` replace the current state for the transform feedback binding `firstBinding + i`, for `i` in `[0, bindingCount)`. The transform feedback binding is updated to start at the offset indicated by `pOffsets[i]` from the start of the buffer `pBuffers[i]`.

Valid Usage

- VUID-vkCmdBindTransformFeedbackBuffersEXT-transformFeedback-02355
`VkPhysicalDeviceTransformFeedbackFeaturesEXT::transformFeedback` **must** be enabled
- VUID-vkCmdBindTransformFeedbackBuffersEXT-firstBinding-02356
`firstBinding` **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBuffers`
- VUID-vkCmdBindTransformFeedbackBuffersEXT-firstBinding-02357
The sum of `firstBinding` and `bindingCount` **must** be less than or equal to `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBuffers`
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pOffsets-02358
All elements of `pOffsets` **must** be less than the size of the corresponding element in `pBuffers`
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pOffsets-02359
All elements of `pOffsets` **must** be a multiple of 4
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pBuffers-02360
All elements of `pBuffers` **must** have been created with the `VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT` flag
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pSize-02361
If the optional `pSize` array is specified, each element of `pSizes` **must** either be `VK_WHOLE_SIZE`, or be less than or equal to `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBufferSize`
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pSizes-02362
All elements of `pSizes` **must** be either `VK_WHOLE_SIZE`, or less than or equal to the size of the corresponding buffer in `pBuffers`
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pOffsets-02363
All elements of `pOffsets` plus `pSizes`, where the `pSizes`, element is not `VK_WHOLE_SIZE`, **must** be less than or equal to the size of the corresponding buffer in `pBuffers`
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pBuffers-02364
Each element of `pBuffers` that is non-sparse **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBindTransformFeedbackBuffersEXT-None-02365
Transform feedback **must** not be active when the `vkCmdBindTransformFeedbackBuffersEXT` command is recorded

Valid Usage (Implicit)

- VUID-vkCmdBindTransformFeedbackBuffersEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBindTransformFeedbackBuffersEXT-pBuffers-parameter
`pBuffers` **must** be a valid pointer to an array of `bindingCount` valid `VkBuffer` handles

- VUID-vkCmdBindTransformFeedbackBuffersEXT-pOffsets-parameter
`pOffsets` **must** be a valid pointer to an array of `bindingCount` `VkDeviceSize` values
- VUID-vkCmdBindTransformFeedbackBuffersEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdBindTransformFeedbackBuffersEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBindTransformFeedbackBuffersEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBindTransformFeedbackBuffersEXT-bindingCount-arraylength
`bindingCount` **must** be greater than 0
- VUID-vkCmdBindTransformFeedbackBuffersEXT-commonparent
Both of `commandBuffer`, and the elements of `pBuffers` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Transform feedback for specific transform feedback buffers is made active by calling:

```
// Provided by VK_EXT_transform_feedback
void vkCmdBeginTransformFeedbackEXT(
    VkCommandBuffer                commandBuffer,
    uint32_t                       firstCounterBuffer,
    uint32_t                       counterBufferCount,
    const VkBuffer*                pCounterBuffers,
    const VkDeviceSize*            pCounterBufferOffsets);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `firstCounterBuffer` is the index of the first transform feedback buffer corresponding to `pCounterBuffers[0]` and `pCounterBufferOffsets[0]`.

- `counterBufferCount` is the size of the `pCounterBuffers` and `pCounterBufferOffsets` arrays.
- `pCounterBuffers` is `NULL` or a pointer to an array of `VkBuffer` handles to counter buffers. Each buffer contains a 4 byte integer value representing the byte offset from the start of the corresponding transform feedback buffer from where to start capturing vertex data. If the byte offset stored to the counter buffer location was done using `vkCmdEndTransformFeedbackEXT` it can be used to resume transform feedback from the previous location. If `pCounterBuffers` is `NULL`, then transform feedback will start capturing vertex data to byte offset zero in all bound transform feedback buffers. For each element of `pCounterBuffers` that is `VK_NULL_HANDLE`, transform feedback will start capturing vertex data to byte zero in the corresponding bound transform feedback buffer.
- `pCounterBufferOffsets` is `NULL` or a pointer to an array of `VkDeviceSize` values specifying offsets within each of the `pCounterBuffers` where the counter values were previously written. The location in each counter buffer at these offsets **must** be large enough to contain 4 bytes of data. This data is the number of bytes captured by the previous transform feedback to this buffer. If `pCounterBufferOffsets` is `NULL`, then it is assumed the offsets are zero.

The active transform feedback buffers will capture primitives emitted from the corresponding `XfbBuffer` in the bound graphics pipeline. Any `XfbBuffer` emitted that does not output to an active transform feedback buffer will not be captured.

Valid Usage

- VUID-vkCmdBeginTransformFeedbackEXT-transformFeedback-02366
`VkPhysicalDeviceTransformFeedbackFeaturesEXT::transformFeedback` **must** be enabled
- VUID-vkCmdBeginTransformFeedbackEXT-None-02367
Transform feedback **must** not be active
- VUID-vkCmdBeginTransformFeedbackEXT-firstCounterBuffer-02368
`firstCounterBuffer` **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBuffers`
- VUID-vkCmdBeginTransformFeedbackEXT-firstCounterBuffer-02369
The sum of `firstCounterBuffer` and `counterBufferCount` **must** be less than or equal to `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBuffers`
- VUID-vkCmdBeginTransformFeedbackEXT-counterBufferCount-02607
If `counterBufferCount` is not 0, and `pCounterBuffers` is not `NULL`, `pCounterBuffers` **must** be a valid pointer to an array of `counterBufferCount` `VkBuffer` handles that are either valid or `VK_NULL_HANDLE`
- VUID-vkCmdBeginTransformFeedbackEXT-pCounterBufferOffsets-02370
For each buffer handle in the array, if it is not `VK_NULL_HANDLE` it **must** reference a buffer large enough to hold 4 bytes at the corresponding offset from the `pCounterBufferOffsets` array
- VUID-vkCmdBeginTransformFeedbackEXT-pCounterBuffer-02371
If `pCounterBuffer` is `NULL`, then `pCounterBufferOffsets` **must** also be `NULL`
- VUID-vkCmdBeginTransformFeedbackEXT-pCounterBuffers-02372
For each buffer handle in the `pCounterBuffers` array that is not `VK_NULL_HANDLE` it **must**

have been created with a `usage` value containing `VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT`

- VUID-vkCmdBeginTransformFeedbackEXT-firstCounterBuffer-09630
The sum of `firstCounterBuffer` and `counterBufferCount` **must** be less than or equal to the number of transform feedback buffers currently bound by `vkCmdBindTransformFeedbackBuffersEXT`
- VUID-vkCmdBeginTransformFeedbackEXT-None-06233
If the `shaderObject` feature is not enabled, a valid graphics pipeline **must** be bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`
- VUID-vkCmdBeginTransformFeedbackEXT-None-04128
The last `pre-rasterization shader stage` of the bound graphics pipeline **must** have been declared with the `Xfb` execution mode
- VUID-vkCmdBeginTransformFeedbackEXT-None-02373
Transform feedback **must** not be made active in a render pass instance with multiview enabled

Valid Usage (Implicit)

- VUID-vkCmdBeginTransformFeedbackEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBeginTransformFeedbackEXT-pCounterBufferOffsets-parameter
If `counterBufferCount` is not 0, and `pCounterBufferOffsets` is not `NULL`, `pCounterBufferOffsets` **must** be a valid pointer to an array of `counterBufferCount` `VkDeviceSize` values
- VUID-vkCmdBeginTransformFeedbackEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdBeginTransformFeedbackEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdBeginTransformFeedbackEXT-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdBeginTransformFeedbackEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBeginTransformFeedbackEXT-commonparent
Both of `commandBuffer`, and the elements of `pCounterBuffers` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	State

Transform feedback for specific transform feedback buffers is made inactive by calling:

```
// Provided by VK_EXT_transform_feedback
void vkCmdEndTransformFeedbackEXT(
    VkCommandBuffer                commandBuffer,
    uint32_t                       firstCounterBuffer,
    uint32_t                       counterBufferCount,
    const VkBuffer*                 pCounterBuffers,
    const VkDeviceSize*             pCounterBufferOffsets);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `firstCounterBuffer` is the index of the first transform feedback buffer corresponding to `pCounterBuffers[0]` and `pCounterBufferOffsets[0]`.
- `counterBufferCount` is the size of the `pCounterBuffers` and `pCounterBufferOffsets` arrays.
- `pCounterBuffers` is `NULL` or a pointer to an array of `VkBuffer` handles to counter buffers. The counter buffers are used to record the current byte positions of each transform feedback buffer where the next vertex output data would be captured. This **can** be used by a subsequent `vkCmdBeginTransformFeedbackEXT` call to resume transform feedback capture from this position. It can also be used by `vkCmdDrawIndirectByteCountEXT` to determine the vertex count of the draw call.
- `pCounterBufferOffsets` is `NULL` or a pointer to an array of `VkDeviceSize` values specifying offsets within each of the `pCounterBuffers` where the counter values can be written. The location in each counter buffer at these offsets **must** be large enough to contain 4 bytes of data. The data stored at this location is the byte offset from the start of the transform feedback buffer binding where the next vertex data would be written. If `pCounterBufferOffsets` is `NULL`, then it is assumed the offsets are zero.

Valid Usage

- VUID-vkCmdEndTransformFeedbackEXT-transformFeedback-02374
`VkPhysicalDeviceTransformFeedbackFeaturesEXT::transformFeedback` **must** be enabled
- VUID-vkCmdEndTransformFeedbackEXT-None-02375
Transform feedback **must** be active

- VUID-vkCmdEndTransformFeedbackEXT-firstCounterBuffer-02376
`firstCounterBuffer` **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBuffers`
- VUID-vkCmdEndTransformFeedbackEXT-firstCounterBuffer-02377
The sum of `firstCounterBuffer` and `counterBufferCount` **must** be less than or equal to `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBuffers`
- VUID-vkCmdEndTransformFeedbackEXT-counterBufferCount-02608
If `counterBufferCount` is not 0, and `pCounterBuffers` is not NULL, `pCounterBuffers` **must** be a valid pointer to an array of `counterBufferCount` `VkBuffer` handles that are either valid or `VK_NULL_HANDLE`
- VUID-vkCmdEndTransformFeedbackEXT-pCounterBufferOffsets-02378
For each buffer handle in the array, if it is not `VK_NULL_HANDLE` it **must** reference a buffer large enough to hold 4 bytes at the corresponding offset from the `pCounterBufferOffsets` array
- VUID-vkCmdEndTransformFeedbackEXT-pCounterBuffer-02379
If `pCounterBuffer` is NULL, then `pCounterBufferOffsets` **must** also be NULL
- VUID-vkCmdEndTransformFeedbackEXT-pCounterBuffers-02380
For each buffer handle in the `pCounterBuffers` array that is not `VK_NULL_HANDLE` it **must** have been created with a `usage` value containing `VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT`

Valid Usage (Implicit)

- VUID-vkCmdEndTransformFeedbackEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdEndTransformFeedbackEXT-pCounterBufferOffsets-parameter
If `counterBufferCount` is not 0, and `pCounterBufferOffsets` is not NULL, `pCounterBufferOffsets` **must** be a valid pointer to an array of `counterBufferCount` `VkDeviceSize` values
- VUID-vkCmdEndTransformFeedbackEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdEndTransformFeedbackEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdEndTransformFeedbackEXT-renderpass
This command **must** only be called inside of a render pass instance
- VUID-vkCmdEndTransformFeedbackEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdEndTransformFeedbackEXT-commonparent
Both of `commandBuffer`, and the elements of `pCounterBuffers` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Inside	Outside	Graphics	State

24.2. Flat Shading

Flat shading a vertex output attribute means to assign all vertices of the primitive the same value for that output. The output values assigned are those of the *provoking vertex* of the primitive. Flat shading is applied to those vertex attributes that `match` fragment input attributes which are decorated as `Flat`.

If neither `geometry` nor `tessellation shading` is active, the provoking vertex is determined by the `primitive topology` defined by `VkPipelineInputAssemblyStateCreateInfo::topology` used to execute the `drawing command`.

If `geometry shading` is active, the provoking vertex is determined by the `primitive topology` defined by the `OutputPoints`, `OutputLineStrip`, or `OutputTriangleStrip` execution mode.

If `tessellation shading` is active but `geometry shading` is not, the provoking vertex **may** be any of the vertices in each primitive.

For a given primitive topology, the pipeline's provoking vertex mode determines which vertex is the provoking vertex. To specify the provoking vertex mode, include a `VkPipelineRasterizationProvokingVertexStateCreateInfoEXT` structure in the `VkPipelineRasterizationStateCreateInfo::pNext` chain when creating the pipeline.

The `VkPipelineRasterizationProvokingVertexStateCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_provoking_vertex
typedef struct VkPipelineRasterizationProvokingVertexStateCreateInfoEXT {
    VkStructureType           sType;
    const void*               pNext;
    VkProvokingVertexModeEXT  provokingVertexMode;
} VkPipelineRasterizationProvokingVertexStateCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `provokingVertexMode` is a `VkProvokingVertexModeEXT` value selecting the provoking vertex mode.

If this struct is not provided when creating the pipeline, the pipeline will use the `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT` mode.

If the `provokingVertexModePerPipeline` limit is `VK_FALSE`, then all pipelines bound within a render pass instance **must** have the same `provokingVertexMode`.

Valid Usage

- VUID-VkPipelineRasterizationProvokingVertexStateCreateInfoEXT-provokingVertexMode-04883

If `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, then the `provokingVertexLast` feature **must** be enabled

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationProvokingVertexStateCreateInfoEXT-sType-sType **must** be `VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_PROVOKING_VERTEX_STATE_CREATE_INFO_EXT`
- VUID-VkPipelineRasterizationProvokingVertexStateCreateInfoEXT-provokingVertexMode-parameter `provokingVertexMode` **must** be a valid `VkProvokingVertexModeEXT` value

Possible values of `VkPipelineRasterizationProvokingVertexStateCreateInfoEXT::provokingVertexMode` are:

```
// Provided by VK_EXT_provoking_vertex
typedef enum VkProvokingVertexModeEXT {
    VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT = 0,
    VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT = 1,
} VkProvokingVertexModeEXT;
```

- `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT` specifies that the provoking vertex is the first non-adjacency vertex in the list of vertices used by a primitive.
- `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT` specifies that the provoking vertex is the last non-adjacency vertex in the list of vertices used by a primitive.

These modes are described more precisely in [Primitive Topologies](#).

To [dynamically set](#) the `provokingVertexMode` state, call:

```
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_provoking_vertex,
VK_EXT_provoking_vertex with VK_EXT_shader_object
void vkCmdSetProvokingVertexModeEXT(
    VkCommandBuffer                      commandBuffer,
    VkProvokingVertexModeEXT            provokingVertexMode);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `provokingVertexMode` specifies the `provokingVertexMode` state.

This command sets the `provokingVertexMode` state for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationProvokingVertexStateCreateInfoEXT::provokingVertexMode` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetProvokingVertexModeEXT-None-09423
At least one of the following **must** be true:
 - The `extendedDynamicState3ProvokingVertexMode` feature is enabled
 - The `shaderObject` feature is enabled
- VUID-vkCmdSetProvokingVertexModeEXT-provokingVertexMode-07447
If `provokingVertexMode` is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, then the `provokingVertexLast` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdSetProvokingVertexModeEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetProvokingVertexModeEXT-provokingVertexMode-parameter
`provokingVertexMode` **must** be a valid `VkProvokingVertexModeEXT` value
- VUID-vkCmdSetProvokingVertexModeEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetProvokingVertexModeEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetProvokingVertexModeEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

24.3. Primitive Clipping

Primitives are culled against the *cull volume* and then clipped to the *clip volume*. In clip coordinates, the *view volume* is defined by:

$$\begin{aligned} -w_c &\leq x_c \leq w_c \\ -w_c &\leq y_c \leq w_c \\ z_m &\leq z_c \leq w_c \end{aligned}$$

where if `VkPipelineViewportDepthClipControlCreateInfoEXT::negativeOneToOne` is `VK_TRUE` z_m is equal to $-w_c$ otherwise z_m is equal to zero.

This view volume **can** be further restricted by as many as `VkPhysicalDeviceLimits::maxClipDistances` application-defined half-spaces.

The cull volume is the intersection of up to `VkPhysicalDeviceLimits::maxCullDistances` application-defined half-spaces (if no application-defined cull half-spaces are enabled, culling against the cull volume is skipped).

A shader **must** write a single cull distance for each enabled cull half-space to elements of the `CullDistance` array. If the cull distance for any enabled cull half-space is negative for all of the vertices of the primitive under consideration, the primitive is discarded. Otherwise the primitive is clipped against the clip volume as defined below.

The clip volume is the intersection of up to `VkPhysicalDeviceLimits::maxClipDistances` application-defined half-spaces with the view volume (if no application-defined clip half-spaces are enabled, the clip volume is the view volume).

A shader **must** write a single clip distance for each enabled clip half-space to elements of the `ClipDistance` array. Clip half-space i is then given by the set of points satisfying the inequality

$$c_i(\mathbf{P}) \geq 0$$

where $c_i(\mathbf{P})$ is the clip distance i at point $\mathbf{P}$. For point primitives, $c_i(\mathbf{P})$ is simply the clip distance for the vertex in question. For line and triangle primitives, per-vertex clip distances are interpolated using a weighted mean, with weights derived according to the algorithms described in sections [Basic Line Segment Rasterization](#) and [Basic Polygon Rasterization](#), using the perspective interpolation equations.

The number of application-defined clip and cull half-spaces that are enabled is determined by the explicit size of the built-in arrays `ClipDistance` and `CullDistance`, respectively, declared as an output in the interface of the entry point of the final shader stage before clipping.

If `VkPipelineRasterizationDepthClipStateCreateInfoEXT` is present in the graphics pipeline state then depth clipping is disabled if `VkPipelineRasterizationDepthClipStateCreateInfoEXT::depthClipEnable` is `VK_FALSE`. Otherwise, if `VkPipelineRasterizationDepthClipStateCreateInfoEXT` is not present, depth clipping is disabled when `VkPipelineRasterizationStateCreateInfo::depthClampEnable` is `VK_TRUE`.

To [dynamically set](#) enable or disable depth clamping, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetDepthClampEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                 depthClampEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthClampEnable` specifies whether depth clamping is enabled.

This command sets whether depth clamping is enabled or disabled for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationStateCreateInfo::depthClampEnable` value used to create the currently active pipeline.

If the depth clamping state is changed dynamically, and the pipeline was not created with `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT` enabled, then depth clipping is enabled when depth clamping is disabled and vice versa.

Valid Usage

- VUID-vkCmdSetDepthClampEnableEXT-None-09423
At least one of the following **must** be true:
 - The `extendedDynamicState3DepthClampEnable` feature is enabled
 - The `shaderObject` feature is enabled
- VUID-vkCmdSetDepthClampEnableEXT-depthClamp-07449
If the `depthClamp` feature is not enabled, `depthClampEnable` **must** be `VK_FALSE`

Valid Usage (Implicit)

- VUID-vkCmdSetDepthClampEnableEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetDepthClampEnableEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetDepthClampEnableEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetDepthClampEnableEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To [dynamically set](#) enable or disable depth clipping, call:

```
// Provided by VK_EXT_depth_clip_enable with VK_EXT_extended_dynamic_state3,  
VK_EXT_depth_clip_enable with VK_EXT_shader_object  
void vkCmdSetDepthClipEnableEXT(  
    VkCommandBuffer          commandBuffer,  
    VkBool32                 depthClipEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthClipEnable` specifies whether depth clipping is enabled.

This command sets whether depth clipping is enabled or disabled for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationDepthClipStateCreateInfoEXT::depthClipEnable` value used to create the currently active pipeline, or by the inverse of

`VkPipelineRasterizationStateCreateInfo::depthClampEnable` if
`VkPipelineRasterizationDepthClipStateCreateInfoEXT` is not specified.

Valid Usage

- VUID-vkCmdSetDepthClipEnableEXT-None-09423
At least one of the following **must** be true:
 - The `extendedDynamicState3DepthClipEnable` feature is enabled
 - The `shaderObject` feature is enabled
- VUID-vkCmdSetDepthClipEnableEXT-depthClipEnable-07451
The `depthClipEnable` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdSetDepthClipEnableEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetDepthClipEnableEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetDepthClipEnableEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetDepthClipEnableEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

When depth clipping is disabled, the plane equation

$$Z_m \leq Z_c \leq W_c$$

(see the clip volume definition above) is ignored by view volume clipping (effectively, there is no near or far plane clipping).

If the primitive under consideration is a point or line segment, then clipping passes it unchanged if its vertices lie entirely within the clip volume.

Possible values of [VkPhysicalDevicePointClippingProperties::pointClippingBehavior](#), specifying clipping behavior of a point primitive whose vertex lies outside the clip volume, are:

```
typedef enum VkPointClippingBehavior {
    VK_POINT_CLIPPING_BEHAVIOR_ALL_CLIP_PLANES = 0,
    VK_POINT_CLIPPING_BEHAVIOR_USER_CLIP_PLANES_ONLY = 1,
    // Provided by VK_KHR_maintenance2
    VK_POINT_CLIPPING_BEHAVIOR_ALL_CLIP_PLANES_KHR =
VK_POINT_CLIPPING_BEHAVIOR_ALL_CLIP_PLANES,
    // Provided by VK_KHR_maintenance2
    VK_POINT_CLIPPING_BEHAVIOR_USER_CLIP_PLANES_ONLY_KHR =
VK_POINT_CLIPPING_BEHAVIOR_USER_CLIP_PLANES_ONLY,
} VkPointClippingBehavior;
```

or the equivalent

```
// Provided by VK_KHR_maintenance2
typedef VkPointClippingBehavior VkPointClippingBehaviorKHR;
```

- [VK_POINT_CLIPPING_BEHAVIOR_ALL_CLIP_PLANES](#) specifies that the primitive is discarded if the vertex lies outside any clip plane, including the planes bounding the view volume.
- [VK_POINT_CLIPPING_BEHAVIOR_USER_CLIP_PLANES_ONLY](#) specifies that the primitive is discarded only if the vertex lies outside any user clip plane.

If either of a line segment's vertices lie outside of the clip volume, the line segment **may** be clipped, with new vertex coordinates computed for each vertex that lies outside the clip volume. A clipped line segment endpoint lies on both the original line segment and the boundary of the clip volume.

This clipping produces a value, $0 \leq t \leq 1$, for each clipped vertex. If the coordinates of a clipped vertex are **P** and the unclipped line segment's vertex coordinates are **P**₁ and **P**₂, then t satisfies the following equation

$$\mathbf{P} = t \mathbf{P}_1 + (1-t) \mathbf{P}_2.$$

t is used to clip vertex output attributes as described in [Clipping Shader Outputs](#).

If the primitive is a polygon, it passes unchanged if every one of its edges lies entirely inside the clip volume, and is either clipped or discarded otherwise. If the edges of the polygon intersect the

boundary of the clip volume, the intersecting edges are reconnected by new edges that lie along the boundary of the clip volume - in some cases requiring the introduction of new vertices into a polygon.

If a polygon intersects an edge of the clip volume's boundary, the clipped polygon **must** include a point on this boundary edge.

Primitives rendered with application-defined half-spaces **must** satisfy a complementarity criterion. Suppose a series of primitives is drawn where each vertex i has a single specified clip distance d_i (or a number of similarly specified clip distances, if multiple half-spaces are enabled). Next, suppose that the same series of primitives are drawn again with each such clip distance replaced by $-d_i$ (and the graphics pipeline is otherwise the same). In this case, primitives **must** not be missing any pixels, and pixels **must** not be drawn twice in regions where those primitives are cut by the clip planes.

The `VkPipelineViewportDepthClipControlCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_depth_clip_control
typedef struct VkPipelineViewportDepthClipControlCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           negativeOneToOne;
} VkPipelineViewportDepthClipControlCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `negativeOneToOne` sets the z_m in the *view volume* to $-w_c$

Valid Usage

- VUID-VkPipelineViewportDepthClipControlCreateInfoEXT-negativeOneToOne-06470
If `depthClipControl` is not enabled, `negativeOneToOne` **must** be `VK_FALSE`

Valid Usage (Implicit)

- VUID-VkPipelineViewportDepthClipControlCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLIP_CONTROL_CREATE_INFO_EXT`

To [dynamically set](#) `negativeOneToOne`, call:

```
// Provided by VK_EXT_depth_clip_control with VK_EXT_extended_dynamic_state3,
VK_EXT_depth_clip_control with VK_EXT_shader_object
void vkCmdSetDepthClipNegativeOneToOneEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                 negativeOneToOne);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `negativeOneToOne` specifies the `negativeOneToOne` state.

This command sets the `negativeOneToOne` state for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineViewportDepthClipControlCreateInfoEXT::negativeOneToOne` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-None-09423
At least one of the following **must** be true:
 - The `extendedDynamicState3DepthClipNegativeOneToOne` feature is enabled
 - The `shaderObject` feature is enabled
- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-depthClipControl-07453
The `depthClipControl` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetDepthClipNegativeOneToOneEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

24.4. Clipping Shader Outputs

Next, vertex output attributes are clipped. The output values associated with a vertex that lies within the clip volume are unaffected by clipping. If a primitive is clipped, however, the output values assigned to vertices produced by clipping are clipped.

Let the output values assigned to the two vertices $\mathbf{P}_1$ and $\mathbf{P}_2$ of an unclipped edge be $\mathbf{c}_1$ and $\mathbf{c}_2$. The value of t (see [Primitive Clipping](#)) for a clipped point $\mathbf{P}$ is used to obtain the output value associated with $\mathbf{P}$ as

$$\mathbf{c} = t \mathbf{c}_1 + (1-t) \mathbf{c}_2.$$

(Multiplying an output value by a scalar means multiplying each of x , y , z , and w by the scalar.)

Since this computation is performed in clip space before division by w_c , clipped output values are perspective-correct.

Polygon clipping creates a clipped vertex along an edge of the clip volume's boundary. This situation is handled by noting that polygon clipping proceeds by clipping against one half-space at a time. Output value clipping is done in the same way, so that clipped points always occur at the intersection of polygon edges (possibly already clipped) with the clip volume's boundary.

For vertex output attributes whose matching fragment input attributes are decorated with **NoPerspective**, the value of t used to obtain the output value associated with $\mathbf{P}$ will be adjusted to produce results that vary linearly in framebuffer space.

Output attributes of integer or unsigned integer type **must** always be flat shaded. Flat shaded attributes are constant over the primitive being rasterized (see [Basic Line Segment Rasterization](#) and [Basic Polygon Rasterization](#)), and no interpolation is performed. The output value $\mathbf{c}$ is taken from either $\mathbf{c}_1$ or $\mathbf{c}_2$, since flat shading has already occurred and the two values are identical.

24.5. Coordinate Transformations

Clip coordinates for a vertex result from shader execution, which yields a vertex coordinate **Position**.

Perspective division on clip coordinates yields *normalized device coordinates*, followed by a *viewport* transformation (see [Controlling the Viewport](#)) to convert these coordinates into *framebuffer coordinates*.

If a vertex in clip coordinates has a position given by

$$\begin{pmatrix} x_c \\ y_c \\ z_c \\ w_c \end{pmatrix}$$

then the vertex's normalized device coordinates are

$$\begin{pmatrix} x_d \\ y_d \\ z_d \end{pmatrix} = \begin{pmatrix} \frac{x_c}{w_c} \\ \frac{y_c}{w_c} \\ \frac{z_c}{w_c} \end{pmatrix}$$

24.6. Controlling the Viewport

The viewport transformation is determined by the selected viewport's width and height in pixels, p_x and p_y , respectively, and its center (o_x, o_y) (also in pixels), as well as its depth range min and max determining a depth range scale value p_z and a depth range bias value o_z (defined below). The vertex's framebuffer coordinates (x_f, y_f, z_f) are given by

$$x_f = (p_x / 2) x_d + o_x$$

$$y_f = (p_y / 2) y_d + o_y$$

$$z_f = p_z \times z_d + o_z$$

Multiple viewports are available, numbered zero up to `VkPhysicalDeviceLimits::maxViewports` minus one. The number of viewports used by a pipeline is controlled by the `viewportCount` member of the `VkPipelineViewportStateCreateInfo` structure used in pipeline creation.

x_f and y_f have limited precision, where the number of fractional bits retained is specified by `VkPhysicalDeviceLimits::subPixelPrecisionBits`. When rasterizing [line segments](#), the number of fractional bits is specified by `VkPhysicalDeviceLineRasterizationPropertiesKHR::lineSubPixelPrecisionBits`.

The `VkPipelineViewportStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineViewportStateCreateInfo {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineViewportStateCreateFlags flags;
    uint32_t                 viewportCount;
    const VkViewport*        pViewports;
    uint32_t                 scissorCount;
    const VkRect2D*          pScissors;
} VkPipelineViewportStateCreateInfo;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is reserved for future use.
- **viewportCount** is the number of viewports used by the pipeline.
- **pViewports** is a pointer to an array of **VkViewport** structures, defining the viewport transforms. If the viewport state is dynamic, this member is ignored.
- **scissorCount** is the number of **scissors** and **must** match the number of viewports.
- **pScissors** is a pointer to an array of **VkRect2D** structures defining the rectangular bounds of the scissor for the corresponding viewport. If the scissor state is dynamic, this member is ignored.

Valid Usage

- VUID-VkPipelineViewportStateCreateInfo-viewportCount-01216
If the **multiViewport** feature is not enabled, **viewportCount** **must** not be greater than **1**
- VUID-VkPipelineViewportStateCreateInfo-scissorCount-01217
If the **multiViewport** feature is not enabled, **scissorCount** **must** not be greater than **1**
- VUID-VkPipelineViewportStateCreateInfo-viewportCount-01218
viewportCount **must** be less than or equal to **VkPhysicalDeviceLimits::maxViewports**
- VUID-VkPipelineViewportStateCreateInfo-scissorCount-01219
scissorCount **must** be less than or equal to **VkPhysicalDeviceLimits::maxViewports**
- VUID-VkPipelineViewportStateCreateInfo-x-02821
The **x** and **y** members of **offset** member of any element of **pScissors** **must** be greater than or equal to **0**
- VUID-VkPipelineViewportStateCreateInfo-offset-02822
Evaluation of (**offset.x** + **extent.width**) **must** not cause a signed integer addition overflow for any element of **pScissors**
- VUID-VkPipelineViewportStateCreateInfo-offset-02823
Evaluation of (**offset.y** + **extent.height**) **must** not cause a signed integer addition overflow for any element of **pScissors**
- VUID-VkPipelineViewportStateCreateInfo-scissorCount-04134
If **scissorCount** and **viewportCount** are both not dynamic, then **scissorCount** and

`viewportCount` **must** be identical

- VUID-VkPipelineViewportStateCreateInfo-viewportCount-04135
`viewportCount` **must** be greater than 0
- VUID-VkPipelineViewportStateCreateInfo-scissorCount-04136
`scissorCount` **must** be greater than 0

Valid Usage (Implicit)

- VUID-VkPipelineViewportStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_STATE_CREATE_INFO`
- VUID-VkPipelineViewportStateCreateInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkPipelineViewportDepthClampControlCreateInfoEXT` or `VkPipelineViewportDepthClipControlCreateInfoEXT`
- VUID-VkPipelineViewportStateCreateInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkPipelineViewportStateCreateInfo-flags-zero-bitmask
`flags` **must** be 0

To dynamically set the viewport count and viewports, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetViewportWithCountEXT(
    VkCommandBuffer          commandBuffer,
    uint32_t                 viewportCount,
    const VkViewport*        pViewports);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `viewportCount` specifies the viewport count.
- `pViewports` specifies the viewports to use for drawing.

This command sets the viewport count and viewports state for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the corresponding `VkPipelineViewportStateCreateInfo::viewportCount` and `pViewports` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetViewportWithCount-None-08971
At least one of the following **must** be true:
 - the `shaderObject` feature is enabled

- VUID-vkCmdSetViewportWithCount-viewportCount-03394
viewportCount **must** be between 1 and VkPhysicalDeviceLimits::maxViewports, inclusive
- VUID-vkCmdSetViewportWithCount-viewportCount-03395
If the multiViewport feature is not enabled, viewportCount **must** be 1

Valid Usage (Implicit)

- VUID-vkCmdSetViewportWithCount-commandBuffer-parameter
commandBuffer **must** be a valid VkCommandBuffer handle
- VUID-vkCmdSetViewportWithCount-pViewports-parameter
pViewports **must** be a valid pointer to an array of viewportCount valid VkViewport structures
- VUID-vkCmdSetViewportWithCount-commandBuffer-recording
commandBuffer **must** be in the recording state
- VUID-vkCmdSetViewportWithCount-commandBuffer-cmdpool
The VkCommandPool that commandBuffer was allocated from **must** support graphics operations
- VUID-vkCmdSetViewportWithCount-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetViewportWithCount-viewportCount-arraylength
viewportCount **must** be greater than 0

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the VkCommandPool that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To dynamically set the scissor count and scissor rectangular bounds, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetScissorWithCountEXT(
    VkCommandBuffer                      commandBuffer,
    uint32_t                             scissorCount,
    const VkRect2D*                      pScissors);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `scissorCount` specifies the scissor count.
- `pScissors` specifies the scissors to use for drawing.

This command sets the scissor count and scissor rectangular bounds state for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the corresponding `VkPipelineViewportStateCreateInfo::scissorCount` and `pScissors` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetScissorWithCount-None-08971
At least one of the following **must** be true:
 - the `shaderObject` feature is enabled
- VUID-vkCmdSetScissorWithCount-scissorCount-03397
`scissorCount` **must** be between 1 and `VkPhysicalDeviceLimits::maxViewports`, inclusive
- VUID-vkCmdSetScissorWithCount-scissorCount-03398
If the `multiViewport` feature is not enabled, `scissorCount` **must** be 1
- VUID-vkCmdSetScissorWithCount-x-03399
The `x` and `y` members of `offset` member of any element of `pScissors` **must** be greater than or equal to 0
- VUID-vkCmdSetScissorWithCount-offset-03400
Evaluation of `(offset.x + extent.width)` **must** not cause a signed integer addition overflow for any element of `pScissors`
- VUID-vkCmdSetScissorWithCount-offset-03401
Evaluation of `(offset.y + extent.height)` **must** not cause a signed integer addition overflow for any element of `pScissors`

Valid Usage (Implicit)

- VUID-vkCmdSetScissorWithCount-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetScissorWithCount-pScissors-parameter
`pScissors` **must** be a valid pointer to an array of `scissorCount` `VkRect2D` structures
- VUID-vkCmdSetScissorWithCount-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdSetScissorWithCount-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetScissorWithCount-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetScissorWithCount-scissorCount-arraylength
`scissorCount` **must** be greater than 0

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineViewportStateCreateFlags;
```

`VkPipelineViewportStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

If a geometry shader is active and has an output variable decorated with `ViewportIndex`, the viewport transformation uses the viewport corresponding to the value assigned to `ViewportIndex` taken from an implementation-dependent vertex of each primitive. If `ViewportIndex` is outside the range zero to `viewportCount` minus one for a primitive, or if the geometry shader did not assign a value to `ViewportIndex` for all vertices of a primitive due to flow control, the values resulting from the viewport transformation of the vertices of such primitives are undefined. If no geometry shader is active, or if the geometry shader does not have an output decorated with `ViewportIndex`, the viewport numbered zero is used by the viewport transformation.

A single vertex **can** be used in more than one individual primitive, in primitives such as `VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP`. In this case, the viewport transformation is applied separately for each primitive.

To **dynamically set** the viewport transformation parameters, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetViewport(
    VkCommandBuffer                commandBuffer,
    uint32_t                       firstViewport,
    uint32_t                       viewportCount,
    const VkViewport*              pViewports);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstViewport` is the index of the first viewport whose parameters are updated by the command.
- `viewportCount` is the number of viewports whose parameters are updated by the command.
- `pViewports` is a pointer to an array of `VkViewport` structures specifying viewport parameters.

This command sets the viewport transformation parameters state for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_VIEWPORT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineViewportStateCreateInfo::pViewports` values used to create the currently active pipeline.

The viewport parameters taken from element `i` of `pViewports` replace the current state for the viewport index `firstViewport + i`, for `i` in `[0, viewportCount)`.

Valid Usage

- VUID-vkCmdSetViewport-firstViewport-01223
The sum of `firstViewport` and `viewportCount` **must** be between `1` and `VkPhysicalDeviceLimits::maxViewports`, inclusive
- VUID-vkCmdSetViewport-firstViewport-01224
If the `multiViewport` feature is not enabled, `firstViewport` **must** be `0`
- VUID-vkCmdSetViewport-viewportCount-01225
If the `multiViewport` feature is not enabled, `viewportCount` **must** be `1`

Valid Usage (Implicit)

- VUID-vkCmdSetViewport-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetViewport-pViewports-parameter
`pViewports` **must** be a valid pointer to an array of `viewportCount` valid `VkViewport` structures
- VUID-vkCmdSetViewport-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetViewport-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdSetViewport-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetViewport-viewportCount-arraylength
`viewportCount` **must** be greater than 0

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Both `VkPipelineViewportStateCreateInfo` and `vkCmdSetViewport` use `VkViewport` to set the viewport transformation parameters.

The `VkViewport` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkViewport {
    float    x;
    float    y;
    float    width;
    float    height;
    float    minDepth;
    float    maxDepth;
} VkViewport;
```

- `x` and `y` are the viewport's upper left corner (x,y).
- `width` and `height` are the viewport's width and height, respectively.
- `minDepth` and `maxDepth` are the depth range for the viewport.

NOTE | Despite their names, `minDepth` **can** be less than, equal to, or greater than `maxDepth`.

The framebuffer depth coordinate `zf` **may** be represented using either a fixed-point or floating-point representation. However, a floating-point representation **must** be used if the depth/stencil attachment has a floating-point depth component. If an m-bit fixed-point representation is used, we

assume that it represents each value $\frac{k}{2^m-1}$, where $k \in \{0, 1, \dots, 2^m-1\}$, as k (e.g. 1.0 is represented in binary as a string of all ones).

The viewport parameters shown in the above equations are found from these values as

$$o_x = x + \text{width} / 2$$

$$o_y = y + \text{height} / 2$$

$$o_z = \text{minDepth} \text{ (or } (\text{maxDepth} + \text{minDepth}) / 2 \text{ if } \text{VkPipelineViewportDepthClipControlCreateInfoEXT}::\text{negativeOneToOne} \text{ is } \text{VK_TRUE})$$

$$p_x = \text{width}$$

$$p_y = \text{height}$$

$$p_z = \text{maxDepth} - \text{minDepth} \text{ (or } (\text{maxDepth} - \text{minDepth}) / 2 \text{ if } \text{VkPipelineViewportDepthClipControlCreateInfoEXT}::\text{negativeOneToOne} \text{ is } \text{VK_TRUE})$$

The application **can** specify a negative term for **height**, which has the effect of negating the y coordinate in clip space before performing the transform. When using a negative **height**, the application **should** also adjust the y value to point to the lower left corner of the viewport instead of the upper left corner. Using the negative **height** allows the application to avoid having to negate the y component of the **Position** output from the last **pre-rasterization shader stage**.

The width and height of the **implementation-dependent maximum viewport dimensions** **must** be greater than or equal to the width and height of the largest image which **can** be created and attached to a framebuffer.

The floating-point viewport bounds are represented with an **implementation-dependent precision**.

Valid Usage

- VUID-VkViewport-width-01770
width must be greater than 0.0
- VUID-VkViewport-width-01771
width must be less than or equal to `VkPhysicalDeviceLimits::maxViewportDimensions[0]`
- VUID-VkViewport-apiVersion-07917
If the `VK_KHR_maintenance1` extension is not enabled, the `VK_AMD_negative_viewport_height` extension is not enabled, and `VkPhysicalDeviceProperties::apiVersion` is less than Vulkan 1.1, **height must** be greater than 0.0

- VUID-VkViewport-height-01773
The absolute value of **height** **must** be less than or equal to `VkPhysicalDeviceLimits::maxViewportDimensions[1]`
- VUID-VkViewport-x-01774
x **must** be greater than or equal to `viewportBoundsRange[0]`
- VUID-VkViewport-x-01232
(**x** + **width**) **must** be less than or equal to `viewportBoundsRange[1]`
- VUID-VkViewport-y-01775
y **must** be greater than or equal to `viewportBoundsRange[0]`
- VUID-VkViewport-y-01776
y **must** be less than or equal to `viewportBoundsRange[1]`
- VUID-VkViewport-y-01777
(**y** + **height**) **must** be greater than or equal to `viewportBoundsRange[0]`
- VUID-VkViewport-y-01233
(**y** + **height**) **must** be less than or equal to `viewportBoundsRange[1]`
- VUID-VkViewport-minDepth-01234
If the `VK_EXT_depth_range_unrestricted` extension is not enabled, **minDepth** **must** be between `0.0` and `1.0`, inclusive
- VUID-VkViewport-maxDepth-01235
If the `VK_EXT_depth_range_unrestricted` extension is not enabled, **maxDepth** **must** be between `0.0` and `1.0`, inclusive

Chapter 25. Rasterization

Rasterization is the process by which a primitive is converted to a two-dimensional image. Each discrete location of this image contains associated data such as depth, color, or other attributes.

Rasterizing a primitive begins by determining which squares of an integer grid in framebuffer coordinates are occupied by the primitive, and assigning one or more depth values to each such square. This process is described below for points, lines, and polygons.

A grid square, including its (x,y) framebuffer coordinates, z (depth), and associated data added by fragment shaders, is called a fragment. A fragment is located by its upper left corner, which lies on integer grid coordinates.

Rasterization operations also refer to a fragment's sample locations, which are offset by fractional values from its upper left corner. The rasterization rules for points, lines, and triangles involve testing whether each sample location is inside the primitive. Fragments need not actually be square, and rasterization rules are not affected by the aspect ratio of fragments. Display of non-square grids, however, will cause rasterized points and line segments to appear fatter in one direction than the other.

We assume that fragments are square, since it simplifies antialiasing and texturing. After rasterization, fragments are processed by [fragment operations](#).

Several factors affect rasterization, including the members of [VkPipelineRasterizationStateCreateInfo](#) and [VkPipelineMultisampleStateCreateInfo](#).

The [VkPipelineRasterizationStateCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineRasterizationStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineRasterizationStateCreateFlags flags;
    VkBool32                  depthClampEnable;
    VkBool32                  rasterizerDiscardEnable;
    VkPolygonMode              polygonMode;
    VkCullModeFlags            cullMode;
    VkFrontFace                 frontFace;
    VkBool32                   depthBiasEnable;
    float                      depthBiasConstantFactor;
    float                      depthBiasClamp;
    float                      depthBiasSlopeFactor;
    float                      lineWidth;
} VkPipelineRasterizationStateCreateInfo;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

- **depthClampEnable** controls whether to clamp the fragment's depth values as described in [Depth Test](#). If the pipeline is not created with [VkPipelineRasterizationDepthClipStateCreateInfoEXT](#) present then enabling depth clamp will also disable clipping primitives to the z planes of the frustum as described in [Primitive Clipping](#). Otherwise depth clipping is controlled by the state set in [VkPipelineRasterizationDepthClipStateCreateInfoEXT](#).
- **rasterizerDiscardEnable** controls whether primitives are discarded immediately before the rasterization stage.
- **polygonMode** is the triangle rendering mode. See [VkPolygonMode](#).
- **cullMode** is the triangle facing direction used for primitive culling. See [VkCullModeFlagBits](#).
- **frontFace** is a [VkFrontFace](#) value specifying the front-facing triangle orientation to be used for culling.
- **depthBiasEnable** controls whether to bias fragment depth values.
- **depthBiasConstantFactor** is a scalar factor controlling the constant depth value added to each fragment.
- **depthBiasClamp** is the maximum (or minimum) depth bias of a fragment.
- **depthBiasSlopeFactor** is a scalar factor applied to a fragment's slope in depth bias calculations.
- **lineWidth** is the width of rasterized line segments.

Valid Usage

- VUID-VkPipelineRasterizationStateCreateInfo-depthClampEnable-00782
If the **depthClamp** feature is not enabled, **depthClampEnable** **must** be **VK_FALSE**
- VUID-VkPipelineRasterizationStateCreateInfo-polygonMode-01507
If the **fillModeNonSolid** feature is not enabled, **polygonMode** **must** be **VK_POLYGON_MODE_FILL**
- VUID-VkPipelineRasterizationStateCreateInfo-pointPolygons-04458
If the [VK_KHR_portability_subset](#) extension is enabled, and [VkPhysicalDevicePortabilitySubsetFeaturesKHR::pointPolygons](#) is **VK_FALSE**, and **rasterizerDiscardEnable** is **VK_FALSE**, **polygonMode** **must** not be **VK_POLYGON_MODE_POINT**

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationStateCreateInfo-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_CREATE_INFO**
- VUID-VkPipelineRasterizationStateCreateInfo-pNext-pNext
Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either **NULL** or a pointer to a valid instance of [VkDepthBiasRepresentationInfoEXT](#), [VkPipelineRasterizationConservativeStateCreateInfoEXT](#), [VkPipelineRasterizationDepthClipStateCreateInfoEXT](#), [VkPipelineRasterizationLineStateCreateInfoKHR](#), [VkPipelineRasterizationProvokingVertexStateCreateInfoEXT](#), or [VkPipelineRasterizationStateStreamCreateInfoEXT](#)

- VUID-VkPipelineRasterizationStateCreateInfo-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkPipelineRasterizationStateCreateInfo-flags-zero-bitmask
flags **must** be 0
- VUID-VkPipelineRasterizationStateCreateInfo-polygonMode-parameter
polygonMode **must** be a valid **VkPolygonMode** value
- VUID-VkPipelineRasterizationStateCreateInfo-cullMode-parameter
cullMode **must** be a valid combination of **VkCullModeFlagBits** values
- VUID-VkPipelineRasterizationStateCreateInfo-frontFace-parameter
frontFace **must** be a valid **VkFrontFace** value

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineRasterizationStateCreateFlags;
```

VkPipelineRasterizationStateCreateFlags is a bitmask type for setting a mask, but is currently reserved for future use.

If the **pNext** chain of **VkPipelineRasterizationStateCreateInfo** includes a **VkPipelineRasterizationDepthClipStateCreateInfoEXT** structure, then that structure controls whether depth clipping is enabled or disabled.

The **VkPipelineRasterizationDepthClipStateCreateInfoEXT** structure is defined as:

```
// Provided by VK_EXT_depth_clip_enable
typedef struct VkPipelineRasterizationDepthClipStateCreateInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineRasterizationDepthClipStateCreateFlagsEXT flags;
    VkBool32                 depthClipEnable;
} VkPipelineRasterizationDepthClipStateCreateInfoEXT;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is reserved for future use.
- **depthClipEnable** controls whether depth clipping is enabled as described in **Primitive Clipping**.

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationDepthClipStateCreateInfoEXT-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_DEPTH_CLIP_STATE_CREATE_INFO_EXT**
- VUID-VkPipelineRasterizationDepthClipStateCreateInfoEXT-flags-zero-bitmask
flags **must** be 0

```
// Provided by VK_EXT_depth_clip_enable
typedef VkFlags VkPipelineRasterizationDepthClipStateCreateFlagsEXT;
```

`VkPipelineRasterizationDepthClipStateCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkPipelineMultisampleStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineMultisampleStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineMultisampleStateCreateFlags flags;
    VkSampleCountFlagBits    rasterizationSamples;
    VkBool32                 sampleShadingEnable;
    float                    minSampleShading;
    const VkSampleMask*      pSampleMask;
    VkBool32                 alphaToCoverageEnable;
    VkBool32                 alphaToOneEnable;
} VkPipelineMultisampleStateCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `rasterizationSamples` is a `VkSampleCountFlagBits` value specifying the number of samples used in rasterization. This value is ignored for the purposes of setting the number of samples used in rasterization if the pipeline is created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` dynamic state set, but if `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` dynamic state is not set, it is still used to define the size of the `pSampleMask` array as described below.
- `sampleShadingEnable` can be used to enable [Sample Shading](#).
- `minSampleShading` specifies a minimum fraction of sample shading if `sampleShadingEnable` is `VK_TRUE`.
- `pSampleMask` is a pointer to an array of `VkSampleMask` values used in the [sample mask test](#).
- `alphaToCoverageEnable` controls whether a temporary coverage value is generated based on the alpha component of the fragment's first color output as specified in the [Multisample Coverage](#) section.
- `alphaToOneEnable` controls whether the alpha component of the fragment's first color output is replaced with one as described in [Multisample Coverage](#).

Each bit in the sample mask is associated with a unique [sample index](#) as defined for the [coverage mask](#). Each bit `b` for mask word `w` in the sample mask corresponds to sample index `i`, where $i = 32 \times w + b$. `pSampleMask` has a length equal to $\lceil \text{rasterizationSamples} / 32 \rceil$ words.

If `pSampleMask` is `NULL`, it is treated as if the mask has all bits set to 1.

Valid Usage

- VUID-VkPipelineMultisampleStateCreateInfo-sampleShadingEnable-00784
If the `sampleRateShading` feature is not enabled, `sampleShadingEnable` **must** be `VK_FALSE`
- VUID-VkPipelineMultisampleStateCreateInfo-alphaToOneEnable-00785
If the `alphaToOne` feature is not enabled, `alphaToOneEnable` **must** be `VK_FALSE`
- VUID-VkPipelineMultisampleStateCreateInfo-minSampleShading-00786
`minSampleShading` **must** be in the range [0,1]

Valid Usage (Implicit)

- VUID-VkPipelineMultisampleStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_MULTISAMPLE_STATE_CREATE_INFO`
- VUID-VkPipelineMultisampleStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkPipelineSampleLocationsStateCreateInfoEXT`
- VUID-VkPipelineMultisampleStateCreateInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkPipelineMultisampleStateCreateInfo-flags-zerobitmask
`flags` **must** be 0
- VUID-VkPipelineMultisampleStateCreateInfo-rasterizationSamples-parameter
`rasterizationSamples` **must** be a valid `VkSampleCountFlagBits` value
- VUID-VkPipelineMultisampleStateCreateInfo-pSampleMask-parameter
If `pSampleMask` is not `NULL`, `pSampleMask` **must** be a valid pointer to an array of $\lceil \frac{\text{rasterizationSamples}}{32} \rceil$ `VkSampleMask` values

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineMultisampleStateCreateFlags;
```

`VkPipelineMultisampleStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The elements of the sample mask array are of type `VkSampleMask`, each representing 32 bits of coverage information:

```
// Provided by VK_VERSION_1_0
typedef uint32_t VkSampleMask;
```

Rasterization only generates fragments which cover one or more pixels inside the framebuffer. Pixels outside the framebuffer are never considered covered in the fragment. Fragments which would be produced by application of any of the primitive rasterization rules described below but

which lie outside the framebuffer are not produced, nor are they processed by any later stage of the pipeline, including any of the [fragment operations](#).

Surviving fragments are processed by fragment shaders. Fragment shaders determine associated data for fragments, and **can** also modify or replace their assigned depth values.

25.1. Discarding Primitives Before Rasterization

Primitives are discarded before rasterization if the `rasterizerDiscardEnable` member of `VkPipelineRasterizationStateCreateInfo` is enabled. When enabled, primitives are discarded after they are processed by the last active shader stage in the pipeline before rasterization.

To [dynamically enable](#) whether primitives are discarded before the rasterization stage, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetRasterizerDiscardEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                 rasterizerDiscardEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `rasterizerDiscardEnable` controls whether primitives are discarded immediately before the rasterization stage.

This command sets the discard enable for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the `VkPipelineRasterizationStateCreateInfo::rasterizerDiscardEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetRasterizerDiscardEnable-None-08970
At least one of the following **must** be true:
 - the `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetRasterizerDiscardEnable-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetRasterizerDiscardEnable-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdSetRasterizerDiscardEnable-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetRasterizerDiscardEnable-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

25.2. Controlling the Vertex Stream Used for Rasterization

By default vertex data output from the last `pre-rasterization shader stage` are directed to vertex stream zero. Geometry shaders **can** emit primitives to multiple independent vertex streams. Each vertex emitted by the geometry shader is directed at one of the vertex streams. As vertices are received on each vertex stream, they are arranged into primitives of the type specified by the geometry shader output primitive type. The shading language instructions `OpEndPrimitive` and `OpEndStreamPrimitive` **can** be used to end the primitive being assembled on a given vertex stream and start a new empty primitive of the same type. An implementation supports up to `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams` streams, which is at least 1. The individual streams are numbered 0 through `maxTransformFeedbackStreams` minus 1. There is no requirement on the order of the streams to which vertices are emitted, and the number of vertices emitted to each vertex stream **can** be completely independent, subject only to the `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreamDataSize` and `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBufferDataSize` limits. The primitives output from all vertex streams are passed to the transform feedback stage to be captured to transform feedback buffers in the manner specified by the last `pre-rasterization shader stage` shader's `XfbBuffer`, `XfbStride`, and `Offsets` decorations on the output interface variables in the graphics pipeline. To use a vertex stream other than zero, or to use multiple streams, the `GeometryStreams` capability **must** be specified.

By default, the primitives output from vertex stream zero are rasterized. If the implementation supports the `VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackRasterizationStreamSelect` property it is possible to rasterize a vertex stream other than zero.

By default, geometry shaders that emit vertices to multiple vertex streams are limited to using only the `OutputPoints` output primitive type. If the implementation supports the `VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackStreamsLinesTriangles` property it is possible to emit `OutputLineStrip` or `OutputTriangleStrip` in addition to `OutputPoints`.

The vertex stream used for rasterization is specified by adding a `VkPipelineRasterizationStateStreamCreateInfoEXT` structure to the `pNext` chain of a `VkPipelineRasterizationStateCreateInfo` structure.

The `VkPipelineRasterizationStateStreamCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_transform_feedback
typedef struct VkPipelineRasterizationStateStreamCreateInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineRasterizationStateStreamCreateFlagsEXT flags;
    uint32_t                 rasterizationStream;
} VkPipelineRasterizationStateStreamCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `rasterizationStream` is the vertex stream selected for rasterization.

If this structure is not present, `rasterizationStream` is assumed to be zero.

Valid Usage

- VUID-VkPipelineRasterizationStateStreamCreateInfoEXT-geometryStreams-02324 `VkPhysicalDeviceTransformFeedbackFeaturesEXT::geometryStreams` **must** be enabled
- VUID-VkPipelineRasterizationStateStreamCreateInfoEXT-rasterizationStream-02325 `rasterizationStream` **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams`
- VUID-VkPipelineRasterizationStateStreamCreateInfoEXT-rasterizationStream-02326 `rasterizationStream` **must** be zero if `VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackRasterizationStreamSelect` is `VK_FALSE`

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationStateStreamCreateInfoEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_STREAM_CREATE_INFO_EXT`
- VUID-VkPipelineRasterizationStateStreamCreateInfoEXT-flags-zero bitmask `flags` **must** be 0

```
// Provided by VK_EXT_transform_feedback
typedef VkFlags VkPipelineRasterizationStateStreamCreateFlagsEXT;
```

`VkPipelineRasterizationStateStreamCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

To **dynamically** set the `rasterizationStream` state, call:

```
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_transform_feedback,  
VK_EXT_shader_object with VK_EXT_transform_feedback  
void vkCmdSetRasterizationStreamEXT(  
    VkCommandBuffer                commandBuffer,  
    uint32_t                        rasterizationStream);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `rasterizationStream` specifies the `rasterizationStream` state.

This command sets the `rasterizationStream` state for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetRasterizationStreamEXT-None-09423
At least one of the following **must** be true:
 - The `extendedDynamicState3RasterizationStream` feature is enabled
 - The `shaderObject` feature is enabled
- VUID-vkCmdSetRasterizationStreamEXT-transformFeedback-07411
The `transformFeedback` feature **must** be enabled
- VUID-vkCmdSetRasterizationStreamEXT-rasterizationStream-07412
`rasterizationStream` **must** be less than `VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams`
- VUID-vkCmdSetRasterizationStreamEXT-rasterizationStream-07413
`rasterizationStream` **must** be zero if `VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackRasterizationStreamSelect` is `VK_FALSE`

Valid Usage (Implicit)

- VUID-vkCmdSetRasterizationStreamEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetRasterizationStreamEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetRasterizationStreamEXT-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- `VK_CMD_SET_RASTERIZATION_STREAM_EXT`-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

25.3. Rasterization Order

Within a subpass of a [render pass instance](#), for a given (x,y,layer,sample) sample location, the following operations are guaranteed to execute in *rasterization order*, for each separate primitive that includes that sample location:

1. [Fragment operations](#), in the order defined
2. [Blending](#), [logic operations](#), and color writes

Execution of these operations for each primitive in a subpass occurs in [primitive order](#).

25.4. Multisampling

Multisampling is a mechanism to antialias all Vulkan primitives: points, lines, and polygons. The technique is to sample all primitives multiple times at each pixel. Each sample in each framebuffer attachment has storage for a color, depth, and/or stencil value, such that per-fragment operations apply to each sample independently. The color sample values **can** be later *resolved* to a single color (see [Resolving Multisample Images](#) and the [Render Pass](#) chapter for more details on how to resolve multisample images to non-multisample images).

Vulkan defines rasterization rules for single-sample modes in a way that is equivalent to a multisample mode with a single sample in the center of each fragment.

Each fragment includes a [coverage mask](#) with a single bit for each sample in the fragment, and a number of depth values and associated data for each sample.

It is understood that each pixel has `rasterizationSamples` locations associated with it. These locations are exact positions, rather than regions or areas, and each is referred to as a sample point. The sample points associated with a pixel **must** be located inside or on the boundary of the unit square that is considered to bound the pixel. Furthermore, the relative locations of sample points **may** be identical for each pixel in the framebuffer, or they **may** differ.

If the current pipeline includes a fragment shader with one or more variables in its interface decorated with `Sample` and `Input`, the data associated with those variables will be assigned independently for each sample. The values for each sample **must** be evaluated at the location of the sample. The data associated with any other variables not decorated with `Sample` and `Input` need not be evaluated independently for each sample.

A *coverage mask* is generated for each fragment, based on which samples within that fragment are determined to be within the area of the primitive that generated the fragment.

Single pixel fragments have one set of samples. Multi-pixel fragments defined by setting the `fragment shading rate` have one set of samples per pixel. Each set of samples has a number of samples determined by `VkPipelineMultisampleStateCreateInfo::rasterizationSamples`. Each sample in a set is assigned a unique *sample index* `i` in the range `[0, rasterizationSamples)`.

To `dynamically set` the `rasterizationSamples`, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetRasterizationSamplesEXT(
    VkCommandBuffer                commandBuffer,
    VkSampleCountFlagBits          rasterizationSamples);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `rasterizationSamples` specifies `rasterizationSamples`.

This command sets the `rasterizationSamples` for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineMultisampleStateCreateInfo::rasterizationSamples` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetRasterizationSamplesEXT-None-09423

At least one of the following **must** be true:

- The `extendedDynamicState3RasterizationSamples` feature is enabled
- The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetRasterizationSamplesEXT-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdSetRasterizationSamplesEXT-rasterizationSamples-parameter `rasterizationSamples` **must** be a valid `VkSampleCountFlagBits` value
- VUID-vkCmdSetRasterizationSamplesEXT-commandBuffer-recording `commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetRasterizationSamplesEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetRasterizationSamplesEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Each sample in a fragment is also assigned a unique *coverage index* j in the range $[0, n \times \text{rasterizationSamples})$, where n is the number of sets in the fragment. If the fragment contains a single set of samples, the *coverage index* is always equal to the *sample index*.

If the `fragment shading rate` is set, the coverage index j is determined as a function of the *pixel index* p , the *sample index* i , and the number of rasterization samples r as:

$$j = i + r \times ((f_w \times f_h) - 1 - p)$$

where the pixel index p is determined as a function of the pixel's framebuffer location (x,y) and the fragment size (f_w, f_h) :

$$p_x = x \% f_w$$

$$p_y = y \% f_h$$

$$p = p_x + (p_y \times f_w)$$

The tables below illustrate the pixel index for multi-pixel fragments:

Table 29. Pixel Indices - 1 Wide

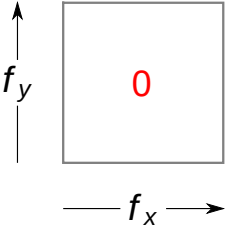
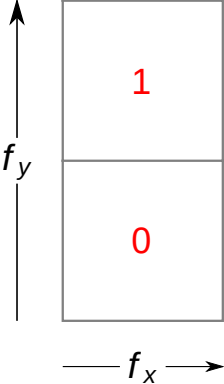
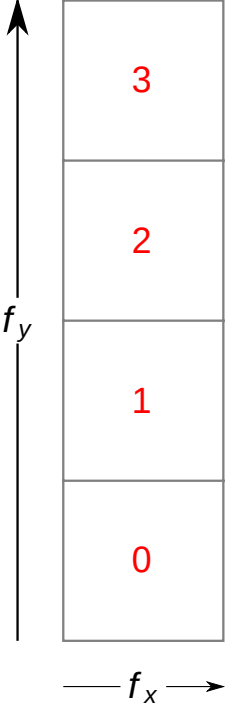
1x1	1x2	1x4
		

Table 30. Pixel Indices - 2 Wide

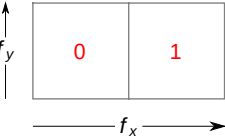
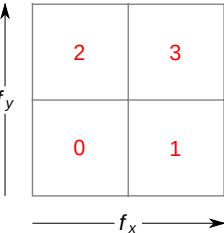
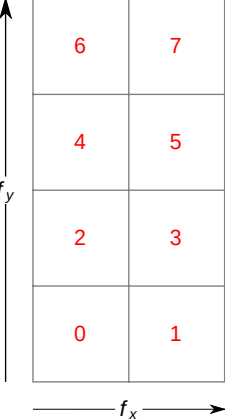
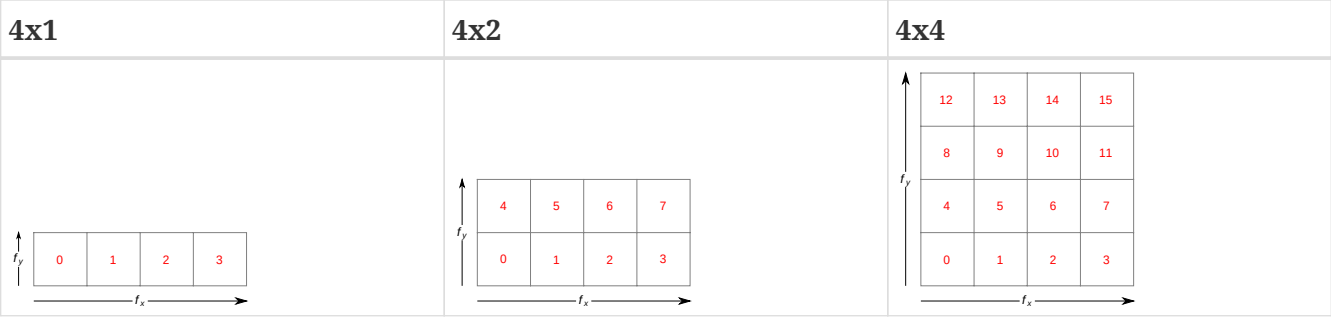
2x1	2x2	2x4
		

Table 31. Pixel Indices - 4 Wide



The coverage mask includes B bits packed into W words, defined as:

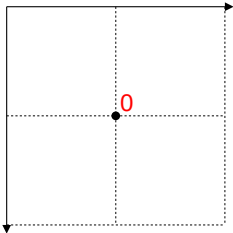
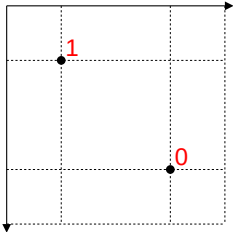
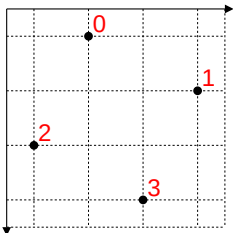
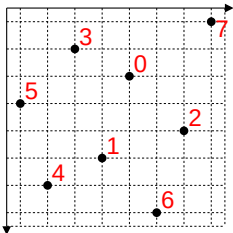
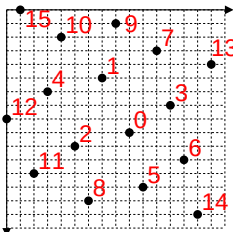
$B = n \times \text{rasterizationSamples}$

$W = \lceil B/32 \rceil$

Bit b in coverage mask word w is 1 if the sample with coverage index $j = 32 \times w + b$ is covered, and 0 otherwise.

If the `standardSampleLocations` member of `VkPhysicalDeviceLimits` is `VK_TRUE`, then the sample counts `VK_SAMPLE_COUNT_1_BIT`, `VK_SAMPLE_COUNT_2_BIT`, `VK_SAMPLE_COUNT_4_BIT`, `VK_SAMPLE_COUNT_8_BIT`, and `VK_SAMPLE_COUNT_16_BIT` have sample locations as listed in the following table, with the *i*th entry in the table corresponding to sample index *i*. `VK_SAMPLE_COUNT_32_BIT` and `VK_SAMPLE_COUNT_64_BIT` do not have standard sample locations. Locations are defined relative to an origin in the upper left corner of the fragment.

Table 32. Standard Sample Locations

Sample count	Sample Locations	
VK_SAMPLE_COUNT_1_BIT	(0.5,0.5)	
VK_SAMPLE_COUNT_2_BIT	(0.75,0.75) (0.25,0.25)	
VK_SAMPLE_COUNT_4_BIT	(0.375, 0.125) (0.875, 0.375) (0.125, 0.625) (0.625, 0.875)	
VK_SAMPLE_COUNT_8_BIT	(0.5625, 0.3125) (0.4375, 0.6875) (0.8125, 0.5625) (0.3125, 0.1875) (0.1875, 0.8125) (0.0625, 0.4375) (0.6875, 0.9375) (0.9375, 0.0625)	
VK_SAMPLE_COUNT_16_BIT	(0.5625, 0.5625) (0.4375, 0.3125) (0.3125, 0.625) (0.75, 0.4375) (0.1875, 0.375) (0.625, 0.8125) (0.8125, 0.6875) (0.6875, 0.1875) (0.375, 0.875) (0.5, 0.0625) (0.25, 0.125) (0.125, 0.75) (0.0, 0.5) (0.9375, 0.25) (0.875, 0.9375) (0.0625, 0.0)	

25.5. Custom Sample Locations

Applications **can** also control the sample locations used for rasterization.

If the `pNext` chain of the `VkPipelineMultisampleStateCreateInfo` structure specified at pipeline creation time includes a `VkPipelineSampleLocationsStateCreateInfoEXT` structure, then that structure controls the sample locations used when rasterizing primitives with the pipeline.

The `VkPipelineSampleLocationsStateCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkPipelineSampleLocationsStateCreateInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkBool32             sampleLocationsEnable;
    VkSampleLocationsInfoEXT sampleLocationsInfo;
} VkPipelineSampleLocationsStateCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `sampleLocationsEnable` controls whether custom sample locations are used. If `sampleLocationsEnable` is `VK_FALSE`, the default sample locations are used and the values specified in `sampleLocationsInfo` are ignored.
- `sampleLocationsInfo` is the sample locations to use during rasterization if `sampleLocationsEnable` is `VK_TRUE` and the graphics pipeline is not created with `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT`.

Valid Usage (Implicit)

- VUID-VkPipelineSampleLocationsStateCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_SAMPLE_LOCATIONS_STATE_CREATE_INFO_EXT`
- VUID-VkPipelineSampleLocationsStateCreateInfoEXT-sampleLocationsInfo-parameter
`sampleLocationsInfo` **must** be a valid `VkSampleLocationsInfoEXT` structure

The `VkSampleLocationsInfoEXT` structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkSampleLocationsInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkSampleCountFlagBits sampleLocationsPerPixel;
    VkExtent2D           sampleLocationGridSize;
    uint32_t             sampleLocationsCount;
    const VkSampleLocationEXT* pSampleLocations;
} VkSampleLocationsInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `sampleLocationsPerPixel` is a `VkSampleCountFlagBits` value specifying the number of sample locations per pixel.
- `sampleLocationGridSize` is the size of the sample location grid to select custom sample locations for.
- `sampleLocationsCount` is the number of sample locations in `pSampleLocations`.
- `pSampleLocations` is a pointer to an array of `sampleLocationsCount` `VkSampleLocationEXT` structures.

This structure **can** be used either to specify the sample locations to be used for rendering or to specify the set of sample locations an image subresource has been last rendered with for the purposes of layout transitions of depth/stencil images created with `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT`.

The sample locations in `pSampleLocations` specify `sampleLocationsPerPixel` number of sample locations for each pixel in the grid of the size specified in `sampleLocationGridSize`. The sample location for sample `i` at the pixel grid location `(x,y)` is taken from `pSampleLocations[(x + y × sampleLocationGridSize.width) × sampleLocationsPerPixel + i]`.

Valid Usage

- VUID-VkSampleLocationsInfoEXT-sampleLocationsPerPixel-01526
`sampleLocationsPerPixel` **must** be a valid `VkSampleCountFlagBits` value that is set in `VkPhysicalDeviceSampleLocationsPropertiesEXT::sampleLocationSampleCounts`
- VUID-VkSampleLocationsInfoEXT-sampleLocationsCount-01527
`sampleLocationsCount` **must** equal `sampleLocationsPerPixel × sampleLocationGridSize.width × sampleLocationGridSize.height`

Valid Usage (Implicit)

- VUID-VkSampleLocationsInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SAMPLE_LOCATIONS_INFO_EXT`
- VUID-VkSampleLocationsInfoEXT-pSampleLocations-parameter
If `sampleLocationsCount` is not 0, `pSampleLocations` **must** be a valid pointer to an array of `sampleLocationsCount` `VkSampleLocationEXT` structures

The `VkSampleLocationEXT` structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkSampleLocationEXT {
    float    x;
    float    y;
} VkSampleLocationEXT;
```

- `x` is the horizontal coordinate of the sample's location.
- `y` is the vertical coordinate of the sample's location.

The domain space of the sample location coordinates has an upper-left origin within the pixel in framebuffer space.

The values specified in a `VkSampleLocationEXT` structure are always clamped to the implementation-dependent sample location coordinate range `[sampleLocationCoordinateRange[0],sampleLocationCoordinateRange[1]]` that **can** be queried using `VkPhysicalDeviceSampleLocationsPropertiesEXT`.

To **dynamically** set the `sampleLocationsEnable` state, call:

```
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_sample_locations,
VK_EXT_sample_locations with VK_EXT_shader_object
void vkCmdSetSampleLocationsEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                 sampleLocationsEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `sampleLocationsEnable` specifies the `sampleLocationsEnable` state.

This command sets the `sampleLocationsEnable` state for subsequent drawing commands when drawing using **shader objects**, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetSampleLocationsEnableEXT-None-09423

At least one of the following **must** be true:

- The `extendedDynamicState3SampleLocationsEnable` feature is enabled
- The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetSampleLocationsEnableEXT-commandBuffer-parameter `commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetSampleLocationsEnableEXT-commandBuffer-recording `commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetSampleLocationsEnableEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetSampleLocationsEnableEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To [dynamically set](#) the sample locations used for rasterization, call:

```
// Provided by VK_EXT_sample_locations
void vkCmdSetSampleLocationsEXT(
    VkCommandBuffer                commandBuffer,
    const VkSampleLocationsInfoEXT* pSampleLocationsInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pSampleLocationsInfo` is the sample locations state to set.

This command sets the custom sample locations for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and when the `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable` property of the bound graphics pipeline is `VK_TRUE`. Otherwise, this state is specified by the `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsInfo` values used to create the

currently active pipeline.

Valid Usage

- VUID-vkCmdSetSampleLocationsEXT-variableSampleLocations-01530
If `VkPhysicalDeviceSampleLocationsPropertiesEXT::variableSampleLocations` is `VK_FALSE` then the current render pass **must** have been begun by specifying a `VkRenderPassSampleLocationsBeginInfoEXT` structure whose `pPostSubpassSampleLocations` member contains an element with a `subpassIndex` matching the current subpass index and the `sampleLocationsInfo` member of that element **must** match the sample locations state pointed to by `pSampleLocationsInfo`

Valid Usage (Implicit)

- VUID-vkCmdSetSampleLocationsEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetSampleLocationsEXT-pSampleLocationsInfo-parameter
`pSampleLocationsInfo` **must** be a valid pointer to a valid `VkSampleLocationsInfoEXT` structure
- VUID-vkCmdSetSampleLocationsEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetSampleLocationsEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetSampleLocationsEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

25.6. Fragment Shading Rates

The features advertised by `VkPhysicalDeviceFragmentShadingRateFeaturesKHR` allow an application to control the [shading rate](#) of a given fragment shader invocation.

The fragment shading rate strongly interacts with [Multisampling](#), and the set of available rates for an implementation **may** be restricted by sample rate.

To query available shading rates, call:

```
// Provided by VK_KHR_fragment_shading_rate
VkResult vkGetPhysicalDeviceFragmentShadingRatesKHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t*                 pFragmentShadingRateCount,
    VkPhysicalDeviceFragmentShadingRateKHR* pFragmentShadingRates);
```

- `physicalDevice` is the handle to the physical device whose properties will be queried.
- `pFragmentShadingRateCount` is a pointer to an integer related to the number of fragment shading rates available or queried, as described below.
- `pFragmentShadingRates` is either `NULL` or a pointer to an array of `VkPhysicalDeviceFragmentShadingRateKHR` structures.

If `pFragmentShadingRates` is `NULL`, then the number of fragment shading rates available is returned in `pFragmentShadingRateCount`. Otherwise, `pFragmentShadingRateCount` **must** point to a variable set by the application to the number of elements in the `pFragmentShadingRates` array, and on return the variable is overwritten with the number of structures actually written to `pFragmentShadingRates`. If `pFragmentShadingRateCount` is less than the number of fragment shading rates available, at most `pFragmentShadingRateCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available fragment shading rates were returned.

The returned array of fragment shading rates **must** be ordered from largest `fragmentSize.width` value to smallest, and each set of fragment shading rates with the same `fragmentSize.width` value **must** be ordered from largest `fragmentSize.height` to smallest. Any two entries in the array **must** not have the same `fragmentSize` values.

For any entry in the array, the following rules also apply:

- The value of `fragmentSize.width` **must** be less than or equal to `maxFragmentSize.width`.
- The value of `fragmentSize.width` **must** be greater than or equal to 1.
- The value of `fragmentSize.width` **must** be a power-of-two.
- The value of `fragmentSize.height` **must** be less than or equal to `maxFragmentSize.height`.
- The value of `fragmentSize.height` **must** be greater than or equal to 1.
- The value of `fragmentSize.height` **must** be a power-of-two.
- The highest sample count in `sampleCounts` **must** be less than or equal to `maxFragmentShadingRateRasterizationSamples`.

- The product of `fragmentSize.width`, `fragmentSize.height`, and the highest sample count in `sampleCounts` **must** be less than or equal to `maxFragmentShadingRateCoverageSamples`.

Implementations **must** support at least the following shading rates:

<code>sampleCounts</code>	<code>fragmentSize</code>
<code>VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT</code>	<code>{2,2}</code>
<code>VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT</code>	<code>{2,1}</code>
<code>~0</code>	<code>{1,1}</code>

If `framebufferColorSampleCounts`, includes `VK_SAMPLE_COUNT_2_BIT`, the required rates **must** also include `VK_SAMPLE_COUNT_2_BIT`.

NOTE

Including the `{1,1}` fragment size is done for completeness; it has no actual effect on the support of rendering without setting the fragment size. All sample counts are supported for this rate.

Valid Usage (Implicit)

- `VUID-vkGetPhysicalDeviceFragmentShadingRatesKHR-physicalDevice-parameter` `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- `VUID-vkGetPhysicalDeviceFragmentShadingRatesKHR-pFragmentShadingRateCount-parameter` `pFragmentShadingRateCount` **must** be a valid pointer to a `uint32_t` value
- `VUID-vkGetPhysicalDeviceFragmentShadingRatesKHR-pFragmentShadingRates-parameter`
If the value referenced by `pFragmentShadingRateCount` is not `0`, and `pFragmentShadingRates` is not `NULL`, `pFragmentShadingRates` **must** be a valid pointer to an array of `pFragmentShadingRateCount` `VkPhysicalDeviceFragmentShadingRateKHR` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`

The `VkPhysicalDeviceFragmentShadingRateKHR` structure is defined as

```
// Provided by VK_KHR_fragment_shading_rate
typedef struct VkPhysicalDeviceFragmentShadingRateKHR {
    VkStructureType    sType;
    void*              pNext;
    VkSampleCountFlags sampleCounts;
    VkExtent2D         fragmentSize;
} VkPhysicalDeviceFragmentShadingRateKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **sampleCounts** is a bitmask of sample counts for which the shading rate described by **fragmentSize** is supported.
- **fragmentSize** is a [VkExtent2D](#) describing the width and height of a supported shading rate.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShadingRateKHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_KHR**
- VUID-VkPhysicalDeviceFragmentShadingRateKHR-pNext-pNext
pNext must be **NULL**

Fragment shading rates **can** be set at three points, with the three rates combined to determine the final shading rate.

25.6.1. Pipeline Fragment Shading Rate

The *pipeline fragment shading rate* **can** be set on a per-draw basis by either setting the rate in a graphics pipeline, or dynamically via [vkCmdSetFragmentShadingRateKHR](#).

The [VkPipelineFragmentShadingRateStateCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_fragment_shading_rate
typedef struct VkPipelineFragmentShadingRateStateCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkExtent2D         fragmentSize;
    VkFragmentShadingRateCombinerOpKHR combinerOps[2];
} VkPipelineFragmentShadingRateStateCreateInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **fragmentSize** specifies a [VkExtent2D](#) structure containing the fragment size used to define the pipeline fragment shading rate for drawing commands using this pipeline.

- `combinerOps` specifies a `VkFragmentShadingRateCombinerOpKHR` value determining how the `pipeline`, `primitive`, and `attachment shading rates` are `combined` for fragments generated by drawing commands using the created pipeline.

If the `pNext` chain of `VkGraphicsPipelineCreateInfo` includes a `VkPipelineFragmentShadingRateStateCreateInfoKHR` structure, then that structure includes parameters controlling the pipeline fragment shading rate.

If this structure is not present, `fragmentSize` is considered to be equal to (1,1), and both elements of `combinerOps` are considered to be equal to `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR`.

Valid Usage (Implicit)

- VUID-VkPipelineFragmentShadingRateStateCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_FRAGMENT_SHADING_RATE_STATE_CREATE_INFO_KHR`

To [dynamically set](#) the pipeline fragment shading rate and combiner operation, call:

```
// Provided by VK_KHR_fragment_shading_rate
void vkCmdSetFragmentShadingRateKHR(
    VkCommandBuffer          commandBuffer,
    const VkExtent2D*        pFragmentSize,
    const VkFragmentShadingRateCombinerOpKHR combinerOps[2]);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pFragmentSize` specifies the pipeline fragment shading rate for subsequent drawing commands.
- `combinerOps` specifies a `VkFragmentShadingRateCombinerOpKHR` determining how the `pipeline`, `primitive`, and `attachment shading rates` are `combined` for fragments generated by subsequent drawing commands.

This command sets the pipeline fragment shading rate and combiner operation for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineFragmentShadingRateStateCreateInfoKHR` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetFragmentShadingRateKHR-pipelineFragmentShadingRate-04507
If `pipelineFragmentShadingRate` is not enabled, `pFragmentSize->width` **must** be 1
- VUID-vkCmdSetFragmentShadingRateKHR-pipelineFragmentShadingRate-04508
If `pipelineFragmentShadingRate` is not enabled, `pFragmentSize->height` **must** be 1
- VUID-vkCmdSetFragmentShadingRateKHR-pipelineFragmentShadingRate-04509
One of `pipelineFragmentShadingRate`, `primitiveFragmentShadingRate`, or

`attachmentFragmentShadingRate` **must** be enabled

- VUID-vkCmdSetFragmentShadingRateKHR-primitiveFragmentShadingRate-04510
If the `primitiveFragmentShadingRate` feature is not enabled, `combinerOps[0]` **must** be `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR`
- VUID-vkCmdSetFragmentShadingRateKHR-attachmentFragmentShadingRate-04511
If the `attachmentFragmentShadingRate` feature is not enabled, `combinerOps[1]` **must** be `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR`
- VUID-vkCmdSetFragmentShadingRateKHR-fragmentSizeNonTrivialCombinerOps-04512
If the `fragmentSizeNonTrivialCombinerOps` limit is not supported, elements of `combinerOps` **must** be either `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR` or `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_REPLACE_KHR`
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04513
`pFragmentSize->width` **must** be greater than or equal to 1
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04514
`pFragmentSize->height` **must** be greater than or equal to 1
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04515
`pFragmentSize->width` **must** be a power-of-two value
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04516
`pFragmentSize->height` **must** be a power-of-two value
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04517
`pFragmentSize->width` **must** be less than or equal to 4
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-04518
`pFragmentSize->height` **must** be less than or equal to 4

Valid Usage (Implicit)

- VUID-vkCmdSetFragmentShadingRateKHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetFragmentShadingRateKHR-pFragmentSize-parameter
`pFragmentSize` **must** be a valid pointer to a valid `VkExtent2D` structure
- VUID-vkCmdSetFragmentShadingRateKHR-combinerOps-parameter
Each element of `combinerOps` **must** be a valid `VkFragmentShadingRateCombinerOpKHR` value
- VUID-vkCmdSetFragmentShadingRateKHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetFragmentShadingRateKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetFragmentShadingRateKHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

25.6.2. Primitive Fragment Shading Rate

The *primitive fragment shading rate* **can** be set via the `PrimitiveShadingRateKHR` built-in in the last active `pre-rasterization shader stage`. The rate associated with a given primitive is sourced from the value written to `PrimitiveShadingRateKHR` by that primitive's `provoking vertex`.

25.6.3. Attachment Fragment Shading Rate

The *attachment shading rate* **can** be set by including `VkFragmentShadingRateAttachmentInfoKHR` in a subpass to define a *fragment shading rate attachment*. Each pixel in the framebuffer is assigned an attachment fragment shading rate by the corresponding texel in the fragment shading rate attachment, according to:

$$x' = \text{floor}(x / \text{region}_x)$$

$$y' = \text{floor}(y / \text{region}_y)$$

where x' and y' are the coordinates of a texel in the fragment shading rate attachment, x and y are the coordinates of the pixel in the framebuffer, and region_x and region_y are the size of the region each texel corresponds to, as defined by the `shadingRateAttachmentTexelSize` member of `VkFragmentShadingRateAttachmentInfoKHR`.

If `multiview is enabled` and the shading rate attachment has multiple layers, the shading rate attachment texel is selected using `layer = ViewIndex`. If `multiview is disabled`, and both the shading rate attachment and the framebuffer have multiple layers, the shading rate attachment texel is selected using `layer = Layer`. Otherwise, `layer = 0`.

The texel is read from the fragment shading rate attachment image as a `texture input operation` without a sampler, using integer coordinates $i = x'$, $j = y'$, $k = 0$, $l = \text{layer}$, and $s = 0$. The fragment size is encoded into the first component of the result of that operation as follows:

$$\text{size}_w = 2^{((\text{texel}/4)\&3)}$$

$$\text{size}_h = 2^{(\text{texel}\&3)}$$

where `texel` is the value in the first component of the returned value, and `sizew` and `sizeh` are the width and height of the fragment size, decoded from the `texel`.

If no fragment shading rate attachment is specified, this size is calculated as `sizew = sizeh = 1`. Applications **must** not specify a width or height greater than 4 by this method.

The *Fragment Shading Rate* enumeration in SPIR-V adheres to the above encoding.

25.6.4. Combining the Fragment Shading Rates

The final rate (`Cxy`) used for fragment shading **must** be one of the rates returned by `vkGetPhysicalDeviceFragmentShadingRatesKHR` for the sample count used by rasterization.

If any of the following conditions are met, `Cxy` is set to {1,1} by the implementation:

- If [Sample Shading](#) is enabled.
- The `fragmentShadingRateWithSampleMask` limit is not supported, and `VkPipelineMultisampleStateCreateInfo::pSampleMask` contains a zero value in any bit used by fragment operations.
- The `fragmentShadingRateWithShaderSampleMask` is not supported, and the fragment shader has `SampleMask` in the input or output interface.
- The `fragmentShadingRateWithShaderDepthStencilWrites` limit is not supported, and the fragment shader declares the `FragDepth` or `FragStencilRefEXT` built-in.
- The `fragmentShadingRateWithConservativeRasterization` limit is not supported, and `VkPipelineRasterizationConservativeStateCreateInfoEXT::conservativeRasterizationMode` is not `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`.
- The `fragmentShadingRateWithFragmentShaderInterlock` limit is not supported, and the fragment shader declares any of the [fragment shader interlock](#) execution modes.
- The `fragmentShadingRateWithCustomSampleLocations` limit is not supported, and `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable` is `VK_TRUE`.
- The fragment shader declares any of the `TileImageColorReadAccessEXT`, `TileImageDepthReadAccessEXT`, or `TileImageStencilReadAccessEXT` capabilities.

Otherwise, each of the specified shading rates are combined and then used to derive the value of `Cxy`. As there are three ways to specify shading rates, two combiner operations are specified - between the [pipeline](#) and [primitive](#) shading rates, and between the result of that and the [attachment shading rate](#).

The equation used for each combiner operation is defined by `VkFragmentShadingRateCombinerOpKHR`:

```
// Provided by VK_KHR_fragment_shading_rate
typedef enum VkFragmentShadingRateCombinerOpKHR {
    VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR = 0,
    VK_FRAGMENT_SHADING_RATE_COMBINER_OP_REPLACE_KHR = 1,
    VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MIN_KHR = 2,
    VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MAX_KHR = 3,
    VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MUL_KHR = 4,
} VkFragmentShadingRateCombinerOpKHR;
```

- `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR` specifies a combiner operation of $\text{combine}(A_{xy}, B_{xy}) = A_{xy}$.
- `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_REPLACE_KHR` specifies a combiner operation of $\text{combine}(A_{xy}, B_{xy}) = B_{xy}$.
- `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MIN_KHR` specifies a combiner operation of $\text{combine}(A_{xy}, B_{xy}) = \min(A_{xy}, B_{xy})$.
- `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MAX_KHR` specifies a combiner operation of $\text{combine}(A_{xy}, B_{xy}) = \max(A_{xy}, B_{xy})$.
- `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MUL_KHR` specifies a combiner operation of $\text{combine}(A_{xy}, B_{xy}) = A_{xy} * B_{xy}$.

where $\text{combine}(A_{xy}, B_{xy})$ is the combine operation, and A_{xy} and B_{xy} are the inputs to the operation.

If `fragmentShadingRateStrictMultiplyCombiner` is `VK_FALSE`, using `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MUL_KHR` with values of 1 for both A and B in the same dimension results in the value 2 being produced for that dimension. See the definition of `fragmentShadingRateStrictMultiplyCombiner` for more information.

These operations are performed in a component-wise fashion.

This is used to generate a combined fragment area using the equation:

$$C_{xy} = \text{combine}(A_{xy}, B_{xy})$$

where C_{xy} is the combined fragment area result, and A_{xy} and B_{xy} are the fragment areas of the fragment shading rates being combined.

Two combine operations are performed, first with A_{xy} equal to the [pipeline fragment shading rate](#) and B_{xy} equal to the [primitive fragment shading rate](#), with the combine() operation selected by `combinerOps[0]`. A second combination is then performed, with A_{xy} equal to the result of the first combination and B_{xy} equal to the [attachment fragment shading rate](#), with the combine() operation selected by `combinerOps[1]`. The result of the second combination is used as the final fragment shading rate, reported via the [ShadingRateKHR built-in](#).

Implementations **should** clamp the inputs to the combiner operations A_{xy} and B_{xy} , and **must** do so if `VkPhysicalDeviceMaintenance6PropertiesKHR::fragmentShadingRateClampCombinerInputs` is `VK_TRUE`. All implementations **must** clamp the result of the second combiner operation.

A fragment shading rate R_{xy} representing any of A_{xy} , B_{xy} or C_{xy} is clamped as follows. If R_{xy} is one of the rates returned by `vkGetPhysicalDeviceFragmentShadingRatesKHR` for the sample count used by rasterization, the clamped shading rate R_{xy}' is R_{xy} . Otherwise, the clamped shading rate is selected from the rates returned by `vkGetPhysicalDeviceFragmentShadingRatesKHR` for the sample count used by rasterization. From this list of supported rates, the following steps are applied in order, to select a single value:

1. Keep only rates where $R_x' \leq R_x$ and $R_y' \leq R_y$.
 - Implementations **may** also keep rates where $R_x' \leq R_y$ and $R_y' \leq R_x$.
2. Keep only rates with the highest area ($R_x' \times R_y'$).
3. Keep only rates with the lowest aspect ratio ($R_x' + R_y'$).
4. In cases where a wide (e.g. 4x1) and tall (e.g. 1x4) rate remain, the implementation **may** choose either rate. However, it **must** choose this rate consistently for the same shading rates, and combiner operations for the lifetime of the `VkDevice`.

25.7. Sample Shading

Sample shading **can** be used to specify a minimum number of unique samples to process for each fragment. If sample shading is enabled, an implementation **must** invoke the fragment shader at least $\max(\lceil \frac{\text{VkPipelineMultisampleStateCreateInfo::minSampleShading}}{\text{VkPipelineMultisampleStateCreateInfo::rasterizationSamples}} \rceil, 1)$ times per fragment. If `VkPipelineMultisampleStateCreateInfo::sampleShadingEnable` is `VK_TRUE`, sample shading is enabled.

If a fragment shader entry point **statically uses** an input variable decorated with a `BuiltIn` of `SampleId` or `SamplePosition`, sample shading is enabled and a value of `1.0` is used instead of `minSampleShading`. If a fragment shader entry point **statically uses** an input variable decorated with `Sample`, sample shading **may** be enabled and a value of `1.0` will be used instead of `minSampleShading` if it is.

NOTE

If a shader decorates an input variable with `Sample` and that value meaningfully impacts the output of a shader, sample shading will be enabled to ensure that the input is in fact interpolated per-sample. This is inherent to the specification and not spelled out here - if an application simply declares such a variable it is implementation-defined whether sample shading is enabled or not. It is possible to see the effects of this by using atomics in the shader or using a pipeline statistics query to query the number of fragment invocations, even if the shader itself does not use any per-sample variables.

If there are fewer fragment invocations than **covered samples**, implementations **may** include those samples in fragment shader invocations in any manner as long as covered samples are all shaded at least once, and each invocation that is not a **helper invocation** covers at least one sample.

25.8. Barycentric Interpolation

When the `fragmentShaderBarycentric` feature is enabled, the `PerVertexKHR interpolation decoration` **can** be used with fragment shader inputs to indicate that the decorated inputs do not have

associated data in the fragment. Such inputs **can** only be accessed in a fragment shader using an array index whose value (0, 1, or 2) identifies one of the vertices of the primitive that produced the fragment. Reads of per-vertex values for missing vertices, such as the third vertex of a line primitive, will return values from the valid vertex with the highest index. This means that the per-vertex values of indices 1 and 2 for point primitives will be equal to those of index 0, and the per-vertex values of index 2 for line primitives will be equal to those of index 1.

When [tessellation](#) and [geometry shading](#) are not active, fragment shader inputs decorated with [PerVertexKHR](#) will take values from one of the vertices of the primitive that produced the fragment, identified by the extra index provided in SPIR-V code accessing the input. If the n vertices passed to a draw call are numbered 0 through $n-1$, and the point, line, and triangle primitives produced by the draw call are numbered with consecutive integers beginning with zero, the following table indicates the original vertex numbers used when the [provoking vertex mode](#) is [VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT](#) for index values of 0, 1, and 2. If an input decorated with [PerVertexKHR](#) is accessed with any other vertex index value, or is accessed while rasterizing a polygon when the [VkPipelineRasterizationStateCreateInfo::polygonMode](#) property of the currently active pipeline is not [VK_POLYGON_MODE_FILL](#), an undefined value is returned.

Primitive Topology	Vertex 0	Vertex 1	Vertex 2
VK_PRIMITIVE_TOPOLOGY_POINT_LIST	i	i	i
VK_PRIMITIVE_TOPOLOGY_LINE_LIST	$2i$	$2i+1$	$2i+1$
VK_PRIMITIVE_TOPOLOGY_LINE_STRIP	i	$i+1$	$i+1$
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST	$3i$	$3i+1$	$3i+2$
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP (even)	i	$i+1$	$i+2$
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP (odd)	i	$i+2$	$i+1$
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN	$i+1$	$i+2$	0
VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY	$4i+1$	$4i+2$	$4i+2$
VK_PRIMITIVE_TOPOLOGY_LINE_STRIP_WITH_ADJACENCY	$i+1$	$i+2$	$i+2$
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY	$6i$	$6i+2$	$6i+4$
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY (even)	$2i$	$2i+2$	$2i+4$

Primitive Topology	Vertex 0	Vertex 1	Vertex 2
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY (odd)	2i	2i+4	2i+2

When the provoking vertex mode is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, the original vertex numbers used are the same as above except as indicated in the table below.

Primitive Topology	Vertex 0	Vertex 1	Vertex 2
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP (odd, and <code>triStripVertexOrderIndependentOfProvokingVertex</code> of <code>VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR</code> is <code>VK_FALSE</code>)	i+1	i	i+2
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_FAN	0	i+1	i+2
VK_PRIMITIVE_TOPOLOGY_TRIANGLE_STRIP_WITH_ADJACENCY (odd)	2i+2	2i	2i+4

When geometry shading is active, primitives processed by fragment shaders are assembled from the vertices emitted by the geometry shader. In this case, the vertices used for fragment shader inputs decorated with `PerVertexKHR` are derived by treating the primitives produced by the shader as though they were specified by a draw call and consulting [the table above](#).

When using tessellation without geometry shading, the tessellator produces primitives in an implementation-dependent manner. While there is no defined vertex ordering for inputs decorated with `PerVertexKHR`, the vertex ordering used in this case will be consistent with the ordering used to derive the values of inputs decorated with `BaryCoordKHR` or `BaryCoordNoPerspKHR`.

Fragment shader inputs decorated with `BaryCoordKHR` or `BaryCoordNoPerspKHR` hold three-component vectors with barycentric weights that indicate the location of the fragment relative to the screen-space locations of vertices of its primitive. For point primitives, such variables are always assigned the value (1,0,0). For [line](#) primitives, the built-ins are obtained by interpolating an attribute whose values for the vertices numbered 0 and 1 are (1,0,0) and (0,1,0), respectively. For [polygon](#) primitives, the built-ins are obtained by interpolating an attribute whose values for the vertices numbered 0, 1, and 2 are (1,0,0), (0,1,0), and (0,0,1), respectively. For `BaryCoordKHR`, the values are obtained using perspective interpolation. For `BaryCoordNoPerspKHR`, the values are obtained using linear interpolation. The values of `BaryCoordKHR` and `BaryCoordNoPerspKHR` are undefined while rasterizing a polygon when the `VkPipelineRasterizationStateCreateInfo::polygonMode` property of the currently active pipeline is not `VK_POLYGON_MODE_FILL`.

25.9. Points

A point is drawn by generating a set of fragments in the shape of a square centered around the vertex of the point. Each vertex has an associated point size controlling the width/height of that square. The point size is taken from the (potentially clipped) shader built-in `PointSize` written by:

- the geometry shader, if active;
- the tessellation evaluation shader, if active and no geometry shader is active;
- the vertex shader, otherwise

and clamped to the implementation-dependent point size range [`pointSizeRange[0]`, `pointSizeRange[1]`]. The value written to `PointSize` **must** be greater than zero. If `maintenance5` is enabled, and a value is not written to `PointSize`, the point size takes a default value of 1.0.

Not all point sizes need be supported, but the size 1.0 **must** be supported. The range of supported sizes and the size of evenly-spaced gradations within that range are implementation-dependent. The range and gradations are obtained from the `pointSizeRange` and `pointSizeGranularity` members of `VkPhysicalDeviceLimits`. If, for instance, the size range is from 0.1 to 2.0 and the gradation size is 0.1, then the sizes 0.1, 0.2, ..., 1.9, 2.0 are supported. Additional point sizes **may** also be supported. There is no requirement that these sizes be equally spaced. If an unsupported size is requested, the nearest supported size is used instead.

25.9.1. Basic Point Rasterization

Point rasterization produces a fragment for each fragment area group of framebuffer pixels with one or more sample points that intersect a region centered at the point's (x_p, y_p) . This region is a square with side equal to the current point size. Coverage bits that correspond to sample points that intersect the region are 1, other coverage bits are 0. All fragments produced in rasterizing a point are assigned the same associated data, which are those of the vertex corresponding to the point. However, the fragment shader built-in `PointCoord` contains point sprite texture coordinates. The s and t point sprite texture coordinates vary from zero to one across the point horizontally left-to-right and vertically top-to-bottom, respectively. The following formulas are used to evaluate s and t :

$$s = \frac{1}{2} + \frac{(x_p - x_f)}{\text{size}}$$
$$t = \frac{1}{2} + \frac{(y_p - y_f)}{\text{size}}$$

where `size` is the point's size; (x_p, y_p) is the location at which the point sprite coordinates are evaluated - this **may** be the framebuffer coordinates of the fragment center, or the location of a sample; and (x_f, y_f) is the exact, unrounded framebuffer coordinate of the vertex for the point.

25.10. Line Segments

Line segment rasterization options are controlled by the `VkPipelineRasterizationLineStateCreateInfoKHR` structure.

The `VkPipelineRasterizationLineStateCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_line_rasterization
typedef struct VkPipelineRasterizationLineStateCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkLineRasterizationModeKHR lineRasterizationMode;
    VkBool32           stippledLineEnable;
    uint32_t           lineStippleFactor;
    uint16_t           lineStipplePattern;
} VkPipelineRasterizationLineStateCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `lineRasterizationMode` is a `VkLineRasterizationModeKHR` value selecting the style of line rasterization.
- `stippledLineEnable` enables `stippled line rasterization`.
- `lineStippleFactor` is the repeat factor used in stippled line rasterization.
- `lineStipplePattern` is the bit pattern used in stippled line rasterization.

If `stippledLineEnable` is `VK_FALSE`, the values of `lineStippleFactor` and `lineStipplePattern` are ignored.

Valid Usage

- VUID-VkPipelineRasterizationLineStateCreateInfoKHR-lineRasterizationMode-02768
If `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR`, then the `rectangularLines` feature **must** be enabled
- VUID-VkPipelineRasterizationLineStateCreateInfoKHR-lineRasterizationMode-02769
If `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR`, then the `bresenhamLines` feature **must** be enabled
- VUID-VkPipelineRasterizationLineStateCreateInfoKHR-lineRasterizationMode-02770
If `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, then the `smoothLines` feature **must** be enabled
- VUID-VkPipelineRasterizationLineStateCreateInfoKHR-stippledLineEnable-02771
If `stippledLineEnable` is `VK_TRUE` and `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR`, then the `stippledRectangularLines` feature **must** be enabled
- VUID-VkPipelineRasterizationLineStateCreateInfoKHR-stippledLineEnable-02772
If `stippledLineEnable` is `VK_TRUE` and `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR`, then the `stippledBresenhamLines` feature **must** be enabled
- VUID-VkPipelineRasterizationLineStateCreateInfoKHR-stippledLineEnable-02773

If `stippledLineEnable` is `VK_TRUE` and `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, then the `stippledSmoothLines` feature **must** be enabled

- VUID-VkPipelineRasterizationLineStateCreateInfoKHR-stippledLineEnable-02774

If `stippledLineEnable` is `VK_TRUE` and `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR`, then the `stippledRectangularLines` feature **must** be enabled and `VkPhysicalDeviceLimits::strictLines` **must** be `VK_TRUE`

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationLineStateCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_LINE_STATE_CREATE_INFO_KHR`
- VUID-VkPipelineRasterizationLineStateCreateInfoKHR-lineRasterizationMode-parameter
`lineRasterizationMode` **must** be a valid `VkLineRasterizationModeKHR` value

Possible values of `VkPipelineRasterizationLineStateCreateInfoKHR::lineRasterizationMode` are:

```
// Provided by VK_KHR_line_rasterization
typedef enum VkLineRasterizationModeKHR {
    VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR = 0,
    VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR = 1,
    VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR = 2,
    VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR = 3,
    VK_LINE_RASTERIZATION_MODE_DEFAULT_EXT = VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR,
    VK_LINE_RASTERIZATION_MODE_RECTANGULAR_EXT =
VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR,
    VK_LINE_RASTERIZATION_MODE_BRESENHAM_EXT =
VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR,
    VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_EXT =
VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR,
} VkLineRasterizationModeKHR;
```

- `VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR` is equivalent to `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR` if `VkPhysicalDeviceLimits::strictLines` is `VK_TRUE`, otherwise lines are drawn as non-`strictLines` parallelograms. Both of these modes are defined in [Basic Line Segment Rasterization](#).
- `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR` specifies lines drawn as if they were rectangles extruded from the line
- `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR` specifies lines drawn by determining which pixel diamonds the line intersects and exits, as defined in [Bresenham Line Segment Rasterization](#).
- `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR` specifies lines drawn if they were rectangles extruded from the line, with alpha falloff, as defined in [Smooth Lines](#).

To [dynamically set](#) the `lineRasterizationMode` state, call:

```
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_line_rasterization,
VK_EXT_line_rasterization with VK_EXT_shader_object
void vkCmdSetLineRasterizationModeEXT(
    VkCommandBuffer                      commandBuffer,
    VkLineRasterizationModeEXT          lineRasterizationMode);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `lineRasterizationMode` specifies the `lineRasterizationMode` state.

This command sets the `lineRasterizationMode` state for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationLineStateCreateInfoKHR::lineRasterizationMode` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetLineRasterizationModeEXT-None-09423
At least one of the following **must** be true:
 - The `extendedDynamicState3LineRasterizationMode` feature is enabled
 - The `shaderObject` feature is enabled
- VUID-vkCmdSetLineRasterizationModeEXT-lineRasterizationMode-07418
If `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR`, then the `rectangularLines` feature **must** be enabled
- VUID-vkCmdSetLineRasterizationModeEXT-lineRasterizationMode-07419
If `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR`, then the `bresenhamLines` feature **must** be enabled
- VUID-vkCmdSetLineRasterizationModeEXT-lineRasterizationMode-07420
If `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, then the `smoothLines` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdSetLineRasterizationModeEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetLineRasterizationModeEXT-lineRasterizationMode-parameter
`lineRasterizationMode` **must** be a valid `VkLineRasterizationModeEXT` value
- VUID-vkCmdSetLineRasterizationModeEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetLineRasterizationModeEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics

operations

- VUID-vkCmdSetLineRasterizationModeEXT-videocoding

This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To **dynamically set** the `stippledLineEnable` state, call:

```
// Provided by VK_EXT_extended_dynamic_state3 with VK_EXT_line_rasterization,  
VK_EXT_line_rasterization with VK_EXT_shader_object  
void vkCmdSetLineStippleEnableEXT(  
    VkCommandBuffer          commandBuffer,  
    VkBool32                 stippledLineEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `stippledLineEnable` specifies the `stippledLineEnable` state.

This command sets the `stippledLineEnable` state for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationLineStateCreateInfoKHR::stippledLineEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetLineStippleEnableEXT-None-09423

At least one of the following **must** be true:

- The `extendedDynamicState3LineStippleEnable` feature is enabled
- The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetLineStippleEnableEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetLineStippleEnableEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdSetLineStippleEnableEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetLineStippleEnableEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To [dynamically set](#) the line width, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetLineWidth(
    VkCommandBuffer          commandBuffer,
    float                    lineWidth);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `lineWidth` is the width of rasterized line segments.

This command sets the line width for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_LINE_WIDTH` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationStateCreateInfo::lineWidth` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetLineWidth-lineWidth-00788
If the `wideLines` feature is not enabled, `lineWidth` **must** be `1.0`

Valid Usage (Implicit)

- VUID-vkCmdSetLineWidth-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetLineWidth-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdSetLineWidth-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetLineWidth-vidcoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Not all line widths need be supported for line segment rasterization, but width `1.0` antialiased segments **must** be provided. The range and gradations are obtained from the `lineWidthRange` and `lineWidthGranularity` members of `VkPhysicalDeviceLimits`. If, for instance, the size range is from `0.1` to `2.0` and the gradation size is `0.1`, then the sizes `0.1`, `0.2`, ..., `1.9`, `2.0` are supported. Additional line widths **may** also be supported. There is no requirement that these widths be equally spaced. If an unsupported width is requested, the nearest supported width is used instead.

25.10.1. Basic Line Segment Rasterization

If the `lineRasterizationMode` member of `VkPipelineRasterizationLineStateCreateInfoKHR` is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR`, rasterized line segments produce fragments which

intersect a rectangle centered on the line segment. Two of the edges are parallel to the specified line segment; each is at a distance of one-half the current width from that segment in directions perpendicular to the direction of the line. The other two edges pass through the line endpoints and are perpendicular to the direction of the specified line segment. Coverage bits that correspond to sample points that intersect the rectangle are 1, other coverage bits are 0.

Next we specify how the data associated with each rasterized fragment are obtained. Let $\mathbf{p}_r = (x_d, y_d)$ be the framebuffer coordinates at which associated data are evaluated. This **may** be the center of a fragment or the location of a sample within the fragment. When `rasterizationSamples` is `VK_SAMPLE_COUNT_1_BIT`, the fragment center **must** be used. Let $\mathbf{p}_a = (x_a, y_a)$ and $\mathbf{p}_b = (x_b, y_b)$ be initial and final endpoints of the line segment, respectively. Set

$$t = \frac{(\mathbf{p}_r - \mathbf{p}_a) \cdot (\mathbf{p}_b - \mathbf{p}_a)}{\|\mathbf{p}_b - \mathbf{p}_a\|^2}$$

(Note that $t = 0$ at $\mathbf{p}_a$ and $t = 1$ at $\mathbf{p}_b$. Also note that this calculation projects the vector from $\mathbf{p}_a$ to $\mathbf{p}_r$ onto the line, and thus computes the normalized distance of the fragment along the line.)

If `strictLines` is `VK_TRUE`, line segments are rasterized using perspective or linear interpolation.

Perspective interpolation for a line segment interpolates two values in a manner that is correct when taking the perspective of the viewport into consideration, by way of the line segment's clip coordinates. An interpolated value f can be determined by

$$f = \frac{(1 - t)f_a / w_a + tf_b / w_b}{(1 - t) / w_a + t / w_b}$$

where f_a and f_b are the data associated with the starting and ending endpoints of the segment, respectively; w_a and w_b are the clip w coordinates of the starting and ending endpoints of the segment, respectively.

Linear interpolation for a line segment directly interpolates two values, and an interpolated value f can be determined by

$$f = (1 - t) f_a + t f_b$$

where f_a and f_b are the data associated with the starting and ending endpoints of the segment, respectively.

The clip coordinate w for a sample is determined using perspective interpolation. The depth value z for a sample is determined using linear interpolation. Interpolation of fragment shader input values are determined by [Interpolation decorations](#).

The above description documents the preferred method of line rasterization, and **must** be used when `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR`.

By default, when `strictLines` is `VK_FALSE`, and when the `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR`, the edges of the lines are generated as a parallelogram surrounding the original line. The major axis is chosen by noting the axis in which there is the

greatest distance between the line start and end points. If the difference is equal in both directions then the X axis is chosen as the major axis. Edges 2 and 3 are aligned to the minor axis and are centered on the endpoints of the line as in [Non strict lines](#), and each is `LineWidth` long. Edges 0 and 1 are parallel to the line and connect the endpoints of edges 2 and 3. Coverage bits that correspond to sample points that intersect the parallelogram are 1, other coverage bits are 0.

Samples that fall exactly on the edge of the parallelogram follow the polygon rasterization rules.

Interpolation occurs as if the parallelogram was decomposed into two triangles where each pair of vertices at each end of the line has identical attributes.

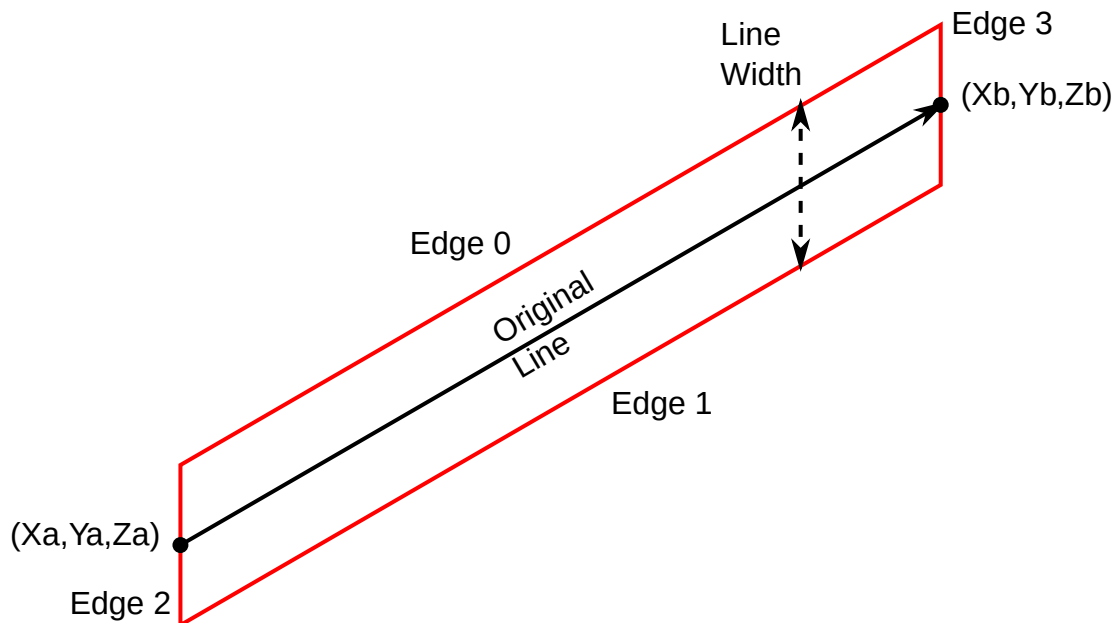


Figure 15. Non strict lines

Only when `strictLines` is `VK_FALSE` implementations **may** deviate from the non-strict line algorithm described above in the following ways:

- Implementations **may** instead interpolate each fragment according to the formula in [Basic Line Segment Rasterization](#) using the original line segment endpoints.
- Rasterization of non-antialiased non-strict line segments **may** be performed using the rules defined in [Bresenham Line Segment Rasterization](#).

If `VkPhysicalDeviceMaintenance5PropertiesKHR::nonStrictSinglePixelWideLinesUseParallelogram` is `VK_TRUE`, and `strictLines` is `VK_FALSE`, non-strict lines of width 1.0 are rasterized as parallelograms, otherwise they are rasterized using Bresenham's algorithm.

If `VkPhysicalDeviceMaintenance5PropertiesKHR::nonStrictWideLinesUseParallelogram` is `VK_TRUE`, and `strictLines` is `VK_FALSE`, non-strict lines of width greater than 1.0 are rasterized as parallelograms, otherwise they are rasterized using Bresenham's algorithm.

25.10.2. Bresenham Line Segment Rasterization

If `lineRasterizationMode` is `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR`, then the following rules replace the line rasterization rules defined in [Basic Line Segment Rasterization](#).

Non-strict lines **may** also follow these rasterization rules for non-antialiased lines.

Line segment rasterization begins by characterizing the segment as either *x-major* or *y-major*. *x-major* line segments have slope in the closed interval $[-1,1]$; all other line segments are *y-major* (slope is determined by the segment's endpoints). We specify rasterization only for *x-major* segments except in cases where the modifications for *y-major* segments are not self-evident.

Ideally, Vulkan uses a *diamond-exit* rule to determine those fragments that are produced by rasterizing a line segment. For each fragment f with center at framebuffer coordinates x_f and y_f , define a diamond-shaped region that is the intersection of four half planes:

$$R_f = \{(x, y) \mid |x - x_f| + |y - y_f| < \frac{1}{2}\}$$

Essentially, a line segment starting at p_a and ending at p_b produces those fragments f for which the segment intersects R_f , except if p_b is contained in R_f .

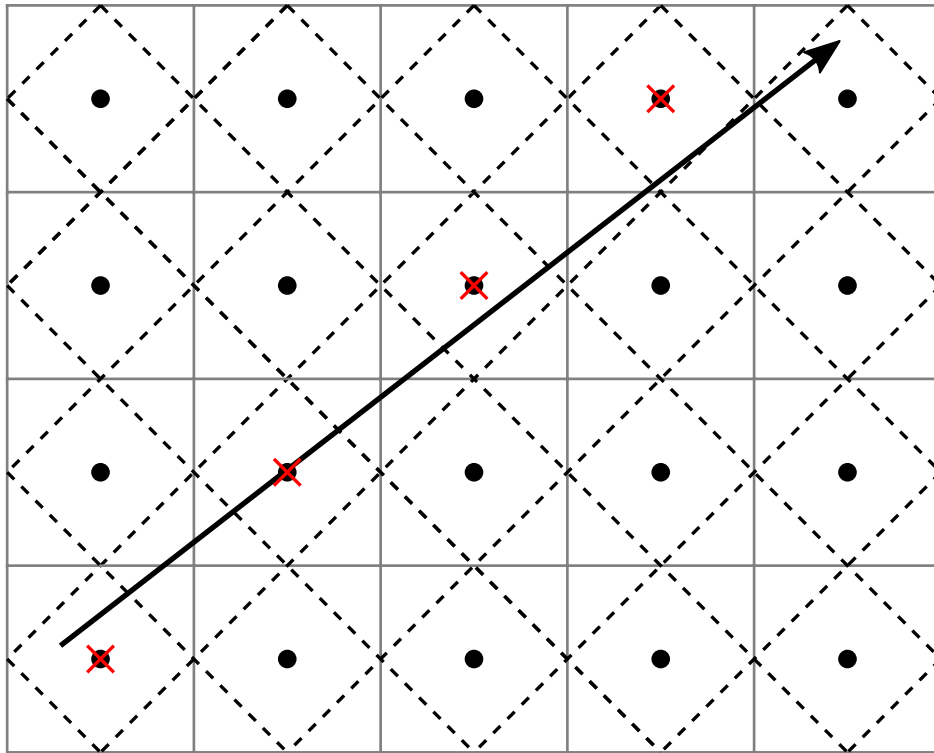


Figure 16. Visualization of Bresenham's algorithm

To avoid difficulties when an endpoint lies on a boundary of R_f we (in principle) perturb the supplied endpoints by a tiny amount. Let p_a and p_b have framebuffer coordinates (x_a, y_a) and (x_b, y_b) , respectively. Obtain the perturbed endpoints p_a' given by $(x_a, y_a) - (\epsilon, \epsilon^2)$ and p_b' given by $(x_b, y_b) - (\epsilon, \epsilon^2)$. Rasterizing the line segment starting at p_a and ending at p_b produces those fragments f for which the segment starting at p_a' and ending on p_b' intersects R_f , except if p_b' is contained in R_f . ϵ is chosen to be so small that rasterizing the line segment produces the same fragments when δ is substituted for ϵ for any $0 < \delta \leq \epsilon$.

When p_a and p_b lie on fragment centers, this characterization of fragments reduces to Bresenham's algorithm with one modification: lines produced in this description are "half-open", meaning that

the final fragment (corresponding to p_b) is not drawn. This means that when rasterizing a series of connected line segments, shared endpoints will be produced only once rather than twice (as would occur with Bresenham's algorithm).

Implementations **may** use other line segment rasterization algorithms, subject to the following rules:

- The coordinates of a fragment produced by the algorithm **must** not deviate by more than one unit in either x or y framebuffer coordinates from a corresponding fragment produced by the diamond-exit rule.
- The total number of fragments produced by the algorithm **must** not differ from that produced by the diamond-exit rule by more than one.
- For an x-major line, two fragments that lie in the same framebuffer-coordinate column **must** not be produced (for a y-major line, two fragments that lie in the same framebuffer-coordinate row **must** not be produced).
- If two line segments share a common endpoint, and both segments are either x-major (both left-to-right or both right-to-left) or y-major (both bottom-to-top or both top-to-bottom), then rasterizing both segments **must** not produce duplicate fragments. Fragments also **must** not be omitted so as to interrupt continuity of the connected segments.

The actual width w of Bresenham lines is determined by rounding the line width to the nearest integer, clamping it to the implementation-dependent `LineWidthRange` (with both values rounded to the nearest integer), then clamping it to be no less than 1.

Bresenham line segments of width other than one are rasterized by offsetting them in the minor direction (for an x-major line, the minor direction is y, and for a y-major line, the minor direction is x) and producing a row or column of fragments in the minor direction. If the line segment has endpoints given by (x_0, y_0) and (x_1, y_1) in framebuffer coordinates, the segment with endpoints $(x_0, y_0 - \frac{w-1}{2})$ and $(x_1, y_1 - \frac{w-1}{2})$ is rasterized, but instead of a single fragment, a column of fragments of height w (a row of fragments of length w for a y-major segment) is produced at each x (y for y-major) location. The lowest fragment of this column is the fragment that would be produced by rasterizing the segment of width 1 with the modified coordinates.

The preferred method of attribute interpolation for a wide line is to generate the same attribute values for all fragments in the row or column described above, as if the adjusted line was used for interpolation and those values replicated to the other fragments, except for `FragCoord` which is interpolated as usual. Implementations **may** instead interpolate each fragment according to the formula in [Basic Line Segment Rasterization](#), using the original line segment endpoints.

When Bresenham lines are being rasterized, sample locations **may** all be treated as being at the pixel center (this **may** affect attribute and depth interpolation).

NOTE

The sample locations described above are **not** used for determining coverage, they are only used for things like attribute interpolation. The rasterization rules that determine coverage are defined in terms of whether the line intersects **pixels**, as opposed to the point sampling rules used for other primitive types. So these rules are independent of the sample locations. One consequence of this is that Bresenham lines cover the same pixels regardless of the number of rasterization samples, and

cover all samples in those pixels (unless masked out or killed).

25.10.3. Line Stipple

If the `stippledLineEnable` member of `VkPipelineRasterizationLineStateCreateInfoKHR` is `VK_TRUE`, then lines are rasterized with a *line stipple* determined by `lineStippleFactor` and `lineStipplePattern`. `lineStipplePattern` is an unsigned 16-bit integer that determines which fragments are to be drawn or discarded when the line is rasterized. `lineStippleFactor` is a count that is used to modify the effective line stipple by causing each bit in `lineStipplePattern` to be used `lineStippleFactor` times.

Line stippling discards certain fragments that are produced by rasterization. The masking is achieved using three parameters: the 16-bit line stipple pattern p , the line stipple factor r , and an integer stipple counter s . Let

$$b = \lfloor \frac{s}{r} \rfloor \bmod 16$$

Then a fragment is produced if the b 'th bit of p is 1, and discarded otherwise. The bits of p are numbered with 0 being the least significant and 15 being the most significant.

The initial value of s is zero. For `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR` lines, s is incremented after production of each fragment of a line segment (fragments are produced in order, beginning at the starting point and working towards the ending point). For `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR` and `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR` lines, the rectangular region is subdivided into adjacent unit-length rectangles, and s is incremented once for each rectangle. Rectangles with a value of s such that the b 'th bit of p is zero are discarded. If the last rectangle in a line segment is shorter than unit-length, then the remainder **may** carry over to the next line segment in the line strip using the same value of s (this is the preferred behavior, for the stipple pattern to appear more consistent through the strip).

s is reset to 0 at the start of each strip (for line strips), and before every line segment in a group of independent segments.

If the line segment has been clipped, then the value of s at the beginning of the line segment is implementation-dependent.

To [dynamically set](#) the line stipple state, call:

```
// Provided by VK_KHR_line_rasterization
void vkCmdSetLineStippleKHR(
    VkCommandBuffer          commandBuffer,
    uint32_t                 lineStippleFactor,
    uint16_t                 lineStipplePattern);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `lineStippleFactor` is the repeat factor used in stippled line rasterization.

- `lineStipplePattern` is the bit pattern used in stippled line rasterization.

This command sets the line stipple state for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_LINE_STIPPLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationLineStateCreateInfoKHR::lineStippleFactor` and `VkPipelineRasterizationLineStateCreateInfoKHR::lineStipplePattern` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetLineStippleKHR-lineStippleFactor-02776
`lineStippleFactor` **must** be in the range [1,256]

Valid Usage (Implicit)

- VUID-vkCmdSetLineStippleKHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetLineStippleKHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdSetLineStippleKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetLineStippleKHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

25.10.4. Smooth Lines

If the `lineRasterizationMode` member of `VkPipelineRasterizationLineStateCreateInfoKHR` is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, then lines are considered to be rectangles using the same geometry as for `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR` lines. The rules for determining which pixels are covered are implementation-dependent, and **may** include nearby pixels where no sample locations are covered or where the rectangle does not intersect the pixel at all. For each pixel that is considered covered, the fragment computes a coverage value that approximates the area of the intersection of the rectangle with the pixel square, and this coverage value is multiplied into the color location 0's alpha value after fragment shading, as described in [Multisample Coverage](#).

NOTE

The details of the rasterization rules and area calculation are left intentionally vague, to allow implementations to generate coverage and values that are aesthetically pleasing.

25.11. Polygons

A polygon results from the decomposition of a triangle strip, triangle fan or a series of independent triangles. Like points and line segments, polygon rasterization is controlled by several variables in the `VkPipelineRasterizationStateCreateInfo` structure.

25.11.1. Basic Polygon Rasterization

The first step of polygon rasterization is to determine whether the triangle is *back-facing* or *front-facing*. This determination is made based on the sign of the (clipped or unclipped) polygon's area computed in framebuffer coordinates. One way to compute this area is:

$$a = -\frac{1}{2} \sum_{i=0}^{n-1} x_f^i y_f^{i+1} - x_f^{i+1} y_f^i$$

where x_f^i and y_f^i are the x and y framebuffer coordinates of the i th vertex of the n -vertex polygon (vertices are numbered starting at zero for the purposes of this computation) and $i + 1$ is $(i + 1) \bmod n$.

The interpretation of the sign of a is determined by the `VkPipelineRasterizationStateCreateInfo::frontFace` property of the currently active pipeline. Possible values are:

```
// Provided by VK_VERSION_1_0
typedef enum VkFrontFace {
    VK_FRONT_FACE_COUNTER_CLOCKWISE = 0,
    VK_FRONT_FACE_CLOCKWISE = 1,
} VkFrontFace;
```

- `VK_FRONT_FACE_COUNTER_CLOCKWISE` specifies that a triangle with positive area is considered front-facing.
- `VK_FRONT_FACE_CLOCKWISE` specifies that a triangle with negative area is considered front-facing.

Any triangle which is not front-facing is back-facing, including zero-area triangles.

To [dynamically set](#) the front face orientation, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetFrontFaceEXT(
    VkCommandBuffer                commandBuffer,
    VkFrontFace                    frontFace);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `frontFace` is a `VkFrontFace` value specifying the front-facing triangle orientation to be used for culling.

This command sets the front face orientation for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the `VkPipelineRasterizationStateCreateInfo::frontFace` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetFrontFace-None-08971
At least one of the following **must** be true:
 - the `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetFrontFace-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetFrontFace-frontFace-parameter
`frontFace` **must** be a valid `VkFrontFace` value
- VUID-vkCmdSetFrontFace-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetFrontFace-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetFrontFace-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Once the orientation of triangles is determined, they are culled according to the [VkPipelineRasterizationStateCreateInfo::cullMode](#) property of the currently active pipeline. Possible values are:

```
// Provided by VK_VERSION_1_0
typedef enum VkCullModeFlagBits {
    VK_CULL_MODE_NONE = 0,
    VK_CULL_MODE_FRONT_BIT = 0x00000001,
    VK_CULL_MODE_BACK_BIT = 0x00000002,
    VK_CULL_MODE_FRONT_AND_BACK = 0x00000003,
} VkCullModeFlagBits;
```

- [VK_CULL_MODE_NONE](#) specifies that no triangles are discarded
- [VK_CULL_MODE_FRONT_BIT](#) specifies that front-facing triangles are discarded
- [VK_CULL_MODE_BACK_BIT](#) specifies that back-facing triangles are discarded
- [VK_CULL_MODE_FRONT_AND_BACK](#) specifies that all triangles are discarded.

Following culling, fragments are produced for any triangles which have not been discarded.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkCullModeFlags;
```

[VkCullModeFlags](#) is a bitmask type for setting a mask of zero or more [VkCullModeFlagBits](#).

To [dynamically set](#) the cull mode, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetCullModeEXT(
    VkCommandBuffer          commandBuffer,
    VkCullModeFlags          cullMode);
```

- [commandBuffer](#) is the command buffer into which the command will be recorded.
- [cullMode](#) specifies the cull mode property to use for drawing.

This command sets the cull mode for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the [VkPipelineRasterizationStateCreateInfo::cullMode](#) value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetCullMode-None-08971
At least one of the following **must** be true:
 - the [shaderObject](#) feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetCullMode-commandBuffer-parameter
[commandBuffer](#) **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetCullMode-cullMode-parameter
[cullMode](#) **must** be a valid combination of [VkCullModeFlagBits](#) values
- VUID-vkCmdSetCullMode-commandBuffer-recording
[commandBuffer](#) **must** be in the [recording state](#)
- VUID-vkCmdSetCullMode-commandBuffer-cmdpool
The [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support graphics operations
- VUID-vkCmdSetCullMode-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to [commandBuffer](#) **must** be externally synchronized
- Host access to the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

The rule for determining which fragments are produced by polygon rasterization is called *point sampling*. The two-dimensional projection obtained by taking the x and y framebuffer coordinates of the polygon's vertices is formed. Fragments are produced for any fragment area groups of pixels

for which any sample points lie inside of this polygon. Coverage bits that correspond to sample points that satisfy the point sampling criteria are 1, other coverage bits are 0. Special treatment is given to a sample whose sample location lies on a polygon edge. In such a case, if two polygons lie on either side of a common edge (with identical endpoints) on which a sample point lies, then exactly one of the polygons **must** result in a covered sample for that fragment during rasterization. As for the data associated with each fragment produced by rasterizing a polygon, we begin by specifying how these values are produced for fragments in a triangle.

Barycentric coordinates are a set of three numbers, a , b , and c , each in the range $[0,1]$, with $a + b + c = 1$. These coordinates uniquely specify any point p within the triangle or on the triangle's boundary as

$$p = a p_a + b p_b + c p_c$$

where p_a , p_b , and p_c are the vertices of the triangle. a , b , and c are determined by:

$$a = \frac{A(pp_b p_c)}{A(p_a p_b p_c)}, \quad b = \frac{A(pp_a p_c)}{A(p_a p_b p_c)}, \quad c = \frac{A(pp_a p_b)}{A(p_a p_b p_c)},$$

where $A(lmn)$ denotes the area in framebuffer coordinates of the triangle with vertices l , m , and n .

Denote an associated datum at p_a , p_b , or p_c as f_a , f_b , or f_c , respectively.

Perspective interpolation for a triangle interpolates three values in a manner that is correct when taking the perspective of the viewport into consideration, by way of the triangle's clip coordinates. An interpolated value f can be determined by

$$f = \frac{a f_a / w_a + b f_b / w_b + c f_c / w_c}{a / w_a + b / w_b + c / w_c}$$

where w_a , w_b , and w_c are the clip w coordinates of p_a , p_b , and p_c , respectively. a , b , and c are the barycentric coordinates of the location at which the data are produced.

Linear interpolation for a triangle directly interpolates three values, and an interpolated value f can be determined by

$$f = a f_a + b f_b + c f_c$$

where f_a , f_b , and f_c are the data associated with p_a , p_b , and p_c , respectively.

The clip coordinate w for a sample is determined using perspective interpolation. The depth value z for a sample is determined using linear interpolation. Interpolation of fragment shader input values are determined by [Interpolation decorations](#).

For a polygon with more than three edges, such as are produced by clipping a triangle, a convex combination of the values of the datum at the polygon's vertices **must** be used to obtain the value assigned to each fragment produced by the rasterization algorithm. That is, it **must** be the case that at every fragment

$$f = \sum_{i=1}^n a_i f_i$$

where n is the number of vertices in the polygon and f_i is the value of f at vertex i . For each i , $0 \leq a_i \leq 1$ and $\sum_{i=1}^n a_i = 1$. The values of a_i **may** differ from fragment to fragment, but at vertex i , $a_i = 1$ and $a_j = 0$ for $j \neq i$.

NOTE

One algorithm that achieves the required behavior is to triangulate a polygon (without adding any vertices) and then treat each triangle individually as already discussed. A scan-line rasterizer that linearly interpolates data along each edge and then linearly interpolates data across each horizontal span from edge to edge also satisfies the restrictions (in this case the numerator and denominator of [perspective interpolation](#) are iterated independently, and a division is performed for each fragment).

25.11.2. Polygon Mode

Possible values of the [VkPipelineRasterizationStateCreateInfo::polygonMode](#) property of the currently active pipeline, specifying the method of rasterization for polygons, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkPolygonMode {
    VK_POLYGON_MODE_FILL = 0,
    VK_POLYGON_MODE_LINE = 1,
    VK_POLYGON_MODE_POINT = 2,
} VkPolygonMode;
```

- [VK_POLYGON_MODE_POINT](#) specifies that polygon vertices are drawn as points.
- [VK_POLYGON_MODE_LINE](#) specifies that polygon edges are drawn as line segments.
- [VK_POLYGON_MODE_FILL](#) specifies that polygons are rendered using the polygon rasterization rules in this section.

These modes affect only the final rasterization of polygons: in particular, a polygon's vertices are shaded and the polygon is clipped and possibly culled before these modes are applied.

If [VkPhysicalDeviceMaintenance5PropertiesKHR::polygonModePointSize](#) is [VK_TRUE](#), the point size of the final rasterization of polygons is taken from [PointSize](#) when [polygon mode](#) is [VK_POLYGON_MODE_POINT](#).

Otherwise, if [VkPhysicalDeviceMaintenance5PropertiesKHR::polygonModePointSize](#) is [VK_FALSE](#), the point size of the final rasterization of polygons is 1.0 when [polygon mode](#) is [VK_POLYGON_MODE_POINT](#).

To [dynamically set](#) the polygon mode, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetPolygonModeEXT(
    VkCommandBuffer          commandBuffer,
    VkPolygonMode             polygonMode);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `polygonMode` specifies polygon mode.

This command sets the polygon mode for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_POLYGON_MODE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationStateCreateInfo::polygonMode` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetPolygonModeEXT-None-09423
At least one of the following **must** be true:
 - The `extendedDynamicState3PolygonMode` feature is enabled
 - The `shaderObject` feature is enabled
- VUID-vkCmdSetPolygonModeEXT-fillModeNonSolid-07424
If the `fillModeNonSolid` feature is not enabled, `polygonMode` **must** be `VK_POLYGON_MODE_FILL`

Valid Usage (Implicit)

- VUID-vkCmdSetPolygonModeEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetPolygonModeEXT-polygonMode-parameter
`polygonMode` **must** be a valid `VkPolygonMode` value
- VUID-vkCmdSetPolygonModeEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetPolygonModeEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetPolygonModeEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally

synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

25.11.3. Depth Bias

The depth values of all fragments generated by the rasterization of a polygon **can** be biased (offset) by a single depth bias value o that is computed for that polygon.

Depth Bias Enable

The depth bias computation is enabled by the [VkPipelineRasterizationStateCreateInfo::depthBiasEnable](#) value used to create the currently active pipeline. If the depth bias enable is [VK_FALSE](#), no bias is applied and the fragment's depth values are unchanged.

To [dynamically enable](#) whether to bias fragment depth values, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetDepthBiasEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                  depthBiasEnable);
```

- [commandBuffer](#) is the command buffer into which the command will be recorded.
- [depthBiasEnable](#) controls whether to bias fragment depth values.

This command sets the depth bias enable for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the [VkPipelineRasterizationStateCreateInfo::depthBiasEnable](#) value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthBiasEnable-None-08970
At least one of the following **must** be true:
 - the [shaderObject](#) feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBiasEnable-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdSetDepthBiasEnable-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdSetDepthBiasEnable-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdSetDepthBiasEnable-videocoding

This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Depth Bias Computation

The depth bias depends on three parameters:

- `depthBiasSlopeFactor` scales the maximum depth slope m of the polygon
- `depthBiasConstantFactor` scales the parameter r of the depth attachment
- the scaled terms are summed to produce a value which is then clamped to a minimum or maximum value specified by `depthBiasClamp`

`depthBiasSlopeFactor`, `depthBiasConstantFactor`, and `depthBiasClamp` **can** each be positive, negative, or zero. These parameters are set as described for `vkCmdSetDepthBias` and `vkCmdSetDepthBias2EXT` below.

The maximum depth slope m of a triangle is

$$m = \sqrt{\left(\frac{\partial z_f}{\partial x_f}\right)^2 + \left(\frac{\partial z_f}{\partial y_f}\right)^2}$$

where (x_f, y_f, z_f) is a point on the triangle. m **may** be approximated as

$$m = \max\left(\left|\frac{\partial z_f}{\partial x_f}\right|, \left|\frac{\partial z_f}{\partial y_f}\right|\right).$$

In a pipeline with a depth bias representation of `VK_DEPTH_BIAS_REPRESENTATION_FLOAT_EXT`, r , for the given primitive is defined as

$$r = 1$$

Otherwise r is the minimum resolvable difference that depends on the depth attachment representation. If `VkDepthBiasRepresentationInfoEXT::depthBiasExact` is `VK_FALSE` it is the smallest difference in framebuffer coordinate z values that is guaranteed to remain distinct throughout polygon rasterization and in the depth attachment. All pairs of fragments generated by the rasterization of two polygons with otherwise identical vertices, but z_f values that differ by r , will have distinct depth values.

For fixed-point depth attachment representations, or in a pipeline with a depth bias representation of `VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORCE_UNORM_EXT`, r is constant throughout the range of the entire depth attachment. If `VkDepthBiasRepresentationInfoEXT::depthBiasExact` is `VK_TRUE`, then its value **must** be

$$r = 2^{-n}$$

Otherwise its value is implementation-dependent but **must** be at most

$$r = 2 \times 2^{-n}$$

where n is the number of bits used for the depth aspect when using a fixed-point attachment, or the number of mantissa bits plus one when using a floating-point attachment.

Otherwise for floating-point depth attachment, there is no single minimum resolvable difference. In this case, the minimum resolvable difference for a given polygon is dependent on the maximum exponent, e , in the range of z values spanned by the primitive. If n is the number of bits in the floating-point mantissa, the minimum resolvable difference, r , for the given primitive is defined as

$$r = 2^{e-n}$$

If no depth attachment is present, r is undefined.

The bias value o for a polygon is

$$o = \text{dbclamp}(m \times \text{depthBiasSlopeFactor} + r \times \text{depthBiasConstantFactor})$$

$$\text{where } \text{dbclamp}(x) = \begin{cases} x & \text{depthBiasClamp} = 0 \text{ or NaN} \\ \min(x, \text{depthBiasClamp}) & \text{depthBiasClamp} > 0 \\ \max(x, \text{depthBiasClamp}) & \text{depthBiasClamp} < 0 \end{cases}$$

m is computed as described above. If the depth attachment uses a fixed-point representation, m is a function of depth values in the range [0,1], and o is applied to depth values in the same range.

Depth bias is applied to triangle topology primitives received by the rasterizer regardless of [polygon mode](#). Depth bias **may** also be applied to line and point topology primitives received by the rasterizer.

To [dynamically set](#) the depth bias parameters, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetDepthBias(
    VkCommandBuffer          commandBuffer,
    float                    depthBiasConstantFactor,
    float                    depthBiasClamp,
    float                    depthBiasSlopeFactor);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthBiasConstantFactor` is a scalar factor controlling the constant depth value added to each fragment.
- `depthBiasClamp` is the maximum (or minimum) depth bias of a fragment.
- `depthBiasSlopeFactor` is a scalar factor applied to a fragment's slope in depth bias calculations.

This command sets the depth bias parameters for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_DEPTH_BIAS` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the corresponding `VkPipelineRasterizationStateCreateInfo::depthBiasConstantFactor`, `depthBiasClamp`, and `depthBiasSlopeFactor` values used to create the currently active pipeline.

Calling this function is equivalent to calling `vkCmdSetDepthBias2EXT` without a `VkDepthBiasRepresentationInfoEXT` in the pNext chain of `VkDepthBiasInfoEXT`.

Valid Usage

- VUID-vkCmdSetDepthBias-depthBiasClamp-00790
If the `depthBiasClamp` feature is not enabled, `depthBiasClamp` **must** be `0.0`

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBias-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetDepthBias-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdSetDepthBias-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdSetDepthBias-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

The `VkDepthBiasRepresentationInfoEXT` structure is defined as:

```
// Provided by VK_EXT_depth_bias_control
typedef struct VkDepthBiasRepresentationInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkDepthBiasRepresentationEXT depthBiasRepresentation;
    VkBool32                 depthBiasExact;
} VkDepthBiasRepresentationInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthBiasRepresentation` is a `VkDepthBiasRepresentationEXT` value specifying the depth bias representation.
- `depthBiasExact` specifies that the implementation is not allowed to scale the depth bias value to ensure a minimum resolvable distance.

Valid Usage

- VUID-VkDepthBiasRepresentationInfoEXT-leastRepresentableValueForceUnormRepresentation-08947
If the `leastRepresentableValueForceUnormRepresentation` feature is not enabled, `depthBiasRepresentation` **must** not be `VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORCE_UNORM_EXT`
- VUID-VkDepthBiasRepresentationInfoEXT-floatRepresentation-08948

If the `floatRepresentation` feature is not enabled, `depthBiasRepresentation` **must** not be `VK_DEPTH_BIAS_REPRESENTATION_FLOAT_EXT`

- VUID-VkDepthBiasRepresentationInfoEXT-depthBiasExact-08949
If the `depthBiasExact` feature is not enabled, `depthBiasExact` **must** be `VK_FALSE`

Valid Usage (Implicit)

- VUID-VkDepthBiasRepresentationInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEPTH_BIAS_REPRESENTATION_INFO_EXT`
- VUID-VkDepthBiasRepresentationInfoEXT-depthBiasRepresentation-parameter
`depthBiasRepresentation` **must** be a valid `VkDepthBiasRepresentationEXT` value

Possible values of `VkDepthBiasRepresentationInfoEXT::depthBiasRepresentation`, specifying the depth bias representation are:

```
// Provided by VK_EXT_depth_bias_control
typedef enum VkDepthBiasRepresentationEXT {
    VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORMAT_EXT = 0,
    VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORCE_UNORM_EXT = 1,
    VK_DEPTH_BIAS_REPRESENTATION_FLOAT_EXT = 2,
} VkDepthBiasRepresentationEXT;
```

- `VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORMAT_EXT` specifies that the depth bias representation is a factor of the format's `r` as described in [Depth Bias Computation](#).
- `VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORCE_UNORM_EXT` specifies that the depth bias representation is a factor of a constant `r` defined by the bit-size or mantissa of the format as described in [Depth Bias Computation](#).
- `VK_DEPTH_BIAS_REPRESENTATION_FLOAT_EXT` specifies that the depth bias representation is a factor of constant `r` equal to 1.

The `VkDepthBiasInfoEXT` structure is defined as:

```
// Provided by VK_EXT_depth_bias_control
typedef struct VkDepthBiasInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    float              depthBiasConstantFactor;
    float              depthBiasClamp;
    float              depthBiasSlopeFactor;
} VkDepthBiasInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `depthBiasConstantFactor` is a scalar factor controlling the constant depth value added to each fragment.
- `depthBiasClamp` is the maximum (or minimum) depth bias of a fragment.
- `depthBiasSlopeFactor` is a scalar factor applied to a fragment's slope in depth bias calculations.

If `pNext` does not contain a `VkDepthBiasRepresentationInfoEXT` structure, then this command is equivalent to including a `VkDepthBiasRepresentationInfoEXT` with `depthBiasExact` set to `VK_FALSE` and `depthBiasRepresentation` set to `VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORMAT_EXT`.

Valid Usage

- VUID-VkDepthBiasInfoEXT-depthBiasClamp-08950
If the `depthBiasClamp` feature is not enabled, `depthBiasClamp` **must** be `0.0`

Valid Usage (Implicit)

- VUID-VkDepthBiasInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEPTH_BIAS_INFO_EXT`
- VUID-VkDepthBiasInfoEXT-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkDepthBiasRepresentationInfoEXT`
- VUID-VkDepthBiasInfoEXT-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique

To [dynamically set](#) the depth bias parameters, call:

```
// Provided by VK_EXT_depth_bias_control
void vkCmdSetDepthBias2EXT(
    VkCommandBuffer          commandBuffer,
    const VkDepthBiasInfoEXT* pDepthBiasInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pDepthBiasInfo` is a pointer to a `VkDepthBiasInfoEXT` structure specifying depth bias parameters.

This command is functionally identical to `vkCmdSetDepthBias`, but includes extensible sub-structures that include `sType` and `pNext` parameters, allowing them to be more easily extended.

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBias2EXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetDepthBias2EXT-pDepthBiasInfo-parameter

`pDepthBiasInfo` **must** be a valid pointer to a valid `VkDepthBiasInfoEXT` structure

- VUID-vkCmdSetDepthBias2EXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdSetDepthBias2EXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetDepthBias2EXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

25.11.4. Conservative Rasterization

If the `pNext` chain of `VkPipelineRasterizationStateCreateInfo` includes a `VkPipelineRasterizationConservativeStateCreateInfoEXT` structure, then that structure includes parameters controlling conservative rasterization.

`VkPipelineRasterizationConservativeStateCreateInfoEXT` is defined as:

```
// Provided by VK_EXT_conservative_rasterization
typedef struct VkPipelineRasterizationConservativeStateCreateInfoEXT {
    VkStructureType             sType;
    const void*                 pNext;
    VkPipelineRasterizationConservativeStateCreateFlagsEXT flags;
    VkConservativeRasterizationModeEXT
conservativeRasterizationMode;
    float
extraPrimitiveOverestimationSize;
} VkPipelineRasterizationConservativeStateCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `conservativeRasterizationMode` is the conservative rasterization mode to use.
- `extraPrimitiveOverestimationSize` is the extra size in pixels to increase the generating primitive during conservative rasterization at each of its edges in `X` and `Y` equally in screen space beyond the base overestimation specified in `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::primitiveOverestimationSize`. If `conservativeRasterizationMode` is not `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`, this value is ignored.

If this structure is not included in the `pNext` chain, `conservativeRasterizationMode` is considered to be `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`, and conservative rasterization is disabled.

Polygon rasterization **can** be made conservative by setting `conservativeRasterizationMode` to `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT` or `VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT` in `VkPipelineRasterizationConservativeStateCreateInfoEXT`.

NOTE

If `conservativePointAndLineRasterization` is supported, conservative rasterization can be applied to line and point primitives, otherwise it must be disabled.

Valid Usage

- VUID-VkPipelineRasterizationConservativeStateCreateInfoEXT-extraPrimitiveOverestimationSize-01769
`extraPrimitiveOverestimationSize` **must** be in the range of `0.0` to `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::maxExtraPrimitiveOverestimationSize` inclusive

Valid Usage (Implicit)

- VUID-VkPipelineRasterizationConservativeStateCreateInfoEXT-sType-sType **must** be `VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_CONSERVATIVE_STATE_CREATE_INFO_EXT`
- VUID-VkPipelineRasterizationConservativeStateCreateInfoEXT-flags-zero bitmask
`flags` **must** be `0`
- VUID-VkPipelineRasterizationConservativeStateCreateInfoEXT-conservativeRasterizationMode-parameter
`conservativeRasterizationMode` **must** be a valid `VkConservativeRasterizationModeEXT` value

```
// Provided by VK_EXT_conservative_rasterization
typedef VkFlags VkPipelineRasterizationConservativeStateCreateFlagsEXT;
```


`VkPipelineRasterizationConservativeStateCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

Possible values of `VkPipelineRasterizationConservativeStateCreateInfoEXT::conservativeRasterizationMode`, specifying the conservative rasterization mode are:

```
// Provided by VK_EXT_conservative_rasterization
typedef enum VkConservativeRasterizationModeEXT {
    VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT = 0,
    VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT = 1,
    VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT = 2,
} VkConservativeRasterizationModeEXT;
```

- `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT` specifies that conservative rasterization is disabled and rasterization proceeds as normal.
- `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT` specifies that conservative rasterization is enabled in overestimation mode.
- `VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT` specifies that conservative rasterization is enabled in underestimation mode.

To dynamically set the `conservativeRasterizationMode`, call:

```
// Provided by VK_EXT_conservative_rasterization with VK_EXT_extended_dynamic_state3,
// VK_EXT_conservative_rasterization with VK_EXT_shader_object
void vkCmdSetConservativeRasterizationModeEXT(
    VkCommandBuffer                commandBuffer,
    VkConservativeRasterizationModeEXT conservativeRasterizationMode);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `conservativeRasterizationMode` specifies the `conservativeRasterizationMode` state.

This command sets the `conservativeRasterizationMode` state for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationConservativeStateCreateInfoEXT::conservativeRasterizationMode` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetConservativeRasterizationModeEXT-None-09423

At least one of the following **must** be true:

- The `extendedDynamicState3ConservativeRasterizationMode` feature is enabled
- The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetConservativeRasterizationModeEXT-commandBuffer-parameter **commandBuffer** **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetConservativeRasterizationModeEXT-conservativeRasterizationMode-parameter **conservativeRasterizationMode** **must** be a valid [VkConservativeRasterizationModeEXT](#) value
- VUID-vkCmdSetConservativeRasterizationModeEXT-commandBuffer-recording **commandBuffer** **must** be in the [recording state](#)
- VUID-vkCmdSetConservativeRasterizationModeEXT-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdSetConservativeRasterizationModeEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To [dynamically set](#) the [extraPrimitiveOverestimationSize](#), call:

```
// Provided by VK_EXT_conservative_rasterization with VK_EXT_extended_dynamic_state3,  
VK_EXT_conservative_rasterization with VK_EXT_shader_object  
void vkCmdSetExtraPrimitiveOverestimationSizeEXT(  
    VkCommandBuffer          commandBuffer,  
    float                    extraPrimitiveOverestimationSize);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **extraPrimitiveOverestimationSize** specifies the [extraPrimitiveOverestimationSize](#).

This command sets the [extraPrimitiveOverestimationSize](#) for subsequent drawing commands when

drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineRasterizationConservativeStateCreateInfoEXT::extraPrimitiveOverestimationSize` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-None-09423

At least one of the following **must** be true:

- The `extendedDynamicState3ExtraPrimitiveOverestimationSize` feature is enabled
- The `shaderObject` feature is enabled

- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-extraPrimitiveOverestimationSize-07428

`extraPrimitiveOverestimationSize` **must** be in the range of `0.0` to `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::maxExtraPrimitiveOverestimationSize` inclusive

Valid Usage (Implicit)

- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-commandBuffer-parameter `commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-commandBuffer-recording `commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetExtraPrimitiveOverestimationSizeEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

When overestimate conservative rasterization is enabled, rather than evaluating coverage at individual sample locations, a determination is made whether any portion of the pixel (including its edges and corners) is covered by the primitive. If any portion of the pixel is covered, then all bits of the [coverage mask](#) for the fragment corresponding to that pixel are enabled.

For the purposes of evaluating which pixels are covered by the primitive, implementations **can** increase the size of the primitive by up to `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::primitiveOverestimationSize` pixels at each of the primitive edges. This **may** increase the number of fragments generated by this primitive and represents an overestimation of the pixel coverage.

This overestimation size can be increased further by setting the `extraPrimitiveOverestimationSize` value above `0.0` in steps of `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::extraPrimitiveOverestimationSizeGranularity` up to and including `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::extraPrimitiveOverestimationSize`. This **may** further increase the number of fragments generated by this primitive.

The actual precision of the overestimation size used for conservative rasterization **may** vary between implementations and produce results that only approximate the `primitiveOverestimationSize` and `extraPrimitiveOverestimationSizeGranularity` properties.

For triangles if `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT` is enabled, fragments will be generated if the primitive area covers any portion of any pixel inside the fragment area, including their edges or corners. The tie-breaking rule described in [Basic Polygon Rasterization](#) does not apply during conservative rasterization and coverage is set for all fragments generated from shared edges of polygons. Degenerate triangles that evaluate to zero area after rasterization, even for pixels containing a vertex or edge of the zero-area polygon, will be culled if `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::degenerateTrianglesRasterized` is `VK_FALSE` or will generate fragments if `degenerateTrianglesRasterized` is `VK_TRUE`. The fragment input values for these degenerate triangles take their attribute and depth values from the provoking vertex. Degenerate triangles are considered backfacing and the application **can** enable backface culling if desired. Triangles that are zero area before rasterization **may** be culled regardless.

For lines if `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT` is enabled, and the implementation sets `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::conservativePointAndLineRasterization` to `VK_TRUE`, fragments will be generated if the line covers any portion of any pixel inside the fragment area, including their edges or corners. Degenerate lines that evaluate to zero length after rasterization will be culled if `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::degenerateLinesRasterized` is `VK_FALSE` or will generate fragments if `degenerateLinesRasterized` is `VK_TRUE`. The fragments input values for these degenerate lines take their attribute and depth values from the provoking vertex. Lines that

are zero length before rasterization **may** be culled regardless.

For points if `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT` is enabled, and the implementation sets `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::conservativePointAndLineRasterization` to `VK_TRUE`, fragments will be generated if the point square covers any portion of any pixel inside the fragment area, including their edges or corners.

When underestimate conservative rasterization is enabled, rather than evaluating coverage at individual sample locations, a determination is made whether all of the pixel (including its edges and corners) is covered by the primitive. If the entire pixel is covered, then a fragment is generated with all bits of its `coverage mask` corresponding to the pixel enabled, otherwise the pixel is not considered covered even if some portion of the pixel is covered. The fragment is discarded if no pixels inside the fragment area are considered covered.

For triangles, if `VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT` is enabled, fragments will only be generated if any pixel inside the fragment area is fully covered by the generating primitive, including its edges and corners.

For lines, if `VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT` is enabled, fragments will be generated if any pixel inside the fragment area, including its edges and corners, are entirely covered by the line.

For points, if `VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT` is enabled, fragments will only be generated if the point square covers the entirety of any pixel square inside the fragment area, including its edges or corners.

For both overestimate and underestimate conservative rasterization modes a fragment has all of its pixel squares fully covered by the generating primitive **must** set `FullyCoveredEXT` to `VK_TRUE` if the implementation enables the `VkPhysicalDeviceConservativeRasterizationPropertiesEXT::fullyCoveredFragmentShaderInputVariable` feature.

When setting the `fragment shading rate` results in fragments covering multiple pixels, coverage for conservative rasterization is still evaluated on a per-pixel basis and may result in fragments with partial coverage. For fragment shader inputs decorated with `FullyCoveredEXT`, a fragment is considered fully covered if and only if all pixels in the fragment are fully covered by the generating primitive.

Chapter 26. Fragment Operations

Fragments produced by rasterization go through a number of operations to determine whether or how values produced by fragment shading are written to the framebuffer.

The following fragment operations adhere to [rasterization order](#), and are typically performed in this order:

1. [Discard rectangles test](#)
2. [Scissor test](#)
3. [Sample mask test](#)
4. Certain [Fragment shading](#) operations:
 - [Sample Mask Accesses](#)
 - [Tile Image Reads](#)
 - [Depth Replacement](#)
 - [Stencil Reference Replacement](#)
 - [Interlocked Operations](#)
5. [Multisample coverage](#)
6. [Depth bounds test](#)
7. [Stencil test](#)
8. [Depth test](#)
9. [Sample counting](#)
10. [Coverage reduction](#)

The [coverage mask](#) generated by rasterization describes the initial coverage of each sample covered by the fragment. Fragment operations will update the coverage mask to add or subtract coverage where appropriate. If a fragment operation results in all bits of the coverage mask being 0, the fragment is discarded, and no further operations are performed. Fragments can also be programmatically discarded in a fragment shader by executing one of

- [OpTerminateInvocation](#)
- [OpKill](#).

When one of the fragment operations in this chapter is described as “replacing” a fragment shader output, that output is replaced unconditionally, even if no fragment shader previously wrote to that output.

If there is a [fragment shader](#) and it declares the [PostDepthCoverage](#) execution mode, the [sample mask test](#) is instead performed after the [depth test](#).

If [VkPhysicalDeviceMaintenance5PropertiesKHR::earlyFragmentMultisampleCoverageAfterSampleCounting](#) is [VK_TRUE](#) and there is a [fragment shader](#) which declares the [EarlyFragmentTests](#) execution mode, [fragment shading](#) and [multisample](#)

coverage operations **must** be performed after sample counting.

Otherwise, if `VkPhysicalDeviceMaintenance5PropertiesKHR::earlyFragmentMultisampleCoverageAfterSampleCounting` is `VK_FALSE` and there is a fragment shader which declares the `EarlyFragmentTests` execution mode, fragment shading and multisample coverage operations **should** instead be performed after sample counting, but **may** be performed before sample counting.

If `VkPhysicalDeviceMaintenance5PropertiesKHR::earlyFragmentSampleMaskTestBeforeSampleCounting` is `VK_TRUE` and there is a fragment shader which declares the `EarlyFragmentTests` execution mode sample mask test operations **must** follow the order of fragment operations from above.

Otherwise, if `VkPhysicalDeviceMaintenance5PropertiesKHR::earlyFragmentSampleMaskTestBeforeSampleCounting` is `VK_FALSE` and there is a fragment shader which declares the `EarlyFragmentTests` execution mode, sample mask test operations **should** follow the order of fragment operations from above but **may** instead be performed after sample counting.

For a pipeline with the following properties:

- a fragment shader is specified
- the fragment shader does not write to storage resources;
- the fragment shader specifies the `DepthReplacing` execution mode; and
- either
 - the fragment shader specifies the `DepthUnchanged` execution mode;
 - the fragment shader specifies the `DepthLess` execution mode and the pipeline uses a `VkPipelineDepthStencilStateCreateInfo::depthCompareOp` of `VK_COMPARE_OP_GREATER` or `VK_COMPARE_OP_GREATER_OR_EQUAL`; or
 - the fragment shader specifies the `DepthGreater` execution mode and the pipeline uses a `VkPipelineDepthStencilStateCreateInfo::depthCompareOp` of `VK_COMPARE_OP_LESS` or `VK_COMPARE_OP_LESS_OR_EQUAL`

the implementation **may** perform depth bounds test before fragment shading and perform an additional depth test immediately after that using the interpolated depth value generated by rasterization.

Once all fragment operations have completed, fragment shader outputs for covered color attachment samples pass through framebuffer operations.

26.1. Discard Rectangles Test

The discard rectangle test compares the framebuffer coordinates (x_b, y_f) of each sample covered by a fragment against a set of *discard rectangles*.

Each discard rectangle is defined by a `VkRect2D`. These values are either set by the `VkPipelineDiscardRectangleStateCreateInfoEXT` structure during pipeline creation, or dynamically by the `vkCmdSetDiscardRectangleEXT` command.

A given sample is considered inside a discard rectangle if the x_f is in the range `[VkRect2D::offset.x, VkRect2D::offset.x + VkRect2D::extent.x)`, and y_f is in the range `[VkRect2D::offset.y, VkRect2D::offset.y + VkRect2D::extent.y)`. If the test is set to be inclusive, samples that are not inside any of the discard rectangles will have their coverage set to 0. If the test is set to be exclusive, samples that are inside any of the discard rectangles will have their coverage set to 0.

If no discard rectangles are specified, the coverage mask is unmodified by this operation.

The `VkPipelineDiscardRectangleStateCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_discard_rectangles
typedef struct VkPipelineDiscardRectangleStateCreateInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkPipelineDiscardRectangleStateCreateFlagsEXT flags;
    VkDiscardRectangleModeEXT discardRectangleMode;
    uint32_t                 discardRectangleCount;
    const VkRect2D*          pDiscardRectangles;
} VkPipelineDiscardRectangleStateCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `discardRectangleMode` is a `VkDiscardRectangleModeEXT` value determining whether the discard rectangle test is inclusive or exclusive.
- `discardRectangleCount` is the number of discard rectangles to use.
- `pDiscardRectangles` is a pointer to an array of `VkRect2D` structures defining discard rectangles.

If the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state is enabled for a pipeline, the `pDiscardRectangles` member is ignored. If the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` dynamic state is not enabled for the pipeline the presence of this structure in the `VkGraphicsPipelineCreateInfo` chain, and a `discardRectangleCount` greater than zero, implicitly enables discard rectangles in the pipeline, otherwise discard rectangles **must** be enabled or disabled by `vkCmdSetDiscardRectangleEnableEXT`. If the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` dynamic state is enabled for the pipeline, the `discardRectangleMode` member is ignored, and the discard rectangle mode **must** be set by `vkCmdSetDiscardRectangleModeEXT`.

When this structure is included in the `pNext` chain of `VkGraphicsPipelineCreateInfo`, it defines parameters of the discard rectangle test. If the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state is not enabled, and this structure is not included in the `pNext` chain, it is equivalent to specifying this structure with a `discardRectangleCount` of 0.

Valid Usage

- VUID-VkPipelineDiscardRectangleStateCreateInfoEXT-discardRectangleCount-00582
`discardRectangleCount` **must** be less than or equal to

Valid Usage (Implicit)

- VUID-VkPipelineDiscardRectangleStateCreateInfoEXT-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PIPELINE_DISCARD_RECTANGLE_STATE_CREATE_INFO_EXT
- VUID-VkPipelineDiscardRectangleStateCreateInfoEXT-flags-zero bitmask
flags **must** be 0
- VUID-VkPipelineDiscardRectangleStateCreateInfoEXT-discardRectangleMode-parameter
discardRectangleMode **must** be a valid VkDiscardRectangleModeEXT value

```
// Provided by VK_EXT_discard_rectangles
typedef VkFlags VkPipelineDiscardRectangleStateCreateFlagsEXT;
```

VkPipelineDiscardRectangleStateCreateFlagsEXT is a bitmask type for setting a mask, but is currently reserved for future use.

VkDiscardRectangleModeEXT values are:

```
// Provided by VK_EXT_discard_rectangles
typedef enum VkDiscardRectangleModeEXT {
    VK_DISCARD_RECTANGLE_MODE_INCLUSIVE_EXT = 0,
    VK_DISCARD_RECTANGLE_MODE_EXCLUSIVE_EXT = 1,
} VkDiscardRectangleModeEXT;
```

- VK_DISCARD_RECTANGLE_MODE_INCLUSIVE_EXT specifies that the discard rectangle test is inclusive.
- VK_DISCARD_RECTANGLE_MODE_EXCLUSIVE_EXT specifies that the discard rectangle test is exclusive.

To dynamically set the discard rectangles, call:

```
// Provided by VK_EXT_discard_rectangles
void vkCmdSetDiscardRectangleEXT(
    VkCommandBuffer          commandBuffer,
    uint32_t                 firstDiscardRectangle,
    uint32_t                 discardRectangleCount,
    const VkRect2D*          pDiscardRectangles);
```

- commandBuffer is the command buffer into which the command will be recorded.
- firstDiscardRectangle is the index of the first discard rectangle whose state is updated by the command.
- discardRectangleCount is the number of discard rectangles whose state are updated by the command.

- `pDiscardRectangles` is a pointer to an array of `VkRect2D` structures specifying discard rectangles.

The discard rectangle taken from element `i` of `pDiscardRectangles` replace the current state for the discard rectangle at index `firstDiscardRectangle + i`, for `i` in `[0, discardRectangleCount)`.

This command sets the discard rectangles for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineDiscardRectangleStateCreateInfoEXT::pDiscardRectangles` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDiscardRectangleEXT-firstDiscardRectangle-00585
The sum of `firstDiscardRectangle` and `discardRectangleCount` **must** be less than or equal to `VkPhysicalDeviceDiscardRectanglePropertiesEXT::maxDiscardRectangles`
- VUID-vkCmdSetDiscardRectangleEXT-x-00587
The `x` and `y` member of `offset` in each `VkRect2D` element of `pDiscardRectangles` **must** be greater than or equal to `0`
- VUID-vkCmdSetDiscardRectangleEXT-offset-00588
Evaluation of `(offset.x + extent.width)` in each `VkRect2D` element of `pDiscardRectangles` **must** not cause a signed integer addition overflow
- VUID-vkCmdSetDiscardRectangleEXT-offset-00589
Evaluation of `(offset.y + extent.height)` in each `VkRect2D` element of `pDiscardRectangles` **must** not cause a signed integer addition overflow

Valid Usage (Implicit)

- VUID-vkCmdSetDiscardRectangleEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetDiscardRectangleEXT-pDiscardRectangles-parameter
`pDiscardRectangles` **must** be a valid pointer to an array of `discardRectangleCount` `VkRect2D` structures
- VUID-vkCmdSetDiscardRectangleEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetDiscardRectangleEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetDiscardRectangleEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetDiscardRectangleEXT-discardRectangleCount-arraylength
`discardRectangleCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To [dynamically set](#) whether discard rectangles are enabled, call:

```
// Provided by VK_EXT_discard_rectangles
void vkCmdSetDiscardRectangleEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                 discardRectangleEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `discardRectangleEnable` specifies whether discard rectangles are enabled or not.

This command sets the discard rectangle enable for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is implied by the `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount` value used to create the currently active pipeline, where a non-zero `discardRectangleCount` implicitly enables discard rectangles, otherwise they are disabled.

Valid Usage

- VUID-vkCmdSetDiscardRectangleEnableEXT-specVersion-07851
The [VK_EXT_discard_rectangles](#) extension **must** be enabled, and the implementation **must** support at least [specVersion 2](#) of this extension

Valid Usage (Implicit)

- VUID-vkCmdSetDiscardRectangleEnableEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetDiscardRectangleEnableEXT-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdSetDiscardRectangleEnableEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetDiscardRectangleEnableEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To `dynamically set` the discard rectangle mode, call:

```
// Provided by VK_EXT_discard_rectangles
void vkCmdSetDiscardRectangleModeEXT(
    VkCommandBuffer          commandBuffer,
    VkDiscardRectangleModeEXT discardRectangleMode);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `discardRectangleMode` specifies the discard rectangle mode for all discard rectangles, either inclusive or exclusive.

This command sets the discard rectangle mode for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleMode` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDiscardRectangleModeEXT-specVersion-07852
The `VK_EXT_discard_rectangles` extension **must** be enabled, and the implementation **must**

support at least **specVersion 2** of this extension

Valid Usage (Implicit)

- VUID-vkCmdSetDiscardRectangleModeEXT-commandBuffer-parameter **commandBuffer** **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdSetDiscardRectangleModeEXT-discardRectangleMode-parameter **discardRectangleMode** **must** be a valid **VkDiscardRectangleModeEXT** value
- VUID-vkCmdSetDiscardRectangleModeEXT-commandBuffer-recording **commandBuffer** **must** be in the **recording state**
- VUID-vkCmdSetDiscardRectangleModeEXT-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdSetDiscardRectangleModeEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

26.2. Scissor Test

The scissor test compares the framebuffer coordinates (x_f, y_f) of each sample covered by a fragment against a *scissor rectangle* at the index equal to the fragment's **ViewportIndex**.

Each scissor rectangle is defined by a **VkRect2D**. These values are either set by the **VkPipelineViewportStateCreateInfo** structure during pipeline creation, or dynamically by the **vkCmdSetScissor** command.

A given sample is considered inside a scissor rectangle if x_f is in the range [**VkRect2D::offset.x**, **VkRect2D::offset.x** + **VkRect2D::extent.x**), and y_f is in the range [**VkRect2D::offset.y**, **VkRect2D::offset.y** + **VkRect2D::extent.y**). Samples with coordinates outside the scissor rectangle at the

corresponding `ViewportIndex` will have their coverage set to 0.

To dynamically set the scissor rectangles, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetScissor(
    VkCommandBuffer          commandBuffer,
    uint32_t                 firstScissor,
    uint32_t                 scissorCount,
    const VkRect2D*          pScissors);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstScissor` is the index of the first scissor whose state is updated by the command.
- `scissorCount` is the number of scissors whose rectangles are updated by the command.
- `pScissors` is a pointer to an array of `VkRect2D` structures defining scissor rectangles.

The scissor rectangles taken from element `i` of `pScissors` replace the current state for the scissor index `firstScissor + i`, for `i` in `[0, scissorCount)`.

This command sets the scissor rectangles for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_SCISSOR` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineViewportStateCreateInfo::pScissors` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetScissor-firstScissor-00592
The sum of `firstScissor` and `scissorCount` **must** be between 1 and `VkPhysicalDeviceLimits::maxViewports`, inclusive
- VUID-vkCmdSetScissor-firstScissor-00593
If the `multiViewport` feature is not enabled, `firstScissor` **must** be 0
- VUID-vkCmdSetScissor-scissorCount-00594
If the `multiViewport` feature is not enabled, `scissorCount` **must** be 1
- VUID-vkCmdSetScissor-x-00595
The `x` and `y` members of `offset` member of any element of `pScissors` **must** be greater than or equal to 0
- VUID-vkCmdSetScissor-offset-00596
Evaluation of `(offset.x + extent.width)` **must** not cause a signed integer addition overflow for any element of `pScissors`
- VUID-vkCmdSetScissor-offset-00597
Evaluation of `(offset.y + extent.height)` **must** not cause a signed integer addition overflow for any element of `pScissors`

Valid Usage (Implicit)

- VUID-vkCmdSetScissor-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetScissor-pScissors-parameter
`pScissors` **must** be a valid pointer to an array of `scissorCount` `VkRect2D` structures
- VUID-vkCmdSetScissor-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdSetScissor-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetScissor-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetScissor-scissorCount-arraylength
`scissorCount` **must** be greater than 0

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

26.3. Sample Mask Test

The sample mask test compares the `coverage mask` for a fragment with the *sample mask* defined by `VkPipelineMultisampleStateCreateInfo::pSampleMask`.

To `dynamically set` the sample mask, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetSampleMaskEXT(
    VkCommandBuffer                commandBuffer,
    VkSampleCountFlagBits          samples,
    const VkSampleMask*            pSampleMask);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `samples` specifies the number of sample bits in the `pSampleMask`.
- `pSampleMask` is a pointer to an array of `VkSampleMask` values, where the array size is based on the `samples` parameter.

This command sets the sample mask for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineMultisampleStateCreateInfo::pSampleMask` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetSampleMaskEXT-None-09423
At least one of the following **must** be true:
 - The `extendedDynamicState3SampleMask` feature is enabled
 - The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetSampleMaskEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetSampleMaskEXT-samples-parameter
`samples` **must** be a valid `VkSampleCountFlagBits` value
- VUID-vkCmdSetSampleMaskEXT-pSampleMask-parameter
`pSampleMask` **must** be a valid pointer to an array of $\lceil \frac{\text{samples}}{32} \rceil$ `VkSampleMask` values
- VUID-vkCmdSetSampleMaskEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetSampleMaskEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetSampleMaskEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Each bit of the coverage mask is associated with a sample index as described in the [rasterization chapter](#). If the bit in `VkPipelineMultisampleStateCreateInfo::pSampleMask` which is associated with that same sample index is set to 0, the coverage mask bit is set to 0.

26.4. Fragment Shading

[Fragment shaders](#) are invoked for each fragment, or as [helper invocations](#).

Most operations in the fragment shader are not performed in [rasterization order](#), with exceptions called out in the following sections.

For fragment shaders invoked by fragments, the following rules apply:

- A fragment shader **must** not be executed if a [fragment operation](#) that executes before fragment shading discards the fragment.
- A fragment shader **may** not be executed if:
 - An implementation determines that another fragment shader, invoked by a subsequent primitive in [primitive order](#), overwrites all results computed by the shader (including writes to storage resources).
 - Any other [fragment operation](#) discards the fragment, and the shader does not write to any storage resources.
 - If a fragment shader statically computes the same values for different framebuffer locations, and does not write to any storage resources, multiple fragments **may** be shaded by one fragment shader invocation. This **may** affect `VK_QUERY_PIPELINE_STATISTIC_FRAGMENT_SHADER_INVOCATIONS_BIT` results, but **must** otherwise not be visible behavior to applications.
- Otherwise, at least one fragment shader **must** be executed.
 - If [sample shading](#) is enabled and multiple invocations per fragment are **required**, additional invocations **must** be executed as specified.

- Each covered sample **must** be included in at least one fragment shader invocation.

If no fragment shader is included in the pipeline, no fragment shader is executed, and undefined values **may** be written to all color attachment outputs during this fragment operation.

NOTE

Multiple fragment shader invocations may be executed for the same fragment for any number of implementation-dependent reasons. When there is more than one fragment shader invocation per fragment, the association of samples to invocations is implementation-dependent. Stores and atomics performed by these additional invocations have the normal effect.

For example, if the subpass includes multiple views in its view mask, a fragment shader may be invoked separately for each view.

26.4.1. Sample Mask

Reading from the `SampleMask` built-in in the `Input` storage class will return the coverage mask for the current fragment as calculated by fragment operations that executed prior to fragment shading.

If `sample shading` is enabled, fragment shaders will only see values of `1` for samples being shaded - other bits will be `0`.

Each bit of the coverage mask is associated with a sample index as described in the [rasterization chapter](#). If the bit in `SampleMask` which is associated with that same sample index is set to `0`, that coverage mask bit is set to `0`.

Values written to the `SampleMask` built-in in the `Output` storage class will be used by the `multisample coverage` operation, with the same encoding as the input built-in.

26.4.2. Fragment Shader Tile Image Reads

If the `VK_EXT_shader_tile_image` extension is enabled, implementations divide the framebuffer into a grid of tiles. A `tile image` is a view of a framebuffer attachment tile for fragments with locations within the tile.

Within a render pass instance initiated by `vkCmdBeginRenderingKHR`, fragment shader invocations **can** read the framebuffer color, depth, and stencil values at the fragment location via tile images.

NOTE

Even though fragment shader invocation can only read from the corresponding fragment location, the abstraction of a tile image is introduced for the following reasons:

- Tile dimensions will be exposed in a future extension
- Future functionality such as executing compute dispatches within render passes via tile shaders can leverage tile images.

Enabling `shaderTileImageColorReadAccess`, `shaderTileImageDepthReadAccess`, `shaderTileImageStencilReadAccess` enables fragment shader invocations to read from color, depth,

and stencil, respectively.

Color values are read from tile image variables with `OpColorAttachmentReadEXT`. Tile image variables are linked to specific color attachments using `Location` decoration. See [Fragment Tile Image Interface](#) for more details.

Depth values are read with `OpDepthAttachmentReadEXT`.

Stencil values are read with `OpStencilAttachmentReadEXT`.

The sample to read is specified by a `sample index` value specified as the `Sample` operand to `OpColorAttachmentReadEXT`, `OpDepthAttachmentReadEXT`, or `OpStencilAttachmentReadEXT`.

If `sample shading` is disabled, a fragment invocation **can** read from all sample locations associated with the fragment regardless of the fragment's coverage. This functionality is supported for `VkPipelineMultisampleStateCreateInfo::rasterizationSamples` > 1 when `VkPhysicalDeviceShaderTileImagePropertiesEXT::shaderTileImageReadSampleFromPixelRateInvocation` is `VK_TRUE`.

If `sample shading` is enabled, and `minSampleShading` is 1.0, a fragment invocation **must** only read from the `coverage index` sample. Tile image access **must** not be used if the value of `minSampleShading` is not 1.0.

If the `fragment shader` declares the `EarlyFragmentTests` execution mode, depth reads are allowed only if depth writes are disabled and stencil reads are allowed only if stencil writes are disabled.

If `VkPhysicalDeviceShaderTileImagePropertiesEXT::shaderTileImageReadFromHelperInvocation` is `VK_FALSE`, values read from helper invocations are undefined otherwise the values read are subject to the coherency guarantees described below.

`OpDepthAttachmentReadEXT` returns an undefined value if no depth attachment is present.

`OpStencilAttachmentReadEXT` returns an undefined value if no stencil attachment is present.

Tile image reads from color, depth and stencil attachments are said to be coherent when the accesses happen in raster order and without `data race` with respect to accesses to the attachments from framebuffer-space pipeline stages. The samples which qualify for coherent access and the enabling conditions are described below.

- Let R_c be the set of components being read from an attachment A in a draw call
- Let W_c be the set of components being written to A by the draw call

The samples which qualify for coherent tile image reads from an attachment A are:

- All samples in a pixel when R_c is disjoint with W_c .
- The samples with coverage in a fragment when R_c is not disjoint with W_c . The samples with coverage are determined by the coverage mask for the fragment as calculated by fragment operations that executed prior to fragment shading, including early fragment tests if enabled for the draw call.

A fragment shader **can** declare `NonCoherentColorAttachmentReadEXT`,

`NonCoherentDepthAttachmentReadEXT`, or `NonCoherentStencilAttachmentReadEXT` execution modes to enable non-coherent tile image reads which require an explicit `vkCmdPipelineBarrier2KHR` call for the writes to an attachment to be made visible via tile image reads.

When `VkPhysicalDeviceShaderTileImagePropertiesEXT::shaderTileImageCoherentReadAccelerated` is `VK_TRUE`, the implementation prefers that coherent tile image reads are used, otherwise the implementation prefers that non-coherent tile image reads are used.

NOTE

In practice, the most common tile image reads usage patterns fall under one of the following:

- Programmable blending - each fragment reads from a single sample (`SampleID`) at its location. Per-sample shading is typically enabled when multisampled rendertargets are used.
- G-buffer generation and shading in one render pass - in the shading phase a fragment reads from a single sample at its location.
- Programmable resolve - a fragment reads from all samples at its location (per-sample shading is disabled). This requires the use of a "full-screen triangle" instead of a rectangle composed of two triangles in order to avoid data races along the shared edge of the triangles.
- 1:1 texturing with LOD - in use cases such a deferred screen space decals a fragment reads a single sample (`SampleID`) from depth buffer, but requires being able to read from helper threads to derive the texture LOD. This use case is supported as long as the attachment components being read are not overwritten by color, depth, or stencil attachment writes.

All of the above use cases are supported by coherent tile image reads, but only the latter three are supported when non-coherent reads are used as there is no mechanism to synchronize non-coherent reads with writes within a draw call.

26.4.3. Depth Replacement

Writing to the `FragDepth` built-in will replace the fragment's calculated depth values for each sample in the input `SampleMask`. `Depth testing` performed after the fragment shader for this fragment will use this new value as z_f .

26.4.4. Stencil Reference Replacement

Writing to the `FragStencilRefEXT` built-in will replace the fragment's stencil reference value for each sample in the input `SampleMask`. `Stencil testing` performed after the fragment shader for this fragment will use this new value as s_r .

26.4.5. Interlocked Operations

`OpBeginInvocationInterlockEXT` and `OpEndInvocationInterlockEXT` define a section of a fragment shader which imposes additional ordering constraints on operations performed within them. These operations are defined as *interlocked operations*. How interlocked operations are ordered against

other fragment shader invocations depends on the specified execution modes.

If the `ShadingRateInterlockOrderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before interlocked operations in fragment shader invocations that execute later in `rasterization order` and cover at least one sample in the same fragment area, and **must** happen after interlocked operations in a fragment shader that executes earlier in `rasterization order` and cover at least one sample in the same fragment area.

If the `ShadingRateInterlockUnorderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before or after interlocked operations in fragment shader invocations that execute earlier or later in `rasterization order` and cover at least one sample in the same fragment area.

If the `PixelInterlockOrderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before interlocked operations in fragment shader invocations that execute later in `rasterization order` and cover at least one sample in the same pixel, and **must** happen after interlocked operations in a fragment shader that executes earlier in `rasterization order` and cover at least one sample in the same pixel.

If the `PixelInterlockUnorderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before or after interlocked operations in fragment shader invocations that execute earlier or later in `rasterization order` and cover at least one sample in the same pixel.

If the `SampleInterlockOrderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before interlocked operations in fragment shader invocations that execute later in `rasterization order` and cover at least one of the same samples, and **must** happen after interlocked operations in a fragment shader that executes earlier in `rasterization order` and cover at least one of the same samples.

If the `SampleInterlockUnorderedEXT` execution mode is specified, any interlocked operations in a fragment shader **must** happen before or after interlocked operations in fragment shader invocations that execute earlier or later in `rasterization order` and cover at least one of the same samples.

26.5. Multisample Coverage

If a fragment shader is active and its entry point's interface includes a built-in output variable decorated with `SampleMask`, the coverage mask is **ANDed** with the bits of the `SampleMask` built-in to generate a new coverage mask. If `sample shading` is enabled, bits written to `SampleMask` corresponding to samples that are not being shaded by the fragment shader invocation are ignored. If no fragment shader is active, or if the active fragment shader does not include `SampleMask` in its interface, the coverage mask is not modified.

Next, the fragment alpha value and coverage mask are modified based on the line coverage factor if the `lineRasterizationMode` member of the `VkPipelineRasterizationStateCreateInfo` structure is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, and the `alphaToCoverageEnable` and `alphaToOneEnable` members of the `VkPipelineMultisampleStateCreateInfo` structure.

To **dynamically** set the **alphaToCoverageEnable** state, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetAlphaToCoverageEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                 alphaToCoverageEnable);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **alphaToCoverageEnable** specifies the **alphaToCoverageEnable** state.

This command sets the **alphaToCoverageEnable** state for subsequent drawing commands when drawing using **shader objects**, or when the graphics pipeline is created with **VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT** set in **VkPipelineDynamicStateCreateInfo::pDynamicStates**. Otherwise, this state is specified by the **VkPipelineMultisampleStateCreateInfo::alphaToCoverageEnable** value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetAlphaToCoverageEnableEXT-None-09423
At least one of the following **must** be true:
 - The **extendedDynamicState3AlphaToCoverageEnable** feature is enabled
 - The **shaderObject** feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetAlphaToCoverageEnableEXT-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdSetAlphaToCoverageEnableEXT-commandBuffer-recording
commandBuffer **must** be in the **recording state**
- VUID-vkCmdSetAlphaToCoverageEnableEXT-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdSetAlphaToCoverageEnableEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To **dynamically set** the **alphaToOneEnable** state, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetAlphaToOneEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                 alphaToOneEnable);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **alphaToOneEnable** specifies the **alphaToOneEnable** state.

This command sets the **alphaToOneEnable** state for subsequent drawing commands when drawing using **shader objects**, or when the graphics pipeline is created with **VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT** set in **VkPipelineDynamicStateCreateInfo::pDynamicStates**. Otherwise, this state is specified by the **VkPipelineMultisampleStateCreateInfo::alphaToOneEnable** value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetAlphaToOneEnableEXT-None-09423
At least one of the following **must** be true:
 - The **extendedDynamicState3AlphaToOneEnable** feature is enabled
 - The **shaderObject** feature is enabled
- VUID-vkCmdSetAlphaToOneEnableEXT-alphaToOne-07607
If the **alphaToOne** feature is not enabled, **alphaToOneEnable** **must** be **VK_FALSE**

Valid Usage (Implicit)

- VUID-vkCmdSetAlphaToOneEnableEXT-commandBuffer-parameter
commandBuffer **must** be a valid **VkCommandBuffer** handle
- VUID-vkCmdSetAlphaToOneEnableEXT-commandBuffer-recording
commandBuffer **must** be in the **recording state**
- VUID-vkCmdSetAlphaToOneEnableEXT-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support graphics operations

- VUID-vkCmdSetAlphaToOneEnableEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

All alpha values in this section refer only to the alpha component of the fragment shader output that has a `Location` and `Index` decoration of zero (see the [Fragment Output Interface](#) section). If that shader output has an integer or unsigned integer type, then these operations are skipped.

If the `lineRasterizationMode` member of the `VkPipelineRasterizationStateCreateInfo` structure is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR` and the fragment came from a line segment, then the alpha value is replaced by multiplying it by the coverage factor for the fragment computed during [smooth line rasterization](#).

If `alphaToCoverageEnable` is enabled, a temporary coverage mask is generated where each bit is determined by the fragment's alpha value, which is ANDed with the fragment coverage mask.

No specific algorithm is specified for converting the alpha value to a temporary coverage mask. It is intended that the number of 1's in this value be proportional to the alpha value (clamped to [0,1]), with all 1's corresponding to a value of 1.0 and all 0's corresponding to 0.0. The algorithm **may** be different at different framebuffer coordinates.

NOTE

Using different algorithms at different framebuffer coordinates **may** help to avoid artifacts caused by regular coverage sample locations.

Finally, if `alphaToOneEnable` is enabled, each alpha value is replaced by the maximum representable alpha value for fixed-point color attachments, or by 1.0 for floating-point attachments. Otherwise, the alpha values are not changed.

26.6. Depth and Stencil Operations

Pipeline state controlling the [depth bounds tests](#), [stencil test](#), and [depth test](#) is specified through the members of the `VkPipelineDepthStencilStateCreateInfo` structure.

The `VkPipelineDepthStencilStateCreateInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineDepthStencilStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineDepthStencilStateCreateFlags flags;
    VkBool32                  depthTestEnable;
    VkBool32                  depthWriteEnable;
    VkCompareOp               depthCompareOp;
    VkBool32                  depthBoundsTestEnable;
    VkBool32                  stencilTestEnable;
    VkStencilOpState          front;
    VkStencilOpState          back;
    float                     minDepthBounds;
    float                     maxDepthBounds;
} VkPipelineDepthStencilStateCreateInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `depthTestEnable` controls whether `depth testing` is enabled.
- `depthWriteEnable` controls whether `depth writes` are enabled when `depthTestEnable` is `VK_TRUE`. Depth writes are always disabled when `depthTestEnable` is `VK_FALSE`.
- `depthCompareOp` is a `VkCompareOp` value specifying the comparison operator to use in the `Depth Comparison` step of the `depth test`.
- `depthBoundsTestEnable` controls whether `depth bounds testing` is enabled.
- `stencilTestEnable` controls whether `stencil testing` is enabled.
- `front` and `back` are `VkStencilOpState` values controlling the corresponding parameters of the `stencil test`.
- `minDepthBounds` is the minimum depth bound used in the `depth bounds test`.
- `maxDepthBounds` is the maximum depth bound used in the `depth bounds test`.

Valid Usage

- VUID-VkPipelineDepthStencilStateCreateInfo-depthBoundsTestEnable-00598
If the `depthBounds` feature is not enabled, `depthBoundsTestEnable` **must** be `VK_FALSE`
- VUID-VkPipelineDepthStencilStateCreateInfo-separateStencilMaskRef-04453
If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::separateStencilMaskRef` is `VK_FALSE`, and the value of `VkPipelineDepthStencilStateCreateInfo::stencilTestEnable` is `VK_TRUE`, and the value of `VkPipelineRasterizationStateCreateInfo::cullMode` is `VK_CULL_MODE_NONE`, the value of `reference` in each of the `VkStencilOpState` structs in `front` and `back` **must** be the same

Valid Usage (Implicit)

- VUID-VkPipelineDepthStencilStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_DEPTH_STENCIL_STATE_CREATE_INFO`
- VUID-VkPipelineDepthStencilStateCreateInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPipelineDepthStencilStateCreateInfo-flags-zerobitmask
`flags` **must** be `0`
- VUID-VkPipelineDepthStencilStateCreateInfo-depthCompareOp-parameter
`depthCompareOp` **must** be a valid `VkCompareOp` value
- VUID-VkPipelineDepthStencilStateCreateInfo-front-parameter
`front` **must** be a valid `VkStencilOpState` structure
- VUID-VkPipelineDepthStencilStateCreateInfo-back-parameter
`back` **must** be a valid `VkStencilOpState` structure

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineDepthStencilStateCreateFlags;
```

`VkPipelineDepthStencilStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

26.7. Depth Bounds Test

The depth bounds test compares the depth value z_a in the depth/stencil attachment at each sample's framebuffer coordinates (x_f, y_f) and `sample index` i against a set of *depth bounds*.

The depth bounds are determined by two floating-point values defining a minimum (`minDepthBounds`) and maximum (`maxDepthBounds`) depth value. These values are either set by the `VkPipelineDepthStencilStateCreateInfo` structure during pipeline creation, or dynamically by `vkCmdSetDepthBoundsTestEnableEXT` and `vkCmdSetDepthBounds`.

A given sample is considered within the depth bounds if z_a is in the range `[minDepthBounds, maxDepthBounds]`. Samples with depth attachment values outside of the depth bounds will have their coverage set to `0`.

If the depth bounds test is disabled, or if there is no depth attachment, the coverage mask is unmodified by this operation.

To [dynamically enable or disable](#) the depth bounds test, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetDepthBoundsTestEnableEXT(
    VkCommandBuffer                      commandBuffer,
    VkBool32                             depthBoundsTestEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthBoundsTestEnable` specifies if the depth bounds test is enabled.

This command sets the depth bounds enable for subsequent drawing commands when drawing using `shader objects`. Otherwise, this state is specified by the `VkPipelineDepthStencilStateCreateInfo::depthBoundsTestEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthBoundsTestEnable-None-08971
At least one of the following **must** be true:
 - the `shaderObject` feature is enabled
- VUID-vkCmdSetDepthBoundsTestEnable-depthBounds-10010
If the `depthBounds` feature is not enabled, `depthBoundsTestEnable` **must** be `VK_FALSE`

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBoundsTestEnable-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetDepthBoundsTestEnable-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetDepthBoundsTestEnable-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetDepthBoundsTestEnable-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To [dynamically set](#) the depth bounds range, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetDepthBounds(
    VkCommandBuffer          commandBuffer,
    float                    minDepthBounds,
    float                    maxDepthBounds);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `minDepthBounds` is the minimum depth bound.
- `maxDepthBounds` is the maximum depth bound.

This command sets the depth bounds range for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_DEPTH_BOUNDS` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineDepthStencilStateCreateInfo::minDepthBounds` and `VkPipelineDepthStencilStateCreateInfo::maxDepthBounds` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthBounds-minDepthBounds-00600
If the `VK_EXT_depth_range_unrestricted` extension is not enabled `minDepthBounds` **must** be between `0.0` and `1.0`, inclusive
- VUID-vkCmdSetDepthBounds-maxDepthBounds-00601
If the `VK_EXT_depth_range_unrestricted` extension is not enabled `maxDepthBounds` **must** be between `0.0` and `1.0`, inclusive

Valid Usage (Implicit)

- VUID-vkCmdSetDepthBounds-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetDepthBounds-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdSetDepthBounds-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations

- VUID-vkCmdSetDepthBounds-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

26.8. Stencil Test

The stencil test compares the stencil attachment value s_a in the depth/stencil attachment at each sample's framebuffer coordinates (x_f, y_f) and `sample index` i against a *stencil reference value*.

If the stencil test is not enabled, as specified by `VkPipelineDepthStencilStateCreateInfo::stencilTestEnable`, or if there is no stencil attachment, the coverage mask is unmodified by this operation.

The stencil test is controlled by one of two sets of stencil-related state, the front stencil state and the back stencil state. Stencil tests and writes use the back stencil state when processing fragments generated by `back-facing polygons`, and the front stencil state when processing fragments generated by `front-facing polygons` or any other primitives.

The comparison operation performed is determined by the `VkCompareOp` value set by `VkStencilOpState::compareOp` during pipeline creation.

The compare mask s_c and stencil reference value s_r of the front or the back stencil state set determine arguments of the comparison operation. s_c is set by the `VkPipelineDepthStencilStateCreateInfo` structure during pipeline creation, or by the `vkCmdSetStencilCompareMask` command. s_r is set by `VkPipelineDepthStencilStateCreateInfo` or by `vkCmdSetStencilReference`.

s_r and s_a are each independently combined with s_c using a bitwise **AND** operation to create masked reference and attachment values s'_r and s'_a . s'_r and s'_a are used as the *reference* and *test* values, respectively, in the operation specified by the `VkCompareOp`.

If the comparison evaluates to false, the coverage for the sample is set to 0.

A new stencil value s_g is generated according to a stencil operation defined by `VkStencilOp` parameters set by `VkPipelineDepthStencilStateCreateInfo`. If the stencil test fails, `failOp` defines the stencil operation used. If the stencil test passes however, the stencil op used is based on the `depth test` - if it passes, `VkPipelineDepthStencilStateCreateInfo::passOp` is used, otherwise `VkPipelineDepthStencilStateCreateInfo::depthFailOp` is used.

The stencil attachment value s_a is then updated with the generated stencil value s_g according to the write mask s_w defined by `writeMask` in `VkPipelineDepthStencilStateCreateInfo::front` and `VkPipelineDepthStencilStateCreateInfo::back` as:

$$s_a = (s_a \& \neg s_w) \mid (s_g \& s_w)$$

To [dynamically enable or disable](#) the stencil test, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetStencilTestEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                 stencilTestEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `stencilTestEnable` specifies if the stencil test is enabled.

This command sets the stencil test enable for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the `VkPipelineDepthStencilStateCreateInfo::stencilTestEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetStencilTestEnable-None-08971
At least one of the following **must** be true:
 - the `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetStencilTestEnable-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetStencilTestEnable-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdSetStencilTestEnable-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetStencilTestEnable-vertexIndexing

This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To **dynamically set** the stencil operation, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetStencilOpEXT(
    VkCommandBuffer          commandBuffer,
    VkStencilFaceFlags       faceMask,
    VkStencilOp              failOp,
    VkStencilOp              passOp,
    VkStencilOp              depthFailOp,
    VkCompareOp              compareOp);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **faceMask** is a bitmask of **VkStencilFaceFlagBits** specifying the set of stencil state for which to update the stencil operation.
- **failOp** is a **VkStencilOp** value specifying the action performed on samples that fail the stencil test.
- **passOp** is a **VkStencilOp** value specifying the action performed on samples that pass both the depth and stencil tests.
- **depthFailOp** is a **VkStencilOp** value specifying the action performed on samples that pass the stencil test and fail the depth test.
- **compareOp** is a **VkCompareOp** value specifying the comparison operator used in the stencil test.

This command sets the stencil operation for subsequent drawing commands when when drawing using **shader objects**. Otherwise, this state is specified by the corresponding **VkPipelineDepthStencilStateCreateInfo::failOp**, **passOp**, **depthFailOp**, and **compareOp** values used to create the currently active pipeline, for both front and back faces.

Valid Usage

- VUID-vkCmdSetStencilOp-None-08971

At least one of the following **must** be true:

- the `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetStencilOp-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetStencilOp-faceMask-parameter
`faceMask` **must** be a valid combination of `VkStencilFaceFlagBits` values
- VUID-vkCmdSetStencilOp-faceMask-requiredbitmask
`faceMask` **must** not be 0
- VUID-vkCmdSetStencilOp-failOp-parameter
`failOp` **must** be a valid `VkStencilOp` value
- VUID-vkCmdSetStencilOp-passOp-parameter
`passOp` **must** be a valid `VkStencilOp` value
- VUID-vkCmdSetStencilOp-depthFailOp-parameter
`depthFailOp` **must** be a valid `VkStencilOp` value
- VUID-vkCmdSetStencilOp-compareOp-parameter
`compareOp` **must** be a valid `VkCompareOp` value
- VUID-vkCmdSetStencilOp-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdSetStencilOp-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetStencilOp-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

The `VkStencilOpState` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkStencilOpState {
    VkStencilOp    failOp;
    VkStencilOp    passOp;
    VkStencilOp    depthFailOp;
    VkCompareOp    compareOp;
    uint32_t       compareMask;
    uint32_t       writeMask;
    uint32_t       reference;
} VkStencilOpState;
```

- `failOp` is a `VkStencilOp` value specifying the action performed on samples that fail the stencil test.
- `passOp` is a `VkStencilOp` value specifying the action performed on samples that pass both the depth and stencil tests.
- `depthFailOp` is a `VkStencilOp` value specifying the action performed on samples that pass the stencil test and fail the depth test.
- `compareOp` is a `VkCompareOp` value specifying the comparison operator used in the stencil test.
- `compareMask` selects the bits of the unsigned integer stencil values participating in the stencil test.
- `writeMask` selects the bits of the unsigned integer stencil values updated by the stencil test in the stencil framebuffer attachment.
- `reference` is an integer stencil reference value that is used in the unsigned stencil comparison.

Valid Usage (Implicit)

- VUID-VkStencilOpState-failOp-parameter `failOp` **must** be a valid `VkStencilOp` value
- VUID-VkStencilOpState-passOp-parameter `passOp` **must** be a valid `VkStencilOp` value
- VUID-VkStencilOpState-depthFailOp-parameter `depthFailOp` **must** be a valid `VkStencilOp` value
- VUID-VkStencilOpState-compareOp-parameter `compareOp` **must** be a valid `VkCompareOp` value

To [dynamically set](#) the stencil compare mask, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetStencilCompareMask(
    VkCommandBuffer          commandBuffer,
    VkStencilFaceFlags       faceMask,
    uint32_t                 compareMask);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `faceMask` is a bitmask of [VkStencilFaceFlagBits](#) specifying the set of stencil state for which to update the compare mask.
- `compareMask` is the new value to use as the stencil compare mask.

This command sets the stencil compare mask for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` set in [VkPipelineDynamicStateCreateInfo::pDynamicStates](#). Otherwise, this state is specified by the [VkStencilOpState::compareMask](#) value used to create the currently active pipeline, for both front and back faces.

Valid Usage (Implicit)

- VUID-vkCmdSetStencilCompareMask-commandBuffer-parameter `commandBuffer` **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetStencilCompareMask-faceMask-parameter `faceMask` **must** be a valid combination of [VkStencilFaceFlagBits](#) values
- VUID-vkCmdSetStencilCompareMask-faceMask-requiredbitmask `faceMask` **must** not be 0
- VUID-vkCmdSetStencilCompareMask-commandBuffer-recording `commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdSetStencilCompareMask-commandBuffer-cmdpool
The [VkCommandPool](#) that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetStencilCompareMask-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the [VkCommandPool](#) that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

`VkStencilFaceFlagBits` values are:

```
// Provided by VK_VERSION_1_0
typedef enum VkStencilFaceFlagBits {
    VK_STENCIL_FACE_FRONT_BIT = 0x00000001,
    VK_STENCIL_FACE_BACK_BIT = 0x00000002,
    VK_STENCIL_FACE_FRONT_AND_BACK = 0x00000003,
    // VK_STENCIL_FACE_FRONT_AND_BACK is a deprecated alias
    VK_STENCIL_FACE_FRONT_AND_BACK = VK_STENCIL_FACE_FRONT_AND_BACK,
} VkStencilFaceFlagBits;
```

- `VK_STENCIL_FACE_FRONT_BIT` specifies that only the front set of stencil state is updated.
- `VK_STENCIL_FACE_BACK_BIT` specifies that only the back set of stencil state is updated.
- `VK_STENCIL_FACE_FRONT_AND_BACK` is the combination of `VK_STENCIL_FACE_FRONT_BIT` and `VK_STENCIL_FACE_BACK_BIT`, and specifies that both sets of stencil state are updated.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkStencilFaceFlags;
```

`VkStencilFaceFlags` is a bitmask type for setting a mask of zero or more `VkStencilFaceFlagBits`.

To [dynamically set](#) the stencil write mask, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetStencilWriteMask(
    VkCommandBuffer          commandBuffer,
    VkStencilFaceFlags        faceMask,
    uint32_t                  writeMask);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `faceMask` is a bitmask of `VkStencilFaceFlagBits` specifying the set of stencil state for which to update the write mask, as described above for `vkCmdSetStencilCompareMask`.
- `writeMask` is the new value to use as the stencil write mask.

This command sets the stencil write mask for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK`

set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `writeMask` value used to create the currently active pipeline, for both `VkPipelineDepthStencilStateCreateInfo::front` and `VkPipelineDepthStencilStateCreateInfo::back` faces.

Valid Usage (Implicit)

- VUID-vkCmdSetStencilWriteMask-commandBuffer-parameter `commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetStencilWriteMask-faceMask-parameter `faceMask` **must** be a valid combination of `VkStencilFaceFlagBits` values
- VUID-vkCmdSetStencilWriteMask-faceMask-requiredbitmask `faceMask` **must** not be 0
- VUID-vkCmdSetStencilWriteMask-commandBuffer-recording `commandBuffer` **must** be in the `recording` state
- VUID-vkCmdSetStencilWriteMask-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetStencilWriteMask-vidcoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To [dynamically](#) set the stencil reference value, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetStencilReference(
    VkCommandBuffer          commandBuffer,
    VkStencilFaceFlags       faceMask,
    uint32_t                 reference);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `faceMask` is a bitmask of `VkStencilFaceFlagBits` specifying the set of stencil state for which to update the reference value, as described above for `vkCmdSetStencilCompareMask`.
- `reference` is the new value to use as the stencil reference value.

This command sets the stencil reference value for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_STENCIL_REFERENCE` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineDepthStencilStateCreateInfo::reference` value used to create the currently active pipeline, for both front and back faces.

Valid Usage (Implicit)

- VUID-vkCmdSetStencilReference-commandBuffer-parameter `commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetStencilReference-faceMask-parameter `faceMask` **must** be a valid combination of `VkStencilFaceFlagBits` values
- VUID-vkCmdSetStencilReference-faceMask-requiredbitmask `faceMask` **must** not be 0
- VUID-vkCmdSetStencilReference-commandBuffer-recording `commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetStencilReference-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetStencilReference-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Possible values of the `failOp`, `passOp`, and `depthFailOp` members of `VkStencilOpState`, specifying what

happens to the stored stencil value if this or certain subsequent tests fail or pass, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkStencilOp {
    VK_STENCIL_OP_KEEP = 0,
    VK_STENCIL_OP_ZERO = 1,
    VK_STENCIL_OP_REPLACE = 2,
    VK_STENCIL_OP_INCREMENT_AND_CLAMP = 3,
    VK_STENCIL_OP_DECREMENT_AND_CLAMP = 4,
    VK_STENCIL_OP_INVERT = 5,
    VK_STENCIL_OP_INCREMENT_AND_WRAP = 6,
    VK_STENCIL_OP_DECREMENT_AND_WRAP = 7,
} VkStencilOp;
```

- **VK_STENCIL_OP_KEEP** keeps the current value.
- **VK_STENCIL_OP_ZERO** sets the value to 0.
- **VK_STENCIL_OP_REPLACE** sets the value to **reference**.
- **VK_STENCIL_OP_INCREMENT_AND_CLAMP** increments the current value and clamps to the maximum representable unsigned value.
- **VK_STENCIL_OP_DECREMENT_AND_CLAMP** decrements the current value and clamps to 0.
- **VK_STENCIL_OP_INVERT** bitwise-inverts the current value.
- **VK_STENCIL_OP_INCREMENT_AND_WRAP** increments the current value and wraps to 0 when the maximum value would have been exceeded.
- **VK_STENCIL_OP_DECREMENT_AND_WRAP** decrements the current value and wraps to the maximum possible value when the value would go below 0.

For purposes of increment and decrement, the stencil bits are considered as an unsigned integer.

26.9. Depth Test

The depth test compares the depth value z_a in the depth/stencil attachment at each sample's framebuffer coordinates (x_f, y_f) and **sample index** i against the sample's depth value z_f . If there is no depth attachment then the depth test is skipped.

The depth test occurs in three stages, as detailed in the following sections.

26.9.1. Depth Clamping and Range Adjustment

If **VkPipelineRasterizationStateCreateInfo::depthClampEnable** is enabled, z_f is clamped to $[z_{\min}, z_{\max}]$, where $z_{\min} = \min(n, f)$, $z_{\max} = \max(n, f)$, and n and f are the **minDepth** and **maxDepth** depth range values of the viewport used by this fragment, respectively. If **depthClampControl** is enabled and **VkPipelineViewportDepthClampControlCreateInfoEXT::depthClampMode** is **VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT**, then $z_{\min}$ and $z_{\max}$ are equal to the **minDepthClamp** and **maxDepthClamp** values of the **VkDepthClampRangeEXT** structure pointed to by **pDepthClampRange**.

Following depth clamping:

- If z_f is not in the range $[z_{\min}, z_{\max}]$, then z_f is undefined following this step.
- If the depth attachment has a fixed-point format and z_f is not in the range $[0, 1]$, then z_f is undefined following this step.

The `VkPipelineViewportDepthClampControlCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_depth_clamp_control
typedef struct VkPipelineViewportDepthClampControlCreateInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkDepthClampModeEXT  depthClampMode;
    const VkDepthClampRangeEXT* pDepthClampRange;
} VkPipelineViewportDepthClampControlCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthClampMode` determines how the clamp range is determined for each viewport.
- `pDepthClampRange` sets the depth clamp range for all viewports if `depthClampMode` is `VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT`.

This structure extends `VkPipelineViewportStateCreateInfo` and specifies the depth clamp range used in the pipeline. If this structure is not provided in the next chain then `depthClampMode` defaults to `VK_DEPTH_CLAMP_MODE_VIEWPORT_RANGE_EXT`.

Valid Usage

- VUID-VkPipelineViewportDepthClampControlCreateInfoEXT-pDepthClampRange-09646
If `depthClampMode` is `VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT`, and the pipeline is not created with `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT`, then `pDepthClampRange` **must** be a valid pointer to a valid `VkDepthClampRangeEXT` structure

Valid Usage (Implicit)

- VUID-VkPipelineViewportDepthClampControlCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLAMP_CONTROL_CREATE_INFO_EXT`
- VUID-VkPipelineViewportDepthClampControlCreateInfoEXT-depthClampMode-parameter
`depthClampMode` **must** be a valid `VkDepthClampModeEXT` value
- VUID-VkPipelineViewportDepthClampControlCreateInfoEXT-pDepthClampRange-parameter
If `pDepthClampRange` is not `NULL`, `pDepthClampRange` **must** be a valid pointer to a valid `VkDepthClampRangeEXT` structure

Possible values of `VkPipelineViewportDepthClampControlCreateInfoEXT::depthClampMode`, specifying which range should be used for depth clamping, are:

```
// Provided by VK_EXT_depth_clamp_control
typedef enum VkDepthClampModeEXT {
    VK_DEPTH_CLAMP_MODE_VIEWPORT_RANGE_EXT = 0,
    VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT = 1,
} VkDepthClampModeEXT;
```

- `VK_DEPTH_CLAMP_MODE_VIEWPORT_RANGE_EXT` specifies that the depth clamp range follows the viewport depth range. The depth clamp range of each viewport will implicitly be set to $z_{\min} = \min(n, f)$ and $z_{\max} = \max(n, f)$, where `n` and `f` are the `minDepth` and `maxDepth` depth range values of the viewport.
- `VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT` specifies that a single user-defined depth clamp range will be used for all viewports. The user-defined depth clamp range is defined by the `minDepthClamp` and `maxDepthClamp` members of `VkDepthClampRangeEXT`.

To dynamically set the viewport depth clamp range parameters, call:

```
// Provided by VK_EXT_depth_clamp_control, VK_EXT_depth_clamp_control with
VK_EXT_shader_object
void vkCmdSetDepthClampRangeEXT(
    VkCommandBuffer                commandBuffer,
    VkDepthClampModeEXT            depthClampMode,
    const VkDepthClampRangeEXT*    pDepthClampRange);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthClampMode` determines how the clamp range is determined for each viewport.
- `pDepthClampRange` sets the depth clamp range for all viewports if `depthClampMode` is `VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT`.

This command sets the viewport depth clamp range for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineViewportDepthClampControlCreateInfoEXT::depthClampMode` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthClampRangeEXT-pDepthClampRange-09647
If `depthClampMode` is `VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT`, then `pDepthClampRange` must be a valid pointer to a valid `VkDepthClampRangeEXT` structure

Valid Usage (Implicit)

- VUID-vkCmdSetDepthClampRangeEXT-commandBuffer-parameter **commandBuffer** **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetDepthClampRangeEXT-depthClampMode-parameter **depthClampMode** **must** be a valid [VkDepthClampModeEXT](#) value
- VUID-vkCmdSetDepthClampRangeEXT-pDepthClampRange-parameter
If **pDepthClampRange** is not **NULL**, **pDepthClampRange** **must** be a valid pointer to a valid [VkDepthClampRangeEXT](#) structure
- VUID-vkCmdSetDepthClampRangeEXT-commandBuffer-recording **commandBuffer** **must** be in the [recording state](#)
- VUID-vkCmdSetDepthClampRangeEXT-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support graphics operations
- VUID-vkCmdSetDepthClampRangeEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Both [VkPipelineViewportDepthClampControlCreateInfoEXT](#) and [vkCmdSetDepthClampRangeEXT](#) use [VkDepthClampRangeEXT](#) to set the viewport depth clamp range parameters.

The [VkDepthClampRangeEXT](#) structure is defined as:

```
// Provided by VK_EXT_depth_clamp_control
typedef struct VkDepthClampRangeEXT {
    float    minDepthClamp;
    float    maxDepthClamp;
} VkDepthClampRangeEXT;
```

- `minDepthClamp` sets $z_{\min}$ in the depth clamp range of the viewport.
- `maxDepthClamp` sets $z_{\max}$ in the depth clamp range of the viewport.

Valid Usage

- VUID-VkDepthClampRangeEXT-pDepthClampRange-00999
`minDepthClamp` **must** be less than or equal to `maxDepthClamp`
- VUID-VkDepthClampRangeEXT-pDepthClampRange-09648
If the `VK_EXT_depth_range_unrestricted` extension is not enabled, `minDepthClamp` **must** be greater than or equal to `0.0`
- VUID-VkDepthClampRangeEXT-pDepthClampRange-09649
If the `VK_EXT_depth_range_unrestricted` extension is not enabled, `maxDepthClamp` **must** be less than or equal to `1.0`

26.9.2. Depth Comparison

If the depth test is not enabled, as specified by `vkCmdSetDepthTestEnableEXT` or `VkPipelineDepthStencilStateCreateInfo::depthTestEnable`, then this step is skipped.

The comparison operation performed is determined by the `VkCompareOp` value set by `vkCmdSetDepthCompareOpEXT`, or by `VkPipelineDepthStencilStateCreateInfo::depthCompareOp` during pipeline creation. z_f and z_a are used as the *reference* and *test* values, respectively, in the operation specified by the `VkCompareOp`.

If the comparison evaluates to false, the coverage for the sample is set to `0`.

26.9.3. Depth Attachment Writes

If depth writes are enabled, as specified by `vkCmdSetDepthWriteEnableEXT` or `VkPipelineDepthStencilStateCreateInfo::depthWriteEnable`, and the comparison evaluated to true, the depth attachment value z_a is set to the sample's depth value z_f . If there is no depth attachment, no value is written.

To **dynamically enable or disable** the depth test, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetDepthTestEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                 depthTestEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthTestEnable` specifies if the depth test is enabled.

This command sets the depth test enable for subsequent drawing commands when drawing using **shader objects**. Otherwise, this state is specified by the `VkPipelineDepthStencilStateCreateInfo::depthTestEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthTestEnable-None-08971

At least one of the following **must** be true:

- the `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetDepthTestEnable-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetDepthTestEnable-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdSetDepthTestEnable-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetDepthTestEnable-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To [dynamically set](#) the depth compare operator, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetDepthCompareOpEXT(
    VkCommandBuffer          commandBuffer,
    VkCompareOp              depthCompareOp);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthCompareOp` is a `VkCompareOp` value specifying the comparison operator used for the [Depth](#)

Comparison step of the depth test.

This command sets the depth comparison operator for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the [VkPipelineDepthStencilStateCreateInfo::depthCompareOp](#) value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthCompareOp-None-08971

At least one of the following **must** be true:

- the [shaderObject](#) feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetDepthCompareOp-commandBuffer-parameter
[commandBuffer](#) **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetDepthCompareOp-depthCompareOp-parameter
[depthCompareOp](#) **must** be a valid [VkCompareOp](#) value
- VUID-vkCmdSetDepthCompareOp-commandBuffer-recording
[commandBuffer](#) **must** be in the [recording state](#)
- VUID-vkCmdSetDepthCompareOp-commandBuffer-cmdpool
The [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support graphics operations
- VUID-vkCmdSetDepthCompareOp-vidcoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to [commandBuffer](#) **must** be externally synchronized
- Host access to the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To [dynamically set](#) the depth write enable, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetDepthWriteEnableEXT(
    VkCommandBuffer                      commandBuffer,
    VkBool32                             depthWriteEnable);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `depthWriteEnable` specifies if depth writes are enabled.

This command sets the depth write enable for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the [VkPipelineDepthStencilStateCreateInfo::depthWriteEnable](#) value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetDepthWriteEnable-None-08971
At least one of the following **must** be true:
 - the `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetDepthWriteEnable-commandBuffer-parameter
`commandBuffer` **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetDepthWriteEnable-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdSetDepthWriteEnable-commandBuffer-cmdpool
The [VkCommandPool](#) that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetDepthWriteEnable-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the [VkCommandPool](#) that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

26.10. Sample Counting

Occlusion queries use query pool entries to track the number of samples that pass all the per-fragment tests. The mechanism of collecting an occlusion query value is described in [Occlusion Queries](#).

The occlusion query sample counter increments by one for each sample with a coverage value of 1 in each fragment that survives all the per-fragment tests, including scissor, sample mask, alpha to coverage, stencil, and depth tests.

26.11. Coverage Reduction

Coverage reduction takes the coverage information for a fragment and converts that to a boolean coverage value for each color sample in each pixel covered by the fragment.

26.11.1. Pixel Coverage

Coverage for each pixel is first extracted from the total fragment coverage mask. This consists of `rasterizationSamples` unique coverage samples for each pixel in the fragment area, each with a unique `sample index`. If the fragment only contains a single pixel, coverage for the pixel is equivalent to the fragment coverage.

If the `fragment shading rate` is set, and the fragment covers multiple pixels, each pixel's coverage consists of the coverage samples with a `pixel index` matching that pixel, and each sample retains its unique `sample index i`.

26.11.2. Color Sample Coverage

Once pixel coverage is determined, coverage for each individual color sample corresponding to that pixel is determined.

The number of `rasterizationSamples` is identical to the number of samples in the color attachments. A color sample is covered if the pixel coverage sample with the same `sample index i` is covered.

Chapter 27. The Framebuffer

27.1. Blending

Blending combines the incoming *source* fragment's R, G, B, and A values with the *destination* R, G, B, and A values of each sample stored in the framebuffer at the fragment's (x_f, y_f) location. Blending is performed for each color sample covered by the fragment, rather than just once for each fragment.

Source and destination values are combined according to the [blend operation](#), quadruplets of source and destination weighting factors determined by the [blend factors](#), and a [blend constant](#), to obtain a new set of R, G, B, and A values, as described below.

Blending is computed and applied separately to each color attachment used by the subpass, with separate controls for each attachment.

Prior to performing the blend operation, signed and unsigned normalized fixed-point color components undergo an implied conversion to floating-point as specified by [Conversion from Normalized Fixed-Point to Floating-Point](#). Blending computations are treated as if carried out in floating-point, and basic blend operations are performed with a precision and dynamic range no lower than that used to represent destination components. [Advanced blending operations](#) are performed with a precision and dynamic range no lower than the smaller of that used to represent destination components or that used to represent 16-bit floating-point values.

NOTE

Blending is only defined for floating-point, UNORM, SNORM, and sRGB formats. Within those formats, the implementation may only support blending on some subset of them. Which formats support blending is indicated by [VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT](#).

The pipeline blend state is included in the [VkPipelineColorBlendStateCreateInfo](#) structure during graphics pipeline creation:

The [VkPipelineColorBlendStateCreateInfo](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineColorBlendStateCreateInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkPipelineColorBlendStateCreateFlags flags;
    VkBool32                  logicOpEnable;
    VkLogicOp                 logicOp;
    uint32_t                  attachmentCount;
    const VkPipelineColorBlendAttachmentState* pAttachments;
    float                     blendConstants[4];
} VkPipelineColorBlendStateCreateInfo;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `flags` is reserved for future use.
- `logicOpEnable` controls whether to apply [Logical Operations](#).
- `logicOp` selects which logical operation to apply.
- `attachmentCount` is the number of [VkPipelineColorBlendAttachmentState](#) elements in `pAttachments`. It is ignored if the pipeline is created with `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT`, `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT`, and `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic states set, and either `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` set or [advancedBlendCoherentOperations](#) is not enabled on the device.
- `pAttachments` is a pointer to an array of [VkPipelineColorBlendAttachmentState](#) structures defining blend state for each color attachment. It is ignored if the pipeline is created with `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT`, `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT`, and `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic states set, and either `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` set or [advancedBlendCoherentOperations](#) is not enabled on the device.
- `blendConstants` is a pointer to an array of four values used as the R, G, B, and A components of the blend constant that are used in blending, depending on the [blend factor](#).

Valid Usage

- VUID-VkPipelineColorBlendStateCreateInfo-pAttachments-00605
If the `independentBlend` feature is not enabled, all elements of `pAttachments` **must** be identical
- VUID-VkPipelineColorBlendStateCreateInfo-logicOpEnable-00606
If the `logicOp` feature is not enabled, `logicOpEnable` **must** be `VK_FALSE`
- VUID-VkPipelineColorBlendStateCreateInfo-logicOpEnable-00607
If `logicOpEnable` is `VK_TRUE`, `logicOp` **must** be a valid [VkLogicOp](#) value
- VUID-VkPipelineColorBlendStateCreateInfo-pAttachments-07353
If `attachmentCount` is not `0`, and any of `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT`, `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT`, `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT`, or `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` are not set, `pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid [VkPipelineColorBlendAttachmentState](#) structures

Valid Usage (Implicit)

- VUID-VkPipelineColorBlendStateCreateInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_STATE_CREATE_INFO`
- VUID-VkPipelineColorBlendStateCreateInfo-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of [VkPipelineColorBlendAdvancedStateCreateInfoEXT](#) or [VkPipelineColorWriteCreateInfoEXT](#)
- VUID-VkPipelineColorBlendStateCreateInfo-sType-unique

The `sType` value of each struct in the `pNext` chain **must** be unique

- `VOID-VkPipelineColorBlendStateCreateInfo-flags-zeroBitmask`
`flags` **must** be 0

- `VOID-VkPipelineColorBlendStateCreateInfo-pAttachments-parameter`

If `attachmentCount` is not 0, and `pAttachments` is not NULL, `pAttachments` **must** be a valid pointer to an array of `attachmentCount` valid `VkPipelineColorBlendAttachmentState` structures

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkPipelineColorBlendStateCreateFlags;
```

`VkPipelineColorBlendStateCreateFlags` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkPipelineColorBlendAttachmentState` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPipelineColorBlendAttachmentState {
    VkBool32                blendEnable;
    VkBlendFactor            srcColorBlendFactor;
    VkBlendFactor            dstColorBlendFactor;
    VkBlendOp                colorBlendOp;
    VkBlendFactor            srcAlphaBlendFactor;
    VkBlendFactor            dstAlphaBlendFactor;
    VkBlendOp                alphaBlendOp;
    VkColorComponentFlags    colorWriteMask;
} VkPipelineColorBlendAttachmentState;
```

- `blendEnable` controls whether blending is enabled for the corresponding color attachment. If blending is not enabled, the source fragment's color for that attachment is passed through unmodified.
- `srcColorBlendFactor` selects which blend factor is used to determine the source factors (S_r, S_g, S_b).
- `dstColorBlendFactor` selects which blend factor is used to determine the destination factors (D_r, D_g, D_b).
- `colorBlendOp` selects which blend operation is used to calculate the RGB values to write to the color attachment.
- `srcAlphaBlendFactor` selects which blend factor is used to determine the source factor S_a .
- `dstAlphaBlendFactor` selects which blend factor is used to determine the destination factor D_a .
- `alphaBlendOp` selects which blend operation is used to calculate the alpha values to write to the color attachment.
- `colorWriteMask` is a bitmask of `VkColorComponentFlagBits` specifying which of the R, G, B, and/or A components are enabled for writing, as described for the [Color Write Mask](#).

Valid Usage

- VUID-VkPipelineColorBlendAttachmentState-srcColorBlendFactor-00608
If the `dualSrcBlend` feature is not enabled, `srcColorBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`
- VUID-VkPipelineColorBlendAttachmentState-dstColorBlendFactor-00609
If the `dualSrcBlend` feature is not enabled, `dstColorBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`
- VUID-VkPipelineColorBlendAttachmentState-srcAlphaBlendFactor-00610
If the `dualSrcBlend` feature is not enabled, `srcAlphaBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`
- VUID-VkPipelineColorBlendAttachmentState-dstAlphaBlendFactor-00611
If the `dualSrcBlend` feature is not enabled, `dstAlphaBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`
- VUID-VkPipelineColorBlendAttachmentState-colorBlendOp-01406
If either of `colorBlendOp` or `alphaBlendOp` is an **advanced blend operation**, then `colorBlendOp` **must** equal `alphaBlendOp`
- VUID-VkPipelineColorBlendAttachmentState-advancedBlendIndependentBlend-01407
If `VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT::advancedBlendIndependentBlend` is `VK_FALSE` and `colorBlendOp` is an **advanced blend operation**, then `colorBlendOp` **must** be the same for all attachments
- VUID-VkPipelineColorBlendAttachmentState-advancedBlendIndependentBlend-01408
If `VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT::advancedBlendIndependentBlend` is `VK_FALSE` and `alphaBlendOp` is an **advanced blend operation**, then `alphaBlendOp` **must** be the same for all attachments
- VUID-VkPipelineColorBlendAttachmentState-advancedBlendAllOperations-01409
If `VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT::advancedBlendAllOperations` is `VK_FALSE`, then `colorBlendOp` **must** not be `VK_BLEND_OP_ZERO_EXT`, `VK_BLEND_OP_SRC_EXT`, `VK_BLEND_OP_DST_EXT`, `VK_BLEND_OP_SRC_OVER_EXT`, `VK_BLEND_OP_DST_OVER_EXT`, `VK_BLEND_OP_SRC_IN_EXT`, `VK_BLEND_OP_DST_IN_EXT`, `VK_BLEND_OP_SRC_OUT_EXT`, `VK_BLEND_OP_DST_OUT_EXT`, `VK_BLEND_OP_SRC_ATOP_EXT`, `VK_BLEND_OP_DST_ATOP_EXT`, `VK_BLEND_OP_XOR_EXT`, `VK_BLEND_OP_INVERT_EXT`, `VK_BLEND_OP_INVERT_RGB_EXT`, `VK_BLEND_OP_LINEARDOUDGE_EXT`, `VK_BLEND_OP_LINEARBURN_EXT`, `VK_BLEND_OP_VIVIDLIGHT_EXT`, `VK_BLEND_OP_LINEARLIGHT_EXT`, `VK_BLEND_OP_PINLIGHT_EXT`, `VK_BLEND_OP_HARDMIX_EXT`, `VK_BLEND_OP_PLUS_EXT`, `VK_BLEND_OP_PLUS_CLAMPED_EXT`, `VK_BLEND_OP_PLUS_CLAMPED_ALPHA_EXT`, `VK_BLEND_OP_PLUS_DARKER_EXT`, `VK_BLEND_OP_MINUS_EXT`, `VK_BLEND_OP_MINUS_CLAMPED_EXT`, `VK_BLEND_OP_CONTRAST_EXT`, `VK_BLEND_OP_INVERT_OVG_EXT`, `VK_BLEND_OP_RED_EXT`, `VK_BLEND_OP_GREEN_EXT`, or `VK_BLEND_OP_BLUE_EXT`
- VUID-VkPipelineColorBlendAttachmentState-colorBlendOp-01410
If `colorBlendOp` or `alphaBlendOp` is an **advanced blend operation**, then `colorAttachmentCount`

of the subpass this pipeline is compiled against **must** be less than or equal to `VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT::advancedBlendMaxColorAttachments`

- VUID-VkPipelineColorBlendAttachmentState-constantAlphaColorBlendFactors-04454
If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::constantAlphaColorBlendFactors` is `VK_FALSE`, `srcColorBlendFactor` **must** not be `VK_BLEND_FACTOR_CONSTANT_ALPHA` or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`
- VUID-VkPipelineColorBlendAttachmentState-constantAlphaColorBlendFactors-04455
If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::constantAlphaColorBlendFactors` is `VK_FALSE`, `dstColorBlendFactor` **must** not be `VK_BLEND_FACTOR_CONSTANT_ALPHA` or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`

Valid Usage (Implicit)

- VUID-VkPipelineColorBlendAttachmentState-srcColorBlendFactor-parameter
`srcColorBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkPipelineColorBlendAttachmentState-dstColorBlendFactor-parameter
`dstColorBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkPipelineColorBlendAttachmentState-colorBlendOp-parameter
`colorBlendOp` **must** be a valid `VkBlendOp` value
- VUID-VkPipelineColorBlendAttachmentState-srcAlphaBlendFactor-parameter
`srcAlphaBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkPipelineColorBlendAttachmentState-dstAlphaBlendFactor-parameter
`dstAlphaBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkPipelineColorBlendAttachmentState-alphaBlendOp-parameter
`alphaBlendOp` **must** be a valid `VkBlendOp` value
- VUID-VkPipelineColorBlendAttachmentState-colorWriteMask-parameter
`colorWriteMask` **must** be a valid combination of `VkColorComponentFlagBits` values

To dynamically set `blendEnable`, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetColorBlendEnableEXT(
    VkCommandBuffer          commandBuffer,
    uint32_t                 firstAttachment,
    uint32_t                 attachmentCount,
    const VkBool32*          pColorBlendEnables);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstAttachment` the first color attachment the color blending enable applies.

- `attachmentCount` the number of color blending enables in the `pColorBlendEnables` array.
- `pColorBlendEnables` an array of booleans to indicate whether color blending is enabled for the corresponding attachment.

This command sets the color blending enable of the specified color attachments for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineColorBlendAttachmentState::blendEnable` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetColorBlendEnableEXT-None-09423
At least one of the following **must** be true:
 - The `extendedDynamicState3ColorBlendEnable` feature is enabled
 - The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetColorBlendEnableEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetColorBlendEnableEXT-pColorBlendEnables-parameter
`pColorBlendEnables` **must** be a valid pointer to an array of `attachmentCount` `VkBool32` values
- VUID-vkCmdSetColorBlendEnableEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetColorBlendEnableEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetColorBlendEnableEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetColorBlendEnableEXT-attachmentCount-arraylength
`attachmentCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To [dynamically set](#) color blend factors and operations, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetColorBlendEquationEXT(
    VkCommandBuffer          commandBuffer,
    uint32_t                 firstAttachment,
    uint32_t                 attachmentCount,
    const VkColorBlendEquationEXT* pColorBlendEquations);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstAttachment` the first color attachment the color blend factors and operations apply to.
- `attachmentCount` the number of `VkColorBlendEquationEXT` elements in the `pColorBlendEquations` array.
- `pColorBlendEquations` an array of `VkColorBlendEquationEXT` structs that specify the color blend factors and operations for the corresponding attachments.

This command sets the color blending factors and operations of the specified attachments for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineColorBlendAttachmentState::srcColorBlendFactor`, `VkPipelineColorBlendAttachmentState::dstColorBlendFactor`, `VkPipelineColorBlendAttachmentState::colorBlendOp`, `VkPipelineColorBlendAttachmentState::srcAlphaBlendFactor`, `VkPipelineColorBlendAttachmentState::dstAlphaBlendFactor`, and `VkPipelineColorBlendAttachmentState::alphaBlendOp` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetColorBlendEquationEXT-None-09423

At least one of the following **must** be true:

- The `extendedDynamicState3ColorBlendEquation` feature is enabled
- The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetColorBlendEquationEXT-commandBuffer-parameter `commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetColorBlendEquationEXT-pColorBlendEquations-parameter `pColorBlendEquations` **must** be a valid pointer to an array of `attachmentCount` valid `VkColorBlendEquationEXT` structures
- VUID-vkCmdSetColorBlendEquationEXT-commandBuffer-recording `commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetColorBlendEquationEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetColorBlendEquationEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetColorBlendEquationEXT-attachmentCount-arraylength
`attachmentCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

The `VkColorBlendEquationEXT` structure is defined as:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
typedef struct VkColorBlendEquationEXT {
    VkBlendFactor    srcColorBlendFactor;
    VkBlendFactor    dstColorBlendFactor;
    VkBlendOp        colorBlendOp;
    VkBlendFactor    srcAlphaBlendFactor;
    VkBlendFactor    dstAlphaBlendFactor;
    VkBlendOp        alphaBlendOp;
} VkColorBlendEquationEXT;
```

- `srcColorBlendFactor` selects which blend factor is used to determine the source factors (S_r, S_g, S_b).
- `dstColorBlendFactor` selects which blend factor is used to determine the destination factors (D_r, D_g, D_b).
- `colorBlendOp` selects which blend operation is used to calculate the RGB values to write to the color attachment.
- `srcAlphaBlendFactor` selects which blend factor is used to determine the source factor S_a .
- `dstAlphaBlendFactor` selects which blend factor is used to determine the destination factor D_a .
- `alphaBlendOp` selects which blend operation is use to calculate the alpha values to write to the color attachment.

Valid Usage

- VUID-VkColorBlendEquationEXT-dualSrcBlend-07357
If the `dualSrcBlend` feature is not enabled, `srcColorBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`
- VUID-VkColorBlendEquationEXT-dualSrcBlend-07358
If the `dualSrcBlend` feature is not enabled, `dstColorBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`
- VUID-VkColorBlendEquationEXT-dualSrcBlend-07359
If the `dualSrcBlend` feature is not enabled, `srcAlphaBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`
- VUID-VkColorBlendEquationEXT-dualSrcBlend-07360
If the `dualSrcBlend` feature is not enabled, `dstAlphaBlendFactor` **must** not be `VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`
- VUID-VkColorBlendEquationEXT-colorBlendOp-07361
`colorBlendOp` and `alphaBlendOp` **must** not be `VK_BLEND_OP_ZERO_EXT`, `VK_BLEND_OP_SRC_EXT`, `VK_BLEND_OP_DST_EXT`, `VK_BLEND_OP_SRC_OVER_EXT`, `VK_BLEND_OP_DST_OVER_EXT`, `VK_BLEND_OP_SRC_IN_EXT`, `VK_BLEND_OP_DST_IN_EXT`, `VK_BLEND_OP_SRC_OUT_EXT`, `VK_BLEND_OP_DST_OUT_EXT`, `VK_BLEND_OP_SRC_ATOP_EXT`, `VK_BLEND_OP_DST_ATOP_EXT`, `VK_BLEND_OP_XOR_EXT`, `VK_BLEND_OP_MULTIPLY_EXT`, `VK_BLEND_OP_SCREEN_EXT`, `VK_BLEND_OP_OVERLAY_EXT`, `VK_BLEND_OP_DARKEN_EXT`, `VK_BLEND_OP_LIGHTEN_EXT`, `VK_BLEND_OP_COLORDODGE_EXT`, `VK_BLEND_OP_COLORBURN_EXT`, `VK_BLEND_OP_HARDLIGHT_EXT`, `VK_BLEND_OP_SOFTLIGHT_EXT`, `VK_BLEND_OP_DIFFERENCE_EXT`, `VK_BLEND_OP_EXCLUSION_EXT`, `VK_BLEND_OP_INVERT_EXT`, `VK_BLEND_OP_INVERT_RGB_EXT`, `VK_BLEND_OP_LINEARODDGE_EXT`, `VK_BLEND_OP_LINEARBURN_EXT`, `VK_BLEND_OP_VIVIDLIGHT_EXT`, `VK_BLEND_OP_LINEARLIGHT_EXT`, `VK_BLEND_OP_PINLIGHT_EXT`, `VK_BLEND_OP_HARDMIX_EXT`, `VK_BLEND_OP_HSL_HUE_EXT`, `VK_BLEND_OP_HSL_SATURATION_EXT`, `VK_BLEND_OP_HSL_COLOR_EXT`, `VK_BLEND_OP_HSL_LUMINOSITY_EXT`, `VK_BLEND_OP_PLUS_EXT`, `VK_BLEND_OP_PLUS_CLAMPED_EXT`, `VK_BLEND_OP_PLUS_CLAMPED_ALPHA_EXT`, `VK_BLEND_OP_PLUS_DARKER_EXT`, `VK_BLEND_OP_MINUS_EXT`, `VK_BLEND_OP_MINUS_CLAMPED_EXT`, `VK_BLEND_OP_CONTRAST_EXT`, `VK_BLEND_OP_INVERT_OVG_EXT`,

VK_BLEND_OP_RED_EXT, VK_BLEND_OP_GREEN_EXT, or VK_BLEND_OP_BLUE_EXT

- VUID-VkColorBlendEquationEXT-constantAlphaColorBlendFactors-07362

If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::constantAlphaColorBlendFactors` is `VK_FALSE`, `srcColorBlendFactor` **must** not be `VK_BLEND_FACTOR_CONSTANT_ALPHA` or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`

- VUID-VkColorBlendEquationEXT-constantAlphaColorBlendFactors-07363

If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::constantAlphaColorBlendFactors` is `VK_FALSE`, `dstColorBlendFactor` **must** not be `VK_BLEND_FACTOR_CONSTANT_ALPHA` or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`

Valid Usage (Implicit)

- VUID-VkColorBlendEquationEXT-srcColorBlendFactor-parameter `srcColorBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkColorBlendEquationEXT-dstColorBlendFactor-parameter `dstColorBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkColorBlendEquationEXT-colorBlendOp-parameter `colorBlendOp` **must** be a valid `VkBlendOp` value
- VUID-VkColorBlendEquationEXT-srcAlphaBlendFactor-parameter `srcAlphaBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkColorBlendEquationEXT-dstAlphaBlendFactor-parameter `dstAlphaBlendFactor` **must** be a valid `VkBlendFactor` value
- VUID-VkColorBlendEquationEXT-alphaBlendOp-parameter `alphaBlendOp` **must** be a valid `VkBlendOp` value

To [dynamically set](#) the color write masks, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetColorWriteMaskEXT(
    VkCommandBuffer          commandBuffer,
    uint32_t                 firstAttachment,
    uint32_t                 attachmentCount,
    const VkColorComponentFlags* pColorWriteMasks);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `firstAttachment` the first color attachment the color write masks apply to.
- `attachmentCount` the number of `VkColorComponentFlags` values in the `pColorWriteMasks` array.
- `pColorWriteMasks` an array of `VkColorComponentFlags` values that specify the color write masks of the corresponding attachments.

This command sets the color write masks of the specified attachments for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with [VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT](#) set in [VkPipelineDynamicStateCreateInfo::pDynamicStates](#). Otherwise, this state is specified by the [VkPipelineColorBlendAttachmentState::colorWriteMask](#) values used to create the currently active pipeline.

NOTE

Formats with bits that are shared between components specified by [VkColorComponentFlagBits](#), such as [VK_FORMAT_E5B9G9R9_UFLOAT_PACK32](#), cannot have their channels individually masked by this functionality; either all components that share bits have to be enabled, or none of them.

Valid Usage

- VUID-vkCmdSetColorWriteMaskEXT-None-09423

At least one of the following **must** be true:

- The [extendedDynamicState3ColorWriteMask](#) feature is enabled
- The [shaderObject](#) feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetColorWriteMaskEXT-commandBuffer-parameter [commandBuffer](#) **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetColorWriteMaskEXT-pColorWriteMasks-parameter [pColorWriteMasks](#) **must** be a valid pointer to an array of [attachmentCount](#) valid combinations of [VkColorComponentFlagBits](#) values
- VUID-vkCmdSetColorWriteMaskEXT-commandBuffer-recording [commandBuffer](#) **must** be in the [recording state](#)
- VUID-vkCmdSetColorWriteMaskEXT-commandBuffer-cmdpool
The [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support graphics operations
- VUID-vkCmdSetColorWriteMaskEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetColorWriteMaskEXT-attachmentCount-arraylength
[attachmentCount](#) **must** be greater than 0

Host Synchronization

- Host access to [commandBuffer](#) **must** be externally synchronized
- Host access to the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

27.1.1. Blend Factors

The source and destination color and alpha blending factors are selected from the enum:

```
// Provided by VK_VERSION_1_0
typedef enum VkBlendFactor {
    VK_BLEND_FACTOR_ZERO = 0,
    VK_BLEND_FACTOR_ONE = 1,
    VK_BLEND_FACTOR_SRC_COLOR = 2,
    VK_BLEND_FACTOR_ONE_MINUS_SRC_COLOR = 3,
    VK_BLEND_FACTOR_DST_COLOR = 4,
    VK_BLEND_FACTOR_ONE_MINUS_DST_COLOR = 5,
    VK_BLEND_FACTOR_SRC_ALPHA = 6,
    VK_BLEND_FACTOR_ONE_MINUS_SRC_ALPHA = 7,
    VK_BLEND_FACTOR_DST_ALPHA = 8,
    VK_BLEND_FACTOR_ONE_MINUS_DST_ALPHA = 9,
    VK_BLEND_FACTOR_CONSTANT_COLOR = 10,
    VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR = 11,
    VK_BLEND_FACTOR_CONSTANT_ALPHA = 12,
    VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA = 13,
    VK_BLEND_FACTOR_SRC_ALPHA_SATURATE = 14,
    VK_BLEND_FACTOR_SRC1_COLOR = 15,
    VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR = 16,
    VK_BLEND_FACTOR_SRC1_ALPHA = 17,
    VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA = 18,
} VkBlendFactor;
```

The semantics of the enum values are described in the table below:

Table 33. Blend Factors

VkBlendFactor	RGB Blend Factors (S_r, S_g, S_b) or (D_r, D_g, D_b)	Alpha Blend Factor (S_a or D_a)
VK_BLEND_FACTOR_ZERO	(0,0,0)	0
VK_BLEND_FACTOR_ONE	(1,1,1)	1
VK_BLEND_FACTOR_SRC_COLOR	(R_{s0}, G_{s0}, B_{s0})	A_{s0}

VkBlendFactor	RGB Blend Factors (S_r, S_g, S_b) or (D_r, D_g, D_b)	Alpha Blend Factor (S_a or D_a)
VK_BLEND_FACTOR_ONE_MINUS_SRC_COLOR	$(1-R_{s0}, 1-G_{s0}, 1-B_{s0})$	$1-A_{s0}$
VK_BLEND_FACTOR_DST_COLOR	(R_d, G_d, B_d)	A_d
VK_BLEND_FACTOR_ONE_MINUS_DST_COLOR	$(1-R_d, 1-G_d, 1-B_d)$	$1-A_d$
VK_BLEND_FACTOR_SRC_ALPHA	(A_{s0}, A_{s0}, A_{s0})	A_{s0}
VK_BLEND_FACTOR_ONE_MINUS_SRC_ALPHA	$(1-A_{s0}, 1-A_{s0}, 1-A_{s0})$	$1-A_{s0}$
VK_BLEND_FACTOR_DST_ALPHA	(A_d, A_d, A_d)	A_d
VK_BLEND_FACTOR_ONE_MINUS_DST_ALPHA	$(1-A_d, 1-A_d, 1-A_d)$	$1-A_d$
VK_BLEND_FACTOR_CONSTANT_COLOR	(R_c, G_c, B_c)	A_c
VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR	$(1-R_c, 1-G_c, 1-B_c)$	$1-A_c$
VK_BLEND_FACTOR_CONSTANT_ALPHA	(A_c, A_c, A_c)	A_c
VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA	$(1-A_c, 1-A_c, 1-A_c)$	$1-A_c$
VK_BLEND_FACTOR_SRC_ALPHA_SATURATE	$(f, f, f); f = \min(A_{s0}, 1-A_d)$	1
VK_BLEND_FACTOR_SRC1_COLOR	(R_{s1}, G_{s1}, B_{s1})	A_{s1}
VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR	$(1-R_{s1}, 1-G_{s1}, 1-B_{s1})$	$1-A_{s1}$
VK_BLEND_FACTOR_SRC1_ALPHA	(A_{s1}, A_{s1}, A_{s1})	A_{s1}
VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA	$(1-A_{s1}, 1-A_{s1}, 1-A_{s1})$	$1-A_{s1}$

In this table, the following conventions are used:

- R_{s0}, G_{s0}, B_{s0} and A_{s0} represent the first source color R, G, B, and A components, respectively, for the fragment output location corresponding to the color attachment being blended.
- R_{s1}, G_{s1}, B_{s1} and A_{s1} represent the second source color R, G, B, and A components, respectively, used in dual source blending modes, for the fragment output location corresponding to the color attachment being blended.
- R_d, G_d, B_d and A_d represent the R, G, B, and A components of the destination color. That is, the color currently in the corresponding color attachment for this fragment/sample.
- R_c, G_c, B_c and A_c represent the blend constant R, G, B, and A components, respectively.

To [dynamically set and change](#) the blend constants, call:

```
// Provided by VK_VERSION_1_0
void vkCmdSetBlendConstants(
    VkCommandBuffer          commandBuffer,
    const float               blendConstants[4]);
```

- `commandBuffer` is the command buffer into which the command will be recorded.

- `blendConstants` is a pointer to an array of four values specifying the R_c , G_c , B_c , and A_c components of the blend constant color used in blending, depending on the `blend factor`.

This command sets blend constants for subsequent drawing commands when when drawing using `shader objects`, or the graphics pipeline is created with `VK_DYNAMIC_STATE_BLEND_CONSTANTS` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineColorBlendStateCreateInfo::blendConstants` values used to create the currently active pipeline.

Valid Usage (Implicit)

- VUID-vkCmdSetBlendConstants-commandBuffer-parameter `commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetBlendConstants-commandBuffer-recording `commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetBlendConstants-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetBlendConstants-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

27.1.2. Dual-Source Blending

Blend factors that use the secondary color input (R_{s1} , G_{s1} , B_{s1} , A_{s1}) (`VK_BLEND_FACTOR_SRC1_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR`, `VK_BLEND_FACTOR_SRC1_ALPHA`, and `VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA`) **may** consume implementation resources that could otherwise be used for rendering to multiple color attachments. Therefore, the number of color attachments that **can** be used in a framebuffer **may** be lower when using dual-source blending.

Dual-source blending is only supported if the `dualSrcBlend` feature is enabled.

The maximum number of color attachments that **can** be used in a subpass when using dual-source blending functions is implementation-dependent and is reported as the `maxFragmentDualSrcAttachments` member of `VkPhysicalDeviceLimits`.

Color outputs **can** be bound to the first and second inputs of the blender using the `Index` decoration, as described in [Fragment Output Interface](#). If the second color input to the blender is not written in the shader, or if no output is bound to the second input of a blender, the value of the second input is undefined.

27.1.3. Blend Operations

Once the source and destination blend factors have been selected, they along with the source and destination components are passed to the blending operations. RGB and alpha components **can** use different operations. Possible values of `VkBlendOp`, specifying the operations, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkBlendOp {
    VK_BLEND_OP_ADD = 0,
    VK_BLEND_OP_SUBTRACT = 1,
    VK_BLEND_OP_REVERSE_SUBTRACT = 2,
    VK_BLEND_OP_MIN = 3,
    VK_BLEND_OP_MAX = 4,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_ZERO_EXT = 1000148000,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SRC_EXT = 1000148001,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DST_EXT = 1000148002,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SRC_OVER_EXT = 1000148003,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DST_OVER_EXT = 1000148004,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SRC_IN_EXT = 1000148005,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DST_IN_EXT = 1000148006,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SRC_OUT_EXT = 1000148007,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DST_OUT_EXT = 1000148008,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SRC_ATOP_EXT = 1000148009,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_DST_ATOP_EXT = 1000148010,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_XOR_EXT = 1000148011,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_MULTIPLY_EXT = 1000148012,
// Provided by VK_EXT_blend_operation_advanced
    VK_BLEND_OP_SCREEN_EXT = 1000148013,
```

```
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_OVERLAY_EXT = 1000148014,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_DARKEN_EXT = 1000148015,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_LIGHTEN_EXT = 1000148016,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_COLORODDGE_EXT = 1000148017,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_COLORBURN_EXT = 1000148018,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_HARDLIGHT_EXT = 1000148019,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_SOFTLIGHT_EXT = 1000148020,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_DIFFERENCE_EXT = 1000148021,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_EXCLUSION_EXT = 1000148022,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_INVERT_EXT = 1000148023,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_INVERT_RGB_EXT = 1000148024,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_LINEARODDGE_EXT = 1000148025,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_LINEARBURN_EXT = 1000148026,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_VIVIDLIGHT_EXT = 1000148027,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_LINEARLIGHT_EXT = 1000148028,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_PINLIGHT_EXT = 1000148029,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_HARDMIX_EXT = 1000148030,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_HSL_HUE_EXT = 1000148031,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_HSL_SATURATION_EXT = 1000148032,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_HSL_COLOR_EXT = 1000148033,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_HSL_LUMINOSITY_EXT = 1000148034,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_PLUS_EXT = 1000148035,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_PLUS_CLAMPED_EXT = 1000148036,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_PLUS_CLAMPED_ALPHA_EXT = 1000148037,
// Provided by VK_EXT_blend_operation_advanced
VK_BLEND_OP_PLUS_DARKER_EXT = 1000148038,
// Provided by VK_EXT_blend_operation_advanced
```

```
VK_BLEND_OP_MINUS_EXT = 1000148039,  
// Provided by VK_EXT_blend_operation_advanced  
VK_BLEND_OP_MINUS_CLAMPED_EXT = 1000148040,  
// Provided by VK_EXT_blend_operation_advanced  
VK_BLEND_OP_CONTRAST_EXT = 1000148041,  
// Provided by VK_EXT_blend_operation_advanced  
VK_BLEND_OP_INVERT_OVG_EXT = 1000148042,  
// Provided by VK_EXT_blend_operation_advanced  
VK_BLEND_OP_RED_EXT = 1000148043,  
// Provided by VK_EXT_blend_operation_advanced  
VK_BLEND_OP_GREEN_EXT = 1000148044,  
// Provided by VK_EXT_blend_operation_advanced  
VK_BLEND_OP_BLUE_EXT = 1000148045,  
} VkBlendOp;
```


The semantics of the basic blend operations are described in the table below:

Table 34. Basic Blend Operations

VkBlendOp	RGB Components	Alpha Component
VK_BLEND_OP_ADD	$R = R_{s0} \times S_r + R_d \times D_r$ $G = G_{s0} \times S_g + G_d \times D_g$ $B = B_{s0} \times S_b + B_d \times D_b$	$A = A_{s0} \times S_a + A_d \times D_a$
VK_BLEND_OP_SUBTRACT	$R = R_{s0} \times S_r - R_d \times D_r$ $G = G_{s0} \times S_g - G_d \times D_g$ $B = B_{s0} \times S_b - B_d \times D_b$	$A = A_{s0} \times S_a - A_d \times D_a$
VK_BLEND_OP_REVERSE_SUBTRACT	$R = R_d \times D_r - R_{s0} \times S_r$ $G = G_d \times D_g - G_{s0} \times S_g$ $B = B_d \times D_b - B_{s0} \times S_b$	$A = A_d \times D_a - A_{s0} \times S_a$
VK_BLEND_OP_MIN	$R = \min(R_{s0}, R_d)$ $G = \min(G_{s0}, G_d)$ $B = \min(B_{s0}, B_d)$	$A = \min(A_{s0}, A_d)$
VK_BLEND_OP_MAX	$R = \max(R_{s0}, R_d)$ $G = \max(G_{s0}, G_d)$ $B = \max(B_{s0}, B_d)$	$A = \max(A_{s0}, A_d)$

In this table, the following conventions are used:

- R_{s0} , G_{s0} , B_{s0} and A_{s0} represent the first source color R, G, B, and A components, respectively.
- R_d , G_d , B_d and A_d represent the R, G, B, and A components of the destination color. That is, the color currently in the corresponding color attachment for this fragment/sample.
- S_r , S_g , S_b and S_a represent the source blend factor R, G, B, and A components, respectively.
- D_r , D_g , D_b and D_a represent the destination blend factor R, G, B, and A components, respectively.

The blending operation produces a new set of values R, G, B and A, which are written to the framebuffer attachment. If blending is not enabled for this attachment, then R, G, B and A are assigned R_{s0} , G_{s0} , B_{s0} and A_{s0} , respectively.

If the color attachment is fixed-point, the components of the source and destination values and blend factors are each clamped to [0,1] or [-1,1] respectively for an unsigned normalized or signed normalized color attachment prior to evaluating the blend operations. If the color attachment is floating-point, no clamping occurs.

If the [numeric format](#) of a framebuffer attachment uses sRGB encoding, the R, G, and B destination color values (after conversion from fixed-point to floating-point) are considered to be encoded for the sRGB color space and hence are linearized prior to their use in blending. Each R, G, and B component is converted from nonlinear to linear as described in the “sRGB EOTF” section of the [Khronos Data Format Specification](#). If the format is not sRGB, no linearization is performed.

If the [numeric format](#) of a framebuffer attachment uses sRGB encoding, then the final R, G and B values are converted into the nonlinear sRGB representation before being written to the framebuffer attachment as described in the “sRGB EOTF⁻¹” section of the Khronos Data Format

Specification.

If the [numeric format](#) of a framebuffer color attachment is not sRGB encoded then the resulting c_s values for R, G and B are unmodified. The value of A is never sRGB encoded. That is, the alpha component is always stored in memory as linear.

If the framebuffer color attachment is `VK_ATTACHMENT_UNUSED`, no writes are performed through that attachment. Writes are not performed to framebuffer color attachments greater than or equal to the `VkSubpassDescription::colorAttachmentCount` or `VkSubpassDescription2::colorAttachmentCount` value.

27.1.4. Advanced Blend Operations

The *advanced blend operations* are those listed in tables [f/X/Y/Z Advanced Blend Operations](#), [Hue-Saturation-Luminosity Advanced Blend Operations](#), and [Additional RGB Blend Operations](#).

If the `pNext` chain of `VkPipelineColorBlendStateCreateInfo` includes a `VkPipelineColorBlendAdvancedStateCreateInfoEXT` structure, then that structure includes parameters that affect advanced blend operations.

The `VkPipelineColorBlendAdvancedStateCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_blend_operation_advanced
typedef struct VkPipelineColorBlendAdvancedStateCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           srcPremultiplied;
    VkBool32           dstPremultiplied;
    VkBlendOverlapEXT  blendOverlap;
} VkPipelineColorBlendAdvancedStateCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `srcPremultiplied` specifies whether the source color of the blend operation is treated as premultiplied.
- `dstPremultiplied` specifies whether the destination color of the blend operation is treated as premultiplied.
- `blendOverlap` is a `VkBlendOverlapEXT` value specifying how the source and destination sample's coverage is correlated.

If this structure is not present, `srcPremultiplied` and `dstPremultiplied` are both considered to be `VK_TRUE`, and `blendOverlap` is considered to be `VK_BLEND_OVERLAP_UNCORRELATED_EXT`.

Valid Usage

- VUID-VkPipelineColorBlendAdvancedStateCreateInfoEXT-srcPremultiplied-01424
If the [non-premultiplied source color](#) property is not supported, `srcPremultiplied` **must** be

VK_TRUE

- VUID-VkPipelineColorBlendAdvancedStateCreateInfoEXT-dstPremultiplied-01425
If the [non-premultiplied destination color](#) property is not supported, [dstPremultiplied](#) **must** be [VK_TRUE](#)
- VUID-VkPipelineColorBlendAdvancedStateCreateInfoEXT-blendOverlap-01426
If the [correlated overlap](#) property is not supported, [blendOverlap](#) **must** be [VK_BLEND_OVERLAP_UNCORRELATED_EXT](#)

Valid Usage (Implicit)

- VUID-VkPipelineColorBlendAdvancedStateCreateInfoEXT-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_ADVANCED_STATE_CREATE_INFO_EXT](#)
- VUID-VkPipelineColorBlendAdvancedStateCreateInfoEXT-blendOverlap-parameter
[blendOverlap](#) **must** be a valid [VkBlendOverlapEXT](#) value

To [dynamically set](#) the advanced blend state, call:

```
// Provided by VK_EXT_blend_operation_advanced with VK_EXT_extended_dynamic_state3,  
VK_EXT_blend_operation_advanced with VK_EXT_shader_object  
void vkCmdSetColorBlendAdvancedEXT(  
    VkCommandBuffer                commandBuffer,  
    uint32_t                       firstAttachment,  
    uint32_t                       attachmentCount,  
    const VkColorBlendAdvancedEXT* pColorBlendAdvanced);
```

- [commandBuffer](#) is the command buffer into which the command will be recorded.
- [firstAttachment](#) the first color attachment the advanced blend parameters apply to.
- [attachmentCount](#) the number of [VkColorBlendAdvancedEXT](#) elements in the [pColorBlendAdvanced](#) array.
- [pColorBlendAdvanced](#) an array of [VkColorBlendAdvancedEXT](#) structs that specify the advanced color blend parameters for the corresponding attachments.

This command sets the advanced blend operation parameters of the specified attachments for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with [VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT](#) set in [VkPipelineDynamicStateCreateInfo::pDynamicStates](#). Otherwise, this state is specified by the [VkPipelineColorBlendAdvancedStateCreateInfoEXT::srcPremultiplied](#), [VkPipelineColorBlendAdvancedStateCreateInfoEXT::dstPremultiplied](#), and [VkPipelineColorBlendAdvancedStateCreateInfoEXT::blendOverlap](#) values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetColorBlendAdvancedEXT-None-09423

At least one of the following **must** be true:

- The `extendedDynamicState3ColorBlendAdvanced` feature is enabled
- The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetColorBlendAdvancedEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetColorBlendAdvancedEXT-pColorBlendAdvanced-parameter
`pColorBlendAdvanced` **must** be a valid pointer to an array of `attachmentCount` valid `VkColorBlendAdvancedEXT` structures
- VUID-vkCmdSetColorBlendAdvancedEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetColorBlendAdvancedEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetColorBlendAdvancedEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetColorBlendAdvancedEXT-attachmentCount-arraylength
`attachmentCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

The `VkColorBlendAdvancedEXT` structure is defined as:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
typedef struct VkColorBlendAdvancedEXT {
    VkBlendOp          advancedBlendOp;
    VkBool32           srcPremultiplied;
    VkBool32           dstPremultiplied;
    VkBlendOverlapEXT  blendOverlap;
    VkBool32           clampResults;
} VkColorBlendAdvancedEXT;
```

- **advancedBlendOp** selects which blend operation is used to calculate the RGB values to write to the color attachment.
- **srcPremultiplied** specifies whether the source color of the blend operation is treated as premultiplied.
- **dstPremultiplied** specifies whether the destination color of the blend operation is treated as premultiplied.
- **blendOverlap** is a **VkBlendOverlapEXT** value specifying how the source and destination sample's coverage is correlated.
- **clampResults** specifies that results **must** be clamped to the [0,1] range before writing to the attachment, which is useful when the attachment format is not normalized fixed-point.

Valid Usage

- VUID-VkColorBlendAdvancedEXT-srcPremultiplied-07505
If the **non-premultiplied source color** property is not supported, **srcPremultiplied** **must** be **VK_TRUE**
- VUID-VkColorBlendAdvancedEXT-dstPremultiplied-07506
If the **non-premultiplied destination color** property is not supported, **dstPremultiplied** **must** be **VK_TRUE**
- VUID-VkColorBlendAdvancedEXT-blendOverlap-07507
If the **correlated overlap** property is not supported, **blendOverlap** **must** be **VK_BLEND_OVERLAP_UNCORRELATED_EXT**

Valid Usage (Implicit)

- VUID-VkColorBlendAdvancedEXT-advancedBlendOp-parameter
advancedBlendOp **must** be a valid **VkBlendOp** value
- VUID-VkColorBlendAdvancedEXT-blendOverlap-parameter
blendOverlap **must** be a valid **VkBlendOverlapEXT** value

When using one of the operations in table [f/X/Y/Z Advanced Blend Operations](#) or [Hue-Saturation-Luminosity Advanced Blend Operations](#), blending is performed according to the following equations:

$$\begin{aligned}
R &= f(R_s', R_d') * p_0(A_s, A_d) + Y * R_s' * p_1(A_s, A_d) + Z * R_d' * p_2(A_s, A_d) \\
G &= f(G_s', G_d') * p_0(A_s, A_d) + Y * G_s' * p_1(A_s, A_d) + Z * G_d' * p_2(A_s, A_d) \\
B &= f(B_s', B_d') * p_0(A_s, A_d) + Y * B_s' * p_1(A_s, A_d) + Z * B_d' * p_2(A_s, A_d) \\
A &= X * p_0(A_s, A_d) + Y * p_1(A_s, A_d) + Z * p_2(A_s, A_d)
\end{aligned}$$

where the function f and terms X , Y , and Z are specified in the table. The R , G , and B components of the source color used for blending are derived according to `srcPremultiplied`. If `srcPremultiplied` is `VK_TRUE`, the fragment color components are considered to have been premultiplied by the A component prior to blending. The base source color (R_s', G_s', B_s') is obtained by dividing through by the A component:

$$(R_s', G_s', B_s') = \begin{cases} (0, 0, 0) & A_s = 0 \\ (\frac{R_s}{A_s}, \frac{G_s}{A_s}, \frac{B_s}{A_s}) & \text{otherwise} \end{cases}$$

If `srcPremultiplied` is `VK_FALSE`, the fragment color components are used as the base color:

$$(R_s', G_s', B_s') = (R_s, G_s, B_s)$$

The R , G , and B components of the destination color used for blending are derived according to `dstPremultiplied`. If `dstPremultiplied` is `VK_TRUE`, the destination components are considered to have been premultiplied by the A component prior to blending. The base destination color (R_d', G_d', B_d') is obtained by dividing through by the A component:

$$(R_d', G_d', B_d') = \begin{cases} (0, 0, 0) & A_d = 0 \\ (\frac{R_d}{A_d}, \frac{G_d}{A_d}, \frac{B_d}{A_d}) & \text{otherwise} \end{cases}$$

If `dstPremultiplied` is `VK_FALSE`, the destination color components are used as the base color:

$$(R_d', G_d', B_d') = (R_d, G_d, B_d)$$

When blending using advanced blend operations, we expect that the R , G , and B components of premultiplied source and destination color inputs be stored as the product of non-premultiplied R , G , and B component values and the A component of the color. If any R , G , or B component of a premultiplied input color is non-zero and the A component is zero, the color is considered ill-formed, and the corresponding component of the blend result is undefined.

All of the advanced blend operation formulas in this chapter compute the result as a premultiplied color. If `dstPremultiplied` is `VK_FALSE`, that result color's R , G , and B components are divided by the A component before being written to the framebuffer. If any R , G , or B component of the color is non-zero and the A component is zero, the result is considered ill-formed, and the corresponding component of the blend result is undefined. If all components are zero, that value is unchanged.

If the A component of any input or result color is less than zero, the color is considered ill-formed, and all components of the blend result are undefined.

The weighting functions p_0 , p_1 , and p_2 are defined in table [Advanced Blend Overlap Modes](#). In these

functions, the A components of the source and destination colors are taken to indicate the portion of the pixel covered by the fragment (source) and the fragments previously accumulated in the pixel (destination). The functions p_0 , p_1 , and p_2 approximate the relative portion of the pixel covered by the intersection of the source and destination, covered only by the source, and covered only by the destination, respectively.

Possible values of `VkPipelineColorBlendAdvancedStateCreateInfoEXT::blendOverlap`, specifying the blend overlap functions, are:

```
// Provided by VK_EXT_blend_operation_advanced
typedef enum VkBlendOverlapEXT {
    VK_BLEND_OVERLAP_UNCORRELATED_EXT = 0,
    VK_BLEND_OVERLAP_DISJOINT_EXT = 1,
    VK_BLEND_OVERLAP_CONJOINT_EXT = 2,
} VkBlendOverlapEXT;
```

- `VK_BLEND_OVERLAP_UNCORRELATED_EXT` specifies that there is no correlation between the source and destination coverage.
- `VK_BLEND_OVERLAP_CONJOINT_EXT` specifies that the source and destination coverage are considered to have maximal overlap.
- `VK_BLEND_OVERLAP_DISJOINT_EXT` specifies that the source and destination coverage are considered to have minimal overlap.

Table 35. Advanced Blend Overlap Modes

Overlap Mode	Weighting Equations
<code>VK_BLEND_OVERLAP_UNCORRELATED_EXT</code>	$p_0(A_s, A_d) = A_s A_d$ $p_1(A_s, A_d) = A_s (1 - A_d)$ $p_2(A_s, A_d) = A_d (1 - A_s)$
<code>VK_BLEND_OVERLAP_CONJOINT_EXT</code>	$p_0(A_s, A_d) = \min(A_s, A_d)$ $p_1(A_s, A_d) = \max(A_s - A_d, 0)$ $p_2(A_s, A_d) = \max(A_d - A_s, 0)$
<code>VK_BLEND_OVERLAP_DISJOINT_EXT</code>	$p_0(A_s, A_d) = \max(A_s + A_d - 1, 0)$ $p_1(A_s, A_d) = \min(A_s, 1 - A_d)$ $p_2(A_s, A_d) = \min(A_d, 1 - A_s)$

Table 36. f/X/Y/Z Advanced Blend Operations

Mode	Blend Coefficients
<code>VK_BLEND_OP_ZERO_EXT</code>	$(X, Y, Z) = (0, 0, 0)$ $f(C_s, C_d) = 0$
<code>VK_BLEND_OP_SRC_EXT</code>	$(X, Y, Z) = (1, 1, 0)$ $f(C_s, C_d) = C_s$

Mode	Blend Coefficients
VK_BLEND_OP_DST_EXT	$(X, Y, Z) = (1, 0, 1)$ $f(C_s, C_d) = C_d$
VK_BLEND_OP_SRC_OVER_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = C_s$
VK_BLEND_OP_DST_OVER_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = C_d$
VK_BLEND_OP_SRC_IN_EXT	$(X, Y, Z) = (1, 0, 0)$ $f(C_s, C_d) = C_s$
VK_BLEND_OP_DST_IN_EXT	$(X, Y, Z) = (1, 0, 0)$ $f(C_s, C_d) = C_d$
VK_BLEND_OP_SRC_OUT_EXT	$(X, Y, Z) = (0, 1, 0)$ $f(C_s, C_d) = 0$
VK_BLEND_OP_DST_OUT_EXT	$(X, Y, Z) = (0, 0, 1)$ $f(C_s, C_d) = 0$
VK_BLEND_OP_SRC_ATOP_EXT	$(X, Y, Z) = (1, 0, 1)$ $f(C_s, C_d) = C_s$
VK_BLEND_OP_DST_ATOP_EXT	$(X, Y, Z) = (1, 1, 0)$ $f(C_s, C_d) = C_d$
VK_BLEND_OP_XOR_EXT	$(X, Y, Z) = (0, 1, 1)$ $f(C_s, C_d) = 0$
VK_BLEND_OP_MULTIPLY_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = C_s C_d$
VK_BLEND_OP_SCREEN_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = C_s + C_d - C_s C_d$
VK_BLEND_OP_OVERLAY_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 2C_s C_d & C_d \leq 0.5 \\ 1 - 2(1 - C_s)(1 - C_d) & \text{otherwise} \end{cases}$
VK_BLEND_OP_DARKEN_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \min(C_s, C_d)$
VK_BLEND_OP_LIGHTEN_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \max(C_s, C_d)$
VK_BLEND_OP_COLOR Dodge_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 0 & C_d \leq 0 \\ \min(1, \frac{C_d}{1 - C_s}) & C_d > 0 \text{ and } C_s < 1 \\ 1 & C_d > 0 \text{ and } C_s \geq 1 \end{cases}$

Mode	Blend Coefficients
VK_BLEND_OP_COLORBURN_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 1 & C_d \geq 1 \\ 1 - \min(1, \frac{1 - C_d}{C_s}) & C_d < 1 \text{ and } C_s > 0 \\ 0 & C_d < 1 \text{ and } C_s \leq 0 \end{cases}$
VK_BLEND_OP_HARDLIGHT_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 2C_s C_d & C_s \leq 0.5 \\ 1 - 2(1 - C_s)(1 - C_d) & \text{otherwise} \end{cases}$
VK_BLEND_OP_SOFTLIGHT_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} C_d - (1 - 2C_s)C_d(1 - C_d) & C_s \leq 0.5 \\ C_d + (2C_s - 1)C_d((16C_d - 12)C_d + 3) & C_s > 0.5 \text{ and } C_d \leq 0.25 \\ C_d + (2C_s - 1)(\sqrt{C_d} - C_d) & C_s > 0.5 \text{ and } C_d > 0.25 \end{cases}$
VK_BLEND_OP_DIFFERENCE_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = C_d - C_s $
VK_BLEND_OP_EXCLUSION_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = C_s + C_d - 2C_s C_d$
VK_BLEND_OP_INVERT_EXT	$(X, Y, Z) = (1, 0, 1)$ $f(C_s, C_d) = 1 - C_d$
VK_BLEND_OP_INVERT_RGB_EXT	$(X, Y, Z) = (1, 0, 1)$ $f(C_s, C_d) = C_s(1 - C_d)$
VK_BLEND_OP_LINEARODDGE_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} C_s + C_d & C_s + C_d \leq 1 \\ 1 & \text{otherwise} \end{cases}$
VK_BLEND_OP_LINEARBURN_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} C_s + C_d - 1 & C_s + C_d > 1 \\ 0 & \text{otherwise} \end{cases}$
VK_BLEND_OP_VIVIDLIGHT_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 1 - \min(1, \frac{1 - C_d}{2C_s}) & 0 < C_s < 0.5 \\ 0 & C_s \leq 0 \\ \min(1, \frac{C_d}{2(1 - C_s)}) & 0.5 \leq C_s < 1 \\ 1 & C_s \geq 1 \end{cases}$
VK_BLEND_OP_LINEARLIGHT_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 1 & 2C_s + C_d > 2 \\ 2C_s + C_d - 1 & 1 < 2C_s + C_d \leq 2 \\ 0 & 2C_s + C_d \leq 1 \end{cases}$
VK_BLEND_OP_PINLIGHT_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 0 & 2C_s - 1 > C_d \text{ and } C_s < 0.5 \\ 2C_s - 1 & 2C_s - 1 > C_d \text{ and } C_s \geq 0.5 \\ 2C_s & 2C_s - 1 \leq C_d \text{ and } C_s < 0.5C_d \\ C_d & 2C_s - 1 \leq C_d \text{ and } C_s \geq 0.5C_d \end{cases}$
VK_BLEND_OP_HARDMIX_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \begin{cases} 0 & C_s + C_d < 1 \\ 1 & \text{otherwise} \end{cases}$

When using one of the HSL blend operations in table [Hue-Saturation-Luminosity Advanced Blend](#)

Operations as the blend operation, the RGB color components produced by the function *f* are effectively obtained by converting both the non-premultiplied source and destination colors to the HSL (hue, saturation, luminosity) color space, generating a new HSL color by selecting H, S, and L components from the source or destination according to the blend operation, and then converting the result back to RGB. In the equations below, a blended RGB color is produced according to the following pseudocode:

```
float minv3(vec3 c) {
    return min(min(c.r, c.g), c.b);
}
float maxv3(vec3 c) {
    return max(max(c.r, c.g), c.b);
}
float lumv3(vec3 c) {
    return dot(c, vec3(0.30, 0.59, 0.11));
}
float satv3(vec3 c) {
    return maxv3(c) - minv3(c);
}

// If any color components are outside [0,1], adjust the color to
// get the components in range.
vec3 ClipColor(vec3 color) {
    float lum = lumv3(color);
    float mincol = minv3(color);
    float maxcol = maxv3(color);
    if (mincol < 0.0) {
        color = lum + ((color-lum)*lum) / (lum-mincol);
    }
    if (maxcol > 1.0) {
        color = lum + ((color-lum)*(1-lum)) / (maxcol-lum);
    }
    return color;
}

// Take the base RGB color <cbase> and override its luminosity
// with that of the RGB color <clum>.
vec3 SetLum(vec3 cbase, vec3 clum) {
    float lbase = lumv3(cbase);
    float llum = lumv3(clum);
    float ldiff = llum - lbase;
    vec3 color = cbase + vec3(ldiff);
    return ClipColor(color);
}

// Take the base RGB color <cbase> and override its saturation with
// that of the RGB color <csat>. The override the luminosity of the
// result with that of the RGB color <clum>.
vec3 SetLumSat(vec3 cbase, vec3 csat, vec3 clum)
{
```

```

float minbase = minv3(cbase);
float sbase = satv3(cbase);
float ssat = satv3(csat);
vec3 color;
if (sbase > 0) {
    // Equivalent (modulo rounding errors) to setting the
    // smallest (R,G,B) component to 0, the largest to <ssat>,
    // and interpolating the "middle" component based on its
    // original value relative to the smallest/largest.
    color = (cbase - minbase) * ssat / sbase;
} else {
    color = vec3(0.0);
}
return SetLum(color, clum);
}

```

Table 37. Hue-Saturation-Luminosity Advanced Blend Operations

Mode	Result
VK_BLEND_OP_HSL_HUE_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \text{SetLumSat}(C_s, C_d, C_d)$
VK_BLEND_OP_HSL_SATURATION_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \text{SetLumSat}(C_d, C_s, C_d)$
VK_BLEND_OP_HSL_COLOR_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \text{SetLum}(C_s, C_d)$
VK_BLEND_OP_HSL_LUMINOSITY_EXT	$(X, Y, Z) = (1, 1, 1)$ $f(C_s, C_d) = \text{SetLum}(C_d, C_s)$

When using one of the operations in table [Additional RGB Blend Operations](#) as the blend operation, the source and destination colors used by these blending operations are interpreted according to `srcPremultiplied` and `dstPremultiplied`. The blending operations below are evaluated where the RGB source and destination color components are both considered to have been premultiplied by the corresponding A component.

$$\begin{aligned}
 (R_s', G_s', B_s') &= \begin{cases} (R_s, G_s, B_s) & \text{if srcPremultiplied is VK_TRUE} \\ (R_s A_s, G_s A_s, B_s A_s) & \text{if srcPremultiplied is VK_FALSE} \end{cases} \\
 (R_d', G_d', B_d') &= \begin{cases} (R_d, G_d, B_d) & \text{if dstPremultiplied is VK_TRUE} \\ (R_d A_d, G_d A_d, B_d A_d) & \text{if dstPremultiplied is VK_FALSE} \end{cases}
 \end{aligned}$$

Table 38. Additional RGB Blend Operations

Mode	Result
VK_BLEND_OP_PLUS_EXT	$(R, G, B, A) = (R_s' + R_d', \\ G_s' + G_d', \\ B_s' + B_d', \\ A_s + A_d)$
VK_BLEND_OP_PLUS_CLAMPED_EXT	$(R, G, B, A) = (\min(1, R_s' + R_d'), \\ \min(1, G_s' + G_d'), \\ \min(1, B_s' + B_d'), \\ \min(1, A_s + A_d))$
VK_BLEND_OP_PLUS_CLAMPED_ALPHA_EXT	$(R, G, B, A) = (\min(\min(1, A_s + A_d), R_s' + R_d'), \\ \min(\min(1, A_s + A_d), G_s' + G_d'), \\ \min(\min(1, A_s + A_d), B_s' + B_d'), \\ \min(1, A_s + A_d))$
VK_BLEND_OP_PLUS_DARKER_EXT	$(R, G, B, A) = (\max(0, \min(1, A_s + A_d) - ((A_s - R_s') + (A_d - R_d'))), \\ \max(0, \min(1, A_s + A_d) - ((A_s - G_s') + (A_d - G_d'))), \\ \max(0, \min(1, A_s + A_d) - ((A_s - B_s') + (A_d - B_d'))), \\ \min(1, A_s + A_d))$
VK_BLEND_OP_MINUS_EXT	$(R, G, B, A) = (R_d' - R_s', \\ G_d' - G_s', \\ B_d' - B_s', \\ A_d - A_s)$
VK_BLEND_OP_MINUS_CLAMPED_EXT	$(R, G, B, A) = (\max(0, R_d' - R_s'), \\ \max(0, G_d' - G_s'), \\ \max(0, B_d' - B_s'), \\ \max(0, A_d - A_s))$
VK_BLEND_OP_CONTRAST_EXT	$(R, G, B, A) = (\frac{A_d}{2} + 2(R_d' - \frac{A_d}{2})(R_s' - \frac{A_s}{2}), \\ \frac{A_d}{2} + 2(G_d' - \frac{A_d}{2})(G_s' - \frac{A_s}{2}), \\ \frac{A_d}{2} + 2(B_d' - \frac{A_d}{2})(B_s' - \frac{A_s}{2}), \\ A_d)$
VK_BLEND_OP_INVERT_OVG_EXT	$(R, G, B, A) = (A_s(1 - R_d') + (1 - A_s)R_d', \\ A_s(1 - G_d') + (1 - A_s)G_d', \\ A_s(1 - B_d') + (1 - A_s)B_d', \\ A_s + A_d - A_sA_d)$
VK_BLEND_OP_RED_EXT	$(R, G, B, A) = (R_s', G_d', B_d', A_d)$
VK_BLEND_OP_GREEN_EXT	$(R, G, B, A) = (R_d', G_s', B_d', A_d)$
VK_BLEND_OP_BLUE_EXT	$(R, G, B, A) = (R_d', G_d', B_s', A_d)$

27.2. Logical Operations

The application **can** enable a *logical operation* between the fragment's color values and the existing value in the framebuffer attachment. This logical operation is applied prior to updating the framebuffer attachment. Logical operations are applied only for signed and unsigned integer and normalized integer framebuffers. Logical operations are not applied to floating-point or sRGB format color attachments.

Logical operations are controlled by the `logicOpEnable` and `logicOp` members of `VkPipelineColorBlendStateCreateInfo`. The `logicOpEnable` state can also be controlled by `vkCmdSetLogicOpEnableEXT` if graphics pipeline is created with `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. If `logicOpEnable` is `VK_TRUE`, then a logical operation selected by `logicOp` is applied between each color attachment and the fragment's corresponding output value, and blending of all attachments is treated as if it were disabled. Any attachments using color formats for which logical operations are not supported simply pass through the color values unmodified. The logical operation is applied independently for each of the red, green, blue, and alpha components. The `logicOp` is selected from the following operations:

```
// Provided by VK_VERSION_1_0
typedef enum VkLogicOp {
    VK_LOGIC_OP_CLEAR = 0,
    VK_LOGIC_OP_AND = 1,
    VK_LOGIC_OP_AND_REVERSE = 2,
    VK_LOGIC_OP_COPY = 3,
    VK_LOGIC_OP_AND_INVERTED = 4,
    VK_LOGIC_OP_NO_OP = 5,
    VK_LOGIC_OP_XOR = 6,
    VK_LOGIC_OP_OR = 7,
    VK_LOGIC_OP_NOR = 8,
    VK_LOGIC_OP_EQUIVALENT = 9,
    VK_LOGIC_OP_INVERT = 10,
    VK_LOGIC_OP_OR_REVERSE = 11,
    VK_LOGIC_OP_COPY_INVERTED = 12,
    VK_LOGIC_OP_OR_INVERTED = 13,
    VK_LOGIC_OP_NAND = 14,
    VK_LOGIC_OP_SET = 15,
} VkLogicOp;
```

The logical operations supported by Vulkan are summarized in the following table in which

- $\neg$ is bitwise invert,
- $\square$ is bitwise and,
- $\sqcup$ is bitwise or,
- $\oplus$ is bitwise exclusive or,
- s is the fragment's R_{s0} , G_{s0} , B_{s0} or A_{s0} component value for the fragment output corresponding to the color attachment being updated, and
- d is the color attachment's R, G, B or A component value:

Table 39. Logical Operations

Mode	Operation
VK_LOGIC_OP_CLEAR	0
VK_LOGIC_OP_AND	$s \square d$
VK_LOGIC_OP_AND_REVERSE	$s \square \neg d$
VK_LOGIC_OP_COPY	s
VK_LOGIC_OP_AND_INVERTED	$\neg s \square d$
VK_LOGIC_OP_NO_OP	d
VK_LOGIC_OP_XOR	$s \oplus d$
VK_LOGIC_OP_OR	$s \sqcup d$
VK_LOGIC_OP_NOR	$\neg (s \sqcup d)$
VK_LOGIC_OP_EQUIVALENT	$\neg (s \square d)$
VK_LOGIC_OP_INVERT	$\neg d$
VK_LOGIC_OP_OR_REVERSE	$s \sqcup \neg d$
VK_LOGIC_OP_COPY_INVERTED	$\neg s$
VK_LOGIC_OP_OR_INVERTED	$\neg s \sqcup d$
VK_LOGIC_OP_NAND	$\neg (s \square d)$
VK_LOGIC_OP_SET	all 1s

The result of the logical operation is then written to the color attachment as controlled by the component write mask, described in [Blend Operations](#).

To [dynamically set](#) whether logical operations are enabled, call:

```
// Provided by VK_EXT_extended_dynamic_state3, VK_EXT_shader_object
void vkCmdSetLogicOpEnableEXT(
    VkCommandBuffer          commandBuffer,
    VkBool32                  logicOpEnable);
```


- `commandBuffer` is the command buffer into which the command will be recorded.
- `logicOpEnable` specifies whether logical operations are enabled.

This command sets whether logical operations are enabled for subsequent drawing commands when drawing using `shader objects`, or when the graphics pipeline is created with `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineColorBlendStateCreateInfo::logicOpEnable` value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetLogicOpEnableEXT-None-09423
At least one of the following **must** be true:
 - The `extendedDynamicState3LogicOpEnable` feature is enabled
 - The `shaderObject` feature is enabled
- VUID-vkCmdSetLogicOpEnableEXT-logicOp-07366
If the `logicOp` feature is not enabled, `logicOpEnable` **must** be `VK_FALSE`

Valid Usage (Implicit)

- VUID-vkCmdSetLogicOpEnableEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetLogicOpEnableEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetLogicOpEnableEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetLogicOpEnableEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

To [dynamically set](#) the logical operation to apply for blend state, call:

```
// Provided by VK_EXT_shader_object
void vkCmdSetLogicOpEXT(
    VkCommandBuffer          commandBuffer,
    VkLogicOp                logicOp);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `logicOp` specifies the logical operation to apply for blend state.

This command sets the logical operation for blend state for subsequent drawing commands when drawing using [shader objects](#). Otherwise, this state is specified by the [VkPipelineColorBlendStateCreateInfo::logicOp](#) value used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetLogicOpEXT-None-09422
At least one of the following **must** be true:
 - The `shaderObject` feature is enabled

Valid Usage (Implicit)

- VUID-vkCmdSetLogicOpEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdSetLogicOpEXT-logicOp-parameter
`logicOp` **must** be a valid [VkLogicOp](#) value
- VUID-vkCmdSetLogicOpEXT-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdSetLogicOpEXT-commandBuffer-cmdpool
The [VkCommandPool](#) that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetLogicOpEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

27.3. Color Write Mask

Bits which **can** be set in `VkPipelineColorBlendAttachmentState::colorWriteMask`, determining whether the final color values R, G, B and A are written to the framebuffer attachment, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkColorComponentFlagBits {
    VK_COLOR_COMPONENT_R_BIT = 0x00000001,
    VK_COLOR_COMPONENT_G_BIT = 0x00000002,
    VK_COLOR_COMPONENT_B_BIT = 0x00000004,
    VK_COLOR_COMPONENT_A_BIT = 0x00000008,
} VkColorComponentFlagBits;
```

- `VK_COLOR_COMPONENT_R_BIT` specifies that the R value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.
- `VK_COLOR_COMPONENT_G_BIT` specifies that the G value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.
- `VK_COLOR_COMPONENT_B_BIT` specifies that the B value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.
- `VK_COLOR_COMPONENT_A_BIT` specifies that the A value is written to the color attachment for the appropriate sample. Otherwise, the value in memory is unmodified.

The color write mask operation is applied regardless of whether blending is enabled.

The color write mask operation is applied only if `Color Write Enable` is enabled for the respective attachment. Otherwise the color write mask is ignored and writes to all components of the attachment are disabled.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkColorComponentFlags;
```

`VkColorComponentFlags` is a bitmask type for setting a mask of zero or more `VkColorComponentFlagBits`.

27.4. Color Write Enable

The `VkPipelineColorWriteCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_color_write_enable
typedef struct VkPipelineColorWriteCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           attachmentCount;
    const VkBool32*     pColorWriteEnables;
} VkPipelineColorWriteCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `attachmentCount` is the number of `VkBool32` elements in `pColorWriteEnables`.
- `pColorWriteEnables` is a pointer to an array of per target attachment boolean values specifying whether color writes are enabled for the given attachment.

When this structure is included in the `pNext` chain of `VkPipelineColorBlendStateCreateInfo`, it defines per-attachment color write state. If this structure is not included in the `pNext` chain, it is equivalent to specifying this structure with `attachmentCount` equal to the `attachmentCount` member of `VkPipelineColorBlendStateCreateInfo`, and `pColorWriteEnables` pointing to an array of as many `VK_TRUE` values.

If the `colorWriteEnable` feature is not enabled on the device, all `VkBool32` elements in the `pColorWriteEnables` array **must** be `VK_TRUE`.

Color Write Enable interacts with the [Color Write Mask](#) as follows:

- If `colorWriteEnable` is `VK_TRUE`, writes to the attachment are determined by the `colorWriteMask`.
- If `colorWriteEnable` is `VK_FALSE`, the `colorWriteMask` is ignored and writes to all components of the attachment are disabled. This is equivalent to specifying a `colorWriteMask` of 0.

Valid Usage

- VUID-VkPipelineColorWriteCreateInfoEXT-pAttachments-04801
If the `colorWriteEnable` feature is not enabled, all elements of `pColorWriteEnables` **must** be `VK_TRUE`
- VUID-VkPipelineColorWriteCreateInfoEXT-attachmentCount-07608

If the pipeline is being created with `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT`, `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT`, `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT`, or `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic states not set, `attachmentCount` **must** be equal to the `attachmentCount` member of the `VkPipelineColorBlendStateCreateInfo` structure specified during pipeline creation

- VUID-VkPipelineColorWriteCreateInfoEXT-attachmentCount-06655
`attachmentCount` **must** be less than or equal to the `maxColorAttachments` member of `VkPhysicalDeviceLimits`

Valid Usage (Implicit)

- VUID-VkPipelineColorWriteCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PIPELINE_COLOR_WRITE_CREATE_INFO_EXT`
- VUID-VkPipelineColorWriteCreateInfoEXT-pColorWriteEnables-parameter
If `attachmentCount` is not 0, `pColorWriteEnables` **must** be a valid pointer to an array of `attachmentCount` `VkBool32` values

To [dynamically enable or disable](#) writes to a color attachment, call:

```
// Provided by VK_EXT_color_write_enable
void vkCmdSetColorWriteEnableEXT(
    VkCommandBuffer          commandBuffer,
    uint32_t                 attachmentCount,
    const VkBool32*          pColorWriteEnables);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `attachmentCount` is the number of `VkBool32` elements in `pColorWriteEnables`.
- `pColorWriteEnables` is a pointer to an array of per target attachment boolean values specifying whether color writes are enabled for the given attachment.

This command sets the color write enables for subsequent drawing commands when drawing using [shader objects](#), or when the graphics pipeline is created with `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`. Otherwise, this state is specified by the `VkPipelineColorWriteCreateInfoEXT::pColorWriteEnables` values used to create the currently active pipeline.

Valid Usage

- VUID-vkCmdSetColorWriteEnableEXT-None-04803
The `colorWriteEnable` feature **must** be enabled
- VUID-vkCmdSetColorWriteEnableEXT-attachmentCount-06656
`attachmentCount` **must** be less than or equal to the `maxColorAttachments` member of `VkPhysicalDeviceLimits`

Valid Usage (Implicit)

- VUID-vkCmdSetColorWriteEnableEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdSetColorWriteEnableEXT-pColorWriteEnables-parameter
`pColorWriteEnables` **must** be a valid pointer to an array of `attachmentCount` `VkBool32` values
- VUID-vkCmdSetColorWriteEnableEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdSetColorWriteEnableEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics operations
- VUID-vkCmdSetColorWriteEnableEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdSetColorWriteEnableEXT-attachmentCount-arraylength
`attachmentCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics	State

Chapter 28. Dispatching Commands

Dispatching commands (commands with **Dispatch** in the name) provoke work in a compute pipeline. Dispatching commands are recorded into a command buffer and when executed by a queue, will produce work which executes according to the bound compute pipeline. A compute pipeline **must** be bound to a command buffer before any dispatching commands are recorded in that command buffer.

To record a dispatch, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDispatch(
    VkCommandBuffer          commandBuffer,
    uint32_t                 groupCountX,
    uint32_t                 groupCountY,
    uint32_t                 groupCountZ);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **groupCountX** is the number of local workgroups to dispatch in the X dimension.
- **groupCountY** is the number of local workgroups to dispatch in the Y dimension.
- **groupCountZ** is the number of local workgroups to dispatch in the Z dimension.

When the command is executed, a global workgroup consisting of **groupCountX** × **groupCountY** × **groupCountZ** local workgroups is assembled.

Valid Usage

- VUID-vkCmdDispatch-magFilter-04553
If a **VkSampler** created with **magFilter** or **minFilter** equal to **VK_FILTER_LINEAR**, and **compareEnable** equal to **VK_FALSE** is used to sample a **VkImageView** as a result of this command, then the image view's **format features** **must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT**
- VUID-vkCmdDispatch-mipmapMode-04770
If a **VkSampler** created with **mipmapMode** equal to **VK_SAMPLER_MIPMAP_MODE_LINEAR**, and **compareEnable** equal to **VK_FALSE** is used to sample a **VkImageView** as a result of this command, then the image view's **format features** **must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT**
- VUID-vkCmdDispatch-unnormalizedCoordinates-09635
If a **VkSampler** created with **unnormalizedCoordinates** equal to **VK_TRUE** is used to sample a **VkImageView** as a result of this command, then the image view's **levelCount** and **layerCount** **must** be 1
- VUID-vkCmdDispatch-unnormalizedCoordinates-09636
If a **VkSampler** created with **unnormalizedCoordinates** equal to **VK_TRUE** is used to sample a **VkImageView** as a result of this command, then the image view's **viewType** **must** be

VK_IMAGE_VIEW_TYPE_1D or VK_IMAGE_VIEW_TYPE_2D

- VUID-vkCmdDispatch-None-06479

If a [VkImageView](#) is sampled with [depth comparison](#), the image view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT](#)

- VUID-vkCmdDispatch-None-02691

If a [VkImageView](#) is accessed using atomic operations as a result of this command, then the image view's [format features](#) **must** contain [VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT](#)

- VUID-vkCmdDispatch-None-07888

If a [VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER](#) descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's [format features](#) **must** contain [VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT](#)

- VUID-vkCmdDispatch-None-02692

If a [VkImageView](#) is sampled with [VK_FILTER_CUBIC_EXT](#) as a result of this command, then the image view's [format features](#) **must** contain [VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT](#)

- VUID-vkCmdDispatch-None-02693

If the [VK_EXT_filter_cubic](#) extension is not enabled and any [VkImageView](#) is sampled with [VK_FILTER_CUBIC_EXT](#) as a result of this command, it **must** not have a [VkImageViewType](#) of [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#)

- VUID-vkCmdDispatch-filterCubic-02694

Any [VkImageView](#) being sampled with [VK_FILTER_CUBIC_EXT](#) as a result of this command **must** have a [VkImageViewType](#) and format that supports cubic filtering, as specified by [VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic](#) returned by [vkGetPhysicalDeviceImageFormatProperties2KHR](#)

- VUID-vkCmdDispatch-filterCubicMinmax-02695

Any [VkImageView](#) being sampled with [VK_FILTER_CUBIC_EXT](#) with a reduction mode of either [VK_SAMPLER_REDUCTION_MODE_MIN](#) or [VK_SAMPLER_REDUCTION_MODE_MAX](#) as a result of this command **must** have a [VkImageViewType](#) and format that supports cubic filtering together with minmax filtering, as specified by [VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax](#) returned by [vkGetPhysicalDeviceImageFormatProperties2KHR](#)

- VUID-vkCmdDispatch-OpTypeImage-07027

For any [VkImageView](#) being written as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDispatch-OpTypeImage-07028

For any [VkImageView](#) being read as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDispatch-OpTypeImage-07029

For any [VkBufferView](#) being written as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDispatch-OpTypeImage-07030
Any [VkBufferView](#) being read as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#) then the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdDispatch-None-08600
For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDispatch-None-08601
For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDispatch-None-10068
For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdDispatch-maintenance4-08602
If the [maintenance4](#) feature is not enabled, then for each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) and [VkPushConstantRange](#) arrays used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdDispatch-None-08114
Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)
- VUID-vkCmdDispatch-None-08606
If the [shaderObject](#) feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdDispatch-None-08608
If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdDispatch-None-08609
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDispatch-None-08610
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage
- VUID-vkCmdDispatch-None-08611
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage
- VUID-vkCmdDispatch-None-08607
If the [shaderObject](#) feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and [VK_NULL_HANDLE](#) shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command
- VUID-vkCmdDispatch-uniformBuffers-06935
If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatch-None-08612
If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatch-storageBuffers-06936
If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatch-None-08613
If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatch-None-06550
If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions
- VUID-vkCmdDispatch-ConstOffset-06551
If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands

- VUID-vkCmdDispatch-viewType-07752
If a [VkImageView](#) is accessed as a result of this command, then the image view's **viewType** **must** match the **Dim** operand of the **OpTypeImage** as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)
- VUID-vkCmdDispatch-format-07753
If a [VkImageView](#) is accessed as a result of this command, then the **numeric type** of the image view's **format** and the **Sampled Type** operand of the **OpTypeImage** **must** match
- VUID-vkCmdDispatch-OpImageWrite-08795
If a [VkImageView](#) created with a format other than **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDispatch-OpImageWrite-08796
If a [VkImageView](#) created with the format **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have four components
- VUID-vkCmdDispatch-OpImageWrite-04469
If a [VkBufferView](#) is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDispatch-SampledType-04470
If a [VkImageView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 64
- VUID-vkCmdDispatch-SampledType-04471
If a [VkImageView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 32
- VUID-vkCmdDispatch-SampledType-04472
If a [VkBufferView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 64
- VUID-vkCmdDispatch-SampledType-04473
If a [VkBufferView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 32
- VUID-vkCmdDispatch-sparseImageInt64Atomics-04474
If the **sparseImageInt64Atomics** feature is not enabled, [VkImage](#) objects created with the **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command
- VUID-vkCmdDispatch-sparseImageInt64Atomics-04475
If the **sparseImageInt64Atomics** feature is not enabled, [VkBuffer](#) objects created with the **VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command

- VUID-vkCmdDispatch-None-07288
Any shader invocation executed by this command **must** [terminate](#)
- VUID-vkCmdDispatch-None-09600
If a descriptor with type equal to any of [VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE](#), [VK_DESCRIPTOR_TYPE_STORAGE_IMAGE](#), or [VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT](#) is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDispatch-groupCountX-00386
[groupCountX](#) **must** be less than or equal to [VkPhysicalDeviceLimits::maxComputeWorkGroupCount\[0\]](#)
- VUID-vkCmdDispatch-groupCountY-00387
[groupCountY](#) **must** be less than or equal to [VkPhysicalDeviceLimits::maxComputeWorkGroupCount\[1\]](#)
- VUID-vkCmdDispatch-groupCountZ-00388
[groupCountZ](#) **must** be less than or equal to [VkPhysicalDeviceLimits::maxComputeWorkGroupCount\[2\]](#)

Valid Usage (Implicit)

- VUID-vkCmdDispatch-commandBuffer-parameter
[commandBuffer](#) **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdDispatch-commandBuffer-recording
[commandBuffer](#) **must** be in the [recording state](#)
- VUID-vkCmdDispatch-commandBuffer-cmdpool
The [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support compute operations
- VUID-vkCmdDispatch-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdDispatch-vidcoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to [commandBuffer](#) **must** be externally synchronized
- Host access to the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

To record an indirect dispatching command, call:

```
// Provided by VK_VERSION_1_0
void vkCmdDispatchIndirect(
    VkCommandBuffer          commandBuffer,
    VkBuffer                  buffer,
    VkDeviceSize              offset);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **buffer** is the buffer containing dispatch parameters.
- **offset** is the byte offset into **buffer** where parameters begin.

vkCmdDispatchIndirect behaves similarly to **vkCmdDispatch** except that the parameters are read by the device from a buffer during execution. The parameters of the dispatch are encoded in a **VkDispatchIndirectCommand** structure taken from **buffer** starting at **offset**.

Valid Usage

- VUID-vkCmdDispatchIndirect-magFilter-04553
If a **VkSampler** created with **magFilter** or **minFilter** equal to **VK_FILTER_LINEAR**, and **compareEnable** equal to **VK_FALSE** is used to sample a **VkImageView** as a result of this command, then the image view's **format features** **must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT**
- VUID-vkCmdDispatchIndirect-mipmapMode-04770
If a **VkSampler** created with **mipmapMode** equal to **VK_SAMPLER_MIPMAP_MODE_LINEAR**, and **compareEnable** equal to **VK_FALSE** is used to sample a **VkImageView** as a result of this command, then the image view's **format features** **must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT**
- VUID-vkCmdDispatchIndirect-unnormalizedCoordinates-09635
If a **VkSampler** created with **unnormalizedCoordinates** equal to **VK_TRUE** is used to sample a **VkImageView** as a result of this command, then the image view's **levelCount** and **layerCount** **must** be 1
- VUID-vkCmdDispatchIndirect-unnormalizedCoordinates-09636
If a **VkSampler** created with **unnormalizedCoordinates** equal to **VK_TRUE** is used to sample a **VkImageView** as a result of this command, then the image view's **viewType** **must** be **VK_IMAGE_VIEW_TYPE_1D** or **VK_IMAGE_VIEW_TYPE_2D**

- VUID-vkCmdDispatchIndirect-None-06479
If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`
- VUID-vkCmdDispatchIndirect-None-02691
If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`
- VUID-vkCmdDispatchIndirect-None-07888
If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`
- VUID-vkCmdDispatchIndirect-None-02692
If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`
- VUID-vkCmdDispatchIndirect-None-02693
If the `VK_EXT_filter_cubic` extension is not enabled and any `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, it **must** not have a `VkImageViewType` of `VK_IMAGE_VIEW_TYPE_3D`, `VK_IMAGE_VIEW_TYPE_CUBE`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`
- VUID-vkCmdDispatchIndirect-filterCubic-02694
Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`
- VUID-vkCmdDispatchIndirect-filterCubicMinmax-02695
Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` with a reduction mode of either `VK_SAMPLER_REDUCTION_MODE_MIN` or `VK_SAMPLER_REDUCTION_MODE_MAX` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering together with minmax filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`
- VUID-vkCmdDispatchIndirect-OpTypeImage-07027
For any `VkImageView` being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`
- VUID-vkCmdDispatchIndirect-OpTypeImage-07028
For any `VkImageView` being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`
- VUID-vkCmdDispatchIndirect-OpTypeImage-07029
For any `VkBufferView` being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's `buffer features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`
- VUID-vkCmdDispatchIndirect-OpTypeImage-07030

Any [VkBufferView](#) being read as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#) then the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDispatchIndirect-None-08600

For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDispatchIndirect-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDispatchIndirect-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDispatchIndirect-maintenance4-08602

If the [maintenance4](#) feature is not enabled, then for each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) and [VkPushConstantRange](#) arrays used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDispatchIndirect-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDispatchIndirect-None-08606

If the [shaderObject](#) feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDispatchIndirect-None-08608

If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDispatchIndirect-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDispatchIndirect-None-08610
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage
- VUID-vkCmdDispatchIndirect-None-08611
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage
- VUID-vkCmdDispatchIndirect-None-08607
If the [shaderObject](#) feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and [VK_NULL_HANDLE](#) shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command
- VUID-vkCmdDispatchIndirect-uniformBuffers-06935
If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchIndirect-None-08612
If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchIndirect-storageBuffers-06936
If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchIndirect-None-08613
If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchIndirect-None-06550
If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions
- VUID-vkCmdDispatchIndirect-ConstOffset-06551
If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands

- VUID-vkCmdDispatchIndirect-viewType-07752
If a [VkImageView](#) is accessed as a result of this command, then the image view's **viewType** **must** match the **Dim** operand of the **OpTypeImage** as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)
- VUID-vkCmdDispatchIndirect-format-07753
If a [VkImageView](#) is accessed as a result of this command, then the **numeric type** of the image view's **format** and the **Sampled Type** operand of the **OpTypeImage** **must** match
- VUID-vkCmdDispatchIndirect-OpImageWrite-08795
If a [VkImageView](#) created with a format other than **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDispatchIndirect-OpImageWrite-08796
If a [VkImageView](#) created with the format **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have four components
- VUID-vkCmdDispatchIndirect-OpImageWrite-04469
If a [VkBufferView](#) is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDispatchIndirect-SampledType-04470
If a [VkImageView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 64
- VUID-vkCmdDispatchIndirect-SampledType-04471
If a [VkImageView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 32
- VUID-vkCmdDispatchIndirect-SampledType-04472
If a [VkBufferView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 64
- VUID-vkCmdDispatchIndirect-SampledType-04473
If a [VkBufferView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 32
- VUID-vkCmdDispatchIndirect-sparseImageInt64Atomics-04474
If the **sparseImageInt64Atomics** feature is not enabled, [VkImage](#) objects created with the **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command
- VUID-vkCmdDispatchIndirect-sparseImageInt64Atomics-04475
If the **sparseImageInt64Atomics** feature is not enabled, [VkBuffer](#) objects created with the **VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command

- VUID-vkCmdDispatchIndirect-None-07288
Any shader invocation executed by this command **must** [terminate](#)
- VUID-vkCmdDispatchIndirect-None-09600
If a descriptor with type equal to any of [VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE](#), [VK_DESCRIPTOR_TYPE_STORAGE_IMAGE](#), or [VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT](#) is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDispatchIndirect-buffer-02708
If [buffer](#) is non-sparse then it **must** be bound completely and contiguously to a single [VkDeviceMemory](#) object
- VUID-vkCmdDispatchIndirect-buffer-02709
[buffer](#) **must** have been created with the [VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT](#) bit set
- VUID-vkCmdDispatchIndirect-offset-02710
[offset](#) **must** be a multiple of 4
- VUID-vkCmdDispatchIndirect-offset-00407
The sum of [offset](#) and the size of [VkDispatchIndirectCommand](#) **must** be less than or equal to the size of [buffer](#)

Valid Usage (Implicit)

- VUID-vkCmdDispatchIndirect-commandBuffer-parameter
[commandBuffer](#) **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdDispatchIndirect-buffer-parameter
[buffer](#) **must** be a valid [VkBuffer](#) handle
- VUID-vkCmdDispatchIndirect-commandBuffer-recording
[commandBuffer](#) **must** be in the [recording state](#)
- VUID-vkCmdDispatchIndirect-commandBuffer-cmdpool
The [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support compute operations
- VUID-vkCmdDispatchIndirect-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdDispatchIndirect-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdDispatchIndirect-commonparent
Both of [buffer](#), and [commandBuffer](#) **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to [commandBuffer](#) **must** be externally synchronized

- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

The `VkDispatchIndirectCommand` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkDispatchIndirectCommand {
    uint32_t    x;
    uint32_t    y;
    uint32_t    z;
} VkDispatchIndirectCommand;
```

- `x` is the number of local workgroups to dispatch in the X dimension.
- `y` is the number of local workgroups to dispatch in the Y dimension.
- `z` is the number of local workgroups to dispatch in the Z dimension.

The members of `VkDispatchIndirectCommand` have the same meaning as the corresponding parameters of `vkCmdDispatch`.

Valid Usage

- VUID-VkDispatchIndirectCommand-x-00417
`x` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0]`
- VUID-VkDispatchIndirectCommand-y-00418
`y` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1]`
- VUID-VkDispatchIndirectCommand-z-00419
`z` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2]`

To record a dispatch using non-zero base values for the components of `WorkgroupId`, call:

```
// Provided by VK_KHR_device_group
void vkCmdDispatchBaseKHR(
    VkCommandBuffer                commandBuffer,
    uint32_t                       baseGroupX,
    uint32_t                       baseGroupY,
    uint32_t                       baseGroupZ,
    uint32_t                       groupCountX,
    uint32_t                       groupCountY,
    uint32_t                       groupCountZ);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `baseGroupX` is the start value for the X component of `WorkgroupId`.
- `baseGroupY` is the start value for the Y component of `WorkgroupId`.
- `baseGroupZ` is the start value for the Z component of `WorkgroupId`.
- `groupCountX` is the number of local workgroups to dispatch in the X dimension.
- `groupCountY` is the number of local workgroups to dispatch in the Y dimension.
- `groupCountZ` is the number of local workgroups to dispatch in the Z dimension.

When the command is executed, a global workgroup consisting of `groupCountX` × `groupCountY` × `groupCountZ` local workgroups is assembled, with `WorkgroupId` values ranging from [`baseGroup*`, `baseGroup* + groupCount*`) in each component. `vkCmdDispatch` is equivalent to `vkCmdDispatchBase(0,0,0,groupCountX,groupCountY,groupCountZ)`.

Valid Usage

- VUID-vkCmdDispatchBase-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDispatchBase-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdDispatchBase-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdDispatchBase-unnormalizedCoordinates-09636
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`

- VUID-vkCmdDispatchBase-None-06479
If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`
- VUID-vkCmdDispatchBase-None-02691
If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`
- VUID-vkCmdDispatchBase-None-07888
If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`
- VUID-vkCmdDispatchBase-None-02692
If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`
- VUID-vkCmdDispatchBase-None-02693
If the `VK_EXT_filter_cubic` extension is not enabled and any `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, it **must** not have a `VkImageViewType` of `VK_IMAGE_VIEW_TYPE_3D`, `VK_IMAGE_VIEW_TYPE_CUBE`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`
- VUID-vkCmdDispatchBase-filterCubic-02694
Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`
- VUID-vkCmdDispatchBase-filterCubicMinmax-02695
Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` with a reduction mode of either `VK_SAMPLER_REDUCTION_MODE_MIN` or `VK_SAMPLER_REDUCTION_MODE_MAX` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering together with minmax filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`
- VUID-vkCmdDispatchBase-OpTypeImage-07027
For any `VkImageView` being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`
- VUID-vkCmdDispatchBase-OpTypeImage-07028
For any `VkImageView` being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`
- VUID-vkCmdDispatchBase-OpTypeImage-07029
For any `VkBufferView` being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's `buffer features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`
- VUID-vkCmdDispatchBase-OpTypeImage-07030

Any [VkBufferView](#) being read as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#) then the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdDispatchBase-None-08600

For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDispatchBase-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDispatchBase-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdDispatchBase-maintenance4-08602

If the [maintenance4](#) feature is not enabled, then for each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) and [VkPushConstantRange](#) arrays used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdDispatchBase-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdDispatchBase-None-08606

If the [shaderObject](#) feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdDispatchBase-None-08608

If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdDispatchBase-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdDispatchBase-None-08610
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage
- VUID-vkCmdDispatchBase-None-08611
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage
- VUID-vkCmdDispatchBase-None-08607
If the [shaderObject](#) feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and [VK_NULL_HANDLE](#) shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command
- VUID-vkCmdDispatchBase-uniformBuffers-06935
If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchBase-None-08612
If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchBase-storageBuffers-06936
If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchBase-None-08613
If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdDispatchBase-None-06550
If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions
- VUID-vkCmdDispatchBase-ConstOffset-06551
If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands

- VUID-vkCmdDispatchBase-viewType-07752
If a [VkImageView](#) is accessed as a result of this command, then the image view's **viewType** **must** match the **Dim** operand of the **OpTypeImage** as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)
- VUID-vkCmdDispatchBase-format-07753
If a [VkImageView](#) is accessed as a result of this command, then the **numeric type** of the image view's **format** and the **Sampled Type** operand of the **OpTypeImage** **must** match
- VUID-vkCmdDispatchBase-OpImageWrite-08795
If a [VkImageView](#) created with a format other than **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdDispatchBase-OpImageWrite-08796
If a [VkImageView](#) created with the format **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have four components
- VUID-vkCmdDispatchBase-OpImageWrite-04469
If a [VkBufferView](#) is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdDispatchBase-SampledType-04470
If a [VkImageView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 64
- VUID-vkCmdDispatchBase-SampledType-04471
If a [VkImageView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 32
- VUID-vkCmdDispatchBase-SampledType-04472
If a [VkBufferView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 64
- VUID-vkCmdDispatchBase-SampledType-04473
If a [VkBufferView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the **SampledType** of the **OpTypeImage** operand of that instruction **must** have a **Width** of 32
- VUID-vkCmdDispatchBase-sparseImageInt64Atomics-04474
If the **sparseImageInt64Atomics** feature is not enabled, [VkImage](#) objects created with the **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command
- VUID-vkCmdDispatchBase-sparseImageInt64Atomics-04475
If the **sparseImageInt64Atomics** feature is not enabled, [VkBuffer](#) objects created with the **VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command

- VUID-vkCmdDispatchBase-None-07288
Any shader invocation executed by this command **must** [terminate](#)
- VUID-vkCmdDispatchBase-None-09600
If a descriptor with type equal to any of [VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE](#), [VK_DESCRIPTOR_TYPE_STORAGE_IMAGE](#), or [VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT](#) is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdDispatchBase-baseGroupX-00421
[baseGroupX](#) **must** be less than [VkPhysicalDeviceLimits::maxComputeWorkGroupCount\[0\]](#)
- VUID-vkCmdDispatchBase-baseGroupY-00422
[baseGroupY](#) **must** be less than [VkPhysicalDeviceLimits::maxComputeWorkGroupCount\[1\]](#)
- VUID-vkCmdDispatchBase-baseGroupZ-00423
[baseGroupZ](#) **must** be less than [VkPhysicalDeviceLimits::maxComputeWorkGroupCount\[2\]](#)
- VUID-vkCmdDispatchBase-groupCountX-00424
[groupCountX](#) **must** be less than or equal to [VkPhysicalDeviceLimits::maxComputeWorkGroupCount\[0\]](#) minus [baseGroupX](#)
- VUID-vkCmdDispatchBase-groupCountY-00425
[groupCountY](#) **must** be less than or equal to [VkPhysicalDeviceLimits::maxComputeWorkGroupCount\[1\]](#) minus [baseGroupY](#)
- VUID-vkCmdDispatchBase-groupCountZ-00426
[groupCountZ](#) **must** be less than or equal to [VkPhysicalDeviceLimits::maxComputeWorkGroupCount\[2\]](#) minus [baseGroupZ](#)
- VUID-vkCmdDispatchBase-baseGroupX-00427
If any of [baseGroupX](#), [baseGroupY](#), or [baseGroupZ](#) are not zero, then the bound compute pipeline **must** have been created with the [VK_PIPELINE_CREATE_DISPATCH_BASE](#) flag or the bound compute shader object **must** have been created with the [VK_SHADER_CREATE_DISPATCH_BASE_BIT_EXT](#) flag

Valid Usage (Implicit)

- VUID-vkCmdDispatchBase-commandBuffer-parameter
[commandBuffer](#) **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdDispatchBase-commandBuffer-recording
[commandBuffer](#) **must** be in the [recording state](#)
- VUID-vkCmdDispatchBase-commandBuffer-cmdpool
The [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support compute operations
- VUID-vkCmdDispatchBase-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdDispatchBase-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

Chapter 29. Device-Generated Commands

This chapter discusses the generation of command buffer content on the device, for which these principle steps are to be taken:

- Define a layout describing the sequence of commands which should be generated.
- Optionally set up device-bindable shaders.
- Retrieve device addresses by `vkGetBufferDeviceAddressEXT` for setting buffers on the device.
- Fill one or more `VkBuffer` with the appropriate content that gets interpreted by the command layout.
- Create a `preprocess VkBuffer` using the device-queried allocation information.
- Optionally preprocess the input data in a separate action.
- Generate and execute the actual commands.

The preprocessing step executes in a separate logical pipeline from either graphics or compute. When preprocessing commands in a separate step they **must** be explicitly synchronized against the command execution. When not preprocessing in a separate step, the preprocessing is automatically synchronized against the command execution.

29.1. Indirect Commands Layout

The device-side command generation happens through an iterative processing of an atomic sequence comprised of command tokens, which are represented by:

```
// Provided by VK_EXT_device_generated_commands
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkIndirectCommandsLayoutEXT)
```

Each indirect command layout **must** have exactly one action command token and it **must** be the last token in the sequence.

NOTE

If the indirect commands layout contains only 1 token, it will be an action command token, and the contents of the indirect buffer will be a sequence of indirect command structures, similar to the ones used for indirect draws and dispatches. On some implementations, using indirect draws and dispatches for these cases will result in increased performance compared to using device-generated commands, due to the overhead that results from using the latter.

29.1.1. Creation and Deletion

Indirect command layouts for `VK_EXT_device_generated_commands` are created by:

```
// Provided by VK_EXT_device_generated_commands
VkResult vkCreateIndirectCommandsLayoutEXT(
    VkDevice device,
    const VkIndirectCommandsLayoutCreateInfoEXT* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkIndirectCommandsLayoutEXT* pIndirectCommandsLayout);
```

- **device** is the logical device that creates the indirect command layout.
- **pCreateInfo** is a pointer to a [VkIndirectCommandsLayoutCreateInfoEXT](#) structure containing parameters affecting creation of the indirect command layout.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pIndirectCommandsLayout** is a pointer to a [VkIndirectCommandsLayoutEXT](#) handle in which the resulting indirect command layout is returned.

Valid Usage

- VUID-vkCreateIndirectCommandsLayoutEXT-deviceGeneratedCommands-11089
The [VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands](#) feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateIndirectCommandsLayoutEXT-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCreateIndirectCommandsLayoutEXT-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkIndirectCommandsLayoutCreateInfoEXT](#) structure
- VUID-vkCreateIndirectCommandsLayoutEXT-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateIndirectCommandsLayoutEXT-pIndirectCommandsLayout-parameter
pIndirectCommandsLayout **must** be a valid pointer to a [VkIndirectCommandsLayoutEXT](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

The `VkIndirectCommandsLayoutCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsLayoutCreateInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkIndirectCommandsLayoutUsageFlagsEXT flags;
    VkShaderStageFlags       shaderStages;
    uint32_t                 indirectStride;
    VkPipelineLayout          pipelineLayout;
    uint32_t                 tokenCount;
    const VkIndirectCommandsLayoutTokenEXT* pTokens;
} VkIndirectCommandsLayoutCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkIndirectCommandsLayoutUsageFlagBitsEXT` specifying usage rules for this layout.
- `shaderStages` is the `VkShaderStageFlags` that this layout supports.
- `indirectStride` is the distance in bytes between sequences in the indirect buffer
- `pipelineLayout` is the optional `VkPipelineLayout` that tokens in this layout use. If the `dynamicGeneratedPipelineLayout` feature is enabled, `pipelineLayout` **can** be `VK_NULL_HANDLE` and the layout **must** be specified by chaining the `VkPipelineLayoutCreateInfo` structure off the `pNext`
- `tokenCount` is the length of the individual command sequence.
- `pTokens` is a pointer to an array of `VkIndirectCommandsLayoutTokenEXT` describing each command token in detail.

The following code illustrates some of the flags:

```
void cmdProcessAllSequences(cmd, indirectExecutionSet, indirectCommandsLayout,
    indirectAddress, sequencesCount)
{
    for (s = 0; s < sequencesCount; s++)
    {
        sUsed = s;

        if (indirectCommandsLayout.flags &
VK_INDIRECT_COMMANDS_LAYOUT_USAGE_UNORDERED_SEQUENCES_BIT_EXT) {
            sUsed = incoherent_implementation_dependent_permutation[ sUsed ];
        }

        cmdProcessSequence( cmd, indirectExecutionSet, indirectCommandsLayout,
            indirectAddress, sUsed );
    }
}
```

```
}
```

When tokens are consumed, an offset is computed based on token offset and stream stride. The resulting offset is required to be aligned. The alignment for a specific token is equal to the scalar alignment of the data type as defined in [Alignment Requirements](#), or 4, whichever is lower.

Valid Usage

- VUID-VkIndirectCommandsLayoutCreateInfoEXT-indirectStride-11090
`indirectStride` **must** be less than or equal to `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::maxIndirectCommandsIndirectStride`
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11091
`shaderStages` **must** only contain stages supported by `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStages`
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-tokenCount-11092
`tokenCount` **must** be less than or equal to `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::maxIndirectCommandsTokenCount`
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11093
The number of tokens in the `pTokens` array with `type` equal to `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` **must** be less than or equal to 1
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11145
The number of tokens in the `pTokens` array with `type` equal to `VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT` **must** be less than or equal to 1
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11094
The number of tokens in the `pTokens` array with `type` equal to `VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT` **must** be less than or equal to 1
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11095
If the action command token in the `pTokens` array is not an indexed draw token, then `pTokens` **must** not contain a member with `type` set to `VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT`
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11096
If the action command token in the `pTokens` array is not a non-mesh draw token, then `pTokens` **must** not contain a member with `type` set to `VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT`
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11097
If the `pTokens` array contains multiple tokens with `type` equal to `VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT`, then there **must** be no duplicate `VkIndirectCommandsVertexBufferTokenEXT::vertexBindingUnit` values
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11099
For all `VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT` and

VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT type tokens in pTokens, there **must** be no overlapping ranges between any specified push constant ranges

- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11100
The action command token **must** be the last token in the pTokens array
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11139
If the pTokens array contains a VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, then this token **must** be the first token in the array
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11101
For any element of pTokens, if type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT and the dynamicGeneratedPipelineLayout is not enabled, then the pipelineLayout **must** not be VK_NULL_HANDLE
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11102
For any element of pTokens, if type is either VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT or VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT and pipelineLayout is VK_NULL_HANDLE, then the pNext chain **must** include a VkPipelineLayoutCreateInfo struct
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11103
For any element of pTokens, the offset **must** be greater than or equal to the offset member of the previous tokens
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11104
For any element of pTokens, if type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_COUNT_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_EXT, or VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_EXT, then shaderStages **must** contain graphics stages
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11105
For any element of pTokens, if type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT, then shaderStages **must** be VK_SHADER_STAGE_COMPUTE_BIT
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-11108
For any element of pTokens, if type is VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT, then shaderStages **must** contain ray tracing stages
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11109
If shaderStages contains graphics stages then the state tokens in pTokens **must** not include VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11110
If shaderStages is VK_SHADER_STAGE_COMPUTE_BIT then the state tokens in pTokens **must** only include VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT, VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT, or VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT

- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11111
If **shaderStages** contains ray tracing stages then the state tokens in **pTokens** **must** only include
VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT,
VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT,
VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT, or
VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11112
The **shaderStages** **must** only contain stages from one of the following:
 - graphics stages
 - VK_SHADER_STAGE_COMPUTE_BIT
 - ray tracing stages
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-11113
If **shaderStages** contains VK_SHADER_STAGE_FRAGMENT_BIT, then **shaderStages** **must** also contain VK_SHADER_STAGE_VERTEX_BIT

Valid Usage (Implicit)

- VUID-VkIndirectCommandsLayoutCreateInfoEXT-sType-sType
sType **must** be VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_CREATE_INFO_EXT
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pNext-pNext
pNext **must** be NULL or a pointer to a valid instance of [VkPipelineLayoutCreateInfo](#)
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-flags-parameter
flags **must** be a valid combination of [VkIndirectCommandsLayoutUsageFlagBitsEXT](#) values
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-parameter
shaderStages **must** be a valid combination of [VkShaderStageFlagBits](#) values
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-shaderStages-requiredbitmask
shaderStages **must** not be 0
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pipelineLayout-parameter
If **pipelineLayout** is not VK_NULL_HANDLE, **pipelineLayout** **must** be a valid [VkPipelineLayout](#) handle
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-pTokens-parameter
pTokens **must** be a valid pointer to an array of **tokenCount** valid [VkIndirectCommandsLayoutTokenEXT](#) structures
- VUID-VkIndirectCommandsLayoutCreateInfoEXT-tokenCount-arraylength
tokenCount **must** be greater than 0

Bits which **can** be set in [VkIndirectCommandsLayoutCreateInfoEXT::flags](#), specifying usage hints of an indirect command layout, are:

```
// Provided by VK_EXT_device_generated_commands
typedef enum VkIndirectCommandsLayoutUsageFlagBitsEXT {
    VK_INDIRECT_COMMANDS_LAYOUT_USAGE_EXPLICIT_PREPROCESS_BIT_EXT = 0x00000001,
    VK_INDIRECT_COMMANDS_LAYOUT_USAGE_UNORDERED_SEQUENCES_BIT_EXT = 0x00000002,
} VkIndirectCommandsLayoutUsageFlagBitsEXT;
```

- `VK_INDIRECT_COMMANDS_LAYOUT_USAGE_EXPLICIT_PREPROCESS_BIT_EXT` specifies that the layout is always used with the manual preprocessing step through calling `vkCmdPreprocessGeneratedCommandsEXT` and executed by `vkCmdExecuteGeneratedCommandsEXT` with `isPreprocessed` set to `VK_TRUE`.
- `VK_INDIRECT_COMMANDS_LAYOUT_USAGE_UNORDERED_SEQUENCES_BIT_EXT` specifies that the processing of sequences will happen at an implementation-dependent order, which is not guaranteed to be deterministic using the same input data. This flag is ignored when the `shaderStages` is `VK_SHADER_STAGE_COMPUTE_BIT` as it is implied that the dispatch sequence is always unordered.

```
// Provided by VK_EXT_device_generated_commands
typedef VkFlags VkIndirectCommandsLayoutUsageFlagsEXT;
```

`VkIndirectCommandsLayoutUsageFlagsEXT` is a bitmask type for setting a mask of zero or more `VkIndirectCommandsLayoutUsageFlagBitsEXT`.

Indirect command layouts for `VK_EXT_device_generated_commands` are destroyed by:

```
// Provided by VK_EXT_device_generated_commands
void vkDestroyIndirectCommandsLayoutEXT(
    VkDevice device,
    VkIndirectCommandsLayoutEXT indirectCommandsLayout,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the layout.
- `indirectCommandsLayout` is the layout to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyIndirectCommandsLayoutEXT-indirectCommandsLayout-11114
All submitted commands that refer to `indirectCommandsLayout` **must** have completed execution
- VUID-vkDestroyIndirectCommandsLayoutEXT-indirectCommandsLayout-11115
If `VkAllocationCallbacks` were provided when `indirectCommandsLayout` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyIndirectCommandsLayoutEXT-indirectCommandsLayout-11116
If no `VkAllocationCallbacks` were provided when `indirectCommandsLayout` was created,

pAllocator must be NULL

Valid Usage (Implicit)

- VUID-vkDestroyIndirectCommandsLayoutEXT-device-parameter
device must be a valid [VkDevice](#) handle
- VUID-vkDestroyIndirectCommandsLayoutEXT-indirectCommandsLayout-parameter
If **indirectCommandsLayout** is not [VK_NULL_HANDLE](#), **indirectCommandsLayout must** be a valid [VkIndirectCommandsLayoutEXT](#) handle
- VUID-vkDestroyIndirectCommandsLayoutEXT-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkDestroyIndirectCommandsLayoutEXT-indirectCommandsLayout-parent
If **indirectCommandsLayout** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Host Synchronization

- Host access to **indirectCommandsLayout must** be externally synchronized

29.1.2. Token Input Streams

For [VK_EXT_device_generated_commands](#), the input streams **can** contain raw [uint32_t](#) values, existing indirect commands such as:

- [VkDrawIndirectCommand](#)
- [VkDrawIndexedIndirectCommand](#)
- [VkDispatchIndirectCommand](#)
- [VkTraceRaysIndirectCommandKHR](#)
- [VkTraceRaysIndirectCommand2KHR](#)

or additional commands as listed below. How the data is used is described in the next section.

The [VkBindIndexBufferIndirectCommandEXT](#) structure specifies the input data for the [VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT](#) token.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkBindIndexBufferIndirectCommandEXT {
    VkDeviceAddress    bufferAddress;
    uint32_t           size;
    VkIndexType         indexType;
} VkBindIndexBufferIndirectCommandEXT;
```

- `bufferAddress` specifies a physical address of the `VkBuffer` used as index buffer.
- `size` is the byte size range which is available for this operation from the provided address.
- `indexType` is a `VkIndexType` value specifying how indices are treated.

Valid Usage

- VUID-VkBindIndexBufferIndirectCommandEXT-None-11117
The buffer's usage flags from which the address was acquired **must** have the `VK_BUFFER_USAGE_INDEX_BUFFER_BIT` bit set
- VUID-VkBindIndexBufferIndirectCommandEXT-bufferAddress-11118
The `bufferAddress` **must** be aligned to the `VkIndexType` of the `indexType` used
- VUID-VkBindIndexBufferIndirectCommandEXT-None-11119
Each element of the buffer from which the address was acquired and that is non-sparse **must** be bound completely and contiguously to a single `VkDeviceMemory` object

Valid Usage (Implicit)

- VUID-VkBindIndexBufferIndirectCommandEXT-indexType-parameter
`indexType` **must** be a valid `VkIndexType` value

The `VkBindVertexBufferIndirectCommandEXT` structure specifies the input data for the `VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT` token.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkBindVertexBufferIndirectCommandEXT {
    VkDeviceAddress    bufferAddress;
    uint32_t           size;
    uint32_t           stride;
} VkBindVertexBufferIndirectCommandEXT;
```

- `bufferAddress` specifies a physical address of the `VkBuffer` used as vertex input binding.
- `size` is the byte size range which is available for this operation from the provided address.
- `stride` is the byte size stride for this vertex input binding as in `VkVertexInputBindingDescription::stride`.

Valid Usage

- VUID-VkBindVertexBufferIndirectCommandEXT-None-11120
The buffer's usage flag from which the address was acquired **must** have the `VK_BUFFER_USAGE_VERTEX_BUFFER_BIT` bit set
- VUID-VkBindVertexBufferIndirectCommandEXT-None-11121
Each element of the buffer from which the address was acquired and that is non-sparse

must be bound completely and contiguously to a single `VkDeviceMemory` object

The `VkDrawIndirectCountIndirectCommandEXT` structure specifies the input data for all draw-type tokens.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkDrawIndirectCountIndirectCommandEXT {
    VkDeviceAddress    bufferAddress;
    uint32_t           stride;
    uint32_t           commandCount;
} VkDrawIndirectCountIndirectCommandEXT;
```

- `bufferAddress` specifies a physical address of the `VkBuffer` used for draw commands.
- `stride` is the byte size stride for the command arguments
- `commandCount` is the number of commands to execute

The corresponding indirect draw struct data will be read from the buffer address.

Valid Usage

- VUID-VkDrawIndirectCountIndirectCommandEXT-None-11122
The buffer's usage flag from which the address was acquired **must** have the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set
- VUID-VkDrawIndirectCountIndirectCommandEXT-None-11123
Each element of the buffer from which the address was acquired and that is non-sparse **must** be bound completely and contiguously to a single `VkDeviceMemory` object

29.1.3. Tokenized Command Processing

The processing for `VK_EXT_device_generated_commands` is in principle illustrated below:

```
void cmdProcessSequence(cmd, indirectExecutionSet, indirectCommandsLayout,
indirectAddress, s)
{
    for (t = 0; t < indirectCommandsLayout.tokenCount; t++)
    {
        uint32_t offset = indirectCommandsLayout.pTokens[t].offset;
        uint32_t stride = indirectCommandsLayout.indirectStride;
        VkDeviceAddress streamData = indirectAddress;
        const void* input = streamData + stride * s + offset;

        // further details later
        indirectCommandsLayout.pTokens[t].command (cmd, indirectExecutionSet, input, s);
    }
}
```

```

void cmdProcessAllSequences(cmd, indirectExecutionSet, indirectCommandsLayout,
indirectAddress, sequencesCount)
{
    for (s = 0; s < sequencesCount; s++)
    {
        sUsed = s;

        if (indirectCommandsLayout.flags &
VK_INDIRECT_COMMANDS_LAYOUT_USAGE_UNORDERED_SEQUENCES_BIT_EXT) {
            sUsed = incoherent_implementation_dependent_permutation[ sUsed ];
        }

        cmdProcessSequence( cmd, indirectExecutionSet, indirectCommandsLayout,
indirectAddress, sUsed );
    }
}

```

The processing of each sequence is considered stateless, therefore all state changes **must** occur prior to action commands within the sequence. A single sequence is strictly targeting the [VkShaderStageFlags](#) it was created with.

The primary input data for each token is provided through [VkBuffer](#) content at preprocessing using [vkCmdPreprocessGeneratedCommandsEXT](#) or execution time using [vkCmdExecuteGeneratedCommandsEXT](#), however some functional arguments, for example push constant layouts, are specified at layout creation time. The input size is different for each token.

Possible values of those elements of the [VkIndirectCommandsLayoutCreateInfoEXT::pTokens](#) array specifying command tokens (other elements of the array specify command parameters) are:

```

// Provided by VK_EXT_device_generated_commands
typedef enum VkIndirectCommandsTokenTypeEXT {
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT = 0,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT = 1,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT = 2,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT = 3,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT = 4,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_EXT = 5,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_EXT = 6,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_COUNT_EXT = 7,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_COUNT_EXT = 8,
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT = 9,
    // Provided by VK_KHR_ray_tracing_maintenance1 with VK_EXT_device_generated_commands
    VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT = 1000386004,
} VkIndirectCommandsTokenTypeEXT;

```

Table 40. Supported Indirect Command Tokens

Common Tokens	Command Data
VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT	u32[] array of indices into the indirect execution set
VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT	u32[] raw data
VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT	u32 placeholder data (not accessed by shader)
Compute Tokens	
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT	VkDispatchIndirectCommand
Ray Tracing Tokens	
VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT	VkTraceRaysIndirectCommand2KHR
Graphics State Tokens	
VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT	VkBindIndexBufferIndirectCommandEXT
VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT	VkBindVertexBufferIndirectCommandEXT
Graphics Draw Tokens	
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_EXT	VkDrawIndexedIndirectCommand
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_EXT	VkDrawIndirectCommand
Graphics Draw Count Tokens	
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_COUNT_EXT	VkDrawIndirectCountIndirectCommandEXT with VkDrawIndexedIndirectCommand
VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_COUNT_EXT	VkDrawIndirectCountIndirectCommandEXT with VkDrawIndirectCommand

The `VkIndirectCommandsLayoutTokenEXT` structure specifies details to the function arguments that need to be known at layout creation time:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsLayoutTokenEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkIndirectCommandsTokenTypeEXT type;
    VkIndirectCommandsTokenDataEXT data;
    uint32_t                 offset;
} VkIndirectCommandsLayoutTokenEXT;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **type** specifies the **VkIndirectCommandsTokenTypeEXT** for **data**.
- **data** specifies a **VkIndirectCommandsTokenDataEXT** containing token-specific details for command execution. It is ignored if **type** does not match any member of the **VkIndirectCommandsTokenDataEXT** union.
- **offset** is the relative byte offset for the token within one sequence of the indirect buffer. The data stored at that offset is the command data for the token, e.g. **VkDispatchIndirectCommand**.

Valid Usage

- VUID-VkIndirectCommandsLayoutTokenEXT-offset-11124
offset **must** be less than or equal to **VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::maxIndirectCommandsTokenOffset**
- VUID-VkIndirectCommandsLayoutTokenEXT-offset-11125
offset **must** be aligned to 4
- VUID-VkIndirectCommandsLayoutTokenEXT-rayTracingMaintenance1-11128
If **rayTracingMaintenance1** is not enabled, **type** **must** not be **VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT**
- VUID-VkIndirectCommandsLayoutTokenEXT-deviceGeneratedCommandsMultiDrawIndirectCount-11129
If **VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::deviceGeneratedCommandsMultiDrawIndirectCount** is not supported, **type** **must** not be **VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_COUNT_EXT** or **VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_COUNT_EXT**

Valid Usage (Implicit)

- VUID-VkIndirectCommandsLayoutTokenEXT-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_TOKEN_EXT**
- VUID-VkIndirectCommandsLayoutTokenEXT-type-parameter
type **must** be a valid **VkIndirectCommandsTokenTypeEXT** value

- VUID-VkIndirectCommandsLayoutTokenEXT-pPushConstant-parameter
If `type` is `VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT`, `VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT`, the `pPushConstant` member of `data` **must** be a valid pointer to a valid `VkIndirectCommandsPushConstantTokenEXT` structure
- VUID-VkIndirectCommandsLayoutTokenEXT-pVertexBuffer-parameter
If `type` is `VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT`, the `pVertexBuffer` member of `data` **must** be a valid pointer to a valid `VkIndirectCommandsVertexBufferTokenEXT` structure
- VUID-VkIndirectCommandsLayoutTokenEXT-pIndexBuffer-parameter
If `type` is `VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT`, the `pIndexBuffer` member of `data` **must** be a valid pointer to a valid `VkIndirectCommandsIndexBufferTokenEXT` structure
- VUID-VkIndirectCommandsLayoutTokenEXT-pExecutionSet-parameter
If `type` is `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT`, the `pExecutionSet` member of `data` **must** be a valid pointer to a valid `VkIndirectCommandsExecutionSetTokenEXT` structure

The `VkIndirectCommandsTokenDataEXT` structure provides token-specific details used to generate the indirect execution layout.

```
// Provided by VK_EXT_device_generated_commands
typedef union VkIndirectCommandsTokenDataEXT {
    const VkIndirectCommandsPushConstantTokenEXT*    pPushConstant;
    const VkIndirectCommandsVertexBufferTokenEXT*    pVertexBuffer;
    const VkIndirectCommandsIndexBufferTokenEXT*     pIndexBuffer;
    const VkIndirectCommandsExecutionSetTokenEXT*     pExecutionSet;
} VkIndirectCommandsTokenDataEXT;
```

- `pPushConstant` is a pointer to a `VkIndirectCommandsPushConstantTokenEXT` struct needed for `VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT` and `VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT` tokens
- `pVertexBuffer` is a pointer to a `VkIndirectCommandsVertexBufferTokenEXT` struct needed for `VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT` tokens
- `pIndexBuffer` is a pointer to a `VkIndirectCommandsIndexBufferTokenEXT` struct needed for `VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT` tokens
- `pExecutionSet` is a pointer to a `VkIndirectCommandsExecutionSetTokenEXT` struct needed for `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` tokens

The appropriate member of the union **must** be set for each token.

The following code provides detailed information on how an individual sequence is processed. For valid usage, all restrictions from the regular commands apply.

```

void cmdProcessSequence(cmd, indirectExecutionSet, indirectCommandsLayout,
indirectAddress, s)
{
    for (uint32_t t = 0; t < indirectCommandsLayout.tokenCount; t++) {
        VkIndirectCommandsLayoutTokenEXT *token = &indirectCommandsLayout.pTokens[t];

        uint32_t offset = token->offset;
        uint32_t stride = indirectCommandsLayout.indirectStride;
        VkDeviceAddress streamData = indirectAddress;
        const void* input = streamData + stride * s + offset;

        switch (token->tokenType) {
        case VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT:
            uint32_t *bind = input;
            VkIndirectCommandsExecutionSetTokenEXT *info = token->data.pExecutionSet;

            if (info->type == VK_INDIRECT_EXECUTION_SET_INFO_TYPE_PIPELINES_EXT) {
                vkCmdBindPipeline(cmd, indirectExecutionSet.pipelineBindPoint,
indirectExecutionSet.pipelines[*bind]);
            } else {
                VkShaderStageFlagBits stages[];
                VkShaderEXT shaders[];
                uint32_t i = 0;
                IterateBitmaskLSBToMSB(iter, info->shaderStages) {
                    stages[i] = iter;
                    shaders[i] = indirectExecutionSet.shaders[bind[i]].shaderObject;
                    i++;
                }
                vkCmdBindShadersEXT(cmd, i, stages, shaders);
            }
            break;

        case VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT:
            uint32_t* data = input;
            VkPushConstantsInfoKHR info = {
                VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO_KHR,
                // this can also use 'dynamicGeneratedPipelineLayout' to pass a
VkPipelineLayoutCreateInfo from pNext
                indirectCommandsLayout.pipelineLayout,
                token->token.pushConstant.updateRange.shaderStages,
                token->token.pushConstant.updateRange.offset,
                token->token.pushConstant.updateRange.size,
                data
            };

            vkCmdPushConstants2KHR(cmd, &info);
            break;

        case VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT:
            VkPushConstantsInfoKHR info = {

```

```

        VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO_KHR,
        // this can also use 'dynamicGeneratedPipelineLayout' to pass a
        VkPipelineLayoutCreateInfo from pNext
        indirectCommandsLayout.pipelineLayout,
        token->token.pushConstant.updateRange.shaderStages,
        token->token.pushConstant.updateRange.offset,
        // this must be 4
        token->token.pushConstant.updateRange.size,
        // this just updates the sequence index
        &s
    };

    vkCmdPushConstants2KHR(cmd, &info);
    break;

case VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT:
    VkBindIndexBufferIndirectCommandEXT* data = input;

    vkCmdBindIndexBuffer(cmd, deriveBuffer(data->bufferAddress), deriveOffset(data->bufferAddress), data->indexType);
    break;

case VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT:
    VkBindVertexBufferIndirectCommandEXT* data = input;

    vkCmdBindVertexBuffers2(cmd, token->token.vertexBuffer->vertexBindingUnit, 1,
    &deriveBuffer(data->bufferAddress),
    &deriveOffset(data->bufferAddress), data->size, data->stride);
    break;

case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_EXT:
    VkDrawIndexedIndirectCommand *data = input;

    vkCmdDrawIndexed(cmd, data->indexCount, data->instanceCount, data->firstIndex,
    data->vertexOffset, data->firstInstance);
    break;
case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_INDEXED_COUNT_EXT:
    VkDrawIndirectCountIndirectCommandEXT* data = input;

    vkCmdDrawIndexedIndirect(cmd, deriveBuffer(data->bufferAddress), deriveoffset
    (data->bufferAddress), min(data->commandCount, indirectCommandsLayout.maxDrawCount),
    data->stride);
    break;

case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_EXT:
    VkDrawIndirectCommand* data = input;

    vkCmdDraw(cmd, data->vertex_count, data->instanceCount, data->firstVertex, data->firstIndex);
    break;

```



```

case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_COUNT_EXT:
    VkDrawIndirectCountIndirectCommandEXT* data = input;

    vkCmdDrawIndirect(cmd, deriveBuffer(data->bufferAddress), deriveoffset(data->bufferAddress), min(data->commandCount, indirectCommandsLayout.maxDrawCount), data->stride);
    break;

// only available if VK_NV_mesh_shader is enabled
case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_NV_EXT:
    VkDrawMeshTasksIndirectCommandNV *data = input;

    vkCmdDrawMeshTasksNV(cmd, data->taskCount, data->firstTask);
    break;

// only available if VK_NV_mesh_shader is enabled
case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_NV_EXT:
    VkDrawIndirectCountIndirectCommandEXT* data = input;

    vkCmdDrawMeshTasksIndirectCountNV(cmd, deriveBuffer(data->bufferAddress), deriveoffset(data->bufferAddress), min(data->commandCount, indirectCommandsLayout.maxDrawCount), data->stride);
    break;

// only available if VK_EXT_mesh_shader is enabled
case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_EXT:
    VkDrawMeshTasksIndirectCommandEXT *data = input;

    vkCmdDrawMeshTasksEXT(cmd, data->groupCountX, data->groupCountY, data->groupCountZ);
    break;

// only available if VK_EXT_mesh_shader is enabled
case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DRAW_MESH_TASKS_COUNT_EXT:
    VkDrawIndirectCountIndirectCommandEXT* data = input;

    vkCmdDrawMeshTasksIndirectCountEXT(cmd, deriveBuffer(data->bufferAddress), deriveoffset(data->bufferAddress), min(data->commandCount, indirectCommandsLayout.maxDrawCount), data->stride);
    break;

case VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT:
    VkDispatchIndirectCommand *data = input;

    vkCmdDispatch(cmd, data->x, data->y, data->z);
    break;

// only available if VK_KHR_ray_tracing_maintenance1 is enabled
case VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT:
    vkCmdTraceRaysIndirect2KHR(cmd, deriveBuffer(input));

```

```

        break;
    }
}

```

The `VkIndirectCommandsPushConstantTokenEXT` structure specifies the layout token info for `VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT` and `VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT` tokens.

```

// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsPushConstantTokenEXT {
    VkPushConstantRange    updateRange;
} VkIndirectCommandsPushConstantTokenEXT;

```

- `updateRange` is the push constant range that will be updated by the token.

The `stageFlags` member of `updateRange` is ignored.

Valid Usage

- VUID-VkIndirectCommandsPushConstantTokenEXT-updateRange-11132
`updateRange` **must** be contained within the push constant info used by `VkIndirectCommandsLayoutCreateInfoEXT`
- VUID-VkIndirectCommandsPushConstantTokenEXT-size-11133
If the token type is `VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT`, the `size` member of `updateRange` **must** be 4

Valid Usage (Implicit)

- VUID-VkIndirectCommandsPushConstantTokenEXT-updateRange-parameter
`updateRange` **must** be a valid `VkPushConstantRange` structure

The `VkIndirectCommandsVertexBufferTokenEXT` structure specifies the layout token info for the `VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT` token.

```

// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsVertexBufferTokenEXT {
    uint32_t    vertexBindingUnit;
} VkIndirectCommandsVertexBufferTokenEXT;

```

- `vertexBindingUnit` is the vertex input binding number to be bound.

Valid Usage

- VUID-VkIndirectCommandsVertexBufferTokenEXT-vertexBindingUnit-11134
`vertexBindingUnit` **must** be less than the total number of vertex input bindings in use by the current graphics state

The `VkIndirectCommandsIndexBufferTokenEXT` structure specifies the layout token info for the `VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT` token.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsIndexBufferTokenEXT {
    VkIndirectCommandsInputModeFlagBitsEXT    mode;
} VkIndirectCommandsIndexBufferTokenEXT;
```

- `mode` specifies the mode to use with this token.

This allows for easy layering of Vulkan atop other APIs. When `VK_INDIRECT_COMMANDS_INPUT_MODE_DXGI_INDEX_BUFFER_EXT` is specified, the indirect buffer can contain a `D3D12_INDEX_BUFFER_VIEW` instead of `VkBindIndexBufferIndirectCommandEXT` as D3D's DXGI format value is mapped to the `VkIndexType`. It works as both structs are otherwise binary compatible.

Valid Usage

- VUID-VkIndirectCommandsIndexBufferTokenEXT-mode-11135
`mode` **must** be non-zero
- VUID-VkIndirectCommandsIndexBufferTokenEXT-mode-11136
`mode` **must** be one of the bits supported in `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsInputModes`

Valid Usage (Implicit)

- VUID-VkIndirectCommandsIndexBufferTokenEXT-mode-parameter
`mode` **must** be a valid `VkIndirectCommandsInputModeFlagBitsEXT` value

Bits which are set in `VkIndirectCommandsIndexBufferTokenEXT::mode`, specifying how an index buffer is used, are:

```
// Provided by VK_EXT_device_generated_commands
typedef enum VkIndirectCommandsInputModeFlagBitsEXT {
    VK_INDIRECT_COMMANDS_INPUT_MODE_VULKAN_INDEX_BUFFER_EXT = 0x00000001,
    VK_INDIRECT_COMMANDS_INPUT_MODE_DXGI_INDEX_BUFFER_EXT = 0x00000002,
} VkIndirectCommandsInputModeFlagBitsEXT;
```

- `VK_INDIRECT_COMMANDS_INPUT_MODE_VULKAN_INDEX_BUFFER_EXT` indicates that the indirect buffer contains `VkBindIndexBufferIndirectCommandEXT`.
- `VK_INDIRECT_COMMANDS_INPUT_MODE_DXGI_INDEX_BUFFER_EXT` indicates that the indirect buffer contains `D3D12_INDEX_BUFFER_VIEW`.

```
// Provided by VK_EXT_device_generated_commands
typedef VkFlags VkIndirectCommandsInputModeFlagsEXT;
```

`VkIndirectCommandsInputModeFlagsEXT` is a bitmask type for setting a mask of zero or more `VkIndirectCommandsInputModeFlagBitsEXT`.

The `VkIndirectCommandsExecutionSetTokenEXT` structure specifies the input data for the `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token.

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectCommandsExecutionSetTokenEXT {
    VkIndirectExecutionSetInfoTypeEXT    type;
    VkShaderStageFlags                   shaderStages;
} VkIndirectCommandsExecutionSetTokenEXT;
```

- `type` describes the type of indirect execution set in use.
- `shaderStages` specifies the shaders that will be changed by this token.

Valid Usage

- VUID-VkIndirectCommandsExecutionSetTokenEXT-shaderStages-11137
Each bit in `shaderStages` must be supported by `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesPipelineBinding` or `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesShaderBinding`

Valid Usage (Implicit)

- VUID-VkIndirectCommandsExecutionSetTokenEXT-type-parameter `type` must be a valid `VkIndirectExecutionSetInfoTypeEXT` value
- VUID-VkIndirectCommandsExecutionSetTokenEXT-shaderStages-parameter

shaderStages **must** be a valid combination of **VkShaderStageFlagBits** values

- VUID-VkIndirectCommandsExecutionSetTokenEXT-shaderStages-requiredbitmask **shaderStages** **must** not be 0

29.2. Indirect Commands Generation and Execution

The generation of commands on the device requires a **preprocess** buffer.

With **VK_EXT_device_generated_commands**, to retrieve the memory size and alignment requirements of a particular execution state call:

```
// Provided by VK_EXT_device_generated_commands
void vkGetGeneratedCommandsMemoryRequirementsEXT(
    VkDevice device,
    const VkGeneratedCommandsMemoryRequirementsInfoEXT* pInfo,
    VkMemoryRequirements2* pMemoryRequirements);
```

- **device** is the logical device that owns the buffer.
- **pInfo** is a pointer to a **VkGeneratedCommandsMemoryRequirementsInfoEXT** structure containing parameters required for the memory requirements query.
- **pMemoryRequirements** is a pointer to a **VkMemoryRequirements2** structure in which the memory requirements of the buffer object are returned.

If the size returned is zero, the preprocessing step can be skipped for this layout.

Valid Usage (Implicit)

- VUID-vkGetGeneratedCommandsMemoryRequirementsEXT-device-parameter **device** **must** be a valid **VkDevice** handle
- VUID-vkGetGeneratedCommandsMemoryRequirementsEXT-pInfo-parameter **pInfo** **must** be a valid pointer to a valid **VkGeneratedCommandsMemoryRequirementsInfoEXT** structure
- VUID-vkGetGeneratedCommandsMemoryRequirementsEXT-pMemoryRequirements-parameter **pMemoryRequirements** **must** be a valid pointer to a **VkMemoryRequirements2** structure

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkGeneratedCommandsMemoryRequirementsInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkIndirectExecutionSetEXT indirectExecutionSet;
    VkIndirectCommandsLayoutEXT indirectCommandsLayout;
    uint32_t             maxSequenceCount;
    uint32_t             maxDrawCount;
} VkGeneratedCommandsMemoryRequirementsInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `indirectExecutionSet` is the indirect execution set to be used for binding shaders.
- `indirectCommandsLayout` is the `VkIndirectCommandsLayoutEXT` that this buffer memory is intended to be used with.
- `maxSequenceCount` is the maximum number of sequences that this buffer memory can be used with.
- `maxDrawCount` is the maximum number of indirect draws that can be executed by any COUNT-type multi-draw indirect tokens. The draw count in the indirect buffer is clamped to this value for these token types.

If the action command token for the layout is not a COUNT-type multi-draw indirect token, `maxDrawCount` is ignored.

Valid Usage

- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-maxSequencesCount-11009
`maxSequencesCount` **must** be less or equal to `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::maxIndirectSequenceCount`
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectCommandsLayout-11010
 If `indirectCommandsLayout` was created with a token sequence that contained the `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token, `indirectExecutionSet` **must** not be `VK_NULL_HANDLE`
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectCommandsLayout-11151
 If `indirectCommandsLayout` was created with a token sequence that contained the `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token, the shader stages used to create the initial shader state of `indirectExecutionSet` **must** equal the `VkIndirectCommandsExecutionSetTokenEXT::shaderStages` used to create `indirectCommandsLayout`
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectCommandsLayout-11011
 If `indirectCommandsLayout` was not created with a token sequence that contained the

VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT token, `indirectExecutionSet` **must** be VK_NULL_HANDLE

- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-maxDrawCount-11146
When not ignored, `maxDrawCount` × `maxSequenceCount` **must** be less than 2^{24}
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectExecutionSet-11012
If `indirectExecutionSet` is VK_NULL_HANDLE, either a `VkGeneratedCommandsPipelineInfoEXT` or a `VkGeneratedCommandsShaderInfoEXT` **must** be included in the `pNext` chain

Valid Usage (Implicit)

- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-sType-sType
`sType` **must** be VK_STRUCTURE_TYPE_GENERATED_COMMANDS_MEMORY_REQUIREMENTS_INFO_EXT
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either NULL or a pointer to a valid instance of `VkGeneratedCommandsPipelineInfoEXT` or `VkGeneratedCommandsShaderInfoEXT`
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectExecutionSet-parameter
If `indirectExecutionSet` is not VK_NULL_HANDLE, `indirectExecutionSet` **must** be a valid `VkIndirectExecutionSetEXT` handle
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-indirectCommandsLayout-parameter
`indirectCommandsLayout` **must** be a valid `VkIndirectCommandsLayoutEXT` handle
- VUID-VkGeneratedCommandsMemoryRequirementsInfoEXT-commonparent
Both of `indirectCommandsLayout`, and `indirectExecutionSet` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkGeneratedCommandsPipelineInfoEXT {
    VkStructureType    sType;
    void*              pNext;
    VkPipeline          pipeline;
} VkGeneratedCommandsPipelineInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is NULL or a pointer to a structure extending this structure.
- `pipeline` is a valid pipeline object.

Valid Usage (Implicit)

- VUID-VkGeneratedCommandsPipelineInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_GENERATED_COMMANDS_PIPELINE_INFO_EXT`
- VUID-VkGeneratedCommandsPipelineInfoEXT-pipeline-parameter
`pipeline` **must** be a valid `VkPipeline` handle

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkGeneratedCommandsShaderInfoEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           shaderCount;
    const VkShaderEXT* pShaders;
} VkGeneratedCommandsShaderInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderCount` is the size of the `pShaders` array.
- `pShaders` is a pointer to an array of shader objects.

Valid Usage

- VUID-VkGeneratedCommandsShaderInfoEXT-pShaders-11127
`pShaders` **must** not contain more than one shader object for a given `VkShaderStageFlagBits` stage

Valid Usage (Implicit)

- VUID-VkGeneratedCommandsShaderInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_GENERATED_COMMANDS_SHADER_INFO_EXT`
- VUID-VkGeneratedCommandsShaderInfoEXT-pShaders-parameter
`pShaders` **must** be a valid pointer to an array of `shaderCount` valid `VkShaderEXT` handles
- VUID-VkGeneratedCommandsShaderInfoEXT-shaderCount-arraylength
`shaderCount` **must** be greater than 0

29.2.1. Indirect Execution Sets

Indirect Execution Sets contain sets of pipelines or shader objects which can be bound individually.

```
// Provided by VK_EXT_device_generated_commands
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkIndirectExecutionSetEXT)
```

Indirect Execution Sets allow the device to bind different shaders and pipeline states using [Device-Generated Commands](#).

Indirect Execution Sets are created by calling:

```
// Provided by VK_EXT_device_generated_commands
VkResult vkCreateIndirectExecutionSetEXT(
    VkDevice device,
    const VkIndirectExecutionSetCreateInfoEXT* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkIndirectExecutionSetEXT* pIndirectExecutionSet);
```

- **device** is the logical device that creates the indirect execution set.
- **pCreateInfo** is a pointer to a [VkIndirectExecutionSetCreateInfoEXT](#) structure containing parameters affecting creation of the indirect execution set.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pIndirectExecutionSet** is a pointer to a [VkIndirectExecutionSetEXT](#) handle in which the resulting indirect execution set is returned.

Valid Usage

- VUID-vkCreateIndirectExecutionSetEXT-deviceGeneratedCommands-11013
The **VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands** feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCreateIndirectExecutionSetEXT-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCreateIndirectExecutionSetEXT-pCreateInfo-parameter
pCreateInfo **must** be a valid pointer to a valid [VkIndirectExecutionSetCreateInfoEXT](#) structure
- VUID-vkCreateIndirectExecutionSetEXT-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateIndirectExecutionSetEXT-pIndirectExecutionSet-parameter
pIndirectExecutionSet **must** be a valid pointer to a [VkIndirectExecutionSetEXT](#) handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkIndirectExecutionSetCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectExecutionSetCreateInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkIndirectExecutionSetInfoTypeEXT type;
    VkIndirectExecutionSetInfoEXT info;
} VkIndirectExecutionSetCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `type` is a `VkIndirectExecutionSetInfoTypeEXT` describing the type of set being created and determining which field of the `info` union will be used.
- `info` is a `VkIndirectExecutionSetInfoEXT` union containing layout information for the set.

Valid Usage

- VUID-VkIndirectExecutionSetCreateInfoEXT-maxIndirectShaderObjectCount-11014
If `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::maxIndirectShaderObjectCount` is zero or `shaderObject` is not enabled `type` must not be `VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT`

Valid Usage (Implicit)

- VUID-VkIndirectExecutionSetCreateInfoEXT-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_CREATE_INFO_EXT`
- VUID-VkIndirectExecutionSetCreateInfoEXT-type-parameter
`type` must be a valid `VkIndirectExecutionSetInfoTypeEXT` value
- VUID-VkIndirectExecutionSetCreateInfoEXT-pPipelineInfo-parameter
If `type` is `VK_INDIRECT_EXECUTION_SET_INFO_TYPE_PIPELINES_EXT`, the `pPipelineInfo` member of `info` must be a valid pointer to a valid `VkIndirectExecutionSetPipelineInfoEXT` structure

- VUID-VkIndirectExecutionSetCreateInfoEXT-pShaderInfo-parameter

If `type` is `VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT`, the `pShaderInfo` member of `info` **must** be a valid pointer to a valid `VkIndirectExecutionSetShaderInfoEXT` structure

Values which **can** be set in `VkIndirectExecutionSetCreateInfoEXT::type`, specifying contents of an indirect execution set, are:

```
// Provided by VK_EXT_device_generated_commands
typedef enum VkIndirectExecutionSetInfoTypeEXT {
    VK_INDIRECT_EXECUTION_SET_INFO_TYPE_PIPELINES_EXT = 0,
    VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT = 1,
} VkIndirectExecutionSetInfoTypeEXT;
```

- `VK_INDIRECT_EXECUTION_SET_INFO_TYPE_PIPELINES_EXT` specifies that the indirect execution set contains `VkPipeline` objects.
- `VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT` specifies that the indirect execution set contains `VkShaderEXT` objects.

The `VkIndirectExecutionSetInfoEXT` union is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef union VkIndirectExecutionSetInfoEXT {
    const VkIndirectExecutionSetPipelineInfoEXT* pPipelineInfo;
    const VkIndirectExecutionSetShaderInfoEXT* pShaderInfo;
} VkIndirectExecutionSetInfoEXT;
```

- `pPipelineInfo` is a pointer to a `VkIndirectExecutionSetPipelineInfoEXT` struct containing pipeline layout information for the set.
- `pShaderInfo` is a pointer to a `VkIndirectExecutionSetShaderInfoEXT` struct containing shader object layout information for the set.

The `VkIndirectExecutionSetPipelineInfoEXT` structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectExecutionSetPipelineInfoEXT {
    VkStructureType sType;
    const void* pNext;
    VkPipeline initialPipeline;
    uint32_t maxPipelineCount;
} VkIndirectExecutionSetPipelineInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `initialPipeline` is the initial pipeline for the set. This pipeline will be automatically added to the set at index `0`.
- `maxPipelineCount` is the maximum number of pipelines stored in the set.

The characteristics of `initialPipeline` will be used to validate all pipelines added to the set even if they are removed from the set or destroyed.

When an Indirect Execution Set created with pipelines is used, `initialPipeline` constitutes the initial shader state.

Valid Usage

- VUID-VkIndirectExecutionSetPipelineInfoEXT-supportedIndirectCommandsShaderStagesPipelineBinding-11015
If `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesPipelineBinding` does not contain `VK_SHADER_STAGE_COMPUTE_BIT`, the `VkPipelineBindPoint` of `initialPipeline` **must** not be `VK_PIPELINE_BIND_POINT_COMPUTE`
- VUID-VkIndirectExecutionSetPipelineInfoEXT-supportedIndirectCommandsShaderStagesPipelineBinding-11016
If `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesPipelineBinding` does not contain `VK_SHADER_STAGE_FRAGMENT_BIT`, the `VkPipelineBindPoint` of `initialPipeline` **must** not be `VK_PIPELINE_BIND_POINT_GRAPHICS`
- VUID-VkIndirectExecutionSetPipelineInfoEXT-supportedIndirectCommandsShaderStagesPipelineBinding-11017
If `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesPipelineBinding` does not contain ray tracing stages, the `VkPipelineBindPoint` of `initialPipeline` **must** not be `VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR`
- VUID-VkIndirectExecutionSetPipelineInfoEXT-maxPipelineCount-11018
`maxPipelineCount` **must** be between `1` and `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::maxIndirectPipelineCount`
- VUID-VkIndirectExecutionSetPipelineInfoEXT-initialPipeline-11019
`initialPipeline` **must** not use descriptors of type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`
- VUID-VkIndirectExecutionSetPipelineInfoEXT-initialPipeline-11153
`initialPipeline` **must** have been created with `VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT`

Valid Usage (Implicit)

- VUID-VkIndirectExecutionSetPipelineInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_PIPELINE_INFO_EXT`

- VUID-VkIndirectExecutionSetPipelineInfoEXT-initialPipeline-parameter **initialPipeline** **must** be a valid [VkPipeline](#) handle

The [VkIndirectExecutionSetShaderInfoEXT](#) structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectExecutionSetShaderInfoEXT {
    VkStructureType           sType;
    const void*               pNext;
    uint32_t                  shaderCount;
    const VkShaderEXT*         pInitialShaders;
    const VkIndirectExecutionSetShaderLayoutInfoEXT* pSetLayoutInfos;
    uint32_t                   maxShaderCount;
    uint32_t                   pushConstantRangeCount;
    const VkPushConstantRange* pPushConstantRanges;
} VkIndirectExecutionSetShaderInfoEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **shaderCount** is the number of members in the **pInitialShaders** and **pSetLayoutInfos** arrays.
- **pInitialShaders** is a pointer to an array containing a [VkShaderEXT](#) object for each shader stage that will be used in the set. These shaders will be automatically added to the set beginning at index 0.
- **pSetLayoutInfos** is a pointer to an array containing a [VkIndirectExecutionSetShaderLayoutInfoEXT](#) used by each corresponding **pInitialShaders** shader stage in the set.
- **maxShaderCount** is the maximum number of shader objects stored in the set.
- **pushConstantRangeCount** is the number of members in the **pPushConstantRanges** array.
- **pPushConstantRanges** is a pointer to the array of [VkPushConstantRange](#) ranges used by all shaders in the set.

The characteristics of **pInitialShaders** will be used to validate all shaders added to the set even if they are removed from the set or destroyed.

When an Indirect Execution Set created with shader objects is used, **pInitialShaders** constitutes the initial shader state.

Valid Usage

- VUID-VkIndirectExecutionSetShaderInfoEXT-pInitialShaders-11020
All members of **pInitialShaders** **must** have a **stage** supported by [VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesShaderBinding](#)
- VUID-VkIndirectExecutionSetShaderInfoEXT-maxShaderCount-11021

`maxShaderCount` **must** not be zero

- VUID-VkIndirectExecutionSetShaderInfoEXT-maxShaderCount-11022
`maxShaderCount` **must** be less than or equal to `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::maxIndirectShaderObjectCount`
- VUID-VkIndirectExecutionSetShaderInfoEXT-maxShaderCount-11036
`maxShaderCount` **must** be greater than or equal to `shaderCount`
- VUID-VkIndirectExecutionSetShaderInfoEXT-stage-11023
The `stage` of each element in the `pInitialShaders` array **must** be unique
- VUID-VkIndirectExecutionSetShaderInfoEXT-pInitialShaders-11154
Each member of `pInitialShaders` **must** have been created with `VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT`

Valid Usage (Implicit)

- VUID-VkIndirectExecutionSetShaderInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_INFO_EXT`
- VUID-VkIndirectExecutionSetShaderInfoEXT-pInitialShaders-parameter
`pInitialShaders` **must** be a valid pointer to an array of `shaderCount` valid `VkShaderEXT` handles
- VUID-VkIndirectExecutionSetShaderInfoEXT-pSetLayoutInfos-parameter
If `pSetLayoutInfos` is not `NULL`, `pSetLayoutInfos` **must** be a valid pointer to an array of `shaderCount` valid `VkIndirectExecutionSetShaderLayoutInfoEXT` structures
- VUID-VkIndirectExecutionSetShaderInfoEXT-pPushConstantRanges-parameter
If `pushConstantRangeCount` is not `0`, `pPushConstantRanges` **must** be a valid pointer to an array of `pushConstantRangeCount` valid `VkPushConstantRange` structures
- VUID-VkIndirectExecutionSetShaderInfoEXT-shaderCount-arraylength
`shaderCount` **must** be greater than `0`

The `VkIndirectExecutionSetShaderLayoutInfoEXT` structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkIndirectExecutionSetShaderLayoutInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    uint32_t                 setLayoutCount;
    const VkDescriptorSetLayout* pSetLayouts;
} VkIndirectExecutionSetShaderLayoutInfoEXT;
```

- `setLayoutCount` is the number of members in the `pSetLayouts` array
- `pSetLayouts` is a pointer to an array containing `VkDescriptorSetLayout` objects used by the shader stage.

Valid Usage

- VUID-VkIndirectExecutionSetShaderLayoutInfoEXT-pSetLayouts-11024

All members of `pSetLayouts` **must** not contain descriptors of type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`

Valid Usage (Implicit)

- VUID-VkIndirectExecutionSetShaderLayoutInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_LAYOUT_INFO_EXT`
- VUID-VkIndirectExecutionSetShaderLayoutInfoEXT-pSetLayouts-parameter
If `setLayoutCount` is not 0, `pSetLayouts` **must** be a valid pointer to an array of `setLayoutCount` valid or `VK_NULL_HANDLE` `VkDescriptorSetLayout` handles

Destroy an Indirect Execution Set by calling:

```
// Provided by VK_EXT_device_generated_commands
void vkDestroyIndirectExecutionSetEXT(
    VkDevice device,
    VkIndirectExecutionSetEXT indirectExecutionSet,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that owns the indirect execution set.
- `indirectExecutionSet` is the indirect execution set to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyIndirectExecutionSetEXT-indirectExecutionSet-11025

All submitted commands that refer to `indirectExecutionSet` **must** have completed execution

Valid Usage (Implicit)

- VUID-vkDestroyIndirectExecutionSetEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyIndirectExecutionSetEXT-indirectExecutionSet-parameter
If `indirectExecutionSet` is not `VK_NULL_HANDLE`, `indirectExecutionSet` **must** be a valid `VkIndirectExecutionSetEXT` handle
- VUID-vkDestroyIndirectExecutionSetEXT-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid

[VkAllocationCallbacks](#) structure

- VUID-vkDestroyIndirectExecutionSetEXT-indirectExecutionSet-parent

If `indirectExecutionSet` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `indirectExecutionSet` **must** be externally synchronized

The `VkWriteIndirectExecutionSetPipelineEXT` struct is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkWriteIndirectExecutionSetPipelineEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           index;
    VkPipeline          pipeline;
} VkWriteIndirectExecutionSetPipelineEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `index` is the element of the set to update
- `pipeline` is the pipeline to store in the indirect execution set

Valid Usage

- VUID-VkWriteIndirectExecutionSetPipelineEXT-index-11026
`index` **must** be less than the value of `VkIndirectExecutionSetPipelineInfoEXT::maxPipelineCount` used to create the set
- VUID-VkWriteIndirectExecutionSetPipelineEXT-pipeline-11027
`pipeline` **must** have been created with `VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT`
- VUID-VkWriteIndirectExecutionSetPipelineEXT-index-11029
`index` **must** not be referenced by submitted command buffers
- VUID-VkWriteIndirectExecutionSetPipelineEXT-pipeline-11030
The shader stages contained in `pipeline` **must** be supported by `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStagesPipelineBinding`

Valid Usage (Implicit)

- VUID-VkWriteIndirectExecutionSetPipelineEXT-sType-sType

sType **must** be `VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_PIPELINE_EXT`

- VUID-VkWriteIndirectExecutionSetPipelineEXT-pipeline-parameter **pipeline** **must** be a valid `VkPipeline` handle

The `VkWriteIndirectExecutionSetShaderEXT` struct is defined as:

```
// Provided by VK_EXT_device_generated_commands with VK_EXT_shader_object
typedef struct VkWriteIndirectExecutionSetShaderEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           index;
    VkShaderEXT         shader;
} VkWriteIndirectExecutionSetShaderEXT;
```

- **sType** is a `VkStructureType` value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **index** is the element of the set to update
- **shader** is the shader to store in the indirect execution set

Shaders need not be stored in the Indirect Execution Set according to their stage. The only restriction for shader indices within a set is that the value of the index **must** be less than the maximum number of shaders in the set.

Valid Usage

- VUID-VkWriteIndirectExecutionSetShaderEXT-index-11031
index **must** be less than `VkIndirectExecutionSetShaderInfoEXT::maxShaderCount`
- VUID-VkWriteIndirectExecutionSetShaderEXT-shader-11032
shader **must** have been created with `VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT`
- VUID-VkWriteIndirectExecutionSetShaderEXT-pInitialShaders-11033
A shader created with the same `VkShaderStageFlagBits` **must** have been passed in the `VkIndirectExecutionSetShaderInfoEXT::pInitialShaders` array
- VUID-VkWriteIndirectExecutionSetShaderEXT-index-11034
index **must** not be in use by submitted command buffers

Valid Usage (Implicit)

- VUID-VkWriteIndirectExecutionSetShaderEXT-sType-sType
sType **must** be `VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_SHADER_EXT`
- VUID-VkWriteIndirectExecutionSetShaderEXT-shader-parameter
shader **must** be a valid `VkShaderEXT` handle

Pipeline elements in an Indirect Execution Set can be updated by calling:

```
// Provided by VK_EXT_device_generated_commands
void vkUpdateIndirectExecutionSetPipelineEXT(
    VkDevice                                device,
    VkIndirectExecutionSetEXT               indirectExecutionSet,
    uint32_t                               executionSetWriteCount,
    const VkWriteIndirectExecutionSetPipelineEXT* pExecutionSetWrites);
```

- **device** is the logical device that owns the indirect execution set.
- **indirectExecutionSet** is the indirect execution set being updated.
- **executionSetWriteCount** is the number of elements in the **pExecutionSetWrites** array.
- **pExecutionSetWrites** is a pointer to an array of **VkWriteIndirectExecutionSetPipelineEXT** structures describing the elements to update.

Valid Usage

- VUID-vkUpdateIndirectExecutionSetPipelineEXT-indirectExecutionSet-11035
indirectExecutionSet **must** have been created with type **VK_INDIRECT_EXECUTION_SET_INFO_TYPE_PIPELINES_EXT**
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-executionSetWriteCount-11037
executionSetWriteCount **must** be less than or equal to **VkIndirectExecutionSetPipelineInfoEXT::maxPipelineCount**
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-pExecutionSetWrites-11042
Each element in the **pExecutionSetWrites** array must have a unique **VkWriteIndirectExecutionSetPipelineEXT::index**
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-None-11038
Each member of the Indirect Execution Set referenced by the update command **must** not be in use by the device
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-None-11039
The layout of each pipeline in **pExecutionSetWrites** **must** be **compatible** with the **initialPipeline** used to create the Indirect Execution Set
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-None-11040
Each pipeline in the Indirect Execution Set **must** have identically defined static and dynamic state values to the **initialPipeline** used to create the Indirect Execution Set
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11147
Each pipeline in the Indirect Execution Set **must** have identically defined **fragment outputs interface** to the **initialPipeline** used to create the Indirect Execution Set
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11152
Each pipeline in the Indirect Execution Set **must** match the **initialPipeline** used to create the Indirect Execution Set in its included shader stages
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11098
Each pipeline in the Indirect Execution Set **must** match the **initialPipeline** used to create

the Indirect Execution Set in its use of `FragDepth`

- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11086
Each pipeline in the Indirect Execution Set **must** match the `initialPipeline` used to create the Indirect Execution Set in its use of `SampleMask`
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-initialPipeline-11085
Each pipeline in the Indirect Execution Set **must** match the `initialPipeline` used to create the Indirect Execution Set in its use of `StencilExportEXT`

Valid Usage (Implicit)

- VUID-vkUpdateIndirectExecutionSetPipelineEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-indirectExecutionSet-parameter
`indirectExecutionSet` **must** be a valid `VkIndirectExecutionSetEXT` handle
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-pExecutionSetWrites-parameter
`pExecutionSetWrites` **must** be a valid pointer to an array of `executionSetWriteCount` valid `VkWriteIndirectExecutionSetPipelineEXT` structures
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-executionSetWriteCount-arraylength
`executionSetWriteCount` **must** be greater than 0
- VUID-vkUpdateIndirectExecutionSetPipelineEXT-indirectExecutionSet-parent
`indirectExecutionSet` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `indirectExecutionSet` **must** be externally synchronized

Shader object elements in an Indirect Execution Set can be updated by calling:

```
// Provided by VK_EXT_device_generated_commands
void vkUpdateIndirectExecutionSetShaderEXT(
    VkDevice                device,
    VkIndirectExecutionSetEXT indirectExecutionSet,
    uint32_t                executionSetWriteCount,
    const VkWriteIndirectExecutionSetShaderEXT* pExecutionSetWrites);
```

- `device` is the logical device that owns the indirect execution set.
- `indirectExecutionSet` is the indirect execution set being updated.
- `executionSetWriteCount` is the number of elements in the `pExecutionSetWrites` array.
- `pExecutionSetWrites` is a pointer to an array of `VkWriteIndirectExecutionSetShaderEXT` structures describing the elements to update.

Valid Usage

- VUID-vkUpdateIndirectExecutionSetShaderEXT-indirectExecutionSet-11041
`indirectExecutionSet` **must** have been created with type `VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT`
- VUID-vkUpdateIndirectExecutionSetShaderEXT-pExecutionSetWrites-11043
Each element in the `pExecutionSetWrites` array must have a unique `VkWriteIndirectExecutionSetShaderEXT::index`
- VUID-vkUpdateIndirectExecutionSetShaderEXT-None-11044
Each member of the Indirect Execution Set referenced by the update command **must** not be in use by the device
- VUID-vkUpdateIndirectExecutionSetShaderEXT-pExecutionSetWrites-11140
The descriptor layout of each shader in `pExecutionSetWrites` **must** be `compatible` with the initial layout info used to create the Indirect Execution Set
- VUID-vkUpdateIndirectExecutionSetShaderEXT-None-11148
Each fragment shader element in the Indirect Execution Set **must** have identically defined `fragment outputs interface` to the initial shader state used to create the Indirect Execution Set
- VUID-vkUpdateIndirectExecutionSetShaderEXT-FragDepth-11054
Each fragment shader element in the Indirect Execution Set **must** match the initial shader state used to create the Indirect Execution Set in its use of `FragDepth`
- VUID-vkUpdateIndirectExecutionSetShaderEXT-SampleMask-11050
Each fragment shader element in the Indirect Execution Set **must** match the initial shader state used to create the Indirect Execution Set in its use of `SampleMask`
- VUID-vkUpdateIndirectExecutionSetShaderEXT-StencilExportEXT-11003
Each fragment shader element in the Indirect Execution Set **must** match the initial shader state used to create the Indirect Execution Set in its use of `StencilExportEXT`

Valid Usage (Implicit)

- VUID-vkUpdateIndirectExecutionSetShaderEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkUpdateIndirectExecutionSetShaderEXT-indirectExecutionSet-parameter
`indirectExecutionSet` **must** be a valid `VkIndirectExecutionSetEXT` handle
- VUID-vkUpdateIndirectExecutionSetShaderEXT-pExecutionSetWrites-parameter
`pExecutionSetWrites` **must** be a valid pointer to an array of `executionSetWriteCount` valid `VkWriteIndirectExecutionSetShaderEXT` structures
- VUID-vkUpdateIndirectExecutionSetShaderEXT-executionSetWriteCount-arraylength
`executionSetWriteCount` **must** be greater than 0
- VUID-vkUpdateIndirectExecutionSetShaderEXT-indirectExecutionSet-parent
`indirectExecutionSet` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `indirectExecutionSet` **must** be externally synchronized

It is legal to update an Indirect Execution Set that is in flight as long as the element indices in `pExecutionSetWrites` are not in use. Any change to an indirect execution set requires recalculating memory requirements by calling `vkGetGeneratedCommandsMemoryRequirementsEXT` for commands that use that modified state. Commands that are in flight or those not using updated elements require no changes.

The lifetimes of pipelines and shader objects contained in a set **must** match or exceed the lifetime of the set.

With `VK_EXT_device_generated_commands`, the actual generation of commands as well as their execution on the device is handled as single action with:

```
// Provided by VK_EXT_device_generated_commands
void vkCmdExecuteGeneratedCommandsEXT(
    VkCommandBuffer                commandBuffer,
    VkBool32                       isPreprocessed,
    const VkGeneratedCommandsInfoEXT* pGeneratedCommandsInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `isPreprocessed` represents whether the input data has already been preprocessed on the device. If it is `VK_FALSE` this command will implicitly trigger the preprocessing step, otherwise not.
- `pGeneratedCommandsInfo` is a pointer to a `VkGeneratedCommandsInfoEXT` structure containing parameters affecting the generation of commands.

If the `VK_INDIRECT_COMMANDS_LAYOUT_USAGE_UNORDERED_SEQUENCES_BIT_EXT` flag was used to create the `VkGeneratedCommandsInfoEXT::indirectCommandsLayout` then the execution of sequences through this command **may** use implementation-defined ordering which is not guaranteed to be coherent using the same input data. It does not affect the order of token processing within a sequence. This is the implied ordering with `VK_INDIRECT_COMMANDS_TOKEN_TYPE_DISPATCH_EXT`.

After a call to `vkCmdExecuteGeneratedCommandsEXT`, command buffer state will become undefined according to the tokens executed. This table specifies the relationship between tokens used and state invalidation.

Table 41. Indirect Execution State Invalidation

Common Tokens	States Invalidated
<code>VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT</code>	Bound shaders and pipelines
<code>VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT</code>	Push constant data
<code>VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT</code>	Push constant data

Common Tokens	States Invalidated
VK_INDIRECT_COMMANDS_TOKEN_TYPE_INDEX_BUFFER_EXT	Index buffer
VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT	Vertex buffer

Valid Usage

- VUID-vkCmdExecuteGeneratedCommandsEXT-magFilter-04553

If a [VkSampler](#) created with [magFilter](#) or [minFilter](#) equal to [VK_FILTER_LINEAR](#), and [compareEnable](#) equal to [VK_FALSE](#) is used to sample a [VkImageView](#) as a result of this command, then the image view's [format features](#) **must** contain [VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-mipmapMode-04770

If a [VkSampler](#) created with [mipmapMode](#) equal to [VK_SAMPLER_MIPMAP_MODE_LINEAR](#), and [compareEnable](#) equal to [VK_FALSE](#) is used to sample a [VkImageView](#) as a result of this command, then the image view's [format features](#) **must** contain [VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-unnormlizedCoordinates-09635

If a [VkSampler](#) created with [unnormlizedCoordinates](#) equal to [VK_TRUE](#) is used to sample a [VkImageView](#) as a result of this command, then the image view's [levelCount](#) and [layerCount](#) **must** be 1

- VUID-vkCmdExecuteGeneratedCommandsEXT-unnormlizedCoordinates-09636

If a [VkSampler](#) created with [unnormlizedCoordinates](#) equal to [VK_TRUE](#) is used to sample a [VkImageView](#) as a result of this command, then the image view's [viewType](#) **must** be [VK_IMAGE_VIEW_TYPE_1D](#) or [VK_IMAGE_VIEW_TYPE_2D](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06479

If a [VkImageView](#) is sampled with [depth comparison](#), the image view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-02691

If a [VkImageView](#) is accessed using atomic operations as a result of this command, then the image view's [format features](#) **must** contain [VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07888

If a [VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER](#) descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's [format features](#) **must** contain [VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-02692

If a [VkImageView](#) is sampled with [VK_FILTER_CUBIC_EXT](#) as a result of this command, then the image view's [format features](#) **must** contain [VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-02693

If the [VK_EXT_filter_cubic](#) extension is not enabled and any [VkImageView](#) is sampled with [VK_FILTER_CUBIC_EXT](#) as a result of this command, it **must** not have a [VkImageViewType](#) of [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-filterCubic-02694
Any [VkImageView](#) being sampled with [VK_FILTER_CUBIC_EXT](#) as a result of this command **must** have a [VkImageViewType](#) and format that supports cubic filtering, as specified by [VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic](#) returned by [vkGetPhysicalDeviceImageFormatProperties2KHR](#)
- VUID-vkCmdExecuteGeneratedCommandsEXT-filterCubicMinmax-02695
Any [VkImageView](#) being sampled with [VK_FILTER_CUBIC_EXT](#) with a reduction mode of either [VK_SAMPLER_REDUCTION_MODE_MIN](#) or [VK_SAMPLER_REDUCTION_MODE_MAX](#) as a result of this command **must** have a [VkImageViewType](#) and format that supports cubic filtering together with minmax filtering, as specified by [VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax](#) returned by [vkGetPhysicalDeviceImageFormatProperties2KHR](#)
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpTypeImage-07027
For any [VkImageView](#) being written as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpTypeImage-07028
For any [VkImageView](#) being read as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpTypeImage-07029
For any [VkBufferView](#) being written as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpTypeImage-07030
Any [VkBufferView](#) being read as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#) then the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08600
For each set *n* that is statically used by a [bound shader](#), a descriptor set **must** have been bound to *n* at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set *n*, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08601
For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-10068
For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdExecuteGeneratedCommandsEXT-maintenance4-08602
If the **maintenance4** feature is not enabled, then for each push constant that is statically used by a **bound shader**, a push constant value **must** have been set for the same pipeline bind point, with a **VkPipelineLayout** that is compatible for push constants, with the **VkPipelineLayout** used to create the current **VkPipeline** or the **VkDescriptorSetLayout** and **VkPushConstantRange** arrays used to create the current **VkShaderEXT**, as described in **Pipeline Layout Compatibility**
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08114
Descriptors in each bound descriptor set, specified via **vkCmdBindDescriptorSets**, **must** be valid as described by **descriptor validity** if they are statically used by a **bound shader**
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08606
If the **shaderObject** feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08608
If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the **VkPipeline** object bound to the pipeline bind point used by this command, since that pipeline was bound
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08609
If the **VkPipeline** object bound to the pipeline bind point used by this command or any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a **VkSampler** object that uses unnormalized coordinates, that sampler **must** not be used to sample from any **VkImage** with a **VkImageView** of the type **VK_IMAGE_VIEW_TYPE_3D**, **VK_IMAGE_VIEW_TYPE_CUBE**, **VK_IMAGE_VIEW_TYPE_1D_ARRAY**, **VK_IMAGE_VIEW_TYPE_2D_ARRAY** or **VK_IMAGE_VIEW_TYPE_CUBE_ARRAY**, in any shader stage
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08610
If the **VkPipeline** object bound to the pipeline bind point used by this command or any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a **VkSampler** object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V **OpImageSample*** or **OpImageSparseSample*** instructions with **ImplicitLod**, **Dref** or **Proj** in their name, in any shader stage
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08611
If the **VkPipeline** object bound to the pipeline bind point used by this command or any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a **VkSampler** object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V **OpImageSample*** or **OpImageSparseSample*** instructions that includes a LOD bias or any offset values, in any shader stage
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08607
If the **shaderObject** feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and **VK_NULL_HANDLE** shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-uniformBuffers-06935
If any stage of the **VkPipeline** object bound to the pipeline bind point used by this command accesses a uniform buffer, and the **robustBufferAccess** feature is not enabled,

that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08612

If the **robustBufferAccess** feature is not enabled, and any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdExecuteGeneratedCommandsEXT-storageBuffers-06936

If any stage of the **VkPipeline** object bound to the pipeline bind point used by this command accesses a storage buffer, and the **robustBufferAccess** feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08613

If the **robustBufferAccess** feature is not enabled, and any **VkShaderEXT** bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06550

If a **bound shader** accesses a **VkSampler** or **VkImageView** object that enables **sampler Y_BC_R conversion**, that object **must** only be used with **OpImageSample*** or **OpImageSparseSample*** instructions

- VUID-vkCmdExecuteGeneratedCommandsEXT-ConstOffset-06551

If a **bound shader** accesses a **VkSampler** or **VkImageView** object that enables **sampler Y_BC_R conversion**, that object **must** not use the **ConstOffset** and **Offset** operands

- VUID-vkCmdExecuteGeneratedCommandsEXT-viewType-07752

If a **VkImageView** is accessed as a result of this command, then the image view's **viewType** **must** match the **Dim** operand of the **OpTypeImage** as described in **Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types**

- VUID-vkCmdExecuteGeneratedCommandsEXT-format-07753

If a **VkImageView** is accessed as a result of this command, then the **numeric type** of the image view's **format** and the **Sampled Type** operand of the **OpTypeImage** **must** match

- VUID-vkCmdExecuteGeneratedCommandsEXT-OpImageWrite-08795

If a **VkImageView** created with a format other than **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdExecuteGeneratedCommandsEXT-OpImageWrite-08796

If a **VkImageView** created with the format **VK_FORMAT_A8_UNORM_KHR** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have four components

- VUID-vkCmdExecuteGeneratedCommandsEXT-OpImageWrite-04469

If a **VkBufferView** is accessed using **OpImageWrite** as a result of this command, then the **Type** of the **Texel** operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdExecuteGeneratedCommandsEXT-SampledType-04470
If a `VkImageView` with a `VkFormat` that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdExecuteGeneratedCommandsEXT-SampledType-04471
If a `VkImageView` with a `VkFormat` that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdExecuteGeneratedCommandsEXT-SampledType-04472
If a `VkBufferView` with a `VkFormat` that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdExecuteGeneratedCommandsEXT-SampledType-04473
If a `VkBufferView` with a `VkFormat` that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdExecuteGeneratedCommandsEXT-sparseImageInt64Atomics-04474
If the `sparseImageInt64Atomics` feature is not enabled, `VkImage` objects created with the `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-sparseImageInt64Atomics-04475
If the `sparseImageInt64Atomics` feature is not enabled, `VkBuffer` objects created with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07288
Any shader invocation executed by this command **must** `terminate`
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09600
If a descriptor with type equal to any of `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written
- VUID-vkCmdExecuteGeneratedCommandsEXT-renderPass-02684
The current render pass **must** be `compatible` with the `renderPass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`
- VUID-vkCmdExecuteGeneratedCommandsEXT-subpass-02685
The subpass index of the current render pass **must** be equal to the `subpass` member of the `VkGraphicsPipelineCreateInfo` structure specified when creating the `VkPipeline` bound to `VK_PIPELINE_BIND_POINT_GRAPHICS`
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07748
If any shader statically accesses an input attachment, a valid descriptor **must** be bound to the pipeline via a descriptor set
- VUID-vkCmdExecuteGeneratedCommandsEXT-OpTypeImage-07468

If any shader executed by this pipeline accesses an `OpTypeImage` variable with a `Dim` operand of `SubpassData`, it **must** be decorated with an `InputAttachmentIndex` that corresponds to a valid input attachment in the current subpass

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07469

Input attachment views accessed in a subpass **must** be created with the same `VkFormat` as the corresponding subpass definition, and be created with a `VkImageView` that is compatible with the attachment referenced by the subpass' `pInputAttachments` [`InputAttachmentIndex`] in the currently bound `VkFramebuffer` as specified by [Fragment Input Attachment Compatibility](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-pDepthInputAttachmentIndex-09595

Input attachment views accessed in a dynamic render pass with a `InputAttachmentIndex` referenced by `VkRenderingInputAttachmentIndexInfoKHR`, or no `InputAttachmentIndex` if `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` or `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are `NULL`, **must** be created with a `VkImageView` that is compatible with the corresponding color, depth, or stencil attachment in `VkRenderingInfo`

- VUID-vkCmdExecuteGeneratedCommandsEXT-pDepthInputAttachmentIndex-09596

Input attachment views accessed in a dynamic render pass via a shader object **must** have an `InputAttachmentIndex` if both `VkRenderingInputAttachmentIndexInfoKHR:pDepthInputAttachmentIndex` and `VkRenderingInputAttachmentIndexInfoKHR:pStencilInputAttachmentIndex` are non-`NULL`

- VUID-vkCmdExecuteGeneratedCommandsEXT-InputAttachmentIndex-09597

If an input attachment view accessed in a dynamic render pass via a shader object has an `InputAttachmentIndex`, the `InputAttachmentIndex` **must** match an index in `VkRenderingInputAttachmentIndexInfoKHR`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06537

Memory backing image subresources used as attachments in the current render pass **must** not be written in any way other than as an attachment by this command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09000

If a color attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_COLOR_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09001

If a depth attachment is written by any prior command in this subpass or by the load,

store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_DEPTH_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09002

If a stencil attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it is not in the `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout, and either:

- the `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT` is set on the currently bound pipeline or
- the last call to `vkCmdSetAttachmentFeedbackLoopEnableEXT` included `VK_IMAGE_ASPECT_STENCIL_BIT` and
 - there is no currently bound graphics pipeline or
 - the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT`

it **must** not be accessed in any way other than as an attachment by this command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09003

If an attachment is written by any prior command in this subpass or by the load, store, or resolve operations for this subpass, it **must** not be accessed in any way other than as an attachment, storage image, or sampled image by this command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06539

If any previously recorded command in the current subpass accessed an image subresource used as an attachment in this subpass in any way other than as an attachment, this command **must** not write to that image subresource as an attachment

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06886

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the depth aspect, `depth writes` **must** be disabled

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06887

If the current render pass instance uses a depth/stencil attachment with a read-only layout for the stencil aspect, both front and back `writeMask` are not zero, and stencil test is enabled, `all stencil ops` **must** be `VK_STENCIL_OP_KEEP`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07831

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_VIEWPORT` dynamic state enabled then `vkCmdSetViewport` **must** have been called and not

subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07832

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_SCISSOR** dynamic state enabled then **vkCmdSetScissor** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07833

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_LINE_WIDTH** dynamic state enabled then **vkCmdSetLineWidth** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08617

If a shader object is bound to any graphics stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to **VK_FALSE**, and the most recent call to **vkCmdSetPolygonModeEXT** in the current command buffer set **polygonMode** to **VK_POLYGON_MODE_LINE**, **vkCmdSetLineWidth** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08618

If a shader object is bound to any graphics stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to **VK_FALSE**, and the most recent call to **vkCmdSetPrimitiveTopologyEXT** in the current command buffer set **primitiveTopology** to any line topology, **vkCmdSetLineWidth** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08619

If a shader object that outputs line primitives is bound to the **VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT** or **VK_SHADER_STAGE_GEOMETRY_BIT** stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to **VK_FALSE**, **vkCmdSetLineWidth** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07834

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the **VK_DYNAMIC_STATE_DEPTH_BIAS** dynamic state enabled, the **current value** of **rasterizerDiscardEnable** is **VK_FALSE**, and the **current value** of **depthBiasEnable** is **VK_TRUE**, then **vkCmdSetDepthBounds** or **vkCmdSetDepthBias2EXT** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07835

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_BLEND_CONSTANTS** dynamic state enabled then **vkCmdSetBlendConstants** **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08621

If a shader object is bound to the **VK_SHADER_STAGE_FRAGMENT_BIT** stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set

`rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer set any element of `pColorBlendEnables` to `VK_TRUE`, and the most recent call to `vkCmdSetColorBlendEquationEXT` in the current command buffer set the same element of `pColorBlendEquations` to a `VkColorBlendEquationEXT` structure with any `VkBlendFactor` member with a value of `VK_BLEND_FACTOR_CONSTANT_COLOR`, `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_COLOR`, `VK_BLEND_FACTOR_CONSTANT_ALPHA`, or `VK_BLEND_FACTOR_ONE_MINUS_CONSTANT_ALPHA`, `vkCmdSetBlendConstants` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07836

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_BOUNDS` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `depthBoundsTestEnable` is `VK_TRUE`, then `vkCmdSetDepthBounds` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07837

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_COMPARE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilCompareMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07838

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilWriteMask` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07839

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_STENCIL_REFERENCE` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, the `current value` of `stencilTestEnable` is `VK_TRUE`, then `vkCmdSetStencilReference` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-maxMultiviewInstanceIndex-02688

If the draw is recorded in a render pass instance with multiview enabled, the maximum instance index **must** be less than or equal to `VkPhysicalDeviceMultiviewProperties::maxMultiviewInstanceIndex`

- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-02689

If the bound graphics pipeline was created with `VkPipelineSampleLocationsStateCreateInfoEXT::sampleLocationsEnable` set to `VK_TRUE` and

the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07634

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` dynamic state enabled, and the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleLocationsEnableEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-06666

If the `VK_EXT_sample_locations` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` dynamic state enabled, the **current value** of `rasterizerDiscardEnable` is `VK_FALSE`, and the **current value** of `sampleLocationsEnable` is `VK_TRUE`, then `vkCmdSetSampleLocationsEXT` **must** have been called and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-blendEnable-04727

If rasterization is not disabled in the bound graphics pipeline, then for each color attachment in the subpass, if the corresponding image view's **format features** do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the `blendEnable` member of the corresponding element of the `pAttachments` member of `pColorBlendState` **must** be `VK_FALSE`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08643

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then for each color attachment in the render pass, if the corresponding image view's **format features** do not contain `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT`, then the corresponding member of `pColorBlendEnables` in the most recent call to `vkCmdSetColorBlendEnableEXT` in the current command buffer that affected that attachment index **must** have been `VK_FALSE`

- VUID-vkCmdExecuteGeneratedCommandsEXT-multisampledRenderToSingleSampled-07284

If rasterization is not disabled in the bound graphics pipeline,

then `rasterizationSamples` for the currently bound graphics pipeline **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08644

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`,

then the most recent call to `vkCmdSetRasterizationSamplesEXT` in the current command buffer **must** have set `rasterizationSamples` to be the same as the number of samples for the current render pass color and/or depth/stencil attachments

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08876

If a shader object is bound to any graphics stage, the current render pass instance **must** have been begun with [vkCmdBeginRenderingKHR](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-imageView-06172

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pDepthAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pDepthAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the depth attachment

- VUID-vkCmdExecuteGeneratedCommandsEXT-imageView-06173

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pStencilAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pStencilAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the stencil attachment

- VUID-vkCmdExecuteGeneratedCommandsEXT-imageView-06174

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pDepthAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pDepthAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL](#), this command **must** not write any values to the depth attachment

- VUID-vkCmdExecuteGeneratedCommandsEXT-imageView-06175

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pStencilAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pStencilAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the stencil attachment

- VUID-vkCmdExecuteGeneratedCommandsEXT-imageView-06176

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pDepthAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pDepthAttachment](#) is [VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the depth attachment

- VUID-vkCmdExecuteGeneratedCommandsEXT-imageView-06177

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the [imageView](#) member of [pStencilAttachment](#) is not [VK_NULL_HANDLE](#), and the [layout](#) member of [pStencilAttachment](#) is [VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL](#), this command **must** not write any values to the stencil attachment

- VUID-vkCmdExecuteGeneratedCommandsEXT-viewMask-06178

If the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the currently bound graphics pipeline **must** have been created with a [VkPipelineRenderingCreateInfo::viewMask](#) equal to [VkRenderingInfo::viewMask](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-colorAttachmentCount-06179

If the [dynamicRenderingUnusedAttachments](#) feature is not enabled and the current render pass instance was begun with [vkCmdBeginRenderingKHR](#), the currently bound graphics pipeline **must** have been created with a [VkPipelineRenderingCreateInfo::colorAttachmentCount](#) equal to [VkRenderingInfo::colorAttachmentCount](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08910

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline

- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08912

If the `dynamicRenderingUnusedAttachments` feature is not enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` equal to `VK_NULL_HANDLE` **must** have the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound pipeline equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08911

If the `dynamicRenderingUnusedAttachments` feature is enabled, and the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingInfo::colorAttachmentCount` greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with an `imageView` not equal to `VK_NULL_HANDLE` **must** have been created with a `VkFormat` equal to the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the currently bound graphics pipeline, or the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats`, if it exists, **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07749

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08646

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-attachmentCount-07750

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT` dynamic state enabled then the `attachmentCount` parameter of `vkCmdSetColorWriteEnableEXT` **must** be greater than or equal to the `VkPipelineColorBlendStateCreateInfo::attachmentCount` of the currently bound graphics pipeline

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08647

If the `colorWriteEnable` feature is enabled on the device, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then the `attachmentCount` parameter of most recent call to `vkCmdSetColorWriteEnableEXT` in the current command buffer **must** be greater than or equal to the number of color attachments in the current render pass instance

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07751

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT` dynamic state enabled then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command for each discard rectangle in `VkPipelineDiscardRectangleStateCreateInfoEXT::discardRectangleCount`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07880

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDiscardRectangleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07881

If the `VK_EXT_discard_rectangles` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `discardRectangleEnable` is `VK_TRUE`, then `vkCmdSetDiscardRectangleModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-rasterizerDiscardEnable-09236

If the `VK_EXT_discard_rectangles` extension is enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetDiscardRectangleEnableEXT` in the current command buffer set `discardRectangleEnable` to `VK_TRUE`, then `vkCmdSetDiscardRectangleEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08913

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pDepthAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08914

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo`

`::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08915

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pDepthAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08916

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to `VK_FORMAT_UNDEFINED`

- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08917

If current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is not enabled, and `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline **must** be equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdExecuteGeneratedCommandsEXT-dynamicRenderingUnusedAttachments-08918

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the `dynamicRenderingUnusedAttachments` feature is enabled, `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, and the value of `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the currently bound graphics pipeline was not equal to the `VkFormat` used to create `VkRenderingInfo::pStencilAttachment->imageView`, the value of the format **must** be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdExecuteGeneratedCommandsEXT-imageView-06183

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` and `VkRenderingFragmentShadingRateAttachmentInfoKHR::imageView` was not `VK_NULL_HANDLE`, the currently bound graphics pipeline **must** have been created with `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

- VUID-vkCmdExecuteGeneratedCommandsEXT-multisampledRenderToSingleSampled-07285

If the current render pass instance was begun with `vkCmdBeginRenderingKHR` with a `VkRenderingInfo::colorAttachmentCount` parameter greater than 0, then each element of the `VkRenderingInfo::pColorAttachments` array with a `imageView` not equal to

`VK_NULL_HANDLE` **must** have been created with a sample count equal to the value of `rasterizationSamples` for the currently bound graphics pipeline

- VUID-vkCmdExecuteGeneratedCommandsEXT-multisampledRenderToSingleSampled-07286

If `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pDepthAttachment->imageView`

- VUID-vkCmdExecuteGeneratedCommandsEXT-multisampledRenderToSingleSampled-07287

If `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, the value of `rasterizationSamples` for the currently bound graphics pipeline **must** be equal to the sample count used to create `VkRenderingInfo::pStencilAttachment->imageView`

- VUID-vkCmdExecuteGeneratedCommandsEXT-renderPass-06198

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, the currently bound pipeline **must** have been created with a `VkGraphicsPipelineCreateInfo::renderPass` equal to `VK_NULL_HANDLE`

- VUID-vkCmdExecuteGeneratedCommandsEXT-pColorAttachments-08963

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound with a fragment shader that statically writes to a color attachment, the color write mask is not zero, color writes are enabled, and the corresponding element of the `VkRenderingInfo::pColorAttachments->imageView` was not `VK_NULL_HANDLE`, then the corresponding element of `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdExecuteGeneratedCommandsEXT-pDepthAttachment-08964

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, depth test is enabled, depth write is enabled, and the `VkRenderingInfo::pDepthAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::depthAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdExecuteGeneratedCommandsEXT-pStencilAttachment-08965

If the current render pass instance was begun with `vkCmdBeginRenderingKHR`, there is a graphics pipeline bound, stencil test is enabled and the `VkRenderingInfo::pStencilAttachment->imageView` was not `VK_NULL_HANDLE`, then the `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` used to create the pipeline **must** not be `VK_FORMAT_UNDEFINED`

- VUID-vkCmdExecuteGeneratedCommandsEXT-primitivesGeneratedQueryWithRasterizerDiscard-06708

If the `primitivesGeneratedQueryWithRasterizerDiscard` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, `rasterization discard` **must** not be enabled

- VUID-vkCmdExecuteGeneratedCommandsEXT-primitivesGeneratedQueryWithNonZeroStreams-06709

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, the bound graphics pipeline **must** not have been created with a non-zero value in

VkPipelineRasterizationStateStreamCreateInfoEXT::rasterizationStream

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07619

If a shader object is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT` dynamic state enabled, then `vkCmdSetTessellationDomainOriginEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07620

If the `depthClamp` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetDepthClampEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07621

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_POLYGON_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetPolygonModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07622

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetRasterizationSamplesEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07623

If a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetSampleMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-alphaToCoverageEnable-08919

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and `alphaToCoverageEnable` was `VK_TRUE` in the last call to `vkCmdSetAlphaToCoverageEnableEXT`, then the `Fragment Output Interface` **must** contain a variable for the alpha `Component` word in `Location 0` at `Index 0`

- VUID-vkCmdExecuteGeneratedCommandsEXT-alphaToCoverageEnable-08920

If a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetAlphaToCoverageEnableEXT` in the current command buffer set `alphaToCoverageEnable` to `VK_TRUE`, then the `Fragment Output Interface` **must** contain a variable for the alpha `Component` word in `Location 0` at `Index 0`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07624

If a shader object is bound to any graphics stage or a graphics pipeline is bound which

was created with the `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToCoverageEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07625

If the `alphaToOne` feature is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAlphaToOneEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07626

If the `logicOp` feature is enabled, a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetLogicOpEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07627

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic state enabled then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08657

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07628

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08658

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetColorBlendEnableEXT` for any attachment set that attachment's value in `pColorBlendEnables` to `VK_TRUE`, then `vkCmdSetColorBlendEquationEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07629

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then

`vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08659

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorWriteMaskEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07630

If the `geometryStreams` feature is enabled, and a shader object is bound to the `VK_SHADER_STAGE_GEOMETRY_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` dynamic state enabled, then `vkCmdSetRasterizationStreamEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07631

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetConservativeRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07632

If the `VK_EXT_conservative_rasterization` extension is enabled, a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT` dynamic state enabled, the current value of `rasterizerDiscardEnable` is `VK_FALSE`, and the current value of `conservativeRasterizationMode` is `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`, then `vkCmdSetExtraPrimitiveOverestimationSizeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07633

If the `depthClipEnable` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT` dynamic state, then `vkCmdSetDepthClipEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07635

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-rasterizerDiscardEnable-09416

If the `VK_EXT_blend_operation_advanced` extension is enabled, and a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then at least one of

`vkCmdSetColorBlendEquationEXT` and `vkCmdSetColorBlendAdvancedEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07636

If the `VK_EXT_provoking_vertex` extension is enabled, a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT` dynamic state enabled, and the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetProvokingVertexModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07637

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic state enabled then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08666

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08667

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08668

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineRasterizationModeEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07638

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` dynamic state enabled then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated`

in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08669

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to any graphics stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPolygonModeEXT` in the current command buffer set `polygonMode` to `VK_POLYGON_MODE_LINE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08670

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object is bound to the `VK_SHADER_STAGE_VERTEX_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, and the most recent call to `vkCmdSetPrimitiveTopologyEXT` in the current command buffer set `primitiveTopology` to any line topology, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08671

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled, and a shader object that outputs line primitives is bound to the `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT` or `VK_SHADER_STAGE_GEOMETRY_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetLineStippleEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07849

If any of the `stippledRectangularLines`, `stippledBresenhamLines` or `stippledSmoothLines` features are enabled and a shader object is bound to any graphics stage, or a bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_KHR` dynamic state enabled, the `current value` of `rasterizerDiscardEnable` is `VK_FALSE`, and the `current value` of `stippledLineEnable` is `VK_TRUE`, then `vkCmdSetLineStippleKHR` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-07639

If the `depthClipControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT` dynamic state enabled, then `vkCmdSetDepthClipNegativeOneToOneEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09650

If the `depthClampControl` feature is enabled, and a shader object is bound to any graphics stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT` dynamic state enabled, and the `current value` of `depthClampEnable` is `VK_TRUE`, then `vkCmdSetDepthClampRangeEXT` **must** have been called

and not subsequently **invalidated** in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-pColorBlendEnables-07470

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT** state enabled and the last call to **vkCmdSetColorBlendEnableEXT** set **pColorBlendEnables** for any attachment to **VK_TRUE**, then for those attachments in the subpass the corresponding image view's **format features** **must** contain **VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT**

- VUID-vkCmdExecuteGeneratedCommandsEXT-rasterizationSamples-07471

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT** state enabled, and the current subpass does not use any color and/or depth/stencil attachments, then the **rasterizationSamples** in the last call to **vkCmdSetRasterizationSamplesEXT** **must** follow the rules for a **zero-attachment** subpass

- VUID-vkCmdExecuteGeneratedCommandsEXT-samples-07472

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_SAMPLE_MASK_EXT** state enabled and the **VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT** state disabled, then the **samples** parameter in the last call to **vkCmdSetSampleMaskEXT** **must** be greater or equal to the **VkPipelineMultisampleStateCreateInfo::rasterizationSamples** parameter used to create the bound graphics pipeline

- VUID-vkCmdExecuteGeneratedCommandsEXT-samples-07473

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_SAMPLE_MASK_EXT** state and **VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT** states enabled, then the **samples** parameter in the last call to **vkCmdSetSampleMaskEXT** **must** be greater or equal to the **rasterizationSamples** parameter in the last call to **vkCmdSetRasterizationSamplesEXT**

- VUID-vkCmdExecuteGeneratedCommandsEXT-rasterizationSamples-07474

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT** state enabled, and neither the **VK_AMD_mixed_attachment_samples** nor the **VK_NV_framebuffer_mixed_samples** extensions are enabled, then the **rasterizationSamples** in the last call to **vkCmdSetRasterizationSamplesEXT** **must** be the same as the current subpass color and/or depth/stencil attachments

- VUID-vkCmdExecuteGeneratedCommandsEXT-firstAttachment-07476

If the bound graphics pipeline state was created with the **VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT** dynamic state enabled then **vkCmdSetColorBlendEnableEXT** **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the **firstAttachment** and **attachmentCount** parameters of **vkCmdSetColorBlendEnableEXT** calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdExecuteGeneratedCommandsEXT-rasterizerDiscardEnable-09417

If a shader object is bound to the **VK_SHADER_STAGE_FRAGMENT_BIT** stage, and the most recent call to **vkCmdSetRasterizerDiscardEnableEXT** in the current command buffer set **rasterizerDiscardEnable** to **VK_FALSE**, then **vkCmdSetColorBlendEnableEXT** **must** have

been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEnableEXT` calls **must** specify an enable for all active color attachments in the current subpass

- VUID-vkCmdExecuteGeneratedCommandsEXT-firstAttachment-07477

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT` dynamic state enabled then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdExecuteGeneratedCommandsEXT-rasterizerDiscardEnable-09418

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and both the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE` and there are color attachments bound, then `vkCmdSetColorBlendEquationEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendEquationEXT` calls **must** specify the blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdExecuteGeneratedCommandsEXT-firstAttachment-07478

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT` dynamic state enabled then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdExecuteGeneratedCommandsEXT-rasterizerDiscardEnable-09419

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, and the most recent call to `vkCmdSetRasterizerDiscardEnableEXT` in the current command buffer set `rasterizerDiscardEnable` to `VK_FALSE`, then `vkCmdSetColorWriteMaskEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorWriteMaskEXT` calls **must** specify the color write mask for all active color attachments in the current subpass

- VUID-vkCmdExecuteGeneratedCommandsEXT-firstAttachment-07479

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` dynamic state enabled then `vkCmdSetColorBlendAdvancedEXT` **must** have been called in the current command buffer prior to this drawing command, and the attachments specified by the `firstAttachment` and `attachmentCount` parameters of `vkCmdSetColorBlendAdvancedEXT` calls **must** specify the advanced blend equations for all active color attachments in the current subpass where blending is enabled

- VUID-vkCmdExecuteGeneratedCommandsEXT-advancedBlendMaxColorAttachments-

07480

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT` and `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT` dynamic states enabled and the last calls to `vkCmdSetColorBlendEnableEXT` and `vkCmdSetColorBlendAdvancedEXT` have enabled advanced blending, then the number of active color attachments in the current subpass **must** not exceed `advancedBlendMaxColorAttachments`

- VUID-vkCmdExecuteGeneratedCommandsEXT-

`primitivesGeneratedQueryWithNonZeroStreams-07481`

If the `primitivesGeneratedQueryWithNonZeroStreams` feature is not enabled and the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query is active, and the bound graphics pipeline was created with `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT` state enabled, the last call to `vkCmdSetRasterizationStreamEXT` **must** have set the `rasterizationStream` to zero

- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsPerPixel-07482

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state disabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` member of the `VkPipelineMultisampleStateCreateInfo` structure the bound graphics pipeline has been created with

- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsPerPixel-07483

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then the `sampleLocationsPerPixel` member of `pSampleLocationsInfo` in the last call to `vkCmdSetSampleLocationsEXT` **must** equal the `rasterizationSamples` parameter of the last call to `vkCmdSetRasterizationSamplesEXT`

- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07484

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, and the current subpass has a depth/stencil attachment, then that attachment **must** have been created with the `VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT` bit set

- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07485

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.width` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`

- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07486
If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state enabled and the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, then the `sampleLocationsInfo.sampleLocationGridSize.height` in the last call to `vkCmdSetSampleLocationsEXT` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling `rasterizationSamples`
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07487
If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage, or the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT` state enabled, and if `sampleLocationsEnable` was `VK_TRUE` in the last call to `vkCmdSetSampleLocationsEnableEXT`, the fragment shader code **must** not statically use the extended instruction `InterpolateAtSample`
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07936
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.width` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.width` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07937
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationGridSize.height` **must** evenly divide `VkMultisamplePropertiesEXT::sampleLocationGridSize.height` as returned by `vkGetPhysicalDeviceMultisamplePropertiesEXT` with a `samples` parameter equaling the value of `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`
- VUID-vkCmdExecuteGeneratedCommandsEXT-sampleLocationsEnable-07938
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT` state disabled and the `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT` state enabled, and the `current value` of `sampleLocationsEnable` is `VK_TRUE`, then `sampleLocationsInfo.sampleLocationsPerPixel` **must** equal `rasterizationSamples` in the last call to `vkCmdSetRasterizationSamplesEXT`
- VUID-vkCmdExecuteGeneratedCommandsEXT-stippledLineEnable-07495
If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the `current stippledLineEnable` state is `VK_TRUE` and the `current lineRasterizationMode` state is

VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR, then the `stippledRectangularLines` feature **must** be enabled

- VUID-vkCmdExecuteGeneratedCommandsEXT-stippledLineEnable-07496

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR`, then the `stippledBresenhamLines` feature **must** be enabled

- VUID-vkCmdExecuteGeneratedCommandsEXT-stippledLineEnable-07497

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR`, then the `stippledSmoothLines` feature **must** be enabled

- VUID-vkCmdExecuteGeneratedCommandsEXT-stippledLineEnable-07498

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT` or `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT` dynamic states enabled, and if the current `stippledLineEnable` state is `VK_TRUE` and the current `lineRasterizationMode` state is `VK_LINE_RASTERIZATION_MODE_DEFAULT_KHR`, then the `stippledRectangularLines` feature **must** be enabled and `VkPhysicalDeviceLimits::strictLines` **must** be `VK_TRUE`

- VUID-vkCmdExecuteGeneratedCommandsEXT-conservativePointAndLineRasterization-07499

If the bound graphics pipeline state was created with the `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT` dynamic state enabled, `conservativePointAndLineRasterization` is not supported, and the effective primitive topology output by the last pre-rasterization shader stage is a line or point, then the `conservativeRasterizationMode` set by the last call to `vkCmdSetConservativeRasterizationModeEXT` **must** be `VK_CONSERVATIVE_RASTERIZATION_MODE_DISABLED_EXT`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08877

If a shader object is bound to the `VK_SHADER_STAGE_FRAGMENT_BIT` stage or a graphics pipeline is bound which was created with the `VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT` dynamic state enabled, and the current value of `rasterizerDiscardEnable` is `VK_FALSE`, then `vkCmdSetAttachmentFeedbackLoopEnableEXT` **must** have been called and not subsequently `invalidated` in the current command buffer prior to this drawing command

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08684

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_VERTEX_BIT`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08685

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages`

with an element of `VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08686

If there is no bound graphics pipeline, and the `tessellationShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08687

If there is no bound graphics pipeline, and the `geometryShader` feature is enabled, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_GEOMETRY_BIT`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08688

If there is no bound graphics pipeline, `vkCmdBindShadersEXT` **must** have been called in the current command buffer with `pStages` with an element of `VK_SHADER_STAGE_FRAGMENT_BIT`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08698

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, then all shaders created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag in the same `vkCreateShadersEXT` call **must** also be bound

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08699

If any graphics shader is bound which was created with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag, any stages in between stages whose shaders which did not create a shader with the `VK_SHADER_CREATE_LINK_STAGE_BIT_EXT` flag as part of the same `vkCreateShadersEXT` call **must** not have any `VkShaderEXT` bound

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08878

All bound graphics shader objects **must** have been created with identical or identically defined push constant ranges

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-08879

All bound graphics shader objects **must** have been created with identical or identically defined arrays of descriptor set layouts

- VUID-vkCmdExecuteGeneratedCommandsEXT-pDynamicStates-08715

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_DEPTH_WRITE_ENABLE` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpDepthAttachmentReadEXT`, the `depthWriteEnable` parameter in the last call to `vkCmdSetDepthWriteEnableEXT` **must** be `VK_FALSE`

- VUID-vkCmdExecuteGeneratedCommandsEXT-pDynamicStates-08716

If the bound graphics pipeline state includes a fragment shader stage, was created with `VK_DYNAMIC_STATE_STENCIL_WRITE_MASK` set in `VkPipelineDynamicStateCreateInfo::pDynamicStates`, and the fragment shader declares the `EarlyFragmentTests` execution mode and uses `OpStencilAttachmentReadEXT`, the `writeMask` parameter in the last call to `vkCmdSetStencilWriteMask` **must** be `0`

- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09116

If a shader object is bound to any graphics stage or the currently bound graphics pipeline was created with `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`, and the format of any color attachment is `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32`, the corresponding element of the

`pColorWriteMasks` parameter of `vkCmdSetColorWriteMaskEXT` **must** either include all of `VK_COLOR_COMPONENT_R_BIT`, `VK_COLOR_COMPONENT_G_BIT`, and `VK_COLOR_COMPONENT_B_BIT`, or none of them

- VUID-vkCmdExecuteGeneratedCommandsEXT-maxFragmentDualSrcAttachments-09239
If **blending** is enabled for any attachment where either the source or destination blend factors for that attachment **use the secondary color input**, the maximum value of `Location` for any output attachment **statically used** in the `Fragment Execution Model` executed by this command **must** be less than `maxFragmentDualSrcAttachments`
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09548
If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, the value of each element of `VkRenderingAttachmentLocationInfoKHR::pColorAttachmentLocations` set by `vkCmdSetRenderingAttachmentLocationsKHR` **must** match the value set for the corresponding element in the currently bound pipeline
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-09549
If the current render pass was begun with `vkCmdBeginRenderingKHR`, and there is no shader object bound to any graphics stage, input attachment index mappings in the currently bound pipeline **must** match those set for the current render pass instance via `VkRenderingInputAttachmentIndexInfoKHR`
- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11046
If `isPreprocessed` is `VK_TRUE` and `vkGetGeneratedCommandsMemoryRequirementsEXT` did not return a required size of zero then `vkCmdPreprocessGeneratedCommandsEXT` **must** have already been executed on the device before this command executes, and the preprocessing command **must** have used the same `pGeneratedCommandsInfo` content as well as the content of the input buffers it references (all except `VkGeneratedCommandsInfoEXT::preprocessBuffer`)
- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11047
If `isPreprocessed` is `VK_TRUE` then the `indirectCommandsLayout` member of `pGeneratedCommandsInfo` **must** have been created with the `VK_INDIRECT_COMMANDS_LAYOUT_USAGE_EXPLICIT_PREPROCESS_BIT_EXT` bit set
- VUID-vkCmdExecuteGeneratedCommandsEXT-indirectCommandsLayout-11141
If the `indirectCommandsLayout` member of `pGeneratedCommandsInfo` was created with the `VK_INDIRECT_COMMANDS_LAYOUT_USAGE_EXPLICIT_PREPROCESS_BIT_EXT` bit set, then `isPreprocessed` **must** be `VK_TRUE`
- VUID-vkCmdExecuteGeneratedCommandsEXT-preprocessAddress-11142
The contents of the `preprocessAddress` member of `pGeneratedCommandsInfo` **must** not have been previously used to record another `vkCmdExecuteGeneratedCommandsEXT`
- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11048
If `isPreprocessed` is `VK_TRUE` then the bound descriptor sets and push constants **must** match identically with those bound during recording of the corresponding call to `vkCmdPreprocessGeneratedCommandsEXT`
- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-10198
If `isPreprocessed` is `VK_TRUE` then the conditional render state and its predicate value **must** match identically with the state and value set during execution of the corresponding call

to [vkCmdPreprocessGeneratedCommandsEXT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11049

If `isPreprocessed` is `VK_TRUE` and the `indirectCommandsLayout` member of `pGeneratedCommandsInfo` contains a draw token, then the graphics state bound on `commandBuffer` **must** match identically with the graphics state bound on the `stateCommandBuffer` passed to [vkCmdPreprocessGeneratedCommandsEXT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11149

If `isPreprocessed` is `VK_TRUE`, then the queue family index of `commandBuffer` **must** be the same as the queue family index used to allocate the `stateCommandBuffer` passed to [vkCmdPreprocessGeneratedCommandsEXT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11051

If `isPreprocessed` is `VK_TRUE` and the `indirectCommandsLayout` member of `pGeneratedCommandsInfo` contains a dispatch token, then the compute state bound on `commandBuffer` **must** match identically with the compute state bound on the `stateCommandBuffer` passed to [vkCmdPreprocessGeneratedCommandsEXT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11052

If `isPreprocessed` is `VK_TRUE` and the `indirectCommandsLayout` member of `pGeneratedCommandsInfo` contains a ray tracing token, then the ray tracing state bound on `commandBuffer` **must** match identically with the ray tracing state bound on the `stateCommandBuffer` passed to [vkCmdPreprocessGeneratedCommandsEXT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11150

If `isPreprocessed` is `VK_TRUE` and the `indirectCommandsLayout` member of `pGeneratedCommandsInfo` contains a ray tracing token, the queue family index `commandBuffer` was allocated from **must** be the same queue family index used to allocate the `stateCommandBuffer` passed to [vkCmdPreprocessGeneratedCommandsEXT](#)

- VUID-vkCmdExecuteGeneratedCommandsEXT-indirectCommandsLayout-11053

If the token sequence of the passed [VkGeneratedCommandsInfoEXT::indirectCommandsLayout](#) contains a `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token, the initial shader state of [VkGeneratedCommandsInfoEXT::indirectExecutionSet](#) **must** be bound on `commandBuffer`

- VUID-vkCmdExecuteGeneratedCommandsEXT-indirectCommandsLayout-11004

If `indirectCommandsLayout` was created with a token sequence that contained the `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token and `indirectExecutionSet` was created using `VK_INDIRECT_EXECUTION_SET_INFO_TYPE_SHADER_OBJECTS_EXT`, every executed `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token **must** bind all the shader stages set in the [VkIndirectCommandsExecutionSetTokenEXT::shaderStages](#) used to create `indirectCommandsLayout`

- VUID-vkCmdExecuteGeneratedCommandsEXT-isPreprocessed-11055

If `isPreprocessed` is `VK_TRUE` and the token sequence of the passed [VkGeneratedCommandsInfoEXT::indirectCommandsLayout](#) contains a `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token, the members of [VkGeneratedCommandsInfoEXT::indirectExecutionSet](#) accessed by this command **must** not have been modified since the preprocess buffer was generated

- VUID-vkCmdExecuteGeneratedCommandsEXT-

deviceGeneratedCommandsTransformFeedback-11057

If `deviceGeneratedCommandsTransformFeedback` is not supported on device, transform feedback **must** not be active

- VUID-vkCmdExecuteGeneratedCommandsEXT-indirectExecutionSet-11058
If transform feedback is active, `VkGeneratedCommandsInfoEXT::indirectExecutionSet` **must** be `VK_NULL_HANDLE`
- VUID-vkCmdExecuteGeneratedCommandsEXT-deviceGeneratedCommands-11059
The `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands` feature **must** be enabled
- VUID-vkCmdExecuteGeneratedCommandsEXT-supportedIndirectCommandsShaderStages-11060
The currently bound shader stages **must** be supported by `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStages`
- VUID-vkCmdExecuteGeneratedCommandsEXT-supportedIndirectCommandsShaderStages-11061
Only stages specified in `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStages` **can** be set in `pGeneratedCommandsInfo->shaderStages`
- VUID-vkCmdExecuteGeneratedCommandsEXT-None-11062
If a rendering pass is currently active, the view mask **must** be 0
- VUID-vkCmdExecuteGeneratedCommandsEXT-commandBuffer-11143
`commandBuffer` **must** not have been created with `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT`

Valid Usage (Implicit)

- VUID-vkCmdExecuteGeneratedCommandsEXT-commandBuffer-parameter-`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdExecuteGeneratedCommandsEXT-pGeneratedCommandsInfo-parameter-`pGeneratedCommandsInfo` **must** be a valid pointer to a valid `VkGeneratedCommandsInfoEXT` structure
- VUID-vkCmdExecuteGeneratedCommandsEXT-commandBuffer-recording-`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdExecuteGeneratedCommandsEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdExecuteGeneratedCommandsEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdExecuteGeneratedCommandsEXT-bufferlevel-`commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Both	Outside	Graphics Compute	Action Indirection

The `VkGeneratedCommandsInfoEXT` is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkGeneratedCommandsInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkShaderStageFlags       shaderStages;
    VkIndirectExecutionSetEXT indirectExecutionSet;
    VkIndirectCommandsLayoutEXT indirectCommandsLayout;
    VkDeviceAddress          indirectAddress;
    VkDeviceSize             indirectAddressSize;
    VkDeviceAddress          preprocessAddress;
    VkDeviceSize             preprocessSize;
    uint32_t                 maxSequenceCount;
    VkDeviceAddress          sequenceCountAddress;
    uint32_t                 maxDrawCount;
} VkGeneratedCommandsInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderStages` is the mask of shader stages used by the commands.
- `indirectExecutionSet` is the indirect execution set to be used for binding shaders.
- `indirectCommandsLayout` is the `VkIndirectCommandsLayoutEXT` that specifies the command sequence data.
- `indirectAddress` is an address that holds the indirect buffer data.
- `indirectAddressSize` is the size in bytes of indirect buffer data starting at `indirectAddress`.
- `preprocessAddress` specifies a physical address of the `VkBuffer` used for preprocessing the input data for execution. If this structure is used with `vkCmdExecuteGeneratedCommandsEXT` with its `isPreprocessed` set to `VK_TRUE`, then the preprocessing step is skipped but data in this address

may still be modified. The contents and the layout of this address are opaque to applications and **must** not be modified outside functions related to device-generated commands or copied to another buffer for reuse.

- `preprocessSize` is the maximum byte size within `preprocessAddress` that is available for preprocessing.
- `maxSequenceCount` is used to determine the number of sequences to execute.
- `sequenceCountAddress` specifies an optional physical address of a single `uint32_t` value containing the requested number of sequences to execute.
- `maxDrawCount` is the maximum number of indirect draws that can be executed by any COUNT-type multi-draw indirect tokens. The draw count in the indirect buffer is clamped to this value for these token types.

If `sequenceCountAddress` is not `NULL`, then `maxSequenceCount` is the maximum number of sequences that can be executed. The actual number is `min(maxSequenceCount, *sequenceCountAddress)`. If `sequenceCountAddress` is `NULL`, then `maxSequenceCount` is the exact number of sequences to execute.

If the action command token for the layout is not a COUNT-type multi-draw indirect token, `maxDrawCount` is ignored.

Valid Usage

- VUID-VkGeneratedCommandsInfoEXT-preprocessAddress-11063
If `vkGetGeneratedCommandsMemoryRequirementsEXT` returns a non-zero size, `preprocessAddress` **must** not be `NULL`
- VUID-VkGeneratedCommandsInfoEXT-preprocessAddress-11064
`VkDeviceMemory` objects bound to the underlying buffer for `preprocessAddress` **must** have been allocated using one of the memory types allowed in the `memoryTypeBits` member of the `VkMemoryRequirements` structure returned by `vkGetGeneratedCommandsMemoryRequirementsEXT`
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11065
If the `indirectCommandsLayout` uses a token of `VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT`, then the `indirectExecutionSet`'s push constant layout **must** contain the `updateRange` specified in `VkIndirectCommandsPushConstantTokenEXT`
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11066
If the `indirectCommandsLayout` uses a token of `VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT`, then the `indirectExecutionSet`'s push constant layout **must** contain the `updateRange` specified in `VkIndirectCommandsPushConstantTokenEXT`
- VUID-VkGeneratedCommandsInfoEXT-maxSequenceCount-11067
`maxSequenceCount` **must** be less or equal to `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::maxIndirectSequenceCount` and `VkGeneratedCommandsMemoryRequirementsInfoEXT::maxSequencesCount` that was used to determine the `preprocessSize`

- VUID-VkGeneratedCommandsInfoEXT-sequenceCountAddress-11068
If `sequenceCountAddress` is not `NULL`, the value contained in the address **must** be less or equal to `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::maxIndirectSequenceCount` and `VkGeneratedCommandsMemoryRequirementsInfoEXT::maxSequencesCount` that was used to determine the `preprocessSize`
- VUID-VkGeneratedCommandsInfoEXT-maxSequenceCount-10246
`maxSequenceCount` **must** not be zero
- VUID-VkGeneratedCommandsInfoEXT-preprocessAddress-11069
The underlying buffer for `preprocessAddress` **must** have the `VK_BUFFER_USAGE_2_PREPROCESS_BUFFER_BIT_EXT` bit set in its usage flag
- VUID-VkGeneratedCommandsInfoEXT-preprocessAddress-11070
If the underlying buffer for `preprocessAddress` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11144
If the `indirectCommandsLayout` contains a `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token, then the descriptor and push constant layout info provided either by `pipelineLayout` or through a `VkPipelineLayoutCreateInfo` in `pNext` of the `VkIndirectCommandsLayoutCreateInfoEXT` used to create `indirectCommandsLayout` **must** be `compatible` with the descriptor and push constant layout info used by `indirectExecutionSet`
- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11002
If `indirectCommandsLayout` was created with a token sequence that contained the `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token, the shader stages used to create the initial shader state of `indirectExecutionSet` **must** equal the `VkIndirectCommandsExecutionSetTokenEXT::shaderStages` used to create `indirectCommandsLayout`
- VUID-VkGeneratedCommandsInfoEXT-preprocessSize-11071
`preprocessSize` **must** be greater than or equal to the memory requirement's size returned by `vkGetGeneratedCommandsMemoryRequirementsEXT` using the matching inputs (`indirectCommandsLayout`, ...) as within this structure
- VUID-VkGeneratedCommandsInfoEXT-sequenceCountAddress-11072
The underlying buffer for `sequenceCountAddress` **must** have the `VK_BUFFER_USAGE_2_INDIRECT_BUFFER_BIT_KHR` bit set in its usage flag
- VUID-VkGeneratedCommandsInfoEXT-sequenceCountAddress-11073
If `sequenceCountAddress` is not `NULL`, `sequenceCountAddress` **must** be aligned to 4
- VUID-VkGeneratedCommandsInfoEXT-indirectAddress-11074
`indirectAddress` **must** be aligned to 4
- VUID-VkGeneratedCommandsInfoEXT-sequenceCountAddress-11075
If the underlying buffer for `sequenceCountAddress` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkGeneratedCommandsInfoEXT-indirectAddress-11076
`indirectAddress` **must** not be `NULL`
- VUID-VkGeneratedCommandsInfoEXT-indirectAddressSize-11077

`indirectAddressSize` **must** be greater than zero

- VUID-VkGeneratedCommandsInfoEXT-maxDrawCount-11078

When not ignored, `maxDrawCount` × `maxSequenceCount` **must** be less than 2^{24}

- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11079

If `indirectCommandsLayout` was created using a `VK_INDIRECT_COMMANDS_TOKEN_TYPE_VERTEX_BUFFER_EXT` token and shader objects are not bound then the currently bound graphics pipeline **must** have been created with `VK_DYNAMIC_STATE_VERTEX_INPUT_BINDING_STRIDE` in `pDynamicStates`

- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-11083

If the token sequence of the passed `indirectCommandsLayout` contains a `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token, the `indirectExecutionSet` **must** not be `VK_NULL_HANDLE`

- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-10241

If the token sequence of the passed `indirectCommandsLayout` does not contains a `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token, the `indirectExecutionSet` **must** be `VK_NULL_HANDLE`

- VUID-VkGeneratedCommandsInfoEXT-indirectExecutionSet-11080

If `indirectExecutionSet` is `VK_NULL_HANDLE`, a `VkGeneratedCommandsPipelineInfoEXT` or `VkGeneratedCommandsShaderInfoEXT` **must** be included in the `pNext` chain

Valid Usage (Implicit)

- VUID-VkGeneratedCommandsInfoEXT-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_GENERATED_COMMANDS_INFO_EXT`

- VUID-VkGeneratedCommandsInfoEXT-shaderStages-parameter

`shaderStages` **must** be a valid combination of `VkShaderStageFlagBits` values

- VUID-VkGeneratedCommandsInfoEXT-shaderStages-requiredbitmask

`shaderStages` **must** not be 0

- VUID-VkGeneratedCommandsInfoEXT-indirectExecutionSet-parameter

If `indirectExecutionSet` is not `VK_NULL_HANDLE`, `indirectExecutionSet` **must** be a valid `VkIndirectExecutionSetEXT` handle

- VUID-VkGeneratedCommandsInfoEXT-indirectCommandsLayout-parameter

`indirectCommandsLayout` **must** be a valid `VkIndirectCommandsLayoutEXT` handle

- VUID-VkGeneratedCommandsInfoEXT-commonparent

Both of `indirectCommandsLayout`, and `indirectExecutionSet` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Referencing the functions defined in `Indirect Commands Layout`, `vkCmdExecuteGeneratedCommandsEXT` behaves as:

```
uint32_t sequencesCount = sequenceCountAddress ?
```

```
min(maxSequenceCount, sequenceCountAddress.load_uint32()) :
maxSequenceCount;
```

```
cmdProcessAllSequences(commandBuffer, indirectExecutionSet,
                        indirectCommandsLayout, indirectAddress,
                        sequencesCount);
```

```
// The stateful commands within indirectCommandsLayout will not
// affect the state of subsequent commands in the target
// command buffer (cmd)
```

Note

NOTE

It is important to note that the affected values of all state related to the **shaderStages** used are undefined after this command. This means that e.g., if this command indirectly alters push constants, the push constant state becomes undefined.

Commands **can** be preprocessed prior execution using the following command:

```
// Provided by VK_EXT_device_generated_commands
void vkCmdPreprocessGeneratedCommandsEXT(
    VkCommandBuffer                commandBuffer,
    const VkGeneratedCommandsInfoEXT* pGeneratedCommandsInfo,
    VkCommandBuffer                stateCommandBuffer);
```

- **commandBuffer** is the command buffer which does the preprocessing.
- **pGeneratedCommandsInfo** is a pointer to a **VkGeneratedCommandsInfoEXT** structure containing parameters affecting the preprocessing step.
- **stateCommandBuffer** is a command buffer from which to snapshot current states affecting the preprocessing step. When a graphics command action token is used, graphics state is snapshotted. When a compute action command token is used, compute state is snapshotted. When a ray tracing action command token is used, ray tracing state is snapshotted. It can be deleted at any time after this command has been recorded.

Note

NOTE

stateCommandBuffer access is not synchronized by the driver, meaning that this command buffer **must** not be modified between threads in an unsafe manner.

Valid Usage

- VUID-vkCmdPreprocessGeneratedCommandsEXT-pGeneratedCommandsInfo-11082
pGeneratedCommandsInfo's indirectCommandsLayout must have been created with the **VK_INDIRECT_COMMANDS_LAYOUT_USAGE_EXPLICIT_PREPROCESS_BIT_EXT** bit set
- VUID-vkCmdPreprocessGeneratedCommandsEXT-indirectCommandsLayout-11084
If the token sequence of the passed **VkGeneratedCommandsInfoEXT**

`::indirectCommandsLayout` contains a `VK_INDIRECT_COMMANDS_TOKEN_TYPE_EXECUTION_SET_EXT` token, the initial shader state of `VkGeneratedCommandsInfoEXT::indirectExecutionSet` **must** be bound on `stateCommandBuffer`

- VUID-vkCmdPreprocessGeneratedCommandsEXT-stateCommandBuffer-11138
`stateCommandBuffer` **must** be in the recording state
- VUID-vkCmdPreprocessGeneratedCommandsEXT-deviceGeneratedCommands-11087
The `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT::deviceGeneratedCommands` feature **must** be enabled
- VUID-vkCmdPreprocessGeneratedCommandsEXT-supportedIndirectCommandsShaderStages-11088
Only stages specified in `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT::supportedIndirectCommandsShaderStages` **can** be set in `pGeneratedCommandsInfo->shaderStages`

Valid Usage (Implicit)

- VUID-vkCmdPreprocessGeneratedCommandsEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdPreprocessGeneratedCommandsEXT-pGeneratedCommandsInfo-parameter
`pGeneratedCommandsInfo` **must** be a valid pointer to a valid `VkGeneratedCommandsInfoEXT` structure
- VUID-vkCmdPreprocessGeneratedCommandsEXT-stateCommandBuffer-parameter
`stateCommandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdPreprocessGeneratedCommandsEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdPreprocessGeneratedCommandsEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdPreprocessGeneratedCommandsEXT-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdPreprocessGeneratedCommandsEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdPreprocessGeneratedCommandsEXT-bufferlevel
`commandBuffer` **must** be a primary `VkCommandBuffer`
- VUID-vkCmdPreprocessGeneratedCommandsEXT-commonparent
Both of `commandBuffer`, and `stateCommandBuffer` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized

- Host access to `stateCommandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Outside	Graphics Compute	Action

The bound descriptor sets and push constants that will be used with indirect command generation **must** already be specified on `stateCommandBuffer` at the time of preprocessing commands with `vkCmdPreprocessGeneratedCommandsEXT`. They **must** match the bound descriptor sets and push constants used in the execution of indirect commands with `vkCmdExecuteGeneratedCommandsEXT`.

If push constants for shader stages are also specified in the `VkGeneratedCommandsInfoEXT::indirectCommandsLayout` with a `VK_INDIRECT_COMMANDS_TOKEN_TYPE_PUSH_CONSTANT_EXT` or `VK_INDIRECT_COMMANDS_TOKEN_TYPE_SEQUENCE_INDEX_EXT` token, then those values override the push constants that were previously pushed.

All state bound on `stateCommandBuffer` will be used. All state bound on `stateCommandBuffer` **must** be identical to the state bound at the time `vkCmdExecuteGeneratedCommandsEXT` is recorded. The queue family index `stateCommandBuffer` was allocated from **must** be the same as the queue family index of the command buffer used in `vkCmdExecuteGeneratedCommandsEXT`.

On some implementations, preprocessing **may** have no effect on performance.

`vkCmdExecuteGeneratedCommandsEXT` **may** write to the preprocess buffer, no matter the `isPreprocess` parameter. In this case, the implementation **must** insert appropriate synchronization automatically, which corresponds to the following pseudocode:

- Barrier
 - `srcStageMask` = `DRAW_INDIRECT`
 - `srcAccessMask` = 0
 - `dstStageMask` = `COMMAND_PREPROCESS_BIT`
 - `dstAccessMask` = `COMMAND_PREPROCESS_WRITE_BIT` | `COMMAND_PREPROCESS_READ_BIT`
- Do internal writes
- Barrier
 - `srcStageMask` = `COMMAND_PREPROCESS_BIT`
 - `srcAccessMask` = `COMMAND_PREPROCESS_WRITE_BIT`

- `dstStageMask = DRAW_INDIRECT`
- `dstAccessMask = INDIRECT_COMMAND_READ_BIT`
- Execute

Chapter 30. Sparse Resources

As documented in [Resource Memory Association](#), `VkBuffer` and `VkImage` resources in Vulkan **must** be bound completely and contiguously to a single `VkDeviceMemory` object. This binding **must** be done before the resource is used, and the binding is immutable for the lifetime of the resource.

Sparse resources relax these restrictions and provide these additional features:

- Sparse resources **can** be bound non-contiguously to one or more `VkDeviceMemory` allocations.
- Sparse resources **can** be re-bound to different memory allocations over the lifetime of the resource.
- Sparse resources **can** have descriptors generated and used orthogonally with memory binding commands.

30.1. Sparse Resource Features

Sparse resources have several features that **must** be enabled explicitly at resource creation time. The features are enabled by including bits in the `flags` parameter of `VkImageCreateInfo` or `VkBufferCreateInfo`. Each feature also has one or more corresponding feature enables specified in `VkPhysicalDeviceFeatures`.

- The `sparseBinding` feature is the base, and provides the following capabilities:
 - Resources **can** be bound at some defined (sparse block) granularity.
 - The entire resource **must** be bound to memory before use regardless of regions actually accessed.
 - No specific mapping of image region to memory offset is defined, i.e. the location that each texel corresponds to in memory is implementation-dependent.
 - Sparse buffers have a well-defined mapping of buffer range to memory range, where an offset into a range of the buffer that is bound to a single contiguous range of memory corresponds to an identical offset within that range of memory.
 - Requested via the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` and `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` bits.
 - A sparse image created using `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` (but not `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`) supports all formats that non-sparse usage supports, and supports both `VK_IMAGE_TILING_OPTIMAL` and `VK_IMAGE_TILING_LINEAR` tiling.
- *Sparse Residency* builds on (and requires) the `sparseBinding` feature. It includes the following capabilities:
 - Resources do not have to be completely bound to memory before use on the device.
 - Images have a prescribed sparse image block layout, allowing specific rectangular regions of the image to be bound to specific offsets in memory allocations.
 - Consistency of access to unbound regions of the resource is defined by the absence or presence of `VkPhysicalDeviceSparseProperties::residencyNonResidentStrict`. If this property is present, accesses to unbound regions of the resource are well defined and behave as if the

data bound is populated with all zeros; writes are discarded. When this property is absent, accesses are considered safe, but reads will return undefined values.

- Requested via the `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` and `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` bits.
- Sparse residency support is advertised on a finer grain via the following features:
 - The `sparseResidencyBuffer` feature provides support for creating `VkBuffer` objects with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidencyImage2D` feature provides support for creating 2D single-sampled `VkImage` objects with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidencyImage3D` feature provides support for creating 3D `VkImage` objects with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidency2Samples` feature provides support for creating 2D `VkImage` objects with 2 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidency4Samples` feature provides support for creating 2D `VkImage` objects with 4 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidency8Samples` feature provides support for creating 2D `VkImage` objects with 8 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
 - The `sparseResidency16Samples` feature provides support for creating 2D `VkImage` objects with 16 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.

Implementations supporting `sparseResidencyImage2D` are only **required** to support sparse 2D, single-sampled images. Support for sparse 3D and MSAA images is **optional** and **can** be enabled via `sparseResidencyImage3D`, `sparseResidency2Samples`, `sparseResidency4Samples`, `sparseResidency8Samples`, and `sparseResidency16Samples`.

- A sparse image created using `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` supports all non-compressed color formats with power-of-two element size that non-sparse usage supports. Additional formats **may** also be supported and **can** be queried via `vkGetPhysicalDeviceSparseImageFormatProperties`. `VK_IMAGE_TILING_LINEAR` tiling is not supported.
- The `sparseResidencyAliased` feature provides the following capability that **can** be enabled per resource:

Allows physical memory ranges to be shared between multiple locations in the same sparse resource or between multiple sparse resources, with each binding of a memory location observing a consistent interpretation of the memory contents.

See [Sparse Memory Aliasing](#) for more information.

30.2. Sparse Buffers and Fully-Resident Images

Both `VkBuffer` and `VkImage` objects created with the `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` or `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` bits **can** be thought of as a linear region of address space. In the `VkImage` case if `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` is not used, this linear region is entirely

opaque, meaning that there is no application-visible mapping between texel location and memory offset.

Unless `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` or `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` are also used, the entire resource **must** be bound to one or more `VkDeviceMemory` objects before use.

30.2.1. Sparse Buffer and Fully-Resident Image Block Size

The sparse block size in bytes for sparse buffers and fully-resident images is reported as `VkMemoryRequirements::alignment`. `alignment` represents both the memory alignment requirement and the binding granularity (in bytes) for sparse resources.

30.3. Sparse Partially-Resident Buffers

`VkBuffer` objects created with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` bit allow the buffer to be made only partially resident. Partially resident `VkBuffer` objects are allocated and bound identically to `VkBuffer` objects using only the `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` feature. The only difference is the ability for some regions of the buffer to be unbound during device use.

30.4. Sparse Partially-Resident Images

`VkImage` objects created with the `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` bit allow specific rectangular regions of the image called sparse image blocks to be bound to specific ranges of memory. This allows the application to manage residency at either image subresource or sparse image block granularity. Each image subresource (outside of the [mip tail](#)) starts on a sparse block boundary and has dimensions that are integer multiples of the corresponding dimensions of the sparse image block.

NOTE

Applications **can** use these types of images to control LOD based on total memory consumption. If memory pressure becomes an issue the application **can** unbind and disable specific mipmap levels of images without having to recreate resources or modify texel data of unaffected levels.

The application **can** also use this functionality to access subregions of the image in a “megatexture” fashion. The application **can** create a large image and only populate the region of the image that is currently being used in the scene.

30.4.1. Accessing Unbound Regions

The following member of `VkPhysicalDeviceSparseProperties` affects how data in unbound regions of sparse resources are handled by the implementation:

- `residencyNonResidentStrict`

If this property is not present, reads of unbound regions of the image will return undefined values. Both reads and writes are still considered *safe* and will not affect other resources or populated regions of the image.

If this property is present, all reads of unbound regions of the image will behave as if the region was bound to memory populated with all zeros; writes will be discarded.

[Image operations](#) performed on unbound memory **may** still alter some component values in the natural way for those accesses, e.g. substituting a value of one for alpha in formats that do not have an alpha component.

Example: Reading the alpha component of an unbacked `VK_FORMAT_R8_UNORM` image will return a value of 1.0f.

See [Physical Device Enumeration](#) for instructions for retrieving physical device properties.

Implementor's Note

For implementations that **cannot** natively handle access to unbound regions of a resource, the implementation **may** allocate and bind memory to the unbound regions. Reads and writes to unbound regions will access the implementation-managed memory instead.

Given that the values resulting from reads of unbound regions are undefined in this scenario, implementations **may** use the same physical memory for all unbound regions of multiple resources within the same process.

30.4.2. Mip Tail Regions

Sparse images created using `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` (without also using `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`) have no specific mapping of image region or image subresource to memory offset defined, so the entire image **can** be thought of as a linear opaque address region. However, images created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` do have a prescribed sparse image block layout, and hence each image subresource **must** start on a sparse block boundary. Within each array layer, the set of mip levels that have a smaller size than the sparse block size in bytes are grouped together into a *mip tail region*.

If the `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` flag is present in the `flags` member of `VkSparseImageFormatProperties`, for the image's `format`, then any mip level which has dimensions that are not integer multiples of the corresponding dimensions of the sparse image block, and all subsequent mip levels, are also included in the mip tail region.

The following member of `VkPhysicalDeviceSparseProperties` **may** affect how the implementation places mip levels in the mip tail region:

- `residencyAlignedMipSize`

Each mip tail region is bound to memory as an opaque region (i.e. **must** be bound using a `VkSparseImageOpaqueMemoryBindInfo` structure) and **may** be of a size greater than or equal to the sparse block size in bytes. This size is guaranteed to be an integer multiple of the sparse block size in bytes.

An implementation **may** choose to allow each array-layer's mip tail region to be bound to memory independently or require that all array-layer's mip tail regions be treated as one. This is dictated by `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` in `VkSparseImageMemoryRequirements::flags`.

The following diagrams depict how `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` and `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` alter memory usage and requirements.

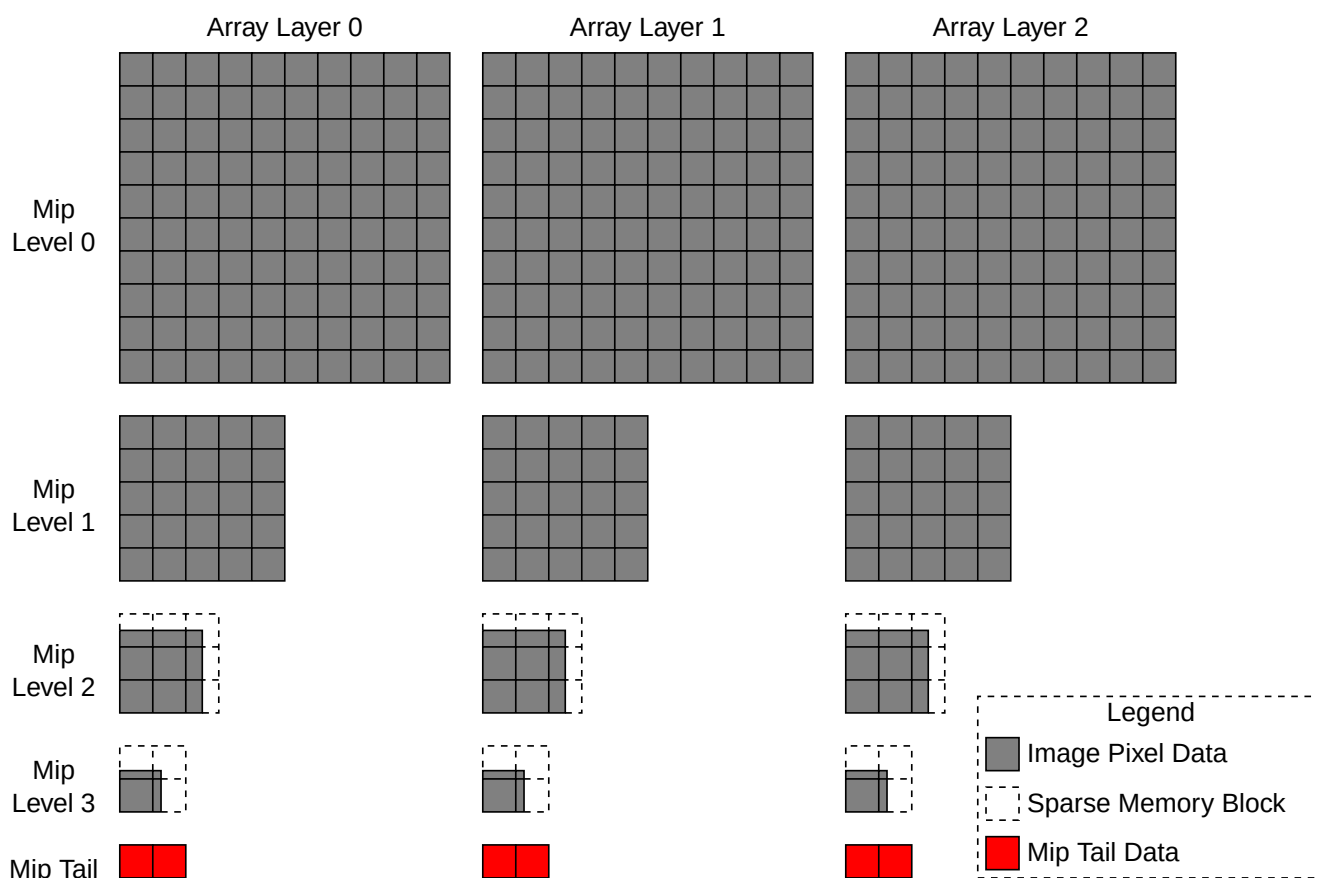


Figure 17. Sparse Image

In the absence of `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` and `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT`, each array layer contains a mip tail region containing texel data for all mip levels smaller than the sparse image block in any dimension.

Mip levels that are as large or larger than a sparse image block in all dimensions **can** be bound individually. Right-edges and bottom-edges of each level are allowed to have partially used sparse blocks. Any bound partially-used-sparse-blocks **must** still have their full sparse block size in bytes allocated in memory.

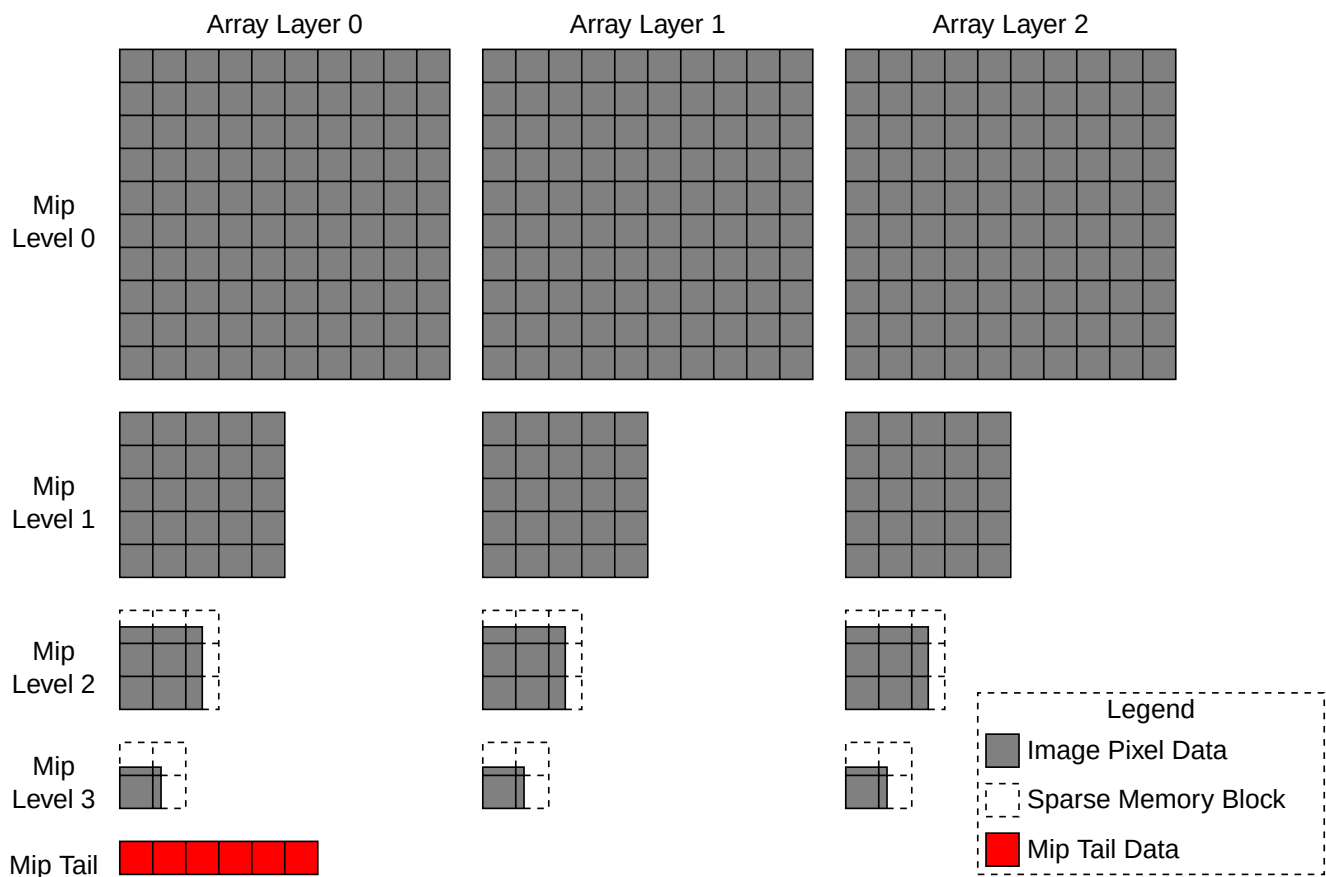


Figure 18. Sparse Image with Single Mip Tail

When `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` is present all array layers will share a single mip tail region.

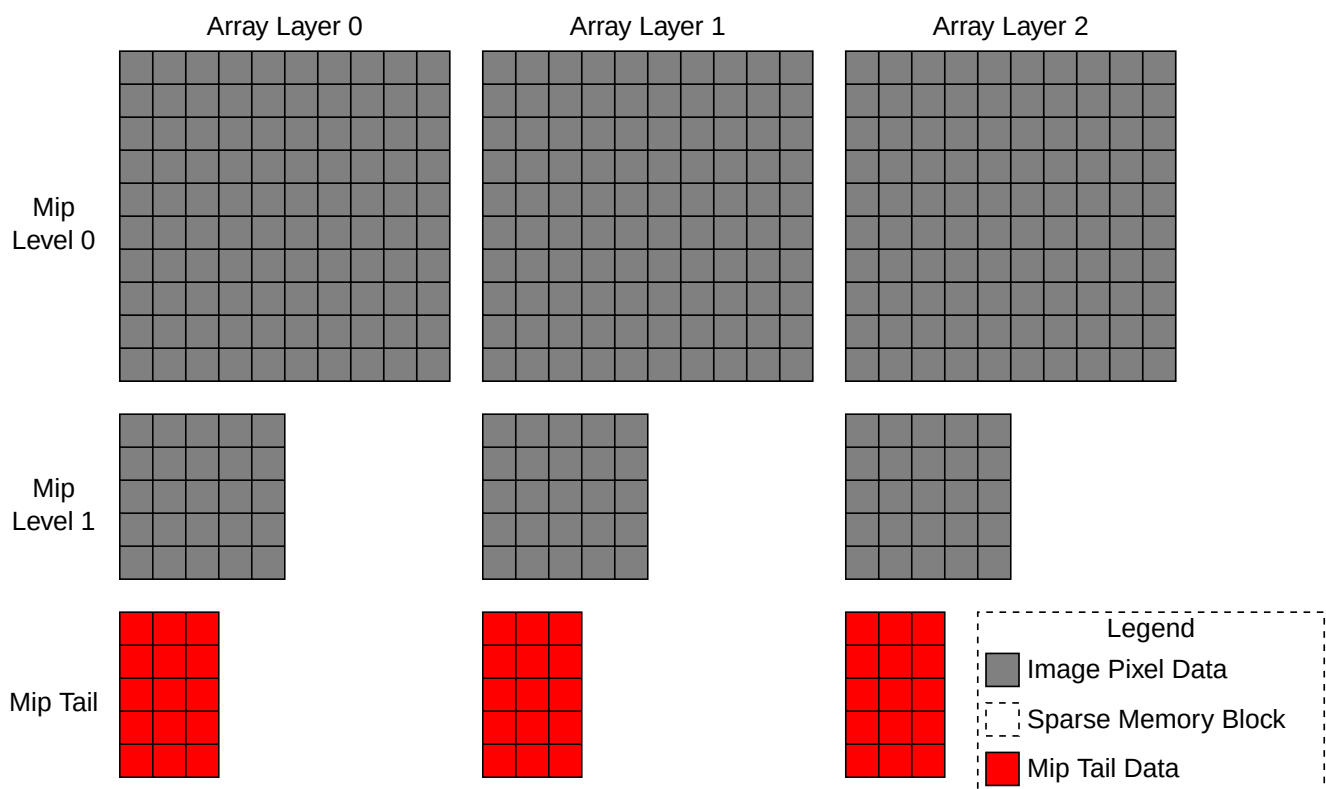


Figure 19. Sparse Image with Aligned Mip Size

NOTE The mip tail regions are presented here in 2D arrays simply for figure size reasons.

Each mip tail is logically a single array of sparse blocks with an implementation-dependent mapping of texels or compressed texel blocks to sparse blocks.

When `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` is present the first mip level that would contain partially used sparse blocks begins the mip tail region. This level and all subsequent levels are placed in the mip tail. Only the first N mip levels whose dimensions are an exact multiple of the sparse image block dimensions **can** be bound and unbound on a sparse block basis.

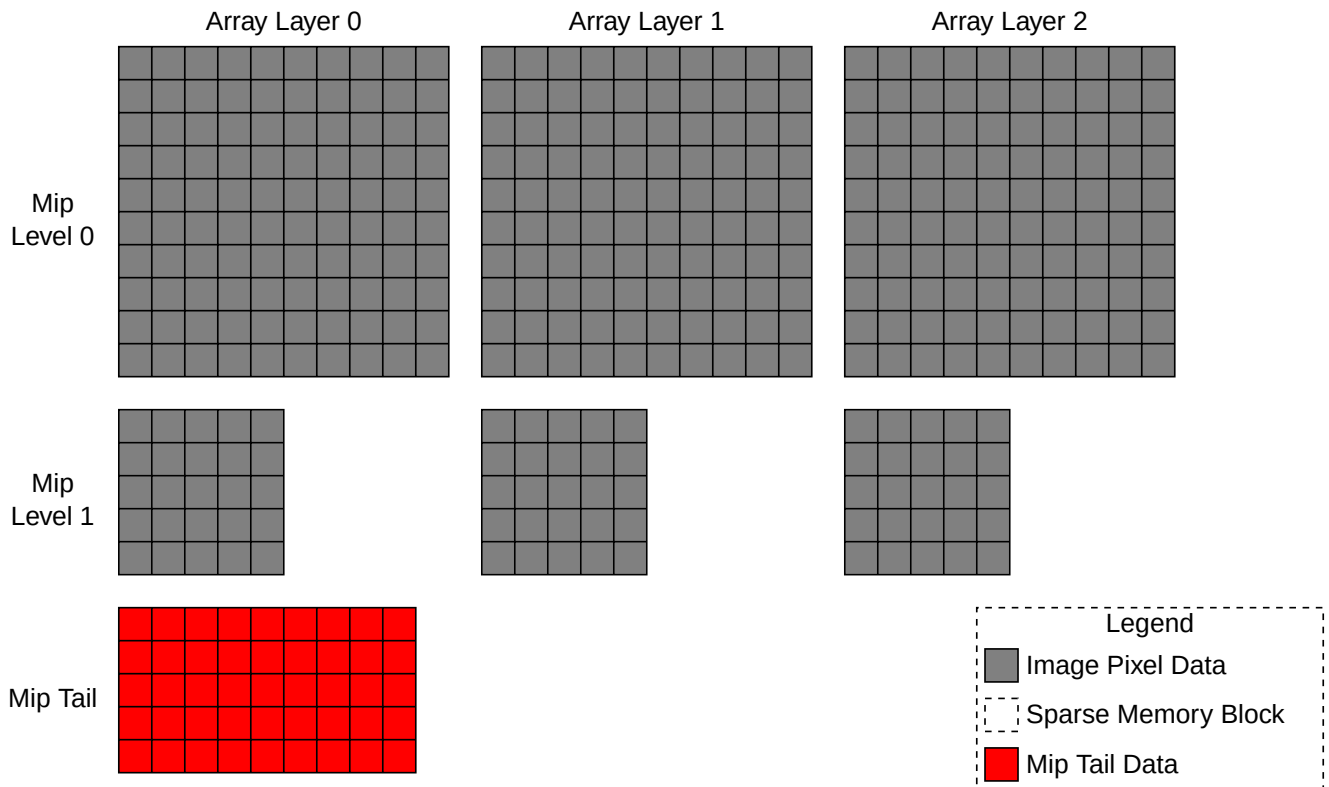


Figure 20. Sparse Image with Aligned Mip Size and Single Mip Tail

NOTE

The mip tail region is presented here in a 2D array simply for figure size reasons. It is logically a single array of sparse blocks with an implementation-dependent mapping of texels or compressed texel blocks to sparse blocks.

When both `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` and `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` are present the constraints from each of these flags are in effect.

30.4.3. Standard Sparse Image Block Shapes

Standard sparse image block shapes define a standard set of dimensions for sparse image blocks that depend on the format of the image. Layout of texels or compressed texel blocks within a sparse image block is implementation-dependent. All currently defined standard sparse image block shapes are 64 KB in size.

For block-compressed formats (e.g. `VK_FORMAT_BC5_UNORM_BLOCK`), the texel size is the size of the compressed texel block (e.g. 128-bit for `BC5`) thus the dimensions of the standard sparse image block shapes apply in terms of compressed texel blocks.

NOTE

For block-compressed formats, the dimensions of a sparse image block in terms of texels **can** be calculated by multiplying the sparse image block dimensions by the compressed texel block dimensions.

Table 42. Standard Sparse Image Block Shapes (Single Sample)

TEXEL SIZE (bits)	Block Shape (2D)	Block Shape (3D)
8-Bit	$256 \times 256 \times 1$	$64 \times 32 \times 32$
16-Bit	$256 \times 128 \times 1$	$32 \times 32 \times 32$
32-Bit	$128 \times 128 \times 1$	$32 \times 32 \times 16$
64-Bit	$128 \times 64 \times 1$	$32 \times 16 \times 16$
128-Bit	$64 \times 64 \times 1$	$16 \times 16 \times 16$

Table 43. Standard Sparse Image Block Shapes (MSAA)

TEXEL SIZE (bits)	Block Shape (2X)	Block Shape (4X)	Block Shape (8X)	Block Shape (16X)
8-Bit	$128 \times 256 \times 1$	$128 \times 128 \times 1$	$64 \times 128 \times 1$	$64 \times 64 \times 1$
16-Bit	$128 \times 128 \times 1$	$128 \times 64 \times 1$	$64 \times 64 \times 1$	$64 \times 32 \times 1$
32-Bit	$64 \times 128 \times 1$	$64 \times 64 \times 1$	$32 \times 64 \times 1$	$32 \times 32 \times 1$
64-Bit	$64 \times 64 \times 1$	$64 \times 32 \times 1$	$32 \times 32 \times 1$	$32 \times 16 \times 1$
128-Bit	$32 \times 64 \times 1$	$32 \times 32 \times 1$	$16 \times 32 \times 1$	$16 \times 16 \times 1$

Implementations that support the standard sparse image block shape for all formats listed in the [Standard Sparse Image Block Shapes \(Single Sample\)](#) and [Standard Sparse Image Block Shapes \(MSAA\)](#) tables **may** advertise the following `VkPhysicalDeviceSparseProperties`:

- `residencyStandard2DBlockShape`
- `residencyStandard2DMultisampleBlockShape`
- `residencyStandard3DBlockShape`

Reporting each of these features does *not* imply that all possible image types are supported as sparse. Instead, this indicates that no supported sparse image of the corresponding type will use custom sparse image block dimensions for any formats that have a corresponding standard sparse image block shape.

30.4.4. Custom Sparse Image Block Shapes

An implementation that does not support a standard image block shape for a particular sparse partially-resident image **may** choose to support a custom sparse image block shape for it instead. The dimensions of such a custom sparse image block shape are reported in `VkSparseImageFormatProperties::imageGranularity`. As with standard sparse image block shapes, the size in bytes of the custom sparse image block shape will be reported in `VkMemoryRequirements::alignment`.

Custom sparse image block dimensions are reported through `vkGetPhysicalDeviceSparseImageFormatProperties` and `vkGetImageSparseMemoryRequirements`.

An implementation **must** not support both the standard sparse image block shape and a custom

sparse image block shape for the same image. The standard sparse image block shape **must** be used if it is supported.

30.4.5. Multiple Aspects

Partially resident images are allowed to report separate sparse properties for different aspects of the image. One example is for depth/stencil images where the implementation separates the depth and stencil data into separate planes. Another reason for multiple aspects is to allow the application to manage memory allocation for implementation-private *metadata* associated with the image. See the figure below:

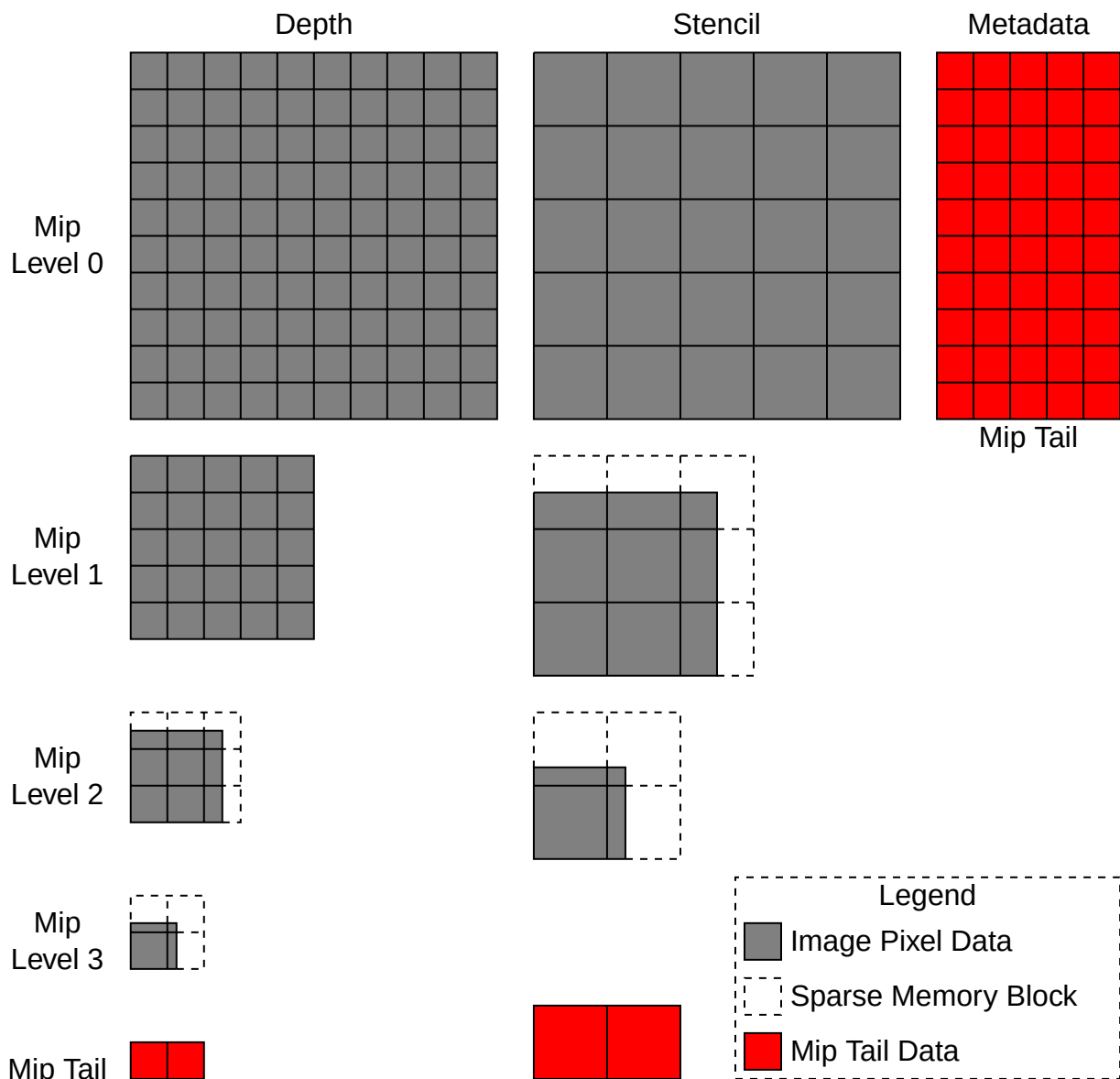


Figure 21. Multiple Aspect Sparse Image

NOTE

The mip tail regions are presented here in 2D arrays simply for figure size reasons. Each mip tail is logically a single array of sparse blocks with an implementation-dependent mapping of texels or compressed texel blocks to sparse blocks.

In the figure above the depth, stencil, and metadata aspects all have unique sparse properties. The per-texel stencil data is $\frac{1}{4}$ the size of the depth data, hence the stencil sparse blocks include $4 \times$ the number of texels. The sparse block size in bytes for all of the aspects is identical and defined by

VkMemoryRequirements::alignment.

Metadata

The metadata aspect of an image has the following constraints:

- All metadata is reported in the mip tail region of the metadata aspect.
- All metadata **must** be bound prior to device use of the sparse image.

30.5. Sparse Memory Aliasing

By default sparse resources have the same aliasing rules as non-sparse resources. See [Memory Aliasing](#) for more information.

VkDevice objects that have the `sparseResidencyAliased` feature enabled are able to use the `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT` and `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` flags for resource creation. These flags allow resources to access physical memory bound into multiple locations within one or more sparse resources in a *data consistent* fashion. This means that reading physical memory from multiple aliased locations will return the same value.

Care **must** be taken when performing a write operation to aliased physical memory. Memory dependencies **must** be used to separate writes to one alias from reads or writes to another alias. Writes to aliased memory that are not properly guarded against accesses to different aliases will have undefined results for all accesses to the aliased memory.

Applications that wish to make use of data consistent sparse memory aliasing **must** abide by the following guidelines:

- All sparse resources that are bound to aliased physical memory **must** be created with the `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT` / `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` flag.
- All resources that access aliased physical memory **must** interpret the memory in the same way. This implies the following:
 - Buffers and images **cannot** alias the same physical memory in a data consistent fashion. The physical memory ranges **must** be used exclusively by buffers or used exclusively by images for data consistency to be guaranteed.
 - Memory in sparse image mip tail regions **cannot** access aliased memory in a data consistent fashion.
 - Sparse images that alias the same physical memory **must** have compatible formats and be using the same sparse image block shape in order to access aliased memory in a data consistent fashion.

Failure to follow any of the above guidelines will require the application to abide by the normal, non-sparse resource [aliasing rules](#). In this case memory **cannot** be accessed in a data consistent fashion.

NOTE

Enabling sparse resource memory aliasing **can** be a way to lower physical memory use, but it **may** reduce performance on some implementations. An application

developer **can** test on their target HW and balance the memory / performance trade-offs measured.

30.6. Sparse Resource Implementation Guidelines (Informative)

This section is Informative. It is included to aid in implementors' understanding of sparse resources.

Device Virtual Address

The basic **sparseBinding** feature allows the resource to reserve its own device virtual address range at resource creation time rather than relying on a bind operation to set this. Without any other creation flags, no other constraints are relaxed compared to normal resources. All pages **must** be bound to physical memory before the device accesses the resource.

The **sparseResidency** features allow sparse resources to be used even when not all pages are bound to memory. Implementations that support access to unbound pages without causing a fault **may** support **residencyNonResidentStrict**.

Not faulting on access to unbound pages is not enough to support **residencyNonResidentStrict**. An implementation **must** also guarantee that reads after writes to unbound regions of the resource always return data for the read as if the memory contains zeros. Depending on any caching hierarchy of the implementation this **may** not always be possible.

Any implementation that does not fault, but does not guarantee correct read values **must** not support **residencyNonResidentStrict**.

Any implementation that **cannot** access unbound pages without causing a fault will require the implementation to bind the entire device virtual address range to physical memory. Any pages that the application does not bind to memory **may** be bound to one (or more) "placeholder" physical page(s) allocated by the implementation. Given the following properties:

- A process **must** not access memory from another process
- Reads return undefined values

It is sufficient for each host process to allocate these placeholder pages and use them for all resources in that process. Implementations **may** allocate more often (per instance, per device, or per resource).

Binding Memory

The byte size reported in **VkMemoryRequirements::size** **must** be greater than or equal to the amount of physical memory **required** to fully populate the resource. Some implementations require "holes" in the device virtual address range that are never accessed. These holes **may** be included in the **size** reported for the resource.

Including or not including the device virtual address holes in the resource size will alter how

the implementation provides support for `VkSparseImageOpaqueMemoryBindInfo`. This operation **must** be supported for all sparse images, even ones created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.

- If the holes are included in the size, this bind function becomes very easy. In most cases the `resourceOffset` is simply a device virtual address offset and the implementation can easily determine what device virtual address to bind. The cost is that the application **may** allocate more physical memory for the resource than it needs.
- If the holes are not included in the size, the application **can** allocate less physical memory than otherwise for the resource. However, in this case the implementation **must** account for the holes when mapping `resourceOffset` to the actual device virtual address intended to be mapped.

If the application always uses `VkSparseImageMemoryBindInfo` to bind memory for the non-tail mip levels, any holes that are present in the resource size **may** never be bound.

NOTE

Since `VkSparseImageMemoryBindInfo` uses texel locations to determine which device virtual addresses to bind, it is impossible to bind device virtual address holes with this operation.

Binding Metadata Memory

All metadata for sparse images have their own sparse properties and are embedded in the mip tail region for said properties. See the [Multiaspect](#) section for details.

Given that metadata is in a mip tail region, and the mip tail region **must** be reported as contiguous (either globally or per-array-layer), some implementations will have to resort to complicated offset → device virtual address mapping for handling `VkSparseImageOpaqueMemoryBindInfo`.

To make this easier on the implementation, the `VK_SPARSE_MEMORY_BIND_METADATA_BIT` explicitly specifies when metadata is bound with `VkSparseImageOpaqueMemoryBindInfo`. When this flag is not present, the `resourceOffset` **may** be treated as a strict device virtual address offset.

When `VK_SPARSE_MEMORY_BIND_METADATA_BIT` is present, the `resourceOffset` **must** have been derived explicitly from the `imageMipTailOffset` in the sparse resource properties returned for the metadata aspect. By manipulating the value returned for `imageMipTailOffset`, the `resourceOffset` does not have to correlate directly to a device virtual address offset, and **may** instead be whatever value makes it easiest for the implementation to derive the correct device virtual address.

30.7. Sparse Resource API

The APIs related to sparse resources are grouped into the following categories:

- [Physical Device Features](#)
- [Physical Device Sparse Properties](#)

- [Sparse Image Format Properties](#)
- [Sparse Resource Creation](#)
- [Sparse Resource Memory Requirements](#)
- [Binding Resource Memory](#)

30.7.1. Physical Device Features

Some sparse-resource related features are reported and enabled in `VkPhysicalDeviceFeatures`. These features **must** be supported and enabled on the `VkDevice` object before applications **can** use them. See [Physical Device Features](#) for information on how to get and set enabled device features, and for more detailed explanations of these features.

Sparse Physical Device Features

- `sparseBinding`: Support for creating `VkBuffer` and `VkImage` objects with the `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` and `VK_IMAGE_CREATE_SPARSE_BINDING_BIT` flags, respectively.
- `sparseResidencyBuffer`: Support for creating `VkBuffer` objects with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` flag.
- `sparseResidencyImage2D`: Support for creating 2D single-sampled `VkImage` objects with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidencyImage3D`: Support for creating 3D `VkImage` objects with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidency2Samples`: Support for creating 2D `VkImage` objects with 2 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidency4Samples`: Support for creating 2D `VkImage` objects with 4 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidency8Samples`: Support for creating 2D `VkImage` objects with 8 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidency16Samples`: Support for creating 2D `VkImage` objects with 16 samples and `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`.
- `sparseResidencyAliased`: Support for creating `VkBuffer` and `VkImage` objects with the `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT` and `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` flags, respectively.

30.7.2. Physical Device Sparse Properties

Some features of the implementation are not possible to disable, and are reported to allow applications to alter their sparse resource usage accordingly. These read-only capabilities are reported in the `VkPhysicalDeviceProperties::sparseProperties` member, which is a `VkPhysicalDeviceSparseProperties` structure.

The `VkPhysicalDeviceSparseProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceSparseProperties {
    VkBool32    residencyStandard2DBlockShape;
    VkBool32    residencyStandard2DMultisampleBlockShape;
    VkBool32    residencyStandard3DBlockShape;
    VkBool32    residencyAlignedMipSize;
    VkBool32    residencyNonResidentStrict;
} VkPhysicalDeviceSparseProperties;
```

- `residencyStandard2DBlockShape` is `VK_TRUE` if the physical device will access all single-sample 2D sparse resources using the standard sparse image block shapes (based on image format), as described in the [Standard Sparse Image Block Shapes \(Single Sample\)](#) table. If this property is not supported the value returned in the `imageGranularity` member of the `VkSparseImageFormatProperties` structure for single-sample 2D images is not **required** to match the standard sparse image block dimensions listed in the table.
- `residencyStandard2DMultisampleBlockShape` is `VK_TRUE` if the physical device will access all multisample 2D sparse resources using the standard sparse image block shapes (based on image format), as described in the [Standard Sparse Image Block Shapes \(MSAA\)](#) table. If this property is not supported, the value returned in the `imageGranularity` member of the `VkSparseImageFormatProperties` structure for multisample 2D images is not **required** to match the standard sparse image block dimensions listed in the table.
- `residencyStandard3DBlockShape` is `VK_TRUE` if the physical device will access all 3D sparse resources using the standard sparse image block shapes (based on image format), as described in the [Standard Sparse Image Block Shapes \(Single Sample\)](#) table. If this property is not supported, the value returned in the `imageGranularity` member of the `VkSparseImageFormatProperties` structure for 3D images is not **required** to match the standard sparse image block dimensions listed in the table.
- `residencyAlignedMipSize` is `VK_TRUE` if images with mip level dimensions that are not integer multiples of the corresponding dimensions of the sparse image block **may** be placed in the mip tail. If this property is not reported, only mip levels with dimensions smaller than the `imageGranularity` member of the `VkSparseImageFormatProperties` structure will be placed in the mip tail. If this property is reported the implementation is allowed to return `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` in the `flags` member of `VkSparseImageFormatProperties`, indicating that mip level dimensions that are not integer multiples of the corresponding dimensions of the sparse image block will be placed in the mip tail.
- `residencyNonResidentStrict` specifies whether the physical device **can** consistently access non-resident regions of a resource. If this property is `VK_TRUE`, access to non-resident regions of resources will be guaranteed to return values as if the resource was populated with 0; writes to non-resident regions will be discarded.

30.7.3. Sparse Image Format Properties

Given that certain aspects of sparse image support, including the sparse image block dimensions, **may** be implementation-dependent, `vkGetPhysicalDeviceSparseImageFormatProperties` **can** be used to query for sparse image format properties prior to resource creation. This command is used

to check whether a given set of sparse image parameters is supported and what the sparse image block shape will be.

Sparse Image Format Properties API

The `VkSparseImageFormatProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageFormatProperties {
    VkImageAspectFlags    aspectMask;
    VkExtent3D            imageGranularity;
    VkSparseImageFormatFlags flags;
} VkSparseImageFormatProperties;
```

- `aspectMask` is a bitmask `VkImageAspectFlagBits` specifying which aspects of the image the properties apply to.
- `imageGranularity` is the width, height, and depth of the sparse image block in texels or compressed texel blocks.
- `flags` is a bitmask of `VkSparseImageFormatFlagBits` specifying additional information about the sparse resource.

Bits which **may** be set in `VkSparseImageFormatProperties::flags`, specifying additional information about the sparse resource, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSparseImageFormatFlagBits {
    VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT = 0x00000001,
    VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT = 0x00000002,
    VK_SPARSE_IMAGE_FORMAT_NONSTANDARD_BLOCK_SIZE_BIT = 0x00000004,
} VkSparseImageFormatFlagBits;
```

- `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` specifies that the image uses a single mip tail region for all array layers.
- `VK_SPARSE_IMAGE_FORMAT_ALIGNED_MIP_SIZE_BIT` specifies that the first mip level whose dimensions are not integer multiples of the corresponding dimensions of the sparse image block begins the mip tail region.
- `VK_SPARSE_IMAGE_FORMAT_NONSTANDARD_BLOCK_SIZE_BIT` specifies that the image uses non-standard sparse image block dimensions, and the `imageGranularity` values do not match the standard sparse image block dimensions for the given format.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSparseImageFormatFlags;
```

`VkSparseImageFormatFlags` is a bitmask type for setting a mask of zero or more `VkSparseImageFormatFlagBits`.

`vkGetPhysicalDeviceSparseImageFormatProperties` returns an array of `VkSparseImageFormatProperties`. Each element describes properties for one set of image aspects that are bound simultaneously for a `VkImage` created with the provided image creation parameters. This is usually one element for each aspect in the image, but for interleaved depth/stencil images there is only one element describing the combined aspects.

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceSparseImageFormatProperties(
    VkPhysicalDevice    physicalDevice,
    VkFormat            format,
    VkImageType         type,
    VkSampleCountFlagBits samples,
    VkImageUsageFlags   usage,
    VkImageTiling       tiling,
    uint32_t*           pPropertyCount,
    VkSparseImageFormatProperties* pProperties);
```

- `physicalDevice` is the physical device from which to query the sparse image format properties.
- `format` is the image format.
- `type` is the dimensionality of the image.
- `samples` is a `VkSampleCountFlagBits` value specifying the number of samples per texel.
- `usage` is a bitmask describing the intended usage of the image.
- `tiling` is the tiling arrangement of the texel blocks in memory.
- `pPropertyCount` is a pointer to an integer related to the number of sparse format properties available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkSparseImageFormatProperties` structures.

If `pProperties` is `NULL`, then the number of sparse format properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of sparse format properties available, at most `pPropertyCount` structures will be written.

If `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` is not supported for the given arguments, `pPropertyCount` will be zero upon return, and no data will be written to `pProperties`.

Multiple aspects are returned for depth/stencil images that are implemented as separate planes by the implementation. The depth and stencil data planes each have unique `VkSparseImageFormatProperties` data.

Depth/stencil images with depth and stencil data interleaved into a single plane will return a single `VkSparseImageFormatProperties` structure with the `aspectMask` set to `VK_IMAGE_ASPECT_DEPTH_BIT | VK_IMAGE_ASPECT_STENCIL_BIT`.

Valid Usage

- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-samples-01094
`samples` **must** be a valid `VkSampleCountFlagBits` value that is set in `VkImageFormatProperties::sampleCounts` returned by `vkGetPhysicalDeviceImageFormatProperties` with `format`, `type`, `tiling`, and `usage` equal to those in this command

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-format-parameter
`format` **must** be a valid `VkFormat` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-type-parameter
`type` **must** be a valid `VkImageType` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-samples-parameter
`samples` **must** be a valid `VkSampleCountFlagBits` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-usage-parameter
`usage` **must** be a valid combination of `VkImageUsageFlagBits` values
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-usage-requiredbitmask
`usage` **must** not be `0`
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-tiling-parameter
`tiling` **must** be a valid `VkImageTiling` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-pPropertyCount-parameter
`pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties-pProperties-parameter
If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkSparseImageFormatProperties` structures

`vkGetPhysicalDeviceSparseImageFormatProperties2KHR` returns an array of `VkSparseImageFormatProperties2`. Each element describes properties for one set of image aspects that are bound simultaneously for a `VkImage` created with the provided image creation parameters. This is usually one element for each aspect in the image, but for interleaved depth/stencil images there is only one element describing the combined aspects.

```
// Provided by VK_KHR_get_physical_device_properties2
void vkGetPhysicalDeviceSparseImageFormatProperties2KHR(
    VkPhysicalDevice                physicalDevice,
    const VkPhysicalDeviceSparseImageFormatInfo2* pFormatInfo,
    uint32_t*                        pPropertyCount,
    VkSparseImageFormatProperties2*  pProperties);
```

- `physicalDevice` is the physical device from which to query the sparse image format properties.
- `pFormatInfo` is a pointer to a `VkPhysicalDeviceSparseImageFormatInfo2` structure containing input parameters to the command.
- `pPropertyCount` is a pointer to an integer related to the number of sparse format properties available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkSparseImageFormatProperties2` structures.

`vkGetPhysicalDeviceSparseImageFormatProperties2KHR` behaves identically to `vkGetPhysicalDeviceSparseImageFormatProperties`, with the ability to return extended information by adding extending structures to the `pNext` chain of its `pProperties` parameter.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSparseImageFormatProperties2-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties2-pFormatInfo-parameter `pFormatInfo` **must** be a valid pointer to a valid `VkPhysicalDeviceSparseImageFormatInfo2` structure
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties2-pPropertyCount-parameter `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceSparseImageFormatProperties2-pProperties-parameter
If the value referenced by `pPropertyCount` is not 0, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkSparseImageFormatProperties2` structures

The `VkPhysicalDeviceSparseImageFormatInfo2` structure is defined as:


```
typedef struct VkPhysicalDeviceSparseImageFormatInfo2 {
    VkStructureType      sType;
    const void*          pNext;
    VkFormat              format;
    VkImageType           type;
    VkSampleCountFlagBits samples;
    VkImageUsageFlags     usage;
    VkImageTiling          tiling;
} VkPhysicalDeviceSparseImageFormatInfo2;
```

or the equivalent

```
// Provided by VK_KHR_get_physical_device_properties2
typedef VkPhysicalDeviceSparseImageFormatInfo2
VkPhysicalDeviceSparseImageFormatInfo2KHR;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **format** is the image format.
- **type** is the dimensionality of the image.
- **samples** is a **VkSampleCountFlagBits** value specifying the number of samples per texel.
- **usage** is a bitmask describing the intended usage of the image.
- **tiling** is the tiling arrangement of the texel blocks in memory.

Valid Usage

- VUID-VkPhysicalDeviceSparseImageFormatInfo2-samples-01095
samples must be a valid **VkSampleCountFlagBits** value that is set in **VkImageFormatProperties::sampleCounts** returned by **vkGetPhysicalDeviceImageFormatProperties** with **format**, **type**, **tiling**, and **usage** equal to those in this command

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSparseImageFormatInfo2-sType-sType
sType must be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SPARSE_IMAGE_FORMAT_INFO_2**
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-pNext-pNext
pNext must be **NULL**
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-format-parameter
format must be a valid **VkFormat** value
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-type-parameter

type must be a valid [VkImageType](#) value

- VUID-VkPhysicalDeviceSparseImageFormatInfo2-samples-parameter **samples** must be a valid [VkSampleCountFlagBits](#) value
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-usage-parameter **usage** must be a valid combination of [VkImageUsageFlagBits](#) values
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-usage-requiredbitmask **usage** must not be 0
- VUID-VkPhysicalDeviceSparseImageFormatInfo2-tiling-parameter **tiling** must be a valid [VkImageTiling](#) value

The [VkSparseImageFormatProperties2](#) structure is defined as:

```
typedef struct VkSparseImageFormatProperties2 {  
    VkStructureType      sType;  
    void*                 pNext;  
    VkSparseImageFormatProperties properties;  
} VkSparseImageFormatProperties2;
```

or the equivalent

```
// Provided by VK_KHR_get_physical_device_properties2  
typedef VkSparseImageFormatProperties2 VkSparseImageFormatProperties2KHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is [NULL](#) or a pointer to a structure extending this structure.
- **properties** is a [VkSparseImageFormatProperties](#) structure which is populated with the same values as in [vkGetPhysicalDeviceSparseImageFormatProperties](#).

Valid Usage (Implicit)

- VUID-VkSparseImageFormatProperties2-sType-sType **sType** must be [VK_STRUCTURE_TYPE_SPARSE_IMAGE_FORMAT_PROPERTIES_2](#)
- VUID-VkSparseImageFormatProperties2-pNext-pNext **pNext** must be [NULL](#)

30.7.4. Sparse Resource Creation

Sparse resources require that one or more sparse feature flags be specified (as part of the [VkPhysicalDeviceFeatures](#) structure described previously in the [Physical Device Features](#) section) when calling [vkCreateDevice](#). When the appropriate device features are enabled, the [VK_BUFFER_CREATE_SPARSE_*](#) and [VK_IMAGE_CREATE_SPARSE_*](#) flags **can** be used. See [vkCreateBuffer](#) and [vkCreateImage](#) for details of the resource creation APIs.

NOTE Specifying `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` or `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` requires specifying `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` or `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`, respectively, as well. This means that resources **must** be created with the appropriate `*_SPARSE_BINDING_BIT` to be used with the sparse binding command (`vkQueueBindSparse`).

30.7.5. Sparse Resource Memory Requirements

Sparse resources have specific memory requirements related to binding sparse memory. These memory requirements are reported differently for `VkBuffer` objects and `VkImage` objects.

Buffer and Fully-Resident Images

Buffers (both fully and partially resident) and fully-resident images **can** be bound to memory using only the data from `VkMemoryRequirements`. For all sparse resources the `VkMemoryRequirements::alignment` member specifies both the binding granularity in bytes and the **required** alignment of `VkDeviceMemory`.

Partially Resident Images

Partially resident images have a different method for binding memory. As with buffers and fully resident images, the `VkMemoryRequirements::alignment` field specifies the binding granularity in bytes for the image.

Requesting sparse memory requirements for `VkImage` objects using `vkGetImageSparseMemoryRequirements` will return an array of one or more `VkSparseImageMemoryRequirements` structures. Each structure describes the sparse memory requirements for a group of aspects of the image.

The sparse image **must** have been created using the `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` flag to retrieve valid sparse image memory requirements.

Sparse Image Memory Requirements

The `VkSparseImageMemoryRequirements` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageMemoryRequirements {
    VkSparseImageFormatProperties    formatProperties;
    uint32_t                        imageMipTailFirstLod;
    VkDeviceSize                    imageMipTailSize;
    VkDeviceSize                    imageMipTailOffset;
    VkDeviceSize                    imageMipTailStride;
} VkSparseImageMemoryRequirements;
```

- `formatProperties` is a `VkSparseImageFormatProperties` structure specifying properties of the image format.

- `imageMipTailFirstLod` is the first mip level at which image subresources are included in the mip tail region.
- `imageMipTailSize` is the memory size (in bytes) of the mip tail region. If `formatProperties.flags` contains `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT`, this is the size of the whole mip tail, otherwise this is the size of the mip tail of a single array layer. This value is guaranteed to be a multiple of the sparse block size in bytes.
- `imageMipTailOffset` is the opaque memory offset used with `VkSparseImageOpaqueMemoryBindInfo` to bind the mip tail region(s).
- `imageMipTailStride` is the offset stride between each array-layer's mip tail, if `formatProperties.flags` does not contain `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` (otherwise the value is undefined).

To query sparse memory requirements for an image, call:

```
// Provided by VK_VERSION_1_0
void vkGetImageSparseMemoryRequirements(
    VkDevice          device,
    VkImage           image,
    uint32_t*         pSparseMemoryRequirementCount,
    VkSparseImageMemoryRequirements* pSparseMemoryRequirements);
```

- `device` is the logical device that owns the image.
- `image` is the `VkImage` object to get the memory requirements for.
- `pSparseMemoryRequirementCount` is a pointer to an integer related to the number of sparse memory requirements available or queried, as described below.
- `pSparseMemoryRequirements` is either `NULL` or a pointer to an array of `VkSparseImageMemoryRequirements` structures.

If `pSparseMemoryRequirements` is `NULL`, then the number of sparse memory requirements available is returned in `pSparseMemoryRequirementCount`. Otherwise, `pSparseMemoryRequirementCount` **must** point to a variable set by the application to the number of elements in the `pSparseMemoryRequirements` array, and on return the variable is overwritten with the number of structures actually written to `pSparseMemoryRequirements`. If `pSparseMemoryRequirementCount` is less than the number of sparse memory requirements available, at most `pSparseMemoryRequirementCount` structures will be written.

If the image was not created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` then `pSparseMemoryRequirementCount` will be zero and `pSparseMemoryRequirements` will not be written to.

NOTE

It is legal for an implementation to report a larger value in `VkMemoryRequirements::size` than would be obtained by adding together memory sizes for all `VkSparseImageMemoryRequirements` returned by `vkGetImageSparseMemoryRequirements`. This **may** occur when the implementation requires unused padding in the address range describing the resource.

Valid Usage (Implicit)

- VUID-vkGetImageSparseMemoryRequirements-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkGetImageSparseMemoryRequirements-image-parameter **image** **must** be a valid [VkImage](#) handle
- VUID-vkGetImageSparseMemoryRequirements-pSparseMemoryRequirementCount-parameter **pSparseMemoryRequirementCount** **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetImageSparseMemoryRequirements-pSparseMemoryRequirements-parameter If the value referenced by **pSparseMemoryRequirementCount** is not `0`, and **pSparseMemoryRequirements** is not `NULL`, **pSparseMemoryRequirements** **must** be a valid pointer to an array of **pSparseMemoryRequirementCount** [VkSparseImageMemoryRequirements](#) structures
- VUID-vkGetImageSparseMemoryRequirements-image-parent **image** **must** have been created, allocated, or retrieved from **device**

To query sparse memory requirements for an image, call:

```
// Provided by VK_KHR_get_memory_requirements2
void vkGetImageSparseMemoryRequirements2KHR(
    VkDevice device,
    const VkImageSparseMemoryRequirementsInfo2* pInfo,
    uint32_t* pSparseMemoryRequirementCount,
    VkSparseImageMemoryRequirements2* pSparseMemoryRequirements);
```

- **device** is the logical device that owns the image.
- **pInfo** is a pointer to a [VkImageSparseMemoryRequirementsInfo2](#) structure containing parameters required for the memory requirements query.
- **pSparseMemoryRequirementCount** is a pointer to an integer related to the number of sparse memory requirements available or queried, as described below.
- **pSparseMemoryRequirements** is either `NULL` or a pointer to an array of [VkSparseImageMemoryRequirements2](#) structures.

Valid Usage (Implicit)

- VUID-vkGetImageSparseMemoryRequirements2-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkGetImageSparseMemoryRequirements2-pInfo-parameter **pInfo** **must** be a valid pointer to a valid [VkImageSparseMemoryRequirementsInfo2](#) structure
- VUID-vkGetImageSparseMemoryRequirements2-pSparseMemoryRequirementCount-

parameter

`pSparseMemoryRequirementCount` **must** be a valid pointer to a `uint32_t` value

- VUID-vkGetImageSparseMemoryRequirements2-pSparseMemoryRequirements-parameter

If the value referenced by `pSparseMemoryRequirementCount` is not `0`, and `pSparseMemoryRequirements` is not `NULL`, `pSparseMemoryRequirements` **must** be a valid pointer to an array of `pSparseMemoryRequirementCount` `VkSparseImageMemoryRequirements2` structures

To determine the sparse memory requirements for an image resource without creating an object, call:

```
// Provided by VK_KHR_maintenance4
void vkGetDeviceImageSparseMemoryRequirementsKHR(
    VkDevice                device,
    const VkDeviceImageMemoryRequirements* pInfo,
    uint32_t*                pSparseMemoryRequirementCount,
    VkSparseImageMemoryRequirements2*      pSparseMemoryRequirements);
```

- `device` is the logical device intended to own the image.
- `pInfo` is a pointer to a `VkDeviceImageMemoryRequirements` structure containing parameters required for the memory requirements query.
- `pSparseMemoryRequirementCount` is a pointer to an integer related to the number of sparse memory requirements available or queried, as described below.
- `pSparseMemoryRequirements` is either `NULL` or a pointer to an array of `VkSparseImageMemoryRequirements2` structures.

Valid Usage (Implicit)

- VUID-vkGetDeviceImageSparseMemoryRequirements-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetDeviceImageSparseMemoryRequirements-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkDeviceImageMemoryRequirements` structure
- VUID-vkGetDeviceImageSparseMemoryRequirements-pSparseMemoryRequirementCount-parameter
`pSparseMemoryRequirementCount` **must** be a valid pointer to a `uint32_t` value

- VUID-vkGetDeviceImageSparseMemoryRequirements-pSparseMemoryRequirements-parameter

If the value referenced by `pSparseMemoryRequirementCount` is not `0`, and `pSparseMemoryRequirements` is not `NULL`, `pSparseMemoryRequirements` **must** be a valid pointer to an array of `pSparseMemoryRequirementCount` `VkSparseImageMemoryRequirements2` structures

The `VkImageSparseMemoryRequirementsInfo2` structure is defined as:

```
typedef struct VkImageSparseMemoryRequirementsInfo2 {
    VkStructureType    sType;
    const void*        pNext;
    VkImage             image;
} VkImageSparseMemoryRequirementsInfo2;
```

or the equivalent

```
// Provided by VK_KHR_get_memory_requirements2
typedef VkImageSparseMemoryRequirementsInfo2 VkImageSparseMemoryRequirementsInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `image` is the image to query.

Valid Usage (Implicit)

- VUID-VkImageSparseMemoryRequirementsInfo2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_SPARSE_MEMORY_REQUIREMENTS_INFO_2`
- VUID-VkImageSparseMemoryRequirementsInfo2-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkImageSparseMemoryRequirementsInfo2-image-parameter
`image` **must** be a valid `VkImage` handle

The `VkSparseImageMemoryRequirements2` structure is defined as:

```
typedef struct VkSparseImageMemoryRequirements2 {
    VkStructureType    sType;
    void*              pNext;
    VkSparseImageMemoryRequirements    memoryRequirements;
} VkSparseImageMemoryRequirements2;
```

or the equivalent

```
// Provided by VK_KHR_get_memory_requirements2
typedef VkSparseImageMemoryRequirements2 VkSparseImageMemoryRequirements2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryRequirements` is a `VkSparseImageMemoryRequirements` structure describing the memory

requirements of the sparse image.

Valid Usage (Implicit)

- `VUID-VkSparseImageMemoryRequirements2-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_SPARSE_IMAGE_MEMORY_REQUIREMENTS_2`
- `VUID-VkSparseImageMemoryRequirements2-pNext-pNext`
`pNext` **must** be `NULL`

30.7.6. Binding Resource Memory

Non-sparse resources are backed by a single physical allocation prior to device use (via `vkBindImageMemory` or `vkBindBufferMemory`), and their backing **must** not be changed. On the other hand, sparse resources **can** be bound to memory non-contiguously and these bindings **can** be altered during the lifetime of the resource.

NOTE

It is important to note that freeing a `VkDeviceMemory` object with `vkFreeMemory` will not cause resources (or resource regions) bound to the memory object to become unbound. Applications **must** not access resources bound to memory that has been freed.

Sparse memory bindings execute on a queue that includes the `VK_QUEUE_SPARSE_BINDING_BIT` bit. Applications **must** use [synchronization primitives](#) to guarantee that other queues do not access ranges of memory concurrently with a binding change. Applications **can** access other ranges of the same resource while a bind operation is executing.

NOTE

Implementations **must** provide a guarantee that simultaneously binding sparse blocks while another queue accesses those same sparse blocks via a sparse resource **must** not access memory owned by another process or otherwise corrupt the system.

While some implementations **may** include `VK_QUEUE_SPARSE_BINDING_BIT` support in queue families that also include graphics and compute support, other implementations **may** only expose a `VK_QUEUE_SPARSE_BINDING_BIT`-only queue family. In either case, applications **must** use [synchronization primitives](#) to explicitly request any ordering dependencies between sparse memory binding operations and other graphics/compute/transfer operations, as sparse binding operations are not automatically ordered against command buffer execution, even within a single queue.

When binding memory explicitly for the `VK_IMAGE_ASPECT_METADATA_BIT` the application **must** use the `VK_SPARSE_MEMORY_BIND_METADATA_BIT` in the `VkSparseMemoryBind::flags` field when binding memory. Binding memory for metadata is done the same way as binding memory for the mip tail, with the addition of the `VK_SPARSE_MEMORY_BIND_METADATA_BIT` flag.

Binding the mip tail for any aspect **must** only be performed using `VkSparseImageOpaqueMemoryBindInfo`. If `formatProperties.flags` contains `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT`, then it **can** be bound with a single

`VkSparseMemoryBind` structure, with `resourceOffset = imageMipTailOffset` and `size = imageMipTailSize`.

If `formatProperties.flags` does not contain `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT` then the offset for the mip tail in each array layer is given as:

```
arrayMipTailOffset = imageMipTailOffset + arrayLayer * imageMipTailStride;
```

and the mip tail **can** be bound with `layerCount` `VkSparseMemoryBind` structures, each using `size = imageMipTailSize` and `resourceOffset = arrayMipTailOffset` as defined above.

Sparse memory binding is handled by the following APIs and related data structures.

Sparse Memory Binding Functions

The `VkSparseMemoryBind` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseMemoryBind {
    VkDeviceSize      resourceOffset;
    VkDeviceSize      size;
    VkDeviceMemory     memory;
    VkDeviceSize      memoryOffset;
    VkSparseMemoryBindFlags flags;
} VkSparseMemoryBind;
```

- `resourceOffset` is the offset into the resource.
- `size` is the size of the memory region to be bound.
- `memory` is the `VkDeviceMemory` object that the range of the resource is bound to. If `memory` is `VK_NULL_HANDLE`, the range is unbound.
- `memoryOffset` is the offset into the `VkDeviceMemory` object to bind the resource range to. If `memory` is `VK_NULL_HANDLE`, this value is ignored.
- `flags` is a bitmask of `VkSparseMemoryBindFlagBits` specifying usage of the binding operation.

The *binding range* [`resourceOffset`, `resourceOffset + size`) has different constraints based on `flags`. If `flags` contains `VK_SPARSE_MEMORY_BIND_METADATA_BIT`, the binding range **must** be within the mip tail region of the metadata aspect. This metadata region is defined by:

$$\text{metadataRegion} = [\text{base}, \text{base} + \text{imageMipTailSize})$$
$$\text{base} = \text{imageMipTailOffset} + \text{imageMipTailStride} \times n$$

and `imageMipTailOffset`, `imageMipTailSize`, and `imageMipTailStride` values are from the `VkSparseImageMemoryRequirements` corresponding to the metadata aspect of the image, and `n` is a

valid array layer index for the image,

`imageMipTailStride` is considered to be zero for aspects where `VkSparseImageMemoryRequirements::formatProperties.flags` contains `VK_SPARSE_IMAGE_FORMAT_SINGLE_MIPTAIL_BIT`.

If `flags` does not contain `VK_SPARSE_MEMORY_BIND_METADATA_BIT`, the binding range **must** be within the range `[0, VkMemoryRequirements::size)`.

Valid Usage

- VUID-VkSparseMemoryBind-memory-01096
If `memory` is not `VK_NULL_HANDLE`, `memory` and `memoryOffset` **must** match the memory requirements of the resource, as described in section [Resource Memory Association](#)
- VUID-VkSparseMemoryBind-resourceOffset-09491
If the resource being bound is a `VkBuffer`, `resourceOffset`, `memoryOffset` and `size` **must** be an integer multiple of the `alignment` of the `VkMemoryRequirements` structure returned from a call to `vkGetBufferMemoryRequirements` with the buffer resource
- VUID-VkSparseMemoryBind-resourceOffset-09492
If the resource being bound is a `VkImage`, `resourceOffset` and `memoryOffset` **must** be an integer multiple of the `alignment` of the `VkMemoryRequirements` structure returned from a call to `vkGetImageMemoryRequirements` with the image resource
- VUID-VkSparseMemoryBind-memory-01097
If `memory` is not `VK_NULL_HANDLE`, `memory` **must** not have been created with a memory type that reports `VK_MEMORY_PROPERTY_LAZILY_ALLOCATED_BIT` bit set
- VUID-VkSparseMemoryBind-size-01098
`size` **must** be greater than 0
- VUID-VkSparseMemoryBind-resourceOffset-01099
`resourceOffset` **must** be less than the size of the resource
- VUID-VkSparseMemoryBind-size-01100
`size` **must** be less than or equal to the size of the resource minus `resourceOffset`
- VUID-VkSparseMemoryBind-memoryOffset-01101
`memoryOffset` **must** be less than the size of `memory`
- VUID-VkSparseMemoryBind-size-01102
`size` **must** be less than or equal to the size of `memory` minus `memoryOffset`
- VUID-VkSparseMemoryBind-memory-02730
If `memory` was created with `VkExportMemoryAllocateInfo::handleTypes` not equal to 0, at least one handle type it contained **must** also have been set in `VkExternalMemoryBufferCreateInfo::handleTypes` or `VkExternalMemoryImageCreateInfo::handleTypes` when the resource was created
- VUID-VkSparseMemoryBind-memory-02731
If `memory` was created by a memory import operation, the external handle type of the imported memory **must** also have been set in `VkExternalMemoryBufferCreateInfo::handleTypes` or `VkExternalMemoryImageCreateInfo::handleTypes` when the resource was created

Valid Usage (Implicit)

- VUID-VkSparseMemoryBind-memory-parameter
If **memory** is not `VK_NULL_HANDLE`, **memory** **must** be a valid `VkDeviceMemory` handle
- VUID-VkSparseMemoryBind-flags-parameter
flags **must** be a valid combination of `VkSparseMemoryBindFlagBits` values

Bits which **can** be set in `VkSparseMemoryBind::flags`, specifying usage of a sparse memory binding operation, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSparseMemoryBindFlagBits {
    VK_SPARSE_MEMORY_BIND_METADATA_BIT = 0x00000001,
} VkSparseMemoryBindFlagBits;
```

- `VK_SPARSE_MEMORY_BIND_METADATA_BIT` specifies that the memory being bound is only for the metadata aspect.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSparseMemoryBindFlags;
```

`VkSparseMemoryBindFlags` is a bitmask type for setting a mask of zero or more `VkSparseMemoryBindFlagBits`.

Memory is bound to `VkBuffer` objects created with the `VK_BUFFER_CREATE_SPARSE_BINDING_BIT` flag using the following structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseBufferMemoryBindInfo {
    VkBuffer          buffer;
    uint32_t          bindCount;
    const VkSparseMemoryBind* pBinds;
} VkSparseBufferMemoryBindInfo;
```

- **buffer** is the `VkBuffer` object to be bound.
- **bindCount** is the number of `VkSparseMemoryBind` structures in the **pBinds** array.
- **pBinds** is a pointer to an array of `VkSparseMemoryBind` structures.

Valid Usage (Implicit)

- VUID-VkSparseBufferMemoryBindInfo-buffer-parameter
buffer **must** be a valid `VkBuffer` handle
- VUID-VkSparseBufferMemoryBindInfo-pBinds-parameter

pBinds **must** be a valid pointer to an array of **bindCount** valid [VkSparseMemoryBind](#) structures

- VUID-VkSparseBufferMemoryBindInfo-bindCount-arraylength
bindCount **must** be greater than 0

Memory is bound to opaque regions of [VkImage](#) objects created with the [VK_IMAGE_CREATE_SPARSE_BINDING_BIT](#) flag using the following structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageOpaqueMemoryBindInfo {
    VkImage          image;
    uint32_t         bindCount;
    const VkSparseMemoryBind* pBinds;
} VkSparseImageOpaqueMemoryBindInfo;
```

- **image** is the [VkImage](#) object to be bound.
- **bindCount** is the number of [VkSparseMemoryBind](#) structures in the **pBinds** array.
- **pBinds** is a pointer to an array of [VkSparseMemoryBind](#) structures.

NOTE

This structure is normally used to bind memory to fully-resident sparse images or for mip tail regions of partially resident images. However, it **can** also be used to bind memory for the entire binding range of partially resident images.

If the **pBinds[i].flags** of an element *i* of **pBinds** does not contain [VK_SPARSE_MEMORY_BIND_METADATA_BIT](#), the **resourceOffset** is in the range [0, [VkMemoryRequirements::size](#)), This range includes data from all aspects of the image, including metadata. For most implementations this will probably mean that the **resourceOffset** is a simple device address offset within the resource. It is possible for an application to bind a range of memory that includes both resource data and metadata. However, the application would not know what part of the image the memory is used for, or if any range is being used for metadata.

If the **pBinds[i].flags** of an element *i* of **pBinds** contains [VK_SPARSE_MEMORY_BIND_METADATA_BIT](#), the binding range specified **must** be within the mip tail region of the metadata aspect. In this case the **resourceOffset** is not **required** to be a simple device address offset within the resource. However, it is defined to be within [**imageMipTailOffset**, **imageMipTailOffset** + **imageMipTailSize**) for the metadata aspect. See [VkSparseMemoryBind](#) for the full constraints on binding region with this flag present.

Valid Usage

- VUID-VkSparseImageOpaqueMemoryBindInfo-pBinds-01103
If the **flags** member of any element of **pBinds** contains [VK_SPARSE_MEMORY_BIND_METADATA_BIT](#), the binding range defined **must** be within the mip

tail region of the metadata aspect of `image`

Valid Usage (Implicit)

- VUID-VkSparseImageOpaqueMemoryBindInfo-image-parameter
`image` **must** be a valid `VkImage` handle
- VUID-VkSparseImageOpaqueMemoryBindInfo-pBinds-parameter
`pBinds` **must** be a valid pointer to an array of `bindCount` valid `VkSparseMemoryBind` structures
- VUID-VkSparseImageOpaqueMemoryBindInfo-bindCount-arraylength
`bindCount` **must** be greater than 0

Memory **can** be bound to sparse image blocks of `VkImage` objects created with the `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` flag using the following structure:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageMemoryBindInfo {
    VkImage          image;
    uint32_t         bindCount;
    const VkSparseImageMemoryBind* pBinds;
} VkSparseImageMemoryBindInfo;
```

- `image` is the `VkImage` object to be bound
- `bindCount` is the number of `VkSparseImageMemoryBind` structures in `pBinds` array
- `pBinds` is a pointer to an array of `VkSparseImageMemoryBind` structures

Valid Usage

- VUID-VkSparseImageMemoryBindInfo-subresource-01722
The `subresource.mipLevel` member of each element of `pBinds` **must** be less than the `mipLevels` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkSparseImageMemoryBindInfo-subresource-01723
The `subresource.arrayLayer` member of each element of `pBinds` **must** be less than the `arrayLayers` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkSparseImageMemoryBindInfo-subresource-01106
The `subresource.aspectMask` member of each element of `pBinds` **must** be valid for the `format` specified in `VkImageCreateInfo` when `image` was created
- VUID-VkSparseImageMemoryBindInfo-image-02901
`image` **must** have been created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` set

Valid Usage (Implicit)

- VUID-VkSparseImageMemoryBindInfo-image-parameter
image **must** be a valid [VkImage](#) handle
- VUID-VkSparseImageMemoryBindInfo-pBinds-parameter
pBinds **must** be a valid pointer to an array of **bindCount** valid [VkSparseImageMemoryBind](#) structures
- VUID-VkSparseImageMemoryBindInfo-bindCount-arraylength
bindCount **must** be greater than 0

The [VkSparseImageMemoryBind](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkSparseImageMemoryBind {
    VkImageSubresource      subresource;
    VkOffset3D              offset;
    VkExtent3D              extent;
    VkDeviceMemory          memory;
    VkDeviceSize            memoryOffset;
    VkSparseMemoryBindFlags flags;
} VkSparseImageMemoryBind;
```

- **subresource** is the image *aspect* and region of interest in the image.
- **offset** are the coordinates of the first texel within the image subresource to bind.
- **extent** is the size in texels of the region within the image subresource to bind. The extent **must** be a multiple of the sparse image block dimensions, except when binding sparse image blocks along the edge of an image subresource it **can** instead be such that any coordinate of **offset** + **extent** equals the corresponding dimensions of the image subresource.
- **memory** is the [VkDeviceMemory](#) object that the sparse image blocks of the image are bound to. If **memory** is [VK_NULL_HANDLE](#), the sparse image blocks are unbound.
- **memoryOffset** is an offset into [VkDeviceMemory](#) object. If **memory** is [VK_NULL_HANDLE](#), this value is ignored.
- **flags** are sparse memory binding flags.

Valid Usage

- VUID-VkSparseImageMemoryBind-memory-01104
If the [sparseResidencyAliased](#) feature is not enabled, and if any other resources are bound to ranges of **memory**, the range of **memory** being bound **must** not overlap with those bound ranges
- VUID-VkSparseImageMemoryBind-memory-01105
memory and **memoryOffset** **must** match the memory requirements of the calling command's **image**, as described in section [Resource Memory Association](#)

- VUID-VkSparseImageMemoryBind-offset-01107
`offset.x` **must** be a multiple of the sparse image block width (`VkSparseImageFormatProperties::imageGranularity.width`) of the image
- VUID-VkSparseImageMemoryBind-extent-09388
`extent.width` **must** be greater than 0
- VUID-VkSparseImageMemoryBind-extent-01108
`extent.width` **must** either be a multiple of the sparse image block width of the image, or else (`extent.width + offset.x`) **must** equal the width of the image subresource
- VUID-VkSparseImageMemoryBind-offset-01109
`offset.y` **must** be a multiple of the sparse image block height (`VkSparseImageFormatProperties::imageGranularity.height`) of the image
- VUID-VkSparseImageMemoryBind-extent-09389
`extent.height` **must** be greater than 0
- VUID-VkSparseImageMemoryBind-extent-01110
`extent.height` **must** either be a multiple of the sparse image block height of the image, or else (`extent.height + offset.y`) **must** equal the height of the image subresource
- VUID-VkSparseImageMemoryBind-offset-01111
`offset.z` **must** be a multiple of the sparse image block depth (`VkSparseImageFormatProperties::imageGranularity.depth`) of the image
- VUID-VkSparseImageMemoryBind-extent-09390
`extent.depth` **must** be greater than 0
- VUID-VkSparseImageMemoryBind-extent-01112
`extent.depth` **must** either be a multiple of the sparse image block depth of the image, or else (`extent.depth + offset.z`) **must** equal the depth of the image subresource
- VUID-VkSparseImageMemoryBind-memory-02732
If `memory` was created with `VkExportMemoryAllocateInfo::handleTypes` not equal to 0, at least one handle type it contained **must** also have been set in `VkExternalMemoryImageCreateInfo::handleTypes` when the image was created
- VUID-VkSparseImageMemoryBind-memory-02733
If `memory` was created by a memory import operation, the external handle type of the imported memory **must** also have been set in `VkExternalMemoryImageCreateInfo::handleTypes` when `image` was created

Valid Usage (Implicit)

- VUID-VkSparseImageMemoryBind-subresource-parameter
`subresource` **must** be a valid `VkImageSubresource` structure
- VUID-VkSparseImageMemoryBind-memory-parameter
If `memory` is not `VK_NULL_HANDLE`, `memory` **must** be a valid `VkDeviceMemory` handle
- VUID-VkSparseImageMemoryBind-flags-parameter
`flags` **must** be a valid combination of `VkSparseMemoryBindFlagBits` values

To submit sparse binding operations to a queue, call:

```
// Provided by VK_VERSION_1_0
VkResult vkQueueBindSparse(
    VkQueue          queue,
    uint32_t         bindInfoCount,
    const VkBindSparseInfo* pBindInfo,
    VkFence          fence);
```

- **queue** is the queue that the sparse binding operations will be submitted to.
- **bindInfoCount** is the number of elements in the **pBindInfo** array.
- **pBindInfo** is a pointer to an array of **VkBindSparseInfo** structures, each specifying a sparse binding submission batch.
- **fence** is an **optional** handle to a fence to be signaled. If **fence** is not **VK_NULL_HANDLE**, it defines a **fence signal operation**.

vkQueueBindSparse is a **queue submission command**, with each batch defined by an element of **pBindInfo** as a **VkBindSparseInfo** structure. Batches begin execution in the order they appear in **pBindInfo**, but **may** complete out of order.

Within a batch, a given range of a resource **must** not be bound more than once. Across batches, if a range is to be bound to one allocation and offset and then to another allocation and offset, then the application **must** guarantee (usually using semaphores) that the binding operations are executed in the correct order, as well as to order binding operations against the execution of command buffer submissions.

As no operation to **vkQueueBindSparse** causes any pipeline stage to access memory, synchronization primitives used in this command effectively only define execution dependencies.

Additional information about fence and semaphore operation is described in [the synchronization chapter](#).

Valid Usage

- VUID-vkQueueBindSparse-fence-01113
If **fence** is not **VK_NULL_HANDLE**, **fence** **must** be unsignaled
- VUID-vkQueueBindSparse-fence-01114
If **fence** is not **VK_NULL_HANDLE**, **fence** **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-vkQueueBindSparse-pSignalSemaphores-01115
Each element of the **pSignalSemaphores** member of each element of **pBindInfo** **must** be unsignaled when the semaphore signal operation it defines is executed on the device
- VUID-vkQueueBindSparse-pWaitSemaphores-01116
When a semaphore wait operation referring to a binary semaphore defined by any element of the **pWaitSemaphores** member of any element of **pBindInfo** executes on **queue**,

there **must** be no other queues waiting on the same semaphore

- VUID-vkQueueBindSparse-pWaitSemaphores-03245

All elements of the `pWaitSemaphores` member of all elements of `pBindInfo` referring to a semaphore created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_BINARY` **must** reference a semaphore signal operation that has been submitted for execution and any [semaphore signal operations](#) on which it depends **must** have also been submitted for execution

Valid Usage (Implicit)

- VUID-vkQueueBindSparse-queue-parameter
`queue` **must** be a valid `VkQueue` handle
- VUID-vkQueueBindSparse-pBindInfo-parameter
If `bindInfoCount` is not 0, `pBindInfo` **must** be a valid pointer to an array of `bindInfoCount` valid `VkBindSparseInfo` structures
- VUID-vkQueueBindSparse-fence-parameter
If `fence` is not `VK_NULL_HANDLE`, `fence` **must** be a valid `VkFence` handle
- VUID-vkQueueBindSparse-queue-type
The `queue` **must** support sparse binding operations
- VUID-vkQueueBindSparse-commonparent
Both of `fence`, and `queue` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `queue` **must** be externally synchronized
- Host access to `fence` **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	SPARSE_BINDING	-

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`

- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`

The `VkBindSparseInfo` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkBindSparseInfo {
    VkStructureType           sType;
    const void*               pNext;
    uint32_t                  waitSemaphoreCount;
    const VkSemaphore*        pWaitSemaphores;
    uint32_t                  bufferBindCount;
    const VkSparseBufferMemoryBindInfo* pBufferBinds;
    uint32_t                  imageOpaqueBindCount;
    const VkSparseImageOpaqueMemoryBindInfo* pImageOpaqueBinds;
    uint32_t                  imageBindCount;
    const VkSparseImageMemoryBindInfo* pImageBinds;
    uint32_t                  signalSemaphoreCount;
    const VkSemaphore*        pSignalSemaphores;
} VkBindSparseInfo;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `waitSemaphoreCount` is the number of semaphores upon which to wait before executing the sparse binding operations for the batch.
- `pWaitSemaphores` is a pointer to an array of semaphores upon which to wait on before the sparse binding operations for this batch begin execution. If semaphores to wait on are provided, they define a [semaphore wait operation](#).
- `bufferBindCount` is the number of sparse buffer bindings to perform in the batch.
- `pBufferBinds` is a pointer to an array of `VkSparseBufferMemoryBindInfo` structures.
- `imageOpaqueBindCount` is the number of opaque sparse image bindings to perform.
- `pImageOpaqueBinds` is a pointer to an array of `VkSparseImageOpaqueMemoryBindInfo` structures, indicating opaque sparse image bindings to perform.
- `imageBindCount` is the number of sparse image bindings to perform.
- `pImageBinds` is a pointer to an array of `VkSparseImageMemoryBindInfo` structures, indicating sparse image bindings to perform.
- `signalSemaphoreCount` is the number of semaphores to be signaled once the sparse binding operations specified by the structure have completed execution.
- `pSignalSemaphores` is a pointer to an array of semaphores which will be signaled when the sparse binding operations for this batch have completed execution. If semaphores to be signaled are provided, they define a [semaphore signal operation](#).

Valid Usage

- VUID-VkBindSparseInfo-pWaitSemaphores-03246

If any element of `pWaitSemaphores` or `pSignalSemaphores` was created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE` then the `pNext` chain **must** include a `VkTimelineSemaphoreSubmitInfo` structure

- VUID-VkBindSparseInfo-pNext-03247

If the `pNext` chain of this structure includes a `VkTimelineSemaphoreSubmitInfo` structure and any element of `pWaitSemaphores` was created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE` then its `waitSemaphoreValueCount` member **must** equal `waitSemaphoreCount`

- VUID-VkBindSparseInfo-pNext-03248

If the `pNext` chain of this structure includes a `VkTimelineSemaphoreSubmitInfo` structure and any element of `pSignalSemaphores` was created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE` then its `signalSemaphoreValueCount` member **must** equal `signalSemaphoreCount`

- VUID-VkBindSparseInfo-pSignalSemaphores-03249

For each element of `pSignalSemaphores` created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE` the corresponding element of `VkTimelineSemaphoreSubmitInfo::pSignalSemaphoreValues` **must** have a value greater than the current value of the semaphore when the semaphore signal operation is executed

- VUID-VkBindSparseInfo-pWaitSemaphores-03250

For each element of `pWaitSemaphores` created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE` the corresponding element of `VkTimelineSemaphoreSubmitInfo::pWaitSemaphoreValues` **must** have a value which does not differ from the current value of the semaphore or from the value of any outstanding semaphore wait or signal operation on that semaphore by more than `maxTimelineSemaphoreValueDifference`

- VUID-VkBindSparseInfo-pSignalSemaphores-03251

For each element of `pSignalSemaphores` created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE` the corresponding element of `VkTimelineSemaphoreSubmitInfo::pSignalSemaphoreValues` **must** have a value which does not differ from the current value of the semaphore or from the value of any outstanding semaphore wait or signal operation on that semaphore by more than `maxTimelineSemaphoreValueDifference`

Valid Usage (Implicit)

- VUID-VkBindSparseInfo-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_BIND_SPARSE_INFO`

- VUID-VkBindSparseInfo-pNext-pNext

Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkDeviceGroupBindSparseInfo`,

[VkFrameBoundaryEXT](#), or [VkTimelineSemaphoreSubmitInfo](#)

- VUID-VkBindSparseInfo-sType-unique
The [sType](#) value of each struct in the [pNext](#) chain **must** be unique
- VUID-VkBindSparseInfo-pWaitSemaphores-parameter
If [waitSemaphoreCount](#) is not 0, [pWaitSemaphores](#) **must** be a valid pointer to an array of [waitSemaphoreCount](#) valid [VkSemaphore](#) handles
- VUID-VkBindSparseInfo-pBufferBinds-parameter
If [bufferBindCount](#) is not 0, [pBufferBinds](#) **must** be a valid pointer to an array of [bufferBindCount](#) valid [VkSparseBufferMemoryBindInfo](#) structures
- VUID-VkBindSparseInfo-pImageOpaqueBinds-parameter
If [imageOpaqueBindCount](#) is not 0, [pImageOpaqueBinds](#) **must** be a valid pointer to an array of [imageOpaqueBindCount](#) valid [VkSparseImageOpaqueMemoryBindInfo](#) structures
- VUID-VkBindSparseInfo-pImageBinds-parameter
If [imageBindCount](#) is not 0, [pImageBinds](#) **must** be a valid pointer to an array of [imageBindCount](#) valid [VkSparseImageMemoryBindInfo](#) structures
- VUID-VkBindSparseInfo-pSignalSemaphores-parameter
If [signalSemaphoreCount](#) is not 0, [pSignalSemaphores](#) **must** be a valid pointer to an array of [signalSemaphoreCount](#) valid [VkSemaphore](#) handles
- VUID-VkBindSparseInfo-commonparent
Both of the elements of [pSignalSemaphores](#), and the elements of [pWaitSemaphores](#) that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same [VkDevice](#)

To specify the values to use when waiting for and signaling semaphores created with a [VkSemaphoreType](#) of [VK_SEMAPHORE_TYPE_TIMELINE](#), add a [VkTimelineSemaphoreSubmitInfo](#) structure to the [pNext](#) chain of the [VkBindSparseInfo](#) structure.

If the [pNext](#) chain of [VkBindSparseInfo](#) includes a [VkDeviceGroupBindSparseInfo](#) structure, then that structure includes device indices specifying which instance of the resources and memory are bound.

The [VkDeviceGroupBindSparseInfo](#) structure is defined as:

```
typedef struct VkDeviceGroupBindSparseInfo {  
    VkStructureType    sType;  
    const void*        pNext;  
    uint32_t           resourceDeviceIndex;  
    uint32_t           memoryDeviceIndex;  
} VkDeviceGroupBindSparseInfo;
```

or the equivalent

```
// Provided by VK_KHR_device_group
typedef VkDeviceGroupBindSparseInfo VkDeviceGroupBindSparseInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **resourceDeviceIndex** is a device index indicating which instance of the resource is bound.
- **memoryDeviceIndex** is a device index indicating which instance of the memory the resource instance is bound to.

These device indices apply to all buffer and image memory binds included in the batch pointing to this structure. The semaphore waits and signals for the batch are executed only by the physical device specified by the **resourceDeviceIndex**.

If this structure is not present, **resourceDeviceIndex** and **memoryDeviceIndex** are assumed to be zero.

Valid Usage

- VUID-VkDeviceGroupBindSparseInfo-resourceDeviceIndex-01118
resourceDeviceIndex and **memoryDeviceIndex** **must** both be valid device indices
- VUID-VkDeviceGroupBindSparseInfo-memoryDeviceIndex-01119
Each memory allocation bound in this batch **must** have allocated an instance for **memoryDeviceIndex**

Valid Usage (Implicit)

- VUID-VkDeviceGroupBindSparseInfo-sType-sType
sType **must** be `VK_STRUCTURE_TYPE_DEVICE_GROUP_BIND_SPARSE_INFO`

Chapter 31. Window System Integration (WSI)

This chapter discusses the window system integration (WSI) between the Vulkan API and the various forms of displaying the results of rendering to a user. Since the Vulkan API **can** be used without displaying results, WSI is provided through the use of optional Vulkan extensions. This chapter provides an overview of WSI. See the appendix for additional details of each WSI extension, including which extensions **must** be enabled in order to use each of the functions described in this chapter.

31.1. WSI Platform

A platform is an abstraction for a window system, OS, etc. Some examples include MS Windows, Android, and Wayland. The Vulkan API **may** be integrated in a unique manner for each platform.

The Vulkan API does not define any type of platform object. Platform-specific WSI extensions are defined, each containing platform-specific functions for using WSI. Use of these extensions is guarded by preprocessor symbols as defined in the [Window System-Specific Header Control](#) appendix.

In order for an application to be compiled to use WSI with a given platform, it **must** either:

- `#define` the appropriate preprocessor symbol prior to including the `vulkan.h` header file, or
- include `vulkan_core.h` and any native platform headers, followed by the appropriate platform-specific header.

The preprocessor symbols and platform-specific headers are defined in the [Window System Extensions and Headers](#) table.

Each platform-specific extension is an instance extension. The application **must** enable instance extensions with `vkCreateInstance` before using them.

31.2. WSI Surface

Native platform surface or window objects are abstracted by surface objects, which are represented by `VkSurfaceKHR` handles:

```
// Provided by VK_KHR_surface
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSurfaceKHR)
```

The `VK_KHR_surface` extension declares the `VkSurfaceKHR` object, and provides a function for destroying `VkSurfaceKHR` objects. Separate platform-specific extensions each provide a function for creating a `VkSurfaceKHR` object for the respective platform. From the application's perspective this is an opaque handle, just like the handles of other Vulkan objects.

NOTE

On certain platforms, the Vulkan loader and ICDs **may** have conventions that treat

the handle as a pointer to a structure containing the platform-specific information about the surface. This will be described in the documentation for the loader-ICD interface, and in the `vk_icd.h` header file of the LoaderAndTools source-code repository. This does not affect the loader-layer interface; layers **may** wrap `VkSurfaceKHR` objects.

31.2.1. Android Platform

To create a `VkSurfaceKHR` object for an Android native window, call:

```
// Provided by VK_KHR_android_surface
VkResult vkCreateAndroidSurfaceKHR(
    VkInstance                instance,
    const VkAndroidSurfaceCreateInfoKHR* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkSurfaceKHR*             pSurface);
```

- `instance` is the instance to associate the surface with.
- `pCreateInfo` is a pointer to a `VkAndroidSurfaceCreateInfoKHR` structure containing parameters affecting the creation of the surface object.
- `pAllocator` is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see [Memory Allocation](#)).
- `pSurface` is a pointer to a `VkSurfaceKHR` handle in which the created surface object is returned.

During the lifetime of a surface created using a particular `ANativeWindow` handle any attempts to create another surface for the same `ANativeWindow` and any attempts to connect to the same `ANativeWindow` through other platform mechanisms will fail.

NOTE In particular, only one `VkSurfaceKHR` **can** exist at a time for a given window. Similarly, a native window **cannot** be used by both a `VkSurfaceKHR` and `EGLSurface` simultaneously.

If successful, `vkCreateAndroidSurfaceKHR` increments the `ANativeWindow`'s reference count, and `vkDestroySurfaceKHR` will decrement it.

On Android, when a swapchain's `imageExtent` does not match the surface's `currentExtent`, the presentable images will be scaled to the surface's dimensions during presentation. `minImageExtent` is (1,1), and `maxImageExtent` is the maximum image size supported by the consumer. For the system compositor, `currentExtent` is the window size (i.e. the consumer's preferred size).

Valid Usage (Implicit)

- VUID-vkCreateAndroidSurfaceKHR-instance-parameter `instance` **must** be a valid `VkInstance` handle
- VUID-vkCreateAndroidSurfaceKHR-pCreateInfo-parameter `pCreateInfo` **must** be a valid pointer to a valid `VkAndroidSurfaceCreateInfoKHR` structure

- VUID-vkCreateAndroidSurfaceKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateAndroidSurfaceKHR-pSurface-parameter
`pSurface` **must** be a valid pointer to a `VkSurfaceKHR` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_NATIVE_WINDOW_IN_USE_KHR`

The `VkAndroidSurfaceCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_android_surface
typedef struct VkAndroidSurfaceCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkAndroidSurfaceCreateFlagsKHR flags;
    struct ANativeWindow* window;
} VkAndroidSurfaceCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `window` is a pointer to the `ANativeWindow` to associate the surface with.

Valid Usage

- VUID-VkAndroidSurfaceCreateInfoKHR-window-01248
`window` **must** point to a valid Android `ANativeWindow`

Valid Usage (Implicit)

- VUID-VkAndroidSurfaceCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_ANDROID_SURFACE_CREATE_INFO_KHR`
- VUID-VkAndroidSurfaceCreateInfoKHR-pNext-pNext
`pNext` **must** be `NULL`

- VUID-VkAndroidSurfaceCreateInfoKHR-flags-zero-bitmask
flags must be 0

To remove an unnecessary compile time dependency, an incomplete type definition of [ANativeWindow](#) is provided in the Vulkan headers:

```
// Provided by VK_KHR_android_surface
struct ANativeWindow;
```

The actual [ANativeWindow](#) type is defined in Android NDK headers.

```
// Provided by VK_KHR_android_surface
typedef VkFlags VkAndroidSurfaceCreateFlagsKHR;
```

[VkAndroidSurfaceCreateFlagsKHR](#) is a bitmask type for setting a mask, but is currently reserved for future use.

31.2.2. Wayland Platform

To create a [VkSurfaceKHR](#) object for a Wayland surface, call:

```
// Provided by VK_KHR_wayland_surface
VkResult vkCreateWaylandSurfaceKHR(
    VkInstance          instance,
    const VkWaylandSurfaceCreateInfoKHR* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkSurfaceKHR* pSurface);
```

- **instance** is the instance to associate the surface with.
- **pCreateInfo** is a pointer to a [VkWaylandSurfaceCreateInfoKHR](#) structure containing parameters affecting the creation of the surface object.
- **pAllocator** is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see [Memory Allocation](#)).
- **pSurface** is a pointer to a [VkSurfaceKHR](#) handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateWaylandSurfaceKHR-instance-parameter
instance must be a valid [VkInstance](#) handle
- VUID-vkCreateWaylandSurfaceKHR-pCreateInfo-parameter
pCreateInfo must be a valid pointer to a valid [VkWaylandSurfaceCreateInfoKHR](#) structure
- VUID-vkCreateWaylandSurfaceKHR-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator must** be a valid pointer to a valid

[VkAllocationCallbacks](#) structure

- VUID-vkCreateWaylandSurfaceKHR-pSurface-parameter
[pSurface](#) **must** be a valid pointer to a [VkSurfaceKHR](#) handle

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

The [VkWaylandSurfaceCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_wayland_surface
typedef struct VkWaylandSurfaceCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkWaylandSurfaceCreateFlagsKHR flags;
    struct wl_display*  display;
    struct wl_surface*  surface;
} VkWaylandSurfaceCreateInfoKHR;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [flags](#) is reserved for future use.
- [display](#) and [surface](#) are pointers to the Wayland [wl_display](#) and [wl_surface](#) to associate the surface with.

Valid Usage

- VUID-VkWaylandSurfaceCreateInfoKHR-display-01304
[display](#) **must** point to a valid Wayland [wl_display](#)
- VUID-VkWaylandSurfaceCreateInfoKHR-surface-01305
[surface](#) **must** point to a valid Wayland [wl_surface](#)

Valid Usage (Implicit)

- VUID-VkWaylandSurfaceCreateInfoKHR-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_WAYLAND_SURFACE_CREATE_INFO_KHR](#)
- VUID-VkWaylandSurfaceCreateInfoKHR-pNext-pNext

`pNext` **must** be `NULL`

- `VOID-VkWaylandSurfaceCreateInfoKHR-flags-zero-bitmask`
`flags` **must** be `0`

On Wayland, `currentExtent` is the special value (`0xFFFFFFFF`, `0xFFFFFFFF`), indicating that the surface size will be determined by the extent of a swapchain targeting the surface. Whatever the application sets a swapchain's `imageExtent` to will be the size of the window, after the first image is presented. `minImageExtent` is (1,1), and `maxImageExtent` is the maximum supported surface size. Any calls to `vkGetPhysicalDeviceSurfacePresentModesKHR` on a surface created with `vkCreateWaylandSurfaceKHR` are **required** to return `VK_PRESENT_MODE_MAILBOX_KHR` as one of the valid present modes.

Some Vulkan functions **may** send protocol over the specified `wl_display` connection when using a swapchain or presentable images created from a `VkSurfaceKHR` referring to a `wl_surface`. Applications **must** therefore ensure that both the `wl_display` and the `wl_surface` remain valid for the lifetime of any `VkSwapchainKHR` objects created from a particular `wl_display` and `wl_surface`. Also, calling `vkQueuePresentKHR` will result in Vulkan sending `wl_surface.commit` requests to the underlying `wl_surface` of each. The `wl_surface.attach`, `wl_surface.damage`, and `wl_surface.commit` requests **must** be issued by the implementation during the call to `vkQueuePresentKHR` and **must** not be issued by the implementation outside of `vkQueuePresentKHR`. This ensures that any Wayland requests sent by the client after the call to `vkQueuePresentKHR` returns will be received by the compositor after the `wl_surface.commit`. Regardless of the mode of swapchain creation, a new `wl_event_queue` **must** be created for each successful `vkCreateWaylandSurfaceKHR` call, and every Wayland object created by the implementation **must** be assigned to this event queue. If the platform provides Wayland 1.11 or greater, this **must** be implemented by the use of Wayland proxy object wrappers, to avoid race conditions.

If the application wishes to synchronize any window changes with a particular frame, such requests **must** be sent to the Wayland display server prior to calling `vkQueuePresentKHR`.

```
// Provided by VK_KHR_wayland_surface
typedef VkFlags VkWaylandSurfaceCreateFlagsKHR;
```

`VkWaylandSurfaceCreateFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

31.2.3. Win32 Platform

To create a `VkSurfaceKHR` object for a Win32 window, call:

```
// Provided by VK_KHR_win32_surface
VkResult vkCreateWin32SurfaceKHR(
    VkInstance                instance,
    const VkWin32SurfaceCreateInfoKHR* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkSurfaceKHR* pSurface);
```

- **instance** is the instance to associate the surface with.
- **pCreateInfo** is a pointer to a **VkWin32SurfaceCreateInfoKHR** structure containing parameters affecting the creation of the surface object.
- **pAllocator** is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see [Memory Allocation](#)).
- **pSurface** is a pointer to a **VkSurfaceKHR** handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateWin32SurfaceKHR-instance-parameter **instance** **must** be a valid **VkInstance** handle
- VUID-vkCreateWin32SurfaceKHR-pCreateInfo-parameter **pCreateInfo** **must** be a valid pointer to a valid **VkWin32SurfaceCreateInfoKHR** structure
- VUID-vkCreateWin32SurfaceKHR-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid **VkAllocationCallbacks** structure
- VUID-vkCreateWin32SurfaceKHR-pSurface-parameter **pSurface** **must** be a valid pointer to a **VkSurfaceKHR** handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

Some Vulkan functions **may** call the **SendMessage** system API when interacting with a **VkSurfaceKHR** through a **VkSwapchainKHR**. In a multithreaded environment, calling **SendMessage** from a thread that is not the thread associated with **pCreateInfo->hwnd** will block until the application has processed the window message. Thus, applications **should** either call these Vulkan functions on the message pump thread, or make sure their message pump is actively running. Failing to do so **may** result in deadlocks.

The functions subject to this requirement are:

- **vkCreateSwapchainKHR**
- **vkDestroySwapchainKHR**
- **vkAcquireNextImageKHR** and **vkAcquireNextImage2KHR**
- **vkQueuePresentKHR**
- **vkAcquireFullScreenExclusiveModeEXT**

- [vkReleaseFullScreenExclusiveModeEXT](#)
- [vkSetHdrMetadataEXT](#)

The [VkWin32SurfaceCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_win32_surface
typedef struct VkWin32SurfaceCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkWin32SurfaceCreateFlagsKHR flags;
    HINSTANCE           hinstance;
    HWND               hwnd;
} VkWin32SurfaceCreateInfoKHR;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [flags](#) is reserved for future use.
- [hinstance](#) is the Win32 [HINSTANCE](#) for the window to associate the surface with.
- [hwnd](#) is the Win32 [HWND](#) for the window to associate the surface with.

Valid Usage

- VUID-VkWin32SurfaceCreateInfoKHR-hinstance-01307
[hinstance](#) **must** be a valid Win32 [HINSTANCE](#)
- VUID-VkWin32SurfaceCreateInfoKHR-hwnd-01308
[hwnd](#) **must** be a valid Win32 [HWND](#)

Valid Usage (Implicit)

- VUID-VkWin32SurfaceCreateInfoKHR-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_WIN32_SURFACE_CREATE_INFO_KHR](#)
- VUID-VkWin32SurfaceCreateInfoKHR-pNext-pNext
[pNext](#) **must** be [NULL](#)
- VUID-VkWin32SurfaceCreateInfoKHR-flags-zerobitmask
[flags](#) **must** be 0

With Win32, [minImageExtent](#), [maxImageExtent](#), and [currentExtent](#) **must** always equal the window size.

The [currentExtent](#) of a Win32 surface **must** have both [width](#) and [height](#) greater than 0, or both of them 0.

NOTE

Due to above restrictions, it is only possible to create a new swapchain on this platform with [imageExtent](#) being equal to the current size of the window, as

reported in [VkSurfaceCapabilitiesKHR::currentExtent](#).

The window size **may** become (0, 0) on this platform (e.g. when the window is minimized), and so a swapchain **cannot** be created until the size changes.

```
// Provided by VK_KHR_win32_surface
typedef VkFlags VkWin32SurfaceCreateFlagsKHR;
```

[VkWin32SurfaceCreateFlagsKHR](#) is a bitmask type for setting a mask, but is currently reserved for future use.

31.2.4. XCB Platform

To create a [VkSurfaceKHR](#) object for an X11 window, using the XCB client-side library, call:

```
// Provided by VK_KHR_xcb_surface
VkResult vkCreateXcbSurfaceKHR(
    VkInstance          instance,
    const VkXcbSurfaceCreateInfoKHR* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkSurfaceKHR*       pSurface);
```

- [instance](#) is the instance to associate the surface with.
- [pCreateInfo](#) is a pointer to a [VkXcbSurfaceCreateInfoKHR](#) structure containing parameters affecting the creation of the surface object.
- [pAllocator](#) is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see [Memory Allocation](#)).
- [pSurface](#) is a pointer to a [VkSurfaceKHR](#) handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateXcbSurfaceKHR-instance-parameter
[instance](#) **must** be a valid [VkInstance](#) handle
- VUID-vkCreateXcbSurfaceKHR-pCreateInfo-parameter
[pCreateInfo](#) **must** be a valid pointer to a valid [VkXcbSurfaceCreateInfoKHR](#) structure
- VUID-vkCreateXcbSurfaceKHR-pAllocator-parameter
If [pAllocator](#) is not `NULL`, [pAllocator](#) **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateXcbSurfaceKHR-pSurface-parameter
[pSurface](#) **must** be a valid pointer to a [VkSurfaceKHR](#) handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkXcbSurfaceCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_xcb_surface
typedef struct VkXcbSurfaceCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkXcbSurfaceCreateFlagsKHR flags;
    xcb_connection_t*   connection;
    xcb_window_t        window;
} VkXcbSurfaceCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `connection` is a pointer to an `xcb_connection_t` to the X server.
- `window` is the `xcb_window_t` for the X11 window to associate the surface with.

Valid Usage

- VUID-VkXcbSurfaceCreateInfoKHR-connection-01310
`connection` **must** point to a valid X11 `xcb_connection_t`
- VUID-VkXcbSurfaceCreateInfoKHR-window-01311
`window` **must** be a valid X11 `xcb_window_t`

Valid Usage (Implicit)

- VUID-VkXcbSurfaceCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_XCB_SURFACE_CREATE_INFO_KHR`
- VUID-VkXcbSurfaceCreateInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkXcbSurfaceCreateInfoKHR-flags-zero-bitmask
`flags` **must** be `0`

With Xcb, `minImageExtent`, `maxImageExtent`, and `currentExtent` **must** always equal the window size.

The `currentExtent` of an Xcb surface **must** have both `width` and `height` greater than 0, or both of them 0.

NOTE

Due to above restrictions, it is only possible to create a new swapchain on this platform with `imageExtent` being equal to the current size of the window, as reported in `VkSurfaceCapabilitiesKHR::currentExtent`.

The window size **may** become (0, 0) on this platform (e.g. when the window is minimized), and so a swapchain **cannot** be created until the size changes.

Some Vulkan functions **may** send protocol over the specified xcb connection when using a swapchain or presentable images created from a `VkSurfaceKHR` referring to an xcb window. Applications **must** therefore ensure the xcb connection is available to Vulkan for the duration of any functions that manipulate such swapchains or their presentable images, and any functions that build or queue command buffers that operate on such presentable images. Specifically, applications using Vulkan with xcb-based swapchains **must**

- Avoid holding a server grab on an xcb connection while waiting for Vulkan operations to complete using a swapchain derived from a different xcb connection referring to the same X server instance. Failing to do so **may** result in deadlock.

```
// Provided by VK_KHR_xcb_surface
typedef VkFlags VkXcbSurfaceCreateFlagsKHR;
```

`VkXcbSurfaceCreateFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

31.2.5. Xlib Platform

To create a `VkSurfaceKHR` object for an X11 window, using the Xlib client-side library, call:

```
// Provided by VK_KHR_xlib_surface
VkResult vkCreateXlibSurfaceKHR(
    VkInstance                instance,
    const VkXlibSurfaceCreateInfoKHR* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkSurfaceKHR*             pSurface);
```

- `instance` is the instance to associate the surface with.
- `pCreateInfo` is a pointer to a `VkXlibSurfaceCreateInfoKHR` structure containing the parameters affecting the creation of the surface object.
- `pAllocator` is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see [Memory Allocation](#)).
- `pSurface` is a pointer to a `VkSurfaceKHR` handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateXlibSurfaceKHR-instance-parameter
`instance` **must** be a valid `VkInstance` handle
- VUID-vkCreateXlibSurfaceKHR-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkXlibSurfaceCreateInfoKHR` structure
- VUID-vkCreateXlibSurfaceKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateXlibSurfaceKHR-pSurface-parameter
`pSurface` **must** be a valid pointer to a `VkSurfaceKHR` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkXlibSurfaceCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_xlib_surface
typedef struct VkXlibSurfaceCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkXlibSurfaceCreateFlagsKHR flags;
    Display*            dpy;
    Window              window;
} VkXlibSurfaceCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `dpy` is a pointer to an Xlib `Display` connection to the X server.
- `window` is an Xlib `Window` to associate the surface with.

Valid Usage

- VUID-VkXlibSurfaceCreateInfoKHR-dpy-01313
`dpy` **must** point to a valid Xlib `Display`

- VUID-VkXlibSurfaceCreateInfoKHR-window-01314
window must be a valid Xlib **Window**

Valid Usage (Implicit)

- VUID-VkXlibSurfaceCreateInfoKHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_XLIB_SURFACE_CREATE_INFO_KHR**
- VUID-VkXlibSurfaceCreateInfoKHR-pNext-pNext
pNext must be **NULL**
- VUID-VkXlibSurfaceCreateInfoKHR-flags-zeroBitmask
flags must be **0**

With Xlib, **minImageExtent**, **maxImageExtent**, and **currentExtent must** always equal the window size.

The **currentExtent** of an Xlib surface **must** have both **width** and **height** greater than 0, or both of them 0.

NOTE

Due to above restrictions, it is only possible to create a new swapchain on this platform with **imageExtent** being equal to the current size of the window, as reported in **VkSurfaceCapabilitiesKHR::currentExtent**.

The window size **may** become (0, 0) on this platform (e.g. when the window is minimized), and so a swapchain **cannot** be created until the size changes.

Some Vulkan functions **may** send protocol over the specified Xlib **Display** connection when using a swapchain or presentable images created from a **VkSurfaceKHR** referring to an Xlib window. Applications **must** therefore ensure the display connection is available to Vulkan for the duration of any functions that manipulate such swapchains or their presentable images, and any functions that build or queue command buffers that operate on such presentable images. Specifically, applications using Vulkan with Xlib-based swapchains **must**

- Avoid holding a server grab on a display connection while waiting for Vulkan operations to complete using a swapchain derived from a different display connection referring to the same X server instance. Failing to do so **may** result in deadlock.

Some implementations may require threads to implement some presentation modes so applications **must** call **XInitThreads()** before calling any other Xlib functions.

```
// Provided by VK_KHR_xlib_surface
typedef VkFlags VkXlibSurfaceCreateFlagsKHR;
```

VkXlibSurfaceCreateFlagsKHR is a bitmask type for setting a mask, but is currently reserved for future use.

31.2.6. Platform-Independent Information

Once created, `VkSurfaceKHR` objects **can** be used in this and other extensions, in particular the `VK_KHR_swapchain` extension.

Several WSI functions return `VK_ERROR_SURFACE_LOST_KHR` if the surface becomes no longer available. After such an error, the surface (and any child swapchain, if one exists) **should** be destroyed, as there is no way to restore them to a not-lost state. Applications **may** attempt to create a new `VkSurfaceKHR` using the same native platform window object, but whether such re-creation will succeed is platform-dependent and **may** depend on the reason the surface became unavailable. A lost surface does not otherwise cause devices to be [lost](#).

To destroy a `VkSurfaceKHR` object, call:

```
// Provided by VK_KHR_surface
void vkDestroySurfaceKHR(
    VkInstance          instance,
    VkSurfaceKHR        surface,
    const VkAllocationCallbacks* pAllocator);
```

- `instance` is the instance used to create the surface.
- `surface` is the surface to destroy.
- `pAllocator` is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see [Memory Allocation](#)).

Destroying a `VkSurfaceKHR` merely severs the connection between Vulkan and the native surface, and does not imply destroying the native surface, closing a window, or similar behavior.

Valid Usage

- VUID-vkDestroySurfaceKHR-surface-01266
All `VkSwapchainKHR` objects created for `surface` **must** have been destroyed prior to destroying `surface`
- VUID-vkDestroySurfaceKHR-surface-01267
If `VkAllocationCallbacks` were provided when `surface` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroySurfaceKHR-surface-01268
If no `VkAllocationCallbacks` were provided when `surface` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroySurfaceKHR-instance-parameter
`instance` **must** be a valid `VkInstance` handle
- VUID-vkDestroySurfaceKHR-surface-parameter

If `surface` is not `VK_NULL_HANDLE`, `surface` **must** be a valid `VkSurfaceKHR` handle

- `VOID-vkDestroySurfaceKHR-pAllocator-parameter`

If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

- `VOID-vkDestroySurfaceKHR-surface-parent`

If `surface` is a valid handle, it **must** have been created, allocated, or retrieved from `instance`

Host Synchronization

- Host access to `surface` **must** be externally synchronized

31.3. Presenting Directly to Display Devices

In some environments applications **can** also present Vulkan rendering directly to display devices without using an intermediate windowing system. This **can** be useful for embedded applications, or implementing the rendering/presentation backend of a windowing system using Vulkan. The `VK_KHR_display` extension provides the functionality necessary to enumerate display devices and create `VkSurfaceKHR` objects that target displays.

31.3.1. Display Enumeration

Displays are represented by `VkDisplayKHR` handles:

```
// Provided by VK_KHR_display
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDisplayKHR)
```

Various functions are provided for enumerating the available display devices present on a Vulkan physical device. To query information about the available displays, call:

```
// Provided by VK_KHR_display
VkResult vkGetPhysicalDeviceDisplayPropertiesKHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t*                 pPropertyCount,
    VkDisplayPropertiesKHR*    pProperties);
```

- `physicalDevice` is a physical device.
- `pPropertyCount` is a pointer to an integer related to the number of display devices available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkDisplayPropertiesKHR` structures.

If `pProperties` is `NULL`, then the number of display devices available for `physicalDevice` is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the

number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If the value of `pPropertyCount` is less than the number of display devices for `physicalDevice`, at most `pPropertyCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available properties were returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceDisplayPropertiesKHR-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceDisplayPropertiesKHR-pPropertyCount-parameter `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceDisplayPropertiesKHR-pProperties-parameter
If the value referenced by `pPropertyCount` is not 0, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkDisplayPropertiesKHR` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkDisplayPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayPropertiesKHR {
    VkDisplayKHR          display;
    const char*           displayName;
    VkExtent2D            physicalDimensions;
    VkExtent2D            physicalResolution;
    VkSurfaceTransformFlagsKHR supportedTransforms;
    VkBool32              planeReorderPossible;
    VkBool32              persistentContent;
} VkDisplayPropertiesKHR;
```

- `display` is a handle that is used to refer to the display described here. This handle will be valid for the lifetime of the Vulkan instance.
- `displayName` is `NULL` or a pointer to a null-terminated UTF-8 string containing the name of the display. Generally, this will be the name provided by the display's EDID. If `NULL`, no suitable name is available. If not `NULL`, the string pointed to **must** remain accessible and unmodified as

long as **display** is valid.

- **physicalDimensions** describes the physical width and height of the visible portion of the display, in millimeters.
- **physicalResolution** describes the physical, native, or preferred resolution of the display.

NOTE

For devices which have no natural value to return here, implementations **should** return the maximum resolution supported.

- **supportedTransforms** is a bitmask of **VkSurfaceTransformFlagBitsKHR** describing which transforms are supported by this display.
- **planeReorderPossible** tells whether the planes on this display **can** have their z order changed. If this is **VK_TRUE**, the application **can** re-arrange the planes on this display in any order relative to each other.
- **persistentContent** tells whether the display supports self-refresh/internal buffering. If this is true, the application **can** submit persistent present operations on swapchains created against this display.

NOTE

Persistent presents **may** have higher latency, and **may** use less power when the screen content is updated infrequently, or when only a portion of the screen needs to be updated in most frames.

To query information about the available displays, call:

```
// Provided by VK_KHR_get_display_properties2
VkResult vkGetPhysicalDeviceDisplayProperties2KHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t*                 pPropertyCount,
    VkDisplayProperties2KHR*   pProperties);
```

- **physicalDevice** is a physical device.
- **pPropertyCount** is a pointer to an integer related to the number of display devices available or queried, as described below.
- **pProperties** is either **NULL** or a pointer to an array of **VkDisplayProperties2KHR** structures.

vkGetPhysicalDeviceDisplayProperties2KHR behaves similarly to **vkGetPhysicalDeviceDisplayPropertiesKHR**, with the ability to return extended information via chained output structures.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceDisplayProperties2KHR-physicalDevice-parameter **physicalDevice** **must** be a valid **VkPhysicalDevice** handle
- VUID-vkGetPhysicalDeviceDisplayProperties2KHR-pPropertyCount-parameter **pPropertyCount** **must** be a valid pointer to a **uint32_t** value

- VUID-vkGetPhysicalDeviceDisplayProperties2KHR-pProperties-parameter
If the value referenced by `pPropertyCount` is not 0, and `pProperties` is not NULL, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkDisplayProperties2KHR` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkDisplayProperties2KHR` structure is defined as:

```
// Provided by VK_KHR_get_display_properties2
typedef struct VkDisplayProperties2KHR {
    VkStructureType    sType;
    void*              pNext;
    VkDisplayPropertiesKHR displayProperties;
} VkDisplayProperties2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is NULL or a pointer to a structure extending this structure.
- `displayProperties` is a `VkDisplayPropertiesKHR` structure.

Valid Usage (Implicit)

- VUID-VkDisplayProperties2KHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DISPLAY_PROPERTIES_2_KHR`
- VUID-VkDisplayProperties2KHR-pNext-pNext
`pNext` **must** be NULL

Acquiring and Releasing Displays

On some platforms, access to displays is limited to a single process or native driver instance. On such platforms, some or all of the displays may not be available to Vulkan if they are already in use by a native windowing system or other application.

To release a previously acquired display, call:

```
// Provided by VK_EXT_direct_mode_display
VkResult vkReleaseDisplayEXT(
    VkPhysicalDevice          physicalDevice,
    VkDisplayKHR              display);
```

- **physicalDevice** The physical device the display is on.
- **display** The display to release control of.

Valid Usage (Implicit)

- VUID-vkReleaseDisplayEXT-physicalDevice-parameter **physicalDevice** **must** be a valid **VkPhysicalDevice** handle
- VUID-vkReleaseDisplayEXT-display-parameter **display** **must** be a valid **VkDisplayKHR** handle
- VUID-vkReleaseDisplayEXT-display-parent **display** **must** have been created, allocated, or retrieved from **physicalDevice**

Return Codes

Success

- **VK_SUCCESS**

Failure

None

Display Planes

Images are presented to individual planes on a display. Devices **must** support at least one plane on each display. Planes **can** be stacked and blended to composite multiple images on one display. Devices **may** support only a fixed stacking order and fixed mapping between planes and displays, or they **may** allow arbitrary application-specified stacking orders and mappings between planes and displays. To query the properties of device display planes, call:

```
// Provided by VK_KHR_display
VkResult vkGetPhysicalDeviceDisplayPlanePropertiesKHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t*                 pPropertyCount,
    VkDisplayPlanePropertiesKHR* pProperties);
```

- **physicalDevice** is a physical device.
- **pPropertyCount** is a pointer to an integer related to the number of display planes available or queried, as described below.
- **pProperties** is either **NULL** or a pointer to an array of **VkDisplayPlanePropertiesKHR** structures.

If `pProperties` is `NULL`, then the number of display planes available for `physicalDevice` is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If the value of `pPropertyCount` is less than the number of display planes for `physicalDevice`, at most `pPropertyCount` structures will be written.

Valid Usage (Implicit)

- `VUID-vkGetPhysicalDeviceDisplayPlanePropertiesKHR-physicalDevice-parameter` `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- `VUID-vkGetPhysicalDeviceDisplayPlanePropertiesKHR-pPropertyCount-parameter` `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- `VUID-vkGetPhysicalDeviceDisplayPlanePropertiesKHR-pProperties-parameter`
If the value referenced by `pPropertyCount` is not 0, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkDisplayPlanePropertiesKHR` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkDisplayPlanePropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayPlanePropertiesKHR {
    VkDisplayKHR    currentDisplay;
    uint32_t        currentStackIndex;
} VkDisplayPlanePropertiesKHR;
```

- `currentDisplay` is the handle of the display the plane is currently associated with. If the plane is not currently attached to any displays, this will be `VK_NULL_HANDLE`.
- `currentStackIndex` is the current z-order of the plane. This will be between 0 and the value returned by `vkGetPhysicalDeviceDisplayPlanePropertiesKHR` in `pPropertyCount`.

To query the properties of a device's display planes, call:

```
// Provided by VK_KHR_get_display_properties2
VkResult vkGetPhysicalDeviceDisplayPlaneProperties2KHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t*                 pPropertyCount,
    VkDisplayPlaneProperties2KHR* pProperties);
```

- `physicalDevice` is a physical device.
- `pPropertyCount` is a pointer to an integer related to the number of display planes available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkDisplayPlaneProperties2KHR` structures.

`vkGetPhysicalDeviceDisplayPlaneProperties2KHR` behaves similarly to `vkGetPhysicalDeviceDisplayPlanePropertiesKHR`, with the ability to return extended information via chained output structures.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceDisplayPlaneProperties2KHR-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceDisplayPlaneProperties2KHR-pPropertyCount-parameter `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceDisplayPlaneProperties2KHR-pProperties-parameter
If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkDisplayPlaneProperties2KHR` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkDisplayPlaneProperties2KHR` structure is defined as:


```
// Provided by VK_KHR_get_display_properties2
typedef struct VkDisplayPlaneProperties2KHR {
    VkStructureType    sType;
    void*              pNext;
    VkDisplayPlanePropertiesKHR displayPlaneProperties;
} VkDisplayPlaneProperties2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `displayPlaneProperties` is a `VkDisplayPlanePropertiesKHR` structure.

Valid Usage (Implicit)

- VUID-VkDisplayPlaneProperties2KHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DISPLAY_PLANE_PROPERTIES_2_KHR`
- VUID-VkDisplayPlaneProperties2KHR-pNext-pNext
`pNext` **must** be `NULL`

To determine which displays a plane is usable with, call

```
// Provided by VK_KHR_display
VkResult vkGetDisplayPlaneSupportedDisplaysKHR(
    VkPhysicalDevice    physicalDevice,
    uint32_t            planeIndex,
    uint32_t*           pDisplayCount,
    VkDisplayKHR*       pDisplays);
```

- `physicalDevice` is a physical device.
- `planeIndex` is the plane which the application wishes to use, and **must** be in the range `[0, physical device plane count - 1]`.
- `pDisplayCount` is a pointer to an integer related to the number of displays available or queried, as described below.
- `pDisplays` is either `NULL` or a pointer to an array of `VkDisplayKHR` handles.

If `pDisplays` is `NULL`, then the number of displays usable with the specified `planeIndex` for `physicalDevice` is returned in `pDisplayCount`. Otherwise, `pDisplayCount` **must** point to a variable set by the application to the number of elements in the `pDisplays` array, and on return the variable is overwritten with the number of handles actually written to `pDisplays`. If the value of `pDisplayCount` is less than the number of usable display-plane pairs for `physicalDevice`, at most `pDisplayCount` handles will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available pairs were returned.

Valid Usage

- VUID-vkGetDisplayPlaneSupportedDisplaysKHR-planeIndex-01249
`planeIndex` **must** be less than the number of display planes supported by the device as determined by calling `vkGetPhysicalDeviceDisplayPlanePropertiesKHR`

Valid Usage (Implicit)

- VUID-vkGetDisplayPlaneSupportedDisplaysKHR-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetDisplayPlaneSupportedDisplaysKHR-pDisplayCount-parameter
`pDisplayCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetDisplayPlaneSupportedDisplaysKHR-pDisplays-parameter
If the value referenced by `pDisplayCount` is not 0, and `pDisplays` is not `NULL`, `pDisplays` **must** be a valid pointer to an array of `pDisplayCount` `VkDisplayKHR` handles

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

Additional properties of displays are queried using specialized query functions.

Display Modes

Display modes are represented by `VkDisplayModeKHR` handles:

```
// Provided by VK_KHR_display
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDisplayModeKHR)
```

Each display has one or more supported modes associated with it by default. These built-in modes are queried by calling:

```
// Provided by VK_KHR_display
VkResult vkGetDisplayModePropertiesKHR(
    VkPhysicalDevice          physicalDevice,
    VkDisplayKHR              display,
    uint32_t*                 pPropertyCount,
    VkDisplayModePropertiesKHR pProperties);
```

- **physicalDevice** is the physical device associated with **display**.
- **display** is the display to query.
- **pPropertyCount** is a pointer to an integer related to the number of display modes available or queried, as described below.
- **pProperties** is either **NULL** or a pointer to an array of **VkDisplayModePropertiesKHR** structures.

If **pProperties** is **NULL**, then the number of display modes available on the specified **display** for **physicalDevice** is returned in **pPropertyCount**. Otherwise, **pPropertyCount** **must** point to a variable set by the application to the number of elements in the **pProperties** array, and on return the variable is overwritten with the number of structures actually written to **pProperties**. If the value of **pPropertyCount** is less than the number of display modes for **physicalDevice**, at most **pPropertyCount** structures will be written, and **VK_INCOMPLETE** will be returned instead of **VK_SUCCESS**, to indicate that not all the available display modes were returned.

Valid Usage (Implicit)

- VUID-vkGetDisplayModePropertiesKHR-physicalDevice-parameter
physicalDevice **must** be a valid **VkPhysicalDevice** handle
- VUID-vkGetDisplayModePropertiesKHR-display-parameter
display **must** be a valid **VkDisplayKHR** handle
- VUID-vkGetDisplayModePropertiesKHR-pPropertyCount-parameter
pPropertyCount **must** be a valid pointer to a **uint32_t** value
- VUID-vkGetDisplayModePropertiesKHR-pProperties-parameter
If the value referenced by **pPropertyCount** is not 0, and **pProperties** is not **NULL**, **pProperties** **must** be a valid pointer to an array of **pPropertyCount** **VkDisplayModePropertiesKHR** structures
- VUID-vkGetDisplayModePropertiesKHR-display-parent
display **must** have been created, allocated, or retrieved from **physicalDevice**

Return Codes

Success

- **VK_SUCCESS**
- **VK_INCOMPLETE**

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkDisplayModePropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayModePropertiesKHR {
    VkDisplayModeKHR          displayMode;
    VkDisplayModeParametersKHR parameters;
} VkDisplayModePropertiesKHR;
```

- `displayMode` is a handle to the display mode described in this structure. This handle will be valid for the lifetime of the Vulkan instance.
- `parameters` is a `VkDisplayModeParametersKHR` structure describing the display parameters associated with `displayMode`.

```
// Provided by VK_KHR_display
typedef VkFlags VkDisplayModeCreateFlagsKHR;
```

`VkDisplayModeCreateFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

To query the properties of a device's built-in display modes, call:

```
// Provided by VK_KHR_get_display_properties2
VkResult vkGetDisplayModeProperties2KHR(
    VkPhysicalDevice          physicalDevice,
    VkDisplayKHR              display,
    uint32_t*                 pPropertyCount,
    VkDisplayModeProperties2KHR* pProperties);
```

- `physicalDevice` is the physical device associated with `display`.
- `display` is the display to query.
- `pPropertyCount` is a pointer to an integer related to the number of display modes available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkDisplayModeProperties2KHR` structures.

`vkGetDisplayModeProperties2KHR` behaves similarly to `vkGetDisplayModePropertiesKHR`, with the ability to return extended information via chained output structures.

Valid Usage (Implicit)

- VUID-vkGetDisplayModeProperties2KHR-physicalDevice-parameter **physicalDevice** **must** be a valid [VkPhysicalDevice](#) handle
- VUID-vkGetDisplayModeProperties2KHR-display-parameter **display** **must** be a valid [VkDisplayKHR](#) handle
- VUID-vkGetDisplayModeProperties2KHR-pPropertyCount-parameter **pPropertyCount** **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetDisplayModeProperties2KHR-pProperties-parameter
If the value referenced by **pPropertyCount** is not 0, and **pProperties** is not NULL, **pProperties** **must** be a valid pointer to an array of **pPropertyCount** [VkDisplayModeProperties2KHR](#) structures
- VUID-vkGetDisplayModeProperties2KHR-display-parent **display** **must** have been created, allocated, or retrieved from **physicalDevice**

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkDisplayModeProperties2KHR` structure is defined as:

```
// Provided by VK_KHR_get_display_properties2
typedef struct VkDisplayModeProperties2KHR {
    VkStructureType    sType;
    void*              pNext;
    VkDisplayModePropertiesKHR displayModeProperties;
} VkDisplayModeProperties2KHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is NULL or a pointer to a structure extending this structure.
- **displayModeProperties** is a [VkDisplayModePropertiesKHR](#) structure.

Valid Usage (Implicit)

- VUID-VkDisplayModeProperties2KHR-sType-sType
sType **must** be `VK_STRUCTURE_TYPE_DISPLAY_MODE_PROPERTIES_2_KHR`

- VUID-VkDisplayModeProperties2KHR-pNext-pNext
pNext must be **NULL**

The **VkDisplayModeParametersKHR** structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayModeParametersKHR {
    VkExtent2D    visibleRegion;
    uint32_t      refreshRate;
} VkDisplayModeParametersKHR;
```

- **visibleRegion** is the 2D extents of the visible region.
- **refreshRate** is a **uint32_t** that is the number of times the display is refreshed each second multiplied by 1000.

NOTE For example, a 60Hz display mode would report a **refreshRate** of 60,000.

Valid Usage

- VUID-VkDisplayModeParametersKHR-width-01990
The **width** member of **visibleRegion** **must** be greater than 0
- VUID-VkDisplayModeParametersKHR-height-01991
The **height** member of **visibleRegion** **must** be greater than 0
- VUID-VkDisplayModeParametersKHR-refreshRate-01992
refreshRate **must** be greater than 0

Additional modes **may** also be created by calling:

```
// Provided by VK_KHR_display
VkResult vkCreateDisplayModeKHR(
    VkPhysicalDevice          physicalDevice,
    VkDisplayKHR              display,
    const VkDisplayModeCreateInfoKHR* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkDisplayModeKHR*         pMode);
```

- **physicalDevice** is the physical device associated with **display**.
- **display** is the display to create an additional mode for.
- **pCreateInfo** is a pointer to a **VkDisplayModeCreateInfoKHR** structure describing the new mode to create.
- **pAllocator** is the allocator used for host memory allocated for the display mode object when there is no more specific allocator available (see [Memory Allocation](#)).

- `pMode` is a pointer to a `VkDisplayModeKHR` handle in which the mode created is returned.

Valid Usage (Implicit)

- VUID-vkCreateDisplayModeKHR-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkCreateDisplayModeKHR-display-parameter `display` **must** be a valid `VkDisplayKHR` handle
- VUID-vkCreateDisplayModeKHR-pCreateInfo-parameter `pCreateInfo` **must** be a valid pointer to a valid `VkDisplayModeCreateInfoKHR` structure
- VUID-vkCreateDisplayModeKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateDisplayModeKHR-pMode-parameter `pMode` **must** be a valid pointer to a `VkDisplayModeKHR` handle
- VUID-vkCreateDisplayModeKHR-display-parent `display` **must** have been created, allocated, or retrieved from `physicalDevice`

Host Synchronization

- Host access to `display` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`

The `VkDisplayModeCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayModeCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkDisplayModeCreateFlagsKHR flags;
    VkDisplayModeParametersKHR parameters;
} VkDisplayModeCreateInfoKHR;
```


- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use, and **must** be zero.
- `parameters` is a `VkDisplayModeParametersKHR` structure describing the display parameters to use in creating the new mode. If the parameters are not compatible with the specified display, the implementation **must** return `VK_ERROR_INITIALIZATION_FAILED`.

Valid Usage (Implicit)

- VUID-VkDisplayModeCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DISPLAY_MODE_CREATE_INFO_KHR`
- VUID-VkDisplayModeCreateInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDisplayModeCreateInfoKHR-flags-zeroBitmask
`flags` **must** be 0
- VUID-VkDisplayModeCreateInfoKHR-parameters-parameter
`parameters` **must** be a valid `VkDisplayModeParametersKHR` structure

Applications that wish to present directly to a display **must** select which layer, or “plane” of the display they wish to target, and a mode to use with the display. Each display supports at least one plane. The capabilities of a given mode and plane combination are determined by calling:

```
// Provided by VK_KHR_display
VkResult vkGetDisplayPlaneCapabilitiesKHR(
    VkPhysicalDevice      physicalDevice,
    VkDisplayModeKHR      mode,
    uint32_t              planeIndex,
    VkDisplayPlaneCapabilitiesKHR* pCapabilities);
```

- `physicalDevice` is the physical device associated with the display specified by `mode`
- `mode` is the display mode the application intends to program when using the specified plane. Note this parameter also implicitly specifies a display.
- `planeIndex` is the plane which the application intends to use with the display, and is less than the number of display planes supported by the device.
- `pCapabilities` is a pointer to a `VkDisplayPlaneCapabilitiesKHR` structure in which the capabilities are returned.

Valid Usage (Implicit)

- VUID-vkGetDisplayPlaneCapabilitiesKHR-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetDisplayPlaneCapabilitiesKHR-mode-parameter

mode **must** be a valid [VkDisplayModeKHR](#) handle

- VUID-vkGetDisplayPlaneCapabilitiesKHR-pCapabilities-parameter
pCapabilities **must** be a valid pointer to a [VkDisplayPlaneCapabilitiesKHR](#) structure
- VUID-vkGetDisplayPlaneCapabilitiesKHR-mode-parent
mode **must** have been created, allocated, or retrieved from [physicalDevice](#)

Host Synchronization

- Host access to **mode** **must** be externally synchronized

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

The [VkDisplayPlaneCapabilitiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplayPlaneCapabilitiesKHR {
    VkDisplayPlaneAlphaFlagsKHR    supportedAlpha;
    VkOffset2D                     minSrcPosition;
    VkOffset2D                     maxSrcPosition;
    VkExtent2D                     minSrcExtent;
    VkExtent2D                     maxSrcExtent;
    VkOffset2D                     minDstPosition;
    VkOffset2D                     maxDstPosition;
    VkExtent2D                     minDstExtent;
    VkExtent2D                     maxDstExtent;
} VkDisplayPlaneCapabilitiesKHR;
```

- **supportedAlpha** is a bitmask of [VkDisplayPlaneAlphaFlagBitsKHR](#) describing the supported alpha blending modes.
- **minSrcPosition** is the minimum source rectangle offset supported by this plane using the specified mode.
- **maxSrcPosition** is the maximum source rectangle offset supported by this plane using the specified mode. The **x** and **y** components of **maxSrcPosition** **must** each be greater than or equal to the **x** and **y** components of **minSrcPosition**, respectively.
- **minSrcExtent** is the minimum source rectangle size supported by this plane using the specified mode.

- `maxSrcExtent` is the maximum source rectangle size supported by this plane using the specified mode.
- `minDstPosition`, `maxDstPosition`, `minDstExtent`, `maxDstExtent` all have similar semantics to their corresponding `*Src*` equivalents, but apply to the output region within the mode rather than the input region within the source image. Unlike the `*Src*` offsets, `minDstPosition` and `maxDstPosition` **may** contain negative values.

The minimum and maximum position and extent fields describe the implementation limits, if any, as they apply to the specified display mode and plane. Vendors **may** support displaying a subset of a swapchain's presentable images on the specified display plane. This is expressed by returning `minSrcPosition`, `maxSrcPosition`, `minSrcExtent`, and `maxSrcExtent` values that indicate a range of possible positions and sizes which **may** be used to specify the region within the presentable images that source pixels will be read from when creating a swapchain on the specified display mode and plane.

Vendors **may** also support mapping the presentable images' content to a subset or superset of the visible region in the specified display mode. This is expressed by returning `minDstPosition`, `maxDstPosition`, `minDstExtent` and `maxDstExtent` values that indicate a range of possible positions and sizes which **may** be used to describe the region within the display mode that the source pixels will be mapped to.

Other vendors **may** support only a 1-1 mapping between pixels in the presentable images and the display mode. This **may** be indicated by returning (0,0) for `minSrcPosition`, `maxSrcPosition`, `minDstPosition`, and `maxDstPosition`, and (display mode width, display mode height) for `minSrcExtent`, `maxSrcExtent`, `minDstExtent`, and `maxDstExtent`.

The value `supportedAlpha` **must** contain at least one valid `VkDisplayPlaneAlphaFlagBitsKHR` bit.

These values indicate the limits of the implementation's individual fields. Not all combinations of values within the offset and extent ranges returned in `VkDisplayPlaneCapabilitiesKHR` are guaranteed to be supported. Presentation requests specifying unsupported combinations **may** fail.

To query the capabilities of a given mode and plane combination, call:

```
// Provided by VK_KHR_get_display_properties2
VkResult vkGetDisplayPlaneCapabilities2KHR(
    VkPhysicalDevice          physicalDevice,
    const VkDisplayPlaneInfo2KHR* pDisplayPlaneInfo,
    VkDisplayPlaneCapabilities2KHR* pCapabilities);
```

- `physicalDevice` is the physical device associated with `pDisplayPlaneInfo`.
- `pDisplayPlaneInfo` is a pointer to a `VkDisplayPlaneInfo2KHR` structure describing the plane and mode.
- `pCapabilities` is a pointer to a `VkDisplayPlaneCapabilities2KHR` structure in which the capabilities are returned.

`vkGetDisplayPlaneCapabilities2KHR` behaves similarly to `vkGetDisplayPlaneCapabilitiesKHR`, with the ability to specify extended inputs via chained input structures, and to return extended

information via chained output structures.

Valid Usage (Implicit)

- VUID-vkGetDisplayPlaneCapabilities2KHR-physicalDevice-parameter **physicalDevice** must be a valid [VkPhysicalDevice](#) handle
- VUID-vkGetDisplayPlaneCapabilities2KHR-pDisplayPlaneInfo-parameter **pDisplayPlaneInfo** must be a valid pointer to a valid [VkDisplayPlaneInfo2KHR](#) structure
- VUID-vkGetDisplayPlaneCapabilities2KHR-pCapabilities-parameter **pCapabilities** must be a valid pointer to a [VkDisplayPlaneCapabilities2KHR](#) structure

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

The [VkDisplayPlaneInfo2KHR](#) structure is defined as:

```
// Provided by VK_KHR_get_display_properties2
typedef struct VkDisplayPlaneInfo2KHR {
    VkStructureType    sType;
    const void*        pNext;
    VkDisplayModeKHR    mode;
    uint32_t            planeIndex;
} VkDisplayPlaneInfo2KHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is [NULL](#) or a pointer to a structure extending this structure.
- **mode** is the display mode the application intends to program when using the specified plane.

NOTE This parameter also implicitly specifies a display.

- **planeIndex** is the plane which the application intends to use with the display.

The members of [VkDisplayPlaneInfo2KHR](#) correspond to the arguments to [vkGetDisplayPlaneCapabilitiesKHR](#), with **sType** and **pNext** added for extensibility.

Valid Usage (Implicit)

- VUID-VkDisplayPlaneInfo2KHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DISPLAY_PLANE_INFO_2_KHR`
- VUID-VkDisplayPlaneInfo2KHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDisplayPlaneInfo2KHR-mode-parameter
`mode` **must** be a valid `VkDisplayModeKHR` handle

Host Synchronization

- Host access to `mode` **must** be externally synchronized

The `VkDisplayPlaneCapabilities2KHR` structure is defined as:

```
// Provided by VK_KHR_get_display_properties2
typedef struct VkDisplayPlaneCapabilities2KHR {
    VkStructureType    sType;
    void*              pNext;
    VkDisplayPlaneCapabilitiesKHR capabilities;
} VkDisplayPlaneCapabilities2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `capabilities` is a `VkDisplayPlaneCapabilitiesKHR` structure.

Valid Usage (Implicit)

- VUID-VkDisplayPlaneCapabilities2KHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DISPLAY_PLANE_CAPABILITIES_2_KHR`
- VUID-VkDisplayPlaneCapabilities2KHR-pNext-pNext
`pNext` **must** be `NULL`

31.3.2. Display Control

To set the power state of a display, call:

```
// Provided by VK_EXT_display_control
VkResult vkDisplayPowerControlEXT(
    VkDevice          device,
    VkDisplayKHR      display,
    const VkDisplayPowerInfoEXT* pDisplayPowerInfo);
```

- **device** is a logical device associated with **display**.
- **display** is the display whose power state is modified.
- **pDisplayPowerInfo** is a pointer to a **VkDisplayPowerInfoEXT** structure specifying the new power state of **display**.

Valid Usage (Implicit)

- VUID-vkDisplayPowerControlEXT-device-parameter **device** **must** be a valid **VkDevice** handle
- VUID-vkDisplayPowerControlEXT-display-parameter **display** **must** be a valid **VkDisplayKHR** handle
- VUID-vkDisplayPowerControlEXT-pDisplayPowerInfo-parameter **pDisplayPowerInfo** **must** be a valid pointer to a valid **VkDisplayPowerInfoEXT** structure
- VUID-vkDisplayPowerControlEXT-commonparent
Both of **device**, and **display** **must** have been created, allocated, or retrieved from the same **VkPhysicalDevice**

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**

The **VkDisplayPowerInfoEXT** structure is defined as:

```
// Provided by VK_EXT_display_control
typedef struct VkDisplayPowerInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkDisplayPowerStateEXT powerState;
} VkDisplayPowerInfoEXT;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.

- `powerState` is a `VkDisplayPowerStateEXT` value specifying the new power state of the display.

Valid Usage (Implicit)

- VUID-VkDisplayPowerInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DISPLAY_POWER_INFO_EXT`
- VUID-VkDisplayPowerInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkDisplayPowerInfoEXT-powerState-parameter
`powerState` **must** be a valid `VkDisplayPowerStateEXT` value

Possible values of `VkDisplayPowerInfoEXT::powerState`, specifying the new power state of a display, are:

```
// Provided by VK_EXT_display_control
typedef enum VkDisplayPowerStateEXT {
    VK_DISPLAY_POWER_STATE_OFF_EXT = 0,
    VK_DISPLAY_POWER_STATE_SUSPEND_EXT = 1,
    VK_DISPLAY_POWER_STATE_ON_EXT = 2,
} VkDisplayPowerStateEXT;
```

- `VK_DISPLAY_POWER_STATE_OFF_EXT` specifies that the display is powered down.
- `VK_DISPLAY_POWER_STATE_SUSPEND_EXT` specifies that the display is put into a low power mode, from which it **may** be able to transition back to `VK_DISPLAY_POWER_STATE_ON_EXT` more quickly than if it were in `VK_DISPLAY_POWER_STATE_OFF_EXT`. This state **may** be the same as `VK_DISPLAY_POWER_STATE_OFF_EXT`.
- `VK_DISPLAY_POWER_STATE_ON_EXT` specifies that the display is powered on.

31.3.3. Display Surfaces

A complete display configuration includes a mode, one or more display planes and any parameters describing their behavior, and parameters describing some aspects of the images associated with those planes. Display surfaces describe the configuration of a single plane within a complete display configuration. To create a `VkSurfaceKHR` object for a display plane, call:

```
// Provided by VK_KHR_display
VkResult vkCreateDisplayPlaneSurfaceKHR(
    VkInstance                                instance,
    const VkDisplaySurfaceCreateInfoKHR*      pCreateInfo,
    const VkAllocationCallbacks*             pAllocator,
    VkSurfaceKHR*                             pSurface);
```

- `instance` is the instance corresponding to the physical device the targeted display is on.
- `pCreateInfo` is a pointer to a `VkDisplaySurfaceCreateInfoKHR` structure specifying which mode,

plane, and other parameters to use, as described below.

- **pAllocator** is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see [Memory Allocation](#)).
- **pSurface** is a pointer to a [VkSurfaceKHR](#) handle in which the created surface is returned.

Valid Usage (Implicit)

- VUID-vkCreateDisplayPlaneSurfaceKHR-instance-parameter **instance** **must** be a valid [VkInstance](#) handle
- VUID-vkCreateDisplayPlaneSurfaceKHR-pCreateInfo-parameter **pCreateInfo** **must** be a valid pointer to a valid [VkDisplaySurfaceCreateInfoKHR](#) structure
- VUID-vkCreateDisplayPlaneSurfaceKHR-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateDisplayPlaneSurfaceKHR-pSurface-parameter
pSurface **must** be a valid pointer to a [VkSurfaceKHR](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**

The [VkDisplaySurfaceCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_display
typedef struct VkDisplaySurfaceCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkDisplaySurfaceCreateFlagsKHR flags;
    VkDisplayModeKHR    displayMode;
    uint32_t            planeIndex;
    uint32_t            planeStackIndex;
    VkSurfaceTransformFlagBitsKHR transform;
    float               globalAlpha;
    VkDisplayPlaneAlphaFlagBitsKHR alphaMode;
    VkExtent2D          imageExtent;
} VkDisplaySurfaceCreateInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use, and **must** be zero.
- `displayMode` is a `VkDisplayModeKHR` handle specifying the mode to use when displaying this surface.
- `planeIndex` is the plane on which this surface appears.
- `planeStackIndex` is the z-order of the plane.
- `transform` is a `VkSurfaceTransformFlagBitsKHR` value specifying the transformation to apply to images as part of the scanout operation.
- `globalAlpha` is the global alpha value. This value is ignored if `alphaMode` is not `VK_DISPLAY_PLANE_ALPHA_GLOBAL_BIT_KHR`.
- `alphaMode` is a `VkDisplayPlaneAlphaFlagBitsKHR` value specifying the type of alpha blending to use.
- `imageExtent` is the size of the presentable images to use with the surface.

NOTE

Creating a display surface **must** not modify the state of the displays, planes, or other resources it names. For example, it **must** not apply the specified mode to be set on the associated display. Application of display configuration occurs as a side effect of presenting to a display surface.

Valid Usage

- VUID-VkDisplaySurfaceCreateInfoKHR-planeIndex-01252
`planeIndex` **must** be less than the number of display planes supported by the device as determined by calling `vkGetPhysicalDeviceDisplayPlanePropertiesKHR`
- VUID-VkDisplaySurfaceCreateInfoKHR-planeReorderPossible-01253
If the `planeReorderPossible` member of the `VkDisplayPropertiesKHR` structure returned by `vkGetPhysicalDeviceDisplayPlanePropertiesKHR` for the display corresponding to `displayMode` is `VK_TRUE` then `planeStackIndex` **must** be less than the number of display planes supported by the device as determined by calling `vkGetPhysicalDeviceDisplayPlanePropertiesKHR`; otherwise `planeStackIndex` **must** equal the `currentStackIndex` member of `VkDisplayPlanePropertiesKHR` returned by `vkGetPhysicalDeviceDisplayPlanePropertiesKHR` for the display plane corresponding to `displayMode`
- VUID-VkDisplaySurfaceCreateInfoKHR-alphaMode-01254
If `alphaMode` is `VK_DISPLAY_PLANE_ALPHA_GLOBAL_BIT_KHR` then `globalAlpha` **must** be between 0 and 1, inclusive
- VUID-VkDisplaySurfaceCreateInfoKHR-alphaMode-01255
`alphaMode` **must** be one of the bits present in the `supportedAlpha` member of `VkDisplayPlaneCapabilitiesKHR` for the display plane corresponding to `displayMode`
- VUID-VkDisplaySurfaceCreateInfoKHR-transform-06740
`transform` **must** be one of the bits present in the `supportedTransforms` member of `VkDisplayPropertiesKHR` for the display corresponding to `displayMode`
- VUID-VkDisplaySurfaceCreateInfoKHR-width-01256

The `width` and `height` members of `imageExtent` **must** be less than or equal to `VkPhysicalDeviceLimits::maxImageDimension2D`

Valid Usage (Implicit)

- `VUID-VkDisplaySurfaceCreateInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_DISPLAY_SURFACE_CREATE_INFO_KHR`
- `VUID-VkDisplaySurfaceCreateInfoKHR-pNext-pNext`
`pNext` **must** be `NULL`
- `VUID-VkDisplaySurfaceCreateInfoKHR-flags-zeroBitmask`
`flags` **must** be `0`
- `VUID-VkDisplaySurfaceCreateInfoKHR-displayMode-parameter`
`displayMode` **must** be a valid `VkDisplayModeKHR` handle
- `VUID-VkDisplaySurfaceCreateInfoKHR-transform-parameter`
`transform` **must** be a valid `VkSurfaceTransformFlagBitsKHR` value
- `VUID-VkDisplaySurfaceCreateInfoKHR-alphaMode-parameter`
`alphaMode` **must** be a valid `VkDisplayPlaneAlphaFlagBitsKHR` value

```
// Provided by VK_KHR_display
typedef VkFlags VkDisplaySurfaceCreateFlagsKHR;
```

`VkDisplaySurfaceCreateFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

Bits which **can** be set in `VkDisplaySurfaceCreateInfoKHR::alphaMode`, specifying the type of alpha blending to use on a display, are:

```
// Provided by VK_KHR_display
typedef enum VkDisplayPlaneAlphaFlagBitsKHR {
    VK_DISPLAY_PLANE_ALPHA_OPAQUE_BIT_KHR = 0x00000001,
    VK_DISPLAY_PLANE_ALPHA_GLOBAL_BIT_KHR = 0x00000002,
    VK_DISPLAY_PLANE_ALPHA_PER_PIXEL_BIT_KHR = 0x00000004,
    VK_DISPLAY_PLANE_ALPHA_PER_PIXEL_PREMULTIPLIED_BIT_KHR = 0x00000008,
} VkDisplayPlaneAlphaFlagBitsKHR;
```

- `VK_DISPLAY_PLANE_ALPHA_OPAQUE_BIT_KHR` specifies that the source image will be treated as opaque.
- `VK_DISPLAY_PLANE_ALPHA_GLOBAL_BIT_KHR` specifies that a global alpha value **must** be specified that will be applied to all pixels in the source image.
- `VK_DISPLAY_PLANE_ALPHA_PER_PIXEL_BIT_KHR` specifies that the alpha value will be determined by the alpha component of the source image's pixels. If the source format contains no alpha values, no blending will be applied. The source alpha values are not premultiplied into the source

image's other color components.

- `VK_DISPLAY_PLANE_ALPHA_PER_PIXEL_PREMULTIPLIED_BIT_KHR` is equivalent to `VK_DISPLAY_PLANE_ALPHA_PER_PIXEL_BIT_KHR`, except the source alpha values are assumed to be premultiplied into the source image's other color components.

```
// Provided by VK_KHR_display
typedef VkFlags VkDisplayPlaneAlphaFlagsKHR;
```

`VkDisplayPlaneAlphaFlagsKHR` is a bitmask type for setting a mask of zero or more `VkDisplayPlaneAlphaFlagBitsKHR`.

31.3.4. Presenting to Headless Surfaces

Vulkan rendering can be presented to a headless surface, where the presentation operation is a no-op producing no externally-visible result.

NOTE

Because there is no real presentation target, the headless presentation engine may be extended to impose an arbitrary or customizable set of restrictions and features. This makes it a useful portable test target for applications targeting a wide range of presentation engines where the actual target presentation engines might be scarce, unavailable or otherwise undesirable or inconvenient to use for general Vulkan application development.

The usual surface query mechanisms must be used to determine the actual restrictions and features of the implementation.

To create a headless `VkSurfaceKHR` object, call:

```
// Provided by VK_EXT_headless_surface
VkResult vkCreateHeadlessSurfaceEXT(
    VkInstance                instance,
    const VkHeadlessSurfaceCreateInfoEXT* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkSurfaceKHR*              pSurface);
```

- `instance` is the instance to associate the surface with.
- `pCreateInfo` is a pointer to a `VkHeadlessSurfaceCreateInfoEXT` structure containing parameters affecting the creation of the surface object.
- `pAllocator` is the allocator used for host memory allocated for the surface object when there is no more specific allocator available (see [Memory Allocation](#)).
- `pSurface` is a pointer to a `VkSurfaceKHR` handle in which the created surface object is returned.

Valid Usage (Implicit)

- VUID-vkCreateHeadlessSurfaceEXT-instance-parameter

`instance` **must** be a valid `VkInstance` handle

- VUID-vkCreateHeadlessSurfaceEXT-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkHeadlessSurfaceCreateInfoEXT` structure
- VUID-vkCreateHeadlessSurfaceEXT-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateHeadlessSurfaceEXT-pSurface-parameter
`pSurface` **must** be a valid pointer to a `VkSurfaceKHR` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkHeadlessSurfaceCreateInfoEXT` structure is defined as:

```
// Provided by VK_EXT_headless_surface
typedef struct VkHeadlessSurfaceCreateInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkHeadlessSurfaceCreateFlagsEXT flags;
} VkHeadlessSurfaceCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

Valid Usage (Implicit)

- VUID-VkHeadlessSurfaceCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_HEADLESS_SURFACE_CREATE_INFO_EXT`
- VUID-VkHeadlessSurfaceCreateInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkHeadlessSurfaceCreateInfoEXT-flags-zerobitmask
`flags` **must** be `0`

For headless surfaces, `currentExtent` is the reserved value (`0xFFFFFFFF`, `0xFFFFFFFF`). Whatever the application sets a swapchain's `imageExtent` to will be the size of the surface, after the first image is

presented.

```
// Provided by VK_EXT_headless_surface
typedef VkFlags VkHeadlessSurfaceCreateFlagsEXT;
```

`VkHeadlessSurfaceCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

31.4. Querying for WSI Support

Not all physical devices will include WSI support. Within a physical device, not all queue families will support presentation. WSI support and compatibility **can** be determined in a platform-neutral manner (which determines support for presentation to a particular surface object) and additionally **may** be determined in platform-specific manners (which determine support for presentation on the specified physical device but do not guarantee support for presentation to a particular surface object).

To determine whether a queue family of a physical device supports presentation to a given surface, call:

```
// Provided by VK_KHR_surface
VkResult vkGetPhysicalDeviceSurfaceSupportKHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t                  queueFamilyIndex,
    VkSurfaceKHR              surface,
    VkBool32*                  pSupported);
```

- `physicalDevice` is the physical device.
- `queueFamilyIndex` is the queue family.
- `surface` is the surface.
- `pSupported` is a pointer to a `VkBool32`. `VK_TRUE` indicates support, and `VK_FALSE` indicates no support.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceSupportKHR-queueFamilyIndex-01269 `queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the given `physicalDevice`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceSupportKHR-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle

- VUID-vkGetPhysicalDeviceSurfaceSupportKHR-surface-parameter **surface** **must** be a valid [VkSurfaceKHR](#) handle
- VUID-vkGetPhysicalDeviceSurfaceSupportKHR-pSupported-parameter **pSupported** **must** be a valid pointer to a [VkBool32](#) value
- VUID-vkGetPhysicalDeviceSurfaceSupportKHR-commonparent Both of **physicalDevice**, and **surface** **must** have been created, allocated, or retrieved from the same [VkInstance](#)

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)
- [VK_ERROR_SURFACE_LOST_KHR](#)

31.4.1. Android Platform

On Android, all physical devices and queue families **must** be capable of presentation with any native window. As a result there is no Android-specific query for these capabilities.

31.4.2. Wayland Platform

To determine whether a queue family of a physical device supports presentation to a Wayland compositor, call:

```
// Provided by VK_KHR_wayland_surface
VkBool32 vkGetPhysicalDeviceWaylandPresentationSupportKHR(
    VkPhysicalDevice      physicalDevice,
    uint32_t              queueFamilyIndex,
    struct wl_display*    display);
```

- **physicalDevice** is the physical device.
- **queueFamilyIndex** is the queue family index.
- **display** is a pointer to the **wl_display** associated with a Wayland compositor.

This platform-specific function **can** be called prior to creating a surface.

Valid Usage

- VUID-vkGetPhysicalDeviceWaylandPresentationSupportKHR-queueFamilyIndex-01306

`queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the given `physicalDevice`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceWaylandPresentationSupportKHR-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceWaylandPresentationSupportKHR-display-parameter `display` **must** be a valid pointer to a `wl_display` value

31.4.3. Win32 Platform

To determine whether a queue family of a physical device supports presentation to the Microsoft Windows desktop, call:

```
// Provided by VK_KHR_win32_surface
VkBool32 vkGetPhysicalDeviceWin32PresentationSupportKHR(
    VkPhysicalDevice    physicalDevice,
    uint32_t            queueFamilyIndex);
```

- `physicalDevice` is the physical device.
- `queueFamilyIndex` is the queue family index.

This platform-specific function **can** be called prior to creating a surface.

Valid Usage

- VUID-vkGetPhysicalDeviceWin32PresentationSupportKHR-queueFamilyIndex-01309 `queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the given `physicalDevice`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceWin32PresentationSupportKHR-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle

31.4.4. XCB Platform

To determine whether a queue family of a physical device supports presentation to an X11 server, using the XCB client-side library, call:

```
// Provided by VK_KHR_xcb_surface
VkBool32 vkGetPhysicalDeviceXcbPresentationSupportKHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t                  queueFamilyIndex,
    xcb_connection_t*          connection,
    xcb_visualid_t             visual_id);
```

- `physicalDevice` is the physical device.
- `queueFamilyIndex` is the queue family index.
- `connection` is a pointer to an `xcb_connection_t` to the X server.
- `visual_id` is an X11 visual (`xcb_visualid_t`).

This platform-specific function **can** be called prior to creating a surface.

Valid Usage

- VUID-vkGetPhysicalDeviceXcbPresentationSupportKHR-queueFamilyIndex-01312
`queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the given `physicalDevice`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceXcbPresentationSupportKHR-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceXcbPresentationSupportKHR-connection-parameter
`connection` **must** be a valid pointer to an `xcb_connection_t` value

31.4.5. Xlib Platform

To determine whether a queue family of a physical device supports presentation to an X11 server, using the Xlib client-side library, call:

```
// Provided by VK_KHR_xlib_surface
VkBool32 vkGetPhysicalDeviceXlibPresentationSupportKHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t                  queueFamilyIndex,
    Display*                  dpy,
    VisualID                   visualID);
```

- `physicalDevice` is the physical device.
- `queueFamilyIndex` is the queue family index.
- `dpy` is a pointer to an Xlib `Display` connection to the server.

- `visualId` is an X11 visual (`VisualID`).

This platform-specific function **can** be called prior to creating a surface.

Valid Usage

- VUID-vkGetPhysicalDeviceXlibPresentationSupportKHR-queueFamilyIndex-01315
`queueFamilyIndex` **must** be less than `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties` for the given `physicalDevice`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceXlibPresentationSupportKHR-physicalDevice-parameter-`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceXlibPresentationSupportKHR-dpy-parameter-`dpy` **must** be a valid pointer to a `Display` value

31.5. Surface Queries

The capabilities of a swapchain targeting a surface are the intersection of the capabilities of the WSI platform, the native window or display, and the physical device. The resulting capabilities **can** be obtained with the queries listed below in this section.

NOTE

In addition to the surface capabilities as obtained by surface queries below, swapchain images are also subject to ordinary image creation limits as reported by `vkGetPhysicalDeviceImageFormatProperties`. As an application is instructed by the appropriate Valid Usage sections, both the surface capabilities and the image creation limits have to be satisfied whenever swapchain images are created.

31.5.1. Surface Capabilities

To query the basic capabilities of a surface, needed in order to create a swapchain, call:

```
// Provided by VK_KHR_surface
VkResult vkGetPhysicalDeviceSurfaceCapabilitiesKHR(
    VkPhysicalDevice    physicalDevice,
    VkSurfaceKHR        surface,
    VkSurfaceCapabilitiesKHR* pSurfaceCapabilities);
```

- `physicalDevice` is the physical device that will be associated with the swapchain to be created, as described for `vkCreateSwapchainKHR`.
- `surface` is the surface that will be associated with the swapchain.
- `pSurfaceCapabilities` is a pointer to a `VkSurfaceCapabilitiesKHR` structure in which the

capabilities are returned.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceCapabilitiesKHR-surface-06523
surface **must** be a valid [VkSurfaceKHR](#) handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilitiesKHR-surface-06211
surface **must** be supported by **physicalDevice**, as reported by [vkGetPhysicalDeviceSurfaceSupportKHR](#) or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceCapabilitiesKHR-physicalDevice-parameter
physicalDevice **must** be a valid [VkPhysicalDevice](#) handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilitiesKHR-surface-parameter
surface **must** be a valid [VkSurfaceKHR](#) handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilitiesKHR-pSurfaceCapabilities-parameter
pSurfaceCapabilities **must** be a valid pointer to a [VkSurfaceCapabilitiesKHR](#) structure
- VUID-vkGetPhysicalDeviceSurfaceCapabilitiesKHR-commonparent
Both of **physicalDevice**, and **surface** **must** have been created, allocated, or retrieved from the same [VkInstance](#)

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)
- [VK_ERROR_SURFACE_LOST_KHR](#)

The [VkSurfaceCapabilitiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_surface
typedef struct VkSurfaceCapabilitiesKHR {
    uint32_t          minImageCount;
    uint32_t          maxImageCount;
    VkExtent2D        currentExtent;
    VkExtent2D        minImageExtent;
    VkExtent2D        maxImageExtent;
    uint32_t          maxImageArrayLayers;
    VkSurfaceTransformFlagsKHR supportedTransforms;
    VkSurfaceTransformFlagBitsKHR currentTransform;
    VkCompositeAlphaFlagsKHR supportedCompositeAlpha;
    VkImageUsageFlags supportedUsageFlags;
} VkSurfaceCapabilitiesKHR;
```

- **minImageCount** is the minimum number of images the specified device supports for a swapchain created for the surface, and will be at least one.
- **maxImageCount** is the maximum number of images the specified device supports for a swapchain created for the surface, and will be either 0, or greater than or equal to **minImageCount**. A value of 0 means that there is no limit on the number of images, though there **may** be limits related to the total amount of memory used by presentable images.
- **currentExtent** is the current width and height of the surface, or the special value (0xFFFFFFFF, 0xFFFFFFFF) indicating that the surface size will be determined by the extent of a swapchain targeting the surface.
- **minImageExtent** contains the smallest valid swapchain extent for the surface on the specified device. The **width** and **height** of the extent will each be less than or equal to the corresponding **width** and **height** of **currentExtent**, unless **currentExtent** has the special value described above.
- **maxImageExtent** contains the largest valid swapchain extent for the surface on the specified device. The **width** and **height** of the extent will each be greater than or equal to the corresponding **width** and **height** of **minImageExtent**. The **width** and **height** of the extent will each be greater than or equal to the corresponding **width** and **height** of **currentExtent**, unless **currentExtent** has the special value described above.
- **maxImageArrayLayers** is the maximum number of layers presentable images **can** have for a swapchain created for this device and surface, and will be at least one.
- **supportedTransforms** is a bitmask of **VkSurfaceTransformFlagBitsKHR** indicating the presentation transforms supported for the surface on the specified device. At least one bit will be set.
- **currentTransform** is **VkSurfaceTransformFlagBitsKHR** value indicating the surface's current transform relative to the presentation engine's natural orientation.
- **supportedCompositeAlpha** is a bitmask of **VkCompositeAlphaFlagBitsKHR**, representing the alpha compositing modes supported by the presentation engine for the surface on the specified device, and at least one bit will be set. Opaque composition **can** be achieved in any alpha compositing mode by either using an image format that has no alpha component, or by ensuring that all pixels in the presentable images have an alpha value of 1.0.
- **supportedUsageFlags** is a bitmask of **VkImageUsageFlagBits** representing the ways the

application **can** use the presentable images of a swapchain created with [VkPresentModeKHR](#) set to [VK_PRESENT_MODE_FIFO_LATEST_READY_EXT](#), [VK_PRESENT_MODE_IMMEDIATE_KHR](#), [VK_PRESENT_MODE_MAILBOX_KHR](#), [VK_PRESENT_MODE_FIFO_KHR](#) or [VK_PRESENT_MODE_FIFO_RELAXED_KHR](#) for the surface on the specified device. [VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT](#) **must** be included in the set. Implementations **may** support additional usages.

- | | |
|-------------|--|
| NOTE | Supported usage flags of a presentable image when using VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR presentation mode are provided by VkSharedPresentSurfaceCapabilitiesKHR::sharedPresentSupportedUsageFlags . |
| NOTE | Formulas such as $\min(N, \text{maxImageCount})$ are not correct, since maxImageCount may be zero. |

To query the basic capabilities of a surface defined by the core or extensions, call:

```
// Provided by VK_KHR_get_surface_capabilities2
VkResult vkGetPhysicalDeviceSurfaceCapabilities2KHR(
    VkPhysicalDevice          physicalDevice,
    const VkPhysicalDeviceSurfaceInfo2KHR* pSurfaceInfo,
    VkSurfaceCapabilities2KHR* pSurfaceCapabilities);
```

- [physicalDevice](#) is the physical device that will be associated with the swapchain to be created, as described for [vkCreateSwapchainKHR](#).
- [pSurfaceInfo](#) is a pointer to a [VkPhysicalDeviceSurfaceInfo2KHR](#) structure describing the surface and other fixed parameters that would be consumed by [vkCreateSwapchainKHR](#).
- [pSurfaceCapabilities](#) is a pointer to a [VkSurfaceCapabilities2KHR](#) structure in which the capabilities are returned.

[vkGetPhysicalDeviceSurfaceCapabilities2KHR](#) behaves similarly to [vkGetPhysicalDeviceSurfaceCapabilitiesKHR](#), with the ability to specify extended inputs via chained input structures, and to return extended information via chained output structures.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pSurfaceInfo-06521
[pSurfaceInfo->surface](#) **must** be a valid [VkSurfaceKHR](#) handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pSurfaceInfo-06522
[pSurfaceInfo->surface](#) **must** be supported by [physicalDevice](#), as reported by [vkGetPhysicalDeviceSurfaceSupportKHR](#) or an equivalent platform-specific mechanism
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pNext-02671
If a [VkSurfaceCapabilitiesFullScreenExclusiveEXT](#) structure is included in the [pNext](#) chain of [pSurfaceCapabilities](#), a [VkSurfaceFullScreenExclusiveWin32InfoEXT](#) structure **must** be included in the [pNext](#) chain of [pSurfaceInfo](#)

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-physicalDevice-parameter **physicalDevice** **must** be a valid [VkPhysicalDevice](#) handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pSurfaceInfo-parameter **pSurfaceInfo** **must** be a valid pointer to a valid [VkPhysicalDeviceSurfaceInfo2KHR](#) structure
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2KHR-pSurfaceCapabilities-parameter **pSurfaceCapabilities** **must** be a valid pointer to a [VkSurfaceCapabilities2KHR](#) structure

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)
- [VK_ERROR_SURFACE_LOST_KHR](#)

The [VkPhysicalDeviceSurfaceInfo2KHR](#) structure is defined as:

```
// Provided by VK_KHR_get_surface_capabilities2
typedef struct VkPhysicalDeviceSurfaceInfo2KHR {
    VkStructureType    sType;
    const void*        pNext;
    VkSurfaceKHR        surface;
} VkPhysicalDeviceSurfaceInfo2KHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **surface** is the surface that will be associated with the swapchain.

The members of [VkPhysicalDeviceSurfaceInfo2KHR](#) correspond to the arguments to [vkGetPhysicalDeviceSurfaceCapabilitiesKHR](#), with **sType** and **pNext** added for extensibility.

Additional capabilities of a surface **may** be available to swapchains created with different full-screen exclusive settings - particularly if exclusive full-screen access is application controlled. These additional capabilities **can** be queried by adding a [VkSurfaceFullScreenExclusiveInfoEXT](#) structure to the **pNext** chain of this structure when used to query surface properties. Additionally, for Win32 surfaces with application controlled exclusive full-screen access, chaining a [VkSurfaceFullScreenExclusiveWin32InfoEXT](#) structure **may** also report additional surface capabilities. These additional capabilities only apply to swapchains created with the same

parameters included in the `pNext` chain of `VkSwapchainCreateInfoKHR`.

Valid Usage

- VUID-VkPhysicalDeviceSurfaceInfo2KHR-pNext-02672
If the `pNext` chain includes a `VkSurfaceFullScreenExclusiveInfoEXT` structure with its `fullScreenExclusive` member set to `VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT`, and `surface` was created using `vkCreateWin32SurfaceKHR`, a `VkSurfaceFullScreenExclusiveWin32InfoEXT` structure **must** be included in the `pNext` chain
- VUID-VkPhysicalDeviceSurfaceInfo2KHR-surface-07919
If `surface` is not `VK_NULL_HANDLE`, `surface` **must** be a valid `VkSurfaceKHR` handle

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSurfaceInfo2KHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SURFACE_INFO_2_KHR`
- VUID-VkPhysicalDeviceSurfaceInfo2KHR-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkSurfaceFullScreenExclusiveInfoEXT` or `VkSurfaceFullScreenExclusiveWin32InfoEXT`
- VUID-VkPhysicalDeviceSurfaceInfo2KHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique

If the `pNext` chain of `VkSwapchainCreateInfoKHR` includes a `VkSurfaceFullScreenExclusiveInfoEXT` structure, then that structure specifies the application's preferred full-screen transition behavior.

The `VkSurfaceFullScreenExclusiveInfoEXT` structure is defined as:

```
// Provided by VK_EXT_full_screen_exclusive
typedef struct VkSurfaceFullScreenExclusiveInfoEXT {
    VkStructureType    sType;
    void*              pNext;
    VkFullScreenExclusiveEXT fullScreenExclusive;
} VkSurfaceFullScreenExclusiveInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fullScreenExclusive` is a `VkFullScreenExclusiveEXT` value specifying the preferred full-screen transition behavior.

If this structure is not present, `fullScreenExclusive` is considered to be `VK_FULL_SCREEN_EXCLUSIVE_DEFAULT_EXT`.

Valid Usage (Implicit)

- VUID-VkSurfaceFullScreenExclusiveInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_INFO_EXT`
- VUID-VkSurfaceFullScreenExclusiveInfoEXT-fullScreenExclusive-parameter
`fullScreenExclusive` **must** be a valid `VkFullScreenExclusiveEXT` value

Possible values of `VkSurfaceFullScreenExclusiveInfoEXT::fullScreenExclusive` are:

```
// Provided by VK_EXT_full_screen_exclusive
typedef enum VkFullScreenExclusiveEXT {
    VK_FULL_SCREEN_EXCLUSIVE_DEFAULT_EXT = 0,
    VK_FULL_SCREEN_EXCLUSIVE_ALLOWED_EXT = 1,
    VK_FULL_SCREEN_EXCLUSIVE_DISALLOWED_EXT = 2,
    VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT = 3,
} VkFullScreenExclusiveEXT;
```

- `VK_FULL_SCREEN_EXCLUSIVE_DEFAULT_EXT` indicates the implementation **should** determine the appropriate full-screen method by whatever means it deems appropriate.
- `VK_FULL_SCREEN_EXCLUSIVE_ALLOWED_EXT` indicates the implementation **may** use full-screen exclusive mechanisms when available. Such mechanisms **may** result in better performance and/or the availability of different presentation capabilities, but **may** require a more disruptive transition during swapchain initialization, first presentation and/or destruction.
- `VK_FULL_SCREEN_EXCLUSIVE_DISALLOWED_EXT` indicates the implementation **should** avoid using full-screen mechanisms which rely on disruptive transitions.
- `VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT` indicates the application will manage full-screen exclusive mode by using the `vkAcquireFullScreenExclusiveModeEXT` and `vkReleaseFullScreenExclusiveModeEXT` commands.

The `VkSurfaceFullScreenExclusiveWin32InfoEXT` structure is defined as:

```
// Provided by VK_KHR_win32_surface with VK_EXT_full_screen_exclusive
typedef struct VkSurfaceFullScreenExclusiveWin32InfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    HMONITOR            hmonitor;
} VkSurfaceFullScreenExclusiveWin32InfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `hmonitor` is the Win32 `HMONITOR` handle identifying the display to create the surface with.

NOTE

If `hmonitor` is invalidated (e.g. the monitor is unplugged) during the lifetime of a

swapchain created with this structure, operations on that swapchain will return `VK_ERROR_OUT_OF_DATE_KHR`.

NOTE

It is the responsibility of the application to change the display settings of the targeted Win32 display using the appropriate platform APIs. Such changes **may** alter the surface capabilities reported for the created surface.

Valid Usage

- VUID-VkSurfaceFullScreenExclusiveWin32InfoEXT-hmonitor-02673
`hmonitor` **must** be a valid `HMONITOR`

Valid Usage (Implicit)

- VUID-VkSurfaceFullScreenExclusiveWin32InfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_WIN32_INFO_EXT`

The `VkSurfaceCapabilities2KHR` structure is defined as:

```
// Provided by VK_KHR_get_surface_capabilities2
typedef struct VkSurfaceCapabilities2KHR {
    VkStructureType    sType;
    void*              pNext;
    VkSurfaceCapabilitiesKHR surfaceCapabilities;
} VkSurfaceCapabilities2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `surfaceCapabilities` is a `VkSurfaceCapabilitiesKHR` structure describing the capabilities of the specified surface.

Valid Usage (Implicit)

- VUID-VkSurfaceCapabilities2KHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_KHR`
- VUID-VkSurfaceCapabilities2KHR-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkSharedPresentSurfaceCapabilitiesKHR`, `VkSurfaceCapabilitiesFullScreenExclusiveEXT`, or `VkSurfaceProtectedCapabilitiesKHR`
- VUID-VkSurfaceCapabilities2KHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique

An application queries if a protected [VkSurfaceKHR](#) is displayable on a specific windowing system using [VkSurfaceProtectedCapabilitiesKHR](#), which **can** be passed in [pNext](#) parameter of [VkSurfaceCapabilities2KHR](#).

The [VkSurfaceProtectedCapabilitiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_surface_protected_capabilities
typedef struct VkSurfaceProtectedCapabilitiesKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           supportsProtected;
} VkSurfaceProtectedCapabilitiesKHR;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [supportsProtected](#) specifies whether a protected swapchain created from [VkPhysicalDeviceSurfaceInfo2KHR::surface](#) for a particular windowing system **can** be displayed on screen or not. If [supportsProtected](#) is [VK_TRUE](#), then creation of swapchains with the [VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR](#) flag set **must** be supported for [surface](#).

Valid Usage (Implicit)

- VUID-VkSurfaceProtectedCapabilitiesKHR-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_SURFACE_PROTECTED_CAPABILITIES_KHR](#)

The [VkSharedPresentSurfaceCapabilitiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_shared_presentable_image
typedef struct VkSharedPresentSurfaceCapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkImageUsageFlags  sharedPresentSupportedUsageFlags;
} VkSharedPresentSurfaceCapabilitiesKHR;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [sharedPresentSupportedUsageFlags](#) is a bitmask of [VkImageUsageFlagBits](#) representing the ways the application **can** use the shared presentable image from a swapchain created with [VkPresentModeKHR](#) set to [VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR](#) or [VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR](#) for the surface on the specified device. [VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT](#) **must** be included in the set but implementations **may** support additional usages.

Valid Usage (Implicit)

- VUID-VkSharedPresentSurfaceCapabilitiesKHR-sType-sType
sType must be `VK_STRUCTURE_TYPE_SHARED_PRESENT_SURFACE_CAPABILITIES_KHR`

The `VkSurfaceCapabilitiesFullScreenExclusiveEXT` structure is defined as:

```
// Provided by VK_EXT_full_screen_exclusive
typedef struct VkSurfaceCapabilitiesFullScreenExclusiveEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           fullScreenExclusiveSupported;
} VkSurfaceCapabilitiesFullScreenExclusiveEXT;
```

- **sType** is a `VkStructureType` value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **fullScreenExclusiveControlSupported** is a boolean describing whether the surface is able to make use of exclusive full-screen access.

This structure **can** be included in the **pNext** chain of `VkSurfaceCapabilities2KHR` to determine support for exclusive full-screen access. If **fullScreenExclusiveSupported** is `VK_FALSE`, it indicates that exclusive full-screen access is not obtainable for this surface.

Applications **must** not attempt to create swapchains with `VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT` set if **fullScreenExclusiveSupported** is `VK_FALSE`.

Valid Usage (Implicit)

- VUID-VkSurfaceCapabilitiesFullScreenExclusiveEXT-sType-sType
sType must be `VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_FULL_SCREEN_EXCLUSIVE_EXT`

To query the basic capabilities of a surface, needed in order to create a swapchain, call:

```
// Provided by VK_EXT_display_surface_counter
VkResult vkGetPhysicalDeviceSurfaceCapabilities2EXT(
    VkPhysicalDevice    physicalDevice,
    VkSurfaceKHR        surface,
    VkSurfaceCapabilities2EXT* pSurfaceCapabilities);
```

- **physicalDevice** is the physical device that will be associated with the swapchain to be created, as described for `vkCreateSwapchainKHR`.
- **surface** is the surface that will be associated with the swapchain.

- `pSurfaceCapabilities` is a pointer to a `VkSurfaceCapabilities2EXT` structure in which the capabilities are returned.

`vkGetPhysicalDeviceSurfaceCapabilities2EXT` behaves similarly to `vkGetPhysicalDeviceSurfaceCapabilitiesKHR`, with the ability to return extended information by adding extending structures to the `pNext` chain of its `pSurfaceCapabilities` parameter.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceCapabilities2EXT-surface-06523
`surface` **must** be a valid `VkSurfaceKHR` handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2EXT-surface-06211
`surface` **must** be supported by `physicalDevice`, as reported by `vkGetPhysicalDeviceSurfaceSupportKHR` or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceCapabilities2EXT-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2EXT-surface-parameter
`surface` **must** be a valid `VkSurfaceKHR` handle
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2EXT-pSurfaceCapabilities-parameter
`pSurfaceCapabilities` **must** be a valid pointer to a `VkSurfaceCapabilities2EXT` structure
- VUID-vkGetPhysicalDeviceSurfaceCapabilities2EXT-commonparent
Both of `physicalDevice`, and `surface` **must** have been created, allocated, or retrieved from the same `VkInstance`

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_SURFACE_LOST_KHR`

The `VkSurfaceCapabilities2EXT` structure is defined as:

```
// Provided by VK_EXT_display_surface_counter
typedef struct VkSurfaceCapabilities2EXT {
    VkStructureType      sType;
    void*                 pNext;
    uint32_t              minImageCount;
    uint32_t              maxImageCount;
    VkExtent2D            currentExtent;
    VkExtent2D            minImageExtent;
    VkExtent2D            maxImageExtent;
    uint32_t              maxImageArrayLayers;
    VkSurfaceTransformFlagsKHR supportedTransforms;
    VkSurfaceTransformFlagBitsKHR currentTransform;
    VkCompositeAlphaFlagsKHR supportedCompositeAlpha;
    VkImageUsageFlags      supportedUsageFlags;
    VkSurfaceCounterFlagsEXT supportedSurfaceCounters;
} VkSurfaceCapabilities2EXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **minImageCount** is the minimum number of images the specified device supports for a swapchain created for the surface, and will be at least one.
- **maxImageCount** is the maximum number of images the specified device supports for a swapchain created for the surface, and will be either 0, or greater than or equal to **minImageCount**. A value of 0 means that there is no limit on the number of images, though there **may** be limits related to the total amount of memory used by presentable images.
- **currentExtent** is the current width and height of the surface, or the special value (0xFFFFFFFF, 0xFFFFFFFF) indicating that the surface size will be determined by the extent of a swapchain targeting the surface.
- **minImageExtent** contains the smallest valid swapchain extent for the surface on the specified device. The **width** and **height** of the extent will each be less than or equal to the corresponding **width** and **height** of **currentExtent**, unless **currentExtent** has the special value described above.
- **maxImageExtent** contains the largest valid swapchain extent for the surface on the specified device. The **width** and **height** of the extent will each be greater than or equal to the corresponding **width** and **height** of **minImageExtent**. The **width** and **height** of the extent will each be greater than or equal to the corresponding **width** and **height** of **currentExtent**, unless **currentExtent** has the special value described above.
- **maxImageArrayLayers** is the maximum number of layers presentable images **can** have for a swapchain created for this device and surface, and will be at least one.
- **supportedTransforms** is a bitmask of [VkSurfaceTransformFlagBitsKHR](#) indicating the presentation transforms supported for the surface on the specified device. At least one bit will be set.
- **currentTransform** is [VkSurfaceTransformFlagBitsKHR](#) value indicating the surface's current transform relative to the presentation engine's natural orientation.
- **supportedCompositeAlpha** is a bitmask of [VkCompositeAlphaFlagBitsKHR](#), representing the alpha

compositing modes supported by the presentation engine for the surface on the specified device, and at least one bit will be set. Opaque composition **can** be achieved in any alpha compositing mode by either using an image format that has no alpha component, or by ensuring that all pixels in the presentable images have an alpha value of 1.0.

- **supportedUsageFlags** is a bitmask of **VkImageUsageFlagBits** representing the ways the application **can** use the presentable images of a swapchain created with **VkPresentModeKHR** set to **VK_PRESENT_MODE_FIFO_LATEST_READY_EXT**, **VK_PRESENT_MODE_IMMEDIATE_KHR**, **VK_PRESENT_MODE_MAILBOX_KHR**, **VK_PRESENT_MODE_FIFO_KHR** or **VK_PRESENT_MODE_FIFO_RELAXED_KHR** for the surface on the specified device. **VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT** **must** be included in the set. Implementations **may** support additional usages.
- **supportedSurfaceCounters** is a bitmask of **VkSurfaceCounterFlagBitsEXT** indicating the supported surface counter types.

Valid Usage

- VUID-VkSurfaceCapabilities2EXT-supportedSurfaceCounters-01246
supportedSurfaceCounters **must** not include **VK_SURFACE_COUNTER_VBLANK_BIT_EXT** unless the surface queried is a **display surface**

Valid Usage (Implicit)

- VUID-VkSurfaceCapabilities2EXT-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_EXT**
- VUID-VkSurfaceCapabilities2EXT-pNext-pNext
pNext **must** be **NULL**

Bits which **can** be set in **VkSurfaceCapabilities2EXT::supportedSurfaceCounters**, indicating supported surface counter types, are:

```
// Provided by VK_EXT_display_surface_counter
typedef enum VkSurfaceCounterFlagBitsEXT {
    VK_SURFACE_COUNTER_VBLANK_BIT_EXT = 0x00000001,
    // VK_SURFACE_COUNTER_VBLANK_EXT is a deprecated alias
    VK_SURFACE_COUNTER_VBLANK_EXT = VK_SURFACE_COUNTER_VBLANK_BIT_EXT,
} VkSurfaceCounterFlagBitsEXT;
```

- **VK_SURFACE_COUNTER_VBLANK_BIT_EXT** specifies a counter incrementing once every time a vertical blanking period occurs on the display associated with the surface.

```
// Provided by VK_EXT_display_surface_counter
typedef VkFlags VkSurfaceCounterFlagsEXT;
```

VkSurfaceCounterFlagsEXT is a bitmask type for setting a mask of zero or more

VkSurfaceCounterFlagBitsEXT.

Bits which **may** be set in [VkSurfaceCapabilitiesKHR::supportedTransforms](#) indicating the presentation transforms supported for the surface on the specified device, and possible values of [VkSurfaceCapabilitiesKHR::currentTransform](#) indicating the surface's current transform relative to the presentation engine's natural orientation, are:

```
// Provided by VK_KHR_surface
typedef enum VkSurfaceTransformFlagBitsKHR {
    VK_SURFACE_TRANSFORM_IDENTITY_BIT_KHR = 0x00000001,
    VK_SURFACE_TRANSFORM_ROTATE_90_BIT_KHR = 0x00000002,
    VK_SURFACE_TRANSFORM_ROTATE_180_BIT_KHR = 0x00000004,
    VK_SURFACE_TRANSFORM_ROTATE_270_BIT_KHR = 0x00000008,
    VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_BIT_KHR = 0x00000010,
    VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_90_BIT_KHR = 0x00000020,
    VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_180_BIT_KHR = 0x00000040,
    VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_270_BIT_KHR = 0x00000080,
    VK_SURFACE_TRANSFORM_INHERIT_BIT_KHR = 0x00000100,
} VkSurfaceTransformFlagBitsKHR;
```

- [VK_SURFACE_TRANSFORM_IDENTITY_BIT_KHR](#) specifies that image content is presented without being transformed.
- [VK_SURFACE_TRANSFORM_ROTATE_90_BIT_KHR](#) specifies that image content is rotated 90 degrees clockwise.
- [VK_SURFACE_TRANSFORM_ROTATE_180_BIT_KHR](#) specifies that image content is rotated 180 degrees clockwise.
- [VK_SURFACE_TRANSFORM_ROTATE_270_BIT_KHR](#) specifies that image content is rotated 270 degrees clockwise.
- [VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_BIT_KHR](#) specifies that image content is mirrored horizontally.
- [VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_90_BIT_KHR](#) specifies that image content is mirrored horizontally, then rotated 90 degrees clockwise.
- [VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_180_BIT_KHR](#) specifies that image content is mirrored horizontally, then rotated 180 degrees clockwise.
- [VK_SURFACE_TRANSFORM_HORIZONTAL_MIRROR_ROTATE_270_BIT_KHR](#) specifies that image content is mirrored horizontally, then rotated 270 degrees clockwise.
- [VK_SURFACE_TRANSFORM_INHERIT_BIT_KHR](#) specifies that the presentation transform is not specified, and is instead determined by platform-specific considerations and mechanisms outside Vulkan.

```
// Provided by VK_KHR_display
typedef VkFlags VkSurfaceTransformFlagsKHR;
```

[VkSurfaceTransformFlagsKHR](#) is a bitmask type for setting a mask of zero or more [VkSurfaceTransformFlagBitsKHR](#).

The `supportedCompositeAlpha` member is of type `VkCompositeAlphaFlagBitsKHR`, containing the following values:

```
// Provided by VK_KHR_surface
typedef enum VkCompositeAlphaFlagBitsKHR {
    VK_COMPOSITE_ALPHA_OPAQUE_BIT_KHR = 0x00000001,
    VK_COMPOSITE_ALPHA_PRE_MULTIPLIED_BIT_KHR = 0x00000002,
    VK_COMPOSITE_ALPHA_POST_MULTIPLIED_BIT_KHR = 0x00000004,
    VK_COMPOSITE_ALPHA_INHERIT_BIT_KHR = 0x00000008,
} VkCompositeAlphaFlagBitsKHR;
```

These values are described as follows:

- `VK_COMPOSITE_ALPHA_OPAQUE_BIT_KHR`: The alpha component, if it exists, of the images is ignored in the compositing process. Instead, the image is treated as if it has a constant alpha of 1.0.
- `VK_COMPOSITE_ALPHA_PRE_MULTIPLIED_BIT_KHR`: The alpha component, if it exists, of the images is respected in the compositing process. The non-alpha components of the image are expected to already be multiplied by the alpha component by the application.
- `VK_COMPOSITE_ALPHA_POST_MULTIPLIED_BIT_KHR`: The alpha component, if it exists, of the images is respected in the compositing process. The non-alpha components of the image are not expected to already be multiplied by the alpha component by the application; instead, the compositor will multiply the non-alpha components of the image by the alpha component during compositing.
- `VK_COMPOSITE_ALPHA_INHERIT_BIT_KHR`: The way in which the presentation engine treats the alpha component in the images is unknown to the Vulkan API. Instead, the application is responsible for setting the composite alpha blending mode using native window system commands. If the application does not set the blending mode using native window system commands, then a platform-specific default will be used.

```
// Provided by VK_KHR_surface
typedef VkFlags VkCompositeAlphaFlagsKHR;
```

`VkCompositeAlphaFlagsKHR` is a bitmask type for setting a mask of zero or more `VkCompositeAlphaFlagBitsKHR`.

31.5.2. Surface Format Support

To query the supported swapchain format-color space pairs for a surface, call:

```
// Provided by VK_KHR_surface
VkResult vkGetPhysicalDeviceSurfaceFormatsKHR(
    VkPhysicalDevice          physicalDevice,
    VkSurfaceKHR              surface,
    uint32_t*                 pSurfaceFormatCount,
    VkSurfaceFormatKHR*       pSurfaceFormats);
```

- `physicalDevice` is the physical device that will be associated with the swapchain to be created, as described for `vkCreateSwapchainKHR`.
- `surface` is the surface that will be associated with the swapchain.
- `pSurfaceFormatCount` is a pointer to an integer related to the number of format pairs available or queried, as described below.
- `pSurfaceFormats` is either `NULL` or a pointer to an array of `VkSurfaceFormatKHR` structures.

If `pSurfaceFormats` is `NULL`, then the number of format pairs supported for the given `surface` is returned in `pSurfaceFormatCount`. Otherwise, `pSurfaceFormatCount` **must** point to a variable set by the application to the number of elements in the `pSurfaceFormats` array, and on return the variable is overwritten with the number of structures actually written to `pSurfaceFormats`. If the value of `pSurfaceFormatCount` is less than the number of format pairs supported, at most `pSurfaceFormatCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available format pairs were returned.

The number of format pairs supported **must** be greater than or equal to 1. `pSurfaceFormats` **must** not contain an entry whose value for `format` is `VK_FORMAT_UNDEFINED`.

If `pSurfaceFormats` includes an entry whose value for `colorSpace` is `VK_COLOR_SPACE_SRGB_NONLINEAR_KHR` and whose value for `format` is a UNORM (or SRGB) format and the corresponding SRGB (or UNORM) format is a color renderable format for `VK_IMAGE_TILING_OPTIMAL`, then `pSurfaceFormats` **must** also contain an entry with the same value for `colorSpace` and `format` equal to the corresponding SRGB (or UNORM) format.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-surface-06524
`surface` **must** be a valid `VkSurfaceKHR` handle
- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-surface-06525
`surface` **must** be supported by `physicalDevice`, as reported by `vkGetPhysicalDeviceSurfaceSupportKHR` or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-surface-parameter
If `surface` is not `VK_NULL_HANDLE`, `surface` **must** be a valid `VkSurfaceKHR` handle
- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-pSurfaceFormatCount-parameter
`pSurfaceFormatCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-pSurfaceFormats-parameter
If the value referenced by `pSurfaceFormatCount` is not 0, and `pSurfaceFormats` is not `NULL`, `pSurfaceFormats` **must** be a valid pointer to an array of `pSurfaceFormatCount` `VkSurfaceFormatKHR` structures

- VUID-vkGetPhysicalDeviceSurfaceFormatsKHR-commonparent
Both of `physicalDevice`, and `surface` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkInstance`

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_SURFACE_LOST_KHR`

The `VkSurfaceFormatKHR` structure is defined as:

```
// Provided by VK_KHR_surface
typedef struct VkSurfaceFormatKHR {
    VkFormat          format;
    VkColorSpaceKHR   colorSpace;
} VkSurfaceFormatKHR;
```

- `format` is a `VkFormat` that is compatible with the specified surface.
- `colorSpace` is a presentation `VkColorSpaceKHR` that is compatible with the surface.

To query the supported swapchain format tuples for a surface, call:

```
// Provided by VK_KHR_get_surface_capabilities2
VkResult vkGetPhysicalDeviceSurfaceFormats2KHR(
    VkPhysicalDevice          physicalDevice,
    const VkPhysicalDeviceSurfaceInfo2KHR* pSurfaceInfo,
    uint32_t*                 pSurfaceFormatCount,
    VkSurfaceFormat2KHR*      pSurfaceFormats);
```

- `physicalDevice` is the physical device that will be associated with the swapchain to be created, as described for `vkCreateSwapchainKHR`.
- `pSurfaceInfo` is a pointer to a `VkPhysicalDeviceSurfaceInfo2KHR` structure describing the surface and other fixed parameters that would be consumed by `vkCreateSwapchainKHR`.
- `pSurfaceFormatCount` is a pointer to an integer related to the number of format tuples available or queried, as described below.
- `pSurfaceFormats` is either `NULL` or a pointer to an array of `VkSurfaceFormat2KHR` structures.

`vkGetPhysicalDeviceSurfaceFormats2KHR` behaves similarly to `vkGetPhysicalDeviceSurfaceFormatsKHR`, with the ability to be extended via `pNext` chains.

If `pSurfaceFormats` is `NULL`, then the number of format tuples supported for the given `surface` is returned in `pSurfaceFormatCount`. Otherwise, `pSurfaceFormatCount` **must** point to a variable set by the application to the number of elements in the `pSurfaceFormats` array, and on return the variable is overwritten with the number of structures actually written to `pSurfaceFormats`. If the value of `pSurfaceFormatCount` is less than the number of format tuples supported, at most `pSurfaceFormatCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available values were returned.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-pSurfaceInfo-06521
`pSurfaceInfo->surface` **must** be a valid `VkSurfaceKHR` handle
- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-pSurfaceInfo-06522
`pSurfaceInfo->surface` **must** be supported by `physicalDevice`, as reported by `vkGetPhysicalDeviceSurfaceSupportKHR` or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-pSurfaceInfo-parameter
`pSurfaceInfo` **must** be a valid pointer to a valid `VkPhysicalDeviceSurfaceInfo2KHR` structure
- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-pSurfaceFormatCount-parameter
`pSurfaceFormatCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceSurfaceFormats2KHR-pSurfaceFormats-parameter
If the value referenced by `pSurfaceFormatCount` is not 0, and `pSurfaceFormats` is not `NULL`, `pSurfaceFormats` **must** be a valid pointer to an array of `pSurfaceFormatCount` `VkSurfaceFormat2KHR` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_SURFACE_LOST_KHR`

The `VkSurfaceFormat2KHR` structure is defined as:

```
// Provided by VK_KHR_get_surface_capabilities2
typedef struct VkSurfaceFormat2KHR {
    VkStructureType    sType;
    void*              pNext;
    VkSurfaceFormatKHR surfaceFormat;
} VkSurfaceFormat2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `surfaceFormat` is a `VkSurfaceFormatKHR` structure describing a format-color space pair that is compatible with the specified surface.

Valid Usage (Implicit)

- VUID-VkSurfaceFormat2KHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SURFACE_FORMAT_2_KHR`
- VUID-VkSurfaceFormat2KHR-pNext-pNext
`pNext` **must** be `NULL`

While the `format` of a presentable image refers to the encoding of each pixel, the `colorSpace` determines how the presentation engine interprets the pixel values. A color space in this document refers to a specific color space (defined by the chromaticities of its primaries and a white point in CIE Lab), and transfer functions indicating the mapping between the image data and the colorimetry with respect to the given color space.

Possible values of `VkSurfaceFormatKHR::colorSpace`, specifying the color spaces that a presentation engine can accept, are:


```

// Provided by VK_KHR_surface
typedef enum VkColorSpaceKHR {
    VK_COLOR_SPACE_SRGB_NONLINEAR_KHR = 0,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_DISPLAY_P3_NONLINEAR_EXT = 1000104001,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_EXTENDED_SRGB_LINEAR_EXT = 1000104002,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_DISPLAY_P3_LINEAR_EXT = 1000104003,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_DCI_P3_NONLINEAR_EXT = 1000104004,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_BT709_LINEAR_EXT = 1000104005,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_BT709_NONLINEAR_EXT = 1000104006,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_BT2020_LINEAR_EXT = 1000104007,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_HDR10_ST2084_EXT = 1000104008,
    // Provided by VK_EXT_swapchain_colorspace
    // VK_COLOR_SPACE_DOLBYVISION_EXT is deprecated, but no reason was given in the API
XML
    VK_COLOR_SPACE_DOLBYVISION_EXT = 1000104009,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_HDR10_HLG_EXT = 1000104010,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_ADOBERGB_LINEAR_EXT = 1000104011,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_ADOBERGB_NONLINEAR_EXT = 1000104012,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_PASS_THROUGH_EXT = 1000104013,
    // Provided by VK_EXT_swapchain_colorspace
    VK_COLOR_SPACE_EXTENDED_SRGB_NONLINEAR_EXT = 1000104014,
    // VK_COLORSPACE_SRGB_NONLINEAR_KHR is a deprecated alias
    VK_COLORSPACE_SRGB_NONLINEAR_KHR = VK_COLOR_SPACE_SRGB_NONLINEAR_KHR,
    // Provided by VK_EXT_swapchain_colorspace
    // VK_COLOR_SPACE_DCI_P3_LINEAR_EXT is a deprecated alias
    VK_COLOR_SPACE_DCI_P3_LINEAR_EXT = VK_COLOR_SPACE_DISPLAY_P3_LINEAR_EXT,
} VkColorSpaceKHR;

```

- **VK_COLOR_SPACE_SRGB_NONLINEAR_KHR** specifies support for the images in sRGB color space, encoded according to the sRGB specification.
- **VK_COLOR_SPACE_DISPLAY_P3_NONLINEAR_EXT** specifies support for the images in Display-P3 color space, encoded using a Display-P3 transfer function.
- **VK_COLOR_SPACE_EXTENDED_SRGB_LINEAR_EXT** specifies support for the images in extended sRGB color space, encoded using a linear transfer function.
- **VK_COLOR_SPACE_EXTENDED_SRGB_NONLINEAR_EXT** specifies support for the images in extended sRGB color space, encoded according to the scRGB specification.

- `VK_COLOR_SPACE_DISPLAY_P3_LINEAR_EXT` specifies support for the images in Display-P3 color space, encoded using a linear transfer function.
- `VK_COLOR_SPACE_DCI_P3_NONLINEAR_EXT` specifies support for the images in DCI-P3 color space, encoded according to the DCI-P3 specification. Note that values in such an image are interpreted as XYZ encoded color data by the presentation engine.
- `VK_COLOR_SPACE_BT709_LINEAR_EXT` specifies support for the images in BT709 color space, encoded using a linear transfer function.
- `VK_COLOR_SPACE_BT709_NONLINEAR_EXT` specifies support for the images in BT709 color space, encoded according to the BT709 specification.
- `VK_COLOR_SPACE_BT2020_LINEAR_EXT` specifies support for the images in BT2020 color space, encoded using a linear transfer function.
- `VK_COLOR_SPACE_HDR10_ST2084_EXT` specifies support for the images in HDR10 (BT2020) color space, encoded according to SMPTE ST2084 Perceptual Quantizer (PQ) specification.
- `VK_COLOR_SPACE_HDR10_HLG_EXT` specifies support for the images in HDR10 (BT2020) color space, encoded according to the Hybrid Log Gamma (HLG) specification.
- `VK_COLOR_SPACE_ADOBERGB_LINEAR_EXT` specifies support for images in Adobe RGB color space, encoded using a linear transfer function.
- `VK_COLOR_SPACE_ADOBERGB_NONLINEAR_EXT` specifies support for the images in Adobe RGB color space, encoded according to the Adobe RGB specification (approximately Gamma 2.2).
- `VK_COLOR_SPACE_PASS_THROUGH_EXT` specifies that color components are used “as is”. This is intended to allow applications to supply data for color spaces not described here.

NOTE

In the initial release of the `VK_KHR_surface` and `VK_KHR_swapchain` extensions, the token `VK_COLORSPACE_SRGB_NONLINEAR_KHR` was used. Starting in the 2016-05-13 updates to the extension branches, matching release 1.0.13 of the core API specification, `VK_COLOR_SPACE_SRGB_NONLINEAR_KHR` is used instead for consistency with Vulkan naming rules. The older enum is still available for backwards compatibility.

NOTE

In older versions of this extension `VK_COLOR_SPACE_DISPLAY_P3_LINEAR_EXT` was misnamed `VK_COLOR_SPACE_DCI_P3_LINEAR_EXT`. This has been updated to indicate that it uses RGB color encoding, not XYZ. The old name is deprecated but is maintained for backwards compatibility.

NOTE

In older versions of the `VK_EXT_swapchain_colorspace` extension, `VK_COLOR_SPACE_DOLBYVISION_EXT` was exposed. The intent was to indicate the presentation engine shall decode an image using the SMPTE ST 2084 Perceptual Quantizer (PQ) EOTF, and then apply a proprietary OOTF to process the image. However, Dolby Vision profile 8.4 describes an encoding using the Hybrid Log Gamma (HLG) OETF, and there is no swapchain extension for signaling Dolby Vision metadata to be used by a proprietary OOTF. This enum is deprecated but is maintained for backwards compatibility.

Note

NOTE

For a traditional “Linear” or non-gamma transfer function color space use `VK_COLOR_SPACE_PASS_THROUGH_EXT`.

The presentation engine interprets the pixel values of the R, G, and B components as having been encoded using an appropriate transfer function. Applications **should** ensure that the appropriate transfer function has been applied. [Textures Output Format Conversion](#) requires that all implementations implicitly apply the sRGB EOTF¹ on R, G, and B components when shaders write to an sRGB pixel format image, which is useful for sRGB color spaces. For sRGB color spaces with other pixel formats, or other non-linear color spaces, applications **can** apply the transfer function explicitly in a shader. The A channel is always interpreted as linearly encoded.

This extension defines enums for [VkColorSpaceKHR](#) that correspond to the following color spaces:

Table 44. Color Spaces and Attributes

Name	Red Primary	Green Primary	Blue Primary	White-point	Transfer function
DCI-P3	1.000, 0.000	0.000, 1.000	0.000, 0.000	0.3333, 0.3333	DCI P3
Display-P3	0.680, 0.320	0.265, 0.690	0.150, 0.060	0.3127, 0.3290 (D65)	Display-P3
BT709	0.640, 0.330	0.300, 0.600	0.150, 0.060	0.3127, 0.3290 (D65)	BT709
sRGB	0.640, 0.330	0.300, 0.600	0.150, 0.060	0.3127, 0.3290 (D65)	sRGB
extended sRGB	0.640, 0.330	0.300, 0.600	0.150, 0.060	0.3127, 0.3290 (D65)	scRGB
HDR10_ST2084	0.708, 0.292	0.170, 0.797	0.131, 0.046	0.3127, 0.3290 (D65)	ST2084 PQ
HDR10_HLG	0.708, 0.292	0.170, 0.797	0.131, 0.046	0.3127, 0.3290 (D65)	HLG
Adobe RGB	0.640, 0.330	0.210, 0.710	0.150, 0.060	0.3127, 0.3290 (D65)	Adobe RGB

The transfer functions are described in the “Transfer Functions” chapter of the [Khronos Data Format Specification](#).

Except Display-P3 OETF, which is:

$$E = \begin{cases} 1.055 \times L^{\frac{1}{2.4}} - 0.055 & \text{for } 0.0030186 \leq L \leq 1 \\ 12.92 \times L & \text{for } 0 \leq L < 0.0030186 \end{cases}$$

where L is the linear value of a color component and E is the encoded value (as stored in the image in memory).

NOTE For most uses, the sRGB OETF is equivalent.

31.5.3. Surface Presentation Mode Support

To query the supported presentation modes for a surface, call:

```
// Provided by VK_KHR_surface
VkResult vkGetPhysicalDeviceSurfacePresentModesKHR(
    VkPhysicalDevice      physicalDevice,
    VkSurfaceKHR          surface,
    uint32_t*             pPresentModeCount,
    VkPresentModeKHR*     pPresentModes);
```

- **physicalDevice** is the physical device that will be associated with the swapchain to be created, as described for [vkCreateSwapchainKHR](#).
- **surface** is the surface that will be associated with the swapchain.
- **pPresentModeCount** is a pointer to an integer related to the number of presentation modes available or queried, as described below.
- **pPresentModes** is either **NULL** or a pointer to an array of [VkPresentModeKHR](#) values, indicating the supported presentation modes.

If **pPresentModes** is **NULL**, then the number of presentation modes supported for the given **surface** is returned in **pPresentModeCount**. Otherwise, **pPresentModeCount** **must** point to a variable set by the application to the number of elements in the **pPresentModes** array, and on return the variable is overwritten with the number of values actually written to **pPresentModes**. If the value of **pPresentModeCount** is less than the number of presentation modes supported, at most **pPresentModeCount** values will be written, and **VK_INCOMPLETE** will be returned instead of **VK_SUCCESS**, to indicate that not all the available modes were returned.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-surface-06524
surface **must** be a valid [VkSurfaceKHR](#) handle
- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-surface-06525
surface **must** be supported by **physicalDevice**, as reported by [vkGetPhysicalDeviceSurfaceSupportKHR](#) or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-physicalDevice-parameter
physicalDevice **must** be a valid [VkPhysicalDevice](#) handle
- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-surface-parameter
If **surface** is not **VK_NULL_HANDLE**, **surface** **must** be a valid [VkSurfaceKHR](#) handle

- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-pPresentModeCount-parameter
`pPresentModeCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-pPresentModes-parameter
If the value referenced by `pPresentModeCount` is not 0, and `pPresentModes` is not `NULL`, `pPresentModes` **must** be a valid pointer to an array of `pPresentModeCount` `VkPresentModeKHR` values
- VUID-vkGetPhysicalDeviceSurfacePresentModesKHR-commonparent
Both of `physicalDevice`, and `surface` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkInstance`

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_SURFACE_LOST_KHR`

Alternatively, to query the supported presentation modes for a surface combined with select other fixed swapchain creation parameters, call:

```
// Provided by VK_EXT_full_screen_exclusive
VkResult vkGetPhysicalDeviceSurfacePresentModes2EXT(
    VkPhysicalDevice          physicalDevice,
    const VkPhysicalDeviceSurfaceInfo2KHR* pSurfaceInfo,
    uint32_t*                 pPresentModeCount,
    VkPresentModeKHR*         pPresentModes);
```

- `physicalDevice` is the physical device that will be associated with the swapchain to be created, as described for `vkCreateSwapchainKHR`.
- `pSurfaceInfo` is a pointer to a `VkPhysicalDeviceSurfaceInfo2KHR` structure describing the surface and other fixed parameters that would be consumed by `vkCreateSwapchainKHR`.
- `pPresentModeCount` is a pointer to an integer related to the number of presentation modes available or queried, as described below.
- `pPresentModes` is either `NULL` or a pointer to an array of `VkPresentModeKHR` values, indicating the supported presentation modes.

`vkGetPhysicalDeviceSurfacePresentModes2EXT` behaves similarly to `vkGetPhysicalDeviceSurfacePresentModesKHR`, with the ability to specify extended inputs via chained input structures.

Valid Usage

- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-pSurfaceInfo-06521
`pSurfaceInfo->surface` **must** be a valid `VkSurfaceKHR` handle
- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-pSurfaceInfo-06522
`pSurfaceInfo->surface` **must** be supported by `physicalDevice`, as reported by `vkGetPhysicalDeviceSurfaceSupportKHR` or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-pSurfaceInfo-parameter
`pSurfaceInfo` **must** be a valid pointer to a valid `VkPhysicalDeviceSurfaceInfo2KHR` structure
- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-pPresentModeCount-parameter
`pPresentModeCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceSurfacePresentModes2EXT-pPresentModes-parameter
If the value referenced by `pPresentModeCount` is not 0, and `pPresentModes` is not NULL, `pPresentModes` **must** be a valid pointer to an array of `pPresentModeCount` `VkPresentModeKHR` values

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_SURFACE_LOST_KHR`

Possible values of elements of the `vkGetPhysicalDeviceSurfacePresentModesKHR::pPresentModes` array, indicating the supported presentation modes for a surface, are:

```
// Provided by VK_KHR_surface
typedef enum VkPresentModeKHR {
    VK_PRESENT_MODE_IMMEDIATE_KHR = 0,
    VK_PRESENT_MODE_MAILBOX_KHR = 1,
    VK_PRESENT_MODE_FIFO_KHR = 2,
    VK_PRESENT_MODE_FIFO_RELAXED_KHR = 3,
    // Provided by VK_KHR_shared_presentable_image
    VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR = 1000111000,
    // Provided by VK_KHR_shared_presentable_image
    VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR = 1000111001,
    // Provided by VK_EXT_present_mode_fifo_latest_ready
    VK_PRESENT_MODE_FIFO_LATEST_READY_EXT = 1000361000,
} VkPresentModeKHR;
```

- **VK_PRESENT_MODE_IMMEDIATE_KHR** specifies that the presentation engine does not wait for a vertical blanking period to update the current image, meaning this mode **may** result in visible tearing. No internal queuing of presentation requests is needed, as the requests are applied immediately.
- **VK_PRESENT_MODE_MAILBOX_KHR** specifies that the presentation engine waits for the next vertical blanking period to update the current image. Tearing **cannot** be observed. An internal single-entry queue is used to hold pending presentation requests. If the queue is full when a new presentation request is received, the new request replaces the existing entry, and any images associated with the prior entry become available for reuse by the application. One request is removed from the queue and processed during each vertical blanking period in which the queue is non-empty.
- **VK_PRESENT_MODE_FIFO_KHR** specifies that the presentation engine waits for the next vertical blanking period to update the current image. Tearing **cannot** be observed. An internal queue is used to hold pending presentation requests. New requests are appended to the end of the queue, and one request is removed from the beginning of the queue and processed during each vertical blanking period in which the queue is non-empty. This is the only value of **presentMode** that is **required** to be supported.
- **VK_PRESENT_MODE_FIFO_RELAXED_KHR** specifies that the presentation engine generally waits for the next vertical blanking period to update the current image. If a vertical blanking period has already passed since the last update of the current image then the presentation engine does not wait for another vertical blanking period for the update, meaning this mode **may** result in visible tearing in this case. This mode is useful for reducing visual stutter with an application that will mostly present a new image before the next vertical blanking period, but may occasionally be late, and present a new image just after the next vertical blanking period. An internal queue is used to hold pending presentation requests. New requests are appended to the end of the queue, and one request is removed from the beginning of the queue and processed during or after each vertical blanking period in which the queue is non-empty.
- **VK_PRESENT_MODE_FIFO_LATEST_READY_EXT** specifies that the presentation engine waits for the next vertical blanking period to update the current image. Tearing **cannot** be observed. An internal queue is used to hold pending presentation requests. New requests are appended to the end of the queue. At each vertical blanking period, the presentation engine dequeues all successive requests that are ready to be presented from the beginning of the queue. The image of the last

dequeued request will be presented. The other dequeued requests will be dropped.

- **VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR** specifies that the presentation engine and application have concurrent access to a single image, which is referred to as a *shared presentable image*. The presentation engine is only required to update the current image after a new presentation request is received. Therefore the application **must** make a presentation request whenever an update is required. However, the presentation engine **may** update the current image at any point, meaning this mode **may** result in visible tearing.
- **VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR** specifies that the presentation engine and application have concurrent access to a single image, which is referred to as a *shared presentable image*. The presentation engine periodically updates the current image on its regular refresh cycle. The application is only required to make one initial presentation request, after which the presentation engine **must** update the current image without any need for further presentation requests. The application **can** indicate the image contents have been updated by making a presentation request, but this does not guarantee the timing of when it will be updated. This mode **may** result in visible tearing if rendering to the image is not timed correctly.

The supported **VkImageUsageFlagBits** of the presentable images of a swapchain created for a surface **may** differ depending on the presentation mode, and can be determined as per the table below:

Table 45. Presentable Image Usage Queries

Presentation mode	Image usage flags
VK_PRESENT_MODE_IMMEDIATE_KHR	VkSurfaceCapabilitiesKHR::supportedUsageFlags
VK_PRESENT_MODE_MAILBOX_KHR	VkSurfaceCapabilitiesKHR::supportedUsageFlags
VK_PRESENT_MODE_FIFO_KHR	VkSurfaceCapabilitiesKHR::supportedUsageFlags
VK_PRESENT_MODE_FIFO_RELAXED_KHR	VkSurfaceCapabilitiesKHR::supportedUsageFlags
VK_PRESENT_MODE_FIFO_LATEST_READY_EXT	VkSurfaceCapabilitiesKHR::supportedUsageFlags
VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR	VkSharedPresentSurfaceCapabilitiesKHR::sharedPresentSupportedUsageFlags
VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR	VkSharedPresentSurfaceCapabilitiesKHR::sharedPresentSupportedUsageFlags

NOTE

For reference, the mode indicated by **VK_PRESENT_MODE_FIFO_KHR** is equivalent to the behavior of {wgl|glX|egl}SwapBuffers with a swap interval of 1, while the mode indicated by **VK_PRESENT_MODE_FIFO_RELAXED_KHR** is equivalent to the behavior of {wgl|glX}SwapBuffers with a swap interval of -1 (from the {WGL|GLX}_EXT_swap_control_tear extensions).

31.6. Full Screen Exclusive Control

Swapchains created with **fullScreenExclusive** set to **VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT** **must** acquire and release exclusive full-screen access explicitly, using the following commands.

To acquire exclusive full-screen access for a swapchain, call:

```
// Provided by VK_EXT_full_screen_exclusive
VkResult vkAcquireFullScreenExclusiveModeEXT(
    VkDevice          device,
    VkSwapchainKHR    swapchain);
```

- **device** is the device associated with **swapchain**.
- **swapchain** is the swapchain to acquire exclusive full-screen access for.

Valid Usage

- VUID-vkAcquireFullScreenExclusiveModeEXT-swapchain-02674
swapchain **must** not be in the retired state
- VUID-vkAcquireFullScreenExclusiveModeEXT-swapchain-02675
swapchain **must** be a swapchain created with a **VkSurfaceFullScreenExclusiveInfoEXT** structure, with **fullScreenExclusive** set to **VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT**
- VUID-vkAcquireFullScreenExclusiveModeEXT-swapchain-02676
swapchain **must** not currently have exclusive full-screen access

A return value of **VK_SUCCESS** indicates that the **swapchain** successfully acquired exclusive full-screen access. The swapchain will retain this exclusivity until either the application releases exclusive full-screen access with **vkReleaseFullScreenExclusiveModeEXT**, destroys the swapchain, or if any of the swapchain commands return **VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT** indicating that the mode was lost because of platform-specific changes.

If the swapchain was unable to acquire exclusive full-screen access to the display then **VK_ERROR_INITIALIZATION_FAILED** is returned. An application **can** attempt to acquire exclusive full-screen access again for the same swapchain even if this command fails, or if **VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT** has been returned by a swapchain command.

Valid Usage (Implicit)

- VUID-vkAcquireFullScreenExclusiveModeEXT-device-parameter
device **must** be a valid **VkDevice** handle
- VUID-vkAcquireFullScreenExclusiveModeEXT-swapchain-parameter
swapchain **must** be a valid **VkSwapchainKHR** handle
- VUID-vkAcquireFullScreenExclusiveModeEXT-swapchain-parent
swapchain **must** have been created, allocated, or retrieved from **device**

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`
- `VK_ERROR_SURFACE_LOST_KHR`

To release exclusive full-screen access from a swapchain, call:

```
// Provided by VK_EXT_full_screen_exclusive
VkResult vkReleaseFullScreenExclusiveModeEXT(
    VkDevice          device,
    VkSwapchainKHR    swapchain);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the swapchain to release exclusive full-screen access from.

NOTE

Applications will not be able to present to `swapchain` after this call until exclusive full-screen access is reacquired. This is usually useful to handle when an application is minimized or otherwise intends to stop presenting for a time.

Valid Usage

- VUID-vkReleaseFullScreenExclusiveModeEXT-swapchain-02677
`swapchain` **must** not be in the retired state
- VUID-vkReleaseFullScreenExclusiveModeEXT-swapchain-02678
`swapchain` **must** be a swapchain created with a `VkSurfaceFullScreenExclusiveInfoEXT` structure, with `fullScreenExclusive` set to `VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT`

Valid Usage (Implicit)

- VUID-vkReleaseFullScreenExclusiveModeEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkReleaseFullScreenExclusiveModeEXT-swapchain-parameter
`swapchain` **must** be a valid `VkSwapchainKHR` handle
- VUID-vkReleaseFullScreenExclusiveModeEXT-swapchain-parent
`swapchain` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_SURFACE_LOST_KHR`

31.7. Device Group Queries

A logical device that represents multiple physical devices **may** support presenting from images on more than one physical device, or combining images from multiple physical devices.

To query these capabilities, call:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
VkResult vkGetDeviceGroupPresentCapabilitiesKHR(
    VkDevice                device,
    VkDeviceGroupPresentCapabilitiesKHR* pDeviceGroupPresentCapabilities);
```

- `device` is the logical device.
- `pDeviceGroupPresentCapabilities` is a pointer to a `VkDeviceGroupPresentCapabilitiesKHR` structure in which the device's capabilities are returned.

Valid Usage (Implicit)

- VUID-vkGetDeviceGroupPresentCapabilitiesKHR-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkGetDeviceGroupPresentCapabilitiesKHR-pDeviceGroupPresentCapabilities-parameter `pDeviceGroupPresentCapabilities` **must** be a valid pointer to a `VkDeviceGroupPresentCapabilitiesKHR` structure

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`

- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkDeviceGroupPresentCapabilitiesKHR` structure is defined as:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
typedef struct VkDeviceGroupPresentCapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           presentMask[VK_MAX_DEVICE_GROUP_SIZE];
    VkDeviceGroupPresentModeFlagsKHR    modes;
} VkDeviceGroupPresentCapabilitiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentMask` is an array of `VK_MAX_DEVICE_GROUP_SIZE` `uint32_t` masks, where the mask at element `i` is non-zero if physical device `i` has a presentation engine, and where bit `j` is set in element `i` if physical device `i` **can** present swapchain images from physical device `j`. If element `i` is non-zero, then bit `i` **must** be set.
- `modes` is a bitmask of `VkDeviceGroupPresentModeFlagBitsKHR` indicating which device group presentation modes are supported.

`modes` always has `VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR` set.

The present mode flags are also used when presenting an image, in `VkDeviceGroupPresentInfoKHR::mode`.

If a device group only includes a single physical device, then `modes` **must** equal `VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR`.

Valid Usage (Implicit)

- VUID-VkDeviceGroupPresentCapabilitiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_CAPABILITIES_KHR`
- VUID-VkDeviceGroupPresentCapabilitiesKHR-pNext-pNext
`pNext` **must** be `NULL`

Bits which **may** be set in `VkDeviceGroupPresentCapabilitiesKHR::modes`, indicating which device group presentation modes are supported, are:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
typedef enum VkDeviceGroupPresentModeFlagBitsKHR {
    VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR = 0x00000001,
    VK_DEVICE_GROUP_PRESENT_MODE_REMOTE_BIT_KHR = 0x00000002,
    VK_DEVICE_GROUP_PRESENT_MODE_SUM_BIT_KHR = 0x00000004,
    VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_MULTI_DEVICE_BIT_KHR = 0x00000008,
} VkDeviceGroupPresentModeFlagBitsKHR;
```

- **VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR** specifies that any physical device with a presentation engine **can** present its own swapchain images.
- **VK_DEVICE_GROUP_PRESENT_MODE_REMOTE_BIT_KHR** specifies that any physical device with a presentation engine **can** present swapchain images from any physical device in its **presentMask**.
- **VK_DEVICE_GROUP_PRESENT_MODE_SUM_BIT_KHR** specifies that any physical device with a presentation engine **can** present the sum of swapchain images from any physical devices in its **presentMask**.
- **VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_MULTI_DEVICE_BIT_KHR** specifies that multiple physical devices with a presentation engine **can** each present their own swapchain images.

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
typedef VkFlags VkDeviceGroupPresentModeFlagsKHR;
```

VkDeviceGroupPresentModeFlagsKHR is a bitmask type for setting a mask of zero or more **VkDeviceGroupPresentModeFlagBitsKHR**.

Some surfaces **may** not be capable of using all the device group present modes.

To query the supported device group present modes for a particular surface, call:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
VkResult vkGetDeviceGroupSurfacePresentModesKHR(
    VkDevice device,
    VkSurfaceKHR surface,
    VkDeviceGroupPresentModeFlagsKHR* pModes);
```

- **device** is the logical device.
- **surface** is the surface.
- **pModes** is a pointer to a **VkDeviceGroupPresentModeFlagsKHR** in which the supported device group present modes for the surface are returned.

The modes returned by this command are not invariant, and **may** change in response to the surface being moved, resized, or occluded. These modes **must** be a subset of the modes returned by **vkGetDeviceGroupPresentCapabilitiesKHR**.

Valid Usage

- VUID-vkGetDeviceGroupSurfacePresentModesKHR-surface-06212
surface **must** be supported by all physical devices associated with **device**, as reported by [vkGetPhysicalDeviceSurfaceSupportKHR](#) or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetDeviceGroupSurfacePresentModesKHR-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkGetDeviceGroupSurfacePresentModesKHR-surface-parameter
surface **must** be a valid [VkSurfaceKHR](#) handle
- VUID-vkGetDeviceGroupSurfacePresentModesKHR-pModes-parameter
pModes **must** be a valid pointer to a [VkDeviceGroupPresentModeFlagsKHR](#) value
- VUID-vkGetDeviceGroupSurfacePresentModesKHR-commonparent
Both of **device**, and **surface** **must** have been created, allocated, or retrieved from the same [VkInstance](#)

Host Synchronization

- Host access to **surface** **must** be externally synchronized

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)
- [VK_ERROR_SURFACE_LOST_KHR](#)

Alternatively, to query the supported device group presentation modes for a surface combined with select other fixed swapchain creation parameters, call:

```
// Provided by VK_EXT_full_screen_exclusive with VK_KHR_device_group or VK_VERSION_1_1
VkResult vkGetDeviceGroupSurfacePresentModes2EXT(
    VkDevice device,
    const VkPhysicalDeviceSurfaceInfo2KHR* pSurfaceInfo,
    VkDeviceGroupPresentModeFlagsKHR* pModes);
```


- **device** is the logical device.
- **pSurfaceInfo** is a pointer to a [VkPhysicalDeviceSurfaceInfo2KHR](#) structure describing the surface and other fixed parameters that would be consumed by [vkCreateSwapchainKHR](#).
- **pModes** is a pointer to a [VkDeviceGroupPresentModeFlagsKHR](#) in which the supported device group present modes for the surface are returned.

[vkGetDeviceGroupSurfacePresentModes2EXT](#) behaves similarly to [vkGetDeviceGroupSurfacePresentModesKHR](#), with the ability to specify extended inputs via chained input structures.

Valid Usage

- VUID-vkGetDeviceGroupSurfacePresentModes2EXT-pSurfaceInfo-06213
pSurfaceInfo->surface **must** be supported by all physical devices associated with **device**, as reported by [vkGetPhysicalDeviceSurfaceSupportKHR](#) or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetDeviceGroupSurfacePresentModes2EXT-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkGetDeviceGroupSurfacePresentModes2EXT-pSurfaceInfo-parameter
pSurfaceInfo **must** be a valid pointer to a valid [VkPhysicalDeviceSurfaceInfo2KHR](#) structure
- VUID-vkGetDeviceGroupSurfacePresentModes2EXT-pModes-parameter
pModes **must** be a valid pointer to a [VkDeviceGroupPresentModeFlagsKHR](#) value

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**
- **VK_ERROR_SURFACE_LOST_KHR**

When using [VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_MULTI_DEVICE_BIT_KHR](#), the application **may** need to know which regions of the surface are used when presenting locally on each physical device. Presentation of swapchain images to this surface need only have valid contents in the regions returned by this command.

To query a set of rectangles used in presentation on the physical device, call:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_surface
VkResult vkGetPhysicalDevicePresentRectanglesKHR(
    VkPhysicalDevice          physicalDevice,
    VkSurfaceKHR              surface,
    uint32_t*                 pRectCount,
    VkRect2D*                 pRects);
```

- `physicalDevice` is the physical device.
- `surface` is the surface.
- `pRectCount` is a pointer to an integer related to the number of rectangles available or queried, as described below.
- `pRects` is either `NULL` or a pointer to an array of `VkRect2D` structures.

If `pRects` is `NULL`, then the number of rectangles used when presenting the given `surface` is returned in `pRectCount`. Otherwise, `pRectCount` **must** point to a variable set by the application to the number of elements in the `pRects` array, and on return the variable is overwritten with the number of structures actually written to `pRects`. If the value of `pRectCount` is less than the number of rectangles, at most `pRectCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available rectangles were returned.

The values returned by this command are not invariant, and **may** change in response to the surface being moved, resized, or occluded.

The rectangles returned by this command **must** not overlap.

Valid Usage

- VUID-vkGetPhysicalDevicePresentRectanglesKHR-surface-06523
`surface` **must** be a valid `VkSurfaceKHR` handle
- VUID-vkGetPhysicalDevicePresentRectanglesKHR-surface-06211
`surface` **must** be supported by `physicalDevice`, as reported by `vkGetPhysicalDeviceSurfaceSupportKHR` or an equivalent platform-specific mechanism

Valid Usage (Implicit)

- VUID-vkGetPhysicalDevicePresentRectanglesKHR-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDevicePresentRectanglesKHR-surface-parameter
`surface` **must** be a valid `VkSurfaceKHR` handle
- VUID-vkGetPhysicalDevicePresentRectanglesKHR-pRectCount-parameter
`pRectCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDevicePresentRectanglesKHR-pRects-parameter
If the value referenced by `pRectCount` is not 0, and `pRects` is not `NULL`, `pRects` **must** be a

valid pointer to an array of `pRectCount` `VkRect2D` structures

- `VUID-vkGetPhysicalDevicePresentRectanglesKHR-commonparent`

Both of `physicalDevice`, and `surface` **must** have been created, allocated, or retrieved from the same `VkInstance`

Host Synchronization

- Host access to `surface` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

31.8. Present Wait

Applications wanting to control the pacing of the application by monitoring when presentation processes have completed to limit the number of outstanding images queued for presentation, need to have a method of being signaled during the presentation process.

Providing a mechanism which allows applications to block, waiting for a specific step of the presentation process to complete allows them to control the amount of outstanding work (and hence the potential lag in responding to user input or changes in the rendering environment).

The `VK_KHR_present_wait` extension allows applications to tell the presentation engine at the `vkQueuePresentKHR` call that it plans on waiting for presentation by passing a `VkPresentIdKHR` structure. The `presentId` passed in that structure may then be passed to a future `vkWaitForPresentKHR` call to cause the application to block until that presentation is finished.

31.9. WSI Swapchain

A swapchain object (a.k.a. swapchain) provides the ability to present rendering results to a surface. Swapchain objects are represented by `VkSwapchainKHR` handles:

```
// Provided by VK_KHR_swapchain
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkSwapchainKHR)
```

A swapchain is an abstraction for an array of presentable images that are associated with a surface.

The presentable images are represented by `VkImage` objects created by the platform. One image (which **can** be an array image for multiview/stereoscopic-3D surfaces) is displayed at a time, but multiple images **can** be queued for presentation. An application renders to the image, and then queues the image for presentation to the surface.

A native window **cannot** be associated with more than one non-retired swapchain at a time. Further, swapchains **cannot** be created for native windows that have a non-Vulkan graphics API surface associated with them.

NOTE

The presentation engine is an abstraction for the platform's compositor or display engine.

The presentation engine **may** be synchronous or asynchronous with respect to the application and/or logical device.

Some implementations **may** use the device's graphics queue or dedicated presentation hardware to perform presentation.

The presentable images of a swapchain are owned by the presentation engine. An application **can** acquire use of a presentable image from the presentation engine. Use of a presentable image **must** occur only after the image is returned by `vkAcquireNextImageKHR`, and before it is released by `vkQueuePresentKHR`. This includes transitioning the image layout and rendering commands.

An application **can** acquire use of a presentable image with `vkAcquireNextImageKHR`. After acquiring a presentable image and before modifying it, the application **must** use a synchronization primitive to ensure that the presentation engine has finished reading from the image. The application **can** then transition the image's layout, queue rendering commands to it, etc. Finally, the application presents the image with `vkQueuePresentKHR`, which releases the acquisition of the image.

The presentation engine controls the order in which presentable images are acquired for use by the application.

NOTE

This allows the platform to handle situations which require out-of-order return of images after presentation. At the same time, it allows the application to generate command buffers referencing all of the images in the swapchain at initialization time, rather than in its main loop.

How this all works is described below.

If a swapchain is created with `presentMode` set to either `VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR` or `VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR`, a single presentable image **can** be acquired, referred to as a shared presentable image. A shared presentable image **may** be concurrently accessed by the application and the presentation engine, without transitioning the image's layout after it is initially presented.

- With `VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR`, the presentation engine is only required to update to the latest contents of a shared presentable image after a present. The application **must** call `vkQueuePresentKHR` to guarantee an update. However, the presentation engine **may**

update from it at any time.

- With `VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR`, the presentation engine will automatically present the latest contents of a shared presentable image during every refresh cycle. The application is only required to make one initial call to `vkQueuePresentKHR`, after which the presentation engine will update from it without any need for further present calls. The application **can** indicate the image contents have been updated by calling `vkQueuePresentKHR`, but this does not guarantee the timing of when updates will occur.

The presentation engine **may** access a shared presentable image at any time after it is first presented. To avoid tearing, an application **should** coordinate access with the presentation engine. This requires presentation engine timing information through platform-specific mechanisms and ensuring that color attachment writes are made available during the portion of the presentation engine's refresh cycle they are intended for.

NOTE

The `VK_KHR_shared_presentable_image` extension does not provide functionality for determining the timing of the presentation engine's refresh cycles.

In order to query a swapchain's status when rendering to a shared presentable image, call:

```
// Provided by VK_KHR_shared_presentable_image
VkResult vkGetSwapchainStatusKHR(
    VkDevice          device,
    VkSwapchainKHR    swapchain);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the swapchain to query.

Valid Usage (Implicit)

- VUID-vkGetSwapchainStatusKHR-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkGetSwapchainStatusKHR-swapchain-parameter `swapchain` **must** be a valid `VkSwapchainKHR` handle
- VUID-vkGetSwapchainStatusKHR-swapchain-parent `swapchain` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`
- `VK_SUBOPTIMAL_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`
- `VK_ERROR_OUT_OF_DATE_KHR`
- `VK_ERROR_SURFACE_LOST_KHR`
- `VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT`

The possible return values for `vkGetSwapchainStatusKHR` **should** be interpreted as follows:

- `VK_SUCCESS` specifies the presentation engine is presenting the contents of the shared presentable image, as per the swapchain's `VkPresentModeKHR`.
- `VK_SUBOPTIMAL_KHR` the swapchain no longer matches the surface properties exactly, but the presentation engine is presenting the contents of the shared presentable image, as per the swapchain's `VkPresentModeKHR`.
- `VK_ERROR_OUT_OF_DATE_KHR` the surface has changed in such a way that it is no longer compatible with the swapchain.
- `VK_ERROR_SURFACE_LOST_KHR` the surface is no longer available.

NOTE

The swapchain state **may** be cached by implementations, so applications **should** regularly call `vkGetSwapchainStatusKHR` when using a swapchain with `VkPresentModeKHR` equal to `VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR`.

To create a swapchain, call:

```
// Provided by VK_KHR_swapchain
VkResult vkCreateSwapchainKHR(
    VkDevice                                device,
    const VkSwapchainCreateInfoKHR*         pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkSwapchainKHR*                         pSwapchain);
```

- `device` is the device to create the swapchain for.
- `pCreateInfo` is a pointer to a `VkSwapchainCreateInfoKHR` structure specifying the parameters of the created swapchain.
- `pAllocator` is the allocator used for host memory allocated for the swapchain object when there

is no more specific allocator available (see [Memory Allocation](#)).

- `pSwapchain` is a pointer to a `VkSwapchainKHR` handle in which the created swapchain object will be returned.

As mentioned above, if `vkCreateSwapchainKHR` succeeds, it will return a handle to a swapchain containing an array of at least `pCreateInfo->minImageCount` presentable images.

While acquired by the application, presentable images **can** be used in any way that equivalent non-presentable images **can** be used. A presentable image is equivalent to a non-presentable image created with the following `VkImageCreateInfo` parameters:

<code>VkImageCreateInfo</code> Field	Value
<code>flags</code>	<code>VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT</code> is set if <code>VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR</code> is set <code>VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT</code> and <code>VK_IMAGE_CREATE_EXTENDED_USAGE_BIT_KHR</code> are both set if <code>VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR</code> is set all other bits are unset
<code>imageType</code>	<code>VK_IMAGE_TYPE_2D</code>
<code>format</code>	<code>pCreateInfo->imageFormat</code>
<code>extent</code>	{ <code>pCreateInfo->imageExtent.width</code> , <code>pCreateInfo->imageExtent.height</code> , 1}
<code>mipLevels</code>	1
<code>arrayLayers</code>	<code>pCreateInfo->imageArrayLayers</code>
<code>samples</code>	<code>VK_SAMPLE_COUNT_1_BIT</code>
<code>tiling</code>	<code>VK_IMAGE_TILING_OPTIMAL</code>
<code>usage</code>	<code>pCreateInfo->imageUsage</code>
<code>sharingMode</code>	<code>pCreateInfo->imageSharingMode</code>
<code>queueFamilyIndexCount</code>	<code>pCreateInfo->queueFamilyIndexCount</code>
<code>pQueueFamilyIndices</code>	<code>pCreateInfo->pQueueFamilyIndices</code>
<code>initialLayout</code>	<code>VK_IMAGE_LAYOUT_UNDEFINED</code>

The `pCreateInfo->surface` **must** not be destroyed until after the swapchain is destroyed.

If `oldSwapchain` is `VK_NULL_HANDLE`, and the native window referred to by `pCreateInfo->surface` is already associated with a Vulkan swapchain, `VK_ERROR_NATIVE_WINDOW_IN_USE_KHR` **must** be returned.

If the native window referred to by `pCreateInfo->surface` is already associated with a non-Vulkan graphics API surface, `VK_ERROR_NATIVE_WINDOW_IN_USE_KHR` **must** be returned.

The native window referred to by `pCreateInfo->surface` **must** not become associated with a non-Vulkan graphics API surface before all associated Vulkan swapchains have been destroyed.

`vkCreateSwapchainKHR` will return `VK_ERROR_DEVICE_LOST` if the logical device was lost. The `VkSwapchainKHR` is a child of the `device`, and **must** be destroyed before the `device`. However, `VkSurfaceKHR` is not a child of any `VkDevice` and is not affected by the lost device. After successfully recreating a `VkDevice`, the same `VkSurfaceKHR` **can** be used to create a new `VkSwapchainKHR`, provided the previous one was destroyed.

If the `oldSwapchain` parameter of `pCreateInfo` is a valid swapchain, which has exclusive full-screen access, that access is released from `pCreateInfo->oldSwapchain`. If the command succeeds in this case, the newly created swapchain will automatically acquire exclusive full-screen access from `pCreateInfo->oldSwapchain`.

NOTE

This implicit transfer is intended to avoid exiting and entering full-screen exclusive mode, which may otherwise cause unwanted visual updates to the display.

In some cases, swapchain creation **may** fail if exclusive full-screen mode is requested for application control, but for some implementation-specific reason exclusive full-screen access is unavailable for the particular combination of parameters provided. If this occurs, `VK_ERROR_INITIALIZATION_FAILED` will be returned.

NOTE

In particular, it will fail if the `imageExtent` member of `pCreateInfo` does not match the extents of the monitor. Other reasons for failure may include the application not being set as high-dpi aware, or if the physical device and monitor are not compatible in this mode.

Valid Usage (Implicit)

- VUID-vkCreateSwapchainKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateSwapchainKHR-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkSwapchainCreateInfoKHR` structure
- VUID-vkCreateSwapchainKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateSwapchainKHR-pSwapchain-parameter
`pSwapchain` **must** be a valid pointer to a `VkSwapchainKHR` handle

Host Synchronization

- Host access to `pCreateInfo->surface` **must** be externally synchronized
- Host access to `pCreateInfo->oldSwapchain` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`
- `VK_ERROR_SURFACE_LOST_KHR`
- `VK_ERROR_NATIVE_WINDOW_IN_USE_KHR`
- `VK_ERROR_INITIALIZATION_FAILED`

The `VkSwapchainCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_swapchain
typedef struct VkSwapchainCreateInfoKHR {
    VkStructureType           sType;
    const void*               pNext;
    VkSwapchainCreateFlagsKHR flags;
    VkSurfaceKHR              surface;
    uint32_t                  minImageCount;
    VkFormat                  imageFormat;
    VkColorSpaceKHR           imageColorSpace;
    VkExtent2D                imageExtent;
    uint32_t                  imageArrayLayers;
    VkImageUsageFlags          imageUsage;
    VkSharingMode              imageSharingMode;
    uint32_t                  queueFamilyIndexCount;
    const uint32_t*            pQueueFamilyIndices;
    VkSurfaceTransformFlagBitsKHR preTransform;
    VkCompositeAlphaFlagBitsKHR  compositeAlpha;
    VkPresentModeKHR           presentMode;
    VkBool32                   clipped;
    VkSwapchainKHR             oldSwapchain;
} VkSwapchainCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkSwapchainCreateFlagBitsKHR` indicating parameters of the swapchain creation.
- `surface` is the surface onto which the swapchain will present images. If the creation succeeds, the swapchain becomes associated with `surface`.
- `minImageCount` is the minimum number of presentable images that the application needs. The

implementation will either create the swapchain with at least that many images, or it will fail to create the swapchain.

- **imageFormat** is a **VkFormat** value specifying the format the swapchain image(s) will be created with.
- **imageColorSpace** is a **VkColorSpaceKHR** value specifying the way the swapchain interprets image data.
- **imageExtent** is the size (in pixels) of the swapchain image(s). The behavior is platform-dependent if the image extent does not match the surface's **currentExtent** as returned by **vkGetPhysicalDeviceSurfaceCapabilitiesKHR**.

NOTE

On some platforms, it is normal that **maxImageExtent** **may** become **(0, 0)**, for example when the window is minimized. In such a case, it is not possible to create a swapchain due to the Valid Usage requirements .

- **imageArrayLayers** is the number of views in a multiview/stereo surface. For non-stereoscopic-3D applications, this value is 1.
- **imageUsage** is a bitmask of **VkImageUsageFlagBits** describing the intended usage of the (acquired) swapchain images.
- **imageSharingMode** is the sharing mode used for the image(s) of the swapchain.
- **queueFamilyIndexCount** is the number of queue families having access to the image(s) of the swapchain when **imageSharingMode** is **VK_SHARING_MODE_CONCURRENT**.
- **pQueueFamilyIndices** is a pointer to an array of queue family indices having access to the images(s) of the swapchain when **imageSharingMode** is **VK_SHARING_MODE_CONCURRENT**.
- **preTransform** is a **VkSurfaceTransformFlagBitsKHR** value describing the transform, relative to the presentation engine's natural orientation, applied to the image content prior to presentation. If it does not match the **currentTransform** value returned by **vkGetPhysicalDeviceSurfaceCapabilitiesKHR**, the presentation engine will transform the image content as part of the presentation operation.
- **compositeAlpha** is a **VkCompositeAlphaFlagBitsKHR** value indicating the alpha compositing mode to use when this surface is composited together with other surfaces on certain window systems.
- **presentMode** is the presentation mode the swapchain will use. A swapchain's present mode determines how incoming present requests will be processed and queued internally.
- **clipped** specifies whether the Vulkan implementation is allowed to discard rendering operations that affect regions of the surface that are not visible.
 - If **clipped** is **VK_TRUE**, the presentable images associated with the swapchain **may** not own all of their pixels. Pixels in the presentable images that correspond to regions of the target surface obscured by another window on the desktop, or subject to some other clipping mechanism will have undefined content when read back. Fragment shaders **may** not execute for these pixels, and thus any side effects they would have had will not occur. Setting **VK_TRUE** does not guarantee any clipping will occur, but allows more efficient presentation methods to be used on some platforms.
 - If **clipped** is **VK_FALSE**, presentable images associated with the swapchain will own all of the pixels they contain.

NOTE

Applications **should** set this value to `VK_TRUE` if they do not expect to read back the content of presentable images before presenting them or after reacquiring them, and if their fragment shaders do not have any side effects that require them to run for all pixels in the presentable image.

- `oldSwapchain` is `VK_NULL_HANDLE`, or the existing non-retired swapchain currently associated with `surface`. Providing a valid `oldSwapchain` **may** aid in the resource reuse, and also allows the application to still present any images that are already acquired from it.

Upon calling `vkCreateSwapchainKHR` with an `oldSwapchain` that is not `VK_NULL_HANDLE`, `oldSwapchain` is retired — even if creation of the new swapchain fails. The new swapchain is created in the non-retired state whether or not `oldSwapchain` is `VK_NULL_HANDLE`.

Upon calling `vkCreateSwapchainKHR` with an `oldSwapchain` that is not `VK_NULL_HANDLE`, any images from `oldSwapchain` that are not acquired by the application **may** be freed by the implementation, which **may** occur even if creation of the new swapchain fails. The application **can** destroy `oldSwapchain` to free all memory associated with `oldSwapchain`.

NOTE

Multiple retired swapchains **can** be associated with the same `VkSurfaceKHR` through multiple uses of `oldSwapchain` that outnumber calls to `vkDestroySwapchainKHR`.

After `oldSwapchain` is retired, the application **can** pass to `vkQueuePresentKHR` any images it had already acquired from `oldSwapchain`. E.g., an application may present an image from the old swapchain before an image from the new swapchain is ready to be presented. As usual, `vkQueuePresentKHR` **may** fail if `oldSwapchain` has entered a state that causes `VK_ERROR_OUT_OF_DATE_KHR` to be returned.

The application **can** continue to use a shared presentable image obtained from `oldSwapchain` until a presentable image is acquired from the new swapchain, as long as it has not entered a state that causes it to return `VK_ERROR_OUT_OF_DATE_KHR`.

Valid Usage

- VUID-VkSwapchainCreateInfoKHR-surface-01270
`surface` **must** be a surface that is supported by the device as determined using `vkGetPhysicalDeviceSurfaceSupportKHR`
- VUID-VkSwapchainCreateInfoKHR-minImageCount-01272
`minImageCount` **must** be less than or equal to the value returned in the `maxImageCount` member of the `VkSurfaceCapabilitiesKHR` structure returned by `vkGetPhysicalDeviceSurfaceCapabilitiesKHR` for the surface if the returned `maxImageCount` is not zero
- VUID-VkSwapchainCreateInfoKHR-presentMode-02839
If `presentMode` is not `VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR` nor `VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR`, then `minImageCount` **must** be greater than or equal to the value returned in the `minImageCount` member of the `VkSurfaceCapabilitiesKHR` structure returned by `vkGetPhysicalDeviceSurfaceCapabilitiesKHR` for the surface

- VUID-VkSwapchainCreateInfoKHR-minImageCount-01383
minImageCount **must** be 1 if presentMode is either VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR
- VUID-VkSwapchainCreateInfoKHR-imageFormat-01273
imageFormat and imageColorSpace **must** match the format and colorSpace members, respectively, of one of the VkSurfaceFormatKHR structures returned by vkGetPhysicalDeviceSurfaceFormatsKHR for the surface
- VUID-VkSwapchainCreateInfoKHR-pNext-07781
imageExtent **must** be between minImageExtent and maxImageExtent, inclusive, where minImageExtent and maxImageExtent are members of the VkSurfaceCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilitiesKHR for the surface
- VUID-VkSwapchainCreateInfoKHR-imageExtent-01689
imageExtent members width and height **must** both be non-zero
- VUID-VkSwapchainCreateInfoKHR-imageArrayLayers-01275
imageArrayLayers **must** be greater than 0 and less than or equal to the maxImageArrayLayers member of the VkSurfaceCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilitiesKHR for the surface
- VUID-VkSwapchainCreateInfoKHR-presentMode-01427
If presentMode is VK_PRESENT_MODE_FIFO_LATEST_READY_EXT, VK_PRESENT_MODE_IMMEDIATE_KHR, VK_PRESENT_MODE_MAILBOX_KHR, VK_PRESENT_MODE_FIFO_KHR or VK_PRESENT_MODE_FIFO_RELAXED_KHR, imageUsage **must** be a subset of the supported usage flags present in the supportedUsageFlags member of the VkSurfaceCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilitiesKHR for surface
- VUID-VkSwapchainCreateInfoKHR-imageUsage-01384
If presentMode is VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR or VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR, imageUsage **must** be a subset of the supported usage flags present in the sharedPresentSupportedUsageFlags member of the VkSharedPresentSurfaceCapabilitiesKHR structure returned by vkGetPhysicalDeviceSurfaceCapabilities2KHR for surface
- VUID-VkSwapchainCreateInfoKHR-imageSharingMode-01277
If imageSharingMode is VK_SHARING_MODE_CONCURRENT, pQueueFamilyIndices **must** be a valid pointer to an array of queueFamilyIndexCount uint32_t values
- VUID-VkSwapchainCreateInfoKHR-imageSharingMode-01278
If imageSharingMode is VK_SHARING_MODE_CONCURRENT, queueFamilyIndexCount **must** be greater than 1
- VUID-VkSwapchainCreateInfoKHR-imageSharingMode-01428
If imageSharingMode is VK_SHARING_MODE_CONCURRENT, each element of pQueueFamilyIndices **must** be unique and **must** be less than pQueueFamilyPropertyCount returned by either vkGetPhysicalDeviceQueueFamilyProperties or vkGetPhysicalDeviceQueueFamilyProperties2KHR for the physicalDevice that was used to create device
- VUID-VkSwapchainCreateInfoKHR-preTransform-01279
preTransform **must** be one of the bits present in the supportedTransforms member of the

`VkSurfaceCapabilitiesKHR` structure returned by `vkGetPhysicalDeviceSurfaceCapabilitiesKHR` for the surface

- VUID-VkSwapchainCreateInfoKHR-compositeAlpha-01280
`compositeAlpha` **must** be one of the bits present in the `supportedCompositeAlpha` member of the `VkSurfaceCapabilitiesKHR` structure returned by `vkGetPhysicalDeviceSurfaceCapabilitiesKHR` for the surface
- VUID-VkSwapchainCreateInfoKHR-presentMode-01281
`presentMode` **must** be one of the `VkPresentModeKHR` values returned by `vkGetPhysicalDeviceSurfacePresentModesKHR` for the surface
- VUID-VkSwapchainCreateInfoKHR-presentModeFifoLatestReady-10161
If the `presentModeFifoLatestReady` feature is not enabled, `presentMode` **must** not be `VK_PRESENT_MODE_FIFO_LATEST_READY_EXT`
- VUID-VkSwapchainCreateInfoKHR-physicalDeviceCount-01429
If the logical device was created with `VkDeviceGroupDeviceCreateInfo::physicalDeviceCount` equal to 1, `flags` **must** not contain `VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR`
- VUID-VkSwapchainCreateInfoKHR-oldSwapchain-01933
If `oldSwapchain` is not `VK_NULL_HANDLE`, `oldSwapchain` **must** be a non-retired swapchain associated with native window referred to by `surface`
- VUID-VkSwapchainCreateInfoKHR-imageFormat-01778
The implied image creation parameters of the swapchain **must** be supported as reported by `vkGetPhysicalDeviceImageFormatProperties`
- VUID-VkSwapchainCreateInfoKHR-flags-03168
If `flags` contains `VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR` then the `pNext` chain **must** include a `VkImageFormatListCreateInfo` structure with a `viewFormatCount` greater than zero and `pViewFormats` **must** have an element equal to `imageFormat`
- VUID-VkSwapchainCreateInfoKHR-pNext-04099
If a `VkImageFormatListCreateInfo` structure was included in the `pNext` chain and `VkImageFormatListCreateInfo::viewFormatCount` is not zero then all of the formats in `VkImageFormatListCreateInfo::pViewFormats` **must** be compatible with the `format` as described in the compatibility table
- VUID-VkSwapchainCreateInfoKHR-flags-04100
If `flags` does not contain `VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR` and the `pNext` chain include a `VkImageFormatListCreateInfo` structure then `VkImageFormatListCreateInfo::viewFormatCount` **must** be 0 or 1
- VUID-VkSwapchainCreateInfoKHR-flags-03187
If `flags` contains `VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR`, then `VkSurfaceProtectedCapabilitiesKHR::supportsProtected` **must** be `VK_TRUE` in the `VkSurfaceProtectedCapabilitiesKHR` structure returned by `vkGetPhysicalDeviceSurfaceCapabilities2KHR` for `surface`
- VUID-VkSwapchainCreateInfoKHR-pNext-02679
If the `pNext` chain includes a `VkSurfaceFullscreenExclusiveInfoEXT` structure with its `fullscreenExclusive` member set to `VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT`,

and `surface` was created using `vkCreateWin32SurfaceKHR`, a `VkSurfaceFullScreenExclusiveWin32InfoEXT` structure **must** be included in the `pNext` chain

Valid Usage (Implicit)

- VUID-VkSwapchainCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SWAPCHAIN_CREATE_INFO_KHR`
- VUID-VkSwapchainCreateInfoKHR-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkDeviceGroupSwapchainCreateInfoKHR`, `VkImageFormatListCreateInfo`, `VkSurfaceFullScreenExclusiveInfoEXT`, `VkSurfaceFullScreenExclusiveWin32InfoEXT`, or `VkSwapchainCounterCreateInfoEXT`
- VUID-VkSwapchainCreateInfoKHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkSwapchainCreateInfoKHR-flags-parameter
`flags` **must** be a valid combination of `VkSwapchainCreateFlagBitsKHR` values
- VUID-VkSwapchainCreateInfoKHR-surface-parameter
`surface` **must** be a valid `VkSurfaceKHR` handle
- VUID-VkSwapchainCreateInfoKHR-imageFormat-parameter
`imageFormat` **must** be a valid `VkFormat` value
- VUID-VkSwapchainCreateInfoKHR-imageColorSpace-parameter
`imageColorSpace` **must** be a valid `VkColorSpaceKHR` value
- VUID-VkSwapchainCreateInfoKHR-imageUsage-parameter
`imageUsage` **must** be a valid combination of `VkImageUsageFlagBits` values
- VUID-VkSwapchainCreateInfoKHR-imageUsage-requiredbitmask
`imageUsage` **must** not be `0`
- VUID-VkSwapchainCreateInfoKHR-imageSharingMode-parameter
`imageSharingMode` **must** be a valid `VkSharingMode` value
- VUID-VkSwapchainCreateInfoKHR-preTransform-parameter
`preTransform` **must** be a valid `VkSurfaceTransformFlagBitsKHR` value
- VUID-VkSwapchainCreateInfoKHR-compositeAlpha-parameter
`compositeAlpha` **must** be a valid `VkCompositeAlphaFlagBitsKHR` value
- VUID-VkSwapchainCreateInfoKHR-presentMode-parameter
`presentMode` **must** be a valid `VkPresentModeKHR` value
- VUID-VkSwapchainCreateInfoKHR-oldSwapchain-parameter
If `oldSwapchain` is not `VK_NULL_HANDLE`, `oldSwapchain` **must** be a valid `VkSwapchainKHR` handle
- VUID-VkSwapchainCreateInfoKHR-commonparent
Both of `oldSwapchain`, and `surface` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkInstance`

Bits which **can** be set in [VkSwapchainCreateInfoKHR::flags](#), specifying parameters of swapchain creation, are:

```
// Provided by VK_KHR_swapchain
typedef enum VkSwapchainCreateFlagBitsKHR {
    // Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
    VK_KHR_swapchain
    VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR = 0x00000001,
    // Provided by VK_VERSION_1_1 with VK_KHR_swapchain
    VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR = 0x00000002,
    // Provided by VK_KHR_swapchain_mutable_format
    VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR = 0x00000004,
} VkSwapchainCreateFlagBitsKHR;
```

- [VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR](#) specifies that images created from the swapchain (i.e. with the [swapchain](#) member of [VkImageSwapchainCreateInfoKHR](#) set to this swapchain's handle) **must** use [VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT](#).
- [VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR](#) specifies that the images of the swapchain **can** be used to create a [VkImageView](#) with a different format than what the swapchain was created with. The list of allowed image view formats is specified by adding a [VkImageFormatListCreateInfo](#) structure to the [pNext](#) chain of [VkSwapchainCreateInfoKHR](#). In addition, this flag also specifies that the swapchain **can** be created with usage flags that are not supported for the format the swapchain is created with but are supported for at least one of the allowed image view formats.

```
// Provided by VK_KHR_swapchain
typedef VkFlags VkSwapchainCreateFlagsKHR;
```

[VkSwapchainCreateFlagsKHR](#) is a bitmask type for setting a mask of zero or more [VkSwapchainCreateFlagBitsKHR](#).

If the [pNext](#) chain of [VkSwapchainCreateInfoKHR](#) includes a [VkDeviceGroupSwapchainCreateInfoKHR](#) structure, then that structure includes a set of device group present modes that the swapchain **can** be used with.

The [VkDeviceGroupSwapchainCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
typedef struct VkDeviceGroupSwapchainCreateInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkDeviceGroupPresentModeFlagsKHR modes;
} VkDeviceGroupSwapchainCreateInfoKHR;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.

- `modes` is a bitfield of modes that the swapchain **can** be used with.

If this structure is not present, `modes` is considered to be `VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR`.

Valid Usage (Implicit)

- VUID-VkDeviceGroupSwapchainCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_GROUP_SWAPCHAIN_CREATE_INFO_KHR`
- VUID-VkDeviceGroupSwapchainCreateInfoKHR-modes-parameter
`modes` **must** be a valid combination of `VkDeviceGroupPresentModeFlagBitsKHR` values
- VUID-VkDeviceGroupSwapchainCreateInfoKHR-modes-requiredbitmask
`modes` **must** not be `0`

If the `pNext` chain of `VkSwapchainCreateInfoKHR` includes a `VkSurfaceFullScreenExclusiveInfoEXT` structure, then that structure specifies the application's preferred full-screen presentation behavior. If this structure is not present, `fullScreenExclusive` is considered to be `VK_FULL_SCREEN_EXCLUSIVE_DEFAULT_EXT`.

To enable surface counters when creating a swapchain, add a `VkSwapchainCounterCreateInfoEXT` structure to the `pNext` chain of `VkSwapchainCreateInfoKHR`. `VkSwapchainCounterCreateInfoEXT` is defined as:

```
// Provided by VK_EXT_display_control
typedef struct VkSwapchainCounterCreateInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkSurfaceCounterFlagsEXT surfaceCounters;
} VkSwapchainCounterCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `surfaceCounters` is a bitmask of `VkSurfaceCounterFlagBitsEXT` specifying surface counters to enable for the swapchain.

Valid Usage

- VUID-VkSwapchainCounterCreateInfoEXT-surfaceCounters-01244
The bits in `surfaceCounters` **must** be supported by `VkSwapchainCreateInfoKHR::surface`, as reported by `vkGetPhysicalDeviceSurfaceCapabilities2EXT`

Valid Usage (Implicit)

- VUID-VkSwapchainCounterCreateInfoEXT-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_SWAPCHAIN_COUNTER_CREATE_INFO_EXT`

- VUID-VkSwapchainCounterCreateInfoEXT-surfaceCounters-parameter
`surfaceCounters` **must** be a valid combination of `VkSurfaceCounterFlagBitsEXT` values

The requested counters become active when the first presentation command for the associated swapchain is processed by the presentation engine. To query the value of an active counter, use:

```
// Provided by VK_EXT_display_control
VkResult vkGetSwapchainCounterEXT(
    VkDevice          device,
    VkSwapchainKHR    swapchain,
    VkSurfaceCounterFlagBitsEXT counter,
    uint64_t*         pCounterValue);
```

- `device` is the `VkDevice` associated with `swapchain`.
- `swapchain` is the swapchain from which to query the counter value.
- `counter` is a `VkSurfaceCounterFlagBitsEXT` value specifying the counter to query.
- `pCounterValue` will return the current value of the counter.

If a counter is not available because the swapchain is out of date, the implementation **may** return `VK_ERROR_OUT_OF_DATE_KHR`.

Valid Usage

- VUID-vkGetSwapchainCounterEXT-swapchain-01245
One or more present commands on `swapchain` **must** have been processed by the presentation engine

Valid Usage (Implicit)

- VUID-vkGetSwapchainCounterEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetSwapchainCounterEXT-swapchain-parameter
`swapchain` **must** be a valid `VkSwapchainKHR` handle
- VUID-vkGetSwapchainCounterEXT-counter-parameter
`counter` **must** be a valid `VkSurfaceCounterFlagBitsEXT` value
- VUID-vkGetSwapchainCounterEXT-pCounterValue-parameter
`pCounterValue` **must** be a valid pointer to a `uint64_t` value
- VUID-vkGetSwapchainCounterEXT-swapchain-parent
`swapchain` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_DEVICE_LOST`
- `VK_ERROR_OUT_OF_DATE_KHR`

To destroy a swapchain object call:

```
// Provided by VK_KHR_swapchain
void vkDestroySwapchainKHR(
    VkDevice          device,
    VkSwapchainKHR    swapchain,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the `VkDevice` associated with `swapchain`.
- `swapchain` is the swapchain to destroy.
- `pAllocator` is the allocator used for host memory allocated for the swapchain object when there is no more specific allocator available (see [Memory Allocation](#)).

The application **must** not destroy a swapchain until after completion of all outstanding operations on images that were acquired from the swapchain. `swapchain` and all associated `VkImage` handles are destroyed, and **must** not be acquired or used any more by the application. The memory of each `VkImage` will only be freed after that image is no longer used by the presentation engine. For example, if one image of the swapchain is being displayed in a window, the memory for that image **may** not be freed until the window is destroyed, or another swapchain is created for the window. Destroying the swapchain does not invalidate the parent `VkSurfaceKHR`, and a new swapchain **can** be created with it.

When a swapchain associated with a display surface is destroyed, if the image most recently presented to the display surface is from the swapchain being destroyed, then either any display resources modified by presenting images from any swapchain associated with the display surface **must** be reverted by the implementation to their state prior to the first present performed on one of these swapchains, or such resources **must** be left in their current state.

If `swapchain` has exclusive full-screen access, it is released before the swapchain is destroyed.

Valid Usage

- VUID-vkDestroySwapchainKHR-swapchain-01282
All uses of presentable images acquired from `swapchain` **must** have completed execution
- VUID-vkDestroySwapchainKHR-swapchain-01283

If `VkAllocationCallbacks` were provided when `swapchain` was created, a compatible set of callbacks **must** be provided here

- VUID-vkDestroySwapchainKHR-swapchain-01284

If no `VkAllocationCallbacks` were provided when `swapchain` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroySwapchainKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroySwapchainKHR-swapchain-parameter
If `swapchain` is not `VK_NULL_HANDLE`, `swapchain` **must** be a valid `VkSwapchainKHR` handle
- VUID-vkDestroySwapchainKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroySwapchainKHR-swapchain-parent
If `swapchain` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

When the `VK_KHR_display_swapchain` extension is enabled, multiple swapchains that share presentable images are created by calling:

```
// Provided by VK_KHR_display_swapchain
VkResult vkCreateSharedSwapchainsKHR(
    VkDevice          device,
    uint32_t          swapchainCount,
    const VkSwapchainCreateInfoKHR* pCreateInfos,
    const VkAllocationCallbacks*   pAllocator,
    VkSwapchainKHR*    pSwapchains);
```

- `device` is the device to create the swapchains for.
- `swapchainCount` is the number of swapchains to create.
- `pCreateInfos` is a pointer to an array of `VkSwapchainCreateInfoKHR` structures specifying the parameters of the created swapchains.
- `pAllocator` is the allocator used for host memory allocated for the swapchain objects when there is no more specific allocator available (see [Memory Allocation](#)).

- `pSwapchains` is a pointer to an array of `VkSwapchainKHR` handles in which the created swapchain objects will be returned.

`vkCreateSharedSwapchainsKHR` is similar to `vkCreateSwapchainKHR`, except that it takes an array of `VkSwapchainCreateInfoKHR` structures, and returns an array of swapchain objects.

The swapchain creation parameters that affect the properties and number of presentable images **must** match between all the swapchains. If the displays used by any of the swapchains do not use the same presentable image layout or are incompatible in a way that prevents sharing images, swapchain creation will fail with the result code `VK_ERROR_INCOMPATIBLE_DISPLAY_KHR`. If any error occurs, no swapchains will be created. Images presented to multiple swapchains **must** be re-acquired from all of them before being modified. After destroying one or more of the swapchains, the remaining swapchains and the presentable images **can** continue to be used.

Valid Usage (Implicit)

- VUID-vkCreateSharedSwapchainsKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateSharedSwapchainsKHR-pCreateInfos-parameter
`pCreateInfos` **must** be a valid pointer to an array of `swapchainCount` valid `VkSwapchainCreateInfoKHR` structures
- VUID-vkCreateSharedSwapchainsKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateSharedSwapchainsKHR-pSwapchains-parameter
`pSwapchains` **must** be a valid pointer to an array of `swapchainCount` `VkSwapchainKHR` handles
- VUID-vkCreateSharedSwapchainsKHR-swapchainCount-arraylength
`swapchainCount` **must** be greater than 0

Host Synchronization

- Host access to `pCreateInfos[i].surface` **must** be externally synchronized
- Host access to `pCreateInfos[i].oldSwapchain` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

- `VK_ERROR_INCOMPATIBLE_DISPLAY_KHR`
- `VK_ERROR_DEVICE_LOST`
- `VK_ERROR_SURFACE_LOST_KHR`

To obtain the array of presentable images associated with a swapchain, call:

```
// Provided by VK_KHR_swapchain
VkResult vkGetSwapchainImagesKHR(
    VkDevice          device,
    VkSwapchainKHR    swapchain,
    uint32_t*         pSwapchainImageCount,
    VkImage*           pSwapchainImages);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the swapchain to query.
- `pSwapchainImageCount` is a pointer to an integer related to the number of presentable images available or queried, as described below.
- `pSwapchainImages` is either `NULL` or a pointer to an array of `VkImage` handles.

If `pSwapchainImages` is `NULL`, then the number of presentable images for `swapchain` is returned in `pSwapchainImageCount`. Otherwise, `pSwapchainImageCount` **must** point to a variable set by the application to the number of elements in the `pSwapchainImages` array, and on return the variable is overwritten with the number of structures actually written to `pSwapchainImages`. If the value of `pSwapchainImageCount` is less than the number of presentable images for `swapchain`, at most `pSwapchainImageCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available presentable images were returned.

Valid Usage (Implicit)

- VUID-vkGetSwapchainImagesKHR-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkGetSwapchainImagesKHR-swapchain-parameter `swapchain` **must** be a valid `VkSwapchainKHR` handle
- VUID-vkGetSwapchainImagesKHR-pSwapchainImageCount-parameter `pSwapchainImageCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetSwapchainImagesKHR-pSwapchainImages-parameter
If the value referenced by `pSwapchainImageCount` is not `0`, and `pSwapchainImages` is not `NULL`, `pSwapchainImages` **must** be a valid pointer to an array of `pSwapchainImageCount` `VkImage` handles
- VUID-vkGetSwapchainImagesKHR-swapchain-parent `swapchain` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

NOTE

By knowing all presentable images used in the swapchain, the application can create command buffers that reference these images prior to entering its main rendering loop.

Images returned by `vkGetSwapchainImagesKHR` are fully backed by memory before they are passed to the application, as if they are each bound completely and contiguously to a single `VkDeviceMemory` object. All presentable images are initially in the `VK_IMAGE_LAYOUT_UNDEFINED` layout, thus before using presentable images, the application **must** transition them to a valid layout for the intended use.

Further, the lifetime of presentable images is controlled by the implementation, so applications **must** not destroy a presentable image. See `vkDestroySwapchainKHR` for further details on the lifetime of presentable images.

Images **can** also be created by using `vkCreateImage` with `VkImageSwapchainCreateInfoKHR` and bound to swapchain memory using `vkBindImageMemory2KHR` with `VkBindImageMemorySwapchainInfoKHR`. These images **can** be used anywhere swapchain images are used, and are useful in logical devices with multiple physical devices to create peer memory bindings of swapchain memory. These images and bindings have no effect on what memory is presented. Unlike images retrieved from `vkGetSwapchainImagesKHR`, these images **must** be destroyed with `vkDestroyImage`.

To acquire an available presentable image to use, and retrieve the index of that image, call:

```
// Provided by VK_KHR_swapchain
VkResult vkAcquireNextImageKHR(
    VkDevice           device,
    VkSwapchainKHR     swapchain,
    uint64_t           timeout,
    VkSemaphore        semaphore,
    VkFence             fence,
    uint32_t*          pImageIndex);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the non-retired swapchain from which an image is being acquired.

- **timeout** specifies how long the function waits, in nanoseconds, if no image is available.
- **semaphore** is `VK_NULL_HANDLE` or a semaphore to signal.
- **fence** is `VK_NULL_HANDLE` or a fence to signal.
- **pImageIndex** is a pointer to a `uint32_t` in which the index of the next image to use (i.e. an index into the array of images returned by `vkGetSwapchainImagesKHR`) is returned.

Valid Usage

- VUID-vkAcquireNextImageKHR-swapchain-01285
swapchain **must** not be in the retired state
- VUID-vkAcquireNextImageKHR-semaphore-01286
If **semaphore** is not `VK_NULL_HANDLE`, it **must** be unsignaled
- VUID-vkAcquireNextImageKHR-semaphore-01779
If **semaphore** is not `VK_NULL_HANDLE`, it **must** not have any uncompleted signal or wait operations pending
- VUID-vkAcquireNextImageKHR-fence-01287
If **fence** is not `VK_NULL_HANDLE`, **fence** **must** be unsignaled
- VUID-vkAcquireNextImageKHR-fence-10066
If **fence** is not `VK_NULL_HANDLE`, **fence** **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-vkAcquireNextImageKHR-semaphore-01780
semaphore and **fence** **must** not both be equal to `VK_NULL_HANDLE`
- VUID-vkAcquireNextImageKHR-surface-07783
If **forward progress** cannot be guaranteed for the **surface** used to create the **swapchain** member of `pAcquireInfo`, **timeout** **must** not be `UINT64_MAX`
- VUID-vkAcquireNextImageKHR-semaphore-03265
semaphore **must** have a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_BINARY`

Valid Usage (Implicit)

- VUID-vkAcquireNextImageKHR-device-parameter
device **must** be a valid `VkDevice` handle
- VUID-vkAcquireNextImageKHR-swapchain-parameter
swapchain **must** be a valid `VkSwapchainKHR` handle
- VUID-vkAcquireNextImageKHR-semaphore-parameter
If **semaphore** is not `VK_NULL_HANDLE`, **semaphore** **must** be a valid `VkSemaphore` handle
- VUID-vkAcquireNextImageKHR-fence-parameter
If **fence** is not `VK_NULL_HANDLE`, **fence** **must** be a valid `VkFence` handle
- VUID-vkAcquireNextImageKHR-pImageIndex-parameter
pImageIndex **must** be a valid pointer to a `uint32_t` value

- VUID-vkAcquireNextImageKHR-swapchain-parent
`swapchain` **must** have been created, allocated, or retrieved from `device`
- VUID-vkAcquireNextImageKHR-semaphore-parent
If `semaphore` is a valid handle, it **must** have been created, allocated, or retrieved from `device`
- VUID-vkAcquireNextImageKHR-fence-parent
If `fence` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized
- Host access to `semaphore` **must** be externally synchronized
- Host access to `fence` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`
- `VK_TIMEOUT`
- `VK_NOT_READY`
- `VK_SUBOPTIMAL_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`
- `VK_ERROR_OUT_OF_DATE_KHR`
- `VK_ERROR_SURFACE_LOST_KHR`
- `VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT`

If an image is acquired successfully, `vkAcquireNextImageKHR` **must** either return `VK_SUCCESS` or `VK_SUBOPTIMAL_KHR`. The implementation **may** return `VK_SUBOPTIMAL_KHR` if the swapchain no longer matches the surface properties exactly, but **can** still be used for presentation.

When successful, `vkAcquireNextImageKHR` acquires a presentable image from `swapchain` that an application **can** use, and sets `pImageIndex` to the index of that image within the swapchain. The presentation engine **may** not have finished reading from the image at the time it is acquired, so the application **must** use `semaphore` and/or `fence` to ensure that the image layout and contents are not modified until the presentation engine reads have completed. Once `vkAcquireNextImageKHR` successfully acquires an image, the semaphore signal operation referenced by `semaphore`, if not `VK_NULL_HANDLE`, and the fence signal operation referenced by `fence`, if not `VK_NULL_HANDLE`,

are submitted for execution. If `vkAcquireNextImageKHR` does not successfully acquire an image, `semaphore` and `fence` are unaffected. The order in which images are acquired is implementation-dependent, and **may** be different than the order the images were presented.

If `timeout` is zero, then `vkAcquireNextImageKHR` does not wait, and will either successfully acquire an image, or fail and return `VK_NOT_READY` if no image is available.

If the specified timeout period expires before an image is acquired, `vkAcquireNextImageKHR` returns `VK_TIMEOUT`. If `timeout` is `UINT64_MAX`, the timeout period is treated as infinite, and `vkAcquireNextImageKHR` will block until an image is acquired or an error occurs.

Let `S` be the number of images in `swapchain`. Let `M` be the value of `VkSurfaceCapabilitiesKHR::minImageCount`.

`vkAcquireNextImageKHR` **should** not be called if the number of images that the application has currently acquired is greater than `S-M`. If `vkAcquireNextImageKHR` is called when the number of images that the application has currently acquired is less than or equal to `S-M`, `vkAcquireNextImageKHR` **must** return in finite time with an allowed `VkResult` code.

NOTE

Returning a result in finite time guarantees that the implementation cannot deadlock an application, or suspend its execution indefinitely with correct API usage. Acquiring too many images at once may block indefinitely, which is covered by valid usage when attempting to use `UINT64_MAX`. For example, a scenario here is when a compositor holds on to images which are currently being presented, and there are not any vacant images left to be acquired.

If the swapchain images no longer match native surface properties, either `VK_SUBOPTIMAL_KHR` or `VK_ERROR_OUT_OF_DATE_KHR` **must** be returned. If `VK_ERROR_OUT_OF_DATE_KHR` is returned, no image is acquired and attempts to present previously acquired images to the swapchain will also fail with `VK_ERROR_OUT_OF_DATE_KHR`. Applications need to create a new swapchain for the surface to continue presenting if `VK_ERROR_OUT_OF_DATE_KHR` is returned.

NOTE

`VK_SUBOPTIMAL_KHR` **may** happen, for example, if the platform surface has been resized but the platform is able to scale the presented images to the new size to produce valid surface updates. It is up to the application to decide whether it prefers to continue using the current swapchain in this state, or to re-create the swapchain to better match the platform surface properties.

If device loss occurs (see [Lost Device](#)) before the timeout has expired, `vkAcquireNextImageKHR` **must** return in finite time with either one of the allowed success codes, or `VK_ERROR_DEVICE_LOST`.

If `semaphore` is not `VK_NULL_HANDLE`, the semaphore **must** be unsignaled, with no signal or wait operations pending. It will become signaled when the application **can** use the image.

NOTE

Use of `semaphore` allows rendering operations to be recorded and submitted before the presentation engine has completed its use of the image.

If `fence` is not equal to `VK_NULL_HANDLE`, the fence **must** be unsignaled, with no signal operations pending. It will become signaled when the application **can** use the image.

NOTE

Applications **should** not rely on `vkAcquireNextImageKHR` blocking in order to meter their rendering speed. The implementation **may** return from this function immediately regardless of how many presentation requests are queued, and regardless of when queued presentation requests will complete relative to the call. Instead, applications **can** use `fence` to meter their frame generation work to match the presentation rate.

An application **must** wait until either the `semaphore` or `fence` is signaled before accessing the image's data.

NOTE

When the presentable image will be accessed by some stage S, the recommended idiom for ensuring correct synchronization is:

- The `VkSubmitInfo` used to submit the image layout transition for execution includes `vkAcquireNextImageKHR::semaphore` in its `pWaitSemaphores` member, with the corresponding element of `pWaitDstStageMask` including S.
- The `synchronization command` that performs any necessary image layout transition includes S in both the `srcStageMask` and `dstStageMask`.

After a successful return, the image indicated by `pImageIndex` and its data will be unmodified compared to when it was presented.

NOTE

Exclusive ownership of presentable images corresponding to a swapchain created with `VK_SHARING_MODE_EXCLUSIVE` as defined in [Resource Sharing](#) is not altered by a call to `vkAcquireNextImageKHR`. That means upon the first acquisition from such a swapchain presentable images are not owned by any queue family, while at subsequent acquisitions the presentable images remain owned by the queue family the image was previously presented on.

The possible return values for `vkAcquireNextImageKHR` depend on the `timeout` provided:

- `VK_SUCCESS` is returned if an image became available.
- `VK_ERROR_SURFACE_LOST_KHR` is returned if the surface becomes no longer available.
- `VK_NOT_READY` is returned if `timeout` is zero and no image was available.
- `VK_TIMEOUT` is returned if `timeout` is greater than zero and less than `UINT64_MAX`, and no image became available within the time allowed.
- `VK_SUBOPTIMAL_KHR` is returned if an image became available, and the swapchain no longer matches the surface properties exactly, but **can** still be used to present to the surface successfully.

NOTE

This **may** happen, for example, if the platform surface has been resized but the platform is able to scale the presented images to the new size to produce valid surface updates. It is up to the application to decide whether it prefers to continue using the current swapchain indefinitely or temporarily in this state, or to re-create the swapchain to better match the platform surface properties.

- `VK_ERROR_OUT_OF_DATE_KHR` is returned if the surface has changed in such a way that it is no longer compatible with the swapchain, and further presentation requests using the swapchain will fail. Applications **must** query the new surface properties and recreate their swapchain if they wish to continue presenting to the surface.

If the native surface and presented image sizes no longer match, presentation **may** fail. If presentation does succeed, the mapping from the presented image to the native surface is implementation-defined. It is the application's responsibility to detect surface size changes and react appropriately. If presentation fails because of a mismatch in the surface and presented image sizes, a `VK_ERROR_OUT_OF_DATE_KHR` error will be returned.

NOTE

For example, consider a 4x3 window/surface that gets resized to be 3x4 (taller than wider). On some window systems, the portion of the window/surface that was previously and still is visible (the 3x3 part) will contain the same contents as before, while the remaining parts of the window will have undefined contents. Other window systems **may** squash/stretch the image to fill the new window size without any undefined contents, or apply some other mapping.

To acquire an available presentable image to use, and retrieve the index of that image, call:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
VkResult vkAcquireNextImage2KHR(
    VkDevice device,
    const VkAcquireNextImageInfoKHR* pAcquireInfo,
    uint32_t* pImageIndex);
```

- `device` is the device associated with `swapchain`.
- `pAcquireInfo` is a pointer to a `VkAcquireNextImageInfoKHR` structure containing parameters of the acquire.
- `pImageIndex` is a pointer to a `uint32_t` value specifying the index of the next image to use.

Valid Usage

- VUID-vkAcquireNextImage2KHR-surface-07784
If `forward progress` cannot be guaranteed for the `surface` used to create `swapchain`, the `timeout` member of `pAcquireInfo` **must** not be `UINT64_MAX`

Valid Usage (Implicit)

- VUID-vkAcquireNextImage2KHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkAcquireNextImage2KHR-pAcquireInfo-parameter
`pAcquireInfo` **must** be a valid pointer to a valid `VkAcquireNextImageInfoKHR` structure

- VUID-vkAcquireNextImage2KHR-pImageIndex-parameter **pImageIndex** must be a valid pointer to a **uint32_t** value

Return Codes

Success

- **VK_SUCCESS**
- **VK_TIMEOUT**
- **VK_NOT_READY**
- **VK_SUBOPTIMAL_KHR**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**
- **VK_ERROR_DEVICE_LOST**
- **VK_ERROR_OUT_OF_DATE_KHR**
- **VK_ERROR_SURFACE_LOST_KHR**
- **VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT**

The **VkAcquireNextImageInfoKHR** structure is defined as:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
typedef struct VkAcquireNextImageInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkSwapchainKHR     swapchain;
    uint64_t           timeout;
    VkSemaphore         semaphore;
    VkFence             fence;
    uint32_t           deviceMask;
} VkAcquireNextImageInfoKHR;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **swapchain** is a non-retired swapchain from which an image is acquired.
- **timeout** specifies how long the function waits, in nanoseconds, if no image is available.
- **semaphore** is **VK_NULL_HANDLE** or a semaphore to signal.
- **fence** is **VK_NULL_HANDLE** or a fence to signal.
- **deviceMask** is a mask of physical devices for which the swapchain image will be ready to use when the semaphore or fence is signaled.

If [vkAcquireNextImageKHR](#) is used, the device mask is considered to include all physical devices in the logical device.

NOTE

[vkAcquireNextImage2KHR](#) signals at most one semaphore, even if the application requests waiting for multiple physical devices to be ready via the [deviceMask](#). However, only a single physical device **can** wait on that semaphore, since the semaphore becomes unsignaled when the wait succeeds. For other physical devices to wait for the image to be ready, it is necessary for the application to submit semaphore signal operation(s) to that first physical device to signal additional semaphore(s) after the wait succeeds, which the other physical device(s) **can** wait upon.

Valid Usage

- VUID-VkAcquireNextImageInfoKHR-swapchain-01675
[swapchain](#) **must** not be in the retired state
- VUID-VkAcquireNextImageInfoKHR-semaphore-01288
If [semaphore](#) is not [VK_NULL_HANDLE](#), it **must** be unsignaled
- VUID-VkAcquireNextImageInfoKHR-semaphore-01781
If [semaphore](#) is not [VK_NULL_HANDLE](#), it **must** not have any uncompleted signal or wait operations pending
- VUID-VkAcquireNextImageInfoKHR-fence-01289
If [fence](#) is not [VK_NULL_HANDLE](#), [fence](#) **must** be unsignaled
- VUID-VkAcquireNextImageInfoKHR-fence-10067
If [fence](#) is not [VK_NULL_HANDLE](#), [fence](#) **must** not be associated with any other queue command that has not yet completed execution on that queue
- VUID-VkAcquireNextImageInfoKHR-semaphore-01782
[semaphore](#) and [fence](#) **must** not both be equal to [VK_NULL_HANDLE](#)
- VUID-VkAcquireNextImageInfoKHR-deviceMask-01290
[deviceMask](#) **must** be a valid device mask
- VUID-VkAcquireNextImageInfoKHR-deviceMask-01291
[deviceMask](#) **must** not be zero
- VUID-VkAcquireNextImageInfoKHR-semaphore-03266
[semaphore](#) **must** have a [VkSemaphoreType](#) of [VK_SEMAPHORE_TYPE_BINARY](#)

Valid Usage (Implicit)

- VUID-VkAcquireNextImageInfoKHR-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_ACQUIRE_NEXT_IMAGE_INFO_KHR](#)
- VUID-VkAcquireNextImageInfoKHR-pNext-pNext
[pNext](#) **must** be [NULL](#)
- VUID-VkAcquireNextImageInfoKHR-swapchain-parameter

`swapchain` **must** be a valid `VkSwapchainKHR` handle

- `VUID-VkAcquireNextImageInfoKHR-semaphore-parameter`

If `semaphore` is not `VK_NULL_HANDLE`, `semaphore` **must** be a valid `VkSemaphore` handle

- `VUID-VkAcquireNextImageInfoKHR-fence-parameter`

If `fence` is not `VK_NULL_HANDLE`, `fence` **must** be a valid `VkFence` handle

- `VUID-VkAcquireNextImageInfoKHR-commonparent`

Each of `fence`, `semaphore`, and `swapchain` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized
- Host access to `semaphore` **must** be externally synchronized
- Host access to `fence` **must** be externally synchronized

After queueing all rendering commands and transitioning the image to the correct layout, to queue an image for presentation, call:

```
// Provided by VK_KHR_swapchain
VkResult vkQueuePresentKHR(
    VkQueue queue,
    const VkPresentInfoKHR* pPresentInfo);
```

- `queue` is a queue that is capable of presentation to the target surface's platform on the same device as the image's swapchain.
- `pPresentInfo` is a pointer to a `VkPresentInfoKHR` structure specifying parameters of the presentation.

NOTE

There is no requirement for an application to present images in the same order that they were acquired - applications can arbitrarily present any image that is currently acquired.

NOTE

The origin of the native orientation of the surface coordinate system is not specified in the Vulkan specification; it depends on the platform. For most platforms the origin is by default upper-left, meaning the pixel of the presented `VkImage` at coordinates (0,0) would appear at the upper left pixel of the platform surface (assuming `VK_SURFACE_TRANSFORM_IDENTITY_BIT_KHR`, and the display standing the right way up).

The result codes `VK_ERROR_OUT_OF_DATE_KHR` and `VK_SUBOPTIMAL_KHR` have the same meaning when returned by `vkQueuePresentKHR` as they do when returned by `vkAcquireNextImageKHR`. If any `swapchain` member of `pPresentInfo` was created with `VK_FULL_SCREEN_EXCLUSIVE_APPLICATION_CONTROLLED_EXT`, `VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT` will be returned if that swapchain does not have

exclusive full-screen access, possibly for implementation-specific reasons outside of the application's control. If multiple swapchains are presented, the result code is determined by applying the following rules in order:

- If the device is lost, `VK_ERROR_DEVICE_LOST` is returned.
- If any of the target surfaces are no longer available the error `VK_ERROR_SURFACE_LOST_KHR` is returned.
- If any of the presents would have a result of `VK_ERROR_OUT_OF_DATE_KHR` if issued separately then `VK_ERROR_OUT_OF_DATE_KHR` is returned.
- If any of the presents would have a result of `VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT` if issued separately then `VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT` is returned.
- If any of the presents would have a result of `VK_SUBOPTIMAL_KHR` if issued separately then `VK_SUBOPTIMAL_KHR` is returned.
- Otherwise `VK_SUCCESS` is returned.

Any writes to memory backing the images referenced by the `pImageIndices` and `pSwapchains` members of `pPresentInfo`, that are available before `vkQueuePresentKHR` is executed, are automatically made visible to the read access performed by the presentation engine. This automatic visibility operation for an image happens-after the semaphore signal operation, and happens-before the presentation engine accesses the image.

Presentation is a read-only operation that will not affect the content of the presentable images. Upon reacquiring the image and transitioning it away from the `VK_IMAGE_LAYOUT_PRESENT_SRC_KHR` layout, the contents will be the same as they were prior to transitioning the image to the present source layout and presenting it. However, if a mechanism other than Vulkan is used to modify the platform window associated with the swapchain, the content of all presentable images in the swapchain becomes undefined.

Calls to `vkQueuePresentKHR` **may** block, but **must** return in finite time. The processing of the presentation happens in issue order with other queue operations, but semaphores **must** be used to ensure that prior rendering and other commands in the specified queue complete before the presentation begins. The presentation command itself does not delay processing of subsequent commands on the queue. However, presentation requests sent to a particular queue are always performed in order. Exact presentation timing is controlled by the semantics of the presentation engine and native platform in use.

If an image is presented to a swapchain created from a display surface, the mode of the associated display will be updated, if necessary, to match the mode specified when creating the display surface. The mode switch and presentation of the specified image will be performed as one atomic operation.

Queueing an image for presentation defines a set of *queue operations*, including waiting on the semaphores and submitting a presentation request to the presentation engine. However, the scope of this set of queue operations does not include the actual processing of the image by the presentation engine.

If `vkQueuePresentKHR` fails to enqueue the corresponding set of queue operations, it **may** return

`VK_ERROR_OUT_OF_HOST_MEMORY` or `VK_ERROR_OUT_OF_DEVICE_MEMORY`. If it does, the implementation **must** ensure that the state and contents of any resources or synchronization primitives referenced is unaffected by the call or its failure.

If `vkQueuePresentKHR` fails in such a way that the implementation is unable to make that guarantee, the implementation **must** return `VK_ERROR_DEVICE_LOST`.

However, if the presentation request is rejected by the presentation engine with an error `VK_ERROR_OUT_OF_DATE_KHR`, `VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT`, or `VK_ERROR_SURFACE_LOST_KHR`, the set of queue operations are still considered to be enqueued and thus any semaphore wait operation specified in `VkPresentInfoKHR` will execute when the corresponding queue operation is complete.

`vkQueuePresentKHR` releases the acquisition of the images referenced by `imageIndices`. The queue family corresponding to the queue `vkQueuePresentKHR` is executed on **must** have ownership of the presented images as defined in [Resource Sharing](#). `vkQueuePresentKHR` does not alter the queue family ownership, but the presented images **must** not be used again before they have been reacquired using `vkAcquireNextImageKHR`.

NOTE

The application **can** continue to present any acquired images from a retired swapchain as long as the swapchain has not entered a state that causes `vkQueuePresentKHR` to return `VK_ERROR_OUT_OF_DATE_KHR`.

Valid Usage

- VUID-vkQueuePresentKHR-pSwapchains-01292
Each element of `pSwapchains` member of `pPresentInfo` **must** be a swapchain that is created for a surface for which presentation is supported from `queue` as determined using a call to `vkGetPhysicalDeviceSurfaceSupportKHR`
- VUID-vkQueuePresentKHR-pSwapchains-01293
If more than one member of `pSwapchains` was created from a display surface, all display surfaces referenced that refer to the same display **must** use the same display mode
- VUID-vkQueuePresentKHR-pWaitSemaphores-01294
When a semaphore wait operation referring to a binary semaphore defined by the elements of the `pWaitSemaphores` member of `pPresentInfo` executes on `queue`, there **must** be no other queues waiting on the same semaphore
- VUID-vkQueuePresentKHR-pWaitSemaphores-03267
All elements of the `pWaitSemaphores` member of `pPresentInfo` **must** be created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_BINARY`
- VUID-vkQueuePresentKHR-pWaitSemaphores-03268
All elements of the `pWaitSemaphores` member of `pPresentInfo` **must** reference a semaphore signal operation that has been submitted for execution and any [semaphore signal operations](#) on which it depends **must** have also been submitted for execution

Valid Usage (Implicit)

- VUID-vkQueuePresentKHR-queue-parameter
`queue` **must** be a valid `VkQueue` handle
- VUID-vkQueuePresentKHR-pPresentInfo-parameter
`pPresentInfo` **must** be a valid pointer to a valid `VkPresentInfoKHR` structure

Host Synchronization

- Host access to `queue` **must** be externally synchronized
- Host access to `pPresentInfo->pWaitSemaphores[]` **must** be externally synchronized
- Host access to `pPresentInfo->pSwapchains[]` **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

Return Codes

Success

- `VK_SUCCESS`
- `VK_SUBOPTIMAL_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`
- `VK_ERROR_OUT_OF_DATE_KHR`
- `VK_ERROR_SURFACE_LOST_KHR`
- `VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT`

The `VkPresentInfoKHR` structure is defined as:

```
// Provided by VK_KHR_swapchain
typedef struct VkPresentInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           waitSemaphoreCount;
    const VkSemaphore* pWaitSemaphores;
    uint32_t           swapchainCount;
    const VkSwapchainKHR* pSwapchains;
    const uint32_t*     pImageIndices;
    VkResult*          pResults;
} VkPresentInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `waitSemaphoreCount` is the number of semaphores to wait for before issuing the present request. The number **may** be zero.
- `pWaitSemaphores` is `NULL` or a pointer to an array of `VkSemaphore` objects with `waitSemaphoreCount` entries, and specifies the semaphores to wait for before issuing the present request.
- `swapchainCount` is the number of swapchains being presented to by this command.
- `pSwapchains` is a pointer to an array of `VkSwapchainKHR` objects with `swapchainCount` entries.
- `pImageIndices` is a pointer to an array of indices into the array of each swapchain's presentable images, with `swapchainCount` entries. Each entry in this array identifies the image to present on the corresponding entry in the `pSwapchains` array.
- `pResults` is a pointer to an array of `VkResult` typed elements with `swapchainCount` entries. Applications that do not need per-swapchain results **can** use `NULL` for `pResults`. If non-`NULL`, each entry in `pResults` will be set to the `VkResult` for presenting the swapchain corresponding to the same index in `pSwapchains`.

Before an application **can** present an image, the image's layout **must** be transitioned to the `VK_IMAGE_LAYOUT_PRESENT_SRC_KHR` layout, or for a shared presentable image the `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR` layout.

NOTE

When transitioning the image to `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR` or `VK_IMAGE_LAYOUT_PRESENT_SRC_KHR`, there is no need to delay subsequent processing, or perform any visibility operations (as `vkQueuePresentKHR` performs automatic visibility operations). To achieve this, the `dstAccessMask` member of the `VkImageMemoryBarrier` **should** be `0`, and the `dstStageMask` parameter **should** be `VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT`.

Valid Usage

- VUID-VkPresentInfoKHR-pSwapchain-09231
Elements of `pSwapchain` **must** be unique
- VUID-VkPresentInfoKHR-pImageIndices-01430

Each element of `pImageIndices` **must** be the index of a presentable image acquired from the swapchain specified by the corresponding element of the `pSwapchains` array, and the presented image subresource **must** be in the `VK_IMAGE_LAYOUT_PRESENT_SRC_KHR` or `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR` layout at the time the operation is executed on a `VkDevice`

- VUID-VkPresentInfoKHR-pNext-06235

If a `VkPresentIdKHR` structure is included in the `pNext` chain, and the `presentId` feature is not enabled, each `presentIds` entry in that structure **must** be NULL

Valid Usage (Implicit)

- VUID-VkPresentInfoKHR-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_PRESENT_INFO_KHR`

- VUID-VkPresentInfoKHR-pNext-pNext

Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either NULL or a pointer to a valid instance of `VkDeviceGroupPresentInfoKHR`, `VkDisplayPresentInfoKHR`, `VkFrameBoundaryEXT`, `VkPresentIdKHR`, or `VkPresentRegionsKHR`

- VUID-VkPresentInfoKHR-sType-unique

The `sType` value of each struct in the `pNext` chain **must** be unique

- VUID-VkPresentInfoKHR-pWaitSemaphores-parameter

If `waitSemaphoreCount` is not 0, `pWaitSemaphores` **must** be a valid pointer to an array of `waitSemaphoreCount` valid `VkSemaphore` handles

- VUID-VkPresentInfoKHR-pSwapchains-parameter

`pSwapchains` **must** be a valid pointer to an array of `swapchainCount` valid `VkSwapchainKHR` handles

- VUID-VkPresentInfoKHR-pImageIndices-parameter

`pImageIndices` **must** be a valid pointer to an array of `swapchainCount` `uint32_t` values

- VUID-VkPresentInfoKHR-pResults-parameter

If `pResults` is not NULL, `pResults` **must** be a valid pointer to an array of `swapchainCount` `VkResult` values

- VUID-VkPresentInfoKHR-swapchainCount-arraylength

`swapchainCount` **must** be greater than 0

- VUID-VkPresentInfoKHR-commonparent

Both of the elements of `pSwapchains`, and the elements of `pWaitSemaphores` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

When the `VK_KHR_incremental_present` extension is enabled, additional fields **can** be specified that allow an application to specify that only certain rectangular regions of the presentable images of a swapchain are changed. This is an optimization hint that a presentation engine **may** use to only update the region of a surface that is actually changing. The application still **must** ensure that all pixels of a presented image contain the desired values, in case the presentation engine ignores this

hint. An application **can** provide this hint by adding a `VkPresentRegionsKHR` structure to the `pNext` chain of the `VkPresentInfoKHR` structure.

The `VkPresentRegionsKHR` structure is defined as:

```
// Provided by VK_KHR_incremental_present
typedef struct VkPresentRegionsKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           swapchainCount;
    const VkPresentRegionKHR* pRegions;
} VkPresentRegionsKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchainCount` is the number of swapchains being presented to by this command.
- `pRegions` is `NULL` or a pointer to an array of `VkPresentRegionKHR` elements with `swapchainCount` entries. If not `NULL`, each element of `pRegions` contains the region that has changed since the last present to the swapchain in the corresponding entry in the `VkPresentInfoKHR::pSwapchains` array.

Valid Usage

- VUID-VkPresentRegionsKHR-swapchainCount-01260
`swapchainCount` **must** be the same value as `VkPresentInfoKHR::swapchainCount`, where `VkPresentInfoKHR` is included in the `pNext` chain of this `VkPresentRegionsKHR` structure

Valid Usage (Implicit)

- VUID-VkPresentRegionsKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PRESENT_REGIONS_KHR`
- VUID-VkPresentRegionsKHR-pRegions-parameter
If `pRegions` is not `NULL`, `pRegions` **must** be a valid pointer to an array of `swapchainCount` valid `VkPresentRegionKHR` structures
- VUID-VkPresentRegionsKHR-swapchainCount-arraylength
`swapchainCount` **must** be greater than 0

For a given image and swapchain, the region to present is specified by the `VkPresentRegionKHR` structure, which is defined as:

```
// Provided by VK_KHR_incremental_present
typedef struct VkPresentRegionKHR {
    uint32_t      rectangleCount;
    const VkRectLayerKHR* pRectangles;
} VkPresentRegionKHR;
```

- **rectangleCount** is the number of rectangles in **pRectangles**, or zero if the entire image has changed and should be presented.
- **pRectangles** is either **NULL** or a pointer to an array of **VkRectLayerKHR** structures. The **VkRectLayerKHR** structure is the framebuffer coordinates, plus layer, of a portion of a presentable image that has changed and **must** be presented. If non-**NULL**, each entry in **pRectangles** is a rectangle of the given image that has changed since the last image was presented to the given swapchain. The rectangles **must** be specified relative to **VkSurfaceCapabilitiesKHR::currentTransform**, regardless of the swapchain's **preTransform**. The presentation engine will apply the **preTransform** transformation to the rectangles, along with any further transformation it applies to the image content.

Valid Usage (Implicit)

- VUID-VkPresentRegionKHR-pRectangles-parameter
If **rectangleCount** is not 0, and **pRectangles** is not **NULL**, **pRectangles** **must** be a valid pointer to an array of **rectangleCount** valid **VkRectLayerKHR** structures

The **VkRectLayerKHR** structure is defined as:

```
// Provided by VK_KHR_incremental_present
typedef struct VkRectLayerKHR {
    VkOffset2D    offset;
    VkExtent2D    extent;
    uint32_t      layer;
} VkRectLayerKHR;
```

- **offset** is the origin of the rectangle, in pixels.
- **extent** is the size of the rectangle, in pixels.
- **layer** is the layer of the image. For images with only one layer, the value of **layer** **must** be 0.

Some platforms allow the size of a surface to change, and then scale the pixels of the image to fit the surface. **VkRectLayerKHR** specifies pixels of the swapchain's image(s), which will be constant for the life of the swapchain.

Valid Usage

- VUID-VkRectLayerKHR-offset-04864
The sum of **offset** and **extent**, after being transformed according to the **preTransform**

member of the [VkSwapchainCreateInfoKHR](#) structure, **must** be no greater than the [imageExtent](#) member of the [VkSwapchainCreateInfoKHR](#) structure passed to [vkCreateSwapchainKHR](#)

- VUID-VkRectLayerKHR-layer-01262

[layer](#) **must** be less than the [imageArrayLayers](#) member of the [VkSwapchainCreateInfoKHR](#) structure passed to [vkCreateSwapchainKHR](#)

When the [VK_KHR_display_swapchain](#) extension is enabled, additional fields **can** be specified when presenting an image to a swapchain by setting [VkPresentInfoKHR::pNext](#) to point to a [VkDisplayPresentInfoKHR](#) structure.

The [VkDisplayPresentInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_display_swapchain
typedef struct VkDisplayPresentInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkRect2D           srcRect;
    VkRect2D           dstRect;
    VkBool32           persistent;
} VkDisplayPresentInfoKHR;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [srcRect](#) is a rectangular region of pixels to present. It **must** be a subset of the image being presented. If [VkDisplayPresentInfoKHR](#) is not specified, this region will be assumed to be the entire presentable image.
- [dstRect](#) is a rectangular region within the visible region of the swapchain's display mode. If [VkDisplayPresentInfoKHR](#) is not specified, this region will be assumed to be the entire visible region of the swapchain's mode. If the specified rectangle is a subset of the display mode's visible region, content from display planes below the swapchain's plane will be visible outside the rectangle. If there are no planes below the swapchain's, the area outside the specified rectangle will be black. If portions of the specified rectangle are outside of the display's visible region, pixels mapping only to those portions of the rectangle will be discarded.
- [persistent](#): If this is [VK_TRUE](#), the display engine will enable buffered mode on displays that support it. This allows the display engine to stop sending content to the display until a new image is presented. The display will instead maintain a copy of the last presented image. This allows less power to be used, but **may** increase presentation latency. If [VkDisplayPresentInfoKHR](#) is not specified, persistent mode will not be used.

If the extent of the [srcRect](#) and [dstRect](#) are not equal, the presented pixels will be scaled accordingly.

Valid Usage

- VUID-VkDisplayPresentInfoKHR-srcRect-01257
`srcRect` **must** specify a rectangular region that is a subset of the image being presented
- VUID-VkDisplayPresentInfoKHR-dstRect-01258
`dstRect` **must** specify a rectangular region that is a subset of the `visibleRegion` parameter of the display mode the swapchain being presented uses
- VUID-VkDisplayPresentInfoKHR-persistentContent-01259
If the `persistentContent` member of the `VkDisplayPropertiesKHR` structure returned by `vkGetPhysicalDeviceDisplayPropertiesKHR` for the display the present operation targets is `VK_FALSE`, then `persistent` **must** be `VK_FALSE`

Valid Usage (Implicit)

- VUID-VkDisplayPresentInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DISPLAY_PRESENT_INFO_KHR`

If the `pNext` chain of `VkPresentInfoKHR` includes a `VkDeviceGroupPresentInfoKHR` structure, then that structure includes an array of device masks and a device group present mode.

The `VkDeviceGroupPresentInfoKHR` structure is defined as:

```
// Provided by VK_VERSION_1_1 with VK_KHR_swapchain, VK_KHR_device_group with
VK_KHR_swapchain
typedef struct VkDeviceGroupPresentInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           swapchainCount;
    const uint32_t*     pDeviceMasks;
    VkDeviceGroupPresentModeFlagBitsKHR mode;
} VkDeviceGroupPresentInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchainCount` is zero or the number of elements in `pDeviceMasks`.
- `pDeviceMasks` is a pointer to an array of device masks, one for each element of `VkPresentInfoKHR::pSwapchains`.
- `mode` is a `VkDeviceGroupPresentModeFlagBitsKHR` value specifying the device group present mode that will be used for this present.

If `mode` is `VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR`, then each element of `pDeviceMasks` selects which instance of the swapchain image is presented. Each element of `pDeviceMasks` **must** have exactly one bit set, and the corresponding physical device **must** have a presentation engine as

reported by [VkDeviceGroupPresentCapabilitiesKHR](#).

If `mode` is `VK_DEVICE_GROUP_PRESENT_MODE_REMOTE_BIT_KHR`, then each element of `pDeviceMasks` selects which instance of the swapchain image is presented. Each element of `pDeviceMasks` **must** have exactly one bit set, and some physical device in the logical device **must** include that bit in its [VkDeviceGroupPresentCapabilitiesKHR::presentMask](#).

If `mode` is `VK_DEVICE_GROUP_PRESENT_MODE_SUM_BIT_KHR`, then each element of `pDeviceMasks` selects which instances of the swapchain image are component-wise summed and the sum of those images is presented. If the sum in any component is outside the representable range, the value of that component is undefined. Each element of `pDeviceMasks` **must** have a value for which all set bits are set in one of the elements of [VkDeviceGroupPresentCapabilitiesKHR::presentMask](#).

If `mode` is `VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_MULTI_DEVICE_BIT_KHR`, then each element of `pDeviceMasks` selects which instance(s) of the swapchain images are presented. For each bit set in each element of `pDeviceMasks`, the corresponding physical device **must** have a presentation engine as reported by [VkDeviceGroupPresentCapabilitiesKHR](#).

If [VkDeviceGroupPresentInfoKHR](#) is not provided or `swapchainCount` is zero then the masks are considered to be 1. If [VkDeviceGroupPresentInfoKHR](#) is not provided, `mode` is considered to be `VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR`.

Valid Usage

- VUID-VkDeviceGroupPresentInfoKHR-swapchainCount-01297
`swapchainCount` **must** equal 0 or [VkPresentInfoKHR::swapchainCount](#)
- VUID-VkDeviceGroupPresentInfoKHR-mode-01298
If `mode` is `VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_BIT_KHR`, then each element of `pDeviceMasks` **must** have exactly one bit set, and the corresponding element of [VkDeviceGroupPresentCapabilitiesKHR::presentMask](#) **must** be non-zero
- VUID-VkDeviceGroupPresentInfoKHR-mode-01299
If `mode` is `VK_DEVICE_GROUP_PRESENT_MODE_REMOTE_BIT_KHR`, then each element of `pDeviceMasks` **must** have exactly one bit set, and some physical device in the logical device **must** include that bit in its [VkDeviceGroupPresentCapabilitiesKHR::presentMask](#)
- VUID-VkDeviceGroupPresentInfoKHR-mode-01300
If `mode` is `VK_DEVICE_GROUP_PRESENT_MODE_SUM_BIT_KHR`, then each element of `pDeviceMasks` **must** have a value for which all set bits are set in one of the elements of [VkDeviceGroupPresentCapabilitiesKHR::presentMask](#)
- VUID-VkDeviceGroupPresentInfoKHR-mode-01301
If `mode` is `VK_DEVICE_GROUP_PRESENT_MODE_LOCAL_MULTI_DEVICE_BIT_KHR`, then for each bit set in each element of `pDeviceMasks`, the corresponding element of [VkDeviceGroupPresentCapabilitiesKHR::presentMask](#) **must** be non-zero
- VUID-VkDeviceGroupPresentInfoKHR-pDeviceMasks-01302
The value of each element of `pDeviceMasks` **must** be equal to the device mask passed in [VkAcquireNextImageInfoKHR::deviceMask](#) when the image index was last acquired
- VUID-VkDeviceGroupPresentInfoKHR-mode-01303

`mode` **must** have exactly one bit set, and that bit **must** have been included in `VkDeviceGroupSwapchainCreateInfoKHR::modes`

Valid Usage (Implicit)

- VUID-VkDeviceGroupPresentInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_INFO_KHR`
- VUID-VkDeviceGroupPresentInfoKHR-pDeviceMasks-parameter
If `swapchainCount` is not 0, `pDeviceMasks` **must** be a valid pointer to an array of `swapchainCount` `uint32_t` values
- VUID-VkDeviceGroupPresentInfoKHR-mode-parameter
`mode` **must** be a valid `VkDeviceGroupPresentModeFlagBitsKHR` value

The `VkPresentIdKHR` structure is defined as:

```
// Provided by VK_KHR_present_id
typedef struct VkPresentIdKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           swapchainCount;
    const uint64_t*     pPresentIds;
} VkPresentIdKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `swapchainCount` is the number of swapchains being presented to the `vkQueuePresentKHR` command.
- `pPresentIds` is `NULL` or a pointer to an array of `uint64_t` with `swapchainCount` entries. If not `NULL`, each non-zero value in `pPresentIds` specifies the present id to be associated with the presentation of the swapchain with the same index in the `vkQueuePresentKHR` call.

For applications to be able to reference specific presentation events queued by a call to `vkQueuePresentKHR`, an identifier needs to be associated with them. When the `presentId` feature is enabled, applications **can** include the `VkPresentIdKHR` structure in the `pNext` chain of the `VkPresentInfoKHR` structure to supply identifiers.

Each `VkSwapchainKHR` has a `presentId` associated with it. This value is initially zero when the `VkSwapchainKHR` is created.

When a `VkPresentIdKHR` structure with a non-`NULL` `pPresentIds` is included in the `pNext` chain of a `VkPresentInfoKHR` structure, each `pSwapchains` entry has a `presentId` associated in the `pPresentIds` array at the same index as the swapchain in the `pSwapchains` array. If this `presentId` is non-zero, then the application **can** later use this value to refer to that image presentation. A value of zero indicates that this presentation has no associated `presentId`. A non-zero `presentId` **must** be greater

than any non-zero `presentId` passed previously by the application for the same swapchain.

There is no requirement for any precise timing relationship between the presentation of the image to the user and the update of the `presentId` value, but implementations **should** make this as close as possible to the presentation of the first pixel in the new image to the user.

Valid Usage

- VUID-VkPresentIdKHR-swapchainCount-04998
`swapchainCount` **must** be the same value as `VkPresentInfoKHR::swapchainCount`, where this `VkPresentIdKHR` is in the `pNext` chain of the `VkPresentInfoKHR` structure
- VUID-VkPresentIdKHR-presentIds-04999
Each `presentIds` entry **must** be greater than any previous `presentIds` entry passed for the associated `pSwapchains` entry

Valid Usage (Implicit)

- VUID-VkPresentIdKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PRESENT_ID_KHR`
- VUID-VkPresentIdKHR-pPresentIds-parameter
If `pPresentIds` is not `NULL`, `pPresentIds` **must** be a valid pointer to an array of `swapchainCount` `uint64_t` values
- VUID-VkPresentIdKHR-swapchainCount-arraylength
`swapchainCount` **must** be greater than `0`

When the `presentWait` feature is enabled, an application **can** wait for an image to be presented to the user by first specifying a `presentId` for the target presentation by adding a `VkPresentIdKHR` structure to the `pNext` chain of the `VkPresentInfoKHR` structure and then waiting for that presentation to complete by calling:

```
// Provided by VK_KHR_present_wait
VkResult vkWaitForPresentKHR(
    VkDevice          device,
    VkSwapchainKHR    swapchain,
    uint64_t          presentId,
    uint64_t          timeout);
```

- `device` is the device associated with `swapchain`.
- `swapchain` is the non-retired swapchain on which an image was queued for presentation.
- `presentId` is the presentation `presentId` to wait for.
- `timeout` is the timeout period in units of nanoseconds. `timeout` is adjusted to the closest value allowed by the implementation-dependent timeout accuracy, which **may** be substantially longer than one nanosecond, and **may** be longer than the requested period.

`vkWaitForPresentKHR` waits for the `presentId` associated with `swapchain` to be increased in value so that it is at least equal to `presentId`.

For `VK_PRESENT_MODE_MAILBOX_KHR` (or other present mode where images may be replaced in the presentation queue) any wait of this type associated with such an image **must** be signaled no later than a wait associated with the replacing image would be signaled.

When the presentation has completed, the `presentId` associated with the related `pSwapchains` entry will be increased in value so that it is at least equal to the value provided in the `VkPresentIdKHR` structure.

There is no requirement for any precise timing relationship between the presentation of the image to the user and the update of the `presentId` value, but implementations **should** make this as close as possible to the presentation of the first pixel in the next image being presented to the user.

The call to `vkWaitForPresentKHR` will block until either the `presentId` associated with `swapchain` is greater than or equal to `presentId`, or `timeout` nanoseconds passes. When the swapchain becomes `OUT_OF_DATE`, the call will either return `VK_SUCCESS` (if the image was delivered to the presentation engine and may have been presented to the user) or will return early with status `VK_ERROR_OUT_OF_DATE_KHR` (if the image was not presented to the user).

As an exception to the normal rules for objects which are externally synchronized, the `swapchain` passed to `vkWaitForPresentKHR` **may** be simultaneously used by other threads in calls to functions other than `vkDestroySwapchainKHR`. Access to the swapchain data associated with this extension **must** be atomic within the implementation.

Valid Usage

- VUID-vkWaitForPresentKHR-swapchain-04997
`swapchain` **must** not be in the retired state
- VUID-vkWaitForPresentKHR-presentWait-06234
The `presentWait` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkWaitForPresentKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkWaitForPresentKHR-swapchain-parameter
`swapchain` **must** be a valid `VkSwapchainKHR` handle
- VUID-vkWaitForPresentKHR-swapchain-parent
`swapchain` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `swapchain` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`
- `VK_TIMEOUT`
- `VK_SUBOPTIMAL_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_DEVICE_LOST`
- `VK_ERROR_OUT_OF_DATE_KHR`
- `VK_ERROR_SURFACE_LOST_KHR`
- `VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT`

31.10. HDR Metadata

This section describes how to improve color reproduction of content to better reproduce colors as seen on the display used to originally optimize the content.

To provide HDR metadata to an implementation, call:

```
// Provided by VK_EXT_hdr_metadata
void vkSetHdrMetadataEXT(
    VkDevice          device,
    uint32_t          swapchainCount,
    const VkSwapchainKHR* pSwapchains,
    const VkHdrMetadataEXT* pMetadata);
```

- `device` is the logical device where the swapchain(s) were created.
- `swapchainCount` is the number of swapchains included in `pSwapchains`.
- `pSwapchains` is a pointer to an array of `swapchainCount` `VkSwapchainKHR` handles.
- `pMetadata` is a pointer to an array of `swapchainCount` `VkHdrMetadataEXT` structures.

The metadata will be applied to the specified `VkSwapchainKHR` objects at the next `vkQueuePresentKHR` call using that `VkSwapchainKHR` object. The metadata will persist until a subsequent `vkSetHdrMetadataEXT` changes it.

Valid Usage (Implicit)

- VUID-vkSetHdrMetadataEXT-device-parameter `device` **must** be a valid `VkDevice` handle

- VUID-vkSetHdrMetadataEXT-pSwapchains-parameter
pSwapchains **must** be a valid pointer to an array of **swapchainCount** valid **VkSwapchainKHR** handles
- VUID-vkSetHdrMetadataEXT-pMetadata-parameter
pMetadata **must** be a valid pointer to an array of **swapchainCount** valid **VkHdrMetadataEXT** structures
- VUID-vkSetHdrMetadataEXT-swapchainCount-arraylength
swapchainCount **must** be greater than 0
- VUID-vkSetHdrMetadataEXT-pSwapchains-parent
Each element of **pSwapchains** **must** have been created, allocated, or retrieved from **device**

The **VkHdrMetadataEXT** structure is defined as:

```
// Provided by VK_EXT_hdr_metadata
typedef struct VkHdrMetadataEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkXYColorEXT       displayPrimaryRed;
    VkXYColorEXT       displayPrimaryGreen;
    VkXYColorEXT       displayPrimaryBlue;
    VkXYColorEXT       whitePoint;
    float              maxLuminance;
    float              minLuminance;
    float              maxContentLightLevel;
    float              maxFrameAverageLightLevel;
} VkHdrMetadataEXT;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **displayPrimaryRed** is a **VkXYColorEXT** structure specifying the red primary of the display used to optimize the content
- **displayPrimaryGreen** is a **VkXYColorEXT** structure specifying the green primary of the display used to optimize the content
- **displayPrimaryBlue** is a **VkXYColorEXT** structure specifying the blue primary of the display used to optimize the content
- **whitePoint** is a **VkXYColorEXT** structure specifying the white-point of the display used to optimize the content
- **maxLuminance** is the maximum luminance of the display used to optimize the content in nits
- **minLuminance** is the minimum luminance of the display used to optimize the content in nits
- **maxContentLightLevel** is the value in nits of the desired luminance for the brightest pixels in the displayed image.
- **maxFrameAverageLightLevel** is the value in nits of the average luminance of the frame which has

the brightest average luminance anywhere in the content.

If any of the above values are unknown, they **can** be set to 0.

NOTE

The meta-data provided here is intended to be used as defined in the SMPTE 2086, CTA 861.3 and CIE 15:2004 specifications. The validity and use of this data is outside the scope of Vulkan.

Valid Usage (Implicit)

- VUID-VkHdrMetadataEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_HDR_METADATA_EXT`
- VUID-VkHdrMetadataEXT-pNext-pNext
`pNext` **must** be `NULL`

The `VkXYColorEXT` structure is defined as:

```
// Provided by VK_EXT_hdr_metadata
typedef struct VkXYColorEXT {
    float    x;
    float    y;
} VkXYColorEXT;
```

- `x` is the x chromaticity coordinate.
- `y` is the y chromaticity coordinate.

Chromaticity coordinates are as specified in CIE 15:2004 “Calculation of chromaticity coordinates” (Section 7.3) and are limited to between 0 and 1 for real colors.

Chapter 32. Deferred Host Operations

Certain Vulkan commands are inherently expensive for the host CPU to execute. It is often desirable to offload such work onto background threads, and to parallelize the work across multiple CPUs. The concept of *deferred operations* allows applications and drivers to coordinate the execution of expensive host commands using an application-managed thread pool.

The [VK_KHR_deferred_host_operations](#) extension defines the infrastructure and usage patterns for *deferrable commands*, but does not specify any commands as deferrable. This is left to additional dependent extensions. Commands **must** not be deferred unless the deferral is specifically allowed by another extension which depends on [VK_KHR_deferred_host_operations](#). This specification will refer to such extensions as *deferral extensions*.

32.1. Requesting Deferral

When an application requests an operation deferral, the implementation **may** defer the operation. When deferral is requested and the implementation defers any operation, the implementation **must** return [VK_OPERATION_DEFERRED_KHR](#) as the success code if no errors occurred. When deferral is requested, the implementation **should** defer the operation when the workload is significant, however if the implementation chooses not to defer any of the requested operations and instead executes all of them immediately, the implementation **must** return [VK_OPERATION_NOT_DEFERRED_KHR](#) as the success code if no errors occurred.

A deferred operation is created *complete* with an initial result value of [VK_SUCCESS](#). The deferred operation becomes *pending* when an operation has been successfully deferred with that deferred operation object.

A deferred operation is considered pending until the deferred operation completes. A pending deferred operation becomes *complete* when it has been fully executed by one or more threads. Pending deferred operations will never complete until they are *joined* by an application thread, using [vkDeferredOperationJoinKHR](#). Applications **can** join multiple threads to the same deferred operation, enabling concurrent execution of subtasks within that operation.

The application **can** query the status of a [VkDeferredOperationKHR](#) using the [vkGetDeferredOperationMaxConcurrencyKHR](#) or [vkGetDeferredOperationResultKHR](#) commands.

Parameters to the command requesting a deferred operation **may** be accessed by the implementation at any time until the deferred operation enters the complete state. The application **must** obey the following rules while a deferred operation is pending:

- Externally synchronized parameters **must** not be accessed.
- Pointer parameters **must** not be modified (e.g. reallocated/freed).
- The contents of pointer parameters which **may** be read by the command **must** not be modified.
- The contents of pointer parameters which **may** be written by the command **must** not be read.
- Vulkan object parameters **must** not be passed as externally synchronized parameters to any other command.

When the deferred operation is complete, the application **should** call [vkGetDeferredOperationResultKHR](#) to obtain the [VkResult](#) indicating success or failure of the operation. The [VkResult](#) value returned will be one of the values that the command requesting the deferred operation is able to return. Writes to output parameters of the requesting command will happen-before the deferred operation is complete.

When a deferral is requested for a command, the implementation **may** perform memory management operations on the allocator supplied to [vkCreateDeferredOperationKHR](#) for the deferred operation object, as described in the [Memory Allocation](#) chapter. Such allocations **must** occur on the thread which requests deferral.

If an allocator was supplied for the deferred command at the time of the deferral request, then the implementation **may** perform memory management operations on this allocator during the execution of [vkDeferredOperationJoinKHR](#). These operations **may** occur concurrently and **may** be performed by any joined thread. The application **must** ensure that the supplied allocator is able to operate correctly under these conditions.

32.2. Deferred Host Operations API

The [VkDeferredOperationKHR](#) handle is defined as:

```
// Provided by VK_KHR_deferred_host_operations
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDeferredOperationKHR)
```

This handle refers to a tracking structure which manages the execution state for a deferred command.

To construct the tracking object for a deferred command, call:

```
// Provided by VK_KHR_deferred_host_operations
VkResult vkCreateDeferredOperationKHR(
    VkDevice device,
    const VkAllocationCallbacks* pAllocator,
    VkDeferredOperationKHR* pDeferredOperation);
```

- [device](#) is the device which owns [operation](#).
- [pAllocator](#) controls host memory allocation as described in the [Memory Allocation](#) chapter.
- [pDeferredOperation](#) is a pointer to a handle in which the created [VkDeferredOperationKHR](#) is returned.

Valid Usage (Implicit)

- VUID-vkCreateDeferredOperationKHR-device-parameter [device](#) **must** be a valid [VkDevice](#) handle
- VUID-vkCreateDeferredOperationKHR-pAllocator-parameter

If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure

- `VUID-vkCreateDeferredOperationKHR-pDeferredOperation-parameter`
`pDeferredOperation` **must** be a valid pointer to a `VkDeferredOperationKHR` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`

To assign a thread to a deferred operation, call:

```
// Provided by VK_KHR_deferred_host_operations
VkResult vkDeferredOperationJoinKHR(
    VkDevice          device,
    VkDeferredOperationKHR operation);
```

- `device` is the device which owns `operation`.
- `operation` is the deferred operation that the calling thread should work on.

The `vkDeferredOperationJoinKHR` command will execute a portion of the deferred operation on the calling thread.

The return value will be one of the following:

- A return value of `VK_SUCCESS` indicates that `operation` is complete. The application **should** use `vkGetDeferredOperationResultKHR` to retrieve the result of `operation`.
- A return value of `VK_THREAD_DONE_KHR` indicates that the deferred operation is not complete, but there is no work remaining to assign to threads. Future calls to `vkDeferredOperationJoinKHR` are not necessary and will simply harm performance. This situation **may** occur when other threads executing `vkDeferredOperationJoinKHR` are about to complete `operation`, and the implementation is unable to partition the workload any further.
- A return value of `VK_THREAD_IDLE_KHR` indicates that the deferred operation is not complete, and there is no work for the thread to do at the time of the call. This situation **may** occur if the operation encounters a temporary reduction in parallelism. By returning `VK_THREAD_IDLE_KHR`, the implementation is signaling that it expects that more opportunities for parallelism will emerge as execution progresses, and that future calls to `vkDeferredOperationJoinKHR` **can** be beneficial. In the meantime, the application **can** perform other work on the calling thread.

Implementations **must** guarantee forward progress by enforcing the following invariants:

1. If only one thread has invoked `vkDeferredOperationJoinKHR` on a given operation, that thread

must execute the operation to completion and return `VK_SUCCESS`.

2. If multiple threads have concurrently invoked `vkDeferredOperationJoinKHR` on the same operation, then at least one of them **must** complete the operation and return `VK_SUCCESS`.

Valid Usage (Implicit)

- VUID-vkDeferredOperationJoinKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDeferredOperationJoinKHR-operation-parameter
`operation` **must** be a valid `VkDeferredOperationKHR` handle
- VUID-vkDeferredOperationJoinKHR-operation-parent
`operation` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_THREAD_DONE_KHR`
- `VK_THREAD_IDLE_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

When a deferred operation is completed, the application **can** destroy the tracking object by calling:

```
// Provided by VK_KHR_deferred_host_operations
void vkDestroyDeferredOperationKHR(
    VkDevice                                device,
    VkDeferredOperationKHR                  operation,
    const VkAllocationCallbacks*            pAllocator);
```

- `device` is the device which owns `operation`.
- `operation` is the completed operation to be destroyed.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyDeferredOperationKHR-operation-03434
If `VkAllocationCallbacks` were provided when `operation` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDeferredOperationKHR-operation-03435

If no `VkAllocationCallbacks` were provided when `operation` was created, `pAllocator` **must** be `NULL`

- `VUID-vkDestroyDeferredOperationKHR-operation-03436`
`operation` **must** be completed

Valid Usage (Implicit)

- `VUID-vkDestroyDeferredOperationKHR-device-parameter`
`device` **must** be a valid `VkDevice` handle
- `VUID-vkDestroyDeferredOperationKHR-operation-parameter`
If `operation` is not `VK_NULL_HANDLE`, `operation` **must** be a valid `VkDeferredOperationKHR` handle
- `VUID-vkDestroyDeferredOperationKHR-pAllocator-parameter`
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- `VUID-vkDestroyDeferredOperationKHR-operation-parent`
If `operation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `operation` **must** be externally synchronized

To query the number of additional threads that can usefully be joined to a deferred operation, call:

```
// Provided by VK_KHR_deferred_host_operations
uint32_t vkGetDeferredOperationMaxConcurrencyKHR(
    VkDevice          device,
    VkDeferredOperationKHR operation);
```

- `device` is the device which owns `operation`.
- `operation` is the deferred operation to be queried.

The returned value is the maximum number of threads that can usefully execute a deferred operation concurrently, reported for the state of the deferred operation at the point this command is called. This value is intended to be used to better schedule work onto available threads. Applications **can** join any number of threads to the deferred operation and expect it to eventually complete, though excessive joins **may** return `VK_THREAD_DONE_KHR` immediately, performing no useful work.

If `operation` is complete, `vkGetDeferredOperationMaxConcurrencyKHR` returns zero.

If `operation` is currently joined to any threads, the value returned by this command **may**

immediately be out of date.

If **operation** is pending, implementations **must** not return zero unless at least one thread is currently executing `vkDeferredOperationJoinKHR` on **operation**. If there are such threads, the implementation **should** return an estimate of the number of additional threads which it could profitably use.

Implementations **may** return $2^{32}-1$ to indicate that the maximum concurrency is unknown and cannot be easily derived. Implementations **may** return values larger than the maximum concurrency available on the host CPU. In these situations, an application **should** clamp the return value rather than oversubscribing the machine.

NOTE

The recommended usage pattern for applications is to query this value once, after deferral, and schedule no more than the specified number of threads to join the operation. Each time a joined thread receives `VK_THREAD_IDLE_KHR`, the application should schedule an additional join at some point in the future, but is not required to do so.

Valid Usage (Implicit)

- `VUID-vkGetDeferredOperationMaxConcurrencyKHR-device-parameter`
device **must** be a valid `VkDevice` handle
- `VUID-vkGetDeferredOperationMaxConcurrencyKHR-operation-parameter`
operation **must** be a valid `VkDeferredOperationKHR` handle
- `VUID-vkGetDeferredOperationMaxConcurrencyKHR-operation-parent`
operation **must** have been created, allocated, or retrieved from **device**

The `vkGetDeferredOperationResultKHR` function is defined as:

```
// Provided by VK_KHR_deferred_host_operations
VkResult vkGetDeferredOperationResultKHR(
    VkDevice          device,
    VkDeferredOperationKHR operation);
```

- **device** is the device which owns **operation**.
- **operation** is the operation whose deferred result is being queried.

If no command has been deferred on **operation**, `vkGetDeferredOperationResultKHR` returns `VK_SUCCESS`.

If the deferred operation is pending, `vkGetDeferredOperationResultKHR` returns `VK_NOT_READY`.

If the deferred operation is complete, it returns the appropriate return value from the original command. This value **must** be one of the `VkResult` values which could have been returned by the original command if the operation had not been deferred.

Valid Usage (Implicit)

- VUID-vkGetDeferredOperationResultKHR-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkGetDeferredOperationResultKHR-operation-parameter **operation** **must** be a valid [VkDeferredOperationKHR](#) handle
- VUID-vkGetDeferredOperationResultKHR-operation-parent **operation** **must** have been created, allocated, or retrieved from **device**

Return Codes

Success

- **VK_SUCCESS**
- **VK_NOT_READY**

Failure

None

Chapter 33. Acceleration Structures

33.1. Acceleration Structures

Acceleration structures are data structures used by the implementation to efficiently manage scene geometry as it is [traversed during a ray tracing query](#). The application is responsible for managing acceleration structure objects (see [Acceleration Structures](#)), including allocation, destruction, executing builds or updates, and synchronizing resources used during ray tracing queries.

There are two types of acceleration structures, *top level acceleration structures* and *bottom level acceleration structures*.

An acceleration structure is considered to be constructed if an [acceleration structure build command](#) or [copy command](#) has been executed with the given acceleration structure as the destination.

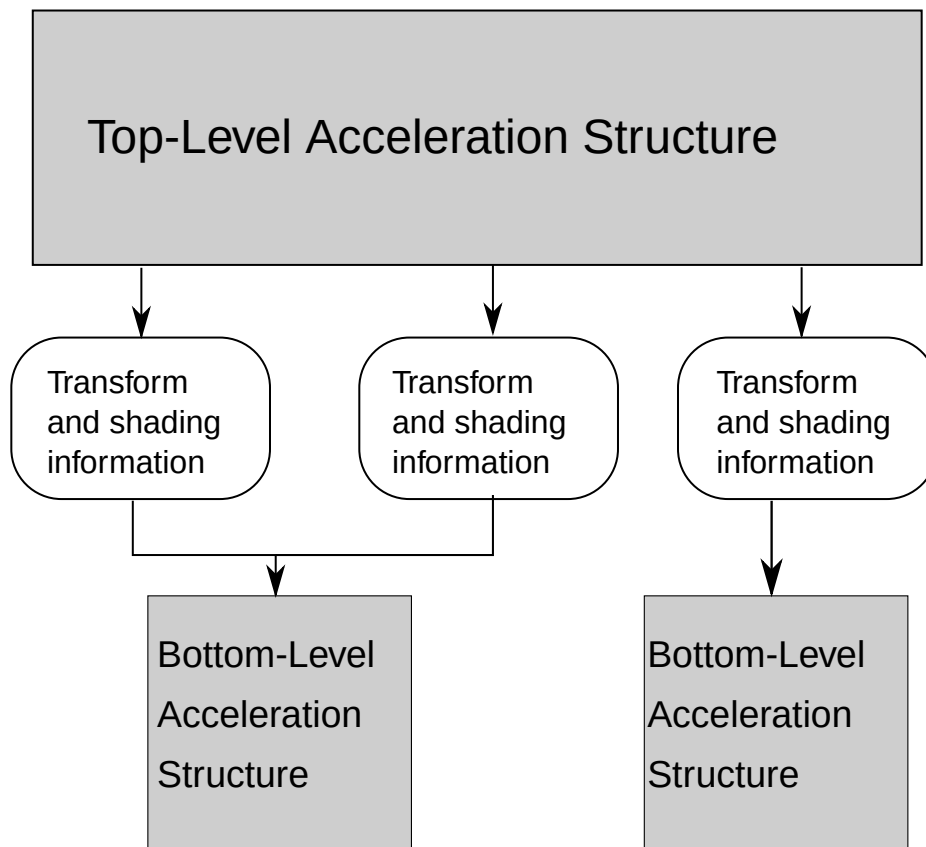


Figure 22. Acceleration Structure

Caption

The diagram shows the relationship between top and bottom level acceleration structures.

33.1.1. Geometry

Geometries refer to a triangle or axis-aligned bounding box.

33.1.2. Top Level Acceleration Structures

Opaque acceleration structure for an array of instances. The descriptor or device address referencing this is the starting point for traversal.

The top level acceleration structure takes a reference to any bottom level acceleration structure referenced by its instances. Those bottom level acceleration structure objects **must** be valid when the top level acceleration structure is accessed.

33.1.3. Bottom Level Acceleration Structures

Opaque acceleration structure for an array of geometries.

33.1.4. Acceleration Structure Update Rules

The API defines two types of operations to produce acceleration structures from geometry:

- A *build* operation is used to construct an acceleration structure.
- An *update* operation is used to modify an existing acceleration structure.

An update operation imposes certain constraints on the input, in exchange for considerably faster execution. When performing an update, the application is required to provide a full description of the acceleration structure, but is prohibited from changing anything other than instance definitions, transform matrices, and vertex or AABB positions. All other aspects of the description **must** exactly match the one from the original build.

More precisely, the application **must** not use an update operation to do any of the following:

- Change primitives or instances from *active* to *inactive*, or vice versa (as defined in [Inactive Primitives and Instances](#)).
- Change the index or vertex formats of triangle geometry.
- Change triangle geometry transform pointers from null to non-null or vice versa.
- Change the number of geometries or instances in the structure.
- Change the geometry flags for any geometry in the structure.
- Change the number of vertices or primitives for any geometry in the structure.

If the original acceleration structure was built using opacity micromaps and `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_EXT` was set in `flags`, the application **must** provide the corresponding micromap information to the update operation. The application is prohibited from changing anything other than the specific opacity values assigned to the triangles.

More precisely, the application **must** not use an update operation to do any of the following:

- Remove micromaps or `VkOpacityMicromapSpecialIndexEXT` values from a geometry which previously had them, or vice versa.
- Change between use of `VkOpacityMicromapSpecialIndexEXT` values and explicit micro-map

triangles.

- Change the subdivision level or format of the micromap triangle associated with any acceleration-structure triangle.

If the original acceleration structure was built using opacity micromaps and `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_EXT` was set in `flags`, the application **must** provide a micromap to the update operation.

If the original acceleration structure was built using opacity micromaps and neither opacity micromap update flag is set the application **must** provide the original micromap to the update operation.

33.1.5. Inactive Primitives and Instances

Acceleration structures allow the use of particular input values to signal *inactive* primitives or instances.

An *inactive* triangle is one for which the first (X) component of any vertex is NaN. If any other vertex component is NaN, and the first is not, the behavior is undefined. If the vertex format does not have a NaN representation, then all triangles are considered active.

An *inactive* instance is one whose acceleration structure reference is 0.

An *inactive* AABB is one for which the minimum X coordinate is NaN. If any other component is NaN, and the first is not, the behavior is undefined.

In the above definitions, “NaN” refers to any type of NaN. Signaling, non-signaling, quiet, loud, or otherwise.

An inactive object is considered invisible to all rays, and **should** not be represented in the acceleration structure. Implementations **should** ensure that the presence of inactive objects does not seriously degrade traversal performance.

Inactive objects are counted in the auto-generated index sequences which are provided to shaders via `InstanceId` and `PrimitiveId` SPIR-V decorations. This allows objects in the scene to change freely between the active and inactive states, without affecting the layout of any arrays which are being indexed using the ID values.

Any transition between the active and inactive states requires a full acceleration structure rebuild. Applications **must** not perform an acceleration structure update where an object is active in the source acceleration structure but would be inactive in the destination, or vice versa.

33.1.6. Degenerate Primitives and Instances

Degenerate primitives and instances behave differently to [inactive primitives and instances](#), and are defined as:

- triangles that have one or more vertices whose respective (X), (Y), (Z) components are identical, or have three vertices that have at least two of the (X), (Y), or (Z) components identical, therefore forming a line or point. Degenerate triangles do not generate any intersections.

- AABBs whose `minX=maxX`, `minY=maxY`, and `minZ=maxZ`. Degenerate AABBs **may** invoke the intersection shader.
- instances that reference bottom level acceleration structures that contain no active primitives. When building an acceleration structure, implementations **should** treat degenerate instances as though they are a point at the instance origin, specified by `VkAccelerationStructureInstanceKHR::transform`.

Unlike *inactive* primitives and instances, *degenerate* primitives and instances **may** transition from the degenerate to the non-degenerate state, or vice versa, when performing an acceleration structure update.

If an acceleration structure is built without `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR` set in `VkAccelerationStructureBuildGeometryInfoKHR::flags`, degenerate primitives **may** be discarded. Primitives that are defined with the same index value for more than one vertex **can** always be discarded.

33.1.7. Building Acceleration Structures

To build acceleration structures call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdBuildAccelerationStructuresKHR(
    VkCommandBuffer                commandBuffer,
    uint32_t                       infoCount,
    const VkAccelerationStructureBuildGeometryInfoKHR* pInfos,
    const VkAccelerationStructureBuildRangeInfoKHR* const* ppBuildRangeInfos);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `infoCount` is the number of acceleration structures to build. It specifies the number of the `pInfos` structures and `ppBuildRangeInfos` pointers that **must** be provided.
- `pInfos` is a pointer to an array of `infoCount` `VkAccelerationStructureBuildGeometryInfoKHR` structures defining the geometry used to build each acceleration structure.
- `ppBuildRangeInfos` is a pointer to an array of `infoCount` pointers to arrays of `VkAccelerationStructureBuildRangeInfoKHR` structures. Each `ppBuildRangeInfos[i]` is a pointer to an array of `pInfos[i].geometryCount` `VkAccelerationStructureBuildRangeInfoKHR` structures defining dynamic offsets to the addresses where geometry data is stored, as defined by `pInfos[i]`.

The `vkCmdBuildAccelerationStructuresKHR` command provides the ability to initiate multiple acceleration structures builds, however there is no ordering or synchronization implied between any of the individual acceleration structure builds.

NOTE

This means that an application **cannot** build a top-level acceleration structure in the same `vkCmdBuildAccelerationStructuresKHR` call as the associated bottom-level or instance acceleration structures are being built. There also **cannot** be any memory aliasing between any acceleration structure memories or scratch memories being used by any of the builds.

Accesses to the acceleration structure scratch buffers as identified by the `VkAccelerationStructureBuildGeometryInfoKHR::scratchData` buffer device addresses **must** be **synchronized** with the `VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` pipeline stage and an access type of `(VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR | VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR)`. Accesses to each `VkAccelerationStructureBuildGeometryInfoKHR::srcAccelerationStructure` and `VkAccelerationStructureBuildGeometryInfoKHR::dstAccelerationStructure` **must** be **synchronized** with the `VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` pipeline stage and an access type of `VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR` or `VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR`, as appropriate.

Accesses to other input buffers as identified by any used values of `VkAccelerationStructureGeometryTrianglesDataKHR::vertexData`, `VkAccelerationStructureGeometryTrianglesDataKHR::indexData`, `VkAccelerationStructureGeometryTrianglesDataKHR::transformData`, `VkAccelerationStructureGeometryAabbsDataKHR::data`, and `VkAccelerationStructureGeometryInstancesDataKHR::data` **must** be **synchronized** with the `VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` pipeline stage and an access type of `VK_ACCESS_SHADER_READ_BIT`.

Valid Usage

- VUID-vkCmdBuildAccelerationStructuresKHR-accelerationStructure-08923
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkCmdBuildAccelerationStructuresKHR-mode-04628
The `mode` member of each element of `pInfos` **must** be a valid `VkBuildAccelerationStructureModeKHR` value
- VUID-vkCmdBuildAccelerationStructuresKHR-srcAccelerationStructure-04629
If the `srcAccelerationStructure` member of any element of `pInfos` is not `VK_NULL_HANDLE`, the `srcAccelerationStructure` member **must** be a valid `VkAccelerationStructureKHR` handle
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-04630
For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `srcAccelerationStructure` member **must** not be `VK_NULL_HANDLE`
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03403
The `srcAccelerationStructure` member of any element of `pInfos` **must** not be the same acceleration structure as the `dstAccelerationStructure` member of any other element of `pInfos`
- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03698
The `dstAccelerationStructure` member of any element of `pInfos` **must** not be the same acceleration structure as the `dstAccelerationStructure` member of any other element of `pInfos`
- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03800

The `dstAccelerationStructure` member of any element of `pInfos` **must** be a valid `VkAccelerationStructureKHR` handle

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03699

For each element of `pInfos`, if its `type` member is `VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR`, its `dstAccelerationStructure` member **must** have been created with a value of `VkAccelerationStructureCreateInfoKHR::type` equal to either `VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR` or `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR`

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03700

For each element of `pInfos`, if its `type` member is `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR`, its `dstAccelerationStructure` member **must** have been created with a value of `VkAccelerationStructureCreateInfoKHR::type` equal to either `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR` or `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR`

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03663

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, `inactive` primitives in its `srcAccelerationStructure` member **must** not be made active

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03664

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, active primitives in its `srcAccelerationStructure` member **must** not be made `inactive`

- VUID-vkCmdBuildAccelerationStructuresKHR-None-03407

The `dstAccelerationStructure` member of any element of `pInfos` **must** not be referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR` in any other element of `pInfos`

- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03701

The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `srcAccelerationStructure` member of any other element of `pInfos` with a `mode` equal to `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, which is accessed by this command

- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03702

The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `dstAccelerationStructure` member of any other element of `pInfos`, which is accessed by this command

- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03703

The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `scratchData` member of any element of `pInfos` (including the same element), which is accessed by this command

- VUID-vkCmdBuildAccelerationStructuresKHR-scratchData-03704

The range of memory backing the `scratchData` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `scratchData`

member of any other element of `pInfos`, which is accessed by this command

- VUID-vkCmdBuildAccelerationStructuresKHR-scratchData-03705

The range of memory backing the `scratchData` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `srcAccelerationStructure` member of any element of `pInfos` with a `mode` equal to `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR` (including the same element), which is accessed by this command

- VUID-vkCmdBuildAccelerationStructuresKHR-dstAccelerationStructure-03706

The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing any acceleration structure referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR` in any other element of `pInfos`, which is accessed by this command

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03667

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `srcAccelerationStructure` member **must** have previously been constructed with `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR` set in `VkAccelerationStructureBuildGeometryInfoKHR::flags` in the build

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03668

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `srcAccelerationStructure` and `dstAccelerationStructure` members **must** either be the same `VkAccelerationStructureKHR`, or not have any `memory aliasing`

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03758

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `geometryCount` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03759

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `flags` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03760

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `type` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03761

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, its `geometryType` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03762

For each element of `pInfos`, if its `mode` member is

VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, its flags member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03763

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.vertexFormat member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03764

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.maxVertex member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03765

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.indexType member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03766

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its geometry.triangles.transformData address was NULL when srcAccelerationStructure was last built, then it **must** be NULL

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03767

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its geometry.triangles.transformData address was not NULL when srcAccelerationStructure was last built, then it **must** not be NULL

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03768

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, and geometry.triangles.indexType is not VK_INDEX_TYPE_NONE_KHR, then the value of each index

referenced **must** be the same as the corresponding index value when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresKHR-primitiveCount-03769

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, the `primitiveCount` member of its corresponding `VkAccelerationStructureBuildRangeInfoKHR` structure **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03801

For each element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, the corresponding `ppBuildRangeInfos[i][j].primitiveCount` **must** be less than or equal to `VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxInstanceCount`

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03707

For each element of `pInfos`, the `buffer` used to create its `dstAccelerationStructure` member **must** be bound to device memory

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03708

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR` the `buffer` used to create its `srcAccelerationStructure` member **must** be bound to device memory

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03709

For each element of `pInfos`, the `buffer` used to create each acceleration structure referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR` **must** be bound to device memory

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03671

If `pInfos[i].mode` is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR`, all addresses between `pInfos[i].scratchData.deviceAddress` and `pInfos[i].scratchData.deviceAddress + N - 1` **must** be in the buffer device address range of the same buffer, where N is given by the `buildScratchSize` member of the `VkAccelerationStructureBuildSizesInfoKHR` structure returned from a call to `vkGetAccelerationStructureBuildSizesKHR` with an identical `VkAccelerationStructureBuildGeometryInfoKHR` structure and primitive count

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03672

If `pInfos[i].mode` is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, all addresses between `pInfos[i].scratchData.deviceAddress` and `pInfos[i].scratchData.deviceAddress + N - 1` **must** be in the buffer device address range of the same buffer, where N is given by the `updateScratchSize` member of the `VkAccelerationStructureBuildSizesInfoKHR` structure returned from a call to `vkGetAccelerationStructureBuildSizesKHR` with an identical `VkAccelerationStructureBuildGeometryInfoKHR` structure and primitive count

- VUID-vkCmdBuildAccelerationStructuresKHR-geometry-03673

The buffers from which the buffer device addresses for all of the `geometry.triangles.vertexData`, `geometry.triangles.indexData`, `geometry.triangles.transformData`, `geometry.aabbs.data`, and `geometry.instances.data`

members of all `pInfos[i].pGeometries` and `pInfos[i].ppGeometries` are queried **must** have been created with the `VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR` usage flag

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03674
The buffer from which the buffer device address `pInfos[i].scratchData.deviceAddress` is queried **must** have been created with `VK_BUFFER_USAGE_STORAGE_BUFFER_BIT` usage flag
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03802
For each element of `pInfos`, its `scratchData.deviceAddress` member **must** be a valid device address obtained from `vkGetBufferDeviceAddressKHR`
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03803
For each element of `pInfos`, if `scratchData.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03710
For each element of `pInfos`, its `scratchData.deviceAddress` member **must** be a multiple of `VkPhysicalDeviceAccelerationStructurePropertiesKHR::minAccelerationStructureScratchOffsetAlignment`
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03804
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, `geometry.triangles.vertexData.deviceAddress` **must** be a valid device address obtained from `vkGetBufferDeviceAddressKHR`
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03805
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.vertexData.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03711
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, `geometry.triangles.vertexData.deviceAddress` **must** be aligned to the size in bytes of the smallest component of the format in `vertexFormat`
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03806
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.indexType` is not `VK_INDEX_TYPE_NONE_KHR`, `geometry.triangles.indexData.deviceAddress` **must** be a valid device address obtained from `vkGetBufferDeviceAddressKHR`
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03807
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.indexType` is not `VK_INDEX_TYPE_NONE_KHR`, if `geometry.triangles.indexData.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03712
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, and with `geometry.triangles.indexType` not equal to

`VK_INDEX_TYPE_NONE_KHR`, `geometry.triangles.indexData.deviceAddress` **must** be aligned to the size in bytes of the type in `indexType`

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03808
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.transformData.deviceAddress` is not `0`, it **must** be a valid device address obtained from `vkGetBufferDeviceAddressKHR`
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03809
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.transformData.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03810
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.transformData.deviceAddress` is not `0`, it **must** be aligned to 16 bytes
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03811
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_AABBS_KHR`, `geometry.aabbs.data.deviceAddress` **must** be a valid device address obtained from `vkGetBufferDeviceAddressKHR`
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03812
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_AABBS_KHR`, if `geometry.aabbs.data.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03714
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_AABBS_KHR`, `geometry.aabbs.data.deviceAddress` **must** be aligned to 8 bytes
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03715
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, if `geometry.arrayOfPointers` is `VK_FALSE`, `geometry.instances.data.deviceAddress` **must** be aligned to 16 bytes
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03716
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, if `geometry.arrayOfPointers` is `VK_TRUE`, `geometry.instances.data.deviceAddress` **must** be aligned to 8 bytes
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03717
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, if `geometry.arrayOfPointers` is `VK_TRUE`, each element of `geometry.instances.data.deviceAddress` in device memory **must** be aligned to 16 bytes
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03813
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, `geometry.instances.data.deviceAddress` **must** be a valid device address obtained from `vkGetBufferDeviceAddressKHR`

- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-03814
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, if `geometry.instances.data.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-06707
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, each `VkAccelerationStructureInstanceKHR::accelerationStructureReference` value in `geometry.instances.data.deviceAddress` **must** be a valid device address containing a value obtained from `vkGetAccelerationStructureDeviceAddressKHR` or `0`
- VUID-vkCmdBuildAccelerationStructuresKHR-commandBuffer-09547
`commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-10126
For each `pInfos[i]`, `dstAccelerationStructure` **must** have been created with a value of `VkAccelerationStructureCreateInfoKHR::size` greater than or equal to either:
 - the memory size required by the build operation, as returned by `vkGetAccelerationStructureBuildSizesKHR` with `pBuildInfo = pInfos[i]` and with each element of the `pMaxPrimitiveCounts` array greater than or equal to the equivalent `ppBuildRangeInfos[i][j].primitiveCount` values for `j` in `[0, pInfos[i].geometryCount)` or,
 - the result of querying the corresponding `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR`, if updating a compacted acceleration structure
- VUID-vkCmdBuildAccelerationStructuresKHR-ppBuildRangeInfos-03676
Each element of `ppBuildRangeInfos[i]` **must** be a valid pointer to an array of `pInfos[i].geometryCount` `VkAccelerationStructureBuildRangeInfoKHR` structures

Valid Usage (Implicit)

- VUID-vkCmdBuildAccelerationStructuresKHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBuildAccelerationStructuresKHR-pInfos-parameter
`pInfos` **must** be a valid pointer to an array of `infoCount` valid `VkAccelerationStructureBuildGeometryInfoKHR` structures
- VUID-vkCmdBuildAccelerationStructuresKHR-ppBuildRangeInfos-parameter
`ppBuildRangeInfos` **must** be a valid pointer to an array of `infoCount` `VkAccelerationStructureBuildRangeInfoKHR` structures
- VUID-vkCmdBuildAccelerationStructuresKHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdBuildAccelerationStructuresKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations

- VUID-vkCmdBuildAccelerationStructuresKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdBuildAccelerationStructuresKHR-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBuildAccelerationStructuresKHR-infoCount-arraylength
infoCount **must** be greater than 0

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

To build acceleration structures with some parameters sourced on the device call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdBuildAccelerationStructuresIndirectKHR(
    VkCommandBuffer          commandBuffer,
    uint32_t                 infoCount,
    const VkAccelerationStructureBuildGeometryInfoKHR* pInfos,
    const VkDeviceAddress*   pIndirectDeviceAddresses,
    const uint32_t*          pIndirectStrides,
    const uint32_t* const*   ppMaxPrimitiveCounts);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **infoCount** is the number of acceleration structures to build.
- **pInfos** is a pointer to an array of **infoCount** **VkAccelerationStructureBuildGeometryInfoKHR** structures defining the geometry used to build each acceleration structure.
- **pIndirectDeviceAddresses** is a pointer to an array of **infoCount** buffer device addresses which point to **pInfos[i].geometryCount** **VkAccelerationStructureBuildRangeInfoKHR** structures defining dynamic offsets to the addresses where geometry data is stored, as defined by **pInfos[i]**.
- **pIndirectStrides** is a pointer to an array of **infoCount** byte strides between elements of **pIndirectDeviceAddresses**.
- **ppMaxPrimitiveCounts** is a pointer to an array of **infoCount** pointers to arrays of **pInfos**

[i].[geometryCount](#) values indicating the maximum number of primitives that will be built by this command for each geometry.

Accesses to acceleration structures, scratch buffers, vertex buffers, index buffers, and instance buffers **must** be synchronized as with [vkCmdBuildAccelerationStructuresKHR](#).

Accesses to any element of [pIndirectDeviceAddresses](#) **must** be [synchronized](#) with the [VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR](#) [pipeline stage](#) and an [access type](#) of [VK_ACCESS_INDIRECT_COMMAND_READ_BIT](#).

Valid Usage

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-accelerationStructureIndirectBuild-03650
The [VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureIndirectBuild](#) feature **must** be enabled
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-mode-04628
The [mode](#) member of each element of [pInfos](#) **must** be a valid [VkBuildAccelerationStructureModeKHR](#) value
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-srcAccelerationStructure-04629
If the [srcAccelerationStructure](#) member of any element of [pInfos](#) is not [VK_NULL_HANDLE](#), the [srcAccelerationStructure](#) member **must** be a valid [VkAccelerationStructureKHR](#) handle
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-04630
For each element of [pInfos](#), if its [mode](#) member is [VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR](#), its [srcAccelerationStructure](#) member **must** not be [VK_NULL_HANDLE](#)
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03403
The [srcAccelerationStructure](#) member of any element of [pInfos](#) **must** not be the same acceleration structure as the [dstAccelerationStructure](#) member of any other element of [pInfos](#)
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03698
The [dstAccelerationStructure](#) member of any element of [pInfos](#) **must** not be the same acceleration structure as the [dstAccelerationStructure](#) member of any other element of [pInfos](#)
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03800
The [dstAccelerationStructure](#) member of any element of [pInfos](#) **must** be a valid [VkAccelerationStructureKHR](#) handle
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03699
For each element of [pInfos](#), if its [type](#) member is [VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR](#), its [dstAccelerationStructure](#) member **must** have been created with a value of [VkAccelerationStructureCreateInfoKHR::type](#) equal to either [VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR](#) or [VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR](#)

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03700
For each element of `pInfos`, if its `type` member is `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR`, its `dstAccelerationStructure` member **must** have been created with a value of `VkAccelerationStructureCreateInfoKHR::type` equal to either `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR` or `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR`
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03663
For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, `inactive` `primitives` in its `srcAccelerationStructure` member **must** not be made active
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03664
For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, active `primitives` in its `srcAccelerationStructure` member **must** not be made `inactive`
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-None-03407
The `dstAccelerationStructure` member of any element of `pInfos` **must** not be referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR` in any other element of `pInfos`
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03701
The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `srcAccelerationStructure` member of any other element of `pInfos` with a `mode` equal to `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03702
The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `dstAccelerationStructure` member of any other element of `pInfos`, which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03703
The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `scratchData` member of any element of `pInfos` (including the same element), which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-scratchData-03704
The range of memory backing the `scratchData` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `scratchData` member of any other element of `pInfos`, which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-scratchData-03705
The range of memory backing the `scratchData` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `srcAccelerationStructure` member of any element of `pInfos` with a `mode` equal to `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR` (including the same element), which is accessed by this command
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-dstAccelerationStructure-03706

The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing any acceleration structure referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR` in any other element of `pInfos`, which is accessed by this command

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03667

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `srcAccelerationStructure` member **must** have previously been constructed with `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR` set in `VkAccelerationStructureBuildGeometryInfoKHR::flags` in the build

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03668

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `srcAccelerationStructure` and `dstAccelerationStructure` members **must** either be the same `VkAccelerationStructureKHR`, or not have any `memory aliasing`

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03758

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `geometryCount` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03759

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `flags` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03760

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `type` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03761

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, its `geometryType` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03762

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, its `flags` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03763

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, if `geometryType` is `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, its

`geometry.triangles.vertexFormat` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03764

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, if `geometryType` is `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, its `geometry.triangles.maxVertex` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03765

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, if `geometryType` is `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, its `geometry.triangles.indexType` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03766

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, if `geometryType` is `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if its `geometry.triangles.transformData` address was `NULL` when `srcAccelerationStructure` was last built, then it **must** be `NULL`

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03767

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, if `geometryType` is `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if its `geometry.triangles.transformData` address was not `NULL` when `srcAccelerationStructure` was last built, then it **must** not be `NULL`

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03768

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, if `geometryType` is `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, and `geometry.triangles.indexType` is not `VK_INDEX_TYPE_NONE_KHR`, then the value of each index referenced **must** be the same as the corresponding index value when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-primitiveCount-03769

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, the `primitiveCount` member of its corresponding `VkAccelerationStructureBuildRangeInfoKHR` structure **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03801
For each element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, the corresponding `ppMaxPrimitiveCounts[i][j]` **must** be less than or equal to `VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxInstanceCount`
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03707
For each element of `pInfos`, the `buffer` used to create its `dstAccelerationStructure` member **must** be bound to device memory
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03708
For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR` the `buffer` used to create its `srcAccelerationStructure` member **must** be bound to device memory
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03709
For each element of `pInfos`, the `buffer` used to create each acceleration structure referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR` **must** be bound to device memory
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03671
If `pInfos[i].mode` is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR`, all addresses between `pInfos[i].scratchData.deviceAddress` and `pInfos[i].scratchData.deviceAddress + N - 1` **must** be in the buffer device address range of the same buffer, where N is given by the `buildScratchSize` member of the `VkAccelerationStructureBuildSizesInfoKHR` structure returned from a call to `vkGetAccelerationStructureBuildSizesKHR` with an identical `VkAccelerationStructureBuildGeometryInfoKHR` structure and primitive count
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03672
If `pInfos[i].mode` is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, all addresses between `pInfos[i].scratchData.deviceAddress` and `pInfos[i].scratchData.deviceAddress + N - 1` **must** be in the buffer device address range of the same buffer, where N is given by the `updateScratchSize` member of the `VkAccelerationStructureBuildSizesInfoKHR` structure returned from a call to `vkGetAccelerationStructureBuildSizesKHR` with an identical `VkAccelerationStructureBuildGeometryInfoKHR` structure and primitive count
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-geometry-03673
The buffers from which the buffer device addresses for all of the `geometry.triangles.vertexData`, `geometry.triangles.indexData`, `geometry.triangles.transformData`, `geometry.aabbs.data`, and `geometry.instances.data` members of all `pInfos[i].pGeometries` and `pInfos[i].ppGeometries` are queried **must** have been created with the `VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR` usage flag
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03674
The buffer from which the buffer device address `pInfos[i].scratchData.deviceAddress` is queried **must** have been created with `VK_BUFFER_USAGE_STORAGE_BUFFER_BIT` usage flag
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03802
For each element of `pInfos`, its `scratchData.deviceAddress` member **must** be a valid device address obtained from `vkGetBufferDeviceAddressKHR`
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03803

For each element of `pInfos`, if `scratchData.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03710

For each element of `pInfos`, its `scratchData.deviceAddress` member **must** be a multiple of `VkPhysicalDeviceAccelerationStructurePropertiesKHR::minAccelerationStructureScratchOffsetAlignment`

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03804

For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, `geometry.triangles.vertexData.deviceAddress` **must** be a valid device address obtained from `vkGetBufferDeviceAddressKHR`

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03805

For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.vertexData.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03711

For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, `geometry.triangles.vertexData.deviceAddress` **must** be aligned to the size in bytes of the smallest component of the format in `vertexFormat`

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03806

For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.indexType` is not `VK_INDEX_TYPE_NONE_KHR`, `geometry.triangles.indexData.deviceAddress` **must** be a valid device address obtained from `vkGetBufferDeviceAddressKHR`

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03807

For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.indexType` is not `VK_INDEX_TYPE_NONE_KHR`, if `geometry.triangles.indexData.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03712

For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, and with `geometry.triangles.indexType` not equal to `VK_INDEX_TYPE_NONE_KHR`, `geometry.triangles.indexData.deviceAddress` **must** be aligned to the size in bytes of the type in `indexType`

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03808

For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.transformData.deviceAddress` is not `0`, it **must** be a valid device address obtained from `vkGetBufferDeviceAddressKHR`

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03809

For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.transformData.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a

single [VkDeviceMemory](#) object

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03810
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.transformData.deviceAddress` is not `0`, it **must** be aligned to 16 bytes
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03811
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_AABBS_KHR`, `geometry.aabbs.data.deviceAddress` **must** be a valid device address obtained from [vkGetBufferDeviceAddressKHR](#)
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03812
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_AABBS_KHR`, if `geometry.aabbs.data.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single [VkDeviceMemory](#) object
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03714
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_AABBS_KHR`, `geometry.aabbs.data.deviceAddress` **must** be aligned to 8 bytes
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03715
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, if `geometry.arrayOfPointers` is `VK_FALSE`, `geometry.instances.data.deviceAddress` **must** be aligned to 16 bytes
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03716
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, if `geometry.arrayOfPointers` is `VK_TRUE`, `geometry.instances.data.deviceAddress` **must** be aligned to 8 bytes
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03717
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, if `geometry.arrayOfPointers` is `VK_TRUE`, each element of `geometry.instances.data.deviceAddress` in device memory **must** be aligned to 16 bytes
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03813
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, `geometry.instances.data.deviceAddress` **must** be a valid device address obtained from [vkGetBufferDeviceAddressKHR](#)
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03814
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, if `geometry.instances.data.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single [VkDeviceMemory](#) object
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-06707
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, each [VkAccelerationStructureInstanceKHR](#) `::accelerationStructureReference` value in `geometry.instances.data.deviceAddress` **must** be a valid device address containing a value obtained from [vkGetAccelerationStructureDeviceAddressKHR](#) or `0`

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-commandBuffer-09547
`commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectDeviceAddresses-03645
For any element of `pIndirectDeviceAddresses`, if the buffer from which it was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectDeviceAddresses-03646
For any element of `pIndirectDeviceAddresses[i]`, all device addresses between `pIndirectDeviceAddresses[i]` and `pIndirectDeviceAddresses[i] + (pInfos[i].geometryCount × pIndirectStrides[i]) - 1` **must** be in the buffer device address range of the same buffer
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectDeviceAddresses-03647
For any element of `pIndirectDeviceAddresses`, the buffer from which it was queried **must** have been created with the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectDeviceAddresses-03648
Each element of `pIndirectDeviceAddresses` **must** be a multiple of 4
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectStrides-03787
Each element of `pIndirectStrides` **must** be a multiple of 4
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectDeviceAddresses-03651
Each `VkAccelerationStructureBuildRangeInfoKHR` structure referenced by any element of `pIndirectDeviceAddresses` **must** be a valid `VkAccelerationStructureBuildRangeInfoKHR` structure
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-03652
`pInfos[i].dstAccelerationStructure` **must** have been created with a value of `VkAccelerationStructureCreateInfoKHR::size` greater than or equal to the memory size required by the build operation, as returned by `vkGetAccelerationStructureBuildSizesKHR` with `pBuildInfo = pInfos[i]` and `pMaxPrimitiveCounts = ppMaxPrimitiveCounts[i]`
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-ppMaxPrimitiveCounts-03653
Each `ppMaxPrimitiveCounts[i][j]` **must** be greater than or equal to the `primitiveCount` value specified by the `VkAccelerationStructureBuildRangeInfoKHR` structure located at `pIndirectDeviceAddresses[i] + (j × pIndirectStrides[i])`

Valid Usage (Implicit)

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pInfos-parameter
`pInfos` **must** be a valid pointer to an array of `infoCount` valid `VkAccelerationStructureBuildGeometryInfoKHR` structures
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectDeviceAddresses-parameter
`pIndirectDeviceAddresses` **must** be a valid pointer to an array of `infoCount` `VkDeviceAddress` values

- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-pIndirectStrides-parameter **pIndirectStrides** **must** be a valid pointer to an array of **infoCount uint32_t** values
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-ppMaxPrimitiveCounts-parameter **ppMaxPrimitiveCounts** **must** be a valid pointer to an array of **infoCount uint32_t** values
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-commandBuffer-recording **commandBuffer** **must** be in the **recording state**
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-commandBuffer-cmdpool
The **VkCommandPool** that **commandBuffer** was allocated from **must** support compute operations
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBuildAccelerationStructuresIndirectKHR-infoCount-arraylength
infoCount **must** be greater than 0

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

The **VkAccelerationStructureBuildGeometryInfoKHR** structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureBuildGeometryInfoKHR {
    VkStructureType                sType;
    const void*                    pNext;
    VkAccelerationStructureTypeKHR type;
    VkBuildAccelerationStructureFlagsKHR flags;
    VkBuildAccelerationStructureModeKHR mode;
    VkAccelerationStructureKHR      srcAccelerationStructure;
    VkAccelerationStructureKHR      dstAccelerationStructure;
    uint32_t                       geometryCount;
    const VkAccelerationStructureGeometryKHR* pGeometries;
    const VkAccelerationStructureGeometryKHR* ppGeometries;
    VkDeviceOrHostAddressKHR        scratchData;
} VkAccelerationStructureBuildGeometryInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `type` is a `VkAccelerationStructureTypeKHR` value specifying the type of acceleration structure being built.
- `flags` is a bitmask of `VkBuildAccelerationStructureFlagBitsKHR` specifying additional parameters of the acceleration structure.
- `mode` is a `VkBuildAccelerationStructureModeKHR` value specifying the type of operation to perform.
- `srcAccelerationStructure` is a pointer to an existing acceleration structure that is to be used to update the `dstAccelerationStructure` acceleration structure when `mode` is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`.
- `dstAccelerationStructure` is a pointer to the target acceleration structure for the build.
- `geometryCount` specifies the number of geometries that will be built into `dstAccelerationStructure`.
- `pGeometries` is a pointer to an array of `VkAccelerationStructureGeometryKHR` structures.
- `ppGeometries` is a pointer to an array of pointers to `VkAccelerationStructureGeometryKHR` structures.
- `scratchData` is the device or host address to memory that will be used as scratch memory for the build.

Only one of `pGeometries` or `ppGeometries` **can** be a valid pointer, the other **must** be `NULL`. Each element of the non-`NULL` array describes the data used to build each acceleration structure geometry.

The index of each element of the `pGeometries` or `ppGeometries` members of `VkAccelerationStructureBuildGeometryInfoKHR` is used as the *geometry index* during ray traversal. The geometry index is available in ray shaders via the `RayGeometryIndexKHR` built-in, and is **used to determine hit and intersection shaders executed during traversal**. The geometry index is available to ray queries via the `OpRayQueryGetIntersectionGeometryIndexKHR` instruction.

Members `srcAccelerationStructure` and `dstAccelerationStructure` **may** be the same or different for an update operation (when `mode` is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`). If they are the same, the update happens in-place. Otherwise, the target acceleration structure is updated and the source is not modified.

Valid Usage

- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03654
`type` **must** not be `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR`
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-pGeometries-03788
If `geometryCount` is not 0, exactly one of `pGeometries` or `ppGeometries` **must** be a valid pointer, the other **must** be `NULL`
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03789
If `type` is `VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR`, the `geometryType` member of elements of either `pGeometries` or `ppGeometries` **must** be `VK_GEOMETRY_TYPE_INSTANCES_KHR`
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03790
If `type` is `VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR`, `geometryCount` **must** be 1
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03791
If `type` is `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR` the `geometryType` member of elements of either `pGeometries` or `ppGeometries` **must** not be `VK_GEOMETRY_TYPE_INSTANCES_KHR`
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03792
If `type` is `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR` then the `geometryType` member of each geometry in either `pGeometries` or `ppGeometries` **must** be the same
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03793
If `type` is `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR` then `geometryCount` **must** be less than or equal to `VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxGeometryCount`
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03794
If `type` is `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR` and the `geometryType` member of either `pGeometries` or `ppGeometries` is `VK_GEOMETRY_TYPE_AABBS_KHR`, the total number of AABBs in all geometries **must** be less than or equal to `VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxPrimitiveCount`
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-03795
If `type` is `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR` and the `geometryType` member of either `pGeometries` or `ppGeometries` is `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, the total number of triangles in all geometries **must** be less than or equal to `VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxPrimitiveCount`
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-flags-03796
If `flags` has the `VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_TRACE_BIT_KHR` bit set, then it **must** not have the `VK_BUILD_ACCELERATION_STRUCTURE_PREFER_FAST_BUILD_BIT_KHR` bit set
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-flags-07334
If `flags` has the `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_EXT` bit

set then it **must** not have the
VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_EXT bit set

Valid Usage (Implicit)

- VUID-VkAccelerationStructureBuildGeometryInfoKHR-sType-sType
sType **must** be VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_GEOMETRY_INFO_KHR
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-pNext-pNext
pNext **must** be NULL
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-type-parameter
type **must** be a valid VkAccelerationStructureTypeKHR value
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-flags-parameter
flags **must** be a valid combination of VkBuildAccelerationStructureFlagBitsKHR values
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-pGeometries-parameter
If geometryCount is not 0, and pGeometries is not NULL, pGeometries **must** be a valid pointer to an array of geometryCount valid VkAccelerationStructureGeometryKHR structures
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-ppGeometries-parameter
If geometryCount is not 0, and ppGeometries is not NULL, ppGeometries **must** be a valid pointer to an array of geometryCount valid pointers to valid VkAccelerationStructureGeometryKHR structures
- VUID-VkAccelerationStructureBuildGeometryInfoKHR-commonparent
Both of dstAccelerationStructure, and srcAccelerationStructure that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same VkDevice

The VkBuildAccelerationStructureModeKHR enumeration is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkBuildAccelerationStructureModeKHR {
    VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR = 0,
    VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR = 1,
} VkBuildAccelerationStructureModeKHR;
```

- VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR specifies that the destination acceleration structure will be built using the specified geometries.
- VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR specifies that the destination acceleration structure will be built using data in a source acceleration structure, updated by the specified geometries.

The VkDeviceOrHostAddressKHR union is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef union VkDeviceOrHostAddressKHR {
    VkDeviceAddress    deviceAddress;
    void*              hostAddress;
} VkDeviceOrHostAddressKHR;
```

- **deviceAddress** is a buffer device address as returned by the [vkGetBufferDeviceAddressKHR](#) command.
- **hostAddress** is a host memory address.

The **VkDeviceOrHostAddressConstKHR** union is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef union VkDeviceOrHostAddressConstKHR {
    VkDeviceAddress    deviceAddress;
    const void*        hostAddress;
} VkDeviceOrHostAddressConstKHR;
```

- **deviceAddress** is a buffer device address as returned by the [vkGetBufferDeviceAddressKHR](#) command.
- **hostAddress** is a const host memory address.

The **VkAccelerationStructureGeometryKHR** structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureGeometryKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkGeometryTypeKHR        geometryType;
    VkAccelerationStructureGeometryDataKHR geometry;
    VkGeometryFlagsKHR        flags;
} VkAccelerationStructureGeometryKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **geometryType** describes which type of geometry this **VkAccelerationStructureGeometryKHR** refers to.
- **geometry** is a [VkAccelerationStructureGeometryDataKHR](#) union describing the geometry data for the relevant geometry type.
- **flags** is a bitmask of [VkGeometryFlagBitsKHR](#) values describing additional properties of how the geometry should be built.

Valid Usage (Implicit)

- VUID-VkAccelerationStructureGeometryKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_KHR`
- VUID-VkAccelerationStructureGeometryKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkAccelerationStructureGeometryKHR-geometryType-parameter
`geometryType` **must** be a valid `VkGeometryTypeKHR` value
- VUID-VkAccelerationStructureGeometryKHR-triangles-parameter
If `geometryType` is `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, the `triangles` member of `geometry` **must** be a valid `VkAccelerationStructureGeometryTrianglesDataKHR` structure
- VUID-VkAccelerationStructureGeometryKHR-aabbs-parameter
If `geometryType` is `VK_GEOMETRY_TYPE_AABBS_KHR`, the `aabbs` member of `geometry` **must** be a valid `VkAccelerationStructureGeometryAabbsDataKHR` structure
- VUID-VkAccelerationStructureGeometryKHR-instances-parameter
If `geometryType` is `VK_GEOMETRY_TYPE_INSTANCES_KHR`, the `instances` member of `geometry` **must** be a valid `VkAccelerationStructureGeometryInstancesDataKHR` structure
- VUID-VkAccelerationStructureGeometryKHR-flags-parameter
`flags` **must** be a valid combination of `VkGeometryFlagBitsKHR` values

The `VkAccelerationStructureGeometryDataKHR` union is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef union VkAccelerationStructureGeometryDataKHR {
    VkAccelerationStructureGeometryTrianglesDataKHR    triangles;
    VkAccelerationStructureGeometryAabbsDataKHR        aabbs;
    VkAccelerationStructureGeometryInstancesDataKHR    instances;
} VkAccelerationStructureGeometryDataKHR;
```

- `triangles` is a `VkAccelerationStructureGeometryTrianglesDataKHR` structure.
- `aabbs` is a `VkAccelerationStructureGeometryAabbsDataKHR` structure.
- `instances` is a `VkAccelerationStructureGeometryInstancesDataKHR` structure.

The `VkAccelerationStructureGeometryTrianglesDataKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureGeometryTrianglesDataKHR {
    VkStructureType      sType;
    const void*          pNext;
    VkFormat              vertexFormat;
    VkDeviceOrHostAddressConstKHR vertexData;
    VkDeviceSize          vertexStride;
    uint32_t              maxVertex;
    VkIndexType           indexType;
    VkDeviceOrHostAddressConstKHR indexData;
    VkDeviceOrHostAddressConstKHR transformData;
} VkAccelerationStructureGeometryTrianglesDataKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **vertexFormat** is the [VkFormat](#) of each vertex element.
- **vertexData** is a device or host address to memory containing vertex data for this geometry.
- **maxVertex** is the number of vertices in **vertexData** minus one.
- **vertexStride** is the stride in bytes between each vertex.
- **indexType** is the [VkIndexType](#) of each index element.
- **indexData** is a device or host address to memory containing index data for this geometry.
- **transformData** is a device or host address to memory containing an optional reference to a [VkTransformMatrixKHR](#) structure describing a transformation from the space in which the vertices in this geometry are described to the space in which the acceleration structure is defined.

NOTE

Unlike the stride for vertex buffers in [VkVertexInputBindingDescription](#) for graphics pipelines which must not exceed **maxVertexInputBindingStride**, **vertexStride** for acceleration structure geometry is instead restricted to being a 32-bit value.

Valid Usage

- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-vertexStride-03735
vertexStride **must** be a multiple of the size in bytes of the smallest component of **vertexFormat**
- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-vertexStride-03819
vertexStride **must** be less than or equal to $2^{32}-1$
- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-vertexFormat-03797
The **format** **features** of **vertexFormat** **must** contain **VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR**
- VUID-VkAccelerationStructureGeometryTrianglesDataKHR-indexType-03798

`indexType` **must** be `VK_INDEX_TYPE_UINT16`, `VK_INDEX_TYPE_UINT32`, or `VK_INDEX_TYPE_NONE_KHR`

Valid Usage (Implicit)

- `VUID-VkAccelerationStructureGeometryTrianglesDataKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_TRIANGLES_DATA_KHR`
- `VUID-VkAccelerationStructureGeometryTrianglesDataKHR-pNext-pNext`
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkAccelerationStructureTrianglesOpacityMicromapEXT`
- `VUID-VkAccelerationStructureGeometryTrianglesDataKHR-sType-unique`
The `sType` value of each struct in the `pNext` chain **must** be unique
- `VUID-VkAccelerationStructureGeometryTrianglesDataKHR-vertexFormat-parameter`
`vertexFormat` **must** be a valid `VkFormat` value
- `VUID-VkAccelerationStructureGeometryTrianglesDataKHR-indexType-parameter`
`indexType` **must** be a valid `VkIndexType` value

The `VkAccelerationStructureTrianglesOpacityMicromapEXT` structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkAccelerationStructureTrianglesOpacityMicromapEXT {
    VkStructureType      sType;
    void*                pNext;
    VkIndexType          indexType;
    VkDeviceOrHostAddressConstKHR indexBuffer;
    VkDeviceSize         indexStride;
    uint32_t             baseTriangle;
    uint32_t             usageCountsCount;
    const VkMicromapUsageEXT* pUsageCounts;
    const VkMicromapUsageEXT* const* ppUsageCounts;
    VkMicromapEXT        micromap;
} VkAccelerationStructureTrianglesOpacityMicromapEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `indexType` is the type of triangle indices used when indexing this micromap
- `indexBuffer` is the address containing the triangle indices
- `indexStride` is the byte stride between triangle indices
- `baseTriangle` is the base value added to the non-negative triangle indices
- `usageCountsCount` specifies the number of usage counts structures that will be used to determine the size of this micromap.
- `pUsageCounts` is a pointer to an array of `VkMicromapUsageEXT` structures.

- `ppUsageCounts` is a pointer to an array of pointers to `VkMicromapUsageEXT` structures.
- `micromap` is the handle to the micromap object to include in this geometry

If `VkAccelerationStructureTrianglesOpacityMicromapEXT` is included in the `pNext` chain of a `VkAccelerationStructureGeometryTrianglesDataKHR` structure, that geometry will reference that micromap.

For each triangle in the geometry, the acceleration structure build fetches an index from `indexBuffer` using `indexType` and `indexStride`. If that value is the unsigned cast of one of the values from `VkOpacityMicromapSpecialIndexEXT` then that triangle behaves as described for that special value in `Ray Opacity Micromap`. Otherwise that triangle uses the opacity micromap information from `micromap` at that index plus `baseTriangle`.

Only one of `pUsageCounts` or `ppUsageCounts` **can** be a valid pointer, the other **must** be `NULL`. The elements of the non-`NULL` array describe the total count used to build this geometry. For a given `format` and `subdivisionLevel` the number of triangles in this geometry matching those values after indirection and special index handling **must** be equal to the sum of matching `count` provided.

If `micromap` is `VK_NULL_HANDLE`, then every value read from `indexBuffer` **must** be one of the values in `VkOpacityMicromapSpecialIndexEXT`.

Valid Usage

- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-pUsageCounts-07335
Only one of `pUsageCounts` or `ppUsageCounts` **can** be a valid pointer, the other **must** be `NULL`

Valid Usage (Implicit)

- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_TRIANGLES_OPACITY_MICROMAP_EXT`
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-indexType-parameter
`indexType` **must** be a valid `VkIndexType` value
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-pUsageCounts-parameter
If `usageCountsCount` is not 0, and `pUsageCounts` is not `NULL`, `pUsageCounts` **must** be a valid pointer to an array of `usageCountsCount` `VkMicromapUsageEXT` structures
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-ppUsageCounts-parameter
If `usageCountsCount` is not 0, and `ppUsageCounts` is not `NULL`, `ppUsageCounts` **must** be a valid pointer to an array of `usageCountsCount` valid pointers to `VkMicromapUsageEXT` structures
- VUID-VkAccelerationStructureTrianglesOpacityMicromapEXT-micromap-parameter
If `micromap` is not `VK_NULL_HANDLE`, `micromap` **must** be a valid `VkMicromapEXT` handle

The `VkOpacityMicromapSpecialIndexEXT` enumeration is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkOpacityMicromapSpecialIndexEXT {
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_TRANSPARENT_EXT = -1,
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_EXT = -2,
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_EXT = -3,
    VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_EXT = -4,
} VkOpacityMicromapSpecialIndexEXT;
```

- `VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_TRANSPARENT_EXT` specifies that the entire triangle is fully transparent.
- `VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_EXT` specifies that the entire triangle is fully opaque.
- `VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_EXT` specifies that the entire triangle is unknown-transparent.
- `VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_EXT` specifies that the entire triangle is unknown-opaque.

The `VkTransformMatrixKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkTransformMatrixKHR {
    float    matrix[3][4];
} VkTransformMatrixKHR;
```

- `matrix` is a 3x4 row-major affine transformation matrix.

Valid Usage

- VUID-VkTransformMatrixKHR-matrix-03799
The first three columns of `matrix` **must** define an invertible 3x3 matrix

The `VkAccelerationStructureGeometryAabbsDataKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureGeometryAabbsDataKHR {
    VkStructureType      sType;
    const void*          pNext;
    VkDeviceOrHostAddressConstKHR data;
    VkDeviceSize         stride;
} VkAccelerationStructureGeometryAabbsDataKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- **data** is a device or host address to memory containing [VkAabbPositionsKHR](#) structures containing position data for each axis-aligned bounding box in the geometry.
- **stride** is the stride in bytes between each entry in **data**. The stride **must** be a multiple of 8.

Valid Usage

- VUID-VkAccelerationStructureGeometryAabbsDataKHR-stride-03545
stride must be a multiple of 8
- VUID-VkAccelerationStructureGeometryAabbsDataKHR-stride-03820
stride must be less than or equal to $2^{32}-1$

Valid Usage (Implicit)

- VUID-VkAccelerationStructureGeometryAabbsDataKHR-sType-sType
sType must be `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_AABBS_DATA_KHR`
- VUID-VkAccelerationStructureGeometryAabbsDataKHR-pNext-pNext
pNext must be `NULL`

The [VkAabbPositionsKHR](#) structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAabbPositionsKHR {
    float    minX;
    float    minY;
    float    minZ;
    float    maxX;
    float    maxY;
    float    maxZ;
} VkAabbPositionsKHR;
```

- **minX** is the x position of one opposing corner of a bounding box.
- **minY** is the y position of one opposing corner of a bounding box.
- **minZ** is the z position of one opposing corner of a bounding box.
- **maxX** is the x position of the other opposing corner of a bounding box.
- **maxY** is the y position of the other opposing corner of a bounding box.
- **maxZ** is the z position of the other opposing corner of a bounding box.

Valid Usage

- VUID-VkAabbPositionsKHR-minX-03546
minX must be less than or equal to **maxX**
- VUID-VkAabbPositionsKHR-minY-03547

minY must be less than or equal to **maxY**

- VUID-VkAabbPositionsKHR-minZ-03548

minZ must be less than or equal to **maxZ**

The `VkAccelerationStructureGeometryInstancesDataKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureGeometryInstancesDataKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkBool32                 arrayOfPointers;
    VkDeviceOrHostAddressConstKHR data;
} VkAccelerationStructureGeometryInstancesDataKHR;
```

- **sType** is a `VkStructureType` value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **arrayOfPointers** specifies whether **data** is used as an array of addresses or just an array.
- **data** is either the address of an array of device or host addresses referencing individual `VkAccelerationStructureInstanceKHR` structures if **arrayOfPointers** is `VK_TRUE`, or the address of an array of `VkAccelerationStructureInstanceKHR` structures. Addresses and `VkAccelerationStructureInstanceKHR` structures are tightly packed.

Valid Usage (Implicit)

- VUID-VkAccelerationStructureGeometryInstancesDataKHR-sType-sType
sType must be `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_INSTANCES_DATA_KHR`
- VUID-VkAccelerationStructureGeometryInstancesDataKHR-pNext-pNext
pNext must be `NULL`

Acceleration structure instances **can** be built into top-level acceleration structures. Each acceleration structure instance is a separate entry in the top-level acceleration structure which includes all the geometry of a bottom-level acceleration structure at a transformed location. Multiple instances **can** point to the same bottom level acceleration structure.

An acceleration structure instance is defined by the structure:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureInstanceKHR {
    VkTransformMatrixKHR      transform;
    uint32_t                  instanceCustomIndex:24;
    uint32_t                  mask:8;
    uint32_t                  instanceShaderBindingTableRecordOffset:24;
    VkGeometryInstanceFlagsKHR flags:8;
    uint64_t                  accelerationStructureReference;
} VkAccelerationStructureInstanceKHR;
```

- **transform** is a [VkTransformMatrixKHR](#) structure describing a transformation to be applied to the acceleration structure.
- **instanceCustomIndex** is a 24-bit application-specified index value accessible to ray shaders in the [InstanceCustomIndexKHR](#) built-in.
- **mask** is an 8-bit visibility mask for the geometry. The instance **may** only be hit if **Cull Mask & instance.mask != 0**
- **instanceShaderBindingTableRecordOffset** is a 24-bit offset used in calculating the hit shader binding table index.
- **flags** is an 8-bit mask of [VkGeometryInstanceFlagBitsKHR](#) values to apply to this instance.
- **accelerationStructureReference** is either :
 - a device address containing the value obtained from [vkGetAccelerationStructureDeviceAddressKHR](#) (used by device operations which reference acceleration structures)
 - a [VkAccelerationStructureKHR](#) object (used by host operations which reference acceleration structures).

The C language specification does not define the ordering of bit-fields, but in practice, this struct produces the correct layout with existing compilers. The intended bit pattern is for the following:

- **instanceCustomIndex** and **mask** occupy the same memory as if a single **uint32_t** was specified in their place
 - **instanceCustomIndex** occupies the 24 least significant bits of that memory
 - **mask** occupies the 8 most significant bits of that memory
- **instanceShaderBindingTableRecordOffset** and **flags** occupy the same memory as if a single **uint32_t** was specified in their place
 - **instanceShaderBindingTableRecordOffset** occupies the 24 least significant bits of that memory
 - **flags** occupies the 8 most significant bits of that memory

If a compiler produces code that diverges from that pattern, applications **must** employ another method to set values according to the correct bit pattern.

Valid Usage (Implicit)

- VUID-VkAccelerationStructureInstanceKHR-flags-parameter
flags must be a valid combination of [VkGeometryInstanceFlagBitsKHR](#) values

Possible values of **flags** in the instance modifying the behavior of that instance are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkGeometryInstanceFlagBitsKHR {
    VK_GEOMETRY_INSTANCE_TRIANGLE_FACING_CULL_DISABLE_BIT_KHR = 0x00000001,
    VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR = 0x00000002,
    VK_GEOMETRY_INSTANCE_FORCE_OPAQUE_BIT_KHR = 0x00000004,
    VK_GEOMETRY_INSTANCE_FORCE_NO_OPAQUE_BIT_KHR = 0x00000008,
    // Provided by VK_EXT_opacity_micromap
    VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_EXT = 0x00000010,
    // Provided by VK_EXT_opacity_micromap
    VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_EXT = 0x00000020,
    VK_GEOMETRY_INSTANCE_TRIANGLE_FRONT_COUNTERCLOCKWISE_BIT_KHR =
VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR,
} VkGeometryInstanceFlagBitsKHR;
```

- [VK_GEOMETRY_INSTANCE_TRIANGLE_FACING_CULL_DISABLE_BIT_KHR](#) disables face culling for this instance.
- [VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR](#) indicates that the [facing determination](#) for geometry in this instance is inverted. Because the facing is determined in object space, an instance transform does not change the winding, but a geometry transform does.
- [VK_GEOMETRY_INSTANCE_FORCE_OPAQUE_BIT_KHR](#) causes this instance to act as though [VK_GEOMETRY_OPAQUE_BIT_KHR](#) were specified on all geometries referenced by this instance. This behavior **can** be overridden by the SPIR-V [NoOpaqueKHR](#) ray flag.
- [VK_GEOMETRY_INSTANCE_FORCE_NO_OPAQUE_BIT_KHR](#) causes this instance to act as though [VK_GEOMETRY_OPAQUE_BIT_KHR](#) were not specified on all geometries referenced by this instance. This behavior **can** be overridden by the SPIR-V [OpaqueKHR](#) ray flag.

[VK_GEOMETRY_INSTANCE_FORCE_NO_OPAQUE_BIT_KHR](#) and [VK_GEOMETRY_INSTANCE_FORCE_OPAQUE_BIT_KHR](#) **must** not be used in the same flag.

```
// Provided by VK_KHR_acceleration_structure
typedef VkFlags VkGeometryInstanceFlagsKHR;
```

[VkGeometryInstanceFlagsKHR](#) is a bitmask type for setting a mask of zero or more [VkGeometryInstanceFlagBitsKHR](#).

[VkAccelerationStructureBuildRangeInfoKHR](#) is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureBuildRangeInfoKHR {
    uint32_t    primitiveCount;
    uint32_t    primitiveOffset;
    uint32_t    firstVertex;
    uint32_t    transformOffset;
} VkAccelerationStructureBuildRangeInfoKHR;
```

- **primitiveCount** defines the number of primitives for a corresponding acceleration structure geometry.
- **primitiveOffset** defines an offset in bytes into the memory where primitive data is defined.
- **firstVertex** is the index of the first vertex to build from for triangle geometry.
- **transformOffset** defines an offset in bytes into the memory where a transform matrix is defined.

The primitive count and primitive offset are interpreted differently depending on the [VkGeometryTypeKHR](#) used:

- For geometries of type [VK_GEOMETRY_TYPE_TRIANGLES_KHR](#), **primitiveCount** is the number of triangles to be built, where each triangle is treated as 3 vertices.
 - If the geometry uses indices, **primitiveCount** × 3 indices are consumed from [VkAccelerationStructureGeometryTrianglesDataKHR::indexData](#), starting at an offset of **primitiveOffset**. The value of **firstVertex** is added to the index values before fetching vertices.
 - If the geometry does not use indices, **primitiveCount** × 3 vertices are consumed from [VkAccelerationStructureGeometryTrianglesDataKHR::vertexData](#), starting at an offset of **primitiveOffset** + [VkAccelerationStructureGeometryTrianglesDataKHR::vertexStride](#) × **firstVertex**.
 - If [VkAccelerationStructureGeometryTrianglesDataKHR::transformData](#) is not **NULL**, a single [VkTransformMatrixKHR](#) structure is consumed from [VkAccelerationStructureGeometryTrianglesDataKHR::transformData](#), at an offset of **transformOffset**. This matrix describes a transformation from the space in which the vertices for all triangles in this geometry are described to the space in which the acceleration structure is defined.
- For geometries of type [VK_GEOMETRY_TYPE_AABBS_KHR](#), **primitiveCount** is the number of axis-aligned bounding boxes. **primitiveCount** [VkAabbPositionsKHR](#) structures are consumed from [VkAccelerationStructureGeometryAabbsDataKHR::data](#), starting at an offset of **primitiveOffset**.
- For geometries of type [VK_GEOMETRY_TYPE_INSTANCES_KHR](#), **primitiveCount** is the number of acceleration structures. **primitiveCount** [VkAccelerationStructureInstanceKHR](#) structures are consumed from [VkAccelerationStructureGeometryInstancesDataKHR::data](#), starting at an offset of **primitiveOffset**.

Valid Usage

- VUID-VkAccelerationStructureBuildRangeInfoKHR-primitiveOffset-03656

For geometries of type `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if the geometry uses indices, the offset `primitiveOffset` from `VkAccelerationStructureGeometryTrianglesDataKHR::indexData` **must** be a multiple of the element size of `VkAccelerationStructureGeometryTrianglesDataKHR::indexType`

- VUID-VkAccelerationStructureBuildRangeInfoKHR-primitiveOffset-03657

For geometries of type `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if the geometry does not use indices, the offset `primitiveOffset` from `VkAccelerationStructureGeometryTrianglesDataKHR::vertexData` **must** be a multiple of the component size of `VkAccelerationStructureGeometryTrianglesDataKHR::vertexFormat`

- VUID-VkAccelerationStructureBuildRangeInfoKHR-transformOffset-03658

For geometries of type `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, the offset `transformOffset` from `VkAccelerationStructureGeometryTrianglesDataKHR::transformData` **must** be a multiple of 16

- VUID-VkAccelerationStructureBuildRangeInfoKHR-primitiveOffset-03659

For geometries of type `VK_GEOMETRY_TYPE_AABBS_KHR`, the offset `primitiveOffset` from `VkAccelerationStructureGeometryAabbsDataKHR::data` **must** be a multiple of 8

- VUID-VkAccelerationStructureBuildRangeInfoKHR-primitiveOffset-03660

For geometries of type `VK_GEOMETRY_TYPE_INSTANCES_KHR`, the offset `primitiveOffset` from `VkAccelerationStructureGeometryInstancesDataKHR::data` **must** be a multiple of 16

33.1.8. Copying Acceleration Structures

An additional command exists for copying acceleration structures without updating their contents. The acceleration structure object **can** be compacted in order to improve performance. Before copying, an application **must** query the size of the resulting acceleration structure.

To query acceleration structure size parameters call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdWriteAccelerationStructuresPropertiesKHR(
    VkCommandBuffer                commandBuffer,
    uint32_t                       accelerationStructureCount,
    const VkAccelerationStructureKHR* pAccelerationStructures,
    VkQueryType                    queryType,
    VkQueryPool                    queryPool,
    uint32_t                       firstQuery);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `accelerationStructureCount` is the count of acceleration structures for which to query the property.
- `pAccelerationStructures` is a pointer to an array of existing previously built acceleration structures.
- `queryType` is a `VkQueryType` value specifying the type of queries managed by the pool.
- `queryPool` is the query pool that will manage the results of the query.

- `firstQuery` is the first query index within the query pool that will contain the `accelerationStructureCount` number of results.

Accesses to any of the acceleration structures listed in `pAccelerationStructures` **must** be `synchronized` with the `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR` pipeline stage or the `VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` pipeline stage, and an access type of `VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR`.

- If `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR`, then the value written out is the number of bytes required by a compacted acceleration structure.
- If `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR`, then the value written out is the number of bytes required by a serialized acceleration structure.

Valid Usage

- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-accelerationStructure-08924
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-queryPool-02493
`queryPool` **must** have been created with a `queryType` matching `queryType`
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-queryPool-02494
The queries identified by `queryPool` and `firstQuery` **must** be *unavailable*
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-buffer-03736
The `buffer` used to create each acceleration structure in `pAccelerationStructures` **must** be bound to device memory
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-query-04880
The sum of `firstQuery` plus `accelerationStructureCount` **must** be less than or equal to the number of queries in `queryPool`
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-04964
All acceleration structures in `pAccelerationStructures` **must** have been built prior to the execution of this command
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-accelerationStructures-03431
All acceleration structures in `pAccelerationStructures` **must** have been built with `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR` if `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR`
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-queryType-06742
`queryType` **must** be `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR`, `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR`, `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR`, or `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR`

Valid Usage (Implicit)

- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-commandBuffer-parameter **commandBuffer** **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-parameter **pAccelerationStructures** **must** be a valid pointer to an array of [accelerationStructureCount](#) valid [VkAccelerationStructureKHR](#) handles
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-queryType-parameter **queryType** **must** be a valid [VkQueryType](#) value
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-queryPool-parameter **queryPool** **must** be a valid [VkQueryPool](#) handle
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-commandBuffer-recording **commandBuffer** **must** be in the [recording state](#)
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support compute operations
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-accelerationStructureCount-arraylength
accelerationStructureCount **must** be greater than 0
- VUID-vkCmdWriteAccelerationStructuresPropertiesKHR-commonparent
Each of **commandBuffer**, **queryPool**, and the elements of **pAccelerationStructures** **must** have been created, allocated, or retrieved from the same [VkDevice](#)

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

To copy an acceleration structure call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdCopyAccelerationStructureKHR(
    VkCommandBuffer                      commandBuffer,
    const VkCopyAccelerationStructureInfoKHR* pInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pInfo` is a pointer to a `VkCopyAccelerationStructureInfoKHR` structure defining the copy operation.

This command copies the `pInfo->src` acceleration structure to the `pInfo->dst` acceleration structure in the manner specified by `pInfo->mode`.

Accesses to `pInfo->src` and `pInfo->dst` **must** be `synchronized` with the `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR` `pipeline stage` or the `VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` `pipeline stage`, and an `access type` of `VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR` or `VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR` as appropriate.

Valid Usage

- VUID-vkCmdCopyAccelerationStructureKHR-accelerationStructure-08925
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkCmdCopyAccelerationStructureKHR-buffer-03737
The `buffer` used to create `pInfo->src` **must** be bound to device memory
- VUID-vkCmdCopyAccelerationStructureKHR-buffer-03738
The `buffer` used to create `pInfo->dst` **must** be bound to device memory

Valid Usage (Implicit)

- VUID-vkCmdCopyAccelerationStructureKHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyAccelerationStructureKHR-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkCopyAccelerationStructureInfoKHR` structure
- VUID-vkCmdCopyAccelerationStructureKHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdCopyAccelerationStructureKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdCopyAccelerationStructureKHR-renderpass
This command **must** only be called outside of a render pass instance

- VUID-vkCmdCopyAccelerationStructureKHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

The `VkCopyAccelerationStructureInfoKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkCopyAccelerationStructureInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkAccelerationStructureKHR src;
    VkAccelerationStructureKHR dst;
    VkCopyAccelerationStructureModeKHR mode;
} VkCopyAccelerationStructureInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `src` is the source acceleration structure for the copy.
- `dst` is the target acceleration structure for the copy.
- `mode` is a `VkCopyAccelerationStructureModeKHR` value specifying additional operations to perform during the copy.

Valid Usage

- VUID-VkCopyAccelerationStructureInfoKHR-mode-03410
`mode` **must** be `VK_COPY_ACCELERATION_STRUCTURE_MODE_COMPACT_KHR` or `VK_COPY_ACCELERATION_STRUCTURE_MODE_CLONE_KHR`
- VUID-VkCopyAccelerationStructureInfoKHR-src-04963
The source acceleration structure `src` **must** have been constructed prior to the execution

of this command

- VUID-VkCopyAccelerationStructureInfoKHR-src-03411

If `mode` is `VK_COPY_ACCELERATION_STRUCTURE_MODE_COMPACT_KHR`, `src` **must** have been constructed with `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR` in the build

- VUID-VkCopyAccelerationStructureInfoKHR-buffer-03718

The `buffer` used to create `src` **must** be bound to device memory

- VUID-VkCopyAccelerationStructureInfoKHR-buffer-03719

The `buffer` used to create `dst` **must** be bound to device memory

- VUID-VkCopyAccelerationStructureInfoKHR-dst-07791

The range of memory backing `dst` that is accessed by this command **must** not overlap the memory backing `src` that is accessed by this command

Valid Usage (Implicit)

- VUID-VkCopyAccelerationStructureInfoKHR-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_INFO_KHR`

- VUID-VkCopyAccelerationStructureInfoKHR-pNext-pNext

`pNext` **must** be `NULL`

- VUID-VkCopyAccelerationStructureInfoKHR-src-parameter

`src` **must** be a valid `VkAccelerationStructureKHR` handle

- VUID-VkCopyAccelerationStructureInfoKHR-dst-parameter

`dst` **must** be a valid `VkAccelerationStructureKHR` handle

- VUID-VkCopyAccelerationStructureInfoKHR-mode-parameter

`mode` **must** be a valid `VkCopyAccelerationStructureModeKHR` value

- VUID-VkCopyAccelerationStructureInfoKHR-commonparent

Both of `dst`, and `src` **must** have been created, allocated, or retrieved from the same `VkDevice`

Possible values of `mode` specifying additional operations to perform during the copy, are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkCopyAccelerationStructureModeKHR {
    VK_COPY_ACCELERATION_STRUCTURE_MODE_CLONE_KHR = 0,
    VK_COPY_ACCELERATION_STRUCTURE_MODE_COMPACT_KHR = 1,
    // Provided by VK_KHR_acceleration_structure
    VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR = 2,
    // Provided by VK_KHR_acceleration_structure
    VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR = 3,
} VkCopyAccelerationStructureModeKHR;
```

- `VK_COPY_ACCELERATION_STRUCTURE_MODE_CLONE_KHR` creates a direct copy of the acceleration structure specified in `src` into the one specified by `dst`. The `dst` acceleration structure **must** have

been created with the same parameters as `src`. If `src` contains references to other acceleration structures, `dst` will reference the same acceleration structures.

- `VK_COPY_ACCELERATION_STRUCTURE_MODE_COMPACT_KHR` creates a more compact version of an acceleration structure `src` into `dst`. The acceleration structure `dst` **must** have been created with a size at least as large as that returned by `vkCmdWriteAccelerationStructuresPropertiesKHR`, or `vkWriteAccelerationStructuresPropertiesKHR` after the build of the acceleration structure specified by `src`. If `src` contains references to other acceleration structures, `dst` will reference the same acceleration structures.
- `VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR` serializes the acceleration structure to a semi-opaque format which can be reloaded on a compatible implementation.
- `VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR` deserializes the semi-opaque serialization format in the buffer to the acceleration structure.

To copy an acceleration structure to device memory call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdCopyAccelerationStructureToMemoryKHR(
    VkCommandBuffer                      commandBuffer,
    const VkCopyAccelerationStructureToMemoryInfoKHR* pInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pInfo` is a pointer to a `VkCopyAccelerationStructureToMemoryInfoKHR` structure defining the copy operation.

Accesses to `pInfo->src` **must** be `synchronized` with the `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR` pipeline stage or the `VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` pipeline stage, and an access type of `VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR`. Accesses to the buffer indicated by `pInfo->dst.deviceAddress` **must** be `synchronized` with the `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR` pipeline stage or the `VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR` pipeline stage, and an access type of `VK_ACCESS_TRANSFER_WRITE_BIT`.

This command produces the same results as `vkCopyAccelerationStructureToMemoryKHR`, but writes its result to a device address, and is executed on the device rather than the host. The output **may** not necessarily be bit-for-bit identical, but it can be equally used by either `vkCmdCopyMemoryToAccelerationStructureKHR` or `vkCopyMemoryToAccelerationStructureKHR`.

The defined header structure for the serialized data consists of:

- `VK_UUID_SIZE` bytes of data matching `VkPhysicalDeviceIDProperties::driverUUID`
- `VK_UUID_SIZE` bytes of data identifying the compatibility for comparison using `vkGetDeviceAccelerationStructureCompatibilityKHR`
- A 64-bit integer of the total size matching the value queried using `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR`

- A 64-bit integer of the deserialized size to be passed in to `VkAccelerationStructureCreateInfoKHR::size`
- A 64-bit integer of the count of the number of acceleration structure handles following. This value matches the value queried using `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR`. This will be zero for a bottom-level acceleration structure. For top-level acceleration structures this number is implementation-dependent; the number of and ordering of the handles may not match the instance descriptions which were used to build the acceleration structure.

The corresponding handles matching the values returned by `vkGetAccelerationStructureDeviceAddressKHR` are tightly packed in the buffer following the count. The application is expected to store a mapping between those handles and the original application-generated bottom-level acceleration structures to provide when deserializing. The serialized data is written to the buffer (or read from the buffer) according to the host endianness.

Valid Usage

- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-accelerationStructure-08926
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-pInfo-03739
`pInfo->dst.deviceAddress` **must** be a valid device address for a buffer bound to device memory
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-pInfo-03740
`pInfo->dst.deviceAddress` **must** be aligned to 256 bytes
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-pInfo-03741
If the buffer pointed to by `pInfo->dst.deviceAddress` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-None-03559
The `buffer` used to create `pInfo->src` **must** be bound to device memory

Valid Usage (Implicit)

- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkCopyAccelerationStructureToMemoryInfoKHR` structure
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations

- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyAccelerationStructureToMemoryKHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkCopyAccelerationStructureToMemoryInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkAccelerationStructureKHR src;
    VkDeviceOrHostAddressKHR dst;
    VkCopyAccelerationStructureModeKHR mode;
} VkCopyAccelerationStructureToMemoryInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `src` is the source acceleration structure for the copy
- `dst` is the device or host address to memory which is the target for the copy
- `mode` is a `VkCopyAccelerationStructureModeKHR` value specifying additional operations to perform during the copy.

Valid Usage

- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-src-04959
The source acceleration structure `src` **must** have been constructed prior to the execution of this command
- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-dst-03561

The memory pointed to by **dst** **must** be at least as large as the serialization size of **src**, as reported by [vkWriteAccelerationStructuresPropertiesKHR](#) or [vkCmdWriteAccelerationStructuresPropertiesKHR](#) with a query type of [VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR](#)

- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-mode-03412
mode **must** be [VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR](#)

Valid Usage (Implicit)

- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-sType-sType
sType **must** be [VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_TO_MEMORY_INFO_KHR](#)
- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-pNext-pNext
pNext **must** be [NULL](#)
- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-src-parameter
src **must** be a valid [VkAccelerationStructureKHR](#) handle
- VUID-VkCopyAccelerationStructureToMemoryInfoKHR-mode-parameter
mode **must** be a valid [VkCopyAccelerationStructureModeKHR](#) value

To copy device memory to an acceleration structure call:

```
// Provided by VK_KHR_acceleration_structure
void vkCmdCopyMemoryToAccelerationStructureKHR(
    VkCommandBuffer                      commandBuffer,
    const VkCopyMemoryToAccelerationStructureInfoKHR* pInfo);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **pInfo** is a pointer to a [VkCopyMemoryToAccelerationStructureInfoKHR](#) structure defining the copy operation.

Accesses to **pInfo->dst** **must** be [synchronized](#) with the [VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR](#) [pipeline stage](#) or the [VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR](#) [pipeline stage](#), and an [access type](#) of [VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR](#). Accesses to the buffer indicated by **pInfo->src.deviceAddress** **must** be [synchronized](#) with the [VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR](#) [pipeline stage](#) or the [VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR](#) [pipeline stage](#), and an [access type](#) of [VK_ACCESS_TRANSFER_READ_BIT](#).

This command can accept acceleration structures produced by either [vkCmdCopyAccelerationStructureToMemoryKHR](#) or [vkCopyAccelerationStructureToMemoryKHR](#).

The structure provided as input to deserialize is as described in [vkCmdCopyAccelerationStructureToMemoryKHR](#), with any acceleration structure handles filled in with the newly-queried handles to bottom level acceleration structures created before

deserialization. These do not need to be built at deserialize time, but **must** be created.

Valid Usage

- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-accelerationStructure-08927
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-pInfo-03742
`pInfo->src.deviceAddress` **must** be a valid device address for a buffer bound to device memory
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-pInfo-03743
`pInfo->src.deviceAddress` **must** be aligned to 256 bytes
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-pInfo-03744
If the buffer pointed to by `pInfo->src.deviceAddress` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-buffer-03745
The `buffer` used to create `pInfo->dst` **must** be bound to device memory

Valid Usage (Implicit)

- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkCopyMemoryToAccelerationStructureInfoKHR` structure
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMemoryToAccelerationStructureKHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

The `VkCopyMemoryToAccelerationStructureInfoKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkCopyMemoryToAccelerationStructureInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkDeviceOrHostAddressConstKHR src;
    VkAccelerationStructureKHR dst;
    VkCopyAccelerationStructureModeKHR mode;
} VkCopyMemoryToAccelerationStructureInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `src` is the device or host address to memory containing the source data for the copy.
- `dst` is the target acceleration structure for the copy.
- `mode` is a `VkCopyAccelerationStructureModeKHR` value specifying additional operations to perform during the copy.

Valid Usage

- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-src-04960
The source memory pointed to by `src` **must** contain data previously serialized using `vkCmdCopyAccelerationStructureToMemoryKHR`, potentially modified to relocate acceleration structure references as described in that command
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-mode-03413
`mode` **must** be `VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR`
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-pInfo-03414
The data in `src` **must** have a format compatible with the destination physical device as returned by `vkGetDeviceAccelerationStructureCompatibilityKHR`
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-dst-03746
`dst` **must** have been created with a `size` greater than or equal to that used to serialize the data in `src`

Valid Usage (Implicit)

- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-sType-sType
sType must be `VK_STRUCTURE_TYPE_COPY_MEMORY_TO_ACCELERATION_STRUCTURE_INFO_KHR`
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-pNext-pNext
pNext must be `NULL`
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-dst-parameter
dst must be a valid `VkAccelerationStructureKHR` handle
- VUID-VkCopyMemoryToAccelerationStructureInfoKHR-mode-parameter
mode must be a valid `VkCopyAccelerationStructureModeKHR` value

To check if a serialized acceleration structure is compatible with the current device call:

```
// Provided by VK_KHR_acceleration_structure
void vkGetDeviceAccelerationStructureCompatibilityKHR(
    VkDevice device,
    const VkAccelerationStructureVersionInfoKHR* pVersionInfo,
    VkAccelerationStructureCompatibilityKHR* pCompatibility);
```

- **device** is the device to check the version against.
- **pVersionInfo** is a pointer to a `VkAccelerationStructureVersionInfoKHR` structure specifying version information to check against the device.
- **pCompatibility** is a pointer to a `VkAccelerationStructureCompatibilityKHR` value in which compatibility information is returned.

Valid Usage

- VUID-vkGetDeviceAccelerationStructureCompatibilityKHR-accelerationStructure-08928
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructure` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetDeviceAccelerationStructureCompatibilityKHR-device-parameter
device must be a valid `VkDevice` handle
- VUID-vkGetDeviceAccelerationStructureCompatibilityKHR-pVersionInfo-parameter
pVersionInfo must be a valid pointer to a valid `VkAccelerationStructureVersionInfoKHR` structure
- VUID-vkGetDeviceAccelerationStructureCompatibilityKHR-pCompatibility-parameter
pCompatibility must be a valid pointer to a `VkAccelerationStructureCompatibilityKHR` value

The `VkAccelerationStructureVersionInfoKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkAccelerationStructureVersionInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    const uint8_t*     pVersionData;
} VkAccelerationStructureVersionInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pVersionData` is a pointer to the version header of an acceleration structure as defined in `vkCmdCopyAccelerationStructureToMemoryKHR`

NOTE

`pVersionData` is a *pointer* to an array of $2 \times \text{VK_UUID_SIZE}$ `uint8_t` values instead of two `VK_UUID_SIZE` arrays as the expected use case for this member is to be pointed at the header of a previously serialized acceleration structure (via `vkCmdCopyAccelerationStructureToMemoryKHR` or `vkCopyAccelerationStructureToMemoryKHR`) that is loaded in memory. Using arrays would necessitate extra memory copies of the UUIDs.

Valid Usage (Implicit)

- VUID-VkAccelerationStructureVersionInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_VERSION_INFO_KHR`
- VUID-VkAccelerationStructureVersionInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkAccelerationStructureVersionInfoKHR-pVersionData-parameter
`pVersionData` **must** be a valid pointer to an array of $2 \times \text{VK_UUID_SIZE}$ `uint8_t` values

Possible values of `pCompatibility` returned by `vkGetDeviceAccelerationStructureCompatibilityKHR` are:

```
// Provided by VK_KHR_acceleration_structure
typedef enum VkAccelerationStructureCompatibilityKHR {
    VK_ACCELERATION_STRUCTURE_COMPATIBILITY_COMPATIBLE_KHR = 0,
    VK_ACCELERATION_STRUCTURE_COMPATIBILITY_INCOMPATIBLE_KHR = 1,
} VkAccelerationStructureCompatibilityKHR;
```

- `VK_ACCELERATION_STRUCTURE_COMPATIBILITY_COMPATIBLE_KHR` if the `pVersionData` version acceleration structure is compatible with `device`.
- `VK_ACCELERATION_STRUCTURE_COMPATIBILITY_INCOMPATIBLE_KHR` if the `pVersionData` version acceleration structure is not compatible with `device`.

33.2. Host Acceleration Structure Operations

Implementations are also required to provide host implementations of the acceleration structure operations if the `accelerationStructureHostCommands` feature is enabled:

- `vkBuildAccelerationStructuresKHR` corresponding to `vkCmdBuildAccelerationStructuresKHR`
- `vkCopyAccelerationStructureKHR` corresponding to `vkCmdCopyAccelerationStructureKHR`
- `vkCopyAccelerationStructureToMemoryKHR` corresponding to `vkCmdCopyAccelerationStructureToMemoryKHR`
- `vkCopyMemoryToAccelerationStructureKHR` corresponding to `vkCmdCopyMemoryToAccelerationStructureKHR`
- `vkWriteAccelerationStructuresPropertiesKHR` corresponding to `vkCmdWriteAccelerationStructuresPropertiesKHR`

These commands are functionally equivalent to their device counterparts, except that they are executed on the host timeline, rather than being enqueued into command buffers.

All acceleration structures used by the host commands **must** be bound to host-visible memory, and all input data for acceleration structure builds **must** be referenced using host addresses instead of device addresses. Applications are not required to map acceleration structure memory when using the host commands.

NOTE

The `vkBuildAccelerationStructuresKHR` and `vkCmdBuildAccelerationStructuresKHR` **may** use different algorithms, and thus are not required to produce identical structures. The structures produced by these two commands **may** exhibit different memory footprints or traversal performance, but should strive to be similar where possible.

Apart from these details, the host and device operations are interchangeable. For example, an application **can** use `vkBuildAccelerationStructuresKHR` to build a structure, compact it on the device using `vkCmdCopyAccelerationStructureKHR`, and serialize the result using `vkCopyAccelerationStructureToMemoryKHR`.

NOTE

For efficient execution, acceleration structures manipulated using these commands should always be bound to host cached memory, as the implementation may need to repeatedly read and write this memory during the execution of the command.

To build acceleration structures on the host, call:

```
// Provided by VK_KHR_acceleration_structure
VkResult vkBuildAccelerationStructuresKHR(
    VkDevice                                device,
    VkDeferredOperationKHR                  deferredOperation,
    uint32_t                                infoCount,
    const VkAccelerationStructureBuildGeometryInfoKHR* pInfos,
    const VkAccelerationStructureBuildRangeInfoKHR* const* ppBuildRangeInfos);
```

- `device` is the `VkDevice` for which the acceleration structures are being built.
- `deferredOperation` is an optional `VkDeferredOperationKHR` to request deferral for this command.
- `infoCount` is the number of acceleration structures to build. It specifies the number of the `pInfos` structures and `ppBuildRangeInfos` pointers that **must** be provided.
- `pInfos` is a pointer to an array of `infoCount` `VkAccelerationStructureBuildGeometryInfoKHR` structures defining the geometry used to build each acceleration structure.
- `ppBuildRangeInfos` is a pointer to an array of `infoCount` pointers to arrays of `VkAccelerationStructureBuildRangeInfoKHR` structures. Each `ppBuildRangeInfos[i]` is a pointer to an array of `pInfos[i].geometryCount` `VkAccelerationStructureBuildRangeInfoKHR` structures defining dynamic offsets to the addresses where geometry data is stored, as defined by `pInfos[i]`.

This command fulfills the same task as `vkCmdBuildAccelerationStructuresKHR` but is executed by the host.

The `vkBuildAccelerationStructuresKHR` command provides the ability to initiate multiple acceleration structures builds, however there is no ordering or synchronization implied between any of the individual acceleration structure builds.

NOTE

This means that an application **cannot** build a top-level acceleration structure in the same `vkBuildAccelerationStructuresKHR` call as the associated bottom-level or instance acceleration structures are being built. There also **cannot** be any memory aliasing between any acceleration structure memories or scratch memories being used by any of the builds.

Valid Usage

- VUID-vkBuildAccelerationStructuresKHR-accelerationStructureHostCommands-03581
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureHostCommands` feature **must** be enabled
- VUID-vkBuildAccelerationStructuresKHR-mode-04628
The `mode` member of each element of `pInfos` **must** be a valid `VkBuildAccelerationStructureModeKHR` value
- VUID-vkBuildAccelerationStructuresKHR-srcAccelerationStructure-04629
If the `srcAccelerationStructure` member of any element of `pInfos` is not `VK_NULL_HANDLE`, the `srcAccelerationStructure` member **must** be a valid `VkAccelerationStructureKHR` handle
- VUID-vkBuildAccelerationStructuresKHR-pInfos-04630
For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `srcAccelerationStructure` member **must** not be `VK_NULL_HANDLE`
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03403
The `srcAccelerationStructure` member of any element of `pInfos` **must** not be the same acceleration structure as the `dstAccelerationStructure` member of any other element of

pInfos

- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03698

The `dstAccelerationStructure` member of any element of `pInfos` **must** not be the same acceleration structure as the `dstAccelerationStructure` member of any other element of `pInfos`

- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03800

The `dstAccelerationStructure` member of any element of `pInfos` **must** be a valid `VkAccelerationStructureKHR` handle

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03699

For each element of `pInfos`, if its `type` member is `VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR`, its `dstAccelerationStructure` member **must** have been created with a value of `VkAccelerationStructureCreateInfoKHR::type` equal to either `VK_ACCELERATION_STRUCTURE_TYPE_TOP_LEVEL_KHR` or `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR`

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03700

For each element of `pInfos`, if its `type` member is `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR`, its `dstAccelerationStructure` member **must** have been created with a value of `VkAccelerationStructureCreateInfoKHR::type` equal to either `VK_ACCELERATION_STRUCTURE_TYPE_BOTTOM_LEVEL_KHR` or `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR`

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03663

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, `inactive primitives` in its `srcAccelerationStructure` member **must** not be made active

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03664

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, `active primitives` in its `srcAccelerationStructure` member **must** not be made `inactive`

- VUID-vkBuildAccelerationStructuresKHR-None-03407

The `dstAccelerationStructure` member of any element of `pInfos` **must** not be referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR` in any other element of `pInfos`

- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03701

The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `srcAccelerationStructure` member of any other element of `pInfos` with a `mode` equal to `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, which is accessed by this command

- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03702

The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `dstAccelerationStructure` member of any other element of `pInfos`, which is accessed by this command

- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03703

The range of memory backing the `dstAccelerationStructure` member of any element of

`pInfos` that is accessed by this command **must** not overlap the memory backing the `scratchData` member of any element of `pInfos` (including the same element), which is accessed by this command

- VUID-vkBuildAccelerationStructuresKHR-scratchData-03704

The range of memory backing the `scratchData` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `scratchData` member of any other element of `pInfos`, which is accessed by this command

- VUID-vkBuildAccelerationStructuresKHR-scratchData-03705

The range of memory backing the `scratchData` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `srcAccelerationStructure` member of any element of `pInfos` with a `mode` equal to `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR` (including the same element), which is accessed by this command

- VUID-vkBuildAccelerationStructuresKHR-dstAccelerationStructure-03706

The range of memory backing the `dstAccelerationStructure` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing any acceleration structure referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR` in any other element of `pInfos`, which is accessed by this command

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03667

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `srcAccelerationStructure` member **must** have previously been constructed with `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_UPDATE_BIT_KHR` set in `VkAccelerationStructureBuildGeometryInfoKHR::flags` in the build

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03668

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `srcAccelerationStructure` and `dstAccelerationStructure` members **must** either be the same `VkAccelerationStructureKHR`, or not have any `memory aliasing`

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03758

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `geometryCount` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03759

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `flags` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03760

For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, its `type` member **must** have the same value which was specified when `srcAccelerationStructure` was last built

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03761

For each element of `pInfos`, if its `mode` member is

VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, its geometryType member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03762

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, its flags member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03763

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.vertexFormat member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03764

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.maxVertex member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03765

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, its geometry.triangles.indexType member **must** have the same value which was specified when srcAccelerationStructure was last built

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03766

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its geometry.triangles.transformData address was NULL when srcAccelerationStructure was last built, then it **must** be NULL

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03767

For each element of pInfos, if its mode member is VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR, then for each VkAccelerationStructureGeometryKHR structure referred to by its pGeometries or ppGeometries members, if geometryType is VK_GEOMETRY_TYPE_TRIANGLES_KHR, if its geometry.triangles.transformData address was not NULL when srcAccelerationStructure was last built, then it **must** not be NULL

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03768
For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, if `geometryType` is `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, and `geometry.triangles.indexType` is not `VK_INDEX_TYPE_NONE_KHR`, then the value of each index referenced **must** be the same as the corresponding index value when `srcAccelerationStructure` was last built
- VUID-vkBuildAccelerationStructuresKHR-primitiveCount-03769
For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, then for each `VkAccelerationStructureGeometryKHR` structure referred to by its `pGeometries` or `ppGeometries` members, the `primitiveCount` member of its corresponding `VkAccelerationStructureBuildRangeInfoKHR` structure **must** have the same value which was specified when `srcAccelerationStructure` was last built
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03801
For each element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, the corresponding `ppBuildRangeInfos[i][j].primitiveCount` **must** be less than or equal to `VkPhysicalDeviceAccelerationStructurePropertiesKHR::maxInstanceCount`
- VUID-vkBuildAccelerationStructuresKHR-pInfos-10126
For each `pInfos[i]`, `dstAccelerationStructure` **must** have been created with a value of `VkAccelerationStructureCreateInfoKHR::size` greater than or equal to either:
 - the memory size required by the build operation, as returned by `vkGetAccelerationStructureBuildSizesKHR` with `pBuildInfo = pInfos[i]` and with each element of the `pMaxPrimitiveCounts` array greater than or equal to the equivalent `ppBuildRangeInfos[i][j].primitiveCount` values for `j` in `[0,pInfos[i].geometryCount)` or,
 - the result of querying the corresponding `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR`, if updating a compacted acceleration structure
- VUID-vkBuildAccelerationStructuresKHR-ppBuildRangeInfos-03676
Each element of `ppBuildRangeInfos[i]` **must** be a valid pointer to an array of `pInfos[i].geometryCount` `VkAccelerationStructureBuildRangeInfoKHR` structures
- VUID-vkBuildAccelerationStructuresKHR-deferredOperation-03678
Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03722
For each element of `pInfos`, the `buffer` used to create its `dstAccelerationStructure` member **must** be bound to host-visible device memory
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03723
For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR` the `buffer` used to create its `srcAccelerationStructure` member **must** be bound to host-visible device memory

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03724
For each element of `pInfos`, the `buffer` used to create each acceleration structure referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR` **must** be bound to host-visible device memory
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03725
If `pInfos[i].mode` is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_BUILD_KHR`, all addresses between `pInfos[i].scratchData.hostAddress` and `pInfos[i].scratchData.hostAddress + N - 1` **must** be valid host memory, where `N` is given by the `buildScratchSize` member of the `VkAccelerationStructureBuildSizesInfoKHR` structure returned from a call to `vkGetAccelerationStructureBuildSizesKHR` with an identical `VkAccelerationStructureBuildGeometryInfoKHR` structure and primitive count
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03726
If `pInfos[i].mode` is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR`, all addresses between `pInfos[i].scratchData.hostAddress` and `pInfos[i].scratchData.hostAddress + N - 1` **must** be valid host memory, where `N` is given by the `updateScratchSize` member of the `VkAccelerationStructureBuildSizesInfoKHR` structure returned from a call to `vkGetAccelerationStructureBuildSizesKHR` with an identical `VkAccelerationStructureBuildGeometryInfoKHR` structure and primitive count
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03771
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, `geometry.triangles.vertexData.hostAddress` **must** be a valid host address
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03772
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.indexType` is not `VK_INDEX_TYPE_NONE_KHR`, `geometry.triangles.indexData.hostAddress` **must** be a valid host address
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03773
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR`, if `geometry.triangles.transformData.hostAddress` is not `0`, it **must** be a valid host address
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03774
For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_AABBS_KHR`, `geometry.aabbs.data.hostAddress` **must** be a valid host address
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03775
For each element of `pInfos`, the `buffer` used to create its `dstAccelerationStructure` member **must** be bound to memory that was not allocated with multiple instances
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03776
For each element of `pInfos`, if its `mode` member is `VK_BUILD_ACCELERATION_STRUCTURE_MODE_UPDATE_KHR` the `buffer` used to create its `srcAccelerationStructure` member **must** be bound to memory that was not allocated with multiple instances
- VUID-vkBuildAccelerationStructuresKHR-pInfos-03777

For each element of `pInfos`, the `buffer` used to create each acceleration structure referenced by the `geometry.instances.data` member of any element of `pGeometries` or `ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR` **must** be bound to memory that was not allocated with multiple instances

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03778

For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, `geometry.instances.data.hostAddress` **must** be a valid host address

- VUID-vkBuildAccelerationStructuresKHR-pInfos-03779

For any element of `pInfos[i].pGeometries` or `pInfos[i].ppGeometries` with a `geometryType` of `VK_GEOMETRY_TYPE_INSTANCES_KHR`, each `VkAccelerationStructureInstanceKHR::accelerationStructureReference` value in `geometry.instances.data.hostAddress` **must** be a valid `VkAccelerationStructureKHR` object

Valid Usage (Implicit)

- VUID-vkBuildAccelerationStructuresKHR-device-parameter

`device` **must** be a valid `VkDevice` handle

- VUID-vkBuildAccelerationStructuresKHR-deferredOperation-parameter

If `deferredOperation` is not `VK_NULL_HANDLE`, `deferredOperation` **must** be a valid `VkDeferredOperationKHR` handle

- VUID-vkBuildAccelerationStructuresKHR-pInfos-parameter

`pInfos` **must** be a valid pointer to an array of `infoCount` valid `VkAccelerationStructureBuildGeometryInfoKHR` structures

- VUID-vkBuildAccelerationStructuresKHR-ppBuildRangeInfos-parameter

`ppBuildRangeInfos` **must** be a valid pointer to an array of `infoCount` `VkAccelerationStructureBuildRangeInfoKHR` structures

- VUID-vkBuildAccelerationStructuresKHR-infoCount-arraylength

`infoCount` **must** be greater than 0

- VUID-vkBuildAccelerationStructuresKHR-deferredOperation-parent

If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_OPERATION_DEFERRED_KHR`
- `VK_OPERATION_NOT_DEFERRED_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`

- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

To copy or compact an acceleration structure on the host, call:

```
// Provided by VK_KHR_acceleration_structure
VkResult vkCopyAccelerationStructureKHR(
    VkDevice device,
    VkDeferredOperationKHR deferredOperation,
    const VkCopyAccelerationStructureInfoKHR* pInfo);
```

- `device` is the device which owns the acceleration structures.
- `deferredOperation` is an optional `VkDeferredOperationKHR` to request deferral for this command.
- `pInfo` is a pointer to a `VkCopyAccelerationStructureInfoKHR` structure defining the copy operation.

This command fulfills the same task as `vkCmdCopyAccelerationStructureKHR` but is executed by the host.

Valid Usage

- VUID-vkCopyAccelerationStructureKHR-accelerationStructureHostCommands-03582
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureHostCommands` feature **must** be enabled
- VUID-vkCopyAccelerationStructureKHR-deferredOperation-03678
Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyAccelerationStructureKHR-buffer-03727
The `buffer` used to create `pInfo->src` **must** be bound to host-visible device memory
- VUID-vkCopyAccelerationStructureKHR-buffer-03728
The `buffer` used to create `pInfo->dst` **must** be bound to host-visible device memory
- VUID-vkCopyAccelerationStructureKHR-buffer-03780
The `buffer` used to create `pInfo->src` **must** be bound to memory that was not allocated with multiple instances
- VUID-vkCopyAccelerationStructureKHR-buffer-03781
The `buffer` used to create `pInfo->dst` **must** be bound to memory that was not allocated with multiple instances

Valid Usage (Implicit)

- VUID-vkCopyAccelerationStructureKHR-device-parameter
`device` **must** be a valid `VkDevice` handle

- VUID-vkCopyAccelerationStructureKHR-deferredOperation-parameter
If `deferredOperation` is not `VK_NULL_HANDLE`, `deferredOperation` **must** be a valid `VkDeferredOperationKHR` handle
- VUID-vkCopyAccelerationStructureKHR-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkCopyAccelerationStructureInfoKHR` structure
- VUID-vkCopyAccelerationStructureKHR-deferredOperation-parent
If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_OPERATION_DEFERRED_KHR`
- `VK_OPERATION_NOT_DEFERRED_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

To copy host accessible memory to an acceleration structure, call:

```
// Provided by VK_KHR_acceleration_structure
VkResult vkCopyMemoryToAccelerationStructureKHR(
    VkDevice device,
    VkDeferredOperationKHR deferredOperation,
    const VkCopyMemoryToAccelerationStructureInfoKHR* pInfo);
```

- `device` is the device which owns `pInfo->dst`.
- `deferredOperation` is an optional `VkDeferredOperationKHR` to request deferral for this command.
- `pInfo` is a pointer to a `VkCopyMemoryToAccelerationStructureInfoKHR` structure defining the copy operation.

This command fulfills the same task as `vkCmdCopyMemoryToAccelerationStructureKHR` but is executed by the host.

This command can accept acceleration structures produced by either `vkCmdCopyAccelerationStructureToMemoryKHR` or `vkCopyAccelerationStructureToMemoryKHR`.

Valid Usage

- VUID-vkCopyMemoryToAccelerationStructureKHR-accelerationStructureHostCommands-

03583

The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureHostCommands` feature **must** be enabled

- VUID-vkCopyMemoryToAccelerationStructureKHR-deferredOperation-03678
Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyMemoryToAccelerationStructureKHR-pInfo-03729
`pInfo->src.hostAddress` **must** be a valid host pointer
- VUID-vkCopyMemoryToAccelerationStructureKHR-pInfo-03750
`pInfo->src.hostAddress` **must** be aligned to 16 bytes
- VUID-vkCopyMemoryToAccelerationStructureKHR-buffer-03730
The `buffer` used to create `pInfo->dst` **must** be bound to host-visible device memory
- VUID-vkCopyMemoryToAccelerationStructureKHR-buffer-03782
The `buffer` used to create `pInfo->dst` **must** be bound to memory that was not allocated with multiple instances

Valid Usage (Implicit)

- VUID-vkCopyMemoryToAccelerationStructureKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCopyMemoryToAccelerationStructureKHR-deferredOperation-parameter
If `deferredOperation` is not `VK_NULL_HANDLE`, `deferredOperation` **must** be a valid `VkDeferredOperationKHR` handle
- VUID-vkCopyMemoryToAccelerationStructureKHR-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkCopyMemoryToAccelerationStructureInfoKHR` structure
- VUID-vkCopyMemoryToAccelerationStructureKHR-deferredOperation-parent
If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_OPERATION_DEFERRED_KHR`
- `VK_OPERATION_NOT_DEFERRED_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

To copy an acceleration structure to host accessible memory, call:

```
// Provided by VK_KHR_acceleration_structure
VkResult vkCopyAccelerationStructureToMemoryKHR(
    VkDevice device,
    VkDeferredOperationKHR deferredOperation,
    const VkCopyAccelerationStructureToMemoryInfoKHR* pInfo);
```

- `device` is the device which owns `pInfo->src`.
- `deferredOperation` is an optional `VkDeferredOperationKHR` to request deferral for this command.
- `pInfo` is a pointer to a `VkCopyAccelerationStructureToMemoryInfoKHR` structure defining the copy operation.

This command fulfills the same task as `vkCmdCopyAccelerationStructureToMemoryKHR` but is executed by the host.

This command produces the same results as `vkCmdCopyAccelerationStructureToMemoryKHR`, but writes its result directly to a host pointer, and is executed on the host rather than the device. The output **may** not necessarily be bit-for-bit identical, but it can be equally used by either `vkCmdCopyMemoryToAccelerationStructureKHR` or `vkCopyMemoryToAccelerationStructureKHR`.

Valid Usage

- VUID-vkCopyAccelerationStructureToMemoryKHR-accelerationStructureHostCommands-03584
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureHostCommands` feature **must** be enabled
- VUID-vkCopyAccelerationStructureToMemoryKHR-deferredOperation-03678
Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyAccelerationStructureToMemoryKHR-buffer-03731
The `buffer` used to create `pInfo->src` **must** be bound to host-visible device memory
- VUID-vkCopyAccelerationStructureToMemoryKHR-pInfo-03732
`pInfo->dst.hostAddress` **must** be a valid host pointer
- VUID-vkCopyAccelerationStructureToMemoryKHR-pInfo-03751
`pInfo->dst.hostAddress` **must** be aligned to 16 bytes
- VUID-vkCopyAccelerationStructureToMemoryKHR-buffer-03783
The `buffer` used to create `pInfo->src` **must** be bound to memory that was not allocated with multiple instances

Valid Usage (Implicit)

- VUID-vkCopyAccelerationStructureToMemoryKHR-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkCopyAccelerationStructureToMemoryKHR-deferredOperation-parameter
If **deferredOperation** is not [VK_NULL_HANDLE](#), **deferredOperation** **must** be a valid [VkDeferredOperationKHR](#) handle
- VUID-vkCopyAccelerationStructureToMemoryKHR-pInfo-parameter
pInfo **must** be a valid pointer to a valid [VkCopyAccelerationStructureToMemoryInfoKHR](#) structure
- VUID-vkCopyAccelerationStructureToMemoryKHR-deferredOperation-parent
If **deferredOperation** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Return Codes

Success

- [VK_SUCCESS](#)
- [VK_OPERATION_DEFERRED_KHR](#)
- [VK_OPERATION_NOT_DEFERRED_KHR](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

To query acceleration structure size parameters on the host, call:

```
// Provided by VK_KHR_acceleration_structure
VkResult vkWriteAccelerationStructuresPropertiesKHR(
    VkDevice          device,
    uint32_t          accelerationStructureCount,
    const VkAccelerationStructureKHR* pAccelerationStructures,
    VkQueryType        queryType,
    size_t            dataSize,
    void*              pData,
    size_t            stride);
```

- **device** is the device which owns the acceleration structures in **pAccelerationStructures**.
- **accelerationStructureCount** is the count of acceleration structures for which to query the property.
- **pAccelerationStructures** is a pointer to an array of existing previously built acceleration structures.

- `queryType` is a `VkQueryType` value specifying the property to be queried.
- `dataSize` is the size in bytes of the buffer pointed to by `pData`.
- `pData` is a pointer to an application-allocated buffer where the results will be written.
- `stride` is the stride in bytes between results for individual queries within `pData`.

This command fulfills the same task as `vkCmdWriteAccelerationStructuresPropertiesKHR` but is executed by the host.

Valid Usage

- VUID-vkWriteAccelerationStructuresPropertiesKHR-accelerationStructureHostCommands-03585
The `VkPhysicalDeviceAccelerationStructureFeaturesKHR::accelerationStructureHostCommands` feature **must** be enabled
- VUID-vkWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-04964
All acceleration structures in `pAccelerationStructures` **must** have been built prior to the execution of this command
- VUID-vkWriteAccelerationStructuresPropertiesKHR-accelerationStructures-03431
All acceleration structures in `pAccelerationStructures` **must** have been built with `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_COMPACTION_BIT_KHR` if `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-06742
`queryType` **must** be `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR`, `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR`, `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR`, or `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-03448
If `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR`, then `stride` **must** be a multiple of the size of `VkDeviceSize`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-03449
If `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR`, then `pData` **must** point to a `VkDeviceSize`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-03450
If `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR`, then `stride` **must** be a multiple of the size of `VkDeviceSize`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-03451
If `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR`, then `pData` **must** point to a `VkDeviceSize`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-06731
If `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR`, then `stride` **must** be a multiple of the size of `VkDeviceSize`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-06732
If `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR`, then `pData` **must** point to a

VkDeviceSize

- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-06733
If `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR`, then `stride` **must** be a multiple of the size of `VkDeviceSize`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-06734
If `queryType` is `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR`, then `pData` **must** point to a `VkDeviceSize`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-dataSize-03452
`dataSize` **must** be greater than or equal to `accelerationStructureCount*stride`
- VUID-vkWriteAccelerationStructuresPropertiesKHR-buffer-03733
The `buffer` used to create each acceleration structure in `pAccelerationStructures` **must** be bound to host-visible device memory
- VUID-vkWriteAccelerationStructuresPropertiesKHR-buffer-03784
The `buffer` used to create each acceleration structure in `pAccelerationStructures` **must** be bound to memory that was not allocated with multiple instances

Valid Usage (Implicit)

- VUID-vkWriteAccelerationStructuresPropertiesKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-parameter
`pAccelerationStructures` **must** be a valid pointer to an array of `accelerationStructureCount` valid `VkAccelerationStructureKHR` handles
- VUID-vkWriteAccelerationStructuresPropertiesKHR-queryType-parameter
`queryType` **must** be a valid `VkQueryType` value
- VUID-vkWriteAccelerationStructuresPropertiesKHR-pData-parameter
`pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkWriteAccelerationStructuresPropertiesKHR-accelerationStructureCount-arraylength
`accelerationStructureCount` **must** be greater than 0
- VUID-vkWriteAccelerationStructuresPropertiesKHR-dataSize-arraylength
`dataSize` **must** be greater than 0
- VUID-vkWriteAccelerationStructuresPropertiesKHR-pAccelerationStructures-parent
Each element of `pAccelerationStructures` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

Chapter 34. Micromap

34.1. Micromaps

Acceleration structures store and organize geometry for ray tracing, but in some cases it is beneficial to include some information within the geometry, particularly for triangles. A *micromap* organizes this data around a map of values corresponding to subdivided microtriangles which can be added to a triangle geometry when building a *bottom level acceleration structure*.

An *opacity micromap* is a type of micromap which stores information to control intersection opacity as described in [Ray Opacity Micromap](#).

A micromap is considered to be constructed if a [micromap build command](#) or [copy command](#) has been executed with the given acceleration structure as the destination.

34.1.1. Building Micromaps

To build micromaps call:

```
// Provided by VK_EXT_opacity_micromap
void vkCmdBuildMicromapsEXT(
    VkCommandBuffer          commandBuffer,
    uint32_t                 infoCount,
    const VkMicromapBuildInfoEXT* pInfos);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `infoCount` is the number of micromaps to build. It specifies the number of the `pInfos` structures that **must** be provided.
- `pInfos` is a pointer to an array of `infoCount` `VkMicromapBuildInfoEXT` structures defining the data used to build each micromap.

The `vkCmdBuildMicromapsEXT` command provides the ability to initiate multiple micromaps builds, however there is no ordering or synchronization implied between any of the individual micromap builds.

NOTE

This means that there **cannot** be any memory aliasing between any micromap memories or scratch memories being used by any of the builds.

Accesses to the micromap scratch buffers as identified by the `VkMicromapBuildInfoEXT::scratchData` buffer device addresses **must** be [synchronized](#) with the `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` pipeline stage and an [access type](#) of (`VK_ACCESS_2_MICROMAP_READ_BIT_EXT` | `VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT`). Accesses to `VkMicromapBuildInfoEXT::dstMicromap` **must** be [synchronized](#) with the `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` pipeline stage and an [access type](#) of `VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT`.

Accesses to other input buffers as identified by any used values of `VkMicromapBuildInfoEXT::data` or `VkMicromapBuildInfoEXT::triangleArray` **must** be **synchronized** with the `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` pipeline stage and an access type of `VK_ACCESS_SHADER_READ_BIT`.

Valid Usage

- VUID-vkCmdBuildMicromapsEXT-pInfos-07461
For each `pInfos[i]`, `dstMicromap` **must** have been created with a value of `VkMicromapCreateInfoEXT::size` greater than or equal to the memory size required by the build operation, as returned by `vkGetMicromapBuildSizesEXT` with `pBuildInfo = pInfos[i]`
- VUID-vkCmdBuildMicromapsEXT-mode-07462
The `mode` member of each element of `pInfos` **must** be a valid `VkBuildMicromapModeEXT` value
- VUID-vkCmdBuildMicromapsEXT-dstMicromap-07463
The `dstMicromap` member of any element of `pInfos` **must** be a valid `VkMicromapEXT` handle
- VUID-vkCmdBuildMicromapsEXT-pInfos-07464
For each element of `pInfos` its `type` member **must** match the value of `VkMicromapCreateInfoEXT::type` when its `dstMicromap` was created
- VUID-vkCmdBuildMicromapsEXT-dstMicromap-07465
The range of memory backing the `dstMicromap` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `dstMicromap` member of any other element of `pInfos`, which is accessed by this command
- VUID-vkCmdBuildMicromapsEXT-dstMicromap-07466
The range of memory backing the `dstMicromap` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `scratchData` member of any element of `pInfos` (including the same element), which is accessed by this command
- VUID-vkCmdBuildMicromapsEXT-scratchData-07467
The range of memory backing the `scratchData` member of any element of `pInfos` that is accessed by this command **must** not overlap the memory backing the `scratchData` member of any other element of `pInfos`, which is accessed by this command
- VUID-vkCmdBuildMicromapsEXT-pInfos-07508
For each element of `pInfos`, the `buffer` used to create its `dstMicromap` member **must** be bound to device memory
- VUID-vkCmdBuildMicromapsEXT-pInfos-07509
If `pInfos[i].mode` is `VK_BUILD_MICROMAP_MODE_BUILD_EXT`, all addresses between `pInfos[i].scratchData.deviceAddress` and `pInfos[i].scratchData.deviceAddress + N - 1` **must** be in the buffer device address range of the same buffer, where N is given by the `buildScratchSize` member of the `VkMicromapBuildSizesInfoEXT` structure returned from a call to `vkGetMicromapBuildSizesEXT` with an identical `VkMicromapBuildInfoEXT` structure and primitive count

- VUID-vkCmdBuildMicromapsEXT-data-07510
The buffers from which the buffer device addresses for all of the `data` and `triangleArray` members of all `pInfos[i]` are queried **must** have been created with the `VK_BUFFER_USAGE_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT` usage flag
- VUID-vkCmdBuildMicromapsEXT-pInfos-07511
For each element of `pInfos[i]` the buffer from which the buffer device address `pInfos[i].scratchData.deviceAddress` is queried **must** have been created with `VK_BUFFER_USAGE_STORAGE_BUFFER_BIT` usage flag
- VUID-vkCmdBuildMicromapsEXT-pInfos-07512
For each element of `pInfos`, its `scratchData.deviceAddress`, `data.deviceAddress`, and `triangleArray.deviceAddress` members **must** be valid device addresses obtained from `vkGetBufferDeviceAddressKHR`
- VUID-vkCmdBuildMicromapsEXT-pInfos-07513
For each element of `pInfos`, if `scratchData.deviceAddress`, `data.deviceAddress`, or `triangleArray.deviceAddress` is the address of a non-sparse buffer then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdBuildMicromapsEXT-pInfos-07514
For each element of `pInfos`, its `scratchData.deviceAddress` member **must** be a multiple of `VkPhysicalDeviceAccelerationStructurePropertiesKHR::minAccelerationStructureScratchOffsetAlignment`
- VUID-vkCmdBuildMicromapsEXT-pInfos-07515
For each element of `pInfos`, its `triangleArray.deviceAddress` and `data.deviceAddress` members **must** be a multiple of `256`

Valid Usage (Implicit)

- VUID-vkCmdBuildMicromapsEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdBuildMicromapsEXT-pInfos-parameter
`pInfos` **must** be a valid pointer to an array of `infoCount` valid `VkMicromapBuildInfoEXT` structures
- VUID-vkCmdBuildMicromapsEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording` state
- VUID-vkCmdBuildMicromapsEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdBuildMicromapsEXT-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdBuildMicromapsEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdBuildMicromapsEXT-infoCount-arraylength
`infoCount` **must** be greater than `0`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

Formats which **can** be set in `VkMicromapUsageEXT::format` and `VkMicromapTriangleEXT::format` for micromap builds, are:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkOpacityMicromapFormatEXT {
    VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT = 1,
    VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT = 2,
} VkOpacityMicromapFormatEXT;
```

- `VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT` indicates that the given micromap format has one bit per subtriangle encoding either fully opaque or fully transparent.
- `VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT` indicates that the given micromap format has two bits per subtriangle encoding four modes which can be interpreted as described in [ray traversal](#).

NOTE For compactness, these values are stored as 16-bit in some structures.

The `VkMicromapBuildInfoEXT` structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkMicromapBuildInfoEXT {
    VkStructureType          sType;
    const void*              pNext;
    VkMicromapTypeEXT        type;
    VkBuildMicromapFlagsEXT   flags;
    VkBuildMicromapModeEXT    mode;
    VkMicromapEXT             dstMicromap;
    uint32_t                 usageCountsCount;
    const VkMicromapUsageEXT* pUsageCounts;
    const VkMicromapUsageEXT* const* ppUsageCounts;
    VkDeviceOrHostAddressConstKHR data;
    VkDeviceOrHostAddressKHR    scratchData;
    VkDeviceOrHostAddressConstKHR triangleArray;
    VkDeviceSize                triangleArrayStride;
} VkMicromapBuildInfoEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **type** is a [VkMicromapTypeEXT](#) value specifying the type of micromap being built.
- **flags** is a bitmask of [VkBuildMicromapFlagBitsEXT](#) specifying additional parameters of the micromap.
- **mode** is a [VkBuildMicromapModeEXT](#) value specifying the type of operation to perform.
- **dstMicromap** is a pointer to the target micromap for the build.
- **usageCountsCount** specifies the number of usage counts structures that will be used to determine the size of this micromap.
- **pUsageCounts** is a pointer to an array of [VkMicromapUsageEXT](#) structures.
- **ppUsageCounts** is a pointer to an array of pointers to [VkMicromapUsageEXT](#) structures.
- **data** is the device or host address to memory which contains the data for the micromap.
- **scratchData** is the device or host address to memory that will be used as scratch memory for the build.
- **triangleArray** is the device or host address to memory containing the [VkMicromapTriangleEXT](#) data
- **triangleArrayStride** is the stride in bytes between each element of **triangleArray**

Only one of **pUsageCounts** or **ppUsageCounts** **can** be a valid pointer, the other **must** be **NULL**. The elements of the non-**NULL** array describe the total counts used to build each micromap. Each element contains a **count** which is the number of micromap triangles of that **format** and **subdivisionLevel** contained in the micromap. Multiple elements with the same **format** and **subdivisionLevel** are allowed and the total count for that **format** and **subdivisionLevel** is the sum of the **count** for each element.

Each micromap triangle refers to one element in **triangleArray** which contains the **format** and

`subdivisionLevel` for that particular triangle as well as a `dataOffset` in bytes which is the location relative to `data` where that triangle's micromap data begins. The data at `triangleArray` is laid out as a 4 byte unsigned integer for the `dataOffset` followed by a 2 byte unsigned integer for the subdivision level then a 2 byte unsigned integer for the format. In practice, compilers compile `VkMicromapTriangleEXT` to match this pattern.

For opacity micromaps, the data at `data` is packed as either one bit per element for `VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT` or two bits per element for `VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT` and is packed from LSB to MSB in each byte. The data at each index in those bytes is interpreted as discussed in [Ray Opacity Micromap](#).

Valid Usage

- VUID-VkMicromapBuildInfoEXT-pUsageCounts-07516
Only one of `pUsageCounts` or `ppUsageCounts` can be a valid pointer, the other **must** be `NULL`
- VUID-VkMicromapBuildInfoEXT-type-07517
If `type` is `VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT` the `format` member of `VkMicromapUsageEXT` **must** be a valid value from `VkOpacityMicromapFormatEXT`
- VUID-VkMicromapBuildInfoEXT-type-07518
If `type` is `VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT` the `format` member of `VkMicromapTriangleEXT` **must** be a valid value from `VkOpacityMicromapFormatEXT`

Valid Usage (Implicit)

- VUID-VkMicromapBuildInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MICROMAP_BUILD_INFO_EXT`
- VUID-VkMicromapBuildInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkMicromapBuildInfoEXT-type-parameter
`type` **must** be a valid `VkMicromapTypeEXT` value
- VUID-VkMicromapBuildInfoEXT-flags-parameter
`flags` **must** be a valid combination of `VkBuildMicromapFlagBitsEXT` values
- VUID-VkMicromapBuildInfoEXT-pUsageCounts-parameter
If `usageCountsCount` is not 0, and `pUsageCounts` is not `NULL`, `pUsageCounts` **must** be a valid pointer to an array of `usageCountsCount` `VkMicromapUsageEXT` structures
- VUID-VkMicromapBuildInfoEXT-ppUsageCounts-parameter
If `usageCountsCount` is not 0, and `ppUsageCounts` is not `NULL`, `ppUsageCounts` **must** be a valid pointer to an array of `usageCountsCount` valid pointers to `VkMicromapUsageEXT` structures

The `VkBuildMicromapModeEXT` enumeration is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkBuildMicromapModeEXT {
    VK_BUILD_MICROMAP_MODE_BUILD_EXT = 0,
} VkBuildMicromapModeEXT;
```

- `VK_BUILD_MICROMAP_MODE_BUILD_EXT` specifies that the destination micromap will be built using the specified data.

The `VkMicromapUsageEXT` structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkMicromapUsageEXT {
    uint32_t    count;
    uint32_t    subdivisionLevel;
    uint32_t    format;
} VkMicromapUsageEXT;
```

- `count` is the number of triangles in the usage format defined by the `subdivisionLevel` and `format` below in the micromap
- `subdivisionLevel` is the subdivision level of this usage format
- `format` is the format of this usage format

Valid Usage

- VUID-VkMicromapUsageEXT-format-07519
If the `VkMicromapTypeEXT` of the micromap is `VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT` then `format` **must** be `VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT` or `VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT`
- VUID-VkMicromapUsageEXT-format-07520
If the `VkMicromapTypeEXT` of the micromap is `VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT` and `format` is `VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT` then `subdivisionLevel` **must** be less than or equal to `VkPhysicalDeviceOpacityMicromapPropertiesEXT::maxOpacity2StateSubdivisionLevel`
- VUID-VkMicromapUsageEXT-format-07521
If the `VkMicromapTypeEXT` of the micromap is `VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT` and `format` is `VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT` then `subdivisionLevel` **must** be less than or equal to `VkPhysicalDeviceOpacityMicromapPropertiesEXT::maxOpacity4StateSubdivisionLevel`

The `format` is interpreted based on the `type` of the micromap using it.

The `VkMicromapTriangleEXT` structure is defined as:


```
// Provided by VK_EXT_opacity_micromap
typedef struct VkMicromapTriangleEXT {
    uint32_t    dataOffset;
    uint16_t    subdivisionLevel;
    uint16_t    format;
} VkMicromapTriangleEXT;
```

- `dataOffset` is the offset in bytes of the start of the data for this triangle. This is a byte aligned value.
- `subdivisionLevel` is the subdivision level of this triangle
- `format` is the format of this triangle

Valid Usage

- VUID-VkMicromapTriangleEXT-format-07522
If the `VkMicromapTypeEXT` of the micromap is `VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT` then `format` **must** be `VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT` or `VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT`
- VUID-VkMicromapTriangleEXT-format-07523
If the `VkMicromapTypeEXT` of the micromap is `VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT` and `format` is `VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT` then `subdivisionLevel` **must** be less than or equal to `VkPhysicalDeviceOpacityMicromapPropertiesEXT::maxOpacity2StateSubdivisionLevel`
- VUID-VkMicromapTriangleEXT-format-07524
If the `VkMicromapTypeEXT` of the micromap is `VK_MICROMAP_TYPE_OPACITY_MICROMAP_EXT` and `format` is `VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT` then `subdivisionLevel` **must** be less than or equal to `VkPhysicalDeviceOpacityMicromapPropertiesEXT::maxOpacity4StateSubdivisionLevel`

The `format` is interpreted based on the `type` of the micromap using it.

34.1.2. Copying Micromaps

An additional command exists for copying micromaps without updating their contents. Before copying, an application **must** query the size of the resulting micromap.

To query micromap size parameters call:

```
// Provided by VK_EXT_opacity_micromap
void vkCmdWriteMicromapsPropertiesEXT(
    VkCommandBuffer                commandBuffer,
    uint32_t                       micromapCount,
    const VkMicromapEXT*           pMicromaps,
    VkQueryType                    queryType,
    VkQueryPool                    queryPool,
    uint32_t                       firstQuery);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **micromapCount** is the count of micromaps for which to query the property.
- **pMicromaps** is a pointer to an array of existing previously built micromaps.
- **queryType** is a **VkQueryType** value specifying the type of queries managed by the pool.
- **queryPool** is the query pool that will manage the results of the query.
- **firstQuery** is the first query index within the query pool that will contain the **micromapCount** number of results.

Accesses to any of the micromaps listed in **pMicromaps** **must** be **synchronized** with the **VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT** **pipeline stage** and an **access type** of **VK_ACCESS_2_MICROMAP_READ_BIT_EXT**.

- If **queryType** is **VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT**, then the value written out is the number of bytes required by a serialized micromap.
- If **queryType** is **VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT**, then the value written out is the number of bytes required by a compacted micromap.

Valid Usage

- VUID-vkCmdWriteMicromapsPropertiesEXT-queryPool-07525
queryPool **must** have been created with a **queryType** matching **queryType**
- VUID-vkCmdWriteMicromapsPropertiesEXT-queryPool-07526
The queries identified by **queryPool** and **firstQuery** **must** be *unavailable*
- VUID-vkCmdWriteMicromapsPropertiesEXT-buffer-07527
The **buffer** used to create each micromap in **pMicromaps** **must** be bound to device memory
- VUID-vkCmdWriteMicromapsPropertiesEXT-query-07528
The sum of **query** plus **micromapCount** **must** be less than or equal to the number of queries in **queryPool**
- VUID-vkCmdWriteMicromapsPropertiesEXT-pMicromaps-07501
All micromaps in **pMicromaps** **must** have been constructed prior to the execution of this command
- VUID-vkCmdWriteMicromapsPropertiesEXT-pMicromaps-07502
All micromaps in **pMicromaps** **must** have been constructed with **VK_BUILD_MICROMAP_ALLOW_COMPACTION_BIT_EXT** if **queryType** is

VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT

- VUID-vkCmdWriteMicromapsPropertiesEXT-queryType-07503

`queryType` **must** be `VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT` or `VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT`

Valid Usage (Implicit)

- VUID-vkCmdWriteMicromapsPropertiesEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdWriteMicromapsPropertiesEXT-pMicromaps-parameter
`pMicromaps` **must** be a valid pointer to an array of `micromapCount` valid `VkMicromapEXT` handles
- VUID-vkCmdWriteMicromapsPropertiesEXT-queryType-parameter
`queryType` **must** be a valid `VkQueryType` value
- VUID-vkCmdWriteMicromapsPropertiesEXT-queryPool-parameter
`queryPool` **must** be a valid `VkQueryPool` handle
- VUID-vkCmdWriteMicromapsPropertiesEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdWriteMicromapsPropertiesEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdWriteMicromapsPropertiesEXT-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdWriteMicromapsPropertiesEXT-videocoding
This command **must** only be called outside of a video coding scope
- VUID-vkCmdWriteMicromapsPropertiesEXT-micromapCount-arraylength
`micromapCount` **must** be greater than 0
- VUID-vkCmdWriteMicromapsPropertiesEXT-commonparent
Each of `commandBuffer`, `queryPool`, and the elements of `pMicromaps` **must** have been created, allocated, or retrieved from the same `VkDevice`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

To copy a micromap call:

```
// Provided by VK_EXT_opacity_micromap
void vkCmdCopyMicromapEXT(
    VkCommandBuffer                      commandBuffer,
    const VkCopyMicromapInfoEXT*        pInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pInfo` is a pointer to a `VkCopyMicromapInfoEXT` structure defining the copy operation.

This command copies the `pInfo->src` micromap to the `pInfo->dst` micromap in the manner specified by `pInfo->mode`.

Accesses to `pInfo->src` and `pInfo->dst` **must** be `synchronized` with the `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` pipeline stage and an `access type` of `VK_ACCESS_2_MICROMAP_READ_BIT_EXT` or `VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT` as appropriate.

Valid Usage

- VUID-vkCmdCopyMicromapEXT-buffer-07529
The `buffer` used to create `pInfo->src` **must** be bound to device memory
- VUID-vkCmdCopyMicromapEXT-buffer-07530
The `buffer` used to create `pInfo->dst` **must** be bound to device memory

Valid Usage (Implicit)

- VUID-vkCmdCopyMicromapEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdCopyMicromapEXT-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkCopyMicromapInfoEXT` structure
- VUID-vkCmdCopyMicromapEXT-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdCopyMicromapEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations

- VUID-vkCmdCopyMicromapEXT-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMicromapEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

The `VkCopyMicromapInfoEXT` structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkCopyMicromapInfoEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkMicromapEXT        src;
    VkMicromapEXT        dst;
    VkCopyMicromapModeEXT mode;
} VkCopyMicromapInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `src` is the source micromap for the copy.
- `dst` is the target micromap for the copy.
- `mode` is a `VkCopyMicromapModeEXT` value specifying additional operations to perform during the copy.

Valid Usage

- VUID-VkCopyMicromapInfoEXT-mode-07531
`mode` **must** be `VK_COPY_MICROMAP_MODE_COMPACT_EXT` or `VK_COPY_MICROMAP_MODE_CLONE_EXT`
- VUID-VkCopyMicromapInfoEXT-src-07532

The source acceleration structure **src** **must** have been constructed prior to the execution of this command

- VUID-VkCopyMicromapInfoEXT-mode-07533

If **mode** is **VK_COPY_MICROMAP_MODE_COMPACT_EXT**, **src** **must** have been constructed with **VK_BUILD_MICROMAP_ALLOW_COMPACTION_BIT_EXT** in the build

- VUID-VkCopyMicromapInfoEXT-buffer-07534

The **buffer** used to create **src** **must** be bound to device memory

- VUID-VkCopyMicromapInfoEXT-buffer-07535

The **buffer** used to create **dst** **must** be bound to device memory

Valid Usage (Implicit)

- VUID-VkCopyMicromapInfoEXT-sType-sType

sType **must** be **VK_STRUCTURE_TYPE_COPY_MICROMAP_INFO_EXT**

- VUID-VkCopyMicromapInfoEXT-pNext-pNext

pNext **must** be **NULL**

- VUID-VkCopyMicromapInfoEXT-src-parameter

src **must** be a valid **VkMicromapEXT** handle

- VUID-VkCopyMicromapInfoEXT-dst-parameter

dst **must** be a valid **VkMicromapEXT** handle

- VUID-VkCopyMicromapInfoEXT-mode-parameter

mode **must** be a valid **VkCopyMicromapModeEXT** value

- VUID-VkCopyMicromapInfoEXT-commonparent

Both of **dst**, and **src** **must** have been created, allocated, or retrieved from the same **VkDevice**

Possible values of **mode** specifying additional operations to perform during the copy, are:

```
// Provided by VK_EXT_opacity_micromap
typedef enum VkCopyMicromapModeEXT {
    VK_COPY_MICROMAP_MODE_CLONE_EXT = 0,
    VK_COPY_MICROMAP_MODE_SERIALIZE_EXT = 1,
    VK_COPY_MICROMAP_MODE_DESERIALIZE_EXT = 2,
    VK_COPY_MICROMAP_MODE_COMPACT_EXT = 3,
} VkCopyMicromapModeEXT;
```

- **VK_COPY_MICROMAP_MODE_CLONE_EXT** creates a direct copy of the micromap specified in **src** into the one specified by **dst**. The **dst** micromap **must** have been created with the same parameters as **src**.
- **VK_COPY_MICROMAP_MODE_SERIALIZE_EXT** serializes the micromap to a semi-opaque format which can be reloaded on a compatible implementation.
- **VK_COPY_MICROMAP_MODE_DESERIALIZE_EXT** deserializes the semi-opaque serialization format in the

buffer to the micromap.

- `VK_COPY_MICROMAP_MODE_COMPACT_EXT` creates a more compact version of a micromap `src` into `dst`. The micromap `dst` **must** have been created with a size at least as large as that returned by `vkCmdWriteMicromapsPropertiesEXT` after the build of the micromap specified by `src`.

To copy a micromap to device memory call:

```
// Provided by VK_EXT_opacity_micromap
void vkCmdCopyMicromapToMemoryEXT(
    VkCommandBuffer                      commandBuffer,
    const VkCopyMicromapToMemoryInfoEXT* pInfo);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pInfo` is an a pointer to a `VkCopyMicromapToMemoryInfoEXT` structure defining the copy operation.

Accesses to `pInfo->src` **must** be `synchronized` with the `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` pipeline stage and an access type of `VK_ACCESS_2_MICROMAP_READ_BIT_EXT`. Accesses to the buffer indicated by `pInfo->dst.deviceAddress` **must** be synchronized with the `VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT` pipeline stage and an access type of `VK_ACCESS_TRANSFER_WRITE_BIT`.

This command produces the same results as `vkCopyMicromapToMemoryEXT`, but writes its result to a device address, and is executed on the device rather than the host. The output **may** not necessarily be bit-for-bit identical, but it can be equally used by either `vkCmdCopyMemoryToMicromapEXT` or `vkCopyMemoryToMicromapEXT`.

The defined header structure for the serialized data consists of:

- `VK_UUID_SIZE` bytes of data matching `VkPhysicalDeviceIDProperties::driverUUID`
- `VK_UUID_SIZE` bytes of data identifying the compatibility for comparison using `vkGetDeviceMicromapCompatibilityEXT` The serialized data is written to the buffer (or read from the buffer) according to the host endianness.

Valid Usage

- VUID-vkCmdCopyMicromapToMemoryEXT-pInfo-07536
`pInfo->dst.deviceAddress` **must** be a valid device address for a buffer bound to device memory
- VUID-vkCmdCopyMicromapToMemoryEXT-pInfo-07537
`pInfo->dst.deviceAddress` **must** be aligned to 256 bytes
- VUID-vkCmdCopyMicromapToMemoryEXT-pInfo-07538
If the buffer pointed to by `pInfo->dst.deviceAddress` is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdCopyMicromapToMemoryEXT-buffer-07539
The `buffer` used to create `pInfo->src` **must** be bound to device memory

Valid Usage (Implicit)

- VUID-vkCmdCopyMicromapToMemoryEXT-commandBuffer-parameter **commandBuffer** **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdCopyMicromapToMemoryEXT-pInfo-parameter **pInfo** **must** be a valid pointer to a valid [VkCopyMicromapToMemoryInfoEXT](#) structure
- VUID-vkCmdCopyMicromapToMemoryEXT-commandBuffer-recording **commandBuffer** **must** be in the [recording state](#)
- VUID-vkCmdCopyMicromapToMemoryEXT-commandBuffer-cmdpool
The [VkCommandPool](#) that **commandBuffer** was allocated from **must** support compute operations
- VUID-vkCmdCopyMicromapToMemoryEXT-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMicromapToMemoryEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the [VkCommandPool](#) that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkCopyMicromapToMemoryInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkMicromapEXT      src;
    VkDeviceOrHostAddressKHR dst;
    VkCopyMicromapModeEXT mode;
} VkCopyMicromapToMemoryInfoEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.

- **src** is the source micromap for the copy
- **dst** is the device or host address to memory which is the target for the copy
- **mode** is a [VkCopyMicromapModeEXT](#) value specifying additional operations to perform during the copy.

Valid Usage

- VUID-VkCopyMicromapToMemoryInfoEXT-src-07540
The source micromap **src** **must** have been constructed prior to the execution of this command
- VUID-VkCopyMicromapToMemoryInfoEXT-dst-07541
The memory pointed to by **dst** **must** be at least as large as the serialization size of **src**, as reported by [vkWriteMicromapsPropertiesEXT](#) or [vkCmdWriteMicromapsPropertiesEXT](#) with a query type of [VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT](#)
- VUID-VkCopyMicromapToMemoryInfoEXT-mode-07542
mode **must** be [VK_COPY_MICROMAP_MODE_SERIALIZE_EXT](#)

Valid Usage (Implicit)

- VUID-VkCopyMicromapToMemoryInfoEXT-sType-sType
sType **must** be [VK_STRUCTURE_TYPE_COPY_MICROMAP_TO_MEMORY_INFO_EXT](#)
- VUID-VkCopyMicromapToMemoryInfoEXT-pNext-pNext
pNext **must** be `NULL`
- VUID-VkCopyMicromapToMemoryInfoEXT-src-parameter
src **must** be a valid [VkMicromapEXT](#) handle
- VUID-VkCopyMicromapToMemoryInfoEXT-mode-parameter
mode **must** be a valid [VkCopyMicromapModeEXT](#) value

To copy device memory to a micromap call:

```
// Provided by VK_EXT_opacity_micromap
void vkCmdCopyMemoryToMicromapEXT(
    VkCommandBuffer                      commandBuffer,
    const VkCopyMemoryToMicromapInfoEXT* pInfo);
```

- **commandBuffer** is the command buffer into which the command will be recorded.
- **pInfo** is a pointer to a [VkCopyMemoryToMicromapInfoEXT](#) structure defining the copy operation.

Accesses to **pInfo->dst** **must** be [synchronized](#) with the [VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT](#) [pipeline stage](#) and an [access type](#) of [VK_ACCESS_2_MICROMAP_READ_BIT_EXT](#). Accesses to the buffer indicated by **pInfo->src.deviceAddress** **must** be synchronized with the

VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT pipeline stage and an access type of VK_ACCESS_TRANSFER_READ_BIT.

This command can accept micromaps produced by either [vkCmdCopyMicromapToMemoryEXT](#) or [vkCopyMicromapToMemoryEXT](#).

Valid Usage

- VUID-vkCmdCopyMemoryToMicromapEXT-pInfo-07543
`pInfo->src.deviceAddress` **must** be a valid device address for a buffer bound to device memory
- VUID-vkCmdCopyMemoryToMicromapEXT-pInfo-07544
`pInfo->src.deviceAddress` **must** be aligned to 256 bytes
- VUID-vkCmdCopyMemoryToMicromapEXT-pInfo-07545
If the buffer pointed to by `pInfo->src.deviceAddress` is non-sparse then it **must** be bound completely and contiguously to a single [VkDeviceMemory](#) object
- VUID-vkCmdCopyMemoryToMicromapEXT-buffer-07546
The `buffer` used to create `pInfo->dst` **must** be bound to device memory

Valid Usage (Implicit)

- VUID-vkCmdCopyMemoryToMicromapEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdCopyMemoryToMicromapEXT-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid [VkCopyMemoryToMicromapInfoEXT](#) structure
- VUID-vkCmdCopyMemoryToMicromapEXT-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdCopyMemoryToMicromapEXT-commandBuffer-cmdpool
The [VkCommandPool](#) that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdCopyMemoryToMicromapEXT-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdCopyMemoryToMicromapEXT-vidcoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the [VkCommandPool](#) that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

The `VkCopyMemoryToMicromapInfoEXT` structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkCopyMemoryToMicromapInfoEXT {
    VkStructureType           sType;
    const void*               pNext;
    VkDeviceOrHostAddressConstKHR src;
    VkMicromapEXT             dst;
    VkCopyMicromapModeEXT     mode;
} VkCopyMemoryToMicromapInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `src` is the device or host address to memory containing the source data for the copy.
- `dst` is the target micromap for the copy.
- `mode` is a `VkCopyMicromapModeEXT` value specifying additional operations to perform during the copy.

Valid Usage

- VUID-VkCopyMemoryToMicromapInfoEXT-src-07547
The source memory pointed to by `src` **must** contain data previously serialized using `vkCmdCopyMicromapToMemoryEXT`
- VUID-VkCopyMemoryToMicromapInfoEXT-mode-07548
`mode` **must** be `VK_COPY_MICROMAP_MODE_DESERIALIZE_EXT`
- VUID-VkCopyMemoryToMicromapInfoEXT-src-07549
The data in `src` **must** have a format compatible with the destination physical device as returned by `vkGetDeviceMicromapCompatibilityEXT`
- VUID-VkCopyMemoryToMicromapInfoEXT-dst-07550
`dst` **must** have been created with a `size` greater than or equal to that used to serialize the data in `src`

Valid Usage (Implicit)

- VUID-VkCopyMemoryToMicromapInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_COPY_MEMORY_TO_MICROMAP_INFO_EXT`
- VUID-VkCopyMemoryToMicromapInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkCopyMemoryToMicromapInfoEXT-dst-parameter
`dst` **must** be a valid `VkMicromapEXT` handle
- VUID-VkCopyMemoryToMicromapInfoEXT-mode-parameter
`mode` **must** be a valid `VkCopyMicromapModeEXT` value

To check if a serialized micromap is compatible with the current device call:

```
// Provided by VK_EXT_opacity_micromap
void vkGetDeviceMicromapCompatibilityEXT(
    VkDevice device,
    const VkMicromapVersionInfoEXT* pVersionInfo,
    VkAccelerationStructureCompatibilityKHR* pCompatibility);
```

- `device` is the device to check the version against.
- `pVersionInfo` is a pointer to a `VkMicromapVersionInfoEXT` structure specifying version information to check against the device.
- `pCompatibility` is a pointer to a `VkAccelerationStructureCompatibilityKHR` value in which compatibility information is returned.

Valid Usage

- VUID-vkGetDeviceMicromapCompatibilityEXT-micromap-07551
The `micromap` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkGetDeviceMicromapCompatibilityEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetDeviceMicromapCompatibilityEXT-pVersionInfo-parameter
`pVersionInfo` **must** be a valid pointer to a valid `VkMicromapVersionInfoEXT` structure
- VUID-vkGetDeviceMicromapCompatibilityEXT-pCompatibility-parameter
`pCompatibility` **must** be a valid pointer to a `VkAccelerationStructureCompatibilityKHR` value

The `VkMicromapVersionInfoEXT` structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkMicromapVersionInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    const uint8_t*     pVersionData;
} VkMicromapVersionInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pVersionData` is a pointer to the version header of a micromap as defined in `vkCmdCopyMicromapToMemoryEXT`

NOTE

`pVersionData` is a *pointer* to an array of $2 \times \text{VK_UUID_SIZE}$ `uint8_t` values instead of two `VK\_UUID\_SIZE` arrays as the expected use case for this member is to be pointed at the header of a previously serialized micromap (via `vkCmdCopyMicromapToMemoryEXT` or `vkCopyMicromapToMemoryEXT`) that is loaded in memory. Using arrays would necessitate extra memory copies of the UUIDs.

Valid Usage (Implicit)

- VUID-VkMicromapVersionInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_MICROMAP_VERSION_INFO_EXT`
- VUID-VkMicromapVersionInfoEXT-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkMicromapVersionInfoEXT-pVersionData-parameter
`pVersionData` **must** be a valid pointer to an array of $2 \times \text{VK_UUID_SIZE}$ `uint8_t` values

34.2. Host Micromap Operations

Implementations are also required to provide host implementations of the micromap operations if the `micromapHostCommands` feature is enabled:

- `vkBuildMicromapsEXT` corresponding to `vkCmdBuildMicromapsEXT`
- `vkCopyMicromapEXT` corresponding to `vkCmdCopyMicromapEXT`
- `vkCopyMicromapToMemoryEXT` corresponding to `vkCmdCopyMicromapToMemoryEXT`
- `vkCopyMemoryToMicromapEXT` corresponding to `vkCmdCopyMemoryToMicromapEXT`
- `vkWriteMicromapsPropertiesEXT` corresponding to `vkCmdWriteMicromapsPropertiesEXT`

These commands are functionally equivalent to their device counterparts, except that they are executed on the host timeline, rather than being enqueued into command buffers.

All micromaps used by the host commands **must** be bound to host-visible memory, and all input

data for micromap builds **must** be referenced using host addresses instead of device addresses. Applications are not required to map micromap memory when using the host commands.

NOTE

The `vkBuildMicromapsEXT` and `vkCmdBuildMicromapsEXT` **may** use different algorithms, and thus are not required to produce identical structures.

Apart from these details, the host and device operations are interchangeable.

NOTE

For efficient execution, micromaps manipulated using these commands should always be bound to host cached memory, as the implementation may need to repeatedly read and write this memory during the execution of the command.

To build micromaps on the host, call:

```
// Provided by VK_EXT_opacity_micromap
VkResult vkBuildMicromapsEXT(
    VkDevice          device,
    VkDeferredOperationKHR deferredOperation,
    uint32_t          infoCount,
    const VkMicromapBuildInfoEXT* pInfos);
```

- `device` is the `VkDevice` for which the micromaps are being built.
- `deferredOperation` is an optional `VkDeferredOperationKHR` to request deferral for this command.
- `infoCount` is the number of micromaps to build. It specifies the number of the `pInfos` that **must** be provided.
- `pInfos` is a pointer to an array of `infoCount` `VkMicromapBuildInfoEXT` structures defining the geometry used to build each micromap.

This command fulfills the same task as `vkCmdBuildMicromapsEXT` but is executed by the host.

The `vkBuildMicromapsEXT` command provides the ability to initiate multiple micromaps builds, however there is no ordering or synchronization implied between any of the individual micromap builds.

NOTE

This means that there **cannot** be any memory aliasing between any micromap memories or scratch memories being used by any of the builds.

Valid Usage

- VUID-vkBuildMicromapsEXT-pInfos-07461

For each `pInfos[i]`, `dstMicromap` **must** have been created with a value of `VkMicromapCreateInfoEXT::size` greater than or equal to the memory size required by the build operation, as returned by `vkGetMicromapBuildSizesEXT` with `pBuildInfo = pInfos[i]`

- VUID-vkBuildMicromapsEXT-mode-07462
The **mode** member of each element of **pInfos** **must** be a valid **VkBuildMicromapModeEXT** value
- VUID-vkBuildMicromapsEXT-dstMicromap-07463
The **dstMicromap** member of any element of **pInfos** **must** be a valid **VkMicromapEXT** handle
- VUID-vkBuildMicromapsEXT-pInfos-07464
For each element of **pInfos** its **type** member **must** match the value of **VkMicromapCreateInfoEXT::type** when its **dstMicromap** was created
- VUID-vkBuildMicromapsEXT-dstMicromap-07465
The range of memory backing the **dstMicromap** member of any element of **pInfos** that is accessed by this command **must** not overlap the memory backing the **dstMicromap** member of any other element of **pInfos**, which is accessed by this command
- VUID-vkBuildMicromapsEXT-dstMicromap-07466
The range of memory backing the **dstMicromap** member of any element of **pInfos** that is accessed by this command **must** not overlap the memory backing the **scratchData** member of any element of **pInfos** (including the same element), which is accessed by this command
- VUID-vkBuildMicromapsEXT-scratchData-07467
The range of memory backing the **scratchData** member of any element of **pInfos** that is accessed by this command **must** not overlap the memory backing the **scratchData** member of any other element of **pInfos**, which is accessed by this command
- VUID-vkBuildMicromapsEXT-pInfos-07552
For each element of **pInfos**, the **buffer** used to create its **dstMicromap** member **must** be bound to host-visible device memory
- VUID-vkBuildMicromapsEXT-pInfos-07553
For each element of **pInfos**, all referenced addresses of **pInfos[i].data.hostAddress** **must** be valid host memory
- VUID-vkBuildMicromapsEXT-pInfos-07554
For each element of **pInfos**, all referenced addresses of **pInfos[i].triangleArray.hostAddress** **must** be valid host memory
- VUID-vkBuildMicromapsEXT-micromapHostCommands-07555
The **VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapHostCommands** feature **must** be enabled
- VUID-vkBuildMicromapsEXT-pInfos-07556
If **pInfos[i].mode** is **VK_BUILD_MICROMAP_MODE_BUILD_EXT**, all addresses between **pInfos[i].scratchData.hostAddress** and **pInfos[i].scratchData.hostAddress + N - 1** **must** be valid host memory, where **N** is given by the **buildScratchSize** member of the **VkMicromapBuildSizesInfoEXT** structure returned from a call to **vkGetMicromapBuildSizesEXT** with an identical **VkMicromapBuildInfoEXT** structure and primitive count
- VUID-vkBuildMicromapsEXT-pInfos-07557
For each element of **pInfos**, the **buffer** used to create its **dstMicromap** member **must** be

bound to memory that was not allocated with multiple instances

Valid Usage (Implicit)

- VUID-vkBuildMicromapsEXT-device-parameter
device **must** be a valid [VkDevice](#) handle
- VUID-vkBuildMicromapsEXT-deferredOperation-parameter
If **deferredOperation** is not [VK_NULL_HANDLE](#), **deferredOperation** **must** be a valid [VkDeferredOperationKHR](#) handle
- VUID-vkBuildMicromapsEXT-pInfos-parameter
pInfos **must** be a valid pointer to an array of **infoCount** valid [VkMicromapBuildInfoEXT](#) structures
- VUID-vkBuildMicromapsEXT-infoCount-arraylength
infoCount **must** be greater than 0
- VUID-vkBuildMicromapsEXT-deferredOperation-parent
If **deferredOperation** is a valid handle, it **must** have been created, allocated, or retrieved from **device**

Return Codes

Success

- [VK_SUCCESS](#)
- [VK_OPERATION_DEFERRED_KHR](#)
- [VK_OPERATION_NOT_DEFERRED_KHR](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)

To copy or compact a micromap on the host, call:

```
// Provided by VK_EXT_opacity_micromap
VkResult vkCopyMicromapEXT(
    VkDevice                                device,
    VkDeferredOperationKHR                  deferredOperation,
    const VkCopyMicromapInfoEXT*            pInfo);
```

- **device** is the device which owns the micromaps.
- **deferredOperation** is an optional [VkDeferredOperationKHR](#) to [request deferral](#) for this command.
- **pInfo** is a pointer to a [VkCopyMicromapInfoEXT](#) structure defining the copy operation.

This command fulfills the same task as [vkCmdCopyMicromapEXT](#) but is executed by the host.

Valid Usage

- VUID-vkCopyMicromapEXT-deferredOperation-03678
Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyMicromapEXT-buffer-07558
The `buffer` used to create `pInfo->src` **must** be bound to host-visible device memory
- VUID-vkCopyMicromapEXT-buffer-07559
The `buffer` used to create `pInfo->dst` **must** be bound to host-visible device memory
- VUID-vkCopyMicromapEXT-micromapHostCommands-07560
The `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapHostCommands` feature **must** be enabled
- VUID-vkCopyMicromapEXT-buffer-07561
The `buffer` used to create `pInfo->src` **must** be bound to memory that was not allocated with multiple instances
- VUID-vkCopyMicromapEXT-buffer-07562
The `buffer` used to create `pInfo->dst` **must** be bound to memory that was not allocated with multiple instances

Valid Usage (Implicit)

- VUID-vkCopyMicromapEXT-device-parameter
`device` **must** be a valid [VkDevice](#) handle
- VUID-vkCopyMicromapEXT-deferredOperation-parameter
If `deferredOperation` is not `VK_NULL_HANDLE`, `deferredOperation` **must** be a valid [VkDeferredOperationKHR](#) handle
- VUID-vkCopyMicromapEXT-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid [VkCopyMicromapInfoEXT](#) structure
- VUID-vkCopyMicromapEXT-deferredOperation-parent
If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_OPERATION_DEFERRED_KHR`
- `VK_OPERATION_NOT_DEFERRED_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

To copy host accessible memory to a micromap, call:

```
// Provided by VK_EXT_opacity_micromap
VkResult vkCopyMemoryToMicromapEXT(
    VkDevice                                device,
    VkDeferredOperationKHR                  deferredOperation,
    const VkCopyMemoryToMicromapInfoEXT*    pInfo);
```

- `device` is the device which owns `pInfo->dst`.
- `deferredOperation` is an optional `VkDeferredOperationKHR` to request deferral for this command.
- `pInfo` is a pointer to a `VkCopyMemoryToMicromapInfoEXT` structure defining the copy operation.

This command fulfills the same task as `vkCmdCopyMemoryToMicromapEXT` but is executed by the host.

This command can accept micromaps produced by either `vkCmdCopyMicromapToMemoryEXT` or `vkCopyMicromapToMemoryEXT`.

Valid Usage

- VUID-vkCopyMemoryToMicromapEXT-deferredOperation-03678
Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyMemoryToMicromapEXT-pInfo-07563
`pInfo->src.hostAddress` **must** be a valid host pointer
- VUID-vkCopyMemoryToMicromapEXT-pInfo-07564
`pInfo->src.hostAddress` **must** be aligned to 16 bytes
- VUID-vkCopyMemoryToMicromapEXT-buffer-07565
The `buffer` used to create `pInfo->dst` **must** be bound to host-visible device memory
- VUID-vkCopyMemoryToMicromapEXT-micromapHostCommands-07566
The `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapHostCommands` feature **must** be enabled
- VUID-vkCopyMemoryToMicromapEXT-buffer-07567
The `buffer` used to create `pInfo->dst` **must** be bound to memory that was not allocated with multiple instances

Valid Usage (Implicit)

- VUID-vkCopyMemoryToMicromapEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCopyMemoryToMicromapEXT-deferredOperation-parameter
If `deferredOperation` is not `VK_NULL_HANDLE`, `deferredOperation` **must** be a valid `VkDeferredOperationKHR` handle
- VUID-vkCopyMemoryToMicromapEXT-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid `VkCopyMemoryToMicromapInfoEXT` structure
- VUID-vkCopyMemoryToMicromapEXT-deferredOperation-parent
If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_OPERATION_DEFERRED_KHR`
- `VK_OPERATION_NOT_DEFERRED_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

To copy a micromap to host accessible memory, call:

```
// Provided by VK_EXT_opacity_micromap
VkResult vkCopyMicromapToMemoryEXT(
    VkDevice                                device,
    VkDeferredOperationKHR                  deferredOperation,
    const VkCopyMicromapToMemoryInfoEXT*    pInfo);
```

- `device` is the device which owns `pInfo->src`.
- `deferredOperation` is an optional `VkDeferredOperationKHR` to request deferral for this command.
- `pInfo` is a pointer to a `VkCopyMicromapToMemoryInfoEXT` structure defining the copy operation.

This command fulfills the same task as `vkCmdCopyMicromapToMemoryEXT` but is executed by the host.

This command produces the same results as `vkCmdCopyMicromapToMemoryEXT`, but writes its

result directly to a host pointer, and is executed on the host rather than the device. The output **may** not necessarily be bit-for-bit identical, but it can be equally used by either [vkCmdCopyMemoryToMicromapEXT](#) or [vkCopyMemoryToMicromapEXT](#).

Valid Usage

- VUID-vkCopyMicromapToMemoryEXT-deferredOperation-03678
Any previous deferred operation that was associated with `deferredOperation` **must** be complete
- VUID-vkCopyMicromapToMemoryEXT-buffer-07568
The `buffer` used to create `pInfo->src` **must** be bound to host-visible device memory
- VUID-vkCopyMicromapToMemoryEXT-pInfo-07569
`pInfo->dst.hostAddress` **must** be a valid host pointer
- VUID-vkCopyMicromapToMemoryEXT-pInfo-07570
`pInfo->dst.hostAddress` **must** be aligned to 16 bytes
- VUID-vkCopyMicromapToMemoryEXT-micromapHostCommands-07571
The `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapHostCommands` feature **must** be enabled
- VUID-vkCopyMicromapToMemoryEXT-buffer-07572
The `buffer` used to create `pInfo->src` **must** be bound to memory that was not allocated with multiple instances

Valid Usage (Implicit)

- VUID-vkCopyMicromapToMemoryEXT-device-parameter
`device` **must** be a valid [VkDevice](#) handle
- VUID-vkCopyMicromapToMemoryEXT-deferredOperation-parameter
If `deferredOperation` is not `VK_NULL_HANDLE`, `deferredOperation` **must** be a valid [VkDeferredOperationKHR](#) handle
- VUID-vkCopyMicromapToMemoryEXT-pInfo-parameter
`pInfo` **must** be a valid pointer to a valid [VkCopyMicromapToMemoryInfoEXT](#) structure
- VUID-vkCopyMicromapToMemoryEXT-deferredOperation-parent
If `deferredOperation` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_OPERATION_DEFERRED_KHR`
- `VK_OPERATION_NOT_DEFERRED_KHR`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

To query micromap size parameters on the host, call:

```
// Provided by VK_EXT_opacity_micromap
VkResult vkWriteMicromapsPropertiesEXT(
    VkDevice          device,
    uint32_t          micromapCount,
    const VkMicromapEXT* pMicromaps,
    VkQueryType        queryType,
    size_t             dataSize,
    void*              pData,
    size_t             stride);
```

- `device` is the device which owns the micromaps in `pMicromaps`.
- `micromapCount` is the count of micromaps for which to query the property.
- `pMicromaps` is a pointer to an array of existing previously built micromaps.
- `queryType` is a `VkQueryType` value specifying the property to be queried.
- `dataSize` is the size in bytes of the buffer pointed to by `pData`.
- `pData` is a pointer to an application-allocated buffer where the results will be written.
- `stride` is the stride in bytes between results for individual queries within `pData`.

This command fulfills the same task as `vkCmdWriteMicromapsPropertiesEXT` but is executed by the host.

Valid Usage

- VUID-vkWriteMicromapsPropertiesEXT-pMicromaps-07501
All micromaps in `pMicromaps` **must** have been constructed prior to the execution of this command
- VUID-vkWriteMicromapsPropertiesEXT-pMicromaps-07502
All micromaps in `pMicromaps` **must** have been constructed with `VK_BUILD_MICROMAP_ALLOW_COMPACTION_BIT_EXT` if `queryType` is `VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT`
- VUID-vkWriteMicromapsPropertiesEXT-queryType-07503
`queryType` **must** be `VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT` or `VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT`
- VUID-vkWriteMicromapsPropertiesEXT-queryType-10071
If `queryType` is `VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT` or `VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT` then `stride` **must** be a multiple of the size of `VkDeviceSize`

- VUID-vkWriteMicromapsPropertiesEXT-queryType-10072
If `queryType` is `VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT` or `VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT` then `pData` **must** point to a `VkDeviceSize`
- VUID-vkWriteMicromapsPropertiesEXT-dataSize-07576
`dataSize` **must** be greater than or equal to `micromapCount*stride`
- VUID-vkWriteMicromapsPropertiesEXT-buffer-07577
The `buffer` used to create each micromap in `pMicromaps` **must** be bound to host-visible device memory
- VUID-vkWriteMicromapsPropertiesEXT-micromapHostCommands-07578
The `VkPhysicalDeviceOpacityMicromapFeaturesEXT::micromapHostCommands` feature **must** be enabled
- VUID-vkWriteMicromapsPropertiesEXT-buffer-07579
The `buffer` used to create each micromap in `pMicromaps` **must** be bound to memory that was not allocated with multiple instances

Valid Usage (Implicit)

- VUID-vkWriteMicromapsPropertiesEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkWriteMicromapsPropertiesEXT-pMicromaps-parameter
`pMicromaps` **must** be a valid pointer to an array of `micromapCount` valid `VkMicromapEXT` handles
- VUID-vkWriteMicromapsPropertiesEXT-queryType-parameter
`queryType` **must** be a valid `VkQueryType` value
- VUID-vkWriteMicromapsPropertiesEXT-pData-parameter
`pData` **must** be a valid pointer to an array of `dataSize` bytes
- VUID-vkWriteMicromapsPropertiesEXT-micromapCount-arraylength
`micromapCount` **must** be greater than 0
- VUID-vkWriteMicromapsPropertiesEXT-dataSize-arraylength
`dataSize` **must** be greater than 0
- VUID-vkWriteMicromapsPropertiesEXT-pMicromaps-parent
Each element of `pMicromaps` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

Chapter 35. Ray Traversal

The ray traversal process identifies and handles intersections between a ray and geometries in an acceleration structure.

Ray traversal cannot be started by a Vulkan API command directly - a shader **must** execute `OpRayQueryProceedKHR` or a `pipeline trace ray` instruction. When the `rayTracingPipeline` feature is enabled, `OpTraceRayKHR` **can** be used for `ray tracing` in a `ray tracing pipeline`. When the `rayQuery` feature is enabled, `OpRayQueryProceedKHR` **can** be used in any shader stage.

35.1. Ray Intersection Candidate Determination

Once tracing begins, rays are first tested against instances in a top-level acceleration structure. A ray that intersects an instance will be transformed into the space of the instance to continue traversal within that instance; therefore the transform matrix stored in the instance **must** be invertible.

In case multiple instances are intersected by a ray, the ray transformation into the space of the instance is invariant under the order in which these instances are encountered in the top-level acceleration structure.

NOTE

Applying multiple forward and reverse transforms to a ray to transition from one instance to another could result in accumulated errors. Thus an implementation should behave as if the ray is transformed from the origin for each instance independently.

Next, rays are tested against geometries in a bottom-level acceleration structure to determine if a hit occurred between them, initially based only on their geometric properties (i.e. their vertices). The implementation performs similar operations to that of rasterization, but with the effective viewport determined by the parameters of the ray, and the geometry transformed into a space determined by that viewport.

The vertices of each primitive are transformed from acceleration structure space $\mathbf{as}$ to ray space $\mathbf{r}$ according to the ray origin and direction as follows:

$$\begin{pmatrix} x_r \\ y_r \\ z_r \end{pmatrix} = \begin{pmatrix} a_x^2(1-c) + c & a_x a_y(1-c) - s a_z & a_x a_z(1-c) + s a_y \\ a_x a_y(1-c) + s a_z & a_y^2(1-c) + c & a_y a_z(1-c) - s a_x \\ a_x a_z(1-c) - s a_y & a_y a_z(1-c) + s a_x & a_z^2(1-c) + c \end{pmatrix} \begin{pmatrix} x_{as} - o_x \\ y_{as} - o_y \\ z_{as} - o_z \end{pmatrix}$$

$\mathbf{a}$ is the axis of rotation from the unnormalized ray direction vector $\mathbf{d}$ to the axis vector $\mathbf{k}$:

$$\mathbf{a} = \begin{cases} \frac{\mathbf{d} \times \mathbf{k}}{\|\mathbf{d} \times \mathbf{k}\|} & \text{if } \|\mathbf{d} \times \mathbf{k}\| \neq 0 \\ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} & \text{if } \|\mathbf{d} \times \mathbf{k}\| = 0 \end{cases}$$

s and c are the sine and cosine of the angle of rotation about $\mathbf{a}$ from $\mathbf{d}$ to $\mathbf{k}$:

$$c = \frac{\mathbf{d} \cdot \mathbf{k}}{\|\mathbf{d}\|}$$

$$s = \sqrt{1 - c^2}$$

$\mathbf{k}$ is the unit vector:

$$\mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}$$

$\mathbf{o}$ and $\mathbf{d}$ are the ray origin and unnormalized direction, respectively; the vector described by x_{as} , y_{as} , and z_{as} is any position in acceleration structure space; and the vector described by x_r , y_r , and z_r is the same position in ray space.

An *intersection candidate* is a unique point of intersection between a ray and a geometric primitive. For any primitive that has within its bounds a position $\mathbf{x}_r \mathbf{y}_r \mathbf{z}_r$ such that

$$\begin{aligned} x_r &= 0 \\ y_r &= 0 \\ t_{min} &< -\frac{z_r}{\|\mathbf{d}\|} < t_{max} \quad \text{if the primitive is a triangle,} \\ t_{min} &\leq -\frac{z_r}{\|\mathbf{d}\|} \leq t_{max} \quad \text{otherwise} \end{aligned}$$

(where $t = -\frac{z_r}{\|\mathbf{d}\|}$), an intersection candidate exists.

Triangle primitive bounds consist of all points on the plane formed by the three vertices and within the bounds of the edges between the vertices, subject to the watertightness constraints below. AABB primitive bounds consist of all points within an implementation-defined bound which includes the specified box.

NOTE

The bounds of the AABB including all points internal to the bound implies that a ray started within the AABB will hit that AABB.

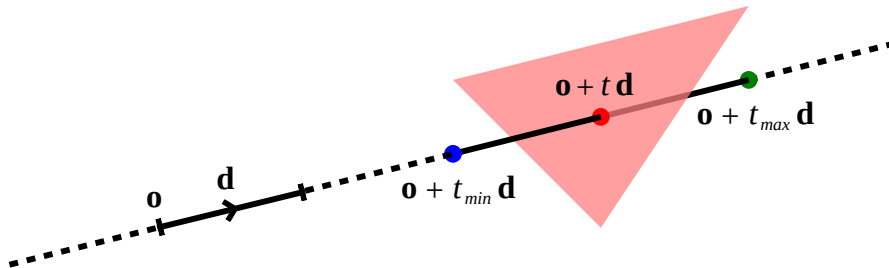


Figure 23. Ray intersection candidate

The determination of this condition is performed in an implementation specific manner, and **may** be performed with floating-point operations. Due to the complexity and number of operations involved, inaccuracies are expected, particularly as the scale of values involved begins to diverge.

Implementations **should** take efforts to maintain as much precision as possible.

NOTE

One very common case is when geometries are close to each other at some distance from the origin in acceleration structure space, where an effect similar to “z-fighting” is likely to be observed. Applications can mitigate this by ensuring their detailed geometries remain close to the origin.

Another likely case is when the origin of a ray is set to a position on a previously intersected surface, and its $t_{\min}$ is zero or near zero; an intersection may be detected on the emitting surface. This case can usually be mitigated by offsetting $t_{\min}$ slightly.

In the case of AABB geometries, implementations **may** increase their size in an acceleration structure in order to mitigate precision issues. This **may** result in false positive intersections being reported to the application.

For triangle intersection candidates, the b and c [barycentric coordinates](#) on the triangle where the above condition is met are made available to future shading. If the ray was traced with a [pipeline trace ray](#) instruction, these values are available as a vector of 2 32-bit floating-point values in the `HitAttributeKHR` storage class.

Once an intersection candidate is determined, it proceeds through the following operations, in order:

1. [Ray Intersection Culling](#)
2. [Ray Intersection Confirmation](#)
3. [Ray Closest Hit Determination](#)
4. [Ray Result Determination](#)

The sections below describe the exact details of these tests. There is no ordering guarantee between operations performed on different intersection candidates.

35.1.1. Watertightness

For a set of triangles with identical transforms, within a single instance:

- Any set of two or more triangles where all triangles have one vertex with an identical position value, that vertex is a *shared vertex*.
- Any set of two triangles with two shared vertices that were specified in the same [winding order](#) in each triangle have a *shared edge* defined by those vertices.

A *closed fan* is a set of three or more triangles where:

- All triangles in the set have the same shared vertex as one of their vertices.
- All edges that include the above vertex are shared edges.
- All above shared edges are shared by exactly two triangles from the set.
- No two triangles in the set intersect, except at shared edges.
- Every triangle in the set is joined to every other triangle in the set by a series of the above

shared edges.

Implementations **should** not double-hit or miss when a ray intersects a shared edge, or a shared vertex of a closed fan.

35.2. Ray Intersection Culling

Candidate intersections go through several phases of culling before confirmation as an actual hit. There is no particular ordering dependency between the different culling operations.

35.2.1. Ray Primitive Culling

If the `rayTraversalPrimitiveCulling` or `rayQuery` features are enabled, the `SkipTrianglesKHR` and `SkipAABBsKHR` ray flags **can** be specified when tracing a ray. `SkipTrianglesKHR` and `SkipAABBsKHR` are mutually exclusive. `SkipTrianglesKHR` is also mutually exclusive with `CullBackFacingTrianglesKHR` and `CullFrontFacingTrianglesKHR`.

If `SkipTrianglesKHR` was included in the `Ray Flags` operand of the ray trace instruction, and the intersection is with a triangle primitive, the intersection is dropped, and no further processing of this intersection occurs. If `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR` was included in the pipeline, traversal with `pipeline trace ray` instructions will all behave as if `SkipTrianglesKHR` was included in their `Ray Flags` operand.

If `SkipAABBsKHR` was included in the `Ray Flags` operand of the ray trace instruction, and the intersection is with an AABB primitive, the intersection is dropped, and no further processing of this intersection occurs. If `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR` was included in the pipeline, traversal with `pipeline trace ray` instructions will all behave as if `SkipAABBsKHR` was included in their `Ray Flags` operand.

35.2.2. Ray Mask Culling

Instances **can** be made invisible to particular rays based on the value of `VkAccelerationStructureInstanceKHR::mask` used to add that instance to a top-level acceleration structure, and the `Cull Mask` parameter used to trace the ray.

For the instance which is intersected, if `mask & Cull Mask == 0`, the intersection is dropped, and no further processing occurs.

35.2.3. Ray Face Culling

As in `polygon rasterization`, one of the stages of ray traversal is to determine if a triangle primitive is back- or front-facing, and primitives **can** be culled based on that facing.

If the intersection candidate is with an AABB primitive, this operation is skipped.

Determination

When a ray intersects a triangle primitive, the order that vertices are specified for the polygon affects whether the ray intersects the front or back face. Front or back facing is determined in the same way as they are for `rasterization`, based on the sign of the polygon's area but using the ray

space coordinates instead of framebuffer coordinates. One way to compute this area is:

$$a = -\frac{1}{2} \sum_{i=0}^{n-1} x_r^i y_r^{i \oplus 1} - x_r^{i \oplus 1} y_r^i$$

where x_r^i and y_r^i are the x and y [ray space coordinates](#) of the i th vertex of the n -vertex polygon (vertices are numbered starting at zero for the purposes of this computation) and $i \oplus 1$ is $(i + 1) \bmod n$.

By default, if a is negative then the intersection is with the front face of the triangle, otherwise it is with the back face. If `VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR` is included in `VkAccelerationStructureInstanceKHR::flags` for the instance containing the intersected triangle, this determination is reversed. Additionally, if a is 0, the intersection candidate is treated as not intersecting with any face, irrespective of the sign.

NOTE

In a left-handed coordinate system, an intersection will be with the front face of a triangle if the vertices of the triangle, as defined in index order, appear from the ray's perspective in a clockwise rotation order. `VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR` was previously annotated as `VK_GEOMETRY_INSTANCE_TRIANGLE_FRONT_COUNTERCLOCKWISE_BIT_KHR` because of this.

If the ray was traced with a [pipeline trace ray](#) instruction, the `HitKindKHR` built-in is set to `HitKindFrontFacingTriangleKHR` if the intersection is with front-facing geometry, and `HitKindBackFacingTriangleKHR` if the intersection is with back-facing geometry, for shader stages considering this intersection.

If the ray was traced with `OpRayQueryProceedKHR`, `OpRayQueryGetIntersectionFrontFaceKHR` will return true for intersection candidates with front faces, or false for back faces.

Culling

If `CullBackFacingTrianglesKHR` was included in the `Ray Flags` parameter of the ray trace instruction, and the intersection is determined as with the back face of a triangle primitive, the intersection is dropped, and no further processing of this intersection occurs.

If `CullFrontFacingTrianglesKHR` was included in the `Ray Flags` parameter of the ray trace instruction, and the intersection is determined as with the front face of a triangle primitive, the intersection is dropped, and no further processing of this intersection occurs.

This culling is disabled if `VK_GEOMETRY_INSTANCE_TRIANGLE_FACING_CULL_DISABLE_BIT_KHR` was included in `VkAccelerationStructureInstanceKHR::flags` for the instance which the intersected geometry belongs to.

Intersection candidates that have not intersected with any face ($a \neq 0$) are unconditionally culled, irrespective of ray flags and geometry instance flags.

The `CullBackFacingTrianglesKHR` and `CullFrontFacingTrianglesKHR` `Ray Flags` are mutually exclusive.

35.2.4. Ray Opacity Culling

Each geometry in the acceleration structure **may** be considered either opaque or not. Opaque geometries continue through traversal as normal, whereas non-opaque geometries need to be either confirmed or discarded by shader code. Intersection candidates **can** also be culled based on their opacity.

Determination

Each individual intersection candidate is initially determined as opaque if `VK_GEOMETRY_OPAQUE_BIT_KHR` was included in the `VkAccelerationStructureGeometryKHR::flags` when the geometry it intersected with was built, otherwise it is considered non-opaque.

If the geometry includes an opacity micromap, the opacity of the intersection at this point is instead derived as described in [Ray Opacity Micromap](#).

If the intersection candidate was generated by an [intersection shader](#), the intersection is initially considered to have opacity matching the AABB candidate that it was generated from.

However, this opacity can be overridden when it is built into an instance. Setting `VK_GEOMETRY_INSTANCE_FORCE_OPAQUE_BIT_KHR` in `VkAccelerationStructureInstanceKHR::flags` will force all geometries in the instance to be considered opaque. Similarly, setting `VK_GEOMETRY_INSTANCE_FORCE_NO_OPAQUE_BIT_KHR` will force all geometries in the instance to be considered non-opaque.

This can again be overridden by including `OpaqueKHR` or `NoOpaqueKHR` in the `Ray Flags` parameter when tracing a ray. `OpaqueKHR` forces all geometries to behave as if they are opaque, regardless of their build parameters. Similarly, `NoOpaqueKHR` forces all geometries to behave as if they are non-opaque.

If the ray was traced with `OpRayQueryProceedKHR`, to determine the opacity of AABB intersection candidates, `OpRayQueryGetIntersectionCandidateAABBOpaqueKHR` **can** be used. This instruction will return `true` for opaque intersection candidates, and `false` for non-opaque intersection candidates.

Culling

If `CullOpaqueKHR` is included in the `Ray Flags` parameter when tracing a ray, an intersection with a geometry that is considered opaque is dropped, and no further processing occurs.

If `CullNoOpaqueKHR` is included in the `Ray Flags` parameter when tracing a ray, an intersection with a geometry that is considered non-opaque is dropped, and no further processing occurs.

The `OpaqueKHR`, `NoOpaqueKHR`, `CullOpaqueKHR`, and `CullNoOpaqueKHR` `Ray Flags` are mutually exclusive.

35.2.5. Ray Opacity Micromap

A `VK_GEOMETRY_TYPE_TRIANGLES_KHR` geometry in the acceleration structure **may** have an opacity micromap associated with it to give finer-grained opacity information.

If the intersection candidate is with a geometry with an associated opacity micromap and `VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_EXT` is not set in its instance then the micromap is used to determine geometry opacity instead of the `VK_GEOMETRY_OPAQUE_BIT_KHR` flag in the geometry.

The opacity information in the micromap object is accessed using the candidate intersection u and v coordinates. The integer u and v are computed from $\lceil u \rceil + \lceil v \rceil$, clamping $\lceil u \rceil$ as needed to keep the sum less than or equal to $1 \ll \text{subdivisionlevel}$. These values are mapped into a linear index with a space filling curve which is defined recursively by traversing into the sub-triangle nearest vertex 0, then the middle triangle with ordering flipped, then nearest vertex 1 then nearest vertex 2.

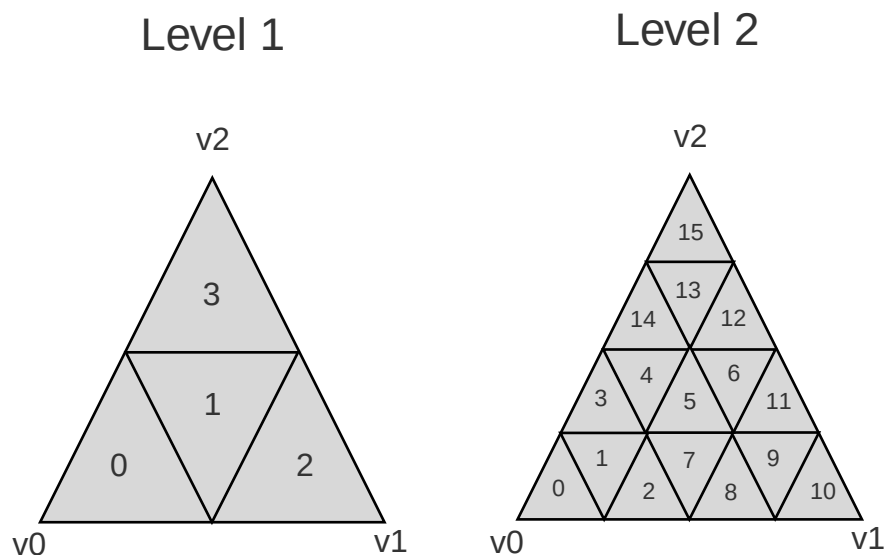


Figure 24. Example ordering for micromap data

NOTE

This encoding is spatially coherent, purely hierarchical, and allows a bit-parallel conversion between barycentric address and index values.

See the appendix for reference code implementing this mapping.

The result of the opacity micromap lookup and operations is to treat the intersection as opaque, non-opaque, or ignored. The interpretation of the values depends on `VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_EXT` in the instance of the candidate intersection or `ForceOpacityMicromap2StateEXT` ray flags on the ray. If either is set, the opacity micromap information is interpreted in 2 state override mode. If the result of the micromap lookup is to treat the intersection candidate as ignored, no further processing of that candidate is done.

If the associated opacity micromap has format `VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT`, each element of the micromap is represented by a single bit at the index derived above.

If the associated opacity micromap has format `VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT`, each element is represented by a two bit value at the index derived above.

4 State value	2 State value	Special index value	2 State override	Result
0	0	<code>VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT</code>	Y	Ignored

4 State value	2 State value	Special index value	2 State override	Result
0	0	VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_TRANSPARENT_EXT	N	Ignored
1	1	VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_EXT	Y	Opaque
1	1	VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_OPAQUE_EXT	N	Opaque
2		VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_EXT	Y	Ignored
2		VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_TRANSPARENT_EXT	N	Non-opaque
3		VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_EXT	Y	Opaque
3		VK_OPACITY_MICROMAP_SPECIAL_INDEX_FULLY_UNKNOWN_OPAQUE_EXT	N	Non-opaque

35.3. Ray Intersection Confirmation

Depending on the opacity of intersected geometry and whether it is a triangle or an AABB, candidate intersections are further processed to determine the eventual hit result. Candidates generated from AABB intersections run through the same confirmation process as triangle hits.

35.3.1. AABB Intersection Candidates

For an intersection candidate with an AABB geometry generated by [Ray Intersection Candidate Determination](#), shader code is executed to determine whether any hits should be reported to the traversal infrastructure; no further processing of this intersection candidate occurs. The occurrence of an AABB intersection candidate does not guarantee the ray intersects the primitive bounds. To avoid propagating false intersections the application **should** verify the intersection candidate before reporting any hits and only report intersections within the bounds of the AABB. Reporting an intersection outside the AABB either through the implementation giving a conservative bound or reporting a t out of range is legal but **may** result in unpredictable closest hit results.

If the ray was traced with a [pipeline trace ray](#) instruction, an [intersection shader](#) is invoked from the [Shader Binding Table](#) according to the [specified indexing](#) for the intersected geometry. If this

shader calls `OpReportIntersectionKHR`, a new intersection candidate is generated as described [below](#). If the intersection shader is `VK_SHADER_UNUSED_KHR` (which is only allowed for a zero shader group) then no further processing of the intersection candidate occurs.

Each new candidate generated as a result of this processing is a generated intersection candidate from the intersection with AABB geometry, with a `t` value equal to the `Hit` parameter of the `OpReportIntersectionKHR` instruction. The new generated candidate is then independently run through [Ray Intersection Confirmation](#) as a [generated intersection](#).

If the ray was traced with `OpRayQueryProceedKHR`, control is returned to the shader which executed `OpRayQueryProceedKHR`, returning `true`. The resulting ray query has a candidate intersection type of `RayQueryCandidateIntersectionAABBKHR`. `OpRayQueryGenerateIntersectionKHR` **can** be called to commit a new intersection candidate with committed intersection type of `RayQueryCommittedIntersectionGeneratedKHR`. Further ray query processing **can** be continued by executing `OpRayQueryProceedKHR` with the same ray query, or intersection **can** be terminated with `OpRayQueryTerminateKHR`. Unlike rays traced with a [pipeline trace ray](#) instruction, candidates generated in this way skip generated intersection candidate confirmation; applications **should** make this determination before generating the intersection.

This operation **may** be executed multiple times for the same intersection candidate.

35.3.2. Triangle and Generated Intersection Candidates

For triangle and [generated intersection candidates](#), additional shader code **may** be executed based on the intersection's opacity.

If the intersection is opaque, the candidate is immediately confirmed as a valid hit and passes to the next stage of processing.

For non-opaque intersection candidates, shader code is executed to determine whether a hit occurred or not.

If the ray was traced with a [pipeline trace ray](#) instruction, an [any-hit shader](#) is invoked from the [Shader Binding Table](#) according to the specified indexing. If this shader calls `OpIgnoreIntersectionKHR`, the candidate is dropped and no further processing of the candidate occurs. If the [any-hit shader](#) identified is `VK_SHADER_UNUSED_KHR`, the candidate is immediately confirmed as a valid hit and passes to the next stage of processing.

If the ray was traced with `OpRayQueryProceedKHR`, control is returned to the shader which executed `OpRayQueryProceedKHR`, returning `true`. As only triangle candidates participate in this operation with ray queries, the resulting candidate intersection type is always `RayQueryCandidateIntersectionTriangleKHR`. `OpRayQueryConfirmIntersectionKHR` **can** be called on the ray query to confirm the candidate as a hit with committed intersection type of `RayQueryCommittedIntersectionTriangleKHR`. Further ray query processing **can** be continued by executing `OpRayQueryProceedKHR` with the same ray query, or intersection **can** be terminated with `OpRayQueryTerminateKHR`. If `OpRayQueryConfirmIntersectionKHR` has not been executed, the candidate is dropped and no further processing of the candidate occurs.

This operation **may** be executed multiple times for the same intersection candidate unless `VK_GEOMETRY_NO_DUPLICATE_ANY_HIT_INVOCATION_BIT_KHR` was specified for the intersected geometry.

35.4. Ray Closest Hit Determination

Unless the ray was traced with the `TerminateOnFirstHitKHR` ray flag, the implementation **must** track the closest confirmed hit until all geometries have been tested and either confirmed or dropped.

After an intersection candidate is confirmed, its t value is compared to $t_{\max}$ to determine which intersection is closer, where t is the parametric distance along the ray at which the intersection occurred.

- If $t < t_{\max}$, $t_{\max}$ is set to t and the candidate is set as the current closest hit.
- If $t > t_{\max}$, the candidate is dropped and no further processing of that candidate occurs.
- If $t = t_{\max}$, the candidate **may** be set as the current closest hit or dropped.

If `TerminateOnFirstHitKHR` was included in the `Ray Flags` used to trace the ray, once the first hit is confirmed, the ray trace is terminated.

35.5. Ray Result Determination

Once all candidates have finished processing the prior stages, or if the ray is forcibly terminated, the final result of the ray trace is determined.

If a closest hit result was identified by [Ray Closest Hit Determination](#), a closest hit has occurred, otherwise the final result is a miss.

For rays traced with `pipeline trace ray` instructions which **can** invoke a closest hit shader, if a closest hit result was identified, a `closest hit shader` is invoked from the `Shader Binding Table` according to the `specified indexing` for the intersected geometry. Control returns to the shader that executed the `pipeline trace ray` instruction once this shader returns. This shader is skipped if either the ray flags included `SkipClosestHitShaderKHR`, or if the `closest hit shader` identified is `VK_SHADER_UNUSED_KHR`.

For rays traced with a `pipeline trace ray` instruction where no hit result was identified, the `miss shader` identified by the `Miss Index` parameter of the instruction is invoked. Control returns to the shader that executed the `pipeline trace ray` instruction once this shader returns. This shader is skipped if the miss shader identified is `VK_SHADER_UNUSED_KHR`.

If the ray was traced with `OpRayQueryProceedKHR`, control is returned to the shader which executed `OpRayQueryProceedKHR`, returning `false`. If a closest hit was identified by [Ray Closest Hit Determination](#), the ray query will now have a committed intersection type of `RayQueryCommittedIntersectionGeneratedKHR` or `RayQueryCommittedIntersectionTriangleKHR`. If no closest hit was identified, the committed intersection type will be `RayQueryCommittedIntersectionNoneKHR`.

No further processing of a ray query occurs after this result is determined.

Chapter 36. Ray Tracing

Ray tracing uses a separate rendering pipeline from both the graphics and compute pipelines (see [Ray Tracing Pipeline](#)).

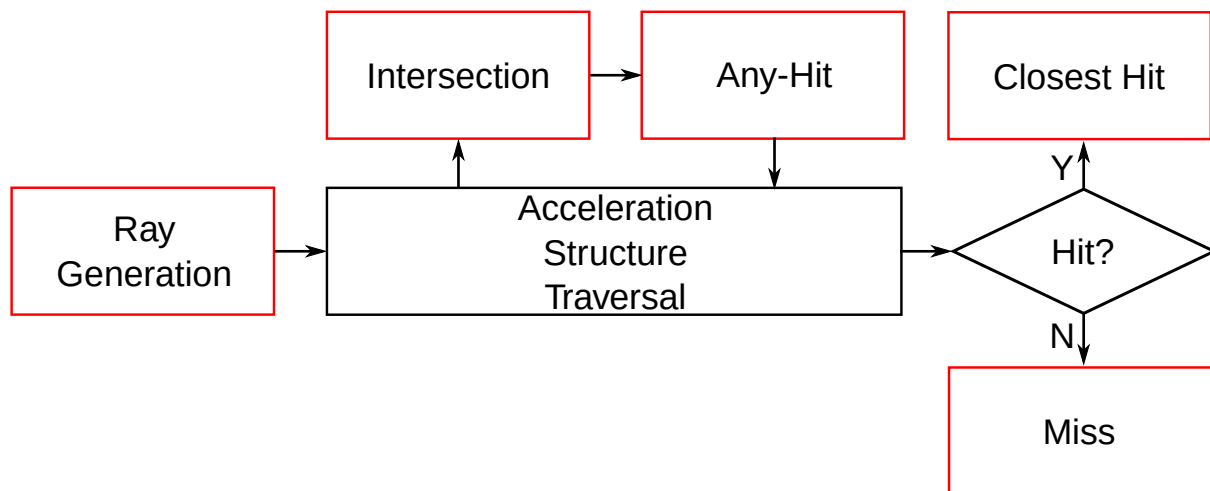


Figure 25. Ray tracing pipeline execution

Caption

Interaction between the different shader stages in the ray tracing pipeline

Within the ray tracing pipeline, a [pipeline trace ray](#) instruction **can** be called to perform a [ray traversal](#) that invokes the various ray tracing shader stages during its execution. The relationship between the ray tracing pipeline object and the geometries present in the acceleration structure traversed is passed into the ray tracing command in a [VkBuffer](#) object known as a *shader binding table*. [OpExecuteCallableKHR](#) can also be used in ray tracing pipelines to invoke a [callable shader](#).

During execution, control alternates between scheduling and other operations. The scheduling functionality is implementation-specific and is responsible for workload execution. The shader stages are programmable. [Traversal](#), which refers to the process of traversing acceleration structures to find potential intersections of rays with geometry, is fixed function.

The programmable portions of the pipeline are exposed in a single-ray programming model, with each invocation handling one ray at a time. Memory operations **can** be synchronized using standard memory barriers. The [Workgroup](#) scope and variables with a storage class of [Workgroup](#) **must** not be used in the ray tracing pipeline.

36.1. Shader Call Instructions

A *shader call* is an instruction which **may** cause execution to continue elsewhere by creating one or more invocations that execute a different shader stage.

The following table lists all shader call instructions and which stages each one **can** directly call.

Instruction	Intersection	Any-Hit	Closest Hit	Miss	Callable
OpTraceRayKHR	X	X	X	X	
OpReportIntersectionKHR		X			
OpExecuteCallableKHR					X

Pipeline trace ray instructions **can** be used recursively; invoked shaders **can** themselves execute pipeline trace ray instructions, to a maximum depth defined by the `maxRayRecursionDepth` limit.

Shaders directly invoked from the API always have a recursion depth of 0; each shader executed by a pipeline trace ray instruction has a recursion depth one higher than the recursion depth of the shader which invoked it. Applications **must** not invoke a shader with a recursion depth greater than the value of `maxPipelineRayRecursionDepth` specified in the pipeline.

There is no explicit recursion limit for other shader call instructions which may recurse (e.g. `OpExecuteCallableKHR`) but there is an upper bound determined by the [stack size](#).

An *invocation repack instruction* is a ray tracing instruction where the implementation **may** change the set of invocations that are executing. When a repack instruction is encountered, the invocation is suspended and a new invocation begins and executes the instruction. After executing the repack instruction (which **may** result in other ray tracing shader stages executing) the new invocation ends and the original invocation is resumed, but it **may** be resumed in a different subgroup or at a different `SubgroupLocalInvocationId` within the same subgroup. When a subset of invocations in a subgroup execute the invocation repack instruction, those that do not execute it remain in the same subgroup at the same `SubgroupLocalInvocationId`.

The `OpTraceRayKHR`, `OpReportIntersectionKHR`, and `OpExecuteCallableKHR` instructions are invocation repack instructions.

When a ray tracing shader executes a dynamic instance of an invocation repack instruction which results in another ray tracing shader being invoked, their instructions are related by [shader-call-order](#).

For ray tracing invocations that are [shader-call-related](#):

- [memory operations](#) on `StorageBuffer`, `Image`, and `ShaderRecordBufferKHR` storage classes **can** be synchronized using the `ShaderCallKHR` scope.
- the `CallableDataKHR`, `IncomingCallableDataKHR`, `RayPayloadKHR`, `HitAttributeKHR`, and `IncomingRayPayloadKHR` storage classes are [system-synchronized](#) and no application availability and visibility operations are required.
- memory operations within a single invocation before and after the shader call instruction are ordered by [program-order](#) and do not require explicit synchronization.

36.2. Ray Tracing Commands

Ray tracing commands provoke work in the ray tracing pipeline. Ray tracing commands are recorded into a command buffer and when executed by a queue will produce work that executes

according to the currently bound ray tracing pipeline. A ray tracing pipeline **must** be bound to a command buffer before any ray tracing commands are recorded in that command buffer.

To dispatch ray tracing use:

```
// Provided by VK_KHR_ray_tracing_pipeline
void vkCmdTraceRaysKHR(
    VkCommandBuffer                commandBuffer,
    const VkStridedDeviceAddressRegionKHR* pRaygenShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR* pMissShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR* pHitShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR* pCallableShaderBindingTable,
    uint32_t                        width,
    uint32_t                        height,
    uint32_t                        depth);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pRaygenShaderBindingTable` is a `VkStridedDeviceAddressRegionKHR` that holds the shader binding table data for the ray generation shader stage.
- `pMissShaderBindingTable` is a `VkStridedDeviceAddressRegionKHR` that holds the shader binding table data for the miss shader stage.
- `pHitShaderBindingTable` is a `VkStridedDeviceAddressRegionKHR` that holds the shader binding table data for the hit shader stage.
- `pCallableShaderBindingTable` is a `VkStridedDeviceAddressRegionKHR` that holds the shader binding table data for the callable shader stage.
- `width` is the width of the ray trace query dimensions.
- `height` is height of the ray trace query dimensions.
- `depth` is depth of the ray trace query dimensions.

When the command is executed, a ray generation group of `width × height × depth` rays is assembled.

Valid Usage

- VUID-vkCmdTraceRaysKHR-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdTraceRaysKHR-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdTraceRaysKHR-unnormalizedCoordinates-09635

If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1

- VUID-vkCmdTraceRaysKHR-unnormalizedCoordinates-09636

If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`

- VUID-vkCmdTraceRaysKHR-None-06479

If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`

- VUID-vkCmdTraceRaysKHR-None-02691

If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`

- VUID-vkCmdTraceRaysKHR-None-07888

If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`

- VUID-vkCmdTraceRaysKHR-None-02692

If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`

- VUID-vkCmdTraceRaysKHR-None-02693

If the `VK_EXT_filter_cubic` extension is not enabled and any `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, it **must** not have a `VkImageViewType` of `VK_IMAGE_VIEW_TYPE_3D`, `VK_IMAGE_VIEW_TYPE_CUBE`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`

- VUID-vkCmdTraceRaysKHR-filterCubic-02694

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdTraceRaysKHR-filterCubicMinmax-02695

Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` with a reduction mode of either `VK_SAMPLER_REDUCTION_MODE_MIN` or `VK_SAMPLER_REDUCTION_MODE_MAX` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering together with minmax filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`

- VUID-vkCmdTraceRaysKHR-OpTypeImage-07027

For any `VkImageView` being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`

- VUID-vkCmdTraceRaysKHR-OpTypeImage-07028

For any `VkImageView` being read as a storage image where the image format field of the

`OpTypeImage` is `Unknown`, the view's `format` features **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`

- VUID-vkCmdTraceRaysKHR-OpTypeImage-07029
For any `VkBufferView` being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's `buffer` features **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`
- VUID-vkCmdTraceRaysKHR-OpTypeImage-07030
Any `VkBufferView` being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown` then the view's `buffer` features **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`
- VUID-vkCmdTraceRaysKHR-None-08600
For each set n that is statically used by a `bound shader`, a descriptor set **must** have been bound to n at the same pipeline bind point, with a `VkPipelineLayout` that is compatible for set n , with the `VkPipelineLayout` used to create the current `VkPipeline` or the `VkDescriptorSetLayout` array used to create the current `VkShaderEXT`, as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdTraceRaysKHR-None-08601
For each push constant that is statically used by a `bound shader`, a push constant value **must** have been set for the same pipeline bind point, with a `VkPipelineLayout` that is compatible for push constants, with the `VkPipelineLayout` used to create the current `VkPipeline` or the `VkDescriptorSetLayout` array used to create the current `VkShaderEXT`, as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdTraceRaysKHR-None-10068
For each array of resources that is used by a `bound shader`, the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command
- VUID-vkCmdTraceRaysKHR-maintenance4-08602
If the `maintenance4` feature is not enabled, then for each push constant that is statically used by a `bound shader`, a push constant value **must** have been set for the same pipeline bind point, with a `VkPipelineLayout` that is compatible for push constants, with the `VkPipelineLayout` used to create the current `VkPipeline` or the `VkDescriptorSetLayout` and `VkPushConstantRange` arrays used to create the current `VkShaderEXT`, as described in [Pipeline Layout Compatibility](#)
- VUID-vkCmdTraceRaysKHR-None-08114
Descriptors in each bound descriptor set, specified via `vkCmdBindDescriptorSets`, **must** be valid as described by [descriptor validity](#) if they are statically used by a `bound shader`
- VUID-vkCmdTraceRaysKHR-None-08606
If the `shaderObject` feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command
- VUID-vkCmdTraceRaysKHR-None-08608
If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the `VkPipeline` object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdTraceRaysKHR-None-08609
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage
- VUID-vkCmdTraceRaysKHR-None-08610
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage
- VUID-vkCmdTraceRaysKHR-None-08611
If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage
- VUID-vkCmdTraceRaysKHR-None-08607
If the [shaderObject](#) feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and [VK_NULL_HANDLE](#) shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command
- VUID-vkCmdTraceRaysKHR-uniformBuffers-06935
If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysKHR-None-08612
If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysKHR-storageBuffers-06936
If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysKHR-None-08613
If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point
- VUID-vkCmdTraceRaysKHR-None-06550

If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions

- VUID-vkCmdTraceRaysKHR-ConstOffset-06551

If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands

- VUID-vkCmdTraceRaysKHR-viewType-07752

If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdTraceRaysKHR-format-07753

If a [VkImageView](#) is accessed as a result of this command, then the [numeric type](#) of the image view's [format](#) and the [Sampled Type](#) operand of the [OpTypeImage](#) **must** match

- VUID-vkCmdTraceRaysKHR-OpImageWrite-08795

If a [VkImageView](#) created with a format other than [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdTraceRaysKHR-OpImageWrite-08796

If a [VkImageView](#) created with the format [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have four components

- VUID-vkCmdTraceRaysKHR-OpImageWrite-04469

If a [VkBufferView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdTraceRaysKHR-SampledType-04470

If a [VkImageView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64

- VUID-vkCmdTraceRaysKHR-SampledType-04471

If a [VkImageView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32

- VUID-vkCmdTraceRaysKHR-SampledType-04472

If a [VkBufferView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64

- VUID-vkCmdTraceRaysKHR-SampledType-04473

If a [VkBufferView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32

- VUID-vkCmdTraceRaysKHR-sparseImageInt64Atomics-04474

If the [sparseImageInt64Atomics](#) feature is not enabled, [VkImage](#) objects created with the

`VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command

- VUID-vkCmdTraceRaysKHR-sparseImageInt64Atomics-04475

If the `sparseImageInt64Atomics` feature is not enabled, `VkBuffer` objects created with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command

- VUID-vkCmdTraceRaysKHR-None-07288

Any shader invocation executed by this command **must** `terminate`

- VUID-vkCmdTraceRaysKHR-None-09600

If a descriptor with type equal to any of `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdTraceRaysKHR-None-03429

Any shader group handle referenced by this call **must** have been queried from the currently bound ray tracing pipeline

- VUID-vkCmdTraceRaysKHR-None-09458

If the bound ray tracing pipeline state was created with the `VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR` dynamic state enabled then `vkCmdSetRayTracingPipelineStackSizeKHR` **must** have been called in the current command buffer prior to this trace command

- VUID-vkCmdTraceRaysKHR-maxPipelineRayRecursionDepth-03679

This command **must** not cause a shader call instruction to be executed from a shader invocation with a `recursion depth` greater than the value of `maxPipelineRayRecursionDepth` used to create the bound ray tracing pipeline

- VUID-vkCmdTraceRaysKHR-commandBuffer-03635

`commandBuffer` **must** not be a protected command buffer

- VUID-vkCmdTraceRaysKHR-size-04023

The `size` member of `pRayGenShaderBindingTable` **must** be equal to its `stride` member

- VUID-vkCmdTraceRaysKHR-pRayGenShaderBindingTable-03680

If the buffer from which `pRayGenShaderBindingTable->deviceAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdTraceRaysKHR-pRayGenShaderBindingTable-03681

The buffer from which the `pRayGenShaderBindingTable->deviceAddress` is queried **must** have been created with the `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` usage flag

- VUID-vkCmdTraceRaysKHR-pRayGenShaderBindingTable-03682

`pRayGenShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`

- VUID-vkCmdTraceRaysKHR-pMissShaderBindingTable-03683

If the buffer from which `pMissShaderBindingTable->deviceAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory`

object

- VUID-vkCmdTraceRaysKHR-pMissShaderBindingTable-03684
The buffer from which the `pMissShaderBindingTable->deviceAddress` is queried **must** have been created with the `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` usage flag
- VUID-vkCmdTraceRaysKHR-pMissShaderBindingTable-03685
`pMissShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-vkCmdTraceRaysKHR-stride-03686
`pMissShaderBindingTable->stride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment`
- VUID-vkCmdTraceRaysKHR-stride-04029
`pMissShaderBindingTable->stride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride`
- VUID-vkCmdTraceRaysKHR-pHitShaderBindingTable-03687
If the buffer from which `pHitShaderBindingTable->deviceAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdTraceRaysKHR-pHitShaderBindingTable-03688
The buffer from which the `pHitShaderBindingTable->deviceAddress` is queried **must** have been created with the `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` usage flag
- VUID-vkCmdTraceRaysKHR-pHitShaderBindingTable-03689
`pHitShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-vkCmdTraceRaysKHR-stride-03690
`pHitShaderBindingTable->stride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment`
- VUID-vkCmdTraceRaysKHR-stride-04035
`pHitShaderBindingTable->stride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride`
- VUID-vkCmdTraceRaysKHR-pCallableShaderBindingTable-03691
If the buffer from which `pCallableShaderBindingTable->deviceAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdTraceRaysKHR-pCallableShaderBindingTable-03692
The buffer from which the `pCallableShaderBindingTable->deviceAddress` is queried **must** have been created with the `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` usage flag
- VUID-vkCmdTraceRaysKHR-pCallableShaderBindingTable-03693
`pCallableShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-vkCmdTraceRaysKHR-stride-03694
`pCallableShaderBindingTable->stride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment`

- VUID-vkCmdTraceRaysKHR-stride-04041
`pCallableShaderBindingTable->stride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride`
- VUID-vkCmdTraceRaysKHR-flags-03696
 If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR`, `pHitShaderBindingTable->deviceAddress` **must** not be zero
- VUID-vkCmdTraceRaysKHR-flags-03697
 If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR`, `pHitShaderBindingTable->deviceAddress` **must** not be zero
- VUID-vkCmdTraceRaysKHR-flags-03511
 If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR`, the shader group handle identified by `pMissShaderBindingTable->deviceAddress` **must** not be zero
- VUID-vkCmdTraceRaysKHR-flags-03512
 If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR`, entries in the table identified by `pHitShaderBindingTable->deviceAddress` accessed as a result of this command in order to execute an any-hit shader **must** not be zero
- VUID-vkCmdTraceRaysKHR-flags-03513
 If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR`, entries in the table identified by `pHitShaderBindingTable->deviceAddress` accessed as a result of this command in order to execute a closest hit shader **must** not be zero
- VUID-vkCmdTraceRaysKHR-flags-03514
 If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR`, entries in the table identified by `pHitShaderBindingTable->deviceAddress` accessed as a result of this command in order to execute an intersection shader **must** not be zero
- VUID-vkCmdTraceRaysKHR-pHitShaderBindingTable-04735
 Any non-zero hit shader group entries in the table identified by `pHitShaderBindingTable->deviceAddress` accessed by this call from a geometry with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR` **must** have been created with `VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR`
- VUID-vkCmdTraceRaysKHR-pHitShaderBindingTable-04736
 Any non-zero hit shader group entries in the table identified by `pHitShaderBindingTable->deviceAddress` accessed by this call from a geometry with a `geometryType` of `VK_GEOMETRY_TYPE_AABBS_KHR` **must** have been created with `VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR`
- VUID-vkCmdTraceRaysKHR-width-03638
`width` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[0]`

- VUID-vkCmdTraceRaysKHR-height-03639
height **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1]` × `VkPhysicalDeviceLimits::maxComputeWorkGroupSize[1]`
- VUID-vkCmdTraceRaysKHR-depth-03640
depth **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2]` × `VkPhysicalDeviceLimits::maxComputeWorkGroupSize[2]`
- VUID-vkCmdTraceRaysKHR-width-03641
width × height × depth **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxRayDispatchInvocationCount`

Valid Usage (Implicit)

- VUID-vkCmdTraceRaysKHR-commandBuffer-parameter
commandBuffer **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdTraceRaysKHR-pRaygenShaderBindingTable-parameter
pRaygenShaderBindingTable **must** be a valid pointer to a valid `VkStridedDeviceAddressRegionKHR` structure
- VUID-vkCmdTraceRaysKHR-pMissShaderBindingTable-parameter
pMissShaderBindingTable **must** be a valid pointer to a valid `VkStridedDeviceAddressRegionKHR` structure
- VUID-vkCmdTraceRaysKHR-pHitShaderBindingTable-parameter
pHitShaderBindingTable **must** be a valid pointer to a valid `VkStridedDeviceAddressRegionKHR` structure
- VUID-vkCmdTraceRaysKHR-pCallableShaderBindingTable-parameter
pCallableShaderBindingTable **must** be a valid pointer to a valid `VkStridedDeviceAddressRegionKHR` structure
- VUID-vkCmdTraceRaysKHR-commandBuffer-recording
commandBuffer **must** be in the `recording` state
- VUID-vkCmdTraceRaysKHR-commandBuffer-cmdpool
The `VkCommandPool` that commandBuffer was allocated from **must** support compute operations
- VUID-vkCmdTraceRaysKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdTraceRaysKHR-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to commandBuffer **must** be externally synchronized
- Host access to the `VkCommandPool` that commandBuffer was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

To dispatch ray tracing, with some parameters sourced on the device, use:

```
// Provided by VK_KHR_ray_tracing_pipeline
void vkCmdTraceRaysIndirectKHR(
    VkCommandBuffer                commandBuffer,
    const VkStridedDeviceAddressRegionKHR* pRaygenShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR* pMissShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR* pHitShaderBindingTable,
    const VkStridedDeviceAddressRegionKHR* pCallableShaderBindingTable,
    VkDeviceAddress                 indirectDeviceAddress);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `pRaygenShaderBindingTable` is a `VkStridedDeviceAddressRegionKHR` that holds the shader binding table data for the ray generation shader stage.
- `pMissShaderBindingTable` is a `VkStridedDeviceAddressRegionKHR` that holds the shader binding table data for the miss shader stage.
- `pHitShaderBindingTable` is a `VkStridedDeviceAddressRegionKHR` that holds the shader binding table data for the hit shader stage.
- `pCallableShaderBindingTable` is a `VkStridedDeviceAddressRegionKHR` that holds the shader binding table data for the callable shader stage.
- `indirectDeviceAddress` is a buffer device address which is a pointer to a `VkTraceRaysIndirectCommandKHR` structure containing the trace ray parameters.

`vkCmdTraceRaysIndirectKHR` behaves similarly to `vkCmdTraceRaysKHR` except that the ray trace query dimensions are read by the device from `indirectDeviceAddress` during execution.

Valid Usage

- VUID-vkCmdTraceRaysIndirectKHR-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format` `features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdTraceRaysIndirectKHR-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this

command, then the image view's **format features must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT**

- VUID-vkCmdTraceRaysIndirectKHR-unnormalizedCoordinates-09635

If a **VkSampler** created with **unnormalizedCoordinates** equal to **VK_TRUE** is used to sample a **VkImageView** as a result of this command, then the image view's **levelCount** and **layerCount must** be 1

- VUID-vkCmdTraceRaysIndirectKHR-unnormalizedCoordinates-09636

If a **VkSampler** created with **unnormalizedCoordinates** equal to **VK_TRUE** is used to sample a **VkImageView** as a result of this command, then the image view's **viewType must** be **VK_IMAGE_VIEW_TYPE_1D** or **VK_IMAGE_VIEW_TYPE_2D**

- VUID-vkCmdTraceRaysIndirectKHR-None-06479

If a **VkImageView** is sampled with **depth comparison**, the image view's **format features must** contain **VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT**

- VUID-vkCmdTraceRaysIndirectKHR-None-02691

If a **VkImageView** is accessed using atomic operations as a result of this command, then the image view's **format features must** contain **VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT**

- VUID-vkCmdTraceRaysIndirectKHR-None-07888

If a **VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER** descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's **format features must** contain **VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT**

- VUID-vkCmdTraceRaysIndirectKHR-None-02692

If a **VkImageView** is sampled with **VK_FILTER_CUBIC_EXT** as a result of this command, then the image view's **format features must** contain **VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT**

- VUID-vkCmdTraceRaysIndirectKHR-None-02693

If the **VK_EXT_filter_cubic** extension is not enabled and any **VkImageView** is sampled with **VK_FILTER_CUBIC_EXT** as a result of this command, it **must** not have a **VkImageViewType** of **VK_IMAGE_VIEW_TYPE_3D**, **VK_IMAGE_VIEW_TYPE_CUBE**, or **VK_IMAGE_VIEW_TYPE_CUBE_ARRAY**

- VUID-vkCmdTraceRaysIndirectKHR-filterCubic-02694

Any **VkImageView** being sampled with **VK_FILTER_CUBIC_EXT** as a result of this command **must** have a **VkImageViewType** and format that supports cubic filtering, as specified by **VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic** returned by **vkGetPhysicalDeviceImageFormatProperties2KHR**

- VUID-vkCmdTraceRaysIndirectKHR-filterCubicMinmax-02695

Any **VkImageView** being sampled with **VK_FILTER_CUBIC_EXT** with a reduction mode of either **VK_SAMPLER_REDUCTION_MODE_MIN** or **VK_SAMPLER_REDUCTION_MODE_MAX** as a result of this command **must** have a **VkImageViewType** and format that supports cubic filtering together with minmax filtering, as specified by **VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax** returned by **vkGetPhysicalDeviceImageFormatProperties2KHR**

- VUID-vkCmdTraceRaysIndirectKHR-OpTypeImage-07027

For any **VkImageView** being written as a storage image where the image format field of the **OpTypeImage** is **Unknown**, the view's **format features must** contain

VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT

- VUID-vkCmdTraceRaysIndirectKHR-OpTypeImage-07028

For any [VkImageView](#) being read as a storage image where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [format features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdTraceRaysIndirectKHR-OpTypeImage-07029

For any [VkBufferView](#) being written as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#), the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdTraceRaysIndirectKHR-OpTypeImage-07030

Any [VkBufferView](#) being read as a storage texel buffer where the image format field of the [OpTypeImage](#) is [Unknown](#) then the view's [buffer features](#) **must** contain [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT](#)

- VUID-vkCmdTraceRaysIndirectKHR-None-08600

For each set n that is statically used by a [bound shader](#), a descriptor set **must** have been bound to n at the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for set n , with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdTraceRaysIndirectKHR-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdTraceRaysIndirectKHR-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdTraceRaysIndirectKHR-maintenance4-08602

If the [maintenance4](#) feature is not enabled, then for each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) and [VkPushConstantRange](#) arrays used to create the current [VkShaderEXT](#), as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdTraceRaysIndirectKHR-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdTraceRaysIndirectKHR-None-08606

If the [shaderObject](#) feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdTraceRaysIndirectKHR-None-08608

If a pipeline is bound to the pipeline bind point used by this command, there **must** not

have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdTraceRaysIndirectKHR-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdTraceRaysIndirectKHR-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage

- VUID-vkCmdTraceRaysIndirectKHR-None-08611

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdTraceRaysIndirectKHR-None-08607

If the [shaderObject](#) feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and [VK_NULL_HANDLE](#) shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command

- VUID-vkCmdTraceRaysIndirectKHR-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdTraceRaysIndirectKHR-None-08612

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdTraceRaysIndirectKHR-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdTraceRaysIndirectKHR-None-08613

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer,

it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdTraceRaysIndirectKHR-None-06550

If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions

- VUID-vkCmdTraceRaysIndirectKHR-ConstOffset-06551

If a [bound shader](#) accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands

- VUID-vkCmdTraceRaysIndirectKHR-viewType-07752

If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdTraceRaysIndirectKHR-format-07753

If a [VkImageView](#) is accessed as a result of this command, then the [numeric type](#) of the image view's [format](#) and the [Sampled Type](#) operand of the [OpTypeImage](#) **must** match

- VUID-vkCmdTraceRaysIndirectKHR-OpImageWrite-08795

If a [VkImageView](#) created with a format other than [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the image view's format

- VUID-vkCmdTraceRaysIndirectKHR-OpImageWrite-08796

If a [VkImageView](#) created with the format [VK_FORMAT_A8_UNORM_KHR](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have four components

- VUID-vkCmdTraceRaysIndirectKHR-OpImageWrite-04469

If a [VkBufferView](#) is accessed using [OpImageWrite](#) as a result of this command, then the [Type](#) of the [Texel](#) operand of that instruction **must** have at least as many components as the buffer view's format

- VUID-vkCmdTraceRaysIndirectKHR-SampledType-04470

If a [VkImageView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64

- VUID-vkCmdTraceRaysIndirectKHR-SampledType-04471

If a [VkImageView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 32

- VUID-vkCmdTraceRaysIndirectKHR-SampledType-04472

If a [VkBufferView](#) with a [VkFormat](#) that has a 64-bit component width is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that instruction **must** have a [Width](#) of 64

- VUID-vkCmdTraceRaysIndirectKHR-SampledType-04473

If a [VkBufferView](#) with a [VkFormat](#) that has a component width less than 64-bit is accessed as a result of this command, the [SampledType](#) of the [OpTypeImage](#) operand of that

instruction **must** have a **Width** of 32

- VUID-vkCmdTraceRaysIndirectKHR-sparseImageInt64Atomics-04474

If the **sparseImageInt64Atomics** feature is not enabled, **VkImage** objects created with the **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command

- VUID-vkCmdTraceRaysIndirectKHR-sparseImageInt64Atomics-04475

If the **sparseImageInt64Atomics** feature is not enabled, **VkBuffer** objects created with the **VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT** flag **must** not be accessed by atomic instructions through an **OpTypeImage** with a **SampledType** with a **Width** of 64 by this command

- VUID-vkCmdTraceRaysIndirectKHR-None-07288

Any shader invocation executed by this command **must** **terminate**

- VUID-vkCmdTraceRaysIndirectKHR-None-09600

If a descriptor with type equal to any of **VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE**, **VK_DESCRIPTOR_TYPE_STORAGE_IMAGE**, or **VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT** is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdTraceRaysIndirectKHR-None-03429

Any shader group handle referenced by this call **must** have been queried from the currently bound ray tracing pipeline

- VUID-vkCmdTraceRaysIndirectKHR-None-09458

If the bound ray tracing pipeline state was created with the **VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR** dynamic state enabled then **vkCmdSetRayTracingPipelineStackSizeKHR** **must** have been called in the current command buffer prior to this trace command

- VUID-vkCmdTraceRaysIndirectKHR-maxPipelineRayRecursionDepth-03679

This command **must** not cause a shader call instruction to be executed from a shader invocation with a **recursion depth** greater than the value of **maxPipelineRayRecursionDepth** used to create the bound ray tracing pipeline

- VUID-vkCmdTraceRaysIndirectKHR-commandBuffer-03635

commandBuffer **must** not be a protected command buffer

- VUID-vkCmdTraceRaysIndirectKHR-size-04023

The **size** member of **pRayGenShaderBindingTable** **must** be equal to its **stride** member

- VUID-vkCmdTraceRaysIndirectKHR-pRayGenShaderBindingTable-03680

If the buffer from which **pRayGenShaderBindingTable->deviceAddress** was queried is non-sparse then it **must** be bound completely and contiguously to a single **VkDeviceMemory** object

- VUID-vkCmdTraceRaysIndirectKHR-pRayGenShaderBindingTable-03681

The buffer from which the **pRayGenShaderBindingTable->deviceAddress** is queried **must** have been created with the **VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR** usage flag

- VUID-vkCmdTraceRaysIndirectKHR-pRayGenShaderBindingTable-03682

pRayGenShaderBindingTable->deviceAddress **must** be a multiple of **VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment**

- VUID-vkCmdTraceRaysIndirectKHR-pMissShaderBindingTable-03683
If the buffer from which `pMissShaderBindingTable->deviceAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdTraceRaysIndirectKHR-pMissShaderBindingTable-03684
The buffer from which the `pMissShaderBindingTable->deviceAddress` is queried **must** have been created with the `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` usage flag
- VUID-vkCmdTraceRaysIndirectKHR-pMissShaderBindingTable-03685
`pMissShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-stride-03686
`pMissShaderBindingTable->stride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-stride-04029
`pMissShaderBindingTable->stride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride`
- VUID-vkCmdTraceRaysIndirectKHR-pHitShaderBindingTable-03687
If the buffer from which `pHitShaderBindingTable->deviceAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdTraceRaysIndirectKHR-pHitShaderBindingTable-03688
The buffer from which the `pHitShaderBindingTable->deviceAddress` is queried **must** have been created with the `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` usage flag
- VUID-vkCmdTraceRaysIndirectKHR-pHitShaderBindingTable-03689
`pHitShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-stride-03690
`pHitShaderBindingTable->stride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-stride-04035
`pHitShaderBindingTable->stride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride`
- VUID-vkCmdTraceRaysIndirectKHR-pCallableShaderBindingTable-03691
If the buffer from which `pCallableShaderBindingTable->deviceAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdTraceRaysIndirectKHR-pCallableShaderBindingTable-03692
The buffer from which the `pCallableShaderBindingTable->deviceAddress` is queried **must** have been created with the `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` usage flag
- VUID-vkCmdTraceRaysIndirectKHR-pCallableShaderBindingTable-03693
`pCallableShaderBindingTable->deviceAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-vkCmdTraceRaysIndirectKHR-stride-03694

`pCallableShaderBindingTable->stride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment`

- VUID-vkCmdTraceRaysIndirectKHR-stride-04041

`pCallableShaderBindingTable->stride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride`

- VUID-vkCmdTraceRaysIndirectKHR-flags-03696

If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR`, `pHitShaderBindingTable->deviceAddress` **must** not be zero

- VUID-vkCmdTraceRaysIndirectKHR-flags-03697

If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR`, `pHitShaderBindingTable->deviceAddress` **must** not be zero

- VUID-vkCmdTraceRaysIndirectKHR-flags-03511

If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR`, the shader group handle identified by `pMissShaderBindingTable->deviceAddress` **must** not be zero

- VUID-vkCmdTraceRaysIndirectKHR-flags-03512

If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR`, entries in the table identified by `pHitShaderBindingTable->deviceAddress` accessed as a result of this command in order to execute an any-hit shader **must** not be zero

- VUID-vkCmdTraceRaysIndirectKHR-flags-03513

If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR`, entries in the table identified by `pHitShaderBindingTable->deviceAddress` accessed as a result of this command in order to execute a closest hit shader **must** not be zero

- VUID-vkCmdTraceRaysIndirectKHR-flags-03514

If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR`, entries in the table identified by `pHitShaderBindingTable->deviceAddress` accessed as a result of this command in order to execute an intersection shader **must** not be zero

- VUID-vkCmdTraceRaysIndirectKHR-pHitShaderBindingTable-04735

Any non-zero hit shader group entries in the table identified by `pHitShaderBindingTable->deviceAddress` accessed by this call from a geometry with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR` **must** have been created with `VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR`

- VUID-vkCmdTraceRaysIndirectKHR-pHitShaderBindingTable-04736

Any non-zero hit shader group entries in the table identified by `pHitShaderBindingTable->deviceAddress` accessed by this call from a geometry with a `geometryType` of `VK_GEOMETRY_TYPE_AABBS_KHR` **must** have been created with `VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR`

- VUID-vkCmdTraceRaysIndirectKHR-indirectDeviceAddress-03632

If the buffer from which `indirectDeviceAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-vkCmdTraceRaysIndirectKHR-indirectDeviceAddress-03633
The buffer from which `indirectDeviceAddress` was queried **must** have been created with the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set
- VUID-vkCmdTraceRaysIndirectKHR-indirectDeviceAddress-03634
`indirectDeviceAddress` **must** be a multiple of 4
- VUID-vkCmdTraceRaysIndirectKHR-indirectDeviceAddress-03636
All device addresses between `indirectDeviceAddress` and `indirectDeviceAddress + sizeof(VkTraceRaysIndirectCommandKHR) - 1` **must** be in the buffer device address range of the same buffer
- VUID-vkCmdTraceRaysIndirectKHR-rayTracingPipelineTraceRaysIndirect-03637
The `rayTracingPipelineTraceRaysIndirect` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdTraceRaysIndirectKHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdTraceRaysIndirectKHR-pRaygenShaderBindingTable-parameter
`pRaygenShaderBindingTable` **must** be a valid pointer to a valid `VkStridedDeviceAddressRegionKHR` structure
- VUID-vkCmdTraceRaysIndirectKHR-pMissShaderBindingTable-parameter
`pMissShaderBindingTable` **must** be a valid pointer to a valid `VkStridedDeviceAddressRegionKHR` structure
- VUID-vkCmdTraceRaysIndirectKHR-pHitShaderBindingTable-parameter
`pHitShaderBindingTable` **must** be a valid pointer to a valid `VkStridedDeviceAddressRegionKHR` structure
- VUID-vkCmdTraceRaysIndirectKHR-pCallableShaderBindingTable-parameter
`pCallableShaderBindingTable` **must** be a valid pointer to a valid `VkStridedDeviceAddressRegionKHR` structure
- VUID-vkCmdTraceRaysIndirectKHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdTraceRaysIndirectKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdTraceRaysIndirectKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdTraceRaysIndirectKHR-vidcoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

The `VkTraceRaysIndirectCommandKHR` structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkTraceRaysIndirectCommandKHR {
    uint32_t    width;
    uint32_t    height;
    uint32_t    depth;
} VkTraceRaysIndirectCommandKHR;
```

- `width` is the width of the ray trace query dimensions.
- `height` is height of the ray trace query dimensions.
- `depth` is depth of the ray trace query dimensions.

The members of `VkTraceRaysIndirectCommandKHR` have the same meaning as the similarly named parameters of `vkCmdTraceRaysKHR`.

Valid Usage

- VUID-VkTraceRaysIndirectCommandKHR-width-03638
`width` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[0]`
- VUID-VkTraceRaysIndirectCommandKHR-height-03639
`height` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[1]`
- VUID-VkTraceRaysIndirectCommandKHR-depth-03640
`depth` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[2]`
- VUID-VkTraceRaysIndirectCommandKHR-width-03641
`width × height × depth` **must** be less than or equal to

To dispatch ray tracing, with some parameters sourced on the device, use:

```
// Provided by VK_KHR_ray_tracing_maintenance1 with VK_KHR_ray_tracing_pipeline
void vkCmdTraceRaysIndirect2KHR(
    VkCommandBuffer          commandBuffer,
    VkDeviceAddress          indirectDeviceAddress);
```

- `commandBuffer` is the command buffer into which the command will be recorded.
- `indirectDeviceAddress` is a buffer device address which is a pointer to a `VkTraceRaysIndirectCommand2KHR` structure containing the trace ray parameters.

`vkCmdTraceRaysIndirect2KHR` behaves similarly to `vkCmdTraceRaysIndirectKHR` except that shader binding table parameters as well as dispatch dimensions are read by the device from `indirectDeviceAddress` during execution.

Valid Usage

- VUID-vkCmdTraceRaysIndirect2KHR-magFilter-04553
If a `VkSampler` created with `magFilter` or `minFilter` equal to `VK_FILTER_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdTraceRaysIndirect2KHR-mipmapMode-04770
If a `VkSampler` created with `mipmapMode` equal to `VK_SAMPLER_MIPMAP_MODE_LINEAR`, and `compareEnable` equal to `VK_FALSE` is used to sample a `VkImageView` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT`
- VUID-vkCmdTraceRaysIndirect2KHR-unnormalizedCoordinates-09635
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `levelCount` and `layerCount` **must** be 1
- VUID-vkCmdTraceRaysIndirect2KHR-unnormalizedCoordinates-09636
If a `VkSampler` created with `unnormalizedCoordinates` equal to `VK_TRUE` is used to sample a `VkImageView` as a result of this command, then the image view's `viewType` **must** be `VK_IMAGE_VIEW_TYPE_1D` or `VK_IMAGE_VIEW_TYPE_2D`
- VUID-vkCmdTraceRaysIndirect2KHR-None-06479
If a `VkImageView` is sampled with `depth comparison`, the image view's `format features` **must** contain `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`
- VUID-vkCmdTraceRaysIndirect2KHR-None-02691
If a `VkImageView` is accessed using atomic operations as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT`

- VUID-vkCmdTraceRaysIndirect2KHR-None-07888
If a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor is accessed using atomic operations as a result of this command, then the storage texel buffer's `format features` **must** contain `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT`
- VUID-vkCmdTraceRaysIndirect2KHR-None-02692
If a `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, then the image view's `format features` **must** contain `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT`
- VUID-vkCmdTraceRaysIndirect2KHR-None-02693
If the `VK_EXT_filter_cubic` extension is not enabled and any `VkImageView` is sampled with `VK_FILTER_CUBIC_EXT` as a result of this command, it **must** not have a `VkImageViewType` of `VK_IMAGE_VIEW_TYPE_3D`, `VK_IMAGE_VIEW_TYPE_CUBE`, or `VK_IMAGE_VIEW_TYPE_CUBE_ARRAY`
- VUID-vkCmdTraceRaysIndirect2KHR-filterCubic-02694
Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubic` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`
- VUID-vkCmdTraceRaysIndirect2KHR-filterCubicMinmax-02695
Any `VkImageView` being sampled with `VK_FILTER_CUBIC_EXT` with a reduction mode of either `VK_SAMPLER_REDUCTION_MODE_MIN` or `VK_SAMPLER_REDUCTION_MODE_MAX` as a result of this command **must** have a `VkImageViewType` and format that supports cubic filtering together with minmax filtering, as specified by `VkFilterCubicImageViewImageFormatPropertiesEXT::filterCubicMinmax` returned by `vkGetPhysicalDeviceImageFormatProperties2KHR`
- VUID-vkCmdTraceRaysIndirect2KHR-OpTypeImage-07027
For any `VkImageView` being written as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`
- VUID-vkCmdTraceRaysIndirect2KHR-OpTypeImage-07028
For any `VkImageView` being read as a storage image where the image format field of the `OpTypeImage` is `Unknown`, the view's `format features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`
- VUID-vkCmdTraceRaysIndirect2KHR-OpTypeImage-07029
For any `VkBufferView` being written as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown`, the view's `buffer features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT`
- VUID-vkCmdTraceRaysIndirect2KHR-OpTypeImage-07030
Any `VkBufferView` being read as a storage texel buffer where the image format field of the `OpTypeImage` is `Unknown` then the view's `buffer features` **must** contain `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT`
- VUID-vkCmdTraceRaysIndirect2KHR-None-08600
For each set n that is statically used by a `bound shader`, a descriptor set **must** have been bound to n at the same pipeline bind point, with a `VkPipelineLayout` that is compatible for set n , with the `VkPipelineLayout` used to create the current `VkPipeline` or the

[VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#) , as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdTraceRaysIndirect2KHR-None-08601

For each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) array used to create the current [VkShaderEXT](#) , as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdTraceRaysIndirect2KHR-None-10068

For each array of resources that is used by a [bound shader](#), the indices used to access members of the array **must** be less than the descriptor count for the identified binding in the descriptor sets used by this command

- VUID-vkCmdTraceRaysIndirect2KHR-maintenance4-08602

If the [maintenance4](#) feature is not enabled, then for each push constant that is statically used by a [bound shader](#), a push constant value **must** have been set for the same pipeline bind point, with a [VkPipelineLayout](#) that is compatible for push constants, with the [VkPipelineLayout](#) used to create the current [VkPipeline](#) or the [VkDescriptorSetLayout](#) and [VkPushConstantRange](#) arrays used to create the current [VkShaderEXT](#) , as described in [Pipeline Layout Compatibility](#)

- VUID-vkCmdTraceRaysIndirect2KHR-None-08114

Descriptors in each bound descriptor set, specified via [vkCmdBindDescriptorSets](#), **must** be valid as described by [descriptor validity](#) if they are statically used by a [bound shader](#)

- VUID-vkCmdTraceRaysIndirect2KHR-None-08606

If the [shaderObject](#) feature is not enabled, a valid pipeline **must** be bound to the pipeline bind point used by this command

- VUID-vkCmdTraceRaysIndirect2KHR-None-08608

If a pipeline is bound to the pipeline bind point used by this command, there **must** not have been any calls to dynamic state setting commands for any state specified statically in the [VkPipeline](#) object bound to the pipeline bind point used by this command, since that pipeline was bound

- VUID-vkCmdTraceRaysIndirect2KHR-None-08609

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used to sample from any [VkImage](#) with a [VkImageView](#) of the type [VK_IMAGE_VIEW_TYPE_3D](#), [VK_IMAGE_VIEW_TYPE_CUBE](#), [VK_IMAGE_VIEW_TYPE_1D_ARRAY](#), [VK_IMAGE_VIEW_TYPE_2D_ARRAY](#) or [VK_IMAGE_VIEW_TYPE_CUBE_ARRAY](#), in any shader stage

- VUID-vkCmdTraceRaysIndirect2KHR-None-08610

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions with [ImplicitLod](#), [Dref](#) or [Proj](#) in their name, in any shader stage

- VUID-vkCmdTraceRaysIndirect2KHR-None-08611

If the [VkPipeline](#) object bound to the pipeline bind point used by this command or any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a [VkSampler](#) object that uses unnormalized coordinates, that sampler **must** not be used with any of the SPIR-V [OpImageSample*](#) or [OpImageSparseSample*](#) instructions that includes a LOD bias or any offset values, in any shader stage

- VUID-vkCmdTraceRaysIndirect2KHR-None-08607

If the [shaderObject](#) feature is enabled, either a valid pipeline **must** be bound to the pipeline bind point used by this command, or a valid combination of valid and [VK_NULL_HANDLE](#) shader objects **must** be bound to every supported shader stage corresponding to the pipeline bind point used by this command

- VUID-vkCmdTraceRaysIndirect2KHR-uniformBuffers-06935

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a uniform buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdTraceRaysIndirect2KHR-None-08612

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a uniform buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdTraceRaysIndirect2KHR-storageBuffers-06936

If any stage of the [VkPipeline](#) object bound to the pipeline bind point used by this command accesses a storage buffer, and the [robustBufferAccess](#) feature is not enabled, that stage **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdTraceRaysIndirect2KHR-None-08613

If the [robustBufferAccess](#) feature is not enabled, and any [VkShaderEXT](#) bound to a stage corresponding to the pipeline bind point used by this command accesses a storage buffer, it **must** not access values outside of the range of the buffer as specified in the descriptor set bound to the same pipeline bind point

- VUID-vkCmdTraceRaysIndirect2KHR-None-06550

If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** only be used with [OpImageSample*](#) or [OpImageSparseSample*](#) instructions

- VUID-vkCmdTraceRaysIndirect2KHR-ConstOffset-06551

If a bound shader accesses a [VkSampler](#) or [VkImageView](#) object that enables [sampler Y'C_BC_R conversion](#), that object **must** not use the [ConstOffset](#) and [Offset](#) operands

- VUID-vkCmdTraceRaysIndirect2KHR-viewType-07752

If a [VkImageView](#) is accessed as a result of this command, then the image view's [viewType](#) **must** match the [Dim](#) operand of the [OpTypeImage](#) as described in [Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types](#)

- VUID-vkCmdTraceRaysIndirect2KHR-format-07753

If a [VkImageView](#) is accessed as a result of this command, then the [numeric type](#) of the image view's [format](#) and the [Sampled Type](#) operand of the [OpTypeImage](#) **must** match

- VUID-vkCmdTraceRaysIndirect2KHR-OpImageWrite-08795
If a `VkImageView` created with a format other than `VK_FORMAT_A8_UNORM_KHR` is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the image view's format
- VUID-vkCmdTraceRaysIndirect2KHR-OpImageWrite-08796
If a `VkImageView` created with the format `VK_FORMAT_A8_UNORM_KHR` is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have four components
- VUID-vkCmdTraceRaysIndirect2KHR-OpImageWrite-04469
If a `VkBufferView` is accessed using `OpImageWrite` as a result of this command, then the `Type` of the `Texel` operand of that instruction **must** have at least as many components as the buffer view's format
- VUID-vkCmdTraceRaysIndirect2KHR-SampledType-04470
If a `VkImageView` with a `VkFormat` that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdTraceRaysIndirect2KHR-SampledType-04471
If a `VkImageView` with a `VkFormat` that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdTraceRaysIndirect2KHR-SampledType-04472
If a `VkBufferView` with a `VkFormat` that has a 64-bit component width is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 64
- VUID-vkCmdTraceRaysIndirect2KHR-SampledType-04473
If a `VkBufferView` with a `VkFormat` that has a component width less than 64-bit is accessed as a result of this command, the `SampledType` of the `OpTypeImage` operand of that instruction **must** have a `Width` of 32
- VUID-vkCmdTraceRaysIndirect2KHR-sparseImageInt64Atomics-04474
If the `sparseImageInt64Atomics` feature is not enabled, `VkImage` objects created with the `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdTraceRaysIndirect2KHR-sparseImageInt64Atomics-04475
If the `sparseImageInt64Atomics` feature is not enabled, `VkBuffer` objects created with the `VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT` flag **must** not be accessed by atomic instructions through an `OpTypeImage` with a `SampledType` with a `Width` of 64 by this command
- VUID-vkCmdTraceRaysIndirect2KHR-None-07288
Any shader invocation executed by this command **must** `terminate`
- VUID-vkCmdTraceRaysIndirect2KHR-None-09600
If a descriptor with type equal to any of `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` is accessed as a result of this command, the image subresource identified by that descriptor **must** be in the image layout identified when the descriptor was written

- VUID-vkCmdTraceRaysIndirect2KHR-None-03429
Any shader group handle referenced by this call **must** have been queried from the currently bound ray tracing pipeline
- VUID-vkCmdTraceRaysIndirect2KHR-None-09458
If the bound ray tracing pipeline state was created with the `VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR` dynamic state enabled then `vkCmdSetRayTracingPipelineStackSizeKHR` **must** have been called in the current command buffer prior to this trace command
- VUID-vkCmdTraceRaysIndirect2KHR-maxPipelineRayRecursionDepth-03679
This command **must** not cause a shader call instruction to be executed from a shader invocation with a `recursion depth` greater than the value of `maxPipelineRayRecursionDepth` used to create the bound ray tracing pipeline
- VUID-vkCmdTraceRaysIndirect2KHR-commandBuffer-03635
`commandBuffer` **must** not be a protected command buffer
- VUID-vkCmdTraceRaysIndirect2KHR-indirectDeviceAddress-03632
If the buffer from which `indirectDeviceAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-vkCmdTraceRaysIndirect2KHR-indirectDeviceAddress-03633
The buffer from which `indirectDeviceAddress` was queried **must** have been created with the `VK_BUFFER_USAGE_INDIRECT_BUFFER_BIT` bit set
- VUID-vkCmdTraceRaysIndirect2KHR-indirectDeviceAddress-03634
`indirectDeviceAddress` **must** be a multiple of 4
- VUID-vkCmdTraceRaysIndirect2KHR-indirectDeviceAddress-03636
All device addresses between `indirectDeviceAddress` and `indirectDeviceAddress + sizeof(VkTraceRaysIndirectCommand2KHR) - 1` **must** be in the buffer device address range of the same buffer
- VUID-vkCmdTraceRaysIndirect2KHR-rayTracingPipelineTraceRaysIndirect2-03637
The `rayTracingPipelineTraceRaysIndirect2` feature **must** be enabled

Valid Usage (Implicit)

- VUID-vkCmdTraceRaysIndirect2KHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdTraceRaysIndirect2KHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdTraceRaysIndirect2KHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support compute operations
- VUID-vkCmdTraceRaysIndirect2KHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdTraceRaysIndirect2KHR-videocoding

This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Outside	Outside	Compute	Action

The `VkTraceRaysIndirectCommand2KHR` structure is defined as:

```
// Provided by VK_KHR_ray_tracing_maintenance1 with VK_KHR_ray_tracing_pipeline
typedef struct VkTraceRaysIndirectCommand2KHR {
    VkDeviceAddress    raygenShaderRecordAddress;
    VkDeviceSize       raygenShaderRecordSize;
    VkDeviceAddress    missShaderBindingTableAddress;
    VkDeviceSize       missShaderBindingTableSize;
    VkDeviceSize       missShaderBindingTableStride;
    VkDeviceAddress    hitShaderBindingTableAddress;
    VkDeviceSize       hitShaderBindingTableSize;
    VkDeviceSize       hitShaderBindingTableStride;
    VkDeviceAddress    callableShaderBindingTableAddress;
    VkDeviceSize       callableShaderBindingTableSize;
    VkDeviceSize       callableShaderBindingTableStride;
    uint32_t           width;
    uint32_t           height;
    uint32_t           depth;
} VkTraceRaysIndirectCommand2KHR;
```

- `raygenShaderRecordAddress` is a `VkDeviceAddress` of the ray generation shader binding table record used by this command.
- `raygenShaderRecordSize` is a `VkDeviceSize` number of bytes corresponding to the ray generation shader binding table record at base address `raygenShaderRecordAddress`.
- `missShaderBindingTableAddress` is a `VkDeviceAddress` of the first record in the miss shader binding table used by this command.
- `missShaderBindingTableSize` is a `VkDeviceSize` number of bytes corresponding to the total size of the miss shader binding table at `missShaderBindingTableAddress` that may be accessed by this

command.

- `missShaderBindingTableStride` is a `VkDeviceSize` number of bytes between records of the miss shader binding table.
- `hitShaderBindingTableAddress` is a `VkDeviceAddress` of the first record in the hit shader binding table used by this command.
- `hitShaderBindingTableSize` is a `VkDeviceSize` number of bytes corresponding to the total size of the hit shader binding table at `hitShaderBindingTableAddress` that may be accessed by this command.
- `hitShaderBindingTableStride` is a `VkDeviceSize` number of bytes between records of the hit shader binding table.
- `callableShaderBindingTableAddress` is a `VkDeviceAddress` of the first record in the callable shader binding table used by this command.
- `callableShaderBindingTableSize` is a `VkDeviceSize` number of bytes corresponding to the total size of the callable shader binding table at `callableShaderBindingTableAddress` that may be accessed by this command.
- `callableShaderBindingTableStride` is a `VkDeviceSize` number of bytes between records of the callable shader binding table.
- `width` is the width of the ray trace query dimensions.
- `height` is height of the ray trace query dimensions.
- `depth` is depth of the ray trace query dimensions.

The members of `VkTraceRaysIndirectCommand2KHR` have the same meaning as the similarly named parameters of `vkCmdTraceRaysKHR`.

Indirect shader binding table buffer parameters **must** satisfy the same memory alignment and binding requirements as their counterparts in `vkCmdTraceRaysIndirectKHR` and `vkCmdTraceRaysKHR`.

Valid Usage

- VUID-VkTraceRaysIndirectCommand2KHR-pRayGenShaderBindingTable-03680
If the buffer from which `raygenShaderRecordAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkTraceRaysIndirectCommand2KHR-pRayGenShaderBindingTable-03681
The buffer from which the `raygenShaderRecordAddress` is queried **must** have been created with the `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` usage flag
- VUID-VkTraceRaysIndirectCommand2KHR-pRayGenShaderBindingTable-03682
`raygenShaderRecordAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-pMissShaderBindingTable-03683
If the buffer from which `missShaderBindingTableAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object

- VUID-VkTraceRaysIndirectCommand2KHR-pMissShaderBindingTable-03684
The buffer from which the `missShaderBindingTableAddress` is queried **must** have been created with the `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` usage flag
- VUID-VkTraceRaysIndirectCommand2KHR-pMissShaderBindingTable-03685
`missShaderBindingTableAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-stride-03686
`missShaderBindingTableStride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-stride-04029
`missShaderBindingTableStride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride`
- VUID-VkTraceRaysIndirectCommand2KHR-pHitShaderBindingTable-03687
If the buffer from which `hitShaderBindingTableAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkTraceRaysIndirectCommand2KHR-pHitShaderBindingTable-03688
The buffer from which the `hitShaderBindingTableAddress` is queried **must** have been created with the `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` usage flag
- VUID-VkTraceRaysIndirectCommand2KHR-pHitShaderBindingTable-03689
`hitShaderBindingTableAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-stride-03690
`hitShaderBindingTableStride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-stride-04035
`hitShaderBindingTableStride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride`
- VUID-VkTraceRaysIndirectCommand2KHR-pCallableShaderBindingTable-03691
If the buffer from which `callableShaderBindingTableAddress` was queried is non-sparse then it **must** be bound completely and contiguously to a single `VkDeviceMemory` object
- VUID-VkTraceRaysIndirectCommand2KHR-pCallableShaderBindingTable-03692
The buffer from which the `callableShaderBindingTableAddress` is queried **must** have been created with the `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` usage flag
- VUID-VkTraceRaysIndirectCommand2KHR-pCallableShaderBindingTable-03693
`callableShaderBindingTableAddress` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupBaseAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-stride-03694
`callableShaderBindingTableStride` **must** be a multiple of `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleAlignment`
- VUID-VkTraceRaysIndirectCommand2KHR-stride-04041
`callableShaderBindingTableStride` **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxShaderGroupStride`

- VUID-VkTraceRaysIndirectCommand2KHR-flags-03696
If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR`, `hitShaderBindingTableAddress` **must** not be zero
- VUID-VkTraceRaysIndirectCommand2KHR-flags-03697
If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR`, `hitShaderBindingTableAddress` **must** not be zero
- VUID-VkTraceRaysIndirectCommand2KHR-flags-03511
If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR`, the shader group handle identified by `missShaderBindingTableAddress` **must** not be zero
- VUID-VkTraceRaysIndirectCommand2KHR-flags-03512
If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR`, entries in the table identified by `hitShaderBindingTableAddress` accessed as a result of this command in order to execute an any-hit shader **must** not be zero
- VUID-VkTraceRaysIndirectCommand2KHR-flags-03513
If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR`, entries in the table identified by `hitShaderBindingTableAddress` accessed as a result of this command in order to execute a closest hit shader **must** not be zero
- VUID-VkTraceRaysIndirectCommand2KHR-flags-03514
If the currently bound ray tracing pipeline was created with `flags` that included `VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR`, entries in the table identified by `hitShaderBindingTableAddress` accessed as a result of this command in order to execute an intersection shader **must** not be zero
- VUID-VkTraceRaysIndirectCommand2KHR-pHitShaderBindingTable-04735
Any non-zero hit shader group entries in the table identified by `hitShaderBindingTableAddress` accessed by this call from a geometry with a `geometryType` of `VK_GEOMETRY_TYPE_TRIANGLES_KHR` **must** have been created with `VK_RAY_TRACING_SHADER_GROUP_TYPE_TRIANGLES_HIT_GROUP_KHR`
- VUID-VkTraceRaysIndirectCommand2KHR-pHitShaderBindingTable-04736
Any non-zero hit shader group entries in the table identified by `hitShaderBindingTableAddress` accessed by this call from a geometry with a `geometryType` of `VK_GEOMETRY_TYPE_AABBS_KHR` **must** have been created with `VK_RAY_TRACING_SHADER_GROUP_TYPE_PROCEDURAL_HIT_GROUP_KHR`
- VUID-VkTraceRaysIndirectCommand2KHR-width-03638
`width` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[0] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[0]`
- VUID-VkTraceRaysIndirectCommand2KHR-height-03639
`height` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[1] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[1]`

- VUID-VkTraceRaysIndirectCommand2KHR-depth-03640
depth **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupCount[2] × VkPhysicalDeviceLimits::maxComputeWorkGroupSize[2]`
- VUID-VkTraceRaysIndirectCommand2KHR-width-03641
width × height × depth **must** be less than or equal to `VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxRayDispatchInvocationCount`

36.3. Shader Binding Table

A *shader binding table* is a resource which establishes the relationship between the ray tracing pipeline and the acceleration structures that were built for the ray tracing pipeline. It indicates the shaders that operate on each geometry in an acceleration structure. In addition, it contains the resources accessed by each shader, including indices of textures, buffer device addresses, and constants. The application allocates and manages *shader binding tables* as `VkBuffer` objects.

Each entry in the shader binding table consists of `shaderGroupHandleSize` bytes of data, either as queried by `vkGetRayTracingShaderGroupHandlesKHR` to refer to those specified shaders, or all zeros to refer to a zero shader group. A zero shader group behaves as though it is a shader group consisting entirely of `VK_SHADER_UNUSED_KHR`. The remainder of the data specified by the stride is application-visible data that can be referenced by a `ShaderRecordBufferKHR` block in the shader.

The shader binding tables to use in a ray tracing pipeline are passed to the `vkCmdTraceRaysKHR`, or `vkCmdTraceRaysIndirectKHR` commands. Shader binding tables are read-only in shaders that are executing on the ray tracing pipeline.

Shader variables identified with the `ShaderRecordBufferKHR` storage class are used to access the provided shader binding table. Such variables **must** be:

- typed as `OpTypeStruct`, or an array of this type,
- identified with a `Block` decoration, and
- laid out explicitly using the `Offset`, `ArrayStride`, and `MatrixStride` decorations as specified in [Offset and Stride Assignment](#).

The `Offset` decoration for any member of a `Block`-decorated variable in the `ShaderRecordBufferKHR` storage class **must** not cause the space required for that variable to extend outside the range `[0, maxStorageBufferRange)`.

Accesses to the shader binding table from ray tracing pipelines **must** be `synchronized` with the `VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR` pipeline stage and an access type of `VK_ACCESS_SHADER_READ_BIT`.

NOTE

Because different shader record buffers can be associated with the same shader, a shader variable with `ShaderRecordBufferKHR` storage class will not be dynamically uniform if different invocations of the same shader can reference different data in the shader record buffer, such as if the same shader occurs twice in the shader binding table with a different shader record buffer. In this case, indexing resources based on values in the `ShaderRecordBufferKHR` storage class, the index should be

decorated as **NonUniform**.

36.3.1. Indexing Rules

In order to execute the correct shaders and access the correct resources during a ray tracing dispatch, the implementation **must** be able to locate shader binding table entries at various stages of execution. This is accomplished by defining a set of indexing rules that compute shader binding table record positions relative to the buffer's base address in memory. The application **must** organize the contents of the shader binding table's memory in a way that application of the indexing rules will lead to correct records.

Ray Generation Shaders

Only one ray generation shader is executed per ray tracing dispatch.

For `vkCmdTraceRaysKHR`, the location of the ray generation shader is specified by the `pRaygenShaderBindingTable->deviceAddress` parameter—there is no indexing. All data accessed **must** be less than `pRaygenShaderBindingTable->size` bytes from `deviceAddress`. `pRaygenShaderBindingTable->stride` is unused, and **must** be equal to `pRaygenShaderBindingTable->size`.

Hit Shaders

The base for the computation of intersection, any-hit, and closest hit shader locations is the `instanceShaderBindingTableRecordOffset` value stored with each instance of a top-level acceleration structure (`VkAccelerationStructureInstanceKHR`). This value determines the beginning of the shader binding table records for a given instance.

In the following rule, `geometryIndex` refers to the `geometry index` of the intersected geometry within the instance.

The `sbtRecordOffset` and `sbtRecordStride` values are passed in as parameters to `traceRayEXT()` calls made in the shaders. See Section 8.19 (Ray Tracing Functions) of the OpenGL Shading Language Specification for more details. In SPIR-V, these correspond to the `SBTOffset` and `SBTStride` parameters to the `pipeline trace ray` instructions.

The result of this computation is then added to `pHitShaderBindingTable->deviceAddress`, a device address passed to `vkCmdTraceRaysKHR`.

For `vkCmdTraceRaysKHR`, the complete rule to compute a hit shader binding table record address in the `pHitShaderBindingTable` is:

$$\text{pHitShaderBindingTable->deviceAddress} + \text{pHitShaderBindingTable->stride} \times (\text{instanceShaderBindingTableRecordOffset} + \text{geometryIndex} \times \text{sbtRecordStride} + \text{sbtRecordOffset})$$

All data accessed **must** be less than `pHitShaderBindingTable->size` bytes from the base address.

Miss Shaders

A miss shader is executed whenever a ray query fails to find an intersection for the given scene geometry. Multiple miss shaders **may** be executed throughout a ray tracing dispatch.

The base for the computation of miss shader locations is `pMissShaderBindingTable->deviceAddress`, a device address passed into `vkCmdTraceRaysKHR`.

The `missIndex` value is passed in as a parameter to `traceRayEXT()` calls made in the shaders. See Section 8.19 (Ray Tracing Functions) of the OpenGL Shading Language Specification for more details. In SPIR-V, this corresponds to the `MissIndex` parameter to the `pipeline trace ray` instructions.

For `vkCmdTraceRaysKHR`, the complete rule to compute a miss shader binding table record address in the `pMissShaderBindingTable` is:

$$\text{pMissShaderBindingTable->deviceAddress} + \text{pMissShaderBindingTable->stride} \times \text{missIndex}$$

All data accessed **must** be less than `pMissShaderBindingTable->size` bytes from the base address.

Callable Shaders

A callable shader is executed when requested by a ray tracing shader. Multiple callable shaders **may** be executed throughout a ray tracing dispatch.

The base for the computation of callable shader locations is `pCallableShaderBindingTable->deviceAddress`, a device address passed into `vkCmdTraceRaysKHR`.

The `sbtRecordIndex` value is passed in as a parameter to `executeCallableEXT()` calls made in the shaders. See Section 8.19 (Ray Tracing Functions) of the OpenGL Shading Language Specification for more details. In SPIR-V, this corresponds to the `SBTIndex` parameter to the `OpExecuteCallableKHR` instruction.

For `vkCmdTraceRaysKHR`, the complete rule to compute a callable shader binding table record address in the `pCallableShaderBindingTable` is:

$$\text{pCallableShaderBindingTable->deviceAddress} + \text{pCallableShaderBindingTable->stride} \times \text{sbtRecordIndex}$$

All data accessed **must** be less than `pCallableShaderBindingTable->size` bytes from the base address.

36.4. Ray Tracing Pipeline Stack

Ray tracing pipelines have a potentially large set of shaders which **may** be invoked in various call chain combinations to perform ray tracing. To store parameters for a given shader execution, an implementation **may** use a stack of data in memory. This stack **must** be sized to the sum of the stack sizes of all shaders in any call chain executed by the application.

If the stack size is not set explicitly, the stack size for a pipeline is:

$$\text{rayGenStackMax} + \min(1, \text{maxPipelineRayRecursionDepth}) \times \max(\text{closestHitStackMax}, \text{missStackMax}, \text{intersectionStackMax} + \text{anyHitStackMax}) + \max(0, \text{maxPipelineRayRecursionDepth} - 1) \times \max(\text{closestHitStackMax}, \text{missStackMax}) + 2 \times \text{callableStackMax}$$

where `rayGenStackMax`, `closestHitStackMax`, `missStackMax`, `anyHitStackMax`, `intersectionStackMax`, and `callableStackMax` are the maximum stack values queried by the respective shader stages for any shaders in any shader groups defined by the pipeline.

This stack size is potentially significant, so an application **may** want to provide a more accurate stack size after pipeline compilation. The value that the application provides is the maximum value of the sum of all shaders in a call chain across all possible call chains, taking into account any application specific knowledge about the properties of the call chains.

For example, if an application has two types of closest hit and miss shaders that it can use but the first level of rays will only use the first kind (possibly reflection) and the second level will only use the second kind (occlusion or shadow ray, for example) then the application can compute the stack size by something similar to:

NOTE

$$\text{rayGenStack} + \max(\text{closestHit1Stack}, \text{miss1Stack}) + \max(\text{closestHit2Stack}, \text{miss2Stack})$$

This is guaranteed to be no larger than the default stack size computation which assumes that both call levels may be the larger of the two.

36.5. Ray Tracing Capture Replay

In a similar way to `bufferDeviceAddressCaptureReplay`, the `rayTracingPipelineShaderGroupHandleCaptureReplay` feature allows the querying of opaque data which **can** be used in a future replay.

During the capture phase, capture/replay tools are expected to query opaque data for shader group handle replay using `vkGetRayTracingCaptureReplayShaderGroupHandlesKHR`.

Providing the opaque data during replay, using `VkRayTracingShaderGroupCreateInfoKHR::pShaderGroupCaptureReplayHandle` at pipeline creation time, causes the implementation to generate identical shader group handles to those in the capture phase, allowing capture/replay tools to reuse previously recorded shader binding table buffer contents or to obtain the same handles by calling `vkGetRayTracingCaptureReplayShaderGroupHandlesKHR` again.

Chapter 37. Video Coding

Vulkan implementations **may** expose one or more queue families supporting video coding operations. These operations are performed by recording them into a command buffer within a [video coding scope](#), and submitting them to queues with compatible video coding capabilities.

The Vulkan video functionalities are designed to be made available through a set of APIs built on top of each other, consisting of:

- A core API providing common video coding functionalities,
- APIs providing codec-independent video decode and video encode related functionalities, respectively,
- Additional codec-specific APIs built on top of those.

This chapter details the fundamental components and operations of these.

37.1. Video Picture Resources

In the context of video coding, multidimensional arrays of image data that **can** be used as the source or target of video coding operations are referred to as *video picture resources*. They **may** store additional metadata that includes implementation-private information used during the execution of video coding operations, as discussed later.

Video picture resources are backed by [VkImage](#) objects. Individual subregions of [VkImageView](#) objects created from such resources **can** be used as [decode output pictures](#), [encode input pictures](#), [reconstructed pictures](#), and/or [reference pictures](#).

The parameters of a video picture resource are specified using a [VkVideoPictureResourceInfoKHR](#) structure.

The [VkVideoPictureResourceInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoPictureResourceInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkOffset2D         codedOffset;
    VkExtent2D         codedExtent;
    uint32_t           baseArrayLayer;
    VkImageView         imageViewBinding;
} VkVideoPictureResourceInfoKHR;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `codedOffset` is the offset in texels of the image subregion to use.
- `codedExtent` is the size in pixels of the coded image data.

- `baseArrayLayer` is the array layer of the image view specified in `imageViewBinding` to use as the video picture resource.
- `imageViewBinding` is an image view representing the video picture resource.

The image subresource referred to by such a structure is defined as the image array layer index specified in `baseArrayLayer` relative to the image subresource range the image view specified in `imageViewBinding` was created with.

The meaning of the `codedOffset` and `codedExtent` depends on the command and context the video picture resource is used in, as well as on the used `video profile` and corresponding codec-specific semantics, as described later.

A video picture resource is uniquely defined by the image subresource referred to by an instance of this structure, together with the `codedOffset` and `codedExtent` members that identify the image subregion within the image subresource referenced corresponding to the video picture resource according to the particular codec-specific semantics.

Accesses to image data within a video picture resource happen at the granularity indicated by `VkVideoCapabilitiesKHR::pictureAccessGranularity`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used `video profile`. As a result, given an effective image subregion corresponding to a video picture resource, the actual image subregion accessed **may** be larger than that as it **may** include additional padding texels due to the picture access granularity. Any writes performed by video coding operations to such padding texels will result in undefined texel values.

Two video picture resources match if they refer to the same image subresource and they specify identical `codedOffset` and `codedExtent` values.

Valid Usage

- VUID-VkVideoPictureResourceInfoKHR-baseArrayLayer-07175
`baseArrayLayer` **must** be less than the `VkImageViewCreateInfo::subresourceRange.layerCount` specified when the image view `imageViewBinding` was created

Valid Usage (Implicit)

- VUID-VkVideoPictureResourceInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_PICTURE_RESOURCE_INFO_KHR`
- VUID-VkVideoPictureResourceInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkVideoPictureResourceInfoKHR-imageViewBinding-parameter
`imageViewBinding` **must** be a valid `VkImageView` handle

37.2. Decoded Picture Buffer

An integral part of video coding pipelines is the reconstruction of pictures from a compressed video bitstream. A *reconstructed picture* is a [video picture resource](#) resulting from this process.

Such reconstructed pictures **can** be used as *reference pictures* in subsequent video coding operations to provide predictions of the values of samples of subsequently decoded or encoded pictures. The correct use of such reconstructed pictures as reference pictures is driven by the video compression standard, the implementation, and the application-specific use cases.

The list of reference pictures used to provide such predictions within a single video coding operation is referred to as the list of *active reference pictures*.

The *decoded picture buffer (DPB)* is an indexed data structure that maintains the set of reference pictures available to be used in video coding operations. Individual indexed entries of the DPB are referred to as the *decoded picture buffer (DPB) slots*. The range of valid DPB slot indices is between zero and $N-1$, where N is the capacity of the DPB. Each DPB slot **can** refer to a reference picture containing a *video frame* or **can** refer to up to two reference pictures containing the top and/or bottom *fields* that, when both present, together represent a full *video frame*.

In Vulkan, the state and the backing store of the DPB is separated as follows:

- The state of individual DPB slots is maintained by [video session](#) objects.
- The backing store of DPB slots is provided by subregions of [VkImage](#) objects used as [video picture resources](#).

In addition, the implementation **may** also maintain opaque metadata associated with DPB slots, including:

- *Reference picture metadata* corresponding to the video picture resource associated with the DPB slot.

Such metadata **may** be stored by the implementation as part of the DPB slot state maintained by the video session, or as part of the video picture resource backing the DPB slot.

Any metadata stored in the video picture resources backing DPB slots are independent of the video session used to store it, hence such video picture resources **can** be shared with other video sessions. Correspondingly, any metadata that is dependent on the video session will always be stored as part of the DPB slot state maintained by that video session.

The responsibility of managing the DPB is split between the application and the implementation as follows:

- The application maintains the association between [DPB slot](#) indices and corresponding [video picture resources](#).
- The implementation maintains global and per-slot opaque [reference picture metadata](#).

In addition, the application is also responsible for managing the mapping between the codec-specific picture IDs and DPB slots, and any other codec-specific states unless otherwise specified.

37.2.1. DPB Slot States

At a given time, each DPB slot is either in *active* or *inactive* state. Initially, all DPB slots managed by a [video session](#) are in *inactive* state.

A DPB slot **can** be *activated* by using it as the target of picture reconstruction in a video coding operation with the reconstructed picture requested to be set up as a reference picture, according to the codec-specific semantics, changing its state to *active* and associating it with a *picture reference* to the [reconstructed pictures](#).

Some video coding standards allow multiple picture references to be associated with a single DPB slot. In this case the state of the individual picture references **can** be independently updated.

NOTE

As an example, H.264 decoding allows associating a separate top field and bottom field picture with the same DPB slot.

As part of reference picture setup, the implementation **may** also generate [reference picture metadata](#). Such reference picture metadata is specific to each picture reference associated with the DPB slot.

If such a video coding operation completes successfully, the activated DPB slot will have a *valid picture reference* and the [reconstructed picture](#) is associated with the DPB slot. This is true even if the DPB slot is used as the target of a picture reconstruction that only sets up a top field or bottom field reference picture and thus does not yet refer to a complete frame. However, if any data provided as input to such a video coding operation is not compliant with the video compression standard used, that video coding operation **may** complete unsuccessfully, in which case the activated DPB slot will have an *invalid picture reference*. This is true even if the DPB slot previously had a valid picture reference to a top field or bottom field reference picture, but the reconstruction of the other field corresponding to the DPB slot failed.

The application **can** use [queries](#) to get feedback about the outcome of video coding operations and use the resulting [VkQueryResultStatusKHR](#) value to determine whether the video coding operation completed successfully (result status is positive) or unsuccessfully (result status is negative).

Using a [reference picture](#) associated with a DPB slot that has an *invalid picture reference* as an [active reference picture](#) in subsequent video coding operations is legal, however, the contents of the outputs of such operations are undefined, and any DPB slots activated by such video coding operations will also have an *invalid picture reference*. This is true even if such video coding operations **may** otherwise complete successfully.

A DPB slot **can** also be *deactivated* by the application, changing its state to *inactive* and invalidating any picture references and [reference picture metadata](#) associated with the DPB slot.

If an already active DPB slot is used as the target of picture reconstruction in a video coding operation, but the decoded picture is not requested to be set up as a reference picture, according to the codec-specific semantics, no reference picture setup happens and the corresponding picture reference and [reference picture metadata](#) is invalidated within the DPB slot. If the DPB slot no longer has any associated picture references after such an operation, the DPB slot is implicitly deactivated.

If an already active DPB slot is used as the target of picture reconstruction when decoding a field picture that is not marked as reference, then the behavior is as follows:

- If the DPB slot is currently associated with a frame, then the DPB slot is deactivated.
- If the DPB slot is not currently associated with a top field picture and the decoded picture is a top field picture, or if the DPB slot is not currently associated with a bottom field picture and the decoded picture is a bottom field picture, then the other field picture association of the DPB slot, if any, is not disturbed.
- If the DPB slot is currently associated with a top field picture and the decoded picture is a top field picture, or if the DPB slot is currently associated with a bottom field picture and the decoded picture is a bottom field picture, then that picture association is invalidated, without disturbing the other field picture association, if any. If the DPB slot no longer has any associated picture references after such an operation, the DPB slot is implicitly deactivated.

A DPB slot **can** be activated with a new frame even if it is already active. In this case all previous associations of the DPB slots with [reference pictures](#) are replaced with an association with the [reconstructed picture](#) used to activate it.

If an already active DPB slot is activated with a reconstructed field picture, then the behavior is as follows:

- If the DPB slot is currently associated with a frame, then that association is replaced with an association with the reconstructed field picture used to activate it.
- If the DPB slot is not currently associated with a top field picture and the DPB slot is activated with a top field picture, or if the DPB slot is not currently associated with a bottom field picture and the DPB slot is activated with a bottom field picture, then the DPB slot is associated with the reconstructed field picture used to activate it, without disturbing the other field picture association, if any.
- If the DPB slot is currently associated with a top field picture and the DPB slot is activated with a new top field picture, or if the DPB slot is currently associated with a bottom field picture and the DPB slot is activated with a new bottom field picture, then that association is replaced with an association with the reconstructed field picture used to activate it, without disturbing the other field picture association, if any.

37.3. Video Profiles

The `VkVideoProfileInfoKHR` structure is defined as follows:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoProfileInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkVideoCodecOperationFlagBitsKHR videoCodecOperation;
    VkVideoChromaSubsamplingFlagsKHR chromaSubsampling;
    VkVideoComponentBitDepthFlagsKHR lumaBitDepth;
    VkVideoComponentBitDepthFlagsKHR chromaBitDepth;
} VkVideoProfileInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `videoCodecOperation` is a `VkVideoCodecOperationFlagBitsKHR` value specifying a video codec operation.
- `chromaSubsampling` is a bitmask of `VkVideoChromaSubsamplingFlagBitsKHR` specifying video chroma subsampling information.
- `lumaBitDepth` is a bitmask of `VkVideoComponentBitDepthFlagBitsKHR` specifying video luma bit depth information.
- `chromaBitDepth` is a bitmask of `VkVideoComponentBitDepthFlagBitsKHR` specifying video chroma bit depth information.

Video profiles are provided as input to video capability queries such as `vkGetPhysicalDeviceVideoCapabilitiesKHR` or `vkGetPhysicalDeviceVideoFormatPropertiesKHR`, as well as when creating resources to be used by video coding operations such as images, buffers, query pools, and video sessions.

The full description of a video profile is specified by an instance of this structure, and the codec-specific and auxiliary structures provided in its `pNext` chain.

When this structure is specified as an input parameter to `vkGetPhysicalDeviceVideoCapabilitiesKHR`, or through the `pProfiles` member of a `VkVideoProfileListInfoKHR` structure in the `pNext` chain of the input parameter of a query command such as `vkGetPhysicalDeviceVideoFormatPropertiesKHR` or `vkGetPhysicalDeviceImageFormatProperties2KHR`, the following error codes indicate specific causes of the failure of the query operation:

- `VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR` indicates that the requested video picture layout (e.g. through the `pictureLayout` member of a `VkVideoDecodeH264ProfileInfoKHR` structure included in the `pNext` chain of `VkVideoProfileInfoKHR`) is not supported.
- `VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR` indicates that a video profile operation specified by `videoCodecOperation` is not supported.
- `VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR` indicates that video format parameters specified by `chromaSubsampling`, `lumaBitDepth`, or `chromaBitDepth` are not supported.
- `VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR` indicates that the codec-specific parameters corresponding to the video codec operation are not supported.

Valid Usage

- VUID-VkVideoProfileInfoKHR-chromaSubsampling-07013
`chromaSubsampling` **must** have a single bit set
- VUID-VkVideoProfileInfoKHR-lumaBitDepth-07014
`lumaBitDepth` **must** have a single bit set
- VUID-VkVideoProfileInfoKHR-chromaSubsampling-07015
If `chromaSubsampling` is not `VK_VIDEO_CHROMA_SUBSAMPLING_MONOCHROME_BIT_KHR`, then `chromaBitDepth` **must** have a single bit set
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-07179
If `videoCodecOperation` is `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoDecodeH264ProfileInfoKHR` structure
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-07180
If `videoCodecOperation` is `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoDecodeH265ProfileInfoKHR` structure
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-09256
If `videoCodecOperation` is `VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoDecodeAV1ProfileInfoKHR` structure
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-07181
If `videoCodecOperation` is `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoEncodeH264ProfileInfoKHR` structure
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-07182
If `videoCodecOperation` is `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoEncodeH265ProfileInfoKHR` structure

Valid Usage (Implicit)

- VUID-VkVideoProfileInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_PROFILE_INFO_KHR`
- VUID-VkVideoProfileInfoKHR-videoCodecOperation-parameter
`videoCodecOperation` **must** be a valid `VkVideoCodecOperationFlagBitsKHR` value
- VUID-VkVideoProfileInfoKHR-chromaSubsampling-parameter
`chromaSubsampling` **must** be a valid combination of `VkVideoChromaSubsamplingFlagBitsKHR` values
- VUID-VkVideoProfileInfoKHR-chromaSubsampling-requiredbitmask
`chromaSubsampling` **must** not be 0
- VUID-VkVideoProfileInfoKHR-lumaBitDepth-parameter
`lumaBitDepth` **must** be a valid combination of `VkVideoComponentBitDepthFlagBitsKHR` values
- VUID-VkVideoProfileInfoKHR-lumaBitDepth-requiredbitmask
`lumaBitDepth` **must** not be 0

- VUID-VkVideoProfileInfoKHR-chromaBitDepth-parameter
chromaBitDepth **must** be a valid combination of [VkVideoComponentBitDepthFlagBitsKHR](#) values

Possible values of [VkVideoProfileInfoKHR::videoCodecOperation](#), specifying the type of video coding operation and video compression standard used by a video profile, are:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoCodecOperationFlagBitsKHR {
    VK_VIDEO_CODEC_OPERATION_NONE_KHR = 0,
    // Provided by VK_KHR_video_encode_h264
    VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR = 0x00010000,
    // Provided by VK_KHR_video_encode_h265
    VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR = 0x00020000,
    // Provided by VK_KHR_video_decode_h264
    VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR = 0x00000001,
    // Provided by VK_KHR_video_decode_h265
    VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR = 0x00000002,
    // Provided by VK_KHR_video_decode_av1
    VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR = 0x00000004,
} VkVideoCodecOperationFlagBitsKHR;
```

- **VK_VIDEO_CODEC_OPERATION_NONE_KHR** indicates no support for any video codec operations.
- **VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR** specifies support for [H.264 decode operations](#).
- **VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR** specifies support for [H.265 decode operations](#).
- **VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR** specifies support for [AV1 decode operations](#).
- **VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR** specifies support for [H.264 encode operations](#).
- **VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR** specifies support for [H.265 encode operations](#).

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoCodecOperationFlagsKHR;
```

VkVideoCodecOperationFlagsKHR is a bitmask type for setting a mask of zero or more [VkVideoCodecOperationFlagBitsKHR](#).

The video format chroma subsampling is defined with the following enums:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoChromaSubsamplingFlagBitsKHR {
    VK_VIDEO_CHROMA_SUBSAMPLING_INVALID_KHR = 0,
    VK_VIDEO_CHROMA_SUBSAMPLING_MONOCHROME_BIT_KHR = 0x00000001,
    VK_VIDEO_CHROMA_SUBSAMPLING_420_BIT_KHR = 0x00000002,
    VK_VIDEO_CHROMA_SUBSAMPLING_422_BIT_KHR = 0x00000004,
    VK_VIDEO_CHROMA_SUBSAMPLING_444_BIT_KHR = 0x00000008,
} VkVideoChromaSubsamplingFlagBitsKHR;
```

- **VK_VIDEO_CHROMA_SUBSAMPLING_MONOCHROME_BIT_KHR** specifies that the format is monochrome.
- **VK_VIDEO_CHROMA_SUBSAMPLING_420_BIT_KHR** specified that the format is 4:2:0 chroma subsampled, i.e. the two chroma components are sampled horizontally and vertically at half the sample rate of the luma component.
- **VK_VIDEO_CHROMA_SUBSAMPLING_422_BIT_KHR** - the format is 4:2:2 chroma subsampled, i.e. the two chroma components are sampled horizontally at half the sample rate of luma component.
- **VK_VIDEO_CHROMA_SUBSAMPLING_444_BIT_KHR** - the format is 4:4:4 chroma sampled, i.e. all three components of the Y'C_BC_R format are sampled at the same rate, thus there is no chroma subsampling.

Chroma subsampling is described in more detail in the [Chroma Reconstruction](#) section.

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoChromaSubsamplingFlagsKHR;
```

VkVideoChromaSubsamplingFlagsKHR is a bitmask type for setting a mask of zero or more **VkVideoChromaSubsamplingFlagBitsKHR**.

Possible values for the video format component bit depth are:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoComponentBitDepthFlagBitsKHR {
    VK_VIDEO_COMPONENT_BIT_DEPTH_INVALID_KHR = 0,
    VK_VIDEO_COMPONENT_BIT_DEPTH_8_BIT_KHR = 0x00000001,
    VK_VIDEO_COMPONENT_BIT_DEPTH_10_BIT_KHR = 0x00000004,
    VK_VIDEO_COMPONENT_BIT_DEPTH_12_BIT_KHR = 0x00000010,
} VkVideoComponentBitDepthFlagBitsKHR;
```

- **VK_VIDEO_COMPONENT_BIT_DEPTH_8_BIT_KHR** specifies a component bit depth of 8 bits.
- **VK_VIDEO_COMPONENT_BIT_DEPTH_10_BIT_KHR** specifies a component bit depth of 10 bits.
- **VK_VIDEO_COMPONENT_BIT_DEPTH_12_BIT_KHR** specifies a component bit depth of 12 bits.

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoComponentBitDepthFlagsKHR;
```

`VkVideoComponentBitDepthFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoComponentBitDepthFlagBitsKHR`.

Additional information about the video decode use case **can** be provided by adding a `VkVideoDecodeUsageInfoKHR` structure to the `pNext` chain of `VkVideoProfileInfoKHR`.

The `VkVideoDecodeUsageInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_queue
typedef struct VkVideoDecodeUsageInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkVideoDecodeUsageFlagsKHR  videoUsageHints;
} VkVideoDecodeUsageInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `videoUsageHints` is a bitmask of `VkVideoDecodeUsageFlagBitsKHR` specifying hints about the intended use of the video decode profile.

Valid Usage (Implicit)

- VUID-VkVideoDecodeUsageInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_USAGE_INFO_KHR`
- VUID-VkVideoDecodeUsageInfoKHR-videoUsageHints-parameter
`videoUsageHints` **must** be a valid combination of `VkVideoDecodeUsageFlagBitsKHR` values

The following bits **can** be specified in `VkVideoDecodeUsageInfoKHR::videoUsageHints` as a hint about the video decode use case:

```
// Provided by VK_KHR_video_decode_queue
typedef enum VkVideoDecodeUsageFlagBitsKHR {
    VK_VIDEO_DECODE_USAGE_DEFAULT_KHR = 0,
    VK_VIDEO_DECODE_USAGE_TRANSCODING_BIT_KHR = 0x00000001,
    VK_VIDEO_DECODE_USAGE_OFFLINE_BIT_KHR = 0x00000002,
    VK_VIDEO_DECODE_USAGE_STREAMING_BIT_KHR = 0x00000004,
} VkVideoDecodeUsageFlagBitsKHR;
```

- `VK_VIDEO_DECODE_USAGE_TRANSCODING_BIT_KHR` specifies that video decoding is intended to be used in conjunction with video encoding to transcode a video bitstream with the same and/or different codecs.
- `VK_VIDEO_DECODE_USAGE_OFFLINE_BIT_KHR` specifies that video decoding is intended to be used to consume a local video bitstream.
- `VK_VIDEO_DECODE_USAGE_STREAMING_BIT_KHR` specifies that video decoding is intended to be used to

consume a video bitstream received as a continuous flow over network.

NOTE

There are no restrictions on the combination of bits that **can** be specified by the application. However, applications **should** use reasonable combinations in order for the implementation to be able to select the most appropriate mode of operation for the particular use case.

```
// Provided by VK_KHR_video_decode_queue
typedef VkFlags VkVideoDecodeUsageFlagsKHR;
```

`VkVideoDecodeUsageFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoDecodeUsageFlagBitsKHR`.

Additional information about the video encode use case **can** be provided by adding a `VkVideoEncodeUsageInfoKHR` structure to the `pNext` chain of `VkVideoProfileInfoKHR`.

The `VkVideoEncodeUsageInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeUsageInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    VkVideoEncodeUsageFlagsKHR videoUsageHints;
    VkVideoEncodeContentFlagsKHR videoContentHints;
    VkVideoEncodeTuningModeKHR tuningMode;
} VkVideoEncodeUsageInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `videoUsageHints` is a bitmask of `VkVideoEncodeUsageFlagBitsKHR` specifying hints about the intended use of the video encode profile.
- `videoContentHints` is a bitmask of `VkVideoEncodeContentFlagBitsKHR` specifying hints about the content to be encoded using the video encode profile.
- `tuningMode` is a `VkVideoEncodeTuningModeKHR` value specifying the tuning mode to use when encoding with the video profile.

Valid Usage (Implicit)

- VUID-VkVideoEncodeUsageInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_USAGE_INFO_KHR`
- VUID-VkVideoEncodeUsageInfoKHR-videoUsageHints-parameter
`videoUsageHints` **must** be a valid combination of `VkVideoEncodeUsageFlagBitsKHR` values
- VUID-VkVideoEncodeUsageInfoKHR-videoContentHints-parameter
`videoContentHints` **must** be a valid combination of `VkVideoEncodeContentFlagBitsKHR`

values

- VUID-VkVideoEncodeUsageInfoKHR-tuningMode-parameter

If **tuningMode** is not 0, **tuningMode** **must** be a valid [VkVideoEncodeTuningModeKHR](#) value

The following bits **can** be specified in [VkVideoEncodeUsageInfoKHR::videoUsageHints](#) as a hint about the video encode use case:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeUsageFlagBitsKHR {
    VK_VIDEO_ENCODE_USAGE_DEFAULT_KHR = 0,
    VK_VIDEO_ENCODE_USAGE_TRANSCODING_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_USAGE_STREAMING_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_USAGE_RECORDING_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_USAGE_CONFERENCING_BIT_KHR = 0x00000008,
} VkVideoEncodeUsageFlagBitsKHR;
```

- [VK_VIDEO_ENCODE_USAGE_TRANSCODING_BIT_KHR](#) specifies that video encoding is intended to be used in conjunction with video decoding to transcode a video bitstream with the same and/or different codecs.
- [VK_VIDEO_ENCODE_USAGE_STREAMING_BIT_KHR](#) specifies that video encoding is intended to be used to produce a video bitstream that is expected to be sent as a continuous flow over network.
- [VK_VIDEO_ENCODE_USAGE_RECORDING_BIT_KHR](#) specifies that video encoding is intended to be used for real-time recording for offline consumption.
- [VK_VIDEO_ENCODE_USAGE_CONFERENCING_BIT_KHR](#) specifies that video encoding is intended to be used in a video conferencing scenario.

NOTE

There are no restrictions on the combination of bits that **can** be specified by the application. However, applications **should** use reasonable combinations in order for the implementation to be able to select the most appropriate mode of operation for the particular use case.

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeUsageFlagsKHR;
```

[VkVideoEncodeUsageFlagsKHR](#) is a bitmask type for setting a mask of zero or more [VkVideoEncodeUsageFlagBitsKHR](#).

The following bits **can** be specified in [VkVideoEncodeUsageInfoKHR::videoContentHints](#) as a hint about the encoded video content:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeContentFlagBitsKHR {
    VK_VIDEO_ENCODE_CONTENT_DEFAULT_KHR = 0,
    VK_VIDEO_ENCODE_CONTENT_CAMERA_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_CONTENT_DESKTOP_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_CONTENT_RENDERED_BIT_KHR = 0x00000004,
} VkVideoEncodeContentFlagBitsKHR;
```

- **VK_VIDEO_ENCODE_CONTENT_CAMERA_BIT_KHR** specifies that video encoding is intended to be used to encode camera content.
- **VK_VIDEO_ENCODE_CONTENT_DESKTOP_BIT_KHR** specifies that video encoding is intended to be used to encode desktop content.
- **VK_VIDEO_ENCODE_CONTENT_RENDERED_BIT_KHR** specified that video encoding is intended to be used to encode rendered (e.g. game) content.

NOTE

There are no restrictions on the combination of bits that **can** be specified by the application. However, applications **should** use reasonable combinations in order for the implementation to be able to select the most appropriate mode of operation for the particular content type.

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeContentFlagsKHR;
```

VkVideoEncodeContentFlagsKHR is a bitmask type for setting a mask of zero or more **VkVideoEncodeContentFlagBitsKHR**.

Possible video encode tuning mode values are as follows:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeTuningModeKHR {
    VK_VIDEO_ENCODE_TUNING_MODE_DEFAULT_KHR = 0,
    VK_VIDEO_ENCODE_TUNING_MODE_HIGH_QUALITY_KHR = 1,
    VK_VIDEO_ENCODE_TUNING_MODE_LOW_LATENCY_KHR = 2,
    VK_VIDEO_ENCODE_TUNING_MODE_ULTRA_LOW_LATENCY_KHR = 3,
    VK_VIDEO_ENCODE_TUNING_MODE_LOSSLESS_KHR = 4,
} VkVideoEncodeTuningModeKHR;
```

- **VK_VIDEO_ENCODE_TUNING_MODE_DEFAULT_KHR** specifies the default tuning mode.
- **VK_VIDEO_ENCODE_TUNING_MODE_HIGH_QUALITY_KHR** specifies that video encoding is tuned for high quality. When using this tuning mode, the implementation **may** compromise the latency of video encoding operations to improve quality.
- **VK_VIDEO_ENCODE_TUNING_MODE_LOW_LATENCY_KHR** specifies that video encoding is tuned for low latency. When using this tuning mode, the implementation **may** compromise quality to increase the performance and lower the latency of video encode operations.

- `VK_VIDEO_ENCODE_TUNING_MODE_ULTRA_LOW_LATENCY_KHR` specifies that video encoding is tuned for ultra-low latency. When using this tuning mode, the implementation **may** compromise quality to maximize the performance and minimize the latency of video encoding operations.
- `VK_VIDEO_ENCODE_TUNING_MODE_LOSSLESS_KHR` specifies that video encoding is tuned for lossless encoding. When using this tuning mode, video encode operations produce lossless output.

The `VkVideoProfileListInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoProfileListInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           profileCount;
    const VkVideoProfileInfoKHR* pProfiles;
} VkVideoProfileListInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `profileCount` is the number of elements in the `pProfiles` array.
- `pProfiles` is a pointer to an array of `VkVideoProfileInfoKHR` structures.

NOTE

Video transcoding is an example of a use case that necessitates the specification of multiple profiles in various contexts.

When the application provides a video decode profile and one or more video encode profiles in the profile list, the implementation ensures that any capabilities returned or resources created are suitable for the video transcoding use cases without the need for manual data transformations.

Valid Usage

- VUID-VkVideoProfileListInfoKHR-pProfiles-06813
`pProfiles` **must** not contain more than one element whose `videoCodecOperation` member specifies a decode operation

Valid Usage (Implicit)

- VUID-VkVideoProfileListInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_PROFILE_LIST_INFO_KHR`
- VUID-VkVideoProfileListInfoKHR-pProfiles-parameter
If `profileCount` is not 0, `pProfiles` **must** be a valid pointer to an array of `profileCount` valid `VkVideoProfileInfoKHR` structures

37.4. Video Capabilities

37.4.1. Video Coding Capabilities

To query video coding capabilities for a specific video profile, call:

```
// Provided by VK_KHR_video_queue
VkResult vkGetPhysicalDeviceVideoCapabilitiesKHR(
    VkPhysicalDevice          physicalDevice,
    const VkVideoProfileInfoKHR* pVideoProfile,
    VkVideoCapabilitiesKHR*    pCapabilities);
```

- **physicalDevice** is the physical device from which to query the video decode or encode capabilities.
- **pVideoProfile** is a pointer to a [VkVideoProfileInfoKHR](#) structure.
- **pCapabilities** is a pointer to a [VkVideoCapabilitiesKHR](#) structure in which the capabilities are returned.

If the [video profile](#) described by **pVideoProfile** is supported by the implementation, then this command returns **VK_SUCCESS** and **pCapabilities** is filled with the capabilities supported with the specified video profile. Otherwise, one of the [video-profile-specific error codes](#) are returned.

Valid Usage

- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07183
If **pVideoProfile->videoCodecOperation** specifies a decode operation, then the **pNext** chain of **pCapabilities** **must** include a [VkVideoDecodeCapabilitiesKHR](#) structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07184
If **pVideoProfile->videoCodecOperation** is **VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR**, then the **pNext** chain of **pCapabilities** **must** include a [VkVideoDecodeH264CapabilitiesKHR](#) structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07185
If **pVideoProfile->videoCodecOperation** is **VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR**, then the **pNext** chain of **pCapabilities** **must** include a [VkVideoDecodeH265CapabilitiesKHR](#) structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-09257
If **pVideoProfile->videoCodecOperation** is **VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR**, then the **pNext** chain of **pCapabilities** **must** include a [VkVideoDecodeAV1CapabilitiesKHR](#) structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07186
If **pVideoProfile->videoCodecOperation** specifies an encode operation, then the **pNext** chain of **pCapabilities** **must** include a [VkVideoEncodeCapabilitiesKHR](#) structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07187
If **pVideoProfile->videoCodecOperation** is **VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR**,

then the **pNext** chain of **pCapabilities** **must** include a [VkVideoEncodeH264CapabilitiesKHR](#) structure

- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-07188

If **pVideoProfile->videoCodecOperation** is **VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR**, then the **pNext** chain of **pCapabilities** **must** include a [VkVideoEncodeH265CapabilitiesKHR](#) structure

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-physicalDevice-parameter **physicalDevice** **must** be a valid [VkPhysicalDevice](#) handle
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pVideoProfile-parameter **pVideoProfile** **must** be a valid pointer to a valid [VkVideoProfileInfoKHR](#) structure
- VUID-vkGetPhysicalDeviceVideoCapabilitiesKHR-pCapabilities-parameter **pCapabilities** **must** be a valid pointer to a [VkVideoCapabilitiesKHR](#) structure

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**
- **VK_ERROR_OUT_OF_DEVICE_MEMORY**
- **VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR**
- **VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR**
- **VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR**
- **VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR**

The [VkVideoCapabilitiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoCapabilitiesKHR {
    VkStructureType      sType;
    void*                pNext;
    VkVideoCapabilityFlagsKHR flags;
    VkDeviceSize          minBitstreamBufferOffsetAlignment;
    VkDeviceSize          minBitstreamBufferSizeAlignment;
    VkExtent2D            pictureAccessGranularity;
    VkExtent2D            minCodedExtent;
    VkExtent2D            maxCodedExtent;
    uint32_t              maxDpbSlots;
    uint32_t              maxActiveReferencePictures;
    VkExtensionProperties  stdHeaderVersion;
} VkVideoCapabilitiesKHR;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is a bitmask of **VkVideoCapabilityFlagBitsKHR** specifying capability flags.
- **minBitstreamBufferOffsetAlignment** is the minimum alignment for bitstream buffer offsets.
- **minBitstreamBufferSizeAlignment** is the minimum alignment for bitstream buffer range sizes.
- **pictureAccessGranularity** is the granularity at which image access to video picture resources happen.
- **minCodedExtent** is the minimum width and height of the coded frames.
- **maxCodedExtent** is the maximum width and height of the coded frames.
- **maxDpbSlots** is the maximum number of **DPB slots** supported by a single video session.
- **maxActiveReferencePictures** is the maximum number of **active reference pictures** a single video coding operation **can** use.
- **stdHeaderVersion** is a **VkExtensionProperties** structure reporting the Video Std header name and version supported for the video profile.

NOTE

It is common for video compression standards to allow using all reference pictures associated with active DPB slots as active reference pictures, hence for video decode profiles the values returned in **maxDpbSlots** and **maxActiveReferencePictures** are often equal. Similarly, in case of video decode profiles supporting field pictures the value of **maxActiveReferencePictures** often equals **maxDpbSlots** × 2.

Valid Usage (Implicit)

- VUID-VkVideoCapabilitiesKHR-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_VIDEO_CAPABILITIES_KHR**
- VUID-VkVideoCapabilitiesKHR-pNext-pNext
Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either **NULL** or a pointer to a valid instance of **VkVideoDecodeAV1CapabilitiesKHR**,

VkVideoDecodeCapabilitiesKHR, VkVideoDecodeH264CapabilitiesKHR, VkVideoDecodeH265CapabilitiesKHR, VkVideoEncodeCapabilitiesKHR, VkVideoEncodeH264CapabilitiesKHR, or VkVideoEncodeH265CapabilitiesKHR

- VUID-VkVideoCapabilitiesKHR-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique

Bits which **can** be set in **VkVideoCapabilitiesKHR::flags** are:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoCapabilityFlagBitsKHR {
    VK_VIDEO_CAPABILITY_PROTECTED_CONTENT_BIT_KHR = 0x00000001,
    VK_VIDEO_CAPABILITY_SEPARATE_REFERENCE_IMAGES_BIT_KHR = 0x00000002,
} VkVideoCapabilityFlagBitsKHR;
```

- **VK_VIDEO_CAPABILITY_PROTECTED_CONTENT_BIT_KHR** indicates that video sessions support producing and consuming protected content.
- **VK_VIDEO_CAPABILITY_SEPARATE_REFERENCE_IMAGES_BIT_KHR** indicates that the **video picture resources** associated with the **DPB slots** of a video session **can** be backed by separate **VkImage** objects. If this capability flag is not present, then all DPB slots of a video session **must** be associated with video picture resources backed by the same **VkImage** object (e.g. using different layers of the same image).

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoCapabilityFlagsKHR;
```

VkVideoCapabilityFlagsKHR is a bitmask type for setting a mask of zero or more **VkVideoCapabilityFlagBitsKHR**.

37.4.2. Video Format Capabilities

To enumerate the supported output, input and DPB image formats and corresponding capabilities for a specific video profile, call:

```
// Provided by VK_KHR_video_queue
VkResult vkGetPhysicalDeviceVideoFormatPropertiesKHR(
    VkPhysicalDevice          physicalDevice,
    const VkPhysicalDeviceVideoFormatInfoKHR* pVideoFormatInfo,
    uint32_t*                 pVideoFormatPropertyCount,
    VkVideoFormatPropertiesKHR* pVideoFormatProperties);
```

- **physicalDevice** is the physical device from which to query the video format properties.
- **pVideoFormatInfo** is a pointer to a **VkPhysicalDeviceVideoFormatInfoKHR** structure specifying the usage and video profiles for which supported image formats and capabilities are returned.
- **pVideoFormatPropertyCount** is a pointer to an integer related to the number of video format

properties available or queried, as described below.

- `pVideoFormatProperties` is a pointer to an array of `VkVideoFormatPropertiesKHR` structures in which supported image formats and capabilities are returned.

If `pVideoFormatProperties` is `NULL`, then the number of video format properties supported for the given `physicalDevice` is returned in `pVideoFormatPropertyCount`. Otherwise, `pVideoFormatPropertyCount` **must** point to a variable set by the application to the number of elements in the `pVideoFormatProperties` array, and on return the variable is overwritten with the number of values actually written to `pVideoFormatProperties`. If the value of `pVideoFormatPropertyCount` is less than the number of video format properties supported, at most `pVideoFormatPropertyCount` values will be written to `pVideoFormatProperties`, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available values were returned.

Video format properties are always queried with respect to a specific set of video profiles. These are specified by chaining the `VkVideoProfileListInfoKHR` structure to `pVideoFormatInfo`.

For most use cases, the images are used by a single video session and a single video profile is provided. For a use case such as video transcoding, where a decode session output image **can** be used as encode input in one or more encode sessions, multiple video profiles corresponding to the video sessions that will share the image **must** be provided.

If any of the [video profiles](#) specified via `VkVideoProfileListInfoKHR::pProfiles` are not supported, then this command returns one of the [video-profile-specific error codes](#). Furthermore, if `VkPhysicalDeviceVideoFormatInfoKHR::imageUsage` includes any image usage flags not supported by the specified video profiles, then this command returns `VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR`.

This command also returns `VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR` if `VkPhysicalDeviceVideoFormatInfoKHR::imageUsage` does not include the appropriate flags as dictated by the decode capability flags returned in `VkVideoDecodeCapabilitiesKHR::flags` for any of the profiles specified in the `VkVideoProfileListInfoKHR` structure provided in the `pNext` chain of `pVideoFormatInfo`.

If the decode capability flags include `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR` but not `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR`, then in order to query video format properties for decode DPB and output usage, `VkPhysicalDeviceVideoFormatInfoKHR::imageUsage` **must** include both `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR` and `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`. Otherwise, the call will fail with `VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR`.

If the decode capability flags include `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR` but not `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR`, then in order to query video format properties for decode DPB usage, `VkPhysicalDeviceVideoFormatInfoKHR::imageUsage` **must** include `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`, but not `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`. Otherwise, the call will fail with `VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR`. Similarly, to query video format properties for decode output usage, `VkPhysicalDeviceVideoFormatInfoKHR::imageUsage` **must** include `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`, but not `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`. Otherwise, the call will fail with `VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR`.

The `imageUsage` member of the `VkPhysicalDeviceVideoFormatInfoKHR` structure specifies the expected video usage flags that the returned video formats **must** support. Correspondingly, the `imageUsageFlags` member of each `VkVideoFormatPropertiesKHR` structure returned will contain at least the same set of image usage flags.

If the implementation supports using video input, output, or DPB images of a particular format in operations other than video decode/encode then the `imageUsageFlags` member of the corresponding `VkVideoFormatPropertiesKHR` structure returned will include additional image usage flags indicating that.

NOTE

For most use cases, only decode or encode related usage flags are going to be specified. For a use case such as transcode, if the image were to be shared between decode and encode session(s), then both decode and encode related usage flags **can** be set.

Multiple `VkVideoFormatPropertiesKHR` entries **may** be returned with the same `format` member with different `componentMapping`, `imageType`, or `imageTiling` values, as described later.

In addition, a different set of `VkVideoFormatPropertiesKHR` entries **may** be returned depending on the `imageUsage` member of the `VkPhysicalDeviceVideoFormatInfoKHR` structure, even for the same set of video profiles, for example, based on whether encode input, encode DPB, decode output, and/or decode DPB usage is requested.

The application **can** select the parameters returned in the `VkVideoFormatPropertiesKHR` entries and use compatible parameters when creating the input, output, and DPB images. The implementation will report all image creation and usage flags that are valid for images used with the requested video profiles but applications **should** create images only with those that are necessary for the particular use case.

Before creating an image, the application **can** obtain the complete set of supported image format features by calling `vkGetPhysicalDeviceImageFormatProperties2KHR` using parameters derived from the members of one of the reported `VkVideoFormatPropertiesKHR` entries and adding the same `VkVideoProfileListInfoKHR` structure to the `pNext` chain of `VkPhysicalDeviceImageFormatInfo2`.

The following applies to all `VkVideoFormatPropertiesKHR` entries returned by `vkGetPhysicalDeviceVideoFormatPropertiesKHR`:

- `vkGetPhysicalDeviceFormatProperties2KHR` **must** succeed when called with `VkVideoFormatPropertiesKHR::format`
- If `VkVideoFormatPropertiesKHR::imageTiling` is `VK_IMAGE_TILING_OPTIMAL`, then the `optimalTilingFeatures` returned by `vkGetPhysicalDeviceFormatProperties2KHR` **must** include all format features required by the image usage flags reported in `VkVideoFormatPropertiesKHR::imageUsageFlags` for the format, as indicated in the [Format Feature Dependent Usage Flags](#) section.
- If `VkVideoFormatPropertiesKHR::imageTiling` is `VK_IMAGE_TILING_LINEAR`, then the `linearTilingFeatures` returned by `vkGetPhysicalDeviceFormatProperties2KHR` **must** include all format features required by the image usage flags reported in `VkVideoFormatPropertiesKHR::imageUsageFlags` for the format, as indicated in the [Format Feature Dependent Usage Flags](#)

section.

- `vkGetPhysicalDeviceImageFormatProperties2KHR` **must** succeed when called with a `VkPhysicalDeviceImageFormatInfo2` structure containing the following information:
 - The `pNext` chain including the same `VkVideoProfileListInfoKHR` structure used to call `vkGetPhysicalDeviceVideoFormatPropertiesKHR`.
 - `format` set to the value of `VkVideoFormatPropertiesKHR::format`.
 - `type` set to the value of `VkVideoFormatPropertiesKHR::imageType`.
 - `tiling` set to the value of `VkVideoFormatPropertiesKHR::imageTiling`.
 - `usage` set to the value of `VkVideoFormatPropertiesKHR::imageUsageFlags`.
 - `flags` set to the value of `VkVideoFormatPropertiesKHR::imageCreateFlags`.

The `componentMapping` member of `VkVideoFormatPropertiesKHR` defines the ordering of the Y'C_BC_R color channels from the perspective of the video codec operations specified in `VkVideoProfileListInfoKHR`. For example, if the implementation produces video decode output with the format `VK_FORMAT_G8_B8R8_2PLANE_420_UNORM` where the blue and red chrominance channels are swapped then the `componentMapping` member of the corresponding `VkVideoFormatPropertiesKHR` structure will have the following member values:

```
components.r = VK_COMPONENT_SWIZZLE_B;      // Cb component
components.g = VK_COMPONENT_SWIZZLE_IDENTITY; // Y component
components.b = VK_COMPONENT_SWIZZLE_R;      // Cr component
components.a = VK_COMPONENT_SWIZZLE_IDENTITY; // unused, defaults to 1.0
```

Valid Usage

- VUID-vkGetPhysicalDeviceVideoFormatPropertiesKHR-pNext-06812
The `pNext` chain of `pVideoFormatInfo` **must** include a `VkVideoProfileListInfoKHR` structure with `profileCount` greater than 0

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceVideoFormatPropertiesKHR-physicalDevice-parameter-`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceVideoFormatPropertiesKHR-pVideoFormatInfo-parameter-`pVideoFormatInfo` **must** be a valid pointer to a valid `VkPhysicalDeviceVideoFormatInfoKHR` structure
- VUID-vkGetPhysicalDeviceVideoFormatPropertiesKHR-pVideoFormatPropertyCount-parameter-`pVideoFormatPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceVideoFormatPropertiesKHR-pVideoFormatProperties-parameter-
If the value referenced by `pVideoFormatPropertyCount` is not 0, and `pVideoFormatProperties`

is not `NULL`, `pVideoFormatProperties` **must** be a valid pointer to an array of `pVideoFormatPropertyCount` `VkVideoFormatPropertiesKHR` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR`
- `VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR`
- `VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR`
- `VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR`
- `VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR`

The `VkPhysicalDeviceVideoFormatInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkPhysicalDeviceVideoFormatInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkImageUsageFlags  imageUsage;
} VkPhysicalDeviceVideoFormatInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imageUsage` is a bitmask of `VkImageUsageFlagBits` specifying the intended usage of the video images.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoFormatInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_FORMAT_INFO_KHR`
- VUID-VkPhysicalDeviceVideoFormatInfoKHR-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkVideoProfileListInfoKHR`
- VUID-VkPhysicalDeviceVideoFormatInfoKHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkPhysicalDeviceVideoFormatInfoKHR-imageUsage-parameter

imageUsage must be a valid combination of [VkImageUsageFlagBits](#) values

- VUID-VkPhysicalDeviceVideoFormatInfoKHR-imageUsage-requiredbitmask

imageUsage must not be 0

The [VkVideoFormatPropertiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoFormatPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkFormat            format;
    VkComponentMapping  componentMapping;
    VkImageCreateFlags  imageCreateFlags;
    VkImageType         imageType;
    VkImageTiling       imageTiling;
    VkImageUsageFlags   imageUsageFlags;
} VkVideoFormatPropertiesKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is NULL or a pointer to a structure extending this structure.
- **format** is a [VkFormat](#) that specifies the format that **can** be used with the specified video profiles and image usages.
- **componentMapping** defines the color channel order used for the format. **format** along with **componentMapping** describe how the color channels are ordered when producing video decoder output or are expected to be ordered in video encoder input, when applicable. If the **format** reported does not require component swizzling then all members of **componentMapping** will be set to [VK_COMPONENT_SWIZZLE_IDENTITY](#).
- **imageCreateFlags** is a bitmask of [VkImageCreateFlagBits](#) specifying the supported image creation flags for the format.
- **imageType** is a [VkImageType](#) that specifies the image type the format **can** be used with.
- **imageTiling** is a [VkImageTiling](#) that specifies the image tiling the format **can** be used with.
- **imageUsageFlags** is a bitmask of [VkImageUsageFlagBits](#) specifying the supported image usage flags for the format.

Valid Usage (Implicit)

- VUID-VkVideoFormatPropertiesKHR-sType-sType
sType must be [VK_STRUCTURE_TYPE_VIDEO_FORMAT_PROPERTIES_KHR](#)
- VUID-VkVideoFormatPropertiesKHR-pNext-pNext
pNext must be NULL

37.5. Video Sessions

Video sessions are objects that represent and maintain the state needed to perform video decode or encode operations using a specific video profile.

In case of video encode profiles this includes the current [rate control](#) configuration and the currently set [video encode quality level](#).

Video sessions are represented by `VkVideoSessionKHR` handles:

```
// Provided by VK_KHR_video_queue
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkVideoSessionKHR)
```

37.5.1. Creating a Video Session

To create a video session object, call:

```
// Provided by VK_KHR_video_queue
VkResult vkCreateVideoSessionKHR(
    VkDevice                                device,
    const VkVideoSessionCreateInfoKHR*      pCreateInfo,
    const VkAllocationCallbacks*           pAllocator,
    VkVideoSessionKHR*                     pVideoSession);
```

- `device` is the logical device that creates the video session.
- `pCreateInfo` is a pointer to a [VkVideoSessionCreateInfoKHR](#) structure containing parameters to be used to create the video session.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pVideoSession` is a pointer to a [VkVideoSessionKHR](#) handle in which the resulting video session object is returned.

The resulting video session object is said to be created with the video codec operation specified in `pCreateInfo->pVideoProfile->videoCodecOperation`.

The name and version of the codec-specific Video Std header to be used with the video session is specified by the [VkExtensionProperties](#) structure pointed to by `pCreateInfo->pStdHeaderVersion`. If a non-existent or unsupported Video Std header version is specified in `pCreateInfo->pStdHeaderVersion->specVersion`, then this command returns `VK_ERROR_VIDEO_STD_VERSION_NOT_SUPPORTED_KHR`.

Video session objects are created in *uninitialized* state. In order to transition the video session into *initial* state, the application **must** issue a [vkCmdControlVideoCodingKHR](#) command with [VkVideoCodingControlInfoKHR::flags](#) including `VK_VIDEO_CODING_CONTROL_RESET_BIT_KHR`.

Video session objects also maintain the [state](#) of the DPB. The number of DPB slots usable with the created video session is specified in `pCreateInfo->maxDpbSlots`, and each slot is initially in the

inactive state.

Each DPB slot maintained by the created video session **can** refer to a reference picture representing a video frame.

In addition, if the `videoCodecOperation` member of the `VkVideoProfileInfoKHR` structure pointed to by `pCreateInfo->pVideoProfile` is `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR` and the `pictureLayout` member of the `VkVideoDecodeH264ProfileInfoKHR` structure provided in the `VkVideoProfileInfoKHR::pNext` chain is not `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_PROGRESSIVE_KHR`, then the created video session supports *interlaced* frames and each DPB slot maintained by the created video session **can** instead refer to separate top field and bottom field reference pictures that together **can** represent a full video frame. In this case, it is up to the application, driven by the video content, whether it associates any individual DPB slot with separate top and/or bottom field pictures or a single picture representing a full frame.

The created video session **can** be used to perform video coding operations using video frames up to the maximum size specified in `pCreateInfo->maxCodedExtent`. The minimum frame size allowed is implicitly derived from `VkVideoCapabilitiesKHR::minCodedExtent`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile specified by `pCreateInfo->pVideoProfile`. Accordingly, the created video session is said to be created with a `minCodedExtent` equal to that.

In case of video session objects created with a video encode operation, implementations **may** return the `VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR` error if any of the specified Video Std parameters do not adhere to the syntactic or semantic requirements of the used video compression standard, or if values derived from parameters according to the rules defined by the used video compression standard do not adhere to the capabilities of the video compression standard or the implementation.

NOTE	Applications should not rely on the <code>VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR</code> error being returned by any command as a means to verify Video Std parameters, as implementations are not required to report the error in any specific set of cases.
-------------	--

Valid Usage (Implicit)

- VUID-vkCreateVideoSessionKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkCreateVideoSessionKHR-pCreateInfo-parameter
`pCreateInfo` **must** be a valid pointer to a valid `VkVideoSessionCreateInfoKHR` structure
- VUID-vkCreateVideoSessionKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkCreateVideoSessionKHR-pVideoSession-parameter
`pVideoSession` **must** be a valid pointer to a `VkVideoSessionKHR` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`
- `VK_ERROR_VIDEO_STD_VERSION_NOT_SUPPORTED_KHR`
- `VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR`

The `VkVideoSessionCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoSessionCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           queueFamilyIndex;
    VkVideoSessionCreateFlagsKHR flags;
    const VkVideoProfileInfoKHR* pVideoProfile;
    VkFormat           pictureFormat;
    VkExtent2D         maxCodedExtent;
    VkFormat           referencePictureFormat;
    uint32_t           maxDpbSlots;
    uint32_t           maxActiveReferencePictures;
    const VkExtensionProperties* pStdHeaderVersion;
} VkVideoSessionCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `queueFamilyIndex` is the index of the queue family the created video session will be used with.
- `flags` is a bitmask of `VkVideoSessionCreateFlagBitsKHR` specifying creation flags.
- `pVideoProfile` is a pointer to a `VkVideoProfileInfoKHR` structure specifying the video profile the created video session will be used with.
- `pictureFormat` is the image format the created video session will be used with. If `pVideoProfile->videoCodecOperation` specifies a decode operation, then `pictureFormat` is the image format of `decode output pictures` usable with the created video session. If `pVideoProfile->videoCodecOperation` specifies an encode operation, then `pictureFormat` is the image format of `encode input pictures` usable with the created video session.
- `maxCodedExtent` is the maximum width and height of the coded frames the created video session will be used with.
- `referencePictureFormat` is the image format of `reference pictures` stored in the `DPB` the created

video session will be used with.

- `maxDpbSlots` is the maximum number of `DPB Slots` that **can** be used with the created video session.
- `maxActiveReferencePictures` is the maximum number of `active reference pictures` that **can** be used in a single video coding operation using the created video session.
- `pStdHeaderVersion` is a pointer to a `VkExtensionProperties` structure requesting the Video Std header version to use for the `videoCodecOperation` specified in `pVideoProfile`.

Valid Usage

- VUID-VkVideoSessionCreateInfoKHR-protectedMemory-07189
If the `protectedMemory` feature is not enabled or if `VkVideoCapabilitiesKHR::flags` does not include `VK_VIDEO_CAPABILITY_PROTECTED_CONTENT_BIT_KHR`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile specified by `pVideoProfile`, then `flags` **must** not include `VK_VIDEO_SESSION_CREATE_PROTECTED_CONTENT_BIT_KHR`
- VUID-VkVideoSessionCreateInfoKHR-flags-08371
If `flags` includes `VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR`, then `videoMaintenance1` **must** be enabled
- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-04845
`pVideoProfile` **must** be a `supported video profile`
- VUID-VkVideoSessionCreateInfoKHR-maxDpbSlots-04847
`maxDpbSlots` **must** be less than or equal to `VkVideoCapabilitiesKHR::maxDpbSlots`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile specified by `pVideoProfile`
- VUID-VkVideoSessionCreateInfoKHR-maxActiveReferencePictures-04849
`maxActiveReferencePictures` **must** be less than or equal to `VkVideoCapabilitiesKHR::maxActiveReferencePictures`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile specified by `pVideoProfile`
- VUID-VkVideoSessionCreateInfoKHR-maxDpbSlots-04850
If either `maxDpbSlots` or `maxActiveReferencePictures` is `0`, then both **must** be `0`
- VUID-VkVideoSessionCreateInfoKHR-maxCodedExtent-04851
`maxCodedExtent` **must** be between `VkVideoCapabilitiesKHR::minCodedExtent` and `VkVideoCapabilitiesKHR::maxCodedExtent`, inclusive, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile specified by `pVideoProfile`
- VUID-VkVideoSessionCreateInfoKHR-referencePictureFormat-04852
If `pVideoProfile->videoCodecOperation` specifies a decode operation and `maxActiveReferencePictures` is greater than `0`, then `referencePictureFormat` **must** be one of the supported decode DPB formats, as returned by `vkGetPhysicalDeviceVideoFormatPropertiesKHR` in `VkVideoFormatPropertiesKHR::format` when called with the `imageUsage` member of its `pVideoFormatInfo` parameter containing `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`, and with a `VkVideoProfileListInfoKHR` structure specified in the `pNext` chain of its `pVideoFormatInfo` parameter whose `pProfiles`

member contains an element matching `pVideoProfile`

- VUID-VkVideoSessionCreateInfoKHR-referencePictureFormat-06814

If `pVideoProfile->videoCodecOperation` specifies an encode operation and `maxActiveReferencePictures` is greater than 0, then `referencePictureFormat` **must** be one of the supported decode DPB formats, as returned by then `referencePictureFormat` **must** be one of the supported encode DPB formats, as returned by `vkGetPhysicalDeviceVideoFormatPropertiesKHR` in `VkVideoFormatPropertiesKHR::format` when called with the `imageUsage` member of its `pVideoFormatInfo` parameter containing `VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR`, and with a `VkVideoProfileListInfoKHR` structure specified in the `pNext` chain of its `pVideoFormatInfo` parameter whose `pProfiles` member contains an element matching `pVideoProfile`

- VUID-VkVideoSessionCreateInfoKHR-pictureFormat-04853

If `pVideoProfile->videoCodecOperation` specifies a decode operation, then `pictureFormat` **must** be one of the supported decode output formats, as returned by `vkGetPhysicalDeviceVideoFormatPropertiesKHR` in `VkVideoFormatPropertiesKHR::format` when called with the `imageUsage` member of its `pVideoFormatInfo` parameter containing `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`, and with a `VkVideoProfileListInfoKHR` structure specified in the `pNext` chain of its `pVideoFormatInfo` parameter whose `pProfiles` member contains an element matching `pVideoProfile`

- VUID-VkVideoSessionCreateInfoKHR-pictureFormat-04854

If `pVideoProfile->videoCodecOperation` specifies an encode operation, then `pictureFormat` **must** be one of the supported encode input formats, as returned by `vkGetPhysicalDeviceVideoFormatPropertiesKHR` in `VkVideoFormatPropertiesKHR::format` when called with the `imageUsage` member of its `pVideoFormatInfo` parameter containing `VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR`, and with a `VkVideoProfileListInfoKHR` structure specified in the `pNext` chain of its `pVideoFormatInfo` parameter whose `pProfiles` member contains an element matching `pVideoProfile`

- VUID-VkVideoSessionCreateInfoKHR-pStdHeaderVersion-07190

`pStdHeaderVersion->extensionName` **must** match `VkVideoCapabilitiesKHR::stdHeaderVersion.extensionName`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile specified by `pVideoProfile`

- VUID-VkVideoSessionCreateInfoKHR-pStdHeaderVersion-07191

`pStdHeaderVersion->specVersion` **must** be less than or equal to `VkVideoCapabilitiesKHR::stdHeaderVersion.specVersion`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile specified by `pVideoProfile`

- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-08251

If `pVideoProfile->videoCodecOperation` is `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and the `pNext` chain of this structure includes a `VkVideoEncodeH264SessionCreateInfoKHR` structure, then its `maxLevelIdc` member **must** be less than or equal to `VkVideoEncodeH264CapabilitiesKHR::maxLevelIdc`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile specified in `pVideoProfile`

- VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-08252

If `pVideoProfile->videoCodecOperation` is `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and the `pNext` chain of this structure includes a `VkVideoEncodeH265SessionCreateInfoKHR` structure, then its `maxLevelIdc` member **must**

be less than or equal to `VkVideoEncodeH265CapabilitiesKHR::maxLevelIdc`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile specified in `pVideoProfile`

Valid Usage (Implicit)

- `VUID-VkVideoSessionCreateInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_SESSION_CREATE_INFO_KHR`
- `VUID-VkVideoSessionCreateInfoKHR-pNext-pNext`
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkVideoEncodeH264SessionCreateInfoKHR` or `VkVideoEncodeH265SessionCreateInfoKHR`
- `VUID-VkVideoSessionCreateInfoKHR-sType-unique`
The `sType` value of each struct in the `pNext` chain **must** be unique
- `VUID-VkVideoSessionCreateInfoKHR-flags-parameter`
`flags` **must** be a valid combination of `VkVideoSessionCreateFlagBitsKHR` values
- `VUID-VkVideoSessionCreateInfoKHR-pVideoProfile-parameter`
`pVideoProfile` **must** be a valid pointer to a valid `VkVideoProfileInfoKHR` structure
- `VUID-VkVideoSessionCreateInfoKHR-pictureFormat-parameter`
`pictureFormat` **must** be a valid `VkFormat` value
- `VUID-VkVideoSessionCreateInfoKHR-referencePictureFormat-parameter`
`referencePictureFormat` **must** be a valid `VkFormat` value
- `VUID-VkVideoSessionCreateInfoKHR-pStdHeaderVersion-parameter`
`pStdHeaderVersion` **must** be a valid pointer to a valid `VkExtensionProperties` structure

Bits which **can** be set in `VkVideoSessionCreateInfoKHR::flags` are:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoSessionCreateFlagBitsKHR {
    VK_VIDEO_SESSION_CREATE_PROTECTED_CONTENT_BIT_KHR = 0x00000001,
    // Provided by VK_KHR_video_encode_queue
    VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_PARAMETER_OPTIMIZATIONS_BIT_KHR = 0x00000002,
    // Provided by VK_KHR_video_maintenance1
    VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR = 0x00000004,
} VkVideoSessionCreateFlagBitsKHR;
```

- `VK_VIDEO_SESSION_CREATE_PROTECTED_CONTENT_BIT_KHR` specifies that the video session uses protected video content.
- `VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_PARAMETER_OPTIMIZATIONS_BIT_KHR` specifies that the implementation is allowed to **override** video session parameters and other codec-specific encoding parameters to optimize video encode operations based on the use case information specified in the **video profile** and the used **video encode quality level**.

NOTE

Not

specifying

`VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_PARAMETER_OPTIMIZATIONS_BIT_KHR` does not guarantee that the implementation will not do any codec-specific parameter overrides, as certain overrides are necessary for the correct operation of the video encoder implementation due to limitations to the available encoding tools on that implementation. This flag, however, enables the implementation to apply further optimizing overrides.

- `VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR` specifies that queries within video coding scopes using the created video session are [executed inline](#) with video coding operations.

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoSessionCreateFlagsKHR;
```

`VkVideoSessionCreateFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoSessionCreateFlagBitsKHR`.

37.5.2. Destroying a Video Session

To destroy a video session, call:

```
// Provided by VK_KHR_video_queue
void vkDestroyVideoSessionKHR(
    VkDevice                device,
    VkVideoSessionKHR       videoSession,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the video session.
- `videoSession` is the video session to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyVideoSessionKHR-videoSession-07192
All submitted commands that refer to `videoSession` **must** have completed execution
- VUID-vkDestroyVideoSessionKHR-videoSession-07193
If `VkAllocationCallbacks` were provided when `videoSession` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyVideoSessionKHR-videoSession-07194
If no `VkAllocationCallbacks` were provided when `videoSession` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyVideoSessionKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkDestroyVideoSessionKHR-videoSession-parameter
If `videoSession` is not `VK_NULL_HANDLE`, `videoSession` **must** be a valid `VkVideoSessionKHR` handle
- VUID-vkDestroyVideoSessionKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyVideoSessionKHR-videoSession-parent
If `videoSession` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `videoSession` **must** be externally synchronized

37.5.3. Video Session Memory Association

After creating a video session object, and before the object **can** be used to record video coding operations into command buffers using it, the application **must** allocate and bind device memory to the video session. Device memory is allocated separately (see [Device Memory](#)) and then associated with the video session.

Video sessions **may** have multiple memory bindings identified by unique unsigned integer values. Appropriate device memory **must** be bound to each such memory binding before using the video session to record command buffer commands with it.

To determine the memory requirements for a video session object, call:

```
// Provided by VK_KHR_video_queue
VkResult vkGetVideoSessionMemoryRequirementsKHR(
    VkDevice                device,
    VkVideoSessionKHR       videoSession,
    uint32_t*               pMemoryRequirementsCount,
    VkVideoSessionMemoryRequirementsKHR* pMemoryRequirements);
```

- `device` is the logical device that owns the video session.
- `videoSession` is the video session to query.
- `pMemoryRequirementsCount` is a pointer to an integer related to the number of memory binding requirements available or queried, as described below.
- `pMemoryRequirements` is `NULL` or a pointer to an array of

`VkVideoSessionMemoryRequirementsKHR` structures in which the memory binding requirements of the video session are returned.

If `pMemoryRequirements` is `NULL`, then the number of memory bindings required for the video session is returned in `pMemoryRequirementsCount`. Otherwise, `pMemoryRequirementsCount` **must** point to a variable set by the application to the number of elements in the `pMemoryRequirements` array, and on return the variable is overwritten with the number of memory binding requirements actually written to `pMemoryRequirements`. If `pMemoryRequirementsCount` is less than the number of memory bindings required for the video session, then at most `pMemoryRequirementsCount` elements will be written to `pMemoryRequirements`, and `VK_INCOMPLETE` will be returned, instead of `VK_SUCCESS`, to indicate that not all required memory binding requirements were returned.

Valid Usage (Implicit)

- VUID-vkGetVideoSessionMemoryRequirementsKHR-device-parameter `device` **must** be a valid `VkDevice` handle
- VUID-vkGetVideoSessionMemoryRequirementsKHR-videoSession-parameter `videoSession` **must** be a valid `VkVideoSessionKHR` handle
- VUID-vkGetVideoSessionMemoryRequirementsKHR-pMemoryRequirementsCount-parameter `pMemoryRequirementsCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetVideoSessionMemoryRequirementsKHR-pMemoryRequirements-parameter
If the value referenced by `pMemoryRequirementsCount` is not `0`, and `pMemoryRequirements` is not `NULL`, `pMemoryRequirements` **must** be a valid pointer to an array of `pMemoryRequirementsCount` `VkVideoSessionMemoryRequirementsKHR` structures
- VUID-vkGetVideoSessionMemoryRequirementsKHR-videoSession-parent `videoSession` **must** have been created, allocated, or retrieved from `device`

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

None

The `VkVideoSessionMemoryRequirementsKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoSessionMemoryRequirementsKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           memoryBindIndex;
    VkMemoryRequirements memoryRequirements;
} VkVideoSessionMemoryRequirementsKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **memoryBindIndex** is the index of the memory binding.
- **memoryRequirements** is a [VkMemoryRequirements](#) structure in which the requested memory binding requirements for the binding index specified by **memoryBindIndex** are returned.

Valid Usage (Implicit)

- VUID-VkVideoSessionMemoryRequirementsKHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_VIDEO_SESSION_MEMORY_REQUIREMENTS_KHR**
- VUID-VkVideoSessionMemoryRequirementsKHR-pNext-pNext
pNext must be **NULL**

To attach memory to a video session object, call:

```
// Provided by VK_KHR_video_queue
VkResult vkBindVideoSessionMemoryKHR(
    VkDevice                device,
    VkVideoSessionKHR       videoSession,
    uint32_t                bindSessionMemoryInfoCount,
    const VkBindVideoSessionMemoryInfoKHR* pBindSessionMemoryInfos);
```

- **device** is the logical device that owns the video session.
- **videoSession** is the video session to be bound with device memory.
- **bindSessionMemoryInfoCount** is the number of elements in **pBindSessionMemoryInfos**.
- **pBindSessionMemoryInfos** is a pointer to an array of **bindSessionMemoryInfoCount** [VkBindVideoSessionMemoryInfoKHR](#) structures specifying memory regions to be bound to specific memory bindings of the video session.

The valid usage statements below refer to the [VkMemoryRequirements](#) structure corresponding to a specific element of **pBindSessionMemoryInfos**, which is defined as follows:

- If the **memoryBindIndex** member of the element of **pBindSessionMemoryInfos** in question matches the **memoryBindIndex** member of one of the elements returned in **pMemoryRequirements** when [vkGetVideoSessionMemoryRequirementsKHR](#) is called with the same **videoSession** and with

`pMemoryRequirementsCount` equal to `bindSessionMemoryInfoCount`, then the `memoryRequirements` member of that element of `pMemoryRequirements` is the `VkMemoryRequirements` structure corresponding to the element of `pBindSessionMemoryInfos` in question.

- Otherwise the element of `pBindSessionMemoryInfos` in question is said to not have a corresponding `VkMemoryRequirements` structure.

Valid Usage

- VUID-vkBindVideoSessionMemoryKHR-videoSession-07195
The memory binding of `videoSession` identified by the `memoryBindIndex` member of any element of `pBindSessionMemoryInfos` **must** not already be backed by a memory object
- VUID-vkBindVideoSessionMemoryKHR-memoryBindIndex-07196
The `memoryBindIndex` member of each element of `pBindSessionMemoryInfos` **must** be unique within `pBindSessionMemoryInfos`
- VUID-vkBindVideoSessionMemoryKHR-pBindSessionMemoryInfos-07197
Each element of `pBindSessionMemoryInfos` **must** have a corresponding `VkMemoryRequirements` structure
- VUID-vkBindVideoSessionMemoryKHR-pBindSessionMemoryInfos-07198
If an element of `pBindSessionMemoryInfos` has a corresponding `VkMemoryRequirements` structure, then the `memory` member of that element of `pBindSessionMemoryInfos` **must** have been allocated using one of the memory types allowed in the `memoryTypeBits` member of the corresponding `VkMemoryRequirements` structure
- VUID-vkBindVideoSessionMemoryKHR-pBindSessionMemoryInfos-07199
If an element of `pBindSessionMemoryInfos` has a corresponding `VkMemoryRequirements` structure, then the `memoryOffset` member of that element of `pBindSessionMemoryInfos` **must** be an integer multiple of the `alignment` member of the corresponding `VkMemoryRequirements` structure
- VUID-vkBindVideoSessionMemoryKHR-pBindSessionMemoryInfos-07200
If an element of `pBindSessionMemoryInfos` has a corresponding `VkMemoryRequirements` structure, then the `memorySize` member of that element of `pBindSessionMemoryInfos` **must** equal the `size` member of the corresponding `VkMemoryRequirements` structure

Valid Usage (Implicit)

- VUID-vkBindVideoSessionMemoryKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkBindVideoSessionMemoryKHR-videoSession-parameter
`videoSession` **must** be a valid `VkVideoSessionKHR` handle
- VUID-vkBindVideoSessionMemoryKHR-pBindSessionMemoryInfos-parameter
`pBindSessionMemoryInfos` **must** be a valid pointer to an array of `bindSessionMemoryInfoCount` valid `VkBindVideoSessionMemoryInfoKHR` structures
- VUID-vkBindVideoSessionMemoryKHR-bindSessionMemoryInfoCount-arraylength
`bindSessionMemoryInfoCount` **must** be greater than 0

- VUID-vkBindVideoSessionMemoryKHR-videoSession-parent
`videoSession` **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `videoSession` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkBindVideoSessionMemoryInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkBindVideoSessionMemoryInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           memoryBindIndex;
    VkDeviceMemory      memory;
    VkDeviceSize        memoryOffset;
    VkDeviceSize        memorySize;
} VkBindVideoSessionMemoryInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `memoryBindIndex` is the memory binding index to bind memory to.
- `memory` is the allocated device memory to be bound to the video session's memory binding with index `memoryBindIndex`.
- `memoryOffset` is the start offset of the region of `memory` which is to be bound.
- `memorySize` is the size in bytes of the region of `memory`, starting from `memoryOffset` bytes, to be bound.

Valid Usage

- VUID-VkBindVideoSessionMemoryInfoKHR-memoryOffset-07201
`memoryOffset` **must** be less than the size of `memory`
- VUID-VkBindVideoSessionMemoryInfoKHR-memorySize-07202

`memorySize` **must** be less than or equal to the size of `memory` minus `memoryOffset`

Valid Usage (Implicit)

- `VUID-VkBindVideoSessionMemoryInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_BIND_VIDEO_SESSION_MEMORY_INFO_KHR`
- `VUID-VkBindVideoSessionMemoryInfoKHR-pNext-pNext`
`pNext` **must** be `NULL`
- `VUID-VkBindVideoSessionMemoryInfoKHR-memory-parameter`
`memory` **must** be a valid `VkDeviceMemory` handle

37.6. Video Profile Compatibility

Resources and query pools used with a particular video session **must** be compatible with the `video profile` the video session was created with.

A `VkBuffer` is compatible with a video profile if it was created with the `VkBufferCreateInfo::pNext` chain including a `VkVideoProfileListInfoKHR` structure with its `pProfiles` member containing an element matching the `VkVideoProfileInfoKHR` structure chain describing the video profile, and `VkBufferCreateInfo::usage` including at least one bit specific to video coding usage.

- `VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR`
- `VK_BUFFER_USAGE_VIDEO_DECODE_DST_BIT_KHR`
- `VK_BUFFER_USAGE_VIDEO_ENCODE_SRC_BIT_KHR`
- `VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR`

A `VkBuffer` is also compatible with a video profile if it was created with `VkBufferCreateInfo::flags` including `VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`.

A `VkImage` is compatible with a video profile if it was created with the `VkImageCreateInfo::pNext` chain including a `VkVideoProfileListInfoKHR` structure with its `pProfiles` member containing an element matching the `VkVideoProfileInfoKHR` structure chain describing the video profile, and `VkImageCreateInfo::usage` including at least one bit specific to video coding usage.

- `VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR`
- `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`
- `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`
- `VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR`
- `VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR`
- `VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR`

A `VkImage` is also compatible with a video profile if all of the following conditions are true for the `VkImageCreateInfo` structure the image was created with:

- `VkImageCreateInfo::flags` included `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`.
- The list of `VkVideoFormatPropertiesKHR` structures, obtained by calling `vkGetPhysicalDeviceVideoFormatPropertiesKHR` with `VkPhysicalDeviceVideoFormatInfoKHR::imageUsage` equal to the `VkImageCreateInfo::usage` the image was created with and the `VkPhysicalDeviceVideoFormatInfoKHR::pNext` chain including a `VkVideoProfileListInfoKHR` structure with its `pProfiles` member containing a single array element specifying the `VkVideoProfileInfoKHR` structure chain describing the video profile in question, contains an element for which all of the following conditions are true with respect to the `VkImageCreateInfo` structure the image was created with:
 - `VkImageCreateInfo::format` equals `VkVideoFormatPropertiesKHR::format`.
 - `VkImageCreateInfo::flags` only contains `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR` and/or bits also set in `VkVideoFormatPropertiesKHR::imageCreateFlags`.

NOTE

Specifying `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR` when creating `decode output pictures` or `encode input pictures` is always supported when `videoMaintenance1` is enabled, regardless of the supported `VkImageCreateFlags` reported in `VkVideoFormatPropertiesKHR::imageCreateFlags`. Accordingly, implementations **should** not report `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR` in `VkVideoFormatPropertiesKHR::imageCreateFlags` for any video format.

- `VkImageCreateInfo::imageType` equals `VkVideoFormatPropertiesKHR::imageType`.
- `VkImageCreateInfo::tiling` equals `VkVideoFormatPropertiesKHR::imageTiling`.
- `VkImageCreateInfo::usage` only contains bits also set in `VkVideoFormatPropertiesKHR::imageUsageFlags`, or `VkImageCreateInfo::flags` includes `VK_IMAGE_CREATE_EXTENDED_USAGE_BIT`.

NOTE

While some of these rules allow creating buffer or image resources that **may** be compatible with any video profile, applications **should** still prefer to include the specific video profiles the buffer or image resource is expected to be used with (through a `VkVideoProfileListInfoKHR` structure included in the `pNext` chain of the corresponding create info structure) whenever the information about the complete set of video profiles is available at resource creation time, to enable the implementation to optimize the created resource for the specific use case. In the absence of that information, the implementation **may** have to make conservative decisions about the memory requirements or representation of the resource.

A `VkImageView` is compatible with a video profile if the `VkImage` it was created from is also compatible with that video profile.

A `VkQueryPool` is compatible with a video profile if it was created with the `VkQueryPoolCreateInfo::pNext` chain including a `VkVideoProfileInfoKHR` structure chain describing the same video profile, and `VkQueryPoolCreateInfo::queryType` having one of the following values:

- `VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR`
- `VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR`

37.7. Video Session Parameters

Video session parameters objects **can** store preprocessed codec-specific parameters used with a compatible video session, and enable reducing the number of parameters needed to be provided and processed by the implementation while recording video coding operations into command buffers.

Parameters stored in such objects are *immutable* to facilitate the concurrent use of the stored parameters in multiple threads. At the same time, new parameters **can** be added to existing objects using the [vkUpdateVideoSessionParametersKHR](#) command.

In order to support concurrent use of the stored immutable parameters while also allowing the video session parameters object to be extended with new parameters, each video session parameters object maintains an *update sequence counter* that is set to **0** at object creation time and **must** be incremented by each subsequent update operation.

Certain video sequences that adhere to particular video compression standards permit updating previously supplied parameters. If a parameter update is necessary, the application has the following options:

- Cache the set of parameters on the application side and create a new video session parameters object adding all the parameters with appropriate changes, as necessary; or
- Create a new video session parameters object providing only the updated parameters and the previously used object as the template, which ensures that parameters not specified at creation time will be copied unmodified from the template object.

The actual types of parameters that **can** be stored and the capacity for individual parameter types, and the methods of initializing, updating, and referring to individual parameters are specific to the video codec operation the video session parameters object was created with.

- For [VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR](#) these are defined in the [H.264 Decode Parameter Sets](#) section.
- For [VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR](#) these are defined in the [H.265 Decode Parameter Sets](#) section.
- For [VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR](#) these are defined in the [AV1 Decode Parameter Sets](#) section.
- For [VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR](#) these are defined in the [H.264 Encode Parameter Sets](#) section.
- For [VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR](#) these are defined in the [H.265 Encode Parameter Sets](#) section.

Video session parameters objects created with an encode operation are further specialized based on the [video encode quality level](#) the video session parameters are used with, as implementations **may** apply different sets of [parameter overrides](#) depending on the used quality level. This enables implementations to store the potentially optimized set of parameters in these objects, further limiting the necessary processing required while recording video encode operations into command buffers.

Video session parameters are represented by `VkVideoSessionParametersKHR` handles:

```
// Provided by VK_KHR_video_queue
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkVideoSessionParametersKHR)
```

37.7.1. Creating Video Session Parameters

To create a video session parameters object, call:

```
// Provided by VK_KHR_video_queue
VkResult vkCreateVideoSessionParametersKHR(
    VkDevice device,
    const VkVideoSessionParametersCreateInfoKHR* pCreateInfo,
    const VkAllocationCallbacks* pAllocator,
    VkVideoSessionParametersKHR* pVideoSessionParameters);
```

- `device` is the logical device that creates the video session parameters object.
- `pCreateInfo` is a pointer to `VkVideoSessionParametersCreateInfoKHR` structure containing parameters to be used to create the video session parameters object.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.
- `pVideoSessionParameters` is a pointer to a `VkVideoSessionParametersKHR` handle in which the resulting video session parameters object is returned.

The resulting video session parameters object is said to be created with the video codec operation `pCreateInfo->videoSession` was created with.

Video session parameters objects created with an encode operation are always created with respect to a [video encode quality level](#). By default, the created video session parameters objects are created with quality level zero, unless otherwise specified by including a `VkVideoEncodeQualityLevelInfoKHR` structure in the `pCreateInfo->pNext` chain, in which case the video session parameters object is created with the quality level specified in `VkVideoEncodeQualityLevelInfoKHR::qualityLevel`.

If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then it will be used as a template for constructing the new video session parameters object. This happens by first adding any parameters according to the additional creation parameters provided in the `pCreateInfo->pNext` chain, followed by adding any parameters from the template object that have a key that does not match the key of any of the already added parameters.

For video session parameters objects created with an encode operation, the template object specified in `pCreateInfo->videoSessionParametersTemplate` **must** have been created with the same [video encode quality level](#) as the newly created object.

NOTE

This means that codec-specific parameters stored in video session parameters objects **can** only be reused across different video encode quality levels by re-specifying them, as previously created video session parameters against other

quality levels **cannot** be used as template because the original codec-specific parameters (before the implementation **may** have applied [parameter overrides](#)) **may** no longer be available in them for the purposes of constructing the derived object.

If `pCreateInfo->videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then the created video session parameters object will initially contain the following sets of parameter entries:

- `StdVideoH264SequenceParameterSet` structures representing [H.264 SPS](#) entries, as follows:
 - If the `pParametersAddInfo` member of the `VkVideoDecodeH264SessionParametersCreateInfoKHR` structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH264SequenceParameterSet` entries specified in `pParametersAddInfo->pStdSPSs` are added first;
 - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH264SequenceParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `seq_parameter_set_id`.
- `StdVideoH264PictureParameterSet` structures representing [H.264 PPS](#) entries, as follows:
 - If the `pParametersAddInfo` member of the `VkVideoDecodeH264SessionParametersCreateInfoKHR` structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH264PictureParameterSet` entries specified in `pParametersAddInfo->pStdPPSs` are added first;
 - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH264PictureParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `seq_parameter_set_id` and `pic_parameter_set_id`.

If `pCreateInfo->videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then the created video session parameters object will initially contain the following sets of parameter entries:

- `StdVideoH265VideoParameterSet` structures representing [H.265 VPS](#) entries, as follows:
 - If the `pParametersAddInfo` member of the `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH265VideoParameterSet` entries specified in `pParametersAddInfo->pStdVPSs` are added first;
 - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265VideoParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `vps_video_parameter_set_id`.
- `StdVideoH265SequenceParameterSet` structures representing [H.265 SPS](#) entries, as follows:
 - If the `pParametersAddInfo` member of the `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH265SequenceParameterSet`

entries specified in `pParametersAddInfo->pStdSPSs` are added first;

- If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265SequenceParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `sps_video_parameter_set_id` and `sps_seq_parameter_set_id`.
- `StdVideoH265PictureParameterSet` structures representing [H.265 PPS](#) entries, as follows:
 - If the `pParametersAddInfo` member of the `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH265PictureParameterSet` entries specified in `pParametersAddInfo->pStdPPSs` are added first;
 - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265PictureParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id`.

If `pCreateInfo->videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR`, then the created video session parameters object will contain a single [AV1 sequence header](#) represented by a `StdVideoAV1SequenceHeader` structure specified through the `pStdSequenceHeader` member of the `VkVideoDecodeAV1SessionParametersCreateInfoKHR` structure provided in the `pCreateInfo->pNext` chain. As such video session parameters objects **can** only contain a single [AV1 sequence header](#), it is not possible to use a previously created object as a template or subsequently update the created video session parameters object.

If `pCreateInfo->videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the created video session parameters object will initially contain the following sets of parameter entries:

- `StdVideoH264SequenceParameterSet` structures representing [H.264 SPS](#) entries, as follows:
 - If the `pParametersAddInfo` member of the `VkVideoEncodeH264SessionParametersCreateInfoKHR` structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH264SequenceParameterSet` entries specified in `pParametersAddInfo->pStdSPSs` are added first;
 - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH264SequenceParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `seq_parameter_set_id`.
- `StdVideoH264PictureParameterSet` structures representing [H.264 PPS](#) entries, as follows:
 - If the `pParametersAddInfo` member of the `VkVideoEncodeH264SessionParametersCreateInfoKHR` structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH264PictureParameterSet` entries specified in `pParametersAddInfo->pStdPPSs` are added first;
 - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH264PictureParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same

seq_parameter_set_id and pic_parameter_set_id.

If `pCreateInfo->videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the created video session parameters object will initially contain the following sets of parameter entries:

- `StdVideoH265VideoParameterSet` structures representing [H.265 VPS](#) entries, as follows:
 - If the `pParametersAddInfo` member of the `VkVideoEncodeH265SessionParametersCreateInfoKHR` structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH265VideoParameterSet` entries specified in `pParametersAddInfo->pStdVPSs` are added first;
 - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265VideoParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `vps_video_parameter_set_id`.
- `StdVideoH265SequenceParameterSet` structures representing [H.265 SPS](#) entries, as follows:
 - If the `pParametersAddInfo` member of the `VkVideoEncodeH265SessionParametersCreateInfoKHR` structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH265SequenceParameterSet` entries specified in `pParametersAddInfo->pStdSPSs` are added first;
 - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265SequenceParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `sps_video_parameter_set_id` and `sps_seq_parameter_set_id`.
- `StdVideoH265PictureParameterSet` structures representing [H.265 PPS](#) entries, as follows:
 - If the `pParametersAddInfo` member of the `VkVideoEncodeH265SessionParametersCreateInfoKHR` structure provided in the `pCreateInfo->pNext` chain is not `NULL`, then the set of `StdVideoH265PictureParameterSet` entries specified in `pParametersAddInfo->pStdPPSs` are added first;
 - If `pCreateInfo->videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265PictureParameterSet` entry stored in it is copied to the created video session parameters object if the created object does not already contain such an entry with the same `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id`.

In case of video session parameters objects created with a video encode operation, implementations **may** return the `VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR` error if any of the specified Video Std parameters do not adhere to the syntactic or semantic requirements of the used video compression standard, or if values derived from parameters according to the rules defined by the used video compression standard do not adhere to the capabilities of the video compression standard or the implementation.

NOTE

Applications **should** not rely on the `VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR` error being returned by any command as a means to verify Video Std parameters, as implementations are not required to report the error in any specific set of cases.

Valid Usage (Implicit)

- `VUID-vkCreateVideoSessionParametersKHR-device-parameter`
`device` **must** be a valid `VkDevice` handle
- `VUID-vkCreateVideoSessionParametersKHR-pCreateInfo-parameter`
`pCreateInfo` **must** be a valid pointer to a valid `VkVideoSessionParametersCreateInfoKHR` structure
- `VUID-vkCreateVideoSessionParametersKHR-pAllocator-parameter`
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- `VUID-vkCreateVideoSessionParametersKHR-pVideoSessionParameters-parameter`
`pVideoSessionParameters` **must** be a valid pointer to a `VkVideoSessionParametersKHR` handle

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_INITIALIZATION_FAILED`
- `VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR`

The `VkVideoSessionParametersCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoSessionParametersCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkVideoSessionParametersCreateFlagsKHR  flags;
    VkVideoSessionParametersKHR              videoSessionParametersTemplate;
    VkVideoSessionKHR                        videoSession;
} VkVideoSessionParametersCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `videoSessionParametersTemplate` is `VK_NULL_HANDLE` or a valid handle to a `VkVideoSessionParametersKHR` object used as a template for constructing the new video session parameters object.

- `videoSession` is the video session object against which the video session parameters object is going to be created.

Limiting values are defined below that are referenced by the relevant valid usage statements of this structure.

- If `videoSession` was created with the codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then let `StdVideoH264SequenceParameterSet` `spsAddList[]` be the list of `H.264 SPS` entries to add to the created video session parameters object, defined as follows:
 - If the `pParametersAddInfo` member of the `VkVideoDecodeH264SessionParametersCreateInfoKHR` structure provided in the `pNext` chain is not `NULL`, then the set of `StdVideoH264SequenceParameterSet` entries specified in `pParametersAddInfo->pStdSPSs` are added to `spsAddList`;
 - If `videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH264SequenceParameterSet` entry stored in it with `seq_parameter_set_id` not matching any of the entries already in `spsAddList` is added to `spsAddList`.
- If `videoSession` was created with the codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then let `StdVideoH264PictureParameterSet` `ppsAddList[]` be the list of `H.264 PPS` entries to add to the created video session parameters object, defined as follows:
 - If the `pParametersAddInfo` member of the `VkVideoDecodeH264SessionParametersCreateInfoKHR` structure provided in the `pNext` chain is not `NULL`, then the set of `StdVideoH264PictureParameterSet` entries specified in `pParametersAddInfo->pStdPPSs` are added to `ppsAddList`;
 - If `videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH264PictureParameterSet` entry stored in it with `seq_parameter_set_id` or `pic_parameter_set_id` not matching any of the entries already in `ppsAddList` is added to `ppsAddList`.
- If `videoSession` was created with the codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then let `StdVideoH265VideoParameterSet` `vpsAddList[]` be the list of `H.265 VPS` entries to add to the created video session parameters object, defined as follows:
 - If the `pParametersAddInfo` member of the `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure provided in the `pNext` chain is not `NULL`, then the set of `StdVideoH265VideoParameterSet` entries specified in `pParametersAddInfo->pStdVPSs` are added to `vpsAddList`;
 - If `videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265VideoParameterSet` entry stored in it with `vps_video_parameter_set_id` not matching any of the entries already in `vpsAddList` is added to `vpsAddList`.
- If `videoSession` was created with the codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then let `StdVideoH265SequenceParameterSet` `spsAddList[]` be the list of `H.265 SPS` entries to add to the created video session parameters object, defined as follows:

- If the `pParametersAddInfo` member of the `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure provided in the `pNext` chain is not `NULL`, then the set of `StdVideoH265SequenceParameterSet` entries specified in `pParametersAddInfo->pStdSPSs` are added to `spsAddList`;
- If `videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265SequenceParameterSet` entry stored in it with `sps_video_parameter_set_id` or `sps_seq_parameter_set_id` not matching any of the entries already in `spsAddList` is added to `spsAddList`.
- If `videoSession` was created with the codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then let `StdVideoH265PictureParameterSet ppsAddList[]` be the list of `H.265 PPS` entries to add to the created video session parameters object, defined as follows:
 - If the `pParametersAddInfo` member of the `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure provided in the `pNext` chain is not `NULL`, then the set of `StdVideoH265PictureParameterSet` entries specified in `pParametersAddInfo->pStdPPSs` are added to `ppsAddList`;
 - If `videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH265PictureParameterSet` entry stored in it with `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, or `pps_pic_parameter_set_id` not matching any of the entries already in `ppsAddList` is added to `ppsAddList`.
- If `videoSession` was created with an encode operation, then let `uint32_t qualityLevel` be the `video encode quality level` of the created video session parameters object, defined as follows:
 - If the `pNext` chain of this structure includes a `VkVideoEncodeQualityLevelInfoKHR` structure, then `qualityLevel` is equal to `VkVideoEncodeQualityLevelInfoKHR::qualityLevel`.
 - Otherwise `qualityLevel` is `0`
- If `videoSession` was created with the codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then let `StdVideoH264SequenceParameterSet spsAddList[]` be the list of `H.264 SPS` entries to add to the created video session parameters object, defined as follows:
 - If the `pParametersAddInfo` member of the `VkVideoEncodeH264SessionParametersCreateInfoKHR` structure provided in the `pNext` chain is not `NULL`, then the set of `StdVideoH264SequenceParameterSet` entries specified in `pParametersAddInfo->pStdSPSs` are added to `spsAddList`;
 - If `videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, then each `StdVideoH264SequenceParameterSet` entry stored in it with `seq_parameter_set_id` not matching any of the entries already in `spsAddList` is added to `spsAddList`.
- If `videoSession` was created with the codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then let `StdVideoH264PictureParameterSet ppsAddList[]` be the list of `H.264 PPS` entries to add to the created video session parameters object, defined as follows:
 - If the `pParametersAddInfo` member of the `VkVideoEncodeH264SessionParametersCreateInfoKHR` structure provided in the `pNext` chain is not `NULL`, then the set of `StdVideoH264PictureParameterSet` entries specified in

pParametersAddInfo->pStdPPSs are added to ppsAddList;

- If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH264PictureParameterSet entry stored in it with seq_parameter_set_id or pic_parameter_set_id not matching any of the entries already in ppsAddList is added to ppsAddList.
- If videoSession was created with the codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then let StdVideoH265VideoParameterSet vpsAddList[] be the list of H.265 VPS entries to add to the created video session parameters object, defined as follows:
 - If the pParametersAddInfo member of the VkVideoEncodeH265SessionParametersCreateInfoKHR structure provided in the pNext chain is not NULL, then the set of StdVideoH265VideoParameterSet entries specified in pParametersAddInfo->pStdVPSs are added to vpsAddList;
 - If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265VideoParameterSet entry stored in it with vps_video_parameter_set_id not matching any of the entries already in vpsAddList is added to vpsAddList.
- If videoSession was created with the codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then let StdVideoH265SequenceParameterSet spsAddList[] be the list of H.265 SPS entries to add to the created video session parameters object, defined as follows:
 - If the pParametersAddInfo member of the VkVideoEncodeH265SessionParametersCreateInfoKHR structure provided in the pNext chain is not NULL, then the set of StdVideoH265SequenceParameterSet entries specified in pParametersAddInfo->pStdSPSs are added to spsAddList;
 - If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265SequenceParameterSet entry stored in it with sps_video_parameter_set_id or sps_seq_parameter_set_id not matching any of the entries already in spsAddList is added to spsAddList.
- If videoSession was created with the codec operation VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR, then let StdVideoH265PictureParameterSet ppsAddList[] be the list of H.265 PPS entries to add to the created video session parameters object, defined as follows:
 - If the pParametersAddInfo member of the VkVideoEncodeH265SessionParametersCreateInfoKHR structure provided in the pNext chain is not NULL, then the set of StdVideoH265PictureParameterSet entries specified in pParametersAddInfo->pStdPPSs are added to ppsAddList;
 - If videoSessionParametersTemplate is not VK_NULL_HANDLE, then each StdVideoH265PictureParameterSet entry stored in it with sps_video_parameter_set_id, pps_seq_parameter_set_id, or pps_pic_parameter_set_id not matching any of the entries already in ppsAddList is added to ppsAddList.

Valid Usage

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSessionParametersTemplate-04855
If `videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, it **must** have been created against `videoSession`
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSessionParametersTemplate-08310
If `videoSessionParametersTemplate` is not `VK_NULL_HANDLE` and `videoSession` was created with an encode operation, then `qualityLevel` **must** equal the `video encode quality` level `videoSessionParametersTemplate` was created with
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07203
If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoDecodeH264SessionParametersCreateInfoKHR` structure
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07204
If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then the number of elements of `spsAddList` **must** be less than or equal to the `maxStdSPSCount` specified in the `VkVideoDecodeH264SessionParametersCreateInfoKHR` structure included in the `pNext` chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07205
If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then the number of elements of `ppsAddList` **must** be less than or equal to the `maxStdPPSCount` specified in the `VkVideoDecodeH264SessionParametersCreateInfoKHR` structure included in the `pNext` chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07206
If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07207
If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then the number of elements of `vpsAddList` **must** be less than or equal to the `maxStdVPSCount` specified in the `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure included in the `pNext` chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07208
If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then the number of elements of `spsAddList` **must** be less than or equal to the `maxStdSPSCount` specified in the `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure included in the `pNext` chain
- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07209
If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then the number of elements of `ppsAddList`

must be less than or equal to the `maxStdPPSCount` specified in the `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure included in the `pNext` chain

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-09258

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR`, then `videoSessionParametersTemplate` **must** be `VK_NULL_HANDLE`

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-09259

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoDecodeAV1SessionParametersCreateInfoKHR` structure

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07210

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoEncodeH264SessionParametersCreateInfoKHR` structure

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-04839

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the number of elements of `spsAddList` **must** be less than or equal to the `maxStdSPSCount` specified in the `VkVideoEncodeH264SessionParametersCreateInfoKHR` structure included in the `pNext` chain

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-04840

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the number of elements of `ppsAddList` **must** be less than or equal to the `maxStdPPSCount` specified in the `VkVideoEncodeH264SessionParametersCreateInfoKHR` structure included in the `pNext` chain

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-07211

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoEncodeH265SessionParametersCreateInfoKHR` structure

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-04841

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the number of elements of `vpsAddList` **must** be less than or equal to the `maxStdVPSCount` specified in the `VkVideoEncodeH265SessionParametersCreateInfoKHR` structure included in the `pNext` chain

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-04842

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the number of elements of `spsAddList` **must** be less than or equal to the `maxStdSPSCount` specified in the `VkVideoEncodeH265SessionParametersCreateInfoKHR` structure included in the `pNext` chain

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-04843

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the number of elements of `ppsAddList` **must** be less than or equal to the `maxStdPPSCount` specified in the `VkVideoEncodeH265SessionParametersCreateInfoKHR` structure included in the `pNext` chain

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-08319

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then `num_tile_columns_minus1` **must** be less than `VkVideoEncodeH265CapabilitiesKHR::maxTiles.width`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile `videoSession` was created with, for each element of `ppsAddList`

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-08320

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then `num_tile_rows_minus1` **must** be less than `VkVideoEncodeH265CapabilitiesKHR::maxTiles.height`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile `videoSession` was created with, for each element of `ppsAddList`

Valid Usage (Implicit)

- VUID-VkVideoSessionParametersCreateInfoKHR-sType-sType

`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_CREATE_INFO_KHR`

- VUID-VkVideoSessionParametersCreateInfoKHR-pNext-pNext

Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkVideoDecodeAV1SessionParametersCreateInfoKHR`, `VkVideoDecodeH264SessionParametersCreateInfoKHR`, `VkVideoDecodeH265SessionParametersCreateInfoKHR`, `VkVideoEncodeH264SessionParametersCreateInfoKHR`, `VkVideoEncodeH265SessionParametersCreateInfoKHR`, or `VkVideoEncodeQualityLevelInfoKHR`

- VUID-VkVideoSessionParametersCreateInfoKHR-sType-unique

The `sType` value of each struct in the `pNext` chain **must** be unique

- VUID-VkVideoSessionParametersCreateInfoKHR-flags-zerobitmask

`flags` **must** be 0

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSessionParametersTemplate-parameter

If `videoSessionParametersTemplate` is not `VK_NULL_HANDLE`, `videoSessionParametersTemplate` **must** be a valid `VkVideoSessionParametersKHR` handle

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSession-parameter

`videoSession` **must** be a valid `VkVideoSessionKHR` handle

- VUID-VkVideoSessionParametersCreateInfoKHR-videoSessionParametersTemplate-parent

If `videoSessionParametersTemplate` is a valid handle, it **must** have been created, allocated,

or retrieved from `videoSession`

- VUID-VkVideoSessionParametersCreateInfoKHR-commonparent

Both of `videoSession`, and `videoSessionParametersTemplate` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoSessionParametersCreateFlagsKHR;
```

`VkVideoSessionParametersCreateFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

37.7.2. Destroying Video Session Parameters

To destroy a video session parameters object, call:

```
// Provided by VK_KHR_video_queue
void vkDestroyVideoSessionParametersKHR(
    VkDevice                device,
    VkVideoSessionParametersKHR videoSessionParameters,
    const VkAllocationCallbacks* pAllocator);
```

- `device` is the logical device that destroys the video session parameters object.
- `videoSessionParameters` is the video session parameters object to destroy.
- `pAllocator` controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyVideoSessionParametersKHR-videoSessionParameters-07212
All submitted commands that refer to `videoSessionParameters` **must** have completed execution
- VUID-vkDestroyVideoSessionParametersKHR-videoSessionParameters-07213
If `VkAllocationCallbacks` were provided when `videoSessionParameters` was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyVideoSessionParametersKHR-videoSessionParameters-07214
If no `VkAllocationCallbacks` were provided when `videoSessionParameters` was created, `pAllocator` **must** be `NULL`

Valid Usage (Implicit)

- VUID-vkDestroyVideoSessionParametersKHR-device-parameter
`device` **must** be a valid `VkDevice` handle

- VUID-vkDestroyVideoSessionParametersKHR-videoSessionParameters-parameter
If `videoSessionParameters` is not `VK_NULL_HANDLE`, `videoSessionParameters` **must** be a valid `VkVideoSessionParametersKHR` handle
- VUID-vkDestroyVideoSessionParametersKHR-pAllocator-parameter
If `pAllocator` is not `NULL`, `pAllocator` **must** be a valid pointer to a valid `VkAllocationCallbacks` structure
- VUID-vkDestroyVideoSessionParametersKHR-videoSessionParameters-parent
If `videoSessionParameters` is a valid handle, it **must** have been created, allocated, or retrieved from `device`

Host Synchronization

- Host access to `videoSessionParameters` **must** be externally synchronized

37.7.3. Updating Video Session Parameters

To update video session parameters object with new parameters, call:

```
// Provided by VK_KHR_video_queue
VkResult vkUpdateVideoSessionParametersKHR(
    VkDevice          device,
    VkVideoSessionParametersKHR videoSessionParameters,
    const VkVideoSessionParametersUpdateInfoKHR* pUpdateInfo);
```

- `device` is the logical device that updates the video session parameters.
- `videoSessionParameters` is the video session parameters object to update.
- `pUpdateInfo` is a pointer to a `VkVideoSessionParametersUpdateInfoKHR` structure specifying the parameter update information.

After a successful call to this command, the `update sequence counter` of `videoSessionParameters` is changed to the value specified in `pUpdateInfo->updateSequenceCount`.

NOTE

As each update issued to a video session parameters object needs to specify the next available update sequence count value, concurrent updates of the same video session parameters object are inherently disallowed. However, recording video coding operations to command buffers referring to parameters previously added to the video session parameters object is allowed, even if there is a concurrent update in progress adding some new entries to the object.

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR` and the `pUpdateInfo->pNext` chain includes a `VkVideoDecodeH264SessionParametersAddInfoKHR` structure, then this command adds the following parameter entries to `videoSessionParameters`:

- The H.264 SPS entries specified in `VkVideoDecodeH264SessionParametersAddInfoKHR::pStdSPSs`.
- The H.264 PPS entries specified in `VkVideoDecodeH264SessionParametersAddInfoKHR::pStdPPSs`.

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR` and the `pUpdateInfo->pNext` chain includes a `VkVideoDecodeH265SessionParametersAddInfoKHR` structure, then this command adds the following parameter entries to `videoSessionParameters`:

- The H.265 VPS entries specified in `VkVideoDecodeH265SessionParametersAddInfoKHR::pStdVPSs`.
- The H.265 SPS entries specified in `VkVideoDecodeH265SessionParametersAddInfoKHR::pStdSPSs`.
- The H.265 PPS entries specified in `VkVideoDecodeH265SessionParametersAddInfoKHR::pStdPPSs`.

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and the `pUpdateInfo->pNext` chain includes a `VkVideoEncodeH264SessionParametersAddInfoKHR` structure, then this command adds the following parameter entries to `videoSessionParameters`:

- The H.264 SPS entries specified in `VkVideoEncodeH264SessionParametersAddInfoKHR::pStdSPSs`.
- The H.264 PPS entries specified in `VkVideoEncodeH264SessionParametersAddInfoKHR::pStdPPSs`.

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and the `pUpdateInfo->pNext` chain includes a `VkVideoEncodeH265SessionParametersAddInfoKHR` structure, then this command adds the following parameter entries to `videoSessionParameters`:

- The H.265 VPS entries specified in `VkVideoEncodeH265SessionParametersAddInfoKHR::pStdVPSs`.
- The H.265 SPS entries specified in `VkVideoEncodeH265SessionParametersAddInfoKHR::pStdSPSs`.
- The H.265 PPS entries specified in `VkVideoEncodeH265SessionParametersAddInfoKHR::pStdPPSs`.

In case of video session parameters objects created with a video encode operation, implementations **may** return the `VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR` error if any of the specified Video Std parameters do not adhere to the syntactic or semantic requirements of the used video compression standard, or if values derived from parameters according to the rules defined by the used video compression standard do not adhere to the capabilities of the video compression standard or the implementation.

NOTE Applications **should** not rely on the `VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR`

error being returned by any command as a means to verify Video Std parameters, as implementations are not required to report the error in any specific set of cases.

Valid Usage

- VUID-vkUpdateVideoSessionParametersKHR-pUpdateInfo-07215
`pUpdateInfo->updateSequenceCount` **must** equal the current `update sequence counter` of `videoSessionParameters` plus one
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07216
If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes a `VkVideoDecodeH264SessionParametersAddInfoKHR` structure, then `videoSessionParameters` **must** not already contain a `StdVideoH264SequenceParameterSet` entry with `seq_parameter_set_id` matching any of the elements of `VkVideoDecodeH264SessionParametersAddInfoKHR::pStdSPSs`
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07217
If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then the number of `StdVideoH264SequenceParameterSet` entries already stored in it plus the value of the `stdSPSCount` member of the `VkVideoDecodeH264SessionParametersAddInfoKHR` structure included in the `pUpdateInfo->pNext` chain **must** be less than or equal to the `VkVideoDecodeH264SessionParametersCreateInfoKHR::maxStdSPSCount` `videoSessionParameters` was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07218
If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes a `VkVideoDecodeH264SessionParametersAddInfoKHR` structure, then `videoSessionParameters` **must** not already contain a `StdVideoH264PictureParameterSet` entry with both `seq_parameter_set_id` and `pic_parameter_set_id` matching any of the elements of `VkVideoDecodeH264SessionParametersAddInfoKHR::pStdPPSs`
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07219
If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then the number of `StdVideoH264PictureParameterSet` entries already stored in it plus the value of the `stdPPSCount` member of the `VkVideoDecodeH264SessionParametersAddInfoKHR` structure included in the `pUpdateInfo->pNext` chain **must** be less than or equal to the `VkVideoDecodeH264SessionParametersCreateInfoKHR::maxStdPPSCount` `videoSessionParameters` was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07220
If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes a `VkVideoDecodeH265SessionParametersAddInfoKHR` structure, then `videoSessionParameters` **must** not already contain a `StdVideoH265VideoParameterSet` entry with `vps_video_parameter_set_id` matching any of the elements of `VkVideoDecodeH265SessionParametersAddInfoKHR::pStdVPSs`

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07221
If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then the number of `StdVideoH265VideoParameterSet` entries already stored in it plus the value of the `stdVPSCount` member of the `VkVideoDecodeH265SessionParametersAddInfoKHR` structure included in the `pUpdateInfo->pNext` chain **must** be less than or equal to the `VkVideoDecodeH265SessionParametersCreateInfoKHR::maxStdVPSCount` `videoSessionParameters` was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07222
If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes a `VkVideoDecodeH265SessionParametersAddInfoKHR` structure, then `videoSessionParameters` **must** not already contain a `StdVideoH265SequenceParameterSet` entry with both `sps_video_parameter_set_id` and `sps_seq_parameter_set_id` matching any of the elements of `VkVideoDecodeH265SessionParametersAddInfoKHR::pStdSPSs`
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07223
If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then the number of `StdVideoH265SequenceParameterSet` entries already stored in it plus the value of the `stdSPSCount` member of the `VkVideoDecodeH265SessionParametersAddInfoKHR` structure included in the `pUpdateInfo->pNext` chain **must** be less than or equal to the `VkVideoDecodeH265SessionParametersCreateInfoKHR::maxStdSPSCount` `videoSessionParameters` was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07224
If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes a `VkVideoDecodeH265SessionParametersAddInfoKHR` structure, then `videoSessionParameters` **must** not already contain a `StdVideoH265PictureParameterSet` entry with `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` all matching any of the elements of `VkVideoDecodeH265SessionParametersAddInfoKHR::pStdPPSs`
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07225
If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then the number of `StdVideoH265PictureParameterSet` entries already stored in it plus the value of the `stdPPSCount` member of the `VkVideoDecodeH265SessionParametersAddInfoKHR` structure included in the `pUpdateInfo->pNext` chain **must** be less than or equal to the `VkVideoDecodeH265SessionParametersCreateInfoKHR::maxStdPPSCount` `videoSessionParameters` was created with
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-09260
`videoSessionParameters` **must** not have been created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR`
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07226
If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes

a `VkVideoEncodeH264SessionParametersAddInfoKHR` structure, then `videoSessionParameters` **must** not already contain a `StdVideoH264SequenceParameterSet` entry with `seq_parameter_set_id` matching any of the elements of `VkVideoEncodeH264SessionParametersAddInfoKHR::pStdSPSs`

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-06441

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the number of `StdVideoH264SequenceParameterSet` entries already stored in it plus the value of the `stdSPSCount` member of the `VkVideoEncodeH264SessionParametersAddInfoKHR` structure included in the `pUpdateInfo->pNext` chain **must** be less than or equal to the `VkVideoEncodeH264SessionParametersCreateInfoKHR::maxStdSPSCount` `videoSessionParameters` was created with

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07227

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes a `VkVideoEncodeH264SessionParametersAddInfoKHR` structure, then `videoSessionParameters` **must** not already contain a `StdVideoH264PictureParameterSet` entry with both `seq_parameter_set_id` and `pic_parameter_set_id` matching any of the elements of `VkVideoEncodeH264SessionParametersAddInfoKHR::pStdPPSs`

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-06442

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the number of `StdVideoH264PictureParameterSet` entries already stored in it plus the value of the `stdPPSCount` member of the `VkVideoEncodeH264SessionParametersAddInfoKHR` structure included in the `pUpdateInfo->pNext` chain **must** be less than or equal to the `VkVideoEncodeH264SessionParametersCreateInfoKHR::maxStdPPSCount` `videoSessionParameters` was created with

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07228

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes a `VkVideoEncodeH265SessionParametersAddInfoKHR` structure, then `videoSessionParameters` **must** not already contain a `StdVideoH265VideoParameterSet` entry with `vps_video_parameter_set_id` matching any of the elements of `VkVideoEncodeH265SessionParametersAddInfoKHR::pStdVPSs`

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-06443

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the number of `StdVideoH265VideoParameterSet` entries already stored in it plus the value of the `stdVPSCount` member of the `VkVideoEncodeH265SessionParametersAddInfoKHR` structure included in the `pUpdateInfo->pNext` chain **must** be less than or equal to the `VkVideoEncodeH265SessionParametersCreateInfoKHR::maxStdVPSCount` `videoSessionParameters` was created with

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07229

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes

a `VkVideoEncodeH265SessionParametersAddInfoKHR` structure, then `videoSessionParameters` **must** not already contain a `StdVideoH265SequenceParameterSet` entry with both `sps_video_parameter_set_id` and `sps_seq_parameter_set_id` matching any of the elements of `VkVideoEncodeH265SessionParametersAddInfoKHR::pStdSPSs`

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-06444

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the number of `StdVideoH265SequenceParameterSet` entries already stored in it plus the value of the `stdSPSCount` member of the `VkVideoEncodeH265SessionParametersAddInfoKHR` structure included in the `pUpdateInfo->pNext` chain **must** be less than or equal to the `VkVideoEncodeH265SessionParametersCreateInfoKHR::maxStdSPSCount` `videoSessionParameters` was created with

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-07230

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes a `VkVideoEncodeH265SessionParametersAddInfoKHR` structure, then `videoSessionParameters` **must** not already contain a `StdVideoH265PictureParameterSet` entry with `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` all matching any of the elements of `VkVideoEncodeH265SessionParametersAddInfoKHR::pStdPPSs`

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-06445

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the number of `StdVideoH265PictureParameterSet` entries already stored in it plus the value of the `stdPPSCount` member of the `VkVideoEncodeH265SessionParametersAddInfoKHR` structure included in the `pUpdateInfo->pNext` chain **must** be less than or equal to the `VkVideoEncodeH265SessionParametersCreateInfoKHR::maxStdPPSCount` `videoSessionParameters` was created with

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-08321

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes a `VkVideoEncodeH265SessionParametersAddInfoKHR` structure, then `num_tile_columns_minus1` **must** be less than `VkVideoEncodeH265CapabilitiesKHR::maxTiles.width`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile `videoSessionParameters` was created with, for each element of `VkVideoEncodeH265SessionParametersAddInfoKHR::pStdPPSs`

- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-08322

If `videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and the `pNext` chain of `pUpdateInfo` includes a `VkVideoEncodeH265SessionParametersAddInfoKHR` structure, then `num_tile_rows_minus1` **must** be less than `VkVideoEncodeH265CapabilitiesKHR::maxTiles.height`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile `videoSessionParameters` was created with, for each element of `VkVideoEncodeH265SessionParametersAddInfoKHR::pStdPPSs`

Valid Usage (Implicit)

- VUID-vkUpdateVideoSessionParametersKHR-device-parameter **device** **must** be a valid [VkDevice](#) handle
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-parameter **videoSessionParameters** **must** be a valid [VkVideoSessionParametersKHR](#) handle
- VUID-vkUpdateVideoSessionParametersKHR-pUpdateInfo-parameter **pUpdateInfo** **must** be a valid pointer to a valid [VkVideoSessionParametersUpdateInfoKHR](#) structure
- VUID-vkUpdateVideoSessionParametersKHR-videoSessionParameters-parent **videoSessionParameters** **must** have been created, allocated, or retrieved from **device**

Return Codes

Success

- [VK_SUCCESS](#)

Failure

- [VK_ERROR_OUT_OF_HOST_MEMORY](#)
- [VK_ERROR_OUT_OF_DEVICE_MEMORY](#)
- [VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR](#)

The [VkVideoSessionParametersUpdateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoSessionParametersUpdateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           updateSequenceCount;
} VkVideoSessionParametersUpdateInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is [NULL](#) or a pointer to a structure extending this structure.
- **updateSequenceCount** is the new [update sequence count](#) to set for the video session parameters object.

Valid Usage (Implicit)

- VUID-VkVideoSessionParametersUpdateInfoKHR-sType-sType **sType** **must** be [VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_UPDATE_INFO_KHR](#)
- VUID-VkVideoSessionParametersUpdateInfoKHR-pNext-pNext
Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either

NULL or a pointer to a valid instance of
VkVideoDecodeH264SessionParametersAddInfoKHR,
VkVideoDecodeH265SessionParametersAddInfoKHR,
VkVideoEncodeH264SessionParametersAddInfoKHR, or
VkVideoEncodeH265SessionParametersAddInfoKHR

- VUID-VkVideoSessionParametersUpdateInfoKHR-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique

37.8. Video Coding Scope

Applications **can** record video coding commands for a video session only within a video coding scope.

To begin a video coding scope, call:

```
// Provided by VK_KHR_video_queue
void vkCmdBeginVideoCodingKHR(
    VkCommandBuffer                commandBuffer,
    const VkVideoBeginCodingInfoKHR* pBeginInfo);
```

- **commandBuffer** is the command buffer in which to record the command.
- **pBeginInfo** is a pointer to a [VkVideoBeginCodingInfoKHR](#) structure specifying the parameters of the video coding scope, including the video session and video session parameters object to use.

After beginning a video coding scope, the video session object specified in **pBeginInfo->videoSession** is *bound* to the command buffer, and the command buffer is ready to record video coding operations. Similarly, if **pBeginInfo->videoSessionParameters** is not **VK_NULL_HANDLE**, it is also *bound* to the command buffer, and video coding operations **can** refer to the codec-specific parameters stored in it.

This command also establishes the set of *bound reference picture resources* that **can** be used as [reconstructed pictures](#) or [reference pictures](#) within the video coding scope. Each element of this set consists of a [video picture resource](#) and the [DPB slot](#) index associated with it, if there is one.

The set of bound reference picture resources is immutable within a video coding scope, however, the DPB slot index associated with any of the bound reference picture resources **can** change during the video coding scope in response to video coding operations.

The [VkVideoReferenceSlotInfoKHR](#) structures provided as the elements of **pBeginInfo->pReferenceSlots** are interpreted by this command as follows:

- If **slotIndex** is non-negative and **pPictureResource** is not **NULL**, then the [video picture resource](#) defined by the [VkVideoPictureResourceInfoKHR](#) structure pointed to by **pPictureResource** is added to the set of bound reference picture resources and is associated with the DPB slot index specified in **slotIndex**.
- If **slotIndex** is non-negative and **pPictureResource** is **NULL**, then the DPB slot with index **slotIndex**

is **deactivated** by this command.

- If **slotIndex** is negative and **pPictureResource** is not **NULL**, then the **video picture resource** defined by the **VkVideoPictureResourceInfoKHR** structure pointed to by **pPictureResource** is added to the set of bound reference picture resources without an associated DPB slot. Such a picture resource **can** be subsequently used as a **reconstructed picture** to associate it with a DPB slot.
- If **slotIndex** is negative and **pPictureResource** is **NULL**, then the element is ignored.

NOTE

It is possible for multiple bound reference picture resources to be associated with the same DPB slot index, or for a single bound reference picture to refer to multiple separate reference pictures. For example, in case of an **H.264 decode profile** with **interlaced frame support** a single DPB slot can refer to two separate pictures for the top and bottom fields. Depending on the picture layout used by the **H.264 decode profile**, the following special cases **may** arise:

- If the picture layout is **VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR**, then the top and bottom field pictures are physically co-located in the same video picture resource with even scanlines corresponding to the top field and odd scanlines corresponding to the bottom field, respectively.
- If the picture layout is **VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR**, then the top and bottom field pictures are stored in separate video picture resources (in separate subregions of the same image layer, in separate layers of the same image, or in entirely separate images), hence two elements of **VkVideoBeginCodingInfoKHR::pReferenceSlots** **can** contain the same **slotIndex** but specify different video picture resources in their **pPictureResource** members.

All non-negative **slotIndex** values specified in the elements of **pBeginInfo->pReferenceSlots** **must** identify DPB slots of the video session that are in the **active state** at the time this command is executed on the device.

NOTE

The application does not have to specify an entry in **pBeginInfo->pReferenceSlots** corresponding to all active DPB slots of the video session, but only for those which are intended to be used in the video coding scope. This way the application can avoid any potential runtime cost associated with binding the corresponding picture resources to the command buffer.

In case of a video encode session, the application is also responsible for providing information about the current **rate control state** configured for the video session by including an instance of the **VkVideoEncodeRateControlInfoKHR** structure in the **pNext** chain of **pBeginInfo**. If no **VkVideoEncodeRateControlInfoKHR** is included, then the presence of an empty **VkVideoEncodeRateControlInfoKHR** structure is implied which indicates that the current **rate control mode** is **VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR**. The specified state **must match** the effective rate control state configured for the video session at the time the recorded command is executed on the device.

NOTE

Including an instance of the **VkVideoEncodeRateControlInfoKHR** structure in the

`pNext` chain of `pBeginInfo` does not change the rate control state configured for the video session, but only specifies the expected rate control state configured at the time the recorded command is executed on the device which allows the implementation to have information about the configured rate control state at command buffer recording time. In order to change the current rate control state of a video session, the application has to issue an appropriate `vkCmdControlVideoCodingKHR` command as described in the [Video Coding Control](#) and [Rate Control State](#) sections.

Valid Usage

- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-07231
The `VkCommandPool` that `commandBuffer` was allocated from **must** support the video codec operation `pBeginInfo->videoSession` was created with, as returned by `vkGetPhysicalDeviceQueueFamilyProperties2KHR` in `VkQueueFamilyVideoPropertiesKHR::videoCodecOperations`
- VUID-vkCmdBeginVideoCodingKHR-None-07232
There **must** be no `active` queries
- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-07233
If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, then `pBeginInfo->videoSession` **must** not have been created with `VK_VIDEO_SESSION_CREATE_PROTECTED_CONTENT_BIT_KHR`
- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-07234
If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, then `pBeginInfo->videoSession` **must** have been created with `VK_VIDEO_SESSION_CREATE_PROTECTED_CONTENT_BIT_KHR`
- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-07235
If `commandBuffer` is an unprotected command buffer, `protectedNoFault` is not supported, and the `pPictureResource` member of any element of `pBeginInfo->pReferenceSlots` is not `NULL`, then `pPictureResource->imageViewBinding` for that element **must** not specify an image view created from a protected image
- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-07236
If `commandBuffer` is a protected command buffer `protectedNoFault` is not supported, and the `pPictureResource` member of any element of `pBeginInfo->pReferenceSlots` is not `NULL`, then `pPictureResource->imageViewBinding` for that element **must** specify an image view created from a protected image
- VUID-vkCmdBeginVideoCodingKHR-slotIndex-07239
If the `slotIndex` member of any element of `pBeginInfo->pReferenceSlots` is not negative, then it **must** specify the index of a DPB slot that is in the `active` state in `pBeginInfo->videoSession` at the time the command is executed on the device
- VUID-vkCmdBeginVideoCodingKHR-pPictureResource-07265
Each video picture resource specified by any non-`NULL` `pPictureResource` member specified in the elements of `pBeginInfo->pReferenceSlots` for which `slotIndex` is not negative **must** `match` one of the video picture resources currently associated with the DPB slot index of

`pBeginInfo->videoSession` specified by `slotIndex` at the time the command is executed on the device

- VUID-vkCmdBeginVideoCodingKHR-pBeginInfo-08253

If `pBeginInfo->videoSession` was created with a video encode operation and the `pNext` chain of `pBeginInfo` does not include an instance of the `VkVideoEncodeRateControlInfoKHR` structure, then the `rate control mode` configured for `pBeginInfo->videoSession` at the time the command is executed on the device **must** be `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR`

- VUID-vkCmdBeginVideoCodingKHR-pBeginInfo-08254

If `pBeginInfo->videoSession` was created with a video encode operation and the `pNext` chain of `pBeginInfo` includes an instance of the `VkVideoEncodeRateControlInfoKHR` structure, then it **must match** the rate control state configured for `pBeginInfo->videoSession` at the time the command is executed on the device

- VUID-vkCmdBeginVideoCodingKHR-pBeginInfo-08255

If `pBeginInfo->videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, the current `rate control mode` is not `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR` or `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`, and `VkVideoEncodeH264CapabilitiesKHR::requiresGopRemainingFrames` is `VK_TRUE`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the `pBeginInfo->videoSession` was created with, then the `pNext` chain of `pBeginInfo` **must** include an instance of the `VkVideoEncodeH264GopRemainingFrameInfoKHR` with its `useGopRemainingFrames` member set to `VK_TRUE`

- VUID-vkCmdBeginVideoCodingKHR-pBeginInfo-08256

If `pBeginInfo->videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, the current `rate control mode` is not `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR` or `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`, and `VkVideoEncodeH265CapabilitiesKHR::requiresGopRemainingFrames` is `VK_TRUE`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the `pBeginInfo->videoSession` was created with, then the `pNext` chain of `pBeginInfo` **must** include an instance of the `VkVideoEncodeH265GopRemainingFrameInfoKHR` with its `useGopRemainingFrames` member set to `VK_TRUE`

Valid Usage (Implicit)

- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-parameter

`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdBeginVideoCodingKHR-pBeginInfo-parameter

`pBeginInfo` **must** be a valid pointer to a valid `VkVideoBeginCodingInfoKHR` structure

- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-recording

`commandBuffer` **must** be in the `recording state`

- VUID-vkCmdBeginVideoCodingKHR-commandBuffer-cmdpool

The `VkCommandPool` that `commandBuffer` was allocated from **must** support decode, or encode

operations

- VUID-vkCmdBeginVideoCodingKHR-renderpass

This command **must** only be called outside of a render pass instance

- VUID-vkCmdBeginVideoCodingKHR-videocoding

This command **must** only be called outside of a video coding scope

- VUID-vkCmdBeginVideoCodingKHR-bufferlevel

commandBuffer **must** be a primary **VkCommandBuffer**

Host Synchronization

- Host access to **commandBuffer** **must** be externally synchronized
- Host access to the **VkCommandPool** that **commandBuffer** was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Outside	Decode Encode	Action State

The **VkVideoBeginCodingInfoKHR** structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoBeginCodingInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkVideoBeginCodingFlagsKHR flags;
    VkVideoSessionKHR        videoSession;
    VkVideoSessionParametersKHR videoSessionParameters;
    uint32_t                 referenceSlotCount;
    const VkVideoReferenceSlotInfoKHR* pReferenceSlots;
} VkVideoBeginCodingInfoKHR;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **flags** is reserved for future use.
- **videoSession** is the video session object to be bound for the processing of the video commands.
- **videoSessionParameters** is **VK_NULL_HANDLE** or a handle of a **VkVideoSessionParametersKHR** object to be used for the processing of the video commands. If **VK_NULL_HANDLE**, then no video session parameters object is bound for the duration of the video coding scope.

- `referenceSlotCount` is the number of elements in the `pReferenceSlots` array.
- `pReferenceSlots` is a pointer to an array of `VkVideoReferenceSlotInfoKHR` structures specifying the information used to determine the set of `bound reference picture resources` for the video coding scope and their initial association with `DPB slot` indices.

Limiting values are defined below that are referenced by the relevant valid usage statements of this structure.

- Let `VkOffset2D codedOffsetGranularity` be the minimum alignment requirement for the coded offset of video picture resources. Unless otherwise defined, the value of the `x` and `y` members of `codedOffsetGranularity` are 0.
 - If `videoSession` was created with an `H.264 decode profile` with a `VkVideoDecodeH264ProfileInfoKHR::pictureLayout` of `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR`, then `codedOffsetGranularity` is equal to `VkVideoDecodeH264CapabilitiesKHR::fieldOffsetGranularity`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for that video profile.

Valid Usage

- VUID-VkVideoBeginCodingInfoKHR-videoSession-07237
`videoSession` **must** have memory bound to all of its memory bindings returned by `vkGetVideoSessionMemoryRequirementsKHR` for `videoSession`
- VUID-VkVideoBeginCodingInfoKHR-slotIndex-04856
Each non-negative `VkVideoReferenceSlotInfoKHR::slotIndex` specified in the elements of `pReferenceSlots` **must** be less than the `VkVideoSessionCreateInfoKHR::maxDpbSlots` specified when `videoSession` was created
- VUID-VkVideoBeginCodingInfoKHR-pPictureResource-07238
Each video picture resource corresponding to any non-NULL `pPictureResource` member specified in the elements of `pReferenceSlots` **must** be `unique` within `pReferenceSlots`
- VUID-VkVideoBeginCodingInfoKHR-pPictureResource-07240
If the `pPictureResource` member of any element of `pReferenceSlots` is not `NULL`, then the image view specified in `pPictureResource->imageViewBinding` for that element **must** be `compatible` with the video profile `videoSession` was created with
- VUID-VkVideoBeginCodingInfoKHR-pPictureResource-07241
If the `pPictureResource` member of any element of `pReferenceSlots` is not `NULL`, then the format of the image view specified in `pPictureResource->imageViewBinding` for that element **must** match the `VkVideoSessionCreateInfoKHR::referencePictureFormat` `videoSession` was created with
- VUID-VkVideoBeginCodingInfoKHR-pPictureResource-07242
If the `pPictureResource` member of any element of `pReferenceSlots` is not `NULL`, then its `codedOffset` member **must** be an integer multiple of `codedOffsetGranularity`
- VUID-VkVideoBeginCodingInfoKHR-pPictureResource-07243
If the `pPictureResource` member of any element of `pReferenceSlots` is not `NULL`, then its `codedExtent` member **must** be between `minCodedExtent` and `maxCodedExtent`, inclusive,

`videoSession` was created with

- VUID-VkVideoBeginCodingInfoKHR-flags-07244

If `VkVideoCapabilitiesKHR::flags` does not include `VK_VIDEO_CAPABILITY_SEPARATE_REFERENCE_IMAGES_BIT_KHR`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile `videoSession` was created with, then `pPictureResource->imageViewBinding` of all elements of `pReferenceSlots` with a non-NULL `pPictureResource` member **must** specify image views created from the same image

- VUID-VkVideoBeginCodingInfoKHR-slotIndex-07245

If `videoSession` was created with a decode operation and the `slotIndex` member of any element of `pReferenceSlots` is not negative, then the image view specified in `pPictureResource->imageViewBinding` for that element **must** have been created with `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`

- VUID-VkVideoBeginCodingInfoKHR-slotIndex-07246

If `videoSession` was created with an encode operation and the `slotIndex` member of any element of `pReferenceSlots` is not negative, then the image view specified in `pPictureResource->imageViewBinding` for that element **must** have been created with `VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR`

- VUID-VkVideoBeginCodingInfoKHR-videoSession-07247

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then `videoSessionParameters` **must** not be `VK_NULL_HANDLE`

- VUID-VkVideoBeginCodingInfoKHR-videoSession-07248

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then `videoSessionParameters` **must** not be `VK_NULL_HANDLE`

- VUID-VkVideoBeginCodingInfoKHR-videoSession-09261

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR`, then `videoSessionParameters` **must** not be `VK_NULL_HANDLE`

- VUID-VkVideoBeginCodingInfoKHR-videoSession-07249

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then `videoSessionParameters` **must** not be `VK_NULL_HANDLE`

- VUID-VkVideoBeginCodingInfoKHR-videoSession-07250

If `videoSession` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then `videoSessionParameters` **must** not be `VK_NULL_HANDLE`

- VUID-VkVideoBeginCodingInfoKHR-videoSessionParameters-04857

If `videoSessionParameters` is not `VK_NULL_HANDLE`, it **must** have been created with `videoSession` specified in `VkVideoSessionParametersCreateInfoKHR::videoSession`

Valid Usage (Implicit)

- VUID-VkVideoBeginCodingInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_BEGIN_CODING_INFO_KHR`
- VUID-VkVideoBeginCodingInfoKHR-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkVideoEncodeH264GopRemainingFrameInfoKHR`, `VkVideoEncodeH264RateControlInfoKHR`, `VkVideoEncodeH265GopRemainingFrameInfoKHR`, `VkVideoEncodeH265RateControlInfoKHR`, or `VkVideoEncodeRateControlInfoKHR`
- VUID-VkVideoBeginCodingInfoKHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkVideoBeginCodingInfoKHR-flags-zero bitmask
`flags` **must** be `0`
- VUID-VkVideoBeginCodingInfoKHR-videoSession-parameter
`videoSession` **must** be a valid `VkVideoSessionKHR` handle
- VUID-VkVideoBeginCodingInfoKHR-videoSessionParameters-parameter
If `videoSessionParameters` is not `VK_NULL_HANDLE`, `videoSessionParameters` **must** be a valid `VkVideoSessionParametersKHR` handle
- VUID-VkVideoBeginCodingInfoKHR-pReferenceSlots-parameter
If `referenceSlotCount` is not `0`, `pReferenceSlots` **must** be a valid pointer to an array of `referenceSlotCount` valid `VkVideoReferenceSlotInfoKHR` structures
- VUID-VkVideoBeginCodingInfoKHR-videoSessionParameters-parent
If `videoSessionParameters` is a valid handle, it **must** have been created, allocated, or retrieved from `videoSession`
- VUID-VkVideoBeginCodingInfoKHR-commonparent
Both of `videoSession`, and `videoSessionParameters` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoBeginCodingFlagsKHR;
```

`VkVideoBeginCodingFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

The `VkVideoReferenceSlotInfoKHR` structure is defined as:


```
// Provided by VK_KHR_video_queue
typedef struct VkVideoReferenceSlotInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    int32_t                  slotIndex;
    const VkVideoPictureResourceInfoKHR* pPictureResource;
} VkVideoReferenceSlotInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `slotIndex` is the index of the `DPB slot` or a negative integer value.
- `pPictureResource` is `NULL` or a pointer to a `VkVideoPictureResourceInfoKHR` structure describing the `video picture resource` associated with the DPB slot index specified by `slotIndex`.

Valid Usage (Implicit)

- VUID-VkVideoReferenceSlotInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_REFERENCE_SLOT_INFO_KHR`
- VUID-VkVideoReferenceSlotInfoKHR-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkVideoDecodeAV1DpbSlotInfoKHR`, `VkVideoDecodeH264DpbSlotInfoKHR`, `VkVideoDecodeH265DpbSlotInfoKHR`, `VkVideoEncodeH264DpbSlotInfoKHR`, or `VkVideoEncodeH265DpbSlotInfoKHR`
- VUID-VkVideoReferenceSlotInfoKHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkVideoReferenceSlotInfoKHR-pPictureResource-parameter
If `pPictureResource` is not `NULL`, `pPictureResource` **must** be a valid pointer to a valid `VkVideoPictureResourceInfoKHR` structure

To end a video coding scope, call:

```
// Provided by VK_KHR_video_queue
void vkCmdEndVideoCodingKHR(
    VkCommandBuffer          commandBuffer,
    const VkVideoEndCodingInfoKHR* pEndCodingInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pEndCodingInfo` is a pointer to a `VkVideoEndCodingInfoKHR` structure specifying the parameters for ending the video coding scope.

After ending a video coding scope, the video session object, the optional video session parameters object, and all `reference picture resources` previously bound by the corresponding `vkCmdBeginVideoCodingKHR` command are *unbound*.

Valid Usage

- VUID-vkCmdEndVideoCodingKHR-None-07251
There **must** be no [active](#) queries

Valid Usage (Implicit)

- VUID-vkCmdEndVideoCodingKHR-commandBuffer-parameter
[commandBuffer](#) **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdEndVideoCodingKHR-pEndCodingInfo-parameter
[pEndCodingInfo](#) **must** be a valid pointer to a valid [VkVideoEndCodingInfoKHR](#) structure
- VUID-vkCmdEndVideoCodingKHR-commandBuffer-recording
[commandBuffer](#) **must** be in the [recording state](#)
- VUID-vkCmdEndVideoCodingKHR-commandBuffer-cmdpool
The [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** support decode, or encode operations
- VUID-vkCmdEndVideoCodingKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdEndVideoCodingKHR-videocoding
This command **must** only be called inside of a video coding scope
- VUID-vkCmdEndVideoCodingKHR-bufferlevel
[commandBuffer](#) **must** be a primary [VkCommandBuffer](#)

Host Synchronization

- Host access to [commandBuffer](#) **must** be externally synchronized
- Host access to the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Inside	Decode Encode	Action State

The [VkVideoEndCodingInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoEndCodingInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    VkVideoEndCodingFlagsKHR flags;
} VkVideoEndCodingInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

Valid Usage (Implicit)

- VUID-VkVideoEndCodingInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_END_CODING_INFO_KHR`
- VUID-VkVideoEndCodingInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkVideoEndCodingInfoKHR-flags-zero bitmask
`flags` **must** be 0

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoEndCodingFlagsKHR;
```

`VkVideoEndCodingFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

37.9. Video Coding Control

To apply dynamic controls to the currently bound video session object, call:

```
// Provided by VK_KHR_video_queue
void vkCmdControlVideoCodingKHR(
    VkCommandBuffer          commandBuffer,
    const VkVideoCodingControlInfoKHR* pCodingControlInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pCodingControlInfo` is a pointer to a `VkVideoCodingControlInfoKHR` structure specifying the control parameters.

The control parameters provided in this call are applied to the video session at the time the command executes on the device and are in effect until a subsequent call to this command with the same video session bound changes the corresponding control parameters.

A newly created video session **must** be reset before performing video coding operations using it by including `VK_VIDEO_CODING_CONTROL_RESET_BIT_KHR` in `pCodingControlInfo->flags`. The reset operation also returns all DPB slots of the video session to the `inactive state`. Correspondingly, any DPB slot index associated with the `bound reference picture resources` is removed.

For encode sessions, the reset operation returns `rate control` configuration to implementation default settings and sets the `video encode quality level` to zero.

After video coding operations are performed using a video session, the reset operation **can** be used to return the video session to the same *initial* state as after the reset of a newly created video session. This **can** be used, for example, when different video sequences are needed to be processed with the same video session object.

If `pCodingControlInfo->flags` includes `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR`, then the command replaces the `rate control` configuration maintained by the video session with the configuration specified in the `VkVideoEncodeRateControlInfoKHR` structure included in the `pCodingControlInfo->pNext` chain.

If `pCodingControlInfo->flags` includes `VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR`, then the command changes the current `video encode quality level` to the value specified in the `qualityLevel` member of the `VkVideoEncodeQualityLevelInfoKHR` structure included in the `pCodingControlInfo->pNext` chain.

Valid Usage

- VUID-vkCmdControlVideoCodingKHR-flags-07017
If `pCodingControlInfo->flags` does not include `VK_VIDEO_CODING_CONTROL_RESET_BIT_KHR`, then the bound video session **must** not be in `uninitialized` state at the time the command is executed on the device
- VUID-vkCmdControlVideoCodingKHR-pCodingControlInfo-08243
If the bound video session was not created with an encode operation, then

<code>pCodingControlInfo->flags</code>	must	not	include
<code>VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR</code>			or
<code>VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR</code>			

Valid Usage (Implicit)

- VUID-vkCmdControlVideoCodingKHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdControlVideoCodingKHR-pCodingControlInfo-parameter
`pCodingControlInfo` **must** be a valid pointer to a valid `VkVideoCodingControlInfoKHR` structure
- VUID-vkCmdControlVideoCodingKHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdControlVideoCodingKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support decode, or encode

operations

- VUID-vkCmdControlVideoCodingKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdControlVideoCodingKHR-videocoding
This command **must** only be called inside of a video coding scope
- VUID-vkCmdControlVideoCodingKHR-bufferlevel
`commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Inside	Decode Encode	Action

The `VkVideoCodingControlInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_queue
typedef struct VkVideoCodingControlInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    VkVideoCodingControlFlagsKHR flags;
} VkVideoCodingControlInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkVideoCodingControlFlagsKHR` specifying control flags.

Valid Usage

- VUID-VkVideoCodingControlInfoKHR-flags-07018
If `flags` includes `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR`, then the `pNext` chain **must** include a `VkVideoEncodeRateControlInfoKHR` structure
- VUID-VkVideoCodingControlInfoKHR-flags-08349
If `flags` includes `VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR`, then the `pNext`

chain **must** include a [VkVideoEncodeQualityLevelInfoKHR](#) structure

Valid Usage (Implicit)

- VUID-VkVideoCodingControlInfoKHR-sType-sType
sType **must** be [VK_STRUCTURE_TYPE_VIDEO_CODING_CONTROL_INFO_KHR](#)
- VUID-VkVideoCodingControlInfoKHR-pNext-pNext
Each **pNext** member of any structure (including this one) in the **pNext** chain **must** be either [NULL](#) or a pointer to a valid instance of [VkVideoEncodeH264RateControlInfoKHR](#), [VkVideoEncodeH265RateControlInfoKHR](#), [VkVideoEncodeQualityLevelInfoKHR](#), or [VkVideoEncodeRateControlInfoKHR](#)
- VUID-VkVideoCodingControlInfoKHR-sType-unique
The **sType** value of each struct in the **pNext** chain **must** be unique
- VUID-VkVideoCodingControlInfoKHR-flags-parameter
flags **must** be a valid combination of [VkVideoCodingControlFlagBitsKHR](#) values
- VUID-VkVideoCodingControlInfoKHR-flags-requiredbitmask
flags **must** not be 0

Bits which **can** be set in [VkVideoCodingControlInfoKHR::flags](#), specifying the video coding control parameters to be modified, are:

```
// Provided by VK_KHR_video_queue
typedef enum VkVideoCodingControlFlagBitsKHR {
    VK_VIDEO_CODING_CONTROL_RESET_BIT_KHR = 0x00000001,
    // Provided by VK_KHR_video_encode_queue
    VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR = 0x00000002,
    // Provided by VK_KHR_video_encode_queue
    VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR = 0x00000004,
} VkVideoCodingControlFlagBitsKHR;
```

- [VK_VIDEO_CODING_CONTROL_RESET_BIT_KHR](#) indicates a request for the bound video session to be reset before other coding control parameters are applied.
- [VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR](#) indicates that the coding control parameters include video encode rate control parameters (see [VkVideoEncodeRateControlInfoKHR](#)).
- [VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR](#) indicates that the coding control parameters include video encode quality level parameters (see [VkVideoEncodeQualityLevelInfoKHR](#)).

```
// Provided by VK_KHR_video_queue
typedef VkFlags VkVideoCodingControlFlagsKHR;
```

[VkVideoCodingControlFlagsKHR](#) is a bitmask type for setting a mask of zero or more

37.10. Inline Queries

If a video session was created with `VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR`, beginning queries using commands such as `vkCmdBeginQuery` within a video coding scope is not allowed. Instead, queries are executed inline by including an instance of the `VkVideoInlineQueryInfoKHR` structure in the `pNext` chain of the parameters of one of the video coding commands, with its `queryPool` member set to a valid `VkQueryPool` handle.

The `VkVideoInlineQueryInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_maintenance1
typedef struct VkVideoInlineQueryInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkQueryPool         queryPool;
    uint32_t            firstQuery;
    uint32_t            queryCount;
} VkVideoInlineQueryInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `queryPool` is `VK_NULL_HANDLE` or a valid handle to a `VkQueryPool` object that will manage the results of the queries.
- `firstQuery` is the query index within the query pool that will contain the query results for the first video coding operation. The query results of subsequent video coding operations will be contained by subsequent query indices.
- `queryCount` is the number of queries to execute.

NOTE

In practice, if `queryPool` is not `VK_NULL_HANDLE`, then `queryCount` will always have to match the number of video coding operations issued by the video coding command this structure is specified to, meaning that using inline queries in a video coding command will always execute a query for each issued video coding operation.

This structure **can** be included in the `pNext` chain of the input parameter structure of video coding commands.

- In the `pNext` chain of the `pDecodeInfo` parameter of the `vkCmdDecodeVideoKHR` command to execute a query for each video decode operation issued by the command.
- In the `pNext` chain of the `pEncodeInfo` parameter of the `vkCmdEncodeVideoKHR` command to execute a query for each video encode operation issued by the command.

Valid Usage

- VUID-VkVideoInlineQueryInfoKHR-queryPool-08372
If `queryPool` is not `VK_NULL_HANDLE`, then `firstQuery` **must** be less than the number of queries in `queryPool`
- VUID-VkVideoInlineQueryInfoKHR-queryPool-08373
If `queryPool` is not `VK_NULL_HANDLE`, then the sum of `firstQuery` and `queryCount` **must** be less than or equal to the number of queries in `queryPool`

Valid Usage (Implicit)

- VUID-VkVideoInlineQueryInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_INLINE_QUERY_INFO_KHR`
- VUID-VkVideoInlineQueryInfoKHR-queryPool-parameter
If `queryPool` is not `VK_NULL_HANDLE`, `queryPool` **must** be a valid `VkQueryPool` handle

37.11. Video Decode Operations

Video decode operations consume compressed video data from a video bitstream buffer and zero or more reference pictures, and produce a *decode output picture* and an optional *reconstructed picture*.

NOTE

Such decode output pictures can be shared with the *Decoded Picture Buffer*, and can also be used as the *input* of video encode operations, with graphics or compute operations, or with *Window System Integration* APIs, depending on the capabilities of the implementation.

Video decode operations **may** access the following resources in the `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR` stage:

- The source video bitstream buffer range and the image subregions corresponding to the list of *active reference pictures* with access `VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR`.
- The image subregions corresponding to the target *decode output picture* and *reconstructed picture* with access `VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR`.

The image subresource of each *video picture resource* accessed by the video decode operation is specified using a corresponding `VkVideoPictureResourceInfoKHR` structure. Each such image subresource **must** be in the appropriate image layout as follows:

- If the image subresource is used in the video decode operation only as *decode output picture*, then it **must** be in the `VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR` layout.
- If the image subresource is used in the video decode operation both as *decode output picture* and *reconstructed picture*, then it **must** be in the `VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR` layout.
- If the image subresource is used in the video decode operation only as *reconstructed picture*,

then it **must** be in the `VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR` layout.

- If the image subresource is used in the video decode operation as a [reference picture](#), then it **must** be in the `VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR` layout.

A video decode operation **may** complete unsuccessfully. In this case the [decode output picture](#) will have undefined contents. Similarly, if [reference picture setup](#) is requested, the [reconstructed picture](#) will also have undefined contents, and the activated DPB slot will have an [invalid picture reference](#).

37.11.1. Codec-Specific Semantics

The following aspects of video decode operations are codec-specific:

- The interpretation of the contents of the source video bitstream buffer range.
- The construction and interpretation of the list of [active reference pictures](#) and the interpretation of the picture data referred to by the corresponding image subregions.
- The construction and interpretation of information related to the [decode output picture](#) and the generation of picture data to the corresponding image subregion.
- The decision on [reference picture setup](#).
- The construction and interpretation of information related to the optional [reconstructed picture](#) and the generation of picture data to the corresponding image subregion.

These codec-specific behaviors are defined for each video codec operation separately.

- If the used video codec operation is `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then the codec-specific aspects of the video decoding process are performed as defined in the [H.264 Decode Operations](#) section.
- If the used video codec operation is `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then the codec-specific aspects of the video decoding process are performed as defined in the [H.265 Decode Operations](#) section.
- If the used video codec operation is `VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR`, then the codec-specific aspects of the video decoding process are performed as defined in the [AV1 Decode Operations](#) section.

37.11.2. Video Decode Operation Steps

Each video decode operation performs the following steps in the `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR` stage:

1. Reads the encoded video data from the source video bitstream buffer range.
2. Performs picture reconstruction of the encoded video data according to the [codec-specific semantics](#), applying any prediction data read from the [active reference pictures](#) in the process;
3. Writes the decoded picture data to the [decode output picture](#), and optionally to the [reconstructed picture](#), if one is specified and is different from the decode output picture, according to the [codec-specific semantics](#);

4. If [reference picture setup](#) is requested, the [DPB slot index](#) specified in the [reconstructed picture information](#) is [activated](#) with the [reconstructed picture](#).

When [reconstructed picture information](#) is provided, the specified [DPB slot](#) index is associated with the corresponding [bound reference picture resource](#), indifferent of whether [reference picture setup](#) is requested.

37.11.3. Capabilities

When calling [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) with [pVideoProfile->videoCodecOperation](#) specifying a decode operation, the [VkVideoDecodeCapabilitiesKHR](#) structure **must** be included in the [pNext](#) chain of the [VkVideoCapabilitiesKHR](#) structure to retrieve capabilities specific to video decoding.

The [VkVideoDecodeCapabilitiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_decode_queue
typedef struct VkVideoDecodeCapabilitiesKHR {
    VkStructureType      sType;
    void*                 pNext;
    VkVideoDecodeCapabilityFlagsKHR flags;
} VkVideoDecodeCapabilitiesKHR;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [flags](#) is a bitmask of [VkVideoDecodeCapabilityFlagBitsKHR](#) describing the supported video decoding capabilities.

Valid Usage (Implicit)

- VUID-VkVideoDecodeCapabilitiesKHR-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_VIDEO_DECODE_CAPABILITIES_KHR](#)

Bits which **may** be set in [VkVideoDecodeCapabilitiesKHR::flags](#), indicating the decoding capabilities supported, are:

```
// Provided by VK_KHR_video_decode_queue
typedef enum VkVideoDecodeCapabilityFlagBitsKHR {
    VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR = 0x00000001,
    VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR = 0x00000002,
} VkVideoDecodeCapabilityFlagBitsKHR;
```

- [VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR](#) indicates support for using the same video picture resource as the [reconstructed picture](#) and [decode output picture](#) in a video decode operation.

- `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR` indicates support for using distinct video picture resources as the [reconstructed picture](#) and [decode output picture](#) in a video decode operation.

NOTE

Some video profiles allow using distinct video picture resources as the reconstructed picture and decode output picture in specific video decode operations even when the video decode profile does not support `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR`. Even if the implementation only reports coincide, the decode output picture for [film grain](#) enabled frames must be a different video picture resource from the reconstructed picture because film grain is applied outside of the coding loop.

Implementations are only **required** to support one of `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR` and `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR`. Accordingly, applications **should** handle both cases to maximize portability.

NOTE

If both `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR` and `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR` are supported, an application can choose to create separate images for decode DPB and decode output. E.g. in cases when linear tiling is preferred (and supported) for the decode output picture and the DPB requires optimal tiling, this avoids the need for a separate copy at the expense of additional memory bandwidth requirements during decoding.

```
// Provided by VK_KHR_video_decode_queue
typedef VkFlags VkVideoDecodeCapabilityFlagsKHR;
```

`VkVideoDecodeCapabilityFlagsKHR` is a bitmask type for setting a mask of zero or more [VkVideoDecodeCapabilityFlagBitsKHR](#).

37.11.4. Video Decode Commands

To launch video decode operations, call:

```
// Provided by VK_KHR_video_decode_queue
void vkCmdDecodeVideoKHR(
    VkCommandBuffer                commandBuffer,
    const VkVideoDecodeInfoKHR*    pDecodeInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pDecodeInfo` is a pointer to a [VkVideoDecodeInfoKHR](#) structure specifying the parameters of the video decode operations.

Each call issues one or more video decode operations. The implicit parameter `opCount` corresponds to the number of video decode operations issued by the command. After calling this command, the

active query index of each active query is incremented by `opCount`.

Currently each call to this command results in the issue of a single video decode operation.

If the bound video session was created with `VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR` and the `pNext` chain of `pDecodeInfo` includes a `VkVideoInlineQueryInfoKHR` structure with its `queryPool` member specifying a valid `VkQueryPool` handle, then this command will execute a query for each video decode operation issued by it.

Active Reference Picture Information

The list of active reference pictures used by a video decode operation is a list of image subregions used as the source of reference picture data and related parameters, and is derived from the `VkVideoReferenceSlotInfoKHR` structures provided as the elements of the `pDecodeInfo->pReferenceSlots` array. For each element of `pDecodeInfo->pReferenceSlots`, one or more elements are added to the active reference picture list, as defined by the codec-specific semantics. Each element of this list contains the following information:

- The image subregion within the image subresource referred to by the video picture resource used as the reference picture.
- The DPB slot index the reference picture is associated with.
- The codec-specific reference information related to the reference picture.

Reconstructed Picture Information

Information related to the optional reconstructed picture used by a video decode operation is derived from the `VkVideoReferenceSlotInfoKHR` structure pointed to by `pDecodeInfo->pSetupReferenceSlot`, if not `NULL`, as defined by the codec-specific semantics, and consists of the following:

- The image subregion within the image subresource referred to by the video picture resource used as the reconstructed picture.
- The DPB slot index to use for picture reconstruction.
- The codec-specific reference information related to the reconstructed picture.

Specifying a valid `VkVideoReferenceSlotInfoKHR` structure in `pDecodeInfo->pSetupReferenceSlot` is always required, unless the video session was created with `VkVideoSessionCreateInfoKHR::maxDpbSlot` equal to zero. However, the DPB slot identified by `pDecodeInfo->pSetupReferenceSlot->slotIndex` is only activated with the reconstructed picture specified in `pDecodeInfo->pSetupReferenceSlot->pPictureResource` if reference picture setup is requested according to the codec-specific semantics.

If reconstructed picture information is specified, and `pDecodeInfo->pSetupReferenceSlot->pPictureResource` refers to a video picture resource different than that of the decode output picture, but reference picture setup is not requested, the contents of the video picture resource corresponding to the reconstructed picture will be undefined after the video decode operation.

NOTE

Some implementations may always output the reconstructed picture or use it as temporary storage during the video decode operation even when the reconstructed

picture is not marked for future reference.

Decode Output Picture Information

Information related to the [decode output picture](#) used by a video decode operation is derived from `pDecodeInfo->dstPictureResource` and any codec-specific parameters provided in the `pDecodeInfo->pNext` chain, as defined by the [codec-specific semantics](#), and consists of the following:

- The image subregion within the image subresource [referred](#) to by the [video picture resource](#) used as the decode output picture.
- The codec-specific picture information related to the decode output picture.

Several limiting values are defined below that are referenced by the relevant valid usage statements of this command.

- Let `uint32_t activeReferencePictureCount` be the size of the list of active reference pictures used by the video decode operation. Unless otherwise defined, `activeReferencePictureCount` is set to the value of `pDecodeInfo->referenceSlotCount`.
 - If the bound video session was created with an [H.264 decode profile](#), then let `activeReferencePictureCount` be the value of `pDecodeInfo->referenceSlotCount` plus the number of elements of the `pDecodeInfo->pReferenceSlots` array that have a [VkVideoDecodeH264DpbSlotInfoKHR](#) structure included in their `pNext` chain with both `pStdReferenceInfo->flags.top_field_flag` and `pStdReferenceInfo->flags.bottom_field_flag` set.

NOTE

This means that the elements of `pDecodeInfo->pReferenceSlots` that include both a top and bottom field reference are counted as two separate active reference pictures, as described in the [active reference picture list construction rules for H.264 decode operations](#).

- Let `VkOffset2D codedOffsetGranularity` be the minimum alignment requirement for the coded offset of video picture resources. Unless otherwise defined, the value of the `x` and `y` members of `codedOffsetGranularity` are 0.
 - If the bound video session was created with an [H.264 decode profile](#) with a `VkVideoDecodeH264ProfileInfoKHR::pictureLayout` of `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR`, then `codedOffsetGranularity` is equal to `VkVideoDecodeH264CapabilitiesKHR::fieldOffsetGranularity`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for that video profile.
- Let `uint32_t dpbFrameUseCount[]` be an array of size `maxDpbSlots`, where `maxDpbSlots` is the `VkVideoSessionCreateInfoKHR::maxDpbSlots` the bound video session was created with, with each element indicating the number of times a frame associated with the corresponding DPB slot index is referred to by the video coding operation. Let the initial value of each element of the array be 0.
 - If `pDecodeInfo->pSetupReferenceSlot` is not `NULL`, then `dpbFrameUseCount[i]` is incremented by one, where `i` equals `pDecodeInfo->pSetupReferenceSlot->slotIndex`. If the bound video

session object was created with an [H.264 decode profile](#), then `dpbFrameUseCount[i]` is decremented by one if either `pStdReferenceInfo->flags.top_field_flag` or `pStdReferenceInfo->flags.bottom_field_flag` is set in the [VkVideoDecodeH264DpbSlotInfoKHR](#) structure in the `pDecodeInfo->pSetupReferenceSlot->pNext` chain.

- For each element of `pDecodeInfo->pReferenceSlots`, `dpbFrameUseCount[i]` is incremented by one, where `i` equals the `slotIndex` member of the corresponding element. If the bound video session object was created with an [H.264 decode profile](#), then `dpbFrameUseCount[i]` is decremented by one if either `pStdReferenceInfo->flags.top_field_flag` or `pStdReferenceInfo->flags.bottom_field_flag` is set in the [VkVideoDecodeH264DpbSlotInfoKHR](#) structure in the `pNext` chain of the corresponding element of `pDecodeInfo->pReferenceSlots`.
- Let `uint32_t dpbTopFieldUseCount[]` and `uint32_t dpbBottomFieldUseCount[]` be arrays of size `maxDpbSlots`, where `maxDpbSlots` is the [VkVideoSessionCreateInfoKHR::maxDpbSlots](#) the bound video session was created with, with each element indicating the number of times the top field or the bottom field, respectively, associated with the corresponding DPB slot index is referred to by the video coding operation. Let the initial value of each element of the arrays be 0.
 - If the bound video session object was created with an [H.264 decode profile](#) and `pDecodeInfo->pSetupReferenceSlot` is not `NULL`, then perform the following:
 - If `pStdReferenceInfo->flags.top_field_flag` is set in the [VkVideoDecodeH264DpbSlotInfoKHR](#) structure in the `pDecodeInfo->pSetupReferenceSlot->pNext` chain, then `dpbTopFieldUseCount[i]` is incremented by one, where `i` equals `pDecodeInfo->pSetupReferenceSlot->slotIndex`.
 - If `pStdReferenceInfo->flags.bottom_field_flag` is set in the [VkVideoDecodeH264DpbSlotInfoKHR](#) structure in the `pDecodeInfo->pSetupReferenceSlot->pNext` chain, then `dpbBottomFieldUseCount[i]` is incremented by one, where `i` equals `pDecodeInfo->pSetupReferenceSlot->slotIndex`.
 - If the bound video session object was created with an [H.264 decode profile](#), then perform the following for each element of `pDecodeInfo->pReferenceSlots`:
 - If `pStdReferenceInfo->flags.top_field_flag` is set in the [VkVideoDecodeH264DpbSlotInfoKHR](#) structure in the `pNext` chain of the element, then `dpbTopFieldUseCount[i]` is incremented by one, where `i` equals the `slotIndex` member of the element.
 - If `pStdReferenceInfo->flags.bottom_field_flag` is set in the [VkVideoDecodeH264DpbSlotInfoKHR](#) structure in the `pNext` chain of the element, then `dpbBottomFieldUseCount[i]` is incremented by one, where `i` equals the `slotIndex` member of the element.

Valid Usage

- VUID-vkCmdDecodeVideoKHR-None-08249
The bound video session **must** have been created with a decode operation
- VUID-vkCmdDecodeVideoKHR-None-07011
The bound video session **must** not be in [uninitialized](#) state at the time the command is

executed on the device

- VUID-vkCmdDecodeVideoKHR-opCount-07134

For each **active** query, the **active query index** corresponding to the query type of that query plus **opCount** **must** be less than or equal to the **last activatable query index** corresponding to the query type of that query plus one

- VUID-vkCmdDecodeVideoKHR-pNext-08365

If the bound video session was created with **VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR**, and the **pNext** chain of **pDecodeInfo** includes a **VkVideoInlineQueryInfoKHR** structure with its **queryPool** member specifying a valid **VkQueryPool** handle, then **VkVideoInlineQueryInfoKHR::queryCount** **must** equal **opCount**

- VUID-vkCmdDecodeVideoKHR-pNext-08366

If the bound video session was created with **VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR**, and the **pNext** chain of **pDecodeInfo** includes a **VkVideoInlineQueryInfoKHR** structure with its **queryPool** member specifying a valid **VkQueryPool** handle, then all the queries used by the command, as specified by the **VkVideoInlineQueryInfoKHR** structure, **must** be *unavailable*

- VUID-vkCmdDecodeVideoKHR-queryType-08367

If the bound video session was created with **VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR**, then the **queryType** used to create the **queryPool** specified in the **VkVideoInlineQueryInfoKHR** structure included in the **pNext** chain of **pDecodeInfo** **must** be **VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR**

- VUID-vkCmdDecodeVideoKHR-queryPool-08368

If the bound video session was created with **VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR**, then the **queryPool** specified in the **VkVideoInlineQueryInfoKHR** structure included in the **pNext** chain of **pDecodeInfo** **must** have been created with a **VkVideoProfileInfoKHR** structure included in the **pNext** chain of **VkQueryPoolCreateInfo** identical to the one specified in **VkVideoSessionCreateInfoKHR::pVideoProfile** the bound video session was created with

- VUID-vkCmdDecodeVideoKHR-queryType-08369

If the bound video session was created with **VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR**, and the **queryType** used to create the **queryPool** specified in the **VkVideoInlineQueryInfoKHR** structure included in the **pNext** chain of **pDecodeInfo** is **VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR**, then the **VkCommandPool** that **commandBuffer** was allocated from **must** have been created with a queue family index that supports **result status queries**, as indicated by **VkQueueFamilyQueryResultStatusPropertiesKHR::queryResultStatusSupport**

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07135

pDecodeInfo->srcBuffer **must** be **compatible** with the video profile the bound video session was created with

- VUID-vkCmdDecodeVideoKHR-commandBuffer-07136

If **commandBuffer** is an unprotected command buffer and **protectedNoFault** is not supported, then **pDecodeInfo->srcBuffer** **must** not be a protected buffer

- VUID-vkCmdDecodeVideoKHR-commandBuffer-07137

If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, then `pDecodeInfo->srcBuffer` **must** be a protected buffer

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07138

`pDecodeInfo->srcBufferOffset` **must** be an integer multiple of `VkVideoCapabilitiesKHR::minBitstreamBufferOffsetAlignment`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07139

`pDecodeInfo->srcBufferRange` **must** be an integer multiple of `VkVideoCapabilitiesKHR::minBitstreamBufferSizeAlignment`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07140

If `pDecodeInfo->pSetupReferenceSlot` is not `NULL` and `VkVideoDecodeCapabilitiesKHR::flags` does not include `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, then the video picture resources specified by `pDecodeInfo->dstPictureResource` and `pDecodeInfo->pSetupReferenceSlot->pPictureResource` **must not match**

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07141

If `pDecodeInfo->pSetupReferenceSlot` is not `NULL` and none of the following is true:

- `VkVideoDecodeCapabilitiesKHR::flags` includes `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with
- the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR` and `VkVideoDecodeAV1ProfileInfoKHR::filmGrainSupport` set to `VK_TRUE`, and `film grain` is enabled for the decoded picture

then the video picture resources specified by `pDecodeInfo->dstPictureResource` and `pDecodeInfo->pSetupReferenceSlot->pPictureResource` **must match**

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07142

`pDecodeInfo->dstPictureResource.imageViewBinding` **must** be `compatible` with the video profile the bound video session was created with

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07143

The format of `pDecodeInfo->dstPictureResource.imageViewBinding` **must** match the `VkVideoSessionCreateInfoKHR::pictureFormat` the bound video session was created with

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07144

`pDecodeInfo->dstPictureResource.codedOffset` **must** be an integer multiple of `codedOffsetGranularity`

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07145

`pDecodeInfo->dstPictureResource.codedExtent` **must** be between `minCodedExtent` and `maxCodedExtent`, inclusive, the bound video session was created with

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07146
`pDecodeInfo->dstPictureResource.imageViewBinding` **must** have been created with `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`
- VUID-vkCmdDecodeVideoKHR-commandBuffer-07147
 If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, then `pDecodeInfo->dstPictureResource.imageViewBinding` **must** not have been created from a protected image
- VUID-vkCmdDecodeVideoKHR-commandBuffer-07148
 If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, then `pDecodeInfo->dstPictureResource.imageViewBinding` **must** have been created from a protected image
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-08376
`pDecodeInfo->pSetupReferenceSlot` **must** not be `NULL` unless the bound video session was created with `VkVideoSessionCreateInfoKHR::maxDpbSlots` equal to zero
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07170
 If `pDecodeInfo->pSetupReferenceSlot` is not `NULL`, then `pDecodeInfo->pSetupReferenceSlot->slotIndex` **must** be less than the `VkVideoSessionCreateInfoKHR::maxDpbSlots` specified when the bound video session was created
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07173
 If `pDecodeInfo->pSetupReferenceSlot` is not `NULL`, then `pDecodeInfo->pSetupReferenceSlot->pPictureResource->codedOffset` **must** be an integer multiple of `codedOffsetGranularity`
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07149
 If `pDecodeInfo->pSetupReferenceSlot` is not `NULL`, then `pDecodeInfo->pSetupReferenceSlot->pPictureResource` **must** `match` one of the `bound reference picture resource`
- VUID-vkCmdDecodeVideoKHR-activeReferencePictureCount-07150
`activeReferencePictureCount` **must** be less than or equal to the `VkVideoSessionCreateInfoKHR::maxActiveReferencePictures` specified when the bound video session was created
- VUID-vkCmdDecodeVideoKHR-slotIndex-07256
 The `slotIndex` member of each element of `pDecodeInfo->pReferenceSlots` **must** be less than the `VkVideoSessionCreateInfoKHR::maxDpbSlots` specified when the bound video session was created
- VUID-vkCmdDecodeVideoKHR-codedOffset-07257
 The `codedOffset` member of the `VkVideoPictureResourceInfoKHR` structure pointed to by the `pPictureResource` member of each element of `pDecodeInfo->pReferenceSlots` **must** be an integer multiple of `codedOffsetGranularity`
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07151
 The `pPictureResource` member of each element of `pDecodeInfo->pReferenceSlots` **must** `match` one of the `bound reference picture resource` associated with the DPB slot index specified in the `slotIndex` member of that element
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07264
 Each video picture resource corresponding to the `pPictureResource` member specified in the elements of `pDecodeInfo->pReferenceSlots` **must** be `unique` within `pDecodeInfo-`

>pReferenceSlots

- VUID-vkCmdDecodeVideoKHR-dpbFrameUseCount-07176
All elements of `dpbFrameUseCount` **must** be less than or equal to 1
- VUID-vkCmdDecodeVideoKHR-dpbTopFieldUseCount-07177
All elements of `dpbTopFieldUseCount` **must** be less than or equal to 1
- VUID-vkCmdDecodeVideoKHR-dpbBottomFieldUseCount-07178
All elements of `dpbBottomFieldUseCount` **must** be less than or equal to 1
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07252
If `pDecodeInfo->pSetupReferenceSlot` is `NULL` or `pDecodeInfo->pSetupReferenceSlot->pPictureResource` does not `refer` to the same image subresource as `pDecodeInfo->dstPictureResource`, then the image subresource `referred` to by `pDecodeInfo->dstPictureResource` **must** be in the `VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR` layout at the time the video decode operation is executed on the device
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07253
If `pDecodeInfo->pSetupReferenceSlot` is not `NULL` and `pDecodeInfo->pSetupReferenceSlot->pPictureResource` `refers` to the same image subresource as `pDecodeInfo->dstPictureResource`, then the image subresource `referred` to by `pDecodeInfo->dstPictureResource` **must** be in the `VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR` layout at the time the video decode operation is executed on the device
- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07254
If `pDecodeInfo->pSetupReferenceSlot` is not `NULL`, then the image subresource `referred` to by `pDecodeInfo->pSetupReferenceSlot->pPictureResource` **must** be in the `VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR` layout at the time the video decode operation is executed on the device
- VUID-vkCmdDecodeVideoKHR-pPictureResource-07255
The image subresource `referred` to by the `pPictureResource` member of each element of `pDecodeInfo->pReferenceSlots` **must** be in the `VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR` layout at the time the video decode operation is executed on the device
- VUID-vkCmdDecodeVideoKHR-pNext-07152
If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then the `pNext` chain of `pDecodeInfo` **must** include a `VkVideoDecodeH264PictureInfoKHR` structure
- VUID-vkCmdDecodeVideoKHR-None-07258
If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR` but was not created with `interlaced frame support`, then the `decode output picture` **must** represent a frame
- VUID-vkCmdDecodeVideoKHR-pSliceOffsets-07153
If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then all elements of the `pSliceOffsets` member of the `VkVideoDecodeH264PictureInfoKHR` structure included in the `pNext` chain of `pDecodeInfo` **must** be less than `pDecodeInfo->srcBufferRange`
- VUID-vkCmdDecodeVideoKHR-StdVideoH264SequenceParameterSet-07154
If the bound video session was created with the video codec operation

VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH264SequenceParameterSet entry with seq_parameter_set_id matching StdVideoDecodeH264PictureInfo::seq_parameter_set_id that is provided in the pStdPictureInfo member of the VkVideoDecodeH264PictureInfoKHR structure included in the pNext chain of pDecodeInfo

- VUID-vkCmdDecodeVideoKHR-StdVideoH264PictureParameterSet-07155

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH264PictureParameterSet entry with seq_parameter_set_id and pic_parameter_set_id matching StdVideoDecodeH264PictureInfo::seq_parameter_set_id and StdVideoDecodeH264PictureInfo::pic_parameter_set_id, respectively, that are provided in the pStdPictureInfo member of the VkVideoDecodeH264PictureInfoKHR structure included in the pNext chain of pDecodeInfo

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07156

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and pDecodeInfo->pSetupReferenceSlot is not NULL, then the pNext chain of pDecodeInfo->pSetupReferenceSlot **must** include a VkVideoDecodeH264DpbSlotInfoKHR structure

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07259

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR but was not created with interlaced frame support, and pDecodeInfo->pSetupReferenceSlot is not NULL, then the reconstructed picture **must** represent a frame

- VUID-vkCmdDecodeVideoKHR-pNext-07157

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, then the pNext chain of each element of pDecodeInfo->pReferenceSlots **must** include a VkVideoDecodeH264DpbSlotInfoKHR structure

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07260

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR but was not created with interlaced frame support, then each active reference picture corresponding to the elements of pDecodeInfo->pReferenceSlots **must** represent a frame

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07261

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, pDecodeInfo->pSetupReferenceSlot is not NULL, and the decode output picture represents a frame, then the reconstructed picture **must** also represent a frame

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07262

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, pDecodeInfo->pSetupReferenceSlot is not NULL, and the decode output picture represents a top field, then the reconstructed picture **must** also represent a top field

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07263

If the bound video session was created with the video codec operation

VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR, pDecodeInfo->pSetupReferenceSlot is not NULL, and the decode output picture represents a bottom field, then the reconstructed picture **must** also represent a bottom field

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07266

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and an active reference picture corresponding to any element of pDecodeInfo->pReferenceSlots represents a frame, then the DPB slot index of the bound video session specified by the slotIndex member of that element **must** be currently associated with a frame picture matching the video picture resource specified by the pPictureResource member of the same element at the time the command is executed on the device

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07267

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and an active reference picture corresponding to any element of pDecodeInfo->pReferenceSlots represents a top field, then the DPB slot index of the bound video session specified by the slotIndex member of that element **must** be currently associated with a top field picture matching the video picture resource specified by the pPictureResource member of the same element at the time the command is executed on the device

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07268

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR and an active reference picture corresponding to any element of pDecodeInfo->pReferenceSlots represents a bottom field, then the DPB slot index of the bound video session specified by the slotIndex member of that element **must** be currently associated with a bottom field picture matching the video picture resource specified by the pPictureResource member of the same element at the time the command is executed on the device

- VUID-vkCmdDecodeVideoKHR-pNext-07158

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the pNext chain of pDecodeInfo **must** include a VkVideoDecodeH265PictureInfoKHR structure

- VUID-vkCmdDecodeVideoKHR-pSliceSegmentOffsets-07159

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then all elements of the pSliceSegmentOffsets member of the VkVideoDecodeH265PictureInfoKHR structure included in the pNext chain of pDecodeInfo **must** be less than pDecodeInfo->srcBufferRange

- VUID-vkCmdDecodeVideoKHR-StdVideoH265VideoParameterSet-07160

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH265VideoParameterSet entry with vps_video_parameter_set_id matching StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id that is provided in the pStdPictureInfo member of the VkVideoDecodeH265PictureInfoKHR structure included in the pNext chain of pDecodeInfo

- VUID-vkCmdDecodeVideoKHR-StdVideoH265SequenceParameterSet-07161

If the bound video session was created with the video codec operation

VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH265SequenceParameterSet entry with sps_video_parameter_set_id and sps_seq_parameter_set_id matching StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id and StdVideoDecodeH265PictureInfo::pps_seq_parameter_set_id, respectively, that are provided in the pStdPictureInfo member of the VkVideoDecodeH265PictureInfoKHR structure included in the pNext chain of pDecodeInfo

- VUID-vkCmdDecodeVideoKHR-StdVideoH265PictureParameterSet-07162

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the bound video session parameters object **must** contain a StdVideoH265PictureParameterSet entry with sps_video_parameter_set_id, pps_seq_parameter_set_id, and pps_pic_parameter_set_id matching StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id, StdVideoDecodeH265PictureInfo::pps_seq_parameter_set_id, and StdVideoDecodeH265PictureInfo::pps_pic_parameter_set_id, respectively, that are provided in the pStdPictureInfo member of the VkVideoDecodeH265PictureInfoKHR structure included in the pNext chain of pDecodeInfo

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-07163

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR and pDecodeInfo->pSetupReferenceSlot is not NULL, then the pNext chain of pDecodeInfo->pSetupReferenceSlot **must** include a VkVideoDecodeH265DpbSlotInfoKHR structure

- VUID-vkCmdDecodeVideoKHR-pNext-07164

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR, then the pNext chain of each element of pDecodeInfo->pReferenceSlots **must** include a VkVideoDecodeH265DpbSlotInfoKHR structure

- VUID-vkCmdDecodeVideoKHR-filmGrainSupport-09248

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR and VkVideoDecodeAV1ProfileInfoKHR::filmGrainSupport set to VK_FALSE, then film grain **must** not be enabled for the decoded picture

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-09249

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, pDecodeInfo->pSetupReferenceSlot is not NULL, and film grain is enabled for the decoded picture, then the video picture resources specified by pDecodeInfo->dstPictureResource and pDecodeInfo->pSetupReferenceSlot->pPictureResource **must** not match

- VUID-vkCmdDecodeVideoKHR-pNext-09250

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then the pNext chain of pDecodeInfo **must** include a VkVideoDecodeAV1PictureInfoKHR structure

- VUID-vkCmdDecodeVideoKHR-frameHeaderOffset-09251

If the bound video session was created with the video codec operation VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR, then the frameHeaderOffset member of the

[VkVideoDecodeAV1PictureInfoKHR](#) structure included in the [pNext](#) chain of [pDecodeInfo](#) **must** be less than the minimum of [pDecodeInfo->srcBufferRange](#)

- VUID-vkCmdDecodeVideoKHR-pTileOffsets-09253

If the bound video session was created with the video codec operation [VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR](#), then all elements of the [pTileOffsets](#) member of the [VkVideoDecodeAV1PictureInfoKHR](#) structure included in the [pNext](#) chain of [pDecodeInfo](#) **must** be less than [pDecodeInfo->srcBufferRange](#)

- VUID-vkCmdDecodeVideoKHR-pTileOffsets-09252

If the bound video session was created with the video codec operation [VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR](#), then for each element *i* of the [pTileOffsets](#) and [pTileSizes](#) members of the [VkVideoDecodeAV1PictureInfoKHR](#) structure included in the [pNext](#) chain of [pDecodeInfo](#) the sum of [pTileOffsets\[i\]](#) and [pTileSizes\[i\]](#) **must** be less than or equal to [pDecodeInfo->srcBufferRange](#)

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-09254

If the bound video session was created with the video codec operation [VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR](#) and [pDecodeInfo->pSetupReferenceSlot](#) is not NULL, then the [pNext](#) chain of [pDecodeInfo->pSetupReferenceSlot](#) **must** include a [VkVideoDecodeAV1DpbSlotInfoKHR](#) structure

- VUID-vkCmdDecodeVideoKHR-pNext-09255

If the bound video session was created with the video codec operation [VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR](#), then the [pNext](#) chain of each element of [pDecodeInfo->pReferenceSlots](#) **must** include a [VkVideoDecodeAV1DpbSlotInfoKHR](#) structure

- VUID-vkCmdDecodeVideoKHR-referenceNameSlotIndices-09262

If the bound video session was created with the video codec operation [VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR](#), then each element of the [referenceNameSlotIndices](#) array member of the [VkVideoDecodeAV1PictureInfoKHR](#) structure included in the [pNext](#) chain of [pDecodeInfo](#) **must** either be negative or **must** equal the [slotIndex](#) member of one of the elements of [pDecodeInfo->pReferenceSlots](#)

- VUID-vkCmdDecodeVideoKHR-slotIndex-09263

If the bound video session was created with the video codec operation [VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR](#), then the [slotIndex](#) member of each element of [pDecodeInfo->pReferenceSlots](#) **must** equal one of the elements of the [referenceNameSlotIndices](#) array member of the [VkVideoDecodeAV1PictureInfoKHR](#) structure included in the [pNext](#) chain of [pDecodeInfo](#)

Valid Usage (Implicit)

- VUID-vkCmdDecodeVideoKHR-commandBuffer-parameter

[commandBuffer](#) **must** be a valid [VkCommandBuffer](#) handle

- VUID-vkCmdDecodeVideoKHR-pDecodeInfo-parameter

[pDecodeInfo](#) **must** be a valid pointer to a valid [VkVideoDecodeInfoKHR](#) structure

- VUID-vkCmdDecodeVideoKHR-commandBuffer-recording

`commandBuffer` **must** be in the `recording` state

- VUID-vkCmdDecodeVideoKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support decode operations
- VUID-vkCmdDecodeVideoKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdDecodeVideoKHR-videocoding
This command **must** only be called inside of a video coding scope
- VUID-vkCmdDecodeVideoKHR-bufferlevel
`commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Inside	Decode	Action

The `VkVideoDecodeInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_queue
typedef struct VkVideoDecodeInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkVideoDecodeFlagsKHR    flags;
    VkBuffer                  srcBuffer;
    VkDeviceSize              srcBufferOffset;
    VkDeviceSize              srcBufferRange;
    VkVideoPictureResourceInfoKHR dstPictureResource;
    const VkVideoReferenceSlotInfoKHR* pSetupReferenceSlot;
    uint32_t                  referenceSlotCount;
    const VkVideoReferenceSlotInfoKHR* pReferenceSlots;
} VkVideoDecodeInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.

- `srcBuffer` is the source video bitstream buffer to read the encoded bitstream from.
- `srcBufferOffset` is the starting offset in bytes from the start of `srcBuffer` to read the encoded bitstream from.
- `srcBufferRange` is the size in bytes of the encoded bitstream to decode from `srcBuffer`, starting from `srcBufferOffset`.
- `dstPictureResource` is the video picture resource to use as the [decode output picture](#).
- `pSetupReferenceSlot` is `NULL` or a pointer to a [VkVideoReferenceSlotInfoKHR](#) structure specifying the [reconstructed picture information](#).
- `referenceSlotCount` is the number of elements in the `pReferenceSlots` array.
- `pReferenceSlots` is `NULL` or a pointer to an array of [VkVideoReferenceSlotInfoKHR](#) structures describing the DPB slots and corresponding [reference picture](#) resources to use in this video decode operation (the set of [active reference pictures](#)).

Valid Usage

- VUID-VkVideoDecodeInfoKHR-srcBuffer-07165
`srcBuffer` **must** have been created with `VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR` set
- VUID-VkVideoDecodeInfoKHR-srcBufferOffset-07166
`srcBufferOffset` **must** be less than the size of `srcBuffer`
- VUID-VkVideoDecodeInfoKHR-srcBufferRange-07167
`srcBufferRange` **must** be less than or equal to the size of `srcBuffer` minus `srcBufferOffset`
- VUID-VkVideoDecodeInfoKHR-pSetupReferenceSlot-07168
If `pSetupReferenceSlot` is not `NULL`, then its `slotIndex` member **must** not be negative
- VUID-VkVideoDecodeInfoKHR-pSetupReferenceSlot-07169
If `pSetupReferenceSlot` is not `NULL`, then its `pPictureResource` **must** not be `NULL`
- VUID-VkVideoDecodeInfoKHR-slotIndex-07171
The `slotIndex` member of each element of `pReferenceSlots` **must** not be negative
- VUID-VkVideoDecodeInfoKHR-pPictureResource-07172
The `pPictureResource` member of each element of `pReferenceSlots` **must** not be `NULL`

Valid Usage (Implicit)

- VUID-VkVideoDecodeInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_INFO_KHR`
- VUID-VkVideoDecodeInfoKHR-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of [VkVideoDecodeAV1PictureInfoKHR](#), [VkVideoDecodeH264PictureInfoKHR](#), [VkVideoDecodeH265PictureInfoKHR](#), or [VkVideoInlineQueryInfoKHR](#)
- VUID-VkVideoDecodeInfoKHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique

- VUID-VkVideoDecodeInfoKHR-flags-zero bitmask
flags must be 0
- VUID-VkVideoDecodeInfoKHR-srcBuffer-parameter
srcBuffer must be a valid [VkBuffer](#) handle
- VUID-VkVideoDecodeInfoKHR-dstPictureResource-parameter
dstPictureResource must be a valid [VkVideoPictureResourceInfoKHR](#) structure
- VUID-VkVideoDecodeInfoKHR-pSetupReferenceSlot-parameter
If **pSetupReferenceSlot** is not NULL, **pSetupReferenceSlot must** be a valid pointer to a valid [VkVideoReferenceSlotInfoKHR](#) structure
- VUID-VkVideoDecodeInfoKHR-pReferenceSlots-parameter
If **referenceSlotCount** is not 0, **pReferenceSlots must** be a valid pointer to an array of **referenceSlotCount** valid [VkVideoReferenceSlotInfoKHR](#) structures

```
// Provided by VK_KHR_video_decode_queue
typedef VkFlags VkVideoDecodeFlagsKHR;
```

VkVideoDecodeFlagsKHR is a bitmask type for setting a mask, but is currently reserved for future use.

37.12. H.264 Decode Operations

Video decode operations using an [H.264 decode profile](#) can be used to decode elementary video stream sequences compliant to the [ITU-T H.264 Specification](#).

NOTE

Refer to the [Preamble](#) for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by Khronos.

This process is performed according to the [video decode operation steps](#) with the codec-specific semantics defined in section 8 of the [ITU-T H.264 Specification](#) as follows:

- Syntax elements, derived values, and other parameters are applied from the following structures:
 - The **StdVideoH264SequenceParameterSet** structure corresponding to the [active SPS](#) specifying the [H.264 sequence parameter set](#).
 - The **StdVideoH264PictureParameterSet** structure corresponding to the [active PPS](#) specifying the [H.264 picture parameter set](#).
 - The **StdVideoDecodeH264PictureInfo** structure specifying the [H.264 picture information](#).
 - The **StdVideoDecodeH264ReferenceInfo** structures specifying the [H.264 reference information](#) corresponding to the optional [reconstructed picture](#) and any [active reference pictures](#).
- The contents of the provided video bitstream buffer range are interpreted as defined in the [H.264 Decode Bitstream Data Access](#) section.
- Picture data in the [video picture resources](#) corresponding to the used [active reference pictures](#),

decode output picture, and optional reconstructed picture is accessed as defined in the H.264 Decode Picture Data Access section.

- The decision on reference picture setup is made according to the parameters specified in the H.264 picture information.

If the parameters and the bitstream adhere to the syntactic and semantic requirements defined in the corresponding sections of the ITU-T H.264 Specification, as described above, and the DPB slots associated with the active reference pictures all refer to valid picture references, then the video decode operation will complete successfully. Otherwise, the video decode operation **may** complete unsuccessfully.

37.12.1. H.264 Decode Bitstream Data Access

If the target decode output picture is a frame, then the video bitstream buffer range **should** contain a VCL NAL unit comprised of the slice headers and data of a picture representing an entire frame, as defined in sections 7.3.3 and 7.3.4, and this data is interpreted as defined in sections 7.4.3 and 7.4.4 of the ITU-T H.264 Specification, respectively.

If the target decode output picture is a field, then the video bitstream buffer range should contain a VCL NAL unit comprised of the slice headers and data of a picture representing a field, as defined in sections 7.3.3 and 7.3.4, and this data is interpreted as defined in sections 7.4.3 and 7.4.4 of the ITU-T H.264 Specification, respectively.

The offsets provided in `VkVideoDecodeH264PictureInfoKHR::pSliceOffsets` **should** specify the starting offsets corresponding to each slice header within the video bitstream buffer range.

37.12.2. H.264 Decode Picture Data Access

The effective `imageOffset` and `imageExtent` corresponding to a decode output picture, reference picture, or reconstructed picture used in video decode operations with an H.264 decode profile are defined as follows:

- `imageOffset` is `(codedOffset.x,codedOffset.y)` and `imageExtent` is `(codedExtent.width, codedExtent.height)`, if the picture represents a frame.
- `imageOffset` is `(codedOffset.x,codedOffset.y)` and `imageExtent` is `(codedExtent.width, codedExtent.height)`, if the picture represents a field and the picture layout of the used H.264 decode profile is `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR`.
- `imageOffset` is `(codedOffset.x,codedOffset.y)` and `imageExtent` is `(codedExtent.width, codedExtent.height / 2)`, if the picture represents a field and the picture layout of the used H.264 decode profile is `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR`.

Where `codedOffset` and `codedExtent` are the members of the `VkVideoPictureResourceInfoKHR` structure corresponding to the picture.

However, accesses to image data within a video picture resource happen at the granularity indicated by `VkVideoCapabilitiesKHR::pictureAccessGranularity`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile. This means that the complete image subregion accessed by video coding operations using an H.264 decode profile for the video

picture resource is defined as the set of texels within the coordinate range:

$([startX, endX), [startY, endY))$

Where:

- `startX` equals `imageOffset.x` rounded down to the nearest integer multiple of `pictureAccessGranularity.width`;
- `endX` equals `imageOffset.x + imageExtent.width` rounded up to the nearest integer multiple of `pictureAccessGranularity.width` and clamped to the width of the image subresource `referred` to by the corresponding `VkVideoPictureResourceInfoKHR` structure;
- `startY` equals `imageOffset.y` rounded down to the nearest integer multiple of `pictureAccessGranularity.height`;
- `endY` equals `imageOffset.y + imageExtent.height` rounded up to the nearest integer multiple of `pictureAccessGranularity.height` and clamped to the height of the image subresource `referred` to by the corresponding `VkVideoPictureResourceInfoKHR` structure.

In case of video decode operations using an `H.264 decode profile`, any access to a picture at the coordinates (x, y) , as defined by the [ITU-T H.264 Specification](#), is an access to the image subresource `referred` to by the corresponding `VkVideoPictureResourceInfoKHR` structure at the texel coordinates specified below:

- (x, y) , if the accessed picture represents a frame.
- $(x, y \times 2)$, if the accessed picture represents a top field and the picture layout of the used `H.264 decode profile` is `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR`.
- $(x, y \times 2 + 1)$, if the accessed picture represents a bottom field and the picture layout of the used `H.264 decode profile` is `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR`.
- (x, y) , if the accessed picture represents a top field and the picture layout of the used `H.264 decode profile` is `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR`.
- $(codedOffset.x + x, codedOffset.y + y)$, if the accessed picture represents a bottom field and the picture layout of the used `H.264 decode profile` is `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR`.

Where `codedOffset` is the member of the corresponding `VkVideoPictureResourceInfoKHR` structure.

37.12.3. H.264 Decode Profile

A video profile supporting H.264 video decode operations is specified by setting `VkVideoProfileInfoKHR::videoCodecOperation` to `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR` and adding a `VkVideoDecodeH264ProfileInfoKHR` structure to the `VkVideoProfileInfoKHR::pNext` chain.

The `VkVideoDecodeH264ProfileInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264ProfileInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    StdVideoH264ProfileIdc    stdProfileIdc;
    VkVideoDecodeH264PictureLayoutFlagBitsKHR pictureLayout;
} VkVideoDecodeH264ProfileInfoKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **stdProfileIdc** is a [StdVideoH264ProfileIdc](#) value specifying the H.264 codec profile IDC, as defined in section A.2 of the [ITU-T H.264 Specification](#).
- **pictureLayout** is a [VkVideoDecodeH264PictureLayoutFlagBitsKHR](#) value specifying the picture layout used by the H.264 video sequence to be decoded.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH264ProfileInfoKHR-sType-sType
sType must be [VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PROFILE_INFO_KHR](#)
- VUID-VkVideoDecodeH264ProfileInfoKHR-pictureLayout-parameter
If **pictureLayout** is not **0**, **pictureLayout must** be a valid [VkVideoDecodeH264PictureLayoutFlagBitsKHR](#) value

The H.264 video decode picture layout flags are defined as follows:

```
// Provided by VK_KHR_video_decode_h264
typedef enum VkVideoDecodeH264PictureLayoutFlagBitsKHR {
    VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_PROGRESSIVE_KHR = 0,
    VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR =
0x00000001,
    VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR =
0x00000002,
} VkVideoDecodeH264PictureLayoutFlagBitsKHR;
```

- [VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_PROGRESSIVE_KHR](#) specifies support for progressive content. This flag has the value **0**.
- [VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR](#) specifies support for or use of a picture layout for interlaced content where all lines belonging to the top field are decoded to the even-numbered lines within the picture resource, and all lines belonging to the bottom field are decoded to the odd-numbered lines within the picture resource.
- [VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR](#) specifies support for or use of a picture layout for interlaced content where all lines belonging to a field are grouped together in a single image subregion, and the two fields comprising the frame **can** be stored in

separate image subregions of the same image subresource or in separate image subresources.

```
// Provided by VK_KHR_video_decode_h264
typedef VkFlags VkVideoDecodeH264PictureLayoutFlagsKHR;
```

`VkVideoDecodeH264PictureLayoutFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoDecodeH264PictureLayoutFlagBitsKHR`.

37.12.4. H.264 Decode Capabilities

When calling `vkGetPhysicalDeviceVideoCapabilitiesKHR` to query the capabilities for an H.264 decode profile, the `VkVideoCapabilitiesKHR::pNext` chain **must** include a `VkVideoDecodeH264CapabilitiesKHR` structure that will be filled with the profile-specific capabilities.

The `VkVideoDecodeH264CapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264CapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    StdVideoH264LevelIdc maxLevelIdc;
    VkOffset2D          fieldOffsetGranularity;
} VkVideoDecodeH264CapabilitiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxLevelIdc` is a `StdVideoH264LevelIdc` value indicating the maximum H.264 level supported by the profile, where enum constant `STD_VIDEO_H264_LEVEL_IDC_<major>_<minor>` identifies H.264 level `<major>.<minor>` as defined in section A.3 of the [ITU-T H.264 Specification](#).
- `fieldOffsetGranularity` is the minimum alignment for `VkVideoPictureResourceInfoKHR::codedOffset` specified for a video picture resource when using the picture layout `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR`.

Valid Usage (Implicit)

- VUID-VkVideoDecodeH264CapabilitiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_CAPABILITIES_KHR`

37.12.5. H.264 Decode Parameter Sets

Video session parameters objects created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR` **can** contain the following types of parameters:

H.264 Sequence Parameter Sets (SPS)

Represented by `StdVideoH264SequenceParameterSet` structures and interpreted as follows:

- `reserved1` and `reserved2` are used only for padding purposes and are otherwise ignored;
- `seq_parameter_set_id` is used as the key of the SPS entry;
- `level_idc` is one of the enum constants `STD_VIDEO_H264_LEVEL_IDC_<major>_<minor>` identifying the H.264 level `<major>.<minor>` as defined in section A.3 of the [ITU-T H.264 Specification](#);
- if `flags.seq_scaling_matrix_present_flag` is set, then the `StdVideoH264ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
 - `scaling_list_present_mask` is a bitmask where bit index `i` corresponds to `seq_scaling_list_present_flag[i]` as defined in section 7.4.2.1 of the [ITU-T H.264 Specification](#);
 - `use_default_scaling_matrix_mask` is a bitmask where bit index `i` corresponds to `UseDefaultScalingMatrix4x4Flag[i]`, when `i < 6`, or corresponds to `UseDefaultScalingMatrix8x8Flag[i-6]`, otherwise, as defined in section 7.3.2.1 of the [ITU-T H.264 Specification](#);
 - `ScalingList4x4` and `ScalingList8x8` correspond to the identically named syntax elements defined in section 7.3.2.1 of the [ITU-T H.264 Specification](#);
- if `flags.vui_parameters_present_flag` is set, then `pSequenceParameterSetVui` is a pointer to a `StdVideoH264SequenceParameterSetVui` structure that is interpreted as follows:
 - `reserved1` is used only for padding purposes and is otherwise ignored;
 - `flags.colour_description_present_flag` is interpreted as the value of `colour_description_present_flag`, as defined in section E.2.1 of the [ITU-T H.264 Specification](#);

NOTE

The name of `colour_description_present_flag` was misspelled in the Video Std header.

- if `flags.nal_hrd_parameters_present_flag` or `flags.vcl_hrd_parameters_present_flag` is set, then the `StdVideoH264HrdParameters` structure pointed to by `pHrdParameters` is interpreted as follows:
 - `reserved1` is used only for padding purposes and is otherwise ignored;
 - all other members of `StdVideoH264HrdParameters` are interpreted as defined in section E.2.2 of the [ITU-T H.264 Specification](#);
- all other members of `StdVideoH264SequenceParameterSetVui` are interpreted as defined in section E.2.1 of the [ITU-T H.264 Specification](#);
- all other members of `StdVideoH264SequenceParameterSet` are interpreted as defined in section 7.4.2.1 of the [ITU-T H.264 Specification](#).

H.264 Picture Parameter Sets (PPS)

Represented by `StdVideoH264PictureParameterSet` structures and interpreted as follows:

- the pair constructed from `seq_parameter_set_id` and `pic_parameter_set_id` is used as the key of the PPS entry;
- if `flags.pic_scaling_matrix_present_flag` is set, then the `StdVideoH264ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
 - `scaling_list_present_mask` is a bitmask where bit index `i` corresponds to `pic_scaling_list_present_flag[i]` as defined in section 7.4.2.2 of the [ITU-T H.264 Specification](#);
 - `use_default_scaling_matrix_mask` is a bitmask where bit index `i` corresponds to `UseDefaultScalingMatrix4x4Flag[i]`, when `i < 6`, or corresponds to `UseDefaultScalingMatrix8x8Flag[i-6]`, otherwise, as defined in section 7.3.2.2 of the [ITU-T H.264 Specification](#);
 - `ScalingList4x4` and `ScalingList8x8` correspond to the identically named syntax elements defined in section 7.3.2.2 of the [ITU-T H.264 Specification](#);
- all other members of `StdVideoH264PictureParameterSet` are interpreted as defined in section 7.4.2.2 of the [ITU-T H.264 Specification](#).

When a `video session parameters` object is created with the codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, the `VkVideoSessionParametersCreateInfoKHR::pNext` chain **must** include a `VkVideoDecodeH264SessionParametersCreateInfoKHR` structure specifying the capacity and initial contents of the object.

The `VkVideoDecodeH264SessionParametersCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264SessionParametersCreateInfoKHR {
    VkStructureType              sType;
    const void*                  pNext;
    uint32_t                     maxStdSPSCount;
    uint32_t                     maxStdPPSCount;
    const VkVideoDecodeH264SessionParametersAddInfoKHR* pParametersAddInfo;
} VkVideoDecodeH264SessionParametersCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxStdSPSCount` is the maximum number of [H.264 SPS](#) entries the created `VkVideoSessionParametersKHR` **can** contain.
- `maxStdPPSCount` is the maximum number of [H.264 PPS](#) entries the created `VkVideoSessionParametersKHR` **can** contain.
- `pParametersAddInfo` is `NULL` or a pointer to a `VkVideoDecodeH264SessionParametersAddInfoKHR` structure specifying H.264 parameters to add upon object creation.

Valid Usage (Implicit)

- `VUID-VkVideoDecodeH264SessionParametersCreateInfoKHR-sType-sType`

`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR`

- `VOID-VkVideoDecodeH264SessionParametersCreateInfoKHR-pParametersAddInfo-parameter`

If `pParametersAddInfo` is not `NULL`, `pParametersAddInfo` **must** be a valid pointer to a valid `VkVideoDecodeH264SessionParametersAddInfoKHR` structure

The `VkVideoDecodeH264SessionParametersAddInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264SessionParametersAddInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    uint32_t                 stdSPSCount;
    const StdVideoH264SequenceParameterSet* pStdSPSs;
    uint32_t                 stdPPSCount;
    const StdVideoH264PictureParameterSet* pStdPPSs;
} VkVideoDecodeH264SessionParametersAddInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdSPSCount` is the number of elements in the `pStdSPSs` array.
- `pStdSPSs` is a pointer to an array of `StdVideoH264SequenceParameterSet` structures describing the `H.264 SPS` entries to add.
- `stdPPSCount` is the number of elements in the `pStdPPSs` array.
- `pStdPPSs` is a pointer to an array of `StdVideoH264PictureParameterSet` structures describing the `H.264 PPS` entries to add.

This structure **can** be specified in the following places:

- In the `pParametersAddInfo` member of the `VkVideoDecodeH264SessionParametersCreateInfoKHR` structure specified in the `pNext` chain of `VkVideoSessionParametersCreateInfoKHR` used to create a `video session parameters` object. In this case, if the video codec operation the video session parameters object is created with is `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then it defines the set of initial parameters to add to the created object (see [Creating Video Session Parameters](#)).
- In the `pNext` chain of `VkVideoSessionParametersUpdateInfoKHR`. In this case, if the video codec operation the `video session parameters` object to be updated was created with is `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, then it defines the set of parameters to add to it (see [Updating Video Session Parameters](#)).

Valid Usage

- `VOID-VkVideoDecodeH264SessionParametersAddInfoKHR-None-04825`

The `seq_parameter_set_id` member of each `StdVideoH264SequenceParameterSet` structure

specified in the elements of `pStdSPSs` **must** be unique within `pStdSPSs`

- VUID-VkVideoDecodeH264SessionParametersAddInfoKHR-None-04826

The pair constructed from the `seq_parameter_set_id` and `pic_parameter_set_id` members of each `StdVideoH264PictureParameterSet` structure specified in the elements of `pStdPPSs` **must** be unique within `pStdPPSs`

Valid Usage (Implicit)

- VUID-VkVideoDecodeH264SessionParametersAddInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR`
- VUID-VkVideoDecodeH264SessionParametersAddInfoKHR-pStdSPSs-parameter
If `stdSPSCount` is not 0, `pStdSPSs` **must** be a valid pointer to an array of `stdSPSCount` `StdVideoH264SequenceParameterSet` values
- VUID-VkVideoDecodeH264SessionParametersAddInfoKHR-pStdPPSs-parameter
If `stdPPSCount` is not 0, `pStdPPSs` **must** be a valid pointer to an array of `stdPPSCount` `StdVideoH264PictureParameterSet` values

37.12.6. H.264 Decoding Parameters

The `VkVideoDecodeH264PictureInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264PictureInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    const StdVideoDecodeH264PictureInfo* pStdPictureInfo;
    uint32_t           sliceCount;
    const uint32_t*     pSliceOffsets;
} VkVideoDecodeH264PictureInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdPictureInfo` is a pointer to a `StdVideoDecodeH264PictureInfo` structure specifying [H.264 picture information](#).
- `sliceCount` is the number of elements in `pSliceOffsets`.
- `pSliceOffsets` is a pointer to an array of `sliceCount` offsets specifying the start offset of the slices of the picture within the video bitstream buffer range specified in `VkVideoDecodeInfoKHR`.

This structure is specified in the `pNext` chain of the `VkVideoDecodeInfoKHR` structure passed to `vkCmdDecodeVideoKHR` to specify the codec-specific picture information for an [H.264 decode operation](#).

Decode Output Picture Information

When this structure is specified in the `pNext` chain of the `VkVideoDecodeInfoKHR` structure passed to `vkCmdDecodeVideoKHR`, the information related to the `decode output picture` is defined as follows:

- If `pStdPictureInfo->flags.field_pic_flag` is not set, then the picture represents a frame.
- If `pStdPictureInfo->flags.field_pic_flag` is set, then the picture represents a field. Specifically:
 - If `pStdPictureInfo->flags.bottom_field_flag` is not set, then the picture represents the top field of the frame.
 - If `pStdPictureInfo->flags.bottom_field_flag` is set, then the picture represents the bottom field of the frame.
- The image subregion used is determined according to the [H.264 Decode Picture Data Access](#) section.
- The decode output picture is associated with the [H.264 picture information](#) provided in `pStdPictureInfo`.

Std Picture Information

The members of the `StdVideoDecodeH264PictureInfo` structure pointed to by `pStdPictureInfo` are interpreted as follows:

- `reserved1` and `reserved2` are used only for padding purposes and are otherwise ignored;
- `flags.is_intra` as defined in section 3.73 of the [ITU-T H.264 Specification](#);
- `flags.is_reference` as defined in section 3.136 of the [ITU-T H.264 Specification](#);
- `flags.complementary_field_pair` as defined in section 3.35 of the [ITU-T H.264 Specification](#);
- `seq_parameter_set_id` and `pic_parameter_set_id` are used to identify the active parameter sets, as described below;
- all other members are interpreted as defined in section 7.4.3 of the [ITU-T H.264 Specification](#).

Reference picture setup is controlled by the value of `StdVideoDecodeH264PictureInfo::flags.is_reference`. If it is set and a `reconstructed picture` is specified, then the latter is used as the target of picture reconstruction to `activate` the `DPB slot` specified in `pDecodeInfo->pSetupReferenceSlot->slotIndex`. If `StdVideoDecodeH264PictureInfo::flags.is_reference` is not set, but a `reconstructed picture` is specified, then the corresponding picture reference associated with the `DPB slot` is invalidated, as described in the [DPB Slot States](#) section.

Active Parameter Sets

The members of the `StdVideoDecodeH264PictureInfo` structure pointed to by `pStdPictureInfo` are used to select the active parameter sets to use from the bound video session parameters object, as follows:

- The *active SPS* is the `SPS` identified by the key specified in `StdVideoDecodeH264PictureInfo::seq_parameter_set_id`.
- The *active PPS* is the `PPS` identified by the key specified by the pair constructed from

`StdVideoDecodeH264PictureInfo::seq_parameter_set_id` and `StdVideoDecodeH264PictureInfo::pic_parameter_set_id`.

Valid Usage (Implicit)

- `VUID-VkVideoDecodeH264PictureInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PICTURE_INFO_KHR`
- `VUID-VkVideoDecodeH264PictureInfoKHR-pStdPictureInfo-parameter`
`pStdPictureInfo` **must** be a valid pointer to a valid `StdVideoDecodeH264PictureInfo` value
- `VUID-VkVideoDecodeH264PictureInfoKHR-pSliceOffsets-parameter`
`pSliceOffsets` **must** be a valid pointer to an array of `sliceCount uint32_t` values
- `VUID-VkVideoDecodeH264PictureInfoKHR-sliceCount-arraylength`
`sliceCount` **must** be greater than 0

The `VkVideoDecodeH264DpbSlotInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h264
typedef struct VkVideoDecodeH264DpbSlotInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    const StdVideoDecodeH264ReferenceInfo* pStdReferenceInfo;
} VkVideoDecodeH264DpbSlotInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdReferenceInfo` is a pointer to a `StdVideoDecodeH264ReferenceInfo` structure specifying [H.264 reference information](#).

This structure is specified in the `pNext` chain of `VkVideoDecodeInfoKHR::pSetupReferenceSlot`, if not `NULL`, and the `pNext` chain of the elements of `VkVideoDecodeInfoKHR::pReferenceSlots` to specify the codec-specific reference picture information for an [H.264 decode operation](#).

Active Reference Picture Information

When this structure is specified in the `pNext` chain of the elements of `VkVideoDecodeInfoKHR::pReferenceSlots`, one or two elements are added to the list of [active reference pictures](#) used by the video decode operation for each element of `VkVideoDecodeInfoKHR::pReferenceSlots` as follows:

- If neither `pStdReferenceInfo->flags.top_field_flag` nor `pStdReferenceInfo->flags.bottom_field_flag` is set, then the picture is added as a frame reference to the list of active reference pictures.
- If `pStdReferenceInfo->flags.top_field_flag` is set, then the picture is added as a top field reference to the list of active reference pictures.
- If `pStdReferenceInfo->flags.bottom_field_flag` is set, then the picture is added as a bottom

field reference to the list of active reference pictures.

- For each added reference picture, the corresponding image subregion used is determined according to the [H.264 Decode Picture Data Access](#) section.
- Each added reference picture is associated with the [DPB slot](#) index specified in the `slotIndex` member of the corresponding element of `VkVideoDecodeInfoKHR::pReferenceSlots`.
- Each added reference picture is associated with the [H.264 reference information](#) provided in `pStdReferenceInfo`.

NOTE

When both the top and bottom field of an interlaced frame currently associated with a DPB slot is intended to be used as an active reference picture and both fields are stored in the same image subregion (which is the case when using `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_INTERLEAVED_LINES_BIT_KHR` which stores the two fields at even and odd scanlines of the same image subregion), both references have to be provided through a single `VkVideoReferenceSlotInfoKHR` structure that has both `flags.top_field_flag` and `flags.bottom_field_flag` set in the `StdVideoDecodeH264ReferenceInfo` structure pointed to by the `pStdReferenceInfo` member of the `VkVideoDecodeH264DpbSlotInfoKHR` structure included in the corresponding `VkVideoReferenceSlotInfoKHR` structure's `pNext` chain. However, this approach can only be used when both fields are stored in the same image subregion. If that is not the case (e.g. when using `VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR` which requires separate `codedOffset` values for the two fields and also allows storing the two fields of a frame in separate image layers or entirely separate images), then a separate `VkVideoReferenceSlotInfoKHR` structure needs to be provided for referencing the two fields, each only setting one of `flags.top_field_flag` or `flags.bottom_field_flag`, and providing the appropriate video picture resource information in `VkVideoReferenceSlotInfoKHR::pPictureResource`.

Reconstructed Picture Information

When this structure is specified in the `pNext` chain of `VkVideoDecodeInfoKHR::pSetupReferenceSlot`, the information related to the [reconstructed picture](#) is defined as follows:

- If neither `pStdReferenceInfo->flags.top_field_flag` nor `pStdReferenceInfo->flags.bottom_field_flag` is set, then the picture represents a frame.
- If `pStdReferenceInfo->flags.top_field_flag` is set, then the picture represents a field, specifically, the top field of the frame.
- If `pStdReferenceInfo->flags.bottom_field_flag` is set, then the picture represents a field, specifically, the bottom field of the frame.
- The image subregion used is determined according to the [H.264 Decode Picture Data Access](#) section.
- If [reference picture setup](#) is requested, then the reconstructed picture is used to [activate](#) the [DPB slot](#) with the index specified in `VkVideoDecodeInfoKHR::pSetupReferenceSlot->slotIndex`.
- The reconstructed picture is associated with the [H.264 reference information](#) provided in `pStdReferenceInfo`.

Std Reference Information

The members of the `StdVideoDecodeH264ReferenceInfo` structure pointed to by `pStdReferenceInfo` are interpreted as follows:

- `flags.top_field_flag` is used to indicate whether the reference is used as top field reference;
- `flags.bottom_field_flag` is used to indicate whether the reference is used as bottom field reference;
- `flags.used_for_long_term_reference` is used to indicate whether the picture is marked as “used for long-term reference” as defined in section 8.2.5.1 of the [ITU-T H.264 Specification](#);
- `flags.is_non_existing` is used to indicate whether the picture is marked as “non-existing” as defined in section 8.2.5.2 of the [ITU-T H.264 Specification](#);
- all other members are interpreted as defined in section 8.2 of the [ITU-T H.264 Specification](#).

Valid Usage (Implicit)

- `VUID-VkVideoDecodeH264DpbSlotInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_DPB_SLOT_INFO_KHR`
- `VUID-VkVideoDecodeH264DpbSlotInfoKHR-pStdReferenceInfo-parameter`
`pStdReferenceInfo` **must** be a valid pointer to a valid `StdVideoDecodeH264ReferenceInfo` value

37.12.7. H.264 Decode Requirements

This section describes the **required** H.264 decoding capabilities for physical devices that have at least one queue family that supports the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`, as returned by `vkGetPhysicalDeviceQueueFamilyProperties2KHR` in `VkQueueFamilyVideoPropertiesKHR::videoCodecOperations`.

Table 46. Required [Video Std Header Versions](#)

Video Std Header Name	Version
<code>vulkan_video_codec_h264std_decode</code>	1.0.0

Table 47. Required Video Capabilities

Video Capability	Requirement	Requirement Type ¹
VkVideoCapabilitiesKHR		
<code>flags</code>	-	min
<code>minBitstreamBufferOffsetAlignment</code>	4096	max
<code>minBitstreamBufferSizeAlignment</code>	4096	max
<code>pictureAccessGranularity</code>	(64,64)	max

Video Capability	Requirement	Requirement Type ¹
<code>minCodedExtent</code>	-	max
<code>maxCodedExtent</code>	-	min
<code>maxDpbSlots</code>	0	min
<code>maxActiveReferencePictures</code>	0	min
VkVideoDecodeCapabilitiesKHR		
<code>flags</code>	VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR or VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR	min
VkVideoDecodeH264CapabilitiesKHR		
<code>maxLevelIdc</code>	STD_VIDEO_H264_LEVEL_IDC_1_0	min
<code>fieldOffsetGranularity</code>	(0,0) except for profiles using VK_VIDEO_DECODE_H264_PICTURE_LAYOUT_INTERLACED_SEPARATE_PLANES_BIT_KHR	implementation-dependent

1

The **Requirement Type** column specifies the requirement is either the minimum value all implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

37.13. H.265 Decode Operations

Video decode operations using an [H.265 decode profile](#) can be used to decode elementary video stream sequences compliant to the [ITU-T H.265 Specification](#).

NOTE

Refer to the [Preamble](#) for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by Khronos.

This process is performed according to the [video decode operation steps](#) with the codec-specific semantics defined in section 8 of [ITU-T H.265 Specification](#):

- Syntax elements, derived values, and other parameters are applied from the following structures:
 - The `StdVideoH265VideoParameterSet` structure corresponding to the [active VPS](#) specifying the [H.265 video parameter set](#).

- The `StdVideoH265SequenceParameterSet` structure corresponding to the `active SPS` specifying the `H.265 sequence parameter set`.
- The `StdVideoH265PictureParameterSet` structure corresponding to the `active PPS` specifying the `H.265 picture parameter set`.
- The `StdVideoDecodeH265PictureInfo` structure specifying the `H.265 picture information`.
- The `StdVideoDecodeH265ReferenceInfo` structures specifying the `H.265 reference information` corresponding to the optional `reconstructed picture` and any `active reference pictures`.
- The contents of the provided video bitstream buffer range are interpreted as defined in the `H.265 Decode Bitstream Data Access` section.
- Picture data in the `video picture resources` corresponding to the used `active reference pictures`, `decode output picture`, and optional `reconstructed picture` is accessed as defined in the `H.265 Decode Picture Data Access` section.
- The decision on `reference picture setup` is made according to the parameters specified in the `H.265 picture information`.

If the parameters and the bitstream adhere to the syntactic and semantic requirements defined in the corresponding sections of the `ITU-T H.265 Specification`, as described above, and the `DPB slots` associated with the `active reference pictures` all refer to `valid picture references`, then the video decode operation will complete successfully. Otherwise, the video decode operation **may** complete `unsuccessfully`.

37.13.1. H.265 Decode Bitstream Data Access

The video bitstream buffer range **should** contain a VCL NAL unit comprised of the slice segment headers and data of a picture representing a frame, as defined in sections 7.3.6 and 7.3.8, and this data is interpreted as defined in sections 7.4.7 and 7.4.9 of the `ITU-T H.265 Specification`, respectively.

The offsets provided in `VkVideoDecodeH265PictureInfoKHR::pSliceSegmentOffsets` **should** specify the starting offsets corresponding to each slice segment header within the video bitstream buffer range.

37.13.2. H.265 Decode Picture Data Access

Accesses to image data within a video picture resource happen at the granularity indicated by `VkVideoCapabilitiesKHR::pictureAccessGranularity`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used `video profile`. Accordingly, the complete image subregion of a `decode output picture`, `reference picture`, or `reconstructed picture` accessed by video coding operations using an `H.265 decode profile` is defined as the set of texels within the coordinate range:

$([0, \text{endX}), [0, \text{endY}))$

Where:

- `endX` equals `codedExtent.width` rounded up to the nearest integer multiple of

`pictureAccessGranularity.width` and clamped to the width of the image subresource [referred](#) to by the corresponding [VkVideoPictureResourceInfoKHR](#) structure;

- `endY` equals `codedExtent.height` rounded up to the nearest integer multiple of `pictureAccessGranularity.height` and clamped to the height of the image subresource [referred](#) to by the corresponding [VkVideoPictureResourceInfoKHR](#) structure;

Where `codedExtent` is the member of the [VkVideoPictureResourceInfoKHR](#) structure corresponding to the picture.

In case of video decode operations using an [H.265 decode profile](#), any access to a picture at the coordinates (x,y), as defined by the [ITU-T H.265 Specification](#), is an access to the image subresource [referred](#) to by the corresponding [VkVideoPictureResourceInfoKHR](#) structure at the texel coordinates (x,y).

37.13.3. H.265 Decode Profile

A video profile supporting H.265 video decode operations is specified by setting [VkVideoProfileInfoKHR::videoCodecOperation](#) to `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR` and adding a [VkVideoDecodeH265ProfileInfoKHR](#) structure to the [VkVideoProfileInfoKHR::pNext](#) chain.

The [VkVideoDecodeH265ProfileInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265ProfileInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    StdVideoH265ProfileIdc stdProfileIdc;
} VkVideoDecodeH265ProfileInfoKHR;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdProfileIdc` is a [StdVideoH265ProfileIdc](#) value specifying the H.265 codec profile IDC, as defined in section A.3 of the [ITU-T H.265 Specification](#).

Valid Usage (Implicit)

- VUID-VkVideoDecodeH265ProfileInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PROFILE_INFO_KHR`

37.13.4. H.265 Decode Capabilities

When calling [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) to query the capabilities for an [H.265 decode profile](#), the [VkVideoCapabilitiesKHR::pNext](#) chain **must** include a [VkVideoDecodeH265CapabilitiesKHR](#) structure that will be filled with the profile-specific capabilities.

The [VkVideoDecodeH265CapabilitiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265CapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    StdVideoH265LevelIdc maxLevelIdc;
} VkVideoDecodeH265CapabilitiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxLevelIdc` is a `StdVideoH265LevelIdc` value indicating the maximum H.265 level supported by the profile, where enum constant `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifies H.265 level `<major>.<minor>` as defined in section A.4 of the [ITU-T H.265 Specification](#).

Valid Usage (Implicit)

- VUID-VkVideoDecodeH265CapabilitiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_CAPABILITIES_KHR`

37.13.5. H.265 Decode Parameter Sets

`Video session parameters` objects created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR` can contain the following types of parameters:

H.265 Video Parameter Sets (VPS)

Represented by `StdVideoH265VideoParameterSet` structures and interpreted as follows:

- `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
- `vps_video_parameter_set_id` is used as the key of the VPS entry;
- the `max_latency_increase_plus1`, `max_dec_pic_buffering_minus1`, and `max_num_reorder_pics` members of the `StdVideoH265DecPicBufMgr` structure pointed to by `pDecPicBufMgr` correspond to `vps_max_latency_increase_plus1`, `vps_max_dec_pic_buffering_minus1`, and `vps_max_num_reorder_pics`, respectively, as defined in section 7.4.3.1 of the [ITU-T H.265 Specification](#);
- the `StdVideoH265HrdParameters` structure pointed to by `pHrdParameters` is interpreted as follows:
 - `reserved` is used only for padding purposes and is otherwise ignored;
 - `flags.fixed_pic_rate_general_flag` is a bitmask where bit index `i` corresponds to `fixed_pic_rate_general_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
 - `flags.fixed_pic_rate_within_cvs_flag` is a bitmask where bit index `i` corresponds to

`fixed_pic_rate_within_cvs_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);

- `flags.low_delay_hrd_flag` is a bitmask where bit index `i` corresponds to `low_delay_hrd_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
- if `flags.nal_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersNal` is a pointer to an array of `vps_max_sub_layers_minus1 + 1` number of `StdVideoH265SubLayerHrdParameters` structures where `vps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265VideoParameterSet` structure and each element is interpreted as follows:
 - `cbr_flag` is a bitmask where bit index `i` corresponds to `cbr_flag[i]` as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
- if `flags.vcl_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersVcl` is a pointer to an array of `vps_max_sub_layers_minus1 + 1` number of `StdVideoH265SubLayerHrdParameters` structures where `vps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265VideoParameterSet` structure and each element is interpreted as follows:
 - `cbr_flag` is a bitmask where bit index `i` corresponds to `cbr_flag[i]` as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
- all other members of `StdVideoH265HrdParameters` are interpreted as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
- the `StdVideoH265ProfileTierLevel` structure pointed to by `pProfileTierLevel` are interpreted as follows:
 - `general_level_idc` is one of the enum constants `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifying the H.265 level `<major>.<minor>` as defined in section A.4 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoH265ProfileTierLevel` are interpreted as defined in section 7.4.4 of the [ITU-T H.265 Specification](#);
- all other members of `StdVideoH265VideoParameterSet` are interpreted as defined in section 7.4.3.1 of the [ITU-T H.265 Specification](#).

H.265 Sequence Parameter Sets (SPS)

Represented by `StdVideoH265SequenceParameterSet` structures and interpreted as follows:

- `reserved1` and `reserved2` are used only for padding purposes and are otherwise ignored;
- the pair constructed from `sps_video_parameter_set_id` and `sps_seq_parameter_set_id` is used as the key of the SPS entry;
- the `StdVideoH265ProfileTierLevel` structure pointed to by `pProfileTierLevel` are interpreted

as follows:

- `general_level_idc` is one of the enum constants `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifying the H.265 level `<major>.<minor>` as defined in section A.4 of the [ITU-T H.265 Specification](#);
- all other members of `StdVideoH265ProfileTierLevel` are interpreted as defined in section 7.4.4 of the [ITU-T H.265 Specification](#);
- the `max_latency_increase_plus1`, `max_dec_pic_buffering_minus1`, and `max_num_reorder_pics` members of the `StdVideoH265DecPicBufMgr` structure pointed to by `pDecPicBufMgr` correspond to `sps_max_latency_increase_plus1`, `sps_max_dec_pic_buffering_minus1`, and `sps_max_num_reorder_pics`, respectively, as defined in section 7.4.3.2 of the [ITU-T H.265 Specification](#);
- if `flags.sps_scaling_list_data_present_flag` is set, then the `StdVideoH265ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
 - `ScalingList4x4`, `ScalingList8x8`, `ScalingList16x16`, and `ScalingList32x32` correspond to `ScalingList[0]`, `ScalingList[1]`, `ScalingList[2]`, and `ScalingList[3]`, respectively, as defined in section 7.3.4 of the [ITU-T H.265 Specification](#);
 - `ScalingListDCCoef16x16` and `ScalingListDCCoef32x32` correspond to `scaling_list_dc_coef_minus8[0]` and `scaling_list_dc_coef_minus8[1]`, respectively, as defined in section 7.3.4 of the [ITU-T H.265 Specification](#);
- `pShortTermRefPicSet` is a pointer to an array of `num_short_term_ref_pic_sets` number of `StdVideoH265ShortTermRefPicSet` structures where each element is interpreted as follows:
 - `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
 - `used_by_curr_pic_flag` is a bitmask where bit index `i` corresponds to `used_by_curr_pic_flag[i]` as defined in section 7.4.8 of the [ITU-T H.265 Specification](#);
 - `use_delta_flag` is a bitmask where bit index `i` corresponds to `use_delta_flag[i]` as defined in section 7.4.8 of the [ITU-T H.265 Specification](#);
 - `used_by_curr_pic_s0_flag` is a bitmask where bit index `i` corresponds to `used_by_curr_pic_s0_flag[i]` as defined in section 7.4.8 of the [ITU-T H.265 Specification](#);
 - `used_by_curr_pic_s1_flag` is a bitmask where bit index `i` corresponds to `used_by_curr_pic_s1_flag[i]` as defined in section 7.4.8 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoH265ShortTermRefPicSet` are interpreted as defined in section 7.4.8 of the [ITU-T H.265 Specification](#);
- if `flags.long_term_ref_pics_present_flag` is set then the `StdVideoH265LongTermRefPicsSps` structure pointed to by `pLongTermRefPicsSps` is interpreted as follows:
 - `used_by_curr_pic_lt_sps_flag` is a bitmask where bit index `i` corresponds to `used_by_curr_pic_lt_sps_flag[i]` as defined in section 7.4.3.2 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoH265LongTermRefPicsSps` are interpreted as defined in section 7.4.3.2 of the [ITU-T H.265 Specification](#);
- if `flags.vui_parameters_present_flag` is set, then the `StdVideoH265SequenceParameterSetVui`

structure pointed to by `pSequenceParameterSetVui` is interpreted as follows:

- `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
 - the `StdVideoH265HrdParameters` structure pointed to by `pHrdParameters` is interpreted as follows:
 - `flags.fixed_pic_rate_general_flag` is a bitmask where bit index `i` corresponds to `fixed_pic_rate_general_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
 - `flags.fixed_pic_rate_within_cvs_flag` is a bitmask where bit index `i` corresponds to `fixed_pic_rate_within_cvs_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
 - `flags.low_delay_hrd_flag` is a bitmask where bit index `i` corresponds to `low_delay_hrd_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
 - if `flags.nal_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersNal` is a pointer to an array of `sps_max_sub_layers_minus1 + 1` number of `StdVideoH265SubLayerHrdParameters` structures where `sps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265SequenceParameterSet` structure and each element is interpreted as follows:
 - `cbr_flag` is a bitmask where bit index `i` corresponds to `cbr_flag[i]` as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - if `flags.vcl_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersVcl` is a pointer to an array of `sps_max_sub_layers_minus1 + 1` number of `StdVideoH265SubLayerHrdParameters` structures where `sps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265SequenceParameterSet` structure and each element is interpreted as follows:
 - `cbr_flag` is a bitmask where bit index `i` corresponds to `cbr_flag[i]` as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoH265HrdParameters` are interpreted as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
 - all other members of `pSequenceParameterSetVui` are interpreted as defined in section E.3.1 of the [ITU-T H.265 Specification](#);
- if `flags.sps_palette_predictor_initializer_present_flag` is set, then the `PredictorPaletteEntries` member of the `StdVideoH265PredictorPaletteEntries` structure pointed to by `pPredictorPaletteEntries` is interpreted as defined in section 7.4.9.13 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoH265SequenceParameterSet` are interpreted as defined in section 7.4.3.1 of the [ITU-T H.265 Specification](#).

H.265 Picture Parameter Sets (PPS)

Represented by `StdVideoH265PictureParameterSet` structures and interpreted as follows:

- `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
- the triplet constructed from `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` is used as the key of the PPS entry;
- if `flags.pps_scaling_list_data_present_flag` is set, then the `StdVideoH265ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
 - `ScalingList4x4`, `ScalingList8x8`, `ScalingList16x16`, and `ScalingList32x32` correspond to `ScalingList[0]`, `ScalingList[1]`, `ScalingList[2]`, and `ScalingList[3]`, respectively, as defined in section 7.3.4 of the [ITU-T H.265 Specification](#);
 - `ScalingListDCCoef16x16` and `ScalingListDCCoef32x32` correspond to `scaling_list_dc_coef_minus8[0]` and `scaling_list_dc_coef_minus8[1]`, respectively, as defined in section 7.3.4 of the [ITU-T H.265 Specification](#);
- if `flags.pps_palette_predictor_initializer_present_flag` is set, then the `PredictorPaletteEntries` member of the `StdVideoH265PredictorPaletteEntries` structure pointed to by `pPredictorPaletteEntries` is interpreted as defined in section 7.4.9.13 of the [ITU-T H.265 Specification](#);
- all other members of `StdVideoH265PictureParameterSet` are interpreted as defined in section 7.4.3.3 of the [ITU-T H.265 Specification](#).

When a `video session parameters` object is created with the codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, the `VkVideoSessionParametersCreateInfoKHR::pNext` chain **must** include a `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure specifying the capacity and initial contents of the object.

The `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265SessionParametersCreateInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    uint32_t                 maxStdVPSCount;
    uint32_t                 maxStdSPSCount;
    uint32_t                 maxStdPPSCount;
    const VkVideoDecodeH265SessionParametersAddInfoKHR* pParametersAddInfo;
} VkVideoDecodeH265SessionParametersCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxStdVPSCount` is the maximum number of `H.265 VPS` entries the created `VkVideoSessionParametersKHR` **can** contain.

- `maxStdSPSCount` is the maximum number of [H.265 SPS](#) entries the created `VkVideoSessionParametersKHR` **can** contain.
- `maxStdPPSCount` is the maximum number of [H.265 PPS](#) entries the created `VkVideoSessionParametersKHR` **can** contain.
- `pParametersAddInfo` is `NULL` or a pointer to a `VkVideoDecodeH265SessionParametersAddInfoKHR` structure specifying H.265 parameters to add upon object creation.

Valid Usage (Implicit)

- `VUID-VkVideoDecodeH265SessionParametersCreateInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR`
- `VUID-VkVideoDecodeH265SessionParametersCreateInfoKHR-pParametersAddInfo-parameter`
If `pParametersAddInfo` is not `NULL`, `pParametersAddInfo` **must** be a valid pointer to a valid `VkVideoDecodeH265SessionParametersAddInfoKHR` structure

The `VkVideoDecodeH265SessionParametersAddInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265SessionParametersAddInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           stdVPSCount;
    const StdVideoH265VideoParameterSet* pStdVPSS;
    uint32_t           stdSPSCount;
    const StdVideoH265SequenceParameterSet* pStdSPSS;
    uint32_t           stdPPSCount;
    const StdVideoH265PictureParameterSet* pStdPPSS;
} VkVideoDecodeH265SessionParametersAddInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdVPSCount` is the number of elements in the `pStdVPSS` array.
- `pStdVPSS` is a pointer to an array of `StdVideoH265VideoParameterSet` structures describing the [H.265 VPS](#) entries to add.
- `stdSPSCount` is the number of elements in the `pStdSPSS` array.
- `pStdSPSS` is a pointer to an array of `StdVideoH265SequenceParameterSet` structures describing the [H.265 SPS](#) entries to add.
- `stdPPSCount` is the number of elements in the `pStdPPSS` array.
- `pStdPPSS` is a pointer to an array of `StdVideoH265PictureParameterSet` structures describing the [H.265 PPS](#) entries to add.

This structure **can** be specified in the following places:

- In the `pParametersAddInfo` member of the `VkVideoDecodeH265SessionParametersCreateInfoKHR` structure specified in the `pNext` chain of `VkVideoSessionParametersCreateInfoKHR` used to create a `video session parameters` object. In this case, if the video codec operation the video session parameters object is created with is `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then it defines the set of initial parameters to add to the created object (see [Creating Video Session Parameters](#)).
- In the `pNext` chain of `VkVideoSessionParametersUpdateInfoKHR`. In this case, if the video codec operation the `video session parameters` object to be updated was created with is `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, then it defines the set of parameters to add to it (see [Updating Video Session Parameters](#)).

Valid Usage

- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-None-04833
The `vps_video_parameter_set_id` member of each `StdVideoH265VideoParameterSet` structure specified in the elements of `pStdVPSS` **must** be unique within `pStdVPSS`
- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-None-04834
The pair constructed from the `sps_video_parameter_set_id` and `sps_seq_parameter_set_id` members of each `StdVideoH265SequenceParameterSet` structure specified in the elements of `pStdSPSS` **must** be unique within `pStdSPSS`
- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-None-04835
The triplet constructed from the `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` members of each `StdVideoH265PictureParameterSet` structure specified in the elements of `pStdPPSS` **must** be unique within `pStdPPSS`

Valid Usage (Implicit)

- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR`
- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-pStdVPSS-parameter
If `stdVPSCount` is not 0, `pStdVPSS` **must** be a valid pointer to an array of `stdVPSCount` `StdVideoH265VideoParameterSet` values
- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-pStdSPSS-parameter
If `stdSPSCount` is not 0, `pStdSPSS` **must** be a valid pointer to an array of `stdSPSCount` `StdVideoH265SequenceParameterSet` values
- VUID-VkVideoDecodeH265SessionParametersAddInfoKHR-pStdPPSS-parameter
If `stdPPSCount` is not 0, `pStdPPSS` **must** be a valid pointer to an array of `stdPPSCount` `StdVideoH265PictureParameterSet` values

37.13.6. H.265 Decoding Parameters

The `VkVideoDecodeH265PictureInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265PictureInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    const StdVideoDecodeH265PictureInfo* pStdPictureInfo;
    uint32_t             sliceSegmentCount;
    const uint32_t*       pSliceSegmentOffsets;
} VkVideoDecodeH265PictureInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdPictureInfo` is a pointer to a `StdVideoDecodeH265PictureInfo` structure specifying [H.265 picture information](#).
- `sliceSegmentCount` is the number of elements in `pSliceSegmentOffsets`.
- `pSliceSegmentOffsets` is a pointer to an array of `sliceSegmentCount` offsets specifying the start offset of the slice segments of the picture within the video bitstream buffer range specified in [VkVideoDecodeInfoKHR](#).

This structure is specified in the `pNext` chain of the [VkVideoDecodeInfoKHR](#) structure passed to [vkCmdDecodeVideoKHR](#) to specify the codec-specific picture information for an [H.265 decode operation](#).

Decode Output Picture Information

When this structure is specified in the `pNext` chain of the [VkVideoDecodeInfoKHR](#) structure passed to [vkCmdDecodeVideoKHR](#), the information related to the [decode output picture](#) is defined as follows:

- The image subregion used is determined according to the [H.265 Decode Picture Data Access](#) section.
- The decode output picture is associated with the [H.265 picture information](#) provided in `pStdPictureInfo`.

Std Picture Information

The members of the `StdVideoDecodeH265PictureInfo` structure pointed to by `pStdPictureInfo` are interpreted as follows:

- `reserved` is used only for padding purposes and is otherwise ignored;
- `flags.IrapPicFlag` as defined in section 3.73 of the [ITU-T H.265 Specification](#);
- `flags.IdrPicFlag` as defined in section 3.67 of the [ITU-T H.265 Specification](#);
- `flags.IsReference` as defined in section 3.132 of the [ITU-T H.265 Specification](#);
- `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` are used to identify the active parameter sets, as described below;
- `PicOrderCntVal` as defined in section 8.3.1 of the [ITU-T H.265 Specification](#);
- `NumBitsForSTRefPicSetInSlice` is the number of bits used in `st_ref_pic_set` when

`short_term_ref_pic_set_sps_flag` is 0, or 0 otherwise, as defined in sections 7.4.7 and 7.4.8 of the [ITU-T H.265 Specification](#);

- `NumDeltaPocsOfRefRpsIdx` is the value of `NumDeltaPocs[RefRpsIdx]` when `short_term_ref_pic_set_sps_flag` is 1, or 0 otherwise, as defined in sections 7.4.7 and 7.4.8 of the [ITU-T H.265 Specification](#);
- `RefPicSetStCurrBefore`, `RefPicSetStCurrAfter`, and `RefPicSetLtCurr` are interpreted as defined in section 8.3.2 of the [ITU-T H.265 Specification](#) where each element of these arrays either identifies an [active reference picture](#) using its [DPB slot](#) index or contains the value `STD_VIDEO_H265_NO_REFERENCE_PICTURE` to indicate “no reference picture”;
- all other members are interpreted as defined in section 8.3.2 of the [ITU-T H.265 Specification](#).

Reference picture setup is controlled by the value of `StdVideoDecodeH265PictureInfo::flags.IsReference`. If it is set and a [reconstructed picture](#) is specified, then the latter is used as the target of picture reconstruction to [activate](#) the corresponding [DPB slot](#). If `StdVideoDecodeH265PictureInfo::flags.IsReference` is not set, but a [reconstructed picture](#) is specified, then the corresponding picture reference associated with the [DPB slot](#) is invalidated, as described in the [DPB Slot States](#) section.

Active Parameter Sets

The members of the `StdVideoDecodeH265PictureInfo` structure pointed to by `pStdPictureInfo` are used to select the active parameter sets to use from the bound video session parameters object, as follows:

- The *active VPS* is the [VPS](#) identified by the key specified in `StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id`.
- The *active SPS* is the [SPS](#) identified by the key specified by the pair constructed from `StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id` and `StdVideoDecodeH265PictureInfo::pps_seq_parameter_set_id`.
- The *active PPS* is the [PPS](#) identified by the key specified by the triplet constructed from `StdVideoDecodeH265PictureInfo::sps_video_parameter_set_id`, `StdVideoDecodeH265PictureInfo::pps_seq_parameter_set_id`, and `StdVideoDecodeH265PictureInfo::pps_pic_parameter_set_id`.

Valid Usage (Implicit)

- `VOID-VkVideoDecodeH265PictureInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PICTURE_INFO_KHR`
- `VOID-VkVideoDecodeH265PictureInfoKHR-pStdPictureInfo-parameter`
`pStdPictureInfo` **must** be a valid pointer to a valid `StdVideoDecodeH265PictureInfo` value
- `VOID-VkVideoDecodeH265PictureInfoKHR-pSliceSegmentOffsets-parameter`
`pSliceSegmentOffsets` **must** be a valid pointer to an array of `sliceSegmentCount` `uint32_t` values
- `VOID-VkVideoDecodeH265PictureInfoKHR-sliceSegmentCount-arraylength`
`sliceSegmentCount` **must** be greater than 0

The `VkVideoDecodeH265DpbSlotInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_h265
typedef struct VkVideoDecodeH265DpbSlotInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    const StdVideoDecodeH265ReferenceInfo* pStdReferenceInfo;
} VkVideoDecodeH265DpbSlotInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdReferenceInfo` is a pointer to a `StdVideoDecodeH265ReferenceInfo` structure specifying reference picture information described in section 8.3 of the [ITU-T H.265 Specification](#).

This structure is specified in the `pNext` chain of `VkVideoDecodeInfoKHR::pSetupReferenceSlot`, if not `NULL`, and the `pNext` chain of the elements of `VkVideoDecodeInfoKHR::pReferenceSlots` to specify the codec-specific reference picture information for an [H.265 decode operation](#).

Active Reference Picture Information

When this structure is specified in the `pNext` chain of the elements of `VkVideoDecodeInfoKHR::pReferenceSlots`, one element is added to the list of [active reference pictures](#) used by the video decode operation for each element of `VkVideoDecodeInfoKHR::pReferenceSlots` as follows:

- The image subregion used is determined according to the [H.265 Decode Picture Data Access](#) section.
- The reference picture is associated with the [DPB slot](#) index specified in the `slotIndex` member of the corresponding element of `VkVideoDecodeInfoKHR::pReferenceSlots`.
- The reference picture is associated with the [H.265 reference information](#) provided in `pStdReferenceInfo`.

Reconstructed Picture Information

When this structure is specified in the `pNext` chain of `VkVideoDecodeInfoKHR::pSetupReferenceSlot`, the information related to the [reconstructed picture](#) is defined as follows:

- The image subregion used is determined according to the [H.265 Decode Picture Data Access](#) section.
- If [reference picture setup](#) is requested, then the reconstructed picture is used to [activate](#) the [DPB slot](#) with the index specified in `VkVideoDecodeInfoKHR::pSetupReferenceSlot->slotIndex`.
- The reconstructed picture is associated with the [H.265 reference information](#) provided in `pStdReferenceInfo`.

Std Reference Information

The members of the `StdVideoDecodeH265ReferenceInfo` structure pointed to by `pStdReferenceInfo` are interpreted as follows:

- `flags.used_for_long_term_reference` is used to indicate whether the picture is marked as “used for long-term reference” as defined in section 8.3.2 of the [ITU-T H.265 Specification](#);
- `flags.unused_for_reference` is used to indicate whether the picture is marked as “unused for reference” as defined in section 8.3.2 of the [ITU-T H.265 Specification](#);
- all other members are interpreted as defined in section 8.3 of the [ITU-T H.265 Specification](#).

Valid Usage (Implicit)

- `VUID-VkVideoDecodeH265DpbSlotInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_DPB_SLOT_INFO_KHR`
- `VUID-VkVideoDecodeH265DpbSlotInfoKHR-pStdReferenceInfo-parameter`
`pStdReferenceInfo` **must** be a valid pointer to a valid `StdVideoDecodeH265ReferenceInfo` value

37.13.7. H.265 Decode Requirements

This section describes the **required** H.265 decoding capabilities for physical devices that have at least one queue family that supports the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`, as returned by `vkGetPhysicalDeviceQueueFamilyProperties2KHR` in `VkQueueFamilyVideoPropertiesKHR::videoCodecOperations`.

Table 48. Required *Video Std Header Versions*

Video Std Header Name	Version
<code>vulkan_video_codec_h265std_decode</code>	1.0.0

Table 49. Required Video Capabilities

Video Capability	Requirement	Requirement Type ¹
VkVideoCapabilitiesKHR		
<code>flags</code>	-	min
<code>minBitstreamBufferOffsetAlignment</code>	4096	max
<code>minBitstreamBufferSizeAlignment</code>	4096	max
<code>pictureAccessGranularity</code>	(64,64)	max
<code>minCodedExtent</code>	-	max
<code>maxCodedExtent</code>	-	min
<code>maxDpbSlots</code>	0	min
<code>maxActiveReferencePictures</code>	0	min
VkVideoDecodeCapabilitiesKHR		

Video Capability	Requirement	Requirement Type ¹
flags	VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR or VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR	min
VkVideoDecodeH265CapabilitiesKHR		
maxLevelIdc	STD_VIDEO_H265_LEVEL_IDC_1_0	min

1

The **Requirement Type** column specifies the requirement is either the minimum value all implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

37.14. AV1 Decode Operations

Video decode operations using an [AV1 decode profile](#) can be used to decode elementary video stream sequences compliant with the [AV1 Specification](#).

NOTE

Refer to the [Preamble](#) for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by Khronos.

This process is performed according to the [video decode operation steps](#) with the codec-specific semantics defined in section 7 of the [AV1 Specification](#):

- Syntax elements, derived values, and other parameters are applied from the following structures:
 - The `StdVideoAV1SequenceHeader` structure stored in the bound video session parameters object specifying the [active sequence header](#).
 - The `StdVideoDecodeAV1PictureInfo` structure specifying the [AV1 picture information](#).
 - The `StdVideoDecodeAV1ReferenceInfo` structures specifying the [AV1 reference information](#) corresponding to the optional [reconstructed picture](#) and any [active reference pictures](#).
- The contents of the provided video bitstream buffer range are interpreted as defined in the [AV1 Decode Bitstream Data Access](#) section.
- Picture data in the [video picture resources](#) corresponding to the used [active reference pictures](#), [decode output picture](#), and optional [reconstructed picture](#) is accessed as defined in the [AV1 Decode Picture Data Access](#) section.
- The decision on [reference picture setup](#) is made according to the parameters specified in the [AV1 picture information](#).

If the parameters and the bitstream adhere to the syntactic and semantic requirements defined in the corresponding sections of the [AV1 Specification](#), as described above, and the [DPB slots](#) associated with the [active reference pictures](#) all refer to [valid picture references](#), then the video decode operation will complete successfully. Otherwise, the video decode operation **may** complete [unsuccessfully](#).

37.14.1. AV1 Decode Bitstream Data Access

The video bitstream buffer range **should** contain one or more frame OBUs, comprised of a frame header OBU and tile group OBU, that together represent an entire frame, as defined in sections 5.10, 5.9, and 5.11, and this data is interpreted as defined in sections 6.9, 6.8, and 6.10 of the [AV1 Specification](#), respectively.

The offset specified in [VkVideoDecodeAV1PictureInfoKHR::frameHeaderOffset](#) **should** specify the starting offset of the frame header OBU of the frame.

NOTE

When the tiles of the frame are encoded into multiple tile groups, each frame OBU has a separate frame header OBU but their content is expected to match per the requirements of the [AV1 Specification](#). Accordingly, the offset specified in [frameHeaderOffset](#) **can** be the offset of any of the otherwise identical frame header OBUs when multiple tile groups are present.

The offsets and sizes provided in [VkVideoDecodeAV1PictureInfoKHR::pTileOffsets](#) and [VkVideoDecodeAV1PictureInfoKHR::pTileSizes](#), respectively, **should** specify the starting offsets and sizes corresponding to each tile within the video bitstream buffer range.

37.14.2. AV1 Decode Picture Data Access

Accesses to image data within a video picture resource happen at the granularity indicated by [VkVideoCapabilitiesKHR::pictureAccessGranularity](#), as returned by [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) for the used [video profile](#). Accordingly, the complete image subregion of a [decode output picture](#), [reference picture](#), or [reconstructed picture](#) accessed by video coding operations using an [AV1 decode profile](#) is defined as the set of texels within the coordinate range:

$([0, \text{endX}), [0, \text{endY}))$

Where:

- `endX` equals `codedExtent.width` rounded up to the nearest integer multiple of `pictureAccessGranularity.width` and clamped to the width of the image subresource [referred](#) to by the corresponding [VkVideoPictureResourceInfoKHR](#) structure;
- `endY` equals `codedExtent.height` rounded up to the nearest integer multiple of `pictureAccessGranularity.height` and clamped to the height of the image subresource [referred](#) to by the corresponding [VkVideoPictureResourceInfoKHR](#) structure;

Where `codedExtent` is the member of the [VkVideoPictureResourceInfoKHR](#) structure corresponding

to the picture.

In case of video decode operations using an [AV1 decode profile](#), any access to a picture at the coordinates (x,y), as defined by the [AV1 Specification](#), is an access to the image subresource referred to by the corresponding [VkVideoPictureResourceInfoKHR](#) structure at the texel coordinates (x,y).

37.14.3. AV1 Reference Names and Semantics

Individual reference frames used in the decoding process have different semantics, as defined in section 6.10.24 of the [AV1 Specification](#). The AV1 semantics associated with a reference picture are indicated by the corresponding enumeration constant defined in the Video Std enumeration type [StdVideoAV1ReferenceName](#):

- [STD_VIDEO_AV1_REFERENCE_NAME_INTRA_FRAME](#) identifies the reference used for intra coding ([INTRA_FRAME](#)), as defined in sections 2 and 7.11.2 of the [AV1 Specification](#).
- All other enumeration constants refer to backward or forward references used for inter coding, as defined in sections 2 and 7.11.3 of the [AV1 Specification](#):
 - [STD_VIDEO_AV1_REFERENCE_NAME_LAST_FRAME](#) identifies the [LAST_FRAME](#) reference
 - [STD_VIDEO_AV1_REFERENCE_NAME_LAST2_FRAME](#) identifies the [LAST2_FRAME](#) reference
 - [STD_VIDEO_AV1_REFERENCE_NAME_LAST3_FRAME](#) identifies the [LAST3_FRAME](#) reference
 - [STD_VIDEO_AV1_REFERENCE_NAME_GOLDEN_FRAME](#) identifies the [GOLDEN_FRAME](#) reference
 - [STD_VIDEO_AV1_REFERENCE_NAME_BWDREF_FRAME](#) identifies the [BWDREF_FRAME](#) reference
 - [STD_VIDEO_AV1_REFERENCE_NAME_ALTREF2_FRAME](#) identifies the [ALTREF2_FRAME](#) reference
 - [STD_VIDEO_AV1_REFERENCE_NAME_ALTREF_FRAME](#) identifies the [ALTREF_FRAME](#) reference

These enumeration constants are not directly used in any APIs but are used to indirectly index into certain Video Std and Vulkan API parameter arrays.

37.14.4. AV1 Decode Profile

A video profile supporting AV1 video decode operations is specified by setting [VkVideoProfileInfoKHR::videoCodecOperation](#) to [VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR](#) and adding a [VkVideoDecodeAV1ProfileInfoKHR](#) structure to the [VkVideoProfileInfoKHR::pNext](#) chain.

The [VkVideoDecodeAV1ProfileInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_decode_av1
typedef struct VkVideoDecodeAV1ProfileInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    StdVideoAV1Profile stdProfile;
    VkBool32           filmGrainSupport;
} VkVideoDecodeAV1ProfileInfoKHR;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdProfile` is a `StdVideoAV1Profile` value specifying the AV1 codec profile, as defined in section A.2 of the [AV1 Specification](#).
- `filmGrainSupport` specifies whether AV1 film grain, as defined in section 7.8.3 of the [AV1 Specification](#), **can** be used with the video profile. When this member is `VK_TRUE`, video session objects created against the video profile will be able to decode pictures that have [film grain](#) enabled.

NOTE

Enabling `filmGrainSupport` **may** increase the memory requirements of video sessions and/or video picture resources on some implementations.

Valid Usage (Implicit)

- VUID-VkVideoDecodeAV1ProfileInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PROFILE_INFO_KHR`

37.14.5. AV1 Decode Capabilities

When calling `vkGetPhysicalDeviceVideoCapabilitiesKHR` to query the capabilities for an [AV1 decode profile](#), the `VkVideoCapabilitiesKHR::pNext` chain **must** include a `VkVideoDecodeAV1CapabilitiesKHR` structure that will be filled with the profile-specific capabilities.

The `VkVideoDecodeAV1CapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_av1
typedef struct VkVideoDecodeAV1CapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    StdVideoAV1Level    maxLevel;
} VkVideoDecodeAV1CapabilitiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxLevel` is a `StdVideoAV1Level` value specifying the maximum AV1 level supported by the profile, as defined in section A.3 of the [AV1 Specification](#).

Valid Usage (Implicit)

- VUID-VkVideoDecodeAV1CapabilitiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_CAPABILITIES_KHR`

37.14.6. AV1 Decode Parameter Sets

Video session parameters objects created with the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR` contain a single instance of the following parameter set:

AV1 Sequence Header

Represented by `StdVideoAV1SequenceHeader` structures and interpreted as follows:

- `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
- the `StdVideoAV1ColorConfig` structure pointed to by `pColorConfig` is interpreted as follows:
 - `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
 - all other members of `StdVideoAV1ColorConfig` are interpreted as defined in section 6.4.2 of the [AV1 Specification](#);
- if `flags.timing_info_present_flag` is set, then the `StdVideoAV1TimingInfo` structure pointed to by `pTimingInfo` is interpreted as follows:
 - `flags.reserved` is used only for padding purposes and is otherwise ignored;
 - all other members of `StdVideoAV1TimingInfo` are interpreted as defined in section 6.4.3 of the [AV1 Specification](#);
- all other members of `StdVideoAV1SequenceHeader` are interpreted as defined in section 6.4 of the [AV1 Specification](#).

When a video session parameters object is created with the codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR`, the `VkVideoSessionParametersCreateInfoKHR::pNext` chain **must** include a `VkVideoDecodeAV1SessionParametersCreateInfoKHR` structure specifying the contents of the object.

The `VkVideoDecodeAV1SessionParametersCreateInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_av1
typedef struct VkVideoDecodeAV1SessionParametersCreateInfoKHR {
    VkStructureType    sType;
    const void*         pNext;
    const StdVideoAV1SequenceHeader* pStdSequenceHeader;
} VkVideoDecodeAV1SessionParametersCreateInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdSequenceHeader` is a pointer to a `StdVideoAV1SequenceHeader` structure describing the [AV1 sequence header](#) entry to store in the created object.

NOTE | As AV1 video session parameters objects will only ever contain a single AV1

sequence header, this has to be specified at object creation time and such video session parameters objects cannot be updated using the [vkUpdateVideoSessionParametersKHR](#) command. When a new AV1 sequence header is decoded from the input video bitstream the application needs to create a new video session parameters object to store it.

Valid Usage (Implicit)

- VUID-VkVideoDecodeAV1SessionParametersCreateInfoKHR-sType-sType **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_SESSION_PARAMETERS_CREATE_INFO_KHR`
- VUID-VkVideoDecodeAV1SessionParametersCreateInfoKHR-pStdSequenceHeader-parameter **must** be a valid pointer to a valid `StdVideoAV1SequenceHeader` value

37.14.7. AV1 Decoding Parameters

The `VkVideoDecodeAV1PictureInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_av1
typedef struct VkVideoDecodeAV1PictureInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    const StdVideoDecodeAV1PictureInfo* pStdPictureInfo;
    int32_t             referenceNameSlotIndices
[VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR];
    uint32_t            frameHeaderOffset;
    uint32_t            tileCount;
    const uint32_t*     pTileOffsets;
    const uint32_t*     pTileSizes;
} VkVideoDecodeAV1PictureInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdPictureInfo` is a pointer to a `StdVideoDecodeAV1PictureInfo` structure specifying [AV1 picture information](#).
- `referenceNameSlotIndices` is an array of seven (`VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR`, which is equal to the Video Std definition `STD_VIDEO_AV1_REFS_PER_FRAME`) signed integer values specifying the index of the [DPB slot](#) or a negative integer value for each [AV1 reference name](#) used for inter coding. In particular, the DPB slot index for the AV1 reference name `frame` is specified in `referenceNameSlotIndices[frame - STD_VIDEO_AV1_REFERENCE_NAME_LAST_FRAME]`.
- `frameHeaderOffset` is the byte offset of the AV1 frame header OBU, as defined in section 5.9 of the [AV1 Specification](#), within the video bitstream buffer range specified in `VkVideoDecodeInfoKHR`.
- `tileCount` is the number of elements in `pTileOffsets` and `pTileSizes`.
- `pTileOffsets` is a pointer to an array of `tileCount` integers specifying the byte offset of the tiles of

the picture within the video bitstream buffer range specified in [VkVideoDecodeInfoKHR](#).

- [pTileSizes](#) is a pointer to an array of [tileCount](#) integers specifying the byte size of the tiles of the picture within the video bitstream buffer range specified in [VkVideoDecodeInfoKHR](#).

This structure is specified in the [pNext](#) chain of the [VkVideoDecodeInfoKHR](#) structure passed to [vkCmdDecodeVideoKHR](#) to specify the codec-specific picture information for an [AV1 decode operation](#).

Decode Output Picture Information

When this structure is specified in the [pNext](#) chain of the [VkVideoDecodeInfoKHR](#) structure passed to [vkCmdDecodeVideoKHR](#), the information related to the [decode output picture](#) is defined as follows:

- The image subregion used is determined according to the [AV1 Decode Picture Data Access](#) section.
- The decode output picture is associated with the [AV1 picture information](#) provided in [pStdPictureInfo](#).

Std Picture Information

The members of the [StdVideoDecodeAV1PictureInfo](#) structure pointed to by [pStdPictureInfo](#) are interpreted as follows:

- [flags.reserved](#), [reserved1](#), and [reserved2](#) are used only for padding purposes and are otherwise ignored;
- [flags.apply_grain](#) indicates that film grain is enabled for the decoded picture, as defined in section 6.8.20 of the [AV1 Specification](#);
- [tg_start](#) and [tg_end](#) are interpreted as defined in section 6.10.1 of the [AV1 Specification](#);
- [OrderHint](#), [OrderHints](#), and [expectedFrameId](#) are interpreted as defined in section 6.8.2 of the [AV1 Specification](#);
- the [StdVideoAV1TileInfo](#) structure pointed to by [pTileInfo](#) is interpreted as follows:
 - [flags.reserved](#) and [reserved1](#) are used only for padding purposes and are otherwise ignored;
 - [pMiColStarts](#) is a pointer to an array of [TileCols](#) number of unsigned integers that corresponds to [MiColStarts](#) defined in section 6.8.14 of the [AV1 Specification](#);
 - [pMiRowStarts](#) is a pointer to an array of [TileRows](#) number of unsigned integers that corresponds to [MiRowStarts](#) defined in section 6.8.14 of the [AV1 Specification](#);
 - [pWidthInSbsMinus1](#) is a pointer to an array of [TileCols](#) number of unsigned integers that corresponds to [width_in_sbs_minus_1](#) defined in section 6.8.14 of the [AV1 Specification](#);
 - [pHeightInSbsMinus1](#) is a pointer to an array of [TileRows](#) number of unsigned integers that corresponds to [height_in_sbs_minus_1](#) defined in section 6.8.14 of the [AV1 Specification](#);
 - all other members of [StdVideoAV1TileInfo](#) are interpreted as defined in section 6.8.14 of the [AV1 Specification](#);
- the [StdVideoAV1Quantization](#) structure pointed to by [pQuantization](#) is interpreted as follows:

- `flags.reserved` is used only for padding purposes and is otherwise ignored;
- all other members of `StdVideoAV1Quantization` are interpreted as defined in section 6.8.11 of the [AV1 Specification](#);
- if `flags.segmentation_enabled` is set, then the `StdVideoAV1Segmentation` structure pointed to by `pSegmentation` is interpreted as follows:
 - the elements of `FeatureEnabled` are bitmasks where bit index j of element i corresponds to `FeatureEnabled[i][j]` as defined in section 6.8.13 of the [AV1 Specification](#);
 - `FeatureData` is interpreted as defined in section 6.8.13 of the [AV1 Specification](#);
- the `StdVideoAV1LoopFilter` structure pointed to by `pLoopFilter` is interpreted as follows:
 - `flags.reserved` is used only for padding purposes and is otherwise ignored;
 - `update_ref_delta` is a bitmask where bit index i is interpreted as the value of `update_ref_delta` corresponding to element i of `loop_filter_ref_deltas` as defined in section 6.8.10 of the [AV1 Specification](#);
 - `update_mode_delta` is a bitmask where bit index i is interpreted as the value of `update_mode_delta` corresponding to element i of `loop_filter_mode_deltas` as defined in section 6.8.10 of the [AV1 Specification](#);
 - all other members of `StdVideoAV1LoopFilter` are interpreted as defined in section 6.8.10 of the [AV1 Specification](#);
- if `flags.enable_cdef` is set in the [active sequence header](#), then the members of the `StdVideoAV1CDEF` structure pointed to by `pCDEF` are interpreted as follows:
 - `cdef_y_sec_strength` and `cdef_uv_sec_strength` are the bitstream values of the corresponding syntax elements defined in section 5.9.19 of the [AV1 Specification](#);
 - all other members of `StdVideoAV1CDEF` are interpreted as defined in section 6.10.14 of the [AV1 Specification](#);
- the `StdVideoAV1LoopRestoration` structure pointed to by `pLoopRestoration` is interpreted as follows:
 - `LoopRestorationSize[plane]` is interpreted as $\log_2(\text{size}) - 5$, where `size` is the value of `LoopRestorationSize[plane]` as defined in section 6.10.15 of the [AV1 Specification](#).
 - all other members of `StdVideoAV1LoopRestoration` are defined as in section 6.10.15 of the [AV1 Specification](#);
- the members of the `StdVideoAV1GlobalMotion` structure provided in `global_motion` are interpreted as defined in section 7.10 of the [AV1 Specification](#);
- if `flags.film_grain_params_present` is set in the [active sequence header](#), then the `StdVideoAV1FilmGrain` structure pointed to by `pFilmGrain` is interpreted as follows:
 - `flags.reserved` is used only for padding purposes and is otherwise ignored;
 - all other members of `StdVideoAV1FilmGrain` are interpreted as defined in section 6.8.20 of the [AV1 Specification](#);
- all other members are interpreted as defined in section 6.8 of the [AV1 Specification](#).

When [film grain is enabled](#) for the decoded frame, the `flags.update_grain` and

`film_grain_params_ref_idx` values specified in `StdVideoAV1FilmGrain` are ignored by AV1 decode operations and the `load_grain_params` function, as defined in section 6.8.20 of the [AV1 Specification](#), is not executed. Instead, the application is responsible for specifying the effective film grain parameters for the frame in `StdVideoAV1FilmGrain`.

When [film grain is enabled](#) for the decoded frame, the application is required to specify a different decode output picture resource in `VkVideoDecodeInfoKHR::dstPictureResource` compared to the reconstructed picture specified in `VkVideoDecodeInfoKHR::pSetupReferenceSlot->pPictureResource` even if the implementation does not report support for `VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR` in `VkVideoDecodeCapabilitiesKHR::flags` for the video decode profile.

Reference picture setup is controlled by the value of `StdVideoDecodeAV1PictureInfo::refresh_frame_flags`. If it is not zero and a [reconstructed picture](#) is specified, then the latter is used as the target of picture reconstruction to [activate](#) the [DPB slot](#) specified in `pDecodeInfo->pSetupReferenceSlot->slotIndex`. If `StdVideoDecodeAV1PictureInfo::refresh_frame_flags` is zero, but a [reconstructed picture](#) is specified, then the corresponding picture reference associated with the [DPB slot](#) is invalidated, as described in the [DPB Slot States](#) section.

Active Parameter Sets

The *active sequence header* is the [AV1 sequence header](#) stored in the bound video session parameters object.

Valid Usage (Implicit)

- VUID-VkVideoDecodeAV1PictureInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PICTURE_INFO_KHR`
- VUID-VkVideoDecodeAV1PictureInfoKHR-pStdPictureInfo-parameter
`pStdPictureInfo` **must** be a valid pointer to a valid `StdVideoDecodeAV1PictureInfo` value
- VUID-VkVideoDecodeAV1PictureInfoKHR-pTileOffsets-parameter
`pTileOffsets` **must** be a valid pointer to an array of `tileCount uint32_t` values
- VUID-VkVideoDecodeAV1PictureInfoKHR-pTileSizes-parameter
`pTileSizes` **must** be a valid pointer to an array of `tileCount uint32_t` values
- VUID-VkVideoDecodeAV1PictureInfoKHR-tileCount-arraylength
`tileCount` **must** be greater than 0

The `VkVideoDecodeAV1DpbSlotInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_decode_av1
typedef struct VkVideoDecodeAV1DpbSlotInfoKHR {
    VkStructureType             sType;
    const void*                 pNext;
    const StdVideoDecodeAV1ReferenceInfo* pStdReferenceInfo;
} VkVideoDecodeAV1DpbSlotInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdReferenceInfo` is a pointer to a `StdVideoDecodeAV1ReferenceInfo` structure specifying [AV1 reference information](#).

This structure is specified in the `pNext` chain of `VkVideoDecodeInfoKHR::pSetupReferenceSlot`, if not `NULL`, and the `pNext` chain of the elements of `VkVideoDecodeInfoKHR::pReferenceSlots` to specify the codec-specific reference picture information for an [AV1 decode operation](#).

Active Reference Picture Information

When this structure is specified in the `pNext` chain of the elements of `VkVideoDecodeInfoKHR::pReferenceSlots`, one element is added to the list of [active reference pictures](#) used by the video decode operation for each element of `VkVideoDecodeInfoKHR::pReferenceSlots` as follows:

- The image subregion used is determined according to the [AV1 Decode Picture Data Access](#) section.
- The reference picture is associated with the [DPB slot](#) index specified in the `slotIndex` member of the corresponding element of `VkVideoDecodeInfoKHR::pReferenceSlots`.
- The reference picture is associated with the [AV1 reference information](#) provided in `pStdReferenceInfo`.

Reconstructed Picture Information

When this structure is specified in the `pNext` chain of `VkVideoDecodeInfoKHR::pSetupReferenceSlot`, the information related to the [reconstructed picture](#) is defined as follows:

- The image subregion used is determined according to the [AV1 Decode Picture Data Access](#) section.
- If [reference picture setup](#) is requested, then the reconstructed picture is used to [activate](#) the [DPB slot](#) with the index specified in `VkVideoDecodeInfoKHR::pSetupReferenceSlot->slotIndex`.
- The reconstructed picture is associated with the [AV1 reference information](#) provided in `pStdReferenceInfo`.

Std Reference Information

The members of the `StdVideoDecodeAV1ReferenceInfo` structure pointed to by `pStdReferenceInfo` are interpreted as follows:

- `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
- `flags.disable_frame_end_update_cdf` is interpreted as defined in section 6.8.2 of the [AV1 Specification](#);
- `flags.segmentation_enabled` is interpreted as defined in section 6.8.13 of the [AV1 Specification](#);
- `frame_type` is interpreted as defined in section 6.8.2 of the [AV1 Specification](#);

NOTE

The `frame_type` member is defined with the type `uint8_t`, but it takes the same

values defined in the `StdVideoAV1FrameType` enumeration type as `StdVideoDecodeAV1PictureInfo::frame_type`.

- `RefFrameSignBias` is a bitmask where bit index `i` corresponds to `RefFrameSignBias[i]` as defined in section 6.8.2 of the [AV1 Specification](#);
- `OrderHint` is interpreted as defined in section 6.8.2 of the [AV1 Specification](#);
- `SavedOrderHints` is interpreted as defined in section 7.20 of the [AV1 Specification](#).

NOTE

When the AV1 reference information is provided for the reconstructed picture, certain parameters (e.g. `frame_type`) are specified both in the [AV1 picture information](#) and in the AV1 reference information. This is necessary because unlike the AV1 picture information, which is only used for the purposes of the video decode operation in question, the AV1 reference information specified for the reconstructed picture **may** be associated with the activated DPB slot, meaning that some implementations **may** maintain it as part of the [reference picture metadata](#) corresponding to the video picture resource associated with the DPB slot. When the AV1 reference information is provided for an active reference picture, the specified parameters correspond to the parameters specified when the DPB slot was activated (set up) with the reference picture, as usual, in order to communicate these parameters for implementations that do not maintain any subset of these parameters as part of the DPB slot's [reference picture metadata](#).

Valid Usage (Implicit)

- `VUID-VkVideoDecodeAV1DpbSlotInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_DPB_SLOT_INFO_KHR`
- `VUID-VkVideoDecodeAV1DpbSlotInfoKHR-pStdReferenceInfo-parameter`
`pStdReferenceInfo` **must** be a valid pointer to a valid `StdVideoDecodeAV1ReferenceInfo` value

37.14.8. AV1 Decode Requirements

This section describes the **required** AV1 decoding capabilities for physical devices that have at least one queue family that supports the video codec operation `VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR`, as returned by `vkGetPhysicalDeviceQueueFamilyProperties2KHR` in `VkQueueFamilyVideoPropertiesKHR::videoCodecOperations`.

Table 50. Required [Video Std Header Versions](#)

Video Std Header Name	Version
<code>vulkan_video_codec_av1std_decode</code>	1.0.0

Table 51. Required Video Capabilities

Video Capability	Requirement	Requirement Type ¹
VkVideoCapabilitiesKHR		
flags	-	min
minBitstreamBufferOffsetAlignment	4096	max
minBitstreamBufferSizeAlignment	4096	max
pictureAccessGranularity	(64,64)	max
minCodedExtent	-	max
maxCodedExtent	-	min
maxDpbSlots	0	min
maxActiveReferencePictures	0	min
VkVideoDecodeCapabilitiesKHR		
flags	VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_COINCIDE_BIT_KHR or VK_VIDEO_DECODE_CAPABILITY_DPB_AND_OUTPUT_DISTINCT_BIT_KHR	min
VkVideoDecodeAV1CapabilitiesKHR		
maxLevel	STD_VIDEO_AV1_LEVEL_2_0	min

1

The **Requirement Type** column specifies the requirement is either the minimum value all implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

37.15. Video Encode Operations

Video encode operations consume an *encode input picture* and zero or more reference pictures, and produce compressed video data to a video bitstream buffer and an optional [reconstructed picture](#).

NOTE

Such encode input pictures can be used as the [output](#) of video decode operations, with graphics or compute operations, or with [Window System Integration](#) APIs, depending on the capabilities of the implementation.

Video encode operations **may** access the following resources in the `VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR` stage:

- The image subregions corresponding to the source [encode input picture](#) and [active reference pictures](#) with access `VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR`.
- The destination video bitstream buffer range and the optional [reconstructed picture](#) with access

VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR.

The image subresource of each [video picture resource](#) accessed by the video encode operation is specified using a corresponding [VkVideoPictureResourceInfoKHR](#) structure. Each such image subresource **must** be in the appropriate image layout as follows:

- If the image subresource is used in the video encode operation as an [encode input picture](#), then it **must** be in the [VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR](#) layout.
- If the image subresource is used in the video encode operation as a [reconstructed picture](#) or [reference picture](#), then it **must** be in the [VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR](#) layout.

A video encode operation **may** complete unsuccessfully. In this case the target video bitstream buffer will have undefined contents. Similarly, if [reference picture setup](#) is requested, the [reconstructed-picture](#) will also have undefined contents, and the activated DPB slot will have an [invalid picture reference](#).

If a video encode operation completes successfully and the codec-specific parameters provided by the application adhere to the syntactic and semantic requirements defined in the corresponding video compression standard, then the target video bitstream buffer will contain compressed video data after the execution of the video encode operation according to the respective [codec-specific semantics](#).

37.15.1. Codec-Specific Semantics

The following aspects of video encode operations are codec-specific:

- The compressed video data written to the target video bitstream buffer range.
- The construction and interpretation of the list of [active reference pictures](#) and the interpretation of the picture data referred to by the corresponding image subregions.
- The construction and interpretation of information related to the [encode input picture](#) and the interpretation of the picture data referred to by the corresponding image subregion.
- The decision on [reference picture setup](#).
- The construction and interpretation of information related to the optional [reconstructed picture](#) and the generation of picture data to the corresponding image subregion.
- Certain aspects of [rate control](#).

These codec-specific behaviors are defined for each video codec operation separately.

- If the used video codec operation is [VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR](#), then the codec-specific aspects of the video encoding process are performed as defined in the [H.264 Encode Operations](#) section.
- If the used video codec operation is [VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR](#), then the codec-specific aspects of the video encoding process are performed as defined in the [H.265 Encode Operations](#) section.

Video Encode Parameter Overrides

Implementations supporting video encode operations for any particular video codec operation often support only a subset of the available encoding tools defined by the corresponding video compression standards. Accordingly, certain implementation-dependent limitations **may** apply to codec-specific parameters provided through the structures defined in the Video Std headers corresponding to the used video codec operation.

Exposing all of these restrictions on particular codec-specific parameter values or combinations thereof in the form of application-queryable capabilities is impractical, hence this specification allows implementations to *override* the value of any of the codec-specific parameters, unless otherwise specified, as long as all of the following conditions are met:

- If the application-provided codec-specific parameters adhere to the syntactic and semantic requirements and rules defined by the used video compression standard, and thus would be usable to produce a video bitstream compliant with that standard, then the codec-specific parameters resulting from the process of implementation overrides **must** also adhere to the same requirements and rules, and any video bitstream produced using the overridden parameters **must** also be compliant.
- The overridden codec-specific parameter values **must** not have an impact on the codec-independent behaviors defined for video encode operations.
- The implementation **must** not override any codec-specific parameters specified to a command that **may** cause application-provided codec-specific parameters specified to subsequent commands to no longer adhere to the semantic requirements and rules defined by the used video compression standard, unless the implementation also overrides those parameters to adhere to any such requirements and rules.
- The overridden codec-specific parameter values **must** not have an impact on the codec-specific picture data access semantics.
- The overridden codec-specific parameter values **may** change the contents of the codec-specific bitstream elements produced by video encode operations or otherwise retrieved by the application (e.g. using the [vkGetEncodedVideoSessionParametersKHR](#) command) but **must** still adhere to the codec-specific semantics defined for that video codec operation, including, but not limited to, the number, type, and order of the encoded codec-specific bitstream elements.

Besides codec-specific parameter overrides performed for implementation-dependent reasons, applications **can** enable the implementation to apply additional [optimizing overrides](#) that **may** improve the efficiency or performance of video encoding operations. However, implementations **must** meet the conditions listed above even in case of such optimizing overrides.

NOTE

Unless the application opts in for optimizing overrides, implementations are not expected to override any of the codec-specific parameters, except when such overrides are necessary for the correct operation of video encoder implementation due to limitations to the available encoding tools on that implementation.

37.15.2. Video Encode Operation Steps

Each video encode operation performs the following steps in the

VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR stage:

1. Reads the input picture data from the [encode input picture](#);
2. Determine derived encoding quality parameters according to the [codec-specific semantics](#) and the current [rate control](#) state;
3. Compresses the input picture data according to the [codec-specific semantics](#), applying any prediction data read from the [active reference pictures](#) and rate control restrictions in the process;
4. Writes the encoded bitstream data to the destination video bitstream buffer range;
5. Performs picture reconstruction of the encoded video data according to the [codec-specific semantics](#), applying any prediction data read from the [active reference pictures](#) in the process, if a [reconstructed picture](#) is specified and [reference picture setup](#) is requested;
6. If [reference picture setup](#) is requested, the [DPB slot index](#) specified in the [reconstructed picture information](#) is [activated](#) with the [reconstructed picture](#);
7. Writes the reconstructed picture data to the [reconstructed picture](#), if one is specified, according to the [codec-specific semantics](#).

When [reconstructed picture information](#) is provided, the specified [DPB slot](#) index is associated with the corresponding [bound reference picture resource](#), indifferent of whether [reference picture setup](#) is requested.

37.15.3. Capabilities

When calling [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) with [pVideoProfile->videoCodecOperation](#) specifying an encode operation, the [VkVideoEncodeCapabilitiesKHR](#) structure **must** be included in the [pNext](#) chain of the [VkVideoCapabilitiesKHR](#) structure to retrieve capabilities specific to video encoding.

The [VkVideoEncodeCapabilitiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeCapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkVideoEncodeCapabilityFlagsKHR    flags;
    VkVideoEncodeRateControlModeFlagsKHR    rateControlModes;
    uint32_t            maxRateControlLayers;
    uint64_t            maxBitrate;
    uint32_t            maxQualityLevels;
    VkExtent2D          encodeInputPictureGranularity;
    VkVideoEncodeFeedbackFlagsKHR    supportedEncodeFeedbackFlags;
} VkVideoEncodeCapabilitiesKHR;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.

- **flags** is a bitmask of [VkVideoEncodeCapabilityFlagBitsKHR](#) describing supported encoding features.
- **rateControlModes** is a bitmask of [VkVideoEncodeRateControlModeFlagBitsKHR](#) indicating supported [rate control modes](#).
- **maxRateControlLayers** indicates the maximum number of [rate control layers](#) supported.
- **maxBitrate** indicates the maximum supported bitrate.
- **maxQualityLevels** indicates the number of discrete [video encode quality levels](#) supported. Implementations **must** support at least one quality level.
- **encodeInputPictureGranularity** indicates the granularity at which [encode input picture](#) data is encoded and **may** indicate a texel granularity up to the size of the largest supported codec-specific coding block. This capability does not impose any valid usage constraints on the application, however, depending on the contents of the encode input picture, it **may** have effects on the encoded bitstream, as described in more detail below.
- **supportedEncodeFeedbackFlags** is a bitmask of [VkVideoEncodeFeedbackFlagBitsKHR](#) values specifying the supported flags for [video encode feedback queries](#).

Implementations **must** include support for at least [VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR](#) and [VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR](#) in **supportedEncodeFeedbackFlags**.

encodeInputPictureGranularity provides information about the way [encode input picture](#) data is used as input to video encode operations. In particular, some implementations **may** not be able to limit the set of texels used to encode the output video bitstream to the image subregion specified in the [VkVideoPictureResourceInfoKHR](#) structure corresponding to the encode input picture (i.e. to the resolution of the image data to encode specified in its **codedExtent** member).

NOTE

For example, the application requests the coded extent to be 1920x1080, but the implementation is only able to source the encode input picture data at the granularity of the codec-specific coding block size which is 16x16 pixels (or as otherwise indicated in **encodeInputPictureGranularity**). In this example, the content is horizontally aligned with the coding block size, but not vertically aligned with it. Thus encoding of the last row of coding blocks will be impacted by the contents of the input image at texel rows 1080 to 1087 (the latter being the next row which is vertically aligned with the coding block size, assuming a zero-based texel row index).

If **codedExtent** rounded up to the next integer multiple of **encodeInputPictureGranularity** is greater than the extent of the image subresource specified for the [encode input picture](#), then the texel values corresponding to texel coordinates outside of the bounds of the image subresource **may** be undefined. However, implementations **should** use well-defined default values for such texels in order to maximize the encoding efficiency for the last coding block row/column, and/or to ensure consistent encoding results across repeated encoding of the same input content. Nonetheless, the values used for such texels **must** not have an effect on whether the video encode operation produces a compliant bitstream, and **must** not have any other effects on the encoded picture data beyond what **may** otherwise result from using these texel values as input to any compression algorithm, as defined in the used video compression standard.

NOTE

While not required, it is generally a good practice for applications to make sure that the image subresource used for the encode input picture has an extent that is an integer multiple of the codec-specific coding block size (or at least `encodeInputPictureGranularity`) and that this padding area is filled with known values in order to improve encoding efficiency, portability, and reproducibility.

Valid Usage (Implicit)

- VUID-VkVideoEncodeCapabilitiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_CAPABILITIES_KHR`

Bits which **may** be set in `VkVideoEncodeCapabilitiesKHR::flags`, indicating the encoding tools supported, are:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeCapabilityFlagBitsKHR {
    VK_VIDEO_ENCODE_CAPABILITY_PRECEDING_EXTERNALLY_ENCODED_BYTES_BIT_KHR =
0x00000001,
    VK_VIDEO_ENCODE_CAPABILITY_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_DETECTION_BIT_KHR =
0x00000002,
} VkVideoEncodeCapabilityFlagBitsKHR;
```

- `VK_VIDEO_ENCODE_CAPABILITY_PRECEDING_EXTERNALLY_ENCODED_BYTES_BIT_KHR` indicates that the implementation supports the use of `VkVideoEncodeInfoKHR::precedingExternallyEncodedBytes`.
- `VK_VIDEO_ENCODE_CAPABILITY_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_DETECTION_BIT_KHR` indicates that the implementation is able to detect and report when the destination video bitstream buffer range provided by the application is not sufficiently large to fit the encoded bitstream data produced by a video encode operation by reporting the `VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR` query result status code.

NOTE

Some implementations **may** not be able to reliably detect insufficient bitstream buffer range conditions in all situations. Such implementations will not report support for the `VK_VIDEO_ENCODE_CAPABILITY_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_DETECTION_BIT_KHR` encode capability flag for the video profile, but **may** still report the `VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR` query result status code in certain cases. Applications **should** always check for the specific query result status code `VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR` even when this encode capability flag is not supported by the implementation for the video profile in question. However, applications **must** not assume that a different negative query result status code indicating an unsuccessful completion of a video encode operation is not the result of an insufficient bitstream buffer condition unless this encode capability flag is supported.

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeCapabilityFlagsKHR;
```

`VkVideoEncodeCapabilityFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoEncodeCapabilityFlagBitsKHR`.

37.15.4. Video Encode Quality Levels

Implementations **can** support more than one video encode quality levels for a video encode profile, which control the number and type of implementation-specific encoding tools and algorithms utilized in the encoding process.

NOTE

Generally, using higher video encode quality levels **may** produce higher quality video streams at the cost of additional processing time. However, as the final quality of an encoded picture depends on the contents of the [encode input picture](#), the contents of the [active reference pictures](#), the codec-specific encode parameters, and the particular implementation-specific tools used corresponding to the individual video encode quality levels, there are no guarantees that using a higher video encode quality level will always produce a higher quality encoded picture for any given set of inputs.

To query properties for a specific video encode quality level supported by a video encode profile, call:

```
// Provided by VK_KHR_video_encode_queue
VkResult vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR(
    VkPhysicalDevice          physicalDevice,
    const VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR* pQualityLevelInfo,
    VkVideoEncodeQualityLevelPropertiesKHR* pQualityLevelProperties);
```

- `physicalDevice` is the physical device to query the video encode quality level properties for.
- `pQualityLevelInfo` is a pointer to a `VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR` structure specifying the video encode profile and quality level to query properties for.
- `pQualityLevelProperties` is a pointer to a `VkVideoEncodeQualityLevelPropertiesKHR` structure in which the properties are returned.

Valid Usage

- VUID-vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR-pQualityLevelInfo-08257

If `pQualityLevelInfo->pVideoProfile->videoCodecOperation` is `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the `pNext` chain of `pQualityLevelProperties` **must** include a `VkVideoEncodeH264QualityLevelPropertiesKHR` structure

- VUID-vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR-pQualityLevelInfo-

08258

If `pQualityLevelInfo->pVideoProfile->videoCodecOperation` is `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the `pNext` chain of `pQualityLevelProperties` **must** include a `VkVideoEncodeH265QualityLevelPropertiesKHR` structure

Valid Usage (Implicit)

- `VUID-vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR-physicalDevice-parameter`
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- `VUID-vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR-pQualityLevelInfo-parameter`
`pQualityLevelInfo` **must** be a valid pointer to a valid `VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR` structure
- `VUID-vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR-pQualityLevelProperties-parameter`
`pQualityLevelProperties` **must** be a valid pointer to a `VkVideoEncodeQualityLevelPropertiesKHR` structure

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR`
- `VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR`
- `VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR`
- `VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR`

The `VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    const VkVideoProfileInfoKHR* pVideoProfile;
    uint32_t           qualityLevel;
} VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pVideoProfile` is a pointer to a `VkVideoProfileInfoKHR` structure specifying the video profile to query the video encode quality level properties for.
- `qualityLevel` is the video encode quality level to query properties for.

Valid Usage

- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-pVideoProfile-08259 `pVideoProfile` **must** be a `supported video profile`
- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-pVideoProfile-08260 `pVideoProfile->videoCodecOperation` **must** specify an encode operation
- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-qualityLevel-08261 `qualityLevel` **must** be less than `VkVideoEncodeCapabilitiesKHR::maxQualityLevels`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile specified in `pVideoProfile`

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR`
- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-pNext-pNext `pNext` **must** be `NULL`
- VUID-VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR-pVideoProfile-parameter `pVideoProfile` **must** be a valid pointer to a valid `VkVideoProfileInfoKHR` structure

The `VkVideoEncodeQualityLevelPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeQualityLevelPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkVideoEncodeRateControlModeFlagBitsKHR preferredRateControlMode;
    uint32_t            preferredRateControlLayerCount;
} VkVideoEncodeQualityLevelPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `preferredRateControlMode` is a `VkVideoEncodeRateControlModeFlagBitsKHR` value indicating the preferred `rate control mode` to use with the video encode quality level.
- `preferredRateControlLayerCount` indicates the preferred number of `rate control layers` to use

with the video encode quality level.

Valid Usage (Implicit)

- VUID-VkVideoEncodeQualityLevelPropertiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_PROPERTIES_KHR`
- VUID-VkVideoEncodeQualityLevelPropertiesKHR-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkVideoEncodeH264QualityLevelPropertiesKHR` or `VkVideoEncodeH265QualityLevelPropertiesKHR`
- VUID-VkVideoEncodeQualityLevelPropertiesKHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique

The `VkVideoEncodeQualityLevelInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeQualityLevelInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t           qualityLevel;
} VkVideoEncodeQualityLevelInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `qualityLevel` is the used video encode quality level.

This structure **can** be specified in the following places:

- In the `pNext` chain of `VkVideoSessionParametersCreateInfoKHR` to specify the video encode quality level to use for a video session parameters object created for a video encode session. If no instance of this structure is included in the `pNext` chain of `VkVideoSessionParametersCreateInfoKHR`, then the video session parameters object is created with a video encode quality level of zero.
- In the `pNext` chain of `VkVideoCodingControlInfoKHR` to change the video encode quality level state of the bound video session.

Valid Usage

- VUID-VkVideoEncodeQualityLevelInfoKHR-qualityLevel-08311
`qualityLevel` **must** be less than `VkVideoEncodeCapabilitiesKHR::maxQualityLevels`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile

Valid Usage (Implicit)

- VUID-VkVideoEncodeQualityLevelInfoKHR-sType-sType
sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR

37.15.5. Retrieving Encoded Session Parameters

Any codec-specific parameters stored in video session parameters objects **may** need to be separately encoded and included in the final video bitstream data, depending on the used video compression standard. In such cases the application **must** call the [vkGetEncodedVideoSessionParametersKHR](#) command to retrieve the encoded parameter data from the used video session parameters object in order to be able to produce a compliant video bitstream.

NOTE

This is needed because implementations **may** have changed some of the codec-specific parameters stored in the video session parameters object, as defined in the [Video Encode Parameter Overrides](#) section. In addition, the [vkGetEncodedVideoSessionParametersKHR](#) command enables the application to retrieve the encoded parameter data without having to encode these codec-specific parameters manually.

Encoded parameter data **can** be retrieved from a video session parameters object created with a video encode operation using the command:

```
// Provided by VK_KHR_video_encode_queue
VkResult vkGetEncodedVideoSessionParametersKHR(
    VkDevice device,
    const VkVideoEncodeSessionParametersGetInfoKHR* pVideoSessionParametersInfo,
    VkVideoEncodeSessionParametersFeedbackInfoKHR* pFeedbackInfo,
    size_t* pDataSize,
    void* pData);
```

- **device** is the logical device that owns the video session parameters object.
- **pVideoSessionParametersInfo** is a pointer to a [VkVideoEncodeSessionParametersGetInfoKHR](#) structure specifying the parameters of the encoded parameter data to retrieve.
- **pFeedbackInfo** is either **NULL** or a pointer to a [VkVideoEncodeSessionParametersFeedbackInfoKHR](#) structure in which feedback about the requested parameter data is returned.
- **pDataSize** is a pointer to a **size_t** value related to the amount of encode parameter data returned, as described below.
- **pData** is either **NULL** or a pointer to a buffer to write the encoded parameter data to.

If **pData** is **NULL**, then the size of the encoded parameter data, in bytes, that **can** be retrieved is returned in **pDataSize**. Otherwise, **pDataSize** **must** point to a variable set by the application to the size of the buffer, in bytes, pointed to by **pData**, and on return the variable is overwritten with the

number of bytes actually written to `pData`. If `pDataSize` is less than the size of the encoded parameter data that **can** be retrieved, then no data will be written to `pData`, zero will be written to `pDataSize`, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that no encoded parameter data was returned.

If `pFeedbackInfo` is not `NULL` then the members of the `VkVideoEncodeSessionParametersFeedbackInfoKHR` structure and any additional structures included in its `pNext` chain that are applicable to the video session parameters object specified in `pVideoSessionParametersInfo->videoSessionParameters` will be filled with feedback about the requested parameter data on all successful calls to this command.

NOTE

This includes the cases when `pData` is `NULL` or when `VK_INCOMPLETE` is returned by the command, and enables the application to determine whether the implementation `overrode` any of the requested video session parameters without actually needing to retrieve the encoded parameter data itself.

Valid Usage

- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08359
`pVideoSessionParametersInfo->videoSessionParameters` **must** have been created with an encode operation
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08262
If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the `pNext` chain of `pVideoSessionParametersInfo` **must** include a `VkVideoEncodeH264SessionParametersGetInfoKHR` structure
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08263
If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then for the `VkVideoEncodeH264SessionParametersGetInfoKHR` structure included in the `pNext` chain of `pVideoSessionParametersInfo`, if its `writeStdSPS` member is `VK_TRUE`, then `pVideoSessionParametersInfo->videoSessionParameters` **must** contain a `StdVideoH264SequenceParameterSet` entry with `seq_parameter_set_id` matching `VkVideoEncodeH264SessionParametersGetInfoKHR::stdSPSId`
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08264
If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then for the `VkVideoEncodeH264SessionParametersGetInfoKHR` structure included in the `pNext` chain of `pVideoSessionParametersInfo`, if its `writeStdPPS` member is `VK_TRUE`, then `pVideoSessionParametersInfo->videoSessionParameters` **must** contain a `StdVideoH264PictureParameterSet` entry with `seq_parameter_set_id` and `pic_parameter_set_id` matching `VkVideoEncodeH264SessionParametersGetInfoKHR::stdSPSId` and `VkVideoEncodeH264SessionParametersGetInfoKHR::stdPPSId`, respectively
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08265
If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the `pNext` chain of

`pVideoSessionParametersInfo` **must** include a `VkVideoEncodeH265SessionParametersGetInfoKHR` structure

- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08266
If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then for the `VkVideoEncodeH265SessionParametersGetInfoKHR` structure included in the `pNext` chain of `pVideoSessionParametersInfo`, if its `writeStdVPS` member is `VK_TRUE`, then `pVideoSessionParametersInfo->videoSessionParameters` **must** contain a `StdVideoH265VideoParameterSet` entry with `vps_video_parameter_set_id` matching `VkVideoEncodeH265SessionParametersGetInfoKHR::stdVPSId`
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08267
If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then for the `VkVideoEncodeH265SessionParametersGetInfoKHR` structure included in the `pNext` chain of `pVideoSessionParametersInfo`, if its `writeStdSPS` member is `VK_TRUE`, then `pVideoSessionParametersInfo->videoSessionParameters` **must** contain a `StdVideoH265SequenceParameterSet` entry with `sps_video_parameter_set_id` and `sps_seq_parameter_set_id` matching `VkVideoEncodeH265SessionParametersGetInfoKHR::stdVPSId` and `VkVideoEncodeH265SessionParametersGetInfoKHR::stdSPSId`, respectively
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-08268
If `pVideoSessionParametersInfo->videoSessionParameters` was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then for the `VkVideoEncodeH265SessionParametersGetInfoKHR` structure included in the `pNext` chain of `pVideoSessionParametersInfo`, if its `writeStdPPS` member is `VK_TRUE`, then `pVideoSessionParametersInfo->videoSessionParameters` **must** contain a `StdVideoH265PictureParameterSet` entry with `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` matching `VkVideoEncodeH265SessionParametersGetInfoKHR::stdVPSId`, `VkVideoEncodeH265SessionParametersGetInfoKHR::stdSPSId`, and `VkVideoEncodeH265SessionParametersGetInfoKHR::stdPPSId`, respectively

Valid Usage (Implicit)

- VUID-vkGetEncodedVideoSessionParametersKHR-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkGetEncodedVideoSessionParametersKHR-pVideoSessionParametersInfo-parameter
`pVideoSessionParametersInfo` **must** be a valid pointer to a valid `VkVideoEncodeSessionParametersGetInfoKHR` structure
- VUID-vkGetEncodedVideoSessionParametersKHR-pFeedbackInfo-parameter
If `pFeedbackInfo` is not `NULL`, `pFeedbackInfo` **must** be a valid pointer to a `VkVideoEncodeSessionParametersFeedbackInfoKHR` structure
- VUID-vkGetEncodedVideoSessionParametersKHR-pDataSize-parameter
`pDataSize` **must** be a valid pointer to a `size_t` value

- VUID-vkGetEncodedVideoSessionParametersKHR-pData-parameter
If the value referenced by `pDataSize` is not 0, and `pData` is not `NULL`, `pData` **must** be a valid pointer to an array of `pDataSize` bytes

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkVideoEncodeSessionParametersGetInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeSessionParametersGetInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkVideoSessionParametersKHR    videoSessionParameters;
} VkVideoEncodeSessionParametersGetInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `videoSessionParameters` is the `VkVideoSessionParametersKHR` object to retrieve encoded parameter data from.

Depending on the used video encode operation, additional codec-specific structures **may** need to be included in the `pNext` chain of this structure to identify the specific video session parameters to retrieve encoded parameter data for, as described in the corresponding sections.

Valid Usage (Implicit)

- VUID-VkVideoEncodeSessionParametersGetInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_GET_INFO_KHR`
- VUID-VkVideoEncodeSessionParametersGetInfoKHR-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkVideoEncodeH264SessionParametersGetInfoKHR` or `VkVideoEncodeH265SessionParametersGetInfoKHR`
- VUID-VkVideoEncodeSessionParametersGetInfoKHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkVideoEncodeSessionParametersGetInfoKHR-videoSessionParameters-parameter

`videoSessionParameters` **must** be a valid `VkVideoSessionParametersKHR` handle

The `VkVideoEncodeSessionParametersFeedbackInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeSessionParametersFeedbackInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           hasOverrides;
} VkVideoEncodeSessionParametersFeedbackInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `hasOverrides` indicates whether any of the requested parameter data were `overridden` by the implementation.

Depending on the used video encode operation, additional codec-specific structures **can** be included in the `pNext` chain of this structure to capture codec-specific feedback information about the requested parameter data, as described in the corresponding sections.

Valid Usage (Implicit)

- VUID-VkVideoEncodeSessionParametersFeedbackInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_FEEDBACK_INFO_KHR`
- VUID-VkVideoEncodeSessionParametersFeedbackInfoKHR-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkVideoEncodeH264SessionParametersFeedbackInfoKHR` or `VkVideoEncodeH265SessionParametersFeedbackInfoKHR`
- VUID-VkVideoEncodeSessionParametersFeedbackInfoKHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique

37.15.6. Video Encode Commands

To launch video encode operations, call:

```
// Provided by VK_KHR_video_encode_queue
void vkCmdEncodeVideoKHR(
    VkCommandBuffer          commandBuffer,
    const VkVideoEncodeInfoKHR* pEncodeInfo);
```

- `commandBuffer` is the command buffer in which to record the command.
- `pEncodeInfo` is a pointer to a `VkVideoEncodeInfoKHR` structure specifying the parameters of the

video encode operations.

Each call issues one or more video encode operations. The implicit parameter `opCount` corresponds to the number of video encode operations issued by the command. After calling this command, the `active query index` of each `active` query is incremented by `opCount`.

Currently each call to this command results in the issue of a single video encode operation.

If the bound video session was created with `VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR` and the `pNext` chain of `pEncodeInfo` includes a `VkVideoInlineQueryInfoKHR` structure with its `queryPool` member specifying a valid `VkQueryPool` handle, then this command will execute a query for each video encode operation issued by it.

Active Reference Picture Information

The list of `active reference pictures` used by a video encode operation is a list of image subregions used as the source of `reference picture` data and related parameters, and is derived from the `VkVideoReferenceSlotInfoKHR` structures provided as the elements of the `pEncodeInfo->pReferenceSlots` array. For each element of `pEncodeInfo->pReferenceSlots`, one or more elements are added to the active reference picture list, as defined by the `codec-specific semantics`. Each element of this list contains the following information:

- The image subregion within the image subresource `referred` to by the `video picture resource` used as the reference picture.
- The `DPB slot` index the reference picture is associated with.
- The codec-specific reference information related to the reference picture.

Reconstructed Picture Information

Information related to the optional `reconstructed picture` used by a video encode operation is derived from the `VkVideoReferenceSlotInfoKHR` structure pointed to by `pEncodeInfo->pSetupReferenceSlot`, if not `NULL`, as defined by the `codec-specific semantics`, and consists of the following:

- The image subregion within the image subresource `referred` to by the `video picture resource` used as the reconstructed picture.
- The `DPB slot` index to use for picture reconstruction.
- The codec-specific reference information related to the reconstructed picture.

Specifying a valid `VkVideoReferenceSlotInfoKHR` structure in `pEncodeInfo->pSetupReferenceSlot` is always required, unless the video session was created with `VkVideoSessionCreateInfoKHR::maxDpbSlot` equal to zero. However, the DPB slot identified by `pEncodeInfo->pSetupReferenceSlot->slotIndex` is only `activated` with the `reconstructed picture` specified in `pEncodeInfo->pSetupReferenceSlot->pPictureResource` if reference picture setup is requested according to the `codec-specific semantics`.

If reconstructed picture information is specified, but reference picture setup is not requested, according to the codec-specific semantics, the contents of the `video picture resource` corresponding to the reconstructed picture will be undefined after the video encode operation.

NOTE

Some implementations may always output the reconstructed picture or use it as temporary storage during the video encode operation even when the reconstructed picture is not marked for future reference.

Encode Input Picture Information

Information related to the [encode input picture](#) used by a video encode operation is derived from `pEncodeInfo->srcPictureResource` and any codec-specific parameters provided in the `pEncodeInfo->pNext` chain, as defined by the [codec-specific semantics](#), and consists of the following:

- The image subregion within the image subresource [referred](#) to by the [video picture resource](#) used as the encode input picture.
- The codec-specific picture information related to the encoded picture.

Several limiting values are defined below that are referenced by the relevant valid usage statements of this command.

- Let `uint32_t activeReferencePictureCount` be the size of the list of active reference pictures used by the video encode operation. Unless otherwise defined, `activeReferencePictureCount` is set to the value of `pEncodeInfo->referenceSlotCount`.
- Let `VkOffset2D codedOffsetGranularity` be the minimum alignment requirement for the coded offset of video picture resources. Unless otherwise defined, the value of the `x` and `y` members of `codedOffsetGranularity` are 0.
- Let `uint32_t dpbFrameUseCount[]` be an array of size `maxDpbSlots`, where `maxDpbSlots` is the `VkVideoSessionCreateInfoKHR::maxDpbSlots` the bound video session was created with, with each element indicating the number of times a frame associated with the corresponding DPB slot index is referred to by the video coding operation. Let the initial value of each element of the array be 0.
 - If `pEncodeInfo->pSetupReferenceSlot` is not `NULL`, then `dpbFrameUseCount[i]` is incremented by one, where `i` equals `pEncodeInfo->pSetupReferenceSlot->slotIndex`.
 - For each element of `pEncodeInfo->pReferenceSlots`, `dpbFrameUseCount[i]` is incremented by one, where `i` equals the `slotIndex` member of the corresponding element.
- Let `VkExtent2D maxCodingBlockSize` be the maximum codec-specific coding block size that **may** be used by the video encode operation.
 - If the bound video session object was created with an [H.264 encode profile](#), then let `maxCodingBlockSize` be equal to the size of an H.264 macroblock, i.e. {16,16}.
 - If the bound video session object was created with an [H.265 encode profile](#), then let `maxCodingBlockSize` be equal to the maximum H.265 coding block size that **may** be used by the video encode operation derived as the maximum of the CTB sizes corresponding to the `VkVideoEncodeH265CtbSizeFlagBitsKHR` bits set in `VkVideoEncodeH265CapabilitiesKHR::ctbSizes`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with.
 - Otherwise, `maxCodingBlockSize` is undefined.
- If `maxCodingBlockSize` is defined, then let `VkExtent2D minCodingBlockExtent` be the coded extent of

the `encode input picture` expressed in terms of codec-specific coding blocks, assuming the maximum size of such coding blocks, as defined by `maxCodingBlockSize`, calculated from the value of the `codedExtent` member of `pEncodeInfo->srcPictureResource` as follows:

- `minCodingBlockExtent.width = (codedExtent.width
maxCodingBlockSize.width - 1) / maxCodingBlockSize.width`
- `minCodingBlockExtent.height = (codedExtent.height
maxCodingBlockSize.height - 1) / maxCodingBlockSize.height`
- If the bound video session object was created with an `H.264 encode profile`, then:
 - Let `StdVideoH264PictureType h264PictureType` be the picture type of the encoded picture set to the value of `pStdPictureInfo->primary_pic_type` specified in the `VkVideoEncodeH264PictureInfoKHR` structure included in the `pEncodeInfo->pNext` chain.
 - Let `StdVideoH264PictureType h264L0PictureTypes[]` and `StdVideoH264PictureType h264L1PictureTypes[]` be the picture types of the reference pictures in the L0 and L1 reference lists, respectively. If `pStdPictureInfo->pRefLists` specified in the `VkVideoEncodeH264PictureInfoKHR` structure included in the `pEncodeInfo->pNext` chain is not `NULL`, then for each reference index specified in the elements of the `pStdPictureInfo->pRefLists->RefPicList0` and `pStdPictureInfo->pRefLists->RefPicList1` arrays, if the reference index is not `STD_VIDEO_H264_NO_REFERENCE_PICTURE`, `pStdReferenceInfo->primary_pic_type` is added to `h264L0PictureTypes` or `h264L1PictureTypes`, respectively, where `pStdReferenceInfo` is the member of the `VkVideoEncodeH264DpbSlotInfoKHR` structure included in the `pNext` chain of the element of `pEncodeInfo->pReferenceSlots` for which `slotIndex` equals the reference index in question.
- If the bound video session object was created with an `H.265 encode profile`, then:
 - Let `StdVideoH265PictureType h265PictureType` be the picture type of the encoded picture set to the value of `pStdPictureInfo->pic_type` specified in the `VkVideoEncodeH265PictureInfoKHR` structure included in the `pEncodeInfo->pNext` chain.
 - Let `StdVideoH265PictureType h265L0PictureTypes[]` and `StdVideoH265PictureType h265L1PictureTypes[]` be the picture types of the reference pictures in the L0 and L1 reference lists, respectively. If `pStdPictureInfo->pRefLists` specified in the `VkVideoEncodeH265PictureInfoKHR` structure included in the `pEncodeInfo->pNext` chain is not `NULL`, then for each reference index specified in the elements of the `pStdPictureInfo->pRefLists->RefPicList0` and `pStdPictureInfo->pRefLists->RefPicList1` arrays, if the reference index is not `STD_VIDEO_H265_NO_REFERENCE_PICTURE`, `pStdReferenceInfo->pic_type` is added to `h265L0PictureTypes` or `h265L1PictureTypes`, respectively, where `pStdReferenceInfo` is the member of the `VkVideoEncodeH265DpbSlotInfoKHR` structure included in the `pNext` chain of the element of `pEncodeInfo->pReferenceSlots` for which `slotIndex` equals the reference index in question.

Valid Usage

- VUID-vkCmdEncodeVideoKHR-None-08250
The bound video session **must** have been created with an encode operation
- VUID-vkCmdEncodeVideoKHR-None-07012
The bound video session **must** not be in `uninitialized` state at the time the command is

executed on the device

- VUID-vkCmdEncodeVideoKHR-None-08318

The bound video session parameters object **must** have been created with the currently set [video encode quality level](#) for the bound video session at the time the command is executed on the device

- VUID-vkCmdEncodeVideoKHR-opCount-07174

For each [active](#) query, the [active query index](#) corresponding to the query type of that query plus [opCount](#) **must** be less than or equal to the [last activatable query index](#) corresponding to the query type of that query plus one

- VUID-vkCmdEncodeVideoKHR-pNext-08360

If the bound video session was created with [VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR](#), and the [pNext](#) chain of [pEncodeInfo](#) includes a [VkVideoInlineQueryInfoKHR](#) structure with its [queryPool](#) member specifying a valid [VkQueryPool](#) handle, then [VkVideoInlineQueryInfoKHR::queryCount](#) **must** equal [opCount](#)

- VUID-vkCmdEncodeVideoKHR-pNext-08361

If the bound video session was created with [VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR](#), and the [pNext](#) chain of [pEncodeInfo](#) includes a [VkVideoInlineQueryInfoKHR](#) structure with its [queryPool](#) member specifying a valid [VkQueryPool](#) handle, then all the queries used by the command, as specified by the [VkVideoInlineQueryInfoKHR](#) structure, **must** be *unavailable*

- VUID-vkCmdEncodeVideoKHR-queryType-08362

If the bound video session was created with [VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR](#), then the [queryType](#) used to create the [queryPool](#) specified in the [VkVideoInlineQueryInfoKHR](#) structure included in the [pNext](#) chain of [pEncodeInfo](#) **must** be [VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR](#) or [VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR](#)

- VUID-vkCmdEncodeVideoKHR-queryPool-08363

If the bound video session was created with [VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR](#), then the [queryPool](#) specified in the [VkVideoInlineQueryInfoKHR](#) structure included in the [pNext](#) chain of [pEncodeInfo](#) **must** have been created with a [VkVideoProfileInfoKHR](#) structure included in the [pNext](#) chain of [VkQueryPoolCreateInfo](#) identical to the one specified in [VkVideoSessionCreateInfoKHR::pVideoProfile](#) the bound video session was created with

- VUID-vkCmdEncodeVideoKHR-queryType-08364

If the bound video session was created with [VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR](#), and the [queryType](#) used to create the [queryPool](#) specified in the [VkVideoInlineQueryInfoKHR](#) structure included in the [pNext](#) chain of [pEncodeInfo](#) is [VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR](#), then the [VkCommandPool](#) that [commandBuffer](#) was allocated from **must** have been created with a queue family index that supports [result status queries](#), as indicated by [VkQueueFamilyQueryResultStatusPropertiesKHR::queryResultStatusSupport](#)

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08201

[pEncodeInfo->dstBuffer](#) **must** be [compatible](#) with the video profile the bound video session was created with

- VUID-vkCmdEncodeVideoKHR-commandBuffer-08202
If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, then `pEncodeInfo->dstBuffer` **must** not be a protected buffer
- VUID-vkCmdEncodeVideoKHR-commandBuffer-08203
If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, then `pEncodeInfo->dstBuffer` **must** be a protected buffer
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08204
`pEncodeInfo->dstBufferOffset` **must** be an integer multiple of `VkVideoCapabilitiesKHR::minBitstreamBufferOffsetAlignment`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08205
`pEncodeInfo->dstBufferRange` **must** be an integer multiple of `VkVideoCapabilitiesKHR::minBitstreamBufferSizeAlignment`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08206
`pEncodeInfo->srcPictureResource.imageViewBinding` **must** be `compatible` with the video profile the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08207
The format of `pEncodeInfo->srcPictureResource.imageViewBinding` **must** match the `VkVideoSessionCreateInfoKHR::pictureFormat` the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08208
`pEncodeInfo->srcPictureResource.codedOffset` **must** be an integer multiple of `codedOffsetGranularity`
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08209
`pEncodeInfo->srcPictureResource.codedExtent` **must** be between `minCodedExtent` and `maxCodedExtent`, inclusive, the bound video session was created with
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08210
`pEncodeInfo->srcPictureResource.imageViewBinding` **must** have been created with `VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR`
- VUID-vkCmdEncodeVideoKHR-commandBuffer-08211
If `commandBuffer` is an unprotected command buffer and `protectedNoFault` is not supported, then `pEncodeInfo->srcPictureResource.imageViewBinding` **must** not have been created from a protected image
- VUID-vkCmdEncodeVideoKHR-commandBuffer-08212
If `commandBuffer` is a protected command buffer and `protectedNoFault` is not supported, then `pEncodeInfo->srcPictureResource.imageViewBinding` **must** have been created from a protected image
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08377
`pEncodeInfo->pSetupReferenceSlot` **must** not be `NULL` unless the bound video session was created with `VkVideoSessionCreateInfoKHR::maxDpbSlots` equal to zero
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08213

If `pEncodeInfo->pSetupReferenceSlot` is not `NULL`, then `pEncodeInfo->pSetupReferenceSlot->slotIndex` **must** be less than the `VkVideoSessionCreateInfoKHR::maxDpbSlots` specified when the bound video session was created

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08214

If `pEncodeInfo->pSetupReferenceSlot` is not `NULL`, then `pEncodeInfo->pSetupReferenceSlot->pPictureResource->codedOffset` **must** be an integer multiple of `codedOffsetGranularity`

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08215

If `pEncodeInfo->pSetupReferenceSlot` is not `NULL`, then `pEncodeInfo->pSetupReferenceSlot->pPictureResource` **must match** one of the `bound reference picture resource`

- VUID-vkCmdEncodeVideoKHR-activeReferencePictureCount-08216

`activeReferencePictureCount` **must** be less than or equal to the `VkVideoSessionCreateInfoKHR::maxActiveReferencePictures` specified when the bound video session was created

- VUID-vkCmdEncodeVideoKHR-slotIndex-08217

The `slotIndex` member of each element of `pEncodeInfo->pReferenceSlots` **must** be less than the `VkVideoSessionCreateInfoKHR::maxDpbSlots` specified when the bound video session was created

- VUID-vkCmdEncodeVideoKHR-codedOffset-08218

The `codedOffset` member of the `VkVideoPictureResourceInfoKHR` structure pointed to by the `pPictureResource` member of each element of `pEncodeInfo->pReferenceSlots` **must** be an integer multiple of `codedOffsetGranularity`

- VUID-vkCmdEncodeVideoKHR-pPictureResource-08219

The `pPictureResource` member of each element of `pEncodeInfo->pReferenceSlots` **must match** one of the `bound reference picture resource` associated with the DPB slot index specified in the `slotIndex` member of that element

- VUID-vkCmdEncodeVideoKHR-pPictureResource-08220

Each video picture resource corresponding to the `pPictureResource` member specified in the elements of `pEncodeInfo->pReferenceSlots` **must** be `unique` within `pEncodeInfo->pReferenceSlots`

- VUID-vkCmdEncodeVideoKHR-dpbFrameUseCount-08221

All elements of `dpbFrameUseCount` **must** be less than or equal to `1`

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08222

The image subresource `referred` to by `pEncodeInfo->srcPictureResource` **must** be in the `VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR` layout at the time the video encode operation is executed on the device

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08223

If `pEncodeInfo->pSetupReferenceSlot` is not `NULL`, then the image subresource `referred` to by `pEncodeInfo->pSetupReferenceSlot->pPictureResource` **must** be in the `VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR` layout at the time the video encode operation is executed on the device

- VUID-vkCmdEncodeVideoKHR-pPictureResource-08224

The image subresource `referred` to by the `pPictureResource` member of each element of `pEncodeInfo->pReferenceSlots` **must** be in the `VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR` layout

at the time the video encode operation is executed on the device

- VUID-vkCmdEncodeVideoKHR-pNext-08225

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the `pNext` chain of `pEncodeInfo` **must** include a `VkVideoEncodeH264PictureInfoKHR` structure

- VUID-vkCmdEncodeVideoKHR-StdVideoH264SequenceParameterSet-08226

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the bound video session parameters object **must** contain a `StdVideoH264SequenceParameterSet` entry with `seq_parameter_set_id` matching `StdVideoEncodeH264PictureInfo::seq_parameter_set_id` that is provided in the `pStdPictureInfo` member of the `VkVideoEncodeH264PictureInfoKHR` structure included in the `pNext` chain of `pEncodeInfo`

- VUID-vkCmdEncodeVideoKHR-StdVideoH264PictureParameterSet-08227

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the bound video session parameters object **must** contain a `StdVideoH264PictureParameterSet` entry with `seq_parameter_set_id` and `pic_parameter_set_id` matching `StdVideoEncodeH264PictureInfo::seq_parameter_set_id` and `StdVideoEncodeH264PictureInfo::pic_parameter_set_id`, respectively, that are provided in the `pStdPictureInfo` member of the `VkVideoEncodeH264PictureInfoKHR` structure included in the `pNext` chain of `pEncodeInfo`

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08228

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and `pEncodeInfo->pSetupReferenceSlot` is not `NULL`, then the `pNext` chain of `pEncodeInfo->pSetupReferenceSlot` **must** include a `VkVideoEncodeH264DpbSlotInfoKHR` structure

- VUID-vkCmdEncodeVideoKHR-pNext-08229

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the `pNext` chain of each element of `pEncodeInfo->pReferenceSlots` **must** include a `VkVideoEncodeH264DpbSlotInfoKHR` structure

- VUID-vkCmdEncodeVideoKHR-constantQp-08269

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and the current `rate control mode` is not `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`, then `VkVideoEncodeH264NaluSliceInfoKHR::constantQp` **must** be zero for each element of the `pNaluSliceEntries` member of the `VkVideoEncodeH264PictureInfoKHR` structure included in the `pNext` chain of `pEncodeInfo`

- VUID-vkCmdEncodeVideoKHR-constantQp-08270

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and the current `rate control mode` is `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`, then `VkVideoEncodeH264NaluSliceInfoKHR::constantQp` **must** be between `VkVideoEncodeH264CapabilitiesKHR::minQp` and `VkVideoEncodeH264CapabilitiesKHR::maxQp`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, for each element of the `pNaluSliceEntries` member

of the [VkVideoEncodeH264PictureInfoKHR](#) structure included in the `pNext` chain of `pEncodeInfo`

- VUID-vkCmdEncodeVideoKHR-constantQp-08271

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and [VkVideoEncodeH264CapabilitiesKHR::flags](#) does not include `VK_VIDEO_ENCODE_H264_CAPABILITY_PER_SLICE_CONSTANT_QP_BIT_KHR`, as returned by [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) for the video profile the bound video session was created with, then [VkVideoEncodeH264NaluSliceInfoKHR::constantQp](#) **must** have the same value for each element of the `pNaluSliceEntries` member of the [VkVideoEncodeH264PictureInfoKHR](#) structure included in the `pNext` chain of `pEncodeInfo`

- VUID-vkCmdEncodeVideoKHR-naluSliceEntryCount-08302

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then the `naluSliceEntryCount` member of the [VkVideoEncodeH264PictureInfoKHR](#) structure included in the `pNext` chain of `pEncodeInfo` **must** be less than or equal to `minCodingBlockExtent.width` multiplied by `minCodingBlockExtent.height`

- VUID-vkCmdEncodeVideoKHR-naluSliceEntryCount-08312

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and [VkVideoEncodeH264CapabilitiesKHR::flags](#) does not include `VK_VIDEO_ENCODE_H264_CAPABILITY_ROW_UNALIGNED_SLICE_BIT_KHR`, as returned by [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) for the video profile the bound video session was created with, then the `naluSliceEntryCount` member of the [VkVideoEncodeH264PictureInfoKHR](#) structure included in the `pNext` chain of `pEncodeInfo` **must** be less than or equal to `minCodingBlockExtent.height`

- VUID-vkCmdEncodeVideoKHR-pNext-08352

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, the `pNext` chain of `pEncodeInfo` includes a [VkVideoEncodeH264PictureInfoKHR](#) structure, and `pEncodeInfo->referenceSlotCount` is greater than zero, then [VkVideoEncodeH264PictureInfoKHR::pStdPictureInfo->pRefLists](#) **must** not be `NULL`

- VUID-vkCmdEncodeVideoKHR-pNext-08339

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, the `pNext` chain of `pEncodeInfo` includes a [VkVideoEncodeH264PictureInfoKHR](#) structure, and [VkVideoEncodeH264PictureInfoKHR::pStdPictureInfo->pRefLists](#) is not `NULL`, then each element of the `RefPicList0` and `RefPicList1` array members of the [StdVideoEncodeH264ReferenceListsInfo](#) structure pointed to by [VkVideoEncodeH264PictureInfoKHR::pStdPictureInfo->pRefLists](#) **must** either be `STD_VIDEO_H264_NO_REFERENCE_PICTURE` or **must** equal the `slotIndex` member of one of the elements of `pEncodeInfo->pReferenceSlots`

- VUID-vkCmdEncodeVideoKHR-pNext-08353

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, the `pNext` chain of `pEncodeInfo` includes a [VkVideoEncodeH264PictureInfoKHR](#) structure, and `pEncodeInfo->referenceSlotCount` is greater than zero, then the `slotIndex` member of each element of `pEncodeInfo->pReferenceSlots` **must** equal one of the elements of the `RefPicList0` or `RefPicList1` array

members of the `StdVideoEncodeH264ReferenceListsInfo` structure pointed to by `VkVideoEncodeH264PictureInfoKHR::pStdPictureInfo->pRefLists`

- VUID-vkCmdEncodeVideoKHR-maxPPictureL0ReferenceCount-08340

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and `VkVideoEncodeH264CapabilitiesKHR::maxPPictureL0ReferenceCount` is zero, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, then `h264PictureType` and each element of `h264L0PictureTypes` and `h264L1PictureTypes` **must** not be `STD_VIDEO_H264_PICTURE_TYPE_P`

- VUID-vkCmdEncodeVideoKHR-maxBPictureL0ReferenceCount-08341

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and `VkVideoEncodeH264CapabilitiesKHR::maxBPictureL0ReferenceCount` and `VkVideoEncodeH264CapabilitiesKHR::maxL1ReferenceCount` are both zero, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, then `h264PictureType` and each element of `h264L0PictureTypes` and `h264L1PictureTypes` **must** not be `STD_VIDEO_H264_PICTURE_TYPE_B`

- VUID-vkCmdEncodeVideoKHR-flags-08342

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and `VkVideoEncodeH264CapabilitiesKHR::flags` does not include `VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, then each element of `h264L0PictureTypes` **must** not be `STD_VIDEO_H264_PICTURE_TYPE_B`

- VUID-vkCmdEncodeVideoKHR-flags-08343

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and `VkVideoEncodeH264CapabilitiesKHR::flags` does not include `VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, then each element of `h264L1PictureTypes` **must** not be `STD_VIDEO_H264_PICTURE_TYPE_B`

- VUID-vkCmdEncodeVideoKHR-pNext-08230

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the `pNext` chain of `pEncodeInfo` **must** include a `VkVideoEncodeH265PictureInfoKHR` structure

- VUID-vkCmdEncodeVideoKHR-StdVideoH265VideoParameterSet-08231

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the bound video session parameters object **must** contain a `StdVideoH265VideoParameterSet` entry with `vps_video_parameter_set_id` matching `StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id` that is provided in the `pStdPictureInfo` member of the `VkVideoEncodeH265PictureInfoKHR` structure included in the `pNext` chain of `pEncodeInfo`

- VUID-vkCmdEncodeVideoKHR-StdVideoH265SequenceParameterSet-08232

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the bound video session parameters

object **must** contain a `StdVideoH265SequenceParameterSet` entry with `sps_video_parameter_set_id` and `sps_seq_parameter_set_id` matching `StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id` and `StdVideoEncodeH265PictureInfo::pps_seq_parameter_set_id`, respectively, that are provided in the `pStdPictureInfo` member of the `VkVideoEncodeH265PictureInfoKHR` structure included in the `pNext` chain of `pEncodeInfo`

- VUID-vkCmdEncodeVideoKHR-StdVideoH265PictureParameterSet-08233

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the bound video session parameters object **must** contain a `StdVideoH265PictureParameterSet` entry with `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` matching `StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id`, `StdVideoEncodeH265PictureInfo::pps_seq_parameter_set_id`, and `StdVideoEncodeH265PictureInfo::pps_pic_parameter_set_id`, respectively, that are provided in the `pStdPictureInfo` member of the `VkVideoEncodeH265PictureInfoKHR` structure included in the `pNext` chain of `pEncodeInfo`

- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-08234

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and `pEncodeInfo->pSetupReferenceSlot` is not `NULL`, then the `pNext` chain of `pEncodeInfo->pSetupReferenceSlot` **must** include a `VkVideoEncodeH265DpbSlotInfoKHR` structure

- VUID-vkCmdEncodeVideoKHR-pNext-08235

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the `pNext` chain of each element of `pEncodeInfo->pReferenceSlots` **must** include a `VkVideoEncodeH265DpbSlotInfoKHR` structure

- VUID-vkCmdEncodeVideoKHR-constantQp-08272

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and the current `rate control mode` is not `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`, then `VkVideoEncodeH265NaluSliceSegmentInfoKHR::constantQp` **must** be zero for each element of the `pNaluSliceSegmentEntries` member of the `VkVideoEncodeH265PictureInfoKHR` structure included in the `pNext` chain of `pEncodeInfo`

- VUID-vkCmdEncodeVideoKHR-constantQp-08273

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and the current `rate control mode` is `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`, then `VkVideoEncodeH265NaluSliceSegmentInfoKHR::constantQp` **must** be between `VkVideoEncodeH265CapabilitiesKHR::minQp` and `VkVideoEncodeH265CapabilitiesKHR::maxQp`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, for each element of the `pNaluSliceSegmentEntries` member of the `VkVideoEncodeH265PictureInfoKHR` structure included in the `pNext` chain of `pEncodeInfo`

- VUID-vkCmdEncodeVideoKHR-constantQp-08274

If the bound video session was created with the video codec operation

`VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and `VkVideoEncodeH265CapabilitiesKHR::flags` does not include `VK_VIDEO_ENCODE_H265_CAPABILITY_PER_SLICE_SEGMENT_CONSTANT_QP_BIT_KHR`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, then `VkVideoEncodeH265NaluSliceSegmentInfoKHR::constantQp` **must** have the same value for each element of the `pNaluSliceSegmentEntries` member of the `VkVideoEncodeH264PictureInfoKHR` structure included in the `pNext` chain of `pEncodeInfo`

- VUID-vkCmdEncodeVideoKHR-naluSliceSegmentEntryCount-08307

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then the `naluSliceSegmentEntryCount` member of the `VkVideoEncodeH265PictureInfoKHR` structure included in the `pNext` chain of `pEncodeInfo` **must** be less than or equal to `minCodingBlockExtent.width` multiplied by `minCodingBlockExtent.height`

- VUID-vkCmdEncodeVideoKHR-naluSliceSegmentEntryCount-08313

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and `VkVideoEncodeH265CapabilitiesKHR::flags` does not include `VK_VIDEO_ENCODE_H265_CAPABILITY_ROW_UNALIGNED_SLICE_SEGMENT_BIT_KHR`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, then the `naluSliceSegmentEntryCount` member of the `VkVideoEncodeH265PictureInfoKHR` structure included in the `pNext` chain of `pEncodeInfo` **must** be less than or equal to `minCodingBlockExtent.height`

- VUID-vkCmdEncodeVideoKHR-pNext-08354

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, the `pNext` chain of `pEncodeInfo` includes a `VkVideoEncodeH265PictureInfoKHR` structure, and `pEncodeInfo->referenceSlotCount` is greater than zero, then `VkVideoEncodeH265PictureInfoKHR::pStdPictureInfo->pRefLists` **must** not be `NULL`

- VUID-vkCmdEncodeVideoKHR-pNext-08344

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, the `pNext` chain of `pEncodeInfo` includes a `VkVideoEncodeH265PictureInfoKHR` structure, and `VkVideoEncodeH265PictureInfoKHR::pStdPictureInfo->pRefLists` is not `NULL`, then each element of the `RefPicList0` and `RefPicList1` array members of the `StdVideoEncodeH265ReferenceListsInfo` structure pointed to by `VkVideoEncodeH265PictureInfoKHR::pStdPictureInfo->pRefLists` **must** either be `STD_VIDEO_H265_NO_REFERENCE_PICTURE` or **must** equal the `slotIndex` member of one of the elements of `pEncodeInfo->pReferenceSlots`

- VUID-vkCmdEncodeVideoKHR-pNext-08355

If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, the `pNext` chain of `pEncodeInfo` includes a `VkVideoEncodeH265PictureInfoKHR` structure, and `pEncodeInfo->referenceSlotCount` is greater than zero, then the `slotIndex` member of each element of `pEncodeInfo->pReferenceSlots` **must** equal one of the elements of the `RefPicList0` or `RefPicList1` array members of the `StdVideoEncodeH265ReferenceListsInfo` structure pointed to by `VkVideoEncodeH265PictureInfoKHR::pStdPictureInfo->pRefLists`

- VUID-vkCmdEncodeVideoKHR-maxPPictureL0ReferenceCount-08345
If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and `VkVideoEncodeH265CapabilitiesKHR::maxPPictureL0ReferenceCount` is zero, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, then `h265PictureType` and each element of `h265L0PictureTypes` and `h265L1PictureTypes` **must** not be `STD_VIDEO_H265_PICTURE_TYPE_P`
- VUID-vkCmdEncodeVideoKHR-maxBPictureL0ReferenceCount-08346
If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and `VkVideoEncodeH265CapabilitiesKHR::maxBPictureL0ReferenceCount` and `VkVideoEncodeH265CapabilitiesKHR::maxL1ReferenceCount` are both zero, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, then `h265PictureType` and each element of `h265L0PictureTypes` and `h265L1PictureTypes` **must** not be `STD_VIDEO_H265_PICTURE_TYPE_B`
- VUID-vkCmdEncodeVideoKHR-flags-08347
If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and `VkVideoEncodeH265CapabilitiesKHR::flags` does not include `VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, then each element of `h265L0PictureTypes` **must** not be `STD_VIDEO_H264_PICTURE_TYPE_B`
- VUID-vkCmdEncodeVideoKHR-flags-08348
If the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and `VkVideoEncodeH265CapabilitiesKHR::flags` does not include `VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the video profile the bound video session was created with, then each element of `h265L1PictureTypes` **must** not be `STD_VIDEO_H265_PICTURE_TYPE_B`

Valid Usage (Implicit)

- VUID-vkCmdEncodeVideoKHR-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdEncodeVideoKHR-pEncodeInfo-parameter
`pEncodeInfo` **must** be a valid pointer to a valid `VkVideoEncodeInfoKHR` structure
- VUID-vkCmdEncodeVideoKHR-commandBuffer-recording
`commandBuffer` **must** be in the `recording state`
- VUID-vkCmdEncodeVideoKHR-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support encode operations
- VUID-vkCmdEncodeVideoKHR-renderpass
This command **must** only be called outside of a render pass instance
- VUID-vkCmdEncodeVideoKHR-videocoding

This command **must** only be called inside of a video coding scope

- VUID-vkCmdEncodeVideoKHR-bufferlevel
`commandBuffer` **must** be a primary `VkCommandBuffer`

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary	Outside	Inside	Encode	Action

The `VkVideoEncodeInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeInfoKHR {
    VkStructureType           sType;
    const void*               pNext;
    VkVideoEncodeFlagsKHR     flags;
    VkBuffer                  dstBuffer;
    VkDeviceSize              dstBufferOffset;
    VkDeviceSize              dstBufferRange;
    VkVideoPictureResourceInfoKHR srcPictureResource;
    const VkVideoReferenceSlotInfoKHR* pSetupReferenceSlot;
    uint32_t                  referenceSlotCount;
    const VkVideoReferenceSlotInfoKHR* pReferenceSlots;
    uint32_t                  precedingExternallyEncodedBytes;
} VkVideoEncodeInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `dstBuffer` is the destination video bitstream buffer to write the encoded bitstream to.
- `dstBufferOffset` is the starting offset in bytes from the start of `dstBuffer` to write the encoded bitstream to.
- `dstBufferRange` is the maximum bitstream size in bytes that **can** be written to `dstBuffer`, starting from `dstBufferOffset`.

- `srcPictureResource` is the video picture resource to use as the [encode input picture](#).
- `pSetupReferenceSlot` is `NULL` or a pointer to a [VkVideoReferenceSlotInfoKHR](#) structure specifying the [reconstructed picture information](#).
- `referenceSlotCount` is the number of elements in the `pReferenceSlots` array.
- `pReferenceSlots` is `NULL` or a pointer to an array of [VkVideoReferenceSlotInfoKHR](#) structures describing the DPB slots and corresponding [reference picture](#) resources to use in this video encode operation (the set of [active reference pictures](#)).
- `precedingExternallyEncodedBytes` is the number of bytes externally encoded by the application to the video bitstream and is used to update the internal state of the implementation's [rate control](#) algorithm to account for the bitrate budget consumed by these externally encoded bytes.

Valid Usage

- VUID-VkVideoEncodeInfoKHR-dstBuffer-08236
`dstBuffer` **must** have been created with `VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR` set
- VUID-VkVideoEncodeInfoKHR-dstBufferOffset-08237
`dstBufferOffset` **must** be less than the size of `dstBuffer`
- VUID-VkVideoEncodeInfoKHR-dstBufferRange-08238
`dstBufferRange` **must** be less than or equal to the size of `dstBuffer` minus `dstBufferOffset`
- VUID-VkVideoEncodeInfoKHR-pSetupReferenceSlot-08239
If `pSetupReferenceSlot` is not `NULL`, then its `slotIndex` member **must** not be negative
- VUID-VkVideoEncodeInfoKHR-pSetupReferenceSlot-08240
If `pSetupReferenceSlot` is not `NULL`, then its `pPictureResource` **must** not be `NULL`
- VUID-VkVideoEncodeInfoKHR-slotIndex-08241
The `slotIndex` member of each element of `pReferenceSlots` **must** not be negative
- VUID-VkVideoEncodeInfoKHR-pPictureResource-08242
The `pPictureResource` member of each element of `pReferenceSlots` **must** not be `NULL`

Valid Usage (Implicit)

- VUID-VkVideoEncodeInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_INFO_KHR`
- VUID-VkVideoEncodeInfoKHR-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of [VkVideoEncodeH264PictureInfoKHR](#), [VkVideoEncodeH265PictureInfoKHR](#), or [VkVideoInlineQueryInfoKHR](#)
- VUID-VkVideoEncodeInfoKHR-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkVideoEncodeInfoKHR-flags-zero bitmask
`flags` **must** be 0
- VUID-VkVideoEncodeInfoKHR-dstBuffer-parameter

`dstBuffer` **must** be a valid `VkBuffer` handle

- `VOID-VkVideoEncodeInfoKHR-srcPictureResource-parameter`
`srcPictureResource` **must** be a valid `VkVideoPictureResourceInfoKHR` structure
- `VOID-VkVideoEncodeInfoKHR-pSetupReferenceSlot-parameter`
If `pSetupReferenceSlot` is not `NULL`, `pSetupReferenceSlot` **must** be a valid pointer to a valid `VkVideoReferenceSlotInfoKHR` structure
- `VOID-VkVideoEncodeInfoKHR-pReferenceSlots-parameter`
If `referenceSlotCount` is not `0`, `pReferenceSlots` **must** be a valid pointer to an array of `referenceSlotCount` valid `VkVideoReferenceSlotInfoKHR` structures

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeFlagsKHR;
```

`VkVideoEncodeFlagsKHR` is a bitmask type for setting a mask, but is currently reserved for future use.

37.16. Video Encode Rate Control

The size of the encoded bitstream data produced by video encode operations is a function of the following set of constraints:

- The capabilities of the compression algorithms defined and employed by the used video compression standard;
- Restrictions imposed by the selected `video profile` according to the rules defined by the used video compression standard;
- Further restrictions imposed by the `capabilities` supported by the implementation for the selected `video profile`;
- The image data in the `encode input picture` and the set of `active reference pictures` (as these affect the effectiveness of the compression algorithms employed by the video encode operations);
- The set of codec-specific and codec-independent encoding parameters provided by the application.

These also inherently define the set of decoder capabilities required for reconstructing and processing the picture data in the encoded bitstream.

Video coding uses *bitrate* as the quantitative metric associated with encoded bitstream data size which expresses the rate at which video bitstream data **can** be transferred or processed, measured in number of bits per second. This bitrate is both a function of the encoded bitstream data size of the encoded pictures as well as the *frame rate* used by the video sequence.

Rate control algorithms are used by video encode operations to enable adjusting encoding parameters to achieve a target bitrate, or otherwise directly or indirectly control the bitrate of the generated video bitstream data. These algorithms are usually not defined by the used video

compression standard, although some video compression standards do provide non-normative guidelines for implementations.

Accordingly, this specification does not mandate implementations to produce identical encoded bitstream data outputs in response to video encode operations, however, it does define a set of codec-independent and codec-specific parameters that enable the application to control the behavior of the rate control algorithms supported by the implementation. Some of these parameters guarantee certain implementation behavior while others provide guidance for implementations to apply various rate control heuristics.

NOTE

Applications need to make sure that they configure rate control parameters appropriately and that they follow the promises made to the implementation through parameters providing guidance for the implementation's rate control algorithms and heuristics in order to be able to get the desired rate control behavior and to be able to hit the set bitrate targets. In addition, the behavior of rate control may also differ across implementations even if the capabilities of the used video profile match between those implementations. This may happen due to implementations applying different rate control algorithms or heuristics internally, and thus even the same set of guidance parameter values may have different effects on the rate control behavior across implementations.

37.16.1. Rate Control Modes

After a video session is reset to the [initial state](#), the default behavior and parameters of video encode rate control are entirely implementation-dependent and the application **cannot** affect the bitrate or quality parameters of the encoded bitstream data produced by video encode operations unless the application changes the rate control configuration of the video session, as described in the [Video Coding Control](#) section.

For each supported [video profile](#), the implementation **may** expose a set of *rate control modes* that are available for use by the application when encoding bitstreams targeting that video profile. These modes allow using different rate control algorithms that fall into one of the following two categories:

1. Per-operation rate control
2. Stream-level rate control

In case of *per-operation rate control*, the bitrate of the generated video bitstream data is indirectly controlled by quality, size, or other encoding parameters specified by the application for each individual video encode operation.

In case of *stream-level rate control*, the application **can** directly specify target bitrates besides other encoding parameters to control the behavior of the rate control algorithm used by the implementation across multiple video encode operations.

The rate control modes are defined with the following enums:

```
// Provided by VK_KHR_video_encode_queue
typedef enum VkVideoEncodeRateControlModeFlagBitsKHR {
    VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR = 0,
    VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_RATE_CONTROL_MODE_CBR_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_RATE_CONTROL_MODE_VBR_BIT_KHR = 0x00000004,
} VkVideoEncodeRateControlModeFlagBitsKHR;
```

- **VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR** specifies the use of implementation-specific rate control.
- **VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR** specifies that rate control is disabled and the application will specify per-operation rate control parameters controlling the encoding quality. In this mode implementations will encode pictures independently of the output bitrate of prior video encode operations.
 - When using an [H.264 encode profile](#), implementations will use the QP value specified in [VkVideoEncodeH264NaluSliceInfoKHR::constantQp](#) to control the quality of the encoded picture.
 - When using an [H.265 encode profile](#), implementations will use the QP value specified in [VkVideoEncodeH265NaluSliceSegmentInfoKHR::constantQp](#) to control the quality of the encoded picture.
- **VK_VIDEO_ENCODE_RATE_CONTROL_MODE_CBR_BIT_KHR** specifies the use of constant bitrate (CBR) rate control mode. In this mode the implementation will attempt to produce the encoded bitstream at a constant bitrate while conforming to the constraints of other rate control parameters.
- **VK_VIDEO_ENCODE_RATE_CONTROL_MODE_VBR_BIT_KHR** specifies the use of variable bitrate (VBR) rate control mode. In this mode the implementation will produce the encoded bitstream at a variable bitrate according to the constraints of other rate control parameters.

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeRateControlModeFlagsKHR;
```

VkVideoEncodeRateControlModeFlagsKHR is a bitmask type for setting a mask of zero or more [VkVideoEncodeRateControlModeFlagBitsKHR](#).

37.16.2. Leaky Bucket Model

Video encoding implementations use the *leaky bucket model* for stream-level rate control. The leaky bucket is a concept referring to the interface between the video encoder and the consumer (for example, a network connection), where the video encoder produces encoded bitstream data corresponding to the encoded pictures and adds them in the leaky bucket while its content are drained by the consumer.

Analogously, a similar leaky bucket is considered to exist at the input interface of a video decoder, into which encoded bitstream data is continuously added and is subsequently consumed by the video decoder. It is desirable to avoid overflowing or underflowing this leaky bucket because:

- In case of an underflow, the video decoder will be unable to consume encoded bitstream data in order to decode pictures (and optionally display them).
- In case of an overflow, the leaky bucket will be unable to accommodate more encoded bitstream data and such data **may** need to be thrown away, leading to the loss of the corresponding encoded pictures.

These requirements **can** be satisfied by imposing various constraints on the encoder-side leaky bucket to avoid its overflow or underflow, depending on the used rate control algorithm and codec parameters. However, enumerating these constraints is outside the scope of this specification.

The term *virtual buffer* is often used as an alternative to refer to the leaky bucket.

This virtual buffer model is defined by the following parameters:

- The bitrate (**R**) at which the encoded bitstream is expected to be processed.
- The size (**B**) of the virtual buffer.
- The initial occupancy (**F**) of the virtual buffer.

In this model the virtual buffer is used to smooth out fluctuations in the bitrate of the encoded bitstream over time without experiencing buffer overflow or underflow, as long as the bitrate of the encoded stream does not diverge from the target bitrate for extended periods of time.

This buffering **may** inherently impose a processing delay, as the goal of the model is to enable decoders maintain a consistent processing rate of an encoded bitstream with varying data rate.

The initial or start-up delay (**D**) is computed as:

$$D = F / R$$

NOTE

Applications need to configure the virtual buffer with sufficient size to avoid or minimize buffer overflows and underflows while also keeping it small enough to meet their latency goals.

37.16.3. Rate Control Layers

Some video compression standards and [video profiles](#) allow associating encoded pictures with specific *video coding layers*. The name, identification, and semantics associated with such video coding layers are defined by the corresponding video compression standards.

Analogously, stream-level rate control **can** be configured to use one or more *rate control layers*:

- When a single rate control layer is configured, it is applied to all encoded pictures, regardless of the picture's video coding layer. In this case the distribution of the available bitrate budget across video coding layers is implementation-dependent.
- When multiple rate control layers are configured, each rate control layer is applied to the corresponding video coding layer, i.e. only across encoded pictures pertaining to the corresponding video coding layer.

Individual rate control layers are identified using *layer indices* between zero and **N-1**, where **N** is the number of active rate control layers.

Rate control layers are only applicable when using [stream-level rate control modes](#).

37.16.4. Rate Control State

Rate control state is maintained by the implementation in the [video session](#) objects and its parameters are specified using an instance of the `VkVideoEncodeRateControlInfoKHR` structure. The complete rate control state of a video session is defined by the following set of parameters:

- The values of the members of the `VkVideoEncodeRateControlInfoKHR` structure used to configure the rate control state.
- The values of the members of any `VkVideoEncodeRateControlLayerInfoKHR` structures specified in `VkVideoEncodeRateControlInfoKHR::pLayers` used to configure the state of individual [rate control layers](#).
- If the video session was created with an [H.264 encode profile](#):
 - The values of the members of the `VkVideoEncodeH264RateControlInfoKHR` structure, if one is specified in the `pNext` chain of the `VkVideoEncodeRateControlInfoKHR` used to configure the rate control state.
 - The values of the members of any `VkVideoEncodeH264RateControlLayerInfoKHR` structures included in the `pNext` chain of a `VkVideoEncodeRateControlLayerInfoKHR` structure used to configure the state of a rate control layer.
- If the video session was created with an [H.265 encode profile](#):
 - The values of the members of the `VkVideoEncodeH265RateControlInfoKHR` structure, if one is specified in the `pNext` chain of the `VkVideoEncodeRateControlInfoKHR` used to configure the rate control state.
 - The values of the members of any `VkVideoEncodeH265RateControlLayerInfoKHR` structures included in the `pNext` chain of a `VkVideoEncodeRateControlLayerInfoKHR` structure used to configure the state of a rate control layer.

Two rate control states match if all the parameters listed above match between them.

The `VkVideoEncodeRateControlInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeRateControlInfoKHR {
    VkStructureType             sType;
    const void*                 pNext;
    VkVideoEncodeRateControlFlagsKHR flags;
    VkVideoEncodeRateControlModeFlagBitsKHR rateControlMode;
    uint32_t                    layerCount;
    const VkVideoEncodeRateControlLayerInfoKHR* pLayers;
    uint32_t                    virtualBufferSizeInMs;
    uint32_t                    initialVirtualBufferSizeInMs;
} VkVideoEncodeRateControlInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is reserved for future use.
- `rateControlMode` is a `VkVideoEncodeRateControlModeFlagBitsKHR` value specifying the `rate control mode`.
- `layerCount` specifies the number of `rate control layers` to use.
- `pLayers` is a pointer to an array of `layerCount` `VkVideoEncodeRateControlLayerInfoKHR` structures, each specifying the rate control configuration of the corresponding rate control layer.
- `virtualBufferSizeInMs` is the size in milliseconds of the virtual buffer used by the implementation's rate control algorithm for the `leaky bucket model`, with respect to the average bitrate of the stream calculated by summing the values of the `averageBitrate` members of the elements of the `pLayers` array.
- `initialVirtualBufferSizeInMs` is the initial occupancy in milliseconds of the virtual buffer used by the implementation's rate control algorithm for the `leaky bucket model`.

If `layerCount` is zero then the values of `virtualBufferSizeInMs` and `initialVirtualBufferSizeInMs` are ignored.

This structure **can** be specified in the following places:

- In the `pNext` chain of `VkVideoBeginCodingInfoKHR` to specify the current rate control state expected to be configured when beginning a `video coding scope`.
- In the `pNext` chain of `VkVideoCodingControlInfoKHR` to change the rate control configuration of the bound video session.

Including this structure in the `pNext` chain of `VkVideoCodingControlInfoKHR` and including `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR` in `VkVideoCodingControlInfoKHR::flags` enables updating the rate control configuration of the bound video session. This replaces the entire rate control configuration of the bound video session and **may** reset the state of all enabled rate control layers to an initial state according to the codec-specific rate control semantics defined in the corresponding sections listed below.

When `layerCount` is greater than one, multiple `rate control layers` are configured, and each rate control layer is applied to the corresponding video coding layer identified by the index of the corresponding element of `pLayer`.

- If the video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then this index specifies the H.264 temporal layer ID of the video coding layer the rate control layer is applied to.
- If the video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then this index specifies the H.265 temporal ID of the video coding layer the rate control layer is applied to.

Additional structures providing codec-specific rate control parameters **can** be included in the `pNext` chain of `VkVideoCodingControlInfoKHR` depending on the `video profile` the bound video session was

created. For further details see:

- [Video Coding Control](#)
- [H.264 Encode Rate Control](#)
- [H.265 Encode Rate Control](#)

The new rate control configuration takes effect when the corresponding [vkCmdControlVideoCodingKHR](#) is executed on the device, and only impacts video encode operations that follow in execution order.

Valid Usage

- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-08248
If `rateControlMode` is `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR` or `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`, then `layerCount` **must** be 0
- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-08275
If `rateControlMode` is `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_CBR_BIT_KHR` or `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_VBR_BIT_KHR`, then `layerCount` **must** be greater than 0
- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-08244
If `rateControlMode` is not `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR`, then it **must** specify one of the bits included in [VkVideoEncodeCapabilitiesKHR::rateControlModes](#), as returned by [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) for the used video profile
- VUID-VkVideoEncodeRateControlInfoKHR-layerCount-08245
`layerCount` member **must** be less than or equal to [VkVideoEncodeCapabilitiesKHR::maxRateControlLayers](#), as returned by [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) for the used video profile
- VUID-VkVideoEncodeRateControlInfoKHR-pLayers-08276
For each element of `pLayers`, its `averageBitrate` member **must** be between 1 and [VkVideoEncodeCapabilitiesKHR::maxBitrate](#), as returned by [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) for the used video profile
- VUID-VkVideoEncodeRateControlInfoKHR-pLayers-08277
For each element of `pLayers`, its `maxBitrate` member **must** be between 1 and [VkVideoEncodeCapabilitiesKHR::maxBitrate](#), as returned by [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) for the used video profile
- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-08356
If `rateControlMode` is `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_CBR_BIT_KHR`, then for each element of `pLayers`, its `averageBitrate` member **must** equal its `maxBitrate` member
- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-08278
If `rateControlMode` is `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_VBR_BIT_KHR`, then for each element of `pLayers`, its `averageBitrate` member **must** be less than or equal to its `maxBitrate` member
- VUID-VkVideoEncodeRateControlInfoKHR-layerCount-08357
If `layerCount` is not zero, then `virtualBufferSizeInMs` **must** be greater than zero
- VUID-VkVideoEncodeRateControlInfoKHR-layerCount-08358

If `layerCount` is not zero, then `initialVirtualBufferSizeInMs` **must** be less than or equal to `virtualBufferSizeInMs`

- VUID-VkVideoEncodeRateControlInfoKHR-videoCodecOperation-07022

If the `videoCodecOperation` of the used video profile is `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, the `pNext` chain this structure is included in also includes an instance of the `VkVideoEncodeH264RateControlInfoKHR` structure, and `layerCount` is greater than 1, then `layerCount` **must** equal `VkVideoEncodeH264RateControlInfoKHR::temporalLayerCount`

- VUID-VkVideoEncodeRateControlInfoKHR-videoCodecOperation-07025

If the `videoCodecOperation` of the used video profile is `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, the `pNext` chain this structure is included in also includes an instance of the `VkVideoEncodeH265RateControlInfoKHR` structure, and `layerCount` is greater than 1, then `layerCount` **must** equal `VkVideoEncodeH265RateControlInfoKHR::subLayerCount`

Valid Usage (Implicit)

- VUID-VkVideoEncodeRateControlInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_INFO_KHR`

- VUID-VkVideoEncodeRateControlInfoKHR-flags-zero bitmask
`flags` **must** be 0

- VUID-VkVideoEncodeRateControlInfoKHR-rateControlMode-parameter

If `rateControlMode` is not 0, `rateControlMode` **must** be a valid `VkVideoEncodeRateControlModeFlagBitsKHR` value

- VUID-VkVideoEncodeRateControlInfoKHR-pLayers-parameter

If `layerCount` is not 0, `pLayers` **must** be a valid pointer to an array of `layerCount` valid `VkVideoEncodeRateControlLayerInfoKHR` structures

```
// Provided by VK_KHR_video_encode_queue
typedef VkFlags VkVideoEncodeRateControlFlagsKHR;
```

`VkVideoEncodeRateControlFlagsKHR` is a bitmask type for setting a mask, but currently reserved for future use.

Rate Control Layer State

The configuration of individual rate control layers is specified using an instance of the `VkVideoEncodeRateControlLayerInfoKHR` structure.

The `VkVideoEncodeRateControlLayerInfoKHR` structure is defined as:


```
// Provided by VK_KHR_video_encode_queue
typedef struct VkVideoEncodeRateControlLayerInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint64_t           averageBitrate;
    uint64_t           maxBitrate;
    uint32_t           frameRateNumerator;
    uint32_t           frameRateDenominator;
} VkVideoEncodeRateControlLayerInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is a pointer to a structure extending this structure.
- `averageBitrate` is the average `bitrate` to be targeted by the implementation's rate control algorithm.
- `maxBitrate` is the peak `bitrate` to be targeted by the implementation's rate control algorithm.
- `frameRateNumerator` is the numerator of the frame rate assumed by the implementation's rate control algorithm.
- `frameRateDenominator` is the denominator of the frame rate assumed by the implementation's rate control algorithm.

NOTE

The ability of the implementation's rate control algorithm to be able to match the requested average and/or peak bitrates **may** be limited by the set of other codec-independent and codec-specific rate control parameters specified by the application, the input content, as well as the application conforming to the rate control guidance provided to the implementation, as described [earlier](#).

Additional structures providing codec-specific rate control parameters **can** be included in the `pNext` chain of `VkVideoEncodeRateControlLayerInfoKHR` depending on the `video profile` the bound video session was created with. For further details see:

- [Video Coding Control](#)
- [H.264 Encode Rate Control](#)
- [H.265 Encode Rate Control](#)

Valid Usage

- VUID-VkVideoEncodeRateControlLayerInfoKHR-frameRateNumerator-08350
`frameRateNumerator` **must** be greater than zero
- VUID-VkVideoEncodeRateControlLayerInfoKHR-frameRateDenominator-08351
`frameRateDenominator` **must** be greater than zero

Valid Usage (Implicit)

- VUID-VkVideoEncodeRateControlLayerInfoKHR-sType-sType
sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_LAYER_INFO_KHR
- VUID-VkVideoEncodeRateControlLayerInfoKHR-pNext-pNext
Each pNext member of any structure (including this one) in the pNext chain **must** be either NULL or a pointer to a valid instance of VkVideoEncodeH264RateControlLayerInfoKHR or VkVideoEncodeH265RateControlLayerInfoKHR
- VUID-VkVideoEncodeRateControlLayerInfoKHR-sType-unique
The sType value of each struct in the pNext chain **must** be unique

37.17. H.264 Encode Operations

Video encode operations using an [H.264 encode profile](#) **can** be used to encode elementary video stream sequences compliant to the [ITU-T H.264 Specification](#).

NOTE

Refer to the [Preamble](#) for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by Khronos.

This process is performed according to the [video encode operation steps](#) with the codec-specific semantics defined in section 8 of the [ITU-T H.264 Specification](#) as follows:

- Syntax elements, derived values, and other parameters are applied from the following structures:
 - The StdVideoH264SequenceParameterSet structure corresponding to the [active SPS](#) specifying the [H.264 sequence parameter set](#).
 - The StdVideoH264PictureParameterSet structure corresponding to the [active PPS](#) specifying the [H.264 picture parameter set](#).
 - The StdVideoEncodeH264PictureInfo structure specifying the [H.264 picture information](#).
 - The StdVideoEncodeH264SliceHeader structures specifying the [H.264 slice header parameters](#) for each encoded H.264 slice.
 - The StdVideoEncodeH264ReferenceInfo structures specifying the [H.264 reference information](#) corresponding to the optional [reconstructed picture](#) and any [active reference pictures](#).
- The encoded bitstream data is written to the destination video bitstream buffer range as defined in the [H.264 Encode Bitstream Data Access](#) section.
- Picture data in the [video picture resources](#) corresponding to the used [encode input picture](#), [active reference pictures](#), and optional [reconstructed picture](#) is accessed as defined in the [H.264 Encode Picture Data Access](#) section.
- The decision on [reference picture setup](#) is made according to the parameters specified in the [H.264 picture information](#).

If the parameters adhere to the syntactic and semantic requirements defined in the corresponding

sections of the [ITU-T H.264 Specification](#), as described above, and the [DPB slots](#) associated with the [active reference pictures](#) all refer to [valid picture references](#), then the video encode operation will complete successfully. Otherwise, the video encode operation **may** complete [unsuccessfully](#).

37.17.1. H.264 Encode Parameter Overrides

Implementations **may** override, unless otherwise specified, any of the H.264 encode parameters specified in the following Video Std structures:

- [StdVideoH264SequenceParameterSet](#)
- [StdVideoH264PictureParameterSet](#)
- [StdVideoEncodeH264PictureInfo](#)
- [StdVideoEncodeH264SliceHeader](#)
- [StdVideoEncodeH264ReferenceInfo](#)

All such H.264 encode parameter overrides **must** fulfill the conditions defined in the [Video Encode Parameter Overrides](#) section.

In addition, implementations **must** not override any of the following H.264 encode parameters:

- [StdVideoEncodeH264PictureInfo::primary_pic_type](#)
- [StdVideoEncodeH264SliceHeader::slice_type](#)

In case of H.264 encode parameters stored in [video session parameters](#) objects, applications need to use the [vkGetEncodedVideoSessionParametersKHR](#) command to determine whether any implementation overrides happened. If the query indicates that implementation overrides were applied, then the application needs to retrieve and use the encoded H.264 parameter sets in the bitstream in order to be able to produce a compliant H.264 video bitstream using the H.264 encode parameters stored in the video session parameters object.

In case of any H.264 encode parameters stored in the encoded bitstream produced by video encode operations, if the implementation supports the [VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR](#) [video encode feedback query](#) flag, the application **can** use such queries to retrieve feedback about whether any implementation overrides have been applied to those H.264 encode parameters.

37.17.2. H.264 Encode Bitstream Data Access

Each video encode operation writes one or more VCL NAL units comprising of slice headers and data of the encoded picture, in the format defined in sections 7.3.3 and 7.3.4, according to the semantics defined in sections 7.4.3 and 7.4.4 of the [ITU-T H.264 Specification](#), respectively. The number of VCL NAL units written is specified by [VkVideoEncodeH264PictureInfoKHR::naluSliceEntryCount](#).

In addition, if [VkVideoEncodeH264PictureInfoKHR::generatePrefixNalu](#) is [VK_TRUE](#) for the video encode operation, then an additional prefix NAL unit is written before each VCL NAL unit corresponding to individual slices in the format defined in section 7.3.2.12, according to the semantics defined in section 7.4.2.12 of the [ITU-T H.264 Specification](#), respectively.

37.17.3. H.264 Encode Picture Data Access

Accesses to image data within a video picture resource happen at the granularity indicated by `VkVideoCapabilitiesKHR::pictureAccessGranularity`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used `video profile`. Accordingly, the complete image subregion of a `encode input picture`, `reference picture`, or `reconstructed picture` accessed by video coding operations using an `H.264 encode profile` is defined as the set of texels within the coordinate range:

$$([0, \text{endX}), [0, \text{endY}))$$

Where:

- `endX` equals `codedExtent.width` rounded up to the nearest integer multiple of `pictureAccessGranularity.width` and clamped to the width of the image subresource `referred` to by the corresponding `VkVideoPictureResourceInfoKHR` structure;
- `endY` equals `codedExtent.height` rounded up to the nearest integer multiple of `pictureAccessGranularity.height` and clamped to the height of the image subresource `referred` to by the corresponding `VkVideoPictureResourceInfoKHR` structure;

Where `codedExtent` is the member of the `VkVideoPictureResourceInfoKHR` structure corresponding to the picture.

In case of video encode operations using an `H.264 encode profile`, any access to a picture at the coordinates `(x,y)`, as defined by the `ITU-T H.264 Specification`, is an access to the image subresource `referred` to by the corresponding `VkVideoPictureResourceInfoKHR` structure at the texel coordinates `(x,y)`.

Implementations **may** choose not to access some or all texels within particular `reference pictures` available to a video encode operation (e.g. due to `video encode parameter overrides` restricting the effective set of used reference pictures, or if the encoding algorithm chooses not to use certain subregions of the reference picture data for sample prediction).

37.17.4. H.264 Frame, Picture, and Slice

H.264 pictures are partitioned into slices, as defined in section 6.3 of the `ITU-T H.264 Specification`.

For the purposes of this specification, the H.264 slices comprising a picture are referred to as the *picture partitions* of the picture.

Video encode operations using an `H.264 encode profile` **can** encode slices of different types, as defined in section 7.4.3 of the `ITU-T H.264 Specification`, by specifying the corresponding enumeration constant value in `StdVideoEncodeH264SliceHeader::slice_type` in the `H.264 slice header parameters` from the Video Std enumeration type `StdVideoH264SliceType`:

- `STD_VIDEO_H264_SLICE_TYPE_P` indicates that the slice is a *P slice* as defined in section 3.109 of the `ITU-T H.264 Specification`.
- `STD_VIDEO_H264_SLICE_TYPE_B` indicates that the slice is a *B slice* as defined in section 3.9 of the

[ITU-T H.264 Specification](#).

- `STD_VIDEO_H264_SLICE_TYPE_I` indicates that the slice is an *I slice* as defined in section 3.66 of the [ITU-T H.264 Specification](#).

Pictures constructed from such slices **can** be of different types, as defined in section 7.4.2.4 of the [ITU-T H.264 Specification](#). Video encode operations using an [H.264 encode profile](#) **can** encode pictures of a specific type by specifying the corresponding enumeration constant value in `StdVideoEncodeH264PictureInfo::primary_pic_type` in the [H.264 picture information](#) from the Video Std enumeration type `StdVideoH264PictureType`:

- `STD_VIDEO_H264_PICTURE_TYPE_P` indicates that the picture is a *P picture*. A frame consisting of a P picture is also referred to as a *P frame*.
- `STD_VIDEO_H264_PICTURE_TYPE_B` indicates that the picture is a *B picture*. A frame consisting of a B picture is also referred to as a *B frame*.
- `STD_VIDEO_H264_PICTURE_TYPE_I` indicates that the picture is an *I picture*. A frame consisting of an I picture is also referred to as an *I frame*.
- `STD_VIDEO_H264_PICTURE_TYPE_IDR` indicates that the picture is a special type of I picture called an *IDR picture* as defined in section 3.69 of the [ITU-T H.264 Specification](#). A frame consisting of an IDR picture is also referred to as an *IDR frame*.

37.17.5. H.264 Coding Blocks

H.264 encode supports a single type of coding block called a *macroblock*, as defined in section 3.84 of the [ITU-T H.264 Specification](#).

37.17.6. H.264 Encode Profile

A video profile supporting H.264 video encode operations is specified by setting `VkVideoProfileInfoKHR::videoCodecOperation` to `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` and adding a `VkVideoEncodeH264ProfileInfoKHR` structure to the `VkVideoProfileInfoKHR::pNext` chain.

The `VkVideoEncodeH264ProfileInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264ProfileInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    StdVideoH264ProfileIdc stdProfileIdc;
} VkVideoEncodeH264ProfileInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdProfileIdc` is a `StdVideoH264ProfileIdc` value specifying the H.264 codec profile IDC, as defined in section A.2 of the [ITU-T H.264 Specification](#).

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264ProfileInfoKHR-sType-sType
sType **must** be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PROFILE_INFO_KHR

37.17.7. H.264 Encode Capabilities

When calling [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) to query the capabilities for an [H.264 encode profile](#), the [VkVideoCapabilitiesKHR::pNext](#) chain **must** include a [VkVideoEncodeH264CapabilitiesKHR](#) structure that will be filled with the profile-specific capabilities.

The [VkVideoEncodeH264CapabilitiesKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264CapabilitiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkVideoEncodeH264CapabilityFlagsKHR flags;
    StdVideoH264LevelIdc maxLevelIdc;
    uint32_t            maxSliceCount;
    uint32_t            maxPPictureL0ReferenceCount;
    uint32_t            maxBPictureL0ReferenceCount;
    uint32_t            maxL1ReferenceCount;
    uint32_t            maxTemporalLayerCount;
    VkBool32            expectDyadicTemporalLayerPattern;
    int32_t             minQp;
    int32_t             maxQp;
    VkBool32            prefersGopRemainingFrames;
    VkBool32            requiresGopRemainingFrames;
    VkVideoEncodeH264StdFlagsKHR stdSyntaxFlags;
} VkVideoEncodeH264CapabilitiesKHR;
```

- sType is a [VkStructureType](#) value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- flags is a bitmask of [VkVideoEncodeH264CapabilityFlagBitsKHR](#) indicating supported H.264 encoding capabilities.
- maxLevelIdc is a [StdVideoH264LevelIdc](#) value indicating the maximum H.264 level supported by the profile, where enum constant [STD_VIDEO_H264_LEVEL_IDC_<major>_<minor>](#) identifies H.264 level <major>.<minor> as defined in section A.3 of the [ITU-T H.264 Specification](#).
- maxSliceCount indicates the maximum number of slices that **can** be encoded for a single picture. Further restrictions **may** apply to the number of slices that **can** be encoded for a single picture depending on other capabilities and codec-specific rules.
- maxPPictureL0ReferenceCount indicates the maximum number of reference pictures the implementation supports in the reference list L0 for [P pictures](#).

NOTE

As implementations **may override** the reference lists, `maxPPictureL0ReferenceCount` does not limit the number of elements that the application **can** specify in the L0 reference list for P pictures. However, if `maxPPictureL0ReferenceCount` is zero, then the use of P pictures is not allowed.

- `maxBPictureL0ReferenceCount` indicates the maximum number of reference pictures the implementation supports in the reference list L0 for B pictures.
- `maxL1ReferenceCount` indicates the maximum number of reference pictures the implementation supports in the reference list L1 if encoding of B pictures is supported.

NOTE

As implementations **may override** the reference lists, `maxBPictureL0ReferenceCount` and `maxL1ReferenceCount` does not limit the number of elements that the application **can** specify in the L0 and L1 reference lists for B pictures. However, if `maxBPictureL0ReferenceCount` and `maxL1ReferenceCount` are both zero, then the use of B pictures is not allowed.

- `maxTemporalLayerCount` indicates the maximum number of H.264 temporal layers supported by the implementation.
- `expectDyadicTemporalLayerPattern` indicates that the implementation's rate control algorithms expect the application to use a **dyadic temporal layer pattern** when encoding multiple temporal layers.
- `minQp` indicates the minimum QP value supported.
- `maxQp` indicates the maximum QP value supported.
- `prefersGopRemainingFrames` indicates that the implementation's rate control algorithm prefers the application to specify the number of frames of each type **remaining** in the current **group of pictures** when beginning a **video coding scope**.
- `requiresGopRemainingFrames` indicates that the implementation's rate control algorithm requires the application to specify the number of frames of each type **remaining** in the current **group of pictures** when beginning a **video coding scope**.
- `stdSyntaxFlags` is a bitmask of `VkVideoEncodeH264StdFlagBitsKHR` indicating capabilities related to H.264 syntax elements.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264CapabilitiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_CAPABILITIES_KHR`

Bits which **may** be set in `VkVideoEncodeH264CapabilitiesKHR::flags`, indicating the H.264 encoding capabilities supported, are:


```
// Provided by VK_KHR_video_encode_h264
typedef enum VkVideoEncodeH264CapabilityFlagBitsKHR {
    VK_VIDEO_ENCODE_H264_CAPABILITY_HRD_COMPLIANCE_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H264_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR =
0x00000002,
    VK_VIDEO_ENCODE_H264_CAPABILITY_ROW_UNALIGNED_SLICE_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H264_CAPABILITY_DIFFERENT_SLICE_TYPE_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR = 0x00000010,
    VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR = 0x00000020,
    VK_VIDEO_ENCODE_H264_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR = 0x00000040,
    VK_VIDEO_ENCODE_H264_CAPABILITY_PER_SLICE_CONSTANT_QP_BIT_KHR = 0x00000080,
    VK_VIDEO_ENCODE_H264_CAPABILITY_GENERATE_PREFIX_NALU_BIT_KHR = 0x00000100,
} VkVideoEncodeH264CapabilityFlagBitsKHR;
```

- `VK_VIDEO_ENCODE_H264_CAPABILITY_HRD_COMPLIANCE_BIT_KHR` indicates whether the implementation **may** be able to generate HRD compliant bitstreams if any of the `nal_hrd_parameters_present_flag` or `vcl_hrd_parameters_present_flag` members of `StdVideoH264SpsVuiFlags` are set to 1 in the [active SPS](#).
- `VK_VIDEO_ENCODE_H264_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR` indicates that if `StdVideoH264PpsFlags::weighted_pred_flag` is set to 1 or `StdVideoH264PictureParameterSet::weighted_bipred_idc` is set to `STD_VIDEO_H264_WEIGHTED_BIPRED_IDC_EXPLICIT` in the [active PPS](#) when encoding a [P picture](#) or [B picture](#), respectively, then the implementation is able to internally decide syntax for `pred_weight_table`, as defined in section 7.4.3.2 of the [ITU-T H.264 Specification](#), and the application is not **required** to provide a weight table in the [H.264 slice header parameters](#).
- `VK_VIDEO_ENCODE_H264_CAPABILITY_ROW_UNALIGNED_SLICE_BIT_KHR` indicates that each slice in a frame with multiple slices may begin or finish at any offset in a macroblock row. If not supported, all slices in the frame **must** begin at the start of a macroblock row (and hence each slice **must** finish at the end of a macroblock row).
- `VK_VIDEO_ENCODE_H264_CAPABILITY_DIFFERENT_SLICE_TYPE_BIT_KHR` indicates that when a frame is encoded with multiple slices, the implementation allows encoding each slice with a different `StdVideoEncodeH264SliceHeader::slice_type` specified in the [H.264 slice header parameters](#). If not supported, all slices of the frame **must** be encoded with the same `slice_type` which corresponds to the picture type of the frame.
- `VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR` indicates support for using a [B frame](#) as L0 reference, as specified in `StdVideoEncodeH264ReferenceListsInfo::RefPicList0` in the [H.264 picture information](#).
- `VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR` indicates support for using a [B frame](#) as L1 reference, as specified in `StdVideoEncodeH264ReferenceListsInfo::RefPicList1` in the [H.264 picture information](#).
- `VK_VIDEO_ENCODE_H264_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR` indicates support for specifying different QP values in the members of `VkVideoEncodeH264QpKHR`.
- `VK_VIDEO_ENCODE_H264_CAPABILITY_PER_SLICE_CONSTANT_QP_BIT_KHR` indicates support for specifying different constant QP values for each slice.

- `VK_VIDEO_ENCODE_H264_CAPABILITY_GENERATE_PREFIX_NALU_BIT_KHR` indicates support for generating prefix NAL units by setting `VkVideoEncodeH264PictureInfoKHR::generatePrefixNalu` to `VK_TRUE`.

```
// Provided by VK_KHR_video_encode_h264
typedef VkFlags VkVideoEncodeH264CapabilityFlagsKHR;
```

`VkVideoEncodeH264CapabilityFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoEncodeH264CapabilityFlagBitsKHR`.

Bits which **may** be set in `VkVideoEncodeH264CapabilitiesKHR::stdSyntaxFlags`, indicating the capabilities related to the H.264 syntax elements, are:

```
// Provided by VK_KHR_video_encode_h264
typedef enum VkVideoEncodeH264StdFlagBitsKHR {
    VK_VIDEO_ENCODE_H264_STD_SEPARATE_COLOR_PLANE_FLAG_SET_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H264_STD_QPPRIME_Y_ZERO_TRANSFORM_BYPASS_FLAG_SET_BIT_KHR =
0x00000002,
    VK_VIDEO_ENCODE_H264_STD_SCALING_MATRIX_PRESENT_FLAG_SET_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H264_STD_CHROMA_QP_INDEX_OFFSET_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H264_STD_SECOND_CHROMA_QP_INDEX_OFFSET_BIT_KHR = 0x00000010,
    VK_VIDEO_ENCODE_H264_STD_PIC_INIT_QP_MINUS26_BIT_KHR = 0x00000020,
    VK_VIDEO_ENCODE_H264_STD_WEIGHTED_PRED_FLAG_SET_BIT_KHR = 0x00000040,
    VK_VIDEO_ENCODE_H264_STD_WEIGHTED_BIPRED_IDC_EXPLICIT_BIT_KHR = 0x00000080,
    VK_VIDEO_ENCODE_H264_STD_WEIGHTED_BIPRED_IDC_IMPLICIT_BIT_KHR = 0x00000100,
    VK_VIDEO_ENCODE_H264_STD_TRANSFORM_8X8_MODE_FLAG_SET_BIT_KHR = 0x00000200,
    VK_VIDEO_ENCODE_H264_STD_DIRECT_SPATIAL_MV_PRED_FLAG_UNSET_BIT_KHR = 0x00000400,
    VK_VIDEO_ENCODE_H264_STD_ENTROPY_CODING_MODE_FLAG_UNSET_BIT_KHR = 0x00000800,
    VK_VIDEO_ENCODE_H264_STD_ENTROPY_CODING_MODE_FLAG_SET_BIT_KHR = 0x00001000,
    VK_VIDEO_ENCODE_H264_STD_DIRECT_8X8_INFERENCE_FLAG_UNSET_BIT_KHR = 0x00002000,
    VK_VIDEO_ENCODE_H264_STD_CONSTRAINED_INTRA_PRED_FLAG_SET_BIT_KHR = 0x00004000,
    VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_DISABLED_BIT_KHR = 0x00008000,
    VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_ENABLED_BIT_KHR = 0x00010000,
    VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_PARTIAL_BIT_KHR = 0x00020000,
    VK_VIDEO_ENCODE_H264_STD_SLICE_QP_DELTA_BIT_KHR = 0x00080000,
    VK_VIDEO_ENCODE_H264_STD_DIFFERENT_SLICE_QP_DELTA_BIT_KHR = 0x00100000,
} VkVideoEncodeH264StdFlagBitsKHR;
```

- `VK_VIDEO_ENCODE_H264_STD_SEPARATE_COLOR_PLANE_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH264SpsFlags::separate_colour_plane_flag` in the `SPS` when that value is 1.
- `VK_VIDEO_ENCODE_H264_STD_QPPRIME_Y_ZERO_TRANSFORM_BYPASS_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH264SpsFlags::qpprime_y_zero_transform_bypass_flag` in the `SPS` when that value is 1.
- `VK_VIDEO_ENCODE_H264_STD_SCALING_MATRIX_PRESENT_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided values for `StdVideoH264SpsFlags::seq_scaling_matrix_present_flag` in the `SPS` and `StdVideoH264PpsFlags::pic_scaling_matrix_present_flag` in the `PPS` when any of those values are 1.

- `VK_VIDEO_ENCODE_H264_STD_CHROMA_QP_INDEX_OFFSET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH264PictureParameterSet::chroma_qp_index_offset` in the `PPS` when that value is non-zero.
- `VK_VIDEO_ENCODE_H264_STD_SECOND_CHROMA_QP_INDEX_OFFSET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH264PictureParameterSet::second_chroma_qp_index_offset` in the `PPS` when that value is non-zero.
- `VK_VIDEO_ENCODE_H264_STD_PIC_INIT_QP_MINUS26_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH264PictureParameterSet::pic_init_qp_minus26` in the `PPS` when that value is non-zero.
- `VK_VIDEO_ENCODE_H264_STD_WEIGHTED_PRED_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH264PpsFlags::weighted_pred_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H264_STD_WEIGHTED_BIPRED_IDC_EXPLICIT_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH264PictureParameterSet::weighted_bipred_idc` in the `PPS` when that value is `STD_VIDEO_H264_WEIGHTED_BIPRED_IDC_EXPLICIT`.
- `VK_VIDEO_ENCODE_H264_STD_WEIGHTED_BIPRED_IDC_IMPLICIT_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH264PictureParameterSet::weighted_bipred_idc` in the `PPS` when that value is `STD_VIDEO_H264_WEIGHTED_BIPRED_IDC_IMPLICIT`.
- `VK_VIDEO_ENCODE_H264_STD_TRANSFORM_8X8_MODE_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH264PpsFlags::transform_8x8_mode_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H264_STD_DIRECT_SPATIAL_MV_PRED_FLAG_UNSET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoEncodeH264SliceHeaderFlags::direct_spatial_mv_pred_flag` in the `H.264 slice header parameters` when that value is 0.
- `VK_VIDEO_ENCODE_H264_STD_ENTROPY_CODING_MODE_FLAG_UNSET_BIT_KHR` indicates whether the implementation supports CAVLC entropy coding, as defined in section 9.2 of the [ITU-T H.264 Specification](#), and thus supports using the application-provided value for `StdVideoH264PpsFlags::entropy_coding_mode_flag` in the `PPS` when that value is 0.
- `VK_VIDEO_ENCODE_H264_STD_ENTROPY_CODING_MODE_FLAG_SET_BIT_KHR` indicates whether the implementation supports CABAC entropy coding, as defined in section 9.3 of the [ITU-T H.264 Specification](#), and thus supports using the application-provided value for `StdVideoH264PpsFlags::entropy_coding_mode_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H264_STD_DIRECT_8X8_INFERENCE_FLAG_UNSET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH264SpsFlags::direct_8x8_inference_flag` in the `SPS` when that value is 0.
- `VK_VIDEO_ENCODE_H264_STD_CONSTRAINED_INTRA_PRED_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH264PpsFlags::constrained_intra_pred_flag` in the `PPS` when that value is 1.

- `VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_DISABLED_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoEncodeH264SliceHeader::disable_deblocking_filter_idc` in the [H.264 slice header parameters](#) when that value is `STD_VIDEO_H264_DISABLE_DEBLOCKING_FILTER_IDC_DISABLED`.
- `VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_ENABLED_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoEncodeH264SliceHeader::disable_deblocking_filter_idc` in the [H.264 slice header parameters](#) when that value is `STD_VIDEO_H264_DISABLE_DEBLOCKING_FILTER_IDC_ENABLED`.
- `VK_VIDEO_ENCODE_H264_STD_DEBLOCKING_FILTER_PARTIAL_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoEncodeH264SliceHeader::disable_deblocking_filter_idc` in the [H.264 slice header parameters](#) when that value is `STD_VIDEO_H264_DISABLE_DEBLOCKING_FILTER_IDC_PARTIAL`.
- `VK_VIDEO_ENCODE_H264_STD_SLICE_QP_DELTA_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoEncodeH264SliceHeader::slice_qp_delta` in the [H.264 slice header parameters](#) when that value is identical across the slices of the encoded frame.
- `VK_VIDEO_ENCODE_H264_STD_DIFFERENT_SLICE_QP_DELTA_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoEncodeH264SliceHeader::slice_qp_delta` in the [H.264 slice header parameters](#) when that value is different across the slices of the encoded frame.

These capability flags provide information to the application about specific H.264 syntax element values that the implementation supports without having to [override](#) them and do not otherwise restrict the values that the application **can** specify for any of the mentioned H.264 syntax elements.

```
// Provided by VK_KHR_video_encode_h264
typedef VkFlags VkVideoEncodeH264StdFlagsKHR;
```

`VkVideoEncodeH264StdFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoEncodeH264StdFlagBitsKHR`.

37.17.8. H.264 Encode Quality Level Properties

When calling `vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR` with `pVideoProfile->videoCodecOperation` specified as `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, the `VkVideoEncodeH264QualityLevelPropertiesKHR` structure **must** be included in the `pNext` chain of the `VkVideoEncodeQualityLevelPropertiesKHR` structure to retrieve additional video encode quality level properties specific to H.264 encoding.

The `VkVideoEncodeH264QualityLevelPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264QualityLevelPropertiesKHR {
    VkStructureType          sType;
    void*                    pNext;
    VkVideoEncodeH264RateControlFlagsKHR preferredRateControlFlags;
    uint32_t                 preferredGopFrameCount;
    uint32_t                 preferredIdrPeriod;
    uint32_t                 preferredConsecutiveBFrameCount;
    uint32_t                 preferredTemporalLayerCount;
    VkVideoEncodeH264QpKHR   preferredConstantQp;
    uint32_t                 preferredMaxL0ReferenceCount;
    uint32_t                 preferredMaxL1ReferenceCount;
    VkBool32                 preferredStdEntropyCodingModeFlag;
} VkVideoEncodeH264QualityLevelPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `preferredRateControlFlags` is a bitmask of `VkVideoEncodeH264RateControlFlagBitsKHR` values indicating the preferred flags to use for `VkVideoEncodeH264RateControlInfoKHR::flags`.
- `preferredGopFrameCount` indicates the preferred value to use for `VkVideoEncodeH264RateControlInfoKHR::gopFrameCount`.
- `preferredIdrPeriod` indicates the preferred value to use for `VkVideoEncodeH264RateControlInfoKHR::idrPeriod`.
- `preferredConsecutiveBFrameCount` indicates the preferred value to use for `VkVideoEncodeH264RateControlInfoKHR::consecutiveBFrameCount`.
- `preferredTemporalLayerCount` indicates the preferred value to use for `VkVideoEncodeH264RateControlInfoKHR::temporalLayerCount`.
- `preferredConstantQp` indicates the preferred values to use for `VkVideoEncodeH264NaluSliceInfoKHR::constantQp` for each picture type when using `rate control mode` `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`.
- `preferredMaxL0ReferenceCount` indicates the preferred maximum number of reference pictures to use in the reference list L0.
- `preferredMaxL1ReferenceCount` indicates the preferred maximum number of reference pictures to use in the reference list L1.
- `preferredStdEntropyCodingModeFlag` indicates the preferred value to use for `entropy_coding_mode_flag` in `StdVideoH264PpsFlags`.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264QualityLevelPropertiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_QUALITY_LEVEL_PROPERTIES_KHR`

37.17.9. H.264 Encode Session

Additional parameters can be specified when creating a video session with an H.264 encode profile by including an instance of the [VkVideoEncodeH264SessionCreateInfoKHR](#) structure in the `pNext` chain of [VkVideoSessionCreateInfoKHR](#).

The [VkVideoEncodeH264SessionCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264SessionCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           useMaxLevelIdc;
    StdVideoH264LevelIdc maxLevelIdc;
} VkVideoEncodeH264SessionCreateInfoKHR;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `useMaxLevelIdc` indicates whether the value of `maxLevelIdc` should be used by the implementation. When it is `VK_FALSE`, the implementation ignores the value of `maxLevelIdc` and uses the value of [VkVideoEncodeH264CapabilitiesKHR::maxLevelIdc](#), as reported by [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) for the video profile.
- `maxLevelIdc` is a `StdVideoH264LevelIdc` value specifying the upper bound on the H.264 level for the video bitstreams produced by the created video session, where enum constant `STD_VIDEO_H264_LEVEL_IDC_<major>_<minor>` identifies H.264 level `<major>.<minor>` as defined in section A.3 of the [ITU-T H.264 Specification](#).

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264SessionCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_CREATE_INFO_KHR`

37.17.10. H.264 Encode Parameter Sets

[Video session parameters](#) objects created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR` can contain the following types of parameters:

H.264 Sequence Parameter Sets (SPS)

Represented by [StdVideoH264SequenceParameterSet](#) structures and interpreted as follows:

- `reserved1` and `reserved2` are used only for padding purposes and are otherwise ignored;
- `seq_parameter_set_id` is used as the key of the SPS entry;
- `level_idc` is one of the enum constants `STD_VIDEO_H264_LEVEL_IDC_<major>_<minor>` identifying

the H.264 level <major>.<minor> as defined in section A.3 of the [ITU-T H.264 Specification](#);

- if `flags.seq_scaling_matrix_present_flag` is set, then the `StdVideoH264ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
 - `scaling_list_present_mask` is a bitmask where bit index `i` corresponds to `seq_scaling_list_present_flag[i]` as defined in section 7.4.2.1 of the [ITU-T H.264 Specification](#);
 - `use_default_scaling_matrix_mask` is a bitmask where bit index `i` corresponds to `UseDefaultScalingMatrix4x4Flag[i]`, when `i < 6`, or corresponds to `UseDefaultScalingMatrix8x8Flag[i-6]`, otherwise, as defined in section 7.3.2.1 of the [ITU-T H.264 Specification](#);
 - `ScalingList4x4` and `ScalingList8x8` correspond to the identically named syntax elements defined in section 7.3.2.1 of the [ITU-T H.264 Specification](#);
- if `flags.vui_parameters_present_flag` is set, then `pSequenceParameterSetVui` is a pointer to a `StdVideoH264SequenceParameterSetVui` structure that is interpreted as follows:
 - `reserved1` is used only for padding purposes and is otherwise ignored;
 - `flags.color_description_present_flag` is interpreted as the value of `colour_description_present_flag`, as defined in section E.2.1 of the [ITU-T H.264 Specification](#);

NOTE

The name of `colour_description_present_flag` was misspelled in the Video Std header.

- if `flags.nal_hrd_parameters_present_flag` or `flags.vcl_hrd_parameters_present_flag` is set, then the `StdVideoH264HrdParameters` structure pointed to by `pHrdParameters` is interpreted as follows:
 - `reserved1` is used only for padding purposes and is otherwise ignored;
 - all other members of `StdVideoH264HrdParameters` are interpreted as defined in section E.2.2 of the [ITU-T H.264 Specification](#);
- all other members of `StdVideoH264SequenceParameterSetVui` are interpreted as defined in section E.2.1 of the [ITU-T H.264 Specification](#);
- all other members of `StdVideoH264SequenceParameterSet` are interpreted as defined in section 7.4.2.1 of the [ITU-T H.264 Specification](#).

H.264 Picture Parameter Sets (PPS)

Represented by `StdVideoH264PictureParameterSet` structures and interpreted as follows:

- the pair constructed from `seq_parameter_set_id` and `pic_parameter_set_id` is used as the key of the PPS entry;
- if `flags.pic_scaling_matrix_present_flag` is set, then the `StdVideoH264ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
 - `scaling_list_present_mask` is a bitmask where bit index `i` corresponds to

`pic_scaling_list_present_flag[i]` as defined in section 7.4.2.2 of the [ITU-T H.264 Specification](#);

- `use_default_scaling_matrix_mask` is a bitmask where bit index `i` corresponds to `UseDefaultScalingMatrix4x4Flag[i]`, when `i < 6`, or corresponds to `UseDefaultScalingMatrix8x8Flag[i-6]`, otherwise, as defined in section 7.3.2.2 of the [ITU-T H.264 Specification](#);
- `ScalingList4x4` and `ScalingList8x8` correspond to the identically named syntax elements defined in section 7.3.2.2 of the [ITU-T H.264 Specification](#);
- all other members of `StdVideoH264PictureParameterSet` are interpreted as defined in section 7.4.2.2 of the [ITU-T H.264 Specification](#).

Implementations **may** override any of these parameters according to the semantics defined in the [Video Encode Parameter Overrides](#) section before storing the resulting H.264 parameter sets into the video session parameters object. Applications need to use the [vkGetEncodedVideoSessionParametersKHR](#) command to determine whether any implementation overrides happened and to retrieve the encoded H.264 parameter sets in order to be able to produce a compliant H.264 video bitstream.

Such H.264 parameter set overrides **may** also have cascading effects on the implementation overrides applied to the encoded bitstream produced by video encode operations. If the implementation supports the `VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR` [video encode feedback query](#) flag, then the application **can** use such queries to retrieve feedback about whether any implementation overrides have been applied to the encoded bitstream.

When a [video session parameters](#) object is created with the codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, the [VkVideoSessionParametersCreateInfoKHR::pNext](#) chain **must** include a [VkVideoEncodeH264SessionParametersCreateInfoKHR](#) structure specifying the capacity and initial contents of the object.

The [VkVideoEncodeH264SessionParametersCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264SessionParametersCreateInfoKHR {
    VkStructureType              sType;
    const void*                  pNext;
    uint32_t                     maxStdSPSCount;
    uint32_t                     maxStdPPSCount;
    const VkVideoEncodeH264SessionParametersAddInfoKHR* pParametersAddInfo;
} VkVideoEncodeH264SessionParametersCreateInfoKHR;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxStdSPSCount` is the maximum number of [H.264 SPS](#) entries the created [VkVideoSessionParametersKHR](#) **can** contain.
- `maxStdPPSCount` is the maximum number of [H.264 PPS](#) entries the created [VkVideoSessionParametersKHR](#) **can** contain.

- `pParametersAddInfo` is `NULL` or a pointer to a [VkVideoEncodeH264SessionParametersAddInfoKHR](#) structure specifying H.264 parameters to add upon object creation.

Valid Usage (Implicit)

- `VOID-VkVideoEncodeH264SessionParametersCreateInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR`
- `VOID-VkVideoEncodeH264SessionParametersCreateInfoKHR-pParametersAddInfo-parameter`
If `pParametersAddInfo` is not `NULL`, `pParametersAddInfo` **must** be a valid pointer to a valid [VkVideoEncodeH264SessionParametersAddInfoKHR](#) structure

The `VkVideoEncodeH264SessionParametersAddInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264SessionParametersAddInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    uint32_t                 stdSPSCount;
    const StdVideoH264SequenceParameterSet* pStdSPSs;
    uint32_t                 stdPPSCount;
    const StdVideoH264PictureParameterSet* pStdPPSs;
} VkVideoEncodeH264SessionParametersAddInfoKHR;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdSPSCount` is the number of elements in the `pStdSPSs` array.
- `pStdSPSs` is a pointer to an array of `StdVideoH264SequenceParameterSet` structures describing the [H.264 SPS](#) entries to add.
- `stdPPSCount` is the number of elements in the `pStdPPSs` array.
- `pStdPPSs` is a pointer to an array of `StdVideoH264PictureParameterSet` structures describing the [H.264 PPS](#) entries to add.

This structure **can** be specified in the following places:

- In the `pParametersAddInfo` member of the [VkVideoEncodeH264SessionParametersCreateInfoKHR](#) structure specified in the `pNext` chain of [VkVideoSessionParametersCreateInfoKHR](#) used to create a [video session parameters](#) object. In this case, if the video codec operation the video session parameters object is created with is `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then it defines the set of initial parameters to add to the created object (see [Creating Video Session Parameters](#)).
- In the `pNext` chain of [VkVideoSessionParametersUpdateInfoKHR](#). In this case, if the video codec operation the [video session parameters](#) object to be updated was created with is `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, then it defines the set of parameters to add to it

(see [Updating Video Session Parameters](#)).

Valid Usage

- VUID-VkVideoEncodeH264SessionParametersAddInfoKHR-None-04837
The `seq_parameter_set_id` member of each `StdVideoH264SequenceParameterSet` structure specified in the elements of `pStdSPSs` **must** be unique within `pStdSPSs`
- VUID-VkVideoEncodeH264SessionParametersAddInfoKHR-None-04838
The pair constructed from the `seq_parameter_set_id` and `pic_parameter_set_id` members of each `StdVideoH264PictureParameterSet` structure specified in the elements of `pStdPPSs` **must** be unique within `pStdPPSs`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264SessionParametersAddInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR`
- VUID-VkVideoEncodeH264SessionParametersAddInfoKHR-pStdSPSs-parameter
If `stdSPSCount` is not `0`, and `pStdSPSs` is not `NULL`, `pStdSPSs` **must** be a valid pointer to an array of `stdSPSCount` `StdVideoH264SequenceParameterSet` values
- VUID-VkVideoEncodeH264SessionParametersAddInfoKHR-pStdPPSs-parameter
If `stdPPSCount` is not `0`, and `pStdPPSs` is not `NULL`, `pStdPPSs` **must** be a valid pointer to an array of `stdPPSCount` `StdVideoH264PictureParameterSet` values

The `VkVideoEncodeH264SessionParametersGetInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264SessionParametersGetInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           writeStdSPS;
    VkBool32           writeStdPPS;
    uint32_t           stdSPSId;
    uint32_t           stdPPSId;
} VkVideoEncodeH264SessionParametersGetInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `writeStdSPS` indicates whether the encoded [H.264 sequence parameter set](#) identified by `stdSPSId` is requested to be retrieved.
- `writeStdPPS` indicates whether the encoded [H.264 picture parameter set](#) identified by the pair constructed from `stdSPSId` and `stdPPSId` is requested to be retrieved.
- `stdSPSId` specifies the H.264 sequence parameter set ID used to identify the retrieved H.264 sequence and/or picture parameter set(s).

- `stdPPSId` specifies the H.264 picture parameter set ID used to identify the retrieved H.264 picture parameter set when `writeStdPPS` is `VK_TRUE`.

When this structure is specified in the `pNext` chain of the `VkVideoEncodeSessionParametersGetInfoKHR` structure passed to `vkGetEncodedVideoSessionParametersKHR`, the command will write encoded parameter data to the output buffer in the following order:

1. The H.264 sequence parameter set identified by `stdSPSId`, if `writeStdSPS` is `VK_TRUE`.
2. The H.264 picture parameter set identified by the pair constructed from `stdSPSId` and `stdPPSId`, if `writeStdPPS` is `VK_TRUE`.

Valid Usage

- VUID-VkVideoEncodeH264SessionParametersGetInfoKHR-writeStdSPS-08279
At least one of `writeStdSPS` and `writeStdPPS` must be `VK_TRUE`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264SessionParametersGetInfoKHR-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_GET_INFO_KHR`

The `VkVideoEncodeH264SessionParametersFeedbackInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264SessionParametersFeedbackInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           hasStdSPSOverrides;
    VkBool32           hasStdPPSOverrides;
} VkVideoEncodeH264SessionParametersFeedbackInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `hasStdSPSOverrides` indicates whether any of the parameters of the requested H.264 sequence parameter set, if one was requested via `VkVideoEncodeH264SessionParametersGetInfoKHR::writeStdSPS`, were overridden by the implementation.
- `hasStdPPSOverrides` indicates whether any of the parameters of the requested H.264 picture parameter set, if one was requested via `VkVideoEncodeH264SessionParametersGetInfoKHR::writeStdPPS`, were overridden by the implementation.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264SessionParametersFeedbackInfoKHR-sType-sType

<code>sType</code>	<code>must</code>	<code>be</code>
<code>VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_FEEDBACK_INFO_KHR</code>		

37.17.11. H.264 Encoding Parameters

The `VkVideoEncodeH264PictureInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264PictureInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    uint32_t                 naluSliceEntryCount;
    const VkVideoEncodeH264NaluSliceInfoKHR* pNaluSliceEntries;
    const StdVideoEncodeH264PictureInfo*    pStdPictureInfo;
    VkBool32                 generatePrefixNalu;
} VkVideoEncodeH264PictureInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `naluSliceEntryCount` is the number of elements in `pNaluSliceEntries`.
- `pNaluSliceEntries` is a pointer to an array of `naluSliceEntryCount` `VkVideoEncodeH264NaluSliceInfoKHR` structures specifying the parameters of the individual H.264 slices to encode for the input picture.
- `pStdPictureInfo` is a pointer to a `StdVideoEncodeH264PictureInfo` structure specifying [H.264 picture information](#).
- `generatePrefixNalu` controls whether prefix NALUs are generated before slice NALUs into the target bitstream, as defined in sections 7.3.2.12 and 7.4.2.12 of the [ITU-T H.264 Specification](#).

This structure is specified in the `pNext` chain of the `VkVideoEncodeInfoKHR` structure passed to `vkCmdEncodeVideoKHR` to specify the codec-specific picture information for an [H.264 encode operation](#).

Encode Input Picture Information

When this structure is specified in the `pNext` chain of the `VkVideoEncodeInfoKHR` structure passed to `vkCmdEncodeVideoKHR`, the information related to the [encode input picture](#) is defined as follows:

- The image subregion used is determined according to the [H.264 Encode Picture Data Access](#) section.
- The encode input picture is associated with the [H.264 picture information](#) provided in `pStdPictureInfo`.

Std Picture Information

The members of the `StdVideoEncodeH264PictureInfo` structure pointed to by `pStdPictureInfo` are interpreted as follows:

- `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
- `flags.IdrPicFlag` as defined in section 7.4.1 of the [ITU-T H.264 Specification](#);
- `flags.is_reference` as defined in section 3.136 of the [ITU-T H.264 Specification](#);
- `seq_parameter_set_id` and `pic_parameter_set_id` are used to identify the active parameter sets, as described below;
- `primary_pic_type` as defined in section 7.4.2 of the [ITU-T H.264 Specification](#);
- `PicOrderCnt` as defined in section 8.2 of the [ITU-T H.264 Specification](#);
- `temporal_id` as defined in section G.7.4.1.1 of the [ITU-T H.264 Specification](#);
- if `pRefLists` is not `NULL`, then it is a pointer to a `StdVideoEncodeH264ReferenceListsInfo` structure that is interpreted as follows:
 - `flags.reserved` is used only for padding purposes and is otherwise ignored;
 - `ref_pic_list_modification_flag_l0` and `ref_pic_list_modification_flag_l1` as defined in section 7.4.3.1 of the [ITU-T H.264 Specification](#);
 - `num_ref_idx_l0_active_minus1` and `num_ref_idx_l1_active_minus1` as defined in section 7.4.3 of the [ITU-T H.264 Specification](#);
 - `RefPicList0` and `RefPicList1` as defined in section 8.2.4 of the [ITU-T H.264 Specification](#) where each element of these arrays either identifies an [active reference picture](#) using its [DPB slot](#) index or contains the value `STD_VIDEO_H264_NO_REFERENCE_PICTURE` to indicate “no reference picture”;
 - if `refList0ModOpCount` is not zero, then `pRefList0ModOperations` is a pointer to an array of `refList0ModOpCount` number of `StdVideoEncodeH264RefListModEntry` structures specifying the modification parameters for the reference list L0 as defined in section 7.4.3.1 of the [ITU-T H.264 Specification](#);
 - if `refList1ModOpCount` is not zero, then `pRefList1ModOperations` is a pointer to an array of `refList1ModOpCount` number of `StdVideoEncodeH264RefListModEntry` structures specifying the modification parameters for the reference list L1 as defined in section 7.4.3.1 of the [ITU-T H.264 Specification](#);
 - if `refPicMarkingOpCount` is not zero, then `refPicMarkingOperations` is a pointer to an array of `refPicMarkingOpCount` number of `StdVideoEncodeH264RefPicMarkingEntry` structures specifying the reference picture marking parameters as defined in section 7.4.3.3 of the [ITU-T H.264 Specification](#);
- all other members are interpreted as defined in section 7.4.3 of the [ITU-T H.264 Specification](#).

Reference picture setup is controlled by the value of `StdVideoEncodeH264PictureInfo::flags.is_reference`. If it is set and a [reconstructed picture](#) is specified, then the latter is used as the target of picture reconstruction to [activate](#) the [DPB slot](#) specified in `pEncodeInfo->pSetupReferenceSlot->slotIndex`. If `StdVideoEncodeH264PictureInfo::flags.is_reference` is not set, but a [reconstructed picture](#) is specified, then the corresponding picture reference associated with the [DPB slot](#) is invalidated, as described in the [DPB Slot States](#) section.

Active Parameter Sets

The members of the `StdVideoEncodeH264PictureInfo` structure pointed to by `pStdPictureInfo` are

used to select the active parameter sets to use from the bound video session parameters object, as follows:

- The *active SPS* is the [SPS](#) identified by the key specified in `StdVideoEncodeH264PictureInfo::seq_parameter_set_id`.
- The *active PPS* is the [PPS](#) identified by the key specified by the pair constructed from `StdVideoEncodeH264PictureInfo::seq_parameter_set_id` and `StdVideoEncodeH264PictureInfo::pic_parameter_set_id`.

H.264 encoding uses *explicit weighted sample prediction* for a slice, as defined in section 8.4.2.3 of the [ITU-T H.264 Specification](#), if any of the following conditions are true for the active [PPS](#) and the `pStdSliceHeader` member of the corresponding element of `pNaluSliceEntries`:

- `pStdSliceHeader->slice_type` is `STD_VIDEO_H264_SLICE_TYPE_P` and `weighted_pred_flag` is enabled in the active PPS.
- `pStdSliceHeader->slice_type` is `STD_VIDEO_H264_SLICE_TYPE_B` and `weighted_bipred_idc` in the active PPS equals `STD_VIDEO_H264_WEIGHTED_BIPRED_IDC_EXPLICIT`.

Valid Usage

- VUID-VkVideoEncodeH264PictureInfoKHR-naluSliceEntryCount-08301
`naluSliceEntryCount` **must** be between 1 and `VkVideoEncodeH264CapabilitiesKHR::maxSliceCount`, inclusive, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile
- VUID-VkVideoEncodeH264PictureInfoKHR-flags-08304
If `VkVideoEncodeH264CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H264_CAPABILITY_GENERATE_PREFIX_NALU_BIT_KHR`, then `generatePrefixNalu` **must** be `VK_FALSE`
- VUID-VkVideoEncodeH264PictureInfoKHR-flags-08314
If `VkVideoEncodeH264CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H264_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR` and the slice corresponding to any element of `pNaluSliceEntries` uses *explicit weighted sample prediction*, then `VkVideoEncodeH264NaluSliceInfoKHR::pStdSliceHeader->pWeightTable` **must** not be `NULL` for that element of `pNaluSliceEntries`
- VUID-VkVideoEncodeH264PictureInfoKHR-flags-08315
If `VkVideoEncodeH264CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H264_CAPABILITY_DIFFERENT_SLICE_TYPE_BIT_KHR`, then `VkVideoEncodeH264NaluSliceInfoKHR::pStdSliceHeader->slice_type` **must** be identical for all elements of `pNaluSliceEntries`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264PictureInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PICTURE_INFO_KHR`
- VUID-VkVideoEncodeH264PictureInfoKHR-pNaluSliceEntries-parameter
`pNaluSliceEntries` **must** be a valid pointer to an array of `naluSliceEntryCount` valid `VkVideoEncodeH264NaluSliceInfoKHR` structures
- VUID-VkVideoEncodeH264PictureInfoKHR-pStdPictureInfo-parameter
`pStdPictureInfo` **must** be a valid pointer to a valid `StdVideoEncodeH264PictureInfo` value
- VUID-VkVideoEncodeH264PictureInfoKHR-naluSliceEntryCount-arraylength
`naluSliceEntryCount` **must** be greater than 0

The `VkVideoEncodeH264NaluSliceInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264NaluSliceInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    int32_t            constantQp;
    const StdVideoEncodeH264SliceHeader* pStdSliceHeader;
} VkVideoEncodeH264NaluSliceInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `constantQp` is the QP to use for the slice if the current `rate control mode` configured for the video session is `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`.
- `pStdSliceHeader` is a pointer to a `StdVideoEncodeH264SliceHeader` structure specifying [H.264 slice header parameters](#) for the slice.

Std Slice Header Parameters

The members of the `StdVideoEncodeH264SliceHeader` structure pointed to by `pStdSliceHeader` are interpreted as follows:

- `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
- if `pWeightTable` is not `NULL`, then it is a pointer to a `StdVideoEncodeH264WeightTable` that is interpreted as follows:
 - `flags.reserved` is used only for padding purposes and is otherwise ignored;
 - all other members of `StdVideoEncodeH264WeightTable` are interpreted as defined in section 7.4.3.2 of the [ITU-T H.264 Specification](#);
- all other members are interpreted as defined in section 7.4.3 of the [ITU-T H.264 Specification](#).

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264NaluSliceInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_NALU_SLICE_INFO_KHR`
- VUID-VkVideoEncodeH264NaluSliceInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkVideoEncodeH264NaluSliceInfoKHR-pStdSliceHeader-parameter
`pStdSliceHeader` **must** be a valid pointer to a valid `StdVideoEncodeH264SliceHeader` value

The `VkVideoEncodeH264DpbSlotInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264DpbSlotInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    const StdVideoEncodeH264ReferenceInfo* pStdReferenceInfo;
} VkVideoEncodeH264DpbSlotInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdReferenceInfo` is a pointer to a `StdVideoEncodeH264ReferenceInfo` structure specifying [H.264 reference information](#).

This structure is specified in the `pNext` chain of `VkVideoEncodeInfoKHR::pSetupReferenceSlot`, if not `NULL`, and the `pNext` chain of the elements of `VkVideoEncodeInfoKHR::pReferenceSlots` to specify the codec-specific reference picture information for an [H.264 encode operation](#).

Active Reference Picture Information

When this structure is specified in the `pNext` chain of the elements of `VkVideoEncodeInfoKHR::pReferenceSlots`, one element is added to the list of [active reference pictures](#) used by the video encode operation for each element of `VkVideoEncodeInfoKHR::pReferenceSlots` as follows:

- The image subregion used is determined according to the [H.264 Encode Picture Data Access](#) section.
- The reference picture is associated with the [DPB slot](#) index specified in the `slotIndex` member of the corresponding element of `VkVideoEncodeInfoKHR::pReferenceSlots`.
- The reference picture is associated with the [H.264 reference information](#) provided in `pStdReferenceInfo`.

Reconstructed Picture Information

When this structure is specified in the `pNext` chain of `VkVideoEncodeInfoKHR::pSetupReferenceSlot`, the information related to the [reconstructed picture](#) is defined as follows:

- The image subregion used is determined according to the [H.264 Encode Picture Data Access](#)

section.

- If [reference picture setup](#) is requested, then the reconstructed picture is used to [activate](#) the [DPB slot](#) with the index specified in [VkVideoEncodeInfoKHR::pSetupReferenceSlot->slotIndex](#).
- The reconstructed picture is associated with the [H.264 reference information](#) provided in [pStdReferenceInfo](#).

Std Reference Information

The members of the [StdVideoEncodeH264ReferenceInfo](#) structure pointed to by [pStdReferenceInfo](#) are interpreted as follows:

- [flags.reserved](#) is used only for padding purposes and is otherwise ignored;
- [flags.used_for_long_term_reference](#) is used to indicate whether the picture is marked as “used for long-term reference” as defined in section 8.2.5.1 of the [ITU-T H.264 Specification](#);
- [primary_pic_type](#) as defined in section 7.4.2 of the [ITU-T H.264 Specification](#);
- [long_term_pic_num](#) and [long_term_frame_idx](#) as defined in section 7.4.3 of the [ITU-T H.264 Specification](#);
- [temporal_id](#) as defined in section G.7.4.1.1 of the [ITU-T H.264 Specification](#);
- all other members are interpreted as defined in section 8.2 of the [ITU-T H.264 Specification](#).

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264DpbSlotInfoKHR-sType-sType [sType](#) **must** be [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_DPB_SLOT_INFO_KHR](#)
- VUID-VkVideoEncodeH264DpbSlotInfoKHR-pStdReferenceInfo-parameter [pStdReferenceInfo](#) **must** be a valid pointer to a valid [StdVideoEncodeH264ReferenceInfo](#) value

37.17.12. H.264 Encode Rate Control

Group of Pictures

In case of H.264 encoding it is common practice to follow a regular pattern of different picture types in display order when encoding subsequent frames. This pattern is referred to as the *group of pictures* (GOP).

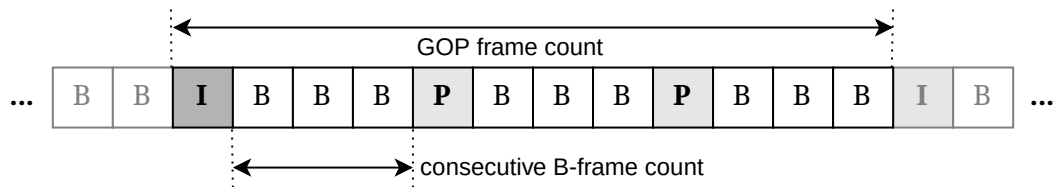
A regular GOP is defined by the following parameters:

- The number of frames in the GOP;
- The number of consecutive B frames between I and/or P frames in display order.

GOPs are further classified as *open* and *closed* GOPs.

Frame types in an open GOP follow each other in display order according to the following algorithm:

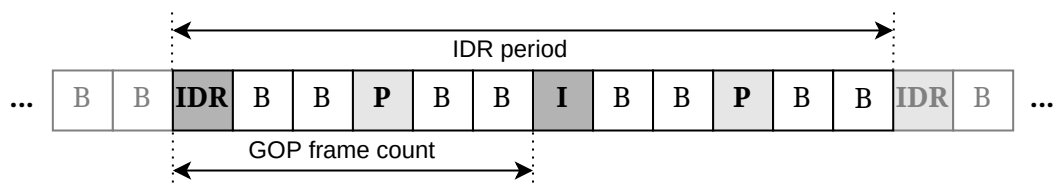
1. The first frame is always an I frame.
2. This is followed by a number of consecutive B frames, as defined above.
3. If the number of frames in the GOP is not reached yet, then the next frame is a P frame and the algorithm continues from step 2.



Text is not SVG - cannot display

Figure 26. H.264 open GOP

In case of a closed GOP, an **IDR frame** is used at a certain period.



Text is not SVG - cannot display

Figure 27. H.264 closed GOP

It is also typical for H.264 encoding to use specific reference picture usage patterns across the frames of the GOP. The two most common reference patterns used are as follows:

Flat Reference Pattern

- Each P frame uses the last non-B frame, in display order, as reference.
- Each B frame uses the last non-B frame, in display order, as its forward reference, and uses the next non-B frame, in display order, as its backward reference.

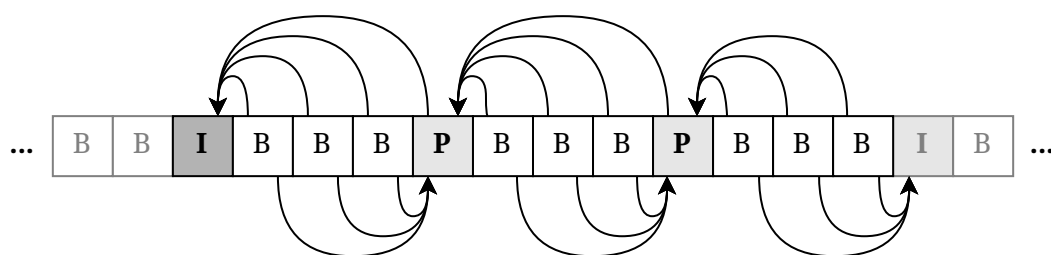


Figure 28. H.264 flat reference pattern

Dyadic Reference Pattern

- Each P frame uses the last non-B frame, in display order, as reference.
- The following algorithm is applied to the sequence of consecutive B frames between I and/or P frames in display order:
 1. The B frame in the middle of this sequence uses the frame preceding the sequence as its forward reference, and uses the frame following the sequence as its backward reference.
 2. The algorithm is executed recursively for the following frame sequences:
 - The B frames of the original sequence preceding the frame in the middle, if any.
 - The B frames of the original sequence following the frame in the middle, if any.

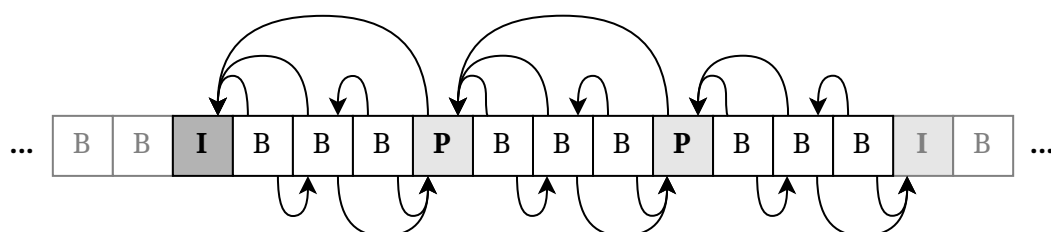


Figure 29. H.264 dyadic reference pattern

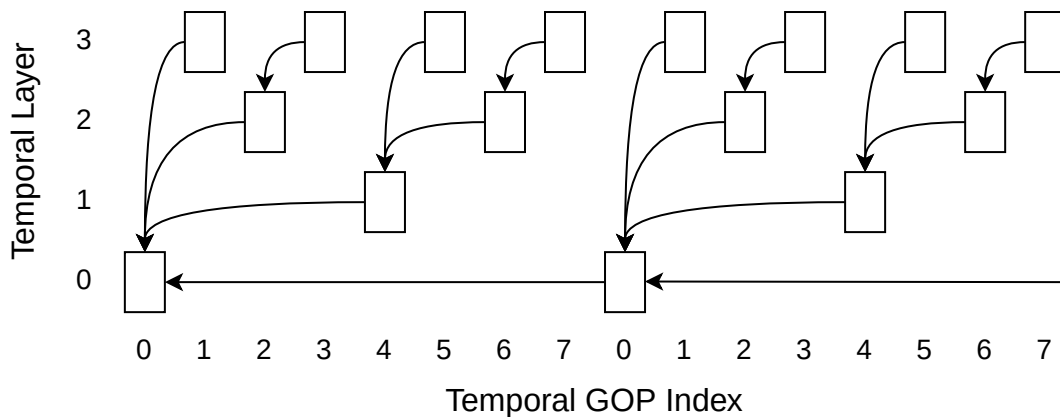
The application **can** provide guidance to the implementation's rate control algorithm about the structure of the GOP used by the application. Any such guidance about the GOP and its structure does not mandate that specific GOP structure to be used by the application, as the picture type of individual encoded pictures is still application-controlled, however, any deviation from the provided guidance **may** result in undesired rate control behavior including, but not limited, to the implementation not being able to conform to the expected average or target bitrates, or other rate control parameters specified by the application.

When an H.264 encode session is used to encode multiple temporal layers, it is also common practice to follow a regular pattern for the H.264 temporal ID for the encoded pictures in display order when encoding subsequent frames. This pattern is referred to as the *temporal GOP*. The most

common temporal layer pattern used is as follows:

Dyadic Temporal Layer Pattern

- The number of frames in the temporal GOP is 2^{n-1} , where n is the number of temporal layers.
- The i^{th} frame in the temporal GOP uses temporal ID t , if and only if the index of the least significant bit set in i equals $n-t-1$, except for the first frame, which is the only frame in the temporal GOP using temporal ID zero.
- The i^{th} frame in the temporal GOP uses the r^{th} frame as reference, where r is calculated from i by clearing the least significant bit set in it, except for the first frame in the temporal GOP, which uses the first frame of the previous temporal GOP, if any, as reference.



Text is not SVG - cannot display

Figure 30. H.264 dyadic temporal layer pattern

NOTE

Multi-layer rate control and multi-layer coding are typically used for streaming cases where low latency is expected, hence B pictures with backward prediction are usually not used.

The `VkVideoEncodeH264RateControlInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264RateControlInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkVideoEncodeH264RateControlFlagsKHR flags;
    uint32_t            gopFrameCount;
    uint32_t            idrPeriod;
    uint32_t            consecutiveBFrameCount;
    uint32_t            temporalLayerCount;
} VkVideoEncodeH264RateControlInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkVideoEncodeH264RateControlFlagBitsKHR` specifying H.264 rate control flags.
- `gopFrameCount` is the number of frames within a [group of pictures \(GOP\)](#) intended to be used by the application. If it is 0, the rate control algorithm **may** assume an implementation-dependent GOP length. If it is `UINT32_MAX`, the GOP length is treated as infinite.
- `idrPeriod` is the interval, in terms of number of frames, between two [IDR frames](#) (see [IDR period](#)). If it is 0, the rate control algorithm **may** assume an implementation-dependent IDR period. If it is `UINT32_MAX`, the IDR period is treated as infinite.
- `consecutiveBFrameCount` is the number of consecutive B frames between I and/or P frames within the [GOP](#).
- `temporalLayerCount` specifies the number of H.264 temporal layers that the application intends to use.

When an instance of this structure is included in the `pNext` chain of the `VkVideoCodingControlInfoKHR` structure passed to the `vkCmdControlVideoCodingKHR` command, and `VkVideoCodingControlInfoKHR::flags` includes `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR`, the parameters in this structure are used as guidance for the implementation's rate control algorithm (see [Video Coding Control](#)).

If `flags` includes `VK_VIDEO_ENCODE_H264_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR`, then the rate control state is reset to an initial state to meet HRD compliance requirements. Otherwise the new rate control state **may** be applied without a reset depending on the implementation and the specified rate control parameters.

NOTE

It would be possible to infer the picture type to be used when encoding a frame, on the basis of the values provided for `consecutiveBFrameCount`, `idrPeriod`, and `gopFrameCount`, but this inferred picture type will not be used by implementations to override the picture type provided to the video encode operation.

Valid Usage

- VUID-VkVideoEncodeH264RateControlInfoKHR-flags-08280
If `VkVideoEncodeH264CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H264_CAPABILITY_HRD_COMPLIANCE_BIT_KHR`, then `flags` **must** not contain `VK_VIDEO_ENCODE_H264_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR`
- VUID-VkVideoEncodeH264RateControlInfoKHR-flags-08281
If `flags` contains `VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR` or `VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR`, then it **must** also contain `VK_VIDEO_ENCODE_H264_RATE_CONTROL_REGULAR_GOP_BIT_KHR`
- VUID-VkVideoEncodeH264RateControlInfoKHR-flags-08282
If `flags` contains `VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR`, then it **must** not also contain `VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR`

- VUID-VkVideoEncodeH264RateControlInfoKHR-flags-08283
If `flags` contains `VK_VIDEO_ENCODE_H264_RATE_CONTROL_REGULAR_GOP_BIT_KHR`, then `gopFrameCount` **must** be greater than 0
- VUID-VkVideoEncodeH264RateControlInfoKHR-idrPeriod-08284
If `idrPeriod` is not 0, then it **must** be greater than or equal to `gopFrameCount`
- VUID-VkVideoEncodeH264RateControlInfoKHR-consecutiveBFrameCount-08285
If `consecutiveBFrameCount` is not 0, then it **must** be less than `gopFrameCount`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264RateControlInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_INFO_KHR`
- VUID-VkVideoEncodeH264RateControlInfoKHR-flags-parameter
`flags` **must** be a valid combination of `VkVideoEncodeH264RateControlFlagBitsKHR` values

Bits which **can** be set in `VkVideoEncodeH264RateControlInfoKHR::flags`, specifying H.264 rate control flags, are:

```
// Provided by VK_KHR_video_encode_h264
typedef enum VkVideoEncodeH264RateControlFlagBitsKHR {
    VK_VIDEO_ENCODE_H264_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H264_RATE_CONTROL_REGULAR_GOP_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H264_RATE_CONTROL_TEMPORAL_LAYER_PATTERN_DYADIC_BIT_KHR =
0x00000010,
} VkVideoEncodeH264RateControlFlagBitsKHR;
```

- `VK_VIDEO_ENCODE_H264_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR` specifies that rate control **should** attempt to produce an HRD compliant bitstream, as defined in annex C of the [ITU-T H.264 Specification](#).
- `VK_VIDEO_ENCODE_H264_RATE_CONTROL_REGULAR_GOP_BIT_KHR` specifies that the application intends to use a [regular GOP structure](#) according to the parameters specified in the `gopFrameCount`, `idrPeriod`, and `consecutiveBFrameCount` members of the `VkVideoEncodeH264RateControlInfoKHR` structure.
- `VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR` specifies that the application intends to follow a [flat reference pattern](#) in the GOP.
- `VK_VIDEO_ENCODE_H264_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR` specifies that the application intends to follow a [dyadic reference pattern](#) in the GOP.
- `VK_VIDEO_ENCODE_H264_RATE_CONTROL_TEMPORAL_LAYER_PATTERN_DYADIC_BIT_KHR` specifies that the application intends to follow a [dyadic temporal layer pattern](#).

```
// Provided by VK_KHR_video_encode_h264
typedef VkFlags VkVideoEncodeH264RateControlFlagsKHR;
```

`VkVideoEncodeH264RateControlFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoEncodeH264RateControlFlagBitsKHR`.

Rate Control Layers

The `VkVideoEncodeH264RateControlLayerInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264RateControlLayerInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    VkBool32                 useMinQp;
    VkVideoEncodeH264QpKHR   minQp;
    VkBool32                 useMaxQp;
    VkVideoEncodeH264QpKHR   maxQp;
    VkBool32                 useMaxFrameSize;
    VkVideoEncodeH264FrameSizeKHR maxFrameSize;
} VkVideoEncodeH264RateControlLayerInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `useMinQp` indicates whether the QP values determined by rate control will be clamped to the lower bounds on the QP values specified in `minQp`.
- `minQp` specifies the lower bounds on the QP values, for each picture type, that the implementation's rate control algorithm will use when `useMinQp` is `VK_TRUE`.
- `useMaxQp` indicates whether the QP values determined by rate control will be clamped to the upper bounds on the QP values specified in `maxQp`.
- `maxQp` specifies the upper bounds on the QP values, for each picture type, that the implementation's rate control algorithm will use when `useMaxQp` is `VK_TRUE`.
- `useMaxFrameSize` indicates whether the implementation's rate control algorithm **should** use the values specified in `maxFrameSize` as the upper bounds on the encoded frame size for each picture type.
- `maxFrameSize` specifies the upper bounds on the encoded frame size, for each picture type, when `useMaxFrameSize` is `VK_TRUE`.

When used, the values in `minQp` and `maxQp` guarantee that the effective QP values used by the implementation will respect those lower and upper bounds, respectively. However, limiting the range of QP values that the implementation is able to use will also limit the capabilities of the implementation's rate control algorithm to comply to other constraints. In particular, the implementation **may** not be able to comply to the following:

- The average and/or peak `bitrate` values to be used for the encoded bitstream specified in the

`averageBitrate` and `maxBitrate` members of the `VkVideoEncodeRateControlLayerInfoKHR` structure.

- The upper bounds on the encoded frame size, for each picture type, specified in the `maxFrameSize` member of `VkVideoEncodeH264RateControlLayerInfoKHR`.

NOTE

In general, applications need to configure rate control parameters appropriately in order to be able to get the desired rate control behavior, as described in the [Video Encode Rate Control](#) section.

When an instance of this structure is included in the `pNext` chain of a `VkVideoEncodeRateControlLayerInfoKHR` structure specified in one of the elements of the `players` array member of the `VkVideoEncodeRateControlInfoKHR` structure passed to the `vkCmdControlVideoCodingKHR` command, `VkVideoCodingControlInfoKHR::flags` includes `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR`, and the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, it specifies the H.264-specific rate control parameters of the rate control layer corresponding to that element of `players`.

Valid Usage

- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-useMinQp-08286

If `useMinQp` is `VK_TRUE`, then the `qpI`, `qpP`, and `qpB` members of `minQp` **must** all be between `VkVideoEncodeH264CapabilitiesKHR::minQp` and `VkVideoEncodeH264CapabilitiesKHR::maxQp`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile

- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-useMaxQp-08287

If `useMaxQp` is `VK_TRUE`, then the `qpI`, `qpP`, and `qpB` members of `maxQp` **must** all be between `VkVideoEncodeH264CapabilitiesKHR::minQp` and `VkVideoEncodeH264CapabilitiesKHR::maxQp`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile

- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-useMinQp-08288

If `useMinQp` is `VK_TRUE` and `VkVideoEncodeH264CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H264_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR`, then the `qpI`, `qpP`, and `qpB` members of `minQp` **must** all specify the same value

- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-useMaxQp-08289

If `useMaxQp` is `VK_TRUE` and `VkVideoEncodeH264CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H264_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR`, then the `qpI`, `qpP`, and `qpB` members of `maxQp` **must** all specify the same value

- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-useMinQp-08374

If `useMinQp` and `useMaxQp` are both `VK_TRUE`, then the `qpI`, `qpP`, and `qpB` members of `minQp` **must** all be less than or equal to the respective members of `maxQp`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-sType-sType
sType must be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_LAYER_INFO_KHR`
- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-minQp-parameter
minQp must be a valid `VkVideoEncodeH264QpKHR` structure
- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-maxQp-parameter
maxQp must be a valid `VkVideoEncodeH264QpKHR` structure
- VUID-VkVideoEncodeH264RateControlLayerInfoKHR-maxFrameSize-parameter
maxFrameSize must be a valid `VkVideoEncodeH264FrameSizeKHR` structure

The `VkVideoEncodeH264QpKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264QpKHR {
    int32_t    qpI;
    int32_t    qpP;
    int32_t    qpB;
} VkVideoEncodeH264QpKHR;
```

- **qpI** is the QP to be used for **I pictures**.
- **qpP** is the QP to be used for **P pictures**.
- **qpB** is the QP to be used for **B pictures**.

The `VkVideoEncodeH264FrameSizeKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264FrameSizeKHR {
    uint32_t    frameISize;
    uint32_t    framePSize;
    uint32_t    frameBSize;
} VkVideoEncodeH264FrameSizeKHR;
```

- **frameISize** is the size in bytes to be used for **I pictures**.
- **framePSize** is the size in bytes to be used for **P pictures**.
- **frameBSize** is the size in bytes to be used for **B pictures**.

GOP Remaining Frames

Besides session level rate control configuration, the application **can** specify the number of frames per frame type remaining in the **group of pictures (GOP)**.

The `VkVideoEncodeH264GopRemainingFrameInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h264
typedef struct VkVideoEncodeH264GopRemainingFrameInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           useGopRemainingFrames;
    uint32_t            gopRemainingI;
    uint32_t            gopRemainingP;
    uint32_t            gopRemainingB;
} VkVideoEncodeH264GopRemainingFrameInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `useGopRemainingFrames` indicates whether the implementation's rate control algorithm **should** use the values specified in `gopRemainingI`, `gopRemainingP`, and `gopRemainingB`. If `useGopRemainingFrames` is `VK_FALSE`, then the values of `gopRemainingI`, `gopRemainingP`, and `gopRemainingB` are ignored.
- `gopRemainingI` specifies the number of `I frames` the implementation's rate control algorithm **should** assume to be remaining in the `GOP` prior to executing the video encode operation.
- `gopRemainingP` specifies the number of `P frames` the implementation's rate control algorithm **should** assume to be remaining in the `GOP` prior to executing the video encode operation.
- `gopRemainingB` specifies the number of `B frames` the implementation's rate control algorithm **should** assume to be remaining in the `GOP` prior to executing the video encode operation.

Setting `useGopRemainingFrames` to `VK_TRUE` and including this structure in the `pNext` chain of `VkVideoBeginCodingInfoKHR` is only mandatory if the `VkVideoEncodeH264CapabilitiesKHR::requiresGopRemainingFrames` reported for the used `video profile` is `VK_TRUE`. However, implementations **may** use these remaining frame counts, when specified, even when it is not required. In particular, when the application does not use a `regular GOP structure`, these values **may** provide additional guidance for the implementation's rate control algorithm.

The `VkVideoEncodeH264CapabilitiesKHR::prefersGopRemainingFrames` capability is also used to indicate that the implementation's rate control algorithm **may** operate more accurately if the application specifies the remaining frame counts using this structure.

As with other rate control guidance values, if the effective order and number of frames encoded by the application are not in line with the remaining frame counts specified in this structure at any given point, then the behavior of the implementation's rate control algorithm **may** deviate from the one expected by the application.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH264GopRemainingFrameInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_GOP_REMAINING_FRAME_INFO_KHR`

37.17.13. H.264 Encode Requirements

This section described the **required** H.264 encoding capabilities for physical devices that have at least one queue family that supports the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR`, as returned by `vkGetPhysicalDeviceQueueFamilyProperties2KHR` in `VkQueueFamilyVideoPropertiesKHR::videoCodecOperations`.

Table 52. Required *Video Std Header Versions*

Video Std Header Name	Version
<code>vulkan_video_codec_h264std_encode</code>	1.0.0

Table 53. Required Video Capabilities

Video Capability	Requirement	Requirement Type ¹
VkVideoCapabilitiesKHR		
<code>flags</code>	-	min
<code>minBitstreamBufferOffsetAlignment</code>	4096	max
<code>minBitstreamBufferSizeAlignment</code>	4096	max
<code>pictureAccessGranularity</code>	(64,64)	max
<code>minCodedExtent</code>	-	max
<code>maxCodedExtent</code>	-	min
<code>maxDpbSlots</code>	0	min
<code>maxActiveReferencePictures</code>	0	min
VkVideoEncodeCapabilitiesKHR		
<code>flags</code>	-	min
<code>rateControlModes</code>	-	min
<code>maxBitrate</code>	64000	min
<code>maxQualityLevels</code>	1	min
<code>encodeInputPictureGranularity</code>	(64,64)	max
<code>supportedEncodeFeedbackFlags</code>	<code>VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR</code> <code>VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR</code>	min
VkVideoEncodeH264CapabilitiesKHR		
<code>flags</code>	-	min
<code>maxLevelIdc</code>	<code>STD_VIDEO_H264_LEVEL_IDC_1_0</code>	min

Video Capability	Requirement	Requirement Type ¹
maxSliceCount	1	min
maxPPictureL0ReferenceCount	0	min
maxBPictureL0ReferenceCount	0	min
maxL1ReferenceCount	0	min
maxTemporalLayerCount	1	min
expectDyadicTemporalLayerPattern	-	implementation-dependent
minQp	-	max
maxQp	-	min
prefersGopRemainingFrames	-	implementation-dependent
requiresGopRemainingFrames	-	implementation-dependent
stdSyntaxFlags	-	min

1

The **Requirement Type** column specifies the requirement is either the minimum value all implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

37.18. H.265 Encode Operations

Video encode operations using an [H.265 encode profile](#) **can** be used to encode elementary video stream sequences compliant to the [ITU-T H.265 Specification](#).

NOTE	Refer to the Preamble for information on how the Khronos Intellectual Property Rights Policy relates to normative references to external materials not created by Khronos.
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This process is performed according to the [video encode operation steps](#) with the codec-specific semantics defined in section 8 of the [ITU-T H.265 Specification](#) as follows:

- Syntax elements, derived values, and other parameters are applied from the following structures:
 - The `StdVideoH265VideoParameterSet` structure corresponding to the [active VPS](#) specifying the [H.265 video parameter set](#).
 - The `StdVideoH265SequenceParameterSet` structure corresponding to the [active SPS](#) specifying the [H.265 sequence parameter set](#).
 - The `StdVideoH265PictureParameterSet` structure corresponding to the [active PPS](#) specifying

the [H.265 picture parameter set](#).

- The `StdVideoEncodeH265PictureInfo` structure specifying the [H.265 picture information](#).
- The `StdVideoEncodeH265SliceSegmentHeader` structures specifying the [H.265 slice segment header parameters](#) for each encoded H.265 slice segment.
- The `StdVideoEncodeH265ReferenceInfo` structures specifying the [H.265 reference information](#) corresponding to the optional [reconstructed picture](#) and any [active reference pictures](#).
- The encoded bitstream data is written to the destination video bitstream buffer range as defined in the [H.265 Encode Bitstream Data Access](#) section.
- Picture data in the [video picture resources](#) corresponding to the used [encode input picture](#), [active reference pictures](#), and optional [reconstructed picture](#) is accessed as defined in the [H.265 Encode Picture Data Access](#) section.
- The decision on [reference picture setup](#) is made according to the parameters specified in the [H.265 picture information](#).

If the parameters adhere to the syntactic and semantic requirements defined in the corresponding sections of the [ITU-T H.265 Specification](#), as described above, and the [DPB slots](#) associated with the [active reference pictures](#) all refer to [valid picture references](#), then the video encode operation will complete successfully. Otherwise, the video encode operation **may** complete [unsuccessfully](#).

37.18.1. H.265 Encode Parameter Overrides

Implementations **may** override, unless otherwise specified, any of the H.265 encode parameters specified in the following Video Std structures:

- `StdVideoH265VideoParameterSet`
- `StdVideoH265SequenceParameterSet`
- `StdVideoH265PictureParameterSet`
- `StdVideoEncodeH265PictureInfo`
- `StdVideoEncodeH265SliceSegmentHeader`
- `StdVideoEncodeH265ReferenceInfo`

All such H.265 encode parameter overrides **must** fulfill the conditions defined in the [Video Encode Parameter Overrides](#) section.

In addition, implementations **must** not override any of the following H.265 encode parameters:

- `StdVideoEncodeH265PictureInfo::pic_type`
- `StdVideoEncodeH265SliceSegmentHeader::slice_type`

In case of H.265 encode parameters stored in [video session parameters](#) objects, applications need to use the `vkGetEncodedVideoSessionParametersKHR` command to determine whether any implementation overrides happened. If the query indicates that implementation overrides were applied, then the application needs to retrieve and use the encoded H.265 parameter sets in the bitstream in order to be able to produce a compliant H.265 video bitstream using the H.265 encode parameters stored in the video session parameters object.

In case of any H.265 encode parameters stored in the encoded bitstream produced by video encode operations, if the implementation supports the `VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR` video encode feedback query flag, the application **can** use such queries to retrieve feedback about whether any implementation overrides have been applied to those H.265 encode parameters.

37.18.2. H.265 Encode Bitstream Data Access

Each video encode operation writes one or more VCL NAL units comprising of slice segment headers and data of the encoded picture, in the format defined in sections 7.3.6 and 7.3.8, according to the semantics defined in sections 7.4.7 and 7.4.9 of the [ITU-T H.265 Specification](#), respectively. The number of VCL NAL units written is specified by `VkVideoEncodeH265PictureInfoKHR::naluSliceSegmentEntryCount`.

37.18.3. H.265 Encode Picture Data Access

Accesses to image data within a video picture resource happen at the granularity indicated by `VkVideoCapabilitiesKHR::pictureAccessGranularity`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile. Accordingly, the complete image subregion of a [encode input picture](#), [reference picture](#), or [reconstructed picture](#) accessed by video coding operations using an [H.265 encode profile](#) is defined as the set of texels within the coordinate range:

$([0, \text{endX}), [0, \text{endY}))$

Where:

- `endX` equals `codedExtent.width` rounded up to the nearest integer multiple of `pictureAccessGranularity.width` and clamped to the width of the image subresource [referred](#) to by the corresponding `VkVideoPictureResourceInfoKHR` structure;
- `endY` equals `codedExtent.height` rounded up to the nearest integer multiple of `pictureAccessGranularity.height` and clamped to the height of the image subresource [referred](#) to by the corresponding `VkVideoPictureResourceInfoKHR` structure;

Where `codedExtent` is the member of the `VkVideoPictureResourceInfoKHR` structure corresponding to the picture.

In case of video encode operations using an [H.265 encode profile](#), any access to a picture at the coordinates (x, y) , as defined by the [ITU-T H.265 Specification](#), is an access to the image subresource [referred](#) to by the corresponding `VkVideoPictureResourceInfoKHR` structure at the texel coordinates (x, y) .

Implementations **may** choose not to access some or all texels within particular [reference pictures](#) available to a video encode operation (e.g. due to [video encode parameter overrides](#) restricting the effective set of used reference pictures, or if the encoding algorithm chooses not to use certain subregions of the reference picture data for sample prediction).

37.18.4. H.265 Frame, Picture, Slice Segments, and Tiles

H.265 pictures consist of one or more slices, slice segments, and tiles, as defined in section 6.3.1 of the [ITU-T H.265 Specification](#).

For the purposes of this specification, the H.265 slice segments comprising a picture are referred to as the *picture partitions* of the picture.

Video encode operations using an [H.265 encode profile](#) **can** encode slice segments of different types, as defined in section 7.4.7.1 of the [ITU-T H.265 Specification](#), by specifying the corresponding enumeration constant value in `StdVideoEncodeH265SliceSegmentHeader::slice_type` in the [H.265 slice segment header parameters](#) from the Video Std enumeration type `StdVideoH265SliceType`:

- `STD_VIDEO_H265_SLICE_TYPE_B` indicates that the slice segment is part of a *B slice* as defined in section 3.12 of the [ITU-T H.265 Specification](#).
- `STD_VIDEO_H265_SLICE_TYPE_P` indicates that the slice segment is part of a *P slice* as defined in section 3.111 of the [ITU-T H.265 Specification](#).
- `STD_VIDEO_H265_SLICE_TYPE_I` indicates that the slice segment is part of an *I slice* as defined in section 3.74 of the [ITU-T H.265 Specification](#).

Pictures constructed from such slice segments **can** be of different types, as defined in section 7.4.3.5 of the [ITU-T H.265 Specification](#). Video encode operations using an [H.265 encode profile](#) **can** encode pictures of a specific type by specifying the corresponding enumeration constant value in `StdVideoEncodeH265PictureInfo::pic_type` in the [H.265 picture information](#) from the Video Std enumeration type `StdVideoH265PictureType`:

- `STD_VIDEO_H265_PICTURE_TYPE_P` indicates that the picture is a *P picture*. A frame consisting of a P picture is also referred to as a *P frame*.
- `STD_VIDEO_H265_PICTURE_TYPE_B` indicates that the picture is a *B picture*. A frame consisting of a B picture is also referred to as a *B frame*.
- `STD_VIDEO_H265_PICTURE_TYPE_I` indicates that the picture is an *I picture*. A frame consisting of an I picture is also referred to as an *I frame*.
- `STD_VIDEO_H265_PICTURE_TYPE_IDR` indicates that the picture is a special type of I picture called an *IDR picture* as defined in section 3.67 of the [ITU-T H.265 Specification](#). A frame consisting of an IDR picture is also referred to as an *IDR frame*.

37.18.5. H.265 Coding Blocks

H.265 encode supports two types of coding blocks:

- *Coding tree unit*, as defined in section 3.35 of the [ITU-T H.265 Specification](#).
- *Coding unit*, as defined in section 3.36 of the [ITU-T H.265 Specification](#).

37.18.6. H.265 Encode Profile

A video profile supporting H.265 video encode operations is specified by setting `VkVideoProfileInfoKHR::videoCodecOperation` to `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` and

adding a `VkVideoEncodeH265ProfileInfoKHR` structure to the `VkVideoProfileInfoKHR::pNext` chain.

The `VkVideoEncodeH265ProfileInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265ProfileInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    StdVideoH265ProfileIdc stdProfileIdc;
} VkVideoEncodeH265ProfileInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdProfileIdc` is a `StdVideoH265ProfileIdc` value specifying the H.265 codec profile IDC, as defined in section A.3 of the [ITU-T H.265 Specification](#).

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265ProfileInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PROFILE_INFO_KHR`

37.18.7. H.265 Encode Capabilities

When calling `vkGetPhysicalDeviceVideoCapabilitiesKHR` to query the capabilities for an [H.265 encode profile](#), the `VkVideoCapabilitiesKHR::pNext` chain **must** include a `VkVideoEncodeH265CapabilitiesKHR` structure that will be filled with the profile-specific capabilities.

The `VkVideoEncodeH265CapabilitiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265CapabilitiesKHR {
    VkStructureType           sType;
    void*                     pNext;
    VkVideoEncodeH265CapabilityFlagsKHR flags;
    StdVideoH265LevelIdc      maxLevelIdc;
    uint32_t                  maxSliceSegmentCount;
    VkExtent2D                maxTiles;
    VkVideoEncodeH265CtbSizeFlagsKHR ctbSizes;
    VkVideoEncodeH265TransformBlockSizeFlagsKHR transformBlockSizes;
    uint32_t                  maxPPictureL0ReferenceCount;
    uint32_t                  maxBPictureL0ReferenceCount;
    uint32_t                  maxL1ReferenceCount;
    uint32_t                  maxSubLayerCount;
    VkBool32
expectDyadicTemporalSubLayerPattern;
    int32_t                   minQp;
    int32_t                   maxQp;
    VkBool32                  prefersGopRemainingFrames;
    VkBool32                  requiresGopRemainingFrames;
    VkVideoEncodeH265StdFlagsKHR stdSyntaxFlags;
} VkVideoEncodeH265CapabilitiesKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **flags** is a bitmask of [VkVideoEncodeH265CapabilityFlagBitsKHR](#) indicating supported H.265 encoding capabilities.
- **maxLevelIdc** is a [StdVideoH265LevelIdc](#) value indicating the maximum H.265 level supported by the profile, where enum constant `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifies H.265 level `<major>.<minor>` as defined in section A.4 of the [ITU-T H.265 Specification](#).
- **maxSliceSegmentCount** indicates the maximum number of slice segments that **can** be encoded for a single picture. Further restrictions **may** apply to the number of slice segments that **can** be encoded for a single picture depending on other capabilities and codec-specific rules.
- **maxTiles** indicates the maximum number of H.265 tile columns and rows, as defined in sections 3.175 and 3.176 of the [ITU-T H.265 Specification](#) that **can** be encoded for a single picture. Further restrictions **may** apply to the number of H.265 tiles that **can** be encoded for a single picture depending on other capabilities and codec-specific rules.
- **ctbSizes** is a bitmask of [VkVideoEncodeH265CtbSizeFlagBitsKHR](#) describing the supported CTB sizes.
- **transformBlockSizes** is a bitmask of [VkVideoEncodeH265TransformBlockSizeFlagBitsKHR](#) describing the supported transform block sizes.
- **maxPPictureL0ReferenceCount** indicates the maximum number of reference pictures the implementation supports in the reference list L0 for [P pictures](#).

NOTE As implementations **may** [override](#) the reference lists,

`maxPPictureL0ReferenceCount` does not limit the number of elements that the application **can** specify in the L0 reference list for P pictures. However, if `maxPPictureL0ReferenceCount` is zero, then the use of P pictures is not allowed. In case of H.265 encoding, pictures **can** be encoded using only forward prediction even if P pictures are not supported, as the [ITU-T H.265 Specification](#) supports *generalized P & B frames* (also known as low delay B frames) whereas B frames **can** refer to past frames through both the L0 and L1 reference lists.

- `maxBPictureL0ReferenceCount` indicates the maximum number of reference pictures the implementation supports in the reference list L0 for [B pictures](#).
- `maxL1ReferenceCount` indicates the maximum number of reference pictures the implementation supports in the reference list L1 if encoding of [B pictures](#) is supported.

NOTE

As implementations **may** [override](#) the reference lists, `maxBPictureL0ReferenceCount` and `maxL1ReferenceCount` does not limit the number of elements that the application **can** specify in the L0 and L1 reference lists for B pictures. However, if `maxBPictureL0ReferenceCount` and `maxL1ReferenceCount` are both zero, then the use of B pictures is not allowed.

- `maxSubLayerCount` indicates the maximum number of H.265 sub-layers supported by the implementation.
- `expectDyadicTemporalSubLayerPattern` indicates that the implementation's rate control algorithms expect the application to use a [dyadic temporal sub-layer pattern](#) when encoding multiple temporal sub-layers.
- `minQp` indicates the minimum QP value supported.
- `maxQp` indicates the maximum QP value supported.
- `prefersGopRemainingFrames` indicates that the implementation's rate control algorithm prefers the application to specify the number of frames of each type [remaining](#) in the current [group of pictures](#) when beginning a [video coding scope](#).
- `requiresGopRemainingFrames` indicates that the implementation's rate control algorithm requires the application to specify the number of frames of each type [remaining](#) in the current [group of pictures](#) when beginning a [video coding scope](#).
- `stdSyntaxFlags` is a bitmask of [VkVideoEncodeH265StdFlagBitsKHR](#) indicating capabilities related to H.265 syntax elements.

Valid Usage (Implicit)

- `VUID-VkVideoEncodeH265CapabilitiesKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_CAPABILITIES_KHR`

Bits which **may** be set in [VkVideoEncodeH265CapabilitiesKHR::flags](#), indicating the H.265 encoding capabilities supported, are:


```
// Provided by VK_KHR_video_encode_h265
typedef enum VkVideoEncodeH265CapabilityFlagBitsKHR {
    VK_VIDEO_ENCODE_H265_CAPABILITY_HRD_COMPLIANCE_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H265_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR =
0x00000002,
    VK_VIDEO_ENCODE_H265_CAPABILITY_ROW_UNALIGNED_SLICE_SEGMENT_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H265_CAPABILITY_DIFFERENT_SLICE_SEGMENT_TYPE_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR = 0x00000010,
    VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR = 0x00000020,
    VK_VIDEO_ENCODE_H265_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR = 0x00000040,
    VK_VIDEO_ENCODE_H265_CAPABILITY_PER_SLICE_SEGMENT_CONSTANT_QP_BIT_KHR =
0x00000080,
    VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_TILES_PER_SLICE_SEGMENT_BIT_KHR =
0x00000100,
    VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_SLICE_SEGMENTS_PER_TILE_BIT_KHR =
0x00000200,
} VkVideoEncodeH265CapabilityFlagBitsKHR;
```

- `VK_VIDEO_ENCODE_H265_CAPABILITY_HRD_COMPLIANCE_BIT_KHR` indicates if the implementation **may** be able to generate HRD compliant bitstreams if any of the `nal_hrd_parameters_present_flag`, `vcl_hrd_parameters_present_flag`, or `sub_pic_hrd_params_present_flag` members of `StdVideoH265HrdFlags` are set to 1 in the HRD parameters of the `active VPS` or `active SPS`, or if `StdVideoH265SpsVuiFlags::vui_hrd_parameters_present_flag` is set to 1 in the `active SPS`.
- `VK_VIDEO_ENCODE_H265_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR` indicates that if the `weighted_pred_flag` or the `weighted_bipred_flag` member of `StdVideoH265PpsFlags` is set to 1 in the `active PPS` when encoding a `P picture` or `B picture`, respectively, then the implementation is able to internally decide syntax for `pred_weight_table`, as defined in section 7.4.7.3 of the [ITU-T H.265 Specification](#), and the application is not **required** to provide a weight table in the `H.265 slice segment header parameters`.
- `VK_VIDEO_ENCODE_H265_CAPABILITY_ROW_UNALIGNED_SLICE_SEGMENT_BIT_KHR` indicates that each slice segment in a frame with a single or multiple tiles per slice may begin or finish at any offset in a CTB row. If not supported, all slice segments in such a frame **must** begin at the start of a CTB row (and hence each slice segment **must** finish at the end of a CTB row). Also indicates that each slice segment in a frame with multiple slices per tile may begin or finish at any offset within the enclosing tile's CTB row. If not supported, slice segments in such a frame **must** begin at the start of the enclosing tile's CTB row (and hence each slice segment **must** finish at the end of the enclosing tile's CTB row).
- `VK_VIDEO_ENCODE_H265_CAPABILITY_DIFFERENT_SLICE_SEGMENT_TYPE_BIT_KHR` indicates that when a frame is encoded with multiple slice segments, the implementation allows encoding each slice segment with a different `StdVideoEncodeH265SliceSegmentHeader::slice_type` specified in the `H.265 slice segment header parameters`. If not supported, all slice segments of the frame **must** be encoded with the same `slice_type` which corresponds to the picture type of the frame.
- `VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_KHR` indicates support for using a `B frame` as L0 reference, as specified in `StdVideoEncodeH265ReferenceListsInfo::RefPicList0` in the `H.265 picture information`.
- `VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L1_LIST_BIT_KHR` indicates support for using a `B`

[frame](#) as L1 reference, as specified in `StdVideoEncodeH265ReferenceListsInfo::RefPicList1` in the [H.265 picture information](#).

- `VK_VIDEO_ENCODE_H265_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR` indicates support for specifying different QP values in the members of [VkVideoEncodeH265QpKHR](#).
- `VK_VIDEO_ENCODE_H265_CAPABILITY_PER_SLICE_SEGMENT_CONSTANT_QP_BIT_KHR` indicates support for specifying different constant QP values for each slice segment.
- `VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_TILES_PER_SLICE_SEGMENT_BIT_KHR` indicates if encoding multiple tiles per slice segment, as defined in section 6.3.1 of the [ITU-T H.265 Specification](#), is supported. If this capability flag is not present, then the implementation is only able to encode a single tile for each slice segment.
- `VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_SLICE_SEGMENTS_PER_TILE_BIT_KHR` indicates if encoding multiple slice segments per tile, as defined in section 6.3.1 of the [ITU-T H.265 Specification](#), is supported. If this capability flag is not present, then the implementation is only able to encode a single slice segment for each tile.

```
// Provided by VK_KHR_video_encode_h265
typedef VkFlags VkVideoEncodeH265CapabilityFlagsKHR;
```

`VkVideoEncodeH265CapabilityFlagsKHR` is a bitmask type for setting a mask of zero or more [VkVideoEncodeH265CapabilityFlagBitsKHR](#).

Bits which **may** be set in `VkVideoEncodeH265CapabilitiesKHR::stdSyntaxFlags`, indicating the capabilities related to the H.265 syntax elements, are:

```
// Provided by VK_KHR_video_encode_h265
typedef enum VkVideoEncodeH265StdFlagBitsKHR {
    VK_VIDEO_ENCODE_H265_STD_SEPARATE_COLOR_PLANE_FLAG_SET_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H265_STD_SAMPLE_ADAPTIVE_OFFSET_ENABLED_FLAG_SET_BIT_KHR =
0x00000002,
    VK_VIDEO_ENCODE_H265_STD_SCALING_LIST_DATA_PRESENT_FLAG_SET_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H265_STD_PCM_ENABLED_FLAG_SET_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H265_STD_SPS_TEMPORAL_MVP_ENABLED_FLAG_SET_BIT_KHR = 0x00000010,
    VK_VIDEO_ENCODE_H265_STD_INIT_QP_MINUS26_BIT_KHR = 0x00000020,
    VK_VIDEO_ENCODE_H265_STD_WEIGHTED_PRED_FLAG_SET_BIT_KHR = 0x00000040,
    VK_VIDEO_ENCODE_H265_STD_WEIGHTED_BIPRED_FLAG_SET_BIT_KHR = 0x00000080,
    VK_VIDEO_ENCODE_H265_STD_LOG2_PARALLEL_MERGE_LEVEL_MINUS2_BIT_KHR = 0x00000100,
    VK_VIDEO_ENCODE_H265_STD_SIGN_DATA_HIDING_ENABLED_FLAG_SET_BIT_KHR = 0x00000200,
    VK_VIDEO_ENCODE_H265_STD_TRANSFORM_SKIP_ENABLED_FLAG_SET_BIT_KHR = 0x00000400,
    VK_VIDEO_ENCODE_H265_STD_TRANSFORM_SKIP_ENABLED_FLAG_UNSET_BIT_KHR = 0x00000800,
    VK_VIDEO_ENCODE_H265_STD_PPS_SLICE_CHROMA_QP_OFFSETS_PRESENT_FLAG_SET_BIT_KHR =
0x00001000,
    VK_VIDEO_ENCODE_H265_STD_TRANSQUANT_BYPASS_ENABLED_FLAG_SET_BIT_KHR = 0x00002000,
    VK_VIDEO_ENCODE_H265_STD_CONSTRAINED_INTRA_PRED_FLAG_SET_BIT_KHR = 0x00004000,
    VK_VIDEO_ENCODE_H265_STD_ENTROPY_CODING_SYNC_ENABLED_FLAG_SET_BIT_KHR =
0x00008000,
    VK_VIDEO_ENCODE_H265_STD_DEBLOCKING_FILTER_OVERRIDE_ENABLED_FLAG_SET_BIT_KHR =
0x00010000,
    VK_VIDEO_ENCODE_H265_STD_DEPENDENT_SLICE_SEGMENTS_ENABLED_FLAG_SET_BIT_KHR =
0x00020000,
    VK_VIDEO_ENCODE_H265_STD_DEPENDENT_SLICE_SEGMENT_FLAG_SET_BIT_KHR = 0x00040000,
    VK_VIDEO_ENCODE_H265_STD_SLICE_QP_DELTA_BIT_KHR = 0x00080000,
    VK_VIDEO_ENCODE_H265_STD_DIFFERENT_SLICE_QP_DELTA_BIT_KHR = 0x00100000,
} VkVideoEncodeH265StdFlagBitsKHR;
```

- **VK_VIDEO_ENCODE_H265_STD_SEPARATE_COLOR_PLANE_FLAG_SET_BIT_KHR** indicates whether the implementation supports using the application-provided value for **StdVideoH265SpsFlags::separate_colour_plane_flag** in the **SPS** when that value is 1.
- **VK_VIDEO_ENCODE_H265_STD_SAMPLE_ADAPTIVE_OFFSET_ENABLED_FLAG_SET_BIT_KHR** indicates whether the implementation supports using the application-provided value for **StdVideoH265SpsFlags::sample_adaptive_offset_enabled_flag** in the **SPS** when that value is 1.
- **VK_VIDEO_ENCODE_H265_STD_SCALING_LIST_DATA_PRESENT_FLAG_SET_BIT_KHR** indicates whether the implementation supports using the application-provided value for the **scaling_list_enabled_flag** and **sps_scaling_list_data_present_flag** members of **StdVideoH265SpsFlags** in the **SPS**, and the application-provided value for **StdVideoH265PpsFlags::pps_scaling_list_data_present_flag** in the **PPS** when those values are 1.
- **VK_VIDEO_ENCODE_H265_STD_PCM_ENABLED_FLAG_SET_BIT_KHR** indicates whether the implementation supports using the application-provided value for **StdVideoH265SpsFlags::pcm_enable_flag** in the **SPS** when that value is 1.
- **VK_VIDEO_ENCODE_H265_STD_SPS_TEMPORAL_MVP_ENABLED_FLAG_SET_BIT_KHR** indicates whether the implementation supports using the application-provided value for **StdVideoH265SpsFlags::sps_temporal_mvp_enabled_flag** in the **SPS** when that value is 1.

- `VK_VIDEO_ENCODE_H265_STD_INIT_QP_MINUS26_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PictureParameterSet::init_qp_minus26` in the `PPS` when that value is non-zero.
- `VK_VIDEO_ENCODE_H265_STD_WEIGHTED_PRED_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PpsFlags::weighted_pred_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H265_STD_WEIGHTED_BIPRED_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PpsFlags::weighted_bipred_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H265_STD_LOG2_PARALLEL_MERGE_LEVEL_MINUS2_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PictureParameterSet::log2_parallel_merge_level_minus2` in the `PPS` when that value is non-zero.
- `VK_VIDEO_ENCODE_H265_STD_SIGN_DATA_HIDING_ENABLED_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PpsFlags::sign_data_hiding_enabled_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H265_STD_TRANSFORM_SKIP_ENABLED_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PpsFlags::transform_skip_enabled_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H265_STD_TRANSFORM_SKIP_ENABLED_FLAG_UNSET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PpsFlags::transform_skip_enabled_flag` in the `PPS` when that value is 0.
- `VK_VIDEO_ENCODE_H265_STD_PPS_SLICE_CHROMA_QP_OFFSETS_PRESENT_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PpsFlags::pps_slice_chroma_qp_offsets_present_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H265_STD_TRANSQUANT_BYPASS_ENABLED_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PpsFlags::transquant_bypass_enabled_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H265_STD_CONSTRAINED_INTRA_PRED_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PpsFlags::constrained_intra_pred_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H265_STD_ENTROPY_CODING_SYNC_ENABLED_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PpsFlags::entropy_coding_sync_enabled_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H265_STD_DEBLOCKING_FILTER_OVERRIDE_ENABLED_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PpsFlags::deblocking_filter_override_enabled_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H265_STD_DEPENDENT_SLICE_SEGMENTS_ENABLED_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoH265PpsFlags::dependent_slice_segments_enabled_flag` in the `PPS` when that value is 1.
- `VK_VIDEO_ENCODE_H265_STD_DEPENDENT_SLICE_SEGMENT_FLAG_SET_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoEncodeH265SliceSegmentHeader::dependent_slice_segment_flag` in the `H.265 slice segment`

header parameters when that value is 1.

- `VK_VIDEO_ENCODE_H265_STD_SLICE_QP_DELTA_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoEncodeH265SliceSegmentHeader::slice_qp_delta` in the H.265 slice segment header parameters when that value is identical across the slice segments of the encoded frame.
- `VK_VIDEO_ENCODE_H265_STD_DIFFERENT_SLICE_QP_DELTA_BIT_KHR` indicates whether the implementation supports using the application-provided value for `StdVideoEncodeH265SliceSegmentHeader::slice_qp_delta` in the H.265 slice segment header parameters when that value is different across the slice segments of the encoded frame.

These capability flags provide information to the application about specific H.265 syntax element values that the implementation supports without having to [override](#) them and do not otherwise restrict the values that the application **can** specify for any of the mentioned H.265 syntax elements.

```
// Provided by VK_KHR_video_encode_h265
typedef VkFlags VkVideoEncodeH265StdFlagsKHR;
```

`VkVideoEncodeH265StdFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoEncodeH265StdFlagBitsKHR`.

Bits which **may** be set in `VkVideoEncodeH265CapabilitiesKHR::ctbSizes`, indicating the CTB sizes supported by the implementation, are:

```
// Provided by VK_KHR_video_encode_h265
typedef enum VkVideoEncodeH265CtbSizeFlagBitsKHR {
    VK_VIDEO_ENCODE_H265_CTB_SIZE_16_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H265_CTB_SIZE_32_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_H265_CTB_SIZE_64_BIT_KHR = 0x00000004,
} VkVideoEncodeH265CtbSizeFlagBitsKHR;
```

- `VK_VIDEO_ENCODE_H265_CTB_SIZE_16_BIT_KHR` specifies that a CTB size of 16x16 is supported.
- `VK_VIDEO_ENCODE_H265_CTB_SIZE_32_BIT_KHR` specifies that a CTB size of 32x32 is supported.
- `VK_VIDEO_ENCODE_H265_CTB_SIZE_64_BIT_KHR` specifies that a CTB size of 64x64 is supported.

```
// Provided by VK_KHR_video_encode_h265
typedef VkFlags VkVideoEncodeH265CtbSizeFlagsKHR;
```

`VkVideoEncodeH265CtbSizeFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoEncodeH265CtbSizeFlagBitsKHR`.

Implementations **must** support at least one of `VkVideoEncodeH265CtbSizeFlagBitsKHR`.

Bits which **may** be set in `VkVideoEncodeH265CapabilitiesKHR::transformBlockSizes`, indicating the transform block sizes supported by the implementation, are:

```
// Provided by VK_KHR_video_encode_h265
typedef enum VkVideoEncodeH265TransformBlockSizeFlagBitsKHR {
    VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_4_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_8_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_16_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_32_BIT_KHR = 0x00000008,
} VkVideoEncodeH265TransformBlockSizeFlagBitsKHR;
```

- `VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_4_BIT_KHR` specifies that a transform block size of 4x4 is supported.
- `VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_8_BIT_KHR` specifies that a transform block size of 8x8 is supported.
- `VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_16_BIT_KHR` specifies that a transform block size of 16x16 is supported.
- `VK_VIDEO_ENCODE_H265_TRANSFORM_BLOCK_SIZE_32_BIT_KHR` specifies that a transform block size of 32x32 is supported.

```
// Provided by VK_KHR_video_encode_h265
typedef VkFlags VkVideoEncodeH265TransformBlockSizeFlagsKHR;
```

`VkVideoEncodeH265TransformBlockSizeFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoEncodeH265TransformBlockSizeFlagBitsKHR`.

Implementations **must** support at least one of `VkVideoEncodeH265TransformBlockSizeFlagBitsKHR`.

37.18.8. H.265 Encode Quality Level Properties

When calling `vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR` with `pVideoProfile->videoCodecOperation` specified as `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, the `VkVideoEncodeH265QualityLevelPropertiesKHR` structure **must** be included in the `pNext` chain of the `VkVideoEncodeQualityLevelPropertiesKHR` structure to retrieve additional video encode quality level properties specific to H.265 encoding.

The `VkVideoEncodeH265QualityLevelPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265QualityLevelPropertiesKHR {
    VkStructureType           sType;
    void*                     pNext;
    VkVideoEncodeH265RateControlFlagsKHR preferredRateControlFlags;
    uint32_t                  preferredGopFrameCount;
    uint32_t                  preferredIdrPeriod;
    uint32_t                  preferredConsecutiveBFrameCount;
    uint32_t                  preferredSubLayerCount;
    VkVideoEncodeH265QpKHR    preferredConstantQp;
    uint32_t                  preferredMaxL0ReferenceCount;
    uint32_t                  preferredMaxL1ReferenceCount;
} VkVideoEncodeH265QualityLevelPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `preferredRateControlFlags` is a bitmask of `VkVideoEncodeH265RateControlFlagBitsKHR` values indicating the preferred flags to use for `VkVideoEncodeH265RateControlInfoKHR::flags`.
- `preferredGopFrameCount` indicates the preferred value to use for `VkVideoEncodeH265RateControlInfoKHR::gopFrameCount`.
- `preferredIdrPeriod` indicates the preferred value to use for `VkVideoEncodeH265RateControlInfoKHR::idrPeriod`.
- `preferredConsecutiveBFrameCount` indicates the preferred value to use for `VkVideoEncodeH265RateControlInfoKHR::consecutiveBFrameCount`.
- `preferredSubLayerCount` indicates the preferred value to use for `VkVideoEncodeH265RateControlInfoKHR::subLayerCount`.
- `preferredConstantQp` indicates the preferred values to use for `VkVideoEncodeH265NaluSliceSegmentInfoKHR::constantQp` for each picture type when using rate control mode `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`.
- `preferredMaxL0ReferenceCount` indicates the preferred maximum number of reference pictures to use in the reference list L0.
- `preferredMaxL1ReferenceCount` indicates the preferred maximum number of reference pictures to use in the reference list L1.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265QualityLevelPropertiesKHR-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_QUALITY_LEVEL_PROPERTIES_KHR`

37.18.9. H.265 Encode Session

Additional parameters can be specified when creating a video session with an H.265 encode profile by including an instance of the `VkVideoEncodeH265SessionCreateInfoKHR` structure in the `pNext`

chain of [VkVideoSessionCreateInfoKHR](#).

The [VkVideoEncodeH265SessionCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265SessionCreateInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           useMaxLevelIdc;
    StdVideoH265LevelIdc maxLevelIdc;
} VkVideoEncodeH265SessionCreateInfoKHR;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `useMaxLevelIdc` indicates whether the value of `maxLevelIdc` should be used by the implementation. When it is `VK_FALSE`, the implementation ignores the value of `maxLevelIdc` and uses the value of [VkVideoEncodeH265CapabilitiesKHR::maxLevelIdc](#), as reported by [vkGetPhysicalDeviceVideoCapabilitiesKHR](#) for the video profile.
- `maxLevelIdc` is a `StdVideoH265LevelIdc` value specifying the upper bound on the H.265 level for the video bitstreams produced by the created video session, where enum constant `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifies H.265 level `<major>.<minor>` as defined in section A.4 of the [ITU-T H.265 Specification](#).

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265SessionCreateInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_CREATE_INFO_KHR`

37.18.10. H.265 Encode Parameter Sets

[Video session parameters](#) objects created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR` can contain the following types of parameters:

H.265 Video Parameter Sets (VPS)

Represented by [StdVideoH265VideoParameterSet](#) structures and interpreted as follows:

- `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
- `vps_video_parameter_set_id` is used as the key of the VPS entry;
- the `max_latency_increase_plus1`, `max_dec_pic_buffering_minus1`, and `max_num_reorder_pics` members of the [StdVideoH265DecPicBufMgr](#) structure pointed to by `pDecPicBufMgr` correspond to `vps_max_latency_increase_plus1`, `vps_max_dec_pic_buffering_minus1`, and `vps_max_num_reorder_pics`, respectively, as defined in section 7.4.3.1 of the [ITU-T H.265](#)

Specification;

- the `StdVideoH265HrdParameters` structure pointed to by `pHrdParameters` is interpreted as follows:
 - `reserved` is used only for padding purposes and is otherwise ignored;
 - `flags.fixed_pic_rate_general_flag` is a bitmask where bit index `i` corresponds to `fixed_pic_rate_general_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
 - `flags.fixed_pic_rate_within_cvs_flag` is a bitmask where bit index `i` corresponds to `fixed_pic_rate_within_cvs_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
 - `flags.low_delay_hrd_flag` is a bitmask where bit index `i` corresponds to `low_delay_hrd_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
 - if `flags.nal_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersNal` is a pointer to an array of `vps_max_sub_layers_minus1 + 1` number of `StdVideoH265SubLayerHrdParameters` structures where `vps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265VideoParameterSet` structure and each element is interpreted as follows:
 - `cbr_flag` is a bitmask where bit index `i` corresponds to `cbr_flag[i]` as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - if `flags.vcl_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersVcl` is a pointer to an array of `vps_max_sub_layers_minus1 + 1` number of `StdVideoH265SubLayerHrdParameters` structures where `vps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265VideoParameterSet` structure and each element is interpreted as follows:
 - `cbr_flag` is a bitmask where bit index `i` corresponds to `cbr_flag[i]` as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoH265HrdParameters` are interpreted as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
- the `StdVideoH265ProfileTierLevel` structure pointed to by `pProfileTierLevel` are interpreted as follows:
 - `general_level_idc` is one of the enum constants `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifying the H.265 level `<major>.<minor>` as defined in section A.4 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoH265ProfileTierLevel` are interpreted as defined in section 7.4.4 of the [ITU-T H.265 Specification](#);
- all other members of `StdVideoH265VideoParameterSet` are interpreted as defined in section 7.4.3.1 of the [ITU-T H.265 Specification](#).

H.265 Sequence Parameter Sets (SPS)

Represented by `StdVideoH265SequenceParameterSet` structures and interpreted as follows:

- `reserved1` and `reserved2` are used only for padding purposes and are otherwise ignored;
- the pair constructed from `sps_video_parameter_set_id` and `sps_seq_parameter_set_id` is used as the key of the SPS entry;
- the `StdVideoH265ProfileTierLevel` structure pointed to by `pProfileTierLevel` are interpreted as follows:
 - `general_level_idc` is one of the enum constants `STD_VIDEO_H265_LEVEL_IDC_<major>_<minor>` identifying the H.265 level `<major>.<minor>` as defined in section A.4 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoH265ProfileTierLevel` are interpreted as defined in section 7.4.4 of the [ITU-T H.265 Specification](#);
- the `max_latency_increase_plus1`, `max_dec_pic_buffering_minus1`, and `max_num_reorder_pics` members of the `StdVideoH265DecPicBufMgr` structure pointed to by `pDecPicBufMgr` correspond to `sps_max_latency_increase_plus1`, `sps_max_dec_pic_buffering_minus1`, and `sps_max_num_reorder_pics`, respectively, as defined in section 7.4.3.2 of the [ITU-T H.265 Specification](#);
- if `flags.sps_scaling_list_data_present_flag` is set, then the `StdVideoH265ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
 - `ScalingList4x4`, `ScalingList8x8`, `ScalingList16x16`, and `ScalingList32x32` correspond to `ScalingList[0]`, `ScalingList[1]`, `ScalingList[2]`, and `ScalingList[3]`, respectively, as defined in section 7.3.4 of the [ITU-T H.265 Specification](#);
 - `ScalingListDCCoef16x16` and `ScalingListDCCoef32x32` correspond to `scaling_list_dc_coef_minus8[0]` and `scaling_list_dc_coef_minus8[1]`, respectively, as defined in section 7.3.4 of the [ITU-T H.265 Specification](#);
- `pShortTermRefPicSet` is a pointer to an array of `num_short_term_ref_pic_sets` number of `StdVideoH265ShortTermRefPicSet` structures where each element is interpreted as follows:
 - `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
 - `used_by_curr_pic_flag` is a bitmask where bit index `i` corresponds to `used_by_curr_pic_flag[i]` as defined in section 7.4.8 of the [ITU-T H.265 Specification](#);
 - `use_delta_flag` is a bitmask where bit index `i` corresponds to `use_delta_flag[i]` as defined in section 7.4.8 of the [ITU-T H.265 Specification](#);
 - `used_by_curr_pic_s0_flag` is a bitmask where bit index `i` corresponds to `used_by_curr_pic_s0_flag[i]` as defined in section 7.4.8 of the [ITU-T H.265 Specification](#);
 - `used_by_curr_pic_s1_flag` is a bitmask where bit index `i` corresponds to `used_by_curr_pic_s1_flag[i]` as defined in section 7.4.8 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoH265ShortTermRefPicSet` are interpreted as defined in section 7.4.8 of the [ITU-T H.265 Specification](#);

- if `flags.long_term_ref_pics_present_flag` is set then the `StdVideoH265LongTermRefPicsSps` structure pointed to by `pLongTermRefPicsSps` is interpreted as follows:
 - `used_by_curr_pic_lt_sps_flag` is a bitmask where bit index `i` corresponds to `used_by_curr_pic_lt_sps_flag[i]` as defined in section 7.4.3.2 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoH265LongTermRefPicsSps` are interpreted as defined in section 7.4.3.2 of the [ITU-T H.265 Specification](#);
- if `flags.vui_parameters_present_flag` is set, then the `StdVideoH265SequenceParameterSetVui` structure pointed to by `pSequenceParameterSetVui` is interpreted as follows:
 - `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
 - the `StdVideoH265HrdParameters` structure pointed to by `pHrdParameters` is interpreted as follows:
 - `flags.fixed_pic_rate_general_flag` is a bitmask where bit index `i` corresponds to `fixed_pic_rate_general_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
 - `flags.fixed_pic_rate_within_cvs_flag` is a bitmask where bit index `i` corresponds to `fixed_pic_rate_within_cvs_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
 - `flags.low_delay_hrd_flag` is a bitmask where bit index `i` corresponds to `low_delay_hrd_flag[i]` as defined in section E.3.2 of the [ITU-T H.265 Specification](#);
 - if `flags.nal_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersNal` is a pointer to an array of `sps_max_sub_layers_minus1 + 1` number of `StdVideoH265SubLayerHrdParameters` structures where `sps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265SequenceParameterSet` structure and each element is interpreted as follows:
 - `cbr_flag` is a bitmask where bit index `i` corresponds to `cbr_flag[i]` as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - if `flags.vcl_hrd_parameters_present_flag` is set, then `pSubLayerHrdParametersVcl` is a pointer to an array of `sps_max_sub_layers_minus1 + 1` number of `StdVideoH265SubLayerHrdParameters` structures where `sps_max_sub_layers_minus1` is the corresponding member of the encompassing `StdVideoH265SequenceParameterSet` structure and each element is interpreted as follows:
 - `cbr_flag` is a bitmask where bit index `i` corresponds to `cbr_flag[i]` as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - all other members of the `StdVideoH265SubLayerHrdParameters` structure are interpreted as defined in section E.3.3 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoH265HrdParameters` are interpreted as defined in section E.3.2 of the [ITU-T H.265 Specification](#);

- all other members of `pSequenceParameterSetVui` are interpreted as defined in section E.3.1 of the [ITU-T H.265 Specification](#);
- if `flags.sps_palette_predictor_initializer_present_flag` is set, then the `PredictorPaletteEntries` member of the `StdVideoH265PredictorPaletteEntries` structure pointed to by `pPredictorPaletteEntries` is interpreted as defined in section 7.4.9.13 of the [ITU-T H.265 Specification](#);
- all other members of `StdVideoH265SequenceParameterSet` are interpreted as defined in section 7.4.3.1 of the [ITU-T H.265 Specification](#).

H.265 Picture Parameter Sets (PPS)

Represented by `StdVideoH265PictureParameterSet` structures and interpreted as follows:

- `reserved1`, `reserved2`, and `reserved3` are used only for padding purposes and are otherwise ignored;
- the triplet constructed from `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` is used as the key of the PPS entry;
- if `flags.pps_scaling_list_data_present_flag` is set, then the `StdVideoH265ScalingLists` structure pointed to by `pScalingLists` is interpreted as follows:
 - `ScalingList4x4`, `ScalingList8x8`, `ScalingList16x16`, and `ScalingList32x32` correspond to `ScalingList[0]`, `ScalingList[1]`, `ScalingList[2]`, and `ScalingList[3]`, respectively, as defined in section 7.3.4 of the [ITU-T H.265 Specification](#);
 - `ScalingListDCCoef16x16` and `ScalingListDCCoef32x32` correspond to `scaling_list_dc_coef_minus8[0]` and `scaling_list_dc_coef_minus8[1]`, respectively, as defined in section 7.3.4 of the [ITU-T H.265 Specification](#);
- if `flags.pps_palette_predictor_initializer_present_flag` is set, then the `PredictorPaletteEntries` member of the `StdVideoH265PredictorPaletteEntries` structure pointed to by `pPredictorPaletteEntries` is interpreted as defined in section 7.4.9.13 of the [ITU-T H.265 Specification](#);
- all other members of `StdVideoH265PictureParameterSet` are interpreted as defined in section 7.4.3.3 of the [ITU-T H.265 Specification](#).

Implementations **may** override any of these parameters according to the semantics defined in the [Video Encode Parameter Overrides](#) section before storing the resulting H.265 parameter sets into the video session parameters object. Applications need to use the `vkGetEncodedVideoSessionParametersKHR` command to determine whether any implementation overrides happened and to retrieve the encoded H.265 parameter sets in order to be able to produce a compliant H.265 video bitstream.

Such H.265 parameter set overrides **may** also have cascading effects on the implementation overrides applied to the encoded bitstream produced by video encode operations. If the implementation supports the `VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR` [video encode feedback query](#) flag, then the application **can** use such queries to retrieve feedback about whether any implementation overrides have been applied to the encoded bitstream.

When a [video session parameters](#) object is created with the codec operation [VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR](#), the [VkVideoSessionParametersCreateInfoKHR::pNext](#) chain **must** include a [VkVideoEncodeH265SessionParametersCreateInfoKHR](#) structure specifying the capacity and initial contents of the object.

The [VkVideoEncodeH265SessionParametersCreateInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265SessionParametersCreateInfoKHR {
    VkStructureType             sType;
    const void*                 pNext;
    uint32_t                    maxStdVPSCount;
    uint32_t                    maxStdSPSCount;
    uint32_t                    maxStdPPSCount;
    const VkVideoEncodeH265SessionParametersAddInfoKHR* pParametersAddInfo;
} VkVideoEncodeH265SessionParametersCreateInfoKHR;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [maxStdVPSCount](#) is the maximum number of [H.265 VPS](#) entries the created [VkVideoSessionParametersKHR](#) **can** contain.
- [maxStdSPSCount](#) is the maximum number of [H.265 SPS](#) entries the created [VkVideoSessionParametersKHR](#) **can** contain.
- [maxStdPPSCount](#) is the maximum number of [H.265 PPS](#) entries the created [VkVideoSessionParametersKHR](#) **can** contain.
- [pParametersAddInfo](#) is [NULL](#) or a pointer to a [VkVideoEncodeH265SessionParametersAddInfoKHR](#) structure specifying H.265 parameters to add upon object creation.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265SessionParametersCreateInfoKHR-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR](#)
- VUID-VkVideoEncodeH265SessionParametersCreateInfoKHR-pParametersAddInfo-parameter
If [pParametersAddInfo](#) is not [NULL](#), [pParametersAddInfo](#) **must** be a valid pointer to a valid [VkVideoEncodeH265SessionParametersAddInfoKHR](#) structure

The [VkVideoEncodeH265SessionParametersAddInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265SessionParametersAddInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    uint32_t                 stdVPSCount;
    const StdVideoH265VideoParameterSet* pStdVPSSs;
    uint32_t                 stdSPSCount;
    const StdVideoH265SequenceParameterSet* pStdSPSSs;
    uint32_t                 stdPPSCount;
    const StdVideoH265PictureParameterSet* pStdPPSSs;
} VkVideoEncodeH265SessionParametersAddInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `stdVPSCount` is the number of elements in the `pStdVPSSs` array.
- `pStdVPSSs` is a pointer to an array of `StdVideoH265VideoParameterSet` structures describing the H.265 VPS entries to add.
- `stdSPSCount` is the number of elements in the `pStdSPSSs` array.
- `pStdSPSSs` is a pointer to an array of `StdVideoH265SequenceParameterSet` structures describing the H.265 SPS entries to add.
- `stdPPSCount` is the number of elements in the `pStdPPSSs` array.
- `pStdPPSSs` is a pointer to an array of `StdVideoH265PictureParameterSet` structures describing the H.265 PPS entries to add.

This structure **can** be specified in the following places:

- In the `pParametersAddInfo` member of the `VkVideoEncodeH265SessionParametersCreateInfoKHR` structure specified in the `pNext` chain of `VkVideoSessionParametersCreateInfoKHR` used to create a `video session parameters` object. In this case, if the video codec operation the video session parameters object is created with is `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then it defines the set of initial parameters to add to the created object (see [Creating Video Session Parameters](#)).
- In the `pNext` chain of `VkVideoSessionParametersUpdateInfoKHR`. In this case, if the video codec operation the `video session parameters` object to be updated was created with is `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, then it defines the set of parameters to add to it (see [Updating Video Session Parameters](#)).

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR`
- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-pStdVPSSs-parameter
If `stdVPSCount` is not 0, and `pStdVPSSs` is not `NULL`, `pStdVPSSs` **must** be a valid pointer to an array of `stdVPSCount` `StdVideoH265VideoParameterSet` values

- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-pStdSPSs-parameter
If `stdSPSCount` is not 0, and `pStdSPSs` is not NULL, `pStdSPSs` **must** be a valid pointer to an array of `stdSPSCount` `StdVideoH265SequenceParameterSet` values
- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-pStdPPSs-parameter
If `stdPPSCount` is not 0, and `pStdPPSs` is not NULL, `pStdPPSs` **must** be a valid pointer to an array of `stdPPSCount` `StdVideoH265PictureParameterSet` values

Valid Usage

- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-None-06438
The `vps_video_parameter_set_id` member of each `StdVideoH265VideoParameterSet` structure specified in the elements of `pStdVPSs` **must** be unique within `pStdVPSs`
- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-None-06439
The pair constructed from the `sps_video_parameter_set_id` and `sps_seq_parameter_set_id` members of each `StdVideoH265SequenceParameterSet` structure specified in the elements of `pStdSPSs` **must** be unique within `pStdSPSs`
- VUID-VkVideoEncodeH265SessionParametersAddInfoKHR-None-06440
The triplet constructed from the `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` members of each `StdVideoH265PictureParameterSet` structure specified in the elements of `pStdPPSs` **must** be unique within `pStdPPSs`

The `VkVideoEncodeH265SessionParametersGetInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265SessionParametersGetInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           writeStdVPS;
    VkBool32           writeStdSPS;
    VkBool32           writeStdPPS;
    uint32_t           stdVPSId;
    uint32_t           stdSPSId;
    uint32_t           stdPPSId;
} VkVideoEncodeH265SessionParametersGetInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is NULL or a pointer to a structure extending this structure.
- `writeStdVPS` indicates whether the encoded [H.265 video parameter set](#) identified by `stdVPSId` is requested to be retrieved.
- `writeStdSPS` indicates whether the encoded [H.265 sequence parameter set](#) identified by the pair constructed from `stdVPSId` and `stdSPSId` is requested to be retrieved.
- `writeStdPPS` indicates whether the encoded [H.265 picture parameter set](#) identified by the triplet constructed from `stdVPSId`, `stdSPSId`, and `stdPPSId` is requested to be retrieved.

- **stdVPSId** specifies the H.265 video parameter set ID used to identify the retrieved H.265 video, sequence, and/or picture parameter set(s).
- **stdSPSId** specifies the H.265 sequence parameter set ID used to identify the retrieved H.265 sequence and/or picture parameter set(s) when **writeStdSPS** and/or **writeStdPPS** is **VK_TRUE**.
- **stdPPSId** specifies the H.265 picture parameter set ID used to identify the retrieved H.265 picture parameter set when **writeStdPPS** is **VK_TRUE**.

When this structure is specified in the **pNext** chain of the **VkVideoEncodeSessionParametersGetInfoKHR** structure passed to **vkGetEncodedVideoSessionParametersKHR**, the command will write encoded parameter data to the output buffer in the following order:

1. The **H.265 video parameter set** identified by **stdVPSId**, if **writeStdVPS** is **VK_TRUE**.
2. The **H.265 sequence parameter set** identified by the pair constructed from **stdVPSId** and **stdSPSId**, if **writeStdSPS** is **VK_TRUE**.
3. The **H.265 picture parameter set** identified by the triplet constructed from **stdVPSId**, **stdSPSId**, and **stdPPSId**, if **writeStdPPS** is **VK_TRUE**.

Valid Usage

- VUID-VkVideoEncodeH265SessionParametersGetInfoKHR-writeStdVPS-08290
At least one of **writeStdVPS**, **writeStdSPS**, and **writeStdPPS** **must** be **VK_TRUE**

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265SessionParametersGetInfoKHR-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_GET_INFO_KHR**

The **VkVideoEncodeH265SessionParametersFeedbackInfoKHR** structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265SessionParametersFeedbackInfoKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           hasStdVPSOverrides;
    VkBool32           hasStdSPSOverrides;
    VkBool32           hasStdPPSOverrides;
} VkVideoEncodeH265SessionParametersFeedbackInfoKHR;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **hasStdVPSOverrides** indicates whether any of the parameters of the requested **H.265 video parameter set**, if one was requested via **VkVideoEncodeH265SessionParametersGetInfoKHR**

::writeStdVPS, were overridden by the implementation.

- hasStdSPS0overrides indicates whether any of the parameters of the requested H.265 sequence parameter set, if one was requested via VkVideoEncodeH265SessionParametersGetInfoKHR ::writeStdSPS, were overridden by the implementation.
- hasStdPPS0overrides indicates whether any of the parameters of the requested H.265 picture parameter set, if one was requested via VkVideoEncodeH265SessionParametersGetInfoKHR ::writeStdPPS, were overridden by the implementation.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265SessionParametersFeedbackInfoKHR-sType-sType must be VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_FEEDBACK_INFO_KHR

37.18.11. H.265 Encoding Parameters

The VkVideoEncodeH265PictureInfoKHR structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265PictureInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    uint32_t            naluSliceSegmentEntryCount;
    const VkVideoEncodeH265NaluSliceSegmentInfoKHR* pNaluSliceSegmentEntries;
    const StdVideoEncodeH265PictureInfo* pStdPictureInfo;
} VkVideoEncodeH265PictureInfoKHR;
```

- sType is a VkStructureType value identifying this structure.
- pNext is NULL or a pointer to a structure extending this structure.
- naluSliceSegmentEntryCount is the number of elements in pNaluSliceSegmentEntries.
- pNaluSliceSegmentEntries is a pointer to an array of naluSliceSegmentEntryCount VkVideoEncodeH265NaluSliceSegmentInfoKHR structures specifying the parameters of the individual H.265 slice segments to encode for the input picture.
- pStdPictureInfo is a pointer to a StdVideoEncodeH265PictureInfo structure specifying H.265 picture information.

This structure is specified in the pNext chain of the VkVideoEncodeInfoKHR structure passed to vkCmdEncodeVideoKHR to specify the codec-specific picture information for an H.265 encode operation.

Encode Input Picture Information

When this structure is specified in the pNext chain of the VkVideoEncodeInfoKHR structure passed to vkCmdEncodeVideoKHR, the information related to the encode input picture is defined as follows:

- The image subregion used is determined according to the [H.265 Encode Picture Data Access](#) section.
- The encode input picture is associated with the [H.265 picture information](#) provided in `pStdPictureInfo`.

Std Picture Information

The members of the `StdVideoEncodeH265PictureInfo` structure pointed to by `pStdPictureInfo` are interpreted as follows:

- `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
- `flags.is_reference` as defined in section 3.132 of the [ITU-T H.265 Specification](#);
- `flags.IrapPicFlag` as defined in section 3.73 of the [ITU-T H.265 Specification](#);
- `flags.used_for_long_term_reference` is used to indicate whether the picture is marked as “used for long-term reference” as defined in section 8.3.2 of the [ITU-T H.265 Specification](#);
- `flags.discardable_flag` and `cross_layer_bla_flag` as defined in section F.7.4.7.1 of the [ITU-T H.265 Specification](#);
- `pic_type` as defined in section 7.4.3.5 of the [ITU-T H.265 Specification](#);
- `sps_video_parameter_set_id`, `pps_seq_parameter_set_id`, and `pps_pic_parameter_set_id` are used to identify the active parameter sets, as described below;
- `PicOrderCntVal` as defined in section 8.3.1 of the [ITU-T H.265 Specification](#);
- `TemporalId` as defined in section 7.4.2.2 of the [ITU-T H.265 Specification](#);
- if `pRefLists` is not `NULL`, then it is a pointer to a `StdVideoEncodeH265ReferenceListsInfo` structure that is interpreted as follows:
 - `flags.reserved` is used only for padding purposes and is otherwise ignored;
 - `ref_pic_list_modification_flag_l0` and `ref_pic_list_modification_flag_l1` as defined in section 7.4.7.2 of the [ITU-T H.265 Specification](#);
 - `num_ref_idx_l0_active_minus1` and `num_ref_idx_l1_active_minus1` as defined in section 7.4.7.1 of the [ITU-T H.265 Specification](#);
 - `RefPicList0` and `RefPicList1` as defined in section 8.3.4 of the [ITU-T H.265 Specification](#) where each element of these arrays either identifies an [active reference picture](#) using its [DPB slot](#) index or contains the value `STD_VIDEO_H265_NO_REFERENCE_PICTURE` to indicate “no reference picture”;
 - `list_entry_l0` and `list_entry_l1` as defined in section 7.4.7.2 of the [ITU-T H.265 Specification](#);
- if `flags.short_term_ref_pic_set_sps_flag` is set, then the `StdVideoH265ShortTermRefPicSet` structure pointed to by `pShortTermRefPicSet` is interpreted as defined for the elements of the `pShortTermRefPicSet` array specified in [H.265 sequence parameter sets](#).
- if `flags.long_term_ref_pics_present_flag` is set in the [active SPS](#), then the `StdVideoEncodeH265LongTermRefPics` structure pointed to by `pLongTermRefPics` is interpreted as follows:
 - `used_by_curr_pic_lt_flag` is a bitmask where bit index `i` corresponds to

- `used_by_curr_pic_lt_flag[i]` as defined in section 7.4.7.1 of the [ITU-T H.265 Specification](#);
- all other members of `StdVideoEncodeH265LongTermRefPics` are interpreted as defined in section 7.4.7.1 of the [ITU-T H.265 Specification](#);
- all other members are interpreted as defined in section 7.4.7.1 of the [ITU-T H.265 Specification](#).

Reference picture setup is controlled by the value of `StdVideoEncodeH265PictureInfo::flags.is_reference`. If it is set and a [reconstructed picture](#) is specified, then the latter is used as the target of picture reconstruction to [activate](#) the [DPB slot](#) specified in `pEncodeInfo->pSetupReferenceSlot->slotIndex`. If `StdVideoEncodeH265PictureInfo::flags.is_reference` is not set, but a [reconstructed picture](#) is specified, then the corresponding picture reference associated with the [DPB slot](#) is invalidated, as described in the [DPB Slot States](#) section.

Active Parameter Sets

The members of the `StdVideoEncodeH265PictureInfo` structure pointed to by `pStdPictureInfo` are used to select the active parameter sets to use from the bound video session parameters object, as follows:

- The *active VPS* is the [VPS](#) identified by the key specified in `StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id`.
- The *active SPS* is the [SPS](#) identified by the key specified by the pair constructed from `StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id` and `StdVideoEncodeH265PictureInfo::pps_seq_parameter_set_id`.
- The *active PPS* is the [PPS](#) identified by the key specified by the triplet constructed from `StdVideoEncodeH265PictureInfo::sps_video_parameter_set_id`, `StdVideoEncodeH265PictureInfo::pps_seq_parameter_set_id`, and `StdVideoEncodeH265PictureInfo::pps_pic_parameter_set_id`.

H.265 encoding uses *explicit weighted sample prediction* for a slice segment, as defined in section 8.5.3.3.4 of the [ITU-T H.265 Specification](#), if any of the following conditions are true for the active [PPS](#) and the `pStdSliceSegmentHeader` member of the corresponding element of `pNaluSliceSegmentEntries`:

- `pStdSliceSegmentHeader->slice_type` is `STD_VIDEO_H265_SLICE_TYPE_P` and `weighted_pred_flag` is enabled in the active PPS.
- `pStdSliceSegmentHeader->slice_type` is `STD_VIDEO_H265_SLICE_TYPE_B` and `weighted_bipred_flag` is enabled in the active PPS.

The number of H.265 tiles, as defined in section 3.174 of the [ITU-T H.265 Specification](#), is derived from the `num_tile_columns_minus1` and `num_tile_rows_minus1` members of the active [PPS](#) as follows:

$$(\text{num_tile_columns_minus1} + 1) \times (\text{num_tile_rows_minus1} + 1)$$

Valid Usage

- VUID-VkVideoEncodeH265PictureInfoKHR-naluSliceSegmentEntryCount-08306
`naluSliceSegmentEntryCount` **must** be between 1 and `VkVideoEncodeH265CapabilitiesKHR`

`::maxSliceSegmentCount`, inclusive, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile

- VUID-VkVideoEncodeH265PictureInfoKHR-flags-08323
If `VkVideoEncodeH265CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_TILES_PER_SLICE_SEGMENT_BIT_KHR`, then `naluSliceSegmentEntryCount` **must** be greater than or equal to the number of H.265 tiles in the picture
- VUID-VkVideoEncodeH265PictureInfoKHR-flags-08324
If `VkVideoEncodeH265CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_SLICE_SEGMENTS_PER_TILE_BIT_KHR`, then `naluSliceSegmentEntryCount` **must** be less than or equal to the number of H.265 tiles in the picture
- VUID-VkVideoEncodeH265PictureInfoKHR-flags-08316
If `VkVideoEncodeH265CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H265_CAPABILITY_PREDICTION_WEIGHT_TABLE_GENERATED_BIT_KHR` and the slice segment corresponding to any element of `pNaluSliceSegmentEntries` uses `explicit weighted sample prediction`, then `VkVideoEncodeH265NaluSliceSegmentInfoKHR::pStdSliceSegmentHeader->pWeightTable` **must** not be `NULL` for that element of `pNaluSliceSegmentEntries`
- VUID-VkVideoEncodeH265PictureInfoKHR-flags-08317
If `VkVideoEncodeH265CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H265_CAPABILITY_DIFFERENT_SLICE_SEGMENT_TYPE_BIT_KHR`, then `VkVideoEncodeH265NaluSliceSegmentInfoKHR::pStdSliceSegmentHeader->slice_type` **must** be identical for all elements of `pNaluSliceSegmentEntries`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265PictureInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PICTURE_INFO_KHR`
- VUID-VkVideoEncodeH265PictureInfoKHR-pNaluSliceSegmentEntries-parameter
`pNaluSliceSegmentEntries` **must** be a valid pointer to an array of `naluSliceSegmentEntryCount` valid `VkVideoEncodeH265NaluSliceSegmentInfoKHR` structures
- VUID-VkVideoEncodeH265PictureInfoKHR-pStdPictureInfo-parameter
`pStdPictureInfo` **must** be a valid pointer to a valid `StdVideoEncodeH265PictureInfo` value
- VUID-VkVideoEncodeH265PictureInfoKHR-naluSliceSegmentEntryCount-arraylength
`naluSliceSegmentEntryCount` **must** be greater than 0

The `VkVideoEncodeH265NaluSliceSegmentInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265NaluSliceSegmentInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    int32_t                  constantQp;
    const StdVideoEncodeH265SliceSegmentHeader* pStdSliceSegmentHeader;
} VkVideoEncodeH265NaluSliceSegmentInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `constantQp` is the QP to use for the slice segment if the current `rate control mode` configured for the video session is `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`.
- `pStdSliceSegmentHeader` is a pointer to a `StdVideoEncodeH265SliceSegmentHeader` structure specifying [H.265 slice segment header parameters](#) for the slice segment.

Std Slice Segment Header Parameters

The members of the `StdVideoEncodeH265SliceSegmentHeader` structure pointed to by `pStdSliceSegmentHeader` are interpreted as follows:

- `flags.reserved` and `reserved1` are used only for padding purposes and are otherwise ignored;
- if `pWeightTable` is not `NULL`, then it is a pointer to a `StdVideoEncodeH265WeightTable` that is interpreted as follows:
 - `flags.luma_weight_l0_flag`, `flags.chroma_weight_l0_flag`, `flags.luma_weight_l1_flag`, and `flags.chroma_weight_l1_flag` are bitmasks where bit index `i` corresponds to `luma_weight_l0_flag[i]`, `chroma_weight_l0_flag[i]`, `luma_weight_l1_flag[i]`, and `chroma_weight_l1_flag[i]`, respectively, as defined in section 7.4.7.3 of the [ITU-T H.265 Specification](#);
 - all other members of `StdVideoEncodeH265WeightTable` are interpreted as defined in section 7.4.7.3 of the [ITU-T H.265 Specification](#);
- all other members are interpreted as defined in section 7.4.7.1 of the [ITU-T H.265 Specification](#).

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265NaluSliceSegmentInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_NALU_SLICE_SEGMENT_INFO_KHR`
- VUID-VkVideoEncodeH265NaluSliceSegmentInfoKHR-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkVideoEncodeH265NaluSliceSegmentInfoKHR-pStdSliceSegmentHeader-parameter
`pStdSliceSegmentHeader` **must** be a valid pointer to a valid `StdVideoEncodeH265SliceSegmentHeader` value

The [VkVideoEncodeH265DpbSlotInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265DpbSlotInfoKHR {
    VkStructureType          sType;
    const void*              pNext;
    const StdVideoEncodeH265ReferenceInfo* pStdReferenceInfo;
} VkVideoEncodeH265DpbSlotInfoKHR;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pStdReferenceInfo` is a pointer to a [StdVideoEncodeH265ReferenceInfo](#) structure specifying [H.265 reference information](#).

This structure is specified in the `pNext` chain of [VkVideoEncodeInfoKHR::pSetupReferenceSlot](#), if not `NULL`, and the `pNext` chain of the elements of [VkVideoEncodeInfoKHR::pReferenceSlots](#) to specify the codec-specific reference picture information for an [H.265 encode operation](#).

Active Reference Picture Information

When this structure is specified in the `pNext` chain of the elements of [VkVideoEncodeInfoKHR::pReferenceSlots](#), one element is added to the list of [active reference pictures](#) used by the video encode operation for each element of [VkVideoEncodeInfoKHR::pReferenceSlots](#) as follows:

- The image subregion used is determined according to the [H.265 Encode Picture Data Access](#) section.
- The reference picture is associated with the [DPB slot](#) index specified in the `slotIndex` member of the corresponding element of [VkVideoEncodeInfoKHR::pReferenceSlots](#).
- The reference picture is associated with the [H.265 reference information](#) provided in `pStdReferenceInfo`.

Reconstructed Picture Information

When this structure is specified in the `pNext` chain of [VkVideoEncodeInfoKHR::pSetupReferenceSlot](#), the information related to the [reconstructed picture](#) is defined as follows:

- The image subregion used is determined according to the [H.265 Encode Picture Data Access](#) section.
- If [reference picture setup](#) is requested, then the reconstructed picture is used to [activate](#) the [DPB slot](#) with the index specified in [VkVideoEncodeInfoKHR::pSetupReferenceSlot->slotIndex](#).
- The reconstructed picture is associated with the [H.265 reference information](#) provided in `pStdReferenceInfo`.

Std Reference Information

The members of the [StdVideoEncodeH265ReferenceInfo](#) structure pointed to by `pStdReferenceInfo` are interpreted as follows:

- `flags.reserved` is used only for padding purposes and is otherwise ignored;
- `flags.used_for_long_term_reference` is used to indicate whether the picture is marked as “used for long-term reference” as defined in section 8.3.2 of the [ITU-T H.265 Specification](#);
- `flags.unused_for_reference` is used to indicate whether the picture is marked as “unused for reference” as defined in section 8.3.2 of the [ITU-T H.265 Specification](#);
- `pic_type` as defined in section 7.4.3.5 of the [ITU-T H.265 Specification](#);
- `PicOrderCntVal` as defined in section 8.3.1 of the [ITU-T H.265 Specification](#);
- `TemporalId` as defined in section 7.4.2.2 of the [ITU-T H.265 Specification](#).

Valid Usage (Implicit)

- `VUID-VkVideoEncodeH265DpbSlotInfoKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_DPB_SLOT_INFO_KHR`
- `VUID-VkVideoEncodeH265DpbSlotInfoKHR-pStdReferenceInfo-parameter`
`pStdReferenceInfo` **must** be a valid pointer to a valid `StdVideoEncodeH265ReferenceInfo` value

37.18.12. H.265 Encode Rate Control

Group of Pictures

In case of H.265 encoding it is common practice to follow a regular pattern of different picture types in display order when encoding subsequent frames. This pattern is referred to as the *group of pictures* (GOP).

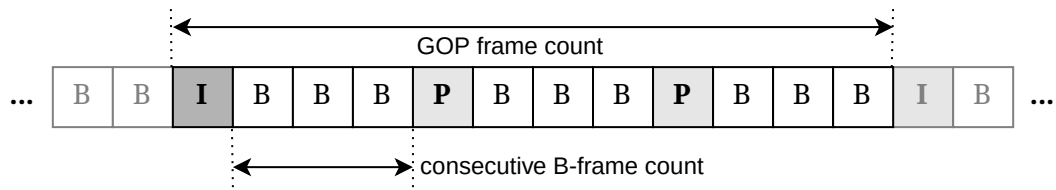
A regular GOP is defined by the following parameters:

- The number of frames in the GOP;
- The number of consecutive B frames between I and/or P frames in display order.

GOPs are further classified as *open* and *closed* GOPs.

Frame types in an open GOP follow each other in display order according to the following algorithm:

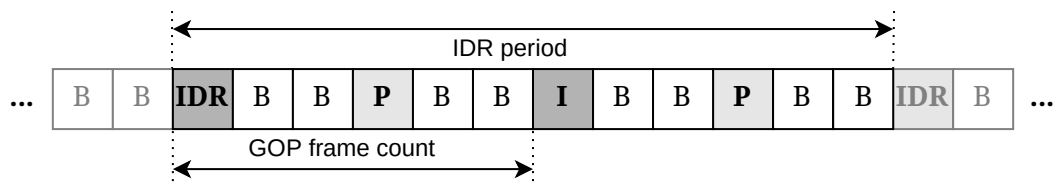
1. The first frame is always an I frame.
2. This is followed by a number of consecutive B frames, as defined above.
3. If the number of frames in the GOP is not reached yet, then the next frame is a P frame and the algorithm continues from step 2.



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Figure 31. H.265 open GOP

In case of a closed GOP, an **IDR frame** is used at a certain period.



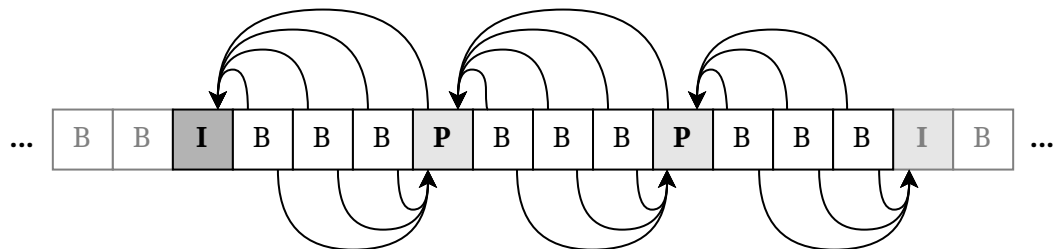
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Figure 32. H.265 closed GOP

It is also typical for H.265 encoding to use specific reference picture usage patterns across the frames of the GOP. The two most common reference patterns used are as follows:

Flat Reference Pattern

- Each P frame uses the last non-B frame, in display order, as reference.
- Each B frame uses the last non-B frame, in display order, as its forward reference, and uses the next non-B frame, in display order, as its backward reference.



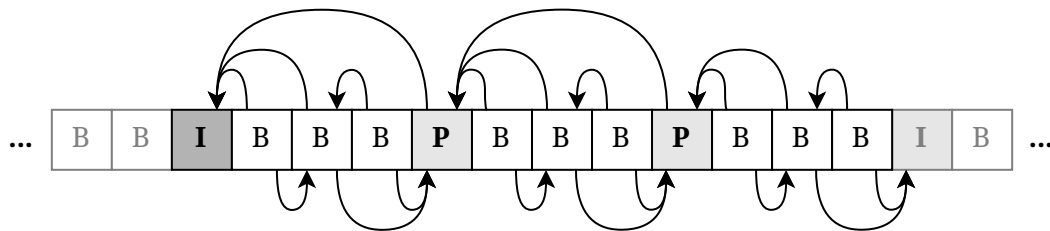
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Figure 33. H.265 flat reference pattern

Dyadic Reference Pattern

- Each P frame uses the last non-B frame, in display order, as reference.

- The following algorithm is applied to the sequence of consecutive B frames between I and/or P frames in display order:
 1. The B frame in the middle of this sequence uses the frame preceding the sequence as its forward reference, and uses the frame following the sequence as its backward reference.
 2. The algorithm is executed recursively for the following frame sequences:
 - The B frames of the original sequence preceding the frame in the middle, if any.
 - The B frames of the original sequence following the frame in the middle, if any.



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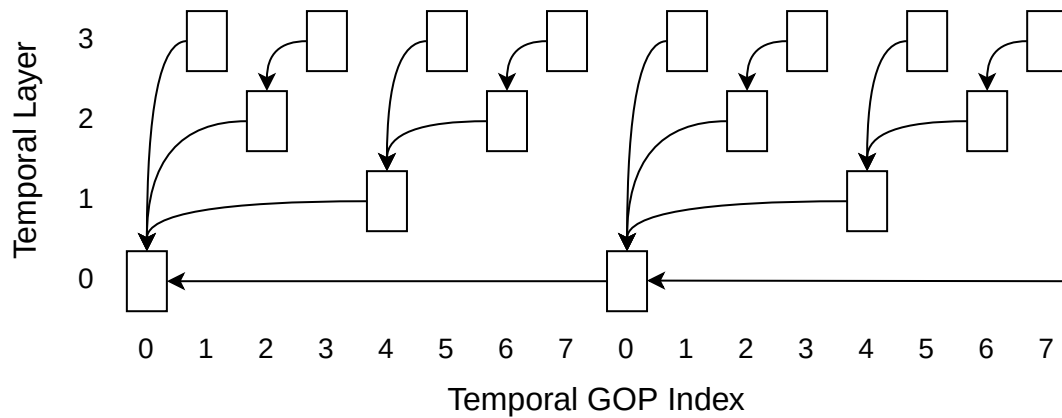
Figure 34. H.265 dyadic reference pattern

The application **can** provide guidance to the implementation's rate control algorithm about the structure of the GOP used by the application. Any such guidance about the GOP and its structure does not mandate that specific GOP structure to be used by the application, as the picture type of individual encoded pictures is still application-controlled, however, any deviation from the provided guidance **may** result in undesired rate control behavior including, but not limited, to the implementation not being able to conform to the expected average or target bitrates, or other rate control parameters specified by the application.

When an H.265 encode session is used to encode multiple temporal sub-layers, it is also common practice to follow a regular pattern for the H.265 temporal ID for the encoded pictures in display order when encoding subsequent frames. This pattern is referred to as the *temporal GOP*. The most common temporal layer pattern used is as follows:

Dyadic Temporal Sub-Layer Pattern

- The number of frames in the temporal GOP is 2^{n-1} , where n is the number of temporal sub-layers.
- The i^{th} frame in the temporal GOP uses temporal ID t , if and only if the index of the least significant bit set in i equals $n-t-1$, except for the first frame, which is the only frame in the temporal GOP using temporal ID zero.
- The i^{th} frame in the temporal GOP uses the r^{th} frame as reference, where r is calculated from i by clearing the least significant bit set in it, except for the first frame in the temporal GOP, which uses the first frame of the previous temporal GOP, if any, as reference.



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Figure 35. H.265 dyadic temporal sub-layer pattern

NOTE

Multi-layer rate control and multi-layer coding are typically used for streaming cases where low latency is expected, hence B pictures with backward prediction are usually not used.

The `VkVideoEncodeH265RateControlInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265RateControlInfoKHR {
    VkStructureType           sType;
    const void*               pNext;
    VkVideoEncodeH265RateControlFlagsKHR flags;
    uint32_t                  gopFrameCount;
    uint32_t                  idrPeriod;
    uint32_t                  consecutiveBFrameCount;
    uint32_t                  subLayerCount;
} VkVideoEncodeH265RateControlInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkVideoEncodeH265RateControlFlagBitsKHR` specifying H.265 rate control flags.
- `gopFrameCount` is the number of frames within a [group of pictures \(GOP\)](#) intended to be used by the application. If it is 0, the rate control algorithm **may** assume an implementation-dependent GOP length. If it is `UINT32_MAX`, the GOP length is treated as infinite.
- `idrPeriod` is the interval, in terms of number of frames, between two [IDR frames](#) (see [IDR period](#)). If it is 0, the rate control algorithm **may** assume an implementation-dependent IDR period. If it is `UINT32_MAX`, the IDR period is treated as infinite.
- `consecutiveBFrameCount` is the number of consecutive B frames between I and/or P frames within

the [GOP](#).

- `subLayerCount` specifies the number of H.265 sub-layers that the application intends to use.

When an instance of this structure is included in the `pNext` chain of the `VkVideoCodingControlInfoKHR` structure passed to the `vkCmdControlVideoCodingKHR` command, and `VkVideoCodingControlInfoKHR::flags` includes `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR`, the parameters in this structure are used as guidance for the implementation's rate control algorithm (see [Video Coding Control](#)).

If `flags` includes `VK_VIDEO_ENCODE_H265_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR`, then the rate control state is reset to an initial state to meet HRD compliance requirements. Otherwise the new rate control state **may** be applied without a reset depending on the implementation and the specified rate control parameters.

NOTE

It would be possible to infer the picture type to be used when encoding a frame, on the basis of the values provided for `consecutiveBFrameCount`, `idrPeriod`, and `gopFrameCount`, but this inferred picture type will not be used by implementations to override the picture type provided to the video encode operation.

Valid Usage

- VUID-VkVideoEncodeH265RateControlInfoKHR-flags-08291
If `VkVideoEncodeH265CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H265_CAPABILITY_HRD_COMPLIANCE_BIT_KHR`, then `flags` **must** not contain `VK_VIDEO_ENCODE_H265_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR`
- VUID-VkVideoEncodeH265RateControlInfoKHR-flags-08292
If `flags` contains `VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR` or `VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR`, then it **must** also contain `VK_VIDEO_ENCODE_H265_RATE_CONTROL_REGULAR_GOP_BIT_KHR`
- VUID-VkVideoEncodeH265RateControlInfoKHR-flags-08293
If `flags` contains `VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR`, then it **must** not also contain `VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR`
- VUID-VkVideoEncodeH265RateControlInfoKHR-flags-08294
If `flags` contains `VK_VIDEO_ENCODE_H265_RATE_CONTROL_REGULAR_GOP_BIT_KHR`, then `gopFrameCount` **must** be greater than 0
- VUID-VkVideoEncodeH265RateControlInfoKHR-idrPeriod-08295
If `idrPeriod` is not 0, then it **must** be greater than or equal to `gopFrameCount`
- VUID-VkVideoEncodeH265RateControlInfoKHR-consecutiveBFrameCount-08296
If `consecutiveBFrameCount` is not 0, then it **must** be less than `gopFrameCount`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265RateControlInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_INFO_KHR`
- VUID-VkVideoEncodeH265RateControlInfoKHR-flags-parameter
`flags` **must** be a valid combination of `VkVideoEncodeH265RateControlFlagBitsKHR` values

Bits which **can** be set in `VkVideoEncodeH265RateControlInfoKHR::flags`, specifying H.265 rate control flags, are:

```
// Provided by VK_KHR_video_encode_h265
typedef enum VkVideoEncodeH265RateControlFlagBitsKHR {
    VK_VIDEO_ENCODE_H265_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR = 0x00000001,
    VK_VIDEO_ENCODE_H265_RATE_CONTROL_REGULAR_GOP_BIT_KHR = 0x00000002,
    VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR = 0x00000004,
    VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR = 0x00000008,
    VK_VIDEO_ENCODE_H265_RATE_CONTROL_TEMPORAL_SUB_LAYER_PATTERN_DYADIC_BIT_KHR =
    0x00000010,
} VkVideoEncodeH265RateControlFlagBitsKHR;
```

- `VK_VIDEO_ENCODE_H265_RATE_CONTROL_ATTEMPT_HRD_COMPLIANCE_BIT_KHR` specifies that rate control **should** attempt to produce an HRD compliant bitstream, as defined in annex C of the [ITU-T H.265 Specification](#).
- `VK_VIDEO_ENCODE_H265_RATE_CONTROL_REGULAR_GOP_BIT_KHR` specifies that the application intends to use a [regular GOP structure](#) according to the parameters specified in the `gopFrameCount`, `idrPeriod`, and `consecutiveBFrameCount` members of the `VkVideoEncodeH265RateControlInfoKHR` structure.
- `VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_FLAT_BIT_KHR` specifies that the application intends to follow a [flat reference pattern](#) in the GOP.
- `VK_VIDEO_ENCODE_H265_RATE_CONTROL_REFERENCE_PATTERN_DYADIC_BIT_KHR` specifies that the application intends to follow a [dyadic reference pattern](#) in the GOP.
- `VK_VIDEO_ENCODE_H265_RATE_CONTROL_TEMPORAL_SUB_LAYER_PATTERN_DYADIC_BIT_KHR` specifies that the application intends to follow a [dyadic temporal sub-layer pattern](#).

```
// Provided by VK_KHR_video_encode_h265
typedef VkFlags VkVideoEncodeH265RateControlFlagsKHR;
```

`VkVideoEncodeH265RateControlFlagsKHR` is a bitmask type for setting a mask of zero or more `VkVideoEncodeH265RateControlFlagBitsKHR`.

Rate Control Layers

The `VkVideoEncodeH265RateControlLayerInfoKHR` structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265RateControlLayerInfoKHR {
    VkStructureType      sType;
    const void*          pNext;
    VkBool32             useMinQp;
    VkVideoEncodeH265QpKHR minQp;
    VkBool32             useMaxQp;
    VkVideoEncodeH265QpKHR maxQp;
    VkBool32             useMaxFrameSize;
    VkVideoEncodeH265FrameSizeKHR maxFrameSize;
} VkVideoEncodeH265RateControlLayerInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `useMinQp` indicates whether the QP values determined by rate control will be clamped to the lower bounds on the QP values specified in `minQp`.
- `minQp` specifies the lower bounds on the QP values, for each picture type, that the implementation's rate control algorithm will use when `useMinQp` is `VK_TRUE`.
- `useMaxQp` indicates whether the QP values determined by rate control will be clamped to the upper bounds on the QP values specified in `maxQp`.
- `maxQp` specifies the upper bounds on the QP values, for each picture type, that the implementation's rate control algorithm will use when `useMaxQp` is `VK_TRUE`.
- `useMaxFrameSize` indicates whether the implementation's rate control algorithm **should** use the values specified in `maxFrameSize` as the upper bounds on the encoded frame size for each picture type.
- `maxFrameSize` specifies the upper bounds on the encoded frame size, for each picture type, when `useMaxFrameSize` is `VK_TRUE`.

When used, the values in `minQp` and `maxQp` guarantee that the effective QP values used by the implementation will respect those lower and upper bounds, respectively. However, limiting the range of QP values that the implementation is able to use will also limit the capabilities of the implementation's rate control algorithm to comply to other constraints. In particular, the implementation **may** not be able to comply to the following:

- The average and/or peak `bitrate` values to be used for the encoded bitstream specified in the `averageBitrate` and `maxBitrate` members of the `VkVideoEncodeRateControlLayerInfoKHR` structure.
- The upper bounds on the encoded frame size, for each picture type, specified in the `maxFrameSize` member of `VkVideoEncodeH265RateControlLayerInfoKHR`.

NOTE

In general, applications need to configure rate control parameters appropriately in order to be able to get the desired rate control behavior, as described in the [Video Encode Rate Control](#) section.

When an instance of this structure is included in the `pNext` chain of a

`VkVideoEncodeRateControlLayerInfoKHR` structure specified in one of the elements of the `players` array member of the `VkVideoEncodeRateControlInfoKHR` structure passed to the `vkCmdControlVideoCodingKHR` command, `VkVideoCodingControlInfoKHR::flags` includes `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR`, and the bound video session was created with the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, it specifies the H.265-specific rate control parameters of the rate control layer corresponding to that element of `players`.

Valid Usage

- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-useMinQp-08297
If `useMinQp` is `VK_TRUE`, then the `qpI`, `qpP`, and `qpB` members of `minQp` **must** all be between `VkVideoEncodeH265CapabilitiesKHR::minQp` and `VkVideoEncodeH265CapabilitiesKHR::maxQp`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-useMaxQp-08298
If `useMaxQp` is `VK_TRUE`, then the `qpI`, `qpP`, and `qpB` members of `maxQp` **must** all be between `VkVideoEncodeH265CapabilitiesKHR::minQp` and `VkVideoEncodeH265CapabilitiesKHR::maxQp`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-useMinQp-08299
If `useMinQp` is `VK_TRUE` and `VkVideoEncodeH265CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H265_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR`, then the `qpI`, `qpP`, and `qpB` members of `minQp` **must** all specify the same value
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-useMaxQp-08300
If `useMaxQp` is `VK_TRUE` and `VkVideoEncodeH265CapabilitiesKHR::flags`, as returned by `vkGetPhysicalDeviceVideoCapabilitiesKHR` for the used video profile, does not include `VK_VIDEO_ENCODE_H265_CAPABILITY_PER_PICTURE_TYPE_MIN_MAX_QP_BIT_KHR`, then the `qpI`, `qpP`, and `qpB` members of `maxQp` **must** all specify the same value
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-useMinQp-08375
If `useMinQp` and `useMaxQp` are both `VK_TRUE`, then the `qpI`, `qpP`, and `qpB` members of `minQp` **must** all be less than or equal to the respective members of `maxQp`

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_LAYER_INFO_KHR`
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-minQp-parameter
`minQp` **must** be a valid `VkVideoEncodeH265QpKHR` structure
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-maxQp-parameter
`maxQp` **must** be a valid `VkVideoEncodeH265QpKHR` structure
- VUID-VkVideoEncodeH265RateControlLayerInfoKHR-maxFrameSize-parameter

maxFrameSize must be a valid [VkVideoEncodeH265FrameSizeKHR](#) structure

The [VkVideoEncodeH265QpKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265QpKHR {
    int32_t    qpI;
    int32_t    qpP;
    int32_t    qpB;
} VkVideoEncodeH265QpKHR;
```

- **qpI** is the QP to be used for [I pictures](#).
- **qpP** is the QP to be used for [P pictures](#).
- **qpB** is the QP to be used for [B pictures](#).

The [VkVideoEncodeH265FrameSizeKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265FrameSizeKHR {
    uint32_t    frameISize;
    uint32_t    framePSize;
    uint32_t    frameBSize;
} VkVideoEncodeH265FrameSizeKHR;
```

- **frameISize** is the size in bytes to be used for [I frames](#).
- **framePSize** is the size in bytes to be used for [P frames](#).
- **frameBSize** is the size in bytes to be used for [B frames](#).

GOP Remaining Frames

Besides session level rate control configuration, the application **can** specify the number of frames per frame type remaining in the [group of pictures \(GOP\)](#).

The [VkVideoEncodeH265GopRemainingFrameInfoKHR](#) structure is defined as:

```
// Provided by VK_KHR_video_encode_h265
typedef struct VkVideoEncodeH265GopRemainingFrameInfoKHR {
    VkStructureType    sType;
    const void*        pNext;
    VkBool32           useGopRemainingFrames;
    uint32_t           gopRemainingI;
    uint32_t           gopRemainingP;
    uint32_t           gopRemainingB;
} VkVideoEncodeH265GopRemainingFrameInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `useGopRemainingFrames` indicates whether the implementation's rate control algorithm **should** use the values specified in `gopRemainingI`, `gopRemainingP`, and `gopRemainingB`. If `useGopRemainingFrames` is `VK_FALSE`, then the values of `gopRemainingI`, `gopRemainingP`, and `gopRemainingB` are ignored.
- `gopRemainingI` specifies the number of `I frames` the implementation's rate control algorithm **should** assume to be remaining in the `GOP` prior to executing the video encode operation.
- `gopRemainingP` specifies the number of `P frames` the implementation's rate control algorithm **should** assume to be remaining in the `GOP` prior to executing the video encode operation.
- `gopRemainingB` specifies the number of `B frames` the implementation's rate control algorithm **should** assume to be remaining in the `GOP` prior to executing the video encode operation.

Setting `useGopRemainingFrames` to `VK_TRUE` and including this structure in the `pNext` chain of `VkVideoBeginCodingInfoKHR` is only mandatory if the `VkVideoEncodeH265CapabilitiesKHR::requiresGopRemainingFrames` reported for the used `video profile` is `VK_TRUE`. However, implementations **may** use these remaining frame counts, when specified, even when it is not required. In particular, when the application does not use a `regular GOP structure`, these values **may** provide additional guidance for the implementation's rate control algorithm.

The `VkVideoEncodeH265CapabilitiesKHR::prefersGopRemainingFrames` capability is also used to indicate that the implementation's rate control algorithm **may** operate more accurately if the application specifies the remaining frame counts using this structure.

As with other rate control guidance values, if the effective order and number of frames encoded by the application are not in line with the remaining frame counts specified in this structure at any given point, then the behavior of the implementation's rate control algorithm **may** deviate from the one expected by the application.

Valid Usage (Implicit)

- VUID-VkVideoEncodeH265GopRemainingFrameInfoKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_GOP_REMAINING_FRAME_INFO_KHR`

37.18.13. H.265 Encode Requirements

This section described the **required** H.265 encoding capabilities for physical devices that have at least one queue family that supports the video codec operation `VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR`, as returned by `vkGetPhysicalDeviceQueueFamilyProperties2KHR` in `VkQueueFamilyVideoPropertiesKHR::videoCodecOperations`.

Table 54. Required *Video Std Header Versions*

Video Std Header Name	Version
<code>vulkan_video_codec_h265std_encode</code>	1.0.0

Table 55. Required Video Capabilities

Video Capability	Requirement	Requirement Type ¹
VkVideoCapabilitiesKHR		
flags	-	min
minBitstreamBufferOffsetAlignment	4096	max
minBitstreamBufferSizeAlignment	4096	max
pictureAccessGranularity	(64,64)	max
minCodedExtent	-	max
maxCodedExtent	-	min
maxDpbSlots	0	min
maxActiveReferencePictures	0	min
VkVideoEncodeCapabilitiesKHR		
flags	-	min
rateControlModes	-	min
maxBitrate	128000	min
maxQualityLevels	1	min
encodeInputPictureGranularity	(64,64)	max
supportedEncodeFeedbackFlags	VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BUFFER_OFFSET_BIT_KHR VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_BYTES_WRITTEN_BIT_KHR	min
VkVideoEncodeH265CapabilitiesKHR		
flags	-	min
maxLevelIdc	STD_VIDEO_H265_LEVEL_IDC_1_0	min
maxSliceSegmentCount	1	min
maxTiles	(1,1)	min
ctbSizes	at least one bit set	implementation-dependent
transformBlockSizes	at least one bit set	implementation-dependent
maxPPictureL0ReferenceCount	0	min
maxBPictureL0ReferenceCount	0	min
maxL1ReferenceCount	0	min
maxSubLayerCount	1	min

Video Capability	Requirement	Requirement Type ¹
expectDyadicTemporalSubLayerPattern	-	implementation-dependent
minQp	-	max
maxQp	-	min
prefersGopRemainingFrames	-	implementation-dependent
requiresGopRemainingFrames	-	implementation-dependent
stdSyntaxFlags	-	min

1

The **Requirement Type** column specifies the requirement is either the minimum value all implementations **must** support, the maximum value all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum value is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

Chapter 38. Extending Vulkan

New functionality **may** be added to Vulkan via either new extensions or new versions of the core, or new versions of an extension in some cases.

This chapter describes how Vulkan is versioned, how compatibility is affected between different versions, and compatibility rules that are followed by the Vulkan Working Group.

38.1. Instance and Device Functionality

Commands that enumerate instance properties, or that accept a [VkInstance](#) object as a parameter, are considered instance-level functionality.

Commands that dispatch from a [VkDevice](#) object or a child object of a [VkDevice](#), or take any of them as a parameter, are considered device-level functionality. Types defined by a [device extension](#) are also considered device-level functionality.

Commands that dispatch from [VkPhysicalDevice](#), or accept a [VkPhysicalDevice](#) object as a parameter, are considered either instance-level or device-level functionality depending if the functionality is specified by an [instance extension](#) or [device extension](#) respectively.

Additionally, commands that enumerate physical device properties are considered device-level functionality.

NOTE

Vulkan 1.0 initially specified new physical device enumeration functionality as instance-level, requiring it to be included in an instance extension. As the capabilities of device-level functionality require discovery via physical device enumeration, this led to the situation where many device extensions required an instance extension as well. To alleviate this extra work, [VK_KHR_get_physical_device_properties2](#) (and subsequently Vulkan 1.1) redefined device-level functionality to include physical device enumeration.

38.2. Core Versions

The Vulkan Specification is regularly updated with bug fixes and clarifications. Occasionally new functionality is added to the core and at some point it is expected that there will be a desire to perform a large, breaking change to the API. In order to indicate to developers how and when these changes are made to the specification, and to provide a way to identify each set of changes, the Vulkan API maintains a version number.

38.2.1. Version Numbers

The Vulkan version number comprises four parts indicating the variant, major, minor and patch version of the Vulkan API Specification.

The *variant* indicates the variant of the Vulkan API supported by the implementation. This is always 0 for the Vulkan API.

NOTE

A non-zero variant indicates the API is a variant of the Vulkan API and applications will typically need to be modified to run against it. The variant field was a later addition to the version number, added in version 1.2.175 of the Specification. As Vulkan uses variant 0, this change is fully backwards compatible with the previous version number format for Vulkan implementations. New version number macros have been added for this change and the old macros deprecated. For existing applications using the older format and macros, an implementation with non-zero variant will decode as a very high Vulkan version. The high version number should be detectable by applications performing suitable version checking.

The *major version* indicates a significant change in the API, which will encompass a wholly new version of the specification.

The *minor version* indicates the incorporation of new functionality into the core specification.

The *patch version* indicates bug fixes, clarifications, and language improvements have been incorporated into the specification.

Compatibility guarantees made about versions of the API sharing any of the same version numbers are documented in [Core Versions](#)

The version number is used in several places in the API. In each such use, the version numbers are packed into a 32-bit integer as follows:

- The variant is a 3-bit integer packed into bits 31-29.
- The major version is a 7-bit integer packed into bits 28-22.
- The minor version number is a 10-bit integer packed into bits 21-12.
- The patch version number is a 12-bit integer packed into bits 11-0.

VK_API_VERSION_VARIANT extracts the API variant number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_VARIANT(version) ((uint32_t)(version) >> 29U)
```

VK_API_VERSION_MAJOR extracts the API major version number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_MAJOR(version) (((uint32_t)(version) >> 22U) & 0x7FU)
```

VK_VERSION_MAJOR extracts the API major version number from a packed version number:

```
// Provided by VK_VERSION_1_0
// VK_VERSION_MAJOR is deprecated, but no reason was given in the API XML
// DEPRECATED: This define is deprecated. VK_API_VERSION_MAJOR should be used instead.
#define VK_VERSION_MAJOR(version) ((uint32_t)(version) >> 22U)
```


VK_API_VERSION_MINOR extracts the API minor version number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_MINOR(version) (((uint32_t)(version) >> 12U) & 0x3FFU)
```

VK_VERSION_MINOR extracts the API minor version number from a packed version number:

```
// Provided by VK_VERSION_1_0
// VK_VERSION_MINOR is deprecated, but no reason was given in the API XML
// DEPRECATED: This define is deprecated. VK_API_VERSION_MINOR should be used instead.
#define VK_VERSION_MINOR(version) (((uint32_t)(version) >> 12U) & 0x3FFU)
```

VK_API_VERSION_PATCH extracts the API patch version number from a packed version number:

```
// Provided by VK_VERSION_1_0
#define VK_API_VERSION_PATCH(version) ((uint32_t)(version) & 0xFFFU)
```

VK_VERSION_PATCH extracts the API patch version number from a packed version number:

```
// Provided by VK_VERSION_1_0
// VK_VERSION_PATCH is deprecated, but no reason was given in the API XML
// DEPRECATED: This define is deprecated. VK_API_VERSION_PATCH should be used instead.
#define VK_VERSION_PATCH(version) ((uint32_t)(version) & 0xFFFU)
```

VK_MAKE_API_VERSION constructs an API version number.

```
// Provided by VK_VERSION_1_0
#define VK_MAKE_API_VERSION(variant, major, minor, patch) \
    (((uint32_t)(variant)) << 29U | (((uint32_t)(major)) << 22U) | \
    (((uint32_t)(minor)) << 12U) | ((uint32_t)(patch)))
```

- **variant** is the variant number.
- **major** is the major version number.
- **minor** is the minor version number.
- **patch** is the patch version number.

VK_MAKE_VERSION constructs an API version number.

```
// Provided by VK_VERSION_1_0
// VK_MAKE_VERSION is deprecated, but no reason was given in the API XML
// DEPRECATED: This define is deprecated. VK_MAKE_API_VERSION should be used instead.
#define VK_MAKE_VERSION(major, minor, patch) \
    (((uint32_t)(major)) << 22U) | (((uint32_t)(minor)) << 12U) | \
    ((uint32_t)(patch)))
```

- **major** is the major version number.
- **minor** is the minor version number.
- **patch** is the patch version number.

VK_API_VERSION_1_0 returns the API version number for Vulkan 1.0.0.

```
// Provided by VK_VERSION_1_0
// Vulkan 1.0 version number
#define VK_API_VERSION_1_0 VK_MAKE_API_VERSION(0, 1, 0, 0)// Patch version should
always be set to 0
```

38.2.2. Querying Version Support

NOTE

In Vulkan 1.0, there is no mechanism to detect the separate versions of [instance-level](#) and [device-level functionality](#) supported. However, the [vkEnumerateInstanceVersion](#) command was added in Vulkan 1.1 to determine the supported version of instance-level functionality - querying for this function via [vkGetInstanceProcAddr](#) will return **NULL** on implementations that only support Vulkan 1.0 functionality. For more information on this, please refer to the Vulkan 1.1 specification.

The version of device-level functionality can be queried by calling [vkGetPhysicalDeviceProperties](#) or [vkGetPhysicalDeviceProperties2KHR](#), and is returned in [VkPhysicalDeviceProperties::apiVersion](#), encoded as described in [Version Numbers](#).

38.3. Layers

When a layer is enabled, it inserts itself into the call chain for Vulkan commands the layer is interested in. Layers **can** be used for a variety of tasks that extend the base behavior of Vulkan beyond what is required by the specification - such as call logging, tracing, validation, or providing additional extensions.

NOTE

For example, an implementation is not expected to check that the value of enums used by the application fall within allowed ranges. Instead, a validation layer would do those checks and flag issues. This avoids a performance penalty during production use of the application because those layers would not be enabled in production.

NOTE

Vulkan layers **may** wrap object handles (i.e. return a different handle value to the application than that generated by the implementation). This is generally discouraged, as it increases the probability of incompatibilities with new extensions. The validation layers wrap handles in order to track the proper use and destruction of each object. See the [“Architecture of the Vulkan Loader Interfaces”](#) document for additional information.

To query the available layers, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateInstanceLayerProperties(
    uint32_t*                pPropertyCount,
    VkLayerProperties*        pProperties);
```

- `pPropertyCount` is a pointer to an integer related to the number of layer properties available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkLayerProperties` structures.

If `pProperties` is `NULL`, then the number of layer properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of layer properties available, at most `pPropertyCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available properties were returned.

The list of available layers may change at any time due to actions outside of the Vulkan implementation, so two calls to `vkEnumerateInstanceLayerProperties` with the same parameters **may** return different results, or retrieve different `pPropertyCount` values or `pProperties` contents. Once an instance has been created, the layers enabled for that instance will continue to be enabled and valid for the lifetime of that instance, even if some of them become unavailable for future instances.

Valid Usage (Implicit)

- VUID-vkEnumerateInstanceLayerProperties-pPropertyCount-parameter `pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkEnumerateInstanceLayerProperties-pProperties-parameter
If the value referenced by `pPropertyCount` is not 0, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkLayerProperties` structures

Return Codes

Success

- `VK_SUCCESS`

- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkLayerProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkLayerProperties {
    char        layerName[VK_MAX_EXTENSION_NAME_SIZE];
    uint32_t    specVersion;
    uint32_t    implementationVersion;
    char        description[VK_MAX_DESCRIPTION_SIZE];
} VkLayerProperties;
```

- `layerName` is an array of `VK_MAX_EXTENSION_NAME_SIZE` `char` containing a null-terminated UTF-8 string which is the name of the layer. Use this name in the `ppEnabledLayerNames` array passed in the `VkInstanceCreateInfo` structure to enable this layer for an instance.
- `specVersion` is the Vulkan version the layer was written to, encoded as described in [Version Numbers](#).
- `implementationVersion` is the version of this layer. It is an integer, increasing with backward compatible changes.
- `description` is an array of `VK_MAX_DESCRIPTION_SIZE` `char` containing a null-terminated UTF-8 string which provides additional details that **can** be used by the application to identify the layer.

`VK_MAX_EXTENSION_NAME_SIZE` is the length in `char` values of an array containing a layer or extension name string, as returned in `VkLayerProperties::layerName`, `VkExtensionProperties::extensionName`, and other queries.

```
#define VK_MAX_EXTENSION_NAME_SIZE    256U
```

`VK_MAX_DESCRIPTION_SIZE` is the length in `char` values of an array containing a string with additional descriptive information about a query, as returned in `VkLayerProperties::description` and other queries.

```
#define VK_MAX_DESCRIPTION_SIZE    256U
```

To enable a layer, the name of the layer **should** be added to the `ppEnabledLayerNames` member of `VkInstanceCreateInfo` when creating a `VkInstance`.

Loader implementations **may** provide mechanisms outside the Vulkan API for enabling specific layers. Layers enabled through such a mechanism are *implicitly enabled*, while layers enabled by including the layer name in the `ppEnabledLayerNames` member of `VkInstanceCreateInfo` are *explicitly*

enabled. Implicitly enabled layers are loaded before explicitly enabled layers, such that implicitly enabled layers are closer to the application, and explicitly enabled layers are closer to the driver. Except where otherwise specified, implicitly enabled and explicitly enabled layers differ only in the way they are enabled, and the order in which they are loaded. Explicitly enabling a layer that is implicitly enabled results in this layer being loaded as an implicitly enabled layer; it has no additional effect.

38.3.1. Device Layer Deprecation

Previous versions of this specification distinguished between instance and device layers. Instance layers were only able to intercept commands that operate on `VkInstance` and `VkPhysicalDevice`, except they were not able to intercept `vkCreateDevice`. Device layers were enabled for individual devices when they were created, and could only intercept commands operating on that device or its child objects.

Device-only layers are now deprecated, and this specification no longer distinguishes between instance and device layers. Layers are enabled during instance creation, and are able to intercept all commands operating on that instance or any of its child objects. At the time of deprecation there were no known device-only layers and no compelling reason to create one.

In order to maintain compatibility with implementations released prior to device-layer deprecation, applications **should** still enumerate and enable device layers. The behavior of `vkEnumerateDeviceLayerProperties` and valid usage of the `ppEnabledLayerNames` member of `VkDeviceCreateInfo` maximizes compatibility with applications written to work with the previous requirements.

To enumerate device layers, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateDeviceLayerProperties(
    VkPhysicalDevice          physicalDevice,
    uint32_t*                 pPropertyCount,
    VkLayerProperties*         pProperties);
```

- `physicalDevice` is the physical device that will be queried.
- `pPropertyCount` is a pointer to an integer related to the number of layer properties available or queried.
- `pProperties` is either `NULL` or a pointer to an array of `VkLayerProperties` structures.

If `pProperties` is `NULL`, then the number of layer properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of layer properties available, at most `pPropertyCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available properties were returned.

The list of layers enumerated by `vkEnumerateDeviceLayerProperties` **must** be exactly the sequence of layers enabled for the instance. The members of `VkLayerProperties` for each enumerated layer **must**

be the same as the properties when the layer was enumerated by `vkEnumerateInstanceLayerProperties`.

NOTE

Due to platform details on Android, `vkEnumerateDeviceLayerProperties` may be called with `physicalDevice` equal to `NULL` during layer discovery. This behavior will only be observed by layer implementations, and not the underlying Vulkan driver.

Valid Usage (Implicit)

- `VOID-vkEnumerateDeviceLayerProperties-physicalDevice-parameter`
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- `VOID-vkEnumerateDeviceLayerProperties-pPropertyCount-parameter`
`pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- `VOID-vkEnumerateDeviceLayerProperties-pProperties-parameter`
If the value referenced by `pPropertyCount` is not `0`, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkLayerProperties` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `ppEnabledLayerNames` and `enabledLayerCount` members of `VkDeviceCreateInfo` are deprecated and their values **must** be ignored by implementations. However, for compatibility, only an empty list of layers or a list that exactly matches the sequence enabled at instance creation time are valid, and validation layers **should** issue diagnostics for other cases.

Regardless of the enabled layer list provided in `VkDeviceCreateInfo`, the sequence of layers active for a device will be exactly the sequence of layers enabled when the parent instance was created.

38.4. Extensions

Extensions **may** define new Vulkan commands, structures, and enumerants. For compilation purposes, the interfaces defined by registered extensions, including new structures and enumerants as well as function pointer types for new commands, are defined in the Khronos-supplied `vulkan_core.h` together with the core API. However, commands defined by extensions **may** not be available for static linking - in which case function pointers to these commands **should** be queried at runtime as described in [Command Function Pointers](#). Extensions **may** be provided by layers as well as by a Vulkan implementation.

Because extensions **may** extend or change the behavior of the Vulkan API, extension authors **should** add support for their extensions to the Khronos validation layers. This is especially important for new commands whose parameters have been wrapped by the validation layers. See the “[Architecture of the Vulkan Loader Interfaces](#)” document for additional information.

NOTE

To enable an instance extension, the name of the extension **can** be added to the `ppEnabledExtensionNames` member of `VkInstanceCreateInfo` when creating a `VkInstance`.

To enable a device extension, the name of the extension **can** be added to the `ppEnabledExtensionNames` member of `VkDeviceCreateInfo` when creating a `VkDevice`.

Physical-Device-Level functionality does not have any enabling mechanism and **can** be used as long as the `VkPhysicalDevice` supports the device extension as determined by `vkEnumerateDeviceExtensionProperties`.

Enabling an extension (with no further use of that extension) does not change the behavior of functionality exposed by the core Vulkan API or any other extension, other than making valid the use of the commands, enums and structures defined by that extension.

Valid Usage sections for individual commands and structures do not currently contain which extensions have to be enabled in order to make their use valid, although they might do so in the future. It is defined only in the [Valid Usage for Extensions](#) section.

38.4.1. Instance Extensions

Instance extensions add new [instance-level functionality](#) to the API, outside of the core specification.

To query the available instance extensions, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateInstanceExtensionProperties(
    const char*          pLayerName,
    uint32_t*            pPropertyCount,
    VkExtensionProperties* pProperties);
```

- `pLayerName` is either `NULL` or a pointer to a null-terminated UTF-8 string naming the layer to retrieve extensions from.
- `pPropertyCount` is a pointer to an integer related to the number of extension properties available or queried, as described below.
- `pProperties` is either `NULL` or a pointer to an array of `VkExtensionProperties` structures.

When `pLayerName` parameter is `NULL`, only extensions provided by the Vulkan implementation or by implicitly enabled layers are returned. When `pLayerName` is the name of a layer, the instance extensions provided by that layer are returned.

If `pProperties` is `NULL`, then the number of extensions properties available is returned in `pPropertyCount`. Otherwise, `pPropertyCount` **must** point to a variable set by the application to the number of elements in the `pProperties` array, and on return the variable is overwritten with the number of structures actually written to `pProperties`. If `pPropertyCount` is less than the number of extension properties available, at most `pPropertyCount` structures will be written, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available properties were returned.

Because the list of available layers may change externally between calls to `vkEnumerateInstanceExtensionProperties`, two calls may retrieve different results if a `pLayerName` is available in one call but not in another. The extensions supported by a layer may also change between two calls, e.g. if the layer implementation is replaced by a different version between those calls.

Implementations **must** not advertise any pair of extensions that cannot be enabled together due to behavioral differences, or any extension that cannot be enabled against the advertised version.

Valid Usage (Implicit)

- VUID-vkEnumerateInstanceExtensionProperties-pLayerName-parameter
If `pLayerName` is not `NULL`, `pLayerName` **must** be a null-terminated UTF-8 string
- VUID-vkEnumerateInstanceExtensionProperties-pPropertyCount-parameter
`pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkEnumerateInstanceExtensionProperties-pProperties-parameter
If the value referenced by `pPropertyCount` is not 0, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkExtensionProperties` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_LAYER_NOT_PRESENT`

38.4.2. Device Extensions

Device extensions add new [device-level functionality](#) to the API, outside of the core specification.

To query the extensions available to a given physical device, call:

```
// Provided by VK_VERSION_1_0
VkResult vkEnumerateDeviceExtensionProperties(
    VkPhysicalDevice          physicalDevice,
    const char*               pLayerName,
    uint32_t*                 pPropertyCount,
    VkExtensionProperties*     pProperties);
```

- `physicalDevice` is the physical device that will be queried.
- `pLayerName` is either `NULL` or a pointer to a null-terminated UTF-8 string naming the layer to retrieve extensions from.
- `pPropertyCount` is a pointer to an integer related to the number of extension properties available or queried, and is treated in the same fashion as the `vkEnumerateInstanceExtensionProperties::pPropertyCount` parameter.
- `pProperties` is either `NULL` or a pointer to an array of `VkExtensionProperties` structures.

When `pLayerName` parameter is `NULL`, only extensions provided by the Vulkan implementation or by implicitly enabled layers are returned. When `pLayerName` is the name of a layer, the device extensions provided by that layer are returned.

Implementations **must** not advertise any pair of extensions that cannot be enabled together due to behavioral differences, or any extension that cannot be enabled against the advertised version.

If `VK_KHR_ray_tracing_pipeline` is advertised as supported by this query, `VK_KHR_pipeline_library` **must** also be supported.

NOTE

Due to platform details on Android, `vkEnumerateDeviceExtensionProperties` may be called with `physicalDevice` equal to `NULL` during layer discovery. This behavior will only be observed by layer implementations, and not the underlying Vulkan driver.

Valid Usage (Implicit)

- VUID-vkEnumerateDeviceExtensionProperties-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkEnumerateDeviceExtensionProperties-pLayerName-parameter
If `pLayerName` is not `NULL`, `pLayerName` **must** be a null-terminated UTF-8 string
- VUID-vkEnumerateDeviceExtensionProperties-pPropertyCount-parameter
`pPropertyCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkEnumerateDeviceExtensionProperties-pProperties-parameter
If the value referenced by `pPropertyCount` is not 0, and `pProperties` is not `NULL`, `pProperties` **must** be a valid pointer to an array of `pPropertyCount` `VkExtensionProperties` structures

Return Codes

Success

- `VK_SUCCESS`
- `VK_INCOMPLETE`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_LAYER_NOT_PRESENT`

The `VkExtensionProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkExtensionProperties {
    char        extensionName[VK_MAX_EXTENSION_NAME_SIZE];
    uint32_t    specVersion;
} VkExtensionProperties;
```

- `extensionName` is an array of `VK_MAX_EXTENSION_NAME_SIZE` `char` containing a null-terminated UTF-8 string which is the name of the extension.
- `specVersion` is the version of this extension. It is an integer, incremented with backward compatible changes.

Accessing Device-Level Functionality From a `VkPhysicalDevice`

Some device extensions also add support for physical-device-level functionality. Physical-device-level functionality **can** be used, if the required extension is supported as advertised by `vkEnumerateDeviceExtensionProperties` for a given `VkPhysicalDevice`.

Accessing Device-Level Functionality From a `VkDevice`

For commands that are dispatched from a `VkDevice`, or from a child object of a `VkDevice`, device extensions **must** be enabled in `vkCreateDevice`.

38.5. Extension Dependencies

Some extensions are dependent on other extensions, or on specific core API versions, to function. To enable extensions with dependencies, any *required extensions* **must** also be enabled through the same API mechanisms when creating an instance with `vkCreateInstance` or a device with `vkCreateDevice`. Each extension which has such dependencies documents them in the [appendix summarizing that extension](#).

If an extension is supported (as queried by `vkEnumerateInstanceExtensionProperties` or `vkEnumerateDeviceExtensionProperties`), then *required extensions* of that extension **must** also be

supported for the same instance or physical device.

Any device extension that has an instance extension dependency that is not enabled by [vkCreateInstance](#) is considered to be unsupported, hence it **must** not be returned by [vkEnumerateDeviceExtensionProperties](#) for any [VkPhysicalDevice](#) child of the instance. Instance extensions do not have dependencies on device extensions.

If a required extension has been [promoted](#) to another extension or to a core API version, then as a *general* rule, the dependency is also satisfied by the promoted extension or core version. This will be true so long as any features required by the original extension are also required or enabled by the promoted extension or core version. However, in some cases an extension is promoted while making some of its features optional in the promoted extension or core version. In this case, the dependency **may** not be satisfied. The only way to be certain is to look at the descriptions of the original dependency and the promoted version in the [Layers & Extensions](#) and [Core Revisions](#) appendices.

NOTE

There is metadata in [vk.xml](#) describing some aspects of promotion, especially [requires](#), [promotedto](#) and [deprecatedby](#) attributes of `<extension>` tags. However, the metadata does not yet fully describe this scenario. In the future, we may extend the XML schema to describe the full set of extensions and versions satisfying a dependency. As discussed in more detail for [Promotion](#) below, when an extension is promoted it does not mean that a mechanical substitution of an extension API by the corresponding promoted API will work in exactly the same fashion; be supported at runtime; or even exist.

38.6. Compatibility Guarantees (Informative)

This section is marked as informal as there is no binding responsibility on implementations of the Vulkan API - these guarantees are however a contract between the Vulkan Working Group and developers using this Specification.

38.6.1. Core Versions

Each of the [major](#), [minor](#), and [patch versions](#) of the Vulkan specification provide different compatibility guarantees.

Patch Versions

A difference in the patch version indicates that a set of bug fixes or clarifications have been made to the Specification. Informative enums returned by Vulkan commands that will not affect the runtime behavior of a valid application may be added in a patch version (e.g. [VkVendorId](#)).

The specification's patch version is strictly increasing for a given major version of the specification; any change to a specification as described above will result in the patch version being increased by 1. Patch versions are applied to all minor versions, even if a given minor version is not affected by the provoking change.

Specifications with different patch versions but the same major and minor version are *fully*

compatible with each other - such that a valid application written against one will work with an implementation of another.

NOTE

If a patch version includes a bug fix or clarification that could have a significant impact on developer expectations, these will be highlighted in the change log. Generally the Vulkan Working Group tries to avoid these kinds of changes, instead fixing them in either an extension or core version.

Minor Versions

Changes in the minor version of the specification indicate that new functionality has been added to the core specification. This will usually include new interfaces in the header, and **may** also include behavior changes and bug fixes. Core functionality **may** be deprecated in a minor version, but will not be obsoleted or removed.

The specification's minor version is strictly increasing for a given major version of the specification; any change to a specification as described above will result in the minor version being increased by 1. Changes that can be accommodated in a patch version will not increase the minor version.

Specifications with a lower minor version are *backwards compatible* with an implementation of a specification with a higher minor version for core functionality and extensions issued with the KHR vendor tag. Vendor and multi-vendor extensions are not guaranteed to remain functional across minor versions, though in general they are with few exceptions - see [Obsolescence](#) for more information.

Major Versions

A difference in the major version of specifications indicates a large set of changes which will likely include interface changes, behavioral changes, removal of [deprecated functionality](#), and the modification, addition, or replacement of other functionality.

The specification's major version is monotonically increasing; any change to the specification as described above will result in the major version being increased. Changes that can be accommodated in a patch or minor version will not increase the major version.

The Vulkan Working Group intends to only issue a new major version of the Specification in order to realize significant improvements to the Vulkan API that will necessarily require breaking compatibility.

A new major version will likely include a wholly new version of the specification to be issued - which could include an overhaul of the versioning semantics for the minor and patch versions. The patch and minor versions of a specification are therefore not meaningful across major versions. If a major version of the specification includes similar versioning semantics, it is expected that the patch and the minor version will be reset to 0 for that major version.

38.6.2. Extensions

A KHR extension **must** be able to be enabled alongside any other KHR extension, and for any minor

or patch version of the core Specification beyond the minimum version it requires. A multi-vendor extension **should** be able to be enabled alongside any KHR extension or other multi-vendor extension, and for any minor or patch version of the core Specification beyond the minimum version it requires. A vendor extension **should** be able to be enabled alongside any KHR extension, multi-vendor extension, or other vendor extension from the same vendor, and for any minor or patch version of the core Specification beyond the minimum version it requires. A vendor extension **may** be able to be enabled alongside vendor extensions from another vendor.

The one other exception to this is if a vendor or multi-vendor extension is [made obsolete](#) by either a core version or another extension, which will be highlighted in the [extension appendix](#).

Promotion

Extensions, or features of an extension, **may** be promoted to a new [core version of the API](#), or a newer extension which an equal or greater number of implementors are in favor of.

When extension functionality is promoted, minor changes **may** be introduced, limited to the following:

- Naming
- Non-intrusive parameter changes
- [Feature advertisement/enablement](#)
- Combining structure parameters into larger structures
- Author ID suffixes changed or removed

NOTE

If extension functionality is promoted, there is no guarantee of direct compatibility, however it should require little effort to port code from the original feature to the promoted one.

The Vulkan Working Group endeavors to ensure that larger changes are marked as either [deprecated](#) or [obsoleted](#) as appropriate, and can do so retroactively if necessary.

Extensions that are promoted are listed as being promoted in their extension appendices, with reference to where they were promoted to.

When an extension is promoted, any backwards compatibility aliases which exist in the extension will **not** be promoted.

NOTE

As a hypothetical example, if the [VK_KHR_surface](#) extension were promoted to part of a future core version, the [VK_COLOR_SPACE_SRGB_NONLINEAR_KHR](#) token defined by that extension would be promoted to [VK_COLOR_SPACE_SRGB_NONLINEAR](#). However, the [VK_COLORSPACE_SRGB_NONLINEAR_KHR](#) token aliases [VK_COLOR_SPACE_SRGB_NONLINEAR_KHR](#). The [VK_COLORSPACE_SRGB_NONLINEAR_KHR](#) would not be promoted, because it is a backwards compatibility alias that exists only due to a naming mistake when the extension was initially published.

Deprecation

Extensions **may** be marked as deprecated when the intended use cases either become irrelevant or can be solved in other ways. Generally, a new feature will become available to solve the use case in another extension or core version of the API, but it is not guaranteed.

NOTE

Features that are intended to replace deprecated functionality have no guarantees of compatibility, and applications may require drastic modification in order to make use of the new features.

Extensions that are deprecated are listed as being deprecated in their extension appendices, with an explanation of the deprecation and any features that are relevant.

Obsolescence

Occasionally, an extension will be marked as obsolete if a new version of the core API or a new extension is fundamentally incompatible with it. An obsoleted extension **must** not be used with the extension or core version that obsoleted it.

Extensions that are obsoleted are listed as being obsoleted in their extension appendices, with reference to what they were obsoleted by.

Aliases

When an extension is promoted or deprecated by a newer feature, some or all of its functionality **may** be replicated into the newer feature. Rather than duplication of all the documentation and definitions, the specification instead identifies the identical commands and types as *aliases* of one another. Each alias is mentioned together with the definition it aliases, with the older aliases marked as “equivalents”. Each alias of the same command has identical behavior, and each alias of the same type has identical meaning - they can be used interchangeably in an application with no compatibility issues.

NOTE

For promoted types, the aliased extension type is semantically identical to the new core type. The C99 headers simply `typedef` the older aliases to the promoted types.

For promoted command aliases, however, there are two separate command definitions, due to the fact that the C99 ABI has no way to alias command definitions without resorting to macros. Calling either command will produce identical behavior within the bounds of the specification, and should still invoke the same path in the implementation. Debug tools may use separate commands with different debug behavior; to write the appropriate command name to an output log, for instance.

Special Use Extensions

Some extensions exist only to support a specific purpose or specific class of application. These are referred to as “special use extensions”. Use of these extensions in applications not meeting the special use criteria is not recommended.

Special use cases are restricted, and only those defined below are used to describe extensions:

Table 56. Extension Special Use Cases

Special Use	XML Tag	Full Description
CAD support	cadsupport	Extension is intended to support specialized functionality used by CAD/CAM applications.
D3D support	d3demulation	Extension is intended to support D3D emulation layers, and applications ported from D3D, by adding functionality specific to D3D.
Developer tools	devtools	Extension is intended to support developer tools such as capture-replay libraries.
Debugging tools	debugging	Extension is intended for use by applications when debugging.
OpenGL / ES support	glemulation	Extension is intended to support OpenGL and/or OpenGL ES emulation layers, and applications ported from those APIs, by adding functionality specific to those APIs.

Special use extensions are identified in the metadata for each such extension in the [Layers & Extensions](#) appendix, using the name in the “Special Use” column above.

Special use extensions are also identified in `vk.xml` with the short name in “XML Tag” column above, as described in the “API Extensions (`extension` tag)” section of the [registry schema documentation](#).

Chapter 39. Features

Features advertise additional functionality which **can** be enabled in the API. If a feature is *supported* by a physical device, it **must** be *enabled* when creating a [VkDevice](#) in order to use its associated functionality with that [VkDevice](#). If a feature is *unsupported*, that functionality **cannot** be used with that [VkDevice](#).

NOTE

Features are reported via the basic [VkPhysicalDeviceFeatures](#) structure, as well as the extensible structure [VkPhysicalDeviceFeatures2](#), which was added in the [VK_KHR_get_physical_device_properties2](#) extension and included in Vulkan 1.1. When new features are added in future Vulkan versions or extensions, each extension **should** introduce one new feature structure, if needed. This structure **can** be added to the [pNext](#) chain of the [VkPhysicalDeviceFeatures2](#) structure.

For convenience, new core versions of Vulkan **may** introduce new unified feature structures for features promoted from extensions. At the same time, the extension's original feature structure (if any) is also promoted to the core API, and is an alias of the extension's structure. This results in multiple names for the same feature: in the original extension's feature structure and the promoted structure alias, in the unified feature structure. When a feature was implicitly supported and enabled in the extension, but an explicit name was added during promotion, then the extension itself acts as an alias for the feature as listed in the table below.

All aliases of the same feature in the core API **must** be reported consistently: either all **must** be reported as supported, or none of them. When a promoted extension is available, any corresponding feature aliases **must** be supported.

Table 57. Extension Feature Aliases

Extension	Feature(s)
-----------	------------

To query supported features, call:

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceFeatures(
    VkPhysicalDevice          physicalDevice,
    VkPhysicalDeviceFeatures* pFeatures);
```

- [physicalDevice](#) is the physical device from which to query the supported features.
- [pFeatures](#) is a pointer to a [VkPhysicalDeviceFeatures](#) structure in which the physical device features are returned. For each feature, a value of [VK_TRUE](#) specifies that the feature is supported on this physical device, and [VK_FALSE](#) specifies that the feature is not supported.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceFeatures-physicalDevice-parameter [physicalDevice](#) **must** be a valid [VkPhysicalDevice](#) handle

- VUID-vkGetPhysicalDeviceFeatures-pFeatures-parameter
pFeatures **must** be a valid pointer to a [VkPhysicalDeviceFeatures](#) structure

Fine-grained features used by a logical device **must** be enabled at **VkDevice** creation time. If a feature is enabled that the physical device does not support, **VkDevice** creation will fail and return **VK_ERROR_FEATURE_NOT_PRESENT**.

The fine-grained features are enabled by passing a pointer to the **VkPhysicalDeviceFeatures** structure via the **pEnabledFeatures** member of the **VkDeviceCreateInfo** structure that is passed into the **vkCreateDevice** call. If a member of **pEnabledFeatures** is **VK_TRUE** or **VK_FALSE**, then the device will be created with the indicated feature enabled or disabled, respectively. Features **can** also be enabled by using the **VkPhysicalDeviceFeatures2** structure.

If an application wishes to enable all features supported by a device, it **can** simply pass in the **VkPhysicalDeviceFeatures** structure that was previously returned by **vkGetPhysicalDeviceFeatures**. To disable an individual feature, the application **can** set the desired member to **VK_FALSE** in the same structure. Setting **pEnabledFeatures** to **NULL** and not including a **VkPhysicalDeviceFeatures2** in the **pNext** chain of **VkDeviceCreateInfo** is equivalent to setting all members of the structure to **VK_FALSE**.

NOTE

Some features, such as **robustBufferAccess**, **may** incur a runtime performance cost. Application writers **should** carefully consider the implications of enabling all supported features.

To query supported features defined by the core or extensions, call:

```
// Provided by VK_KHR_get_physical_device_properties2
void vkGetPhysicalDeviceFeatures2KHR(
    VkPhysicalDevice          physicalDevice,
    VkPhysicalDeviceFeatures2* pFeatures);
```

- **physicalDevice** is the physical device from which to query the supported features.
- **pFeatures** is a pointer to a **VkPhysicalDeviceFeatures2** structure in which the physical device features are returned.

Each structure in **pFeatures** and its **pNext** chain contains members corresponding to fine-grained features. Each structure in **pFeatures** and its **pNext** chain contains **VkBool32** members corresponding to fine-grained features. Each such member is returned with a **VK_TRUE** value indicating that feature is supported on this physical device, or a **VK_FALSE** value indicating it is unsupported.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceFeatures2-physicalDevice-parameter
physicalDevice **must** be a valid **VkPhysicalDevice** handle
- VUID-vkGetPhysicalDeviceFeatures2-pFeatures-parameter
pFeatures **must** be a valid pointer to a **VkPhysicalDeviceFeatures2** structure

The `VkPhysicalDeviceFeatures2` structure is defined as:

```
typedef struct VkPhysicalDeviceFeatures2 {
    VkStructureType    sType;
    void*              pNext;
    VkPhysicalDeviceFeatures  features;
} VkPhysicalDeviceFeatures2;
```

or the equivalent

```
// Provided by VK_KHR_get_physical_device_properties2
typedef VkPhysicalDeviceFeatures2 VkPhysicalDeviceFeatures2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `features` is a `VkPhysicalDeviceFeatures` structure describing the fine-grained features of the Vulkan 1.0 API.

The `pNext` chain of this structure is used to extend the structure with features defined by extensions. This structure **can** be used in `vkGetPhysicalDeviceFeatures2KHR` or **can** be included in the `pNext` chain of a `VkDeviceCreateInfo` structure, in which case it controls which features are enabled on the device in lieu of `pEnabledFeatures`.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFeatures2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2`

The `VkPhysicalDeviceFeatures` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceFeatures {
    VkBool32    robustBufferAccess;
    VkBool32    fullDrawIndexUint32;
    VkBool32    imageCubeArray;
    VkBool32    independentBlend;
    VkBool32    geometryShader;
    VkBool32    tessellationShader;
    VkBool32    sampleRateShading;
    VkBool32    dualSrcBlend;
    VkBool32    logicOp;
    VkBool32    multiDrawIndirect;
    VkBool32    drawIndirectFirstInstance;
    VkBool32    depthClamp;
    VkBool32    depthBiasClamp;
```

```

VkBool32    fillModeNonSolid;
VkBool32    depthBounds;
VkBool32    wideLines;
VkBool32    largePoints;
VkBool32    alphaToOne;
VkBool32    multiViewport;
VkBool32    samplerAnisotropy;
VkBool32    textureCompressionETC2;
VkBool32    textureCompressionASTC_LDR;
VkBool32    textureCompressionBC;
VkBool32    occlusionQueryPrecise;
VkBool32    pipelineStatisticsQuery;
VkBool32    vertexPipelineStoresAndAtomics;
VkBool32    fragmentStoresAndAtomics;
VkBool32    shaderTessellationAndGeometryPointSize;
VkBool32    shaderImageGatherExtended;
VkBool32    shaderStorageImageExtendedFormats;
VkBool32    shaderStorageImageMultisample;
VkBool32    shaderStorageImageReadWithoutFormat;
VkBool32    shaderStorageImageWriteWithoutFormat;
VkBool32    shaderUniformBufferArrayDynamicIndexing;
VkBool32    shaderSampledImageArrayDynamicIndexing;
VkBool32    shaderStorageBufferArrayDynamicIndexing;
VkBool32    shaderStorageImageArrayDynamicIndexing;
VkBool32    shaderClipDistance;
VkBool32    shaderCullDistance;
VkBool32    shaderFloat64;
VkBool32    shaderInt64;
VkBool32    shaderInt16;
VkBool32    shaderResourceResidency;
VkBool32    shaderResourceMinLod;
VkBool32    sparseBinding;
VkBool32    sparseResidencyBuffer;
VkBool32    sparseResidencyImage2D;
VkBool32    sparseResidencyImage3D;
VkBool32    sparseResidency2Samples;
VkBool32    sparseResidency4Samples;
VkBool32    sparseResidency8Samples;
VkBool32    sparseResidency16Samples;
VkBool32    sparseResidencyAliased;
VkBool32    variableMultisampleRate;
VkBool32    inheritedQueries;
} VkPhysicalDeviceFeatures;

```

This structure describes the following features:

- **robustBufferAccess** specifies that accesses to buffers are bounds-checked against the range of the buffer descriptor (as determined by **VkDescriptorBufferInfo::range**, **VkBufferViewCreateInfo::range**, or the size of the buffer). Out of bounds accesses **must** not cause application termination, and the effects of shader loads, stores, and atomics **must**

conform to an implementation-dependent behavior as described below.

- A buffer access is considered to be out of bounds if any of the following are true:
 - The pointer was formed by `OpImageTexelPointer` and the coordinate is less than zero or greater than or equal to the number of whole elements in the bound range.
 - The pointer was not formed by `OpImageTexelPointer` and the object pointed to is not wholly contained within the bound range. This includes accesses performed via *variable pointers* where the buffer descriptor being accessed cannot be statically determined. Uninitialized pointers and pointers equal to `OpConstantNull` are treated as pointing to a zero-sized object, so all accesses through such pointers are considered to be out of bounds. Buffer accesses through buffer device addresses are not bounds-checked.
 - If the `VkPhysicalDeviceCooperativeMatrixFeaturesKHR::cooperativeMatrixRobustBufferAccess` feature is not enabled, then accesses using `OpCooperativeMatrixLoadKHR` and `OpCooperativeMatrixStoreKHR` **may** not be bounds-checked.

NOTE

If a SPIR-V `OpLoad` instruction loads a structure and the tail end of the structure is out of bounds, then all members of the structure are considered out of bounds even if the members at the end are not statically used.

- If `robustBufferAccess2` is not enabled and any buffer access is determined to be out of bounds, then any other access of the same type (load, store, or atomic) to the same buffer that accesses an address less than 16 bytes away from the out of bounds address **may** also be considered out of bounds.
- If the access is a load that reads from the same memory locations as a prior store in the same shader invocation, with no other intervening accesses to the same memory locations in that shader invocation, then the result of the load **may** be the value stored by the store instruction, even if the access is out of bounds. If the load is `Volatile`, then an out of bounds load **must** return the appropriate out of bounds value.
- Accesses to descriptors written with a `VK_NULL_HANDLE` resource or view are not considered to be out of bounds. Instead, each type of descriptor access defines a specific behavior for accesses to a null descriptor.
- Out-of-bounds buffer loads will return any of the following values:
 - If the access is to a uniform buffer and `robustBufferAccess2` is enabled, loads of offsets between the end of the descriptor range and the end of the descriptor range rounded up to a multiple of `robustUniformBufferAccessSizeAlignment` bytes **must** return either zero values or the contents of the memory at the offset being loaded. Loads of offsets past the descriptor range rounded up to a multiple of `robustUniformBufferAccessSizeAlignment` bytes **must** return zero values.
 - If the access is to a storage buffer and `robustBufferAccess2` is enabled, loads of offsets between the end of the descriptor range and the end of the descriptor range rounded up to a multiple of `robustStorageBufferAccessSizeAlignment` bytes **must** return either zero values or the contents of the memory at the offset being loaded. Loads of offsets past the descriptor range rounded up to a multiple of `robustStorageBufferAccessSizeAlignment`

bytes **must** return zero values. Similarly, stores to addresses between the end of the descriptor range and the end of the descriptor range rounded up to a multiple of `robustStorageBufferAccessSizeAlignment` bytes **may** be discarded.

- Non-atomic accesses to storage buffers that are a multiple of 32 bits **may** be decomposed into 32-bit accesses that are individually bounds-checked.
- If the access is to an index buffer and `robustBufferAccess2` is enabled, zero values **must** be returned.
- If the access is to a uniform texel buffer or storage texel buffer and `robustBufferAccess2` is enabled, zero values **must** be returned, and then [Conversion to RGBA](#) is applied based on the buffer view's format.
- Values from anywhere within the memory range(s) bound to the buffer (possibly including bytes of memory past the end of the buffer, up to the end of the bound range).
- Zero values, or (0,0,0,x) vectors for vector reads where x is a valid value represented in the type of the vector components and **may** be any of:
 - 0, 1, or the maximum representable positive integer value, for signed or unsigned integer components
 - 0.0 or 1.0, for floating-point components
- Out-of-bounds writes **may** modify values within the memory range(s) bound to the buffer, but **must** not modify any other memory.
 - If `robustBufferAccess2` is enabled, out of bounds writes **must** not modify any memory.
- Out-of-bounds atomics **may** modify values within the memory range(s) bound to the buffer, but **must** not modify any other memory, and return an undefined value.
 - If `robustBufferAccess2` is enabled, out of bounds atomics **must** not modify any memory, and return an undefined value.
- If `robustBufferAccess2` is disabled, vertex input attributes are considered out of bounds if the offset of the attribute in the bound vertex buffer range plus the size of the attribute is greater than either:
 - `vertexBufferRangeSize`, if `bindingStride == 0`; or
 - $(\text{vertexBufferRangeSize} - (\text{vertexBufferRangeSize} \% \text{bindingStride}))$

where `vertexBufferRangeSize` is the byte size of the memory range bound to the vertex buffer binding and `bindingStride` is the byte stride of the corresponding vertex input binding. Further, if any vertex input attribute using a specific vertex input binding is out of bounds, then all vertex input attributes using that vertex input binding for that vertex shader invocation are considered out of bounds.

- If a vertex input attribute is out of bounds, it will be assigned one of the following values:
 - Values from anywhere within the memory range(s) bound to the buffer, converted according to the format of the attribute.
 - Zero values, format converted according to the format of the attribute.
 - Zero values, or (0,0,0,x) vectors, as described above.

- If **robustBufferAccess2** is enabled, vertex input attributes are considered out of bounds if the offset of the attribute in the bound vertex buffer range plus the size of the attribute is greater than the byte size of the memory range bound to the vertex buffer binding.
 - If a vertex input attribute is out of bounds, the **raw data** extracted are zero values, and missing G, B, or A components are **filled with (0,0,1)**.
- If **robustBufferAccess** is not enabled, applications **must** not perform out of bounds accesses .
- **fullDrawIndexUint32** specifies the full 32-bit range of indices is supported for indexed draw calls when using a **VkIndexType** of **VK_INDEX_TYPE_UINT32**. **maxDrawIndexedIndexValue** is the maximum index value that **may** be used (aside from the primitive restart index, which is always $2^{32}-1$ when the **VkIndexType** is **VK_INDEX_TYPE_UINT32**). If this feature is supported, **maxDrawIndexedIndexValue** **must** be $2^{32}-1$; otherwise it **must** be no smaller than $2^{24}-1$. See **maxDrawIndexedIndexValue**.
- **imageCubeArray** specifies whether image views with a **VkImageViewType** of **VK_IMAGE_VIEW_TYPE_CUBE_ARRAY** **can** be created, and that the corresponding **SampledCubeArray** and **ImageCubeArray** SPIR-V capabilities **can** be used in shader code.
- **independentBlend** specifies whether the **VkPipelineColorBlendAttachmentState** settings are controlled independently per-attachment. If this feature is not enabled, the **VkPipelineColorBlendAttachmentState** settings for all color attachments **must** be identical. Otherwise, a different **VkPipelineColorBlendAttachmentState** **can** be provided for each bound color attachment.
- **geometryShader** specifies whether geometry shaders are supported. If this feature is not enabled, the **VK_SHADER_STAGE_GEOMETRY_BIT** and **VK_PIPELINE_STAGE_GEOMETRY_SHADER_BIT** enum values **must** not be used. This also specifies whether shader modules **can** declare the **Geometry** capability.
- **tessellationShader** specifies whether tessellation control and evaluation shaders are supported. If this feature is not enabled, the **VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT**, **VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT**, **VK_PIPELINE_STAGE_TESSELLATION_CONTROL_SHADER_BIT**, **VK_PIPELINE_STAGE_TESSELLATION_EVALUATION_SHADER_BIT**, and **VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_STATE_CREATE_INFO** enum values **must** not be used. This also specifies whether shader modules **can** declare the **Tessellation** capability.
- **sampleRateShading** specifies whether **Sample Shading** and multisample interpolation are supported. If this feature is not enabled, the **sampleShadingEnable** member of the **VkPipelineMultisampleStateCreateInfo** structure **must** be **VK_FALSE** and the **minSampleShading** member is ignored. This also specifies whether shader modules **can** declare the **SampleRateShading** capability.
- **dualSrcBlend** specifies whether blend operations which take two sources are supported. If this feature is not enabled, the **VK_BLEND_FACTOR_SRC1_COLOR**, **VK_BLEND_FACTOR_ONE_MINUS_SRC1_COLOR**, **VK_BLEND_FACTOR_SRC1_ALPHA**, and **VK_BLEND_FACTOR_ONE_MINUS_SRC1_ALPHA** enum values **must** not be used as source or destination blending factors. See **Dual-Source Blending**.
- **logicOp** specifies whether logic operations are supported. If this feature is not enabled, the **logicOpEnable** member of the **VkPipelineColorBlendStateCreateInfo** structure **must** be **VK_FALSE**, and the **logicOp** member is ignored.
- **multiDrawIndirect** specifies whether multiple draw indirect is supported. If this feature is not

enabled, the `drawCount` parameter to the `vkCmdDrawIndirect` and `vkCmdDrawIndexedIndirect` commands **must** be 0 or 1. The `maxDrawIndirectCount` member of the `VkPhysicalDeviceLimits` structure **must** also be 1 if this feature is not supported. See `maxDrawIndirectCount`.

- `drawIndirectFirstInstance` specifies whether indirect drawing calls support the `firstInstance` parameter. If this feature is not enabled, the `firstInstance` member of all `VkDrawIndirectCommand` and `VkDrawIndexedIndirectCommand` structures that are provided to the `vkCmdDrawIndirect` and `vkCmdDrawIndexedIndirect` commands **must** be 0.
- `depthClamp` specifies whether depth clamping is supported. If this feature is not enabled, the `depthClampEnable` member of the `VkPipelineRasterizationStateCreateInfo` structure **must** be `VK_FALSE`. Otherwise, setting `depthClampEnable` to `VK_TRUE` will enable depth clamping.
- `depthBiasClamp` specifies whether depth bias clamping is supported. If this feature is not enabled, the `depthBiasClamp` member of the `VkPipelineRasterizationStateCreateInfo` structure **must** be 0.0 unless the `VK_DYNAMIC_STATE_DEPTH_BIAS` dynamic state is enabled, in which case the `depthBiasClamp` parameter to `vkCmdSetDepthBias` **must** be 0.0.
- `fillModeNonSolid` specifies whether point and wireframe fill modes are supported. If this feature is not enabled, the `VK_POLYGON_MODE_POINT` and `VK_POLYGON_MODE_LINE` enum values **must** not be used.
- `depthBounds` specifies whether depth bounds tests are supported. If this feature is not enabled, the `depthBoundsTestEnable` member of the `VkPipelineDepthStencilStateCreateInfo` structure **must** be `VK_FALSE` unless the `VK_DYNAMIC_STATE_DEPTH_BOUNDS_TEST_ENABLE` dynamic state is enabled, in which case the `depthBoundsTestEnable` parameter to `vkCmdSetDepthBoundsTestEnableEXT` **must** be `VK_FALSE`. When `depthBoundsTestEnable` is `VK_FALSE`, the `minDepthBounds` and `maxDepthBounds` members of the `VkPipelineDepthStencilStateCreateInfo` structure are ignored.
- `widelines` specifies whether lines with width other than 1.0 are supported. If this feature is not enabled, the `lineWidth` member of the `VkPipelineRasterizationStateCreateInfo` structure **must** be 1.0 unless the `VK_DYNAMIC_STATE_LINE_WIDTH` dynamic state is enabled, in which case the `lineWidth` parameter to `vkCmdSetLineWidth` **must** be 1.0. When this feature is supported, the range and granularity of supported line widths are indicated by the `lineWidthRange` and `lineWidthGranularity` members of the `VkPhysicalDeviceLimits` structure, respectively.
- `largePoints` specifies whether points with size greater than 1.0 are supported. If this feature is not enabled, only a point size of 1.0 written by a shader is supported. The range and granularity of supported point sizes are indicated by the `pointSizeRange` and `pointSizeGranularity` members of the `VkPhysicalDeviceLimits` structure, respectively.
- `alphaToOne` specifies whether the implementation is able to replace the alpha value of the fragment shader color output in the `Multisample Coverage` fragment operation. If this feature is not enabled, then the `alphaToOneEnable` member of the `VkPipelineMultisampleStateCreateInfo` structure **must** be `VK_FALSE`. Otherwise setting `alphaToOneEnable` to `VK_TRUE` will enable alpha-to-one behavior.
- `multiViewport` specifies whether more than one viewport is supported. If this feature is not enabled:
 - The `viewportCount` and `scissorCount` members of the `VkPipelineViewportStateCreateInfo` structure **must** be 1.

- The `firstViewport` and `viewportCount` parameters to the `vkCmdSetViewport` command **must** be 0 and 1, respectively.
- The `firstScissor` and `scissorCount` parameters to the `vkCmdSetScissor` command **must** be 0 and 1, respectively.
- `samplerAnisotropy` specifies whether anisotropic filtering is supported. If this feature is not enabled, the `anisotropyEnable` member of the `VkSamplerCreateInfo` structure **must** be `VK_FALSE`.
- `textureCompressionETC2` specifies whether all of the ETC2 and EAC compressed texture formats are supported. If this feature is enabled, then the `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`, `VK_FORMAT_FEATURE_BLIT_SRC_BIT` and `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT` features **must** be supported in `optimalTilingFeatures` for the following formats:
 - `VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK`
 - `VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK`
 - `VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK`
 - `VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK`
 - `VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK`
 - `VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK`
 - `VK_FORMAT_EAC_R11_UNORM_BLOCK`
 - `VK_FORMAT_EAC_R11_SNORM_BLOCK`
 - `VK_FORMAT_EAC_R11G11_UNORM_BLOCK`
 - `VK_FORMAT_EAC_R11G11_SNORM_BLOCK`

To query for additional properties, or if the feature is not enabled, `vkGetPhysicalDeviceFormatProperties` and `vkGetPhysicalDeviceImageFormatProperties` **can** be used to check for supported properties of individual formats as normal.

- `textureCompressionASTC_LDR` specifies whether all of the ASTC LDR compressed texture formats are supported. If this feature is enabled, then the `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`, `VK_FORMAT_FEATURE_BLIT_SRC_BIT` and `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT` features **must** be supported in `optimalTilingFeatures` for the following formats:
 - `VK_FORMAT_ASTC_4x4_UNORM_BLOCK`
 - `VK_FORMAT_ASTC_4x4_SRGB_BLOCK`
 - `VK_FORMAT_ASTC_5x4_UNORM_BLOCK`
 - `VK_FORMAT_ASTC_5x4_SRGB_BLOCK`
 - `VK_FORMAT_ASTC_5x5_UNORM_BLOCK`
 - `VK_FORMAT_ASTC_5x5_SRGB_BLOCK`
 - `VK_FORMAT_ASTC_6x5_UNORM_BLOCK`
 - `VK_FORMAT_ASTC_6x5_SRGB_BLOCK`
 - `VK_FORMAT_ASTC_6x6_UNORM_BLOCK`
 - `VK_FORMAT_ASTC_6x6_SRGB_BLOCK`

- VK_FORMAT_ASTC_8x5_UNORM_BLOCK
- VK_FORMAT_ASTC_8x5_SRGB_BLOCK
- VK_FORMAT_ASTC_8x6_UNORM_BLOCK
- VK_FORMAT_ASTC_8x6_SRGB_BLOCK
- VK_FORMAT_ASTC_8x8_UNORM_BLOCK
- VK_FORMAT_ASTC_8x8_SRGB_BLOCK
- VK_FORMAT_ASTC_10x5_UNORM_BLOCK
- VK_FORMAT_ASTC_10x5_SRGB_BLOCK
- VK_FORMAT_ASTC_10x6_UNORM_BLOCK
- VK_FORMAT_ASTC_10x6_SRGB_BLOCK
- VK_FORMAT_ASTC_10x8_UNORM_BLOCK
- VK_FORMAT_ASTC_10x8_SRGB_BLOCK
- VK_FORMAT_ASTC_10x10_UNORM_BLOCK
- VK_FORMAT_ASTC_10x10_SRGB_BLOCK
- VK_FORMAT_ASTC_12x10_UNORM_BLOCK
- VK_FORMAT_ASTC_12x10_SRGB_BLOCK
- VK_FORMAT_ASTC_12x12_UNORM_BLOCK
- VK_FORMAT_ASTC_12x12_SRGB_BLOCK

To query for additional properties, or if the feature is not enabled, [vkGetPhysicalDeviceFormatProperties](#) and [vkGetPhysicalDeviceImageFormatProperties](#) can be used to check for supported properties of individual formats as normal.

- `textureCompressionBC` specifies whether all of the BC compressed texture formats are supported. If this feature is enabled, then the `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`, `VK_FORMAT_FEATURE_BLIT_SRC_BIT` and `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT` features **must** be supported in `optimalTilingFeatures` for the following formats:

- VK_FORMAT_BC1_RGB_UNORM_BLOCK
- VK_FORMAT_BC1_RGB_SRGB_BLOCK
- VK_FORMAT_BC1_RGBA_UNORM_BLOCK
- VK_FORMAT_BC1_RGBA_SRGB_BLOCK
- VK_FORMAT_BC2_UNORM_BLOCK
- VK_FORMAT_BC2_SRGB_BLOCK
- VK_FORMAT_BC3_UNORM_BLOCK
- VK_FORMAT_BC3_SRGB_BLOCK
- VK_FORMAT_BC4_UNORM_BLOCK
- VK_FORMAT_BC4_SNORM_BLOCK

- VK_FORMAT_BC5_UNORM_BLOCK
- VK_FORMAT_BC5_SNORM_BLOCK
- VK_FORMAT_BC6H_UFLOAT_BLOCK
- VK_FORMAT_BC6H_SFLOAT_BLOCK
- VK_FORMAT_BC7_UNORM_BLOCK
- VK_FORMAT_BC7_SRGB_BLOCK

To query for additional properties, or if the feature is not enabled, `vkGetPhysicalDeviceFormatProperties` and `vkGetPhysicalDeviceImageFormatProperties` can be used to check for supported properties of individual formats as normal.

- `occlusionQueryPrecise` specifies whether occlusion queries returning actual sample counts are supported. Occlusion queries are created in a `VkQueryPool` by specifying the `queryType` of `VK_QUERY_TYPE_OCCLUSION` in the `VkQueryPoolCreateInfo` structure which is passed to `vkCreateQueryPool`. If this feature is enabled, queries of this type **can** enable `VK_QUERY_CONTROL_PRECISE_BIT` in the `flags` parameter to `vkCmdBeginQuery`. If this feature is not supported, the implementation supports only boolean occlusion queries. When any samples are passed, boolean queries will return a non-zero result value, otherwise a result value of zero is returned. When this feature is enabled and `VK_QUERY_CONTROL_PRECISE_BIT` is set, occlusion queries will report the actual number of samples passed.
- `pipelineStatisticsQuery` specifies whether the pipeline statistics queries are supported. If this feature is not enabled, queries of type `VK_QUERY_TYPE_PIPELINE_STATISTICS` **cannot** be created, and none of the `VkQueryPipelineStatisticFlagBits` bits **can** be set in the `pipelineStatistics` member of the `VkQueryPoolCreateInfo` structure.
- `vertexPipelineStoresAndAtomics` specifies whether storage buffers and images support stores and atomic operations in the vertex, tessellation, and geometry shader stages. If this feature is not enabled, all storage image, storage texel buffer, and storage buffer variables used by these stages in shader modules **must** be decorated with the `NonWritable` decoration (or the `readonly` memory qualifier in GLSL).
- `fragmentStoresAndAtomics` specifies whether storage buffers and images support stores and atomic operations in the fragment shader stage. If this feature is not enabled, all storage image, storage texel buffer, and storage buffer variables used by the fragment stage in shader modules **must** be decorated with the `NonWritable` decoration (or the `readonly` memory qualifier in GLSL).
- `shaderTessellationAndGeometryPointSize` specifies whether the `PointSize` built-in decoration is available in the tessellation control, tessellation evaluation, and geometry shader stages. If this feature is not enabled, members decorated with the `PointSize` built-in decoration **must** not be read from or written to and all points written from a tessellation or geometry shader will have a size of 1.0. This also specifies whether shader modules **can** declare the `TessellationPointSize` capability for tessellation control and evaluation shaders, or if the shader modules **can** declare the `GeometryPointSize` capability for geometry shaders. An implementation supporting this feature **must** also support one or both of the `tessellationShader` or `geometryShader` features.
- `shaderImageGatherExtended` specifies whether the extended set of image gather instructions are available in shader code. If this feature is not enabled, the `OpImage*Gather` instructions do not support the `Offset` and `ConstOffsets` operands. This also specifies whether shader modules **can**

declare the `ImageGatherExtended` capability.

- `shaderStorageImageExtendedFormats` specifies whether all the “storage image extended formats” below are supported; if this feature is supported, then the `VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT` **must** be supported in `optimalTilingFeatures` for the following formats:

- `VK_FORMAT_R16G16_SFLOAT`
- `VK_FORMAT_B10G11R11_UFLOAT_PACK32`
- `VK_FORMAT_R16_SFLOAT`
- `VK_FORMAT_R16G16B16A16_UNORM`
- `VK_FORMAT_A2B10G10R10_UNORM_PACK32`
- `VK_FORMAT_R16G16_UNORM`
- `VK_FORMAT_R8G8_UNORM`
- `VK_FORMAT_R16_UNORM`
- `VK_FORMAT_R8_UNORM`
- `VK_FORMAT_R16G16B16A16_SNORM`
- `VK_FORMAT_R16G16_SNORM`
- `VK_FORMAT_R8G8_SNORM`
- `VK_FORMAT_R16_SNORM`
- `VK_FORMAT_R8_SNORM`
- `VK_FORMAT_R16G16_SINT`
- `VK_FORMAT_R8G8_SINT`
- `VK_FORMAT_R16_SINT`
- `VK_FORMAT_R8_SINT`
- `VK_FORMAT_A2B10G10R10_UINT_PACK32`
- `VK_FORMAT_R16G16_UINT`
- `VK_FORMAT_R8G8_UINT`
- `VK_FORMAT_R16_UINT`
- `VK_FORMAT_R8_UINT`

NOTE

`shaderStorageImageExtendedFormats` feature only adds a guarantee of format support, which is specified for the whole physical device. Therefore enabling or disabling the feature via `vkCreateDevice` has no practical effect.

To query for additional properties, or if the feature is not supported, `vkGetPhysicalDeviceFormatProperties` and `vkGetPhysicalDeviceImageFormatProperties` **can** be used to check for supported properties of individual formats, as usual rules allow.

`VK_FORMAT_R32G32_UINT`, `VK_FORMAT_R32G32_SINT`, and `VK_FORMAT_R32G32_SFLOAT` from `StorageImageExtendedFormats` SPIR-V capability, are already covered by

core Vulkan [mandatory format support](#).

- `shaderStorageImageMultisample` specifies whether multisampled storage images are supported. If this feature is not enabled, images that are created with a `usage` that includes `VK_IMAGE_USAGE_STORAGE_BIT` **must** be created with `samples` equal to `VK_SAMPLE_COUNT_1_BIT`. This also specifies whether shader modules **can** declare the `StorageImageMultisample` and `ImageMSArray` capabilities.
- `shaderStorageImageReadWithoutFormat` specifies whether storage images and storage texel buffers require a format qualifier to be specified when reading. `shaderStorageImageReadWithoutFormat` applies only to formats listed in the [storage without format](#) list.
- `shaderStorageImageWriteWithoutFormat` specifies whether storage images and storage texel buffers require a format qualifier to be specified when writing. `shaderStorageImageWriteWithoutFormat` applies only to formats listed in the [storage without format](#) list.
- `shaderUniformBufferArrayDynamicIndexing` specifies whether arrays of uniform buffers **can** be indexed by integer expressions that are dynamically uniform within the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also specifies whether shader modules **can** declare the `UniformBufferArrayDynamicIndexing` capability.
- `shaderSampledImageArrayDynamicIndexing` specifies whether arrays of samplers or sampled images **can** be indexed by integer expressions that are dynamically uniform within the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of `VK_DESCRIPTOR_TYPE_SAMPLER`, `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, or `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE` **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also specifies whether shader modules **can** declare the `SampledImageArrayDynamicIndexing` capability.
- `shaderStorageBufferArrayDynamicIndexing` specifies whether arrays of storage buffers **can** be indexed by integer expressions that are dynamically uniform within the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also specifies whether shader modules **can** declare the `StorageBufferArrayDynamicIndexing` capability.
- `shaderStorageImageArrayDynamicIndexing` specifies whether arrays of storage images **can** be indexed by integer expressions that are dynamically uniform within the invocation group in shader code. If this feature is not enabled, resources with a descriptor type of `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE` **must** be indexed only by constant integral expressions when aggregated into arrays in shader code. This also specifies whether shader modules **can** declare the `StorageImageArrayDynamicIndexing` capability.
- `shaderClipDistance` specifies whether clip distances are supported in shader code. If this feature is not enabled, any members decorated with the `ClipDistance` built-in decoration **must** not be read from or written to in shader modules. This also specifies whether shader modules **can** declare the `ClipDistance` capability.

- **shaderCullDistance** specifies whether cull distances are supported in shader code. If this feature is not enabled, any members decorated with the **CullDistance** built-in decoration **must** not be read from or written to in shader modules. This also specifies whether shader modules **can** declare the **CullDistance** capability.
- **shaderFloat64** specifies whether 64-bit floats (doubles) are supported in shader code. If this feature is not enabled, 64-bit floating-point types **must** not be used in shader code. This also specifies whether shader modules **can** declare the **Float64** capability. Declaring and using 64-bit floats is enabled for all storage classes that SPIR-V allows with the **Float64** capability.
- **shaderInt64** specifies whether 64-bit integers (signed and unsigned) are supported in shader code. If this feature is not enabled, 64-bit integer types **must** not be used in shader code. This also specifies whether shader modules **can** declare the **Int64** capability. Declaring and using 64-bit integers is enabled for all storage classes that SPIR-V allows with the **Int64** capability.
- **shaderInt16** specifies whether 16-bit integers (signed and unsigned) are supported in shader code. If this feature is not enabled, 16-bit integer types **must** not be used in shader code. This also specifies whether shader modules **can** declare the **Int16** capability. However, this only enables a subset of the storage classes that SPIR-V allows for the **Int16** SPIR-V capability: Declaring and using 16-bit integers in the **Private**, **Workgroup** (for non-Block variables), and **Function** storage classes is enabled, while declaring them in the interface storage classes (e.g., **UniformConstant**, **Uniform**, **StorageBuffer**, **Input**, **Output**, and **PushConstant**) is not enabled.
- **shaderResourceResidency** specifies whether image operations that return resource residency information are supported in shader code. If this feature is not enabled, the **OpImageSparse*** instructions **must** not be used in shader code. This also specifies whether shader modules **can** declare the **SparseResidency** capability. The feature requires at least one of the **sparseResidency*** features to be supported.
- **shaderResourceMinLod** specifies whether image operations specifying the minimum resource LOD are supported in shader code. If this feature is not enabled, the **MinLod** image operand **must** not be used in shader code. This also specifies whether shader modules **can** declare the **MinLod** capability.
- **sparseBinding** specifies whether resource memory **can** be managed at opaque sparse block level instead of at the object level. If this feature is not enabled, resource memory **must** be bound only on a per-object basis using the **vkBindBufferMemory** and **vkBindImageMemory** commands. In this case, buffers and images **must** not be created with **VK_BUFFER_CREATE_SPARSE_BINDING_BIT** and **VK_IMAGE_CREATE_SPARSE_BINDING_BIT** set in the **flags** member of the **VkBufferCreateInfo** and **VkImageCreateInfo** structures, respectively. Otherwise resource memory **can** be managed as described in [Sparse Resource Features](#).
- **sparseResidencyBuffer** specifies whether the device **can** access partially resident buffers. If this feature is not enabled, buffers **must** not be created with **VK_BUFFER_CREATE_SPARSE_RESIDENCY_BIT** set in the **flags** member of the **VkBufferCreateInfo** structure.
- **sparseResidencyImage2D** specifies whether the device **can** access partially resident 2D images with 1 sample per pixel. If this feature is not enabled, images with an **imageType** of **VK_IMAGE_TYPE_2D** and **samples** set to **VK_SAMPLE_COUNT_1_BIT** **must** not be created with **VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT** set in the **flags** member of the **VkImageCreateInfo** structure.
- **sparseResidencyImage3D** specifies whether the device **can** access partially resident 3D images. If

this feature is not enabled, images with an `imageType` of `VK_IMAGE_TYPE_3D` **must** not be created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` set in the `flags` member of the `VkImageCreateInfo` structure.

- `sparseResidency2Samples` specifies whether the physical device **can** access partially resident 2D images with 2 samples per pixel. If this feature is not enabled, images with an `imageType` of `VK_IMAGE_TYPE_2D` and `samples` set to `VK_SAMPLE_COUNT_2_BIT` **must** not be created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` set in the `flags` member of the `VkImageCreateInfo` structure.
- `sparseResidency4Samples` specifies whether the physical device **can** access partially resident 2D images with 4 samples per pixel. If this feature is not enabled, images with an `imageType` of `VK_IMAGE_TYPE_2D` and `samples` set to `VK_SAMPLE_COUNT_4_BIT` **must** not be created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` set in the `flags` member of the `VkImageCreateInfo` structure.
- `sparseResidency8Samples` specifies whether the physical device **can** access partially resident 2D images with 8 samples per pixel. If this feature is not enabled, images with an `imageType` of `VK_IMAGE_TYPE_2D` and `samples` set to `VK_SAMPLE_COUNT_8_BIT` **must** not be created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` set in the `flags` member of the `VkImageCreateInfo` structure.
- `sparseResidency16Samples` specifies whether the physical device **can** access partially resident 2D images with 16 samples per pixel. If this feature is not enabled, images with an `imageType` of `VK_IMAGE_TYPE_2D` and `samples` set to `VK_SAMPLE_COUNT_16_BIT` **must** not be created with `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT` set in the `flags` member of the `VkImageCreateInfo` structure.
- `sparseResidencyAliased` specifies whether the physical device **can** correctly access data aliased into multiple locations. If this feature is not enabled, the `VK_BUFFER_CREATE_SPARSE_ALIASED_BIT` and `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT` enum values **must** not be used in `flags` members of the `VkBufferCreateInfo` and `VkImageCreateInfo` structures, respectively.
- `variableMultisampleRate` specifies whether all pipelines that will be bound to a command buffer during a subpass which uses no attachments **must** have the same value for `VkPipelineMultisampleStateCreateInfo::rasterizationSamples`. If set to `VK_TRUE`, the implementation supports variable multisample rates in a subpass which uses no attachments. If set to `VK_FALSE`, then all pipelines bound in such a subpass **must** have the same multisample rate. This has no effect in situations where a subpass uses any attachments.
- `inheritedQueries` specifies whether a secondary command buffer **may** be executed while a query is active.

The `VkPhysicalDeviceVariablePointersFeatures` structure is defined as:

```
typedef struct VkPhysicalDeviceVariablePointersFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           variablePointersStorageBuffer;
    VkBool32           variablePointers;
} VkPhysicalDeviceVariablePointersFeatures;
```

or the equivalent

```
// Provided by VK_KHR_variable_pointers
typedef VkPhysicalDeviceVariablePointersFeatures
VkPhysicalDeviceVariablePointersFeaturesKHR;
```

```
// Provided by VK_KHR_variable_pointers
typedef VkPhysicalDeviceVariablePointersFeatures
VkPhysicalDeviceVariablePointerFeaturesKHR;
```

This structure describes the following features:

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **variablePointersStorageBuffer** specifies whether the implementation supports the SPIR-V **VariablePointersStorageBuffer** capability. When this feature is not enabled, shader modules **must** not declare the **SPV_KHR_variable_pointers** extension or the **VariablePointersStorageBuffer** capability.
- **variablePointers** specifies whether the implementation supports the SPIR-V **VariablePointers** capability. When this feature is not enabled, shader modules **must** not declare the **VariablePointers** capability.

If the **VkPhysicalDeviceVariablePointersFeatures** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDeviceVariablePointersFeatures** **can** also be used in the **pNext** chain of **VkDeviceCreateInfo** to selectively enable these features.

Valid Usage

- VUID-VkPhysicalDeviceVariablePointersFeatures-variablePointers-01431
If **variablePointers** is enabled then **variablePointersStorageBuffer** **must** also be enabled

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVariablePointersFeatures-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES**

The **VkPhysicalDeviceMultiviewFeatures** structure is defined as:

```
typedef struct VkPhysicalDeviceMultiviewFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           multiview;
    VkBool32           multiviewGeometryShader;
    VkBool32           multiviewTessellationShader;
} VkPhysicalDeviceMultiviewFeatures;
```

or the equivalent

```
// Provided by VK_KHR_multiview
typedef VkPhysicalDeviceMultiviewFeatures VkPhysicalDeviceMultiviewFeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `multiview` specifies whether the implementation supports multiview rendering within a render pass. If this feature is not enabled, the view mask of each subpass **must** always be zero.
- `multiviewGeometryShader` specifies whether the implementation supports multiview rendering within a render pass, with `geometry shaders`. If this feature is not enabled, then a pipeline compiled against a subpass with a non-zero view mask **must** not include a geometry shader.
- `multiviewTessellationShader` specifies whether the implementation supports multiview rendering within a render pass, with `tessellation shaders`. If this feature is not enabled, then a pipeline compiled against a subpass with a non-zero view mask **must** not include any tessellation shaders.

If the `VkPhysicalDeviceMultiviewFeatures` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceMultiviewFeatures` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage

- VUID-VkPhysicalDeviceMultiviewFeatures-multiviewGeometryShader-00580
If `multiviewGeometryShader` is enabled then `multiview` **must** also be enabled
- VUID-VkPhysicalDeviceMultiviewFeatures-multiviewTessellationShader-00581
If `multiviewTessellationShader` is enabled then `multiview` **must** also be enabled

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMultiviewFeatures-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_FEATURES`

The [VkPhysicalDeviceShaderAtomicFloatFeaturesEXT](#) structure is defined as:

```
// Provided by VK_EXT_shader_atomic_float
typedef struct VkPhysicalDeviceShaderAtomicFloatFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderBufferFloat32Atomics;
    VkBool32           shaderBufferFloat32AtomicAdd;
    VkBool32           shaderBufferFloat64Atomics;
    VkBool32           shaderBufferFloat64AtomicAdd;
    VkBool32           shaderSharedFloat32Atomics;
    VkBool32           shaderSharedFloat32AtomicAdd;
    VkBool32           shaderSharedFloat64Atomics;
    VkBool32           shaderSharedFloat64AtomicAdd;
    VkBool32           shaderImageFloat32Atomics;
    VkBool32           shaderImageFloat32AtomicAdd;
    VkBool32           sparseImageFloat32Atomics;
    VkBool32           sparseImageFloat32AtomicAdd;
} VkPhysicalDeviceShaderAtomicFloatFeaturesEXT;
```

This structure describes the following features:

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **shaderBufferFloat32Atomics** indicates whether shaders **can** perform 32-bit floating-point load, store and exchange atomic operations on storage buffers.
- **shaderBufferFloat32AtomicAdd** indicates whether shaders **can** perform 32-bit floating-point add atomic operations on storage buffers.
- **shaderBufferFloat64Atomics** indicates whether shaders **can** perform 64-bit floating-point load, store and exchange atomic operations on storage buffers.
- **shaderBufferFloat64AtomicAdd** indicates whether shaders **can** perform 64-bit floating-point add atomic operations on storage buffers.
- **shaderSharedFloat32Atomics** indicates whether shaders **can** perform 32-bit floating-point load, store and exchange atomic operations on shared memory.
- **shaderSharedFloat32AtomicAdd** indicates whether shaders **can** perform 32-bit floating-point add atomic operations on shared memory.
- **shaderSharedFloat64Atomics** indicates whether shaders **can** perform 64-bit floating-point load, store and exchange atomic operations on shared memory.
- **shaderSharedFloat64AtomicAdd** indicates whether shaders **can** perform 64-bit floating-point add atomic operations on shared memory.
- **shaderImageFloat32Atomics** indicates whether shaders **can** perform 32-bit floating-point load, store and exchange atomic image operations.
- **shaderImageFloat32AtomicAdd** indicates whether shaders **can** perform 32-bit floating-point add atomic operations on shared memory.

atomic image operations.

- `sparseImageFloat32Atomics` indicates whether 32-bit floating-point load, store and exchange atomic operations **can** be used on sparse images.
- `sparseImageFloat32AtomicAdd` indicates whether 32-bit floating-point add atomic operations **can** be used on sparse images.

If the `VkPhysicalDeviceShaderAtomicFloatFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderAtomicFloatFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderAtomicFloatFeaturesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_FEATURES_EXT`

The `VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_shader_atomic_float2
typedef struct VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderBufferFloat16Atomics;
    VkBool32           shaderBufferFloat16AtomicAdd;
    VkBool32           shaderBufferFloat16AtomicMinMax;
    VkBool32           shaderBufferFloat32AtomicMinMax;
    VkBool32           shaderBufferFloat64AtomicMinMax;
    VkBool32           shaderSharedFloat16Atomics;
    VkBool32           shaderSharedFloat16AtomicAdd;
    VkBool32           shaderSharedFloat16AtomicMinMax;
    VkBool32           shaderSharedFloat32AtomicMinMax;
    VkBool32           shaderSharedFloat64AtomicMinMax;
    VkBool32           shaderImageFloat32AtomicMinMax;
    VkBool32           sparseImageFloat32AtomicMinMax;
} VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderBufferFloat16Atomics` indicates whether shaders **can** perform 16-bit floating-point load, store, and exchange atomic operations on storage buffers.
- `shaderBufferFloat16AtomicAdd` indicates whether shaders **can** perform 16-bit floating-point add atomic operations on storage buffers.

- `shaderBufferFloat16AtomicMinMax` indicates whether shaders **can** perform 16-bit floating-point min and max atomic operations on storage buffers.
- `shaderBufferFloat32AtomicMinMax` indicates whether shaders **can** perform 32-bit floating-point min and max atomic operations on storage buffers.
- `shaderBufferFloat64AtomicMinMax` indicates whether shaders **can** perform 64-bit floating-point min and max atomic operations on storage buffers.
- `shaderSharedFloat16Atomics` indicates whether shaders **can** perform 16-bit floating-point load, store and exchange atomic operations on shared memory.
- `shaderSharedFloat16AtomicAdd` indicates whether shaders **can** perform 16-bit floating-point add atomic operations on shared memory.
- `shaderSharedFloat16AtomicMinMax` indicates whether shaders **can** perform 16-bit floating-point min and max atomic operations on shared memory.
- `shaderSharedFloat32AtomicMinMax` indicates whether shaders **can** perform 32-bit floating-point min and max atomic operations on shared memory.
- `shaderSharedFloat64AtomicMinMax` indicates whether shaders **can** perform 64-bit floating-point min and max atomic operations on shared memory.
- `shaderImageFloat32AtomicMinMax` indicates whether shaders **can** perform 32-bit floating-point min and max atomic image operations.
- `sparseImageFloat32AtomicMinMax` indicates whether 32-bit floating-point min and max atomic operations **can** be used on sparse images.

If the `VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_2_FEATURES_EXT`

The `VkPhysicalDeviceShaderAtomicInt64Features` structure is defined as:

```
typedef struct VkPhysicalDeviceShaderAtomicInt64Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderBufferInt64Atomics;
    VkBool32           shaderSharedInt64Atomics;
} VkPhysicalDeviceShaderAtomicInt64Features;
```

or the equivalent


```
// Provided by VK_KHR_shader_atomic_int64
typedef VkPhysicalDeviceShaderAtomicInt64Features
VkPhysicalDeviceShaderAtomicInt64FeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderBufferInt64Atomics` indicates whether shaders **can** perform 64-bit unsigned and signed integer atomic operations on buffers.
- `shaderSharedInt64Atomics` indicates whether shaders **can** perform 64-bit unsigned and signed integer atomic operations on shared memory.

If the `VkPhysicalDeviceShaderAtomicInt64Features` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderAtomicInt64Features` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderAtomicInt64Features-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_INT64_FEATURES`

The `VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_shader_image_atomic_int64
typedef struct VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderImageInt64Atomics;
    VkBool32           sparseImageInt64Atomics;
} VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderImageInt64Atomics` indicates whether shaders **can** support 64-bit unsigned and signed integer atomic operations on images.
- `sparseImageInt64Atomics` indicates whether 64-bit integer atomics **can** be used on sparse images.

If the `VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to

indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderAtomicInt64FeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- `VUID-VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT-sType-sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_IMAGE_ATOMIC_INT64_FEATURES_EXT`

The `VkPhysicalDevice8BitStorageFeatures` structure is defined as:

```
typedef struct VkPhysicalDevice8BitStorageFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           storageBuffer8BitAccess;
    VkBool32           uniformAndStorageBuffer8BitAccess;
    VkBool32           storagePushConstant8;
} VkPhysicalDevice8BitStorageFeatures;
```

or the equivalent

```
// Provided by VK_KHR_8bit_storage
typedef VkPhysicalDevice8BitStorageFeatures VkPhysicalDevice8BitStorageFeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `storageBuffer8BitAccess` indicates whether objects in the `StorageBuffer`, `ShaderRecordBufferKHR`, or `PhysicalStorageBuffer` storage class with the `Block` decoration **can** have 8-bit integer members. If this feature is not enabled, 8-bit integer members **must** not be used in such objects. This also indicates whether shader modules **can** declare the `StorageBuffer8BitAccess` capability.
- `uniformAndStorageBuffer8BitAccess` indicates whether objects in the `Uniform` storage class with the `Block` decoration **can** have 8-bit integer members. If this feature is not enabled, 8-bit integer members **must** not be used in such objects. This also indicates whether shader modules **can** declare the `UniformAndStorageBuffer8BitAccess` capability.
- `storagePushConstant8` indicates whether objects in the `PushConstant` storage class **can** have 8-bit integer members. If this feature is not enabled, 8-bit integer members **must** not be used in such objects. This also indicates whether shader modules **can** declare the `StoragePushConstant8` capability.

If the `VkPhysicalDevice8BitStorageFeatures` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to

indicate whether each corresponding feature is supported. `VkPhysicalDevice8BitStorageFeatures` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- `VUID-VkPhysicalDevice8BitStorageFeatures-sType-sType`
`sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_8BIT_STORAGE_FEATURES`

The `VkPhysicalDevice16BitStorageFeatures` structure is defined as:

```
typedef struct VkPhysicalDevice16BitStorageFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           storageBuffer16BitAccess;
    VkBool32           uniformAndStorageBuffer16BitAccess;
    VkBool32           storagePushConstant16;
    VkBool32           storageInputOutput16;
} VkPhysicalDevice16BitStorageFeatures;
```

or the equivalent

```
// Provided by VK_KHR_16bit_storage
typedef VkPhysicalDevice16BitStorageFeatures VkPhysicalDevice16BitStorageFeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `storageBuffer16BitAccess` specifies whether objects in the `StorageBuffer`, `ShaderRecordBufferKHR`, or `PhysicalStorageBuffer` storage class with the `Block` decoration can have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or 16-bit floating-point members must not be used in such objects. This also specifies whether shader modules can declare the `StorageBuffer16BitAccess` capability.
- `uniformAndStorageBuffer16BitAccess` specifies whether objects in the `Uniform` storage class with the `Block` decoration can have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or 16-bit floating-point members must not be used in such objects. This also specifies whether shader modules can declare the `UniformAndStorageBuffer16BitAccess` capability.
- `storagePushConstant16` specifies whether objects in the `PushConstant` storage class can have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or floating-point members must not be used in such objects. This also specifies whether shader modules can declare the `StoragePushConstant16` capability.
- `storageInputOutput16` specifies whether objects in the `Input` and `Output` storage classes can have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or

16-bit floating-point members **must** not be used in such objects. This also specifies whether shader modules **can** declare the `StorageInputOutput16` capability.

If the `VkPhysicalDevice16BitStorageFeatures` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDevice16BitStorageFeatures` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDevice16BitStorageFeatures-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_16BIT_STORAGE_FEATURES`

The `VkPhysicalDeviceShaderFloat16Int8Features` structure is defined as:

```
typedef struct VkPhysicalDeviceShaderFloat16Int8Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderFloat16;
    VkBool32           shaderInt8;
} VkPhysicalDeviceShaderFloat16Int8Features;
```

or the equivalent

```
// Provided by VK_KHR_shader_float16_int8
typedef VkPhysicalDeviceShaderFloat16Int8Features
VkPhysicalDeviceShaderFloat16Int8FeaturesKHR;
```

```
// Provided by VK_KHR_shader_float16_int8
typedef VkPhysicalDeviceShaderFloat16Int8Features
VkPhysicalDeviceFloat16Int8FeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderFloat16` indicates whether 16-bit floats (halves) are supported in shader code. This also indicates whether shader modules **can** declare the `Float16` capability. However, this only enables a subset of the storage classes that SPIR-V allows for the `Float16` SPIR-V capability: Declaring and using 16-bit floats in the `Private`, `Workgroup` (for non-Block variables), and `Function` storage classes is enabled, while declaring them in the interface storage classes (e.g., `UniformConstant`, `Uniform`, `StorageBuffer`, `Input`, `Output`, and `PushConstant`) is not enabled.
- `shaderInt8` indicates whether 8-bit integers (signed and unsigned) are supported in shader code.

This also indicates whether shader modules **can** declare the **Int8** capability. However, this only enables a subset of the storage classes that SPIR-V allows for the **Int8** SPIR-V capability: Declaring and using 8-bit integers in the **Private**, **Workgroup** (for non-Block variables), and **Function** storage classes is enabled, while declaring them in the interface storage classes (e.g., **UniformConstant**, **Uniform**, **StorageBuffer**, **Input**, **Output**, and **PushConstant**) is not enabled.

If the **VkPhysicalDeviceShaderFloat16Int8Features** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDeviceShaderFloat16Int8Features** **can** also be used in the **pNext** chain of **VkDeviceCreateInfo** to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderFloat16Int8Features-sType-sType
sType must be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES**

The **VkPhysicalDeviceShaderClockFeaturesKHR** structure is defined as:

```
// Provided by VK_KHR_shader_clock
typedef struct VkPhysicalDeviceShaderClockFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderSubgroupClock;
    VkBool32           shaderDeviceClock;
} VkPhysicalDeviceShaderClockFeaturesKHR;
```

This structure describes the following features:

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **shaderSubgroupClock** indicates whether shaders **can** perform **Subgroup** scoped clock reads.
- **shaderDeviceClock** indicates whether shaders **can** perform **Device** scoped clock reads.

If the **VkPhysicalDeviceShaderClockFeaturesKHR** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDeviceShaderClockFeaturesKHR** **can** also be used in the **pNext** chain of **VkDeviceCreateInfo** to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderClockFeaturesKHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_CLOCK_FEATURES_KHR**

The **VkPhysicalDeviceSamplerYcbcrConversionFeatures** structure is defined as:

```
typedef struct VkPhysicalDeviceSamplerYcbcrConversionFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           samplerYcbcrConversion;
} VkPhysicalDeviceSamplerYcbcrConversionFeatures;
```

or the equivalent

```
// Provided by VK_KHR_sampler_ycbcr_conversion
typedef VkPhysicalDeviceSamplerYcbcrConversionFeatures
VkPhysicalDeviceSamplerYcbcrConversionFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `samplerYcbcrConversion` specifies whether the implementation supports `sampler Y'CBCR conversion`. If `samplerYcbcrConversion` is `VK_FALSE`, `sampler Y'CBCR conversion` is not supported, and samplers using `sampler Y'CBCR conversion` **must** not be used.

If the `VkPhysicalDeviceSamplerYcbcrConversionFeatures` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceSamplerYcbcrConversionFeatures` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSamplerYcbcrConversionFeatures-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_YCBCR_CONVERSION_FEATURES`

The `VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_blend_operation_advanced
typedef struct VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           advancedBlendCoherentOperations;
} VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `advancedBlendCoherentOperations` specifies whether blending using `advanced blend operations` is guaranteed to execute atomically and in `primitive order`. If this is `VK_TRUE`, `VK_ACCESS_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT` is treated the same as `VK_ACCESS_COLOR_ATTACHMENT_READ_BIT`, and advanced blending needs no additional synchronization over basic blending. If this is `VK_FALSE`, then memory dependencies are required to guarantee order between two advanced blending operations that occur on the same sample.

If the `VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_FEATURES_EXT`

The `VkPhysicalDeviceVertexAttributeDivisorFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_vertex_attribute_divisor
typedef struct VkPhysicalDeviceVertexAttributeDivisorFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           vertexAttributeInstanceRateDivisor;
    VkBool32           vertexAttributeInstanceRateZeroDivisor;
} VkPhysicalDeviceVertexAttributeDivisorFeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `vertexAttributeInstanceRateDivisor` specifies whether vertex attribute fetching may be repeated in the case of instanced rendering.
- `vertexAttributeInstanceRateZeroDivisor` specifies whether a zero value for `VkVertexInputBindingDivisorDescriptionEXT::divisor` is supported.

If the `VkPhysicalDeviceVertexAttributeDivisorFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceVertexAttributeDivisorFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVertexAttributeDivisorFeaturesKHR-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_FEATURES_KHR

The `VkPhysicalDeviceASTCDecodeFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_astc_decode_mode
typedef struct VkPhysicalDeviceASTCDecodeFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           decodeModeSharedExponent;
} VkPhysicalDeviceASTCDecodeFeaturesEXT;
```

This structure describes the following feature:

- sType is a `VkStructureType` value identifying this structure.
- pNext is `NULL` or a pointer to a structure extending this structure.
- decodeModeSharedExponent indicates whether the implementation supports decoding ASTC compressed formats to `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32` internal precision.

If the `VkPhysicalDeviceASTCDecodeFeaturesEXT` structure is included in the pNext chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceASTCDecodeFeaturesEXT` can also be used in the pNext chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceASTCDecodeFeaturesEXT-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ASTC_DECODE_FEATURES_EXT

The `VkPhysicalDeviceTransformFeedbackFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_transform_feedback
typedef struct VkPhysicalDeviceTransformFeedbackFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           transformFeedback;
    VkBool32           geometryStreams;
} VkPhysicalDeviceTransformFeedbackFeaturesEXT;
```

This structure describes the following features:

- sType is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `transformFeedback` indicates whether the implementation supports transform feedback and shader modules **can** declare the `TransformFeedback` capability.
- `geometryStreams` indicates whether the implementation supports the `GeometryStreams` SPIR-V capability.

If the `VkPhysicalDeviceTransformFeedbackFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceTransformFeedbackFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceTransformFeedbackFeaturesEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_FEATURES_EXT`

The `VkPhysicalDeviceVulkanMemoryModelFeatures` structure is defined as:

```
typedef struct VkPhysicalDeviceVulkanMemoryModelFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           vulkanMemoryModel;
    VkBool32           vulkanMemoryModelDeviceScope;
    VkBool32           vulkanMemoryModelAvailabilityVisibilityChains;
} VkPhysicalDeviceVulkanMemoryModelFeatures;
```

or the equivalent

```
// Provided by VK_KHR_vulkan_memory_model
typedef VkPhysicalDeviceVulkanMemoryModelFeatures
VkPhysicalDeviceVulkanMemoryModelFeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `vulkanMemoryModel` indicates whether shader modules **can** declare the `VulkanMemoryModel` capability.
- `vulkanMemoryModelDeviceScope` indicates whether the Vulkan Memory Model can use `Device` scope synchronization. This also indicates whether shader modules **can** declare the `VulkanMemoryModelDeviceScope` capability.
- `vulkanMemoryModelAvailabilityVisibilityChains` indicates whether the Vulkan Memory Model

can use [availability and visibility chains](#) with more than one element.

If the `VkPhysicalDeviceVulkanMemoryModelFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceVulkanMemoryModelFeaturesKHR` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVulkanMemoryModelFeatures-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_MEMORY_MODEL_FEATURES`

The `VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_compute_shader_derivatives
typedef struct VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           computeDerivativeGroupQuads;
    VkBool32           computeDerivativeGroupLinear;
} VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `computeDerivativeGroupQuads` indicates that the implementation supports the `ComputeDerivativeGroupQuadsKHR` SPIR-V capability.
- `computeDerivativeGroupLinear` indicates that the implementation supports the `ComputeDerivativeGroupLinearKHR` SPIR-V capability.

See [Quad shader scope](#) for more information.

If the `VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_FEATURES_KHR`

The `VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_fragment_shader_barycentric
typedef struct VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           fragmentShaderBarycentric;
} VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fragmentShaderBarycentric` indicates that the implementation supports the `BaryCoordKHR` and `BaryCoordNoPerspKHR` SPIR-V fragment shader built-ins and supports the `PerVertexKHR` SPIR-V decoration on fragment shader input variables.

See [Barycentric Interpolation](#) for more information.

If the `VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR-sType-sType `sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_FEATURES_KHR`

The `VkPhysicalDeviceUniformBufferStandardLayoutFeatures` structure is defined as:

```
typedef struct VkPhysicalDeviceUniformBufferStandardLayoutFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           uniformBufferStandardLayout;
} VkPhysicalDeviceUniformBufferStandardLayoutFeatures;
```

or the equivalent

```
// Provided by VK_KHR_uniform_buffer_standard_layout
typedef VkPhysicalDeviceUniformBufferStandardLayoutFeatures
VkPhysicalDeviceUniformBufferStandardLayoutFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `uniformBufferStandardLayout` indicates that the implementation supports the same layouts for uniform buffers as for storage and other kinds of buffers. See [Standard Buffer Layout](#).

If the `VkPhysicalDeviceUniformBufferStandardLayoutFeatures` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceUniformBufferStandardLayoutFeatures` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceUniformBufferStandardLayoutFeatures-sType-sType `sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFORM_BUFFER_STANDARD_LAYOUT_FEATURES`

The `VkPhysicalDeviceDepthClipEnableFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_depth_clip_enable
typedef struct VkPhysicalDeviceDepthClipEnableFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           depthClipEnable;
} VkPhysicalDeviceDepthClipEnableFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthClipEnable` indicates that the implementation supports setting the depth clipping operation explicitly via the `VkPipelineRasterizationDepthClipStateCreateInfoEXT` pipeline state. Otherwise depth clipping is only enabled when `VkPipelineRasterizationStateCreateInfo::depthClampEnable` is `VK_FALSE`.

If the `VkPhysicalDeviceDepthClipEnableFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceDepthClipEnableFeaturesEXT` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDepthClipEnableFeaturesEXT-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_ENABLE_FEATURES_EXT

The `VkPhysicalDeviceMemoryPriorityFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_memory_priority
typedef struct VkPhysicalDeviceMemoryPriorityFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           memoryPriority;
} VkPhysicalDeviceMemoryPriorityFeaturesEXT;
```

This structure describes the following feature:

- sType is a `VkStructureType` value identifying this structure.
- pNext is `NULL` or a pointer to a structure extending this structure.
- memoryPriority indicates that the implementation supports memory priorities specified at memory allocation time via `VkMemoryPriorityAllocateInfoEXT`.

If the `VkPhysicalDeviceMemoryPriorityFeaturesEXT` structure is included in the pNext chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceMemoryPriorityFeaturesEXT` **can** also be used in the pNext chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMemoryPriorityFeaturesEXT-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PRIORITY_FEATURES_EXT

The `VkPhysicalDeviceBufferDeviceAddressFeatures` structure is defined as:

```
typedef struct VkPhysicalDeviceBufferDeviceAddressFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           bufferDeviceAddress;
    VkBool32           bufferDeviceAddressCaptureReplay;
    VkBool32           bufferDeviceAddressMultiDevice;
} VkPhysicalDeviceBufferDeviceAddressFeatures;
```

or the equivalent

```
// Provided by VK_KHR_buffer_device_address
typedef VkPhysicalDeviceBufferDeviceAddressFeatures
VkPhysicalDeviceBufferDeviceAddressFeaturesKHR;
```

This structure describes the following features:

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **bufferDeviceAddress** indicates that the implementation supports accessing buffer memory in shaders as storage buffers via an address queried from [vkGetBufferDeviceAddressKHR](#).
- **bufferDeviceAddressCaptureReplay** indicates that the implementation supports saving and reusing buffer and device addresses, e.g. for trace capture and replay.
- **bufferDeviceAddressMultiDevice** indicates that the implementation supports the **bufferDeviceAddress**, **rayTracingPipeline** and **rayQuery** features for logical devices created with multiple physical devices. If this feature is not supported, buffer and acceleration structure addresses **must** not be queried on a logical device created with more than one physical device.

NOTE

bufferDeviceAddressMultiDevice exists to allow certain legacy platforms to be able to support **bufferDeviceAddress** without needing to support shared GPU virtual addresses for multi-device configurations.

See [vkGetBufferDeviceAddressKHR](#) for more information.

If the **VkPhysicalDeviceBufferDeviceAddressFeatures** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to [vkGetPhysicalDeviceFeatures2KHR](#), it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDeviceBufferDeviceAddressFeatures** **can** also be used in the **pNext** chain of [VkDeviceCreateInfo](#) to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceBufferDeviceAddressFeatures-sType-sType
sType must be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BUFFER_DEVICE_ADDRESS_FEATURES**

The **VkPhysicalDeviceImagelessFramebufferFeatures** structure is defined as:

```
typedef struct VkPhysicalDeviceImagelessFramebufferFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           imagelessFramebuffer;
} VkPhysicalDeviceImagelessFramebufferFeatures;
```

or the equivalent


```
// Provided by VK_KHR_imageless_framebuffer
typedef VkPhysicalDeviceImagelessFramebufferFeatures
VkPhysicalDeviceImagelessFramebufferFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imagelessFramebuffer` indicates that the implementation supports specifying the image view for attachments at render pass begin time via `VkRenderPassAttachmentBeginInfo`.

If the `VkPhysicalDeviceImagelessFramebufferFeatures` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceImagelessFramebufferFeatures` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImagelessFramebufferFeatures-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGELESS_FRAMEBUFFER_FEATURES`

The `VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_fragment_shader_interlock
typedef struct VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           fragmentShaderSampleInterlock;
    VkBool32           fragmentShaderPixelInterlock;
    VkBool32           fragmentShaderShadingRateInterlock;
} VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `fragmentShaderSampleInterlock` indicates that the implementation supports the `FragmentShaderSampleInterlockEXT` SPIR-V capability.
- `fragmentShaderPixelInterlock` indicates that the implementation supports the `FragmentShaderPixelInterlockEXT` SPIR-V capability.
- `fragmentShaderShadingRateInterlock` indicates that the implementation supports the `FragmentShaderShadingRateInterlockEXT` SPIR-V capability.

If the `VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_INTERLOCK_FEATURES_EXT`

The `VkPhysicalDeviceCooperativeMatrixFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_cooperative_matrix
typedef struct VkPhysicalDeviceCooperativeMatrixFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           cooperativeMatrix;
    VkBool32           cooperativeMatrixRobustBufferAccess;
} VkPhysicalDeviceCooperativeMatrixFeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `cooperativeMatrix` indicates that the implementation supports the `CooperativeMatrixKHR` SPIR-V capability.
- `cooperativeMatrixRobustBufferAccess` indicates that the implementation supports robust buffer access for SPIR-V `OpCooperativeMatrixLoadKHR` and `OpCooperativeMatrixStoreKHR` instructions.

If the `VkPhysicalDeviceCooperativeMatrixFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceCooperativeMatrixFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCooperativeMatrixFeaturesKHR-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_FEATURES_KHR`

The `VkPhysicalDeviceYcbcrImageArraysFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_ycbcr_image_arrays
typedef struct VkPhysicalDeviceYcbcrImageArraysFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           ycbcrImageArrays;
} VkPhysicalDeviceYcbcrImageArraysFeaturesEXT;
```

This structure describes the following feature:

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **ycbcrImageArrays** indicates that the implementation supports creating images with a format that requires [Y'C_BC_R conversion](#) and has multiple array layers.

If the [VkPhysicalDeviceYcbcrImageArraysFeaturesEXT](#) structure is included in the **pNext** chain of the [VkPhysicalDeviceFeatures2](#) structure passed to [vkGetPhysicalDeviceFeatures2KHR](#), it is filled in to indicate whether each corresponding feature is supported. [VkPhysicalDeviceYcbcrImageArraysFeaturesEXT](#) **can** also be used in the **pNext** chain of [VkDeviceCreateInfo](#) to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceYcbcrImageArraysFeaturesEXT-sType-sType
sType must be [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_YCBCR_IMAGE_ARRAYS_FEATURES_EXT](#)

The [VkPhysicalDeviceTimelineSemaphoreFeatures](#) structure is defined as:

```
typedef struct VkPhysicalDeviceTimelineSemaphoreFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           timelineSemaphore;
} VkPhysicalDeviceTimelineSemaphoreFeatures;
```

or the equivalent

```
// Provided by VK_KHR_timeline_semaphore
typedef VkPhysicalDeviceTimelineSemaphoreFeatures
VkPhysicalDeviceTimelineSemaphoreFeaturesKHR;
```

This structure describes the following feature:

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.

- `timelineSemaphore` indicates whether semaphores created with a `VkSemaphoreType` of `VK_SEMAPHORE_TYPE_TIMELINE` are supported.

If the `VkPhysicalDeviceTimelineSemaphoreFeatures` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceTimelineSemaphoreFeatures` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceTimelineSemaphoreFeatures-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_FEATURES`

The `VkPhysicalDeviceIndexTypeUint8FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_index_type_uint8
typedef struct VkPhysicalDeviceIndexTypeUint8FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           indexTypeUint8;
} VkPhysicalDeviceIndexTypeUint8FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `indexTypeUint8` indicates that `VK_INDEX_TYPE_UINT8_KHR` can be used with `vkCmdBindIndexBuffer2KHR` and `vkCmdBindIndexBuffer`.

If the `VkPhysicalDeviceIndexTypeUint8FeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceIndexTypeUint8FeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceIndexTypeUint8FeaturesKHR-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INDEX_TYPE_UINT8_FEATURES_KHR`

The `VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_primitive_topology_list_restart
typedef struct VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           primitiveTopologyListRestart;
    VkBool32           primitiveTopologyPatchListRestart;
} VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT;
```

This structure describes the following features:

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **primitiveTopologyListRestart** indicates that list type primitives, **VK_PRIMITIVE_TOPOLOGY_POINT_LIST**, **VK_PRIMITIVE_TOPOLOGY_LINE_LIST**, **VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST**, **VK_PRIMITIVE_TOPOLOGY_LINE_LIST_WITH_ADJACENCY** and **VK_PRIMITIVE_TOPOLOGY_TRIANGLE_LIST_WITH_ADJACENCY**, **can** use the primitive restart index value in index buffers.
- **primitiveTopologyPatchListRestart** indicates that the **VK_PRIMITIVE_TOPOLOGY_PATCH_LIST** topology **can** use the primitive restart index value in index buffers.

If the **VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT** **can** also be used in the **pNext** chain of **VkDeviceCreateInfo** to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT-sType-sType **must** be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVE_TOPOLOGY_LIST_RESTART_FEATURES_EXT**

The **VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures** structure is defined as:

```
typedef struct VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           separateDepthStencilLayouts;
} VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures;
```

or the equivalent

```
// Provided by VK_KHR_separate_depth_stencil_layouts
typedef VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures
VkPhysicalDeviceSeparateDepthStencilLayoutsFeaturesKHR;
```

This structure describes the following feature:

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **separateDepthStencilLayouts** indicates whether the implementation supports a **VkImageMemoryBarrier** for a depth/stencil image with only one of **VK_IMAGE_ASPECT_DEPTH_BIT** or **VK_IMAGE_ASPECT_STENCIL_BIT** set, and whether **VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL**, **VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL**, **VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL**, or **VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL** can be used.

If the **VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures** can also be used in the **pNext** chain of **VkDeviceCreateInfo** to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSeparateDepthStencilLayoutsFeatures-sType-sType **must** be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SEPARATE_DEPTH_STENCIL_LAYOUTS_FEATURES**

The **VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR** structure is defined as:

```
// Provided by VK_KHR_pipeline_executable_properties
typedef struct VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineExecutableInfo;
} VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR;
```

This structure describes the following feature:

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **pipelineExecutableInfo** indicates that the implementation supports reporting properties and statistics about the pipeline executables associated with a compiled pipeline.

If the **VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is

filled in to indicate whether each corresponding feature is supported. `VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- `VUID-VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR-sType-sType` `sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_EXECUTABLE_PROPERTIES_FEATURES_KHR`

The `VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_attachment_feedback_loop_dynamic_state
typedef struct VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           attachmentFeedbackLoopDynamicState;
} VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `attachmentFeedbackLoopDynamicState` specifies whether dynamic feedback loops are supported.

If the `VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- `VUID-VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT-sType-sType` `sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_DYNAMIC_STATE_FEATURES_EXT`

The `VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT` structure is defined as:


```
// Provided by VK_EXT_legacy_vertex_attributes
typedef struct VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           legacyVertexAttributes;
} VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT;
```

This structure describes the following features:

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **legacyVertexAttributes** specifies whether compatibility features for vertex attributes are supported when using dynamic vertex input state.

If the **VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT** **can** also be used in the **pNext** chain of **VkDeviceCreateInfo** to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT-sType-sType
sType must be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_FEATURES_EXT**

The **VkPhysicalDeviceLineRasterizationFeaturesKHR** structure is defined as:

```
// Provided by VK_KHR_line_rasterization
typedef struct VkPhysicalDeviceLineRasterizationFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rectangularLines;
    VkBool32           bresenhamLines;
    VkBool32           smoothLines;
    VkBool32           stippledRectangularLines;
    VkBool32           stippledBresenhamLines;
    VkBool32           stippledSmoothLines;
} VkPhysicalDeviceLineRasterizationFeaturesKHR;
```

This structure describes the following features:

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **rectangularLines** indicates whether the implementation supports **rectangular line rasterization**.
- **bresenhamLines** indicates whether the implementation supports **Bresenham-style line**

rasterization.

- `smoothLines` indicates whether the implementation supports [smooth line rasterization](#).
- `stippledRectangularLines` indicates whether the implementation supports [stippled line rasterization](#) with `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_KHR` lines.
- `stippledBresenhamLines` indicates whether the implementation supports [stippled line rasterization](#) with `VK_LINE_RASTERIZATION_MODE_BRESENHAM_KHR` lines.
- `stippledSmoothLines` indicates whether the implementation supports [stippled line rasterization](#) with `VK_LINE_RASTERIZATION_MODE_RECTANGULAR_SMOOTH_KHR` lines.

If the `VkPhysicalDeviceLineRasterizationFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceLineRasterizationFeaturesKHR` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLineRasterizationFeaturesKHR-sType-sType `sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_FEATURES_KHR`

The `VkPhysicalDeviceAccelerationStructureFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkPhysicalDeviceAccelerationStructureFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           accelerationStructure;
    VkBool32           accelerationStructureCaptureReplay;
    VkBool32           accelerationStructureIndirectBuild;
    VkBool32           accelerationStructureHostCommands;
    VkBool32           descriptorBindingAccelerationStructureUpdateAfterBind;
} VkPhysicalDeviceAccelerationStructureFeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `accelerationStructure` indicates whether the implementation supports the acceleration structure functionality. See [Acceleration Structures](#).
- `accelerationStructureCaptureReplay` indicates whether the implementation supports saving and reusing acceleration structure device addresses, e.g. for trace capture and replay.
- `accelerationStructureIndirectBuild` indicates whether the implementation supports indirect acceleration structure build commands, e.g. `vkCmdBuildAccelerationStructuresIndirectKHR`.
- `accelerationStructureHostCommands` indicates whether the implementation supports host side

acceleration structure commands, e.g. [vkBuildAccelerationStructuresKHR](#), [vkCopyAccelerationStructureKHR](#), [vkCopyAccelerationStructureToMemoryKHR](#), [vkCopyMemoryToAccelerationStructureKHR](#), [vkWriteAccelerationStructuresPropertiesKHR](#).

- [descriptorBindingAccelerationStructureUpdateAfterBind](#) indicates whether the implementation supports updating acceleration structure descriptors after a set is bound. If this feature is not enabled, [VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT](#) **must** not be used with [VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR](#).

If the [VkPhysicalDeviceAccelerationStructureFeaturesKHR](#) structure is included in the [pNext](#) chain of the [VkPhysicalDeviceFeatures2](#) structure passed to [vkGetPhysicalDeviceFeatures2KHR](#), it is filled in to indicate whether each corresponding feature is supported. [VkPhysicalDeviceAccelerationStructureFeaturesKHR](#) **can** also be used in the [pNext](#) chain of [VkDeviceCreateInfo](#) to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceAccelerationStructureFeaturesKHR-sType-sType **must** be [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_FEATURES_KHR](#)

The [VkPhysicalDeviceRayTracingPipelineFeaturesKHR](#) structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkPhysicalDeviceRayTracingPipelineFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rayTracingPipeline;
    VkBool32           rayTracingPipelineShaderGroupHandleCaptureReplay;
    VkBool32           rayTracingPipelineShaderGroupHandleCaptureReplayMixed;
    VkBool32           rayTracingPipelineTraceRaysIndirect;
    VkBool32           rayTraversalPrimitiveCulling;
} VkPhysicalDeviceRayTracingPipelineFeaturesKHR;
```

This structure describes the following features:

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [rayTracingPipeline](#) indicates whether the implementation supports the ray tracing pipeline functionality. See [Ray Tracing](#).
- [rayTracingPipelineShaderGroupHandleCaptureReplay](#) indicates whether the implementation supports saving and reusing shader group handles, e.g. for trace capture and replay.
- [rayTracingPipelineShaderGroupHandleCaptureReplayMixed](#) indicates whether the implementation supports reuse of shader group handles being arbitrarily mixed with creation of non-reused shader group handles. If this is [VK_FALSE](#), all reused shader group handles **must** be specified before any non-reused handles **may** be created.
- [rayTracingPipelineTraceRaysIndirect](#) indicates whether the implementation supports indirect

ray tracing commands, e.g. [vkCmdTraceRaysIndirectKHR](#).

- [rayTraversalPrimitiveCulling](#) indicates whether the implementation supports [primitive culling](#) during ray traversal.

If the [VkPhysicalDeviceRayTracingPipelineFeaturesKHR](#) structure is included in the [pNext](#) chain of the [VkPhysicalDeviceFeatures2](#) structure passed to [vkGetPhysicalDeviceFeatures2KHR](#), it is filled in to indicate whether each corresponding feature is supported. [VkPhysicalDeviceRayTracingPipelineFeaturesKHR](#) **can** also be used in the [pNext](#) chain of [VkDeviceCreateInfo](#) to selectively enable these features.

Valid Usage

- VUID-VkPhysicalDeviceRayTracingPipelineFeaturesKHR-rayTracingPipelineShaderGroupHandleCaptureReplayMixed-03575
If [rayTracingPipelineShaderGroupHandleCaptureReplayMixed](#) is [VK_TRUE](#), [rayTracingPipelineShaderGroupHandleCaptureReplay](#) **must** also be [VK_TRUE](#)

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayTracingPipelineFeaturesKHR-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_FEATURES_KHR](#)

The [VkPhysicalDeviceRayQueryFeaturesKHR](#) structure is defined as:

```
// Provided by VK_KHR_ray_query
typedef struct VkPhysicalDeviceRayQueryFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rayQuery;
} VkPhysicalDeviceRayQueryFeaturesKHR;
```

This structure describes the following feature:

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [rayQuery](#) indicates whether the implementation supports ray query ([OpRayQueryProceedKHR](#)) functionality.

If the [VkPhysicalDeviceRayQueryFeaturesKHR](#) structure is included in the [pNext](#) chain of the [VkPhysicalDeviceFeatures2](#) structure passed to [vkGetPhysicalDeviceFeatures2KHR](#), it is filled in to indicate whether each corresponding feature is supported. [VkPhysicalDeviceRayQueryFeaturesKHR](#) **can** also be used in the [pNext](#) chain of [VkDeviceCreateInfo](#) to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayQueryFeaturesKHR-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_QUERY_FEATURES_KHR

The `VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_ray_tracing_maintenance1
typedef struct VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rayTracingMaintenance1;
    VkBool32           rayTracingPipelineTraceRaysIndirect2;
} VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR;
```

This structure describes the following features:

- sType is a `VkStructureType` value identifying this structure.
- pNext is `NULL` or a pointer to a structure extending this structure.
- rayTracingMaintenance1 indicates that the implementation supports the following:
 - The `CullMaskKHR` SPIR-V builtin using the `SPV_KHR_ray_cull_mask` SPIR-V extension.
 - Additional acceleration structure property queries: `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR` and `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR`.
 - A new access flag `VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR`.
 - A new pipeline stage flag bit `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`
- rayTracingPipelineTraceRaysIndirect2 indicates whether the implementation supports the extended indirect ray tracing command `vkCmdTraceRaysIndirect2KHR`.

If the `VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR-sType-sType
sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_MAINTENANCE_1_FEATURES_KHR`

The `VkPhysicalDeviceVideoMaintenance1FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_video_maintenance1
typedef struct VkPhysicalDeviceVideoMaintenance1FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           videoMaintenance1;
} VkPhysicalDeviceVideoMaintenance1FeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `videoMaintenance1` indicates that the implementation supports the following:
 - The new buffer creation flag `VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`.
 - The new image creation flag `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`.
 - The new video session creation flag `VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR`.

If the `VkPhysicalDeviceVideoMaintenance1FeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceVideoMaintenance1FeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVideoMaintenance1FeaturesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_MAINTENANCE_1_FEATURES_KHR`

The `VkPhysicalDeviceExtendedDynamicState3FeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_extended_dynamic_state3
typedef struct VkPhysicalDeviceExtendedDynamicState3FeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           extendedDynamicState3TessellationDomainOrigin;
    VkBool32           extendedDynamicState3DepthClampEnable;
    VkBool32           extendedDynamicState3PolygonMode;
    VkBool32           extendedDynamicState3RasterizationSamples;
    VkBool32           extendedDynamicState3SampleMask;
    VkBool32           extendedDynamicState3AlphaToCoverageEnable;
    VkBool32           extendedDynamicState3AlphaToOneEnable;
    VkBool32           extendedDynamicState3LogicOpEnable;
    VkBool32           extendedDynamicState3ColorBlendEnable;
    VkBool32           extendedDynamicState3ColorBlendEquation;
    VkBool32           extendedDynamicState3ColorWriteMask;
    VkBool32           extendedDynamicState3RasterizationStream;
    VkBool32           extendedDynamicState3ConservativeRasterizationMode;
    VkBool32           extendedDynamicState3ExtraPrimitiveOverestimationSize;
    VkBool32           extendedDynamicState3DepthClipEnable;
    VkBool32           extendedDynamicState3SampleLocationsEnable;
    VkBool32           extendedDynamicState3ColorBlendAdvanced;
    VkBool32           extendedDynamicState3ProvokingVertexMode;
    VkBool32           extendedDynamicState3LineRasterizationMode;
    VkBool32           extendedDynamicState3LineStippleEnable;
    VkBool32           extendedDynamicState3DepthClipNegativeOneToOne;
    VkBool32           extendedDynamicState3ViewportWScalingEnable;
    VkBool32           extendedDynamicState3ViewportSwizzle;
    VkBool32           extendedDynamicState3CoverageToColorEnable;
    VkBool32           extendedDynamicState3CoverageToColorLocation;
    VkBool32           extendedDynamicState3CoverageModulationMode;
    VkBool32           extendedDynamicState3CoverageModulationTableEnable;
    VkBool32           extendedDynamicState3CoverageModulationTable;
    VkBool32           extendedDynamicState3CoverageReductionMode;
    VkBool32           extendedDynamicState3RepresentativeFragmentTestEnable;
    VkBool32           extendedDynamicState3ShadingRateImageEnable;
} VkPhysicalDeviceExtendedDynamicState3FeaturesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `extendedDynamicState3TessellationDomainOrigin` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT`
- `extendedDynamicState3DepthClampEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT`

- `extendedDynamicState3PolygonMode` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_POLYGON_MODE_EXT`
- `extendedDynamicState3RasterizationSamples` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT`
- `extendedDynamicState3SampleMask` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT`
- `extendedDynamicState3AlphaToCoverageEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT`
- `extendedDynamicState3AlphaToOneEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT`
- `extendedDynamicState3LogicOpEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT`
- `extendedDynamicState3ColorBlendEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT`
- `extendedDynamicState3ColorBlendEquation` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT`
- `extendedDynamicState3ColorWriteMask` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT`
- `extendedDynamicState3RasterizationStream` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT`
- `extendedDynamicState3ConservativeRasterizationMode` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT`
- `extendedDynamicState3ExtraPrimitiveOverestimationSize` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT`
- `extendedDynamicState3DepthClipEnable` indicates that the implementation supports the following dynamic state:

- `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT`
- `extendedDynamicState3SampleLocationsEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT`
- `extendedDynamicState3ColorBlendAdvanced` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT`
- `extendedDynamicState3ProvokingVertexMode` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT`
- `extendedDynamicState3LineRasterizationMode` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT`
- `extendedDynamicState3LineStippleEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT`
- `extendedDynamicState3DepthClipNegativeOneToOne` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT`
- `extendedDynamicState3ViewportWScalingEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_VIEWPORT_W_SCALING_ENABLE_NV`
- `extendedDynamicState3ViewportSwizzle` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_VIEWPORT_SWIZZLE_NV`
- `extendedDynamicState3CoverageToColorEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_ENABLE_NV`
- `extendedDynamicState3CoverageToColorLocation` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_LOCATION_NV`
- `extendedDynamicState3CoverageModulationMode` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_COVERAGE_MODULATION_MODE_NV`
- `extendedDynamicState3CoverageModulationTableEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_ENABLE_NV`
- `extendedDynamicState3CoverageModulationTable` indicates that the implementation supports the

following dynamic state:

- `VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_NV`
- `extendedDynamicState3CoverageReductionMode` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_COVERAGE_REDUCTION_MODE_NV`
- `extendedDynamicState3RepresentativeFragmentTestEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_REPRESENTATIVE_FRAGMENT_TEST_ENABLE_NV`
- `extendedDynamicState3ShadingRateImageEnable` indicates that the implementation supports the following dynamic state:
 - `VK_DYNAMIC_STATE_SHADING_RATE_IMAGE_ENABLE_NV`

If the `VkPhysicalDeviceExtendedDynamicState3FeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceExtendedDynamicState3FeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- `VUID-VkPhysicalDeviceExtendedDynamicState3FeaturesEXT-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_FEATURES_EXT`

The `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           deviceGeneratedCommands;
    VkBool32           dynamicGeneratedPipelineLayout;
} VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceGeneratedCommands` indicates whether the implementation supports functionality to generate commands on the device.
- `dynamicGeneratedPipelineLayout` indicates the implementation allows the `pipelineLayout` member of `VkIndirectCommandsLayoutCreateInfoEXT` to be `VK_NULL_HANDLE` and `VkPipelineLayoutCreateInfo` **can** be chained off those structures' `pNext` instead.

If the `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- `VOID-VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT-sType-sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_FEATURES_EXT`

The `VkPhysicalDeviceDeviceMemoryReportFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_device_memory_report
typedef struct VkPhysicalDeviceDeviceMemoryReportFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           deviceMemoryReport;
} VkPhysicalDeviceDeviceMemoryReportFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `deviceMemoryReport` indicates whether the implementation supports the ability to register device memory report callbacks.

If the `VkPhysicalDeviceDeviceMemoryReportFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceDeviceMemoryReportFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- `VOID-VkPhysicalDeviceDeviceMemoryReportFeaturesEXT-sType-sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_MEMORY_REPORT_FEATURES_EXT`

The `VkPhysicalDeviceGlobalPriorityQueryFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_global_priority
typedef struct VkPhysicalDeviceGlobalPriorityQueryFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           globalPriorityQuery;
} VkPhysicalDeviceGlobalPriorityQueryFeaturesKHR;
```

This structure describes the following feature:

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **globalPriorityQuery** indicates whether the implementation supports the ability to query global queue priorities.

If the [VkPhysicalDeviceGlobalPriorityQueryFeaturesKHR](#) structure is included in the **pNext** chain of the [VkPhysicalDeviceFeatures2](#) structure passed to [vkGetPhysicalDeviceFeatures2KHR](#), it is filled in to indicate whether each corresponding feature is supported. [VkPhysicalDeviceGlobalPriorityQueryFeaturesKHR](#) **can** also be used in the **pNext** chain of [VkDeviceCreateInfo](#) to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceGlobalPriorityQueryFeaturesKHR-sType-sType
sType must be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GLOBAL_PRIORITY_QUERY_FEATURES_KHR**

The [VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures](#) structure is defined as:

```
typedef struct VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderZeroInitializeWorkgroupMemory;
} VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures;
```

or the equivalent

```
// Provided by VK_KHR_zero_initialize_workgroup_memory
typedef VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures
VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeaturesKHR;
```

This structure describes the following feature:

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.

- `shaderZeroInitializeWorkgroupMemory` specifies whether the implementation supports initializing a variable in Workgroup storage class.

If the `VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeatures-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_WORKGROUP_MEMORY_FEATURES`

The `VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_shader_subgroup_uniform_control_flow
typedef struct VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderSubgroupUniformControlFlow;
} VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderSubgroupUniformControlFlow` specifies whether the implementation supports the shader execution mode `SubgroupUniformControlFlowKHR`

If the `VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_UNIFORM_CONTROL_FLOW_FEATURES_KHR`

The `VkPhysicalDeviceRobustness2FeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_robustness2
typedef struct VkPhysicalDeviceRobustness2FeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32            robustBufferAccess2;
    VkBool32            robustImageAccess2;
    VkBool32            nullDescriptor;
} VkPhysicalDeviceRobustness2FeaturesEXT;
```

This structure describes the following features:

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **robustBufferAccess2** indicates whether buffer accesses are tightly bounds-checked against the range of the descriptor. Uniform buffers **must** be bounds-checked to the range of the descriptor, where the range is rounded up to a multiple of **robustUniformBufferAccessSizeAlignment**. Storage buffers **must** be bounds-checked to the range of the descriptor, where the range is rounded up to a multiple of **robustStorageBufferAccessSizeAlignment**. Out of bounds buffer loads will return zero values, and **image load, sample, and atomic operations** from texel buffers will have (0,0,1) values **inserted for missing G, B, or A components** based on the format.
- **robustImageAccess2** indicates whether image accesses are tightly bounds-checked against the dimensions of the image view. Out of bounds **image load, sample, and atomic operations** from images will return zero values, with (0,0,1) values **inserted for missing G, B, or A components** based on the format.
- **nullDescriptor** indicates whether descriptors **can** be written with a **VK_NULL_HANDLE** resource or view, which are considered valid to access and act as if the descriptor were bound to nothing.

If the **VkPhysicalDeviceRobustness2FeaturesEXT** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDeviceRobustness2FeaturesEXT** **can** also be used in the **pNext** chain of **VkDeviceCreateInfo** to selectively enable these features.

Valid Usage

- VUID-VkPhysicalDeviceRobustness2FeaturesEXT-robustBufferAccess2-04000
If **robustBufferAccess2** is enabled then **robustBufferAccess** **must** also be enabled

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRobustness2FeaturesEXT-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_FEATURES_EXT**

The **VkPhysicalDeviceShaderTerminateInvocationFeatures** structure is defined as:


```
typedef struct VkPhysicalDeviceShaderTerminateInvocationFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderTerminateInvocation;
} VkPhysicalDeviceShaderTerminateInvocationFeatures;
```

or the equivalent

```
// Provided by VK_KHR_shader_terminate_invocation
typedef VkPhysicalDeviceShaderTerminateInvocationFeatures
VkPhysicalDeviceShaderTerminateInvocationFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderTerminateInvocation` specifies whether the implementation supports SPIR-V modules that use the `SPV_KHR_terminate_invocation` extension.

If the `VkPhysicalDeviceShaderTerminateInvocationFeatures` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderTerminateInvocationFeatures` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderTerminateInvocationFeatures-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TERMINATE_INVOCATION_FEATURES`

The `VkPhysicalDeviceCustomBorderColorFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_custom_border_color
typedef struct VkPhysicalDeviceCustomBorderColorFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           customBorderColors;
    VkBool32           customBorderColorWithoutFormat;
} VkPhysicalDeviceCustomBorderColorFeaturesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `customBorderColors` indicates that the implementation supports providing a `borderColor` value with one of the following values at sampler creation time:
 - `VK_BORDER_COLOR_FLOAT_CUSTOM_EXT`
 - `VK_BORDER_COLOR_INT_CUSTOM_EXT`
- `customBorderColorWithoutFormat` indicates that explicit formats are not required for custom border colors and the value of the `format` member of the `VkSamplerCustomBorderColorCreateInfoEXT` structure **may** be `VK_FORMAT_UNDEFINED`. If this feature bit is not set, applications **must** provide the `VkFormat` of the image view(s) being sampled by this sampler in the `format` member of the `VkSamplerCustomBorderColorCreateInfoEXT` structure.

If the `VkPhysicalDeviceCustomBorderColorFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceCustomBorderColorFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCustomBorderColorFeaturesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_FEATURES_EXT`

The `VkPhysicalDeviceBorderColorSwizzleFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_border_color_swizzle
typedef struct VkPhysicalDeviceBorderColorSwizzleFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           borderColorSwizzle;
    VkBool32           borderColorSwizzleFromImage;
} VkPhysicalDeviceBorderColorSwizzleFeaturesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `borderColorSwizzle` indicates that defined values are returned by sampled image operations when used with a sampler that uses a `VK_BORDER_COLOR_FLOAT_OPAQUE_BLACK`, `VK_BORDER_COLOR_INT_OPAQUE_BLACK`, `VK_BORDER_COLOR_FLOAT_CUSTOM_EXT`, or `VK_BORDER_COLOR_INT_CUSTOM_EXT` `borderColor` and an image view that uses a non-identity component mapping, when either `borderColorSwizzleFromImage` is enabled or the `VkSamplerBorderColorComponentMappingCreateInfoEXT` is specified.
- `borderColorSwizzleFromImage` indicates that the implementation will return the correct border color values from sampled image operations under the conditions expressed above, without the

application having to specify the border color component mapping when creating the sampler object. If this feature bit is not set, applications **can** chain a [VkSamplerBorderColorComponentMappingCreateInfoEXT](#) structure when creating samplers for use with image views that do not have an [identity swizzle](#) and, when those samplers are combined with image views using the same component mapping, sampled image operations that use opaque black or custom border colors will return the correct border color values.

If the [VkPhysicalDeviceBorderColorSwizzleFeaturesEXT](#) structure is included in the [pNext](#) chain of the [VkPhysicalDeviceFeatures2](#) structure passed to [vkGetPhysicalDeviceFeatures2KHR](#), it is filled in to indicate whether each corresponding feature is supported. [VkPhysicalDeviceBorderColorSwizzleFeaturesEXT](#) **can** also be used in the [pNext](#) chain of [VkDeviceCreateInfo](#) to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceBorderColorSwizzleFeaturesEXT-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BORDER_COLOR_SWIZZLE_FEATURES_EXT](#)

The [VkPhysicalDevicePortabilitySubsetFeaturesKHR](#) structure is defined as:

```
// Provided by VK_KHR_portability_subset
typedef struct VkPhysicalDevicePortabilitySubsetFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           constantAlphaColorBlendFactors;
    VkBool32           events;
    VkBool32           imageViewFormatReinterpretation;
    VkBool32           imageViewFormatSwizzle;
    VkBool32           imageView2DOn3DImage;
    VkBool32           multisampleArrayImage;
    VkBool32           mutableComparisonSamplers;
    VkBool32           pointPolygons;
    VkBool32           samplerMipLodBias;
    VkBool32           separateStencilMaskRef;
    VkBool32           shaderSampleRateInterpolationFunctions;
    VkBool32           tessellationIsolines;
    VkBool32           tessellationPointMode;
    VkBool32           triangleFans;
    VkBool32           vertexAttributeAccessBeyondStride;
} VkPhysicalDevicePortabilitySubsetFeaturesKHR;
```

This structure describes the following features:

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [constantAlphaColorBlendFactors](#) indicates whether this implementation supports constant *alpha Blend Factors* used as source or destination *color Blending*.

- **events** indicates whether this implementation supports synchronization using **Events**.
- **imageViewFormatReinterpretation** indicates whether this implementation supports a **VkImageView** being created with a texel format containing a different number of components, or a different number of bits in each component, than the texel format of the underlying **VkImage**.
- **imageViewFormatSwizzle** indicates whether this implementation supports remapping format components using **VkImageViewCreateInfo::components**.
- **imageView2DOn3DImage** indicates whether this implementation supports a **VkImage** being created with the **VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT** flag set, permitting a 2D or 2D array image view to be created on a 3D **VkImage**.
- **multisampleArrayImage** indicates whether this implementation supports a **VkImage** being created as a 2D array with multiple samples per texel.
- **mutableComparisonSamplers** indicates whether this implementation allows descriptors with comparison samplers to be **updated**.
- **pointPolygons** indicates whether this implementation supports **Rasterization** using a **point Polygon Mode**.
- **samplerMipLodBias** indicates whether this implementation supports setting a **mipmap LOD bias value** when **creating a sampler**.
- **separateStencilMaskRef** indicates whether this implementation supports separate front and back **Stencil Test** reference values.
- **shaderSampleRateInterpolationFunctions** indicates whether this implementation supports fragment shaders which use the **InterpolationFunction** capability and the extended instructions **InterpolateAtCentroid**, **InterpolateAtOffset**, and **InterpolateAtSample** from the **GLSL.std.450** extended instruction set. This member is only meaningful if the **sampleRateShading** feature is supported.
- **tessellationIsolines** indicates whether this implementation supports **isoline output** from the **Tessellation** stage of a graphics pipeline. This member is only meaningful if **tessellationShader** are supported.
- **tessellationPointMode** indicates whether this implementation supports **point output** from the **Tessellation** stage of a graphics pipeline. This member is only meaningful if **tessellationShader** are supported.
- **triangleFans** indicates whether this implementation supports **Triangle Fans** primitive topology.
- **vertexAttributeAccessBeyondStride** indicates whether this implementation supports accessing a vertex input attribute beyond the stride of the corresponding vertex input binding.

If the **VkPhysicalDevicePortabilitySubsetFeaturesKHR** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDevicePortabilitySubsetFeaturesKHR** can also be used in the **pNext** chain of **VkDeviceCreateInfo** to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePortabilitySubsetFeaturesKHR-sType-sType

sType must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_FEATURES_KHR`

The `VkPhysicalDevicePerformanceQueryFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkPhysicalDevicePerformanceQueryFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           performanceCounterQueryPools;
    VkBool32           performanceCounterMultipleQueryPools;
} VkPhysicalDevicePerformanceQueryFeaturesKHR;
```

This structure describes the following features:

- **sType** is a `VkStructureType` value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **performanceCounterQueryPools** indicates whether the implementation supports performance counter query pools.
- **performanceCounterMultipleQueryPools** indicates whether the implementation supports using multiple performance query pools in a primary command buffer and secondary command buffers executed within it.

If the `VkPhysicalDevicePerformanceQueryFeaturesKHR` structure is included in the **pNext** chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDevicePerformanceQueryFeaturesKHR` can also be used in the **pNext** chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePerformanceQueryFeaturesKHR-sType-sType
sType must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_FEATURES_KHR`

The `VkPhysicalDeviceDepthClipControlFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_depth_clip_control
typedef struct VkPhysicalDeviceDepthClipControlFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           depthClipControl;
} VkPhysicalDeviceDepthClipControlFeaturesEXT;
```

This structure describes the following feature:

- **sType** is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthClipControl` indicates that the implementation supports setting `VkPipelineViewportDepthClipControlCreateInfoEXT::negativeOneToOne` to `VK_TRUE`.

If the `VkPhysicalDeviceDepthClipControlFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceDepthClipControlFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDepthClipControlFeaturesEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_CONTROL_FEATURES_EXT`

The `VkPhysicalDeviceDepthClampControlFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_depth_clamp_control
typedef struct VkPhysicalDeviceDepthClampControlFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           depthClampControl;
} VkPhysicalDeviceDepthClampControlFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthClampControl` indicates that the implementation supports setting `VkPipelineViewportDepthClampControlCreateInfoEXT::depthClampMode` to `VK_DEPTH_CLAMP_MODE_USER_DEFINED_RANGE_EXT`.

If the `VkPhysicalDeviceDepthClampControlFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceDepthClampControlFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDepthClampControlFeaturesEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_CONTROL_FEATURES_EXT`

The `VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_workgroup_memory_explicit_layout
typedef struct VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           workgroupMemoryExplicitLayout;
    VkBool32           workgroupMemoryExplicitLayoutScalarBlockLayout;
    VkBool32           workgroupMemoryExplicitLayout8BitAccess;
    VkBool32           workgroupMemoryExplicitLayout16BitAccess;
} VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `workgroupMemoryExplicitLayout` indicates whether the implementation supports the SPIR-V `WorkgroupMemoryExplicitLayoutKHR` capability.
- `workgroupMemoryExplicitLayoutScalarBlockLayout` indicates whether the implementation supports scalar alignment for laying out Workgroup Blocks.
- `workgroupMemoryExplicitLayout8BitAccess` indicates whether objects in the `Workgroup` storage class with the `Block` decoration **can** have 8-bit integer members. If this feature is not enabled, 8-bit integer members **must** not be used in such objects. This also indicates whether shader modules **can** declare the `WorkgroupMemoryExplicitLayout8BitAccessKHR` capability.
- `workgroupMemoryExplicitLayout16BitAccess` indicates whether objects in the `Workgroup` storage class with the `Block` decoration **can** have 16-bit integer and 16-bit floating-point members. If this feature is not enabled, 16-bit integer or 16-bit floating-point members **must** not be used in such objects. This also indicates whether shader modules **can** declare the `WorkgroupMemoryExplicitLayout16BitAccessKHR` capability.

If the `VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_WORKGROUP_MEMORY_EXPLICIT_LAYOUT_FEATURES_KHR`

The `VkPhysicalDeviceSynchronization2Features` structure is defined as:


```
typedef struct VkPhysicalDeviceSynchronization2Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           synchronization2;
} VkPhysicalDeviceSynchronization2Features;
```

or the equivalent

```
// Provided by VK_KHR_synchronization2
typedef VkPhysicalDeviceSynchronization2Features
VkPhysicalDeviceSynchronization2FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `synchronization2` indicates whether the implementation supports the new set of synchronization commands introduced in `VK_KHR_synchronization2`.

If the `VkPhysicalDeviceSynchronization2Features` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceSynchronization2Features` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSynchronization2Features-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SYNCHRONIZATION_2_FEATURES`

The `VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_vertex_input_dynamic_state
typedef struct VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           vertexInputDynamicState;
} VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.

- `vertexInputDynamicState` indicates that the implementation supports the following dynamic states:
 - `VK_DYNAMIC_STATE_VERTEX_INPUT_EXT`

If the `VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_INPUT_DYNAMIC_STATE_FEATURES_EXT`

The `VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_primitives_generated_query
typedef struct VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           primitivesGeneratedQuery;
    VkBool32           primitivesGeneratedQueryWithRasterizerDiscard;
    VkBool32           primitivesGeneratedQueryWithNonZeroStreams;
} VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `primitivesGeneratedQuery` indicates whether the implementation supports the `VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT` query type.
- `primitivesGeneratedQueryWithRasterizerDiscard` indicates whether the implementation supports this query when `rasterization discard` is enabled.
- `primitivesGeneratedQueryWithNonZeroStreams` indicates whether the implementation supports this query with a non-zero index in `vkCmdBeginQueryIndexedEXT`.

If the `VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVES_GENERATED_QUERY_FEATURES_EXT`

The `VkPhysicalDeviceFragmentShadingRateFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_fragment_shading_rate
typedef struct VkPhysicalDeviceFragmentShadingRateFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineFragmentShadingRate;
    VkBool32           primitiveFragmentShadingRate;
    VkBool32           attachmentFragmentShadingRate;
} VkPhysicalDeviceFragmentShadingRateFeaturesKHR;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pipelineFragmentShadingRate` indicates that the implementation supports the [pipeline fragment shading rate](#).
- `primitiveFragmentShadingRate` indicates that the implementation supports the [primitive fragment shading rate](#).
- `attachmentFragmentShadingRate` indicates that the implementation supports the [attachment fragment shading rate](#).

If the `VkPhysicalDeviceFragmentShadingRateFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceFragmentShadingRateFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShadingRateFeaturesKHR-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_FEATURES_KHR`

The `VkPhysicalDeviceColorWriteEnableFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_color_write_enable
typedef struct VkPhysicalDeviceColorWriteEnableFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           colorWriteEnable;
} VkPhysicalDeviceColorWriteEnableFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `colorWriteEnable` indicates that the implementation supports the dynamic state `VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT`.

If the `VkPhysicalDeviceColorWriteEnableFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceColorWriteEnableFeaturesEXT` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceColorWriteEnableFeaturesEXT-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COLOR_WRITE_ENABLE_FEATURES_EXT`

The `VkPhysicalDeviceProvokingVertexFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_provoking_vertex
typedef struct VkPhysicalDeviceProvokingVertexFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           provokingVertexLast;
    VkBool32           transformFeedbackPreservesProvokingVertex;
} VkPhysicalDeviceProvokingVertexFeaturesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `provokingVertexLast` indicates whether the implementation supports the `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT` provoking vertex mode for flat shading.
- `transformFeedbackPreservesProvokingVertex` indicates that the order of vertices within each primitive written by transform feedback will preserve the provoking vertex. This does not apply to triangle fan primitives when `transformFeedbackPreservesTriangleFanProvokingVertex` is

`VK_FALSE`. `transformFeedbackPreservesProvokingVertex` **must** be `VK_FALSE` when the `VK_EXT_transform_feedback` extension is not supported.

If the `VkPhysicalDeviceProvokingVertexFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceProvokingVertexFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

When `VkPhysicalDeviceProvokingVertexFeaturesEXT` is in the `pNext` chain of `VkDeviceCreateInfo` but the `transformFeedback` feature is not enabled, the value of `transformFeedbackPreservesProvokingVertex` is ignored.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceProvokingVertexFeaturesEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_FEATURES_EXT`

The `VkPhysicalDeviceMultiDrawFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_multi_draw
typedef struct VkPhysicalDeviceMultiDrawFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           multiDraw;
} VkPhysicalDeviceMultiDrawFeaturesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `multiDraw` indicates that the implementation supports `vkCmdDrawMultiEXT` and `vkCmdDrawMultiIndexedEXT`.

If the `VkPhysicalDeviceMultiDrawFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceMultiDrawFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMultiDrawFeaturesEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_FEATURES_EXT`

The `VkPhysicalDeviceOpacityMicromapFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkPhysicalDeviceOpacityMicromapFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           micromap;
    VkBool32           micromapCaptureReplay;
    VkBool32           micromapHostCommands;
} VkPhysicalDeviceOpacityMicromapFeaturesEXT;
```

This structure describes the following feature:

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **micromap** indicates whether the implementation supports the micromap array feature.
- **micromapCaptureReplay** indicates whether the implementation supports capture and replay of addresses for micromap arrays.
- **micromapHostCommands** indicates whether the implementation supports host side micromap array commands.

If the **VkPhysicalDeviceOpacityMicromapFeaturesEXT** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDeviceOpacityMicromapFeaturesEXT** can also be used in the **pNext** chain of **VkDeviceCreateInfo** to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceOpacityMicromapFeaturesEXT-sType-sType
sType must be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_FEATURES_EXT**

The **VkPhysicalDevicePresentIdFeaturesKHR** structure is defined as:

```
// Provided by VK_KHR_present_id
typedef struct VkPhysicalDevicePresentIdFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           presentId;
} VkPhysicalDevicePresentIdFeaturesKHR;
```

This structure describes the following feature:

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **presentId** indicates that the implementation supports specifying present ID values in the

VkPresentIdKHR extension to the VkPresentInfoKHR struct.

If the `VkPhysicalDevicePresentIdFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDevicePresentIdFeaturesKHR` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePresentIdFeaturesKHR-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_ID_FEATURES_KHR`

The `VkPhysicalDevicePresentWaitFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_present_wait
typedef struct VkPhysicalDevicePresentWaitFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           presentWait;
} VkPhysicalDevicePresentWaitFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentWait` indicates that the implementation supports `vkWaitForPresentKHR`.

If the `VkPhysicalDevicePresentWaitFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDevicePresentWaitFeaturesKHR` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePresentWaitFeaturesKHR-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_WAIT_FEATURES_KHR`

The `VkPhysicalDeviceHostImageCopyFeaturesEXT` structure is defined as:


```
// Provided by VK_EXT_host_image_copy
typedef struct VkPhysicalDeviceHostImageCopyFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           hostImageCopy;
} VkPhysicalDeviceHostImageCopyFeaturesEXT;
```

This structure describes the following feature:

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **hostImageCopy** indicates that the implementation supports copying from host memory to images using the [vkCopyMemoryToImageEXT](#) command, copying from images to host memory using the [vkCopyImageToMemoryEXT](#) command, and copying between images using the [vkCopyImageToImageEXT](#) command.

If the [VkPhysicalDeviceHostImageCopyFeaturesEXT](#) structure is included in the **pNext** chain of the [VkPhysicalDeviceFeatures2](#) structure passed to [vkGetPhysicalDeviceFeatures2KHR](#), it is filled in to indicate whether each corresponding feature is supported. [VkPhysicalDeviceHostImageCopyFeaturesEXT](#) can also be used in the **pNext** chain of [VkDeviceCreateInfo](#) to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceHostImageCopyFeaturesEXT-sType-sType
sType must be [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_FEATURES_EXT](#)

The [VkPhysicalDeviceShaderIntegerDotProductFeatures](#) structure is defined as:

```
typedef struct VkPhysicalDeviceShaderIntegerDotProductFeatures {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderIntegerDotProduct;
} VkPhysicalDeviceShaderIntegerDotProductFeatures;
```

or the equivalent

```
// Provided by VK_KHR_shader_integer_dot_product
typedef VkPhysicalDeviceShaderIntegerDotProductFeatures
VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR;
```

This structure describes the following feature:

- **sType** is a [VkStructureType](#) value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderIntegerDotProduct` specifies whether shader modules **can** declare the `DotProductInputAllKHR`, `DotProductInput4x8BitKHR`, `DotProductInput4x8BitPackedKHR` and `DotProductKHR` capabilities.

If the `VkPhysicalDeviceShaderIntegerDotProductFeatures` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderIntegerDotProductFeatures` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderIntegerDotProductFeatures-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_FEATURES`

The `VkPhysicalDeviceMaintenance4Features` structure is defined as:

```
typedef struct VkPhysicalDeviceMaintenance4Features {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance4;
} VkPhysicalDeviceMaintenance4Features;
```

or the equivalent

```
// Provided by VK_KHR_maintenance4
typedef VkPhysicalDeviceMaintenance4Features VkPhysicalDeviceMaintenance4FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maintenance4` indicates that the implementation supports the following:
 - The application **may** destroy a `VkPipelineLayout` object immediately after using it to create another object.
 - `LocalSizeId` **can** be used as an alternative to `LocalSize` to specify the local workgroup size with specialization constants.
 - Images created with identical creation parameters will always have the same alignment requirements.
 - The size memory requirement of a buffer or image is never greater than that of another buffer or image created with a greater or equal size.

- Push constants do not have to be initialized before they are dynamically accessed.
- The interface matching rules allow a larger output vector to match with a smaller input vector, with additional values being discarded.

If the `VkPhysicalDeviceMaintenance4Features` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceMaintenance4Features` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance4Features-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_FEATURES`

The `VkPhysicalDeviceMaintenance5FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance5
typedef struct VkPhysicalDeviceMaintenance5FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance5;
} VkPhysicalDeviceMaintenance5FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maintenance5` indicates that the implementation supports the following:
 - The ability to expose support for the optional format `VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR`.
 - The ability to expose support for the optional format `VK_FORMAT_A8_UNORM_KHR`.
 - A property to indicate that multisample coverage operations are performed after sample counting in EarlyFragmentTests mode.
 - Creating a `VkBufferView` with a subset of the associated `VkBuffer` usage using `VkBufferUsageFlags2CreateInfoKHR`.
 - A new function `vkCmdBindIndexBuffer2KHR`, allowing a range of memory to be bound as an index buffer.
 - `vkGetDeviceProcAddr` will return `NULL` for function pointers of core functions for versions higher than the version requested by the application.
 - `vkCmdBindVertexBuffers2EXT` supports using `VK_WHOLE_SIZE` in the `pSizes` parameter.
 - If `PointSize` is not written, a default value of `1.0` is used for the size of points.
 - `VkShaderModuleCreateInfo` **can** be added as a chained structure to pipeline creation via `VkPipelineShaderStageCreateInfo`, rather than having to create a shader module.

- A function `vkGetRenderingAreaGranularityKHR` to query the optimal render area for a dynamic rendering instance.
- A property to indicate that depth/stencil texturing operations with `VK_COMPONENT_SWIZZLE_ONE` have defined behavior.
- `vkGetDeviceImageSubresourceLayoutKHR` allows an application to perform a `vkGetImageSubresourceLayout` query without having to create an image.
- `VK_REMAINING_ARRAY_LAYERS` as the `layerCount` member of `VkImageSubresourceLayers`.
- A property to indicate whether `PointSize` controls the final rasterization of polygons if `polygon mode` is `VK_POLYGON_MODE_POINT`.
- Two properties to indicate the non-strict line rasterization algorithm used.
- Two new flags words `VkPipelineCreateFlagBits2KHR` and `VkBufferUsageFlagBits2KHR`.
- Physical-device-level functions **can** now be called with any value in the valid range for a type beyond the defined enumerants, such that applications can avoid checking individual features, extensions, or versions before querying supported properties of a particular enumerant.
- Copies between images of any type are allowed, with 1D images treated as 2D images with a height of 1.

If the `VkPhysicalDeviceMaintenance5FeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceMaintenance5FeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- `VUID-VkPhysicalDeviceMaintenance5FeaturesKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_FEATURES_KHR`

The `VkPhysicalDeviceMaintenance6FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance6
typedef struct VkPhysicalDeviceMaintenance6FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance6;
} VkPhysicalDeviceMaintenance6FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maintenance6` indicates that the implementation supports the following:

- `VK_NULL_HANDLE` can be used when binding an index buffer
- `VkBindMemoryStatusKHR` can be included in the `pNext` chain of the `VkBindBufferMemoryInfo` and `VkBindImageMemoryInfo` structures, enabling applications to retrieve `VkResult` values for individual memory binding operations.
- `VkPhysicalDeviceMaintenance6PropertiesKHR::blockTexelViewCompatibleMultipleLayers` property to indicate that the implementation supports creating image views with `VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT` where the `layerCount` member of `subresourceRange` is greater than 1.
- `VkPhysicalDeviceMaintenance6PropertiesKHR::maxCombinedImageSamplerDescriptorCount` property which indicates the maximum descriptor size required for any format that requires a sampler $Y'C_B C_R$ conversion supported by the implementation.
- A `VkPhysicalDeviceMaintenance6PropertiesKHR::fragmentShadingRateClampCombinerInputs` property which indicates whether the implementation clamps the inputs to fragment shading rate combiner operations.

If the `VkPhysicalDeviceMaintenance6FeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceMaintenance6FeaturesKHR` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance6FeaturesKHR-sType-sType `sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_FEATURES_KHR`

The `VkPhysicalDeviceMaintenance7FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance7
typedef struct VkPhysicalDeviceMaintenance7FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           maintenance7;
} VkPhysicalDeviceMaintenance7FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maintenance7` indicates that the implementation supports the following:
 - The `VK_RENDERING_CONTENTS_INLINE_BIT_KHR` and `VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR` flags can be used to record commands in render pass instances both inline and in secondary command buffers executed with `vkCmdExecuteCommands` for dynamic rendering and legacy render passes

respectively.

- Querying information regarding the underlying devices in environments where the Vulkan implementation is provided through layered implementations. This is done by chaining [VkPhysicalDeviceLayeredApiPropertiesListKHR](#) to [VkPhysicalDeviceProperties2](#).
- New limits which indicate the maximum total count of dynamic uniform buffers and dynamic storage buffers that **can** be included in a pipeline layout.
- 32-bit timestamp queries **must** wrap on overflow
- A property that indicates whether a fragment shading rate attachment can have a size that is too small to cover a specified render area.
- A property that indicates support for writing to one aspect of a depth/stencil attachment without performing a read-modify-write operation on the other aspect

If the [VkPhysicalDeviceMaintenance7FeaturesKHR](#) structure is included in the **pNext** chain of the [VkPhysicalDeviceFeatures2](#) structure passed to [vkGetPhysicalDeviceFeatures2KHR](#), it is filled in to indicate whether each corresponding feature is supported. [VkPhysicalDeviceMaintenance7FeaturesKHR](#) **can** also be used in the **pNext** chain of [VkDeviceCreateInfo](#) to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance7FeaturesKHR-sType-sType
sType must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_FEATURES_KHR`

The [VkPhysicalDeviceDynamicRenderingFeatures](#) structure is defined as:

```
typedef struct VkPhysicalDeviceDynamicRenderingFeatures {  
    VkStructureType    sType;  
    void*              pNext;  
    VkBool32           dynamicRendering;  
} VkPhysicalDeviceDynamicRenderingFeatures;
```

or the equivalent

```
// Provided by VK_KHR_dynamic_rendering  
typedef VkPhysicalDeviceDynamicRenderingFeatures  
VkPhysicalDeviceDynamicRenderingFeaturesKHR;
```

This structure describes the following feature:

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is `NULL` or a pointer to a structure extending this structure.
- **dynamicRendering** specifies that the implementation supports dynamic render pass instances using the [vkCmdBeginRenderingKHR](#) command.

If the `VkPhysicalDeviceDynamicRenderingFeatures` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceDynamicRenderingFeatures` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDynamicRenderingFeatures-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_FEATURES`

The `VkPhysicalDeviceImageViewMinLodFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_image_view_min_lod
typedef struct VkPhysicalDeviceImageViewMinLodFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           minLod;
} VkPhysicalDeviceImageViewMinLodFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `minLod` indicates whether the implementation supports clamping the minimum LOD value during `Image Level(s) Selection`, `Texel Gathering` and `Integer Texel Coordinate Operations` with a given `VkImageView` by `VkImageViewMinLodCreateInfoEXT::minLod`.

If the `VkPhysicalDeviceImageViewMinLodFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceImageViewMinLodFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageViewMinLodFeaturesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_MIN_LOD_FEATURES_EXT`

The `VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT` structure is defined as:


```
// Provided by VK_EXT_attachment_feedback_loop_layout
typedef struct VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           attachmentFeedbackLoopLayout;
} VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `attachmentFeedbackLoopLayout` indicates whether the implementation supports using `VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT` image layout for images created with `VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT`.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_LAYOUT_FEATURES_EXT`

The `VkPhysicalDeviceNestedCommandBufferFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_nested_command_buffer
typedef struct VkPhysicalDeviceNestedCommandBufferFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           nestedCommandBuffer;
    VkBool32           nestedCommandBufferRendering;
    VkBool32           nestedCommandBufferSimultaneousUse;
} VkPhysicalDeviceNestedCommandBufferFeaturesEXT;
```

This structure describes the following features:

- `nestedCommandBuffer` indicates the implementation supports nested command buffers, which allows [Secondary Command Buffers](#) to execute other [Secondary Command Buffers](#).
- `nestedCommandBufferRendering` indicates that it is valid to call [vkCmdExecuteCommands](#) inside a [Secondary Command Buffer](#) recorded with `VK_COMMAND_BUFFER_USAGE_RENDER_PASS_CONTINUE_BIT`.
- `nestedCommandBufferSimultaneousUse` indicates that the implementation supports nested command buffers with command buffers that are recorded with `VK_COMMAND_BUFFER_USAGE_SIMULTANEOUS_USE_BIT`.

If the `VkPhysicalDeviceNestedCommandBufferFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to [vkGetPhysicalDeviceFeatures2KHR](#), it is filled in to indicate whether each corresponding feature is supported.

`VkPhysicalDeviceNestedCommandBufferFeaturesEXT` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceNestedCommandBufferFeaturesEXT-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_FEATURES_EXT`

The `VkPhysicalDevicePipelineBinaryFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPhysicalDevicePipelineBinaryFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineBinaries;
} VkPhysicalDevicePipelineBinaryFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pipelineBinaries` indicates that the implementation supports [Pipeline Binaries](#).

If the `VkPhysicalDevicePipelineBinaryFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDevicePipelineBinaryFeaturesKHR` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePipelineBinaryFeaturesKHR-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_FEATURES_KHR`

The `VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_ray_tracing_position_fetch
typedef struct VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           rayTracingPositionFetch;
} VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `rayTracingPositionFetch` indicates that the implementation supports fetching the object space vertex positions of a hit triangle.

If the `VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_POSITION_FETCH_FEATURES_KHR`

The `VkPhysicalDeviceShaderFloatControls2FeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_shader_float_controls2
typedef struct VkPhysicalDeviceShaderFloatControls2FeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderFloatControls2;
} VkPhysicalDeviceShaderFloatControls2FeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderFloatControls2` specifies whether shader modules **can** declare the `FloatControls2` capability.

If the `VkPhysicalDeviceShaderFloatControls2FeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderFloatControls2FeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderFloatControls2FeaturesKHR-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT_CONTROLS_2_FEATURES_KHR`

The `VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_non_seamless_cube_map
typedef struct VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           nonSeamlessCubeMap;
} VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `nonSeamlessCubeMap` indicates that the implementation supports `VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT`.

If the `VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NON_SEAMLESS_CUBE_MAP_FEATURES_EXT`

The `VkPhysicalDeviceShaderTileImageFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_shader_tile_image
typedef struct VkPhysicalDeviceShaderTileImageFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderTileImageColorReadAccess;
    VkBool32           shaderTileImageDepthReadAccess;
    VkBool32           shaderTileImageStencilReadAccess;
} VkPhysicalDeviceShaderTileImageFeaturesEXT;
```

The members of the `VkPhysicalDeviceShaderTileImageFeaturesEXT` structure describe the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderTileImageColorReadAccess` indicates that the implementation supports the `TileImageColorReadAccessEXT` SPIR-V capability.
- `shaderTileImageDepthReadAccess` indicates that the implementation supports the

`TileImageDepthReadAccessEXT` SPIR-V capability.

- `shaderTileImageStencilReadAccess` indicates that the implementation supports the `TileImageStencilReadAccessEXT` SPIR-V capability.

If the `VkPhysicalDeviceShaderTileImageFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderTileImageFeaturesEXT` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderTileImageFeaturesEXT-sType-sType
`sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_FEATURES_EXT`

The `VkPhysicalDeviceDepthBiasControlFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_depth_bias_control
typedef struct VkPhysicalDeviceDepthBiasControlFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           depthBiasControl;
    VkBool32           leastRepresentableValueForceUnormRepresentation;
    VkBool32           floatRepresentation;
    VkBool32           depthBiasExact;
} VkPhysicalDeviceDepthBiasControlFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `depthBiasControl` indicates whether the implementation supports the `vkCmdSetDepthBias2EXT` command and the `VkDepthBiasRepresentationInfoEXT` structure.
- `leastRepresentableValueForceUnormRepresentation` indicates whether the implementation supports using the `VK_DEPTH_BIAS_REPRESENTATION_LEAST_REPRESENTABLE_VALUE_FORCE_UNORM_EXT` depth bias representation.
- `floatRepresentation` indicates whether the implementation supports using the `VK_DEPTH_BIAS_REPRESENTATION_FLOAT_EXT` depth bias representation.
- `depthBiasExact` indicates whether the implementation supports forcing depth bias to not be scaled to ensure a minimum resolvable difference using `VkDepthBiasRepresentationInfoEXT::depthBiasExact`.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDepthBiasControlFeaturesEXT-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_BIAS_CONTROL_FEATURES_EXT

The `VkPhysicalDeviceShaderObjectFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_shader_object
typedef struct VkPhysicalDeviceShaderObjectFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderObject;
} VkPhysicalDeviceShaderObjectFeaturesEXT;
```

This structure describes the following feature:

- `shaderObject` indicates whether the implementation supports [shader objects](#).

If the `VkPhysicalDeviceShaderObjectFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderObjectFeaturesEXT` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderObjectFeaturesEXT-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_FEATURES_EXT

The `VkPhysicalDeviceFrameBoundaryFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_frame_boundary
typedef struct VkPhysicalDeviceFrameBoundaryFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           frameBoundary;
} VkPhysicalDeviceFrameBoundaryFeaturesEXT;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `frameBoundary` indicates whether the implementation supports frame boundary information.

If the `VkPhysicalDeviceFrameBoundaryFeaturesEXT` structure is included in the `pNext` chain of the

`VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceFrameBoundaryFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFrameBoundaryFeaturesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAME_BOUNDARY_FEATURES_EXT`

The `VkPhysicalDevicePresentModeFifoLatestReadyFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_present_mode_fifo_latest_ready
typedef struct VkPhysicalDevicePresentModeFifoLatestReadyFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           presentModeFifoLatestReady;
} VkPhysicalDevicePresentModeFifoLatestReadyFeaturesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `presentModeFifoLatestReady` specifies whether the implementation supports the `VK_PRESENT_MODE_FIFO_LATEST_READY_EXT` present mode.

If the `VkPhysicalDevicePresentModeFifoLatestReadyFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDevicePresentModeFifoLatestReadyFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePresentModeFifoLatestReadyFeaturesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_MODE_FIFO_LATEST_READY_FEATURES_EXT`

The `VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_dynamic_rendering_unused_attachments
typedef struct VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           dynamicRenderingUnusedAttachments;
} VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT;
```


This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `dynamicRenderingUnusedAttachments` indicates that the implementation supports binding graphics pipelines within a render pass instance where any pipeline `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` element with a format other than `VK_FORMAT_UNDEFINED` is allowed with a corresponding `VkRenderingInfo::pColorAttachments` element with an `imageView` equal to `VK_NULL_HANDLE`, or any pipeline `VkPipelineRenderingCreateInfo::pColorAttachmentFormats` element with a `VK_FORMAT_UNDEFINED` format is allowed with a corresponding `VkRenderingInfo::pColorAttachments` element with a non-`VK_NULL_HANDLE` `imageView`. Also a `VkPipelineRenderingCreateInfo::depthAttachmentFormat` other than `VK_FORMAT_UNDEFINED` is allowed with a `VK_NULL_HANDLE` `VkRenderingInfo::pDepthAttachment`, or a `VkPipelineRenderingCreateInfo::depthAttachmentFormat` of `VK_FORMAT_UNDEFINED` is allowed with a non-`VK_NULL_HANDLE` `VkRenderingInfo::pDepthAttachment`. Also a `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` other than `VK_FORMAT_UNDEFINED` is allowed with a `VK_NULL_HANDLE` `VkRenderingInfo::pStencilAttachment`, or a `VkPipelineRenderingCreateInfo::stencilAttachmentFormat` of `VK_FORMAT_UNDEFINED` is allowed with a non-`VK_NULL_HANDLE` `VkRenderingInfo::pStencilAttachment`. Any writes to a `VkRenderingInfo::pColorAttachments`, `VkRenderingInfo::pDepthAttachment`, or `VkRenderingInfo::pStencilAttachment` with `VK_NULL_HANDLE` are discarded.

If the `VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT` can also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- `VUID-VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT-sType-sType` must be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_UNUSED_ATTACHMENTS_FEATURES_EXT`

The `VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_shader_maximal_reconvergence
typedef struct VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderMaximalReconvergence;
} VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderMaximalReconvergence` specifies whether the implementation supports the shader execution mode `MaximallyReconvergesKHR`

If the `VkPhysicalDevicePrivateDataFeaturesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDevicePrivateDataFeaturesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MAXIMAL_RECONVERGENCE_FEATURES_KHR`

The `VkPhysicalDeviceShaderSubgroupRotateFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_shader_subgroup_rotate
typedef struct VkPhysicalDeviceShaderSubgroupRotateFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderSubgroupRotate;
    VkBool32           shaderSubgroupRotateClustered;
} VkPhysicalDeviceShaderSubgroupRotateFeaturesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderSubgroupRotate` specifies whether shader modules **can** declare the `GroupNonUniformRotateKHR` capability.
- `shaderSubgroupRotateClustered` specifies whether shader modules **can** use the `ClusterSize` operand to `OpGroupNonUniformRotateKHR`.

If the `VkPhysicalDeviceShaderSubgroupRotateFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderSubgroupRotateFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderSubgroupRotateFeaturesKHR-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_ROTATE_FEATURES_KHR`

The `VkPhysicalDeviceShaderExpectAssumeFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_shader_expect_assume
typedef struct VkPhysicalDeviceShaderExpectAssumeFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderExpectAssume;
} VkPhysicalDeviceShaderExpectAssumeFeaturesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderExpectAssume` specifies whether shader modules **can** declare the `ExpectAssumeKHR` capability.

If the `VkPhysicalDeviceShaderExpectAssumeFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderExpectAssumeFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderExpectAssumeFeaturesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_EXPECT_ASSUME_FEATURES_KHR`

The `VkPhysicalDeviceDynamicRenderingLocalReadFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_dynamic_rendering_local_read
typedef struct VkPhysicalDeviceDynamicRenderingLocalReadFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           dynamicRenderingLocalRead;
} VkPhysicalDeviceDynamicRenderingLocalReadFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `dynamicRenderingLocalRead` specifies that the implementation supports local reads inside dynamic render pass instances using the `vkCmdBeginRenderingKHR` command.

If the `VkPhysicalDeviceDynamicRenderingLocalReadFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceDynamicRenderingLocalReadFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDynamicRenderingLocalReadFeaturesKHR-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_LOCAL_READ_FEATURES_KHR`

The `VkPhysicalDeviceShaderQuadControlFeaturesKHR` structure is defined as:

```
// Provided by VK_KHR_shader_quad_control
typedef struct VkPhysicalDeviceShaderQuadControlFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderQuadControl;
} VkPhysicalDeviceShaderQuadControlFeaturesKHR;
```

This structure describes the following features:

- `shaderQuadControl` indicates whether the implementation supports shaders with the `QuadControlKHR` capability.

If the `VkPhysicalDeviceShaderQuadControlFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderQuadControlFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderQuadControlFeaturesKHR-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_QUAD_CONTROL_FEATURES_KHR`

The `VkPhysicalDeviceMapMemoryPlacedFeaturesEXT` structure is defined as:

```
// Provided by VK_EXT_map_memory_placed
typedef struct VkPhysicalDeviceMapMemoryPlacedFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           memoryMapPlaced;
    VkBool32           memoryMapRangePlaced;
    VkBool32           memoryUnmapReserve;
} VkPhysicalDeviceMapMemoryPlacedFeaturesEXT;
```

This structure describes the following features:

- `memoryMapPlaced` indicates that the implementation supports placing memory maps at

application-specified virtual addresses.

- **memoryMapRangePlaced** indicates that the implementation supports placing memory maps of a subrange of a memory object at application-specified virtual addresses.
- **memoryUnmapReserve** indicates that the implementation supports leaving the memory range reserved when unmapping a memory object.

If the **VkPhysicalDeviceMapMemoryPlacedFeaturesEXT** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDeviceMapMemoryPlacedFeaturesEXT** **can** also be used in the **pNext** chain of **VkDeviceCreateInfo** to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMapMemoryPlacedFeaturesEXT-sType-sType
sType must be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_FEATURES_EXT**

The **VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT** structure is defined as:

```
// Provided by VK_EXT_shader_replicated_composites
typedef struct VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderReplicatedComposites;
} VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT;
```

- **sType** is a **VkStructureType** value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **shaderReplicatedComposites** specifies whether shader modules **can** declare the **ReplicatedCompositesEXT** capability.

If the **VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT** structure is included in the **pNext** chain of the **VkPhysicalDeviceFeatures2** structure passed to **vkGetPhysicalDeviceFeatures2KHR**, it is filled in to indicate whether each corresponding feature is supported. **VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT** **can** also be used in the **pNext** chain of **VkDeviceCreateInfo** to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT-sType-sType
sType must be **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_REPLICATED_COMPOSITES_FEATURES_EXT**

The **VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR** structure is defined as:

```
// Provided by VK_KHR_shader_relaxed_extended_instruction
typedef struct VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           shaderRelaxedExtendedInstruction;
} VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR;
```

This structure describes the following feature:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderRelaxedExtendedInstruction` specifies whether the implementation supports SPIR-V modules that use the `SPV_KHR_relaxed_extended_instruction` extension.

If the `VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_RELAXED_EXTENDED_INSTRUCTION_FEATURES_KHR`

39.1. Feature Requirements

All Vulkan graphics implementations **must** support the following features:

- If Vulkan 1.0 is supported, `robustBufferAccess` **must** be supported if `VK_KHR_portability_subset` is not advertised
- If `VK_KHR_dynamic_rendering` is supported, `dynamicRendering` **must** be supported
- If `VK_KHR_multiview` is supported, `multiview` **must** be supported
- If `VK_KHR_shader_float16_int8` is supported, `shaderFloat16`, or `shaderInt8` **must** be supported
- If `VK_KHR_16bit_storage` is supported, `storageBuffer16BitAccess` **must** be supported
- If `VK_KHR_imageless_framebuffer` is supported, `imagelessFramebuffer` **must** be supported
- If `VK_KHR_performance_query` is supported, `performanceCounterQueryPools` **must** be supported
- If `VK_KHR_variable_pointers` is supported, `variablePointersStorageBuffer` **must** be supported
- If `VK_KHR_sampler_ycbcr_conversion` is supported, `samplerYcbcrConversion` **must** be supported
- If `VK_KHR_shader_subgroup_extended_types` is supported, `shaderSubgroupExtendedTypes` **must** be supported

- If `VK_KHR_8bit_storage` is supported, `storageBuffer8BitAccess` **must** be supported
- If `VK_KHR_shader_atomic_int64` is supported, `shaderBufferInt64Atomics` **must** be supported
- If `VK_KHR_shader_clock` is supported, `shaderSubgroupClock` **must** be supported
- If `VK_KHR_global_priority` is supported, `globalPriorityQuery` **must** be supported
- If `VK_KHR_timeline_semaphore` is supported, `timelineSemaphore` **must** be supported
- If `VK_KHR_vulkan_memory_model` is supported, `vulkanMemoryModel` **must** be supported
- If `VK_KHR_shader_terminate_invocation` is supported, `shaderTerminateInvocation` **must** be supported
- If `VK_KHR_fragment_shading_rate` is supported, `pipelineFragmentShadingRate` **must** be supported
- If `VK_KHR_dynamic_rendering_local_read` is supported, `dynamicRenderingLocalRead` **must** be supported
- If `VK_KHR_shader_quad_control` is supported, `shaderQuadControl` **must** be supported
- If `VK_KHR_separate_depth_stencil_layouts` is supported, `separateDepthStencilLayouts` **must** be supported
- If `VK_KHR_present_wait` is supported, `presentWait` **must** be supported
- If `VK_KHR_uniform_buffer_standard_layout` is supported, `uniformBufferStandardLayout` **must** be supported
- If `VK_KHR_buffer_device_address` is supported, `bufferDeviceAddress` **must** be supported
- If `VK_KHR_pipeline_executable_properties` is supported, `pipelineExecutableInfo` **must** be supported
- If `VK_KHR_shader_integer_dot_product` is supported, `shaderIntegerDotProduct` **must** be supported
- If `VK_KHR_present_id` is supported, `presentId` **must** be supported
- If `VK_KHR_synchronization2` is supported, `synchronization2` **must** be supported
- If `VK_KHR_shader_subgroup_uniform_control_flow` is supported, `shaderSubgroupUniformControlFlow` **must** be supported
- If `VK_KHR_zero_initialize_workgroup_memory` is supported, `shaderZeroInitializeWorkgroupMemory` **must** be supported
- If `VK_KHR_workgroup_memory_explicit_layout` is supported, `workgroupMemoryExplicitLayout` **must** be supported
- If `VK_KHR_ray_tracing_maintenance1` is supported, `rayTracingMaintenance1` **must** be supported
- If `VK_KHR_maintenance4` is supported, `maintenance4` **must** be supported
- If `VK_KHR_shader_subgroup_rotate` is supported, `shaderSubgroupRotate` **must** be supported
- If `VK_KHR_shader_maximal_reconvergence` is supported, `shaderMaximalReconvergence` **must** be supported
- If `VK_KHR_maintenance5` is supported, `maintenance5` **must** be supported
- If `VK_KHR_ray_tracing_position_fetch` is supported, `rayTracingPositionFetch` **must** be supported
- If `VK_KHR_pipeline_binary` is supported, `pipelineBinaries` **must** be supported

- If `VK_KHR_cooperative_matrix` is supported, `cooperativeMatrix` **must** be supported
- If `VK_KHR_compute_shader_derivatives` is supported, `computeDerivativeGroupLinear` **must** be supported
- If `VK_KHR_video_maintenance1` is supported, `videoMaintenance1` **must** be supported
- If `VK_KHR_vertex_attribute_divisor` is supported, `vertexAttributeInstanceRateDivisor` **must** be supported
- If `VK_KHR_shader_float_controls2` is supported, `shaderFloatControls2` **must** be supported
- If `VK_KHR_index_type_uint8` is supported, `indexTypeUint8` **must** be supported
- If `VK_KHR_line_rasterization` is supported, `rectangularLines`, `bresenhamLines`, `smoothLines`, `stippledRectangularLines`, `stippledBresenhamLines`, or `stippledSmoothLines` **must** be supported
- If `VK_KHR_shader_expect_assume` is supported, `shaderExpectAssume` **must** be supported
- If `VK_KHR_maintenance6` is supported, `maintenance6` **must** be supported
- If `VK_KHR_shader_relaxed_extended_instruction` is supported, `shaderRelaxedExtendedInstruction` **must** be supported
- If `VK_KHR_maintenance7` is supported, `maintenance7` **must** be supported
- If `VK_EXT_transform_feedback` is supported, `transformFeedback` **must** be supported
- If `VK_EXT_depth_clip_enable` is supported, `depthClipEnable` **must** be supported
- If `VK_EXT_shader_image_atomic_int64` is supported, the following features **must** be supported:
 - `shaderInt64`
 - `shaderImageInt64Atomics`
- If `VK_EXT_memory_priority` is supported, `memoryPriority` **must** be supported
- If `VK_EXT_fragment_shader_interlock` is supported, `fragmentShaderSampleInterlock`, `fragmentShaderPixelInterlock`, or `fragmentShaderShadingRateInterlock` **must** be supported
- If `VK_EXT_ycbcr_image_arrays` is supported, `ycbcrImageArrays` **must** be supported
- If `VK_EXT_provoking_vertex` is supported, `provokingVertexLast` **must** be supported
- If `VK_EXT_shader_atomic_float` is supported, the following features **must** be supported:
 - `shaderImageFloat32Atomics` if `sparseImageFloat32Atomics` is supported
 - `shaderImageFloat32AtomicAdd` if `sparseImageFloat32AtomicAdd` is supported
- If `VK_EXT_host_image_copy` is supported, `hostImageCopy` **must** be supported
- If `VK_EXT_map_memory_placed` is supported, `memoryMapPlaced` **must** be supported
- If `VK_EXT_shader_atomic_float2` is supported, `shaderImageFloat32AtomicMinMax` **must** be supported if `sparseImageFloat32AtomicMinMax` is supported
- If `VK_EXT_depth_bias_control` is supported, `depthBiasControl` **must** be supported
- If `VK_EXT_device_memory_report` is supported, `deviceMemoryReport` **must** be supported
- If `VK_EXT_custom_border_color` is supported, `customBorderColors` **must** be supported
- If `VK_EXT_attachment_feedback_loop_layout` is supported, `attachmentFeedbackLoopLayout` **must** be supported

- If `VK_EXT_vertex_input_dynamic_state` is supported, `vertexInputDynamicState` **must** be supported
- If `VK_EXT_depth_clip_control` is supported, `depthClipControl` **must** be supported
- If `VK_EXT_primitive_topology_list_restart` is supported, `primitiveTopologyListRestart` **must** be supported
- If `VK_EXT_present_mode_fifo_latest_ready` is supported, `presentModeFifoLatestReady` **must** be supported
- If `VK_EXT_frame_boundary` is supported, `frameBoundary` **must** be supported
- If `VK_EXT_color_write_enable` is supported, `colorWriteEnable` **must** be supported
- If `VK_EXT_primitives_generated_query` is supported, `primitivesGeneratedQuery` **must** be supported
- If `VK_EXT_image_view_min_lod` is supported, `minLod` **must** be supported
- If `VK_EXT_multi_draw` is supported, `multiDraw` **must** be supported
- If `VK_EXT_shader_tile_image` is supported, `shaderTileImageColorReadAccess` **must** be supported
- If `VK_EXT_opacity_micromap` is supported, `micromap` **must** be supported
- If `VK_EXT_border_color_swizzle` is supported, `borderColorSwizzle` **must** be supported
- If `VK_EXT_non_seamless_cube_map` is supported, `nonSeamlessCubeMap` **must** be supported
- If `VK_EXT_nested_command_buffer` is supported, `nestedCommandBuffer` **must** be supported
- If `VK_EXT_extended_dynamic_state3` is supported, the following features **must** be supported:
 - `tessellationShader` if `extendedDynamicState3TessellationDomainOrigin` is supported
 - `depthClamp` if `extendedDynamicState3DepthClampEnable` is supported
 - `fillModeNonSolid` if `extendedDynamicState3PolygonMode` is supported
 - `alphaToOne` if `extendedDynamicState3AlphaToOneEnable` is supported
 - `logicOp` if `extendedDynamicState3LogicOpEnable` is supported
 - `geometryStreams` if `VK_EXT_transform_feedback` and `extendedDynamicState3RasterizationStream` are supported
- If `VK_EXT_shader_object` is supported, `shaderObject` **must** be supported
- If `VK_EXT_legacy_vertex_attributes` is supported, `legacyVertexAttributes` **must** be supported
- If `VK_EXT_dynamic_rendering_unused_attachments` is supported, `dynamicRenderingUnusedAttachments` **must** be supported
- If `VK_EXT_attachment_feedback_loop_dynamic_state` is supported, `attachmentFeedbackLoopDynamicState` **must** be supported
- If `VK_EXT_shader_replicated_composites` is supported, `shaderReplicatedComposites` **must** be supported
- If `VK_EXT_device_generated_commands` is supported, `deviceGeneratedCommands` **must** be supported
- If `VK_EXT_depth_clamp_control` is supported, `depthClampControl` **must** be supported
- If `VK_KHR_acceleration_structure` is supported, the following features **must** be supported:
 - `accelerationStructure`

- `bufferDeviceAddress`
- `descriptorBindingAccelerationStructureUpdateAfterBind`
- `descriptorIndexing` if Vulkan 1.2 is supported
- If `VK_KHR_ray_tracing_pipeline` is supported, the following features **must** be supported:
 - `rayTracingPipeline`
 - `rayTracingPipelineTraceRaysIndirect`
 - `rayTraversalPrimitiveCulling` if `VK_KHR_ray_query` is supported
- If `VK_KHR_ray_query` is supported, `rayQuery` **must** be supported
- `VK_EXT_transform_feedback`, if the `extendedDynamicState3RasterizationStream` feature is supported.
- `VK_EXT_conservative_rasterization` extension, if the `extendedDynamicState3ConservativeRasterizationMode` feature is supported.
- `VK_EXT_conservative_rasterization` extension, if the `extendedDynamicState3ExtraPrimitiveOverestimationSize` feature is supported.
- `VK_EXT_sample_locations` extension, if the `extendedDynamicState3SampleLocationsEnable` feature is supported.
- `VK_EXT_blend_operation_advanced` extension, if the `extendedDynamicState3ColorBlendAdvanced` feature is supported.
- `VK_EXT_provoking_vertex`, if the `extendedDynamicState3ProvokingVertexMode` feature is supported.
- `VK_KHR_line_rasterization` or `VK_EXT_line_rasterization` extension, if the `extendedDynamicState3LineRasterizationMode` feature is supported.
- `VK_KHR_line_rasterization` or `VK_EXT_line_rasterization` extension, if the `extendedDynamicState3LineStippleEnable` feature is supported.
- `VK_EXT_depth_clip_control`, if the `extendedDynamicState3DepthClipNegativeOneToOne` feature is supported.

All other features defined in the Specification are **optional**.

Chapter 40. Limits

Limits are implementation-dependent minimums, maximums, and other device characteristics that an application **may** need to be aware of.

NOTE

Limits are reported via the basic [VkPhysicalDeviceLimits](#) structure as well as the extensible structure [VkPhysicalDeviceProperties2](#), which was added in [VK_KHR_get_physical_device_properties2](#) and included in Vulkan 1.1. When limits are added in future Vulkan versions or extensions, each extension **should** introduce one new limit structure, if needed. This structure **can** be added to the [pNext](#) chain of the [VkPhysicalDeviceProperties2](#) structure.

The [VkPhysicalDeviceLimits](#) structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkPhysicalDeviceLimits {
    uint32_t          maxImageDimension1D;
    uint32_t          maxImageDimension2D;
    uint32_t          maxImageDimension3D;
    uint32_t          maxImageDimensionCube;
    uint32_t          maxImageArrayLayers;
    uint32_t          maxTexelBufferElements;
    uint32_t          maxUniformBufferRange;
    uint32_t          maxStorageBufferRange;
    uint32_t          maxPushConstantsSize;
    uint32_t          maxMemoryAllocationCount;
    uint32_t          maxSamplerAllocationCount;
    VkDeviceSize      bufferImageGranularity;
    VkDeviceSize      sparseAddressSpaceSize;
    uint32_t          maxBoundDescriptorSets;
    uint32_t          maxPerStageDescriptorSamplers;
    uint32_t          maxPerStageDescriptorUniformBuffers;
    uint32_t          maxPerStageDescriptorStorageBuffers;
    uint32_t          maxPerStageDescriptorSampledImages;
    uint32_t          maxPerStageDescriptorStorageImages;
    uint32_t          maxPerStageDescriptorInputAttachments;
    uint32_t          maxPerStageResources;
    uint32_t          maxDescriptorSetSamplers;
    uint32_t          maxDescriptorSetUniformBuffers;
    uint32_t          maxDescriptorSetUniformBuffersDynamic;
    uint32_t          maxDescriptorSetStorageBuffers;
    uint32_t          maxDescriptorSetStorageBuffersDynamic;
    uint32_t          maxDescriptorSetSampledImages;
    uint32_t          maxDescriptorSetStorageImages;
    uint32_t          maxDescriptorSetInputAttachments;
    uint32_t          maxVertexInputAttributes;
    uint32_t          maxVertexInputBindings;
    uint32_t          maxVertexInputAttributeOffset;
    uint32_t          maxVertexInputBindingStride;
```

uint32_t	maxVertexOutputComponents;
uint32_t	maxTessellationGenerationLevel;
uint32_t	maxTessellationPatchSize;
uint32_t	maxTessellationControlPerVertexInputComponents;
uint32_t	maxTessellationControlPerVertexOutputComponents;
uint32_t	maxTessellationControlPerPatchOutputComponents;
uint32_t	maxTessellationControlTotalOutputComponents;
uint32_t	maxTessellationEvaluationInputComponents;
uint32_t	maxTessellationEvaluationOutputComponents;
uint32_t	maxGeometryShaderInvocations;
uint32_t	maxGeometryInputComponents;
uint32_t	maxGeometryOutputComponents;
uint32_t	maxGeometryOutputVertices;
uint32_t	maxGeometryTotalOutputComponents;
uint32_t	maxFragmentInputComponents;
uint32_t	maxFragmentOutputAttachments;
uint32_t	maxFragmentDualSrcAttachments;
uint32_t	maxFragmentCombinedOutputResources;
uint32_t	maxComputeSharedMemorySize;
uint32_t	maxComputeWorkGroupCount[3];
uint32_t	maxComputeWorkGroupInvocations;
uint32_t	maxComputeWorkGroupSize[3];
uint32_t	subPixelPrecisionBits;
uint32_t	subTexelPrecisionBits;
uint32_t	mipmapPrecisionBits;
uint32_t	maxDrawIndexedIndexValue;
uint32_t	maxDrawIndirectCount;
float	maxSamplerLodBias;
float	maxSamplerAnisotropy;
uint32_t	maxViewports;
uint32_t	maxViewportDimensions[2];
float	viewportBoundsRange[2];
uint32_t	viewportSubPixelBits;
size_t	minMemoryMapAlignment;
VkDeviceSize	minTexelBufferOffsetAlignment;
VkDeviceSize	minUniformBufferOffsetAlignment;
VkDeviceSize	minStorageBufferOffsetAlignment;
int32_t	minTexelOffset;
uint32_t	maxTexelOffset;
int32_t	minTexelGatherOffset;
uint32_t	maxTexelGatherOffset;
float	minInterpolationOffset;
float	maxInterpolationOffset;
uint32_t	subPixelInterpolationOffsetBits;
uint32_t	maxFramebufferWidth;
uint32_t	maxFramebufferHeight;
uint32_t	maxFramebufferLayers;
VkSampleCountFlags	framebufferColorSampleCounts;
VkSampleCountFlags	framebufferDepthSampleCounts;
VkSampleCountFlags	framebufferStencilSampleCounts;
VkSampleCountFlags	framebufferNoAttachmentsSampleCounts;

```

uint32_t      maxColorAttachments;
VkSampleCountFlags sampledImageColorSampleCounts;
VkSampleCountFlags sampledImageIntegerSampleCounts;
VkSampleCountFlags sampledImageDepthSampleCounts;
VkSampleCountFlags sampledImageStencilSampleCounts;
VkSampleCountFlags storageImageSampleCounts;
uint32_t      maxSampleMaskWords;
VkBool32      timestampComputeAndGraphics;
float         timestampPeriod;
uint32_t      maxClipDistances;
uint32_t      maxCullDistances;
uint32_t      maxCombinedClipAndCullDistances;
uint32_t      discreteQueuePriorities;
float         pointSizeRange[2];
float         lineWidthRange[2];
float         pointSizeGranularity;
float         lineWidthGranularity;
VkBool32      strictLines;
VkBool32      standardSampleLocations;
VkDeviceSize  optimalBufferCopyOffsetAlignment;
VkDeviceSize  optimalBufferCopyRowPitchAlignment;
VkDeviceSize  nonCoherentAtomSize;
} VkPhysicalDeviceLimits;

```

The `VkPhysicalDeviceLimits` are properties of the physical device. These are available in the `limits` member of the `VkPhysicalDeviceProperties` structure which is returned from `vkGetPhysicalDeviceProperties`.

- `maxImageDimension1D` is the largest dimension (`width`) that is guaranteed to be supported for all images created with an `imageType` of `VK_IMAGE_TYPE_1D`. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using `vkGetPhysicalDeviceImageFormatProperties`.
- `maxImageDimension2D` is the largest dimension (`width` or `height`) that is guaranteed to be supported for all images created with an `imageType` of `VK_IMAGE_TYPE_2D` and without `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT` set in `flags`. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using `vkGetPhysicalDeviceImageFormatProperties`.
- `maxImageDimension3D` is the largest dimension (`width`, `height`, or `depth`) that is guaranteed to be supported for all images created with an `imageType` of `VK_IMAGE_TYPE_3D`. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using `vkGetPhysicalDeviceImageFormatProperties`.
- `maxImageDimensionCube` is the largest dimension (`width` or `height`) that is guaranteed to be supported for all images created with an `imageType` of `VK_IMAGE_TYPE_2D` and with `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT` set in `flags`. Some combinations of image parameters (format, usage, etc.) **may** allow support for larger dimensions, which **can** be queried using `vkGetPhysicalDeviceImageFormatProperties`.
- `maxImageArrayLayers` is the maximum number of layers (`arrayLayers`) for an image.

- `maxTexelBufferElements` is the maximum number of addressable texels for a buffer view created on a buffer which was created with the `VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT` or `VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT` set in the `usage` member of the `VkBufferCreateInfo` structure.
- `maxUniformBufferRange` is the maximum value that **can** be specified in the `range` member of a `VkDescriptorBufferInfo` structure passed to `vkUpdateDescriptorSets` for descriptors of type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`.
- `maxStorageBufferRange` is the maximum value that **can** be specified in the `range` member of a `VkDescriptorBufferInfo` structure passed to `vkUpdateDescriptorSets` for descriptors of type `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC`.
- `maxPushConstantsSize` is the maximum size, in bytes, of the pool of push constant memory. For each of the push constant ranges indicated by the `pPushConstantRanges` member of the `VkPipelineLayoutCreateInfo` structure, $(\text{offset} + \text{size})$ **must** be less than or equal to this limit.
- `maxMemoryAllocationCount` is the maximum number of device memory allocations, as created by `vkAllocateMemory`, which **can** simultaneously exist.
- `maxSamplerAllocationCount` is the maximum number of sampler objects, as created by `vkCreateSampler`, which **can** simultaneously exist on a device.
- `bufferImageGranularity` is the granularity, in bytes, at which buffer or linear image resources, and optimal image resources **can** be bound to adjacent offsets in the same `VkDeviceMemory` object without aliasing. See [Buffer-Image Granularity](#) for more details.
- `sparseAddressSpaceSize` is the total amount of address space available, in bytes, for sparse memory resources. This is an upper bound on the sum of the sizes of all sparse resources, regardless of whether any memory is bound to them.
- `maxBoundDescriptorSets` is the maximum number of descriptor sets that **can** be simultaneously used by a pipeline. All `DescriptorSet` decorations in shader modules **must** have a value less than `maxBoundDescriptorSets`. See [Descriptor Sets](#).
- `maxPerStageDescriptorSamplers` is the maximum number of samplers that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` count against this limit. A descriptor is accessible to a shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. See [Sampler](#) and [Combined Image Sampler](#).
- `maxPerStageDescriptorUniformBuffers` is the maximum number of uniform buffers that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` count against this limit. A descriptor is accessible to a shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. See [Uniform Buffer](#) and [Dynamic Uniform Buffer](#).
- `maxPerStageDescriptorStorageBuffers` is the maximum number of storage buffers that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` count against this limit. A descriptor is accessible to a pipeline shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. See [Storage Buffer](#) and [Dynamic Storage Buffer](#).

- `maxPerStageDescriptorSampledImages` is the maximum number of sampled images that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, or `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` count against this limit. A descriptor is accessible to a pipeline shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. See [Combined Image Sampler](#), [Sampled Image](#), and [Uniform Texel Buffer](#).
- `maxPerStageDescriptorStorageImages` is the maximum number of storage images that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` count against this limit. A descriptor is accessible to a pipeline shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. See [Storage Image](#), and [Storage Texel Buffer](#).
- `maxPerStageDescriptorInputAttachments` is the maximum number of input attachments that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` count against this limit. A descriptor is accessible to a pipeline shader stage when the `stageFlags` member of the `VkDescriptorSetLayoutBinding` structure has the bit for that shader stage set. These are only supported for the fragment stage. See [Input Attachment](#).
- `maxPerStageResources` is the maximum number of resources that **can** be accessible to a single shader stage in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER`, `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER`, `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER`, `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER`, `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC`, or `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` count against this limit. For the fragment shader stage the framebuffer color attachments also count against this limit.
- `maxDescriptorSetSamplers` is the maximum number of samplers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_SAMPLER` or `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER` count against this limit. See [Sampler](#) and [Combined Image Sampler](#).
- `maxDescriptorSetUniformBuffers` is the maximum number of uniform buffers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` count against this limit. See [Uniform Buffer](#) and [Dynamic Uniform Buffer](#).
- `maxDescriptorSetUniformBuffersDynamic` is the maximum number of dynamic uniform buffers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` count against this limit. See [Dynamic Uniform Buffer](#).
- `maxDescriptorSetStorageBuffers` is the maximum number of storage buffers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` count against this limit. See [Storage Buffer](#) and [Dynamic Storage Buffer](#).
- `maxDescriptorSetStorageBuffersDynamic` is the maximum number of dynamic storage buffers that

can be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` count against this limit. See [Dynamic Storage Buffer](#).

- `maxDescriptorSetSampledImages` is the maximum number of sampled images that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_COMBINED_IMAGE_SAMPLER`, `VK_DESCRIPTOR_TYPE_SAMPLED_IMAGE`, or `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` count against this limit. See [Combined Image Sampler](#), [Sampled Image](#), and [Uniform Texel Buffer](#).
- `maxDescriptorSetStorageImages` is the maximum number of storage images that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_STORAGE_IMAGE`, or `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` count against this limit. See [Storage Image](#), and [Storage Texel Buffer](#).
- `maxDescriptorSetInputAttachments` is the maximum number of input attachments that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_INPUT_ATTACHMENT` count against this limit. See [Input Attachment](#).
- `maxVertexInputAttributes` is the maximum number of vertex input attributes that **can** be specified for a graphics pipeline. These are described in the array of `VkVertexInputAttributeDescription` structures that are provided at graphics pipeline creation time via the `pVertexAttributeDescriptions` member of the `VkPipelineVertexInputStateCreateInfo` structure. See [Vertex Attributes](#) and [Vertex Input Description](#).
- `maxVertexInputBindings` is the maximum number of vertex buffers that **can** be specified for providing vertex attributes to a graphics pipeline. These are described in the array of `VkVertexInputBindingDescription` structures that are provided at graphics pipeline creation time via the `pVertexBindingDescriptions` member of the `VkPipelineVertexInputStateCreateInfo` structure. The `binding` member of `VkVertexInputBindingDescription` **must** be less than this limit. See [Vertex Input Description](#).
- `maxVertexInputAttributeOffset` is the maximum vertex input attribute offset that **can** be added to the vertex input binding stride. The `offset` member of the `VkVertexInputAttributeDescription` structure **must** be less than or equal to this limit. See [Vertex Input Description](#).
- `maxVertexInputBindingStride` is the maximum vertex input binding stride that **can** be specified in a vertex input binding. The `stride` member of the `VkVertexInputBindingDescription` structure **must** be less than or equal to this limit. See [Vertex Input Description](#).
- `maxVertexOutputComponents` is the maximum number of components of output variables which **can** be output by a vertex shader. See [Vertex Shaders](#).
- `maxTessellationGenerationLevel` is the maximum tessellation generation level supported by the fixed-function tessellation primitive generator. See [Tessellation](#).
- `maxTessellationPatchSize` is the maximum patch size, in vertices, of patches that **can** be processed by the tessellation control shader and tessellation primitive generator. The `patchControlPoints` member of the `VkPipelineTessellationStateCreateInfo` structure specified at pipeline creation time and the value provided in the `OutputVertices` execution mode of shader modules **must** be less than or equal to this limit. See [Tessellation](#).
- `maxTessellationControlPerVertexInputComponents` is the maximum number of components of input variables which **can** be provided as per-vertex inputs to the tessellation control shader stage.
- `maxTessellationControlPerVertexOutputComponents` is the maximum number of components of

per-vertex output variables which **can** be output from the tessellation control shader stage.

- **maxTessellationControlPerPatchOutputComponents** is the maximum number of components of per-patch output variables which **can** be output from the tessellation control shader stage.
- **maxTessellationControlTotalOutputComponents** is the maximum total number of components of per-vertex and per-patch output variables which **can** be output from the tessellation control shader stage.
- **maxTessellationEvaluationInputComponents** is the maximum number of components of input variables which **can** be provided as per-vertex inputs to the tessellation evaluation shader stage.
- **maxTessellationEvaluationOutputComponents** is the maximum number of components of per-vertex output variables which **can** be output from the tessellation evaluation shader stage.
- **maxGeometryShaderInvocations** is the maximum invocation count supported for instanced geometry shaders. The value provided in the **Invocations** execution mode of shader modules **must** be less than or equal to this limit. See [Geometry Shading](#).
- **maxGeometryInputComponents** is the maximum number of components of input variables which **can** be provided as inputs to the geometry shader stage.
- **maxGeometryOutputComponents** is the maximum number of components of output variables which **can** be output from the geometry shader stage.
- **maxGeometryOutputVertices** is the maximum number of vertices which **can** be emitted by any geometry shader.
- **maxGeometryTotalOutputComponents** is the maximum total number of components of output variables, across all emitted vertices, which **can** be output from the geometry shader stage.
- **maxFragmentInputComponents** is the maximum number of components of input variables which **can** be provided as inputs to the fragment shader stage.
- **maxFragmentOutputAttachments** is the maximum number of output attachments which **can** be written to by the fragment shader stage.
- **maxFragmentDualSrcAttachments** is the maximum number of output attachments which **can** be written to by the fragment shader stage when blending is enabled and one of the dual source blend modes is in use. See [Dual-Source Blending](#) and [dualSrcBlend](#).
- **maxFragmentCombinedOutputResources** is the total number of storage buffers, storage images, and output **Location** decorated color attachments (described in [Fragment Output Interface](#)) which **can** be used in the fragment shader stage.
- **maxComputeSharedMemorySize** is the maximum total storage size, in bytes, available for variables declared with the **Workgroup** storage class in shader modules (or with the **shared** storage qualifier in GLSL) in the compute shader stage.
- **maxComputeWorkGroupCount[3]** is the maximum number of local workgroups that **can** be dispatched by a single dispatching command. These three values represent the maximum number of local workgroups for the X, Y, and Z dimensions, respectively. The workgroup count parameters to the dispatching commands **must** be less than or equal to the corresponding limit. See [Dispatching Commands](#).
- **maxComputeWorkGroupInvocations** is the maximum total number of compute shader invocations in

a single local workgroup. The product of the X, Y, and Z sizes, as specified by the `LocalSize` or `LocalSizeId` execution mode in shader modules or by the object decorated by the `WorkgroupSize` decoration, **must** be less than or equal to this limit.

- `maxComputeWorkGroupSize[3]` is the maximum size of a local compute workgroup, per dimension. These three values represent the maximum local workgroup size in the X, Y, and Z dimensions, respectively. The `x`, `y`, and `z` sizes, as specified by the `LocalSize` or `LocalSizeId` execution mode or by the object decorated by the `WorkgroupSize` decoration in shader modules, **must** be less than or equal to the corresponding limit.
- `subPixelPrecisionBits` is the number of bits of subpixel precision in framebuffer coordinates x_f and y_f . See [Rasterization](#).
- `subTexelPrecisionBits` is the number of bits of precision in the division along an axis of an image used for minification and magnification filters. $2^{\text{subTexelPrecisionBits}}$ is the actual number of divisions along each axis of the image represented. Sub-texel values calculated during image sampling will snap to these locations when generating the filtered results.
- `mipmapPrecisionBits` is the number of bits of division that the LOD calculation for mipmap fetching get snapped to when determining the contribution from each mip level to the mip filtered results. $2^{\text{mipmapPrecisionBits}}$ is the actual number of divisions.
- `maxDrawIndexedIndexValue` is the maximum index value that **can** be used for indexed draw calls when using 32-bit indices. This excludes the primitive restart index value of 0xFFFFFFFF. See [fullDrawIndexUint32](#).
- `maxDrawIndirectCount` is the maximum draw count that is supported for indirect drawing calls. See [multiDrawIndirect](#).
- `maxSamplerLodBias` is the maximum absolute sampler LOD bias. The sum of the `mipLodBias` member of the [VkSamplerCreateInfo](#) structure and the `Bias` operand of image sampling operations in shader modules (or 0 if no `Bias` operand is provided to an image sampling operation) are clamped to the range $[-\text{maxSamplerLodBias}, +\text{maxSamplerLodBias}]$. See [\[samplers-mipLodBias\]](#).
- `maxSamplerAnisotropy` is the maximum degree of sampler anisotropy. The maximum degree of anisotropic filtering used for an image sampling operation is the minimum of the `maxAnisotropy` member of the [VkSamplerCreateInfo](#) structure and this limit. See [\[samplers-maxAnisotropy\]](#).
- `maxViewports` is the maximum number of active viewports. The `viewportCount` member of the [VkPipelineViewportStateCreateInfo](#) structure that is provided at pipeline creation **must** be less than or equal to this limit.
- `maxViewportDimensions[2]` are the maximum viewport dimensions in the X (width) and Y (height) dimensions, respectively. The maximum viewport dimensions **must** be greater than or equal to the largest image which **can** be created and used as a framebuffer attachment. See [Controlling the Viewport](#).
- `viewportBoundsRange[2]` is the [minimum, maximum] range that the corners of a viewport **must** be contained in. This range **must** be at least $[-2 \times \text{size}, 2 \times \text{size} - 1]$, where $\text{size} = \max(\text{maxViewportDimensions}[0], \text{maxViewportDimensions}[1])$. See [Controlling the Viewport](#).

NOTE

The intent of the `viewportBoundsRange` limit is to allow a maximum sized viewport to be arbitrarily shifted relative to the output target as long as at least

some portion intersects. This would give a bounds limit of $[-size + 1, 2 \times size - 1]$ which would allow all possible non-empty-set intersections of the output target and the viewport. Since these numbers are typically powers of two, picking the signed number range using the smallest possible number of bits ends up with the specified range.

- `viewportSubPixelBits` is the number of bits of subpixel precision for viewport bounds. The subpixel precision that floating-point viewport bounds are interpreted at is given by this limit.
- `minMemoryMapAlignment` is the minimum **required** alignment, in bytes, of host visible memory allocations within the host address space. When mapping a memory allocation with `vkMapMemory`, subtracting `offset` bytes from the returned pointer will always produce an integer multiple of this limit. See [Host Access to Device Memory Objects](#). The value **must** be a power of two.
- `minTexelBufferOffsetAlignment` is the minimum **required** alignment, in bytes, for the `offset` member of the `VkBufferViewCreateInfo` structure for texel buffers. The value **must** be a power of two. `VkBufferViewCreateInfo::offset` **must** be a multiple of this value.
- `minUniformBufferOffsetAlignment` is the minimum **required** alignment, in bytes, for the `offset` member of the `VkDescriptorBufferInfo` structure for uniform buffers. When a descriptor of type `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER` or `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` is updated, the `offset` **must** be an integer multiple of this limit. Similarly, dynamic offsets for uniform buffers **must** be multiples of this limit. The value **must** be a power of two.
- `minStorageBufferOffsetAlignment` is the minimum **required** alignment, in bytes, for the `offset` member of the `VkDescriptorBufferInfo` structure for storage buffers. When a descriptor of type `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` is updated, the `offset` **must** be an integer multiple of this limit. Similarly, dynamic offsets for storage buffers **must** be multiples of this limit. The value **must** be a power of two.
- `minTexelOffset` is the minimum offset value for the `ConstOffset` image operand of any of the `OpImageSample*` or `OpImageFetch*` image instructions.
- `maxTexelOffset` is the maximum offset value for the `ConstOffset` image operand of any of the `OpImageSample*` or `OpImageFetch*` image instructions.
- `minTexelGatherOffset` is the minimum offset value for the `Offset`, `ConstOffset`, or `ConstOffsets` image operands of any of the `OpImage*Gather` image instructions.
- `maxTexelGatherOffset` is the maximum offset value for the `Offset`, `ConstOffset`, or `ConstOffsets` image operands of any of the `OpImage*Gather` image instructions.
- `minInterpolationOffset` is the base minimum (inclusive) negative offset value for the `Offset` operand of the `InterpolateAtOffset` extended instruction.
- `maxInterpolationOffset` is the base maximum (inclusive) positive offset value for the `Offset` operand of the `InterpolateAtOffset` extended instruction.
- `subPixelInterpolationOffsetBits` is the number of fractional bits that the `x` and `y` offsets to the `InterpolateAtOffset` extended instruction **may** be rounded to as fixed-point values.
- `maxFramebufferWidth` is the maximum width for a framebuffer. The `width` member of the `VkFramebufferCreateInfo` structure **must** be less than or equal to this limit.
- `maxFramebufferHeight` is the maximum height for a framebuffer. The `height` member of the

[VkFramebufferCreateInfo](#) structure **must** be less than or equal to this limit.

- `maxFramebufferLayers` is the maximum layer count for a layered framebuffer. The `layers` member of the [VkFramebufferCreateInfo](#) structure **must** be less than or equal to this limit.
- `framebufferColorSampleCounts` is a bitmask¹ of [VkSampleCountFlagBits](#) indicating the color sample counts that are supported for all framebuffer color attachments with floating- or fixed-point formats. There is no limit specifying the color sample counts that are supported for all color attachments with integer formats.
- `framebufferDepthSampleCounts` is a bitmask¹ of [VkSampleCountFlagBits](#) indicating the supported depth sample counts for all framebuffer depth/stencil attachments, when the format includes a depth component.
- `framebufferStencilSampleCounts` is a bitmask¹ of [VkSampleCountFlagBits](#) indicating the supported stencil sample counts for all framebuffer depth/stencil attachments, when the format includes a stencil component.
- `framebufferNoAttachmentsSampleCounts` is a bitmask¹ of [VkSampleCountFlagBits](#) indicating the supported sample counts for a [subpass which uses no attachments](#).
- `maxColorAttachments` is the maximum number of color attachments that **can** be used by a subpass in a render pass. The `colorAttachmentCount` member of the [VkSubpassDescription](#) or [VkSubpassDescription2](#) structure **must** be less than or equal to this limit.
- `sampledImageColorSampleCounts` is a bitmask¹ of [VkSampleCountFlagBits](#) indicating the sample counts supported for all 2D images created with [VK_IMAGE_TILING_OPTIMAL](#), `usage` containing [VK_IMAGE_USAGE_SAMPLED_BIT](#), and a non-integer color format.
- `sampledImageIntegerSampleCounts` is a bitmask¹ of [VkSampleCountFlagBits](#) indicating the sample counts supported for all 2D images created with [VK_IMAGE_TILING_OPTIMAL](#), `usage` containing [VK_IMAGE_USAGE_SAMPLED_BIT](#), and an integer color format.
- `sampledImageDepthSampleCounts` is a bitmask¹ of [VkSampleCountFlagBits](#) indicating the sample counts supported for all 2D images created with [VK_IMAGE_TILING_OPTIMAL](#), `usage` containing [VK_IMAGE_USAGE_SAMPLED_BIT](#), and a depth format.
- `sampledImageStencilSampleCounts` is a bitmask¹ of [VkSampleCountFlagBits](#) indicating the sample counts supported for all 2D images created with [VK_IMAGE_TILING_OPTIMAL](#), `usage` containing [VK_IMAGE_USAGE_SAMPLED_BIT](#), and a stencil format.
- `storageImageSampleCounts` is a bitmask¹ of [VkSampleCountFlagBits](#) indicating the sample counts supported for all 2D images created with [VK_IMAGE_TILING_OPTIMAL](#), and `usage` containing [VK_IMAGE_USAGE_STORAGE_BIT](#).
- `maxSampleMaskWords` is the maximum number of array elements of a variable decorated with the `SampleMask` built-in decoration.
- `timestampComputeAndGraphics` specifies support for timestamps on all graphics and compute queues. If this limit is [VK_TRUE](#), all queues that advertise the [VK_QUEUE_GRAPHICS_BIT](#) or [VK_QUEUE_COMPUTE_BIT](#) in the [VkQueueFamilyProperties::queueFlags](#) support [VkQueueFamilyProperties::timestampValidBits](#) of at least 36. See [Timestamp Queries](#).
- `timestampPeriod` is the number of nanoseconds **required** for a timestamp query to be incremented by 1. See [Timestamp Queries](#).
- `maxClipDistances` is the maximum number of clip distances that **can** be used in a single shader

stage. The size of any array declared with the `ClipDistance` built-in decoration in a shader module **must** be less than or equal to this limit.

- `maxCullDistances` is the maximum number of cull distances that **can** be used in a single shader stage. The size of any array declared with the `CullDistance` built-in decoration in a shader module **must** be less than or equal to this limit.
- `maxCombinedClipAndCullDistances` is the maximum combined number of clip and cull distances that **can** be used in a single shader stage. The sum of the sizes of all arrays declared with the `ClipDistance` and `CullDistance` built-in decoration used by a single shader stage in a shader module **must** be less than or equal to this limit.
- `discreteQueuePriorities` is the number of discrete priorities that **can** be assigned to a queue based on the value of each member of `VkDeviceQueueCreateInfo::pQueuePriorities`. This **must** be at least 2, and levels **must** be spread evenly over the range, with at least one level at 1.0, and another at 0.0. See [Queue Priority](#).
- `pointSizeRange[2]` is the range `[minimum,maximum]` of supported sizes for points. Values written to variables decorated with the `PointSize` built-in decoration are clamped to this range.
- `lineWidthRange[2]` is the range `[minimum,maximum]` of supported widths for lines. Values specified by the `lineWidth` member of the `VkPipelineRasterizationStateCreateInfo` or the `lineWidth` parameter to `vkCmdSetLineWidth` are clamped to this range.
- `pointSizeGranularity` is the granularity of supported point sizes. Not all point sizes in the range defined by `pointSizeRange` are supported. This limit specifies the granularity (or increment) between successive supported point sizes.
- `lineWidthGranularity` is the granularity of supported line widths. Not all line widths in the range defined by `lineWidthRange` are supported. This limit specifies the granularity (or increment) between successive supported line widths.
- `strictLines` specifies whether lines are rasterized according to the preferred method of rasterization. If set to `VK_FALSE`, lines **may** be rasterized under a relaxed set of rules. If set to `VK_TRUE`, lines are rasterized as per the strict definition. See [Basic Line Segment Rasterization](#).
- `standardSampleLocations` specifies whether rasterization uses the standard sample locations as documented in [Multisampling](#). If set to `VK_TRUE`, the implementation uses the documented sample locations. If set to `VK_FALSE`, the implementation **may** use different sample locations.
- `optimalBufferCopyOffsetAlignment` is the optimal buffer offset alignment in bytes for `vkCmdCopyBufferToImage2KHR`, `vkCmdCopyBufferToImage`, `vkCmdCopyImageToBuffer2KHR`, and `vkCmdCopyImageToBuffer`. This value is also the optimal host memory offset alignment in bytes for `vkCopyMemoryToImageEXT` and `vkCopyImageToMemoryEXT`. The per texel alignment requirements are enforced, but applications **should** use the optimal alignment for optimal performance and power use. The value **must** be a power of two.
- `optimalBufferCopyRowPitchAlignment` is the optimal buffer row pitch alignment in bytes for `vkCmdCopyBufferToImage2KHR`, `vkCmdCopyBufferToImage`, `vkCmdCopyImageToBuffer2KHR`, and `vkCmdCopyImageToBuffer`. This value is also the optimal host memory row pitch alignment in bytes for `vkCopyMemoryToImageEXT` and `vkCopyImageToMemoryEXT`. Row pitch is the number of bytes between texels with the same X coordinate in adjacent rows (Y coordinates differ by one). The per texel alignment requirements are enforced, but applications **should** use the optimal alignment for optimal performance and power use. The value **must** be a power of

two.

- **nonCoherentAtomSize** is the size and alignment in bytes that bounds concurrent access to [host-mapped device memory](#). The value **must** be a power of two.

1

For all bitmasks of [VkSampleCountFlagBits](#), the sample count limits defined above represent the minimum supported sample counts for each image type. Individual images **may** support additional sample counts, which are queried using [vkGetPhysicalDeviceImageFormatProperties](#) as described in [Supported Sample Counts](#).

Bits which **may** be set in the sample count limits returned by [VkPhysicalDeviceLimits](#), as well as in other queries and structures representing image sample counts, are:

```
// Provided by VK_VERSION_1_0
typedef enum VkSampleCountFlagBits {
    VK_SAMPLE_COUNT_1_BIT = 0x00000001,
    VK_SAMPLE_COUNT_2_BIT = 0x00000002,
    VK_SAMPLE_COUNT_4_BIT = 0x00000004,
    VK_SAMPLE_COUNT_8_BIT = 0x00000008,
    VK_SAMPLE_COUNT_16_BIT = 0x00000010,
    VK_SAMPLE_COUNT_32_BIT = 0x00000020,
    VK_SAMPLE_COUNT_64_BIT = 0x00000040,
} VkSampleCountFlagBits;
```

- **VK_SAMPLE_COUNT_1_BIT** specifies an image with one sample per pixel.
- **VK_SAMPLE_COUNT_2_BIT** specifies an image with 2 samples per pixel.
- **VK_SAMPLE_COUNT_4_BIT** specifies an image with 4 samples per pixel.
- **VK_SAMPLE_COUNT_8_BIT** specifies an image with 8 samples per pixel.
- **VK_SAMPLE_COUNT_16_BIT** specifies an image with 16 samples per pixel.
- **VK_SAMPLE_COUNT_32_BIT** specifies an image with 32 samples per pixel.
- **VK_SAMPLE_COUNT_64_BIT** specifies an image with 64 samples per pixel.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkSampleCountFlags;
```

VkSampleCountFlags is a bitmask type for setting a mask of zero or more [VkSampleCountFlagBits](#).

The **VkPhysicalDevicePushDescriptorPropertiesKHR** structure is defined as:

```
// Provided by VK_KHR_push_descriptor
typedef struct VkPhysicalDevicePushDescriptorPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxPushDescriptors;
} VkPhysicalDevicePushDescriptorPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxPushDescriptors` is the maximum number of descriptors that **can** be used in a descriptor set layout created with `VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR` set.

If the `VkPhysicalDevicePushDescriptorPropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- `VUID-VkPhysicalDevicePushDescriptorPropertiesKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PUSH_DESCRIPTOR_PROPERTIES_KHR`

The `VkPhysicalDeviceMultiviewProperties` structure is defined as:

```
typedef struct VkPhysicalDeviceMultiviewProperties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxMultiviewViewCount;
    uint32_t           maxMultiviewInstanceIndex;
} VkPhysicalDeviceMultiviewProperties;
```

or the equivalent

```
// Provided by VK_KHR_multiview
typedef VkPhysicalDeviceMultiviewProperties VkPhysicalDeviceMultiviewPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxMultiviewViewCount` is one greater than the maximum view index that **can** be used in a subpass.
- `maxMultiviewInstanceIndex` is the maximum valid value of instance index allowed to be generated by a drawing command recorded within a subpass of a multiview render pass instance.

If the `VkPhysicalDeviceMultiviewProperties` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- `VUID-VkPhysicalDeviceMultiviewProperties-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_PROPERTIES`

The `VkPhysicalDeviceFloatControlsProperties` structure is defined as:

```
typedef struct VkPhysicalDeviceFloatControlsProperties {
    VkStructureType    sType;
    void*              pNext;
    VkShaderFloatControlsIndependence  denormBehaviorIndependence;
    VkShaderFloatControlsIndependence  roundingModeIndependence;
    VkBool32           shaderSignedZeroInfNanPreserveFloat16;
    VkBool32           shaderSignedZeroInfNanPreserveFloat32;
    VkBool32           shaderSignedZeroInfNanPreserveFloat64;
    VkBool32           shaderDenormPreserveFloat16;
    VkBool32           shaderDenormPreserveFloat32;
    VkBool32           shaderDenormPreserveFloat64;
    VkBool32           shaderDenormFlushToZeroFloat16;
    VkBool32           shaderDenormFlushToZeroFloat32;
    VkBool32           shaderDenormFlushToZeroFloat64;
    VkBool32           shaderRoundingModeRTEFloat16;
    VkBool32           shaderRoundingModeRTEFloat32;
    VkBool32           shaderRoundingModeRTEFloat64;
    VkBool32           shaderRoundingModeRTZFloat16;
    VkBool32           shaderRoundingModeRTZFloat32;
    VkBool32           shaderRoundingModeRTZFloat64;
} VkPhysicalDeviceFloatControlsProperties;
```

or the equivalent

```
// Provided by VK_KHR_shader_float_controls
typedef VkPhysicalDeviceFloatControlsProperties
VkPhysicalDeviceFloatControlsPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `denormBehaviorIndependence` is a `VkShaderFloatControlsIndependence` value indicating whether, and how, denorm behavior can be set independently for different bit widths.
- `roundingModeIndependence` is a `VkShaderFloatControlsIndependence` value indicating whether, and how, rounding modes can be set independently for different bit widths.

- `shaderSignedZeroInfNanPreserveFloat16` is a boolean value indicating whether sign of a zero, Nans and $\pm\infty$ **can** be preserved in 16-bit floating-point computations. It also indicates whether the `SignedZeroInfNanPreserve` execution mode **can** be used for 16-bit floating-point types.
- `shaderSignedZeroInfNanPreserveFloat32` is a boolean value indicating whether sign of a zero, Nans and $\pm\infty$ **can** be preserved in 32-bit floating-point computations. It also indicates whether the `SignedZeroInfNanPreserve` execution mode **can** be used for 32-bit floating-point types.
- `shaderSignedZeroInfNanPreserveFloat64` is a boolean value indicating whether sign of a zero, Nans and $\pm\infty$ **can** be preserved in 64-bit floating-point computations. It also indicates whether the `SignedZeroInfNanPreserve` execution mode **can** be used for 64-bit floating-point types.
- `shaderDenormPreserveFloat16` is a boolean value indicating whether denormals **can** be preserved in 16-bit floating-point computations. It also indicates whether the `DenormPreserve` execution mode **can** be used for 16-bit floating-point types.
- `shaderDenormPreserveFloat32` is a boolean value indicating whether denormals **can** be preserved in 32-bit floating-point computations. It also indicates whether the `DenormPreserve` execution mode **can** be used for 32-bit floating-point types.
- `shaderDenormPreserveFloat64` is a boolean value indicating whether denormals **can** be preserved in 64-bit floating-point computations. It also indicates whether the `DenormPreserve` execution mode **can** be used for 64-bit floating-point types.
- `shaderDenormFlushToZeroFloat16` is a boolean value indicating whether denormals **can** be flushed to zero in 16-bit floating-point computations. It also indicates whether the `DenormFlushToZero` execution mode **can** be used for 16-bit floating-point types.
- `shaderDenormFlushToZeroFloat32` is a boolean value indicating whether denormals **can** be flushed to zero in 32-bit floating-point computations. It also indicates whether the `DenormFlushToZero` execution mode **can** be used for 32-bit floating-point types.
- `shaderDenormFlushToZeroFloat64` is a boolean value indicating whether denormals **can** be flushed to zero in 64-bit floating-point computations. It also indicates whether the `DenormFlushToZero` execution mode **can** be used for 64-bit floating-point types.
- `shaderRoundingModeRTEFloat16` is a boolean value indicating whether an implementation supports the round-to-nearest-even rounding mode for 16-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTE` execution mode **can** be used for 16-bit floating-point types.
- `shaderRoundingModeRTEFloat32` is a boolean value indicating whether an implementation supports the round-to-nearest-even rounding mode for 32-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTE` execution mode **can** be used for 32-bit floating-point types.
- `shaderRoundingModeRTEFloat64` is a boolean value indicating whether an implementation supports the round-to-nearest-even rounding mode for 64-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTE` execution mode **can** be used for 64-bit floating-point types.
- `shaderRoundingModeRTZFloat16` is a boolean value indicating whether an implementation supports the round-towards-zero rounding mode for 16-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTZ` execution mode **can** be used for 16-bit floating-point types.

- `shaderRoundingModeRTZFloat32` is a boolean value indicating whether an implementation supports the round-towards-zero rounding mode for 32-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTZ` execution mode **can** be used for 32-bit floating-point types.
- `shaderRoundingModeRTZFloat64` is a boolean value indicating whether an implementation supports the round-towards-zero rounding mode for 64-bit floating-point arithmetic and conversion instructions. It also indicates whether the `RoundingModeRTZ` execution mode **can** be used for 64-bit floating-point types.

If the `VkPhysicalDeviceFloatControlsProperties` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFloatControlsProperties-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT_CONTROLS_PROPERTIES`

Values which **may** be returned in the `denormBehaviorIndependence` and `roundingModeIndependence` fields of `VkPhysicalDeviceFloatControlsProperties` are:

```
typedef enum VkShaderFloatControlsIndependence {
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY = 0,
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_ALL = 1,
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE = 2,
    // Provided by VK_KHR_shader_float_controls
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY_KHR =
VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY,
    // Provided by VK_KHR_shader_float_controls
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_ALL_KHR =
VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_ALL,
    // Provided by VK_KHR_shader_float_controls
    VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE_KHR =
VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE,
} VkShaderFloatControlsIndependence;
```

or the equivalent

```
// Provided by VK_KHR_shader_float_controls
typedef VkShaderFloatControlsIndependence VkShaderFloatControlsIndependenceKHR;
```

- `VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY` specifies that shader float controls for 32-bit floating-point **can** be set independently; other bit widths **must** be set identically to each other.
- `VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_ALL` specifies that shader float controls for all bit widths **can** be set independently.

- `VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE` specifies that shader float controls for all bit widths **must** be set identically.

The `VkPhysicalDeviceDiscardRectanglePropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_discard_rectangles
typedef struct VkPhysicalDeviceDiscardRectanglePropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxDiscardRectangles;
} VkPhysicalDeviceDiscardRectanglePropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxDiscardRectangles` is the maximum number of active discard rectangles that **can** be specified.

If the `VkPhysicalDeviceDiscardRectanglePropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDiscardRectanglePropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DISCARD_RECTANGLE_PROPERTIES_EXT`

The `VkPhysicalDeviceSampleLocationsPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_sample_locations
typedef struct VkPhysicalDeviceSampleLocationsPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkSampleCountFlags sampleLocationSampleCounts;
    VkExtent2D         maxSampleLocationGridSize;
    float              sampleLocationCoordinateRange[2];
    uint32_t           sampleLocationSubPixelBits;
    VkBool32           variableSampleLocations;
} VkPhysicalDeviceSampleLocationsPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `sampleLocationSampleCounts` is a bitmask of `VkSampleCountFlagBits` indicating the sample counts supporting custom sample locations.
- `maxSampleLocationGridSize` is the maximum size of the pixel grid in which sample locations **can** vary that is supported for all sample counts in `sampleLocationSampleCounts`.

- `sampleLocationCoordinateRange[2]` is the range of supported sample location coordinates.
- `sampleLocationSubPixelBits` is the number of bits of subpixel precision for sample locations.
- `variableSampleLocations` specifies whether the sample locations used by all pipelines that will be bound to a command buffer during a subpass **must** match. If set to `VK_TRUE`, the implementation supports variable sample locations in a subpass. If set to `VK_FALSE`, then the sample locations **must** stay constant in each subpass.

If the `VkPhysicalDeviceSampleLocationsPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceSampleLocationsPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLE_LOCATIONS_PROPERTIES_EXT`

The `VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_legacy_vertex_attributes
typedef struct VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           nativeUnalignedPerformance;
} VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT;
```

This structure describes the following features:

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `nativeUnalignedPerformance` specifies whether unaligned vertex fetches do not incur significant performance penalties as compared to aligned fetches.

If the `VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceFeatures2` structure passed to `vkGetPhysicalDeviceFeatures2KHR`, it is filled in to indicate whether each corresponding feature is supported. `VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT` **can** also be used in the `pNext` chain of `VkDeviceCreateInfo` to selectively enable these features.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_PROPERTIES_EXT`

The `VkPhysicalDeviceExternalMemoryHostPropertiesEXT` structure is defined as:


```
// Provided by VK_EXT_external_memory_host
typedef struct VkPhysicalDeviceExternalMemoryHostPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize        minImportedHostPointerAlignment;
} VkPhysicalDeviceExternalMemoryHostPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `minImportedHostPointerAlignment` is the minimum **required** alignment, in bytes, for the base address and size of host pointers that **can** be imported to a Vulkan memory object. The value **must** be a power of two.

If the `VkPhysicalDeviceExternalMemoryHostPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExternalMemoryHostPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_MEMORY_HOST_PROPERTIES_EXT`

The `VkPhysicalDevicePointClippingProperties` structure is defined as:

```
typedef struct VkPhysicalDevicePointClippingProperties {
    VkStructureType    sType;
    void*              pNext;
    VkPointClippingBehavior pointClippingBehavior;
} VkPhysicalDevicePointClippingProperties;
```

or the equivalent

```
// Provided by VK_KHR_maintenance2
typedef VkPhysicalDevicePointClippingProperties
VkPhysicalDevicePointClippingPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pointClippingBehavior` is a `VkPointClippingBehavior` value specifying the point clipping behavior supported by the implementation.

If the `VkPhysicalDevicePointClippingProperties` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in

with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- `VUID-VkPhysicalDevicePointClippingProperties-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_POINT_CLIPPING_PROPERTIES`

The `VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_blend_operation_advanced
typedef struct VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           advancedBlendMaxColorAttachments;
    VkBool32           advancedBlendIndependentBlend;
    VkBool32           advancedBlendNonPremultipliedSrcColor;
    VkBool32           advancedBlendNonPremultipliedDstColor;
    VkBool32           advancedBlendCorrelatedOverlap;
    VkBool32           advancedBlendAllOperations;
} VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `advancedBlendMaxColorAttachments` is one greater than the highest color attachment index that **can** be used in a subpass, for a pipeline that uses an **advanced blend operation**.
- `advancedBlendIndependentBlend` specifies whether advanced blend operations **can** vary per-attachment.
- `advancedBlendNonPremultipliedSrcColor` specifies whether the source color **can** be treated as non-premultiplied. If this is `VK_FALSE`, then `VkPipelineColorBlendAdvancedStateCreateInfoEXT::srcPremultiplied` **must** be `VK_TRUE`.
- `advancedBlendNonPremultipliedDstColor` specifies whether the destination color **can** be treated as non-premultiplied. If this is `VK_FALSE`, then `VkPipelineColorBlendAdvancedStateCreateInfoEXT::dstPremultiplied` **must** be `VK_TRUE`.
- `advancedBlendCorrelatedOverlap` specifies whether the overlap mode **can** be treated as correlated. If this is `VK_FALSE`, then `VkPipelineColorBlendAdvancedStateCreateInfoEXT::blendOverlap` **must** be `VK_BLEND_OVERLAP_UNCORRELATED_EXT`.
- `advancedBlendAllOperations` specifies whether all advanced blend operation enums are supported. See the valid usage of `VkPipelineColorBlendAttachmentState`.

If the `VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT-sType-sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_PROPERTIES_EXT

The `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_vertex_attribute_divisor
typedef struct VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxVertexAttribDivisor;
    VkBool32           supportsNonZeroFirstInstance;
} VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxVertexAttribDivisor` is the maximum value of the number of instances that will repeat the value of vertex attribute data when instanced rendering is enabled.
- `supportsNonZeroFirstInstance` specifies whether a non-zero value for the `firstInstance` parameter of `drawing commands` is supported when `VkVertexInputBindingDivisorDescriptionKHR::divisor` is not 1.

If the `VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR-sType-sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_PROPERTIES_KHR

The `VkPhysicalDeviceMaintenance3Properties` structure is defined as:

```
typedef struct VkPhysicalDeviceMaintenance3Properties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxPerSetDescriptors;
    VkDeviceSize        maxMemoryAllocationSize;
} VkPhysicalDeviceMaintenance3Properties;
```

or the equivalent

```
// Provided by VK_KHR_maintenance3
typedef VkPhysicalDeviceMaintenance3Properties
VkPhysicalDeviceMaintenance3PropertiesKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **maxPerSetDescriptors** is a maximum number of descriptors (summed over all descriptor types) in a single descriptor set that is guaranteed to satisfy any implementation-dependent constraints on the size of a descriptor set itself. Applications **can** query whether a descriptor set that goes beyond this limit is supported using [vkGetDescriptorSetLayoutSupportKHR](#).
- **maxMemoryAllocationSize** is the maximum size of a memory allocation that **can** be created, even if there is more space available in the heap.

If the [VkPhysicalDeviceMaintenance3Properties](#) structure is included in the **pNext** chain of the [VkPhysicalDeviceProperties2](#) structure passed to [vkGetPhysicalDeviceProperties2KHR](#), it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance3Properties-sType-sType **sType must** be [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_3_PROPERTIES](#)

The [VkPhysicalDeviceMaintenance4Properties](#) structure is defined as:

```
typedef struct VkPhysicalDeviceMaintenance4Properties {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize        maxBufferSize;
} VkPhysicalDeviceMaintenance4Properties;
```

or the equivalent

```
// Provided by VK_KHR_maintenance4
typedef VkPhysicalDeviceMaintenance4Properties
VkPhysicalDeviceMaintenance4PropertiesKHR;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **maxBufferSize** is the maximum size [VkBuffer](#) that **can** be created.

If the [VkPhysicalDeviceMaintenance4Properties](#) structure is included in the **pNext** chain of the [VkPhysicalDeviceProperties2](#) structure passed to [vkGetPhysicalDeviceProperties2KHR](#), it is filled in

with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance4Properties-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_PROPERTIES`

The `VkPhysicalDeviceMaintenance5PropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance5
typedef struct VkPhysicalDeviceMaintenance5PropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           earlyFragmentMultisampleCoverageAfterSampleCounting;
    VkBool32           earlyFragmentSampleMaskTestBeforeSampleCounting;
    VkBool32           depthStencilSwizzleOneSupport;
    VkBool32           polygonModePointSize;
    VkBool32           nonStrictSinglePixelWideLinesUseParallelogram;
    VkBool32           nonStrictWideLinesUseParallelogram;
} VkPhysicalDeviceMaintenance5PropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `earlyFragmentMultisampleCoverageAfterSampleCounting` is a boolean value indicating whether the `fragment shading` and `multisample coverage` operations are performed after `sample counting` for `fragment shaders` with `EarlyFragmentTests` execution mode.
- `earlyFragmentSampleMaskTestBeforeSampleCounting` is a boolean value indicating whether the `sample mask test` operation is performed before `sample counting` for `fragment shaders` using the `EarlyFragmentTests` execution mode.
- `depthStencilSwizzleOneSupport` is a boolean indicating that depth/stencil texturing operations with `VK_COMPONENT_SWIZZLE_ONE` have defined behavior.
- `polygonModePointSize` is a boolean value indicating whether the point size of the final rasterization of polygons with `VK_POLYGON_MODE_POINT` is controlled by `PointSize`.
- `nonStrictSinglePixelWideLinesUseParallelogram` is a boolean value indicating whether non-strict lines with a width of 1.0 are rasterized as parallelograms or using Bresenham's algorithm.
- `nonStrictWideLinesUseParallelogram` is a boolean value indicating whether non-strict lines with a width greater than 1.0 are rasterized as parallelograms or using Bresenham's algorithm.

If the `VkPhysicalDeviceMaintenance5PropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance5PropertiesKHR-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_PROPERTIES_KHR

The `VkPhysicalDeviceMaintenance6PropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance6
typedef struct VkPhysicalDeviceMaintenance6PropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           blockTexelViewCompatibleMultipleLayers;
    uint32_t           maxCombinedImageSamplerDescriptorCount;
    VkBool32           fragmentShadingRateClampCombinerInputs;
} VkPhysicalDeviceMaintenance6PropertiesKHR;
```

- sType is a `VkStructureType` value identifying this structure.
- pNext is `NULL` or a pointer to a structure extending this structure.
- blockTexelViewCompatibleMultipleLayers is a boolean value indicating that an implementation supports creating image views with `VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT` where the `layerCount` member of `subresourceRange` is greater than 1.
- maxCombinedImageSamplerDescriptorCount is the maximum number of combined image sampler descriptors that the implementation uses to access any of the [formats that require a sampler Y'C_BC_R conversion](#) supported by the implementation.
- fragmentShadingRateClampCombinerInputs is a boolean value indicating that an implementation clamps the inputs to [combiner operations](#).

If the `VkPhysicalDeviceMaintenance6PropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMaintenance6PropertiesKHR-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_PROPERTIES_KHR

The `VkPhysicalDeviceMaintenance7PropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance7
typedef struct VkPhysicalDeviceMaintenance7PropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           robustFragmentShadingRateAttachmentAccess;
    VkBool32           separateDepthStencilAttachmentAccess;
    uint32_t           maxDescriptorSetTotalUniformBuffersDynamic;
    uint32_t           maxDescriptorSetTotalStorageBuffersDynamic;
    uint32_t           maxDescriptorSetTotalBuffersDynamic;
    uint32_t           maxDescriptorSetUpdateAfterBindTotalUniformBuffersDynamic;
    uint32_t           maxDescriptorSetUpdateAfterBindTotalStorageBuffersDynamic;
    uint32_t           maxDescriptorSetUpdateAfterBindTotalBuffersDynamic;
} VkPhysicalDeviceMaintenance7PropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `robustFragmentShadingRateAttachmentAccess` indicates whether the scaled size of a fragment shading rate attachment **can** be less than the size of the render area. If `robustFragmentShadingRateAttachmentAccess` is `VK_FALSE`, the size of the attachment multiplied by the texel size **must** be greater than or equal to the size of the render area. If it is `VK_TRUE` and the fragment shading rate attachment was created with `VkImageSubresourceRange::baseMipLevel` equal to 0, the scaled size **can** be smaller than the render area, and shading rates for missing texels are defined by [texel replacement for invalid texels](#).
- `separateDepthStencilAttachmentAccess` indicates support for writing to one aspect of a depth/stencil attachment without performing read-modify-write operations on the other aspect. If this property is `VK_TRUE`, writes to one aspect **must** not result in read-modify-write operations on the other aspect. If `VK_FALSE`, writes to one aspect **may** result in writes to the other aspect as defined by [render pass load operations](#), [render pass store operations](#) and [render pass resolve operations](#).
- `maxDescriptorSetTotalUniformBuffersDynamic` is the maximum total count of dynamic uniform buffers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` count against this limit. Only descriptors in descriptor set layouts created without the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set count against this limit. See [Dynamic Uniform Buffer](#).
- `maxDescriptorSetTotalStorageBuffersDynamic` is the maximum total count of dynamic storage buffers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` count against this limit. Only descriptors in descriptor set layouts created without the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set count against this limit. See [Dynamic Storage Buffer](#).
- `maxDescriptorSetTotalBuffersDynamic` is the maximum total count of dynamic uniform buffers and storage buffers that **can** be included in a pipeline layout. Descriptors with a type of `VK_DESCRIPTOR_TYPE_UNIFORM_BUFFER_DYNAMIC` or `VK_DESCRIPTOR_TYPE_STORAGE_BUFFER_DYNAMIC` count against this limit. Only descriptors in descriptor set layouts created without the

`VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set count against this limit.

- `maxDescriptorSetUpdateAfterBindTotalUniformBuffersDynamic` is similar to `maxDescriptorSetTotalUniformBuffersDynamic` but counts descriptors from descriptor sets created with or without the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set.
- `maxDescriptorSetUpdateAfterBindTotalStorageBuffersDynamic` is similar to `maxDescriptorSetTotalStorageBuffersDynamic` but counts descriptors from descriptor sets created with or without the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set.
- `maxDescriptorSetUpdateAfterBindTotalBuffersDynamic` is similar to `maxDescriptorSetTotalBuffersDynamic` but counts descriptors from descriptor sets created with or without the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set. While an application **can** allocate dynamic storage buffer descriptors from a pool created with the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT`, bindings for these descriptors **must** not be present in any descriptor set layout that includes bindings created with `VK_DESCRIPTOR_BINDING_UPDATE_AFTER_BIND_BIT`.

If the `VkPhysicalDeviceMaintenance7PropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- `VUID-VkPhysicalDeviceMaintenance7PropertiesKHR-sType-sType` `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_PROPERTIES_KHR`

The `VkPhysicalDeviceLayeredApiPropertiesListKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance7
typedef struct VkPhysicalDeviceLayeredApiPropertiesListKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           layeredApiCount;
    VkPhysicalDeviceLayeredApiPropertiesKHR* pLayeredApis;
} VkPhysicalDeviceLayeredApiPropertiesListKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `layeredApiCount` is an integer related to the number of layered implementations underneath the Vulkan physical device, as described below.
- `pLayeredApis` is a pointer to an array of `VkPhysicalDeviceLayeredApiPropertiesKHR` in which information regarding the layered implementations underneath the Vulkan physical device are returned.

If `pLayeredApis` is `NULL`, then the number of layered implementations that are underneath the top-most Vulkan physical device (i.e. the one returned by `vkGetPhysicalDeviceProperties2KHR`) is

returned in `layeredApiCount`. Otherwise, `layeredApiCount` **must** be set by the application to the number of elements in the `pLayeredApis` array, and on return the variable is overwritten with the number of values actually written to `pLayeredApis`. If the value of `layeredApiCount` is less than the number of layered implementations underneath the Vulkan physical device, at most `layeredApiCount` values will be written to `pLayeredApis`. An implementation that is not a layer will return 0 in `layeredApiCount`.

In the presence of multiple layered implementations, each element of `pLayeredApis` corresponds to an API implementation that is implemented on top of the API at the previous index. If there are layered implementations underneath a non-Vulkan implementation, they may not be visible in this query as the corresponding APIs may lack such a query.

If the `VkPhysicalDeviceLayeredApiPropertiesListKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLayeredApiPropertiesListKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_LIST_KHR`
- VUID-VkPhysicalDeviceLayeredApiPropertiesListKHR-pLayeredApis-parameter
If `layeredApiCount` is not 0, and `pLayeredApis` is not NULL, `pLayeredApis` **must** be a valid pointer to an array of `layeredApiCount` `VkPhysicalDeviceLayeredApiPropertiesKHR` structures

The `VkPhysicalDeviceLayeredApiPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance7
typedef struct VkPhysicalDeviceLayeredApiPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t            vendorID;
    uint32_t            deviceID;
    VkPhysicalDeviceLayeredApiKHR    layeredAPI;
    char                deviceName[VK_MAX_PHYSICAL_DEVICE_NAME_SIZE];
} VkPhysicalDeviceLayeredApiPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is NULL or a pointer to a structure extending this structure.
- `vendorID` is a unique identifier for the vendor of the layered implementation.
- `deviceID` is a unique identifier for the layered implementation among devices available from the vendor.
- `layeredAPI` is a `VkPhysicalDeviceLayeredApiKHR` specifying the API implemented by the layered implementation.
- `deviceName` is an array of `VK_MAX_PHYSICAL_DEVICE_NAME_SIZE` `char` containing a null-terminated

UTF-8 string which is the name of the device.

If `layeredAPI` is `VK_PHYSICAL_DEVICE_LAYERED_API_VULKAN_KHR`, additional Vulkan-specific information can be queried by including the `VkPhysicalDeviceLayeredApiVulkanPropertiesKHR` structure in the `pNext` chain. Otherwise if such a structure is included in the `pNext` chain, it is ignored.

Valid Usage (Implicit)

- `VOID-VkPhysicalDeviceLayeredApiPropertiesKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_KHR`
- `VOID-VkPhysicalDeviceLayeredApiPropertiesKHR-pNext-pNext`
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkPhysicalDeviceLayeredApiVulkanPropertiesKHR`
- `VOID-VkPhysicalDeviceLayeredApiPropertiesKHR-sType-unique`
The `sType` value of each struct in the `pNext` chain **must** be unique

The list of possible API implementations of a layered implementation underneath the Vulkan physical device, as returned in `VkPhysicalDeviceLayeredApiPropertiesKHR::layeredAPI`, are:

```
// Provided by VK_KHR_maintenance7
typedef enum VkPhysicalDeviceLayeredApiKHR {
    VK_PHYSICAL_DEVICE_LAYERED_API_VULKAN_KHR = 0,
    VK_PHYSICAL_DEVICE_LAYERED_API_D3D12_KHR = 1,
    VK_PHYSICAL_DEVICE_LAYERED_API_METAL_KHR = 2,
    VK_PHYSICAL_DEVICE_LAYERED_API_OPENGL_KHR = 3,
    VK_PHYSICAL_DEVICE_LAYERED_API_OPENGL_ES_KHR = 4,
} VkPhysicalDeviceLayeredApiKHR;
```

- `VK_PHYSICAL_DEVICE_LAYERED_API_VULKAN_KHR` - the device implements the Vulkan API.
- `VK_PHYSICAL_DEVICE_LAYERED_API_D3D12_KHR` - the device implements the D3D12 API.
- `VK_PHYSICAL_DEVICE_LAYERED_API_METAL_KHR` - the device implements the Metal API.
- `VK_PHYSICAL_DEVICE_LAYERED_API_OPENGL_KHR` - the device implements the OpenGL API.
- `VK_PHYSICAL_DEVICE_LAYERED_API_OPENGL_ES_KHR` - the device implements the OpenGL ES API.

The `VkPhysicalDeviceLayeredApiVulkanPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_maintenance7
typedef struct VkPhysicalDeviceLayeredApiVulkanPropertiesKHR {
    VkStructureType      sType;
    void*                pNext;
    VkPhysicalDeviceProperties2 properties;
} VkPhysicalDeviceLayeredApiVulkanPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `properties` is a `VkPhysicalDeviceProperties2` in which properties of the underlying layered Vulkan implementation are returned.

The implementation **must** zero-fill the contents of `properties.properties.limits` and `properties.properties.sparseProperties`.

Valid Usage

- VUID-VkPhysicalDeviceLayeredApiVulkanPropertiesKHR-pNext-10011
Only `VkPhysicalDeviceDriverProperties` and `VkPhysicalDeviceIDProperties` are allowed in the `pNext` chain of `properties`

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLayeredApiVulkanPropertiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_VULKAN_PROPERTIES_KHR`

The `VkPhysicalDeviceConservativeRasterizationPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_conservative_rasterization
typedef struct VkPhysicalDeviceConservativeRasterizationPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    float               primitiveOverestimationSize;
    float               maxExtraPrimitiveOverestimationSize;
    float               extraPrimitiveOverestimationSizeGranularity;
    VkBool32            primitiveUnderestimation;
    VkBool32            conservativePointAndLineRasterization;
    VkBool32            degenerateTrianglesRasterized;
    VkBool32            degenerateLinesRasterized;
    VkBool32            fullyCoveredFragmentShaderInputVariable;
    VkBool32            conservativeRasterizationPostDepthCoverage;
} VkPhysicalDeviceConservativeRasterizationPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `primitiveOverestimationSize` is the size in pixels the generating primitive is increased at each of its edges during conservative rasterization overestimation mode. Even with a size of 0.0, conservative rasterization overestimation rules still apply and if any part of the pixel rectangle is covered by the generating primitive, fragments are generated for the entire pixel. However implementations **may** make the pixel coverage area even more conservative by increasing the size of the generating primitive.
- `maxExtraPrimitiveOverestimationSize` is the maximum size in pixels of extra overestimation the

implementation supports in the pipeline state. A value of 0.0 means the implementation does not support any additional overestimation of the generating primitive during conservative rasterization. A value above 0.0 allows the application to further increase the size of the generating primitive during conservative rasterization overestimation.

- `extraPrimitiveOverestimationSizeGranularity` is the granularity of extra overestimation that can be specified in the pipeline state between 0.0 and `maxExtraPrimitiveOverestimationSize` inclusive. A value of 0.0 means the implementation can use the smallest representable non-zero value in the screen space pixel fixed-point grid.
- `primitiveUnderestimation` is `VK_TRUE` if the implementation supports the `VK_CONSERVATIVE_RASTERIZATION_MODE_UNDERESTIMATE_EXT` conservative rasterization mode in addition to `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`. Otherwise the implementation only supports `VK_CONSERVATIVE_RASTERIZATION_MODE_OVERESTIMATE_EXT`.
- `conservativePointAndLineRasterization` is `VK_TRUE` if the implementation supports conservative rasterization of point and line primitives as well as triangle primitives. Otherwise the implementation only supports triangle primitives.
- `degenerateTrianglesRasterized` is `VK_FALSE` if the implementation culls primitives generated from triangles that become zero area after they are quantized to the fixed-point rasterization pixel grid. `degenerateTrianglesRasterized` is `VK_TRUE` if these primitives are not culled and the provoking vertex attributes and depth value are used for the fragments. The primitive area calculation is done on the primitive generated from the clipped triangle if applicable. Zero area primitives are backfacing and the application **can** enable backface culling if desired.
- `degenerateLinesRasterized` is `VK_FALSE` if the implementation culls lines that become zero length after they are quantized to the fixed-point rasterization pixel grid. `degenerateLinesRasterized` is `VK_TRUE` if zero length lines are not culled and the provoking vertex attributes and depth value are used for the fragments.
- `fullyCoveredFragmentShaderInputVariable` is `VK_TRUE` if the implementation supports the SPIR-V builtin fragment shader input variable `FullyCoveredEXT` specifying that conservative rasterization is enabled and the fragment area is fully covered by the generating primitive.
- `conservativeRasterizationPostDepthCoverage` is `VK_TRUE` if the implementation supports conservative rasterization with the `PostDepthCoverage` execution mode enabled. Otherwise the `PostDepthCoverage` execution mode **must** not be used when conservative rasterization is enabled.

If the `VkPhysicalDeviceConservativeRasterizationPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- `VUID-VkPhysicalDeviceConservativeRasterizationPropertiesEXT-sType-sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CONSERVATIVE_RASTERIZATION_PROPERTIES_EXT`

The `VkPhysicalDeviceDepthStencilResolveProperties` structure is defined as:

```
typedef struct VkPhysicalDeviceDepthStencilResolveProperties {
    VkStructureType    sType;
    void*              pNext;
    VkResolveModeFlags supportedDepthResolveModes;
    VkResolveModeFlags supportedStencilResolveModes;
    VkBool32           independentResolveNone;
    VkBool32           independentResolve;
} VkPhysicalDeviceDepthStencilResolveProperties;
```

or the equivalent

```
// Provided by VK_KHR_depth_stencil_resolve
typedef VkPhysicalDeviceDepthStencilResolveProperties
VkPhysicalDeviceDepthStencilResolvePropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `supportedDepthResolveModes` is a bitmask of `VkResolveModeFlagBits` indicating the set of supported depth resolve modes. `VK_RESOLVE_MODE_SAMPLE_ZERO_BIT` **must** be included in the set but implementations **may** support additional modes.
- `supportedStencilResolveModes` is a bitmask of `VkResolveModeFlagBits` indicating the set of supported stencil resolve modes. `VK_RESOLVE_MODE_SAMPLE_ZERO_BIT` **must** be included in the set but implementations **may** support additional modes. `VK_RESOLVE_MODE_AVERAGE_BIT` **must** not be included in the set.
- `independentResolveNone` is `VK_TRUE` if the implementation supports setting the depth and stencil resolve modes to different values when one of those modes is `VK_RESOLVE_MODE_NONE`. Otherwise the implementation only supports setting both modes to the same value.
- `independentResolve` is `VK_TRUE` if the implementation supports all combinations of the supported depth and stencil resolve modes, including setting either depth or stencil resolve mode to `VK_RESOLVE_MODE_NONE`. An implementation that supports `independentResolve` **must** also support `independentResolveNone`.

If the `VkPhysicalDeviceDepthStencilResolveProperties` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDepthStencilResolveProperties-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_STENCIL_RESOLVE_PROPERTIES`

The `VkPhysicalDevicePerformanceQueryPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_performance_query
typedef struct VkPhysicalDevicePerformanceQueryPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           allowCommandBufferQueryCopies;
} VkPhysicalDevicePerformanceQueryPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `allowCommandBufferQueryCopies` is `VK_TRUE` if the performance query pools are allowed to be used with `vkCmdCopyQueryPoolResults`.

If the `VkPhysicalDevicePerformanceQueryPropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePerformanceQueryPropertiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_PROPERTIES_KHR`

The `VkPhysicalDeviceTransformFeedbackPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_transform_feedback
typedef struct VkPhysicalDeviceTransformFeedbackPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxTransformFeedbackStreams;
    uint32_t           maxTransformFeedbackBuffers;
    VkDeviceSize       maxTransformFeedbackBufferSize;
    uint32_t           maxTransformFeedbackStreamDataSize;
    uint32_t           maxTransformFeedbackBufferDataSize;
    uint32_t           maxTransformFeedbackBufferDataStride;
    VkBool32           transformFeedbackQueries;
    VkBool32           transformFeedbackStreamsLinesTriangles;
    VkBool32           transformFeedbackRasterizationStreamSelect;
    VkBool32           transformFeedbackDraw;
} VkPhysicalDeviceTransformFeedbackPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxTransformFeedbackStreams` is the maximum number of vertex streams that can be output from geometry shaders declared with the `GeometryStreams` capability. If the implementation does not support `VkPhysicalDeviceTransformFeedbackFeaturesEXT::geometryStreams` then `maxTransformFeedbackStreams` **must** be 1.

- `maxTransformFeedbackBuffers` is the maximum number of transform feedback buffers that can be bound for capturing shader outputs from the last [pre-rasterization shader stage](#).
- `maxTransformFeedbackBufferSize` is the maximum size that can be specified when binding a buffer for transform feedback in [vkCmdBindTransformFeedbackBuffersEXT](#).
- `maxTransformFeedbackStreamDataSize` is the maximum amount of data in bytes for each vertex that captured to one or more transform feedback buffers associated with a specific vertex stream.
- `maxTransformFeedbackBufferDataSize` is the maximum amount of data in bytes for each vertex that can be captured to a specific transform feedback buffer.
- `maxTransformFeedbackBufferDataStride` is the maximum stride between each capture of vertex data to the buffer.
- `transformFeedbackQueries` is `VK_TRUE` if the implementation supports the `VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT` query type. `transformFeedbackQueries` is `VK_FALSE` if queries of this type **cannot** be created.
- `transformFeedbackStreamsLinesTriangles` is `VK_TRUE` if the implementation supports the geometry shader `OpExecutionMode` of `OutputLineStrip` and `OutputTriangleStrip` in addition to `OutputPoints` when more than one vertex stream is output. If `transformFeedbackStreamsLinesTriangles` is `VK_FALSE` the implementation only supports an `OpExecutionMode` of `OutputPoints` when more than one vertex stream is output from the geometry shader.
- `transformFeedbackRasterizationStreamSelect` is `VK_TRUE` if the implementation supports the `GeometryStreams` SPIR-V capability and the application can use [VkPipelineRasterizationStateStreamCreateInfoEXT](#) to modify which vertex stream output is used for rasterization. Otherwise vertex stream `0` **must** always be used for rasterization.
- `transformFeedbackDraw` is `VK_TRUE` if the implementation supports the [vkCmdDrawIndirectByteCountEXT](#) function otherwise the function **must** not be called.

If the `VkPhysicalDeviceTransformFeedbackPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to [vkGetPhysicalDeviceProperties2KHR](#), it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- `VOID-VkPhysicalDeviceTransformFeedbackPropertiesEXT-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_PROPERTIES_EXT`

The `VkPhysicalDeviceAccelerationStructurePropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_acceleration_structure
typedef struct VkPhysicalDeviceAccelerationStructurePropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint64_t           maxGeometryCount;
    uint64_t           maxInstanceCount;
    uint64_t           maxPrimitiveCount;
    uint32_t           maxPerStageDescriptorAccelerationStructures;
    uint32_t           maxPerStageDescriptorUpdateAfterBindAccelerationStructures;
    uint32_t           maxDescriptorSetAccelerationStructures;
    uint32_t           maxDescriptorSetUpdateAfterBindAccelerationStructures;
    uint32_t           minAccelerationStructureScratchOffsetAlignment;
} VkPhysicalDeviceAccelerationStructurePropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxGeometryCount` is the maximum number of geometries in the bottom level acceleration structure.
- `maxInstanceCount` is the maximum number of instances in the top level acceleration structure.
- `maxPrimitiveCount` is the maximum number of triangles or AABBs in all geometries in the bottom level acceleration structure.
- `maxPerStageDescriptorAccelerationStructures` is the maximum number of acceleration structure bindings that **can** be accessible to a single shader stage in a pipeline layout. Descriptor bindings with a descriptor type of `VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR` count against this limit. Only descriptor bindings in descriptor set layouts created without the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set count against this limit.
- `maxPerStageDescriptorUpdateAfterBindAccelerationStructures` is similar to `maxPerStageDescriptorAccelerationStructures` but counts descriptor bindings from descriptor sets created with or without the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set.
- `maxDescriptorSetAccelerationStructures` is the maximum number of acceleration structure descriptors that **can** be included in descriptor bindings in a pipeline layout across all pipeline shader stages and descriptor set numbers. Descriptor bindings with a descriptor type of `VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR` count against this limit. Only descriptor bindings in descriptor set layouts created without the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set count against this limit.
- `maxDescriptorSetUpdateAfterBindAccelerationStructures` is similar to `maxDescriptorSetAccelerationStructures` but counts descriptor bindings from descriptor sets created with or without the `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` bit set.
- `minAccelerationStructureScratchOffsetAlignment` is the minimum **required** alignment, in bytes, for scratch data passed in to an acceleration structure build command. The value **must** be a power of two.

Due to the fact that the geometry, instance, and primitive counts are specified at acceleration

structure creation as 32-bit values, `maxGeometryCount`, `maxInstanceCount`, and `maxPrimitiveCount` **must** not exceed $2^{32}-1$.

If the `VkPhysicalDeviceAccelerationStructurePropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- `VUID-VkPhysicalDeviceAccelerationStructurePropertiesKHR-sType-sType`
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_PROPERTIES_KHR`

The `VkPhysicalDeviceRayTracingPipelinePropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_ray_tracing_pipeline
typedef struct VkPhysicalDeviceRayTracingPipelinePropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           shaderGroupHandleSize;
    uint32_t           maxRayRecursionDepth;
    uint32_t           maxShaderGroupStride;
    uint32_t           shaderGroupBaseAlignment;
    uint32_t           shaderGroupHandleCaptureReplaySize;
    uint32_t           maxRayDispatchInvocationCount;
    uint32_t           shaderGroupHandleAlignment;
    uint32_t           maxRayHitAttributeSize;
} VkPhysicalDeviceRayTracingPipelinePropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderGroupHandleSize` is the size in bytes of the shader header.
- `maxRayRecursionDepth` is the maximum number of levels of ray recursion allowed in a trace command.
- `maxShaderGroupStride` is the maximum stride in bytes allowed between shader groups in the shader binding table.
- `shaderGroupBaseAlignment` is the **required** alignment in bytes for the base of the shader binding table.
- `shaderGroupHandleCaptureReplaySize` is the number of bytes for the information required to do capture and replay for shader group handles.
- `maxRayDispatchInvocationCount` is the maximum number of ray generation shader invocations which **may** be produced by a single `vkCmdTraceRaysIndirectKHR` or `vkCmdTraceRaysKHR` command.
- `shaderGroupHandleAlignment` is the **required** alignment in bytes for each entry in a shader binding table. The value **must** be a power of two.

- `maxRayHitAttributeSize` is the maximum size in bytes for a ray attribute structure

If the `VkPhysicalDeviceRayTracingPipelinePropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRayTracingPipelinePropertiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_PROPERTIES_KHR`

The `VkPhysicalDeviceCooperativeMatrixPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_cooperative_matrix
typedef struct VkPhysicalDeviceCooperativeMatrixPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkShaderStageFlags cooperativeMatrixSupportedStages;
} VkPhysicalDeviceCooperativeMatrixPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `cooperativeMatrixSupportedStages` is a bitfield of `VkShaderStageFlagBits` describing the shader stages that cooperative matrix instructions are supported in. `cooperativeMatrixSupportedStages` will have the `VK_SHADER_STAGE_COMPUTE_BIT` bit set if any of the physical device's queues support `VK_QUEUE_COMPUTE_BIT`.

`cooperativeMatrixSupportedStages` **must** not have any bits other than `VK_SHADER_STAGE_COMPUTE_BIT` set.

If the `VkPhysicalDeviceCooperativeMatrixPropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCooperativeMatrixPropertiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_PROPERTIES_KHR`

The `VkPhysicalDeviceTimelineSemaphoreProperties` structure is defined as:

```
typedef struct VkPhysicalDeviceTimelineSemaphoreProperties {
    VkStructureType    sType;
    void*              pNext;
    uint64_t           maxTimelineSemaphoreValueDifference;
} VkPhysicalDeviceTimelineSemaphoreProperties;
```

or the equivalent

```
// Provided by VK_KHR_timeline_semaphore
typedef VkPhysicalDeviceTimelineSemaphoreProperties
VkPhysicalDeviceTimelineSemaphorePropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxTimelineSemaphoreValueDifference` indicates the maximum difference allowed by the implementation between the current value of a timeline semaphore and any pending signal or wait operations.

If the `VkPhysicalDeviceTimelineSemaphoreProperties` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceTimelineSemaphoreProperties-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_PROPERTIES`

The `VkPhysicalDeviceLineRasterizationPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_line_rasterization
typedef struct VkPhysicalDeviceLineRasterizationPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           lineSubPixelPrecisionBits;
} VkPhysicalDeviceLineRasterizationPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `lineSubPixelPrecisionBits` is the number of bits of subpixel precision in framebuffer coordinates x_f and y_f when rasterizing [line segments](#).

If the `VkPhysicalDeviceLineRasterizationPropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is

filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceLineRasterizationPropertiesKHR-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_PROPERTIES_KHR

The `VkPhysicalDeviceRobustness2PropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_robustness2
typedef struct VkPhysicalDeviceRobustness2PropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize        robustStorageBufferAccessSizeAlignment;
    VkDeviceSize        robustUniformBufferAccessSizeAlignment;
} VkPhysicalDeviceRobustness2PropertiesEXT;
```

- sType is a `VkStructureType` value identifying this structure.
- pNext is `NULL` or a pointer to a structure extending this structure.
- robustStorageBufferAccessSizeAlignment is the number of bytes that the range of a storage buffer descriptor is rounded up to when used for bounds-checking when the `robustBufferAccess2` feature is enabled. This value **must** be either 1 or 4.
- robustUniformBufferAccessSizeAlignment is the number of bytes that the range of a uniform buffer descriptor is rounded up to when used for bounds-checking when the `robustBufferAccess2` feature is enabled. This value **must** be a power of two in the range [1, 256].

If the `VkPhysicalDeviceRobustness2PropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceRobustness2PropertiesEXT-sType-sType
sType **must** be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_PROPERTIES_EXT

The `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_device_generated_commands
typedef struct VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT {
    VkStructureType          sType;
    void*                    pNext;
    uint32_t                 maxIndirectPipelineCount;
    uint32_t                 maxIndirectShaderObjectCount;
    uint32_t                 maxIndirectSequenceCount;
    uint32_t                 maxIndirectCommandsTokenCount;
    uint32_t                 maxIndirectCommandsTokenOffset;
    uint32_t                 maxIndirectCommandsIndirectStride;
    VkIndirectCommandsInputModeFlagsEXT supportedIndirectCommandsInputModes;
    VkShaderStageFlags       supportedIndirectCommandsShaderStages;
    VkShaderStageFlags
supportedIndirectCommandsShaderStagesPipelineBinding;
    VkShaderStageFlags
supportedIndirectCommandsShaderStagesShaderBinding;
    VkBool32                deviceGeneratedCommandsTransformFeedback;
    VkBool32
deviceGeneratedCommandsMultiDrawIndirectCount;
} VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxIndirectPipelineCount` is the maximum number of pipelines passed to `vkCreateIndirectExecutionSetEXT`.
- `maxIndirectShaderObjectCount` is the maximum number of shader objects passed to `vkCreateIndirectExecutionSetEXT`. If this value is zero, binding shader objects indirectly is not supported.
- `maxIndirectSequenceCount` is the maximum number of sequences in `VkGeneratedCommandsInfoEXT` and in `VkGeneratedCommandsMemoryRequirementsInfoEXT`.
- `maxIndirectCommandsTokenCount` is the maximum number of tokens in `VkIndirectCommandsLayoutCreateInfoEXT`.
- `maxIndirectCommandsTokenOffset` is the maximum offset in `VkIndirectCommandsLayoutTokenEXT`.
- `maxIndirectCommandsIndirectStride` is the maximum stream stride in `VkIndirectCommandsLayoutCreateInfoEXT`.
- `supportedIndirectCommandsInputModes` indicates the supported input modes.
- `supportedIndirectCommandsShaderStages` indicates the stages which **can** be used to generate indirect commands. Implementations are required to support, at minimum: `VK_SHADER_STAGE_VERTEX_BIT`, `VK_SHADER_STAGE_FRAGMENT_BIT`, `VK_SHADER_STAGE_COMPUTE_BIT`.
- `supportedIndirectCommandsShaderStagesPipelineBinding` indicates the stages which **can** be used within indirect execution sets for indirectly binding shader stages using pipelines.
- `supportedIndirectCommandsShaderStagesShaderBinding` indicates the stages which **can** be used within indirect execution sets for indirectly binding shader stages using shader objects.

- `deviceGeneratedCommandsTransformFeedback` indicates whether the implementation supports interactions with `VK_EXT_transform_feedback` for pipelines not created with `VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT`.
- `deviceGeneratedCommandsMultiDrawIndirectCount` indicates whether the implementation supports COUNT variants of multi-draw indirect tokens.

If the `VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_PROPERTIES_EXT`

The `VkPhysicalDevicePortabilitySubsetPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_portability_subset
typedef struct VkPhysicalDevicePortabilitySubsetPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           minVertexInputBindingStrideAlignment;
} VkPhysicalDevicePortabilitySubsetPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `minVertexInputBindingStrideAlignment` indicates the minimum alignment for vertex input strides. `VkVertexInputBindingDescription::stride` **must** be a multiple of, and at least as large as, this value. The value **must** be a power of two.

If the `VkPhysicalDevicePortabilitySubsetPropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePortabilitySubsetPropertiesKHR-sType-sType **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_PROPERTIES_KHR`

The `VkPhysicalDeviceFragmentShadingRatePropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_fragment_shading_rate
typedef struct VkPhysicalDeviceFragmentShadingRatePropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkExtent2D         minFragmentShadingRateAttachmentTexelSize;
    VkExtent2D         maxFragmentShadingRateAttachmentTexelSize;
    uint32_t           maxFragmentShadingRateAttachmentTexelSizeAspectRatio;
    VkBool32           primitiveFragmentShadingRateWithMultipleViewports;
    VkBool32           layeredShadingRateAttachments;
    VkBool32           fragmentShadingRateNonTrivialCombinerOps;
    VkExtent2D         maxFragmentSize;
    uint32_t           maxFragmentSizeAspectRatio;
    uint32_t           maxFragmentShadingRateCoverageSamples;
    VkSampleCountFlagBits maxFragmentShadingRateRasterizationSamples;
    VkBool32           fragmentShadingRateWithShaderDepthStencilWrites;
    VkBool32           fragmentShadingRateWithSampleMask;
    VkBool32           fragmentShadingRateWithShaderSampleMask;
    VkBool32           fragmentShadingRateWithConservativeRasterization;
    VkBool32           fragmentShadingRateWithFragmentShaderInterlock;
    VkBool32           fragmentShadingRateWithCustomSampleLocations;
    VkBool32           fragmentShadingRateStrictMultiplyCombiner;
} VkPhysicalDeviceFragmentShadingRatePropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `minFragmentShadingRateAttachmentTexelSize` indicates minimum supported width and height of the portion of the framebuffer corresponding to each texel in a fragment shading rate attachment. Each value **must** be less than or equal to the values in `maxFragmentShadingRateAttachmentTexelSize`. Each value **must** be a power-of-two. It **must** be (0,0) if the `attachmentFragmentShadingRate` feature is not supported.
- `maxFragmentShadingRateAttachmentTexelSize` indicates maximum supported width and height of the portion of the framebuffer corresponding to each texel in a fragment shading rate attachment. Each value **must** be greater than or equal to the values in `minFragmentShadingRateAttachmentTexelSize`. Each value **must** be a power-of-two. It **must** be (0,0) if the `attachmentFragmentShadingRate` feature is not supported.
- `maxFragmentShadingRateAttachmentTexelSizeAspectRatio` indicates the maximum ratio between the width and height of the portion of the framebuffer corresponding to each texel in a fragment shading rate attachment. `maxFragmentShadingRateAttachmentTexelSizeAspectRatio` **must** be a power-of-two value, and **must** be less than or equal to $\frac{\max(\maxFragmentShadingRateAttachmentTexelSize.width}{\minFragmentShadingRateAttachmentTexelSize.height}, \frac{\maxFragmentShadingRateAttachmentTexelSize.height}{\minFragmentShadingRateAttachmentTexelSize.width})$. It **must** be 0 if the `attachmentFragmentShadingRate` feature is not supported.
- `primitiveFragmentShadingRateWithMultipleViewports` specifies whether the `primitive fragment shading rate` can be used when multiple viewports are used. If this value is `VK_FALSE`, only a

single viewport **must** be used, and applications **must** not write to the `ViewportIndex` built-in when setting `PrimitiveShadingRateKHR`. It **must** be `VK_FALSE` if the `geometryShader` feature is not supported, or if the `primitiveFragmentShadingRate` feature is not supported.

- `layeredShadingRateAttachments` specifies whether a shading rate attachment image view **can** be created with multiple layers. If this value is `VK_FALSE`, when creating an image view with a `usage` that includes `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`, `layerCount` **must** be 1. It **must** be `VK_FALSE` if the `multiview` feature, or the `geometryShader` feature is not supported, or if the `attachmentFragmentShadingRate` feature is not supported.
- `fragmentShadingRateNonTrivialCombinerOps` specifies whether `VkFragmentShadingRateCombinerOpKHR` enums other than `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_KEEP_KHR` or `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_REPLACE_KHR` **can** be used. It **must** be `VK_FALSE` unless either the `primitiveFragmentShadingRate` or `attachmentFragmentShadingRate` feature is supported.
- `maxFragmentSize` indicates the maximum supported width and height of a fragment. Its `width` and `height` members **must** both be power-of-two values. This limit is purely informational, and is not validated.
- `maxFragmentSizeAspectRatio` indicates the maximum ratio between the width and height of a fragment. `maxFragmentSizeAspectRatio` **must** be a power-of-two value, and **must** be less than or equal to the maximum of the `width` and `height` members of `maxFragmentSize`. This limit is purely informational, and is not validated.
- `maxFragmentShadingRateCoverageSamples` specifies the maximum number of coverage samples supported in a single fragment. `maxFragmentShadingRateCoverageSamples` **must** be less than or equal to the product of the `width` and `height` members of `maxFragmentSize`, and the sample count reported by `maxFragmentShadingRateRasterizationSamples`. `maxFragmentShadingRateCoverageSamples` **must** be less than or equal to `maxSampleMaskWords` × 32 if `fragmentShadingRateWithShaderSampleMask` is supported. This limit is purely informational, and is not validated.
- `maxFragmentShadingRateRasterizationSamples` is a `VkSampleCountFlagBits` value specifying the maximum sample rate supported when a fragment covers multiple pixels. This limit is purely informational, and is not validated.
- `fragmentShadingRateWithShaderDepthStencilWrites` specifies whether the implementation supports writing `FragDepth` or `FragStencilRefEXT` from a fragment shader for multi-pixel fragments. If this value is `VK_FALSE`, writing to those built-ins will clamp the fragment shading rate to (1,1).
- `fragmentShadingRateWithSampleMask` specifies whether the implementation supports setting valid bits of `VkPipelineMultisampleStateCreateInfo::pSampleMask` to 0 for multi-pixel fragments. If this value is `VK_FALSE`, zeroing valid bits in the sample mask will clamp the fragment shading rate to (1,1).
- `fragmentShadingRateWithShaderSampleMask` specifies whether the implementation supports reading or writing `SampleMask` for multi-pixel fragments. If this value is `VK_FALSE`, using that built-in will clamp the fragment shading rate to (1,1).
- `fragmentShadingRateWithConservativeRasterization` specifies whether `conservative rasterization` is supported for multi-pixel fragments. It **must** be `VK_FALSE` if `VK_EXT_conservative_rasterization` is not supported. If this value is `VK_FALSE`, using `conservative rasterization` will clamp the

fragment shading rate to (1,1).

- `fragmentShadingRateWithFragmentShaderInterlock` specifies whether `fragment shader interlock` is supported for multi-pixel fragments. It **must** be `VK_FALSE` if `VK_EXT_fragment_shader_interlock` is not supported. If this value is `VK_FALSE`, using `fragment shader interlock` will clamp the fragment shading rate to (1,1).
- `fragmentShadingRateWithCustomSampleLocations` specifies whether `custom sample locations` are supported for multi-pixel fragments. It **must** be `VK_FALSE` if `VK_EXT_sample_locations` is not supported. If this value is `VK_FALSE`, using `custom sample locations` will clamp the fragment shading rate to (1,1).
- `fragmentShadingRateStrictMultiplyCombiner` specifies whether `VK_FRAGMENT_SHADING_RATE_COMBINER_OP_MUL_KHR` accurately performs a multiplication or not. Implementations where this value is `VK_FALSE` will instead combine rates with an addition. If `fragmentShadingRateNonTrivialCombinerOps` is `VK_FALSE`, implementations **must** report this as `VK_FALSE`. If `fragmentShadingRateNonTrivialCombinerOps` is `VK_TRUE`, implementations **should** report this as `VK_TRUE`.

NOTE

Multiplication of the combiner rates using the fragment width/height in linear space is equivalent to an addition of those values in log2 space. Some implementations inadvertently implemented an addition in linear space due to unclear requirements originating outside of this specification. This resulted in `fragmentShadingRateStrictMultiplyCombiner` being added. Fortunately, this only affects situations where a rate of 1 in either dimension is combined with another rate of 1. All other combinations result in the exact same result as if multiplication was performed in linear space due to the clamping logic, and the fact that both the sum and product of 2 and 2 are equal. In many cases, this limit will not affect the correct operation of applications.

If the `VkPhysicalDeviceFragmentShadingRatePropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

These properties are related to `fragment shading rates`.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShadingRatePropertiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_PROPERTIES_KHR`

The `VkPhysicalDeviceCustomBorderColorPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_custom_border_color
typedef struct VkPhysicalDeviceCustomBorderColorPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxCustomBorderColorSamplers;
} VkPhysicalDeviceCustomBorderColorPropertiesEXT;
```

- `maxCustomBorderColorSamplers` indicates the maximum number of samplers with custom border colors which **can** simultaneously exist on a device.

If the `VkPhysicalDeviceCustomBorderColorPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceCustomBorderColorPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_PROPERTIES_EXT`

The `VkPhysicalDeviceProvokingVertexPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_provoking_vertex
typedef struct VkPhysicalDeviceProvokingVertexPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           provokingVertexModePerPipeline;
    VkBool32           transformFeedbackPreservesTriangleFanProvokingVertex;
} VkPhysicalDeviceProvokingVertexPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `provokingVertexModePerPipeline` indicates whether the implementation supports graphics pipelines with different provoking vertex modes within the same render pass instance.
- `transformFeedbackPreservesTriangleFanProvokingVertex` indicates whether the implementation can preserve the provoking vertex order when writing triangle fan vertices to transform feedback.

If the `VkPhysicalDeviceProvokingVertexPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceProvokingVertexPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_PROPERTIES_EXT`

The `VkPhysicalDeviceHostImageCopyPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_host_image_copy
typedef struct VkPhysicalDeviceHostImageCopyPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           copySrcLayoutCount;
    VkImageLayout*     pCopySrcLayouts;
    uint32_t           copyDstLayoutCount;
    VkImageLayout*     pCopyDstLayouts;
    uint8_t            optimalTilingLayoutUUID[VK_UUID_SIZE];
    VkBool32           identicalMemoryTypeRequirements;
} VkPhysicalDeviceHostImageCopyPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `copySrcLayoutCount` is an integer related to the number of image layouts for host copies from images available or queried, as described below.
- `pCopySrcLayouts` is a pointer to an array of `VkImageLayout` in which supported image layouts for use with host copy operations from images are returned.
- `copyDstLayoutCount` is an integer related to the number of image layouts for host copies to images available or queried, as described below.
- `pCopyDstLayouts` is a pointer to an array of `VkImageLayout` in which supported image layouts for use with host copy operations to images are returned.
- `optimalTilingLayoutUUID` is an array of `VK_UUID_SIZE` `uint8_t` values representing a universally unique identifier for the implementation's swizzling layout of images created with `VK_IMAGE_TILING_OPTIMAL`.
- `identicalMemoryTypeRequirements` indicates that specifying the `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` flag in `VkImageCreateInfo::usage` does not affect the memory type requirements of the image.

If the `VkPhysicalDeviceHostImageCopyPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

If `pCopyDstLayouts` is `NULL`, then the number of image layouts that are supported in `VkCopyMemoryToImageInfoEXT::dstImageLayout` and `VkCopyImageToImageInfoEXT::dstImageLayout` is returned in `copyDstLayoutCount`. Otherwise, `copyDstLayoutCount` **must** be set by the application to the number of elements in the `pCopyDstLayouts` array, and on return the variable is overwritten with the number of values actually written to `pCopyDstLayouts`. If the value of `copyDstLayoutCount` is less than the number of image layouts that are supported, at most `copyDstLayoutCount` values will be written to `pCopyDstLayouts`. The implementation **must** include the `VK_IMAGE_LAYOUT_GENERAL` image layout in `pCopyDstLayouts`.

If `pCopySrcLayouts` is `NULL`, then the number of image layouts that are supported in

`VkCopyImageToMemoryInfoEXT::srcImageLayout` and `VkCopyImageToImageInfoEXT::srcImageLayout` is returned in `copySrcLayoutCount`. Otherwise, `copySrcLayoutCount` **must** be set by the application to the number of elements in the `pCopySrcLayouts` array, and on return the variable is overwritten with the number of values actually written to `pCopySrcLayouts`. If the value of `copySrcLayoutCount` is less than the number of image layouts that are supported, at most `copySrcLayoutCount` values will be written to `pCopySrcLayouts`. The implementation **must** include the `VK_IMAGE_LAYOUT_GENERAL` image layout in `pCopySrcLayouts`.

The `optimalTilingLayoutUUID` value can be used to ensure compatible data layouts when using the `VK_HOST_IMAGE_COPY_MEMCPY_EXT` flag in `vkCopyMemoryToImageEXT` and `vkCopyImageToMemoryEXT`.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceHostImageCopyPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_PROPERTIES_EXT`
- VUID-VkPhysicalDeviceHostImageCopyPropertiesEXT-pCopySrcLayouts-parameter
If `copySrcLayoutCount` is not 0, and `pCopySrcLayouts` is not `NULL`, `pCopySrcLayouts` **must** be a valid pointer to an array of `copySrcLayoutCount` `VkImageLayout` values
- VUID-VkPhysicalDeviceHostImageCopyPropertiesEXT-pCopyDstLayouts-parameter
If `copyDstLayoutCount` is not 0, and `pCopyDstLayouts` is not `NULL`, `pCopyDstLayouts` **must** be a valid pointer to an array of `copyDstLayoutCount` `VkImageLayout` values

The `VkPhysicalDeviceMultiDrawPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_multi_draw
typedef struct VkPhysicalDeviceMultiDrawPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxMultiDrawCount;
} VkPhysicalDeviceMultiDrawPropertiesEXT;
```

The members of the `VkPhysicalDeviceMultiDrawPropertiesEXT` structure describe the following features:

- `maxMultiDrawCount` indicates the maximum number of draw calls which **can** be batched into a single multidraw.

If the `VkPhysicalDeviceMultiDrawPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMultiDrawPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_PROPERTIES_EXT`

The `VkPhysicalDeviceNestedCommandBufferPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_nested_command_buffer
typedef struct VkPhysicalDeviceNestedCommandBufferPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxCommandBufferNestingLevel;
} VkPhysicalDeviceNestedCommandBufferPropertiesEXT;
```

The members of the `VkPhysicalDeviceNestedCommandBufferPropertiesEXT` structure describe the following features:

- `maxCommandBufferNestingLevel` indicates the maximum nesting level of calls to `vkCmdExecuteCommands` from [Secondary Command Buffers](#). A `maxCommandBufferNestingLevel` of `UINT32_MAX` means there is no limit to the nesting level.

If the `VkPhysicalDeviceNestedCommandBufferPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceNestedCommandBufferPropertiesEXT-sType-sType `sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_PROPERTIES_EXT`

The `VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_fragment_shader_barycentric
typedef struct VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           triStripVertexOrderIndependentOfProvokingVertex;
} VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR;
```

- When the [provoking vertex mode](#) is `VK_PROVOKING_VERTEX_MODE_LAST_VERTEX_EXT`, and the primitive order is odd in a triangle strip, the ordering of vertices is defined in [last vertex table](#). `triStripVertexOrderIndependentOfProvokingVertex` equal to `VK_TRUE` indicates that the implementation ignores this and uses the vertex order defined by `VK_PROVOKING_VERTEX_MODE_FIRST_VERTEX_EXT` instead.

If the `VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR-sType-sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_PROPERTIES_KHR

The `VkPhysicalDeviceExtendedDynamicState3PropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_extended_dynamic_state3
typedef struct VkPhysicalDeviceExtendedDynamicState3PropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           dynamicPrimitiveTopologyUnrestricted;
} VkPhysicalDeviceExtendedDynamicState3PropertiesEXT;
```

- `dynamicPrimitiveTopologyUnrestricted` indicates that the implementation allows `vkCmdSetPrimitiveTopologyEXT` to use a different `primitive topology class` to the one specified in the active graphics pipeline.

If the `VkPhysicalDeviceExtendedDynamicState3PropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExtendedDynamicState3PropertiesEXT-sType-sType must be VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_PROPERTIES_EXT

The `VkPhysicalDeviceOpacityMicromapPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_opacity_micromap
typedef struct VkPhysicalDeviceOpacityMicromapPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           maxOpacity2StateSubdivisionLevel;
    uint32_t           maxOpacity4StateSubdivisionLevel;
} VkPhysicalDeviceOpacityMicromapPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `maxOpacity2StateSubdivisionLevel` is the maximum allowed `subdivisionLevel` when `format` is `VK_OPACITY_MICROMAP_FORMAT_2_STATE_EXT`
- `maxOpacity4StateSubdivisionLevel` is the maximum allowed `subdivisionLevel` when `format` is

VK_OPACITY_MICROMAP_FORMAT_4_STATE_EXT

If the `VkPhysicalDeviceOpacityMicromapPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceOpacityMicromapPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_PROPERTIES_EXT`

The `VkPhysicalDeviceShaderObjectPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_shader_object
typedef struct VkPhysicalDeviceShaderObjectPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    uint8_t             shaderBinaryUUID[VK_UUID_SIZE];
    uint32_t            shaderBinaryVersion;
} VkPhysicalDeviceShaderObjectPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `shaderBinaryUUID` is an array of `VK_UUID_SIZE` `uint8_t` values representing a universally unique identifier for one or more implementations whose shader binaries are guaranteed to be compatible with each other.
- `shaderBinaryVersion` is an unsigned integer incremented to represent backwards compatible differences between implementations with the same `shaderBinaryUUID`.

The purpose and usage of the values of this structure are described in greater detail in [Binary Shader Compatibility](#).

If the `VkPhysicalDeviceShaderObjectPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceShaderObjectPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_PROPERTIES_EXT`

The `VkPhysicalDevicePipelineBinaryPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_pipeline_binary
typedef struct VkPhysicalDevicePipelineBinaryPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           pipelineBinaryInternalCache;
    VkBool32           pipelineBinaryInternalCacheControl;
    VkBool32           pipelineBinaryPrefersInternalCache;
    VkBool32           pipelineBinaryPrecompiledInternalCache;
    VkBool32           pipelineBinaryCompressedData;
} VkPhysicalDevicePipelineBinaryPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `pipelineBinaryInternalCache` specifies that the implementation maintains a pipeline cache internal to the implementation. If this is `VK_TRUE`, applications **can** create pipeline binaries with only a pipeline create info, and in this case, an implementation **may** be able to create a pipeline binary directly without application needing to capture the binary itself.
- `pipelineBinaryInternalCacheControl` specifies whether the driver's internal cache **can** be disabled. If this property is `VK_TRUE` `VkDevicePipelineBinaryInternalCacheControlKHR::disableInternalCache` **can** be used to disable the driver's internal cache, allowing an application to take full control of both memory and disk usage.
- `pipelineBinaryPrefersInternalCache` specifies that the implementation prefers to maintain an internal cache, and applications **should** not store pipeline binaries in their own on-disk caches to avoid increased on-disk storage requirements. Applications are encouraged to only store pipeline keys instead, and aim to create pipeline binaries from key alone on subsequent runs of the application.
- `pipelineBinaryPrecompiledInternalCache` specifies that the implementation **may** have pipeline binaries in its internal cache, which is populated without the application ever having generated that pipeline itself. Applications **can** attempt to create binaries without extracting pipeline binary data from the pipeline prior for a set of pipeline keys, including from previous runs of the application.
- `pipelineBinaryCompressedData` specifies that the binary data is already compressed and so applications **should** not attempt to compress it.

NOTE

These properties tend to be platform specific and may change depending on external configuration which is outside the scope of this specification. These properties are intended to guide applications when implementations have dedicated caching solutions available. In particular, if the `pipelineBinaryPrefersInternalCache` limit is exposed, relying on the internal cache may provide some advantage compared to an application-specific solution. An application with its own dedicated solution may still use its own caching system even with this limit exposed.

If the `VkPhysicalDevicePipelineBinaryPropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in

with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDevicePipelineBinaryPropertiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_PROPERTIES_KHR`

The `VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR` structure is defined as:

```
// Provided by VK_KHR_compute_shader_derivatives
typedef struct VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           meshAndTaskShaderDerivatives;
} VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR;
```

The members of the `VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR` structure describe the following:

- `meshAndTaskShaderDerivatives` indicates whether the mesh and task shader stages support the `ComputeDerivativeGroupQuadsKHR` and `ComputeDerivativeGroupLinearKHR` SPIR-V capabilities.

If the `VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_PROPERTIES_KHR`

The `VkPhysicalDeviceMapMemoryPlacedPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_map_memory_placed
typedef struct VkPhysicalDeviceMapMemoryPlacedPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkDeviceSize        minPlacedMemoryMapAlignment;
} VkPhysicalDeviceMapMemoryPlacedPropertiesEXT;
```

The members of the `VkPhysicalDeviceMapMemoryPlacedPropertiesEXT` structure describe the following:

- `minPlacedMemoryMapAlignment` is the minimum alignment required for memory object offsets and virtual address ranges when using placed memory mapping.

If the `VkPhysicalDeviceMapMemoryPlacedPropertiesEXT` structure is included in the `pNext` chain of the `VkPhysicalDeviceProperties2` structure passed to `vkGetPhysicalDeviceProperties2KHR`, it is filled in with each corresponding implementation-dependent property.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceMapMemoryPlacedPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_PROPERTIES_EXT`

40.1. Limit Requirements

The following table specifies the **required** minimum/maximum for all Vulkan graphics implementations. Where a limit corresponds to a fine-grained device feature which is **optional**, the feature name is listed with two **required** limits, one when the feature is supported and one when it is not supported. If an implementation supports a feature, the limits reported are the same whether or not the feature is enabled.

Table 58. Required Limit Types

Type	Limit	Feature
<code>uint32_t</code>	<code>maxImageDimension1D</code>	-
<code>uint32_t</code>	<code>maxImageDimension2D</code>	-
<code>uint32_t</code>	<code>maxImageDimension3D</code>	-
<code>uint32_t</code>	<code>maxImageDimensionCube</code>	-
<code>uint32_t</code>	<code>maxImageArrayLayers</code>	-
<code>uint32_t</code>	<code>maxTexelBufferElements</code>	-
<code>uint32_t</code>	<code>maxUniformBufferRange</code>	-
<code>uint32_t</code>	<code>maxStorageBufferRange</code>	-
<code>uint32_t</code>	<code>maxPushConstantsSize</code>	-
<code>uint32_t</code>	<code>maxMemoryAllocationCount</code>	-
<code>uint32_t</code>	<code>maxSamplerAllocationCount</code>	-
<code>VkDeviceSize</code>	<code>bufferImageGranularity</code>	-
<code>VkDeviceSize</code>	<code>sparseAddressSpaceSize</code>	<code>sparseBinding</code>
<code>uint32_t</code>	<code>maxBoundDescriptorSets</code>	-
<code>uint32_t</code>	<code>maxPerStageDescriptorSamplers</code>	-
<code>uint32_t</code>	<code>maxPerStageDescriptorUniformBuffers</code>	-
<code>uint32_t</code>	<code>maxPerStageDescriptorStorageBuffers</code>	-
<code>uint32_t</code>	<code>maxPerStageDescriptorSampledImages</code>	-
<code>uint32_t</code>	<code>maxPerStageDescriptorStorageImages</code>	-

Type	Limit	Feature
uint32_t	maxPerStageDescriptorInputAttachments	-
uint32_t	maxPerStageResources	-
uint32_t	maxDescriptorSetSamplers	-
uint32_t	maxDescriptorSetUniformBuffers	-
uint32_t	maxDescriptorSetUniformBuffersDynamic	-
uint32_t	maxDescriptorSetStorageBuffers	-
uint32_t	maxDescriptorSetStorageBuffersDynamic	-
uint32_t	maxDescriptorSetSampledImages	-
uint32_t	maxDescriptorSetStorageImages	-
uint32_t	maxDescriptorSetInputAttachments	-
uint32_t	maxVertexInputAttributes	-
uint32_t	maxVertexInputBindings	-
uint32_t	maxVertexInputAttributeOffset	-
uint32_t	maxVertexInputBindingStride	-
uint32_t	maxVertexOutputComponents	-
uint32_t	maxTessellationGenerationLevel	tessellationShader
uint32_t	maxTessellationPatchSize	tessellationShader
uint32_t	maxTessellationControlPerVertexInputComponents	tessellationShader
uint32_t	maxTessellationControlPerVertexOutputComponents	tessellationShader
uint32_t	maxTessellationControlPerPatchOutputComponents	tessellationShader
uint32_t	maxTessellationControlTotalOutputComponents	tessellationShader
uint32_t	maxTessellationEvaluationInputComponents	tessellationShader
uint32_t	maxTessellationEvaluationOutputComponents	tessellationShader
uint32_t	maxGeometryShaderInvocations	geometryShader
uint32_t	maxGeometryInputComponents	geometryShader
uint32_t	maxGeometryOutputComponents	geometryShader
uint32_t	maxGeometryOutputVertices	geometryShader
uint32_t	maxGeometryTotalOutputComponents	geometryShader
uint32_t	maxFragmentInputComponents	-
uint32_t	maxFragmentOutputAttachments	-
uint32_t	maxFragmentDualSrcAttachments	dualSrcBlend
uint32_t	maxFragmentCombinedOutputResources	-
uint32_t	maxComputeSharedMemorySize	-
3 × uint32_t	maxComputeWorkGroupCount	-
uint32_t	maxComputeWorkGroupInvocations	-

Type	Limit	Feature
$3 \times \text{uint32_t}$	maxComputeWorkGroupSize	-
uint32_t	subPixelPrecisionBits	-
uint32_t	subTexelPrecisionBits	-
uint32_t	mipmapPrecisionBits	-
uint32_t	maxDrawIndexedIndexValue	fullDrawIndexUint32
uint32_t	maxDrawIndirectCount	multiDrawIndirect
float	maxSamplerLodBias	-
float	maxSamplerAnisotropy	samplerAnisotropy
uint32_t	maxViewports	multiViewport
$2 \times \text{uint32_t}$	maxViewportDimensions	-
$2 \times \text{float}$	viewportBoundsRange	-
uint32_t	viewportSubPixelBits	-
size_t	minMemoryMapAlignment	-
VkDeviceSize	minTexelBufferOffsetAlignment	-
VkDeviceSize	minUniformBufferOffsetAlignment	-
VkDeviceSize	minStorageBufferOffsetAlignment	-
int32_t	minTexelOffset	-
uint32_t	maxTexelOffset	-
int32_t	minTexelGatherOffset	shaderImageGatherExtended
uint32_t	maxTexelGatherOffset	shaderImageGatherExtended
float	minInterpolationOffset	sampleRateShading
float	maxInterpolationOffset	sampleRateShading
uint32_t	subPixelInterpolationOffsetBits	sampleRateShading
uint32_t	maxFramebufferWidth	-
uint32_t	maxFramebufferHeight	-
uint32_t	maxFramebufferLayers	-
VkSampleCountFlags	framebufferColorSampleCounts	-
VkSampleCountFlags	framebufferDepthSampleCounts	-
VkSampleCountFlags	framebufferStencilSampleCounts	-
VkSampleCountFlags	framebufferNoAttachmentsSampleCounts	-
uint32_t	maxColorAttachments	-

Type	Limit	Feature
VkSampleCountFlags	sampledImageColorSampleCounts	-
VkSampleCountFlags	sampledImageIntegerSampleCounts	-
VkSampleCountFlags	sampledImageDepthSampleCounts	-
VkSampleCountFlags	sampledImageStencilSampleCounts	-
VkSampleCountFlags	storageImageSampleCounts	shaderStorageImageMultisample
uint32_t	maxSampleMaskWords	-
VkBool32	timestampComputeAndGraphics	-
float	timestampPeriod	-
uint32_t	maxClipDistances	shaderClipDistance
uint32_t	maxCullDistances	shaderCullDistance
uint32_t	maxCombinedClipAndCullDistances	shaderCullDistance
uint32_t	discreteQueuePriorities	-
2 × float	pointSizeRange	largePoints
2 × float	lineWidthRange	wideLines
float	pointSizeGranularity	largePoints
float	lineWidthGranularity	wideLines
VkBool32	strictLines	-
VkBool32	standardSampleLocations	-
VkDeviceSize	optimalBufferCopyOffsetAlignment	-
VkDeviceSize	optimalBufferCopyRowPitchAlignment	-
VkDeviceSize	nonCoherentAtomSize	-
uint32_t	maxDiscardRectangles	VK_EXT_discard_rectangles
VkDeviceSize	maxBufferSize	maintenance4
float	primitiveOverestimationSize	VK_EXT_conservative_rasterization
VkBool32	maxExtraPrimitiveOverestimationSize	VK_EXT_conservative_rasterization
float	extraPrimitiveOverestimationSizeGranularity	VK_EXT_conservative_rasterization
VkBool32	degenerateTriangleRasterized	VK_EXT_conservative_rasterization
float	degenerateLinesRasterized	VK_EXT_conservative_rasterization

Type	Limit	Feature
VkBool32	fullyCoveredFragmentShaderInputVariable	VK_EXT_conservative_rasterization
VkBool32	conservativeRasterizationPostDepthCoverage	VK_EXT_conservative_rasterization
uint32_t	maxVertexAttribDivisor	VK_KHR_vertex_attribute_divisor
uint32_t	maxTransformFeedbackStreams	VK_EXT_transform_feedback
uint32_t	maxTransformFeedbackBuffers	VK_EXT_transform_feedback
VkDeviceSize	maxTransformFeedbackBufferSize	VK_EXT_transform_feedback
uint32_t	maxTransformFeedbackStreamDataSize	VK_EXT_transform_feedback
uint32_t	maxTransformFeedbackBufferDataSize	VK_EXT_transform_feedback
uint32_t	maxTransformFeedbackBufferDataStride	VK_EXT_transform_feedback
VkBool32	transformFeedbackQueries	VK_EXT_transform_feedback
VkBool32	transformFeedbackStreamsLinesTriangles	VK_EXT_transform_feedback
VkBool32	transformFeedbackRasterizationStreamSelect	VK_EXT_transform_feedback
VkBool32	transformFeedbackDraw	VK_EXT_transform_feedback
uint32_t	maxGeometryCount	VK_NV_ray_tracing, VK_KHR_acceleration_structure
uint32_t	maxInstanceCount	VK_NV_ray_tracing, VK_KHR_acceleration_structure
uint32_t	shaderGroupHandleSize	VK_NV_ray_tracing, VK_KHR_ray_tracing_pipeline
uint32_t	maxShaderGroupStride	VK_NV_ray_tracing, VK_KHR_ray_tracing_pipeline
uint32_t	shaderGroupBaseAlignment	VK_NV_ray_tracing, VK_KHR_ray_tracing_pipeline
uint32_t	maxPerStageDescriptorAccelerationStructures	VK_KHR_acceleration_structure
uint32_t	maxPerStageDescriptorUpdateAfterBindAccelerationStructures	VK_KHR_acceleration_structure
uint32_t	maxDescriptorSetAccelerationStructures	VK_NV_ray_tracing, VK_KHR_acceleration_structure
uint32_t	maxDescriptorSetUpdateAfterBindAccelerationStructures	VK_KHR_acceleration_structure
uint32_t	minAccelerationStructureScratchOffsetAlignment	VK_KHR_acceleration_structure
uint32_t	maxRayRecursionDepth	VK_KHR_ray_tracing_pipeline
uint32_t	shaderGroupHandleCaptureReplaySize	VK_KHR_ray_tracing_pipeline
uint32_t	maxRayDispatchInvocationCount	VK_KHR_ray_tracing_pipeline

Type	Limit	Feature
uint32_t	shaderGroupHandleAlignment	VK_KHR_ray_tracing_pipeline
uint32_t	maxRayHitAttributeSize	VK_KHR_ray_tracing_pipeline
uint64_t	maxTimelineSemaphoreValueDifference	timelineSemaphore
uint32_t	lineSubPixelPrecisionBits	VK_KHR_line_rasterization, VK_EXT_line_rasterization
uint32_t	maxCustomBorderColorSamplers	VK_EXT_custom_border_color
VkDeviceSize	robustStorageBufferAccessSizeAlignment	VK_EXT_robustness2
VkDeviceSize	robustUniformBufferAccessSizeAlignment	VK_EXT_robustness2
2 × uint32_t	minFragmentShadingRateAttachmentTexelSize	attachmentFragmentShadingRate
2 × uint32_t	maxFragmentShadingRateAttachmentTexelSize	attachmentFragmentShadingRate
uint32_t	maxFragmentShadingRateAttachmentTexelSizeAspectRatio	attachmentFragmentShadingRate
VkBool32	primitiveFragmentShadingRateWithMultipleViewports	primitiveFragmentShadingRate
VkBool32	layeredShadingRateAttachments	attachmentFragmentShadingRate
VkBool32	fragmentShadingRateNonTrivialCombinerOps	pipelineFragmentShadingRate
2 × uint32_t	maxFragmentSize	pipelineFragmentShadingRate
uint32_t	maxFragmentSizeAspectRatio	pipelineFragmentShadingRate
uint32_t	maxFragmentShadingRateCoverageSamples	pipelineFragmentShadingRate
VkSampleCountFlagsBits	maxFragmentShadingRateRasterizationSamples	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithShaderDepthStencilWrites	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithSampleMask	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithShaderSampleMask	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithConservativeRasterization	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithFragmentShaderInterlock	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateWithCustomSampleLocations	pipelineFragmentShadingRate
VkBool32	fragmentShadingRateStrictMultiplyCombiner	pipelineFragmentShadingRate
VkBool32	triStripVertexOrderIndependentOfProvokingVertex	-
VkBool32	dynamicPrimitiveTopologyUnrestricted	VK_EXT_extended_dynamic_state3
uint32_t	maxOpacity2StateSubdivisionLevel	VK_EXT_opacity_micromap
uint32_t	maxOpacity4StateSubdivisionLevel	VK_EXT_opacity_micromap
uint32_t	maxIndirectShaderObjectCount	shaderObject

Type	Limit	Feature
VkBool32	separateDepthStencilAttachmentAccess	maintenance7
uint32_t	maxDescriptorSetTotalUniformBuffersDynamic	maintenance7
uint32_t	maxDescriptorSetTotalStorageBuffersDynamic	maintenance7
uint32_t	maxDescriptorSetTotalBuffersDynamic	maintenance7
uint32_t	maxDescriptorSetUpdateAfterBindTotalUniformBuffersDynamic	maintenance7
uint32_t	maxDescriptorSetUpdateAfterBindTotalStorageBuffersDynamic	maintenance7
uint32_t	maxDescriptorSetUpdateAfterBindTotalBuffersDynamic	maintenance7

Table 59. Required Limits

Limit	Unsupported Limit	Supported Limit	Limit Type ¹
maxImageDimension1D	-	4096	min
maxImageDimension2D	-	4096	min
maxImageDimension3D	-	256	min
maxImageDimensionCube	-	4096	min
maxImageArrayLayers	-	256	min
maxTexelBufferElements	-	65536	min
maxUniformBufferRange	-	16384	min
maxStorageBufferRange	-	2 ²⁷	min
maxPushConstantsSize	-	128	min
maxMemoryAllocationCount	-	4096	min
maxSamplerAllocationCount	-	4000	min
bufferImageGranularity	-	131072	max
sparseAddressSpaceSize	0	2 ³¹	min
maxBoundDescriptorSets	-	4	min
maxPerStageDescriptorSamplers	-	16	min
maxPerStageDescriptorUniformBuffers	-	12	min
maxPerStageDescriptorStorageBuffers	-	4	min
maxPerStageDescriptorSampledImages	-	16	min
maxPerStageDescriptorStorageImages	-	4	min
maxPerStageDescriptorInputAttachments	-	4	min
maxPerStageResources	-	128 ²	min

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
maxDescriptorSetSamplers	-	96 ⁸	min, $n \times$ PerStage
maxDescriptorSetUniformBuffers	-	72 ⁸	min, $n \times$ PerStage
maxDescriptorSetUniformBuffersDynamic	-	8	min
maxDescriptorSetStorageBuffers	-	24 ⁸	min, $n \times$ PerStage
maxDescriptorSetStorageBuffersDynamic	-	4	min
maxDescriptorSetTotalUniformBuffersDynamic	-	maxDescriptorSetUn iformBuffersDynami c	min
maxDescriptorSetTotalStorageBuffersDynamic	-	maxDescriptorSetSt orageBuffersDynami c	min
maxDescriptorSetTotalBuffersDynamic	-	maxDescriptorSetUn iformBuffersDynami c + maxDescriptorSetSt orageBuffersDynami c	min
maxDescriptorSetSampledImages	-	96 ⁸	min, $n \times$ PerStage
maxDescriptorSetStorageImages	-	24 ⁸	min, $n \times$ PerStage
maxDescriptorSetInputAttachments	-	4	min
maxVertexInputAttributes	-	16	min
maxVertexInputBindings	-	16 ¹⁰	min
maxVertexInputAttributeOffset	-	2047	min
maxVertexInputBindingStride	-	2048	min
maxVertexOutputComponents	-	64	min
maxTessellationGenerationLevel	0	64	min
maxTessellationPatchSize	0	32	min
maxTessellationControlPerVertexInputComponents	0	64	min
maxTessellationControlPerVertexOutputComponents	0	64	min
maxTessellationControlPerPatchOutputComponents	0	120	min
maxTessellationControlTotalOutputComponents	0	2048	min
maxTessellationEvaluationInputComponents	0	64	min
maxTessellationEvaluationOutputComponents	0	64	min

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
maxGeometryShaderInvocations	0	32	min
maxGeometryInputComponents	0	64	min
maxGeometryOutputComponents	0	64	min
maxGeometryOutputVertices	0	256	min
maxGeometryTotalOutputComponents	0	1024	min
maxFragmentInputComponents	-	64	min
maxFragmentOutputAttachments	-	4	min
maxFragmentDualSrcAttachments	0	1	min
maxFragmentCombinedOutputResources	-	4	min
maxComputeSharedMemorySize	-	16384	min
maxComputeWorkGroupCount	-	(65535,65535,65535)	min
maxComputeWorkGroupInvocations	-	128	min
maxComputeWorkGroupSize	-	(128,128,64)	min
subPixelPrecisionBits	-	4	min
subTexelPrecisionBits	-	4	min
mipmapPrecisionBits	-	4	min
maxDrawIndexedIndexValue	$2^{24}-1$	$2^{32}-1$	min
maxDrawIndirectCount	1	$2^{16}-1$	min
maxSamplerLodBias	-	2	min
maxSamplerAnisotropy	1	16	min
maxViewports	1	16	min
maxViewportDimensions	-	(4096,4096) ³	min
viewportBoundsRange	-	(-8192,8191) ⁴	(max,min)
viewportSubPixelBits	-	0	min
minMemoryMapAlignment	-	64	min
minTexelBufferOffsetAlignment	-	256	max
minUniformBufferOffsetAlignment	-	256	max
minStorageBufferOffsetAlignment	-	256	max
minTexelOffset	-	-8	max
maxTexelOffset	-	7	min
minTexelGatherOffset	0	-8	max
maxTexelGatherOffset	0	7	min

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
minInterpolationOffset	0.0	-0.5 ⁵	max
maxInterpolationOffset	0.0	0.5 - (1 ULP) ⁵	min
subPixelInterpolationOffsetBits	0	4 ⁵	min
maxFramebufferWidth	-	4096	min
maxFramebufferHeight	-	4096	min
maxFramebufferLayers	-	256	min
framebufferColorSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
framebufferDepthSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
framebufferStencilSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
framebufferNoAttachmentsSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
maxColorAttachments	-	4	min
sampledImageColorSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
sampledImageIntegerSampleCounts	-	VK_SAMPLE_COUNT_1_BIT	min
sampledImageDepthSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
sampledImageStencilSampleCounts	-	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min
storageImageSampleCounts	VK_SAMPLE_COUNT_1_BIT	(VK_SAMPLE_COUNT_1_BIT VK_SAMPLE_COUNT_4_BIT)	min

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
maxSampleMaskWords	-	1	min
timestampComputeAndGraphics	-	-	implementatio n-dependent
timestampPeriod	-	-	duration
maxClipDistances	0	8	min
maxCullDistances	0	8	min
maxCombinedClipAndCullDistances	0	8	min
discreteQueuePriorities	-	2	min
pointSizeRange	(1.0,1.0)	(1.0,64.0 - ULP) ⁶	(max,min)
lineWidthRange	(1.0,1.0)	(1.0,8.0 - ULP) ⁷	(max,min)
pointSizeGranularity	0.0	1.0 ⁶	max, fixed point increment
lineWidthGranularity	0.0	1.0 ⁷	max, fixed point increment
strictLines	-	-	implementatio n-dependent
standardSampleLocations	-	-	implementatio n-dependent
optimalBufferCopyOffsetAlignment	-	-	recommendati on
optimalBufferCopyRowPitchAlignment	-	-	recommendati on
nonCoherentAtomSize	-	256	max
maxPushDescriptors	-	32	min
maxMultiviewViewCount	-	6	min
maxMultiviewInstanceIndex	-	2 ²⁷ -1	min
maxDiscardRectangles	0	4	min
sampleLocationSampleCounts	-	VK_SAMPLE_COUNT_4_ BIT	min
maxSampleLocationGridSize	-	(1,1)	min
sampleLocationCoordinateRange	-	(0.0, 0.9375)	(max,min)
sampleLocationSubPixelBits	-	4	min
variableSampleLocations	-	false	implementatio n-dependent

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
nativeUnalignedPerformance	-	false	implementatio n-dependent
minImportedHostPointerAlignment	-	65536	max
advancedBlendMaxColorAttachments	-	1	min
advancedBlendIndependentBlend	-	false	implementatio n-dependent
advancedBlendNonPremultipliedSrcColor	-	false	implementatio n-dependent
advancedBlendNonPremultipliedDstColor	-	false	implementatio n-dependent
advancedBlendCorrelatedOverlap	-	false	implementatio n-dependent
advancedBlendAllOperations	-	false	implementatio n-dependent
maxPerSetDescriptors	-	1024	min
maxMemoryAllocationSize	-	2 ³⁰	min
maxBufferSize	-	2 ³⁰	min
primitiveOverestimationSize	-	0.0	min
maxExtraPrimitiveOverestimationSize	-	0.0	min
extraPrimitiveOverestimationSizeGranularity	-	0.0	min
primitiveUnderestimation	-	false	implementatio n-dependent
conservativePointAndLineRasterization	-	false	implementatio n-dependent
degenerateTrianglesRasterized	-	false	implementatio n-dependent
degenerateLinesRasterized	-	false	implementatio n-dependent
fullyCoveredFragmentShaderInputVariable	-	false	implementatio n-dependent
conservativeRasterizationPostDepthCoverage	-	false	implementatio n-dependent
maxVertexAttribDivisor	-	2 ¹⁶ -1	min
maxTransformFeedbackStreams	-	1	min
maxTransformFeedbackBuffers	-	1	min
maxTransformFeedbackBufferSize	-	2 ²⁷	min

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
maxTransformFeedbackStreamDataSize	-	512	min
maxTransformFeedbackBufferDataSize	-	512	min
maxTransformFeedbackBufferDataStride	-	512	min
transformFeedbackQueries	-	false	implementatio n-dependent
transformFeedbackStreamsLinesTriangles	-	false	implementatio n-dependent
transformFeedbackRasterizationStreamSelect	-	false	implementatio n-dependent
transformFeedbackDraw	-	false	implementatio n-dependent
VkPhysicalDeviceRayTracingPipelinePropertiesKHR::shaderGroupHandleSize	-	32	exact
VkPhysicalDeviceRayTracingPipelinePropertiesKHR::maxRayRecursionDepth	-	1	min
maxShaderGroupStride	-	4096	min
shaderGroupBaseAlignment	-	64	max
maxGeometryCount	-	$2^{24}-1$	min
maxInstanceCount	-	$2^{24}-1$	min
maxPrimitiveCount	-	$2^{29}-1$	min
maxPerStageDescriptorAccelerationStructures	-	16	min
maxPerStageDescriptorUpdateAfterBindAccelerationStructures	-	500000 ⁹	min
maxDescriptorSetAccelerationStructures	-	16	min
maxDescriptorSetUpdateAfterBindAccelerationStructures	-	500000 ⁹	min
minAccelerationStructureScratchOffsetAlignment	-	256	max
shaderGroupHandleCaptureReplaySize	-	64	max
maxRayDispatchInvocationCount	-	2^{30}	min
shaderGroupHandleAlignment	-	32	max
maxRayHitAttributeSize	-	32	min
maxTimelineSemaphoreValueDifference	-	$2^{31}-1$	min
lineSubPixelPrecisionBits	-	4	min
maxIndirectSequenceCount	-	2^{20}	min
maxIndirectCommandsTokenCount	-	16	min

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
maxIndirectCommandsTokenOffset	-	2047	min
maxIndirectPipelineCount	-	2 ¹²	min
deviceGeneratedCommandsTransformFeedback	-	false	implementatio n-dependent
deviceGeneratedCommandsMultiDrawIndirectCount	-	false	implementatio n-dependent
maxIndirectShaderObjectCount	0	2 ¹²	implementatio n-dependent
maxIndirectCommandsIndirectStride	-	2048	min
supportedIndirectCommandsInputModes	-	VK_INDIRECT_COMMAN DS_INPUT_MODE_VULK AN_INDEX_BUFFER_EX T	min
supportedIndirectCommandsShaderStages	-	(VK_SHADER_STAGE_C OMPUTE_BIT VK_SHADER_STAGE_VE RTEX_BIT VK_SHADER_STAGE_FR AGMENT_BIT)	min
supportedIndirectCommandsShaderStagesPipelineBi nding	-	0	min
supportedIndirectCommandsShaderStagesShaderBind ing	-	0	min
maxCustomBorderColorSamplers	-	32	min
robustStorageBufferAccessSizeAlignment	-	4	max
robustUniformBufferAccessSizeAlignment	-	256	max
minFragmentShadingRateAttachmentTexelSize	(0,0)	(32,32)	max
maxFragmentShadingRateAttachmentTexelSize	(0,0)	(8,8)	min
maxFragmentShadingRateAttachmentTexelSizeAspect Ratio	0	1	min
primitiveFragmentShadingRateWithMultipleViewpor ts	false	false	implementatio n-dependent
layeredShadingRateAttachments	false	false	implementatio n-dependent
fragmentShadingRateNonTrivialCombinerOps	-	false	implementatio n-dependent
maxFragmentSize	-	(2,2)	min
maxFragmentSizeAspectRatio	-	2	min
maxFragmentShadingRateCoverageSamples	-	16	min

Limit	Unsupport ed Limit	Supported Limit	Limit Type ¹
maxFragmentShadingRateRasterizationSamples	-	VK_SAMPLE_COUNT_4_BIT	min
fragmentShadingRateWithShaderDepthStencilWrites	-	false	implementatio n-dependent
fragmentShadingRateWithSampleMask	-	false	implementatio n-dependent
fragmentShadingRateWithShaderSampleMask	-	false	implementatio n-dependent
fragmentShadingRateWithConservativeRasterization	-	false	implementatio n-dependent
fragmentShadingRateWithFragmentShaderInterlock	-	false	implementatio n-dependent
fragmentShadingRateWithCustomSampleLocations	-	false	implementatio n-dependent
fragmentShadingRateStrictMultiplyCombiner	-	false	implementatio n-dependent
maxMultiDrawCount	-	1024	min
maxCommandBufferNestingLevel	-	1	min
triStripVertexOrderIndependentOfProvokingVertex	-	false	implementatio n-dependent
dynamicPrimitiveTopologyUnrestricted	-	-	implementatio n-dependent
maxOpacity2StateSubdivisionLevel	-	3	min
maxOpacity4StateSubdivisionLevel	-	3	min
minPlacedMemoryMapAlignment	-	65536	max
separateDepthStencilAttachmentAccess	false	-	implementatio n-dependent

1

The **Limit Type** column specifies the limit is either the minimum limit all implementations **must** support, the maximum limit all implementations **must** support, or the exact value all implementations **must** support. For bitmasks a minimum limit is the least bits all implementations **must** set, but they **may** have additional bits set beyond this minimum.

2

The **maxPerStageResources** **must** be at least the smallest of the following:

- the sum of the **maxPerStageDescriptorUniformBuffers**, **maxPerStageDescriptorStorageBuffers**, **maxPerStageDescriptorSampledImages**, **maxPerStageDescriptorStorageImages**, **maxPerStageDescriptorInputAttachments**, **maxColorAttachments** limits, or

- 128.

It **may** not be possible to reach this limit in every stage.

3

See `maxViewportDimensions` for the **required** relationship to other limits.

4

See `viewportBoundsRange` for the **required** relationship to other limits.

5

The values `minInterpolationOffset` and `maxInterpolationOffset` describe the closed interval of supported interpolation offsets: `[minInterpolationOffset, maxInterpolationOffset]`. The ULP is determined by `subPixelInterpolationOffsetBits`. If `subPixelInterpolationOffsetBits` is 4, this provides increments of $(1/2^4) = 0.0625$, and thus the range of supported interpolation offsets would be `[-0.5, 0.4375]`.

6

The point size ULP is determined by `pointSizeGranularity`. If the `pointSizeGranularity` is 0.125, the range of supported point sizes **must** be at least `[1.0, 63.875]`.

7

The line width ULP is determined by `lineWidthGranularity`. If the `lineWidthGranularity` is 0.0625, the range of supported line widths **must** be at least `[1.0, 7.9375]`.

8

The minimum `maxDescriptorSet*` limit is n times the corresponding *specification* minimum `maxPerStageDescriptor*` limit, where n is the number of shader stages supported by the `VkPhysicalDevice`. If all shader stages are supported, $n = 6$ (vertex, tessellation control, tessellation evaluation, geometry, fragment, compute).

10

If the `VK_KHR_portability_subset` extension is enabled, the required minimum value of `maxVertexInputBindings` is 8.

40.2. Additional Multisampling Capabilities

To query additional multisampling capabilities which **may** be supported for a specific sample count, beyond the minimum capabilities described for [Limits](#) above, call:

```
// Provided by VK_EXT_sample_locations
void vkGetPhysicalDeviceMultisamplePropertiesEXT(
    VkPhysicalDevice          physicalDevice,
    VkSampleCountFlagBits     samples,
    VkMultisamplePropertiesEXT* pMultisampleProperties);
```

- `physicalDevice` is the physical device from which to query the additional multisampling

capabilities.

- **samples** is a [VkSampleCountFlagBits](#) value specifying the sample count to query capabilities for.
- **pMultisampleProperties** is a pointer to a [VkMultisamplePropertiesEXT](#) structure in which information about additional multisampling capabilities specific to the sample count is returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceMultisamplePropertiesEXT-physicalDevice-parameter **physicalDevice** **must** be a valid [VkPhysicalDevice](#) handle
- VUID-vkGetPhysicalDeviceMultisamplePropertiesEXT-samples-parameter **samples** **must** be a valid [VkSampleCountFlagBits](#) value
- VUID-vkGetPhysicalDeviceMultisamplePropertiesEXT-pMultisampleProperties-parameter **pMultisampleProperties** **must** be a valid pointer to a [VkMultisamplePropertiesEXT](#) structure

The [VkMultisamplePropertiesEXT](#) structure is defined as

```
// Provided by VK_EXT_sample_locations
typedef struct VkMultisamplePropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkExtent2D         maxSampleLocationGridSize;
} VkMultisamplePropertiesEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **maxSampleLocationGridSize** is the maximum size of the pixel grid in which sample locations **can** vary.

Valid Usage (Implicit)

- VUID-VkMultisamplePropertiesEXT-sType-sType **sType** **must** be [VK_STRUCTURE_TYPE_MULTISAMPLE_PROPERTIES_EXT](#)
- VUID-VkMultisamplePropertiesEXT-pNext-pNext **pNext** **must** be **NULL**

If the sample count for which additional multisampling capabilities are requested using [vkGetPhysicalDeviceMultisamplePropertiesEXT](#) is set in [sampleLocationSampleCounts](#) the **width** and **height** members of [VkMultisamplePropertiesEXT::maxSampleLocationGridSize](#) **must** be greater than or equal to the corresponding members of [maxSampleLocationGridSize](#), respectively, otherwise both members **must** be 0.

Chapter 41. Formats

Supported buffer and image formats **may** vary across implementations. A minimum set of format features are guaranteed, but others **must** be explicitly queried before use to ensure they are supported by the implementation.

The features for the set of formats ([VkFormat](#)) supported by the implementation are queried individually using the [vkGetPhysicalDeviceFormatProperties](#) command.

41.1. Format Definition

The following image formats **can** be passed to, and **may** be returned from Vulkan commands. The memory required to store each format is discussed with that format, and also summarized in the [Representation and Texel Block Size](#) section and the [Compatible formats](#) table.

```
// Provided by VK_VERSION_1_0
typedef enum VkFormat {
    VK_FORMAT_UNDEFINED = 0,
    VK_FORMAT_R4G4_UNORM_PACK8 = 1,
    VK_FORMAT_R4G4B4A4_UNORM_PACK16 = 2,
    VK_FORMAT_B4G4R4A4_UNORM_PACK16 = 3,
    VK_FORMAT_R5G6B5_UNORM_PACK16 = 4,
    VK_FORMAT_B5G6R5_UNORM_PACK16 = 5,
    VK_FORMAT_R5G5B5A1_UNORM_PACK16 = 6,
    VK_FORMAT_B5G5R5A1_UNORM_PACK16 = 7,
    VK_FORMAT_A1R5G5B5_UNORM_PACK16 = 8,
    VK_FORMAT_R8_UNORM = 9,
    VK_FORMAT_R8_SNORM = 10,
    VK_FORMAT_R8_USCALED = 11,
    VK_FORMAT_R8_SSCALED = 12,
    VK_FORMAT_R8_UINT = 13,
    VK_FORMAT_R8_SINT = 14,
    VK_FORMAT_R8_SRGB = 15,
    VK_FORMAT_R8G8_UNORM = 16,
    VK_FORMAT_R8G8_SNORM = 17,
    VK_FORMAT_R8G8_USCALED = 18,
    VK_FORMAT_R8G8_SSCALED = 19,
    VK_FORMAT_R8G8_UINT = 20,
    VK_FORMAT_R8G8_SINT = 21,
    VK_FORMAT_R8G8_SRGB = 22,
    VK_FORMAT_R8G8B8_UNORM = 23,
    VK_FORMAT_R8G8B8_SNORM = 24,
    VK_FORMAT_R8G8B8_USCALED = 25,
    VK_FORMAT_R8G8B8_SSCALED = 26,
    VK_FORMAT_R8G8B8_UINT = 27,
    VK_FORMAT_R8G8B8_SINT = 28,
    VK_FORMAT_R8G8B8_SRGB = 29,
    VK_FORMAT_B8G8R8_UNORM = 30,
    VK_FORMAT_B8G8R8_SNORM = 31,
```

```
VK_FORMAT_B8G8R8_USCALED = 32,  
VK_FORMAT_B8G8R8_SSCALED = 33,  
VK_FORMAT_B8G8R8_UINT = 34,  
VK_FORMAT_B8G8R8_SINT = 35,  
VK_FORMAT_B8G8R8_SRGB = 36,  
VK_FORMAT_R8G8B8A8_UNORM = 37,  
VK_FORMAT_R8G8B8A8_SNORM = 38,  
VK_FORMAT_R8G8B8A8_USCALED = 39,  
VK_FORMAT_R8G8B8A8_SSCALED = 40,  
VK_FORMAT_R8G8B8A8_UINT = 41,  
VK_FORMAT_R8G8B8A8_SINT = 42,  
VK_FORMAT_R8G8B8A8_SRGB = 43,  
VK_FORMAT_B8G8R8A8_UNORM = 44,  
VK_FORMAT_B8G8R8A8_SNORM = 45,  
VK_FORMAT_B8G8R8A8_USCALED = 46,  
VK_FORMAT_B8G8R8A8_SSCALED = 47,  
VK_FORMAT_B8G8R8A8_UINT = 48,  
VK_FORMAT_B8G8R8A8_SINT = 49,  
VK_FORMAT_B8G8R8A8_SRGB = 50,  
VK_FORMAT_A8B8G8R8_UNORM_PACK32 = 51,  
VK_FORMAT_A8B8G8R8_SNORM_PACK32 = 52,  
VK_FORMAT_A8B8G8R8_USCALED_PACK32 = 53,  
VK_FORMAT_A8B8G8R8_SSCALED_PACK32 = 54,  
VK_FORMAT_A8B8G8R8_UINT_PACK32 = 55,  
VK_FORMAT_A8B8G8R8_SINT_PACK32 = 56,  
VK_FORMAT_A8B8G8R8_SRGB_PACK32 = 57,  
VK_FORMAT_A2R10G10B10_UNORM_PACK32 = 58,  
VK_FORMAT_A2R10G10B10_SNORM_PACK32 = 59,  
VK_FORMAT_A2R10G10B10_USCALED_PACK32 = 60,  
VK_FORMAT_A2R10G10B10_SSCALED_PACK32 = 61,  
VK_FORMAT_A2R10G10B10_UINT_PACK32 = 62,  
VK_FORMAT_A2R10G10B10_SINT_PACK32 = 63,  
VK_FORMAT_A2B10G10R10_UNORM_PACK32 = 64,  
VK_FORMAT_A2B10G10R10_SNORM_PACK32 = 65,  
VK_FORMAT_A2B10G10R10_USCALED_PACK32 = 66,  
VK_FORMAT_A2B10G10R10_SSCALED_PACK32 = 67,  
VK_FORMAT_A2B10G10R10_UINT_PACK32 = 68,  
VK_FORMAT_A2B10G10R10_SINT_PACK32 = 69,  
VK_FORMAT_R16_UNORM = 70,  
VK_FORMAT_R16_SNORM = 71,  
VK_FORMAT_R16_USCALED = 72,  
VK_FORMAT_R16_SSCALED = 73,  
VK_FORMAT_R16_UINT = 74,  
VK_FORMAT_R16_SINT = 75,  
VK_FORMAT_R16_SFLOAT = 76,  
VK_FORMAT_R16G16_UNORM = 77,  
VK_FORMAT_R16G16_SNORM = 78,  
VK_FORMAT_R16G16_USCALED = 79,  
VK_FORMAT_R16G16_SSCALED = 80,  
VK_FORMAT_R16G16_UINT = 81,  
VK_FORMAT_R16G16_SINT = 82,
```

```
VK_FORMAT_R16G16_SFLOAT = 83,  
VK_FORMAT_R16G16B16_UNORM = 84,  
VK_FORMAT_R16G16B16_SNORM = 85,  
VK_FORMAT_R16G16B16_USCALED = 86,  
VK_FORMAT_R16G16B16_SSCALED = 87,  
VK_FORMAT_R16G16B16_UINT = 88,  
VK_FORMAT_R16G16B16_SINT = 89,  
VK_FORMAT_R16G16B16_SFLOAT = 90,  
VK_FORMAT_R16G16B16A16_UNORM = 91,  
VK_FORMAT_R16G16B16A16_SNORM = 92,  
VK_FORMAT_R16G16B16A16_USCALED = 93,  
VK_FORMAT_R16G16B16A16_SSCALED = 94,  
VK_FORMAT_R16G16B16A16_UINT = 95,  
VK_FORMAT_R16G16B16A16_SINT = 96,  
VK_FORMAT_R16G16B16A16_SFLOAT = 97,  
VK_FORMAT_R32_UINT = 98,  
VK_FORMAT_R32_SINT = 99,  
VK_FORMAT_R32_SFLOAT = 100,  
VK_FORMAT_R32G32_UINT = 101,  
VK_FORMAT_R32G32_SINT = 102,  
VK_FORMAT_R32G32_SFLOAT = 103,  
VK_FORMAT_R32G32B32_UINT = 104,  
VK_FORMAT_R32G32B32_SINT = 105,  
VK_FORMAT_R32G32B32_SFLOAT = 106,  
VK_FORMAT_R32G32B32A32_UINT = 107,  
VK_FORMAT_R32G32B32A32_SINT = 108,  
VK_FORMAT_R32G32B32A32_SFLOAT = 109,  
VK_FORMAT_R64_UINT = 110,  
VK_FORMAT_R64_SINT = 111,  
VK_FORMAT_R64_SFLOAT = 112,  
VK_FORMAT_R64G64_UINT = 113,  
VK_FORMAT_R64G64_SINT = 114,  
VK_FORMAT_R64G64_SFLOAT = 115,  
VK_FORMAT_R64G64B64_UINT = 116,  
VK_FORMAT_R64G64B64_SINT = 117,  
VK_FORMAT_R64G64B64_SFLOAT = 118,  
VK_FORMAT_R64G64B64A64_UINT = 119,  
VK_FORMAT_R64G64B64A64_SINT = 120,  
VK_FORMAT_R64G64B64A64_SFLOAT = 121,  
VK_FORMAT_B10G11R11_UFLOAT_PACK32 = 122,  
VK_FORMAT_E5B9G9R9_UFLOAT_PACK32 = 123,  
VK_FORMAT_D16_UNORM = 124,  
VK_FORMAT_X8_D24_UNORM_PACK32 = 125,  
VK_FORMAT_D32_SFLOAT = 126,  
VK_FORMAT_S8_UINT = 127,  
VK_FORMAT_D16_UNORM_S8_UINT = 128,  
VK_FORMAT_D24_UNORM_S8_UINT = 129,  
VK_FORMAT_D32_SFLOAT_S8_UINT = 130,  
VK_FORMAT_BC1_RGB_UNORM_BLOCK = 131,  
VK_FORMAT_BC1_RGB_SRGB_BLOCK = 132,  
VK_FORMAT_BC1_RGBA_UNORM_BLOCK = 133,
```

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VK_FORMAT_BC1_RGBA_SRGB_BLOCK = 134,  
VK_FORMAT_BC2_UNORM_BLOCK = 135,  
VK_FORMAT_BC2_SRGB_BLOCK = 136,  
VK_FORMAT_BC3_UNORM_BLOCK = 137,  
VK_FORMAT_BC3_SRGB_BLOCK = 138,  
VK_FORMAT_BC4_UNORM_BLOCK = 139,  
VK_FORMAT_BC4_SNORM_BLOCK = 140,  
VK_FORMAT_BC5_UNORM_BLOCK = 141,  
VK_FORMAT_BC5_SNORM_BLOCK = 142,  
VK_FORMAT_BC6H_UFLOAT_BLOCK = 143,  
VK_FORMAT_BC6H_SFLOAT_BLOCK = 144,  
VK_FORMAT_BC7_UNORM_BLOCK = 145,  
VK_FORMAT_BC7_SRGB_BLOCK = 146,  
VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK = 147,  
VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK = 148,  
VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK = 149,  
VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK = 150,  
VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK = 151,  
VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK = 152,  
VK_FORMAT_EAC_R11_UNORM_BLOCK = 153,  
VK_FORMAT_EAC_R11_SNORM_BLOCK = 154,  
VK_FORMAT_EAC_R11G11_UNORM_BLOCK = 155,  
VK_FORMAT_EAC_R11G11_SNORM_BLOCK = 156,  
VK_FORMAT_ASTC_4x4_UNORM_BLOCK = 157,  
VK_FORMAT_ASTC_4x4_SRGB_BLOCK = 158,  
VK_FORMAT_ASTC_5x4_UNORM_BLOCK = 159,  
VK_FORMAT_ASTC_5x4_SRGB_BLOCK = 160,  
VK_FORMAT_ASTC_5x5_UNORM_BLOCK = 161,  
VK_FORMAT_ASTC_5x5_SRGB_BLOCK = 162,  
VK_FORMAT_ASTC_6x5_UNORM_BLOCK = 163,  
VK_FORMAT_ASTC_6x5_SRGB_BLOCK = 164,  
VK_FORMAT_ASTC_6x6_UNORM_BLOCK = 165,  
VK_FORMAT_ASTC_6x6_SRGB_BLOCK = 166,  
VK_FORMAT_ASTC_8x5_UNORM_BLOCK = 167,  
VK_FORMAT_ASTC_8x5_SRGB_BLOCK = 168,  
VK_FORMAT_ASTC_8x6_UNORM_BLOCK = 169,  
VK_FORMAT_ASTC_8x6_SRGB_BLOCK = 170,  
VK_FORMAT_ASTC_8x8_UNORM_BLOCK = 171,  
VK_FORMAT_ASTC_8x8_SRGB_BLOCK = 172,  
VK_FORMAT_ASTC_10x5_UNORM_BLOCK = 173,  
VK_FORMAT_ASTC_10x5_SRGB_BLOCK = 174,  
VK_FORMAT_ASTC_10x6_UNORM_BLOCK = 175,  
VK_FORMAT_ASTC_10x6_SRGB_BLOCK = 176,  
VK_FORMAT_ASTC_10x8_UNORM_BLOCK = 177,  
VK_FORMAT_ASTC_10x8_SRGB_BLOCK = 178,  
VK_FORMAT_ASTC_10x10_UNORM_BLOCK = 179,  
VK_FORMAT_ASTC_10x10_SRGB_BLOCK = 180,  
VK_FORMAT_ASTC_12x10_UNORM_BLOCK = 181,  
VK_FORMAT_ASTC_12x10_SRGB_BLOCK = 182,  
VK_FORMAT_ASTC_12x12_UNORM_BLOCK = 183,  
VK_FORMAT_ASTC_12x12_SRGB_BLOCK = 184,
```

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VK_FORMAT_G8B8G8R8_422_UNORM = 1000156000,
VK_FORMAT_B8G8R8G8_422_UNORM = 1000156001,
VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM = 1000156002,
VK_FORMAT_G8_B8R8_2PLANE_420_UNORM = 1000156003,
VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM = 1000156004,
VK_FORMAT_G8_B8R8_2PLANE_422_UNORM = 1000156005,
VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM = 1000156006,
VK_FORMAT_R10X6_UNORM_PACK16 = 1000156007,
VK_FORMAT_R10X6G10X6_UNORM_2PACK16 = 1000156008,
VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16 = 1000156009,
VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16 = 1000156010,
VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16 = 1000156011,
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16 = 1000156012,
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16 = 1000156013,
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16 = 1000156014,
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16 = 1000156015,
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16 = 1000156016,
VK_FORMAT_R12X4_UNORM_PACK16 = 1000156017,
VK_FORMAT_R12X4G12X4_UNORM_2PACK16 = 1000156018,
VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16 = 1000156019,
VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16 = 1000156020,
VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16 = 1000156021,
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16 = 1000156022,
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16 = 1000156023,
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16 = 1000156024,
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16 = 1000156025,
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16 = 1000156026,
VK_FORMAT_G16B16G16R16_422_UNORM = 1000156027,
VK_FORMAT_B16G16R16G16_422_UNORM = 1000156028,
VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM = 1000156029,
VK_FORMAT_G16_B16R16_2PLANE_420_UNORM = 1000156030,
VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM = 1000156031,
VK_FORMAT_G16_B16R16_2PLANE_422_UNORM = 1000156032,
VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM = 1000156033,
// Provided by VK_KHR_maintenance5
VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR = 1000470000,
// Provided by VK_KHR_maintenance5
VK_FORMAT_A8_UNORM_KHR = 1000470001,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G8B8G8R8_422_UNORM_KHR = VK_FORMAT_G8B8G8R8_422_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_B8G8R8G8_422_UNORM_KHR = VK_FORMAT_B8G8R8G8_422_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM_KHR = VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G8_B8R8_2PLANE_420_UNORM_KHR = VK_FORMAT_G8_B8R8_2PLANE_420_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM_KHR = VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G8_B8R8_2PLANE_422_UNORM_KHR = VK_FORMAT_G8_B8R8_2PLANE_422_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion

```



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VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM_KHR = VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_R10X6_UNORM_PACK16_KHR = VK_FORMAT_R10X6_UNORM_PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_R10X6G10X6_UNORM_2PACK16_KHR = VK_FORMAT_R10X6G10X6_UNORM_2PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16_KHR =
VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16_KHR =
VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16_KHR =
VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16_KHR =
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16_KHR =
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16_KHR =
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16_KHR =
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16_KHR =
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_R12X4_UNORM_PACK16_KHR = VK_FORMAT_R12X4_UNORM_PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_R12X4G12X4_UNORM_2PACK16_KHR = VK_FORMAT_R12X4G12X4_UNORM_2PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16_KHR =
VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16_KHR =
VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16_KHR =
VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16_KHR =
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16_KHR =
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16_KHR =
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16,

```



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// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16_KHR =
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16_KHR =
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G16B16G16R16_422_UNORM_KHR = VK_FORMAT_G16B16G16R16_422_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_B16G16R16G16_422_UNORM_KHR = VK_FORMAT_B16G16R16G16_422_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM_KHR =
VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G16_B16R16_2PLANE_420_UNORM_KHR = VK_FORMAT_G16_B16R16_2PLANE_420_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM_KHR =
VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G16_B16R16_2PLANE_422_UNORM_KHR = VK_FORMAT_G16_B16R16_2PLANE_422_UNORM,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM_KHR =
VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM,
} VkFormat;

```

- **VK_FORMAT_UNDEFINED** specifies that the format is not specified.
- **VK_FORMAT_R4G4_UNORM_PACK8** specifies a two-component, 8-bit packed unsigned normalized format that has a 4-bit R component in bits 4..7, and a 4-bit G component in bits 0..3.
- **VK_FORMAT_R4G4B4A4_UNORM_PACK16** specifies a four-component, 16-bit packed unsigned normalized format that has a 4-bit R component in bits 12..15, a 4-bit G component in bits 8..11, a 4-bit B component in bits 4..7, and a 4-bit A component in bits 0..3.
- **VK_FORMAT_B4G4R4A4_UNORM_PACK16** specifies a four-component, 16-bit packed unsigned normalized format that has a 4-bit B component in bits 12..15, a 4-bit G component in bits 8..11, a 4-bit R component in bits 4..7, and a 4-bit A component in bits 0..3.
- **VK_FORMAT_R5G6B5_UNORM_PACK16** specifies a three-component, 16-bit packed unsigned normalized format that has a 5-bit R component in bits 11..15, a 6-bit G component in bits 5..10, and a 5-bit B component in bits 0..4.
- **VK_FORMAT_B5G6R5_UNORM_PACK16** specifies a three-component, 16-bit packed unsigned normalized format that has a 5-bit B component in bits 11..15, a 6-bit G component in bits 5..10, and a 5-bit R component in bits 0..4.
- **VK_FORMAT_R5G5B5A1_UNORM_PACK16** specifies a four-component, 16-bit packed unsigned normalized format that has a 5-bit R component in bits 11..15, a 5-bit G component in bits 6..10, a 5-bit B component in bits 1..5, and a 1-bit A component in bit 0.
- **VK_FORMAT_B5G5R5A1_UNORM_PACK16** specifies a four-component, 16-bit packed unsigned normalized format that has a 5-bit B component in bits 11..15, a 5-bit G component in bits 6..10, a 5-bit R component in bits 1..5, and a 1-bit A component in bit 0.

- `VK_FORMAT_A1R5G5B5_UNORM_PACK16` specifies a four-component, 16-bit packed unsigned normalized format that has a 1-bit A component in bit 15, a 5-bit R component in bits 10..14, a 5-bit G component in bits 5..9, and a 5-bit B component in bits 0..4.
- `VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR` specifies a four-component, 16-bit packed unsigned normalized format that has a 1-bit A component in bit 15, a 5-bit B component in bits 10..14, a 5-bit G component in bits 5..9, and a 5-bit R component in bits 0..4.
- `VK_FORMAT_A8_UNORM_KHR` specifies a one-component, 8-bit unsigned normalized format that has a single 8-bit A component.
- `VK_FORMAT_R8_UNORM` specifies a one-component, 8-bit unsigned normalized format that has a single 8-bit R component.
- `VK_FORMAT_R8_SNORM` specifies a one-component, 8-bit signed normalized format that has a single 8-bit R component.
- `VK_FORMAT_R8_USCALED` specifies a one-component, 8-bit unsigned scaled integer format that has a single 8-bit R component.
- `VK_FORMAT_R8_SSCALED` specifies a one-component, 8-bit signed scaled integer format that has a single 8-bit R component.
- `VK_FORMAT_R8_UINT` specifies a one-component, 8-bit unsigned integer format that has a single 8-bit R component.
- `VK_FORMAT_R8_SINT` specifies a one-component, 8-bit signed integer format that has a single 8-bit R component.
- `VK_FORMAT_R8_SRGB` specifies a one-component, 8-bit unsigned normalized format that has a single 8-bit R component stored with sRGB nonlinear encoding.
- `VK_FORMAT_R8G8_UNORM` specifies a two-component, 16-bit unsigned normalized format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- `VK_FORMAT_R8G8_SNORM` specifies a two-component, 16-bit signed normalized format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- `VK_FORMAT_R8G8_USCALED` specifies a two-component, 16-bit unsigned scaled integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- `VK_FORMAT_R8G8_SSCALED` specifies a two-component, 16-bit signed scaled integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- `VK_FORMAT_R8G8_UINT` specifies a two-component, 16-bit unsigned integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- `VK_FORMAT_R8G8_SINT` specifies a two-component, 16-bit signed integer format that has an 8-bit R component in byte 0, and an 8-bit G component in byte 1.
- `VK_FORMAT_R8G8_SRGB` specifies a two-component, 16-bit unsigned normalized format that has an 8-bit R component stored with sRGB nonlinear encoding in byte 0, and an 8-bit G component stored with sRGB nonlinear encoding in byte 1.
- `VK_FORMAT_R8G8B8_UNORM` specifies a three-component, 24-bit unsigned normalized format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- `VK_FORMAT_R8G8B8_SNORM` specifies a three-component, 24-bit signed normalized format that has

an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.

- **VK_FORMAT_R8G8B8_USCALED** specifies a three-component, 24-bit unsigned scaled format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- **VK_FORMAT_R8G8B8_SSCALED** specifies a three-component, 24-bit signed scaled format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- **VK_FORMAT_R8G8B8_UINT** specifies a three-component, 24-bit unsigned integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- **VK_FORMAT_R8G8B8_SINT** specifies a three-component, 24-bit signed integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, and an 8-bit B component in byte 2.
- **VK_FORMAT_R8G8B8_SRGB** specifies a three-component, 24-bit unsigned normalized format that has an 8-bit R component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, and an 8-bit B component stored with sRGB nonlinear encoding in byte 2.
- **VK_FORMAT_B8G8R8_UNORM** specifies a three-component, 24-bit unsigned normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- **VK_FORMAT_B8G8R8_SNORM** specifies a three-component, 24-bit signed normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- **VK_FORMAT_B8G8R8_USCALED** specifies a three-component, 24-bit unsigned scaled format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- **VK_FORMAT_B8G8R8_SSCALED** specifies a three-component, 24-bit signed scaled format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- **VK_FORMAT_B8G8R8_UINT** specifies a three-component, 24-bit unsigned integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- **VK_FORMAT_B8G8R8_SINT** specifies a three-component, 24-bit signed integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, and an 8-bit R component in byte 2.
- **VK_FORMAT_B8G8R8_SRGB** specifies a three-component, 24-bit unsigned normalized format that has an 8-bit B component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, and an 8-bit R component stored with sRGB nonlinear encoding in byte 2.
- **VK_FORMAT_R8G8B8A8_UNORM** specifies a four-component, 32-bit unsigned normalized format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_R8G8B8A8_SNORM** specifies a four-component, 32-bit signed normalized format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_R8G8B8A8_USCALED** specifies a four-component, 32-bit unsigned scaled format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.

- **VK_FORMAT_R8G8B8A8_SSCALED** specifies a four-component, 32-bit signed scaled format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_R8G8B8A8_UINT** specifies a four-component, 32-bit unsigned integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_R8G8B8A8_SINT** specifies a four-component, 32-bit signed integer format that has an 8-bit R component in byte 0, an 8-bit G component in byte 1, an 8-bit B component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_R8G8B8A8_SRGB** specifies a four-component, 32-bit unsigned normalized format that has an 8-bit R component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, an 8-bit B component stored with sRGB nonlinear encoding in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_UNORM** specifies a four-component, 32-bit unsigned normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_SNORM** specifies a four-component, 32-bit signed normalized format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_USCALED** specifies a four-component, 32-bit unsigned scaled format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_SSCALED** specifies a four-component, 32-bit signed scaled format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_UINT** specifies a four-component, 32-bit unsigned integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_SINT** specifies a four-component, 32-bit signed integer format that has an 8-bit B component in byte 0, an 8-bit G component in byte 1, an 8-bit R component in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_B8G8R8A8_SRGB** specifies a four-component, 32-bit unsigned normalized format that has an 8-bit B component stored with sRGB nonlinear encoding in byte 0, an 8-bit G component stored with sRGB nonlinear encoding in byte 1, an 8-bit R component stored with sRGB nonlinear encoding in byte 2, and an 8-bit A component in byte 3.
- **VK_FORMAT_A8B8G8R8_UNORM_PACK32** specifies a four-component, 32-bit packed unsigned normalized format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- **VK_FORMAT_A8B8G8R8_SNORM_PACK32** specifies a four-component, 32-bit packed signed normalized format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- **VK_FORMAT_A8B8G8R8_USCALED_PACK32** specifies a four-component, 32-bit packed unsigned scaled integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23,

an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.

- **VK_FORMAT_A8B8G8R8_SSCALED_PACK32** specifies a four-component, 32-bit packed signed scaled integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- **VK_FORMAT_A8B8G8R8_UINT_PACK32** specifies a four-component, 32-bit packed unsigned integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- **VK_FORMAT_A8B8G8R8_SINT_PACK32** specifies a four-component, 32-bit packed signed integer format that has an 8-bit A component in bits 24..31, an 8-bit B component in bits 16..23, an 8-bit G component in bits 8..15, and an 8-bit R component in bits 0..7.
- **VK_FORMAT_A8B8G8R8_SRGB_PACK32** specifies a four-component, 32-bit packed unsigned normalized format that has an 8-bit A component in bits 24..31, an 8-bit B component stored with sRGB nonlinear encoding in bits 16..23, an 8-bit G component stored with sRGB nonlinear encoding in bits 8..15, and an 8-bit R component stored with sRGB nonlinear encoding in bits 0..7.
- **VK_FORMAT_A2R10G10B10_UNORM_PACK32** specifies a four-component, 32-bit packed unsigned normalized format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- **VK_FORMAT_A2R10G10B10_SNORM_PACK32** specifies a four-component, 32-bit packed signed normalized format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- **VK_FORMAT_A2R10G10B10_USCALED_PACK32** specifies a four-component, 32-bit packed unsigned scaled integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- **VK_FORMAT_A2R10G10B10_SSCALED_PACK32** specifies a four-component, 32-bit packed signed scaled integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- **VK_FORMAT_A2R10G10B10_UINT_PACK32** specifies a four-component, 32-bit packed unsigned integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- **VK_FORMAT_A2R10G10B10_SINT_PACK32** specifies a four-component, 32-bit packed signed integer format that has a 2-bit A component in bits 30..31, a 10-bit R component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit B component in bits 0..9.
- **VK_FORMAT_A2B10G10R10_UNORM_PACK32** specifies a four-component, 32-bit packed unsigned normalized format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- **VK_FORMAT_A2B10G10R10_SNORM_PACK32** specifies a four-component, 32-bit packed signed normalized format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- **VK_FORMAT_A2B10G10R10_USCALED_PACK32** specifies a four-component, 32-bit packed unsigned scaled integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- **VK_FORMAT_A2B10G10R10_SSCALED_PACK32** specifies a four-component, 32-bit packed signed scaled

integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.

- **VK_FORMAT_A2B10G10R10_UINT_PACK32** specifies a four-component, 32-bit packed unsigned integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- **VK_FORMAT_A2B10G10R10_SINT_PACK32** specifies a four-component, 32-bit packed signed integer format that has a 2-bit A component in bits 30..31, a 10-bit B component in bits 20..29, a 10-bit G component in bits 10..19, and a 10-bit R component in bits 0..9.
- **VK_FORMAT_R16_UNORM** specifies a one-component, 16-bit unsigned normalized format that has a single 16-bit R component.
- **VK_FORMAT_R16_SNORM** specifies a one-component, 16-bit signed normalized format that has a single 16-bit R component.
- **VK_FORMAT_R16_USCALED** specifies a one-component, 16-bit unsigned scaled integer format that has a single 16-bit R component.
- **VK_FORMAT_R16_SSCALED** specifies a one-component, 16-bit signed scaled integer format that has a single 16-bit R component.
- **VK_FORMAT_R16_UINT** specifies a one-component, 16-bit unsigned integer format that has a single 16-bit R component.
- **VK_FORMAT_R16_SINT** specifies a one-component, 16-bit signed integer format that has a single 16-bit R component.
- **VK_FORMAT_R16_SFLOAT** specifies a one-component, 16-bit signed floating-point format that has a single 16-bit R component.
- **VK_FORMAT_R16G16_UNORM** specifies a two-component, 32-bit unsigned normalized format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_SNORM** specifies a two-component, 32-bit signed normalized format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_USCALED** specifies a two-component, 32-bit unsigned scaled integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_SSCALED** specifies a two-component, 32-bit signed scaled integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_UINT** specifies a two-component, 32-bit unsigned integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_SINT** specifies a two-component, 32-bit signed integer format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16_SFLOAT** specifies a two-component, 32-bit signed floating-point format that has a 16-bit R component in bytes 0..1, and a 16-bit G component in bytes 2..3.
- **VK_FORMAT_R16G16B16_UNORM** specifies a three-component, 48-bit unsigned normalized format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16_SNORM** specifies a three-component, 48-bit signed normalized format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.

component in bytes 4..5.

- **VK_FORMAT_R16G16B16_USCALED** specifies a three-component, 48-bit unsigned scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16_SSCALED** specifies a three-component, 48-bit signed scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16_UINT** specifies a three-component, 48-bit unsigned integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16_SINT** specifies a three-component, 48-bit signed integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16_SFLOAT** specifies a three-component, 48-bit signed floating-point format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, and a 16-bit B component in bytes 4..5.
- **VK_FORMAT_R16G16B16A16_UNORM** specifies a four-component, 64-bit unsigned normalized format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R16G16B16A16_SNORM** specifies a four-component, 64-bit signed normalized format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R16G16B16A16_USCALED** specifies a four-component, 64-bit unsigned scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R16G16B16A16_SSCALED** specifies a four-component, 64-bit signed scaled integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R16G16B16A16_UINT** specifies a four-component, 64-bit unsigned integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R16G16B16A16_SINT** specifies a four-component, 64-bit signed integer format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R16G16B16A16_SFLOAT** specifies a four-component, 64-bit signed floating-point format that has a 16-bit R component in bytes 0..1, a 16-bit G component in bytes 2..3, a 16-bit B component in bytes 4..5, and a 16-bit A component in bytes 6..7.
- **VK_FORMAT_R32_UINT** specifies a one-component, 32-bit unsigned integer format that has a single 32-bit R component.
- **VK_FORMAT_R32_SINT** specifies a one-component, 32-bit signed integer format that has a single 32-bit R component.
- **VK_FORMAT_R32_SFLOAT** specifies a one-component, 32-bit signed floating-point format that has a

single 32-bit R component.

- **VK_FORMAT_R32G32_UINT** specifies a two-component, 64-bit unsigned integer format that has a 32-bit R component in bytes 0..3, and a 32-bit G component in bytes 4..7.
- **VK_FORMAT_R32G32_SINT** specifies a two-component, 64-bit signed integer format that has a 32-bit R component in bytes 0..3, and a 32-bit G component in bytes 4..7.
- **VK_FORMAT_R32G32_SFLOAT** specifies a two-component, 64-bit signed floating-point format that has a 32-bit R component in bytes 0..3, and a 32-bit G component in bytes 4..7.
- **VK_FORMAT_R32G32B32_UINT** specifies a three-component, 96-bit unsigned integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, and a 32-bit B component in bytes 8..11.
- **VK_FORMAT_R32G32B32_SINT** specifies a three-component, 96-bit signed integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, and a 32-bit B component in bytes 8..11.
- **VK_FORMAT_R32G32B32_SFLOAT** specifies a three-component, 96-bit signed floating-point format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, and a 32-bit B component in bytes 8..11.
- **VK_FORMAT_R32G32B32A32_UINT** specifies a four-component, 128-bit unsigned integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, a 32-bit B component in bytes 8..11, and a 32-bit A component in bytes 12..15.
- **VK_FORMAT_R32G32B32A32_SINT** specifies a four-component, 128-bit signed integer format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, a 32-bit B component in bytes 8..11, and a 32-bit A component in bytes 12..15.
- **VK_FORMAT_R32G32B32A32_SFLOAT** specifies a four-component, 128-bit signed floating-point format that has a 32-bit R component in bytes 0..3, a 32-bit G component in bytes 4..7, a 32-bit B component in bytes 8..11, and a 32-bit A component in bytes 12..15.
- **VK_FORMAT_R64_UINT** specifies a one-component, 64-bit unsigned integer format that has a single 64-bit R component.
- **VK_FORMAT_R64_SINT** specifies a one-component, 64-bit signed integer format that has a single 64-bit R component.
- **VK_FORMAT_R64_SFLOAT** specifies a one-component, 64-bit signed floating-point format that has a single 64-bit R component.
- **VK_FORMAT_R64G64_UINT** specifies a two-component, 128-bit unsigned integer format that has a 64-bit R component in bytes 0..7, and a 64-bit G component in bytes 8..15.
- **VK_FORMAT_R64G64_SINT** specifies a two-component, 128-bit signed integer format that has a 64-bit R component in bytes 0..7, and a 64-bit G component in bytes 8..15.
- **VK_FORMAT_R64G64_SFLOAT** specifies a two-component, 128-bit signed floating-point format that has a 64-bit R component in bytes 0..7, and a 64-bit G component in bytes 8..15.
- **VK_FORMAT_R64G64B64_UINT** specifies a three-component, 192-bit unsigned integer format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, and a 64-bit B component in bytes 16..23.
- **VK_FORMAT_R64G64B64_SINT** specifies a three-component, 192-bit signed integer format that has a

64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, and a 64-bit B component in bytes 16..23.

- **VK_FORMAT_R64G64B64_SFLOAT** specifies a three-component, 192-bit signed floating-point format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, and a 64-bit B component in bytes 16..23.
- **VK_FORMAT_R64G64B64A64_UINT** specifies a four-component, 256-bit unsigned integer format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, a 64-bit B component in bytes 16..23, and a 64-bit A component in bytes 24..31.
- **VK_FORMAT_R64G64B64A64_SINT** specifies a four-component, 256-bit signed integer format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, a 64-bit B component in bytes 16..23, and a 64-bit A component in bytes 24..31.
- **VK_FORMAT_R64G64B64A64_SFLOAT** specifies a four-component, 256-bit signed floating-point format that has a 64-bit R component in bytes 0..7, a 64-bit G component in bytes 8..15, a 64-bit B component in bytes 16..23, and a 64-bit A component in bytes 24..31.
- **VK_FORMAT_B10G11R11_UFLOAT_PACK32** specifies a three-component, 32-bit packed unsigned floating-point format that has a 10-bit B component in bits 22..31, an 11-bit G component in bits 11..21, an 11-bit R component in bits 0..10. See [Unsigned 10-Bit Floating-Point Numbers](#) and [Unsigned 11-Bit Floating-Point Numbers](#).
- **VK_FORMAT_E5B9G9R9_UFLOAT_PACK32** specifies a three-component, 32-bit packed unsigned floating-point format that has a 5-bit shared exponent in bits 27..31, a 9-bit B component mantissa in bits 18..26, a 9-bit G component mantissa in bits 9..17, and a 9-bit R component mantissa in bits 0..8.
- **VK_FORMAT_D16_UNORM** specifies a one-component, 16-bit unsigned normalized format that has a single 16-bit depth component.
- **VK_FORMAT_X8_D24_UNORM_PACK32** specifies a two-component, 32-bit format that has 24 unsigned normalized bits in the depth component and, **optionally**, 8 bits that are unused.
- **VK_FORMAT_D32_SFLOAT** specifies a one-component, 32-bit signed floating-point format that has 32 bits in the depth component.
- **VK_FORMAT_S8_UINT** specifies a one-component, 8-bit unsigned integer format that has 8 bits in the stencil component.
- **VK_FORMAT_D16_UNORM_S8_UINT** specifies a two-component, 24-bit format that has 16 unsigned normalized bits in the depth component and 8 unsigned integer bits in the stencil component.
- **VK_FORMAT_D24_UNORM_S8_UINT** specifies a two-component, 32-bit packed format that has 8 unsigned integer bits in the stencil component, and 24 unsigned normalized bits in the depth component.
- **VK_FORMAT_D32_SFLOAT_S8_UINT** specifies a two-component format that has 32 signed float bits in the depth component and 8 unsigned integer bits in the stencil component. There are **optionally** 24 bits that are unused.
- **VK_FORMAT_BC1_RGB_UNORM_BLOCK** specifies a three-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data. This format has no alpha and is considered opaque.
- **VK_FORMAT_BC1_RGB_SRGB_BLOCK** specifies a three-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel

data with sRGB nonlinear encoding. This format has no alpha and is considered opaque.

- **VK_FORMAT_BC1_RGBA_UNORM_BLOCK** specifies a four-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data, and provides 1 bit of alpha.
- **VK_FORMAT_BC1_RGBA_SRGB_BLOCK** specifies a four-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data with sRGB nonlinear encoding, and provides 1 bit of alpha.
- **VK_FORMAT_BC2_UNORM_BLOCK** specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values.
- **VK_FORMAT_BC2_SRGB_BLOCK** specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values with sRGB nonlinear encoding.
- **VK_FORMAT_BC3_UNORM_BLOCK** specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values.
- **VK_FORMAT_BC3_SRGB_BLOCK** specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values with sRGB nonlinear encoding.
- **VK_FORMAT_BC4_UNORM_BLOCK** specifies a one-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized red texel data.
- **VK_FORMAT_BC4_SNORM_BLOCK** specifies a one-component, block-compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of signed normalized red texel data.
- **VK_FORMAT_BC5_UNORM_BLOCK** specifies a two-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- **VK_FORMAT_BC5_SNORM_BLOCK** specifies a two-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of signed normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- **VK_FORMAT_BC6H_UFLOAT_BLOCK** specifies a three-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned floating-point RGB texel data.
- **VK_FORMAT_BC6H_SFLOAT_BLOCK** specifies a three-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of signed floating-point RGB texel data.
- **VK_FORMAT_BC7_UNORM_BLOCK** specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_BC7_SRGB_BLOCK** specifies a four-component, block-compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.

- **VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK** specifies a three-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data. This format has no alpha and is considered opaque.
- **VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK** specifies a three-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data with sRGB nonlinear encoding. This format has no alpha and is considered opaque.
- **VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK** specifies a four-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data, and provides 1 bit of alpha.
- **VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK** specifies a four-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGB texel data with sRGB nonlinear encoding, and provides 1 bit of alpha.
- **VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK** specifies a four-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values.
- **VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK** specifies a four-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with the first 64 bits encoding alpha values followed by 64 bits encoding RGB values with sRGB nonlinear encoding applied.
- **VK_FORMAT_EAC_R11_UNORM_BLOCK** specifies a one-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized red texel data.
- **VK_FORMAT_EAC_R11_SNORM_BLOCK** specifies a one-component, ETC2 compressed format where each 64-bit compressed texel block encodes a 4×4 rectangle of signed normalized red texel data.
- **VK_FORMAT_EAC_R11G11_UNORM_BLOCK** specifies a two-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- **VK_FORMAT_EAC_R11G11_SNORM_BLOCK** specifies a two-component, ETC2 compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of signed normalized RG texel data with the first 64 bits encoding red values followed by 64 bits encoding green values.
- **VK_FORMAT_ASTC_4x4_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_4x4_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 4×4 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_5x4_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×4 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_5x4_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×4 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.

- **VK_FORMAT_ASTC_5x5_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_5x5_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 5×5 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_6x5_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×5 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_6x5_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×5 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_6x6_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_6x6_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 6×6 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_8x5_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×5 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_8x5_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×5 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_8x6_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×6 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_8x6_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×6 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_8x8_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×8 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_8x8_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes an 8×8 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_10x5_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×5 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_10x5_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×5 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.

- **VK_FORMAT_ASTC_10x6_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×6 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_10x6_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×6 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_10x8_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×8 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_10x8_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×8 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_10x10_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×10 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_10x10_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 10×10 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_12x10_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×10 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_12x10_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×10 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_ASTC_12x12_UNORM_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×12 rectangle of unsigned normalized RGBA texel data.
- **VK_FORMAT_ASTC_12x12_SRGB_BLOCK** specifies a four-component, ASTC compressed format where each 128-bit compressed texel block encodes a 12×12 rectangle of unsigned normalized RGBA texel data with sRGB nonlinear encoding applied to the RGB components.
- **VK_FORMAT_G8B8G8R8_422_UNORM** specifies a four-component, 32-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each *i* coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has an 8-bit G component for the even *i* coordinate in byte 0, an 8-bit B component in byte 1, an 8-bit G component for the odd *i* coordinate in byte 2, and an 8-bit R component in byte 3. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- **VK_FORMAT_B8G8R8G8_422_UNORM** specifies a four-component, 32-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each *i* coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the

image. This format has an 8-bit B component in byte 0, an 8-bit G component for the even i coordinate in byte 1, an 8-bit R component in byte 2, and an 8-bit G component for the odd i coordinate in byte 3. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.

- **VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM** specifies an unsigned normalized *multi-planar format* that has an 8-bit G component in plane 0, an 8-bit B component in plane 1, and an 8-bit R component in plane 2. The horizontal and vertical dimensions of the R and B planes are halved relative to the image dimensions, and each R and B component is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, **VK_IMAGE_ASPECT_PLANE_1_BIT** for the B plane, and **VK_IMAGE_ASPECT_PLANE_2_BIT** for the R plane. This format only supports images with a width and height that is a multiple of two.
- **VK_FORMAT_G8_B8R8_2PLANE_420_UNORM** specifies an unsigned normalized *multi-planar format* that has an 8-bit G component in plane 0, and a two-component, 16-bit BR plane 1 consisting of an 8-bit B component in byte 0 and an 8-bit R component in byte 1. The horizontal and vertical dimensions of the BR plane are halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, and **VK_IMAGE_ASPECT_PLANE_1_BIT** for the BR plane. This format only supports images with a width and height that is a multiple of two.
- **VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM** specifies an unsigned normalized *multi-planar format* that has an 8-bit G component in plane 0, an 8-bit B component in plane 1, and an 8-bit R component in plane 2. The horizontal dimension of the R and B plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, **VK_IMAGE_ASPECT_PLANE_1_BIT** for the B plane, and **VK_IMAGE_ASPECT_PLANE_2_BIT** for the R plane. This format only supports images with a width that is a multiple of two.
- **VK_FORMAT_G8_B8R8_2PLANE_422_UNORM** specifies an unsigned normalized *multi-planar format* that has an 8-bit G component in plane 0, and a two-component, 16-bit BR plane 1 consisting of an 8-bit B component in byte 0 and an 8-bit R component in byte 1. The horizontal dimension of the BR plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, and **VK_IMAGE_ASPECT_PLANE_1_BIT** for the BR plane. This format only supports images with a width that is a multiple of two.
- **VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM** specifies an unsigned normalized *multi-planar format* that has an 8-bit G component in plane 0, an 8-bit B component in plane 1, and an 8-bit R component in plane 2. Each plane has the same dimensions and each R, G and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, **VK_IMAGE_ASPECT_PLANE_1_BIT** for the B plane, and **VK_IMAGE_ASPECT_PLANE_2_BIT** for the R plane.

- **VK_FORMAT_R10X6_UNORM_PACK16** specifies a one-component, 16-bit unsigned normalized format that has a single 10-bit R component in the top 10 bits of a 16-bit word, with the bottom 6 bits unused.
- **VK_FORMAT_R10X6G10X6_UNORM_2PACK16** specifies a two-component, 32-bit unsigned normalized format that has a 10-bit R component in the top 10 bits of the word in bytes 0..1, and a 10-bit G component in the top 10 bits of the word in bytes 2..3, with the bottom 6 bits of each word unused.
- **VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16** specifies a four-component, 64-bit unsigned normalized format that has a 10-bit R component in the top 10 bits of the word in bytes 0..1, a 10-bit G component in the top 10 bits of the word in bytes 2..3, a 10-bit B component in the top 10 bits of the word in bytes 4..5, and a 10-bit A component in the top 10 bits of the word in bytes 6..7, with the bottom 6 bits of each word unused.
- **VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16** specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 10-bit G component for the even i coordinate in the top 10 bits of the word in bytes 0..1, a 10-bit B component in the top 10 bits of the word in bytes 2..3, a 10-bit G component for the odd i coordinate in the top 10 bits of the word in bytes 4..5, and a 10-bit R component in the top 10 bits of the word in bytes 6..7, with the bottom 6 bits of each word unused. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- **VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16** specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 10-bit B component in the top 10 bits of the word in bytes 0..1, a 10-bit G component for the even i coordinate in the top 10 bits of the word in bytes 2..3, a 10-bit R component in the top 10 bits of the word in bytes 4..5, and a 10-bit G component for the odd i coordinate in the top 10 bits of the word in bytes 6..7, with the bottom 6 bits of each word unused. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- **VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16** specifies an unsigned normalized *multi-planar format* that has a 10-bit G component in the top 10 bits of each 16-bit word of plane 0, a 10-bit B component in the top 10 bits of each 16-bit word of plane 1, and a 10-bit R component in the top 10 bits of each 16-bit word of plane 2, with the bottom 6 bits of each word unused. The horizontal and vertical dimensions of the R and B planes are halved relative to the image dimensions, and each R and B component is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, **VK_IMAGE_ASPECT_PLANE_1_BIT** for the B plane, and **VK_IMAGE_ASPECT_PLANE_2_BIT** for the R plane. This format only supports images with a width and height that is a multiple of two.

- **VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16** specifies an unsigned normalized *multi-planar format* that has a 10-bit G component in the top 10 bits of each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 10-bit B component in the top 10 bits of the word in bytes 0..1, and a 10-bit R component in the top 10 bits of the word in bytes 2..3, with the bottom 6 bits of each word unused. The horizontal and vertical dimensions of the BR plane are halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, and **VK_IMAGE_ASPECT_PLANE_1_BIT** for the BR plane. This format only supports images with a width and height that is a multiple of two.
- **VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16** specifies an unsigned normalized *multi-planar format* that has a 10-bit G component in the top 10 bits of each 16-bit word of plane 0, a 10-bit B component in the top 10 bits of each 16-bit word of plane 1, and a 10-bit R component in the top 10 bits of each 16-bit word of plane 2, with the bottom 6 bits of each word unused. The horizontal dimension of the R and B plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, **VK_IMAGE_ASPECT_PLANE_1_BIT** for the B plane, and **VK_IMAGE_ASPECT_PLANE_2_BIT** for the R plane. This format only supports images with a width that is a multiple of two.
- **VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16** specifies an unsigned normalized *multi-planar format* that has a 10-bit G component in the top 10 bits of each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 10-bit B component in the top 10 bits of the word in bytes 0..1, and a 10-bit R component in the top 10 bits of the word in bytes 2..3, with the bottom 6 bits of each word unused. The horizontal dimension of the BR plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, and **VK_IMAGE_ASPECT_PLANE_1_BIT** for the BR plane. This format only supports images with a width that is a multiple of two.
- **VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16** specifies an unsigned normalized *multi-planar format* that has a 10-bit G component in the top 10 bits of each 16-bit word of plane 0, a 10-bit B component in the top 10 bits of each 16-bit word of plane 1, and a 10-bit R component in the top 10 bits of each 16-bit word of plane 2, with the bottom 6 bits of each word unused. Each plane has the same dimensions and each R, G and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, **VK_IMAGE_ASPECT_PLANE_1_BIT** for the B plane, and **VK_IMAGE_ASPECT_PLANE_2_BIT** for the R plane.
- **VK_FORMAT_R12X4_UNORM_PACK16** specifies a one-component, 16-bit unsigned normalized format that has a single 12-bit R component in the top 12 bits of a 16-bit word, with the bottom 4 bits unused.
- **VK_FORMAT_R12X4G12X4_UNORM_2PACK16** specifies a two-component, 32-bit unsigned normalized format that has a 12-bit R component in the top 12 bits of the word in bytes 0..1, and a 12-bit G component in the top 12 bits of the word in bytes 2..3, with the bottom 4 bits of each word unused.

- **VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16** specifies a four-component, 64-bit unsigned normalized format that has a 12-bit R component in the top 12 bits of the word in bytes 0..1, a 12-bit G component in the top 12 bits of the word in bytes 2..3, a 12-bit B component in the top 12 bits of the word in bytes 4..5, and a 12-bit A component in the top 12 bits of the word in bytes 6..7, with the bottom 4 bits of each word unused.
- **VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16** specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 12-bit G component for the even i coordinate in the top 12 bits of the word in bytes 0..1, a 12-bit B component in the top 12 bits of the word in bytes 2..3, a 12-bit G component for the odd i coordinate in the top 12 bits of the word in bytes 4..5, and a 12-bit R component in the top 12 bits of the word in bytes 6..7, with the bottom 4 bits of each word unused. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- **VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16** specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 12-bit B component in the top 12 bits of the word in bytes 0..1, a 12-bit G component for the even i coordinate in the top 12 bits of the word in bytes 2..3, a 12-bit R component in the top 12 bits of the word in bytes 4..5, and a 12-bit G component for the odd i coordinate in the top 12 bits of the word in bytes 6..7, with the bottom 4 bits of each word unused. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- **VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16** specifies an unsigned normalized *multi-planar format* that has a 12-bit G component in the top 12 bits of each 16-bit word of plane 0, a 12-bit B component in the top 12 bits of each 16-bit word of plane 1, and a 12-bit R component in the top 12 bits of each 16-bit word of plane 2, with the bottom 4 bits of each word unused. The horizontal and vertical dimensions of the R and B planes are halved relative to the image dimensions, and each R and B component is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, **VK_IMAGE_ASPECT_PLANE_1_BIT** for the B plane, and **VK_IMAGE_ASPECT_PLANE_2_BIT** for the R plane. This format only supports images with a width and height that is a multiple of two.
- **VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16** specifies an unsigned normalized *multi-planar format* that has a 12-bit G component in the top 12 bits of each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 12-bit B component in the top 12 bits of the word in bytes 0..1, and a 12-bit R component in the top 12 bits of the word in bytes 2..3, with the bottom 4 bits of each word unused. The horizontal and vertical dimensions of the BR plane are halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using

`VK_IMAGE_ASPECT_PLANE_0_BIT` for the G plane, and `VK_IMAGE_ASPECT_PLANE_1_BIT` for the BR plane. This format only supports images with a width and height that is a multiple of two.

- `VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16` specifies an unsigned normalized *multi-planar format* that has a 12-bit G component in the top 12 bits of each 16-bit word of plane 0, a 12-bit B component in the top 12 bits of each 16-bit word of plane 1, and a 12-bit R component in the top 12 bits of each 16-bit word of plane 2, with the bottom 4 bits of each word unused. The horizontal dimension of the R and B plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using `VK_IMAGE_ASPECT_PLANE_0_BIT` for the G plane, `VK_IMAGE_ASPECT_PLANE_1_BIT` for the B plane, and `VK_IMAGE_ASPECT_PLANE_2_BIT` for the R plane. This format only supports images with a width that is a multiple of two.
- `VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16` specifies an unsigned normalized *multi-planar format* that has a 12-bit G component in the top 12 bits of each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 12-bit B component in the top 12 bits of the word in bytes 0..1, and a 12-bit R component in the top 12 bits of the word in bytes 2..3, with the bottom 4 bits of each word unused. The horizontal dimension of the BR plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using `VK_IMAGE_ASPECT_PLANE_0_BIT` for the G plane, and `VK_IMAGE_ASPECT_PLANE_1_BIT` for the BR plane. This format only supports images with a width that is a multiple of two.
- `VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16` specifies an unsigned normalized *multi-planar format* that has a 12-bit G component in the top 12 bits of each 16-bit word of plane 0, a 12-bit B component in the top 12 bits of each 16-bit word of plane 1, and a 12-bit R component in the top 12 bits of each 16-bit word of plane 2, with the bottom 4 bits of each word unused. Each plane has the same dimensions and each R, G and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using `VK_IMAGE_ASPECT_PLANE_0_BIT` for the G plane, `VK_IMAGE_ASPECT_PLANE_1_BIT` for the B plane, and `VK_IMAGE_ASPECT_PLANE_2_BIT` for the R plane.
- `VK_FORMAT_G16B16G16R16_422_UNORM` specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 16-bit G component for the even i coordinate in the word in bytes 0..1, a 16-bit B component in the word in bytes 2..3, a 16-bit G component for the odd i coordinate in the word in bytes 4..5, and a 16-bit R component in the word in bytes 6..7. This format only supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.
- `VK_FORMAT_B16G16R16G16_422_UNORM` specifies a four-component, 64-bit format containing a pair of G components, an R component, and a B component, collectively encoding a 2×1 rectangle of unsigned normalized RGB texel data. One G value is present at each i coordinate, with the B and R values shared across both G values and thus recorded at half the horizontal resolution of the image. This format has a 16-bit B component in the word in bytes 0..1, a 16-bit G component for the even i coordinate in the word in bytes 2..3, a 16-bit R component in the word in bytes 4..5, and a 16-bit G component for the odd i coordinate in the word in bytes 6..7. This format only

supports images with a width that is a multiple of two. For the purposes of the constraints on copy extents, this format is treated as a compressed format with a 2×1 compressed texel block.

- **VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM** specifies an unsigned normalized *multi-planar format* that has a 16-bit G component in each 16-bit word of plane 0, a 16-bit B component in each 16-bit word of plane 1, and a 16-bit R component in each 16-bit word of plane 2. The horizontal and vertical dimensions of the R and B planes are halved relative to the image dimensions, and each R and B component is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, **VK_IMAGE_ASPECT_PLANE_1_BIT** for the B plane, and **VK_IMAGE_ASPECT_PLANE_2_BIT** for the R plane. This format only supports images with a width and height that is a multiple of two.
- **VK_FORMAT_G16_B16R16_2PLANE_420_UNORM** specifies an unsigned normalized *multi-planar format* that has a 16-bit G component in each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 16-bit B component in the word in bytes 0..1, and a 16-bit R component in the word in bytes 2..3. The horizontal and vertical dimensions of the BR plane are halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$ and $\lfloor j_G \times 0.5 \rfloor = j_B = j_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, and **VK_IMAGE_ASPECT_PLANE_1_BIT** for the BR plane. This format only supports images with a width and height that is a multiple of two.
- **VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM** specifies an unsigned normalized *multi-planar format* that has a 16-bit G component in each 16-bit word of plane 0, a 16-bit B component in each 16-bit word of plane 1, and a 16-bit R component in each 16-bit word of plane 2. The horizontal dimension of the R and B plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, **VK_IMAGE_ASPECT_PLANE_1_BIT** for the B plane, and **VK_IMAGE_ASPECT_PLANE_2_BIT** for the R plane. This format only supports images with a width that is a multiple of two.
- **VK_FORMAT_G16_B16R16_2PLANE_422_UNORM** specifies an unsigned normalized *multi-planar format* that has a 16-bit G component in each 16-bit word of plane 0, and a two-component, 32-bit BR plane 1 consisting of a 16-bit B component in the word in bytes 0..1, and a 16-bit R component in the word in bytes 2..3. The horizontal dimension of the BR plane is halved relative to the image dimensions, and each R and B value is shared with the G components for which $\lfloor i_G \times 0.5 \rfloor = i_B = i_R$. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, and **VK_IMAGE_ASPECT_PLANE_1_BIT** for the BR plane. This format only supports images with a width that is a multiple of two.
- **VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM** specifies an unsigned normalized *multi-planar format* that has a 16-bit G component in each 16-bit word of plane 0, a 16-bit B component in each 16-bit word of plane 1, and a 16-bit R component in each 16-bit word of plane 2. Each plane has the same dimensions and each R, G and B component contributes to a single texel. The location of each plane when this image is in linear layout can be determined via [vkGetImageSubresourceLayout](#), using **VK_IMAGE_ASPECT_PLANE_0_BIT** for the G plane, **VK_IMAGE_ASPECT_PLANE_1_BIT** for the B plane, and **VK_IMAGE_ASPECT_PLANE_2_BIT** for the R plane.

41.1.1. Compatible Formats of Planes of Multi-Planar Formats

Individual planes of multi-planar formats are size-compatible with single-plane color formats if they occupy the same number of bits per texel block, and are compatible with those formats if they have the same block extent.

In the following table, individual planes of a *multi-planar* format are compatible with the format listed against the relevant plane index for that multi-planar format, and any format compatible with the listed single-plane format according to [Format Compatibility Classes](#). These planes are also [size-compatible](#) with any format that is [size-compatible](#) with the listed single-plane format.

Table 60. Plane Format Compatibility Table

Plane	Compatible format for plane	Width relative to the width w of the plane with the largest dimensions	Height relative to the height h of the plane with the largest dimensions
VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM			
0	VK_FORMAT_R8_UNORM	w	h
1	VK_FORMAT_R8_UNORM	$w/2$	$h/2$
2	VK_FORMAT_R8_UNORM	$w/2$	$h/2$
VK_FORMAT_G8_B8R8_2PLANE_420_UNORM			
0	VK_FORMAT_R8_UNORM	w	h
1	VK_FORMAT_R8G8_UNORM	$w/2$	$h/2$
VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM			
0	VK_FORMAT_R8_UNORM	w	h
1	VK_FORMAT_R8_UNORM	$w/2$	h
2	VK_FORMAT_R8_UNORM	$w/2$	h
VK_FORMAT_G8_B8R8_2PLANE_422_UNORM			
0	VK_FORMAT_R8_UNORM	w	h
1	VK_FORMAT_R8G8_UNORM	$w/2$	h
VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM			
0	VK_FORMAT_R8_UNORM	w	h
1	VK_FORMAT_R8_UNORM	w	h
2	VK_FORMAT_R8_UNORM	w	h
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16			
0	VK_FORMAT_R10X6_UNORM_PACK16	w	h
1	VK_FORMAT_R10X6_UNORM_PACK16	$w/2$	$h/2$
2	VK_FORMAT_R10X6_UNORM_PACK16	$w/2$	$h/2$
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16			
0	VK_FORMAT_R10X6_UNORM_PACK16	w	h

Plane	Compatible format for plane	Width relative to the width w of the plane with the largest dimensions	Height relative to the height h of the plane with the largest dimensions
1	VK_FORMAT_R10X6G10X6_UNORM_2PACK16	$w/2$	$h/2$
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16			
0	VK_FORMAT_R10X6_UNORM_PACK16	w	h
1	VK_FORMAT_R10X6_UNORM_PACK16	$w/2$	h
2	VK_FORMAT_R10X6_UNORM_PACK16	$w/2$	h
VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16			
0	VK_FORMAT_R10X6_UNORM_PACK16	w	h
1	VK_FORMAT_R10X6G10X6_UNORM_2PACK16	$w/2$	h
VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16			
0	VK_FORMAT_R10X6_UNORM_PACK16	w	h
1	VK_FORMAT_R10X6_UNORM_PACK16	w	h
2	VK_FORMAT_R10X6_UNORM_PACK16	w	h
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16			
0	VK_FORMAT_R12X4_UNORM_PACK16	w	h
1	VK_FORMAT_R12X4_UNORM_PACK16	$w/2$	$h/2$
2	VK_FORMAT_R12X4_UNORM_PACK16	$w/2$	$h/2$
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16			
0	VK_FORMAT_R12X4_UNORM_PACK16	w	h
1	VK_FORMAT_R12X4G12X4_UNORM_2PACK16	$w/2$	$h/2$
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16			
0	VK_FORMAT_R12X4_UNORM_PACK16	w	h
1	VK_FORMAT_R12X4_UNORM_PACK16	$w/2$	h
2	VK_FORMAT_R12X4_UNORM_PACK16	$w/2$	h
VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16			
0	VK_FORMAT_R12X4_UNORM_PACK16	w	h
1	VK_FORMAT_R12X4G12X4_UNORM_2PACK16	$w/2$	h
VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16			
0	VK_FORMAT_R12X4_UNORM_PACK16	w	h
1	VK_FORMAT_R12X4_UNORM_PACK16	w	h
2	VK_FORMAT_R12X4_UNORM_PACK16	w	h
VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM			
0	VK_FORMAT_R16_UNORM	w	h

Plane	Compatible format for plane	Width relative to the width w of the plane with the largest dimensions	Height relative to the height h of the plane with the largest dimensions
1	VK_FORMAT_R16_UNORM	$w/2$	$h/2$
2	VK_FORMAT_R16_UNORM	$w/2$	$h/2$
VK_FORMAT_G16_B16R16_2PLANE_420_UNORM			
0	VK_FORMAT_R16_UNORM	w	h
1	VK_FORMAT_R16G16_UNORM	$w/2$	$h/2$
VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM			
0	VK_FORMAT_R16_UNORM	w	h
1	VK_FORMAT_R16_UNORM	$w/2$	h
2	VK_FORMAT_R16_UNORM	$w/2$	h
VK_FORMAT_G16_B16R16_2PLANE_422_UNORM			
0	VK_FORMAT_R16_UNORM	w	h
1	VK_FORMAT_R16G16_UNORM	$w/2$	h
VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM			
0	VK_FORMAT_R16_UNORM	w	h
1	VK_FORMAT_R16_UNORM	w	h
2	VK_FORMAT_R16_UNORM	w	h

41.1.2. Multi-planar Format Image Aspect

When using [VkImageAspectFlagBits](#) to select a plane of a multi-planar format, the following are the valid options:

- Two planes
 - VK_IMAGE_ASPECT_PLANE_0_BIT
 - VK_IMAGE_ASPECT_PLANE_1_BIT
- Three planes
 - VK_IMAGE_ASPECT_PLANE_0_BIT
 - VK_IMAGE_ASPECT_PLANE_1_BIT
 - VK_IMAGE_ASPECT_PLANE_2_BIT

41.1.3. Packed Formats

For the purposes of address alignment when accessing buffer memory containing vertex attribute or texel data, the following formats are considered *packed* - components of the texels or attributes are stored in bitfields packed into one or more 8-, 16-, or 32-bit fundamental data type.

- Packed into 8-bit data types:

- VK_FORMAT_R4G4_UNORM_PACK8

- Packed into 16-bit data types:

- VK_FORMAT_R4G4B4A4_UNORM_PACK16
- VK_FORMAT_B4G4R4A4_UNORM_PACK16
- VK_FORMAT_R5G6B5_UNORM_PACK16
- VK_FORMAT_B5G6R5_UNORM_PACK16
- VK_FORMAT_R5G5B5A1_UNORM_PACK16
- VK_FORMAT_B5G5R5A1_UNORM_PACK16
- VK_FORMAT_A1R5G5B5_UNORM_PACK16
- VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR
- VK_FORMAT_R10X6_UNORM_PACK16
- VK_FORMAT_R10X6G10X6_UNORM_2PACK16
- VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16
- VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16
- VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16
- VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16
- VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16
- VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16
- VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16
- VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16
- VK_FORMAT_R12X4_UNORM_PACK16
- VK_FORMAT_R12X4G12X4_UNORM_2PACK16
- VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16
- VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16
- VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16
- VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16
- VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16
- VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16
- VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16
- VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16

- Packed into 32-bit data types:

- VK_FORMAT_A8B8G8R8_UNORM_PACK32
- VK_FORMAT_A8B8G8R8_SNORM_PACK32
- VK_FORMAT_A8B8G8R8_USCALED_PACK32

- VK_FORMAT_A8B8G8R8_SSCALED_PACK32
- VK_FORMAT_A8B8G8R8_UINT_PACK32
- VK_FORMAT_A8B8G8R8_SINT_PACK32
- VK_FORMAT_A8B8G8R8_SRGB_PACK32
- VK_FORMAT_A2R10G10B10_UNORM_PACK32
- VK_FORMAT_A2R10G10B10_SNORM_PACK32
- VK_FORMAT_A2R10G10B10_USCALED_PACK32
- VK_FORMAT_A2R10G10B10_SSCALED_PACK32
- VK_FORMAT_A2R10G10B10_UINT_PACK32
- VK_FORMAT_A2R10G10B10_SINT_PACK32
- VK_FORMAT_A2B10G10R10_UNORM_PACK32
- VK_FORMAT_A2B10G10R10_SNORM_PACK32
- VK_FORMAT_A2B10G10R10_USCALED_PACK32
- VK_FORMAT_A2B10G10R10_SSCALED_PACK32
- VK_FORMAT_A2B10G10R10_UINT_PACK32
- VK_FORMAT_A2B10G10R10_SINT_PACK32
- VK_FORMAT_B10G11R11_UFLOAT_PACK32
- VK_FORMAT_E5B9G9R9_UFLOAT_PACK32
- VK_FORMAT_X8_D24_UNORM_PACK32

41.1.4. Identification of Formats

A “format” is represented by a single enum value. The name of a format is usually built up by using the following pattern:

```
VK_FORMAT_{component-format|compression-scheme}_{numeric-format}
```

The component-format indicates either the size of the R, G, B, and A components (if they are present) in the case of a color format, or the size of the depth (D) and stencil (S) components (if they are present) in the case of a depth/stencil format (see below). An X indicates a component that is unused, but **may** be present for padding.

Table 61. Interpretation of Numeric Format

Numeric format	Type-Declaration instructions	Numeric type	Description
UNORM	OpTypeFloat	floating-point	The components are unsigned normalized values in the range [0,1]
SNORM	OpTypeFloat	floating-point	The components are signed normalized values in the range [-1,1]
USCALED	OpTypeFloat	floating-point	The components are unsigned integer values that get converted to floating-point in the range [0,2 ⁿ -1]
SSCALED	OpTypeFloat	floating-point	The components are signed integer values that get converted to floating-point in the range [-2 ⁿ⁻¹ ,2 ⁿ⁻¹ -1]
UINT	OpTypeInt	unsigned integer	The components are unsigned integer values in the range [0,2 ⁿ -1]
SINT	OpTypeInt	signed integer	The components are signed integer values in the range [-2 ⁿ⁻¹ ,2 ⁿ⁻¹ -1]
UFLOAT	OpTypeFloat	floating-point	The components are unsigned floating-point numbers (used by packed, shared exponent, and some compressed formats)
SFLOAT	OpTypeFloat	floating-point	The components are signed floating-point numbers
SRGB	OpTypeFloat	floating-point	The R, G, and B components are unsigned normalized values that represent values using sRGB nonlinear encoding, while the A component (if one exists) is a regular unsigned normalized value
n is the number of bits in the component.			

The suffix **_PACKnn** indicates that the format is packed into an underlying type with **nn** bits. The suffix **_mPACKnn** is a short-hand that indicates that the format has **m** groups of components (which may or may not be stored in separate *planes*) that are each packed into an underlying type with **nn** bits.

The suffix **_BLOCK** indicates that the format is a block-compressed format, with the representation of multiple pixels encoded interdependently within a region.

Table 62. Interpretation of Compression Scheme

Compression scheme	Description
BC	Block Compression. See Block-Compressed Image Formats .
ETC2	Ericsson Texture Compression. See ETC Compressed Image Formats .
EAC	ETC2 Alpha Compression. See ETC Compressed Image Formats .

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	← Byte	
R		VK_FORMAT_R16_*															
R		G		VK_FORMAT_R16G16_*													
R		G		B		VK_FORMAT_R16G16B16_*											
R		G		B		A		VK_FORMAT_R16G16B16A16_*									
G ₀		B		G ₁		R		VK_FORMAT_G10X6B10X6G10X6R10X6_4PACK16_422_UNORM VK_FORMAT_G12X4B12X4G12X4R12X4_4PACK16_422_UNORM VK_FORMAT_G16B16G16R16_UNORM									
B		G ₀		R		G ₁		VK_FORMAT_B10X6G10X6R10X6G10X6_4PACK16_422_UNORM VK_FORMAT_B12X4G12X4R12X4G12X4_4PACK16_422_UNORM VK_FORMAT_B16G16R16G16_422_UNORM									
R		VK_FORMAT_R32_*															
R		G				VK_FORMAT_R32G32_*											
R		G				B				VK_FORMAT_R32G32B32_*							
R		G				B				A			VK_FORMAT_R32G32B32A32_*				
R						VK_FORMAT_R64_*											
R						G						VK_FORMAT_R64G64_*					
VK_FORMAT_R64G64B64_* as VK_FORMAT_R64G64_* but with B in bytes 16-23																	
VK_FORMAT_R64G64B64A64_* as VK_FORMAT_R64G64B64_* but with A in bytes 24-31																	

Packed formats store multiple components within one underlying type. The bit representation is that the first component specified in the name of the format is in the most-significant bits and the last component specified is in the least-significant bits of the underlying type. The in-memory ordering of bytes comprising the underlying type is determined by the host endianness.

Table 64. Bit Mappings for Packed 8-Bit Formats

Bit							
7	6	5	4	3	2	1	0
VK_FORMAT_R4G4_UNORM_PACK8							
R				G			
3	2	1	0	3	2	1	0

Table 65. Bit Mappings for Packed 16-Bit Formats

Bit															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VK_FORMAT_R4G4B4A4_UNORM_PACK16															
R				G				B				A			
3	2	1	0	3	2	1	0	3	2	1	0	3	2	1	0
VK_FORMAT_B4G4R4A4_UNORM_PACK16															
B				G				R				A			

Bit																
3	2	1	0	3	2	1	0	3	2	1	0	3	2	1	0	
VK_FORMAT_R5G6B5_UNORM_PACK16																
R					G						B					
4	3	2	1	0	5	4	3	2	1	0	4	3	2	1	0	
VK_FORMAT_B5G6R5_UNORM_PACK16																
B					G						R					
4	3	2	1	0	5	4	3	2	1	0	4	3	2	1	0	
VK_FORMAT_R5G5B5A1_UNORM_PACK16																
R					G						B					A
4	3	2	1	0	4	3	2	1	0	4	3	2	1	0	0	
VK_FORMAT_B5G5R5A1_UNORM_PACK16																
B					G						R					A
4	3	2	1	0	4	3	2	1	0	4	3	2	1	0	0	
VK_FORMAT_A1R5G5B5_UNORM_PACK16																
A	R					G					B					
0	4	3	2	1	0	4	3	2	1	0	4	3	2	1	0	
VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR																
A	B					G					R					
0	4	3	2	1	0	4	3	2	1	0	4	3	2	1	0	
VK_FORMAT_R10X6_UNORM_PACK16																
R										X						
9	8	7	6	5	4	3	2	1	0	5	4	3	2	1	0	
VK_FORMAT_R12X4_UNORM_PACK16																
R												X				
11	10	9	8	7	6	5	4	3	2	1	0	3	2	1	0	

Table 66. Bit Mappings for Packed 32-Bit Formats

Bit																															
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VK_FORMAT_A8B8G8R8*_PACK32																															
A								B								G								R							
7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
VK_FORMAT_A2R10G10B10*_PACK32																															
A		R								G								B													
1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0
VK_FORMAT_A2B10G10R10*_PACK32																															
A		B								G								R													
1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0
VK_FORMAT_B10G11R11_UFLOAT_PACK32																															

Bit																																
B										G										R												
9	8	7	6	5	4	3	2	1	0	10	9	8	7	6	5	4	3	2	1	0	10	9	8	7	6	5	4	3	2	1	0	
VK_FORMAT_E5B9G9R9_UFLOAT_PACK32																																
E					B									G										R								
4	3	2	1	0	8	7	6	5	4	3	2	1	0	8	7	6	5	4	3	2	1	0	8	7	6	5	4	3	2	1	0	
VK_FORMAT_X8_D24_UNORM_PACK32																																
X									D																							
7	6	5	4	3	2	1	0		23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

41.1.6. Depth/Stencil Formats

Depth/stencil formats are considered opaque and need not be stored in the exact number of bits per texel or component ordering indicated by the format enum. However, implementations **must** not substitute a different depth or stencil precision than is described in the format (e.g. D16 **must** not be implemented as D24 or D32).

41.1.7. Format Compatibility Classes

Uncompressed color formats are *compatible* with each other if they occupy the same number of bits per texel block as long as neither or both are alpha formats (e.g., `VK_FORMAT_A8_UNORM_KHR`). Compressed color formats are compatible with each other if the only difference between them is the [numeric format](#) of the uncompressed pixels. Each depth/stencil format is only compatible with itself. In the [following](#) table, all the formats in the same row are compatible. Each format has a defined *texel block extent* specifying how many texels each texel block represents in each dimension.

Table 67. Compatible Formats

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
8-bit Block size 1 byte 1x1x1 block extent 1 texel/block	<code>VK_FORMAT_R4G4_UNORM_PACK8</code> , <code>VK_FORMAT_R8_UNORM</code> , <code>VK_FORMAT_R8_SNORM</code> , <code>VK_FORMAT_R8_USCALED</code> , <code>VK_FORMAT_R8_SSCALED</code> , <code>VK_FORMAT_R8_UINT</code> , <code>VK_FORMAT_R8_SINT</code> , <code>VK_FORMAT_R8_SRGB</code>

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
16-bit Block size 2 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR, VK_FORMAT_R10X6_UNORM_PACK16, VK_FORMAT_R12X4_UNORM_PACK16, VK_FORMAT_R4G4B4A4_UNORM_PACK16, VK_FORMAT_B4G4R4A4_UNORM_PACK16, VK_FORMAT_R5G6B5_UNORM_PACK16, VK_FORMAT_B5G6R5_UNORM_PACK16, VK_FORMAT_R5G5B5A1_UNORM_PACK16, VK_FORMAT_B5G5R5A1_UNORM_PACK16, VK_FORMAT_A1R5G5B5_UNORM_PACK16, VK_FORMAT_R8G8_UNORM, VK_FORMAT_R8G8_SNORM, VK_FORMAT_R8G8_USCALED, VK_FORMAT_R8G8_SSCALED, VK_FORMAT_R8G8_UINT, VK_FORMAT_R8G8_SINT, VK_FORMAT_R8G8_SRGB, VK_FORMAT_R16_UNORM, VK_FORMAT_R16_SNORM, VK_FORMAT_R16_USCALED, VK_FORMAT_R16_SSCALED, VK_FORMAT_R16_UINT, VK_FORMAT_R16_SINT, VK_FORMAT_R16_SFLOAT
8-bit alpha Block size 1 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_A8_UNORM_KHR
24-bit Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R8G8B8_UNORM, VK_FORMAT_R8G8B8_SNORM, VK_FORMAT_R8G8B8_USCALED, VK_FORMAT_R8G8B8_SSCALED, VK_FORMAT_R8G8B8_UINT, VK_FORMAT_R8G8B8_SINT, VK_FORMAT_R8G8B8_SRGB, VK_FORMAT_B8G8R8_UNORM, VK_FORMAT_B8G8R8_SNORM, VK_FORMAT_B8G8R8_USCALED, VK_FORMAT_B8G8R8_SSCALED, VK_FORMAT_B8G8R8_UINT, VK_FORMAT_B8G8R8_SINT, VK_FORMAT_B8G8R8_SRGB

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
32-bit Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R10X6G10X6_UNORM_2PACK16, VK_FORMAT_R12X4G12X4_UNORM_2PACK16, VK_FORMAT_R8G8B8A8_UNORM, VK_FORMAT_R8G8B8A8_SNORM, VK_FORMAT_R8G8B8A8_USCALED, VK_FORMAT_R8G8B8A8_SSCALED, VK_FORMAT_R8G8B8A8_UINT, VK_FORMAT_R8G8B8A8_SINT, VK_FORMAT_R8G8B8A8_SRGB, VK_FORMAT_B8G8R8A8_UNORM, VK_FORMAT_B8G8R8A8_SNORM, VK_FORMAT_B8G8R8A8_USCALED, VK_FORMAT_B8G8R8A8_SSCALED, VK_FORMAT_B8G8R8A8_UINT, VK_FORMAT_B8G8R8A8_SINT, VK_FORMAT_B8G8R8A8_SRGB, VK_FORMAT_A8B8G8R8_UNORM_PACK32, VK_FORMAT_A8B8G8R8_SNORM_PACK32, VK_FORMAT_A8B8G8R8_USCALED_PACK32, VK_FORMAT_A8B8G8R8_SSCALED_PACK32, VK_FORMAT_A8B8G8R8_UINT_PACK32, VK_FORMAT_A8B8G8R8_SINT_PACK32, VK_FORMAT_A8B8G8R8_SRGB_PACK32, VK_FORMAT_A2R10G10B10_UNORM_PACK32, VK_FORMAT_A2R10G10B10_SNORM_PACK32, VK_FORMAT_A2R10G10B10_USCALED_PACK32, VK_FORMAT_A2R10G10B10_SSCALED_PACK32, VK_FORMAT_A2R10G10B10_UINT_PACK32, VK_FORMAT_A2R10G10B10_SINT_PACK32, VK_FORMAT_A2B10G10R10_UNORM_PACK32, VK_FORMAT_A2B10G10R10_SNORM_PACK32, VK_FORMAT_A2B10G10R10_USCALED_PACK32, VK_FORMAT_A2B10G10R10_SSCALED_PACK32, VK_FORMAT_A2B10G10R10_UINT_PACK32, VK_FORMAT_A2B10G10R10_SINT_PACK32, VK_FORMAT_R16G16_UNORM, VK_FORMAT_R16G16_SNORM, VK_FORMAT_R16G16_USCALED, VK_FORMAT_R16G16_SSCALED, VK_FORMAT_R16G16_UINT, VK_FORMAT_R16G16_SINT, VK_FORMAT_R16G16_SFLOAT, VK_FORMAT_R32_UINT, VK_FORMAT_R32_SINT, VK_FORMAT_R32_SFLOAT, VK_FORMAT_B10G11R11_UFLOAT_PACK32, VK_FORMAT_E5B9G9R9_UFLOAT_PACK32

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
48-bit Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R16G16B16_UNORM, VK_FORMAT_R16G16B16_SNORM, VK_FORMAT_R16G16B16_USCALED, VK_FORMAT_R16G16B16_SSCALED, VK_FORMAT_R16G16B16_UINT, VK_FORMAT_R16G16B16_SINT, VK_FORMAT_R16G16B16_SFLOAT
64-bit Block size 8 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R16G16B16A16_UNORM, VK_FORMAT_R16G16B16A16_SNORM, VK_FORMAT_R16G16B16A16_USCALED, VK_FORMAT_R16G16B16A16_SSCALED, VK_FORMAT_R16G16B16A16_UINT, VK_FORMAT_R16G16B16A16_SINT, VK_FORMAT_R16G16B16A16_SFLOAT, VK_FORMAT_R32G32_UINT, VK_FORMAT_R32G32_SINT, VK_FORMAT_R32G32_SFLOAT, VK_FORMAT_R64_UINT, VK_FORMAT_R64_SINT, VK_FORMAT_R64_SFLOAT
96-bit Block size 12 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R32G32B32_UINT, VK_FORMAT_R32G32B32_SINT, VK_FORMAT_R32G32B32_SFLOAT
128-bit Block size 16 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R32G32B32A32_UINT, VK_FORMAT_R32G32B32A32_SINT, VK_FORMAT_R32G32B32A32_SFLOAT, VK_FORMAT_R64G64_UINT, VK_FORMAT_R64G64_SINT, VK_FORMAT_R64G64_SFLOAT
192-bit Block size 24 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R64G64B64_UINT, VK_FORMAT_R64G64B64_SINT, VK_FORMAT_R64G64B64_SFLOAT
256-bit Block size 32 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R64G64B64A64_UINT, VK_FORMAT_R64G64B64A64_SINT, VK_FORMAT_R64G64B64A64_SFLOAT

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
D16 Block size 2 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D16_UNORM
D24 Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_X8_D24_UNORM_PACK32
D32 Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D32_SFLOAT
S8 Block size 1 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_S8_UINT
D16S8 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D16_UNORM_S8_UINT
D24S8 Block size 4 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D24_UNORM_S8_UINT
D32S8 Block size 5 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_D32_SFLOAT_S8_UINT
BC1_RGB Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC1_RGB_UNORM_BLOCK, VK_FORMAT_BC1_RGB_SRGB_BLOCK
BC1_RGBA Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC1_RGBA_UNORM_BLOCK, VK_FORMAT_BC1_RGBA_SRGB_BLOCK
BC2 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC2_UNORM_BLOCK, VK_FORMAT_BC2_SRGB_BLOCK

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
BC3 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC3_UNORM_BLOCK, VK_FORMAT_BC3_SRGB_BLOCK
BC4 Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC4_UNORM_BLOCK, VK_FORMAT_BC4_SNORM_BLOCK
BC5 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC5_UNORM_BLOCK, VK_FORMAT_BC5_SNORM_BLOCK
BC6H Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC6H_UFLOAT_BLOCK, VK_FORMAT_BC6H_SFLOAT_BLOCK
BC7 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_BC7_UNORM_BLOCK, VK_FORMAT_BC7_SRGB_BLOCK
ETC2_RGB Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK, VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK
ETC2_RGBA Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK, VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK
ETC2_EAC_RGBA Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK, VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK
EAC_R Block size 8 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_EAC_R11_UNORM_BLOCK, VK_FORMAT_EAC_R11_SNORM_BLOCK
EAC_RG Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_EAC_R11G11_UNORM_BLOCK, VK_FORMAT_EAC_R11G11_SNORM_BLOCK

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
ASTC_4x4 Block size 16 byte 4x4x1 block extent 16 texel/block	VK_FORMAT_ASTC_4x4_UNORM_BLOCK, VK_FORMAT_ASTC_4x4_SRGB_BLOCK
ASTC_5x4 Block size 16 byte 5x4x1 block extent 20 texel/block	VK_FORMAT_ASTC_5x4_UNORM_BLOCK, VK_FORMAT_ASTC_5x4_SRGB_BLOCK
ASTC_5x5 Block size 16 byte 5x5x1 block extent 25 texel/block	VK_FORMAT_ASTC_5x5_UNORM_BLOCK, VK_FORMAT_ASTC_5x5_SRGB_BLOCK
ASTC_6x5 Block size 16 byte 6x5x1 block extent 30 texel/block	VK_FORMAT_ASTC_6x5_UNORM_BLOCK, VK_FORMAT_ASTC_6x5_SRGB_BLOCK
ASTC_6x6 Block size 16 byte 6x6x1 block extent 36 texel/block	VK_FORMAT_ASTC_6x6_UNORM_BLOCK, VK_FORMAT_ASTC_6x6_SRGB_BLOCK
ASTC_8x5 Block size 16 byte 8x5x1 block extent 40 texel/block	VK_FORMAT_ASTC_8x5_UNORM_BLOCK, VK_FORMAT_ASTC_8x5_SRGB_BLOCK
ASTC_8x6 Block size 16 byte 8x6x1 block extent 48 texel/block	VK_FORMAT_ASTC_8x6_UNORM_BLOCK, VK_FORMAT_ASTC_8x6_SRGB_BLOCK
ASTC_8x8 Block size 16 byte 8x8x1 block extent 64 texel/block	VK_FORMAT_ASTC_8x8_UNORM_BLOCK, VK_FORMAT_ASTC_8x8_SRGB_BLOCK
ASTC_10x5 Block size 16 byte 10x5x1 block extent 50 texel/block	VK_FORMAT_ASTC_10x5_UNORM_BLOCK, VK_FORMAT_ASTC_10x5_SRGB_BLOCK
ASTC_10x6 Block size 16 byte 10x6x1 block extent 60 texel/block	VK_FORMAT_ASTC_10x6_UNORM_BLOCK, VK_FORMAT_ASTC_10x6_SRGB_BLOCK

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
ASTC_10x8 Block size 16 byte 10x8x1 block extent 80 texel/block	VK_FORMAT_ASTC_10x8_UNORM_BLOCK, VK_FORMAT_ASTC_10x8_SRGB_BLOCK
ASTC_10x10 Block size 16 byte 10x10x1 block extent 100 texel/block	VK_FORMAT_ASTC_10x10_UNORM_BLOCK, VK_FORMAT_ASTC_10x10_SRGB_BLOCK
ASTC_12x10 Block size 16 byte 12x10x1 block extent 120 texel/block	VK_FORMAT_ASTC_12x10_UNORM_BLOCK, VK_FORMAT_ASTC_12x10_SRGB_BLOCK
ASTC_12x12 Block size 16 byte 12x12x1 block extent 144 texel/block	VK_FORMAT_ASTC_12x12_UNORM_BLOCK, VK_FORMAT_ASTC_12x12_SRGB_BLOCK
32-bit G8B8G8R8 Block size 4 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_G8B8G8R8_422_UNORM
32-bit B8G8R8G8 Block size 4 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_B8G8R8G8_422_UNORM
8-bit 3-plane 420 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM
8-bit 2-plane 420 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G8_B8R8_2PLANE_420_UNORM
8-bit 3-plane 422 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM
8-bit 2-plane 422 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G8_B8R8_2PLANE_422_UNORM

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
8-bit 3-plane 444 Block size 3 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM
64-bit R10G10B10A10 Block size 8 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16
64-bit G10B10G10R10 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16
64-bit B10G10R10G10 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16
10-bit 3-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16
10-bit 2-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16
10-bit 3-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16
10-bit 2-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16
10-bit 3-plane 444 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16
64-bit R12G12B12A12 Block size 8 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
64-bit G12B12G12R12 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16
64-bit B12G12R12G12 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16
12-bit 3-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16
12-bit 2-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16
12-bit 3-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16
12-bit 2-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16
12-bit 3-plane 444 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16
64-bit G16B16G16R16 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_G16B16G16R16_422_UNORM
64-bit B16G16R16G16 Block size 8 byte 2x1x1 block extent 1 texel/block	VK_FORMAT_B16G16R16G16_422_UNORM
16-bit 3-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM

Class, Texel Block Size, Texel Block Extent, # Texels/Block	Formats
16-bit 2-plane 420 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G16_B16R16_2PLANE_420_UNORM
16-bit 3-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM
16-bit 2-plane 422 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G16_B16R16_2PLANE_422_UNORM
16-bit 3-plane 444 Block size 6 byte 1x1x1 block extent 1 texel/block	VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM

Size Compatibility

Color formats with the same texel block size are considered *size-compatible* as long as neither or both are alpha formats (e.g., VK_FORMAT_A8_UNORM_KHR). If two size-compatible formats have different block extents (i.e. for compressed formats), then an image with size $A \times B \times C$ in one format with a block extent of $a \times b \times c$ can be represented as an image with size $X \times Y \times Z$ in the other format with block extent $x \times y \times z$ at the ratio between the block extents for each format, where

$$\lceil A/a \rceil = \lceil X/x \rceil$$

$$\lceil B/b \rceil = \lceil Y/y \rceil$$

$$\lceil C/c \rceil = \lceil Z/z \rceil$$

NOTE

For example, a 7x3 image in the VK_FORMAT_ASTC_8x5_UNORM_BLOCK format can be represented as a 1x1 VK_FORMAT_R64G64_UINT image.

Images created with the VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT flag can have size-compatible views created from them to enable access via different size-compatible formats. Image views created in this way will be sized to match the expectations of the block extents noted above.

Copy operations are able to copy between size-compatible formats in different resources to enable manipulation of data in different formats. The extent used in these copy operations always matches

the source image, and is resized to the expectations of the block extents noted above for the destination image.

41.2. Format Properties

To query supported format features which are properties of the physical device, call:

```
// Provided by VK_VERSION_1_0
void vkGetPhysicalDeviceFormatProperties(
    VkPhysicalDevice    physicalDevice,
    VkFormat            format,
    VkFormatProperties* pFormatProperties);
```

- `physicalDevice` is the physical device from which to query the format properties.
- `format` is the format whose properties are queried.
- `pFormatProperties` is a pointer to a `VkFormatProperties` structure in which physical device properties for `format` are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceFormatProperties-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceFormatProperties-format-parameter `format` **must** be a valid `VkFormat` value
- VUID-vkGetPhysicalDeviceFormatProperties-pFormatProperties-parameter `pFormatProperties` **must** be a valid pointer to a `VkFormatProperties` structure

The `VkFormatProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkFormatProperties {
    VkFormatFeatureFlags    linearTilingFeatures;
    VkFormatFeatureFlags    optimalTilingFeatures;
    VkFormatFeatureFlags    bufferFeatures;
} VkFormatProperties;
```

- `linearTilingFeatures` is a bitmask of `VkFormatFeatureFlagBits` specifying features supported by images created with a `tiling` parameter of `VK_IMAGE_TILING_LINEAR`.
- `optimalTilingFeatures` is a bitmask of `VkFormatFeatureFlagBits` specifying features supported by images created with a `tiling` parameter of `VK_IMAGE_TILING_OPTIMAL`.
- `bufferFeatures` is a bitmask of `VkFormatFeatureFlagBits` specifying features supported by buffers.

NOTE

If no format feature flags are supported, the format itself is not supported, and images of that format cannot be created.

If **format** is a block-compressed format, then **bufferFeatures** **must** not support any features for the format.

If **format** is not a multi-plane format then **linearTilingFeatures** and **optimalTilingFeatures** **must** not contain **VK_FORMAT_FEATURE_DISJOINT_BIT**.

Bits which **can** be set in the **VkFormatProperties** features **linearTilingFeatures**, **optimalTilingFeatures**, **VkDrmFormatModifierPropertiesEXT::drmFormatModifierTilingFeatures**, and **bufferFeatures** are:

```
// Provided by VK_VERSION_1_0
typedef enum VkFormatFeatureFlagBits {
    VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT = 0x00000001,
    VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT = 0x00000002,
    VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT = 0x00000004,
    VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT = 0x00000008,
    VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT = 0x00000010,
    VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT = 0x00000020,
    VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT = 0x00000040,
    VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT = 0x00000080,
    VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT = 0x00000100,
    VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT = 0x00000200,
    VK_FORMAT_FEATURE_BLIT_SRC_BIT = 0x00000400,
    VK_FORMAT_FEATURE_BLIT_DST_BIT = 0x00000800,
    VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT = 0x00001000,
    VK_FORMAT_FEATURE_TRANSFER_SRC_BIT = 0x00004000,
    VK_FORMAT_FEATURE_TRANSFER_DST_BIT = 0x00008000,
    VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT = 0x00020000,
    VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT = 0x00040000,

    VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT =
    0x00080000,

    VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT =
    0x00100000,

    VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT =
    0x00200000,
    VK_FORMAT_FEATURE_DISJOINT_BIT = 0x00400000,
    VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT = 0x00800000,
    // Provided by VK_KHR_video_decode_queue
    VK_FORMAT_FEATURE_VIDEO_DECODE_OUTPUT_BIT_KHR = 0x02000000,
    // Provided by VK_KHR_video_decode_queue
    VK_FORMAT_FEATURE_VIDEO_DECODE_DPB_BIT_KHR = 0x04000000,
    // Provided by VK_KHR_acceleration_structure
    VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR = 0x20000000,
    // Provided by VK_EXT_filter_cubic
```

```

VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT = 0x00002000,
// Provided by VK_KHR_fragment_shading_rate
VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR = 0x40000000,
// Provided by VK_KHR_video_encode_queue
VK_FORMAT_FEATURE_VIDEO_ENCODE_INPUT_BIT_KHR = 0x08000000,
// Provided by VK_KHR_video_encode_queue
VK_FORMAT_FEATURE_VIDEO_ENCODE_DPB_BIT_KHR = 0x10000000,
// Provided by VK_KHR_maintenance1
VK_FORMAT_FEATURE_TRANSFER_SRC_BIT_KHR = VK_FORMAT_FEATURE_TRANSFER_SRC_BIT,
// Provided by VK_KHR_maintenance1
VK_FORMAT_FEATURE_TRANSFER_DST_BIT_KHR = VK_FORMAT_FEATURE_TRANSFER_DST_BIT,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT_KHR =
VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT_KHR =
VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT,
// Provided by VK_KHR_sampler_ycbcr_conversion

VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT_KH
R =
VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT,
// Provided by VK_KHR_sampler_ycbcr_conversion

VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT_KH
R =
VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT,
// Provided by VK_KHR_sampler_ycbcr_conversion

VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEA
BLE_BIT_KHR =
VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEA
BLE_BIT,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_FEATURE_DISJOINT_BIT_KHR = VK_FORMAT_FEATURE_DISJOINT_BIT,
// Provided by VK_KHR_sampler_ycbcr_conversion
VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT_KHR =
VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT,
} VkFormatFeatureFlagBits;

```

These values all have the same meaning as the equivalently named values for [VkFormatFeatureFlags2](#) and **may** be set in [linearTilingFeatures](#), [optimalTilingFeatures](#), and [VkDrmFormatModifierPropertiesEXT::drmFormatModifierTilingFeatures](#), specifying that the features are supported by [images](#) or [image views](#) or [sampler Y'C_BC_R conversion objects](#) created with the queried [vkGetPhysicalDeviceFormatProperties::format](#):

- [VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT](#) specifies that an image view **can** be [sampled from](#).
- [VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT](#) specifies that an image view **can** be used as a [storage image](#).

- `VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT` specifies that an image view **can** be used as storage image that supports atomic operations.
- `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT` specifies that an image view **can** be used as a framebuffer color attachment and as an input attachment.
- `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT` specifies that an image view **can** be used as a framebuffer color attachment that supports blending.
- `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT` specifies that an image view **can** be used as a framebuffer depth/stencil attachment and as an input attachment.
- `VK_FORMAT_FEATURE_BLIT_SRC_BIT` specifies that an image **can** be used as `srcImage` for the `vkCmdBlitImage2KHR` and `vkCmdBlitImage` commands.
- `VK_FORMAT_FEATURE_BLIT_DST_BIT` specifies that an image **can** be used as `dstImage` for the `vkCmdBlitImage2KHR` and `vkCmdBlitImage` commands.
- `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT` specifies that if `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT` is also set, an image view **can** be used with a sampler that has either of `magFilter` or `minFilter` set to `VK_FILTER_LINEAR`, or `mipmapMode` set to `VK_SAMPLER_MIPMAP_MODE_LINEAR`. If `VK_FORMAT_FEATURE_BLIT_SRC_BIT` is also set, an image can be used as the `srcImage` to `vkCmdBlitImage2KHR` and `vkCmdBlitImage` with a `filter` of `VK_FILTER_LINEAR`. This bit **must** only be exposed for formats that also support the `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT` or `VK_FORMAT_FEATURE_BLIT_SRC_BIT`.

If the format being queried is a depth/stencil format, this bit only specifies that the depth aspect (not the stencil aspect) of an image of this format supports linear filtering, and that linear filtering of the depth aspect is supported whether depth compare is enabled in the sampler or not. Where depth comparison is supported it **may** be linear filtered whether this bit is present or not, but where this bit is not present the filtered value **may** be computed in an implementation-dependent manner which differs from the normal rules of linear filtering. The resulting value **must** be in the range [0,1] and **should** be proportional to, or a weighted average of, the number of comparison passes or failures.

- `VK_FORMAT_FEATURE_TRANSFER_SRC_BIT` specifies that an image **can** be used as a source image for [copy commands](#). If the application `apiVersion` is Vulkan 1.0 and `VK_KHR_maintenance1` is not supported, `VK_FORMAT_FEATURE_TRANSFER_SRC_BIT` is implied to be set when the format feature flag is not 0.
- `VK_FORMAT_FEATURE_TRANSFER_DST_BIT` specifies that an image **can** be used as a destination image for [copy commands](#) and [clear commands](#). If the application `apiVersion` is Vulkan 1.0 and `VK_KHR_maintenance1` is not supported, `VK_FORMAT_FEATURE_TRANSFER_DST_BIT` is implied to be set when the format feature flag is not 0.
- `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT` specifies that `VkImage` **can** be used with a sampler that has either of `magFilter` or `minFilter` set to `VK_FILTER_CUBIC_EXT`, or be the source image for a blit with `filter` set to `VK_FILTER_CUBIC_EXT`. This bit **must** only be exposed for formats that also support the `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`. If the format being queried is a depth/stencil format, this only specifies that the depth aspect is cubic filterable.
- `VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT` specifies that an application **can** define a [sampler Y'C_BC_R conversion](#) using this format as a source, and that an image of this format **can** be used with a `VkSamplerYcbcrConversionCreateInfo` `xChromaOffset` and/or `yChromaOffset` of

VK_CHROMA_LOCATION_MIDPOINT. Otherwise both `xChromaOffset` and `yChromaOffset` **must** be VK_CHROMA_LOCATION_COSITED_EVEN. If a format does not incorporate chroma downsampling (it is not a “422” or “420” format) but the implementation supports sampler Y'C_BC_R conversion for this format, the implementation **must** set VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT.

- VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT specifies that an application **can** define a [sampler Y'C_BC_R conversion](#) using this format as a source, and that an image of this format **can** be used with a [VkSamplerYcbcrConversionCreateInfo](#) `xChromaOffset` and/or `yChromaOffset` of VK_CHROMA_LOCATION_COSITED_EVEN. Otherwise both `xChromaOffset` and `yChromaOffset` **must** be VK_CHROMA_LOCATION_MIDPOINT. If neither VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT nor VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT is set, the application **must** not define a [sampler Y'C_BC_R conversion](#) using this format as a source.
- VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT specifies that an application **can** define a [sampler Y'C_BC_R conversion](#) using this format as a source with `chromaFilter` set to VK_FILTER_LINEAR.
- VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT specifies that the format can have different chroma, min, and mag filters.
- VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT specifies that reconstruction is explicit, as described in [Chroma Reconstruction](#). If this bit is not present, reconstruction is implicit by default.
- VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT specifies that reconstruction **can** be forcibly made explicit by setting [VkSamplerYcbcrConversionCreateInfo::forceExplicitReconstruction](#) to VK_TRUE. If the format being queried supports VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT it **must** also support VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT.
- VK_FORMAT_FEATURE_DISJOINT_BIT specifies that a multi-planar image **can** have the VK_IMAGE_CREATE_DISJOINT_BIT set during image creation. An implementation **must** not set VK_FORMAT_FEATURE_DISJOINT_BIT for *single-plane formats*.
- VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR specifies that an image view **can** be used as a [fragment shading rate attachment](#). An implementation **must** not set this feature for formats with a [numeric format](#) other than UINT, or set it as a buffer feature.
- VK_FORMAT_FEATURE_VIDEO_DECODE_OUTPUT_BIT_KHR specifies that an image view with this format **can** be used as a [decode output picture](#) in [video decode operations](#).
- VK_FORMAT_FEATURE_VIDEO_DECODE_DPB_BIT_KHR specifies that an image view with this format **can** be used as an output [reconstructed picture](#) or an input [reference picture](#) in [video decode operations](#).
- VK_FORMAT_FEATURE_VIDEO_ENCODE_INPUT_BIT_KHR specifies that an image view with this format **can** be used as an [encode input picture](#) in [video encode operations](#).
- VK_FORMAT_FEATURE_VIDEO_ENCODE_DPB_BIT_KHR specifies that an image view with this format **can** be used as an output [reconstructed picture](#) or an input [reference picture](#) in [video encode operations](#).

NOTE

Specific [video profiles](#) **may** have additional restrictions on the format and other image creation parameters corresponding to image views used by video coding operations that **can** be enumerated using the [vkGetPhysicalDeviceVideoFormatPropertiesKHR](#) command.

The following bits **may** be set in [bufferFeatures](#), specifying that the features are supported by [buffers](#) or [buffer views](#) created with the queried [vkGetPhysicalDeviceFormatProperties::format](#):

- [VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT](#) specifies that the format **can** be used to create a buffer view that **can** be bound to a [VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER](#) descriptor.
- [VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT](#) specifies that the format **can** be used to create a buffer view that **can** be bound to a [VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER](#) descriptor.
- [VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT](#) specifies that atomic operations are supported on [VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER](#) with this format.
- [VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT](#) specifies that the format **can** be used as a vertex attribute format ([VkVertexInputAttributeDescription::format](#)).
- [VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR](#) specifies that the format **can** be used as the vertex format when creating an [acceleration structure](#) ([VkAccelerationStructureGeometryTrianglesDataKHR::vertexFormat](#)). This format **can** also be used as the vertex format in host memory when doing [host acceleration structure](#) builds.

NOTE

[VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT](#) and [VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT](#) are only intended to be advertised for single-component formats, since SPIR-V atomic operations require a scalar type.

```
// Provided by VK_VERSION_1_0
typedef VkFlags VkFormatFeatureFlags;
```

[VkFormatFeatureFlags](#) is a bitmask type for setting a mask of zero or more [VkFormatFeatureFlagBits](#).

To query supported format features which are properties of the physical device, call:

```
// Provided by VK_KHR_get_physical_device_properties2
void vkGetPhysicalDeviceFormatProperties2KHR(
    VkPhysicalDevice          physicalDevice,
    VkFormat                  format,
    VkFormatProperties2*      pFormatProperties);
```

- [physicalDevice](#) is the physical device from which to query the format properties.
- [format](#) is the format whose properties are queried.
- [pFormatProperties](#) is a pointer to a [VkFormatProperties2](#) structure in which physical device properties for [format](#) are returned.

`vkGetPhysicalDeviceFormatProperties2KHR` behaves similarly to `vkGetPhysicalDeviceFormatProperties`, with the ability to return extended information in a `pNext` chain of output structures.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceFormatProperties2-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceFormatProperties2-format-parameter `format` **must** be a valid `VkFormat` value
- VUID-vkGetPhysicalDeviceFormatProperties2-pFormatProperties-parameter `pFormatProperties` **must** be a valid pointer to a `VkFormatProperties2` structure

The `VkFormatProperties2` structure is defined as:

```
typedef struct VkFormatProperties2 {  
    VkStructureType    sType;  
    void*              pNext;  
    VkFormatProperties  formatProperties;  
} VkFormatProperties2;
```

or the equivalent

```
// Provided by VK_KHR_get_physical_device_properties2  
typedef VkFormatProperties2 VkFormatProperties2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `formatProperties` is a `VkFormatProperties` structure describing features supported by the requested format.

Valid Usage (Implicit)

- VUID-VkFormatProperties2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_2`
- VUID-VkFormatProperties2-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkDrmFormatModifierPropertiesList2EXT`, `VkDrmFormatModifierPropertiesListEXT`, or `VkFormatProperties3`
- VUID-VkFormatProperties2-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique

To obtain the list of [Linux DRM format modifiers](#) compatible with a [VkFormat](#), add a [VkDrmFormatModifierPropertiesListEXT](#) structure to the [pNext](#) chain of [VkFormatProperties2](#).

The [VkDrmFormatModifierPropertiesListEXT](#) structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkDrmFormatModifierPropertiesListEXT {
    VkStructureType      sType;
    void*                pNext;
    uint32_t             drmFormatModifierCount;
    VkDrmFormatModifierPropertiesEXT* pDrmFormatModifierProperties;
} VkDrmFormatModifierPropertiesListEXT;
```

- [sType](#) is a [VkStructureType](#) value identifying this structure.
- [pNext](#) is [NULL](#) or a pointer to a structure extending this structure.
- [drmFormatModifierCount](#) is an inout parameter related to the number of modifiers compatible with the [format](#), as described below.
- [pDrmFormatModifierProperties](#) is either [NULL](#) or a pointer to an array of [VkDrmFormatModifierPropertiesEXT](#) structures.

If [pDrmFormatModifierProperties](#) is [NULL](#), then the function returns in [drmFormatModifierCount](#) the number of modifiers compatible with the queried [format](#). Otherwise, the application **must** set [drmFormatModifierCount](#) to the length of the array [pDrmFormatModifierProperties](#); the function will write at most [drmFormatModifierCount](#) elements to the array, and will return in [drmFormatModifierCount](#) the number of elements written.

Among the elements in array [pDrmFormatModifierProperties](#), each returned [drmFormatModifier](#) **must** be unique.

Valid Usage (Implicit)

- VUID-VkDrmFormatModifierPropertiesListEXT-sType-sType
[sType](#) **must** be [VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_EXT](#)

The [VkDrmFormatModifierPropertiesEXT](#) structure describes properties of a [VkFormat](#) when that format is combined with a [Linux DRM format modifier](#). These properties, like those of [VkFormatProperties2](#), are independent of any particular image.

The [VkDrmFormatModifierPropertiesEXT](#) structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkDrmFormatModifierPropertiesEXT {
    uint64_t          drmFormatModifier;
    uint32_t          drmFormatModifierPlaneCount;
    VkFormatFeatureFlags  drmFormatModifierTilingFeatures;
} VkDrmFormatModifierPropertiesEXT;
```

- `drmFormatModifier` is a *Linux DRM format modifier*.
- `drmFormatModifierPlaneCount` is the number of *memory planes* in any image created with `format` and `drmFormatModifier`. An image's *memory plane count* is distinct from its *format plane count*, as explained below.
- `drmFormatModifierTilingFeatures` is a bitmask of `VkFormatFeatureFlagBits` that are supported by any image created with `format` and `drmFormatModifier`.

The returned `drmFormatModifierTilingFeatures` **must** contain at least one bit.

The implementation **must** not return `DRM_FORMAT_MOD_INVALID` in `drmFormatModifier`.

An image's *memory plane count* (as returned by `drmFormatModifierPlaneCount`) is distinct from its *format plane count* (in the sense of `multi-planar` Y'C_BC_R formats). In `VkImageAspectFlags`, each `VK_IMAGE_ASPECT_MEMORY_PLANE_i_BIT_EXT` represents a *memory plane* and each `VK_IMAGE_ASPECT_PLANE_i_BIT` a *format plane*.

An image's set of *format planes* is an ordered partition of the image's **content** into separable groups of format components. The ordered partition is encoded in the name of each `VkFormat`. For example, `VK_FORMAT_G8_B8R8_2PLANE_420_UNORM` contains two *format planes*; the first plane contains the green component and the second plane contains the blue component and red component. If the format name does not contain `PLANE`, then the format contains a single plane; for example, `VK_FORMAT_R8G8B8A8_UNORM`. Some commands, such as `vkCmdCopyBufferToImage`, do not operate on all format components in the image, but instead operate only on the *format planes* explicitly chosen by the application and operate on each *format plane* independently.

An image's set of *memory planes* is an ordered partition of the image's **memory** rather than the image's **content**. Each *memory plane* is a contiguous range of memory. The union of an image's *memory planes* is not necessarily contiguous.

If an image is `linear`, then the partition is the same for *memory planes* and for *format planes*. Therefore, if the returned `drmFormatModifier` is `DRM_FORMAT_MOD_LINEAR`, then `drmFormatModifierPlaneCount` **must** equal the *format plane count*, and `drmFormatModifierTilingFeatures` **must** be identical to the `VkFormatProperties2::linearTilingFeatures` returned in the same `pNext` chain.

If an image is `non-linear`, then the partition of the image's **memory** into *memory planes* is implementation-specific and **may** be unrelated to the partition of the image's **content** into *format planes*. For example, consider an image whose `format` is `VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM`, `tiling` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, whose `drmFormatModifier` is not `DRM_FORMAT_MOD_LINEAR`, and `flags` lacks `VK_IMAGE_CREATE_DISJOINT_BIT`. The image has 3 *format planes*, and commands such as `vkCmdCopyBufferToImage` act on each *format plane* independently as

if the data of each *format plane* were separable from the data of the other planes. In a straightforward implementation, the implementation **may** store the image's content in 3 adjacent *memory planes* where each *memory plane* corresponds exactly to a *format plane*. However, the implementation **may** also store the image's content in a single *memory plane* where all format components are combined using an implementation-private block-compressed format; or the implementation **may** store the image's content in a collection of 7 adjacent *memory planes* using an implementation-private sharding technique. Because the image is non-linear and non-disjoint, the implementation has much freedom when choosing the image's placement in memory.

The *memory plane count* applies to function parameters and structures only when the API specifies an explicit requirement on `drmFormatModifierPlaneCount`. In all other cases, the *memory plane count* is ignored.

The list of [Linux DRM format modifiers](#) compatible with a `VkFormat` can be obtained by adding a `VkDrmFormatModifierPropertiesList2EXT` structure to the `pNext` chain of `VkFormatProperties2`.

The `VkDrmFormatModifierPropertiesList2EXT` structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
typedef struct VkDrmFormatModifierPropertiesList2EXT {
    VkStructureType      sType;
    void*                pNext;
    uint32_t             drmFormatModifierCount;
    VkDrmFormatModifierProperties2EXT* pDrmFormatModifierProperties;
} VkDrmFormatModifierPropertiesList2EXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `drmFormatModifierCount` is an inout parameter related to the number of modifiers compatible with the `format`, as described below.
- `pDrmFormatModifierProperties` is either `NULL` or a pointer to an array of `VkDrmFormatModifierProperties2EXT` structures.

If `pDrmFormatModifierProperties` is `NULL`, the number of modifiers compatible with the queried `format` is returned in `drmFormatModifierCount`. Otherwise, the application **must** set `drmFormatModifierCount` to the length of the array `pDrmFormatModifierProperties`; the function will write at most `drmFormatModifierCount` elements to the array, and will return in `drmFormatModifierCount` the number of elements written.

Among the elements in array `pDrmFormatModifierProperties`, each returned `drmFormatModifier` **must** be unique.

Among the elements in array `pDrmFormatModifierProperties`, the bits reported in `drmFormatModifierTilingFeatures` **must** include the bits reported in the corresponding element of `VkDrmFormatModifierPropertiesListEXT::pDrmFormatModifierProperties`.

Valid Usage (Implicit)

- VUID-VkDrmFormatModifierPropertiesList2EXT-sType-sType
sType **must** be VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_2_EXT

The [VkDrmFormatModifierProperties2EXT](#) structure describes properties of a [VkFormat](#) when that format is combined with a [Linux DRM format modifier](#). These properties, like those of [VkFormatProperties2](#), are independent of any particular image.

The [VkDrmFormatModifierPropertiesEXT](#) structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
typedef struct VkDrmFormatModifierProperties2EXT {
    uint64_t          drmFormatModifier;
    uint32_t          drmFormatModifierPlaneCount;
    VkFormatFeatureFlags2  drmFormatModifierTilingFeatures;
} VkDrmFormatModifierProperties2EXT;
```

- `drmFormatModifier` is a *Linux DRM format modifier*.
- `drmFormatModifierPlaneCount` is the number of *memory planes* in any image created with `format` and `drmFormatModifier`. An image's *memory plane count* is distinct from its *format plane count*, as explained below.
- `drmFormatModifierTilingFeatures` is a bitmask of [VkFormatFeatureFlagBits2](#) that are supported by any image created with `format` and `drmFormatModifier`.

To query supported format extended features which are properties of the physical device, add [VkFormatProperties3](#) structure to the `pNext` chain of [VkFormatProperties2](#).

The [VkFormatProperties3](#) structure is defined as:

```
typedef struct VkFormatProperties3 {
    VkStructureType    sType;
    void*              pNext;
    VkFormatFeatureFlags2  linearTilingFeatures;
    VkFormatFeatureFlags2  optimalTilingFeatures;
    VkFormatFeatureFlags2  bufferFeatures;
} VkFormatProperties3;
```

or the equivalent

```
// Provided by VK_KHR_format_feature_flags2
typedef VkFormatProperties3 VkFormatProperties3KHR;
```

- `linearTilingFeatures` is a bitmask of [VkFormatFeatureFlagBits2](#) specifying features supported

by images created with a `tiling` parameter of `VK_IMAGE_TILING_LINEAR`.

- `optimalTilingFeatures` is a bitmask of `VkFormatFeatureFlagBits2` specifying features supported by images created with a `tiling` parameter of `VK_IMAGE_TILING_OPTIMAL`.
- `bufferFeatures` is a bitmask of `VkFormatFeatureFlagBits2` specifying features supported by buffers.

The bits reported in `linearTilingFeatures`, `optimalTilingFeatures` and `bufferFeatures` **must** include the bits reported in the corresponding fields of `VkFormatProperties2::formatProperties`.

Valid Usage (Implicit)

- VUID-VkFormatProperties3-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_3`

Bits which **can** be set in the `VkFormatProperties3` features `linearTilingFeatures`, `optimalTilingFeatures`, and `bufferFeatures` are:

```
// Flag bits for VkFormatFeatureFlagBits2
typedef VkFlags64 VkFormatFeatureFlagBits2;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT =
0x00000001ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT_KHR =
0x00000001ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_IMAGE_BIT =
0x00000002ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_IMAGE_BIT_KHR =
0x00000002ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_IMAGE_ATOMIC_BIT =
0x00000004ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_IMAGE_ATOMIC_BIT_KHR =
0x00000004ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_UNIFORM_TEXEL_BUFFER_BIT =
0x00000008ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_UNIFORM_TEXEL_BUFFER_BIT_KHR =
0x00000008ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_BIT =
0x00000010ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_BIT_KHR =
0x00000010ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_ATOMIC_BIT = 0x00000020ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_ATOMIC_BIT_KHR = 0x00000020ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VERTEX_BUFFER_BIT =
0x00000040ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VERTEX_BUFFER_BIT_KHR =
0x00000040ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BIT =
```

```

0x00000080ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BIT_KHR =
0x00000080ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BLEND_BIT =
0x00000100ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BLEND_BIT_KHR = 0x00000100ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_DEPTH_STENCIL_ATTACHMENT_BIT
= 0x00000200ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_DEPTH_STENCIL_ATTACHMENT_BIT_KHR = 0x00000200ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_BLIT_SRC_BIT =
0x00000400ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_BLIT_SRC_BIT_KHR =
0x00000400ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_BLIT_DST_BIT =
0x00000800ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_BLIT_DST_BIT_KHR =
0x00000800ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_LINEAR_BIT = 0x00001000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_LINEAR_BIT_KHR = 0x00001000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_CUBIC_BIT = 0x00002000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT = 0x00002000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_TRANSFER_SRC_BIT =
0x00004000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_TRANSFER_SRC_BIT_KHR =
0x00004000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_TRANSFER_DST_BIT =
0x00008000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_TRANSFER_DST_BIT_KHR =
0x00008000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_MINMAX_BIT = 0x00010000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_MINMAX_BIT_KHR = 0x00010000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_MIDPOINT_CHROMA_SAMPLES_BIT
= 0x00020000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_MIDPOINT_CHROMA_SAMPLES_BIT_KHR = 0x00020000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT = 0x00040000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT_KHR =
0x00040000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT
= 0x00080000ULL;

```

```

static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT_
KHR = 0x00080000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT
= 0x00100000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT_
KHR = 0x00100000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORC
EABLE_BIT = 0x00200000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORC
EABLE_BIT_KHR = 0x00200000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_DISJOINT_BIT =
0x00400000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_DISJOINT_BIT_KHR =
0x00400000ULL;
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_COSITED_CHROMA_SAMPLES_BIT =
0x00800000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_COSITED_CHROMA_SAMPLES_BIT_KHR = 0x00800000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT = 0x80000000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT_KHR = 0x80000000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT = 0x100000000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT_KHR = 0x100000000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT = 0x200000000ULL;
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT_KHR = 0x200000000ULL;
// Provided by VK_KHR_video_decode_queue with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VIDEO_DECODE_OUTPUT_BIT_KHR
= 0x02000000ULL;
// Provided by VK_KHR_video_decode_queue with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VIDEO_DECODE_DPB_BIT_KHR =
0x04000000ULL;
// Provided by VK_KHR_acceleration_structure with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR = 0x200000000ULL;
// Provided by VK_KHR_fragment_shading_rate with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2
VK_FORMAT_FEATURE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR = 0x400000000ULL;

```

```
// Provided by VK_EXT_host_image_copy
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_HOST_IMAGE_TRANSFER_BIT_EXT
= 0x400000000000ULL;
// Provided by VK_KHR_video_encode_queue with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VIDEO_ENCODE_INPUT_BIT_KHR =
0x080000000ULL;
// Provided by VK_KHR_video_encode_queue with VK_KHR_format_feature_flags2 or
VK_VERSION_1_3
static const VkFormatFeatureFlagBits2 VK_FORMAT_FEATURE_2_VIDEO_ENCODE_DPB_BIT_KHR =
0x100000000ULL;
```

or the equivalent

```
// Provided by VK_KHR_format_feature_flags2
typedef VkFormatFeatureFlagBits2 VkFormatFeatureFlagBits2KHR;
```

The following bits **may** be set in `linearTilingFeatures` and `optimalTilingFeatures`, specifying that the features are supported by `images` or `image views` or `sampler Y'CBCR conversion objects` created with the queried `vkGetPhysicalDeviceFormatProperties2KHR::format`:

- `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT` specifies that an image view **can** be `sampled from`.
- `VK_FORMAT_FEATURE_2_STORAGE_IMAGE_BIT` specifies that an image view **can** be used as a `storage image`.
- `VK_FORMAT_FEATURE_2_STORAGE_IMAGE_ATOMIC_BIT` specifies that an image view **can** be used as storage image that supports atomic operations.
- `VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BIT` specifies that an image view **can** be used as a framebuffer color attachment and as an input attachment.
- `VK_FORMAT_FEATURE_2_COLOR_ATTACHMENT_BLEND_BIT` specifies that an image view **can** be used as a framebuffer color attachment that supports blending.
- `VK_FORMAT_FEATURE_2_DEPTH_STENCIL_ATTACHMENT_BIT` specifies that an image view **can** be used as a framebuffer depth/stencil attachment and as an input attachment.
- `VK_FORMAT_FEATURE_2_BLIT_SRC_BIT` specifies that an image **can** be used as the `srcImage` for `vkCmdBlitImage2KHR` and `vkCmdBlitImage`.
- `VK_FORMAT_FEATURE_2_BLIT_DST_BIT` specifies that an image **can** be used as the `dstImage` for `vkCmdBlitImage2KHR` and `vkCmdBlitImage`.
- `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_LINEAR_BIT` specifies that if `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT` is also set, an image view **can** be used with a sampler that has either of `magFilter` or `minFilter` set to `VK_FILTER_LINEAR`, or `mipmapMode` set to `VK_SAMPLER_MIPMAP_MODE_LINEAR`. If `VK_FORMAT_FEATURE_2_BLIT_SRC_BIT` is also set, an image can be used as the `srcImage` for `vkCmdBlitImage2KHR` and `vkCmdBlitImage` with a `filter` of `VK_FILTER_LINEAR`. This bit **must** only be exposed for formats that also support the `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT` or `VK_FORMAT_FEATURE_2_BLIT_SRC_BIT`.

If the format being queried is a depth/stencil format, this bit only specifies that the depth aspect (not the stencil aspect) of an image of this format supports linear filtering. Where depth comparison is supported it **may** be linear filtered whether this bit is present or not, but where this bit is not present the filtered value **may** be computed in an implementation-dependent manner which differs from the normal rules of linear filtering. The resulting value **must** be in the range [0,1] and **should** be proportional to, or a weighted average of, the number of comparison passes or failures.

- `VK_FORMAT_FEATURE_2_TRANSFER_SRC_BIT` specifies that an image **can** be used as a source image for copy commands.
- `VK_FORMAT_FEATURE_2_TRANSFER_DST_BIT` specifies that an image **can** be used as a destination image for copy commands and clear commands.
- `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_FILTER_CUBIC_BIT` specifies that `VkImage` **can** be used with a sampler that has either of `magFilter` or `minFilter` set to `VK_FILTER_CUBIC_EXT`, or be the source image for a blit with `filter` set to `VK_FILTER_CUBIC_EXT`. This bit **must** only be exposed for formats that also support the `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_BIT`. If the format being queried is a depth/stencil format, this only specifies that the depth aspect is cubic filterable.
- `VK_FORMAT_FEATURE_2_MIDPOINT_CHROMA_SAMPLES_BIT` specifies that an application **can** define a sampler `Y'CBCR` conversion using this format as a source, and that an image of this format **can** be used with a `VkSamplerYcbcrConversionCreateInfo` `xChromaOffset` and/or `yChromaOffset` of `VK_CHROMA_LOCATION_MIDPOINT`. Otherwise both `xChromaOffset` and `yChromaOffset` **must** be `VK_CHROMA_LOCATION_COSITED_EVEN`. If a format does not incorporate chroma downsampling (it is not a “422” or “420” format) but the implementation supports sampler `Y'CBCR` conversion for this format, the implementation **must** set `VK_FORMAT_FEATURE_2_MIDPOINT_CHROMA_SAMPLES_BIT`.
- `VK_FORMAT_FEATURE_2_COSITED_CHROMA_SAMPLES_BIT` specifies that an application **can** define a sampler `Y'CBCR` conversion using this format as a source, and that an image of this format **can** be used with a `VkSamplerYcbcrConversionCreateInfo` `xChromaOffset` and/or `yChromaOffset` of `VK_CHROMA_LOCATION_COSITED_EVEN`. Otherwise both `xChromaOffset` and `yChromaOffset` **must** be `VK_CHROMA_LOCATION_MIDPOINT`. If neither `VK_FORMAT_FEATURE_2_COSITED_CHROMA_SAMPLES_BIT` nor `VK_FORMAT_FEATURE_2_MIDPOINT_CHROMA_SAMPLES_BIT` is set, the application **must** not define a sampler `Y'CBCR` conversion using this format as a source.
- `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT` specifies that an application **can** define a sampler `Y'CBCR` conversion using this format as a source with `chromaFilter` set to `VK_FILTER_LINEAR`.
- `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT` specifies that the format can have different chroma, min, and mag filters.
- `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT` specifies that reconstruction is explicit, as described in [Chroma Reconstruction](#). If this bit is not present, reconstruction is implicit by default.
- `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT` specifies that reconstruction **can** be forcibly made explicit by setting `VkSamplerYcbcrConversionCreateInfo::forceExplicitReconstruction` to `VK_TRUE`. If the format being queried supports `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT` it **must** also support

VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT.

- VK_FORMAT_FEATURE_2_DISJOINT_BIT specifies that a multi-planar image **can** have the VK_IMAGE_CREATE_DISJOINT_BIT set during image creation. An implementation **must** not set VK_FORMAT_FEATURE_2_DISJOINT_BIT for *single-plane formats*.
- VK_FORMAT_FEATURE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR specifies that an image view **can** be used as a [fragment shading rate attachment](#). An implementation **must** not set this feature for formats with a [numeric format](#) other than `UINT`, or set it as a buffer feature.
- VK_FORMAT_FEATURE_2_VIDEO_DECODE_OUTPUT_BIT_KHR specifies that an image view with this format **can** be used as a [decode output picture](#) in [video decode operations](#).
- VK_FORMAT_FEATURE_2_VIDEO_DECODE_DPB_BIT_KHR specifies that an image view with this format **can** be used as an output [reconstructed picture](#) or an input [reference picture](#) in [video decode operations](#).
- VK_FORMAT_FEATURE_2_VIDEO_ENCODE_INPUT_BIT_KHR specifies that an image view with this format **can** be used as an [encode input picture](#) in [video encode operations](#).
- VK_FORMAT_FEATURE_2_VIDEO_ENCODE_DPB_BIT_KHR specifies that an image view with this format **can** be used as an output [reconstructed picture](#) or an input [reference picture](#) in [video encode operations](#).

NOTE

Specific [video profiles](#) **may** have additional restrictions on the format and other image creation parameters corresponding to image views used by video coding operations that **can** be enumerated using the [vkGetPhysicalDeviceVideoFormatPropertiesKHR](#) command.

- VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT specifies that image views or buffer views created with this format **can** be used as [storage images](#) or [storage texel buffers](#) respectively for read operations without specifying a format.
- VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT specifies that image views or buffer views created with this format **can** be used as [storage images](#) or [storage texel buffers](#) respectively for write operations without specifying a format.
- VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT specifies that image views created with this format **can** be used for depth comparison performed by `OpImage*Dref*` instructions.
- VK_FORMAT_FEATURE_2_HOST_IMAGE_TRANSFER_BIT_EXT specifies that an image **can** be created with VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT.

The following bits **may** be set in [bufferFeatures](#), specifying that the features are supported by [buffers](#) or [buffer views](#) created with the queried [vkGetPhysicalDeviceFormatProperties2KHR::format](#):

- VK_FORMAT_FEATURE_2_UNIFORM_TEXEL_BUFFER_BIT specifies that the format **can** be used to create a buffer view that **can** be bound to a `VK_DESCRIPTOR_TYPE_UNIFORM_TEXEL_BUFFER` descriptor.
- VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_BIT specifies that the format **can** be used to create a buffer view that **can** be bound to a `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` descriptor.
- VK_FORMAT_FEATURE_2_STORAGE_TEXEL_BUFFER_ATOMIC_BIT specifies that atomic operations are

supported on `VK_DESCRIPTOR_TYPE_STORAGE_TEXEL_BUFFER` with this format.

- `VK_FORMAT_FEATURE_2_VERTEX_BUFFER_BIT` specifies that the format **can** be used as a vertex attribute format (`VkVertexInputAttributeDescription::format`).
- `VK_FORMAT_FEATURE_2_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR` specifies that the format **can** be used as the vertex format when creating an [acceleration structure](#) (`VkAccelerationStructureGeometryTrianglesDataKHR::vertexFormat`). This format **can** also be used as the vertex format in host memory when doing [host acceleration structure](#) builds.
- `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT` specifies that buffer views created with this format **can** be used as [storage texel buffers](#) for read operations without specifying a format.
- `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT` specifies that buffer views created with this format **can** be used as [storage texel buffers](#) for write operations without specifying a format.

```
typedef VkFlags64 VkFormatFeatureFlags2;
```

or the equivalent

```
// Provided by VK_KHR_format_feature_flags2
typedef VkFormatFeatureFlags2 VkFormatFeatureFlags2KHR;
```

`VkFormatFeatureFlags2` is a bitmask type for setting a mask of zero or more [VkFormatFeatureFlagBits2](#).

41.2.1. Potential Format Features

Some [valid usage conditions](#) depend on the format features supported by a [VkImage](#) whose [VkImageTiling](#) is unknown. In such cases the exact [VkFormatFeatureFlagBits](#) supported by the [VkImage](#) cannot be determined, so the valid usage conditions are expressed in terms of the *potential format features* of the [VkImage](#) format.

The *potential format features* of a [VkFormat](#) are defined as follows:

- The union of [VkFormatFeatureFlagBits](#) and [VkFormatFeatureFlagBits2](#), supported when the [VkImageTiling](#) is `VK_IMAGE_TILING_OPTIMAL` , `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, or `VK_IMAGE_TILING_LINEAR`

41.3. Required Format Support

Implementations **must** support at least the following set of features on the listed formats. For images, these features **must** be supported for every [VkImageType](#) (including arrayed and cube variants) unless otherwise noted. These features are supported on existing formats without needing to advertise an extension or needing to explicitly enable them. Support for additional functionality beyond the requirements listed here is queried using the [vkGetPhysicalDeviceFormatProperties](#) command.

NOTE

Unless otherwise excluded below, the required formats are supported for all `VkImageCreateFlags` values as long as those flag values are otherwise allowed.

The following tables show which feature bits **must** be supported for each format. Formats that are required to support `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT` **must** also support `VK_FORMAT_FEATURE_TRANSFER_SRC_BIT` and `VK_FORMAT_FEATURE_TRANSFER_DST_BIT`.

Table 68. Key for Format Feature Tables

□	This feature must be supported on the named format
†	This feature must be supported on at least some of the named formats, with more information in the table where the symbol appears
‡	This feature must be supported with some caveats or preconditions, with more information in the table where the symbol appears
§	This feature must be supported with some caveats or preconditions, with more information in the table where the symbol appears

Table 69. Feature Bits in `optimalTilingFeatures`

<code>VK_FORMAT_FEATURE_TRANSFER_SRC_BIT</code>
<code>VK_FORMAT_FEATURE_TRANSFER_DST_BIT</code>
<code>VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT</code>
<code>VK_FORMAT_FEATURE_BLIT_SRC_BIT</code>
<code>VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT</code>
<code>VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT</code>
<code>VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT</code>
<code>VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT</code>
<code>VK_FORMAT_FEATURE_BLIT_DST_BIT</code>
<code>VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT</code>
<code>VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT</code>

Table 70. Feature Bits in `bufferFeatures`

<code>VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT</code>
<code>VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT</code>
<code>VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT</code>
<code>VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT</code>

Table 71. Mandatory Format Support: Sub-Byte Components

[illegible]

Table 72. Mandatory Format Support: 1-3 Byte-Sized Components

[illegible]

VK_FORMAT_B8G8R8_USCALED																			
VK_FORMAT_B8G8R8_SSCALED																			
VK_FORMAT_B8G8R8_UINT																			
VK_FORMAT_B8G8R8_SINT																			
VK_FORMAT_B8G8R8_SRGB																			
VK_FORMAT_A8_UNORM_KHR																			
Format features marked with ‡ must be supported for optimalTilingFeatures if the VkPhysicalDevice supports the shaderStorageImageExtendedFormats feature.																			

Table 73. Mandatory Format Support: 4 Byte-Sized Components

[illegible]

Table 74. Mandatory Format Support: 10- and 12-Bit Components

VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT														
VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT														
VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT														
VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT														
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT														
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT														
VK_FORMAT_FEATURE_BLIT_DST_BIT														
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT														
VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT														
VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT														
VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT														
VK_FORMAT_FEATURE_BLIT_SRC_BIT														
VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT														
Format														
VK_FORMAT_A2R10G10B10_UNORM_PACK32														
VK_FORMAT_A2R10G10B10_SNORM_PACK32														
VK_FORMAT_A2R10G10B10_USCALED_PACK32														
VK_FORMAT_A2R10G10B10_SSCALED_PACK32														
VK_FORMAT_A2R10G10B10_UINT_PACK32														
VK_FORMAT_A2R10G10B10_SINT_PACK32														
VK_FORMAT_A2B10G10R10_UNORM_PACK32	‡	‡	‡	‡		‡	‡	‡		‡	‡			
VK_FORMAT_A2B10G10R10_SNORM_PACK32														
VK_FORMAT_A2B10G10R10_USCALED_PACK32														
VK_FORMAT_A2B10G10R10_SSCALED_PACK32														
VK_FORMAT_A2B10G10R10_UINT_PACK32	‡	‡		‡		‡	‡			‡				
VK_FORMAT_A2B10G10R10_SINT_PACK32														
VK_FORMAT_R10X6_UNORM_PACK16														
VK_FORMAT_R10X6G10X6_UNORM_2PACK16														
VK_FORMAT_R12X4_UNORM_PACK16														
VK_FORMAT_R12X4G12X4_UNORM_2PACK16														
Format features marked with ‡ must be supported for optimalTilingFeatures if the VkPhysicalDevice supports the shaderStorageImageExtendedFormats feature.														

Table 75. Mandatory Format Support: 16-bit Components

	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT											
	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT											
	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT											
	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT											
	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT											
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT											
	VK_FORMAT_FEATURE_BLIT_DST_BIT											
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT											
	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT											
	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT											
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT											
	VK_FORMAT_FEATURE_BLIT_SRC_BIT											
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT											
Format												
VK_FORMAT_R16_UNORM				‡					□			
VK_FORMAT_R16_SNORM				‡					□			
VK_FORMAT_R16_USCALED												
VK_FORMAT_R16_SSCALED												
VK_FORMAT_R16_UINT	□	□		‡		□	□		□	□		
VK_FORMAT_R16_SINT	□	□		‡		□	□		□	□		
VK_FORMAT_R16_SFLOAT	□	□	□	‡		□	□	□	□	□		
VK_FORMAT_R16G16_UNORM				‡					□			
VK_FORMAT_R16G16_SNORM				‡					□			
VK_FORMAT_R16G16_USCALED												
VK_FORMAT_R16G16_SSCALED												
VK_FORMAT_R16G16_UINT	□	□		‡		□	□		□	□		
VK_FORMAT_R16G16_SINT	□	□		‡		□	□		□	□		
VK_FORMAT_R16G16_SFLOAT	□	□	□	‡	§	□	□	□	□	□	§	§
VK_FORMAT_R16G16B16_UNORM												
VK_FORMAT_R16G16B16_SNORM												
VK_FORMAT_R16G16B16_USCALED												
VK_FORMAT_R16G16B16_SSCALED												
VK_FORMAT_R16G16B16_UINT												
VK_FORMAT_R16G16B16_SINT												
VK_FORMAT_R16G16B16_SFLOAT												
VK_FORMAT_R16G16B16A16_UNORM				‡					□			
VK_FORMAT_R16G16B16A16_SNORM				‡					□			

VK_FORMAT_R16G16B16A16_USCALED													
VK_FORMAT_R16G16B16A16_SSCALED													
VK_FORMAT_R16G16B16A16_UINT	□	□		□		□	□			□	□	□	
VK_FORMAT_R16G16B16A16_SINT	□	□		□		□	□			□	□	□	
VK_FORMAT_R16G16B16A16_SFLOAT	□	□	□	□	§	□	□	□		□	□	□	§

Format features marked with § **must** be supported for `optimalTilingFeatures` if the `VkPhysicalDevice` supports the `shaderStorageImageExtendedFormats` feature.

Table 76. Mandatory Format Support: 32-bit Components

VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT													
VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT													
VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT													
VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT													
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT													
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT													
VK_FORMAT_FEATURE_BLIT_DST_BIT													
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT													
VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT													
VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT													
VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT													
VK_FORMAT_FEATURE_BLIT_SRC_BIT													
VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT													
Format													
VK_FORMAT_R32_UINT	☐	☐		☐	☐	☐	☐			☐	☐	☐	☐
VK_FORMAT_R32_SINT	☐	☐		☐	☐	☐	☐			☐	☐	☐	☐
VK_FORMAT_R32_SFLOAT	☐	☐		☐	†	☐	☐			☐	☐	☐	
VK_FORMAT_R32G32_UINT	☐	☐		☐		☐	☐			☐	☐	☐	
VK_FORMAT_R32G32_SINT	☐	☐		☐		☐	☐			☐	☐	☐	
VK_FORMAT_R32G32_SFLOAT	☐	☐		☐		☐	☐			☐	☐	☐	
VK_FORMAT_R32G32B32_UINT										☐			
VK_FORMAT_R32G32B32_SINT										☐			
VK_FORMAT_R32G32B32_SFLOAT										☐			
VK_FORMAT_R32G32B32A32_UINT	☐	☐		☐		☐	☐			☐	☐	☐	
VK_FORMAT_R32G32B32A32_SINT	☐	☐		☐		☐	☐			☐	☐	☐	
VK_FORMAT_R32G32B32A32_SFLOAT	☐	☐		☐		☐	☐			☐	☐	☐	
Format features marked with † must be supported for optimalTilingFeatures if the VkPhysicalDevice supports the shaderImageFloat32Atomics or the shaderImageFloat32AtomicAdd or the shaderImageFloat32AtomicMinMax feature.													

Table 77. Mandatory Format Support: 64-bit/uneven Components

	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT												
	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT												
	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT												
	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT												
	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT												
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT												
	VK_FORMAT_FEATURE_BLIT_DST_BIT												
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT												
	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT												
	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT												
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT												
	VK_FORMAT_FEATURE_BLIT_SRC_BIT												
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT												
Format													
VK_FORMAT_R64_UINT				†	†								
VK_FORMAT_R64_SINT				†	†								
VK_FORMAT_R64_SFLOAT													
VK_FORMAT_R64G64_UINT													
VK_FORMAT_R64G64_SINT													
VK_FORMAT_R64G64_SFLOAT													
VK_FORMAT_R64G64B64_UINT													
VK_FORMAT_R64G64B64_SINT													
VK_FORMAT_R64G64B64_SFLOAT													
VK_FORMAT_R64G64B64A64_UINT													
VK_FORMAT_R64G64B64A64_SINT													
VK_FORMAT_R64G64B64A64_SFLOAT													
VK_FORMAT_B10G11R11_UFLOAT_PACK32	□	□	□	‡						□			
VK_FORMAT_E5B9G9R9_UFLOAT_PACK32	□	□	□										
Format features marked with ‡ must be supported for <code>optimalTilingFeatures</code> if the <code>VkPhysicalDevice</code> supports the <code>shaderStorageImageExtendedFormats</code> feature.													
If the <code>shaderImageInt64Atomics</code> feature is supported, <code>VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT</code> and <code>VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT</code> must be advertised in <code>optimalTilingFeatures</code> for both <code>VK_FORMAT_R64_UINT</code> and <code>VK_FORMAT_R64_SINT</code> .													

Table 78. Mandatory Format Support: Depth/Stencil With `VkImageType VK_IMAGE_TYPE_2D`

VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT													
VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT													
VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT													
VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT													
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT													
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT													
VK_FORMAT_FEATURE_BLIT_DST_BIT													
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT													
VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT													
VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT													
VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT													
VK_FORMAT_FEATURE_BLIT_SRC_BIT													
VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT													
Format													
VK_FORMAT_D16_UNORM													
VK_FORMAT_X8_D24_UNORM_PACK32													
VK_FORMAT_D32_SFLOAT													
VK_FORMAT_S8_UINT													
VK_FORMAT_D16_UNORM_S8_UINT													
VK_FORMAT_D24_UNORM_S8_UINT													
VK_FORMAT_D32_SFLOAT_S8_UINT													
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT feature must be supported for at least one of VK_FORMAT_X8_D24_UNORM_PACK32 and VK_FORMAT_D32_SFLOAT, and must be supported for at least one of VK_FORMAT_D24_UNORM_S8_UINT and VK_FORMAT_D32_SFLOAT_S8_UINT.													
bufferFeatures must not support any features for these formats													

Table 79. Mandatory Format Support: BC Compressed Formats With `VkImageType VK_IMAGE_TYPE_2D` and `VK_IMAGE_TYPE_3D`

[illegible]

Table 80. Mandatory Format Support: ETC2 and EAC Compressed Formats With VkImageType VK_IMAGE_TYPE_2D

	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT											
	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT											
	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT											
	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT											
	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT											
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT											
	VK_FORMAT_FEATURE_BLIT_DST_BIT											
	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT											
	VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT											
	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT											
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT											
	VK_FORMAT_FEATURE_BLIT_SRC_BIT											
	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT											
Format												
VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK	†	†	†									
VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK	†	†	†									
VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK	†	†	†									
VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK	†	†	†									
VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK	†	†	†									
VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK	†	†	†									
VK_FORMAT_EAC_R11_UNORM_BLOCK	†	†	†									
VK_FORMAT_EAC_R11_SNORM_BLOCK	†	†	†									
VK_FORMAT_EAC_R11G11_UNORM_BLOCK	†	†	†									
VK_FORMAT_EAC_R11G11_SNORM_BLOCK	†	†	†									

The VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT, VK_FORMAT_FEATURE_BLIT_SRC_BIT and VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT features **must** be supported in optimalTilingFeatures for all the formats in at least one of: this table, [Mandatory Format Support: BC Compressed Formats With VkImageType VK_IMAGE_TYPE_2D and VK_IMAGE_TYPE_3D](#), or [Mandatory Format Support: ASTC LDR Compressed Formats With VkImageType VK_IMAGE_TYPE_2D](#).

Table 81. Mandatory Format Support: ASTC LDR Compressed Formats With `VkImageType VK_IMAGE_TYPE_2D`

VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_ATOMIC_BIT													
VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT													
VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT													
VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT													
VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT													
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BLEND_BIT													
VK_FORMAT_FEATURE_BLIT_DST_BIT													
VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT													
VK_FORMAT_FEATURE_STORAGE_IMAGE_ATOMIC_BIT													
VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT													
VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT													
VK_FORMAT_FEATURE_BLIT_SRC_BIT													
VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT													
Format													
VK_FORMAT_ASTC_4x4_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_4x4_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_5x4_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_5x4_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_5x5_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_5x5_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_6x5_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_6x5_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_6x6_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_6x6_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x5_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x5_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x6_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x6_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x8_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_8x8_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x5_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x5_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x6_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x6_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x8_UNORM_BLOCK	†	†	†										

VK_FORMAT_ASTC_10x8_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x10_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_10x10_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x10_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x10_SRGB_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x12_UNORM_BLOCK	†	†	†										
VK_FORMAT_ASTC_12x12_SRGB_BLOCK	†	†	†										

The `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT`, `VK_FORMAT_FEATURE_BLIT_SRC_BIT` and `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_LINEAR_BIT` features **must** be supported in optimalTilingFeatures for all the formats in at least one of: this table, [Mandatory Format Support: BC Compressed Formats With VkImageType VK_IMAGE_TYPE_2D](#) and `VK_IMAGE_TYPE_3D`, or [Mandatory Format Support: ETC2 and EAC Compressed Formats With VkImageType VK_IMAGE_TYPE_2D](#).

If cubic filtering is supported, `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT` **must** be supported for the following image view types:

- `VK_IMAGE_VIEW_TYPE_2D`
- `VK_IMAGE_VIEW_TYPE_2D_ARRAY`

for the following formats:

- `VK_FORMAT_R4G4_UNORM_PACK8`
- `VK_FORMAT_R4G4B4A4_UNORM_PACK16`
- `VK_FORMAT_B4G4R4A4_UNORM_PACK16`
- `VK_FORMAT_R5G6B5_UNORM_PACK16`
- `VK_FORMAT_B5G6R5_UNORM_PACK16`
- `VK_FORMAT_R5G5B5A1_UNORM_PACK16`
- `VK_FORMAT_B5G5R5A1_UNORM_PACK16`
- `VK_FORMAT_A1R5G5B5_UNORM_PACK16`
- `VK_FORMAT_R8_UNORM`
- `VK_FORMAT_R8_SNORM`
- `VK_FORMAT_R8_SRGB`
- `VK_FORMAT_R8G8_UNORM`
- `VK_FORMAT_R8G8_SNORM`
- `VK_FORMAT_R8G8_SRGB`
- `VK_FORMAT_R8G8B8_UNORM`
- `VK_FORMAT_R8G8B8_SNORM`
- `VK_FORMAT_R8G8B8_SRGB`
- `VK_FORMAT_B8G8R8_UNORM`

- `VK_FORMAT_B8G8R8_SNORM`
- `VK_FORMAT_B8G8R8_SRGB`
- `VK_FORMAT_R8G8B8A8_UNORM`
- `VK_FORMAT_R8G8B8A8_SNORM`
- `VK_FORMAT_R8G8B8A8_SRGB`
- `VK_FORMAT_B8G8R8A8_UNORM`
- `VK_FORMAT_B8G8R8A8_SNORM`
- `VK_FORMAT_B8G8R8A8_SRGB`
- `VK_FORMAT_A8B8G8R8_UNORM_PACK32`
- `VK_FORMAT_A8B8G8R8_SNORM_PACK32`
- `VK_FORMAT_A8B8G8R8_SRGB_PACK32`

If ETC compressed formats are supported, `VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT` **must** be supported for the following image view types:

- `VK_IMAGE_VIEW_TYPE_2D`
- `VK_IMAGE_VIEW_TYPE_2D_ARRAY`

for the following additional formats:

- `VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK`
- `VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK`
- `VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK`
- `VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK`
- `VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK`
- `VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK`

If cubic filtering is supported for any other formats, the following image view types **must** be supported for those formats:

- `VK_IMAGE_VIEW_TYPE_2D`
- `VK_IMAGE_VIEW_TYPE_2D_ARRAY`

To be used with `VkImageView` with `subresourceRange.aspectMask` equal to `VK_IMAGE_ASPECT_COLOR_BIT`, **sampler Y'C_BC_R conversion** **must** be enabled for the following formats:

Table 82. Formats Requiring Sampler $Y'C_B C_R$ Conversion for `VK_IMAGE_ASPECT_COLOR_BIT` Image Views[illegible]

- VK_FORMAT_R8G8B8A8_SNORM
- VK_FORMAT_R8G8B8A8_UINT
- VK_FORMAT_R8G8B8A8_SINT
- VK_FORMAT_R32_UINT
- VK_FORMAT_R32_SINT
- VK_FORMAT_R32_SFLOAT
- VK_FORMAT_R32G32_UINT
- VK_FORMAT_R32G32_SINT
- VK_FORMAT_R32G32_SFLOAT
- VK_FORMAT_R32G32B32A32_UINT
- VK_FORMAT_R32G32B32A32_SINT
- VK_FORMAT_R32G32B32A32_SFLOAT
- VK_FORMAT_R16G16B16A16_UINT
- VK_FORMAT_R16G16B16A16_SINT
- VK_FORMAT_R16G16B16A16_SFLOAT
- VK_FORMAT_R16G16_SFLOAT
- VK_FORMAT_B10G11R11_UFLOAT_PACK32
- VK_FORMAT_R16_SFLOAT
- VK_FORMAT_R16G16B16A16_UNORM
- VK_FORMAT_A2B10G10R10_UNORM_PACK32
- VK_FORMAT_R16G16_UNORM
- VK_FORMAT_R8G8_UNORM
- VK_FORMAT_R16_UNORM
- VK_FORMAT_R8_UNORM
- VK_FORMAT_R16G16B16A16_SNORM
- VK_FORMAT_R16G16_SNORM
- VK_FORMAT_R8G8_SNORM
- VK_FORMAT_R16_SNORM
- VK_FORMAT_R8_SNORM
- VK_FORMAT_R16G16_SINT
- VK_FORMAT_R8G8_SINT
- VK_FORMAT_R16_SINT
- VK_FORMAT_R8_SINT
- VK_FORMAT_A2B10G10R10_UINT_PACK32
- VK_FORMAT_R16G16_UINT

- `VK_FORMAT_R8G8_UINT`
- `VK_FORMAT_R16_UINT`
- `VK_FORMAT_R8_UINT`
- `VK_FORMAT_A8_UNORM_KHR`

NOTE

This list of formats is the union of required storage formats from [Required Format Support](#) section and formats listed in `shaderStorageImageExtendedFormats`.

An implementation that supports `VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT` for any format from the given list of formats and supports `shaderStorageImageReadWithoutFormat` **must** support `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT` for that same format if Vulkan 1.3 or the `VK_KHR_format_feature_flags2` extension is supported.

An implementation that supports `VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT` for any format from the given list of formats and supports `shaderStorageImageWriteWithoutFormat` **must** support `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT` for that same format if Vulkan 1.3 or the `VK_KHR_format_feature_flags2` extension is supported.

An implementation that does not support either of `VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT` or `VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT` for a format **must** not report support for `VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT` or `VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT` for that format if it is not listed in the [SPIR-V and Vulkan Image Format Compatibility](#) table.

NOTE

Some older implementations do not follow this restriction. They report support for formats as storage images even though they do not support access without the `Format` qualifier and there is no matching `Format` token. Such images cannot be either read from or written to.

Drivers which pass Vulkan conformance test suite version 1.3.9.0, or any subsequent version will conform to the requirement above.

41.3.2. Depth Comparison Format Support

If Vulkan 1.3 or the `VK_KHR_format_feature_flags2` extension is supported, a depth/stencil format with a depth component supporting `VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT` **must** support `VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT`.

41.3.3. Format Feature Dependent Usage Flags

Certain resource usage flags depend on support for the corresponding format feature flag for the format in question. The following tables list the `VkBufferUsageFlagBits` and `VkImageUsageFlagBits` that have such dependencies, and the format feature flags they depend on. Additional restrictions, including, but not limited to, further required format feature flags specific to the particular use of the resource **may** apply, as described in the respective sections of this specification.

Table 83. Format Feature Dependent Buffer Usage Flags

Buffer usage flag	Required format feature flag
VK_BUFFER_USAGE_UNIFORM_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_UNIFORM_TEXEL_BUFFER_BIT
VK_BUFFER_USAGE_STORAGE_TEXEL_BUFFER_BIT	VK_FORMAT_FEATURE_STORAGE_TEXEL_BUFFER_BIT
VK_BUFFER_USAGE_VERTEX_BUFFER_BIT	VK_FORMAT_FEATURE_VERTEX_BUFFER_BIT

Table 84. Format Feature Dependent Image Usage Flags

Image usage flag	Required format feature flag
VK_IMAGE_USAGE_SAMPLED_BIT	VK_FORMAT_FEATURE_SAMPLED_IMAGE_BIT
VK_IMAGE_USAGE_STORAGE_BIT	VK_FORMAT_FEATURE_STORAGE_IMAGE_BIT
VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT
VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT	VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
VK_IMAGE_USAGE_INPUT_ATTACHMENT_BIT	VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT or VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT
VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR	VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR
VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR	VK_FORMAT_FEATURE_VIDEO_DECODE_OUTPUT_BIT_KHR
VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR	VK_FORMAT_FEATURE_VIDEO_DECODE_DPB_BIT_KHR
VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR	VK_FORMAT_FEATURE_VIDEO_ENCODE_INPUT_BIT_KHR
VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR	VK_FORMAT_FEATURE_VIDEO_ENCODE_DPB_BIT_KHR

Chapter 42. Additional Capabilities

This chapter describes additional capabilities beyond the minimum capabilities described in the [Limits](#) and [Formats](#) chapters, including:

- [Additional Image Capabilities](#)
- [Additional Buffer Capabilities](#)
- [Optional Semaphore Capabilities](#)
- [Optional Fence Capabilities](#)
- [Timestamp Calibration Capabilities](#)

42.1. Additional Image Capabilities

Additional image capabilities, such as larger dimensions or additional sample counts for certain image types, or additional capabilities for *linear* tiling format images, are described in this section.

To query additional capabilities specific to image types, call:

```
// Provided by VK_VERSION_1_0
VkResult vkGetPhysicalDeviceImageFormatProperties(
    VkPhysicalDevice    physicalDevice,
    VkFormat            format,
    VkImageType         type,
    VkImageTiling       tiling,
    VkImageUsageFlags   usage,
    VkImageCreateFlags   flags,
    VkImageFormatProperties* pImageFormatProperties);
```

- **physicalDevice** is the physical device from which to query the image capabilities.
- **format** is a [VkFormat](#) value specifying the image format, corresponding to [VkImageCreateInfo::format](#).
- **type** is a [VkImageType](#) value specifying the image type, corresponding to [VkImageCreateInfo::imageType](#).
- **tiling** is a [VkImageTiling](#) value specifying the image tiling, corresponding to [VkImageCreateInfo::tiling](#).
- **usage** is a bitmask of [VkImageUsageFlagBits](#) specifying the intended usage of the image, corresponding to [VkImageCreateInfo::usage](#).
- **flags** is a bitmask of [VkImageCreateFlagBits](#) specifying additional parameters of the image, corresponding to [VkImageCreateInfo::flags](#).
- **pImageFormatProperties** is a pointer to a [VkImageFormatProperties](#) structure in which capabilities are returned.

The **format**, **type**, **tiling**, **usage**, and **flags** parameters correspond to parameters that would be

consumed by `vkCreateImage` (as members of `VkImageCreateInfo`).

If `format` is not a supported image format, or if the combination of `format`, `type`, `tiling`, `usage`, and `flags` is not supported for images, then `vkGetPhysicalDeviceImageFormatProperties` returns `VK_ERROR_FORMAT_NOT_SUPPORTED`.

The limitations on an image format that are reported by `vkGetPhysicalDeviceImageFormatProperties` have the following property: if `usage1` and `usage2` of type `VkImageUsageFlags` are such that the bits set in `usage1` are a subset of the bits set in `usage2`, and `flags1` and `flags2` of type `VkImageCreateFlags` are such that the bits set in `flags1` are a subset of the bits set in `flags2`, then the limitations for `usage1` and `flags1` **must** be no more strict than the limitations for `usage2` and `flags2`, for all values of `format`, `type`, and `tiling`.

If `VK_EXT_host_image_copy` is supported, `usage` includes `VK_IMAGE_USAGE_SAMPLED_BIT`, and `flags` does not include either of `VK_IMAGE_CREATE_SPARSE_BINDING_BIT`, `VK_IMAGE_CREATE_SPARSE_RESIDENCY_BIT`, or `VK_IMAGE_CREATE_SPARSE_ALIASED_BIT`, then the result of calls to `vkGetPhysicalDeviceImageFormatProperties` with identical parameters except for the inclusion of `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` in `usage` **must** be identical.

Valid Usage

- VUID-vkGetPhysicalDeviceImageFormatProperties-tiling-02248
`tiling` **must** not be `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`. (Use `vkGetPhysicalDeviceImageFormatProperties2KHR` instead)

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceImageFormatProperties-physicalDevice-parameter
`physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceImageFormatProperties-format-parameter
`format` **must** be a valid `VkFormat` value
- VUID-vkGetPhysicalDeviceImageFormatProperties-type-parameter
`type` **must** be a valid `VkImageType` value
- VUID-vkGetPhysicalDeviceImageFormatProperties-tiling-parameter
`tiling` **must** be a valid `VkImageTiling` value
- VUID-vkGetPhysicalDeviceImageFormatProperties-usage-parameter
`usage` **must** be a valid combination of `VkImageUsageFlagBits` values
- VUID-vkGetPhysicalDeviceImageFormatProperties-usage-requiredbitmask
`usage` **must** not be 0
- VUID-vkGetPhysicalDeviceImageFormatProperties-flags-parameter
`flags` **must** be a valid combination of `VkImageCreateFlagBits` values
- VUID-vkGetPhysicalDeviceImageFormatProperties-pImageFormatProperties-parameter
`pImageFormatProperties` **must** be a valid pointer to a `VkImageFormatProperties` structure

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_FORMAT_NOT_SUPPORTED`

The `VkImageFormatProperties` structure is defined as:

```
// Provided by VK_VERSION_1_0
typedef struct VkImageFormatProperties {
    VkExtent3D          maxExtent;
    uint32_t            maxMipLevels;
    uint32_t            maxArrayLayers;
    VkSampleCountFlags  sampleCounts;
    VkDeviceSize        maxResourceSize;
} VkImageFormatProperties;
```

- `maxExtent` are the maximum image dimensions. See the [Allowed Extent Values](#) section below for how these values are constrained by `type`.
- `maxMipLevels` is the maximum number of mipmap levels. `maxMipLevels` **must** be equal to the number of levels in the complete mipmap chain based on the `maxExtent.width`, `maxExtent.height`, and `maxExtent.depth`, except when one of the following conditions is true, in which case it **may** instead be 1:
 - `vkGetPhysicalDeviceImageFormatProperties::tiling` was `VK_IMAGE_TILING_LINEAR`
 - `VkPhysicalDeviceImageFormatInfo2::tiling` was `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`
 - the `VkPhysicalDeviceImageFormatInfo2::pNext` chain included a `VkPhysicalDeviceExternalImageFormatInfo` structure with a handle type included in the `handleTypes` member for which mipmap image support is not required
 - image `format` is one of the [formats that require a sampler Y'C_BC_R conversion](#)
- `maxArrayLayers` is the maximum number of array layers. `maxArrayLayers` **must** be no less than `VkPhysicalDeviceLimits::maxImageArrayLayers`, except when one of the following conditions is true, in which case it **may** instead be 1:
 - `tiling` is `VK_IMAGE_TILING_LINEAR`
 - `tiling` is `VK_IMAGE_TILING_OPTIMAL` and `type` is `VK_IMAGE_TYPE_3D`
 - `format` is one of the [formats that require a sampler Y'C_BC_R conversion](#)
- If `tiling` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`, then `maxArrayLayers` **must** not be 0.
- `sampleCounts` is a bitmask of `VkSampleCountFlagBits` specifying all the supported sample counts

for this image as described [below](#).

- `maxResourceSize` is an upper bound on the total image size in bytes, inclusive of all image subresources. Implementations **may** have an address space limit on total size of a resource, which is advertised by this property. `maxResourceSize` **must** be at least 2^{31} .

NOTE

There is no mechanism to query the size of an image before creating it, to compare that size against `maxResourceSize`. If an application attempts to create an image that exceeds this limit, the creation will fail and `vkCreateImage` will return `VK_ERROR_OUT_OF_DEVICE_MEMORY`. While the advertised limit **must** be at least 2^{31} , it **may** not be possible to create an image that approaches that size, particularly for `VK_IMAGE_TYPE_1D`.

If the combination of parameters to `vkGetPhysicalDeviceImageFormatProperties` is not supported by the implementation for use in `vkCreateImage`, then all members of `VkImageFormatProperties` will be filled with zero.

NOTE

Filling `VkImageFormatProperties` with zero for unsupported formats is an exception to the usual rule that output structures have undefined contents on error. This exception was unintentional, but is preserved for backwards compatibility.

To query additional capabilities specific to image types, call:

```
// Provided by VK_KHR_get_physical_device_properties2
VkResult vkGetPhysicalDeviceImageFormatProperties2KHR(
    VkPhysicalDevice          physicalDevice,
    const VkPhysicalDeviceImageFormatInfo2* pImageFormatInfo,
    VkImageFormatProperties2* pImageFormatProperties);
```

- `physicalDevice` is the physical device from which to query the image capabilities.
- `pImageFormatInfo` is a pointer to a `VkPhysicalDeviceImageFormatInfo2` structure describing the parameters that would be consumed by `vkCreateImage`.
- `pImageFormatProperties` is a pointer to a `VkImageFormatProperties2` structure in which capabilities are returned.

`vkGetPhysicalDeviceImageFormatProperties2KHR` behaves similarly to `vkGetPhysicalDeviceImageFormatProperties`, with the ability to return extended information in a `pNext` chain of output structures.

If the `pNext` chain of `pImageFormatInfo` includes a `VkVideoProfileListInfoKHR` structure with a `profileCount` member greater than 0, then this command returns format capabilities specific to image types used in conjunction with the specified [video profiles](#). In this case, this command will return one of the [video-profile-specific error codes](#) if any of the profiles specified via `VkVideoProfileListInfoKHR::pProfiles` are not supported. Furthermore, if `VkPhysicalDeviceImageFormatInfo2::usage` includes any image usage flag not supported by the specified video profiles, then this command returns `VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR`.

Valid Usage

- VUID-vkGetPhysicalDeviceImageFormatProperties2-pNext-09004

If the `pNext` chain of `pImageFormatProperties` includes a `VkHostImageCopyDevicePerformanceQueryEXT` structure, `pImageFormatInfo->usage` **must** contain `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT`

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceImageFormatProperties2-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceImageFormatProperties2-pImageFormatInfo-parameter `pImageFormatInfo` **must** be a valid pointer to a valid `VkPhysicalDeviceImageFormatInfo2` structure
- VUID-vkGetPhysicalDeviceImageFormatProperties2-pImageFormatProperties-parameter `pImageFormatProperties` **must** be a valid pointer to a `VkImageFormatProperties2` structure

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`
- `VK_ERROR_FORMAT_NOT_SUPPORTED`
- `VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR`
- `VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR`
- `VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR`
- `VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR`
- `VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR`

The `VkPhysicalDeviceImageFormatInfo2` structure is defined as:

```
typedef struct VkPhysicalDeviceImageFormatInfo2 {
    VkStructureType    sType;
    const void*        pNext;
    VkFormat            format;
    VkImageType        type;
    VkImageTiling       tiling;
    VkImageUsageFlags  usage;
    VkImageCreateFlags  flags;
} VkPhysicalDeviceImageFormatInfo2;
```

or the equivalent

```
// Provided by VK_KHR_get_physical_device_properties2
typedef VkPhysicalDeviceImageFormatInfo2 VkPhysicalDeviceImageFormatInfo2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure. The `pNext` chain of `VkPhysicalDeviceImageFormatInfo2` is used to provide additional image parameters to `vkGetPhysicalDeviceImageFormatProperties2KHR`.
- `format` is a `VkFormat` value indicating the image format, corresponding to `VkImageCreateInfo::format`.
- `type` is a `VkImageType` value indicating the image type, corresponding to `VkImageCreateInfo::imageType`.
- `tiling` is a `VkImageTiling` value indicating the image tiling, corresponding to `VkImageCreateInfo::tiling`.
- `usage` is a bitmask of `VkImageUsageFlagBits` indicating the intended usage of the image, corresponding to `VkImageCreateInfo::usage`.
- `flags` is a bitmask of `VkImageCreateFlagBits` indicating additional parameters of the image, corresponding to `VkImageCreateInfo::flags`.

The members of `VkPhysicalDeviceImageFormatInfo2` correspond to the arguments to `vkGetPhysicalDeviceImageFormatProperties`, with `sType` and `pNext` added for extensibility.

Valid Usage

- VUID-VkPhysicalDeviceImageFormatInfo2-tiling-02249
`tiling` **must** be `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT` if and only if the `pNext` chain includes `VkPhysicalDeviceImageDrmFormatModifierInfoEXT`
- VUID-VkPhysicalDeviceImageFormatInfo2-tiling-02313
If `tiling` is `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT` and `flags` contains `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT`, then the `pNext` chain **must** include a `VkImageFormatListCreateInfo` structure with non-zero `viewFormatCount`

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageFormatInfo2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_FORMAT_INFO_2`
- VUID-VkPhysicalDeviceImageFormatInfo2-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkImageFormatListCreateInfo`, `VkPhysicalDeviceExternalImageFormatInfo`, `VkPhysicalDeviceImageFormatDrmFormatModifierInfoEXT`, `VkPhysicalDeviceImageViewImageFormatInfoEXT`, or `VkVideoProfileListInfoKHR`
- VUID-VkPhysicalDeviceImageFormatInfo2-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkPhysicalDeviceImageFormatInfo2-format-parameter
`format` **must** be a valid `VkFormat` value
- VUID-VkPhysicalDeviceImageFormatInfo2-type-parameter
`type` **must** be a valid `VkImageType` value
- VUID-VkPhysicalDeviceImageFormatInfo2-tiling-parameter
`tiling` **must** be a valid `VkImageTiling` value
- VUID-VkPhysicalDeviceImageFormatInfo2-usage-parameter
`usage` **must** be a valid combination of `VkImageUsageFlagBits` values
- VUID-VkPhysicalDeviceImageFormatInfo2-usage-requiredbitmask
`usage` **must** not be `0`
- VUID-VkPhysicalDeviceImageFormatInfo2-flags-parameter
`flags` **must** be a valid combination of `VkImageCreateFlagBits` values

The `VkImageFormatProperties2` structure is defined as:

```
typedef struct VkImageFormatProperties2 {  
    VkStructureType    sType;  
    void*              pNext;  
    VkImageFormatProperties    imageFormatProperties;  
} VkImageFormatProperties2;
```

or the equivalent

```
// Provided by VK_KHR_get_physical_device_properties2  
typedef VkImageFormatProperties2 VkImageFormatProperties2KHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure. The `pNext` chain of `VkImageFormatProperties2` is used to allow the specification of additional capabilities to be returned from `vkGetPhysicalDeviceImageFormatProperties2KHR`.

- `imageFormatProperties` is a `VkImageFormatProperties` structure in which capabilities are returned.

If the combination of parameters to `vkGetPhysicalDeviceImageFormatProperties2KHR` is not supported by the implementation for use in `vkCreateImage`, then all members of `imageFormatProperties` will be filled with zero.

NOTE

Filling `imageFormatProperties` with zero for unsupported formats is an exception to the usual rule that output structures have undefined contents on error. This exception was unintentional, but is preserved for backwards compatibility. This exception only applies to `imageFormatProperties`, not `sType`, `pNext`, or any structures chained from `pNext`.

Valid Usage (Implicit)

- VUID-VkImageFormatProperties2-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_IMAGE_FORMAT_PROPERTIES_2`
- VUID-VkImageFormatProperties2-pNext-pNext
Each `pNext` member of any structure (including this one) in the `pNext` chain **must** be either `NULL` or a pointer to a valid instance of `VkExternalImageFormatProperties`, `VkFilterCubicImageViewImageFormatPropertiesEXT`, `VkHostImageCopyDevicePerformanceQueryEXT`, or `VkSamplerYcbcrConversionImageFormatProperties`
- VUID-VkImageFormatProperties2-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique

To determine the image capabilities compatible with an external memory handle type, add a `VkPhysicalDeviceExternalImageFormatInfo` structure to the `pNext` chain of the `VkPhysicalDeviceImageFormatInfo2` structure and a `VkExternalImageFormatProperties` structure to the `pNext` chain of the `VkImageFormatProperties2` structure.

The `VkPhysicalDeviceExternalImageFormatInfo` structure is defined as:

```
typedef struct VkPhysicalDeviceExternalImageFormatInfo {
    VkStructureType           sType;
    const void*               pNext;
    VkExternalMemoryHandleTypeFlagBits handleType;
} VkPhysicalDeviceExternalImageFormatInfo;
```

or the equivalent

```
// Provided by VK_KHR_external_memory_capabilities
typedef VkPhysicalDeviceExternalImageFormatInfo
VkPhysicalDeviceExternalImageFormatInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleType` is a `VkExternalMemoryHandleTypeFlagBits` value specifying the memory handle type that will be used with the memory associated with the image.

If `handleType` is 0, `vkGetPhysicalDeviceImageFormatProperties2KHR` will behave as if `VkPhysicalDeviceExternalImageFormatInfo` was not present, and `VkExternalImageFormatProperties` will be ignored.

If `handleType` is not compatible with the `format`, `type`, `tiling`, `usage`, and `flags` specified in `VkPhysicalDeviceImageFormatInfo2`, then `vkGetPhysicalDeviceImageFormatProperties2KHR` returns `VK_ERROR_FORMAT_NOT_SUPPORTED`.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExternalImageFormatInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_IMAGE_FORMAT_INFO`
- VUID-VkPhysicalDeviceExternalImageFormatInfo-handleType-parameter
If `handleType` is not 0, `handleType` **must** be a valid `VkExternalMemoryHandleTypeFlagBits` value

Possible values of `VkPhysicalDeviceExternalImageFormatInfo::handleType`, specifying an external memory handle type, are:

```

typedef enum VkExternalMemoryHandleTypeFlagBits {
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT = 0x00000001,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT = 0x00000002,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT = 0x00000004,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT = 0x00000008,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT = 0x00000010,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT = 0x00000020,
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT = 0x00000040,
    // Provided by VK_EXT_external_memory_dma_buf
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT = 0x00000200,
    // Provided by VK_EXT_external_memory_host
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT = 0x00000080,
    // Provided by VK_EXT_external_memory_host
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT = 0x00000100,
    // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT,
    // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT,
    // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT,
    // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT,
    // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT,
    // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT,
    // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT_KHR =
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT,
} VkExternalMemoryHandleTypeFlagBits;

```

or the equivalent

```

// Provided by VK_KHR_external_memory_capabilities
typedef VkExternalMemoryHandleTypeFlagBits VkExternalMemoryHandleTypeFlagBitsKHR;

```

- **VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT** specifies a POSIX file descriptor handle that has only limited valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the POSIX system calls **dup**, **dup2**, **close**, and the non-standard system call **dup3**. Additionally, it **must** be transportable over a socket using an **SCM_RIGHTS** control message. It owns a reference to the underlying memory resource represented by its Vulkan memory object.
- **VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT** specifies an NT handle that has only limited

valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the functions `DuplicateHandle`, `CloseHandle`, `CompareObjectHandles`, `GetHandleInformation`, and `SetHandleInformation`. It owns a reference to the underlying memory resource represented by its Vulkan memory object.

- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT` specifies a global share handle that has only limited valid usage outside of Vulkan and other compatible APIs. It is not compatible with any native APIs. It does not own a reference to the underlying memory resource represented by its Vulkan memory object, and will therefore become invalid when all Vulkan memory objects associated with it are destroyed.
- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT` specifies an NT handle returned by `IDXGIResource1::CreateSharedHandle` referring to a Direct3D 10 or 11 texture resource. It owns a reference to the memory used by the Direct3D resource.
- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT` specifies a global share handle returned by `IDXGIResource::GetSharedHandle` referring to a Direct3D 10 or 11 texture resource. It does not own a reference to the underlying Direct3D resource, and will therefore become invalid when all Vulkan memory objects and Direct3D resources associated with it are destroyed.
- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT` specifies an NT handle returned by `ID3D12Device::CreateSharedHandle` referring to a Direct3D 12 heap resource. It owns a reference to the resources used by the Direct3D heap.
- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT` specifies an NT handle returned by `ID3D12Device::CreateSharedHandle` referring to a Direct3D 12 committed resource. It owns a reference to the memory used by the Direct3D resource.
- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT` specifies a host pointer returned by a host memory allocation command. It does not own a reference to the underlying memory resource, and will therefore become invalid if the host memory is freed.
- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT` specifies a host pointer to *host mapped foreign memory*. It does not own a reference to the underlying memory resource, and will therefore become invalid if the foreign memory is unmapped or otherwise becomes no longer available.
- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT` is a file descriptor for a Linux `dma_buf`. It owns a reference to the underlying memory resource represented by its Vulkan memory object.

Some external memory handle types can only be shared within the same underlying physical device and/or the same driver version, as defined in the following table:

Table 85. External Memory Handle Types Compatibility

Handle type	VkPhysicalDeviceIDProperties::driverUUID	VkPhysicalDeviceIDProperties::deviceUUID
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT	Must match	Must match
VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT	No restriction	No restriction
VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT	No restriction	No restriction
VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT	No restriction	No restriction

NOTE The above table does not restrict the drivers and devices with which `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT` and `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT` **may** be shared, as these handle types inherently mean memory that does not come from the same device, as they import memory from the host or a foreign device, respectively.

NOTE Even though the above table does not restrict the drivers and devices with which `VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT` **may** be shared, query mechanisms exist in the Vulkan API that prevent the import of incompatible dma-bufs (such as `vkGetMemoryFdPropertiesKHR`) and that prevent incompatible usage of dma-bufs (such as `VkPhysicalDeviceExternalBufferInfo` and `VkPhysicalDeviceExternalImageFormatInfo`).

```
typedef VkFlags VkExternalMemoryHandleTypeFlags;
```

or the equivalent

```
// Provided by VK_KHR_external_memory_capabilities
typedef VkExternalMemoryHandleTypeFlags VkExternalMemoryHandleTypeFlagsKHR;
```

`VkExternalMemoryHandleTypeFlags` is a bitmask type for setting a mask of zero or more `VkExternalMemoryHandleTypeFlagBits`.

The `VkExternalImageFormatProperties` structure is defined as:

```
typedef struct VkExternalImageFormatProperties {
    VkStructureType      sType;
    void*                pNext;
    VkExternalMemoryProperties externalMemoryProperties;
} VkExternalImageFormatProperties;
```

or the equivalent

```
// Provided by VK_KHR_external_memory_capabilities
typedef VkExternalImageFormatProperties VkExternalImageFormatPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `externalMemoryProperties` is a `VkExternalMemoryProperties` structure specifying various capabilities of the external handle type when used with the specified image creation parameters.

Valid Usage (Implicit)

- VUID-VkExternalImageFormatProperties-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EXTERNAL_IMAGE_FORMAT_PROPERTIES`

The `VkExternalMemoryProperties` structure is defined as:

```
typedef struct VkExternalMemoryProperties {
    VkExternalMemoryFeatureFlags      externalMemoryFeatures;
    VkExternalMemoryHandleTypeFlags   exportFromImportedHandleTypes;
    VkExternalMemoryHandleTypeFlags   compatibleHandleTypes;
} VkExternalMemoryProperties;
```

or the equivalent

```
// Provided by VK_KHR_external_memory_capabilities
typedef VkExternalMemoryProperties VkExternalMemoryPropertiesKHR;
```


- `externalMemoryFeatures` is a bitmask of `VkExternalMemoryFeatureFlagBits` specifying the features of `handleType`.
- `exportFromImportedHandleTypes` is a bitmask of `VkExternalMemoryHandleTypeFlagBits` specifying which types of imported handle `handleType` **can** be exported from.
- `compatibleHandleTypes` is a bitmask of `VkExternalMemoryHandleTypeFlagBits` specifying handle types which **can** be specified at the same time as `handleType` when creating an image compatible with external memory.

`compatibleHandleTypes` **must** include at least `handleType`. Inclusion of a handle type in `compatibleHandleTypes` does not imply the values returned in `VkImageFormatProperties2` will be the same when `VkPhysicalDeviceExternalImageFormatInfo::handleType` is set to that type. The application is responsible for querying the capabilities of all handle types intended for concurrent use in a single image and intersecting them to obtain the compatible set of capabilities.

Bits which **may** be set in `VkExternalMemoryProperties::externalMemoryFeatures`, specifying features of an external memory handle type, are:

```
typedef enum VkExternalMemoryFeatureFlagBits {
    VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT = 0x00000001,
    VK_EXTERNAL_MEMORY_FEATURE_EXPORTABLE_BIT = 0x00000002,
    VK_EXTERNAL_MEMORY_FEATURE_IMPORTABLE_BIT = 0x00000004,
    // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT_KHR =
VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT,
    // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_FEATURE_EXPORTABLE_BIT_KHR =
VK_EXTERNAL_MEMORY_FEATURE_EXPORTABLE_BIT,
    // Provided by VK_KHR_external_memory_capabilities
    VK_EXTERNAL_MEMORY_FEATURE_IMPORTABLE_BIT_KHR =
VK_EXTERNAL_MEMORY_FEATURE_IMPORTABLE_BIT,
} VkExternalMemoryFeatureFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_external_memory_capabilities
typedef VkExternalMemoryFeatureFlagBits VkExternalMemoryFeatureFlagBitsKHR;
```

- `VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT` specifies that images or buffers created with the specified parameters and handle type **must** use the mechanisms defined by `VkMemoryDedicatedRequirements` and `VkMemoryDedicatedAllocateInfo` to create (or import) a dedicated allocation for the image or buffer.
- `VK_EXTERNAL_MEMORY_FEATURE_EXPORTABLE_BIT` specifies that handles of this type **can** be exported from Vulkan memory objects.
- `VK_EXTERNAL_MEMORY_FEATURE_IMPORTABLE_BIT` specifies that handles of this type **can** be imported as Vulkan memory objects.

Because their semantics in external APIs roughly align with that of an image or buffer with a dedicated allocation in Vulkan, implementations are **required** to report `VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT` for the following external handle types:

- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT`
- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT`
- `VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT`

Implementations **must** not report `VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT` for images or buffers with external handle type `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT`, or `VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT`.

```
typedef VkFlags VkExternalMemoryFeatureFlags;
```

or the equivalent

```
// Provided by VK_KHR_external_memory_capabilities
typedef VkExternalMemoryFeatureFlags VkExternalMemoryFeatureFlagsKHR;
```

`VkExternalMemoryFeatureFlags` is a bitmask type for setting a mask of zero or more `VkExternalMemoryFeatureFlagBits`.

To query the image capabilities that are compatible with a [Linux DRM format modifier](#), set `VkPhysicalDeviceImageFormatInfo2::tiling` to `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT` and add a `VkPhysicalDeviceImageDrmFormatModifierInfoEXT` structure to the `pNext` chain of `VkPhysicalDeviceImageFormatInfo2`.

The `VkPhysicalDeviceImageDrmFormatModifierInfoEXT` structure is defined as:

```
// Provided by VK_EXT_image_drm_format_modifier
typedef struct VkPhysicalDeviceImageDrmFormatModifierInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    uint64_t            drmFormatModifier;
    VkSharingMode       sharingMode;
    uint32_t            queueFamilyIndexCount;
    const uint32_t*     pQueueFamilyIndices;
} VkPhysicalDeviceImageDrmFormatModifierInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `drmFormatModifier` is the image's *Linux DRM format modifier*, corresponding to `VkImageDrmFormatModifierExplicitCreateInfoEXT::modifier` or `VkImageDrmFormatModifierListCreateInfoEXT::pModifiers`.
- `sharingMode` specifies how the image will be accessed by multiple queue families.

- `queueFamilyIndexCount` is the number of entries in the `pQueueFamilyIndices` array.
- `pQueueFamilyIndices` is a pointer to an array of queue families that will access the image. It is ignored if `sharingMode` is not `VK_SHARING_MODE_CONCURRENT`.

If the `drmFormatModifier` is incompatible with the parameters specified in `VkPhysicalDeviceImageFormatInfo2` and its `pNext` chain, then `vkGetPhysicalDeviceImageFormatProperties2KHR` returns `VK_ERROR_FORMAT_NOT_SUPPORTED`. The implementation **must** support the query of any `drmFormatModifier`, including unknown and invalid modifier values.

Valid Usage

- VUID-VkPhysicalDeviceImageDrmFormatModifierInfoEXT-sharingMode-02314
If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, then `pQueueFamilyIndices` **must** be a valid pointer to an array of `queueFamilyIndexCount` `uint32_t` values
- VUID-VkPhysicalDeviceImageDrmFormatModifierInfoEXT-sharingMode-02315
If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, then `queueFamilyIndexCount` **must** be greater than 1
- VUID-VkPhysicalDeviceImageDrmFormatModifierInfoEXT-sharingMode-02316
If `sharingMode` is `VK_SHARING_MODE_CONCURRENT`, each element of `pQueueFamilyIndices` **must** be unique and **must** be less than the `pQueueFamilyPropertyCount` returned by `vkGetPhysicalDeviceQueueFamilyProperties2KHR` for the `physicalDevice` that was used to create `device`

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageDrmFormatModifierInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_DRM_FORMAT_MODIFIER_INFO_EXT`
- VUID-VkPhysicalDeviceImageDrmFormatModifierInfoEXT-sharingMode-parameter
`sharingMode` **must** be a valid `VkSharingMode` value

To determine the number of combined image samplers required to support a multi-planar format, add `VkSamplerYcbcrConversionImageFormatProperties` to the `pNext` chain of the `VkImageFormatProperties2` structure in a call to `vkGetPhysicalDeviceImageFormatProperties2KHR`.

The `VkSamplerYcbcrConversionImageFormatProperties` structure is defined as:

```
typedef struct VkSamplerYcbcrConversionImageFormatProperties {
    VkStructureType    sType;
    void*              pNext;
    uint32_t           combinedImageSamplerDescriptorCount;
} VkSamplerYcbcrConversionImageFormatProperties;
```

or the equivalent

```
// Provided by VK_KHR_sampler_ycbcr_conversion
typedef VkSamplerYcbcrConversionImageFormatProperties
VkSamplerYcbcrConversionImageFormatPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `combinedImageSamplerDescriptorCount` is the number of combined image sampler descriptors that the implementation uses to access the format.

Valid Usage (Implicit)

- VUID-VkSamplerYcbcrConversionImageFormatProperties-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_IMAGE_FORMAT_PROPERTIES`

`combinedImageSamplerDescriptorCount` is a number between 1 and the number of planes in the format. A descriptor set layout binding with immutable Y'C_BC_R conversion samplers will have a maximum `combinedImageSamplerDescriptorCount` which is the maximum across all formats supported by its samplers of the `combinedImageSamplerDescriptorCount` for each format. Descriptor sets with that layout will internally use that maximum `combinedImageSamplerDescriptorCount` descriptors for each descriptor in the binding. This expanded number of descriptors will be consumed from the descriptor pool when a descriptor set is allocated, and counts towards the `maxDescriptorSetSamplers`, `maxDescriptorSetSampledImages`, `maxPerStageDescriptorSamplers`, and `maxPerStageDescriptorSampledImages` limits.

NOTE

All descriptors in a binding use the same maximum `combinedImageSamplerDescriptorCount` descriptors to allow implementations to use a uniform stride for dynamic indexing of the descriptors in the binding.

For example, consider a descriptor set layout binding with two descriptors and immutable samplers for multi-planar formats that have `VkSamplerYcbcrConversionImageFormatProperties::combinedImageSamplerDescriptorCount` values of 2 and 3 respectively. There are two descriptors in the binding and the maximum `combinedImageSamplerDescriptorCount` is 3, so descriptor sets with this layout consume 6 descriptors from the descriptor pool. To create a descriptor pool that allows allocating four descriptor sets with this layout, `descriptorCount` must be at least 24.

Instead of querying all the potential formats that the application might use in the descriptor layout, the application **can** use the `VkPhysicalDeviceMaintenance6PropertiesKHR::maxCombinedImageSamplerDescriptorCount` property to determine the maximum descriptor size that will accommodate any and all `formats that require a sampler Y'CBCR conversion` supported by the implementation.

To query if using `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` has a negative impact on device performance when accessing an image, add `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` to

`VkPhysicalDeviceImageFormatInfo2::usage`, and add a `VkHostImageCopyDevicePerformanceQueryEXT` structure to the `pNext` chain of a `VkImageFormatProperties2` structure passed to `vkGetPhysicalDeviceImageFormatProperties2KHR`. This structure is defined as:

```
// Provided by VK_EXT_host_image_copy
typedef struct VkHostImageCopyDevicePerformanceQueryEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           optimalDeviceAccess;
    VkBool32           identicalMemoryLayout;
} VkHostImageCopyDevicePerformanceQueryEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `optimalDeviceAccess` returns `VK_TRUE` if use of host image copy has no adverse effect on device access performance, compared to an image that is created with exact same creation parameters, and bound to the same `VkDeviceMemory`, except that `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` is replaced with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` and `VK_IMAGE_USAGE_TRANSFER_DST_BIT`.
- `identicalMemoryLayout` returns `VK_TRUE` if use of host image copy has no impact on memory layout compared to an image that is created with exact same creation parameters, and bound to the same `VkDeviceMemory`, except that `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT` is replaced with `VK_IMAGE_USAGE_TRANSFER_SRC_BIT` and `VK_IMAGE_USAGE_TRANSFER_DST_BIT`.

The implementation **may** return `VK_FALSE` in `optimalDeviceAccess` if `identicalMemoryLayout` is `VK_FALSE`. If `identicalMemoryLayout` is `VK_TRUE`, `optimalDeviceAccess` **must** be `VK_TRUE`.

The implementation **may** return `VK_TRUE` in `optimalDeviceAccess` while `identicalMemoryLayout` is `VK_FALSE`. In this situation, any device performance impact **should** not be measurable.

If `VkPhysicalDeviceImageFormatInfo2::format` is a block-compressed format and `vkGetPhysicalDeviceImageFormatProperties2KHR` returns `VK_SUCCESS`, the implementation **must** return `VK_TRUE` in `optimalDeviceAccess`.

NOTE	Applications can make use of <code>optimalDeviceAccess</code> to determine their resource copying strategy. If a resource is expected to be accessed more on device than on the host, and the implementation considers the resource sub-optimally accessed, it is likely better to use device copies instead.
------	---

NOTE	Layout not being identical yet still considered optimal for device access could happen if the implementation has different memory layout patterns, some of which are easier to access on the host.
------	--

NOTE	The most practical reason for <code>optimalDeviceAccess</code> to be <code>VK_FALSE</code> is that host image access may disable framebuffer compression where it would otherwise have been enabled. This represents far more efficient host image access since no compression algorithm is required to read or write to the image, but it would impact device
------	--

access performance. Some implementations may only set `optimalDeviceAccess` to `VK_FALSE` if certain conditions are met, such as specific image usage flags or creation flags.

Valid Usage (Implicit)

- VUID-VkHostImageCopyDevicePerformanceQueryEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_HOST_IMAGE_COPY_DEVICE_PERFORMANCE_QUERY_EXT`

To determine if cubic filtering can be used with a given image format and a given image view type add a `VkPhysicalDeviceImageViewImageFormatInfoEXT` structure to the `pNext` chain of the `VkPhysicalDeviceImageFormatInfo2` structure, and a `VkFilterCubicImageViewImageFormatPropertiesEXT` structure to the `pNext` chain of the `VkImageFormatProperties2` structure.

The `VkPhysicalDeviceImageViewImageFormatInfoEXT` structure is defined as:

```
// Provided by VK_EXT_filter_cubic
typedef struct VkPhysicalDeviceImageViewImageFormatInfoEXT {
    VkStructureType    sType;
    void*              pNext;
    VkImageViewType     imageViewType;
} VkPhysicalDeviceImageViewImageFormatInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `imageViewType` is a `VkImageViewType` value specifying the type of the image view.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceImageViewImageFormatInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_IMAGE_FORMAT_INFO_EXT`
- VUID-VkPhysicalDeviceImageViewImageFormatInfoEXT-imageViewType-parameter
`imageViewType` **must** be a valid `VkImageViewType` value

The `VkFilterCubicImageViewImageFormatPropertiesEXT` structure is defined as:

```
// Provided by VK_EXT_filter_cubic
typedef struct VkFilterCubicImageViewImageFormatPropertiesEXT {
    VkStructureType    sType;
    void*              pNext;
    VkBool32           filterCubic;
    VkBool32           filterCubicMinmax;
} VkFilterCubicImageViewImageFormatPropertiesEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `filterCubic` tells if image format, image type and image view type **can** be used with cubic filtering. This field is set by the implementation. An application-specified value is ignored.
- `filterCubicMinmax` tells if image format, image type and image view type **can** be used with cubic filtering and minmax filtering. This field is set by the implementation. An application-specified value is ignored.

Valid Usage (Implicit)

- VUID-VkFilterCubicImageViewImageFormatPropertiesEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_FILTER_CUBIC_IMAGE_VIEW_IMAGE_FORMAT_PROPERTIES_EXT`

Valid Usage

- VUID-VkFilterCubicImageViewImageFormatPropertiesEXT-pNext-02627
If the `pNext` chain of the `VkImageFormatProperties2` structure includes a `VkFilterCubicImageViewImageFormatPropertiesEXT` structure, the `pNext` chain of the `VkPhysicalDeviceImageFormatInfo2` structure **must** include a `VkPhysicalDeviceImageViewImageFormatInfoEXT` structure with an `imageViewType` that is compatible with `imageType`

42.1.1. Supported Sample Counts

`vkGetPhysicalDeviceImageFormatProperties` returns a bitmask of `VkSampleCountFlagBits` in `sampleCounts` specifying the supported sample counts for the image parameters.

`sampleCounts` will be `VK_SAMPLE_COUNT_1_BIT` if at least one of the following conditions is true:

- `tiling` is `VK_IMAGE_TILING_LINEAR`
- `type` is not `VK_IMAGE_TYPE_2D`
- `flags` contains `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`
- Neither the `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT` flag nor the `VK_FORMAT_FEATURE_DEPTH_STENCIL_ATTACHMENT_BIT` flag in `VkFormatProperties::optimalTilingFeatures` returned by `vkGetPhysicalDeviceFormatProperties` is set

- `VkPhysicalDeviceExternalImageFormatInfo::handleType` is an external handle type for which multisampled image support is not required.
- `format` is one of the [formats that require a sampler Y'C_BC_R conversion](#)
- `usage` contains `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

Otherwise, the bits set in `sampleCounts` will be the sample counts supported for the specified values of `usage` and `format`. For each bit set in `usage`, the supported sample counts relate to the limits in `VkPhysicalDeviceLimits` as follows:

- If `usage` includes `VK_IMAGE_USAGE_COLOR_ATTACHMENT_BIT` and `format` is a floating- or fixed-point color format, a superset of `VkPhysicalDeviceLimits::framebufferColorSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, and `format` includes a depth component, a superset of `VkPhysicalDeviceLimits::framebufferDepthSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_DEPTH_STENCIL_ATTACHMENT_BIT`, and `format` includes a stencil component, a superset of `VkPhysicalDeviceLimits::framebufferStencilSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_SAMPLED_BIT`, and `format` includes a color component, a superset of `VkPhysicalDeviceLimits::sampledImageColorSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_SAMPLED_BIT`, and `format` includes a depth component, a superset of `VkPhysicalDeviceLimits::sampledImageDepthSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_SAMPLED_BIT`, and `format` is an integer format, a superset of `VkPhysicalDeviceLimits::sampledImageIntegerSampleCounts`
- If `usage` includes `VK_IMAGE_USAGE_STORAGE_BIT`, a superset of `VkPhysicalDeviceLimits::storageImageSampleCounts`

If multiple bits are set in `usage`, `sampleCounts` will be the intersection of the per-usage values described above.

If none of the bits described above are set in `usage`, then there is no corresponding limit in `VkPhysicalDeviceLimits`. In this case, `sampleCounts` **must** include at least `VK_SAMPLE_COUNT_1_BIT`.

42.1.2. Allowed Extent Values Based on Image Type

Implementations **may** support extent values larger than the [required minimum/maximum values](#) for certain types of images. `VkImageFormatProperties::maxExtent` for each type is subject to the constraints below.

NOTE

Implementations **must** support images with dimensions up to the [required minimum/maximum values](#) for all types of images. It follows that the query for additional capabilities **must** return extent values that are at least as large as the required values.

For `VK_IMAGE_TYPE_1D`:

- `maxExtent.width` ≥ `VkPhysicalDeviceLimits::maxImageDimension1D`
- `maxExtent.height` = 1

- `maxExtent.depth = 1`

For `VK_IMAGE_TYPE_2D` when `flags` does not contain `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`:

- `maxExtent.width ≥ VkPhysicalDeviceLimits::maxImageDimension2D`
- `maxExtent.height ≥ VkPhysicalDeviceLimits::maxImageDimension2D`
- `maxExtent.depth = 1`

For `VK_IMAGE_TYPE_2D` when `flags` contains `VK_IMAGE_CREATE_CUBE_COMPATIBLE_BIT`:

- `maxExtent.width ≥ VkPhysicalDeviceLimits::maxImageDimensionCube`
- `maxExtent.height ≥ VkPhysicalDeviceLimits::maxImageDimensionCube`
- `maxExtent.depth = 1`

For `VK_IMAGE_TYPE_3D`:

- `maxExtent.width ≥ VkPhysicalDeviceLimits::maxImageDimension3D`
- `maxExtent.height ≥ VkPhysicalDeviceLimits::maxImageDimension3D`
- `maxExtent.depth ≥ VkPhysicalDeviceLimits::maxImageDimension3D`

42.2. Additional Buffer Capabilities

To query the external handle types supported by buffers, call:

```
// Provided by VK_KHR_external_memory_capabilities
void vkGetPhysicalDeviceExternalBufferPropertiesKHR(
    VkPhysicalDevice          physicalDevice,
    const VkPhysicalDeviceExternalBufferInfo* pExternalBufferInfo,
    VkExternalBufferProperties* pExternalBufferProperties);
```

- `physicalDevice` is the physical device from which to query the buffer capabilities.
- `pExternalBufferInfo` is a pointer to a `VkPhysicalDeviceExternalBufferInfo` structure describing the parameters that would be consumed by `vkCreateBuffer`.
- `pExternalBufferProperties` is a pointer to a `VkExternalBufferProperties` structure in which capabilities are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceExternalBufferProperties-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceExternalBufferProperties-pExternalBufferInfo-parameter `pExternalBufferInfo` **must** be a valid pointer to a valid `VkPhysicalDeviceExternalBufferInfo` structure
- VUID-vkGetPhysicalDeviceExternalBufferProperties-pExternalBufferProperties-

parameter
`pExternalBufferProperties` **must** be a valid pointer to a `VkExternalBufferProperties` structure

The `VkPhysicalDeviceExternalBufferInfo` structure is defined as:

```
typedef struct VkPhysicalDeviceExternalBufferInfo {  
    VkStructureType          sType;  
    const void*              pNext;  
    VkBufferCreateFlags      flags;  
    VkBufferUsageFlags       usage;  
    VkExternalMemoryHandleTypeFlagBits handleType;  
} VkPhysicalDeviceExternalBufferInfo;
```

or the equivalent

```
// Provided by VK_KHR_external_memory_capabilities  
typedef VkPhysicalDeviceExternalBufferInfo VkPhysicalDeviceExternalBufferInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkBufferCreateFlagBits` describing additional parameters of the buffer, corresponding to `VkBufferCreateInfo::flags`.
- `usage` is a bitmask of `VkBufferUsageFlagBits` describing the intended usage of the buffer, corresponding to `VkBufferCreateInfo::usage`.
- `handleType` is a `VkExternalMemoryHandleTypeFlagBits` value specifying the memory handle type that will be used with the memory associated with the buffer.

Only usage flags representable in `VkBufferUsageFlagBits` are returned in this structure's `usage`. If the `pNext` chain includes a `VkBufferUsageFlags2CreateInfoKHR` structure, all usage flags of the buffer are returned in `VkBufferUsageFlags2CreateInfoKHR::usage`.

Valid Usage

- VUID-VkPhysicalDeviceExternalBufferInfo-None-09499
If the `pNext` chain does not include a `VkBufferUsageFlags2CreateInfoKHR` structure, `usage` **must** be a valid combination of `VkBufferUsageFlagBits` values
- VUID-VkPhysicalDeviceExternalBufferInfo-None-09500
If the `pNext` chain does not include a `VkBufferUsageFlags2CreateInfoKHR` structure, `usage` **must** not be 0

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExternalBufferInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_BUFFER_INFO`
- VUID-VkPhysicalDeviceExternalBufferInfo-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkBufferUsageFlags2CreateInfoKHR`
- VUID-VkPhysicalDeviceExternalBufferInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkPhysicalDeviceExternalBufferInfo-flags-parameter
`flags` **must** be a valid combination of `VkBufferCreateFlagBits` values
- VUID-VkPhysicalDeviceExternalBufferInfo-handleType-parameter
`handleType` **must** be a valid `VkExternalMemoryHandleTypeFlagBits` value

The `VkExternalBufferProperties` structure is defined as:

```
typedef struct VkExternalBufferProperties {  
    VkStructureType      sType;  
    void*                pNext;  
    VkExternalMemoryProperties externalMemoryProperties;  
} VkExternalBufferProperties;
```

or the equivalent

```
// Provided by VK_KHR_external_memory_capabilities  
typedef VkExternalBufferProperties VkExternalBufferPropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `externalMemoryProperties` is a `VkExternalMemoryProperties` structure specifying various capabilities of the external handle type when used with the specified buffer creation parameters.

Valid Usage (Implicit)

- VUID-VkExternalBufferProperties-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EXTERNAL_BUFFER_PROPERTIES`
- VUID-VkExternalBufferProperties-pNext-pNext
`pNext` **must** be `NULL`

42.3. Optional Semaphore Capabilities

Semaphores **may** support import and export of their [payload](#) to external handles. To query the external handle types supported by semaphores, call:

```
// Provided by VK_KHR_external_semaphore_capabilities
void vkGetPhysicalDeviceExternalSemaphorePropertiesKHR(
    VkPhysicalDevice          physicalDevice,
    const VkPhysicalDeviceExternalSemaphoreInfo* pExternalSemaphoreInfo,
    VkExternalSemaphoreProperties* pExternalSemaphoreProperties);
```

- `physicalDevice` is the physical device from which to query the semaphore capabilities.
- `pExternalSemaphoreInfo` is a pointer to a [VkPhysicalDeviceExternalSemaphoreInfo](#) structure describing the parameters that would be consumed by [vkCreateSemaphore](#).
- `pExternalSemaphoreProperties` is a pointer to a [VkExternalSemaphoreProperties](#) structure in which capabilities are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceExternalSemaphoreProperties-physicalDevice-parameter `physicalDevice` **must** be a valid [VkPhysicalDevice](#) handle
- VUID-vkGetPhysicalDeviceExternalSemaphoreProperties-pExternalSemaphoreInfo-parameter `pExternalSemaphoreInfo` **must** be a valid pointer to a valid [VkPhysicalDeviceExternalSemaphoreInfo](#) structure
- VUID-vkGetPhysicalDeviceExternalSemaphoreProperties-pExternalSemaphoreProperties-parameter `pExternalSemaphoreProperties` **must** be a valid pointer to a [VkExternalSemaphoreProperties](#) structure

The [VkPhysicalDeviceExternalSemaphoreInfo](#) structure is defined as:

```
typedef struct VkPhysicalDeviceExternalSemaphoreInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkExternalSemaphoreHandleTypeFlagBits handleType;
} VkPhysicalDeviceExternalSemaphoreInfo;
```

or the equivalent

```
// Provided by VK_KHR_external_semaphore_capabilities
typedef VkPhysicalDeviceExternalSemaphoreInfo
VkPhysicalDeviceExternalSemaphoreInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleType` is a `VkExternalSemaphoreHandleTypeFlagBits` value specifying the external semaphore handle type for which capabilities will be returned.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExternalSemaphoreInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_SEMAPHORE_INFO`
- VUID-VkPhysicalDeviceExternalSemaphoreInfo-pNext-pNext
`pNext` **must** be `NULL` or a pointer to a valid instance of `VkSemaphoreCreateInfo`
- VUID-VkPhysicalDeviceExternalSemaphoreInfo-sType-unique
The `sType` value of each struct in the `pNext` chain **must** be unique
- VUID-VkPhysicalDeviceExternalSemaphoreInfo-handleType-parameter
`handleType` **must** be a valid `VkExternalSemaphoreHandleTypeFlagBits` value

Bits which **may** be set in `VkPhysicalDeviceExternalSemaphoreInfo::handleType`, specifying an external semaphore handle type, are:

```
typedef enum VkExternalSemaphoreHandleTypeFlagBits {
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT = 0x00000001,
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT = 0x00000002,
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT = 0x00000004,
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT = 0x00000008,
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT = 0x00000010,
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D11_FENCE_BIT =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT,
    // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT,
    // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT,
    // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT,
    // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT,
    // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT,
} VkExternalSemaphoreHandleTypeFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_external_semaphore_capabilities
typedef VkExternalSemaphoreHandleTypeFlagBits
VkExternalSemaphoreHandleTypeFlagBitsKHR;
```

- `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT` specifies a POSIX file descriptor handle that has only limited valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the POSIX system calls `dup`, `dup2`, `close`, and the non-standard system call `dup3`. Additionally, it **must** be transportable over a socket using an `SCM_RIGHTS` control message. It owns a reference to the underlying synchronization primitive represented by its Vulkan semaphore object.
- `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT` specifies an NT handle that has only limited valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the functions `DuplicateHandle`, `CloseHandle`, `CompareObjectHandles`, `GetHandleInformation`, and `SetHandleInformation`. It owns a reference to the underlying synchronization primitive represented by its Vulkan semaphore object.
- `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT` specifies a global share handle that has only limited valid usage outside of Vulkan and other compatible APIs. It is not compatible with any native APIs. It does not own a reference to the underlying synchronization primitive represented by its Vulkan semaphore object, and will therefore become invalid when all Vulkan semaphore objects associated with it are destroyed.
- `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT` specifies an NT handle returned by `ID3D12Device::CreateSharedHandle` referring to a Direct3D 12 fence, or `ID3D11Device5::CreateFence` referring to a Direct3D 11 fence. It owns a reference to the underlying synchronization primitive associated with the Direct3D fence.
- `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D11_FENCE_BIT` is an alias of `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT` with the same meaning. It is provided for convenience and code clarity when interacting with D3D11 fences.
- `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT` specifies a POSIX file descriptor handle to a Linux Sync File or Android Fence object. It can be used with any native API accepting a valid sync file or fence as input. It owns a reference to the underlying synchronization primitive associated with the file descriptor. Implementations which support importing this handle type **must** accept any type of sync or fence FD supported by the native system they are running on.

NOTE

Handles of type `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT` generated by the implementation may represent either Linux Sync Files or Android Fences at the implementation's discretion. Applications **should** only use operations defined for both types of file descriptors, unless they know via means external to Vulkan the type of the file descriptor, or are prepared to deal with the system-defined operation failures resulting from using the wrong type.

Some external semaphore handle types can only be shared within the same underlying physical device and/or the same driver version, as defined in the following table:

Table 86. External Semaphore Handle Types Compatibility

Handle type	VkPhysicalDeviceIDProperties::driverUUID	VkPhysicalDeviceIDProperties::deviceUUID
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT	Must match	Must match
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT	Must match	Must match
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT	Must match	Must match
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT	Must match	Must match
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT	No restriction	No restriction
VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_ZIRCON_EVENT_BIT_FUCHSIA	No restriction	No restriction

```
typedef VkFlags VkExternalSemaphoreHandleTypeFlags;
```

or the equivalent

```
// Provided by VK_KHR_external_semaphore_capabilities
typedef VkExternalSemaphoreHandleTypeFlags VkExternalSemaphoreHandleTypeFlagsKHR;
```

`VkExternalSemaphoreHandleTypeFlags` is a bitmask type for setting a mask of zero or more `VkExternalSemaphoreHandleTypeFlagBits`.

The `VkExternalSemaphoreProperties` structure is defined as:

```
typedef struct VkExternalSemaphoreProperties {
    VkStructureType           sType;
    void*                     pNext;
    VkExternalSemaphoreHandleTypeFlags exportFromImportedHandleTypes;
    VkExternalSemaphoreHandleTypeFlags compatibleHandleTypes;
    VkExternalSemaphoreFeatureFlags externalSemaphoreFeatures;
} VkExternalSemaphoreProperties;
```

or the equivalent

```
// Provided by VK_KHR_external_semaphore_capabilities
typedef VkExternalSemaphoreProperties VkExternalSemaphorePropertiesKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.

- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `exportFromImportedHandleTypes` is a bitmask of `VkExternalSemaphoreHandleTypeFlagBits` specifying which types of imported handle `handleType` **can** be exported from.
- `compatibleHandleTypes` is a bitmask of `VkExternalSemaphoreHandleTypeFlagBits` specifying handle types which **can** be specified at the same time as `handleType` when creating a semaphore.
- `externalSemaphoreFeatures` is a bitmask of `VkExternalSemaphoreFeatureFlagBits` describing the features of `handleType`.

If `handleType` is not supported by the implementation, then `VkExternalSemaphoreProperties::externalSemaphoreFeatures` will be zero.

Valid Usage (Implicit)

- VUID-VkExternalSemaphoreProperties-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EXTERNAL_SEMAPHORE_PROPERTIES`
- VUID-VkExternalSemaphoreProperties-pNext-pNext
`pNext` **must** be `NULL`

Bits which **may** be set in `VkExternalSemaphoreProperties::externalSemaphoreFeatures`, specifying the features of an external semaphore handle type, are:

```
typedef enum VkExternalSemaphoreFeatureFlagBits {
    VK_EXTERNAL_SEMAPHORE_FEATURE_EXPORTABLE_BIT = 0x00000001,
    VK_EXTERNAL_SEMAPHORE_FEATURE_IMPORTABLE_BIT = 0x00000002,
    // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_FEATURE_EXPORTABLE_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_FEATURE_EXPORTABLE_BIT,
    // Provided by VK_KHR_external_semaphore_capabilities
    VK_EXTERNAL_SEMAPHORE_FEATURE_IMPORTABLE_BIT_KHR =
VK_EXTERNAL_SEMAPHORE_FEATURE_IMPORTABLE_BIT,
} VkExternalSemaphoreFeatureFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_external_semaphore_capabilities
typedef VkExternalSemaphoreFeatureFlagBits VkExternalSemaphoreFeatureFlagBitsKHR;
```

- `VK_EXTERNAL_SEMAPHORE_FEATURE_EXPORTABLE_BIT` specifies that handles of this type **can** be exported from Vulkan semaphore objects.
- `VK_EXTERNAL_SEMAPHORE_FEATURE_IMPORTABLE_BIT` specifies that handles of this type **can** be imported as Vulkan semaphore objects.

```
typedef VkFlags VkExternalSemaphoreFeatureFlags;
```

or the equivalent

```
// Provided by VK_KHR_external_semaphore_capabilities
typedef VkExternalSemaphoreFeatureFlags VkExternalSemaphoreFeatureFlagsKHR;
```

`VkExternalSemaphoreFeatureFlags` is a bitmask type for setting a mask of zero or more `VkExternalSemaphoreFeatureFlagBits`.

42.4. Optional Fence Capabilities

Fences **may** support import and export of their `payload` to external handles. To query the external handle types supported by fences, call:

```
// Provided by VK_KHR_external_fence_capabilities
void vkGetPhysicalDeviceExternalFencePropertiesKHR(
    VkPhysicalDevice          physicalDevice,
    const VkPhysicalDeviceExternalFenceInfo* pExternalFenceInfo,
    VkExternalFenceProperties* pExternalFenceProperties);
```

- `physicalDevice` is the physical device from which to query the fence capabilities.
- `pExternalFenceInfo` is a pointer to a `VkPhysicalDeviceExternalFenceInfo` structure describing the parameters that would be consumed by `vkCreateFence`.
- `pExternalFenceProperties` is a pointer to a `VkExternalFenceProperties` structure in which capabilities are returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceExternalFenceProperties-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceExternalFenceProperties-pExternalFenceInfo-parameter `pExternalFenceInfo` **must** be a valid pointer to a valid `VkPhysicalDeviceExternalFenceInfo` structure
- VUID-vkGetPhysicalDeviceExternalFenceProperties-pExternalFenceProperties-parameter `pExternalFenceProperties` **must** be a valid pointer to a `VkExternalFenceProperties` structure

The `VkPhysicalDeviceExternalFenceInfo` structure is defined as:

```
typedef struct VkPhysicalDeviceExternalFenceInfo {
    VkStructureType    sType;
    const void*        pNext;
    VkExternalFenceHandleTypeFlagBits handleType;
} VkPhysicalDeviceExternalFenceInfo;
```

or the equivalent

```
// Provided by VK_KHR_external_fence_capabilities
typedef VkPhysicalDeviceExternalFenceInfo VkPhysicalDeviceExternalFenceInfoKHR;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `handleType` is a `VkExternalFenceHandleTypeFlagBits` value specifying an external fence handle type for which capabilities will be returned.

NOTE

Handles of type `VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT` generated by the implementation may represent either Linux Sync Files or Android Fences at the implementation's discretion. Applications **should** only use operations defined for both types of file descriptors, unless they know via means external to Vulkan the type of the file descriptor, or are prepared to deal with the system-defined operation failures resulting from using the wrong type.

Valid Usage (Implicit)

- VUID-VkPhysicalDeviceExternalFenceInfo-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_FENCE_INFO`
- VUID-VkPhysicalDeviceExternalFenceInfo-pNext-pNext
`pNext` **must** be `NULL`
- VUID-VkPhysicalDeviceExternalFenceInfo-handleType-parameter
`handleType` **must** be a valid `VkExternalFenceHandleTypeFlagBits` value

Bits which **may** be set in

- `VkPhysicalDeviceExternalFenceInfo::handleType`
- `VkExternalFenceProperties::exportFromImportedHandleTypes`
- `VkExternalFenceProperties::compatibleHandleTypes`

indicate external fence handle types, and are:

```
typedef enum VkExternalFenceHandleTypeFlagBits {
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT = 0x00000001,
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT = 0x00000002,
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT = 0x00000004,
    VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT = 0x00000008,
    // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT_KHR =
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT,
    // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR =
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT,
    // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR =
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT,
    // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT_KHR =
VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT,
} VkExternalFenceHandleTypeFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_external_fence_capabilities
typedef VkExternalFenceHandleTypeFlagBits VkExternalFenceHandleTypeFlagBitsKHR;
```

- **VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT** specifies a POSIX file descriptor handle that has only limited valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the POSIX system calls **dup**, **dup2**, **close**, and the non-standard system call **dup3**. Additionally, it **must** be transportable over a socket using an **SCM_RIGHTS** control message. It owns a reference to the underlying synchronization primitive represented by its Vulkan fence object.
- **VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT** specifies an NT handle that has only limited valid usage outside of Vulkan and other compatible APIs. It **must** be compatible with the functions **DuplicateHandle**, **CloseHandle**, **CompareObjectHandles**, **GetHandleInformation**, and **SetHandleInformation**. It owns a reference to the underlying synchronization primitive represented by its Vulkan fence object.
- **VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT** specifies a global share handle that has only limited valid usage outside of Vulkan and other compatible APIs. It is not compatible with any native APIs. It does not own a reference to the underlying synchronization primitive represented by its Vulkan fence object, and will therefore become invalid when all Vulkan fence objects associated with it are destroyed.
- **VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT** specifies a POSIX file descriptor handle to a Linux Sync File or Android Fence. It can be used with any native API accepting a valid sync file or fence as input. It owns a reference to the underlying synchronization primitive associated with the file descriptor. Implementations which support importing this handle type **must** accept any type of sync or fence FD supported by the native system they are running on.

Some external fence handle types can only be shared within the same underlying physical device and/or the same driver version, as defined in the following table:

Table 87. External Fence Handle Types Compatibility

Handle type	VkPhysicalDeviceIDProperties::driverUUID	VkPhysicalDeviceIDProperties::deviceUUID
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT	Must match	Must match
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT	Must match	Must match
VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT	Must match	Must match
VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT	No restriction	No restriction

```
typedef VkFlags VkExternalFenceHandleTypeFlags;
```

or the equivalent

```
// Provided by VK_KHR_external_fence_capabilities
typedef VkExternalFenceHandleTypeFlags VkExternalFenceHandleTypeFlagsKHR;
```

`VkExternalFenceHandleTypeFlags` is a bitmask type for setting a mask of zero or more `VkExternalFenceHandleTypeFlagBits`.

The `VkExternalFenceProperties` structure is defined as:

```
typedef struct VkExternalFenceProperties {
    VkStructureType           sType;
    void*                     pNext;
    VkExternalFenceHandleTypeFlags exportFromImportedHandleTypes;
    VkExternalFenceHandleTypeFlags compatibleHandleTypes;
    VkExternalFenceFeatureFlags externalFenceFeatures;
} VkExternalFenceProperties;
```

or the equivalent

```
// Provided by VK_KHR_external_fence_capabilities
typedef VkExternalFenceProperties VkExternalFencePropertiesKHR;
```

- `exportFromImportedHandleTypes` is a bitmask of `VkExternalFenceHandleTypeFlagBits` indicating which types of imported handle `handleType` **can** be exported from.
- `compatibleHandleTypes` is a bitmask of `VkExternalFenceHandleTypeFlagBits` specifying handle types which **can** be specified at the same time as `handleType` when creating a fence.

- `externalFenceFeatures` is a bitmask of `VkExternalFenceFeatureFlagBits` indicating the features of `handleType`.

If `handleType` is not supported by the implementation, then `VkExternalFenceProperties::externalFenceFeatures` will be zero.

Valid Usage (Implicit)

- VUID-VkExternalFenceProperties-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_EXTERNAL_FENCE_PROPERTIES`
- VUID-VkExternalFenceProperties-pNext-pNext
`pNext` **must** be `NULL`

Bits which **may** be set in `VkExternalFenceProperties::externalFenceFeatures`, indicating features of a fence external handle type, are:

```
typedef enum VkExternalFenceFeatureFlagBits {
    VK_EXTERNAL_FENCE_FEATURE_EXPORTABLE_BIT = 0x00000001,
    VK_EXTERNAL_FENCE_FEATURE_IMPORTABLE_BIT = 0x00000002,
    // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_FEATURE_EXPORTABLE_BIT_KHR =
VK_EXTERNAL_FENCE_FEATURE_EXPORTABLE_BIT,
    // Provided by VK_KHR_external_fence_capabilities
    VK_EXTERNAL_FENCE_FEATURE_IMPORTABLE_BIT_KHR =
VK_EXTERNAL_FENCE_FEATURE_IMPORTABLE_BIT,
} VkExternalFenceFeatureFlagBits;
```

or the equivalent

```
// Provided by VK_KHR_external_fence_capabilities
typedef VkExternalFenceFeatureFlagBits VkExternalFenceFeatureFlagBitsKHR;
```

- `VK_EXTERNAL_FENCE_FEATURE_EXPORTABLE_BIT` specifies handles of this type **can** be exported from Vulkan fence objects.
- `VK_EXTERNAL_FENCE_FEATURE_IMPORTABLE_BIT` specifies handles of this type **can** be imported to Vulkan fence objects.

```
typedef VkFlags VkExternalFenceFeatureFlags;
```

or the equivalent

```
// Provided by VK_KHR_external_fence_capabilities
typedef VkExternalFenceFeatureFlags VkExternalFenceFeatureFlagsKHR;
```


`VkExternalFenceFeatureFlags` is a bitmask type for setting a mask of zero or more `VkExternalFenceFeatureFlagBits`.

42.5. Timestamp Calibration Capabilities

To query the set of time domains for which a physical device supports timestamp calibration, call:

```
// Provided by VK_KHR_calibrated_timestamps
VkResult vkGetPhysicalDeviceCalibrateableTimeDomainsKHR(
    VkPhysicalDevice          physicalDevice,
    uint32_t*                 pTimeDomainCount,
    VkTimeDomainKHR*          pTimeDomains);
```

- `physicalDevice` is the physical device from which to query the set of calibrateable time domains.
- `pTimeDomainCount` is a pointer to an integer related to the number of calibrateable time domains available or queried, as described below.
- `pTimeDomains` is either `NULL` or a pointer to an array of `VkTimeDomainKHR` values, indicating the supported calibrateable time domains.

If `pTimeDomains` is `NULL`, then the number of calibrateable time domains supported for the given `physicalDevice` is returned in `pTimeDomainCount`. Otherwise, `pTimeDomainCount` **must** point to a variable set by the application to the number of elements in the `pTimeDomains` array, and on return the variable is overwritten with the number of values actually written to `pTimeDomains`. If the value of `pTimeDomainCount` is less than the number of calibrateable time domains supported, at most `pTimeDomainCount` values will be written to `pTimeDomains`, and `VK_INCOMPLETE` will be returned instead of `VK_SUCCESS`, to indicate that not all the available time domains were returned.

Valid Usage (Implicit)

- VUID-vkGetPhysicalDeviceCalibrateableTimeDomainsKHR-physicalDevice-parameter `physicalDevice` **must** be a valid `VkPhysicalDevice` handle
- VUID-vkGetPhysicalDeviceCalibrateableTimeDomainsKHR-pTimeDomainCount-parameter `pTimeDomainCount` **must** be a valid pointer to a `uint32_t` value
- VUID-vkGetPhysicalDeviceCalibrateableTimeDomainsKHR-pTimeDomains-parameter
If the value referenced by `pTimeDomainCount` is not `0`, and `pTimeDomains` is not `NULL`, `pTimeDomains` **must** be a valid pointer to an array of `pTimeDomainCount` `VkTimeDomainKHR` values

Return Codes

Success

- `VK_SUCCESS`

- VK_INCOMPLETE

Failure

- VK_ERROR_OUT_OF_HOST_MEMORY
- VK_ERROR_OUT_OF_DEVICE_MEMORY

Chapter 43. Debugging

To aid developers in tracking down errors in the application's use of Vulkan, particularly in combination with an external debugger or profiler, *debugging extensions* may be available.

The `VkObjectType` enumeration defines values, each of which corresponds to a specific Vulkan handle type. These values **can** be used to associate debug information with a particular type of object through one or more extensions.

```
// Provided by VK_VERSION_1_0
typedef enum VkObjectType {
    VK_OBJECT_TYPE_UNKNOWN = 0,
    VK_OBJECT_TYPE_INSTANCE = 1,
    VK_OBJECT_TYPE_PHYSICAL_DEVICE = 2,
    VK_OBJECT_TYPE_DEVICE = 3,
    VK_OBJECT_TYPE_QUEUE = 4,
    VK_OBJECT_TYPE_SEMAPHORE = 5,
    VK_OBJECT_TYPE_COMMAND_BUFFER = 6,
    VK_OBJECT_TYPE_FENCE = 7,
    VK_OBJECT_TYPE_DEVICE_MEMORY = 8,
    VK_OBJECT_TYPE_BUFFER = 9,
    VK_OBJECT_TYPE_IMAGE = 10,
    VK_OBJECT_TYPE_EVENT = 11,
    VK_OBJECT_TYPE_QUERY_POOL = 12,
    VK_OBJECT_TYPE_BUFFER_VIEW = 13,
    VK_OBJECT_TYPE_IMAGE_VIEW = 14,
    VK_OBJECT_TYPE_SHADER_MODULE = 15,
    VK_OBJECT_TYPE_PIPELINE_CACHE = 16,
    VK_OBJECT_TYPE_PIPELINE_LAYOUT = 17,
    VK_OBJECT_TYPE_RENDER_PASS = 18,
    VK_OBJECT_TYPE_PIPELINE = 19,
    VK_OBJECT_TYPE_DESCRIPTOR_SET_LAYOUT = 20,
    VK_OBJECT_TYPE_SAMPLER = 21,
    VK_OBJECT_TYPE_DESCRIPTOR_POOL = 22,
    VK_OBJECT_TYPE_DESCRIPTOR_SET = 23,
    VK_OBJECT_TYPE_FRAMEBUFFER = 24,
    VK_OBJECT_TYPE_COMMAND_POOL = 25,
    VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION = 1000156000,
    VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE = 1000085000,
// Provided by VK_KHR_surface
    VK_OBJECT_TYPE_SURFACE_KHR = 1000000000,
// Provided by VK_KHR_swapchain
    VK_OBJECT_TYPE_SWAPCHAIN_KHR = 1000001000,
// Provided by VK_KHR_display
    VK_OBJECT_TYPE_DISPLAY_KHR = 1000002000,
// Provided by VK_KHR_display
    VK_OBJECT_TYPE_DISPLAY_MODE_KHR = 1000002001,
// Provided by VK_KHR_video_queue
    VK_OBJECT_TYPE_VIDEO_SESSION_KHR = 1000023000,
// Provided by VK_KHR_video_queue
```

```

    VK_OBJECT_TYPE_VIDEO_SESSION_PARAMETERS_KHR = 1000023001,
// Provided by VK_EXT_debug_utils
    VK_OBJECT_TYPE_DEBUG_UTILS_MESSENGER_EXT = 1000128000,
// Provided by VK_KHR_acceleration_structure
    VK_OBJECT_TYPE_ACCELERATION_STRUCTURE_KHR = 1000150000,
// Provided by VK_KHR_deferred_host_operations
    VK_OBJECT_TYPE_DEFERRED_OPERATION_KHR = 1000268000,
// Provided by VK_EXT_opacity_micromap
    VK_OBJECT_TYPE_MICROMAP_EXT = 1000396000,
// Provided by VK_EXT_shader_object
    VK_OBJECT_TYPE_SHADER_EXT = 1000482000,
// Provided by VK_KHR_pipeline_binary
    VK_OBJECT_TYPE_PIPELINE_BINARY_KHR = 1000483000,
// Provided by VK_EXT_device_generated_commands
    VK_OBJECT_TYPE_INDIRECT_COMMANDS_LAYOUT_EXT = 1000572000,
// Provided by VK_EXT_device_generated_commands
    VK_OBJECT_TYPE_INDIRECT_EXECUTION_SET_EXT = 1000572001,
// Provided by VK_KHR_descriptor_update_template
    VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_KHR =
VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE,
// Provided by VK_KHR_sampler_ycbcr_conversion
    VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION_KHR =
VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION,
} VkObjectType;

```

Table 88. *VkObjectType* and Vulkan Handle Relationship

VkObjectType	Vulkan Handle Type
VK_OBJECT_TYPE_UNKNOWN	Unknown/Undefined Handle
VK_OBJECT_TYPE_INSTANCE	VkInstance
VK_OBJECT_TYPE_PHYSICAL_DEVICE	VkPhysicalDevice
VK_OBJECT_TYPE_DEVICE	VkDevice
VK_OBJECT_TYPE_QUEUE	VkQueue
VK_OBJECT_TYPE_SEMAPHORE	VkSemaphore
VK_OBJECT_TYPE_COMMAND_BUFFER	VkCommandBuffer
VK_OBJECT_TYPE_FENCE	VkFence
VK_OBJECT_TYPE_DEVICE_MEMORY	VkDeviceMemory
VK_OBJECT_TYPE_BUFFER	VkBuffer
VK_OBJECT_TYPE_IMAGE	VkImage
VK_OBJECT_TYPE_EVENT	VkEvent
VK_OBJECT_TYPE_QUERY_POOL	VkQueryPool
VK_OBJECT_TYPE_BUFFER_VIEW	VkBufferView
VK_OBJECT_TYPE_IMAGE_VIEW	VkImageView

VkObjectType	Vulkan Handle Type
VK_OBJECT_TYPE_SHADER_MODULE	VkShaderModule
VK_OBJECT_TYPE_PIPELINE_CACHE	VkPipelineCache
VK_OBJECT_TYPE_PIPELINE_LAYOUT	VkPipelineLayout
VK_OBJECT_TYPE_RENDER_PASS	VkRenderPass
VK_OBJECT_TYPE_PIPELINE	VkPipeline
VK_OBJECT_TYPE_DESCRIPTOR_SET_LAYOUT	VkDescriptorSetLayout
VK_OBJECT_TYPE_SAMPLER	VkSampler
VK_OBJECT_TYPE_DESCRIPTOR_POOL	VkDescriptorPool
VK_OBJECT_TYPE_DESCRIPTOR_SET	VkDescriptorSet
VK_OBJECT_TYPE_FRAMEBUFFER	VkFramebuffer
VK_OBJECT_TYPE_COMMAND_POOL	VkCommandPool
VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION	VkSamplerYcbcrConversion
VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE	VkDescriptorUpdateTemplate
VK_OBJECT_TYPE_SURFACE_KHR	VkSurfaceKHR
VK_OBJECT_TYPE_SWAPCHAIN_KHR	VkSwapchainKHR
VK_OBJECT_TYPE_DISPLAY_KHR	VkDisplayKHR
VK_OBJECT_TYPE_DISPLAY_MODE_KHR	VkDisplayModeKHR
VK_OBJECT_TYPE_VIDEO_SESSION_KHR	VkVideoSessionKHR
VK_OBJECT_TYPE_VIDEO_SESSION_PARAMETERS_KHR	VkVideoSessionParametersKHR
VK_OBJECT_TYPE_DEBUG_UTILS_MESSENGER_EXT	VkDebugUtilsMessengerEXT
VK_OBJECT_TYPE_ACCELERATION_STRUCTURE_KHR	VkAccelerationStructureKHR
VK_OBJECT_TYPE_DEFERRED_OPERATION_KHR	VkDeferredOperationKHR
VK_OBJECT_TYPE_INDIRECT_COMMANDS_LAYOUT_EXT	VkIndirectCommandsLayoutEXT
VK_OBJECT_TYPE_INDIRECT_EXECUTION_SET_EXT	VkIndirectExecutionSetEXT
VK_OBJECT_TYPE_MICROMAP_EXT	VkMicromapEXT
VK_OBJECT_TYPE_SHADER_EXT	VkShaderEXT

If this Specification was generated with any such extensions included, they will be described in the remainder of this chapter.

43.1. Debug Utilities

Vulkan provides flexible debugging utilities for debugging an application.

The [Object Debug Annotation](#) section describes how to associate either a name or binary data with a specific Vulkan object.

The [Queue Labels](#) section describes how to annotate and group the work submitted to a queue.

The [Command Buffer Labels](#) section describes how to associate logical elements of the scene with commands in a [VkCommandBuffer](#).

The [Debug Messengers](#) section describes how to create debug messenger objects associated with an application supplied callback to capture debug messages from a variety of Vulkan components.

43.1.1. Object Debug Annotation

It can be useful for an application to provide its own content relative to a specific Vulkan object.

The following commands allow application developers to associate application-defined information with Vulkan objects. These commands are device-level commands but they **may** reference instance-level objects (such as [VkInstance](#)) and physical device-level objects (such as [VkPhysicalDevice](#)) with a few restrictions: * The data for the corresponding object **may** still be available after the [VkDevice](#) used in the corresponding API call to set it is destroyed, but access to this data is not guaranteed and should be avoided. * Subsequent calls to change the data of the same object across multiple [VkDevice](#) objects, **may** result in the data being changed to the most recent version for all [VkDevice](#) objects and not just the [VkDevice](#) used in the most recent API call.

Object Naming

An object can be given an application-defined name by calling:

```
// Provided by VK_EXT_debug_utils
VkResult vkSetDebugUtilsObjectNameEXT(
    VkDevice device,
    const VkDebugUtilsObjectNameInfoEXT* pNameInfo);
```

- [device](#) is the device that is associated with the named object passed in via [objectHandle](#).
- [pNameInfo](#) is a pointer to a [VkDebugUtilsObjectNameInfoEXT](#) structure specifying parameters of the name to set on the object.

Valid Usage

- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-02587
[pNameInfo->objectType](#) **must** not be [VK_OBJECT_TYPE_UNKNOWN](#)
- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-02588
[pNameInfo->objectHandle](#) **must** not be [VK_NULL_HANDLE](#)
- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-07872
If [pNameInfo->objectHandle](#) is the valid handle of an instance-level object, the [VkDevice](#) identified by [device](#) **must** be a descendent of the same [VkInstance](#) as the object identified by [pNameInfo->objectHandle](#)

- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-07873

If `pNameInfo->objectHandle` is the valid handle of a physical-device-level object, the `VkDevice` identified by `device` **must** be a descendant of the same `VkPhysicalDevice` as the object identified by `pNameInfo->objectHandle`

- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-07874

If `pNameInfo->objectHandle` is the valid handle of a device-level object, that object **must** be a descendent of the `VkDevice` identified by `device`

Valid Usage (Implicit)

- VUID-vkSetDebugUtilsObjectNameEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkSetDebugUtilsObjectNameEXT-pNameInfo-parameter
`pNameInfo` **must** be a valid pointer to a valid `VkDebugUtilsObjectNameInfoEXT` structure

Host Synchronization

- Host access to `pNameInfo->objectHandle` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkDebugUtilsObjectNameInfoEXT` structure is defined as:

```
// Provided by VK_EXT_debug_utils
typedef struct VkDebugUtilsObjectNameInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkObjectType        objectType;
    uint64_t            objectHandle;
    const char*         pObjectName;
} VkDebugUtilsObjectNameInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `objectType` is a `VkObjectType` specifying the type of the object to be named.

- `objectHandle` is the object to be named.
- `pObjectName` is either `NULL` or a null-terminated UTF-8 string specifying the name to apply to `objectHandle`.

Applications **may** change the name associated with an object simply by calling `vkSetDebugUtilsObjectNameEXT` again with a new string. If `pObjectName` is either `NULL` or an empty string, then any previously set name is removed.

Valid Usage

- VUID-VkDebugUtilsObjectNameInfoEXT-objectType-02589
If `objectType` is `VK_OBJECT_TYPE_UNKNOWN`, `objectHandle` **must** not be `VK_NULL_HANDLE`
- VUID-VkDebugUtilsObjectNameInfoEXT-objectType-02590
If `objectType` is not `VK_OBJECT_TYPE_UNKNOWN`, `objectHandle` **must** be `VK_NULL_HANDLE` or a valid Vulkan handle of the type associated with `objectType` as defined in the [VkObjectType and Vulkan Handle Relationship](#) table

Valid Usage (Implicit)

- VUID-VkDebugUtilsObjectNameInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_NAME_INFO_EXT`
- VUID-VkDebugUtilsObjectNameInfoEXT-objectType-parameter
`objectType` **must** be a valid `VkObjectType` value
- VUID-VkDebugUtilsObjectNameInfoEXT-pObjectName-parameter
If `pObjectName` is not `NULL`, `pObjectName` **must** be a null-terminated UTF-8 string

Object Data Association

In addition to setting a name for an object, debugging and validation layers **may** have uses for additional binary data on a per-object basis that have no other place in the Vulkan API.

For example, a `VkShaderModule` could have additional debugging data attached to it to aid in offline shader tracing.

Additional data can be attached to an object by calling `vkSetDebugUtilsObjectTagEXT` as defined below.

```
// Provided by VK_EXT_debug_utils
VkResult vkSetDebugUtilsObjectTagEXT(
    VkDevice device,
    const VkDebugUtilsObjectTagInfoEXT* pTagInfo);
```

- `device` is the device that created the object.
- `pTagInfo` is a pointer to a `VkDebugUtilsObjectTagInfoEXT` structure specifying parameters of the

tag to attach to the object.

Valid Usage

- VUID-vkSetDebugUtilsObjectTagEXT-pNameInfo-07875
If `pNameInfo->objectHandle` is the valid handle of an instance-level object, the `VkDevice` identified by `device` **must** be a descendent of the same `VkInstance` as the object identified by `pNameInfo->objectHandle`
- VUID-vkSetDebugUtilsObjectTagEXT-pNameInfo-07876
If `pNameInfo->objectHandle` is the valid handle of a physical-device-level object, the `VkDevice` identified by `device` **must** be a descendant of the same `VkPhysicalDevice` as the object identified by `pNameInfo->objectHandle`
- VUID-vkSetDebugUtilsObjectTagEXT-pNameInfo-07877
If `pNameInfo->objectHandle` is the valid handle of a device-level object, that object **must** be a descendent of the `VkDevice` identified by `device`

Valid Usage (Implicit)

- VUID-vkSetDebugUtilsObjectTagEXT-device-parameter
`device` **must** be a valid `VkDevice` handle
- VUID-vkSetDebugUtilsObjectTagEXT-pTagInfo-parameter
`pTagInfo` **must** be a valid pointer to a valid `VkDebugUtilsObjectTagInfoEXT` structure

Host Synchronization

- Host access to `pTagInfo->objectHandle` **must** be externally synchronized

Return Codes

Success

- `VK_SUCCESS`

Failure

- `VK_ERROR_OUT_OF_HOST_MEMORY`
- `VK_ERROR_OUT_OF_DEVICE_MEMORY`

The `VkDebugUtilsObjectTagInfoEXT` structure is defined as:

```
// Provided by VK_EXT_debug_utils
typedef struct VkDebugUtilsObjectTagInfoEXT {
    VkStructureType    sType;
    const void*        pNext;
    VkObjectType        objectType;
    uint64_t            objectHandle;
    uint64_t            tagName;
    size_t              tagSize;
    const void*        pTag;
} VkDebugUtilsObjectTagInfoEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **objectType** is a [VkObjectType](#) specifying the type of the object to be named.
- **objectHandle** is the object to be tagged.
- **tagName** is a numerical identifier of the tag.
- **tagSize** is the number of bytes of data to attach to the object.
- **pTag** is a pointer to an array of **tagSize** bytes containing the data to be associated with the object.

The **tagName** parameter gives a name or identifier to the type of data being tagged. This can be used by debugging layers to easily filter for only data that can be used by that implementation.

Valid Usage

- VUID-VkDebugUtilsObjectTagInfoEXT-objectType-01908
objectType **must** not be **VK_OBJECT_TYPE_UNKNOWN**
- VUID-VkDebugUtilsObjectTagInfoEXT-objectHandle-01910
objectHandle **must** be a valid Vulkan handle of the type associated with **objectType** as defined in the [VkObjectType](#) and [Vulkan Handle Relationship](#) table

Valid Usage (Implicit)

- VUID-VkDebugUtilsObjectTagInfoEXT-sType-sType
sType **must** be **VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_TAG_INFO_EXT**
- VUID-VkDebugUtilsObjectTagInfoEXT-pNext-pNext
pNext **must** be **NULL**
- VUID-VkDebugUtilsObjectTagInfoEXT-objectType-parameter
objectType **must** be a valid [VkObjectType](#) value
- VUID-VkDebugUtilsObjectTagInfoEXT-pTag-parameter
pTag **must** be a valid pointer to an array of **tagSize** bytes
- VUID-VkDebugUtilsObjectTagInfoEXT-tagSize-arraylength
tagSize **must** be greater than **0**

43.1.2. Queue Labels

All Vulkan work is submitted using queues. It is possible for an application to use multiple queues, each containing multiple command buffers, when performing work. It can be useful to identify which queue, or even where in a queue, something has occurred.

To begin identifying a region using a debug label inside a queue, you may use the [vkQueueBeginDebugUtilsLabelEXT](#) command.

Then, when the region of interest has passed, you may end the label region using [vkQueueEndDebugUtilsLabelEXT](#).

Additionally, a single debug label may be inserted at any time using [vkQueueInsertDebugUtilsLabelEXT](#).

A queue debug label region is opened by calling:

```
// Provided by VK_EXT_debug_utils
void vkQueueBeginDebugUtilsLabelEXT(
    VkQueue queue,
    const VkDebugUtilsLabelEXT* pLabelInfo);
```

- `queue` is the queue in which to start a debug label region.
- `pLabelInfo` is a pointer to a [VkDebugUtilsLabelEXT](#) structure specifying parameters of the label region to open.

Valid Usage (Implicit)

- VUID-vkQueueBeginDebugUtilsLabelEXT-queue-parameter
`queue` **must** be a valid [VkQueue](#) handle
- VUID-vkQueueBeginDebugUtilsLabelEXT-pLabelInfo-parameter
`pLabelInfo` **must** be a valid pointer to a valid [VkDebugUtilsLabelEXT](#) structure

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

The [VkDebugUtilsLabelEXT](#) structure is defined as:

```
// Provided by VK_EXT_debug_utils
typedef struct VkDebugUtilsLabelEXT {
    VkStructureType    sType;
    const void*        pNext;
    const char*        pLabelName;
    float              color[4];
} VkDebugUtilsLabelEXT;
```

- **sType** is a [VkStructureType](#) value identifying this structure.
- **pNext** is **NULL** or a pointer to a structure extending this structure.
- **pLabelName** is a pointer to a null-terminated UTF-8 string containing the name of the label.
- **color** is an optional RGBA color value that can be associated with the label. A particular implementation **may** choose to ignore this color value. The values contain RGBA values in order, in the range 0.0 to 1.0. If all elements in **color** are 0.0, then it is ignored.

Valid Usage (Implicit)

- VUID-VkDebugUtilsLabelEXT-sType-sType
sType must be [VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT](#)
- VUID-VkDebugUtilsLabelEXT-pNext-pNext
pNext must be **NULL**
- VUID-VkDebugUtilsLabelEXT-pLabelName-parameter
pLabelName must be a null-terminated UTF-8 string

A queue debug label region is closed by calling:

```
// Provided by VK_EXT_debug_utils
void vkQueueEndDebugUtilsLabelEXT(
    VkQueue queue);
```

- **queue** is the queue in which a debug label region should be closed.

The calls to [vkQueueBeginDebugUtilsLabelEXT](#) and [vkQueueEndDebugUtilsLabelEXT](#) **must** be matched and balanced.

Valid Usage

- VUID-vkQueueEndDebugUtilsLabelEXT-None-01911
There **must** be an outstanding [vkQueueBeginDebugUtilsLabelEXT](#) command prior to the [vkQueueEndDebugUtilsLabelEXT](#) on the queue

Valid Usage (Implicit)

- VUID-vkQueueEndDebugUtilsLabelEXT-queue-parameter
queue must be a valid [VkQueue](#) handle

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

A single label can be inserted into a queue by calling:

```
// Provided by VK_EXT_debug_utils
void vkQueueInsertDebugUtilsLabelEXT(
    VkQueue queue,
    const VkDebugUtilsLabelEXT* pLabelInfo);
```

- **queue** is the queue into which a debug label will be inserted.
- **pLabelInfo** is a pointer to a [VkDebugUtilsLabelEXT](#) structure specifying parameters of the label to insert.

Valid Usage (Implicit)

- VUID-vkQueueInsertDebugUtilsLabelEXT-queue-parameter
queue must be a valid [VkQueue](#) handle
- VUID-vkQueueInsertDebugUtilsLabelEXT-pLabelInfo-parameter
pLabelInfo must be a valid pointer to a valid [VkDebugUtilsLabelEXT](#) structure

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
-	-	-	Any	-

43.1.3. Command Buffer Labels

Typical Vulkan applications will submit many command buffers in each frame, with each command buffer containing a large number of individual commands. Being able to logically annotate regions of command buffers that belong together as well as hierarchically subdivide the

frame is important to a developer's ability to navigate the commands viewed holistically.

To identify the beginning of a debug label region in a command buffer, [vkCmdBeginDebugUtilsLabelEXT](#) can be used as defined below.

To indicate the end of a debug label region in a command buffer, [vkCmdEndDebugUtilsLabelEXT](#) can be used.

To insert a single command buffer debug label inside of a command buffer, [vkCmdInsertDebugUtilsLabelEXT](#) can be used as defined below.

A command buffer debug label region can be opened by calling:

```
// Provided by VK_EXT_debug_utils
void vkCmdBeginDebugUtilsLabelEXT(
    VkCommandBuffer                      commandBuffer,
    const VkDebugUtilsLabelEXT*          pLabelInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pLabelInfo` is a pointer to a [VkDebugUtilsLabelEXT](#) structure specifying parameters of the label region to open.

Valid Usage (Implicit)

- VUID-vkCmdBeginDebugUtilsLabelEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid [VkCommandBuffer](#) handle
- VUID-vkCmdBeginDebugUtilsLabelEXT-pLabelInfo-parameter
`pLabelInfo` **must** be a valid pointer to a valid [VkDebugUtilsLabelEXT](#) structure
- VUID-vkCmdBeginDebugUtilsLabelEXT-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdBeginDebugUtilsLabelEXT-commandBuffer-cmdpool
The [VkCommandPool](#) that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdBeginDebugUtilsLabelEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the [VkCommandPool](#) that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	Action State

A command buffer label region can be closed by calling:

```
// Provided by VK_EXT_debug_utils
void vkCmdEndDebugUtilsLabelEXT(
    VkCommandBuffer                      commandBuffer);
```

- `commandBuffer` is the command buffer into which the command is recorded.

An application **may** open a debug label region in one command buffer and close it in another, or otherwise split debug label regions across multiple command buffers or multiple queue submissions. When viewed from the linear series of submissions to a single queue, the calls to `vkCmdBeginDebugUtilsLabelEXT` and `vkCmdEndDebugUtilsLabelEXT` **must** be matched and balanced.

There **can** be problems reporting command buffer debug labels during the recording process because command buffers **may** be recorded out of sequence with the resulting execution order. Since the recording order **may** be different, a solitary command buffer **may** have an inconsistent view of the debug label regions by itself. Therefore, if an issue occurs during the recording of a command buffer, and the environment requires returning debug labels, the implementation **may** return only those labels it is aware of. This is true even if the implementation is aware of only the debug labels within the command buffer being actively recorded.

Valid Usage

- VUID-vkCmdEndDebugUtilsLabelEXT-commandBuffer-01912
There **must** be an outstanding `vkCmdBeginDebugUtilsLabelEXT` command prior to the `vkCmdEndDebugUtilsLabelEXT` on the queue that `commandBuffer` is submitted to
- VUID-vkCmdEndDebugUtilsLabelEXT-commandBuffer-01913
If `commandBuffer` is a secondary command buffer, there **must** be an outstanding `vkCmdBeginDebugUtilsLabelEXT` command recorded to `commandBuffer` that has not previously been ended by a call to `vkCmdEndDebugUtilsLabelEXT`

Valid Usage (Implicit)

- VUID-vkCmdEndDebugUtilsLabelEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle

- VUID-vkCmdEndDebugUtilsLabelEXT-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)
- VUID-vkCmdEndDebugUtilsLabelEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdEndDebugUtilsLabelEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	Action State

A single debug label can be inserted into a command buffer by calling:

```
// Provided by VK_EXT_debug_utils
void vkCmdInsertDebugUtilsLabelEXT(
    VkCommandBuffer                commandBuffer,
    const VkDebugUtilsLabelEXT*    pLabelInfo);
```

- `commandBuffer` is the command buffer into which the command is recorded.
- `pInfo` is a pointer to a `VkDebugUtilsLabelEXT` structure specifying parameters of the label to insert.

Valid Usage (Implicit)

- VUID-vkCmdInsertDebugUtilsLabelEXT-commandBuffer-parameter
`commandBuffer` **must** be a valid `VkCommandBuffer` handle
- VUID-vkCmdInsertDebugUtilsLabelEXT-pLabelInfo-parameter
`pLabelInfo` **must** be a valid pointer to a valid `VkDebugUtilsLabelEXT` structure
- VUID-vkCmdInsertDebugUtilsLabelEXT-commandBuffer-recording
`commandBuffer` **must** be in the [recording state](#)

- VUID-vkCmdInsertDebugUtilsLabelEXT-commandBuffer-cmdpool
The `VkCommandPool` that `commandBuffer` was allocated from **must** support graphics, or compute operations
- VUID-vkCmdInsertDebugUtilsLabelEXT-videocoding
This command **must** only be called outside of a video coding scope

Host Synchronization

- Host access to `commandBuffer` **must** be externally synchronized
- Host access to the `VkCommandPool` that `commandBuffer` was allocated from **must** be externally synchronized

Command Properties

Command Buffer Levels	Render Pass Scope	Video Coding Scope	Supported Queue Types	Command Type
Primary Secondary	Both	Outside	Graphics Compute	Action

43.1.4. Debug Messengers

Vulkan allows an application to register multiple callbacks with any Vulkan component wishing to report debug information. Some callbacks may log the information to a file, others may cause a debug break point or other application-defined behavior. A primary producer of callback messages are the validation layers. An application **can** register callbacks even when no validation layers are enabled, but they will only be called for the Vulkan loader and, if implemented, other layer and driver events.

A `VkDebugUtilsMessengerEXT` is a messenger object which handles passing along debug messages to a provided debug callback.

```
// Provided by VK_EXT_debug_utils
VK_DEFINE_NON_DISPATCHABLE_HANDLE(VkDebugUtilsMessengerEXT)
```

The debug messenger will provide detailed feedback on the application's use of Vulkan when events of interest occur. When an event of interest does occur, the debug messenger will submit a debug message to the debug callback that was provided during its creation. Additionally, the debug messenger is responsible with filtering out debug messages that the callback is not interested in and will only provide desired debug messages.

A debug messenger triggers a debug callback with a debug message when an event of interest occurs. To create a debug messenger which will trigger a debug callback, call:

```
// Provided by VK_EXT_debug_utils
VkResult vkCreateDebugUtilsMessengerEXT(
    VkInstance                                instance,
    const VkDebugUtilsMessengerCreateInfoEXT* pCreateInfo,
    const VkAllocationCallbacks*             pAllocator,
    VkDebugUtilsMessengerEXT*                 pMessenger);
```

- **instance** is the instance the messenger will be used with.
- **pCreateInfo** is a pointer to a [VkDebugUtilsMessengerCreateInfoEXT](#) structure containing the callback pointer, as well as defining conditions under which this messenger will trigger the callback.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.
- **pMessenger** is a pointer to a [VkDebugUtilsMessengerEXT](#) handle in which the created object is returned.

Valid Usage (Implicit)

- VUID-vkCreateDebugUtilsMessengerEXT-instance-parameter **instance** **must** be a valid [VkInstance](#) handle
- VUID-vkCreateDebugUtilsMessengerEXT-pCreateInfo-parameter **pCreateInfo** **must** be a valid pointer to a valid [VkDebugUtilsMessengerCreateInfoEXT](#) structure
- VUID-vkCreateDebugUtilsMessengerEXT-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkCreateDebugUtilsMessengerEXT-pMessenger-parameter **pMessenger** **must** be a valid pointer to a [VkDebugUtilsMessengerEXT](#) handle

Return Codes

Success

- **VK_SUCCESS**

Failure

- **VK_ERROR_OUT_OF_HOST_MEMORY**

The application **must** ensure that [vkCreateDebugUtilsMessengerEXT](#) is not executed in parallel with any Vulkan command that is also called with **instance** or child of **instance** as the dispatchable argument.

The definition of [VkDebugUtilsMessengerCreateInfoEXT](#) is:

```
// Provided by VK_EXT_debug_utils
typedef struct VkDebugUtilsMessengerCreateInfoEXT {
    VkStructureType             sType;
    const void*                 pNext;
    VkDebugUtilsMessengerCreateFlagsEXT flags;
    VkDebugUtilsMessageSeverityFlagsEXT messageSeverity;
    VkDebugUtilsMessageTypeFlagsEXT messageType;
    PFN_vkDebugUtilsMessengerCallbackEXT pfnUserCallback;
    void*                       pUserData;
} VkDebugUtilsMessengerCreateInfoEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is `0` and is reserved for future use.
- `messageSeverity` is a bitmask of `VkDebugUtilsMessageSeverityFlagBitsEXT` specifying which severity of event(s) will cause this callback to be called.
- `messageType` is a bitmask of `VkDebugUtilsMessageTypeFlagBitsEXT` specifying which type of event(s) will cause this callback to be called.
- `pfnUserCallback` is the application callback function to call.
- `pUserData` is user data to be passed to the callback.

For each `VkDebugUtilsMessengerEXT` that is created the `VkDebugUtilsMessengerCreateInfoEXT::messageSeverity` and `VkDebugUtilsMessengerCreateInfoEXT::messageType` determine when that `VkDebugUtilsMessengerCreateInfoEXT::pfnUserCallback` is called. The process to determine if the user's `pfnUserCallback` is triggered when an event occurs is as follows:

1. The implementation will perform a bitwise AND of the event's `VkDebugUtilsMessageSeverityFlagBitsEXT` with the `messageSeverity` provided during creation of the `VkDebugUtilsMessengerEXT` object.
 - a. If the value is 0, the message is skipped.
2. The implementation will perform bitwise AND of the event's `VkDebugUtilsMessageTypeFlagBitsEXT` with the `messageType` provided during the creation of the `VkDebugUtilsMessengerEXT` object.
 - a. If the value is 0, the message is skipped.
3. The callback will trigger a debug message for the current event

The callback will come directly from the component that detected the event, unless some other layer intercepts the calls for its own purposes (filter them in a different way, log to a system error log, etc.).

An application **can** receive multiple callbacks if multiple `VkDebugUtilsMessengerEXT` objects are created. A callback will always be executed in the same thread as the originating Vulkan call.

A callback **can** be called from multiple threads simultaneously (if the application is making Vulkan

calls from multiple threads).

Valid Usage (Implicit)

- VUID-VkDebugUtilsMessengerCreateInfoEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CREATE_INFO_EXT`
- VUID-VkDebugUtilsMessengerCreateInfoEXT-flags-zeroBitmask
`flags` **must** be `0`
- VUID-VkDebugUtilsMessengerCreateInfoEXT-messageSeverity-parameter
`messageSeverity` **must** be a valid combination of `VkDebugUtilsMessageSeverityFlagBitsEXT` values
- VUID-VkDebugUtilsMessengerCreateInfoEXT-messageSeverity-requiredBitmask
`messageSeverity` **must** not be `0`
- VUID-VkDebugUtilsMessengerCreateInfoEXT-messageType-parameter
`messageType` **must** be a valid combination of `VkDebugUtilsMessageTypeFlagBitsEXT` values
- VUID-VkDebugUtilsMessengerCreateInfoEXT-messageType-requiredBitmask
`messageType` **must** not be `0`
- VUID-VkDebugUtilsMessengerCreateInfoEXT-pfnUserCallback-parameter
`pfnUserCallback` **must** be a valid `PFN_vkDebugUtilsMessengerCallbackEXT` value

```
// Provided by VK_EXT_debug_utils
typedef VkFlags VkDebugUtilsMessengerCreateFlagsEXT;
```

`VkDebugUtilsMessengerCreateFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

Bits which **can** be set in `VkDebugUtilsMessengerCreateInfoEXT::messageSeverity`, specifying event severities which cause a debug messenger to call the callback, are:

```
// Provided by VK_EXT_debug_utils
typedef enum VkDebugUtilsMessageSeverityFlagBitsEXT {
    VK_DEBUG_UTILS_MESSAGE_SEVERITY_VERBOSE_BIT_EXT = 0x00000001,
    VK_DEBUG_UTILS_MESSAGE_SEVERITY_INFO_BIT_EXT = 0x00000010,
    VK_DEBUG_UTILS_MESSAGE_SEVERITY_WARNING_BIT_EXT = 0x00000100,
    VK_DEBUG_UTILS_MESSAGE_SEVERITY_ERROR_BIT_EXT = 0x00001000,
} VkDebugUtilsMessageSeverityFlagBitsEXT;
```

- `VK_DEBUG_UTILS_MESSAGE_SEVERITY_VERBOSE_BIT_EXT` specifies the most verbose output indicating all diagnostic messages from the Vulkan loader, layers, and drivers should be captured.
- `VK_DEBUG_UTILS_MESSAGE_SEVERITY_INFO_BIT_EXT` specifies an informational message such as resource details that may be handy when debugging an application.
- `VK_DEBUG_UTILS_MESSAGE_SEVERITY_WARNING_BIT_EXT` specifies use of Vulkan that **may** expose an

application bug. Such cases may not be immediately harmful, such as a fragment shader outputting to a location with no attachment. Other cases **may** point to behavior that is almost certainly bad when unintended such as using an image whose memory has not been filled. In general if you see a warning but you know that the behavior is intended/desired, then simply ignore the warning.

- `VK_DEBUG_UTILS_MESSAGE_SEVERITY_ERROR_BIT_EXT` specifies that the application has violated a valid usage condition of the specification.

The values of `VkDebugUtilsMessageSeverityFlagBitsEXT` are sorted based on severity. The higher the flag value, the more severe the message. This allows for simple boolean operation comparisons when looking at `VkDebugUtilsMessageSeverityFlagBitsEXT` values.

For example:

NOTE

```
if (messageSeverity >=
VK_DEBUG_UTILS_MESSAGE_SEVERITY_WARNING_BIT_EXT) {
    // Do something for warnings and errors
}
```

In addition, space has been left between the enums to allow for later addition of new severities in between the existing values.

```
// Provided by VK_EXT_debug_utils
typedef VkFlags VkDebugUtilsMessageSeverityFlagsEXT;
```

`VkDebugUtilsMessageSeverityFlagsEXT` is a bitmask type for setting a mask of zero or more `VkDebugUtilsMessageSeverityFlagBitsEXT`.

Bits which **can** be set in `VkDebugUtilsMessengerCreateInfoEXT::messageType`, specifying event types which cause a debug messenger to call the callback, are:

```
// Provided by VK_EXT_debug_utils
typedef enum VkDebugUtilsMessageTypeFlagBitsEXT {
    VK_DEBUG_UTILS_MESSAGE_TYPE_GENERAL_BIT_EXT = 0x00000001,
    VK_DEBUG_UTILS_MESSAGE_TYPE_VALIDATION_BIT_EXT = 0x00000002,
    VK_DEBUG_UTILS_MESSAGE_TYPE_PERFORMANCE_BIT_EXT = 0x00000004,
} VkDebugUtilsMessageTypeFlagBitsEXT;
```

- `VK_DEBUG_UTILS_MESSAGE_TYPE_GENERAL_BIT_EXT` specifies that some general event has occurred. This is typically a non-specification, non-performance event.
- `VK_DEBUG_UTILS_MESSAGE_TYPE_VALIDATION_BIT_EXT` specifies that something has occurred during validation against the Vulkan specification that may indicate invalid behavior.
- `VK_DEBUG_UTILS_MESSAGE_TYPE_PERFORMANCE_BIT_EXT` specifies a potentially non-optimal use of Vulkan, e.g. using `vkCmdClearColorImage` when setting `VkAttachmentDescription::loadOp` to

`VK_ATTACHMENT_LOAD_OP_CLEAR` would have worked.

```
// Provided by VK_EXT_debug_utils
typedef VkFlags VkDebugUtilsMessageTypeFlagsEXT;
```

`VkDebugUtilsMessageTypeFlagsEXT` is a bitmask type for setting a mask of zero or more `VkDebugUtilsMessageTypeFlagBitsEXT`.

The prototype for the `VkDebugUtilsMessengerCreateInfoEXT::pfnUserCallback` function implemented by the application is:

```
// Provided by VK_EXT_debug_utils
typedef VkBool32 (VKAPI_PTR *PFN_vkDebugUtilsMessengerCallbackEXT)(
    VkDebugUtilsMessageSeverityFlagBitsEXT    messageSeverity,
    VkDebugUtilsMessageTypeFlagsEXT           messageTypes,
    const VkDebugUtilsMessengerCallbackDataEXT* pCallbackData,
    void*                                      pUserData);
```

- `messageSeverity` specifies the `VkDebugUtilsMessageSeverityFlagBitsEXT` that triggered this callback.
- `messageTypes` is a bitmask of `VkDebugUtilsMessageTypeFlagBitsEXT` specifying which type of event(s) triggered this callback.
- `pCallbackData` contains all the callback related data in the `VkDebugUtilsMessengerCallbackDataEXT` structure.
- `pUserData` is the user data provided when the `VkDebugUtilsMessengerEXT` was created.

The callback returns a `VkBool32`, which is interpreted in a layer-specified manner. The application **should** always return `VK_FALSE`. The `VK_TRUE` value is reserved for use in layer development.

Valid Usage

- VUID-PFN_vkDebugUtilsMessengerCallbackEXT-None-04769
The callback **must** not make calls to any Vulkan commands

The definition of `VkDebugUtilsMessengerCallbackDataEXT` is:

```
// Provided by VK_EXT_debug_utils
typedef struct VkDebugUtilsMessengerCallbackDataEXT {
    VkStructureType           sType;
    const void*               pNext;
    VkDebugUtilsMessengerCallbackDataFlagsEXT flags;
    const char*               pMessageIdName;
    int32_t                   messageIdNumber;
    const char*               pMessage;
    uint32_t                  queueLabelCount;
    const VkDebugUtilsLabelEXT* pQueueLabels;
    uint32_t                  cmdBufLabelCount;
    const VkDebugUtilsLabelEXT* pCmdBufLabels;
    uint32_t                  objectCount;
    const VkDebugUtilsObjectNameInfoEXT* pObjects;
} VkDebugUtilsMessengerCallbackDataEXT;
```

- `sType` is a [VkStructureType](#) value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is `0` and is reserved for future use.
- `pMessageIdName` is `NULL` or a null-terminated UTF-8 string that identifies the particular message ID that is associated with the provided message. If the message corresponds to a validation layer message, then this string may contain the portion of the Vulkan specification that is believed to have been violated.
- `messageIdNumber` is the ID number of the triggering message. If the message corresponds to a validation layer message, then this number is related to the internal number associated with the message being triggered.
- `pMessage` is a null-terminated UTF-8 string detailing the trigger conditions.
- `queueLabelCount` is a count of items contained in the `pQueueLabels` array.
- `pQueueLabels` is `NULL` or a pointer to an array of [VkDebugUtilsLabelEXT](#) active in the current [VkQueue](#) at the time the callback was triggered. Refer to [Queue Labels](#) for more information.
- `cmdBufLabelCount` is a count of items contained in the `pCmdBufLabels` array.
- `pCmdBufLabels` is `NULL` or a pointer to an array of [VkDebugUtilsLabelEXT](#) active in the current [VkCommandBuffer](#) at the time the callback was triggered. Refer to [Command Buffer Labels](#) for more information.
- `objectCount` is a count of items contained in the `pObjects` array.
- `pObjects` is a pointer to an array of [VkDebugUtilsObjectNameInfoEXT](#) objects related to the detected issue. The array is roughly in order of importance, but the 0th element is always guaranteed to be the most important object for this message.

NOTE

This structure should only be considered valid during the lifetime of the triggered callback.

Since adding queue and command buffer labels behaves like pushing and popping onto a stack, the

order of both `pQueueLabels` and `pCmdBufLabels` is based on the order the labels were defined. The result is that the first label in either `pQueueLabels` or `pCmdBufLabels` will be the first defined (and therefore the oldest) while the last label in each list will be the most recent.

NOTE

`pQueueLabels` will only be non-NULL if one of the objects in `pObjects` can be related directly to a defined `VkQueue` which has had one or more labels associated with it.

Likewise, `pCmdBufLabels` will only be non-NULL if one of the objects in `pObjects` can be related directly to a defined `VkCommandBuffer` which has had one or more labels associated with it. Additionally, while command buffer labels allow for beginning and ending across different command buffers, the debug messaging framework **cannot** guarantee that labels in `pCmdBufLabels` will contain those defined outside of the associated command buffer. This is partially due to the fact that the association of one command buffer with another may not have been defined at the time the debug message is triggered.

Valid Usage (Implicit)

- VUID-VkDebugUtilsMessengerCallbackDataEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CALLBACK_DATA_EXT`
- VUID-VkDebugUtilsMessengerCallbackDataEXT-pNext-pNext
`pNext` **must** be NULL
- VUID-VkDebugUtilsMessengerCallbackDataEXT-flags-zerobitmask
`flags` **must** be 0
- VUID-VkDebugUtilsMessengerCallbackDataEXT-pMessageIdName-parameter
If `pMessageIdName` is not NULL, `pMessageIdName` **must** be a null-terminated UTF-8 string
- VUID-VkDebugUtilsMessengerCallbackDataEXT-pMessage-parameter
If `pMessage` is not NULL, `pMessage` **must** be a null-terminated UTF-8 string
- VUID-VkDebugUtilsMessengerCallbackDataEXT-pQueueLabels-parameter
If `queueLabelCount` is not 0, `pQueueLabels` **must** be a valid pointer to an array of `queueLabelCount` valid `VkDebugUtilsLabelEXT` structures
- VUID-VkDebugUtilsMessengerCallbackDataEXT-pCmdBufLabels-parameter
If `cmdBufLabelCount` is not 0, `pCmdBufLabels` **must** be a valid pointer to an array of `cmdBufLabelCount` valid `VkDebugUtilsLabelEXT` structures
- VUID-VkDebugUtilsMessengerCallbackDataEXT-pObjects-parameter
If `objectCount` is not 0, `pObjects` **must** be a valid pointer to an array of `objectCount` valid `VkDebugUtilsObjectNameInfoEXT` structures

```
// Provided by VK_EXT_debug_utils
typedef VkFlags VkDebugUtilsMessengerCallbackDataFlagsEXT;
```

`VkDebugUtilsMessengerCallbackDataFlagsEXT` is a bitmask type for setting a mask, but is currently reserved for future use.

To intentionally submit a debug message, call:

```
// Provided by VK_EXT_debug_utils
void vkSubmitDebugUtilsMessageEXT(
    VkInstance                                instance,
    VkDebugUtilsMessageSeverityFlagBitsEXT    messageSeverity,
    VkDebugUtilsMessageTypeFlagsEXT           messageTypes,
    const VkDebugUtilsMessengerCallbackDataEXT* pCallbackData);
```

- **instance** is the debug stream's **VkInstance**.
- **messageSeverity** is a **VkDebugUtilsMessageSeverityFlagBitsEXT** value specifying the severity of this event/message.
- **messageTypes** is a bitmask of **VkDebugUtilsMessageTypeFlagBitsEXT** specifying which type of event(s) to identify with this message.
- **pCallbackData** contains all the callback related data in the **VkDebugUtilsMessengerCallbackDataEXT** structure.

The call will propagate through the layers and generate callback(s) as indicated by the message's flags. The parameters are passed on to the callback in addition to the **pUserData** value that was defined at the time the messenger was registered.

Valid Usage

- VUID-vkSubmitDebugUtilsMessageEXT-objectType-02591
The **objectType** member of each element of **pCallbackData->pObjects** **must** not be **VK_OBJECT_TYPE_UNKNOWN**

Valid Usage (Implicit)

- VUID-vkSubmitDebugUtilsMessageEXT-instance-parameter
instance **must** be a valid **VkInstance** handle
- VUID-vkSubmitDebugUtilsMessageEXT-messageSeverity-parameter
messageSeverity **must** be a valid **VkDebugUtilsMessageSeverityFlagBitsEXT** value
- VUID-vkSubmitDebugUtilsMessageEXT-messageTypes-parameter
messageTypes **must** be a valid combination of **VkDebugUtilsMessageTypeFlagBitsEXT** values
- VUID-vkSubmitDebugUtilsMessageEXT-messageTypes-requiredbitmask
messageTypes **must** not be **0**
- VUID-vkSubmitDebugUtilsMessageEXT-pCallbackData-parameter
pCallbackData **must** be a valid pointer to a valid **VkDebugUtilsMessengerCallbackDataEXT** structure

To destroy a **VkDebugUtilsMessengerEXT** object, call:

```
// Provided by VK_EXT_debug_utils
void vkDestroyDebugUtilsMessengerEXT(
    VkInstance          instance,
    VkDebugUtilsMessengerEXT messenger,
    const VkAllocationCallbacks* pAllocator);
```

- **instance** is the instance where the callback was created.
- **messenger** is the [VkDebugUtilsMessengerEXT](#) object to destroy. **messenger** is an externally synchronized object and **must** not be used on more than one thread at a time. This means that [vkDestroyDebugUtilsMessengerEXT](#) **must** not be called when a callback is active.
- **pAllocator** controls host memory allocation as described in the [Memory Allocation](#) chapter.

Valid Usage

- VUID-vkDestroyDebugUtilsMessengerEXT-messenger-01915
If [VkAllocationCallbacks](#) were provided when **messenger** was created, a compatible set of callbacks **must** be provided here
- VUID-vkDestroyDebugUtilsMessengerEXT-messenger-01916
If no [VkAllocationCallbacks](#) were provided when **messenger** was created, **pAllocator** **must** be **NULL**

Valid Usage (Implicit)

- VUID-vkDestroyDebugUtilsMessengerEXT-instance-parameter
instance **must** be a valid [VkInstance](#) handle
- VUID-vkDestroyDebugUtilsMessengerEXT-messenger-parameter
If **messenger** is not [VK_NULL_HANDLE](#), **messenger** **must** be a valid [VkDebugUtilsMessengerEXT](#) handle
- VUID-vkDestroyDebugUtilsMessengerEXT-pAllocator-parameter
If **pAllocator** is not **NULL**, **pAllocator** **must** be a valid pointer to a valid [VkAllocationCallbacks](#) structure
- VUID-vkDestroyDebugUtilsMessengerEXT-messenger-parent
If **messenger** is a valid handle, it **must** have been created, allocated, or retrieved from **instance**

Host Synchronization

- Host access to **messenger** **must** be externally synchronized

The application **must** ensure that [vkDestroyDebugUtilsMessengerEXT](#) is not executed in parallel with any Vulkan command that is also called with **instance** or child of **instance** as the dispatchable

argument.

43.2. Frame Boundary

The `VkFrameBoundaryEXT` structure is defined as:

```
// Provided by VK_EXT_frame_boundary
typedef struct VkFrameBoundaryEXT {
    VkStructureType      sType;
    const void*          pNext;
    VkFrameBoundaryFlagsEXT flags;
    uint64_t             frameID;
    uint32_t             imageCount;
    const VkImage*        pImages;
    uint32_t             bufferCount;
    const VkBuffer*        pBuffer;
    uint64_t             tagName;
    size_t               tagSize;
    const void*          pTag;
} VkFrameBoundaryEXT;
```

- `sType` is a `VkStructureType` value identifying this structure.
- `pNext` is `NULL` or a pointer to a structure extending this structure.
- `flags` is a bitmask of `VkFrameBoundaryFlagBitsEXT` that can flag the last submission of a frame identifier.
- `frameID` is the frame identifier.
- `imageCount` is the number of images that store frame results.
- `pImages` is a pointer to an array of `VkImage` objects with `imageCount` entries.
- `bufferCount` is the number of buffers the store the frame results.
- `pBuffer` is a pointer to an array of `VkBuffer` objects with `bufferCount` entries.
- `tagName` is a numerical identifier for tag data.
- `tagSize` is the number of bytes of tag data.
- `pTag` is a pointer to an array of `tagSize` bytes containing tag data.

The application **can** associate frame boundary information to a queue submission call by adding a `VkFrameBoundaryEXT` structure to the `pNext` chain of `queue submission`, `VkPresentInfoKHR`, or `VkBindSparseInfo`.

The frame identifier is used to associate one or more queue submission to a frame, it is thus meant to be unique within a frame lifetime, i.e. it is possible (but not recommended) to reuse frame identifiers, as long as any two frames with any chance of having overlapping queue submissions (as in the example above) use two different frame identifiers.

NOTE Since the concept of frame is application-dependent, there is no way to validate the

use of frame identifier. It is good practice to use a monotonically increasing counter as the frame identifier and not reuse identifiers between frames.

The `pImages` and `pBuffers` arrays contain a list of images and buffers which store the "end result" of the frame. As the concept of frame is application-dependent, not all frames **may** produce their results in images or buffers, yet this is a sufficiently common case to be handled by `VkFrameBoundaryEXT`. Note that no extra information, such as image layout is being provided, since the images are meant to be used by tools which would already be tracking this required information. Having the possibility of passing a list of end-result images makes `VkFrameBoundaryEXT` as expressive as `vkQueuePresentKHR`, which is often the default frame boundary delimiter.

The application **can** also associate arbitrary extra information via tag data using `tagName`, `tagSize` and `pTag`. This extra information is typically tool-specific.

Valid Usage (Implicit)

- VUID-VkFrameBoundaryEXT-sType-sType
`sType` **must** be `VK_STRUCTURE_TYPE_FRAME_BOUNDARY_EXT`
- VUID-VkFrameBoundaryEXT-flags-parameter
`flags` **must** be a valid combination of `VkFrameBoundaryFlagBitsEXT` values
- VUID-VkFrameBoundaryEXT-pImages-parameter
If `imageCount` is not 0, and `pImages` is not NULL, `pImages` **must** be a valid pointer to an array of `imageCount` valid `VkImage` handles
- VUID-VkFrameBoundaryEXT-pBuffers-parameter
If `bufferCount` is not 0, and `pBuffers` is not NULL, `pBuffers` **must** be a valid pointer to an array of `bufferCount` valid `VkBuffer` handles
- VUID-VkFrameBoundaryEXT-pTag-parameter
If `tagSize` is not 0, and `pTag` is not NULL, `pTag` **must** be a valid pointer to an array of `tagSize` bytes
- VUID-VkFrameBoundaryEXT-commonparent
Both of the elements of `pBuffers`, and the elements of `pImages` that are valid handles of non-ignored parameters **must** have been created, allocated, or retrieved from the same `VkDevice`

The bit which **can** be set in `VkFrameBoundaryEXT::flags` is:

```
// Provided by VK_EXT_frame_boundary
typedef enum VkFrameBoundaryFlagBitsEXT {
    VK_FRAME_BOUNDARY_FRAME_END_BIT_EXT = 0x00000001,
} VkFrameBoundaryFlagBitsEXT;
```

- `VK_FRAME_BOUNDARY_FRAME_END_BIT_EXT` specifies that this queue submission is the last one for this frame, i.e. once this queue submission has terminated, then the work for this frame is completed.

Note that in the presence of timeline semaphores, the last queue submission might not be the last one to be submitted, as timeline semaphores allow for wait-before-signal submissions. In the context of frame boundary, the queue submission that should be done flagged as the last one is the one that is meant to be executed last, even if it **may** not be the last one to be submitted.

```
// Provided by VK_EXT_frame_boundary
typedef VkFlags VkFrameBoundaryFlagsEXT;
```

[VkFrameBoundaryFlagsEXT](#) is a bitmask type for setting a mask of zero or more [VkFrameBoundaryFlagBitsEXT](#).

Appendix A: Vulkan Environment for SPIR-V

Shaders for Vulkan are defined by the [Khronos SPIR-V Specification](#) as well as the [Khronos SPIR-V Extended Instructions for GLSL](#) Specification. This appendix defines additional SPIR-V requirements applying to Vulkan shaders.

Versions and Formats

A Vulkan 1.0 implementation **must** support the 1.0 version of SPIR-V and the 1.0 version of the SPIR-V Extended Instructions for GLSL. If the [VK_KHR_spirv_1_4](#) extension is enabled, the implementation **must** additionally support the 1.4 version of SPIR-V.

A SPIR-V module passed into [vkCreateShaderModule](#) is interpreted as a series of 32-bit words in host endianness, with literal strings packed as described in section 2.2 of the SPIR-V Specification. The first few words of the SPIR-V module **must** be a magic number and a SPIR-V version number, as described in section 2.3 of the SPIR-V Specification.

Capabilities

The [table below](#) lists the set of SPIR-V capabilities that **may** be supported in Vulkan implementations. The application **must** not use any of these capabilities in SPIR-V passed to [vkCreateShaderModule](#) unless one of the following conditions is met for the [VkDevice](#) specified in the [device](#) parameter of [vkCreateShaderModule](#):

- The corresponding field in the table is blank.
- Any corresponding Vulkan feature is enabled.
- Any corresponding Vulkan extension is enabled.
- Any corresponding Vulkan property is supported.
- The corresponding core version is supported (as returned by [VkPhysicalDeviceProperties::apiVersion](#)).

Table 89. List of SPIR-V Capabilities and Corresponding Vulkan Features, Extensions, or Core Version

SPIR-V OpCapability	Vulkan feature, extension, or core version
Matrix	VK_VERSION_1_0
Shader	VK_VERSION_1_0
InputAttachment	VK_VERSION_1_0
Sampled1D	VK_VERSION_1_0

SPIR-V OpCapability Vulkan feature, extension, or core version
Image1D VK_VERSION_1_0
SampledBuffer VK_VERSION_1_0
ImageBuffer VK_VERSION_1_0
ImageQuery VK_VERSION_1_0
DerivativeControl VK_VERSION_1_0
Geometry VkPhysicalDeviceFeatures::geometryShader
Tessellation VkPhysicalDeviceFeatures::tessellationShader
Float64 VkPhysicalDeviceFeatures::shaderFloat64
Int64 VkPhysicalDeviceFeatures::shaderInt64
Int64Atomics VkPhysicalDeviceVulkan12Features::shaderBufferInt64Atomics VkPhysicalDeviceVulkan12Features::shaderSharedInt64Atomics VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT::shaderImageInt64Atomics
AtomicFloat16AddEXT VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderBufferFloat16AtomicAdd VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderSharedFloat16AtomicAdd
AtomicFloat32AddEXT VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::shaderBufferFloat32AtomicAdd VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::shaderSharedFloat32AtomicAdd VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::shaderImageFloat32AtomicAdd
AtomicFloat64AddEXT VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::shaderBufferFloat64AtomicAdd VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::shaderSharedFloat64AtomicAdd
AtomicFloat16MinMaxEXT VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderBufferFloat16AtomicMinMax VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderSharedFloat16AtomicMinMax
AtomicFloat32MinMaxEXT VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderBufferFloat32AtomicMinMax VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderSharedFloat32AtomicMinMax VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderImageFloat32AtomicMinMax

SPIR-V OpCapability Vulkan feature, extension, or core version
AtomicFloat64MinMaxEXT VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderBufferFloat64AtomicMinMax VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::shaderSharedFloat64AtomicMinMax
Int64ImageEXT VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT::shaderImageInt64Atomics
Int16 VkPhysicalDeviceFeatures::shaderInt16
TessellationPointSize VkPhysicalDeviceFeatures::shaderTessellationAndGeometryPointSize
GeometryPointSize VkPhysicalDeviceFeatures::shaderTessellationAndGeometryPointSize
ImageGatherExtended VkPhysicalDeviceFeatures::shaderImageGatherExtended
StorageImageMultisample VkPhysicalDeviceFeatures::shaderStorageImageMultisample
UniformBufferArrayDynamicIndexing VkPhysicalDeviceFeatures::shaderUniformBufferArrayDynamicIndexing
SampledImageArrayDynamicIndexing VkPhysicalDeviceFeatures::shaderSampledImageArrayDynamicIndexing
StorageBufferArrayDynamicIndexing VkPhysicalDeviceFeatures::shaderStorageBufferArrayDynamicIndexing
StorageImageArrayDynamicIndexing VkPhysicalDeviceFeatures::shaderStorageImageArrayDynamicIndexing
ClipDistance VkPhysicalDeviceFeatures::shaderClipDistance
CullDistance VkPhysicalDeviceFeatures::shaderCullDistance
ImageCubeArray VkPhysicalDeviceFeatures::imageCubeArray
SampleRateShading VkPhysicalDeviceFeatures::sampleRateShading
SparseResidency VkPhysicalDeviceFeatures::shaderResourceResidency
MinLod VkPhysicalDeviceFeatures::shaderResourceMinLod
SampledCubeArray VkPhysicalDeviceFeatures::imageCubeArray

SPIR-V OpCapability Vulkan feature, extension, or core version
ImageMSArray VkPhysicalDeviceFeatures::shaderStorageImageMultisample
StorageImageExtendedFormats VK_VERSION_1_0
InterpolationFunction VkPhysicalDeviceFeatures::sampleRateShading
StorageImageReadWithoutFormat VkPhysicalDeviceFeatures::shaderStorageImageReadWithoutFormat VK_KHR_format_feature_flags2
StorageImageWriteWithoutFormat VkPhysicalDeviceFeatures::shaderStorageImageWriteWithoutFormat VK_KHR_format_feature_flags2
MultiViewport VkPhysicalDeviceFeatures::multiViewport
DrawParameters VK_KHR_shader_draw_parameters
MultiView VkPhysicalDeviceMultiviewFeatures::multiview
DeviceGroup VK_KHR_device_group
VariablePointersStorageBuffer VkPhysicalDeviceVariablePointersFeatures::variablePointersStorageBuffer
VariablePointers VkPhysicalDeviceVariablePointersFeatures::variablePointers
ShaderClockKHR VK_KHR_shader_clock
StencilExportEXT VK_EXT_shader_stencil_export
StorageBuffer16BitAccess VkPhysicalDevice16BitStorageFeatures::storageBuffer16BitAccess
UniformAndStorageBuffer16BitAccess VkPhysicalDevice16BitStorageFeatures::uniformAndStorageBuffer16BitAccess
StoragePushConstant16 VkPhysicalDevice16BitStorageFeatures::storagePushConstant16
StorageInputOutput16 VkPhysicalDevice16BitStorageFeatures::storageInputOutput16
SampleMaskPostDepthCoverage VK_EXT_post_depth_coverage

SPIR-V OpCapability Vulkan feature, extension, or core version
FragmentFullyCoveredEXT VK_EXT_conservative_rasterization
Float16 VkPhysicalDeviceVulkan12Features::shaderFloat16
Int8 VkPhysicalDeviceVulkan12Features::shaderInt8
StorageBuffer8BitAccess VkPhysicalDeviceVulkan12Features::storageBuffer8BitAccess
UniformAndStorageBuffer8BitAccess VkPhysicalDeviceVulkan12Features::uniformAndStorageBuffer8BitAccess
StoragePushConstant8 VkPhysicalDeviceVulkan12Features::storagePushConstant8
VulkanMemoryModel VkPhysicalDeviceVulkan12Features::vulkanMemoryModel
VulkanMemoryModelDeviceScope VkPhysicalDeviceVulkan12Features::vulkanMemoryModelDeviceScope
DenormPreserve VkPhysicalDeviceVulkan12Properties::shaderDenormPreserveFloat16 VkPhysicalDeviceVulkan12Properties::shaderDenormPreserveFloat32 VkPhysicalDeviceVulkan12Properties::shaderDenormPreserveFloat64
DenormFlushToZero VkPhysicalDeviceVulkan12Properties::shaderDenormFlushToZeroFloat16 VkPhysicalDeviceVulkan12Properties::shaderDenormFlushToZeroFloat32 VkPhysicalDeviceVulkan12Properties::shaderDenormFlushToZeroFloat64
SignedZeroInfNanPreserve VkPhysicalDeviceVulkan12Properties::shaderSignedZeroInfNanPreserveFloat16 VkPhysicalDeviceVulkan12Properties::shaderSignedZeroInfNanPreserveFloat32 VkPhysicalDeviceVulkan12Properties::shaderSignedZeroInfNanPreserveFloat64
RoundingModeRTE VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTEFloat16 VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTEFloat32 VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTEFloat64
RoundingModeRTZ VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTZFloat16 VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTZFloat32 VkPhysicalDeviceVulkan12Properties::shaderRoundingModeRTZFloat64
ComputeDerivativeGroupQuadsKHR VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR::computeDerivativeGroupQuads

SPIR-V OpCapability Vulkan feature, extension, or core version
ComputeDerivativeGroupLinearKHR VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR::computeDerivativeGroupLinear
RayTracingKHR VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTracingPipeline
RayQueryKHR VkPhysicalDeviceRayQueryFeaturesKHR::rayQuery
RayTraversalPrimitiveCullingKHR VkPhysicalDeviceRayTracingPipelineFeaturesKHR::rayTraversalPrimitiveCulling VkPhysicalDeviceRayQueryFeaturesKHR::rayQuery
RayCullMaskKHR VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR::rayTracingMaintenance1
TransformFeedback VkPhysicalDeviceTransformFeedbackFeaturesEXT::transformFeedback
GeometryStreams VkPhysicalDeviceTransformFeedbackFeaturesEXT::geometryStreams
PhysicalStorageBufferAddresses VkPhysicalDeviceVulkan12Features::bufferDeviceAddress
FragmentShaderSampleInterlockEXT VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT::fragmentShaderSampleInterlock
FragmentShaderPixelInterlockEXT VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT::fragmentShaderPixelInterlock
FragmentShaderShadingRateInterlockEXT VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT::fragmentShaderShadingRateInterlock
FragmentShadingRateKHR VkPhysicalDeviceFragmentShadingRateFeaturesKHR::pipelineFragmentShadingRate VkPhysicalDeviceFragmentShadingRateFeaturesKHR::primitiveFragmentShadingRate VkPhysicalDeviceFragmentShadingRateFeaturesKHR::attachmentFragmentShadingRate
WorkgroupMemoryExplicitLayoutKHR VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR ::workgroupMemoryExplicitLayout
WorkgroupMemoryExplicitLayout8BitAccessKHR VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR ::workgroupMemoryExplicitLayout8BitAccess
WorkgroupMemoryExplicitLayout16BitAccessKHR VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR ::workgroupMemoryExplicitLayout16BitAccess
DotProductInputAll VkPhysicalDeviceVulkan13Features::shaderIntegerDotProduct VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR::shaderIntegerDotProduct

SPIR-V OpCapability Vulkan feature, extension, or core version
DotProductInput4x8Bit VkPhysicalDeviceVulkan13Features::shaderIntegerDotProduct VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR::shaderIntegerDotProduct
DotProductInput4x8BitPacked VkPhysicalDeviceVulkan13Features::shaderIntegerDotProduct VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR::shaderIntegerDotProduct
DotProduct VkPhysicalDeviceVulkan13Features::shaderIntegerDotProduct VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR::shaderIntegerDotProduct
FragmentBarycentricKHR VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR::fragmentShaderBarycentric
RayTracingOpacityMicromapEXT VK_EXT_opacity_micromap
RayTracingPositionFetchKHR VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR::rayTracingPositionFetch
RayQueryPositionFetchKHR VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR::rayTracingPositionFetch
TileImageColorReadAccessEXT VkPhysicalDeviceShaderTileImageFeaturesEXT::shaderTileImageColorReadAccess
TileImageDepthReadAccessEXT VkPhysicalDeviceShaderTileImageFeaturesEXT::shaderTileImageDepthReadAccess
TileImageStencilReadAccessEXT VkPhysicalDeviceShaderTileImageFeaturesEXT::shaderTileImageStencilReadAccess
CooperativeMatrixKHR VkPhysicalDeviceCooperativeMatrixFeaturesKHR::cooperativeMatrix
GroupNonUniformRotateKHR VkPhysicalDeviceShaderSubgroupRotateFeaturesKHR::shaderSubgroupRotate
ExpectAssumeKHR VkPhysicalDeviceShaderExpectAssumeFeaturesKHR::shaderExpectAssume
FloatControls2 VkPhysicalDeviceShaderFloatControls2FeaturesKHR::shaderFloatControls2
QuadControlKHR VkPhysicalDeviceShaderQuadControlFeaturesKHR::shaderQuadControl
ReplicatedCompositesEXT VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT::shaderReplicatedComposites

The application **must** not pass a SPIR-V module containing any of the following to `vkCreateShaderModule`:

- any OpCapability not listed above,

- an unsupported capability, or
- a capability which corresponds to a Vulkan feature or extension which has not been enabled.

SPIR-V Extensions

The [following table](#) lists SPIR-V extensions that implementations **may** support. The application **must** not pass a SPIR-V module to [vkCreateShaderModule](#) that uses the following SPIR-V extensions unless one of the following conditions is met for the [VkDevice](#) specified in the [device](#) parameter of [vkCreateShaderModule](#):

- Any corresponding Vulkan extension is enabled.
- The corresponding core version is supported (as returned by [VkPhysicalDeviceProperties::apiVersion](#)).

Table 90. List of SPIR-V Extensions and Corresponding Vulkan Extensions or Core Version

SPIR-V OpExtension	Vulkan extension or core version
SPV_KHR_variable_pointers	VK_KHR_variable_pointers
SPV_KHR_shader_draw_parameters	VK_KHR_shader_draw_parameters
SPV_KHR_8bit_storage	VK_KHR_8bit_storage
SPV_KHR_16bit_storage	VK_KHR_16bit_storage
SPV_KHR_shader_clock	VK_KHR_shader_clock
SPV_KHR_float_controls	VK_KHR_shader_float_controls
SPV_KHR_storage_buffer_storage_class	VK_KHR_storage_buffer_storage_class
SPV_KHR_post_depth_coverage	VK_EXT_post_depth_coverage
SPV_EXT_shader_stencil_export	VK_EXT_shader_stencil_export
SPV_KHR_vulkan_memory_model	VK_KHR_vulkan_memory_model
SPV_KHR_ray_tracing	VK_KHR_ray_tracing_pipeline
SPV_KHR_ray_query	VK_KHR_ray_query

SPIR-V OpExtension
Vulkan extension or core version
SPV_KHR_ray_cull_mask VK_KHR_ray_tracing_maintenance1
SPV_KHR_physical_storage_buffer VK_KHR_buffer_device_address
SPV_EXT_fragment_shader_interlock VK_EXT_fragment_shader_interlock
SPV_KHR_fragment_shading_rate VK_KHR_fragment_shading_rate
SPV_KHR_non_semantic_info VK_KHR_shader_non_semantic_info
SPV_EXT_shader_image_int64 VK_EXT_shader_image_atomic_int64
SPV_KHR_terminate_invocation VK_KHR_shader_terminate_invocation
SPV_KHR_multiview VK_KHR_multiview
SPV_KHR_workgroup_memory_explicit_layout VK_KHR_workgroup_memory_explicit_layout
SPV_EXT_shader_atomic_float_add VK_EXT_shader_atomic_float
SPV_KHR_fragment_shader_barycentric VK_KHR_fragment_shader_barycentric
SPV_KHR_subgroup_uniform_control_flow VK_KHR_shader_subgroup_uniform_control_flow
SPV_EXT_shader_atomic_float_min_max VK_EXT_shader_atomic_float2
SPV_EXT_shader_atomic_float16_add VK_EXT_shader_atomic_float2
SPV_EXT_fragment_fully_covered VK_EXT_conservative_rasterization
SPV_KHR_integer_dot_product VK_KHR_shader_integer_dot_product
SPV_KHR_device_group VK_KHR_device_group
SPV_KHR_ray_tracing_position_fetch VK_KHR_ray_tracing_position_fetch
SPV_EXT_shader_tile_image VK_EXT_shader_tile_image

SPIR-V OpExtension
Vulkan extension or core version
SPV_EXT_opacity_micromap VK_EXT_opacity_micromap
SPV_KHR_cooperative_matrix VK_KHR_cooperative_matrix
SPV_KHR_maximal_reconvergence VK_KHR_shader_maximal_reconvergence
SPV_KHR_subgroup_rotate VK_KHR_shader_subgroup_rotate
SPV_KHR_expect_assume VK_KHR_shader_expect_assume
SPV_KHR_float_controls2 VK_KHR_shader_float_controls2
SPV_KHR_quad_control VK_KHR_shader_quad_control
SPV_KHR_compute_shader_derivatives VK_KHR_compute_shader_derivatives
SPV_EXT_replicated_composites VK_EXT_shader_replicated_composites
SPV_KHR_relaxed_extended_instruction VK_KHR_shader_relaxed_extended_instruction

Validation Rules Within a Module

A SPIR-V module passed to `vkCreateShaderModule` **must** conform to the following rules:

Standalone SPIR-V Validation

The following rules **can** be validated with only the SPIR-V module itself. They do not depend on knowledge of the implementation and its capabilities or knowledge of runtime information, such as enabled features.

Valid Usage

- VUID-StandaloneSpirv-None-04633
Every entry point **must** have no return value and accept no arguments
- VUID-StandaloneSpirv-None-04634
The static function-call graph for an entry point **must** not contain cycles; that is, static recursion is not allowed
- VUID-StandaloneSpirv-None-04635
The `Logical` or `PhysicalStorageBuffer64` addressing model **must** be selected

- VUID-StandaloneSpirv-None-04636
Scope for execution **must** be limited to Workgroup or Subgroup
- VUID-StandaloneSpirv-None-04637
If the Scope for execution is Workgroup, then it **must** only be used in the task, mesh, tessellation control, or compute Execution Model
- VUID-StandaloneSpirv-None-04638
Scope for memory **must** be limited to Device, QueueFamily, Workgroup, ShaderCallKHR, Subgroup, or Invocation
- VUID-StandaloneSpirv-ExecutionModel-07320
If the Execution Model is TessellationControl, and the MemoryModel is GLSL450, the Scope for memory **must** not be Workgroup
- VUID-StandaloneSpirv-None-07321
If the Scope for memory is Workgroup, then it **must** only be used in the task, mesh, tessellation control, or compute Execution Model
- VUID-StandaloneSpirv-None-04640
If the Scope for memory is ShaderCallKHR, then it **must** only be used in ray generation, intersection, closest hit, any-hit, miss, and callable Execution Model
- VUID-StandaloneSpirv-None-04641
If the Scope for memory is Invocation, then memory semantics **must** be None
- VUID-StandaloneSpirv-None-04642
Scope for group operations **must** be limited to Subgroup
- VUID-StandaloneSpirv-SubgroupVoteKHR-07951
If none of the SubgroupVoteKHR, GroupNonUniform, or SubgroupBallotKHR capabilities are declared, Scope for memory **must** not be Subgroup
- VUID-StandaloneSpirv-None-04643
Storage Class **must** be limited to UniformConstant, Input, Uniform, Output, Workgroup, Private, Function, PushConstant, Image, StorageBuffer, RayPayloadKHR, IncomingRayPayloadKHR, HitAttributeKHR, CallableDataKHR, IncomingCallableDataKHR, ShaderRecordBufferKHR, PhysicalStorageBuffer, or TileImageEXT
- VUID-StandaloneSpirv-None-04644
If the Storage Class is Output, then it **must** not be used in the GLCompute, RayGenerationKHR, IntersectionKHR, AnyHitKHR, ClosestHitKHR, MissKHR, or CallableKHR Execution Model
- VUID-StandaloneSpirv-None-04645
If the Storage Class is Workgroup, then it **must** only be used in the task, mesh, or compute Execution Model
- VUID-StandaloneSpirv-None-08720
If the Storage Class is TileImageEXT, then it **must** only be used in the fragment execution model
- VUID-StandaloneSpirv-OpAtomicStore-04730
OpAtomicStore **must** not use Acquire, AcquireRelease, or SequentiallyConsistent memory semantics
- VUID-StandaloneSpirv-OpAtomicLoad-04731

`OpAtomicLoad` **must** not use `Release`, `AcquireRelease`, or `SequentiallyConsistent` memory semantics

- VUID-StandaloneSpirv-OpMemoryBarrier-04732
`OpMemoryBarrier` **must** use one of `Acquire`, `Release`, `AcquireRelease`, or `SequentiallyConsistent` memory semantics
- VUID-StandaloneSpirv-OpMemoryBarrier-04733
`OpMemoryBarrier` **must** include at least one `Storage Class`
- VUID-StandaloneSpirv-OpControlBarrier-04650
If the semantics for `OpControlBarrier` includes one of `Acquire`, `Release`, `AcquireRelease`, or `SequentiallyConsistent` memory semantics, then it **must** include at least one `Storage Class`
- VUID-StandaloneSpirv-OpVariable-04651
Any `OpVariable` with an `Initializer` operand **must** have `Output`, `Private`, `Function`, or `Workgroup` as its `Storage Class` operand
- VUID-StandaloneSpirv-OpVariable-04734
Any `OpVariable` with an `Initializer` operand and `Workgroup` as its `Storage Class` operand **must** use `OpConstantNull` as the initializer
- VUID-StandaloneSpirv-OpReadClockKHR-04652
`Scope` for `OpReadClockKHR` **must** be limited to `Subgroup` or `Device`
- VUID-StandaloneSpirv-OriginLowerLeft-04653
The `OriginLowerLeft Execution Mode` **must** not be used; fragment entry points **must** declare `OriginUpperLeft`
- VUID-StandaloneSpirv-PixelCenterInteger-04654
The `PixelCenterInteger Execution Mode` **must** not be used (pixels are always centered at half-integer coordinates)
- VUID-StandaloneSpirv-UniformConstant-04655
Any variable in the `UniformConstant Storage Class` **must** be typed as either `OpTypeImage`, `OpTypeSampler`, `OpTypeSampledImage`, `OpTypeAccelerationStructureKHR`, or an array of one of these types
- VUID-StandaloneSpirv-Uniform-06807
Any variable in the `Uniform` or `StorageBuffer Storage Class` **must** be typed as `OpTypeStruct` or an array of this type
- VUID-StandaloneSpirv-PushConstant-06808
Any variable in the `PushConstant Storage Class` **must** be typed as `OpTypeStruct`
- VUID-StandaloneSpirv-OpTypeImage-04656
`OpTypeImage` **must** declare a scalar 32-bit float, 64-bit integer, or 32-bit integer type for the “Sampled Type” (`RelaxedPrecision` **can** be applied to a sampling instruction and to the variable holding the result of a sampling instruction)
- VUID-StandaloneSpirv-OpTypeImage-04657
`OpTypeImage` **must** have a “Sampled” operand of 1 (sampled image) or 2 (storage image)
- VUID-StandaloneSpirv-OpTypeSampledImage-06671
`OpTypeSampledImage` **must** have a `OpTypeImage` with a “Sampled” operand of 1 (sampled image)

- VUID-StandaloneSpirv-Image-04965
The **SPIR-V Type** of the **Image Format** operand of an **OpTypeImage** **must** match the **Sampled Type**, as defined in **Image Format and Type Matching**
- VUID-StandaloneSpirv-OpImageTexelPointer-04658
If an **OpImageTexelPointer** is used in an atomic operation, the image type of the **image** parameter to **OpImageTexelPointer** **must** have an image format of **R64i**, **R64ui**, **R32f**, **R32i**, or **R32ui**
- VUID-StandaloneSpirv-OpImageQuerySizeLod-04659
OpImageQuerySizeLod, **OpImageQueryLod**, and **OpImageQueryLevels** **must** only consume an “Image” operand whose type has its “Sampled” operand set to 1
- VUID-StandaloneSpirv-OpTypeImage-09638
An **OpTypeImage** **must** not have a “Dim” operand of **Rect**
- VUID-StandaloneSpirv-OpTypeImage-06214
An **OpTypeImage** with a “Dim” operand of **SubpassData** **must** have an “Arrayed” operand of 0 (non-arrayed) and a “Sampled” operand of 2 (storage image)
- VUID-StandaloneSpirv-SubpassData-04660
The (u,v) coordinates used for a **SubpassData** **must** be the <id> of a constant vector (0,0)
- VUID-StandaloneSpirv-OpTypeImage-06924
Objects of types **OpTypeImage**, **OpTypeSampler**, **OpTypeSampledImage**, **OpTypeAccelerationStructureKHR**, and arrays of these types **must** not be stored to or modified
- VUID-StandaloneSpirv-Uniform-06925
Any variable in the **Uniform Storage Class** decorated as **Block** **must** not be stored to or modified
- VUID-StandaloneSpirv-Offset-04663
Image operand **Offset** **must** only be used with **OpImage*Gather** instructions
- VUID-StandaloneSpirv-Offset-04865
Any image instruction which uses an **Offset**, **ConstOffset**, or **ConstOffsets** image operand, **must** only consume a “Sampled Image” operand whose type has its “Sampled” operand set to 1
- VUID-StandaloneSpirv-OpImageGather-04664
The “Component” operand of **OpImageGather**, and **OpImageSparseGather** **must** be the <id> of a constant instruction
- VUID-StandaloneSpirv-OpImage-04777
OpImage*Dref* instructions **must** not consume an image whose **Dim** is 3D
- VUID-StandaloneSpirv-None-04667
Structure types **must** not contain opaque types
- VUID-StandaloneSpirv-BuiltIn-04668
Any **BuiltIn** decoration not listed in **Built-In Variables** **must** not be used
- VUID-StandaloneSpirv-OpEntryPoint-09658
For a given **OpEntryPoint**, any **BuiltIn** decoration **must** not be used more than once by the **Input** interface

- VUID-StandaloneSpirv-OpEntryPoint-09659
For a given **OpEntryPoint**, any **BuiltIn** decoration **must** not be used more than once by the **Output** interface
- VUID-StandaloneSpirv-Location-06672
The **Location** or **Component** decorations **must** only be used with the **Input**, **Output**, **RayPayloadKHR**, **IncomingRayPayloadKHR**, **HitAttributeKHR**, **HitObjectAttributeNV**, **CallableDataKHR**, **IncomingCallableDataKHR**, or **ShaderRecordBufferKHR** storage classes
- VUID-StandaloneSpirv-Location-04915
The **Location** or **Component** decorations **must** not be used with **BuiltIn**
- VUID-StandaloneSpirv-Location-04916
The **Location** decorations **must** be used on **user-defined variables**
- VUID-StandaloneSpirv-Location-04917
If a **user-defined variable** is not a pointer to a **Block** decorated **OpTypeStruct**, then the **OpVariable** **must** have a **Location** decoration
- VUID-StandaloneSpirv-Location-04918
If a **user-defined variable** has a **Location** decoration, and the variable is a pointer to a **OpTypeStruct**, then the members of that structure **must** not have **Location** decorations
- VUID-StandaloneSpirv-Location-04919
If a **user-defined variable** does not have a **Location** decoration, and the variable is a pointer to a **Block** decorated **OpTypeStruct**, then each member of the struct **must** have a **Location** decoration
- VUID-StandaloneSpirv-Component-04920
The **Component** decoration value **must** not be greater than 3
- VUID-StandaloneSpirv-Component-04921
If the **Component** decoration is used on an **OpVariable** that has a **OpTypeVector** type with a **Component Type** with a **Width** that is less than or equal to 32, the sum of its **Component Count** and the **Component** decoration value **must** be less than or equal to 4
- VUID-StandaloneSpirv-Component-04922
If the **Component** decoration is used on an **OpVariable** that has a **OpTypeVector** type with a **Component Type** with a **Width** that is equal to 64, the sum of two times its **Component Count** and the **Component** decoration value **must** be less than or equal to 4
- VUID-StandaloneSpirv-Component-04923
The **Component** decorations value **must** not be 1 or 3 for scalar or two-component 64-bit data types
- VUID-StandaloneSpirv-Component-04924
The **Component** decorations **must** not be used with any type that is not a scalar or vector, or an array of such a type
- VUID-StandaloneSpirv-Component-07703
The **Component** decorations **must** not be used for a 64-bit vector type with more than two components
- VUID-StandaloneSpirv-Input-09557
The pointers of any **Input** or **Output Interface user-defined variables** **must** not contain any **PhysicalStorageBuffer Storage Class** pointers

- VUID-StandaloneSpirv-GLSLShared-04669
The **GLSLShared** and **GLSLPacked** decorations **must** not be used
- VUID-StandaloneSpirv-Flat-04670
The **Flat**, **NoPerspective**, **Sample**, and **Centroid** decorations **must** only be used on variables with the **Output** or **Input Storage Class**
- VUID-StandaloneSpirv-Flat-06201
The **Flat**, **NoPerspective**, **Sample**, and **Centroid** decorations **must** not be used on variables with the **Output** storage class in a fragment shader
- VUID-StandaloneSpirv-Flat-06202
The **Flat**, **NoPerspective**, **Sample**, and **Centroid** decorations **must** not be used on variables with the **Input** storage class in a vertex shader
- VUID-StandaloneSpirv-PerVertexKHR-06777
The **PerVertexKHR** decoration **must** only be used on variables with the **Input Storage Class** in a fragment shader
- VUID-StandaloneSpirv-Flat-04744
Any variable with integer or double-precision floating-point type and with **Input Storage Class** in a fragment shader, **must** be decorated **Flat**
- VUID-StandaloneSpirv-ViewportRelativeNV-04672
The **ViewportRelativeNV** decoration **must** only be used on a variable decorated with **Layer** in the vertex, tessellation evaluation, or geometry shader stages
- VUID-StandaloneSpirv-ViewportRelativeNV-04673
The **ViewportRelativeNV** decoration **must** not be used unless a variable decorated with one of **ViewportIndex** or **ViewportMaskNV** is also statically used by the same **OpEntryPoint**
- VUID-StandaloneSpirv-ViewportMaskNV-04674
The **ViewportMaskNV** and **ViewportIndex** decorations **must** not both be statically used by one or more **OpEntryPoint**'s that form the **pre-rasterization shader stages** of a graphics pipeline
- VUID-StandaloneSpirv-FPRoundingMode-04675
Rounding modes other than round-to-nearest-even and round-towards-zero **must** not be used for the **FPRoundingMode** decoration
- VUID-StandaloneSpirv-Invariant-04677
Variables decorated with **Invariant** and variables with structure types that have any members decorated with **Invariant** **must** be in the **Output** or **Input Storage Class**, **Invariant** used on an **Input Storage Class** variable or structure member has no effect
- VUID-StandaloneSpirv-VulkanMemoryModel-04678
If the **VulkanMemoryModel** capability is not declared, the **Volatile** decoration **must** be used on any variable declaration that includes one of the **SMIDNV**, **WarpIDNV**, **SubgroupSize**, **SubgroupLocalInvocationId**, **SubgroupEqMask**, **SubgroupGeMask**, **SubgroupGtMask**, **SubgroupLeMask**, or **SubgroupLtMask Builtin** decorations when used in the ray generation, closest hit, miss, intersection, or callable shaders, or with the **RayTmaxKHR Builtin** decoration when used in an intersection shader
- VUID-StandaloneSpirv-VulkanMemoryModel-04679
If the **VulkanMemoryModel** capability is declared, the **OpLoad** instruction **must** use the **Volatile** memory semantics when it accesses into any variable that includes one of the

SMIDNV, WarpIDNV, SubgroupSize, SubgroupLocalInvocationId, SubgroupEqMask, SubgroupGeMask, SubgroupGtMask, SubgroupLeMask, or SubgroupLtMask BuiltIn decorations when used in the ray generation, closest hit, miss, intersection, or callable shaders, or with the RayTmaxKHR BuiltIn decoration when used in an intersection shader

- VUID-StandaloneSpirv-OpTypeRuntimeArray-04680

OpTypeRuntimeArray **must** only be used for:

- the last member of a Block-decorated OpTypeStruct in StorageBuffer or PhysicalStorageBuffer storage Storage Class
- BufferBlock-decorated OpTypeStruct in the Uniform storage Storage Class
- the outermost dimension of an arrayed variable in the StorageBuffer, Uniform, or UniformConstant storage Storage Class
- variables in the NodePayloadAMD extension storage Storage Class when the CoalescingAMD extension Execution Mode is specified

- VUID-StandaloneSpirv-Function-04681

A type T that is an array sized with a specialization constant **must** neither be, nor be contained in, the type T_2 of a variable V , unless either: a) T is equal to T_2 , b) V is declared in the Function, or Private Storage Class, c) V is a non-Block variable in the Workgroup Storage Class, or d) V is an interface variable with an additional level of arrayness, as described in interface matching, and T is the member type of the array type T_2

- VUID-StandaloneSpirv-OpControlBarrier-04682

If OpControlBarrier is used in ray generation, intersection, any-hit, closest hit, miss, fragment, vertex, tessellation evaluation, or geometry shaders, the execution Scope **must** be Subgroup

- VUID-StandaloneSpirv-LocalSize-06426

For each compute shader entry point, either a LocalSize or LocalSizeId Execution Mode, or an object decorated with the WorkgroupSize decoration **must** be specified

- VUID-StandaloneSpirv-DerivativeGroupQuadsNV-04684

For compute shaders using the DerivativeGroupQuadsNV execution mode, the first two dimensions of the local workgroup size **must** be a multiple of two

- VUID-StandaloneSpirv-DerivativeGroupLinearNV-04778

For compute shaders using the DerivativeGroupLinearNV execution mode, the product of the dimensions of the local workgroup size **must** be a multiple of four

- VUID-StandaloneSpirv-DerivativeGroupQuadsKHR-10151

For compute, mesh, or task shaders using the DerivativeGroupQuadsKHR execution mode, the first two dimensions of the local workgroup size **must** be a multiple of two

- VUID-StandaloneSpirv-DerivativeGroupLinearKHR-10152

For compute, mesh, or task shaders using the DerivativeGroupLinearKHR execution mode, the product of the dimensions of the local workgroup size **must** be a multiple of four

- VUID-StandaloneSpirv-OpGroupNonUniformBallotBitCount-04685

If OpGroupNonUniformBallotBitCount is used, the group operation **must** be limited to Reduce, InclusiveScan, or ExclusiveScan

- VUID-StandaloneSpirv-None-04686

The *Pointer* operand of all atomic instructions **must** have a *Storage Class* limited to *Uniform*, *Workgroup*, *Image*, *StorageBuffer*, *PhysicalStorageBuffer*, or *TaskPayloadWorkgroupEXT*

- VUID-StandaloneSpirv-Offset-04687
Output variables or block members decorated with *Offset* that have a 64-bit type, or a composite type containing a 64-bit type, **must** specify an *Offset* value aligned to a 8 byte boundary
- VUID-StandaloneSpirv-Offset-04689
The size of any output block containing any member decorated with *Offset* that is a 64-bit type **must** be a multiple of 8
- VUID-StandaloneSpirv-Offset-04690
The first member of an output block specifying a *Offset* decoration **must** specify a *Offset* value that is aligned to an 8 byte boundary if that block contains any member decorated with *Offset* and is a 64-bit type
- VUID-StandaloneSpirv-Offset-04691
Output variables or block members decorated with *Offset* that have a 32-bit type, or a composite type contains a 32-bit type, **must** specify an *Offset* value aligned to a 4 byte boundary
- VUID-StandaloneSpirv-Offset-04692
Output variables, blocks or block members decorated with *Offset* **must** only contain base types that have components that are either 32-bit or 64-bit in size
- VUID-StandaloneSpirv-Offset-04716
Only variables or block members in the output interface decorated with *Offset* **can** be captured for transform feedback, and those variables or block members **must** also be decorated with *XfbBuffer* and *XfbStride*, or inherit *XfbBuffer* and *XfbStride* decorations from a block containing them
- VUID-StandaloneSpirv-XfbBuffer-04693
All variables or block members in the output interface of the entry point being compiled decorated with a specific *XfbBuffer* value **must** all be decorated with identical *XfbStride* values
- VUID-StandaloneSpirv-Stream-04694
If any variables or block members in the output interface of the entry point being compiled are decorated with *Stream*, then all variables belonging to the same *XfbBuffer* **must** specify the same *Stream* value
- VUID-StandaloneSpirv-XfbBuffer-04696
For any two variables or block members in the output interface of the entry point being compiled with the same *XfbBuffer* value, the ranges determined by the *Offset* decoration and the size of the type **must** not overlap
- VUID-StandaloneSpirv-XfbBuffer-04697
All block members in the output interface of the entry point being compiled that are in the same block and have a declared or inherited *XfbBuffer* decoration **must** specify the same *XfbBuffer* value
- VUID-StandaloneSpirv-RayPayloadKHR-04698
RayPayloadKHR Storage Class **must** only be used in ray generation, closest hit or miss

shaders

- VUID-StandaloneSpirv-IncomingRayPayloadKHR-04699
IncomingRayPayloadKHR Storage Class **must** only be used in closest hit, any-hit, or miss shaders
- VUID-StandaloneSpirv-IncomingRayPayloadKHR-04700
There **must** be at most one variable with the **IncomingRayPayloadKHR Storage Class** in the input interface of an entry point
- VUID-StandaloneSpirv-HitAttributeKHR-04701
HitAttributeKHR Storage Class **must** only be used in intersection, any-hit, or closest hit shaders
- VUID-StandaloneSpirv-HitAttributeKHR-04702
There **must** be at most one variable with the **HitAttributeKHR Storage Class** in the input interface of an entry point
- VUID-StandaloneSpirv-HitAttributeKHR-04703
A variable with **HitAttributeKHR Storage Class** **must** only be written to in an intersection shader
- VUID-StandaloneSpirv-CallableDataKHR-04704
CallableDataKHR Storage Class **must** only be used in ray generation, closest hit, miss, and callable shaders
- VUID-StandaloneSpirv-IncomingCallableDataKHR-04705
IncomingCallableDataKHR Storage Class **must** only be used in callable shaders
- VUID-StandaloneSpirv-IncomingCallableDataKHR-04706
There **must** be at most one variable with the **IncomingCallableDataKHR Storage Class** in the input interface of an entry point
- VUID-StandaloneSpirv-ShaderRecordBufferKHR-07119
ShaderRecordBufferKHR Storage Class **must** only be used in ray generation, intersection, any-hit, closest hit, callable, or miss shaders
- VUID-StandaloneSpirv-Base-07650
The **Base** operand of **OpPtrAccessChain** **must** have a storage class of **Workgroup**, **StorageBuffer**, or **PhysicalStorageBuffer**
- VUID-StandaloneSpirv-Base-07651
If the **Base** operand of **OpPtrAccessChain** has a **Workgroup Storage Class**, then the **VariablePointers** capability **must** be declared
- VUID-StandaloneSpirv-Base-07652
If the **Base** operand of **OpPtrAccessChain** has a **StorageBuffer Storage Class**, then the **VariablePointers** or **VariablePointersStorageBuffer** capability **must** be declared
- VUID-StandaloneSpirv-PhysicalStorageBuffer64-04708
If the **PhysicalStorageBuffer64** addressing model is enabled, all instructions that support memory access operands and that use a physical pointer **must** include the **Aligned** operand
- VUID-StandaloneSpirv-PhysicalStorageBuffer64-04709
If the **PhysicalStorageBuffer64** addressing model is enabled, any access chain instruction

that accesses into a **RowMajor** matrix **must** only be used as the **Pointer** operand to **OpLoad** or **OpStore**

- VUID-StandaloneSpirv-PhysicalStorageBuffer64-04710
If the **PhysicalStorageBuffer64** addressing model is enabled, **OpConvertUToPtr** and **OpConvertPtrToU** **must** use an integer type whose **Width** is 64
- VUID-StandaloneSpirv-OpTypeForwardPointer-04711
OpTypeForwardPointer **must** have a **Storage Class** of **PhysicalStorageBuffer**
- VUID-StandaloneSpirv-None-04745
All block members in a variable with a **Storage Class** of **PushConstant** declared as an array **must** only be accessed by dynamically uniform indices
- VUID-StandaloneSpirv-OpVariable-06673
There **must** not be more than one **OpVariable** in the **PushConstant Storage Class** listed in the **Interface** for each **OpEntryPoint**
- VUID-StandaloneSpirv-OpEntryPoint-06674
Each **OpEntryPoint** **must** not statically use more than one **OpVariable** in the **PushConstant Storage Class**
- VUID-StandaloneSpirv-OpEntryPoint-08721
Each **OpEntryPoint** **must** not have more than one **Input** variable assigned the same **Component** word inside a **Location** slot, either explicitly or implicitly
- VUID-StandaloneSpirv-OpEntryPoint-08722
Each **OpEntryPoint** **must** not have more than one **Output** variable assigned the same **Component** word inside a **Location** slot, either explicitly or implicitly
- VUID-StandaloneSpirv-Result-04780
The **Result Type** operand of any **OpImageRead** or **OpImageSparseRead** instruction **must** be a vector of four components
- VUID-StandaloneSpirv-Base-04781
The **Base** operand of any **OpBitCount**, **OpBitReverse**, **OpBitFieldInsert**, **OpBitFieldSExtract**, or **OpBitFieldUExtract** instruction **must** be a 32-bit integer scalar or a vector of 32-bit integers
- VUID-StandaloneSpirv-PushConstant-06675
Any variable in the **PushConstant** or **StorageBuffer** storage class **must** be decorated as **Block**
- VUID-StandaloneSpirv-Uniform-06676
Any variable in the **Uniform Storage Class** **must** be decorated as **Block** or **BufferBlock**
- VUID-StandaloneSpirv-UniformConstant-06677
Any variable in the **UniformConstant**, **StorageBuffer**, or **Uniform Storage Class** **must** be decorated with **DescriptorSet** and **Binding**
- VUID-StandaloneSpirv-InputAttachmentIndex-06678
Variables decorated with **InputAttachmentIndex** **must** be in the **UniformConstant Storage Class**
- VUID-StandaloneSpirv-DescriptorSet-06491
If a variable is decorated by **DescriptorSet** or **Binding**, the **Storage Class** **must** correspond to an entry in [Shader Resource and Storage Class Correspondence](#)

- VUID-StandaloneSpirv-Input-06778
Variables with a **Storage Class** of **Input** in a fragment shader stage that are decorated with **PerVertexKHR** **must** be declared as arrays
- VUID-StandaloneSpirv-MeshEXT-07102
The module **must** not contain both an entry point that uses the **TaskEXT** or **MeshEXT Execution Model** and an entry point that uses the **TaskNV** or **MeshNV Execution Model**
- VUID-StandaloneSpirv-MeshEXT-07106
In mesh shaders using the **MeshEXT Execution Model** **OpSetMeshOutputsEXT** **must** be called before any outputs are written
- VUID-StandaloneSpirv-MeshEXT-07107
In mesh shaders using the **MeshEXT Execution Model** all variables declared as output **must** not be read from
- VUID-StandaloneSpirv-MeshEXT-07108
In mesh shaders using the **MeshEXT Execution Model** for **OpSetMeshOutputsEXT** instructions, the “Vertex Count” and “Primitive Count” operands **must** not depend on **ViewIndex**
- VUID-StandaloneSpirv-MeshEXT-07109
In mesh shaders using the **MeshEXT Execution Model** variables decorated with **PrimitivePointIndicesEXT**, **PrimitiveLineIndicesEXT**, or **PrimitiveTriangleIndicesEXT** declared as an array **must** not be accessed by indices that depend on **ViewIndex**
- VUID-StandaloneSpirv-MeshEXT-07110
In mesh shaders using the **MeshEXT Execution Model** any values stored in variables decorated with **PrimitivePointIndicesEXT**, **PrimitiveLineIndicesEXT**, or **PrimitiveTriangleIndicesEXT** **must** not depend on **ViewIndex**
- VUID-StandaloneSpirv-MeshEXT-07111
In mesh shaders using the **MeshEXT Execution Model** variables in workgroup or private **Storage Class** declared as or containing a composite type **must** not be accessed by indices that depend on **ViewIndex**
- VUID-StandaloneSpirv-MeshEXT-07330
In mesh shaders using the **MeshEXT Execution Model** the **OutputVertices Execution Mode** **must** be greater than 0
- VUID-StandaloneSpirv-MeshEXT-07331
In mesh shaders using the **MeshEXT Execution Model** the **OutputPrimitivesEXT Execution Mode** **must** be greater than 0
- VUID-StandaloneSpirv-Input-07290
Variables with a **Storage Class** of **Input** or **Output** and a type of **OpTypeBool** **must** be decorated with the **BuiltIn** decoration
- VUID-StandaloneSpirv-TileImageEXT-08723
The tile image variable declarations **must** obey the constraints on the **TileImageEXT Storage Class** and the **Location** decoration described in **Fragment Tile Image Interface**
- VUID-StandaloneSpirv-None-08724
The **TileImageEXT Storage Class** **must** only be used for declaring tile image variables
- VUID-StandaloneSpirv-Pointer-08973
The **Storage Class** of the **Pointer** operand to **OpCooperativeMatrixLoadKHR** or

`OpCooperativeMatrixStoreKHR` **must** be limited to `Workgroup`, `StorageBuffer`, or `PhysicalStorageBuffer`

- VUID-StandaloneSpirv-UniformBufferArrayDynamicIndexing-10127
If the `UniformBufferArrayDynamicIndexing` capability is not declared, and an instruction accesses memory through a uniform buffer, the uniform buffer through which that memory is accessed **must** be determined by [constant integral expressions](#)
- VUID-StandaloneSpirv-SampledImageArrayDynamicIndexing-10128
If the `SampledImageArrayDynamicIndexing` capability is not declared, and an instruction accesses memory through a sampled image or sampler, the sampled image or sampler through which that memory is accessed **must** be determined by [constant integral expressions](#)
- VUID-StandaloneSpirv-StorageBufferArrayDynamicIndexing-10129
If the `StorageBufferArrayDynamicIndexing` capability is not declared, and an instruction accesses memory through a storage buffer, the storage buffer through which that memory is accessed **must** be determined by [constant integral expressions](#)
- VUID-StandaloneSpirv-StorageImageArrayDynamicIndexing-10130
If the `StorageImageArrayDynamicIndexing` capability is not declared, and an instruction accesses memory through a storage image, the storage image through which that memory is accessed **must** be determined by [constant integral expressions](#)
- VUID-StandaloneSpirv-InputAttachmentArrayDynamicIndexing-10131
If the `InputAttachmentArrayDynamicIndexing` capability is not declared, and an instruction accesses memory through an input attachment, the input attachment through which that memory is accessed **must** be determined by [constant integral expressions](#)
- VUID-StandaloneSpirv-UniformTexelBufferArrayDynamicIndexing-10132
If the `UniformTexelBufferArrayDynamicIndexing` capability is not declared, and an instruction accesses memory through a uniform texel buffer, the uniform texel buffer through which that memory is accessed **must** be determined by [constant integral expressions](#)
- VUID-StandaloneSpirv-StorageTexelBufferArrayDynamicIndexing-10133
If the `StorageTexelBufferArrayDynamicIndexing` capability is not declared, and an instruction accesses memory through a storage texel buffer, the storage texel buffer through which that memory is accessed **must** be determined by [constant integral expressions](#)

Runtime SPIR-V Validation

The following rules **must** be validated at runtime. These rules depend on knowledge of the implementation and its capabilities and knowledge of runtime information, such as enabled features.

Valid Usage

- VUID-RuntimeSpirv-vulkanMemoryModel-06265
If `vulkanMemoryModel` is enabled and `vulkanMemoryModelDeviceScope` is not enabled, `Device`

memory scope **must** not be used

- VUID-RuntimeSpirv-vulkanMemoryModel-06266

If `vulkanMemoryModel` is not enabled, `QueueFamily` memory scope **must** not be used

- VUID-RuntimeSpirv-shaderSubgroupClock-06267

If `shaderSubgroupClock` is not enabled, the `Subgroup` scope **must** not be used for `OpReadClockKHR`

- VUID-RuntimeSpirv-shaderDeviceClock-06268

If `shaderDeviceClock` is not enabled, the `Device` scope **must** not be used for `OpReadClockKHR`

- VUID-RuntimeSpirv-None-09558

If `dynamicRenderingLocalRead` is not enabled, any variable created with a “Type” of `OpTypeImage` that has a “Dim” operand of `SubpassData` **must** be decorated with `InputAttachmentIndex`

- VUID-RuntimeSpirv-OpTypeImage-09644

Any variable declared as an `OpTypeArray` where the `Element Type` is an `OpTypeImage` with a “Dim” operand of `SubpassData` **must** be decorated with `InputAttachmentIndex`

- VUID-RuntimeSpirv-shaderStorageImageWriteWithoutFormat-07956

If the `VK_KHR_format_feature_flags2` extension is not enabled, and `shaderStorageImageWriteWithoutFormat` is not enabled, any variable created with a “Type” of `OpTypeImage` that has a “Sampled” operand of 2 and an “Image Format” operand of `Unknown` **must** be decorated with `NonWritable`

- VUID-RuntimeSpirv-shaderStorageImageReadWithoutFormat-07957

If the `VK_KHR_format_feature_flags2` extension is not enabled, and `shaderStorageImageReadWithoutFormat` is not enabled, any variable created with a “Type” of `OpTypeImage` that has a “Sampled” operand of 2 and an “Image Format” operand of `Unknown` **must** be decorated with `NonReadable`

- VUID-RuntimeSpirv-OpImageWrite-07112

`OpImageWrite` to any `Image` whose `Image Format` is not `Unknown` **must** have the `Texel` operand contain at least as many components as the corresponding `VkFormat` as given in the [SPIR-V Image Format compatibility table](#)

- VUID-RuntimeSpirv-Location-06272

The sum of `Location` and the number of locations the variable it decorates consumes **must** be less than or equal to the value for the matching `Execution Model` defined in [Shader Input and Output Locations](#)

- VUID-RuntimeSpirv-Location-06428

The maximum number of storage buffers, storage images, and output `Location` decorated color attachments written to in the `Fragment Execution Model` **must** be less than or equal to `maxFragmentCombinedOutputResources`

- VUID-RuntimeSpirv-None-06278

`shaderBufferInt64Atomics` **must** be enabled for 64-bit integer atomic operations to be supported on a *Pointer* with a `Storage Class` of `StorageBuffer` or `Uniform`

- VUID-RuntimeSpirv-None-06279

`shaderSharedInt64Atomics` **must** be enabled for 64-bit integer atomic operations to be supported on a *Pointer* with a `Storage Class` of `Workgroup`

- VUID-RuntimeSpirv-None-06284
`shaderBufferFloat32Atomics,` or `shaderBufferFloat32AtomicAdd,` or
`shaderBufferFloat64Atomics,` or `shaderBufferFloat64AtomicAdd,` or
`shaderBufferFloat16Atomics,` or `shaderBufferFloat16AtomicAdd,` or
`shaderBufferFloat16AtomicMinMax,` or `shaderBufferFloat32AtomicMinMax,` or
`shaderBufferFloat64AtomicMinMax,` **must** be enabled for floating-point atomic operations to be supported on a *Pointer* with a *Storage Class* of *StorageBuffer*
- VUID-RuntimeSpirv-None-06285
`shaderSharedFloat32Atomics,` or `shaderSharedFloat32AtomicAdd,` or
`shaderSharedFloat64Atomics,` or `shaderSharedFloat64AtomicAdd,` or
`shaderSharedFloat16Atomics,` or `shaderSharedFloat16AtomicAdd,` or
`shaderSharedFloat16AtomicMinMax,` or `shaderSharedFloat32AtomicMinMax,` or
`shaderSharedFloat64AtomicMinMax,` **must** be enabled for floating-point atomic operations to be supported on a *Pointer* with a *Storage Class* of *Workgroup*
- VUID-RuntimeSpirv-None-06286
`shaderImageFloat32Atomics,` or `shaderImageFloat32AtomicAdd,` or
`shaderImageFloat32AtomicMinMax,` **must** be enabled for 32-bit floating-point atomic operations to be supported on a *Pointer* with a *Storage Class* of *Image*
- VUID-RuntimeSpirv-None-06287
`sparseImageFloat32Atomics,` or `sparseImageFloat32AtomicAdd,` or
`sparseImageFloat32AtomicMinMax,` **must** be enabled for 32-bit floating-point atomics to be supported on sparse images
- VUID-RuntimeSpirv-None-06288
`shaderImageInt64Atomics` **must** be enabled for 64-bit integer atomic operations to be supported on a *Pointer* with a *Storage Class* of *Image*
- VUID-RuntimeSpirv-denormBehaviorIndependence-06289
If `denormBehaviorIndependence` is `VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY`, then the entry point **must** use the same denormals *Execution Mode* for both 16-bit and 64-bit floating-point types
- VUID-RuntimeSpirv-denormBehaviorIndependence-06290
If `denormBehaviorIndependence` is `VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE`, then the entry point **must** use the same denormals *Execution Mode* for all floating-point types
- VUID-RuntimeSpirv-roundingModeIndependence-06291
If `roundingModeIndependence` is `VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY`, then the entry point **must** use the same rounding *Execution Mode* for both 16-bit and 64-bit floating-point types
- VUID-RuntimeSpirv-roundingModeIndependence-06292
If `roundingModeIndependence` is `VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE`, then the entry point **must** use the same rounding *Execution Mode* for all floating-point types
- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat16-06293
If `shaderSignedZeroInfNanPreserveFloat16` is `VK_FALSE`, then *SignedZeroInfNanPreserve* for 16-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat32-06294
If `shaderSignedZeroInfNanPreserveFloat32` is `VK_FALSE`, then *SignedZeroInfNanPreserve* for

32-bit floating-point type **must** not be used

- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat64-06295
If `shaderSignedZeroInfNanPreserveFloat64` is `VK_FALSE`, then `SignedZeroInfNanPreserve` for 64-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormPreserveFloat16-06296
If `shaderDenormPreserveFloat16` is `VK_FALSE`, then `DenormPreserve` for 16-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormPreserveFloat32-06297
If `shaderDenormPreserveFloat32` is `VK_FALSE`, then `DenormPreserve` for 32-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormPreserveFloat64-06298
If `shaderDenormPreserveFloat64` is `VK_FALSE`, then `DenormPreserve` for 64-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormFlushToZeroFloat16-06299
If `shaderDenormFlushToZeroFloat16` is `VK_FALSE`, then `DenormFlushToZero` for 16-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormFlushToZeroFloat32-06300
If `shaderDenormFlushToZeroFloat32` is `VK_FALSE`, then `DenormFlushToZero` for 32-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderDenormFlushToZeroFloat64-06301
If `shaderDenormFlushToZeroFloat64` is `VK_FALSE`, then `DenormFlushToZero` for 64-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderRoundingModeRTEFloat16-06302
If `shaderRoundingModeRTEFloat16` is `VK_FALSE`, then `RoundingModeRTE` for 16-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderRoundingModeRTEFloat32-06303
If `shaderRoundingModeRTEFloat32` is `VK_FALSE`, then `RoundingModeRTE` for 32-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderRoundingModeRTEFloat64-06304
If `shaderRoundingModeRTEFloat64` is `VK_FALSE`, then `RoundingModeRTE` for 64-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderRoundingModeRTZFloat16-06305
If `shaderRoundingModeRTZFloat16` is `VK_FALSE`, then `RoundingModeRTZ` for 16-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderRoundingModeRTZFloat32-06306
If `shaderRoundingModeRTZFloat32` is `VK_FALSE`, then `RoundingModeRTZ` for 32-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderRoundingModeRTZFloat64-06307
If `shaderRoundingModeRTZFloat64` is `VK_FALSE`, then `RoundingModeRTZ` for 64-bit floating-point type **must** not be used
- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat16-09559
If `shaderSignedZeroInfNanPreserveFloat16` is `VK_FALSE` then any `FPFastMathDefault` execution

mode with a type of 16-bit float **must** include the **NSZ**, **NotInf**, and **NotNaN** flags

- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat16-09560

If **shaderSignedZeroInfNanPreserveFloat16** is **VK_FALSE** then any **FPFastMathMode** decoration on an instruction with result type or any operand type that includes a 16-bit float **must** include the **NSZ**, **NotInf**, and **NotNaN** flags

- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat32-09561

If **shaderSignedZeroInfNanPreserveFloat32** is **VK_FALSE** then any **FPFastMathDefault** execution mode with a type of 32-bit float **must** include the **NSZ**, **NotInf**, and **NotNaN** flags

- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat32-09562

If **shaderSignedZeroInfNanPreserveFloat32** is **VK_FALSE** then any **FPFastMathMode** decoration on an instruction with result type or any operand type that includes a 32-bit float **must** include the **NSZ**, **NotInf**, and **NotNaN** flags

- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat64-09563

If **shaderSignedZeroInfNanPreserveFloat64** is **VK_FALSE** then any **FPFastMathDefault** execution mode with a type of 64-bit float **must** include the **NSZ**, **NotInf**, and **NotNaN** flags

- VUID-RuntimeSpirv-shaderSignedZeroInfNanPreserveFloat64-09564

If **shaderSignedZeroInfNanPreserveFloat64** is **VK_FALSE** then any **FPFastMathMode** decoration on an instruction with result type or any operand type that includes a 64-bit float **must** include the **NSZ**, **NotInf**, and **NotNaN** flags

- VUID-RuntimeSpirv-Offset-06308

The **Offset** plus size of the type of each variable, in the output interface of the entry point being compiled, decorated with **XfbBuffer** **must** not be greater than **VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBufferDataSize**

- VUID-RuntimeSpirv-XfbBuffer-06309

For any given **XfbBuffer** value, define the buffer data size to be smallest number of bytes such that, for all outputs decorated with the same **XfbBuffer** value, the size of the output interface variable plus the **Offset** is less than or equal to the buffer data size. For a given **Stream**, the sum of all the buffer data sizes for all buffers writing to that stream the **must** not exceed **VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreamDataSize**

- VUID-RuntimeSpirv-OpEmitStreamVertex-06310

The **Stream** value to **OpEmitStreamVertex** and **OpEndStreamPrimitive** **must** be less than **VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams**

- VUID-RuntimeSpirv-transformFeedbackStreamsLinesTriangles-06311

If the geometry shader emits to more than one vertex stream and **VkPhysicalDeviceTransformFeedbackPropertiesEXT::transformFeedbackStreamsLinesTriangles** is **VK_FALSE**, then **Execution Mode** **must** be **OutputPoints**

- VUID-RuntimeSpirv-Stream-06312

The **stream** number value to **Stream** **must** be less than **VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackStreams**

- VUID-RuntimeSpirv-XfbStride-06313

The **XFB Stride** value to **XfbStride** **must** be less than or equal to **VkPhysicalDeviceTransformFeedbackPropertiesEXT::maxTransformFeedbackBufferDataStrid**

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- VUID-RuntimeSpirv-PhysicalStorageBuffer64-06314

If the `PhysicalStorageBuffer64` addressing model is enabled any load or store through a physical pointer type **must** be aligned to a multiple of the size of the largest scalar type in the pointed-to type

- VUID-RuntimeSpirv-PhysicalStorageBuffer64-06315

If the `PhysicalStorageBuffer64` addressing model is enabled the pointer value of a memory access instruction **must** be at least as aligned as specified by the `Aligned` memory access operand

- VUID-RuntimeSpirv-OpTypeCooperativeMatrixKHR-10163

For `OpTypeCooperativeMatrixKHR`, the component type, scope, number of rows, and number of columns **must** match one of the matrices in any of the supported `VkCooperativeMatrixPropertiesKHR`, where

- for `Use` of `MatrixA`, the number of rows must match `VkCooperativeMatrixPropertiesKHR::MSize` and the number of columns must match `VkCooperativeMatrixPropertiesKHR::KSize` and the type **must** match `VkCooperativeMatrixPropertiesKHR::AType`
- for `Use` of `MatrixB`, the number of rows must match `VkCooperativeMatrixPropertiesKHR::KSize` and the number of columns must match `VkCooperativeMatrixPropertiesKHR::NSize` and the type **must** match `VkCooperativeMatrixPropertiesKHR::BType`
- for `Use` of `MatrixAccumulator`, the number of rows must match `VkCooperativeMatrixPropertiesKHR::MSize` and the number of columns must match `VkCooperativeMatrixPropertiesKHR::NSize` and the type **must** match `VkCooperativeMatrixPropertiesKHR::CType` or `VkCooperativeMatrixPropertiesKHR::ResultType`

- VUID-RuntimeSpirv-OpCooperativeMatrixMulAddKHR-10060

For `OpCooperativeMatrixMulAddKHR`, the operands **must** match a supported `VkCooperativeMatrixPropertiesKHR`, such that:

- The type of `A` **must** have `Rows` match `VkCooperativeMatrixPropertiesKHR::MSize`, `Columns` match `VkCooperativeMatrixPropertiesKHR::KSize`, `Use` be `MatrixAKHR`, and `ComponentType` match `VkCooperativeMatrixPropertiesKHR::AType`.
- The type of `B` **must** have `Rows` match `VkCooperativeMatrixPropertiesKHR::KSize`, `Columns` match `VkCooperativeMatrixPropertiesKHR::NSize`, `Use` be `MatrixBKHR`, and `ComponentType` match `VkCooperativeMatrixPropertiesKHR::BType`.
- The type of `C` **must** have `Rows` match `VkCooperativeMatrixPropertiesKHR::MSize`, `Columns` match `VkCooperativeMatrixPropertiesKHR::NSize`, `Use` be `MatrixAccumulatorKHR`, and `ComponentType` match `VkCooperativeMatrixPropertiesKHR::CType`.
- The type of `Result` **must** have `Rows` match `VkCooperativeMatrixPropertiesKHR::MSize`, `Columns` match `VkCooperativeMatrixPropertiesKHR::NSize`, `Use` be `MatrixAccumulatorKHR`, and `ComponentType` match `VkCooperativeMatrixPropertiesKHR::ResultType`.

- If and only if `VkCooperativeMatrixPropertiesKHR::AType` is a signed integer type, `MatrixASignedComponents` **must** be used.
 - If and only if `VkCooperativeMatrixPropertiesKHR::BType` is a signed integer type, `MatrixBSignedComponents` **must** be used.
 - If and only if `VkCooperativeMatrixPropertiesKHR::CType` is a signed integer type, `MatrixCSignedComponents` **must** be used.
 - If and only if `VkCooperativeMatrixPropertiesKHR::ResultType` is a signed integer type, `MatrixResultSignedComponents` **must** be used.
 - If and only if `VkCooperativeMatrixPropertiesKHR::saturatingAccumulation` is `VK_TRUE`, `SaturatingAccumulationKHR` **must** be used.
 - If and only if `VkCooperativeMatrixPropertiesKHR::saturatingAccumulation` is `VK_FALSE`, `SaturatingAccumulationKHR` **must** not be used.
 - The scope of all cooperative matrix operands **must** match `VkCooperativeMatrixPropertiesKHR::scope`.
- VUID-RuntimeSpirv-cooperativeMatrixSupportedStages-08985
`OpTypeCooperativeMatrixKHR` and `OpCooperativeMatrix*` instructions **must** not be used in shader stages not included in `VkPhysicalDeviceCooperativeMatrixPropertiesKHR::cooperativeMatrixSupportedStages`
 - VUID-RuntimeSpirv-DescriptorSet-06323
`DescriptorSet` and `Binding` decorations **must** obey the constraints on `Storage Class`, type, and descriptor type described in `DescriptorSet and Binding Assignment`
 - VUID-RuntimeSpirv-OpCooperativeMatrixLoadKHR-08986
For `OpCooperativeMatrixLoadKHR` and `OpCooperativeMatrixStoreKHR` instructions, the `Pointer` and `Stride` operands **must** be aligned to at least the lesser of 16 bytes or the natural alignment of a row or column (depending on `ColumnMajor`) of the matrix (where the natural alignment is the number of columns/rows multiplied by the component size)
 - VUID-RuntimeSpirv-shaderSampleRateInterpolationFunctions-06325
If the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::shaderSampleRateInterpolationFunctions` is `VK_FALSE`, then `GLSL.std.450` fragment interpolation functions are not supported by the implementation and `OpCapability` **must** not be `InterpolationFunction`
 - VUID-RuntimeSpirv-tessellationShader-06326
If `tessellationShader` is enabled, and the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::tessellationIsolines` is `VK_FALSE`, then `OpExecutionMode` **must** not be `Isolines`
 - VUID-RuntimeSpirv-tessellationShader-06327
If `tessellationShader` is enabled, and the `VK_KHR_portability_subset` extension is enabled, and `VkPhysicalDevicePortabilitySubsetFeaturesKHR::tessellationPointMode` is `VK_FALSE`, then `OpExecutionMode` **must** not be `PointMode`
 - VUID-RuntimeSpirv-storageBuffer8BitAccess-06328
If `storageBuffer8BitAccess` is `VK_FALSE`, then objects containing an 8-bit integer element **must** not have `Storage Class` of `StorageBuffer`, `ShaderRecordBufferKHR`, or `PhysicalStorageBuffer`

- VUID-RuntimeSpirv-uniformAndStorageBuffer8BitAccess-06329
If `uniformAndStorageBuffer8BitAccess` is `VK_FALSE`, then objects in the `Uniform Storage Class` with the `Block` decoration **must** not have an 8-bit integer member
- VUID-RuntimeSpirv-storagePushConstant8-06330
If `storagePushConstant8` is `VK_FALSE`, then objects containing an 8-bit integer element **must** not have `Storage Class` of `PushConstant`
- VUID-RuntimeSpirv-storageBuffer16BitAccess-06331
If `storageBuffer16BitAccess` is `VK_FALSE`, then objects containing 16-bit integer or 16-bit floating-point elements **must** not have `Storage Class` of `StorageBuffer`, `ShaderRecordBufferKHR`, or `PhysicalStorageBuffer`
- VUID-RuntimeSpirv-uniformAndStorageBuffer16BitAccess-06332
If `uniformAndStorageBuffer16BitAccess` is `VK_FALSE`, then objects in the `Uniform Storage Class` with the `Block` decoration **must** not have 16-bit integer or 16-bit floating-point members
- VUID-RuntimeSpirv-storagePushConstant16-06333
If `storagePushConstant16` is `VK_FALSE`, then objects containing 16-bit integer or 16-bit floating-point elements **must** not have `Storage Class` of `PushConstant`
- VUID-RuntimeSpirv-storageInputOutput16-06334
If `storageInputOutput16` is `VK_FALSE`, then objects containing 16-bit integer or 16-bit floating-point elements **must** not have `Storage Class` of `Input` or `Output`
- VUID-RuntimeSpirv-None-06337
`shaderBufferFloat16Atomics`, or `shaderBufferFloat16AtomicAdd`, or `shaderBufferFloat16AtomicMinMax`, or `shaderSharedFloat16Atomics`, or `shaderSharedFloat16AtomicAdd`, or `shaderSharedFloat16AtomicMinMax` **must** be enabled for 16-bit floating-point atomic operations
- VUID-RuntimeSpirv-None-06338
`shaderBufferFloat32Atomics`, or `shaderBufferFloat32AtomicAdd`, or `shaderSharedFloat32Atomics`, or `shaderSharedFloat32AtomicAdd`, or `shaderImageFloat32Atomics`, or `shaderImageFloat32AtomicAdd`, or `shaderBufferFloat32AtomicMinMax`, or `shaderSharedFloat32AtomicMinMax`, or `shaderImageFloat32AtomicMinMax` **must** be enabled for 32-bit floating-point atomic operations
- VUID-RuntimeSpirv-None-06339
`shaderBufferFloat64Atomics`, or `shaderBufferFloat64AtomicAdd`, or `shaderSharedFloat64Atomics`, or `shaderSharedFloat64AtomicAdd`, or `shaderBufferFloat64AtomicMinMax`, or `shaderSharedFloat64AtomicMinMax`, **must** be enabled for 64-bit floating-point atomic operations
- VUID-RuntimeSpirv-NonWritable-06340
If `fragmentStoresAndAtomics` is not enabled, then all storage image, storage texel buffer, and storage buffer variables in the fragment stage **must** be decorated with the `NonWritable` decoration
- VUID-RuntimeSpirv-NonWritable-06341
If `vertexPipelineStoresAndAtomics` is not enabled, then all storage image, storage texel buffer, and storage buffer variables in the vertex, tessellation, and geometry stages **must** be decorated with the `NonWritable` decoration

- VUID-RuntimeSpirv-None-06342
If `subgroupQuadOperationsInAllStages` is `VK_FALSE`, then `quad subgroup operations` **must** not be used except for in fragment and compute stages
- VUID-RuntimeSpirv-Offset-06344
The first element of the `Offset` operand of `InterpolateAtOffset` **must** be greater than or equal to:
$$\text{frag}_{\text{width}} \times \text{minInterpolationOffset}$$

where $\text{frag}_{\text{width}}$ is the width of the current fragment in pixels
- VUID-RuntimeSpirv-Offset-06345
The first element of the `Offset` operand of `InterpolateAtOffset` **must** be less than or equal to
$$\text{frag}_{\text{width}} \times (\text{maxInterpolationOffset} + \text{ULP}) - \text{ULP}$$

where $\text{frag}_{\text{width}}$ is the width of the current fragment in pixels and $\text{ULP} = 1 / 2^{\text{subPixelInterpolationOffsetBits}}$
- VUID-RuntimeSpirv-Offset-06346
The second element of the `Offset` operand of `InterpolateAtOffset` **must** be greater than or equal to
$$\text{frag}_{\text{height}} \times \text{minInterpolationOffset}$$

where $\text{frag}_{\text{height}}$ is the height of the current fragment in pixels
- VUID-RuntimeSpirv-Offset-06347
The second element of the `Offset` operand of `InterpolateAtOffset` **must** be less than or equal to
$$\text{frag}_{\text{height}} \times (\text{maxInterpolationOffset} + \text{ULP}) - \text{ULP}$$

where $\text{frag}_{\text{height}}$ is the height of the current fragment in pixels and $\text{ULP} = 1 / 2^{\text{subPixelInterpolationOffsetBits}}$
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06348
For `OpRayQueryInitializeKHR` instructions, all components of the `RayOrigin` and `RayDirection` operands **must** be finite floating-point values
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06349
For `OpRayQueryInitializeKHR` instructions, the `RayTmin` and `RayTmax` operands **must** be non-negative floating-point values
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06350
For `OpRayQueryInitializeKHR` instructions, the `RayTmin` operand **must** be less than or equal to the `RayTmax` operand
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06351
For `OpRayQueryInitializeKHR` instructions, `RayOrigin`, `RayDirection`, `RayTmin`, and `RayTmax` operands **must** not contain NaNs
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06352
For `OpRayQueryInitializeKHR` instructions, `Acceleration Structure` **must** be an acceleration structure built as a `top-level acceleration structure`
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06889
For `OpRayQueryInitializeKHR` instructions, the `Rayflags` operand **must** not contain both `SkipTrianglesKHR` and `SkipAABBsKHR`

- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06890
For `OpRayQueryInitializeKHR` instructions, the `Rayflags` operand **must** not contain more than one of `SkipTrianglesKHR`, `CullBackFacingTrianglesKHR`, and `CullFrontFacingTrianglesKHR`
- VUID-RuntimeSpirv-OpRayQueryInitializeKHR-06891
For `OpRayQueryInitializeKHR` instructions, the `Rayflags` operand **must** not contain more than one of `OpaqueKHR`, `NoOpaqueKHR`, `CullOpaqueKHR`, and `CullNoOpaqueKHR`
- VUID-RuntimeSpirv-OpRayQueryGenerateIntersectionKHR-06353
For `OpRayQueryGenerateIntersectionKHR` instructions, `Hit T` **must** satisfy the condition $\text{RayTmin} \leq \text{Hit } T \leq \text{RayTmax}$, where `RayTmin` is equal to the value returned by `OpRayQueryGetRayTminKHR` with the same ray query object, and `RayTmax` is equal to the value of `OpRayQueryGetIntersectionTKHR` for the current committed intersection with the same ray query object
- VUID-RuntimeSpirv-flags-08761
For `OpRayQueryGetIntersectionTriangleVertexPositionsKHR` instructions, `Acceleration Structure` **must** have been built with `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_KHR` in `flags`
- VUID-RuntimeSpirv-OpTraceRayKHR-06355
For `OpTraceRayKHR` instructions, all components of the `RayOrigin` and `RayDirection` operands **must** be finite floating-point values
- VUID-RuntimeSpirv-OpTraceRayKHR-06356
For `OpTraceRayKHR` instructions, the `RayTmin` and `RayTmax` operands **must** be non-negative floating-point values
- VUID-RuntimeSpirv-OpTraceRayKHR-06552
For `OpTraceRayKHR` instructions, the `Rayflags` operand **must** not contain both `SkipTrianglesKHR` and `SkipAABBsKHR`
- VUID-RuntimeSpirv-OpTraceRayKHR-06892
For `OpTraceRayKHR` instructions, the `Rayflags` operand **must** not contain more than one of `SkipTrianglesKHR`, `CullBackFacingTrianglesKHR`, and `CullFrontFacingTrianglesKHR`
- VUID-RuntimeSpirv-OpTraceRayKHR-06893
For `OpTraceRayKHR` instructions, the `Rayflags` operand **must** not contain more than one of `OpaqueKHR`, `NoOpaqueKHR`, `CullOpaqueKHR`, and `CullNoOpaqueKHR`
- VUID-RuntimeSpirv-OpTraceRayKHR-06553
For `OpTraceRayKHR` instructions, if the `Rayflags` operand contains `SkipTrianglesKHR`, the pipeline **must** not have been created with `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR` set
- VUID-RuntimeSpirv-OpTraceRayKHR-06554
For `OpTraceRayKHR` instructions, if the `Rayflags` operand contains `SkipAABBsKHR`, the pipeline **must** not have been created with `VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR` set
- VUID-RuntimeSpirv-OpTraceRayKHR-06357
For `OpTraceRayKHR` instructions, the `RayTmin` operand **must** be less than or equal to the `RayTmax` operand

- VUID-RuntimeSpirv-OpTraceRayKHR-06358
For `OpTraceRayKHR` instructions, `RayOrigin`, `RayDirection`, `RayTmin`, and `RayTmax` operands **must** not contain NaNs
- VUID-RuntimeSpirv-OpTraceRayKHR-06359
For `OpTraceRayKHR` instructions, `Acceleration Structure` **must** be an acceleration structure built as a `top-level acceleration structure`
- VUID-RuntimeSpirv-OpReportIntersectionKHR-06998
The value of the “Hit Kind” operand of `OpReportIntersectionKHR` **must** be in the range [0,127]
- VUID-RuntimeSpirv-x-06429
In compute shaders using the `GLCompute Execution Model` the `x` size in `LocalSize` or `LocalSizeId` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupSize[0]`
- VUID-RuntimeSpirv-y-06430
In compute shaders using the `GLCompute Execution Model` the `y` size in `LocalSize` or `LocalSizeId` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupSize[1]`
- VUID-RuntimeSpirv-z-06431
In compute shaders using the `GLCompute Execution Model` the `z` size in `LocalSize` or `LocalSizeId` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupSize[2]`
- VUID-RuntimeSpirv-x-06432
In compute shaders using the `GLCompute Execution Model` the product of `x` size, `y` size, and `z` size in `LocalSize` or `LocalSizeId` **must** be less than or equal to `VkPhysicalDeviceLimits::maxComputeWorkGroupInvocations`
- VUID-RuntimeSpirv-LocalSizeId-06434
If `Execution Mode LocalSizeId` is used, `maintenance4` **must** be enabled
- VUID-RuntimeSpirv-maintenance4-06817
If `maintenance4` is not enabled, any `OpTypeVector` output interface variables **must** not have a higher `Component Count` than a matching `OpTypeVector` input interface variable
- VUID-RuntimeSpirv-OpEntryPoint-08743
Any `user-defined variables` shared between the `OpEntryPoint` of two shader stages, and declared with `Input` as its `Storage Class` for the subsequent shader stage, **must** have all `Location` slots and `Component` words declared in the preceding shader stage’s `OpEntryPoint` with `Output` as the `Storage Class`
- VUID-RuntimeSpirv-OpEntryPoint-07754
Any `user-defined variables` between the `OpEntryPoint` of two shader stages **must** have the same type and width for each `Component`
- VUID-RuntimeSpirv-OpVariable-08746
Any `OpVariable`, `Block`-decorated `OpTypeStruct`, or `Block`-decorated `OpTypeStruct` members shared between the `OpEntryPoint` of two shader stages **must** have matching decorations as defined in `interface matching`
- VUID-RuntimeSpirv-Workgroup-06530

The sum of size in bytes for variables and [padding](#) in the [Workgroup Storage Class](#) in the [GLCompute Execution Model](#) **must** be less than or equal to [maxComputeSharedMemorySize](#)

- VUID-RuntimeSpirv-shaderZeroInitializeWorkgroupMemory-06372
If [shaderZeroInitializeWorkgroupMemory](#) is not enabled, any [OpVariable](#) with [Workgroup](#) as its [Storage Class](#) **must** not have an [Initializer](#) operand
- VUID-RuntimeSpirv-OpImage-06376
If an [OpImage*Gather](#) operation has an image operand of [Offset](#), [ConstOffset](#), or [ConstOffsets](#) the offset value **must** be greater than or equal to [minTexelGatherOffset](#)
- VUID-RuntimeSpirv-OpImage-06377
If an [OpImage*Gather](#) operation has an image operand of [Offset](#), [ConstOffset](#), or [ConstOffsets](#) the offset value **must** be less than or equal to [maxTexelGatherOffset](#)
- VUID-RuntimeSpirv-OpImageSample-06435
If an [OpImageSample*](#) or [OpImageFetch*](#) operation has an image operand of [ConstOffset](#) then the offset value **must** be greater than or equal to [minTexelOffset](#)
- VUID-RuntimeSpirv-OpImageSample-06436
If an [OpImageSample*](#) or [OpImageFetch*](#) operation has an image operand of [ConstOffset](#) then the offset value **must** be less than or equal to [maxTexelOffset](#)
- VUID-RuntimeSpirv-samples-08725
If an [OpTypeImage](#) has an [MS](#) operand 0, its bound image **must** have been created with [VkImageCreateInfo::samples](#) as [VK_SAMPLE_COUNT_1_BIT](#)
- VUID-RuntimeSpirv-samples-08726
If an [OpTypeImage](#) has an [MS](#) operand 1, its bound image **must** not have been created with [VkImageCreateInfo::samples](#) as [VK_SAMPLE_COUNT_1_BIT](#)
- VUID-RuntimeSpirv-SubgroupUniformControlFlowKHR-06379
The [Execution Mode SubgroupUniformControlFlowKHR](#) **must** not be applied to an entry point unless [shaderSubgroupUniformControlFlow](#) is enabled, the corresponding shader stage bit is set in [subgroupSupportedStages](#), and the entry point does not execute any [invocation repack instructions](#)
- VUID-RuntimeSpirv-OpEntryPoint-08727
Each [OpEntryPoint](#) **must** not have more than one variable decorated with [InputAttachmentIndex](#) per image aspect of the attachment image bound to it, either explicitly or implicitly as described by [input attachment interface](#)
- VUID-RuntimeSpirv-shaderTileImageColorReadAccess-08728
If [shaderTileImageColorReadAccess](#) is not enabled, [OpColorAttachmentReadEXT](#) operation **must** not be used
- VUID-RuntimeSpirv-shaderTileImageDepthReadAccess-08729
If [shaderTileImageDepthReadAccess](#) is not enabled, [OpDepthAttachmentReadEXT](#) operation **must** not be used
- VUID-RuntimeSpirv-shaderTileImageStencilReadAccess-08730
If [shaderTileImageStencilReadAccess](#) is not enabled, [OpStencilAttachmentReadEXT](#) operation **must** not be used
- VUID-RuntimeSpirv-minSampleShading-08731
If [sample shading](#) is enabled and [minSampleShading](#) is 1.0, the [sample](#) operand of any

`OpColorAttachmentReadEXT`, `OpDepthAttachmentReadEXT`, or `OpStencilAttachmentReadEXT` operation **must** evaluate to the value of the [coverage index](#) for any given fragment invocation

- VUID-RuntimeSpirv-minSampleShading-08732

If [sample shading](#) is enabled and any of the `OpColorAttachmentReadEXT`, `OpDepthAttachmentReadEXT`, or `OpStencilAttachmentReadEXT` operations are used, then `minSampleShading` **must** be 1.0

- VUID-RuntimeSpirv-MeshEXT-09218

In mesh shaders using the `MeshEXT` or `MeshNV Execution Model` and the `OutputPoints Execution Mode`, if `maintenance5` is not enabled, and if the number of output points is greater than 0, a `PointSize` decorated variable **must** be written to for each output point

- VUID-RuntimeSpirv-maintenance5-09190

If `maintenance5` is enabled and a `PointSize` decorated variable is written to, all execution paths **must** write to a `PointSize` decorated variable

- VUID-RuntimeSpirv-MaximallyReconvergesKHR-09565

The execution mode `MaximallyReconvergesKHR` **must** not be applied to an entry point unless the entry point does not execute any [invocation repack instructions](#)

- VUID-RuntimeSpirv-shaderSubgroupRotateClustered-09566

If `shaderSubgroupRotateClustered` is `VK_FALSE`, then the `ClusterSize` operand to `OpGroupNonUniformRotateKHR` **must** not be used

- VUID-RuntimeSpirv-protectedNoFault-09645

If `protectedNoFault` is not supported, the `Storage Class` of the `PhysicalStorageBuffer` **must** not be used if the buffer being accessed is [protected](#)

- VUID-RuntimeSpirv-meshAndTaskShaderDerivatives-10153

If `meshAndTaskShaderDerivatives` is `VK_FALSE`, the `DerivativeGroupLinearKHR` and `DerivativeGroupQuadsKHR` execution modes **must** not be used in the `MeshEXT`, `MeshNV`, `TaskEXT`, or `TaskNV Execution Model`

Precision and Operation of SPIR-V Instructions

The following rules apply to half, single, and double-precision floating-point instructions:

- Positive and negative infinities and positive and negative zeros are generated as dictated by [IEEE 754](#), but subject to the precisions allowed in the [table below](#).
- Signaling NaNs are not required to be generated and exceptions are never raised. Signaling NaN **may** be converted to quiet NaNs values by any floating-point instruction.
- The set of operations `OpPhi`, `OpSelect`, `OpFunctionCall`, `OpReturnValue`, `OpVectorExtractDynamic`, `OpVectorInsertDynamic`, `OpVectorShuffle`, `OpCompositeConstruct`, `OpCompositeExtract`, `OpCompositeInsert`, `OpTranspose`, `OpCopyObject`, `OpCopyLogical`, `OpCopyMemory`, `OpGroupNonUniformRotateKHR`, `OpCooperativeMatrixLoadKHR`, `OpCooperativeMatrixStoreKHR`, `OpAtomicLoad`, `OpAtomicStore`, `OpAtomicExchange`, `OpStore`, and `OpLoad` are referred to as *bit-preserving operations*.
- The floating-point environment used for an instruction can be determined as follows:

- If the SPIR-V specifies it explicitly using the `FPFastMath` decoration or `FPFastMathDefault Execution Mode` then that is used.
- If the environment is not specified in the SPIR-V then it is determined as follows:
 - If the operation is not decorated `NoContraction` then the flags `AllowContract`, `AllowReassoc`, `AllowRecip`, and `AllowTransform` are assumed.
 - If any of the following conditions are true then the flags `NSZ`, `NotInf`, and `NotNaN` are assumed:
 - The entry point does not use the `Execution Mode SignedZeroInfNanPreserve` with a bit-width corresponding to one of the operands or to the result type.
 - The operation is not a bit-preserving operation and is not one of `OpFConvert`, `OpFNegate`, `OpFAdd`, `OpFSub`, `OpFMul`, `OpFDiv`, `OpIsNan`, or `OpIsInf`.
 - The operation is an `OpLoad` from the `Input Storage Class` in the fragment shader stage.
- All bit-preserving operations and the following instructions **must** not flush denormalized values: `OpConstant`, `OpConstantComposite`, `OpSpecConstant`, `OpSpecConstantComposite`, and `OpBitcast`.
- Denormalized values are supported.
 - By default, any half, single, or double-precision denormalized value input into a shader or potentially generated by any instruction (except those listed above) or any extended instructions for GLSL in a shader **may** be flushed to zero.
 - If the entry point is declared with the `DenormFlushToZero Execution Mode` then for the affected instructions the denormalized result **must** be flushed to zero and the denormalized operands **may** be flushed to zero. Denormalized values obtained via unpacking an integer into a vector of values with smaller bit width and interpreting those values as floating-point numbers **must** be flushed to zero.
 - The following core SPIR-V instructions **must** respect the `DenormFlushToZero Execution Mode`: `OpSpecConstantOp` (with opcode `OpFConvert`), `OpFConvert`, `OpFNegate`, `OpFAdd`, `OpFSub`, `OpFMul`, `OpFDiv`, `OpFRem`, `OpFMod`, `OpVectorTimesScalar`, `OpMatrixTimesScalar`, `OpVectorTimesMatrix`, `OpMatrixTimesVector`, `OpMatrixTimesMatrix`, `OpOuterProduct`, `OpDot`; and the following extended instructions for GLSL: `Round`, `RoundEven`, `Trunc`, `FAbs`, `Floor`, `Ceil`, `Fract`, `Radians`, `Degrees`, `Sin`, `Cos`, `Tan`, `Asin`, `Acos`, `Atan`, `Sinh`, `Cosh`, `Tanh`, `Asinh`, `Acosh`, `Atanh`, `Atan2`, `Pow`, `Exp`, `Log`, `Exp2`, `Log2`, `Sqrt`, `InverseSqrt`, `Determinant`, `MatrixInverse`, `Modf`, `ModfStruct`, `FMin`, `FMax`, `FClamp`, `FMix`, `Step`, `SmoothStep`, `Fma`, `UnpackHalf2x16`, `Length`, `Distance`, `Cross`, `Normalize`, `FaceForward`, `Reflect`, `Refract`, `NMin`, `NMax`, and `NClamp`.
 - The following core SPIR-V instructions **must** respect the `DenormPreserve Execution Mode`: `OpSpecConstantOp`, `OpFConvert`, `OpFNegate`, `OpFAdd`, `OpFSub`, `OpFMul`, `OpVectorTimesScalar`, `OpMatrixTimesScalar`, `OpVectorTimesMatrix`, `OpMatrixTimesVector`, `OpMatrixTimesMatrix`, `OpOuterProduct`, `OpDot`, `OpFOrdEqual`, `OpFUnordEqual`, `OpFOrdNotEqual`, `OpFUnordNotEqual`, `OpFOrdLessThan`, `OpFUnordLessThan`, `OpFOrdGreaterThan`, `OpFUnordGreaterThan`, `OpFOrdLessThanEqual`, `OpFUnordLessThanEqual`, `OpFOrdGreaterThanEqual`, `OpFUnordGreaterThanEqual`, `OpGroupNonUniformAllEqual`; and the following extended instructions for GLSL: `FAbs`, `FSign`, `Radians`, `Degrees`, `FMin`, `FMax`, `FClamp`, `FMix`, `Fma`, `PackHalf2x16`, `PackDouble2x32`, `UnpackHalf2x16`, `UnpackDouble2x32`, `NMin`, `NMax`, and `NClamp`.

The precision of double-precision instructions is at least that of single precision.

The precision of individual operations is defined in [Precision of Individual Operations](#). Subject to the constraints below, however, implementations **may** reorder or combine operations, resulting in expressions exhibiting different precisions than might be expected from the constituent operations.

Evaluation of Expressions

Implementations **may** rearrange floating-point operations using any of the mathematical properties governing the expressions in precise arithmetic, even where the floating-point operations do not share these properties. This includes, but is not limited to, associativity and distributivity, and **may** involve a different number of rounding steps than would occur if the operations were not rearranged. In shaders that use the [SignedZeroInfNanPreserve Execution Mode](#) the values **must** be preserved if they are generated after any rearrangement but the [Execution Mode](#) does not change which rearrangements are valid. This rearrangement **can** be prevented for particular operations by using the [NoContraction](#) decoration.

For example, in the absence of the [NoContraction](#) decoration implementations are allowed to implement $a + b - a$ and $\frac{a \times b}{a}$ as b . The [SignedZeroInfNanPreserve](#) does not prevent these transformations, even though they may overflow to infinity or NaN when evaluated in floating-point.

NOTE

If the [NoContraction](#) decoration is applied then operations may not be rearranged, so, for example, $a + a - a$ must account for possible overflow to infinity. If infinities are not preserved then the expression may be replaced with a , since the replacement is exact when overflow does not occur and infinities may be replaced with undefined values. If both [NoContraction](#) and [SignedZeroInfNanPreserve](#) are used then the result must be infinity for sufficiently large a .

Precision of Individual Operations

The precision of individual operations is defined either in terms of rounding (correctly rounded), as an error bound in ULP, or as inherited from a formula as follows:

Correct Result

Operations that are described as returning the “correct result” will return the infinitely precise result which, due to the nature of the operation, will not need rounding.

Correctly Rounded

Operations described as “correctly rounded” will return the infinitely precise result, x , rounded so as to be representable in floating-point. If the entry point is declared with the [RoundingModeRTE](#) or the [RoundingModeRTZ Execution Mode](#) then this is done according to [IEEE 754](#) “roundTiesToEven” or “roundTowardZero” rounding directions, respectively. Otherwise, they are rounded with [implementation-defined rounding mode](#).

Correctly Rounded with Implementation-Defined Rounding Mode

Operations described as “correctly rounded with implementation-defined rounding mode” will return the infinitely precise result, x , rounded so as to be representable in floating-point. If x is exactly representable then x will be returned. Otherwise, either the floating-point value closest to and no less than x or the value closest to and no greater than x will be returned. Which value is

chosen is implementation-defined.

ULP

Where an error bound of n ULP (units in the last place) is given, for an operation with infinitely precise result x the value returned **must** be in the range $[x - n \times \text{ulp}(x), x + n \times \text{ulp}(x)]$. The function $\text{ulp}(x)$ is defined as follows:

If there exist non-equal, finite floating-point numbers a and b such that $a \leq x \leq b$ then $\text{ulp}(x)$ is the minimum possible distance between such numbers, $\text{ulp}(x) = \min_{a, b} |b - a|$. If such numbers do not exist then $\text{ulp}(x)$ is defined to be the difference between the two non-equal, finite floating-point numbers nearest to x .

Where the range of allowed return values includes any value of magnitude larger than that of the largest representable finite floating-point number, operations **may**, additionally, return either an infinity of the appropriate sign or the finite number with the largest magnitude of the appropriate sign. If the infinitely precise result of the operation is not mathematically defined then the value returned is undefined.

Inherited From ...

Where an operation's precision is described as being inherited from a formula, the result returned **must** be at least as accurate as the result of computing an approximation to x using a formula equivalent to the given formula applied to the supplied inputs. Specifically, the formula given may be transformed using the mathematical associativity, commutativity and distributivity of the operators involved to yield an equivalent formula. The SPIR-V precision rules, when applied to each such formula and the given input values, define a range of permitted values. If NaN is one of the permitted values then the operation may return any result, otherwise let the largest permitted value in any of the ranges be $F_{\max}$ and the smallest be $F_{\min}$. The operation **must** return a value in the range $[x - E, x + E]$ where $E = \max(|x - F_{\min}|, |x - F_{\max}|)$. If the entry point is declared with the **DenormFlushToZero** execution mode, then any intermediate denormal value(s) while evaluating the formula **may** be flushed to zero. Denormal final results **must** be flushed to zero. If the entry point is declared with the **DenormPreserve Execution Mode**, then denormals **must** be preserved throughout the formula.

For half- (16 bit) and single- (32 bit) precision instructions, precisions are **required** to be at least as follows:

Table 91. Precision of Core SPIR-V Instructions

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
OpFNegate	Correct result.	
OpFAdd	Correctly rounded.	
OpFSub	Correctly rounded.	
OpFMul, OpVectorTimesScalar, OpMatrixTimesScalar	Correctly rounded.	

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
OpMatrixTimesVector	Inherited from $\sum_{j=0}^{n-1} M_{ij} \times v_j$.	
OpVectorTimesMatrix	Inherited from $\sum_{j=0}^{n-1} v_j \times M_{ji}$.	
OpMatrixTimesMatrix	Inherited from $\sum_{k=0}^{n-1} M_{ik} \times N_{kj}$.	
OpOuterProduct	Correctly rounded.	
OpDot(x, y)	Inherited from $\sum_{i=0}^{n-1} x_i \times y_i$.	
OpIsNan, OpIsInf	Correct result.	
OpFOrdEqual, OpFUnordEqual	Correct result.	
OpFOrdNotEqual, OpFUnordNotEqual	Correct result.	
OpFOrdLessThan, OpFUnordLessThan	Correct result.	
OpFOrdGreaterThan, OpFUnordGreaterThan	Correct result.	
OpFOrdLessThanEqual, OpFUnordLessThanEqual	Correct result.	
OpFOrdGreaterThanEqual, OpFUnordGreaterThanEqual	Correct result.	
OpFDiv(x,y)	2.5 ULP for $ y = 0$ or $ y $ in the range $[2^{-126}, 2^{126}]$.	2.5 ULP for $ y = 0$ or $ y $ in the range $[2^{-14}, 2^{14}]$.
OpFRem(x,y)	Inherited from $x - y \times \text{trunc}(x/y)$.	
OpFMod(x,y)	Inherited from $x - y \times \text{floor}(x/y)$.	
OpQuantizeToF16	Correctly rounded with implementation defined rounding mode.	
conversions between types	Correctly rounded.	

NOTE

The **OpFRem** and **OpFMod** instructions use cheap approximations of remainder, and the error can be large due to the discontinuity in `trunc()` and `floor()`. This can produce mathematically unexpected results in some cases, such as `FMod(x,x)` computing `x` rather than `0`, and can also cause the result to have a different sign than the infinitely precise result.

Table 92. Precision of GLSL.std.450 Instructions

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
fma()	Inherited from OpFMul followed by OpFAdd .	
exp(x), exp2(x)	$3 + 2 \times x $ ULP.	$1 + 2 \times x $ ULP.

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
<code>log()</code> , <code>log2()</code>	3 ULP outside the range [0.5, 2.0]. Absolute error $< 2^{-21}$ inside the range [0.5, 2.0].	3 ULP outside the range [0.5, 2.0]. Absolute error $< 2^{-7}$ inside the range [0.5, 2.0].
<code>pow(x, y)</code>	Inherited from <code>exp2(y × log2(x))</code> .	
<code>sqrt()</code>	Inherited from 1.0 / <code>inversesqrt()</code> .	
<code>inversesqrt()</code>	2 ULP.	
<code>radians(x)</code>	Inherited from $x \times C_{\pi_180}$, where C_{π_180} is a correctly rounded approximation to $\frac{\pi}{180}$.	
<code>degrees(x)</code>	Inherited from $x \times C_{180_pi}$, where C_{180_pi} is a correctly rounded approximation to $\frac{180}{\pi}$.	
<code>sin()</code>	Absolute error $\leq 2^{-11}$ inside the range $[-\pi, \pi]$.	Absolute error $\leq 2^{-7}$ inside the range $[-\pi, \pi]$.
<code>cos()</code>	Absolute error $\leq 2^{-11}$ inside the range $[-\pi, \pi]$.	Absolute error $\leq 2^{-7}$ inside the range $[-\pi, \pi]$.
<code>tan()</code>	Inherited from $\frac{\sin()}{\cos()}$.	
<code>asin(x)</code>	Inherited from <code>atan2(x, sqrt(1.0 - x × x))</code> .	
<code>acos(x)</code>	Inherited from <code>atan2(sqrt(1.0 - x × x), x)</code> .	
<code>atan()</code> , <code>atan2()</code>	4096 ULP	5 ULP.
<code>sinh(x)</code>	Inherited from $(\exp(x) - \exp(-x)) \times 0.5$.	
<code>cosh(x)</code>	Inherited from $(\exp(x) + \exp(-x)) \times 0.5$.	
<code>tanh()</code>	Inherited from $\frac{\sinh()}{\cosh()}$.	
<code>asinh(x)</code>	Inherited from <code>log(x + sqrt(x × x + 1.0))</code> .	
<code>acosh(x)</code>	Inherited from <code>log(x + sqrt(x × x - 1.0))</code> .	
<code>atanh(x)</code>	Inherited from $\log(\frac{1.0 + x}{1.0 - x}) \times 0.5$.	
<code>frexp()</code>	Correct result.	
<code>ldexp()</code>	Correctly rounded.	
<code>length(x)</code>	Inherited from <code>sqrt(dot(x, x))</code> .	
<code>distance(x, y)</code>	Inherited from <code>length(x - y)</code> .	
<code>cross()</code>	Inherited from <code>OpFSub(OpFMul, OpFMul)</code> .	
<code>normalize(x)</code>	Inherited from $x \times \text{inversesqrt}(\text{dot}(x, x))$.	
<code>faceforward(N, I, NRef)</code>	Inherited from <code>dot(NRef, I) < 0.0 ? N : -N</code> .	
<code>reflect(x, y)</code>	Inherited from $x - 2.0 \times \text{dot}(y, x) \times y$.	
<code>refract(I, N, eta)</code>	Inherited from $k < 0.0 ? 0.0 : \text{eta} \times I - (\text{eta} \times \text{dot}(N, I) + \text{sqrt}(k)) \times N$, where $k = 1 - \text{eta} \times \text{eta} \times (1.0 - \text{dot}(N, I) \times \text{dot}(N, I))$.	

Instruction	Single precision, unless decorated with RelaxedPrecision	Half precision
round	Correctly rounded.	
roundEven	Correctly rounded.	
trunc	Correctly rounded.	
fabs	Correct result.	
fsign	Correct result.	
floor	Correctly rounded.	
ceil	Correctly rounded.	
fract	Correctly rounded.	
modf	Correctly rounded.	
fmin	Correct result.	
fmax	Correct result.	
fclamp	Correct result.	
fmix(x, y, a)	Inherited from $x \times (1.0 - a) + y \times a$.	
step	Correctly rounded.	
smoothStep(edge0, edge1, x)	Inherited from $t \times t \times (3.0 - 2.0 \times t)$, where $t = clamp(\frac{x - edge0}{edge1 - edge0}, 0.0, 1.0)$.	
nmin	Correct result.	
nmax	Correct result.	
nclamp	Correct result.	
packHalf2x16	Correctly rounded with implementation defined rounding mode.	

GLSL.std.450 extended instructions specifically defined in terms of the above instructions inherit the above errors. GLSL.std.450 extended instructions not listed above and not defined in terms of the above have undefined precision.

For the **OpSRem** and **OpSMod** instructions, if either operand is negative the result is undefined.

NOTE

While the **OpSRem** and **OpSMod** instructions are supported by the Vulkan environment, they require non-negative values and thus do not enable additional functionality beyond what **OpUMod** provides.

OpCooperativeMatrixMulAddKHR performs its operations in an implementation-dependent order and internal precision.

Signedness of SPIR-V Image Accesses

SPIR-V associates a signedness with all integer image accesses. This is required in certain parts of

the SPIR-V and the Vulkan image access pipeline to ensure defined results. The signedness is determined from a combination of the access instruction's **Image Operands** and the underlying image's **Sampled Type** as follows:

1. If the instruction's **Image Operands** contains the **SignExtend** operand then the access is signed.
2. If the instruction's **Image Operands** contains the **ZeroExtend** operand then the access is unsigned.
3. Otherwise, the image accesses signedness matches that of the **Sampled Type** of the **OpTypeImage** being accessed.

Image Format and Type Matching

When specifying the **Image Format** of an **OpTypeImage**, the converted bit width and type, as shown in the table below, **must** match the **Sampled Type**. The signedness **must** match the **signedness of any access** to the image.

NOTE

Formatted accesses are always converted from a shader readable type to the resource's format or vice versa via **Format Conversion** for reads and **Texel Output Format Conversion** for writes. As such, the bit width and format below do not necessarily match 1:1 with what might be expected for some formats.

For a given **Image Format**, the **Sampled Type** **must** be the type described in the *Type* column of the below table, with its **Literal Width** set to that in the *Bit Width* column. Every access that is made to the image **must** have a signedness equal to that in the *Signedness* column (where applicable).

Image Format	Type-Declaration instructions	Bit Width	Signedness
Unknown	Any	Any	Any

Image Format	Type-Declaration instructions	Bit Width	Signedness
Rgba32f	OpTypeFloat	32	N/A
Rg32f			
R32f			
Rgba16f			
Rg16f			
R16f			
Rgba16			
Rg16			
R16			
Rgba16Snorm			
Rg16Snorm			
R16Snorm			
Rgb10A2			
R11fG11fB10f			
Rgba8			
Rg8			
R8			
Rgba8Snorm			
Rg8Snorm			
R8Snorm			

Image Format	Type-Declaration instructions	Bit Width	Signedness
Rgba32i	OpTypeInt	32	1
Rg32i			
R32i			
Rgba16i			
Rg16i			
R16i			
Rgba8i			
Rg8i			
R8i			
Rgba32ui			0
Rg32ui			
R32ui			
Rgba16ui			
Rg16ui			
R16ui			
Rgb10a2ui			
Rgba8ui			
Rg8ui			
R8ui			
R64i	OpTypeInt	64	1
R64ui			0

The *SPIR-V Type* is defined by an instruction in SPIR-V, declared with the Type-Declaration Instruction, Bit Width, and Signedness from above.

Compatibility Between SPIR-V Image Dimensions and Vulkan ImageView Types

SPIR-V **Image Dim** values are compatible with **VkImageView viewType** values as defined below:

Table 93. SPIR-V and Vulkan ImageView Dimension Compatibility

SPIR-V Image Dim	Compatible Vulkan ImageView viewType
1D	VK_IMAGE_VIEW_TYPE_1D, VK_IMAGE_VIEW_TYPE_1D_ARRAY
2D	VK_IMAGE_VIEW_TYPE_2D, VK_IMAGE_VIEW_TYPE_2D_ARRAY
3D	VK_IMAGE_VIEW_TYPE_3D
Cube	VK_IMAGE_VIEW_TYPE_CUBE, VK_IMAGE_VIEW_TYPE_CUBE_ARRAY

Compatibility Between SPIR-V Image Formats and Vulkan Formats

SPIR-V **Image Format** values are compatible with **VkFormat** values as defined below:

Table 94. SPIR-V and Vulkan Image Format Compatibility

SPIR-V Image Format	Compatible Vulkan Format
Unknown	Any
R8	VK_FORMAT_R8_UNORM
R8Snorm	VK_FORMAT_R8_SNORM
R8ui	VK_FORMAT_R8_UINT
R8i	VK_FORMAT_R8_SINT
Rg8	VK_FORMAT_R8G8_UNORM
Rg8Snorm	VK_FORMAT_R8G8_SNORM
Rg8ui	VK_FORMAT_R8G8_UINT
Rg8i	VK_FORMAT_R8G8_SINT
Rgba8	VK_FORMAT_R8G8B8A8_UNORM
Rgba8Snorm	VK_FORMAT_R8G8B8A8_SNORM
Rgba8ui	VK_FORMAT_R8G8B8A8_UINT
Rgba8i	VK_FORMAT_R8G8B8A8_SINT
Rgb10A2	VK_FORMAT_A2B10G10R10_UNORM_PACK32
Rgb10a2ui	VK_FORMAT_A2B10G10R10_UINT_PACK32
R16	VK_FORMAT_R16_UNORM
R16Snorm	VK_FORMAT_R16_SNORM
R16ui	VK_FORMAT_R16_UINT
R16i	VK_FORMAT_R16_SINT
R16f	VK_FORMAT_R16_SFLOAT
Rg16	VK_FORMAT_R16G16_UNORM
Rg16Snorm	VK_FORMAT_R16G16_SNORM
Rg16ui	VK_FORMAT_R16G16_UINT
Rg16i	VK_FORMAT_R16G16_SINT
Rg16f	VK_FORMAT_R16G16_SFLOAT
Rgba16	VK_FORMAT_R16G16B16A16_UNORM
Rgba16Snorm	VK_FORMAT_R16G16B16A16_SNORM
Rgba16ui	VK_FORMAT_R16G16B16A16_UINT
Rgba16i	VK_FORMAT_R16G16B16A16_SINT
Rgba16f	VK_FORMAT_R16G16B16A16_SFLOAT
R32ui	VK_FORMAT_R32_UINT
R32i	VK_FORMAT_R32_SINT
R32f	VK_FORMAT_R32_SFLOAT

SPIR-V Image Format	Compatible Vulkan Format
Rg32ui	VK_FORMAT_R32G32_UINT
Rg32i	VK_FORMAT_R32G32_SINT
Rg32f	VK_FORMAT_R32G32_SFLOAT
Rgba32ui	VK_FORMAT_R32G32B32A32_UINT
Rgba32i	VK_FORMAT_R32G32B32A32_SINT
Rgba32f	VK_FORMAT_R32G32B32A32_SFLOAT
R64ui	VK_FORMAT_R64_UINT
R64i	VK_FORMAT_R64_SINT
R11fG11fB10f	VK_FORMAT_B10G11R11_UFLOAT_PACK32

Ray Query Precision and Operation

The values returned by `OpRayQueryGetIntersectionTriangleVertexPositionsKHR` are transformed by the geometry transform, which is performed at standard [floating-point](#) precision, but without a specifically defined order of floating-point operations to perform the matrix multiplication.

Appendix B: Memory Model

NOTE

This memory model describes synchronizations provided by all implementations; however, some of the synchronizations defined require extra features to be supported by the implementation. See [VkPhysicalDeviceVulkanMemoryModelFeatures](#).

Agent

Operation is a general term for any task that is executed on the system.

NOTE

An operation is by definition something that is executed. Thus if an instruction is skipped due to control flow, it does not constitute an operation.

Each operation is executed by a particular *agent*. Possible agents include each shader invocation, each host thread, and each fixed-function stage of the pipeline.

Memory Location

A *memory location* identifies unique storage for 8 bits of data. Memory operations access a *set of memory locations* consisting of one or more memory locations at a time, e.g. an operation accessing a 32-bit integer in memory would read/write a set of four memory locations. Memory operations that access whole aggregates **may** access any padding bytes between elements or members, but no padding bytes at the end of the aggregate. Two sets of memory locations *overlap* if the intersection of their sets of memory locations is non-empty. A memory operation **must** not affect memory at a memory location not within its set of memory locations.

Memory locations for buffers and images are explicitly allocated in [VkDeviceMemory](#) objects, and are implicitly allocated for SPIR-V variables in each shader invocation.

Variables with [Workgroup](#) storage class that point to a block-decorated type share a set of memory locations.

Allocation

The values stored in newly allocated memory locations are determined by a SPIR-V variable's initializer, if present, or else are undefined. At the time an allocation is created there have been no [memory operations](#) to any of its memory locations. The initialization is not considered to be a memory operation.

NOTE

For tessellation control shader output variables, a consequence of initialization not being considered a memory operation is that some implementations may need to insert a barrier between the initialization of the output variables and any reads of those variables.

Memory Operation

For an operation A and memory location M:

- A *reads* M if and only if the data stored in M is an input to A.
- A *writes* M if and only if the data output from A is stored to M.
- A *accesses* M if and only if it either reads or writes (or both) M.

NOTE

A write whose value is the same as what was already in those memory locations is still considered to be a write and has all the same effects.

Reference

A *reference* is an object that a particular agent **can** use to access a set of memory locations. On the host, a reference is a host virtual address. On the device, a reference is:

- The descriptor that a variable is bound to, for variables in Image, Uniform, or StorageBuffer storage classes. If the variable is an array (or array of arrays, etc.) then each element of the array **may** be a unique reference.
- The address range for a buffer in **PhysicalStorageBuffer** storage class, where the base of the address range is queried with **vkGetBufferDeviceAddressKHR** and the length of the range is the size of the buffer.
- A single common reference for all variables with **Workgroup** storage class that point to a block-decorated type.
- The variable itself for non-block-decorated type variables in **Workgroup** storage class.
- The variable itself for variables in other storage classes.

Two memory accesses through distinct references **may** require availability and visibility operations as defined [below](#).

Program-Order

A *dynamic instance* of an instruction is defined in SPIR-V (<https://registry.khronos.org/spir-v/specs/unified1/SPIRV.html#DynamicInstance>) as a way of referring to a particular execution of a static instruction. Program-order is an ordering on dynamic instances of instructions executed by a single shader invocation:

- (Basic block): If instructions A and B are in the same basic block, and A is listed in the module before B, then the n'th dynamic instance of A is program-ordered before the n'th dynamic instance of B.
- (Branch): The dynamic instance of a branch or switch instruction is program-ordered before the dynamic instance of the OpLabel instruction to which it transfers control.
- (Call entry): The dynamic instance of an **OpFunctionCall** instruction is program-ordered before the dynamic instances of the **OpFunctionParameter** instructions and the body of the called

function.

- (Call exit): The dynamic instance of the instruction following an `OpFunctionCall` instruction is program-ordered after the dynamic instance of the return instruction executed by the called function.
- (Transitive Closure): If dynamic instance A of any instruction is program-ordered before dynamic instance B of any instruction and B is program-ordered before dynamic instance C of any instruction then A is program-ordered before C.
- (Complete definition): No other dynamic instances are program-ordered.

For instructions executed on the host, the source language defines the program-order relation (e.g. as “sequenced-before”).

Shader Call Related

Shader-call-related is an equivalence relation on invocations defined as the symmetric and transitive closure of:

- A is shader-call-related to B if A is created by an `shader call` instruction executed by B.

Shader Call Order

Shader-call-order is a partial order on dynamic instances of instructions executed by invocations that are shader-call-related:

- (Program order): If dynamic instance A is program-ordered before B, then A is shader-call-ordered before B.
- (Shader call entry): If A is a dynamic instance of an `shader call` instruction and B is a dynamic instance executed by an invocation that is created by A, then A is shader-call-ordered before B.
- (Shader call exit): If A is a dynamic instance of an `shader call` instruction, B is the next dynamic instance executed by the same invocation, and C is a dynamic instance executed by an invocation that is created by A, then C is shader-call-ordered before B.
- (Transitive closure): If A is shader-call-ordered-before B and B is shader-call-ordered-before C, then A is shader-call-ordered-before C.
- (Complete definition): No other dynamic instances are shader-call-ordered.

Scope

Atomic and barrier instructions include scopes which identify sets of shader invocations that **must** obey the requested ordering and atomicity rules of the operation, as defined below.

The various scopes are described in detail in [the Shaders chapter](#).

Atomic Operation

An *atomic operation* on the device is any SPIR-V operation whose name begins with **OpAtomic**. An atomic operation on the host is any operation performed with an `std::atomic` typed object.

Each atomic operation has a memory **scope** and a **semantics**. Informally, the scope determines which other agents it is atomic with respect to, and the **semantics** constrains its ordering against other memory accesses. Device atomic operations have explicit scopes and semantics. Each host atomic operation implicitly uses the **CrossDevice** scope, and uses a memory semantics equivalent to a C++ `std::memory_order` value of `relaxed`, `acquire`, `release`, `acq_rel`, or `seq_cst`.

Two atomic operations A and B are *potentially-mutually-ordered* if and only if all of the following are true:

- They access the same set of memory locations.
- They use the same reference.
- A is in the instance of B's memory scope.
- B is in the instance of A's memory scope.
- A and B are not the same operation (irreflexive).

Two atomic operations A and B are *mutually-ordered* if and only if they are potentially-mutually-ordered and any of the following are true:

- A and B are both device operations.
- A and B are both host operations.

NOTE

If two atomic operations are not mutually-ordered, and if their sets of memory locations overlap, then each **must** be synchronized against the other as if they were non-atomic operations.

Scoped Modification Order

For a given atomic write A, all atomic writes that are mutually-ordered with A occur in an order known as A's *scoped modification order*. A's scoped modification order relates no other operations.

NOTE

Invocations outside the instance of A's memory scope **may** observe the values at A's set of memory locations becoming visible to it in an order that disagrees with the scoped modification order.

NOTE

It is valid to have non-atomic operations or atomics in a different scope instance to the same set of memory locations, as long as they are synchronized against each other as if they were non-atomic (if they are not, it is treated as a **data race**). That means this definition of A's scoped modification order could include atomic operations that occur much later, after intervening non-atomics. That is a bit non-intuitive, but it helps to keep this definition simple and non-circular.

Memory Semantics

Non-atomic memory operations, by default, **may** be observed by one agent in a different order than they were written by another agent.

Atomics and some synchronization operations include *memory semantics*, which are flags that constrain the order in which other memory accesses (including non-atomic memory accesses and [availability and visibility operations](#)) performed by the same agent **can** be observed by other agents, or **can** observe accesses by other agents.

Device instructions that include semantics are `OpAtomic*`, `OpControlBarrier`, `OpMemoryBarrier`, and `OpMemoryNamedBarrier`. Host instructions that include semantics are some `std::atomic` methods and memory fences.

SPIR-V supports the following memory semantics:

- Relaxed: No constraints on order of other memory accesses.
- Acquire: A memory read with this semantic performs an *acquire operation*. A memory barrier with this semantic is an *acquire barrier*.
- Release: A memory write with this semantic performs a *release operation*. A memory barrier with this semantic is a *release barrier*.
- AcquireRelease: A memory read-modify-write operation with this semantic performs both an acquire operation and a release operation, and inherits the limitations on ordering from both of those operations. A memory barrier with this semantic is both a release and acquire barrier.

NOTE	SPIR-V does not support “consume” semantics on the device.
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The memory semantics operand also includes *storage class semantics* which indicate which storage classes are constrained by the synchronization. SPIR-V storage class semantics include:

- UniformMemory
- WorkgroupMemory
- ImageMemory
- OutputMemory

Each SPIR-V memory operation accesses a single storage class. Semantics in synchronization operations can include a combination of storage classes.

The UniformMemory storage class semantic applies to accesses to memory in the PhysicalStorageBuffer, `ShaderRecordBufferKHR`, Uniform and StorageBuffer storage classes. The WorkgroupMemory storage class semantic applies to accesses to memory in the Workgroup storage class. The ImageMemory storage class semantic applies to accesses to memory in the Image storage class. The OutputMemory storage class semantic applies to accesses to memory in the Output storage class.

NOTE	Informally, these constraints limit how memory operations can be reordered, and these limits apply not only to the order of accesses as performed in the agent that
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executes the instruction, but also to the order the effects of writes become visible to all other agents within the same instance of the instruction’s memory scope.

NOTE

Release and acquire operations in different threads **can** act as synchronization operations, to guarantee that writes that happened before the release are visible after the acquire. (This is not a formal definition, just an Informative forward reference.)

NOTE

The OutputMemory storage class semantic is only useful in tessellation control shaders, which is the only execution model where output variables are shared between invocations.

The memory semantics operand **can** also include availability and visibility flags, which apply availability and visibility operations as described in [availability and visibility](#). The availability/visibility flags are:

- MakeAvailable: Semantics **must** be Release or AcquireRelease. Performs an availability operation before the release operation or barrier.
- MakeVisible: Semantics **must** be Acquire or AcquireRelease. Performs a visibility operation after the acquire operation or barrier.

The specifics of these operations are defined in [Availability and Visibility Semantics](#).

Host atomic operations **may** support a different list of memory semantics and synchronization operations, depending on the host architecture and source language.

Release Sequence

After an atomic operation A performs a release operation on a set of memory locations M, the *release sequence headed by A* is the longest continuous subsequence of A’s scoped modification order that consists of:

- the atomic operation A as its first element
- atomic read-modify-write operations on M by any agent

NOTE

The atomics in the last bullet **must** be mutually-ordered with A by virtue of being in A’s scoped modification order.

NOTE

This intentionally omits “atomic writes to M performed by the same agent that performed A”, which is present in the corresponding C++ definition.

Synchronizes-With

Synchronizes-with is a relation between operations, where each operation is either an atomic operation or a memory barrier (aka fence on the host).

If A and B are atomic operations, then A synchronizes-with B if and only if all of the following are true:

- A performs a release operation
- B performs an acquire operation
- A and B are mutually-ordered
- B reads a value written by A or by an operation in the release sequence headed by A

`OpControlBarrier`, `OpMemoryBarrier`, and `OpMemoryNamedBarrier` are *memory barrier* instructions in SPIR-V.

If A is a release barrier and B is an atomic operation that performs an acquire operation, then A synchronizes-with B if and only if all of the following are true:

- there exists an atomic write X (with any memory semantics)
- A is program-ordered before X
- X and B are mutually-ordered
- B reads a value written by X or by an operation in the release sequence headed by X
 - If X is relaxed, it is still considered to head a hypothetical release sequence for this rule
- A and B are in the instance of each other's memory scopes
- X's storage class is in A's semantics.

If A is an atomic operation that performs a release operation and B is an acquire barrier, then A synchronizes-with B if and only if all of the following are true:

- there exists an atomic read X (with any memory semantics)
- X is program-ordered before B
- X and A are mutually-ordered
- X reads a value written by A or by an operation in the release sequence headed by A
- A and B are in the instance of each other's memory scopes
- X's storage class is in B's semantics.

If A is a release barrier and B is an acquire barrier, then A synchronizes-with B if all of the following are true:

- there exists an atomic write X (with any memory semantics)
- A is program-ordered before X
- there exists an atomic read Y (with any memory semantics)
- Y is program-ordered before B
- X and Y are mutually-ordered
- Y reads the value written by X or by an operation in the release sequence headed by X
 - If X is relaxed, it is still considered to head a hypothetical release sequence for this rule

- A and B are in the instance of each other's memory scopes
- X's and Y's storage class is in A's and B's semantics.
 - NOTE: X and Y **must** have the same storage class, because they are mutually ordered.

If A is a release barrier, B is an acquire barrier, and C is a control barrier (where A **can** equal C, and B **can** equal C), then A synchronizes-with B if all of the following are true:

- A is program-ordered before (or equals) C
- C is program-ordered before (or equals) B
- A and B are in the instance of each other's memory scopes
- A and B are in the instance of C's execution scope

NOTE This is similar to the barrier-barrier synchronization above, but with a control barrier filling the role of the relaxed atomics.

Let F be an ordering of fragment shader invocations, such that invocation F_1 is ordered before invocation F_2 if and only if F_1 and F_2 overlap as described in [Fragment Shader Interlock](#) and F_1 executes the interlocked code before F_2 .

If A is an `OpEndInvocationInterlockEXT` instruction and B is an `OpBeginInvocationInterlockEXT` instruction, then A synchronizes-with B if the agent that executes A is ordered before the agent that executes B in F. A and B are both considered to have `FragmentInterlock` memory scope and semantics of UniformMemory and ImageMemory, and A is considered to have Release semantics and B is considered to have Acquire semantics.

NOTE `OpBeginInvocationInterlockEXT` and `OpBeginInvocationInterlockEXT` do not perform implicit availability or visibility operations. Usually, shaders using fragment shader interlock will declare the relevant resources as `coherent` to get implicit [per-instruction availability and visibility operations](#).

If A is a release barrier and B is an acquire barrier, then A synchronizes-with B if all of the following are true:

- A is shader-call-ordered-before B
- A and B are in the instance of each other's memory scopes

No other release and acquire barriers synchronize-with each other.

System-Synchronizes-With

System-synchronizes-with is a relation between arbitrary operations on the device or host. Certain operations system-synchronize-with each other, which informally means the first operation occurs before the second and that the synchronization is performed without using application-visible memory accesses.

If there is an [execution dependency](#) between two operations A and B, then the operation in the first

synchronization scope `system-synchronizes-with` the operation in the second synchronization scope.

NOTE

This covers all Vulkan synchronization primitives, including device operations executing before a synchronization primitive is signaled, wait operations happening before subsequent device operations, signal operations happening before host operations that wait on them, and host operations happening before `vkQueueSubmit`. The list is spread throughout the synchronization chapter, and is not repeated here.

`System-synchronizes-with` implicitly includes all storage class semantics and has `CrossDevice` scope.

If A `system-synchronizes-with` B, we also say A is *system-synchronized-before* B and B is *system-synchronized-after* A.

Private vs. Non-Private

By default, non-atomic memory operations are treated as *private*, meaning such a memory operation is not intended to be used for communication with other agents. Memory operations with the `NonPrivatePointer/NonPrivateTexel` bit set are treated as *non-private*, and are intended to be used for communication with other agents.

More precisely, for private memory operations to be `Location-Ordered` between distinct agents requires using `system-synchronizes-with` rather than shader-based synchronization. Private memory operations still obey program-order.

Atomic operations are always considered non-private.

Inter-Thread-Happens-Before

Let SC be a non-empty set of storage class semantics. Then (using template syntax) operation A *inter-thread-happens-before*<SC> operation B if and only if any of the following is true:

- A `system-synchronizes-with` B
- A `synchronizes-with` B, and both A and B have all of SC in their semantics
- A is an operation on memory in a storage class in SC or that has all of SC in its semantics, B is a release barrier or release atomic with all of SC in its semantics, and A is program-ordered before B
- A is an acquire barrier or acquire atomic with all of SC in its semantics, B is an operation on memory in a storage class in SC or that has all of SC in its semantics, and A is program-ordered before B
- A and B are both host operations and A *inter-thread-happens-before* B as defined in the host language specification
- A *inter-thread-happens-before*<SC> some X and X *inter-thread-happens-before*<SC> B

Happens-Before

Operation A *happens-before* operation B if and only if any of the following is true:

- A is program-ordered before B
- A inter-thread-happens-before<SC> B for some set of storage classes SC

Happens-after is defined similarly.

NOTE

Unlike C++, happens-before is not always sufficient for a write to be visible to a read. Additional [availability and visibility](#) operations **may** be required for writes to be [visible-to](#) other memory accesses.

NOTE

Happens-before is not transitive, but each of program-order and inter-thread-happens-before<SC> are transitive. These can be thought of as covering the “single-threaded” case and the “multi-threaded” case, and it is not necessary (and not valid) to form chains between the two.

Availability and Visibility

Availability and *visibility* are states of a write operation, which (informally) track how far the write has permeated the system, i.e. which agents and references are able to observe the write. Availability state is per *memory domain*. Visibility state is per (agent,reference) pair. Availability and visibility states are per-memory location for each write.

Memory domains are named according to the agents whose memory accesses use the domain. Domains used by shader invocations are organized hierarchically into multiple smaller memory domains which correspond to the different [scopes](#). Each memory domain is considered the *dual* of a scope, and vice versa. The memory domains defined in Vulkan include:

- *host* - accessible by host agents
- *device* - accessible by all device agents for a particular device
- *shader* - accessible by shader agents for a particular device, corresponding to the [Device](#) scope
- *queue family instance* - accessible by shader agents in a single queue family, corresponding to the [QueueFamily](#) scope.
- *fragment interlock instance* - accessible by fragment shader agents that [overlap](#), corresponding to the [FragmentInterlock](#) scope.
- *shader call instance* - accessible by shader agents that are [shader-call-related](#), corresponding to the [ShaderCallKHR](#) scope.
- *workgroup instance* - accessible by shader agents in the same workgroup, corresponding to the [Workgroup](#) scope.
- *subgroup instance* - accessible by shader agents in the same subgroup, corresponding to the [Subgroup](#) scope.

The memory domains are nested in the order listed above, except for shader call instance domain,

with memory domains later in the list nested in the domains earlier in the list. The shader call instance domain is at an implementation-dependent location in the list, and is nested according to that location. The shader call instance domain is not broader than the queue family instance domain.

NOTE

Memory domains do not correspond to storage classes or device-local and host-local [VkDeviceMemory](#) allocations, rather they indicate whether a write can be made visible only to agents in the same subgroup, same workgroup, overlapping fragment shader invocation, shader-call-related ray tracing invocation, in any shader invocation, or anywhere on the device, or host. The shader, queue family instance, fragment interlock instance, shader call instance, workgroup instance, and subgroup instance domains are only used for shader-based availability/visibility operations, in other cases writes can be made available from/visible to the shader via the device domain.

Availability operations, visibility operations, and memory domain operations alter the state of the write operations that happen-before them, and which are included in their *source scope* to be available or visible to their *destination scope*.

- For an availability operation, the source scope is a set of (agent,reference,memory location) tuples, and the destination scope is a set of memory domains.
- For a memory domain operation, the source scope is a memory domain and the destination scope is a memory domain.
- For a visibility operation, the source scope is a set of memory domains and the destination scope is a set of (agent,reference,memory location) tuples.

How the scopes are determined depends on the specific operation. Availability and memory domain operations expand the set of memory domains to which the write is available. Visibility operations expand the set of (agent,reference,memory location) tuples to which the write is visible.

Recall that availability and visibility states are per-memory location, and let W be a write operation to one or more locations performed by agent A via reference R . Let L be one of the locations written. (W,L) (the write W to L), is initially not available to any memory domain and only visible to (A,R,L) . An availability operation AV that happens-after W and that includes (A,R,L) in its source scope makes (W,L) *available* to the memory domains in its destination scope.

A memory domain operation DOM that happens-after AV and for which (W,L) is available in the source scope makes (W,L) available in the destination memory domain.

A visibility operation VIS that happens-after AV (or DOM) and for which (W,L) is available in any domain in the source scope makes (W,L) *visible* to all (agent,reference, L) tuples included in its destination scope.

If write W_2 happens-after W , and their sets of memory locations overlap, then W will not be available/visible to all agents/references for those memory locations that overlap (and future $AV/DOM/VIS$ ops cannot revive W 's write to those locations).

Availability, memory domain, and visibility operations are treated like other non-atomic memory

accesses for the purpose of [memory semantics](#), meaning they can be ordered by release-acquire sequences or memory barriers.

An *availability chain* is a sequence of availability operations to increasingly broad memory domains, where element N+1 of the chain is performed in the dual scope instance of the destination memory domain of element N and element N happens-before element N+1. An example is an availability operation with destination scope of the workgroup instance domain that happens-before an availability operation to the shader domain performed by an invocation in the same workgroup. An availability chain AVC that happens-after W and that includes (A,R,L) in the source scope makes (W,L) *available* to the memory domains in its final destination scope. An availability chain with a single element is just the availability operation.

Similarly, a *visibility chain* is a sequence of visibility operations from increasingly narrow memory domains, where element N of the chain is performed in the dual scope instance of the source memory domain of element N+1 and element N happens-before element N+1. An example is a visibility operation with source scope of the shader domain that happens-before a visibility operation with source scope of the workgroup instance domain performed by an invocation in the same workgroup. A visibility chain VISC that happens-after AVC (or DOM) and for which (W,L) is available in any domain in the source scope makes (W,L) *visible* to all (agent,reference,L) tuples included in its final destination scope. A visibility chain with a single element is just the visibility operation.

Availability, Visibility, and Domain Operations

The following operations generate availability, visibility, and domain operations. When multiple availability/visibility/domain operations are described, they are system-synchronized-with each other in the order listed.

An operation that performs a [memory dependency](#) generates:

- If the source access mask includes `VK_ACCESS_HOST_WRITE_BIT`, then the dependency includes a memory domain operation from host domain to device domain.
- An availability operation with source scope of all writes in the first [access scope](#) of the dependency and a destination scope of the device domain.
- A visibility operation with source scope of the device domain and destination scope of the second access scope of the dependency.
- If the destination access mask includes `VK_ACCESS_HOST_READ_BIT` or `VK_ACCESS_HOST_WRITE_BIT`, then the dependency includes a memory domain operation from device domain to host domain.

[vkFlushMappedMemoryRanges](#) performs an availability operation, with a source scope of (agents,references) = (all host threads, all mapped memory ranges passed to the command), and destination scope of the host domain.

[vkInvalidateMappedMemoryRanges](#) performs a visibility operation, with a source scope of the host domain and a destination scope of (agents,references) = (all host threads, all mapped memory ranges passed to the command).

[vkQueueSubmit](#) performs a memory domain operation from host to device, and a visibility

operation with source scope of the device domain and destination scope of all agents and references on the device.

Availability and Visibility Semantics

A memory barrier or atomic operation via agent A that includes `MakeAvailable` in its semantics performs an availability operation whose source scope includes agent A and all references in the storage classes in that instruction's storage class semantics, and all memory locations, and whose destination scope is a set of memory domains selected as specified below. The implicit availability operation is program-ordered between the barrier or atomic and all other operations program-ordered before the barrier or atomic.

A memory barrier or atomic operation via agent A that includes `MakeVisible` in its semantics performs a visibility operation whose source scope is a set of memory domains selected as specified below, and whose destination scope includes agent A and all references in the storage classes in that instruction's storage class semantics, and all memory locations. The implicit visibility operation is program-ordered between the barrier or atomic and all other operations program-ordered after the barrier or atomic.

The memory domains are selected based on the memory scope of the instruction as follows:

- `Device` scope uses the shader domain
- `QueueFamily` scope uses the queue family instance domain
- `FragmentInterlock` scope uses the fragment interlock instance domain
- `ShaderCallKHR` scope uses the shader call instance domain
- `Workgroup` scope uses the workgroup instance domain
- `Subgroup` uses the subgroup instance domain
- `Invocation` perform no availability/visibility operations.

When an availability operation performed by an agent A includes a memory domain D in its destination scope, where D corresponds to scope instance S, it also includes the memory domains that correspond to each smaller scope instance S' that is a subset of S and that includes A. Similarly for visibility operations.

Per-Instruction Availability and Visibility Semantics

A memory write instruction that includes `MakePointerAvailable`, or an image write instruction that includes `MakeTexelAvailable`, performs an availability operation whose source scope includes the agent and reference used to perform the write and the memory locations written by the instruction, and whose destination scope is a set of memory domains selected by the Scope operand specified in [Availability and Visibility Semantics](#). The implicit availability operation is program-ordered between the write and all other operations program-ordered after the write.

A memory read instruction that includes `MakePointerVisible`, or an image read instruction that includes `MakeTexelVisible`, performs a visibility operation whose source scope is a set of memory domains selected by the Scope operand as specified in [Availability and Visibility Semantics](#), and

whose destination scope includes the agent and reference used to perform the read and the memory locations read by the instruction. The implicit visibility operation is program-ordered between read and all other operations program-ordered before the read.

NOTE

Although reads with per-instruction visibility only perform visibility ops from the shader or fragment interlock instance or shader call instance or workgroup instance or subgroup instance domain, they will also see writes that were made visible via the device domain, i.e. those writes previously performed by non-shader agents and made visible via API commands.

NOTE

It is expected that all invocations in a subgroup execute on the same processor with the same path to memory, and thus availability and visibility operations with subgroup scope can be expected to be “free”.

Location-Ordered

Let X and Y be memory accesses to overlapping sets of memory locations M , where $X \neq Y$. Let (A_x, R_x) be the agent and reference used for X , and (A_y, R_y) be the agent and reference used for Y . For now, let “ $\rightarrow$ ” denote happens-before and “ $\rightarrow^{rcpo}$ ” denote the reflexive closure of program-ordered before.

If D_1 and D_2 are different memory domains, then let $DOM(D_1, D_2)$ be a memory domain operation from D_1 to D_2 . Otherwise, let $DOM(D, D)$ be a placeholder such that $X \rightarrow DOM(D, D) \rightarrow Y$ if and only if $X \rightarrow Y$.

X is *location-ordered* before Y for a location L in M if and only if any of the following is true:

- $A_x == A_y$ and $R_x == R_y$ and $X \rightarrow Y$
 - NOTE: this case means no availability/visibility ops are required when it is the same (agent, reference).
- X is a read, both X and Y are non-private, and $X \rightarrow Y$
- X is a read, and X (transitively) system-synchronizes with Y
- If $R_x == R_y$ and A_x and A_y access a common memory domain D (e.g. are in the same workgroup instance if D is the workgroup instance domain), and both X and Y are non-private:
 - X is a write, Y is a write, $AVC(A_x, R_x, D, L)$ is an availability chain making (X, L) available to domain D , and $X \rightarrow^{rcpo} AVC(A_x, R_x, D, L) \rightarrow Y$
 - X is a write, Y is a read, $AVC(A_x, R_x, D, L)$ is an availability chain making (X, L) available to domain D , $VISC(A_y, R_y, D, L)$ is a visibility chain making writes to L available in domain D visible to Y , and $X \rightarrow^{rcpo} AVC(A_x, R_x, D, L) \rightarrow VISC(A_y, R_y, D, L) \rightarrow^{rcpo} Y$
 - If [VkPhysicalDeviceVulkanMemoryModelFeatures::vulkanMemoryModelAvailabilityVisibilityChains](#) is **VK_FALSE**, then **AVC** and **VISC** **must** each only have a single element in the chain, in each sub-bullet above.
- Let D_x and D_y each be either the device domain or the host domain, depending on whether A_x and A_y execute on the device or host:
 - X is a write and Y is a write, and $X \rightarrow AV(A_x, R_x, D_x, L) \rightarrow DOM(D_x, D_y) \rightarrow Y$

- X is a write and Y is a read, and $X \rightarrow \text{AV}(A_X, R_X, D_X, L) \rightarrow \text{DOM}(D_X, D_Y) \rightarrow \text{VIS}(A_Y, R_Y, D_Y, L) \rightarrow Y$

NOTE

The final bullet (synchronization through device/host domain) requires API-level synchronization operations, since the device/host domains are not accessible via shader instructions. And “device domain” is not to be confused with “device scope”, which synchronizes through the “shader domain”.

Data Race

Let X and Y be operations that access overlapping sets of memory locations M , where $X \neq Y$, and at least one of X and Y is a write, and X and Y are not mutually-ordered atomic operations. If there does not exist a location-ordered relation between X and Y for each location in M , then there is a *data race*.

Applications **must** ensure that no data races occur during the execution of their application.

NOTE

Data races can only occur due to instructions that are actually executed. For example, an instruction skipped due to control flow must not contribute to a data race.

Visible-To

Let X be a write and Y be a read whose sets of memory locations overlap, and let M be the set of memory locations that overlap. Let M_2 be a non-empty subset of M . Then X is *visible-to* Y for memory locations M_2 if and only if all of the following are true:

- X is location-ordered before Y for each location L in M_2 .
- There does not exist another write Z to any location L in M_2 such that X is location-ordered before Z for location L and Z is location-ordered before Y for location L .

If X is visible-to Y , then Y reads the value written by X for locations M_2 .

NOTE

It is possible for there to be a write between X and Y that overwrites a subset of the memory locations, but the remaining memory locations (M_2) will still be visible-to Y .

Acyclicity

Reads-from is a relation between operations, where the first operation is a write, the second operation is a read, and the second operation reads the value written by the first operation. *From-reads* is a relation between operations, where the first operation is a read, the second operation is a write, and the first operation reads a value written earlier than the second operation in the second operation’s scoped modification order or location order (or the first operation reads from the initial value, and the second operation is any write to the same locations).

Then the implementation **must** guarantee that no cycles exist in the union of the following relations:

- location-ordered
- scoped modification order (over all atomic writes)
- reads-from
- from-reads

NOTE

This is a “consistency” axiom, which informally guarantees that sequences of operations cannot violate causality.

Scoped Modification Order Coherence

Let A and B be mutually-ordered atomic operations, where A is location-ordered before B. Then the following rules are a consequence of acyclicity:

- If A and B are both reads and A does not read the initial value, then the write that A takes its value from **must** be earlier in its own scoped modification order than (or the same as) the write that B takes its value from (no cycles between location-order, reads-from, and from-reads).
- If A is a read and B is a write and A does not read the initial value, then A **must** take its value from a write earlier than B in B’s scoped modification order (no cycles between location-order, scope modification order, and reads-from).
- If A is a write and B is a read, then B **must** take its value from A or a write later than A in A’s scoped modification order (no cycles between location-order, scoped modification order, and from-reads).
- If A and B are both writes, then A **must** be earlier than B in A’s scoped modification order (no cycles between location-order and scoped modification order).
- If A is a write and B is a read-modify-write and B reads the value written by A, then B comes immediately after A in A’s scoped modification order (no cycles between scoped modification order and from-reads).

Shader I/O

If a shader invocation A in a shader stage other than **Vertex** performs a memory read operation X from an object in storage class **CallableDataKHR**, **IncomingCallableDataKHR**, **RayPayloadKHR**, **HitAttributeKHR**, **IncomingRayPayloadKHR**, or **Input**, then X is system-synchronized-after all writes to the corresponding **CallableDataKHR**, **IncomingCallableDataKHR**, **RayPayloadKHR**, **HitAttributeKHR**, **IncomingRayPayloadKHR**, or **Output** storage variable(s) in the shader invocation(s) that contribute to generating invocation A, and those writes are all visible-to X.

NOTE

It is not necessary for the upstream shader invocations to have completed execution, they only need to have generated the output that is being read.

Deallocation

A call to **vkFreeMemory** **must** happen-after all memory operations on all memory locations in that **VkDeviceMemory** object.

NOTE

Normally, device memory operations in a given queue are synchronized with `vkFreeMemory` by having a host thread wait on a fence signaled by that queue, and the wait happens-before the call to `vkFreeMemory` on the host.

The deallocation of SPIR-V variables is managed by the system and happens-after all operations on those variables.

Descriptions (Informative)

This subsection offers more easily understandable consequences of the memory model for app/compiler developers.

Let SC be the storage class(es) specified by a release or acquire operation or barrier.

- An atomic write with release semantics must not be reordered against any read or write to SC that is program-ordered before it (regardless of the storage class the atomic is in).
- An atomic read with acquire semantics must not be reordered against any read or write to SC that is program-ordered after it (regardless of the storage class the atomic is in).
- Any write to SC program-ordered after a release barrier must not be reordered against any read or write to SC program-ordered before that barrier.
- Any read from SC program-ordered before an acquire barrier must not be reordered against any read or write to SC program-ordered after the barrier.

A control barrier (even if it has no memory semantics) must not be reordered against any memory barriers.

This memory model allows memory accesses with and without availability and visibility operations, as well as atomic operations, all to be performed on the same memory location. This is critical to allow it to reason about memory that is reused in multiple ways, e.g. across the lifetime of different shader invocations or draw calls. While GLSL (and legacy SPIR-V) applies the “coherent” decoration to variables (for historical reasons), this model treats each memory access instruction as having optional implicit availability/visibility operations. GLSL to SPIR-V compilers should map all (non-atomic) operations on a coherent variable to `Make{Pointer, Texel}{Available}{Visible}` flags in this model.

Atomic operations implicitly have availability/visibility operations, and the scope of those operations is taken from the atomic operation’s scope.

Tessellation Output Ordering

For SPIR-V that uses the Vulkan Memory Model, the `OutputMemory` storage class is used to synchronize accesses to tessellation control output variables. For legacy SPIR-V that does not enable the Vulkan Memory Model via `OpMemoryModel1`, tessellation outputs can be ordered using a control barrier with no particular memory scope or semantics, as defined below.

Let X and Y be memory operations performed by shader invocations A_x and A_y . Operation X is *tessellation-output-ordered* before operation Y if and only if all of the following are true:

- There is a dynamic instance of an **OpControlBarrier** instruction C such that X is program-ordered before C in A_x and C is program-ordered before Y in A_y .
- A_x and A_y are in the same instance of C's execution scope.

If shader invocations A_x and A_y in the **TessellationControl** execution model execute memory operations X and Y, respectively, on the **Output** storage class, and X is tessellation-output-ordered before Y with a scope of **Workgroup**, then X is location-ordered before Y, and if X is a write and Y is a read then X is visible-to Y.

Cooperative Matrix Memory Access

For each dynamic instance of a cooperative matrix load instruction (**OpCooperativeMatrixLoadKHR**), some implementation-dependent invocation(s) within the instance of the matrix's scope perform a non-atomic load from each memory location that is defined to be accessed by the instruction.

For each memory location accessed by a dynamic instance of a cooperative matrix store instruction (**OpCooperativeMatrixStoreKHR**), a single implementation-dependent invocation within the instance of the matrix's scope performs a non-atomic store to that memory location.

Appendix C: Compressed Image Formats

The compressed texture formats used by Vulkan are described in the specifically identified sections of the [Khronos Data Format Specification](#), version 1.3.

Unless otherwise described, the quantities encoded in these compressed formats are treated as normalized, unsigned values.

Those formats listed as sRGB-encoded have in-memory representations of R, G and B components which are nonlinearly-encoded as R', G', and B'; any alpha component is unchanged. As part of filtering, the nonlinear R', G', and B' values are converted to linear R, G, and B components; any alpha component is unchanged. The conversion between linear and nonlinear encoding is performed as described in the “KHR_DF_TRANSFER_SRGB” section of the Khronos Data Format Specification.

Block-Compressed Image Formats

BC1, BC2 and BC3 formats are described in “S3TC Compressed Texture Image Formats” chapter of the [Khronos Data Format Specification](#). BC4 and BC5 are described in the “RGTC Compressed Texture Image Formats” chapter. BC6H and BC7 are described in the “BPTC Compressed Texture Image Formats” chapter.

Table 95. Mapping of Vulkan BC Formats to Descriptions

VkFormat	Khronos Data Format Specification description
Formats described in the “S3TC Compressed Texture Image Formats” chapter	
VK_FORMAT_BC1_RGB_UNORM_BLOCK	BC1 with no alpha
VK_FORMAT_BC1_RGB_SRGB_BLOCK	BC1 with no alpha, sRGB-encoded
VK_FORMAT_BC1_RGBA_UNORM_BLOCK	BC1 with alpha
VK_FORMAT_BC1_RGBA_SRGB_BLOCK	BC1 with alpha, sRGB-encoded
VK_FORMAT_BC2_UNORM_BLOCK	BC2
VK_FORMAT_BC2_SRGB_BLOCK	BC2, sRGB-encoded
VK_FORMAT_BC3_UNORM_BLOCK	BC3
VK_FORMAT_BC3_SRGB_BLOCK	BC3, sRGB-encoded
Formats described in the “RGTC Compressed Texture Image Formats” chapter	
VK_FORMAT_BC4_UNORM_BLOCK	BC4 unsigned
VK_FORMAT_BC4_SNORM_BLOCK	BC4 signed
VK_FORMAT_BC5_UNORM_BLOCK	BC5 unsigned
VK_FORMAT_BC5_SNORM_BLOCK	BC5 signed
Formats described in the “BPTC Compressed Texture Image Formats” chapter	
VK_FORMAT_BC6H_UFLOAT_BLOCK	BC6H (unsigned version)
VK_FORMAT_BC6H_SFLOAT_BLOCK	BC6H (signed version)
VK_FORMAT_BC7_UNORM_BLOCK	BC7
VK_FORMAT_BC7_SRGB_BLOCK	BC7, sRGB-encoded

ETC Compressed Image Formats

The following formats are described in the “ETC2 Compressed Texture Image Formats” chapter of the [Khronos Data Format Specification](#).

Table 96. Mapping of Vulkan ETC Formats to Descriptions

VkFormat	Khronos Data Format Specification description
VK_FORMAT_ETC2_R8G8B8_UNORM_BLOCK	RGB ETC2
VK_FORMAT_ETC2_R8G8B8_SRGB_BLOCK	RGB ETC2 with sRGB encoding
VK_FORMAT_ETC2_R8G8B8A1_UNORM_BLOCK	RGB ETC2 with punch-through alpha
VK_FORMAT_ETC2_R8G8B8A1_SRGB_BLOCK	RGB ETC2 with punch-through alpha and sRGB
VK_FORMAT_ETC2_R8G8B8A8_UNORM_BLOCK	RGBA ETC2
VK_FORMAT_ETC2_R8G8B8A8_SRGB_BLOCK	RGBA ETC2 with sRGB encoding
VK_FORMAT_EAC_R11_UNORM_BLOCK	Unsigned R11 EAC
VK_FORMAT_EAC_R11_SNORM_BLOCK	Signed R11 EAC
VK_FORMAT_EAC_R11G11_UNORM_BLOCK	Unsigned RG11 EAC
VK_FORMAT_EAC_R11G11_SNORM_BLOCK	Signed RG11 EAC

ASTC Compressed Image Formats

ASTC formats are described in the “ASTC Compressed Texture Image Formats” chapter of the [Khronos Data Format Specification](#).

Table 97. Mapping of Vulkan ASTC Formats to Descriptions

VkFormat	Compressed texel block dimensions	Requested mode
VK_FORMAT_ASTC_4x4_UNORM_BLOCK	4 × 4	Linear LDR
VK_FORMAT_ASTC_4x4_SRGB_BLOCK	4 × 4	sRGB
VK_FORMAT_ASTC_5x4_UNORM_BLOCK	5 × 4	Linear LDR
VK_FORMAT_ASTC_5x4_SRGB_BLOCK	5 × 4	sRGB
VK_FORMAT_ASTC_5x5_UNORM_BLOCK	5 × 5	Linear LDR
VK_FORMAT_ASTC_5x5_SRGB_BLOCK	5 × 5	sRGB
VK_FORMAT_ASTC_6x5_UNORM_BLOCK	6 × 5	Linear LDR
VK_FORMAT_ASTC_6x5_SRGB_BLOCK	6 × 5	sRGB
VK_FORMAT_ASTC_6x6_UNORM_BLOCK	6 × 6	Linear LDR
VK_FORMAT_ASTC_6x6_SRGB_BLOCK	6 × 6	sRGB
VK_FORMAT_ASTC_8x5_UNORM_BLOCK	8 × 5	Linear LDR
VK_FORMAT_ASTC_8x5_SRGB_BLOCK	8 × 5	sRGB
VK_FORMAT_ASTC_8x6_UNORM_BLOCK	8 × 6	Linear LDR
VK_FORMAT_ASTC_8x6_SRGB_BLOCK	8 × 6	sRGB
VK_FORMAT_ASTC_8x8_UNORM_BLOCK	8 × 8	Linear LDR
VK_FORMAT_ASTC_8x8_SRGB_BLOCK	8 × 8	sRGB
VK_FORMAT_ASTC_10x5_UNORM_BLOCK	10 × 5	Linear LDR
VK_FORMAT_ASTC_10x5_SRGB_BLOCK	10 × 5	sRGB
VK_FORMAT_ASTC_10x6_UNORM_BLOCK	10 × 6	Linear LDR
VK_FORMAT_ASTC_10x6_SRGB_BLOCK	10 × 6	sRGB
VK_FORMAT_ASTC_10x8_UNORM_BLOCK	10 × 8	Linear LDR
VK_FORMAT_ASTC_10x8_SRGB_BLOCK	10 × 8	sRGB
VK_FORMAT_ASTC_10x10_UNORM_BLOCK	10 × 10	Linear LDR
VK_FORMAT_ASTC_10x10_SRGB_BLOCK	10 × 10	sRGB
VK_FORMAT_ASTC_12x10_UNORM_BLOCK	12 × 10	Linear LDR
VK_FORMAT_ASTC_12x10_SRGB_BLOCK	12 × 10	sRGB
VK_FORMAT_ASTC_12x12_UNORM_BLOCK	12 × 12	Linear LDR

VkFormat	Compressed texel block dimensions	Requested mode
VK_FORMAT_ASTC_12x12_SRGB_BLOCK	12 × 12	sRGB

ASTC textures containing any HDR blocks **should** not be passed into the API using an sRGB or UNORM texture format.

NOTE

An HDR block in a texture passed using a LDR UNORM format will return the appropriate ASTC error color if the implementation supports only the ASTC LDR profile, but may result in either the error color or a decompressed HDR color if the implementation supports HDR decoding.

ASTC Decode Mode

If the `VK_EXT_astc_decode_mode` extension is enabled, the decode mode is determined as follows:

Table 98. Mapping of Vulkan ASTC Decoding Format to ASTC Decoding Modes

VkFormat	Decoding mode
VK_FORMAT_R16G16B16A16_SFLOAT	decode_float16
VK_FORMAT_R8G8B8A8_UNORM	decode_unorm8
VK_FORMAT_E5B9G9R9_UFLOAT_PACK32	decode_rgb9e5

Otherwise, the ASTC decode mode is `decode_float16`.

Note that an implementation **may** use HDR mode when linear LDR mode is requested unless the decode mode is `decode_unorm8`.

Appendix D: Core Revisions (Informative)

New minor versions of the Vulkan API are defined periodically by the Khronos Vulkan Working Group. These consist of some amount of additional functionality added to the core API, potentially including both new functionality and functionality [promoted](#) from extensions.

Vulkan Version 1.0

Vulkan Version 1.0 was the initial release of the Vulkan API.

New Macros

- [VK_API_VERSION](#)
- [VK_API_VERSION_1_0](#)
- [VK_API_VERSION_MAJOR](#)
- [VK_API_VERSION_MINOR](#)
- [VK_API_VERSION_PATCH](#)
- [VK_API_VERSION_VARIANT](#)
- [VK_DEFINE_HANDLE](#)
- [VK_DEFINE_NON_DISPATCHABLE_HANDLE](#)
- [VK_HEADER_VERSION](#)
- [VK_HEADER_VERSION_COMPLETE](#)
- [VK_MAKE_API_VERSION](#)
- [VK_MAKE_VERSION](#)
- [VK_NULL_HANDLE](#)
- [VK_USE_64_BIT_PTR_DEFINES](#)
- [VK_VERSION_MAJOR](#)
- [VK_VERSION_MINOR](#)
- [VK_VERSION_PATCH](#)

New Base Types

- [VkBool32](#)
- [VkDeviceAddress](#)
- [VkDeviceSize](#)
- [VkFlags](#)
- [VkSampleMask](#)

New Object Types

- [VkBuffer](#)
- [VkBufferView](#)
- [VkCommandBuffer](#)
- [VkCommandPool](#)
- [VkDescriptorPool](#)
- [VkDescriptorSet](#)
- [VkDescriptorSetLayout](#)
- [VkDevice](#)
- [VkDeviceMemory](#)
- [VkEvent](#)
- [VkFence](#)
- [VkFramebuffer](#)
- [VkImage](#)
- [VkImageView](#)
- [VkInstance](#)
- [VkPhysicalDevice](#)
- [VkPipeline](#)
- [VkPipelineCache](#)
- [VkPipelineLayout](#)
- [VkQueryPool](#)
- [VkQueue](#)
- [VkRenderPass](#)
- [VkSampler](#)
- [VkSemaphore](#)
- [VkShaderModule](#)

New Commands

- [vkAllocateCommandBuffers](#)
- [vkAllocateDescriptorSets](#)
- [vkAllocateMemory](#)
- [vkBeginCommandBuffer](#)
- [vkBindBufferMemory](#)
- [vkBindImageMemory](#)

- [vkCmdBeginQuery](#)
- [vkCmdBeginRenderPass](#)
- [vkCmdBindDescriptorSets](#)
- [vkCmdBindIndexBuffer](#)
- [vkCmdBindPipeline](#)
- [vkCmdBindVertexBuffers](#)
- [vkCmdBlitImage](#)
- [vkCmdClearAttachments](#)
- [vkCmdClearColorImage](#)
- [vkCmdClearDepthStencilImage](#)
- [vkCmdCopyBuffer](#)
- [vkCmdCopyBufferToImage](#)
- [vkCmdCopyImage](#)
- [vkCmdCopyImageToBuffer](#)
- [vkCmdCopyQueryPoolResults](#)
- [vkCmdDispatch](#)
- [vkCmdDispatchIndirect](#)
- [vkCmdDraw](#)
- [vkCmdDrawIndexed](#)
- [vkCmdDrawIndexedIndirect](#)
- [vkCmdDrawIndirect](#)
- [vkCmdEndQuery](#)
- [vkCmdEndRenderPass](#)
- [vkCmdExecuteCommands](#)
- [vkCmdFillBuffer](#)
- [vkCmdNextSubpass](#)
- [vkCmdPipelineBarrier](#)
- [vkCmdPushConstants](#)
- [vkCmdResetEvent](#)
- [vkCmdResetQueryPool](#)
- [vkCmdResolveImage](#)
- [vkCmdSetBlendConstants](#)
- [vkCmdSetDepthBias](#)
- [vkCmdSetDepthBounds](#)
- [vkCmdSetEvent](#)

- [vkCmdSetLineWidth](#)
- [vkCmdSetScissor](#)
- [vkCmdSetStencilCompareMask](#)
- [vkCmdSetStencilReference](#)
- [vkCmdSetStencilWriteMask](#)
- [vkCmdSetViewport](#)
- [vkCmdUpdateBuffer](#)
- [vkCmdWaitEvents](#)
- [vkCmdWriteTimestamp](#)
- [vkCreateBuffer](#)
- [vkCreateBufferView](#)
- [vkCreateCommandPool](#)
- [vkCreateComputePipelines](#)
- [vkCreateDescriptorPool](#)
- [vkCreateDescriptorSetLayout](#)
- [vkCreateDevice](#)
- [vkCreateEvent](#)
- [vkCreateFence](#)
- [vkCreateFramebuffer](#)
- [vkCreateGraphicsPipelines](#)
- [vkCreateImage](#)
- [vkCreateImageView](#)
- [vkCreateInstance](#)
- [vkCreatePipelineCache](#)
- [vkCreatePipelineLayout](#)
- [vkCreateQueryPool](#)
- [vkCreateRenderPass](#)
- [vkCreateSampler](#)
- [vkCreateSemaphore](#)
- [vkCreateShaderModule](#)
- [vkDestroyBuffer](#)
- [vkDestroyBufferView](#)
- [vkDestroyCommandPool](#)
- [vkDestroyDescriptorPool](#)
- [vkDestroyDescriptorSetLayout](#)

- [vkDestroyDevice](#)
- [vkDestroyEvent](#)
- [vkDestroyFence](#)
- [vkDestroyFramebuffer](#)
- [vkDestroyImage](#)
- [vkDestroyImageView](#)
- [vkDestroyInstance](#)
- [vkDestroyPipeline](#)
- [vkDestroyPipelineCache](#)
- [vkDestroyPipelineLayout](#)
- [vkDestroyQueryPool](#)
- [vkDestroyRenderPass](#)
- [vkDestroySampler](#)
- [vkDestroySemaphore](#)
- [vkDestroyShaderModule](#)
- [vkDeviceWaitIdle](#)
- [vkEndCommandBuffer](#)
- [vkEnumerateDeviceExtensionProperties](#)
- [vkEnumerateDeviceLayerProperties](#)
- [vkEnumerateInstanceExtensionProperties](#)
- [vkEnumerateInstanceLayerProperties](#)
- [vkEnumeratePhysicalDevices](#)
- [vkFlushMappedMemoryRanges](#)
- [vkFreeCommandBuffers](#)
- [vkFreeDescriptorSets](#)
- [vkFreeMemory](#)
- [vkGetBufferMemoryRequirements](#)
- [vkGetDeviceMemoryCommitment](#)
- [vkGetDeviceProcAddr](#)
- [vkGetDeviceQueue](#)
- [vkGetEventStatus](#)
- [vkGetFenceStatus](#)
- [vkGetImageMemoryRequirements](#)
- [vkGetImageSparseMemoryRequirements](#)
- [vkGetImageSubresourceLayout](#)

- [vkGetInstanceProcAddr](#)
- [vkGetPhysicalDeviceFeatures](#)
- [vkGetPhysicalDeviceFormatProperties](#)
- [vkGetPhysicalDeviceImageFormatProperties](#)
- [vkGetPhysicalDeviceMemoryProperties](#)
- [vkGetPhysicalDeviceProperties](#)
- [vkGetPhysicalDeviceQueueFamilyProperties](#)
- [vkGetPhysicalDeviceSparseImageFormatProperties](#)
- [vkGetPipelineCacheData](#)
- [vkGetQueryPoolResults](#)
- [vkGetRenderAreaGranularity](#)
- [vkInvalidateMappedMemoryRanges](#)
- [vkMapMemory](#)
- [vkMergePipelineCaches](#)
- [vkQueueBindSparse](#)
- [vkQueueSubmit](#)
- [vkQueueWaitIdle](#)
- [vkResetCommandBuffer](#)
- [vkResetCommandPool](#)
- [vkResetDescriptorPool](#)
- [vkResetEvent](#)
- [vkResetFences](#)
- [vkSetEvent](#)
- [vkUnmapMemory](#)
- [vkUpdateDescriptorSets](#)
- [vkWaitForFences](#)

New Structures

- [VkAllocationCallbacks](#)
- [VkApplicationInfo](#)
- [VkAttachmentDescription](#)
- [VkAttachmentReference](#)
- [VkBaseInStructure](#)
- [VkBaseOutStructure](#)
- [VkBindSparseInfo](#)

- [VkBufferCopy](#)
- [VkBufferCreateInfo](#)
- [VkBufferImageCopy](#)
- [VkBufferMemoryBarrier](#)
- [VkBufferViewCreateInfo](#)
- [VkClearAttachment](#)
- [VkClearDepthStencilValue](#)
- [VkClearRect](#)
- [VkCommandBufferAllocateInfo](#)
- [VkCommandBufferBeginInfo](#)
- [VkCommandBufferInheritanceInfo](#)
- [VkCommandPoolCreateInfo](#)
- [VkComponentMapping](#)
- [VkComputePipelineCreateInfo](#)
- [VkCopyDescriptorSet](#)
- [VkDescriptorBufferInfo](#)
- [VkDescriptorImageInfo](#)
- [VkDescriptorPoolCreateInfo](#)
- [VkDescriptorPoolSize](#)
- [VkDescriptorSetAllocateInfo](#)
- [VkDescriptorSetLayoutBinding](#)
- [VkDescriptorSetLayoutCreateInfo](#)
- [VkDeviceCreateInfo](#)
- [VkDeviceQueueCreateInfo](#)
- [VkDispatchIndirectCommand](#)
- [VkDrawIndexedIndirectCommand](#)
- [VkDrawIndirectCommand](#)
- [VkEventCreateInfo](#)
- [VkExtensionProperties](#)
- [VkExtent2D](#)
- [VkExtent3D](#)
- [VkFenceCreateInfo](#)
- [VkFormatProperties](#)
- [VkFramebufferCreateInfo](#)
- [VkGraphicsPipelineCreateInfo](#)

- [VkImageBlit](#)
- [VkImageCopy](#)
- [VkImageCreateInfo](#)
- [VkImageFormatProperties](#)
- [VkImageMemoryBarrier](#)
- [VkImageResolve](#)
- [VkImageSubresource](#)
- [VkImageSubresourceLayers](#)
- [VkImageSubresourceRange](#)
- [VkImageViewCreateInfo](#)
- [VkInstanceCreateInfo](#)
- [VkLayerProperties](#)
- [VkMappedMemoryRange](#)
- [VkMemoryAllocateInfo](#)
- [VkMemoryBarrier](#)
- [VkMemoryHeap](#)
- [VkMemoryRequirements](#)
- [VkMemoryType](#)
- [VkOffset2D](#)
- [VkOffset3D](#)
- [VkPhysicalDeviceFeatures](#)
- [VkPhysicalDeviceLimits](#)
- [VkPhysicalDeviceMemoryProperties](#)
- [VkPhysicalDeviceProperties](#)
- [VkPhysicalDeviceSparseProperties](#)
- [VkPipelineCacheCreateInfo](#)
- [VkPipelineCacheHeaderVersionOne](#)
- [VkPipelineColorBlendAttachmentState](#)
- [VkPipelineColorBlendStateCreateInfo](#)
- [VkPipelineDepthStencilStateCreateInfo](#)
- [VkPipelineDynamicStateCreateInfo](#)
- [VkPipelineInputAssemblyStateCreateInfo](#)
- [VkPipelineMultisampleStateCreateInfo](#)
- [VkPipelineRasterizationStateCreateInfo](#)
- [VkPipelineShaderStageCreateInfo](#)

- [VkPipelineTessellationStateCreateInfo](#)
- [VkPipelineVertexInputStateCreateInfo](#)
- [VkPipelineViewportStateCreateInfo](#)
- [VkPushConstantRange](#)
- [VkQueryPoolCreateInfo](#)
- [VkQueueFamilyProperties](#)
- [VkRect2D](#)
- [VkRenderPassBeginInfo](#)
- [VkRenderPassCreateInfo](#)
- [VkSamplerCreateInfo](#)
- [VkSemaphoreCreateInfo](#)
- [VkSparseBufferMemoryBindInfo](#)
- [VkSparseImageFormatProperties](#)
- [VkSparseImageMemoryBind](#)
- [VkSparseImageMemoryBindInfo](#)
- [VkSparseImageMemoryRequirements](#)
- [VkSparseImageOpaqueMemoryBindInfo](#)
- [VkSparseMemoryBind](#)
- [VkSpecializationInfo](#)
- [VkSpecializationMapEntry](#)
- [VkStencilOpState](#)
- [VkSubmitInfo](#)
- [VkSubpassDependency](#)
- [VkSubpassDescription](#)
- [VkSubresourceLayout](#)
- [VkVertexInputAttributeDescription](#)
- [VkVertexInputBindingDescription](#)
- [VkViewport](#)
- [VkWriteDescriptorSet](#)
- Extending [VkBindDescriptorSetsInfoKHR](#), [VkPushConstantsInfoKHR](#), [VkPushDescriptorSetInfoKHR](#), [VkPushDescriptorSetWithTemplateInfoKHR](#), [VkSetDescriptorBufferOffsetsInfoEXT](#), [VkBindDescriptorBufferEmbeddedSamplersInfoEXT](#), [VkIndirectCommandsLayoutCreateInfoEXT](#):
 - [VkPipelineLayoutCreateInfo](#)
- Extending [VkPipelineShaderStageCreateInfo](#):

- [VkShaderModuleCreateInfo](#)

New Unions

- [VkClearColorValue](#)
- [VkClearValue](#)

New Function Pointers

- [PFN_vkAllocationFunction](#)
- [PFN_vkFreeFunction](#)
- [PFN_vkInternalAllocationNotification](#)
- [PFN_vkInternalFreeNotification](#)
- [PFN_vkReallocationFunction](#)
- [PFN_vkVoidFunction](#)

New Enums

- [VkAccessFlagBits](#)
- [VkAttachmentDescriptionFlagBits](#)
- [VkAttachmentLoadOp](#)
- [VkAttachmentStoreOp](#)
- [VkBlendFactor](#)
- [VkBlendOp](#)
- [VkBorderColor](#)
- [VkBufferCreateFlagBits](#)
- [VkBufferUsageFlagBits](#)
- [VkColorComponentFlagBits](#)
- [VkCommandBufferLevel](#)
- [VkCommandBufferResetFlagBits](#)
- [VkCommandBufferUsageFlagBits](#)
- [VkCommandPoolCreateFlagBits](#)
- [VkCommandPoolResetFlagBits](#)
- [VkCompareOp](#)
- [VkComponentSwizzle](#)
- [VkCullModeFlagBits](#)
- [VkDependencyFlagBits](#)
- [VkDescriptorPoolCreateFlagBits](#)

- [VkDescriptorSetLayoutCreateFlagBits](#)
- [VkDescriptorType](#)
- [VkDynamicState](#)
- [VkEventCreateFlagBits](#)
- [VkFenceCreateFlagBits](#)
- [VkFilter](#)
- [VkFormat](#)
- [VkFormatFeatureFlagBits](#)
- [VkFramebufferCreateFlagBits](#)
- [VkFrontFace](#)
- [VkImageAspectFlagBits](#)
- [VkImageCreateFlagBits](#)
- [VkImageLayout](#)
- [VkImageTiling](#)
- [VkImageType](#)
- [VkImageUsageFlagBits](#)
- [VkImageViewCreateFlagBits](#)
- [VkImageViewType](#)
- [VkIndexType](#)
- [VkInstanceCreateFlagBits](#)
- [VkInternalAllocationType](#)
- [VkLogicOp](#)
- [VkMemoryHeapFlagBits](#)
- [VkMemoryMapFlagBits](#)
- [VkMemoryPropertyFlagBits](#)
- [VkObjectType](#)
- [VkPhysicalDeviceType](#)
- [VkPipelineBindPoint](#)
- [VkPipelineCacheHeaderVersion](#)
- [VkPipelineCreateFlagBits](#)
- [VkPipelineShaderStageCreateFlagBits](#)
- [VkPipelineStageFlagBits](#)
- [VkPolygonMode](#)
- [VkPrimitiveTopology](#)
- [VkQueryControlFlagBits](#)

- [VkQueryPipelineStatisticFlagBits](#)
- [VkQueryResultFlagBits](#)
- [VkQueryType](#)
- [VkQueueFlagBits](#)
- [VkRenderPassCreateFlagBits](#)
- [VkResult](#)
- [VkSampleCountFlagBits](#)
- [VkSamplerAddressMode](#)
- [VkSamplerCreateFlagBits](#)
- [VkSamplerMipmapMode](#)
- [VkShaderStageFlagBits](#)
- [VkSharingMode](#)
- [VkSparseImageFormatFlagBits](#)
- [VkSparseMemoryBindFlagBits](#)
- [VkStencilFaceFlagBits](#)
- [VkStencilOp](#)
- [VkStructureType](#)
- [VkSubpassContents](#)
- [VkSubpassDescriptionFlagBits](#)
- [VkSystemAllocationScope](#)
- [VkVendorId](#)
- [VkVertexInputRate](#)

New Bitmasks

- [VkAccessFlags](#)
- [VkAttachmentDescriptionFlags](#)
- [VkBufferCreateFlags](#)
- [VkBufferUsageFlags](#)
- [VkBufferViewCreateFlags](#)
- [VkColorComponentFlags](#)
- [VkCommandBufferResetFlags](#)
- [VkCommandBufferUsageFlags](#)
- [VkCommandPoolCreateFlags](#)
- [VkCommandPoolResetFlags](#)
- [VkCullModeFlags](#)

- [VkDependencyFlags](#)
- [VkDescriptorPoolCreateFlags](#)
- [VkDescriptorPoolResetFlags](#)
- [VkDescriptorSetLayoutCreateFlags](#)
- [VkDeviceCreateFlags](#)
- [VkDeviceQueueCreateFlags](#)
- [VkEventCreateFlags](#)
- [VkFenceCreateFlags](#)
- [VkFormatFeatureFlags](#)
- [VkFramebufferCreateFlags](#)
- [VkImageAspectFlags](#)
- [VkImageCreateFlags](#)
- [VkImageUsageFlags](#)
- [VkImageViewCreateFlags](#)
- [VkInstanceCreateFlags](#)
- [VkMemoryHeapFlags](#)
- [VkMemoryMapFlags](#)
- [VkMemoryPropertyFlags](#)
- [VkPipelineCacheCreateFlags](#)
- [VkPipelineColorBlendStateCreateFlags](#)
- [VkPipelineCreateFlags](#)
- [VkPipelineDepthStencilStateCreateFlags](#)
- [VkPipelineDynamicStateCreateFlags](#)
- [VkPipelineInputAssemblyStateCreateFlags](#)
- [VkPipelineLayoutCreateFlags](#)
- [VkPipelineMultisampleStateCreateFlags](#)
- [VkPipelineRasterizationStateCreateFlags](#)
- [VkPipelineShaderStageCreateFlags](#)
- [VkPipelineStageFlags](#)
- [VkPipelineTessellationStateCreateFlags](#)
- [VkPipelineVertexInputStateCreateFlags](#)
- [VkPipelineViewportStateCreateFlags](#)
- [VkQueryControlFlags](#)
- [VkQueryPipelineStatisticFlags](#)
- [VkQueryPoolCreateFlags](#)

- [VkQueryResultFlags](#)
- [VkQueueFlags](#)
- [VkRenderPassCreateFlags](#)
- [VkSampleCountFlags](#)
- [VkSamplerCreateFlags](#)
- [VkSemaphoreCreateFlags](#)
- [VkShaderModuleCreateFlags](#)
- [VkShaderStageFlags](#)
- [VkSparseImageFormatFlags](#)
- [VkSparseMemoryBindFlags](#)
- [VkStencilFaceFlags](#)
- [VkSubpassDescriptionFlags](#)

New Headers

- [vk_platform](#)

New Enum Constants

- [VK_ATTACHMENT_UNUSED](#)
- [VK_FALSE](#)
- [VK_LOD_CLAMP_NONE](#)
- [VK_MAX_DESCRIPTION_SIZE](#)
- [VK_MAX_EXTENSION_NAME_SIZE](#)
- [VK_MAX_MEMORY_HEAPS](#)
- [VK_MAX_MEMORY_TYPES](#)
- [VK_MAX_PHYSICAL_DEVICE_NAME_SIZE](#)
- [VK_QUEUE_FAMILY_IGNORED](#)
- [VK_REMAINING_ARRAY_LAYERS](#)
- [VK_REMAINING_MIP_LEVELS](#)
- [VK_SUBPASS_EXTERNAL](#)
- [VK_TRUE](#)
- [VK_UUID_SIZE](#)
- [VK_WHOLE_SIZE](#)

Appendix E: Layers & Extensions

(Informative)

Extensions to the Vulkan API **can** be defined by authors, groups of authors, and the Khronos Vulkan Working Group. In order not to compromise the readability of the Vulkan Specification, the core Specification does not incorporate most extensions. The online Registry of extensions is available at URL

<https://registry.khronos.org/vulkan/>

and allows generating versions of the Specification incorporating different extensions.

Authors creating extensions and layers **must** follow the mandatory procedures described in the [Vulkan Documentation and Extensions](#) document when creating extensions and layers.

The remainder of this appendix documents a set of extensions chosen when this document was built. Versions of the Specification published in the Registry include:

- Core API + mandatory extensions required of all Vulkan implementations.
- Core API + all registered and published Khronos (**KHR**) extensions.
- Core API + all registered and published extensions.

Extensions are grouped as Khronos **KHR**, multivendor **EXT**, and then alphabetically by author ID. Within each group, extensions are listed in alphabetical order by their name.

Extension Dependencies

Extensions which have dependencies on specific core versions or on other extensions will list such dependencies.

For core versions, the specified version **must** be supported at runtime. All extensions implicitly require support for Vulkan 1.0.

For a device extension, use of any device-level functionality defined by that extension requires that any extensions that extension depends on be enabled.

For any extension, use of any instance-level functionality defined by that extension requires only that any extensions that extension depends on be supported at runtime.

Extension Interactions

Some extensions define APIs which are only supported when other extensions or core versions are supported at runtime. Such interactions are noted as “API Interactions”.

List of Current Extensions

- [VK_KHR_16bit_storage](#) (promoted to core)
- [VK_KHR_8bit_storage](#) (promoted to core)
- [VK_KHR_acceleration_structure](#)
- [VK_KHR_android_surface](#)
- [VK_KHR_bind_memory2](#) (promoted to core)
- [VK_KHR_buffer_device_address](#) (promoted to core)
- [VK_KHR_calibrated_timestamps](#)
- [VK_KHR_compute_shader_derivatives](#)
- [VK_KHR_cooperative_matrix](#)
- [VK_KHR_copy_commands2](#) (promoted to core)
- [VK_KHR_create_renderpass2](#) (promoted to core)
- [VK_KHR_dedicated_allocation](#) (promoted to core)
- [VK_KHR_deferred_host_operations](#)
- [VK_KHR_depth_stencil_resolve](#) (promoted to core)
- [VK_KHR_descriptor_update_template](#) (promoted to core)
- [VK_KHR_device_group](#) (promoted to core)
- [VK_KHR_device_group_creation](#) (promoted to core)
- [VK_KHR_display](#)
- [VK_KHR_display_swapchain](#)
- [VK_KHR_draw_indirect_count](#) (promoted to core)
- [VK_KHR_driver_properties](#) (promoted to core)
- [VK_KHR_dynamic_rendering](#) (promoted to core)
- [VK_KHR_dynamic_rendering_local_read](#)
- [VK_KHR_external_fence](#) (promoted to core)
- [VK_KHR_external_fence_capabilities](#) (promoted to core)
- [VK_KHR_external_fence_fd](#)
- [VK_KHR_external_fence_win32](#)
- [VK_KHR_external_memory](#) (promoted to core)
- [VK_KHR_external_memory_capabilities](#) (promoted to core)
- [VK_KHR_external_memory_fd](#)
- [VK_KHR_external_memory_win32](#)
- [VK_KHR_external_semaphore](#) (promoted to core)
- [VK_KHR_external_semaphore_capabilities](#) (promoted to core)

- [VK_KHR_external_semaphore_fd](#)
- [VK_KHR_external_semaphore_win32](#)
- [VK_KHR_format_feature_flags2](#) (promoted to core)
- [VK_KHR_fragment_shader_barycentric](#)
- [VK_KHR_fragment_shading_rate](#)
- [VK_KHR_get_display_properties2](#)
- [VK_KHR_get_memory_requirements2](#) (promoted to core)
- [VK_KHR_get_physical_device_properties2](#) (promoted to core)
- [VK_KHR_get_surface_capabilities2](#)
- [VK_KHR_global_priority](#)
- [VK_KHR_image_format_list](#) (promoted to core)
- [VK_KHR_imageless_framebuffer](#) (promoted to core)
- [VK_KHR_incremental_present](#)
- [VK_KHR_index_type_uint8](#)
- [VK_KHR_line_rasterization](#)
- [VK_KHR_load_store_op_none](#)
- [VK_KHR_maintenance1](#) (promoted to core)
- [VK_KHR_maintenance2](#) (promoted to core)
- [VK_KHR_maintenance3](#) (promoted to core)
- [VK_KHR_maintenance4](#) (promoted to core)
- [VK_KHR_maintenance5](#)
- [VK_KHR_maintenance6](#)
- [VK_KHR_maintenance7](#)
- [VK_KHR_map_memory2](#)
- [VK_KHR_multiview](#) (promoted to core)
- [VK_KHR_performance_query](#)
- [VK_KHR_pipeline_binary](#)
- [VK_KHR_pipeline_executable_properties](#)
- [VK_KHR_pipeline_library](#)
- [VK_KHR_portability_enumeration](#)
- [VK_KHR_present_id](#)
- [VK_KHR_present_wait](#)
- [VK_KHR_push_descriptor](#)
- [VK_KHR_ray_query](#)
- [VK_KHR_ray_tracing_maintenance1](#)

- [VK_KHR_ray_tracing_pipeline](#)
- [VK_KHR_ray_tracing_position_fetch](#)
- [VK_KHR_relaxed_block_layout](#) (promoted to core)
- [VK_KHR_sampler_mirror_clamp_to_edge](#) (promoted to core)
- [VK_KHR_sampler_ycbcr_conversion](#) (promoted to core)
- [VK_KHR_separate_depth_stencil_layouts](#) (promoted to core)
- [VK_KHR_shader_atomic_int64](#) (promoted to core)
- [VK_KHR_shader_clock](#)
- [VK_KHR_shader_draw_parameters](#) (promoted to core)
- [VK_KHR_shader_expect_assume](#)
- [VK_KHR_shader_float16_int8](#) (promoted to core)
- [VK_KHR_shader_float_controls](#) (promoted to core)
- [VK_KHR_shader_float_controls2](#)
- [VK_KHR_shader_integer_dot_product](#) (promoted to core)
- [VK_KHR_shader_maximal_reconvergence](#)
- [VK_KHR_shader_non_semantic_info](#) (promoted to core)
- [VK_KHR_shader_quad_control](#)
- [VK_KHR_shader_relaxed_extended_instruction](#)
- [VK_KHR_shader_subgroup_extended_types](#) (promoted to core)
- [VK_KHR_shader_subgroup_rotate](#)
- [VK_KHR_shader_subgroup_uniform_control_flow](#)
- [VK_KHR_shader_terminate_invocation](#) (promoted to core)
- [VK_KHR_shared_presentable_image](#)
- [VK_KHR_spirv_1_4](#) (promoted to core)
- [VK_KHR_storage_buffer_storage_class](#) (promoted to core)
- [VK_KHR_surface](#)
- [VK_KHR_surface_protected_capabilities](#)
- [VK_KHR_swapchain](#)
- [VK_KHR_swapchain_mutable_format](#)
- [VK_KHR_synchronization2](#) (promoted to core)
- [VK_KHR_timeline_semaphore](#) (promoted to core)
- [VK_KHR_uniform_buffer_standard_layout](#) (promoted to core)
- [VK_KHR_variable_pointers](#) (promoted to core)
- [VK_KHR_vertex_attribute_divisor](#)
- [VK_KHR_video_decode_av1](#)

- [VK_KHR_video_decode_h264](#)
- [VK_KHR_video_decode_h265](#)
- [VK_KHR_video_decode_queue](#)
- [VK_KHR_video_encode_h264](#)
- [VK_KHR_video_encode_h265](#)
- [VK_KHR_video_encode_queue](#)
- [VK_KHR_video_maintenance1](#)
- [VK_KHR_video_queue](#)
- [VK_KHR_vulkan_memory_model](#) (promoted to core)
- [VK_KHR_wayland_surface](#)
- [VK_KHR_win32_keyed_mutex](#)
- [VK_KHR_win32_surface](#)
- [VK_KHR_workgroup_memory_explicit_layout](#)
- [VK_KHR_xcb_surface](#)
- [VK_KHR_xlib_surface](#)
- [VK_KHR_zero_initialize_workgroup_memory](#) (promoted to core)
- [VK_EXT_astc_decode_mode](#)
- [VK_EXT_attachment_feedback_loop_dynamic_state](#)
- [VK_EXT_attachment_feedback_loop_layout](#)
- [VK_EXT_blend_operation_advanced](#)
- [VK_EXT_border_color_swizzle](#)
- [VK_EXT_color_write_enable](#)
- [VK_EXT_conservative_rasterization](#)
- [VK_EXT_custom_border_color](#)
- [VK_EXT_debug_utils](#)
- [VK_EXT_depth_bias_control](#)
- [VK_EXT_depth_clamp_control](#)
- [VK_EXT_depth_clip_control](#)
- [VK_EXT_depth_clip_enable](#)
- [VK_EXT_depth_range_unrestricted](#)
- [VK_EXT_device_generated_commands](#)
- [VK_EXT_device_memory_report](#)
- [VK_EXT_direct_mode_display](#)
- [VK_EXT_discard_rectangles](#)
- [VK_EXT_display_control](#)

- `VK_EXT_display_surface_counter`
- `VK_EXT_dynamic_rendering_unused_attachments`
- `VK_EXT_extended_dynamic_state3`
- `VK_EXT_external_memory_acquire_unmodified`
- `VK_EXT_external_memory_dma_buf`
- `VK_EXT_external_memory_host`
- `VK_EXT_filter_cubic`
- `VK_EXT_fragment_shader_interlock`
- `VK_EXT_frame_boundary`
- `VK_EXT_full_screen_exclusive`
- `VK_EXT_hdr_metadata`
- `VK_EXT_headless_surface`
- `VK_EXT_host_image_copy`
- `VK_EXT_image_drm_format_modifier`
- `VK_EXT_image_view_min_lod`
- `VK_EXT_layer_settings`
- `VK_EXT_legacy_vertex_attributes`
- `VK_EXT_map_memory_placed`
- `VK_EXT_memory_budget`
- `VK_EXT_memory_priority`
- `VK_EXT_multi_draw`
- `VK_EXT_nested_command_buffer`
- `VK_EXT_non_seamless_cube_map`
- `VK_EXT_opacity_micromap`
- `VK_EXT_pci_bus_info`
- `VK_EXT_physical_device_drm`
- `VK_EXT_post_depth_coverage`
- `VK_EXT_present_mode_fifo_latest_ready`
- `VK_EXT_primitive_topology_list_restart`
- `VK_EXT_primitives_generated_query`
- `VK_EXT_provoking_vertex`
- `VK_EXT_queue_family_foreign`
- `VK_EXT_robustness2`
- `VK_EXT_sample_locations`
- `VK_EXT_shader_atomic_float`

- [VK_EXT_shader_atomic_float2](#)
- [VK_EXT_shader_image_atomic_int64](#)
- [VK_EXT_shader_object](#)
- [VK_EXT_shader_replicated_composites](#)
- [VK_EXT_shader_stencil_export](#)
- [VK_EXT_shader_tile_image](#)
- [VK_EXT_swapchain_colorspace](#)
- [VK_EXT_transform_feedback](#)
- [VK_EXT_vertex_input_dynamic_state](#)
- [VK_EXT_ycbcr_image_arrays](#)

VK_KHR_16bit_storage

Name String

VK_KHR_16bit_storage

Extension Type

Device extension

Registered Extension Number

84

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)
and
[VK_KHR_storage_buffer_storage_class](#)
or
[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_16bit_storage](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

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Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for [GL_EXT_shader_16bit_storage](#)

Contributors

- Alexander Galazin, ARM
- Jan-Harald Fredriksen, ARM
- Joerg Wagner, ARM

- Neil Henning, Codeplay
- Jeff Bolz, Nvidia
- Daniel Koch, Nvidia
- David Neto, Google
- John Kessenich, Google

Description

The `VK_KHR_16bit_storage` extension allows use of 16-bit types in shader input and output interfaces, and push constant blocks. This extension introduces several new optional features which map to SPIR-V capabilities and allow access to 16-bit data in `Block`-decorated objects in the `Uniform` and the `StorageBuffer` storage classes, and objects in the `PushConstant` storage class. This extension allows 16-bit variables to be declared and used as user-defined shader inputs and outputs but does not change location assignment and component assignment rules.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. However, if Vulkan 1.1 is supported and this extension is not, the `storageBuffer16BitAccess` capability is optional. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDevice16BitStorageFeaturesKHR`

New Enum Constants

- `VK_KHR_16BIT_STORAGE_EXTENSION_NAME`
- `VK_KHR_16BIT_STORAGE_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_16BIT_STORAGE_FEATURES_KHR`

New SPIR-V Capabilities

- `StorageBuffer16BitAccess`
- `UniformAndStorageBuffer16BitAccess`
- `StoragePushConstant16`
- `StorageInputOutput16`

Version History

- Revision 1, 2017-03-23 (Alexander Galazin)
 - Initial draft

VK_KHR_8bit_storage

Name String

VK_KHR_8bit_storage

Extension Type

Device extension

Registered Extension Number

178

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)
and
[VK_KHR_storage_buffer_storage_class](#)
or
[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_8bit_storage](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Alexander Galazin [\[GitHub\]](#)[alegal-arm](#)

Other Extension Metadata

Last Modified Date

2018-02-05

Interactions and External Dependencies

- This extension provides API support for [GL_EXT_shader_16bit_storage](#)

IP Status

No known IP claims.

Contributors

- Alexander Galazin, Arm

Description

The [VK_KHR_8bit_storage](#) extension allows use of 8-bit types in uniform and storage buffers, and

push constant blocks. This extension introduces several new optional features which map to SPIR-V capabilities and allow access to 8-bit data in **Block**-decorated objects in the **Uniform** and the **StorageBuffer** storage classes, and objects in the **PushConstant** storage class.

The **StorageBuffer8BitAccess** capability **must** be supported by all implementations of this extension. The other capabilities are optional.

Promotion to Vulkan 1.2

Vulkan APIs in this extension are included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, the **StorageBuffer8BitAccess** capability is optional. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

New Structures

- Extending **VkPhysicalDeviceFeatures2**, **VkDeviceCreateInfo**:
 - **VkPhysicalDevice8BitStorageFeaturesKHR**

New Enum Constants

- **VK_KHR_8BIT_STORAGE_EXTENSION_NAME**
- **VK_KHR_8BIT_STORAGE_SPEC_VERSION**
- Extending **VkStructureType**:
 - **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_8BIT_STORAGE_FEATURES_KHR**

New SPIR-V Capabilities

- **StorageBuffer8BitAccess**
- **UniformAndStorageBuffer8BitAccess**
- **StoragePushConstant8**

Version History

- Revision 1, 2018-02-05 (Alexander Galazin)
 - Initial draft

VK_KHR_acceleration_structure

Name String

VK_KHR_acceleration_structure

Extension Type

Device extension

Registered Extension Number

151

Revision

13

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

and

[VK_EXT_descriptor_indexing](#)

and

[VK_KHR_buffer_device_address](#)

or

[Vulkan Version 1.2](#)

and

[VK_KHR_deferred_host_operations](#)

API Interactions

- Interacts with [VK_VERSION_1_2](#)
- Interacts with [VK_VERSION_1_3](#)
- Interacts with [VK_EXT_debug_report](#)
- Interacts with [VK_KHR_format_feature_flags2](#)

Contact

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Other Extension Metadata

Last Modified Date

2021-09-30

Contributors

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- Patrick Kelly, Epic Games
- Yuriy O'Donnell, Epic Games
- Michael Doggett, Facebook/Oculus
- Ricardo Garcia, Igalia
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- Don Scorgie, Imagination
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- Ralph Potter, Samsung
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- Jesse Barker, Unity
- Baldur Karlsson, Valve

Description

In order to be efficient, rendering techniques such as ray tracing need a quick way to identify which primitives may be intersected by a ray traversing the geometries. Acceleration structures are the most common way to represent the geometry spatially sorted, in order to quickly identify such potential intersections.

This extension adds new functionalities:

- Acceleration structure objects and build commands
- Structures to describe geometry inputs to acceleration structure builds
- Acceleration structure copy commands

New Object Types

- [VkAccelerationStructureKHR](#)

New Commands

- [vkBuildAccelerationStructuresKHR](#)
- [vkCmdBuildAccelerationStructuresIndirectKHR](#)
- [vkCmdBuildAccelerationStructuresKHR](#)
- [vkCmdCopyAccelerationStructureKHR](#)
- [vkCmdCopyAccelerationStructureToMemoryKHR](#)
- [vkCmdCopyMemoryToAccelerationStructureKHR](#)
- [vkCmdWriteAccelerationStructuresPropertiesKHR](#)
- [vkCopyAccelerationStructureKHR](#)
- [vkCopyAccelerationStructureToMemoryKHR](#)
- [vkCopyMemoryToAccelerationStructureKHR](#)
- [vkCreateAccelerationStructureKHR](#)
- [vkDestroyAccelerationStructureKHR](#)
- [vkGetAccelerationStructureBuildSizesKHR](#)
- [vkGetAccelerationStructureDeviceAddressKHR](#)
- [vkGetDeviceAccelerationStructureCompatibilityKHR](#)
- [vkWriteAccelerationStructuresPropertiesKHR](#)

New Structures

- [VkAabbPositionsKHR](#)
- [VkAccelerationStructureBuildGeometryInfoKHR](#)
- [VkAccelerationStructureBuildRangeInfoKHR](#)
- [VkAccelerationStructureBuildSizesInfoKHR](#)

- [VkAccelerationStructureCreateInfoKHR](#)
- [VkAccelerationStructureDeviceAddressInfoKHR](#)
- [VkAccelerationStructureGeometryAabbsDataKHR](#)
- [VkAccelerationStructureGeometryInstancesDataKHR](#)
- [VkAccelerationStructureGeometryKHR](#)
- [VkAccelerationStructureGeometryTrianglesDataKHR](#)
- [VkAccelerationStructureInstanceKHR](#)
- [VkAccelerationStructureVersionInfoKHR](#)
- [VkCopyAccelerationStructureInfoKHR](#)
- [VkCopyAccelerationStructureToMemoryInfoKHR](#)
- [VkCopyMemoryToAccelerationStructureInfoKHR](#)
- [VkTransformMatrixKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceAccelerationStructureFeaturesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceAccelerationStructurePropertiesKHR](#)
- Extending [VkWriteDescriptorSet](#):
 - [VkWriteDescriptorSetAccelerationStructureKHR](#)

New Unions

- [VkAccelerationStructureGeometryDataKHR](#)
- [VkDeviceOrHostAddressConstKHR](#)
- [VkDeviceOrHostAddressKHR](#)

New Enums

- [VkAccelerationStructureBuildTypeKHR](#)
- [VkAccelerationStructureCompatibilityKHR](#)
- [VkAccelerationStructureCreateFlagBitsKHR](#)
- [VkAccelerationStructureTypeKHR](#)
- [VkBuildAccelerationStructureFlagBitsKHR](#)
- [VkBuildAccelerationStructureModeKHR](#)
- [VkCopyAccelerationStructureModeKHR](#)
- [VkGeometryFlagBitsKHR](#)
- [VkGeometryInstanceFlagBitsKHR](#)
- [VkGeometryTypeKHR](#)

New Bitmasks

- [VkAccelerationStructureCreateFlagsKHR](#)
- [VkBuildAccelerationStructureFlagsKHR](#)
- [VkGeometryFlagsKHR](#)
- [VkGeometryInstanceFlagsKHR](#)

New Enum Constants

- [VK_KHR_ACCELERATION_STRUCTURE_EXTENSION_NAME](#)
- [VK_KHR_ACCELERATION_STRUCTURE_SPEC_VERSION](#)
- Extending [VkAccessFlagBits](#):
 - [VK_ACCESS_ACCELERATION_STRUCTURE_READ_BIT_KHR](#)
 - [VK_ACCESS_ACCELERATION_STRUCTURE_WRITE_BIT_KHR](#)
- Extending [VkBufferUsageFlagBits](#):
 - [VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR](#)
 - [VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR](#)
- Extending [VkCopyAccelerationStructureModeKHR](#):
 - [VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR](#)
 - [VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR](#)
- Extending [VkDescriptorType](#):
 - [VK_DESCRIPTOR_TYPE_ACCELERATION_STRUCTURE_KHR](#)
- Extending [VkFormatFeatureFlagBits](#):
 - [VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR](#)
- Extending [VkIndexType](#):
 - [VK_INDEX_TYPE_NONE_KHR](#)
- Extending [VkObjectType](#):
 - [VK_OBJECT_TYPE_ACCELERATION_STRUCTURE_KHR](#)
- Extending [VkPipelineStageFlagBits](#):
 - [VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR](#)
- Extending [VkQueryType](#):
 - [VK_QUERY_TYPE_ACCELERATION_STRUCTURE_COMPACTED_SIZE_KHR](#)
 - [VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_SIZE_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_GEOMETRY_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_SIZES_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_INFO_KHR](#)

- `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_DEVICE_ADDRESS_INFO_KHR`
- `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_AABBS_DATA_KHR`
- `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_INSTANCES_DATA_KHR`
- `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_KHR`
- `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_GEOMETRY_TRIANGLES_DATA_KHR`
- `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_VERSION_INFO_KHR`
- `VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_INFO_KHR`
- `VK_STRUCTURE_TYPE_COPY_ACCELERATION_STRUCTURE_TO_MEMORY_INFO_KHR`
- `VK_STRUCTURE_TYPE_COPY_MEMORY_TO_ACCELERATION_STRUCTURE_INFO_KHR`
- `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_FEATURES_KHR`
- `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ACCELERATION_STRUCTURE_PROPERTIES_KHR`
- `VK_STRUCTURE_TYPE_WRITE_DESCRIPTOR_SET_ACCELERATION_STRUCTURE_KHR`

If `VK_EXT_debug_report` is supported:

- Extending `VkDebugReportObjectTypeEXT`:
 - `VK_DEBUG_REPORT_OBJECT_TYPE_ACCELERATION_STRUCTURE_KHR_EXT`

If `VK_KHR_format_feature_flags2` or Vulkan Version 1.3 is supported:

- Extending `VkFormatFeatureFlagBits2`:
 - `VK_FORMAT_FEATURE_2_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR`

Issues

(1) How does this extension differ from `VK_NV_ray_tracing`?

DISCUSSION:

The following is a summary of the main functional differences between `VK_KHR_acceleration_structure` and `VK_NV_ray_tracing`:

- added acceleration structure serialization / deserialization (`VK_COPY_ACCELERATION_STRUCTURE_MODE_SERIALIZE_KHR`, `VK_COPY_ACCELERATION_STRUCTURE_MODE_DESERIALIZE_KHR`, `vkCmdCopyAccelerationStructureToMemoryKHR`, `vkCmdCopyMemoryToAccelerationStructureKHR`)
- document inactive primitives and instances
- added `VkPhysicalDeviceAccelerationStructureFeaturesKHR` structure
- added indirect and batched acceleration structure builds (`vkCmdBuildAccelerationStructuresIndirectKHR`)
- added host acceleration structure commands
- reworked geometry structures so they could be better shared between device, host, and indirect

builds

- explicitly made [VkAccelerationStructureKHR](#) use device addresses
- added acceleration structure compatibility check function ([vkGetDeviceAccelerationStructureCompatibilityKHR](#))
- add parameter for requesting memory requirements for host and/or device build
- added format feature for acceleration structure build vertex formats ([VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR](#))

(3) What are the changes between the public provisional (VK_KHR_ray_tracing v8) release and the internal provisional (VK_KHR_ray_tracing v9) release?

- added [geometryFlags](#) to [VkAccelerationStructureCreateGeometryTypeInfoKHR](#) (later reworked to obsolete this)
- added [minAccelerationStructureScratchOffsetAlignment](#) property to [VkPhysicalDeviceRayTracingPropertiesKHR](#)
- fix naming and return enum from [vkGetDeviceAccelerationStructureCompatibilityKHR](#)
 - renamed [VkAccelerationStructureVersionKHR](#) to [VkAccelerationStructureVersionInfoKHR](#)
 - renamed [VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_VERSION_KHR](#) to [VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_VERSION_INFO_KHR](#)
 - removed [VK_ERROR_INCOMPATIBLE_VERSION_KHR](#)
 - added [VkAccelerationStructureCompatibilityKHR](#) enum
 - remove return value from [vkGetDeviceAccelerationStructureCompatibilityKHR](#) and added return enum parameter
- Require Vulkan 1.1
- added creation time capture and replay flags
 - added [VkAccelerationStructureCreateFlagBitsKHR](#) and [VkAccelerationStructureCreateFlagsKHR](#)
 - renamed the [flags](#) member of [VkAccelerationStructureCreateInfoKHR](#) to [buildFlags](#) (later removed) and added the [createFlags](#) member
- change [vkCmdBuildAccelerationStructuresIndirectKHR](#) to use buffer device address for indirect parameter
- make [VK_KHR_deferred_host_operations](#) an interaction instead of a required extension (later went back on this)
- renamed [VkAccelerationStructureBuildOffsetInfoKHR](#) to [VkAccelerationStructureBuildRangeInfoKHR](#)
 - renamed the [ppOffsetInfos](#) parameter of [vkCmdBuildAccelerationStructuresKHR](#) to [ppBuildRangeInfos](#)
- Re-unify geometry description between build and create
 - remove [VkAccelerationStructureCreateGeometryTypeInfoKHR](#) and [VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_GEOMETRY_TYPE_INFO_KHR](#)

- added `VkAccelerationStructureCreateInfoKHR` structure (later removed)
- change type of the `pGeometryInfos` member of `VkAccelerationStructureCreateInfoKHR` from `VkAccelerationStructureCreateGeometryTypeInfoKHR` to `VkAccelerationStructureGeometryKHR` (later removed)
- added `pCreateInfoInfos` member to `VkAccelerationStructureCreateInfoKHR` (later removed)
- Fix ppGeometries ambiguity, add pGeometries
 - remove `geometryArrayOfPointers` member of `VkAccelerationStructureBuildGeometryInfoKHR`
 - disambiguate two meanings of `ppGeometries` by explicitly adding `pGeometries` to the `VkAccelerationStructureBuildGeometryInfoKHR` structure and require one of them be `NULL`
- added `nullDescriptor` support for acceleration structures
- changed the `update` member of `VkAccelerationStructureBuildGeometryInfoKHR` from a bool to the `mode` `VkBuildAccelerationStructureModeKHR` enum which allows future extensibility in update types
- Clarify deferred host ops for pipeline creation
 - `VkDeferredOperationKHR` is now a top-level parameter for `vkBuildAccelerationStructuresKHR`, `vkCreateRayTracingPipelinesKHR`, `vkCopyAccelerationStructureToMemoryKHR`, `vkCopyAccelerationStructureKHR`, and `vkCopyMemoryToAccelerationStructureKHR`
 - removed `VkDeferredOperationInfoKHR` structure
 - change deferred host creation/return parameter behavior such that the implementation can modify such parameters until the deferred host operation completes
 - `VK_KHR_deferred_host_operations` is required again
- Change acceleration structure build to always be sized
 - de-alias `VkAccelerationStructureMemoryRequirementsTypeNV` and `VkAccelerationStructureMemoryRequirementsTypeKHR`, and remove `VkAccelerationStructureMemoryRequirementsTypeKHR`
 - add `vkGetAccelerationStructureBuildSizesKHR` command and `VkAccelerationStructureBuildSizesInfoKHR` structure and `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_BUILD_SIZES_INFO_KHR` enum to query sizes for acceleration structures and scratch storage
 - move size queries for scratch space to `vkGetAccelerationStructureBuildSizesKHR`
 - remove `compactedSize`, `buildFlags`, `maxGeometryCount`, `pGeometryInfos`, `pCreateInfoInfos` members of `VkAccelerationStructureCreateInfoKHR` and add the `size` member
 - add `maxVertex` member to `VkAccelerationStructureGeometryTrianglesDataKHR` structure
 - remove `VkAccelerationStructureCreateInfoKHR` structure

(4) What are the changes between the internal provisional (VK_KHR_ray_tracing v9) release and the final (VK_KHR_acceleration_structure v11) release?

- refactor `VK_KHR_ray_tracing` into 3 extensions, enabling implementation flexibility and

decoupling ray query support from ray pipelines:

- `VK_KHR_acceleration_structure` (for acceleration structure operations)
- `VK_KHR_ray_tracing_pipeline` (for ray tracing pipeline and shader stages)
- `VK_KHR_ray_query` (for ray queries in existing shader stages)
- clarify buffer usage flags for ray tracing
 - `VK_BUFFER_USAGE_RAY_TRACING_BIT_NV` is left alone in `VK_NV_ray_tracing` (required on `scratch` and `instanceData`)
 - `VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR` is added as an alias of `VK_BUFFER_USAGE_RAY_TRACING_BIT_NV` in `VK_KHR_ray_tracing_pipeline` and is required on shader binding table buffers
 - `VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR` is added in `VK_KHR_acceleration_structure` for all vertex, index, transform, aabb, and instance buffer data referenced by device build commands
 - `VK_BUFFER_USAGE_STORAGE_BUFFER_BIT` is used for `scratchData`
- add `max primitive counts` (`ppMaxPrimitiveCounts`) to `vkCmdBuildAccelerationStructuresIndirectKHR`
- Allocate acceleration structures from `VkBuffers` and add a mode to constrain the device address
 - de-alias `VkBindAccelerationStructureMemoryInfoNV` and `vkBindAccelerationStructureMemoryNV`, and remove `VkBindAccelerationStructureMemoryInfoKHR`, `VkAccelerationStructureMemoryRequirementsInfoKHR`, and `vkGetAccelerationStructureMemoryRequirementsKHR`
 - acceleration structures now take a `VkBuffer` and offset at creation time for memory placement
 - add a new `VK_BUFFER_USAGE_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR` buffer usage for such buffers
 - add a new `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR` acceleration structure type for layering
- move `VK_GEOMETRY_TYPE_INSTANCES_KHR` to main enum instead of being added via extension
- make build commands more consistent - all now build multiple acceleration structures and are named `plurally` (`vkCmdBuildAccelerationStructuresIndirectKHR`, `vkCmdBuildAccelerationStructuresKHR`, `vkBuildAccelerationStructuresKHR`)
- add interactions with `VK_DESCRIPTOR_SET_LAYOUT_CREATE_UPDATE_AFTER_BIND_POOL_BIT` for acceleration structures, including a new feature (`descriptorBindingAccelerationStructureUpdateAfterBind`) and 3 new properties (`maxPerStageDescriptorAccelerationStructures`, `maxPerStageDescriptorUpdateAfterBindAccelerationStructures`, `maxDescriptorSetUpdateAfterBindAccelerationStructures`)
- extension is no longer provisional
- define synchronization requirements for builds, traces, and copies
- define synchronization requirements for AS build inputs and indirect build buffer

(5) What is `VK_ACCELERATION_STRUCTURE_TYPE_GENERIC_KHR` for?

RESOLVED: It is primarily intended for API layering. In DXR, the acceleration structure is basically just a buffer in a special layout, and you do not know at creation time whether it will be used as a top or bottom level acceleration structure. We thus added a generic acceleration structure type whose type is unknown at creation time, but is specified at build time instead. Applications which are written directly for Vulkan should not use it.

Version History

- Revision 1, 2019-12-05 (Members of the Vulkan Ray Tracing TSG)
 - Internal revisions (forked from `VK_NV_ray_tracing`)
- Revision 2, 2019-12-20 (Daniel Koch, Eric Werness)
 - Add const version of `DeviceOrHostAddress` (!3515)
 - Add VU to clarify that only handles in the current pipeline are valid (!3518)
 - Restore some missing VUs and add in-place update language (#1902, !3522)
 - rename `VkAccelerationStructureInstanceKHR` member from `accelerationStructure` to `accelerationStructureReference` to better match its type (!3523)
 - Allow `VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS` for pipeline creation if shader group handles cannot be reused (!3523)
 - update documentation for the `VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS` error code and add missing documentation for new return codes from `VK_KHR_deferred_host_operations` (!3523)
 - list new query types for `VK_KHR_ray_tracing` (!3523)
 - Fix VU statements for `VkAccelerationStructureGeometryKHR` referring to correct union members and update to use more current wording (!3523)
- Revision 3, 2020-01-10 (Daniel Koch, Jon Leech, Christoph Kubisch)
 - Fix 'instance of' and 'that/which contains/defines' markup issues (!3528)
 - factor out `VK_KHR_pipeline_library` as stand-alone extension (!3540)
 - Resolve Vulkan-hpp issues (!3543)
 - add missing require for `VkGeometryInstanceFlagsKHR`
 - de-alias `VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_CREATE_INFO_NV` since the KHR structure is no longer equivalent
 - add `len` to `pDataSize` attribute for `vkWriteAccelerationStructuresPropertiesKHR`
- Revision 4, 2020-01-23 (Daniel Koch, Eric Werness)
 - Improve `vkWriteAccelerationStructuresPropertiesKHR`, add return value and VUs (#1947)
 - Clarify language to allow multiple raygen shaders (#1959)
 - Various editorial feedback (!3556)
 - Add language to help deal with looped self-intersecting fans (#1901)

- Change vkCmdTraceRays{,Indirect}KHR args to pointers (!3559)
- Add scratch address validation language (#1941, !3551)
- Fix definition and add hierarchy information for shader call scope (#1977, !3571)
- Revision 5, 2020-02-04 (Eric Werness, Jeff Bolz, Daniel Koch)
 - remove vestigial accelerationStructureUUID (!3582)
 - update definition of repack instructions and improve memory model interactions (#1910, #1913, !3584)
 - Fix wrong sType for VkPhysicalDeviceRayTracingFeaturesKHR (#1988)
 - Use provisional SPIR-V capabilities (#1987)
 - require rayTraversalPrimitiveCulling if rayQuery is supported (#1927)
 - Miss shaders do not have object parameters (!3592)
 - Fix missing required types in XML (!3592)
 - clarify matching conditions for update (!3592)
 - add goal that host and device builds be similar (!3592)
 - clarify that `maxPrimitiveCount` limit should apply to triangles and AABBs (!3592)
 - Require alignment for instance arrayOfPointers (!3592)
 - Zero is a valid value for instance flags (!3592)
 - Add some alignment VUs that got lost in refactoring (!3592)
 - Recommend TMin epsilon rather than culling (!3592)
 - Get angle from dot product not cross product (!3592)
 - Clarify that AH can access the payload and attributes (!3592)
 - Match DXR behavior for inactive primitive definition (!3592)
 - Use a more generic term than degenerate for inactive to avoid confusion (!3592)
- Revision 6, 2020-02-20 (Daniel Koch)
 - fix some dangling NV references (#1996)
 - rename `VkCmdTraceRaysIndirectCommandKHR` to `VkTraceRaysIndirectCommandKHR` (!3607)
 - update contributor list (!3611)
 - use `uint64_t` instead of `VkAccelerationStructureReferenceKHR` in `VkAccelerationStructureInstanceKHR` (#2004)
- Revision 7, 2020-02-28 (Tobias Hector)
 - remove HitTKHR SPIR-V builtin (spirv/spirv-extensions#7)
- Revision 8, 2020-03-06 (Tobias Hector, Dae Kim, Daniel Koch, Jeff Bolz, Eric Werness)
 - explicitly state that Tmax is updated when new closest intersection is accepted (#2020,!3536)
 - Made references to min and max t values consistent (!3644)

- finish enumerating differences relative to VK_NV_ray_tracing in issues (1) and (2) (#1974,!3642)
- fix formatting in some math equations (!3642)
- Restrict the Hit Kind operand of `OpReportIntersectionKHR` to 7-bits (spirv/spirv-extensions#8,!3646)
- Say ray tracing '**should**' be watertight (#2008,!3631)
- Clarify memory requirements for ray tracing buffers (#2005,!3649)
- Add callable size limits (#1997,!3652)
- Revision 9, 2020-04-15 (Eric Werness, Daniel Koch, Tobias Hector, Joshua Barczak)
 - Add geometry flags to acceleration structure creation (!3672)
 - add build scratch memory alignment (`minAccelerationStructureScratchOffsetAlignment`) (#2065,!3725)
 - fix naming and return enum from `vkGetDeviceAccelerationStructureCompatibilityKHR` (#2051,!3726)
 - require SPIR-V 1.4 (#2096,!3777)
 - added creation time capture/replay flags (#2104,!3774)
 - require Vulkan 1.1 (#2133,!3806)
 - use device addresses instead of `VkBuffers` for ray tracing commands (#2074,!3815)
 - add interactions with Vulkan 1.2 and `VK_KHR_vulkan_memory_model` (#2133,!3830)
 - make `VK_KHR_pipeline_library` an interaction instead of required (#2045,#2108,!3830)
 - make `VK_KHR_deferred_host_operations` an interaction instead of required (#2045,!3830)
 - removed `maxCallableSize` and added explicit stack size management for ray pipelines (#1997,!3817,!3772,!3844)
 - improved documentation for `VkAccelerationStructureVersionInfoKHR` (#2135,3835)
 - rename `VkAccelerationStructureBuildOffsetInfoKHR` to `VkAccelerationStructureBuildRangeInfoKHR` (#2058,!3754)
 - Re-unify geometry description between build and create (!3754)
 - Fix ppGeometries ambiguity, add pGeometries (#2032,!3811)
 - add interactions with `VK_EXT_robustness2` and allow `nullDescriptor` support for acceleration structures (#1920,!3848)
 - added future extensibility for AS updates (#2114,!3849)
 - Fix VU for dispatchrays and add a limit on the size of the full grid (#2160,!3851)
 - Add `shaderGroupHandleAlignment` property (#2180,!3875)
 - Clarify deferred host ops for pipeline creation (#2067,!3813)
 - Change acceleration structure build to always be sized (#2131,#2197,#2198,!3854,!3883,!3880)
- Revision 10, 2020-07-03 (Mathieu Robart, Daniel Koch, Eric Werness, Tobias Hector)

- Decomposition of the specification, from VK_KHR_ray_tracing to VK_KHR_acceleration_structure (#1918,!3912)
- clarify buffer usage flags for ray tracing (#2181,!3939)
- add max primitive counts to build indirect command (#2233,!3944)
- Allocate acceleration structures from VkBuffers and add a mode to constrain the device address (#2131,!3936)
- Move VK_GEOMETRY_TYPE_INSTANCES_KHR to main enum (#2243,!3952)
- make build commands more consistent (#2247,!3958)
- add interactions with UPDATE_AFTER_BIND (#2128,!3986)
- correct and expand build command VUs (!4020)
- fix copy command VUs (!4018)
- added various alignment requirements (#2229,!3943)
- fix valid usage for arrays of geometryCount items (#2198,!4010)
- define what is allowed to change on RTAS updates and relevant VUs (#2177,!3961)
- Revision 11, 2020-11-12 (Eric Werness, Josh Barczak, Daniel Koch, Tobias Hector)
 - de-alias NV and KHR acceleration structure types and associated commands (#2271,!4035)
 - specify alignment for host copy commands (#2273,!4037)
 - document VK_FORMAT_FEATURE_ACCELERATION_STRUCTURE_VERTEX_BUFFER_BIT_KHR
 - specify that acceleration structures are non-linear (#2289,!4068)
 - add several missing VUs for strides, vertexFormat, and indexType (#2315,!4069)
 - restore VUs for VkAccelerationStructureBuildGeometryInfoKHR (#2337,!4098)
 - ban multi-instance memory for host operations (#2324,!4102)
 - allow dstAccelerationStructure to be null for vkGetAccelerationStructureBuildSizesKHR (#2330,!4111)
 - more build VU cleanup (#2138,#4130)
 - specify host endianness for AS serialization (#2261,!4136)
 - add invertible transform matrix VU (#1710,!4140)
 - require geometryCount to be 1 for TLAS builds (!4145)
 - improved validity conditions for build addresses (#4142)
 - add single statement SPIR-V VUs, build limit VUs (!4158)
 - document limits for vertex and aabb strides (#2390,!4184)
 - specify that VK_PIPELINE_STAGE_ACCELERATION_STRUCTURE_BUILD_BIT_KHR applies to AS copies (#2382,#4173)
 - define sync for AS build inputs and indirect buffer (#2407,!4208)
- Revision 12, 2021-08-06 (Samuel Bourasseau)
 - rename VK_GEOMETRY_INSTANCE_TRIANGLE_FRONT_COUNTERCLOCKWISE_BIT_KHR to

VK_GEOMETRY_INSTANCE_TRIANGLE_FLIP_FACING_BIT_KHR (keep previous as alias).

- Clarify description and add note.
- Revision 13, 2021-09-30 (Jon Leech)
 - Add interaction with [VK_KHR_format_feature_flags2](#) to [vk.xml](#)

VK_KHR_android_surface

Name String

VK_KHR_android_surface

Extension Type

Instance extension

Registered Extension Number

9

Revision

6

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_surface](#)

Contact

- Jesse Hall [\[GitHub\]](#)[critsec](#)

Other Extension Metadata

Last Modified Date

2016-01-14

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Faith Ekstrand, Intel
- Ian Elliott, LunarG
- Courtney Goeltzenleuchter, LunarG
- Jesse Hall, Google
- James Jones, NVIDIA
- Antoine Labour, Google

- Jon Leech, Khronos
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung
- Daniel Rakos, AMD
- Graham Sellers, AMD
- Ray Smith, ARM
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG

Description

The `VK_KHR_android_surface` extension is an instance extension. It provides a mechanism to create a `VkSurfaceKHR` object (defined by the `VK_KHR_surface` extension) that refers to an `ANativeWindow`, Android's native surface type. The `ANativeWindow` represents the producer endpoint of any buffer queue, regardless of consumer endpoint. Common consumer endpoints for `ANativeWindows` are the system window compositor, video encoders, and application-specific compositors importing the images through a `SurfaceTexture`.

New Base Types

- `ANativeWindow`

New Commands

- `vkCreateAndroidSurfaceKHR`

New Structures

- `VkAndroidSurfaceCreateInfoKHR`

New Bitmasks

- `VkAndroidSurfaceCreateFlagsKHR`

New Enum Constants

- `VK_KHR_ANDROID_SURFACE_EXTENSION_NAME`
- `VK_KHR_ANDROID_SURFACE_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_ANDROID_SURFACE_CREATE_INFO_KHR`

Issues

1) Does Android need a way to query for compatibility between a particular physical device (and queue family?) and a specific Android display?

RESOLVED: No. Currently on Android, any physical device is expected to be able to present to the system compositor, and all queue families must support the necessary image layout transitions and synchronization operations.

Version History

- Revision 1, 2015-09-23 (Jesse Hall)
 - Initial draft.
- Revision 2, 2015-10-26 (Ian Elliott)
 - Renamed from VK_EXT_KHR_android_surface to VK_KHR_android_surface.
- Revision 3, 2015-11-03 (Daniel Rakos)
 - Added allocation callbacks to surface creation function.
- Revision 4, 2015-11-10 (Jesse Hall)
 - Removed VK_ERROR_INVALID_ANDROID_WINDOW_KHR.
- Revision 5, 2015-11-28 (Daniel Rakos)
 - Updated the surface create function to take a pCreateInfo structure.
- Revision 6, 2016-01-14 (James Jones)
 - Moved VK_ERROR_NATIVE_WINDOW_IN_USE_KHR from the VK_KHR_android_surface to the VK_KHR_surface extension.

VK_KHR_bind_memory2

Name String

VK_KHR_bind_memory2

Extension Type

Device extension

Registered Extension Number

158

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Tobias Hector [\[GitHub\]tobski](#)

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Tobias Hector, Imagination Technologies

Description

This extension provides versions of [vkBindBufferMemory](#) and [vkBindImageMemory](#) that allow multiple bindings to be performed at once, and are extensible.

This extension also introduces [VK_IMAGE_CREATE_ALIAS_BIT_KHR](#), which allows “identical” images that alias the same memory to interpret the contents consistently, even across image layout changes.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- [vkBindBufferMemory2KHR](#)
- [vkBindImageMemory2KHR](#)

New Structures

- [VkBindBufferMemoryInfoKHR](#)
- [VkBindImageMemoryInfoKHR](#)

New Enum Constants

- [VK_KHR_BIND_MEMORY_2_EXTENSION_NAME](#)
- [VK_KHR_BIND_MEMORY_2_SPEC_VERSION](#)
- Extending [VkImageCreateFlagBits](#):
 - [VK_IMAGE_CREATE_ALIAS_BIT_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_INFO_KHR](#)

- [VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_INFO_KHR](#)

Version History

- Revision 1, 2017-05-19 (Tobias Hector)
 - Pulled bind memory functions into their own extension

VK_KHR_buffer_device_address

Name String

[VK_KHR_buffer_device_address](#)

Extension Type

Device extension

Registered Extension Number

258

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)
and

[VK_KHR_device_group](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_physical_storage_buffer](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Jeff Bolz [\[GitHub\]jeffbolznv](#)

Other Extension Metadata

Last Modified Date

2019-06-24

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_EXT_buffer_reference` and `GL_EXT_buffer_reference2` and `GL_EXT_buffer_reference_uvec2`

Contributors

- Jeff Bolz, NVIDIA
- Neil Henning, AMD
- Tobias Hector, AMD
- Faith Ekstrand, Intel
- Baldur Karlsson, Valve
- Jan-Harald Fredriksen, Arm

Description

This extension allows the application to query a 64-bit buffer device address value for a buffer, which can be used to access the buffer memory via the `PhysicalStorageBuffer` storage class in the `GL_EXT_buffer_reference` GLSL extension and `SPV_KHR_physical_storage_buffer` SPIR-V extension.

Another way to describe this extension is that it adds “pointers to buffer memory in shaders”. By calling `vkGetBufferDeviceAddressKHR` with a `VkBuffer`, it will return a `VkDeviceAddress` value which represents the address of the start of the buffer.

`vkGetBufferOpaqueCaptureAddressKHR` and `vkGetDeviceMemoryOpaqueCaptureAddressKHR` allow opaque addresses for buffers and memory objects to be queried for the current process. A trace capture and replay tool can then supply these addresses to be used at replay time to match the addresses used when the trace was captured. To enable tools to insert these queries, new memory allocation flags must be specified for memory objects that will be bound to buffers accessed via the `PhysicalStorageBuffer` storage class. **Note that this mechanism is intended only to support capture/replay tools, and is not recommended for use in other applications.**

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, the `bufferDeviceAddress` feature is optional. The original type, enum, and command names are still available as aliases of the core functionality.

Promotion to Vulkan 1.3

Support for the `bufferDeviceAddress` feature is mandatory in Vulkan 1.3, regardless of whether this extension is supported.

New Commands

- `vkGetBufferDeviceAddressKHR`
- `vkGetBufferOpaqueCaptureAddressKHR`
- `vkGetDeviceMemoryOpaqueCaptureAddressKHR`

New Structures

- [VkBufferDeviceAddressInfoKHR](#)
- [VkDeviceMemoryOpaqueCaptureAddressInfoKHR](#)
- Extending [VkBufferCreateInfo](#):
 - [VkBufferOpaqueCaptureAddressCreateInfoKHR](#)
- Extending [VkMemoryAllocateInfo](#):
 - [VkMemoryOpaqueCaptureAddressAllocateInfoKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceBufferDeviceAddressFeaturesKHR](#)

New Enum Constants

- [VK_KHR_BUFFER_DEVICE_ADDRESS_EXTENSION_NAME](#)
- [VK_KHR_BUFFER_DEVICE_ADDRESS_SPEC_VERSION](#)
- Extending [VkBufferCreateFlagBits](#):
 - [VK_BUFFER_CREATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR](#)
- Extending [VkBufferUsageFlagBits](#):
 - [VK_BUFFER_USAGE_SHADER_DEVICE_ADDRESS_BIT_KHR](#)
- Extending [VkMemoryAllocateFlagBits](#):
 - [VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_BIT_KHR](#)
 - [VK_MEMORY_ALLOCATE_DEVICE_ADDRESS_CAPTURE_REPLAY_BIT_KHR](#)
- Extending [VkResult](#):
 - [VK_ERROR_INVALID_OPAQUE_CAPTURE_ADDRESS_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_BUFFER_DEVICE_ADDRESS_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_BUFFER_OPAQUE_CAPTURE_ADDRESS_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_DEVICE_MEMORY_OPAQUE_CAPTURE_ADDRESS_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_MEMORY_OPAQUE_CAPTURE_ADDRESS_ALLOCATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BUFFER_DEVICE_ADDRESS_FEATURES_KHR](#)

New SPIR-V Capabilities

- [PhysicalStorageBufferAddresses](#)

Version History

- Revision 1, 2019-06-24 (Jan-Harald Fredriksen)
 - Internal revisions based on [VK_EXT_buffer_device_address](#)

VK_KHR_calibrated_timestamps

Name String

VK_KHR_calibrated_timestamps

Extension Type

Device extension

Registered Extension Number

544

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Daniel Rakos [\[GitHub\]](#)[aqnuep](#)

Extension Proposal

[VK_EXT_calibrated_timestamps](#)

Other Extension Metadata

Last Modified Date

2023-07-12

IP Status

No known IP claims.

Contributors

- Matthaeus G. Chajdas, AMD
- Alan Harrison, AMD
- Derrick Owens, AMD
- Daniel Rakos, RasterGrid
- Faith Ekstrand, Intel
- Keith Packard, Valve

Description

This extension provides an interface to query calibrated timestamps obtained quasi simultaneously

from two time domains.

New Commands

- [vkGetCalibratedTimestampsKHR](#)
- [vkGetPhysicalDeviceCalibrateableTimeDomainsKHR](#)

New Structures

- [VkCalibratedTimestampInfoKHR](#)

New Enums

- [VkTimeDomainKHR](#)

New Enum Constants

- [VK_KHR_CALIBRATED_TIMESTAMPS_EXTENSION_NAME](#)
- [VK_KHR_CALIBRATED_TIMESTAMPS_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_CALIBRATED_TIMESTAMP_INFO_KHR](#)

Version History

- Revision 1, 2023-07-12 (Daniel Rakos)
 - Initial draft.

VK_KHR_compute_shader_derivatives

Name String

[VK_KHR_compute_shader_derivatives](#)

Extension Type

Device extension

Registered Extension Number

512

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

SPIR-V Dependencies

- [SPV_KHR_compute_shader_derivatives](#)

Contact

- Jean-Noe Morissette [\[GitHub\]](#)[MagicPoncho](#)

Extension Proposal

[VK_KHR_compute_shader_derivatives](#)

Other Extension Metadata

Last Modified Date

2024-06-26

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension requires [SPV_KHR_compute_shader_derivatives](#)
- This extension provides API support for [GL_KHR_compute_shader_derivatives](#)

Contributors

- Jean-Noe Morissette, Epic Games
- Daniel Koch, NVIDIA
- Pat Brown, NVIDIA
- Stu Smith, AMD
- Jan-Harald Fredriksen, Arm
- Tobias Hector, AMD
- Ralph Potter, Samsung
- Pan Gao, Huawei
- Samuel (Sheng-Wen) Huang, MediaTek
- Graeme Leese, Broadcom
- Hans-Kristian Arntzen, Valve
- Matthew Netsh, Qualcomm

Description

This extension adds Vulkan support for the [SPV_KHR_compute_shader_derivatives](#) SPIR-V extension.

The SPIR-V extension provides two new execution modes, both of which allow execution models with defined workgroups to use built-ins that evaluate derivatives explicitly or implicitly. Derivatives will be computed via differencing over a 2x2 group of shader invocations. The [DerivativeGroupQuadsKHR](#) execution mode assembles shader invocations into 2x2 groups, where each group has x and y coordinates of the local invocation ID of the form (2m+{0,1}, 2n+{0,1}). The [DerivativeGroupLinearKHR](#) execution mode assembles shader invocations into 2x2 groups, where

each group has local invocation index values of the form $4m + \{0,1,2,3\}$.

The new execution modes are supported in compute shaders and optionally (see [meshAndTaskShaderDerivatives](#)) in mesh and task shaders.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceComputeShaderDerivativesFeaturesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceComputeShaderDerivativesPropertiesKHR](#)

New Enum Constants

- [VK_KHR_COMPUTE_SHADER_DERIVATIVES_EXTENSION_NAME](#)
- [VK_KHR_COMPUTE_SHADER_DERIVATIVES_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COMPUTE_SHADER_DERIVATIVES_PROPERTIES_KHR](#)

New SPIR-V Capability

- [ComputeDerivativeGroupQuadsKHR](#)
- [ComputeDerivativeGroupLinearKHR](#)

Examples

None.

Version History

- Revision 1, 2023-02-27 (Jean-Noe Morisette)
 - Initial draft
 - Add properties and clarify mesh and task support (Daniel Koch)

VK_KHR_cooperative_matrix

Name String

[VK_KHR_cooperative_matrix](#)

Extension Type

Device extension

Registered Extension Number

507

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_cooperative_matrix](#)

Contact

- Kevin Petit [\[GitHub\]](#)kpet

Extension Proposal

[VK_KHR_cooperative_matrix](#)

Other Extension Metadata

Last Modified Date

2023-05-03

Interactions and External Dependencies

- This extension provides API support for [GLSL_KHR_cooperative_matrix](#)

Contributors

- Jeff Bolz, NVIDIA
- Markus Tavenrath, NVIDIA
- Daniel Koch, NVIDIA
- Kevin Petit, Arm Ltd.
- Boris Zanin, AMD

Description

This extension adds support for using cooperative matrix types in SPIR-V. Cooperative matrix types are medium-sized matrices that are primarily supported in compute shaders, where the storage for the matrix is spread across all invocations in some scope (usually a subgroup) and those invocations cooperate to efficiently perform matrix multiplies.

Cooperative matrix types are defined by the [SPV_KHR_cooperative_matrix](#) SPIR-V extension and can be used with the [GLSL_KHR_cooperative_matrix](#) GLSL extension.

This extension includes support for enumerating the matrix types and dimensions that are supported by the implementation.

New Commands

- [vkGetPhysicalDeviceCooperativeMatrixPropertiesKHR](#)

New Structures

- [VkCooperativeMatrixPropertiesKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceCooperativeMatrixFeaturesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceCooperativeMatrixPropertiesKHR](#)

New Enums

- [VkComponentTypeKHR](#)
- [VkScopeKHR](#)

New Enum Constants

- [VK_KHR_COOPERATIVE_MATRIX_EXTENSION_NAME](#)
- [VK_KHR_COOPERATIVE_MATRIX_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_COOPERATIVE_MATRIX_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COOPERATIVE_MATRIX_PROPERTIES_KHR](#)

New SPIR-V Capabilities

- [CooperativeMatrixKHR](#)

Issues

Version History

- Revision 2, 2023-05-03 (Kevin Petit)
 - First KHR revision
- Revision 1, 2019-02-05 (Jeff Bolz)
 - NVIDIA vendor extension

VK_KHR_copy_commands2

Name String

[VK_KHR_copy_commands2](#)

Extension Type

Device extension

Registered Extension Number

338

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Matthew Netsch [\[GitHub\]](#)[mnetsch](#)

Other Extension Metadata

Last Modified Date

2020-07-06

Contributors

- Jeff Leger, Qualcomm
- Tobias Hector, AMD
- Jan-Harald Fredriksen, ARM
- Tom Olson, ARM

Description

This extension provides extensible versions of the Vulkan buffer and image copy commands. The new commands are functionally identical to the core commands, except that their copy parameters are specified using extensible structures that can be used to pass extension-specific information.

The following extensible copy commands are introduced with this extension: [vkCmdCopyBuffer2KHR](#), [vkCmdCopyImage2KHR](#), [vkCmdCopyBufferToImage2KHR](#), [vkCmdCopyImageToBuffer2KHR](#), [vkCmdBlitImage2KHR](#), and [vkCmdResolveImage2KHR](#). Each command contains an **Info2KHR* structure parameter that includes *sType/pNext* members. Lower level structures describing each region to be copied are also extended with *sType/pNext* members.

New Commands

- [vkCmdBlitImage2KHR](#)
- [vkCmdCopyBuffer2KHR](#)
- [vkCmdCopyBufferToImage2KHR](#)
- [vkCmdCopyImage2KHR](#)
- [vkCmdCopyImageToBuffer2KHR](#)
- [vkCmdResolveImage2KHR](#)

New Structures

- [VkBlitImageInfo2KHR](#)
- [VkBufferCopy2KHR](#)
- [VkBufferImageCopy2KHR](#)
- [VkCopyBufferInfo2KHR](#)
- [VkCopyBufferToImageInfo2KHR](#)
- [VkCopyImageInfo2KHR](#)
- [VkCopyImageToBufferInfo2KHR](#)
- [VkImageBlit2KHR](#)
- [VkImageCopy2KHR](#)
- [VkImageResolve2KHR](#)
- [VkResolveImageInfo2KHR](#)

New Enum Constants

- [VK_KHR_COPY_COMMANDS_2_EXTENSION_NAME](#)
- [VK_KHR_COPY_COMMANDS_2_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_BLIT_IMAGE_INFO_2_KHR](#)
 - [VK_STRUCTURE_TYPE_BUFFER_COPY_2_KHR](#)
 - [VK_STRUCTURE_TYPE_BUFFER_IMAGE_COPY_2_KHR](#)
 - [VK_STRUCTURE_TYPE_COPY_BUFFER_INFO_2_KHR](#)
 - [VK_STRUCTURE_TYPE_COPY_BUFFER_TO_IMAGE_INFO_2_KHR](#)
 - [VK_STRUCTURE_TYPE_COPY_IMAGE_INFO_2_KHR](#)
 - [VK_STRUCTURE_TYPE_COPY_IMAGE_TO_BUFFER_INFO_2_KHR](#)
 - [VK_STRUCTURE_TYPE_IMAGE_BLIT_2_KHR](#)
 - [VK_STRUCTURE_TYPE_IMAGE_COPY_2_KHR](#)
 - [VK_STRUCTURE_TYPE_IMAGE_RESOLVE_2_KHR](#)

- `VK_STRUCTURE_TYPE_RESOLVE_IMAGE_INFO_2_KHR`

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Version History

- Revision 1, 2020-07-06 (Jeff Leger)
 - Internal revisions

VK_KHR_create_renderpass2

Name String

`VK_KHR_create_renderpass2`

Extension Type

Device extension

Registered Extension Number

110

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_multiview](#)

and

[VK_KHR_maintenance2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Tobias Hector [\[GitHub\]](#)[tobias](#)

Other Extension Metadata

Last Modified Date

2018-02-07

Contributors

- Tobias Hector
- Jeff Bolz

Description

This extension provides a new command to create render passes in a way that can be easily extended by other extensions through the substructures of render pass creation. The Vulkan 1.0 render pass creation sub-structures do not include `sType/pNext` members. Additionally, the render pass begin/next/end commands have been augmented with new extensible structures for passing additional subpass information.

The `VkRenderPassMultiviewCreateInfo` and `VkInputAttachmentAspectReference` structures that extended the original `VkRenderPassCreateInfo` are not accepted into the new creation functions, and instead their parameters are folded into this extension as follows:

- Elements of `VkRenderPassMultiviewCreateInfo::pViewMasks` are now specified in `VkSubpassDescription2KHR::viewMask`.
- Elements of `VkRenderPassMultiviewCreateInfo::pViewOffsets` are now specified in `VkSubpassDependency2KHR::viewOffset`.
- `VkRenderPassMultiviewCreateInfo::correlationMaskCount` and `VkRenderPassMultiviewCreateInfo::pCorrelationMasks` are directly specified in `VkRenderPassCreateInfo2KHR`.
- `VkInputAttachmentAspectReference::aspectMask` is now specified in the relevant input attachment reference in `VkAttachmentReference2KHR::aspectMask`

The details of these mappings are explained fully in the new structures.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- `vkCmdBeginRenderPass2KHR`
- `vkCmdEndRenderPass2KHR`
- `vkCmdNextSubpass2KHR`
- `vkCreateRenderPass2KHR`

New Structures

- `VkAttachmentDescription2KHR`
- `VkAttachmentReference2KHR`
- `VkRenderPassCreateInfo2KHR`
- `VkSubpassBeginInfoKHR`

- [VkSubpassDependency2KHR](#)
- [VkSubpassDescription2KHR](#)
- [VkSubpassEndInfoKHR](#)

New Enum Constants

- `VK_KHR_CREATE_RENDERPASS_2_EXTENSION_NAME`
- `VK_KHR_CREATE_RENDERPASS_2_SPEC_VERSION`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_2_KHR`
 - `VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_2_KHR`
 - `VK_STRUCTURE_TYPE_RENDER_PASS_CREATE_INFO_2_KHR`
 - `VK_STRUCTURE_TYPE_SUBPASS_BEGIN_INFO_KHR`
 - `VK_STRUCTURE_TYPE_SUBPASS_DEPENDENCY_2_KHR`
 - `VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_2_KHR`
 - `VK_STRUCTURE_TYPE_SUBPASS_END_INFO_KHR`

Version History

- Revision 1, 2018-02-07 (Tobias Hector)
 - Internal revisions

VK_KHR_dedicated_allocation

Name String

`VK_KHR_dedicated_allocation`

Extension Type

Device extension

Registered Extension Number

128

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_memory_requirements2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Faith Ekstrand, Intel

Description

This extension enables resources to be bound to a dedicated allocation, rather than suballocated. For any particular resource, applications **can** query whether a dedicated allocation is recommended, in which case using a dedicated allocation **may** improve the performance of access to that resource. Normal device memory allocations must support multiple resources per allocation, memory aliasing and sparse binding, which could interfere with some optimizations. Applications should query the implementation for when a dedicated allocation **may** be beneficial by adding a `VkMemoryDedicatedRequirementsKHR` structure to the `pNext` chain of the `VkMemoryRequirements2` structure passed as the `pMemoryRequirements` parameter of a call to `vkGetBufferMemoryRequirements2KHR` or `vkGetImageMemoryRequirements2KHR`. Certain external handle types and external images or buffers **may** also depend on dedicated allocations on implementations that associate image or buffer metadata with OS-level memory objects.

This extension adds a two small structures to memory requirements querying and memory allocation: a new structure that flags whether an image/buffer should have a dedicated allocation, and a structure indicating the image or buffer that an allocation will be bound to.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending [VkMemoryAllocateInfo](#):
 - [VkMemoryDedicatedAllocateInfoKHR](#)
- Extending [VkMemoryRequirements2](#):
 - [VkMemoryDedicatedRequirementsKHR](#)

New Enum Constants

- `VK_KHR_DEDICATED_ALLOCATION_EXTENSION_NAME`
- `VK_KHR_DEDICATED_ALLOCATION_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS_KHR`

Examples

```
// Create an image with a dedicated allocation based on the
// implementation's preference

VkImageCreateInfo imageCreateInfo =
{
    // Image creation parameters
};

VkImage image;
VkResult result = vkCreateImage(
    device,
    &imageCreateInfo,
    NULL,          // pAllocator
    &image);

VkMemoryDedicatedRequirementsKHR dedicatedRequirements =
{
    .sType = VK_STRUCTURE_TYPE_MEMORY_DEDICATED_REQUIREMENTS_KHR,
    .pNext = NULL,
};

VkMemoryRequirements2 memoryRequirements =
{
    .sType = VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2,
    .pNext = &dedicatedRequirements,
};

const VkImageMemoryRequirementsInfo2 imageRequirementsInfo =
{
    .sType = VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2,
    .pNext = NULL,
    .image = image
};

vkGetImageMemoryRequirements2(
    device,
    &imageRequirementsInfo,
    &memoryRequirements);
```



```

if (dedicatedRequirements.prefersDedicatedAllocation) {
    // Allocate memory with VkMemoryDedicatedAllocateInfoKHR::image
    // pointing to the image we are allocating the memory for

    VkMemoryDedicatedAllocateInfoKHR dedicatedInfo =
    {
        .sType = VK_STRUCTURE_TYPE_MEMORY_DEDICATED_ALLOCATE_INFO_KHR,
        .pNext = NULL,
        .image = image,
        .buffer = VK_NULL_HANDLE,
    };

    VkMemoryAllocateInfo memoryAllocateInfo =
    {
        .sType = VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_INFO,
        .pNext = &dedicatedInfo,
        .allocationSize = memoryRequirements.size,
        .memoryTypeIndex = FindMemoryTypeIndex(memoryRequirements.memoryTypeBits),
    };

    VkDeviceMemory memory;
    vkAllocateMemory(
        device,
        &memoryAllocateInfo,
        NULL, // pAllocator
        &memory);

    // Bind the image to the memory

    vkBindImageMemory(
        device,
        image,
        memory,
        0);
} else {
    // Take the normal memory sub-allocation path
}

```

Version History

- Revision 1, 2017-02-27 (James Jones)
 - Copy content from VK_NV_dedicated_allocation
 - Add some references to external object interactions to the overview.
- Revision 2, 2017-03-27 (Faith Ekstrand)
 - Rework the extension to be query-based
- Revision 3, 2017-07-31 (Faith Ekstrand)
 - Clarify that memory objects allocated with `VkMemoryDedicatedAllocateInfoKHR` can only

have the specified resource bound and no others.

VK_KHR_deferred_host_operations

Name String

VK_KHR_deferred_host_operations

Extension Type

Device extension

Registered Extension Number

269

Revision

4

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Josh Barczak [\[GitHub\]](#)jbarczak

Other Extension Metadata

Last Modified Date

2020-11-12

IP Status

No known IP claims.

Contributors

- Joshua Barczak, Intel
- Jeff Bolz, NVIDIA
- Daniel Koch, NVIDIA
- Slawek Grajewski, Intel
- Tobias Hector, AMD
- Yuriy O'Donnell, Epic
- Eric Werness, NVIDIA
- Baldur Karlsson, Valve
- Jesse Barker, Unity
- Contributors to VK_KHR_acceleration_structure, VK_KHR_ray_tracing_pipeline

Description

The [VK_KHR_deferred_host_operations](#) extension defines the infrastructure and usage patterns for deferrable commands, but does not specify any commands as deferrable. This is left to additional dependent extensions. Commands **must** not be deferred unless the deferral is specifically allowed by another extension which depends on [VK_KHR_deferred_host_operations](#).

New Object Types

- [VkDeferredOperationKHR](#)

New Commands

- [vkCreateDeferredOperationKHR](#)
- [vkDeferredOperationJoinKHR](#)
- [vkDestroyDeferredOperationKHR](#)
- [vkGetDeferredOperationMaxConcurrencyKHR](#)
- [vkGetDeferredOperationResultKHR](#)

New Enum Constants

- [VK_KHR_DEFERRED_HOST_OPERATIONS_EXTENSION_NAME](#)
- [VK_KHR_DEFERRED_HOST_OPERATIONS_SPEC_VERSION](#)
- Extending [VkObjectType](#):
 - [VK_OBJECT_TYPE_DEFERRED_OPERATION_KHR](#)
- Extending [VkResult](#):
 - [VK_OPERATION_DEFERRED_KHR](#)
 - [VK_OPERATION_NOT_DEFERRED_KHR](#)
 - [VK_THREAD_DONE_KHR](#)
 - [VK_THREAD_IDLE_KHR](#)

Code Examples

The following examples will illustrate the concept of deferrable operations using a hypothetical example. The command [vkDoSomethingExpensive](#) denotes a deferrable command.

The following example illustrates how a vulkan application might request deferral of an expensive operation:

```
// create a deferred operation
VkDeferredOperationKHR hOp;
VkResult result = vkCreateDeferredOperationKHR(device, pCallbacks, &hOp);
assert(result == VK_SUCCESS);

result = vkDoSomethingExpensive(device, hOp, ...);
```

```

assert( result == VK_OPERATION_DEFERRED_KHR );

// operation was deferred. Execute it asynchronously
std::async::launch(
    [ hOp ] ( )
    {
        vkDeferredOperationJoinKHR(device, hOp);

        result = vkGetDeferredOperationResultKHR(device, hOp);

        // deferred operation is now complete. 'result' indicates success or failure

        vkDestroyDeferredOperationKHR(device, hOp, pCallbacks);
    }
);

```

The following example illustrates extracting concurrency from a single deferred operation:

```

// create a deferred operation
VkDeferredOperationKHR hOp;
VkResult result = vkCreateDeferredOperationKHR(device, pCallbacks, &hOp);
assert(result == VK_SUCCESS);

result = vkDoSomethingExpensive(device, hOp, ...);
assert( result == VK_OPERATION_DEFERRED_KHR );

// Query the maximum amount of concurrency and clamp to the desired maximum
uint32_t numLaunches = std::min(vkGetDeferredOperationMaxConcurrencyKHR(device, hOp),
maxThreads);

std::vector<std::future<void> > joins;

for (uint32_t i = 0; i < numLaunches; i++) {
    joins.emplace_back(std::async::launch(
        [ hOp ] ( )
        {
            vkDeferredOperationJoinKHR(device, hOp);
            // in a job system, a return of VK_THREAD_IDLE_KHR should queue
another
            // job, but it is not functionally required
        }
    ));
}

for (auto &f : joins) {
    f.get();
}

result = vkGetDeferredOperationResultKHR(device, hOp);

```

```
// deferred operation is now complete. 'result' indicates success or failure
```

```
vkDestroyDeferredOperationKHR(device, hOp, pCallbacks);
```

The following example shows a subroutine which guarantees completion of a deferred operation, in the presence of multiple worker threads, and returns the result of the operation.

```
VkResult FinishDeferredOperation(VkDeferredOperationKHR hOp)
{
    // Attempt to join the operation until the implementation indicates that we should
    stop

    VkResult result = vkDeferredOperationJoinKHR(device, hOp);
    while( result == VK_THREAD_IDLE_KHR )
    {
        std::this_thread::yield();
        result = vkDeferredOperationJoinKHR(device, hOp);
    }

    switch( result )
    {
    case VK_SUCCESS:
        {
            // deferred operation has finished. Query its result.
            result = vkGetDeferredOperationResultKHR(device, hOp);
        }
        break;

    case VK_THREAD_DONE_KHR:
        {
            // deferred operation is being wrapped up by another thread
            // wait for that thread to finish
            do
            {
                std::this_thread::yield();
                result = vkGetDeferredOperationResultKHR(device, hOp);
            } while( result == VK_NOT_READY );
        }
        break;

    default:
        assert(false); // other conditions are illegal.
        break;
    }

    return result;
}
```

Issues

1. Should this extension have a `VkPhysicalDevice*FeaturesKHR` structure?

RESOLVED: No. This extension does not add any functionality on its own and requires a dependent extension to actually enable functionality and thus there is no value in adding a feature structure. If necessary, any dependent extension could add a feature boolean if it wanted to indicate that it is adding optional deferral support.

Version History

- Revision 1, 2019-12-05 (Josh Barczak, Daniel Koch)
 - Initial draft.
- Revision 2, 2020-03-06 (Daniel Koch, Tobias Hector)
 - Add missing `VK_OBJECT_TYPE_DEFERRED_OPERATION_KHR` enum
 - fix sample code
 - Clarified deferred operation parameter lifetimes (#2018,!3647)
- Revision 3, 2020-05-15 (Josh Barczak)
 - Clarify behavior of `vkGetDeferredOperationMaxConcurrencyKHR`, allowing it to return 0 if the operation is complete (#2036,!3850)
- Revision 4, 2020-11-12 (Tobias Hector, Daniel Koch)
 - Remove `VkDeferredOperationInfoKHR` and change return value semantics when deferred host operations are in use (#2067,3813)
 - clarify return value of `vkGetDeferredOperationResultKHR` (#2339,!4110)

VK_KHR_depth_stencil_resolve

Name String

`VK_KHR_depth_stencil_resolve`

Extension Type

Device extension

Registered Extension Number

200

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_create_renderpass2](#)

or

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Jan-Harald Fredriksen [\[GitHub\]janharald](#)

Other Extension Metadata

Last Modified Date

2018-04-09

Contributors

- Jan-Harald Fredriksen, Arm
- Andrew Garrard, Samsung Electronics
- Soowan Park, Samsung Electronics
- Jeff Bolz, NVIDIA
- Daniel Rakos, AMD

Description

This extension adds support for automatically resolving multisampled depth/stencil attachments in a subpass in a similar manner as for color attachments.

Multisampled color attachments can be resolved at the end of a subpass by specifying `pResolveAttachments` entries corresponding to the `pColorAttachments` array entries. This does not allow for a way to map the resolve attachments to the depth/stencil attachment. The `vkCmdResolveImage` command does not allow for depth/stencil images. While there are other ways to resolve the depth/stencil attachment, they can give sub-optimal performance. Extending the `VkSubpassDescription2` in this extension allows an application to add a `pDepthStencilResolveAttachment`, that is similar to the color `pResolveAttachments`, that the `pDepthStencilAttachment` can be resolved into.

Depth and stencil samples are resolved to a single value based on the resolve mode. The set of possible resolve modes is defined in the `VkResolveModeFlagBits` enum. The `VK_RESOLVE_MODE_SAMPLE_ZERO_BIT` mode is the only mode that is required of all implementations (that support the extension or support Vulkan 1.2 or higher). Some implementations may also support averaging (the same as color sample resolve) or taking the minimum or maximum sample, which may be more suitable for depth/stencil resolve.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceDepthStencilResolvePropertiesKHR](#)
- Extending [VkSubpassDescription2](#):
 - [VkSubpassDescriptionDepthStencilResolveKHR](#)

New Enums

- [VkResolveModeFlagBitsKHR](#)

New Bitmasks

- [VkResolveModeFlagsKHR](#)

New Enum Constants

- [VK_KHR_DEPTH_STENCIL_RESOLVE_EXTENSION_NAME](#)
- [VK_KHR_DEPTH_STENCIL_RESOLVE_SPEC_VERSION](#)
- Extending [VkResolveModeFlagBits](#):
 - [VK_RESOLVE_MODE_AVERAGE_BIT_KHR](#)
 - [VK_RESOLVE_MODE_MAX_BIT_KHR](#)
 - [VK_RESOLVE_MODE_MIN_BIT_KHR](#)
 - [VK_RESOLVE_MODE_NONE_KHR](#)
 - [VK_RESOLVE_MODE_SAMPLE_ZERO_BIT_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_STENCIL_RESOLVE_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_SUBPASS_DESCRIPTION_DEPTH_STENCIL_RESOLVE_KHR](#)

Version History

- Revision 1, 2018-04-09 (Jan-Harald Fredriksen)
 - Initial revision

VK_KHR_descriptor_update_template

Name String

[VK_KHR_descriptor_update_template](#)

Extension Type

Device extension

Registered Extension Number

86

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

API Interactions

- Interacts with `VK_EXT_debug_report`
- Interacts with `VK_KHR_push_descriptor`

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Markus Tavenrath [\[GitHub\]mtavenrath](#)

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with `VK_KHR_push_descriptor`

Contributors

- Jeff Bolz, NVIDIA
- Michael Worcester, Imagination Technologies

Description

Applications may wish to update a fixed set of descriptors in a large number of descriptor sets very frequently, i.e. during initialization phase or if it is required to rebuild descriptor sets for each frame. For those cases it is also not unlikely that all information required to update a single descriptor set is stored in a single struct. This extension provides a way to update a fixed set of descriptors in a single `VkDescriptorSet` with a pointer to an application-defined data structure describing the new descriptors.

Promotion to Vulkan 1.1

`vkCmdPushDescriptorSetWithTemplateKHR` is included as an interaction with `VK_KHR_push_descriptor`. If Vulkan 1.1 and `VK_KHR_push_descriptor` are supported, this is included by `VK_KHR_push_descriptor`.

The base functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Object Types

- [VkDescriptorUpdateTemplateKHR](#)

New Commands

- [vkCreateDescriptorUpdateTemplateKHR](#)
- [vkDestroyDescriptorUpdateTemplateKHR](#)
- [vkUpdateDescriptorSetWithTemplateKHR](#)

If [VK_KHR_push_descriptor](#) is supported:

- [vkCmdPushDescriptorSetWithTemplateKHR](#)

New Structures

- [VkDescriptorUpdateTemplateCreateInfoKHR](#)
- [VkDescriptorUpdateTemplateEntryKHR](#)

New Enums

- [VkDescriptorUpdateTemplateTypeKHR](#)

New Bitmasks

- [VkDescriptorUpdateTemplateCreateFlagsKHR](#)

New Enum Constants

- [VK_KHR_DESCRIPTOR_UPDATE_TEMPLATE_EXTENSION_NAME](#)
- [VK_KHR_DESCRIPTOR_UPDATE_TEMPLATE_SPEC_VERSION](#)
- Extending [VkDescriptorUpdateTemplateType](#):
 - [VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_DESCRIPTOR_SET_KHR](#)
- Extending [VkObjectType](#):
 - [VK_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_CREATE_INFO_KHR](#)

If [VK_EXT_debug_report](#) is supported:

- Extending [VkDebugReportObjectTypeEXT](#):
 - [VK_DEBUG_REPORT_OBJECT_TYPE_DESCRIPTOR_UPDATE_TEMPLATE_KHR_EXT](#)

If [VK_KHR_push_descriptor](#) is supported:

- Extending [VkDescriptorUpdateTemplateType](#):
 - [VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_KHR](#)

Version History

- Revision 1, 2016-01-11 (Markus Tavenrath)
 - Initial draft

VK_KHR_device_group

Name String

[VK_KHR_device_group](#)

Extension Type

Device extension

Registered Extension Number

61

Revision

4

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_device_group_creation](#)

API Interactions

- Interacts with [VK_KHR_bind_memory2](#)
- Interacts with [VK_KHR_surface](#)
- Interacts with [VK_KHR_swapchain](#)

SPIR-V Dependencies

- [SPV_KHR_device_group](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jeff Bolz [\[GitHub\]jeffbolznv](#)

Other Extension Metadata

Last Modified Date

2017-10-10

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Tobias Hector, Imagination Technologies

Description

This extension provides functionality to use a logical device that consists of multiple physical devices, as created with the [VK_KHR_device_group_creation](#) extension. A device group can allocate memory across the subdevices, bind memory from one subdevice to a resource on another subdevice, record command buffers where some work executes on an arbitrary subset of the subdevices, and potentially present a swapchain image from one or more subdevices.

Promotion to Vulkan 1.1

The following enums, types and commands are included as interactions with [VK_KHR_swapchain](#):

- [VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_CAPABILITIES_KHR](#)
- [VK_STRUCTURE_TYPE_IMAGE_SWAPCHAIN_CREATE_INFO_KHR](#)
- [VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_SWAPCHAIN_INFO_KHR](#)
- [VK_STRUCTURE_TYPE_ACQUIRE_NEXT_IMAGE_INFO_KHR](#)
- [VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_INFO_KHR](#)
- [VK_STRUCTURE_TYPE_DEVICE_GROUP_SWAPCHAIN_CREATE_INFO_KHR](#)
- [VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR](#)
- [VkDeviceGroupPresentModeFlagBitsKHR](#)
- [VkDeviceGroupPresentCapabilitiesKHR](#)
- [VkImageSwapchainCreateInfoKHR](#)
- [VkBindImageMemorySwapchainInfoKHR](#)
- [VkAcquireNextImageInfoKHR](#)
- [VkDeviceGroupPresentInfoKHR](#)
- [VkDeviceGroupSwapchainCreateInfoKHR](#)
- [vkGetDeviceGroupPresentCapabilitiesKHR](#)
- [vkGetDeviceGroupSurfacePresentModesKHR](#)
- [vkGetPhysicalDevicePresentRectanglesKHR](#)
- [vkAcquireNextImage2KHR](#)

If Vulkan 1.1 and [VK_KHR_swapchain](#) are supported, these are included by [VK_KHR_swapchain](#).

The base functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- [vkCmdDispatchBaseKHR](#)
- [vkCmdSetDeviceMaskKHR](#)
- [vkGetDeviceGroupPeerMemoryFeaturesKHR](#)

If [VK_KHR_surface](#) is supported:

- [vkGetDeviceGroupPresentCapabilitiesKHR](#)
- [vkGetDeviceGroupSurfacePresentModesKHR](#)
- [vkGetPhysicalDevicePresentRectanglesKHR](#)

If [VK_KHR_swapchain](#) is supported:

- [vkAcquireNextImage2KHR](#)

New Structures

- Extending [VkBindSparseInfo](#):
 - [VkDeviceGroupBindSparseInfoKHR](#)
- Extending [VkCommandBufferBeginInfo](#):
 - [VkDeviceGroupCommandBufferBeginInfoKHR](#)
- Extending [VkMemoryAllocateInfo](#):
 - [VkMemoryAllocateFlagsInfoKHR](#)
- Extending [VkRenderPassBeginInfo](#), [VkRenderingInfo](#):
 - [VkDeviceGroupRenderPassBeginInfoKHR](#)
- Extending [VkSubmitInfo](#):
 - [VkDeviceGroupSubmitInfoKHR](#)

If [VK_KHR_bind_memory2](#) is supported:

- Extending [VkBindBufferMemoryInfo](#):
 - [VkBindBufferMemoryDeviceGroupInfoKHR](#)
- Extending [VkBindImageMemoryInfo](#):
 - [VkBindImageMemoryDeviceGroupInfoKHR](#)

If [VK_KHR_surface](#) is supported:

- [VkDeviceGroupPresentCapabilitiesKHR](#)

If [VK_KHR_swapchain](#) is supported:

- [VkAcquireNextImageInfoKHR](#)
- Extending [VkBindImageMemoryInfo](#):

- [VkBindImageMemorySwapchainInfoKHR](#)
- Extending [VkImageCreateInfo](#):
 - [VkImageSwapchainCreateInfoKHR](#)
- Extending [VkPresentInfoKHR](#):
 - [VkDeviceGroupPresentInfoKHR](#)
- Extending [VkSwapchainCreateInfoKHR](#):
 - [VkDeviceGroupSwapchainCreateInfoKHR](#)

New Enums

- [VkMemoryAllocateFlagBitsKHR](#)
- [VkPeerMemoryFeatureFlagBitsKHR](#)

If [VK_KHR_surface](#) is supported:

- [VkDeviceGroupPresentModeFlagBitsKHR](#)

New Bitmasks

- [VkMemoryAllocateFlagsKHR](#)
- [VkPeerMemoryFeatureFlagsKHR](#)

If [VK_KHR_surface](#) is supported:

- [VkDeviceGroupPresentModeFlagsKHR](#)

New Enum Constants

- [VK_KHR_DEVICE_GROUP_EXTENSION_NAME](#)
- [VK_KHR_DEVICE_GROUP_SPEC_VERSION](#)
- Extending [VkDependencyFlagBits](#):
 - [VK_DEPENDENCY_DEVICE_GROUP_BIT_KHR](#)
- Extending [VkMemoryAllocateFlagBits](#):
 - [VK_MEMORY_ALLOCATE_DEVICE_MASK_BIT_KHR](#)
- Extending [VkPeerMemoryFeatureFlagBits](#):
 - [VK_PEER_MEMORY_FEATURE_COPY_DST_BIT_KHR](#)
 - [VK_PEER_MEMORY_FEATURE_COPY_SRC_BIT_KHR](#)
 - [VK_PEER_MEMORY_FEATURE_GENERIC_DST_BIT_KHR](#)
 - [VK_PEER_MEMORY_FEATURE_GENERIC_SRC_BIT_KHR](#)
- Extending [VkPipelineCreateFlagBits](#):
 - [VK_PIPELINE_CREATE_DISPATCH_BASE_KHR](#)
 - [VK_PIPELINE_CREATE_VIEW_INDEX_FROM_DEVICE_INDEX_BIT_KHR](#)

- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_DEVICE_GROUP_BIND_SPARSE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_DEVICE_GROUP_COMMAND_BUFFER_BEGIN_INFO_KHR`
 - `VK_STRUCTURE_TYPE_DEVICE_GROUP_RENDER_PASS_BEGIN_INFO_KHR`
 - `VK_STRUCTURE_TYPE_DEVICE_GROUP_SUBMIT_INFO_KHR`
 - `VK_STRUCTURE_TYPE_MEMORY_ALLOCATE_FLAGS_INFO_KHR`

If [VK_KHR_bind_memory2](#) is supported:

- Extending [VkImageCreateFlagBits](#):
 - `VK_IMAGE_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_BIND_BUFFER_MEMORY_DEVICE_GROUP_INFO_KHR`
 - `VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_DEVICE_GROUP_INFO_KHR`

If [VK_KHR_surface](#) is supported:

- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_CAPABILITIES_KHR`

If [VK_KHR_swapchain](#) is supported:

- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_ACQUIRE_NEXT_IMAGE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_SWAPCHAIN_INFO_KHR`
 - `VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_INFO_KHR`
 - `VK_STRUCTURE_TYPE_DEVICE_GROUP_SWAPCHAIN_CREATE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_IMAGE_SWAPCHAIN_CREATE_INFO_KHR`
- Extending [VkSwapchainCreateFlagBitsKHR](#):
 - `VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR`

New Built-in Variables

- [DeviceIndex](#)

New SPIR-V Capabilities

- [DeviceGroup](#)

Version History

- Revision 1, 2016-10-19 (Jeff Bolz)
 - Internal revisions

- Revision 2, 2017-05-19 (Tobias Hector)
 - Removed extended memory bind functions to VK_KHR_bind_memory2, added dependency on that extension, and device-group-specific structs for those functions.
- Revision 3, 2017-10-06 (Ian Elliott)
 - Corrected Vulkan 1.1 interactions with the WSI extensions. All Vulkan 1.1 WSI interactions are with the VK_KHR_swapchain extension.
- Revision 4, 2017-10-10 (Jeff Bolz)
 - Rename “SFR” bits and structure members to use the phrase “split instance bind regions”.

VK_KHR_device_group_creation

Name String

VK_KHR_device_group_creation

Extension Type

Instance extension

Registered Extension Number

71

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jeff Bolz [\[GitHub\]jeffbolznv](#)

Other Extension Metadata

Last Modified Date

2016-10-19

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA

Description

This extension provides instance-level commands to enumerate groups of physical devices, and to create a logical device from a subset of one of those groups. Such a logical device can then be used with new features in the [VK_KHR_device_group](#) extension.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- [vkEnumeratePhysicalDeviceGroupsKHR](#)

New Structures

- [VkPhysicalDeviceGroupPropertiesKHR](#)
- Extending [VkDeviceCreateInfo](#):
 - [VkDeviceGroupDeviceCreateInfoKHR](#)

New Enum Constants

- [VK_KHR_DEVICE_GROUP_CREATION_EXTENSION_NAME](#)
- [VK_KHR_DEVICE_GROUP_CREATION_SPEC_VERSION](#)
- [VK_MAX_DEVICE_GROUP_SIZE_KHR](#)
- Extending [VkMemoryHeapFlagBits](#):
 - [VK_MEMORY_HEAP_MULTI_INSTANCE_BIT_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES_KHR](#)

Examples

```
VkDeviceCreateInfo devCreateInfo = { VK_STRUCTURE_TYPE_DEVICE_CREATE_INFO };
// (not shown) fill out devCreateInfo as usual.
uint32_t deviceGroupCount = 0;
VkPhysicalDeviceGroupPropertiesKHR *props = NULL;

// Query the number of device groups
vkEnumeratePhysicalDeviceGroupsKHR(g_vkInstance, &deviceGroupCount, NULL);

// Allocate and initialize structures to query the device groups
props = (VkPhysicalDeviceGroupPropertiesKHR *)malloc(deviceGroupCount*sizeof
(VkPhysicalDeviceGroupPropertiesKHR));
for (i = 0; i < deviceGroupCount; ++i) {
```

```

        props[i].sType = VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GROUP_PROPERTIES_KHR;
        props[i].pNext = NULL;
    }
    vkEnumeratePhysicalDeviceGroupsKHR(g_vkInstance, &deviceGroupCount, props);

    // If the first device group has more than one physical device. create
    // a logical device using all of the physical devices.
    VkDeviceGroupDeviceCreateInfoKHR deviceGroupInfo = {
VK_STRUCTURE_TYPE_DEVICE_GROUP_DEVICE_CREATE_INFO_KHR };
    if (props[0].physicalDeviceCount > 1) {
        deviceGroupInfo.physicalDeviceCount = props[0].physicalDeviceCount;
        deviceGroupInfo.pPhysicalDevices = props[0].physicalDevices;
        devCreateInfo.pNext = &deviceGroupInfo;
    }

    vkCreateDevice(props[0].physicalDevices[0], &devCreateInfo, NULL, &g_vkDevice);
    free(props);

```

Version History

- Revision 1, 2016-10-19 (Jeff Bolz)
 - Internal revisions

VK_KHR_display

Name String

VK_KHR_display

Extension Type

Instance extension

Registered Extension Number

3

Revision

23

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_surface](#)

Contact

- James Jones [\[GitHub\]](#)cubanismo
- Norbert Nopper [\[GitHub\]](#)FslNopper

Other Extension Metadata

Last Modified Date

2017-03-13

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Norbert Nopper, Freescale
- Jeff Vigil, Qualcomm
- Daniel Rakos, AMD

Description

This extension provides the API to enumerate displays and available modes on a given device.

New Object Types

- [VkDisplayKHR](#)
- [VkDisplayModeKHR](#)

New Commands

- [vkCreateDisplayModeKHR](#)
- [vkCreateDisplayPlaneSurfaceKHR](#)
- [vkGetDisplayModePropertiesKHR](#)
- [vkGetDisplayPlaneCapabilitiesKHR](#)
- [vkGetDisplayPlaneSupportedDisplaysKHR](#)
- [vkGetPhysicalDeviceDisplayPlanePropertiesKHR](#)
- [vkGetPhysicalDeviceDisplayPropertiesKHR](#)

New Structures

- [VkDisplayModeCreateInfoKHR](#)
- [VkDisplayModeParametersKHR](#)
- [VkDisplayModePropertiesKHR](#)
- [VkDisplayPlaneCapabilitiesKHR](#)
- [VkDisplayPlanePropertiesKHR](#)
- [VkDisplayPropertiesKHR](#)
- [VkDisplaySurfaceCreateInfoKHR](#)

New Enums

- [VkDisplayPlaneAlphaFlagBitsKHR](#)

New Bitmasks

- [VkDisplayModeCreateFlagsKHR](#)
- [VkDisplayPlaneAlphaFlagsKHR](#)
- [VkDisplaySurfaceCreateFlagsKHR](#)
- [VkSurfaceTransformFlagsKHR](#)

New Enum Constants

- [VK_KHR_DISPLAY_EXTENSION_NAME](#)
- [VK_KHR_DISPLAY_SPEC_VERSION](#)
- Extending [VkObjectType](#):
 - [VK_OBJECT_TYPE_DISPLAY_KHR](#)
 - [VK_OBJECT_TYPE_DISPLAY_MODE_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_DISPLAY_MODE_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_DISPLAY_SURFACE_CREATE_INFO_KHR](#)

Issues

1) Which properties of a mode should be fixed in the mode information vs. settable in some other function when setting the mode? E.g., do we need to double the size of the mode pool to include both stereo and non-stereo modes? YUV and RGB scanout even if they both take RGB input images? BGR vs. RGB input? etc.

RESOLVED: Many modern displays support at most a handful of resolutions and timings natively. Other “modes” are expected to be supported using scaling hardware on the display engine or GPU. Other properties, such as rotation and mirroring should not require duplicating hardware modes just to express all combinations. Further, these properties may be implemented on a per-display or per-overlay granularity.

To avoid the exponential growth of modes as mutable properties are added, as was the case with [EGLConfig](#)/[WGL](#) pixel formats/[GLXFBConfig](#), this specification should separate out hardware properties and configurable state into separate objects. Modes and overlay planes will express capabilities of the hardware, while a separate structure will allow applications to configure scaling, rotation, mirroring, color keys, LUT values, alpha masks, etc. for a given swapchain independent of the mode in use. Constraints on these settings will be established by properties of the immutable objects.

Note the resolution of this issue may affect issue 5 as well.

2) What properties of a display itself are useful?

RESOLVED: This issue is too broad. It was meant to prompt general discussion, but resolving this issue amounts to completing this specification. All interesting properties should be included. The issue will remain as a placeholder since removing it would make it hard to parse existing discussion notes that refer to issues by number.

3) How are multiple overlay planes within a display or mode enumerated?

RESOLVED: They are referred to by an index. Each display will report the number of overlay planes it contains.

4) Should swapchains be created relative to a mode or a display?

RESOLVED: When using this extension, swapchains are created relative to a mode and a plane. The mode implies the display object the swapchain will present to. If the specified mode is not the display's current mode, the new mode will be applied when the first image is presented to the swapchain, and the default operating system mode, if any, will be restored when the swapchain is destroyed.

5) Should users query generic ranges from displays and construct their own modes explicitly using those constraints rather than querying a fixed set of modes (Most monitors only have one real "mode" these days, even though many support relatively arbitrary scaling, either on the monitor side or in the GPU display engine, making "modes" something of a relic/compatibility construct).

RESOLVED: Expose both. Display information structures will expose a set of predefined modes, as well as any attributes necessary to construct a customized mode.

6) Is it fine if we return the display and display mode handles in the structure used to query their properties?

RESOLVED: Yes.

7) Is there a possibility that not all displays of a device work with all of the present queues of a device? If yes, how do we determine which displays work with which present queues?

RESOLVED: No known hardware has such limitations, but determining such limitations is supported automatically using the existing [VK_KHR_surface](#) and [VK_KHR_swapchain](#) query mechanisms.

8) Should all presentation need to be done relative to an overlay plane, or can a display mode + display be used alone to target an output?

RESOLVED: Require specifying a plane explicitly.

9) Should displays have an associated window system display, such as an [HDC](#) or [Display*](#)?

RESOLVED: No. Displays are independent of any windowing system in use on the system. Further, neither [HDC](#) nor [Display*](#) refer to a physical display object.

10) Are displays queried from a physical GPU or from a device instance?

RESOLVED: Developers prefer to query modes directly from the physical GPU so they can use display information as an input to their device selection algorithms prior to device creation. This

avoids the need to create placeholder device instances to enumerate displays.

This preference must be weighed against the extra initialization that must be done by driver vendors prior to device instance creation to support this usage.

11) Should displays and/or modes be dispatchable objects? If functions are to take displays, overlays, or modes as their first parameter, they must be dispatchable objects as defined in Khronos bug 13529. If they are not added to the list of dispatchable objects, functions operating on them must take some higher-level object as their first parameter. There is no performance case against making them dispatchable objects, but they would be the first extension objects to be dispatchable.

RESOLVED: Do not make displays or modes dispatchable. They will dispatch based on their associated physical device.

12) Should hardware cursor capabilities be exposed?

RESOLVED: Defer. This could be a separate extension on top of the base WSI specs.

13) How many display objects should be enumerated for "tiled" display devices? There are ongoing design discussions among lower-level display API authors regarding how to expose displays if they are one physical display device to an end user, but may internally be implemented as two side-by-side displays using the same display engine (and sometimes cabling) resources as two physically separate display devices.

RESOLVED: Tiled displays will appear as a single display object in this API.

14) Should the raw EDID data be included in the display information?

RESOLVED: No. A future extension could be added which reports the EDID if necessary. This may be complicated by the outcome of issue 13.

15) Should min and max scaling factor capabilities of overlays be exposed?

RESOLVED: Yes. This is exposed indirectly by allowing applications to query the min/max position and extent of the source and destination regions from which image contents are fetched by the display engine when using a particular mode and overlay pair.

16) Should devices be able to expose planes that can be moved between displays? If so, how?

RESOLVED: Yes. Applications can determine which displays a given plane supports using [vkGetDisplayPlaneSupportedDisplaysKHR](#).

17) Should there be a way to destroy display modes? If so, does it support destroying "built in" modes?

RESOLVED: Not in this extension. A future extension could add this functionality.

18) What should the lifetime of display and built-in display mode objects be?

RESOLVED: The lifetime of the instance. These objects cannot be destroyed. A future extension may be added to expose a way to destroy these objects and/or support display hotplug.

19) Should persistent mode for smart panels be enabled/disabled at swapchain creation time, or on a per-present basis.

RESOLVED: On a per-present basis.

Examples

NOTE

The example code for the `VK_KHR_display` and `VK_KHR_display_swapchain` extensions was removed from the appendix after revision 1.0.43. The display enumeration example code was ported to the cube demo that is shipped with the official Khronos SDK, and is being kept up-to-date in that location (see: <https://github.com/KhronosGroup/Vulkan-Tools/blob/main/cube/cube.c>).

Version History

- Revision 1, 2015-02-24 (James Jones)
 - Initial draft
- Revision 2, 2015-03-12 (Norbert Nopper)
 - Added overlay enumeration for a display.
- Revision 3, 2015-03-17 (Norbert Nopper)
 - Fixed typos and namings as discussed in Bugzilla.
 - Reordered and grouped functions.
 - Added functions to query count of display, mode and overlay.
 - Added native display handle, which may be needed on some platforms to create a native Window.
- Revision 4, 2015-03-18 (Norbert Nopper)
 - Removed primary and virtualPostion members (see comment of James Jones in Bugzilla).
 - Added native overlay handle to information structure.
 - Replaced , with ; in struct.
- Revision 6, 2015-03-18 (Daniel Rakos)
 - Added WSI extension suffix to all items.
 - Made the whole API more “Vulkanish”.
 - Replaced all functions with a single `vkGetDisplayInfoKHR` function to better match the rest of the API.
 - Made the display, display mode, and overlay objects be first class objects, not subclasses of `VkBaseObject` as they do not support the common functions anyways.
 - Renamed `*Info` structures to `*Properties`.
 - Removed `overlayIndex` field from `VkOverlayProperties` as there is an implicit index already as a result of moving to a “Vulkanish” API.
 - Displays are not get through device, but through physical GPU to match the rest of the

Vulkan API. Also this is something ISVs explicitly requested.

- Added issue (6) and (7).
- Revision 7, 2015-03-25 (James Jones)
 - Added an issues section
 - Added rotation and mirroring flags
- Revision 8, 2015-03-25 (James Jones)
 - Combined the duplicate issues sections introduced in last change.
 - Added proposed resolutions to several issues.
- Revision 9, 2015-04-01 (Daniel Rakos)
 - Rebased extension against Vulkan 0.82.0
- Revision 10, 2015-04-01 (James Jones)
 - Added issues (10) and (11).
 - Added more straw-man issue resolutions, and cleaned up the proposed resolution for issue (4).
 - Updated the rotation and mirroring enums to have proper bitmask semantics.
- Revision 11, 2015-04-15 (James Jones)
 - Added proposed resolution for issues (1) and (2).
 - Added issues (12), (13), (14), and (15)
 - Removed pNativeHandle field from overlay structure.
 - Fixed small compilation errors in example code.
- Revision 12, 2015-07-29 (James Jones)
 - Rewrote the guts of the extension against the latest WSI swapchain specifications and the latest Vulkan API.
 - Address overlay planes by their index rather than an object handle and refer to them as “planes” rather than “overlays” to make it slightly clearer that even a display with no “overlays” still has at least one base “plane” that images can be displayed on.
 - Updated most of the issues.
 - Added an “extension type” section to the specification header.
 - Reused the VK_EXT_KHR_surface surface transform enumerations rather than redefining them here.
 - Updated the example code to use the new semantics.
- Revision 13, 2015-08-21 (Ian Elliott)
 - Renamed this extension and all of its enumerations, types, functions, etc. This makes it compliant with the proposed standard for Vulkan extensions.
 - Switched from “revision” to “version”, including use of the VK_MAKE_VERSION macro in the header file.
- Revision 14, 2015-09-01 (James Jones)

- Restore single-field revision number.
- Revision 15, 2015-09-08 (James Jones)
 - Added alpha flags enum.
 - Added premultiplied alpha support.
- Revision 16, 2015-09-08 (James Jones)
 - Added description section to the spec.
 - Added issues 16 - 18.
- Revision 17, 2015-10-02 (James Jones)
 - Planes are now a property of the entire device rather than individual displays. This allows planes to be moved between multiple displays on devices that support it.
 - Added a function to create a `VkSurfaceKHR` object describing a display plane and mode to align with the new per-platform surface creation conventions.
 - Removed detailed mode timing data. It was agreed that the mode extents and refresh rate are sufficient for current use cases. Other information could be added back in as an extension if it is needed in the future.
 - Added support for smart/persistent/buffered display devices.
- Revision 18, 2015-10-26 (Ian Elliott)
 - Renamed from `VK_EXT_KHR_display` to `VK_KHR_display`.
- Revision 19, 2015-11-02 (James Jones)
 - Updated example code to match revision 17 changes.
- Revision 20, 2015-11-03 (Daniel Rakos)
 - Added allocation callbacks to creation functions.
- Revision 21, 2015-11-10 (Jesse Hall)
 - Added `VK_DISPLAY_PLANE_ALPHA_OPAQUE_BIT_KHR`, and use `VkDisplayPlaneAlphaFlagBitsKHR` for `VkDisplayPlanePropertiesKHR::alphaMode` instead of `VkDisplayPlaneAlphaFlagsKHR`, since it only represents one mode.
 - Added reserved flags bitmask to `VkDisplayPlanePropertiesKHR`.
 - Use `VkSurfaceTransformFlagBitsKHR` instead of obsolete `VkSurfaceTransformKHR`.
 - Renamed `vkGetDisplayPlaneSupportedDisplaysKHR` parameters for clarity.
- Revision 22, 2015-12-18 (James Jones)
 - Added missing “planeIndex” parameter to `vkGetDisplayPlaneSupportedDisplaysKHR()`
- Revision 23, 2017-03-13 (James Jones)
 - Closed all remaining issues. The specification and implementations have been shipping with the proposed resolutions for some time now.
 - Removed the sample code and noted it has been integrated into the official Vulkan SDK cube demo.

VK_KHR_display_swapchain

Name String

VK_KHR_display_swapchain

Extension Type

Device extension

Registered Extension Number

4

Revision

10

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_swapchain](#)

and

[VK_KHR_display](#)

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2017-03-13

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Jeff Vigil, Qualcomm
- Jesse Hall, Google

Description

This extension provides an API to create a swapchain directly on a device's display without any underlying window system.

New Commands

- [vkCreateSharedSwapchainsKHR](#)

New Structures

- Extending [VkPresentInfoKHR](#):
 - [VkDisplayPresentInfoKHR](#)

New Enum Constants

- [VK_KHR_DISPLAY_SWAPCHAIN_EXTENSION_NAME](#)
- [VK_KHR_DISPLAY_SWAPCHAIN_SPEC_VERSION](#)
- Extending [VkResult](#):
 - [VK_ERROR_INCOMPATIBLE_DISPLAY_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_DISPLAY_PRESENT_INFO_KHR](#)

Issues

1) Should swapchains sharing images each hold a reference to the images, or should it be up to the application to destroy the swapchains and images in an order that avoids the need for reference counting?

RESOLVED: Take a reference. The lifetime of presentable images is already complex enough.

2) Should the [srcRect](#) and [dstRect](#) parameters be specified as part of the presentation command, or at swapchain creation time?

RESOLVED: As part of the presentation command. This allows moving and scaling the image on the screen without the need to respecify the mode or create a new swapchain and presentable images.

3) Should [srcRect](#) and [dstRect](#) be specified as rects, or separate offset/extent values?

RESOLVED: As rects. Specifying them separately might make it easier for hardware to expose support for one but not the other, but in such cases applications must just take care to obey the reported capabilities and not use non-zero offsets or extents that require scaling, as appropriate.

4) How can applications create multiple swapchains that use the same images?

RESOLVED: By calling [vkCreateSharedSwapchainsKHR](#).

An earlier resolution used [vkCreateSwapchainKHR](#), chaining multiple [VkSwapchainCreateInfoKHR](#) structures through [pNext](#). In order to allow each swapchain to also allow other extension structs, a level of indirection was used: [VkSwapchainCreateInfoKHR::pNext](#) pointed to a different structure, which had both [sType](#) and [pNext](#) members for additional extensions, and also had a pointer to the next [VkSwapchainCreateInfoKHR](#) structure. The number of swapchains to be created could only be found by walking this linked list of alternating structures, and the [pSwapchains](#) out parameter was reinterpreted to be an array of [VkSwapchainKHR](#) handles.

Another option considered was a method to specify a “shared” swapchain when creating a new swapchain, such that groups of swapchains using the same images could be built up one at a time. This was deemed unusable because drivers need to know all of the displays an image will be used

on when determining which internal formats and layouts to use for that image.

Examples

NOTE

The example code for the `VK_KHR_display` and `VK_KHR_display_swapchain` extensions was removed from the appendix after revision 1.0.43. The display swapchain creation example code was ported to the cube demo that is shipped with the official Khronos SDK, and is being kept up-to-date in that location (see: <https://github.com/KhronosGroup/Vulkan-Tools/blob/main/cube/cube.c>).

Version History

- Revision 1, 2015-07-29 (James Jones)
 - Initial draft
- Revision 2, 2015-08-21 (Ian Elliott)
 - Renamed this extension and all of its enumerations, types, functions, etc. This makes it compliant with the proposed standard for Vulkan extensions.
 - Switched from “revision” to “version”, including use of the `VK_MAKE_VERSION` macro in the header file.
- Revision 3, 2015-09-01 (James Jones)
 - Restore single-field revision number.
- Revision 4, 2015-09-08 (James Jones)
 - Allow creating multiple swapchains that share the same images using a single call to `vkCreateSwapchainKHR()`.
- Revision 5, 2015-09-10 (Alon Or-bach)
 - Removed underscores from `SWAP_CHAIN` in two enums.
- Revision 6, 2015-10-02 (James Jones)
 - Added support for smart panels/buffered displays.
- Revision 7, 2015-10-26 (Ian Elliott)
 - Renamed from `VK_EXT_KHR_display_swapchain` to `VK_KHR_display_swapchain`.
- Revision 8, 2015-11-03 (Daniel Rakos)
 - Updated sample code based on the changes to `VK_KHR_swapchain`.
- Revision 9, 2015-11-10 (Jesse Hall)
 - Replaced `VkDisplaySwapchainCreateInfoKHR` with `vkCreateSharedSwapchainsKHR`, changing resolution of issue #4.
- Revision 10, 2017-03-13 (James Jones)
 - Closed all remaining issues. The specification and implementations have been shipping with the proposed resolutions for some time now.
 - Removed the sample code and noted it has been integrated into the official Vulkan SDK cube demo.

VK_KHR_draw_indirect_count

Name String

VK_KHR_draw_indirect_count

Extension Type

Device extension

Registered Extension Number

170

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Piers Daniell [[GitHub](#)]pdaniell-nv

Other Extension Metadata

Last Modified Date

2017-08-25

IP Status

No known IP claims.

Contributors

- Matthaeus G. Chajdas, AMD
- Derrick Owens, AMD
- Graham Sellers, AMD
- Daniel Rakos, AMD
- Dominik Witczak, AMD
- Piers Daniell, NVIDIA

Description

This extension is based on the [VK_AMD_draw_indirect_count](#) extension. This extension allows an application to source the number of draws for indirect drawing calls from a buffer.

Applications might want to do culling on the GPU via a compute shader prior to drawing. This enables the application to generate an arbitrary number of drawing commands and execute them without host intervention.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, the commands [vkCmdDrawIndirectCountKHR](#) and [vkCmdDrawIndexedIndirectCountKHR](#) are optional. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- [vkCmdDrawIndexedIndirectCountKHR](#)
- [vkCmdDrawIndirectCountKHR](#)

New Enum Constants

- [VK_KHR_DRAW_INDIRECT_COUNT_EXTENSION_NAME](#)
- [VK_KHR_DRAW_INDIRECT_COUNT_SPEC_VERSION](#)

Version History

- Revision 1, 2017-08-25 (Piers Daniell)
 - Initial draft based on [VK_AMD_draw_indirect_count](#)

VK_KHR_driver_properties

Name String

[VK_KHR_driver_properties](#)

Extension Type

Device extension

Registered Extension Number

197

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Daniel Rakos [\[GitHub\]drakos-amd](#)

Other Extension Metadata

Last Modified Date

2018-04-11

IP Status

No known IP claims.

Contributors

- Baldur Karlsson
- Matthaeus G. Chajdas, AMD
- Piers Daniell, NVIDIA
- Alexander Galazin, Arm
- Jesse Hall, Google
- Daniel Rakos, AMD

Description

This extension provides a new physical device query which allows retrieving information about the driver implementation, allowing applications to determine which physical device corresponds to which particular vendor's driver, and which conformance test suite version the driver implementation is compliant with.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- [VkConformanceVersionKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceDriverPropertiesKHR](#)

New Enums

- [VkDriverIdKHR](#)

New Enum Constants

- [VK_KHR_DRIVER_PROPERTIES_EXTENSION_NAME](#)

- `VK_KHR_DRIVER_PROPERTIES_SPEC_VERSION`
- `VK_MAX_DRIVER_INFO_SIZE_KHR`
- `VK_MAX_DRIVER_NAME_SIZE_KHR`
- Extending `VkDriverId`:
 - `VK_DRIVER_ID_AMD_OPEN_SOURCE_KHR`
 - `VK_DRIVER_ID_AMD_PROPRIETARY_KHR`
 - `VK_DRIVER_ID_ARM_PROPRIETARY_KHR`
 - `VK_DRIVER_ID_BROADCOM_PROPRIETARY_KHR`
 - `VK_DRIVER_ID_GGP_PROPRIETARY_KHR`
 - `VK_DRIVER_ID_GOOGLE_SWIFTSHADER_KHR`
 - `VK_DRIVER_ID_IMAGINATION_PROPRIETARY_KHR`
 - `VK_DRIVER_ID_INTEL_OPEN_SOURCE_MESA_KHR`
 - `VK_DRIVER_ID_INTEL_PROPRIETARY_WINDOWS_KHR`
 - `VK_DRIVER_ID_MESA_RADV_KHR`
 - `VK_DRIVER_ID_NVIDIA_PROPRIETARY_KHR`
 - `VK_DRIVER_ID_QUALCOMM_PROPRIETARY_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRIVER_PROPERTIES_KHR`

Version History

- Revision 1, 2018-04-11 (Daniel Rakos)
 - Internal revisions

VK_KHR_dynamic_rendering

Name String

`VK_KHR_dynamic_rendering`

Extension Type

Device extension

Registered Extension Number

45

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

and

[VK_KHR_depth_stencil_resolve](#)

or

[Vulkan Version 1.2](#)

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Tobias Hector [\[GitHub\]](#)[tobski](#)

Extension Proposal

[VK_KHR_dynamic_rendering](#)

Other Extension Metadata

Last Modified Date

2021-10-06

Contributors

- Tobias Hector, AMD
- Arseny Kapoulkine, Roblox
- François Duranleau, Gameloft
- Stuart Smith, AMD
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- James Fitzpatrick, Imagination
- Piotr Byszewski, Mobica
- Jesse Hall, Google
- Mike Blumenkrantz, Valve

Description

This extension allows applications to create single-pass render pass instances without needing to create render pass objects or framebuffers. Dynamic render passes can also span across multiple primary command buffers, rather than relying on secondary command buffers.

This extension also incorporates [VK_ATTACHMENT_STORE_OP_NONE_KHR](#) from

`VK_QCOM_render_pass_store_ops`, enabling applications to avoid unnecessary synchronization when an attachment is not written during a render pass.

New Commands

- [vkCmdBeginRenderingKHR](#)
- [vkCmdEndRenderingKHR](#)

New Structures

- [VkRenderingAttachmentInfoKHR](#)
- [VkRenderingInfoKHR](#)
- Extending [VkCommandBufferInheritanceInfo](#):
 - [VkCommandBufferInheritanceRenderingInfoKHR](#)
- Extending [VkGraphicsPipelineCreateInfo](#):
 - [VkPipelineRenderingCreateInfoKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceDynamicRenderingFeaturesKHR](#)

New Enums

- [VkRenderingFlagBitsKHR](#)

New Bitmasks

- [VkRenderingFlagsKHR](#)

New Enum Constants

- [VK_KHR_DYNAMIC_RENDERING_EXTENSION_NAME](#)
- [VK_KHR_DYNAMIC_RENDERING_SPEC_VERSION](#)
- Extending [VkAttachmentStoreOp](#):
 - [VK_ATTACHMENT_STORE_OP_NONE_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_COMMAND_BUFFER_INHERITANCE_RENDERING_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_RENDERING_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_RENDERING_INFO_KHR](#)

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External

interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Version History

- Revision 1, 2021-10-06 (Tobias Hector)
 - Initial revision

VK_KHR_dynamic_rendering_local_read

Name String

VK_KHR_dynamic_rendering_local_read

Extension Type

Device extension

Registered Extension Number

233

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_dynamic_rendering](#)

or

[Vulkan Version 1.3](#)

Contact

- Tobias Hector [\[GitHub\]](#)tobski

Extension Proposal

[VK_KHR_dynamic_rendering_local_read](#)

Other Extension Metadata

Last Modified Date

2023-11-03

Contributors

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- Pan Gao, Huawei
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- Shahbaz Youssefi, Google
- Alyssa Rosenzweig, Valve
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- Graeme Leese, Broadcom
- Piers Daniell, Nvidia
- Stuart Smith, AMD
- Daniel Story, Nintendo
- James Fitzpatrick, Imagination
- Piotr Byszewski, Mobica
- Spencer Fricke, LunarG
- Tom Olson, Arm
- Michal Pietrasiuk, Intel
- Matthew Netsch, Qualcomm
- Marty Johnson, Khronos
- Wyvern Wang, Huawei
- Jeff Bolz, Nvidia
- Samuel (Sheng-Wen) Huang, MediaTek

Description

This extension enables reads from attachments and resources written by previous fragment shaders within a dynamic render pass.

New Commands

- [vkCmdSetRenderingAttachmentLocationsKHR](#)
- [vkCmdSetRenderingInputAttachmentIndicesKHR](#)

New Structures

- Extending [VkGraphicsPipelineCreateInfo](#), [VkCommandBufferInheritanceInfo](#):
 - [VkRenderingAttachmentLocationInfoKHR](#)
 - [VkRenderingInputAttachmentIndexInfoKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceDynamicRenderingLocalReadFeaturesKHR](#)

New Enum Constants

- [VK_KHR_DYNAMIC_RENDERING_LOCAL_READ_EXTENSION_NAME](#)

- [VK_KHR_DYNAMIC_RENDERING_LOCAL_READ_SPEC_VERSION](#)
- Extending [VkImageLayout](#):
 - [VK_IMAGE_LAYOUT_RENDERING_LOCAL_READ_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_LOCAL_READ_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_RENDERING_ATTACHMENT_LOCATION_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_RENDERING_INPUT_ATTACHMENT_INDEX_INFO_KHR](#)

Version History

- Revision 1, 2023-11-03 (Tobias Hector)
 - Initial revision

VK_KHR_external_fence

Name String

[VK_KHR_external_fence](#)

Extension Type

Device extension

Registered Extension Number

114

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_fence_capabilities](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jesse Hall [\[GitHub\]](#)[critsec](#)

Other Extension Metadata

Last Modified Date

2017-05-08

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Cass Everitt, Oculus
- Contributors to [VK_KHR_external_semaphore](#)

Description

An application using external memory may wish to synchronize access to that memory using fences. This extension enables an application to create fences from which non-Vulkan handles that reference the underlying synchronization primitive can be exported.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending [VkFenceCreateInfo](#):
 - [VkExportFenceCreateInfoKHR](#)

New Enums

- [VkFenceImportFlagBitsKHR](#)

New Bitmasks

- [VkFenceImportFlagsKHR](#)

New Enum Constants

- [VK_KHR_EXTERNAL_FENCE_EXTENSION_NAME](#)
- [VK_KHR_EXTERNAL_FENCE_SPEC_VERSION](#)
- Extending [VkFenceImportFlagBits](#):
 - [VK_FENCE_IMPORT_TEMPORARY_BIT_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_EXPORT_FENCE_CREATE_INFO_KHR](#)

Issues

This extension borrows concepts, semantics, and language from [VK_KHR_external_semaphore](#). That extension's issues apply equally to this extension.

Version History

- Revision 1, 2017-05-08 (Jesse Hall)
 - Initial revision

VK_KHR_external_fence_capabilities

Name String

VK_KHR_external_fence_capabilities

Extension Type

Instance extension

Registered Extension Number

113

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jesse Hall [\[GitHub\]](#)[critsec](#)

Other Extension Metadata

Last Modified Date

2017-05-08

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Cass Everitt, Oculus
- Contributors to [VK_KHR_external_semaphore_capabilities](#)

Description

An application may wish to reference device fences in multiple Vulkan logical devices or instances, in multiple processes, and/or in multiple APIs. This extension provides a set of capability queries and handle definitions that allow an application to determine what types of “external” fence handles an implementation supports for a given set of use cases.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- [vkGetPhysicalDeviceExternalFencePropertiesKHR](#)

New Structures

- [VkExternalFencePropertiesKHR](#)
- [VkPhysicalDeviceExternalFenceInfoKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceIDPropertiesKHR](#)

New Enums

- [VkExternalFenceFeatureFlagBitsKHR](#)
- [VkExternalFenceHandleTypeFlagBitsKHR](#)

New Bitmasks

- [VkExternalFenceFeatureFlagsKHR](#)
- [VkExternalFenceHandleTypeFlagsKHR](#)

New Enum Constants

- [VK_KHR_EXTERNAL_FENCE_CAPABILITIES_EXTENSION_NAME](#)
- [VK_KHR_EXTERNAL_FENCE_CAPABILITIES_SPEC_VERSION](#)
- [VK_LUID_SIZE_KHR](#)
- Extending [VkExternalFenceFeatureFlagBits](#):
 - [VK_EXTERNAL_FENCE_FEATURE_EXPORTABLE_BIT_KHR](#)
 - [VK_EXTERNAL_FENCE_FEATURE_IMPORTABLE_BIT_KHR](#)
- Extending [VkExternalFenceHandleTypeFlagBits](#):
 - [VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_FD_BIT_KHR](#)
 - [VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR](#)
 - [VK_EXTERNAL_FENCE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR](#)

- `VK_EXTERNAL_FENCE_HANDLE_TYPE_SYNC_FD_BIT_KHR`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_EXTERNAL_FENCE_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_FENCE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES_KHR`

Version History

- Revision 1, 2017-05-08 (Jesse Hall)
 - Initial version

VK_KHR_external_fence_fd

Name String

`VK_KHR_external_fence_fd`

Extension Type

Device extension

Registered Extension Number

116

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_fence](#)

or

[Vulkan Version 1.1](#)

Contact

- Jesse Hall [\[GitHub\]](#)[critsec](#)

Other Extension Metadata

Last Modified Date

2017-05-08

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Cass Everitt, Oculus
- Contributors to [VK_KHR_external_semaphore_fd](#)

Description

An application using external memory may wish to synchronize access to that memory using fences. This extension enables an application to export fence payload to and import fence payload from POSIX file descriptors.

New Commands

- [vkGetFenceFdKHR](#)
- [vkImportFenceFdKHR](#)

New Structures

- [VkFenceGetFdInfoKHR](#)
- [VkImportFenceFdInfoKHR](#)

New Enum Constants

- [VK_KHR_EXTERNAL_FENCE_FD_EXTENSION_NAME](#)
- [VK_KHR_EXTERNAL_FENCE_FD_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_FENCE_GET_FD_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_IMPORT_FENCE_FD_INFO_KHR](#)

Issues

This extension borrows concepts, semantics, and language from [VK_KHR_external_semaphore_fd](#). That extension's issues apply equally to this extension.

Version History

- Revision 1, 2017-05-08 (Jesse Hall)
 - Initial revision

VK_KHR_external_fence_win32

Name String

[VK_KHR_external_fence_win32](#)

Extension Type

Device extension

Registered Extension Number

115

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_fence](#)

Contact

- Jesse Hall [\[GitHub\]](#)[critsec](#)

Other Extension Metadata

Last Modified Date

2017-05-08

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Cass Everitt, Oculus
- Contributors to [VK_KHR_external_semaphore_win32](#)

Description

An application using external memory may wish to synchronize access to that memory using fences. This extension enables an application to export fence payload to and import fence payload from Windows handles.

New Commands

- [vkGetFenceWin32HandleKHR](#)
- [vkImportFenceWin32HandleKHR](#)

New Structures

- [VkFenceGetWin32HandleInfoKHR](#)
- [VkImportFenceWin32HandleInfoKHR](#)
- Extending [VkFenceCreateInfo](#):

- [VkExportFenceWin32HandleInfoKHR](#)

New Enum Constants

- `VK_KHR_EXTERNAL_FENCE_WIN32_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_FENCE_WIN32_SPEC_VERSION`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_EXPORT_FENCE_WIN32_HANDLE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_FENCE_GET_WIN32_HANDLE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_IMPORT_FENCE_WIN32_HANDLE_INFO_KHR`

Issues

This extension borrows concepts, semantics, and language from [VK_KHR_external_semaphore_win32](#). That extension's issues apply equally to this extension.

1) Should D3D12 fence handle types be supported, like they are for semaphores?

RESOLVED: No. Doing so would require extending the fence signal and wait operations to provide values to signal / wait for, like [VkD3D12FenceSubmitInfoKHR](#) does. A D3D12 fence can be signaled by importing it into a [VkSemaphore](#) instead of a [VkFence](#), and applications can check status or wait on the D3D12 fence using non-Vulkan APIs. The convenience of being able to do these operations on [VkFence](#) objects does not justify the extra API complexity.

Version History

- Revision 1, 2017-05-08 (Jesse Hall)
 - Initial revision

VK_KHR_external_memory

Name String

`VK_KHR_external_memory`

Extension Type

Device extension

Registered Extension Number

73

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_memory_capabilities](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2016-10-20

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with [VK_KHR_dedicated_allocation](#).
- Interacts with [VK_NV_dedicated_allocation](#).

Contributors

- Faith Ekstrand, Intel
- Ian Elliott, Google
- Jesse Hall, Google
- Tobias Hector, Imagination Technologies
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Matthew Netsch, Qualcomm Technologies, Inc.
- Daniel Rakos, AMD
- Carsten Rohde, NVIDIA
- Ray Smith, ARM
- Lina Versace, Google

Description

An application may wish to reference device memory in multiple Vulkan logical devices or instances, in multiple processes, and/or in multiple APIs. This extension enables an application to export non-Vulkan handles from Vulkan memory objects such that the underlying resources can be referenced outside the scope of the Vulkan logical device that created them.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending [VkBufferCreateInfo](#):
 - [VkExternalMemoryBufferCreateInfoKHR](#)
- Extending [VkImageCreateInfo](#):
 - [VkExternalMemoryImageCreateInfoKHR](#)
- Extending [VkMemoryAllocateInfo](#):
 - [VkExportMemoryAllocateInfoKHR](#)

New Enum Constants

- [VK_KHR_EXTERNAL_MEMORY_EXTENSION_NAME](#)
- [VK_KHR_EXTERNAL_MEMORY_SPEC_VERSION](#)
- [VK_QUEUE_FAMILY_EXTERNAL_KHR](#)
- Extending [VkResult](#):
 - [VK_ERROR_INVALID_EXTERNAL_HANDLE_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_EXPORT_MEMORY_ALLOCATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_BUFFER_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_IMAGE_CREATE_INFO_KHR](#)

Issues

1) How do applications correlate two physical devices across process or Vulkan instance boundaries?

RESOLVED: New device ID fields have been introduced by [VK_KHR_external_memory_capabilities](#). These fields, combined with the existing [VkPhysicalDeviceProperties::driverVersion](#) field can be used to identify compatible devices across processes, drivers, and APIs. [VkPhysicalDeviceProperties::pipelineCacheUUID](#) is not sufficient for this purpose because despite its description in the specification, it need only identify a unique pipeline cache format in practice. Multiple devices may be able to use the same pipeline cache data, and hence it would be desirable for all of them to have the same pipeline cache UUID. However, only the same concrete physical device can be used when sharing memory, so an actual unique device ID was introduced. Further, the pipeline cache UUID was specific to Vulkan, but correlation with other, non-extensible APIs is required to enable interoperation with those APIs.

2) If memory objects are shared between processes and APIs, is this considered aliasing according to the rules outlined in the [Memory Aliasing](#) section?

RESOLVED: Yes. Applications must take care to obey all restrictions imposed on aliased resources when using memory across multiple Vulkan instances or other APIs.

3) Are new image layouts or metadata required to specify image layouts and layout transitions compatible with non-Vulkan APIs, or with other instances of the same Vulkan driver?

RESOLVED: Separate instances of the same Vulkan driver running on the same GPU should have identical internal layout semantics, so applications have the tools they need to ensure views of images are consistent between the two instances. Other APIs will fall into two categories: Those that are Vulkan-compatible, and those that are Vulkan-incompatible. Vulkan-incompatible APIs will require the image to be in the GENERAL layout whenever they are accessing them.

Note this does not attempt to address cross-device transitions, nor transitions to engines on the same device which are not visible within the Vulkan API. Both of these are beyond the scope of this extension.

4) Is a new barrier flag or operation of some type needed to prepare external memory for handoff to another Vulkan instance or API and/or receive it from another instance or API?

RESOLVED: Yes. Some implementations need to perform additional cache management when transitioning memory between address spaces and other APIs, instances, or processes which may operate in a separate address space. Options for defining this transition include:

- A new structure that can be added to the `pNext` list in `VkMemoryBarrier`, `VkBufferMemoryBarrier`, and `VkImageMemoryBarrier`.
- A new bit in `VkAccessFlags` that can be set to indicate an “external” access.
- A new bit in `VkDependencyFlags`
- A new special queue family that represents an “external” queue.

A new structure has the advantage that the type of external transition can be described in as much detail as necessary. However, there is not currently a known need for anything beyond differentiating between external and internal accesses, so this is likely an over-engineered solution. The access flag bit has the advantage that it can be applied at buffer, image, or global granularity, and semantically it maps pretty well to the operation being described. Additionally, the API already includes `VK_ACCESS_MEMORY_READ_BIT` and `VK_ACCESS_MEMORY_WRITE_BIT` which appear to be intended for this purpose. However, there is no obvious pipeline stage that would correspond to an external access, and therefore no clear way to use `VK_ACCESS_MEMORY_READ_BIT` or `VK_ACCESS_MEMORY_WRITE_BIT`. `VkDependencyFlags` and `VkPipelineStageFlags` operate at command granularity rather than image or buffer granularity, which would make an entire pipeline barrier an internal→external or external→internal barrier. This may not be a problem in practice, but seems like the wrong scope. Another downside of `VkDependencyFlags` is that it lacks inherent directionality: there are no `src` and `dst` variants of it in the barrier or dependency description semantics, so two bits might need to be added to describe both internal→external and external→internal transitions. Transitioning a resource to a special queue family corresponds well with the operation of transitioning to a separate Vulkan instance, in that both operations ideally include scheduling a barrier on both sides of the transition: Both the releasing and the acquiring queue or process. Using a special queue family requires adding an additional reserved queue family index. Re-using `VK_QUEUE_FAMILY_IGNORED` would have left it unclear how to transition a concurrent usage resource

from one process to another, since the semantics would have likely been equivalent to the currently-ignored transition of `VK_QUEUE_FAMILY_IGNORED` → `VK_QUEUE_FAMILY_IGNORED`. Fortunately, creating a new reserved queue family index is not invasive.

Based on the above analysis, the approach of transitioning to a special “external” queue family was chosen.

5) Do internal driver memory arrangements and/or other internal driver image properties need to be exported and imported when sharing images across processes or APIs.

RESOLVED: Some vendors claim this is necessary on their implementations, but it was determined that the security risks of allowing opaque metadata to be passed from applications to the driver were too high. Therefore, implementations which require metadata will need to associate it with the objects represented by the external handles, and rely on the dedicated allocation mechanism to associate the exported and imported memory objects with a single image or buffer.

6) Most prior interoperation and cross-process sharing APIs have been based on image-level sharing. Should Vulkan sharing be based on memory-object sharing or image sharing?

RESOLVED: These extensions have assumed memory-level sharing is the correct granularity. Vulkan is a lower-level API than most prior APIs, and as such attempts to closely align with to the underlying primitives of the hardware and system-level drivers it abstracts. In general, the resource that holds the backing store for both images and buffers of various types is memory. Images and buffers are merely metadata containing brief descriptions of the layout of bits within that memory.

Because memory object-based sharing is aligned with the overall Vulkan API design, it enables the full range of Vulkan capabilities with external objects. External memory can be used as backing for sparse images, for example, whereas such usage would be awkward at best with a sharing mechanism based on higher-level primitives such as images. Further, aligning the mechanism with the API in this way provides some hope of trivial compatibility with future API enhancements. If new objects backed by memory objects are added to the API, they too can be used across processes with minimal additions to the base external memory APIs.

Earlier APIs implemented interop at a higher level, and this necessitated entirely separate sharing APIs for images and buffers. To co-exist and interoperate with those APIs, the Vulkan external sharing mechanism must accommodate their model. However, if it can be agreed that memory-based sharing is the more desirable and forward-looking design, legacy interoperation constraints can be considered another reason to favor memory-based sharing: while native and legacy driver primitives that may be used to implement sharing may not be as low-level as the API here suggests, raw memory is still the least common denominator among the types. Image-based sharing can be cleanly derived from a set of base memory- object sharing APIs with minimal effort, whereas image-based sharing does not generalize well to buffer or raw-memory sharing. Therefore, following the general Vulkan design principle of minimalism, it is better to expose interoperability with image-based native and external primitives via the memory sharing API, and place sufficient limits on their usage to ensure they can be used only as backing for equivalent Vulkan images. This provides a consistent API for applications regardless of which platform or external API they are targeting, which makes development of multi-API and multi-platform applications simpler.

7) Should Vulkan define a common external handle type and provide Vulkan functions to facilitate cross-process sharing of such handles rather than relying on native handles to define the external objects?

RESOLVED: No. Cross-process sharing of resources is best left to native platforms. There are myriad security and extensibility issues with such a mechanism, and attempting to re-solve all those issues within Vulkan does not align with Vulkan's purpose as a graphics API. If desired, such a mechanism could be built as a layer or helper library on top of the opaque native handle defined in this family of extensions.

8) Must implementations provide additional guarantees about state implicitly included in memory objects for those memory objects that may be exported?

RESOLVED: Implementations must ensure that sharing memory objects does not transfer any information between the exporting and importing instances and APIs other than that required to share the data contained in the memory objects explicitly shared. As specific examples, data from previously freed memory objects that used the same underlying physical memory, and data from memory objects using adjacent physical memory must not be visible to applications importing an exported memory object.

9) Must implementations validate external handles the application provides as inputs to memory import operations?

RESOLVED: Implementations must return an error to the application if the provided memory handle cannot be used to complete the requested import operation. However, implementations need not validate handles are of the exact type specified by the application.

Version History

- Revision 1, 2016-10-20 (James Jones)
 - Initial version

VK_KHR_external_memory_capabilities

Name String

VK_KHR_external_memory_capabilities

Extension Type

Instance extension

Registered Extension Number

72

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2016-10-17

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with [VK_KHR_dedicated_allocation](#).
- Interacts with [VK_NV_dedicated_allocation](#).

Contributors

- Ian Elliott, Google
- Jesse Hall, Google
- James Jones, NVIDIA

Description

An application may wish to reference device memory in multiple Vulkan logical devices or instances, in multiple processes, and/or in multiple APIs. This extension provides a set of capability queries and handle definitions that allow an application to determine what types of “external” memory handles an implementation supports for a given set of use cases.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- [vkGetPhysicalDeviceExternalBufferPropertiesKHR](#)

New Structures

- [VkExternalBufferPropertiesKHR](#)
- [VkExternalMemoryPropertiesKHR](#)

- [VkPhysicalDeviceExternalBufferInfoKHR](#)
- Extending [VkImageFormatProperties2](#):
 - [VkExternalImageFormatPropertiesKHR](#)
- Extending [VkPhysicalDeviceImageFormatInfo2](#):
 - [VkPhysicalDeviceExternalImageFormatInfoKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceIDPropertiesKHR](#)

New Enums

- [VkExternalMemoryFeatureFlagBitsKHR](#)
- [VkExternalMemoryHandleTypeFlagBitsKHR](#)

New Bitmasks

- [VkExternalMemoryFeatureFlagsKHR](#)
- [VkExternalMemoryHandleTypeFlagsKHR](#)

New Enum Constants

- [VK_KHR_EXTERNAL_MEMORY_CAPABILITIES_EXTENSION_NAME](#)
- [VK_KHR_EXTERNAL_MEMORY_CAPABILITIES_SPEC_VERSION](#)
- [VK_LUID_SIZE_KHR](#)
- Extending [VkExternalMemoryFeatureFlagBits](#):
 - [VK_EXTERNAL_MEMORY_FEATURE_DEDICATED_ONLY_BIT_KHR](#)
 - [VK_EXTERNAL_MEMORY_FEATURE_EXPORTABLE_BIT_KHR](#)
 - [VK_EXTERNAL_MEMORY_FEATURE_IMPORTABLE_BIT_KHR](#)
- Extending [VkExternalMemoryHandleTypeFlagBits](#):
 - [VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_BIT_KHR](#)
 - [VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D11_TEXTURE_KMT_BIT_KHR](#)
 - [VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_HEAP_BIT_KHR](#)
 - [VK_EXTERNAL_MEMORY_HANDLE_TYPE_D3D12_RESOURCE_BIT_KHR](#)
 - [VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_FD_BIT_KHR](#)
 - [VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR](#)
 - [VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_EXTERNAL_BUFFER_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_EXTERNAL_IMAGE_FORMAT_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_BUFFER_INFO_KHR](#)

- `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_IMAGE_FORMAT_INFO_KHR`
- `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES_KHR`

Issues

1) Why do so many external memory capabilities need to be queried on a per-memory-handle-type basis?

PROPOSED RESOLUTION: This is because some handle types are based on OS-native objects that have far more limited capabilities than the very generic Vulkan memory objects. Not all memory handle types can name memory objects that support 3D images, for example. Some handle types cannot even support the deferred image and memory binding behavior of Vulkan and require specifying the image when allocating or importing the memory object.

2) Do the [VkExternalImageFormatPropertiesKHR](#) and [VkExternalBufferPropertiesKHR](#) structs need to include a list of memory type bits that support the given handle type?

PROPOSED RESOLUTION: No. The memory types that do not support the handle types will simply be filtered out of the results returned by [vkGetImageMemoryRequirements](#) and [vkGetBufferMemoryRequirements](#) when a set of handle types was specified at image or buffer creation time.

3) Should the non-opaque handle types be moved to their own extension?

PROPOSED RESOLUTION: Perhaps. However, defining the handle type bits does very little and does not require any platform-specific types on its own, and it is easier to maintain the bitfield values in a single extension for now. Presumably more handle types could be added by separate extensions though, and it would be midly weird to have some platform-specific ones defined in the core spec and some in extensions

4) Do we need a `D3D11_TILEPOOL` type?

PROPOSED RESOLUTION: No. This is technically possible, but the synchronization is awkward. D3D11 surfaces must be synchronized using shared mutexes, and these synchronization primitives are shared by the entire memory object, so D3D11 shared allocations divided among multiple buffer and image bindings may be difficult to synchronize.

5) Should the Windows 7-compatible handle types be named “KMT” handles or “GLOBAL_SHARE” handles?

PROPOSED RESOLUTION: KMT, simply because it is more concise.

6) How do applications identify compatible devices and drivers across instance, process, and API boundaries when sharing memory?

PROPOSED RESOLUTION: New device properties are exposed that allow applications to correctly correlate devices and drivers. A device and driver UUID that must both match to ensure sharing compatibility between two Vulkan instances, or a Vulkan instance and an extensible external API are added. To allow correlating with Direct3D devices, a device LUID is added that corresponds to a DXGI adapter LUID. A driver ID is not needed for Direct3D because mismatched driver component

versions are not currently supported on the Windows OS. Should support for such configurations be introduced at the OS level, further Vulkan extensions would be needed to correlate userspace component builds.

Version History

- Revision 1, 2016-10-17 (James Jones)
 - Initial version

VK_KHR_external_memory_fd

Name String

VK_KHR_external_memory_fd

Extension Type

Device extension

Registered Extension Number

75

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_memory](#)

or

[Vulkan Version 1.1](#)

Contact

- James Jones [\[GitHub\]cubanismo](#)

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA

Description

An application may wish to reference device memory in multiple Vulkan logical devices or instances, in multiple processes, and/or in multiple APIs. This extension enables an application to export POSIX file descriptor handles from Vulkan memory objects and to import Vulkan memory objects from POSIX file descriptor handles exported from other Vulkan memory objects or from similar resources in other APIs.

New Commands

- [vkGetMemoryFdKHR](#)
- [vkGetMemoryFdPropertiesKHR](#)

New Structures

- [VkMemoryFdPropertiesKHR](#)
- [VkMemoryGetFdInfoKHR](#)
- Extending [VkMemoryAllocateInfo](#):
 - [VkImportMemoryFdInfoKHR](#)

New Enum Constants

- [VK_KHR_EXTERNAL_MEMORY_FD_EXTENSION_NAME](#)
- [VK_KHR_EXTERNAL_MEMORY_FD_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_IMPORT_MEMORY_FD_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_MEMORY_FD_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_MEMORY_GET_FD_INFO_KHR](#)

Issues

1) Does the application need to close the file descriptor returned by [vkGetMemoryFdKHR](#)?

RESOLVED: Yes, unless it is passed back in to a driver instance to import the memory. A successful get call transfers ownership of the file descriptor to the application, and a successful import transfers it back to the driver. Destroying the original memory object will not close the file descriptor or remove its reference to the underlying memory resource associated with it.

2) Do drivers ever need to expose multiple file descriptors per memory object?

RESOLVED: No. This would indicate there are actually multiple memory objects, rather than a single memory object.

3) How should the valid size and memory type for POSIX file descriptor memory handles created outside of Vulkan be specified?

RESOLVED: The valid memory types are queried directly from the external handle. The size will be

specified by future extensions that introduce such external memory handle types.

Version History

- Revision 1, 2016-10-21 (James Jones)
 - Initial revision

VK_KHR_external_memory_win32

Name String

VK_KHR_external_memory_win32

Extension Type

Device extension

Registered Extension Number

74

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_memory](#)

or

[Vulkan Version 1.1](#)

Contact

- James Jones [\[GitHub\]](#)cubanismo

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Carsten Rohde, NVIDIA

Description

An application may wish to reference device memory in multiple Vulkan logical devices or instances, in multiple processes, and/or in multiple APIs. This extension enables an application to export Windows handles from Vulkan memory objects and to import Vulkan memory objects from Windows handles exported from other Vulkan memory objects or from similar resources in other APIs.

New Commands

- [vkGetMemoryWin32HandleKHR](#)
- [vkGetMemoryWin32HandlePropertiesKHR](#)

New Structures

- [VkMemoryGetWin32HandleInfoKHR](#)
- [VkMemoryWin32HandlePropertiesKHR](#)
- Extending [VkMemoryAllocateInfo](#):
 - [VkExportMemoryWin32HandleInfoKHR](#)
 - [VkImportMemoryWin32HandleInfoKHR](#)

New Enum Constants

- [VK_KHR_EXTERNAL_MEMORY_WIN32_EXTENSION_NAME](#)
- [VK_KHR_EXTERNAL_MEMORY_WIN32_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_EXPORT_MEMORY_WIN32_HANDLE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_IMPORT_MEMORY_WIN32_HANDLE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_MEMORY_GET_WIN32_HANDLE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_MEMORY_WIN32_HANDLE_PROPERTIES_KHR](#)

Issues

1) Do applications need to call [CloseHandle\(\)](#) on the values returned from [vkGetMemoryWin32HandleKHR](#) when [handleType](#) is [VK_EXTERNAL_MEMORY_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR](#)?

RESOLVED: Yes. A successful get call transfers ownership of the handle to the application. Destroying the memory object will not destroy the handle or the handle's reference to the underlying memory resource. Unlike file descriptor opaque handles, win32 opaque handle ownership can not be transferred back to a driver by an import operation.

2) Should the language regarding KMT/Windows 7 handles be moved to a separate extension so that it can be deprecated over time?

RESOLVED: No. Support for them can be deprecated by drivers if they choose, by no longer

returning them in the supported handle types of the instance level queries.

3) How should the valid size and memory type for windows memory handles created outside of Vulkan be specified?

RESOLVED: The valid memory types are queried directly from the external handle. The size is determined by the associated image or buffer memory requirements for external handle types that require dedicated allocations, and by the size specified when creating the object from which the handle was exported for other external handle types.

Version History

- Revision 1, 2016-10-21 (James Jones)
 - Initial revision

VK_KHR_external_semaphore

Name String

VK_KHR_external_semaphore

Extension Type

Device extension

Registered Extension Number

78

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_semaphore_capabilities](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- James Jones [\[GitHub\]cubanismo](#)

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- Faith Ekstrand, Intel
- Jesse Hall, Google
- Tobias Hector, Imagination Technologies
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Matthew Netsch, Qualcomm Technologies, Inc.
- Ray Smith, ARM
- Lina Versace, Google

Description

An application using external memory may wish to synchronize access to that memory using semaphores. This extension enables an application to create semaphores from which non-Vulkan handles that reference the underlying synchronization primitive can be exported.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending [VkSemaphoreCreateInfo](#):
 - [VkExportSemaphoreCreateInfoKHR](#)

New Enums

- [VkSemaphoreImportFlagBitsKHR](#)

New Bitmasks

- [VkSemaphoreImportFlagsKHR](#)

New Enum Constants

- [VK_KHR_EXTERNAL_SEMAPHORE_EXTENSION_NAME](#)
- [VK_KHR_EXTERNAL_SEMAPHORE_SPEC_VERSION](#)
- Extending [VkSemaphoreImportFlagBits](#):
 - [VK_SEMAPHORE_IMPORT_TEMPORARY_BIT_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_CREATE_INFO_KHR](#)

Issues

1) Should there be restrictions on what side effects can occur when waiting on imported semaphores that are in an invalid state?

RESOLVED: Yes. Normally, validating such state would be the responsibility of the application, and the implementation would be free to enter an undefined state if valid usage rules were violated. However, this could cause security concerns when using imported semaphores, as it would require the importing application to trust the exporting application to ensure the state is valid. Requiring this level of trust is undesirable for many potential use cases.

2) Must implementations validate external handles the application provides as input to semaphore state import operations?

RESOLVED: Implementations must return an error to the application if the provided semaphore state handle cannot be used to complete the requested import operation. However, implementations need not validate handles are of the exact type specified by the application.

Version History

- Revision 1, 2016-10-21 (James Jones)
 - Initial revision

VK_KHR_external_semaphore_capabilities

Name String

VK_KHR_external_semaphore_capabilities

Extension Type

Instance extension

Registered Extension Number

77

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2016-10-20

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA

Description

An application may wish to reference device semaphores in multiple Vulkan logical devices or instances, in multiple processes, and/or in multiple APIs. This extension provides a set of capability queries and handle definitions that allow an application to determine what types of “external” semaphore handles an implementation supports for a given set of use cases.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- [vkGetPhysicalDeviceExternalSemaphorePropertiesKHR](#)

New Structures

- [VkExternalSemaphorePropertiesKHR](#)
- [VkPhysicalDeviceExternalSemaphoreInfoKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceIDPropertiesKHR](#)

New Enums

- [VkExternalSemaphoreFeatureFlagBitsKHR](#)
- [VkExternalSemaphoreHandleTypeFlagBitsKHR](#)

New Bitmasks

- [VkExternalSemaphoreFeatureFlagsKHR](#)

- [VkExternalSemaphoreHandleTypeFlagsKHR](#)

New Enum Constants

- `VK_KHR_EXTERNAL_SEMAPHORE_CAPABILITIES_EXTENSION_NAME`
- `VK_KHR_EXTERNAL_SEMAPHORE_CAPABILITIES_SPEC_VERSION`
- `VK_LUID_SIZE_KHR`
- Extending [VkExternalSemaphoreFeatureFlagBits](#):
 - `VK_EXTERNAL_SEMAPHORE_FEATURE_EXPORTABLE_BIT_KHR`
 - `VK_EXTERNAL_SEMAPHORE_FEATURE_IMPORTABLE_BIT_KHR`
- Extending [VkExternalSemaphoreHandleTypeFlagBits](#):
 - `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_D3D12_FENCE_BIT_KHR`
 - `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_FD_BIT_KHR`
 - `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR`
 - `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_KMT_BIT_KHR`
 - `VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_SYNC_FD_BIT_KHR`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_EXTERNAL_SEMAPHORE_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_SEMAPHORE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ID_PROPERTIES_KHR`

Version History

- Revision 1, 2016-10-20 (James Jones)
 - Initial revision

VK_KHR_external_semaphore_fd

Name String

`VK_KHR_external_semaphore_fd`

Extension Type

Device extension

Registered Extension Number

80

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_semaphore](#)

or

[Vulkan Version 1.1](#)

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Carsten Rohde, NVIDIA

Description

An application using external memory may wish to synchronize access to that memory using semaphores. This extension enables an application to export semaphore payload to and import semaphore payload from POSIX file descriptors.

New Commands

- [vkGetSemaphoreFdKHR](#)
- [vkImportSemaphoreFdKHR](#)

New Structures

- [VkImportSemaphoreFdInfoKHR](#)
- [VkSemaphoreGetFdInfoKHR](#)

New Enum Constants

- [VK_KHR_EXTERNAL_SEMAPHORE_FD_EXTENSION_NAME](#)
- [VK_KHR_EXTERNAL_SEMAPHORE_FD_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_FD_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_SEMAPHORE_GET_FD_INFO_KHR](#)

Issues

1) Does the application need to close the file descriptor returned by [vkGetSemaphoreFdKHR](#)?

RESOLVED: Yes, unless it is passed back in to a driver instance to import the semaphore. A successful get call transfers ownership of the file descriptor to the application, and a successful import transfers it back to the driver. Destroying the original semaphore object will not close the file descriptor or remove its reference to the underlying semaphore resource associated with it.

Version History

- Revision 1, 2016-10-21 (Jesse Hall)
 - Initial revision

VK_KHR_external_semaphore_win32

Name String

VK_KHR_external_semaphore_win32

Extension Type

Device extension

Registered Extension Number

79

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_semaphore](#)

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA

- Carsten Rohde, NVIDIA

Description

An application using external memory may wish to synchronize access to that memory using semaphores. This extension enables an application to export semaphore payload to and import semaphore payload from Windows handles.

New Commands

- [vkGetSemaphoreWin32HandleKHR](#)
- [vkImportSemaphoreWin32HandleKHR](#)

New Structures

- [VkImportSemaphoreWin32HandleInfoKHR](#)
- [VkSemaphoreGetWin32HandleInfoKHR](#)
- Extending [VkSemaphoreCreateInfo](#):
 - [VkExportSemaphoreWin32HandleInfoKHR](#)
- Extending [VkSubmitInfo](#):
 - [VkD3D12FenceSubmitInfoKHR](#)

New Enum Constants

- [VK_KHR_EXTERNAL_SEMAPHORE_WIN32_EXTENSION_NAME](#)
- [VK_KHR_EXTERNAL_SEMAPHORE_WIN32_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_D3D12_FENCE_SUBMIT_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_EXPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_IMPORT_SEMAPHORE_WIN32_HANDLE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_SEMAPHORE_GET_WIN32_HANDLE_INFO_KHR](#)

Issues

1) Do applications need to call [CloseHandle\(\)](#) on the values returned from [vkGetSemaphoreWin32HandleKHR](#) when [handleType](#) is [VK_EXTERNAL_SEMAPHORE_HANDLE_TYPE_OPAQUE_WIN32_BIT_KHR](#)?

RESOLVED: Yes. A successful get call transfers ownership of the handle to the application. Destroying the semaphore object will not destroy the handle or the handle's reference to the underlying semaphore resource. Unlike file descriptor opaque handles, win32 opaque handle ownership can not be transferred back to a driver by an import operation.

2) Should the language regarding KMT/Windows 7 handles be moved to a separate extension so that it can be deprecated over time?

RESOLVED: No. Support for them can be deprecated by drivers if they choose, by no longer returning them in the supported handle types of the instance level queries.

3) Should applications be allowed to specify additional object attributes for shared handles?

RESOLVED: Yes. Applications will be allowed to provide similar attributes to those they would to any other handle creation API.

4) How do applications communicate the desired fence values to use with **D3D12_FENCE**-based Vulkan semaphores?

RESOLVED: There are a couple of options. The values for the signaled and reset states could be communicated up front when creating the object and remain static for the life of the Vulkan semaphore, or they could be specified using auxiliary structures when submitting semaphore signal and wait operations, similar to what is done with the keyed mutex extensions. The latter is more flexible and consistent with the keyed mutex usage, but the former is a much simpler API.

Since Vulkan tends to favor flexibility and consistency over simplicity, a new structure specifying D3D12 fence acquire and release values is added to the [vkQueueSubmit](#) function.

Version History

- Revision 1, 2016-10-21 (James Jones)
 - Initial revision

VK_KHR_format_feature_flags2

Name String

VK_KHR_format_feature_flags2

Extension Type

Device extension

Registered Extension Number

361

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Lionel Landwerlin [\[GitHub\]](#)llandwerlin

Other Extension Metadata

Last Modified Date

2021-07-01

IP Status

No known IP claims.

Contributors

- Lionel Landwerlin, Intel
- Faith Ekstrand, Intel
- Tobias Hector, AMD
- Spencer Fricke, Samsung Electronics
- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, ARM

Description

This extension adds a new [VkFormatFeatureFlagBits2KHR](#) 64bits format feature flag type to extend the existing [VkFormatFeatureFlagBits](#) which is limited to 31 flags. At the time of this writing 29 bits of [VkFormatFeatureFlagBits](#) are already used.

Because [VkFormatProperties2](#) is already defined to extend the Vulkan 1.0 [vkGetPhysicalDeviceFormatProperties](#) command, this extension defines a new [VkFormatProperties3KHR](#) to extend the [VkFormatProperties](#).

On top of replicating all the bits from [VkFormatFeatureFlagBits](#), [VkFormatFeatureFlagBits2KHR](#) adds the following bits :

- [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT_KHR](#) and [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT_KHR](#) indicate that an implementation supports respectively reading and writing a given [VkFormat](#) through storage operations without specifying the format in the shader.
- [VK_FORMAT_FEATURE_2_SAMPLED_IMAGE_DEPTH_COMPARISON_BIT_KHR](#) indicates that an implementation supports depth comparison performed by [OpImage*Dref*](#) instructions on a given [VkFormat](#). Previously the result of executing a [OpImage*Dref*](#) instruction on an image view, where the [format](#) was not one of the depth/stencil formats with a depth component, was undefined. This bit clarifies on which formats such instructions can be used.

Prior to version 2 of this extension, implementations exposing the [shaderStorageImageReadWithoutFormat](#) and [shaderStorageImageWriteWithoutFormat](#) features may not report [VK_FORMAT_FEATURE_2_STORAGE_READ_WITHOUT_FORMAT_BIT_KHR](#) and [VK_FORMAT_FEATURE_2_STORAGE_WRITE_WITHOUT_FORMAT_BIT_KHR](#) in [VkFormatProperties3KHR::bufferFeatures](#). Despite this, buffer reads/writes are supported as intended by the original

features.

New Structures

- Extending [VkFormatProperties2](#):
 - [VkFormatProperties3KHR](#)

New Enums

- [VkFormatFeatureFlagBits2KHR](#)

New Bitmasks

- [VkFormatFeatureFlags2KHR](#)

New Enum Constants

- [VK_KHR_FORMAT_FEATURE_FLAGS_2_EXTENSION_NAME](#)
- [VK_KHR_FORMAT_FEATURE_FLAGS_2_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_3_KHR](#)

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Version History

- Revision 2, 2022-07-20 (Lionel Landwerlin)
 - Clarify that [VK_FORMAT_FEATURE_2_STORAGE_\(READ|WRITE\)_WITHOUT_FORMAT_BIT](#) also apply to buffer views.
- Revision 1, 2020-07-21 (Lionel Landwerlin)
 - Initial draft

VK_KHR_fragment_shader_barycentric

Name String

[VK_KHR_fragment_shader_barycentric](#)

Extension Type

Device extension

Registered Extension Number

323

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_fragment_shader_barycentric](#)

Contact

- Stu Smith

Extension Proposal

[VK_KHR_fragment_shader_barycentric](#)

Other Extension Metadata

Last Modified Date

2022-03-10

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for [GL_EXT_fragment_shader_barycentric](#)

Contributors

- Stu Smith, AMD
- Tobias Hector, AMD
- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, Arm
- Slawek Grajewski, Intel
- Pat Brown, NVIDIA
- Hans-Kristian Arntzen, Valve
- Contributors to the [VK_NV_fragment_shader_barycentric](#) specification

Description

This extension is based on the [VK_NV_fragment_shader_barycentric](#) extension, and adds support for the following SPIR-V extension in Vulkan:

- [SPV_KHR_fragment_shader_barycentric](#)

The extension provides access to three additional fragment shader variable decorations in SPIR-V:

- **PerVertexKHR**, which indicates that a fragment shader input will not have interpolated values, but instead must be accessed with an extra array index that identifies one of the vertices of the primitive producing the fragment
- **BaryCoordKHR**, which indicates that the variable is a three-component floating-point vector holding barycentric weights for the fragment produced using perspective interpolation
- **BaryCoordNoPerspKHR**, which indicates that the variable is a three-component floating-point vector holding barycentric weights for the fragment produced using linear interpolation

When using GLSL source-based shader languages, the following variables from **GL_EXT_fragment_shader_barycentric** map to these SPIR-V built-in decorations:

- `in vec3 gl_BaryCoordEXT;` → **BaryCoordKHR**
- `in vec3 gl_BaryCoordNoPerspEXT;` → **BaryCoordNoPerspKHR**

GLSL variables declared using the **pervertexEXT** GLSL qualifier are expected to be decorated with **PerVertexKHR** in SPIR-V.

New Structures

- Extending **VkPhysicalDeviceFeatures2**, **VkDeviceCreateInfo**:
 - **VkPhysicalDeviceFragmentShaderBarycentricFeaturesKHR**
- Extending **VkPhysicalDeviceProperties2**:
 - **VkPhysicalDeviceFragmentShaderBarycentricPropertiesKHR**

New Enum Constants

- **VK_KHR_FRAGMENT_SHADER_BARYCENTRIC_EXTENSION_NAME**
- **VK_KHR_FRAGMENT_SHADER_BARYCENTRIC_SPEC_VERSION**
- Extending **VkStructureType**:
 - **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_FEATURES_KHR**
 - **VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_BARYCENTRIC_PROPERTIES_KHR**

New Built-In Variables

- **BaryCoordKHR**
- **BaryCoordNoPerspKHR**

New SPIR-V Decorations

- **PerVertexKHR**

New SPIR-V Capabilities

- **FragmentBarycentricKHR**

Issues

1) What are the interactions with MSAA and how are `BaryCoordKHR` and `BaryCoordNoPerspKHR` interpolated?

RESOLVED: The inputs decorated with `BaryCoordKHR` or `BaryCoordNoPerspKHR` **may** also be decorated with the `Centroid` or `Sample` qualifiers to specify interpolation, like any other fragment shader input. If `shaderSampleRateInterpolationFunctions` is enabled, the extended instructions `InterpolateAtCentroid`, `InterpolateAtOffset`, and `InterpolateAtSample` from the GLSL.std.450 **may** also be used with inputs decorated with `BaryCoordKHR` or `BaryCoordNoPerspKHR`.

Version History

- Revision 1, 2022-03-10 (Stu Smith)
 - Initial revision

VK_KHR_fragment_shading_rate

Name String

`VK_KHR_fragment_shading_rate`

Extension Type

Device extension

Registered Extension Number

227

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

and

[VK_KHR_create_renderpass2](#)

or

[Vulkan Version 1.2](#)

API Interactions

- Interacts with `VK_VERSION_1_3`
- Interacts with `VK_KHR_dynamic_rendering`
- Interacts with `VK_KHR_format_feature_flags2`

SPIR-V Dependencies

- [SPV_KHR_fragment_shading_rate](#)

Contact

- Tobias Hector [\[GitHub\]](#)tobski

Extension Proposal

[VK_KHR_fragment_shading_rate](#)

Other Extension Metadata

Last Modified Date

2021-09-30

Interactions and External Dependencies

- This extension provides API support for [GL_EXT_fragment_shading_rate](#)

Contributors

- Tobias Hector, AMD
- Guennadi Riguer, AMD
- Matthaeus Chajdas, AMD
- Pat Brown, Nvidia
- Matthew Netsch, Qualcomm
- Slawomir Grajewski, Intel
- Jan-Harald Fredriksen, Arm
- Jeff Bolz, Nvidia
- Arseny Kapoulkine, Roblox
- Contributors to the [VK_NV_shading_rate_image](#) specification
- Contributors to the [VK_EXT_fragment_density_map](#) specification

Description

This extension adds the ability to change the rate at which fragments are shaded. Rather than the usual single fragment invocation for each pixel covered by a primitive, multiple pixels can be shaded by a single fragment shader invocation.

Up to three methods are available to the application to change the fragment shading rate:

- [Pipeline Fragment Shading Rate](#), which allows the specification of a rate per-draw.
- [Primitive Fragment Shading Rate](#), which allows the specification of a rate per primitive, specified during shading.
- [Attachment Fragment Shading Rate](#), which allows the specification of a rate per-region of the framebuffer, specified in a specialized image attachment.

Additionally, these rates can all be specified and combined in order to adjust the overall detail in

the image at each point.

This functionality can be used to focus shading efforts where higher levels of detail are needed in some parts of a scene compared to others. This can be particularly useful in high resolution rendering, or for XR contexts.

This extension also adds support for the `SPV_KHR_fragment_shading_rate` extension which enables setting the `primitive fragment shading rate`, and allows querying the final shading rate from a fragment shader.

New Commands

- `vkCmdSetFragmentShadingRateKHR`
- `vkGetPhysicalDeviceFragmentShadingRatesKHR`

New Structures

- `VkPhysicalDeviceFragmentShadingRateKHR`
- Extending `VkGraphicsPipelineCreateInfo`:
 - `VkPipelineFragmentShadingRateStateCreateInfoKHR`
- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceFragmentShadingRateFeaturesKHR`
- Extending `VkPhysicalDeviceProperties2`:
 - `VkPhysicalDeviceFragmentShadingRatePropertiesKHR`
- Extending `VkSubpassDescription2`:
 - `VkFragmentShadingRateAttachmentInfoKHR`

If `Vulkan Version 1.3` or `VK_KHR_dynamic_rendering` is supported:

- Extending `VkRenderingInfo`:
 - `VkRenderingFragmentShadingRateAttachmentInfoKHR`

New Enums

- `VkFragmentShadingRateCombinerOpKHR`

New Enum Constants

- `VK_KHR_FRAGMENT_SHADING_RATE_EXTENSION_NAME`
- `VK_KHR_FRAGMENT_SHADING_RATE_SPEC_VERSION`
- Extending `VkAccessFlagBits`:
 - `VK_ACCESS_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR`
- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_FRAGMENT_SHADING_RATE_KHR`

- Extending [VkFormatFeatureFlagBits](#):
 - `VK_FORMAT_FEATURE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- Extending [VkImageLayout](#):
 - `VK_IMAGE_LAYOUT_FRAGMENT_SHADING_RATE_ATTACHMENT_OPTIMAL_KHR`
- Extending [VkImageUsageFlagBits](#):
 - `VK_IMAGE_USAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- Extending [VkPipelineStageFlagBits](#):
 - `VK_PIPELINE_STAGE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_FEATURES_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADING_RATE_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_PIPELINE_FRAGMENT_SHADING_RATE_STATE_CREATE_INFO_KHR`

If [VK_KHR_format_feature_flags2](#) or [Vulkan Version 1.3](#) is supported:

- Extending [VkFormatFeatureFlagBits2](#):
 - `VK_FORMAT_FEATURE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

If [Vulkan Version 1.3](#) or [VK_KHR_dynamic_rendering](#) is supported:

- Extending [VkPipelineCreateFlagBits](#):
 - `VK_PIPELINE_CREATE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
 - `VK_PIPELINE_RASTERIZATION_STATE_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_INFO_KHR`

Version History

- Revision 1, 2020-05-06 (Tobias Hector)
 - Initial revision
- Revision 2, 2021-09-30 (Jon Leech)
 - Add interaction with [VK_KHR_format_feature_flags2](#) to `vk.xml`

VK_KHR_get_display_properties2

Name String

`VK_KHR_get_display_properties2`

Extension Type

Instance extension

Registered Extension Number

122

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_display](#)

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2017-02-21

IP Status

No known IP claims.

Contributors

- Ian Elliott, Google
- James Jones, NVIDIA

Description

This extension provides new queries for device display properties and capabilities that can be easily extended by other extensions, without introducing any further queries. This extension can be considered the [VK_KHR_display](#) equivalent of the [VK_KHR_get_physical_device_properties2](#) extension.

New Commands

- [vkGetDisplayModeProperties2KHR](#)
- [vkGetDisplayPlaneCapabilities2KHR](#)
- [vkGetPhysicalDeviceDisplayPlaneProperties2KHR](#)
- [vkGetPhysicalDeviceDisplayProperties2KHR](#)

New Structures

- [VkDisplayModeProperties2KHR](#)
- [VkDisplayPlaneCapabilities2KHR](#)

- [VkDisplayPlaneInfo2KHR](#)
- [VkDisplayPlaneProperties2KHR](#)
- [VkDisplayProperties2KHR](#)

New Enum Constants

- `VK_KHR_GET_DISPLAY_PROPERTIES_2_EXTENSION_NAME`
- `VK_KHR_GET_DISPLAY_PROPERTIES_2_SPEC_VERSION`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_DISPLAY_MODE_PROPERTIES_2_KHR`
 - `VK_STRUCTURE_TYPE_DISPLAY_PLANE_CAPABILITIES_2_KHR`
 - `VK_STRUCTURE_TYPE_DISPLAY_PLANE_INFO_2_KHR`
 - `VK_STRUCTURE_TYPE_DISPLAY_PLANE_PROPERTIES_2_KHR`
 - `VK_STRUCTURE_TYPE_DISPLAY_PROPERTIES_2_KHR`

Issues

1) What should this extension be named?

RESOLVED: `VK_KHR_get_display_properties2`. Other alternatives:

- `VK_KHR_display2`
- One extension, combined with `VK_KHR_surface_capabilities2`.

2) Should extensible input structs be added for these new functions:

RESOLVED:

- `vkGetPhysicalDeviceDisplayProperties2KHR`: No. The only current input is a [VkPhysicalDevice](#). Other inputs would not make sense.
- `vkGetPhysicalDeviceDisplayPlaneProperties2KHR`: No. The only current input is a [VkPhysicalDevice](#). Other inputs would not make sense.
- `vkGetDisplayModeProperties2KHR`: No. The only current inputs are a [VkPhysicalDevice](#) and a [VkDisplayModeKHR](#). Other inputs would not make sense.

3) Should additional display query functions be extended?

RESOLVED:

- `vkGetDisplayPlaneSupportedDisplaysKHR`: No. Extensions should instead extend `vkGetDisplayPlaneCapabilitiesKHR()`.

Version History

- Revision 1, 2017-02-21 (James Jones)
 - Initial draft.

VK_KHR_get_memory_requirements2

Name String

VK_KHR_get_memory_requirements2

Extension Type

Device extension

Registered Extension Number

147

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Faith Ekstrand [\[GitHub\]gfxstrand](#)

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- Faith Ekstrand, Intel
- Jeff Bolz, NVIDIA
- Jesse Hall, Google

Description

This extension provides new queries for memory requirements of images and buffers that can be easily extended by other extensions, without introducing any additional commands. The Vulkan 1.0 [VkMemoryRequirements](#) and [VkSparseImageMemoryRequirements](#) structures do not include **sType** and **pNext** members. This extension wraps them in new structures with these members, so an application can query a chain of memory requirements structures by constructing the chain and letting the implementation fill them in. A new command is added for each **vkGet*MemoryRequirements** command in core Vulkan 1.0.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- [vkGetBufferMemoryRequirements2KHR](#)
- [vkGetImageMemoryRequirements2KHR](#)
- [vkGetImageSparseMemoryRequirements2KHR](#)

New Structures

- [VkBufferMemoryRequirementsInfo2KHR](#)
- [VkImageMemoryRequirementsInfo2KHR](#)
- [VkImageSparseMemoryRequirementsInfo2KHR](#)
- [VkMemoryRequirements2KHR](#)
- [VkSparseImageMemoryRequirements2KHR](#)

New Enum Constants

- [VK_KHR_GET_MEMORY_REQUIREMENTS_2_EXTENSION_NAME](#)
- [VK_KHR_GET_MEMORY_REQUIREMENTS_2_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_BUFFER_MEMORY_REQUIREMENTS_INFO_2_KHR](#)
 - [VK_STRUCTURE_TYPE_IMAGE_MEMORY_REQUIREMENTS_INFO_2_KHR](#)
 - [VK_STRUCTURE_TYPE_IMAGE_SPARSE_MEMORY_REQUIREMENTS_INFO_2_KHR](#)
 - [VK_STRUCTURE_TYPE_MEMORY_REQUIREMENTS_2_KHR](#)
 - [VK_STRUCTURE_TYPE_SPARSE_IMAGE_MEMORY_REQUIREMENTS_2_KHR](#)

Version History

- Revision 1, 2017-03-23 (Faith Ekstrand)
 - Internal revisions

VK_KHR_get_physical_device_properties2

Name String

[VK_KHR_get_physical_device_properties2](#)

Extension Type

Instance extension

Registered Extension Number

60

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jeff Bolz [\[GitHub\]jeffbolznv](#)

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Ian Elliott, Google

Description

This extension provides new queries for device features, device properties, and format properties that can be easily extended by other extensions, without introducing any further queries. The Vulkan 1.0 feature/limit/formatproperty structures do not include **sType/pNext** members. This extension wraps them in new structures with **sType/pNext** members, so an application can query a chain of feature/limit/formatproperty structures by constructing the chain and letting the implementation fill them in. A new command is added for each **vkGetPhysicalDevice*** command in core Vulkan 1.0. The new feature structure (and a **pNext** chain of extending structures) can also be passed in to device creation to enable features.

This extension also allows applications to use the physical-device components of device extensions before **vkCreateDevice** is called.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- [vkGetPhysicalDeviceFeatures2KHR](#)
- [vkGetPhysicalDeviceFormatProperties2KHR](#)
- [vkGetPhysicalDeviceImageFormatProperties2KHR](#)
- [vkGetPhysicalDeviceMemoryProperties2KHR](#)
- [vkGetPhysicalDeviceProperties2KHR](#)
- [vkGetPhysicalDeviceQueueFamilyProperties2KHR](#)
- [vkGetPhysicalDeviceSparseImageFormatProperties2KHR](#)

New Structures

- [VkFormatProperties2KHR](#)
- [VkImageFormatProperties2KHR](#)
- [VkPhysicalDeviceImageFormatInfo2KHR](#)
- [VkPhysicalDeviceMemoryProperties2KHR](#)
- [VkPhysicalDeviceProperties2KHR](#)
- [VkPhysicalDeviceSparseImageFormatInfo2KHR](#)
- [VkQueueFamilyProperties2KHR](#)
- [VkSparseImageFormatProperties2KHR](#)
- Extending [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceFeatures2KHR](#)

New Enum Constants

- [VK_KHR_GET_PHYSICAL_DEVICE_PROPERTIES_2_EXTENSION_NAME](#)
- [VK_KHR_GET_PHYSICAL_DEVICE_PROPERTIES_2_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_FORMAT_PROPERTIES_2_KHR](#)
 - [VK_STRUCTURE_TYPE_IMAGE_FORMAT_PROPERTIES_2_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_FORMAT_INFO_2_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PROPERTIES_2_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROPERTIES_2_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SPARSE_IMAGE_FORMAT_INFO_2_KHR](#)
 - [VK_STRUCTURE_TYPE_QUEUE_FAMILY_PROPERTIES_2_KHR](#)
 - [VK_STRUCTURE_TYPE_SPARSE_IMAGE_FORMAT_PROPERTIES_2_KHR](#)

Examples

```
// Get features with a hypothetical future extension.
VkHypotheticalExtensionFeaturesKHR hypotheticalFeatures =
{
    .sType = VK_STRUCTURE_TYPE_HYPOTHETICAL_FEATURES_KHR,
    .pNext = NULL,
};

VkPhysicalDeviceFeatures2KHR features =
{
    .sType = VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2_KHR,
    .pNext = &hypotheticalFeatures,
};

// After this call, features and hypotheticalFeatures have been filled out.
vkGetPhysicalDeviceFeatures2KHR(physicalDevice, &features);

// Properties/limits can be chained and queried similarly.

// Enable some features:
VkHypotheticalExtensionFeaturesKHR enabledHypotheticalFeatures =
{
    .sType = VK_STRUCTURE_TYPE_HYPOTHETICAL_FEATURES_KHR,
    .pNext = NULL,
};

VkPhysicalDeviceFeatures2KHR enabledFeatures =
{
    .sType = VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FEATURES_2_KHR,
    .pNext = &enabledHypotheticalFeatures,
};

enabledFeatures.features.xyz = VK_TRUE;
enabledHypotheticalFeatures.abc = VK_TRUE;

VkDeviceCreateInfo deviceCreateInfo =
{
    .sType = VK_STRUCTURE_TYPE_DEVICE_CREATE_INFO,
    .pNext = &enabledFeatures,
    ...
    .pEnabledFeatures = NULL,
};

VkDevice device;
vkCreateDevice(physicalDevice, &deviceCreateInfo, NULL, &device);
```

Version History

- Revision 1, 2016-09-12 (Jeff Bolz)

- Internal revisions
- Revision 2, 2016-11-02 (Ian Elliott)
 - Added ability for applications to use the physical-device components of device extensions before `vkCreateDevice` is called.

VK_KHR_get_surface_capabilities2

Name String

VK_KHR_get_surface_capabilities2

Extension Type

Instance extension

Registered Extension Number

120

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_surface](#)

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2017-02-27

IP Status

No known IP claims.

Contributors

- Ian Elliott, Google
- James Jones, NVIDIA
- Alon Or-bach, Samsung

Description

This extension provides new queries for device surface capabilities that can be easily extended by other extensions, without introducing any further queries. This extension can be considered the [VK_KHR_surface](#) equivalent of the [VK_KHR_get_physical_device_properties2](#) extension.

New Commands

- [vkGetPhysicalDeviceSurfaceCapabilities2KHR](#)
- [vkGetPhysicalDeviceSurfaceFormats2KHR](#)

New Structures

- [VkPhysicalDeviceSurfaceInfo2KHR](#)
- [VkSurfaceCapabilities2KHR](#)
- [VkSurfaceFormat2KHR](#)

New Enum Constants

- [VK_KHR_GET_SURFACE_CAPABILITIES_2_EXTENSION_NAME](#)
- [VK_KHR_GET_SURFACE_CAPABILITIES_2_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SURFACE_INFO_2_KHR](#)
 - [VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_KHR](#)
 - [VK_STRUCTURE_TYPE_SURFACE_FORMAT_2_KHR](#)

Issues

1) What should this extension be named?

RESOLVED: [VK_KHR_get_surface_capabilities2](#). Other alternatives:

- [VK_KHR_surface2](#)
- One extension, combining a separate display-specific query extension.

2) Should additional WSI query functions be extended?

RESOLVED:

- [vkGetPhysicalDeviceSurfaceCapabilitiesKHR](#): Yes. The need for this motivated the extension.
- [vkGetPhysicalDeviceSurfaceSupportKHR](#): No. Currently only has boolean output. Extensions should instead extend [vkGetPhysicalDeviceSurfaceCapabilities2KHR](#).
- [vkGetPhysicalDeviceSurfaceFormatsKHR](#): Yes.
- [vkGetPhysicalDeviceSurfacePresentModesKHR](#): No. Recent discussion concluded this introduced too much variability for applications to deal with. Extensions should instead extend [vkGetPhysicalDeviceSurfaceCapabilities2KHR](#).
- [vkGetPhysicalDeviceXlibPresentationSupportKHR](#): Not in this extension.
- [vkGetPhysicalDeviceXcbPresentationSupportKHR](#): Not in this extension.
- [vkGetPhysicalDeviceWaylandPresentationSupportKHR](#): Not in this extension.
- [vkGetPhysicalDeviceWin32PresentationSupportKHR](#): Not in this extension.

Version History

- Revision 1, 2017-02-27 (James Jones)
 - Initial draft.

VK_KHR_global_priority

Name String

VK_KHR_global_priority

Extension Type

Device extension

Registered Extension Number

189

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Tobias Hector [\[GitHub\]](#)tobski

Other Extension Metadata

Last Modified Date

2021-10-22

Contributors

- Tobias Hector, AMD
- Contributors to [VK_EXT_global_priority](#)
- Contributors to [VK_EXT_global_priority_query](#)

Description

In Vulkan, users can specify device-scope queue priorities. In some cases it may be useful to extend this concept to a system-wide scope. This device extension allows applications to query the global queue priorities supported by a queue family, and then set a priority when creating queues. The default queue priority is [VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_EXT](#).

Implementations can report which global priority levels are treated differently by the

implementation. It is intended primarily for use in system integration along with certain platform-specific priority enforcement rules.

The driver implementation will attempt to skew hardware resource allocation in favor of the higher-priority task. Therefore, higher-priority work may retain similar latency and throughput characteristics even if the system is congested with lower priority work.

The global priority level of a queue shall take precedence over the per-process queue priority (`VkDeviceQueueCreateInfo::pQueuePriorities`).

Abuse of this feature may result in starving the rest of the system from hardware resources. Therefore, the driver implementation may deny requests to acquire a priority above the default priority (`VK_QUEUE_GLOBAL_PRIORITY_MEDIUM_EXT`) if the caller does not have sufficient privileges. In this scenario `VK_ERROR_NOT_PERMITTED_EXT` is returned.

The driver implementation may fail the queue allocation request if resources required to complete the operation have been exhausted (either by the same process or a different process). In this scenario `VK_ERROR_INITIALIZATION_FAILED` is returned.

New Structures

- Extending `VkDeviceQueueCreateInfo`:
 - `VkDeviceQueueGlobalPriorityCreateInfoKHR`
- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceGlobalPriorityQueryFeaturesKHR`
- Extending `VkQueueFamilyProperties2`:
 - `VkQueueFamilyGlobalPriorityPropertiesKHR`

New Enums

- `VkQueueGlobalPriorityKHR`

New Enum Constants

- `VK_KHR_GLOBAL_PRIORITY_EXTENSION_NAME`
- `VK_KHR_GLOBAL_PRIORITY_SPEC_VERSION`
- `VK_MAX_GLOBAL_PRIORITY_SIZE_KHR`
- Extending `VkResult`:
 - `VK_ERROR_NOT_PERMITTED_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_DEVICE_QUEUE_GLOBAL_PRIORITY_CREATE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_GLOBAL_PRIORITY_QUERY_FEATURES_KHR`
 - `VK_STRUCTURE_TYPE_QUEUE_FAMILY_GLOBAL_PRIORITY_PROPERTIES_KHR`

Issues

1) Can we additionally query whether a caller is permitted to acquire a specific global queue priority in this extension?

RESOLVED: No. Whether a caller has enough privilege goes with the OS, and the Vulkan driver cannot really guarantee that the privilege will not change in between this query and the actual queue creation call.

2) If more than 1 queue using global priority is requested, is there a good way to know which queue is failing the device creation?

RESOLVED: No. There is not a good way at this moment, and it is also not quite actionable for the applications to know that because the information may not be accurate. Queue creation can fail because of runtime constraints like insufficient privilege or lack of resource, and the failure is not necessarily tied to that particular queue configuration requested.

Version History

- Revision 1, 2021-10-22 (Tobias Hector)
 - Initial draft

VK_KHR_image_format_list

Name String

VK_KHR_image_format_list

Extension Type

Device extension

Registered Extension Number

148

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Faith Ekstrand [\[GitHub\]gfxstrand](#)

Other Extension Metadata

Last Modified Date

2017-03-20

IP Status

No known IP claims.

Contributors

- Faith Ekstrand, Intel
- Jan-Harald Fredriksen, ARM
- Jeff Bolz, NVIDIA
- Jeff Leger, Qualcomm
- Neil Henning, Codeplay

Description

On some implementations, setting the `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT` on image creation can cause access to that image to perform worse than an equivalent image created without `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT` because the implementation does not know what view formats will be paired with the image.

This extension allows an application to provide the list of all formats that **can** be used with an image when it is created. The implementation may then be able to create a more efficient image that supports the subset of formats required by the application without having to support all formats in the format compatibility class of the image format.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending `VkImageCreateInfo`, `VkSwapchainCreateInfoKHR`, `VkPhysicalDeviceImageFormatInfo2`:
 - `VkImageFormatListCreateInfoKHR`

New Enum Constants

- `VK_KHR_IMAGE_FORMAT_LIST_EXTENSION_NAME`
- `VK_KHR_IMAGE_FORMAT_LIST_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_IMAGE_FORMAT_LIST_CREATE_INFO_KHR`

Version History

- Revision 1, 2017-03-20 (Faith Ekstrand)
 - Initial revision

VK_KHR_imageless_framebuffer

Name String

VK_KHR_imageless_framebuffer

Extension Type

Device extension

Registered Extension Number

109

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)
and
[VK_KHR_maintenance2](#)
or
[Vulkan Version 1.1](#)
and
[VK_KHR_image_format_list](#)
or
[Vulkan Version 1.2](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Tobias Hector [[GitHub](#)][tobias](#)

Other Extension Metadata

Last Modified Date

2018-12-14

Contributors

- Tobias Hector
- Graham Wihlidal

Description

This extension allows framebuffers to be created without the need for creating images first, allowing more flexibility in how they are used, and avoiding the need for many of the confusing compatibility rules.

Framebuffers are now created with a small amount of additional metadata about the image views that will be used in [VkFramebufferAttachmentsCreateInfoKHR](#), and the actual image views are provided at render pass begin time via [VkRenderPassAttachmentBeginInfoKHR](#).

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- [VkFramebufferAttachmentImageInfoKHR](#)
- Extending [VkFramebufferCreateInfo](#):
 - [VkFramebufferAttachmentsCreateInfoKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceImagelessFramebufferFeaturesKHR](#)
- Extending [VkRenderPassBeginInfo](#):
 - [VkRenderPassAttachmentBeginInfoKHR](#)

New Enum Constants

- [VK_KHR_IMAGELESS_FRAMEBUFFER_EXTENSION_NAME](#)
- [VK_KHR_IMAGELESS_FRAMEBUFFER_SPEC_VERSION](#)
- Extending [VkFramebufferCreateFlagBits](#):
 - [VK_FRAMEBUFFER_CREATE_IMAGELESS_BIT_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENTS_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_FRAMEBUFFER_ATTACHMENT_IMAGE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGELESS_FRAMEBUFFER_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_RENDER_PASS_ATTACHMENT_BEGIN_INFO_KHR](#)

Version History

- Revision 1, 2018-12-14 (Tobias Hector)
 - Internal revisions

VK_KHR_incremental_present

Name String

VK_KHR_incremental_present

Extension Type

Device extension

Registered Extension Number

85

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_swapchain](#)

Contact

- Ian Elliott [\[GitHub\]](#)[ianelliottus](#)

Other Extension Metadata

Last Modified Date

2016-11-02

IP Status

No known IP claims.

Contributors

- Ian Elliott, Google
- Jesse Hall, Google
- Alon Or-bach, Samsung
- James Jones, NVIDIA
- Daniel Rakos, AMD
- Ray Smith, ARM
- Mika Isojarvi, Google
- Jeff Juliano, NVIDIA
- Jeff Bolz, NVIDIA

Description

This device extension extends [vkQueuePresentKHR](#), from the [VK_KHR_swapchain](#) extension, allowing

an application to specify a list of rectangular, modified regions of each image to present. This should be used in situations where an application is only changing a small portion of the presentable images within a swapchain, since it enables the presentation engine to avoid wasting time presenting parts of the surface that have not changed.

This extension is leveraged from the [EGL_KHR_swap_buffers_with_damage](#) extension.

New Structures

- [VkPresentRegionKHR](#)
- [VkRectLayerKHR](#)
- Extending [VkPresentInfoKHR](#):
 - [VkPresentRegionsKHR](#)

New Enum Constants

- [VK_KHR_INCREMENTAL_PRESENT_EXTENSION_NAME](#)
- [VK_KHR_INCREMENTAL_PRESENT_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PRESENT_REGIONS_KHR](#)

Issues

1) How should we handle stereoscopic-3D swapchains? We need to add a layer for each rectangle. One approach is to create another struct containing the [VkRect2D](#) plus layer, and have [VkPresentRegionsKHR](#) point to an array of that struct. Another approach is to have two parallel arrays, [pRectangles](#) and [pLayers](#), where [pRectangles\[i\]](#) and [pLayers\[i\]](#) must be used together. Which approach should we use, and if the array of a new structure, what should that be called?

RESOLVED: Create a new structure, which is a [VkRect2D](#) plus a layer, and will be called [VkRectLayerKHR](#).

2) Where is the origin of the [VkRectLayerKHR](#)?

RESOLVED: The upper left corner of the presentable image(s) of the swapchain, per the definition of framebuffer coordinates.

3) Does the rectangular region, [VkRectLayerKHR](#), specify pixels of the swapchain's image(s), or of the surface?

RESOLVED: Of the image(s). Some presentation engines may scale the pixels of a swapchain's image(s) to the size of the surface. The size of the swapchain's image(s) will be consistent, where the size of the surface may vary over time.

4) What if all of the rectangles for a given swapchain contain a width and/or height of zero?

RESOLVED: The application is indicating that no pixels changed since the last present. The presentation engine may use such a hint and not update any pixels for the swapchain. However, all

other semantics of [vkQueuePresentKHR](#) must still be honored, including waiting for semaphores to signal.

5) When the swapchain is created with [VkSwapchainCreateInfoKHR::preTransform](#) set to a value other than [VK_SURFACE_TRANSFORM_IDENTITY_BIT_KHR](#), should the rectangular region, [VkRectLayerKHR](#), be transformed to align with the [preTransform](#)?

RESOLVED: No. The rectangular region in [VkRectLayerKHR](#) should not be transformed. As such, it may not align with the extents of the swapchain's image(s). It is the responsibility of the presentation engine to transform the rectangular region. This matches the behavior of the Android presentation engine, which set the precedent.

Version History

- Revision 1, 2016-11-02 (Ian Elliott)
 - Internal revisions
- Revision 2, 2021-03-18 (Ian Elliott)
 - Clarified alignment of rectangles for presentation engines that support transformed swapchains.

VK_KHR_index_type_uint8

Name String

[VK_KHR_index_type_uint8](#)

Extension Type

Device extension

Registered Extension Number

534

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Piers Daniell [\[GitHub\]](#)[pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2023-06-06

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA

Description

This extension allows `uint8_t` indices to be used with `vkCmdBindIndexBuffer`.

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceIndexTypeUint8FeaturesKHR`

New Enum Constants

- `VK_KHR_INDEX_TYPE_UINT8_EXTENSION_NAME`
- `VK_KHR_INDEX_TYPE_UINT8_SPEC_VERSION`
- Extending `VkIndexType`:
 - `VK_INDEX_TYPE_UINT8_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_INDEX_TYPE_UINT8_FEATURES_KHR`

Version History

- Revision 1, 2023-06-06 (Piers Daniell)
 - Internal revisions

VK_KHR_line_rasterization

Name String

`VK_KHR_line_rasterization`

Extension Type

Device extension

Registered Extension Number

535

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Piers Daniell [[GitHub](#)][pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2023-06-08

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Allen Jensen, NVIDIA
- Faith Ekstrand, Intel

Description

This extension adds some line rasterization features that are commonly used in CAD applications and supported in other APIs like OpenGL. Bresenham-style line rasterization is supported, smooth rectangular lines (coverage to alpha) are supported, and stippled lines are supported for all three line rasterization modes.

New Commands

- [vkCmdSetLineStippleKHR](#)

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceLineRasterizationFeaturesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceLineRasterizationPropertiesKHR](#)
- Extending [VkPipelineRasterizationStateCreateInfo](#):
 - [VkPipelineRasterizationLineStateCreateInfoKHR](#)

New Enums

- [VkLineRasterizationModeKHR](#)

New Enum Constants

- `VK_KHR_LINE_RASTERIZATION_EXTENSION_NAME`
- `VK_KHR_LINE_RASTERIZATION_SPEC_VERSION`
- Extending [VkDynamicState](#):
 - `VK_DYNAMIC_STATE_LINE_STIPPLE_KHR`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_FEATURES_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LINE_RASTERIZATION_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_LINE_STATE_CREATE_INFO_KHR`

Issues

1) Do we need to support Bresenham-style and smooth lines with more than one rasterization sample? i.e. the equivalent of `glDisable(GL_MULTISAMPLE)` in OpenGL when the framebuffer has more than one sample?

RESOLVED: Yes. For simplicity, Bresenham line rasterization carries forward a few restrictions from OpenGL, such as not supporting per-sample shading, alpha to coverage, or alpha to one.

Version History

- Revision 1, 2019-05-09 (Jeff Bolz)
 - Initial draft

VK_KHR_load_store_op_none

Name String

`VK_KHR_load_store_op_none`

Extension Type

Device extension

Registered Extension Number

527

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Shahbaz Youssefi [\[GitHub\]syoussefi](#)

Extension Proposal

[VK_KHR_load_store_op_none](#)

Other Extension Metadata

Last Modified Date

2023-05-16

Contributors

- Shahbaz Youssefi, Google
- Bill Licea-Kane, Qualcomm Technologies, Inc.
- Tobias Hector, AMD

Description

This extension provides `VK_ATTACHMENT_LOAD_OP_NONE_KHR` and `VK_ATTACHMENT_STORE_OP_NONE_KHR`, which are identically promoted from the [VK_EXT_load_store_op_none](#) extension.

New Enum Constants

- `VK_KHR_LOAD_STORE_OP_NONE_EXTENSION_NAME`
- `VK_KHR_LOAD_STORE_OP_NONE_SPEC_VERSION`
- Extending [VkAttachmentLoadOp](#):
 - `VK_ATTACHMENT_LOAD_OP_NONE_KHR`
- Extending [VkAttachmentStoreOp](#):
 - `VK_ATTACHMENT_STORE_OP_NONE_KHR`

NOTE

While `VK_ATTACHMENT_STORE_OP_NONE` is part of Vulkan 1.3, this extension was not promoted to core either in whole or in part. This functionality was promoted from [VK_KHR_dynamic_rendering](#).

Version History

- Revision 1, 2023-05-16 (Shahbaz Youssefi)
 - Initial revision, based on [VK_EXT_load_store_op_none](#).

VK_KHR_maintenance1

Name String

`VK_KHR_maintenance1`

Extension Type

Device extension

Registered Extension Number

70

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Piers Daniell [[GitHub](#)][pdaniell-nv](#)

Other Extension Metadata**Last Modified Date**

2018-03-13

Contributors

- Dan Ginsburg, Valve
- Daniel Koch, NVIDIA
- Daniel Rakos, AMD
- Jan-Harald Fredriksen, ARM
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- Jesse Hall, Google
- John Kessenich, Google
- Michael Worcester, Imagination Technologies
- Neil Henning, Codeplay Software Ltd.
- Piers Daniell, NVIDIA
- Slawomir Grajewski, Intel
- Tobias Hector, Imagination Technologies
- Tom Olson, ARM

Description

`VK_KHR_maintenance1` adds a collection of minor features that were intentionally left out or overlooked from the original Vulkan 1.0 release.

The new features are as follows:

- Allow 2D and 2D array image views to be created from 3D images, which can then be used as color framebuffer attachments. This allows applications to render to slices of a 3D image.
- Support `vkCmdCopyImage` between 2D array layers and 3D slices. This extension allows copying from layers of a 2D array image to slices of a 3D image and vice versa.
- Allow negative height to be specified in the `VkViewport::height` field to perform y-inversion of the clip-space to framebuffer-space transform. This allows apps to avoid having to use `gl_Position.y = -gl_Position.y` in shaders also targeting other APIs.
- Allow implementations to express support for doing just transfers and clears of image formats that they otherwise support no other format features for. This is done by adding new format feature flags `VK_FORMAT_FEATURE_TRANSFER_SRC_BIT_KHR` and `VK_FORMAT_FEATURE_TRANSFER_DST_BIT_KHR`.
- Support `vkCmdFillBuffer` on transfer-only queues. Previously `vkCmdFillBuffer` was defined to only work on command buffers allocated from command pools which support graphics or compute queues. It is now allowed on queues that just support transfer operations.
- Fix the inconsistency of how error conditions are returned between the `vkCreateGraphicsPipelines` and `vkCreateComputePipelines` functions and the `vkAllocateDescriptorSets` and `vkAllocateCommandBuffers` functions.
- Add new `VK_ERROR_OUT_OF_POOL_MEMORY_KHR` error so implementations can give a more precise reason for `vkAllocateDescriptorSets` failures.
- Add a new command `vkTrimCommandPoolKHR` which gives the implementation an opportunity to release any unused command pool memory back to the system.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- `vkTrimCommandPoolKHR`

New Bitmasks

- `VkCommandPoolTrimFlagsKHR`

New Enum Constants

- `VK_KHR_MAINTENANCE1_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE1_SPEC_VERSION`

- `VK_KHR_MAINTENANCE_1_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_1_SPEC_VERSION`
- Extending `VkFormatFeatureFlagBits`:
 - `VK_FORMAT_FEATURE_TRANSFER_DST_BIT_KHR`
 - `VK_FORMAT_FEATURE_TRANSFER_SRC_BIT_KHR`
- Extending `VkImageCreateFlagBits`:
 - `VK_IMAGE_CREATE_2D_ARRAY_COMPATIBLE_BIT_KHR`
- Extending `VkResult`:
 - `VK_ERROR_OUT_OF_POOL_MEMORY_KHR`

Issues

1. Are viewports with zero height allowed?

RESOLVED: Yes, although they have low utility.

Version History

- Revision 1, 2016-10-26 (Piers Daniell)
 - Internal revisions
- Revision 2, 2018-03-13 (Jon Leech)
 - Add issue for zero-height viewports

VK_KHR_maintenance2

Name String

`VK_KHR_maintenance2`

Extension Type

Device extension

Registered Extension Number

118

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Michael Worcester [\[GitHub\]](#)[michaelworcester](#)

Other Extension Metadata

Last Modified Date

2017-09-05

Contributors

- Michael Worcester, Imagination Technologies
- Stuart Smith, Imagination Technologies
- Jeff Bolz, NVIDIA
- Daniel Koch, NVIDIA
- Jan-Harald Fredriksen, ARM
- Daniel Rakos, AMD
- Neil Henning, Codeplay
- Piers Daniell, NVIDIA

Description

VK_KHR_maintenance2 adds a collection of minor features that were intentionally left out or overlooked from the original Vulkan 1.0 release.

The new features are as follows:

- Allow the application to specify which aspect of an input attachment might be read for a given subpass.
- Allow implementations to express the clipping behavior of points.
- Allow creating images with usage flags that may not be supported for the base image's format, but are supported for image views of the image that have a different but compatible format.
- Allow creating uncompressed image views of compressed images.
- Allow the application to select between an upper-left and lower-left origin for the tessellation domain space.
- Adds two new image layouts for depth stencil images to allow either the depth or stencil aspect to be read-only while the other aspect is writable.

Input Attachment Specification

Input attachment specification allows an application to specify which aspect of a multi-aspect image (e.g. a depth/stencil format) will be accessed via a **subpassLoad** operation.

On some implementations there **may** be a performance penalty if the implementation does not

know (at [vkCreateRenderPass](#) time) which aspect(s) of multi-aspect images **can** be accessed as input attachments.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- [VkInputAttachmentAspectReferenceKHR](#)
- Extending [VkImageViewCreateInfo](#):
 - [VkImageViewUsageCreateInfoKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDevicePointClippingPropertiesKHR](#)
- Extending [VkPipelineTessellationStateCreateInfo](#):
 - [VkPipelineTessellationDomainOriginStateCreateInfoKHR](#)
- Extending [VkRenderPassCreateInfo](#):
 - [VkRenderPassInputAttachmentAspectCreateInfoKHR](#)

New Enums

- [VkPointClippingBehaviorKHR](#)
- [VkTessellationDomainOriginKHR](#)

New Enum Constants

- [VK_KHR_MAINTENANCE2_EXTENSION_NAME](#)
- [VK_KHR_MAINTENANCE2_SPEC_VERSION](#)
- [VK_KHR_MAINTENANCE_2_EXTENSION_NAME](#)
- [VK_KHR_MAINTENANCE_2_SPEC_VERSION](#)
- Extending [VkImageCreateFlagBits](#):
 - [VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT_KHR](#)
 - [VK_IMAGE_CREATE_EXTENDED_USAGE_BIT_KHR](#)
- Extending [VkImageLayout](#):
 - [VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_STENCIL_READ_ONLY_OPTIMAL_KHR](#)
 - [VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_STENCIL_ATTACHMENT_OPTIMAL_KHR](#)
- Extending [VkPointClippingBehavior](#):
 - [VK_POINT_CLIPPING_BEHAVIOR_ALL_CLIP_PLANES_KHR](#)
 - [VK_POINT_CLIPPING_BEHAVIOR_USER_CLIP_PLANES_ONLY_KHR](#)
- Extending [VkStructureType](#):

- `VK_STRUCTURE_TYPE_IMAGE_VIEW_USAGE_CREATE_INFO_KHR`
- `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_POINT_CLIPPING_PROPERTIES_KHR`
- `VK_STRUCTURE_TYPE_PIPELINE_TESSELLATION_DOMAIN_ORIGIN_STATE_CREATE_INFO_KHR`
- `VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO_KHR`
- Extending `VkTessellationDomainOrigin`:
 - `VK_TESSELLATION_DOMAIN_ORIGIN_LOWER_LEFT_KHR`
 - `VK_TESSELLATION_DOMAIN_ORIGIN_UPPER_LEFT_KHR`

Input Attachment Specification Example

Consider the case where a render pass has two subpasses and two attachments.

Attachment 0 has the format `VK_FORMAT_D24_UNORM_S8_UINT`, attachment 1 has some color format.

Subpass 0 writes to attachment 0, subpass 1 reads only the depth information from attachment 0 (using `inputAttachmentRead`) and writes to attachment 1.

```
VkInputAttachmentAspectReferenceKHR references[] = {
    {
        .subpass = 1,
        .inputAttachmentIndex = 0,
        .aspectMask = VK_IMAGE_ASPECT_DEPTH_BIT
    }
};

VkRenderPassInputAttachmentAspectCreateInfoKHR specifyAspects = {
    .sType =
VK_STRUCTURE_TYPE_RENDER_PASS_INPUT_ATTACHMENT_ASPECT_CREATE_INFO_KHR,
    .pNext = NULL,
    .aspectReferenceCount = 1,
    .pAspectReferences = references
};

VkRenderPassCreateInfo createInfo = {
    ...
    .pNext = &specifyAspects,
    ...
};

vkCreateRenderPass(...);
```

Issues

1) What is the default tessellation domain origin?

RESOLVED: Vulkan 1.0 originally inadvertently documented a lower-left origin, but the

conformance tests and all implementations implemented an upper-left origin. This extension adds a control to select between lower-left (for compatibility with OpenGL) and upper-left, and we retroactively fix unextended Vulkan to have a default of an upper-left origin.

Version History

- Revision 1, 2017-04-28

VK_KHR_maintenance3

Name String

VK_KHR_maintenance3

Extension Type

Device extension

Registered Extension Number

169

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jeff Bolz [\[GitHub\]jeffbolznv](#)

Other Extension Metadata

Last Modified Date

2017-09-05

Contributors

- Jeff Bolz, NVIDIA

Description

VK_KHR_maintenance3 adds a collection of minor features that were intentionally left out or overlooked from the original Vulkan 1.0 release.

The new features are as follows:

- A limit on the maximum number of descriptors that are supported in a single descriptor set layout. Some implementations have a limit on the total size of descriptors in a set, which cannot be expressed in terms of the limits in Vulkan 1.0.
- A limit on the maximum size of a single memory allocation. Some platforms have kernel interfaces that limit the maximum size of an allocation.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- [vkGetDescriptorSetLayoutSupportKHR](#)

New Structures

- [VkDescriptorSetLayoutSupportKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceMaintenance3PropertiesKHR](#)

New Enum Constants

- [VK_KHR_MAINTENANCE3_EXTENSION_NAME](#)
- [VK_KHR_MAINTENANCE3_SPEC_VERSION](#)
- [VK_KHR_MAINTENANCE_3_EXTENSION_NAME](#)
- [VK_KHR_MAINTENANCE_3_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_DESCRIPTOR_SET_LAYOUT_SUPPORT_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_3_PROPERTIES_KHR](#)

Version History

- Revision 1, 2017-08-22

VK_KHR_maintenance4

Name String

[VK_KHR_maintenance4](#)

Extension Type

Device extension

Registered Extension Number

414

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Piers Daniell [\[GitHub\]](#)[pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2021-10-25

Interactions and External Dependencies

- Requires SPIR-V 1.2 for [LocalSizeId](#)

Contributors

- Lionel Duc, NVIDIA
- Faith Ekstrand, Intel
- Spencer Fricke, Samsung
- Tobias Hector, AMD
- Lionel Landwerlin, Intel
- Graeme Leese, Broadcom
- Tom Olson, Arm
- Stu Smith, AMD
- Yiwei Zhang, Google

Description

[VK_KHR_maintenance4](#) adds a collection of minor features, none of which would warrant an entire extension of their own.

The new features are as follows:

- Allow the application to destroy their [VkPipelineLayout](#) object immediately after it was used to create another object. It is no longer necessary to keep its handle valid while the created object

is in use.

- Add a new `maxBufferSize` implementation-defined limit for the maximum size `VkBuffer` that **can** be created.
- Add support for the SPIR-V 1.2 `LocalSizeId` execution mode, which can be used as an alternative to `LocalSize` to specify the local workgroup size with specialization constants.
- Add a guarantee that images created with identical creation parameters will always have the same alignment requirements.
- Add new `vkGetDeviceBufferMemoryRequirementsKHR`, `vkGetDeviceImageMemoryRequirementsKHR`, and `vkGetDeviceImageSparseMemoryRequirementsKHR` to allow the application to query the image memory requirements without having to create an image object and query it.
- Relax the requirement that push constants must be initialized before they are dynamically accessed.
- Relax the interface matching rules to allow a larger output vector to match with a smaller input vector, with additional values being discarded.
- Add a guarantee for buffer memory requirement that the size memory requirement is never greater than the result of aligning create size with the alignment memory requirement.

New Commands

- `vkGetDeviceBufferMemoryRequirementsKHR`
- `vkGetDeviceImageMemoryRequirementsKHR`
- `vkGetDeviceImageSparseMemoryRequirementsKHR`

New Structures

- `VkDeviceBufferMemoryRequirementsKHR`
- `VkDeviceImageMemoryRequirementsKHR`
- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceMaintenance4FeaturesKHR`
- Extending `VkPhysicalDeviceProperties2`:
 - `VkPhysicalDeviceMaintenance4PropertiesKHR`

New Enum Constants

- `VK_KHR_MAINTENANCE_4_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_4_SPEC_VERSION`
- Extending `VkImageAspectFlagBits`:
 - `VK_IMAGE_ASPECT_NONE_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_DEVICE_BUFFER_MEMORY_REQUIREMENTS_KHR`

- `VK_STRUCTURE_TYPE_DEVICE_IMAGE_MEMORY_REQUIREMENTS_KHR`
- `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_FEATURES_KHR`
- `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_4_PROPERTIES_KHR`

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Issues

None.

Version History

- Revision 1, 2021-08-18 (Piers Daniell)
 - Internal revisions
- Revision 2, 2021-10-25 (Yiwei Zhang)
 - More guarantees on buffer memory requirements

VK_KHR_maintenance5

Name String

`VK_KHR_maintenance5`

Extension Type

Device extension

Registered Extension Number

471

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

and

[VK_KHR_dynamic_rendering](#)

or

[Vulkan Version 1.3](#)

API Interactions

- Interacts with `VK_VERSION_1_1`

- Interacts with VK_VERSION_1_2
- Interacts with VK_VERSION_1_3
- Interacts with VK_EXT_attachment_feedback_loop_layout
- Interacts with VK_EXT_buffer_device_address
- Interacts with VK_EXT_conditional_rendering
- Interacts with VK_EXT_descriptor_buffer
- Interacts with VK_EXT_fragment_density_map
- Interacts with VK_EXT_graphics_pipeline_library
- Interacts with VK_EXT_opacity_micromap
- Interacts with VK_EXT_pipeline_creation_cache_control
- Interacts with VK_EXT_pipeline_protected_access
- Interacts with VK_EXT_transform_feedback
- Interacts with VK_KHR_acceleration_structure
- Interacts with VK_KHR_buffer_device_address
- Interacts with VK_KHR_device_group
- Interacts with VK_KHR_dynamic_rendering
- Interacts with VK_KHR_fragment_shading_rate
- Interacts with VK_KHR_pipeline_executable_properties
- Interacts with VK_KHR_pipeline_library
- Interacts with VK_KHR_ray_tracing_pipeline
- Interacts with VK_KHR_video_decode_queue
- Interacts with VK_KHR_video_encode_queue
- Interacts with VK_NV_device_generated_commands
- Interacts with VK_NV_displacement_micromap
- Interacts with VK_NV_ray_tracing
- Interacts with VK_NV_ray_tracing_motion_blur

Contact

- Stu Smith [[GitHub](#)][stu-s](#)

Extension Proposal

[VK_KHR_maintenance5](#)

Other Extension Metadata

Last Modified Date

2023-05-02

Interactions and External Dependencies

Contributors

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- Slawomir Cygan, Intel
- Lionel Landwerlin, Intel
- James Fitzpatrick, Imagination Technologies
- Andrew Garrard, Imagination Technologies
- Ralph Potter, Samsung
- Pan Gao, Huawei
- Jan-Harald Fredriksen, ARM
- Jon Leech, Khronos
- Mike Blumenkrantz, Valve

Description

`VK_KHR_maintenance5` adds a collection of minor features, none of which would warrant an entire extension of their own.

The new features are as follows:

- A new `VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR` format
- A new `VK_FORMAT_A8_UNORM_KHR` format
- A property to indicate that multisample coverage operations are performed after sample counting in EarlyFragmentTests mode
- Relax `VkBufferView` creation requirements by allowing subsets of the associated `VkBuffer` usage using `VkBufferUsageFlags2CreateInfoKHR`
- A new command `vkCmdBindIndexBuffer2KHR`, allowing a range of memory to be bound as an index buffer
- `vkGetDeviceProcAddr` must return `NULL` for supported core functions beyond the version requested by the application.
- A property to indicate that the sample mask test is performed after sample counting in EarlyFragmentTests mode
- `vkCmdBindVertexBuffers2` now supports using `VK_WHOLE_SIZE` in the `pSizes` parameter.
- A default size of 1.0 is used if `PointSize` is not written
- Shader modules are deprecated - applications can now pass `VkShaderModuleCreateInfo` as a chained struct to pipeline creation via `VkPipelineShaderStageCreateInfo`
- A function `vkGetRenderingAreaGranularityKHR` to query the optimal render area for a dynamic rendering instance.

- A property to indicate that depth/stencil texturing operations with `VK_COMPONENT_SWIZZLE_ONE` have defined behavior
- Add `vkGetImageSubresourceLayout2KHR` and a new function `vkGetDeviceImageSubresourceLayoutKHR` to allow the application to query the image memory layout without having to create an image object and query it.
- Allow `VK_REMAINING_ARRAY_LAYERS` as the `layerCount` member of `VkImageSubresourceLayers`
- Adds stronger guarantees for propagation of `VK_ERROR_DEVICE_LOST` return values
- A property to indicate whether `PointSize` controls the final rasterization of polygons if `polygon mode` is `VK_POLYGON_MODE_POINT`
- Two properties to indicate the non-strict line rasterization algorithm used
- Two new flags words `VkPipelineCreateFlagBits2KHR` and `VkBufferUsageFlagBits2KHR`
- Physical-device-level functions can now be called with any value in the valid range for a type beyond the defined enumerants, such that applications can avoid checking individual features, extensions, or versions before querying supported properties of a particular enumerant.
- Clarification that copies between images of any type are allowed, treating 1D images as 2D images with a height of 1.

New Commands

- `vkCmdBindIndexBuffer2KHR`
- `vkGetDeviceImageSubresourceLayoutKHR`
- `vkGetImageSubresourceLayout2KHR`
- `vkGetRenderingAreaGranularityKHR`

New Structures

- `VkDeviceImageSubresourceInfoKHR`
- `VkImageSubresource2KHR`
- `VkRenderingAreaInfoKHR`
- `VkSubresourceLayout2KHR`
- Extending `VkBufferViewCreateInfo`, `VkBufferCreateInfo`, `VkPhysicalDeviceExternalBufferInfo`, `VkDescriptorBufferBindingInfoEXT`:
 - `VkBufferUsageFlags2CreateInfoKHR`
- Extending `VkComputePipelineCreateInfo`, `VkGraphicsPipelineCreateInfo`, `VkRayTracingPipelineCreateInfoNV`, `VkRayTracingPipelineCreateInfoKHR`:
 - `VkPipelineCreateFlags2CreateInfoKHR`
- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceMaintenance5FeaturesKHR`
- Extending `VkPhysicalDeviceProperties2`:
 - `VkPhysicalDeviceMaintenance5PropertiesKHR`

New Enums

- [VkBufferUsageFlagBits2KHR](#)
- [VkPipelineCreateFlagBits2KHR](#)

New Bitmasks

- [VkBufferUsageFlags2KHR](#)
- [VkPipelineCreateFlags2KHR](#)

New Enum Constants

- [VK_KHR_MAINTENANCE_5_EXTENSION_NAME](#)
- [VK_KHR_MAINTENANCE_5_SPEC_VERSION](#)
- Extending [VkFormat](#):
 - [VK_FORMAT_A1B5G5R5_UNORM_PACK16_KHR](#)
 - [VK_FORMAT_A8_UNORM_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_BUFFER_USAGE_FLAGS_2_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_DEVICE_IMAGE_SUBRESOURCE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_IMAGE_SUBRESOURCE_2_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_5_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_CREATE_FLAGS_2_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_RENDERING_AREA_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_SUBRESOURCE_LAYOUT_2_KHR](#)

If [VK_KHR_dynamic_rendering](#) or [Vulkan Version 1.3](#) and [VK_EXT_fragment_density_map](#) is supported:

- Extending [VkPipelineCreateFlagBits2KHR](#):
 - [VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT](#)

If [VK_KHR_dynamic_rendering](#) or [Vulkan Version 1.3](#) and [VK_KHR_fragment_shading_rate](#) is supported:

- Extending [VkPipelineCreateFlagBits2KHR](#):
 - [VK_PIPELINE_CREATE_2_RENDERING_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR](#)

If [VK_EXT_attachment_feedback_loop_layout](#) is supported:

- Extending [VkPipelineCreateFlagBits2KHR](#):
 - [VK_PIPELINE_CREATE_2_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT](#)

- `VK_PIPELINE_CREATE_2_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT`

If `VK_EXT_conditional_rendering` is supported:

- Extending `VkBufferUsageFlagBits2KHR`:
 - `VK_BUFFER_USAGE_2_CONDITIONAL_RENDERING_BIT_EXT`

If `VK_EXT_descriptor_buffer` is supported:

- Extending `VkBufferUsageFlagBits2KHR`:
 - `VK_BUFFER_USAGE_2_PUSH_DESCRIPTOR_DESCRIPTOR_BUFFER_BIT_EXT`
 - `VK_BUFFER_USAGE_2_RESOURCE_DESCRIPTOR_BUFFER_BIT_EXT`
 - `VK_BUFFER_USAGE_2_SAMPLER_DESCRIPTOR_BUFFER_BIT_EXT`
- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_DESCRIPTOR_BUFFER_BIT_EXT`

If `VK_EXT_graphics_pipeline_library` is supported:

- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_LINK_TIME_OPTIMIZATION_BIT_EXT`
 - `VK_PIPELINE_CREATE_2_RETAIN_LINK_TIME_OPTIMIZATION_INFO_BIT_EXT`

If `VK_EXT_opacity_micromap` is supported:

- Extending `VkBufferUsageFlagBits2KHR`:
 - `VK_BUFFER_USAGE_2_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT`
 - `VK_BUFFER_USAGE_2_MICROMAP_STORAGE_BIT_EXT`
- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT`

If `VK_EXT_pipeline_protected_access` is supported:

- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_NO_PROTECTED_ACCESS_BIT_EXT`
 - `VK_PIPELINE_CREATE_2_PROTECTED_ACCESS_ONLY_BIT_EXT`

If `VK_EXT_transform_feedback` is supported:

- Extending `VkBufferUsageFlagBits2KHR`:
 - `VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT`
 - `VK_BUFFER_USAGE_2_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT`

If `VK_KHR_acceleration_structure` is supported:

- Extending `VkBufferUsageFlagBits2KHR`:

- `VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_BUILD_INPUT_READ_ONLY_BIT_KHR`
- `VK_BUFFER_USAGE_2_ACCELERATION_STRUCTURE_STORAGE_BIT_KHR`

If `VK_KHR_pipeline_executable_properties` is supported:

- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR`
 - `VK_PIPELINE_CREATE_2_CAPTURE_STATISTICS_BIT_KHR`

If `VK_KHR_pipeline_library` is supported:

- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_LIBRARY_BIT_KHR`

If `VK_KHR_ray_tracing_pipeline` is supported:

- Extending `VkBufferUsageFlagBits2KHR`:
 - `VK_BUFFER_USAGE_2_SHADER_BINDING_TABLE_BIT_KHR`
- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR`
 - `VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR`
 - `VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR`
 - `VK_PIPELINE_CREATE_2_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR`
 - `VK_PIPELINE_CREATE_2_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR`
 - `VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_AABBS_BIT_KHR`
 - `VK_PIPELINE_CREATE_2_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR`

If `VK_KHR_video_decode_queue` is supported:

- Extending `VkBufferUsageFlagBits2KHR`:
 - `VK_BUFFER_USAGE_2_VIDEO_DECODE_DST_BIT_KHR`
 - `VK_BUFFER_USAGE_2_VIDEO_DECODE_SRC_BIT_KHR`

If `VK_KHR_video_encode_queue` is supported:

- Extending `VkBufferUsageFlagBits2KHR`:
 - `VK_BUFFER_USAGE_2_VIDEO_ENCODE_DST_BIT_KHR`
 - `VK_BUFFER_USAGE_2_VIDEO_ENCODE_SRC_BIT_KHR`

If `VK_NV_device_generated_commands` is supported:

- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_NV`

If `VK_NV_displacement_micromap` is supported:

- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_RAY_TRACING_DISPLACEMENT_MICROMAP_BIT_NV`

If `VK_NV_ray_tracing` is supported:

- Extending `VkBufferUsageFlagBits2KHR`:
 - `VK_BUFFER_USAGE_2_RAY_TRACING_BIT_NV`
- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_DEFER_COMPILE_BIT_NV`

If `VK_NV_ray_tracing_motion_blur` is supported:

- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_RAY_TRACING_ALLOW_MOTION_BIT_NV`

If `Vulkan Version 1.1` or `VK_KHR_device_group` is supported:

- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_DISPATCH_BASE_BIT_KHR`
 - `VK_PIPELINE_CREATE_2_VIEW_INDEX_FROM_DEVICE_INDEX_BIT_KHR`

If `Vulkan Version 1.2` or `VK_KHR_buffer_device_address` or `VK_EXT_buffer_device_address` is supported:

- Extending `VkBufferUsageFlagBits2KHR`:
 - `VK_BUFFER_USAGE_2_SHADER_DEVICE_ADDRESS_BIT_KHR`

If `Vulkan Version 1.3` or `VK_EXT_pipeline_creation_cache_control` is supported:

- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_EARLY_RETURN_ON_FAILURE_BIT_KHR`
 - `VK_PIPELINE_CREATE_2_FAIL_ON_PIPELINE_COMPILE_REQUIRED_BIT_KHR`

Issues

None.

Version History

- Revision 1, 2022-12-12 (Stu Smith)
 - Initial revision

VK_KHR_maintenance6

Name String

VK_KHR_maintenance6

Extension Type

Device extension

Registered Extension Number

546

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

API Interactions

- Interacts with VK_EXT_descriptor_buffer
- Interacts with VK_KHR_push_descriptor

Contact

- Jon Leech [\[GitHub\]oddhack](#)

Extension Proposal

[VK_KHR_maintenance6](#)

Other Extension Metadata

Last Modified Date

2023-08-03

Interactions and External Dependencies

- Interacts with [VK_EXT_robustness2](#)

Contributors

- Jon Leech, Khronos
- Stu Smith, AMD
- Mike Blumenkrantz, Valve
- Ralph Potter, Samsung
- James Fitzpatrick, Imagination Technologies
- Piers Daniell, NVIDIA
- Daniel Story, Nintendo

Description

[VK_KHR_maintenance6](#) adds a collection of minor features, none of which would warrant an entire extension of their own.

The new features are as follows:

- [VkBindMemoryStatusKHR](#) may be included in the [pNext](#) chain of [VkBindBufferMemoryInfo](#) and [VkBindImageMemoryInfo](#), allowing applications to identify individual resources for which memory binding failed during calls to [vkBindBufferMemory2KHR](#) and [vkBindImageMemory2KHR](#).
- A new property [fragmentShadingRateClampCombinerInputs](#) to indicate if an implementation clamps the inputs to fragment shading rate combiner operations.
- [VK_NULL_HANDLE](#) is allowed to be used when binding an index buffer, instead of a valid [VkBuffer](#) handle. When the [nullDescriptor](#) feature is enabled, every index fetched results in a value of zero.
- A new property [maxCombinedImageSamplerDescriptorCount](#) to indicate the maximum number of descriptors needed for any of the [formats that require a sampler Y'C_BC_R conversion](#) supported by the implementation.
- A new property [blockTexelViewCompatibleMultipleLayers](#) indicating whether [VK_IMAGE_CREATE_BLOCK_TEXEL_VIEW_COMPATIBLE_BIT](#) is allowed to be used with [layerCount](#) > 1
- [pNext](#) extensible *2 versions of all descriptor binding commands.

New Commands

- [vkCmdBindDescriptorSets2KHR](#)
- [vkCmdPushConstants2KHR](#)

If [VK_EXT_descriptor_buffer](#) is supported:

- [vkCmdBindDescriptorBufferEmbeddedSamplers2EXT](#)
- [vkCmdSetDescriptorBufferOffsets2EXT](#)

If [VK_KHR_push_descriptor](#) is supported:

- [vkCmdPushDescriptorSet2KHR](#)
- [vkCmdPushDescriptorSetWithTemplate2KHR](#)

New Structures

- [VkBindDescriptorSetsInfoKHR](#)
- [VkPushConstantsInfoKHR](#)
- Extending [VkBindBufferMemoryInfo](#), [VkBindImageMemoryInfo](#):
 - [VkBindMemoryStatusKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):

- [VkPhysicalDeviceMaintenance6FeaturesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceMaintenance6PropertiesKHR](#)

If [VK_EXT_descriptor_buffer](#) is supported:

- [VkBindDescriptorBufferEmbeddedSamplersInfoEXT](#)
- [VkSetDescriptorBufferOffsetsInfoEXT](#)

If [VK_KHR_push_descriptor](#) is supported:

- [VkPushDescriptorSetInfoKHR](#)
- [VkPushDescriptorSetWithTemplateInfoKHR](#)

New Enum Constants

- [VK_KHR_MAINTENANCE_6_EXTENSION_NAME](#)
- [VK_KHR_MAINTENANCE_6_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_SETS_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_BIND_MEMORY_STATUS_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_6_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_PUSH_CONSTANTS_INFO_KHR](#)

If [VK_EXT_descriptor_buffer](#) is supported:

- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_BIND_DESCRIPTOR_BUFFER_EMBEDDED_SAMPLERS_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_SET_DESCRIPTOR_BUFFER_OFFSETS_INFO_EXT](#)

If [VK_KHR_push_descriptor](#) is supported:

- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_PUSH_DESCRIPTOR_SET_WITH_TEMPLATE_INFO_KHR](#)

Issues

None.

Version History

- Revision 1, 2023-08-01 (Jon Leech)

- Initial revision

VK_KHR_maintenance7

Name String

VK_KHR_maintenance7

Extension Type

Device extension

Registered Extension Number

563

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

Contact

- Mike Blumenkrantz [\[GitHub\]](#)zmike

Extension Proposal

[VK_KHR_maintenance7](#)

Other Extension Metadata

Last Modified Date

2024-01-30

Interactions and External Dependencies

Contributors

- Mike Blumenkrantz, Valve
- Hans-Kristian Arntzen, Valve
- Pan Gao, Huawei
- Tobias Hector, AMD
- Jon Leech, Khronos
- Daniel Story, Nintendo
- Shahbaz Youssefi, Google
- Yiwei Zhang, Google
- Matthew Netsch, Qualcomm

Description

[VK_KHR_maintenance7](#) adds a collection of minor features, none of which would warrant an entire extension of their own.

The proposed new features are as follows:

- Add a property query to determine if a framebuffer writes to depth or stencil aspect does not trigger a write access in the sibling aspect. For example, this allows sampling stencil aspect as a texture while rendering to the sibling depth attachment and vice-versa given appropriate image layouts.
- Add a way to query information regarding the underlying devices in environments where the Vulkan implementation is provided through layered implementations. For example, running on Mesa/Venus, driver ID is returned as `VK_DRIVER_ID_MESA_VENUS`, but it can be necessary to know what the real driver under the hood is. The new [VkPhysicalDeviceLayeredApiPropertiesKHR](#) structure can be used to gather information regarding layers underneath the top-level physical device.
- Promote `VK_RENDERING_CONTENTS_INLINE_BIT_EXT` and `VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_EXT` to KHR
- Add a limit to report the maximum total count of dynamic uniform buffers and dynamic storage buffers that can be included in a pipeline layout.
- Require that for an unsigned integer query, the 32-bit result value **must** be equal to the 32 least significant bits of the equivalent 64-bit result value.
- Add query for robust access support when using fragment shading rate attachments

New Structures

- [VkPhysicalDeviceLayeredApiPropertiesKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceMaintenance7FeaturesKHR](#)
- Extending [VkPhysicalDeviceLayeredApiPropertiesKHR](#):
 - [VkPhysicalDeviceLayeredApiVulkanPropertiesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceLayeredApiPropertiesListKHR](#)
 - [VkPhysicalDeviceMaintenance7PropertiesKHR](#)

New Enums

- [VkPhysicalDeviceLayeredApiKHR](#)

New Enum Constants

- `VK_KHR_MAINTENANCE_7_EXTENSION_NAME`
- `VK_KHR_MAINTENANCE_7_SPEC_VERSION`

- Extending [VkRenderingFlagBits](#):
 - `VK_RENDERING_CONTENTS_INLINE_BIT_KHR`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_PROPERTIES_LIST_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LAYERED_API_VULKAN_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_FEATURES_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAINTENANCE_7_PROPERTIES_KHR`
- Extending [VkSubpassContents](#):
 - `VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_KHR`

Issues

None.

Version History

- Revision 1, 2024-01-30 (Jon Leech)
 - Initial revision

VK_KHR_map_memory2

Name String

`VK_KHR_map_memory2`

Extension Type

Device extension

Registered Extension Number

272

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Faith Ekstrand [\[GitHub\]gfxstrand](#)

Extension Proposal

[VK_KHR_map_memory2](#)

Other Extension Metadata

Last Modified Date

2023-03-14

Interactions and External Dependencies

- None

Contributors

- Faith Ekstrand, Collabora
- Tobias Hector, AMD

Description

This extension provides extensible versions of the Vulkan memory map and unmap commands. The new commands are functionally identical to the core commands, except that their parameters are specified using extensible structures that can be used to pass extension-specific information.

New Commands

- [vkMapMemory2KHR](#)
- [vkUnmapMemory2KHR](#)

New Structures

- [VkMemoryMapInfoKHR](#)
- [VkMemoryUnmapInfoKHR](#)

New Enums

- [VkMemoryUnmapFlagBitsKHR](#)

New Bitmasks

- [VkMemoryUnmapFlagsKHR](#)

New Enum Constants

- [VK_KHR_MAP_MEMORY_2_EXTENSION_NAME](#)
- [VK_KHR_MAP_MEMORY_2_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_MEMORY_MAP_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_MEMORY_UNMAP_INFO_KHR](#)

Version History

- Revision 0, 2022-08-03 (Faith Ekstrand)
 - Internal revisions

- Revision 1, 2023-03-14
 - Public release

VK_KHR_multiview

Name String

VK_KHR_multiview

Extension Type

Device extension

Registered Extension Number

54

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_multiview](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jeff Bolz [\[GitHub\]jeffbolznv](#)

Other Extension Metadata

Last Modified Date

2016-10-28

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for [GL_EXT_multiview](#)

Contributors

- Jeff Bolz, NVIDIA

Description

This extension has the same goal as the OpenGL ES `GL_OVR_multiview` extension. Multiview is a rendering technique originally designed for VR where it is more efficient to record a single set of commands to be executed with slightly different behavior for each “view”.

It includes a concise way to declare a render pass with multiple views, and gives implementations freedom to render the views in the most efficient way possible. This is done with a multiview configuration specified during `render pass` creation with the `VkRenderPassMultiviewCreateInfo` passed into `VkRenderPassCreateInfo::pNext`.

This extension enables the use of the `SPV_KHR_multiview` shader extension, which adds a new `ViewIndex` built-in type that allows shaders to control what to do for each view. If using GLSL there is also the `GL_EXT_multiview` extension that introduces a `highp int gl_ViewIndex;` built-in variable for vertex, tessellation, geometry, and fragment shaders.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceMultiviewFeaturesKHR`
- Extending `VkPhysicalDeviceProperties2`:
 - `VkPhysicalDeviceMultiviewPropertiesKHR`
- Extending `VkRenderPassCreateInfo`:
 - `VkRenderPassMultiviewCreateInfoKHR`

New Enum Constants

- `VK_KHR_MULTIVIEW_EXTENSION_NAME`
- `VK_KHR_MULTIVIEW_SPEC_VERSION`
- Extending `VkDependencyFlagBits`:
 - `VK_DEPENDENCY_VIEW_LOCAL_BIT_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_FEATURES_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTIVIEW_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_RENDER_PASS_MULTIVIEW_CREATE_INFO_KHR`

New Built-In Variables

- `ViewIndex`

New SPIR-V Capabilities

- [MultiView](#)

Version History

- Revision 1, 2016-10-28 (Jeff Bolz)
 - Internal revisions

VK_KHR_performance_query

Name String

[VK_KHR_performance_query](#)

Extension Type

Device extension

Registered Extension Number

117

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Use

- [Developer tools](#)

Contact

- Alon Or-bach [\[GitHub\]](#)[alonorbach](#)

Other Extension Metadata

Last Modified Date

2019-10-08

IP Status

No known IP claims.

Contributors

- Jesse Barker, Unity Technologies
- Kenneth Benzie, Codeplay

- Jan-Harald Fredriksen, ARM
- Jeff Leger, Qualcomm
- Jesse Hall, Google
- Tobias Hector, AMD
- Neil Henning, Codeplay
- Baldur Karlsson
- Lionel Landwerlin, Intel
- Peter Lohrmann, AMD
- Alon Or-bach, Samsung
- Daniel Rakos, AMD
- Niklas Smedberg, Unity Technologies
- Igor Ostrowski, Intel

Description

The `VK_KHR_performance_query` extension adds a mechanism to allow querying of performance counters for use in applications and by profiling tools.

Each queue family **may** expose counters that **can** be enabled on a queue of that family. We extend `VkQueryType` to add a new query type for performance queries, and chain a structure on `VkQueryPoolCreateInfo` to specify the performance queries to enable.

New Commands

- `vkAcquireProfilingLockKHR`
- `vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR`
- `vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR`
- `vkReleaseProfilingLockKHR`

New Structures

- `VkAcquireProfilingLockInfoKHR`
- `VkPerformanceCounterDescriptionKHR`
- `VkPerformanceCounterKHR`
- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDevicePerformanceQueryFeaturesKHR`
- Extending `VkPhysicalDeviceProperties2`:
 - `VkPhysicalDevicePerformanceQueryPropertiesKHR`
- Extending `VkQueryPoolCreateInfo`:
 - `VkQueryPoolPerformanceCreateInfoKHR`

- Extending [VkSubmitInfo](#), [VkSubmitInfo2](#):
 - [VkPerformanceQuerySubmitInfoKHR](#)

New Unions

- [VkPerformanceCounterResultKHR](#)

New Enums

- [VkAcquireProfilingLockFlagBitsKHR](#)
- [VkPerformanceCounterDescriptionFlagBitsKHR](#)
- [VkPerformanceCounterScopeKHR](#)
- [VkPerformanceCounterStorageKHR](#)
- [VkPerformanceCounterUnitKHR](#)

New Bitmasks

- [VkAcquireProfilingLockFlagsKHR](#)
- [VkPerformanceCounterDescriptionFlagsKHR](#)

New Enum Constants

- [VK_KHR_PERFORMANCE_QUERY_EXTENSION_NAME](#)
- [VK_KHR_PERFORMANCE_QUERY_SPEC_VERSION](#)
- Extending [VkQueryType](#):
 - [VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_ACQUIRE_PROFILING_LOCK_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_DESCRIPTION_KHR](#)
 - [VK_STRUCTURE_TYPE_PERFORMANCE_COUNTER_KHR](#)
 - [VK_STRUCTURE_TYPE_PERFORMANCE_QUERY_SUBMIT_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PERFORMANCE_QUERY_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_QUERY_POOL_PERFORMANCE_CREATE_INFO_KHR](#)

Issues

1) Should this extension include a mechanism to begin a query in command buffer *A* and end the query in command buffer *B*?

RESOLVED No - queries are tied to command buffer creation and thus have to be encapsulated within a single command buffer.

2) Should this extension include a mechanism to begin and end queries globally on the queue, not using the existing command buffer commands?

RESOLVED No - for the same reasoning as the resolution of 1).

3) Should this extension expose counters that require multiple passes?

RESOLVED Yes - users should re-submit a command buffer with the same commands in it multiple times, specifying the pass to count as the query parameter in `VkPerformanceQuerySubmitInfoKHR`.

4) How to handle counters across parallel workloads?

RESOLVED In the spirit of Vulkan, a counter description flag `VK_PERFORMANCE_COUNTER_DESCRIPTION_CONCURRENTLY_IMPACTED_BIT_KHR` denotes that the accuracy of a counter result is affected by parallel workloads.

5) How to handle secondary command buffers?

RESOLVED Secondary command buffers inherit any counter pass index specified in the parent primary command buffer. Note: this is no longer an issue after change from issue 10 resolution

6) What commands does the profiling lock have to be held for?

RESOLVED For any command buffer that is being queried with a performance query pool, the profiling lock **must** be held while that command buffer is in the *recording*, *executable*, or *pending state*.

7) Should we support `vkCmdCopyQueryPoolResults`?

RESOLVED Yes.

8) Should we allow performance queries to interact with multiview?

RESOLVED Yes, but the performance queries must be performed once for each pass per view.

9) Should a `queryCount > 1` be usable for performance queries?

RESOLVED Yes. Some vendors will have costly performance counter query pool creation, and would rather if a certain set of counters were to be used multiple times that a `queryCount > 1` can be used to amortize the instantiation cost.

10) Should we introduce an indirect mechanism to set the counter pass index?

RESOLVED Specify the counter pass index at submit time instead, to avoid requiring re-recording of command buffers when multiple counter passes are needed.

Examples

The following example shows how to find what performance counters a queue family supports, setup a query pool to record these performance counters, how to add the query pool to the command buffer to record information, and how to get the results from the query pool.


```

// A previously created physical device
VkPhysicalDevice physicalDevice;

// One of the queue families our device supports
uint32_t queueFamilyIndex;

uint32_t counterCount;

// Get the count of counters supported
vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR(
    physicalDevice,
    queueFamilyIndex,
    &counterCount,
    NULL,
    NULL);

VkPerformanceCounterKHR* counters =
    malloc(sizeof(VkPerformanceCounterKHR) * counterCount);
VkPerformanceCounterDescriptionKHR* counterDescriptions =
    malloc(sizeof(VkPerformanceCounterDescriptionKHR) * counterCount);

// Get the counters supported
vkEnumeratePhysicalDeviceQueueFamilyPerformanceQueryCountersKHR(
    physicalDevice,
    queueFamilyIndex,
    &counterCount,
    counters,
    counterDescriptions);

// Try to enable the first 8 counters
uint32_t enabledCounters[8];

const uint32_t enabledCounterCount = min(counterCount, 8);

for (uint32_t i = 0; i < enabledCounterCount; i++) {
    enabledCounters[i] = i;
}

// A previously created device that had the performanceCounterQueryPools feature
// set to VK_TRUE
VkDevice device;

VkQueryPoolPerformanceCreateInfoKHR performanceQueryCreateInfo = {
    .sType = VK_STRUCTURE_TYPE_QUERY_POOL_PERFORMANCE_CREATE_INFO_KHR,
    .pNext = NULL,

    // Specify the queue family that this performance query is performed on
    .queueFamilyIndex = queueFamilyIndex,

    // The number of counters to enable

```

```

.counterIndexCount = enabledCounterCount,

// The array of indices of counters to enable
.pCounterIndices = enabledCounters
};

// Get the number of passes our counters will require.
uint32_t numPasses;

vkGetPhysicalDeviceQueueFamilyPerformanceQueryPassesKHR(
    physicalDevice,
    &performanceQueryCreateInfo,
    &numPasses);

VkQueryPoolCreateInfo queryPoolCreateInfo = {
    .sType = VK_STRUCTURE_TYPE_QUERY_POOL_CREATE_INFO,
    .pNext = &performanceQueryCreateInfo,
    .flags = 0,
    // Using our new query type here
    .queryType = VK_QUERY_TYPE_PERFORMANCE_QUERY_KHR,
    .queryCount = 1,
    .pipelineStatistics = 0
};

VkQueryPool queryPool;

VkResult result = vkCreateQueryPool(
    device,
    &queryPoolCreateInfo,
    NULL,
    &queryPool);

assert(VK_SUCCESS == result);

// A queue from queueFamilyIndex
VkQueue queue;

// A command buffer we want to record counters on
VkCommandBuffer commandBuffer;

VkCommandBufferBeginInfo commandBufferBeginInfo = {
    .sType = VK_STRUCTURE_TYPE_COMMAND_BUFFER_BEGIN_INFO,
    .pNext = NULL,
    .flags = 0,
    .pInheritanceInfo = NULL
};

VkAcquireProfilingLockInfoKHR lockInfo = {
    .sType = VK_STRUCTURE_TYPE_ACQUIRE_PROFILING_LOCK_INFO_KHR,
    .pNext = NULL,

```

```

    .flags = 0,
    .timeout = UINT64_MAX // Wait forever for the lock
};

// Acquire the profiling lock before we record command buffers
// that will use performance queries

result = vkAcquireProfilingLockKHR(device, &lockInfo);

assert(VK_SUCCESS == result);

result = vkBeginCommandBuffer(commandBuffer, &commandBufferBeginInfo);

assert(VK_SUCCESS == result);

vkCmdResetQueryPool(
    commandBuffer,
    queryPool,
    0,
    1);

vkCmdBeginQuery(
    commandBuffer,
    queryPool,
    0,
    0);

// Perform the commands you want to get performance information on
// ...

// Perform a barrier to ensure all previous commands were complete before
// ending the query
vkCmdPipelineBarrier(commandBuffer,
    VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT,
    VK_PIPELINE_STAGE_BOTTOM_OF_PIPE_BIT,
    0,
    0,
    NULL,
    0,
    NULL,
    0,
    NULL);

vkCmdEndQuery(
    commandBuffer,
    queryPool,
    0);

result = vkEndCommandBuffer(commandBuffer);

assert(VK_SUCCESS == result);

```

```

for (uint32_t counterPass = 0; counterPass < numPasses; counterPass++) {

    VkPerformanceQuerySubmitInfoKHR performanceQuerySubmitInfo = {
        VK_STRUCTURE_TYPE_PERFORMANCE_QUERY_SUBMIT_INFO_KHR,
        NULL,
        counterPass
    };

    // Submit the command buffer and wait for its completion
    // ...
}

// Release the profiling lock after the command buffer is no longer in the
// pending state.
vkReleaseProfilingLockKHR(device);

result = vkResetCommandBuffer(commandBuffer, 0);

assert(VK_SUCCESS == result);

// Create an array to hold the results of all counters
VkPerformanceCounterResultKHR* recordedCounters = malloc(
    sizeof(VkPerformanceCounterResultKHR) * enabledCounterCount);

result = vkGetQueryPoolResults(
    device,
    queryPool,
    0,
    1,
    sizeof(VkPerformanceCounterResultKHR) * enabledCounterCount,
    recordedCounters,
    sizeof(VkPerformanceCounterResultKHR) * enabledCounterCount,
    NULL);

// recordedCounters is filled with our counters, we will look at one for posterity
switch (counters[0].storage) {
    case VK_PERFORMANCE_COUNTER_STORAGE_INT32:
        // use recordCounters[0].int32 to get at the counter result!
        break;
    case VK_PERFORMANCE_COUNTER_STORAGE_INT64:
        // use recordCounters[0].int64 to get at the counter result!
        break;
    case VK_PERFORMANCE_COUNTER_STORAGE_UINT32:
        // use recordCounters[0].uint32 to get at the counter result!
        break;
    case VK_PERFORMANCE_COUNTER_STORAGE_UINT64:
        // use recordCounters[0].uint64 to get at the counter result!
        break;
    case VK_PERFORMANCE_COUNTER_STORAGE_FLOAT32:

```

```
    // use recordCounters[0].float32 to get at the counter result!
    break;
case VK_PERFORMANCE_COUNTER_STORAGE_FLOAT64:
    // use recordCounters[0].float64 to get at the counter result!
    break;
}
```

Version History

- Revision 1, 2019-10-08

VK_KHR_pipeline_binary

Name String

VK_KHR_pipeline_binary

Extension Type

Device extension

Registered Extension Number

484

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_maintenance5](#)

Contact

- Stu Smith [\[GitHub\]](#)stu-s

Extension Proposal

[VK_KHR_pipeline_binary](#)

Other Extension Metadata

Last Modified Date

2024-07-01

Contributors

- Stu Smith, AMD
- Tobias Hector, AMD
- Alan Harrison, AMD
- Maciej Jesionowski, AMD

- Younggwan Kim, Arm
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- Jakub Kuderski, Google
- Piotr Byszewski, Mobica
- Piers Daniell, NVIDIA
- Ralph Potter, Samsung
- Matthew Netsch, Qualcomm
- Hans-Kristian Arntzen, Valve
- Samuel Pitoiset, Valve
- Tatsuyuki Ishi, Valve

Description

This extension provides a method to obtain binary data associated with individual pipelines such that applications can manage caching themselves instead of using `VkPipelineCache` objects.

New Object Types

- [VkPipelineBinaryKHR](#)

New Commands

- [vkCreatePipelineBinariesKHR](#)
- [vkDestroyPipelineBinaryKHR](#)
- [vkGetPipelineBinaryDataKHR](#)
- [vkGetPipelineKeyKHR](#)
- [vkReleaseCapturedPipelineDataKHR](#)

New Structures

- [VkPipelineBinaryCreateInfoKHR](#)
- [VkPipelineBinaryDataInfoKHR](#)
- [VkPipelineBinaryDataKHR](#)
- [VkPipelineBinaryHandlesInfoKHR](#)
- [VkPipelineBinaryKeyKHR](#)
- [VkPipelineBinaryKeysAndDataKHR](#)
- [VkPipelineCreateInfoKHR](#)

- [VkReleaseCapturedPipelineDataInfoKHR](#)
- Extending [VkDeviceCreateInfo](#):
 - [VkDevicePipelineBinaryInternalCacheControlKHR](#)
- Extending [VkGraphicsPipelineCreateInfo](#), [VkComputePipelineCreateInfo](#), [VkRayTracingPipelineCreateInfoKHR](#):
 - [VkPipelineBinaryInfoKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDevicePipelineBinaryFeaturesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDevicePipelineBinaryPropertiesKHR](#)

New Enum Constants

- [VK_KHR_PIPELINE_BINARY_EXTENSION_NAME](#)
- [VK_KHR_PIPELINE_BINARY_SPEC_VERSION](#)
- [VK_MAX_PIPELINE_BINARY_KEY_SIZE_KHR](#)
- Extending [VkObjectType](#):
 - [VK_OBJECT_TYPE_PIPELINE_BINARY_KHR](#)
- Extending [VkPipelineCreateFlagBits2KHR](#):
 - [VK_PIPELINE_CREATE_2_CAPTURE_DATA_BIT_KHR](#)
- Extending [VkResult](#):
 - [VK_ERROR_NOT_ENOUGH_SPACE_KHR](#)
 - [VK_PIPELINE_BINARY_MISSING_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_DEVICE_PIPELINE_BINARY_INTERNAL_CACHE_CONTROL_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_BINARY_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_BINARY_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_BINARY_DATA_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_BINARY_HANDLES_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_BINARY_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_BINARY_KEY_KHR](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_RELEASE_CAPTURED_PIPELINE_DATA_INFO_KHR](#)

Version History

- Revision 1, 2021-12-10 (Chris Glover)
 - Initial draft.

VK_KHR_pipeline_executable_properties

Name String

VK_KHR_pipeline_executable_properties

Extension Type

Device extension

Registered Extension Number

270

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Use

- [Developer tools](#)

Contact

- Faith Ekstrand [\[GitHub\]gfxstrand](#)

Other Extension Metadata

Last Modified Date

2019-05-28

IP Status

No known IP claims.

Interactions and External Dependencies

Contributors

- Faith Ekstrand, Intel
- Ian Romanick, Intel
- Kenneth Graunke, Intel
- Baldur Karlsson, Valve

- Jesse Hall, Google
- Jeff Bolz, Nvidia
- Piers Daniel, Nvidia
- Tobias Hector, AMD
- Jan-Harald Fredriksen, ARM
- Tom Olson, ARM
- Daniel Koch, Nvidia
- Spencer Fricke, Samsung

Description

When a pipeline is created, its state and shaders are compiled into zero or more device-specific executables, which are used when executing commands against that pipeline. This extension adds a mechanism to query properties and statistics about the different executables produced by the pipeline compilation process. This is intended to be used by debugging and performance tools to allow them to provide more detailed information to the user. Certain compile time shader statistics provided through this extension may be useful to developers for debugging or performance analysis.

New Commands

- [vkGetPipelineExecutableInternalRepresentationsKHR](#)
- [vkGetPipelineExecutablePropertiesKHR](#)
- [vkGetPipelineExecutableStatisticsKHR](#)

New Structures

- [VkPipelineExecutableInfoKHR](#)
- [VkPipelineExecutableInternalRepresentationKHR](#)
- [VkPipelineExecutablePropertiesKHR](#)
- [VkPipelineExecutableStatisticKHR](#)
- [VkPipelineInfoKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDevicePipelineExecutablePropertiesFeaturesKHR](#)

New Unions

- [VkPipelineExecutableStatisticValueKHR](#)

New Enums

- [VkPipelineExecutableStatisticFormatKHR](#)

New Enum Constants

- `VK_KHR_PIPELINE_EXECUTABLE_PROPERTIES_EXTENSION_NAME`
- `VK_KHR_PIPELINE_EXECUTABLE_PROPERTIES_SPEC_VERSION`
- Extending `VkPipelineCreateFlagBits`:
 - `VK_PIPELINE_CREATE_CAPTURE_INTERNAL_REPRESENTATIONS_BIT_KHR`
 - `VK_PIPELINE_CREATE_CAPTURE_STATISTICS_BIT_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PIPELINE_EXECUTABLE_PROPERTIES_FEATURES_KHR`
 - `VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_INTERNAL_REPRESENTATION_KHR`
 - `VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_PIPELINE_EXECUTABLE_STATISTIC_KHR`
 - `VK_STRUCTURE_TYPE_PIPELINE_INFO_KHR`

Issues

1) What should we call the pieces of the pipeline which are produced by the compilation process and about which you can query properties and statistics?

RESOLVED: Call them “executables”. The name “binary” was used in early drafts of the extension but it was determined that “pipeline binary” could have a fairly broad meaning (such as a binary serialized form of an entire pipeline) and was too big of a namespace for the very specific needs of this extension.

Version History

- Revision 1, 2019-05-28 (Faith Ekstrand)
 - Initial draft

VK_KHR_pipeline_library

Name String

`VK_KHR_pipeline_library`

Extension Type

Device extension

Registered Extension Number

291

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Christoph Kubisch [\[GitHub\]](#)[pixeljetstream](#)

Other Extension Metadata

Last Modified Date

2020-01-08

IP Status

No known IP claims.

Contributors

- See contributors to [VK_KHR_ray_tracing_pipeline](#)

Description

A pipeline library is a special pipeline that cannot be bound, instead it defines a set of shaders and shader groups which can be linked into other pipelines. This extension defines the infrastructure for pipeline libraries, but does not specify the creation or usage of pipeline libraries. This is left to additional dependent extensions.

New Structures

- Extending [VkGraphicsPipelineCreateInfo](#):
 - [VkPipelineLibraryCreateInfoKHR](#)

New Enum Constants

- [VK_KHR_PIPELINE_LIBRARY_EXTENSION_NAME](#)
- [VK_KHR_PIPELINE_LIBRARY_SPEC_VERSION](#)
- Extending [VkPipelineCreateFlagBits](#):
 - [VK_PIPELINE_CREATE_LIBRARY_BIT_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PIPELINE_LIBRARY_CREATE_INFO_KHR](#)

Version History

- Revision 1, 2020-01-08 (Christoph Kubisch)
 - Initial draft.

VK_KHR_portability_enumeration

Name String

VK_KHR_portability_enumeration

Extension Type

Instance extension

Registered Extension Number

395

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Charles Giessen [\[GitHub\]charles-lunarg](#)

Other Extension Metadata

Last Modified Date

2021-06-02

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with [VK_KHR_portability_subset](#)

Contributors

- Lenny Komow, LunarG
- Charles Giessen, LunarG

Description

This extension allows applications to control whether devices that expose the [VK_KHR_portability_subset](#) extension are included in the results of physical device enumeration. Since devices which support the [VK_KHR_portability_subset](#) extension are not fully conformant Vulkan implementations, the Vulkan loader does not report those devices unless the application explicitly asks for them. This prevents applications which may not be aware of non-conformant devices from accidentally using them, as any device which supports the [VK_KHR_portability_subset](#) extension mandates that the extension must be enabled if that device is used.

This extension is implemented in the loader.

New Enum Constants

- `VK_KHR_PORTABILITY_ENUMERATION_EXTENSION_NAME`
- `VK_KHR_PORTABILITY_ENUMERATION_SPEC_VERSION`
- Extending [VkInstanceCreateFlagBits](#):
 - `VK_INSTANCE_CREATE_ENUMERATE_PORTABILITY_BIT_KHR`

Version History

- Revision 1, 2021-06-02 (Lenny Komow)
 - Initial version

VK_KHR_present_id

Name String

`VK_KHR_present_id`

Extension Type

Device extension

Registered Extension Number

295

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_swapchain](#)

and

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Keith Packard [\[GitHub\]keithp](#)

Other Extension Metadata

Last Modified Date

2019-05-15

IP Status

No known IP claims.

Contributors

- Keith Packard, Valve
- Ian Elliott, Google
- Alon Or-bach, Samsung

Description

This device extension allows an application that uses the [VK_KHR_swapchain](#) extension to provide an identifier for present operations on a swapchain. An application **can** use this to reference specific present operations in other extensions.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDevicePresentIdFeaturesKHR](#)
- Extending [VkPresentInfoKHR](#):
 - [VkPresentIdKHR](#)

New Enum Constants

- [VK_KHR_PRESENT_ID_EXTENSION_NAME](#)
- [VK_KHR_PRESENT_ID_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_ID_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_PRESENT_ID_KHR](#)

Issues

None.

Examples

Version History

- Revision 1, 2019-05-15 (Keith Packard)
 - Initial version

VK_KHR_present_wait

Name String

[VK_KHR_present_wait](#)

Extension Type

Device extension

Registered Extension Number

249

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_swapchain](#)
and
[VK_KHR_present_id](#)

Contact

- Keith Packard [[GitHub](#)][keithp](#)

Other Extension Metadata

Last Modified Date

2019-05-15

IP Status

No known IP claims.

Contributors

- Keith Packard, Valve
- Ian Elliott, Google
- Tobias Hector, AMD
- Daniel Stone, Collabora

Description

This device extension allows an application that uses the [VK_KHR_swapchain](#) extension to wait for present operations to complete. An application can use this to monitor and control the pacing of the application by managing the number of outstanding images yet to be presented.

New Commands

- [vkWaitForPresentKHR](#)

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDevicePresentWaitFeaturesKHR](#)

New Enum Constants

- `VK_KHR_PRESENT_WAIT_EXTENSION_NAME`
- `VK_KHR_PRESENT_WAIT_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_WAIT_FEATURES_KHR`

Issues

1) When does the wait finish?

RESOLVED. The wait will finish when the present is visible to the user. There is no requirement for any precise timing relationship between the presentation of the image to the user, but implementations **should** signal the wait as close as possible to the presentation of the first pixel in the new image to the user.

2) Should this use fences or other existing synchronization mechanism.

RESOLVED. Because display and rendering are often implemented in separate drivers, this extension will provide a separate synchronization API.

3) Should this extension share present identification with other extensions?

RESOLVED. Yes. A new extension, `VK_KHR_present_id`, should be created to provide a shared structure for presentation identifiers.

4) What happens when presentations complete out of order wrt calls to `vkQueuePresent`? This could happen if the semaphores for the presentations were ready out of order.

OPTION A: Require that when a `PresentId` is set that the driver ensure that images are always presented in the order of calls to `vkQueuePresent`.

OPTION B: Finish both waits when the earliest present completes. This will complete the later present wait earlier than the actual presentation. This should be the easiest to implement as the driver need only track the largest present ID completed. This is also the 'natural' consequence of interpreting the existing `vkWaitForPresentKHR` specification.

OPTION C: Finish both waits when both have completed. This will complete the earlier presentation later than the actual presentation time. This is allowed by the current specification as there is no precise timing requirement for when the `presentId` value is updated. This requires slightly more complexity in the driver as it will need to track all outstanding `presentId` values.

Examples

Version History

- Revision 1, 2019-02-19 (Keith Packard)
 - Initial version

VK_KHR_push_descriptor

Name String

VK_KHR_push_descriptor

Extension Type

Device extension

Registered Extension Number

81

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

API Interactions

- Interacts with [VK_VERSION_1_1](#)
- Interacts with [VK_KHR_descriptor_update_template](#)

Contact

- Jeff Bolz [\[GitHub\]jeffbolznv](#)

Other Extension Metadata

Last Modified Date

2017-09-12

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Michael Worcester, Imagination Technologies

Description

This extension allows descriptors to be written into the command buffer, while the implementation is responsible for managing their memory. Push descriptors may enable easier porting from older APIs and in some cases can be more efficient than writing descriptors into descriptor sets.

New Commands

- [vkCmdPushDescriptorSetKHR](#)

If [Vulkan Version 1.1](#) or [VK_KHR_descriptor_update_template](#) is supported:

- [vkCmdPushDescriptorSetWithTemplateKHR](#)

New Structures

- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDevicePushDescriptorPropertiesKHR](#)

New Enum Constants

- [VK_KHR_PUSH_DESCRIPTOR_EXTENSION_NAME](#)
- [VK_KHR_PUSH_DESCRIPTOR_SPEC_VERSION](#)
- Extending [VkDescriptorSetLayoutCreateFlagBits](#):
 - [VK_DESCRIPTOR_SET_LAYOUT_CREATE_PUSH_DESCRIPTOR_BIT_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PUSH_DESCRIPTOR_PROPERTIES_KHR](#)

If [Vulkan Version 1.1](#) or [VK_KHR_descriptor_update_template](#) is supported:

- Extending [VkDescriptorUpdateTemplateType](#):
 - [VK_DESCRIPTOR_UPDATE_TEMPLATE_TYPE_PUSH_DESCRIPTOR_KHR](#)

Version History

- Revision 1, 2016-10-15 (Jeff Bolz)
 - Internal revisions
- Revision 2, 2017-09-12 (Tobias Hector)
 - Added interactions with Vulkan 1.1

VK_KHR_ray_query

Name String

[VK_KHR_ray_query](#)

Extension Type

Device extension

Registered Extension Number

349

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_spirv_1_4](#)

and

[VK_KHR_acceleration_structure](#)

SPIR-V Dependencies

- [SPV_KHR_ray_query](#)

Contact

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Other Extension Metadata

Last Modified Date

2020-11-12

Interactions and External Dependencies

- This extension provides API support for [GLSL_EXT_ray_query](#)

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- Jesse Barker, Unity
- Baldur Karlsson, Valve

Description

Rasterization has been the dominant method to produce interactive graphics, but increasing performance of graphics hardware has made ray tracing a viable option for interactive rendering. Being able to integrate ray tracing with traditional rasterization makes it easier for applications to incrementally add ray traced effects to existing applications or to do hybrid approaches with rasterization for primary visibility and ray tracing for secondary queries.

Ray queries are available to all shader types, including graphics, compute and ray tracing pipelines. Ray queries are not able to launch additional shaders, instead returning traversal results to the calling shader.

This extension adds support for the following SPIR-V extension in Vulkan:

- `SPV_KHR_ray_query`

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceRayQueryFeaturesKHR](#)

New Enum Constants

- [VK_KHR_RAY_QUERY_EXTENSION_NAME](#)
- [VK_KHR_RAY_QUERY_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_QUERY_FEATURES_KHR](#)

New SPIR-V Capabilities

- [RayQueryKHR](#)
- [RayTraversalPrimitiveCullingKHR](#)

Sample Code

Example of ray query in a GLSL shader, illustrating how to use ray queries to determine whether a given position (at ray origin) is in shadow or not, by tracing a ray towards the light, and checking for any intersections with geometry occluding the light.

```
rayQueryEXT rq;

rayQueryInitializeEXT(rq, accStruct, gl_RayFlagsTerminateOnFirstHitEXT, cullMask,
origin, tMin, direction, tMax);

// Traverse the acceleration structure and store information about the first
// intersection (if any)
rayQueryProceedEXT(rq);

if (rayQueryGetIntersectionTypeEXT(rq, true) ==
gl_RayQueryCommittedIntersectionNoneEXT) {
    // Not in shadow
}
```

Issues

(1) What are the changes between the public provisional (VK_KHR_ray_tracing v8) release and the final (VK_KHR_acceleration_structure v11 / VK_KHR_ray_query v1) release?

- refactor VK_KHR_ray_tracing into 3 extensions, enabling implementation flexibility and decoupling ray query support from ray pipelines:
 - [VK_KHR_acceleration_structure](#) (for acceleration structure operations)
 - [VK_KHR_ray_tracing_pipeline](#) (for ray tracing pipeline and shader stages)

- [VK_KHR_ray_query](#) (for ray queries in existing shader stages)
- Update SPIRV capabilities to use [RayQueryKHR](#)
- extension is no longer provisional

Version History

- Revision 1, 2020-11-12 (Mathieu Robart, Daniel Koch, Andrew Garrard)
 - Decomposition of the specification, from [VK_KHR_ray_tracing](#) to [VK_KHR_ray_query](#) (#1918,!3912)
 - update to use [RayQueryKHR](#) SPIR-V capability
 - add numerical limits for ray parameters (#2235,!3960)
 - relax formula for ray intersection candidate determination (#2322,!4080)
 - restrict traces to TLAS (#2239,!4141)
 - require [HitT](#) to be in ray interval for [OpRayQueryGenerateIntersectionKHR](#) (#2359,!4146)
 - add ray query shader stages for AS read bit (#2407,!4203)

VK_KHR_ray_tracing_maintenance1

Name String

[VK_KHR_ray_tracing_maintenance1](#)

Extension Type

Device extension

Registered Extension Number

387

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_acceleration_structure](#)

API Interactions

- Interacts with [VK_VERSION_1_3](#)
- Interacts with [VK_EXT_device_generated_commands](#)
- Interacts with [VK_KHR_ray_tracing_pipeline](#)
- Interacts with [VK_KHR_synchronization2](#)

SPIR-V Dependencies

- [SPV_KHR_ray_cull_mask](#)

Contact

- Daniel Koch [\[GitHub\]dgkoch](#)

Other Extension Metadata

Last Modified Date

2022-02-21

Interactions and External Dependencies

- This extension provides API support for [GLSL_EXT_ray_cull_mask](#)
- Interacts with [VK_KHR_ray_tracing_pipeline](#)
- Interacts with [VK_KHR_synchronization2](#)

Contributors

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- Dae Kim, Imagination
- Joshua Barczak, Intel
- Lionel Landwerlin, Intel
- Daniel Koch, NVIDIA
- Eric Werness, NVIDIA
- Spencer Fricke, Samsung

Description

[VK_KHR_ray_tracing_maintenance1](#) adds a collection of minor ray tracing features, none of which would warrant an entire extension of their own.

The new features are as follows:

- Adds support for the [SPV_KHR_ray_cull_mask](#) SPIR-V extension in Vulkan. This extension provides access to built-in [CullMaskKHR](#) shader variable which contains the value of the [OpTrace* Cull Mask](#) parameter. This new shader variable is accessible in the intersection, any-hit, closest hit and miss shader stages.
- Adds support for a new pipeline stage and access mask built on top of [VK_KHR_synchronization2](#):
 - [VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR](#) to specify execution of [acceleration structure copy commands](#)

- `VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR` to specify read access to a [shader binding table](#) in any shader pipeline stage
- Adds two new acceleration structure query parameters:
 - `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR` to query the acceleration structure size on the device timeline
 - `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR` to query the number of bottom level acceleration structure pointers for serialization
- Adds an optional new indirect ray tracing dispatch command, [vkCmdTraceRaysIndirect2KHR](#), which sources the shader binding table parameters as well as the dispatch dimensions from the device. The `rayTracingPipelineTraceRaysIndirect2` feature indicates whether this functionality is supported.

New Commands

If [VK_KHR_ray_tracing_pipeline](#) is supported:

- [vkCmdTraceRaysIndirect2KHR](#)

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceRayTracingMaintenance1FeaturesKHR](#)

If [VK_KHR_ray_tracing_pipeline](#) is supported:

- [VkTraceRaysIndirectCommand2KHR](#)

New Enum Constants

- `VK_KHR_RAY_TRACING_MAINTENANCE_1_EXTENSION_NAME`
- `VK_KHR_RAY_TRACING_MAINTENANCE_1_SPEC_VERSION`
- Extending [VkQueryType](#):
 - `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SERIALIZATION_BOTTOM_LEVEL_POINTERS_KHR`
 - `VK_QUERY_TYPE_ACCELERATION_STRUCTURE_SIZE_KHR`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_MAINTENANCE_1_FEATURES_KHR`

If [VK_KHR_synchronization2](#) or [Vulkan Version 1.3](#) and [VK_KHR_ray_tracing_pipeline](#) is supported:

- Extending [VkAccessFlagBits2](#):
 - `VK_ACCESS_2_SHADER_BINDING_TABLE_READ_BIT_KHR`

If [VK_EXT_device_generated_commands](#) is supported:

- Extending [VkIndirectCommandsTokenTypeEXT](#):

- `VK_INDIRECT_COMMANDS_TOKEN_TYPE_TRACE_RAYS2_EXT`

If [VK_KHR_synchronization2](#) or [Vulkan Version 1.3](#) is supported:

- Extending [VkPipelineStageFlagBits2](#):
 - `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_COPY_BIT_KHR`

New Built-In Variables

- `CullMaskKHR`

New SPIR-V Capabilities

- `RayCullMaskKHR`

Issues

None Yet!

Version History

- Revision 1, 2022-02-21 (Members of the Vulkan Ray Tracing TSG)
 - internal revisions

VK_KHR_ray_tracing_pipeline

Name String

`VK_KHR_ray_tracing_pipeline`

Extension Type

Device extension

Registered Extension Number

348

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_spirv_1_4](#)

and

[VK_KHR_acceleration_structure](#)

API Interactions

- Interacts with `VK_KHR_ray_query`

SPIR-V Dependencies

- [SPV_KHR_ray_tracing](#)

Contact

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Other Extension Metadata

Last Modified Date

2020-11-12

Interactions and External Dependencies

- This extension provides API support for [GLSL_EXT_ray_tracing](#)
- This extension interacts with [Vulkan 1.2](#) and [VK_KHR_vulkan_memory_model](#), adding the [shader-call-related](#) relation of invocations, [shader-call-order](#) partial order of dynamic instances of instructions, and the [ShaderCallKHR](#) scope.
- This extension interacts with [VK_KHR_pipeline_library](#), enabling pipeline libraries to be used with ray tracing pipelines and enabling usage of [VkRayTracingPipelineInterfaceCreateInfoKHR](#).

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- Jesse Barker, Unity
- Baldur Karlsson, Valve

Description

Rasterization has been the dominant method to produce interactive graphics, but increasing performance of graphics hardware has made ray tracing a viable option for interactive rendering. Being able to integrate ray tracing with traditional rasterization makes it easier for applications to incrementally add ray traced effects to existing applications or to do hybrid approaches with rasterization for primary visibility and ray tracing for secondary queries.

To enable ray tracing, this extension adds a few different categories of new functionality:

- A new ray tracing pipeline type with new shader domains: ray generation, intersection, any-hit, closest hit, miss, and callable
- A shader binding indirection table to link shader groups with acceleration structure items
- Ray tracing commands which initiate the ray pipeline traversal and invocation of the various new shader domains depending on which traversal conditions are met

This extension adds support for the following SPIR-V extension in Vulkan:

- [SPV_KHR_ray_tracing](#)

New Commands

- [vkCmdSetRayTracingPipelineStackSizeKHR](#)
- [vkCmdTraceRaysIndirectKHR](#)
- [vkCmdTraceRaysKHR](#)
- [vkCreateRayTracingPipelinesKHR](#)
- [vkGetRayTracingCaptureReplayShaderGroupHandlesKHR](#)
- [vkGetRayTracingShaderGroupHandlesKHR](#)
- [vkGetRayTracingShaderGroupStackSizeKHR](#)

New Structures

- [VkRayTracingPipelineCreateInfoKHR](#)
- [VkRayTracingPipelineInterfaceCreateInfoKHR](#)
- [VkRayTracingShaderGroupCreateInfoKHR](#)
- [VkStridedDeviceAddressRegionKHR](#)
- [VkTraceRaysIndirectCommandKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceRayTracingPipelineFeaturesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceRayTracingPipelinePropertiesKHR](#)

New Enums

- [VkRayTracingShaderGroupTypeKHR](#)
- [VkShaderGroupShaderKHR](#)

New Enum Constants

- [VK_KHR_RAY_TRACING_PIPELINE_EXTENSION_NAME](#)
- [VK_KHR_RAY_TRACING_PIPELINE_SPEC_VERSION](#)
- [VK_SHADER_UNUSED_KHR](#)
- Extending [VkBufferUsageFlagBits](#):
 - [VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR](#)
- Extending [VkDynamicState](#):
 - [VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR](#)
- Extending [VkPipelineBindPoint](#):
 - [VK_PIPELINE_BIND_POINT_RAY_TRACING_KHR](#)

- Extending [VkPipelineCreateFlagBits](#):
 - VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_ANY_HIT_SHADERS_BIT_KHR
 - VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_CLOSEST_HIT_SHADERS_BIT_KHR
 - VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_INTERSECTION_SHADERS_BIT_KHR
 - VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_MISS_SHADERS_BIT_KHR
 - VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR
 - VK_PIPELINE_CREATE_RAY_TRACING_SKIP_AABBS_BIT_KHR
 - VK_PIPELINE_CREATE_RAY_TRACING_SKIP_TRIANGLES_BIT_KHR
- Extending [VkPipelineStageFlagBits](#):
 - VK_PIPELINE_STAGE_RAY_TRACING_SHADER_BIT_KHR
- Extending [VkShaderStageFlagBits](#):
 - VK_SHADER_STAGE_ANY_HIT_BIT_KHR
 - VK_SHADER_STAGE_CALLABLE_BIT_KHR
 - VK_SHADER_STAGE_CLOSEST_HIT_BIT_KHR
 - VK_SHADER_STAGE_INTERSECTION_BIT_KHR
 - VK_SHADER_STAGE_MISS_BIT_KHR
 - VK_SHADER_STAGE_RAYGEN_BIT_KHR
- Extending [VkStructureType](#):
 - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_FEATURES_KHR
 - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_PIPELINE_PROPERTIES_KHR
 - VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_CREATE_INFO_KHR
 - VK_STRUCTURE_TYPE_RAY_TRACING_PIPELINE_INTERFACE_CREATE_INFO_KHR
 - VK_STRUCTURE_TYPE_RAY_TRACING_SHADER_GROUP_CREATE_INFO_KHR

New or Modified Built-In Variables

- LaunchIdKHR
- LaunchSizeKHR
- WorldRayOriginKHR
- WorldRayDirectionKHR
- ObjectRayOriginKHR
- ObjectRayDirectionKHR
- RayTminKHR
- RayTmaxKHR
- InstanceCustomIndexKHR
- InstanceId

- `ObjectToWorldKHR`
- `WorldToObjectKHR`
- `HitKindKHR`
- `IncomingRayFlagsKHR`
- `RayGeometryIndexKHR`
- (modified) `PrimitiveId`

New SPIR-V Capabilities

- `RayTracingKHR`
- `RayTraversalPrimitiveCullingKHR`

Issues

(1) How does this extension differ from `VK_NV_ray_tracing`?

DISCUSSION:

The following is a summary of the main functional differences between `VK_KHR_ray_tracing_pipeline` and `VK_NV_ray_tracing`:

- added support for indirect ray tracing (`vkCmdTraceRaysIndirectKHR`)
- uses `SPV_KHR_ray_tracing` instead of `SPV_NV_ray_tracing`
 - refer to KHR SPIR-V enums instead of NV SPIR-V enums (which are functionally equivalent and aliased to the same values).
 - added `RayGeometryIndexKHR` built-in
- removed `vkCompileDeferredNV` compilation functionality and replaced with `deferred host operations` interactions for ray tracing
- added `VkPhysicalDeviceRayTracingPipelineFeaturesKHR` structure
- extended `VkPhysicalDeviceRayTracingPipelinePropertiesKHR` structure
 - renamed `maxRecursionDepth` to `maxRayRecursionDepth` and it has a minimum of 1 instead of 31
 - require `shaderGroupHandleSize` to be 32 bytes
 - added `maxRayDispatchInvocationCount`, `shaderGroupHandleAlignment` and `maxRayHitAttributeSize`
- reworked geometry structures so they could be better shared between device, host, and indirect builds
- changed SBT parameters to a structure and added size (`VkStridedDeviceAddressRegionKHR`)
- add parameter for requesting memory requirements for host and/or device build
- added `pipeline library` support for ray tracing
- added `watertightness guarantees`
- added no-null-shader pipeline flags (`VK_PIPELINE_CREATE_RAY_TRACING_NO_NULL_*_SHADERS_BIT_KHR`)

- added [memory model interactions](#) with ray tracing and define how subgroups work and can be repacked

(3) What are the changes between the public provisional (VK_KHR_ray_tracing v8) release and the internal provisional (VK_KHR_ray_tracing v9) release?

- Require Vulkan 1.1 and SPIR-V 1.4
- Added interactions with Vulkan 1.2 and [VK_KHR_vulkan_memory_model](#)
- added creation time capture and replay flags
 - added [VK_PIPELINE_CREATE_RAY_TRACING_SHADER_GROUP_HANDLE_CAPTURE_REPLAY_BIT_KHR](#) to [VkPipelineCreateFlagBits](#)
- replace [VkStridedBufferRegionKHR](#) with [VkStridedDeviceAddressRegionKHR](#) and change [vkCmdTraceRaysKHR](#), [vkCmdTraceRaysIndirectKHR](#), to take these for the shader binding table and use device addresses instead of buffers.
- require the shader binding table buffers to have the [VK_BUFFER_USAGE_RAY_TRACING_BIT_KHR](#) set
- make [VK_KHR_pipeline_library](#) an interaction instead of required extension
- rename the [libraries](#) member of [VkRayTracingPipelineCreateInfoKHR](#) to [pLibraryInfo](#) and make it a pointer
- make [VK_KHR_deferred_host_operations](#) an interaction instead of a required extension (later went back on this)
- added explicit stack size management for ray tracing pipelines
 - removed the [maxCallableSize](#) member of [VkRayTracingPipelineInterfaceCreateInfoKHR](#)
 - added the [pDynamicState](#) member to [VkRayTracingPipelineCreateInfoKHR](#)
 - added [VK_DYNAMIC_STATE_RAY_TRACING_PIPELINE_STACK_SIZE_KHR](#) dynamic state for ray tracing pipelines
 - added [vkGetRayTracingShaderGroupStackSizeKHR](#) and [vkCmdSetRayTracingPipelineStackSizeKHR](#) commands
 - added [VkShaderGroupShaderKHR](#) enum
- Added [maxRayDispatchInvocationCount](#) limit to [VkPhysicalDeviceRayTracingPipelinePropertiesKHR](#)
- Added [shaderGroupHandleAlignment](#) property to [VkPhysicalDeviceRayTracingPipelinePropertiesKHR](#)
- Added [maxRayHitAttributeSize](#) property to [VkPhysicalDeviceRayTracingPipelinePropertiesKHR](#)
- Clarify deferred host ops for pipeline creation
 - [VkDeferredOperationKHR](#) is now a top-level parameter for [vkCreateRayTracingPipelinesKHR](#)
 - removed [VkDeferredOperationInfoKHR](#) structure
 - change deferred host creation/return parameter behavior such that the implementation can modify such parameters until the deferred host operation completes
 - [VK_KHR_deferred_host_operations](#) is required again

(4) What are the changes between the internal provisional (VK_KHR_ray_tracing v9) release and the final (VK_KHR_acceleration_structure v11 / VK_KHR_ray_tracing_pipeline v1) release?

- refactor VK_KHR_ray_tracing into 3 extensions, enabling implementation flexibility and decoupling ray query support from ray pipelines:
 - VK_KHR_acceleration_structure (for acceleration structure operations)
 - VK_KHR_ray_tracing_pipeline (for ray tracing pipeline and shader stages)
 - VK_KHR_ray_query (for ray queries in existing shader stages)
- Require **Volatile** for the following builtins in the ray generation, closest hit, miss, intersection, and callable shader stages:
 - SubgroupSize, SubgroupLocalInvocationId, SubgroupEqMask, SubgroupGeMask, SubgroupGtMask, SubgroupLeMask, SubgroupLtMask
 - SMIDNV, WarpIDNV
- clarify buffer usage flags for ray tracing
 - VK_BUFFER_USAGE_SHADER_BINDING_TABLE_BIT_KHR is added as an alias of VK_BUFFER_USAGE_RAY_TRACING_BIT_NV and is required on shader binding table buffers
 - VK_BUFFER_USAGE_STORAGE_BUFFER_BIT is used in VK_KHR_acceleration_structure for scratchData
- rename maxRecursionDepth to maxRayPipelineRecursionDepth (pipeline creation) and maxRayRecursionDepth (limit) to reduce confusion
- Add queryable maxRayHitAttributeSize limit and rename members of VkRayTracingPipelineInterfaceCreateInfoKHR to maxPipelineRayPayloadSize and maxPipelineRayHitAttributeSize for clarity
- Update SPIRV capabilities to use RayTracingKHR
- extension is no longer provisional
- define synchronization requirements for indirect trace rays and indirect buffer

(5) This extension adds gl_InstanceID for the intersection, any-hit, and closest hit shaders, but in KHR_vulkan_glsl, gl_InstanceID is replaced with gl_InstanceIndex. Which should be used for Vulkan in this extension?

RESOLVED: This extension uses gl_InstanceID and maps it to InstanceId in SPIR-V. It is acknowledged that this is different than other shader stages in Vulkan. There are two main reasons for the difference here:

- symmetry with gl_PrimitiveID which is also available in these shaders
- there is no “baseInstance” relevant for these shaders, and so ID makes it more obvious that this is zero-based.

(6) Why is VK_KHR_pipeline_library an interaction instead of a required dependency, particularly when the “Feature Requirements” section says it is required to be supported anyhow?

RESOLVED: If VK_KHR_pipeline_library were a required extension dependency, then every application would need to enable the extension whether or not they actually want to use the

pipeline library functionality. Developers found this to be annoying and unfriendly behavior. We do wish to require all **implementations** to support it though, and thus it is listed in the feature requirements section.

Sample Code

Example ray generation GLSL shader

```
#version 450 core
#extension GL_EXT_ray_tracing : require
layout(set = 0, binding = 0, rgba8) uniform image2D image;
layout(set = 0, binding = 1) uniform accelerationStructureEXT as;
layout(location = 0) rayPayloadEXT float payload;

void main()
{
    vec4 col = vec4(0, 0, 0, 1);

    vec3 origin = vec3(float(gl_LaunchIDEXT.x)/float(gl_LaunchSizeEXT.x), float
(gl_LaunchIDEXT.y)/float(gl_LaunchSizeEXT.y), 1.0);
    vec3 dir = vec3(0.0, 0.0, -1.0);

    traceRayEXT(as, 0, 0xff, 0, 1, 0, origin, 0.0, dir, 1000.0, 0);

    col.y = payload;

    imageStore(image, ivec2(gl_LaunchIDEXT.xy), col);
}
```

Version History

- Revision 1, 2020-11-12 (Mathieu Robart, Daniel Koch, Eric Werness, Tobias Hector)
 - Decomposition of the specification, from VK_KHR_ray_tracing to VK_KHR_ray_tracing_pipeline (#1918,!3912)
 - require certain subgroup and sm_shader_builtin shader builtins to be decorated as volatile in the ray generation, closest hit, miss, intersection, and callable stages (#1924,!3903,!3954)
 - clarify buffer usage flags for ray tracing (#2181,!3939)
 - rename maxRecursionDepth to maxRayPipelineRecursionDepth and maxRayRecursionDepth (#2203,!3937)
 - add queryable maxRayHitAttributeSize and rename members of VkRayTracingPipelineInterfaceCreateInfoKHR (#2102,!3966)
 - update to use RayTracingKHR SPIR-V capability
 - add VUs for matching hit group type against geometry type (#2245,!3994)
 - require RayTMaxKHR be volatile in intersection shaders (#2268,!4030)
 - add numerical limits for ray parameters (#2235,!3960)

- fix SBT indexing rules for device addresses (#2308,!4079)
- relax formula for ray intersection candidate determination (#2322,!4080)
- add more details on [ShaderRecordBufferKHR](#) variables (#2230,!4083)
- clarify valid bits for [InstanceCustomIndexKHR](#) (GLSL/GLSL#19,!4128)
- allow at most one [IncomingRayPayloadKHR](#), [IncomingCallableDataKHR](#), and [HitAttributeKHR](#) (!4129)
- add minimum for maxShaderGroupStride (#2353,!4131)
- require VK_KHR_pipeline_library extension to be supported (#2348,!4135)
- clarify meaning of 'geometry index' (#2272,!4137)
- restrict traces to TLAS (#2239,!4141)
- add note about maxPipelineRayPayloadSize (#2383,!4172)
- do not require raygen shader in pipeline libraries (!4185)
- define sync for indirect trace rays and indirect buffer (#2407,!4208)

VK_KHR_ray_tracing_position_fetch

Name String

[VK_KHR_ray_tracing_position_fetch](#)

Extension Type

Device extension

Registered Extension Number

482

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_acceleration_structure](#)

SPIR-V Dependencies

- [SPV_KHR_ray_tracing_position_fetch](#)

Contact

- Eric Werness

Extension Proposal

[VK_KHR_ray_tracing_position_fetch](#)

Other Extension Metadata

Last Modified Date

2023-02-17

Interactions and External Dependencies

- This extension provides API support for [GLSL_EXT_ray_tracing_position_fetch](#)
- Interacts with [VK_KHR_ray_tracing_pipeline](#)
- Interacts with [VK_KHR_ray_query](#)

Contributors

- Eric Werness, NVIDIA
- Stu Smith, AMD
- Yuriy O'Donnell, Epic Games
- Ralph Potter, Samsung
- Joshua Barczak, Intel
- Lionel Landwerlin, Intel
- Andrew Garrard, Imagination Technologies
- Alex Bourd, Qualcomm
- Yunpeng Zhu, Huawei Technologies
- Marius Bjorge, Arm
- Daniel Koch, NVIDIA

Description

[VK_KHR_ray_tracing_position_fetch](#) adds the ability to fetch the vertex positions in the shader from a hit triangle as stored in the acceleration structure.

An application adds [VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_KHR](#) to the acceleration structure at build time. Then, if the hit is a triangle geometry, the shader (any-hit or closest hit for ray pipelines or using ray query) **can** fetch the three, three-component vertex positions in object space, of the triangle which was hit.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceRayTracingPositionFetchFeaturesKHR](#)

New Enum Constants

- [VK_KHR_RAY_TRACING_POSITION_FETCH_EXTENSION_NAME](#)
- [VK_KHR_RAY_TRACING_POSITION_FETCH_SPEC_VERSION](#)
- Extending [VkBuildAccelerationStructureFlagBitsKHR](#):

- `VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DATA_ACCESS_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_RAY_TRACING_POSITION_FETCH_FEATURES_KHR`

New Built-In Variables

- `HitTriangleVertexPositionsKHR`

New SPIR-V Capabilities

- `RayTracingPositionFetchKHR`
- `RayQueryPositionFetchKHR`

Issues

None Yet!

Version History

- Revision 1, 2023-02-17 (Eric Werness)
 - internal revisions

VK_KHR_relaxed_block_layout

Name String

`VK_KHR_relaxed_block_layout`

Extension Type

Device extension

Registered Extension Number

145

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- John Kessenich [[GitHub](#)]johnkslang

Other Extension Metadata

Last Modified Date

2017-03-26

IP Status

No known IP claims.

Contributors

- John Kessenich, Google

Description

The `VK_KHR_relaxed_block_layout` extension allows implementations to indicate they can support more variation in block `Offset` decorations. For example, placing a vector of three floats at an offset of $16 \times N + 4$.

See [Offset and Stride Assignment](#) for details.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Enum Constants

- `VK_KHR_RELAXED_BLOCK_LAYOUT_EXTENSION_NAME`
- `VK_KHR_RELAXED_BLOCK_LAYOUT_SPEC_VERSION`

Version History

- Revision 1, 2017-03-26 (JohnK)

VK_KHR_sampler_mirror_clamp_to_edge

Name String

`VK_KHR_sampler_mirror_clamp_to_edge`

Extension Type

Device extension

Registered Extension Number

15

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Tobias Hector [\[GitHub\]](#)tobski

Other Extension Metadata

Last Modified Date

2019-08-17

Contributors

- Tobias Hector, Imagination Technologies
- Jon Leech, Khronos

Description

`VK_KHR_sampler_mirror_clamp_to_edge` extends the set of sampler address modes to include an additional mode (`VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE`) that effectively uses a texture map twice as large as the original image in which the additional half of the new image is a mirror image of the original image.

This new mode relaxes the need to generate images whose opposite edges match by using the original image to generate a matching “mirror image”. This mode allows the texture to be mirrored only once in the negative s, t, and r directions.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2. However, if Vulkan 1.2 is supported and this extension is not, the [VkSamplerAddressMode](#) `VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE` is optional. Since the original extension did not use an author suffix on the enum `VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE`, it is used by both core and extension implementations.

New Enum Constants

- `VK_KHR_SAMPLER_MIRROR_CLAMP_TO_EDGE_EXTENSION_NAME`
- `VK_KHR_SAMPLER_MIRROR_CLAMP_TO_EDGE_SPEC_VERSION`
- Extending [VkSamplerAddressMode](#):
 - `VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE`
 - `VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE_KHR`

Example

Creating a sampler with the new address mode in each dimension

```

VkSamplerCreateInfo createInfo =
{
    .sType = VK_STRUCTURE_TYPE_SAMPLER_CREATE_INFO,
    // Other members set to application-desired values
};

createInfo.addressModeU = VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE;
createInfo.addressModeV = VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE;
createInfo.addressModeW = VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE;

VkSampler sampler;
VkResult result = vkCreateSampler(
    device,
    &createInfo,
    &sampler);

```

Issues

1) Why are both KHR and core versions of the `VK_SAMPLER_ADDRESS_MODE_MIRROR_CLAMP_TO_EDGE` token present?

RESOLVED: This functionality was intended to be required in Vulkan 1.0. We realized shortly before public release that not all implementations could support it, and moved the functionality into an optional extension, but did not apply the KHR extension suffix. Adding a KHR-suffixed alias of the non-suffixed enum has been done to comply with our own naming rules.

In a related change, before spec revision 1.1.121 this extension was hardwired into the spec Makefile so it was always included with the Specification, even in the core-only versions. This has now been reverted, and it is treated as any other extension.

Version History

- Revision 1, 2016-02-16 (Tobias Hector)
 - Initial draft
- Revision 2, 2019-08-14 (Jon Leech)
 - Add KHR-suffixed alias of non-suffixed enum.
- Revision 3, 2019-08-17 (Jon Leech)
 - Add an issue explaining the reason for the extension API not being suffixed with KHR.

VK_KHR_sampler_ycbcr_conversion

Name String

`VK_KHR_sampler_ycbcr_conversion`

Extension Type

Device extension

Registered Extension Number

157

Revision

14

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_maintenance1](#)

and

[VK_KHR_bind_memory2](#)

and

[VK_KHR_get_memory_requirements2](#)

and

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

API Interactions

- Interacts with [VK_EXT_debug_report](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Andrew Garrard [\[GitHub\]](#)[fluppeteer](#)

Other Extension Metadata**Last Modified Date**

2017-08-11

IP Status

No known IP claims.

Contributors

- Andrew Garrard, Samsung Electronics
- Tobias Hector, Imagination Technologies
- James Jones, NVIDIA
- Daniel Koch, NVIDIA
- Daniel Rakos, AMD
- Romain Guy, Google
- Jesse Hall, Google
- Tom Cooksey, ARM Ltd

- Jeff Leger, Qualcomm Technologies, Inc
- Jan-Harald Fredriksen, ARM Ltd
- Jan Outters, Samsung Electronics
- Alon Or-bach, Samsung Electronics
- Michael Worcester, Imagination Technologies
- Jeff Bolz, NVIDIA
- Tony Zlatinski, NVIDIA
- Matthew Netsch, Qualcomm Technologies, Inc

Description

The use of $Y'C_B C_R$ sampler conversion is an area in 3D graphics not used by most Vulkan developers. It is mainly used for processing inputs from video decoders and cameras. The use of the extension assumes basic knowledge of $Y'C_B C_R$ concepts.

This extension provides the ability to perform specified color space conversions during texture sampling operations for the $Y'C_B C_R$ color space natively. It also adds a selection of multi-planar formats, image aspect plane, and the ability to bind memory to the planes of an image collectively or separately.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted. However, if Vulkan 1.1 is supported and this extension is not, the `samplerYcbcrConversion` capability is optional. The original type, enum, and command names are still available as aliases of the core functionality.

New Object Types

- [VkSamplerYcbcrConversionKHR](#)

New Commands

- [vkCreateSamplerYcbcrConversionKHR](#)
- [vkDestroySamplerYcbcrConversionKHR](#)

New Structures

- [VkSamplerYcbcrConversionCreateInfoKHR](#)
- Extending [VkBindImageMemoryInfo](#):
 - [VkBindImagePlaneMemoryInfoKHR](#)
- Extending [VkImageFormatProperties2](#):
 - [VkSamplerYcbcrConversionImageFormatPropertiesKHR](#)
- Extending [VkImageMemoryRequirementsInfo2](#):

- [VkImagePlaneMemoryRequirementsInfoKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceSamplerYcbcrConversionFeaturesKHR](#)
- Extending [VkSamplerCreateInfo](#), [VkImageViewCreateInfo](#):
 - [VkSamplerYcbcrConversionInfoKHR](#)

New Enums

- [VkChromaLocationKHR](#)
- [VkSamplerYcbcrModelConversionKHR](#)
- [VkSamplerYcbcrRangeKHR](#)

New Enum Constants

- [VK_KHR_SAMPLER_YCBCR_CONVERSION_EXTENSION_NAME](#)
- [VK_KHR_SAMPLER_YCBCR_CONVERSION_SPEC_VERSION](#)
- Extending [VkChromaLocation](#):
 - [VK_CHROMA_LOCATION_COSITED_EVEN_KHR](#)
 - [VK_CHROMA_LOCATION_MIDPOINT_KHR](#)
- Extending [VkFormat](#):
 - [VK_FORMAT_B10X6G10X6R10X6G10X6_422_UNORM_4PACK16_KHR](#)
 - [VK_FORMAT_B12X4G12X4R12X4G12X4_422_UNORM_4PACK16_KHR](#)
 - [VK_FORMAT_B16G16R16G16_422_UNORM_KHR](#)
 - [VK_FORMAT_B8G8R8G8_422_UNORM_KHR](#)
 - [VK_FORMAT_G10X6B10X6G10X6R10X6_422_UNORM_4PACK16_KHR](#)
 - [VK_FORMAT_G10X6_B10X6R10X6_2PLANE_420_UNORM_3PACK16_KHR](#)
 - [VK_FORMAT_G10X6_B10X6R10X6_2PLANE_422_UNORM_3PACK16_KHR](#)
 - [VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_420_UNORM_3PACK16_KHR](#)
 - [VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_422_UNORM_3PACK16_KHR](#)
 - [VK_FORMAT_G10X6_B10X6_R10X6_3PLANE_444_UNORM_3PACK16_KHR](#)
 - [VK_FORMAT_G12X4B12X4G12X4R12X4_422_UNORM_4PACK16_KHR](#)
 - [VK_FORMAT_G12X4_B12X4R12X4_2PLANE_420_UNORM_3PACK16_KHR](#)
 - [VK_FORMAT_G12X4_B12X4R12X4_2PLANE_422_UNORM_3PACK16_KHR](#)
 - [VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_420_UNORM_3PACK16_KHR](#)
 - [VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_422_UNORM_3PACK16_KHR](#)
 - [VK_FORMAT_G12X4_B12X4_R12X4_3PLANE_444_UNORM_3PACK16_KHR](#)
 - [VK_FORMAT_G16B16G16R16_422_UNORM_KHR](#)

- `VK_FORMAT_G16_B16R16_2PLANE_420_UNORM_KHR`
- `VK_FORMAT_G16_B16R16_2PLANE_422_UNORM_KHR`
- `VK_FORMAT_G16_B16_R16_3PLANE_420_UNORM_KHR`
- `VK_FORMAT_G16_B16_R16_3PLANE_422_UNORM_KHR`
- `VK_FORMAT_G16_B16_R16_3PLANE_444_UNORM_KHR`
- `VK_FORMAT_G8B8G8R8_422_UNORM_KHR`
- `VK_FORMAT_G8_B8R8_2PLANE_420_UNORM_KHR`
- `VK_FORMAT_G8_B8R8_2PLANE_422_UNORM_KHR`
- `VK_FORMAT_G8_B8_R8_3PLANE_420_UNORM_KHR`
- `VK_FORMAT_G8_B8_R8_3PLANE_422_UNORM_KHR`
- `VK_FORMAT_G8_B8_R8_3PLANE_444_UNORM_KHR`
- `VK_FORMAT_R10X6G10X6B10X6A10X6_UNORM_4PACK16_KHR`
- `VK_FORMAT_R10X6G10X6_UNORM_2PACK16_KHR`
- `VK_FORMAT_R10X6_UNORM_PACK16_KHR`
- `VK_FORMAT_R12X4G12X4B12X4A12X4_UNORM_4PACK16_KHR`
- `VK_FORMAT_R12X4G12X4_UNORM_2PACK16_KHR`
- `VK_FORMAT_R12X4_UNORM_PACK16_KHR`
- Extending `VkFormatFeatureFlagBits`:
 - `VK_FORMAT_FEATURE_COSITED_CHROMA_SAMPLES_BIT_KHR`
 - `VK_FORMAT_FEATURE_DISJOINT_BIT_KHR`
 - `VK_FORMAT_FEATURE_MIDPOINT_CHROMA_SAMPLES_BIT_KHR`
 - `VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_BIT_KHR`
 - `VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_CHROMA_RECONSTRUCTION_EXPLICIT_FORCEABLE_BIT_KHR`
 - `VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_LINEAR_FILTER_BIT_KHR`
 - `VK_FORMAT_FEATURE_SAMPLED_IMAGE_YCBCR_CONVERSION_SEPARATE_RECONSTRUCTION_FILTER_BIT_KHR`
- Extending `VkImageAspectFlagBits`:
 - `VK_IMAGE_ASPECT_PLANE_0_BIT_KHR`
 - `VK_IMAGE_ASPECT_PLANE_1_BIT_KHR`
 - `VK_IMAGE_ASPECT_PLANE_2_BIT_KHR`
- Extending `VkImageCreateFlagBits`:
 - `VK_IMAGE_CREATE_DISJOINT_BIT_KHR`
- Extending `VkObjectType`:
 - `VK_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION_KHR`
- Extending `VkSamplerYcbcrModelConversion`:

- VK_SAMPLER_YCBCR_MODEL_CONVERSION_RGB_IDENTITY_KHR
- VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_2020_KHR
- VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_601_KHR
- VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_709_KHR
- VK_SAMPLER_YCBCR_MODEL_CONVERSION_YCBCR_IDENTITY_KHR
- Extending [VkSamplerYcbcrRange](#):
 - VK_SAMPLER_YCBCR_RANGE_ITU_FULL_KHR
 - VK_SAMPLER_YCBCR_RANGE_ITU_NARROW_KHR
- Extending [VkStructureType](#):
 - VK_STRUCTURE_TYPE_BIND_IMAGE_PLANE_MEMORY_INFO_KHR
 - VK_STRUCTURE_TYPE_IMAGE_PLANE_MEMORY_REQUIREMENTS_INFO_KHR
 - VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLER_YCBCR_CONVERSION_FEATURES_KHR
 - VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_CREATE_INFO_KHR
 - VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_IMAGE_FORMAT_PROPERTIES_KHR
 - VK_STRUCTURE_TYPE_SAMPLER_YCBCR_CONVERSION_INFO_KHR

If `VK_EXT_debug_report` is supported:

- Extending [VkDebugReportObjectTypeEXT](#):
 - VK_DEBUG_REPORT_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION_EXT
 - VK_DEBUG_REPORT_OBJECT_TYPE_SAMPLER_YCBCR_CONVERSION_KHR_EXT

Version History

- Revision 1, 2017-01-24 (Andrew Garrard)
 - Initial draft
- Revision 2, 2017-01-25 (Andrew Garrard)
 - After initial feedback
- Revision 3, 2017-01-27 (Andrew Garrard)
 - Higher bit depth formats, renaming, swizzle
- Revision 4, 2017-02-22 (Andrew Garrard)
 - Added query function, formats as RGB, clarifications
- Revision 5, 2017-04-?? (Andrew Garrard)
 - Simplified query and removed output conversions
- Revision 6, 2017-04-24 (Andrew Garrard)
 - Tidying, incorporated new image query, restored transfer functions
- Revision 7, 2017-04-25 (Andrew Garrard)

- Added cosited option/midpoint requirement for formats, “bypassConversion”
- Revision 8, 2017-04-25 (Andrew Garrard)
 - Simplified further
- Revision 9, 2017-04-27 (Andrew Garrard)
 - Disjoint no more
- Revision 10, 2017-04-28 (Andrew Garrard)
 - Restored disjoint
- Revision 11, 2017-04-29 (Andrew Garrard)
 - Now Ycbcr conversion, and KHR
- Revision 12, 2017-06-06 (Andrew Garrard)
 - Added conversion to image view creation
- Revision 13, 2017-07-13 (Andrew Garrard)
 - Allowed cosited-only chroma samples for formats
- Revision 14, 2017-08-11 (Andrew Garrard)
 - Reflected quantization changes in BT.2100-1

VK_KHR_separate_depth_stencil_layouts

Name String

VK_KHR_separate_depth_stencil_layouts

Extension Type

Device extension

Registered Extension Number

242

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

and

[VK_KHR_create_renderpass2](#)

or

[Vulkan Version 1.2](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Piers Daniell [[GitHub](#)][pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2019-06-25

Contributors

- Daniel Koch, NVIDIA
- Jeff Bolz, NVIDIA
- Jesse Barker, Unity
- Tobias Hector, AMD

Description

This extension allows image memory barriers for depth/stencil images to have just one of the `VK_IMAGE_ASPECT_DEPTH_BIT` or `VK_IMAGE_ASPECT_STENCIL_BIT` aspect bits set, rather than require both. This allows their layouts to be set independently. To support depth/stencil images with different layouts for the depth and stencil aspects, the depth/stencil attachment interface has been updated to support a separate layout for stencil.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending [VkAttachmentDescription2](#):
 - [VkAttachmentDescriptionStencilLayoutKHR](#)
- Extending [VkAttachmentReference2](#):
 - [VkAttachmentReferenceStencilLayoutKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceSeparateDepthStencilLayoutsFeaturesKHR](#)

New Enum Constants

- `VK_KHR_SEPARATE_DEPTH_STENCIL_LAYOUTS_EXTENSION_NAME`
- `VK_KHR_SEPARATE_DEPTH_STENCIL_LAYOUTS_SPEC_VERSION`
- Extending [VkImageLayout](#):
 - `VK_IMAGE_LAYOUT_DEPTH_ATTACHMENT_OPTIMAL_KHR`

- `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL_KHR`
- `VK_IMAGE_LAYOUT_STENCIL_ATTACHMENT_OPTIMAL_KHR`
- `VK_IMAGE_LAYOUT_STENCIL_READ_ONLY_OPTIMAL_KHR`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_ATTACHMENT_DESCRIPTION_STENCIL_LAYOUT_KHR`
 - `VK_STRUCTURE_TYPE_ATTACHMENT_REFERENCE_STENCIL_LAYOUT_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SEPARATE_DEPTH_STENCIL_LAYOUTS_FEATURES_KHR`

Version History

- Revision 1, 2019-06-25 (Piers Daniell)
 - Internal revisions

VK_KHR_shader_atomic_int64

Name String

`VK_KHR_shader_atomic_int64`

Extension Type

Device extension

Registered Extension Number

181

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Aaron Hagan [\[GitHub\]](#)[ahagan](#)

Other Extension Metadata

Last Modified Date

2018-07-05

Interactions and External Dependencies

- This extension provides API support for `GL_ARB_gpu_shader_int64` and `GL_EXT_shader_atomic_int64`

Contributors

- Aaron Hagan, AMD
- Daniel Rakos, AMD
- Jeff Bolz, NVIDIA
- Neil Henning, Codeplay

Description

This extension advertises the SPIR-V **Int64Atomics** capability for Vulkan, which allows a shader to contain 64-bit atomic operations on signed and unsigned integers. The supported operations include `OpAtomicMin`, `OpAtomicMax`, `OpAtomicAnd`, `OpAtomicOr`, `OpAtomicXor`, `OpAtomicAdd`, `OpAtomicExchange`, and `OpAtomicCompareExchange`.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, the `shaderBufferInt64Atomics` capability is optional. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceShaderAtomicInt64FeaturesKHR`

New Enum Constants

- `VK_KHR_SHADER_ATOMIC_INT64_EXTENSION_NAME`
- `VK_KHR_SHADER_ATOMIC_INT64_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_INT64_FEATURES_KHR`

New SPIR-V Capabilities

- `Int64Atomics`

Version History

- Revision 1, 2018-07-05 (Aaron Hagan)
 - Internal revisions

VK_KHR_shader_clock

Name String

VK_KHR_shader_clock

Extension Type

Device extension

Registered Extension Number

182

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_shader_clock](#)

Contact

- Aaron Hagan [\[GitHub\]ahagan](#)

Other Extension Metadata

Last Modified Date

2019-4-25

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for [GL_ARB_shader_clock](#) and [GL_EXT_shader_realtime_clock](#)

Contributors

- Aaron Hagan, AMD
- Daniel Koch, NVIDIA

Description

This extension advertises the SPIR-V [ShaderClockKHR](#) capability for Vulkan, which allows a shader to query a real-time or monotonically incrementing counter at the subgroup level or across the device level. The two valid SPIR-V scopes for [OpReadClockKHR](#) are [Subgroup](#) and [Device](#).

When using GLSL source-based shading languages, the `clockRealtime*EXT()` timing functions map to the `OpReadClockKHR` instruction with a scope of `Device`, and the `clock*ARB()` timing functions map to the `OpReadClockKHR` instruction with a scope of `Subgroup`.

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceShaderClockFeaturesKHR`

New Enum Constants

- `VK_KHR_SHADER_CLOCK_EXTENSION_NAME`
- `VK_KHR_SHADER_CLOCK_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_CLOCK_FEATURES_KHR`

New SPIR-V Capabilities

- `ShaderClockKHR`

Version History

- Revision 1, 2019-4-25 (Aaron Hagan)
 - Initial revision

VK_KHR_shader_draw_parameters

Name String

`VK_KHR_shader_draw_parameters`

Extension Type

Device extension

Registered Extension Number

64

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- `SPV_KHR_shader_draw_parameters`

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Daniel Koch [\[GitHub\]dgkoch](#)

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for `GL_ARB_shader_draw_parameters`

Contributors

- Daniel Koch, NVIDIA Corporation
- Jeff Bolz, NVIDIA
- Daniel Rakos, AMD
- Jan-Harald Fredriksen, ARM
- John Kessenich, Google
- Stuart Smith, IMG

Description

This extension adds support for the following SPIR-V extension in Vulkan:

- `SPV_KHR_shader_draw_parameters`

The extension provides access to three additional built-in shader variables in Vulkan:

- `BaseInstance`, containing the `firstInstance` parameter passed to drawing commands,
- `BaseVertex`, containing the `firstVertex` or `vertexOffset` parameter passed to drawing commands, and
- `DrawIndex`, containing the index of the draw call currently being processed from an indirect drawing call.

When using GLSL source-based shader languages, the following variables from `GL_ARB_shader_draw_parameters` can map to these SPIR-V built-in decorations:

- `in int gl_BaseInstanceARB;` → `BaseInstance`,
- `in int gl_BaseVertexARB;` → `BaseVertex`, and
- `in int gl_DrawIDARB;` → `DrawIndex`.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1. However, the `shaderDrawParameters` feature bit was added to distinguish whether it is actually available or not.

New Enum Constants

- `VK_KHR_SHADER_DRAW_PARAMETERS_EXTENSION_NAME`
- `VK_KHR_SHADER_DRAW_PARAMETERS_SPEC_VERSION`

New Built-In Variables

- `BaseInstance`
- `BaseVertex`
- `DrawIndex`

New SPIR-V Capabilities

- `DrawParameters`

Issues

1) Is this the same functionality as `GL_ARB_shader_draw_parameters`?

RESOLVED: It is actually a superset, as it also adds in support for arrayed drawing commands.

In GL for `GL_ARB_shader_draw_parameters`, `gl_BaseVertexARB` holds the integer value passed to the parameter to the command that resulted in the current shader invocation. In the case where the command has no `baseVertex` parameter, the value of `gl_BaseVertexARB` is zero. This means that `gl_BaseVertexARB = baseVertex` (for `glDrawElements` commands with `baseVertex`) or 0. In particular there are no `glDrawArrays` commands that take a `baseVertex` parameter.

Now in Vulkan, we have `BaseVertex = vertexOffset` (for indexed drawing commands) or `firstVertex` (for arrayed drawing commands), and so Vulkan's version is really a superset of GL functionality.

Version History

- Revision 1, 2016-10-05 (Daniel Koch)
 - Internal revisions

VK_KHR_shader_expect_assume

Name String

`VK_KHR_shader_expect_assume`

Extension Type

Device extension

Registered Extension Number

545

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_expect_assume](#)

Contact

- Kevin Petit [\[GitHub\]](#)[kpet](#)

Extension Proposal

[VK_KHR_shader_expect_assume](#)

Other Extension Metadata

Last Modified Date

2023-12-06

IP Status

No known IP claims.

Contributors

- Kevin Petit, Arm
- Tobias Hector, AMD
- James Fitzpatrick, Imagination Technologies

Description

This extension allows the use of the [SPV_KHR_expect_assume](#) extension in SPIR-V shader modules which enables SPIR-V producers to provide optimization hints to the Vulkan implementation.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderExpectAssumeFeaturesKHR](#)

New Enum Constants

- `VK_KHR_SHADER_EXPECT_ASSUME_EXTENSION_NAME`
- `VK_KHR_SHADER_EXPECT_ASSUME_SPEC_VERSION`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_EXPECT_ASSUME_FEATURES_KHR`

New SPIR-V Capabilities

- [ExpectAssumeKHR](#)

Version History

- Revision 1, 2023-12-06 (Kevin Petit)
 - Initial revision

VK_KHR_shader_float16_int8

Name String

`VK_KHR_shader_float16_int8`

Extension Type

Device extension

Registered Extension Number

83

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Alexander Galazin [[GitHub](#)][alegal-arm](#)

Other Extension Metadata

Last Modified Date

2018-03-07

Interactions and External Dependencies

- This extension interacts with [VK_KHR_8bit_storage](#)
- This extension interacts with [VK_KHR_16bit_storage](#)
- This extension interacts with [VK_KHR_shader_float_controls](#)
- This extension provides API support for [GL_EXT_shader_explicit_arithmetic_types](#)

IP Status

No known IP claims.

Contributors

- Alexander Galazin, Arm
- Jan-Harald Fredriksen, Arm
- Jeff Bolz, NVIDIA
- Graeme Leese, Broadcom
- Daniel Rakos, AMD

Description

The [VK_KHR_shader_float16_int8](#) extension allows use of 16-bit floating-point types and 8-bit integer types in shaders for arithmetic operations.

It introduces two new optional features [shaderFloat16](#) and [shaderInt8](#) which directly map to the [Float16](#) and the [Int8](#) SPIR-V capabilities. The [VK_KHR_shader_float16_int8](#) extension also specifies precision requirements for half-precision floating-point SPIR-V operations. This extension does not enable use of 8-bit integer types or 16-bit floating-point types in any [shader input and output interfaces](#) and therefore does not supersede the [VK_KHR_8bit_storage](#) or [VK_KHR_16bit_storage](#) extensions.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, both the [shaderFloat16](#) and [shaderInt8](#) capabilities are optional. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceFloat16Int8FeaturesKHR](#)
 - [VkPhysicalDeviceShaderFloat16Int8FeaturesKHR](#)

New Enum Constants

- [VK_KHR_SHADER_FLOAT16_INT8_EXTENSION_NAME](#)
- [VK_KHR_SHADER_FLOAT16_INT8_SPEC_VERSION](#)

- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT16_INT8_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT16_INT8_FEATURES_KHR](#)

Version History

- Revision 1, 2018-03-07 (Alexander Galazin)
 - Initial draft

VK_KHR_shader_float_controls

Name String

[VK_KHR_shader_float_controls](#)

Extension Type

Device extension

Registered Extension Number

198

Revision

4

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_float_controls](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Alexander Galazin [[GitHub](#)][alegal-arm](#)

Other Extension Metadata

Last Modified Date

2018-09-11

IP Status

No known IP claims.

Contributors

- Alexander Galazin, Arm
- Jan-Harald Fredriksen, Arm
- Jeff Bolz, NVIDIA
- Graeme Leese, Broadcom
- Daniel Rakos, AMD

Description

The `VK_KHR_shader_float_controls` extension enables efficient use of floating-point computations through the ability to query and override the implementation's default behavior for rounding modes, denormals, signed zero, and infinity.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending `VkPhysicalDeviceProperties2`:
 - `VkPhysicalDeviceFloatControlsPropertiesKHR`

New Enums

- `VkShaderFloatControlsIndependenceKHR`

New Enum Constants

- `VK_KHR_SHADER_FLOAT_CONTROLS_EXTENSION_NAME`
- `VK_KHR_SHADER_FLOAT_CONTROLS_SPEC_VERSION`
- Extending `VkShaderFloatControlsIndependence`:
 - `VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_32_BIT_ONLY_KHR`
 - `VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_ALL_KHR`
 - `VK_SHADER_FLOAT_CONTROLS_INDEPENDENCE_NONE_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FLOAT_CONTROLS_PROPERTIES_KHR`

New SPIR-V Capabilities

- `DenormPreserve`
- `DenormFlushToZero`
- `SignedZeroInfNanPreserve`
- `RoundingModeRTE`

- `RoundingModeRTZ`

Issues

1) Which instructions must flush denorms?

RESOLVED: Only floating-point conversion, floating-point arithmetic, floating-point relational (except `OpIsNaN`, `OpIsInf`), and floating-point GLSL.std.450 extended instructions must flush denormals.

2) What is the denorm behavior for intermediate results?

RESOLVED: When a SPIR-V instruction is implemented as a sequence of other instructions:

- in the `DenormFlushToZero` execution mode, the intermediate instructions may flush denormals, the final result of the sequence **must** not be denormal.
- in the `DenormPreserve` execution mode, denormals must be preserved throughout the whole sequence.

3) Do denorm and rounding mode controls apply to `OpSpecConstantOp`?

RESOLVED: Yes, except when the opcode is `OpQuantizeToF16`.

4) The SPIR-V specification says that `OpConvertFToU` and `OpConvertFToS` unconditionally round towards zero. Do the rounding mode controls specified through the execution modes apply to them?

RESOLVED: No, these instructions unconditionally round towards zero.

5) Do any of the “Pack” GLSL.std.450 instructions count as conversion instructions and have the rounding mode applied?

RESOLVED: No, only instructions listed in “section 3.32.11. Conversion Instructions” of the SPIR-V specification count as conversion instructions.

6) When using inf/nan-ignore mode, what is expected of `OpIsNaN` and `OpIsInf`?

RESOLVED: These instructions must always accurately detect inf/nan if it is passed to them.

Version 4 API Incompatibility

The original versions of `VK_KHR_shader_float_controls` shipped with booleans named “`separateDenormSettings`” and “`separateRoundingModeSettings`”, which at first glance could have indicated “they can all be set independently, or not”. However the spec language as written indicated that the 32-bit value could always be set independently, and only the 16- and 64-bit controls needed to be the same if these values were `VK_FALSE`.

As a result of this slight disparity, and lack of test coverage for this facet of the extension, we ended up with two different behaviors in the wild, where some implementations worked as written, and others worked based on the naming. As these are hard limits in hardware with reasons for exposure as written, it was not possible to standardize on a single way to make this work within the

existing API.

No known users of this part of the extension exist in the wild, and as such the Vulkan WG took the unusual step of retroactively changing the once boolean value into a tri-state enum, breaking source compatibility. This was however done in such a way as to retain ABI compatibility, in case any code using this did exist; with the numerical values 0 and 1 retaining their original specified meaning, and a new value signifying the additional “all need to be set together” state. If any applications exist today, compiled binaries will continue to work as written in most cases, but will need changes before the code can be recompiled.

Version History

- Revision 4, 2019-06-18 (Tobias Hector)
 - Modified settings restrictions, see [Version 4 API incompatibility](#)
- Revision 3, 2018-09-11 (Alexander Galazin)
 - Minor restructuring
- Revision 2, 2018-04-17 (Alexander Galazin)
 - Added issues and resolutions
- Revision 1, 2018-04-11 (Alexander Galazin)
 - Initial draft

VK_KHR_shader_float_controls2

Name String

VK_KHR_shader_float_controls2

Extension Type

Device extension

Registered Extension Number

529

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

and

[VK_KHR_shader_float_controls](#)

SPIR-V Dependencies

- [SPV_KHR_float_controls2](#)

Contact

- Graeme Leese [\[GitHub\]gnl21](#)

Extension Proposal

[VK_KHR_shader_float_controls2](#)

Other Extension Metadata

Last Modified Date

2023-05-16

Interactions and External Dependencies

- This extension requires [SPV_KHR_float_controls2](#).

Contributors

- Graeme Leese, Broadcom

Description

This extension enables use of the more expressive fast floating-point math flags in the [SPV_KHR_float_controls2](#) extension. These flags give finer-grained control over which optimizations compilers may apply, potentially speeding up execution while retaining correct results.

The extension also adds control over the fast-math modes to the GLSL extended instruction set, making these operations more consistent with SPIR-V and allowing their use in situations where floating-point conformance is important.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderFloatControls2FeaturesKHR](#)

New Enum Constants

- [VK_KHR_SHADER_FLOAT_CONTROLS_2_EXTENSION_NAME](#)
- [VK_KHR_SHADER_FLOAT_CONTROLS_2_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_FLOAT_CONTROLS_2_FEATURES_KHR](#)

New SPIR-V Capabilities

- [FloatControls2](#)

Version History

- Revision 1, 2023-05-16 (Graeme Leese)
 - Initial draft

VK_KHR_shader_integer_dot_product

Name String

VK_KHR_shader_integer_dot_product

Extension Type

Device extension

Registered Extension Number

281

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_integer_dot_product](#)

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Kevin Petit [\[GitHub\]](#)kpet

Extension Proposal

[VK_KHR_shader_integer_dot_product](#)

Other Extension Metadata

Last Modified Date

2021-06-16

Interactions and External Dependencies

- This extension interacts with [VK_KHR_shader_float16_int8](#).

IP Status

No known IP claims.

Contributors

- Kévin Petit, Arm Ltd.
- Jeff Bolz, NVidia
- Spencer Fricke, Samsung

- Jesse Hall, Google
- John Kessenich, Google
- Graeme Leese, Broadcom
- Einar Hov, Arm Ltd.
- Stuart Brady, Arm Ltd.
- Pablo Cascon, Arm Ltd.
- Tobias Hector, AMD
- Jeff Leger, Qualcomm
- Ruihao Zhang, Qualcomm
- Pierre Boudier, NVidia
- Jon Leech, The Khronos Group
- Tom Olson, Arm Ltd.

Description

This extension adds support for the integer dot product SPIR-V instructions defined in `SPV_KHR_integer_dot_product`. These instructions are particularly useful for neural network inference and training but find uses in other general-purpose compute applications as well.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderIntegerDotProductFeaturesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceShaderIntegerDotProductPropertiesKHR](#)

New Enum Constants

- `VK_KHR_SHADER_INTEGER_DOT_PRODUCT_EXTENSION_NAME`
- `VK_KHR_SHADER_INTEGER_DOT_PRODUCT_SPEC_VERSION`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_FEATURES_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_INTEGER_DOT_PRODUCT_PROPERTIES_KHR`

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

New SPIR-V Capabilities

- [DotProductInputAllKHR](#)
- [DotProductInput4x8BitKHR](#)
- [DotProductInput4x8BitPackedKHR](#)
- [DotProductKHR](#)

Version History

- Revision 1, 2021-06-16 (Kévin Petit)
 - Initial revision

VK_KHR_shader_maximal_reconvergence

Name String

[VK_KHR_shader_maximal_reconvergence](#)

Extension Type

Device extension

Registered Extension Number

435

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_maximal_reconvergence](#)

Contact

- Alan Baker [\[GitHub\]alan-baker](#)

Extension Proposal

[VK_KHR_shader_maximal_reconvergence](#)

Other Extension Metadata

Last Modified Date

2021-11-12

IP Status

No known IP claims.

Interactions and External Dependencies

- Requires SPIR-V 1.3.
- This extension requires [SPV_KHR_maximal_reconvergence](#)

Contributors

- Alan Baker, Google

Description

This extension allows the use of the [SPV_KHR_maximal_reconvergence](#) SPIR-V extension in shader modules. [SPV_KHR_maximal_reconvergence](#) provides stronger guarantees that diverged subgroups will reconverge. These guarantees should match shader author intuition about divergence and reconvergence of invocations based on the structure of the code in the HLL.

Developers should utilize this extension if they require stronger guarantees about reconvergence than either the core spec or [SPV_KHR_subgroup_uniform_control_flow](#). This extension will define the rules that govern how invocations diverge and reconverge in a way that should match developer intuition. It allows robust programs to be written relying on subgroup operations and other tangled instructions.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderMaximalReconvergenceFeaturesKHR](#)

New Enum Constants

- [VK_KHR_SHADER_MAXIMAL_RECONVERGENCE_EXTENSION_NAME](#)
- [VK_KHR_SHADER_MAXIMAL_RECONVERGENCE_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_MAXIMAL_RECONVERGENCE_FEATURES_KHR](#)

Version History

- Revision 1, 2021-11-12 (Alan Baker)
 - Internal draft version

VK_KHR_shader_non_semantic_info

Name String

[VK_KHR_shader_non_semantic_info](#)

Extension Type

Device extension

Registered Extension Number

294

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- [SPV_KHR_non_semantic_info](#)

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Baldur Karlsson [\[GitHub\]](#)[baldurk](#)

Other Extension Metadata

Last Modified Date

2019-10-16

IP Status

No known IP claims.

Contributors

- Baldur Karlsson, Valve

Description

This extension allows the use of the [SPV_KHR_non_semantic_info](#) extension in SPIR-V shader modules.

New Enum Constants

- [VK_KHR_SHADER_NON_SEMANTIC_INFO_EXTENSION_NAME](#)
- [VK_KHR_SHADER_NON_SEMANTIC_INFO_SPEC_VERSION](#)

Promotion to Vulkan 1.3

Functionality in this extension is included in core Vulkan 1.3. Because the extension has no API controlling its functionality, this results only in a change to the [SPIR-V Extensions table](#).

Version History

- Revision 1, 2019-10-16 (Baldur Karlsson)
 - Initial revision

VK_KHR_shader_quad_control

Name String

VK_KHR_shader_quad_control

Extension Type

Device extension

Registered Extension Number

236

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

and

[VK_KHR_vulkan_memory_model](#)

and

[VK_KHR_shader_maximal_reconvergence](#)

SPIR-V Dependencies

- [SPV_KHR_quad_control](#)

Contact

- Tobias Hector [[GitHub](#)]tobski

Extension Proposal

[VK_KHR_shader_quad_control](#)

Other Extension Metadata

Last Modified Date

2023-11-01

IP Status

No known IP claims.

Contributors

- Tobias Hector, AMD
- Bill Licea-Kane, Qualcomm
- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, Arm
- Nicolai Hähnle, AMD

- Jeff Bolz, NVidia
- Alan Baker, Google
- Hans-Kristian Arntzen, Valve

Description

This extension adds new quad any/all operations, requires that derivatives are well-defined in quad-uniform control flow, and adds the ability to require helper invocations participate in group operations.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderQuadControlFeaturesKHR](#)

New Enum Constants

- [VK_KHR_SHADER_QUAD_CONTROL_EXTENSION_NAME](#)
- [VK_KHR_SHADER_QUAD_CONTROL_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_QUAD_CONTROL_FEATURES_KHR](#)

New SPIR-V Capabilities

- [QuadControlKHR](#)

Version History

- Revision 1, 2023-11-01 (Tobias Hector)
 - Initial draft

VK_KHR_shader_relaxed_extended_instruction

Name String

[VK_KHR_shader_relaxed_extended_instruction](#)

Extension Type

Device extension

Registered Extension Number

559

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- [SPV_KHR_relaxed_extended_instruction](#)

Contact

- Nathan Gauër [[GitHub](#)][Keenuts](#)

Extension Proposal

[VK_KHR_shader_relaxed_extended_instruction](#)

Other Extension Metadata

Last Modified Date

2024-01-24

IP Status

No known IP claims.

Contributors

- Alan Baker, Google LLC
- Nathan Gauër, Google LLC

Description

This extension allows the use of the [SPV_KHR_relaxed_extended_instruction](#) extension in SPIR-V shader modules.

It adds a new SPIR-V instruction, which allows some usage of forward references in non-semantic instruction sets. This extensions interacts with the [SPV_KHR_non_semantic_info](#) extension, hence with [VK_KHR_shader_non_semantic_info](#).

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderRelaxedExtendedInstructionFeaturesKHR](#)

New Enum Constants

- [VK_KHR_SHADER_RELAXED_EXTENDED_INSTRUCTION_EXTENSION_NAME](#)
- [VK_KHR_SHADER_RELAXED_EXTENDED_INSTRUCTION_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_RELAXED_EXTENDED_INSTRUCTION_FEATURES_KHR](#)

Version History

- Revision 1, 2024-01-24 (Nathan Gauër)

- Initial revision

VK_KHR_shader_subgroup_extended_types

Name String

VK_KHR_shader_subgroup_extended_types

Extension Type

Device extension

Registered Extension Number

176

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Neil Henning [\[GitHub\]sheredom](#)

Other Extension Metadata

Last Modified Date

2019-01-08

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for [GLSL_EXT_shader_subgroup_extended_types](#)

Contributors

- Jeff Bolz, NVIDIA
- Jan-Harald Fredriksen, Arm
- Neil Henning, AMD
- Daniel Koch, NVIDIA
- Jeff Leger, Qualcomm
- Graeme Leese, Broadcom
- David Neto, Google

- Daniel Rakos, AMD

Description

This extension enables the Non Uniform Group Operations in SPIR-V to support 8-bit integer, 16-bit integer, 64-bit integer, 16-bit floating-point, and vectors of these types.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderSubgroupExtendedTypesFeaturesKHR](#)

New Enum Constants

- [VK_KHR_SHADER_SUBGROUP_EXTENDED_TYPES_EXTENSION_NAME](#)
- [VK_KHR_SHADER_SUBGROUP_EXTENDED_TYPES_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_EXTENDED_TYPES_FEATURES_KHR](#)

Version History

- Revision 1, 2019-01-08 (Neil Henning)
 - Initial draft

VK_KHR_shader_subgroup_rotate

Name String

[VK_KHR_shader_subgroup_rotate](#)

Extension Type

Device extension

Registered Extension Number

417

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- [SPV_KHR_subgroup_rotate](#)

Contact

- Kevin Petit [\[GitHub\]](#)kpet

Extension Proposal

[VK_KHR_shader_subgroup_rotate](#)

Last Modified Date

2024-01-29

IP Status

No known IP claims.

Contributors

- Kévin Petit, Arm Ltd.
- Tobias Hector, AMD
- John Leech, Khronos
- Matthew Netsch, Qualcomm
- Jan-Harald Fredriksen, Arm Ltd.
- Graeme Leese, Broadcom
- Tom Olson, Arm Ltd.
- Spencer Fricke, LunarG Inc.

This extension adds support for the subgroup rotate instruction defined in [SPV_KHR_subgroup_rotate](#).

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderSubgroupRotateFeaturesKHR](#)

New Enum Constants

- [VK_KHR_SHADER_SUBGROUP_ROTATE_EXTENSION_NAME](#)
- [VK_KHR_SHADER_SUBGROUP_ROTATE_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_ROTATE_FEATURES_KHR](#)
- Extending [VkSubgroupFeatureFlagBits](#):
 - [VK_SUBGROUP_FEATURE_ROTATE_BIT_KHR](#)
 - [VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT_KHR](#)

New SPIR-V Capabilities

- [GroupNonUniformRotateKHR](#)

Version History

- Revision 2, 2024-01-29 (Kévin Petit)
 - Add `VK_SUBGROUP_FEATURE_ROTATE_BIT_KHR` and `VK_SUBGROUP_FEATURE_ROTATE_CLUSTERED_BIT_KHR`
- Revision 1, 2023-06-20 (Kévin Petit)
 - Initial revision

VK_KHR_shader_subgroup_uniform_control_flow

Name String

`VK_KHR_shader_subgroup_uniform_control_flow`

Extension Type

Device extension

Registered Extension Number

324

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_subgroup_uniform_control_flow](#)

Contact

- Alan Baker [\[GitHub\]alan-baker](#)

Other Extension Metadata

Last Modified Date

2020-08-27

IP Status

No known IP claims.

Interactions and External Dependencies

- Requires SPIR-V 1.3.
- This extension provides API support for `GL_EXT_subgroupuniform_qualifier`

Contributors

- Alan Baker, Google
- Jeff Bolz, NVIDIA

Description

This extension allows the use of the `SPV_KHR_subgroup_uniform_control_flow` SPIR-V extension in shader modules. `SPV_KHR_subgroup_uniform_control_flow` provides stronger guarantees that diverged subgroups will reconverge.

Developers should utilize this extension if they use subgroup operations to reduce the work performed by a uniform subgroup. This extension will guarantee that uniform subgroup will reconverge in the same manner as invocation groups (see “Uniform Control Flow” in the [Khronos SPIR-V Specification](#)).

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderSubgroupUniformControlFlowFeaturesKHR](#)

New Enum Constants

- `VK_KHR_SHADER_SUBGROUP_UNIFORM_CONTROL_FLOW_EXTENSION_NAME`
- `VK_KHR_SHADER_SUBGROUP_UNIFORM_CONTROL_FLOW_SPEC_VERSION`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_SUBGROUP_UNIFORM_CONTROL_FLOW_FEATURES_KHR`

Version History

- Revision 1, 2020-08-27 (Alan Baker)
 - Internal draft version

VK_KHR_shader_terminate_invocation

Name String

`VK_KHR_shader_terminate_invocation`

Extension Type

Device extension

Registered Extension Number

216

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_terminate_invocation](#)

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Jesse Hall [\[GitHub\]](#)[critsec](#)

Other Extension Metadata

Last Modified Date

2020-08-11

IP Status

No known IP claims.

Contributors

- Alan Baker, Google
- Jeff Bolz, NVIDIA
- Jesse Hall, Google
- Ralph Potter, Samsung
- Tom Olson, Arm

Description

This extension adds Vulkan support for the [SPV_KHR_terminate_invocation](#) SPIR-V extension. That SPIR-V extension provides a new instruction, [OpTerminateInvocation](#), which causes a shader invocation to immediately terminate and sets the coverage of shaded samples to 0; only previously executed instructions will have observable effects. The [OpTerminateInvocation](#) instruction, along with the [OpDemoteToHelperInvocation](#) instruction from the [VK_EXT_shader_demote_to_helper_invocation](#) extension, together replace the [OpKill](#) instruction, which could behave like either of these instructions. [OpTerminateInvocation](#) provides the behavior required by the GLSL [discard](#) statement, and should be used when available by GLSL compilers and applications that need the GLSL [discard](#) behavior.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):

- [VkPhysicalDeviceShaderTerminateInvocationFeaturesKHR](#)

New Enum Constants

- `VK_KHR_SHADER_TERMINATE_INVOCATION_EXTENSION_NAME`
- `VK_KHR_SHADER_TERMINATE_INVOCATION_SPEC_VERSION`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TERMINATE_INVOCATION_FEATURES_KHR`

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Version History

- Revision 1, 2020-08-11 (Jesse Hall)

VK_KHR_shared_presentable_image

Name String

`VK_KHR_shared_presentable_image`

Extension Type

Device extension

Registered Extension Number

112

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_swapchain](#)

and

[VK_KHR_get_surface_capabilities2](#)

and

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Alon Or-bach [\[GitHub\]alonorbach](#)

Other Extension Metadata

Last Modified Date

2017-03-20

IP Status

No known IP claims.

Contributors

- Alon Or-bach, Samsung Electronics
- Ian Elliott, Google
- Jesse Hall, Google
- Pablo Ceballos, Google
- Chris Forbes, Google
- Jeff Juliano, NVIDIA
- James Jones, NVIDIA
- Daniel Rakos, AMD
- Tobias Hector, Imagination Technologies
- Graham Connor, Imagination Technologies
- Michael Worcester, Imagination Technologies
- Cass Everitt, Oculus
- Johannes Van Waveren, Oculus

Description

This extension extends [VK_KHR_swapchain](#) to enable creation of a shared presentable image. This allows the application to use the image while the presentation engine is accessing it, in order to reduce the latency between rendering and presentation.

New Commands

- [vkGetSwapchainStatusKHR](#)

New Structures

- Extending [VkSurfaceCapabilities2KHR](#):
 - [VkSharedPresentSurfaceCapabilitiesKHR](#)

New Enum Constants

- [VK_KHR_SHARED_PRESENTABLE_IMAGE_EXTENSION_NAME](#)
- [VK_KHR_SHARED_PRESENTABLE_IMAGE_SPEC_VERSION](#)
- Extending [VkImageLayout](#):

- `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR`
- Extending `VkPresentModeKHR`:
 - `VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR`
 - `VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_SHARED_PRESENT_SURFACE_CAPABILITIES_KHR`

Issues

1) Should we allow a Vulkan WSI swapchain to toggle between normal usage and shared presentation usage?

RESOLVED: No. WSI swapchains are typically recreated with new properties instead of having their properties changed. This can also save resources, assuming that fewer images are needed for shared presentation, and assuming that most VR applications do not need to switch between normal and shared usage.

2) Should we have a query for determining how the presentation engine refresh is triggered?

RESOLVED: Yes. This is done via which presentation modes a surface supports.

3) Should the object representing a shared presentable image be an extension of a `VkSwapchainKHR` or a separate object?

RESOLVED: Extension of a swapchain due to overlap in creation properties and to allow common functionality between shared and normal presentable images and swapchains.

4) What should we call the extension and the new structures it creates?

RESOLVED: Shared presentable image / shared present.

5) Should the `minImageCount` and `presentMode` values of the `VkSwapchainCreateInfoKHR` be ignored, or required to be compatible values?

RESOLVED: `minImageCount` must be 1, and `presentMode` should be set to either `VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR` or `VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR`.

6) What should the layout of the shared presentable image be?

RESOLVED: After acquiring the shared presentable image, the application must transition it to the `VK_IMAGE_LAYOUT_SHARED_PRESENT_KHR` layout prior to it being used. After this initial transition, any image usage that was requested during swapchain creation **can** be performed on the image without layout transitions being performed.

7) Do we need a new API for the trigger to refresh new content?

RESOLVED: `vkQueuePresentKHR` to act as API to trigger a refresh, as will allow combination with other compatible extensions to `vkQueuePresentKHR`.

8) How should an application detect a `VK_ERROR_OUT_OF_DATE_KHR` error on a swapchain using the `VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR` present mode?

RESOLVED: Introduce `vkGetSwapchainStatusKHR` to allow applications to query the status of a swapchain using a shared presentation mode.

9) What should subsequent calls to `vkQueuePresentKHR` for `VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR` swapchains be defined to do?

RESOLVED: State that implementations may use it as a hint for updated content.

10) Can the ownership of a shared presentable image be transferred to a different queue?

RESOLVED: No. It is not possible to transfer ownership of a shared presentable image obtained from a swapchain created using `VK_SHARING_MODE_EXCLUSIVE` after it has been presented.

11) How should `vkQueueSubmit` behave if a command buffer uses an image from a `VK_ERROR_OUT_OF_DATE_KHR` swapchain?

RESOLVED: `vkQueueSubmit` is expected to return the `VK_ERROR_DEVICE_LOST` error.

12) Can Vulkan provide any guarantee on the order of rendering, to enable beam chasing?

RESOLVED: This could be achieved via use of render passes to ensure strip rendering.

Version History

- Revision 1, 2017-03-20 (Alon Or-bach)
 - Internal revisions

VK_KHR_spirv_1_4

Name String

`VK_KHR_spirv_1_4`

Extension Type

Device extension

Registered Extension Number

237

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

and

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Jesse Hall [\[GitHub\]](#)[critsec](#)

Other Extension Metadata

Last Modified Date

2019-04-01

IP Status

No known IP claims.

Interactions and External Dependencies

- Requires SPIR-V 1.4.

Contributors

- Alexander Galazin, Arm
- David Neto, Google
- Jesse Hall, Google
- John Kessenich, Google
- Neil Henning, AMD
- Tom Olson, Arm

Description

This extension allows the use of SPIR-V 1.4 shader modules. SPIR-V 1.4's new features primarily make it an easier target for compilers from high-level languages, rather than exposing new hardware functionality.

SPIR-V 1.4 incorporates features that are also available separately as extensions. SPIR-V 1.4 shader modules do not need to enable those extensions with the `OpExtension` opcode, since they are integral parts of SPIR-V 1.4.

SPIR-V 1.4 introduces new floating-point execution mode capabilities, also available via `SPV_KHR_float_controls`. Implementations are not required to support all of these new capabilities; support can be queried using `VkPhysicalDeviceFloatControlsPropertiesKHR` from the `VK_KHR_shader_float_controls` extension.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Enum Constants

- `VK_KHR_SPIRV_1_4_EXTENSION_NAME`
- `VK_KHR_SPIRV_1_4_SPEC_VERSION`

Issues

1. Should we have an extension specific to this SPIR-V version, or add a version-generic query for SPIR-V version? SPIR-V 1.4 does not need any other API changes.

RESOLVED: Just expose SPIR-V 1.4.

Most new SPIR-V versions introduce optionally-required capabilities or have implementation-defined limits, and would need more API and specification changes specific to that version to make them available in Vulkan. While we could expose the parts of a new SPIR-V version that do not need accompanying changes generically, we will still end up writing extensions specific to each version for the remaining parts. Thus the generic mechanism will not reduce future spec-writing effort. In addition, making it clear which parts of a future version are supported by the generic mechanism and which cannot be used without specific support would be difficult to get right ahead of time.

2. Can different stages of the same pipeline use shaders with different SPIR-V versions?

RESOLVED: Yes.

Mixing SPIR-V versions 1.0-1.3 in the same pipeline has not been disallowed, so it would be inconsistent to disallow mixing 1.4 with previous versions. SPIR-V 1.4 does not introduce anything that should cause new difficulties here.

3. Must Vulkan extensions corresponding to SPIR-V extensions that were promoted to core in 1.4 be enabled in order to use that functionality in a SPIR-V 1.4 module?

RESOLVED: No, with caveats.

The SPIR-V 1.4 module does not need to declare the SPIR-V extensions, since the functionality is now part of core, so there is no need to enable the Vulkan extension that allows SPIR-V modules to declare the SPIR-V extension. However, when the functionality that is now core in SPIR-V 1.4 is optionally supported, the query for support is provided by a Vulkan extension, and that query can only be used if the extension is enabled.

This applies to any SPIR-V version; specifically for SPIR-V 1.4 this only applies to the functionality from `SPV_KHR_float_controls`, which was made available in Vulkan by `VK_KHR_shader_float_controls`. Even though the extension was promoted in SPIR-V 1.4, the capabilities are still optional in implementations that support `VK_KHR_spirv_1_4`.

A SPIR-V 1.4 module does not need to enable `SPV_KHR_float_controls` in order to use the capabilities, so if the application has *a priori* knowledge that the implementation supports the capabilities, it does not need to enable `VK_KHR_shader_float_controls`. However, if it does not have this knowledge and has to query for support at runtime, it must enable `VK_KHR_shader_float_controls` in order to use `VkPhysicalDeviceFloatControlsPropertiesKHR`.

Version History

- Revision 1, 2019-04-01 (Jesse Hall)
 - Internal draft versions

VK_KHR_storage_buffer_storage_class

Name String

VK_KHR_storage_buffer_storage_class

Extension Type

Device extension

Registered Extension Number

132

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- [SPV_KHR_storage_buffer_storage_class](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Alexander Galazin [\[GitHub\]](#)[alegal-arm](#)

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- Alexander Galazin, ARM
- David Neto, Google

Description

This extension adds support for the following SPIR-V extension in Vulkan:

- `SPV_KHR_storage_buffer_storage_class`

This extension provides a new SPIR-V `StorageBuffer` storage class. A `Block`-decorated object in this class is equivalent to a `BufferBlock`-decorated object in the `Uniform` storage class.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1.

New Enum Constants

- `VK_KHR_STORAGE_BUFFER_STORAGE_CLASS_EXTENSION_NAME`
- `VK_KHR_STORAGE_BUFFER_STORAGE_CLASS_SPEC_VERSION`

Version History

- Revision 1, 2017-03-23 (Alexander Galazin)
 - Initial draft

VK_KHR_surface

Name String

`VK_KHR_surface`

Extension Type

Instance extension

Registered Extension Number

1

Revision

25

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- James Jones [\[GitHub\]cubanismo](#)
- Ian Elliott [\[GitHub\]ianelliottus](#)

Other Extension Metadata

Last Modified Date

2016-08-25

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Ian Elliott, LunarG
- Jesse Hall, Google
- James Jones, NVIDIA
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung
- Daniel Rakos, AMD
- Graham Sellers, AMD
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG
- Faith Ekstrand, Intel

Description

The `VK_KHR_surface` extension is an instance extension. It introduces `VkSurfaceKHR` objects, which abstract native platform surface or window objects for use with Vulkan. It also provides a way to determine whether a queue family in a physical device supports presenting to particular surface.

Separate extensions for each platform provide the mechanisms for creating `VkSurfaceKHR` objects, but once created they may be used in this and other platform-independent extensions, in particular the `VK_KHR_swapchain` extension.

New Object Types

- `VkSurfaceKHR`

New Commands

- `vkDestroySurfaceKHR`
- `vkGetPhysicalDeviceSurfaceCapabilitiesKHR`
- `vkGetPhysicalDeviceSurfaceFormatsKHR`
- `vkGetPhysicalDeviceSurfacePresentModesKHR`
- `vkGetPhysicalDeviceSurfaceSupportKHR`

New Structures

- `VkSurfaceCapabilitiesKHR`
- `VkSurfaceFormatKHR`

New Enums

- [VkColorSpaceKHR](#)
- [VkCompositeAlphaFlagBitsKHR](#)
- [VkPresentModeKHR](#)
- [VkSurfaceTransformFlagBitsKHR](#)

New Bitmasks

- [VkCompositeAlphaFlagsKHR](#)

New Enum Constants

- [VK_KHR_SURFACE_EXTENSION_NAME](#)
- [VK_KHR_SURFACE_SPEC_VERSION](#)
- Extending [VkObjectType](#):
 - [VK_OBJECT_TYPE_SURFACE_KHR](#)
- Extending [VkResult](#):
 - [VK_ERROR_NATIVE_WINDOW_IN_USE_KHR](#)
 - [VK_ERROR_SURFACE_LOST_KHR](#)

Examples

NOTE

The example code for the [VK_KHR_surface](#) and [VK_KHR_swapchain](#) extensions was removed from the appendix after revision 1.0.29. This WSI example code was ported to the cube demo that is shipped with the official Khronos SDK, and is being kept up-to-date in that location (see: <https://github.com/KhronosGroup/Vulkan-Tools/blob/main/cube/cube.c>).

Issues

1) Should this extension include a method to query whether a physical device supports presenting to a specific window or native surface on a given platform?

RESOLVED: Yes. Without this, applications would need to create a device instance to determine whether a particular window can be presented to. Knowing that a device supports presentation to a platform in general is not sufficient, as a single machine might support multiple seats, or instances of the platform that each use different underlying physical devices. Additionally, on some platforms, such as the X Window System, different drivers and devices might be used for different windows depending on which section of the desktop they exist on.

2) Should the [vkGetPhysicalDeviceSurfaceCapabilitiesKHR](#), [vkGetPhysicalDeviceSurfaceFormatsKHR](#), and [vkGetPhysicalDeviceSurfacePresentModesKHR](#) functions be in this extension and operate on physical devices, rather than being in [VK_KHR_swapchain](#) (i.e. device extension) and being dependent on [VkDevice](#)?

RESOLVED: Yes. While it might be useful to depend on `VkDevice` (and therefore on enabled extensions and features) for the queries, Vulkan was released only with the `VkPhysicalDevice` versions. Many cases can be resolved by a Valid Usage statement, and/or by a separate `pNext` chain version of the query struct specific to a given extension or parameters, via extensible versions of the queries: `vkGetPhysicalDeviceSurfacePresentModes2EXT`, `vkGetPhysicalDeviceSurfaceCapabilities2KHR`, and `vkGetPhysicalDeviceSurfaceFormats2KHR`.

3) Should Vulkan support Xlib or XCB as the API for accessing the X Window System platform?

RESOLVED: Both. XCB is a more modern and efficient API, but Xlib usage is deeply ingrained in many applications and likely will remain in use for the foreseeable future. Not all drivers necessarily need to support both, but including both as options in the core specification will probably encourage support, which should in turn ease adoption of the Vulkan API in older codebases. Additionally, the performance improvements possible with XCB likely will not have a measurable impact on the performance of Vulkan presentation and other minimal window system interactions defined here.

4) Should the GBM platform be included in the list of platform enums?

RESOLVED: Deferred, and will be addressed with a platform-specific extension to be written in the future.

Version History

- Revision 1, 2015-05-20 (James Jones)
 - Initial draft, based on LunarG KHR spec, other KHR specs, patches attached to bugs.
- Revision 2, 2015-05-22 (Ian Elliott)
 - Created initial Description section.
 - Removed query for whether a platform requires the use of a queue for presentation, since it was decided that presentation will always be modeled as being part of the queue.
 - Fixed typos and other minor mistakes.
- Revision 3, 2015-05-26 (Ian Elliott)
 - Improved the Description section.
- Revision 4, 2015-05-27 (James Jones)
 - Fixed compilation errors in example code.
- Revision 5, 2015-06-01 (James Jones)
 - Added issues 1 and 2 and made related spec updates.
- Revision 6, 2015-06-01 (James Jones)
 - Merged the platform type mappings table previously removed from `VK_KHR_swapchain` with the platform description table in this spec.
 - Added issues 3 and 4 documenting choices made when building the initial list of native platforms supported.
- Revision 7, 2015-06-11 (Ian Elliott)

- Updated table 1 per input from the KHR TSG.
- Updated issue 4 (GBM) per discussion with Daniel Stone. He will create a platform-specific extension sometime in the future.
- Revision 8, 2015-06-17 (James Jones)
 - Updated enum-extending values using new convention.
 - Fixed the value of `VK_SURFACE_PLATFORM_INFO_TYPE_SUPPORTED_KHR`.
- Revision 9, 2015-06-17 (James Jones)
 - Rebased on Vulkan API version 126.
- Revision 10, 2015-06-18 (James Jones)
 - Marked issues 2 and 3 resolved.
- Revision 11, 2015-06-23 (Ian Elliott)
 - Examples now show use of function pointers for extension functions.
 - Eliminated extraneous whitespace.
- Revision 12, 2015-07-07 (Daniel Rakos)
 - Added error section describing when each error is expected to be reported.
 - Replaced the term “queue node index” with “queue family index” in the spec as that is the agreed term to be used in the latest version of the core header and spec.
 - Replaced `bool32_t` with `VkBool32`.
- Revision 13, 2015-08-06 (Daniel Rakos)
 - Updated spec against latest core API header version.
- Revision 14, 2015-08-20 (Ian Elliott)
 - Renamed this extension and all of its enumerations, types, functions, etc. This makes it compliant with the proposed standard for Vulkan extensions.
 - Switched from “revision” to “version”, including use of the `VK_MAKE_VERSION` macro in the header file.
 - Did miscellaneous cleanup, etc.
- Revision 15, 2015-08-20 (Ian Elliott—porting a 2015-07-29 change from James Jones)
 - Moved the surface transform enums here from `VK_WSI_swapchain` so they could be reused by `VK_WSI_display`.
- Revision 16, 2015-09-01 (James Jones)
 - Restore single-field revision number.
- Revision 17, 2015-09-01 (James Jones)
 - Fix example code compilation errors.
- Revision 18, 2015-09-26 (Jesse Hall)
 - Replaced `VkSurfaceDescriptionKHR` with the `VkSurfaceKHR` object, which is created via layered extensions. Added `VkDestroySurfaceKHR`.

- Revision 19, 2015-09-28 (Jesse Hall)
 - Renamed from VK_EXT_KHR_swapchain to VK_EXT_KHR_surface.
- Revision 20, 2015-09-30 (Jeff Vigil)
 - Add error result VK_ERROR_SURFACE_LOST_KHR.
- Revision 21, 2015-10-15 (Daniel Rakos)
 - Updated the resolution of issue #2 and include the surface capability queries in this extension.
 - Renamed SurfaceProperties to SurfaceCapabilities as it better reflects that the values returned are the capabilities of the surface on a particular device.
 - Other minor cleanup and consistency changes.
- Revision 22, 2015-10-26 (Ian Elliott)
 - Renamed from VK_EXT_KHR_surface to VK_KHR_surface.
- Revision 23, 2015-11-03 (Daniel Rakos)
 - Added allocation callbacks to vkDestroySurfaceKHR.
- Revision 24, 2015-11-10 (Jesse Hall)
 - Removed VkSurfaceTransformKHR. Use VkSurfaceTransformFlagBitsKHR instead.
 - Rename VkSurfaceCapabilitiesKHR member maxImageArraySize to maxImageArrayLayers.
- Revision 25, 2016-01-14 (James Jones)
 - Moved VK_ERROR_NATIVE_WINDOW_IN_USE_KHR from the VK_KHR_android_surface to the VK_KHR_surface extension.
- 2016-08-23 (Ian Elliott)
 - Update the example code, to not have so many characters per line, and to split out a new example to show how to obtain function pointers.
- 2016-08-25 (Ian Elliott)
 - A note was added at the beginning of the example code, stating that it will be removed from future versions of the appendix.

VK_KHR_surface_protected_capabilities

Name String

VK_KHR_surface_protected_capabilities

Extension Type

Instance extension

Registered Extension Number

240

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

and

[VK_KHR_get_surface_capabilities2](#)

Contact

- Sandeep Shinde [\[GitHub\]](#)sashinde

Other Extension Metadata

Last Modified Date

2018-12-18

IP Status

No known IP claims.

Contributors

- Sandeep Shinde, NVIDIA
- James Jones, NVIDIA
- Daniel Koch, NVIDIA

Description

This extension extends [VkSurfaceCapabilities2KHR](#), providing applications a way to query whether swapchains **can** be created with the [VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR](#) flag set.

Vulkan 1.1 added (optional) support for protect memory and protected resources including buffers ([VK_BUFFER_CREATE_PROTECTED_BIT](#)), images ([VK_IMAGE_CREATE_PROTECTED_BIT](#)), and swapchains ([VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR](#)). However, on implementations which support multiple windowing systems, not all window systems **may** be able to provide a protected display path.

This extension provides a way to query if a protected swapchain created for a surface (and thus a specific windowing system) **can** be displayed on screen. It extends the existing [VkSurfaceCapabilities2KHR](#) structure with a new [VkSurfaceProtectedCapabilitiesKHR](#) structure from which the application **can** obtain information about support for protected swapchain creation through [vkGetPhysicalDeviceSurfaceCapabilities2KHR](#).

New Structures

- Extending [VkSurfaceCapabilities2KHR](#):
 - [VkSurfaceProtectedCapabilitiesKHR](#)

New Enum Constants

- [VK_KHR_SURFACE_PROTECTED_CAPABILITIES_EXTENSION_NAME](#)

- `VK_KHR_SURFACE_PROTECTED_CAPABILITIES_SPEC_VERSION`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_SURFACE_PROTECTED_CAPABILITIES_KHR`

Version History

- Revision 1, 2018-12-18 (Sandeep Shinde, Daniel Koch)
 - Internal revisions.

VK_KHR_swapchain

Name String

`VK_KHR_swapchain`

Extension Type

Device extension

Registered Extension Number

2

Revision

70

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_surface](#)

API Interactions

- Interacts with `VK_VERSION_1_1`

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)
- Ian Elliott [\[GitHub\]](#)[ianelliottus](#)

Other Extension Metadata

Last Modified Date

2017-10-06

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with Vulkan 1.1

Contributors

- Patrick Doane, Blizzard
- Ian Elliott, LunarG
- Jesse Hall, Google
- Mathias Heyer, NVIDIA
- James Jones, NVIDIA
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung
- Daniel Rakos, AMD
- Graham Sellers, AMD
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG
- Faith Ekstrand, Intel
- Matthaeus G. Chajdas, AMD
- Ray Smith, ARM

Description

The [VK_KHR_swapchain](#) extension is the device-level companion to the [VK_KHR_surface](#) extension. It introduces [VkSwapchainKHR](#) objects, which provide the ability to present rendering results to a surface.

New Object Types

- [VkSwapchainKHR](#)

New Commands

- [vkAcquireNextImageKHR](#)
- [vkCreateSwapchainKHR](#)
- [vkDestroySwapchainKHR](#)
- [vkGetSwapchainImagesKHR](#)
- [vkQueuePresentKHR](#)

If [Vulkan Version 1.1](#) is supported:

- [vkAcquireNextImage2KHR](#)
- [vkGetDeviceGroupPresentCapabilitiesKHR](#)
- [vkGetDeviceGroupSurfacePresentModesKHR](#)
- [vkGetPhysicalDevicePresentRectanglesKHR](#)

New Structures

- [VkPresentInfoKHR](#)
- [VkSwapchainCreateInfoKHR](#)

If [Vulkan Version 1.1](#) is supported:

- [VkAcquireNextImageInfoKHR](#)
- [VkDeviceGroupPresentCapabilitiesKHR](#)
- Extending [VkBindImageMemoryInfo](#):
 - [VkBindImageMemorySwapchainInfoKHR](#)
- Extending [VkImageCreateInfo](#):
 - [VkImageSwapchainCreateInfoKHR](#)
- Extending [VkPresentInfoKHR](#):
 - [VkDeviceGroupPresentInfoKHR](#)
- Extending [VkSwapchainCreateInfoKHR](#):
 - [VkDeviceGroupSwapchainCreateInfoKHR](#)

New Enums

- [VkSwapchainCreateFlagBitsKHR](#)

If [Vulkan Version 1.1](#) is supported:

- [VkDeviceGroupPresentModeFlagBitsKHR](#)

New Bitmasks

- [VkSwapchainCreateFlagsKHR](#)

If [Vulkan Version 1.1](#) is supported:

- [VkDeviceGroupPresentModeFlagsKHR](#)

New Enum Constants

- [VK_KHR_SWAPCHAIN_EXTENSION_NAME](#)
- [VK_KHR_SWAPCHAIN_SPEC_VERSION](#)
- Extending [VkImageLayout](#):
 - [VK_IMAGE_LAYOUT_PRESENT_SRC_KHR](#)
- Extending [VkObjectType](#):
 - [VK_OBJECT_TYPE_SWAPCHAIN_KHR](#)
- Extending [VkResult](#):
 - [VK_ERROR_OUT_OF_DATE_KHR](#)

- `VK_SUBOPTIMAL_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PRESENT_INFO_KHR`
 - `VK_STRUCTURE_TYPE_SWAPCHAIN_CREATE_INFO_KHR`

If `Vulkan Version 1.1` is supported:

- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_ACQUIRE_NEXT_IMAGE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_BIND_IMAGE_MEMORY_SWAPCHAIN_INFO_KHR`
 - `VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_CAPABILITIES_KHR`
 - `VK_STRUCTURE_TYPE_DEVICE_GROUP_PRESENT_INFO_KHR`
 - `VK_STRUCTURE_TYPE_DEVICE_GROUP_SWAPCHAIN_CREATE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_IMAGE_SWAPCHAIN_CREATE_INFO_KHR`
- Extending `VkSwapchainCreateFlagBitsKHR`:
 - `VK_SWAPCHAIN_CREATE_PROTECTED_BIT_KHR`
 - `VK_SWAPCHAIN_CREATE_SPLIT_INSTANCE_BIND_REGIONS_BIT_KHR`

Issues

1) Does this extension allow the application to specify the memory backing of the presentable images?

RESOLVED: No. Unlike standard images, the implementation will allocate the memory backing of the presentable image.

2) What operations are allowed on presentable images?

RESOLVED: This is determined by the image usage flags specified when creating the presentable image's swapchain.

3) Does this extension support MSAA presentable images?

RESOLVED: No. Presentable images are always single-sampled. Multi-sampled rendering must use regular images. To present the rendering results the application must manually resolve the multi-sampled image to a single-sampled presentable image prior to presentation.

4) Does this extension support stereo/multi-view presentable images?

RESOLVED: Yes. The number of views associated with a presentable image is determined by the `imageArrayLayers` specified when creating a swapchain. All presentable images in a given swapchain use the same array size.

5) Are the layers of stereo presentable images half-sized?

RESOLVED: No. The image extents always match those requested by the application.

6) Do the “present” and “acquire next image” commands operate on a queue? If not, do they need to include explicit semaphore objects to interlock them with queue operations?

RESOLVED: The present command operates on a queue. The image ownership operation it represents happens in order with other operations on the queue, so no explicit semaphore object is required to synchronize its actions.

Applications may want to acquire the next image in separate threads from those in which they manage their queue, or in multiple threads. To make such usage easier, the acquire next image command takes a semaphore to signal as a method of explicit synchronization. The application must later queue a wait for this semaphore before queuing execution of any commands using the image.

7) Does [vkAcquireNextImageKHR](#) block if no images are available?

RESOLVED: The command takes a timeout parameter. Special values for the timeout are 0, which makes the call a non-blocking operation, and [UINT64_MAX](#), which blocks indefinitely. Values in between will block for up to the specified time. The call will return when an image becomes available or an error occurs. It may, but is not required to, return before the specified timeout expires if the swapchain becomes out of date.

8) Can multiple presents be queued using one [vkQueuePresentKHR](#) call?

RESOLVED: Yes. [VkPresentInfoKHR](#) contains a list of swapchains and corresponding image indices that will be presented. When supported, all presentations queued with a single [vkQueuePresentKHR](#) call will be applied atomically as one operation. The same swapchain must not appear in the list more than once. Later extensions may provide applications stronger guarantees of atomicity for such present operations, and/or allow them to query whether atomic presentation of a particular group of swapchains is possible.

9) How do the presentation and acquire next image functions notify the application the targeted surface has changed?

RESOLVED: Two new result codes are introduced for this purpose:

- [VK_SUBOPTIMAL_KHR](#) - Presentation will still succeed, subject to the window resize behavior, but the swapchain is no longer configured optimally for the surface it targets. Applications should query updated surface information and recreate their swapchain at the next convenient opportunity.
- [VK_ERROR_OUT_OF_DATE_KHR](#) - Failure. The swapchain is no longer compatible with the surface it targets. The application must query updated surface information and recreate the swapchain before presentation will succeed.

These can be returned by both [vkAcquireNextImageKHR](#) and [vkQueuePresentKHR](#).

10) Does the [vkAcquireNextImageKHR](#) command return a semaphore to the application via an output parameter, or accept a semaphore to signal from the application as an object handle parameter?

RESOLVED: Accept a semaphore to signal as an object handle. This avoids the need to specify

whether the application must destroy the semaphore or whether it is owned by the swapchain, and if the latter, what its lifetime is and whether it can be reused for other operations once it is received from [vkAcquireNextImageKHR](#).

11) What types of swapchain queuing behavior should be exposed? Options include swap interval specification, mailbox/most recent vs. FIFO queue management, targeting specific vertical blank intervals or absolute times for a given present operation, and probably others. For some of these, whether they are specified at swapchain creation time or as per-present parameters needs to be decided as well.

RESOLVED: The base swapchain extension will expose 3 possible behaviors (of which, FIFO will always be supported):

- Immediate present: Does not wait for vertical blanking period to update the current image, likely resulting in visible tearing. No internal queue is used. Present requests are applied immediately.
- Mailbox queue: Waits for the next vertical blanking period to update the current image. No tearing should be observed. An internal single-entry queue is used to hold pending presentation requests. If the queue is full when a new presentation request is received, the new request replaces the existing entry, and any images associated with the prior entry become available for reuse by the application.
- FIFO queue: Waits for the next vertical blanking period to update the current image. No tearing should be observed. An internal queue containing `numSwapchainImages` - 1 entries is used to hold pending presentation requests. New requests are appended to the end of the queue, and one request is removed from the beginning of the queue and processed during each vertical blanking period in which the queue is non-empty

Not all surfaces will support all of these modes, so the modes supported will be returned using a surface information query. All surfaces must support the FIFO queue mode. Applications must choose one of these modes up front when creating a swapchain. Switching modes can be accomplished by recreating the swapchain.

12) Can `VK_PRESENT_MODE_MAILBOX_KHR` provide non-blocking guarantees for [vkAcquireNextImageKHR](#)? If so, what is the proper criteria?

RESOLVED: Yes. The difficulty is not immediately obvious here. Naively, if at least 3 images are requested, mailbox mode should always have an image available for the application if the application does not own any images when the call to [vkAcquireNextImageKHR](#) was made. However, some presentation engines may have more than one “current” image, and would still need to block in some cases. The right requirement appears to be that if the application allocates the surface’s minimum number of images + 1 then it is guaranteed non-blocking behavior when it does not currently own any images.

13) Is there a way to create and initialize a new swapchain for a surface that has generated a `VK_SUBOPTIMAL_KHR` return code while still using the old swapchain?

RESOLVED: Not as part of this specification. This could be useful to allow the application to create an “optimal” replacement swapchain and rebuild all its command buffers using it in a background thread at a low priority while continuing to use the “suboptimal” swapchain in the main thread. It

could probably use the same “atomic replace” semantics proposed for recreating direct-to-device swapchains without incurring a mode switch. However, after discussion, it was determined some platforms probably could not support concurrent swapchains for the same surface though, so this will be left out of the base KHR extensions. A future extension could add this for platforms where it is supported.

14) Should there be a special value for `VkSurfaceCapabilitiesKHR::maxImageCount` to indicate there are no practical limits on the number of images in a swapchain?

RESOLVED: Yes. There will often be cases where there is no practical limit to the number of images in a swapchain other than the amount of available resources (i.e., memory) in the system. Trying to derive a hard limit from things like memory size is prone to failure. It is better in such cases to leave it to applications to figure such soft limits out via trial/failure iterations.

15) Should there be a special value for `VkSurfaceCapabilitiesKHR::currentExtent` to indicate the size of the platform surface is undefined?

RESOLVED: Yes. On some platforms (Wayland, for example), the surface size is defined by the images presented to it rather than the other way around.

16) Should there be a special value for `VkSurfaceCapabilitiesKHR::maxImageExtent` to indicate there is no practical limit on the surface size?

RESOLVED: No. It seems unlikely such a system would exist. 0 could be used to indicate the platform places no limits on the extents beyond those imposed by Vulkan for normal images, but this query could just as easily return those same limits, so a special “unlimited” value does not seem useful for this field.

17) How should surface rotation and mirroring be exposed to applications? How do they specify rotation and mirroring transforms applied prior to presentation?

RESOLVED: Applications can query both the supported and current transforms of a surface. Both are specified relative to the device’s “natural” display rotation and direction. The supported transforms indicate which orientations the presentation engine accepts images in. For example, a presentation engine that does not support transforming surfaces as part of presentation, and which is presenting to a surface that is displayed with a 90-degree rotation, would return only one supported transform bit: `VK_SURFACE_TRANSFORM_ROTATE_90_BIT_KHR`. Applications must transform their rendering by the transform they specify when creating the swapchain in `preTransform` field.

18) Can surfaces ever not support `VK_MIRROR_NONE`? Can they support vertical and horizontal mirroring simultaneously? Relatedly, should `VK_MIRROR_NONE_BIT` be zero, or bit one, and should applications be allowed to specify multiple pre and current mirror transform bits, or exactly one?

RESOLVED: Since some platforms may not support presenting with a transform other than the native window’s current transform, and prerotation/mirroring are specified relative to the device’s natural rotation and direction, rather than relative to the surface’s current rotation and direction, it is necessary to express lack of support for no mirroring. To allow this, the `MIRROR_NONE` enum must occupy a bit in the flags. Since `MIRROR_NONE` must be a bit in the bitmask rather than a bitmask with no values set, allowing more than one bit to be set in the bitmask would make it possible to describe undefined transforms such as `VK_MIRROR_NONE_BIT | VK_MIRROR_HORIZONTAL_BIT`, or a

transform that includes both “no mirroring” and “horizontal mirroring” simultaneously. Therefore, it is desirable to allow specifying all supported mirroring transforms using only one bit. The question then becomes, should there be a `VK_MIRROR_HORIZONTAL_AND_VERTICAL_BIT` to represent a simultaneous horizontal and vertical mirror transform? However, such a transform is equivalent to a 180 degree rotation, so presentation engines and applications that wish to support or use such a transform can express it through rotation instead. Therefore, 3 exclusive bits are sufficient to express all needed mirroring transforms.

19) Should support for sRGB be required?

RESOLVED: In the advent of UHD and HDR display devices, proper color space information is vital to the display pipeline represented by the swapchain. The application can discover the supported format/color-space pairs and select a pair most suited to its rendering needs. Currently only the sRGB color space is supported, future extensions may provide support for more color spaces. See issues 23 and 24.

20) Is there a mechanism to modify or replace an existing swapchain with one targeting the same surface?

RESOLVED: Yes. This is described above in the text.

21) Should there be a way to set prerotation and mirroring using native APIs when presenting using a Vulkan swapchain?

RESOLVED: Yes. The transforms that can be expressed in this extension are a subset of those possible on native platforms. If a platform exposes a method to specify the transform of presented images for a given surface using native methods and exposes more transforms or other properties for surfaces than Vulkan supports, it might be impossible, difficult, or inconvenient to set some of those properties using Vulkan KHR extensions and some using the native interfaces. To avoid overwriting properties set using native commands when presenting using a Vulkan swapchain, the application can set the pretransform to “inherit”, in which case the current native properties will be used, or if none are available, a platform-specific default will be used. Platforms that do not specify a reasonable default or do not provide native mechanisms to specify such transforms should not include the inherit bits in the `supportedTransforms` bitmask they return in `VkSurfaceCapabilitiesKHR`.

22) Should the content of presentable images be clipped by objects obscuring their target surface?

RESOLVED: Applications can choose which behavior they prefer. Allowing the content to be clipped could enable more efficient presentation methods on some platforms, but some applications might rely on the content of presentable images to perform techniques such as partial updates or motion blurs.

23) What is the purpose of specifying a `VkColorSpaceKHR` along with `VkFormat` when creating a swapchain?

RESOLVED: While Vulkan itself is color space agnostic (e.g. even the meaning of R, G, B and A can be freely defined by the rendering application), the swapchain eventually will have to present the images on a display device with specific color reproduction characteristics. If any color space transformations are necessary before an image can be displayed, the color space of the presented

image must be known to the swapchain. A swapchain will only support a restricted set of color format and -space pairs. This set can be discovered via [vkGetPhysicalDeviceSurfaceFormatsKHR](#). As it can be expected that most display devices support the sRGB color space, at least one format/color-space pair has to be exposed, where the color space is [VK_COLOR_SPACE_SRGB_NONLINEAR_KHR](#).

24) How are sRGB formats and the sRGB color space related?

RESOLVED: While Vulkan exposes a number of SRGB texture formats, using such formats does not guarantee working in a specific color space. It merely means that the hardware can directly support applying the non-linear transfer functions defined by the sRGB standard color space when reading from or writing to images of those formats. Still, it is unlikely that a swapchain will expose a [*_SRGB](#) format along with any color space other than [VK_COLOR_SPACE_SRGB_NONLINEAR_KHR](#).

On the other hand, non-[*_SRGB](#) formats will be very likely exposed in pair with a SRGB color space. This means, the hardware will not apply any transfer function when reading from or writing to such images, yet they will still be presented on a device with sRGB display characteristics. In this case the application is responsible for applying the transfer function, for instance by using shader math.

25) How are the lifetimes of surfaces and swapchains targeting them related?

RESOLVED: A surface must outlive any swapchains targeting it. A [VkSurfaceKHR](#) owns the binding of the native window to the Vulkan driver.

26) How can the application control the way the alpha component of swapchain images is treated by the presentation engine during compositing?

RESOLVED: We should add new enum values to allow the application to negotiate with the presentation engine on how to treat image alpha values during the compositing process. Since not all platforms can practically control this through the Vulkan driver, a value of [VK_COMPOSITE_ALPHA_INHERIT_BIT_KHR](#) is provided like for surface transforms.

27) Is [vkCreateSwapchainKHR](#) the right function to return [VK_ERROR_NATIVE_WINDOW_IN_USE_KHR](#), or should the various platform-specific [VkSurfaceKHR](#) factory functions catch this error earlier?

RESOLVED: For most platforms, the [VkSurfaceKHR](#) structure is a simple container holding the data that identifies a native window or other object representing a surface on a particular platform. For the surface factory functions to return this error, they would likely need to register a reference on the native objects with the native display server somehow, and ensure no other such references exist. Surfaces were not intended to be that heavyweight.

Swapchains are intended to be the objects that directly manipulate native windows and communicate with the native presentation mechanisms. Swapchains will already need to communicate with the native display server to negotiate allocation and/or presentation of presentable images for a native surface. Therefore, it makes more sense for swapchain creation to be the point at which native object exclusivity is enforced. Platforms may choose to enforce further restrictions on the number of [VkSurfaceKHR](#) objects that may be created for the same native window if such a requirement makes sense on a particular platform, but a global requirement is only sensible at the swapchain level.

Examples

NOTE

The example code for the `VK_KHR_surface` and `VK_KHR_swapchain` extensions was removed from the appendix after revision 1.0.29. This WSI example code was ported to the cube demo that is shipped with the official Khronos SDK, and is being kept up-to-date in that location (see: <https://github.com/KhronosGroup/Vulkan-Tools/blob/main/cube/cube.c>).

Version History

- Revision 1, 2015-05-20 (James Jones)
 - Initial draft, based on LunarG KHR spec, other KHR specs, patches attached to bugs.
- Revision 2, 2015-05-22 (Ian Elliott)
 - Made many agreed-upon changes from 2015-05-21 KHR TSG meeting. This includes using only a queue for presentation, and having an explicit function to acquire the next image.
 - Fixed typos and other minor mistakes.
- Revision 3, 2015-05-26 (Ian Elliott)
 - Improved the Description section.
 - Added or resolved issues that were found in improving the Description. For example, `pSurfaceDescription` is used consistently, instead of sometimes using `pSurface`.
- Revision 4, 2015-05-27 (James Jones)
 - Fixed some grammatical errors and typos
 - Filled in the description of `imageUseFlags` when creating a swapchain.
 - Added a description of `swapInterval`.
 - Replaced the paragraph describing the order of operations on a queue for image ownership and presentation.
- Revision 5, 2015-05-27 (James Jones)
 - Imported relevant issues from the (abandoned) `vk_wsi_persistent_swapchain_images` extension.
 - Added issues 6 and 7, regarding behavior of the acquire next image and present commands with respect to queues.
 - Updated spec language and examples to align with proposed resolutions to issues 6 and 7.
- Revision 6, 2015-05-27 (James Jones)
 - Added issue 8, regarding atomic presentation of multiple swapchains
 - Updated spec language and examples to align with proposed resolution to issue 8.
- Revision 7, 2015-05-27 (James Jones)
 - Fixed compilation errors in example code, and made related spec fixes.
- Revision 8, 2015-05-27 (James Jones)
 - Added issue 9, and the related `VK_SUBOPTIMAL_KHR` result code.

- Renamed VK_OUT_OF_DATE_KHR to VK_ERROR_OUT_OF_DATE_KHR.
- Revision 9, 2015-05-27 (James Jones)
 - Added inline proposed resolutions (marked with [JRJ]) to some XXX questions/issues. These should be moved to the issues section in a subsequent update if the proposals are adopted.
- Revision 10, 2015-05-28 (James Jones)
 - Converted vkAcquireNextImageKHR back to a non-queue operation that uses a VkSemaphore object for explicit synchronization.
 - Added issue 10 to determine whether vkAcquireNextImageKHR generates or returns semaphores, or whether it operates on a semaphore provided by the application.
- Revision 11, 2015-05-28 (James Jones)
 - Marked issues 6, 7, and 8 resolved.
 - Renamed VkSurfaceCapabilityPropertiesKHR to VkSurfacePropertiesKHR to better convey the mutable nature of the information it contains.
- Revision 12, 2015-05-28 (James Jones)
 - Added issue 11 with a proposed resolution, and the related issue 12.
 - Updated various sections of the spec to match the proposed resolution to issue 11.
- Revision 13, 2015-06-01 (James Jones)
 - Moved some structures to VK_EXT_KHR_swap_chain to resolve the specification's issues 1 and 2.
- Revision 14, 2015-06-01 (James Jones)
 - Added code for example 4 demonstrating how an application might make use of the two different present and acquire next image KHR result codes.
 - Added issue 13.
- Revision 15, 2015-06-01 (James Jones)
 - Added issues 14 - 16 and related spec language.
 - Fixed some spelling errors.
 - Added language describing the meaningful return values for vkAcquireNextImageKHR and vkQueuePresentKHR.
- Revision 16, 2015-06-02 (James Jones)
 - Added issues 17 and 18, as well as related spec language.
 - Removed some erroneous text added by mistake in the last update.
- Revision 17, 2015-06-15 (Ian Elliott)
 - Changed special value from "-1" to "0" so that the data types can be unsigned.
- Revision 18, 2015-06-15 (Ian Elliott)
 - Clarified the values of VkSurfacePropertiesKHR::minImageCount and the timeout parameter of the vkAcquireNextImageKHR function.
- Revision 19, 2015-06-17 (James Jones)

- Misc. cleanup. Removed resolved inline issues and fixed typos.
- Fixed clarification of `VkSurfacePropertiesKHR::minImageCount` made in version 18.
- Added a brief “Image Ownership” definition to the list of terms used in the spec.
- Revision 20, 2015-06-17 (James Jones)
 - Updated enum-extending values using new convention.
- Revision 21, 2015-06-17 (James Jones)
 - Added language describing how to use `VK_IMAGE_LAYOUT_PRESENT_SOURCE_KHR`.
 - Cleaned up an XXX comment regarding the description of which queues `vkQueuePresentKHR` can be used on.
- Revision 22, 2015-06-17 (James Jones)
 - Rebased on Vulkan API version 126.
- Revision 23, 2015-06-18 (James Jones)
 - Updated language for issue 12 to read as a proposed resolution.
 - Marked issues 11, 12, 13, 16, and 17 resolved.
 - Temporarily added links to the relevant bugs under the remaining unresolved issues.
 - Added issues 19 and 20 as well as proposed resolutions.
- Revision 24, 2015-06-19 (Ian Elliott)
 - Changed special value for `VkSurfacePropertiesKHR::currentExtent` back to “-1” from “0”. This value will never need to be unsigned, and “0” is actually a legal value.
- Revision 25, 2015-06-23 (Ian Elliott)
 - Examples now show use of function pointers for extension functions.
 - Eliminated extraneous whitespace.
- Revision 26, 2015-06-25 (Ian Elliott)
 - Resolved Issues 9 & 10 per KHR TSG meeting.
- Revision 27, 2015-06-25 (James Jones)
 - Added `oldSwapchain` member to `VkSwapchainCreateInfoKHR`.
- Revision 28, 2015-06-25 (James Jones)
 - Added the “inherit” bits to the rotation and mirroring flags and the associated issue 21.
- Revision 29, 2015-06-25 (James Jones)
 - Added the “clipped” flag to `VkSwapchainCreateInfoKHR`, and the associated issue 22.
 - Specified that presenting an image does not modify it.
- Revision 30, 2015-06-25 (James Jones)
 - Added language to the spec that clarifies the behavior of `vkCreateSwapchainKHR()` when the `oldSwapchain` field of `VkSwapchainCreateInfoKHR` is not NULL.
- Revision 31, 2015-06-26 (Ian Elliott)

- Example of new `VkSwapchainCreateInfoKHR` members, “oldSwapchain” and “clipped”.
- Example of using `VkSurfacePropertiesKHR::min|maxImageCount` to set `VkSwapchainCreateInfoKHR::minImageCount`.
- Rename `vkGetSurfaceInfoKHR()`'s 4th parameter to “pDataSize”, for consistency with other functions.
- Add macro with C-string name of extension (just to header file).
- Revision 32, 2015-06-26 (James Jones)
 - Minor adjustments to the language describing the behavior of “oldSwapchain”
 - Fixed the version date on my previous two updates.
- Revision 33, 2015-06-26 (Jesse Hall)
 - Add usage flags to `VkSwapchainCreateInfoKHR`
- Revision 34, 2015-06-26 (Ian Elliott)
 - Rename `vkQueuePresentKHR()`'s 2nd parameter to “pPresentInfo”, for consistency with other functions.
- Revision 35, 2015-06-26 (Faith Ekstrand)
 - Merged the `VkRotationFlagBitsKHR` and `VkMirrorFlagBitsKHR` enums into a single `VkSurfaceTransformFlagBitsKHR` enum.
- Revision 36, 2015-06-26 (Faith Ekstrand)
 - Added a `VkSurfaceTransformKHR` enum that is not a bitmask. Each value in `VkSurfaceTransformKHR` corresponds directly to one of the bits in `VkSurfaceTransformFlagBitsKHR` so transforming from one to the other is easy. Having a separate enum means that `currentTransform` and `preTransform` are now unambiguous by definition.
- Revision 37, 2015-06-29 (Ian Elliott)
 - Corrected one of the signatures of `vkAcquireNextImageKHR`, which had the last two parameters switched from what it is elsewhere in the specification and header files.
- Revision 38, 2015-06-30 (Ian Elliott)
 - Corrected a typo in description of the `vkGetSwapchainInfoKHR()` function.
 - Corrected a typo in header file comment for `VkPresentInfoKHR::sType`.
- Revision 39, 2015-07-07 (Daniel Rakos)
 - Added error section describing when each error is expected to be reported.
 - Replaced `bool32_t` with `VkBool32`.
- Revision 40, 2015-07-10 (Ian Elliott)
 - Updated to work with version 138 of the `vulkan.h` header. This includes declaring the `VkSwapchainKHR` type using the new `VK_DEFINE_NONDISP_HANDLE` macro, and no longer extending `VkObjectType` (which was eliminated).
- Revision 41 2015-07-09 (Mathias Heyer)
 - Added color space language.

- Revision 42, 2015-07-10 (Daniel Rakos)
 - Updated query mechanism to reflect the convention changes done in the core spec.
 - Removed “queue” from the name of `VK_STRUCTURE_TYPE_QUEUE_PRESENT_INFO_KHR` to be consistent with the established naming convention.
 - Removed reference to the no longer existing `VkObjectType` enum.
- Revision 43, 2015-07-17 (Daniel Rakos)
 - Added support for concurrent sharing of swapchain images across queue families.
 - Updated sample code based on recent changes
- Revision 44, 2015-07-27 (Ian Elliott)
 - Noted that support for `VK_PRESENT_MODE_FIFO_KHR` is required. That is ICDs may optionally support `IMMEDIATE` and `MAILBOX`, but must support `FIFO`.
- Revision 45, 2015-08-07 (Ian Elliott)
 - Corrected a typo in spec file (type and variable name had wrong case for the `imageColorSpace` member of the `VkSwapchainCreateInfoKHR` struct).
 - Corrected a typo in header file (last parameter in `PFN_vkGetSurfacePropertiesKHR` was missing “KHR” at the end of type: `VkSurfacePropertiesKHR`).
- Revision 46, 2015-08-20 (Ian Elliott)
 - Renamed this extension and all of its enumerations, types, functions, etc. This makes it compliant with the proposed standard for Vulkan extensions.
 - Switched from “revision” to “version”, including use of the `VK_MAKE_VERSION` macro in the header file.
 - Made improvements to several descriptions.
 - Changed the status of several issues from `PROPOSED` to `RESOLVED`, leaving no unresolved issues.
 - Resolved several TODOs, did miscellaneous cleanup, etc.
- Revision 47, 2015-08-20 (Ian Elliott—porting a 2015-07-29 change from James Jones)
 - Moved the surface transform enums to `VK_WSI_swapchain` so they could be reused by `VK_WSI_display`.
- Revision 48, 2015-09-01 (James Jones)
 - Various minor cleanups.
- Revision 49, 2015-09-01 (James Jones)
 - Restore single-field revision number.
- Revision 50, 2015-09-01 (James Jones)
 - Update Example #4 to include code that illustrates how to use the `oldSwapchain` field.
- Revision 51, 2015-09-01 (James Jones)
 - Fix example code compilation errors.
- Revision 52, 2015-09-08 (Matthaeus G. Chajdas)

- Corrected a typo.
- Revision 53, 2015-09-10 (Alon Or-bach)
 - Removed underscore from SWAP_CHAIN left in VK_STRUCTURE_TYPE_SWAPCHAIN_CREATE_INFO_KHR.
- Revision 54, 2015-09-11 (Jesse Hall)
 - Described the execution and memory coherence requirements for image transitions to and from VK_IMAGE_LAYOUT_PRESENT_SOURCE_KHR.
- Revision 55, 2015-09-11 (Ray Smith)
 - Added errors for destroying and binding memory to presentable images
- Revision 56, 2015-09-18 (James Jones)
 - Added fence argument to vkAcquireNextImageKHR
 - Added example of how to meter a host thread based on presentation rate.
- Revision 57, 2015-09-26 (Jesse Hall)
 - Replace VkSurfaceDescriptionKHR with VkSurfaceKHR.
 - Added issue 25 with agreed resolution.
- Revision 58, 2015-09-28 (Jesse Hall)
 - Renamed from VK_EXT_KHR_device_swapchain to VK_EXT_KHR_swapchain.
- Revision 59, 2015-09-29 (Ian Elliott)
 - Changed vkDestroySwapchainKHR() to return void.
- Revision 60, 2015-10-01 (Jeff Vigil)
 - Added error result VK_ERROR_SURFACE_LOST_KHR.
- Revision 61, 2015-10-05 (Faith Ekstrand)
 - Added the VkCompositeAlpha enum and corresponding structure fields.
- Revision 62, 2015-10-12 (Daniel Rakos)
 - Added VK_PRESENT_MODE_FIFO_RELAXED_KHR.
- Revision 63, 2015-10-15 (Daniel Rakos)
 - Moved surface capability queries to VK_EXT_KHR_surface.
- Revision 64, 2015-10-26 (Ian Elliott)
 - Renamed from VK_EXT_KHR_swapchain to VK_KHR_swapchain.
- Revision 65, 2015-10-28 (Ian Elliott)
 - Added optional pResult member to VkPresentInfoKHR, so that per-swapchain results can be obtained from vkQueuePresentKHR().
- Revision 66, 2015-11-03 (Daniel Rakos)
 - Added allocation callbacks to create and destroy functions.
 - Updated resource transition language.

- Updated sample code.
- Revision 67, 2015-11-10 (Jesse Hall)
 - Add reserved flags bitmask to VkSwapchainCreateInfoKHR.
 - Modify naming and member ordering to match API style conventions, and so the VkSwapchainCreateInfoKHR image property members mirror corresponding VkImageCreateInfo members but with an 'image' prefix.
 - Make VkPresentInfoKHR::pResults non-const; it is an output array parameter.
 - Make pPresentInfo parameter to vkQueuePresentKHR const.
- Revision 68, 2016-04-05 (Ian Elliott)
 - Moved the “validity” include for vkAcquireNextImage to be in its proper place, after the prototype and list of parameters.
 - Clarified language about presentable images, including how they are acquired, when applications can and cannot use them, etc. As part of this, removed language about “ownership” of presentable images, and replaced it with more-consistent language about presentable images being “acquired” by the application.
- 2016-08-23 (Ian Elliott)
 - Update the example code, to use the final API command names, to not have so many characters per line, and to split out a new example to show how to obtain function pointers. This code is more similar to the LunarG “cube” demo program.
- 2016-08-25 (Ian Elliott)
 - A note was added at the beginning of the example code, stating that it will be removed from future versions of the appendix.
- Revision 69, 2017-09-07 (Tobias Hector)
 - Added interactions with Vulkan 1.1
- Revision 70, 2017-10-06 (Ian Elliott)
 - Corrected interactions with Vulkan 1.1

VK_KHR_swapchain_mutable_format

Name String

VK_KHR_swapchain_mutable_format

Extension Type

Device extension

Registered Extension Number

201

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_swapchain](#)

and

[VK_KHR_maintenance2](#)

or

[Vulkan Version 1.1](#)

and

[VK_KHR_image_format_list](#)

or

[Vulkan Version 1.2](#)

Contact

- Daniel Rakos [\[GitHub\]](#)[drakos-amd](#)

Other Extension Metadata

Last Modified Date

2018-03-28

IP Status

No known IP claims.

Contributors

- Faith Ekstrand, Intel
- Jan-Harald Fredriksen, ARM
- Jesse Hall, Google
- Daniel Rakos, AMD
- Ray Smith, ARM

Description

This extension allows processing of swapchain images as different formats to that used by the window system, which is particularly useful for switching between sRGB and linear RGB formats.

It adds a new swapchain creation flag that enables creating image views from presentable images with a different format than the one used to create the swapchain.

New Enum Constants

- [VK_KHR_SWAPCHAIN_MUTABLE_FORMAT_EXTENSION_NAME](#)
- [VK_KHR_SWAPCHAIN_MUTABLE_FORMAT_SPEC_VERSION](#)
- Extending [VkSwapchainCreateFlagBitsKHR](#):
 - [VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR](#)

Issues

1) Are there any new capabilities needed?

RESOLVED: No. It is expected that all implementations exposing this extension support swapchain image format mutability.

2) Do we need a separate `VK_SWAPCHAIN_CREATE_EXTENDED_USAGE_BIT_KHR`?

RESOLVED: No. This extension requires `VK_KHR_maintenance2` and presentable images of swapchains created with `VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR` are created internally in a way equivalent to specifying both `VK_IMAGE_CREATE_MUTABLE_FORMAT_BIT` and `VK_IMAGE_CREATE_EXTENDED_USAGE_BIT_KHR`.

3) Do we need a separate structure to allow specifying an image format list for swapchains?

RESOLVED: No. We simply use the same `VkImageFormatListCreateInfoKHR` structure introduced by `VK_KHR_image_format_list`. The structure is required to be included in the `pNext` chain of `VkSwapchainCreateInfoKHR` for swapchains created with `VK_SWAPCHAIN_CREATE_MUTABLE_FORMAT_BIT_KHR`.

Version History

- Revision 1, 2018-03-28 (Daniel Rakos)
 - Internal revisions.

VK_KHR_synchronization2

Name String

`VK_KHR_synchronization2`

Extension Type

Device extension

Registered Extension Number

315

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

`VK_KHR_get_physical_device_properties2`

or

`Vulkan Version 1.1`

API Interactions

- Interacts with `VK_EXT_blend_operation_advanced`

- Interacts with `VK_EXT_conditional_rendering`
- Interacts with `VK_EXT_device_generated_commands`
- Interacts with `VK_EXT_fragment_density_map`
- Interacts with `VK_EXT_mesh_shader`
- Interacts with `VK_EXT_transform_feedback`
- Interacts with `VK_KHR_acceleration_structure`
- Interacts with `VK_KHR_fragment_shading_rate`
- Interacts with `VK_KHR_ray_tracing_pipeline`
- Interacts with `VK_NV_device_generated_commands`
- Interacts with `VK_NV_mesh_shader`
- Interacts with `VK_NV_ray_tracing`
- Interacts with `VK_NV_shading_rate_image`

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Tobias Hector [\[GitHub\]tobski](#)

Other Extension Metadata

Last Modified Date

2020-12-03

Interactions and External Dependencies

- Interacts with [VK_KHR_create_renderpass2](#)

Contributors

- Tobias Hector

Description

This extension modifies the original core synchronization APIs to simplify the interface and improve usability of these APIs. It also adds new pipeline stage and access flag types that extend into the 64-bit range, as we have run out within the 32-bit range. The new flags are identical to the old values within the 32-bit range, with new stages and bits beyond that.

Pipeline stages and access flags are now specified together in memory barrier structures, making the connection between the two more obvious. Additionally, scoping the pipeline stages into the barrier structs allows the use of the `MEMORY_READ` and `MEMORY_WRITE` flags without sacrificing precision. The per-stage access flags should be used to disambiguate specific accesses in a given stage or set of stages - for instance, between uniform reads and sampling operations.

Layout transitions have been simplified as well; rather than requiring a different set of layouts for depth/stencil/color attachments, there are generic `VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR` and `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR` layouts which are contextually applied based on the image

format. For example, for a depth format image, `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR` is equivalent to `VK_IMAGE_LAYOUT_DEPTH_READ_ONLY_OPTIMAL_KHR`. `VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR` also functionally replaces `VK_IMAGE_LAYOUT_SHADER_READ_ONLY_OPTIMAL`.

Events are now more efficient, because they include memory dependency information when you set them on the device. Previously, this information was only known when waiting on an event, so the dependencies could not be satisfied until the wait occurred. That sometimes meant stalling the pipeline when the wait occurred. The new API provides enough information for implementations to satisfy these dependencies in parallel with other tasks.

Queue submission has been changed to wrap command buffers and semaphores in extensible structures, which incorporate changes from Vulkan 1.1, `VK_KHR_device_group`, and `VK_KHR_timeline_semaphore`. This also adds a pipeline stage to the semaphore signal operation, mirroring the existing pipeline stage specification for wait operations.

Other miscellaneous changes include:

- Events can now be specified as interacting only with the device, allowing more efficient access to the underlying object.
- Image memory barriers that do not perform an image layout transition can be specified by setting `oldLayout` equal to `newLayout`.
 - E.g. the old and new layout can both be set to `VK_IMAGE_LAYOUT_UNDEFINED`, without discarding data in the image.
- Queue family ownership transfer parameters are simplified in some cases.
- Extensions with commands or functions with a `VkPipelineStageFlags` or `VkPipelineStageFlagBits` parameter have had those APIs replaced with equivalents using `VkPipelineStageFlags2KHR`.
- The new event and barrier interfaces are now more extensible for future changes.
- Relevant pipeline stage masks can now be specified as empty with the new `VK_PIPELINE_STAGE_NONE_KHR` and `VK_PIPELINE_STAGE_2_NONE_KHR` values.
- `VkMemoryBarrier2KHR` can be chained to `VkSubpassDependency2`, overriding the original 32-bit stage and access masks.

New Base Types

- `VkFlags64`

New Commands

- `vkCmdPipelineBarrier2KHR`
- `vkCmdResetEvent2KHR`
- `vkCmdSetEvent2KHR`
- `vkCmdWaitEvents2KHR`
- `vkCmdWriteTimestamp2KHR`
- `vkQueueSubmit2KHR`

New Structures

- [VkBufferMemoryBarrier2KHR](#)
- [VkCommandBufferSubmitInfoKHR](#)
- [VkDependencyInfoKHR](#)
- [VkImageMemoryBarrier2KHR](#)
- [VkSemaphoreSubmitInfoKHR](#)
- [VkSubmitInfo2KHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceSynchronization2FeaturesKHR](#)
- Extending [VkSubpassDependency2](#):
 - [VkMemoryBarrier2KHR](#)

New Enums

- [VkAccessFlagBits2KHR](#)
- [VkPipelineStageFlagBits2KHR](#)
- [VkSubmitFlagBitsKHR](#)

New Bitmasks

- [VkAccessFlags2KHR](#)
- [VkPipelineStageFlags2KHR](#)
- [VkSubmitFlagsKHR](#)

New Enum Constants

- [VK_KHR_SYNCHRONIZATION_2_EXTENSION_NAME](#)
- [VK_KHR_SYNCHRONIZATION_2_SPEC_VERSION](#)
- Extending [VkAccessFlagBits](#):
 - [VK_ACCESS_NONE_KHR](#)
- Extending [VkEventCreateFlagBits](#):
 - [VK_EVENT_CREATE_DEVICE_ONLY_BIT_KHR](#)
- Extending [VkImageLayout](#):
 - [VK_IMAGE_LAYOUT_ATTACHMENT_OPTIMAL_KHR](#)
 - [VK_IMAGE_LAYOUT_READ_ONLY_OPTIMAL_KHR](#)
- Extending [VkPipelineStageFlagBits](#):
 - [VK_PIPELINE_STAGE_NONE_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_BUFFER_MEMORY_BARRIER_2_KHR](#)

- VK_STRUCTURE_TYPE_COMMAND_BUFFER_SUBMIT_INFO_KHR
- VK_STRUCTURE_TYPE_DEPENDENCY_INFO_KHR
- VK_STRUCTURE_TYPE_IMAGE_MEMORY_BARRIER_2_KHR
- VK_STRUCTURE_TYPE_MEMORY_BARRIER_2_KHR
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SYNCHRONIZATION_2_FEATURES_KHR
- VK_STRUCTURE_TYPE_SEMAPHORE_SUBMIT_INFO_KHR
- VK_STRUCTURE_TYPE_SUBMIT_INFO_2_KHR

If [VK_EXT_blend_operation_advanced](#) is supported:

- Extending [VkAccessFlagBits2](#):
 - VK_ACCESS_2_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT

If [VK_EXT_conditional_rendering](#) is supported:

- Extending [VkAccessFlagBits2](#):
 - VK_ACCESS_2_CONDITIONAL_RENDERING_READ_BIT_EXT
- Extending [VkPipelineStageFlagBits2](#):
 - VK_PIPELINE_STAGE_2_CONDITIONAL_RENDERING_BIT_EXT

If [VK_EXT_device_generated_commands](#) is supported:

- Extending [VkAccessFlagBits2](#):
 - VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_EXT
 - VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_EXT
- Extending [VkPipelineStageFlagBits2](#):
 - VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_EXT

If [VK_EXT_fragment_density_map](#) is supported:

- Extending [VkAccessFlagBits2](#):
 - VK_ACCESS_2_FRAGMENT_DENSITY_MAP_READ_BIT_EXT
- Extending [VkPipelineStageFlagBits2](#):
 - VK_PIPELINE_STAGE_2_FRAGMENT_DENSITY_PROCESS_BIT_EXT

If [VK_EXT_mesh_shader](#) is supported:

- Extending [VkPipelineStageFlagBits2](#):
 - VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_EXT
 - VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_EXT

If [VK_EXT_transform_feedback](#) is supported:

- Extending [VkAccessFlagBits2](#):
 - `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT`
 - `VK_ACCESS_2_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT`
 - `VK_ACCESS_2_TRANSFORM_FEEDBACK_WRITE_BIT_EXT`
- Extending [VkPipelineStageFlagBits2](#):
 - `VK_PIPELINE_STAGE_2_TRANSFORM_FEEDBACK_BIT_EXT`

If [VK_KHR_acceleration_structure](#) is supported:

- Extending [VkAccessFlagBits2](#):
 - `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_KHR`
 - `VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_KHR`
- Extending [VkPipelineStageFlagBits2](#):
 - `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_KHR`

If [VK_KHR_fragment_shading_rate](#) is supported:

- Extending [VkAccessFlagBits2](#):
 - `VK_ACCESS_2_FRAGMENT_SHADING_RATE_ATTACHMENT_READ_BIT_KHR`
- Extending [VkPipelineStageFlagBits2](#):
 - `VK_PIPELINE_STAGE_2_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_KHR`

If [VK_KHR_ray_tracing_pipeline](#) is supported:

- Extending [VkPipelineStageFlagBits2](#):
 - `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_KHR`

If [VK_NV_device_generated_commands](#) is supported:

- Extending [VkAccessFlagBits2](#):
 - `VK_ACCESS_2_COMMAND_PREPROCESS_READ_BIT_NV`
 - `VK_ACCESS_2_COMMAND_PREPROCESS_WRITE_BIT_NV`
- Extending [VkPipelineStageFlagBits2](#):
 - `VK_PIPELINE_STAGE_2_COMMAND_PREPROCESS_BIT_NV`

If [VK_NV_mesh_shader](#) is supported:

- Extending [VkPipelineStageFlagBits2](#):
 - `VK_PIPELINE_STAGE_2_MESH_SHADER_BIT_NV`
 - `VK_PIPELINE_STAGE_2_TASK_SHADER_BIT_NV`

If [VK_NV_ray_tracing](#) is supported:

- Extending `VkAccessFlagBits2`:
 - `VK_ACCESS_2_ACCELERATION_STRUCTURE_READ_BIT_NV`
 - `VK_ACCESS_2_ACCELERATION_STRUCTURE_WRITE_BIT_NV`
- Extending `VkPipelineStageFlagBits2`:
 - `VK_PIPELINE_STAGE_2_ACCELERATION_STRUCTURE_BUILD_BIT_NV`
 - `VK_PIPELINE_STAGE_2_RAY_TRACING_SHADER_BIT_NV`

If `VK_NV_shading_rate_image` is supported:

- Extending `VkAccessFlagBits2`:
 - `VK_ACCESS_2_SHADING_RATE_IMAGE_READ_BIT_NV`
- Extending `VkPipelineStageFlagBits2`:
 - `VK_PIPELINE_STAGE_2_SHADING_RATE_IMAGE_BIT_NV`

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Examples

See <https://github.com/KhronosGroup/Vulkan-Docs/wiki/Synchronization-Examples>

Version History

- Revision 1, 2020-12-03 (Tobias Hector)
 - Internal revisions

VK_KHR_timeline_semaphore

Name String

`VK_KHR_timeline_semaphore`

Extension Type

Device extension

Registered Extension Number

208

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Faith Ekstrand [\[GitHub\]gfxstrand](#)

Other Extension Metadata

Last Modified Date

2019-06-12

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension interacts with [VK_KHR_external_semaphore_capabilities](#)
- This extension interacts with [VK_KHR_external_semaphore](#)
- This extension interacts with [VK_KHR_external_semaphore_win32](#)

Contributors

- Jeff Bolz, NVIDIA
- Yuriy O'Donnell, Epic Games
- Faith Ekstrand, Intel
- Jesse Hall, Google
- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Daniel Rakos, AMD
- Ray Smith, Arm

Description

This extension introduces a new type of semaphore that has an integer payload identifying a point in a timeline. Such timeline semaphores support the following operations:

- Host query - A host operation that allows querying the payload of the timeline semaphore.
- Host wait - A host operation that allows a blocking wait for a timeline semaphore to reach a specified value.
- Host signal - A host operation that allows advancing the timeline semaphore to a specified value.
- Device wait - A device operation that allows waiting for a timeline semaphore to reach a

specified value.

- Device signal - A device operation that allows advancing the timeline semaphore to a specified value.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Commands

- [vkGetSemaphoreCounterValueKHR](#)
- [vkSignalSemaphoreKHR](#)
- [vkWaitSemaphoresKHR](#)

New Structures

- [VkSemaphoreSignalInfoKHR](#)
- [VkSemaphoreWaitInfoKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceTimelineSemaphoreFeaturesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceTimelineSemaphorePropertiesKHR](#)
- Extending [VkSemaphoreCreateInfo](#), [VkPhysicalDeviceExternalSemaphoreInfo](#):
 - [VkSemaphoreTypeCreateInfoKHR](#)
- Extending [VkSubmitInfo](#), [VkBindSparseInfo](#):
 - [VkTimelineSemaphoreSubmitInfoKHR](#)

New Enums

- [VkSemaphoreTypeKHR](#)
- [VkSemaphoreWaitFlagBitsKHR](#)

New Bitmasks

- [VkSemaphoreWaitFlagsKHR](#)

New Enum Constants

- [VK_KHR_TIMELINE_SEMAPHORE_EXTENSION_NAME](#)
- [VK_KHR_TIMELINE_SEMAPHORE_SPEC_VERSION](#)
- Extending [VkSemaphoreType](#):
 - [VK_SEMAPHORE_TYPE_BINARY_KHR](#)

- `VK_SEMAPHORE_TYPE_TIMELINE_KHR`
- Extending `VkSemaphoreWaitFlagBits`:
 - `VK_SEMAPHORE_WAIT_ANY_BIT_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_FEATURES_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TIMELINE_SEMAPHORE_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_SEMAPHORE_SIGNAL_INFO_KHR`
 - `VK_STRUCTURE_TYPE_SEMAPHORE_TYPE_CREATE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_SEMAPHORE_WAIT_INFO_KHR`
 - `VK_STRUCTURE_TYPE_TIMELINE_SEMAPHORE_SUBMIT_INFO_KHR`

Issues

1) Do we need a new object type for this?

RESOLVED: No, we just introduce a new type of semaphore object, as `VK_KHR_external_semaphore_win32` already uses semaphores as the destination for importing D3D12 fence objects, which are semantically close/identical to the proposed synchronization primitive.

2) What type of payload the new synchronization primitive has?

RESOLVED: A 64-bit unsigned integer that can only be set to strictly increasing values by signal operations and is not changed by wait operations.

3) Does the new synchronization primitive have the same signal-before-wait requirement as the existing semaphores do?

RESOLVED: No. Timeline semaphores support signaling and waiting entirely asynchronously. It is the responsibility of the application to avoid deadlock.

4) Does the new synchronization primitive allow resetting its payload?

RESOLVED: No, allowing the payload value to “go backwards” is problematic. Applications looking for reset behavior should create a new instance of the synchronization primitive instead.

5) How do we enable host waits on the synchronization primitive?

RESOLVED: Both a non-blocking query of the current payload value of the synchronization primitive, and a blocking wait operation are provided.

6) How do we enable device waits and signals on the synchronization primitive?

RESOLVED: Similar to `VK_KHR_external_semaphore_win32`, this extension introduces a new structure that can be chained to `VkSubmitInfo` to specify the values signaled semaphores should be set to, and the values waited semaphores need to reach.

7) Can the new synchronization primitive be used to synchronize presentation and swapchain

image acquisition operations?

RESOLVED: Some implementations may have problems with supporting that directly, thus it is not allowed in this extension.

8) Do we want to support external sharing of the new synchronization primitive type?

RESOLVED: Yes. Timeline semaphore specific external sharing capabilities can be queried using [vkGetPhysicalDeviceExternalSemaphorePropertiesKHR](#) by chaining the new [VkSemaphoreTypeCreateInfoKHR](#) structure to its [pExternalSemaphoreInfo](#) structure. This allows having a different set of external semaphore handle types supported for timeline semaphores vs. binary semaphores.

9) Do we need to add a host signal operation for the new synchronization primitive type?

RESOLVED: Yes. This helps in situations where one host thread submits a workload but another host thread has the information on when the workload is ready to be executed.

10) How should the new synchronization primitive interact with the ordering requirements of the original [VkSemaphore](#)?

RESOLVED: Prior to calling any command which **may** cause a wait operation on a binary semaphore, the application **must** ensure that the semaphore signal operation that has been submitted for execution and any semaphore signal operations on which it depends (if any) **must** have also been submitted for execution.

11) Should we have separate feature bits for different sub-features of timeline semaphores?

RESOLVED: No. The only feature which cannot be supported universally is timeline semaphore import/export. For import/export, the application is already required to query available external handle types via [vkGetPhysicalDeviceExternalSemaphorePropertiesKHR](#) and provide the semaphore type by adding a [VkSemaphoreTypeCreateInfoKHR](#) structure to the [pNext](#) chain of [VkPhysicalDeviceExternalSemaphoreInfo](#) so no new feature bit is required.

Version History

- Revision 1, 2018-05-10 (Faith Ekstrand)
 - Initial version
- Revision 2, 2019-06-12 (Faith Ekstrand)
 - Added an `initialValue` parameter to timeline semaphore creation

VK_KHR_uniform_buffer_standard_layout

Name String

[VK_KHR_uniform_buffer_standard_layout](#)

Extension Type

Device extension

Registered Extension Number

254

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Graeme Leese [\[GitHub\]](#)gnl21

Other Extension Metadata

Last Modified Date

2019-01-25

Contributors

- Graeme Leese, Broadcom
- Jeff Bolz, NVIDIA
- Tobias Hector, AMD
- Faith Ekstrand, Intel
- Neil Henning, AMD

Description

This extension enables tighter array and struct packing to be used with uniform buffers.

It modifies the alignment rules for uniform buffers, allowing for tighter packing of arrays and structures. This allows, for example, the std430 layout, as defined in [GLSL](#) to be supported in uniform buffers.

Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):

- [VkPhysicalDeviceUniformBufferStandardLayoutFeaturesKHR](#)

New Enum Constants

- [VK_KHR_UNIFORM_BUFFER_STANDARD_LAYOUT_EXTENSION_NAME](#)
- [VK_KHR_UNIFORM_BUFFER_STANDARD_LAYOUT_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_UNIFORM_BUFFER_STANDARD_LAYOUT_FEATURES_KHR](#)

Version History

- Revision 1, 2019-01-25 (Graeme Leese)
 - Initial draft

VK_KHR_variable_pointers

Name String

[VK_KHR_variable_pointers](#)

Extension Type

Device extension

Registered Extension Number

121

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)
and
[VK_KHR_storage_buffer_storage_class](#)
or
[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_variable_pointers](#)

Deprecation State

- *Promoted* to Vulkan 1.1

Contact

- Jesse Hall [\[GitHub\]](#)[critsec](#)

Other Extension Metadata

Last Modified Date

2017-09-05

IP Status

No known IP claims.

Contributors

- John Kessenich, Google
- Neil Henning, Codeplay
- David Neto, Google
- Daniel Koch, Nvidia
- Graeme Leese, Broadcom
- Weifeng Zhang, Qualcomm
- Stephen Clarke, Imagination Technologies
- Faith Ekstrand, Intel
- Jesse Hall, Google

Description

The `VK_KHR_variable_pointers` extension allows implementations to indicate their level of support for the `SPV_KHR_variable_pointers` SPIR-V extension. The SPIR-V extension allows shader modules to use invocation-private pointers into uniform and/or storage buffers, where the pointer values can be dynamic and non-uniform.

The `SPV_KHR_variable_pointers` extension introduces two capabilities. The first, `VariablePointersStorageBuffer`, **must** be supported by all implementations of this extension. The second, `VariablePointers`, is optional.

Promotion to Vulkan 1.1

All functionality in this extension is included in core Vulkan 1.1, with the KHR suffix omitted, however support for the `variablePointersStorageBuffer` feature is made optional. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceVariablePointerFeaturesKHR`
 - `VkPhysicalDeviceVariablePointersFeaturesKHR`

New Enum Constants

- `VK_KHR_VARIABLE_POINTERS_EXTENSION_NAME`

- `VK_KHR_VARIABLE_POINTERS_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTERS_FEATURES_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VARIABLE_POINTER_FEATURES_KHR`

New SPIR-V Capabilities

- `VariablePointers`
- `VariablePointersStorageBuffer`

Issues

1) Do we need an optional property for the SPIR-V `VariablePointersStorageBuffer` capability or should it be mandatory when this extension is advertised?

RESOLVED: Add it as a distinct feature, but make support mandatory. Adding it as a feature makes the extension easier to include in a future core API version. In the extension, the feature is mandatory, so that presence of the extension guarantees some functionality. When included in a core API version, the feature would be optional.

2) Can support for these capabilities vary between shader stages?

RESOLVED: No, if the capability is supported in any stage it must be supported in all stages.

3) Should the capabilities be features or limits?

RESOLVED: Features, primarily for consistency with other similar extensions.

Version History

- Revision 1, 2017-03-14 (Jesse Hall and John Kessenich)
 - Internal revisions

VK_KHR_vertex_attribute_divisor

Name String

`VK_KHR_vertex_attribute_divisor`

Extension Type

Device extension

Registered Extension Number

526

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Shahbaz Youssefi [\[GitHub\]syoussefi](#)

Extension Proposal

[VK_KHR_vertex_attribute_divisor](#)

Other Extension Metadata

Last Modified Date

2023-09-20

IP Status

No known IP claims.

Contributors

- Shahbaz Youssefi, Google
- Contributors to [VK_EXT_vertex_attribute_divisor](#)

Description

This extension is based on the [VK_EXT_vertex_attribute_divisor](#) extension. The only difference is the new property [supportsNonZeroFirstInstance](#), which indicates support for non-zero values in [firstInstance](#). This allows the extension to be supported on implementations that have traditionally only supported OpenGL ES.

New Structures

- [VkVertexInputBindingDivisorDescriptionKHR](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceVertexAttributeDivisorFeaturesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceVertexAttributeDivisorPropertiesKHR](#)
- Extending [VkPipelineVertexInputStateCreateInfo](#):
 - [VkPipelineVertexInputDivisorStateCreateInfoKHR](#)

New Enum Constants

- [VK_KHR_VERTEX_ATTRIBUTE_DIVISOR_EXTENSION_NAME](#)

- [VK_KHR_VERTEX_ATTRIBUTE_DIVISOR_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_ATTRIBUTE_DIVISOR_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_VERTEX_INPUT_DIVISOR_STATE_CREATE_INFO_KHR](#)

Version History

- Revision 1, 2023-09-20 (Shahbaz Youssefi)
 - First Version, based on [VK_EXT_vertex_attribute_divisor](#)

VK_KHR_video_decode_av1

Name String

[VK_KHR_video_decode_av1](#)

Extension Type

Device extension

Registered Extension Number

513

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_video_decode_queue](#)

Contact

- Daniel Rakos [\[GitHub\]](#)[aqnuep](#)

Extension Proposal

[VK_KHR_video_decode_av1](#)

Other Extension Metadata

Last Modified Date

2024-01-02

IP Status

No known IP claims.

Contributors

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- Benjamin Cheng, AMD
- Ho Hin Lau, AMD
- Lynne Iribarren, Independent
- David Airlie, Red Hat, Inc.
- Ping Liu, Intel
- Srinath Kumarapuram, NVIDIA
- Vassili Nikolaev, NVIDIA
- Tony Zlatinski, NVIDIA
- Charlie Turner, Igalia
- Daniel Almeida, Collabora
- Nicolas Dufresne, Collabora
- Daniel Rakos, RasterGrid

Description

This extension builds upon the [VK_KHR_video_decode_queue](#) extension by adding support for decoding elementary video stream sequences compliant with the AV1 video compression standard.

New Structures

- Extending [VkVideoCapabilitiesKHR](#):
 - [VkVideoDecodeAV1CapabilitiesKHR](#)
- Extending [VkVideoDecodeInfoKHR](#):
 - [VkVideoDecodeAV1PictureInfoKHR](#)
- Extending [VkVideoProfileInfoKHR](#), [VkQueryPoolCreateInfo](#):
 - [VkVideoDecodeAV1ProfileInfoKHR](#)
- Extending [VkVideoReferenceSlotInfoKHR](#):
 - [VkVideoDecodeAV1DpbSlotInfoKHR](#)
- Extending [VkVideoSessionParametersCreateInfoKHR](#):
 - [VkVideoDecodeAV1SessionParametersCreateInfoKHR](#)

New Enum Constants

- [VK_KHR_VIDEO_DECODE_AV1_EXTENSION_NAME](#)
- [VK_KHR_VIDEO_DECODE_AV1_SPEC_VERSION](#)
- [VK_MAX_VIDEO_AV1_REFERENCES_PER_FRAME_KHR](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_CAPABILITIES_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_DPB_SLOT_INFO_KHR](#)

- [VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PICTURE_INFO_KHR](#)
- [VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_PROFILE_INFO_KHR](#)
- [VK_STRUCTURE_TYPE_VIDEO_DECODE_AV1_SESSION_PARAMETERS_CREATE_INFO_KHR](#)
- Extending [VkVideoCodecOperationFlagBitsKHR](#):
 - [VK_VIDEO_CODEC_OPERATION_DECODE_AV1_BIT_KHR](#)

Version History

- Revision 1, 2024-01-02 (Daniel Rakos)
 - Internal revisions

VK_KHR_video_decode_h264

Name String

[VK_KHR_video_decode_h264](#)

Extension Type

Device extension

Registered Extension Number

41

Revision

9

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_video_decode_queue](#)

Contact

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Extension Proposal

[VK_KHR_video_decode_h264](#)

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD

- Chunbo Chen, Intel
- HoHin Lau, AMD
- Jake Beju, AMD
- Peter Fang, AMD
- Ping Liu, Intel
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Daniel Rakos, RasterGrid

Description

This extension builds upon the [VK_KHR_video_decode_queue](#) extension by adding support for decoding elementary video stream sequences compliant with the H.264/AVC video compression standard.

NOTE	This extension was promoted to KHR from the provisional extension VK_EXT_video_decode_h264 .
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New Structures

- Extending [VkVideoCapabilitiesKHR](#):
 - [VkVideoDecodeH264CapabilitiesKHR](#)
- Extending [VkVideoDecodeInfoKHR](#):
 - [VkVideoDecodeH264PictureInfoKHR](#)
- Extending [VkVideoProfileInfoKHR](#), [VkQueryPoolCreateInfo](#):
 - [VkVideoDecodeH264ProfileInfoKHR](#)
- Extending [VkVideoReferenceSlotInfoKHR](#):
 - [VkVideoDecodeH264DpbSlotInfoKHR](#)
- Extending [VkVideoSessionParametersCreateInfoKHR](#):
 - [VkVideoDecodeH264SessionParametersCreateInfoKHR](#)
- Extending [VkVideoSessionParametersUpdateInfoKHR](#):
 - [VkVideoDecodeH264SessionParametersAddInfoKHR](#)

New Enums

- [VkVideoDecodeH264PictureLayoutFlagBitsKHR](#)

New Bitmasks

- [VkVideoDecodeH264PictureLayoutFlagsKHR](#)

New Enum Constants

- `VK_KHR_VIDEO_DECODE_H264_EXTENSION_NAME`
- `VK_KHR_VIDEO_DECODE_H264_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_CAPABILITIES_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_DPB_SLOT_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PICTURE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_PROFILE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_DECODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR`
- Extending `VkVideoCodecOperationFlagBitsKHR`:
 - `VK_VIDEO_CODEC_OPERATION_DECODE_H264_BIT_KHR`

Version History

- Revision 1, 2018-6-11 (Peter Fang)
 - Initial draft
- Revision 2, March 29 2021 (Tony Zlatinski)
 - Spec and API Updates
- Revision 3, August 1 2021 (Srinath Kumarapuram)
 - Rename `VkVideoDecodeH264FieldLayoutFlagsEXT` to `VkVideoDecodeH264PictureLayoutFlagsEXT`, `VkVideoDecodeH264FieldLayoutFlagBitsEXT` to `VkVideoDecodeH264PictureLayoutFlagBitsEXT` (along with the names of enumerants it defines), and `VkVideoDecodeH264ProfileEXT.fieldLayout` to `VkVideoDecodeH264ProfileEXT.pictureLayout`, following Vulkan naming conventions.
- Revision 4, 2022-03-16 (Ahmed Abdelkhalek)
 - Relocate Std header version reporting/requesting from this extension to `VK_KHR_video_queue` extension.
 - Remove the now empty `VkVideoDecodeH264SessionCreateInfoEXT`.
- Revision 5, 2022-03-31 (Ahmed Abdelkhalek)
 - Use type `StdVideoH264Level` for `VkVideoDecodeH264Capabilities.maxLevel`
- Revision 6, 2022-08-09 (Daniel Rakos)
 - Rename `VkVideoDecodeH264ProfileEXT` to `VkVideoDecodeH264ProfileInfoEXT`
 - Rename `VkVideoDecodeH264MvcEXT` to `VkVideoDecodeH264MvcInfoEXT`
- Revision 7, 2022-09-18 (Daniel Rakos)
 - Change type of `VkVideoDecodeH264ProfileInfoEXT::pictureLayout` to `VkVideoDecodeH264PictureLayoutFlagBitsEXT`

- Remove MVC support and related `VkVideoDecodeH264MvcInfoEXT` structure
- Rename `spsStdCount`, `pSpsStd`, `ppsStdCount`, and `pPpsStd` to `stdSPSCount`, `pStdSPSs`, `stdPPSCount`, and `pStdPPSs`, respectively, in `VkVideoDecodeH264SessionParametersAddInfoEXT`
- Rename `maxSpsStdCount` and `maxPpsStdCount` to `maxStdSPSCount` and `maxStdPPSCount`, respectively, in `VkVideoDecodeH264SessionParametersCreateInfoEXT`
- Rename `slicesCount` and `pSlicesDataOffsets` to `sliceCount` and `pSliceOffsets`, respectively, in `VkVideoDecodeH264PictureInfoEXT`
- Revision 8, 2022-09-29 (Daniel Rakos)
 - Change extension from `EXT` to `KHR`
 - Extension is no longer provisional
- Revision 9, 2023-12-05 (Daniel Rakos)
 - Condition reference picture setup based on the value of `StdVideoDecodeH264PictureInfo::flags.is_reference`

VK_KHR_video_decode_h265

Name String

`VK_KHR_video_decode_h265`

Extension Type

Device extension

Registered Extension Number

188

Revision

8

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_video_decode_queue](#)

Contact

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Extension Proposal

[VK_KHR_video_decode_h265](#)

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- HoHin Lau, AMD
- Jake Beju, AMD
- Peter Fang, AMD
- Ping Liu, Intel
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Daniel Rakos, RasterGrid

Description

This extension builds upon the [VK_KHR_video_decode_queue](#) extension by adding support for decoding elementary video stream sequences compliant with the H.265/HEVC video compression standard.

NOTE

This extension was promoted to **KHR** from the provisional extension **VK_EXT_video_decode_h265**.

New Structures

- Extending [VkVideoCapabilitiesKHR](#):
 - [VkVideoDecodeH265CapabilitiesKHR](#)
- Extending [VkVideoDecodeInfoKHR](#):
 - [VkVideoDecodeH265PictureInfoKHR](#)
- Extending [VkVideoProfileInfoKHR](#), [VkQueryPoolCreateInfo](#):
 - [VkVideoDecodeH265ProfileInfoKHR](#)
- Extending [VkVideoReferenceSlotInfoKHR](#):
 - [VkVideoDecodeH265DpbSlotInfoKHR](#)
- Extending [VkVideoSessionParametersCreateInfoKHR](#):
 - [VkVideoDecodeH265SessionParametersCreateInfoKHR](#)
- Extending [VkVideoSessionParametersUpdateInfoKHR](#):
 - [VkVideoDecodeH265SessionParametersAddInfoKHR](#)

New Enum Constants

- **VK_KHR_VIDEO_DECODE_H265_EXTENSION_NAME**
- **VK_KHR_VIDEO_DECODE_H265_SPEC_VERSION**
- Extending [VkStructureType](#):

- `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_CAPABILITIES_KHR`
- `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_DPB_SLOT_INFO_KHR`
- `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PICTURE_INFO_KHR`
- `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_PROFILE_INFO_KHR`
- `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR`
- `VK_STRUCTURE_TYPE_VIDEO_DECODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR`
- Extending `VkVideoCodecOperationFlagBitsKHR`:
 - `VK_VIDEO_CODEC_OPERATION_DECODE_H265_BIT_KHR`

Version History

- Revision 1, 2018-6-11 (Peter Fang)
 - Initial draft
- Revision 1.6, March 29 2021 (Tony Zlatinski)
 - Spec and API updates.
- Revision 2, 2022-03-16 (Ahmed Abdelkhalek)
 - Relocate Std header version reporting/requesting from this extension to `VK_KHR_video_queue` extension.
 - Remove the now empty `VkVideoDecodeH265SessionCreateInfoEXT`.
- Revision 3, 2022-03-31 (Ahmed Abdelkhalek)
 - Use type `StdVideoH265Level` for `VkVideoDecodeH265Capabilities.maxLevel`
- Revision 4, 2022-08-09 (Daniel Rakos)
 - Rename `VkVideoDecodeH265ProfileEXT` to `VkVideoDecodeH265ProfileInfoEXT`
- Revision 5, 2022-09-18 (Daniel Rakos)
 - Rename `vpsStdCount`, `pVpsStd`, `spsStdCount`, `pSpsStd`, `ppsStdCount`, and `pPpsStd` to `stdVPSCount`, `pStdVPSs`, `stdSPSCount`, `pStdSPSs`, `stdPPSCount`, and `pStdPPSs`, respectively, in `VkVideoDecodeH265SessionParametersAddInfoEXT`
 - Rename `maxVpsStdCount`, `maxSpsStdCount`, and `maxPpsStdCount` to `maxStdVPSCount`, `maxStdSPSCount` and `maxStdPPSCount`, respectively, in `VkVideoDecodeH265SessionParametersCreateInfoEXT`
 - Rename `slicesCount` and `pSlicesDataOffsets` to `sliceCount` and `pSliceOffsets`, respectively, in `VkVideoDecodeH265PictureInfoEXT`
- Revision 6, 2022-11-14 (Daniel Rakos)
 - Rename `slice` to `sliceSegment` in the APIs for better clarity
- Revision 7, 2022-11-14 (Daniel Rakos)
 - Change extension from `EXT` to `KHR`
 - Extension is no longer provisional
- Revision 8, 2023-12-05 (Daniel Rakos)

- Condition reference picture setup based on the value of `StdVideoDecodeH265PictureInfo::flags.IsReference`

VK_KHR_video_decode_queue

Name String

`VK_KHR_video_decode_queue`

Extension Type

Device extension

Registered Extension Number

25

Revision

8

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_video_queue](#)

and

[VK_KHR_synchronization2](#)

or

[Vulkan Version 1.3](#)

API Interactions

- Interacts with [VK_VERSION_1_3](#)
- Interacts with [VK_KHR_format_feature_flags2](#)

Contact

- jake.beju@amd.com

Extension Proposal

[VK_KHR_video_decode_queue](#)

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Jake Beju, AMD

- Olivier Lapicque, NVIDIA
- Peter Fang, AMD
- Piers Daniell, NVIDIA
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Daniel Rakos, RasterGrid

Description

This extension builds upon the [VK_KHR_video_queue](#) extension by adding common APIs specific to video decoding and thus enabling implementations to expose queue families supporting video decode operations.

More specifically, it adds video decode specific capabilities and a new command buffer command that allows recording video decode operations against a video session.

This extension is to be used in conjunction with other codec specific video decode extensions that enable decoding video sequences of specific video compression standards.

New Commands

- [vkCmdDecodeVideoKHR](#)

New Structures

- [VkVideoDecodeInfoKHR](#)
- Extending [VkVideoCapabilitiesKHR](#):
 - [VkVideoDecodeCapabilitiesKHR](#)
- Extending [VkVideoProfileInfoKHR](#), [VkQueryPoolCreateInfo](#):
 - [VkVideoDecodeUsageInfoKHR](#)

New Enums

- [VkVideoDecodeCapabilityFlagBitsKHR](#)
- [VkVideoDecodeUsageFlagBitsKHR](#)

New Bitmasks

- [VkVideoDecodeCapabilityFlagsKHR](#)
- [VkVideoDecodeFlagsKHR](#)
- [VkVideoDecodeUsageFlagsKHR](#)

New Enum Constants

- [VK_KHR_VIDEO_DECODE_QUEUE_EXTENSION_NAME](#)

- `VK_KHR_VIDEO_DECODE_QUEUE_SPEC_VERSION`
- Extending `VkAccessFlagBits2`:
 - `VK_ACCESS_2_VIDEO_DECODE_READ_BIT_KHR`
 - `VK_ACCESS_2_VIDEO_DECODE_WRITE_BIT_KHR`
- Extending `VkBufferUsageFlagBits`:
 - `VK_BUFFER_USAGE_VIDEO_DECODE_DST_BIT_KHR`
 - `VK_BUFFER_USAGE_VIDEO_DECODE_SRC_BIT_KHR`
- Extending `VkFormatFeatureFlagBits`:
 - `VK_FORMAT_FEATURE_VIDEO_DECODE_DPB_BIT_KHR`
 - `VK_FORMAT_FEATURE_VIDEO_DECODE_OUTPUT_BIT_KHR`
- Extending `VkImageLayout`:
 - `VK_IMAGE_LAYOUT_VIDEO_DECODE_DPB_KHR`
 - `VK_IMAGE_LAYOUT_VIDEO_DECODE_DST_KHR`
 - `VK_IMAGE_LAYOUT_VIDEO_DECODE_SRC_KHR`
- Extending `VkImageUsageFlagBits`:
 - `VK_IMAGE_USAGE_VIDEO_DECODE_DPB_BIT_KHR`
 - `VK_IMAGE_USAGE_VIDEO_DECODE_DST_BIT_KHR`
 - `VK_IMAGE_USAGE_VIDEO_DECODE_SRC_BIT_KHR`
- Extending `VkPipelineStageFlagBits2`:
 - `VK_PIPELINE_STAGE_2_VIDEO_DECODE_BIT_KHR`
- Extending `VkQueueFlagBits`:
 - `VK_QUEUE_VIDEO_DECODE_BIT_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_VIDEO_DECODE_CAPABILITIES_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_DECODE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_DECODE_USAGE_INFO_KHR`

If `VK_KHR_format_feature_flags2` or Vulkan Version 1.3 is supported:

- Extending `VkFormatFeatureFlagBits2`:
 - `VK_FORMAT_FEATURE_2_VIDEO_DECODE_DPB_BIT_KHR`
 - `VK_FORMAT_FEATURE_2_VIDEO_DECODE_OUTPUT_BIT_KHR`

Version History

- Revision 1, 2018-6-11 (Peter Fang)
 - Initial draft

- Revision 1.5, Nov 09 2018 (Tony Zlatinski)
 - API Updates
- Revision 1.6, Jan 08 2020 (Tony Zlatinski)
 - API unify with the video_encode_queue spec
- Revision 1.7, March 29 2021 (Tony Zlatinski)
 - Spec and API updates.
- Revision 2, September 30 2021 (Jon Leech)
 - Add interaction with `VK_KHR_format_feature_flags2` to `vk.xml`
- Revision 3, 2022-02-25 (Ahmed Abdelkhalek)
 - Add `VkVideoDecodeCapabilitiesKHR` with new flags to report support for decode DPB and output coinciding in the same image, or in distinct images.
- Revision 4, 2022-03-31 (Ahmed Abdelkhalek)
 - Remove redundant `VkVideoDecodeInfoKHR.coded{Offset|Extent}`
- Revision 5, 2022-07-18 (Daniel Rakos)
 - Remove `VkVideoDecodeFlagBitsKHR` as it contains no defined flags for now
- Revision 6, 2022-08-12 (Daniel Rakos)
 - Add `VkVideoDecodeUsageInfoKHR` structure and related flags
- Revision 7, 2022-09-29 (Daniel Rakos)
 - Extension is no longer provisional
- Revision 8, 2023-12-05 (Daniel Rakos)
 - Require the specification of a reconstructed picture in all cases, except when the video session was created with no DPB slots to match shipping implementations
 - Make DPB slot activation behavior codec-specific to continue allowing application control over reference picture setup now that a reconstructed picture is always mandatory

VK_KHR_video_encode_h264

Name String

`VK_KHR_video_encode_h264`

Extension Type

Device extension

Registered Extension Number

39

Revision

14

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_video_encode_queue](#)

Contact

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Extension Proposal

[VK_KHR_video_encode_h264](#)

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

Contributors

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- Tony Zlatinski, NVIDIA
- Ravi Chaudhary, NVIDIA
- Yang Liu, AMD
- Daniel Rakos, RasterGrid
- Aidan Fabius, Core Avionics & Industrial Inc.
- Lynne Iribarren, Independent

Description

This extension builds upon the [VK_KHR_video_encode_queue](#) extension by adding support for encoding elementary video stream sequences compliant with the H.264/AVC video compression standard.

NOTE

This extension was promoted to **KHR** from the provisional extension [VK_EXT_video_encode_h264](#).

New Structures

- [VkVideoEncodeH264FrameSizeKHR](#)
- [VkVideoEncodeH264NaluSliceInfoKHR](#)
- [VkVideoEncodeH264QpKHR](#)
- Extending [VkVideoBeginCodingInfoKHR](#):
 - [VkVideoEncodeH264GopRemainingFrameInfoKHR](#)
- Extending [VkVideoCapabilitiesKHR](#):
 - [VkVideoEncodeH264CapabilitiesKHR](#)
- Extending [VkVideoCodingControlInfoKHR](#), [VkVideoBeginCodingInfoKHR](#):
 - [VkVideoEncodeH264RateControlInfoKHR](#)
- Extending [VkVideoEncodeInfoKHR](#):
 - [VkVideoEncodeH264PictureInfoKHR](#)
- Extending [VkVideoEncodeQualityLevelPropertiesKHR](#):
 - [VkVideoEncodeH264QualityLevelPropertiesKHR](#)
- Extending [VkVideoEncodeRateControlLayerInfoKHR](#):
 - [VkVideoEncodeH264RateControlLayerInfoKHR](#)
- Extending [VkVideoEncodeSessionParametersFeedbackInfoKHR](#):
 - [VkVideoEncodeH264SessionParametersFeedbackInfoKHR](#)
- Extending [VkVideoEncodeSessionParametersGetInfoKHR](#):
 - [VkVideoEncodeH264SessionParametersGetInfoKHR](#)
- Extending [VkVideoProfileInfoKHR](#), [VkQueryPoolCreateInfo](#):
 - [VkVideoEncodeH264ProfileInfoKHR](#)
- Extending [VkVideoReferenceSlotInfoKHR](#):
 - [VkVideoEncodeH264DpbSlotInfoKHR](#)
- Extending [VkVideoSessionCreateInfoKHR](#):
 - [VkVideoEncodeH264SessionCreateInfoKHR](#)
- Extending [VkVideoSessionParametersCreateInfoKHR](#):
 - [VkVideoEncodeH264SessionParametersCreateInfoKHR](#)
- Extending [VkVideoSessionParametersUpdateInfoKHR](#):
 - [VkVideoEncodeH264SessionParametersAddInfoKHR](#)

New Enums

- [VkVideoEncodeH264CapabilityFlagBitsKHR](#)
- [VkVideoEncodeH264RateControlFlagBitsKHR](#)
- [VkVideoEncodeH264StdFlagBitsKHR](#)

New Bitmasks

- [VkVideoEncodeH264CapabilityFlagsKHR](#)
- [VkVideoEncodeH264RateControlFlagsKHR](#)
- [VkVideoEncodeH264StdFlagsKHR](#)

New Enum Constants

- [VK_KHR_VIDEO_ENCODE_H264_EXTENSION_NAME](#)
- [VK_KHR_VIDEO_ENCODE_H264_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_CAPABILITIES_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_DPB_SLOT_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_GOP_REMAINING_FRAME_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_NALU_SLICE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PICTURE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_PROFILE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_QUALITY_LEVEL_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_RATE_CONTROL_LAYER_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_ADD_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_FEEDBACK_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H264_SESSION_PARAMETERS_GET_INFO_KHR](#)
- Extending [VkVideoCodecOperationFlagBitsKHR](#):
 - [VK_VIDEO_CODEC_OPERATION_ENCODE_H264_BIT_KHR](#)

Version History

- Revision 0, 2018-7-23 (Ahmed Abdelkhalek)
 - Initial draft
- Revision 0.5, 2020-02-13 (Tony Zlatinski)
 - General Spec cleanup
 - Added DPB structures
 - Change the VCL frame encode structure
 - Added a common Non-VCL Picture Paramameters structure
- Revision 1, 2021-03-29 (Tony Zlatinski)

- Spec and API updates
- Revision 2, August 1 2021 (Srinath Kumarapuram)
 - Rename `VkVideoEncodeH264CapabilitiesFlagsEXT` to `VkVideoEncodeH264CapabilityFlagsEXT` and `VkVideoEncodeH264CapabilitiesFlagsEXT` to `VkVideoEncodeH264CapabilityFlagsEXT`, following Vulkan naming conventions.
- Revision 3, 2021-12-08 (Ahmed Abdelkhalek)
 - Rate control updates
- Revision 4, 2022-02-04 (Ahmed Abdelkhalek)
 - Align `VkVideoEncodeH264VclFrameInfoEXT` structure to similar one in `VK_EXT_video_encode_h265` extension
- Revision 5, 2022-02-10 (Ahmed Abdelkhalek)
 - Updates to encode capability interface
- Revision 6, 2022-03-16 (Ahmed Abdelkhalek)
 - Relocate Std header version reporting/requesting from this extension to `VK_KHR_video_queue` extension.
 - Remove redundant `maxPictureSizeInMbs` from `VkVideoEncodeH264SessionCreateInfoEXT`.
 - Remove the now empty `VkVideoEncodeH264SessionCreateInfoEXT`.
- Revision 7, 2022-04-06 (Ahmed Abdelkhalek)
 - Add capability flag to report support to use B frame in L1 reference list.
 - Add capability flag to report support for disabling SPS `direct_8x8_inference_flag`.
- Revision 8, 2022-07-18 (Daniel Rakos)
 - Replace `VkVideoEncodeH264RateControlStructureFlagBitsEXT` bit enum with `VkVideoEncodeH264RateControlStructureEXT` enum
 - Rename `VkVideoEncodeH264ProfileEXT` to `VkVideoEncodeH264ProfileInfoEXT`
 - Rename `VkVideoEncodeH264ReferenceListsEXT` to `VkVideoEncodeH264ReferenceListsInfoEXT`
 - Rename `VkVideoEncodeH264EmitPictureParametersEXT` to `VkVideoEncodeH264EmitPictureParametersInfoEXT`
 - Rename `VkVideoEncodeH264NaluSliceEXT` to `VkVideoEncodeH264NaluSliceInfoEXT`
- Revision 9, 2022-09-18 (Daniel Rakos)
 - Rename `spsStdCount`, `pSpsStd`, `ppsStdCount`, and `pPpsStd` to `stdSPSCount`, `pStdSPSs`, `stdPPSCount`, and `pStdPPSs`, respectively, in `VkVideoEncodeH264SessionParametersAddInfoEXT`
 - Rename `maxSpsStdCount` and `maxPpsStdCount` to `maxStdSPSCount` and `maxStdPPSCount`, respectively, in `VkVideoEncodeH264SessionParametersCreateInfoEXT`
- Revision 10, 2023-03-06 (Daniel Rakos)
 - Removed `VkVideoEncodeH264EmitPictureParametersInfoEXT`
 - Changed member types in `VkVideoEncodeH264CapabilitiesEXT` and `VkVideoEncodeH264ReferenceListsInfoEXT` from `uint8_t` to `uint32_t`

- Changed the type of `VkVideoEncodeH264RateControlInfoEXT::temporalLayerCount` and `VkVideoEncodeH264RateControlLayerInfoEXT::temporalLayerId` from `uint8_t` to `uint32_t`
- Removed `VkVideoEncodeH264InputModeFlagsEXT` and `VkVideoEncodeH264OutputModeFlagsEXT` as we only support frame-in-frame-out mode for now
- Rename `pCurrentPictureInfo` in `VkVideoEncodeH264VclFrameInfoEXT` to `pStdPictureInfo`
- Rename `pSliceHeaderStd` in `VkVideoEncodeH264NaluSliceInfoEXT` to `pStdSliceHeader`
- Rename `pReferenceFinalLists` in `VkVideoEncodeH264VclFrameInfoEXT` and `VkVideoEncodeH264NaluSliceInfoEXT` to `pStdReferenceFinalLists`
- Removed the `slotIndex` member of `VkVideoEncodeH264DpbSlotInfoEXT` and changed it to be chained to `VkVideoReferenceSlotInfoKHR`
- Replaced `VkVideoEncodeH264ReferenceListsInfoEXT` with the new Video Std header structure `StdVideoEncodeH264ReferenceLists` that also includes data previously part of the now removed `StdVideoEncodeH264RefMemMgmtCtrlOperations` structure
- Added new capability flag `VK_VIDEO_ENCODE_H264_CAPABILITY_DIFFERENT_REFERENCE_FINAL_LISTS_BIT_EXT`
- Revision 11, 2023-05-22 (Daniel Rakos)
 - Renamed `VkVideoEncodeH264VclFrameInfoEXT` to `VkVideoEncodeH264PictureInfoEXT`
 - Added `VkVideoEncodeH264PictureInfoEXT::generatePrefixNalu` and `VK_VIDEO_ENCODE_H264_CAPABILITY_GENERATE_PREFIX_NALU_BIT_EXT` to enable the generation of H.264 prefix NALUs when supported by the implementation
 - Removed `VkVideoEncodeH264RateControlLayerInfoEXT::temporalLayerId`
 - Added `expectDyadicTemporalLayerPattern` capability
 - Added the `VkVideoEncodeH264SessionParametersGetInfoEXT` structure to identify the H.264 parameter sets to retrieve encoded parameter data for, and the `VkVideoEncodeH264SessionParametersFeedbackInfoEXT` structure to retrieve H.264 parameter set override information when using the new `vkGetEncodedVideoSessionParametersKHR` command
 - Added `VkVideoEncodeH264NaluSliceInfoEXT::constantQp` to specify per-slice constant QP when rate control mode is `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`
 - Added `VkVideoEncodeH264QualityLevelPropertiesEXT` for retrieving H.264 specific quality level recommendations
 - Replaced `VkVideoEncodeH264RateControlStructureEXT` enum with the flags type `VkVideoEncodeH264RateControlFlagsEXT` and bits defined in `VkVideoEncodeH264RateControlFlagBitsEXT` and added HRD compliance flag
 - Removed `useInitialRcQp` and `initialRcQp` members of `VkVideoEncodeH264RateControlLayerInfoEXT`
 - Added `prefersGopRemainingFrames` and `requiresGopRemainingFrames`, and the new `VkVideoEncodeH264GopRemainingFrameInfoEXT` structure to allow specifying remaining frames of each type in the rate control GOP
 - Added `maxTemporalLayers`, `maxQp`, and `minQp` capabilities

- Added `maxLevelIdc` capability and new `VkVideoEncodeH264SessionCreateInfoEXT` structure to specify upper bounds on the H.264 level of the produced video bitstream
- Moved capability flags specific to codec syntax restrictions from `VkVideoEncodeH264CapabilityFlagsEXT` to the new `VkVideoEncodeH264StdFlagsEXT` which is now included as a separate `stdSyntaxFlags` member in `VkVideoEncodeH264CapabilitiesEXT`
- Removed codec syntax override values from `VkVideoEncodeH264CapabilitiesEXT`
- Removed `VkVideoEncodeH264NaluSliceInfoEXT::mbCount` and `VK_VIDEO_ENCODE_H264_CAPABILITY_SLICE_MB_COUNT_BIT_EXT`
- Replaced `VK_VIDEO_ENCODE_H264_CAPABILITY_MULTIPLE_SLICES_PER_FRAME_BIT_EXT` with the new `maxSliceCount` capability
- Removed `VkVideoEncodeH264PictureInfoEXT::pStdReferenceFinalLists` capability flag `VK_VIDEO_ENCODE_H264_CAPABILITY_DIFFERENT_REFERENCE_FINAL_LISTS_BIT_EXT` and removed `pStdReferenceFinalLists` members from the `VkVideoEncodeH264PictureInfoEXT` and `VkVideoEncodeH264NaluSliceInfoEXT` structures as reference lists info is now included in `pStdPictureInfo`
- Added capability flag `VK_VIDEO_ENCODE_H264_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_EXT`
- Revision 12, 2023-07-19 (Daniel Rakos)
 - Added video std capability flags `VK_VIDEO_ENCODE_H264_STD_SLICE_QP_DELTA_BIT_EXT` and `VK_VIDEO_ENCODE_H264_STD_DIFFERENT_SLICE_QP_DELTA_BIT_EXT`
 - Fixed optionality of the array members of `VkVideoEncodeH264SessionParametersAddInfoEXT`
 - Fixed optionality of `VkVideoEncodeH264RateControlInfoEXT::flags`
- Revision 13, 2023-09-04 (Daniel Rakos)
 - Change extension from `EXT` to `KHR`
 - Extension is no longer provisional
- Revision 14, 2023-12-05 (Daniel Rakos)
 - Condition reference picture setup based on the value of `StdVideoEncodeH264PictureInfo::flags.is_reference`

VK_KHR_video_encode_h265

Name String

`VK_KHR_video_encode_h265`

Extension Type

Device extension

Registered Extension Number

40

Revision

14

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_video_encode_queue](#)

Contact

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Extension Proposal

[VK_KHR_video_encode_h265](#)

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

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Description

This extension builds upon the [VK_KHR_video_encode_queue](#) extension by adding support for encoding elementary video stream sequences compliant with the H.265/HEVC video compression standard.

NOTE

This extension was promoted to **KHR** from the provisional extension [VK_EXT_video_encode_h265](#).

New Structures

- [VkVideoEncodeH265FrameSizeKHR](#)

- [VkVideoEncodeH265NaluSliceSegmentInfoKHR](#)
- [VkVideoEncodeH265QpKHR](#)
- Extending [VkVideoBeginCodingInfoKHR](#):
 - [VkVideoEncodeH265GopRemainingFrameInfoKHR](#)
- Extending [VkVideoCapabilitiesKHR](#):
 - [VkVideoEncodeH265CapabilitiesKHR](#)
- Extending [VkVideoCodingControlInfoKHR](#), [VkVideoBeginCodingInfoKHR](#):
 - [VkVideoEncodeH265RateControlInfoKHR](#)
- Extending [VkVideoEncodeInfoKHR](#):
 - [VkVideoEncodeH265PictureInfoKHR](#)
- Extending [VkVideoEncodeQualityLevelPropertiesKHR](#):
 - [VkVideoEncodeH265QualityLevelPropertiesKHR](#)
- Extending [VkVideoEncodeRateControlLayerInfoKHR](#):
 - [VkVideoEncodeH265RateControlLayerInfoKHR](#)
- Extending [VkVideoEncodeSessionParametersFeedbackInfoKHR](#):
 - [VkVideoEncodeH265SessionParametersFeedbackInfoKHR](#)
- Extending [VkVideoEncodeSessionParametersGetInfoKHR](#):
 - [VkVideoEncodeH265SessionParametersGetInfoKHR](#)
- Extending [VkVideoProfileInfoKHR](#), [VkQueryPoolCreateInfo](#):
 - [VkVideoEncodeH265ProfileInfoKHR](#)
- Extending [VkVideoReferenceSlotInfoKHR](#):
 - [VkVideoEncodeH265DpbSlotInfoKHR](#)
- Extending [VkVideoSessionCreateInfoKHR](#):
 - [VkVideoEncodeH265SessionCreateInfoKHR](#)
- Extending [VkVideoSessionParametersCreateInfoKHR](#):
 - [VkVideoEncodeH265SessionParametersCreateInfoKHR](#)
- Extending [VkVideoSessionParametersUpdateInfoKHR](#):
 - [VkVideoEncodeH265SessionParametersAddInfoKHR](#)

New Enums

- [VkVideoEncodeH265CapabilityFlagBitsKHR](#)
- [VkVideoEncodeH265CtbSizeFlagBitsKHR](#)
- [VkVideoEncodeH265RateControlFlagBitsKHR](#)
- [VkVideoEncodeH265StdFlagBitsKHR](#)
- [VkVideoEncodeH265TransformBlockSizeFlagBitsKHR](#)

New Bitmasks

- [VkVideoEncodeH265CapabilityFlagsKHR](#)
- [VkVideoEncodeH265CtbSizeFlagsKHR](#)
- [VkVideoEncodeH265RateControlFlagsKHR](#)
- [VkVideoEncodeH265StdFlagsKHR](#)
- [VkVideoEncodeH265TransformBlockSizeFlagsKHR](#)

New Enum Constants

- [VK_KHR_VIDEO_ENCODE_H265_EXTENSION_NAME](#)
- [VK_KHR_VIDEO_ENCODE_H265_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_CAPABILITIES_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_DPB_SLOT_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_GOP_REMAINING_FRAME_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_NALU_SLICE_SEGMENT_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PICTURE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_PROFILE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_QUALITY_LEVEL_PROPERTIES_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_RATE_CONTROL_LAYER_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_ADD_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_CREATE_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_FEEDBACK_INFO_KHR](#)
 - [VK_STRUCTURE_TYPE_VIDEO_ENCODE_H265_SESSION_PARAMETERS_GET_INFO_KHR](#)
- Extending [VkVideoCodecOperationFlagBitsKHR](#):
 - [VK_VIDEO_CODEC_OPERATION_ENCODE_H265_BIT_KHR](#)

Version History

- Revision 0, 2019-11-14 (Ahmed Abdelkhalek)
 - Initial draft
- Revision 0.5, 2020-02-13 (Tony Zlatinski)
 - General Spec cleanup
 - Added DPB structures
 - Change the VCL frame encode structure

- Added a common Non-VCL Picture Paramameters structure
- Revision 2, Oct 10 2021 (Srinath Kumarapuram)
 - Vulkan Video Encode h.265 update and spec edits
- Revision 3, 2021-12-08 (Ahmed Abdelkhalek)
 - Rate control updates
- Revision 4, 2022-01-11 (Ahmed Abdelkhalek)
 - Replace occurrences of “slice” by “slice segment” and rename structures/enums to reflect this.
- Revision 5, 2022-02-10 (Ahmed Abdelkhalek)
 - Updates to encode capability interface
- Revision 6, 2022-03-16 (Ahmed Abdelkhalek)
 - Relocate Std header version reporting/requesting from this extension to VK_KHR_video_queue extension.
 - Remove the now empty VkVideoEncodeH265SessionCreateInfoEXT.
- Revision 7, 2022-03-24 (Ahmed Abdelkhalek)
 - Add capability flags to report support to disable transform skip and support to use B frame in L1 reference list.
- Revision 8, 2022-07-18 (Daniel Rakos)
 - Replace `VkVideoEncodeH265RateControlStructureFlagBitsEXT` bit enum with `VkVideoEncodeH265RateControlStructureEXT` enum
 - Rename `VkVideoEncodeH265ProfileEXT` to `VkVideoEncodeH265ProfileInfoEXT`
 - Rename `VkVideoEncodeH265ReferenceListsEXT` to `VkVideoEncodeH265ReferenceListsInfoEXT`
 - Rename `VkVideoEncodeH265EmitPictureParametersEXT` to `VkVideoEncodeH265EmitPictureParametersInfoEXT`
 - Rename `VkVideoEncodeH265NaluSliceSegmentEXT` to `VkVideoEncodeH265NaluSliceSegmentInfoEXT`
- Revision 9, 2022-09-18 (Daniel Rakos)
 - Rename `vpsStdCount`, `pVpsStd`, `spsStdCount`, `pSpsStd`, `ppsStdCount`, and `pPpsStd` to `stdVPSCount`, `pStdVPSs`, `stdSPSCount`, `pStdSPSs`, `stdPPSCount`, and `pStdPPSs`, respectively, in `VkVideoEncodeH265SessionParametersAddInfoEXT`
 - Rename `maxVpsStdCount`, `maxSpsStdCount`, and `maxPpsStdCount` to `maxStdVPSCount`, `maxStdSPSCount` and `maxStdPPSCount`, respectively, in `VkVideoEncodeH265SessionParametersCreateInfoEXT`
- Revision 10, 2023-03-06 (Daniel Rakos)
 - Removed `VkVideoEncodeH265EmitPictureParametersInfoEXT`
 - Changed member types in `VkVideoEncodeH265CapabilitiesEXT` and `VkVideoEncodeH265ReferenceListsInfoEXT` from `uint8_t` to `uint32_t`
 - Changed the type of `VkVideoEncodeH265RateControlInfoEXT::subLayerCount` and `VkVideoEncodeH265RateControlLayerInfoEXT::temporalId` from `uint8_t` to `uint32_t`

- Removed `VkVideoEncodeH265InputModeFlagsEXT` and `VkVideoEncodeH265OutputModeFlagsEXT` as we only support frame-in-frame-out mode for now
- Rename `pCurrentPictureInfo` in `VkVideoEncodeH265VclFrameInfoEXT` to `pStdPictureInfo`
- Rename `pSliceSegmentHeaderStd` in `VkVideoEncodeH265NaluSliceSegmentInfoEXT` to `pStdSliceSegmentHeader`
- Rename `pReferenceFinalLists` in `VkVideoEncodeH265VclFrameInfoEXT` and `VkVideoEncodeH265NaluSliceSegmentInfoEXT` to `pStdReferenceFinalLists`
- Removed the `slotIndex` member of `VkVideoEncodeH265DpbSlotInfoEXT` and changed it to be chained to `VkVideoReferenceSlotInfoKHR`
- Replaced `VkVideoEncodeH265ReferenceListsInfoEXT` with the new Video Std header structure `StdVideoEncodeH265ReferenceLists`
- Added new capability flag `VK_VIDEO_ENCODE_H265_CAPABILITY_DIFFERENT_REFERENCE_FINAL_LISTS_BIT_EXT`
- Revision 11, 2023-05-26 (Daniel Rakos)
 - Renamed `VkVideoEncodeH265VclFrameInfoEXT` to `VkVideoEncodeH265PictureInfoEXT`
 - Removed `VkVideoEncodeH265RateControlLayerInfoEXT::temporalId`
 - Added `expectDyadicTemporalSubLayerPattern` capability
 - Added the `VkVideoEncodeH265SessionParametersGetInfoEXT` structure to identify the H.265 parameter sets to retrieve encoded parameter data for, and the `VkVideoEncodeH265SessionParametersFeedbackInfoEXT` structure to retrieve H.265 parameter set override information when using the new `vkGetEncodedVideoSessionParametersKHR` command
 - Added `VkVideoEncodeH265NaluSliceSegmentInfoEXT::constantQp` to specify per-slice segment constant QP when rate control mode is `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DISABLED_BIT_KHR`
 - Added `VkVideoEncodeH265QualityLevelPropertiesEXT` for retrieving H.265 specific quality level recommendations
 - Replaced `VkVideoEncodeH265RateControlStructureEXT` enum with the flags type `VkVideoEncodeH265RateControlFlagsEXT` and bits defined in `VkVideoEncodeH265RateControlFlagBitsEXT` and added HRD compliance flag
 - Removed `useInitialRcQp` and `initialRcQp` members of `VkVideoEncodeH265RateControlLayerInfoEXT`
 - Added `prefersGopRemainingFrames` and `requiresGopRemainingFrames`, and the new `VkVideoEncodeH265GopRemainingFrameInfoEXT` structure to allow specifying remaining frames of each type in the rate control GOP
 - Renamed `maxSubLayersCount` capability to `maxSubLayerCount`
 - Added `maxQp`, and `minQp` capabilities
 - Added `maxLevelIdc` capability and new `VkVideoEncodeH265SessionCreateInfoEXT` structure to specify upper bounds on the H.265 level of the produced video bitstream
 - Moved capability flags specific to codec syntax restrictions from `VkVideoEncodeH265CapabilityFlagsEXT` to the new `VkVideoEncodeH265StdFlagsEXT` which is now

included as a separate `stdSyntaxFlags` member in `VkVideoEncodeH265CapabilitiesEXT`

- Added `std` prefix to codec syntax capabilities in `VkVideoEncodeH265CapabilitiesEXT`
- Removed `VkVideoEncodeH265NaluSliceSegmentInfoEXT::ctbCount` and `VK_VIDEO_ENCODE_H265_CAPABILITY_SLICE_SEGMENT_CTB_COUNT_BIT_EXT`
- Replaced `VK_VIDEO_ENCODE_H265_CAPABILITY_MULTIPLE_SLICE_SEGMENTS_PER_FRAME_BIT_EXT` with the new `maxSliceSegmentCount` capability
- Added `maxTiles` capability
- Removed codec syntax min/max capabilities from `VkVideoEncodeH265CapabilitiesEXT`
- Removed `VK_VIDEO_ENCODE_H265_CAPABILITY_DIFFERENT_REFERENCE_FINAL_LISTS_BIT_EXT` capability flag and removed `pStdReferenceFinalLists` members from the `VkVideoEncodeH265PictureInfoEXT` and `VkVideoEncodeH265NaluSliceSegmentInfoEXT` structures as reference lists info is now included in `pStdPictureInfo`
- Added capability flag `VK_VIDEO_ENCODE_H265_CAPABILITY_B_FRAME_IN_L0_LIST_BIT_EXT`
- Revision 12, 2023-07-19 (Daniel Rakos)
 - Added video std capability flags `VK_VIDEO_ENCODE_H265_STD_SLICE_QP_DELTA_BIT_EXT` and `VK_VIDEO_ENCODE_H265_STD_DIFFERENT_SLICE_QP_DELTA_BIT_EXT`
 - Fixed optionality of the array members of `VkVideoEncodeH265SessionParametersAddInfoEXT`
 - Fixed optionality of `VkVideoEncodeH265RateControlInfoEXT::flags`
- Revision 13, 2023-09-04 (Daniel Rakos)
 - Change extension from `EXT` to `KHR`
 - Extension is no longer provisional
- Revision 14, 2023-12-05 (Daniel Rakos)
 - Condition reference picture setup based on the value of `StdVideoEncodeH265PictureInfo::flags.is_reference`

VK_KHR_video_encode_queue

Name String

`VK_KHR_video_encode_queue`

Extension Type

Device extension

Registered Extension Number

300

Revision

12

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_video_queue](#)

and

[VK_KHR_synchronization2](#)

or

[Vulkan Version 1.3](#)

API Interactions

- Interacts with [VK_VERSION_1_3](#)
- Interacts with [VK_KHR_format_feature_flags2](#)

Contact

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Extension Proposal

[VK_KHR_video_encode_queue](#)

Other Extension Metadata

Last Modified Date

2023-12-05

IP Status

No known IP claims.

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- Lynne Iribarren, Independent

Description

This extension builds upon the [VK_KHR_video_queue](#) extension by adding common APIs specific to video encoding and thus enabling implementations to expose queue families supporting video encode operations.

More specifically, it adds video encode specific capabilities and a new command buffer command that allows recording video encode operations against a video session.

This extension is to be used in conjunction with other codec specific video encode extensions that enable encoding video sequences of specific video compression standards.

New Commands

- [vkCmdEncodeVideoKHR](#)
- [vkGetEncodedVideoSessionParametersKHR](#)
- [vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR](#)

New Structures

- [VkPhysicalDeviceVideoEncodeQualityLevelInfoKHR](#)
- [VkVideoEncodeInfoKHR](#)
- [VkVideoEncodeQualityLevelPropertiesKHR](#)
- [VkVideoEncodeRateControlLayerInfoKHR](#)
- [VkVideoEncodeSessionParametersFeedbackInfoKHR](#)
- [VkVideoEncodeSessionParametersGetInfoKHR](#)
- Extending [VkQueryPoolCreateInfo](#):
 - [VkQueryPoolVideoEncodeFeedbackCreateInfoKHR](#)
- Extending [VkVideoCapabilitiesKHR](#):
 - [VkVideoEncodeCapabilitiesKHR](#)
- Extending [VkVideoCodingControlInfoKHR](#), [VkVideoBeginCodingInfoKHR](#):
 - [VkVideoEncodeRateControlInfoKHR](#)
- Extending [VkVideoCodingControlInfoKHR](#), [VkVideoSessionParametersCreateInfoKHR](#):
 - [VkVideoEncodeQualityLevelInfoKHR](#)
- Extending [VkVideoProfileInfoKHR](#), [VkQueryPoolCreateInfo](#):
 - [VkVideoEncodeUsageInfoKHR](#)

New Enums

- [VkVideoEncodeCapabilityFlagBitsKHR](#)
- [VkVideoEncodeContentFlagBitsKHR](#)
- [VkVideoEncodeFeedbackFlagBitsKHR](#)

- [VkVideoEncodeRateControlModeFlagBitsKHR](#)
- [VkVideoEncodeTuningModeKHR](#)
- [VkVideoEncodeUsageFlagBitsKHR](#)

New Bitmasks

- [VkVideoEncodeCapabilityFlagsKHR](#)
- [VkVideoEncodeContentFlagsKHR](#)
- [VkVideoEncodeFeedbackFlagsKHR](#)
- [VkVideoEncodeFlagsKHR](#)
- [VkVideoEncodeRateControlFlagsKHR](#)
- [VkVideoEncodeRateControlModeFlagsKHR](#)
- [VkVideoEncodeUsageFlagsKHR](#)

New Enum Constants

- [VK_KHR_VIDEO_ENCODE_QUEUE_EXTENSION_NAME](#)
- [VK_KHR_VIDEO_ENCODE_QUEUE_SPEC_VERSION](#)
- Extending [VkAccessFlagBits2](#):
 - [VK_ACCESS_2_VIDEO_ENCODE_READ_BIT_KHR](#)
 - [VK_ACCESS_2_VIDEO_ENCODE_WRITE_BIT_KHR](#)
- Extending [VkBufferUsageFlagBits](#):
 - [VK_BUFFER_USAGE_VIDEO_ENCODE_DST_BIT_KHR](#)
 - [VK_BUFFER_USAGE_VIDEO_ENCODE_SRC_BIT_KHR](#)
- Extending [VkFormatFeatureFlagBits](#):
 - [VK_FORMAT_FEATURE_VIDEO_ENCODE_DPB_BIT_KHR](#)
 - [VK_FORMAT_FEATURE_VIDEO_ENCODE_INPUT_BIT_KHR](#)
- Extending [VkImageLayout](#):
 - [VK_IMAGE_LAYOUT_VIDEO_ENCODE_DPB_KHR](#)
 - [VK_IMAGE_LAYOUT_VIDEO_ENCODE_DST_KHR](#)
 - [VK_IMAGE_LAYOUT_VIDEO_ENCODE_SRC_KHR](#)
- Extending [VkImageUsageFlagBits](#):
 - [VK_IMAGE_USAGE_VIDEO_ENCODE_DPB_BIT_KHR](#)
 - [VK_IMAGE_USAGE_VIDEO_ENCODE_DST_BIT_KHR](#)
 - [VK_IMAGE_USAGE_VIDEO_ENCODE_SRC_BIT_KHR](#)
- Extending [VkPipelineStageFlagBits2](#):
 - [VK_PIPELINE_STAGE_2_VIDEO_ENCODE_BIT_KHR](#)

- Extending [VkQueryResultStatusKHR](#):
 - `VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR`
- Extending [VkQueryType](#):
 - `VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR`
- Extending [VkQueueFlagBits](#):
 - `VK_QUEUE_VIDEO_ENCODE_BIT_KHR`
- Extending [VkResult](#):
 - `VK_ERROR_INVALID_VIDEO_STD_PARAMETERS_KHR`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR`
 - `VK_STRUCTURE_TYPE_QUERY_POOL_VIDEO_ENCODE_FEEDBACK_CREATE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_ENCODE_CAPABILITIES_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_ENCODE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_ENCODE_QUALITY_LEVEL_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_ENCODE_RATE_CONTROL_LAYER_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_FEEDBACK_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_ENCODE_SESSION_PARAMETERS_GET_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_ENCODE_USAGE_INFO_KHR`
- Extending [VkVideoCodingControlFlagBitsKHR](#):
 - `VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR`
 - `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR`
- Extending [VkVideoSessionCreateFlagBitsKHR](#):
 - `VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_PARAMETER_OPTIMIZATIONS_BIT_KHR`

If [VK_KHR_format_feature_flags2](#) or [Vulkan Version 1.3](#) is supported:

- Extending [VkFormatFeatureFlagBits2](#):
 - `VK_FORMAT_FEATURE_2_VIDEO_ENCODE_DPB_BIT_KHR`
 - `VK_FORMAT_FEATURE_2_VIDEO_ENCODE_INPUT_BIT_KHR`

Version History

- Revision 1, 2018-07-23 (Ahmed Abdelkhalek)
 - Initial draft
- Revision 1.1, 10/29/2019 (Tony Zlatinski)

- Updated the reserved spec tokens and renamed `VkVideoEncoderKHR` to `VkVideoSessionKHR`
- Revision 1.6, Jan 08 2020 (Tony Zlatinski)
 - API unify with the `video_decode_queue` spec
- Revision 2, March 29 2021 (Tony Zlatinski)
 - Spec and API updates.
- Revision 3, 2021-09-30 (Jon Leech)
 - Add interaction with `VK_KHR_format_feature_flags2` to `vk.xml`
- Revision 4, 2022-02-10 (Ahmed Abdelkhalek)
 - Updates to encode capability interface
- Revision 5, 2022-03-31 (Ahmed Abdelkhalek)
 - Remove redundant `VkVideoEncodeInfoKHR.codedExtent`
- Revision 6, 2022-07-18 (Daniel Rakos)
 - Remove `VkVideoEncodeRateControlFlagBitsKHR` and `VkVideoEncodeFlagBitsKHR` as they contain no defined flags for now
 - Add `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_BIT_KHR` and `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_LAYER_BIT_KHR` to indicate rate control and rate control layer change requests, respectively, in video coding control operations
- Revision 7, 2022-08-12 (Daniel Rakos)
 - Add `VkVideoEncodeUsageInfoKHR` structure and related flags
- Revision 8, 2023-03-06 (Daniel Rakos)
 - Replace `VK_QUERY_TYPE_VIDEO_ENCODE_BITSTREAM_BUFFER_RANGE_KHR` queries with more generic `VK_QUERY_TYPE_VIDEO_ENCODE_FEEDBACK_KHR` queries that can be extended in the future with more feedback values
 - Rename `dstBitstreamBuffer`, `dstBitstreamBufferOffset`, and `dstBitstreamBufferMaxRange` in `VkVideoEncodeInfoKHR` to `dstBuffer`, `dstBufferOffset`, and `dstBufferRange`, respectively, for consistency with the naming convention in the video decode extensions
 - Change the type of `rateControlLayerCount` and `qualityLevelCount` in `VkVideoEncodeCapabilitiesKHR` from `uint8_t` to `uint32_t` and rename them to `maxRateControlLayers` and `maxQualityLevels`, respectively
 - Change the type of `averageBitrate` and `maxBitrate` in `VkVideoEncodeRateControlLayerInfoKHR` from `uint32_t` to `uint64_t`
 - Fixed the definition of rate control flag bits and added the new `VK_VIDEO_ENCODE_RATE_CONTROL_MODE_DEFAULT_KHR` constant to indicate implementation-specific automatic rate control
 - Change the type of `VkVideoEncodeRateControlInfoKHR::layerCount` from `uint8_t` to `uint32_t`
 - Rename `pLayerConfigs` to `pLayers` in `VkVideoEncodeRateControlInfoKHR`
- Revision 9, 2023-03-28 (Daniel Rakos)
 - Removed `VK_VIDEO_CODING_CONTROL_ENCODE_RATE_CONTROL_LAYER_BIT_KHR` and the ability to

change the state of individual rate control layers

- Added new `VK_VIDEO_ENCODE_FEEDBACK_BITSTREAM_HAS_OVERRIDES_BIT_KHR` flag to video encode feedback queries
- Added new video session create flag `VK_VIDEO_SESSION_CREATE_ALLOW_ENCODE_PARAMETER_OPTIMIZATIONS_BIT_KHR` to opt-in to video session and encoding parameter optimizations
- Added the `vkGetEncodedVideoSessionParametersKHR` command to enable retrieving encoded video session parameter data
- Moved `virtualBufferSizeInMs` and `initialVirtualBufferSizeInMs` from `VkVideoEncodeRateControlLayerInfoKHR` to `VkVideoEncodeRateControlInfoKHR`
- Added `maxBitrate` capability
- Renamed `inputImageDataFillAlignment` capability to `encodeInputPictureGranularity` to better reflect its purpose
- Added new `vkGetPhysicalDeviceVideoEncodeQualityLevelPropertiesKHR` command and related structures to enable querying recommended settings for video encode quality levels
- Added `VK_VIDEO_CODING_CONTROL_ENCODE_QUALITY_LEVEL_BIT_KHR` flag and `VkVideoEncodeQualityLevelInfoKHR` structure to allow controlling video encode quality level and removed `qualityLevel` from the encode operation parameters
- Revision 10, 2023-07-19 (Daniel Rakos)
 - Added `VK_QUERY_RESULT_STATUS_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_KHR` query result status code and the related capability flag `VK_VIDEO_ENCODE_CAPABILITY_INSUFFICIENT_BITSTREAM_BUFFER_RANGE_DETECTION_BIT_KHR`
- Revision 11, 2023-09-04 (Daniel Rakos)
 - Extension is no longer provisional
- Revision 12, 2023-12-05 (Daniel Rakos)
 - Require the specification of a reconstructed picture in all cases, except when the video session was created with no DPB slots to match shipping implementations
 - Make DPB slot activation behavior codec-specific to continue allowing application control over reference picture setup now that a reconstructed picture is always mandatory

VK_KHR_video_maintenance1

Name String

`VK_KHR_video_maintenance1`

Extension Type

Device extension

Registered Extension Number

516

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_video_queue](#)

Contact

- Daniel Rakos [\[GitHub\]](#)[aqnuep](#)

Extension Proposal

[VK_KHR_video_maintenance1](#)

Other Extension Metadata

Last Modified Date

2023-07-27

IP Status

No known IP claims.

Contributors

- Ahmed Abdelkhalek, AMD
- Aidan Fabius, Core Avionics & Industrial Inc.
- Ping Liu, Intel
- Lynne Iribarren, Independent
- Srinath Kumarapuram, NVIDIA
- Tony Zlatinski, NVIDIA
- Daniel Rakos, RasterGrid

Description

[VK_KHR_video_maintenance1](#) adds a collection of minor video coding features, none of which would warrant an entire extension of their own.

The new features are as follows:

- Allow creating buffers that can be used in video coding operations, independent of the used video profile, using the new buffer creation flag [VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR](#).
- Allow creating images that can be used as decode output or encode input pictures, independent of the used video profile, using the new image creation flag [VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR](#).
- Allow specifying queries used by video coding operations as part of the video coding command

parameters, instead of using begin/end query when the video session is created using the new video session creation flag `VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR`.

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceVideoMaintenance1FeaturesKHR`
- Extending `VkVideoDecodeInfoKHR`, `VkVideoEncodeInfoKHR`:
 - `VkVideoInlineQueryInfoKHR`

New Enum Constants

- `VK_KHR_VIDEO_MAINTENANCE_1_EXTENSION_NAME`
- `VK_KHR_VIDEO_MAINTENANCE_1_SPEC_VERSION`
- Extending `VkBufferCreateFlagBits`:
 - `VK_BUFFER_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`
- Extending `VkImageCreateFlagBits`:
 - `VK_IMAGE_CREATE_VIDEO_PROFILE_INDEPENDENT_BIT_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_MAINTENANCE_1_FEATURES_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_INLINE_QUERY_INFO_KHR`
- Extending `VkVideoSessionCreateFlagBitsKHR`:
 - `VK_VIDEO_SESSION_CREATE_INLINE_QUERIES_BIT_KHR`

Version History

- Revision 1, 2023-07-27 (Daniel Rakos)
 - internal revisions

VK_KHR_video_queue

Name String

`VK_KHR_video_queue`

Extension Type

Device extension

Registered Extension Number

24

Revision

8

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.1](#)

and

[VK_KHR_synchronization2](#)

or

[Vulkan Version 1.3](#)

Contact

- Tony Zlatinski [\[GitHub\]tzlatinski](#)

Extension Proposal

[VK_KHR_video_queue](#)

Other Extension Metadata

Last Modified Date

2022-09-29

IP Status

No known IP claims.

Contributors

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- George Hao, AMD
- Jake Beju, AMD
- Piers Daniell, NVIDIA
- Srinath Kumarapuram, NVIDIA
- Tobias Hector, AMD
- Tony Zlatinski, NVIDIA
- Daniel Rakos, RasterGrid

Description

This extension provides common APIs to enable exposing queue families with support for video codec operations by introducing the following new object types and related functionalities:

- Video session objects that represent and maintain the state needed to perform video codec operations.
- Video session parameters objects that act as a container for codec specific parameters.

In addition, it also introduces query commands that allow applications to determine video coding related capabilities, and command buffer commands that enable recording video coding operations against a video session.

This extension is to be used in conjunction with other extensions that enable specific video coding operations.

New Object Types

- [VkVideoSessionKHR](#)
- [VkVideoSessionParametersKHR](#)

New Commands

- [vkBindVideoSessionMemoryKHR](#)
- [vkCmdBeginVideoCodingKHR](#)
- [vkCmdControlVideoCodingKHR](#)
- [vkCmdEndVideoCodingKHR](#)
- [vkCreateVideoSessionKHR](#)
- [vkCreateVideoSessionParametersKHR](#)
- [vkDestroyVideoSessionKHR](#)
- [vkDestroyVideoSessionParametersKHR](#)
- [vkGetPhysicalDeviceVideoCapabilitiesKHR](#)
- [vkGetPhysicalDeviceVideoFormatPropertiesKHR](#)
- [vkGetVideoSessionMemoryRequirementsKHR](#)
- [vkUpdateVideoSessionParametersKHR](#)

New Structures

- [VkBindVideoSessionMemoryInfoKHR](#)
- [VkPhysicalDeviceVideoFormatInfoKHR](#)
- [VkVideoBeginCodingInfoKHR](#)
- [VkVideoCapabilitiesKHR](#)
- [VkVideoCodingControlInfoKHR](#)
- [VkVideoEndCodingInfoKHR](#)
- [VkVideoFormatPropertiesKHR](#)
- [VkVideoPictureResourceInfoKHR](#)
- [VkVideoReferenceSlotInfoKHR](#)
- [VkVideoSessionCreateInfoKHR](#)
- [VkVideoSessionMemoryRequirementsKHR](#)
- [VkVideoSessionParametersCreateInfoKHR](#)
- [VkVideoSessionParametersUpdateInfoKHR](#)
- Extending [VkPhysicalDeviceImageFormatInfo2](#), [VkPhysicalDeviceVideoFormatInfoKHR](#),

VkImageCreateInfo, VkBufferCreateInfo:

- [VkVideoProfileListInfoKHR](#)

• Extending [VkQueryPoolCreateInfo](#):

- [VkVideoProfileInfoKHR](#)

• Extending [VkQueueFamilyProperties2](#):

- [VkQueueFamilyQueryResultStatusPropertiesKHR](#)
- [VkQueueFamilyVideoPropertiesKHR](#)

New Enums

- [VkQueryResultStatusKHR](#)
- [VkVideoCapabilityFlagBitsKHR](#)
- [VkVideoChromaSubsamplingFlagBitsKHR](#)
- [VkVideoCodecOperationFlagBitsKHR](#)
- [VkVideoCodingControlFlagBitsKHR](#)
- [VkVideoComponentBitDepthFlagBitsKHR](#)
- [VkVideoSessionCreateFlagBitsKHR](#)

New Bitmasks

- [VkVideoBeginCodingFlagsKHR](#)
- [VkVideoCapabilityFlagsKHR](#)
- [VkVideoChromaSubsamplingFlagsKHR](#)
- [VkVideoCodecOperationFlagsKHR](#)
- [VkVideoCodingControlFlagsKHR](#)
- [VkVideoComponentBitDepthFlagsKHR](#)
- [VkVideoEndCodingFlagsKHR](#)
- [VkVideoSessionCreateFlagsKHR](#)
- [VkVideoSessionParametersCreateFlagsKHR](#)

New Enum Constants

- [VK_KHR_VIDEO_QUEUE_EXTENSION_NAME](#)
- [VK_KHR_VIDEO_QUEUE_SPEC_VERSION](#)
- Extending [VkObjectType](#):
 - [VK_OBJECT_TYPE_VIDEO_SESSION_KHR](#)
 - [VK_OBJECT_TYPE_VIDEO_SESSION_PARAMETERS_KHR](#)
- Extending [VkQueryResultFlagBits](#):
 - [VK_QUERY_RESULT_WITH_STATUS_BIT_KHR](#)

- Extending `VkQueryType`:
 - `VK_QUERY_TYPE_RESULT_STATUS_ONLY_KHR`
- Extending `VkResult`:
 - `VK_ERROR_IMAGE_USAGE_NOT_SUPPORTED_KHR`
 - `VK_ERROR_VIDEO_PICTURE_LAYOUT_NOT_SUPPORTED_KHR`
 - `VK_ERROR_VIDEO_PROFILE_CODEC_NOT_SUPPORTED_KHR`
 - `VK_ERROR_VIDEO_PROFILE_FORMAT_NOT_SUPPORTED_KHR`
 - `VK_ERROR_VIDEO_PROFILE_OPERATION_NOT_SUPPORTED_KHR`
 - `VK_ERROR_VIDEO_STD_VERSION_NOT_SUPPORTED_KHR`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_BIND_VIDEO_SESSION_MEMORY_INFO_KHR`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VIDEO_FORMAT_INFO_KHR`
 - `VK_STRUCTURE_TYPE_QUEUE_FAMILY_QUERY_RESULT_STATUS_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_QUEUE_FAMILY_VIDEO_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_BEGIN_CODING_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_CAPABILITIES_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_CODING_CONTROL_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_END_CODING_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_FORMAT_PROPERTIES_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_PICTURE_RESOURCE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_PROFILE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_PROFILE_LIST_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_REFERENCE_SLOT_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_SESSION_CREATE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_SESSION_MEMORY_REQUIREMENTS_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_CREATE_INFO_KHR`
 - `VK_STRUCTURE_TYPE_VIDEO_SESSION_PARAMETERS_UPDATE_INFO_KHR`

Version History

- Revision 0.1, 2019-11-21 (Tony Zlatinski)
 - Initial draft
- Revision 0.2, 2019-11-27 (Tony Zlatinski)
 - Make vulkan video core common between decode and encode
- Revision 1, March 29 2021 (Tony Zlatinski)
 - Spec and API updates.

- Revision 2, August 1 2021 (Srinath Kumarapuram)
 - Rename `VkVideoCapabilitiesFlagBitsKHR` to `VkVideoCapabilityFlagBitsKHR` (along with the names of enumerants it defines) and `VkVideoCapabilitiesFlagsKHR` to `VkVideoCapabilityFlagsKHR`, following Vulkan naming conventions.
- Revision 3, 2022-03-16 (Ahmed Abdelkhalek)
 - Relocate Std header version reporting/requesting from codec-operation specific extensions to this extension.
 - Make Std header versions codec-operation specific instead of only codec-specific.
- Revision 4, 2022-05-30 (Daniel Rakos)
 - Refactor the video format query APIs and related language
 - Extend `VkResult` with video-specific error codes
- Revision 5, 2022-08-11 (Daniel Rakos)
 - Add `VkVideoSessionParametersCreateFlagsKHR`
 - Remove `VkVideoCodingQualityPresetFlagsKHR`
 - Rename `VkQueueFamilyQueryResultStatusProperties2KHR` to `VkQueueFamilyQueryResultStatusPropertiesKHR`
 - Rename `VkVideoQueueFamilyProperties2KHR` to `VkQueueFamilyVideoPropertiesKHR`
 - Rename `VkVideoProfileKHR` to `VkVideoProfileInfoKHR`
 - Rename `VkVideoProfilesKHR` to `VkVideoProfileListInfoKHR`
 - Rename `VkVideoGetMemoryPropertiesKHR` to `VkVideoSessionMemoryRequirementsKHR`
 - Rename `VkVideoBindMemoryKHR` to `VkBindVideoSessionMemoryInfoKHR`
 - Fix `pNext` constness of `VkPhysicalDeviceVideoFormatInfoKHR` and `VkVideoSessionMemoryRequirementsKHR`
 - Fix incorrectly named value enums in bit enum types `VkVideoCodecOperationFlagBitsKHR` and `VkVideoChromaSubsamplingFlagBitsKHR`
 - Remove unnecessary default values from `VkVideoSessionCreateFlagBitsKHR` and `VkVideoCodingControlFlagBitsKHR`
 - Eliminate nested pointer in `VkVideoSessionMemoryRequirementsKHR`
 - Rename `VkVideoPictureResourceKHR` to `VkVideoPictureResourceInfoKHR`
 - Rename `VkVideoReferenceSlotKHR` to `VkVideoReferenceSlotInfoKHR`
- Revision 6, 2022-09-18 (Daniel Rakos)
 - Rename the `maxReferencePicturesSlotsCount` and `maxReferencePicturesActiveCount` fields of `VkVideoCapabilitiesKHR` and `VkVideoSessionCreateInfoKHR` to `maxDpbSlots` and `maxActiveReferencePictures`, respectively, to clarify their meaning
 - Rename `capabilityFlags` to `flags` in `VkVideoCapabilitiesKHR`
 - Rename `videoPictureExtentGranularity` to `pictureAccessGranularity` in `VkVideoCapabilitiesKHR`

- Rename `minExtent` and `maxExtent` to `minCodedExtent` and `maxCodedExtent`, respectively, in `VkVideoCapabilitiesKHR`
- Rename `referencePicturesFormat` to `referencePictureFormat` in `VkVideoSessionCreateInfoKHR`
- Revision 7, 2022-09-26 (Daniel Rakos)
 - Change type of `VkVideoReferenceSlotInfoKHR::slotIndex` from `int8_t` to `int32_t`
- Revision 8, 2022-09-29 (Daniel Rakos)
 - Extension is no longer provisional

VK_KHR_vulkan_memory_model

Name String

`VK_KHR_vulkan_memory_model`

Extension Type

Device extension

Registered Extension Number

212

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_vulkan_memory_model](#)

Deprecation State

- *Promoted* to Vulkan 1.2

Contact

- Jeff Bolz [\[GitHub\]jeffbolznv](#)

Other Extension Metadata

Last Modified Date

2018-12-10

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Alan Baker, Google
- Tobias Hector, AMD
- David Neto, Google
- Robert Simpson, Qualcomm Technologies, Inc.
- Brian Sumner, AMD

Description

The [VK_KHR_vulkan_memory_model](#) extension allows use of the features guarded by the [VulkanMemoryModel](#), [VulkanMemoryModelDeviceScope](#), and [VulkanMemoryModelAvailabilityVisibilityChains](#) capabilities in shader modules. The [Vulkan Memory Model](#) formally defines how to synchronize memory accesses to the same memory locations performed by multiple shader invocations.

NOTE	Version 3 of the spec added a member (vulkanMemoryModelAvailabilityVisibilityChains) to VkPhysicalDeviceVulkanMemoryModelFeaturesKHR , which is an incompatible change from version 2.
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Promotion to Vulkan 1.2

All functionality in this extension is included in core Vulkan 1.2, with the KHR suffix omitted. However, if Vulkan 1.2 is supported and this extension is not, the [vulkanMemoryModel](#) capability is optional. The original type, enum, and command names are still available as aliases of the core functionality.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceVulkanMemoryModelFeaturesKHR](#)

New Enum Constants

- [VK_KHR_VULKAN_MEMORY_MODEL_EXTENSION_NAME](#)
- [VK_KHR_VULKAN_MEMORY_MODEL_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VULKAN_MEMORY_MODEL_FEATURES_KHR](#)

New SPIR-V Capabilities

- [VulkanMemoryModelKHR](#)

Version History

- Revision 1, 2018-06-24 (Jeff Bolz)
 - Initial draft
- Revision 3, 2018-12-10 (Jeff Bolz)
 - Add `vulkanMemoryModelAvailabilityVisibilityChains` member to the `VkPhysicalDeviceVulkanMemoryModelFeaturesKHR` structure.

VK_KHR_wayland_surface

Name String

`VK_KHR_wayland_surface`

Extension Type

Instance extension

Registered Extension Number

7

Revision

6

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_surface](#)

Contact

- Jesse Hall [\[GitHub\]](#)[critsec](#)
- Ian Elliott [\[GitHub\]](#)[ianelliottus](#)

Other Extension Metadata

Last Modified Date

2015-11-28

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Faith Ekstrand, Intel
- Ian Elliott, LunarG
- Courtney Goeltzenleuchter, LunarG

- Jesse Hall, Google
- James Jones, NVIDIA
- Antoine Labour, Google
- Jon Leech, Khronos
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung
- Daniel Rakos, AMD
- Graham Sellers, AMD
- Ray Smith, ARM
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG

Description

The `VK_KHR_wayland_surface` extension is an instance extension. It provides a mechanism to create a `VkSurfaceKHR` object (defined by the `VK_KHR_surface` extension) that refers to a Wayland `wl_surface`, as well as a query to determine support for rendering to a Wayland compositor.

New Commands

- `vkCreateWaylandSurfaceKHR`
- `vkGetPhysicalDeviceWaylandPresentationSupportKHR`

New Structures

- `VkWaylandSurfaceCreateInfoKHR`

New Bitmasks

- `VkWaylandSurfaceCreateFlagsKHR`

New Enum Constants

- `VK_KHR_WAYLAND_SURFACE_EXTENSION_NAME`
- `VK_KHR_WAYLAND_SURFACE_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_WAYLAND_SURFACE_CREATE_INFO_KHR`

Issues

1) Does Wayland need a way to query for compatibility between a particular physical device and a specific Wayland display? This would be a more general query than `vkGetPhysicalDeviceSurfaceSupportKHR`: if the Wayland-specific query returned `VK_TRUE` for a

([VkPhysicalDevice](#), [struct wl_display*](#)) pair, then the physical device could be assumed to support presentation to any [VkSurfaceKHR](#) for surfaces on the display.

RESOLVED: Yes. [vkGetPhysicalDeviceWaylandPresentationSupportKHR](#) was added to address this issue.

2) Should we require surfaces created with [vkCreateWaylandSurfaceKHR](#) to support the [VK_PRESENT_MODE_MAILBOX_KHR](#) present mode?

RESOLVED: Yes. Wayland is an inherently mailbox window system and mailbox support is required for some Wayland compositor interactions to work as expected. While handling these interactions may be possible with [VK_PRESENT_MODE_FIFO_KHR](#), it is much more difficult to do without deadlock and requiring all Wayland applications to be able to support implementations which only support [VK_PRESENT_MODE_FIFO_KHR](#) would be an onerous restriction on application developers.

Version History

- Revision 1, 2015-09-23 (Jesse Hall)
 - Initial draft, based on the previous contents of [VK_EXT_KHR_swapchain](#) (later renamed [VK_EXT_KHR_surface](#)).
- Revision 2, 2015-10-02 (James Jones)
 - Added [vkGetPhysicalDeviceWaylandPresentationSupportKHR\(\)](#) to resolve issue #1.
 - Adjusted wording of issue #1 to match the agreed-upon solution.
 - Renamed “window” parameters to “surface” to match Wayland conventions.
- Revision 3, 2015-10-26 (Ian Elliott)
 - Renamed from [VK_EXT_KHR_wayland_surface](#) to [VK_KHR_wayland_surface](#).
- Revision 4, 2015-11-03 (Daniel Rakos)
 - Added allocation callbacks to [vkCreateWaylandSurfaceKHR](#).
- Revision 5, 2015-11-28 (Daniel Rakos)
 - Updated the surface create function to take a [pCreateInfo](#) structure.
- Revision 6, 2017-02-08 (Faith Ekstrand)
 - Added the requirement that implementations support [VK_PRESENT_MODE_MAILBOX_KHR](#).
 - Added wording about interactions between [vkQueuePresentKHR](#) and the Wayland requests sent to the compositor.

VK_KHR_win32_keyed_mutex

Name String

[VK_KHR_win32_keyed_mutex](#)

Extension Type

Device extension

Registered Extension Number

76

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_memory_win32](#)

Contact

- Carsten Rohde [\[GitHub\]crohde](#)

Other Extension Metadata

Last Modified Date

2016-10-21

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Jeff Juliano, NVIDIA
- Carsten Rohde, NVIDIA

Description

Applications that wish to import Direct3D 11 memory objects into the Vulkan API may wish to use the native keyed mutex mechanism to synchronize access to the memory between Vulkan and Direct3D. This extension provides a way for an application to access the keyed mutex associated with an imported Vulkan memory object when submitting command buffers to a queue.

New Structures

- Extending [VkSubmitInfo](#), [VkSubmitInfo2](#):
 - [VkWin32KeyedMutexAcquireReleaseInfoKHR](#)

New Enum Constants

- [VK_KHR_WIN32_KEYED_MUTEX_EXTENSION_NAME](#)
- [VK_KHR_WIN32_KEYED_MUTEX_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_WIN32_KEYED_MUTEX_ACQUIRE_RELEASE_INFO_KHR](#)

Version History

- Revision 1, 2016-10-21 (James Jones)
 - Initial revision

VK_KHR_win32_surface

Name String

VK_KHR_win32_surface

Extension Type

Instance extension

Registered Extension Number

10

Revision

6

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_surface](#)

Contact

- Jesse Hall [\[GitHub\]](#)[critsec](#)
- Ian Elliott [\[GitHub\]](#)[ianelliottus](#)

Other Extension Metadata

Last Modified Date

2017-04-24

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Faith Ekstrand, Intel
- Ian Elliott, LunarG
- Courtney Goeltzenleuchter, LunarG
- Jesse Hall, Google
- James Jones, NVIDIA
- Antoine Labour, Google

- Jon Leech, Khronos
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung
- Daniel Rakos, AMD
- Graham Sellers, AMD
- Ray Smith, ARM
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG

Description

The `VK_KHR_win32_surface` extension is an instance extension. It provides a mechanism to create a `VkSurfaceKHR` object (defined by the `VK_KHR_surface` extension) that refers to a Win32 `HWND`, as well as a query to determine support for rendering to the windows desktop.

New Commands

- `vkCreateWin32SurfaceKHR`
- `vkGetPhysicalDeviceWin32PresentationSupportKHR`

New Structures

- `VkWin32SurfaceCreateInfoKHR`

New Bitmasks

- `VkWin32SurfaceCreateFlagsKHR`

New Enum Constants

- `VK_KHR_WIN32_SURFACE_EXTENSION_NAME`
- `VK_KHR_WIN32_SURFACE_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_WIN32_SURFACE_CREATE_INFO_KHR`

Issues

1) Does Win32 need a way to query for compatibility between a particular physical device and a specific screen? Compatibility between a physical device and a window generally only depends on what screen the window is on. However, there is not an obvious way to identify a screen without already having a window on the screen.

RESOLVED: No. While it may be useful, there is not a clear way to do this on Win32. However, a method was added to query support for presenting to the windows desktop as a whole.

2) If a native window object (**HWND**) is used by one graphics API, and then is later used by a different graphics API (one of which is Vulkan), can these uses interfere with each other?

RESOLVED: Yes.

Uses of a window object by multiple graphics APIs results in undefined behavior. Such behavior may succeed when using one Vulkan implementation but fail when using a different Vulkan implementation. Potential failures include:

- Creating then destroying a flip presentation model DXGI swapchain on a window object can prevent [vkCreateSwapchainKHR](#) from succeeding on the same window object.
- Creating then destroying a [VkSwapchainKHR](#) on a window object can prevent creation of a bitblt model DXGI swapchain on the same window object.
- Creating then destroying a [VkSwapchainKHR](#) on a window object can effectively [SetPixelFormat](#) to a different format than the format chosen by an OpenGL application.
- Creating then destroying a [VkSwapchainKHR](#) on a window object on one [VkPhysicalDevice](#) can prevent [vkCreateSwapchainKHR](#) from succeeding on the same window object, but on a different [VkPhysicalDevice](#) that is associated with a different Vulkan ICD.

In all cases the problem can be worked around by creating a new window object.

Technical details include:

- Creating a DXGI swapchain over a window object can alter the object for the remainder of its lifetime. The alteration persists even after the DXGI swapchain has been destroyed. This alteration can make it impossible for a conformant Vulkan implementation to create a [VkSwapchainKHR](#) over the same window object. Mention of this alteration can be found in the remarks section of the MSDN documentation for [DXGI_SWAP_EFFECT](#).
- Calling GDI's [SetPixelFormat](#) (needed by OpenGL's WGL layer) on a window object alters the object for the remainder of its lifetime. The MSDN documentation for [SetPixelFormat](#) explains that a window object's pixel format can be set only one time.
- Creating a [VkSwapchainKHR](#) over a window object can alter the object for its remaining lifetime. Either of the above alterations may occur as a side effect of [vkCreateSwapchainKHR](#).

Version History

- Revision 1, 2015-09-23 (Jesse Hall)
 - Initial draft, based on the previous contents of VK_EXT_KHR_swapchain (later renamed VK_EXT_KHR_surface).
- Revision 2, 2015-10-02 (James Jones)
 - Added presentation support query for win32 desktops.
- Revision 3, 2015-10-26 (Ian Elliott)
 - Renamed from VK_EXT_KHR_win32_surface to VK_KHR_win32_surface.
- Revision 4, 2015-11-03 (Daniel Rakos)
 - Added allocation callbacks to [vkCreateWin32SurfaceKHR](#).

- Revision 5, 2015-11-28 (Daniel Rakos)
 - Updated the surface create function to take a pCreateInfo structure.
- Revision 6, 2017-04-24 (Jeff Juliano)
 - Add issue 2 addressing reuse of a native window object in a different Graphics API, or by a different Vulkan ICD.

VK_KHR_workgroup_memory_explicit_layout

Name String

VK_KHR_workgroup_memory_explicit_layout

Extension Type

Device extension

Registered Extension Number

337

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_KHR_workgroup_memory_explicit_layout](#)

Contact

- Caio Marcelo de Oliveira Filho [\[GitHub\]cmarcelo](#)

Other Extension Metadata

Last Modified Date

2020-06-01

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for [GL_EXT_shared_memory_block](#)

Contributors

- Caio Marcelo de Oliveira Filho, Intel

- Jeff Bolz, NVIDIA
- Graeme Leese, Broadcom
- Faith Ekstrand, Intel
- Daniel Koch, NVIDIA

Description

This extension adds Vulkan support for the `SPV_KHR_workgroup_memory_explicit_layout` SPIR-V extension, which allows shaders to explicitly define the layout of `Workgroup` storage class memory and create aliases between variables from that storage class in a compute shader.

The aliasing feature allows different “views” on the same data, so the shader can bulk copy data from another storage class using one type (e.g. an array of large vectors), and then use the data with a more specific type. It also enables reducing the amount of workgroup memory consumed by allowing the shader to alias data whose lifetimes do not overlap.

The explicit layout support and some form of aliasing is also required for layering OpenCL on top of Vulkan.

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceWorkgroupMemoryExplicitLayoutFeaturesKHR`

New Enum Constants

- `VK_KHR_WORKGROUP_MEMORY_EXPLICIT_LAYOUT_EXTENSION_NAME`
- `VK_KHR_WORKGROUP_MEMORY_EXPLICIT_LAYOUT_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_WORKGROUP_MEMORY_EXPLICIT_LAYOUT_FEATURES_KHR`

New SPIR-V Capabilities

- `WorkgroupMemoryExplicitLayoutKHR`
- `WorkgroupMemoryExplicitLayout8BitAccessKHR`
- `WorkgroupMemoryExplicitLayout16BitAccessKHR`

Version History

- Revision 1, 2020-06-01 (Caio Marcelo de Oliveira Filho)
 - Initial version

VK_KHR_xcb_surface

Name String

`VK_KHR_xcb_surface`

Extension Type

Instance extension

Registered Extension Number

6

Revision

6

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_surface](#)

Contact

- Jesse Hall [\[GitHub\]](#)[critsec](#)
- Ian Elliott [\[GitHub\]](#)[ianelliottus](#)

Other Extension Metadata**Last Modified Date**

2015-11-28

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Faith Ekstrand, Intel
- Ian Elliott, LunarG
- Courtney Goeltzenleuchter, LunarG
- Jesse Hall, Google
- James Jones, NVIDIA
- Antoine Labour, Google
- Jon Leech, Khronos
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung
- Daniel Rakos, AMD
- Graham Sellers, AMD
- Ray Smith, ARM

- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG

Description

The `VK_KHR_xcb_surface` extension is an instance extension. It provides a mechanism to create a `VkSurfaceKHR` object (defined by the `VK_KHR_surface` extension) that refers to an X11 `Window`, using the XCB client-side library, as well as a query to determine support for rendering via XCB.

New Commands

- `vkCreateXcbSurfaceKHR`
- `vkGetPhysicalDeviceXcbPresentationSupportKHR`

New Structures

- `VkXcbSurfaceCreateInfoKHR`

New Bitmasks

- `VkXcbSurfaceCreateFlagsKHR`

New Enum Constants

- `VK_KHR_XCB_SURFACE_EXTENSION_NAME`
- `VK_KHR_XCB_SURFACE_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_XCB_SURFACE_CREATE_INFO_KHR`

Issues

1) Does XCB need a way to query for compatibility between a particular physical device and a specific screen? This would be a more general query than `vkGetPhysicalDeviceSurfaceSupportKHR`: If it returned `VK_TRUE`, then the physical device could be assumed to support presentation to any window on that screen.

RESOLVED: Yes, this is needed for toolkits that want to create a `VkDevice` before creating a window. To ensure the query is reliable, it must be made against a particular X visual rather than the screen in general.

Version History

- Revision 1, 2015-09-23 (Jesse Hall)
 - Initial draft, based on the previous contents of `VK_EXT_KHR_swapchain` (later renamed `VK_EXT_KHR_surface`).
- Revision 2, 2015-10-02 (James Jones)
 - Added presentation support query for an `(xcb_connection_t*, xcb_visualid_t)` pair.

- Removed “root” parameter from CreateXcbSurfaceKHR(), as it is redundant when a window on the same screen is specified as well.
- Adjusted wording of issue #1 and added agreed upon resolution.
- Revision 3, 2015-10-14 (Ian Elliott)
 - Removed “root” parameter from CreateXcbSurfaceKHR() in one more place.
- Revision 4, 2015-10-26 (Ian Elliott)
 - Renamed from VK_EXT_KHR_xcb_surface to VK_KHR_xcb_surface.
- Revision 5, 2015-10-23 (Daniel Rakos)
 - Added allocation callbacks to vkCreateXcbSurfaceKHR.
- Revision 6, 2015-11-28 (Daniel Rakos)
 - Updated the surface create function to take a pCreateInfo structure.

VK_KHR_xlib_surface

Name String

VK_KHR_xlib_surface

Extension Type

Instance extension

Registered Extension Number

5

Revision

6

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_surface](#)

Contact

- Jesse Hall [\[GitHub\]critsec](#)
- Ian Elliott [\[GitHub\]ianelliottus](#)

Other Extension Metadata

Last Modified Date

2015-11-28

IP Status

No known IP claims.

Contributors

- Patrick Doane, Blizzard
- Faith Ekstrand, Intel
- Ian Elliott, LunarG
- Courtney Goeltzenleuchter, LunarG
- Jesse Hall, Google
- James Jones, NVIDIA
- Antoine Labour, Google
- Jon Leech, Khronos
- David Mao, AMD
- Norbert Nopper, Freescale
- Alon Or-bach, Samsung
- Daniel Rakos, AMD
- Graham Sellers, AMD
- Ray Smith, ARM
- Jeff Vigil, Qualcomm
- Chia-I Wu, LunarG

Description

The `VK_KHR_xlib_surface` extension is an instance extension. It provides a mechanism to create a `VkSurfaceKHR` object (defined by the `VK_KHR_surface` extension) that refers to an X11 `Window`, using the Xlib client-side library, as well as a query to determine support for rendering via Xlib.

New Commands

- `vkCreateXlibSurfaceKHR`
- `vkGetPhysicalDeviceXlibPresentationSupportKHR`

New Structures

- `VkXlibSurfaceCreateInfoKHR`

New Bitmasks

- `VkXlibSurfaceCreateFlagsKHR`

New Enum Constants

- `VK_KHR_XLIB_SURFACE_EXTENSION_NAME`
- `VK_KHR_XLIB_SURFACE_SPEC_VERSION`
- Extending `VkStructureType`:

- `VK_STRUCTURE_TYPE_XLIB_SURFACE_CREATE_INFO_KHR`

Issues

1) Does X11 need a way to query for compatibility between a particular physical device and a specific screen? This would be a more general query than `vkGetPhysicalDeviceSurfaceSupportKHR`; if it returned `VK_TRUE`, then the physical device could be assumed to support presentation to any window on that screen.

RESOLVED: Yes, this is needed for toolkits that want to create a `VkDevice` before creating a window. To ensure the query is reliable, it must be made against a particular X visual rather than the screen in general.

Version History

- Revision 1, 2015-09-23 (Jesse Hall)
 - Initial draft, based on the previous contents of `VK_EXT_KHR_swapchain` (later renamed `VK_EXT_KHR_surface`).
- Revision 2, 2015-10-02 (James Jones)
 - Added presentation support query for (Display*, VisualID) pair.
 - Removed “root” parameter from `CreateXlibSurfaceKHR()`, as it is redundant when a window on the same screen is specified as well.
 - Added appropriate X errors.
 - Adjusted wording of issue #1 and added agreed upon resolution.
- Revision 3, 2015-10-14 (Ian Elliott)
 - Renamed this extension from `VK_EXT_KHR_x11_surface` to `VK_EXT_KHR_xlib_surface`.
- Revision 4, 2015-10-26 (Ian Elliott)
 - Renamed from `VK_EXT_KHR_xlib_surface` to `VK_KHR_xlib_surface`.
- Revision 5, 2015-11-03 (Daniel Rakos)
 - Added allocation callbacks to `vkCreateXlibSurfaceKHR`.
- Revision 6, 2015-11-28 (Daniel Rakos)
 - Updated the surface create function to take a `pCreateInfo` structure.

VK_KHR_zero_initialize_workgroup_memory

Name String

`VK_KHR_zero_initialize_workgroup_memory`

Extension Type

Device extension

Registered Extension Number

326

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Deprecation State

- *Promoted* to Vulkan 1.3

Contact

- Alan Baker [\[GitHub\]](#)[alan-baker](#)

Other Extension Metadata

Last Modified Date

2020-11-18

IP Status

No known IP claims.

Contributors

- Alan Baker, Google
- Jeff Bolz, Nvidia
- Faith Ekstrand, Intel

Description

This extension allows the use of a null constant initializer on shader Workgroup memory variables, allowing implementations to expose any special hardware or instructions they may have. Zero initialization is commonly used by applications running untrusted content (e.g. web browsers) as way of defeating memory-scraping attacks.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceZeroInitializeWorkgroupMemoryFeaturesKHR](#)

New Enum Constants

- [VK_KHR_ZERO_INITIALIZE_WORKGROUP_MEMORY_EXTENSION_NAME](#)
- [VK_KHR_ZERO_INITIALIZE_WORKGROUP_MEMORY_SPEC_VERSION](#)
- Extending [VkStructureType](#):

- `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ZERO_INITIALIZE_WORKGROUP_MEMORY_FEATURES_KHR`

Promotion to Vulkan 1.3

Vulkan APIs in this extension are included in core Vulkan 1.3, with the KHR suffix omitted. External interactions defined by this extension, such as SPIR-V token names, retain their original names. The original Vulkan API names are still available as aliases of the core functionality.

Version History

- Revision 1, 2020-11-18 (Alan Baker)
 - Internal draft version

VK_EXT_astc_decode_mode

Name String

`VK_EXT_astc_decode_mode`

Extension Type

Device extension

Registered Extension Number

68

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Jan-Harald Fredriksen [\[GitHub\]janharaldfredriksen-arm](#)

Other Extension Metadata

Last Modified Date

2018-08-07

Contributors

- Jan-Harald Fredriksen, Arm

Description

The existing specification requires that low dynamic range (LDR) ASTC textures are decompressed to FP16 values per component. In many cases, decompressing LDR textures to a lower precision intermediate result gives acceptable image quality. Source material for LDR textures is typically authored as 8-bit UNORM values, so decoding to FP16 values adds little value. On the other hand, reducing precision of the decoded result reduces the size of the decompressed data, potentially improving texture cache performance and saving power.

The goal of this extension is to enable this efficiency gain on existing ASTC texture data. This is achieved by giving the application the ability to select the intermediate decoding precision.

Three decoding options are provided:

- Decode to `VK_FORMAT_R16G16B16A16_SFLOAT` precision: This is the default, and matches the required behavior in the core API.
- Decode to `VK_FORMAT_R8G8B8A8_UNORM` precision: This is provided as an option in LDR mode.
- Decode to `VK_FORMAT_E5B9G9R9_UFLOAT_PACK32` precision: This is provided as an option in both LDR and HDR mode. In this mode, negative values cannot be represented and are clamped to zero. The alpha component is ignored, and the results are as if alpha was 1.0. This decode mode is optional and support can be queried via the physical device properties.

New Structures

- Extending `VkImageViewCreateInfo`:
 - `VkImageViewASTCDecodeModeEXT`
- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceASTCDecodeFeaturesEXT`

New Enum Constants

- `VK_EXT_ASTC_DECODE_MODE_EXTENSION_NAME`
- `VK_EXT_ASTC_DECODE_MODE_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_IMAGE_VIEW_ASTC_DECODE_MODE_EXT`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ASTC_DECODE_FEATURES_EXT`

Issues

1) Are implementations allowed to decode at a higher precision than what is requested?

RESOLUTION: No.

If we allow this, then this extension could be exposed on all implementations that support ASTC.

But developers would have no way of knowing what precision was actually used, and thus whether the image quality is sufficient at reduced

precision.

2) Should the decode mode be image view state and/or sampler state?

RESOLUTION: Image view state only.
Some implementations treat the different decode modes as different texture formats.

Example

Create an image view that decodes to `VK_FORMAT_R8G8B8A8_UNORM` precision:

```
VkImageViewASTCDecodeModeEXT decodeMode =
{
    .sType = VK_STRUCTURE_TYPE_IMAGE_VIEW_ASTC_DECODE_MODE_EXT,
    .pNext = NULL,
    .decodeMode = VK_FORMAT_R8G8B8A8_UNORM
};

VkImageViewCreateInfo createInfo =
{
    .sType = VK_STRUCTURE_TYPE_IMAGE_VIEW_CREATE_INFO,
    .pNext = &decodeMode,
    // flags, image, viewType set to application-desired values
    .format = VK_FORMAT_ASTC_8x8_UNORM_BLOCK,
    // components, subresourceRange set to application-desired values
};

VkImageView imageView;
VkResult result = vkCreateImageView(
    device,
    &createInfo,
    NULL,
    &imageView);
```

Version History

- Revision 1, 2018-08-07 (Jan-Harald Fredriksen)
 - Initial revision

VK_EXT_attachment_feedback_loop_dynamic_state

Name String

`VK_EXT_attachment_feedback_loop_dynamic_state`

Extension Type

Device extension

Registered Extension Number

525

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

and

[VK_EXT_attachment_feedback_loop_layout](#)

Special Uses

- [OpenGL / ES support](#)
- [D3D support](#)

Contact

- Mike Blumenkrantz [\[GitHub\]](#)zmike

Extension Proposal

[VK_EXT_attachment_feedback_loop_dynamic_state](#)

Other Extension Metadata

Last Modified Date

2023-04-28

IP Status

No known IP claims.

Contributors

- Mike Blumenkrantz, Valve
- Daniel Story, Nintendo
- Stu Smith, AMD
- Samuel Pitoiset, Valve
- Ricardo Garcia, Igalia

Description

This extension adds support for setting attachment feedback loops dynamically on command buffers.

New Commands

- [vkCmdSetAttachmentFeedbackLoopEnableEXT](#)

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceAttachmentFeedbackLoopDynamicStateFeaturesEXT](#)

New Enum Constants

- [VK_EXT_ATTACHMENT_FEEDBACK_LOOP_DYNAMIC_STATE_EXTENSION_NAME](#)
- [VK_EXT_ATTACHMENT_FEEDBACK_LOOP_DYNAMIC_STATE_SPEC_VERSION](#)
- Extending [VkDynamicState](#):
 - [VK_DYNAMIC_STATE_ATTACHMENT_FEEDBACK_LOOP_ENABLE_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_DYNAMIC_STATE_FEATURES_EXT](#)

Version History

- Revision 1, 2023-04-28 (Mike Blumenkrantz)
 - Initial revision

VK_EXT_attachment_feedback_loop_layout

Name String

[VK_EXT_attachment_feedback_loop_layout](#)

Extension Type

Device extension

Registered Extension Number

340

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Uses

- [OpenGL / ES support](#)

- [D3D support](#)

Contact

- Joshua Ashton [\[GitHub\]](#)[Joshua-Ashton](#)

Extension Proposal

[VK_EXT_attachment_feedback_loop_layout](#)

Other Extension Metadata

Last Modified Date

2022-04-04

IP Status

No known IP claims.

Contributors

- Joshua Ashton, Valve
- Faith Ekstrand, Collabora
- Bas Nieuwenhuizen, Google
- Samuel Iglesias Gonsálvez, Igalia
- Ralph Potter, Samsung
- Jan-Harald Fredriksen, Arm
- Ricardo Garcia, Igalia

Description

This extension adds a new image layout, [VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT](#), which allows applications to have an image layout in which they are able to both render to and sample/fetch from the same subresource of an image in a given render pass.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceAttachmentFeedbackLoopLayoutFeaturesEXT](#)

New Enum Constants

- [VK_EXT_ATTACHMENT_FEEDBACK_LOOP_LAYOUT_EXTENSION_NAME](#)
- [VK_EXT_ATTACHMENT_FEEDBACK_LOOP_LAYOUT_SPEC_VERSION](#)
- Extending [VkDependencyFlagBits](#):
 - [VK_DEPENDENCY_FEEDBACK_LOOP_BIT_EXT](#)
- Extending [VkImageLayout](#):
 - [VK_IMAGE_LAYOUT_ATTACHMENT_FEEDBACK_LOOP_OPTIMAL_EXT](#)
- Extending [VkImageUsageFlagBits](#):

- `VK_IMAGE_USAGE_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT`
- Extending [VkPipelineCreateFlagBits](#):
 - `VK_PIPELINE_CREATE_COLOR_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT`
 - `VK_PIPELINE_CREATE_DEPTH_STENCIL_ATTACHMENT_FEEDBACK_LOOP_BIT_EXT`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ATTACHMENT_FEEDBACK_LOOP_LAYOUT_FEATURES_EXT`

Version History

- Revision 2, 2022-04-04 (Joshua Ashton)
 - Renamed from VALVE to EXT.
- Revision 1, 2021-03-09 (Joshua Ashton)
 - Initial draft.

VK_EXT_blend_operation_advanced

Name String

`VK_EXT_blend_operation_advanced`

Extension Type

Device extension

Registered Extension Number

149

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Jeff Bolz [\[GitHub\]jeffbolznv](#)

Other Extension Metadata

Last Modified Date

2017-06-12

Contributors

- Jeff Bolz, NVIDIA

Description

This extension adds a number of “advanced” blending operations that **can** be used to perform new color blending operations, many of which are more complex than the standard blend modes provided by unextended Vulkan. This extension requires different styles of usage, depending on the level of hardware support and the enabled features:

- If `VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT::advancedBlendCoherentOperations` is `VK_FALSE`, the new blending operations are supported, but a memory dependency **must** separate each advanced blend operation on a given sample. `VK_ACCESS_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT` is used to synchronize reads using advanced blend operations.
- If `VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT::advancedBlendCoherentOperations` is `VK_TRUE`, advanced blend operations obey primitive order just like basic blend operations.

In unextended Vulkan, the set of blending operations is limited, and **can** be expressed very simply. The `VK_BLEND_OP_MIN` and `VK_BLEND_OP_MAX` blend operations simply compute component-wise minimums or maximums of source and destination color components. The `VK_BLEND_OP_ADD`, `VK_BLEND_OP_SUBTRACT`, and `VK_BLEND_OP_REVERSE_SUBTRACT` modes multiply the source and destination colors by source and destination factors and either add the two products together or subtract one from the other. This limited set of operations supports many common blending operations but precludes the use of more sophisticated transparency and blending operations commonly available in many dedicated imaging APIs.

This extension provides a number of new “advanced” blending operations. Unlike traditional blending operations using `VK_BLEND_OP_ADD`, these blending equations do not use source and destination factors specified by `VkBlendFactor`. Instead, each blend operation specifies a complete equation based on the source and destination colors. These new blend operations are used for both RGB and alpha components; they **must** not be used to perform separate RGB and alpha blending (via different values of color and alpha `VkBlendOp`).

These blending operations are performed using premultiplied colors, where RGB colors **can** be considered premultiplied or non-premultiplied by alpha, according to the `srcPremultiplied` and `dstPremultiplied` members of `VkPipelineColorBlendAdvancedStateCreateInfoEXT`. If a color is considered non-premultiplied, the (R,G,B) color components are multiplied by the alpha component prior to blending. For non-premultiplied color components in the range [0,1], the corresponding premultiplied color component would have values in the range $[0 \times A, 1 \times A]$.

Many of these advanced blending equations are formulated where the result of blending source and destination colors with partial coverage have three separate contributions: from the portions covered by both the source and the destination, from the portion covered only by the source, and from the portion covered only by the destination. The blend parameter `VkPipelineColorBlendAdvancedStateCreateInfoEXT::blendOverlap` **can** be used to specify a correlation between source and destination pixel coverage. If set to `VK_BLEND_OVERLAP_CONJOINT_EXT`, the source and destination are considered to have maximal overlap, as would be the case if drawing two objects on top of each other. If set to `VK_BLEND_OVERLAP_DISJOINT_EXT`, the source and

destination are considered to have minimal overlap, as would be the case when rendering a complex polygon tessellated into individual non-intersecting triangles. If set to `VK_BLEND_OVERLAP_UNCORRELATED_EXT`, the source and destination coverage are assumed to have no spatial correlation within the pixel.

In addition to the coherency issues on implementations not supporting `advancedBlendCoherentOperations`, this extension has several limitations worth noting. First, the new blend operations have a limit on the number of color attachments they **can** be used with, as indicated by `VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT::advancedBlendMaxColorAttachments`. Additionally, blending precision **may** be limited to 16-bit floating-point, which **may** result in a loss of precision and dynamic range for framebuffer formats with 32-bit floating-point components, and in a loss of precision for formats with 12- and 16-bit signed or unsigned normalized integer components.

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceBlendOperationAdvancedFeaturesEXT`
- Extending `VkPhysicalDeviceProperties2`:
 - `VkPhysicalDeviceBlendOperationAdvancedPropertiesEXT`
- Extending `VkPipelineColorBlendStateCreateInfo`:
 - `VkPipelineColorBlendAdvancedStateCreateInfoEXT`

New Enums

- `VkBlendOverlapEXT`

New Enum Constants

- `VK_EXT_BLEND_OPERATION_ADVANCED_EXTENSION_NAME`
- `VK_EXT_BLEND_OPERATION_ADVANCED_SPEC_VERSION`
- Extending `VkAccessFlagBits`:
 - `VK_ACCESS_COLOR_ATTACHMENT_READ_NONCOHERENT_BIT_EXT`
- Extending `VkBlendOp`:
 - `VK_BLEND_OP_BLUE_EXT`
 - `VK_BLEND_OP_COLORBURN_EXT`
 - `VK_BLEND_OP_COLORDODGE_EXT`
 - `VK_BLEND_OP_CONTRAST_EXT`
 - `VK_BLEND_OP_DARKEN_EXT`
 - `VK_BLEND_OP_DIFFERENCE_EXT`
 - `VK_BLEND_OP_DST_ATOP_EXT`
 - `VK_BLEND_OP_DST_EXT`

- VK_BLEND_OP_DST_IN_EXT
- VK_BLEND_OP_DST_OUT_EXT
- VK_BLEND_OP_DST_OVER_EXT
- VK_BLEND_OP_EXCLUSION_EXT
- VK_BLEND_OP_GREEN_EXT
- VK_BLEND_OP_HARDLIGHT_EXT
- VK_BLEND_OP_HARDMIX_EXT
- VK_BLEND_OP_HSL_COLOR_EXT
- VK_BLEND_OP_HSL_HUE_EXT
- VK_BLEND_OP_HSL_LUMINOSITY_EXT
- VK_BLEND_OP_HSL_SATURATION_EXT
- VK_BLEND_OP_INVERT_EXT
- VK_BLEND_OP_INVERT_OVG_EXT
- VK_BLEND_OP_INVERT_RGB_EXT
- VK_BLEND_OP_LIGHTEN_EXT
- VK_BLEND_OP_LINEARBURN_EXT
- VK_BLEND_OP_LINEARDODGE_EXT
- VK_BLEND_OP_LINEARLIGHT_EXT
- VK_BLEND_OP_MINUS_CLAMPED_EXT
- VK_BLEND_OP_MINUS_EXT
- VK_BLEND_OP_MULTIPLY_EXT
- VK_BLEND_OP_OVERLAY_EXT
- VK_BLEND_OP_PINLIGHT_EXT
- VK_BLEND_OP_PLUS_CLAMPED_ALPHA_EXT
- VK_BLEND_OP_PLUS_CLAMPED_EXT
- VK_BLEND_OP_PLUS_DARKER_EXT
- VK_BLEND_OP_PLUS_EXT
- VK_BLEND_OP_RED_EXT
- VK_BLEND_OP_SCREEN_EXT
- VK_BLEND_OP_SOFTLIGHT_EXT
- VK_BLEND_OP_SRC_ATOP_EXT
- VK_BLEND_OP_SRC_EXT
- VK_BLEND_OP_SRC_IN_EXT
- VK_BLEND_OP_SRC_OUT_EXT
- VK_BLEND_OP_SRC_OVER_EXT

- `VK_BLEND_OP_VIVIDLIGHT_EXT`
- `VK_BLEND_OP_XOR_EXT`
- `VK_BLEND_OP_ZERO_EXT`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_FEATURES_EXT`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BLEND_OPERATION_ADVANCED_PROPERTIES_EXT`
 - `VK_STRUCTURE_TYPE_PIPELINE_COLOR_BLEND_ADVANCED_STATE_CREATE_INFO_EXT`

Issues

None.

Version History

- Revision 1, 2017-06-12 (Jeff Bolz)
 - Internal revisions
- Revision 2, 2017-06-12 (Jeff Bolz)
 - Internal revisions

VK_EXT_border_color_swizzle

Name String

`VK_EXT_border_color_swizzle`

Extension Type

Device extension

Registered Extension Number

412

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_EXT_custom_border_color](#)

Special Uses

- [OpenGL / ES support](#)
- [D3D support](#)

Contact

- Piers Daniell [\[GitHub\]](#)[pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2021-10-12

IP Status

No known IP claims.

Contributors

- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, Arm
- Ricardo Garcia, Igalia
- Shahbaz Youssefi, Google
- Stu Smith, AMD

Description

After the publication of `VK_EXT_custom_border_color`, it was discovered that some implementations had undefined behavior when combining a sampler that uses a custom border color with image views whose component mapping is not the identity mapping.

Since `VK_EXT_custom_border_color` has already shipped, this new extension `VK_EXT_border_color_swizzle` was created to define the interaction between custom border colors and non-identity image view swizzles, and provide a work-around for implementations that must pre-swizzle the sampler border color to match the image view component mapping it is combined with.

This extension also defines the behavior between samplers with an opaque black border color and image views with a non-identity component swizzle, which was previously left undefined.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceBorderColorSwizzleFeaturesEXT](#)
- Extending [VkSamplerCreateInfo](#):
 - [VkSamplerBorderColorComponentMappingCreateInfoEXT](#)

New Enum Constants

- `VK_EXT_BORDER_COLOR_SWIZZLE_EXTENSION_NAME`
- `VK_EXT_BORDER_COLOR_SWIZZLE_SPEC_VERSION`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_BORDER_COLOR_SWIZZLE_FEATURES_EXT`
 - `VK_STRUCTURE_TYPE_SAMPLER_BORDER_COLOR_COMPONENT_MAPPING_CREATE_INFO_EXT`

Issues

None.

Version History

- Revision 1, 2021-10-12 (Piers Daniell)
 - Internal revisions.

VK_EXT_color_write_enable

Name String

VK_EXT_color_write_enable

Extension Type

Device extension

Registered Extension Number

382

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Sharif Elcott [\[GitHub\]](#)selcott

Other Extension Metadata

Last Modified Date

2020-02-25

IP Status

No known IP claims.

Contributors

- Sharif Elcott, Google
- Tobias Hector, AMD
- Piers Daniell, NVIDIA

Description

This extension allows for selectively enabling and disabling writes to output color attachments via a pipeline dynamic state.

The intended use cases for this new state are mostly identical to those of `colorWriteMask`, such as selectively disabling writes to avoid feedback loops between subpasses or bandwidth savings for unused outputs. By making the state dynamic, one additional benefit is the ability to reduce pipeline counts and pipeline switching via shaders that write a superset of the desired data of which subsets are selected dynamically. The reason for a new state, `colorWriteEnable`, rather than making `colorWriteMask` dynamic is that, on many implementations, the more flexible per-component semantics of the `colorWriteMask` state cannot be made dynamic in a performant manner.

New Commands

- [vkCmdSetColorWriteEnableEXT](#)

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceColorWriteEnableFeaturesEXT](#)
- Extending [VkPipelineColorBlendStateCreateInfo](#):
 - [VkPipelineColorWriteCreateInfoEXT](#)

New Enum Constants

- [VK_EXT_COLOR_WRITE_ENABLE_EXTENSION_NAME](#)
- [VK_EXT_COLOR_WRITE_ENABLE_SPEC_VERSION](#)
- Extending [VkDynamicState](#):
 - [VK_DYNAMIC_STATE_COLOR_WRITE_ENABLE_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_COLOR_WRITE_ENABLE_FEATURES_EXT](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_COLOR_WRITE_CREATE_INFO_EXT](#)

Version History

- Revision 1, 2020-01-25 (Sharif Elcott)
 - Internal revisions

VK_EXT_conservative_rasterization

Name String

[VK_EXT_conservative_rasterization](#)

Extension Type

Device extension

Registered Extension Number

102

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_EXT_fragment_fully_covered](#)

Contact

- Piers Daniell [\[GitHub\]](#)[pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2020-06-09

Interactions and External Dependencies

- This extension requires [SPV_EXT_fragment_fully_covered](#) if the [VkPhysicalDeviceConservativeRasterizationPropertiesEXT::fullyCoveredFragmentShaderInputVariable](#) feature is used.
- This extension requires [SPV_KHR_post_depth_coverage](#) if the [VkPhysicalDeviceConservativeRasterizationPropertiesEXT::conservativeRasterizationPostDepthCoverage](#) feature is used.
- This extension provides API support for [GL_NV_conservative_raster_underestimation](#) if the [VkPhysicalDeviceConservativeRasterizationPropertiesEXT::fullyCoveredFragmentShaderInputVariable](#) feature is used.

Contributors

- Daniel Koch, NVIDIA
- Daniel Rakos, AMD
- Jeff Bolz, NVIDIA
- Slawomir Grajewski, Intel
- Stu Smith, Imagination Technologies

Description

This extension adds a new rasterization mode called conservative rasterization. There are two modes of conservative rasterization; overestimation and underestimation.

When overestimation is enabled, if any part of the primitive, including its edges, covers any part of the rectangular pixel area, including its sides, then a fragment is generated with all coverage samples turned on. This extension allows for some variation in implementations by accounting for differences in overestimation, where the generating primitive size is increased at each of its edges by some sub-pixel amount to further increase conservative pixel coverage. Implementations can allow the application to specify an extra overestimation beyond the base overestimation the implementation already does. It also allows implementations to either cull degenerate primitives or rasterize them.

When underestimation is enabled, fragments are only generated if the rectangular pixel area is fully covered by the generating primitive. If supported by the implementation, when a pixel rectangle is fully covered the fragment shader input variable builtin called FullyCoveredEXT is set to true. The shader variable works in either overestimation or underestimation mode.

Implementations can process degenerate triangles and lines by either discarding them or generating conservative fragments for them. Degenerate triangles are those that end up with zero area after the rasterizer quantizes them to the fixed-point pixel grid. Degenerate lines are those with zero length after quantization.

New Structures

- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceConservativeRasterizationPropertiesEXT](#)
- Extending [VkPipelineRasterizationStateCreateInfo](#):
 - [VkPipelineRasterizationConservativeStateCreateInfoEXT](#)

New Enums

- [VkConservativeRasterizationModeEXT](#)

New Bitmasks

- [VkPipelineRasterizationConservativeStateCreateFlagsEXT](#)

New Enum Constants

- [VK_EXT_CONSERVATIVE_RASTERIZATION_EXTENSION_NAME](#)
- [VK_EXT_CONSERVATIVE_RASTERIZATION_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CONSERVATIVE_RASTERIZATION_PROPERTIES_EXT](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_CONSERVATIVE_STATE_CREATE_INFO_EXT](#)

New Built-In Variables

- [FullyCoveredEXT](#)

New SPIR-V Capabilities

- [FragmentFullyCoveredEXT](#)

Version History

- Revision 1.1, 2020-09-06 (Piers Daniell)
 - Add missing SPIR-V and GLSL dependencies.
- Revision 1, 2017-08-28 (Piers Daniell)
 - Internal revisions

VK_EXT_custom_border_color

Name String

VK_EXT_custom_border_color

Extension Type

Device extension

Registered Extension Number

288

Revision

12

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Uses

- [OpenGL / ES support](#)
- [D3D support](#)

Contact

- Liam Middlebrook [[GitHub](#)][liam-middlebrook](#)

Other Extension Metadata

Last Modified Date

2020-04-16

IP Status

No known IP claims.

Contributors

- Joshua Ashton, Valve
- Hans-Kristian Arntzen, Valve
- Philip Rebohle, Valve
- Liam Middlebrook, NVIDIA
- Jeff Bolz, NVIDIA
- Tobias Hector, AMD
- Faith Ekstrand, Intel
- Spencer Fricke, Samsung Electronics
- Graeme Leese, Broadcom
- Jesse Hall, Google
- Jan-Harald Fredriksen, ARM
- Tom Olson, ARM
- Stuart Smith, Imagination Technologies
- Donald Scorgie, Imagination Technologies
- Alex Walters, Imagination Technologies
- Peter Quayle, Imagination Technologies

Description

This extension provides cross-vendor functionality to specify a custom border color for use when the sampler address mode `VK_SAMPLER_ADDRESS_MODE_CLAMP_TO_BORDER` is used.

To create a sampler which uses a custom border color set `VkSamplerCreateInfo::borderColor` to one of:

- `VK_BORDER_COLOR_FLOAT_CUSTOM_EXT`
- `VK_BORDER_COLOR_INT_CUSTOM_EXT`

When `VK_BORDER_COLOR_FLOAT_CUSTOM_EXT` or `VK_BORDER_COLOR_INT_CUSTOM_EXT` is used, applications must provide a `VkSamplerCustomBorderColorCreateInfoEXT` in the `pNext` chain for `VkSamplerCreateInfo`.

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceCustomBorderColorFeaturesEXT`
- Extending `VkPhysicalDeviceProperties2`:
 - `VkPhysicalDeviceCustomBorderColorPropertiesEXT`

- Extending [VkSamplerCreateInfo](#):
 - [VkSamplerCustomBorderColorCreateInfoEXT](#)

New Enum Constants

- [VK_EXT_CUSTOM_BORDER_COLOR_EXTENSION_NAME](#)
- [VK_EXT_CUSTOM_BORDER_COLOR_SPEC_VERSION](#)
- Extending [VkBorderColor](#):
 - [VK_BORDER_COLOR_FLOAT_CUSTOM_EXT](#)
 - [VK_BORDER_COLOR_INT_CUSTOM_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_FEATURES_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_CUSTOM_BORDER_COLOR_PROPERTIES_EXT](#)
 - [VK_STRUCTURE_TYPE_SAMPLER_CUSTOM_BORDER_COLOR_CREATE_INFO_EXT](#)

Issues

1) Should `VkClearColorValue` be used for the border color value, or should we have our own struct/union? Do we need to specify the type of the input values for the components? This is more of a concern if `VkClearColorValue` is used here because it provides a union of float,int,uint types.

RESOLVED: Will reuse existing `VkClearColorValue` structure in order to easily take advantage of float,int,uint border color types.

2) For hardware which supports a limited number of border colors what happens if that number is exceeded? Should this be handled by the driver unbeknownst to the application? In Revision 1 we had solved this issue using a new Object type, however that may have lead to additional system resource consumption which would otherwise not be required.

RESOLVED: Added `VkPhysicalDeviceCustomBorderColorPropertiesEXT::maxCustomBorderColorSamplers` for tracking implementation-specific limit, and Valid Usage statement handling overflow.

3) Should this be supported for immutable samplers at all, or by a feature bit? Some implementations may not be able to support custom border colors on immutable samplers — is it worthwhile enabling this to work on them for implementations that can support it, or forbidding it entirely.

RESOLVED: Samplers created with a custom border color are forbidden from being immutable. This resolves concerns for implementations where the custom border color is an index to a LUT instead of being directly embedded into sampler state.

4) Should `UINT` and `SINT` (unsigned integer and signed integer) border color types be separated or should they be combined into one generic `INT` (integer) type?

RESOLVED: Separating these does not make much sense as the existing fixed border color types do not have this distinction, and there is no reason in hardware to do so. This separation would also create unnecessary work and considerations for the application.

Version History

- Revision 1, 2019-10-10 (Joshua Ashton)
 - Internal revisions.
- Revision 2, 2019-10-11 (Liam Middlebrook)
 - Remove `VkCustomBorderColor` object and associated functions
 - Add issues concerning HW limitations for custom border color count
- Revision 3, 2019-10-12 (Joshua Ashton)
 - Re-expose the limits for the maximum number of unique border colors
 - Add extra details about border color tracking
 - Fix typos
- Revision 4, 2019-10-12 (Joshua Ashton)
 - Changed `maxUniqueCustomBorderColors` to a `uint32_t` from a `VkDeviceSize`
- Revision 5, 2019-10-14 (Liam Middlebrook)
 - Added features bit
- Revision 6, 2019-10-15 (Joshua Ashton)
 - Type-ize `VK_BORDER_COLOR_CUSTOM`
 - Fix const-ness on `pNext` of `VkSamplerCustomBorderColorCreateInfoEXT`
- Revision 7, 2019-11-26 (Liam Middlebrook)
 - Renamed `maxUniqueCustomBorderColors` to `maxCustomBorderColors`
- Revision 8, 2019-11-29 (Joshua Ashton)
 - Renamed `borderColor` member of `VkSamplerCustomBorderColorCreateInfoEXT` to `customBorderColor`
- Revision 9, 2020-02-19 (Joshua Ashton)
 - Renamed `maxCustomBorderColors` to `maxCustomBorderColorSamplers`
- Revision 10, 2020-02-21 (Joshua Ashton)
 - Added format to `VkSamplerCustomBorderColorCreateInfoEXT` and feature bit
- Revision 11, 2020-04-07 (Joshua Ashton)
 - Dropped `UINT/SINT` border color differences, consolidated types
- Revision 12, 2020-04-16 (Joshua Ashton)
 - Renamed `VK_BORDER_COLOR_CUSTOM_FLOAT_EXT` to `VK_BORDER_COLOR_FLOAT_CUSTOM_EXT` for consistency

VK_EXT_debug_utils

Name String

`VK_EXT_debug_utils`

Extension Type

Instance extension

Registered Extension Number

129

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

None

Special Use

- [Debugging tools](#)

Contact

- Mark Young [\[GitHub\]marky-lunarg](#)

Other Extension Metadata**Last Modified Date**

2020-04-03

Revision

2

IP Status

No known IP claims.

Dependencies

- This extension is written against version 1.0 of the Vulkan API.
- Requires [VkObjectType](#)

Contributors

- Mark Young, LunarG
- Baldur Karlsson
- Ian Elliott, Google
- Courtney Goeltzenleuchter, Google
- Karl Schultz, LunarG
- Mark Lobodzinski, LunarG
- Mike Schuchardt, LunarG
- Jaakko Konttinen, AMD

- Dan Ginsburg, Valve Software
- Rolando Olivares, Epic Games
- Dan Baker, Oxide Games
- Kyle Spagnoli, NVIDIA
- Jon Ashburn, LunarG
- Piers Daniell, NVIDIA

Description

Due to the nature of the Vulkan interface, there is very little error information available to the developer and application. By using the `VK_EXT_debug_utils` extension, developers **can** obtain more information. When combined with validation layers, even more detailed feedback on the application's use of Vulkan will be provided.

This extension provides the following capabilities:

- The ability to create a debug messenger which will pass along debug messages to an application supplied callback.
- The ability to identify specific Vulkan objects using a name or tag to improve tracking.
- The ability to identify specific sections within a `VkQueue` or `VkCommandBuffer` using labels to aid organization and offline analysis in external tools.

The main difference between this extension and `VK_EXT_debug_report` and `VK_EXT_debug_marker` is that those extensions use `VkDebugReportObjectTypeEXT` to identify objects. This extension uses the core `VkObjectType` in place of `VkDebugReportObjectTypeEXT`. The primary reason for this move is that no future object type handle enumeration values will be added to `VkDebugReportObjectTypeEXT` since the creation of `VkObjectType`.

In addition, this extension combines the functionality of both `VK_EXT_debug_report` and `VK_EXT_debug_marker` by allowing object name and debug markers (now called labels) to be returned to the application's callback function. This should assist in clarifying the details of a debug message including: what objects are involved and potentially which location within a `VkQueue` or `VkCommandBuffer` the message occurred.

New Object Types

- `VkDebugUtilsMessengerEXT`

New Commands

- `vkCmdBeginDebugUtilsLabelEXT`
- `vkCmdEndDebugUtilsLabelEXT`
- `vkCmdInsertDebugUtilsLabelEXT`
- `vkCreateDebugUtilsMessengerEXT`
- `vkDestroyDebugUtilsMessengerEXT`

- [vkQueueBeginDebugUtilsLabelEXT](#)
- [vkQueueEndDebugUtilsLabelEXT](#)
- [vkQueueInsertDebugUtilsLabelEXT](#)
- [vkSetDebugUtilsObjectNameEXT](#)
- [vkSetDebugUtilsObjectTagEXT](#)
- [vkSubmitDebugUtilsMessageEXT](#)

New Structures

- [VkDebugUtilsLabelEXT](#)
- [VkDebugUtilsMessengerCallbackDataEXT](#)
- [VkDebugUtilsObjectTagInfoEXT](#)
- Extending [VkInstanceCreateInfo](#):
 - [VkDebugUtilsMessengerCreateInfoEXT](#)
- Extending [VkPipelineShaderStageCreateInfo](#):
 - [VkDebugUtilsObjectNameInfoEXT](#)

New Function Pointers

- [PFN_vkDebugUtilsMessengerCallbackEXT](#)

New Enums

- [VkDebugUtilsMessageSeverityFlagBitsEXT](#)
- [VkDebugUtilsMessageTypeFlagBitsEXT](#)

New Bitmasks

- [VkDebugUtilsMessageSeverityFlagsEXT](#)
- [VkDebugUtilsMessageTypeFlagsEXT](#)
- [VkDebugUtilsMessengerCallbackDataFlagsEXT](#)
- [VkDebugUtilsMessengerCreateFlagsEXT](#)

New Enum Constants

- [VK_EXT_DEBUG_UTILS_EXTENSION_NAME](#)
- [VK_EXT_DEBUG_UTILS_SPEC_VERSION](#)
- Extending [VkObjectType](#):
 - [VK_OBJECT_TYPE_DEBUG_UTILS_MESSENGER_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT](#)

- `VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CALLBACK_DATA_EXT`
- `VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CREATE_INFO_EXT`
- `VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_NAME_INFO_EXT`
- `VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_TAG_INFO_EXT`

Examples

Example 1

`VK_EXT_debug_utils` allows an application to register multiple callbacks with any Vulkan component wishing to report debug information. Some callbacks may log the information to a file, others may cause a debug break point or other application-defined behavior. An application **can** register callbacks even when no validation layers are enabled, but they will only be called for loader and, if implemented, driver events.

To capture events that occur while creating or destroying an instance an application **can** link a `VkDebugUtilsMessengerCreateInfoEXT` structure to the `pNext` element of the `VkInstanceCreateInfo` structure given to `vkCreateInstance`.

Example uses: Create three callback objects. One will log errors and warnings to the debug console using Windows `OutputDebugString`. The second will cause the debugger to break at that callback when an error happens and the third will log warnings to stdout.

```
extern VkInstance instance;
VkResult res;
VkDebugUtilsMessengerEXT cb1, cb2, cb3;

// Must call extension functions through a function pointer:
PFN_vkCreateDebugUtilsMessengerEXT pfnCreateDebugUtilsMessengerEXT =
(PFN_vkCreateDebugUtilsMessengerEXT)vkGetInstanceProcAddr(instance,
"vkCreateDebugUtilsMessengerEXT");
PFN_vkDestroyDebugUtilsMessengerEXT pfnDestroyDebugUtilsMessengerEXT =
(PFN_vkDestroyDebugUtilsMessengerEXT)vkGetInstanceProcAddr(instance,
"vkDestroyDebugUtilsMessengerEXT");

VkDebugUtilsMessengerCreateInfoEXT callback1 = {
    .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CREATE_INFO_EXT,
    .pNext = NULL,
    .flags = 0,
    .messageSeverity = VK_DEBUG_UTILS_MESSAGE_SEVERITY_ERROR_BIT_EXT |
                      VK_DEBUG_UTILS_MESSAGE_SEVERITY_WARNING_BIT_EXT,
    .messageType = VK_DEBUG_UTILS_MESSAGE_TYPE_GENERAL_BIT_EXT |
                  VK_DEBUG_UTILS_MESSAGE_TYPE_VALIDATION_BIT_EXT,
    .pfnUserCallback = myOutputDebugString,
    .pUserData = NULL
};
res = pfnCreateDebugUtilsMessengerEXT(instance, &callback1, NULL, &cb1);
if (res != VK_SUCCESS) {
    // Do error handling for VK_ERROR_OUT_OF_MEMORY
```

```

}

callback1.messageSeverity = VK_DEBUG_UTILS_MESSAGE_SEVERITY_ERROR_BIT_EXT;
callback1.pfnUserCallback = myDebugBreak;
callback1.pUserData = NULL;
res = pfnCreateDebugUtilsMessengerEXT(instance, &callback1, NULL, &cb2);
if (res != VK_SUCCESS) {
    // Do error handling for VK_ERROR_OUT_OF_MEMORY
}

VkDebugUtilsMessengerCreateInfoEXT callback3 = {
    .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_MESSENGER_CREATE_INFO_EXT,
    .pNext = NULL,
    .flags = 0,
    .messageSeverity = VK_DEBUG_UTILS_MESSAGE_SEVERITY_WARNING_BIT_EXT,
    .messageType = VK_DEBUG_UTILS_MESSAGE_TYPE_GENERAL_BIT_EXT |
        VK_DEBUG_UTILS_MESSAGE_TYPE_VALIDATION_BIT_EXT,
    .pfnUserCallback = myStdOutLogger,
    .pUserData = NULL
};
res = pfnCreateDebugUtilsMessengerEXT(instance, &callback3, NULL, &cb3);
if (res != VK_SUCCESS) {
    // Do error handling for VK_ERROR_OUT_OF_MEMORY
}

...

// Remove callbacks when cleaning up
pfnDestroyDebugUtilsMessengerEXT(instance, cb1, NULL);
pfnDestroyDebugUtilsMessengerEXT(instance, cb2, NULL);
pfnDestroyDebugUtilsMessengerEXT(instance, cb3, NULL);

```

Example 2

Associate a name with an image, for easier debugging in external tools or with validation layers that can print a friendly name when referring to objects in error messages.

```

extern VkInstance instance;
extern VkDevice device;
extern VkImage image;

// Must call extension functions through a function pointer:
PFN_vkSetDebugUtilsObjectNameEXT pfnSetDebugUtilsObjectNameEXT =
(PFN_vkSetDebugUtilsObjectNameEXT)vkGetInstanceProcAddr(instance,
"vkSetDebugUtilsObjectNameEXT");

// Set a name on the image
const VkDebugUtilsObjectNameInfoEXT imageNameInfo =
{
    .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_OBJECT_NAME_INFO_EXT,

```

```

        .pNext = NULL,
        .objectType = VK_OBJECT_TYPE_IMAGE,
        .objectHandle = (uint64_t)image,
        .pObjectName = "Brick Diffuse Texture",
    };

    pfnSetDebugUtilsObjectNameEXT(device, &imageNameInfo);

    // A subsequent error might print:
    //   Image 'Brick Diffuse Texture' (0xc0dec0dedeadbeef) is used in a
    //   command buffer with no memory bound to it.

```

Example 3

Annotating regions of a workload with naming information so that offline analysis tools can display a more usable visualization of the commands submitted.

```

extern VkInstance instance;
extern VkCommandBuffer commandBuffer;

// Must call extension functions through a function pointer:
PFN_vkQueueBeginDebugUtilsLabelEXT pfnQueueBeginDebugUtilsLabelEXT =
(PFN_vkQueueBeginDebugUtilsLabelEXT)vkGetInstanceProcAddr(instance,
"vkQueueBeginDebugUtilsLabelEXT");
PFN_vkQueueEndDebugUtilsLabelEXT pfnQueueEndDebugUtilsLabelEXT =
(PFN_vkQueueEndDebugUtilsLabelEXT)vkGetInstanceProcAddr(instance,
"vkQueueEndDebugUtilsLabelEXT");
PFN_vkCmdBeginDebugUtilsLabelEXT pfnCmdBeginDebugUtilsLabelEXT =
(PFN_vkCmdBeginDebugUtilsLabelEXT)vkGetInstanceProcAddr(instance,
"vkCmdBeginDebugUtilsLabelEXT");
PFN_vkCmdEndDebugUtilsLabelEXT pfnCmdEndDebugUtilsLabelEXT =
(PFN_vkCmdEndDebugUtilsLabelEXT)vkGetInstanceProcAddr(instance,
"vkCmdEndDebugUtilsLabelEXT");
PFN_vkCmdInsertDebugUtilsLabelEXT pfnCmdInsertDebugUtilsLabelEXT =
(PFN_vkCmdInsertDebugUtilsLabelEXT)vkGetInstanceProcAddr(instance,
"vkCmdInsertDebugUtilsLabelEXT");

// Describe the area being rendered
const VkDebugUtilsLabelEXT houseLabel =
{
    .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT,
    .pNext = NULL,
    .pLabelName = "Brick House",
    .color = { 1.0f, 0.0f, 0.0f, 1.0f },
};

// Start an annotated group of calls under the 'Brick House' name
pfnCmdBeginDebugUtilsLabelEXT(commandBuffer, &houseLabel);
{
    // A mutable structure for each part being rendered

```

```

VkDebugUtilsLabelEXT housePartLabel =
{
    .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT,
    .pNext = NULL,
    .pLabelName = NULL,
    .color = { 0.0f, 0.0f, 0.0f, 0.0f },
};

// Set the name and insert the marker
housePartLabel.pLabelName = "Walls";
pfnCmdInsertDebugUtilsLabelEXT(commandBuffer, &housePartLabel);

// Insert the drawcall for the walls
vkCmdDrawIndexed(commandBuffer, 1000, 1, 0, 0, 0);

// Insert a recursive region for two sets of windows
housePartLabel.pLabelName = "Windows";
pfnCmdBeginDebugUtilsLabelEXT(commandBuffer, &housePartLabel);
{
    vkCmdDrawIndexed(commandBuffer, 75, 6, 1000, 0, 0);
    vkCmdDrawIndexed(commandBuffer, 100, 2, 1450, 0, 0);
}
pfnCmdEndDebugUtilsLabelEXT(commandBuffer);

housePartLabel.pLabelName = "Front Door";
pfnCmdInsertDebugUtilsLabelEXT(commandBuffer, &housePartLabel);

vkCmdDrawIndexed(commandBuffer, 350, 1, 1650, 0, 0);

housePartLabel.pLabelName = "Roof";
pfnCmdInsertDebugUtilsLabelEXT(commandBuffer, &housePartLabel);

vkCmdDrawIndexed(commandBuffer, 500, 1, 2000, 0, 0);
}
// End the house annotation started above
pfnCmdEndDebugUtilsLabelEXT(commandBuffer);

// Do other work

vkEndCommandBuffer(commandBuffer);

// Describe the queue being used
const VkDebugUtilsLabelEXT queueLabel =
{
    .sType = VK_STRUCTURE_TYPE_DEBUG_UTILS_LABEL_EXT,
    .pNext = NULL,
    .pLabelName = "Main Render Work",
    .color = { 0.0f, 1.0f, 0.0f, 1.0f },
};

// Identify the queue label region

```

```

pfnQueueBeginDebugUtilsLabelEXT(queue, &queueLabel);

// Submit the work for the main render thread
const VkCommandBuffer cmd_bufs[] = {commandBuffer};
VkSubmitInfo submit_info =
{
    .sType = VK_STRUCTURE_TYPE_SUBMIT_INFO,
    .pNext = NULL,
    .waitSemaphoreCount = 0,
    .pWaitSemaphores = NULL,
    .pWaitDstStageMask = NULL,
    .commandBufferCount = 1,
    .pCommandBuffers = cmd_bufs,
    .signalSemaphoreCount = 0,
    .pSignalSemaphores = NULL
};
vkQueueSubmit(queue, 1, &submit_info, fence);

// End the queue label region
pfnQueueEndDebugUtilsLabelEXT(queue);

```

Issues

1) Should we just name this extension `VK_EXT_debug_report2`

RESOLVED: No. There is enough additional changes to the structures to break backwards compatibility. So, a new name was decided that would not indicate any interaction with the previous extension.

2) Will validation layers immediately support all the new features.

RESOLVED: Not immediately. As one can imagine, there is a lot of work involved with converting the validation layer logging over to the new functionality. Basic logging, as seen in the origin `VK_EXT_debug_report` extension will be made available immediately. However, adding the labels and object names will take time. Since the priority for Khronos at this time is to continue focusing on Valid Usage statements, it may take a while before the new functionality is fully exposed.

3) If the validation layers will not expose the new functionality immediately, then what is the point of this extension?

RESOLVED: We needed a replacement for `VK_EXT_debug_report` because the `VkDebugReportObjectTypeEXT` enumeration will no longer be updated and any new objects will need to be debugged using the new functionality provided by this extension.

4) Should this extension be split into two separate parts (1 extension that is an instance extension providing the callback functionality, and another device extension providing the general debug marker and annotation functionality)?

RESOLVED: No, the functionality for this extension is too closely related. If we did split up the extension, where would the structures and enums live, and how would you define that the device

behavior in the instance extension is really only valid if the device extension is enabled, and the functionality is passed in. It is cleaner to just define this all as an instance extension, plus it allows the application to enable all debug functionality provided with one enable string during [vkCreateInstance](#).

Version History

- Revision 1, 2017-09-14 (Mark Young and all listed Contributors)
 - Initial draft, based on [VK_EXT_debug_report](#) and [VK_EXT_debug_marker](#) in addition to previous feedback supplied from various companies including Valve, Epic, and Oxide games.
- Revision 2, 2020-04-03 (Mark Young and Piers Daniell)
 - Updated to allow either [NULL](#) or an empty string to be passed in for [pObjectName](#) in [VkDebugUtilsObjectNameInfoEXT](#), because the loader and various drivers support [NULL](#) already.

VK_EXT_depth_bias_control

Name String

[VK_EXT_depth_bias_control](#)

Extension Type

Device extension

Registered Extension Number

284

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Use

- [D3D support](#)

Contact

- Joshua Ashton [\[GitHub\]](#)[Joshua-Ashton](#)

Extension Proposal

[VK_EXT_depth_bias_control](#)

Other Extension Metadata

Last Modified Date

2023-02-15

IP Status

No known IP claims.

Contributors

- Joshua Ashton, VALVE
- Hans-Kristian Arntzen, VALVE
- Mike Blumenkrantz, VALVE
- Georg Lehmann, VALVE
- Piers Daniell, NVIDIA
- Lionel Landwerlin, INTEL
- Tobias Hector, AMD
- Ricardo Garcia, IGALIA
- Jan-Harald Fredriksen, ARM
- Shahbaz Youssefi, GOOGLE
- Tom Olson, ARM

Description

This extension adds a new structure, [VkDepthBiasRepresentationInfoEXT](#), that can be added to a [pNext](#) chain of [VkPipelineRasterizationStateCreateInfo](#) and allows setting the scaling and representation of depth bias for a pipeline.

This state can also be set dynamically by using the new structure mentioned above in combination with the new [vkCmdSetDepthBias2EXT](#) command.

New Commands

- [vkCmdSetDepthBias2EXT](#)

New Structures

- [VkDepthBiasInfoEXT](#)
- Extending [VkDepthBiasInfoEXT](#), [VkPipelineRasterizationStateCreateInfo](#):
 - [VkDepthBiasRepresentationInfoEXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceDepthBiasControlFeaturesEXT](#)

New Enums

- [VkDepthBiasRepresentationEXT](#)

New Enum Constants

- `VK_EXT_DEPTH_BIAS_CONTROL_EXTENSION_NAME`
- `VK_EXT_DEPTH_BIAS_CONTROL_SPEC_VERSION`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_DEPTH_BIAS_INFO_EXT`
 - `VK_STRUCTURE_TYPE_DEPTH_BIAS_REPRESENTATION_INFO_EXT`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_BIAS_CONTROL_FEATURES_EXT`

Version History

- Revision 1, 2022-09-22 (Joshua Ashton)
 - Initial draft.

VK_EXT_depth_clamp_control

Name String

`VK_EXT_depth_clamp_control`

Extension Type

Device extension

Registered Extension Number

583

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Jules Blok [\[GitHub\]](#)jules

Extension Proposal

[VK_EXT_depth_clamp_control](#)

Other Extension Metadata

Last Modified Date

2024-07-15

Contributors

- Jules Blok, Independent

Description

This extension allows the application to control the viewport depth clamp range separately from the viewport `minDepth` and `maxDepth`. This gives the ability for the application to restrict depth values to an application-defined range rather than the viewport depth range.

It can be used to set a smaller or larger clamping range than the viewport depth range without affecting the depth mapping of the viewport transform.

New Commands

- [vkCmdSetDepthClampRangeEXT](#)

New Structures

- [VkDepthClampRangeEXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceDepthClampControlFeaturesEXT](#)
- Extending [VkPipelineViewportStateCreateInfo](#):
 - [VkPipelineViewportDepthClampControlCreateInfoEXT](#)

New Enums

- [VkDepthClampModeEXT](#)

New Enum Constants

- [VK_EXT_DEPTH_CLAMP_CONTROL_EXTENSION_NAME](#)
- [VK_EXT_DEPTH_CLAMP_CONTROL_SPEC_VERSION](#)
- Extending [VkDynamicState](#):
 - [VK_DYNAMIC_STATE_DEPTH_CLAMP_RANGE_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLAMP_CONTROL_FEATURES_EXT](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLAMP_CONTROL_CREATE_INFO_EXT](#)

Issues

1) Should the depth clamp range be a per-viewport parameter?

RESOLVED: No. Because the depth clamp range was previously defined to be equal to the viewport depth range, conformant runtimes are already handling the depth clamp range as a per-viewport parameter. However because of complexities from interactions with multiple viewports a per-viewport clamp range is left to a future extensions if a use case arises.

2) Should this pipeline state be dynamic?

RESOLVED: Yes. Since the viewport depth range can already be a dynamic state conformant runtimes are already able to handle the depth clamp range as a dynamic state.

3) Can the depth clamp range be ignored when depth clamping is disabled?

RESOLVED: Yes. This extension overrides the clamping range used only when depth clamping is enabled. The alternative would be highly unintuitive. As a consequence the [VK_EXT_depth_clip_enable](#) extension is required if depth clipping is desired in combination with this extension.

Version History

- Revision 1, 2024-02-13 (Jules Blok)
 - Initial draft

VK_EXT_depth_clip_control

Name String

VK_EXT_depth_clip_control

Extension Type

Device extension

Registered Extension Number

356

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Use

- [OpenGL / ES support](#)

Contact

- Shahbaz Youssefi [\[GitHub\]syoussefi](#)

Other Extension Metadata

Last Modified Date

2021-11-09

Contributors

- Spencer Fricke, Samsung Electronics
- Shahbaz Youssefi, Google
- Ralph Potter, Samsung Electronics

Description

This extension allows the application to use the OpenGL depth range in NDC, i.e. with depth in range [-1, 1], as opposed to Vulkan's default of [0, 1]. The purpose of this extension is to allow efficient layering of OpenGL over Vulkan, by avoiding emulation in the pre-rasterization shader stages. This emulation, which effectively duplicates `gl_Position` but with a different depth value, costs ALU and consumes shader output components that the implementation may not have to spare to meet OpenGL minimum requirements.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceDepthClipControlFeaturesEXT](#)
- Extending [VkPipelineViewportStateCreateInfo](#):
 - [VkPipelineViewportDepthClipControlCreateInfoEXT](#)

New Enum Constants

- [VK_EXT_DEPTH_CLIP_CONTROL_EXTENSION_NAME](#)
- [VK_EXT_DEPTH_CLIP_CONTROL_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_CONTROL_FEATURES_EXT](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_VIEWPORT_DEPTH_CLIP_CONTROL_CREATE_INFO_EXT](#)

Issues

1) Should this extension include an origin control option to match `GL_LOWER_LEFT` found in `ARB_clip_control`?

RESOLVED: No. The fix for porting over the origin is a simple y-axis flip. The depth clip control is a much harder problem to solve than what this extension is aimed to solve. Adding an equivalent to `GL_LOWER_LEFT` would require more testing.

2) Should this pipeline state be dynamic?

RESOLVED: Yes. The purpose of this extension is to emulate the OpenGL depth range, which is expected to be globally fixed (in case of OpenGL ES) or very infrequently changed (with [glClipControl](#) in OpenGL).

3) Should the control provided in this extension be an enum that could be extended in the future?

RESOLVED: No. It is highly unlikely that the depth range is changed to anything other than [0, 1] in

the future. Should that happen a new extension will be required to extend such an enum, and that extension might as well add a new struct to chain to [VkPipelineViewportStateCreateInfo::pNext](#) instead.

Version History

- Revision 0, 2020-10-01 (Spencer Fricke)
 - Internal revisions
- Revision 1, 2020-11-26 (Shahbaz Youssefi)
 - Language fixes

VK_EXT_depth_clip_enable

Name String

[VK_EXT_depth_clip_enable](#)

Extension Type

Device extension

Registered Extension Number

103

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Use

- [D3D support](#)

Contact

- Piers Daniell [\[GitHub\]](#)[pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2018-12-20

Contributors

- Daniel Rakos, AMD
- Henri Verbeet, CodeWeavers

- Jeff Bolz, NVIDIA
- Philip Rebohle, DXVK
- Tobias Hector, AMD

Description

This extension allows the depth clipping operation, that is normally implicitly controlled by `VkPipelineRasterizationStateCreateInfo::depthClampEnable`, to instead be controlled explicitly by `VkPipelineRasterizationDepthClipStateCreateInfoEXT::depthClipEnable`.

This is useful for translating DX content which assumes depth clamping is always enabled, but depth clip can be controlled by the `DepthClipEnable` rasterization state (`D3D12_RASTERIZER_DESC`).

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceDepthClipEnableFeaturesEXT`
- Extending `VkPipelineRasterizationStateCreateInfo`:
 - `VkPipelineRasterizationDepthClipStateCreateInfoEXT`

New Bitmasks

- `VkPipelineRasterizationDepthClipStateCreateFlagsEXT`

New Enum Constants

- `VK_EXT_DEPTH_CLIP_ENABLE_EXTENSION_NAME`
- `VK_EXT_DEPTH_CLIP_ENABLE_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEPTH_CLIP_ENABLE_FEATURES_EXT`
 - `VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_DEPTH_CLIP_STATE_CREATE_INFO_EXT`

Version History

- Revision 1, 2018-12-20 (Piers Daniell)
 - Internal revisions

VK_EXT_depth_range_unrestricted

Name String

`VK_EXT_depth_range_unrestricted`

Extension Type

Device extension

Registered Extension Number

14

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Piers Daniell [[GitHub](#)][pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2017-06-22

Contributors

- Daniel Koch, NVIDIA
- Jeff Bolz, NVIDIA

Description

This extension removes the [VkViewport](#) `minDepth` and `maxDepth` restrictions that the values must be between `0.0` and `1.0`, inclusive. It also removes the same restriction on [VkPipelineDepthStencilStateCreateInfo](#) `minDepthBounds` and `maxDepthBounds`. Finally it removes the restriction on the `depth` value in [VkClearDepthStencilValue](#).

New Enum Constants

- `VK_EXT_DEPTH_RANGE_UNRESTRICTED_EXTENSION_NAME`
- `VK_EXT_DEPTH_RANGE_UNRESTRICTED_SPEC_VERSION`

Issues

1) How do [VkViewport](#) `minDepth` and `maxDepth` values outside of the `0.0` to `1.0` range interact with [Primitive Clipping](#)?

RESOLVED: The behavior described in [Primitive Clipping](#) still applies. If depth clamping is disabled the depth values are still clipped to $0 \leq z_c \leq w_c$ before the viewport transform. If depth clamping is enabled the above equation is ignored and the depth values are instead clamped to the [VkViewport](#) `minDepth` and `maxDepth` values, which in the case of this extension can be outside of the `0.0` to `1.0` range.

2) What happens if a resulting depth fragment is outside of the `0.0` to `1.0` range and the depth buffer is fixed-point rather than floating-point?

RESOLVED: This situation can also arise without this extension (when fragment shaders replace depth values, for example), and this extension does not change the behavior, which is defined in the [Depth Test](#) section of the Fragment Operations chapter.

Version History

- Revision 1, 2017-06-22 (Piers Daniell)
 - Internal revisions

VK_EXT_device_generated_commands

Name String

VK_EXT_device_generated_commands

Extension Type

Device extension

Registered Extension Number

573

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_buffer_device_address](#)

and

[VK_KHR_maintenance5](#)

API Interactions

- Interacts with VK_EXT_shader_object

Contact

- Mike Blumenkrantz [\[GitHub\]zmike](#)

Extension Proposal

[VK_EXT_device_generated_commands](#)

Other Extension Metadata

Last Modified Date

2024-02-23

Interactions and External Dependencies

- This extension requires Vulkan 1.1
- This extension requires [VK_EXT_buffer_device_address](#) or [VK_KHR_buffer_device_address](#) or

Vulkan 1.2 for the ability to bind vertex and index buffers on the device.

- This extension requires `VK_KHR_maintenance5` for the ability to use `VkPipelineCreateFlags2KHR`.
- This extension interacts with `VK_NV_mesh_shader`. If the latter extension is not supported, remove the command tokens to initiate NV mesh tasks drawing in this extension.
- This extension interacts with `VK_EXT_mesh_shader`. If the latter extension is not supported, remove the command tokens to initiate EXT mesh tasks drawing in this extension.
- This extension interacts with `VK_KHR_ray_tracing_pipeline`. If the latter extension is not supported, remove the command tokens to initiate ray tracing in this extension.
- This extension interacts with `VK_EXT_shader_object`. If the latter extension is not supported, remove references to shader objects in this extension.

Contributors

- Mike Blumenkrantz, VALVE
- Hans-Kristian Arntzen, VALVE
- Jan-Harald Fredriksen, ARM
- Spencer Fricke, LunarG
- Ricardo Garcia, Igalia
- Tobias Hector, AMD
- Baldur Karlsson, VALVE
- Christoph Kubisch, NVIDIA
- Lionel Landwerlin, INTEL
- Jon Leech, Khronos
- Ting Wei, ARM
- Ken Shanyi Zhang, AMD
- Faith Ekstrand, Collabora
- Vikram Kushwaha, NVIDIA
- Connor Abbott, VALVE
- Samuel Pitoiset, VALVE

Description

This extension allows the device to generate a number of commands for command buffers. It provides a subset of functionality from both `VK_NV_device_generated_commands` and `VK_NV_device_generated_commands_compute` as well as some new features.

When rendering a large number of objects, the device can be leveraged to implement a number of critical functions, like updating matrices, or implementing occlusion culling, frustum culling, front to back sorting, etc. Implementing those on the device does not require any special extension, since an application is free to define its own data structures, and just process them using shaders.

To render objects which have been processed on the device, Vulkan has several ways to perform indirect rendering, from the most basic `vkCmdDrawIndirect` with one indirect draw to `vkCmdDrawIndirectCount` which supports multiple indirect draws batched together, with a way to determine number of draws at device execution time.

However, if rendering state needs to change between the indirect draws, then unextended Vulkan forces the application to speculatively record a prohibitive number of redundant indirect commands covering all possible state combinations - which could end up processing nothing after culling - or read back the processed stream and issue graphics command from the host. For very large scenes, the synchronization overhead and cost to generate the command buffer can become the bottleneck. This extension allows an application to generate a device side stream of state changes and commands, and convert it efficiently into a command buffer without having to read it back to the host.

Furthermore, it allows incremental changes to such command buffers by manipulating only partial sections of a command stream—for example pipeline and shader object bindings. Unextended Vulkan requires re-creation of entire command buffers in such a scenario, or updates synchronized on the host.

The intended usage for this extension is for the application to:

- create `VkBuffer` objects and retrieve physical addresses from them via `vkGetBufferDeviceAddressKHR`
- create a `VkIndirectExecutionSetEXT` for the ability to change shaders on the device.
- create a `VkIndirectCommandsLayoutEXT`, which lists the `VkIndirectCommandsTokenTypeEXT` it wants to dynamically execute as an atomic command sequence. This step likely involves some internal device code compilation, since the intent is for the GPU to generate the command buffer based on the layout.
- fill the input stream buffers with the data for each of the inputs it needs. Each input is an array that will be filled with token-dependent data.
- set up a preprocess `VkBuffer` that uses memory according to the information retrieved via `vkGetGeneratedCommandsMemoryRequirementsEXT`.
- optionally preprocess the generated content using `vkCmdPreprocessGeneratedCommandsEXT`, for example on an asynchronous compute queue, or for the purpose of re-using the data in multiple executions.
- call `vkCmdExecuteGeneratedCommandsEXT` to create and execute the actual device commands for all sequences based on the inputs provided.

For each draw in a sequence, the following can be specified:

- a number of vertex buffer bindings
- a different index buffer, with an optional dynamic offset and index type
- a number of different push constants
- updates to bound shader stages

For each dispatch in a sequence, the following can be specified:

- a number of different push constants
- updates to bound shader stages

For each trace rays in a sequence, the following can be specified:

- a number of different push constants
- updates to bound shader stages

While the GPU can be faster than a CPU to generate the commands, it will not happen asynchronously to the device, therefore the primary use case is generating “less” total work (occlusion culling, classification to use specialized shaders, etc.).

New Object Types

- [VkIndirectCommandsLayoutEXT](#)
- [VkIndirectExecutionSetEXT](#)

New Commands

- [vkCmdExecuteGeneratedCommandsEXT](#)
- [vkCmdPreprocessGeneratedCommandsEXT](#)
- [vkCreateIndirectCommandsLayoutEXT](#)
- [vkCreateIndirectExecutionSetEXT](#)
- [vkDestroyIndirectCommandsLayoutEXT](#)
- [vkDestroyIndirectExecutionSetEXT](#)
- [vkGetGeneratedCommandsMemoryRequirementsEXT](#)
- [vkUpdateIndirectExecutionSetPipelineEXT](#)
- [vkUpdateIndirectExecutionSetShaderEXT](#)

New Structures

- [VkBindIndexBufferIndirectCommandEXT](#)
- [VkBindVertexBufferIndirectCommandEXT](#)
- [VkDrawIndirectCountIndirectCommandEXT](#)
- [VkGeneratedCommandsInfoEXT](#)
- [VkGeneratedCommandsMemoryRequirementsInfoEXT](#)
- [VkIndirectCommandsExecutionSetTokenEXT](#)
- [VkIndirectCommandsIndexBufferTokenEXT](#)
- [VkIndirectCommandsLayoutCreateInfoEXT](#)
- [VkIndirectCommandsLayoutTokenEXT](#)
- [VkIndirectCommandsPushConstantTokenEXT](#)

- [VkIndirectCommandsVertexBufferTokenEXT](#)
- [VkIndirectExecutionSetCreateInfoEXT](#)
- [VkIndirectExecutionSetPipelineInfoEXT](#)
- [VkIndirectExecutionSetShaderInfoEXT](#)
- [VkIndirectExecutionSetShaderLayoutInfoEXT](#)
- [VkWriteIndirectExecutionSetPipelineEXT](#)
- Extending [VkGeneratedCommandsInfoEXT](#),
[VkGeneratedCommandsMemoryRequirementsInfoEXT](#):
 - [VkGeneratedCommandsPipelineInfoEXT](#)
 - [VkGeneratedCommandsShaderInfoEXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceDeviceGeneratedCommandsFeaturesEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceDeviceGeneratedCommandsPropertiesEXT](#)

If [VK_EXT_shader_object](#) is supported:

- [VkWriteIndirectExecutionSetShaderEXT](#)

New Unions

- [VkIndirectCommandsTokenDataEXT](#)
- [VkIndirectExecutionSetInfoEXT](#)

New Enums

- [VkIndirectCommandsInputModeFlagBitsEXT](#)
- [VkIndirectCommandsLayoutUsageFlagBitsEXT](#)
- [VkIndirectCommandsTokenTypeEXT](#)
- [VkIndirectExecutionSetInfoTypeEXT](#)

New Bitmasks

- [VkIndirectCommandsInputModeFlagsEXT](#)
- [VkIndirectCommandsLayoutUsageFlagsEXT](#)

New Enum Constants

- [VK_EXT_DEVICE_GENERATED_COMMANDS_EXTENSION_NAME](#)
- [VK_EXT_DEVICE_GENERATED_COMMANDS_SPEC_VERSION](#)
- Extending [VkAccessFlagBits](#):
 - [VK_ACCESS_COMMAND_PREPROCESS_READ_BIT_EXT](#)

- `VK_ACCESS_COMMAND_PREPROCESS_WRITE_BIT_EXT`
- Extending `VkBufferUsageFlagBits2KHR`:
 - `VK_BUFFER_USAGE_2_PREPROCESS_BUFFER_BIT_EXT`
- Extending `VkObjectType`:
 - `VK_OBJECT_TYPE_INDIRECT_COMMANDS_LAYOUT_EXT`
 - `VK_OBJECT_TYPE_INDIRECT_EXECUTION_SET_EXT`
- Extending `VkPipelineCreateFlagBits2KHR`:
 - `VK_PIPELINE_CREATE_2_INDIRECT_BINDABLE_BIT_EXT`
- Extending `VkPipelineStageFlagBits`:
 - `VK_PIPELINE_STAGE_COMMAND_PREPROCESS_BIT_EXT`
- Extending `VkShaderCreateFlagBitsEXT`:
 - `VK_SHADER_CREATE_INDIRECT_BINDABLE_BIT_EXT`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_GENERATED_COMMANDS_INFO_EXT`
 - `VK_STRUCTURE_TYPE_GENERATED_COMMANDS_MEMORY_REQUIREMENTS_INFO_EXT`
 - `VK_STRUCTURE_TYPE_GENERATED_COMMANDS_PIPELINE_INFO_EXT`
 - `VK_STRUCTURE_TYPE_GENERATED_COMMANDS_SHADER_INFO_EXT`
 - `VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_CREATE_INFO_EXT`
 - `VK_STRUCTURE_TYPE_INDIRECT_COMMANDS_LAYOUT_TOKEN_EXT`
 - `VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_CREATE_INFO_EXT`
 - `VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_PIPELINE_INFO_EXT`
 - `VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_INFO_EXT`
 - `VK_STRUCTURE_TYPE_INDIRECT_EXECUTION_SET_SHADER_LAYOUT_INFO_EXT`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_FEATURES_EXT`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_GENERATED_COMMANDS_PROPERTIES_EXT`
 - `VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_PIPELINE_EXT`
 - `VK_STRUCTURE_TYPE_WRITE_INDIRECT_EXECUTION_SET_SHADER_EXT`

Example Code

TODO

Version History

- Revision 1, 2024-02-23 (Mike Blumenkrantz)
 - Initial version

VK_EXT_device_memory_report

Name String

VK_EXT_device_memory_report

Extension Type

Device extension

Registered Extension Number

285

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Use

- [Developer tools](#)

Contact

- Yiwei Zhang [\[GitHub\]zhangyiwei](#)

Other Extension Metadata

Last Modified Date

2021-01-06

IP Status

No known IP claims.

Contributors

- Yiwei Zhang, Google
- Jesse Hall, Google

Description

This device extension allows registration of device memory event callbacks upon device creation, so that applications or middleware can obtain detailed information about memory usage and how memory is associated with Vulkan objects. This extension exposes the actual underlying device memory usage, including allocations that are not normally visible to the application, such as memory consumed by [vkCreateGraphicsPipelines](#). It is intended primarily for use by debug tooling rather than for production applications.

New Structures

- [VkDeviceMemoryReportCallbackDataEXT](#)
- Extending [VkDeviceCreateInfo](#):
 - [VkDeviceDeviceMemoryReportCreateInfoEXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceDeviceMemoryReportFeaturesEXT](#)

New Function Pointers

- [PFN_vkDeviceMemoryReportCallbackEXT](#)

New Enums

- [VkDeviceMemoryReportEventTypeEXT](#)

New Bitmasks

- [VkDeviceMemoryReportFlagsEXT](#)

New Enum Constants

- [VK_EXT_DEVICE_MEMORY_REPORT_EXTENSION_NAME](#)
- [VK_EXT_DEVICE_MEMORY_REPORT_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_DEVICE_DEVICE_MEMORY_REPORT_CREATE_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_DEVICE_MEMORY_REPORT_CALLBACK_DATA_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DEVICE_MEMORY_REPORT_FEATURES_EXT](#)

Issues

1) Should this be better expressed as an extension to [VK_EXT_debug_utils](#) and its general-purpose messenger construct?

RESOLVED: No. The intended lifecycle is quite different. We want to make this extension tied to the device's lifecycle. Each ICD just handles its own implementation of this extension, and this extension will only be directly exposed from the ICD. So we can avoid the extra implementation complexity used to accommodate the flexibility of [VK_EXT_debug_utils](#) extension.

2) Can we extend and use the existing internal allocation callbacks instead of adding the new callback structure in this extension?

RESOLVED: No. Our memory reporting layer that combines this information with other memory information it collects directly (e.g. bindings of resources to [VkDeviceMemory](#)) would have to intercept all entry points that take a [VkAllocationCallbacks](#) parameter and inject its own [pfnInternalAllocation](#) and [pfnInternalFree](#). That may be doable for the extensions we know about, but not for ones we do not. The proposal would work fine in the face of most unknown extensions.

But even for ones we know about, since apps can provide a different set of callbacks and userdata and those can be retained by the driver and used later (esp. for pool object, but not just those), we would have to dynamically allocate the interception trampoline every time. That is getting to be an unreasonably large amount of complexity and (possibly) overhead.

We are interested in both alloc/free and import/unimport. The latter is fairly important for tracking (and avoiding double-counting) of swapchain images (still true with “native swapchains” based on external memory) and media/camera interop. Though we might be able to handle this with additional [VkInternalAllocationType](#) values, for import/export we do want to be able to tie this to the external resource, which is one thing that the [memoryObjectId](#) is for.

The internal alloc/free callbacks are not extensible except via new [VkInternalAllocationType](#) values. The [VkDeviceMemoryReportCallbackDataEXT](#) in this extension is extensible. That was deliberate: there is a real possibility we will want to get extra information in the future. As one example, currently this reports only physical allocations, but we believe there are interesting cases for tracking how populated that VA region is.

The callbacks are clearly specified as only callable within the context of a call from the application into Vulkan. We believe there are some cases where drivers can allocate device memory asynchronously. This was one of the sticky issues that derailed the internal device memory allocation reporting design (which is essentially what this extension is trying to do) leading up to 1.0.

[VkAllocationCallbacks](#) is described in a section called “Host memory” and the intro to it is very explicitly about host memory. The other callbacks are all inherently about host memory. But this extension is very focused on device memory.

3) Should the callback be reporting which heap is used?

RESOLVED: Yes. It is important for non-UMA systems to have all the device memory allocations attributed to the corresponding device memory heaps. For internally-allocated device memory, [heapIndex](#) will always correspond to an advertised heap, rather than having a magic value indicating a non-advertised heap. Drivers can advertise heaps that do not have any corresponding memory types if they need to.

4) Should we use an array of callback for the layers to intercept instead of chaining multiple of the [VkDeviceDeviceMemoryReportCreateInfoEXT](#) structures in the [pNext](#) of [VkDeviceCreateInfo](#)?

RESOLVED No. The pointer to the [VkDeviceDeviceMemoryReportCreateInfoEXT](#) structure itself is const and you cannot just cast it away. Thus we cannot update the callback array inside the structure. In addition, we cannot drop this [pNext](#) chain either, so making a copy of this whole structure does not work either.

5) Should we track bulk allocations shared among multiple objects?

RESOLVED No. Take the shader heap as an example. Some implementations will let multiple [VkPipeline](#) objects share the same shader heap. We are not asking the implementation to report [VK_OBJECT_TYPE_PIPELINE](#) along with a [VK_NULL_HANDLE](#) for this bulk allocation. Instead, this bulk allocation is considered as a layer below what this extension is interested in. Later, when the actual [VkPipeline](#) objects are created by suballocating from the bulk allocation, we ask the

implementation to report the valid handles of the [VkPipeline](#) objects along with the actual suballocated sizes and different [memoryObjectId](#).

6) Can we require the callbacks to be always called in the same thread with the Vulkan commands?

RESOLVED No. Some implementations might choose to multiplex work from multiple application threads into a single backend thread and perform JIT allocations as a part of that flow. Since this behavior is theoretically legit, we cannot require the callbacks to be always called in the same thread with the Vulkan commands, and the note is to remind the applications to handle this case properly.

7) Should we add an additional “allocation failed” event type with things like size and heap index reported?

RESOLVED Yes. This fits in well with the callback infrastructure added in this extension, and implementation touches the same code and has the same overheads as the rest of the extension. It could help debugging things like getting a [VK_ERROR_OUT_OF_HOST_MEMORY](#) error when ending a command buffer. Right now the allocation failure could have happened anywhere during recording, and a callback would be really useful to understand where and why.

Version History

- Revision 1, 2020-08-26 (Yiwei Zhang)
 - Initial version
- Revision 2, 2021-01-06 (Yiwei Zhang)
 - Minor description update

VK_EXT_direct_mode_display

Name String

[VK_EXT_direct_mode_display](#)

Extension Type

Instance extension

Registered Extension Number

89

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_display](#)

Contact

- James Jones [\[GitHub\]cubanismo](#)

Other Extension Metadata

Last Modified Date

2016-12-13

IP Status

No known IP claims.

Contributors

- Pierre Boudier, NVIDIA
- James Jones, NVIDIA
- Damien Leone, NVIDIA
- Pierre-Loup Griffais, Valve
- Liam Middlebrook, NVIDIA

Description

This is extension, along with related platform extensions, allows applications to take exclusive control of displays associated with a native windowing system. This is especially useful for virtual reality applications that wish to hide HMDs (head mounted displays) from the native platform's display management system, desktop, and/or other applications.

New Commands

- [vkReleaseDisplayEXT](#)

New Enum Constants

- [VK_EXT_DIRECT_MODE_DISPLAY_EXTENSION_NAME](#)
- [VK_EXT_DIRECT_MODE_DISPLAY_SPEC_VERSION](#)

Issues

1) Should this extension and its related platform-specific extensions leverage [VK_KHR_display](#), or provide separate equivalent interfaces.

RESOLVED: Use [VK_KHR_display](#) concepts and objects. [VK_KHR_display](#) can be used to enumerate all displays on the system, including those attached to/in use by a window system or native platform, but [VK_KHR_display_swapchain](#) will fail to create a swapchain on in-use displays. This extension and its platform-specific children will allow applications to grab in-use displays away from window systems and/or native platforms, allowing them to be used with [VK_KHR_display_swapchain](#).

2) Are separate calls needed to acquire displays and enable direct mode?

RESOLVED: No, these operations happen in one combined command. Acquiring a display puts it

into direct mode.

Version History

- Revision 1, 2016-12-13 (James Jones)
 - Initial draft

VK_EXT_discard_rectangles

Name String

VK_EXT_discard_rectangles

Extension Type

Device extension

Registered Extension Number

100

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Piers Daniell [[GitHub](#)][pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2023-01-18

Interactions and External Dependencies

- Interacts with [VK_KHR_device_group](#)
- Interacts with Vulkan 1.1

Contributors

- Daniel Koch, NVIDIA
- Jeff Bolz, NVIDIA

Description

This extension provides additional orthogonally aligned “discard rectangles” specified in

framebuffer-space coordinates that restrict rasterization of all points, lines and triangles.

From zero to an implementation-dependent limit (specified by `maxDiscardRectangles`) number of discard rectangles can be operational at once. When one or more discard rectangles are active, rasterized fragments can either survive if the fragment is within any of the operational discard rectangles (`VK_DISCARD_RECTANGLE_MODE_INCLUSIVE_EXT` mode) or be rejected if the fragment is within any of the operational discard rectangles (`VK_DISCARD_RECTANGLE_MODE_EXCLUSIVE_EXT` mode).

These discard rectangles operate orthogonally to the existing scissor test functionality. The discard rectangles can be different for each physical device in a device group by specifying the device mask and setting discard rectangle dynamic state.

Version 2 of this extension introduces new dynamic states `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT` and `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT`, and the corresponding functions `vkCmdSetDiscardRectangleEnableEXT` and `vkCmdSetDiscardRectangleModeEXT`. Applications that use these dynamic states must ensure the implementation advertises at least `specVersion 2` of this extension.

New Commands

- `vkCmdSetDiscardRectangleEXT`
- `vkCmdSetDiscardRectangleEnableEXT`
- `vkCmdSetDiscardRectangleModeEXT`

New Structures

- Extending `VkGraphicsPipelineCreateInfo`:
 - `VkPipelineDiscardRectangleStateCreateInfoEXT`
- Extending `VkPhysicalDeviceProperties2`:
 - `VkPhysicalDeviceDiscardRectanglePropertiesEXT`

New Enums

- `VkDiscardRectangleModeEXT`

New Bitmasks

- `VkPipelineDiscardRectangleStateCreateFlagsEXT`

New Enum Constants

- `VK_EXT_DISCARD_RECTANGLES_EXTENSION_NAME`
- `VK_EXT_DISCARD_RECTANGLES_SPEC_VERSION`
- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_ENABLE_EXT`
 - `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_EXT`

- `VK_DYNAMIC_STATE_DISCARD_RECTANGLE_MODE_EXT`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DISCARD_RECTANGLE_PROPERTIES_EXT`
 - `VK_STRUCTURE_TYPE_PIPELINE_DISCARD_RECTANGLE_STATE_CREATE_INFO_EXT`

Version History

- Revision 2, 2023-01-18 (Piers Daniell)
 - Add dynamic states for discard rectangle enable/disable and mode.
- Revision 1, 2016-12-22 (Piers Daniell)
 - Internal revisions

VK_EXT_display_control

Name String

`VK_EXT_display_control`

Extension Type

Device extension

Registered Extension Number

92

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_EXT_display_surface_counter](#)

and

[VK_KHR_swapchain](#)

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2016-12-13

IP Status

No known IP claims.

Contributors

- Pierre Boudier, NVIDIA
- James Jones, NVIDIA
- Damien Leone, NVIDIA
- Pierre-Loup Griffais, Valve
- Daniel Vetter, Intel

Description

This extension defines a set of utility functions for use with the [VK_KHR_display](#) and [VK_KHR_display_swapchain](#) extensions.

New Commands

- [vkDisplayPowerControlEXT](#)
- [vkGetSwapchainCounterEXT](#)
- [vkRegisterDeviceEventEXT](#)
- [vkRegisterDisplayEventEXT](#)

New Structures

- [VkDeviceEventInfoEXT](#)
- [VkDisplayEventInfoEXT](#)
- [VkDisplayPowerInfoEXT](#)
- Extending [VkSwapchainCreateInfoKHR](#):
 - [VkSwapchainCounterCreateInfoEXT](#)

New Enums

- [VkDeviceEventTypeEXT](#)
- [VkDisplayEventTypeEXT](#)
- [VkDisplayPowerStateEXT](#)

New Enum Constants

- [VK_EXT_DISPLAY_CONTROL_EXTENSION_NAME](#)
- [VK_EXT_DISPLAY_CONTROL_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_DEVICE_EVENT_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_DISPLAY_EVENT_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_DISPLAY_POWER_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_SWAPCHAIN_COUNTER_CREATE_INFO_EXT](#)

Issues

1) Should this extension add an explicit “WaitForVsync” API or a fence signaled at vsync that the application can wait on?

RESOLVED: A fence. A separate API could later be provided that allows exporting the fence to a native object that could be inserted into standard run loops on POSIX and Windows systems.

2) Should callbacks be added for a vsync event, or in general to monitor events in Vulkan?

RESOLVED: No, fences should be used. Some events are generated by interrupts which are managed in the kernel. In order to use a callback provided by the application, drivers would need to have the userspace driver spawn threads that would wait on the kernel event, and hence the callbacks could be difficult for the application to synchronize with its other work given they would arrive on a foreign thread.

3) Should vblank or scanline events be exposed?

RESOLVED: Vblank events. Scanline events could be added by a separate extension, but the latency of processing an interrupt and waking up a userspace event is high enough that the accuracy of a scanline event would be rather low. Further, per-scanline interrupts are not supported by all hardware.

Version History

- Revision 1, 2016-12-13 (James Jones)
 - Initial draft

VK_EXT_display_surface_counter

Name String

VK_EXT_display_surface_counter

Extension Type

Instance extension

Registered Extension Number

91

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_display](#)

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2016-12-13

IP Status

No known IP claims.

Contributors

- Pierre Boudier, NVIDIA
- James Jones, NVIDIA
- Damien Leone, NVIDIA
- Pierre-Loup Griffais, Valve
- Daniel Vetter, Intel

Description

This extension defines a vertical blanking period counter associated with display surfaces. It provides a mechanism to query support for such a counter from a [VkSurfaceKHR](#) object.

New Commands

- [vkGetPhysicalDeviceSurfaceCapabilities2EXT](#)

New Structures

- [VkSurfaceCapabilities2EXT](#)

New Enums

- [VkSurfaceCounterFlagBitsEXT](#)

New Bitmasks

- [VkSurfaceCounterFlagsEXT](#)

New Enum Constants

- [VK_EXT_DISPLAY_SURFACE_COUNTER_EXTENSION_NAME](#)
- [VK_EXT_DISPLAY_SURFACE_COUNTER_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES2_EXT](#)
 - [VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_2_EXT](#)

Version History

- Revision 1, 2016-12-13 (James Jones)
 - Initial draft

VK_EXT_dynamic_rendering_unused_attachments

Name String

VK_EXT_dynamic_rendering_unused_attachments

Extension Type

Device extension

Registered Extension Number

500

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

and

[VK_KHR_dynamic_rendering](#)

or

[Vulkan Version 1.3](#)

Contact

- Piers Daniell [[GitHub](#)]pdaniell-nv

Extension Proposal

[VK_EXT_dynamic_rendering_unused_attachments](#)

Other Extension Metadata

Last Modified Date

2023-05-22

IP Status

No known IP claims.

Contributors

- Daniel Story, Nintendo
- Hans-Kristian Arntzen, Valve

- Jan-Harald Fredriksen, Arm
- James Fitzpatrick, Imagination Technologies
- Pan Gao, Huawei Technologies
- Ricardo Garcia, Igalia
- Stu Smith, AMD

Description

This extension lifts some restrictions in the [VK_KHR_dynamic_rendering](#) extension to allow render pass instances and bound pipelines within those render pass instances to have an unused attachment specified in one but not the other. It also allows pipelines to use different formats in a render pass as long the attachment is NULL.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceDynamicRenderingUnusedAttachmentsFeaturesEXT](#)

New Enum Constants

- [VK_EXT_DYNAMIC_RENDERING_UNUSED_ATTACHMENTS_EXTENSION_NAME](#)
- [VK_EXT_DYNAMIC_RENDERING_UNUSED_ATTACHMENTS_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DYNAMIC_RENDERING_UNUSED_ATTACHMENTS_FEATURES_EXT](#)

Issues

None.

Version History

- Revision 1, 2023-05-22 (Piers Daniell)
 - Internal revisions

VK_EXT_extended_dynamic_state3

Name String

[VK_EXT_extended_dynamic_state3](#)

Extension Type

Device extension

Registered Extension Number

456

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

API Interactions

- Interacts with `VK_VERSION_1_1`
- Interacts with `VK_EXT_blend_operation_advanced`
- Interacts with `VK_EXT_conservative_rasterization`
- Interacts with `VK_EXT_depth_clip_control`
- Interacts with `VK_EXT_depth_clip_enable`
- Interacts with `VK_EXT_line_rasterization`
- Interacts with `VK_EXT_provoking_vertex`
- Interacts with `VK_EXT_sample_locations`
- Interacts with `VK_EXT_transform_feedback`
- Interacts with `VK_KHR_maintenance2`
- Interacts with `VK_NV_clip_space_w_scaling`
- Interacts with `VK_NV_coverage_reduction_mode`
- Interacts with `VK_NV_fragment_coverage_to_color`
- Interacts with `VK_NV_framebuffer_mixed_samples`
- Interacts with `VK_NV_representative_fragment_test`
- Interacts with `VK_NV_shading_rate_image`
- Interacts with `VK_NV_viewport_swizzle`
- Interacts with `VkPhysicalDeviceExtendedDynamicState3FeaturesEXT::extendedDynamicState3AlphaToOneEnable`
- Interacts with `VkPhysicalDeviceExtendedDynamicState3FeaturesEXT::extendedDynamicState3DepthClampEnable`
- Interacts with `VkPhysicalDeviceExtendedDynamicState3FeaturesEXT::extendedDynamicState3LogicOpEnable`
- Interacts with `VkPhysicalDeviceExtendedDynamicState3FeaturesEXT::extendedDynamicState3PolygonMod`

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- Interacts with `VkPhysicalDeviceExtendedDynamicState3FeaturesEXT::extendedDynamicState3TessellationDomainOrigin`
- Interacts with `VkPhysicalDeviceMeshShaderFeaturesEXT::extendedDynamicState3RasterizationStream`

Contact

- Piers Daniell [[GitHub](#)][pdaniell-nv](#)

Extension Proposal

[VK_EXT_extended_dynamic_state3](#)

Other Extension Metadata

Last Modified Date

2022-09-02

IP Status

No known IP claims.

Contributors

- Daniel Story, Nintendo
- Jamie Madill, Google
- Jan-Harald Fredriksen, Arm
- Faith Ekstrand, Collabora
- Mike Blumenkrantz, Valve
- Ricardo Garcia, Igalia
- Samuel Pitoiset, Valve
- Shahbaz Youssefi, Google
- Stu Smith, AMD
- Tapani Pälli, Intel

Description

This extension adds almost all of the remaining pipeline state as dynamic state to help applications further reduce the number of monolithic pipelines they need to create and bind.

New Commands

- [vkCmdSetAlphaToCoverageEnableEXT](#)
- [vkCmdSetAlphaToOneEnableEXT](#)
- [vkCmdSetColorBlendEnableEXT](#)
- [vkCmdSetColorBlendEquationEXT](#)

- [vkCmdSetColorWriteMaskEXT](#)
- [vkCmdSetDepthClampEnableEXT](#)
- [vkCmdSetLogicOpEnableEXT](#)
- [vkCmdSetPolygonModeEXT](#)
- [vkCmdSetRasterizationSamplesEXT](#)
- [vkCmdSetSampleMaskEXT](#)

If [VK_EXT_blend_operation_advanced](#) is supported:

- [vkCmdSetColorBlendAdvancedEXT](#)

If [VK_EXT_conservative_rasterization](#) is supported:

- [vkCmdSetConservativeRasterizationModeEXT](#)
- [vkCmdSetExtraPrimitiveOverestimationSizeEXT](#)

If [VK_EXT_depth_clip_control](#) is supported:

- [vkCmdSetDepthClipNegativeOneToOneEXT](#)

If [VK_EXT_depth_clip_enable](#) is supported:

- [vkCmdSetDepthClipEnableEXT](#)

If [VK_EXT_line_rasterization](#) is supported:

- [vkCmdSetLineRasterizationModeEXT](#)
- [vkCmdSetLineStippleEnableEXT](#)

If [VK_EXT_provoking_vertex](#) is supported:

- [vkCmdSetProvokingVertexModeEXT](#)

If [VK_EXT_sample_locations](#) is supported:

- [vkCmdSetSampleLocationsEnableEXT](#)

If [VK_EXT_transform_feedback](#) is supported:

- [vkCmdSetRasterizationStreamEXT](#)

If [VK_KHR_maintenance2](#) or [Vulkan Version 1.1](#) is supported:

- [vkCmdSetTessellationDomainOriginEXT](#)

If [VK_NV_clip_space_w_scaling](#) is supported:

- [vkCmdSetViewportWScalingEnableNV](#)

If [VK_NV_coverage_reduction_mode](#) is supported:

- [vkCmdSetCoverageReductionModeNV](#)

If [VK_NV_fragment_coverage_to_color](#) is supported:

- [vkCmdSetCoverageToColorEnableNV](#)
- [vkCmdSetCoverageToColorLocationNV](#)

If [VK_NV_framebuffer_mixed_samples](#) is supported:

- [vkCmdSetCoverageModulationModeNV](#)
- [vkCmdSetCoverageModulationTableEnableNV](#)
- [vkCmdSetCoverageModulationTableNV](#)

If [VK_NV_representative_fragment_test](#) is supported:

- [vkCmdSetRepresentativeFragmentTestEnableNV](#)

If [VK_NV_shading_rate_image](#) is supported:

- [vkCmdSetShadingRateImageEnableNV](#)

If [VK_NV_viewport_swizzle](#) is supported:

- [vkCmdSetViewportSwizzleNV](#)

New Structures

- [VkColorBlendAdvancedEXT](#)
- [VkColorBlendEquationEXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceExtendedDynamicState3FeaturesEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceExtendedDynamicState3PropertiesEXT](#)

New Enum Constants

- [VK_EXT_EXTENDED_DYNAMIC_STATE_3_EXTENSION_NAME](#)
- [VK_EXT_EXTENDED_DYNAMIC_STATE_3_SPEC_VERSION](#)
- Extending [VkDynamicState](#):
 - [VK_DYNAMIC_STATE_ALPHA_TO_COVERAGE_ENABLE_EXT](#)
 - [VK_DYNAMIC_STATE_ALPHA_TO_ONE_ENABLE_EXT](#)
 - [VK_DYNAMIC_STATE_COLOR_BLEND_ENABLE_EXT](#)
 - [VK_DYNAMIC_STATE_COLOR_BLEND_EQUATION_EXT](#)
 - [VK_DYNAMIC_STATE_COLOR_WRITE_MASK_EXT](#)
 - [VK_DYNAMIC_STATE_DEPTH_CLAMP_ENABLE_EXT](#)

- `VK_DYNAMIC_STATE_LOGIC_OP_ENABLE_EXT`
- `VK_DYNAMIC_STATE_POLYGON_MODE_EXT`
- `VK_DYNAMIC_STATE_RASTERIZATION_SAMPLES_EXT`
- `VK_DYNAMIC_STATE_SAMPLE_MASK_EXT`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_FEATURES_EXT`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTENDED_DYNAMIC_STATE_3_PROPERTIES_EXT`

If `VK_EXT_blend_operation_advanced` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_COLOR_BLEND_ADVANCED_EXT`

If `VK_EXT_conservative_rasterization` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_CONSERVATIVE_RASTERIZATION_MODE_EXT`
 - `VK_DYNAMIC_STATE_EXTRA_PRIMITIVE_OVERESTIMATION_SIZE_EXT`

If `VK_EXT_depth_clip_control` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_DEPTH_CLIP_NEGATIVE_ONE_TO_ONE_EXT`

If `VK_EXT_depth_clip_enable` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_DEPTH_CLIP_ENABLE_EXT`

If `VK_EXT_line_rasterization` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_LINE_RASTERIZATION_MODE_EXT`
 - `VK_DYNAMIC_STATE_LINE_STIPPLE_ENABLE_EXT`

If `VK_EXT_provoking_vertex` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_PROVOKING_VERTEX_MODE_EXT`

If `VK_EXT_sample_locations` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_ENABLE_EXT`

If `VK_EXT_transform_feedback` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_RASTERIZATION_STREAM_EXT`

If `VK_KHR_maintenance2` or `Vulkan Version 1.1` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_TESSELLATION_DOMAIN_ORIGIN_EXT`

If `VK_NV_clip_space_w_scaling` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_VIEWPORT_W_SCALING_ENABLE_NV`

If `VK_NV_coverage_reduction_mode` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_COVERAGE_REDUCTION_MODE_NV`

If `VK_NV_fragment_coverage_to_color` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_ENABLE_NV`
 - `VK_DYNAMIC_STATE_COVERAGE_TO_COLOR_LOCATION_NV`

If `VK_NV_framebuffer_mixed_samples` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_COVERAGE_MODULATION_MODE_NV`
 - `VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_ENABLE_NV`
 - `VK_DYNAMIC_STATE_COVERAGE_MODULATION_TABLE_NV`

If `VK_NV_representative_fragment_test` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_REPRESENTATIVE_FRAGMENT_TEST_ENABLE_NV`

If `VK_NV_shading_rate_image` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_SHADING_RATE_IMAGE_ENABLE_NV`

If `VK_NV_viewport_swizzle` is supported:

- Extending `VkDynamicState`:
 - `VK_DYNAMIC_STATE_VIEWPORT_SWIZZLE_NV`

Issues

1) What about the `VkPipelineMultisampleStateCreateInfo` state `sampleShadingEnable` and `minSampleShading`?

UNRESOLVED

- `sampleShadingEnable` and `minSampleShading` are required when compiling the fragment shader, and it is not meaningful to set them dynamically since they always need to match the fragment shader state, so this hardware state may as well just come from the pipeline with the fragment shader.

Version History

- Revision 2, 2022-07-18 (Piers Daniell)
 - Added `rasterizationSamples`
- Revision 1, 2022-05-18 (Piers Daniell)
 - Internal revisions

VK_EXT_external_memory_acquire_unmodified

Name String

`VK_EXT_external_memory_acquire_unmodified`

Extension Type

Device extension

Registered Extension Number

454

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_memory](#)

or

[Vulkan Version 1.1](#)

Contact

- Lina Versace [\[GitHub\]](#)linyaa-kiwi

Extension Proposal

[VK_EXT_external_memory_acquire_unmodified](#)

Other Extension Metadata

Last Modified Date

2023-03-09

Contributors

- Lina Versace, Google
- Chia-I Wu, Google
- James Jones, NVIDIA
- Yiwei Zhang, Google

Description

A memory barrier **may** have a performance penalty when acquiring ownership of a subresource range from an external queue family. This extension provides API that **may** reduce the performance penalty if ownership of the subresource range was previously released to the external queue family and if the resource's memory has remained unmodified between the release and acquire operations.

New Structures

- Extending [VkBufferMemoryBarrier](#), [VkBufferMemoryBarrier2](#), [VkImageMemoryBarrier](#), [VkImageMemoryBarrier2](#):
 - [VkExternalMemoryAcquireUnmodifiedEXT](#)

New Enum Constants

- [VK_EXT_EXTERNAL_MEMORY_ACQUIRE_UNMODIFIED_EXTENSION_NAME](#)
- [VK_EXT_EXTERNAL_MEMORY_ACQUIRE_UNMODIFIED_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_EXTERNAL_MEMORY_ACQUIRE_UNMODIFIED_EXT](#)

Version History

- Revision 1, 2023-03-09 (Lina Versace)
 - Initial revision

VK_EXT_external_memory_dma_buf

Name String

[VK_EXT_external_memory_dma_buf](#)

Extension Type

Device extension

Registered Extension Number

126

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_memory_fd](#)

Contact

- Lina Versace [\[GitHub\]](#)linyaa-kiwi

Other Extension Metadata

Last Modified Date

2017-10-10

IP Status

No known IP claims.

Contributors

- Lina Versace, Google
- James Jones, NVIDIA
- Faith Ekstrand, Intel

Description

A [dma_buf](#) is a type of file descriptor, defined by the Linux kernel, that allows sharing memory across kernel device drivers and across processes. This extension enables applications to import a [dma_buf](#) as [VkDeviceMemory](#), to export [VkDeviceMemory](#) as a [dma_buf](#), and to create [VkBuffer](#) objects that **can** be bound to that memory.

New Enum Constants

- [VK_EXT_EXTERNAL_MEMORY_DMA_BUF_EXTENSION_NAME](#)
- [VK_EXT_EXTERNAL_MEMORY_DMA_BUF_SPEC_VERSION](#)
- Extending [VkExternalMemoryHandleTypeFlagBits](#):
 - [VK_EXTERNAL_MEMORY_HANDLE_TYPE_DMA_BUF_BIT_EXT](#)

Issues

1) How does the application, when creating a [VkImage](#) that it intends to bind to [dma_buf](#) [VkDeviceMemory](#) containing an externally produced image, specify the memory layout (such as row pitch and DRM format modifier) of the [VkImage](#)? In other words, how does the application

achieve behavior comparable to that provided by `EGL_EXT_image_dma_buf_import` and `EGL_EXT_image_dma_buf_import_modifiers`?

RESOLVED: Features comparable to those in `EGL_EXT_image_dma_buf_import` and `EGL_EXT_image_dma_buf_import_modifiers` will be provided by an extension layered atop this one.

2) Without the ability to specify the memory layout of external `dma_buf` images, how is this extension useful?

RESOLVED: This extension provides exactly one new feature: the ability to import/export between `dma_buf` and `VkDeviceMemory`. This feature, together with features provided by `VK_KHR_external_memory_fd`, is sufficient to bind a `VkBuffer` to `dma_buf`.

Version History

- Revision 1, 2017-10-10 (Lina Versace)
 - Squashed internal revisions

VK_EXT_external_memory_host

Name String

`VK_EXT_external_memory_host`

Extension Type

Device extension

Registered Extension Number

179

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

`VK_KHR_external_memory`

or

`Vulkan Version 1.1`

Contact

- Daniel Rakos [\[GitHub\]drakos-amd](#)

Other Extension Metadata

Last Modified Date

2017-11-10

IP Status

No known IP claims.

Contributors

- Jaakko Konttinen, AMD
- David Mao, AMD
- Daniel Rakos, AMD
- Tobias Hector, Imagination Technologies
- Faith Ekstrand, Intel
- James Jones, NVIDIA

Description

This extension enables an application to import host allocations and host mapped foreign device memory to Vulkan memory objects.

New Commands

- [vkGetMemoryHostPointerPropertiesEXT](#)

New Structures

- [VkMemoryHostPointerPropertiesEXT](#)
- Extending [VkMemoryAllocateInfo](#):
 - [VkImportMemoryHostPointerInfoEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceExternalMemoryHostPropertiesEXT](#)

New Enum Constants

- [VK_EXT_EXTERNAL_MEMORY_HOST_EXTENSION_NAME](#)
- [VK_EXT_EXTERNAL_MEMORY_HOST_SPEC_VERSION](#)
- Extending [VkExternalMemoryHandleTypeFlagBits](#):
 - [VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_ALLOCATION_BIT_EXT](#)
 - [VK_EXTERNAL_MEMORY_HANDLE_TYPE_HOST_MAPPED_FOREIGN_MEMORY_BIT_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_IMPORT_MEMORY_HOST_POINTER_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_MEMORY_HOST_POINTER_PROPERTIES_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_EXTERNAL_MEMORY_HOST_PROPERTIES_EXT](#)

Issues

1) What memory type has to be used to import host pointers?

RESOLVED: Depends on the implementation. Applications have to use the new [vkGetMemoryHostPointerPropertiesEXT](#) command to query the supported memory types for a particular host pointer. The reported memory types may include memory types that come from a memory heap that is otherwise not usable for regular memory object allocation and thus such a heap's size may be zero.

2) Can the application still access the contents of the host allocation after importing?

RESOLVED: Yes. However, usual synchronization requirements apply.

3) Can the application free the host allocation?

RESOLVED: No, it violates valid usage conditions. Using the memory object imported from a host allocation that is already freed thus results in undefined behavior.

4) Is [vkMapMemory](#) expected to return the same host address which was specified when importing it to the memory object?

RESOLVED: No. Implementations are allowed to return the same address but it is not required. Some implementations might return a different virtual mapping of the allocation, although the same physical pages will be used.

5) Is there any limitation on the alignment of the host pointer and/or size?

RESOLVED: Yes. Both the address and the size have to be an integer multiple of [minImportedHostPointerAlignment](#). In addition, some platforms and foreign devices may have additional restrictions.

6) Can the same host allocation be imported multiple times into a given physical device?

RESOLVED: No, at least not guaranteed by this extension. Some platforms do not allow locking the same physical pages for device access multiple times, so attempting to do it may result in undefined behavior.

7) Does this extension support exporting the new handle type?

RESOLVED: No.

8) Should we include the possibility to import host mapped foreign device memory using this API?

RESOLVED: Yes, through a separate handle type. Implementations are still allowed to support only one of the handle types introduced by this extension by not returning import support for a particular handle type as returned in [VkExternalMemoryPropertiesKHR](#).

Version History

- Revision 1, 2017-11-10 (Daniel Rakos)
 - Internal revisions

VK_EXT_filter_cubic

Name String

VK_EXT_filter_cubic

Extension Type

Device extension

Registered Extension Number

171

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- Matthew Netsch [\[GitHub\]mnetsch](#)

Other Extension Metadata

Last Modified Date

2019-12-13

Contributors

- Bill Licea-Kane, Qualcomm Technologies, Inc.
- Andrew Garrard, Samsung
- Daniel Koch, NVIDIA
- Donald Scorgie, Imagination Technologies
- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, ARM
- Jeff Leger, Qualcomm Technologies, Inc.
- Tobias Hector, AMD
- Tom Olson, ARM
- Stuart Smith, Imagination Technologies

Description

VK_EXT_filter_cubic extends VK_IMG_filter_cubic.

It documents cubic filtering of other image view types. It adds new structures that **can** be added to

the **pNext** chain of [VkPhysicalDeviceImageFormatInfo2](#) and [VkImageFormatProperties2](#) that **can** be used to determine which image types and which image view types support cubic filtering.

New Structures

- Extending [VkImageFormatProperties2](#):
 - [VkFilterCubicImageViewImageFormatPropertiesEXT](#)
- Extending [VkPhysicalDeviceImageFormatInfo2](#):
 - [VkPhysicalDeviceImageViewImageFormatInfoEXT](#)

New Enum Constants

- [VK_EXT_FILTER_CUBIC_EXTENSION_NAME](#)
- [VK_EXT_FILTER_CUBIC_SPEC_VERSION](#)
- Extending [VkFilter](#):
 - [VK_FILTER_CUBIC_EXT](#)
- Extending [VkFormatFeatureFlagBits](#):
 - [VK_FORMAT_FEATURE_SAMPLED_IMAGE_FILTER_CUBIC_BIT_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_FILTER_CUBIC_IMAGE_VIEW_IMAGE_FORMAT_PROPERTIES_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_IMAGE_FORMAT_INFO_EXT](#)

Version History

- Revision 3, 2019-12-13 (wwlk)
 - Delete requirement to cubic filter the formats USCALED_PACKED32, SSCALED_PACKED32, UINT_PACK32, and SINT_PACK32 (cut/paste error)
- Revision 2, 2019-06-05 (wwlk)
 - Clarify 1D optional
- Revision 1, 2019-01-24 (wwlk)
 - Initial version

VK_EXT_fragment_shader_interlock

Name String

[VK_EXT_fragment_shader_interlock](#)

Extension Type

Device extension

Registered Extension Number

252

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_EXT_fragment_shader_interlock](#)

Contact

- Piers Daniell [[GitHub](#)][pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2019-05-02

Interactions and External Dependencies

- This extension provides API support for [GL_ARB_fragment_shader_interlock](#)

Contributors

- Daniel Koch, NVIDIA
- Graeme Leese, Broadcom
- Jan-Harald Fredriksen, Arm
- Faith Ekstrand, Intel
- Jeff Bolz, NVIDIA
- Ruihao Zhang, Qualcomm
- Slawomir Grajewski, Intel
- Spencer Fricke, Samsung

Description

This extension adds support for the [FragmentShaderPixelInterlockEXT](#), [FragmentShaderSampleInterlockEXT](#), and [FragmentShaderShadingRateInterlockEXT](#) capabilities from the [SPV_EXT_fragment_shader_interlock](#) extension to Vulkan.

Enabling these capabilities provides a critical section for fragment shaders to avoid overlapping pixels being processed at the same time, and certain guarantees about the ordering of fragment shader invocations of fragments of overlapping pixels.

This extension can be useful for algorithms that need to access per-pixel data structures via shader loads and stores. Algorithms using this extension can access per-pixel data structures in critical

sections without other invocations accessing the same per-pixel data. Additionally, the ordering guarantees are useful for cases where the API ordering of fragments is meaningful. For example, applications may be able to execute programmable blending operations in the fragment shader, where the destination buffer is read via image loads and the final value is written via image stores.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceFragmentShaderInterlockFeaturesEXT](#)

New Enum Constants

- [VK_EXT_FRAGMENT_SHADER_INTERLOCK_EXTENSION_NAME](#)
- [VK_EXT_FRAGMENT_SHADER_INTERLOCK_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAGMENT_SHADER_INTERLOCK_FEATURES_EXT](#)

New SPIR-V Capabilities

- [FragmentShaderInterlockEXT](#)
- [FragmentShaderPixelInterlockEXT](#)
- [FragmentShaderShadingRateInterlockEXT](#)

Version History

- Revision 1, 2019-05-24 (Piers Daniell)
 - Internal revisions

VK_EXT_frame_boundary

Name String

[VK_EXT_frame_boundary](#)

Extension Type

Device extension

Registered Extension Number

376

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

- James Fitzpatrick [[GitHub](#)][jamesfitzpatrick](#)

Extension Proposal

[VK_EXT_frame_boundary](#)

Other Extension Metadata

Last Modified Date

2023-06-14

Contributors

- James Fitzpatrick, Imagination Technologies
- Hugues Evrard, Google
- Melih Yasin Yalcin, Google
- Andrew Garrard, Imagination Technologies
- Jan-Harald Fredriksen, Arm
- Vassili Nikolaev, NVIDIA
- Ting Wei, Huawei

Description

[VK_EXT_frame_boundary](#) is a device extension that helps **tools** (such as debuggers) to group queue submissions per frames in non-trivial scenarios, typically when [vkQueuePresentKHR](#) is not a relevant frame boundary delimiter.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceFrameBoundaryFeaturesEXT](#)
- Extending [VkSubmitInfo](#), [VkSubmitInfo2](#), [VkPresentInfoKHR](#), [VkBindSparseInfo](#):
 - [VkFrameBoundaryEXT](#)

New Enums

- [VkFrameBoundaryFlagBitsEXT](#)

New Bitmasks

- [VkFrameBoundaryFlagsEXT](#)

New Enum Constants

- [VK_EXT_FRAME_BOUNDARY_EXTENSION_NAME](#)
- [VK_EXT_FRAME_BOUNDARY_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_FRAME_BOUNDARY_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_FRAME_BOUNDARY_FEATURES_EXT](#)

Version History

- Revision 0, 2022-01-14 (Hugues Evard)
 - Initial proposal
- Revision 1, 2023-06-14 (James Fitzpatrick)
 - Initial draft

VK_EXT_full_screen_exclusive

Name String

[VK_EXT_full_screen_exclusive](#)

Extension Type

Device extension

Registered Extension Number

256

Revision

4

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

and

[VK_KHR_surface](#)

and

[VK_KHR_get_surface_capabilities2](#)

and

[VK_KHR_swapchain](#)

API Interactions

- Interacts with [VK_VERSION_1_1](#)
- Interacts with [VK_KHR_device_group](#)

- Interacts with [VK_KHR_win32_surface](#)

Contact

- James Jones [\[GitHub\]](#)[cubanismo](#)

Other Extension Metadata

Last Modified Date

2019-03-12

IP Status

No known IP claims.

Interactions and External Dependencies

- Interacts with Vulkan 1.1
- Interacts with [VK_KHR_device_group](#)
- Interacts with [VK_KHR_win32_surface](#)

Contributors

- Hans-Kristian Arntzen, ARM
- Slawomir Grajewski, Intel
- Tobias Hector, AMD
- James Jones, NVIDIA
- Daniel Rakos, AMD
- Jeff Juliano, NVIDIA
- Joshua Schnarr, NVIDIA
- Aaron Hagan, AMD

Description

This extension allows applications to set the policy for swapchain creation and presentation mechanisms relating to full-screen access. Implementations may be able to acquire exclusive access to a particular display for an application window that covers the whole screen. This can increase performance on some systems by bypassing composition, however it can also result in disruptive or expensive transitions in the underlying windowing system when a change occurs.

Applications can choose between explicitly disallowing or allowing this behavior, letting the implementation decide, or managing this mode of operation directly using the new [vkAcquireFullScreenExclusiveModeEXT](#) and [vkReleaseFullScreenExclusiveModeEXT](#) commands.

New Commands

- [vkAcquireFullScreenExclusiveModeEXT](#)
- [vkGetPhysicalDeviceSurfacePresentModes2EXT](#)
- [vkReleaseFullScreenExclusiveModeEXT](#)

If [VK_KHR_device_group](#) or [Vulkan Version 1.1](#) is supported:

- [vkGetDeviceGroupSurfacePresentModes2EXT](#)

New Structures

- Extending [VkPhysicalDeviceSurfaceInfo2KHR](#), [VkSwapchainCreateInfoKHR](#):
 - [VkSurfaceFullScreenExclusiveInfoEXT](#)
- Extending [VkSurfaceCapabilities2KHR](#):
 - [VkSurfaceCapabilitiesFullScreenExclusiveEXT](#)

If [VK_KHR_win32_surface](#) is supported:

- Extending [VkPhysicalDeviceSurfaceInfo2KHR](#), [VkSwapchainCreateInfoKHR](#):
 - [VkSurfaceFullScreenExclusiveWin32InfoEXT](#)

New Enums

- [VkFullScreenExclusiveEXT](#)

New Enum Constants

- [VK_EXT_FULL_SCREEN_EXCLUSIVE_EXTENSION_NAME](#)
- [VK_EXT_FULL_SCREEN_EXCLUSIVE_SPEC_VERSION](#)
- Extending [VkResult](#):
 - [VK_ERROR_FULL_SCREEN_EXCLUSIVE_MODE_LOST_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_SURFACE_CAPABILITIES_FULL_SCREEN_EXCLUSIVE_EXT](#)
 - [VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_INFO_EXT](#)

If [VK_KHR_win32_surface](#) is supported:

- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_SURFACE_FULL_SCREEN_EXCLUSIVE_WIN32_INFO_EXT](#)

Issues

1) What should the extension & flag be called?

RESOLVED: [VK_EXT_full_screen_exclusive](#).

Other options considered (prior to the app-controlled mode) were:

- [VK_EXT_smooth_fullscreen_transition](#)
- [VK_EXT_fullscreen_behavior](#)
- [VK_EXT_fullscreen_preference](#)

- VK_EXT_fullscreen_hint
- VK_EXT_fast_fullscreen_transition
- VK_EXT_avoid_fullscreen_exclusive

2) Do we need more than a boolean toggle?

RESOLVED: Yes.

Using an enum with default/allowed/disallowed/app-controlled enables applications to accept driver default behavior, specifically override it in either direction without implying the driver is ever required to use full-screen exclusive mechanisms, or manage this mode explicitly.

3) Should this be a KHR or EXT extension?

RESOLVED: EXT, in order to allow it to be shipped faster.

4) Can the fullscreen hint affect the surface capabilities, and if so, should the hint also be specified as input when querying the surface capabilities?

RESOLVED: Yes on both accounts.

While the hint does not guarantee a particular fullscreen mode will be used when the swapchain is created, it can sometimes imply particular modes will NOT be used. If the driver determines that it will opt-out of using a particular mode based on the policy, and knows it can only support certain capabilities if that mode is used, it would be confusing at best to the application to report those capabilities in such cases. Not allowing implementations to report this state to applications could result in situations where applications are unable to determine why swapchain creation fails when they specify certain hint values, which could result in never-terminating surface creation loops.

5) Should full-screen be one word or two?

RESOLVED: Two words.

"Fullscreen" is not in my dictionary, and web searches did not turn up definitive proof that it is a colloquially accepted compound word. Documentation for the corresponding Windows API mechanisms dithers. The text consistently uses a hyphen, but none-the-less, there is a SetFullscreenState method in the DXGI swapchain object. Given this inconclusive external guidance, it is best to adhere to the Vulkan style guidelines and avoid inventing new compound words.

Version History

- Revision 4, 2019-03-12 (Tobias Hector)
 - Added application-controlled mode, and related functions
 - Tidied up appendix
- Revision 3, 2019-01-03 (James Jones)
 - Renamed to VK_EXT_full_screen_exclusive
 - Made related adjustments to the tri-state enumerant names.

- Revision 2, 2018-11-27 (James Jones)
 - Renamed to VK_KHR_fullscreen_behavior
 - Switched from boolean flag to tri-state enum
- Revision 1, 2018-11-06 (James Jones)
 - Internal revision

VK_EXT_hdr_metadata

Name String

VK_EXT_hdr_metadata

Extension Type

Device extension

Registered Extension Number

106

Revision

3

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_swapchain](#)

Contact

- Courtney Goeltzenleuchter [\[GitHub\]courtney-g](#)

Other Extension Metadata

Last Modified Date

2024-03-26

IP Status

No known IP claims.

Contributors

- Courtney Goeltzenleuchter, Google
- Sebastian Wick, Red Hat Inc.
- Tobias Hector, AMD

Description

This extension defines two new structures and a function to assign SMPTE (the Society of Motion Picture and Television Engineers) 2086 metadata and CTA (Consumer Technology Association) 861.3

metadata to a swapchain.

SMPTE 2086 metadata defines the color volume of the display on which the content was optimized for viewing and includes the color primaries, white point, and luminance range. When such content is reproduced on another display, this metadata can be used by the presentation engine to improve processing of images. For instance, values in the image can first be clamped to the color volume described in the metadata, and then what remains can be remapped to the color volume of the presentation engine.

CTA 861.3 metadata additionally includes the maximum intended luminance for the content and the maximum average light level across frames.

This extension does not define exactly how this metadata is used, however, it simply provides a mechanism to provide it to the presentation engine. Presentation engines may process the image based on the metadata before displaying it, resulting in the image being modified outside of Vulkan. For example, the clamping of colors in the image to the color volume may change those values in the image itself.

The metadata does not override or otherwise influence the color space and color encoding.

New Commands

- [vkSetHdrMetadataEXT](#)

New Structures

- [VkHdrMetadataEXT](#)
- [VkXYColorEXT](#)

New Enum Constants

- [VK_EXT_HDR_METADATA_EXTENSION_NAME](#)
- [VK_EXT_HDR_METADATA_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_HDR_METADATA_EXT](#)

Issues

1) Do we need a query function for the currently specified metadata?

No, Vulkan does not provide queries for state that the application can track on its own.

2) Should we specify default metadata if not specified by the application?

No, the metadata is optional and the absence of the metadata is well-defined.

Version History

- Revision 1, 2016-12-27 (Courtney Goeltzenleuchter)

- Initial version
- Revision 2, 2018-12-19 (Courtney Goeltzenleuchter)
 - Correct implicit validity for VkHdrMetadataEXT structure
- Revision 3, 2024-03-26 (Tobias Hector & Sebastian Wick)
 - Clarifications and removal of erroneous "reference monitor" term

VK_EXT_headless_surface

Name String

VK_EXT_headless_surface

Extension Type

Instance extension

Registered Extension Number

257

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

VK_KHR_surface

Contact

- Lisa Wu [\[GitHub\]chengtianww](#)

Other Extension Metadata

Last Modified Date

2019-03-21

IP Status

No known IP claims.

Contributors

- Ray Smith, Arm

Description

The **VK_EXT_headless_surface** extension is an instance extension. It provides a mechanism to create **VkSurfaceKHR** objects independently of any window system or display device. The presentation operation for a swapchain created from a headless surface is by default a no-op, resulting in no externally-visible result.

Because there is no real presentation target, future extensions can layer on top of the headless surface to introduce arbitrary or customizable sets of restrictions or features. These could include features like saving to a file or restrictions to emulate a particular presentation target.

This functionality is expected to be useful for application and driver development because it allows any platform to expose an arbitrary or customizable set of restrictions and features of a presentation engine. This makes it a useful portable test target for applications targeting a wide range of presentation engines where the actual target presentation engines might be scarce, unavailable or otherwise undesirable or inconvenient to use for general Vulkan application development.

New Commands

- [vkCreateHeadlessSurfaceEXT](#)

New Structures

- [VkHeadlessSurfaceCreateInfoEXT](#)

New Bitmasks

- [VkHeadlessSurfaceCreateFlagsEXT](#)

New Enum Constants

- [VK_EXT_HEADLESS_SURFACE_EXTENSION_NAME](#)
- [VK_EXT_HEADLESS_SURFACE_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_HEADLESS_SURFACE_CREATE_INFO_EXT](#)

Version History

- Revision 1, 2019-03-21 (Ray Smith)
 - Initial draft

VK_EXT_host_image_copy

Name String

[VK_EXT_host_image_copy](#)

Extension Type

Device extension

Registered Extension Number

271

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

and

[VK_KHR_copy_commands2](#)

and

[VK_KHR_format_feature_flags2](#)

or

[Vulkan Version 1.3](#)

Contact

- Shahbaz Youssefi [\[GitHub\]syoussefi](#)

Extension Proposal

[VK_EXT_host_image_copy](#)

Other Extension Metadata

Last Modified Date

2023-04-26

Contributors

- Shahbaz Youssefi, Google
- Faith Ekstrand, Collabora
- Hans-Kristian Arntzen, Valve
- Piers Daniell, NVIDIA
- Jan-Harald Fredriksen, Arm
- James Fitzpatrick, Imagination
- Daniel Story, Nintendo

Description

This extension allows applications to copy data between host memory and images on the host processor, without staging the data through a GPU-accessible buffer. This removes the need to allocate and manage the buffer and its associated memory. On some architectures it may also eliminate an extra copy operation. This extension additionally allows applications to copy data between images on the host.

To support initializing a new image in preparation for a host copy, it is now possible to transition a new image to [VK_IMAGE_LAYOUT_GENERAL](#) or other host-copyable layouts via [vkTransitionImageLayoutEXT](#). Additionally, it is possible to perform copies that preserve the swizzling layout of the image by using the [VK_HOST_IMAGE_COPY_MEMCPY_EXT](#) flag. In that case, the

memory size needed for copies to or from a buffer can be retrieved by chaining [VkSubresourceHostMemcpySizeEXT](#) to [pLayout](#) in [vkGetImageSubresourceLayout2EXT](#).

New Commands

- [vkCopyImageToImageEXT](#)
- [vkCopyImageToMemoryEXT](#)
- [vkCopyMemoryToImageEXT](#)
- [vkGetImageSubresourceLayout2EXT](#)
- [vkTransitionImageLayoutEXT](#)

New Structures

- [VkCopyImageToImageInfoEXT](#)
- [VkCopyImageToMemoryInfoEXT](#)
- [VkCopyMemoryToImageInfoEXT](#)
- [VkHostImageLayoutTransitionInfoEXT](#)
- [VkImageSubresource2EXT](#)
- [VkImageToMemoryCopyEXT](#)
- [VkMemoryToImageCopyEXT](#)
- [VkSubresourceLayout2EXT](#)
- Extending [VkImageFormatProperties2](#):
 - [VkHostImageCopyDevicePerformanceQueryEXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceHostImageCopyFeaturesEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceHostImageCopyPropertiesEXT](#)
- Extending [VkSubresourceLayout2KHR](#):
 - [VkSubresourceHostMemcpySizeEXT](#)

New Enums

- [VkHostImageCopyFlagBitsEXT](#)

New Bitmasks

- [VkHostImageCopyFlagsEXT](#)

New Enum Constants

- [VK_EXT_HOST_IMAGE_COPY_EXTENSION_NAME](#)
- [VK_EXT_HOST_IMAGE_COPY_SPEC_VERSION](#)

- Extending `VkFormatFeatureFlagBits2`:
 - `VK_FORMAT_FEATURE_2_HOST_IMAGE_TRANSFER_BIT_EXT`
- Extending `VkImageUsageFlagBits`:
 - `VK_IMAGE_USAGE_HOST_TRANSFER_BIT_EXT`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_COPY_IMAGE_TO_IMAGE_INFO_EXT`
 - `VK_STRUCTURE_TYPE_COPY_IMAGE_TO_MEMORY_INFO_EXT`
 - `VK_STRUCTURE_TYPE_COPY_MEMORY_TO_IMAGE_INFO_EXT`
 - `VK_STRUCTURE_TYPE_HOST_IMAGE_COPY_DEVICE_PERFORMANCE_QUERY_EXT`
 - `VK_STRUCTURE_TYPE_HOST_IMAGE_LAYOUT_TRANSITION_INFO_EXT`
 - `VK_STRUCTURE_TYPE_IMAGE_TO_MEMORY_COPY_EXT`
 - `VK_STRUCTURE_TYPE_MEMORY_TO_IMAGE_COPY_EXT`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_FEATURES_EXT`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_HOST_IMAGE_COPY_PROPERTIES_EXT`
 - `VK_STRUCTURE_TYPE_SUBRESOURCE_HOST_MEMCPY_SIZE_EXT`

Issues

1) When uploading data to an image, the data is usually loaded from disk. Why not have the application load the data directly into a `VkDeviceMemory` bound to a buffer (instead of host memory), and use `vkCmdCopyBufferToImage`? The same could be done when downloading data from an image.

RESOLVED: This may not always be possible. Complicated Vulkan applications such as game engines often have decoupled subsystems for streaming data and rendering. It may be unreasonable to require the streaming subsystem to coordinate with the rendering subsystem to allocate memory on its behalf, especially as Vulkan may not be the only API supported by the engine. In emulation layers, the image data is necessarily provided by the application in host memory, so an optimization as suggested is not possible. Most importantly, the device memory may not be mappable by an application, but still accessible to the driver.

2) Are `optimalBufferCopyOffsetAlignment` and `optimalBufferCopyRowPitchAlignment` applicable to host memory as well with the functions introduced by this extension? Or should there be new limits?

RESOLVED: No alignment requirements for the host memory pointer.

3) Should there be granularity requirements for image offsets and extents?

RESOLVED: No granularity requirements, i.e. a granularity of 1 pixel (for non-compressed formats) and 1 texel block (for compressed formats) is assumed.

4) How should the application deal with layout transitions before or after copying to or from images?

RESOLVED: An existing issue with linear images is that when emulating other APIs, it is impossible to know when to transition them as they are written to by the host and then used bindlessly. The copy operations in this extension are affected by the same limitation. A new command is thus introduced by this extension to address this problem by allowing the host to perform an image layout transition between a handful of layouts.

Version History

- Revision 0, 2021-01-20 (Faith Ekstrand)
 - Initial idea and xml
- Revision 1, 2023-04-26 (Shahbaz Youssefi)
 - Initial revision

VK_EXT_image_drm_format_modifier

Name String

`VK_EXT_image_drm_format_modifier`

Extension Type

Device extension

Registered Extension Number

159

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

`VK_KHR_bind_memory2`
and
`VK_KHR_get_physical_device_properties2`
and
`VK_KHR_sampler_ycbcr_conversion`
or
`Vulkan Version 1.1`
and
`VK_KHR_image_format_list`
or
`Vulkan Version 1.2`

API Interactions

- Interacts with `VK_VERSION_1_3`
- Interacts with `VK_KHR_format_feature_flags2`

Contact

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Other Extension Metadata

Last Modified Date

2021-09-30

IP Status

No known IP claims.

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Description

This extension provides the ability to use *DRM format modifiers* with images, enabling Vulkan to better integrate with the Linux ecosystem of graphics, video, and display APIs.

Its functionality closely overlaps with [EGL_EXT_image_dma_buf_import_modifiers](#)² and [EGL_MESA_image_dma_buf_export](#)³. Unlike the EGL extensions, this extension does not require the use of a specific handle type (such as a `dma_buf`) for external memory and provides more explicit control of image creation.

Introduction to DRM Format Modifiers

A *DRM format modifier* is a 64-bit, vendor-prefixed, semi-opaque unsigned integer. Most *modifiers* represent a concrete, vendor-specific tiling format for images. Some exceptions are `DRM_FORMAT_MOD_LINEAR` (which is not vendor-specific); `DRM_FORMAT_MOD_NONE` (which is an alias of `DRM_FORMAT_MOD_LINEAR` due to historical accident); and `DRM_FORMAT_MOD_INVALID` (which does not represent a tiling format). The *modifier*'s vendor prefix consists of the 8 most significant bits. The canonical list of *modifiers* and vendor prefixes is found in `drm_fourcc.h` in the Linux kernel source. The other dominant source of *modifiers* are vendor kernel trees.

One goal of *modifiers* in the Linux ecosystem is to enumerate for each vendor a reasonably sized set of tiling formats that are appropriate for images shared across processes, APIs, and/or devices, where each participating component may possibly be from different vendors. A non-goal is to enumerate all tiling formats supported by all vendors. Some tiling formats used internally by vendors are inappropriate for sharing; no *modifiers* should be assigned to such tiling formats.

Modifier values typically do not *describe* memory layouts. More precisely, a *modifier*'s lower 56 bits usually have no structure. Instead, modifiers *name* memory layouts; they name a small set of vendor-preferred layouts for image sharing. As a consequence, in each vendor namespace the modifier values are often sequentially allocated starting at 1.

Each *modifier* is usually supported by a single vendor and its name matches the pattern `{VENDOR}_FORMAT_MOD_*` or `DRM_FORMAT_MOD_{VENDOR}_*`. Examples are `I915_FORMAT_MOD_X_TILED` and `DRM_FORMAT_MOD_BROADCOM_VC4_T_TILED`. An exception is `DRM_FORMAT_MOD_LINEAR`, which is supported by most vendors.

Many APIs in Linux use *modifiers* to negotiate and specify the memory layout of shared images. For example, a Wayland compositor and Wayland client may, by relaying *modifiers* over the Wayland protocol `zwp_linux_dmabuf_v1`, negotiate a vendor-specific tiling format for a shared `wl_buffer`. The client may allocate the underlying memory for the `wl_buffer` with GBM, providing the chosen *modifier* to `gbm_bo_create_with_modifiers`. The client may then import the `wl_buffer` into Vulkan for producing image content, providing the resource's `dma_buf` to `VkImportMemoryFdInfoKHR` and its *modifier* to `VkImageDrmFormatModifierExplicitCreateInfoEXT`. The compositor may then import the `wl_buffer` into OpenGL for sampling, providing the resource's `dma_buf` and *modifier* to `eglCreateImage`. The compositor may also bypass OpenGL and submit the `wl_buffer` directly to the kernel's display API, providing the `dma_buf` and *modifier* through `drm_mode_fb_cmd2`.

Format Translation

Modifier-capable APIs often pair *modifiers* with DRM formats, which are defined in `drm_fourcc.h`. However, `VK_EXT_image_drm_format_modifier` uses `VkFormat` instead of DRM formats. The application must convert between `VkFormat` and DRM format when it sends or receives a DRM format to or from an external API.

The mapping from `VkFormat` to DRM format is lossy. Therefore, when receiving a DRM format from an external API, often the application must use information from the external API to accurately map the DRM format to a `VkFormat`. For example, DRM formats do not distinguish between RGB and sRGB (as of 2018-03-28); external information is required to identify the image's color space.

The mapping between `VkFormat` and DRM format is also incomplete. For some DRM formats there exist no corresponding Vulkan format, and for some Vulkan formats there exist no corresponding DRM format.

Usage Patterns

Three primary usage patterns are intended for this extension:

- **Negotiation.** The application negotiates with *modifier*-aware, external components to determine sets of image creation parameters supported among all components.

In the Linux ecosystem, the negotiation usually assumes the image is a 2D, single-sampled, non-mipmapped, non-array image; this extension permits that assumption but does not require it. The result of the negotiation usually resembles a set of tuples such as `(drmFormat, drmFormatModifier)`, where each participating component supports all tuples in the set.

Many details of this negotiation - such as the protocol used during negotiation, the set of image

creation parameters expressible in the protocol, and how the protocol chooses which process and which API will create the image - are outside the scope of this specification.

In this extension, [vkGetPhysicalDeviceFormatProperties2KHR](#) with [VkDrmFormatModifierPropertiesListEXT](#) serves a primary role during the negotiation, and [vkGetPhysicalDeviceImageFormatProperties2KHR](#) with [VkPhysicalDeviceImageDrmFormatModifierInfoEXT](#) serves a secondary role.

- **Import.** The application imports an image with a *modifier*.

In this pattern, the application receives from an external source the image's memory and its creation parameters, which are often the result of the negotiation described above. Some image creation parameters are implicitly defined by the external source; for example, [VK_IMAGE_TYPE_2D](#) is often assumed. Some image creation parameters are usually explicit, such as the image's *format*, *drmFormatModifier*, and *extent*; and each plane's *offset* and *rowPitch*.

Before creating the image, the application first verifies that the physical device supports the received creation parameters by querying [vkGetPhysicalDeviceFormatProperties2KHR](#) with [VkDrmFormatModifierPropertiesListEXT](#) and [vkGetPhysicalDeviceImageFormatProperties2KHR](#) with [VkPhysicalDeviceImageDrmFormatModifierInfoEXT](#). Then the application creates the image by chaining [VkImageDrmFormatModifierExplicitCreateInfoEXT](#) and [VkExternalMemoryImageCreateInfo](#) onto [VkImageCreateInfo](#).

- **Export.** The application creates an image and allocates its memory. Then the application exports to *modifier*-aware consumers the image's memory handles; its creation parameters; its *modifier*; and the *offset*, *size*, and *rowPitch* of each *memory plane*.

In this pattern, the Vulkan device is the authority for the image; it is the allocator of the image's memory and the decider of the image's creation parameters. When choosing the image's creation parameters, the application usually chooses a tuple (*format*, *drmFormatModifier*) from the result of the negotiation described above. The negotiation's result often contains multiple tuples that share the same format but differ in their *modifier*. In this case, the application should defer the choice of the image's *modifier* to the Vulkan implementation by providing all such *modifiers* to [VkImageDrmFormatModifierListCreateInfoEXT::pDrmFormatModifiers](#); and the implementation should choose from [pDrmFormatModifiers](#) the optimal *modifier* in consideration with the other image parameters.

The application creates the image by chaining [VkImageDrmFormatModifierListCreateInfoEXT](#) and [VkExternalMemoryImageCreateInfo](#) onto [VkImageCreateInfo](#). The protocol and APIs by which the application will share the image with external consumers will likely determine the value of [VkExternalMemoryImageCreateInfo::handleTypes](#). The implementation chooses for the image an optimal *modifier* from [VkImageDrmFormatModifierListCreateInfoEXT::pDrmFormatModifiers](#). The application then queries the implementation-chosen *modifier* with [vkGetImageDrmFormatModifierPropertiesEXT](#), and queries the memory layout of each plane with [vkGetImageSubresourceLayout](#).

The application then allocates the image's memory with [VkMemoryAllocateInfo](#), adding chained extending structures for external memory; binds it to the image; and exports the

memory, for example, with [vkGetMemoryFdKHR](#).

Finally, the application sends the image's creation parameters, its *modifier*, its per-plane memory layout, and the exported memory handle to the external consumers. The details of how the application transmits this information to external consumers is outside the scope of this specification.

Prior Art

Extension [EGL_EXT_image_dma_buf_import](#)¹ introduced the ability to create an [EGLImage](#) by importing for each plane a dma_buf, offset, and row pitch.

Later, extension [EGL_EXT_image_dma_buf_import_modifiers](#)² introduced the ability to query which combination of formats and *modifiers* the implementation supports and to specify *modifiers* during creation of the [EGLImage](#).

Extension [EGL_MESA_image_dma_buf_export](#)³ is the inverse of [EGL_EXT_image_dma_buf_import_modifiers](#).

The Linux kernel modesetting API (KMS), when configuring the display's framebuffer with [struct drm_mode_fb_cmd2](#)⁴, allows one to specify the framebuffer's *modifier* as well as a per-plane memory handle, offset, and row pitch.

GBM, a graphics buffer manager for Linux, allows creation of a [gbm_bo](#) (that is, a graphics *buffer object*) by importing data similar to that in [EGL_EXT_image_dma_buf_import_modifiers](#)¹; and symmetrically allows exporting the same data from the [gbm_bo](#). See the references to *modifier* and *plane* in [gbm.h](#)⁵.

New Commands

- [vkGetImageDrmFormatModifierPropertiesEXT](#)

New Structures

- [VkDrmFormatModifierPropertiesEXT](#)
- [VkImageDrmFormatModifierPropertiesEXT](#)
- Extending [VkFormatProperties2](#):
 - [VkDrmFormatModifierPropertiesListEXT](#)
- Extending [VkImageCreateInfo](#):
 - [VkImageDrmFormatModifierExplicitCreateInfoEXT](#)
 - [VkImageDrmFormatModifierListCreateInfoEXT](#)
- Extending [VkPhysicalDeviceImageFormatInfo2](#):
 - [VkPhysicalDeviceImageDrmFormatModifierInfoEXT](#)

If [VK_KHR_format_feature_flags2](#) or Vulkan Version 1.3 is supported:

- [VkDrmFormatModifierProperties2EXT](#)
- Extending [VkFormatProperties2](#):

- [VkDrmFormatModifierPropertiesList2EXT](#)

New Enum Constants

- `VK_EXT_IMAGE_DRM_FORMAT_MODIFIER_EXTENSION_NAME`
- `VK_EXT_IMAGE_DRM_FORMAT_MODIFIER_SPEC_VERSION`
- Extending [VkImageAspectFlagBits](#):
 - `VK_IMAGE_ASPECT_MEMORY_PLANE_0_BIT_EXT`
 - `VK_IMAGE_ASPECT_MEMORY_PLANE_1_BIT_EXT`
 - `VK_IMAGE_ASPECT_MEMORY_PLANE_2_BIT_EXT`
 - `VK_IMAGE_ASPECT_MEMORY_PLANE_3_BIT_EXT`
- Extending [VkImageTiling](#):
 - `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`
- Extending [VkResult](#):
 - `VK_ERROR_INVALID_DRM_FORMAT_MODIFIER_PLANE_LAYOUT_EXT`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_EXT`
 - `VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_EXPLICIT_CREATE_INFO_EXT`
 - `VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_LIST_CREATE_INFO_EXT`
 - `VK_STRUCTURE_TYPE_IMAGE_DRM_FORMAT_MODIFIER_PROPERTIES_EXT`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_DRM_FORMAT_MODIFIER_INFO_EXT`

If [VK_KHR_format_feature_flags2](#) or [Vulkan Version 1.3](#) is supported:

- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_DRM_FORMAT_MODIFIER_PROPERTIES_LIST_2_EXT`

Issues

1) Should this extension define a single DRM format modifier per [VkImage](#)? Or define one per plane?

+

RESOLVED: There exists a single DRM format modifier per [VkImage](#).

DISCUSSION: Prior art, such as [EGL_EXT_image_dma_buf_import_modifiers](#)², `struct drm_mode_fb_cmd2`⁴, and `struct gbm_import_fd_modifier_data`⁵, allows defining one *modifier* per plane. However, developers of the GBM and kernel APIs concede it was a mistake. Beginning in Linux 4.10, the kernel requires that the application provide the same DRM format *modifier* for each plane. (See Linux commit [bae781b259269590109e8a4a8227331362b88212](#)). And GBM provides an entry point, `gbm_bo_get_modifier`, for querying the *modifier* of the image but does not provide one to query the modifier of individual planes.

2) When creating an image with `VkImageDrmFormatModifierExplicitCreateInfoEXT`, which is typically used when *importing* an image, should the application explicitly provide the size of each plane?

+

RESOLVED: No. The application **must** not provide the size. To enforce this, the API requires that `VkImageDrmFormatModifierExplicitCreateInfoEXT::pPlaneLayouts->size` **must** be 0.

DISCUSSION: Prior art, such as `EGL_EXT_image_dma_buf_import_modifiers`², `struct drm_mode_fb_cmd2`⁴, and `struct gbm_import_fd_modifier_data`⁵, omits from the API the size of each plane. Instead, the APIs infer each plane's size from the import parameters, which include the image's pixel format and a `dma_buf`, offset, and row pitch for each plane.

However, Vulkan differs from EGL and GBM with regards to image creation in the following ways:

Differences in Image Creation

- **Undedicated allocation by default.** When importing or exporting a set of `dma_bufs` as an `EGLImage` or `gbm_bo`, common practice mandates that each `dma_buf`'s memory be dedicated (in the sense of `VK_KHR_dedicated_allocation`) to the image (though not necessarily dedicated to a single plane). In particular, neither the GBM documentation nor the EGL extension specifications explicitly state this requirement, but in light of common practice this is likely due to under-specification rather than intentional omission. In contrast, `VK_EXT_image_drm_format_modifier` permits, but does not require, the implementation to require dedicated allocations for images created with `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT`.
- **Separation of image creation and memory allocation.** When importing a set of `dma_bufs` as an `EGLImage` or `gbm_bo`, EGL and GBM create the image resource and bind it to memory (the `dma_bufs`) simultaneously. This allows EGL and GBM to query each `dma_buf`'s size during image creation. In Vulkan, image creation and memory allocation are independent unless a dedicated allocation is used (as in `VK_KHR_dedicated_allocation`). Therefore, without requiring dedicated allocation, Vulkan cannot query the size of each `dma_buf` (or other external handle) when calculating the image's memory layout. Even if dedication allocation were required, Vulkan cannot calculate the image's memory layout until after the image is bound to its `dma_bufs`.

The above differences complicate the potential inference of plane size in Vulkan. Consider the following problematic cases:

Problematic Plane Size Calculations

- **Padding.** Some plane of the image may require implementation-dependent padding.
- **Metadata.** For some *modifiers*, the image may have a metadata plane which requires a non-trivial calculation to determine its size.
- **Mipmapped, array, and 3D images.** The implementation may support `VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT` for images whose `mipLevels`, `arrayLayers`, or `depth` is greater than 1. For such images with certain *modifiers*, the calculation of each plane's size may be non-trivial.

However, an application-provided plane size solves none of the above problems.

For simplicity, consider an external image with a single memory plane. The implementation is obviously capable calculating the image's size when its tiling is `VK_IMAGE_TILING_OPTIMAL`. Likewise, any reasonable implementation is capable of calculating the image's size when its tiling uses a supported *modifier*.

Suppose that the external image's size is smaller than the implementation-calculated size. If the application provided the external image's size to `vkCreateImage`, the implementation would observe the mismatched size and recognize its inability to comprehend the external image's layout (unless the implementation used the application-provided size to select a refinement of the tiling layout indicated by the *modifier*, which is strongly discouraged). The implementation would observe the conflict, and reject image creation with `VK_ERROR_INVALID_DRM_FORMAT_MODIFIER_PLANE_LAYOUT_EXT`. On the other hand, if the application did not provide the external image's size to `vkCreateImage`, then the application would observe after calling `vkGetImageMemoryRequirements` that the external image's size is less than the size required by the implementation. The application would observe the conflict and refuse to bind the `VkImage` to the external memory. In both cases, the result is explicit failure.

Suppose that the external image's size is larger than the implementation-calculated size. If the application provided the external image's size to `vkCreateImage`, for reasons similar to above the implementation would observe the mismatched size and recognize its inability to comprehend the image data residing in the extra size. The implementation, however, must assume that image data resides in the entire size provided by the application. The implementation would observe the conflict and reject image creation with `VK_ERROR_INVALID_DRM_FORMAT_MODIFIER_PLANE_LAYOUT_EXT`. On the other hand, if the application did not provide the external image's size to `vkCreateImage`, then the application would observe after calling `vkGetImageMemoryRequirements` that the external image's size is larger than the implementation-usable size. The application would observe the conflict and refuse to bind the `VkImage` to the external memory. In both cases, the result is explicit failure.

Therefore, an application-provided size provides no benefit, and this extension should not require it. This decision renders `VkSubresourceLayout::size` an unused field during image creation, and thus introduces a risk that implementations may require applications to submit sideband creation parameters in the unused field. To prevent implementations from relying on sideband data, this extension *requires* the application to set `size` to 0.

References

1. `EGL_EXT_image_dma_buf_import`
2. `EGL_EXT_image_dma_buf_import_modifiers`
3. `EGL_MESA_image_dma_buf_export`
4. `struct drm_mode_fb_cmd2`
5. `gbm.h`

Version History

- Revision 1, 2018-08-29 (Lina Versace)
 - First stable revision

- Revision 2, 2021-09-30 (Jon Leech)
 - Add interaction with [VK_KHR_format_feature_flags2](#) to [vk.xml](#)

VK_EXT_image_view_min_lod

Name String

VK_EXT_image_view_min_lod

Extension Type

Device extension

Registered Extension Number

392

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

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Other Extension Metadata

Last Modified Date

2021-11-09

IP Status

No known IP claims.

Contributors

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Description

This extension allows applications to clamp the minimum LOD value during [Image Level\(s\) Selection](#), [Texel Gathering](#) and [Integer Texel Coordinate Operations](#) with a given [VkImageView](#) by [VkImageViewMinLodCreateInfoEXT::minLod](#).

This extension may be useful to restrict a [VkImageView](#) to only mips which have been uploaded, and the use of fractional [minLod](#) can be useful for smoothly introducing new mip levels when using linear mipmap filtering.

New Structures

- Extending [VkImageViewCreateInfo](#):
 - [VkImageViewMinLodCreateInfoEXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceImageViewMinLodFeaturesEXT](#)

New Enum Constants

- [VK_EXT_IMAGE_VIEW_MIN_LOD_EXTENSION_NAME](#)
- [VK_EXT_IMAGE_VIEW_MIN_LOD_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_IMAGE_VIEW_MIN_LOD_CREATE_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_IMAGE_VIEW_MIN_LOD_FEATURES_EXT](#)

Version History

- Revision 1, 2021-07-06 (Joshua Ashton)
 - Initial version

VK_EXT_layer_settings

Name String

[VK_EXT_layer_settings](#)

Extension Type

Instance extension

Registered Extension Number

497

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

None

Contact

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Extension Proposal

[VK_EXT_layer_settings](#)

Other Extension Metadata

Last Modified Date

2023-09-23

IP Status

No known IP claims.

Contributors

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Description

This extension provides a mechanism for configuring programmatically through the Vulkan API the behavior of layers.

This extension provides the [VkLayerSettingsCreateInfoEXT](#) struct that can be included in the [pNext](#) chain of the [VkInstanceCreateInfo](#) structure passed as the [pCreateInfo](#) parameter of [vkCreateInstance](#).

The structure contains an array of [VkLayerSettingEXT](#) structure values that configure specific features of layers.

NOTE

The [VK_EXT_layer_settings](#) extension subsumes all the functionality provided in the [VK_EXT_validation_flags](#) extension and the [VK_EXT_validation_features](#) extension.

New Structures

- [VkLayerSettingEXT](#)
- Extending [VkInstanceCreateInfo](#):
 - [VkLayerSettingsCreateInfoEXT](#)

New Enums

- [VkLayerSettingTypeEXT](#)

New Enum Constants

- [VK_EXT_LAYER_SETTINGS_EXTENSION_NAME](#)
- [VK_EXT_LAYER_SETTINGS_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_LAYER_SETTINGS_CREATE_INFO_EXT](#)

Example

One example usage of [VK_EXT_layer_settings](#) is as implemented by the Vulkan Profiles layer.

It allows the profiles layer tests used by the profiles layer C.I. to programmatically configure the layer for each test without affecting the C.I. environment, allowing to run multiple tests concurrently.

```
const char* profile_file_data = JSON_TEST_FILES_PATH "VP_KHR_roadmap_2022.json";
const char* profile_name_data = "VP_KHR_roadmap_2022";
VkBool32 emulate_portability_data = VK_TRUE;
const char* simulate_capabilities[] = {
    "SIMULATE_API_VERSION_BIT",
    "SIMULATE_FEATURES_BIT",
    "SIMULATE_PROPERTIES_BIT",
    "SIMULATE_EXTENSIONS_BIT",
    "SIMULATE_FORMATS_BIT",
    "SIMULATE_QUEUE_FAMILY_PROPERTIES_BIT"
};
const char* debug_reports[] = {
    "DEBUG_REPORT_ERROR_BIT",
    "DEBUG_REPORT_WARNING_BIT",
    "DEBUG_REPORT_NOTIFICATION_BIT",
    "DEBUG_REPORT_DEBUG_BIT"
};

const VkLayerSettingEXT settings[] = {
    {kLayerName, kLayerSettingsProfileFile, VK_LAYER_SETTING_TYPE_STRING_EXT, 1,
    &profile_file_data},
    {kLayerName, kLayerSettingsProfileName, VK_LAYER_SETTING_TYPE_STRING_EXT, 1,
    &profile_name_data},
    {kLayerName, kLayerSettingsEmulatePortability, VK_LAYER_SETTING_TYPE_BOOL32_EXT, 1,
```

```

1, &emulate_portability_data},
    {kLayerName, kLayerSettingsSimulateCapabilities,
VK_LAYER_SETTING_TYPE_STRING_EXT,
    static_cast<uint32_t>(std::size(simulate_capabilities)),
simulate_capabilities},
    {kLayerName, kLayerSettingsDebugReports, VK_LAYER_SETTING_TYPE_STRING_EXT,
    static_cast<uint32_t>(std::size(debug_reports)), debug_reports}
};

const VkLayerSettingsCreateInfoEXT layer_settings_create_info{
    VK_STRUCTURE_TYPE_LAYER_SETTINGS_CREATE_INFO_EXT, nullptr,
    static_cast<uint32_t>(std::size(settings)), settings};

VkInstanceCreateInfo inst_create_info = {};
...
inst_create_info.pNext = &layer_settings_create_info;
vkCreateInstance(&inst_create_info, nullptr, &instances);

```

Issues

- How should application developers figure out the list of available settings?

This extension does not provide a reflection API for layer settings. Layer settings are described in each layer JSON manifest and the documentation of each layer which implements this extension.

Version History

- Revision 1, 2020-06-17 (Mark Lobodzinski)
 - Initial revision for Validation layer internal usages
- Revision 2, 2023-09-26 (Christophe Riccio)
 - Refactor APIs for any layer usages and public release

VK_EXT_legacy_vertex_attributes

Name String

VK_EXT_legacy_vertex_attributes

Extension Type

Device extension

Registered Extension Number

496

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_EXT_vertex_input_dynamic_state](#)

Special Use

- [OpenGL / ES support](#)

Contact

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Extension Proposal

[VK_EXT_legacy_vertex_attributes](#)

Other Extension Metadata

Last Modified Date

2024-02-23

IP Status

No known IP claims.

Contributors

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- Piers Daniell, NVIDIA
- Spencer Fricke, LunarG
- Alyssa Rosenzweig, Valve

Description

This extension adds support for legacy features of (non-64-bit) vertex attributes as found in OpenGL:

- Vertex attributes loaded from arbitrary buffer alignments
- Vertex attributes using arbitrary strides
- Vertex attributes where the component data type of the binding does not match the component numeric type of the shader input

These features are only usable with dynamic vertex input. Unaligned loads of vertex attributes may incur performance penalties, indicated with a property.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceLegacyVertexAttributesFeaturesEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceLegacyVertexAttributesPropertiesEXT](#)

New Enum Constants

- `VK_EXT_LEGACY_VERTEX_ATTRIBUTES_EXTENSION_NAME`
- `VK_EXT_LEGACY_VERTEX_ATTRIBUTES_SPEC_VERSION`
- Extending [VkStructureType](#):
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_FEATURES_EXT`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_LEGACY_VERTEX_ATTRIBUTES_PROPERTIES_EXT`

Issues

1.) Should implementations convert float/integer values?

RESOLVED: No. When fetching an integer data type from float values or float data types from integer values, the resulting shader values are implementation-dependent.

Version History

- Revision 1, 2024-02-16 (Mike Blumenkrantz)
 - Initial revision

VK_EXT_map_memory_placed

Name String

`VK_EXT_map_memory_placed`

Extension Type

Device extension

Registered Extension Number

273

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_map_memory2](#)

Contact

- Faith Ekstrand [\[GitHub\]](#)[gfxstrand](#)

Extension Proposal

[VK_EXT_map_memory_placed](#)

Other Extension Metadata

Last Modified Date

2023-03-21

IP Status

No known IP claims.

Interactions and External Dependencies

- Depends on [apitext:VK_KHR_map_memory2](#)
- Interacts with [apitext:VK_EXT_external_memory_host](#)

Contributors

- Faith Ekstrand, Collabora
- Tobias Hector, AMD
- James Jones, NVIDIA
- Georg Lehmann, Valve
- Derek Lesho, Codeweavers

Description

This extension allows an application to request that [vkMapMemory2KHR](#) attempt to place the memory map at a particular virtual address.

New Structures

- Extending [VkMemoryMapInfoKHR](#):
 - [VkMemoryMapPlacedInfoEXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceMapMemoryPlacedFeaturesEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceMapMemoryPlacedPropertiesEXT](#)

New Enum Constants

- [VK_EXT_MAP_MEMORY_PLACED_EXTENSION_NAME](#)
- [VK_EXT_MAP_MEMORY_PLACED_SPEC_VERSION](#)
- Extending [VkMemoryMapFlagBits](#):
 - [VK_MEMORY_MAP_PLACED_BIT_EXT](#)
- Extending [VkMemoryUnmapFlagBitsKHR](#):
 - [VK_MEMORY_UNMAP_RESERVE_BIT_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_MEMORY_MAP_PLACED_INFO_EXT](#)

- `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_FEATURES_EXT`
- `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MAP_MEMORY_PLACED_PROPERTIES_EXT`

Version History

- Revision 0, 2024-01-14 (Faith Ekstrand)
 - Internal revisions

VK_EXT_memory_budget

Name String

`VK_EXT_memory_budget`

Extension Type

Device extension

Registered Extension Number

238

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Jeff Bolz [\[GitHub\]jeffbolznv](#)

Other Extension Metadata

Last Modified Date

2018-10-08

Contributors

- Jeff Bolz, NVIDIA
- Jeff Juliano, NVIDIA

Description

While running a Vulkan application, other processes on the machine might also be attempting to use the same device memory, which can pose problems. This extension adds support for querying the amount of memory used and the total memory budget for a memory heap. The values returned by this query are implementation-dependent and can depend on a variety of factors including

operating system and system load.

The `VkPhysicalDeviceMemoryBudgetPropertiesEXT::heapBudget` values can be used as a guideline for how much total memory from each heap the **current process** can use at any given time, before allocations may start failing or causing performance degradation. The values may change based on other activity in the system that is outside the scope and control of the Vulkan implementation.

The `VkPhysicalDeviceMemoryBudgetPropertiesEXT::heapUsage` will display the **current process** estimated heap usage.

With this information, the idea is for an application at some interval (once per frame, per few seconds, etc) to query `heapBudget` and `heapUsage`. From here the application can notice if it is over budget and decide how it wants to handle the memory situation (free it, move to host memory, changing mipmap levels, etc). This extension is designed to be used in concert with `VK_EXT_memory_priority` to help with this part of memory management.

New Structures

- Extending `VkPhysicalDeviceMemoryProperties2`:
 - `VkPhysicalDeviceMemoryBudgetPropertiesEXT`

New Enum Constants

- `VK_EXT_MEMORY_BUDGET_EXTENSION_NAME`
- `VK_EXT_MEMORY_BUDGET_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_BUDGET_PROPERTIES_EXT`

Version History

- Revision 1, 2018-10-08 (Jeff Bolz)
 - Initial revision

VK_EXT_memory_priority

Name String

`VK_EXT_memory_priority`

Extension Type

Device extension

Registered Extension Number

239

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Jeff Bolz [[GitHub](#)][jeffbolznv](#)

Other Extension Metadata

Last Modified Date

2018-10-08

Contributors

- Jeff Bolz, NVIDIA
- Jeff Juliano, NVIDIA

Description

This extension adds a **priority** value specified at memory allocation time. On some systems with both device-local and non-device-local memory heaps, the implementation may transparently move memory from one heap to another when a heap becomes full (for example, when the total memory used across all processes exceeds the size of the heap). In such a case, this priority value may be used to determine which allocations are more likely to remain in device-local memory.

New Structures

- Extending [VkMemoryAllocateInfo](#):
 - [VkMemoryPriorityAllocateInfoEXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceMemoryPriorityFeaturesEXT](#)

New Enum Constants

- [VK_EXT_MEMORY_PRIORITY_EXTENSION_NAME](#)
- [VK_EXT_MEMORY_PRIORITY_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_MEMORY_PRIORITY_ALLOCATE_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MEMORY_PRIORITY_FEATURES_EXT](#)

Version History

- Revision 1, 2018-10-08 (Jeff Bolz)

- Initial revision

VK_EXT_multi_draw

Name String

VK_EXT_multi_draw

Extension Type

Device extension

Registered Extension Number

393

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Mike Blumenkrantz [\[GitHub\]](#)zmike

Other Extension Metadata

Last Modified Date

2021-05-19

Interactions and External Dependencies

- Interacts with Vulkan 1.1.
- Interacts with [VK_KHR_shader_draw_parameters](#).

IP Status

No known IP claims.

Contributors

- Mike Blumenkrantz, VALVE
- Piers Daniell, NVIDIA
- Faith Ekstrand, INTEL
- Spencer Fricke, SAMSUNG
- Ricardo Garcia, IGALIA
- Jon Leech, KHRONOS

- Stu Smith, AMD

Description

Processing multiple draw commands in sequence incurs measurable overhead within drivers due to repeated state checks and updates during dispatch. This extension enables passing the entire sequence of draws directly to the driver in order to avoid any such overhead, using an array of [VkMultiDrawInfoEXT](#) or [VkMultiDrawIndexedInfoEXT](#) structs with [vkCmdDrawMultiEXT](#) or [vkCmdDrawMultiIndexedEXT](#), respectively. These functions could be used any time multiple draw commands are being recorded without any state changes between them in order to maximize performance.

New Commands

- [vkCmdDrawMultiEXT](#)
- [vkCmdDrawMultiIndexedEXT](#)

New Structures

- [VkMultiDrawIndexedInfoEXT](#)
- [VkMultiDrawInfoEXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceMultiDrawFeaturesEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceMultiDrawPropertiesEXT](#)

New Enum Constants

- [VK_EXT_MULTI_DRAW_EXTENSION_NAME](#)
- [VK_EXT_MULTI_DRAW_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_FEATURES_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_MULTI_DRAW_PROPERTIES_EXT](#)

New or Modified Built-In Variables

- (modified)[DrawIndex](#)

Version History

- Revision 1, 2021-01-20 (Mike Blumenkrantz)
 - Initial version

VK_EXT_nested_command_buffer

Name String

VK_EXT_nested_command_buffer

Extension Type

Device extension

Registered Extension Number

452

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Piers Daniell [[GitHub](#)][pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2023-09-18

Contributors

- Daniel Story, Nintendo
- Peter Kohaut, NVIDIA
- Shahbaz Youssefi, Google
- Slawomir Grajewski, Intel
- Stu Smith, AMD

Description

With core Vulkan it is not legal to call [vkCmdExecuteCommands](#) when recording a secondary command buffer. This extension relaxes that restriction, allowing secondary command buffers to execute other secondary command buffers.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceNestedCommandBufferFeaturesEXT](#)

- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceNestedCommandBufferPropertiesEXT](#)

New Enum Constants

- [VK_EXT_NESTED_COMMAND_BUFFER_EXTENSION_NAME](#)
- [VK_EXT_NESTED_COMMAND_BUFFER_SPEC_VERSION](#)
- Extending [VkRenderingFlagBits](#):
 - [VK_RENDERING_CONTENTS_INLINE_BIT_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_FEATURES_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NESTED_COMMAND_BUFFER_PROPERTIES_EXT](#)
- Extending [VkSubpassContents](#):
 - [VK_SUBPASS_CONTENTS_INLINE_AND_SECONDARY_COMMAND_BUFFERS_EXT](#)

Issues

1) The Command Buffer Levels property for the Vulkan commands comes from the `cmdbufferlevel` attribute in `vk.xml` for the command, and it is currently not possible to modify this attribute based on whether an extension is enabled. For this extension we want the `cmdbufferlevel` attribute for `vkCmdExecuteCommands` to be `primary,secondary` when this extension is enabled and `primary` otherwise.

RESOLVED: The `cmdbufferlevel` attribute for [vkCmdExecuteCommands](#) has been changed to `primary,secondary` and a new VUID added to prohibit recording this command in a secondary command buffer unless this extension is enabled.

Version History

- Revision 1, 2023-09-18 (Piers Daniell)
 - Internal revisions

VK_EXT_non_seamless_cube_map

Name String

`VK_EXT_non_seamless_cube_map`

Extension Type

Device extension

Registered Extension Number

423

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Uses

- [D3D support](#)
- [OpenGL / ES support](#)

Contact

- Georg Lehmann [[GitHub](#)][DadSchoorse](#)

Extension Proposal

[VK_EXT_non_seamless_cube_map](#)

Other Extension Metadata

Last Modified Date

2021-09-04

IP Status

No known IP claims.

Contributors

- Georg Lehmann

Description

This extension provides functionality to disable [cube map edge handling](#) on a per sampler level which matches the behavior of other graphics APIs.

This extension may be useful for building translation layers for those APIs or for porting applications that rely on this cube map behavior.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceNonSeamlessCubeMapFeaturesEXT](#)

New Enum Constants

- [VK_EXT_NON_SEAMLESS_CUBE_MAP_EXTENSION_NAME](#)
- [VK_EXT_NON_SEAMLESS_CUBE_MAP_SPEC_VERSION](#)
- Extending [VkSamplerCreateFlagBits](#):
 - [VK_SAMPLER_CREATE_NON_SEAMLESS_CUBE_MAP_BIT_EXT](#)

- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_NON_SEAMLESS_CUBE_MAP_FEATURES_EXT](#)

Version History

- Revision 1, 2021-09-04 (Georg Lehmann)
 - First Version

VK_EXT_opacity_micromap

Name String

[VK_EXT_opacity_micromap](#)

Extension Type

Device extension

Registered Extension Number

397

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_acceleration_structure](#)

and

[VK_KHR_synchronization2](#)

or

[Vulkan Version 1.3](#)

SPIR-V Dependencies

- [SPV_EXT_opacity_micromap](#)

Contact

- Christoph Kubisch [\[GitHub\]](#)[pixeljetstream](#)
- Eric Werness

Extension Proposal

[VK_EXT_opacity_micromap](#)

Other Extension Metadata

Last Modified Date

2022-08-24

Interactions and External Dependencies

- This extension provides API support for [GLSL_EXT_opacity_micromap](#)

Contributors

- Christoph Kubisch, NVIDIA
- Eric Werness, NVIDIA
- Josh Barczak, Intel
- Stu Smith, AMD

Description

When adding transparency to a ray traced scene, an application can choose between further tessellating the geometry or using an any-hit shader to allow the ray through specific parts of the geometry. These options have the downside of either significantly increasing memory consumption or adding runtime overhead to run shader code in the middle of traversal, respectively.

This extension adds the ability to add an *opacity micromap* to geometry when building an acceleration structure. The opacity micromap compactly encodes opacity information which can be read by the implementation to mark parts of triangles as opaque or transparent. The format is externally visible to allow the application to compress its internal geometry and surface representations into the compressed format ahead of time. The compressed format subdivides each triangle into a set of subtriangles, each of which can be assigned either two or four opacity values. These opacity values can control if a ray hitting that subtriangle is treated as an opaque hit, complete miss, or possible hit, depending on the controls described in [Ray Opacity Micromap](#).

This extension provides:

- a [VkMicromapEXT](#) structure to store the micromap,
- functions similar to acceleration structure build functions to build the opacity micromap array, and
- a structure to extend [VkAccelerationStructureGeometryTrianglesDataKHR](#) to attach a micromap to the geometry of the acceleration structure.

New Object Types

- [VkMicromapEXT](#)

New Commands

- [vkBuildMicromapsEXT](#)
- [vkCmdBuildMicromapsEXT](#)
- [vkCmdCopyMemoryToMicromapEXT](#)
- [vkCmdCopyMicromapEXT](#)
- [vkCmdCopyMicromapToMemoryEXT](#)
- [vkCmdWriteMicromapsPropertiesEXT](#)

- [vkCopyMemoryToMicromapEXT](#)
- [vkCopyMicromapEXT](#)
- [vkCopyMicromapToMemoryEXT](#)
- [vkCreateMicromapEXT](#)
- [vkDestroyMicromapEXT](#)
- [vkGetDeviceMicromapCompatibilityEXT](#)
- [vkGetMicromapBuildSizesEXT](#)
- [vkWriteMicromapsPropertiesEXT](#)

New Structures

- [VkCopyMemoryToMicromapInfoEXT](#)
- [VkCopyMicromapInfoEXT](#)
- [VkCopyMicromapToMemoryInfoEXT](#)
- [VkMicromapBuildInfoEXT](#)
- [VkMicromapBuildSizesInfoEXT](#)
- [VkMicromapCreateInfoEXT](#)
- [VkMicromapTriangleEXT](#)
- [VkMicromapUsageEXT](#)
- [VkMicromapVersionInfoEXT](#)
- Extending [VkAccelerationStructureGeometryTrianglesDataKHR](#):
 - [VkAccelerationStructureTrianglesOpacityMicromapEXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceOpacityMicromapFeaturesEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceOpacityMicromapPropertiesEXT](#)

New Enums

- [VkBuildMicromapFlagBitsEXT](#)
- [VkBuildMicromapModeEXT](#)
- [VkCopyMicromapModeEXT](#)
- [VkMicromapCreateFlagBitsEXT](#)
- [VkMicromapTypeEXT](#)
- [VkOpacityMicromapFormatEXT](#)
- [VkOpacityMicromapSpecialIndexEXT](#)

New Bitmasks

- [VkBuildMicromapFlagsEXT](#)
- [VkMicromapCreateFlagsEXT](#)

New Enum Constants

- [VK_EXT_OPACITY_MICROMAP_EXTENSION_NAME](#)
- [VK_EXT_OPACITY_MICROMAP_SPEC_VERSION](#)
- Extending [VkAccessFlagBits2](#):
 - [VK_ACCESS_2_MICROMAP_READ_BIT_EXT](#)
 - [VK_ACCESS_2_MICROMAP_WRITE_BIT_EXT](#)
- Extending [VkBufferUsageFlagBits](#):
 - [VK_BUFFER_USAGE_MICROMAP_BUILD_INPUT_READ_ONLY_BIT_EXT](#)
 - [VK_BUFFER_USAGE_MICROMAP_STORAGE_BIT_EXT](#)
- Extending [VkBuildAccelerationStructureFlagBitsKHR](#):
 - [VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_DISABLE_OPACITY_MICROMAPS_EXT](#)
 - [VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_DATA_UPDATE_EXT](#)
 - [VK_BUILD_ACCELERATION_STRUCTURE_ALLOW_OPACITY_MICROMAP_UPDATE_EXT](#)
- Extending [VkGeometryInstanceFlagBitsKHR](#):
 - [VK_GEOMETRY_INSTANCE_DISABLE_OPACITY_MICROMAPS_EXT](#)
 - [VK_GEOMETRY_INSTANCE_FORCE_OPACITY_MICROMAP_2_STATE_EXT](#)
- Extending [VkObjectType](#):
 - [VK_OBJECT_TYPE_MICROMAP_EXT](#)
- Extending [VkPipelineCreateFlagBits](#):
 - [VK_PIPELINE_CREATE_RAY_TRACING_OPACITY_MICROMAP_BIT_EXT](#)
- Extending [VkPipelineStageFlagBits2](#):
 - [VK_PIPELINE_STAGE_2_MICROMAP_BUILD_BIT_EXT](#)
- Extending [VkQueryType](#):
 - [VK_QUERY_TYPE_MICROMAP_COMPACTED_SIZE_EXT](#)
 - [VK_QUERY_TYPE_MICROMAP_SERIALIZATION_SIZE_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_ACCELERATION_STRUCTURE_TRIANGLES_OPACITY_MICROMAP_EXT](#)
 - [VK_STRUCTURE_TYPE_COPY_MEMORY_TO_MICROMAP_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_COPY_MICROMAP_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_COPY_MICROMAP_TO_MEMORY_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_MICROMAP_BUILD_INFO_EXT](#)

- VK_STRUCTURE_TYPE_MICROMAP_BUILD_SIZES_INFO_EXT
- VK_STRUCTURE_TYPE_MICROMAP_CREATE_INFO_EXT
- VK_STRUCTURE_TYPE_MICROMAP_VERSION_INFO_EXT
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_FEATURES_EXT
- VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_OPACITY_MICROMAP_PROPERTIES_EXT

Reference Code

```
uint32_t BarycentricsToSpaceFillingCurveIndex(float u, float v, uint32_t level)
{
    u = clamp(u, 0.0f, 1.0f);
    v = clamp(v, 0.0f, 1.0f);

    uint32_t iu, iv, iw;

    // Quantize barycentric coordinates
    float fu = u * (1u << level);
    float fv = v * (1u << level);

    iu = (uint32_t)fu;
    iv = (uint32_t)fv;

    float uf = fu - float(iu);
    float vf = fv - float(iv);

    if (iu >= (1u << level)) iu = (1u << level) - 1u;
    if (iv >= (1u << level)) iv = (1u << level) - 1u;

    uint32_t iuv = iu + iv;

    if (iuv >= (1u << level))
        iu -= iuv - (1u << level) + 1u;

    iw = ~(iu + iv);

    if (uf + vf >= 1.0f && iuv < (1u << level) - 1u) --iw;

    uint32_t b0 = ~(iu ^ iw);
    b0 &= ((1u << level) - 1u);
    uint32_t t = (iu ^ iv) & b0;

    uint32_t f = t;
    f ^= f >> 1u;
    f ^= f >> 2u;
    f ^= f >> 4u;
    f ^= f >> 8u;
    uint32_t b1 = ((f ^ iu) & ~b0) | t;
```

```

// Interleave bits
b0 = (b0 | (b0 << 8u)) & 0x00ff00ffu;
b0 = (b0 | (b0 << 4u)) & 0x0f0f0f0fu;
b0 = (b0 | (b0 << 2u)) & 0x33333333u;
b0 = (b0 | (b0 << 1u)) & 0x55555555u;
b1 = (b1 | (b1 << 8u)) & 0x00ff00ffu;
b1 = (b1 | (b1 << 4u)) & 0x0f0f0f0fu;
b1 = (b1 | (b1 << 2u)) & 0x33333333u;
b1 = (b1 | (b1 << 1u)) & 0x55555555u;

return b0 | (b1 << 1u);
}

```

Issues

(1) Is the build actually similar to an acceleration structure build?

- Resolved: The build should be much lighter-weight than an acceleration structure build, but the infrastructure is similar enough that it makes sense to keep the concepts compatible.

(2) Why does `VkMicromapUsageEXT` not have `type/pNext`?

- Resolved: There can be a very large number of these structures, so doubling the size of these can be significant memory consumption. Also, an application may be loading these directly from a file which is more compatible with it being a flat structure. The including structures are extensible and are probably a more suitable place to add extensibility.

(3) Why is there a SPIR-V extension?

- Resolved: There is a ray flag. To be consistent with how the existing ray tracing extensions work that ray flag needs its own extension.

(4) Should there be indirect micromap build?

- Resolved: Not for now. There is more in-depth usage metadata required and it seems less likely that something like a GPU culling system would need to change the counts for a micromap.

(5) Should micromaps have a micromap device address?

- Resolved: There is no need right now (can just use the handle) but that is a bit different from acceleration structures, though the two are not completely parallel in their usage.

(6) Why are the alignment requirements defined as a mix of hardcoded values and caps?

- Resolved: This is most parallel with the definition of `VK_KHR_acceleration_structure` and maintaining commonality makes it easier for applications to share memory.

Version History

- Revision 2, 2022-06-22 (Eric Werness)
 - EXTify and clean up for discussion

- Revision 1, 2022-01-01 (Eric Werness)
 - Initial revision

VK_EXT_pci_bus_info

Name String

VK_EXT_pci_bus_info

Extension Type

Device extension

Registered Extension Number

213

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Matthaeus G. Chajdas [\[GitHub\]](#)[anteru](#)

Other Extension Metadata

Last Modified Date

2018-12-10

IP Status

No known IP claims.

Contributors

- Matthaeus G. Chajdas, AMD
- Daniel Rakos, AMD

Description

This extension adds a new query to obtain PCI bus information about a physical device.

Not all physical devices have PCI bus information, either due to the device not being connected to the system through a PCI interface or due to platform specific restrictions and policies. Thus this extension is only expected to be supported by physical devices which can provide the information.

As a consequence, applications should always check for the presence of the extension string for each individual physical device for which they intend to issue the new query for and should not have any assumptions about the availability of the extension on any given platform.

New Structures

- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDevicePCIBusInfoPropertiesEXT](#)

New Enum Constants

- [VK_EXT_PCI_BUS_INFO_EXTENSION_NAME](#)
- [VK_EXT_PCI_BUS_INFO_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PCI_BUS_INFO_PROPERTIES_EXT](#)

Version History

- Revision 2, 2018-12-10 (Daniel Rakos)
 - Changed all members of the new structure to have the `uint32_t` type
- Revision 1, 2018-10-11 (Daniel Rakos)
 - Initial revision

VK_EXT_physical_device_drm

Name String

[VK_EXT_physical_device_drm](#)

Extension Type

Device extension

Registered Extension Number

354

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Simon Ser [\[GitHub\]emersion](#)

Other Extension Metadata

Last Modified Date

2021-06-09

IP Status

No known IP claims.

Contributors

- Simon Ser

Description

This extension provides new facilities to query DRM properties for physical devices, enabling users to match Vulkan physical devices with DRM nodes on Linux.

Its functionality closely overlaps with [EGL_EXT_device_drm](#)¹. Unlike the EGL extension, this extension does not expose a string containing the name of the device file and instead exposes device minor numbers.

DRM defines multiple device node types. Each physical device may have one primary node and one render node associated. Physical devices may have no primary node (e.g. if the device does not have a display subsystem), may have no render node (e.g. if it is a software rendering engine), or may have neither (e.g. if it is a software rendering engine without a display subsystem).

To query DRM properties for a physical device, chain [VkPhysicalDeviceDrmPropertiesEXT](#) to [VkPhysicalDeviceProperties2](#).

New Structures

- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceDrmPropertiesEXT](#)

New Enum Constants

- [VK_EXT_PHYSICAL_DEVICE_DRM_EXTENSION_NAME](#)
- [VK_EXT_PHYSICAL_DEVICE_DRM_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_DRM_PROPERTIES_EXT](#)

References

1. [EGL_EXT_device_drm](#)

Version History

- Revision 1, 2021-06-09
 - First stable revision

VK_EXT_post_depth_coverage

Name String

VK_EXT_post_depth_coverage

Extension Type

Device extension

Registered Extension Number

156

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- [SPV_KHR_post_depth_coverage](#)

Contact

- Daniel Koch [\[GitHub\]dgkoch](#)

Other Extension Metadata

Last Modified Date

2017-07-17

Interactions and External Dependencies

- This extension provides API support for [GL_ARB_post_depth_coverage](#) and [GL_EXT_post_depth_coverage](#)

Contributors

- Jeff Bolz, NVIDIA

Description

This extension adds support for the following SPIR-V extension in Vulkan:

- [SPV_KHR_post_depth_coverage](#)

which allows the fragment shader to control whether values in the [SampleMask](#) built-in input

variable reflect the coverage after early [depth](#) and [stencil](#) tests are applied.

This extension adds a new [PostDepthCoverage](#) execution mode under the [SampleMaskPostDepthCoverage](#) capability. When this mode is specified along with [EarlyFragmentTests](#), the value of an input variable decorated with the [SampleMask](#) built-in reflects the coverage after the early fragment tests are applied. Otherwise, it reflects the coverage before the depth and stencil tests.

When using GLSL source-based shading languages, the [post_depth_coverage](#) layout qualifier from `GL_ARB_post_depth_coverage` or `GL_EXT_post_depth_coverage` maps to the [PostDepthCoverage](#) execution mode.

New Enum Constants

- [VK_EXT_POST_DEPTH_COVERAGE_EXTENSION_NAME](#)
- [VK_EXT_POST_DEPTH_COVERAGE_SPEC_VERSION](#)

New SPIR-V Capabilities

- [SampleMaskPostDepthCoverage](#)

Version History

- Revision 1, 2017-07-17 (Daniel Koch)
 - Internal revisions

VK_EXT_present_mode_fifo_latest_ready

Name String

[VK_EXT_present_mode_fifo_latest_ready](#)

Extension Type

Device extension

Registered Extension Number

362

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_swapchain](#)

Contact

- Lionel Duc [\[GitHub\]](#)[nvlduc](#)

Extension Proposal

[VK_EXT_present_mode_fifo_latest_ready](#)

Other Extension Metadata

Last Modified Date

2024-05-28

IP Status

No known IP claims.

Contributors

- James Jones, NVIDIA
- Lionel Duc, NVIDIA

Description

This device extension adds a new present mode, [VK_PRESENT_MODE_FIFO_LATEST_READY_EXT](#).

This tear-free present mode behaves much like [VK_PRESENT_MODE_FIFO_KHR](#), except that each vertical blanking period dequeues consecutive present requests until the latest ready is found to update the current image.

While this seems similar in concept to [VK_PRESENT_MODE_MAILBOX_KHR](#), the fundamental difference is that the processing of the present requests is done during vblank. From the application perspective, this means for example, that in a flip-based model, a single vblank **may** cause multiple swapchain images to be released at once, while [VK_PRESENT_MODE_MAILBOX_KHR](#) **may** continuously be releasing images as new requests become ready.

This additional present mode is useful when using a time-based present API.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDevicePresentModeFifoLatestReadyFeaturesEXT](#)

New Enum Constants

- [VK_EXT_PRESENT_MODE_FIFO_LATEST_READY_EXTENSION_NAME](#)
- [VK_EXT_PRESENT_MODE_FIFO_LATEST_READY_SPEC_VERSION](#)
- Extending [VkPresentModeKHR](#):
 - [VK_PRESENT_MODE_FIFO_LATEST_READY_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRESENT_MODE_FIFO_LATEST_READY_FEATURES_EXT](#)

Version History

- Revision 1, 2024-05-28 (Lionel Duc)
 - Internal revisions

VK_EXT_primitive_topology_list_restart

Name String

VK_EXT_primitive_topology_list_restart

Extension Type

Device extension

Registered Extension Number

357

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Use

- [OpenGL / ES support](#)

Contact

- Shahbaz Youssefi [\[GitHub\]syoussefi](#)

Other Extension Metadata

Last Modified Date

2021-01-11

IP Status

No known IP claims.

Contributors

- Courtney Goeltzenleuchter, Google
- Shahbaz Youssefi, Google

Description

This extension allows list primitives to use the primitive restart index value. This provides a more

efficient implementation when layering OpenGL functionality on Vulkan by avoiding emulation which incurs data copies.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDevicePrimitiveTopologyListRestartFeaturesEXT](#)

New Enum Constants

- [VK_EXT_PRIMITIVE_TOPOLOGY_LIST_RESTART_EXTENSION_NAME](#)
- [VK_EXT_PRIMITIVE_TOPOLOGY_LIST_RESTART_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVE_TOPOLOGY_LIST_RESTART_FEATURES_EXT](#)

Version History

- Revision 0, 2020-09-14 (Courtney Goeltzenleuchter)
 - Internal revisions
- Revision 1, 2021-01-11 (Shahbaz Youssefi)
 - Add the [primitiveTopologyPatchListRestart](#) feature
 - Internal revisions

VK_EXT_primitives_generated_query

Name String

[VK_EXT_primitives_generated_query](#)

Extension Type

Device extension

Registered Extension Number

383

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_EXT_transform_feedback](#)

Special Use

- [OpenGL / ES support](#)

Contact

- Shahbaz Youssefi [[GitHub](#)][syoussefi](#)

Extension Proposal

[VK_EXT_primitives_generated_query](#)

Other Extension Metadata

Last Modified Date

2022-01-24

Contributors

- Shahbaz Youssefi, Google
- Piers Daniell, NVIDIA
- Faith Ekstrand, Collabora
- Jan-Harald Fredriksen, Arm

Description

This extension adds support for a new query type to match OpenGL's [GL_PRIMITIVES_GENERATED](#) to support layering.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDevicePrimitivesGeneratedQueryFeaturesEXT](#)

New Enum Constants

- [VK_EXT_PRIMITIVES_GENERATED_QUERY_EXTENSION_NAME](#)
- [VK_EXT_PRIMITIVES_GENERATED_QUERY_SPEC_VERSION](#)
- Extending [VkQueryType](#):
 - [VK_QUERY_TYPE_PRIMITIVES_GENERATED_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PRIMITIVES_GENERATED_QUERY_FEATURES_EXT](#)

Issues

1) Can the query from [VK_EXT_transform_feedback](#) be used instead?

RESOLVED: No. While the query from [VK_EXT_transform_feedback](#) can produce the same results as in this extension, it is only available while transform feedback is active. The OpenGL [GL_PRIMITIVES_GENERATED](#) query is independent from transform feedback. Emulation through artificial transform feedback is unnecessarily inefficient.

2) Can [VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT](#) be used instead?

RESOLVED: It could, but we prefer the extension for simplicity. Vulkan requires that only one query be active at a time. If both the `GL_PRIMITIVES_GENERATED` and the `GL_CLIPPING_INPUT_PRIMITIVES_ARB` queries need to be simultaneously enabled, emulation of both through `VK_QUERY_PIPELINE_STATISTIC_CLIPPING_INVOCATIONS_BIT` is inconvenient.

3) On some hardware, this query cannot be implemented if `VkPipelineRasterizationStateCreateInfo::rasterizerDiscardEnable` is enabled. How will this be handled?

RESOLVED: A feature flag is exposed by this extension for this. On said hardware, the GL implementation disables rasterizer-discard and achieves the same effect through other means. It will not be able to do the same in Vulkan due to lack of state information. A feature flag is exposed by this extension so the OpenGL implementation on top of Vulkan would be able to implement a similar workaround.

4) On some hardware, this query cannot be implemented for non-zero query indices. How will this be handled?

RESOLVED: A feature flag is exposed by this extension for this. If this feature is not present, the query from `VK_EXT_transform_feedback` can be used to the same effect.

5) How is the interaction of this extension with `transformFeedbackRasterizationStreamSelect` handled?

RESOLVED: Disallowed for non-zero streams. In OpenGL, the rasterization stream is always stream zero.

Version History

- Revision 1, 2021-06-23 (Shahbaz Youssefi)
 - Internal revisions

VK_EXT_provoking_vertex

Name String

`VK_EXT_provoking_vertex`

Extension Type

Device extension

Registered Extension Number

255

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Use

- [OpenGL / ES support](#)

Contact

- Jesse Hall [\[GitHub\]jessehall](#)

Other Extension Metadata

Last Modified Date

2021-02-22

IP Status

No known IP claims.

Contributors

- Alexis Hétu, Google
- Bill Licea-Kane, Qualcomm
- Daniel Koch, Nvidia
- Jamie Madill, Google
- Jan-Harald Fredriksen, Arm
- Faith Ekstrand, Intel
- Jeff Bolz, Nvidia
- Jeff Leger, Qualcomm
- Jesse Hall, Google
- Jörg Wagner, Arm
- Matthew Netsch, Qualcomm
- Mike Blumenkrantz, Valve
- Piers Daniell, Nvidia
- Tobias Hector, AMD

Description

This extension allows changing the provoking vertex convention between Vulkan's default convention (first vertex) and OpenGL's convention (last vertex).

This extension is intended for use by API-translation layers that implement APIs like OpenGL on top of Vulkan, and need to match the source API's provoking vertex convention. Applications using Vulkan directly should use Vulkan's default convention.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceProvokingVertexFeaturesEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceProvokingVertexPropertiesEXT](#)
- Extending [VkPipelineRasterizationStateCreateInfo](#):
 - [VkPipelineRasterizationProvokingVertexStateCreateInfoEXT](#)

New Enums

- [VkProvokingVertexModeEXT](#)

New Enum Constants

- [VK_EXT_PROVOKING_VERTEX_EXTENSION_NAME](#)
- [VK_EXT_PROVOKING_VERTEX_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_FEATURES_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PROVOKING_VERTEX_PROPERTIES_EXT](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_PROVOKING_VERTEX_STATE_CREATE_INFO_EXT](#)

Issues

1) At what granularity should this state be set?

RESOLVED: At pipeline bind, with an optional per-render pass restriction.

The most natural place to put this state is in the graphics pipeline object. Some implementations require it to be known when creating the pipeline, and pipeline state is convenient for implementing OpenGL 3.2's `glProvokingVertex`, which can change the state between draw calls. However, some implementations can only change it approximately render pass granularity. To accommodate both, provoking vertex will be pipeline state, but implementations can require that only one mode is used within a render pass instance; the render pass's mode is chosen implicitly when the first pipeline is bound.

2) Does the provoking vertex mode affect the order that vertices are written to transform feedback buffers?

RESOLVED: Yes, to enable layered implementations of OpenGL and D3D.

All of OpenGL, OpenGL ES, and Direct3D 11 require that vertices are written to transform feedback buffers such that flat-shaded attributes have the same value when drawing the contents of the transform feedback buffer as they did in the original drawing when the transform feedback buffer was written (assuming the provoking vertex mode has not changed, in APIs that support more than one mode).

Version History

- Revision 1, (1c) 2021-02-22 (Jesse Hall)
 - Added
VkPhysicalDeviceProvokingVertexPropertiesEXT::transformFeedbackPreservesTriangleFanProvokingVertex to accommodate implementations that cannot change the transform feedback vertex order for triangle fans.
- Revision 1, (1b) 2020-06-14 (Jesse Hall)
 - Added
VkPhysicalDeviceProvokingVertexFeaturesEXT::transformFeedbackPreservesProvokingVertex and required that transform feedback write vertices so as to preserve the provoking vertex of each primitive.
- Revision 1, (1a) 2019-10-23 (Jesse Hall)
 - Initial draft, based on a proposal by Alexis Héту

VK_EXT_queue_family_foreign

Name String

VK_EXT_queue_family_foreign

Extension Type

Device extension

Registered Extension Number

127

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_external_memory](#)

or

[Vulkan Version 1.1](#)

Contact

- Lina Versace [\[GitHub\]](#)linyaa-kiwi

Other Extension Metadata

Last Modified Date

2017-11-01

IP Status

No known IP claims.

Contributors

- Lina Versace, Google
- James Jones, NVIDIA
- Faith Ekstrand, Intel
- Jesse Hall, Google
- Daniel Rakos, AMD
- Ray Smith, ARM

Description

This extension defines a special queue family, `VK_QUEUE_FAMILY_FOREIGN_EXT`, which can be used to transfer ownership of resources backed by external memory to foreign, external queues. This is similar to `VK_QUEUE_FAMILY_EXTERNAL_KHR`, defined in `VK_KHR_external_memory`. The key differences between the two are:

- The queues represented by `VK_QUEUE_FAMILY_EXTERNAL_KHR` must share the same physical device and the same driver version as the current `VkInstance`. `VK_QUEUE_FAMILY_FOREIGN_EXT` has no such restrictions. It can represent devices and drivers from other vendors, and can even represent non-Vulkan-capable devices.
- All resources backed by external memory support `VK_QUEUE_FAMILY_EXTERNAL_KHR`. Support for `VK_QUEUE_FAMILY_FOREIGN_EXT` is more restrictive.
- Applications should expect transitions to/from `VK_QUEUE_FAMILY_FOREIGN_EXT` to be more expensive than transitions to/from `VK_QUEUE_FAMILY_EXTERNAL_KHR`.

New Enum Constants

- `VK_EXT_QUEUE_FAMILY_FOREIGN_EXTENSION_NAME`
- `VK_EXT_QUEUE_FAMILY_FOREIGN_SPEC_VERSION`
- `VK_QUEUE_FAMILY_FOREIGN_EXT`

Version History

- Revision 1, 2017-11-01 (Lina Versace)
 - Squashed internal revisions

VK_EXT_robustness2

Name String

`VK_EXT_robustness2`

Extension Type

Device extension

Registered Extension Number

287

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Liam Middlebrook [[GitHub](#)][liam-middlebrook](#)

Other Extension Metadata

Last Modified Date

2020-01-29

IP Status

No known IP claims.

Contributors

- Liam Middlebrook, NVIDIA
- Jeff Bolz, NVIDIA

Description

This extension adds stricter requirements for how out of bounds reads and writes are handled. Most accesses **must** be tightly bounds-checked, out of bounds writes **must** be discarded, out of bound reads **must** return zero. Rather than allowing multiple possible (0,0,0,x) vectors, the out of bounds values are treated as zero, and then missing components are inserted based on the format as described in [Conversion to RGBA](#) and [vertex input attribute extraction](#).

These additional requirements **may** be expensive on some implementations, and should only be enabled when truly necessary.

This extension also adds support for “null descriptors”, where [VK_NULL_HANDLE](#) can be used instead of a valid handle. Accesses to null descriptors have well-defined behavior, and do not rely on robustness.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceRobustness2FeaturesEXT](#)

- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceRobustness2PropertiesEXT](#)

New Enum Constants

- [VK_EXT_ROBUSTNESS_2_EXTENSION_NAME](#)
- [VK_EXT_ROBUSTNESS_2_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_FEATURES_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_ROBUSTNESS_2_PROPERTIES_EXT](#)

Issues

1. Why do [VkPhysicalDeviceRobustness2PropertiesEXT::robustUniformBufferAccessSizeAlignment](#) and [VkPhysicalDeviceRobustness2PropertiesEXT::robustStorageBufferAccessSizeAlignment](#) exist?

RESOLVED: Some implementations cannot efficiently tightly bounds-check all buffer accesses. Rather, the size of the bound range is padded to some power of two multiple, up to 256 bytes for uniform buffers and up to 4 bytes for storage buffers, and that padded size is bounds-checked. This is sufficient to implement D3D-like behavior, because D3D only allows binding whole uniform buffers or ranges that are a multiple of 256 bytes, and D3D raw and structured buffers only support 32-bit accesses.

Examples

None.

Version History

- Revision 1, 2019-11-01 (Jeff Bolz, Liam Middlebrook)
 - Initial draft

VK_EXT_sample_locations

Name String

[VK_EXT_sample_locations](#)

Extension Type

Device extension

Registered Extension Number

144

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Daniel Rakos [\[GitHub\]](#)[drakos-amd](#)

Other Extension Metadata

Last Modified Date

2017-08-02

Contributors

- Mais Alnasser, AMD
- Matthaeus G. Chajdas, AMD
- Maciej Jesionowski, AMD
- Daniel Rakos, AMD
- Slawomir Grajewski, Intel
- Jeff Bolz, NVIDIA
- Bill Licea-Kane, Qualcomm

Description

This extension allows an application to modify the locations of samples within a pixel used in rasterization. Additionally, it allows applications to specify different sample locations for each pixel in a group of adjacent pixels, which **can** increase antialiasing quality (particularly if a custom resolve shader is used that takes advantage of these different locations).

It is common for implementations to optimize the storage of depth values by storing values that **can** be used to reconstruct depth at each sample location, rather than storing separate depth values for each sample. For example, the depth values from a single triangle **may** be represented using plane equations. When the depth value for a sample is needed, it is automatically evaluated at the sample location. Modifying the sample locations causes the reconstruction to no longer evaluate the same depth values as when the samples were originally generated, thus the depth aspect of a depth/stencil attachment **must** be cleared before rendering to it using different sample locations.

Some implementations **may** need to evaluate depth image values while performing image layout transitions. To accommodate this, instances of the [VkSampleLocationsInfoEXT](#) structure **can** be specified for each situation where an explicit or automatic layout transition has to take place. [VkSampleLocationsInfoEXT](#) **can** be chained from [VkImageMemoryBarrier](#) structures to provide sample locations for layout transitions performed by [vkCmdWaitEvents](#) and [vkCmdPipelineBarrier](#) calls, and [VkRenderPassSampleLocationsBeginInfoEXT](#) **can** be chained from

[VkRenderPassBeginInfo](#) to provide sample locations for layout transitions performed implicitly by a render pass instance.

New Commands

- [vkCmdSetSampleLocationsEXT](#)
- [vkGetPhysicalDeviceMultisamplePropertiesEXT](#)

New Structures

- [VkAttachmentSampleLocationsEXT](#)
- [VkMultisamplePropertiesEXT](#)
- [VkSampleLocationEXT](#)
- [VkSubpassSampleLocationsEXT](#)
- Extending [VkImageMemoryBarrier](#), [VkImageMemoryBarrier2](#):
 - [VkSampleLocationsInfoEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceSampleLocationsPropertiesEXT](#)
- Extending [VkPipelineMultisampleStateCreateInfo](#):
 - [VkPipelineSampleLocationsStateCreateInfoEXT](#)
- Extending [VkRenderPassBeginInfo](#):
 - [VkRenderPassSampleLocationsBeginInfoEXT](#)

New Enum Constants

- [VK_EXT_SAMPLE_LOCATIONS_EXTENSION_NAME](#)
- [VK_EXT_SAMPLE_LOCATIONS_SPEC_VERSION](#)
- Extending [VkDynamicState](#):
 - [VK_DYNAMIC_STATE_SAMPLE_LOCATIONS_EXT](#)
- Extending [VkImageCreateFlagBits](#):
 - [VK_IMAGE_CREATE_SAMPLE_LOCATIONS_COMPATIBLE_DEPTH_BIT_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_MULTISAMPLE_PROPERTIES_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SAMPLE_LOCATIONS_PROPERTIES_EXT](#)
 - [VK_STRUCTURE_TYPE_PIPELINE_SAMPLE_LOCATIONS_STATE_CREATE_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_RENDER_PASS_SAMPLE_LOCATIONS_BEGIN_INFO_EXT](#)
 - [VK_STRUCTURE_TYPE_SAMPLE_LOCATIONS_INFO_EXT](#)

Version History

- Revision 1, 2017-08-02 (Daniel Rakos)
 - Internal revisions

VK_EXT_shader_atomic_float

Name String

VK_EXT_shader_atomic_float

Extension Type

Device extension

Registered Extension Number

261

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

API Interactions

- Interacts with
VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::sparseImageFloat32AtomicAdd
- Interacts with VkPhysicalDeviceShaderAtomicFloatFeaturesEXT::sparseImageFloat32Atomics

SPIR-V Dependencies

- [SPV_EXT_shader_atomic_float_add](#)

Contact

- Vikram Kushwaha [\[GitHub\]vkushwaha-nv](#)

Other Extension Metadata

Last Modified Date

2020-07-15

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for [GL_EXT_shader_atomic_float](#)

Contributors

- Vikram Kushwaha, NVIDIA
- Jeff Bolz, NVIDIA

Description

This extension allows a shader to contain floating-point atomic operations on buffer, workgroup, and image memory. It also advertises the SPIR-V `AtomicFloat32AddEXT` and `AtomicFloat64AddEXT` capabilities that allows atomic addition on floating-points numbers. The supported operations include `OpAtomicFAddEXT`, `OpAtomicExchange`, `OpAtomicLoad` and `OpAtomicStore`.

New Structures

- Extending `VkPhysicalDeviceFeatures2`, `VkDeviceCreateInfo`:
 - `VkPhysicalDeviceShaderAtomicFloatFeaturesEXT`

New Enum Constants

- `VK_EXT_SHADER_ATOMIC_FLOAT_EXTENSION_NAME`
- `VK_EXT_SHADER_ATOMIC_FLOAT_SPEC_VERSION`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_FEATURES_EXT`

New SPIR-V Capabilities

- `AtomicFloat32AddEXT`
- `AtomicFloat64AddEXT`

Version History

- Revision 1, 2020-07-15 (Vikram Kushwaha)
 - Internal revisions

VK_EXT_shader_atomic_float2

Name String

`VK_EXT_shader_atomic_float2`

Extension Type

Device extension

Registered Extension Number

274

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_EXT_shader_atomic_float](#)

API Interactions

- Interacts with [VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT::sparseImageFloat32AtomicMinMax](#)

SPIR-V Dependencies

- [SPV_EXT_shader_atomic_float16_add](#)
- [SPV_EXT_shader_atomic_float_min_max](#)

Contact

- Faith Ekstrand [\[GitHub\]](#)[gfxstrand](#)

Other Extension Metadata

Last Modified Date

2020-08-14

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for [GLSL_EXT_shader_atomic_float2](#)

Contributors

- Faith Ekstrand, Intel

Description

This extension allows a shader to perform 16-bit floating-point atomic operations on buffer and workgroup memory as well as floating-point atomic minimum and maximum operations on buffer, workgroup, and image memory. It advertises the SPIR-V [AtomicFloat16AddEXT](#) capability which allows atomic add operations on 16-bit floating-point numbers and the SPIR-V [AtomicFloat16MinMaxEXT](#), [AtomicFloat32MinMaxEXT](#) and [AtomicFloat64MinMaxEXT](#) capabilities which allow atomic minimum and maximum operations on floating-point numbers. The supported operations include [OpAtomicFAddEXT](#), [OpAtomicFMinEXT](#) and [OpAtomicFMaxEXT](#).

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderAtomicFloat2FeaturesEXT](#)

New Enum Constants

- [VK_EXT_SHADER_ATOMIC_FLOAT_2_EXTENSION_NAME](#)

- [VK_EXT_SHADER_ATOMIC_FLOAT_2_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_ATOMIC_FLOAT_2_FEATURES_EXT](#)

Issues

1) Should this extension add support for 16-bit image atomics?

RESOLVED: No. While Vulkan supports creating storage images with [VK_FORMAT_R16_SFLOAT](#) and doing load and store on them, the data in the shader has a 32-bit representation. Vulkan currently has no facility for even basic reading or writing such images using 16-bit float values in the shader. Adding such functionality would be required before 16-bit image atomics would make sense and is outside the scope of this extension.

New SPIR-V Capabilities

- [AtomicFloat32MinMaxEXT](#)
- [AtomicFloat32MinMaxEXT](#)
- [AtomicFloat32MinMaxEXT](#)
- [AtomicFloat64MinMaxEXT](#)

Version History

- Revision 1, 2020-08-14 (Faith Ekstrand)
 - Internal revisions

VK_EXT_shader_image_atomic_int64

Name String

[VK_EXT_shader_image_atomic_int64](#)

Extension Type

Device extension

Registered Extension Number

235

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)
or
[Vulkan Version 1.1](#)

SPIR-V Dependencies

- [SPV_EXT_shader_image_int64](#)

Contact

- Tobias Hector [\[GitHub\]](#)[tobski](#)

Other Extension Metadata

Last Modified Date

2020-07-14

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for [GLSL_EXT_shader_image_int64](#)

Contributors

- Matthaeus Chajdas, AMD
- Graham Wihlidal, Epic Games
- Tobias Hector, AMD
- Jeff Bolz, Nvidia
- Faith Ekstrand, Intel

Description

This extension extends existing 64-bit integer atomic support to enable these operations on images as well.

When working with large 2- or 3-dimensional data sets (e.g. rasterization or screen-space effects), image accesses are generally more efficient than equivalent buffer accesses. This extension allows applications relying on 64-bit integer atomics in this manner to quickly improve performance with only relatively minor code changes.

64-bit integer atomic support is guaranteed for optimally tiled images with the [VK_FORMAT_R64_UINT](#) and [VK_FORMAT_R64_SINT](#) formats.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderImageAtomicInt64FeaturesEXT](#)

New Enum Constants

- [VK_EXT_SHADER_IMAGE_ATOMIC_INT64_EXTENSION_NAME](#)
- [VK_EXT_SHADER_IMAGE_ATOMIC_INT64_SPEC_VERSION](#)
- Extending [VkStructureType](#):

- [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_IMAGE_ATOMIC_INT64_FEATURES_EXT](#)

Version History

- Revision 1, 2020-07-14 (Tobias Hector)
 - Initial draft

VK_EXT_shader_object

Name String

[VK_EXT_shader_object](#)

Extension Type

Device extension

Registered Extension Number

483

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

and

[VK_KHR_dynamic_rendering](#)

or

[Vulkan Version 1.3](#)

API Interactions

- Interacts with [VK_VERSION_1_1](#)
- Interacts with [VK_VERSION_1_3](#)
- Interacts with [VK_EXT_blend_operation_advanced](#)
- Interacts with [VK_EXT_conservative_rasterization](#)
- Interacts with [VK_EXT_depth_clamp_control](#)
- Interacts with [VK_EXT_depth_clip_control](#)
- Interacts with [VK_EXT_depth_clip_enable](#)
- Interacts with [VK_EXT_fragment_density_map](#)
- Interacts with [VK_EXT_line_rasterization](#)
- Interacts with [VK_EXT_mesh_shader](#)

- Interacts with `VK_EXT_provoking_vertex`
- Interacts with `VK_EXT_sample_locations`
- Interacts with `VK_EXT_subgroup_size_control`
- Interacts with `VK_EXT_transform_feedback`
- Interacts with `VK_KHR_device_group`
- Interacts with `VK_KHR_fragment_shading_rate`
- Interacts with `VK_NV_clip_space_w_scaling`
- Interacts with `VK_NV_coverage_reduction_mode`
- Interacts with `VK_NV_fragment_coverage_to_color`
- Interacts with `VK_NV_framebuffer_mixed_samples`
- Interacts with `VK_NV_mesh_shader`
- Interacts with `VK_NV_representative_fragment_test`
- Interacts with `VK_NV_shading_rate_image`
- Interacts with `VK_NV_viewport_swizzle`

Contact

- Daniel Story [\[GitHub\]daniel-story](#)

Extension Proposal

[VK_EXT_shader_object](#)

Other Extension Metadata

Last Modified Date

2023-03-30

Interactions and External Dependencies

- Interacts with `VK_EXT_extended_dynamic_state`
- Interacts with `VK_EXT_extended_dynamic_state2`
- Interacts with `VK_EXT_extended_dynamic_state3`
- Interacts with `VK_EXT_vertex_input_dynamic_state`

IP Status

No known IP claims.

Contributors

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Description

This extension introduces a new [VkShaderEXT](#) object type which represents a single compiled shader stage. Shader objects provide a more flexible alternative to [VkPipeline](#) objects, which may be helpful in certain use cases.

New Object Types

- [VkShaderEXT](#)

New Commands

- [vkCmdBindShadersEXT](#)
- [vkCmdBindVertexBuffers2EXT](#)
- [vkCmdSetAlphaToCoverageEnableEXT](#)
- [vkCmdSetAlphaToOneEnableEXT](#)
- [vkCmdSetColorBlendEnableEXT](#)
- [vkCmdSetColorBlendEquationEXT](#)

- [vkCmdSetColorWriteMaskEXT](#)
- [vkCmdSetCullModeEXT](#)
- [vkCmdSetDepthBiasEnableEXT](#)
- [vkCmdSetDepthBoundsTestEnableEXT](#)
- [vkCmdSetDepthClampEnableEXT](#)
- [vkCmdSetDepthCompareOpEXT](#)
- [vkCmdSetDepthTestEnableEXT](#)
- [vkCmdSetDepthWriteEnableEXT](#)
- [vkCmdSetFrontFaceEXT](#)
- [vkCmdSetLogicOpEXT](#)
- [vkCmdSetLogicOpEnableEXT](#)
- [vkCmdSetPatchControlPointsEXT](#)
- [vkCmdSetPolygonModeEXT](#)
- [vkCmdSetPrimitiveRestartEnableEXT](#)
- [vkCmdSetPrimitiveTopologyEXT](#)
- [vkCmdSetRasterizationSamplesEXT](#)
- [vkCmdSetRasterizerDiscardEnableEXT](#)
- [vkCmdSetSampleMaskEXT](#)
- [vkCmdSetScissorWithCountEXT](#)
- [vkCmdSetStencilOpEXT](#)
- [vkCmdSetStencilTestEnableEXT](#)
- [vkCmdSetTessellationDomainOriginEXT](#)
- [vkCmdSetVertexInputEXT](#)
- [vkCmdSetViewportWithCountEXT](#)
- [vkCreateShadersEXT](#)
- [vkDestroyShaderEXT](#)
- [vkGetShaderBinaryDataEXT](#)

If [VK_EXT_blend_operation_advanced](#) is supported:

- [vkCmdSetColorBlendAdvancedEXT](#)

If [VK_EXT_conservative_rasterization](#) is supported:

- [vkCmdSetConservativeRasterizationModeEXT](#)
- [vkCmdSetExtraPrimitiveOverestimationSizeEXT](#)

If [VK_EXT_depth_clamp_control](#) is supported:

- [vkCmdSetDepthClampRangeEXT](#)

If [VK_EXT_depth_clip_control](#) is supported:

- [vkCmdSetDepthClipNegativeOneToOneEXT](#)

If [VK_EXT_depth_clip_enable](#) is supported:

- [vkCmdSetDepthClipEnableEXT](#)

If [VK_EXT_line_rasterization](#) is supported:

- [vkCmdSetLineRasterizationModeEXT](#)
- [vkCmdSetLineStippleEnableEXT](#)

If [VK_EXT_provoking_vertex](#) is supported:

- [vkCmdSetProvokingVertexModeEXT](#)

If [VK_EXT_sample_locations](#) is supported:

- [vkCmdSetSampleLocationsEnableEXT](#)

If [VK_EXT_transform_feedback](#) is supported:

- [vkCmdSetRasterizationStreamEXT](#)

If [VK_NV_clip_space_w_scaling](#) is supported:

- [vkCmdSetViewportWScalingEnableNV](#)

If [VK_NV_coverage_reduction_mode](#) is supported:

- [vkCmdSetCoverageReductionModeNV](#)

If [VK_NV_fragment_coverage_to_color](#) is supported:

- [vkCmdSetCoverageToColorEnableNV](#)
- [vkCmdSetCoverageToColorLocationNV](#)

If [VK_NV_framebuffer_mixed_samples](#) is supported:

- [vkCmdSetCoverageModulationModeNV](#)
- [vkCmdSetCoverageModulationTableEnableNV](#)
- [vkCmdSetCoverageModulationTableNV](#)

If [VK_NV_representative_fragment_test](#) is supported:

- [vkCmdSetRepresentativeFragmentTestEnableNV](#)

If [VK_NV_shading_rate_image](#) is supported:

- [vkCmdSetShadingRateImageEnableNV](#)

If [VK_NV_viewport_swizzle](#) is supported:

- [vkCmdSetViewportSwizzleNV](#)

New Structures

- [VkColorBlendAdvancedEXT](#)
- [VkColorBlendEquationEXT](#)
- [VkShaderCreateInfoEXT](#)
- [VkVertexInputAttributeDescription2EXT](#)
- [VkVertexInputBindingDescription2EXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderObjectFeaturesEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceShaderObjectPropertiesEXT](#)
- Extending [VkPipelineShaderStageCreateInfo](#), [VkShaderCreateInfoEXT](#):
 - [VkShaderRequiredSubgroupSizeCreateInfoEXT](#)

New Enums

- [VkShaderCodeTypeEXT](#)
- [VkShaderCreateFlagBitsEXT](#)

New Bitmasks

- [VkShaderCreateFlagsEXT](#)

New Enum Constants

- [VK_EXT_SHADER_OBJECT_EXTENSION_NAME](#)
- [VK_EXT_SHADER_OBJECT_SPEC_VERSION](#)
- Extending [VkObjectType](#):
 - [VK_OBJECT_TYPE_SHADER_EXT](#)
- Extending [VkResult](#):
 - [VK_ERROR_INCOMPATIBLE_SHADER_BINARY_EXT](#)
 - [VK_INCOMPATIBLE_SHADER_BINARY_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_FEATURES_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_OBJECT_PROPERTIES_EXT](#)

- `VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT`
- `VK_STRUCTURE_TYPE_SHADER_REQUIRED_SUBGROUP_SIZE_CREATE_INFO_EXT`
- `VK_STRUCTURE_TYPE_VERTEX_INPUT_ATTRIBUTE_DESCRIPTION_2_EXT`
- `VK_STRUCTURE_TYPE_VERTEX_INPUT_BINDING_DESCRIPTION_2_EXT`

If `VK_EXT_fragment_density_map` is supported:

- Extending `VkShaderCreateFlagBitsEXT`:
 - `VK_SHADER_CREATE_FRAGMENT_DENSITY_MAP_ATTACHMENT_BIT_EXT`

If `VK_EXT_mesh_shader` or `VK_NV_mesh_shader` is supported:

- Extending `VkShaderCreateFlagBitsEXT`:
 - `VK_SHADER_CREATE_NO_TASK_SHADER_BIT_EXT`

If `VK_EXT_subgroup_size_control` or `Vulkan Version 1.3` is supported:

- Extending `VkShaderCreateFlagBitsEXT`:
 - `VK_SHADER_CREATE_ALLOW_VARYING_SUBGROUP_SIZE_BIT_EXT`
 - `VK_SHADER_CREATE_REQUIRE_FULL_SUBGROUPS_BIT_EXT`

If `VK_KHR_device_group` or `Vulkan Version 1.1` is supported:

- Extending `VkShaderCreateFlagBitsEXT`:
 - `VK_SHADER_CREATE_DISPATCH_BASE_BIT_EXT`

If `VK_KHR_fragment_shading_rate` is supported:

- Extending `VkShaderCreateFlagBitsEXT`:
 - `VK_SHADER_CREATE_FRAGMENT_SHADING_RATE_ATTACHMENT_BIT_EXT`

Examples

Example 1

Create linked pair of vertex and fragment shaders.

```
// Logical device created with the shaderObject feature enabled
VkDevice device;

// SPIR-V shader code for a vertex shader, along with its size in bytes
void* pVertexSpirv;
size_t vertexSpirvSize;

// SPIR-V shader code for a fragment shader, along with its size in bytes
void* pFragmentSpirv;
size_t fragmentSpirvSize;
```

```

// Descriptor set layout compatible with the shaders
VkDescriptorSetLayout descriptorSetLayout;

VkShaderCreateInfoEXT shaderCreateInfos[2] =
{
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = VK_SHADER_CREATE_LINK_STAGE_BIT_EXT,
        .stage = VK_SHADER_STAGE_VERTEX_BIT,
        .nextStage = VK_SHADER_STAGE_FRAGMENT_BIT,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = vertexSpirvSize,
        .pCode = pVertexSpirv,
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
        .pPushConstantRanges = NULL,
        .pSpecializationInfo = NULL
    },
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = VK_SHADER_CREATE_LINK_STAGE_BIT_EXT,
        .stage = VK_SHADER_STAGE_FRAGMENT_BIT,
        .nextStage = 0,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = fragmentSpirvSize,
        .pCode = pFragmentSpirv,
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
        .pPushConstantRanges = NULL,
        .pSpecializationInfo = NULL
    }
};

VkResult result;
VkShaderEXT shaders[2];

result = vkCreateShadersEXT(device, 2, &shaderCreateInfos, NULL, shaders);
if (result != VK_SUCCESS)
{
    // Handle error
}

```

Later, during command buffer recording, bind the linked shaders and draw.

```

// Command buffer in the recording state
VkCommandBuffer commandBuffer;

// Vertex and fragment shader objects created above
VkShaderEXT shaders[2];

// Assume vertex buffers, descriptor sets, etc. have been bound, and existing
// state setting commands have been called to set all required state

const VkShaderStageFlagBits stages[2] =
{
    VK_SHADER_STAGE_VERTEX_BIT,
    VK_SHADER_STAGE_FRAGMENT_BIT
};

// Bind linked shaders
vkCmdBindShadersEXT(commandBuffer, 2, stages, shaders);

// Equivalent to the previous line. Linked shaders can be bound one at a time,
// in any order:
// vkCmdBindShadersEXT(commandBuffer, 1, &stages[1], &shaders[1]);
// vkCmdBindShadersEXT(commandBuffer, 1, &stages[0], &shaders[0]);

// The above is sufficient to draw if the device was created with the
// tessellationShader and geometryShader features disabled. Otherwise, since
// those stages should not execute, vkCmdBindShadersEXT() must be called at
// least once with each of their stages in pStages before drawing:

const VkShaderStageFlagBits unusedStages[3] =
{
    VK_SHADER_STAGE_TESSELLATION_CONTROL_BIT,
    VK_SHADER_STAGE_TESSELLATION_EVALUATION_BIT,
    VK_SHADER_STAGE_GEOMETRY_BIT
};

// NULL pShaders is equivalent to an array of stageCount VK_NULL_HANDLE values,
// meaning no shaders are bound to those stages, and that any previously bound
// shaders are unbound
vkCmdBindShadersEXT(commandBuffer, 3, unusedStages, NULL);

// Graphics shader objects may only be used to draw inside dynamic render pass
// instances begun with vkCmdBeginRendering(), assume one has already been begun

// Draw a triangle
vkCmdDraw(commandBuffer, 3, 1, 0, 0);

```

Example 2

Create unlinked vertex, geometry, and fragment shaders.

```

// Logical device created with the shaderObject feature enabled
VkDevice device;

// SPIR-V shader code for vertex shaders, along with their sizes in bytes
void* pVertexSpirv[2];
size_t vertexSpirvSize[2];

// SPIR-V shader code for a geometry shader, along with its size in bytes
void pGeometrySpirv;
size_t geometrySpirvSize;

// SPIR-V shader code for fragment shaders, along with their sizes in bytes
void* pFragmentSpirv[2];
size_t fragmentSpirvSize[2];

// Descriptor set layout compatible with the shaders
VkDescriptorSetLayout descriptorSetLayout;

VkShaderCreateInfoEXT shaderCreateInfos[5] =
{
    // Stage order does not matter
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .stage = VK_SHADER_STAGE_GEOMETRY_BIT,
        .nextStage = VK_SHADER_STAGE_FRAGMENT_BIT,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = pGeometrySpirv,
        .pCode = geometrySpirvSize,
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
        .pPushConstantRanges = NULL,
        .pSpecializationInfo = NULL
    },
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .stage = VK_SHADER_STAGE_VERTEX_BIT,
        .nextStage = VK_SHADER_STAGE_GEOMETRY_BIT,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = vertexSpirvSize[0],
        .pCode = pVertexSpirv[0],
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
    }
}

```



```

        .pNext = NULL,
        .pSpecializationInfo = NULL
    },
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .stage = VK_SHADER_STAGE_FRAGMENT_BIT,
        .nextStage = 0,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = fragmentSpirvSize[0],
        .pCode = pFragmentSpirv[0],
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
        .pNext = NULL,
        .pSpecializationInfo = NULL
    },
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .stage = VK_SHADER_STAGE_FRAGMENT_BIT,
        .nextStage = 0,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = fragmentSpirvSize[1],
        .pCode = pFragmentSpirv[1],
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
        .pNext = NULL,
        .pSpecializationInfo = NULL
    },
    {
        .sType = VK_STRUCTURE_TYPE_SHADER_CREATE_INFO_EXT,
        .pNext = NULL,
        .flags = 0,
        .stage = VK_SHADER_STAGE_VERTEX_BIT,
        // Suppose we want this vertex shader to be able to be followed by
        // either a geometry shader or fragment shader:
        .nextStage = VK_SHADER_STAGE_GEOMETRY_BIT | VK_SHADER_STAGE_FRAGMENT_BIT,
        .codeType = VK_SHADER_CODE_TYPE_SPIRV_EXT,
        .codeSize = vertexSpirvSize[1],
        .pCode = pVertexSpirv[1],
        .pName = "main",
        .setLayoutCount = 1,
        .pSetLayouts = &descriptorSetLayout;
        .pushConstantRangeCount = 0,
        .pNext = NULL,

```

```

        .pSpecializationInfo = NULL
    }
};

VkResult result;
VkShaderEXT shaders[5];

result = vkCreateShadersEXT(device, 5, &shaderCreateInfos, NULL, shaders);
if (result != VK_SUCCESS)
{
    // Handle error
}

```

Later, during command buffer recording, bind the linked shaders in different combinations and draw.

```

// Command buffer in the recording state
VkCommandBuffer commandBuffer;

// Vertex, geometry, and fragment shader objects created above
VkShaderEXT shaders[5];

// Assume vertex buffers, descriptor sets, etc. have been bound, and existing
// state setting commands have been called to set all required state

const VkShaderStageFlagBits stages[3] =
{
    // Any order is allowed
    VK_SHADER_STAGE_FRAGMENT_BIT,
    VK_SHADER_STAGE_VERTEX_BIT,
    VK_SHADER_STAGE_GEOMETRY_BIT,
};

VkShaderEXT bindShaders[3] =
{
    shaders[2], // FS
    shaders[1], // VS
    shaders[0]  // GS
};

// Bind unlinked shaders
vkCmdBindShadersEXT(commandBuffer, 3, stages, bindShaders);

// Assume the tessellationShader feature is disabled, so vkCmdBindShadersEXT()
// need not have been called with either tessellation stage

// Graphics shader objects may only be used to draw inside dynamic render pass
// instances begun with vkCmdBeginRendering(), assume one has already been begun

// Draw a triangle

```

```

vkCmdDraw(commandBuffer, 3, 1, 0, 0);

// Bind a different unlinked fragment shader
const VkShaderStageFlagBits fragmentStage = VK_SHADER_STAGE_FRAGMENT_BIT;
vkCmdBindShadersEXT(commandBuffer, 1, &fragmentStage, &shaders[3]);

// Draw another triangle
vkCmdDraw(commandBuffer, 3, 1, 0, 0);

// Bind a different unlinked vertex shader
const VkShaderStageFlagBits vertexStage = VK_SHADER_STAGE_VERTEX_BIT;
vkCmdBindShadersEXT(commandBuffer, 1, &vertexStage, &shaders[4]);

// Draw another triangle
vkCmdDraw(commandBuffer, 3, 1, 0, 0);

```

Version History

- Revision 1, 2023-03-30 (Daniel Story)
 - Initial draft

VK_EXT_shader_replicated_composites

Name String

VK_EXT_shader_replicated_composites

Extension Type

Device extension

Registered Extension Number

565

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- [SPV_EXT_replicated_composites](#)

Contact

- Kevin Petit [\[GitHub\]](#)kpet

Extension Proposal

[VK_EXT_shader_replicated_composites](#)

Last Modified Date

2024-02-08

IP Status

No known IP claims.

Contributors

- Kévin Petit, Arm Ltd.
- Jeff Bolz, NVIDIA
- Piers Daniell, NVIDIA

This extension adds support for creating composites from a single value in SPIR-V modules, as defined by SPV_EXT_replicated_composites.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderReplicatedCompositesFeaturesEXT](#)

New Enum Constants

- [VK_EXT_SHADER_REPLICATED_COMPOSITES_EXTENSION_NAME](#)
- [VK_EXT_SHADER_REPLICATED_COMPOSITES_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_REPLICATED_COMPOSITES_FEATURES_EXT](#)

New SPIR-V Capabilities

- [ReplicatedCompositesEXT](#)

Version History

- Revision 1, 2024-02-08 (Kévin Petit)
 - Initial revision

VK_EXT_shader_stencil_export

Name String

[VK_EXT_shader_stencil_export](#)

Extension Type

Device extension

Registered Extension Number

141

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

SPIR-V Dependencies

- [SPV_EXT_shader_stencil_export](#)

Contact

- Dominik Witczak [\[GitHub\]dominikwiczakamd](#)

Other Extension Metadata

Last Modified Date

2017-07-19

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for [GL_ARB_shader_stencil_export](#)

Contributors

- Dominik Witczak, AMD
- Daniel Rakos, AMD
- Rex Xu, AMD

Description

This extension adds support for the SPIR-V extension [SPV_EXT_shader_stencil_export](#), providing a mechanism whereby a shader may generate the stencil reference value per invocation. When stencil testing is enabled, this allows the test to be performed against the value generated in the shader.

New Enum Constants

- [VK_EXT_SHADER_STENCIL_EXPORT_EXTENSION_NAME](#)
- [VK_EXT_SHADER_STENCIL_EXPORT_SPEC_VERSION](#)

Version History

- Revision 1, 2017-07-19 (Dominik Witczak)
 - Initial draft

VK_EXT_shader_tile_image

Name String

VK_EXT_shader_tile_image

Extension Type

Device extension

Registered Extension Number

396

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[Vulkan Version 1.3](#)

SPIR-V Dependencies

- [SPV_EXT_shader_tile_image](#)

Contact

- Jan-Harald Fredriksen [\[GitHub\]](#)[janharaldfredriksen-arm](#)

Extension Proposal

[VK_EXT_shader_tile_image](#)

Other Extension Metadata

Last Modified Date

2023-03-23

IP Status

No known IP claims.

Interactions and External Dependencies

- This extension provides API support for [GL_EXT_shader_tile_image](#)

Contributors

- Sandeep Kakarlapudi, Arm
- Jan-Harald Fredriksen, Arm
- James Fitzpatrick, Imagination
- Andrew Garrard, Imagination
- Jeff Leger, Qualcomm
- Huilong Wang, Huawei

- Graeme Leese, Broadcom
- Hans-Kristian Arntzen, Valve
- Tobias Hector, AMD
- Jeff Bolz, NVIDIA
- Shahbaz Youssefi, Google

Description

This extension allows fragment shader invocations to read color, depth and stencil values at their pixel location in rasterization order. The functionality is only available when using dynamic render passes introduced by VK_KHR_dynamic_rendering. Example use cases are programmable blending and deferred shading.

See [fragment shader tile image reads](#) for more information.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceShaderTileImageFeaturesEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceShaderTileImagePropertiesEXT](#)

New Enum Constants

- [VK_EXT_SHADER_TILE_IMAGE_EXTENSION_NAME](#)
- [VK_EXT_SHADER_TILE_IMAGE_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_FEATURES_EXT](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_SHADER_TILE_IMAGE_PROPERTIES_EXT](#)

Issues

None.

Examples

Color read example.

```
layout( location = 0 /* aliased to color attachment 0 */ ) tileImageEXT highp
attachmentEXT color0;
layout( location = 1 /* aliased to color attachment 1 */ ) tileImageEXT highp
attachmentEXT color1;

layout( location = 0 ) out vec4 fragColor;
```



```
void main()
{
    vec4 value = colorAttachmentReadEXT(color0) + colorAttachmentReadEXT(color1);
    fragColor = value;
}
```

Depth & Stencil read example.

```
void main()
{
    // read sample 0: works for non-MSAA or MSAA targets
    highp float last_depth = depthAttachmentReadEXT();
    lowp uint last_stencil = stencilAttachmentReadEXT();

    //..
}
```

Version History

- Revision 1, 2023-03-23 (Sandeep Kakarlapudi)
 - Initial version

VK_EXT_swapchain_colorspace

Name String

VK_EXT_swapchain_colorspace

Extension Type

Instance extension

Registered Extension Number

105

Revision

5

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_surface](#)

Contact

- Courtney Goeltzenleuchter [\[GitHub\]](#)courtney-g

Other Extension Metadata

Last Modified Date

2019-04-26

IP Status

No known IP claims.

Contributors

- Courtney Goeltzenleuchter, Google

Description

This extension expands [VkColorSpaceKHR](#) to add support for most standard color spaces beyond [VK_COLOR_SPACE_SRGB_NONLINEAR_KHR](#). This extension also adds support for [VK_COLOR_SPACE_PASS_THROUGH_EXT](#) which allows applications to use color spaces not explicitly enumerated in [VkColorSpaceKHR](#).

New Enum Constants

- [VK_EXT_SWAPCHAIN_COLOR_SPACE_EXTENSION_NAME](#)
- [VK_EXT_SWAPCHAIN_COLOR_SPACE_SPEC_VERSION](#)
- Extending [VkColorSpaceKHR](#):
 - [VK_COLOR_SPACE_ADOBERGB_LINEAR_EXT](#)
 - [VK_COLOR_SPACE_ADOBERGB_NONLINEAR_EXT](#)
 - [VK_COLOR_SPACE_BT2020_LINEAR_EXT](#)
 - [VK_COLOR_SPACE_BT709_LINEAR_EXT](#)
 - [VK_COLOR_SPACE_BT709_NONLINEAR_EXT](#)
 - [VK_COLOR_SPACE_DCI_P3_LINEAR_EXT](#)
 - [VK_COLOR_SPACE_DCI_P3_NONLINEAR_EXT](#)
 - [VK_COLOR_SPACE_DISPLAY_P3_LINEAR_EXT](#)
 - [VK_COLOR_SPACE_DISPLAY_P3_NONLINEAR_EXT](#)
 - [VK_COLOR_SPACE_DOLBYVISION_EXT](#)
 - [VK_COLOR_SPACE_EXTENDED_SRGB_LINEAR_EXT](#)
 - [VK_COLOR_SPACE_EXTENDED_SRGB_NONLINEAR_EXT](#)
 - [VK_COLOR_SPACE_HDR10_HLG_EXT](#)
 - [VK_COLOR_SPACE_HDR10_ST2084_EXT](#)
 - [VK_COLOR_SPACE_PASS_THROUGH_EXT](#)

Issues

- 1) Does the spec need to specify which kinds of image formats support the color spaces?

RESOLVED: Pixel format is independent of color space (though some color spaces really want / need floating-point color components to be useful). Therefore, do not plan on documenting what formats support which color spaces. An application **can** call [vkGetPhysicalDeviceSurfaceFormatsKHR](#) to query what a particular implementation supports.

2) How does application determine if HW supports appropriate transfer function for a color space?

RESOLVED: Extension indicates that implementation **must** not do the OETF encoding if it is not sRGB. That responsibility falls to the application shaders. Any other native OETF / EOTF functions supported by an implementation can be described by separate extension.

Version History

- Revision 1, 2016-12-27 (Courtney Goeltzenleuchter)
 - Initial version
- Revision 2, 2017-01-19 (Courtney Goeltzenleuchter)
 - Add pass through and multiple options for BT2020.
 - Clean up some issues with equations not displaying properly.
- Revision 3, 2017-06-23 (Courtney Goeltzenleuchter)
 - Add extended sRGB non-linear enum.
- Revision 4, 2019-04-26 (Graeme Leese)
 - Clarify color space transfer function usage.
 - Refer to normative definitions in the Data Format Specification.
 - Clarify DCI-P3 and Display P3 usage.
- Revision 5, 2024-03-16 (Zehui Lin)
 - Fix interchanged concepts of EOTF and OETF.
 - Clarify that the presentation engine can accept the color spaces.

VK_EXT_transform_feedback

Name String

VK_EXT_transform_feedback

Extension Type

Device extension

Registered Extension Number

29

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Special Uses

- [OpenGL / ES support](#)
- [D3D support](#)
- [Developer tools](#)

Contact

- Piers Daniell [[GitHub](#)][pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2018-10-09

Contributors

- Baldur Karlsson, Valve
- Boris Zanin, Mobica
- Daniel Rakos, AMD
- Donald Scorgie, Imagination
- Henri Verbeet, CodeWeavers
- Jan-Harald Fredriksen, Arm
- Faith Ekstrand, Intel
- Jeff Bolz, NVIDIA
- Jesse Barker, Unity
- Jesse Hall, Google
- Pierre-Loup Griffais, Valve
- Philip Rebohle, DXVK
- Ruihao Zhang, Qualcomm
- Samuel Pitoiset, Valve
- Slawomir Grajewski, Intel
- Stu Smith, Imagination Technologies

Description

This extension adds transform feedback to the Vulkan API by exposing the SPIR-V [TransformFeedback](#) and [GeometryStreams](#) capabilities to capture vertex, tessellation or geometry shader outputs to one or more buffers. It adds API functionality to bind transform feedback buffers to capture the primitives emitted by the graphics pipeline from SPIR-V outputs decorated for

transform feedback. The transform feedback capture can be paused and resumed by way of storing and retrieving a byte counter. The captured data can be drawn again where the vertex count is derived from the byte counter without CPU intervention. If the implementation is capable, a vertex stream other than zero can be rasterized.

All these features are designed to match the full capabilities of OpenGL core transform feedback functionality and beyond. Many of the features are optional to allow base OpenGL ES GPUs to also implement this extension.

The primary purpose of the functionality exposed by this extension is to support translation layers from other 3D APIs. This functionality is not considered forward looking, and is not expected to be promoted to a KHR extension or to core Vulkan. Unless this is needed for translation, it is recommended that developers use alternative techniques of using the GPU to process and capture vertex data.

New Commands

- [vkCmdBeginQueryIndexedEXT](#)
- [vkCmdBeginTransformFeedbackEXT](#)
- [vkCmdBindTransformFeedbackBuffersEXT](#)
- [vkCmdDrawIndirectByteCountEXT](#)
- [vkCmdEndQueryIndexedEXT](#)
- [vkCmdEndTransformFeedbackEXT](#)

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceTransformFeedbackFeaturesEXT](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDeviceTransformFeedbackPropertiesEXT](#)
- Extending [VkPipelineRasterizationStateCreateInfo](#):
 - [VkPipelineRasterizationStateStreamCreateInfoEXT](#)

New Bitmasks

- [VkPipelineRasterizationStateStreamCreateFlagsEXT](#)

New Enum Constants

- [VK_EXT_TRANSFORM_FEEDBACK_EXTENSION_NAME](#)
- [VK_EXT_TRANSFORM_FEEDBACK_SPEC_VERSION](#)
- Extending [VkAccessFlagBits](#):
 - [VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT](#)
 - [VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT](#)

- `VK_ACCESS_TRANSFORM_FEEDBACK_WRITE_BIT_EXT`
- Extending `VkBufferUsageFlagBits`:
 - `VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_BUFFER_BIT_EXT`
 - `VK_BUFFER_USAGE_TRANSFORM_FEEDBACK_COUNTER_BUFFER_BIT_EXT`
- Extending `VkPipelineStageFlagBits`:
 - `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`
- Extending `VkQueryType`:
 - `VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT`
- Extending `VkStructureType`:
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_FEATURES_EXT`
 - `VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_TRANSFORM_FEEDBACK_PROPERTIES_EXT`
 - `VK_STRUCTURE_TYPE_PIPELINE_RASTERIZATION_STATE_STREAM_CREATE_INFO_EXT`

Issues

1) Should we include pause/resume functionality?

RESOLVED: Yes, this is needed to ease layering other APIs which have this functionality. To pause use `vkCmdEndTransformFeedbackEXT` and provide valid buffer handles in the `pCounterBuffers` array and offsets in the `pCounterBufferOffsets` array for the implementation to save the resume points. Then to resume use `vkCmdBeginTransformFeedbackEXT` with the previous `pCounterBuffers` and `pCounterBufferOffsets` values. Between the pause and resume there needs to be a memory barrier for the counter buffers with a source access of `VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_WRITE_BIT_EXT` at pipeline stage `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT` to a destination access of `VK_ACCESS_TRANSFORM_FEEDBACK_COUNTER_READ_BIT_EXT` at pipeline stage `VK_PIPELINE_STAGE_TRANSFORM_FEEDBACK_BIT_EXT`.

2) How does this interact with multiview?

RESOLVED: Transform feedback cannot be made active in a render pass with multiview enabled.

3) How should queries be done?

RESOLVED: There is a new query type `VK_QUERY_TYPE_TRANSFORM_FEEDBACK_STREAM_EXT`. A query pool created with this type will capture 2 integers - `numPrimitivesWritten` and `numPrimitivesNeeded` - for the specified vertex stream output from the last `pre-rasterization shader stage`. The vertex stream output queried is zero by default, but can be specified with the new `vkCmdBeginQueryIndexedEXT` and `vkCmdEndQueryIndexedEXT` commands.

Version History

- Revision 1, 2018-10-09 (Piers Daniell)
 - Internal revisions

VK_EXT_vertex_input_dynamic_state

Name String

VK_EXT_vertex_input_dynamic_state

Extension Type

Device extension

Registered Extension Number

353

Revision

2

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

Contact

- Piers Daniell [\[GitHub\]pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2020-08-21

IP Status

No known IP claims.

Contributors

- Jeff Bolz, NVIDIA
- Spencer Fricke, Samsung
- Stu Smith, AMD

Description

One of the states that contributes to the combinatorial explosion of pipeline state objects that need to be created, is the vertex input binding and attribute descriptions. By allowing them to be dynamic applications may reduce the number of pipeline objects they need to create.

This extension adds dynamic state support for what is normally static state in [VkPipelineVertexInputStateCreateInfo](#).

New Commands

- [vkCmdSetVertexInputEXT](#)

New Structures

- [VkVertexInputAttributeDescription2EXT](#)
- [VkVertexInputBindingDescription2EXT](#)
- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceVertexInputDynamicStateFeaturesEXT](#)

New Enum Constants

- [VK_EXT_VERTEX_INPUT_DYNAMIC_STATE_EXTENSION_NAME](#)
- [VK_EXT_VERTEX_INPUT_DYNAMIC_STATE_SPEC_VERSION](#)
- Extending [VkDynamicState](#):
 - [VK_DYNAMIC_STATE_VERTEX_INPUT_EXT](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_VERTEX_INPUT_DYNAMIC_STATE_FEATURES_EXT](#)
 - [VK_STRUCTURE_TYPE_VERTEX_INPUT_ATTRIBUTE_DESCRIPTION_2_EXT](#)
 - [VK_STRUCTURE_TYPE_VERTEX_INPUT_BINDING_DESCRIPTION_2_EXT](#)

Version History

- Revision 2, 2020-11-05 (Piers Daniell)
 - Make [VkVertexInputBindingDescription2EXT](#) extensible
 - Add new [VkVertexInputAttributeDescription2EXT](#) struct for the [pVertexAttributeDescriptions](#) parameter to [vkCmdSetVertexInputEXT](#) so it is also extensible
- Revision 1, 2020-08-21 (Piers Daniell)
 - Internal revisions

VK_EXT_ycbcr_image_arrays

Name String

[VK_EXT_ycbcr_image_arrays](#)

Extension Type

Device extension

Registered Extension Number

253

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_sampler_ycbcr_conversion](#)

or

[Vulkan Version 1.1](#)

Contact

- Piers Daniell [\[GitHub\]](#)[pdaniell-nv](#)

Other Extension Metadata

Last Modified Date

2019-01-15

Contributors

- Piers Daniell, NVIDIA

Description

This extension allows images of a format that requires [Y'C_BC_R conversion](#) to be created with multiple array layers, which is otherwise restricted.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDeviceYcbcrImageArraysFeaturesEXT](#)

New Enum Constants

- [VK_EXT_YCBCR_IMAGE_ARRAYS_EXTENSION_NAME](#)
- [VK_EXT_YCBCR_IMAGE_ARRAYS_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_YCBCR_IMAGE_ARRAYS_FEATURES_EXT](#)

Version History

- Revision 1, 2019-01-15 (Piers Daniell)
 - Initial revision

List of Provisional Extensions

- [VK_KHR_portability_subset](#)

VK_KHR_portability_subset

Name String

VK_KHR_portability_subset

Extension Type

Device extension

Registered Extension Number

164

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

[VK_KHR_get_physical_device_properties2](#)

or

[Vulkan Version 1.1](#)

- This is a *provisional* extension and must be used with caution. See the [description](#) of provisional header files for enablement and stability details.

Contact

- Bill Hollings [\[GitHub\]](#)[billhollings](#)

Other Extension Metadata

Last Modified Date

2020-07-21

IP Status

No known IP claims.

Contributors

- Bill Hollings, The Brenwill Workshop Ltd.
- Daniel Koch, NVIDIA
- Dzmitry Malyshau, Mozilla
- Chip Davis, CodeWeavers
- Dan Ginsburg, Valve
- Mike Weiblen, LunarG
- Neil Trevett, NVIDIA
- Alexey Knyazev, Independent

Description

The [VK_KHR_portability_subset](#) extension allows a non-conformant Vulkan implementation to be built on top of another non-Vulkan graphics API, and identifies differences between that implementation and a fully-conformant native Vulkan implementation.

This extension provides Vulkan implementations with the ability to mark otherwise-required capabilities as unsupported, or to establish additional properties and limits that the application should adhere to in order to guarantee portable behavior and operation across platforms, including platforms where Vulkan is not natively supported.

The goal of this specification is to document, and make queryable, capabilities which are required to be supported by a fully-conformant Vulkan 1.0 implementation, but may be optional for an implementation of the Vulkan 1.0 Portability Subset.

The intent is that this extension will be advertised only on implementations of the Vulkan 1.0 Portability Subset, and not on conformant implementations of Vulkan 1.0. Fully-conformant Vulkan implementations provide all the required capabilities, and so will not provide this extension. Therefore, the existence of this extension can be used to determine that an implementation is likely not fully conformant with the Vulkan spec.

If this extension is supported by the Vulkan implementation, the application must enable this extension.

This extension defines several new structures that can be chained to the existing structures used by certain standard Vulkan calls, in order to query for non-conformant portable behavior.

New Structures

- Extending [VkPhysicalDeviceFeatures2](#), [VkDeviceCreateInfo](#):
 - [VkPhysicalDevicePortabilitySubsetFeaturesKHR](#)
- Extending [VkPhysicalDeviceProperties2](#):
 - [VkPhysicalDevicePortabilitySubsetPropertiesKHR](#)

New Enum Constants

- [VK_KHR_PORTABILITY_SUBSET_EXTENSION_NAME](#)
- [VK_KHR_PORTABILITY_SUBSET_SPEC_VERSION](#)
- Extending [VkStructureType](#):
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_FEATURES_KHR](#)
 - [VK_STRUCTURE_TYPE_PHYSICAL_DEVICE_PORTABILITY_SUBSET_PROPERTIES_KHR](#)

Issues

None.

Version History

- Revision 1, 2020-07-21 (Bill Hollings)
 - Initial draft.

List of Deprecated Extensions

- [VK_EXT_load_store_op_none](#)

VK_EXT_load_store_op_none

Name String

VK_EXT_load_store_op_none

Extension Type

Device extension

Registered Extension Number

401

Revision

1

Ratification Status

Ratified

Extension and Version Dependencies

None

Deprecation State

- *Promoted* to [VK_KHR_load_store_op_none](#) extension

Contact

- Shahbaz Youssefi [\[GitHub\]](#)[syoussefi](#)

Other Extension Metadata

Last Modified Date

2021-06-06

Contributors

- Shahbaz Youssefi, Google
- Bill Licea-Kane, Qualcomm Technologies, Inc.
- Tobias Hector, AMD

Description

This extension incorporates [VK_ATTACHMENT_STORE_OP_NONE_EXT](#) from [VK_QCOM_render_pass_store_ops](#), enabling applications to avoid unnecessary synchronization when an attachment is not written during a render pass.

Additionally, [VK_ATTACHMENT_LOAD_OP_NONE_EXT](#) is introduced to avoid unnecessary synchronization when an attachment is not used during a render pass at all. In combination with [VK_ATTACHMENT_STORE_OP_NONE_EXT](#), this is useful as an alternative to preserve attachments in applications that cannot decide if an attachment will be used in a render pass until after the necessary pipelines have been created.

Promotion to `VK_KHR_load_store_op_none`

All functionality in this extension is included in `VK_KHR_load_store_op_none`, with the suffix changed to KHR. The original enum names are still available as aliases of the KHR functionality.

New Enum Constants

- `VK_EXT_LOAD_STORE_OP_NONE_EXTENSION_NAME`
- `VK_EXT_LOAD_STORE_OP_NONE_SPEC_VERSION`
- Extending `VkAttachmentLoadOp`:
 - `VK_ATTACHMENT_LOAD_OP_NONE_EXT`
- Extending `VkAttachmentStoreOp`:
 - `VK_ATTACHMENT_STORE_OP_NONE_EXT`

NOTE

While `VK_ATTACHMENT_STORE_OP_NONE` is part of Vulkan 1.3, this extension was not promoted to core either in whole or in part. This functionality was promoted from `VK_KHR_dynamic_rendering`.

Version History

- Revision 1, 2021-06-06 (Shahbaz Youssefi)
 - Initial revision, based on `VK_QCOM_render_pass_store_ops`.
 - Added `VK_ATTACHMENT_LOAD_OP_NONE_EXT`.

Appendix F: API Boilerplate

This appendix defines Vulkan API features that are infrastructure required for a complete functional description of Vulkan, but do not logically belong elsewhere in the Specification.

Vulkan Header Files

Vulkan is defined as an API in the C99 language. Khronos provides a corresponding set of header files for applications using the API, which may be used in either C or C++ code. The interface descriptions in the specification are the same as the interfaces defined in these header files, and both are derived from the `vk.xml` XML API Registry, which is the canonical machine-readable description of the Vulkan API. The Registry, scripts used for processing it into various forms, and documentation of the registry schema are available as described at <https://registry.khronos.org/vulkan/#apiregistry>.

Language bindings for other languages can be defined using the information in the Specification and the Registry. Khronos does not provide any such bindings, but third-party developers have created some additional bindings.

Vulkan Combined API Header `vulkan.h` (Informative)

Applications normally will include the header `vulkan.h`. In turn, `vulkan.h` always includes the following headers:

- `vk_platform.h`, defining platform-specific macros and headers.
- `vulkan_core.h`, defining APIs for the Vulkan core and all registered extensions *other* than `window system-specific` and `provisional` extensions, which are included in separate header files.

In addition, specific preprocessor macros defined at the time `vulkan.h` is included cause header files for the corresponding window system-specific and provisional interfaces to be included, as described below.

Vulkan Platform-Specific Header `vk_platform.h` (Informative)

Platform-specific macros and interfaces are defined in `vk_platform.h`. These macros are used to control platform-dependent behavior, and their exact definitions are under the control of specific platforms and Vulkan implementations.

Platform-Specific Calling Conventions

On many platforms the following macros are empty strings, causing platform- and compiler-specific default calling conventions to be used.

`VKAPI_ATTR` is a macro placed before the return type in Vulkan API function declarations. This macro controls calling conventions for C++11 and GCC/Clang-style compilers.

`VKAPI_CALL` is a macro placed after the return type in Vulkan API function declarations. This macro controls calling conventions for MSVC-style compilers.

VKAPI_PTR is a macro placed between the '(' and '*' in Vulkan API function pointer declarations. This macro also controls calling conventions, and typically has the same definition as **VKAPI_ATTR** or **VKAPI_CALL**, depending on the compiler.

With these macros, a Vulkan function declaration takes the form of:

```
VKAPI_ATTR <return_type> VKAPI_CALL <command_name>(<command_parameters>);
```

Additionally, a Vulkan function pointer type declaration takes the form of:

```
typedef <return_type> (VKAPI_PTR *PFN_<command_name>)(<command_parameters>);
```

Platform-Specific Header Control

If the **VK_NO_STDINT_H** macro is defined by the application at compile time, extended integer types used by the Vulkan API, such as **uint8_t**, **must** also be defined by the application. Otherwise, the Vulkan headers will not compile. If **VK_NO_STDINT_H** is not defined, the system **<stdint.h>** is used to define these types. There is a fallback path when Microsoft Visual Studio version 2008 and earlier versions are detected at compile time.

If the **VK_NO_STDDEF_H** macro is defined by the application at compile time, **size_t**, **must** also be defined by the application. Otherwise, the Vulkan headers will not compile. If **VK_NO_STDDEF_H** is not defined, the system **<stddef.h>** is used to define this type.

Vulkan Core API Header **vulkan_core.h**

Applications that do not make use of window system-specific extensions may simply include **vulkan_core.h** instead of **vulkan.h**, although there is usually no reason to do so. In addition to the Vulkan API, **vulkan_core.h** also defines a small number of C preprocessor macros that are described below.

Vulkan Header File Version Number

VK_HEADER_VERSION is the version number of the **vulkan_core.h** header. This value is kept synchronized with the patch version of the released Specification.

```
// Provided by VK_VERSION_1_0
// Version of this file
#define VK_HEADER_VERSION 301
```

VK_HEADER_VERSION_COMPLETE is the complete version number of the **vulkan_core.h** header, comprising the major, minor, and patch versions. The major/minor values are kept synchronized with the complete version of the released Specification. This value is intended for use by automated tools to identify exactly which version of the header was used during their generation.

Applications should not use this value as their **VkApplicationInfo::apiVersion**. Instead applications

should explicitly select a specific fixed major/minor API version using, for example, one of the `VK_API_VERSION_*` values.

```
// Provided by VK_VERSION_1_0
// Complete version of this file
#define VK_HEADER_VERSION_COMPLETE VK_MAKE_API_VERSION(0, 1, 3, VK_HEADER_VERSION)
```

`VK_API_VERSION` is now commented out of `vulkan_core.h` and **cannot** be used.

```
// Provided by VK_VERSION_1_0
// DEPRECATED: This define has been removed. Specific version defines (e.g.
VK_API_VERSION_1_0), or the VK_MAKE_VERSION macro, should be used instead.
// #define VK_API_VERSION VK_MAKE_API_VERSION(0, 1, 0, 0) // Patch version should
always be set to 0
```

Vulkan Handle Macros

`VK_DEFINE_HANDLE` defines a [dispatchable handle](#) type.

```
// Provided by VK_VERSION_1_0

#define VK_DEFINE_HANDLE(object) typedef struct object##_T* object;
```

- `object` is the name of the resulting C type.

The only dispatchable handle types are those related to device and instance management, such as [VkDevice](#).

`VK_DEFINE_NON_DISPATCHABLE_HANDLE` defines a [non-dispatchable handle](#) type.

```
// Provided by VK_VERSION_1_0

#ifndef VK_DEFINE_NON_DISPATCHABLE_HANDLE
    #if (VK_USE_64_BIT_PTR_DEFINES==1)
        #define VK_DEFINE_NON_DISPATCHABLE_HANDLE(object) typedef struct object##_T
    *object;
    #else
        #define VK_DEFINE_NON_DISPATCHABLE_HANDLE(object) typedef uint64_t object;
    #endif
#endif
```

- `object` is the name of the resulting C type.

Most Vulkan handle types, such as [VkBuffer](#), are non-dispatchable.

NOTE The `vulkan_core.h` header allows the `VK_DEFINE_NON_DISPATCHABLE_HANDLE`

and `VK_NULL_HANDLE` definitions to be overridden by the application. If `VK_DEFINE_NON_DISPATCHABLE_HANDLE` is already defined when `vulkan_core.h` is compiled, the default definitions for `VK_DEFINE_NON_DISPATCHABLE_HANDLE` and `VK_NULL_HANDLE` are skipped. This allows the application to define a binary-compatible custom handle which **may** provide more type-safety or other features needed by the application. Applications **must** not define handles in a way that is not binary compatible - where binary compatibility is platform dependent.

`VK_NULL_HANDLE` is a reserved value representing a non-valid object handle. It may be passed to and returned from Vulkan commands only when [specifically allowed](#).

```
// Provided by VK_VERSION_1_0

#ifndef VK_DEFINE_NON_DISPATCHABLE_HANDLE
    #if (VK_USE_64_BIT_PTR_DEFINES==1)
        #if (defined(__cplusplus) && (__cplusplus >= 201103L)) || (defined(_MSVC_LANG)
&& (_MSVC_LANG >= 201103L))
            #define VK_NULL_HANDLE nullptr
        #else
            #define VK_NULL_HANDLE ((void*)0)
        #endif
    #else
        #define VK_NULL_HANDLE 0ULL
    #endif
#endif
#ifndef VK_NULL_HANDLE
    #define VK_NULL_HANDLE 0
#endif
```

`VK_USE_64_BIT_PTR_DEFINES` defines whether the default non-dispatchable handles are declared using either a 64-bit pointer type or a 64-bit unsigned integer type.

`VK_USE_64_BIT_PTR_DEFINES` is set to '1' to use a 64-bit pointer type or any other value to use a 64-bit unsigned integer type.

```
// Provided by VK_VERSION_1_0

#ifndef VK_USE_64_BIT_PTR_DEFINES
    #if defined(__LP64__) || defined(WIN64) || (defined(__x86_64__) &&
!defined(__ILP32__) ) || defined(_M_X64) || defined(__ia64) || defined (_M_IA64) ||
defined(__aarch64__) || defined(__powerpc64__) || (defined(__riscv) && __riscv_xlen ==
64)
        #define VK_USE_64_BIT_PTR_DEFINES 1
    #else
        #define VK_USE_64_BIT_PTR_DEFINES 0
    #endif
#endif
```

NOTE

The `vulkan_core.h` header allows the `VK_USE_64_BIT_PTR_DEFINES` definition to be overridden by the application. This allows the application to select either a 64-bit pointer type or a 64-bit unsigned integer type for non-dispatchable handles in the case where the predefined preprocessor check does not identify the desired configuration.

NOTE

This macro was introduced starting with the Vulkan 1.2.174 headers, and its availability can be checked at compile time by requiring `VK_HEADER_VERSION >= 174`.

It is not available if you are using older headers, such as may be shipped with an older Vulkan SDK. Developers requiring this functionality may wish to include a copy of the current Vulkan headers with their project in this case.

Window System-Specific Header Control (Informative)

To use a Vulkan extension supporting a platform-specific window system, header files for that window system **must** be included at compile time, or platform-specific types **must** be forward-declared. The Vulkan header files are unable to determine whether or not an external header is available at compile time, so platform-specific extensions are provided in separate headers from the core API and platform-independent extensions, allowing applications to decide which ones they need to be defined and how the external headers are included.

Extensions dependent on particular sets of platform headers, or that forward-declare platform-specific types, are declared in a header named for that platform. Before including these platform-specific Vulkan headers, applications **must** include both `vulkan_core.h` and any external native headers the platform extensions depend on.

As a convenience for applications that do not need the flexibility of separate platform-specific Vulkan headers, `vulkan.h` includes `vulkan_core.h`, and then conditionally includes platform-specific Vulkan headers and the external headers they depend on. Applications control which platform-specific headers are included by #defining macros before including `vulkan.h`.

The correspondence between platform-specific extensions, external headers they require, the platform-specific header which declares them, and the preprocessor macros which enable inclusion by `vulkan.h` are shown in the [following table](#).

Table 99. Window System Extensions and Headers

Extension Name	Window System Name	Platform-specific Header	Required External Headers	Controlling <code>vulkan.h</code> Macro
VK_KHR_android_surface	Android	<code>vulkan_android.h</code>	None	<code>VK_USE_PLATFORM_ANDROID_KHR</code>
VK_KHR_wayland_surface	Wayland	<code>vulkan_wayland.h</code>	<code><wayland-client.h></code>	<code>VK_USE_PLATFORM_WAYLAND_KHR</code>

Extension Name	Window System Name	Platform-specific Header	Required External Headers	Controlling <code>vulkan.h</code> Macro
VK_KHR_win32_surface , VK_KHR_external_memory_win32 , VK_KHR_win32_keyed_mutex , VK_KHR_external_semaphore_win32 , VK_KHR_external_fence_win32 , VK_NV_external_memory_win32 , VK_NV_win32_keyed_mutex	Microsoft Windows	vulkan_win32.h	<code><windows.h></code>	<code>VK_USE_PLATFORM_WIN32_KHR</code>
VK_KHR_xcb_surface	X11 Xcb	vulkan_xcb.h	<code><xcb/xcb.h></code>	<code>VK_USE_PLATFORM_XCB_KHR</code>
VK_KHR_xlib_surface	X11 Xlib	vulkan_xlib.h	<code><X11/Xlib.h></code>	<code>VK_USE_PLATFORM_XLIB_KHR</code>
VK_EXT_directfb_surface	DirectFB	vulkan_directfb.h	<code><directfb/directfb.h></code>	<code>VK_USE_PLATFORM_DIRECTFB_EXT</code>
VK_EXT_acquire_xlib_display	X11 XRandR	vulkan_xlib_xrandr.h	<code><X11/Xlib.h></code> , <code><X11/extensions/Xrandr.h></code>	<code>VK_USE_PLATFORM_XLIB_XRANDR_EXT</code>
VK_GGP_stream_descriptor_surface , VK_GGP_frame_token	Google Games Platform	vulkan_ggp.h	<code><ggp_c/vulkan_types.h></code>	<code>VK_USE_PLATFORM_GGP</code>
VK_MVK_ios_surface	iOS	vulkan_ios.h	None	<code>VK_USE_PLATFORM_IOS_MVK</code>
VK_MVK_macos_surface	macOS	vulkan_macos.h	None	<code>VK_USE_PLATFORM_MACOS_MVK</code>
VK_NN_vi_surface	VI	vulkan_vi.h	None	<code>VK_USE_PLATFORM_VI_NN</code>
VK_FUCHSIA_imagepipe_surface	Fuchsia	vulkan_fuchsia.h	<code><zircon/types.h></code>	<code>VK_USE_PLATFORM_FUCHSIA</code>
VK_EXT_metal_surface	Metal on CoreAnimation	vulkan_metal.h	None	<code>VK_USE_PLATFORM_METAL_EXT</code>
VK_QNX_screen_surface	QNX Screen	vulkan_screen.h	<code><screen/screen.h></code>	<code>VK_USE_PLATFORM_SCREEN_QNX</code>

NOTE

This section describes the purpose of the headers independently of the specific underlying functionality of the window system extensions themselves. Each extension name will only link to a description of that extension when viewing a specification built with that extension included.

Provisional Extension Header Control (Informative)

Provisional extensions **should** not be used in production applications. The functionality defined by such extensions **may** change in ways that break backwards compatibility between revisions, and before final release of a non-provisional version of that extension.

Provisional extensions are defined in a separate *provisional header*, `vulkan_beta.h`, allowing applications to decide whether or not to include them. The mechanism is similar to [window system-specific headers](#): before including `vulkan_beta.h`, applications **must** include `vulkan_core.h`.

NOTE

Sometimes a provisional extension will include a subset of its interfaces in `vulkan_core.h`. This may occur if the provisional extension is promoted from an existing vendor or EXT extension and some of the existing interfaces are defined as aliases of the provisional extension interfaces. All other interfaces of that provisional extension which are not aliased will be included in `vulkan_beta.h`.

As a convenience for applications, `vulkan.h` conditionally includes `vulkan_beta.h`. Applications **can** control inclusion of `vulkan_beta.h` by #defining the macro `VK_ENABLE_BETA_EXTENSIONS` before including `vulkan.h`.

NOTE

Starting in version 1.2.171 of the Specification, all provisional enumerants are protected by the macro `VK_ENABLE_BETA_EXTENSIONS`. Applications needing to use provisional extensions must always define this macro, even if they are explicitly including `vulkan_beta.h`. This is a minor change to behavior, affecting only provisional extensions.

NOTE

This section describes the purpose of the provisional header independently of the specific provisional extensions which are contained in that header at any given time. The extension appendices for provisional extensions note their provisional status, and link back to this section for more information. Provisional extensions are intended to provide early access for bleeding-edge developers, with the understanding that extension interfaces may change in response to developer feedback. Provisional extensions are very likely to eventually be updated and released as non-provisional extensions, but there is no guarantee this will happen, or how long it will take if it does happen.

Video Std Headers

Performing video coding operations usually involves the application having to provide various parameters, data structures, or other syntax elements specific to the particular video compression standard used, and the associated semantics are covered by the specification of those.

The interface descriptions of these are available in the header files derived from the `video.xml` XML file, which is the canonical machine-readable description of data structures and enumerations that are associated with the externally-provided video compression standards.

Video Std Header Name	Description	Header File	Related Extensions
vulkan_video_codecs_common	Codec-independent common definitions	<vk_video/vulkan_video_codecs_common.h>	-
vulkan_video_codec_h264std	ITU-T H.264 common definitions	<vk_video/vulkan_video_codec_h264std.h>	VK_KHR_video_decode_h264 , VK_KHR_video_encode_h264
vulkan_video_codec_h264std_decode	ITU-T H.264 decode-specific definitions	<vk_video/vulkan_video_codec_h264std_decode.h>	VK_KHR_video_decode_h264
vulkan_video_codec_h264std_encode	ITU-T H.264 encode-specific definitions	<vk_video/vulkan_video_codec_h264std_encode.h>	VK_KHR_video_encode_h264
vulkan_video_codec_h265std	ITU-T H.265 common definitions	<vk_video/vulkan_video_codec_h265std.h>	VK_KHR_video_decode_h265 , VK_KHR_video_encode_h265
vulkan_video_codec_h265std_decode	ITU-T H.265 decode-specific definitions	<vk_video/vulkan_video_codec_h265std_decode.h>	VK_KHR_video_decode_h265
vulkan_video_codec_h265std_encode	ITU-T H.265 encode-specific definitions	<vk_video/vulkan_video_codec_h265std_encode.h>	VK_KHR_video_encode_h265
vulkan_video_codec_av1std	AV1 common definitions	<vk_video/vulkan_video_codec_av1std.h>	VK_KHR_video_decode_av1
vulkan_video_codec_av1std_decode	AV1 decode-specific definitions	<vk_video/vulkan_video_codec_av1std_decode.h>	VK_KHR_video_decode_av1

Appendix G: Invariance

The Vulkan specification is not pixel exact. It therefore does not guarantee an exact match between images produced by different Vulkan implementations. However, the specification does specify exact matches, in some cases, for images produced by the same implementation. The purpose of this appendix is to identify and provide justification for those cases that require exact matches.

Repeatability

The obvious and most fundamental case is repeated issuance of a series of Vulkan commands. For any given Vulkan and framebuffer state vector, and for any Vulkan command, the resulting Vulkan and framebuffer state **must** be identical whenever the command is executed on that initial Vulkan and framebuffer state. This repeatability requirement does not apply when using shaders containing side effects (image and buffer variable stores and atomic operations), because these memory operations are not guaranteed to be processed in a defined order.

One purpose of repeatability is avoidance of visual artifacts when a double-buffered scene is redrawn. If rendering is not repeatable, swapping between two buffers rendered with the same command sequence **may** result in visible changes in the image. Such false motion is distracting to the viewer. Another reason for repeatability is testability.

Repeatability, while important, is a weak requirement. Given only repeatability as a requirement, two scenes rendered with one (small) polygon changed in position might differ at every pixel. Such a difference, while within the law of repeatability, is certainly not within its spirit. Additional invariance rules are desirable to ensure useful operation.

Multi-Pass Algorithms

Invariance is necessary for a whole set of useful multi-pass algorithms. Such algorithms render multiple times, each time with a different Vulkan mode vector, to eventually produce a result in the framebuffer. Examples of these algorithms include:

- “Erasing” a primitive from the framebuffer by redrawing it, either in a different color or using the XOR logical operation.
- Using stencil operations to compute capping planes.

Invariance Rules

For a given Vulkan device:

Rule 1 *For any given Vulkan and framebuffer state vector, and for any given Vulkan command, the resulting Vulkan and framebuffer state **must** be identical each time the command is executed on that initial Vulkan and framebuffer state.*

Rule 2 *Changes to the following state values have no side effects (the use of any other state value is not affected by the change):*

Required:

- *Color and depth/stencil attachment contents*
- *Scissor parameters (other than enable)*
- *Write masks (color, depth, stencil)*
- *Clear values (color, depth, stencil)*

Strongly suggested:

- *Stencil parameters (other than enable)*
- *Depth test parameters (other than enable)*
- *Blend parameters (other than enable)*
- *Logical operation parameters (other than enable)*

Corollary 1 *Fragment generation is invariant with respect to the state values listed in Rule 2.*

Rule 3 *The arithmetic of each per-fragment operation is invariant except with respect to parameters that directly control it.*

Corollary 2 *Images rendered into different color attachments of the same framebuffer, either simultaneously or separately using the same command sequence, are pixel identical.*

Rule 4 *Identical pipelines will produce the same result when run multiple times with the same input. The wording “Identical pipelines” means [VkPipeline](#) objects that have been created with identical SPIR-V binaries and identical state, which are then used by commands executed using the same Vulkan state vector. Invariance is relaxed for shaders with side effects, such as performing stores or atomics.*

Rule 5 *All fragment shaders that either conditionally or unconditionally assign [FragCoord.z](#) to [FragDepth](#) are depth-invariant with respect to each other, for those fragments where the assignment to [FragDepth](#) actually is done.*

If a sequence of Vulkan commands specifies primitives to be rendered with shaders containing side effects (image and buffer variable stores and atomic operations), invariance rules are relaxed. In particular, rule 1, corollary 2, and rule 4 do not apply in the presence of shader side effects.

The following weaker versions of rules 1 and 4 apply to Vulkan commands involving shader side effects:

Rule 6 *For any given Vulkan and framebuffer state vector, and for any given Vulkan command, the contents of any framebuffer state not directly or indirectly affected by results of shader image or buffer variable stores or atomic operations **must** be identical each time the command is executed on that initial Vulkan and framebuffer state.*

Rule 7 *Identical pipelines will produce the same result when run multiple times with the same input as long as:*

- *shader invocations do not use image atomic operations;*

- *no framebuffer memory is written to more than once by image stores, unless all such stores write the same value; and*
- *no shader invocation, or other operation performed to process the sequence of commands, reads memory written to by an image store.*

NOTE

The OpenGL specification has the following invariance rule: Consider a primitive p' obtained by translating a primitive p through an offset (x, y) in window coordinates, where x and y are integers. As long as neither p' nor p is clipped, it **must** be the case that each fragment f' produced from p' is identical to a corresponding fragment f from p except that the center of f' is offset by (x, y) from the center of f .

This rule does not apply to Vulkan and is an intentional difference from OpenGL.

When any sequence of Vulkan commands triggers shader invocations that perform image stores or atomic operations, and subsequent Vulkan commands read the memory written by those shader invocations, these operations **must** be explicitly synchronized.

Tessellation Invariance

When using a pipeline containing tessellation evaluation shaders, the fixed-function tessellation primitive generator consumes the input patch specified by an application and emits a new set of primitives. The following invariance rules are intended to provide repeatability guarantees. Additionally, they are intended to allow an application with a carefully crafted tessellation evaluation shader to ensure that the sets of triangles generated for two adjacent patches have identical vertices along shared patch edges, avoiding “cracks” caused by minor differences in the positions of vertices along shared edges.

Rule 1 *When processing two patches with identical outer and inner tessellation levels, the tessellation primitive generator will emit an identical set of point, line, or triangle primitives as long as the pipeline used to process the patch primitives has tessellation evaluation shaders specifying the same tessellation mode, spacing, vertex order, and point mode decorations. Two sets of primitives are considered identical if and only if they contain the same number and type of primitives and the generated tessellation coordinates for the vertex numbered m of the primitive numbered n are identical for all values of m and n .*

Rule 2 *The set of vertices generated along the outer edge of the subdivided primitive in triangle and quad tessellation, and the tessellation coordinates of each, depend only on the corresponding outer tessellation level and the spacing decorations in the tessellation shaders of the pipeline.*

Rule 3 *The set of vertices generated when subdividing any outer primitive edge is always symmetric. For triangle tessellation, if the subdivision generates a vertex with tessellation coordinates of the form $(0, x, 1-x)$, $(x, 0, 1-x)$, or $(x, 1-x, 0)$, it will also generate a vertex with coordinates of exactly $(0, 1-x, x)$, $(1-x, 0, x)$, or $(1-x, x, 0)$, respectively. For quad tessellation, if the subdivision generates a vertex with coordinates of $(x, 0)$ or $(0, x)$, it will also generate a vertex with coordinates of exactly $(1-x, 0)$ or $(0, 1-x)$, respectively. For isoline tessellation, if it generates vertices at $(0, x)$ and $(1, x)$ where x is not zero, it will also generate vertices at exactly $(0, 1-x)$ and $(1, 1-x)$, respectively.*

Rule 4 *The set of vertices generated when subdividing outer edges in triangular and quad tessellation*

must be independent of the specific edge subdivided, given identical outer tessellation levels and spacing. For example, if vertices at $(x, 1 - x, 0)$ and $(1 - x, x, 0)$ are generated when subdividing the $w = 0$ edge in triangular tessellation, vertices **must** be generated at $(x, 0, 1 - x)$ and $(1 - x, 0, x)$ when subdividing an otherwise identical $v = 0$ edge. For quad tessellation, if vertices at $(x, 0)$ and $(1 - x, 0)$ are generated when subdividing the $v = 0$ edge, vertices **must** be generated at $(0, x)$ and $(0, 1 - x)$ when subdividing an otherwise identical $u = 0$ edge.

Rule 5 When processing two patches that are identical in all respects enumerated in rule 1 except for vertex order, the set of triangles generated for triangle and quad tessellation **must** be identical except for vertex and triangle order. For each triangle $n1$ produced by processing the first patch, there **must** be a triangle $n2$ produced when processing the second patch each of whose vertices has the same tessellation coordinates as one of the vertices in $n1$.

Rule 6 When processing two patches that are identical in all respects enumerated in rule 1 other than matching outer tessellation levels and/or vertex order, the set of interior triangles generated for triangle and quad tessellation **must** be identical in all respects except for vertex and triangle order. For each interior triangle $n1$ produced by processing the first patch, there **must** be a triangle $n2$ produced when processing the second patch each of whose vertices has the same tessellation coordinates as one of the vertices in $n1$. A triangle produced by the tessellator is considered an interior triangle if none of its vertices lie on an outer edge of the subdivided primitive.

Rule 7 For quad and triangle tessellation, the set of triangles connecting an inner and outer edge depends only on the inner and outer tessellation levels corresponding to that edge and the spacing decorations.

Rule 8 The value of all defined components of **TessCoord** will be in the range $[0, 1]$. Additionally, for any defined component x of **TessCoord**, the results of computing $1.0 - x$ in a tessellation evaluation shader will be exact. If any floating-point values in the range $[0, 1]$ fail to satisfy this property, such values **must** not be used as tessellation coordinate components.

Appendix H: Lexicon

This appendix defines terms, abbreviations, and API prefixes used in the Specification.

Glossary

The terms defined in this section are used consistently throughout the Specification and may be used with or without capitalization.

Accessible (Descriptor Binding)

A descriptor binding is accessible to a shader stage if that stage is included in the `stageFlags` of the descriptor binding. Descriptors using that binding **can** only be used by stages in which they are accessible.

Acquire Operation (Resource)

An operation that acquires ownership of an image subresource or buffer range.

Active (Transform Feedback)

Transform feedback is made active after `vkCmdBeginTransformFeedbackEXT` executes and remains active until `vkCmdEndTransformFeedbackEXT` executes. While transform feedback is active, data written to variables in the output interface of the last `pre-rasterization shader stage` of the graphics pipeline are captured to the bound transform feedback buffers if those variables are decorated for transform feedback.

Adjacent Vertex

A vertex in an adjacency primitive topology that is not part of a given primitive, but is accessible in geometry shaders.

Active Object (Ray Tracing)

A primitive or instance in a ray tracing acceleration structure which has a corresponding ID, and is not *inactive* (meaning that it is visible to rays).

Advanced Blend Operation

Blending performed using one of the blend operation enums introduced by the `VK_EXT_blend_operation_advanced` extension. See [Advanced Blending Operations](#).

Alias (API type/command)

An identical definition of another API type/command with the same behavior but a different name.

Aliased Range (Memory)

A range of a device memory allocation that is bound to multiple resources simultaneously.

Allocation Scope

An association of a host memory allocation to a parent object or command, where the allocation's lifetime ends before or at the same time as the parent object is freed or destroyed, or during the parent command.

API command

Any command defined in the Vulkan specification. These entry points all have a vk prefix.

Aspect (Image)

Some image types contain multiple kinds (called “aspects”) of data for each pixel, where each aspect is used in a particular way by the pipeline and **may** be stored differently or separately from other aspects. For example, the color components of an image format make up the color aspect of the image, and **can** be used as a framebuffer color attachment. Some operations, like depth testing, operate only on specific aspects of an image.

Attachment (Render Pass)

A zero-based integer index name used in render pass creation to refer to a framebuffer attachment that is accessed by one or more subpasses. The index also refers to an attachment description which includes information about the properties of the image view that will later be attached.

Availability Operation

An operation that causes the values generated by specified memory write accesses to become available for future access.

Available

A state of values written to memory that allows them to be made visible.

Axis-aligned Bounding Box

A box bounding a region in space defined by extents along each axis and thus representing a box where each edge is aligned to one of the major axes.

Back-Facing

See Facingness.

Backward Prediction

Sample prediction performed during video decode and encode operations based on reference pictures that occur temporally (in display order) after the current picture.

Backward Reference

A reference picture that occurs temporally (in display order) after the current picture.

Batch

A single structure submitted to a queue as part of a [queue submission command](#), describing a set of queue operations to execute.

Backwards Compatibility

A given version of the API is backwards compatible with an earlier version if an application, relying only on valid behavior and functionality defined by the earlier specification, is able to correctly run against each version without any modification. This assumes no active attempt by that application to not run when it detects a different version.

Binary Semaphore

A semaphore with a boolean payload indicating whether the semaphore is signaled or unsignaled. Represented by a [VkSemaphore](#) object created with a semaphore type of [VK_SEMAPHORE_TYPE_BINARY](#).

Binding (Memory)

An association established between a range of a resource object and a range of a memory object. These associations determine the memory locations affected by operations performed on elements of a resource object. Memory bindings are established using the [vkBindBufferMemory](#) command for non-sparse buffer objects, using the [vkBindImageMemory](#) command for non-sparse image objects, and using the [vkQueueBindSparse](#) command for sparse resources.

Blend Constant

Four floating-point (RGBA) values used as an input to blending.

Blending

Arithmetic operations between a fragment color value and a value in a color attachment that produce a final color value to be written to the attachment.

Buffer

A resource that represents a linear array of data in device memory. Represented by a [VkBuffer](#) object.

Buffer Device Address

A 64-bit value used in a shader to access buffer memory through the [PhysicalStorageBuffer](#) storage class.

Buffer View

An object that represents a range of a specific buffer, and state controlling how the contents are interpreted. Represented by a [VkBufferView](#) object.

Built-In Variable

A variable decorated in a shader, where the decoration makes the variable take values provided by the execution environment or values that are generated by fixed-function pipeline stages.

Built-In Interface Block

A block defined in a shader containing only variables decorated with built-in decorations, and is used to match against other shader stages.

Clip Coordinates

The homogeneous coordinate space in which vertex positions ([Position](#) decoration) are written by [pre-rasterization shader stages](#).

Clip Distance

A built-in output from [pre-rasterization shader stages](#) defining a clip half-space against which the primitive is clipped.

Clip Volume

The intersection of the view volume with all clip half-spaces.

Coding Block (Video)

An element of a video picture, comprising a rectangular block of texel values that are coded as a single unit in video coding. Video coding blocks have a corresponding coding block size (width and height) defining the dimensions of these elements in units of texels. These dimensions are usually codec-specific and a single video compression standard **may** support more than one coding block type and size. Video compression standards **may** use multiple coding block types to organize coding blocks of different kinds and sizes into hierarchies.

Color Attachment

A subpass attachment point, or image view, that is the target of fragment color outputs and blending.

Color Renderable Format

A [VkFormat](#) where `VK_FORMAT_FEATURE_COLOR_ATTACHMENT_BIT` is set in one of the following, depending on the image's tiling:

- [VkFormatProperties::linearTilingFeatures](#)
- [VkFormatProperties::optimalTilingFeatures](#)
- [VkDrmFormatModifierPropertiesEXT::drmFormatModifierTilingFeatures](#)

Combined Image Sampler

A descriptor type that includes both a sampled image and a sampler.

Command Buffer

An object that records commands to be submitted to a queue. Represented by a [VkCommandBuffer](#) object.

Command Buffer Nesting Level

The Command Buffer Nesting Level of a secondary command buffer is equal to the maximum nesting level of all secondary command buffers executed by that command buffer plus one, where a secondary command buffer that executes no other secondary command buffers has a nesting level of zero.

Command Pool

An object that command buffer memory is allocated from, and that owns that memory. Command pools aid multithreaded performance by enabling different threads to use different allocators, without internal synchronization on each use. Represented by a [VkCommandPool](#) object.

Compatible Allocator

When allocators are compatible, allocations from each allocator **can** be freed by the other allocator.

Compatible Image Formats

When formats are compatible, images created with one of the formats **can** have image views

created from it using any of the compatible formats. Also see *Size-Compatible Image Formats*.

Compatible Queues

Queues within a queue family. Compatible queues have identical properties.

Complete Mipmap Chain

The entire set of mip levels that can be provided for an image, from the largest application-specified mip level size down to the *minimum mip level size*. See [Image Mip Level Sizing](#).

Completed Operation

A deferred operation whose corresponding command has been executed to completion. See [Deferred Host Operations](#)

Component (Format)

A distinct part of a format. Color components are represented with **R**, **G**, **B**, and **A**. Depth and stencil components are represented with **D** and **S**. Formats **can** have multiple instances of the same component. Some formats have other notations such as **E** or **X** which are not considered a component of the format.

Compressed Texel Block

An element of an image having a block-compressed format, comprising a rectangular block of texel values that are encoded as a single value in memory. Compressed texel blocks of a particular block-compressed format have a corresponding width, height, and depth defining the dimensions of these elements in units of texels, and a size in bytes of the encoding in memory.

Constant Integral Expressions

A SPIR-V constant instruction whose type is **OpTypeInt**. See *Constant Instruction* in section 2.2.1 “Instructions” of the [Khronos SPIR-V Specification](#).

Cooperative Matrix

A SPIR-V type where the storage for and computations performed on the matrix are spread across a set of invocations such as a subgroup.

Coverage Index

The index of a sample in the coverage mask.

Coverage Mask

A bitfield associated with a fragment representing the samples that were determined to be covered based on the result of rasterization, and then subsequently modified by fragment operations or the fragment shader.

Cull Distance

A built-in output from [pre-rasterization shader stages](#) defining a cull half-space where the primitive is rejected if all vertices have a negative value for the same cull distance.

Cull Volume

The intersection of the view volume with all cull half-spaces.

Decode Output Picture

A video picture resource used to store the result of a video decode operation.

Decoded Picture Buffer

An indexed set of reference pictures used by a video session. Abbreviated as DPB.

Decoded Picture Buffer Slot

An entry within a DPB that **can** be associated with a particular reference picture.

Decoded Picture Buffer Slot Index

The index of a DPB slot within its encompassing DPB.

Decoration (SPIR-V)

Auxiliary information such as built-in variables, stream numbers, invariance, interpolation type, relaxed precision, etc., added to variables or structure-type members through decorations.

Deferrable Command

A command which allows deferred execution of host-side work. See [Deferred Host Operations](#).

Deferrable Operation

A single logical item of host-side work which can be deferred. Represented by the [VkDeferredOperationKHR](#) object. See [Deferred Host Operations](#).

Deprecated (feature)

A feature is deprecated if it is no longer recommended as the correct or best way to achieve its intended purpose.

Depth/Stencil Attachment

A subpass attachment point, or image view, that is the target of depth and/or stencil test operations and writes.

Depth/Stencil Format

A [VkFormat](#) that includes depth and/or stencil components.

Depth/Stencil Image (or ImageView)

A [VkImage](#) (or [VkImageView](#)) with a depth/stencil format.

Depth/Stencil Resolve Attachment

A subpass attachment point, or image view, that is the target of a multisample resolve operation from the corresponding depth/stencil attachment at the end of the subpass.

Derivative Group

A set of compute, mesh, task, or fragment shader invocations that cooperate to compute derivatives, including implicit derivatives for sampled image operations.

Descriptor

Information about a resource or resource view written into a descriptor set that is used to access the resource or view from a shader.

Descriptor Binding

An entry in a descriptor set layout corresponding to zero or more descriptors of a single descriptor type in a set. Defined by a [VkDescriptorSetLayoutBinding](#) structure.

Descriptor Pool

An object that descriptor sets are allocated from, and that owns the storage of those descriptor sets. Descriptor pools aid multithreaded performance by enabling different threads to use different allocators, without internal synchronization on each use. Represented by a [VkDescriptorPool](#) object.

Descriptor Set

An object that resource descriptors are written into via the API, and that **can** be bound to a command buffer such that the descriptors contained within it **can** be accessed from shaders. Represented by a [VkDescriptorSet](#) object.

Descriptor Set Layout

An object defining the set of resources (types and counts) and their relative arrangement (in the binding namespace) within a descriptor set. Used when allocating descriptor sets and when creating pipeline layouts. Represented by a [VkDescriptorSetLayout](#) object.

Device

The processor(s) and execution environment that perform tasks requested by the application via the Vulkan API.

Device Group

A set of physical devices that support accessing each other's memory and recording a single command buffer that **can** be executed on all the physical devices.

Device Index

A zero-based integer that identifies one physical device from a logical device. A device index is valid if it is less than the number of physical devices in the logical device.

Device Mask

A bitmask where each bit represents one device index. A device mask value is valid if every bit that is set in the mask is at a bit position that is less than the number of physical devices in the logical device.

Device Memory

Memory accessible to the device. Represented by a [VkDeviceMemory](#) object.

Device-Level Command

Any command that is dispatched from a logical device, or from a child object of a logical device.

Device-Level Functionality

All device-level commands and objects, and their structures, enumerated types, and enumerants. Additionally, physical-device-level functionality defined by a [device extension](#) is also considered device-level functionality.

Device-Level Object

Logical device objects and their child objects. For example, [VkDevice](#), [VkQueue](#), and [VkCommandBuffer](#) objects are device-level objects.

Device-Local Memory

Memory that is connected to the device, and **may** be more performant for device access than host-local memory.

Direct Drawing Commands

Drawing commands that take all their parameters as direct arguments to the command (and not sourced via structures in buffer memory as the *indirect drawing commands*). Includes [vkCmdDrawMultiIndexedEXT](#), [vkCmdDrawMultiEXT](#), [vkCmdDraw](#), and [vkCmdDrawIndexed](#).

Disjoint

Disjoint planes are *image planes* to which memory is bound independently.

A *disjoint image* consists of multiple *disjoint planes*, and is created with the [VK_IMAGE_CREATE_DISJOINT_BIT](#) bit set.

Dispatchable Command

A non-global command. The first argument to each dispatchable command is a dispatchable handle type.

Dispatchable Handle

A handle of a pointer handle type which **may** be used by layers as part of intercepting API commands.

Dispatching Commands

Commands that provoke work using a compute pipeline. Includes [vkCmdDispatch](#) and [vkCmdDispatchIndirect](#).

Drawing Commands

Commands that provoke work using a graphics pipeline. Includes [vkCmdDraw](#), [vkCmdDrawIndexed](#), [vkCmdDrawIndirectCountKHR](#), [vkCmdDrawIndexedIndirectCountKHR](#), [vkCmdDrawMultiIndexedEXT](#), [vkCmdDrawMultiEXT](#), [vkCmdDrawIndirect](#), and [vkCmdDrawIndexedIndirect](#).

Duration (Command)

The *duration* of a Vulkan command refers to the interval between calling the command and its return to the caller.

Dynamic Storage Buffer

A storage buffer whose offset is specified each time the storage buffer is bound to a command buffer via a descriptor set.

Dynamic Uniform Buffer

A uniform buffer whose offset is specified each time the uniform buffer is bound to a command buffer via a descriptor set.

Dynamically Uniform

See *Dynamically Uniform* in section 2.2 “Terms” of the [Khronos SPIR-V Specification](#).

Encode Input Picture

A video picture resource used as the input of a video encode operation.

Element

Arrays are composed of multiple elements, where each element exists at a unique index within that array. Used primarily to describe data passed to or returned from the Vulkan API.

Explicitly-Enabled Layer

A layer enabled by the application by adding it to the enabled layer list in [vkCreateInstance](#) or [vkCreateDevice](#).

Event

A synchronization primitive that is signaled when execution of previous commands completes through a specified set of pipeline stages. Events can be waited on by the device and polled by the host. Represented by a [VkEvent](#) object.

Executable State (Command Buffer)

A command buffer that has ended recording commands and **can** be executed. See also Initial State and Recording State.

Execution Dependency

A dependency that guarantees that certain pipeline stages’ work for a first set of commands has completed execution before certain pipeline stages’ work for a second set of commands begins execution. This is accomplished via pipeline barriers, subpass dependencies, events, or implicit ordering operations.

Execution Dependency Chain

A sequence of execution dependencies that transitively act as a single execution dependency.

Explicit chroma reconstruction

An implementation of sampler $Y'C_B C_R$ conversion which reconstructs reduced-resolution chroma samples to luma resolution and then separately performs texture sample interpolation. This is distinct from an implicit implementation, which incorporates chroma sample reconstruction into texture sample interpolation.

Extension Scope

The set of objects and commands that **can** be affected by an extension. Extensions are either device scope or instance scope.

Extending Structure

A structure type which may appear in the *pNext chain* of another structure, extending the functionality of the other structure. Extending structures may be defined by either core API versions or extensions.

External Handle

A resource handle which has meaning outside of a specific Vulkan device or its parent instance. External handles **may** be used to share resources between multiple Vulkan devices in different instances, or between Vulkan and other APIs. Some external handle types correspond to platform-defined handles, in which case the resource **may** outlive any particular Vulkan device or instance and **may** be transferred between processes, or otherwise manipulated via functionality defined by the platform for that handle type.

External synchronization

A type of synchronization **required** of the application, where parameters defined to be externally synchronized **must** not be used simultaneously in multiple threads.

Facingness (Polygon)

A classification of a polygon as either front-facing or back-facing, depending on the orientation (winding order) of its vertices.

Facingness (Fragment)

A fragment is either front-facing or back-facing, depending on the primitive it was generated from. If the primitive was a polygon (regardless of polygon mode), the fragment inherits the facingness of the polygon. All other fragments are front-facing.

Fence

A synchronization primitive that is signaled when a set of batches or sparse binding operations complete execution on a queue. Fences **can** be waited on by the host. Represented by a [VkFence](#) object.

Field (Video)

Possibly discontinuous subregions of a frame. Frames **may** consist of two fields, a top field and a bottom field.

Flat Shading

A property of a vertex attribute that causes the value from a single vertex (the provoking vertex) to be used for all vertices in a primitive, and for interpolation of that attribute to return that single value unaltered.

Format Features

A set of features from [VkFormatFeatureFlagBits](#) that a [VkFormat](#) is capable of using for various commands. The list is determined by factors such as [VkImageTiling](#).

Forward Prediction

Sample prediction performed during video decode and encode operations based on reference pictures that occur temporally (in display order) before the current picture.

Forward Reference

A reference picture that occurs temporally (in display order) before the current picture.

Fragment

A rectangular framebuffer region with associated data produced by [rasterization](#) and processed

by [fragment operations](#) including the fragment shader.

Fragment Area

The width and height, in pixels, of a fragment.

Fragment Input Attachment Interface

Variables with [UniformConstant](#) storage class and a decoration of [InputAttachmentIndex](#) that are statically used by a fragment shader's entry point, which receive values from input attachments.

Fragment Output Interface

A fragment shader entry point's variables with [Output](#) storage class, which output to color and/or depth/stencil attachments.

Frame (Video)

A multi-dimensional array of luma samples and an optional multi-dimensional array of chroma samples.

Fragment Tile Image Interface

A fragment shader entry point's variables with [TileImageEXT](#) storage class and a decoration of [Location](#), which are used to read values from color attachments.

Framebuffer

A collection of image views and a set of dimensions that, in conjunction with a render pass, define the inputs and outputs used by drawing commands. Represented by a [VkFramebuffer](#) object.

Framebuffer Attachment

One of the image views used in a framebuffer.

Framebuffer Coordinates

A coordinate system in which adjacent pixels' coordinates differ by 1 in x and/or y, with (0,0) in the upper left corner and pixel centers at half-integers.

Framebuffer-Space

Operating with respect to framebuffer coordinates.

Framebuffer-Local

A framebuffer-local dependency guarantees that only for a single framebuffer region, the first set of operations happens-before the second set of operations.

Framebuffer-Global

A framebuffer-global dependency guarantees that for all framebuffer regions, the first set of operations happens-before the second set of operations.

Framebuffer Region

A framebuffer region is a set of sample (x, y, layer, sample) coordinates that is a subset of the entire framebuffer.

Front-Facing

See Facingness.

Full Compatibility

A given version of the API is fully compatible with another version if an application, relying only on valid behavior and functionality defined by either of those specifications, is able to correctly run against each version without any modification. This assumes no active attempt by that application to not run when it detects a different version.

Global Command

A Vulkan command for which the first argument is not a dispatchable handle type.

Global Workgroup

A collection of local workgroups dispatched by a single dispatching command.

Handle

An opaque integer or pointer value used to refer to a Vulkan object. Each object type has a unique handle type.

Happen-after, happens-after

A transitive, irreflexive and antisymmetric ordering relation between operations. An execution dependency with a source of **A** and a destination of **B** enforces that **B** happens-after **A**. The inverse relation of happens-before.

Happen-before, happens-before

A transitive, irreflexive and antisymmetric ordering relation between operations. An execution dependency with a source of **A** and a destination of **B** enforces that **A** happens-before **B**. The inverse relation of happens-after.

Helper Invocation

A fragment shader invocation that is created solely for the purposes of evaluating derivatives for use in non-helper fragment shader invocations, and which does not have side effects.

Host

The processor(s) and execution environment that the application runs on, and that the Vulkan API is exposed on.

Host Mapped Device Memory

Device memory that is mapped for host access using [vkMapMemory](#).

Host Mapped Foreign Memory

Memory owned by a foreign device that is mapped for host access.

Host Memory

Memory not accessible to the device, used to store implementation data structures.

Host-Accessible Subresource

A buffer, or a linear image subresource in either the [VK_IMAGE_LAYOUT_PREINITIALIZED](#) or

`VK_IMAGE_LAYOUT_GENERAL` layout. Host-accessible subresources have a well-defined addressing scheme which can be used by the host.

Host-Local Memory

Memory that is not local to the device, and **may** be less performant for device access than device-local memory.

Host-Visible Memory

Device memory that **can** be mapped on the host and **can** be read and written by the host.

ICD

Installable Client Driver. An ICD is represented as a [VkPhysicalDevice](#).

Identically Defined Objects

Objects of the same type where all arguments to their creation or allocation functions, with the exception of `pAllocator`, are

1. Vulkan handles which refer to the same object or
2. identical scalar or enumeration values or
3. Host pointers which point to an array of values or structures which also satisfy these three constraints.

Image

A resource that represents a multi-dimensional formatted interpretation of device memory. Represented by a [VkImage](#) object.

Image Subresource

A specific mipmap level, layer, and set of aspects of an image.

Image Subresource Range

A set of image subresources that are contiguous mipmap levels and layers.

Image View

An object that represents an image subresource range of a specific image, and state controlling how the contents are interpreted. Represented by a [VkImageView](#) object.

Immutable Sampler

A sampler descriptor provided at descriptor set layout creation time for a specific binding. This sampler is then used for that binding in all descriptor sets allocated with the layout, and it **cannot** be changed.

Implicit chroma reconstruction

An implementation of sampler $Y'C_B C_R$ conversion which reconstructs the reduced-resolution chroma samples directly at the sample point, as part of the normal texture sampling operation. This is distinct from an *explicit chroma reconstruction* implementation, which reconstructs the reduced-resolution chroma samples to the resolution of the luma samples, then filters the result as part of texture sample interpolation.

Implicitly-Enabled Layer

A layer enabled by a loader-defined mechanism outside the Vulkan API, rather than explicitly by the application during instance or device creation.

Inactive Object (Ray Tracing)

A primitive or instance in a ray tracing acceleration structure which has a corresponding ID, but which will never report an intersection with any ray.

Index Buffer

A buffer bound via [vkCmdBindIndexBuffer](#) which is the source of index values used to fetch vertex attributes for a [vkCmdDrawIndexed](#) or [vkCmdDrawIndexedIndirect](#) command.

Indexed Drawing Commands

Drawing commands which use an *index buffer* as the source of index values used to fetch vertex attributes for a drawing command. Includes [vkCmdDrawIndexed](#), [vkCmdDrawIndexedIndirectCountKHR](#), [vkCmdDrawMultiIndexedEXT](#), and [vkCmdDrawIndexedIndirect](#).

Indirect Commands

Drawing or dispatching commands that source some of their parameters from structures in buffer memory. Includes [vkCmdDrawIndirect](#), [vkCmdDrawIndexedIndirect](#), [vkCmdDrawIndirectCountKHR](#), [vkCmdDrawIndexedIndirectCountKHR](#), and [vkCmdDispatchIndirect](#).

Indirect Drawing Commands

Drawing commands that source some of their parameters from structures in buffer memory. Includes [vkCmdDrawIndirect](#), [vkCmdDrawIndirectCountKHR](#), [vkCmdDrawIndexedIndirectCountKHR](#), and [vkCmdDrawIndexedIndirect](#).

Initial State (Command Buffer)

A command buffer that has not begun recording commands. See also Recording State and Executable State.

Input Attachment

A descriptor type that represents an image view, and supports unfiltered read-only access in a shader, only at the fragment's location in the view.

Instance

The top-level Vulkan object, which represents the application's connection to the implementation. Represented by a [VkInstance](#) object.

Instance-Level Command

Any command that is dispatched from an instance, or from a child object of an instance, except for physical devices and their children.

Instance-Level Functionality

All instance-level commands and objects, and their structures, enumerated types, and enumerants.

Instance-Level Object

High-level Vulkan objects, which are not physical devices, nor children of physical devices. For example, [VkInstance](#) is an instance-level object.

Instance (Memory)

In a logical device representing more than one physical device, some device memory allocations have the requested amount of memory allocated multiple times, once for each physical device in a device mask. Each such replicated allocation is an instance of the device memory.

Instance (Resource)

In a logical device representing more than one physical device, buffer and image resources exist on all physical devices but **can** be bound to memory differently on each. Each such replicated resource is an instance of the resource.

Inter Block (Video)

A video coding block that uses inter prediction.

Inter Prediction (Video)

Sample prediction derived from reference picture samples or metadata.

Internal Synchronization

A type of synchronization **required** of the implementation, where parameters not defined to be externally synchronized **may** require internal mutexing to avoid multithreaded race conditions.

Intra Block (Video)

A video coding block that only uses intra prediction.

Intra Prediction (Video)

Sample prediction that is not derived from reference picture samples or metadata.

Invocation (Shader)

A single execution of an entry point in a SPIR-V module. For example, a single vertex's execution of a vertex shader or a single fragment's execution of a fragment shader.

Invocation Group

A set of shader invocations that are executed in parallel and that **must** execute the same control flow path in order for control flow to be considered dynamically uniform.

Invocation Repack Instruction

A ray tracing [instruction](#) where the implementation **may** change the set of invocations that are executing.

Join (Deferred Host Operations)

The act of instructing a thread to participate in the execution of a deferred operation. See [Deferred Host Operations](#).

Linear Resource

A resource is *linear* if it is one of the following:

- a [VkBuffer](#)
- a [VkImage](#) created with [VK_IMAGE_TILING_LINEAR](#)
- a [VkImage](#) created with [VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT](#) and whose [Linux DRM format modifier](#) is [DRM_FORMAT_MOD_LINEAR](#)

Because a [VkAccelerationStructureKHR](#) resource does not have memory bound to it directly, it is considered neither linear nor non-linear. However, the [VkBuffer](#) on which a [VkAccelerationStructureKHR](#) resource is placed is a linear resource.

A resource is *non-linear* if it is one of the following:

- a [VkImage](#) created with [VK_IMAGE_TILING_OPTIMAL](#)
- a [VkImage](#) created with [VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT](#) and whose [Linux DRM format modifier](#) is not [DRM_FORMAT_MOD_LINEAR](#)

Linux DRM Format Modifier

A 64-bit, vendor-prefixed, semi-opaque unsigned integer describing vendor-specific details of an image's memory layout. In Linux graphics APIs, *modifiers* are commonly used to specify the memory layout of externally shared images. An image has a *modifier* if and only if it is created with *tiling* equal to [VK_IMAGE_TILING_DRM_FORMAT_MODIFIER_EXT](#). For more details, refer to the appendix for extension [VK_EXT_image_drm_format_modifier](#).

Local Workgroup

A collection of compute shader invocations invoked by a single dispatching command, which share data via [WorkgroupLocal](#) variables and can synchronize with each other.

Logical Device

An object that represents the application's interface to the physical device. The logical device is the parent of most Vulkan objects. Represented by a [VkDevice](#) object.

Logical Operation

Bitwise operations between a fragment color value and a value in a color attachment, that produce a final color value to be written to the attachment.

Lost Device

A state that a logical device **may** be in as a result of unrecoverable implementation errors, or other exceptional conditions.

Mappable

See Host-Visible Memory.

Memory Dependency

A memory dependency is an execution dependency which includes availability and visibility operations such that:

- The first set of operations happens-before the availability operation
- The availability operation happens-before the visibility operation

- The visibility operation happens before the second set of operations

Memory Domain

A memory domain is an abstract place to which memory writes are made available by availability operations and memory domain operations. The memory domains correspond to the set of agents that the write **can** then be made visible to. The memory domains are *host*, *device*, *shader*, *workgroup instance* (for workgroup instance there is a unique domain for each compute workgroup) and *subgroup instance* (for subgroup instance there is a unique domain for each subgroup).

Memory Domain Operation

An operation that makes the writes that are available to one memory domain available to another memory domain.

Memory Heap

A region of memory from which device memory allocations **can** be made.

Memory Type

An index used to select a set of memory properties (e.g. mappable, cached) for a device memory allocation.

Minimum Mip Level Size

The smallest size that is permitted for a mip level. For conventional images this is 1x1x1. See [Image Mip Level Sizing](#).

Mip Tail Region

The set of mipmap levels of a sparse residency texture that are too small to fill a sparse block, and that **must** all be bound to memory collectively and opaquely.

Multi-planar

A *multi-planar format* (or “planar format”) is an image format consisting of more than one *plane*, identifiable with a `_2PLANE` or `_3PLANE` component to the format name and listed in [Formats Requiring Sampler Y'C_BC_R Conversion for VK_IMAGE_ASPECT_COLOR_BIT Image Views](#). A *multi-planar image* (or “planar image”) is an image of a multi-planar format.

Nested Command Buffers

A nested command buffer is a secondary command buffer that is executed by another secondary command buffer, which may itself execute other secondary command buffers.

Non-Dispatchable Handle

A handle of an integer handle type. Handle values **may** not be unique, even for two objects of the same type.

Non-Indexed Drawing Commands

Drawing commands for which the vertex attributes are sourced in linear order from the vertex input attributes for a drawing command (i.e. they do not use an *index buffer*). Includes [vkCmdDraw](#), [vkCmdDrawIndirectCountKHR](#), [vkCmdDrawMultiEXT](#), and [vkCmdDrawIndirect](#).

Normalized

A value that is interpreted as being in the range [0,1] as a result of being implicitly divided by some other value.

Normalized Device Coordinates

A coordinate space after perspective division is applied to clip coordinates, and before the viewport transformation converts them to framebuffer coordinates.

Obsoleted (feature)

A feature is obsolete if it can no longer be used.

Opaque Capture Address

A 64-bit value representing the device address of a buffer or memory object that is expected to be used by trace capture/replay tools in combination with the [bufferDeviceAddress](#) feature.

Overlapped Range (Aliased Range)

The aliased range of a device memory allocation that intersects a given image subresource of an image or range of a buffer.

Ownership (Resource)

If an entity (e.g. a queue family) has ownership of a resource, access to that resource is well-defined for access by that entity.

Packed Format

A format whose components are stored as a single texel block in memory, with their relative locations defined within that element.

Payload

Portable or exportable reference to the internal data of an object in Vulkan.

Peer Memory

An instance of memory corresponding to a different physical device than the physical device performing the memory access, in a logical device that represents multiple physical devices.

Physical Device

An object that represents a single device in the system. Represented by a [VkPhysicalDevice](#) object.

Physical-Device-Level Command

Any command that is dispatched from a physical device.

Physical-Device-Level Functionality

All physical-device-level commands and objects, and their structures, enumerated types, and enumerants.

Physical-Device-Level Object

Physical device objects. For example, [VkPhysicalDevice](#) is a physical-device-level object.

Picture Partition (Video)

Collective term that refers to a subregion of a video picture resource that **may** be decoded or encoded independently according to the partitioning scheme defined in the corresponding video compression standard. Picture partitions **may** be non-rectangular.

Pipeline

An object controlling how graphics or compute work is executed on the device. A pipeline includes one or more shaders, as well as state controlling any non-programmable stages of the pipeline. Represented by a [VkPipeline](#) object.

Pipeline Barrier

An execution and/or memory dependency recorded as an explicit command in a command buffer, that forms a dependency between the previous and subsequent commands.

Pipeline Cache

An object that **can** be used to collect and retrieve information from pipelines as they are created, and **can** be populated with previously retrieved information in order to accelerate pipeline creation. Represented by a [VkPipelineCache](#) object.

Pipeline Layout

An object defining the set of resources (via a collection of descriptor set layouts) and push constants used by pipelines that are created using the layout. Used when creating a pipeline and when binding descriptor sets and setting push constant values. Represented by a [VkPipelineLayout](#) object.

Pipeline Library

A pipeline that cannot be directly used, instead defining a set of shaders and shader groups which will be [linked into other pipelines](#).

Pipeline Stage

A logically independent execution unit that performs some of the operations defined by an action command.

Pipeline Trace Ray Instruction

A ray tracing instruction which traces a ray into an acceleration structure when using ray tracing pipelines. One of:

- [OpTraceRayKHR](#)

[pNext](#) Chain

A set of structures [chained together](#) through their [pNext](#) members.

Planar

See *multi-planar*.

Plane

An *image plane* is part of the representation of an image, containing a subset of the color components necessary to represent the texels in the image and with a contiguous mapping of

coordinates to bound memory. Most images consist only of a single plane, but some formats spread the components across multiple image planes. The host-accessible properties of each image plane are accessible for a linear layout using [vkGetImageSubresourceLayout](#). If a multi-planar image is created with the [VK_IMAGE_CREATE_DISJOINT_BIT](#) bit set, the image is described as *disjoint*, and its planes are therefore bound to memory independently.

Point Sampling (Rasterization)

A rule that determines whether a fragment sample location is covered by a polygon primitive by testing whether the sample location is in the interior of the polygon in framebuffer-space, or on the boundary of the polygon according to the tie-breaking rules.

Potential Format Features

The union of all [VkFormatFeatureFlagBits](#) that the implementation supports for a specified [VkFormat](#), over all supported image tilings.

Pre-rasterization

Operations that execute before [rasterization](#), and any state associated with those operations.

Presentable image

A [VkImage](#) object obtained from a [VkSwapchainKHR](#) used to present to a [VkSurfaceKHR](#) object.

Preserve Attachment

One of a list of attachments in a subpass description that is not read or written by the subpass, but that is read or written on earlier and later subpasses and whose contents **must** be preserved through this subpass.

Primary Command Buffer

A command buffer that **can** execute secondary command buffers, and **can** be submitted directly to a queue.

Primitive Topology

State controlling how vertices are assembled into primitives, e.g. as lists of triangles, strips of lines, etc.

Promoted (feature)

A feature from an older extension is considered promoted if it is made available as part of a new core version or newer extension with wider support.

Provisional

A feature is released provisionally in order to get wider feedback on the functionality before it is finalized. Provisional features may change in ways that break backwards compatibility, and thus are not recommended for use in production applications.

Provoking Vertex

The vertex in a primitive from which flat shaded attribute values are taken. This is generally the “first” vertex in the primitive, and depends on the primitive topology.

Push Constants

A small bank of values writable via the API and accessible in shaders. Push constants allow the application to set values used in shaders without creating buffers or modifying and binding descriptor sets for each update.

Push Constant Interface

The set of variables with `PushConstant` storage class that are statically used by a shader entry point, and which receive values from push constant commands.

Push Descriptors

Descriptors that are written directly into a command buffer rather than into a descriptor set. Push descriptors allow the application to set descriptors used in shaders without allocating or modifying descriptor sets for each update.

Descriptor Update Template

An object specifying a mapping from descriptor update information in host memory to elements in a descriptor set, which helps enable more efficient descriptor set updates.

Query Pool

An object containing a number of query entries and their associated state and results. Represented by a `VkQueryPool` object.

Queue

An object that executes command buffers and sparse binding operations on a device. Represented by a `VkQueue` object.

Queue Family

A set of queues that have common properties and support the same functionality, as advertised in `VkQueueFamilyProperties`.

Queue Operation

A unit of work to be executed by a specific queue on a device, submitted via a `queue submission command`. Each queue submission command details the specific queue operations that occur as a result of calling that command. Queue operations typically include work that is specific to each command, and synchronization tasks.

Queue Submission

Zero or more batches and an optional fence to be signaled, passed to a command for execution on a queue. See the [Devices and Queues chapter](#) for more information.

Ray Tracing Command

Commands that provoke work using a ray tracing pipeline. Includes `vkCmdTraceRaysKHR`, and `vkCmdTraceRaysIndirectKHR`.

Reconstructed Picture

A video picture resource reconstructed from a compressed bitstream using video decode or encode operations that **can** be used as a reference picture by future video decode or encode operations with the same video session.

Recording State (Command Buffer)

A command buffer that is ready to record commands. See also Initial State and Executable State.

Reference Picture

A video picture resource used by video decode and encode operations to provide predictions of the values of samples in the subsequently decoded or encoded pictures.

Reference Picture Metadata

Opaque state associated with a DPB slot, maintained by a video session.

Release Operation (Resource)

An operation that releases ownership of an image subresource or buffer range.

Render Pass

An object that represents a set of framebuffer attachments and phases of rendering using those attachments. Represented by a [VkRenderPass](#) object.

Render Pass Instance

A use of a render pass in a command buffer.

Required Extensions

Extensions that **must** be enabled alongside extensions dependent on them (see [Extension Dependencies](#)).

Reset (Command Buffer)

Resetting a command buffer discards any previously recorded commands and puts a command buffer in the initial state.

Residency Code

An integer value returned by sparse image instructions, indicating whether any sparse unbound texels were accessed.

Resolve Attachment

A subpass attachment point, or image view, that is the target of a multisample resolve operation from the corresponding color attachment at the end of the subpass.

Retired Swapchain

A swapchain that has been used as the [oldSwapchain](#) parameter to [vkCreateSwapchainKHR](#). Images cannot be acquired from a retired swapchain, however images that were acquired (but not presented) before the swapchain was retired **can** be presented.

Sample Index

The index of a sample within a [single set of samples](#).

Sample Shading

Invoking the fragment shader multiple times per fragment, with the covered samples partitioned among the invocations.

Sampled Image

A descriptor type that represents an image view, and supports filtered (sampled) and unfiltered read-only access in a shader.

Sampler

An object containing state controlling how sampled image data is sampled (or filtered) when accessed in a shader. Also a descriptor type describing the object. Represented by a [VkSampler](#) object.

Secondary Command Buffer

A command buffer that **can** be executed by a primary command buffer, and **must** not be submitted directly to a queue.

Self-Dependency

A subpass dependency from a subpass to itself, i.e. with [srcSubpass](#) equal to [dstSubpass](#). A self-dependency is not automatically performed during a render pass instance, rather a subset of it **can** be performed via [vkCmdPipelineBarrier](#) during the subpass.

Semaphore

A synchronization primitive that supports signal and wait operations, and **can** be used to synchronize operations within a queue or across queues. Represented by a [VkSemaphore](#) object.

Shader

Instructions selected (via an entry point) from a shader module, which are executed in a shader stage.

Shader Call

An [instruction](#) which **may** cause execution to continue in a different shader stage.

Shader Code

A stream of instructions used to describe the operation of a shader.

Shader Group

A set of Shader Stages that are part of a [VkPipeline](#) containing multiple of such sets. This allows the device to make use of all the shader groups from the bound pipeline independently.

Shader Module

A collection of shader code, potentially including several functions and entry points, that is used to create shaders in pipelines. Represented by a [VkShaderModule](#) object.

Shader Stage

A stage of the graphics or compute pipeline that executes shader code.

Shading Rate

The ratio of the number of fragment shader invocations generated in a fully covered framebuffer region to the size (in pixels) of that region.

Shared presentable image

A presentable image created from a swapchain with `VkPresentModeKHR` set to either `VK_PRESENT_MODE_SHARED_DEMAND_REFRESH_KHR` or `VK_PRESENT_MODE_SHARED_CONTINUOUS_REFRESH_KHR`.

Side Effect

A store to memory or atomic operation on memory from a shader invocation.

Single-plane format

A format that is not *multi-planar*.

Size-Compatible Image Formats

When a compressed image format and an uncompressed image format are size-compatible, it means that the texel block size of the uncompressed format **must** equal the texel block size of the compressed format.

Sparse Block

An element of a sparse resource that can be independently bound to memory. Sparse blocks of a particular sparse resource have a corresponding size in bytes that they use in the bound memory.

Sparse Image Block

A sparse block in a sparse partially-resident image. In addition to the sparse block size in bytes, sparse image blocks have a corresponding width, height, and depth defining the dimensions of these elements in units of texels or compressed texel blocks, the latter being used in case of sparse images having a block-compressed format.

Sparse Unbound Texel

A texel read from a region of a sparse texture that does not have memory bound to it.

Static Use

An object in a shader is statically used by a shader entry point if any function in the entry point's call tree contains an instruction using the object. A reference in the entry point's interface list does not constitute a static use. Static use is used to constrain the set of descriptors used by a shader entry point.

Storage Buffer

A descriptor type that represents a buffer, and supports reads, writes, and atomics in a shader.

Storage Image

A descriptor type that represents an image view, and supports unfiltered loads, stores, and atomics in a shader.

Storage Texel Buffer

A descriptor type that represents a buffer view, and supports unfiltered, formatted reads, writes, and atomics in a shader.

Subpass

A phase of rendering within a render pass, that reads and writes a subset of the attachments.

Subpass Dependency

An execution and/or memory dependency between two subpasses described as part of render pass creation, and automatically performed between subpasses in a render pass instance. A subpass dependency limits the overlap of execution of the pair of subpasses, and **can** provide guarantees of memory coherence between accesses in the subpasses.

Subpass Description

Lists of attachment indices for input attachments, color attachments, depth/stencil attachment, resolve attachments, depth/stencil resolve, and preserve attachments used by the subpass in a render pass.

Subset (Self-Dependency)

A subset of a self-dependency is a pipeline barrier performed during the subpass of the self-dependency, and whose stage masks and access masks each contain a subset of the bits set in the identically named mask in the self-dependency.

Texel Block

A single addressable element of an image with an uncompressed [VkFormat](#), or a single compressed block of an image with a compressed [VkFormat](#).

Texel Block Size

The size (in bytes) used to store a texel block of a compressed or uncompressed image.

Texel Coordinate System

One of three coordinate systems (normalized, unnormalized, integer) defining how texel coordinates are interpreted in an image or a specific mipmap level of an image.

Tile Image

A per-tile view of a framebuffer attachment. If the [VK_EXT_shader_tile_image](#) extension is enabled, the framebuffer is considered to be divided into tiles.

Timeline Semaphore

A semaphore with a strictly increasing 64-bit unsigned integer payload indicating whether the semaphore is signaled with respect to a particular reference value. Represented by a [VkSemaphore](#) object created with a semaphore type of [VK_SEMAPHORE_TYPE_TIMELINE](#).

Uniform Texel Buffer

A descriptor type that represents a buffer view, and supports unfiltered, formatted, read-only access in a shader.

Uniform Buffer

A descriptor type that represents a buffer, and supports read-only access in a shader.

Units in the Last Place (ULP)

A measure of floating-point error loosely defined as the smallest representable step in a floating-point format near a given value. For the precise definition see [Precision and Operation of SPIR-V instructions](#) or Jean-Michel Muller, “On the definition of ulp(x)”, RR-5504, INRIA. Other sources may also use the term “unit of least precision”.

Unnormalized

A value that is interpreted according to its conventional interpretation, and is not normalized.

User-Defined Variable Interface

A shader entry point's variables with **Input** or **Output** storage class that are not built-in variables.

Vertex Input Attribute

A graphics pipeline resource that produces input values for the vertex shader by reading data from a vertex input binding and converting it to the attribute's format.

Vertex Stream

A vertex stream is where the last **pre-rasterization shader stages** outputs vertex data, which then goes to the rasterizer, is captured to a transform feedback buffer, or both. Geometry shaders **can** emit primitives to multiple independent vertex streams. Each vertex emitted by the geometry shader is directed at one of the vertex streams.

Variable-Sized Descriptor Binding

A descriptor binding whose size will be specified when a descriptor set is allocated using this layout.

Vertex Input Binding

A graphics pipeline resource that is bound to a buffer and includes state that affects addressing calculations within that buffer.

Vertex Input Interface

A vertex shader entry point's variables with **Input** storage class, which receive values from vertex input attributes.

Video Bitstream Buffer

A resource that represents a linear array of data in device memory storing encoded video data. Represented by a **VkBuffer** object.

Video Coding Scope

A series of subsequent commands recorded into a command buffer starting with a **vkCmdBeginVideoCodingKHR** command and ending with a **vkCmdEndVideoCodingKHR** command that encompasses a set of video decode or encode operations.

Video Coding Operations

Any operations recorded into a command buffer within a video coding scope, including video decode and encode operations.

Video Decode Operation

An operation consuming data from a video bitstream buffer and zero or more reference pictures, and producing data to a decode output picture and an optional reconstructed picture.

Video Encode Operation

An operation consuming data from an encode input picture and zero or more reference pictures, and producing data to a video bitstream buffer and an optional reconstructed picture.

Video Picture Resource

A resource that represents a multi-dimensional formatted interpretation of device memory to be used with a video session as a decode output picture, encode input picture, reconstructed picture, and/or reference picture. It **may** contain metadata associated with a particular video session it is used with. Represented by a [VkImage](#) object and referred to using [VkImageView](#) objects created from it.

Video Session

A resource that represents and maintains the state needed to perform video decode or encode operations. Represented by a [VkVideoSessionKHR](#) object.

Video Session Parameters

A resource that stores preprocessed codec-specific parameters used with a compatible video session in video codec operations. Represented by a [VkVideoSessionParametersKHR](#) object.

Video Transcoding

The process of using the outputs of video decoding operations as inputs in video encoding operations.

View Mask

When multiview is enabled, a view mask is a property of a subpass controlling which views the rendering commands are broadcast to.

View Volume

A subspace in homogeneous coordinates, corresponding to post-projection x and y values between -1 and +1, and z values between 0 and +1.

Viewport Transformation

A transformation from normalized device coordinates to framebuffer coordinates, based on a viewport rectangle and depth range.

Visibility Operation

An operation that causes available values to become visible to specified memory accesses.

Visible

A state of values written to memory that allows them to be accessed by a set of operations.

Common Abbreviations

The abbreviations and acronyms defined in this section are sometimes used in the Specification and the API where they are considered clear and commonplace.

Src

Source

Dst

Destination

Min

Minimum

Max

Maximum

Rect

Rectangle

Info

Information

LOD

Level of Detail

Log

Logarithm

ID

Identifier

UUID

Universally Unique Identifier

Op

Operation

R

Red color component

G

Green color component

B

Blue color component

A

Alpha color component

RTZ

Round towards zero

RTE

Round to nearest even

Video-Specific Abbreviations

The following abbreviations and acronyms are used in the context of video decode and encode

operations to refer to commonly used video compression terms in their usual abbreviated form:

AVC

Advanced Video Coding

Bipred

Bidirectional Prediction

CABAC

Context-Adaptive Binary Arithmetic Coding

CAVLC

Context-Adaptive Variable-Length Coding

CBR

Constant Bit Rate

CTB

Coding Tree Block

Diff

Difference

DPB

Decoded Picture Buffer

GOP

Group Of Pictures

HDR

High Dynamic Range

HEVC

High Efficiency Video Coding

HRD

Hypothetical Reference Decoder

IDC

Indicator

IDR

Instantaneous Decoder Refresh

MB

Macroblock

MV

Motion Vector

NALU

Network Abstraction Layer Unit

OBU

Open Bitstream Unit

PCM

Pulse-Code Modulation

Pic

Picture

Pred

Prediction

PPS

Picture Parameter Set

QP

Quantization Parameter

RC

Rate Control

SPS

Sequence Parameter Set

Std

Standard

VBR

Variable Bit Rate

VCL

Video Coding Layer

VPS

Video Parameter Set

Prefixes

Prefixes are used in the API to denote specific semantic meaning of Vulkan names, or as a label to avoid name clashes, and are explained here:

VK/Vk/vk

Vulkan namespace

All types, commands, enumerants and defines in this specification are prefixed with these two characters.

PFN/pfn

Function Pointer

Denotes that a type is a function pointer, or that a variable is of a pointer type.

p

Pointer

Variable is a pointer.

vkCmd

Commands that record commands in command buffers

These API commands do not result in immediate processing on the device. Instead, they record the requested action in a command buffer for execution when the command buffer is submitted to a queue.

s

Structure

Used to denote the `VK_STRUCTURE_TYPE*` member of each structure in `sType`

Appendix I: Credits (Informative)

Vulkan 1.3 is the result of contributions from many people and companies participating in the Khronos Vulkan Working Group, as well as input from the Vulkan Advisory Panel.

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